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Editors

Handbook of Metrology and Applications

Springer

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Dinesh K. Aswal • Sanjay Yadav •
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Handbook of Metrology and Applications

With 829 Figures and 236 Tables

Springer

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Preface

There is a significant infrastructure of services, resources, transportation, and communication networks in today's society, much of which is invisible. Most of the time, their presence and efficient operation are taken for granted, even though they are necessary for daily living. The science of measurement known as metrology is a part of this lesser unnoticed infrastructure. Metrology includes the theoretical and practical aspects of measurement, instrumentation, and related standards. Metrology aims to ensure that measurements are accurate, reliable, and consistent across different systems and contexts.

Metrology plays an important role in all areas of human endeavors and across all the sectors of industry, trade, regulations, legislation, defense, environment, education, science, food, transport, health, and quality of life. It involves the development and maintenance of measurement standards, the calibration of measurement instruments, and the application of measurement techniques to solve problems and improve processes. Metrology is primarily classified into three sub-fields: scientific or fundamental metrology; applied, technical, or industrial metrology; and legal metrology. Scientific metrology focuses on the development of units of measurement, the advancement of measurement techniques, the realization of measurement standards, and the dissemination of traceability from these standards to users in society. Industrial metrology ensures the applicability of measurement devices, their calibration, and quality control while applying measurement to manufacturing and other processes and their use in society. Legal metrology is concerned with ensuring accuracy and uniformity in weights, measures, and weighing and measuring instruments.

This handbook provides comprehensive and up-to-date information on the topics of scientific, industrial, and legal metrology in 15 parts with 94 chapters. These chapters discuss the state-of-the-art review of various metrological aspects of metrology and quality infrastructure, redefinition of SI Units and their implications, applications of time and frequency metrology, certified reference materials, industrial metrology, Industry 4.0, metrology in additive manufacturing, digital transformations in metrology, metrology for advanced communication, environmental metrology, metrology in biomedical engineering, legal metrology and global trade, ionizing radiation metrology, advanced techniques in the evaluation of measurement uncertainty, etc.

The chapters in Part I provide detailed information about the evolution of metrology and quality infrastructure. The ability of a nation to assess its level of conformity depends on its quality infrastructure, which primarily consists of metrology, standardization, and accreditation. Part II discusses the advancements related to the updated International System of Units (SI) system and its implications. The chapters include the scientific and policy details about the historic decision to update the SI by the world scientific community in November 2018 at the 26th meeting of the General Conference on Weights and Measures (CGPM). The basic units are now defined in terms of the constants that characterize the natural world, which are the most reliable referencing methods. The establishment of time and frequency standards, as well as their dissemination, are both covered in the chapters of Part III. Part IV deals with studies related to Certified Reference Materials/*Bharatiya Nirdeshak Dravyas (BNDs)*. The *Bharatiya Nirdeshak Dravya (BND)* is an Indian trademark of Certified Reference Materials (CRMs) produced by CSIR - National Physical Laboratory (CSIR-NPL), India, independently as well as in association with Reference Material Producers (RMPs). The BNDs are the primary standards, which ensure the reliability and comparability of the measurements as a benchmark for the quality assurance achieved through international networking. Part V elaborates the industrial metrology with several chapters on diverse sub-fields of it. Part VI is dedicated to Industry 4.0 and engineering metrology. As a result of the widespread usage of digital technology, the industry has undergone a revolutionary digital transition known as Industry 4.0. Due to its nearly unlimited design freedom, capacity to make individualized parts locally, and effective material usage, additive manufacturing is a rapidly expanding industry that has the potential to spark a revolution in manufacturing. Part VII deals with the advancement in the metrology for additive manufacturing. Part VIII comprises chapters related to digital transformations in metrology. Digital technology is one of the most fascinating trends in society today and is revolutionizing metrology. Part IX conations chapters related to optics in metrology as well as nanometrology. Part X is dedicated to the metrology for advanced communication. This comprises the development and application of cutting-edge communications technologies through the collection of measurements, information, and study of the metrology and comprehension of physical phenomena, material capabilities, and communication systems and components. The studies centered around both the global issues related to climate change and the local environmental challenges for the quality of the air, water, and soil. Part XI includes chapters related to the diverse issues of environmental metrology. The use of metrological evaluations and conformance assessment, particularly where measurements support health, is becoming more and more prominent. Part XII deals with the studies related to metrology in biomedical engineering. Global commerce of commodities and services is essential for maintaining global financial stability, economic progress, and population welfare. Part XIII consists of chapters related to legal metrology and global trades. Part XIV comprises chapters related to radiation metrology. It includes the chapters focused on measuring of radioactivity and the standards for radiation dosimetry. Ionizing radiation has numerous uses in both industry and healthcare. Ionizing radiation may be controlled and used in a way

that doesn't endanger patients, radiation technicians, the general public, or the environment thanks to accurate metrology. Finally, Part XV discusses the research works related to the introduction and various techniques used in the evaluation of measurement uncertainty. These chapters explain the most used methodologies for the evaluation of measurement uncertainty in metrology with practical examples.

In the present handbook. the internationally recognized team of editors adopts a consistent and systematic approach and writing style, including ample cross-reference among topics, offering readers a user-friendly knowledgebase greater than the sum of its parts, perfect for frequent consultation. Moreover, the content of this handbook is highly interdisciplinary in nature, with insights from not only metrology but also engineering, material science, optics, physics, chemistry, bio-medical, and much more. This handbook acts as a single window source and reference to researchers, metrologists, industrialists, university graduates, masters, academicians, administrators, policymakers, regulators, and other stakeholders for a better understanding of metrology and related decision-making. The sections on different topics are each presented differently in a single window. Data are provided in all of the chapters as tables, block diagrams, pictures, layouts, and line diagrams, which keeps the chapters interesting for readers without compromising the technical depth of the material.

The editors would like to express sincere thanks to all the authors who contributed the chapters to this handbook. We gratefully thank all of the reviewers involved in the chapters' peer-review process. We acknowledge the great contribution made by each section editor. We especially thank Professor William D. Phillips, a Noble Laureate, and Professor Peter J. Mohr, the authorities in the field of metrology for their contribution to this handbook. Finally, we thank the Springer team for their steadfast assistance in bringing out this useful handbook.

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Dr. Toshiyuki Takatsuji was born in 1964, graduated from Kobe University in 1998, finished the master course in 1990, and received PhD in 1999. He started his carrier as a scientist from National Research Laboratory of Metrology (NRLM) in 1990. From 1994 to 1996, he stayed in National Measurement Laboratory (NML), Commonwealth Scientific and Industrial Research Organization (CSIRO) of Australia as a visiting scientist. In 2001 NRLM was restructured to National Metrology Institute of Japan (NMIJ), National Institute of Advanced Industrial Science and Technology (AIST). From 2015 to 2020 he was the director of Research Institute for Engineering Measurement (RIEM), NMIJ.

His research interests are on length and dimensional metrology, in particular laser interferometry and coordinate metrology and he has published numerous research papers.

He was awarded the Best Paper Award by Japan Society of Precision Engineering in 2006, The Commendation for Science and Technology by the Minister of Education, Culture, Sports, Science and Technology, Prizes for Science and Technology, Research Category

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Dr. Harish Kumar received the B.Tech. degree in Mechanical and Automation Engineering and the Ph. D. degree in Mechanical Engineering from Guru Gobind Singh Indraprastha University, New Delhi, India, in 2003 and 2015, respectively. His thesis was on design and development of force transducers. He worked at National Physical Laboratory, New Delhi, India, during 2007 to 2017 as Scientist and worked in the force and mass metrology. He has worked in the design and development of Kibble balance. Since 2017, he has been working at the National Institute of Technology Delhi, India, and currently working in additive manufacturing in addition to metrology. He has published more than 100 papers in peer-reviewed journals/conferences of repute.

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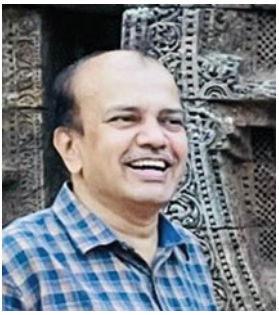

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Part I

Metrology and Quality Infrastructure

International and National Metrology

1

A Brief History of Evolution

Shanay Rab, Meher Wan, and Sanjay Yadav

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Abstract

Metrology, the science of measurement, has evolved significantly throughout human history. Early forms of the measurements were based on human body parts, such as the foot and the hand, which were later standardized in ancient civilizations such as Egypt and Greece. With the invention of more advanced tools, such as the telescope and the microscope, measurements of astronomical and subatomic phenomena became possible. In the eighteenth and nineteenth centuries, the development of the metric system and the use of precision instruments, such as the balances and clocks, have greatly improved the accuracy of the measurements. Today, metrology continues to evolve with the use of computer technology and advanced sensors, allowing for even greater precision and automation in measurement. For that, the International System of Units (SI) plays an important role and is considered as the heart of the metrology and measurement system. Thus, it is of utmost importance to understand how the SI system is evolved historically. This system provides a standardized set of

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units for all types of measurements, making it easier for scientists, engineers, and researchers to communicate and share data across different countries and disciplines. One of the main benefits of the SI system is its consistency and coherence. The SI system also allows for easy updates and improvements. The current SI system is based on fundamental constants of nature; it can easily be updated to reflect new scientific discoveries or technological advancements. The historical development of the measurement system is the primary subject of the present chapter. The chapter focuses on how measurement systems have changed and grown throughout history, from the ancient scriptures to the present-day SI unit system. Various time periods have been taken into consideration as per available literature on measurement systems that show the paradigm shifts in the measurement of different quantities. Also discussed is the current state-of-the-art SI unit system. The current study would provide a succinct report on numerous metrological advancements from the prehistoric to modern eras.

Keywords

Metrology · SI units · Fundamental constant · Measurement · Traceability

Introduction

Humans realized a long ago that they needed to rely on each other as they developed into a civilization, began to cultivate, and lived in communities. Due to the opening up of trade, a measure was definitely needed. Various parts of the body used as points of reference for measurements are clearly evident throughout recorded history. The units that were based on the body parts were arbitrary and unseemly but surely relevant during the period of use. Due to the fact that each individual's body part size is unique, the results of the measurements vary from person to person. This led to issues with both international trade and day-to-day business dealings. The necessity for precise measurement was recognized in order to get over the limitations of using body parts as units and to ensure that the measurement methods remain uniform. To achieve this, the need for the development of universally accepted measurement standards was in focus to improve the reliability and accuracy of the measurements. As a result, measurements performed become coherent and consistent. This resulted in a standardized set of measurement units, known as standard units, for the sake of uniformity (Fanton 2019; Gupta 2009; Rab et al. 2020; Aswal 2020a; Frisinger 2018; Himbert 2009; Fletcher 1968; Shalev 2002).

The International System (SI) of Units for physical quantities is very important in the present era and serves as a tool for any scientific advancement in different countries. Modern metric systems of measurement are almost universally accepted and used in science, engineering, technology, international trade, and commerce. As science progresses, cutting-edge technologies emerge that necessitate measurements

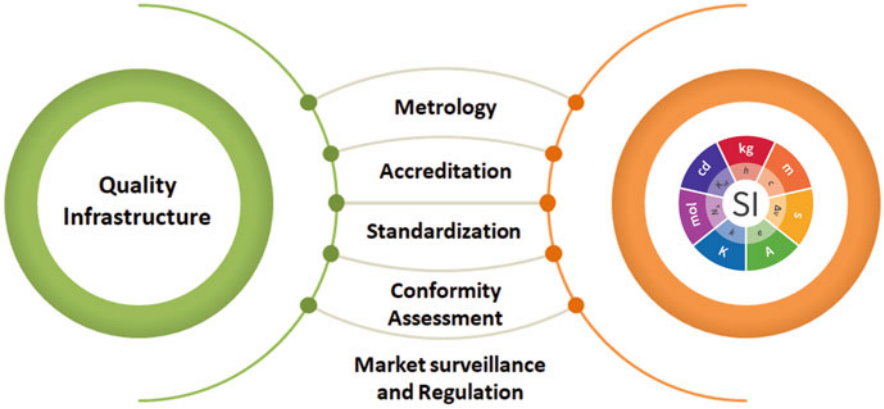

Fig. 1 Linkage of quality infrastructure and SI units

that are ever more exact and accurate. New and improved measurement procedures, measurement standards, and their definitions have been developed as a result of the introduction of new technologies, which have increased the accuracy of the products (Yadav and Aswal 2020; Schlamminger et al. 2020; Stock et al. 2019; Abd Elhady 2020; Göbel and Siegner 2019; Milton and Donnellan 2019).

The foundation of a robust national quality infrastructure (QI), which is necessary for any country's overall growth and success, is profaned by the SI unit system. It consists of the three basic pillars of metrology, standardization, and accreditation that support a strong QI in any nation. Figure 1 shows the basic linkage of QI and SI units. The QI, a fundamental enabling system for providing conformity assessment such as calibration and testing, certification, and inspection, was developed on a technical hierarchy to ensure the accuracy and precision of measurements traceable to SI units (Rab et al. 2020, 2021a, b; Aswal 2020b; ▶ Chap. 5, "Quality Measurements and Relevant Indian Infrastructure"; Kumar and Albashrawi 2022; Mandal et al. 2021; Harmes-Liedtke and Di Matteo 2011).

The advancements in metrology that brought about the current state of the art in the SI unit system are discussed in the section that follows. The early historical method of uniform measurement is discussed in section "Early Historical Evolution of Measurements" of this chapter. The history of measurements was presented through a timeline that concentrated on significant events. The process of uniform measurement is discussed in section "French Revolution and Progress Toward the Uniform Measurement System" in relation to the French Revolution and its impact. At its 26th conference, the CGPM finalized the redefined SI unit system. The reasons for the changes are discussed in section "Current Status of SI Unit System" along with the current status of seven SI units and the 27th CGPM's significant decision on SI unit prefixes. Section "Future of Metrology" provides the future of metrology with the connection of digital transformation and its implications for metrology going forward.

Early Historical Evolution of Measurements

The expansion of agriculture necessitated the establishment of a system of measurements in order to quantify the distribution of crops and the amount of food consumed by families. In order to manage population growth and combat hunger throughout the shift of humankind from nomadic bands to settled agricultural settlements, metrology was essential.

China, India, Egypt, and Mesopotamia (modern Iraq) are the four great antiquity civilizations. All possessed an early understanding of metrology (Maisels 1999; McClellan III and Dorn 2015; Moore and Lewis 2009). The precise measurements of Egyptian pyramids demonstrate a high level of meticulousness even during that period (Procter and Kozak-Holland 2019; Baker and Baker 2001). The earliest known samples from the early Indus Valley Civilization, dating approximately 5000 BCE, are where the history of measurement methods in India began (Rab et al. 2020; Javonillo 2011). Along with Mesopotamia and Egypt, the *Harappa* civilization is regarded as one of the earliest and largest civilizations in recorded history. Standardized weights and measurements were significant innovations of this civilization, which was located along the banks of the *Indus* Valley. It has been discovered that the *Harappan* civilization had well-established length measurement standards. Bricks' length, breadth, and width were all significantly adopted in 4:2:1 regulation ratio. This ratio, which was found to stay constant when brick constructions were being built, indicates that civilization had at least a fundamental understanding of the requirement for a measurement standard in construction. It is considered that *Mohenjo-Daro* was the largest city of the *Harappan* civilization (Jansen 2014). In the city, every building was connected by subterranean drainage pipes as part of sanitation. This perspective demonstrates the significance of measurements in the technological advancements as well as their rise and growth in this civilization. The corbelled drain that serves as the Great Bath's (Vidale 2010; Das 2022) exit at *Mohenjo-Daro* is around 1.8 m tall (Fig. 2), or 108 *Angulas* at a length of 16.764 mm apiece. The reference (Gupta 2009; Rab et al. 2020; Singh 2019; Shrivastava 2017; Sharma et al. 2020a; Kenoyer 2010) gives a list of the various length measurement units used throughout the *Harappan* period.

As previously mentioned, the history of Indian measurement systems indicates that the earliest existing samples belong to the fifth millennium BC, marking the beginning of the early Indus Valley Civilization. Yet, some of the historical statements/mentions on the history of measurement were also witnessed in numerous ancient books like Vedas, Bhagavata Purana, Vishnu Puran, Mahabharata, Suryasidhanta, etc. The ancient Indian scriptures include a wealth of knowledge on the many ways and methods used in ancient India to measure length and time. It was then observed that, interestingly, the speed of light as described in the Vedic literature worked out to be exactly equal to the speed of light as determined by modern measurements when compared to this formerly utilized unit of length measurement used during the Harappa civilization period. Many works of Vedic

Fig. 2 *Mohenjo-daro*'s Great Bath (Public domain)

literature make evident the Yojana's use as a prehistoric length measurement unit (Rab et al. 2020; Gupta 2020a).

In the Indian subcontinent, three important Mughal structures – the Taj Mahal, Moti Masjid, and Jamah Masjid – are described in great detail in the Persian text "*Shah Jahan Nama*." The "*Shah Jahan Nama*" also provides the size of these three structures in "*Gaz*" unit (Andrews 1987). It is important to note that the length of a *Gaz* to be 80.5 cm, which is precisely equal to half of a *Dhanusha*, or 96 *Angulas*, in Taj Mahal and River Front columns of Agra. The "*Angulam*" and its multiples "*Vitasti*" (12 *Angulam*) and "*Dhanusha*" (108 *Angulam*) have been used as the units of measurement from the *Harappan* era up until the premodern era when the Taj Mahal was constructed (Rab et al. 2020; Balasubramaniam 2005, 2009, 2010a). For example, the uniform dimensions in the construction of Mughal structures are shown in Fig. 3.

Further, it is assumed that the weights, which were used in conjunction with a crude balance for commerce, were most likely seeds, beans, or grains. There is no evidence of any attempt to establish a permanent standard before the enormous Khufu Pyramid was built in Egypt around 2900 B.C. Natural standards of length like the hand, span, palm, and digit were employed from the beginning. The first pharaoh to order the establishment of a fixed unit of measurement was Khufu (Moffett et al. 2003). The

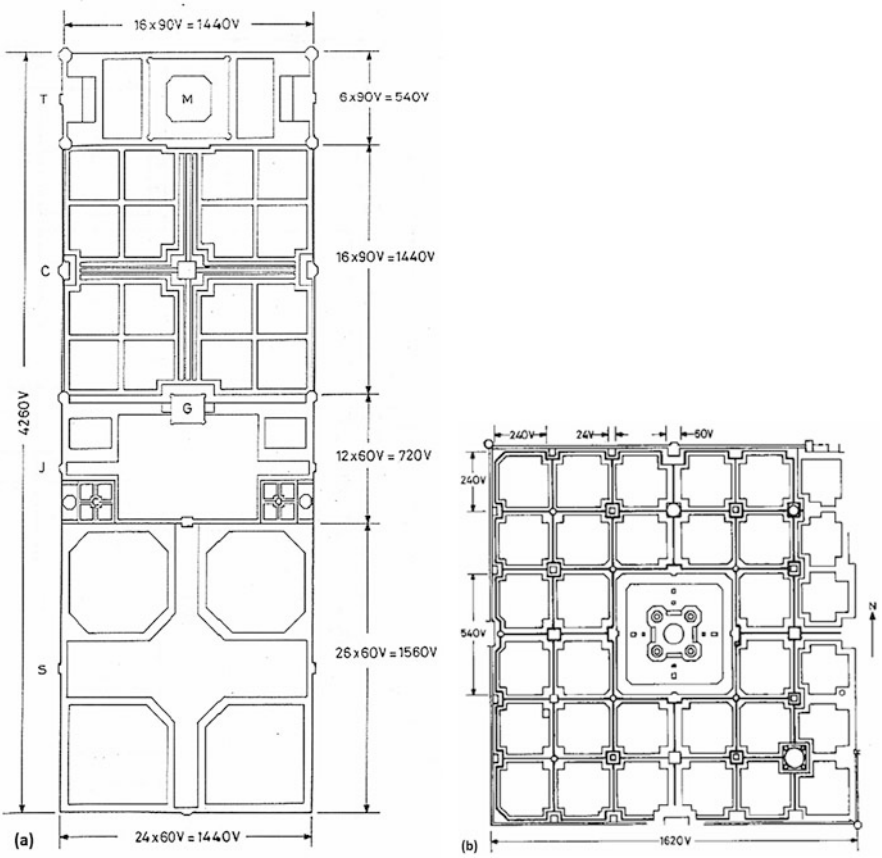

Fig. 3 (**a**) Dimensions of entire complex of Taj Mahal in Agra. (**b**) Dimensions of entire complex of Humayun's tomb at New Delhi; The *Vitasti* (V) equals 12 *Angulams* and each *Angulam* measures 1.763 cm (Balasubramaniam 2010b)

Royal Egyptian Cubit, a standard constructed of black granite, was selected (Hirsch 2013). According to history, it was as long as the forearm and hand of the ruler Pharaoh. For a while, the Egyptians, Greeks, and Romans were only partially successful in creating practical measurement systems and standards that were recognized and followed across their countries (Lewis 2001; Heath 2013; Cuomo 2019). These systems were rather robust and practical despite all of their flaws, but, more importantly, the craftspeople that used them appreciated and accepted them.

As previously indicated, the rise and fall of numerous dynasties and empires occurred in the region over the course of the history of measuring, which spans a period of more than five thousand years. The historical chronology in Table 1 includes significant developments in ancient and medieval metrology.

Table 1 Timeline of the major ancient events related to metrology

Year	Metrological history/Progress	References
~5000 BCE (Before Common Era)	Egypt, Elam (now in Iran), Mesopotamia (currently in Iraq and its environs), and the Indus Valley (now parts of India, Pakistan, and Afghanistan) appear to have employed various standards. The early measurement standards and several measurement instruments are still on display at different museums. Some structures, like the Egyptian pyramids, are still standing and can be measured precisely. Early techniques for determining length relied on using human body parts. It seems that the lengths and widths of fingers, thumbs, hands, hand spans, cubits, and body spans were often used as measurements	Rab et al. (2020), Moore and Lewis (2009), Lyonnet and Dubova (2020), Cooperrider and Gentner (2019)
~3100 to 2181 BCE	During the time known as the Old Kingdom in Egypt, the construction of pyramids began	Bárta (2005), Spence (2000)
~3000 BCE	The people of ancient India are thought to have used length measures such as the *Dhanusha* (or bow suggesting a length based on how far you could shoot an arrow); the *Krosa* (or cow-call suggesting a distance from which you could hear a cow); and the *Yojana* (or stage suggesting a distance of how far you would walk before taking a rest break) The peoples of Egypt, Mesopotamia, the Indus Valley, and in Iran devised the earliest known uniform weights and measures at about this time. Many different cubits of different lengths were used. In addition, the cubit, the length from your elbow to the tip of the middle finger, was divided in different ways	Rab et al. (2020), Gupta (2020a, b), Kher (1965)
~3000 BCE	The Sumerian numeric system was introduced. Living in what is now known as Southern Iraq, the ancient Sumerians had a numeric system with 60 as its base. It is thought to have come from their observations of the stars	Powell Jr (1971), Diakonoff (1983)
~2750 BCE	The cubit is found in ancient Egypt and is considered the first recorded standard of length measurement. The Pharaoh's forearm, measured from the end of his forefinger to the middle of his elbow, served as its standard	Hirsch (2013)

(continued)

Table 1 (continued)

Year	Metrological history/Progress	References
~2600 BCE	The people of the Indus Valley were highly accurate at measuring length, mass, volume, and time	Rab et al. (2020), Jarrige and Meadow (1980)
~2500 BCE	The great pyramid of Giza was built in Egypt. Along each side of the base, 440 royal cubits (about 230 meters) were used to construct this pyramid	Bartlett (2014), Herz-Fischler (2009)
~1800 BCE	At this period, water clocks (clepsydra) were also in use in Mesopotamia. The Babylonians, Assyrians, and Persians were the Sumerian civilization's successors, and they all continued to utilize a variety of standards	Brown et al. (1999), Woods and Woods (2011)
~1766 to 1122 BCE	During the Shang Dynasty, China adopted a decimal-place number system. Ten was written as ten-blank, eleven as ten-one, and zero was left as a blank space for numbers like 405, which was represented as four-blank-five in Chinese characters	Lay-Yong (1986), Lay-Yong and Tian-Se (1986)
~1500 BCE	Egyptians used sundials to keep track of the position and length of the shadows cast on a defined circular surface, which allowed them to measure the passing of time	Sloley (1931), Remijsen (2021)
~800 BCE	The foot was used as a unit of measurement by the ancient Greeks and Romans. In Greece, its size might range from 270–350 mm. An average Roman foot typically measures about 295.7 mm, but in the provinces, a larger length of 334 mm is employed. Today's foot measures 304.8 mm, or 12 inches, which is slightly longer than the original Roman foot	Parvis (2018), Stone (2014)
~425 BCE	Greek philosopher Plato studied under Socrates and instructed Aristotle. All three laid the foundation for our current conceptions of reasoning, philosophy, and mathematics, many of which are frequently used in metrology	Peters (1967), Koutsoyiannis and Mamassis (2021), Dossey (1992)
~321 BCE to 185	The Arthashastra contains extensive documentation on the foundations of legal metrology and other measurement systems	Rab et al. (2020), AM (2003), Rahman et al. (2014)
~300 BCE to 185 AD	One of the longest-reigning dynasties in history was the Chola, a Tamil thalassocratic empire in Southern India. They have well-established water storage	Datta-Ray (2009), Raj et al. (2017)

(continued)

Table 1 (continued)

Year	Metrological history/Progress	References
	tanks, canals, and shipping technology for trade and commerce	
~250 BCE	Archimedes of Syracuse wrote that the earth revolved around the sun. Archimedes is considered to have invented astronomical devices which could identify the positions and motions of the sun, moon, and planets	Pfeffer (n.d.), Wright (2017)
~221 BCE	Chinese weights and measurements were defined by Shih Huang Ti. As part of a plan to unite the country, Emperor Shi Huang Ti intended to implement a significant change in Chinese measurement	Martzloff (2007)
~200 BCE	Before 200 BCE, the Indian mathematician Brahmagupta is credited with inventing "0" in the *Bakhshali* Manuscript. The word "zero" comes from the Sanskrit word *Shunya*, which means "empty" or "blank." By Arab writers, this was rendered as "sifr (empty)," then by Latin writers as "zephirum," and lastly by English speakers as "zero" or "cypher"	Kaye (1919), Puttaswamy (2012), Dutta (2002), Mattessich (1998)
~9 CE (Common Era)	China's measurements were standardized by Emperor Wang-Mang. These are referred to as "the Wang-Mang good measures"	Wang (1996), Hegesh (2020)
~40 CE	A device that Heron of Alexandria described as a "*Dioptra*" may have been invented around 150 BCE. The word "*Dioptra*" is a Greek word that means "to see through." A *Dioptra*, the predecessor of the modern theodolite, was used to measure angles precisely	Papadopoulos (2007)
~830 CE	In his book *Kitab fial-jabrw'al-Muqabala*, the Arab mathematician Abu Ja'far Muhammad ibn Musa al-Khwarizmi discussed the use of 0 (zero). From that term, the area of mathematics where symbols are used to establish fundamental mathematical ideas is known as "algebra"	Sen and Agarwal (2015)
~1196 CE	The first record of a standardized unit of measurement was established during an "*Assize of Measures*" that was declared by King Richard I of England	Scorgie (1997), Williams (2020a)
~1215 CE	The Barons of the English King John, sometimes known as "Lackland," made him sign a contract known as the Magna	McKechnie (2022)

(continued)

Table 1 (continued)

Year	Metrological history/Progress	References
	Carta, which included a provision for standard weights and measures	
~1543 CE	The solar system as proposed by Nicolas Copernicus, with the sun at the center and the planets revolving around it, was published	Zielinska (2007), Sagan (1975)
~1636 CE	In order to determine the speed of sound, Marin Mersenne measured how long it took an echo to travel a predetermined distance. His sound speed was within 10% of the current standard value	Finn (1964), Lindsay (1966)
~1643 CE	The mercury barometer was invented by Evangelista Torricelli	Karwatka (2015)
~1672 CE	Regarding the nature of light and color, Isaac Newton published new theories. When two flat pieces of glass were pressed together, he observed circular bands of rainbow-like colors. It became known as Newton's rings	Naughtin (2012), Greated (2011)
~1687 CE	*Philosophiae Naturalis Principia Mathematica*, also known as *Principia Mathematica*, is a work by Sir Isaac Newton that was published by the Royal Society	Newton (1833), Bussotti and Pisano (2014)
~1742 CE	Anders Celsius built his Celsius thermometer for his meteorological measurements, using 0 °C for the boiling point of water and 100 °C for the freezing point	Beckman (1997)
~1744 CE	Swedish naturalist, Carl Linnaeus suggested reversing the temperature scale of Anders Celsius so that 0 degree represented the freezing point of water (273.15 K) and 100 degrees the boiling point (373.15 K). The centigrade temperature scale was developed from this, and it gradually spread over the globe	Naughtin (2012), Clarke (2017)
~1750	The Industrial Revolution begins. Every facet of daily life is altered by the Industrial Revolution, which transforms rural cultures into ones that are dominated by large-scale industry and urbanization. These advancements are driven by significant coal and iron ore resources, which offer an alternate source of energy to traditional human power. In 1713, the first practical steam engine is developed; by the turn of the century, upgraded versions of the engine are powering	Hill (2018), Hills (1993)

(continued)

Table 1 (continued)

Year	Metrological history/Progress	References
	equipment, trains, and ships. Together with the development of the telegraph, these advancements in transportation have shrunk the planet and sped up communication. Globalization starts, and with it the demand for trustworthy, precise measurement increases	
1790s	The metric system was first implemented during the French Revolution	Frey et al. (2004), Hallerberg (1973)

French Revolution and Progress Toward the Uniform Measurement System

The French Revolution made an impact in the measurement and metrology fields for social and ethical reasons. According to the Republic's motto and the aspirations of the populace as recorded in the *cahiers de doléances* (peoples' claim books), the unit system should be distinct and equal for all. The planned reforms were successful in the majority of areas, and eventually other nations, first in Europe and then beyond, copied them (Hunt 1840). Figure 4 shows the graphical illustration used during the late 1800s to represent the various standards for measurement. During the Revolution in 1789, various quantities connected with each unit might vary from town to town and even from trade to trade; it has been claimed that there were up to a quarter of a million alternative definitions for the 800 or so units of measure in use in France on the eve of the Revolution. Many traders opted to use their own measuring tools, which opened the door to fraud and hampered trade and industry. Local entrenched interests encouraged these discrepancies, which hampered trade and taxation.

The *Académie des sciences* appointed eminent French scientists to explore weights and measures in 1790 (Crosland 2005). After examining numerous options, scientists came up with a number of suggestions for a new system of weights and measures throughout the course of the following year. They proposed that the definition of a unit of weight is a cube of water whose dimension was a decimal fraction of the unit of length and that the definition of a unit of length is based on a fractional arc of a quadrant of the earth's meridian. On March 1791, the French Assembly approved the proposals. Two explorers, Pierre Méchain and Jean Baptiste-Delambre, were given the task of measuring this distance – the earth's quarter circumference – in 1792. After more than 6 years, the meter was established in 1798 (Sharma et al. 2020a; Murdin 2009). The initial step in the establishment of the current International System of Units was the deposit of two platinum standards, one for the kilogram and one for the meter, at the *Archives de la République* in Paris on June 22, 1799 (Fischer and Ullrich 2016; Nelson 1981).

Napoleon III authorized the formation of an international scientific commission in 1869 to spread the new metric measurement system and enable measurement comparisons across nations. The French government extended an invitation for

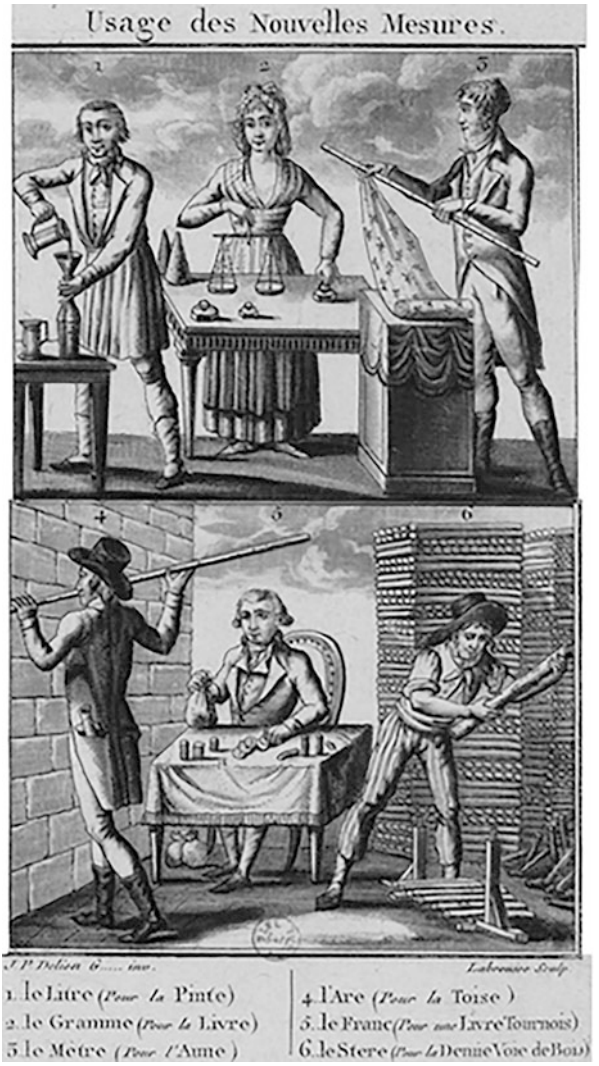

Fig. 4 Illustration of uniform measurement during the French Revolution (Public domain)

nations to join the panel later that year. In 1870, the newly established International Metre Commission held its inaugural meeting. This community ultimately resulted in 17 countries signing the Metre Convention on May 20, 1875. The meter and kilogram, which were determined by the physical artifacts that had been made, and the second, which would be based on astronomical time, were the three units that were agreed upon, regardless of the fact that the agreement was termed the "Metre Convention" (Quinn 2004). The first General Conference on Weights and Measures (CGPM) delegates approved new international prototypes of the meter and kilogram in 1889 (Quinn 2017a; Gupta 2020b; Stock 2018). The time-to-time progress of world metrology is listed in Table 2.

Table 2 Major development in metrology after the French Revolution

Year	Progress in metrology	References
1790s	The meter and kilogram were the only unit of measurement used when the metric system was originally introduced during the French Revolution	Quinn (2004), Smeaton and Ely (2000)
1791	The French Assembly authorized a survey between Dunkirk and Barcelona to determine the length of the meridian and endorsed the suggested guidelines for the new decimal system of measurement made by the expert committee	Williams (2014), Quinn (2011)
1792	The units of length, area, capacity, and mass were given the designations meter, square, liter, and grave, respectively, by the expert's committee	Quinn (2017a)
1795	The eight SI prefixes that were officially adopted are deca, hecto, kilo, myria, deci, centi, milli, and myrio, derived from Greek and Latin numbers. Initially, all were represented by lowercase symbols	Brown (2022, 2019)
1799	The final standard *mètre des Archives* and *kilogramme des Archives* were deposited in the Archives nationales after Pierre Méchain and Jean-Baptiste Delambre finished the meridian survey	Smeaton and Ely (2000), Swindells (1975)
1830s	The principles of a coherent system based on length, mass, and time were established by Carl Friedrich Gauss	Williams (2020b)
1860s	The demand for a coherent system of units comprising base units and derived units was developed by a group working under the auspices of the British Association for the Advancement of Science (BAAS)	Nye (1999)
1874	The CGS system, based on the three mechanical units of the centimeter, gram, and second, and using prefixes ranging from micro to mega to indicate decimal submultiples and multiples, was introduced by the BAAS	Quinn (2001)
1875	The signing of the Metre Convention. At the Pavillon de Breteuil in Sèvres, the International Bureau of Weights and Measures (BIPM) was established, officially initiating the development of new metric prototypes. The obligation for comparing the kilogram and meter to established prototypes was transferred from French to international management by the Treaty of the Metre. The convention originally included standards for only meter and kilogram	Quinn (2017b)
1889	First General Conference on Weights and Measures (CGPM) approved the eight prefixes (deca, hecto, kilo, myria, deci, centi, milli, and myrio) for use	Cardarelli (2003a)
1889	The British company Johnson, Matthey & Co. produced a set of 30 prototype meters and 40 prototype kilograms, which were approved by the CGPM. In each case, the prototypes were built of an alloy consisting of 90% platinum and 10% iridium	Grozier et al. (2017), Ehtesham et al. (2020)

(continued)

Table 2 (continued)

Year	Progress in metrology	References
1921	All physical quantities, including electrical units as originally specified in 1893, are now covered by the Treaty	Nelson (1981)
1946	The magnetic constant, μ_0, also referred to as the permeability of free space, was approved to be fixed exactly as 4×10^{-7} N/A^2 in the new definition of the ampere, which was being used before as mechanical units for measuring length, mass, and time	Goldfarb (2018)
1948	The International Committee for Weights and Measures (CIPM) was commissioned to conduct an international study of the measurement needs of the scientific, technical, and academia by the International Union of Pure and Applied Physics (IUPAP) and by the French Government at the Ninth General Conference on Weights and Measures (CGPM). They also asked the CIPM to make recommendations for a single practical system of units of measurement, suitable for adoption by all countries adhering to the Metre Convention	Stock et al. (2019)
1954	The 10th CGPM included two more base quantities – temperature and luminous intensity – making a total of six base quantities. Electric current was identified as the fourth basis quantity in the practical system of units. The meter, kilogram, second, ampere, degree Kelvin (later renamed kelvin), and candela were suggested as the six base units	Yadav and Aswal (2020), Newell (2014)
1960	The International System of Units, also known as SI, was given as the system's name by the 11th CGPM. The SI is also referred to as "the modern metric system" by the BIPM	Newell and Tiesinga (2019), Petrozzi (2013)
1960	In the SI system, two prefixes were made obsolete (myria and myrio) and six were added, including three for forming multiples (mega, giga, and tera) and three for forming submultiples (micro, nano, and pico)	Petrozzi (2013), Foster (2010)
1964	In the SI system, two prefixes for forming submultiples were added (femto and atto), creating a situation where there were *more* prefixes for small than large quantities	El-Basheer and Mohamed (n.d.)
1971	The 14th CGPM expanded the definition of SI by including a seventh base quantity, the amount of substance represented by the mole	Quinn (2001), Liebisch et al. (2019)
1975	The 15th CGPM introduced the grey (symbol Gy) for ionizing radiation and the becquerel (symbol Bq) for "activity referred to a radionuclide" to the list of named derived units	Newell and Tiesinga (2019)
1975	Two prefixes for forming multiples were added (peta and exa) in the SI unit system	Brown (2022)
1979	The sievert (Sv) was added to the list of named derived units by the 16th CGPM as the unit of dose equivalent in order to distinguish between "absorbed dose" and "dose equivalent"	Glavič (2021), Allisy-Roberts (2005)

(continued)

Table 2 (continued)

Year	Progress in metrology	References
1983	The definition of meter was changed at the 17th CGPM. The length that light would have traveled in a vacuum over a time span of 1/299,792,458 s was predicated on the exact value of the speed of light, c = 299,792,458 m/s	Giacomo (1984, 1983)
1991	Four prefixes were added to the SI unit system. Two for forming multiples (zetta and yotta) and two for forming submultiples (zepto and yocto)	Cardarelli (2003b)
1999	Catalytic activity's symbol kat was added to the list of named derived units in the 21st CGPM	Dybkaer (2000)
2007	In order to make the switch from explicit unit definitions to explicit constant definitions in the 23rd CGPM meeting, the CGPM recommended that the CIPM keep looking into ways to provide exact fixed values for physical constants of nature that could then be used in the definitions of units of measure in place of the International Prototype of Kilogram (IPK)	Milton et al. (2014), Cabiati and Bich (2009)
2018	The redefinitions of SI units have been adopted at the 26th CGPM. The redefined kilogram, the ampere, the mole, and the kelvin in terms of fundamental physical constants	Stock et al. (2019)
2019	Outcome of the CGPM at its 26th meeting in November 2018 has come into force from 20 May 2019, i.e., World Metrology Day	Davis (2019), Quinn (2019)
2022	Four prefixes in SI units were added in the 27th CGPM. Two forming multiples (ronna and quetta) and two for forming submultiples (ronto and quecto) are included in the family of SI unit prefixes	Brown and Milton (2023)

The creation of the decimal metric system during the French Revolution, as well as the subsequent deposit of two platinum standards representing the meter and kilogram in the *Archives de la République* in Paris on June 22, 1799, can be considered the first step in the development of the modern International System of Units. Since the SI is not static in nature, but it evolves several developments in the science of measurement, some decisions have been abrogated or modified; others have been clarified by additions. The development of the SI unit system and the current state of the art of the seven base units are discussed in the subsequent sections. The timeline diagram (Fig. 5) shows the decisions of the CGPM and CIPM in chronological order regarding the development of SI.

Current Status of SI Unit System

As described earlier, modern metrology has its roots in the French Revolution and was initially introduced in 1960 at the 11th CGPM meeting with the redefinition of the SI unit of length. Since then, numerous important and groundbreaking efforts have gone into establishing the SI unit system and relating it to really invariant

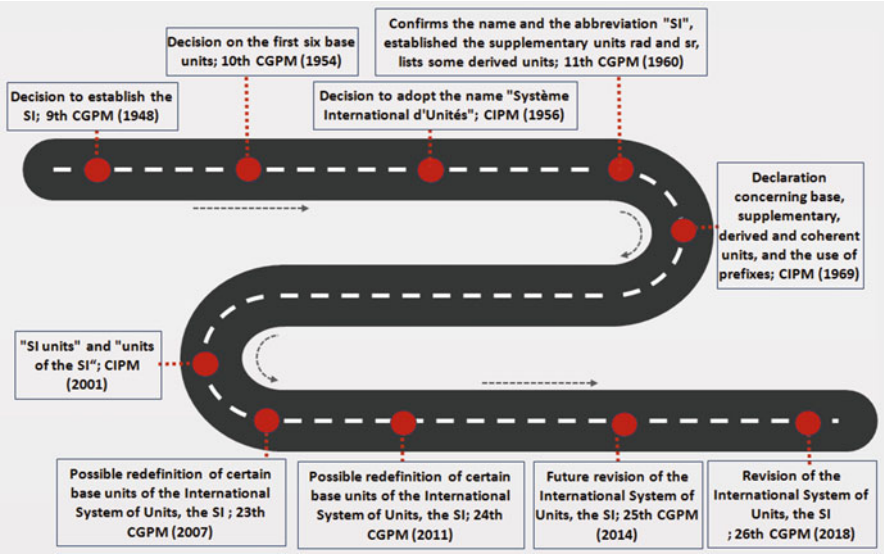

Fig. 5 Timeline development of SI unit system

quantities expressed in terms of natural constants or fundamental constants. In accordance with seven fundamental physical quantities, the SI chooses seven units to act as its base units. The base units are a preferred set for describing or analyzing the relationships between units in the SI because all SI units may be represented in terms of them. The system supports an infinite number of additional units, referred to as derived units, which are always expressed as products of base unit powers, optionally with a nontrivial numerical multiplier. The unit is referred to as a coherent derived unit when that multiplier is 1. A coherent system of units is created by the SI's base and coherent derived units (the set of coherent SI units). The special names and symbols have been given to 22 coherent derived units. The 22 derived units with unique names and symbols can be used in conjunction with the seven base units to represent other derived units that have been chosen to make it easier to measure a variety of quantities.

More notably, the CGPM mandated the CIPM to investigate the adoption of fundamental constants as the basis for all units of measure, rather than the artifacts in use at the time, at its 23rd meeting in 2007. Since then, a considerable measure of work has been conducted around the world to redefine the SI unit system in terms of fundamental constants. The journey toward revision of SI unit is fully documented in the resolution, which was overwhelmingly adopted by CGPM attending 60 member states from around the world.

The CGPM approved the SI redefinition during its 26th meeting in November 2018, and implemented it with effect from 20 May 2019 (World Metrology Day). Table 3 contains all seven definitions for the SI unit system's base units. Various articles provide a detailed overview of how these base units were formed (Rab et al.

Table 3 Definition of SI base units on the basis of physical constants

Base quantity	Name	Symbol	Definition
Length	Meter	m	It is defined by taking the fixed numerical value of the speed of light in vacuum c to be 299,792,458 when expressed in the unit $m \cdot s^{-1}$, where the second is defined in terms of the cesium frequency Δv_{Cs}
Mass	Kilogram	kg	It is defined by taking the fixed numerical value of the Planck constant h to be $6.626,070,15 \times 10^{-34}$ when expressed in the unit J·s, which is equal to $kg \cdot m^2 \cdot s^{-1}$, where the meter and the second are defined in terms of c and Δv_{Cs}
Time	Second	s	It is defined by taking the fixed numerical value of the cesium frequency Δv_{Cs}, the unperturbed ground state hyperfine transition frequency of the cesium-133 atom, to be 9,192,631,770 when expressed in the unit Hz, which is equal to s^{-1}
Electric current	Ampere	A	It is defined by taking the fixed numerical value of the elementary charge e to be $1.602,176,634 \times 10^{-19}$ when expressed in unit C, which is equal to A·s, where the second is defined in terms of Δv_{Cs}
Temperature	Kelvin	K	It is defined by taking the fixed numerical value of the Boltzmann constant k to be $1.380,649 \times 10^{-23}$ when expressed in the unit $J \cdot K^{-1}$, which is equal to $kg \cdot m^2 \cdot s^{-2} \cdot K^{-1}$, where the kilogram, meter, and second are defined in terms of h, c, and Δv_{Cs}
Amount of substance	Mole	mol	One mole contains exactly $6.022,140,76 \times 10^{23}$ elementary entities. This number is the fixed numerical value of the Avogadro constant, N_A, when expressed in the unit mol^{-1} and is called the Avogadro number The amount of substance, in a system, is a measure of the number of specified elementary entities. An elementary entity may be an atom, a molecule, an ion, an electron, or any other particle or a specified group of particles
Luminous intensity	Candela	cd	It is defined by taking the fixed numerical value of the luminous efficacy of monochromatic radiation of frequency 540×10^{12} Hz, K_{cd}, to be 683 when expressed in the unit $lm \cdot W^{-1}$, which is equal to $cd.sr.W^{-1}$, or $cd.sr.kg^{-1}.m^{-2}.s^3$, where the kilogram, meter, and second are defined in terms of h, c, and Δv_{Cs}

2020; Yadav and Aswal 2020; Stock et al. 2019). The new definitions aimed to strengthen the SI without changing the value of any units, ensuring that measurements were consistent. Figure 6 depicts the representation of these constants in SI base units. In November 2022, the world scientific community voted to expand the range of prefixes used within the International System of Units, meaning that four new prefixes will now be used to express measurements worldwide. The progress of the SI unit prefixes is listed in Table 2.

Fig. 6 SI logo, Credit: BIPM reproduced from https://www.bipm.org/en/si-download-area/graphics-files.html under a Creative Commons Attribution 4.0 (Full terms available at https://creativecommons.org/licenses/by-nd/4.0/)

Future of Metrology

As discussed, metrology is an ever-evolving field of science. It has come along far from human body parts–based measurements to fundamental constants-based SI units. However, the metrology fraternity through CIPM keeps upgrading the technology behind metrology. During the 27th meeting of CGPM, it was decided to redefine the second. The CGPM has noted that several National Metrological Institutes (NMIs) have already surpassed the accuracy defined by the present realization of second through the unperturbed ground state hyperfine transition frequency of cesium-133 atom. Several NMIs have already achieved better accuracy in achieving the realization of second by a factor of up to 100 with the help of optical frequency standards. The CGPM invited the member states to support research activities toward the adoption of the new definition of second (Sharma et al. 2020b; ▶ Chap. 18, "Atomic Frequency Standards" Chang et al. 2022).

The CGPM observes that the digital transformation of metrology is the need of the hour. The word "digital metrology" can be used to describe potential advances and substantial modifications to existing metrological services, as well as new metrological services necessitated by digital trends. With the help of digitalization, intelligent data-driven solutions will manage responsive, productive, and efficient production systems, enabling the circular economy. Further, virtual verification and calibration will be possible, thanks to digital calibration certificates (DCCs) stored on global distributed ledgers and paired with digital and machine-readable standards and regulations. Several examples such as process analytics, manufacturing process control, quick demand forecasting, and reliable data through digital certification will all be enabled through measurement (Chang et al. 2022; Rab et al. 2022; Mustapää et al. 2020; Thiel 2018; Eichstädt et al. 2021; Garg et al. 2021).

Artificial intelligence (AI) and machine learning are also supposed to help improve the metrological process. AI and machine learning can help to analyze large sets of data to identify patterns and trends. AI and machine learning can be used

to automate measurements, reducing the need for human intervention and increasing efficiency. This can be useful in identifying sources of error and in improving the accuracy and precision of measurements. AI will be useful in the predictive maintenance of metrological processes. It can help to predict when equipment used for metrology is likely to fail or require maintenance, allowing for proactive maintenance and reducing downtime. During the calibration process, AI and machine learning can help to optimize calibration procedures and reduce measurement uncertainty which can subsequently improve the reliability and accuracy of measurements. It will also impact real-time quality control, identifying defects and errors in products and processes. However, there are several challenges which are important to address in the employment of AI and machine learning in metrological processes. To train the AI and machine learning models, very high quality data is required as accuracy and reliability of AI or machine learning models depend on the primary data through which these are trained. Inaccurate or incomplete data can lead to incorrect predictions and unreliable results. AI and machine learning programs may provide accurate results but it is many times difficult to understand and explain the process of these models to achieve these results. Explainable AI (XAI) techniques need to be developed especially for metrological operations to avoid confusion and maintain transparency. XAI must be the prioritized area of research for NMIs and individuals. Regulatory compliances in the case of critical industries such as healthcare are very critical and challenging for AI and machine learning–based metrological techniques as failure in such industrial processes may lead to loss of life and significant ethical considerations (Eichstädt et al. 2021; Cunha and Santos 2020; Temenos et al. 2022).

The potential for design and maintenance is endless due to cloud computing and digital metrology. Digital measurement equipment is much simpler to use and calibrate. But soon these devices will be able to communicate with one another across entire manufacturing networks. This enables workflows to audit themselves, finding, isolating, and repairing errors, or foreseeing them and preventing them from happening in the first place. Networks that are fully digitalized could automate performance reports and incorporate their insights into the production process. There are several NMIs which are leading the efforts in the direction of the digitalization of metrology (Toro and Lehmann 2021). National Physical Laboratory-UK (NPL-UK) is attempting to embed measurement data into a trusted digital infrastructure that will ensure future traceability chains (Cheney et al. 2021). It is developing techniques to improve data analytics for analyzing the experimental data with the help of statistical learning, machine learning, and clustering. It is working in the field of software verification and validation to process the generation of reference data, design of performance metrics, and automated self-verification of the software. It is also working to develop standards and platforms for quality and provenance for ensuring high-quality reproducibility of it. Overall, NPL-UK is trying to update the available metrological services and creating new techniques for the generation of high-quality data and its assessment of quality. Physikalisch-Technische Bundesanstalt (PTB), Germany, is working for digital transformation of metrological processes and services through big data, communication systems,

simulations, and virtual systems (Thiel 2018). The Federal Institute of Metrology METAS of Switzerland is attempting to create various technologies and platforms toward the digitalization of metrology (Barbosa et al. 2022a). It encompasses the approach digitalization of metrology through metrology for digitalization and vice versa. The National Institute of Standards and Technology (NIST), USA, is taking up the challenge to achieve digital transformation with legacy components and cybersecurity components in the digitalization of metrology (Barbosa et al. 2022b; Barcik et al. 2022). Several other NMIs are taking up the new challenges in the field of metrology presented by the twenty-first-century circumstances. The NMIs aim to shorten the traceability chains and reduce uncertainty in measurements. It is being attempted to embed the traceability of SI directly into measuring instruments including in situ calibration of sensors and local standards.

Technological, social, and industrial progress changes ongoing demands on metrology and measurement systems. To have faith in emerging technologies, stimulate mass use, and ensure a safe rollout, measuring infrastructure must be established in tandem with these advancements. Metrology will become a fundamental step inside industrial processes in the future for inspection and quality control operations. Traditional metrology techniques are not available online, and as a result, involve removing pieces from a production line and transporting them to a quality control department for inspection. The goal is to find quality assurance flaws. As a result, framing metrology as a value-adding process is not always straightforward. This is beginning to change as a result of Industry 4.0 and 5.0 manufacturing technologies combined with automated, 3D, noncontact in-line metrology systems. In-line metrology solutions work alongside the production line as it runs. This allows items to be measured and inspected without the need for human intervention. It will not only rectify any form of default, but it will also assess the impact of any changes on adjacent areas. As a result, metrology will provide flawless control over collateral harm. Furthermore, with the rapid advancement in technologies, metrology has grown to handle vast measuring requirements, including an assortment of sensors to make your production processes more complete.

Metrology will support automated decision-making even in critical cases. Through the digitalization of metrology, decisions made by machine learning algorithms and artificial intelligence will be supported by metrology directly. Even in safety-critical situations, automated decisions will be made accurately with the help of agile and responsive regulation of metrological operations. In future, metrology will also boost confidence during the process of understanding complex and large systems, in general.

Conclusion

Originally, metrology was shaped to ensure that human economic operations such as commerce and exchanges were secure. Because of established measurement standards, humanity has come together. The existing level of metrological performance much exceeds the industry's success standards; further, development is now more

urgently required to promote scientific progress. Specific branches of research, such as geography, astrophysics, medical treatment, and so on, now demand even more precision; all of these fields contribute to human well-being.

The study gives historical viewpoints on the development and expansion of the measurement system. The study emphasizes the development of measurement systems across different eras in ancient times, the legal issues of metrology, the growth of metrological frameworks around the world, and the current status of SI units. The chapter also discusses the need and potential outcomes of the development of a framework for digital metrology.

Our drive to comprehend the world has led us to develop instruments that have benefited humanity for decades. We will only continue to develop and produce increasingly more complicated measures as technology advances. It is necessary to notice that measurement history is as complicated as human history. Every major civilization understood the necessity of a standardized measurement for continuing development. Everything in the world today is measurable, from the speed of the universe's expansion to the tiniest quantum particles that make up all matter. Significant measuring methods, tools, and terminology are employed in disciplines that did not previously exist due to new developments in technology and processing capabilities. Since the advent of the computer, databases of measurements have been compared and processed to identify trends that enable quick and precise forecast. The exponential rate of technological progress we are currently seeing points to a future that is virtually beyond comprehension.

Disclaimer The discussion focused on the various stages of global metrology evolution, which were derived from publicly available data or other reliable sources. Measurement systems have altered drastically from the ancient past to the contemporary metric system. Despite the writers' best efforts to include references and citations from old historical epics, stories, research papers, and publications published in several local languages, spelling and translation errors are inevitable, but they are not done on purpose. Some of the historical information in this chapter is provided for the sole purpose of accumulating knowledge and information, and should not be used or considered in the resolution of any legal difficulties.

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Wake-Up India to Build Self-Reliant Metrology Structure for Modern India

2

R. Sadananda Murthy

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Abstract

Metrology, the science of measurement, is unimaginable without correct measurements in modern era. Modern society cannot function without metrology. A world without measurement tools means there would be no modern medicine, healthcare, pharmacology, or surgery among many other critical services. Metrology is not just the routine of making measurements; it's also about the infrastructure that ensures the accuracy of measurement. It establishes a common understanding of units and processes. Metrology also covers the aspects of accuracy, precision, repeatability, and measurement uncertainties. Quality infrastructure involves metrology, standardization, testing, and calibration. In this aspect, quality management certifications and accreditation play a major role in conformity and compliance. To achieve this, India can capitalize on its long history of scientific excellence and progress in research to implement science education across all layers of society. Due to the rapid industrialization of the world over the past decades, demand for specialists in the field of metrology has been rising very quickly. The supply is not able to keep pace with the demand. If we study the history of world economics, the dramatic relationship between technological invention and economic growth is apparent. The present chapter briefly discusses the role of metrology in modern India. It is clear that economic growth cannot happen without technological innovation and invention. The convergence approach requires the involvement of all stakeholders like government agencies, national institutions, educational institutions, manufacturers, R&D establishments, calibration and testing laboratories, end users, private and public undertakings, and the relevant public on a single platform to identify the needs and establish a road to success for encouraging relevant inventions, innovations, and technology development suitable to take India forward. A proper ecosystem for developing a robust metrology structure and establishments for innovations and inventions in science, technology, and institutions for the promotion of skill and knowledge development in metrology are essential for any economy to progress and compete, survive, and thrive in this competitive progressive world.

Keywords

Measurement · Metrology · Economy · Quality infrastructure · Indian infrastructure

Impact of Measurement on Daily Life

Time, length, area, speed, weight, volume, force, torque, pressure, light, energy, and temperature are just few of the physical properties for which we have defined the accurate measures. Without these measures, we would have great difficulty in conducting our normal lives. Yet we live without paying much special attention to them.

The society in the twenty-first century cannot progress without the indispensable tools of measurement on which our civilization is built. A world without measurement tools means there would be no modern medicine, healthcare, pharmacology, or surgery among many other critical services. In fact, measurement is being used in engineering, construction, and other technical fields, physics, and other sciences as well as psychology, medical, pharmaceutical, chemical, aerospace, etc. For any development requires science and technology, but without metrology there is no science and technology (Gupta et al 2020; Rab et al. 2020; Yadav and Aswal 2020; Aswal 2020a; Rab et al. 2021a).

What Is the Importance of Measurement?

Measurement is a scientific, economic, and social necessity. The results of measurement are used to increase the understanding of a subject, for example, to confirm the design specifications, or to determine the quantity/quality of any commodity, or in control process of any commercial transactions, or to predict future events/developments, etc.

Measurements are key wherever decision-making is involved based on the measurements. If the measurements are not accurate/correct, then it may have grave consequences with respect to money, environment, safety and health of people involved, etc. It is therefore imperative that the measurements on which decision-making is done are totally dependable and reliable (Fanton 2019; Squara et al. 2021).

Measurement Protects People

- Vitally important activities involving public health like dosing of drugs or radiotherapy measurements, food safety, etc. require accurate measuring operations. Similarly, reliability of measuring instruments in intensive care units or operation theatre is highly critical.
- Measurement of quantities like working hours, ambient temperatures, or noise and lighting levels of a professional premises aids in the application of labor laws and rules.
- Measurements also ensure compliance to road safety standards by measuring speed, alcohol levels in blood, vehicle braking efficiency, etc.
- Measurements of quality of air and water help in protection of environment by ensuring that statutory requirements are met.

Measurement Governs Transactions

- All transactions between individuals and companies involve measurements during packing of food and consumables, metering gas consumption of subscribers, counting petrol at the fuel pumps or on pipeline retail agency or bulk weighing transactions, etc.
- Measurement becomes an essential factor between customers and subcontractors. Without reliable measurements, it is not possible to guarantee that the subcontracted parts or items match the customer's requirements.

Measurement Makes Industries to be more Innovative and Competitive

- The more accurately the industry meets the customer's and user requirement defines how competitive the industry is in the market. Meeting this competitiveness is impossible without the measurements which study and meet the customer's expectations. Performance measurements for industrial products play a key role in defining quality metric of the industries.
- Quality is also demonstrated to the customers through industry certifications which itself are based on measurement.
- To be competitive in the market, it is expected that the industry precisely measures and controls the production volumes and performances of tools so that it minimizes the cost of rejections/rework operations.

Measurement Increases Knowledge Base

- Metrology is present at every step in the field of fundamental research. It is used to design the conditions of observations, to build and quality the instruments required for observations, and to qualify the results obtained. In research field, the characterization of force fields, determination of physical constants, and observation of physical phenomena all involve measurements.

Measurement Help Make Decisions

- To continue or end a process.
- To accept or reject an item/product.
- To decide whether to rework or to qualify a design.
- To take corrective actions or not.
- To establish facts, be it scientific or legal.

The measurements become unnecessary if the data from measurements are not utilized in the decision-making process or in establishing the facts. "The more

critical the decision, the more critical the data. The more critical the data, the more critical the measurement."

What Is the Meaning of Measurand?

Measurand is defined as the "particular quantity subject to measurement," Quantity intended to be measured (Kacker 2021).

What Is Not Measurement?

There are some processes seem to be measurement but are not: -

- Comparing two pieces of spring or rod to see which is longer is not a measurement.
- Counting.
- Tests (Yes/No, Pass/Fail).

Where Would the World be Without National and International Standards?

Life in the world would be very difficult if trusted standards, based on measurements, didn't exist. Signs in public places use the same basic symbols wherever we go. We trust our favorite imported or local foods and beverages are safe. Credit cards work almost anywhere in the world. All these are the benefits formulation of standards. Standards established by measurements ensure consistency in terms of quality, ecology, safety, economy, reliability, compatibility, efficiency, interoperability, and effectiveness of goods and services. Standards codify the latest technology and facilitate its transfer.

Relevance of Calibration and Testing Laboratories

The start of the twentieth century brought quality control to the limelight with more emphasis on the quality of the final product, thus enabling customer satisfaction. Implementing ISO 9001 process and procedures and then objectively monitoring the quality of product became significant. To check the quality, different types of equipments were required as per the process requirements. This paved the way for the metrological equipment industry to start spreading its roots.

Today this industry is linked to almost every other industry like oil and gas, automotive and ancillary, plastics, rubber, steel, food, R&D, cement, textile, construction, chemical, mining, etc. With the rapid progress of the industries, QC is considered the most important department of any organization to accept or reject a

product. Organizations are investing millions buying quality instrument brands to ensure their product satisfies the protocols meeting high-quality standards. With these huge potential requirements, metrological equipment's manufacturers have cemented their products in the market (Aswal 2020a, b; Yadav et al. 2021; Rab et al. 2022a, b).

Metrology, National/International Standards, and Regulations

Metrology is not just the routine of making measurements; it's also about the infrastructure that ensures the accuracy of measurement. It establishes a common understanding of units and processes. Metrology also covers the aspects of accuracy, precision, repeatability, and measurement uncertainties. It also involves traceability or verification with a "Standard" or reference measuring systems. Metrology involves all theoretical and practical aspects of measurement, whatever the measurement uncertainty or the field of application. In this modern world, in view of importance of safety of living beings, safety of environment, safety of products and processes, safety of natural resources and the universe, and safety of economic growth and development is very critical. Various standards and statutory and regulatory guidelines are formulated by various national and international agencies in the field of legal metrology, pharmaceutical, medical, automotive, chemical, defense, etc. which are to be adhered to (Yadav et al. 2020; Garg et al. 2021; Rab et al. 2021b).

What Is Metrology?

Metrology establishes a common understanding of units, crucial in linking human activities, or is the science of measurement, embracing both experimental and theoretical determinations at any level of uncertainty in any field of science and technology. It comprises the administration of measuring instruments, application for units of measurement, and testing procedures which are commonly established in either legislation or documented standards. The purpose is to provide accurate and reliable measurements for trade, health, etc. Measurement is fundamental to almost all human activity, and so it is important that the accuracy of any measurement is fit for its intended purpose.

Measurement can be classified into two parts: direct measurement and indirect measurement. Direct measurement – When measurement is done using direct measurement or modern instruments, it is called direct measurement. Examples include length and weight. Indirect measurement – When measurement is done through a formula or other calculations, it is called indirect measurement. Example includes measuring the distance between two stars. Therefore, it's very important to have knowledge of measurement in today's life. According to James Cattell "*The history of measurement is the history of science.*" Current civilization is inconceivable

without the indispensable tools of measurement on which day-to-day life depends (Porter 2001; Czichos et al. 2011; Rab et al. 2020; Aswal 2020a).

Importance of Metrology in Modern World

The quality aspects of products and services have become forerunner of all activities in industries with the introduction of ISO 9000 and in view of globalization and international competitiveness in trade and commerce. Measurement and calibration play a vital role in meeting the requirement of industries in order to produce quality product and services. Measurement is the basis for the development of trade and economy. Measurements also need to be uniform and internationally compatible in order to eliminate technical barriers to trade and promote international cooperation. The validity of precise and accurate measurements across different stakeholders is essential for trade, efficiency, quality, and safety (Stock et al. 2019; Rab and Yadav 2022). Metrology drives and influences much of what we do or experience in our everyday lives. It is often unnoticed or beyond our awareness. Industry, trade, regulations, quality of life, science, and innovations all rely on metrology considerably. Thus, metrology forms a natural and vital part of everyday life.

Globalization of trade requires that the measurements be traceable, comparable, and mutually acceptable across the globe. This is applicable not only in trade of raw materials and manufactured products but also in all aspects of trade. The world demands for confidence in the results of measurements. Decisions made based on measurement results are increasingly seen as having direct influence on the economy and human safety. Even in the areas of science, the measurements can be validated if it is made through a well-defined system of units and verified equipment. Quality infrastructure involves metrology, standardization, testing, and calibration. In this aspect, quality management certifications and accreditation play a major role in conformity and compliance (Rab et al. 2022c).

Who Is a Metrologist?

Metrologist – with Skill and Knowledge *"Understanding the benefits of having sound knowledge, perspective, and the appropriate skill sets in metrology is just as crucial in today's technological world as it was 50 years ago. The same goes for key decision making based on solid processes, procedures, methods, and data analysis. Taking responsibility and ownership of those decisions makes good metrologists vital to a successful organization's reputation, quality, and cost containment." – Keith Bevan.*

Measurement scientist or metrologist is one who studies and practices metrology. He is capable of finding and using measurement science resources. Some metrologists are involved in designing of measurement instruments or systems while others perform calibration of those instruments. Still others may do basic research into the underlying principles of metrology. Metrologists can help in increasing business

productivity and efficiency. They can also improve reliability of a process for internal/external customer satisfaction, to achieve and maintain compliance. They can also help in cost saving through advances in computer technology and by minimizing human errors. Considering the large verity of people in the field, you can imagine how broad the subject matter is. Metrology after all borrows the concepts from physics, mathematics (mostly statistics), chemistry, all kinds of engineering, and even a little biology and medicine at times.

Metrology is everywhere, but metrologists are very few. Curricula in technical fields do not even mention metrology. How do metrologists get education, then? The people who come from diverse backgrounds in engineering, math, and science, and become metrologists, have had very little exposure in the field of metrology. A metrologist with a broad well-rounded education should have a strong knowledge of pure sciences like physics and knowledge of applied statistics. A metrologist should also have specialized knowledge in several engineering areas, the most common being mechanical and dimensional measurements, electrical measurements, and physical areas where measurement of time, mass, temperature, pressure, force, torque, flow, etc. are involved. Not only this, the understanding of more than one field is needed at a time. For example, some physical measurements like mass are strongly influenced by temperature. Therefore, it is important to understand how to measure temperature well in order to measure mass or mass-related parameters.

Finally, one also has to understand specific international and national standards and practices specific to metrology. Comprehending "the guide to expression of uncertainty in measurement," for example, is critical for every professional metrologist. At present, most of metrology education is accomplished through short courses (Fuschino 2021; Priimak and Razina 2021).

Metrology Training Centers

Science and technology are changing so rapidly and irreversibly today. Hence, the fundamental principle for progress must be the convergence between different sciences, technologies, and peoples, while each of them mutually benefiting from the convergence. To achieve this, India can capitalize on its long history of scientific excellence and progress in research to implement science education across all layers of society.

Even though India has a large-scale infrastructure base in science and technology which is fast expanding, it still has a long way to go in realizing its potential in becoming world leader in science and technology. Education in the fields of science and technology is still not at par with international benchmark. Scientific research, education, and innovation are imperative for any nation to achieve continuous development. Measurements are everywhere, but measurement scientists or metrologists with skill and knowledge are very few (Sharma et al. 2020).

Challenges Faced

The laboratory may have the best high accuracy standard equipment that was purchased from reputed manufacturers; the equipment may have traceability to national and international standards and may have the best accommodation and environmental conditions too. The high accuracy standards are not realized unless there is a sound knowledge on basic principles of measurements and influential quantities during calibration or testing, and correct analysis and precautions are taken during measurements. In fact, considerations which are often overlooked, can result in total error of several percent, either in deviation or in measurement uncertainty. "It is like this — a person may own the best race car from the most reputed company but cannot automatically take part in a race until he is qualified enough to participate. Hence, a sound knowledge and expertise is the basics are a necessity in building a credible and reliable laboratory."

With the market evolving rapidly, calibration services of these equipments have also gained importance. Thus, a huge number of calibration and testing laboratories found themselves registered with the NABL accreditation. The accreditation is done as per ISO 17025:2017. Besides other medical laboratories, reference material manufacturers and proficiency test providers are gaining momentum into the industry. Today NABL has accredited more than 6000 different types of laboratories which in itself is a huge network.

This has given rise to many more entrepreneurs entering this market, which was once considered a niche, creating an ecosystem nurturing many families. Our Indian manufacturers have made us proud by taking a share of the international market competing with the multinationals. So, the Indian metrology ecosystem strives in the world and has created a benchmark. Many Indian companies have found their associates, while some have set up their own offices internationally to cater to the growing market as per the organization strategy. Many engineers, senior executives, managers, technical experts, and management staff work in different companies from the metrology industry. Thus, the Indian economy opened itself to a diversified product mix for all types of consumers. Today customers have options like never before right from the higher-end accurate product to lower-end ones. This is decided by the end user as per the application. The overall scenario looks poised for growth in this rapidly evolving industry.

The challenges faced in acquiring skill and knowledge in the field of manufacturing testing equipment and calibration have eventually led to the difficulty in establishment of a ISO17025 accredited laboratory. But there is lack of organized institutions, training centers, technically competent trainers with sound knowledge and skill, effective metrology course materials and books that can be used by workforce at every level, and implementing theory and practice to meet a current day requirement. Upcoming calibration and testing laboratories, as well as the ones that have already been accredited, are facing similar challenges. They are either

unable to acquire sufficiently trained workforce or fulfill the needs of their appointed personnel for upgradation of the required skill and knowledge. Due to rapid industrialization of the world over the past decades, demand for specialists in the field of metrology has been rising very quickly. However, the supply is not able to keep pace with the demand.

Therefore, to meet the ever-growing demand for accurate measurements in line with global requirement, effective quality infrastructure becomes paramount importance in any developing country for robust economic growth. Development of quality infrastructure involves metrology, standardization, testing, and calibration, and quality certification and accreditation are important for assessing conformity. To have an effective quality infrastructure, any country requires technically competent and professionally trained workforce in the fields of metrology, science, and technology.

They should have skill and knowledge (theory and practice) in the field of metrology in the following areas:

(a) Routine operation and maintenance of equipment and conducting calibration and testing activity.
(b) Review and interpretation of results of calibration and testing.
(c) Auditing and assessing in the field of calibration and testing.
(d) Required training for the above workforce.
(e) Development and innovation in the fields of science, technology, and metrology.

The following must be satisfied in order to produce technically valid calibration certificates or test results with the global metrology system:

- Measurements made at one place or country should be acceptable in any other country without technical barriers for free trade, as per the WTO.
- One standard, one test accepted everywhere as per international organization like ILAC, OIML, etc. To produce internationally acceptable measurement and test results depends on the technical competency of the laboratory. In order to achieve technical competency, personnel involved are required to acquire in-depth professional knowledge, expertise, skill, aptitude, and continual training.

Science, Technology, and Engineering of Modern World

It is important to understand the difference between science, technology, and engineering. Science is the methodical study of the natural world, through observation of physical systems, gathering evidence, and performing experimentation. It is concerned with "understanding the fundamental laws of nature and the behaviour of matter and living things." Science explores new knowledge methodically through observation and experimentation. There are many specialized branches of science which deal with different fields of study such as physics, chemistry, biology, or cosmology (Atrens and Atrens 2018; Onalbek et al. 2020; Mehra et al. 2021).

Engineering

Engineering involves the practical application of the principles of science and utilizes technology to create useful products and services for mankind, within economic, environmental, and resource constraints. Engineers specialize in specific branches of engineering such as electronics, electrical, mechanical, chemical, or civil. Engineering metrology is one of the branches of engineering which deals with measurement.

Technology

Technology is the body of knowledge system, process, and artifacts that result from engineering. "Technology" is the application of science to achieve practical results, especially in industry. Technology involves the development of devices, processes, and systems to address society's practical problems.

Impact of Science and Technology on Modern Society

Impact of science and technology in modern society: Science and technology impacts the modern society in wide-ranging and broad ways influencing each and every aspect of society such as politics, diplomacy, defense, economy, healthcare, medicine, transportation, agriculture, and many more. The products of science and technology are found in each and every corner of our lives. Science has served the humanity and will continue to do so until needs are fulfilled.

"*Science creates basic knowledge of the world; engineers apply that knowledge in a practical way to create technology.*" Science and technology have had a big impact on all of humanity's activities. Science and technological inventions and discoveries such as theory of origin of the universe, the theory of evolution, etc. have had an immense role in influencing our understanding of the world and our outlook of nature.

The wide applications of technologies developed due to scientific discoveries made by humanity have led to the development of civilizations in every age. The developments due to science and technology have stimulated economic growth, enhanced standard of living, encouraged cultural development, and have also had impact on religion, thought, and many other human aspects. History is a rich secular nature of science, and technology has meant that scientists from all backgrounds have contributed to different fields of science, free of politics, religion, caste, and region. Every human being has the right to use science and technology for beneficial purposes. However, mutual cooperation and coordination between academia and industries is very important for the growth and evolution of science and technology would no doubt be a boon to the world because human beings will come to know a lot about the world in which they live and carry out their activities:

Development in technology leads to the advancement in science and helps to revolutionise several fields including medicine, agriculture, education, etc, the science of today is the technology of tomorrow. (Edward Teller)

Since science and technology are changing at a rapid pace, there must be a convergence, a creative union of sciences, technologies, and peoples while aiming and mutually benefiting each other. India can capitalize on its long history of excellence in science and its progress in scientific research to further science education to make sure it is implemented across all layers of society. Science and technology have a broad and wide-ranging influence on areas such as politics, diplomacy, defense, economy, health and medicine, transportation, agriculture, and many more. The end results of science and technology are found in every corner of our lives. Science has served the humanity from time immemorial and will continue to do so till human needs exist.

The study of history of economics brings out the dramatic relationship between technology and economic growth, leaving no room to doubt that the latter could not happen without the former. In today's technology-driven world of the twenty-first century, a country is considered powerful only on the basis of developmental parameters, among which the most important is the advancements in science and technology a country has made by virtue of its scientific acumen and human resources.

Glory of Ancient Indian Contribution to the World in Metrology, Science, and Technology

India is one of the oldest civilizations in the world, and it is a fact that India had a strong tradition of science and technology. In the ancient civilization, the growth of the civilization was the result of incredible ancient engineering and technology during those days. Indian heritage and culture is among the oldest and richest among all civilizations. Many of the findings of science which we consider to have been made in the west were made and documented in India many centuries ago. This is the land of Aryabhatta the father of astronomy as listed in ancient India contributions (Gupta 2009; Rab et al. 2020):

We owe a lot to the Indians, who taught us how to count, without which no worthwhile scientific discovery could have been made. (ALBERT EINSTEIN)

Historical research has revealed that from manufacturing the best steel to teaching the world to count digits, India was instrumental in contributing to the advancement of science and technology. These developments were made centuries before modern laboratories were set up. Many theories formulated by ancient Indian scientists have reenforced the fundamentals of modern science and technology. While some of these ancient contributions have been acknowledged today, some are still unknown.

Few of Ancient Indian Contributions to the World in Metrology, Science, and Technology

Here is a list of some of the major contributions made by ancient Indian scientists in the field of science and technology that will make everyone feel proud of their legacy:

- Little needs to be written about the idea of "zero," one of the most important concepts throughout the history of humanity. Indian mathematician Aryabhatta was the first person to create the symbol for zero "0" in 476 AD. Aryabhatta was the first of many mathematician-astronomers from the early age of Indian mathematics and astronomy.
- The decimal system – It was India which gave ingenious method of expressing all numbers in the form of ten base symbols which formed the decimal system.
- Numeral notations – Indians had devised a system of symbols for numbers from 1 to 9 as early as 500 BC.
- *Theory of gravity* – Theory of gravity: Indians knew of gravitation way before Newton. One of the first postulations on gravity was by the astronomer and mathematician Varahamihira (505–587 CE), and later Brahma Gupta exclaimed "bodies fall towards the earth as it is like the earth attracts bodies, just as it is like water to flow."
- *Fibonacci* – Fibonacci numbers and sequence first appeared in Indian mathematics through the Rishi Pingala, in connection with the Sanskrit tradition of prosody. Later mathematicians like Virahanka, Gopala, and Hemachandra also wrote about this sequence much before the Italian mathematician Fibonacci introduced this fascinating sequence to Europe.
- *Binary numbers* – Binary numbers: Binary numbers are the basic language which a computer understands. Binary refers to a set of two digits 1 and 0. The combinations of which are called bits and bytes. This binary number system was first introduced by Vedic scholar Pingala in his book *Chander shastra*.
- *Chakravala method* – This method of obtaining integer solutions was developed by one of the well-known mathematicians of the seventh century, Brahmagupta. Later, another mathematician Jayadeva generalized this method for a wider range of equations. It was further refined by the mathematician Bhaskaracharya.
- *Ruler measurement*. 2400 BC – In Indus Valley civilization. Excavations at Harappan sites have found rulers and linear measures made from ivory and shell.
- *The theory of atom* – The atomic theory developed 2600 years ago in ancient India by one of the noble scientists of ancient India kanad.
- *The Heliocentric theory* of astronomical predictions – Ancient Indian mathematicians often made accurate astronomical predictions by their applied mathematical knowledge, Aryabhatta being the most significant.
- *Wootz steel* – In 200 BC -A pioneers steel alloys matrix developed in India.
- *Smelting of zinc* – India is known to be the first to smelt zinc by distinction process.

- *World's first plastic surgery and cataract surgery* – Book written by Sushruta in the sixth century BC, the Sushruta Samhita details various illnesses, plants, preparations, and cures. The text also describes complex techniques of plastic surgery as well as a cure for cataract.
- *Ayurveda* – 5000 BC. Charaka authored a major foundational treatise Charaka samhita on the ancient science of Ayurveda.
- *Weighing scale* – 2400 BC. It has been revealed that balances were being used to compare and measure goods for trade in the Indus Valley civilization.
- *Pythagorean theorem* – Pythagorean theorem and geometrical proof for isosceles right angle were available 700 BC. The Baudhayana sulba sutra has postulated it in a sutra. The knowledge and teachings of our ancient scientists have taken a beating just because it hasn't been well documented.

Achievements/Success Stories of Modern India

The Indian Space Research Organisation (ISRO) has successfully managed to break the stereotype of an archaic in efficient public sector to emerge as an institution that is giving cutthroat competition to premium space agencies. India's vaccination drive shows its capability and moving ahead with new energy. The independent development of Covid-19 vaccine in less than a year marks an extraordinary achievement, more so for a developing country like India that was disadvantaged on so many fronts. It not only came out with locally produced vaccines but also indigenously developed its own.

Downfall of Development of Science and Technology and Its Education in India

Pre-Independence of India Era: Exploitation of Physical Raw Material and Intellectual Property

Science has been an integral part of the Indian culture from time immemorial. Many theories and techniques developed by the ancient Indian scientists have reenforced the fundamentals of modern science and technology. Some of these path-breaking contributions have been acknowledged, while some are still unknown to most. Gurukul system of education existed where ancient source of science and technology were taught. Nalanda and Takshashila universities are great examples of this. There was a decline in scientific activity in India during the pre-independence era. This was due to the frequent invasions by foreign rulers who caused a lot of destruction, especially of the native iron industry. These invasions also caused loss of many scientific documents. Regional and political instability could also be one of the reasons which disturbed the progress of scientific acumen in the country. "Necessity is the mother of invention," and people in the precolonial era were satisfied with their agro-based lifestyle and did not see the need for innovation and invention. Thus, India lagged behind in scientific advancements. The credit for most of the inventions

and discoveries has so far always rested about them in schoolbooks, heard in televisions or in movies, or talked about tirelessly. It's the ancient science and its knowledge and teaching that have taken a beating just because their achievements were well documented. It has been found time and again in our written and oral literature about the supremely advanced ancient Indian science that is not taught in school.

After Independence of India Era: Exploitation of Physical Raw Material and Intellectual Capability

Even after independence, Indian resources and talents have been exploited during the twentieth and twenty-first centuries. This is due to lack of interest in industrialization and development of science and technology by successive governments and lack of environment for inventions and innovations in the country. Even though India has glorious past in science, we could not use and materialize to continue the tradition. We were the front-runners in science. In the ancient civilization, the growth of the civilization was the result of incredible ancient engineering and technology during those days. This made us to be still developing country rather than making developed country by this time. Raw material and natural resources of India were exploited during the first industrial revolution in the nineteenth and twentieth centuries. Raw material was exploited, and finished products were to India at exorbitant price. In the twentieth century, India also exported its natural resources and raw material to other countries at lesser cost and imported products with value addition at their end at exorbitant cost.

In the twentieth century, India also exported human resources to other nations because of the lack of employment opportunities. In the twenty-first century, our technical skill and knowledge of our engineers were exploited by multinational companies at lesser cost and made huge profits again supplying value-added goods and services. The ability to establish infrastructure to make use of intellectual capability to produce products and services is the way to make self-reliant and progress for the benefit country and to become a global leader of the twenty-first century. However, India is now at a position to rise and thrive once again and prove its worth by making noticeable achievements as a science and technology expert in the coming future. Now we can reverse the process of exploitation and become self-sufficient and make developed country.

Position of India Compared to the World in Science and Technology Development

(a) The number of people in developed countries like DENMARK and SOUTH KOREA more than 8000 people gets into research and more than 5000 people in GERMANY, JAPAN, AUSTRALIA, BELGIUM, FRANCE, the UK, etc. Only 252 people get into research per million people in India.

(b) 2015, India filed only 2015 international patients, whereas Japan nearly 45,000 and South Korea nearly 30,000 and Korea nearly 15,000.
(c) India spends 0.64% of GDP on science and technology, whereas Israel and South Korea spend nearly 5% of GDP. Similarly, the USA, Japan, the UK, Germany, Denmark, Belgium, etc. spend more than 3% of GDP. That is why the USA, Russia, and Japan are considered as developed countries because of this spending on science and technology, while India is still counted among developing nations.

Convergence Approach to Build Modern India Through Development of Metrology, Science, Technology, and Their Education

If we study the history of world economics, the dramatic relationship between technological invention and economic growth is apparent. It is clear that the economic growth cannot happen without technological innovation and invention:

> *Prosperity in a society is the accumulation of solutions to human problems.* (Hanauer and Beinhocker)

Most modern metrology infrastructure, science and technology development institutions, and their skill and knowledge centers are essential for creating ecosystems, to encourage discoveries, inventions, and innovations in India. Because science and technology are changing so rapidly and irreversibly today, the fundamental principle for progress must be the convergence of different sciences, technologies, and people, while each of them mutually benefiting from the convergence. Convergence approach requires involvement of all stakeholders like government agencies, national institutions, educational institutions, manufacturers, R&D establishments, calibration and testing laboratories, end users, private and public undertakings, and relevant public on a single platform to identify the needs and establish road to success for encouraging relevant inventions, innovations, and technology development suitable to take India forward.

India can capitalize on a long history of scientific excellence and its progress in scientific research. This should reflect in science education which needs to be implemented across all layers of society. Robust national metrology institutions should be upgraded based on required needs. Without strong national metrology institutions support, there can be no progress in inventions or innovations or technology development. Substantial investment on science and technology development is made by the developed and developing nations to meet challenges of competitive and progressive world.

In the current twenty-first-century world which is technology oriented, any country is considered powerful purely based on development parameters, most important among which is the technological advancements a country has made with the use of scientific acumen of its human resources. The most powerful

creations of the human mind which are science and technology, driven by an ethical society, must become the engines of progress. They should be able to transport the world from suffering and conflicts to prosperity and harmony.

Development in technology reaffirms and re-enforces advancement in science and helps to revolutionize many fields of human activity like medicine, agriculture, education, etc. Technology is the single greatest factor that creates a distinction between modern economies from the primitive ones. This is because technology lowers the cost of production while innovating new products. It also increases both productivity and allocated efficiency of all industries. The quality aspects of products and services have become forerunner of all activities in industries with the introduction of ISO 9000 and in view of globalization and international competitiveness in trade and commerce. "*For any development requires science and technology, but without metrology there is no science and technology.*"

Results of progress in science and technology can be seen due to spending on metrology in other developed and developing countries compared to India. Most of the developed countries and developing countries have realized that only metrology, science, and technology can drive the economy for progressive development, and hence they spend more than 2–5% of GDP.

Encourage Discovery

A discovery is something that already existed at the time of discovery but had not been identified until then. Nothing changes as a result of the discovery apart from the increase in knowledge that it brings. Discoveries are therefore defined as the first description of a natural law or a law derived from the laws of nature.

Promote Inventions

Invention is the creation of a new idea. Innovation is the integration and/or conversion of an idea. Inventions are creative, innovative achievements that give rise to previously unknown solutions and the application of new technique. "Necessity is the mother of invention," but it is invention that drives the economy. The biggest utility of invention is that it solves real-world problems and brings changes in the way world functions. Innovation shapes the new way of life and it transcends the culture of the world. The modern era is arguably the greatest time in the history for innovation. None of this innovation would have happened without the invention of electricity supplied on demand. So, with this one advancement came the fastest-changing period in history of humanity and the greatest population growth, too. Invention is the process by which we devise and produce something which is useful which didn't previously exist, by independent investigation, experimentation, and mental activity. An invention involves such a high level of mental effort that the inventor is usually acclaimed in the society, even if the invention doesn't become a commercial success. The study of history of economics brings out the dramatic

relationship between technology and economic growth, leaving no room to doubt that the latter could not happen without the former.

Encourage and Promote Innovations

Innovations which may or may not involve invention are a complex process of introducing novel ideas and put them to use or practice while keeping entrepreneurship as an integral part. A curious aspect of innovation is that it's considered noteworthy only if it becomes a commercial success. Society thus benefits from innovation, not from invention alone, and it may happen that there is a significant time lapse from the stage of invention to innovation. Innovation is the integration and/or conversion of an idea into a product/service that creates significant new value in the marketplace. Importance of innovation in this VUCA (volatility, uncertainty, complexity, ambiguity) world is undeniable. Innovation is doing new things or doing things in a new way. Innovation excellence is essential for the survival of any business and organization. Getting innovation mindset should be at most priority.

Innovation is important for higher revenues, increased profits, new customers, and higher market share, increased customer loyalty, to motivate people, and to make a great place to work. Invention can be defined as the process of creating a product or introducing a process for the first time. Practical application of inventions into marketable products or services is extremely necessary. Innovation on the other hand happens when someone improves or makes significant contribution to changing of an existing product, process, or service, which changes lives and makes human race move forward. One of the most important contributions that inventions make is to create entirely new fields of studies.

Key Prerequisites to Invention and Innovation

(a) A skilled workforce educated in the best practices.
(b) Entrepreneurship.
(c) Patent and communications policies and practices while properly balancing intellectual property protection and free flow of information.
(d) Organizations, industries, and associations that create a network which can enable information flow.
(e) The use of economic statistics which can accurately and reliably measure the metrics of invention, innovation, prosperity, research and development, investment, economic benefits of digital technologies, and business finance.
(f) An economy that promotes equality and inclusiveness by seeking to improve living standards for all.
(g) Robust institutions of governance, like democracies, which ensure freedom of expression and civility and give free opportunity for everyone to experiment in life without fear of penalty.

Promote and Encourage Indigenous Technology Development

Technology can be defined as a body of knowledge of techniques, methods, and designs that accomplish a certain work in ways with certain consequences, even when one cannot exactly explain them. Technology may also be defined as an effort to organize the world to solve a problem by developing and producing goods and services for use in problem solving. We can improve things without invention or innovation, through a process of continuous improvement or parameter optimization, to improve value. However, inventions can lead to creation of new technologies which don't exist as of today, thus creating new jobs and improving quality of life in society.

Conclusion

Most modern metrology infrastructure, science and technology development institutions, and their skill and knowledge centers are essential for creating ecosystems, to encourage discoveries, inventions, and innovations in India. The above prerequisites for creating proper ecosystem for developing robust metrology structure and establishments for innovations and inventions in science, technology, and institutions for promotion of skill and knowledge development in metrology are essential to any economy to progress and compete, survive, and thrive in this competitive progressive world.

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Quality: Introduction, Relevance, and Significance for Economic Growth

3

Alok Jain, Shanay Rab, Sanjay Yadav, and Prapti Singh

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Abstract

Quality explicitly relates to "fitness for use"; thus, the quality of one product or service may be lower or higher than that of another. The quality of the goods or services may be accepted or rejected based on the specifications or guidelines/protocols set as per national or international standards. The next important issues are the ways the quality is implemented, maintained, retained, tested, verified, and performed and the identification of the organizations responsible for these

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processes. Even though those things are undoubtedly important, controlling the quality is much more indispensable when producing goods or maintaining services without flaws. Organizations can manage quality efficiently through a strong management system adhering to national and international standards. The whole process is characterized as "Total Quality Management" and covers facets such as developing products and services, enhancing processes and systems, and ensuring that the entire organization(s) is efficient and functional. This chapter introduces in detail the quality concept; various terminologies used; historical aspects and evolution of quality across the globe; benefits, relevance, and significance of quality management, tools, and method; and its need and impact on global trade and economic growth. It is hoped that the chapter would provide a significant resource material and reference for the beginners, students, laboratory personals, researchers, and even for the regulators and quality experts in this terse and concise report.

Keywords

Quality · Quality Management System · Total Quality Management · Standards · Accreditation · Traceability · Metrology · Economy

Introduction

A good place to start a chapter on quality would be by asking what the term "quality" actually means. There have been innumerable definitions assigned to this seemingly simple word, owing to its relativity, but to fulfill our quest of understanding the term, the definition given by the ISO (1994) would suffice: "The totality of features and characteristics of a product or service that bear on its ability to satisfy stated or implied needs" (Shewfelt 1999; https://www.fao.org/3/w7295e/w7295e03.htm#:~:text=The%20term%20%22quality%22%20has%20a,satisfy%20stated%20or%20implied%20needs%22. Accessed 31 Oct 2022). Simply put, a product is said to be of good "quality" when it meets certain parameters that either satiate or exceed customer expectations or requirements. Through this definition, it can also be noted that there are several function-based definitions of quality, all of which help in illustrating the core of each function. They are as follows:

(i) **Transcendent Excellent** – Timeline of evolution of quality is the most recognizable way of using the term. A product or service that meets customer expectations is almost always deemed to be "excellent."
(ii) **Product-based definition** – Quantities of product attributes: This again alludes to the certain attributes of a product, in terms of either its functions or design, and the quantity in which these attributes are present, which makes a product a "quality" product. Example: If a smartphone contains some state-of-the-art functions or is coupled with an attractive and interactive design, it is often said to be of good "quality" by the consumer.

(iii) **User-based definition** – Fitness for intended use: As mentioned earlier, while discussing the ISO definition, a product is deemed to be of good "quality" when it either meets or exceeds customer expectations. This is where the definition "fitness for intended use" comes in. For the smartphone example given previously, it can be said that if a smartphone has a smooth user interface and navigation design, which fulfills the intended purpose, and has been equipped with trendy functions, which exceeds the expectations of a customer, then it will be considered a good "quality" product.

(iv) **Value-based definition** – Quality vs. price: There are a lot of products in a market that are manufactured to fulfill the same functions. This presents the consumer with an overwhelming array of choices to make and decide which products to choose from. What is it that sets a product apart from its identical counterpart, seeing that they both have functions that would make them a "quality" product? A seller can set its product apart through its price. It is imperative that sellers and firms strike a good balance between the rate at which the product is priced, which must be valued at a rate that is affordable, and the quality of the product, which is appropriate to its price, when put against the competition in the market. It is also worth noting that there are products that are deemed to be "basic," simply because there is a lack of extra features in comparison to their more expensive counterparts. However, even though a product may be of "low quality" in terms of style or features, this product still gives good value for money for its "overall quality." This does not mean that this "overall quality" of a product cannot be enhanced or improved upon.

(v) **Manufacturing-based definition** – Conformance to specifications: Philip B. Crosby, the author of *Quality Is Free*, one of the most prominent and influential bodies of work written on quality, says that quality is nothing but a "conformance to requirements" (Crosby 1979) (https://www.bl.uk/people/philip-crosby. Accessed 31 Oct 2022). It then becomes imperative to define what those "requirements" mean. A way of finding this is by identifying and applying the customer requirements as early as can be in the product/service development cycle. Crosby then expanded on this definition in 1992, stating that "Quality meaning getting everyone to do what they have agreed to do and to do it right the first time, is the skeletal structure of an organization, finance is the nourishment, and relationships are the soul." The thing that the organization has "agreed to do" which it intends to "do it right" would here refer to meeting the requirements and demands of the customers in a manner that strives to satiate or even determine the needs of the customers for them. The finances used in order to manufacture said product according to specified requirements and the relationships forged through a trust built upon the promise of quality are aspects that set a firm apart from its other competitors (Shewfelt 1999; https://www.fao.org/3/w7295e/w7295e03.htm#:~:text=The%20term%20%22quality%22%20has%20a,satisfy%20stated%20or%20implied%20needs%22. Accessed 31 Oct 2022; https://www.bl.uk/people/philip-crosby. Accessed 31 Oct 2022; Aswal 2020a; Merrick 1999; Juran et al. 2005; Mitra 2016).

The various key aspects of quality are as follows:

(i) The product must look good and have style.
(ii) The product should have good functionality.
(iii) The product should be reliable and should only have acceptable levels of breakdowns or failures.
(iv) It should have consistency.
(v) It should be durable and last as long as it promises.
(vi) It should have good after-sales service.
(vii) It should possess value for money. This aspect is the most important, as in any market, there are different levels of quality and the customers must be satisfied with the knowledge that the price reflects the quality.

It is important for a business to ensure that their product has a good design, which is fundamental in order to produce a product efficiently, reliably, and at the lowest possible cost. Quality helps in determining a firm's success in a number of ways:

(i) Enhancing the consumer trust worthiness and loyalty vide which the consumers come back and buy repeatedly, endorse and recommend the product or service to other users.
(ii) Quality ensures a firm has a strong brand reputation.
(iii) Retailers will want to purchase and stock the products.
(iv) If a product is perceived to possess a good value for its money, it may then be priced at a premium rate and may become more price inelastic.
(v) A decrease in the returns and replacements of a product or service will lead to reduced costs. It is the duty of a firm to ensure the good quality of their products consistently with minimal errors.

When the definition of quality is projected onto any analytical work, it is defined as the "delivery of reliable information within an agreed span of time under agreed conditions, at agreed costs, and with necessary aftercare." The "agreed conditions" mentioned here should specify the precision and accuracy with which a data must be prepared, which may determine its "fitness of use" that may vary for different applications (https://www.fao.org/3/w7295e/w7295e03.htm#:~:text=The%20term%20%22quality%22%20has%20a,satisfy%20stated%20or%20implied%20needs%22. Accessed 31 Oct 2022; https://www.bl.uk/people/philip-crosby. Accessed 31 Oct 2022; Jackson 1959; Hellsten and Klefsjö 2000; Venkatraman 2007). Despite this, there have been instances where the reliability of a data is left unquestioned and no requests for further specifications are made. Laboratories will continue to operate in accordance with the standard operating procedures and methods that have not changed. What complicates matters more is that no future use of this data can be foreseen, which would mean that it would be extremely difficult to specify the required precision and accuracy in the data. This creates a rather awkward situation where the laboratories are left unsure about their quality as they lack the necessary

documentation to account for the same (Nanda 2005; Wiengarten and Pagell 2012; Isaksson 2006; Aswal 2020b).

There are three levels that help an organization to maintain the quality of their production:

(i) Quality management (QM)
(ii) Quality assurance (QA)
(iii) Quality control (QC)

Before delving into these concepts, it is important to first look back into time at the history and evolution of quality and asks how the concept of ensuring "excellence" actually came about?

Historical Aspects and Evolution of Quality

It is fact universally acknowledged that human beings have been involved in the process of producing goods since the Paleolithic or Old Stone Age, when pots and weapons made of stones fulfilled the purposes of hunting and cooking. It was then, with the creation of a social system or structure, that professions such as masonry, pottery, farming, horticulture, etc., were created, which in turn introduced the system of trade. These traded goods would have to be of an acceptable or excellent "quality" in order to be sold, irrespective of the trading system. So where does this conscious effort of ensuring quality while producing certain goods and services come from? Are there any recorded instances of such practices being prevalent throughout the Ages, and how far must one look back in order to trace the evolution of the concept of quality through time? Figure 1 shows the timeline evolution of quality (Rab et al. 2020; Shah et al. 2011; Schroeder et al. 2005; Yong and Wilkinson 2002).

Going back to thirteenth-century Europe, craftsmen created unions of their own, called "guilds," to create stricter rules that would help in maintaining the quality of their products and services. One of the ways in which these rules were enforced was by

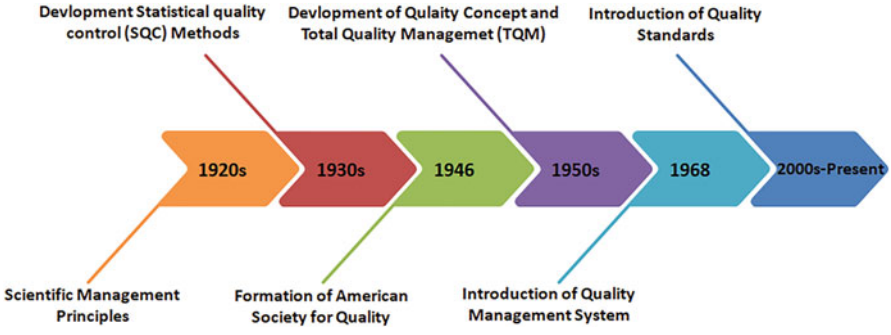


Fig. 1 Timeline evolution of quality

creating an Inspection Committee, which would assign a product of good or flawless quality with a special mark or symbol. In another practice, craftsmen themselves would imprint the symbol on products created by them. This mark originally came in handy when identifying the origin of a faulty item, but over the years, it evolved into a form of peer evaluation and came to represent what good craftsmanship looked like. These markers were the indicators and acted as a proof of the quality of a product sold throughout the medieval European market. This system managed to stay in relevance through several centuries, that is, until the smoke, emblematic of the Industrial Revolution, came sweeping in, changing the quality scene forever.

Up until the seventeenth century, the world of manufacturing was bound by the archaic rules of craftsmanship. The tide of change, however, came in the form of the factory system, which laid a heavier emphasis on product inspection, and its use was popularized in Great Britain, bringing about the onset of the Industrial Revolution starting in the 1800s. While Britain saw a paradigm shift in its quality assurance practices, the same could not be said about its neighbor across the pond. The Americans largely followed the craftsmanship model well into the 1800s, as they were shaped by changes in the predominant production methods. As most craftsmen were selling their goods locally, it meant that they had a personal stake in meeting customers' need of quality, a task in which, if they failed, a loss of customer trust was inevitable. This risk then gave rise to the process of conducting a quality control, wherein the goods were inspected thoroughly before being sold in the market.

But this wasn't the only form of quality assurance that was present. As mentioned above, by the turn of the nineteenth century, the Industrial Revolution had started to grip the United Kingdom and Europe, which brought with it "the factory system." In this system, the trades that were originally particular to their craftsmen were divided into specialized tasks. The craftsmen then became factory workers, and shop owners became production supervisors, changing the way in which commodities were produced. However, this practice initiated the decline of employee autonomy and empowerment at the workplace. The quality in the factory was then ensured through the laborer's work and audits/inspections, and defective products were either removed or scrapped altogether. While the Western world was settling into the factory system, the USA sought to defy tradition and adopt a new management approach, developed by Frederick W. Taylor, called the "the Taylor system," which aimed to increase productivity without expanding the workforce with skilled craftsmen (https://en.wikipedia.org/wiki/Scientific_management#cite_note-Mitcham05p1153-1. Accessed 25 Oct 2022). This was achieved by assigning the task of designing the factories to specialized engineers and making the craftsmen and supervisors the inspectors and managers who helped in executing the engineers' plan. While this led to an increase in productivity, it unfortunately resulted in the decline of the product quality. In order to tackle it amicably, the manufacturers focused their efforts on creating inspection departments exclusively devoted to keep defective products isolated and away from consumers' reach.

During the Second World War, the USA enacted legislation for fulfilling the demands of the civil society and military production. As such, ensuring quality became a paramount safety issue and played a significant role in the war effort. Since the existence

of faulty equipment can result in severe casualties, the US armed forces used to test and inspect each unit of war products for ensuring that it is safe for use. This practice, however, needed a massive pool of testing personnel, causing additional problems of recruiting and retaining well-trained and competent staff. In order to ensure the product quality, safety, and ease in operations, the US defense forces started following the practice of sample, test, inspection, and verification. The Bell Laboratories, in association with industrial experts, introduced the sampling tables, which were also made a part of military standard. MIL-STD-105, as well as integrated into military contracts to get the desired products from the suppliers. Quality, in turn, was ensured by supporting the suppliers and imparting trainings on Walter A. Shewhart's "Statistical Quality Control (SQC)" techniques (https://www.isical.ac.in/~repro/history/public/notepage/Statistical_Quality_Control-F.html. Accessed 25 Oct 2022).

The twentieth century is marked as the beginning of the insertion of "processes" in the quality systems. A "process" in literal sense is defined as a set of actions through certain inputs, which provides outputs with value addition. Later on, Walter A. Shewhart concentrated his efforts on controlling these processes in the mid-1920s, which made quality an integral part for the finished goods as well as the processes used for making them. Shewhart was very clear in his opinion and expertise. He expressed that industrial processes often generate a data bank, which can be easily analyzed statistically to ensure that the processes are stable and in control, or can be fixed if needed. Shewhart then created the "Control Charts," an essential quality tool, used even today. W. Edwards Deming, a statistician and follower of Shewhart's SQC techniques, became the leader of the quality movement in both the USA and Japan (https://asq.org/quality-resources/history-of-quality. Accessed 25 Oct 2022).

Total quality, in America, had started out as retaliation to the Japanese quality revolution following the Second World War, wherein total quality entered into the production of goods for civil society and trade from producing military goods. At first, however, Japanese products were shunned from international markets for their poor quality, and this forced them to re-evaluate their ways of thinking about quality. The Japanese then called upon the expertise of two American quality experts:

(i) W. Edwards Deming, who was frustrated to see the end of the practice of statistical quality control once the war and government contracts came to an end
(ii) Joseph M. Juran, who had foretold that the quality of the products made in Japan would outdo the products made in the USA by the mid-1970s considering the rate of quality improvements in Japan

The Japanese industries were more focused on improving all the organizational, manufacturing, management, and administrative processes, rather than banking on only product inspection and taking feedback from the consumers who used them. This resulted in the production and export of higher-quality goods at competitive prices, benefiting customers across the globe (https://en.wikipedia.org/wiki/Scientific_management#cite_note-Mitcham05p1153-1. Accessed 25 Oct 2022; https://www.isical.ac.in/~repro/history/public/notepage/Statistical_Quality_Control-F.html. Accessed 25 Oct 2022; https://asq.org/quality-resources/history-of-

quality. Accessed 25 Oct 2022; Harris 1995; Martínez-Lorente et al. 1998; Kaur et al. 2021; Kumar and Albashrawi 2022).

Total Quality Management (TQM) in USA

Initially, the manufacturers in the USA were under the impression that the secret of Japanese success lies in a competitive or lower price and accordingly responded with a reduction in domestic manufacturing costs and imposing severe restrictions on imports. However, despite these efforts, they could not get desired positive results for enhancing the competitiveness of the US-made quality goods. Later on, as time passed, the war between competitive prices weakened, while competition in the quality of goods continued to grow. The emergence of the wholesome approach of America, which stressed not only on statistical data but also the entire processes, was called Total Quality Management (TQM), which is still in use all over the world. This was followed by several other quality initiatives, for example, the introduction and implementation of ISO 9000 series standards; the introduction of the Baldrige National Quality Program, and the announcement of the Malcolm Baldrige National Quality Award during the year 1987. Initially, American industries were reluctant to follow the standards but slowly started to adopt them and finally came on board (Nasim et al. 2020; Lim et al. 2022; Dayton 2001). Figure 2 depicts the various elements and features of TQM.

The Practices beyond TQM

At the onset of the twenty-first century, the quality management movement became popular and mellowed. An array of new quality systems was introduced and

Fig. 2 Total quality management (TQM) principles

implemented, and schools of thoughts beyond the past practices established by Deming, Juran, and the earlier Japanese quality experts emerged and are listed as follows:

- The ISO 9000 series of quality management standards was amended, emphasizing customer satisfaction during the year 2000.
- The ISO 9001 series of standards was revised, emphasizing risk management in the year 2015.
- A business results criterion was added in the requirements of the Malcolm Baldrige National Quality Award in 1995 to assess the capabilities of the applicants.
- Motorola, a giant company, introduced the Six Sigma approach for improving business processes and optimizing defects, which achieved breakthroughs and significant bottom-line results.
- Yoji Akao (2014) introduced a new process for quality function based on consumer requirements for change in designs, goods, or services.
- Further revisions were made to the ISO 9000 series of standards based on sector-specific versions like QS-9000 and ISO/TS 16949 for automotive industries, AS9000 for aerospace, TL 9000 for telecommunications, and ISO 14000 for environmental management.
- Thus, quality system moved beyond the manufacturing sector to the services for healthcare, education, and government.

Quality: Perspectives of the Quality Experts

Quality management is not a policy derivative or an idea of an individual, but it is a contribution of several quality experts. The experts have a crucial and significant impact on the world's socioeconomic framework, especially trade and commerce, by improving not only businesses but also several stakeholders from governments, defense, academia, the medical field, and many more (Kher et al. 2010; Rab et al. 2021a, b; Grover et al. 2004). These experts, also sometimes known as Quality Gurus, can be categorized in three phases since the 1940s:

(i) The early 1950s: American quality experts who brought the ideas of quality to Japan.
(ii) The late 1950s: Emergence of Japanese quality experts who brought about new theories on quality in response to the Americans.

During 1970s–1980s:

(iii) American quality experts began to follow the Japanese theories on quality after the Japanese industrial success.

Let's now look at the theories of few of these Gurus. W. Edwards Deming opined that management has greater importance in and upholds the responsibility of

maintaining quality, at both the individual and organizational levels, as he believed that 94% of quality problems are caused by the management itself. He then introduced a philosophy of management for small or large organizations in the public-private or service sectors, emphasizing on 14 points of strategy (https://www.bl.uk/people/w-edwards-deming. Accessed 30 Oct 2022):

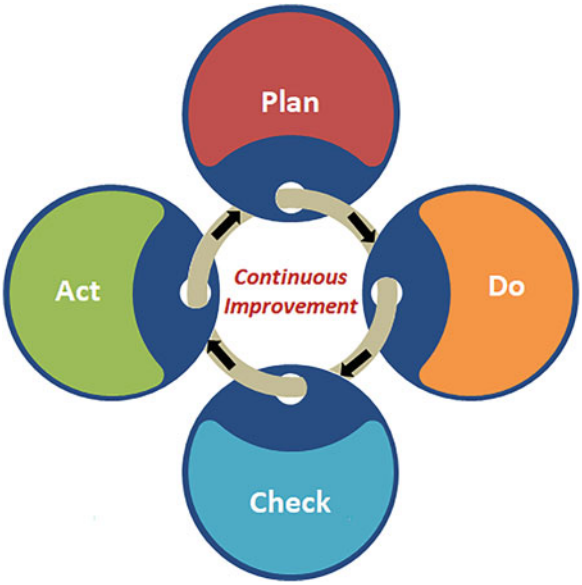
- Make reliable and consistent efforts for the improvement of goods and services.
- Implement the newfangled viewpoints and avoid working with commonly accepted stages of interruption and faulty and defective workmanship.
- Decrease the dependency upon mass inspection and rather focus on work based on data and statistical evidence of quality system.
- Stop the practice of taking decisions based on price but quality.
- Find problems. It is the management's job to work continually on the system.
- Implement contemporary training techniques on the job.
- Implement contemporary supervisory and monitoring techniques for the workers engaged in production. Change the responsibility of foremen from quantity to quality.
- Create a fearless working environment so that everyone works efficiently and effectively for the organization.
- Minimize or stop breaking down barriers between departments.
- Eliminate numerical goals, posters, and slogans for the workforce asking for new levels of productivity without providing methods.
- Eliminate work standards that prescribe numerical quotas.
- Remove barriers that stand between the hourly worker and their right to pride in workmanship.
- Implement a dynamic system of imparting training and educating staff.
- Create a management system of top brass for the implementation, monitoring, and corrective actions on all the above points on a daily basis.

He also encouraged the introduction of a systematic approach of Plan, Do, Check, Act (PDCA) for problem-solving, which is also called the Deming cycle, although it was his colleague, Dr. Shewhart, who was the originator of the idea. The PDCA cycle (Fig. 3) is elaborated as (i) Plan what is needed, (ii) Do it, (iii) Check that it works, and (iv) Act to correct any problem or improve performance.

The PDCA is still a universally well-accepted and implemented method for improvement and reducing the difference between the requirements of the customers and the performance of the process. It is about learning and continuously improving and repeats the cycle when a cycle is completed.

Joseph M. Juran (https://www.juran.com/blog/the-history-of-quality/#:~:text=Juran-Dr,quality%20control%2C%20and%20quality%20improvement. Accessed 30 Oct 2022) developed a methodology based on the quality trilogy. It encompassed tools such as Quality Planning, Quality Control, and Quality Improvement. A good quality management needs first quality actions to be planned, improved, and controlled. He was a firm believer that quality is related to customer satisfaction and dissatisfaction with the product and stressed the need for continuous quality improvement and summarized his

Fig. 3 PDCA cycle



methodology in a 10-point formula: (i) Build awareness of the need and opportunity for improvement, (ii) Set goals for improvement, (iii) Organize to reach the goals, (iv) Provide training, (v) Carry out projects to solve problems, (vi) Report progress, (vii) Give recognition, (viii) Communicate results, (ix) Keep score of improvements achieved, and (x) Maintain momentum.

Kaoru Ishikawa (https://asq.org/quality-resources/articles/quality-nugget-creating-ishikawa-fishbone-diagrams-with-r?id=6972c50861424b36966406e53618b367. Accessed 30 Oct 2022) made several contributions to quality. The Ishikawa fishbone diagram is the most popular tool, used all over the world for quality improvements. The diagram signifies and investigates the actual causes behind a problem or effect. The identified causes, highlighting the problem/effects, are used to minimize or Remove the application of possible or probable corrective actions. It also supports generating systematic ideas and presenting results to stakeholders.

Genichi Taguchi (https://www.qualitygurus.com/genichi-taguchi/. Accessed 30 Oct 2022) believed that quality and reliability should be implemented at the design stage itself where they really belong and describes quality in three stages: (i) system design, (ii) parameter design, and (iii) tolerance design. His developed methodology, popularly known as "Taguchi methodology," is a prototyping method that allows the staff to detect the optimal settings required to manufacture a vigorous product that can endure manufacturing time after time, piece after piece, and provide what the customer wants.

The Single Minute Exchange of Die (SMED) system was introduced by Shigeo Shingo (https://en.wikipedia.org/wiki/Single-minute_exchange_of_die#:~:text=Shingo%20moved%20to%20the%20US,%2C%20less%20than%20ten%

20minutes. Accessed 30 Oct 2022), an inventor known for his Just-in-Time manufacturing concept, wherein set-up times are abridged from hours to minutes. He also worked on the Poka-Yoke (mistake proofing) system where faults are examined, the manufacturing is stopped, and instant feedback is given to find out the root causes of the problem, and preventive and corrective actions are taken to avoid any reoccurrence. The inclusion of a checklist allows for the prevention of any errors. He further emphasized on the evitable errors to become defects at later stages rather than suggesting to stop errors before the product reached the consumer to avoid becoming defects.

The idea of "quality is free" and "zero defects" was proposed by Philip B. Crosby (https://www.bl.uk/people/philip-crosby#:~:text=Throughout%20his%20work%2C%20Crosby's%20thinking,performance%20standard%20is%20zero%20defects. Accessed 30 Oct 2022), describing the following four points for improvement in the quality process:

(i) Quality is conformance to requirements.
(ii) The system of quality is prevention.
(iii) The performance standard is zero defects.
(iv) The measurement of quality is the price of non-conformance.

He further marked and suggested 14 steps as follows:

- Ensure management commitment to framing a suitable and appropriate quality policy.
- Establishing a Quality Improvement Team (QIT) of the management with the responsibility of the planning, implementation, and execution of the quality improvement process.
- Find out the existing and possible glitches in the quality system.
- Make a detailed analysis of the cost of quality and utilize problems as management tools to assess waste.
- Arrange quality awareness programs and tools for the engaged quality staff.
- Take corrective actions using established formal systems to remove the root causes of problems.
- Set up a zero-defects committee and program.
- Impart training to staff in quality improvement.
- Celebrate a Zero Defects Day to transmit the management recommitment and the staff responsibilities.
- Inspire individuals and staff members, setting improvement goals.
- Inspire staff to connect and inform management of any hindrances to achieving their goals.
- Provide official credit to the participating staff.
- Set up quality councils for the exchange of sharing ideas and information about quality management.
- Do and repeat it and make a new QIT.

Quality Beyond – QC, QA, QMS, TQM, Etc.

As time goes by, the concept of quality has undergone a sea of changes, and several terms have emerged, which are multifaceted in practice, intricate and connected with quality, and elaborated as follows:

1. ***Quality Control (QC):*** The QC basically refers to the operational techniques used to fulfill the requirements of quality.
2. ***Quality Assurance (QA):*** It refers to methodical and prearranged actions that are essential to providing adequate confidence that a product or service would satisfy the specified requirements for quality.
3. ***Quality Management Systems (QMS):*** The QMS embodied an assortment of business processes ensuring constantly the fulfillment of customer requirements and enhancement of their satisfaction. It is aligned with an organization's purpose and strategic direction.
4. ***Total Quality Management (TQM):*** It refers to the process of individual and organizational development for ensuring the satisfaction level of all the stakeholders.

At first sight, these terminologies seem simple and easy to understand, but upon delving deeper into the evolution of quality, it is evident that they bring a significant change in the ways the enterprises work. Nowadays, customers are well aware of the quality and often buy quality products or services as well as follow the desired specifications on competitive pricing. In order to meet these demands, the business and trade houses used to adopt well-established management practices and systems that can design, fabricate, manufacture, and produce quality goods, fulfilling customer requirements, specifications, and outlooks with competitive pricing. However, nothing is everlasting and the quality system is too dynamic, and these practices continue to alter from time to time, with the emergence of varied quality concepts and definitions by quality experts, compelling enterprises to execute changed quality practices and systems for overall product quality.

By the end of the twenty-first century, quality was characterized as "conforming to the standard and specifications of a product." W. Edwards Deming stressed that "quality is to fulfill the requirements of customers and satisfy them," and accordingly, the quality remained "customer focused." This allowed enterprises to remain focused on satisfying consumer expectations and retaining their loyalty. The companies used to gather information regarding customer expectations through personal interaction, interviews, service feedback, and market surveys. Recently, the launch of Apple's products revealed that honing the attention only on customer satisfaction is not adequate, but consumers' inherent, latent, and emotional views are equally important to ensure the good quality of a product or service. Utilizing such inputs, Apple kept on introducing continuously new and innovative products (such as iPhone, iPad, Mac, etc.). This resulted in an exponential increase in their sales volumes, providing services with customer delight. As a result, the quality notion further changed to "customer delights." Such modifications/revisions always keep

participating enterprises on their toes and urge them to update their quality systems (Mukherjee 2006; Sroufe and Curkovic 2008; Charantimath 2017; Ibidapo 2022; Kumar et al. 2018; Rab and Yadav 2022).

The Concept of "Standard" and "Inspection" Control

The early twentieth century witnessed the introduction of control practices in industrial processing and manufacturing activities for ensuring quality products. In this period, manufacturers paid attention to the relative productivity and production costs. Accordingly, this period is called "product focused through quality control." The Ford Motor Company, USA, introduced the concept of the Assembly Line, their newly established manufacturing unit that resulted in a production volume manifold. Later on, this idea and concept was adopted and used by several other industries. Since this time was the "product-focused" era, industries focused their attention on control products. Therefore, the quality concept was intended to "conform to the standards and specifications of a product." This encouraged quality engineers to contrive the technique of "inspection," to control the quality of a finished good. This compelled the product designers, engineers, and quality experts to design and formulate the standards and specifications of the products and the standards of the manufacturing process and operations, i.e., the Standard Operating Procedures (SOPs). The workforce involved in manufacturing and quality system was advised to perform their duties and tasks as laid down in the SOPs (Rab and Yadav 2022; https://gacbe.ac.in/pdf/ematerial/18BMA64C-U5.pdf. Accessed 25 Oct 2022; Yadav and Aswal 2020; Lahidji and Tucker 2016; Wruck and Jensen 1994).

Development of Process for Quality Control

As reported earlier, the "full inspection" method is required by manufacturers to ensure product quality. This process was time-consuming, labor-intensive, and had additional high internal quality costs. Walter Shewhart, a quality control engineer in Bell Laboratories, designed a method for ensuring quality by introducing a control chart, suggesting random sampling inspection rather than 100% inspection to cut inspection cost and increase the overall effectiveness of the system (https://gacbe.ac.in/pdf/ematerial/18BMA64C-U5.pdf. Accessed 25 Oct 2022). The combination of the use of random sampling inspection and control charts increases the involvement of statistical tools, probability theory, and random sampling methods for improving the quality levels. Therefore, this method came to be popularly known as "Statistical Process Control (SPC)" or "Statistical Quality Control (SQC)." The sampling inspection ensuring method could not be considered a perfect tool for ascertaining product quality, and random sampling resulted in a few faulty goods being sent to consumers. This imposes additional costs beyond quality costs. Shewhart, however, suggested that additional cost on fewer defective pieces within the acceptable limit is not a big issue, considering the huge savings on inspection cost (Rab and Yadav

2022; https://gacbe.ac.in/pdf/ematerial/18BMA64C-U5.pdf. Accessed 25 Oct 2022; Yadav and Aswal 2020; Lahidji and Tucker 2016).

The QMS During the Mid-twentieth Century

Often, better quality and lower price of a product are normally the deciding factors for a consumer to buy a product. This is the reason why manufacturers focused their attention on the control of quality and prices. As a core issue, Juran (1951) advocated the concept of "quality costs"; bifurcating the quality costs into preventive costs, assessment costs, and internal and external failure costs. As per an estimate, the losses that occurred due to the manufacturing of defective pieces and failures are higher than the costs of quality control, especially the costs due to internal and external failures. Therefore, the implementation of SPC could not produce the desired results to control the quality costs. Several quality experts suggested other "invisible" or "hidden" quality costs. The term "invisible cost" is used to indicate failure costs that are inadequately recorded in company accounts, and/or failure costs that are never actually discovered. Yang (Snieska et al. 2013) addressed the hidden costs through the definition and addition of two new categories: "extra resultant cost" and "estimated hidden cost." As such, the quality costs can be categorized as (i) preventive, (ii) evaluation, (iii) internal failure, (iv) external failure, (v) extra resultant, and (vi) estimated hidden.

As pointed out earlier, several Quality Gurus including Feigenbaum were of the view that implementing an SPC system alone would not control quality costs successfully. During the intervening time, a new quality concept came into existence called "quality assurance," which was "users oriented," indicating that the product is fit for the intended purpose and quality requirements. Therefore, it was inferred that quality is defect-free or possesses zero defects and fulfills the required specifications.

A literature survey revealed that quality assurance may also not be adequate to achieve the desired results by controlling production processes. Therefore, the concept of Total Quality Control (TQC) was proposed in 1959, which stresses that product quality requirements, to be instigated at all the stages of the product life cycle, comprise product design, incoming quality approval, process quality control, product reliability, inventory, delivery, and customer service. The concepts and approaches of SPC, TQC, and "costs of quality" were introduced in Japan in 1960 by Deming and Juran (Aljuboury 2010). The Union of Japanese Scientists and Engineers (JUSE) introduced and developed the concepts, principles, and approaches of Statistical Process Control and Total Quality Control. The JUSE promoted the practices of TQC and the quality concepts, pursuing the zero-defect culture and executing the task right the first time (https://www.qualitygurus.com/genichi-taguchi/. Accessed 30 Oct 2022).

Emergence of Japanese Company-Wide Quality Control (CWQC)

The Japanese enterprises espoused the TQC practices and stressed on imparting education and training on quality for all the staff engaged in establishing a quality culture in the organization. Therefore, the implementation of TQC in Japanese

industries was very different from the original TQC (https://www.juran.com/blog/the-history-of-quality/#:~:text=Juran-Dr,quality%20control%2C%20and%20quality%20improvement. Accessed 30 Oct 2022). The Japanese TQC possessed several critical characteristics listed as follows:

(i) It is customer focused and quality first as the quality policies.
(ii) It advocates full participation and teamwork.
(iii) It emphasizes imparting education and training on quality to all staff members.
(iv) It stresses upon the concept of "do the right thing the first time."
(v) It advocates the concept and implementation of a "zero-defect" culture.
(vi) It stresses upon "continuous improvement" as the key quality activity.
(vii) It holds everyone responsible for achieving high-quality levels.
(viii) It emphasizes on preventive activities and quality assurance.
(ix) It instigates a quality culture regime in the organization.

The Japanese TQC system, when implemented with these crucial characteristics, is called a Company-Wide Quality Control (CWQC) system. The insinuation of CWQC in association with Japanese industrial competitiveness and tactical advantages enabled them to enter into Western markets. Based on this CWQC-driven system, high-quality products at lower prices were sold to consumers, and the Japanese industries enjoyed their enhanced market share globally, providing huge competition to Western and Asian industries and manufacturers.

Development of Total Quality Management

The Japanese competitive domination compelled the American and Western world to devise benchmarking projections, adopting Japanese CWQC performance indicators of quality management, which finally led them to develop a polished CWQC called Total Quality Management (TQM) system with the inclusion of refined characteristics listed as follows:

(i) Customer-focused management.
(ii) Continuous improvement of the key quality activity.
(iii) Top management promises persistence in pursuing quality.
(iv) Full participation and working as a team.
(v) Education and training on quality to staff.
(vi) Employees' good quality concept.
(vii) Quality leadership.
(viii) Long-term supplier relationship.
(ix) Implementation of the quality management system.
(x) Cultivation of quality culture.

The TQM was also fairly influenced by the Western Quality Gurus (Deming, Juran, and Crosby). As mentioned earlier, Deming's concept was that "quality is to

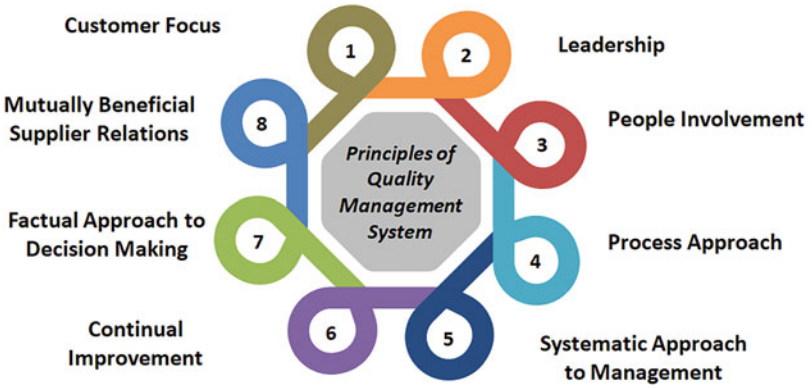


Fig. 4 Quality management system (QMS) principles

fulfill the requirements of customers and satisfy them"; Crosby vehemently propagated the concept of "conformance to customers' requirements," and Juran suggested "fitness for purpose of use, assessed by the users instead of industries, or the merchants." Therefore, the TQM has developed the integration of management viewpoints, quality principles, sets of SOPs, the use of statistical tools, quality control process, standardization, improvement mechanism, quality concept, staff participation, education and training, quality culture, etc. The eight principles of TQM are shown in Fig. 4. The mid-1980s and later periods witnessed the emergence of several important quality programs. Among them, ISO 9000 system and Six Sigma program came into practice in the year 1987. It is important to note here that the ISO system has been revised four times in 1994, 2000, 2008, and 2015. The Six Sigma quality scheme was widely imitated by GE in 1995 (Karthi and et.al. 2012).

Benefits of Quality Management (QM) and QM Tools and Methods

For the business growth of any organization, customer satisfaction and loyalty are extremely important. It is always crucial for industries to have loyal customers as business cannot thrive only on new customers but also requires repeated consumption from loyal customers. The QM safeguards improved quality goods and services for customer satisfaction, which eventually leads to customer loyalty. The quality is assessed through various indicators like performance, consistency, reliability, and durability. It is an important parameter vibe which an organization is compared with its competitors.

The QM tools ensure changes in the systems and processes, which finally results in improved and better-quality goods and services. The QM methods (say, TQM or Six Sigma) are used for delivering a high-quality product fulfilling product requirements, specifications, and customer satisfaction (Permana et al. 2021; Siva et al. 2016). The buyer would not buy and recommend the product if their previous buying experience is not good, in the event of a defective product or service. They would

come back to your organization if they are satisfied with their purchases. It is the responsibility of the organization to make sure that the buyer is satisfied with the product. It is the organization that has to understand what specifications of the product are and what the customer expectations are. It is imperative to gather the pertinent data to provide some insight and vision about requirements and demands. One of the sources for such information is the regular feedback from the customers that needs to be collected, monitored, and analyzed carefully.

The QM ensures high-quality products and services by eliminating defects and incorporating continuous changes and improvements in the system. It also ensures that the promised goods and services are delivered to the clients, for which various promotional activities can be carried out to make the public aware. The QM tools help an organization to design and create a product that the customer actually wants and desires. The QM ensures increased revenues and earnings, increased cash flow, higher productivity, timely disbursal of salary and perks to the staff, employees' well-being and satisfied employees, a healthy workplace, higher customer loyalty, etc. The removal of needless processes leading to waste of time and not helping productivity is always required. The QM enables employees to deliver more work in less time, the reduction of waste and inventory, staff to work closely with suppliers and to incorporate "Just-in-Time" philosophy, close harmony among employees, and a strong feeling of teamwork (Aycock 2003; Canel et al. 2000).

Need for Quality Management and Relevance in Global Trade

As per the detailed discussion in preceding sections of this chapter, one can inarguably and easily infer that quality is virtually or visibly crucial and important in every sphere of our lives, including global trade, and removing trade barriers. It would not be a stretch to conclude that all economies, and all their sectors, rely heavily on the existence of a quality culture. One can say that progress comes off the back of the continuous strive toward achieving good quality. All the countries need to be enabled to enjoy the advantages of globalization while effectively protecting themselves from its risks. The standards and their enforcement should not be considered as trade's new barriers but as a tool to facilitate global trade, ensuring adherence to laid-down standards, procedures, specifications, and regulations for products and services under global trade. For example, in developing countries, expensive certification procedures mean that agricultural smallholders and cooperatives that do not have access to technical assistance are at a serious disadvantage. Public institutions are increasingly unable to defend the interests of producers, especially small-scale ones. The result is that in many instances, small producers in developing countries get completely cut off from the process of standard setting and monitoring. Formerly, only tariffs were in principle recognized as restrictions to export/import and international trading. However, with the globalization of economies, the days when nations could afford to indulge in bilateral tariff-cutting negotiations have largely gone. Attention has shifted to so-called technical, or non-tariff, barriers to trade. A robust and strong national quality infrastructure (NQI) is essential in breaking down

technical barriers to trade. It is thus the key to the greater integration of the partner countries into the international trading system (Kher et al. 2010; Rab et al. 2021a; Rab et al. 2021b; Carvalho et al. 2021; Rab et al. 2022). The quality infrastructure (QI), its main pillars, and other elements are described separately in this section of the handbook in some chapters and other related literature. Readers are encouraged to go through the details of QI in these chapters.

Conclusion

The discussion presented in this chapter clearly reveals that quality infrastructure and quality management system of any country need to be strong, robust, effective, and sustainable for industrial, social, and economic growth. Practically, almost all the industries, organizations, academic institutions, business houses, hospitals, etc., have huge advantages in implementing quality system, accreditation, and conformity assessment schemes. The process, product, and service efficiencies can be significantly improved by imposing such schemes. The implementation of a QMS will aid the company in crucial areas like reducing the number of defective goods, enhancing internal communication, raising customer satisfaction, growing market share, and expanding globally. The winery's drive to continue its quest for quality improvement is driven by grasping these benefits. This report also illustrates the direct or indirect relationship between specific QM practices and other activities that leads to growth and innovation. The report is presented in a concise way, introducing quality subject and its terminologies; evolution of quality and historical perspectives; quality management system; and their applicability, importance, and significance in national and international trade. It is the endeavor of the authors to present a useful resource material at one place for all the stakeholders, including students, quality practitioners in calibration and testing laboratories, academicians and educators, accreditation and quality specialists, legal metrologists, regulators, administrators, and policymakers.

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Accreditation in the Life of General Citizen of India

4

Battal Singh

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Abstract

The concept of accreditation evolved after the Second World War to standardize the products manufactured at different places by different manufacturing units. The accreditation was required to avoid the rejection, retesting, and recalibration and minimize the cost of the production, and accreditation was limited to the industrial goods and products. Initially, there was a problem in the measurement of the dimensions of the product produced by the different manufacturing units. Because of the difference in the dimension in the same product or similar products produced by the two or more than two units, there was rejection in one of the products. To avoid the reproduction due to difference in the products, the standardization was required so that products can be manufactured within the permissible limits. To obtain standardization accreditation is the most reliable tool, as there are various requirements which need to be fulfilled by the testing and calibration laboratories. By getting accreditation of the measurement

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performed at different places of the globe, the measurement can be done within the permissible limits. In India the concept of accreditation was introduced in the early 1980s as it was expected in the future standardization will be the key for the success in the business not only at local level but at international level also. Initially, the growth of accreditation was very slow, as the National Accreditation Board for Testing and Calibration Laboratories (NABL) took around 11 years to accredit the first laboratory and accreditation was limited to the measurement at the production level in the manufacturing industries. Due to globalization the demand for accreditation was increased to export the products, and also to check the quality of imported products accredited laboratories were required. Over a period of time the accreditation was required in every field from manufacturing units to the food products, health products, etc. The growth of accreditation labs in India was remarkable after 2012. Almost in every field the National Accreditation Board for Testing and Calibration Laboratories (NABL) has accredited the laboratories, and many regulators have started recognizing the accredited laboratories for performing the testing and calibration in the required parameters. Today accreditation has entered everywhere; it seems that the laboratories of all fields get accreditation as it is the requirement of the regulators, ministries, various departments from the government and private sectors, and customers from our county as well as from aboard. Here, it has been discussed how the accreditation has impacted the general citizen of the country and how the accreditation entered in the life of the common men.

Keywords

Accreditation · ILAC · APLAC · NABL · Weight and measures · Legal Metrology

Introduction

The term accreditation has great importance in today's scenario, and it was started during the Second World War. The need for accreditation was felt during the Second World War when the bullets produced at different manufacturing units were not fitted in the barrel of gun produced at different manufacturing units. The requirement of standardization in the field of measurement was felt at that time. The big challenge was that how to produce/manufacture standardized products to avoid the retesting and rejection. The term standardization comes into the existence. How measurements done by one organization can be similar or within the permissible limit so that it can be used by the other organization? What mechanism can be used to assure the measurements done by the organization? To assure the measurements done by the manufacturing unit, accreditation comes into the existence. The concept of accreditation was started first time in Australia, with the organization, namely, National Association of Testing Authorities (NATA) in 1947. Slowly the concept of accreditation evolved in other parts of the world.

The Indian government also felt the need of standardization in the measurement. To cater this need the National Coordination of Testing and Calibration Facilities (NCTCF), now the National Accreditation Board for Testing and Calibration Laboratories (NABL) (www.nabl-india.org-seen), India, was established with the objective to provide the government and industry with a scheme for third-party assessment of labs in 1982, and NCTCF was changed into National Accreditation Board for Testing and Calibration Laboratories (NABL) in the year 1993. In the year 1998 NABL was registered under the societies act, as an autonomous body under the Department of Science and Technology (DST). Presently NABL is a constituent board of Quality Council of India. NABL merged with Quality Council of India (QCI) (www.qcin.org-seen) on 04.01.2017. The Quality Council of India is an autonomous body under the Department for Promotion of Industry and Internal Trade (DPIIT), Ministry of Commerce and Industry, Government of India.

The growth of accreditation in India was very slow as NABL took 11 years to accredit the first laboratory. Initially the accreditation was limited to the manufacturing units, but as time passes, it becomes a requirement for everyone. The comparison of the present situation of accreditation with the early 1990 is completely different. Today the number of accredited labs in all the field is more than 7000. Accredited labs include calibration and testing as per the ISO/IEC 17025 (ISO/IEC 17025 2017) (general requirements for the competence of testing and calibration laboratories), medical testing laboratories as per ISO 15189 (ISO 15189 2012) (medical laboratories – requirements for quality and competence), proficiency testing providers laboratories as per ISO/IEC 17043 (ISO/IEC 17043) (conformity assessment-general requirements for proficiency testing), and reference material producers as per ISO 17034 (ISO 17034) (general requirements for the competence of reference material producers).

NABL is signatory to Mutual Recognition Arrangement (MRA) since 2000 as per ISO/IEC 17011 (ISO/IEC 17011 2017). In the year 2000 NABL was evaluated by Asia Pacific Laboratory Accreditation Cooperation (APLAC), now Asia Pacific Accreditation Cooperation (APAC) for testing and calibration laboratories as per ISO/IEC 17025. In year 2004 and 2008 NABL was re-evaluated and reaffirmed for testing and calibration laboratories. In the year 2012 NABL achieved MRA signatory status for the testing of medical laboratories and re-evaluated and reaffirmed for testing and calibration laboratories. During the APLAC evaluation in year 2016, MRA signatory status of proficiency testing providers (PTP) as per ISO/IEC 17043 and reference material producers (RMP) as per ISO 17034 was added with the previous scope of testing, calibration, and medical testing. Due to COVID-19 the APAC evaluation was conducted in 2021, whereas it was due in the year 2020. In the APAC evaluation 2021 NABL was re-valuated and reaffirmed for testing, calibration, medical testing, PTP and RMP.

It has been observed in the last decade the requirements of accreditation were felt in every sector of the country whether it is industry, food, agriculture, pharma, or chemicals and in the field of medical testing. From the industry point of view, the accreditation is very important because the testing done by the NABL accredited

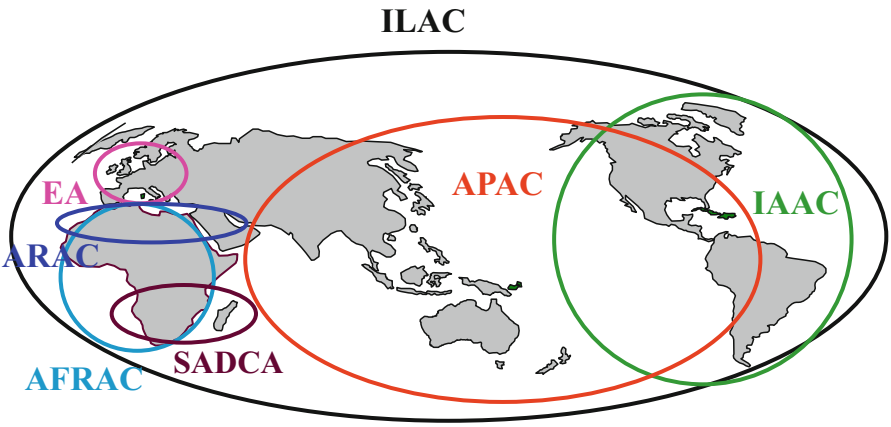

Fig. 1 International linkage of accreditation organizations

laboratory is accepted in the members country of APAC and International Laboratory Accreditation Cooperation (ILAC) and NABL is MRA signatory.

The status of NABL at the international level can be understood by Fig. 1.

EA	European Cooperation for Accreditation
APAC (www. https://www.apac-accreditation.org-seen)	Asia Pacific Accreditation Cooperation
ILAC (www.ilac.org-seen)	International Laboratory Accreditation Cooperation
IAAC	Inter-American Accreditation Cooperation
AFRAC	African Regional Accreditation Cooperation
ARAC	Arab Accreditation Cooperation
SADCA	Southern African Development Community Accreditation
Unaffiliated bodies	Peer-evaluated ABs who are not geographically located in one of the established regions

Here, it can be seen in the figure, the bigger circle is showing the ILAC region which is covering the whole globe. It shows that all the accreditation bodies which are members of the regional bodies are members of ILAC. NABL is MRA signatory in APAC and MRA members of ILAC also.

The accreditation can be understood with the definition "**Third party attestation related to a conformity assessment body conveying formal demonstration of its competence to carry out specific conformity assessment tasks (ISO/IEC 17000** 2004**)**" as per ISO/IEC 17011; here the accreditation body formally recognizing the conformity assessment body to perform specific task based on the assessment done by the assessors on behalf of the accreditation body and the technical competency of the conformity assessment body (CAB) is witnessed by the assessment team. Also, the management system of the laboratory is assessed based on the specific standard

for testing and calibration, medical testing, proficiency testing providers (PTP), and reference material producers (RMP).

The results which are produced by the accredited CABs are believed because the laboratory has quality management systems and various quality assurance activities to be confirmed by the laboratory, trained personnel, and traceable instruments. In a nutshell there is very stringent and rigorous process of accreditation followed by the accreditation body to assess the conformity assessment body for the specific scope of accreditation. The requirements of accreditation can be understood with the help of the parameters of accreditation in the testing calibration and medical testing.

The accreditation services provided by the NABL in the various types of testing are as follows:

In the field of testing, it includes chemical, biological, mechanical, electrical, electronics, fluid flow, forensic, non-destructive (NDT), photometry, radiological, and diagnostic radiology QA testing.

In the field of calibration, it includes mechanical, electro-technical, fluid flow, thermal, optical, medical devices, radiological, etc.

Similarly medical testing laboratories include clinical biochemistry, clinical pathology, cytogenetics, cytopathology, hematology and immuno-hematology, histopathology, microbiological and serology, and nuclear medicine (in vitro).

In the field of PTP, it includes testing, calibration, medical, and inspection, and RMP includes chemical composition, biological and clinical properties, physical properties, and engineering properties.

Now by considering the above scope of accreditation covered by NABL, it is evident that almost all the parameters covered and some of the parameters are directly affecting the common men. For example, food and agriculture product testing, drugs and pharmaceuticals products, water, environment and pollution, marine/aqua culture food products all are covered under scope of accreditation. Similarly, in the field of calibration, the calibration of mass, balance, and volume is required for the scientific purpose.

Most of the regulators require NABL accreditation for the purpose of approval. Some of the regulators are as follows:

- Ministry of Health and Family Welfare (Food Safety and Standards Authority of India (FSSAI))
- Ministry of Consumer Affairs (Weights and Measures)
- Ministry of Drinking Water and Sanitation
- Ministry of Commerce and Industry (Tea Board, Coffee Board, Spices Board, etc.)
- Ministry of Health and Family Welfare (Central Government Health Scheme (CGHS) and Govt. Hospital Laboratories)
- Ministry of Health and Family Welfare (Central Drugs Standard Control Organization)
- Ministry of Power (Bureau of Energy Efficiency (BEE))
- Ministry of Environment and Forests (Central Pollution Control Board (CPCB))
- Ministry of Consumer Affairs (Bureau of Indian Standards (BIS))

- Ministry of Commerce and Industry (Export Inspection Council (EIC))
- Ministry of Commerce and Industry (Agricultural and Processed Food Products Export Development Authority (APEDA), Marine Products Export Development Authority (MPEDA), Export Inspection Council (EIC), etc.)

Any laboratory requiring approval from the regulators to work with them requires NABL accreditation. There are some regulators which are performing integrated assessment for approving the laboratory, for example, FSSAI (https://fssai.gov.in-seen), tea board, AMPEDA, EIC, etc.

The requirement of accreditation is increasing day by day; almost in every field accreditation is required. In early 2000 there were very few laboratories accredited, and the demand of accreditation increased in the coming years. The accreditation is required by the govt. departments, regulators, and exporters.

The need of accreditation can be understood during the COVID-19. The reverse transcription–polymerase chain reaction (RTPCR) testing laboratories accredited by NABL have played a very big role, and still there are many states which are allowing to travel only after RTPCR negative report. Also, in the field of medical testing, the testing done by the accredited labs as per ISO 15189 is more reliable, and there in case of health insurance, the hospital which are having accredited pathology labs are given preference over the hospital which are having non-accredited laboratory.

Today, the international, national, or regional standards are available for each and every commodity which is used by the common men of our country. The common commodities can be food, water, air, soil, etc. which are directly impacting over the health of living things. For the safeguard of human beings, the testing is required in the field of air, water, and soil. The policies can be framed based on the testing results by the various types of the laboratories. Now the question which is always alive is the reliability of the test results produced by the laboratory. How the test results produced by the testing laboratory can be believed? And the best possible answer of this question is "**Accreditation**" as per ISO/IEC 17025, because the laboratory which has accreditation to perform specific task can be considered competent. Also, accreditation laboratory is proving their competence in every accreditation cycle based on the assessment conducted by the accreditation body.

The importance of accreditation is increasing day by day not only in the fields of manufacturing, production, research and development, and medical sciences but in the life of the common citizen of India, as there are many regulators in India which are taking care of quality in their respective fields.

The Legal Metrology Department under the Ministry of Consumer Affairs is taking care of weights and measures. By discussing the structure and working of Legal Metrology Department, the accreditation in the life of general citizens of India can be understood. Also, the Legal Metrology Department can be considered the fourth pillar of quality infrastructure. The term quality infrastructure is very important and can be understood with the help of Fig. 2.

The quality infrastructure of the country depends upon the four pillars, namely, National Physical Laboratory (NPL), New Delhi, which is custodian of physical standards of the country, and these physical standards are traceable to the SI unit by

Fig. 2 Quality infrastructure in India

means of intercomparison with other MNIs of the world. Second is the National Accreditation Board for Testing and Calibration (NABL) which is responsible for providing the accreditation to the testing, calibration, and medical testing laboratories in the country. The Bureau of Indian Standards (Bureau of Indian Standards Act 2016) (BIS) is responsible for developing various types of standards which can be referred for various purposes. The Legal Metrology can be considered the fourth pillar of the quality infrastructure because the weights and measure used for the commercial purposes are controlled by the Legal Metrology department only.

The Department of Legal Metrology takes care about the accurate quantity of weights and measure for the purpose of the various consumers/customers in the country. In fact, it is a direct implementation of quality infrastructure at ground level for the public across India that ensures accurate measurements of weight, length, and volume of the day-to-day commodity quantity purchased by a consumer from a trader. At present, few labs from government are unable to provide good services due to increase in population and variety of commodities with the time. Hence, the role of private labs to augment metrology service is also of paramount interest.

The prime function of the Legal Metrology Department is to ensure the quantity within the permissible limits. The Legal Metrology Department functions at center as well as state level. The Legal Metrology Department at center level is under the Ministry of Consumer Affairs, whereas at state level it is under the different ministries, differing from state to state. There are more than 100 secondary reference standard laboratories across the country, and these laboratories maintain the

Fig. 3 Brass Bullion weights used for trading purpose (https://dir.indiamart.com/impcat/brass-bullion-weight.html-seen)

Fig. 4 Weighing machine used for trading purpose (https://amtech-enterprises.business.site-seen)

traceability through RRSL laboratories; all five RRSL laboratories are NABL accredited and maintain their traceability through NPL-India.

The weights and measures which are commonly calibrated/verified/stamping by the Legal Metrology Department can be understood with the help of Figs. 3, 4, and 5.

Fig. 5 Brass conical measure used to measure the various liquids (https://www.indiamart.com/proddetail/brass-conical-measure-11082205288.html-seen)

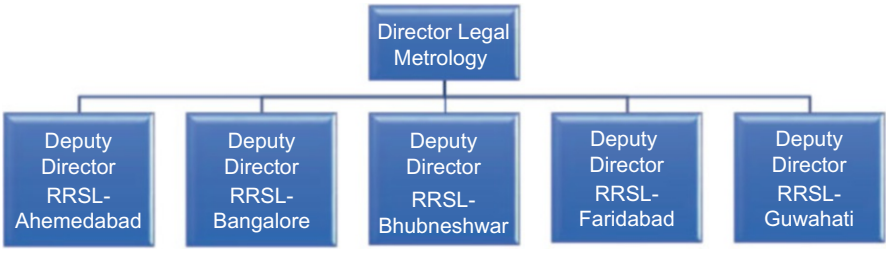

Fig. 6 Hierarchy of RRSL laboratories

There are many other measures, like measuring tape and steel scale, pressure gauges which are used on petrol pumps, etc.

The structure of Legal Metrology can be understood as below:

The Regional Reference Standard Laboratories (RRSL) under the Ministry of Consumer Affairs are under the control of the director of Legal Metrology, and the organization structure of the RRSL can be understood by Fig. 6.

The work of the Legal Metrology RRSL laboratories is to support the secondary reference laboratories of the state level. The traceability of the secondary standard reference laboratory of all the states is maintained through RRSL to the NPL.

The secondary standard reference laboratories are under the control of the controller of Legal Metrology of the particular state, and these labs are under the different ministries in the different states.

There are very few secondary reference laboratories accredited, and to improve the quality of work and their confidence, accreditation can be an effective tool. It has been discussed that all these labs must be under the ambit of accreditation. Also, it was observed that to develop the culture of accreditation among these labs, first recognition program can be initiated based on certain requirements like trained personnel, traceable instrument, and environmental and accommodation facilities of the laboratory. The joint program by NABL and the Legal Metrology Department has planned for the recognition of the secondary reference laboratories of various states. The recognition will be for 2 years and fee will be minimal.

The calibration/verification/stamping is done only by the state government labs under the Department of Legal Metrology, and it has been observed that the workload is very high; therefore initiative has been taken for Government Approved Test Centre (GATC) as per the GATC rule, 2013, from the private sector. The GATC rule is already established, and a Joint approval program can be initiated by NABL and the Legal Metrology Department. For executing the GATC program, the amendments in GATC rules are required so that the same can be executed smoothly. The amendments in the GATC rule can be discussed, and the policy document of Procedure for Approval from Legal Metrology Department for Government Approved Test Centre (GATC) for Calibration/Verification/Stamping can be discussed for executing the program jointly.

The requirements of GATC program were needed because there were many constraints in executing the calibration/verification/stamping of weights and measure across the country; therefore the Ministry of Consumer Affairs, Government of India, has issued a GATC act in 2013. The act was related to the approval of private laboratories for the execution of calibration/verification/stamping of weights and measure. The act was made for the private sector laboratories to perform calibration/verification/stamping of weights and measure because of the shortage of employees in the Legal Metrology Department at state as well as center government level. The state government was not able to verify the stamping of weights and measure with the required time period. Scientifically the weights and measure should be calibrated/verify/stamping within a year, but it was not possible. The reason of not executing the calibration/verification/stamping within the stipulated time period was not only due to shortage of manpower but due to nonavailability of infrastructure for executing the required functions. Therefore, the Indian government decided to approve the GATC (government-approved Test Centre) from the private sector. The rule was made in the year of 2013, but the Government of India was not able to execute this rule. Again, there was shortage of manpower for approving the GATC from the private sector. The GATC rules are well defined regarding the requirements of the laboratories in terms of personnel and the technical competence. As the Legal Metrology Department has constraints about the personnel who can execute the approval of GATC along with funds required for the execution of this scheme for the private sector laboratories. The Legal Metrology department has started discussion

with NABL for jointly executing GATC program, related to the training of personnel with respect to accreditation process, requirements of the accreditation body and measurement techniques in the field of weights and measures. The NBAL has conducted three training programs for the officers of Legal Metrology Departments from the center government and states along with union territories. The first training program was conducted in July 2017 for more than 50 officers of Legal Metrology, and the second training program was conducted in Sept 2018 where more than 50 officers were trained regarding the requirements of ISO/IEC 17025:2017 and techniques of mass metrology. A similar kind of training program was conducted for Legal Metrology Department officers in Feb 2019.

There were two programs agreed between Legal Metrology Department and NABL to upgrade quality of weight and measures:

NABL Legal Metrology Accreditation of Calibration Resources and Entities (NALM-ACRE) Recognition Program

There are many laboratories across the country in the Legal Metrology Department under the center government and the state government. In the states only very few laboratories are accredited, namely, one state level secondary reference laboratory of Kerala state is accredited, and recently one more laboratory from Bihar state got accreditation for the calibration of weights. The legal metrology departments of various states are interested to get the accreditation, but due to certain limitations it is not possible for them to go directly for the accreditation as per the international standard ISO/IEC 17025:2017. To overcome this situation NABL and Legal Metrology mutually decided to launch a recognition program so that in between recognition period or in the second cycle of the recognition, the laboratory may be accredited as per the international standard. There are more than 100 secondary reference standard laboratories in the various states which are providing the traceability to the district and working standard level laboratories. The objective of this program was to bring these laboratories under the ambit of accreditation. The accreditation as per ISO/IEC 17025 in the field of calibration for weights and measure was not possible due to the unavailability of funds with the states and NABL fee for the accreditation is around 2 lakhs per laboratory which includes the application fee, assessment fee, annual accreditation fee, etc. Also, the availability of manpower is not sufficient in most of the states; therefore it was decided that first the state level laboratories will be recognized and fee charges will be minimum. The parameters which can be considered for the recognition were:

(i) Qualification, experience, and competence of the personnel available with the laboratories
(ii) Training as per the ISO/IEC 17025:2017 and relevant standard of the calibration of weights and measures or any other training, for example, as per OIML standards, etc.
(iii) Availability of the instruments for calibration of the weights and measures

(iv) Environmental conditions of the laboratory
(v) Exposure to the measurement of uncertainty in the field of calibration of weights and measures
(vi) Any other requirement of NABL and Legal Metrology Department of center government

It was agreed the program will be launched jointly for the recognition of state level laboratory. The state level laboratory will send the application to NABL and fee will be minimal. Once the application will be accepted by the NABL and found that application is complete in all respect, the onsite visit will be arranged. The team for the onsite visit will include NABL representative and one member from the Legal metrology Department. The team will submit the report of onsite visit along with their recommendation. The report shall be reviewed by the accreditation committee. The accreditation committee consist of the experts of weights and measures and expert of Legal Metrology rules and regulations. Once the accreditation committee recommends the case, the approval shall be given by the CEO-NABL and Director Legal Metrology Department, Ministry of Consumer Affairs. The joint certificate of recognition by NABL and Legal Metrology Department shall be issued to the laboratory for the period of 2 years. To maintain the recognition the laboratory has to face the onsite surveillance visit. Also, after the recognition the laboratory may initiate for the accreditation of the laboratory based on the availability of funds and personnel in between the recognition cycle. In case a laboratory has successfully participated in the PT program of weights and measures, the laboratory may be considered for the accreditation based on the onsite visit for the management system as per the ISO/IEC 17025:2017, or in case laboratory has not participated successfully in PT program, then full-fledged assessment shall be conducted for the accreditation as per the ISO/IEC 17025:2017.

The NALM-ACRE program is under consideration, and hopefully it will be executed to upgrade the quality of the state level laboratories of the state government.

Government Approved Test Center (GATC) Program

The GATC program was already discussed, why this program is required, and it was decided that The GATC program will be in two phases. In phase-I only the government labs shall be considered for the GATC approval. The labs of Government organizations/semi-government/state government/autonomous body/public sector shall be considered, whereas in the later stage phase-II will be implemented and all the private labs will be considered for the GATC approval.

The instruments covered under the scope of GATC are as follows:

1. Water meter
2. Sphygmomanometer
3. Clinical thermometer

4. Automatic rail weighbridges
5. Tape measures
6. Non-automatic instrument of accuracy class-IIII/Class-III (up to 150 Kg)
7. Load cell
8. Beam scale
9. Counter machine
10. Weights of all categories

Here, it can be seen all the above instruments are very important for the daily use, especially weights, weighing balances, and volumetric vessels. The common citizen of the country cannot survive without accurate measurement of weights, volumetric vessels, and accurate performance of the weighing balances.

There were all the requirements given in the GATC act 2013, and all the responsibility and authorities were with Legal Metrology Department of Center Government, Ministry of Consumer Affairs. It has been realized by the government that the Legal Metrology Department cannot execute GATC program independently due to limitation of personnel and funds. To overcome this limitation and to execute this program, it was expected that NABL can be best option because in the last decade NABL has grown like anything and NABL has expertise in third party assessment. The personnel and expertise of NABL can be used for the execution of GATC program. After many meetings with chairman-NABL and higher officials of the Ministry of Consumer Affairs and Legal Metrology Department, it was decided that GATC program will be jointly executed. When joint execution of GATC program was agreed between NABL and Legal Metrology Department, Ministry of Consumer Affairs, it was observed that GATC rules need to be amended so that NABL may work jointly for the approval of GATC from the private sector.

Two meetings between Chairman-NABL, along with other officials of NABL, and higher officials of the Ministry of Consumer Affairs, along with the director and other officials of Legal Metrology Department, were held in the month of November 2020.

The amendments in the GATC act were suggested by the NABL to the Legal Metrology Department, Ministry of Consumer Affairs, and to decide the required amendments in the GATC act two more meetings between the NABL officers and Legal Metrology Department officers were held in the month of December 2020. After these meetings the amendments were finalized and agreed by the Legal Metrology Department, and modus operandi was decided between the NABL and Legal Metrology.

To improve the quality of weight and measures across the country, the Ministry of Consumer Affairs, Government of India, released the gazette notification for the approval of GATC from the private sector on 1 February 2021 (The Legal Metrology 2021), **the gazette notification was as follows**:

MINISTRY OF CONSUMER AFFAIRS, FOOD AND PUBLIC DISTRIBUTION

(Department of Consumer Affairs)

NOTIFICATION

New Delhi, 1st February, 2021

G.S.R. 95(E) (G.S.R 2021).—In exercise of the power conferred by sub-section (1) read with clause (n), (o) and (p) of sub-section (2) of section 52 of the Legal Metrology Act, 2009 (1 of 2010) (hereinafter referred to as the Act), The Central Government hereby makes the following rules further to amend the Legal Metrology (Government Approved Test Centre) Rules, 2013 (hereinafter referred to as the rules), namely: -

1.

(1) 1. Short title and commencement. — (1) These rules may be called the Legal Metrology (Government Approved Test Centre) Amendments Rules, 2021.

(2) They shall come into the force on the date of their publication in the Official Gazette.

2.

In the Legal Metrology (Government Approved Test Centre) Rules, 2013, in the rule 5, after sub-rule (16) the following sub-rule shall be inserted, namely: -

"(17) The National Accreditation Board for Testing and Calibration Laboratories ISO: IEC 17025:2017 accredited laboratories for calibration of weights and measures which are in the conformity with the conditions specified in sub-rule (3) of rule 5 shall be eligible to be notified as Government Approved Test Centre subject to compliance of the provisions of the Act and the rules made thereunder"

[F. No. WM-19(105)/2020]

ANUPAM MISHRA, Jt. Secy.

Note:

The principal rules were published in the Gazette of India, Extraordinary, Part II, Section 3, sub-section(i), *vide* G.S.R. number 593 (E). dated 5th September, 2013 and last amended *vide* notification number G.S.R. 94 (E). dated 20th January, 2016.

After the gazette notification, the Director, Legal Metrology Department, issued a notice on 04.08.2021 for "inviting application for Recognition/notification of NABL accredited testing/calibration laboratories for GATC purpose under legal Metrology." This notice was sent to all the NABL accredited testing/calibration laboratories accredited in the field of weights and measures and the companies engaged in the manufacturing of weights and measures, particularly the manufactures of weights, balances, volumetric vessels, measuring tapes, etc. All the concern laboratories are informed regarding the approval of GATC. The detailed information about the approval of GATC may be obtained from the legal Metrology unit, Department of

Consumer Affairs website https://consumeraffairs.nic.in/organisation-and-units/orders (https://consumeraffairs.nic.in/organisation-and-units/orders-seen) by the legal Metrology. In this notice the applications are invited by the interested parties/laboratories, which are willing to be the GATC.

In the continuation of this notice, the director of Legal Metrology, Ministry of Consumer Affairs, Government of India, New Delhi, issued an office memorandum on 06.08.2021 to all the Controllers of Legal Metrology of all states and union territories, and the subject of office memorandum was "Implementation of GATC." The matter of the memorandum was:

> "It is to inform that there is a provision of Government Approved Test Centre (GATC) in section 24(3) of Legal Metrology Act 2009. It has been implemented by the department. The application may be submitted through CEO, NABL. The details in this regard are available at the department website www.consumeraffairs.nic.in (www.consumeraffairs.nic.in-seen)"

The primary condition for getting GATC approval is NABL accreditation as per ISO/IEC 17025, which is mentioned in the Gazette notification published by the Legal Metrology Department, Ministry of Consumer Affairs. There are two steps which are required for GATC approval. i.e. Accreditation as per ISO/IEC 17025, and specific requirements of the Legal Metrology Department as per GATC Act 2013.

Here, it can be seen that the center government is very serious to upgrade the quality of weights and measure used for trade and commercial purpose. Due to shortage of personnel at center and state level, the work of Calibration/Verification / Stamping can be handed over to the private accredited laboratories.

As the notification has been already issued to all the states that NABL accredited laboratories shall be considered for the approval of GATC, application for GATC approval shall be sent to the NABL and NABL shall conduct the visit for the approval of GATC. There are certain requirements of the Legal Metrology Department which need to be fulfilled by the laboratory, and these requirements shall be verified by the joint team of NABL and Legal Metrology Department. The GATC approval shall be valid only till the validation of accreditation, and we can see now the importance of accreditation in the common citizen of our country. If we look 35–40 years back, the stones and other things were used to weigh the things, but slowly the stones were rooted out. The cast iron weights have taken over, but still there was a question of accuracy of the weight. To verify the mass of the weight within permissible limit, the Legal Metrology Department started calibration/verification/stamping of the weights used for the trade and commercial purpose.

Now the calibration/verification/stamping will be done by the GATC and primary condition to be the GATC is accreditation by NABL as per ISO/IEC 17025:2017 then only the laboratory can get GATC approval. The accreditation has entered in the life of common men of India. The weights which are used by the shopkeeper or any trader shall be verified by the NABL accredited laboratory.

In the case of secondary reference state level and working standard, laboratories in our country are more than 1400, and these laboratories are performing calibration/verification/stamping for the weights and measure. The question can be raised that

very few laboratories are accredited; how do these laboratories perform calibration/verification/stamping for weights and measure? The answer is very simple: the personnel working in these laboratories are well trained as per the requirements of the Legal Metrology Department. Also, all the Legal Metrology laboratories of our county are following the requirements of International Organization of Legal Metrology (OIML). In technical term all the state level laboratories of the country are working as the requirements of the International Organization of Legal Metrology (OIML), but still the government is very serious about the accreditation of these laboratories. The Regional Reference Standard Laboratories (RRSL) of the Legal Metrology and few state level laboratories are already accredited by NABL. As per the requirements of International Organization of Legal Metrology (OIML), all Regional Reference Standard Laboratories (RRSL) maintained their traceability particularly in the case of weights from National Physical Laboratory (NPL), New Delhi, India. The state level laboratories maintain their traceability from the Regional Reference Standard Laboratories (RRSL) and down the line state level laboratories provide traceability to the working standard laboratories at district level. It can be seen that most of the state level laboratories are not accredited, but they are getting traceability from the RRSL laboratories, and these laboratories are NABL accredited; in this way accreditation is impacting the working of state level laboratories.

In the last 40 years the impact of accreditation increased like anything in our country. The accreditation is required for so many things. It is required not only for scientific and research purpose but in daily life also. In the coming future, the impact of accreditation will definitely increase because most of the regulators in India require accreditation for the approval purpose. There are more than 4 lakh laboratories in our country which are engaged in testing, calibration, and medical testing, but the no. of accredited laboratories is around 7000. It can be seen that the 7000 accredited labs are significant, but still there is a huge opportunity in the field of accreditation. The day is not far away when calibration, testing, and medical testing of accredited laboratory will be accepted. The knowledge of common citizen is increasing day by day, and in the future the common citizen will ask for the accreditation for any type of testing and calibration, and in this context the Government of India felt the requirement of accreditation for the common citizen.

The GATC approval could be the revolutionary action by the Government of India. After the approval of GATC from the private sector, the calibration/verification/stamping may be possible for all the weights and measure available for the trade and commercial purpose in the country. The laboratories and manufactures of weights, weighing balances, volumetric vessels, steel scale, measuring tape, etc. will definitely come forward getting GATC approval. The GATC approval can only be obtained after the NABL accreditation. The government has understood the importance of the accreditation in the field of testing and calibration. The accreditation is the guarantee of competence of testing and calibration.

Many times, it has been seen due to poor quality of weights and measures there is a loss for the manufactures/producer and sometimes for the consumer of that weight and measure. A real time example of calibration of weight can be considered. A

sugar mill has sent the cast iron weights of 20 kg, for the calibration, and the results were surprising as one of the 20 kg weight was actually found 20.5 kg. When the management came to know about the actual value of the 20 kg weight, they were shocked and surprised because they have never thought that a single weight can have a difference of 500g. When the management of that sugar mill calculated the loss due to the use of that weight, it was in lakhs of rupees annually. After getting the unacceptable results of that weight, the weight was changed by the management, and calibration was done for the new weight. As the sugar mill has not got verification and stamping of that weight by the Legal Metrology Department due to the nonavailability of Legal Metrology Department in the sugar mill, in the present scenario the sugar mill was in the loss because whatever weight was weighed by that weight was 500 g plus for every 20 kg. The situation can be reversed; just imagine the weight of that weight were 19.5 kg, meaning 500g less as compared to the actual value. In this situation whatever weight measured with the help of that weight 500 g less was weighed for every 20 kg and consumer of sugar was getting less quantity. The situation is not healthy for the producer and for the consumer in either way. The objective of Legal Metrology is to ensure the quantity of weight and measure so that appropriate quantity should be given to the consumer and both manufacturer and consumer should not be at loss.

Another example of dimensional measurement can be taken. The measuring tape was purchased by the customer from the local market. The person who has purchased that measuring tape has taken the measurement for the door and given the order to the carpenter. The carpenter has made the doors based on the measurement given by that person. It was shocking to that person once he found the length of doors were approximately 2 inches short and were not fitting in the frame of the door. When investigation was done, it was found that the measuring tape was not proper. The initial measuring point was approx. 2 inches short. There was a manufacturing defect in that measuring tape. Because of the wrong measurement, that person suffers the loss of approximately 2 lakh rupees. The importance of correct measurement can be understood with the help of two examples discussed.

There are many processes and products which are used in daily life that required accurate measurement. The weights and measure are used daily in everybody's life; therefore the measurement of weights and measure should be accurate, or in other words it should be within the permissible limit so that the producer and consumer both should be in a win-win situation.

Reverse Transcription–Polymerase Chain Reaction (RT-PCR) Laboratories Accreditation

Another importance of accreditation was felt in the health sector specially for the medical testing laboratories. It can be seen that the previous situation the MRA is important for the testing and calibration laboratories as the results produced by the accredited laboratory in the field of testing and calibration are accepted in the member countries of APAC and ILAC. The retesting and re-calibration of the

exported item is not required if the exporting and importing countries are MAR signatory in APAC and ILAC. This situation was rarely faced in the field of medical testing. Because the medical tests performed on human beings and the results are frequently changed due to the complexity of environmental conditions and stability of the human samples. It has been observed that tests performed by one medical testing laboratory for some hospitals are not acceptable to the other hospital if the patient changes the hospital for his or her treatment. The repeatability and reproducibility are the core issue in the human samples for medical testing laboratories. There are various issues which affect the results of the testing. But during the pandemic of coronavirus disease of 2019 (COVID-19), everything has been changed from the technique of the assessment to the acceptability of the results of medical testing. The techniques of assessment changed due to the requirement of the accreditation not only at country level but at global level.

When the COVID-19 hits India in the month of March 2020 and the Government of India was forced to announce the country-wide lockdown, everything was closed except the essential services related to the health and food supply. NABL was also not able to conduct the assessment of the laboratories because there were no means of transportation to travel from one place to another place due to the strict lockdown. On the other hand, the government required more and more RTPCR testing laboratories for the testing of COVID-19 patient. The number of labs available for the RTPCR was less than 100, and to cater the need of COVID testing more and more RTPCR testing laboratories were required. As the matter was very serious and concerned with the human health, the Indian Council of Medical Research (ICMR) was in the search of RTPCR testing laboratories not only from the government sector but also from the private sector. To assure the quality of testing, what tool can be applied, the answer was again accreditation. ICMR (https://www.icmr.gov.in-seen) issues the order on 21.03.2020 in which the guideline laid down for COVID-19 testing in private laboratories in India, that NABL accredited laboratories for RTPCR testing shall be accepted. After this announcement many laboratories have applied for the accreditation, but there was a very serious constraint that during the lockdown NABL was not able to conduct the onsite assessment. The reason for not conducting the assessment were:

Non-availability of mode of transportation, air/train/car/bus, due to strict lockdown all over the country.

Non-availability of accommodation in the hotels, etc.

Non-availability of assessors due to the COVID-19; the assessors were avoiding the onsite assessment.

Most of the laboratories were in the red zone, and they want to apply for the NABL accreditation but not able to reach to the laboratories.

The requirement for the RTPCR testing was on peak, and only test results of NABL accredited laboratories were accepted by the ICMR; therefore every laboratory was in the need of accreditation.

Due to the unavoidable situation for the accreditation of RTPCR laboratories, NABL developed the program for the accreditation of RTPCR applicant laboratory

by means of online/remote assessment. The online/remote assessment was never done by NABL in any of the fields in the past 40 years. The concept of online/remote assessment can be understood by the definition of online/remote assessment "assessment of the physical location or virtual site of a conformity assessment body using electronic means"; here a virtual site is an online environment allowing persons to execute processes, e.g., in a cloud environment (ISO/IEC 17011:2017). It is evident that there was no physical visit of the assessors on the site of the applicant or renewal of the conformity assessment body. There is certain limitation during the online/remote assessment that good quality of Internet connection is required and sometimes it is very difficult to judge the observation of the witness of the test conducted for the competence of the conformity assessment body. The documents can be shown within some limitation, only one type of document can be shown to the assessment team, and in the online/remote assessment one assessor may witness one activity whereas in the physical visit one assessor may witness multiple activities. The quality of the assessment may be under doubt, and there was no past experience in conducting the online/remote assessment for any of the lab. Still NABL has opted the online/remote assessment for the RTPCR testing laboratories in the case of applicant as well as renewal of the RTPCR testing laboratories. The question raised was why did NABL opt for online/remote assessment whereas in the past there was no online/remote assessment conducted? The simple answer for this question can be the requirement of RTPCR testing laboratories to cater the need of testing in the pandemic of COVID-19. Now, another question may be raised why accreditation is required for the RTPCR testing? Why not non-accredited RTPCR testing laboratories are allowed to test the COVID-19 patient? It has been already discussed that in the case of medical testing as per ISO 15189:2012, the MRA signatory is not so much significant which in the case of testing and calibration laboratories as per ISO/IEC 17025, still because of the requirement of ICMR and the safety of the patient, accreditation is required for the acceptance of test results in the RTPCR testing by private laboratories. The accreditation makes sure about the operation of the laboratory with reference to the technical competence, use of traceable instruments, controlled environmental and accommodation conditions of the laboratory, and procedure of collection, transportation, and storage of the COVID-19 sample as per the guideline issued by ICMR. Accreditation makes sure that a specific laboratory is competent to perform the specific testing; therefore risk of wrong diagnosis and treatment is very less or it can be minimized. The requirement of accreditation for the private laboratories was not limited to the laboratories, but it was required to travel from one country to another country. In the recent past there are some states in India which have made RTPCR testing mandatory to enter in the state. Also, the states which have made RTPCR testing mandatory and someone wants to travel to that state, he or she can be allowed to board in the aircraft only after the negative RTPCR test report. In case of positive RTPCT test report, then the passenger will not be allowed to board in the aircraft. Similarly, the person with positive report will not be allowed to enter by road in the state. There are certain states which are conducting RTPCR tests when anyone wants to visit the state. The condition for the doctors,

administrative authorities, and state governments for accepting the result of RTPCR test is NABL accreditation.

Due to the requirement of accreditation for the acceptance of RTPCR test, the growth of accredited laboratories was remarkable, and presently there are more than 1700 laboratories. These laboratories were able to cater the testing need of the COVID-19 patent at the time of peak of COVID-19. There was a time when India was getting more than four lakhs COVID-19 patients and number of tests conducted were more than 15 lakhs to 20 lakh tests. Presently in India more than 16 lakhs RTPCR tests are conducted for COVID-19 patients. Today all the labs which are conducting RTPCR testing for the COVID-19 patients from the private sector are NABL accredited because only after the NABL accreditation the private RTPCR testing laboratory can be recognized and after the recognition the results can be accepted. NABL has done tremendous job during the time of pandemic by accrediting the RTPCR laboratories.

RTPCR testing is not only required in India for travelling from one state to another, but it is mandatory for travelling abroad. Most of the countries have banned the flights due to pandemic, and when flights were started, they required RTPCR test report, and this test report shall be an accreditation body which has MRA signatory status in APAC and ILAC. In international flights a person without negative RTPCR test will not be allowed to board. Here, it can be seen the importance of accreditation in the case of RTPCR testing not only in India but abroad also.

The overall scenario of the medical testing laboratories is that there are more than 80000 medical testing laboratories and the number of accredited laboratories in the field of pathology testing is around 1500. Also, in the case of medical treatment accreditation is not mandatory, and it can be seen that there are many labs which are still working without NABL accreditation but same is not true in the case of RTPCR testing. As the ICMR issued the order only NABL accredited laboratories testing shall be accepted, and the reason was only assurance of the results produced. For other types of the laboratories in the field of medical testing, a similar type of the guidelines may be issued by the other departments so that in the future slowly all labs may come under the ambit of accreditation.

Recognition of Food Testing Laboratories by Safety and Standards Authority of India (FSSAI)

Safety and Standards Authority of India (FSSAI) is responsible for maintaining the quality of food items manufactured, storage, sale, and import of food items for the human consumption. To regulate the food items for the human consumption, the food testing laboratories are required, and there are many laboratories which are engaged in the food testing. FSSAI has recognition scheme for the laboratories based on the certain requirements. The first requirement for the recognition of food testing laboratory is accreditation of the laboratory as per the ISO/IEC 17025:2017 or latest version of the standard and specific requirements/guideline of the NABL and second requirements of the FSSAI related to the qualification and experience of the

personnel in the field of food testing. Here, it can be seen that the laboratory willing to get recognition with FSSAI is required NABL accreditation. It was observed on many food items the mark of FSSAI. The mark of FSSAI on the human consumption food items whether manufactured in India or imported from any other country bearing the mark of FSSAI means that the food item is tested in the recognized laboratory and that the laboratory fulfilled the requirements for the accreditation as well as specific requirement of FSSAI. FSSAI has segregated the laboratories in various types, for example, level 1 laboratory, level 2 laboratory, referral food laboratory, R & D capabilities, training facilities, and other facilities. All these laboratories have specific requirements for the grading.

Recognition of Testing Laboratories by Agricultural and Processed Food Products Export Development Authority (https://www.apeda.gov.in-seen) (APEDA)

Agricultural and Processed Food Products Export Development Authority (APEDA) recognized laboratories for its registered importers. As the demand of Indian agricultural products is increasing day by day in the world, safety and quality of the agricultural product is required for the export purposes. The importing countries have certain requirements in the food products which are imported from India. There are certain international standards which need to be fulfilled by the product, and these food products need to be tested in the testing laboratories. APEDA has a scheme to recognize the food testing laboratories in the field of residual analysis of pesticides, heavy metals, drugs and antibiotics, aflatoxins and microbiological analysis, etc. There are certain requirements for the recognition of performing the testing for the agricultural products. It is stated in the requirement for the recognition that NABL accredited laboratory in the field of residual analysis of pesticides, heavy metals, drugs and antibiotics, aflatoxins and microbiological analysis etc. will be given preference if they follow latest version of ISO/IEC 17025.

Recognition of Testing Laboratories by the Bureau of Indian Standards (BIS)

The Bureau of Indian Standards (BIS), the National Standards Body of India, was established under The Bureau of Indian Standards Act, 1986, now revised to The Bureau of Indian Standards Act 2016. BIS has various conformity schemes for the protection of the interest of the consumers. ISI mark is one of the schemes to protect the interest of the consumers of various products produced by the Indian manufactures. BIS has recognition scheme for the manufactures from India and abroad. There are certain requirements, and accreditation of the laboratories is one of the requirements along with the compliance with various international and national standards of the product testing. Accreditation means the laboratory shall be accredited as per the ISO/IEC 17025:2017 and other requirements of the NABL along with required

product testing standards. BIS also declared in their requirement for the recognition that the accreditation body shall be a full MRA signatory member of ILAC (International Laboratory Accreditation Cooperation) and APAC (Asia Pacific Accreditation Cooperation). MRA signatory means that the accreditation body has been peer evaluated by APAC and eventually gets MRA status with ILAC.

Recognition of Testing Laboratories by Other Regulators

There are many other regulators, departments, and ministries which relay on the test results produced by the NABL accredited laboratories. The few examples are Central Pollution Control Board (CPCB) and State Pollution Control Board (SPCB) which accepts testing results of NABL accredited laboratories for the licensing of the various industries for operating in the particular region. The license is given by the Ministry of Environment, Forest and Climate Change (MoEF) to the various industries to operate in the particular region. Also, Central Pollution Control Board (CPCB) and State Pollution Control Board (SPCB) are endeavoring for getting NABL accreditation for their laboratories. Some of the laboratories of CPCB and SPCB are already accredited in the field of environmental testing.

The Ministry of Water recently started the program for the accreditation of water testing laboratories across the country. Slowly the water testing laboratories are applying for the accreditation in the field of water testing so that policies can be framed based on the results produced by these laboratories situated in the various states of the country.

Export Import Authority of India is also relaying on the NABL accredited laboratories. The EIC is giving recognition to the accredited laboratories only. Accreditation has entered almost in every department and ministry where testing is essential and decisions are taken on the test results produced by the laboratories, whether the laboratory is private or a part of the government organization or department.

Conclusion

It is evident that today accreditation has entered everywhere, from the science and technology to household items of the country. By looking back in the early 1980s, there was no concept of accreditation in the country, although weight and measure department was well established at the center as well as state level. In the 1970s RRSL laboratories were started in the various places in the country to support weights and measure of the country. India is a big country not only area wise but population wise also, and it was desperately required to assure the quality of weights and measures for the general use of the public. The calibration and testing done by the accredited laboratories is acceptable for the scientific, research, and development purpose and for commercial purpose verification and stamping of weight and measure done by the Legal Metrology Department is acceptable by the law. Today

Legal Metrology Department at center and state level is willing to get NABL accreditation for their laboratories. Also, as per the GATC rule 2013, private laboratories can be the part of calibration/verification and stamping. Legal Metrology Department, Ministry of Commerce, decided that only NABL accredited laboratories can apply for the GATC approval. Also, manufactures of the weight and measures may apply for the GATC approval. After the gazette notification many manufactures of the weights, weighing balance, measuring tape, volumetric vessels, etc. are willing to get the GATC approval.

If present scenario of country is considered in terms of accredited laboratories, there are more than 7500 accredited laboratories in the field of testing, calibration, medical testing, PTP and RMP. Out of these 7500 accredited laboratories, the no. of testing laboratories is more than 4000, and calibration laboratories are more than 1100, where medical testing laboratories are more than 1500. The accredited laboratories in the field of testing cover every parameter/product. For the common citizen's point of view, the common parameters can be food and agricultural products which consist of human food like milks and related products and other products produced in agriculture which are suitable for the human consumption, pharma products which consist of various types of medicines and other related products, and textiles which consist of various types of clothes and other products which are manufactured by the various textiles; even building material can be tested for the assurance of their quality, and the soil, water, and air are also tested for the suitability of their use for the establishment of manufacturing units in the particular area.

In the last 40 years, the importance of the accreditation has been increasing day by day. That day is not far away when common men will ask for the testing by the accredited laboratory. Today the common men are not aware of the benefit of the accreditation level. However, government and related departments are well aware of the benefit of the accreditation, and slowly the common men are getting awareness about the accreditation of the testing laboratories. The person or patient prefers the NABL accredited laboratory in the medical testing. Also, whenever any customer wants to purchase any product for the daily use, he prefers ISI mark product. As discussed the ISI mark is given by the BIS for the various types of products used by the customers.

In the future, the number of accredited laboratories will increase because there are many laboratories which are performing testing and calibration without accreditation. However, accreditation has already entered into the life of the common citizen of India.

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Quality Measurements and Relevant Indian Infrastructure

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Abstract

In the modern era, competition is more intense than ever before as modern business becomes more diversified, bigger, and improved. Therefore, it is crucial to confirm that an organization is providing customers with the best selection of products in order to maintain momentum and compete with its rivals. Companies can only guarantee the quality of deliveries by doing rigorous quality checks in advance. Strong national quality infrastructure with high standards is the only way to accomplish this goal. A strong national measurement system (NMS),

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quality infrastructure (QI), and stricter metrological regulations and certification in trade and commerce are required due to the fast industrialization, growing demand for more precise measurements, updated technologies, and highly skilled workforce. Any nation that wants to engage in international trade and the exchange of services, especially developing nations like India, must have a strong QI. Through social, strategic, and industrial expansion, the QI is essential to the country's economic development. The upper organizations in India that are in charge of existing QI are CSIR-National Physical Laboratory (for scientific and industrial metrology), Bhabha Atomic Research Centre (for nuclear radiation metrology), Department of Legal Metrology (for legal metrology), Bureau of Indian Standards (for documentary standards and specifications), and National Accreditation Board for Testing and Calibration Laboratories (NABL) and National Accreditation Board Certification Bodies (NABCB) under Quality Council of India (for accreditation and certification). In this chapter, the role of quality measurements and relevant Indian infrastructure is briefly discussed. For establishing, implementing, and confirming the quality criteria for goods and services, a QI is an effective tool. For domestic markets to operate effectively as well as to have access and to be recognized on the global stage, high-quality infrastructure is essential. Along with promoting and preserving economic development, it is crucial for social and environmental harmony.

Keywords

Quality infrastructure · Metrology · Accreditation · Standards · Certification

Introduction

After World War II, with the rise in demand for various consumer and industrial products, there arose the need to regulate the increase in world trade. Various organizations were formed in different parts of the world. Some countries came together to create the International Trade Organization (ITO) under the UN's Conference on Trade and Employment within the framework of the United Nations. However, they could not reach a common consensus and the talks for the creation of ITO broke down. Following this, around 23 countries met in Geneva in 1947 and agreed to regulate international trade by way of a general agreement. This was the onset of the General Agreement on Tariff and Trade (GATT). After eight rounds of talks and negotiations on trade-related issues, the World Trade Organization (WTO) was created in 1996 with its headquarters in Geneva with more than 95 percent of developing countries as its members. WTO's prime objective is to supervise and liberalize international trade and to negotiate and formalize trade agreements between member countries through their governments, with the aim of achieving free trade between the countries. Though every country wants to protect its animal life and safety, and in this garb sets some barriers to free trade. Technical rules, product standards, requirements for testing and calibration, and different inspection

and certification procedures between nations are examples of these trade barriers, also known as technical barriers to trade (TBT). The prime objective of WTO is to reduce these technical barriers to trade so that the regulations, standards, testing, and certification procedures do not create unnecessary hurdles in free trade (Overview of India's Quality Infrastructure 2018; Aswal 2020a, b; Rab et al. 2020, 2021a).

In the last quarter century, globalization of trade made it possible for the public to enjoy a vast range and quality of products manufactured in other countries. In order to have smoother international trade, it is essential for all products of the same category to conform to the same or similar standards. Thus, a product tested in one country and conforming to the standards of the importing country should not be required to be rechecked after importing. Rather, the result of testing in the exporting country should be acceptable to the importer in the other country. This is one of the objectives of WTO, which translates to "tested once, accepted everywhere." To control conformity assessment activities around member countries, a conference held in 1977 facilitated international trade by accepting accredited test and calibration results. This was formalized in 1996 as cooperation between 44 countries in Amsterdam and was known as International Laboratory Accreditation Cooperation (ILAC) (Hoekman and Kostecki 2009; Grochau et al. 2020; Patra and Mukhopadhayay 2019).

On a regional scale, the Asia Pacific Accreditation Cooperation (APAC) was initiated as cooperation in 1992 and was formally established as a regional forum in April 1995 with the signing of a memorandum of understanding (known as APLAC MOU) by 16 regional economies in the Asia Pacific region. Similarly, in the other parts of the world, such cooperations were established between member economies. The European Cooperation for Accreditation (EA), Inter-American Accreditation Cooperation (IAAC), South African Development Community in Accreditation (SADCA), African Accreditation Cooperation, and Arab Accreditation (ARAC) were later established in other regions. All these regional bodies and their member accreditation bodies operate through various bilateral and multilateral mutual recognition arrangements (MRAs). This helps in fulfilling the requirement "tested once, accepted everywhere" (Rab et al. 2021a, b; Czichos et al. 2011). section "Introduction" of this chapter provides background information and discusses the need for stronger infrastructure (QI). The development of measurement systems was then covered in section "Development of Measurement Systems" from both national and international perspectives. In section "Relevant Indian Infrastructure," a number of pertinent organizations connected to Indian QI are briefly described.

Development of Measurement Systems

Technological advancements after World War II created a need for a coherent system of units that can be used by one and all as different countries followed different systems of units earlier. The Treaty of the Metre (May 20, 1875), known as the Metre Convention as well, established an international conference for designing and

developing units for measurements. This conference is known as the General Conference on Weights and Measures (CGPM). The International Bureau of Weights and Measures (BIPM) was established at Sevŕes near Paris as an organization for custodians of International Standards. In order to facilitate the decisions in CGPM, an International Committee for Weights and Measures (CIPM) was established as an advisory body to CGPM. While CIPM is the body that runs the BIPM activities through its Directorate, CGPM is a diplomatic body to endorse any change in the international standards. On receiving representations from the International Union of Pure and Applied Physics (IUPAP), CGPM asked CIPM to study and come out with a practical system of units that can be adopted and followed by the whole world.

BIPM is the custodian of seven base units of the system called SI unit system which was adopted by the CGPM in 1960. In addition to these seven base units, there are several derived units that are byproducts of powers of the base units, with no factor other than 1. These derived units are known to represent the derived quantities and are expressed in terms of the base units. Dimensionally too, they are expressed in terms of the dimensions of the base units. All the measurable physical quantities can now be expressed in terms of base or derived units (Rab et al. 2020, 2021a; Rab and Yadav 2022; Stock et al. 2019).

The global metrological hierarchal structure is somewhat like as given in Fig. 1. The standards maintained at BIPM derive their reference value directly from SI units. From this, the national standards of various countries, maintained at their

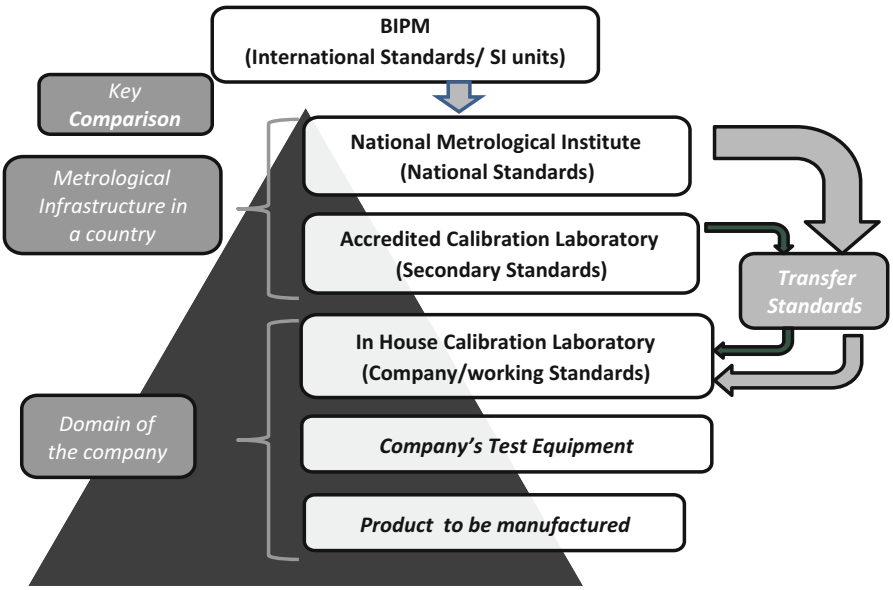

Fig. 1 Traceability transfer from SI units (BIPM) to working level

respective National Metrological Institutes, derive their reference values through comparison, called "key comparison." These national standards, at various levels of accuracy and uncertainty, form the backbone of any nation's traceability structure. NMIs are usually also the custodian of the primary standards unless otherwise mandated by the government. Other standards (secondary standards) draw their reference value and uncertainty from the national standards maintained at NMIs. From these secondary standards, various working standards or shop floor instruments are calibrated in accredited laboratories. In this fashion, the traceability from the SI units maintained at BIPM is transferred to the working level. It is always ensured that the chain of traceability is an unbroken one.

The key comparisons among all permanent members of CGPM are organized by Consultative Committees (CCs) on behalf of CIPM, under the framework of CIPM MRA. Inked in 1999, this MRA is for National Measurement Standards and for Calibration and Measurement Certificates issued by NMIs. This MRA is also a step toward giving an open, transparent, and comparable environment for agreements by concerned governments for fair trade within economies and nations where the calibration results produced in one country are acceptable to another. It is also a way to reduce or demolish trade barriers.

The National Measurement System

The post-World War II period witnessed the application of physical measurements over a vast range of applications in daily life and industry. With the advancement in technology, especially electronic measurements, and information technology, these physical measurements proliferated at a rate like never before. This also resulted in a massive expansion in the measurement-related budgets in every country, as such measurements were essential for public health, safety, and the environment. Major users of such measurements were government and private entities alike. Public welfare programs like a supply of safe drinking water or measurement of air pollution consumed a lot of public money. It necessitated the need for making the right policies and regulations to control the wasteful expenditure related to the measurement system. "If the direct cost of making measurements is large, the indirect cost of making poor measurements must be huge." A nation's measurement requirements are diverse just because the implementation of various economic and social schemes is dealt with by many different regulations, both at the national and sub-national levels. The economic ecosystem provided by industries, traders, consumers, etc. must thrive along with the social ecosystem (Gupta et al. 2020; Yadav and Aswal 2020; Fanton 2019). Such diverse requirements are difficult to weave into a single pattern. Effective policy comes from effective decisions and planning based on the right data. The data is used to create and make policies for the nation's social and daily life (Rab et al. 2022). Such a national measurement system structure can be defined to have five elements:

(i) Conceptual system: where measurement quantities and units are defined.
(ii) Technical infrastructure: to implement the defined quantities and units and to provide techniques for implementation.
(iii) Realization of measurement capabilities: to realize the measurement quantities.
(iv) Enforcement network: to disseminate and enforce the regulations, decisions, and policies.
(v) End use measurement: to support the different levels of system.

Such a national measurement system provides a coherent system that ensures that consistent measurements are made throughout the country. This provides a qualitative basis and quantitative data for taking effective decisions in policy making. Good data for taking effective decision-making not only requires measurement in the laboratory but also requires all other activities to support the measurement for good data. In any country, there are many activities that ensure the production of good data in the laboratory. The measurement infrastructure supports not only the market economy but also social infrastructure. This also requires technical infrastructure to support both economic and social infrastructure at various levels. This is known as "infra-technology," which includes measurement and test methods, applicable regulations and standards, implementation systems, and relevant training. Such "infra-technology" includes regulations, standards, metrology, standardization, accreditation/certification and training, etc.

Thus, a national measurement system must include:

(a) Setting national policy.
(b) Introducing metrology within that policy.
(c) Framing regulations.
(d) Provision for infrastructure for metrology.
(e) Engagement of other national and international bodies.

These five activities are mostly fall in the governmental role. Other roles are:

(a) Developing and maintaining documented standards.
(b) Appropriate metrological controls.
(c) Verification of regulated equipment/instruments.
(d) In-service checks of regulated equipment/instrument.
(e) Inspection and control of packaged materials.
(f) Market surveillance.

These six activities are the role of legal metrology system of the country.

Apart from above, the following roles, played by many governmental and private institutions, also contribute to making of national measurement infrastructure of a country:

(a) Maintaining national standards of measurements and dissemination of units.
(b) Maintaining and disseminating certified reference materials (CRMs).

(c) Technical advice.
(d) Type approval and controls.
(e) Calibration.
(f) Testing.
(g) Training.

These roles fall into the domain of the NMI, standard developing organizations, accreditation bodies, assessment and certification bodies, and training institutes. The government, legal metrology, and other bodies listed above must work in the most coherent manner to thrive the challenges faced by the country for economic, social, and technological requirements.

Relevant Indian Infrastructure

An emerging economy initially requires a trade measuring system with proper legislation for its transition. After that, additional legal metrology controls for global traceability and regulatory measurements can be implemented. A complete metrology system, complete with a thorough traceability infrastructure and national metrological controls, will exist in a developed economy. This section provides details on the quality infrastructure in India, including the current systems for metrology, standardization, accreditation, technical rules, conformity assessment, and support activities. The CSIR-National Physical Laboratory (CSIR-NPL) is the keeper of national measuring standards within the Indian Quality Improvement system; the Department of Legal Metrology makes rules and regulations and enforces metrology for trade and commerce; and the Bureau of Indian Standards (BIS) deals with documentary standards of goods and services, and accreditation bodies operate under the auspices of the Quality Council of India (QCI) in India, according to the country's QI system. Other nations also have their own NQIs, which may or may not be similar systems with numerous acting bodies and agencies operating at various levels (Aswal 2020b, c; Rab et al. 2021a; Mandal et al. 2021). Figure 2 provides an illustration of the general picture of the structures of any NQI. An NQI for conformity assessment is required by each of the government departments/regulators listed in Table 1 in order to implement various schemes, related to manufacturing, food, energy, health, environment, electronics, export, etc.

Legal Framework

Legal Metrology Act, 2009 (replacing the earlier Standards of Weights and Measures Act, 1976), provides for the "establishment and enforcement of standards of weights and measures, regulate trade and commerce in weights, measures and other goods which are sold or distributed by weight, measure or number and for matters connected therewith or incidental thereto." This Act replaces the earlier Act and provides for standard weights and measures, the establishment of an institution for

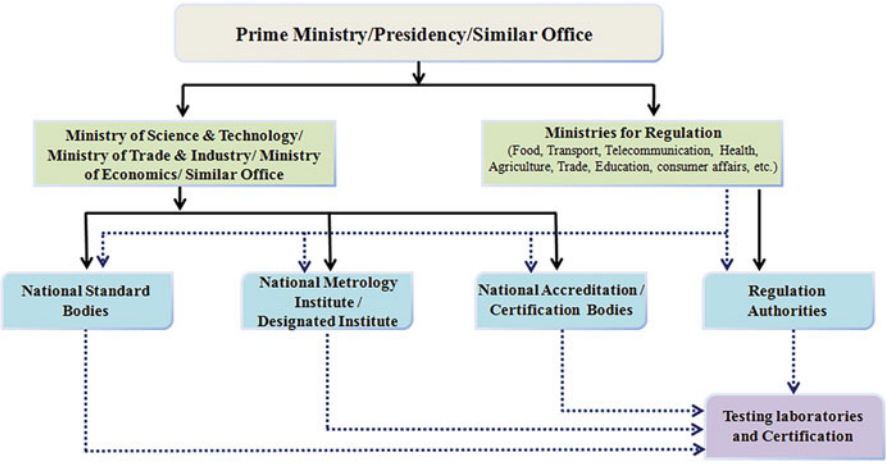

Fig. 2 Linkages of organizational relationships of the national quality infrastructure; continuous lines indicate oversight relationships, and dot-line relationships indicate coordination

controlling legal metrology rules framed under the Act, and provision for verification and approval of weights of measures and offenses and penalties thereon. Thus, this Act and the rules formed there under (Legal Metrology Rules 2011) are the backbones of the legal framework in India. This Act established that "every unit of weight or measure shall be in accordance with the metric system based on the international system of units. The base unit of (i) length shall be the meter; (ii) mass shall be the kilogram; (iii) time shall be the second; (iv) electric current shall be the ampere; (v) thermodynamic temperature shall be the kelvin; (vi) luminous intensity shall be the candela; and (vii) amount of substance shall be the mole." The Act further established that "the reference standards, secondary standards, and working standards of weights and measures shall be such as may be prescribed."

The Act provides clarity on the units further to be operated in the country. It provides the following: (i) the base units of weights and measures specified as above shall be the standard units of weights and measures; (ii) the base unit of numeration (an international form of Indian numerals) specified in shall be the standard unit of enumeration; (iii) for the purpose of deriving the value of base, derived, and other units, the central government shall prepare or cause to be prepared objects or equipment in such manner as may be prescribed; and (iv) the physical characteristics, configuration, constructional details, materials, equipment, performance, tolerances, period of reverification, methods, or procedures of tests shall be such as may be prescribed. These prescriptions are further elaborated in the Legal Metrology (General) Rules 2011 framed under the Act.

Legal Metrology (General) Rules, 2011, rules provide for, amongst other things, that calibration to the various measuring instruments shall be traceable to "National Measurement Standards." It also provides that for any transaction (in trade,

Table 1 Partial list of various bodies in India as a part of countries' conformity assessment

Name of organization/body	Status and control	Scope and subject area	References
Automotive research Association of India (ARAI)	Research Institute of the Automotive Industry with the Ministry of Heavy Industries and Public Enterprises, Govt. of India	Lay down the automotive industry standards and published safety standards	Automotive Research Association of India (ARAI) (2022)
Standardisation testing and quality certification (STQC)	Ministry of Electronics and Information Technology, government of India	Provides quality assurance services in the area of electronics and IT through countrywide network of laboratories and centers. The services include testing, calibration, IT, and e-governance, training, and certification to public and private organizations	Standardisation Testing and Quality Certification (STQC) (2022)
Telecom regulatory Authority of India (TRAI)	Jurisdiction of Department of Telecommunications, Ministry of Communications, government of India	Regulator of the telecommunications sector in India. The goal of the organization is to foster the conditions necessary for the country's telecommunications sector	TRAI, Telecom Regulatory Authority of India (2022)
Bureau of Energy Efficiency (BEE)	Agency of the Government of India, under the Ministry of Power	Provides energy performance standards for appliances' energy conservation building code Regulatory authority for mandatory energy labeling for eight types of appliances, notified by the central government	Bureau of Energy Efficiency (BEE) (2022)
Securities and exchange Board of India (SEBI)	Ministry of Finance, government of India	It is the regulatory body for securities and commodity market in India. It monitors and regulates the Indian capital and securities market while ensuring	Securities and Exchange Board of India (SEBI) (2022)

(continued)

Table 1 (continued)

Name of organization/body	Status and control	Scope and subject area	References
		to protect the interests of the investors, formulating regulations and guidelines	
Central drugs standard control organisation (CDSCO)	Directorate general of health services Ministry of Health and Family Welfare, Govt. of India	Lay down the standards for drugs and healthcare devices/technologies, and approve new drugs under the drugs and cosmetics act	Central Drugs Standard Control Organisation (2022)
Food Safety and Standards Authority of India (FSSAI)	An autonomous body established under the Ministry of Health and Family Welfare, government of India	Lay down standards for articles of food and regulate their manufacture, storage, distribution, sale, and import	Food Safety and Standards Authority of India (FSSAI) (2022)
Oil industry safety Directorate (OISD)	A technical advisory body in India. It was established in 1986 by the ministry of petroleum and natural gas	Provide product design, safety standards, codes of practices, and guidance standards in oil and gas sector	Oil Industry Safety Directorate (2022)
Telecommunication engineering Centre (TEC)	A nodal agency of the Department of Telecommunications, Ministry of Communications and Information technology, government of India	Formulate standards with regard to telecom network equipment, services and interoperability, the associated conformity tests, and fundamental technical plans. Responsible for drawing up of standards, generic requirements, interface requirements, service requirements, and specifications for telecom products, services, and networks	Telecommunication Engineering Center (2022)
Railways design and standards organization (RDSO)	Research and development organization under the Ministry of Railways of India	Provide standards for materials and products specially needed by Indian railways	Railways Design and Standards Organization (RDSO) (2022)

(continued)

Table 1 (continued)

Name of organization/body	Status and control	Scope and subject area	References
Atomic energy regulatory Board (AERB)	Regulatory body, Department of Atomic Energy	Notifying technical regulations and assessment, regulatory inspection and enforcement of sitting, construction, commissioning, operation, and decommissioning of nuclear and radiation facilities including medical and industrial radiography	Atomic Energy Regulatory Board (2022)
University Grants Commission (UGC)	Statutory body, under Ministry of Human Resource Development	Determining and maintaining standards of teaching, examination, and research in universities, framing regulations on minimum standards of education, monitoring developments in the field of collegiate and university education, disbursing grants to the universities and colleges, serving as a vital link between the union and state governments and institutions of higher learning, and advising the central and state governments on the measures necessary for the improvement of university education	University Grants Commission (2022)

commerce, etc.), "the product is required to conform to the specifications laid down by the International Organization of Legal Metrology about the matters aforesaid."

The legal metrology framework in India operates as a federal structure. This Act further established the administrative framework to control the proliferation of the weights and measures in India. In the center, the Department of Legal Metrology is established under the Department of Consumer Affairs, Ministry of Consumer Affairs, Food and Public Distribution, Government of India. This is the centralized

department to control the legal metrology framework requirements centrally. For the purpose of disseminating traceability for standards covered by weights and measures, there are currently five Regional Reference Standards Laboratories (RRSL) in operation. In the states, the Controllers of Legal Metrology are the premier officer to enforce Legal Metrology Act at the ground level. Additional Controller, Joint Controller, Deputy Controller, Assistant Controller, Inspectors, and other employees of the state exercise the powers and discharge the duties conferred or imposed on them by or under this Act in relation to intrastate trade and commerce. This framework is further strengthened by the functioning of more than 100 secondary standards laboratories and 1100 working standards laboratories for the dissemination of metrological traceability. These state authorities have been authorized to "inspect any weight or measure or other goods in relation to which any trade and commerce has taken place or is intended to take place and in respect of which an offense punishable under this Act appears to have been or is likely to be, committed are either kept or concealed in any premises or are in the course of transportation" (Rodrigues Filho and Gonçalves 2015; Velkar 2018; Pasaribu and Sirait 2020).

National Metrology Institute

The CSIR-National Physical Laboratory (CSIR-NPL) located in New Delhi is the National Metrology Institute (NMI) of India, engaged in keeping the National Physical Standards of the country, except for the radiation standards, which are kept by the Bhabha Atomic Research Centre (BARC). NPL functions as a premier laboratory under the Council of Scientific and Industrial Research (CSIR), Government of India. Established in 1947, one of its main functions is to "establish, maintain, and improve continuously by research, for the benefit of the nation, National Standards of Measurements and to realize the Units based on International System (Under the subordinate Legislations of Weights and Measures Act, 1956, reissued in 1988 under the 1976 Act)."

CSIR-NPL is a signatory to the International Bureau of Weights and Measures (BIPM) Mutual Recognition Arrangement (MRA). The National Physical Laboratory has the responsibility of realizing the units of physical measurements based on the international system (SI units) under the subordinate legislations of the Weights and Measures Act, 1956 (reissued under the Legal Metrology Act, 2009). NPL also has the statutory obligation to realize, establish, maintain, reproduce, and update the national standards of measurement and calibration facilities for different parameters.

The laboratory is maintaining at present six out of seven SI base units (except mol). However, some of the related activities related to mol are also being followed. The derived units for physical measurement for force, pressure, vacuum, luminous flux, sound pressure, ultrasonic power and pressure, and the units for electrical and electronic parameters, viz., dc voltage, resistance, current and power, ac voltage, current and power, low-frequency voltage, impedance and power, high-frequency voltage, power, impedance, attenuation and noise, microwave power, and frequency. Impedance, attenuation, and noise are also maintained by CSIR-NPL.

The national standards of physical measurement at CSIR-NPL are traceable to international standards. The laboratory periodically carries out an intercomparison of national standards with the corresponding standards maintained by National Metrology Institutes (NMIs) of other countries under the consultative committees of the International Committee of Weights and Measures (CIPM) and the member nations of the Asia Pacific Metrology Programme (APMP). The major implication of this exercise of establishing equivalence of national standards on measurement at CSIR-NPL with those of other NMIs is that calibration certificates issued by NPL would have global acceptability.

Certified reference materials (CRMs): CRMs are required and used to ensure high-quality measurement as well as to maintain the traceability of analytical measurements to national/international measurement system (SI unit) in order to fulfill the mandatory requirement of quality systems (as per ISO/IEC 17025 standard). The use of appropriate reference materials provides traceability to the measurement in the chemical field, where formal traceability to SI units is not possible. Two types of materials are recognized by ISO Committee on Reference Material (ISO REMCO), (i) certified reference materials (CRMs) and (ii) reference materials (RMs). Certified reference material (CRM) is used whenever the calibration cannot be strictly made in SI units and the traceability could not be established. The CRM should be provided by NMIs or a competent supplier, usually an accredited RMP (recognized material producer). The RMP is accredited according to ISO 17034, produces CRM as per ISO Guide 35, and issues certificates according to ISO Guide 31. Many laboratory-accredited bodies have accredited RMPs as per ISO 17034. The values given to CRM are considered to have metrological traceability.

In India, many RMPs have come up in the recent past, mainly due to development in the field of metrology, and also due to the requirement for the CRMs. The accreditation for the RMP's capability is assessed by National Board for Testing and Calibration Laboratories (NABL), which is APAC (Asia Pacific Accreditation Cooperation) MRA partner. The RMPs are capable of producing the CRMs on a continuous basis, although the values of the CRM are certified by CSIR-NPL. CSIR-NPL is engaged in coordinating an interlaboratory program on planning, preparation, and dissemination of certified Indian reference materials named as Bharatiya Nirdeshak Dravyas (BNDs). Many CRMs for petroleum and petroleum products, pharmaceutical materials, metals and alloys, building materials, etc. are now being made locally, and the values of these CRMs are provided by CSIR-NPL (Rab et al. 2020; Yadav et al. 2021; Mande et al. 2021; Garg et al. 2021; Gupta 2009).

Bureau of Indian Standards (BIS)

The Indian Standards Institution (ISI), as it was formerly known, was founded in 1947 with the aim of fostering the peaceful growth of standardization work in India. The Bureau of Indian Standards (BIS), the national standards body of India, has been actively involved in activities related to international standardization and has represented India's interests at various times as these standards have been developed.

BIS is a founding member of the International Organization for Standardization (ISO) and has been actively taking part in ISO and International Electrotechnical Commission (IEC) operations.

The BIS formulates Indian standards through technical committees, which are constituted of sectional committees, subcommittees, and expert panels established to address particular groups of topics. These committees and panels are represented by industries, consumers, regulators, policy makers, and people from academia. As a matter of policy, BIS's efforts to develop standards have been as consistent as practicable with the pertinent directives established by the International Organization for Standardization (ISO).

The bureau's primary duties include developing, recognizing, and promoting Indian standards, More than 21,500 standards are developed by BIS, covering crucial economic sectors that aid the sector in raising the caliber of its goods and services. The BIS has identified 15 divisions that are crucial to Indian business. Each division has a separate division council to oversee and supervise the work for the preparation of the Indian standard. To ensure compliance with the international standards, the standards are periodically reviewed and revised in accordance with technological advancement (Bantu et al. 2020; Miguel et al. 2021).

Quality Council of India (QCI)

It is the National Body for Accreditation registered as a society and functions under the aegis of the Department of Promotion of Industry and Internal Trade under the Ministry of Commence, Government of India. It functions with the support of the Government of India and the Indian industry represented by the three premier industry associations, (i) Associated Chambers of Commerce and Industry of India (ASSOCHAM), (ii) Confederation of Indian Industry (CII), and (iii) Federation of Indian Chambers of Commerce and Industry (FICCI). QCI is also a signatory to IAF (International Accreditation Forum) and ILAC (International Laboratory Accreditation Cooperation) MRAs. QCI with its accreditation boards, namely, National Accreditation Board for Certification Bodies (NABCB), National Accreditation Board for Education and Training (NABET), National Accreditation Board for Testing and Calibration Laboratories (NABL), National Accreditation Board for Hospitals & Healthcare Providers (NABH), and National Accreditation Board for Quality Promotion (NBQP), is engaged in accreditation activity of various proliferated institutes, laboratories, and hospital services. These five accreditation boards engage in accreditation programs and work under the umbrella of QCI (Fig. 3). Each board functions independently and only in its specific field of competence.

In addition to the above, there are scores of bodies engaged in the field of measurement and have established state-of-the-art laboratories. These are India's premier institutions for research and development. In the field of electrical measurements, Central Power Research Institutes (CPRIs) are engaged in the field of testing as well as research in the field electrical power sector. In addition, the Electrical Research and Development Association (ERDA) is also engaged in the field of

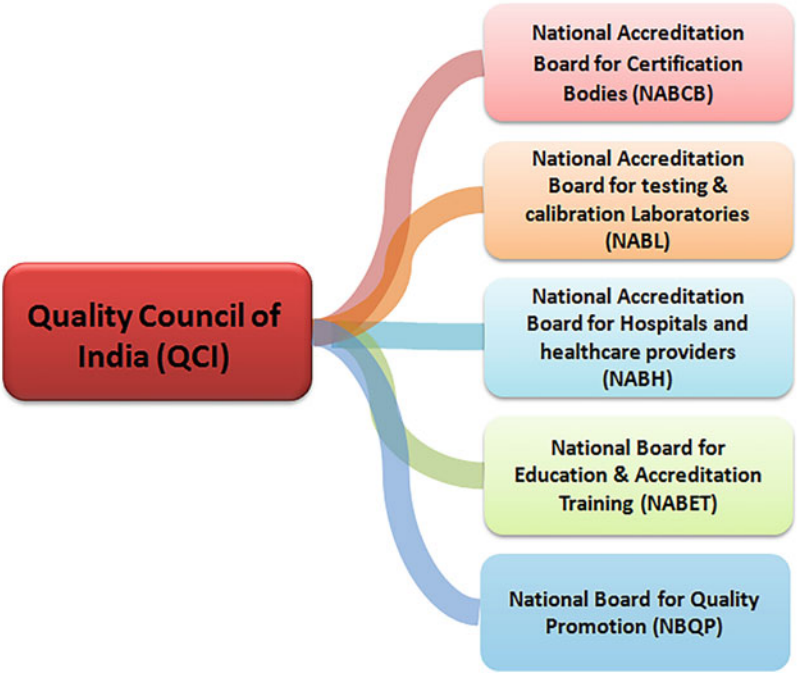

Fig. 3 Structure of quality council of India

research and testing. Another governmental organization, i.e., Electronics Research and Testing Laboratories (ERTLs), has a number of testing laboratories engaged in testing in the field of electronics. This also provides help in product development to the manufacturers.

In addition to the above, there are several organizations engaged in the research and development and testing of mechanical, metallurgical, civil, analytical chemistry, and other fields. These organizations have trained staff, state-of-the-art facilities, and a well-developed management structure to control the transfer of metrological traceability to the shop floor level.

Bhabha Atomic Research Centre (BARC)

The major nuclear research institution in India is the Bhabha Atomic Research Centre (BARC). It was established as Atomic Energy Establishment, Trombay (AEET), in January 1954 as a multidisciplinary research program crucial to India's nuclear program by Dr. Homi Jehangir Bhabha. It operates under the Department of Atomic Energy (DAE), Government of India. After Homi Bhabha passed away in 1966, AEET was renamed Bhabha Atomic Research Centre (BARC).

BARC's primary goal is to maintain and secure nuclear energy activities. It deals with all facets of atomic energy, from computerized modeling and simulation to theoretical reactor technology, novel reactor fuel material development, risk analysis, evaluation, etc. Under the authority of the Legal Metrology Act and in accordance with the CIPM MRA, CSIR-NPL had advised BIPM to designate BARC as the nation's Designated Institute (DI) status for ionizing radiation in 2003. BARC is now responsible for the establishment, maintenance, and dissemination of ionizing radiation in India. The organization has taken part in 52 intercomparison exercises and is currently moving forward with the appropriate steps to apply for the approval and registration of CMCs in ionizing radiation in close collaboration with CSIR-NPL.

Agricultural and Processed Food Products Export Development Authority (APEDA)

In accordance with the Agricultural and Processed Food Products Export Development Authority Act, which was approved by the Parliament in December 1985, the Government of India formed the Agricultural and Processed Food Products Export Development Authority (APEDA). Along with improving infrastructure and quality, APEDA has been actively involved in market development to boost the export of agricultural products. APEDA's plan scheme, the "Agriculture Export Promotion Scheme," aims to promote agricultural exports by offering financial support to registered exporters through the scheme's sub-components of market development, infrastructure development, quality development, and transportation assistance. It operates with 13 virtual offices and five regional offices in Mumbai, Bengaluru, Hyderabad, Kolkata, and Guwahati. APEDA connects the Indian exporters to the international market and offers comprehensive export services. It also offers referral services and makes suggestions for the appropriate joint venture partners.

Central Pollution Control Board (CPCB)

The Central Pollution Control Board (CPCB) of India is a department of the Indian government in charge of monitoring the quality of the air and water as well as any other pollution-related concerns. It is a legally recognized organization that is under the purview of India's Ministry of Environment, Forests, and Climate Change. Founded in 1974 under the Water (Prevention and Control of Pollution Act), it was later given duties and authority under the Air (Prevention and Control of Pollution) Act, which was passed in 1981. With seven zonal offices and five laboratories, the CPCB's headquarters are located in New Delhi. Given the complexity of environmental monitoring, experts from the metrology institute, academia, and industry must cooperate to enhance the measurement of hazardous emissions to protect human health and quantify climate-influencing factors. A nationwide certification program is being developed in India by CSIR-NPL in collaboration with the CPCB and the Ministry of Environment and Climate Change.

Conclusion and Way Forward

There is a growing awareness of quality infrastructure's significance to the economy and to society at large on a global scale. Better quality infrastructure is a requirement for free and equitable trade on a national and international level since it forms the basis of technical rules, documented standards, and legal metrology. Concepts like safety, security, efficiency, reliability, and precision are of utmost importance when creating systems that offer assurances of product quality in any institution, business, or organization. In order to smoothen market transactions, the consumers must have the confidence that purchased goods are of right quantity and quality, and for this accurate and commonly accepted measurements are essential. Importantly, precise and globally recognized measurements provide market access for the export of goods like food and commodities.

A brief description of the necessity for measurement and the national quality infrastructure is presented. The nations with well-established and reliable national quality infrastructure gain from a wide range of factors, such as increased participation in international trade, economic growth, and wealth. Advanced innovations, technologies, and product development are made possible through the knowledge gained and measurements made via standards and measuring procedures, which benefits both consumer protection and domestic commercial activity. A well-established, high-quality infrastructure is a need in any nation because it creates a win-win situation for all the stakeholders involved.

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An Insight on Quality Infrastructure

6

Shanay Rab, Sanjay Yadav, Meher Wan, and Dinesh K. Aswal

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Abstract

Globalization and Industrial Revolutions have not only nourished new opportunities and challenges but also caused unnecessary disruption and polarization within and between countries and society. In this case, the economic competition is fiercer than ever before in the modern world. As a result, for barrier-free import and export, a firm must guarantee that it provides consumers with a high-quality product selection to remain up with its competitors. Companies can ensure the

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quality of deliveries by completing rigorous quality checks ahead of time, and it can only be accomplished with a robust national *Quality Infrastructure (QI)*. Most of the nations have their own QI system, which supports government policy goals in industrial development, federal trade rules and control, food safety, efficient use of natural and human resources, health, the environment and climate change, etc. Several countries having highly strong QI have their industrial and economic growth healthy. On the other hand, some countries lack the infrastructure required to satisfy quality standards, posing concerns and challenges in today's competitive world. As a result, the QI plays a critical role in social, strategic, and industrial progress of any country. This chapter provides an overview of the institutional framework of QI with National and International perspectives. The study also describes the *Strengths, Weaknesses, Opportunities and Threats (SWOT)* analysis of the QI. Further, the *Global Quality Infrastructure Index (GQII)* has been updated with the inclusion of current data up to Dec 31, 2021, and summarized for the member states (63 countries) of the *International Bureau of Weights and Measure (BIPM)*. Hopefully, this study would benefit government agencies, industry, academia, and various organizations as a reference bank for future decisions and policymaking connected to robust QI.

Keywords

Quality infrastructure · Metrology · Accreditation · Standard · Certification · GQII

Introduction

The current Covid-19 pandemic is a watershed moment in world economic policy, and it has caused a number of recessions and has impacted worldwide trade. Unlike industrialized countries, which have been using significant financial resources to limit the damage to their economy, countries in the world do not have these resources – at least not to the same extent. As a result, focusing on long-term economic development has become even more critical for these countries' national economies. Aside from financial assistance, there are slew of long-term technical options for keeping the economy afloat during this downturn. These options include QI initiatives that would assist firms, and other economic stakeholders ensure the quality of their products and processes, allowing them to become more competitive. Quality has become increasingly linked to health, cleanliness, safety, resilience, and sustainability due to the ongoing global Covid-19 outbreak (Aswal 2020a, b; Naidoo and Fisher 2020; Iyke 2020; Aizawa 2019).

In an economy, a quality policy assures that the institutions responsible for metrology, standards, accreditation, and certification are well-established, vigorous, and operate together cohesively and in synergy. Such intuitions are firmly connected and constitute a network in a particular economy at the national level, i.e., *National Quality Infrastructure (NQI)*. The NQI is also linked to various international institutions and organizations, forming an *"International Quality Infrastructure" (IQI)*

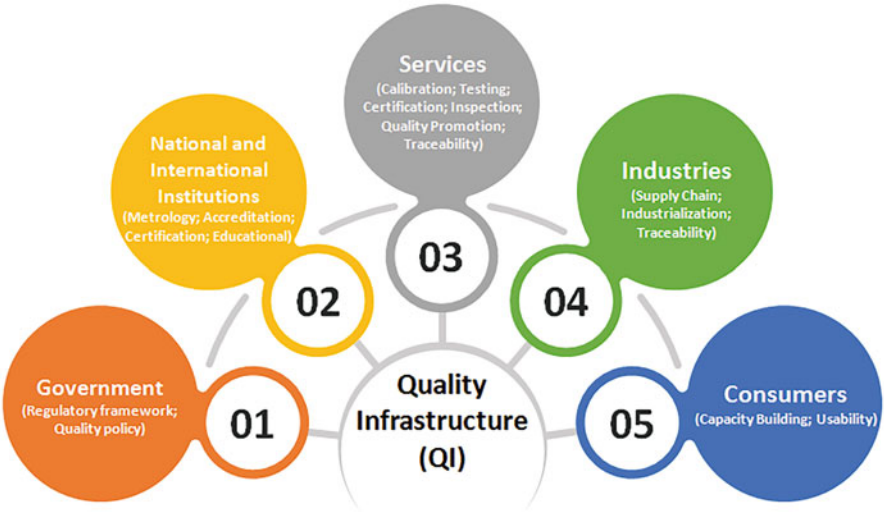

Fig. 1 Basic Quality Infrastructure Framework

network (Rab et al. 2021a, b; Harmes-Liedtke and Di Matteo 2011, 2019). Figure 1 exhibits the main linkages of government to the consumer in a QI system. In order to correlate the QI with well-being and quality of life, a global quality Infrastructure index (GQII) has been introduced (Rab et al. 2021b) including the data up to the end of the year 2020. In the updated version herein, we have used the data up to December 2021. The index uses publicly available sources from official websites of different constituents of IQI. The index is used to rank the global status of various countries in terms of QI. The indicator also shows how QI relates to other developmental indicators including economic performance, GDP per capita, competitiveness, economic complexity, export performance, etc.

The linkages between appropriate national institutes and international institutes in the case of NQI are formed depending on national requirements. The *World Bank* and other donor organizations aid many countries in developing NQIs to promote industrial development, removing trade and entrepreneurial barriers, and facilitate global technical cooperation (Tippmann 2013; UNIDO 2019; Clemens and Kremer 2016). Some of the significant organizations/agencies directly or indirectly connected with QI are summarized in Table 1.

As witnessed in Table 1, there are many organizations working for the IQI, for understanding the harmonization, uniform and barrier-free trade. It is interesting to grasp, examine, and analyze the functions of the QIs to further strengthen them. A SWOT analysis of the QI system is also performed to understand the different aspects of QI, like what are the strengths and weaknesses of the QI system and how these can be utilized into opportunities to make the QI system strong and robust. Further in the implementation, the anticipated threats are identified and highlighted. With the help of the updated QI data in GQII, this chapter also provides a comprehensive overview of the QI of 63 BIPM member states.

Table 1 A list of major organizations/institutes/groups related to QI

Name of the organization	Description
United Nations (UN)	The *United Nations (UN)* is an intergovernmental organization whose mission is to promote international peace and security, foster cordial relations among nations, foster international collaboration, and serve as a focal point for harmonizing nations' actions. The UN was established in 1945 and is currently headquartered in New York City on the international territory; and has offices in Geneva, Nairobi, Vienna, and The Hague, with a current total membership of 193 countries (United Nations 2020)
World Trade Organization (WTO)	The *World Trade Organization (WTO)* is an international organization that regulates and supports international trade. The organization is used by governments to establish, update, and enforce international trade rules. It began operations on January 1, 1995, due to the 1994 *Marrakesh Agreement*, which replaced the 1948-established *General Agreement on Tariffs and Trade (GATT)*. The WTO is headquartered in Geneva, Switzerland, and has 164 member countries (World Trade Organization 2019)
United Nations Industrial Development Organization (UNIDO)	The *United Nations Industrial Development Organization (UNIDO)* is a UN specialized organization that helps countries to improve their economies and industries. The *UN General Assembly* founded UNIDO in 1966, and it is located at the UN Office in Vienna, Austria, with 170 member states (United Nations Industrial Development Organization 2020)
The World Health Organization (WHO)	The *World Health Organization (WHO)* is a UN specialized organization which deals with international public health. The WHO was established in 1948 and is headquartered in Geneva, Switzerland. The WHO has six regional offices and 150 field offices worldwide, currently with 194 members (World Health Organization 2020)
International Organization for Standardization (ISO)	The *International Organization for Standardization (ISO)* is a global standard-setting organization composed of various national standards bodies' representatives. The ISO was founded in 1947 and is headquartered in Geneva, Switzerland, with 167 national standards bodies (International Organization for Standardization 2019)
International Electrotechnical Commission (IEC)	The International Electrotechnical Commission (IEC) is a non-profit organization that develops and publishes the international standards for all electrical, electronic, and related technologies, commonly known as "electrotechnology." The IEC was founded in 1906 and is headquartered in Geneva, Switzerland. There are currently 88 countries that are members of the IEC (2020)

(continued)

Table 1 (continued)

Name of the organization	Description
International Telecommunication Union (ITU)	The International Telecommunication Union (ITU) is a UN-specialized organization in charge of all information and communication technologies issues. It was founded as The International Telegraph Union in 1865, making it the oldest UN agency (The International Telecommunication Union (ITU) 2019). The ITU is a multinational organization with 193 member countries based in Geneva, Switzerland
ASTM International	*ASTM International*, formerly known as the *American Society for Testing and Materials*, founded in 1902, is an international standards organization that develops and publishes voluntary technical standards through consensus for a wide range of materials, products, systems, and services (ASTM International 2021)
The International Atomic Energy Agency (IAEA)	The *International Atomic Energy Agency (IAEA)* is the international organization for nuclear-related research and technological cooperation. The IAEA was founded in 1957 and is currently headquartered in Vienna, Austria, with 173 member states (International Atomic Energy Agency 2021)
International Accreditation Forum (IAF)	The *International Accreditation Forum, Inc. (IAF)* is a global organization of Conformity Assessment Accreditation Bodies and other organizations involved in conformity assessment in the sectors of management systems, products, services, personnel, and other such of the conformity assessment program. The IAF was established in 1993 due to the inaugural meeting of "Organisations that Accredit Quality System Registrars and Certification Programs" in Houston, Texas, on January 28, 1993 (The International Accreditation Forum (IAF) 2020)
International Laboratory Accreditation Cooperation (ILAC)	The *International Laboratory Accreditation Cooperation (ILAC)* first began as a conference held in Copenhagen, Denmark, on October 24–28, 1977, to foster international cooperation to facilitate trade by promoting the acceptance of accredited test and calibration results. The ILAC was founded later on in 1996 as a formal collaboration with a charter to create a network of mutual recognition agreements among accreditation agencies. In calibration, testing, medical testing, inspection, proficiency testing providers, and reference material manufacturer's certification, ILAC manages the international arrangements (International Laboratory Accreditation Cooperation 2021)
World Meteorological Organization (WMO)	The *World Meteorological Organization (WMO)* is a United Nations specialized organization that promotes worldwide collaboration in atmospheric science, climatology, hydrology, and geophysics. The WMO comprises 193 countries and territories, allows the "free and unlimited" exchange of data, information, and

(continued)

Table 1 (continued)

Name of the organization	Description
	research among its members' meteorological and hydrological organizations. The WMO is headquartered in Geneva, Switzerland, and it is governed by the World Meteorological Congress, which meets every 4 years to define policies and goals (The World Meteorological Organization 2020)
International Bureau of Weights and Measures (BIPM)	The *International Bureau of Weights and Measures (BIPM)* is an office formed under the Metre Convention and consultation with The International Committee for Weights and Measures (CIPM) that allows member states to collaborate on measurement science and standards. The BIPM was established on May 20, 1875. Following the signing of the Metre Convention, a treaty among 17 Member States that has since grown to 63 (International Bureau of Weights and Measures 2019)
National Metrology Institutes (NMIs)	Each country has a single *National Metrology Institute (NMI)*, which is responsible for maintaining national measurement standards and distributing the International System of Units. Most of the countries have their own NMI, which is internationally recognized and part of a worldwide framework that ensures comparable measuring standards and capabilities. The directors of the state institutes for metrology from all 48 member countries of the *Metre Convention* were invited to sign the *Mutual Recognition Agreement (MRA)* on the mutual recognition of national measurement standards and calibration and measurement certificates at the 21st *General Conference of Weights and Measures (CGPM)*, held in Paris under the auspices of the BIPM. The phrase *"National Metrology Institute"* was established worldwide in the MRA, and signed by 38 countries represented at the CGPM. The acronym NMI is used all across the world for such an institute (Rab et al. 2021a; Kind and Lübbig 2003)
International Organization of Legal Metrology (OIML)	The *International Organization of Legal Metrology (OIML)* is an intergovernmental organization founded in 1955 to assist economies in establishing effective legal metrology infrastructures that are mutually compatible and internationally recognized in all areas where governments are responsible, such as trade facilitation, mutual confidence, and harmonization of consumer protection levels around the world (International Organization of Legal Metrology 2020)
Group of 20	The G20 consists of 19 countries as well as the European Union. With almost 80% of global GDP, 75% of international trade, and 60% of the world's population, the group plays a key role in ensuring global economic growth through the development in QI (Bradford 2016)

Significance of Sustainable Development Goals (SDGs) and G20 in QI

Protecting the environment, ensuring dignified lifestyles for all people, and achieving inclusive economic growth and prosperity are all related to QI. The UNIDO is assisting QI constituents in several countries across the world. It has helped to create and strengthen numerous QI-related institutes by supplying cutting-edge equipment and building technical skills. The 2030 Agenda for Sustainable Development, which was launched in 2015, is a beacon around which all countries, businesses, and civil society can unite with the common goals of decreasing poverty and establishing a more sustainable and equitable world. The SDGs are a blueprint for a more prosperous and sustainable future for everyone. The SDGs address global issues such as poverty, inequality, climate change, environmental degradation, peace, and justice (THE 17 GOALS; UNIDO).

The QI also helps countries to achieve the SDGs in the most efficient way possible. International commerce is regarded as a source of wealth and poverty reduction, and it improves the efficiency of domestic markets and promotes access to global markets. It is accomplished through quality assurance, standard compliance, and addressing consumer demands at home and overseas. The QI assists in meeting market demands, and it can address social and environmental issues while attempting to avoid excessive trade obstacles. The UNIDO is now developing a QI Scorecard for Sustainable Development, which will make the link between SDGs and QI even more visible and measurable (UNIDO 2019).

The pandemic's economic impact is exacerbated by supply chain disruptions that have put the manufacturing industry on hold and declining commodity prices. This has roiled financial markets, tightened liquidity conditions in many nations, resulted in extraordinary capital outflows from poor countries, and put pressure on foreign exchange markets, with some countries facing dollar shortages. Local currency weakness limits the government's ability to implement fiscal stimulus on the scale required to stabilize the economy and address the health and humanitarian crises.

To deal with such a financial crisis, The G20 was established in 1999 with the goal of discussing policies that would lead to global financial stability. It is a global strategic platform that brings together the world's leading industrialized and emerging economies. The G20 includes Australia, Argentina, Brazil, China, Canada, France, India, Germany, Indonesia, Italy, Japan, Korea, Mexico, Saudi Arabia, Russia, South Africa, the United Kingdom, the United States, Turkey, and the European Union. The G20 has a critical role in ensuring global economic growth and prosperity in the future (Cooper and Thakur 2013; Kirton 2016).

The *G20 Hangzhou Summit* in September 2016 defined *Quality Infrastructure Investment (QII)* as an investment "which aims to ensure economic efficiency in view of life-cycle cost, safety, resilience against natural hazards, job creation, capacity building, and transfer of expertise and know-how on mutually agreed terms and conditions while addressing social and environmental impacts and aligning with economic and development strategies" (The G20 or Group of Twenty 2021; The 2016 G20 Hangzhou Summit 2016).

Fig. 2 The G20 Principles for Promoting Quality Infrastructure Investment

At the G20 Finance Ministers' and Central Bank Governors' Meeting in Fukuoka, Japan (8–9 June 2019), the leaders reaffirmed the need for QI by embracing the Principles for QII. "Maximizing the positive impact of infrastructure to achieve sustainable growth and development while preserving the sustainability of public finances, raising economic efficiency in view of life-cycle cost, integrating environmental and social considerations, including women's economic empowerment, building resilience against natural hazards and other risks, and strengthening infrastructure governance" were among the recommendations in their communiqué (Yoshino et al. 2019; Langston and Crowley 2022; Jaramillo et al. 2021). The G20 Principles for Promoting QI Investment are exhibited in Fig. 2.

Similarly, there are other groups of countries such as the G7, BRICS, APEC, African Union, European Union, and other collaborations for national developments. For example, Canada, France, Germany, Italy, Japan, the United Kingdom, and the United States are members of the Group of Seven (G7), an intergovernmental political forum. By 2020, the collective group will control a little more than half of the global net wealth. The majority of the members are international leaders in politics, economics, social issues, legal issues, environmental issues, military, religious, cultural, and diplomatic connections.

Quality Infrastructure and Its Baselines

QI studies have been carried out in order to better understand and compare economic developments and performance. As previously stated metrology, standardization, accreditation, and certification are the four fundamental foundations or bases of any country's QI. Through calibration, testing, certification, verification, and inspection, these pillars ensure conformity assessment. The national quality policy of a country

ensures that metrology, standards, accreditation, and certification agencies are well-established, robust, and collaborate. Such institutions are interconnected and form a QI network for any country (Aswal 2020a, b).

Figure 3 shows these major bodies associated with the QI system, for example, in Indian NQI, CSIR-NPL is responsible for scientific and industrial metrology (NMI); *Bhabha Atomic Research Centre (BARC)* is responsible for nuclear radiation metrology as a *Designated Institute (DI)*; the Legal Metrology Department is responsible for legal metrology; *Bureau of Indian Standards (BIS)* is working for documentary standards and product specifications, and *Quality Council of India (QCI)* is

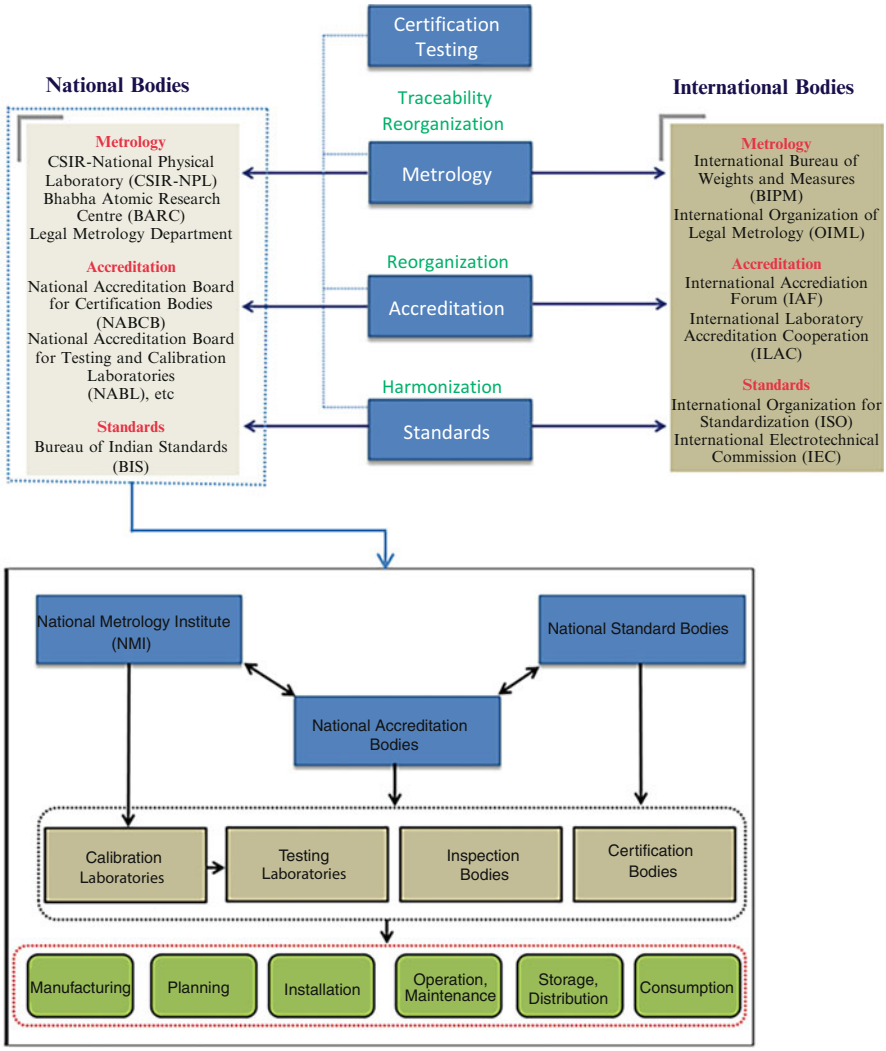

Fig. 3 National and international bodies associated in the QI system; a case of country India

responsible for accreditation and certification (Rab et al. 2021a, b). There are few more institutions or bodies like Agricultural and Processed Food Products Export Development Authority (APEDA), Food Safety and Standards Authority of India (FSSAI), Express Industry Council of India (EICI), Engineering Export Promotional Council of India (EEPCI), Projects Export Council of India (PECI), Manufacturers' Association for Information Technology (MAIT), Organisation of Plastic Processors of India (OPPI), National Association of Software and Service Companies (NASSCOM), National Green Tribunal (NGT), Indian Stainless Steel Development Association (ISSDA), Federation of Indian Export Organisation (FIEO), Central Drugs Standard Control Organisation (CDSCO), Atomic Energy Research Board (AERB), Telecom Regulatory Authority of India (TRAI), etc. apart from several other financial and insurance regulators, which are directly and indirectly connected with Indian NQI.

The major four constituents of the QI system are briefly described in the subsequent section.

Metrology

One of the central pillars of the QI system is metrology. The beginning of modern metrology, which is based on the metric system, may be traced back to France's adoption of the metric system in the 1790s. Following that, the definitions of the SI unit system were developed under the authority of CGPM and supervision of CIPM, as part of an international standards body having its working office known as BIPM in France (International Bureau of Weights and Measures 2019).

Metrology is essential for almost all walks of human life. It is critical for environmental conservation, ensuring dignified and quality lives, and creating inclusive economic growth and prosperity. It is necessary for trade, scientific comparison, innovation and developing technologies, technical cooperation, and simply basic information exchange. Building good metrological traceability leads to better QI, which is a step toward establishing the right policy framework conditions and the rule of law, which are essential for a prosperous economy (Rab et al. 2020; Yadav and Aswal 2020; Zafer et al. 2019).

However, it has been observed that not all countries have the same metrological capabilities. On the other hand, some countries have comprehensive and robust metrology programs, while others either do not have or do not have appropriate QIs. One of the parameters for evaluating such capabilities is documented as CMCs, which have the highest credibility and reliability. After a rigorous process as per CIPM MRA, these CMCs of the NMIs are published on *Key Comparison Database (KCDB)* of BIPM (Rab et al. 2022a, b). The NMIs participate in the key comparisons to evaluate if their measurement standards and procedures are comparable. *Consultative Committees (CCs)* or *Regional Metrology Organizations (RMOs)* conduct such comparisons, referred to as Key or Supplementary Comparisons. As a result, the participation in frequent and the number of comparisons results in better reliability, compatibility of measuring standards, and confidence in the measurement standards.

For the regulatory framework, each country requires a measurement system as well as proper legislations. Legal metrology ensures fair commerce through accurate measurements by transferring the benefits of metrology to ordinary residents of the country through suitable legal enforcement. As a result, legal metrology is an important aspect of the technical regulation system, and such legal legislations are framed to fulfill the country-specific requirements and by and large agree to the WTO's Agreement on Technical Barriers to Trade (TBT Agreement) criteria.

Accreditation

The accreditation is basically started in laboratory services after World War II and was coordinated by the *ILAC*. The accreditation through *Certification Bodies (CBs)* came into existence much later and is being managed by the *International Accreditation Forum (IAF)*. The ILAC and IAF, two international accreditation organizations, are in the process of merging. The key mechanism for cross-border recognition of conformity assessment services is accreditation. The global and regional accreditation co-operations control the acknowledgment of the technical competence and impartiality of *NABs*. The NABs sign the ILAC MRA and the IAF *Multilateral Recognition Agreement (MLA)* for this purpose. The signatories agree to recognize accreditations and conformity assessment.

In most nations, there is only one *Accreditation Body (AB).* On the other hand, some have multiple ABs, like several ABs competing in India in various accreditation areas. In some exceptional cases, bi-national and regional ABs are also present. The *Joint Accreditation System for Australia and New Zealand (JAS-ANZ)*, for example, is in the existence as CB for accreditation for both countries. Furthermore, not all countries or economies have a governing agency for certification. The critical need for developing a NAB is sometimes lacking in smaller and less developed countries. Some of these economies have created *National Accreditation Focal Points (NAFPs)*. These competent bodies operate as information sources in nations that do not have accrediting bodies. These accredited bodies in various areas could lead to the diffusion of their country's competency, authority, and credibility (Cross-Frontier Accreditation 2021; GQII 2021).

Standardization and Certification

As part of the *Technical Barriers to Trade (TBT)* agreement, the *WTO* encourages adopting international standards as best practices. It demands involvement in standardization initiatives, including participation by developing nations. The *ISO* develops international standards in a variety of fields, except for electrics and electronics, which are handled by the *IEC*, and telecommunications, which are regulated by the *ITU* (Rab et al. 2021a, b; World Bank 2019; QI4D 2022).

SWOT Analysis

The SWOT analysis describes the strengths, weaknesses, opportunities, and threats based on the information acquired from the networks studied for QI. The SWOT analysis of QI is examined focusing on a better understanding of the restrictions that countries confront throughout the investment cycle and the inefficiencies that undercut the value-added proposition. Table 2 describes the SWOT analysis of QI.

Table 2 SWOT analysis of QI

Strength	Weakness	Opportunity	Threat
• Reduction of barriers to trade • The cornerstone for the development of the economy and society • Safety of industrial production sites • Facilitation of digitalization and innovation • Protection of consumers from unsafe products • Clean environment and climate change • Sustainable energy • Improving public health	• Lack of coordination between the central bodies of QI (Metrology, Accreditation, Standards, and Conformity assessments) • Lack of adequate government support to the QI bodies • Comparatively frequent interventions of government organizations • Regulations should be for organization booster but not for hindrances • The duality strategy is setting high requirements for exports while lowering them for domestic consumers • Because the private sector is weekly organized, the government is more likely to intervene in the market. In this regard, technical laws are frequently preferred, whereas industry self-regulation is weaker	• Development and harmonization of metrology, standards, accreditation, conformity assessment, and market surveillance • Innovation in standardization of products, processes, and organizational arrangements • Strengthen the corporation between various countries and advanced development of metrological infrastructure • Funds for research and development are scarce, so QI bodies rely frequently on the finances arranged by themselves or depend on international developmental funding • National country-specific standards formation in line with international standards	• Parity exists between the developed and underdeveloped countries, which reflect the less QI in underdeveloped countries • Conflict may arise between technological, innovations, and legal framework • Maintenance of infrastructure and facilities • Funding issues with the developing and underdeveloped countries • Export restrictions • Import dependency • The higher price of domestic products, due to inclusion of cost for metrology, standards, and accreditation

As discussed earlier, the goal of a SWOT analysis is to identify the strengths and weaknesses of the QI, as well as opportunities and threats in various areas of the QI system. The pillars of QI (Metrology, Accreditation, Standardization, and Certification) play a key role in improving technology development and industrialization, affordable healthcare and safety, clean environment and climate change, sustainable energy, international trade facilitation, and the creation of a level playing field in society, according to a SWOT analysis. The analysis also indicates the need of more synchronization and synergy between all the pillars of QI keeping metrology as the focal theme for enhancing the growth of the economy and quality of life.

Updated GQII and Its Significance

The GQII formula includes the primary major component of the QI system. The authors have defined critical metrics for each component to measure the country's QI development state. The GQII formation was extensively explored and reported in our previous publication (Rab et al. 2021b). The present studies provide the GQII metrics on QI and its components for BIPM member states in an updated version. This allows us to evaluate and rank the progress of a country's various QI elements. On January 19, 2021, the Republic of Estonia also became the permanent member state of BIPM (2021). Thus, the GQII ranking has been calculated for all the 63 states compared to the 62 countries in the previous edition. The GQII follows the equation as:

$$GQII_i = (C_1 + C_2 + C_3 + C_4 + C_5 + C_6) \times \frac{100}{6} \tag{1}$$

where C_1, C_2, C_3, C_4, C_5, C_6 are defined as follows:

$$C_1 = \frac{CMC_i/P_i}{(CMC/P)_{max}}; C_2 = \frac{K\&SC_i}{K\&SC_{max}}; C_3 = \frac{ISO_i/P_i}{(ISO/P)_{max}}; C_4 = \frac{TC_{si}}{TC_{max}};$$
$$C_5 = \frac{E_{Edui}\%}{E_{max}}; C_6 = \frac{M_{ITCi}}{M_{max}}$$

$i = Country_1, Country_2, Country_3, \ldots Country_n$

P_i = Country population

CMC_i = Total number of calibartion and measurement capabilities

$K\&SC_i$ = Total number of key and supplementry comparisons

ISO_i = Total number of vaild ISO Standards issued (9001, 14001, 22000, 13485, and 27001)

TC_{si} = Total number of technical committees participations according to ISO

E_{Edui} % = % expenditure of GDP on Education

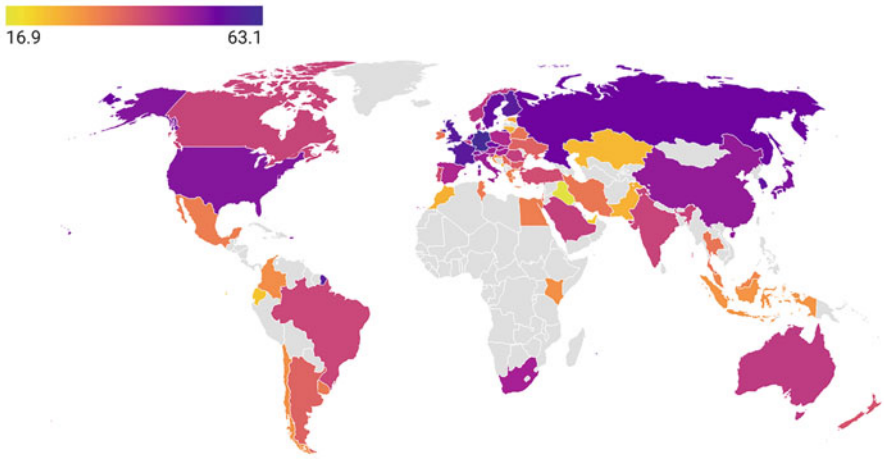

Fig. 4 BIPM member states visualization with GQII (**The world map is only intended to be used as a visual aid and does not indicate any view on the legal position of any country or territory or the delimitation frontiers or boundaries, has been expressed*)

Membership (M_{ITCi}) = Number of memberships of international QI system (WTO, IAF, ILAC, CIPM, OIML, ISO, IEC, ITU)

Figure 4 depicts the current situation of QI from a geographical standpoint. The figure demonstrates that European countries have rather well-developed QIs. In addition, several African countries have limited quality assurance agencies and are attempting to improve their scientific and industrial metrological capabilities.

Figure 5 shows the comparison of nations' GQII rankings with their GDP per capita rankings. The GDP per capita is used as a global metric for determining a country's wealth (GDP). GDP per capita, in its most basic form, shows how much economic production value can be given to each person of a country. It also aids in comprehending how the economy grows in tandem with the population. These QI indicator correlations are critical for determining the role of a country's competitiveness and economic development. As per GQII ranking, a country with a well-developed QI is economically affluent, whereas a poorly developed QI is economically disadvantaged. The ranking clearly shows that higher CMCs, participation in more key comparisons (BIPM 2020) and technical committees' more ISO certifications (The ISO Survey) and higher economic investment in education (World Bank 2020) certainly contribute to a higher GQII score for a country. Only openly available data in metrology, standards, certification, and education have been considered for this indicator. Accreditation data is not included in the methodology since it is either unavailable or provided in a very complex and variable manner from country to country. The ranking and GQII score of each country is shown in the appendix (Table A).

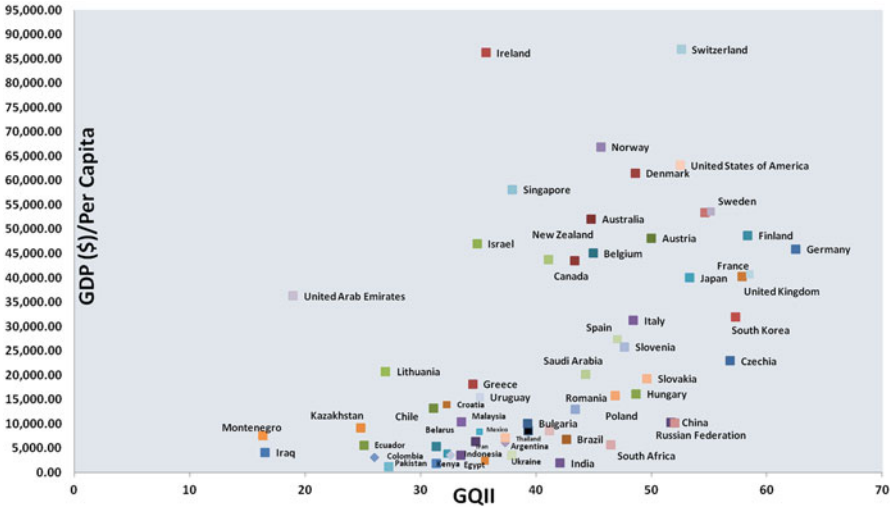

Fig. 5 GDP ($) per capita with GQII

Discussion and Way Forward

The transition to a more sustainable and inclusive economy is one of the current generation's most pressing issues. Climate change, biodiversity loss, and pollution are all becoming increasingly worrisome. These environmental crises are intertwined with the world's numerous development challenges. During the COVID-19 epidemic, digitalization increased substantially. The pace of digital change will continue to pick up. Innovation and the massive volumes of data generated can enable governments to progress considerably on their development agendas. The concentration of power among digital behemoths, on the other hand, produces dependencies that stifle competition.

Meanwhile, new difficulties such as privacy, security, and accountability arise. As each country adopts and improves its NQI, it is also responsible for meeting technology needs, minimizing superfluous technical trade barriers, and addressing environmental, health, and safety concerns. The process of developing an internationally recognized NQI takes time, and policymakers must decide which sort of NQI is most suited to their country's future engagement with the global economy. Indeed, authorities are grappling with transforming the systems they inherited and restructuring the conformity assessment infrastructure in a timely and cost-effective manner.

Germany (1st), France (2nd), and Finland (3rd) are the top three countries in this edition of the GQII ranking. In the case of India, the GQII ranking this year is 29 out of 63 BIPM member states studied, with a GQII score of 43.2. It is also discovered that the development of the individual components of the QI has a high level of coherence. A country with high metrology values is likely to be advanced in other areas as well. The main issue was that accreditation data was processed in very diverse ways from country to country, making it difficult to compare. The GQII would be an excellent monitoring tool for international development programs focusing on QI.

Conclusion

A good QI is the first step toward a stable political environment and the rule of law. It also leads to high-quality goods, which is essential for a healthy economy. As a result, it is critical to research and comprehend the function of QI in any country's development. Further, during and after the COVID-19 pandemic, QI work would be exceedingly more critical to guarantee that products and services, particularly in the medical and food sectors, meet the established criteria. Governments' attempts to provide essential medical items and assure food safety and security in the most efficient, effective, and sustainable manner require advanced QI services and mutual recognition agreements among trading partners. In the middle- and low-income countries, there is increasing potential to assist QI development. It is unavoidable for development partners to promote awareness about the relevance of QI in assisting nations in attaining sustainable growth and development and ensuring that QI systems are aligned with the demands of the private sector and regulatory bodies across trading partners. It is also critical to guarantee that development partners work and coordinate to ensure that our client countries receive the best possible assistance based on available resources, competencies, and capabilities.

This chapter provides a concise review of the QI. The IQI bodies have been studied to strengthen NQI. In addition, a SWOT analysis was performed to understand the QI better. The second edition of GQII has been reported as a measure of QI in various countries based on data from the various QI components. The quality of life in 63 BIPM member nations has been studied. According to the findings of the GQII studies, most countries with high GDP also have high QI or vice versa. Stakeholders and policymakers should find this insightful study to be highly useful as a reference source.

Appendix

Table A GQII ranking of BIPM member states

Country	GQII score	Rank
Germany	63.06	1
France	57.97	2
Finland	57.78	3
United Kingdom	57.69	4
Korea	56.14	5
Sweden	55.24	6
Russian Federation	54.57	7
Netherlands	54.50	8
Czechia	54.37	9
Japan	53.10	10
Switzerland	52.54	11
United States of America	52.54	12
China	51.05	13
Italy	49.67	14
Austria	49.57	15
Slovakia	49.44	16
Denmark	49.43	17
Hungary	48.55	18
South Africa	48.29	19
Slovenia	48.28	20
Spain	47.42	21
Poland	47.22	22
Belgium	45.16	23
Norway	44.95	24
Australia	44.73	25
Romania	44.03	26
Saudi Arabia	43.86	27
India	43.23	28
Canada	43.08	29
Brazil	42.93	30
Portugal	41.83	31
Turkey	41.50	32
New Zealand	40.52	33
Bulgaria	39.39	34

(continued)

Table A (continued)

Country	GQII score	Rank
Ukraine	38.42	35
Argentina	38.29	36
Singapore	37.13	37
Serbia	37.01	38
Israel	36.11	39
Belarus	35.73	40
Thailand	35.43	41
Ireland	35.43	42
Iran	35.14	43
Uruguay	34.90	44
Tunisia	34.49	45
Mexico	34.16	46
Egypt	33.99	47
Greece	33.74	48
Malaysia	32.78	49
Colombia	31.78	50
Kenya	31.49	51
Chile	31.48	52
Croatia	30.98	53
Indonesia	30.87	54
Estonia	28.04	55
Pakistan	26.53	56
Lithuania	26.53	57
Morocco	26.17	58
Montenegro	25.97	59
Kazakhstan	25.24	60
Ecuador	23.51	61
United Arab Emirates	22.98	62
Iraq	16.94	63

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Part II

Redefinition of SI Units and Its Implications

The Quantum Reform of the International System of Units

7

William D. Phillips and Peter J. Mohr

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Abstract

On 20 May 2019, *Le Système internationale d'unités* (SI) or the International System of Units underwent a historic change to become defined in terms of assigned values of constants of nature. This chapter provides a brief historical overview of units and describes the changes that occurred in the redefinition. We dedicate this chapter to the memory of Prof. Ian Mills, OBE, FRS (1930–2022), who played a significant role in the redefinition.

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SI · Units · Redefinition · Measurement standards

Introduction

Le Système internationale d'unités (SI), or International System of Units, came into being in 1960, through the action of the *Conference général des poids et mesures* (CGPM) or General Conference on Weights and Measures (https://www.bipm.org/en/measurement-units; https://www.bipm.org/en/committees/cg/cgpm). It was firmly based on the metric system, whose roots were in the French Revolution of the late eighteenth century (Adler 2003; Quinn 2017) and followed in the tradition set by the international *Convention du Mètre* (the Treaty of the Meter) of 1875, which established the metric system as the official system of measurement units for the world, or at least for those 17 countries who were signatories of the original treaty (https://bipm.org/en/metre-convention). That treaty established the *Bureau International des Poids et Mesures* (BIPM) or International Bureau of Weights and Measures (http://bipm.org/en/home), which became the guardian not only of the spirit of the metric system and of its evolution and growth, but also, as we will see below, of the artifacts that formed the basis for that system of units.

On 20 May 2019, the 144th anniversary of the signing of the original *Convention du Mètre,* the SI underwent what we consider to be the most revolutionary change in the metric system since its inception during the French Revolution. Approved by the CGPM on 16 November 2018 in Versailles (Stock et al. 2019), the result of this revolutionary reform is that today all of the seven base units of the SI (the meter, kilogram, second, ampere, kelvin, mole, and candela), are defined by the adoption of fixed values of seven "constants of nature." As a result, all of the derived units of the SI, which are derived from the base units, are in turn defined by those constants of nature. The International System of Units is now free of those artifacts on which the original metric system was based. In principle, scientists on Earth could communicate to hypothetical scientists on any other planets what we mean by "kilogram," "meter," "second," and so forth, assuming, as seems likely, that the laws of physics and the natural constants are uniform throughout the universe.

Here, we present an abbreviated and incomplete historical review of the units for time and length, as an example of how such units have changed over time, and as an example of how constants of nature can be, and have been, used to define a unit. Then, we present a similar review of the unit of mass, describing the more recent changes in the definition of this unit – why a change was needed and how it was accomplished. We briefly describe the situation with the other base units of the SI.

A Brief History of Time (with Our Apologies to the Late, Great Stephen Hawking (Hawking 1988))

For a long while, and beginning at an indefinite time, the unit of time, the second, has been $1/(24 \times 60 \times 60)$ of a day, an understanding arising from choosing to have 24 h in a day, 60 min in an hour, and 60 s in a minute. But, what is a "day"? In modern times, until the middle of the twentieth century, this was understood to be the "mean solar day." The solar day, the time between successive passages of the sun across the meridian, varies throughout the year, because of factors like the inclination of the earth's axis of rotation relative to the plane of its orbit around the sun and the eccentricity of that orbit. This deviation is as much as 30 s from the mean (https://en.wikipedia.org/wiki/Solar_time). How to determine the mean value, averaged over a year, was never specified by the governing metrological bodies established by the *Convention du Mètre* and was left to the astronomers. The understanding and observation that the mean solar day was in some sense an artifact that was susceptible to change over time (friction due to the tides, changes in ocean currents, and other effects can change the length of a day) led to a redefinition of the second. In 1960, the CGPM defined the second as a fraction of the tropical year 1900. While this provided a stable definition, it was still essentially an artifact, and not one to which one could easily refer.

Meanwhile, the use of transitions between energy levels of atoms or molecules was being discussed as a possible new approach to the definition of the unit of time (Rabi 1945). The idea of using atoms as frequency standards dates back to Maxwell in 1873 (Maxwell 1873) and Thomson (Lord Kelvin) and Tait in 1879 (Thomson and Tait 1879). The history of much of the early development of such atomic clocks is reviewed by Ramsey (Ramsey 1983). The first "atomic" clock operated at a National Metrology Institute (NMI) was the 1949 molecular ammonia clock at the US National Bureau of Standards (see Fig. 1), and the first NMI cesium atomic clock was operated at the U.K. National Physical Laboratory in 1955 (see Fig. 2). With such atomic clocks providing superior performance and convenience compared to the astronomical definitions of the second, in 1967/1968 the CGPM redefined the second as *"the duration of 9192631770 periods of the radiation corresponding to the transition between the two hyperfine levels of the ground state of the caesium 133 atom."* This is equivalent to defining the hyperfine frequency of ^{133}Cs and as such represents an example of defining a unit of measure by defining a constant of nature, a definite improvement over the mean solar day, which is not guaranteed to be stable over time.

The definition of the SI second in terms of ^{133}Cs is the current (2023) definition. It is not likely to remain so for very much longer. The hyperfine frequency, which is a natural constant, is not a "fundamental" constant. There can be, and in fact are, other transitions in other atoms that are superior. For example, Al^+ with 9×10^{-19} relative uncertainty (Brewer et al. 2019) and Sr with 2×10^{-18} relative uncertainty (Nicholson et al. 2015) (more recently, a resolution of 8×10^{-21} relative uncertainty has been achieved (Bothwell et al. 2022)), and today the redefinition of the SI second, in

Fig. 1 Dr. Harold Lyons (right), inventor of the ammonia absorption cell atomic clock, observes, while Dr. Edward U Condon, the director of the National Bureau of Standards, examines a model of the ammonia molecule (1949) (http://ethw.org/Milestones:First_Atomic_Clock,_1948). (Credit: NIST, USA Digital Archives)

terms of some ion or atom with an appropriate transition at an optical frequency, is under active consideration.

That the current and future definitions of the SI second are not in terms of fundamental constants of nature leaves open the possibility that the discovery of better transitions will continue to lead to future redefinitions of the second. The story of length, below, is an example of a unit, the meter, being defined in terms of a "fundamental" constant, so that no future redefinition of length should ever be needed.

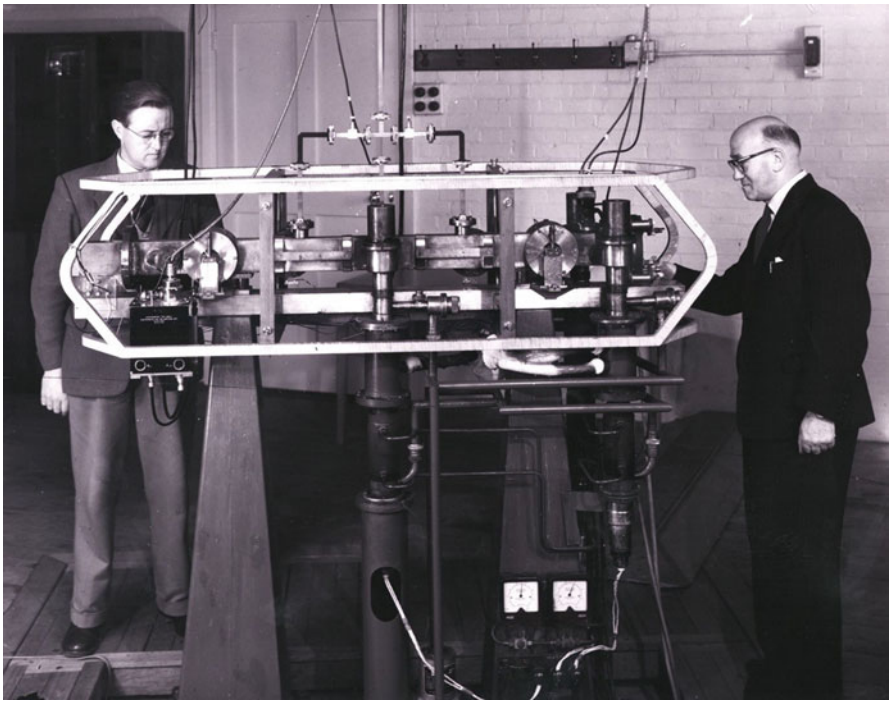

Fig. 2 Jack Parry (left) and Louis Essen (right) with the NPL cesium atomic clock in 1955 (https://en.wikipedia.org/wiki/Atomic_clock). (Credit: NPL)

A Short History of Length

In order to appreciate the need for, and the beauty of, defining a unit by fixing the value of a fundamental constant of nature, we review some of the history of units for length. This history includes various ways of defining the unit of length and illustrates the appeal of these various ways. According to the usual folklore, which is probably more or less accurate, the first standards of length were based on the dimensions of the human body (see Fig. 3). A yard was the distance from one's nose to the end of an extended hand. A foot was the length of a foot. A fathom was the length of one's arm-span – the distance between outstretched hands, a hand was the distance across the palm of that body part, and the cubit was the length from the elbow to the fingertip.

The advantages of such a measuring system are obvious: One always has the standards of measurement "at hand" so to speak, a tremendous convenience. The disadvantages are equally obvious – the variability of the measurement standard from one user to another could be annoying (as in commerce) or disastrous (as in engineering) (https://usma.org/unit-mixups).

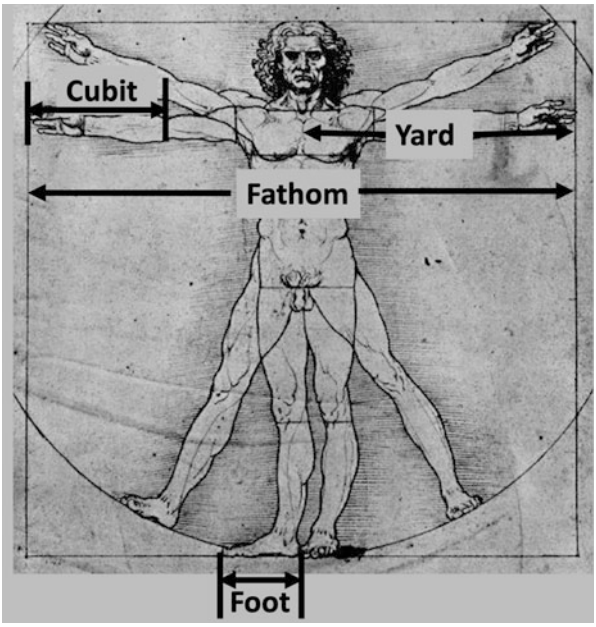

Fig. 3 A few standard lengths referred to the dimensions of the human body (the Vitruvian Man, drawing by Leonardo da Vinci, dated to c. 1490). (Image: Public domain)

The approaches used to deal with the variability of length standards seem at times to have been remarkably modern. One oft-recounted definition of the "foot" as a standard of length was the average of the foot length of 16 randomly chosen men (see Fig. 4) (https://en.wikipedia.org/wiki/Foot(unit)):

> Stand at the door of a church on a Sunday and bid 16 men to stop, tall ones and small ones, as they happen to pass out when the service is finished; then make them put their left feet one behind the other, and the length thus obtained shall be a right and lawful rood to measure and survey the land with, and the 16^{th} part of it shall be the right and lawful foot.

The beauty and modernity of this definition lies in the recognition that random choice and averaging is likely to produce a more consistent result than a single realization of the definition using one's body.

More commonly, at least according to legend, the unit of length was based on the body of a particular person, typically the monarch. In ancient Egypt, the royal cubit was based on the bodily dimensions of the Pharaoh (https://www.bksv.com/en/knowledge/blog/perspectives/egyptian-cubit). The Pharaoh was not readily available for repeated measurements in the field, for the construction of pyramids, for example, so another rather modern approach was used – the creation of an artifact standard that represented the royal cubit, an artifact in stone, as shown in Fig. 5.

Another way to establish a consistent standard of length foreshadowed modern practice in a different way. The inch (1 ft = 12 in) was at one time defined to be the

Fig. 4 Foot length of 16 randomly chosen men. (Woodcut published in the book Geometrei by Jakob Kobel, Frankfurt, c. 1535) (Image: Public domain)

Fig. 5 A replica of the Royal Cubit of Amentop I, from the National Institute of Standards and Technology (NIST) digital archives (https://www.bksv.com/en/knowledge/blog/perspectives/egyptian-cubit). (Credit: NIST, USA Digital Archives)

length of three barleycorns (see Fig. 6) (https://en.wikipedia.org/wiki/Barleycorn (unit)).

The hope for such a definition was that nature itself would guarantee the consistency of the unit – an aspirational "natural constant." Unfortunately, whether due to variations in cultivation conditions, temperature, rainfall, or the like, the length of barleycorns was not reliably constant. (Interestingly, the barleycorn as a unit of length lives on as the difference between successive shoe sizes in some English-speaking countries (https://en.wikipedia.org/wiki/Shoesize).)

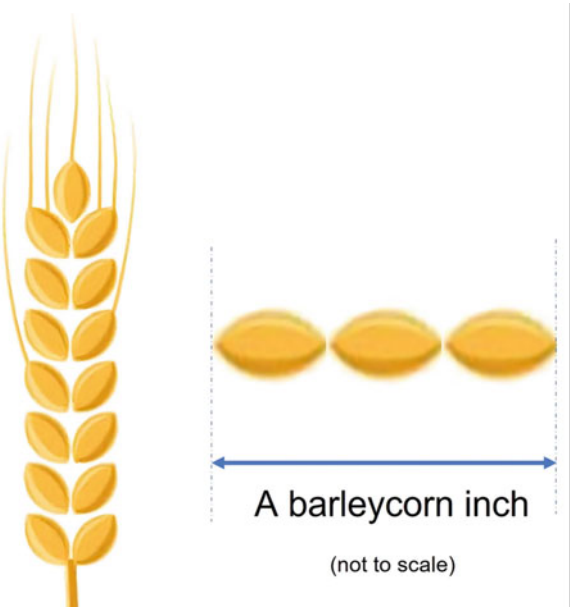

Fig. 6 Inch defined as the length of three barleycorns (https://workingbyhand.wordpress.com/2022/04/18/a-note-on-inches-and-feet)

Such seemingly modern approaches to length standards notwithstanding, it was the ancient approach of an artifact standard that became widespread. Figure 7 shows one such artifact, (literally) fixed in stone at the city hall of the German town of Regensburg, presenting what appears to be a rather generous realization of the fathom. Such artifacts provided clear, reliable, and, at least for the requirements of the day, stable standards of measurement. However, if one were to go to another town, one would typically find another such artifact, with no guarantee that the standard of length would be the same there as it was in Regensburg.

The French Revolution, whose beginning is often associated with the storming of The Bastille in 1789, engendered a spirit of reform that extended to the system of measurement units. The resulting Metric System of units was characterized not only by having the multiples and subdivisions of units by factors of 10, but by definitions of those units that would, at least in principle, be available "for all time, for all people." A commemorative medal containing that sentiment is shown in Fig. 8.

The allegorical creature using dividers to measure the earth evokes the new unit for length, the meter, and its definition, being one ten-millionth (10^{-7}) of the length of the meridian from the earth's pole to the equator (as the dividers are indeed placed), passing through Paris.

The meter was to be the measure of all things (Adler 2003). Not only was the meter and its multiples and submultiples to be the measure of all lengths, areas, and volumes, but it was also to form the basis for the new unit of mass, as we shall see below.

Fig. 7 Artifact imbedded in a wall. (Credit: David Newell; Reproduced by permission)

Fig. 8 Commemorative medal for the Metric System (http://monnaiesanciennes.free.fr/imageselsen104/MED-000576002dc.jpg). (Credit: *Direction Générale des Entreprises*, France)

In order to realize the definition of the new unit of length, two parties, led by astronomers Delambre (who went north from Paris) and Méchain (who went south), surveyed the meridian, measuring the length between Dunkirk and Barcelona, which by appropriate extrapolation gave the measure of the pole-to-equator meridian in the units of the Ancien Régime (or at least some version of it, since nonuniformity of such units was a major motivation for the metric reform).

Measuring the earth had been difficult, taking 7 years and involving many adventures and misadventures. It was not a task likely to be repeated very often. So, to preserve that hard-won measure of all things, an artifact was made, in the same spirit as the stone royal cubit of ancient Egypt. A platinum rod, end-to-end 1 m, was deposited in the Archives of France in 1799 (https://en.wikipedia.org/wiki/Metre). This "meter of the archives" is pictured in Fig. 9, and it became the defining artifact for the meter for France, and ultimately for an increasing number of countries that, in the following years, adopted the metric system.

In 1875, 17 countries met in Paris and adopted, as an international treaty, the *Convention du Mètre.* This treaty established the metric system as the measurement standard for those countries and created the International Bureau of Weights and Measures (*Bureau International des Poids et Mésure*s – the BIPM) as the repository of new artifacts of length and mass yet to be manufactured. The new International Prototype of the Meter was a "line standard" for which the distance between two scratches engraved into a Pt-Ir alloy bar (a distance made as close as possible to the length of the Meter of the Archives) became the standard of length for all those in the world who adopted the metric system (see Fig. 10).

The latter part of the nineteenth century, when the new meter bars were made, experienced a revolution in length metrology. Physicists in France and elsewhere had established that light was a wave phenomenon. Scientists like Babinet in 1827 and Maxwell in 1859 had earlier suggested that the wavelength of light would be a more suitable standard of length than either an artifact or the dimensions of the planet (p. 154 of Ref. (Quinn 2017)). Albert Michelson, the inventor of his eponymous interferometer was the first, with Benoît, to compare the wavelength of light to the

Fig. 9 The meter standard made of Platinum by Lenoir (1799) with its case; Photo Courtesy: National Archives of France (AE/I/23/10) & UNESCO Memory of the World Register, 2005

Fig. 10 Line standard meter. (Credit: NIST, USA Museum Collection)

International Prototype of the meter, in 1892 and 1893 (p. 154 of Ref. (Quinn 2017)). Subsequent improvements by Benoît with Fabry and Perot established optical measurement of length with an uncertainty, a few 100 nanometers, similar to the uncertainty in realizing the meter from the primary meter bar itself, with the definition of the meter being the distance between two engraved lines in a metal surface (p. 155 of Ref. (Quinn 2017)).

The ease with which length could be measured using optical interferometry meant that, soon, people were using the wavelength of light as a *de facto* standard. In fact, the wavelength (in air!) of the red light from a cadmium lamp, which had been used in the Michelson experiments and those that followed, was defined to be a certain number of angstroms, thus establishing a practical length scale that was independent of the length scale determined by the international prototype of the meter. (Today, the angstrom is exactly 10^{-10} m, but in the early part of the twentieth century, it represented a separate length scale to be used for spectroscopy (https://en.wikipedia.org/wiki/Angstrom).)

By the middle of the twentieth century, it was clear that the definition of the meter needed to be changed to reflect the widespread use of interferometry to measure length. There was much discussion about the best choice of a reference wavelength. While the red cadmium line had been in common use for some time, the spread of wavelengths due to the isotopic mixture was of concern. Eventually, the orange line emitted by ^{86}Kr was chosen, and in 1960 the General Conference on Weights and Measures adopted a new definition of the meter which defined the wavelength in vacuum of that spectral emission (see Fig. 11). This definition of the meter realized a dream expressed by, among others, Maxwell, who in the nineteenth century wrote: "...the most universal standard of length...would be the wavelength in vacuum of a

Fig. 11 Krypton lamp used for the realization of the meter as defined in 1960 by the CGPM (https://www.nobelprize.org/uploads/2018/06/advanced-physicsprize2005.pdf). (Credit: NIST, USA Museum Collection)

particular kind of light. . .” (Maxwell 1873). That same conference adopted the *Système International d’Unités* (the SI) as the official system of units, with the meter, kilogram, second, ampere, (degree) kelvin, and candela as its six base units. It was an auspicious year indeed, being also the year when the first laser was demonstrated.

The development of the laser foreshadowed the next, and we expect, the last change in the definition of the meter. Rapid developments in laser technology soon took laser light from the spectrally broad, pulsed output of the first flashlamp-pumped ruby laser to the spectrally narrow output of continuous-wave helium-neon lasers stabilized to absorption lines in a gas of molecular iodine. By the early 1980s, the wavelength of the I_2-stabilized He-Ne laser (See Fig. 12) had become the *de facto* standard of length, just as the wavelength of light from a cadmium lamp had become the *de facto* length standard earlier in the century. Metrologists were once again using a length standard that was not the SI standard, but which represented a

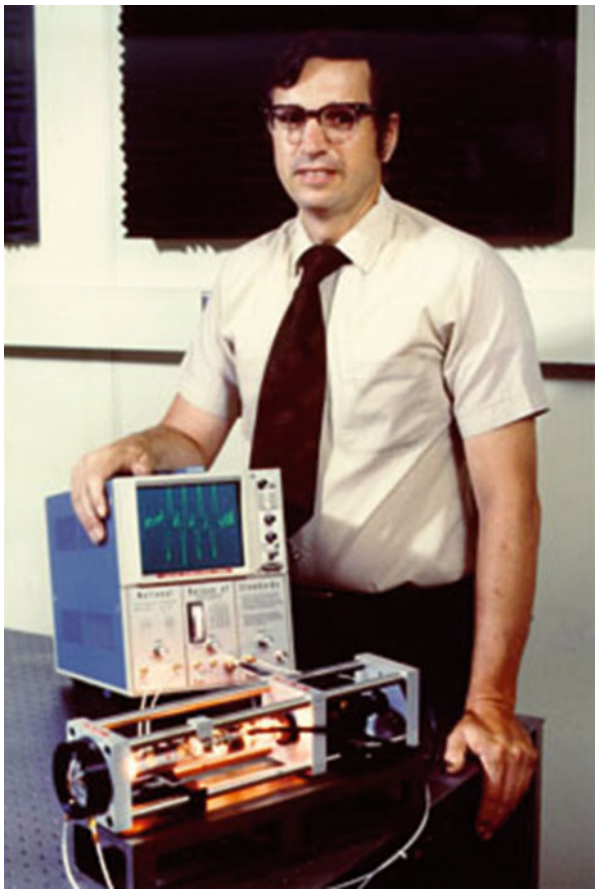

Fig. 12 An I_2-stabilized He-Ne laser (https://www.nist.gov/image-25845) with the spectrum of iodine absorption to which the laser is locked. Dr. Howard Layer is seen in the picture circa 1980 at NIST. (Credit: NIST, USA)

technology that had surpassed the capabilities of the official SI length standard. This same situation has recurred several times in the history of metrology.

The time had come again to redefine the meter. The obvious choice, following historical precedent, would have been to define the wavelength of some laser stabilized to some molecular or atomic transition, with the I_2-stablized He-Ne laser as the obvious candidate. Fortunately, the metrology community did not make the obvious choice, but rather a bold and beautiful choice, one that set the stage for a vast reform of the SI, accomplished in just the past few years. That beautiful choice was to define the speed of light.

This new definition of the meter, indeed this new approach to defining the meter, depended in large part on the development of another new technology – the measurement of the frequency of optical radiation. By means of frequency chains, whereby harmonics of microwave frequencies were used to lock the frequency of far-infrared lasers and those lasers were used to lock the frequencies of shorter

wavelength lasers, it was possible to determine the frequency of visible light with an accuracy at least as good as the accuracy with which optical wavelengths could be compared. Because of the universal relation: $c = f\lambda$, if one measures the frequency f of light, and if one defines the speed of light c, then the wavelength λ is known to the same accuracy as that to which the frequency is measured. The new definition of the meter, adopted by the CGPM in 1983, is that the meter is the distance traveled by light in a certain time, effectively fixing, exactly, the speed of light.

One of the beautiful features of this new definition is that future improvements in technology are automatically incorporated into the existing definition. In the past, the technology of optical interferometry made the International Prototype Meter obsolete. Then, the technology of lasers made the Krypton lamp obsolete. Now, improvements in lasers and improvements in the ability to measure optical frequencies will not necessitate any change in the new definition of the meter. In fact, such improvements have happened and were recognized in part by the award of the 2005 Nobel prize in Physics to John L. Hall and Theodor W. Hänsch (https://www.nobelprize.org/uploads/2018/06/advanced-physicsprize2005.pdf).

While the krypton definition of the meter had the advantage of being based on a constant of nature, that constant, being particular to a certain atomic species, was not, as it turned out, the best choice. Now, the definition of the meter is based on a fundamental or universal constant of nature, the speed of light. This reference to a fundamental constant is the heart of the beauty and the strength of the current definition of the meter. That achievement of 1983 has now, as of 2019, been brought to four other units of the SI. The most revolutionary of the changes of 2019 to the SI units is the change in the definition of the kilogram.

A Light History of Mass

As with standards of length, standards of mass have a long history. Figure 13 shows a set of weights or masses used in ancient Babylonia. Such sets of artifact weights or mass-standards were common throughout the ancient world and were often used to

Fig. 13 Artifact mass standards of ancient Babylonia (https://www.wikiwand.com/en/Ancient_Mesopotamian_units_of_measurement#Media/File:Mesopotamian_weights_made_from_haematite.JPG). (Credit: Geni; Reproduced under CC BY-SA 4.0)

verify the weight of precious metals of the sort that might be used in coinage. The problem with mass artifacts, as with the even more prolific length artifacts, was that they varied from place to place. At least in some cases, the standard of mass varied according to what sort of material was being weighed. In prerevolutionary France, for example, the livre (pound) used to measure the weight of bread was different from that used to measure the weight of metals (Adler 2003).

As with standards of length, the French revolutionaries determined to eliminate the inconsistency of mass standards by referring the new unit of mass, named the kilogram, to the newly minted standard of length, the meter. The kilogram was to be the mass of one cubic decimeter (a cube with 10 cm on each edge or one litre) of water. Originally, the water was to be at its freezing temperature, 0 °C, but the fact that water has a density maximum at 4 °C motivated the definition to be changed accordingly.

The kilogram definition had some obvious advantages. Being based on the new meter as the standard of length, it was consistent with the revolutionary ideal that the meter should be the measure of all things. Water was readily available, and at 4 °C, the density extremum, temperature variations were minimally perturbative. Nevertheless, it did not turn out to be very easy to realize the definition at the level of accuracy with which one could compare masses using a balance. So, as with the unit of length, whose definition based on the size of the earth was difficult to realize, the unit of mass was enshrined as an artifact. The Kilogram of the Archives, a platinum cylinder made to have the mass of one litre of water, as well as could be determined, was deposited in the Archives of France in 1799, just as the Meter of the Archives was. Figure 14 shows one of the authors of this chapter holding the 1799 Kilogram of the Archives, on a visit to that unheated institution in December of 2019.

The 1875 *Convention du Mètre* mandated the creation of a new standard of mass, to become the International Prototype of the Kilogram (IPK). Made of platinum-iridium, the IPK was chosen from among a number of nearly identical copies as the one whose mass was closest to the mass of the Kilogram of the Archives, insuring consistency of the definition upon the adoption of the new standard. The IPK itself, stored under three glass enclosures, open to the atmosphere, is shown in Fig. 15 and has only been taken out for comparison to the working standards a few times since its creation. The working standards have been used from time to time to compare against the national standards distributed to the signatories of the *Convention du Mètre.* All of these kilograms were made at the same time and in the same way so that they are nominally identical except for slight differences in mass, differences that were measured at the time.

While various copies of Pt-Ir kilograms can be compared to one another with a precision on the order of a microgram (a part in 10^9), differences between copies vary over time by tens of micrograms. Figure 16 shows the time variation of a subset of kilogram copies relative to the IPK. Variation of tens of micrograms (parts in 10^8) is typical, and variations of more than a hundred micrograms sometimes occur.

This situation, where time variations of various standard kilograms, all nominally identical, can be many times the precision with which kilograms can be compared, is clearly intolerable. Because the various copies are nominally identical, it is

Fig. 14 William D Phillips holding the case containing the 1799 Kilogram of the Archives, at the Archives of France, in December of 2019

impossible to assert that the IPK, the standard of mass in the SI, is not changing. In fact, an examination of Fig. 16 would lead one to conclude that the IPK is probably changing by tens of micrograms. On the other hand, the "mass" of the IPK cannot change, because, by definition, it is always 1 kg. Presumably the masses of the various Pt-Ir kilograms vary because surface or other contaminants change over time, and change differently in the different copies. Because there is no mass standard that can be guaranteed to be more stable than the IPK and its copies, there is in practice no way of knowing which of the kilograms is changing by how much or in what direction.

Standardized cleaning techniques were established in an attempt to address the apparent change in mass with time, but different (nominally identical) kilogram copies responded differently to such cleaning, compounding the uncertainty about what is meant by a kilogram. (See Fig. 17.1 on p. 343 of Ref. (Quinn 2012), and also (Davis 2003).)

It became clear that the kilogram needed to be redefined, and that redefinition should, by preference, be in the same spirit as the redefinition of the meter. That is, the kilogram should be defined by choosing a fixed value of some constant of nature. To see what that constant might be, consider the equation $E = mc^2$. The meaning of this equation is that E is the energy of an object of mass m at rest. Now consider the equation $E = \hbar\omega$, where $\hbar$ is the reduced Planck constant. The meaning of this

Fig. 15 The international prototype kilogram and its six official copies in a safe at the BIPM (Davis 2003). (© BIPM and IOP Publishing Ltd. Reproduced by permission of IOP Publishing. All rights reserved)

equation is that E is the energy of a single photon of light having frequency ω. If we combine these two equations, we find $E = mc^2 = \hbar\omega$, so that $m = \hbar\omega/c^2$.

The apparent meaning of this final equation is that if an object were to emit a single photon of frequency ω, its mass would change by m. Since we have fixed c in defining the SI meter, and since we have (see above) long ago fixed the frequency of the cesium hyperfine interval in defining the SI second, if we now fix the value of the Planck constant h, we have, at least in principle, a way to define the kilogram via the change in mass of an object that emits a photon of known frequency. In practice the process of thus "weighing" a photon is not currently good enough to form the basis of realizing the new definition of the kilogram better than several parts in 10^7 (Rainville et al. 2005). Fortunately, two other methods exist that allow a sufficiently good realization of the kilogram definition based on choosing a fixed value for h: the

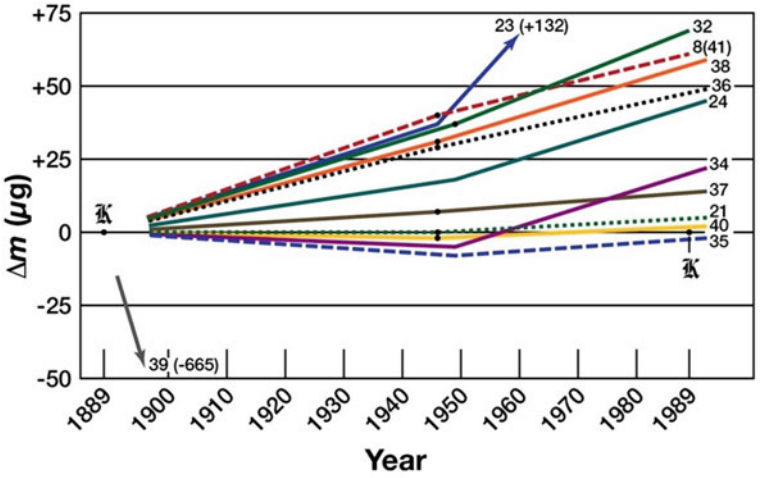

Fig. 16 Time variation of some of the national kilogram standards relative to IPK. (Adapted from Fig. 17-4 (p. 345) of Ref. (Quinn 2012)) (Credit: NIST, USA Digital Archives)

Kibble balance and the X-ray crystal density (XRCD) method (Taylor and Mohr 1999; Mills et al. 2005, 2011; Possolo et al. 2018; Wood and Bettin 2019).

Kibble Balance Method to Realize the Kilogram

In 1975, Bryan Kibble of the UK National Physical Laboratory introduced the idea behind what we now call the Kibble balance or the Watt balance (Kibble 1975). It was originally seen as a way to realize the definition of the SI ampere, or, equivalently, to determine the relationship between a legal, national, definition of the ampere and the SI ampere. The SI definition was: *The ampere is that constant current which, if maintained in two straight parallel conductors of infinite length, of negligible circular cross-section, and placed 1 m apart in vacuum, would produce between these conductors a force equal to* 2×10^{-7} *newton per meter of length.*

Note that this definition is equivalent to fixing the vacuum magnetic permeability as $\mu_0 = 4\pi \times 10^{-7}$ N/A^2. So the SI definition of the ampere, which predated the official establishment of the SI, had already adopted the practice of defining a base unit in a way that fixed the value of a natural constant before this was done for the SI second and well before the recent (May 2019) reform of the SI where all the base units are so defined.

The "legal" or "national" definition of the ampere, mentioned above, existed because of the difficulty in realizing the SI definition. Originally (before the discovery of the Josephson effect and the integer quantum Hall effect), the legal definition of the ampere was that current which would produce a voltage drop of one "legal volt" across a resistance of one "legal ohm." The legal volt was set by assigning a voltage to the average of an ensemble of standard cells (electrochemical voltage

sources), and the legal ohm was set by assigning a resistance to the average of an ensemble of wire-wound resistors. Each major national metrology institute would have its own set of standard cells and resistors. The values assigned to these standard cells and resistors were as close as possible to their SI values.

With the advent of the Josephson and quantum Hall effects, the legal volt and ohm were replaced by fixing the (legal) values of $2e/h$ and h/e^2 in units of hertz per (legal) volt and (legal) ohms. In 1990, the international metrology community agreed upon fixed values for these constants, chosen according to the best known SI values, but they were not in fact the SI values. This eliminated the time variation of legal electrical units and differences from one nation to another. As we will see below, the measurement of current, voltage, and resistance in this way is crucial to the realization of the kilogram with the new definition using the Kibble balance.

Consider now the realization of the SI ampere. One cannot directly realize a definition involving infinitely long wires, so in practice one makes coils, determines their dimensions and locations, and measures the force between the coils when they are energized with current measured in legal/national units. The measurement of the force along with the relevant dimensions gives the current in SI units, and thus one determines the ratio of the legal and SI amperes. A severe problem is that the measurement of dimensions of the current-carrying coils is difficult (one must know the location of every part of every wire). It is this difficulty that Kibble's method addresses.

A schematic diagram of the apparatus is in Fig. 17. The measurement is done in two phases. In one, called the weighing mode, a current I is passed through the coil

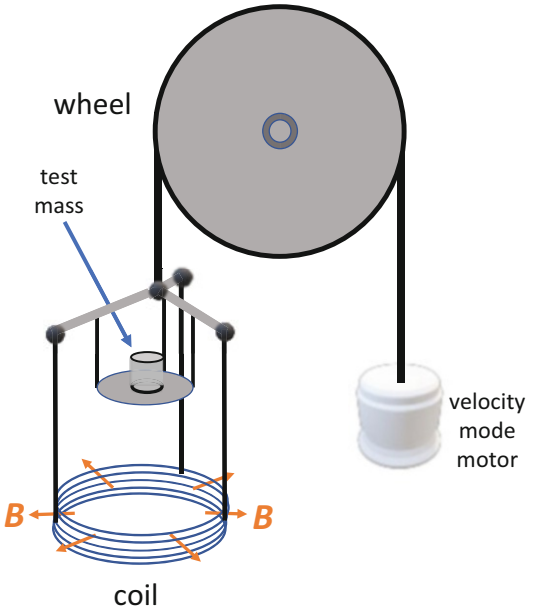

Fig. 17 Schematic diagram of a Kibble balance. (Image: Public domain)

that is suspended from a wheel that is free to turn. The coil is in a radial magnetic flux density B produced by a permanent magnet, and the current is adjusted to provide an upward Lorentz force that balances the force of gravity mg on the test mass. The local acceleration of gravity g is separately measured. The electrical force is ideally IBL, where L is the length of the wire in the coil, so that

$$mg = IBL \tag{1}$$

The product BL depends on the details of the coil winding and the strength of the magnetic field, both of which are difficult to determine accurately. To address this, there is a second phase of the measurement called the velocity mode in which the coil is slowly driven up and down with a velocity v by the velocity mode motor. This produces a potential difference V between the ends of the coil according to Faraday's law, given by

$$V = vBL \tag{2}$$

Elimination of the troublesome product BL between Eqs. (1) and (2) yields

$$m = \frac{IV}{vg} \tag{3}$$

This equation can be written as $mgv = IV$, equating mechanical and electrical power in watts, which is why the Kibble balance was previously known as the watt balance. Moreover, the two-phase nature of the measurement eliminates both mechanical and electrical energy losses. However, Eq. (3) as a way to realize a new definition of the kilogram lacks any reference to the Planck constant, which we say above is the heart of the definition. The Planck constant comes in because of the way that current and voltage are measured using the Josephson effect and the quantum Hall effect. Voltages are measured relative to the Josephson constant $2e/h$ while currents are measured in terms of the voltage across a resistor measured relative to the von Klitzing constant h/e^2. The units of V are proportional to $h/(2e)$ (volts/hertz) while the units of resistance are proportional to h/e^2 (ohms). The current I, measured as voltage divided by resistance is proportional to $(h/2e)/(h/e^2)$, so the unit of mass is proportional to $(h/2e)^2/(h/e^2) = h/4$, that is, m is proportional to h (among other things), but is independent of e.

This explains how fixing the value of h allows a realization of the new definition of the kilogram. The actual process is, naturally, rather complex (Olsen et al. 1980, 1989; Kibble et al. 1983; Robinson and Schlamminger 2016; Thomas et al. 2017; Wood et al. 2017; Ahmedov et al. 2018; Li et al. 2019; Schlamminger and Haddad 2019; Kim et al. 2020; Fang et al. 2020). It is, in fact, far more complex than the realization of the old definition, which simply involved comparing masses to the mass of the IPK. Figure 18 shows a Kibble balance apparatus at the National Institute

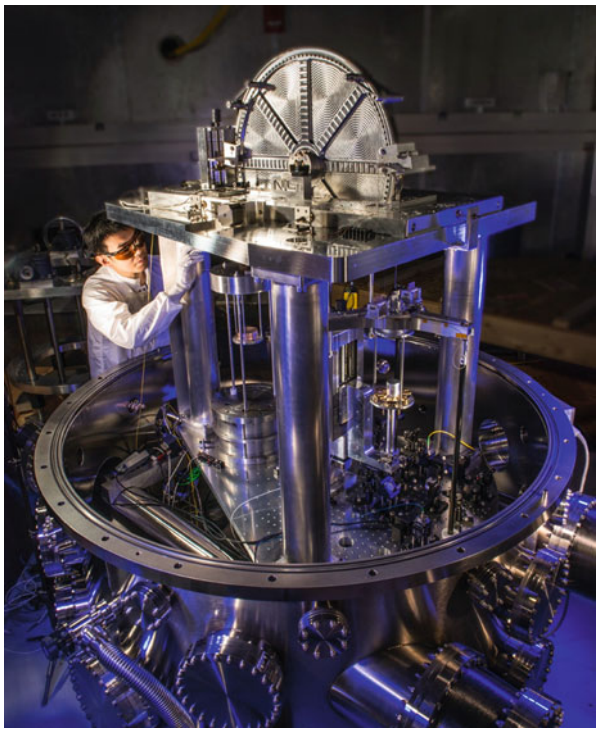

Fig. 18 A Kibble balance at the National Institute of Standards and Technology. The wheel at the top of the image can rotate, allowing the vertical translation of a suspended coil. It also acts as a balance to determine the electromagnetic force on that coil. (Credit: NIST, USA Digital Archives)

of Standards and Technology that is capable of realizing the new definition of the kilogram to parts in 10^8.

Before the redefinition, the relation between the Planck constant and the mass used in the Kibble balance meant that when a known mass was used, the experiment provided a value for the Planck constant. After the redefinition, the roles of the mass and the Planck constant were reversed so that a known value for the Planck constant meant that the Kibble balance experiment was a measurement of the mass.

Silicon Sphere Method to Realize the Kilogram

The basic idea of this method, enabled by fixing the value of h, is to determine, by some means, the mass in kilograms of a single atom of silicon, and also to determine the number of silicon atoms in a crystal of silicon, yielding the mass of that sample of silicon.

This approach to realizing the kilogram has its origins in the Avogadro project (Azuma et al. 2015; Newell et al. 2018). The original goal of that project was to determine the Avogadro constant by measuring the number of atoms in a known mass (known in terms of the IPK) of ^{28}Si. That number, along with the molar mass of

^{28}Si (determined by well-established mass comparison techniques) and the atomic mass unit in kilograms gives N_A, the Avogadro constant. The heart of the Avogadro project is the X-Ray Crystal Density (XRCD) method, which involves making a nearly perfect silicon sphere from a nearly perfect crystal of nearly isotopically pure ^{28}Si, and measuring the dimensions of that sphere, as well as measuring the lattice constant of that perfect crystal (see Figs.19 and 20) (Azuma et al. 2015; Deslattes 1969; https://newscientist.com/article/dn14229-roundest-objects-in-the-world-created/).

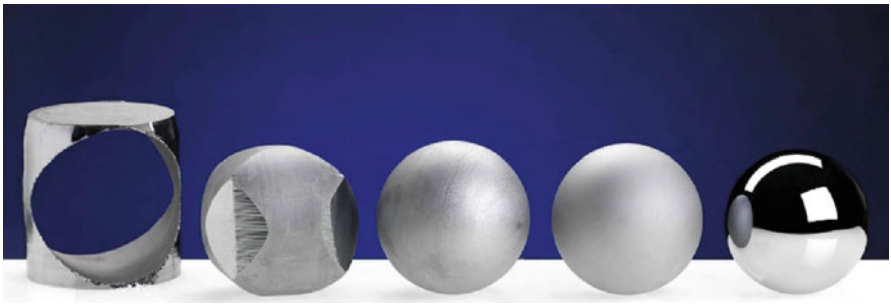

Fig. 19 Steps in the process of manufacturing a near-perfect silicon sphere, from single-crystal boule to finished, polished sphere (Meeβ et al. 2021). The creation of a near-perfect silicon sphere begins with the growing of a suitably large perfect crystal of isotopically pure (enriched) ^{28}Si, using the crystal growth techniques developed over many years by the semiconductor industry for producing silicon chips from large wafers of single-crystal silicon. The grown crystal is then fabricated into a nearly perfect sphere through a process of sequential cutting, polishing, and measuring. The measurement process is performed interferometrically, and the final sphere is round to within some tens of nm with a surface roughness better than a nm. The deviation from sphericity reflects the underlying crystal structure. (Credit: Rudolf Meeß et al. 2021 Meas. Sci. Technol. 32 074004; CC-BY IOP Open Publishing)

Fig. 20 A finished silicon sphere (https://newscientist.com/article/dn14229-roundest-objects-in-the-world-created/). (Credit: CSIRO)

Before the redefinition, the measurement of N_A also provided a value for h, because there is a relation $N_A h = C$, where C is an accurately known combination of constants, including the Rydberg constant and the fine-structure constant, so that measuring either h or N_A effectively determines the other (see Sec. 5 in Mohr 2008a).

With the establishment of the techniques of the Avogadro project, it became possible to invert the goal of the project and imagine a definition of the kilogram based on fixing the value of N_A, although this was not the approach finally taken. Fixing the value of the Planck constant h also allows the redefinition of the kilogram using the XRCD methods of the Avogadro project because the mass in kilograms of a silicon atom is well known once the value of h is fixed.

There are two ways in which the Si mass can be determined. First: with h fixed, the electron mass in kilograms is known from the Rydberg constant and the fine-structure-constant with an uncertainty of parts in 10^{10} (Tiesinga et al. 2021). The ratio of the electron mass to the mass of ^{12}C is determined with an uncertainty of parts in 10^{11} by comparing the electron spin flip frequency of hydrogen-like $^{12}C^{5+}$ to the cyclotron frequency of $^{12}C^{5+}$ in the same magnetic field (Köhler et al. 2015). The ratio of ^{12}C and ^{28}Si masses is known to parts in 10^{11} (Huang et al. 2021; Wang et al. 2021). This chain of measurements gives the mass in kilograms of a silicon atom with an uncertainty of parts in 10^{10}, much better than the parts in 10^8 uncertainty of measurements on the silicon sphere.

Second: With h and c fixed, the recoil velocity $\hbar\omega/mc$ of a Rb (or a similar species like Cs) atom receiving the momentum of an optical photon of frequency ω gives the mass of an Rb atom in kilograms. This velocity measurement, using atom interferometric determination of the Doppler shift associated with the recoil velocity, gives the mass in kilograms of Rb to parts in 10^{10} (Weiss et al. 1993; Browaeys et al. 2005; Cladé et al. 2019; Yu et al. 2019; Morel et al. 2020). With the known ratios of atom masses good to parts in 10^{11} (Huang et al. 2021; Wang et al. 2021), the silicon mass is known to parts in 10^{10} and can be used directly with the XRCD measurements to determine the mass of a silicon sphere.

While it is tempting to think of the silicon sphere as simply another example of an artifact kilogram, it is not that at all. The artifact IPK is a unique object, and independent of a comparison to that object one would not be able to fabricate a replica. With a fixed value assigned to h, one can use the process described to make a kilogram of silicon, without reference to any artifact mass standard.

One of the appealing aspects of the new definition of the kilogram is that it only specifies the value of h, not the technique by which, given that value of h, one would realize the definition. There is no priority given to either the Kibble balance or the silicon sphere method of realizing the kilogram. Only when both methods were in sufficient agreement was the final decision made to adopt the new definition. As of this writing (late 2023) the performances of both methods are similar (a few parts in 10^8) and improvements are expected in both. The new definition puts an end to the unknown and unknowable changes in the standard of mass that had plagued mass metrology since the inception of the IPK.

The Current Story of the Ampere

As mentioned above, the old definition of the ampere, involving the forces between current-carrying wires, has been difficult to implement with high accuracy, which led to the institution of a different, non-SI set of electrical units, the "legal" electrical units. Early in the twentieth century, these legal units were different in each major National Metrology Institute and involved ensembles of electrochemical standard cells to represent the legal standard of voltage and ensembles of wire-wound resistors to represent the legal standard of resistance. Given such legal standards for the ohm and volt, one could realize a legal ampere as the current which produced a voltage of a legal volt fixed across a legal ohm. In 1990, the worldwide metrology community adopted conventional, fixed values of the Josephson constant $2e/h$ and the von Klitzing constant h/e^2 as defining constants for legal electrical metrology (Taylor and Witt 1989). That system existed in parallel to the SI electrical units based on the mechanical definition of the ampere, with the SI values of $2e/h$ and h/e^2 continuing to be measured quantities while the legal values of those constants remained fixed.

With the SI value of h being fixed in order to redefine the kilogram, it is natural to also fix the SI value of the elementary charge e. This defines the ampere, a coulomb/second, as being the current of a certain number of electrons per second, that number being the inverse of the electron charge in coulombs. While in principle one could count electrons, the technology for doing that is at the moment insufficient for the required accuracy and precision for realizing the SI ampere. Instead, one does exactly what was done for the legal ampere, except that now the fixed and exact values of $2e/h$ and h/e^2 are SI values and one realizes the SI ampere. This means that one no longer needs the old "legal" system of electrical units because one has exact SI values of the Josephson and von Klitzing constants. At various times after the 20 May 2019 adoption of the reformed SI, various countries switched their legal electrical metrology from the old 1990 values to the new SI values, so that there is no longer any distinction between the SI and legal electrical units.

In every case of changing the SI definition of a unit of measure, the new definition was chosen to be within the uncertainty of the old definition. That is, with a definition based on fixing the value of a constant of nature like h or e, that fixed value is within the uncertainty of the best measurement of that constant before the change in definition. In this way, the transition from one definition to another is seamless and invisible to even the most demanding metrologist. This is not the case for the system of legal electrical units. Because electrical measurements had advanced significantly between 1990, when the legal values of $2e/h$ and h/e^2 were chosen, and 2018, when the SI values of those constants were chosen, there was an abrupt jump in the legal electrical units when each country made its legal units consistent with the SI. Those jumps amounted to about 1.1×10^{-7} for the Josephson constant and 1.8×10^{-8} for the von Klitzing constant.

As noted in section "Kibble Balance Method to Realize the Kilogram," before the redefinition, the definition of the ampere in terms of parallel conductors had the consequence that the vacuum magnetic permeability μ_0 was exact. Moreover, the

vacuum electric permittivity, $\epsilon_0 = 1/(c^2\mu_0)$, was also exact. Since the redefinition, neither of these constants is exact, but rather determined by experiment through their relation to the fine-structure constant α, which is given by

$$\alpha = \frac{e^2}{4\pi\varepsilon_0\hbar c}$$

The fine-structure constant is the dimensionless constant that characterizes the strength of electromagnetic interactions. It is accurately known (parts in 10^{10}), and being dimensionless, it cannot be affected by any redefinition of units. Before the redefinition $e^2/\hbar$ was determined by the value of α, because both ϵ_0 and c were exact. After the redefinition, since both e and $\hbar$ are exact, the value of ϵ_0 is determined by the value of α. Before the redefinition, even though the ratio $e^2/\hbar$ was precisely determined, values of e and $\hbar$ were less precise individually.

The Rest of the Base Units: The Kelvin, Mole, and Candela

Here we briefly summarize the effect of the redefinition of the SI on the rest of the seven base units of the SI – the kelvin, mole, and candela. Resources for additional information include Ref (Moldover et al. 2016; Guttler et al. 2019; Sperling and Kűuck 2019). For more details about these units, the reader is referred to chapters of this volume on temperature (Chap. 11, "Quantum Definition of New Kelvin and Way Forward"), mole (Chap. 13, "The Mole and the New System of Units (SI)"), and candela (Chap. 12, "Realization of Candela"), which deal with them.

In the past, the kelvin, the SI unit of thermodynamic temperature, was 1/273.16 of the temperature of the triple point of water. In principle, ideal gas thermometers could be used, along with this definition, to measure temperatures in units of the kelvin. Moreover, based on this definition, the value of the molar gas constant R could be determined by measuring acoustic resonant frequencies of a gas, such as argon or helium, in a cavity. The frequencies depend on the speed of sound in the gas, which in turn is related to R. The Boltzmann constant k is related by $k = R/N_A$, where the Avogadro constant N_A was more accurately known than either k or R. As a result, the value of the Boltzmann constant was best determined by the frequency measurements of R.

In the new SI, this relationship is turned around, and the kelvin is defined by fixing the value of the Boltzmann constant as $k = 1.380649 \times 10^{-23}$ J/K. This, together with the exactly defined value of N_A, yields an exact value for $R = kN_A$. As with all of the unit redefinitions, the fixed value of k was chosen to be the best measured value before the redefinition. One of the effects of the redefinition is that the triple point of water is no longer, by definition, 273.16 K. Operationally, it is the same kinds of measurements using acoustic gas thermometers that may be used to realize the kelvin under the redefined SI. However, the spirit of the new definition is significantly different. The kelvin now has a distinctly microscopic, atomic flavor, in contrast to the macroscopic nature of the old definition. In principle, with the value

of k fixed, one could determine the temperature of an atomic gas by measuring the velocity distribution of the constituent atoms, and hence their mean kinetic energy. At present, such measurements are not the best way to determine temperature, but one can imagine that the new definition will encourage the further development and improvement of such measurements, which would cement both the spirit and practice of considering temperature as a microscopic property (Darquié et al. 2013).

The old definition of the mole was that a mole is the amount of substance having as many elementary entities as there are ^{12}C atoms in 12 g of ^{12}C. That number of entities is N_A, the Avogadro constant. In the old SI, the value of N_A was measured using the techniques described above for the silicon sphere method of realizing the kilogram. In the new SI, the mole is redefined by fixing the value of the Avogadro constant. One of the implications of this redefinition is that a mole of ^{12}C is no longer, by definition, exactly 12 g of ^{12}C, but instead is a measured quantity. It is worth noting that this change in the definition uncouples the mole from the kilogram. Without this uncoupling, there would be two different definitions of the kilogram which would be an inconsistency in the SI.

While the mole and kelvin have new definitions in the revised SI, and these definitions have a different, more microscopic and quantum, spirit than did the old definitions, for the immediate future the experimental methods involving these units will remain essentially the same.

The definition of the unit of luminous intensity, the candela, is unchanged by the redefinition of the SI, but it is now expressed in a manner that fixes the value of a constant of nature. The previous definition was that *"The candela is the luminous intensity, in a given direction, of a source that emits monochromatic radiation of frequency* 540×10^{12} *hertz and that has a radiant intensity in that direction of 1/683 watt per steradian."* The current, and completely equivalent definition is: *"The luminous efficacy of monochromatic radiation of frequency* 540×10^{12} *Hz,* K_{cd}*, is* 683 *lm/W."* Luminous efficacy describes how bright light appears to the human eye. While it is not in principle a "constant of nature" in the way that the speed of light is, it has been found to be reasonably constant across the range of demographic groups and so is useful for comparing the effectiveness of various light sources.

Conclusion

On 16 November 2018, in Versailles, France, a group of 60 countries comprising the CGPM made history with a unanimous vote, agreeing to redefine the SI in a profound way to take effect on 20 May 2019. The festive event celebrated scientists' fulfillment of the long-standing dream of a measurement system based entirely on unchanging fundamental properties of nature. The spirit of the day can be seen in Fig. 21, where five of many authors of papers that contributed to the change are seen toasting the successful vote.

The redefinition by the assignment of fixed values to a set of defining constants completely specifies the measurement system in a simple way. In fact, quoting from the SI Brochure, the document that provides the official basis for the SI, the complete

Fig. 21 Celebration of the redefinition of the SI at Versailles: l. to r., Ian Mills, OBE, FRS, former President of the *Comité Consultatif des Unités*, to whom this chapter is dedicated; Terry Quinn, CBE, FRS, Emeritus Director of the BIPM; Barry Taylor, NIST; Peter Mohr, NIST, one of the authors of this chapter; and Edwin Williams, NIST. (Credit: NIST, USA)

definition of the SI system of units is given by the brief listing of constants, exactly as shown here (https://www.bipm.org/en/publications/si-brochure):

"The International System of Units, the SI, is the system of units in which

- **The unperturbed ground state hyperfine transition frequency of the cesium 133 atom, $\Delta\nu_{Cs}$, is 9 192 631 770 Hz.**
- **The speed of light in vacuum, c, is 299 792 458 m/s.**
- **The Planck constant, h, is 6.626 070 15 $\times$ 10^{-34} J s.**
- **The elementary charge, e, is 1.602 176 634 $\times$ 10^{-19} C.**
- **The Boltzmann constant, k, is 1.380 649 $\times$ 10^{-23} J/K.**
- **The Avogadro constant, N_A, is 6.022 140 76 $\times$ 10^{23} mol^{-1}.**
- **The luminous efficacy of monochromatic radiation of frequency 540 $\times$ 10^{12} Hz, K_{cd}, is 683 lm/W.**

where the hertz, joule, coulomb, lumen, and watt, with unit symbols Hz, J, C, lm, and W, respectively, are related to the units second, meter, kilogram, ampere, kelvin, mole, and candela, with unit symbols s, m, kg, A, K, mol, and cd, respectively, according to Hz = s^{-1}, J = kg m^2 s^{-2}, C = A s, lm = cd m^2 m^{-2} = cd sr, and W = kg m^2 s^{-3}."

There it is, the definition of the SI in one sentence. Of course, additional information is needed to implement the definitions, as described in the preceding sections. However, as technology evolves, enabling more precise measurements, there will be no need to change the definitions, with the exception of the particular frequency used to define time, as mentioned in section "A Brief History of Time

(with Our Apologies to the Late, Great Stephen Hawking (Hawking 1988))." But the form of the definition will not need to change.

Numerical values for the constants *h, e, k,* and N_A in the list are based on the results of measurements up to the time of the 107th meeting of the *Comité international des poids et mesures* (CIPM) or the International Committee for Weights and Measures, which met in June 2018 and recommended the redefinition of the SI to the CGPM for consideration at its meeting in November, 2018. The CIPM had earlier requested that the Committee on Data for Science and Technology (CODATA) Task Group on Fundamental Constants carry out a survey and least-squares adjustment to determine the best current values for consideration at their meeting (Newell et al. 2018; Mohr et al. 2018). The other three constants in the list were not changed by the redefinition, although the wordings of the corresponding definitions were modified.

As is evident in the list of values of the SI defining constants, the definitions apply to combinations of the traditional base units of the earlier SI. For example, the definition of c relates the combination of units m/s to a physical phenomenon, namely, the invariant speed of light in vacuum, and the definition of h determines the combination kg m^2/s or J/Hz. The constants have been chosen to determine combinations of units so that all of the traditional base units are defined (Mohr 2008b). As a result, it is no longer necessary to single out base units, as was previously the case because they were defined individually. Here we summarize some features of the redefinition.

- For the kilogram, variations of the mass of an artifact are eliminated, along with the possible associated drift of the definition of the kilogram. Also, the artifact kilogram no longer has the mass of exactly 1 kg.
- Accurate electrical measurements in terms of the Josephson effect and the quantum Hall effect are now directly based on SI units because h and e are exactly defined in the SI. The constants μ_0 and ϵ_0 are no longer exact, although very precise, and still satisfy the relation $\mu_0\epsilon_0 c^2 = 1$.
- The new definition of the mole specifies the exact number of entities in a mole, rather than indirectly specifying it as the number of entities in 12 g of ^{12}C. Now, the molar mass of ^{12}C is not exactly 12 g.
- The Boltzmann constant is exactly defined and is no longer based on the triple point of water, which is affected by the purity and isotopic composition of the water being used as the standard. The triple point of water is no longer exactly 273.16 K.
- Besides the fact that four important fundamental constants have become exact, other physical constants are also either exact or have substantially reduced uncertainties. Important among these are energy-equivalence relations between the joule, volt, hertz, and kelvin, all of which are now exact.

To conclude, we note that the idea of using fundamental constants in equations of physics to define measurement scales has a long history. Many scientists considered finding a suitable set of constants, such as the set c, $\hbar$, e, and m_e, for example. People who contributed in this pursuit include Gauss, Maxwell, Stoney, Planck, Fleming,

Hartree, and others (Tomilin 1998). However, in general, it was found to be more practical to use artifacts to define measurement standards. It is only relatively recently that technology has progressed to the point that it has become a practical reality to apply the laws of physics using fixed values of the constants to define a complete measurement system. The turning point came in 2018 when it was decided that the last artifact, the International Prototype of the Kilogram, could be replaced by mass determinations based on the Kibble balance and silicon spheres to redefine the SI.

Cross-References

- ▶ Quantum Definition of New Kelvin and Way Forward
- ▶ Realization of Candela
- ▶ The Mole and the New System of Units (SI)

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Realization of the New Kilogram Based on the Planck Constant by the X-Ray Crystal Density Method

8

Naoki Kuramoto

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Abstract

The new definition of the kilogram was implemented on May 20, 2019. The kilogram is now defined by fixing the value of the Planck constant. In principle, national metrology institutes can realize the kilogram from the Planck constant individually to establish their primary mass standards. The Planck constant is related to a mass of 1 kg by the X-ray crystal density method. This method can be therefore used to realize the kilogram by the new definition. As an example of such realization using this method, the realization experiment at the National Metrology Institute of Japan is introduced. For this realization, the volume of a 1 kg ^{28}Si-enriched sphere is measured by optical interferometry. The

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characterization of the sphere surface by X-ray photoelectron spectroscopy and ellipsometry is also performed. Details of the realization and future dissemination of mass standards in Japan using the ^{28}Si-enriched sphere are described.

Keywords

New SI · Kilogram · Planck constant · X-ray crystal density method · Mass standards · Silicon crystal · Volume measurement · Surface characterization · X-ray photoelectron spectroscopy · Ellipsometry · Optical interferometry

Introduction

Humans have been trying to understand this world by developing diverse measurement technologies. For seamless sharing of measurement results around the world, the international measurement scale is provided by the International System of Units (SI) (BIPM 2019), and the most advanced technologies in each era have been used to define units in the SI to establish more reliable measurement scales. The SI is not eternal and continues to evolve with the development of science and technology. The kilogram is at the core of the SI, and its definition was also fundamentally revised taking advantage of state-of-the-art technologies in 2019 (Stock et al. 2019, Kuramoto et al. 2020, Kuramoto 2021, 2022).

The origin of the kilogram dates back to the end of the eighteenth century. The original concept was based on the mass of 1 dm^3 of water and was materialized in a cylinder of platinum in 1799. Platinum was chosen to make the cylinder chemically stable and resistant to rust. In 1889, this artifact was replaced by a cylinder made of an alloy of 90 % platinum and 10 % iridium prepared by a state-of-the-art metallurgical technology at the time. This alloy is harder and less likely to wear than pure platinum. This new artefact is known as the international prototype of the kilogram, and the kilogram was defined as the mass of this artifact (Fig. 1), which was stored at the Bureau International des Poids et Mesures (BIPM, France). Its copies were distributed to the signatories of the Meter Convention and were used as their national prototypes, leading to the worldwide unification of the mass scale.

By definition, the mass of the international prototype of the kilogram was exactly 1 kg without any uncertainty associated. In practice, however, the mass of this – and in fact any – artefact cannot be perfectly stable. One hundred years after its manufacture, the long-term stability of the mass of the international prototype of the kilogram was estimated to be several tens of micrograms (Girard 1994, Kuramoto 2021), and its inevitable instability casts a shadow on the accuracy of the international mass scale.

To overcome this issue, the Planck constant was proposed as the new basis of the kilogram because, to the best of our present knowledge, it is invariant in time and space. The unit of the Planck constant is J s, which is equal to kg m^2 s^{-1}. The meter has been defined by fixing the numerical value of the speed of light since 1983, and the second has been defined by fixing the numerical value of the cesium frequency since 1967 (BIPM 2019). Consequently, assigning a fixed numerical value to the Planck constant has the effect of redefining the kilogram. Moreover, the Planck

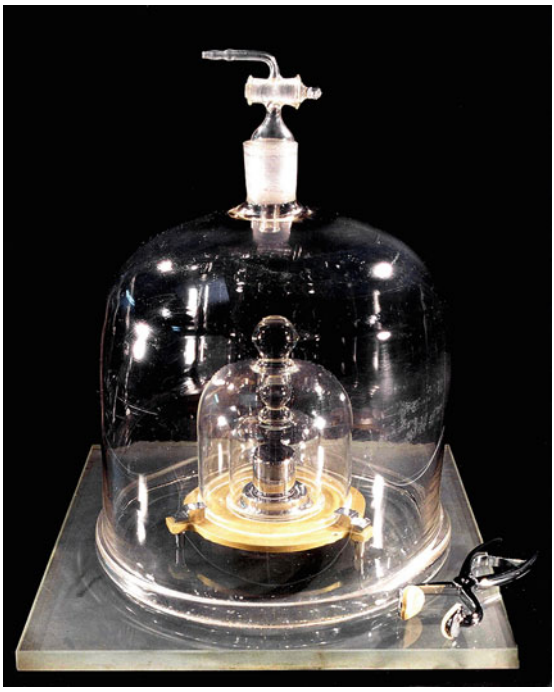

Fig. 1 International prototype of the kilogram enclosed by three bell jars: This cylindrical weight made of a platinum–iridium alloy is kept at the Bureau International des Poids et Mesures (BIPM) in Sèrves, France. The diameter and height are both approximately 39 mm. (Photograph courtesy of the BIPM)

constant can be practically related to a mass of 1 kg by two different methods. One is the X-ray crystal density (XRCD) method (Kuramoto et al. 2020), which counts the number of silicon atoms in a 1 kg single-crystal silicon sphere. The mass of the sphere is then determined from the mass of a silicon atom and the number of atoms. Because the atomic mass can be derived using the Planck constant with the help of other physical constants, this method relates the sphere mass to the Planck constant. The second method is to use the Kibble balance (Robinson and Schlamminger 2016), in which the gravitational force acting on a 1 kg weight is balanced with the electromagnetic force. The electromagnetic force is measured using quantum electrical standards based on the Planck constant. Thus, the mass of the weight is related to the Planck constant by measuring the gravitational acceleration.

In the 2010s, the time was ripe to redefine the kilogram. The two methods were improved by worldwide cooperation between national metrology institutes (NMIs) (Kuramoto 2021), leading to measurements of the Planck constant h based on the mass of the international prototype of the kilogram with uncertainties superior to the long-term stability of the artefact itself. In 2017, the h value for the new definition of the kilogram was determined by the Task Group on Fundamental Constants of the Committee on Data for Science and Technology (CODATA TGFC) from the eight values of h measured using the two methods (Newell et al. 2018). The eight data values used by the task group are summarized in Table 1 and Fig. 2. IAC-11, IAC-15, IAC-17, and NMIJ-17 were measured by the XRCD method (Azuma et al. 2015, Bartl et al. 2017, Kuramoto et al. 2017b). NIST-15, NIST-17, NRC-17,

Table 1 Details of the eight data values used to determine the Planck constant h for the new definition of the kilogram (Kuramoto 2021). See Newell et al. (2018) and Mohr et al. (2018) for a complete list of input data

Source	Data ID [a]	Research institute	Method	Relative standard uncertainty in h
Schlamminger et al. (2015)	NIST-15	National Institute of Standards and Technology (USA)	Kibble balance	5.7×10^{-8}
Wood et al. (2017)	NRC-17	National Research Council (Canada)	Kibble balance	9.1×10^{-9}
Haddad et al. (2017)	NIST-17	National Institute of Standards and Technology (U.S.A)	Kibble balance	1.3×10^{-8}
Thomas et al. (2017)	LNE-17	Laboratoire national de métrologie et d'essais (France)	Kibble balance	5.7×10^{-8}
Azuma et al. (2015)	IAC-11	International Avogadro Coordination project	XRCD method	3.0×10^{-8}
Azuma et al. (2015)	IAC-15	International Avogadro Coordination project	XRCD method	2.0×10^{-8}
Bartl et al. (2017)	IAC-17	International Avogadro Coordination project	XRCD method	1.2×10^{-8}
Kuramoto et al. (2017b)	NMIJ-17	National Metrology Institute of Japan (Japan)	XRCD method	2.4×10^{-8}

[a] Data name given in the CODATA 2017 Special Adjustment of the Fundamental Constants (Newell et al. 2018)

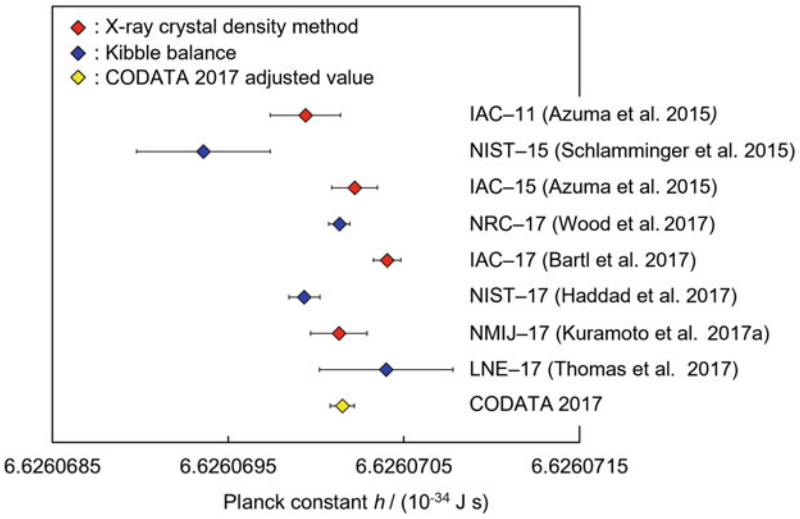

Fig. 2 Eight key data values used to determine the CODATA 2017 adjusted value of the Planck constant (Kuramoto 2021). The numerical value of the CODATA 2017 adjusted value is used in the new definition of the kilogram. A complete list of input data in the CODATA 2017 adjustment is provided by Newell et al. (2018) and Mohr et al. (2018)

and LNE-17 were measured using the Kibble balances (Schlamminger et al. 2015, Haddad et al. 2017, Wood et al. 2017, Thomas et al. 2017). The h value determined by the task group is known as the CODATA 2017 adjusted value and is $6.626\ 070\ 150(69) \times 10^{-34}$ J s, with a relative standard uncertainty of 1.0×10^{-8}. At present, the kilogram is defined by fixing the value of h to $6.626\ 070\ 15 \times 10^{-34}$ J s. This new definition came into effect on May 20, 2019.

In principle, it is now therefore possible for NMIs to realize the kilogram individually from the Planck constant to establish their primary mass standards. However, as of May 2022, all NMIs disseminate mass standards not from their own individual realization experiments but from the consensus value of the kilogram to avoid any possible international inconsistency among mass measurements (Davidson and Stock 2021). The consensus value is an agreed common basis to ensure the international equivalence of mass standards. As of May 2022, the consensus value is 1 kg – 2 μg with a standard uncertainty of 20 μg. The mass of the international prototype of the kilogram is not exactly 1 kg but 1 kg – 2 μg on the basis of the consensus value. NMIs still use their national prototypes of the kilogram as primary mass standards, and the relationships between the national prototypes and the international prototype are known. The masses of the national prototypes are therefore traceable to the Planck constant through the consensus value. The consensus value will be updated after each international comparison of the realizations of the kilogram by NMIs organized by the Consultative Committee for Mass and Related Quantities (CCM) scheduled to take place every 2 years. After the consistency of the realization results from a sufficient number of individual realization experiments is achieved, NMIs can provide mass standards directly traceable to their own realization experiments. Criteria for the transition from the internationally coordinated dissemination with the consensus value to the dissemination from the realizations by NMIs are described by CCM (2019).

The National Metrology Institute of Japan (NMIJ) is the NMI of Japan. After the use of individual realization experiment for the dissemination of mass standards becomes viable, NMIJ will use the XRCD method to establish the national mass standard. The XRCD method can realize the kilogram with a relative uncertainty of a few parts in 10^8. In this method, the kilogram at a nominal mass of 1 kg is realized using a single-crystal silicon sphere (Stock et al. 2018, Davidson and Stock 2021, Kuramoto et al. 2020, 2021). As an example of the realizations using this method, the realization experiment at NMIJ is introduced.

X-Ray Crystal Density Method

The new definition defines only the value of h and does not imply any method for its practical realization (Stock et al. 2019, Kuramoto 2021, 2022). We can realize the kilogram using any method – including the XRCD method and the Kibble balance – that relates the Planck constant to the mass. In the XRCD method, the number N of Si atoms in a 1 kg Si sphere is counted by measuring its volume V_s and lattice constant a (Kuramoto et al. 2020, 2021). Figure 3 shows an example of 1 kg Si spheres used for the realization, manufactured from a silicon crystal highly enriched

Fig. 3 1 kg ^{28}Si-enriched spheres manufactured from the AVO28 crystal. Left, AVO28-S5c; right, AVO28-S8c. (Photograph courtesy of AIST)

with the ^{28}Si isotope produced by the International Avogadro Coordination (IAC) project (Andreas et al. 2011a). The number N is given by

$$N = 8\, V_{\mathrm{s}}/a^3, \tag{1}$$

where 8 is the number of atoms per unit cell of crystalline silicon. The mass of the sphere m_{s} is therefore expressed by

$$m_{\mathrm{s}} = N\, m(\mathrm{Si}), \tag{2}$$

where m(Si) is the mass of a single Si atom. The new definition of the kilogram is based on the Planck constant h, and the mass of a single electron m(e) is given from the Planck constant h by

$$m(\mathrm{e}) = \frac{2hR_\infty}{c\alpha^2}. \tag{3}$$

where c is the speed of light in vacuum, α is the fine-structure constant, and R_∞ is the Rydberg constant (Cladé et al. 2016). The mass of a single Si atom m(Si) is related to m(e) by

$$\frac{A_{\mathrm{r}}(\mathrm{Si})}{A_{\mathrm{r}}(\mathrm{e})} = \frac{m(\mathrm{Si})}{m(\mathrm{e})}, \tag{4}$$

where A_{r}(e) and A_{r}(Si) are the relative atomic masses of electron and Si, respectively. Silicon has three stable isotopes, ^{28}Si, ^{29}Si and ^{30}Si. The value of A_{r}(Si) can therefore

be determined by measuring the isotopic amount-of-substance fraction of each isotope. The relationship between $m(\mathrm{Si})$ and h is given from (3) and (4) by.

$$m(\mathrm{Si}) = \frac{2A_\mathrm{r}(\mathrm{Si})}{A_\mathrm{r}(\mathrm{e})} \frac{R_\infty h}{\alpha^2 c}. \tag{5}$$

By combining Eqs. (1), (2), and (5), we can express the sphere mass m_s in terms of h using

$$m_\mathrm{s} = \frac{2R_\infty h}{c\alpha^2} \frac{A_\mathrm{r}(\mathrm{Si})}{A_\mathrm{r}(\mathrm{e})} \frac{8V_\mathrm{s}}{a^3}. \tag{6}$$

In this equation, $2R_\infty h/(c\alpha^2)$ is the mass of an electron, $A_\mathrm{r}(\mathrm{Si})/A_\mathrm{r}(\mathrm{e})$ is the mass ratio of silicon to electron, and $8V_\mathrm{s}/a^3$ is the number of silicon atoms in the sphere. The mass of the sphere is therefore determined on the basis of h by counting the number of silicon atoms.

Equation (6) describes a perfect Si crystal. That is, the Si sphere is assumed to consist of only Si atom. However, the surface of Si spheres is covered by a surface layer mainly consisting of amorphous SiO_2 (Busch et al. 2011). Figure 4 shows the surface model of the Si sphere under vacuum, derived on the basis of surface characterization by various analysis techniques developed by the IAC project (Andreas et al. 2011a, Busch et al. 2011). In addition to the oxide layer (OL), a chemisorbed water layer (CWL) and a carbonaceous layer (CL) are also present. The CWL is a chemically absorbed water layer that remains under high-vacuum conditions. The CL is a carbonaceous contamination layer formed by different adsorbed gases and contaminants, stemming from the environment during the measurement, handling, storage, and cleaning of the sphere. These layers are hereinafter referred to

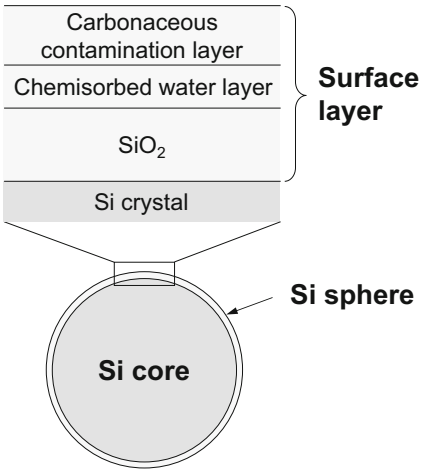

Fig. 4 Model of the surface layer of ^{28}Si-enriched spheres in vacuum. (Kuramoto 2021)

as the surface layer (SL). In addition, point defects are present inside the sphere. Equation (6) should therefore be modified to

$$m_{\mathrm{core}} = \frac{2R_\infty h}{c\alpha^2}\frac{A_{\mathrm{r}}(\mathrm{Si})}{A_{\mathrm{r}}(\mathrm{e})}\frac{8V_{\mathrm{core}}}{a^3} - m_{\mathrm{deficit}}, \tag{7}$$

where m_{core} and V_{core} are the mass and volume of the "Si core," which excludes the surface layer, respectively, and m_{deficit} is the effect of point defects (i.e., impurities and self-point defects in the crystal) on the core mass (Azuma et al. 2015). The mass of the sphere m_{sphere} including the mass of the surface layer m_{SL} is given by

$$m_{\mathrm{sphere}} = m_{\mathrm{core}} + m_{\mathrm{SL}}. \tag{8}$$

The relative uncertainty of $R_\infty/(\alpha^2 A_{\mathrm{r}}(\mathrm{e}))$ is estimated to be 3.0×10^{-10} (Tiesinga et al. 2021), and the values of h and c are exactly defined in the new SI (Stock et al. 2019). For the two 1 kg ^{28}Si-enriched spheres AVO28-S5c and AVO28-S8c shown in Fig. 3, the values of a, $A_{\mathrm{r}}(\mathrm{Si})$, and m_{deficit} were already measured by the IAC project with relative uncertainties at the level of 10^{-9} (Azuma et al. 2015, Bartl et al. 2017). There is no known mechanism that changes those parameters when the Si crystal is kept close to room temperature. The kilogram can therefore be realized by measuring V_{core} and m_{SL} of the ^{28}Si-enriched spheres at NMIJ. Details of the measurements for the realization are described in the following sections.

Procedure of Realization of the Kilogram at NMIJ

The masses of AVO28-S5c and AVO28-S8c are almost 1 kg, and details of the spheres are described by Azuma et al. (2015). To measure the sphere core volume V_{core} in Eq. (7), an optical interferometer for measuring the sphere diameter is used. However, the diameter measured using the interferometer is slightly different from the core diameter owing to the surface layer shown in Fig. 4. To calculate the core diameter from the measured diameter, the thickness of the surface layer is required. For the thickness measurement, an X-ray photoelectron spectroscopy (XPS) system and a spectroscopic ellipsometer are used. The surface characterization is also required to derive the mass of the surface layer, m_{SL} in Eq. (8). Details of each measurement apparatus are presented in the following sections. A realization result using AVO28-S5c is also shown as an example.

Sphere Volume Measurement by Optical Interferometry

Figure 5 shows a schematic drawing of the optical interferometer used to determine the volume of the ^{28}Si-enriched sphere by optical frequency tuning (Kuramoto et al. 2017a, 2019). A ^{28}Si-enriched sphere is placed in a fused-quartz Fabry–Perot etalon.

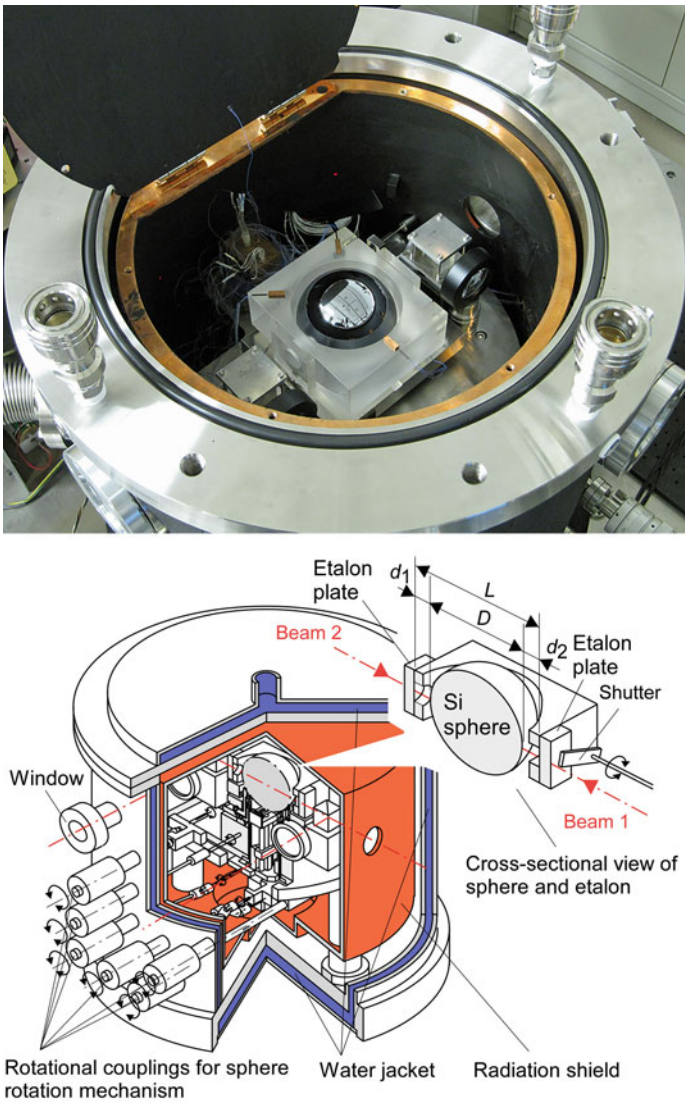

Fig. 5 Upper: Optical interferometer used to measure the volume of the ^{28}Si sphere developed by Kuramoto et al. (Photograph courtesy of AIST). Lower: Schematic of the optical interferometer. The ^{28}Si sphere and etalon are installed in a vacuum chamber equipped with an active radiation shield to control the sphere temperature (Kuramoto et al. 2017a)

The sphere and etalon are installed in a vacuum chamber equipped with an active radiation shield to control the sphere temperature. The pressure in the chamber is reduced to 10^{-2} Pa. Two beams (Beam 1 and Beam 2) from an external cavity diode laser are irradiated onto the sphere from opposite sides of the etalon. The light beams reflected from the inner surface of the etalon plate and the adjacent surface of the

sphere interfere to produce concentric circular fringes. These are projected onto CCD cameras. The fractional fringe orders of interference for the gaps between the sphere and the etalon, d_1 and d_2, are measured by phase-shifting interferometry (Andreas et al. 2011b, Kuramoto et al. 2011). The sphere diameter D is calculated as $D = L-(d_1 + d_2)$, where L is the etalon spacing. To determine L, the sphere is removed from the light path by a lifting device installed underneath the sphere, and Beam 1 is interrupted by a shutter. Beam 2 passes through a hole in the lifting device, and the beams reflected from the two etalon plates produce fringes on a CCD camera. The fringes are also analyzed by phase-shifting interferometry. The sphere volume is determined from the mean diameter calculated on the basis of the diameter measurement from many different directions. The phase shifts required for phase-shifting interferometry are produced by changing the optical frequency of the diode laser. The optical frequency is controlled on the basis of the optical frequency comb, which is used as the national length standard of Japan (Inaba et al. 2006, 2009).

The temperature of the sphere is measured using small platinum resistance thermometers (PRTs) inserted into copper blocks that come to contact with the sphere. The copper blocks are coated with polyether ether ketone (PEEK) to prevent damage of the silicon sphere. The PRTs are calibrated using temperature fixed points in ITS-90 (Preston-Thomas 1990). The measured diameters are converted to those at 20.000 °C using the thermal expansion coefficient of the ^{28}Si-enriched crystal (Bartl et al. 2011).

A sphere rotation mechanism installed under the sphere is used to measure the diameter from many different directions. In a set of diameter measurements, the diameter is measured from 145 directions distributed nearly uniformly on the sphere surface (Kuramoto et al. 2017a). Ten sets of diameter measurement are performed, and between each set, the sphere is rotated to distribute the starting point of each set of measurements to the vertices of a regular dodecahedron. Because the ten directions defined by the vertices of a regular dodecahedron are uniformly distributed, this procedure distributes all of the measurement points as uniformly as possible. The total number of measurement directions is therefore 1450. The sphere volume is calculated from the mean diameter.

As an example of the results of the diameter measurement, the Mollweide map projection of the distribution of the diameter based on the 145 directions for AVO28-S5c is shown in Fig. 6 (Kuramoto et al. 2021). The three-dimensional plot of the diameter is displayed in Fig. 7, where the deviation from the mean diameter is enhanced (Ota et al. 2021).

A different type of optical interferometer for the volume measurement of ^{28}Si-enriched spheres has been developed at Physikalisch-Technische Bundesanstalt (PTB, Germany) (Nicolaus et al. 2017). Figure 8 shows the mean diameters of two ^{28}Si-enriched spheres measured at PTB and NMIJ (Kuramoto 2021). In the interferometer developed at PTB, an etalon with spherical reference surfaces is used. The mean diameters obtained using the two interferometers with different optical configurations and phase-shifting algorithms show excellent agreement within their

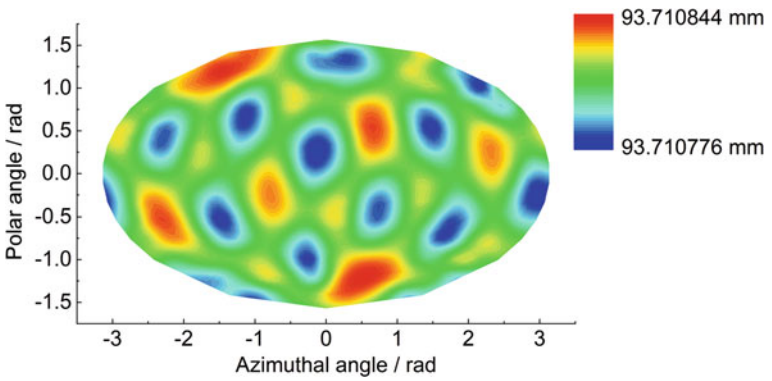

Fig. 6 Mollweide map projection of the distribution of the diameter based on 145 directions for AVO28-S5c. (Kuramoto et al. 2021)

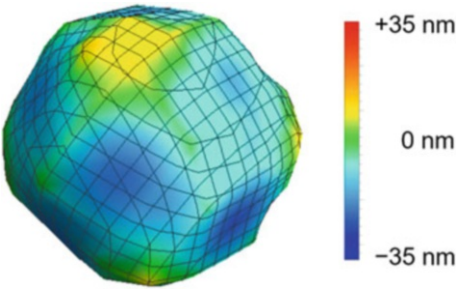

Fig. 7 Three-dimensional plots of the distribution of the diameter of AVO28-S5c. The peak-to-valley value of the diameter is 69 nm. (Ota et al. 2021)

uncertainties. This shows the high reliability of the diameter and volume measurements at the two national metrology institutes.

Sphere Surface Characterization by XPS and Ellipsometry

X-ray photoelectron spectroscopy (XPS) and ellipsometry are used for the sphere surface characterization at NMIJ. The main component of the XPS system is ULVAC-Phi 1600C equipped with a monochromatic Al Kα X-ray source. The pressure in the XPS chamber is reduced to 2.5×10^{-7} Pa. A manipulator with a five-axis freedom is installed in the XPS chamber to realize the rotation of the sphere around the horizontal and vertical axes for the mapping of the entire surface. The sphere is placed on two rollers of the manipulator during the measurement. The rollers are made of polyimide for protecting the sphere surface from damage during the rotation. Details of the XPS system are provided by Zhang et al. (2017).

The main component of the spectroscopic ellipsometer is Semilab GES5E. Its spectral bandwidth ranges from 250 nm to 990 nm. The ellipsometer and an automatic sphere rotation system are integrated to a vacuum chamber to characterize

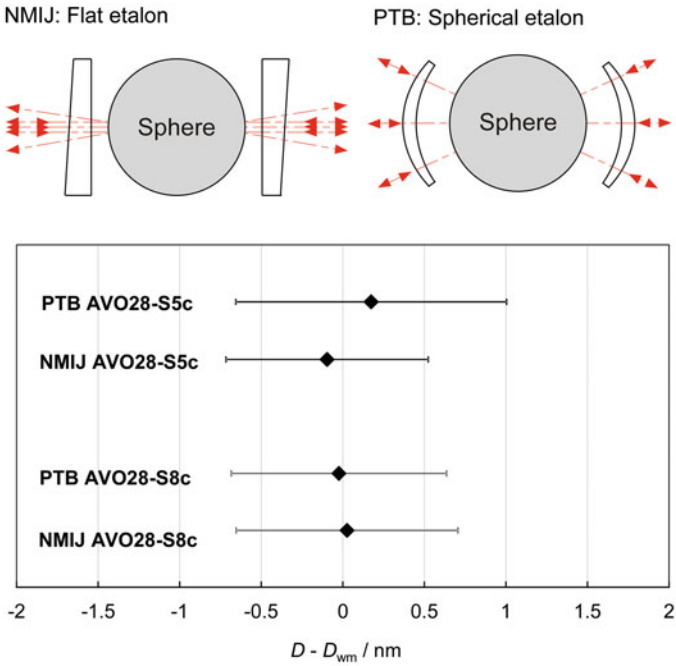

Fig. 8 Top: Optical configurations of the two interferometers of PTB and NMIJ. Bottom: Mean diameters of the ^{28}Si-enriched spheres measured at PTB and NMIJ. The difference from the weighted mean diameter D_{wm} calculated from the diameters measured at the two national metrology institutes is plotted. The horizontal error bars indicate the standard uncertainty of each value (Kuramoto 2021)

the surface layer in vacuum, where the pressure is reduced to 1×10^{-3} Pa. The ellipsometric measurement can be carried out over the entire sphere surface using the automatic sphere rotation system. Details of the ellipsometer are provided by Fujita et al. (2017).

The thicknesses of the CL and OL at 52 points on the sphere surface are measured using the XPS system. The 52 points are distributed nearly uniformly on the sphere surface. The ellipsometer is used to examine the uniformity of the thickness of the OL at 812 points on the sphere surface. Details of the strategy to distribute the measurement point by XPS and ellipsometry on the sphere surface as uniformly as possible are given by Kuramoto et al. (2017c).

Table 2 shows an example of the results of the surface characterization, where the thicknesses of all sublayers of AVO28-S5c measured by Kuramoto et al. (2021) are tabulated. The thickness of the CWL estimated from the water adsorption coefficient to silicon surface reported by Mizushima (2004) is also listed in this table. The total thickness of the surface layer is about 3 nm.

Table 2 Mass, thickness, and density of each sublayer in vacuum of AVO28-S5c (Kuramoto et al. 2021)

Layer	Thickness / nm	Density / (g cm^{-3})	Mass / μg
OL	1.26(7)	2.2(1)	76.5(5.4)
CL	0.81(9)	0.88(14)	19.6(3.8)
CWL	0.28(8)	1.0(1)	7.7(2.3)
SL			103.8(7.0)

Mass of Surface Layer

The mass of the surface layer m_{SL} is given by

$$m_{\mathrm{SL}} = m_{\mathrm{OL}} + m_{\mathrm{CL}} + m_{\mathrm{CWL}}, \tag{9}$$

where m_{OL}, m_{CL}, and m_{CWL} are the masses of the OL, CL, and CWL, respectively. The masses are calculated from the thickness and density of each sublayer. The masses of all sublayers of AVO28-S5c measured by Kuramoto et al. (2021) are summarized in Table 2 with the density and thickness of each sublayer. The mass of the surface layer was determined to be about 120 μg with a standard uncertainty of 8.9 μg.

Volume of Si Core

The volume measured using the optical interferometer is slightly different from the core volume owing to the presence of the surface layer shown in Fig. 4. To determine the Si core volume, the total phase retardation upon reflection at the sphere surface δ is calculated from the thickness and optical constants of each sublayer on the basis of the procedure described by Kuramoto et al. (2011). The value of δ is slightly different from π, and the effect of this phase shift on the gap measurement Δd is calculated using $\Delta d = \lambda\,(\delta - \pi)/(4\pi)$. This means that the actual diameter D_{actual} including the surface layer is larger than the apparent diameter measured by optical interferometry D_{app} by $-2\Delta d$. The phase shift correction to obtain the Si core diameter from the apparent diameter, Δd_0, was determined using $\Delta d_0 = -(2\Delta d + d_{\mathrm{total}})$, where d_{total} is the sum of the thicknesses of all sublayers. The Si core diameter D_{core} is determined using $D_{\mathrm{core}} = D_{\mathrm{app}} + \Delta d_0$.

Mass Deficit

Owing to point defects and impurities, there is a mass difference between a sphere having Si atoms occupying all regular sites and the real sphere. The mass difference, m_{deficit} in Eq. (7), is given by

$$m_{\mathrm{deficit}} = V_{\mathrm{core}} \sum_i (m_{28} - m_i) N_i, \tag{10}$$

where m_{28} and m_i are the mass of a ^{28}Si atom and that of the point defect named i, respectively. For a vacancy, $m_i = m_v = 0$. For interstitial point defects, m_i is the sum of the masses of the defect and a ^{28}Si atom. N_i is the concentration of the point defect i. For example, m_{deficit} of AVO28-S5c was estimated to be 3.6(3.6) μg (Azuma et al. 2015).

Lattice Constant

For the realization, NMIJ uses the value of the lattice constant measured in the IAC project. Thus, only the summary of the measurement procedure is introduced in this section. The lattice constant of silicon crystals can be measured accurately using the combination of X-ray and optical interferometers. Figure 9 shows the measurement principle (Kuramoto 2021). The X-ray interferometer consists of three Si crystal blades that are cut so that the {220} planes are orthogonal to the blade faces. An X-ray beam is split into two beams by the first blade (splitter). The two beams are split again by the second blade (mirror) and recombined by the third blade (analyzer). The X-ray interference fringes produced by the beam recombination are integrated in an X-ray detector. When the analyzer is displaced with respect to the other two blades in a direction orthogonal to the {220} planes, a periodic intensity variation of the X-ray interference signal is observed, as shown in Fig. 9. The period of the variation is the lattice spacing of {220}, d_{220}, and the displacement of the analyzer is measured using the optical interferometer. The fundamental concept of the combination of the X-ray and optical interferometers is to determine the unknown d_{220} on the basis of the known wavelength λ of the laser used in the optical interferometer. The displacement of the analyzer is measured by counting the number of the optical interference fringes having a period of λ/2. Simultaneously, the number of the X-ray fringes having a period of d_{220} is also counted. The lattice spacing d_{220} is determined by comparing the counts of the X-ray and optical interference fringes. The lattice constant a is determined as $a = \sqrt{8}\, d_{220}$, where $\sqrt{8}$ accounts for the different spacings of the {100} and {220} planes. Details of the lattice constant measurement are summarized by Ferroglio et al. (2008).

An X-ray interferometer was manufactured from the AVO28 crystal. The lattice constant of the X-ray interferometer was measured at Istituto Nazionale di Ricerca Metrologica (INRiM, Italy) by Massa et al. (2015). Since contaminants such as carbon and nitrogen strain the crystal lattice, a contamination gradient in the crystal makes the lattice parameters of the sphere different from that of the X-ray interferometer. To measure the contamination gradient, samples were taken from the AVO28 crystal around the sphere. The impurity concentrations in the samples were measured by infrared spectroscopy (Zakel et al. 2011). The lattice constant of AVO28-S5c was then calculated with a relative standard uncertainty of 1.8×10^{-9} by taking into account the different contaminations of the sphere and the X-ray interferometer (Azuma et al. 2015).

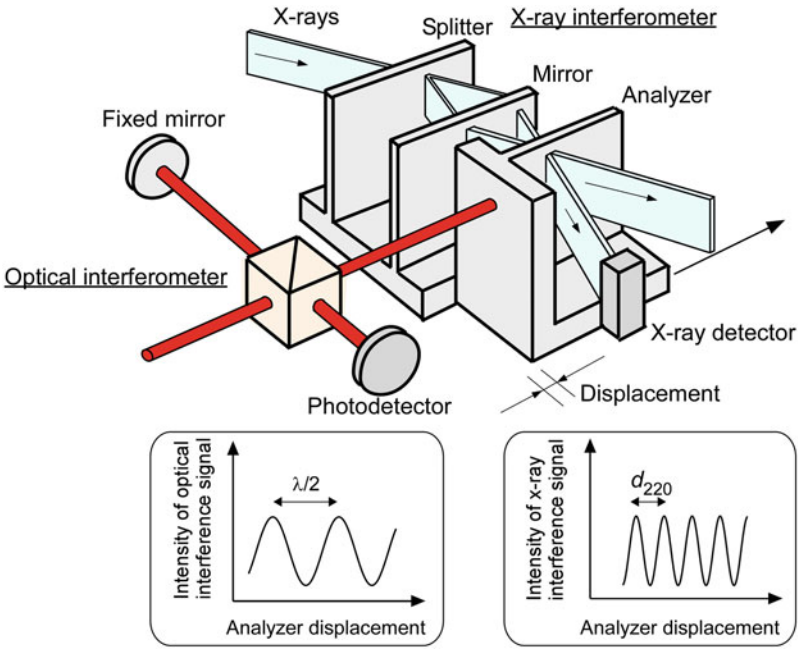

Fig. 9 Principle of the X-ray interferometer to measure the lattice constant. While the analyzer crystal is moved, its displacement is measured using the optical interferometer by counting the optical fringes having a period of λ/2. Simultaneously, the x-ray fringes having a period of d_{220} are also counted. d_{220} is determined on the basis of the laser wavelength λ by comparing the counts of X-ray and optical interference fringes. (Kuramoto 2021)

Molar Mass

For the realization, NMIJ uses the value of the molar mass measured in the IAC project. Thus, only the summary of the measurement procedure is briefly introduced in this section. The molar mass of a silicon crystal material is given by

$$M = \sum_{n=28}^{30} M(^{n}\mathrm{Si})x_n, \tag{11}$$

where $M(^{n}\mathrm{Si})$ is the molar mass of the silicon isotope ^{n}Si and x_n is the amount-of-substance fraction of ^{n}Si. The Institute of Reference Materials and Measurements (IRMM) measured x_n of the ^{28}Si-enriched crystal AVO28 by gas mass spectrometry, where samples prepared from the AVO28 crystal were converted to SiF_4 gas by several chemical processes and the ratios of the ion currents from $^{28}SiF_3^+$, $^{29}SiF_3^+$, and $^{30}SiF_3^+$ were measured (Bulska et al. 2011). However, a small amount of natural silicon was estimated to be contaminated during the conversion processes, and the effect of the contamination on the ^{28}Si fraction measurement could not be accurately corrected. Furthermore, owing to the extremely high isotopic enrichment, the range in which

isotope ratios had to be determined was extremely wide: $1 \times 10^{-6} \leq x(^{n}\mathrm{Si})/x(^{28}\mathrm{Si}) \leq 1$. This caused many difficulties owing to several issues such as detector linearity and dynamic range, limiting the improvement of the measurement uncertainty.

To overcome these difficulties, a new strategy based on isotope dilution mass spectrometry combined with multicollector inductively coupled plasma mass spectrometry was developed at PTB (Rienitz et al. 2010). This strategy does not directly measure $x(^{28}\mathrm{Si})$. Instead, $x(^{29}\mathrm{Si})$ and $x(^{30}\mathrm{Si})$ are measured using an enriched spike of ^{30}Si. Then, $x(^{28}\mathrm{Si})$ is obtained indirectly as $x(^{28}\mathrm{Si}) = 1 - x(^{29}\mathrm{Si}) - x(^{30}\mathrm{Si})$. This significantly narrows the isotope ratio measurement range. In addition, the conversion of the ^{28}Si-enriched crystal to a liquid sample introduced to the mass spectrometer is carried out quantitatively in a single step by using alkaline solutions, avoiding cumulative contaminations by natural silicon. With these improvements, it has become possible to markedly improve the accuracy of the molar mass measurement.

To determine the molar mass of AVO28-S5c, small samples were taken from the AVO28 crystal from both sides of the sphere. The molar masses of the samples were measured at PTB, NMIJ, and the National Institute of Standards and Technology (NIST, U.S.A) and used to estimate the molar mass of AVO28-S5c with a relative standard uncertainty of 5×10^{-9} (Azuma et al. 2015).

Mass of Si Sphere

The sphere mass m_{sphere} is derived from the Planck constant using Eqs. (7) and (8), realizing the new definition of the kilogram. As an example, the uncertainty budget of the determination of m_{sphere} using AVO28-S5c in 2019 is shown in Table 3 (Kuramoto et al. 2021). This determination was performed for the first key comparison of the realizations of the kilogram by NMIs, CCM.M-K8.2019 (Stock et al. 2020). The values of a and $A_{\mathrm{r}}(\mathrm{Si})$ of AVO28-S5c were measured with relative standard uncertainties of 1.8×10^{-9} and 5.4×10^{-9}, respectively, in the IAC project (Azuma et al. 2015, Bartl et al. 2017). The relative standard uncertainty of m_{sphere} was estimated to be 2.1×10^{-8}. This means that the relative standard uncertainty of the realization of the new kilogram at NMIJ was 2.1×10^{-8}, which corresponds to 21 μg for 1 kg.

The largest uncertainty source in the realization is the sphere volume determination by optical interferometry, as shown in Table 3. To enable a more accurate realization at NMIJ, the optical interferometer is being improved (Ota et al. 2021). This improvement will reduce the relative standard uncertainty of the realization of the kilogram to 1.7×10^{-8}, which corresponds to 17 μg for 1 kg.

In CCM.M-K8.2019, the level of agreement between realizations of the kilogram by NMIs was evaluated. This comparison was organized by BIPM and had seven participants including NMIJ (Stock et al. 2020, Davidson and Stock 2021). The participants determined the mass of one or two 1 kg standards by their realization experiment. At BIPM, all mass standards were compared with a reference standard using a mass comparator. These weighings, together with the mass values determined by the participants, allowed a comparison of the consistency of the individual

Table 3 Uncertainty budget for the realization of the new kilogram by the XRCD method using AVO28-S5c at NMIJ (Kuramoto et al. 2021)

Quantity	Relative standard uncertainty in m_{sphere}
Core volume, V_{core}	1.8×10^{-8}
Mass of the surface layer, m_{SL}	7.0×10^{-9}
Relative atomic mass of silicon, A_{r}(Si)	5.4×10^{-9}
Lattice parameter, a	5.5×10^{-9}
Mass deficit due to impurities and vacancies, m_{deficit}	3.8×10^{-9}
Relative combined standard uncertainty	2.1×10^{-8}

realizations. The chi-squared test for consistency using the 95 % cut-off criterion was passed, although the two results with the smallest uncertainty were not in agreement with each other within their standard uncertainties. For this comparison, NMIJ determined the masses two 1 kg platinum-iridium standards on the basis of the mass of AVO28-S5c determined by the XRCD method (Kuramoto et al. 2021). The result of NMIJ was in agreement with the results of the other participants and the key comparison reference value within their standard uncertainties.

Dissemination of Mass Standards Based on the ^{28}Si-Enriched Spheres

Figure 10 shows the traceability chain of mass standards planned at NMIJ after the use of individual realization experiment for the dissemination of mass standards becomes viable (Kuramoto et al. 2020). The masses of the ^{28}Si-enriched spheres are determined by the XRCD method on the basis of the Planck constant, as described in the previous sections. On the basis of the masses of the ^{28}Si-enriched spheres, the masses of 1 kg platinum–iridium weights and 1 kg stainless steel weights in air and vacuum are determined. These 1 kg weights will form an ensemble of reference mass standards and maintain the value of the realized kilogram at NMIJ.

The mass differences between the ^{28}Si-enriched spheres and the 1 kg weights are measured using a mass comparator installed in a vacuum chamber, which enables mass comparisons at a constant atmospheric pressure or under vacuum (Mizushima et al. 2017). Owing to density differences among silicon, stainless steel, and platinum–iridium, the ^{28}Si-enriched spheres and the 1 kg weights have different volumes. This requires air buoyancy corrections in the mass difference measurement. For a precise air buoyancy correction, artifacts are used to accurately determine the air density (Mizushima et al. 2004). The artifacts are a pair of weights that are equal in mass and surface area but different in volume. Furthermore, the masses of the ^{28}Si-enriched spheres and the 1 kg weights in air and those under vacuum are different owing to the difference in the amount of water adsorbed onto their surfaces. For the precise estimation of this sorption effect, a pair of weights that are equal in mass and volume but different in surface area are used (Mizushima et al. 2015).

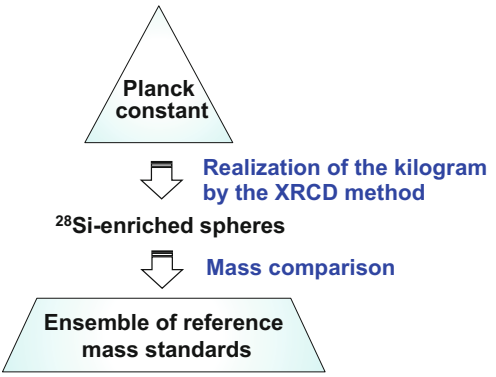

Fig. 10 Traceability chain from the new definition of the kilogram planned at NMIJ. (Kuramoto et al. 2020)

Small Mass Measurement Based on the Planck Constant

Under the previous definition of the kilogram, a reference weight calibrated in a manner traceable to the international prototype of the kilogram was essential to ensure the metrological traceability of mass measurements. Obtaining a reference weight with a mass of less than 1 kg requires submultiples of a primary 1 kg standard. Each time a mass is subdivided, the relative uncertainty of the mass of the weight calibrated by the subdivision process accrues progressively. Figure 11 shows the relative standard uncertainties of the masses of weights with various masses estimated from maximum possible error used in legal metrology (OIML 2004). At the level of 1 kg, the relative standard uncertainty of the mass is 8×10^{-6}. However, by the time the mass is subdivided to 1 mg, the relative standard uncertainty is increased approximately 2,400,000 times (Kuramoto 2021).

On the other hand, the new definition enables us to realize the kilogram using any measurement technology that relates the Planck constant to the kilogram (Kuramoto 2022). Furthermore, 1 kg is not the unique reference point anymore; thus, the kilogram can be directly realized in any range without using a reference weight. These advantages are exploited in the development of the electrostatic force balance for mass measurement in the sub-milligram range (Shaw et al. 2016). In the electrostatic force balance, electrostatic force is balanced with the gravity force acting on a sample. The electromagnetic force is measured using quantum electrical standards based on the Planck constant. The electrostatic force balance can therefore realize the kilogram in the new SI without using the reference weights. Recently, Shaw et al. extended the measurement range by this technique to the microgram range (Shaw et al. 2019). At NMIJ, Yamamoto et al. (2020) and Fujita and Kuramoto (2022) developed an electrostatic force balance for mass measurements in the milligram range. These balances will be also used for the realization of the kilogram at NMIJ in the future.

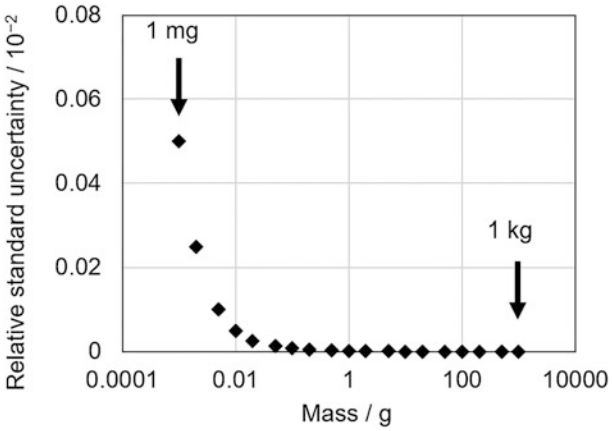

Fig. 11 Relative standard uncertainties of the masses of standard weights estimated from the maximum possible error in OIML R-111 (OIML 2004). Obtaining a standard weight with a mass less than 1 kg usually requires submultiples of a primary 1 kg standard. Each time a mass is subdivided, the relative uncertainties of the masses of the weights calibrated by the subdivision process accrues progressively. (Kuramoto 2021)

Conclusion

Under the new definition of the kilogram based on the Planck constant, it is, in principle, possible for each NMI to realize the kilogram independently. After the confirmation of the international consistency of the individual realizations by NMIs, NMIJ will realize the new kilogram by the XRCD method, where two ^{28}Si-enriched spheres manufactured by the International Avogadro Coordination project are used. The masses of the two spheres are almost 1 kg, and the relative standard uncertainty of the realization is estimated to be 2.1×10^{-8}, which corresponds to 21 μg for 1 kg. On the basis of the spheres, the masses of weights consisting of the NMIJ ensemble of reference mass standards are measured. The ensemble will be used for the dissemination of mass standards to scientific and industrial fields in Japan.

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Quantum Redefinition of Mass

9

The State of the Art

Bushra Ehtesham, Thomas John, H. K. Singh, and Nidhi Singh

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Abstract

This chapter deals with mass measurement and metrological activities from prehistoric civilizations until the present. The importance of the International Prototype of the Kilogram (IPK) and its limitations that created the need to redefine the kilogram in terms of the fundamental constant has been discussed. The redefinition of the kilogram in terms of Planck constant h is a result of the consequent efforts of the metrology community. Some of the significant events that took place in this process are highlighted in the chapter. An in-depth discussion is offered on the four-phase CCM dissemination method. Also, the

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traceability of mass dissemination in India before and after redefinition has been discussed. One of the benefits of the new kilogram definition is that it allows each country to establish its realization capability. The basic principle of different realization techniques adopted by different NMIs to define the kilogram, which includes the Kibble balance, Joule balance, and XRCD method, is also provided. The role of two quantum effects, viz., quantum hall effect and Josephson effect, in defining kilogram using Kibbe balance is explained. Finally, a brief discussion on the present status of different NMI across the globe working on the different primary realization techniques is presented.

Introduction

The mass has been one of the most important physical quantities in day-to-day life for centuries as well as in the scientific study of nature, ranging from celestial to subatomic scales. The mass of an object is measured by weighing, and in popular parlance, mass is often synonymous with weight, but fundamentally they represent two different physical quantities though intimately connected. As the need for weighing different objects was crucial for many daily requirements of humankind, mass measuring tools came into practice at the early stages of human civilization using uniform size objects like stones, seeds, etc. Early developments happened in isolation in different parts of the world, and they primarily served the limited needs of the local communities in their respective areas. Those developments consisted of primitive tools, but they never lacked in ingenuity. Over time, more and more sophisticated techniques were developed that ultimately resulted in the availability of modern ultrahigh-precision balances of various types in use today. Meanwhile, the advent of international trade and commerce demanded the availability of a common set of standards and measuring methods for physical quantities that can facilitate creating a harmonious procedure for the activities concerned. The signing of the Meter Convention and the founding of the BIPM in 1875 heralded a new beginning for more comprehensive international cooperation to establish a common system of units, standards, and measurement methods accepted globally. The kilogram was accepted as the unit of mass, and subsequent events catalyzed its definition in terms of an artifact, which was termed the international prototype kilogram (IPK) in 1889. The kilogram is one of the seven base units in the SI system. Advances in twentieth-century physics paved the way for defining the SI base units in terms of fundamental physical constants, and this led to the abrogation of the artifact-dependent definitions of the SI base units one by one, and the kilogram is the last one that has been redefined in this manner. The IPK remained the primary standard of mass until it was replaced by its new definition, based on the Planck constant h, on 20th May 2019.

This chapter briefly describes the metrological practices prevalent in primitive civilizations and the subsequent evolutions, driven by human curiosity and the thirst to unravel nature. Next, the IPK and its role in ensuring traceability for mass measurements worldwide will be discussed. The shortcomings of the IPK as the primary standard of mass and the consequent efforts of the mass metrology

community in redefining the kilogram will also be addressed. Then the latest developments taking place in mass metrology post-redefinition of the kilogram will be presented in some detail. Finally, the details of different NMIs working on the primary realization method of the kilogram are discussed.

Weights and Measures in Ancient Times

Weighing things is one of the primary and essential needs of human civilization. The use of weights and measures can be traced back to the Egyptian civilization, the earliest civilization known. The most primitive form of metrology appeared around 8000 BC when human society transitioned from nomadic wandering to a settled existence. As a center of world trade, Egypt has a unit of mass that has varied over time, giving the median value of 9.33 g (Willard and Selin 2008; Alberti 2004). The hierarchy of weights in Egypt goes as 10 *qedet* = 1 *deben*; 10 *deben* = 1 *sept*. The Egyptian weights were usually made of stones, but gold and copper deben were also found with mass 12–14 g and 27 g, respectively. Egyptian balances were made of wood or bronze. Usually, loaf-shaped weights and a few animal-shaped weights (18th Dynasty characteristic weights) were used. Authorized reference standards of length, volume, and weight were preserved in places of sanctity like places of worship and royal palaces. In this era, weighing was mainly used for weighing jewels and metals in manufacturing. Figure 1 shows the oldest known Egyptian

Fig. 1 The oldest Egyptian balance is made from pinkish-brown limestone. (Source: Willard and Selin 2008)

balance based on the equal arm principle and made of limestone and belongs to the Predynastic Period (8000 BCE–3000 BCE). Balances were also used for religious purposes, like weighing tributes to the pharaoh and national god *Amun*. Over a while, various improvements were conducted to Egyptian balance, and it served as a standard weighing instrument till Roman times (Willard and Selin 2008).

Babylonia was a state of Mesopotamia, and it was one of the most developed civilizations around 2300 BC. The unit for weight was *shakel*, and 3000 *shakel* = 1 *talent*. The oldest available weights were limestone with a dome shape and a flat bottom. Some bronze specimens also exist. The mean mass of a *shakel* was found to be 11.4 g. *Tyrian shakel* weighs 14 g and was well-known for the pureness of metal and homogeneity of weight (Holland and Selin 2008). In Africa, Akan gold weights were made of bronze or brass (Niangoran-Bouah and Selin 2008). Some weights were made of silver, copper, and solid gold also. Akan weights can be divided into three categories: geometrical weights, figurative weights, and weights with graphic design. Irrespective of their shape and mass, all weights closely depict the Akan cultural beliefs and are called *abrammuo*. For example, gold weight is shown in Fig. 2, which shows a bird looking backward called *Sankofa* preached "to pick it up if it falls behind you," in other words, learn from your experience and never forget your roots. Figure 3 shows different types of weights used in Africa (Niangoran-Bouah and Selin 2008).

Fig. 2 "Sankofa" bird made of brass by Akan's using lost wax cast technique (The British Museum n.d.)

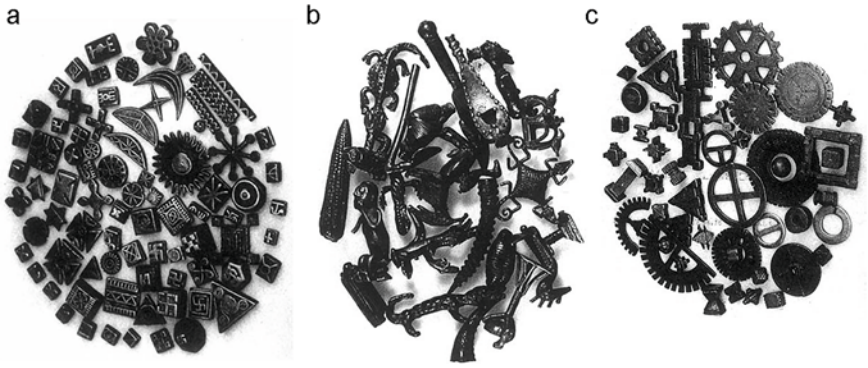

Fig. 3 Weights and measures in Africa: Akan gold weights. (**a**) Weights with graphic designs. (**b**) Weights with figurative elements. (**c**) Weights with geometric elements. (Source: (Niangoran-Bouah and Selin 2008))

Weights and measures for the Indus Valley civilization date back to the fifth millennium BCE. The Indus civilization comprised of Harappa and Mohenjo-Daro regions, and its spread was more than one million square kilometers. In ancient India, the flourishing trade of raw materials, both inland and overseas, led to a golden era of 550 years from 2300 BCE to 1750 BCE (Iwata and Selin 2008a). Weights mined from Mohenjo-Daro, Harappa, and Chanhu-Daro show elaborate systems of ascending weights. The first seven Indus weights double in size in ratio 1:2:4:8::16:32:64. As ratio 16th weight was widely excavated from these sites, which had a mean mass of 13.7 g, it is assumed to be the unit of mass for Indus people. These weights are cubic in shape, and 68% of excavated weights were made of chert, but a few weights made of agate and graphite were also found. The most significant weight found in Mohenjo-Daro is 10,865 g. These weights were used alongside two pan balances made of copper, bronze, ceramic, and sometimes terracotta. Figure 4 shows the weights used in the Indus civilization and excavated from Taxila (Iwata and Selin 2008a). In the Vedic era, different units of weights were used to measure different items. For example, the *drone* was the unit for food-related items, whereas *Aksa* was the unit to weigh gold (Gupta 2010). In the pre-Akbar period, precious metals like gold and silver were measured by the grain of wheat or barleycorn. Akbar standardizes the weight standard as *jau* (barleycorn). Before the arrival of the metric system in India (1853), the terminology being used for weights is given below (Shrivastava 2017):

4 chawal (grain of rice) = 1 dhan (weight of one wheat berry)
4 dhan = 1 ratti = 1.75 grains = 0.11339825 g
8 ratti = 1 masha = 0.9071856 g
12 masha = 96 ratti = 1 tola =180 grains = 11.66375 g
80 tolas = 1 seer = 870.89816 g
40 seers = 1 maund = 8 pasri = 37.32422 kg
1 chattank = 4 kancha = 5 tola 1 pav = 2 adh-pav = 4 chattank = ¼ seer

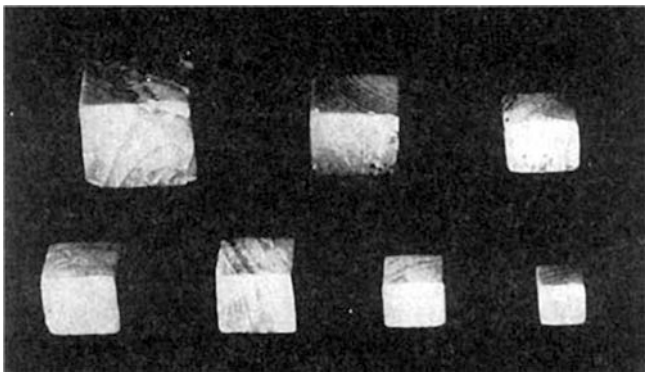

Fig. 4 The weights used in the Indus Valley civilization. (The figure is replicated from (Iwata and Selin 2008a))

1 seer = 4 pav = 16 chattank = 80 tola = 933.1 g
1 paseri = 5 seer

The Arabic physicists extensively studied concepts of weight, density, buoyancy, and their related measurements. They were conversant with Archimedes' principle and the law of the lever. Using these principles, they developed scientific apparatus, i.e., balance, to study in detail on areas such as specific weights. Thabit ibn Qurra (circa 836–901) worked in depth on the equi-armed balance. His ideas subsequently became quite effective in the West. Twelfth-century scientist Al-Khāzini (fl. 1115–1130) wrote *Kitāb Mizān al-Hikmah* (*Book of the Balance of Wisdom*) which encompasses formulae for the specific and absolute weights of various alloys. Further, a book entitled *Control of Weights and Measures* was written by eighth-century present-day Tunisian scientist *Yahyā al Kināni* (Rebstock and Selin 2008; McGrew and Selin 2008; Djebbar and Selin 2008).

There is evidence suggesting the existence of Peruvian's mass standard also; however, it has been challenging to estimate the mass standard precisely in view of the fewer weights discovered in this region as compared with the other civilizations. The notation was based on the decimal system. The largest unit was presumed to be 23.1 kg. In other civilizations like Egypt, Mesopotamia, Indus, and China, 27–30 kg was found to be the largest weight range, probably because of the low concentration of oxygen in the atmosphere in the Andean highlands.

The following are the material and the percentage used in making the weights:

Weight material	Percent
Stone	39
Iron	32
Lead	24
Nonferrous	5

The preferred shape was globe which comprises one-third of the weights; others were conical, cylindrical, or in the spindle and other irregular forms (Iwata and Selin

Fig. 5 Weights of different shapes used in Burma. (Taken from Gear et al. (2008))

2008b). *Mohists* were the followers of philosopher *Mo Zi* (warring states period, BC 480–221) and performed many experiments to determine the properties of levers and balance as discovered from Chinese text on physics. They gave the definition of forces and weights. Also, they used laws of lever and balance to develop a simple version of the "Atwood machine," which was later developed in Europe in 1780 (Smith and Selin 2008).

The Burmese empire used a weight system based on animal-shaped weights (Gear et al. 2008). These mythical animals are mainly in the shape of *leonine* (lion-like), *elephantine*, *ansorine* (goose-like), and *linaceous* (poultry-like), as shown in Fig. 5. These weights are highly influenced by the Buddhist art in India and usually are in the dimension of 70 × 120 mm. These weights were made of copper alloy, lead, and tin. *Kyat* was the unit of mass whose weight varied from 14 g to 16 g from the fifteenth century to the eighteenth century, respectively. One of the main duties of a monarch was to ensure the standardization of weights across the Burmese empire. An early record of nationwide standardization can be found, which dates back to the eleventh century. In the nineteenth century, legal weight was crafted in the palace under the direction of the chief minister. For measuring valuable commodities like seeds, etc., the equal arm balance of Chinese type was used (Gear et al. 2008).

Mass Measurements in the Modern Era

Human civilization has come a long way since 8000 BC, when Egyptians fabricated the first balance, and its illustrations can still be found on the wall of Egyptian tombs. The need to measure mass faster and with higher accuracy leads to the development of modern-era balance with elaborate techniques behind it. Presently there are various ultra-advanced balances to measure a wide range of masses with mind-boggling accuracy. There are different types of balance available these days. Though a complete description of the latest advancement in these types of balance is not

possible here, a few types of modern balances are described. The present-day **equal arm balance/trip balance** is an improved version of ancient Egyptian balance. As in ancient Egyptian balance, it also consists of two pans hung on each side of the lever. The quantity that must be measured is placed in one pan, and calibrated masses are put against it in another pan until both pans are balanced. The value of calibrated mass used to balance the item represents the mass of the measured quantity. This balance is also used when one item needs to be adjusted with respect to the other. Thus, it is helpful in situations like balancing tubes or centrifugation (Zimmerer 1983).

After World War II, **single pan substitution balance** became famous as they were faster in measurement than equal arm balance. Even less skilled operators could use it with relatively higher accuracy than equal arm balance. The second pan was replaced by the fixed calibrated counterweights securely put inside the balance frame. The operator can change these weights by a rotating knob depending on the measured mass. The main advantage of this balance was its more straightforward construction as it requires only two knife edges instead of three, as in the case of equal arm balance. The main knife edge was under constant load irrespective of the mass of the object (Jones and Schoonover 2002).

The fundamental principle of balance is that the gravitational force balances force on one object with the other object. However, there is a class of balance where the gravitational force is balanced by electromagnetic force, called **electronic balance**. A wide variety of principles is being used for electronic balances having the finest precession of 10^7 at full capacity (Guzy 2008). Two major types of electronic balances are **hybrid balance** and **electromotive force balance**. In hybrid balance, the large angular displacement of the balance beam is restricted by servo-controlled electromagnetic force being applied to the beam. In an electromagnetic force, the entire load is balanced either by direct levitation or through a fixed-ratio lever system (Jones and Schoonover 2002). The sensitivity and response of the balance do not depend on the dynamics of the balance beam but are primarily controlled by servo system characteristics. One of the highly accurate electronic balances is the **analytical balance** (Clark 1947). These are widely used in laboratories, especially in chemical analysis labs where precise mass measurement is necessary. Essential components of an analytical balance are a beam arrest, a measurement pan, levelling feet with a spirit level, and an enclosure. The primary function of beam arrest (a mechanical device) is to provide safety to the delicate internal system of balance when it is moved. An analytical balance is so sensitive that even an air current can alter its reading; thus, an enclosure is used around the measurement pan. A lower version of analytical balance called **precision balance** is also used in areas where extreme precision is not required. They are top-loading balance. Though the readability of the top-loading balance is less, they are a more convenient and economical choice as they can withstand harsh chemicals, dust, dirt, and even splashes of liquids.

Another principle widely used for electronic balance is the use of elastic force against gravitational force, e.g., the twisting property of wire. Microbalances and ultramicrobalances that weigh fractional gram values are **torsion balances**. As quarts are used to produce a very stable frequency, they are commonly used in

torsion balance. A thin wafer of quartz with electrodes on both surfaces produces vibrations through the piezoelectric effect (Zimmerer 1983; Gillies and Ritter 1993).

A triple beam balance is a type of balance commonly used in the laboratory to determine sample mass (by weight comparison). It is called a triple beam because of the three beams on the scale used to determine the item's weight. The first beam measures 0–10 g, the middle beam weighs 10 g increments, and the far beam weighs 100 g increments.

Platform scale is another type of balance, using a well-designed scheme of multiplying levers. In this type of balance, the weight of the measuring item is transferred to a moving beam which balances itself against a counterweight. This balance is widely used in veterinarian facilities for weighing animals.

Mass comparators are also a class of balances that offers high resolution and exceptional repeatability in weighing. This is why it is widely used in metrological institutes for the calibration and verification of masses. A window-range mass comparator is used for mass calibration within a range as they provide excellent resolution. Full-range mass comparators are multipurpose and very efficient when samples of different sizes need to be weighed together. Commercially mass comparators are available for masses in the range of 1 mg up to 5000 kg (Borys et al. 2012). There are various types of mass comparator, among which vacuum mass comparator and robotic mass comparator provide exceptionally accurate results. As the name suggests, vacuum mass comparators weigh mass in a vacuum, eliminating the need for buoyancy correction and significantly improving mass measurement uncertainty. Another highly accurate comparator is the robotic mass comparator. When the determination of mass value to the utmost accuracy is required, these comparators are used by laboratories. The entire calibration process is automated, and a robotic arm attached to the comparator transfers weights from magazine to comparator and vice versa. It can also calibrate complete weight sets having different values automatically (Toledo n.d.).

Artifact-Based Definition of Mass: Need to Change

The rapid developments in the seventeenth century onward resulted in the increase in global trade, exchange of social goods, scientific ideas, and technological know-how. This highlighted the need for some sort of universal system of units and measurements. In 1889, at the first General Conference on Weights and Measures (CGPM), the prototype of the kilogram called international prototype of kilogram (IPK) was adopted and declared unit of mass. The official definition of the unit of mass was adopted in the third meeting of the CGPM in 1903. The IPK served as the primary standard for mass from 1889 till 2018. It is an alloy of 90% platinum and 10% iridium alloy cast in the shape of a cylinder of 39 mm in height and diameter by Johnson Matthey in 1879 (Ehtesham et al. 2021). The choice of material required to construct mass standards was constrained by a host of physical and chemical properties which designated materials must possess, e.g., the materials needed to have properties such as high density, good hardness, good electrical and thermal

conductivity, immunity toward corrosion, and ability to minimize the effects of air buoyancy. A good electrical conductivity is necessary to eliminate parasitic forces due to static electricity. Another prerequisite is a low magnetic susceptibility to limit parasitic force due to magnetic field. In that era, platinum was considered to fulfil all these criteria except hardness. Thus, iridium was mixed with platinum to cast the primary standard of mass. Additionally, the technology for melting platinum and purifying it was just being perfected on an industrial level. Initially, three copies of the platinum-iridium prototype were fabricated by Johnson and Matthey, out of which one was made IPK or called Le Grand K, and the rest two were made its official copies. Later on, the number of official copies rose to six. Johnson and Matthey also produced the first 40 Pt-Ir prototypes, which were calibrated against IPK and distributed to the member state of the Meter Convention (Yadav and Aswal 2020). They are called the national prototype of the kilogram (NPK). The kilogram was the last unit until 2018 which was defined by a man-made object. Being an artifact, IPK is subjected to surface contamination or damage while cleaning, washing, or during measurements. The outcome of the second periodic verification held in 1946 shows the mass gain in some of the official copies with respect to IPK. This trend was confirmed by the third periodic verification held from 1939 to 1946. A careful comparison between IPK and its official copies kept at BIPM shows the drift of approximately 50 μg in IPK over 100 years, as shown in Fig. 6 (Ehtesham et al. 2021; Bosse et al. 2017).

Due to these reasons, on 16th November 2018 in the 26th CGPM conference held in Versailles, France, the old definition of mass, which was "The kilogram is the unit of mass; it is equal to the mass of international prototype of the kilogram" was

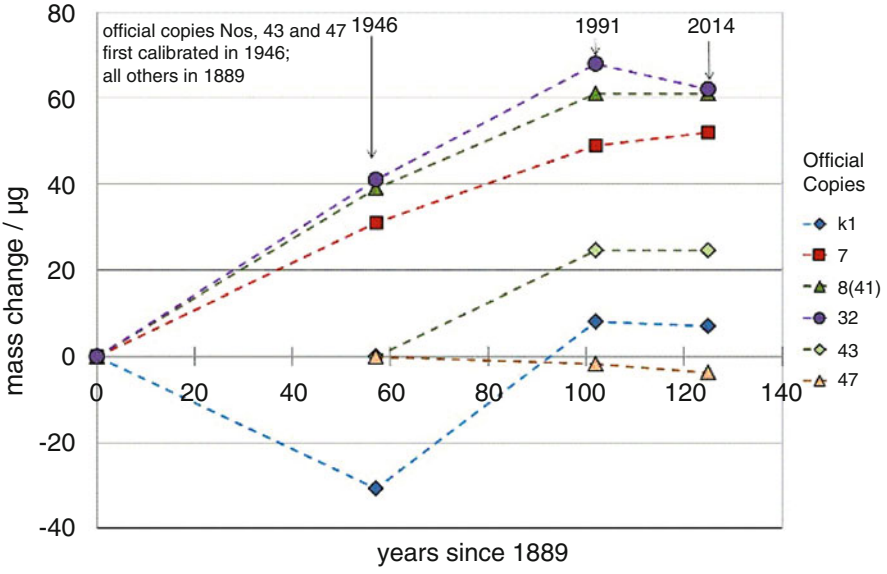

Fig. 6 The behavior of the six official copies over the period with respect to IPK

replaced by the new definition of the kilogram. The new definition of kilogram based on fundamental constant is "The kilogram, symbol kg, is the SI unit of mass. It is defined by taking the fixed numerical value of the Planck constant h to be $6.62607015 \times 10^{-34}$ when expressed in unit J s, which is equal to kg m^2 s^{-1}, where the meter and the second are defined in terms of c and Δv_{Cs}" (Ehtesham et al. 2020). The CSIR-NPL being the national metrology institute (NMI) of India also voted in favor of the redefinition of the kilogram in Versailles. India is also a member state of BIPM since 11th January 1957.

Events Leading to the Redefinition of the Kilogram

The need to express the kilogram in terms of the physical constant was thought around 1791 by King Louis XVI, who also passed a decree emphasizing the definition and realization of measurement units in terms of physical constants. In that era, the kilogram was initially expressed as the mass of 1 l (10^3 cm^3) of distilled water at 0 °C. Later in 1795, the reference temperature of the water was changed to 4 °C. The kilogram was then defined as the mass of 1 l (10^3 cm^3) of distilled water at 4 °C. In 1799, it was the first time when an artifact made of a platinum sponge was fabricated to realize the kilogram. The artifact called kilogram of archives, the **KA**, is cylindrical in shape (Davis et al. 2016). The more refined artifact called IPK was later fabricated with 90% platinum and 10% iridium alloy, and in 1901 at the third CGPM, official definition of mass, viz., "The kilogram is the unit of mass; it is equal to the mass of international prototype of the kilogram," was adopted. During the second (1939–1946) and third periodic verification (1988–1992) of IPK with its official copies, drift in the mass was observed (Davis et al. 2016; Girard 1994). The official copies of mass showed deviation of several tens of micrograms as compared to IPK. The nonconformity in the weights of IPK and its official copies implied that either the mass of IPK or its official copies or both are changing with time. In 1995, at the 20th CGPM, it was recommended to first monitor the stability of IPK and second to find new ways to express the kilogram in terms of fundamental constant. Two foremost methods were put forth and discussed to link mass to fundamental constant, viz., watt balance and X-ray crystal density (XRCD) method. In the watt balance principle, mass is realized in terms of the Planck constant (h), whereas in the XRCD method, mass is linked through the Avogadro constant. In 1990, two revolutionary developments of Josephson voltage standard (JVS) and quantum Hall resistance (QHR) standard linked voltage and resistance measurement to the Planck constant. Though the principle of watt balance was given by Dr. Bryan Kibble way back in 1975, with the advent of JVS and QHR, now the electrical measurement of watt balance can be traceable to "h." In 2011 at the 24th CGPM meeting, it was decided that the Planck constant would be used for the redefinition of mass with unit kg m^2 s^{-1}. There were four necessary conditions laid out by the Consultative Committee for Mass (CCM) to follow before the redefinition could happen. Details of these four conditions can be found in reference (Ehtesham et al. 2020). To redefine the kilogram in terms of the Planck constant, at least

Table 1 List of fixed value of fundamental constant assigned by CODATA in 2017

Quantity	Value
Planck constant (h)	$6.626070\ 15 \times 10^{-34}$ J s
Elementary charge (e)	$1.602176634 \times 10^{-19}$ C
Boltzmann constant (k)	1.380649×10^{-23} J K^{-1}
Avogadro constant (N_A)	$6.022140\ 76 \times 10^{23}$ mol^{-1}

3 independent results from the XRCD experiment or watt balance experiment should yield relative standard uncertainty of less than 5 parts in 10^8 (50 ppb), and at least 1 of the results should be less than 2 parts in 10^8 (20 ppb). In 2014, a pilot study was conducted by CCM called "The Extraordinary Calibration" to check the consistency in results of various ongoing experiments to realize a new definition. LNE, NIST, and NRC participated in the calibration with their results of Kibble balance, and NMIJ and PTB contributed with the results of XRCD experiments. The results of these five NMIs were compared with the results of IPK from BIPM to check the consistency (Stock et al. 2018). In 1966, the Committee on Data for Science and Technology (CODATA) was established with the purpose of providing the most accurate value of fundamental constants and conversion factors that can be used in various streams of research, including the metrology community. In 2017 CODATA provides exact values of fundamental constants, which will be going to use in the redefinition of SI units of mass, current, temperature, and amount of substance. The fixed values of fundamental constants are shown in Table 1 (Newell et al. 2018). Finally, on 16th November 2018, at the 26th CGPM meeting held in Versailles, France, delegates from the entire metrology community unanimously voted for the redefinition of kilogram, ampere, kelvin, and mole. Officially the new definition of the kilogram "The kilogram, symbol kg, is the SI unit of mass. It is defined by taking the fixed numerical value of the Planck constant h to be $6.626\ 070\ 15 \times 10^{-34}$ when expressed in unit J s, which is equal to kg m^2 s^{-1}, where the meter and the second are defined in terms of c and Δv_{Cs}" came into effect globally from 20th May 2019 (World Metrology Day). The IPK which prior has no uncertainty was assigned the uncertainty of 10 μg until the "consensus value" based on the first key comparison between various realization experiments (Kibble balance, joule balance, XRCD method) is completed. On completion of the first key comparison (CCM.M-K8.2019), on 1st February 2021, the standard uncertainty in the "consensus value" of the kilogram is set to be 20 μg (Davidson and Stock 2021).

Traceability of Mass Calibration in India

India signed the Meter Convention in 1957 and got prototype no. 57 (NPK-57). CSIR-National Physical Laboratory is the NMI of India. According to the subordinate legislation of the Weights and Measures Act 1956 passed by parliament, NPL is responsible for realizing the units of physical measurements based on the International System (SI units). It is the mandate of NPL to realize, establish, maintain, reproduce, and update the national standards of measurement and calibration

facilities for different parameters. NPL is presently maintaining standards for meter, kilogram, second, kelvin, ampere, and candela. NPL maintains the traceability of kilogram using NPK-57, which is kept under the custody of the director of CSIR-NPL. Once every 10 years, NPK-57 is sent to BIPM and is calibrated against IPK (CSIR-National Physical Laboratory n.d.). The last calibration occurred in 2012, and after following the CCM decision on implementing the redefinition of the kg, the uncertainty of NPK-57 is 49 μg at $k = 2$.

Within India, CSIR-NPL disseminates traceability to industries, government departments, R&D organizations, legal metrology organizations, regional calibration laboratories, and other customers through its calibration service traceable to NPK-57. The most important customers who avail of the calibration service of CSIR-NPL include the pharmaceutical industries, aviation industries, satellite or aerospace industries, bullion markets, automobile industries of India, Indian army, Indian navy, Indian air force, DRDO, ISRO, etc. CSIR-NPL also provides traceable calibration services to SARC NMIs.

NPK-57 is a very precious artifact, so it can't be used in day-to-day calibration works. For that, NPL has four transfer standards of 1 kg, two of them made of stainless steel and two of them made of Ni-Cr alloy. These transfer standards are calibrated against NPK-57 every 3 years, and they are used to calibrate other working standards that vary from 1 mg to 500 kg. Working standards are made of stainless steel, brass, and Ni-Cr alloy. Working standards of NPL are calibrated against the transfer standards for 1 or 2 years. These weights are used to calibrate customers' weights through the subdivision or substitution method. CSIR-NPL uses a high-precision mass comparator for calibration, such as a 1 kg vacuum mass comparator of Sartorius make, having repeatability in a vacuum less than 0.1 μg. For calibration up to 5 g mass, the manual mass comparator is used with a readability of 1 μg. To provide better accuracy in mass measurement to its user, CSIR-NPL has purchased a robotic mas comparator (Mettler Toledo, e5) for mass up to 5 g with improved readability of 0.1 μg. Mettler Toledo XP604KM mass comparator is used to calibrate masses of a higher order, such as 500 kg, with the readability of 50 mg. The maximum mass calibration capability of CSIR-NPL is 2000 kg which can be calibrated with repeatability of 1 g using the KE2000 mass comparator. Figure 7 depicts the hierarchy and traceability of mass standards starting from IPK in India before the redefinition of the kilogram. Following the kilogram's new definition, there is a change in the traceability pyramid shown in Fig. 8. As the change is in the definition of the kilogram only, the IPK will be replaced in due course by the new realization technique based on the Planck constant as the primary source of traceability in this chart.

Post-Redefinition and the Need for Establishing New Methods for Traceability

With the worldwide implementation of the new definition of the unit of mass, the IPK lost its status as the primary standard and ceased to be the source of traceability for mass calibrations. There is a transition period for which the Consultative

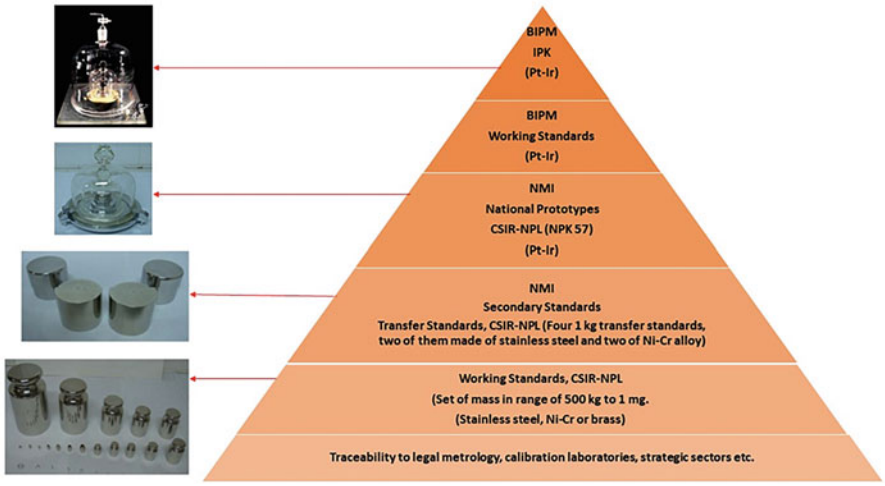

Fig. 7 Hierarchy of mass dissemination in India before redefinition

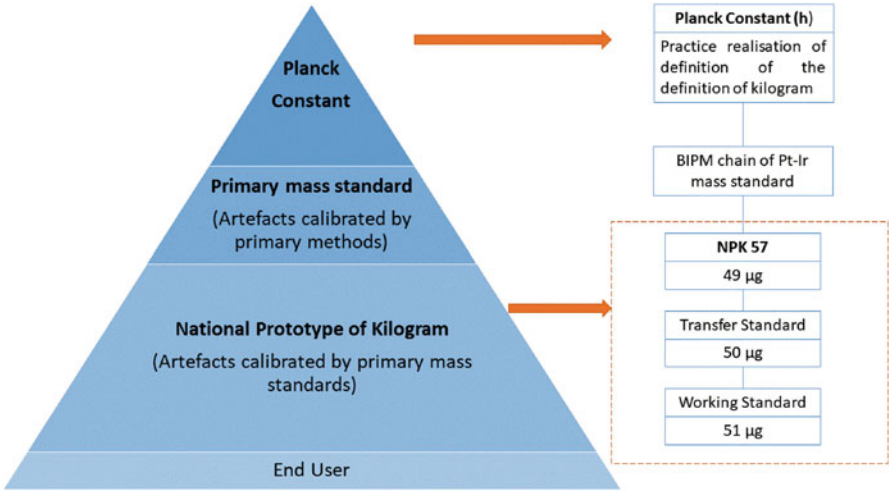

Fig. 8 Hierarchy of mass dissemination in India after redefinition

Committee for Mass and Related Quantities (CCM) has put in place a mechanism that follows a four-phase program that would ensure traceability of the measurements shown in Fig. 9; however, details can be found in (Ehtesham et al. 2020; Davidson and Stock 2021; CCM n.d.; Stock et al. 2020; Bettin et al. 2019). The end of phase 4 will mark the full implementation of the new definition, by which time the new realization methods based on h would become available to provide traceability. In this new scheme of things, there can be many sources for traceability, unlike the

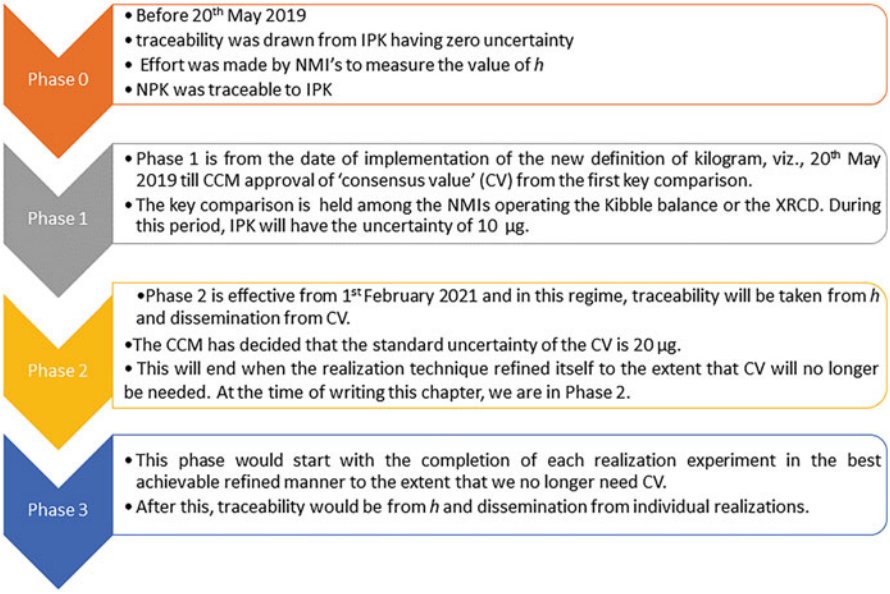

Fig. 9 Four-phase dissemination mechanism for kilogram

case with IPK, as any NMI that has established the new realization method can provide the needed traceability.

Basically, two methods, viz., the Kibble balance and the X-ray crystal density (XRCD) technique, are available to realize the mass unit based on *h* (Bettin et al. 2019; Kibble and Hunt 1979). As of now, a couple of leading NMIs, like NPL, UK, and NIST, USA, have been able to establish the Kibble balance facility in their countries. There are a few other NMIs, like the National Research Council (NRC-Canada), BIPM-France, Swiss Federal Office of Metrology and Accreditation (METAS-Switzerland), Centre de Metrologie Electrique (BNM-LNE/CME-France), National Institute of Metrology (NIM-China), Korea Research Institute of Standards and Science (KRISS-Republic of Korea), etc., which also have either developed their Kibble balance or have reached an advanced stage of the development. Once operational, all these countries will be able to provide traceability for mass measurements both nationally and internationally (Ehtesham et al. 2020).

Methods of Realization of New Definition of Kilogram

The new definition of kilogram eliminates the dependency on an artifact for the global traceability chain of mass; instead, it provides an opportunity for every NMI the possibility to realize a kilogram. There are two foremost techniques to realize kilogram from the atomic or quantum standards, i.e., Kibble balance and the XRCD method. Having redefined the kilogram, the emphasis now is on making an available

number of realization options with the required measurement accuracy and lowest possible uncertainty that would fulfil the CCM requirements envisioned in the four-phased traceability scheme. This will put in place a robust infrastructure necessary to fully implement the new definition by realizing the unit of mass widely. XRCD and Kibble balance are the two independent methods available for relating mass to the Planck constant *h*. Both these methods were used for the experimental determination of the most accurate value of *h* that has been used for the redefinition of the kilogram. Now, after the implementation of the new definition, both these techniques are being used reversely as realization methods for the redefined kilogram. Both of them have been able to realize the kilogram with an uncertainty of a few parts in 10^8. The laboratories that have developed them are working to improve their instruments and measurements to reduce uncertainty further increasingly.

The two laboratories, viz., PTB, Germany, and NMIJ, Japan, realized the kilogram by the XRCD method. The XRCD method involves counting the number of atoms in a crystal where the mass of the atom is related with h through the Avogadro number N_A and the Rydberg constant R_∞. The measurement is performed using a sphere made of silicon atoms of accurately known isotopic composition (^{28}Si), and the number of atoms in the sphere is counted by measuring its volume and lattice constant.

The Kibble balance works by balancing the gravitational force on the mass by an electromagnetic force generated by a current-carrying coil immersed in a magnetic field. Kibble balance was the first experiment that linked microscopic mass to macroscopic physical mass. The connection between the test mass and h is achieved through the two quantum phenomena based on electrical measurements that employ the Josephson effect and the quantum Hall effect. About a dozen NMIs are working on developing the Kibble balance in their countries. As of now, laboratories with operational Kibble balances include NIST, USA; BIPM, France; KRISS, Republic of Korea; NIM, China; and NRC, Canada, and they have participated in the first key comparison of realizations of the kilogram (CCM.M-K8.2019) conducted for verifying the consistency of these individual realizations, an essential step in the planned phase-wise dissemination of the kilogram. The remaining sections of this chapter will present the basic theory underlying both the XRCD and Kibble balance techniques. But before dealing in-depth with primary realization techniques for kilogram, the role of quantum metrology has to be discussed first.

The Josephson Voltage Standard

Before dealing further with the Kibble balance principle, it is important to discuss two quantum phenomena-based standards, viz., Josephson voltage standard (JVS) and quantum Hall effect-based resistance standard (QHRS). The quantum volt is based on the inverse AC variant of the Josephson effect. The Josephson effect,

discovered in 1962 by Brian Josephson, is the tunnelling of Cooper pair condensates through a thin tunnel barrier separating two superconducting electrodes without resistance. The Josephson effect links voltage to frequency, the fundamental constants h and e, and an integer "n" only:

$$V_J = n\frac{h}{2e}f$$

where V_J is the quantized Josephson voltage, f is the frequency of the RF radiation used to irradiate the Josephson junction, e is the electronic charge, and n is an integer. Irradiation of a Josephson junction by electromagnetic radiation of frequency f, the Josephson oscillations get synchronized with the irradiation frequency, and the voltage V across the junction is a stepwise function of the bias current. The height of the voltage step is given by $V = (h/2e)f$. Thus, the voltage V is independent of the materials of the junction and its operating temperature. The phenomenon described above is referred to as the AC Josephson effect or, strictly speaking, the inverse AC Josephson effect. Since the frequency can be measured with an uncertainty of the order of 10^{-15}, therefore, the Josephson effect provides a quantum voltage standard producing a voltage reference V_R with reproduction uncertainty unavailable to classical standards. The voltage V_R is given by the following:

$$V_R = n\frac{h}{2e}f = n\frac{f}{K_J}$$

Here n is the step number in the stepwise current-voltage characteristic. It is interesting to note that the value of the Josephson constant $K_J = 2e/h$ is determinable with an accuracy better than that of e and h separately. The BIPM has fixed the following value of $K_J = 2e/h$ to be 4835979×10^{12} Hz/V. This conventional value of the Josephson constant is denoted by $K_{J\text{-}90}$:

$$V_{90} = n\frac{f}{K_{J\text{-}90}}$$

The above equation represents the unit volt, and $K_{J\text{-}90}$ is taken as a constant with zero uncertainty as there is no comparison to SI quantities made. A realization of the volt is achieved; the SI definition of volt is used. The uncertainty of voltage measurements based on $K_{J\text{-}90}$ is a few parts in 10^{10} or better (Wood and Solve 2009).

The Quantum Hall Resistance Standard

The quantum Hall effect (QHE) is a characteristic of a perfectly quantized two-dimensional electron gas (2DEG) and provides an invariant reference for resistance

linked to the natural constants, "h" the Plank constant and "e" the electronic charge. The 2DEG can be realized in a high-mobility semiconductor device such as silicon MOSFET (metal-oxide-semiconductor field-effect transistor) or GaAs/$Al_xGa_{1-x}As$ heterostructure, monolayer graphene, and some topological insulator thin films. In the GaAs Hall device, the 2DEG is located in the inversion layer formed at the interface between two semiconductors, one of them acting as the insulator. When the Hall device is cooled to a low temperature (T ~ 1 K or lower) and subjected to a high magnetic field (6–12 T), the 2DEG is completely quantized, and for the fixed source to drain current, the Hall voltage V_H shows a series of constant voltage plateaux (called Hall plateau) as a function of the applied field. Simultaneous to these plateaux, the longitudinal transport becomes dissipationless, as shown by the vanishing longitudinal resistance. In the limit of zero dissipation, that is, the longitudinal resistance along the direction of the current remains zero over the plateau region, the Hall resistance of the ith plateau $R_H(i)$ is quantized according to the following relation:

$$R_H = \frac{V_H}{I_{SD}} = \frac{h}{ie^2} = \frac{R_K}{i}$$

Here "i" is an integer and R_K is the von Klitzing constant. It follows that for $i = 1$ R_H (1) = R_K. The QHE has a universal character, i.e., it does not depend on the (1) device type, (2) plateau index, (3) mobility of the charge carriers, and (4) device width at the level of the relative uncertainty of 3×10^{-10}. The $R_K = 25812.807\ \Omega$ was adopted by all member states of the Meter Convention and came into effect as of 1st January 1990. The resistance scaling process linking the quantum Hall resistance (QHR) to decade value resistors uses cryogenic current comparator (CCC), Josephson potentiometer, Hamon network, or direct current comparator (DCC) bridge. The smallest uncertainty can be obtained using a CCC and a 100 Ω standard resistor. The value of R_K agreed upon by the member states of BIPM was referred to as $R_{K\text{-}90}$ and $R_{K\text{-}90} = 25812.807\ \Omega$ in 1990. Just as in the case of the Josephson voltage standard, $R_K = \frac{R_{K\text{-}90}}{i}$ proves a representation of the unit of resistance, viz., Ohm. $R_{K\text{-}90}$ is treated as a constant with zero uncertainty. In the new SI, the elementary charge and the Planck constant are assigned fixed values. Consequently, the von Klitzing constant R_K and the Josephson constant K_J have fixed values with zero uncertainty (Jeckelmann and Jeanneret 2001).

Kibble Balance

The new definition links mass to the Planck constant, and its realization is performed with the Kibble balance. Dr. Bryan Peter Kibble (NPL, UK) in 1975 has given the principle of watt balance. After his death, in his honor, the watt balance was renamed as Kibble balance in 2016.

Dr. Bryan Peter Kibble (1938–2016) https://www.theguardian.com/science/2016/aug/11/bryan-kibble-obituary

Principle of Kibble Balance

The principle of Kibble balance involves balancing of mechanical power with electrical power and takes advantage of the fact that electrical power is linked to the Planck constant by two quantum phenomena, i.e., quantum Hall resistance (QHR) and Josephson voltage standard (JVS).

It operates in two different modes, i.e., static mode/force mode and dynamic mode/velocity mode as shown in Fig. 10. In static mode, a mass in association with coil is hanged from a knife edge in a balance. The coil having wire of length L is positioned in a magnetic field of flux density B. The suspended mass will experience downward gravitational force which is balanced by an equal and opposite electromagnetic force by sending a current I through the coil:

$$mg = ILB \tag{1}$$

In dynamic mode, the coil is moved through the magnetic field at a vertical speed v so that a voltage U is induced:

$$U = BLv \tag{2}$$

Assuming BL to be constant in both the modes and eliminating it from Eqs. (1) and (2):

$$mgv = UI \tag{3}$$

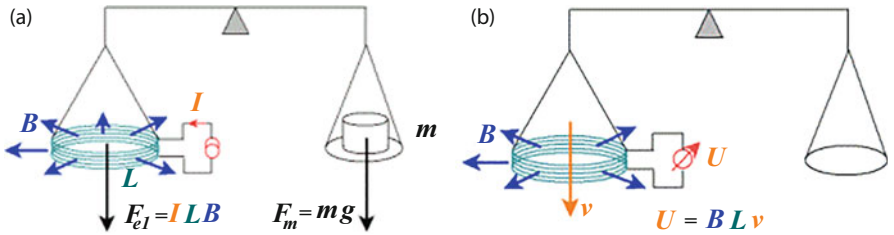

Fig. 10 Schematic depicting basic principle of Kibble balance. (**a**) Static mode of Kibble balance. (**b**) Dynamic mode of Kibble balance. (Adapted from https://www.bipm.org/en/bipm/mass/watt-balance/wb_principle.html)

From Eq. (3) it can be observed that the key quantities to be measured are the voltage (U) and current (I). From the primary standards of the voltage and resistance, i.e., Josephson voltage standard (JVS) and quantum Hall resistance (QHR), respectively, traceability of voltage and current can be achieved. The mass is given by the following:

$$m = \frac{UI}{gv} \tag{4}$$

From Ohm's law, I can be written as $I = V/R$, where V is the voltage drop across standard resistor R and then:

$$m = \frac{U}{gv}\frac{V}{R} \tag{5}$$

The two voltages U and V both are given in terms of the Josephson constant K_J:

$$K_J = \frac{2e}{h} \tag{6}$$

$$U_n = n\frac{hf}{2e} = nfK_J^{-1} \tag{7}$$

The resistance is expressed in terms of the von Klitzing constant (R_K) through the quantized Hall resistance:

$$R = \frac{1}{i}\frac{h}{e^2} = \frac{R_K}{i} \tag{8}$$

Thus, Eq. (5) can be written as follows:

$$m = \frac{in_1f_1n_2f_2K_J^{-2}}{gvR_K} = \frac{in_1f_1n_2f_2}{gv} \times \frac{h^2}{4e^2} \times \frac{e^2}{h} = (in_1n_2) \times \frac{f_1f_2}{4gv} \times h \tag{9}$$

Finally:

$$m = \frac{bf^2}{4gv}h \tag{10}$$

In Eq. (10), m is directly linked to h, where v is measured in m s^{-1} and g is measured in m s^{-2}. The frequency (f) is delivered as a constant in the Josephson standard, and b is determined by the quantized Hall plateau number (generally $i = 2$) and the Shapiro step number.

Planck Balance

The full-fledged Kibble balance is a very sophisticated system and requires a large amount of money and scientific skill to build. Additionally, it cannot be used for the day-to-day calibration purposes. Here the Planck balance provides a good alternate for NMI to carry out regular calibrations of E1 and E2 class weights. PTB-Germany plans to develop two tabletop Planck balance (PB). The first one is called PB2, and the second balance is called PB1 in collaboration with Technische Universität Ilmenau. Both balances will be robust in design and meant for regular calibration of masses of continuous range, unlike many fully developed Kibble balances in the world. Though PB is based on the principle of Kibble balance, it avoids some of the stringent requirements of Kibble balance as it is meant to calibrate E1 and E2 class weights only. Thus the low standard uncertainty set by CGPM for Kibble balance is not mandatory here. The major role of PB is not to accurately determine the value of the Planck constant but to be used in routine calibration to disseminate the unit kilogram to its nominal ranges. Unlike fully matured Kibble balance across the globe, which uses one nominal value for mass, generally 1 kg to measure the value of Planck constant as accurately as possible, PB can calibrate a various range of mass and is made for industrial use (Rothleitner et al. 2017).

PB1 calibrates weights in the range of 1 mg to 1 kg, and PB2 calibrates weights in the range of 1 mg to 100 gm. PB1 is aimed to calibrate E1 class weights with a relative uncertainty of 8.4×10^{-8}, and PB2 is meant to calibrate E2 class weights with a relative uncertainty of 2.7×10^{-7}. In the recent publication (Rothleitner et al. 2020), PTB has mentioned the development of PB1 so far. The balance is constructed mainly using standard industrial equipment such as commercial load cells and interferometers. Mass in the range of 1 kg to 100 g, 100 g to 10 g, and 10 g to 1 mg is tested for static and dynamic mode both in the vacuum chamber. For 1 kg mass, the relative standard deviation for static mode was 0.6 ppm, and for dynamic mode, it is 2.4 ppm (Rothleitner et al. 2020; Lin et al. 2020). For PB2 the latest combined uncertainty for 20 g measurement is reported as 2.47 ppm in which vertical alignment error is a significant factor (Vasilyan et al. 2021). PB2 can realize SI unit of mass for five decade values in range of 1 mg to 50 g.

Joule Balance

The NMI of China, National Institute of Metrology (NIM), had proposed the joule balance experiment back in 2006, and to date, they have developed two joule balances called NIM-1 and NIM-2. The joule balance is inspired by the Kibble balance, but unlike the two modes of measurement in Kibble balance, it has static mode only, with the dynamic mode replaced by inductance measurement, thus

changing the final power balancing equation of Kibble balance to energy balancing, from where it got its name (Li et al. 2015). Joule balance consists of two coils: an inner fixed coil called an exciting coil, which is used to produce a magnetic field when the current (I_{ex}) flows through it (Zhonghua et al. 2014; You et al. 2017). The outer coil is a set of two coils in which current (I_{mo}) is flowing in opposite directions to each other. In the static mode of joule balance, the gravitational force of mass is balanced by the electromagnetic force between the exciting coil and outer coil. It can be represented as follows (Zhonghua et al. 2014):

$$\underbrace{-\left(\frac{\partial M}{\partial z} I_{ex} I_{mo}\right)}_{\text{Force between exciting coil and movable coil}} + \underbrace{mg}_{} = \underbrace{\Delta f_z(z)}_{\text{Residual Force}} \tag{11}$$

Here, M is the mutual inductance between both coils, and z is the position where mutual inductance is being measured. By integrating Eq. (1) for position z_1 and z_2:

$$[M(z_1) - M(z_2)] I_{\text{ex}} I_{\text{mo}} + mg(z_1 - z_2) = \int_{z_1}^{z_2} \Delta f_z(z) dz \tag{12}$$

Equation (12) is the energy balancing equation of joule balance (Zhonghua et al. 2014). Mutual inductance can be measured in terms of change of flux, and change of flux can be accomplished by the integration of induced voltage (U) with respect to time in the coil. Thus Eq. (12) is traceable to h through JVS and QHR. The latest reported relative uncertainty by NIM-2 joule balance in 2019 was 5.2×10^{-8} for 1 kg Pt-Ir mass and 6.5×10^{-8} for 1 kg stainless steel mass (Li et al. 2020).

The XRCD Method

The next technique to realize a new definition is the XRCD method. The basic principle of the XRCD method is to compare the unknown mass to the mass of a single atom by counting atoms in a nearly perfect Si sphere. This method is easy to conceptualize as the quantity of the same type, i.e., counting the atomic mass of an atom to make it equal to mass. Here the mass of the atom is known in terms of the Planck constant:

$$N = \frac{8V_{\text{S}}}{a^3} \tag{13}$$

In Eq. (13), N is the number of Si atoms in the micro-volume of the crystal, V_{S} (a^3) is the volume of Si sphere, a is the lattice constant, and 8 is the number of atoms in one unit cell of Si. The mass of the Si sphere is given by the following:

$$m_{\text{s}} = Nm(\text{Si}) \tag{14}$$

Here m(Si) is the mass of a single Si atom. Planck constant h and mass of single-electron m(e) can be expressed as follows:

$$R_\infty = \frac{m(\mathrm{e})c\alpha^2}{2h} \tag{15}$$

Here c is the speed of light in a vacuum, α is the finite-structure constant, and R_∞ is the Rydberg constant. The relative uncertainty of R_∞ and a is 5.0×10^{-12} and 3.2×10^{-10}, respectively. Again, m(Si) is related to m(e) by the following:

$$\frac{A_\mathrm{r}(\mathrm{Si})}{A_\mathrm{r}(\mathrm{e})} = \frac{m(\mathrm{Si})}{m(\mathrm{e})} \tag{16}$$

Here A_r(e) and A_r(Si) are the relative atomic masses of electron and Si, respectively. The relationship between m(Si) and the Planck constant h is given by the following:

$$\frac{h}{m(\mathrm{Si})} = \frac{1}{2}\frac{A_\mathrm{r}(\mathrm{e})}{A_\mathrm{r}(\mathrm{Si})}\frac{\alpha^2 c}{R_\alpha} \tag{17}$$

Finally, the sphere mass is expressed in terms of h by the following:

$$m_\mathrm{s} = \frac{2R_\infty h}{\alpha^2 c}\frac{A_\mathrm{r}(\mathrm{Si})}{A_\mathrm{r}(\mathrm{e})}\frac{8V_\mathrm{s}}{a^3} \tag{18}$$

Taking into consideration mass deficit (m_d) due to impurities and vacancies and mass of the surface layer (m_SL) on Si sphere, Eq. (18) can be written as follows:

$$m_s = \underbrace{\frac{2R_\infty h}{\alpha^2 c}\frac{A_r(Si)}{A_r(e)}\frac{8V_s}{a^3} - m_d}_{\text{Mass of the Si Core}} + \underbrace{m_{SL}}_{\text{Mass of the surface layer}} \tag{19}$$

The National Metrology Institute of Japan (NMIJ) is realizing kilogram using the XRCD method with the help of two 1 kg ^{28}Si-enriched spheres fabricated by the International Avogadro Coordination project. As per their latest publication, the relative standard uncertainty is estimated as 2.1×10^{-8}, and the highest contributing factor is the core volume of Si sphere, estimated as 1.8×10^{-8} (Kuramoto et al. 2021).

Twin Pressure Balance

Measurement Standard Laboratory (MSL), New Zealand, is implementing the principle of Kibble balance in another unique way using two pressure balances joined together using a differential pressure sensor. In Kibble balance, force in static mode and velocity of the coil in dynamic mode must be in line with the gravitational force at the point of measurement. A difference in alignment between two modes of 75 μrad can cause a cosine error of 3 parts in 10^9, which is quite significant when the accuracy of Kibble balance is a concern (Sutton et al. 2016). Thus to eliminate the misalignment, the twin pressure balance attaches the coil to one of the pistons of

the piston-cylinder unit, and the downward force on the coil can be compared to the second pressure balance also called a reference pressure balance. This eliminates the horizontal movement of coil hence making the vertical movement reproducible and insensitive to the horizontal forces. In the static mode of measurement, the twin pressure balance works as a force comparator where force difference is measured by the pressure difference generated between two pressure balance. When mass is on the equation for static mode (Fung et al. 2020):

$$PA + I\gamma = (m + m_{\text{tare}})g \tag{20}$$

Here, m is the measured mass, mass of tare is m_{tare}, P is the pressure, I is the current in the coil, and γ is the flux linkage of magnetic field B. When mass is off the equation for static mode, it changes to the following:

$$(P + \Delta P)A = I\gamma + m_{\text{tare}}g \tag{21}$$

Combining Eqs. (20) and (21), final equation for static mode of measurement comes out to be:

$$mg = 2I\gamma - \Delta PA \tag{22}$$

For dynamic mode the coil is attached to the piston with the help of three rods and is moved with the constant velocity (v) guided by the second pressure balance in the presence of the constant magnetic field. The equation for induced voltage (U) in the coil for dynamic mode measurement is as follows:

$$U = -\gamma.v \tag{23}$$

Details of the MSL twin pressure balance can be obtained from (Sutton et al. 2016; Fung et al. 2020).

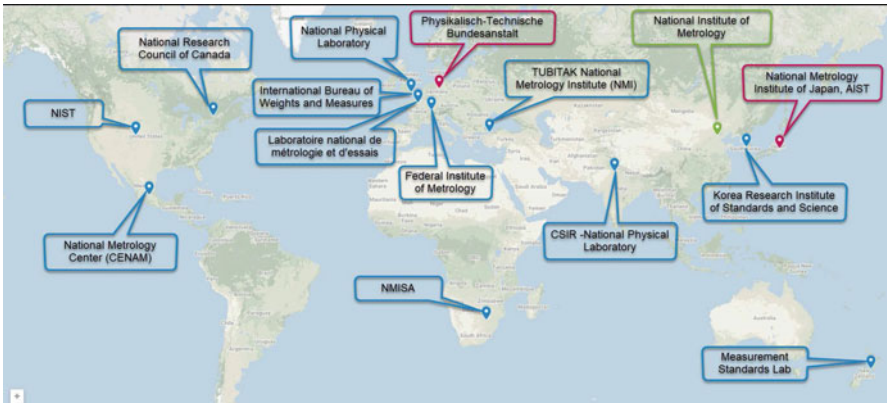

Fig. 11 World map showing the presence of Kibble balance technique (blue), joule balance technique (green), and XRCD method (red)

The International Status of Existing Primary Realization Methods (Fig. 11)

NPL, UK	Dr Bryan Peter Kibble invented Kibble balance in 1976 MARK-I and MARK-II are two Kibble balances of NPL MARK-I was the first Kibble balance having yoke-based permanent magnet system. A flat magnetic profile was achieved by a figure 8-shaped coil hung vertically in the center of the air gap having a magnetic field of 0.68 T The drawback of this design was that because only a fraction of the coil length was used to produce force, thus, to meet requirements, a massive system was fabricated having the weight of the magnet and coil to be 6000 kg and 30 kg, respectively. This significantly increases the heating of the system NPL, UK, is now working on a single-mode two-phase Kibble balance (Robinson 2012; Stock 2012)
NIST, USA	The initial results from NIST's Kibble balances were published in 1989 To date, they have developed four full-fledged Kibble balances The latest Kibble balance called NIST-4 uses Sm_2Co_{17} as a permanent magnet and low-carbon steel as a yoke It works in a vacuum, with a relative uncertainty of 1.3×10^{-8}, published in 2017 (Haddad et al. 2017) NIST has also developed a Lego Kibbe balance for 10 g and achieved uncertainty of the order of 10^{-2} (Chao et al. 2015) One of the recent activities taken up by NIST is developing a tabletop Kibble balance called "KIBB-g1 Kibble balance" for mass up to 10 g with targeted uncertainty of 10^{-6} (Chao et al. 2020; Chao et al. 2019)
NRC, Canada	The Kibble balance project started in 2009 When the MARK-II of NPL, UK, was sent to NRC Canada NRC successfully identified and corrected a systematic error relating to the effect of the test mass's weight on the equipment's construction (Wood et al. 2017) The relative uncertainty reported in 2017 is 9.1×10^{-9}, which is the smallest in the world (Wood et al. 2017)
BIPM, France	The Kibble balance project started in 2005 It operates in one-mode two-measurement scheme technique It operates in a vacuum and uses a bifilar coil The relative uncertainty of 1 kg Pt/Ir artifact published by BIPM in 2020 is 49 μg. Details of the BIPM Kibble balance can be read in (Fang et al. 2016, 2020)
KRISS, Korea	The Kibble balance project started in 2012 Out of four different permanent magnet configurations widely used by different NMIs, KRISS uses closed cylindrical permanent magnet system (Kim et al. 2014, 2017; Choi et al. 2016) The latest relative uncertainty of KRISS Kibble balance published in 2020 is 1.2×10^{-7} (Kim et al. 2020)

(continued)

LNE, France	The Kibble balance project started in 2002 It uses an indigenously designed flexure bearing balance, and in 2017, the balance was operated in air, and h is calculated with the relative uncertainty of 5.7×10^{-8} (Thomas et al. 2017) Recently the balance was operated in a vacuum, and details of the experiment can be read in (Espel et al. 2020)
Measurement Standard Laboratory (MSL), New Zealand	MSL Kibble balance is based on twin pressure balance Details of the MSL twin pressure balance can be obtained from (Sutton et al. 2016; Fung et al. 2020) The latest development on MSL Kibble balance like the development of mass loader with pneumatic control, construction of vacuum chamber for balance, magnet assembly and designing of coil, etc., is present in (Fu 2021)
Federal Institute of Metrology (METAS), Switzerland	METAS has created a 100 g balance called MARK II (Eichenberger et al. 2011; Schlamminger and Haddad 2019) They have also developed a 1 kg Kibble balance which operated in a vacuum, and the first set of measurements was conducted in 2021 (Eichenberger et al. 2022) The reported uncertainty published in 2022, for 1 kg stainless steel test mass at $k = 1$, is 43 parts in 10^9
National Metrology Institute of Japan (NMIJ)	Work on XRCD method The publication in 2021 reported the relative standard uncertainty to be 2.1×10^{-8} Details can be found in Kuramoto et al. (2021)
The NMI of China, National Institute of Metrology (NIM)	In 2006 principle of Joule balance was proposed by NIM China The first joule balance developed by China was called NIM 1 The advanced version of NIM 1 balance was NIM 2, which operates in a vacuum. Details of NIM 2 Joule balance can be read in Li et al. (2020)
The National Metrology Institute of Turkey (UME)	The National Metrology Institute of Turkey (UME) is using the oscillating magnet technique to realize the new definition of kilogram In this technique, the magnet oscillates around the equilibrium position, and the signal obtained from static mode and dynamic mode measurement is read in the frequency domain According to the latest publication (Haci et al. 2020), in this novel technique, the effect of environmental noise on the setup is comparatively lower than the traditional Kibble balance A commercial Michelson interferometer is used for displacement measurement in their Kibble balance
NMISA, South Africa	A Lego Kibble balance for 1–3 g and a modified equal arm Kibble balance for 1–3 g have been created by NMISA in South Africa A two pan equal arm balance has been converted to an equal arm Kibble balance by NMISA

(continued)

	On both sides, the 3D-printed structure has been adjusted to accommodate coil and magnet assembly. To create a magnetic field, a neodymium ring magnet (N42) is employed (Mametja et al. 2018) The coil is made up of copper wire of 0.1 mm diameter and has 3000 turns. The measured BL value for NMISA Kibble balance is 14.15 Tm ± 0.12% The modified equal arm model reported less than 1% relative uncertainty, with the Lego software resolution being a limiting factor as this was fixed to 2 decimal digits (Sonntag et al. 2020)
CSIR-National Physical Laboratory (NPL-India)	Developed demonstrational model of 1 g Kibble balance setup (Kumar et al. 2017; Ehtesham et al. 2022) Measurements were conducted for 1 g and achieved an uncertainty of less than 1% Balance is upgraded for 1–10 g, and measurements are conducted. Publication of these results is underway

Conclusion

This chapter discusses the advancement of the mass standard with respect to primitive as well as the modern era. Primary advances happened locally in different parts of the world, which predominantly attended to the requirements of the local communities in their respective areas. With the expansion and advent of trading, more and more sophisticated techniques were developed that eventually resulted in the modern ultrahigh-precision balances of various types in use today. Further authors also discuss the crucial advances that paved the path for the redefinition of the kilogram which advocate the philosophy of time-invariant measurement. The change in traceability chain after and before the redefinition of kilogram is covered for India as well as globally. The role of quantum standards in the realization of the new definition of kilogram is also explained in detail. Authors have briefly discussed the presently existing realization methods which includes fully established (Kibble balance and the XRCD) as well as methods which are under development. Briefly, the present status of different NMI across the globe working on the different primary realization techniques is also discussed. The advantage of the new definition is that it gives the flexibility to develop methods which are convenient to user for the realization of kg in terms of h.

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Optical Frequency Comb

10

A Novel Ruler of Light for Realization of SI Unit Meter

Mukesh Jewariya

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Abstract

Optical Frequency comb has become a ruler of light nowadays for standard of length. A notable synergy among existing technologies occurs for stabilization of laser frequencies in the rule of the frequency spectrum, called "comb" of sharp and equidistance lines. With such an optical frequency comb, the exact frequencies of the entire comb lines are measured, revolutionizing the measurement of length in more precise way and realizing SI Unit Meter with better accuracy. This capability has transformed optical frequency metrology for better precision. The pulse from Optical Frequency Comb can produce a temporal coherence interference fringe pattern, which can be used as a practical method for realization of SI Unit Meter. The emergence of the optical frequency comb know-how, in 2005 (Nobel Prize in Physics) has eliminated the demand for knotty frequency chains. By straight connecting, the second, using frequency standards of Cs, or Rb, by optical frequency comb, we can set up a novel technique for realistic realization of length of the SI unit Meter (Primary Standard). By means of stabilizing and characterizing the two different constraints – the carrier offset frequency f_0 and the repetition frequency f_r of the optical comb is effectively used for practical realization of SI unit "Metre." The accuracy in the measurement of the length

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increased by many order using optical frequency comb, i.e., 2×10^{-13} instead 10^{-7} by using Iodine stabilized He-Ne Laser. In this chapter we present a review of optical frequency comb as a new quantum standard for realization of SI unit "Metre." A brief review of determining the absolute value of laser frequency in calibration using an optical frequency comb is illustrated here in this chapter.

In this chapter we reviewed the optical frequency comb as a novel tool for realization of SI unit Meter for length metrology. Particularly, an iodine-stabilized He-Ne laser incessantly sweeps to a steady optical frequency comb, where its output frequency is modulated over a wide spectrum to determine the exact frequency using optical interferometry. This method established a practical method to measure length in terms of SI unit meter with minimum uncertainty, which is straight traceable to SI unit second (Time Standard).

Introduction

Length is one of the most vital physical parameter, and its accurate measurement is of most significance for development of science and technology for the benefit of human race. It is the most common measurement system used by human from finding the smallest distance inside a nucleus to farthest interstellar bodies. It is the most familiar unit of measurement in our daily life. In prehistoric Egypt and Mesopotamia, one of the initial standards of measuring the length was the cubit. The royal cubit, as shown in Fig. 1, was used to construct the key structures, which is even found today in Egypt. This cubit was the length, determined by taking the length of the pharaoh's arm, from elbow to the end of the middle finger plus the span of his hand. The royal cubit was made of granite and standardized using a rod. Similarly at other places where length measurements were used, they were determined using the length of the human body parts like foot, hand, and palm. In India and China, they standardized their units based on yard, Gaj, etc., but length measurements in Europe were still mainly using body parts until the eighteenth century.

The development of the metric system in eighteenth century is innovative. A French king tried to create a worldwide organism of measurement, which wasn't depend on human body size that is diverse from one person's body to another person's body or body parts or from one place to another place with different

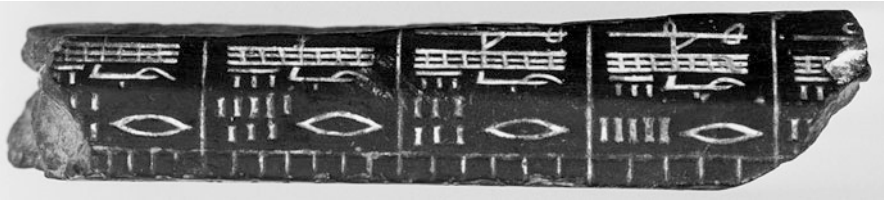

Fig. 1 A Royal Cubit as a standard for length used by rulers of Egypt (https://www.nist.gov/sites/default/files/images/2019/05/09/fragment_of_a_cubit_measuring_rod_met_102082_cropped.jpg)

geographical locations. A committee was formed for this, which wanted to establish a system "for all times, for all peoples." After the French revolution, the idea of a metric system of units initiated. There was a strong will among the people of the committee that the new unit that was going to be defined for length shall be equal to "one ten-millionth of the distance from the North Pole to the Equator (the one fourth part (quadrant) of the Earth's circumference). This was measured along a straight line over the meridian that was passing through Paris" (Larousse 1874). It took more than 6 years (1792–1798) by Pierre Méchain and Jean-Baptiste Delambre and their team for surveying the meridian arc (Alder 2002). In 1799, several platinum bars were made, and based on those bars the definition of the meter was decided and kept in the National Archives in the form of Meter Bar known as the mètre des Archives on 22 June 1799 (Comptes Rendus de la 7e CGPM 1927; Babinet 1829).

The meter was legitimately recognized as an international unit of measurement of length by "Metre Convention" of 1875. In 1889, BIPM introduced an element alloy rod made of platinum/iridium with a cross-shaped section, on which "one meter was fixed as the distance between two center lines, at a 0 °C", as shown in Fig. 2 [dimension of bar are in mm]. Thereafter in the seventh Conférence Générale des Poids et Mesures (CGPM), which took place in 1927, it was defined that this platinum–iridium bar is under atmospheric pressure and supported by two rollers and one meter is the distance between two points under atmospheric pressure (International Committee of Poinds and Measures and the Consultative Committee for the Definition of the Meter 1984; Quinn 1999, 2003; CSIR-National Physical Laboratory).

The first proposal that light could be a means for the realization of the length was probably made by the Babinet (A French natural philosopher) in 1827 (Meggers 1948). In 1893, Albert A Michelson declared in a review article that "light waves [were] now the most convenient and universally employed means we [possessed] for making accurate measurements" (Michelson 1893; Bönsch and Potulski 1998; The International Metre Commission 1870; The BIPM and the evolution of the definition

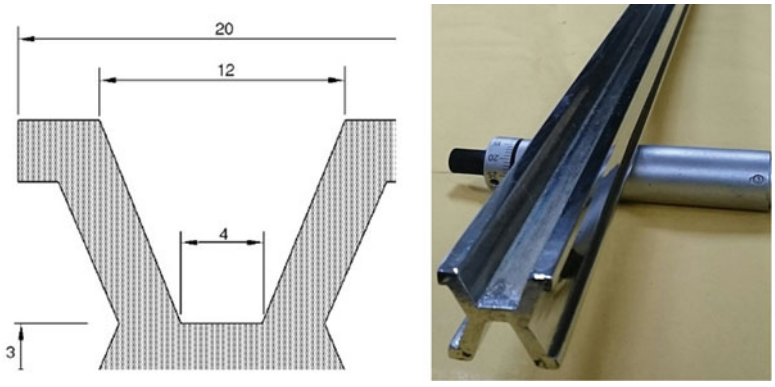

Fig. 2 The Tresca Meter Bar, a replica of National Prototype Metre Bar No. 57, given to India in 1957 for SI unit "metre" (Sharma et al. 2020)

of the metre 2016; Article 3; Barrell 1962; Phelps III 1966; https://en.wikipedia.org/wiki/International_Bureau_of_Weights_and_Measures; Michelson and Benoît 1895). In 1960 the definition of the meter closed relying on meter bar, which was based on an artifact and needed maintenance and continuous monitoring. At the 11th CGPM, in 1960, the meter was defined as the "**length equal to 1 650 763.73 wavelengths in vacuum of the light radiated at the transition between the levels $5d_5$ and $2p_{10}$ of 86Kr**" (International Committee of Poinds and Measures and the Consultative Committee for the Definition of the Meter 1984; Quinn 1999, 2003). This definition was decisively rooted in a krypton-86 spectral lamp with a wavelength of approximately 606 nm, which was realized by the German national metrology institute {Physikalisch-Technische Bundesanstalt (PTB)}, by Johann Georg Ernst Engelhard. With the discovery of laser parallel it was felt that one meter defined based on krypton is not able to stabilize and the definition needsa to be changed based on laser light. In the 12th CGPM, 1983, the Meter was defined again as "**The length of the path travelled by light in vacuum during a time interval of 1/299 792 458 of a second**." Light is understood as a plane electromagnetic wave or a superposition of plane waves and the length is considered along the propagating direction of the light wave. Under vacuum conditions there is a strict relationship between the frequency ν and the wavelength λ_0 of monochromatic light:

$$c_0 = \lambda_0 \cdot \nu = 299{,}792{,}458 \text{ ms}^{-1} \tag{1}$$

in which c_0 represents the speed of light in vacuum. It is having a fixed value by the meter definition in 1983. The realization of SI unit meter based on the 1983 definition needs superposition of at least two lights. Since the definition for the meter is valid for vacuum conditions, in most situations and in laboratory conditions the realization of the length is carried out at atmospheric conditions. Here, the accurate quantification of the air on the speed of light is of importance for correct measurement. Under atmospheric conditions, the air refractive index n gives reduction in the speed of light ($c = c_0/n$) as well as the wavelength. The relative effect of n is around 3×10^{-4} orders corresponding to 0.3 mm per meter of measured length under standard air conditions (Bönsch and Potulski 1998; Phelps III 1966; Michelson and Benoît 1895).

Realization of meter in terms of length traveled by light requires the measurement method. The SI definition for the meter presumes a measuring method based on laser interferometer. The accuracy of the realization of lengths is in perpetual development, triggered by new requirements placed on their accuracy.

The first physical phenomenon used was a multiple of a specific lamp's wavelength (CIPM 1960). The most important point of the post-1983 definition of the meter is that time is included in the definition. The accuracy demanded by the time of 1/299,792,458th of a second is precisely the accuracy in the definition of the meter, and, thus, length. Since time is the inverse of frequency, the precision by which a frequency is found translates into the precision of the length. The model light source that maintains a consistent frequency (wavelength) with high accuracy is the laser light source. This is why, initially, the iodine stabilized helium-neon laser's

wavelength was used as the practical means of realizing the length standard. BIPM recommend use of interferometry. In this we use lasers that need a quantum reference like hyperfine transitions of iodine, which act as reference for stabilization of He Ne lasers.

So for realization of SI Unit "Metre," an iodine-stabilized He Ne laser operating at 633 nm (474 THz) was employed and it is the most used laser in dimension metrology because of its easy operation, maintenance, and accuracy up to 10^{-11} part for calibration of commercial displacement measuring interferometer (see Fig. 3). The beat frequency method is used for calibration at 474 THz (633 nm) frequency/wavelength. This method is used for measurement of frequency difference between two stabilized lasers. The reference as said above in terms of frequency comes from another iodine-stabilized He- Ne laser having offset frequency controlled by a PLL (phase-locked loop) controller (Riehle et al. 2018; Pollock et al. 1983; Jennings et al. 1983a, b; Schweitzer et al. 1973; Layer et al. 1976; Layer 1980; Holzwarth et al. 2001; Yu et al. 2001; Gillespie and Fraser 1936; Ye et al. 1999, 2000; Zhang et al. 2001). In this laser the beat frequency of offset lock He-Ne lasers is controlled corresponding to the offset frequency, which is determined by temperature compensated crystal oscillator (TCXO). The frequency of the offset lock He-Ne laser is stabilized holding the offset frequency. But the stabilized laser frequency has some uncertainty for stabilization so a new definition is introduced and a new method is adopted to realize SI unit meter.

The latest definition of SI unit meter approved by 26th CGPM, put into practice on 20 May 2019 is "***The metre, symbol m, is the SI unit of length. It is defined by taking the fixed numerical value of the speed of light in vacuum c to be 299 792 458 when expressed in the unit m s–1, where the second is defined in terms of the caesium frequency ΔˇCs***."

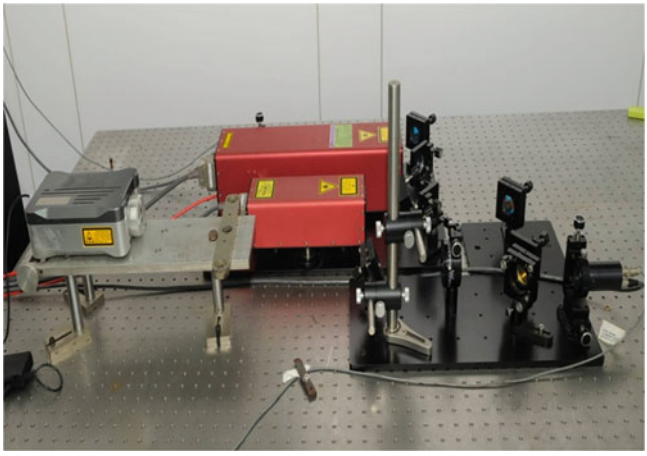

Fig. 3 A primary standard of length in India: Iodine ($^{127}I_2$)- stabilized He-Ne laser at CSIR-NPL (Sharma et al. 2020)

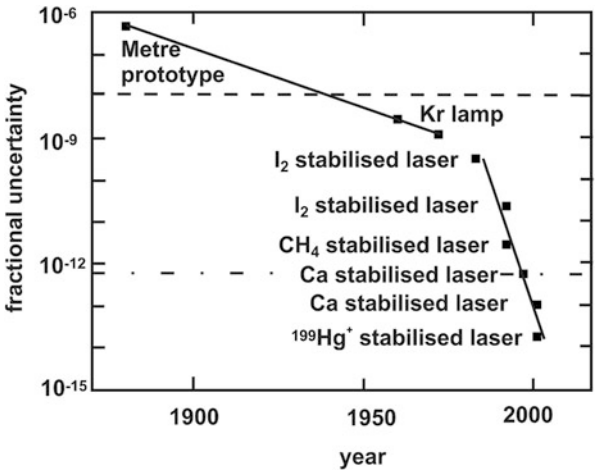

Fig. 4 Evolution of the realization of the meter according to the *mise en pratique* (MeP) (Quinn 2003)

Since there are teething troubles in measuring accurate and precise frequency of reference used, i.e., hyperfine transition in cesium atom, the accurate and precise information of reference frequencies is needed for tying optical frequencies with microwave. The main problem in the measurement of optical frequency is that the frequency in visible region is about a factor of 50,000 more than that of time standard (Cs clock) and there is hardly any technique available for frequency counting for frequencies in visible region. In this regard frequency comb is a new candidate for realization of new definition of SI unit meter (Udem et al. 2002; Ferreira-Barragáns et al. 2011). A pictorial view of realization of SI unit meter is shown in Fig. 4 and below time line in Table 1.

Realizing Meter Using an Intelligent Method by Ruler of Light: mb

As per new the definition, the most convenient method for realization of Si unit meter is to map out the laser frequency of optical frequency comb to the microwave standard, which is linked to the time. In this method, the second is directly linked to the meter. Its practical realization is by taking the fixed value of speed of light, $c_0 =$ 299,792,458 m/s, and the frequency of transition used. The Comité International des Poids et Mesures (CIPM) gave suggestions for the practical realization of the meter, as per methods mentioned in mise en pratique (in later text referenced as **MeP**) to define meter. The three methods to realize SI unit meter are as follows: (Gill 2005; Hollberg et al. 2005; Baklanov and Dmitriev 2002; Ye 2004; Joo et al. 2008; Salvadé et al. 2008; Cui et al. 2008; Minoshima et al. 2007).

(a) By means of the length l of the path traveled in vacuum by a plane electromagnetic wave in a time t; this length is obtained from the measured time t, using the

Table 1 History of "Metre"

History of meter		
100 years	**Years**	**Task by CIPM**
	1791	Commission chooses polar quadrant of Earth as basis for length standard
	1792	Delambre and Méchain commence measurements along the Paris meridian
	1793	National Convention (France) adopts 1 m = 443.44 lignes of Toise de Perou
	1799	Metre des Archives, (platinum), 25.3 mm × 4 mm set to 1/10,000,000 of Earth quadrant
	1872	Construction started of 30 new Prototype Meters 90% Pt, 10% Ir, design by Tresca
	1875	Convention du Metre signed
	1889	Work on 30 prototype meters completed, deposited at the BIPM
	1889	Definition of meter in terms of prototype Meter
100 years	1927	Re-definition of the meter
	1956	Engraved lines on meter bars re-ruled for use at 0 °C and 20 °C
	1960	Re-definition of meter in terms of 86Kr wavelength (4×10^{-9})
	1975	Speed of light fixed at c = 299,792,458 m s^{-1}
	1983	Definition of meter as distance traveled by light in 1/299,792,458 s
	2019	**Revision of the SI base unit definitions to explicit constant format**

relation $l = c_0 \cdot t$ and the value of the speed of light in vacuum c_0 = 299,792,458 m/s.

(b) By means of the wavelength in vacuum l of a plane electromagnetic wave of frequency f; this wavelength is obtained from the measured frequency f using the relation $l = c_0/f$ and the value of the speed of light in vacuum c_0 = 299,792,458 m/s.

(c) By means of one of the radiations from a given list, whose stated wavelength in vacuum or whose stated frequency can be used with the uncertainty shown, provided that the given specifications and accepted good practice are followed.

In method (c) one can use a wave of known wavelength (in vacuum), which can be done by a direct measurement or a frequency given by one of the reference value of the recommended vacuum wavelengths listed by CIPM (Quinn 2003) below. In 2003, the International Committee for Weights and Measures (CIPM) recommended updated values for the wavelengths for the realization of the SI unit meter for practical definition.

CIPM recommended wavelengths for the definition of SI unit meter (Quinn 2003):

Frequency (THz)	Fractional uncertainty	Wavelength (nm)	Laser used
473.612,7	1.5×10^{-6}	632.990,8	HeNe unstabilized
473.612,353,604	2.1×10^{-11}	632.991,212,58	HeNe/I2
551.580,162,400	4.5×10^{-11}	543.515,663,608	HeNe/I2
563.260,223,513	8.9×10^{-12}	532.245,036,104	2*f* (Nd:YAG)/I2

One of the vital wavelength for length metrology is 633 nm (474 THz) from an iodine ($^{127}I_2$) He–Ne-stabilized laser. This $^{127}I_2$ He-Ne laser is used for realization of the SI meter and is nowadays in most NMIs commonly used for calibrating the frequency of lasers as recommended by committee (474 THz).

The International Committee for Weights and Measures (CIPM) Report on CCTF (CCL-CCTF WGFS & WGPSFS) on 6 June 2017 made its recommendations to all national metrology institutes (NMIs) to practice the optical frequency comb technique to the uppermost level of accuracy, for length calibration. The comb techniques are very convenient because it can offer traceability to the SI unit meter through Second and an optical frequency tuneable source as well as measurement technique (Ferreira-Barragáns et al. 2011; Samoudi et al. 2012).

What Is an Optical Frequency Comb and How Does It Work

Optical frequency comb is an optical frequency ruler in terahertz frequency region. It is composed of equally spaced electromagnetic field in frequency domain, generated by a femto-second laser in regular intervals of frequency and having a coherent phase with each other. The frequency of a piece of mode is given as per the following relation:

$$\nu_N = N \times f_{rep} + f_0 \tag{2}$$

Here ν_N: the optical frequency of the mode, N: a large integer (typically of the order of 10^6), f_{rep}: repetition rate of the laser pulses, and f_0: "carrier-envelop offset frequency," which is the difference of group velocity and phase velocity. It is to be noted that the electric field in each mode is not firmly the same, but it maintains a constant separation between two pulses as shown in Fig. 5.

The electromagnetic fields created by a fs laser is mixed with a cw laser whose frequency is ν_{cw} for measurement and detection. The corresponding mixed field is detected using a fast photodiode (whose bandwidth is f_{BW}). This results in a radio-frequency spectrum, consisting of two parts, the repetition rate f_{rep} and optical beat-notes frequency $f_{b,N} = |\nu_{cw} - N \times f_{rep} - f_0|$, where N is integer value such that $f_{b,N} < f_{BW}$. For detection of f_{rep} the photodiode are required. We have some specific operation region for residual noise of f_{rep} that enables the detection. At the current stage one can only rely on particular photodiode that is fully committed for the detection of f_{rep}.

f_0 is determined by the self-referencing technique or there are many other known techniques (like zero offset), which requires at least one octave (Ell et al. 2006; Matos et al. 2006; Fortier et al. 2006; O. l. w. G. r. r. IdeestaQE 2014; Coddington et al. 2008). A highly nonlinear fiber causes the broadening of the spectrum sufficiently to reach the octave spanning condition. These highly nonlinear fibers are engineered such that the pulses that are propagating inside it keeps a short pulse time duration and high peak intensity for creating high nonlinear effects. By four-wave mixing method, the nonlinear fiber gradually broadens the spectrum with a

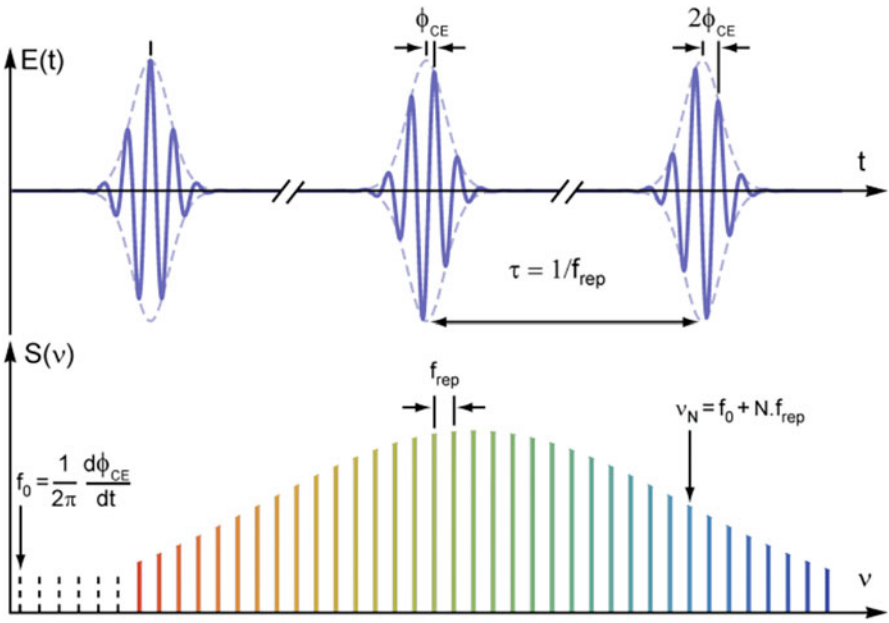

Fig. 5 An optical frequency comb with equally spaced frequency modes

factor equal or higher than two times the f_0. Using this one can calculate optical frequency $\nu_N = N \times f_{rep} + f_0$ and other with $\nu_{2N} = 2N \times f_{rep} + f_0$. By keeping two times the frequency ν_N in a nonlinear crystal, and detecting the optical beat-note frequency $2 \times \nu_N$ and ν_{2N} using a fast photodiode, one can get signal at frequency $|2N \times f_{rep} + 2f_0 - 2N \times f_{rep} - f_0| = f_0$. The frequencies in the mode of $2\nu_N$ and ν_{2N} to the signal are favorable, because these added up coherently to generate a signal of larger amplitude. Since typical optical power per mode is in the pW range, this coherent signal is easy to detect. The set of radio-frequency signals fb, Ncw, f_{rep}, and f_0 give all the parameters that completely define Eq. 1, and therefore provides a mean to compare optical to microwave frequencies.

Practical Realization of Meter

Optical frequency combs, nowadays are known as ruler of light and they are employed to generate optical frequencies with an uncertainty limited to the uncertainty of the master clock. These optical frequency combs are designed in such a way that the round trip time for all longitudinal modes remains constant.

To realize the mode configuration of an optical frequency comb and its stabilization technique, let's consider an idealized case of a pulse in cavity whose length is L and whose carrier frequency is fc. The emitted pulse is basically a repetition of the first pulse separated by the round trip time:

$$T = 2\,L/v_g \tag{3}$$

where v_g: cavity mean group velocity.

Emitted pulses are not symmetrical because of pulse envelope A(t). The pulse is propagating with two different velocities: a group velocity (v_g) and phase velocity (carrier wave) and both have a difference. Consequently, a single round trip of a carrier pulse phase has angle $\Delta\phi$ as shown in Fig. 6. In this case, the electric field at a given place can be written as,

$$E(t) = Re\,[A(t)\exp(-i2\pi f_c)t]. \tag{4}$$

Because the envelope is periodic

$$E(t) = Re\left[\sum\nolimits_n A_n \exp(-i2\pi(f_c + nf_r))t\right], \tag{5}$$

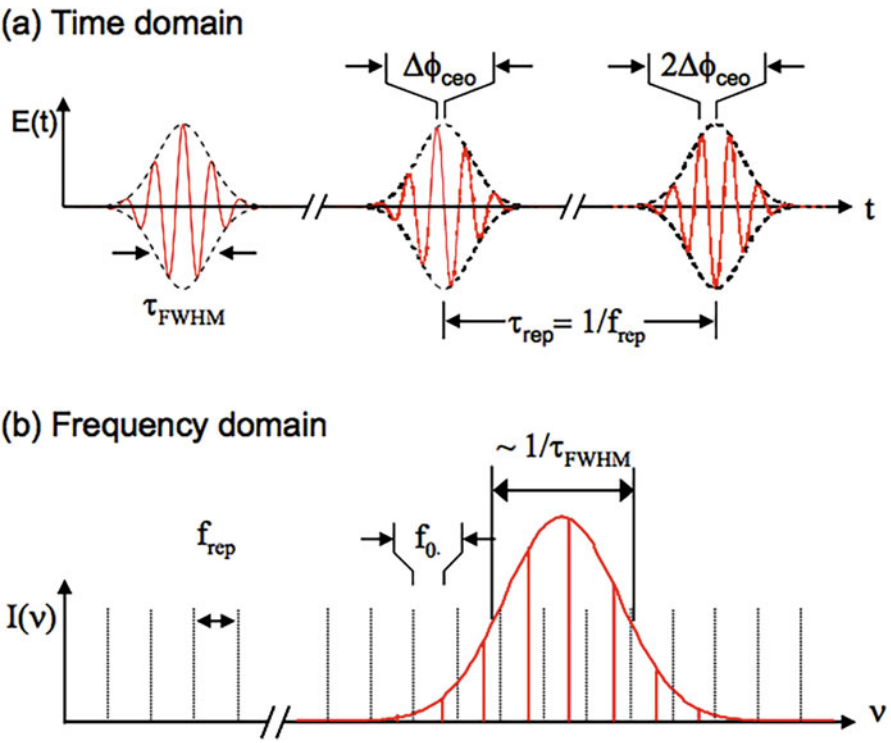

Fig. 6 An optical frequency comb representing time domain and frequency domains for the EM fields

where An are Fourier components of A(t), carrier frequency f_c is not essentially an integer multiple of f_r. The offset are selected by satisfying $f_0 < f_r$, so that modes will shift with respect to the harmonics of the repetition frequency

$$f_n = f_0 + nf_r, \tag{6}$$

where n is the integer value of the order of 10^6. Using above Eq. 6 one can mix two radio frequencies (RF) f_r and f_0 onto an optical frequencies f_n. f_0 is detected by the self-referencing technique as described above with an f–2f interferometer setup in most of the cases; however there are other techniques available (Jones et al. 2000; Holzwarth et al. 2000). If the optical comb gives a complete optical octave, then there is a mode, with the number 2n, which shall oscillate at the same time at

$$f_{2n} = f_0 + 2nf_r. \tag{7}$$

The beat note between n and 2n (the mode and frequency doubled mode at 2n) gives the offset frequency as

$$2(f_0 + nf_r) - (f_0 + 2nf_r) = f_0. \tag{8}$$

The frequency of another laser, which is under calibration (continuous wave), its f_{cw}, is measured by a beat frequency, f_{beat}, with the adjacent mode of the optical comb. In this case, the value of f_{cw} is written as follows:

$$f_{cw} = nf_r \pm f_0 \pm f_{beat}. \tag{9}$$

The polarity of Eq. 9 is decided by measurement conditions, changing f_0 and f_r. If second harmonic generation (SHG) is takes into consideration, the zero offset value f_0 has to be multiplied by a factor of 2 as shown in the following equation

$$f_{cw} = nf_r \pm 2f_0 \pm f_{beat}. \tag{10}$$

The repetition rate stays constant because sum frequency generation is dominant process in SHG.

Experimental Setup

Most of the NMIs are using optical frequency comb with 250 MHz repetition rate mode-locked erbium fiber-ring laser and a nonlinear micro-structured fiber for realization of SI unit meter. The main component of optical frequency comb is a fiber laser with erbium-doped fiber amplifier (EDFA) for high power generation up to 2 mW at 1500 nm central wavelength as shown in Fig. 7. The femtosecond pulse is divided into two parts and sends to the monitor dock and to the external parts of the OFC. One part is amplified in an erbium-doped fiber amplifier and gets broadened using highly nonlinear fiber to cover a full spectrum. The beat signal of offset

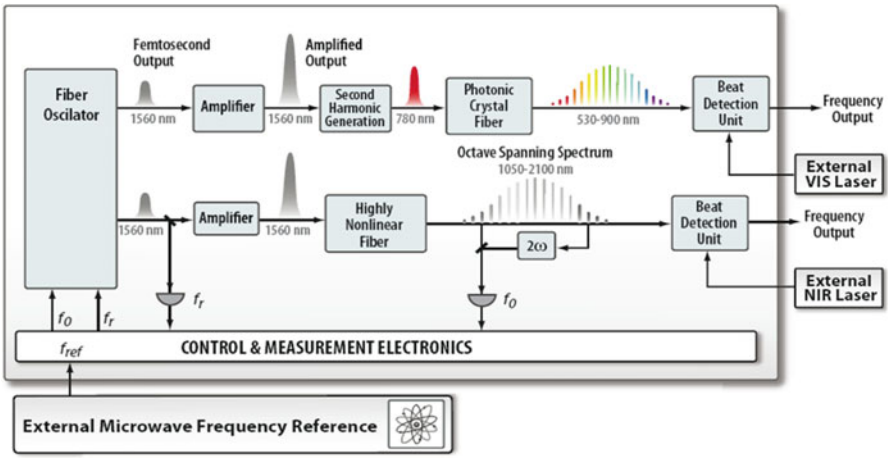

Fig. 7 Experimental setup for an optical frequency comb for calibration of $^{127}I_2$ He–Ne lasers at 633 nm (Samoudi 2017)

frequency f_0 is measured using a f−2f interferometer. Another part of the beam is used for generating high power.

A fast p-i-n junction (PIN) diode is used to measure the carrier offset frequency. The fiber laser head has an inbuilt fast PIN photodiode, which is used to measure the fourth harmonic of fr with better phase sensitivity. Electronic circuits of two phase-locked loops are used to lock the repetition rate and offset frequency. A Frequency counter is used without dead time to detect the beat signals with cw lasers and to control the phase-locked loop. An oscilloscope and a spectrum analyzer are used for monitoring of electronic signals, pulse, and beats.

A beat detection component (Fig. 8) is composed of silver/gold coated mirrors, polarized beam splitters, half or quarter wave plates and is used to direct the beams to a photodetector for measurement. A Cesium (Cs) atomic clock is used to be referenced to a primary frequency standard for repetition rate frequency, fr, and the carrier envelope offset frequency (20 MHz), f_0, and $f_{beat.}$

To realize the SI unit meter Zeeman-stabilized 633 nm He–Ne lasers are used. A grating having 2100 lines/mm is used for spectral filtering. The 633 nm laser is calibrated with optical frequency comb pulse of 633 nm. It is vital to give surety that both beams of 633 nm should overlap with each other, should pass precisely in same path, and have same beam size to get a good beat signal ration for better sensitivity. This beat note frequency is measured by an avalanche photo detector. Most of the significant frequencies are measured and counted by a spectrum analyzer and frequency counters, respectively, which are referenced to Cesium clock. This has good stability value for length measurements. To realize long-term stability it is plausible to use H-Masers and fountains based on cold atoms. During frequency detection, the optical frequency comb as well as the residual phase is very consistently locked. A frequency counter is employed to count the frequency of pulse under measurement continuously for every second without dead time for more than 5 h.

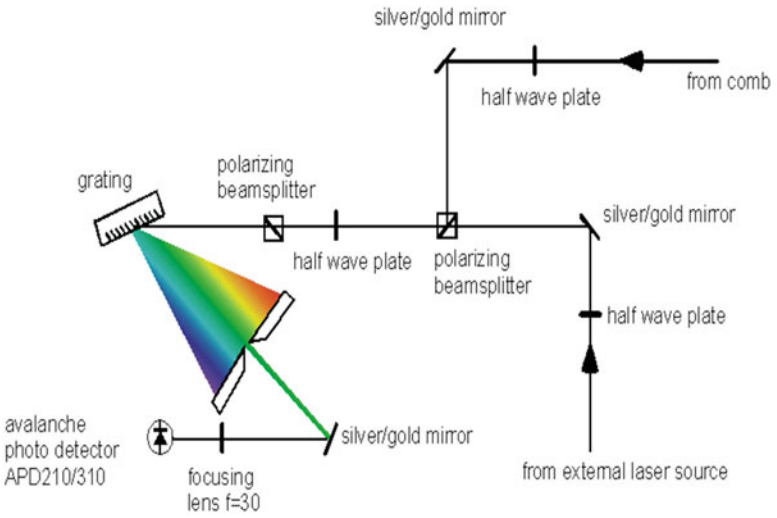

Fig. 8 Beat detection unit used for measurement of 633 nm He–Ne lasers using a optical frequency comb (Samoudi 2017)

As we have seen, to realize the SI unit meter an iodine-stabilized 633 nm (474 THz) He-Ne wavelength/frequency is calibrated against the optical frequency comb frequency of 474 THz. For a known frequency f, the analogous wavelength λ in a medium is represented by $\lambda = c/fn$, where c is the speed of light in vacuum and n is the refractive index of the medium. When measurement is executed in air, the refractive index can be known, so using Edlen's equation, by monitoring the temperature, pressure, humidity, and CO_2 concentration (Hyun et al. 2009; Birch and Downs 1993), the compensation factor for wavelength correction can be applied. In this case the length L to be determined is given as $L = \lambda(m + \varepsilon)$, where m is a positive integer, and ε is an excess fraction ($1 > \varepsilon \geq 0$). The accurate determination of L necessitates both m and ε to be correctly known, but using interferometer method for a single wavelength allows only the value of ε to be known. To determine the absolute value of L, the wavelength needs to be modulated over an extensive range in either hopping or sweeping mode, which can be only possible using optical frequency comb that covers the entire range of spectrum. Measurement uncertainty for optical frequency comb-based 633 nm He-Ne laser calibrations can be done in accordance with GUM analysis (BIPM 2008).

Conclusions

Optical frequency combs have obtained much consideration in modern days because of their huge prospective in length metrology and other applications. The absolute wavelength/frequency measurements 633 nm He–Ne lasers with optical frequency comb are in good agreement. The stability of beat frequency for longer averaging

times is reported by many researchers. Using Allan deviation analysis and Edlen equation, a correction can be applied to measurement for accurate determination of wavelength/frequency. Measurement uncertainty can be calculated in accordance with GUM analysis. The realization of SI unit meter using optical frequency comb improves many order as compared to the present method based on iodine-stabilized 633 nm He-Ne lasers.

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Quantum Definition of New Kelvin and Way Forward

11

Babita, Umesh Pant, and D. D. Shivagan

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Abstract

In the efforts of redefining the SI units with fundamental constants of nature, kelvin (K), the SI unit of thermodynamic temperatures (T), was redefined based on the Boltzmann constant (k) in the 26th meeting of the General Conference on Weights and Measures (CGPM) held in November 2018. The new definitions of four SI base units, kilogram, ampere, kelvin, and mole, have been implemented since May 20, 2019. The redefined kelvin by fixing the value of Boltzmann constant ($k = 1.380649 \times 10^{-23}\,JK^{-1}$) has opened the door for the realization and dissemination of the thermodynamic temperature, independent of any material

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property. Before the redefinition, the temperature dissemination was made through the practical scales, the International Temperature Scale of 1990 (ITS-90) using defined fixed points from 0.65 K to 1357.77 K and Provisional Low Temperature Scale of 2000 (PLTS-2000) in the temperature range from 0.9 mK to 1 K. Now, through the *mise en pratique* (MeP-K) for the realization of new kelvin, the Consultative Committee for Thermometry (CCT) has proposed the primary thermometry methods to realize *k* and dissemination of thermodynamic temperature. The primary methods such as acoustic gas thermometry, dielectric constant gas thermometry, Johnson noise thermometry, and Doppler-broadening thermometry have made immense contributions in low uncertainty realization of *k* and dissemination of thermodynamic temperatures (T) with direct traceability to redefined kelvin. The redefinition approach, various primary thermometric methods and their capabilities, implications of redefinition, and way forward have been discussed in detail in this chapter.

Keywords

SI units · Redefinition of kelvin · Boltzmann constant · Thermodynamic temperature · *MeP*-K

Introduction

International System of Units (SI) was formulated to bring the uniformity in the measurement system, and the ninth edition of SI describes how the definition of the SI units is established in terms of a set of seven defining constants (BIPM 2019). In 2005 Consultative Committee for Thermometry (CCT) expressed about the requirement of redefinition of kelvin in terms of fundamental constant (CCT 2005). Consequently, in 2011, at its 24th meeting, the General Conference on Weights and Measures (CGPM) proposed to redefine kilogram, ampere, kelvin, and mole in terms of the fundamental constants Planck's constant (*h*), elementary charge (e), Boltzmann constant (*k*), and Avogadro constant (N_A), respectively (BIPM 2019; 24th General Conference on Weights and Measures (CGPM) 2011). In its 26th meeting held on November 2018, CGPM has redefined all the SI units in terms of fundamental constant (26th General Conference on Weights and Measures (CGPM) 2018).

Since 1954, kelvin, the unit of thermodynamic temperature, was defined in terms of triple point of water as "the kelvin is the fraction 1/273.16 of the thermodynamic temperature of the triple point of water." The thermodynamic temperature of 273.16 K was assigned to TPW. TPW was realized with the best possible uncertainty of 3.4×10^{-7} relative standard uncertainty and remained as the basis of kelvin definition from ITS-68 to ITS-90. In 2005, it was noticed that the isotopic composition of water affects the measurement and contributes in uncertainty. It was observed in the CCT-K7 intercomparison that the standard deviation of TPW measured data of the transfer TPW cells of several national metrology institutes (NMIs) with BIPM reference TPW cells was 50 μK. Majority of TPW cells were constructed using continental water, which resulted in the temperature corrections of +50 μK and the standard uncertainty

of 35 μK, because of isotopic variation in water (Stock et al. 2006; Babita et al. 2021). Moreover, ITS-90 and PLTS-2000 with temperature range above 0.65 K and 0.9 mK to 1 K, respectively, are formulated for the practical realization of the kelvin (Preston-Thomas 1990; Rusby et al. 2002). These temperature scales are realized using the set of defined fixed points (such as boiling, melting, and freezing, triple point states of high pure substances) and the measuring devices, where TPW temperature is taken as a reference point at 273.16 K. The interpolating polynomial function generated for temperature sensors using these defined fixed points provides the value of T_{90} and helps to estimate T-$T_{90.}$ However, for temperature away from TPW, the T-T_{90} and associated uncertainty are several orders higher than at TPW, because of the propagation of drifts. The above discussion depicts that the definition based on TPW is material dependent and measured T-T_{90} varies with the temperature (Fischer et al. 2011). Hence, the efforts were taken for the redefinition of kelvin in terms of Boltzmann constant (k) for the direct measurement of thermodynamic temperature to make it free from material-dependent property and to maintain the uniformity in measurement throughout the temperature scale, by primary thermometry methods (Fellmuth et al. 2006; Fischer et al. 2007).

In the nineteenth century, Ludwig Boltzmann derived the Maxwell-Boltzmann velocity distribution in terms of microscopic thermal energy (kT) (Fischer et al. 2018). Kelvin, the unit of temperature, is directly linked to the joule, the unit of energy, by fixing the value of the Boltzmann constant k. Therefore, CCT and CIPM had decided to redefine the kelvin in terms of k, but before that, it was essential to realize the value of k with an uncertainty comparable to the uncertainty of the realization of TPW. With the effort of world-leading NMIs, the kelvin is redefined by fixing the numerical value of the Boltzmann constant as 1.380649×10^{-23} J.K^{-1}. Before redefinition, the primary thermometric methods employed to determine Boltzmann constant are acoustic gas thermometry (AGT), dielectric constant gas thermometry (DCGT), Johnson noise thermometry (JNT), and Doppler-broadening thermometry (DBT). The relative standard uncertainty in the determination of Boltzmann constant is 0.37 ppm in CODATA 2017, which satisfies the mise en pratique (MeP-K) criteria for the redefinition of kelvin (Newell et al. 2018). The recent MeP-K for the definition of the kelvin in the SI describes primary thermometry methods for the practical realization of new kelvin based on k (2019). In this chapter, the detailed description of primary thermometry methods defined in MeP-K for measurement and dissemination of thermodynamic temperature and its impact on present international temperature scale are presented.

Historical Prospects of Temperature Measurements

Temperature can be sensed by human touch in terms of hot and cold. Thermodynamically it can be explained as when two bodies are brought into contact, the heat transfers from hotter to cooler until both the bodies come to thermal equilibrium. Temperature is an intensive quantity, which makes it a nonadditive quantity. This implies that the 2T temperature cannot be achieved by mixing two systems maintained at the same temperature, T. The measurement of temperature is known

as thermometry. The zeroth law of thermodynamics is considered as a fundamental law for thermometry, which states that "if two systems are separately in thermal equilibrium with the third, then they are in thermal equilibrium with each other" (Quinn 1990).

The first instrument made up of the glass bulb connected to a long tube immersed in a liquid was invented by Galileo Galilei to measure the degree of hotness and coldness between the year 1592 and 1603. In this device, the liquid height of the liquid column varies as a function of ambient temperature. The first thermometer, which measures temperature, was developed by the Florentine Academy of Sciences around 1650 using a spiral-shaped tube with a closed end and graduation. With the development of thermometers, the necessity of standardizing these thermometers using fixed points has arisen. In seventeenth century, D.G. Fahrenheit employed three fixed points: a mixture of ice, water, and ammonium chloride; a mixture of ice and water; and the human body temperature with the corresponding defined temperature as 0 °F, 32 °F, and 96 °F, respectively, to develop a mercury-in-glass thermometer (Fischer and Fellmuth 2005). In another development, Swedish astronomer and physicist Andres Celsius described the scale of mercury-in-glass thermometer with ice point (0 °C) and boiling point (100 °C) of water at standard atmospheric pressure (101.325 kPa) in 1742. The scale was defined by dividing the gap in 100 equal parts, and unit was termed as degree Celsius (°C). At the start of eighteenth century, two approaches given by Fahrenheit and Amontons became the basis of further advancement in thermometry. The need of temperature measurements in BIPM activities started in 1878 during the preparation of the meter prototype. The temperature and thermal expansion coefficient of the platinum-iridium meter bar need to be measured. Therefore, it was decided that each national meter prototype should be equipped with two mercury-in-glass thermometers calibrated at BIPM. After that, it was crucial to develop a uniform temperature scale for the calibration of thermometers. Initially, the practical temperature scale was described using ice point (0 °C) and steam point (100 °C). Later in 1854, William Thomson (Lord Kelvin) introduced the new temperature scale using triple point of water temperature and absolute zero temperature. Eventually, the thermodynamic temperature of the triple point of water (TPW) was recommended as the primary reference fixed point to define unit kelvin in 1948 at the 9th CGPM. Since the ice and the boiling point of water depends on the atmospheric pressure, the TPW temperature was more stable because it exists at a fixed temperature (273.16 K) and pressure (611.657 Pa). The thermodynamic temperature of the TPW was determined by Gay-Lussac's gas law as 273.16 K (0.01 °C). In 1954, CGPM officially accepted the definition of the kelvin (K), the SI unit of thermodynamic temperature, based on TPW state as "the kelvin is the fraction 1/273.16 of the thermodynamic temperature of the triple point of water (TPW)." In 1967, at the 13th CGPM meeting, kelvin was defined as the unit of temperature, denoted by K, and it was listed as one of the seven SI base units. Apart from impurities, water used to realize TPW state consists of different isotopic compositions of oxygen ($_8O^{16}$, $_8O^{17}$, $_8O^{18}$) and hydrogen ($_1H^1$, $_1H^2$) atoms. Therefore, the isotopic composition of water also affects the accuracy of TPW temperature realization. To address this issue, CIPM in 2005 made the

resolution to use the water standard as VSMOW water for the realization definition of kelvin based on TPW. The isotopic composition defined in VSMOW is as follows (Proceedings of the 23rd meeting of the general conference on weights and measures 2007): "0.00015576 mole of ^{2}H per mole of ^{1}H; 0.0003799 mole of ^{17}O per mole of ^{16}O; and, 0.0020052 mole of ^{18}O per mole of ^{16}O." Hence, it was required to eliminate the material-dependent property from the definition of SI unit of temperature. Consequently, in the 24th CGPM meeting, it was decided to redefine kelvin in terms of the fundamental constant, which is *k* (24th General Conference on Weights and Measures (CGPM) 2011). After many international efforts, in the 26th CGPM meeting held in November 2018, kelvin was successfully redefined in terms of the Boltzmann constant (*k*) (26th General Conference on Weights and Measures (CGPM) 2018; Fischer 2019).

At present, kelvin (K) is defined as follows:

> The kelvin, symbol *K*, is the SI unit of thermodynamic temperature. It is defined by taking the fixed numerical value of the Boltzmann constant '*k*' to be 1.380 649 $\times$ 10^{-23} when expressed in the unit J K^{-1}, which is equal to kg m^2 s^{-2} K^{-1}, where the kilogram, metre and second are defined in terms of *h*, *c* and $\Delta\nu_{Cs}$. Where *h* is the Planck constant, *c* is the speed of light in vacuum and $\Delta\nu_{Cs}$ is the cesium frequency corresponding to the transition between the two hyperfine levels of the unperturbed ground state of the ^{133}Cs atom. (CCT 2019)

The Redefinition of SI Kelvin

Prior to redefinition, the definition of kelvin was based on TPW, which was material-dependent quasi-artefact that involved in definition as well as dissemination in ITS-90 scale. Efforts were made to define all seven base units in terms of the fundamental constant, the constants of nature invariant in time and space. The Boltzmann constant (*k*) appears in the expression of the mean microscopic thermal energy (*k*T) to determine the mean kinetic energy of the particle of an ideal gas (Fellmuth et al. 2006). By fixing the numerical value of *k*, the kelvin is directly related to the unit of thermal energy: the joule. Therefore, thermal metrologist made efforts to realize *k* with lowest possible uncertainty for redefinition of kelvin. The primary thermometric methods were employed to determine the Boltzmann constant, and the following criterion was ensured in the *mise en pratique* (MeP-K) for the redefinition of kelvin (C.C. for T 2014; Fellmuth et al. 2016a): "The relative standard uncertainty of the adjusted value of *k* is less than 1 $\times$ 10^{-6}; and the determination of *k* is based on at least two fundamentally different methods, of which at least one result for each shall have a relative standard uncertainty less than 3 $\times$ 10^{-6}."

Hence, it was the challenging task to determine Boltzmann constant (*k*) with low uncertainty (below 1 ppm) by employing different primary thermometric methods. To make new definition consistent with the earlier definition, *k* was realized at TPW temperature. After fixing the value of *k*, the thermodynamic temperature with minimum possible uncertainty can be determined directly using primary

thermometers (Fellmuth et al. 2006). For a decade, researchers from various NMIs have carried out rigorous research to develop primary thermometers for low uncertainty determination of the Boltzmann constant. The CODATA Task Group on Fundamental Physical Constants (TGFC) periodically reviewed the progress of the state-of-the-art determinations of the values of physical fundamental constants. In CODATA 2017, the adjusted value of Boltzmann constant was determined with a relative standard uncertainty of 0.37 ppm (Newell et al. 2018). This determination of k is based on the three primary thermometry methods: AGT, DCGT, and JNT. Apart from this, the DBT was used to determine k with slightly higher uncertainties for additional confirmation. The status of all determination of k using three primary thermometers (AGT, DCGT, and JNT) and adjusted CODATA value of k is presented in Fig. 1.

This adjusted CODATA 2017 value of k was fixed as 1.380649×10^{-23} J/K with relative standard uncertainty of 0.37 ppm. Consequently, in 2018 CGPM, this value of k was considered for the redefinition of kelvin without assigning any uncertainty to it. This also ensures that the best estimate of the value of T_{TPW} remains 273.16 K, to make ITS-90-based measurements consistent. One of the implications of the 2018 definition is that the determined relative uncertainty in k, 3.7×10^{-7}, is transferred to the temperature of the triple point of water, T_{TPW}. Therefore, the standard uncertainty of T_{TPW} is hence now $u(T_{TPW}) = 0.1$ mK compared to the best TPW CMCs of 0.06 mK. The new definition was implemented worldwide on May 20, 2019.

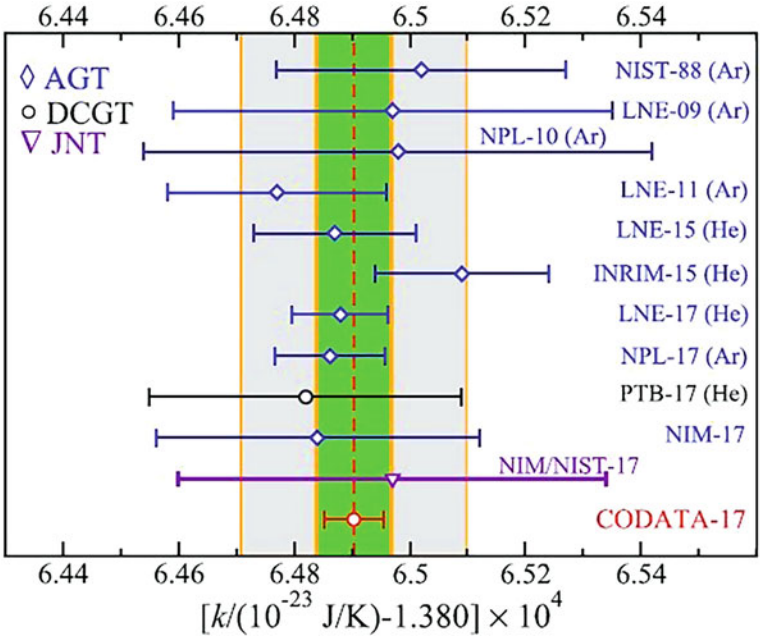

Fig. 1 The final determination of Boltzmann constant (k) in CODATA 2017, highlighting the contribution of AGT, DCGT, and JNT primary thermometry methods (Fischer et al. 2018)

Mise en Pratique (MeP-K) for the Realization of Kelvin

The *mise en pratique* for the definition of the kelvin in the SI (*MeP*-K) is formulated by the Consultative Committee for Thermometry (CCT) of the International Committee for Weights and Measures (CIPM) to describe the methods for the practical realization of redefined kelvin based on the Boltzmann constant (k) (CCT 2019). Primary thermometers based on well-understood physical system are employed for the direct measurement of the thermodynamic temperature. Further, these thermometers are subdivided into two categories: absolute and relative primary thermometry methods. In absolute primary thermometry method, the thermodynamic temperature is measured directly using the defined numerical value of Boltzmann constant. It does not require any fixed point ($n = 0$, $n =$ no. of fixed points), and parameters specified in the equation of state are measured, whereas in relative primary thermometry method, one or more reference fixed points ($n > 0$) are used to determine the one key parameter (speed of sound in AGT, dielectric constant in DCGT, etc.) to measure the thermodynamic temperature. *MeP*-K also includes the practical realization of the kelvin applying defined temperature scales, ITS-90 (0.65 K and higher) and the PLTS-2000 (0.9 mK to 1 K), and their measured temperatures are denoted by T_{90} and T_{2000}, respectively. The *MeP*-K also describes the T-T_{90} and T-T_{2000} for these two temperature scales and their corresponding uncertainties. The present *MeP*-K illustrates the use of the following primary thermometers for direct thermodynamic temperature measurement: acoustic gas thermometry (AGT), spectral-band radiometric thermometry (1235 K and above), polarizing gas thermometry (dielectric constant gas thermometry (DCGT) and refractive-index gas thermometry (RIGT)), and Johnson noise thermometry (JNT). However, there are other primary thermometers based on equation of state-based method, statistical- and quantum-based method can be evolved for the best dissemination method for Boltzmann constant-based new kelvin. No single method or fixed point is preferred in the dissemination of new kelvin. Therefore, it is very coherent, robust, and quantum mechanical-based dissemination process.

Primary Thermometry Methods for the Realization and Dissemination of New Kelvin

Primary thermometers are based on the well-understood physical systems. At the microscopic level, kinetic energy of the gas particles in a close volume is defined by thermal energy of the system, given as k.T. In primary thermometry methods, the equation of state that describes the relation between thermodynamic temperature and independent quantities can be written down explicitly without having any unknown or temperature-dependent constant (Quinn 1990). Mean microscopic energy of gas system cannot be measured directly. By using statistical mechanics, the microscopic average kinetic energy of gas molecules will provide thermal energy k.T of the system and which is related to different macroscopic quantities of primary thermometry methods (Fischer et al. 2007).

Internationally different NMIs used different methods for the determination of k and thermodynamic temperature since the last three decades. Each method is dependent on some temperature-dependent variable; like in gas thermometric methods, temperature is measured by using extensive property of amount of gas in constant volume gas thermometer (CVGT) method and the intensive property of gas such as the speed of sound, the refractive index, and dielectric property of gas in AGT, RIGT, and DCGT methods, respectively. Other than gas thermometry, there are few other methods like JNT, radiation thermometry, and DBT that depend on electronic noise in resistor, Stefan's and Planck's law of radiation, and broadening of absorption line of molecules, respectively. In gas thermometry, kinetic energy is responsible for the increase of the momentum of a gas particles in a closed volume, which causes an increase of the macroscopic quantity, pressure "P." In gas thermometry, statistical thermodynamics relate classical quantity, pressure to the quantum mean thermal energy kT of the gas molecules system, by equation of state.

The primary thermometry methods described in *MeP*-K used for the measurement of thermodynamic temperature are given below (Fischer et al. 2018; CCT 2019; Fischer 2019):

- Acoustic gas thermometry (AGT)
- Dielectric constant gas thermometry (DCGT)
- Refractive-index gas thermometry (RIGT)
- Spectral radiation thermometry (SRT)
- Johnson noise thermometry (JNT)
- Doppler-broadening thermometry (DBT)

Acoustic Gas Thermometry (AGT)

In acoustic gas thermometry, thermodynamic temperature is determined by the measurement of speed of sound in a monoatomic gas medium with average molar mass M (Fellmuth et al. 2006) (Fig. 2).

To determine Boltzmann constant (k) by using AGT method, the relation between k and the limiting low-pressure speed of sound u_0 in an ideal gas medium is derived from an equation given by (De Podesta et al. 2013; Pitre et al. 2011):

$$k = \frac{Mu_o^2}{\gamma_o T N_A} \tag{1}$$

In the above equation, γ_o is the zero-density ratio of the constant-pressure-specific heat to the constant-volume-specific heat, M is the molar mass of the gas, N_A is the Avogadro constant, and T is the thermodynamic temperature.

Similar to CVGT method, to consider the nonideal behavior of the real gases, the speed of sound in a dilute gas medium is expressed in terms of the virial expansion coefficient in pressure (Moldover et al. 2014):

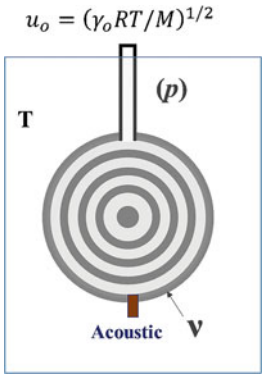

Fig. 2 Principle for acoustic gas thermometry (AGT) method (Fellmuth et al. 2006)

$$u^2 = \frac{\gamma_o k\, T}{m} + A_1(T)p + A_2(T)\, p^2 \ldots \tag{2}$$

where m is the average mass of one atom or molecule in gas medium and $A_1(T)$ and $A_2(T)$ are the virial expansion coefficient in pressure.

In low pressure limit, Eq. (2) reduces to Eq. (1) for the calculation of Boltzmann constant. For the realization of Boltzmann constant with uncertainty below 1 ppm, the measurement of speed of sound was carried out at TPW, reference fixed point, to make it consistent with the earlier definition by using absolute acoustic gas thermometry method.

After fixing the value of k, the unknown thermodynamic temperatures (T) can be measured by using relative AGT method, i.e., by taking the ratio with reference of TPW temperature, and the ratio of speed of sound at both temperature, which is given as

$$\frac{T}{273.16\ \mathrm{K}} = \frac{m_T}{m_{TPW}} \left(\frac{u_{0,T}}{u_{0,TPW}}\right)^2 \tag{3}$$

While using the same gas with same isotopic composition inside the resonator, the above equation reduces to

$$\frac{T}{273.16\ \mathrm{K}} = \left(\frac{u_{0,T}}{u_{0,TPW}}\right)^2 \tag{4}$$

In 1877, the principle of sound wave propagation in cylindrical and spherical resonators was introduced by the Rayleigh in his work "The Theory of Sound." A.R. Colclough has developed the first cylindrical acoustic interferometer with fixed frequency and variable path in helium gas medium to determine the gas constant near TPW temperature (Colclough et al. 1979). In 1988, Moldover et al. determined universal gas constant (R) using AGT with argon gas-filled stainless steel spherical resonator. For that measurement, the volume of resonator was estimated by

pycnometry, which has higher measurement uncertainty in the volume measurement than the recently used microwave measurements. With this stainless steel spherical resonator, Moldover et al. estimated the numerical value of k using AGT with the relative uncertainty of 1.8 ppm as mentioned in Table 1 (Moldover et al. 1988; Moldover and Trusler 1988).

To discuss the scientific and technological aspects involved in the determination of Boltzmann constant, a workshop was held on January 21, 2005, at PTB Germany, and after analyzing state-of-the-art method, M. Moldover of NIST claimed that the Boltzmann constant could be determined with the relative uncertainty of 1 ppm. Later, many groups have worked on this method and could achieve the uncertainty in the determination of k below 1 ppm. The measurement uncertainty for the realization of k was estimated by the different groups, shown in Tables 1 and 2.

Results obtained from AGT method have a major contribution in CODATA value for the realization of k and hence in the redefinition of kelvin. LNE France and NPL UK developed AGT using quasi-spherical resonator and realized k with uncertainty of 0.6 ppm (2017) and 0.7 ppm (2017), respectively. These two results further became basis for CODATA 2017 value of k. The NPL UK acoustic resonator for the measurement of acoustic and microwave frequencies is shown in Fig. 3.

After redefinition of kelvin, the AGT is being used for low uncertainty measurement of thermodynamic temperature for the wide range, i.e., from 7 K to 550 K, having the possibility from LHe (4 K) temperature to freezing point of copper (1350 K) (Moldover et al. 2014). In recent developments, INRiM has used He gas in AGT method for the determination of thermodynamic temperature in low temperature range from 236 K to 430 K, and uncertainty in measurement is from

Table 1 List of work done for the determination of k by using spherical cavity in acoustic gas thermometry (AGT) method

Sr. no.	Group/institute(s)	Experimental description	Year	k [$\times 10^{-23}$ JK^{-1}]	$u(k)/k$ [$\times 10^{-6}$]
1.	NIST (Moldover et al. 1988)	Stainless steel spherical resonator with radius 9 cm, Ar gas	1988	1.38065020	1.8
2.	LNE-INM/CNAM (Pitre et al. 2009)	50 mm radius triaxial ellipsoid copper resonator, He gas	2009	1.38064970	2.7
3.	INRiM Italy (Gavioso et al. 2010)	Triaxial ellipsoid, radius (50.250 mm, 50.29 mm, 50.33 mm), volume 0.5 L, Cu material, diamond finish, Ar gas	2010	1.38064040	7.49
4.	NPL UK, LNE-INM France, IRMM Belgium (Sutton et al. 2010)	50 mm radius, diamond-turned OFHC Cu quasi-spherical resonator, Ar gas	2010	1.38064980	3.1

(continued)

Table 1 (continued)

Sr. no.	Group/institute(s)	Experimental description	Year	k [$\times 10^{-23}$ JK^{-1}]	$u(k)/k$ [$\times 10^{-6}$]
5.	LNE-Cnam (LCM) France (Pitre et al. 2011)	50 mm, OFHC Cu quasi- spherical resonator, 10 mm thickness, Ar gas	2011	1.38064770	1.24
6.	NPL UK (De Podesta et al. 2013)	OFHC Cu quasi-spherical resonator of radius 62 mm radius, volume 1 L, Ar gas	2013	1.38065156	0.71
7.	LNE-Cnam France, INRiM Italy (Pitre et al. 2015)	50 mm radius Cu ellipsoid resonator, 0.5 L volume, He gas	2015	1.38064870	1.02
8.	INRiM Italy, PTB Germany, LNE-Cnam (LCM) France, Scottish Universities Environmental Research Centre UK (Gavioso et al. 2015)	90 mm Cu resonator triaxial ellipsoid, 3 L volume, He gas	2015	1.38065090	1.06
9.	CEM-UVa Spain (Pérez-Sanz et al. 2015)	Misaligned stainless steel spherical cavity of 40 mm radius and volume 268 cm^3, Ar gas	2015	1.38065600	16
10.	LCM-LNE Cnam France, LNE France, CRPG Nancy INRiM Italy et al. (Pitre et al. 2017)	90 mm OFHC Cu triaxial resonator of 3 L volume, helium-4 gas	2017	1.38064878	0.6
11.	NPL UK et al. (De Podesta et al. 2017)	OFHC Cu quasi-spherical resonator of radius 62 mm radius, volume 1 L, Ar gas	2017	1.38064862	0.7
12.	UVa-CEM Spain (Segovia et al. 2017)	Triaxial ellipsoidal copper resonator of radius 40 mm, Ar gas	2017	1.38064670	6.7
13.	NMIJ, AIST Japan (Misawa et al. 2018)	Copper quasi-spherical resonator of 62 mm inner radius and 10 mm thickness, 1 L volume, Ar gas	2017	1.380631	

0.25 mK to 0.89 mK, respectively, in 2019 (Gavioso et al. 2019). In 2020, VNIIFTRI Russia has performed the ITS-90 realization in the temperature range from 79 K to 83.80 K and obtained value of T-T_{90} as −4.47 mK at 79 K and − 4.81 mK at 83.80 K with an uncertainty 0.97 mK and 1.02 mK, respectively (Kytin et al. 2020). In the recent advances, J.V. Widiatmo et al. have developed acoustic gas thermometer for

Table 2 List of work done for the determination of k by using cylindrical acoustic gas thermometry (cAGT) method

Sr. no.	Group/ institute	Experimental description	Year of publication	k [$\times 10^{-23}$ J K^{-1}]	$u(k)/k$ [$\times 10^{-6}$]
1.	NPL UK (Colclough et al. 1979)	Cylindrical cavity resonator made up of brass coated with nickel-tin, Ar gas	1979	1.38065600	8.5
2.	NIM China (Zhang et al. 2011)	Cylindrical cavity resonator (length 129.39 mm, diameter 80 mm, thickness 30 mm) made up of stainless steel, Ar gas	2011	1.38065060	7.9
3.	NIM China (Lin et al. 2013)	(80 mm long, 80 mm diameter) cylindrical steel cavity, Ar gas	2013	1.38064760	3.7
4.	NIM China (Feng et al. 2017)	(80 mm long, 80 mm diameter) cylindrical copper cavity, Ar gas	2017	1.38064840	2

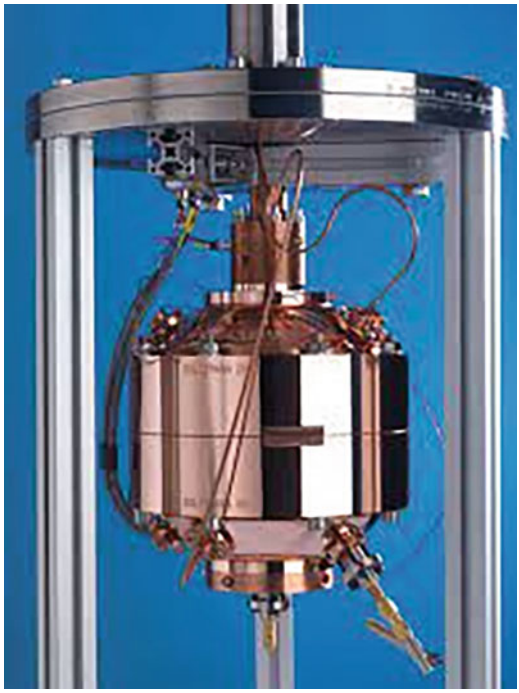

Fig. 3 NPL UK acoustic gas thermometer using diamond-turned OFHC quasi-spherical cavity resonator (Moldover et al. 2016)

the measurement of T-T_{90} in 5N pure neon gas using OFHC copper quasi-spherical resonator. They found the combined uncertainty of T-T_{90} measurement as 2.6 mK at M.P. of Ga temperature (Widiatmo et al. 2021).

Dielectric Constant Gas Thermometry (DCGT)

The principle of DCGT method describes the thermodynamic temperature estimation based on the measurement of intensive property, the dielectric constant. But this method is different from AGT method as in this method, a reference temperature fixed point is always required for the molar polarizability.

In DCGT, the density in the state of equation of a gas was replaced by the dielectric constant (ε), which is measured by the change in capacitance of a capacitor filled with gas at different pressures and constant temperature (measurement of isotherms). If the gas is ideal or there are no interactions between gas particles, the working equation for DCGT method can be derived simply by combining the classical ideal gas law and Clausius-Mossotti Eq. (15) (Fig. 4).

In DCGT method from the known value of gas pressure p, change of the capacitance C(p) in a capacitor will be calculated.

For an ideal gas, the dielectric constant is given by the relation

$$\varepsilon = \varepsilon_o + \frac{\alpha_o N}{V} \tag{5}$$

Here, $\varepsilon_{0\ \mathrm{is}}$ permittivity of vacuum; ε is permittivity of medium; $\varepsilon/\varepsilon_0$ will provide dielectric constant; α_0 = static electric dipole polarizability of atoms, which is $2.28151331(23) \times 10^{-41}$ C m^2 V^{-1} for He atoms with uncertainty of 0.1 ppm; and $n = N/V$ is the number density.

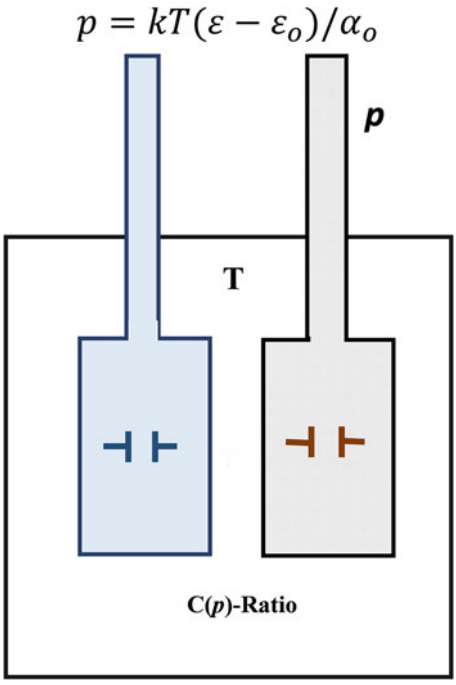

Fig. 4 Principle of dielectric constant gas thermometry (DCGT) method (Fellmuth et al. 2006)

Therefore, state equation (pV = NRT) becomes

$$p = \frac{kT(\varepsilon - \varepsilon_0)}{\alpha_0} \tag{6}$$

In the above equation, pressure of filled gas and dielectric properties are known; then temperature can be calculated.

Mathematically relation between molar polarizability A_ε and Boltzmann constant is related as

$$k = \frac{\alpha_o}{3\varepsilon_o} / \frac{A_\varepsilon}{R} \tag{7}$$

$$A_\varepsilon = \frac{N_A \mathrm{A} \alpha_0}{3\varepsilon_0} \tag{8}$$

N_A is the Avogadro constant in Eq. (8).

For a real gas, based on virial expansion and Clausius-Mossotti equation, p is given by

$$p \approx \frac{\chi}{\frac{3A_\varepsilon}{RT} + k_{\mathrm{eff}}} \left[1 + \frac{B(T)}{3A_\varepsilon}\chi + \frac{C(T)}{(3A_\varepsilon)^2}\chi^2 + \ldots \right] \tag{9}$$

where $\chi = \varepsilon/\varepsilon_0 - 1$ is the dielectric susceptibility and k_{eff} is effective compressibility of capacitor calculated from the material parameters of the capacitor.

The relative change in capacitance $(C(p) - C(0))/C(0) = \chi + (\varepsilon/\varepsilon_0)\, k_{\mathrm{eff}} p$ of the gas-filled capacitor is determined as a function of pressure p of the gas. C(p) is measured for various gas pressures between the capacitor electrodes and at evacuated condition.

A polynomial fit to the resulting p vs relative change in capacitance, i.e., $(C(p) - C(0))/C(0)$ yields $3A_\varepsilon/RT$, which is a macroscopic quantity.

In 1970s, the DCGT method was first developed in H.H. Wills Physics Laboratory, University of Bristol, UK, for the precise measurement of thermodynamic temperature (Quinn 1990). PTB Germany developed DCGT method with He gas for the determination of k. Because of low uncertainty in theoretically calculated value of static electric dipole polarizability of He gas, it was preferred while taking measurement (Gaiser et al. 2015). To measure k by DCGT, the temperature of gas and capacitor requires the low uncertainty measurement of the temperature, and the temperature must be traceable to TPW with a standard uncertainty better than 0.1 mK. By applying virial expansion, the pressure and relative capacitance change data pairs are fitted, which reflects the interaction of the atoms of the real helium gas having bosonic interaction and takes account of thermodynamic temperature as one of the parameters (Fellmuth et al. 2009).

In 2017, PTB Germany published results obtained for the realization of Boltzmann constant, and the results were enhanced by few changes in measurement

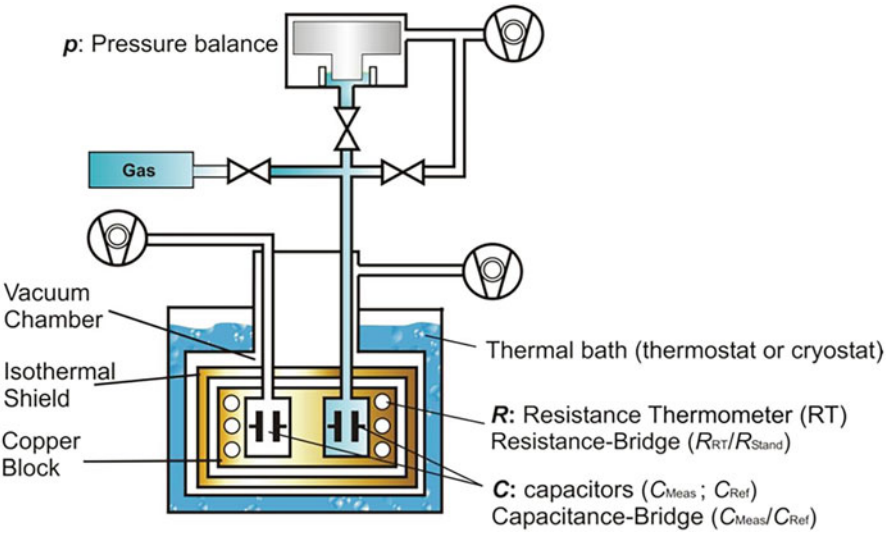

Fig. 5 Schematic diagram of the PTB Germany DCGT setup (Gaiser et al. 2015)

setup, which includes the use of three capacitors for experiment, reducing uncertainty in compressibility of electrodes due to individual determination of adiabatic compressibility of electrodes and additional shielding of capacitor electrodes for better stability (Gaiser et al. 2017) (Fig. 5).

Since 2011, PTB Germany has been continuously working on the development and advancement of DCGT method, which is summarized in Table 3 and achieved the best uncertainty as 1.9 ppm in the determination of *k*. This estimation using DCGT has contributed in CODATA 2017 to estimate the numerical value of *k*.

For thermodynamic temperature measurement, DCGT method was used by PTB for the calculation of standard uncertainty in the thermodynamic temperature range from 4.8 K to 25 K and uncertainty lies between 0.23 mK to 0.29 mK in 2015 (Gaiser et al. 2015).

In 2020, PTB has measured thermodynamic temperature from 30 K to 200 K, which was done by using three different gases (He, Ne, and Ar) in DCGT setup, and combined the results for T-T_{90} with the results obtained from RIGT and AGT methods (Gaiser et al. 2020).

Refractive-Index Gas Thermometry

The relation between refractive index of gas and density of gas medium is used to measure thermodynamic temperature. The refractive index of any gas medium is calculated by using the relation

Table 3 List of work done for the determination of k by using DCGT method

Sr. no.	Group/ institute	Experimental description	Year of publication	k [$\times 10^{-23}$ J K^{-1}]	$u(k)/k$ [$\times 10^{-6}$]
1.	PTB Germany (Zandt et al. 2010; Fellmuth et al. 2011)	Two stainless steel cylindrical capacitors (10 pF), pressure range (1 MPa to 7 MPa), He gas	2011	1.38065500	7.9
2.	PTB Germany (Lin et al. 2013; Gaiser and Fellmuth 2012)	Tungsten carbide (TC) cylindrical capacitor (10 pF), pressure range (1 MPa to 7 MPa), He gas	2013	1.38065090	4.3
3.	PTB Germany (Zandt et al. 2015)	Two tungsten carbide (TC) cylindrical cross capacitors (10 pF), pressure range (1 MPa to 7 MPa) with improved pressure measurement system, He gas	2015	1.38065090	4
4.	PTB Germany (Gaiser et al. 2017)	Three tungsten carbide (TC) cylindrical cross capacitors (10 pF), pressure range (1 MPa to 7 MPa), He gas	2017	1.38064820	1.9

$$n = \frac{c_o}{c} = \sqrt{\varepsilon_r \mu_r} \tag{10}$$

where ε_r is the relative dielectric permittivity, μ_r is the relative magnetic permeability, c_o is the speed of light in vacuum, and c is the speed of light in gas medium (Rourke et al. 2019).

The principle of RIGT is similar to DCGT and can easily be derived from the Clausius-Mossotti equation by replacing dielectric constant with n^2 where n is the refractive index of gas as shown in Fig. 6.

The equation in terms of virial expansion coefficient can be written as follows:

$$\frac{n^2 - 1}{n^2 + 2} = A_R \rho \left(1 + b_R \rho + c_R \rho^2 + \ldots\right) \tag{11}$$

where A_R, b_R, and c_R are refractivity virial coefficients. This equation is called as Lorentz-Lorenz equation (Fellmuth et al. 2006; Colclough 1974).

RIGT was first proposed by A.R. Colclough in 1974 (Colclough 1974). The experimental setup for RIGT is similar to AGT method. In 2004, May et al. has evaluated relative dielectric permittivity in He and Ar gas medium using oxygen-free high thermal conductivity (OFHC) copper quasi-spherical cavity resonator (QSCR) by measuring resonance frequencies (May et al. 2004). Nowadays, the realization of

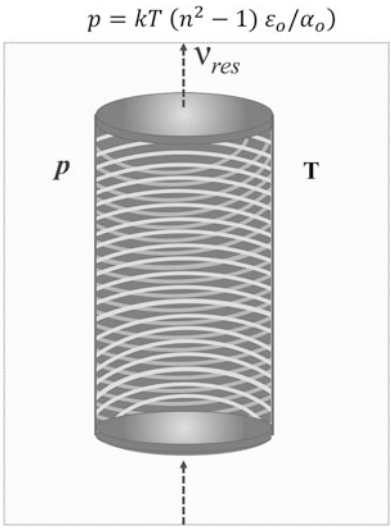

Fig. 6 Principle of refractive-index gas thermometry (RIGT) method (Fellmuth et al. 2006)

RIGT is generally done using the gas-filled quasi-spherical cavity resonator (QSCR) by measuring the microwave frequencies. The measurement of microwave frequencies is similar as in AGT.

In 2017, Bo Gao et al. at NIMs China developed single-pressure refractive-index gas thermometry (SPRIGT) method for the measurement of thermodynamic temperature. At a single pressure of gas inside the cavity, refractive index at one reference point (TPW) was measured, and by taking the ratio with refractive index at other temperatures, the thermodynamic temperature can be calculated (Gao et al. 2017). In 2018, SPRIGT working group improved the stability of pressure from 30 kPa to 90 kPa below 0.1 ppm (Gao et al. 2018).

NIST USA has developed RIGT method for the determination of k, and the measured value and relative standard uncertainty are presented in Table 4.

In 2016, NRC Canada used RIGT method for the measurement of thermodynamic temperature in the temperature range 24.5 K to 84 K by using copper quasi-spherical resonator filled with He gas and obtained the standard uncertainty in T-T_{90} in the range from 0.56 mK to 2.9 mK (Rourke 2017). Later on, the group has published a review article on the development of RIGT in 2019 (Rourke et al. 2019).

In 2020, Gao et al. carried out the thermodynamic temperature measurement from 5 K to 24.5 K by using the same methodology. The accuracy in thermodynamic temperature measurement is in the range 0.05 K to 0.17 mK and which is comparable to the previous measurement taken by using other methods (Gao et al. 2020). In 2021, D. Madonna Ripa et al. measured refractive index of helium and neon gas at five different temperatures in the interval between the triple point of hydrogen at 13.8 K and the triple point of xenon at 161.4 K for pressures up to 380 kPa (Ripa et al. 2021). The T-T_{90} was determined with different methods of analysis with improved uncertainties. The measurement of thermodynamic temperature can

Table 4 List of work done for the determination of k by using RIGT method

Sr. no.	Group/ institute	Experimental description	Year of publication	k [$\times 10^{-23}$ J K^{-1}]	$u(k)/k$ [$\times 10^{-6}$]
1.	NIST USA (Schmidt et al. 2007)	Quasi-spherical resonator of steel coated with copper having radius 4.82 cm, He gas	2007	1.38065300	9.1
2.	NIST USA (Egan et al. 2017)	Quasi-monolithic heterodyne interferometer, He gas	2017	1.38065200	12.5

be done in the temperature range 2.6 K to 1000 K in the present RIGT developed setup.

Radiometric Thermometry

The absolute measurement of the blackbody radiation is based on the Planck's law of thermal radiation. According to the Planck's law, the spectral radiance of a blackbody cavity is the radiative power emitted per unit area per unit solid angle per unit wavelength, given as follows (Todd et al. 2017):

$$L_{b,\lambda}(\lambda, T) = \frac{2hc^2}{n^2\lambda^5} \frac{1}{\exp\left(\frac{hc}{n.\lambda.k.T}\right) - 1} \tag{12}$$

where the spectral radiance concentration $L_{b,\lambda}(\lambda, T)$ is independent of the material and shape of the cavity wall and function of the vacuum wavelength λ and temperature T. The h stands for the Planck's constant, c is the speed of light, n stands for the refractive index of gas inside cavity, and k is the Boltzmann constant. The unit of the spectral radiance is $Wm^{-2}\ sr^{-1}\ nm^{-1}$.

Equation (12) can be written in the simplified form:

$$L_{b,\lambda}(\lambda, T) = \frac{c_1}{n^2\lambda^5} \frac{1}{\exp\left(\frac{c_2}{n\lambda T}\right) - 1} \tag{13}$$

where c_1 and c_2 are the first and second radiation constants, respectively:

$$c_1 = 2hc^2 = 3.7417749\ \text{W.m}^2$$

$$c_2 = hc/k = 0.014388\ \text{K.m}$$

The total radiation (L) emitted by the blackbody cavity can be calculated by integrating spectral power over all wavelengths over the solid angle 2π, which gives the Stefan-Boltzmann law of total radiation:

$$L_b(T) = \int_0^\infty L_\lambda d\lambda = \frac{2\pi^5 k^4 n^2}{15h^3 c^2} T^4 \tag{14}$$

Total radiation thermometers are based on the Stefan-Boltzmann law of thermal radiation. Total radiant energy emitted by a surface per unit area of a blackbody at temperature T is given by Eq. (15):

$$L_b(T) = \frac{2\pi^5 k^4 n^2}{15h^3 c^2} T^4 = \sigma T^4 \tag{15}$$

where σ is the Stefan-Boltzmann constant ($5.670374419 \times 10^{-8}$ W m^{-2} K^{-4}).

The ratio of radiation power can estimate the unknown temperature at temperature T and at triple point of water temperature T_{TPW}. The equation can be written as follows:

$$\frac{L_b(T)}{L_b\left(T_{TPW}\right)} = \frac{T^4}{{T_{TPW}}^4} \tag{16}$$

Absolute radiometric measurement has been carried out to measure thermodynamic temperature by a total radiation thermometer. The absolute total radiometric thermometry and spectral-band radiometric thermometry will play a vital role in the dissemination of new kelvin in high-temperature range above 1235 K.

The spectral-band radiometry has advantage over total radiation thermometer that it can be operated over the spectral range chosen via filter, whereas in total radiation thermometer, spectral power was calculated over the overall wavelength. The limited spectral range provides better signal-to-noise ratio. The recent *mise en pratique* include "spectral-band radiometric thermometry" for the determination of thermodynamic temperature above 1235 K (CCT 2019). This method includes two approaches: absolute primary radiometric thermometry and relative primary radiometric thermometry. In absolute primary radiometric thermometry, the spectral power of the blackbody source is determined using a detector of known (absolute) spectral responsivity in a particular waveband and in a defined solid angle. The measurement of the spectral power requires the knowledge of detector and the spectral responsivity of the filter. Also, the geometric factor such as A1, A2, and distance (d) between blackbody source should be known to define solid angle (Anhalt and Machin 2016). A typical schematic of the measurement of spectral radiance or irradiance is shown in Fig. 7.

In the case of relative primary radiometric thermometry, the spectral responsivity of the filter is not required to be known. The optical power is measured on the fixed-point blackbodies with known thermodynamic temperatures. There are three approaches to measure thermodynamic temperature using relative primary radiometric thermometry: (1) extrapolation from one fixed point, which requires only knowledge of the relative spectral responsivity of the detector and filter; (2) interpolation or extrapolation from two fixed points, which requires only the bandwidth of

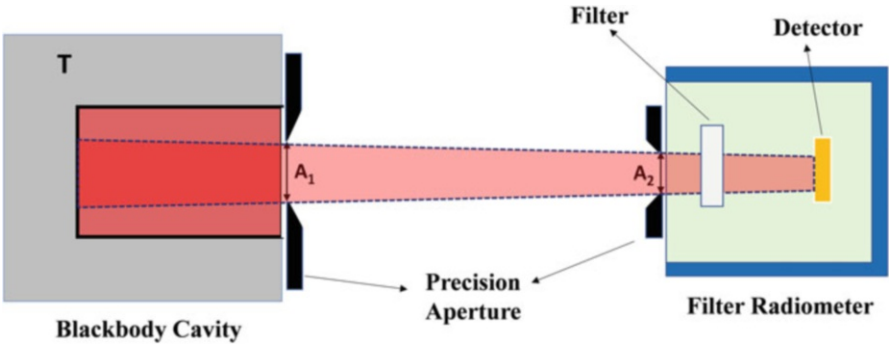

Fig. 7 Schematic for the measurement of thermal radiation using filter radiometer, where A_1 and A_2 are the precision apertures (Fischer 2019)

the responsivity; and (3) interpolation or extrapolation from three or more fixed points, for which detailed measurements of responsivity are not required (CCT 2019). The Planck's form of the Sakuma-Hattori equation, a well-understood parametric approximation of the integral expression of the radiative power, is used for the simplification of interpolation and extrapolation. The relative primary radiometric thermometry has slightly higher uncertainties than absolute primary radiometric thermometry. However, absolute primary radiometric thermometry required a sound knowledge of the absolute spectral responsivity of filter lens. The calibration facility of spectral responsivity of a filter lens is expensive and time consuming. Therefore, relative primary radiometric thermometry using multiple fixed points has an advantage over absolute primary radiometric thermometry, where calibration of the filter is not required. The high-temperature fixed points (HTFP), metal-carbon (M-C) eutectic fixed points in combination with pure metal fixed point (Ag, Au, or Cu), are employed to disseminate thermodynamic temperature above 1235 K using relative radiometric thermometry (Sadli et al. 2016; Woolliams et al. 2016). The M-C eutectic fixed points ranging up to 3200 °C, as shown in Fig. 8, are being developed and realized for the last two decades for the dissemination of thermodynamic temperature in high-temperature range at NMIs (Woolliams et al. 2006; Machin 2013; Pant et al. 2018, 2019, 2020).

Johnson Noise Thermometry (JNT)

Johnson noise thermometry (JNT) is based on the electrical noise generated in conductors due to the zigzag momentum (similar to Brownian motion invented by Einstein in 1906) of charge carriers in it as shown in Fig. 9. In this method, microscopic quantity is electrical noise, related to the temperature, and macroscopic quantity is electrical resistance.

JNT method was first introduced by J.B. Johnson in 1927, and later on, in 1928, H. Nyquist reported theoretical prospectus of it and derived expression from Planck's blackbody law. They provided a different outlook toward the determination

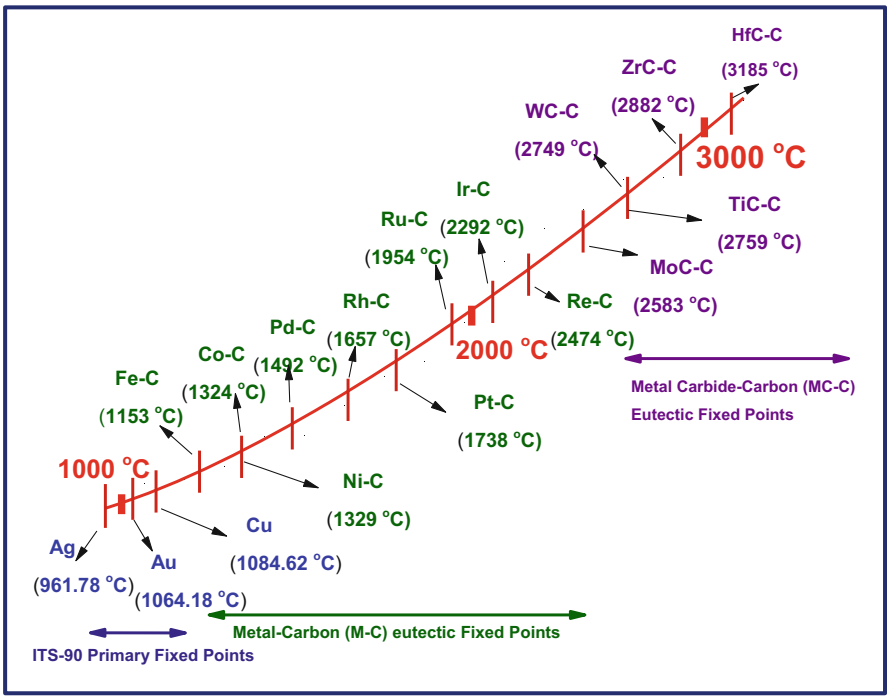

Fig. 8 Metal-carbon (M-C) and metal carbide-carbon (MC-C) fixed points

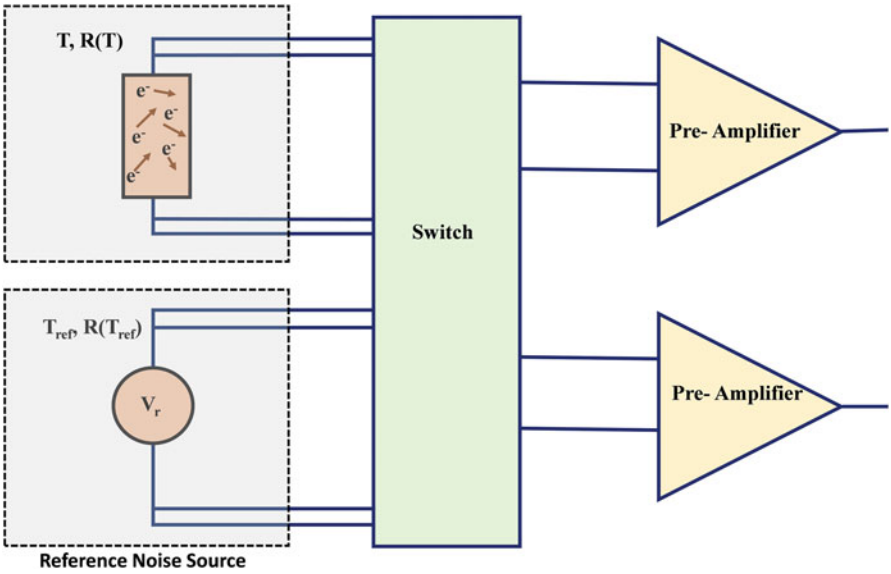

Fig. 9 Block diagram for Johnson noise thermometry method where thermal noise in electrical resistor at T is compared with the reference voltage source at T_{ref} (Fischer 2019)

of k by taking the concept of electromotive force that is responsible for the flow of electrons (electricity) through conductors, which creates thermal effect (Johnson 1928; White et al. 1996). After 1950s, JNT method was developed internationally for the measurement of thermodynamic temperature at different fixed points (Qu et al. 2019).

In JNT method, the thermodynamic temperature measurement is based on the mean square noise voltage U, caused due to the statistical motion of particles in a resistor.

The temperature dependence of the developed voltage is given by Nyquist formula:

$$\left\langle U^2 \right\rangle = 4kT.R\Delta\nu \tag{17}$$

valid for frequencies, $\nu << kT/h$

Electrical resistance R, a frequency-independent quantity, and $\Delta\nu$ are measured at TPW (Nam et al. 2003, 2004).

In noise thermometry method, Boltzmann constant can be determined directly from the ratio of mean square of the noise voltage and resistance. If the sensing resistor is replaced with an AC Josephson voltage standard, the absolute measurement at TPW would directly provide a determination of k.

As the noise voltages are extremely small, one has to use special electronic circuit and arrangements, with reference voltage source traceable to quantum standard of voltage. The main challenges in this method are to optimize the performance of the electronic system such as drift in the gain and bandwidth and the parameters, viz., long measurement time, nonlinearity, and transmission-line effects. The initial efforts brought the uncertainty value to 1×10^{-5}, when measured at zinc fixed point. NIST and the Measurement Standards Laboratory of New Zealand proposed the perfect quantization of voltages from the Josephson effect (Tew et al. 2007).

In 2003, for improving JNT method, a reference voltage source was used called quantized voltage noise source (QVNS). With this improved setup, the expected relative uncertainty is below 10 ppm in the thermodynamic temperature range 270 K to 1000 K (Nam et al. 2003).

In this direction after 2003, many improvements have been made in the experimental setup of JNT specially in circuit design for the thermodynamic temperature measurement with low uncertainty in the last two decades (Nam et al. 2004; Sun et al. 2011; Qu et al. 2013).

In 2014, a systematic error that arises due to software part instead of hardware part was considered for the measurement of k. The experiment setup includes two different resistors, and their noise ratio contributed 12 ppm systemic error (Pollarolo et al. 2014). For the wide range of temperature measurement, i.e., from 3 K to 300 K with high sensitivity 110 ppm (5.5 mK), boron nitride-encapsulated monolayer graphene device is used for thermal conductance in 2015 by Jesse Crossno et al. (Crossno et al. 2015).

In 2016, a reference point of water was chosen for the determination of k, and systematic error in JNT method was resolved at NIST USA (Bull 2016). At NIST

USA, a new approach was introduced for improving spectral irregularity in JNT method; electromagnetic interference (EMI) was examined by two QVNS chips, in 2016 (Pollarolo et al. 2016). In 2017, the NIM-NIST reported a relative standard uncertainty of 2.71×10^{-6} in Boltzmann constant by using purely electronic approach and contributed in CODATA 2017 value (Flowers-Jacobs et al. 2017a). Internationally, there are few groups that published their work on JNT method in the direction of Boltzmann constant realization and redefinition of kelvin summarized in Table 5.

In 2007, for the realization of T-$T_{90,}$ NIST used JNT method for the temperature value close to freezing point of zinc and obtained the uncertainty in measurement −14.24 mK. In 2010, by using QVNS, thermodynamic temperature measurements at freezing point of tin (505 K) and freezing point of zinc 693 K were taken. By using absolute JNT method, obtained values of T-T_{90} are −2.5 mK and + 6.7 mK for FP of Zn and FP of Sn, respectively, whereas from relative JNT method, the measurement of T-T_{90} was found to be +4.18 mK for FP of Zn by taking FP of Sn as reference point (Tew et al. 2010). The measurement of thermodynamic temperature using JNT was further extended for the temperature range 505 K to 800 K in 2013 after the 24th CGPM meeting, and the measurement data was compared with AGT and radiation thermometry method (Yamazawa et al. 2013).

In 2019, a review article on JNT method was publish by J.F. Qu in which authors explained the historical development of AGT, its contribution in redefinition of kelvin and thermodynamic temperature measurement (Qu et al. 2019).

JNT is purely electronic-based method, and it can be used for satellites, industrial applications, and high-accuracy thermodynamic temperature measurement in metrology. The main advantage of JNT method over other primary thermometry methods in present situation is that this method can be used for thermodynamic temperature measurement in the broadest anticipated temperature range, i.e., from 50 nK to 2473 K, and provide best uncertainty in measurement as compared to other methods (Dedyulin et al. 2022).

Table 5 List of work done for the determination of k by using JNT method

Sr. no.	Group/institute	Experimental description	Year of publication	k [$\times 10^{-23}$ J K^{-1}]	$u(k)/k$ [$\times 10^{-6}$]
1.	NIST USA et al. (Benz et al. 2011)	QVNS,100 Ω resistor, TPW cell	2011	1.38065200	12
2.	NIM China (Qu et al. 2015)	QVNS, 200 Ω, TPW cell	2015	1.38065160	3.9
3.	NIM China (Qu et al. 2017)	QVNS,100 Ω resistor, TPW cell	2017	1.38064970	2.71
4.	NIST USA (Flowers-Jacobs et al. 2017b)	QVNS, 200 Ω resistor (two 100 Ω resistors connected in series), TPW cell	2017	1.38064300	5
5.	AIST Japan (Yamada et al. 2016; Urano et al. 2017)	IQVNS, 100 Ω resistor (two 50 Ω resistors connected in series), TPW cell	2017	1.38064360	10.2

Doppler-Broadening Thermometry

DBT method provides Maxwell-Boltzmann velocity distribution along the axis of laser beam. In 2000, this method was first time purposed by Ch.J. Borde for the realization of Boltzmann constant (Dzcke 1953; Galatry 1961; Bordé 2002). This is the primary thermometry utilizing EM radiation measurements (LASERs), it depends on the Doppler shift of the recurrence of an electromagnetic wave in a moving frame of reference when contrasted with a rest frame. Doppler shift contains a linear and a quadratic part in terms of velocity of moving frame.

For atomic thermal velocities at room temperature, it is sufficient to consider only the linear dependence of Doppler shift, given as

$$\nu = \nu_0 \, (1 - \beta . K/k) \tag{18}$$

with small corrections in Doppler shift. Here, ν_0 is resonance frequency at rest.

Doppler broadening $\Delta\nu = \nu - \nu_0$, $\beta = v/c_0$, K is wave vector in laboratory frame of reference, and k is Boltzmann constant. If a laser is used to irradiate the atom in laboratory moving with velocity v, for which ν_0 is the resonance frequency for the atom at rest, the absorption will occur at a laser frequency of $\nu = \nu_0 \, (1 + \beta . K/k)$.

The thermodynamic temperature is related to Doppler-broadened absorption line as $\Delta\nu = [2kT/(mc_0^2)]$ when laser beam is propagated through an absorption cell containing an ideal gas of atoms or molecules at temperature T as shown in Fig. 10.

The main advantage of this method is that the Doppler profile and hence the Boltzmann constant can be determined by relative radiation measurements. Laser frequency can be easily controlled up to a high precision with small uncertainties (Gianfrani 2016).

For the determination of k, the experiment was performed in an ammonia line at 30 THz with a CO_2 laser. The uncertainty value in the Boltzmann constant inferred from the line width from this experiment was found to be within 3.6×10^{-5}. Since then, it was proposed to consider as an effective method to determine k with a reduced uncertainty of the level of 10^{-6}. A related joint project of universities of Naples, Milan, and INRiM was being carried out where they determined Doppler broadening of absorption line of water vapor with a diode laser in the near infrared spectral range using a high-resolution spectroscopy. The lowest uncertainty value achieved was 2.4×10^{-6}. However, it was still a tedious task to separate the effect of

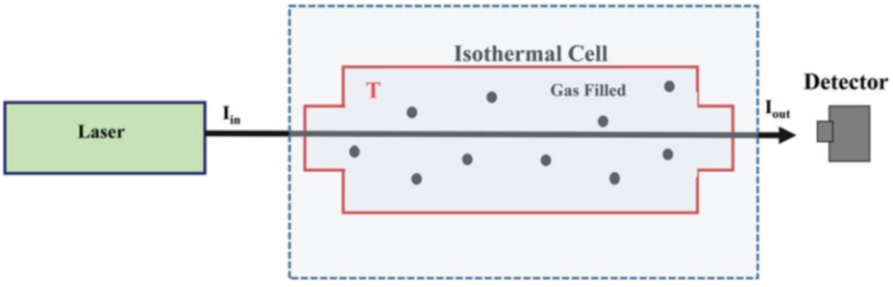

Fig. 10 Principle of Doppler-broadening thermometry method (Fischer 2019)

Table 6 List of work done for the determination of k by using DBT method

Sr. no.	Group/institute	Experimental description	Year of publication	k [$\times 10^{-23}$ J K^{-1}]	$u(k)/k$ [$\times 10^{-6}$]
1.	UMR CNRS France (Daussy et al. 2007)	CO_2 laser (8–12 μm) spectrometer, ammonia gas	2007	1.38065000	200
2.	SUN Italy (Casa et al. 2008)	CO_2 laser (single mode 2.006 μm) spectrometer, CO_2 gas	2008	1.38058000	160
3.	UMR CNRS France (Djerroud et al. 2009)	CO_2 laser (≈10 μm) spectrometer, ammonia gas	2009	1.38066900	38
4.	UMR France, LNE-Cnam France (Lemarchand et al. 2010)	CO_2 laser (8–2 μm) spectrometer, ammonia gas	2010	1.38071600	37
5.	UMR CNRS France (Lemarchand et al. 2011)	CO_2 laser (8–12 μm) spectrometer, ammonia gas	2011	1.38070400	-
6.	INRIM Italy et al. (Moretti et al. 2013)	CO_2 laser (10.34 μm) spectrometer, $H_2{}^{18}O$ molecules	2013	1.38063100	24
7.	IPAS Australia et al. (Truong et al. 2015)	Cesium and rubidium laser, multiple noble gases	2015	1.38054500	6

Doppler broadening from the other effects causing the line broadening. Hence, this method is still to reach the criteria to fulfil the condition of the determination of k with the required uncertainty.

The values of Boltzmann constant with uncertainty obtained from DBT method for the corresponding laboratories are summarized in Table 6.

At ASI Italy, a new approach in primary thermometry was used to estimate the T-T_{90} in the temperature range 9 K to 700 K, under mK. $YLiF_4$ crystal doped with erbium (Er), and CO molecules are used as the transition medium in temperature range 9 K to 100 K and 80 K to 700 K, respectively (Amato 2019). DBT thermometry has a major contribution in astrophysical observation, plasma thermometry, and gas detection (Truong et al. 2015).

Implications and Way Forward

Before the redefinition of the kelvin, the practical realization and dissemination of temperature is achieved using the practical temperature scales (ITS-90 and PLTS-2000). Post redefinition, the ways for practical realization and dissemination of thermodynamics temperature are described in the *mise en pratique* for the definition of the kelvin (MePK-19), as shown in Fig. 11 (CCT 2019; Fellmuth et al. 2016b). The practical realization of new kelvin is not linked to a specific or single preferred

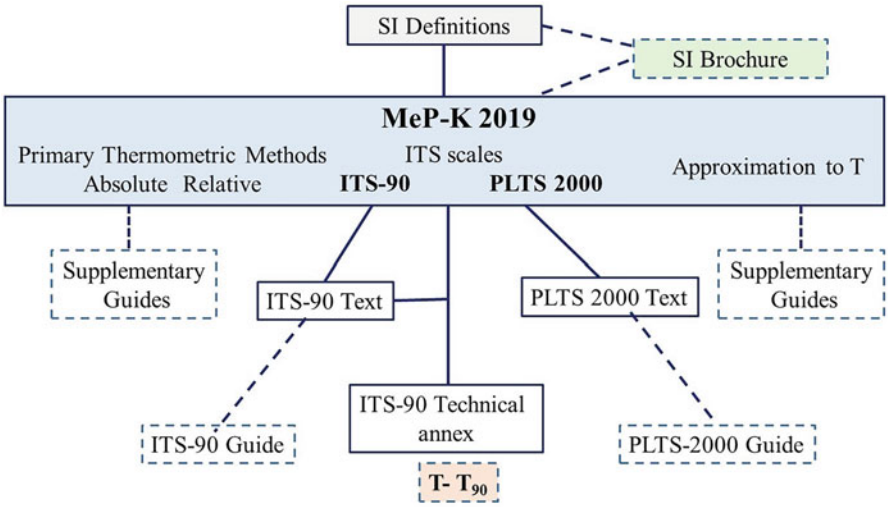

Fig. 11 MeP-K 2019 for the definition of the kelvin (Fellmuth et al. 2016b)

method, as before for TPW in ITS-90. Any method that is traceable to fundamental constant to derive thermodynamic temperature can be used for the practical realization. In this section, we describe the progress in realizing redefined kelvin and its impact on the current temperature scale. The MePK-19 describes methods that are easiest to implement and provides low uncertainty in temperature measurement.

The implications of the redefined kelvin based on Boltzmann constant are as follows:

- Before the new definition (till May 19, 2019):
  - $u_r(k) = 3.7 \times 10^{-7}$.
  - $T_{TPW} = 273.16$ K was exact without any uncertainty; $u(T_{TPW}) = 0$.
- After the new definition (from May 20, 2019).
  - Boltzmann constant is fixed as $k = 1.380649 \times 10^{-23}$ JK^{-1} without any uncertainty; $u(k) = 0$.
  - $u_r(T_{TPW}) = 3.7 \times 10^{-7}$, which makes $u(T_{TPW}) = 0.1$ mK.
- However, ITS-90 still remains as unchanged.
  - Within the use of ITS-90, $u(T_{90}(TPW)) = 0$, and $T_{90}(TPW) = 273.16$ K exactly.
- The new definition based on Boltzmann constant links the kelvin, the unit of thermodynamic temperature, to the thermal energy of the atoms or molecules of the system.
- The thermodynamic temperature can be realized and disseminated using primary thermometers based on a well-understood physical system where temperature is determined from measurements of other quantities.
- Primary thermometric methods for the measurement of thermodynamic temperature described in the MePK-19 are acoustic gas thermometry (AGT), dielectric

constant gas thermometry (DCGT), refractive-index gas thermometry (RIGT), Johnson noise thermometry (JNT), and spectral-band radiometric thermometry.
- Both absolute thermometry and relative thermometry are successfully used to measure and disseminate thermodynamic temperature, without linking to any specific fixed point.
- The redefined kelvin has no immediate effect on the current status of the ITS-90 and the PLTS-2000. However, the temperature range below 20 K and above 1300 K can be significantly benefited as the measurement of thermodynamic temperature in this can provide lower uncertainty than currently measured T_{90} and T_{2000}.
- The newly developed metal-carbon eutectic high-temperature fixed-point blackbodies can be used along with radiometers for direct thermodynamic temperature realization with improved uncertainties in a very high-temperature range above 1300 K.
- Coulomb blockade thermometry (CBT), Johnson noise thermometry (JNT), and low-temperature acoustic gas thermometry (AGT) are being employed for the realization of low temperature scale below 25 K to replace the current complex ITS-90 and PLTS-2000 temperature scale.
- Efforts are being made to improve the uncertainty and applicability of primary thermometers for the measurement and dissemination of thermodynamic temperature and better estimation of T-T_{90} (Machin 2023). Consequently, the traceability of temperature measurement will have direct linkage to redefined kelvin using primary thermometry methods, which is independent of any temperature scale.

Conclusion

The kelvin, the unit of thermodynamic temperature, is redefined in terms of the Boltzmann constant in Nov 2018 and implemented from May 20, 2019. The primary thermometry methods involved to determine Boltzmann constant with low uncertainty for redefinition are acoustic gas thermometry (AGT), dielectric constant gas thermometry (DCGT), Johnson noise thermometry (JNT), and Doppler-broadening thermometry (DBT). Further, these primary thermometric methods in combination with radiometric thermometry and present ITS-90 and PLTS-2000 scale are employed for the practical realization of redefined kelvin as described in the *mise en pratique* for the definition of the kelvin (MePK-19). The primary thermometry methods (absolute and relative) are used for the direct measurement of thermodynamic temperature as well as the dissemination of new kelvin based on Boltzmann constant.

This chapter describes the progress in the development of primary thermometers defined in MePK-19. The national metrology institutes (NMIs) across the world are developing different primary thermometers for the direct measurement and dissemination of thermodynamic temperature. Methods such as AGT, DCGT, RIGT, JNT, DBT, and radiometric thermometry and their status toward the redefined kelvin are presented. This fundamental constant quantum standard-based redefinition will give

the stability to the temperature measuring system without propagating the uncertainties, as happens with the triple point of water-based definition. New pathways have been opened for the realization of absolute scale, by using the primary thermometry methods, without any material-dependent properties like TPW and ITS-90 fixed points. Boltzmann constant is fixed as $k = 1.380649 \times 10^{-23}$ JK^{-1} without any uncertainty, u(k) =0, and the realization uncertainty of k has been transferred to TPW as $u_r(T_{TPW}) = 3.7 \times 10^{-7}$, which makes $u(T_{TPW}) = 0.1$ mK. The new definition has greater opportunities in reducing the uncertainties below 25 K and above 1200 K. Also the measurement of thermodynamic temperature (T) and estimates of T-T_{90} will bring in more thermodynamically close assignments to defined fixed points and a robust new practical temperature of ITS-20XX.

Cross-References

- ▶ Progress of Quantum Hall Research for Disseminating the Redefined SI
- ▶ Quantum Redefinition of Mass
- ▶ Realization of Candela
- ▶ Realization of the New Kilogram Based on the Planck Constant by the X-Ray Crystal Density Method
- ▶ The Mole and the New System of Units (SI)
- ▶ Time and Frequency Metrology

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Realization of Candela

12

Past, Present, and Future

Shibu Saha, Vijeta Jangra, V. K. Jaiswal, and Parag Sharma

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Abstract

The new definitions of seven SI base units, which are now based on their respective defining constants, were approved and adopted by the 26th CGPM. Candela, being one of the seven SI base units, was also redefined in terms of a defining constant, K_{cd} (maximum luminous efficacy of photopic vision). It is the only SI base unit which is derived from a physiological phenomenon of human vision. Candela has seen a long route of evolution from being defined on the basis

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of combustion candles to being now defined on the basis of a defining constant linked to a physiological phenomenon. This chapter traces the path of evolution of candela and future prospects of its definition.

Keywords

Candela · Photometry · Metrology · Quantum photometry

Introduction

Measurement is the process of quantifying some property of a phenomenon, or a physical entity, which can be expressed in terms of numbers, and assigning the number to that property. This facilitates quantitative comparison of that property. Though this sounds simple, a lot of efforts are involved in making a precise scientific measurement. This branch of science dealing with measurement is called metrology.

At present, scientific metrology is carried out under the umbrella of the General Conference on Weights and Measures (CGPM) and revolves around the seven SI base units. Out of these seven base units, candela (cd) is the one used for the measurement of light. The measurement of light is always associated with a geometrical aspect and hence, manifests as more than one parameter for measuring light for different applications, candela being one of them. Candela is the unit of luminous intensity which is the measure of the power (as perceived by the human eye) of a light source, along a particular direction, within one steradian (Roychoudhury 2014).

However, apart from the geometrical consideration, measurement of light is also associated with physiological phenomenon of vision. Thus, candela is the only base unit which gives a physical basis to a physiological phenomenon. Photometry deals with measurement of light that is perceived by the human eye. Thus, photometry is limited only in the visible range of the electromagnetic spectrum. On the other hand, when the physiological aspect of human vision is removed from consideration, the measurement of light is termed as radiometry. The radiometric counterpart of luminous intensity is radiant intensity having the unit of watt per steradian.

Candela has been a unit that has evolved with the evolution of scientific endeavor of mankind. As human beings evolved, the artificial lighting became a basic requirement. Though the requirement of light measurement came up from a field of science, industrial development propelled the standardization of light measurement that would encourage seamless exchange. During the evolution of light measurement, initially the unit and the ways of realization were different across region which are finally adopted to be known as candela. However, the method of realization of candela has kept evolving, even till recently. The evolution of light measurement is pictorially shown in Fig. 1 (Zwinkels et al. 2010; International metrology in the field of Photometry and Radiometry BIPM 2020; Johnston 2001; Resolution 6 of the 10th CGPM 1954; Resolution 12 of the 11th CGPM 1960; Resolution 5 of the 13th CGPM 1967; Resolution 3 of the 16th CGPM 1979; Resolution 1 of the 24th CGPM 2011a; Resolution 1 of the 26th CGPM 2018).

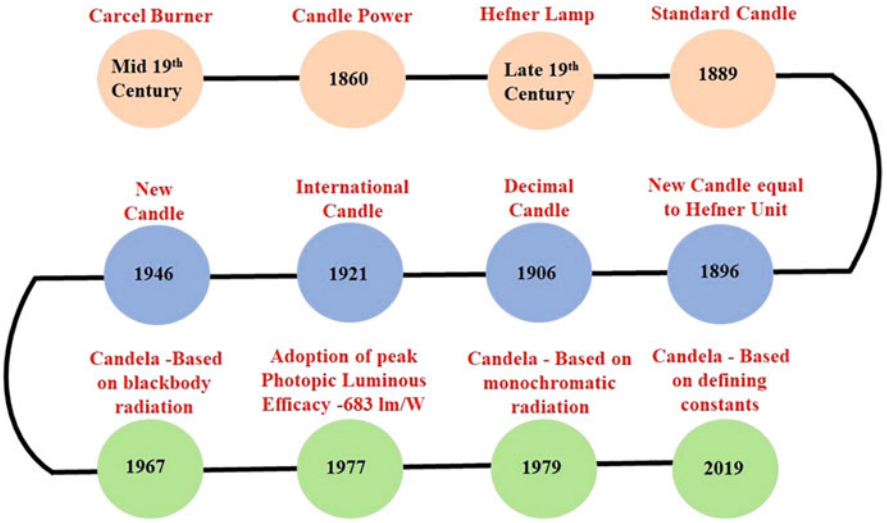

Fig. 1 Timeline of realization of candela (Zwinkels et al. 2010; International metrology in the field of Photometry and Radiometry BIPM 2020; Johnston 2001; Resolution 6 of the 10th CGPM 1954; Resolution 12 of the 11th CGPM 1960; Resolution 5 of the 13th CGPM 1967; Resolution 3 of the 16th CGPM 1979; Resolution 1 of the 24th CGPM 2011a; Resolution 1 of the 26th CGPM 2018)

A Brief History of Measurement of Light

Light has been indispensable for mankind even before the present-day societal arrangement. As the scientific knowledge and capability of the human species progressed, the artificial illumination became a basic requirement for the humanity. Though initially illumination may have been a social requirement, slowly with rising scientific queries, light became a quantity requiring measurement.

The modern strategy of quantification owes its origin to the interests of the stargazers. This prompted a gentle progress from subjective to quantitative strategies for light measurement in the late nineteenth century (Johnston 2001). The industrial requirement of measuring brightness in the gas-lighting sector became the initial driving factor for quantification of light for commercial purposes (Roychoudhury 2014). As the electric illumination industry started maturing by the end of the 1800s, requirement for quantification of illumination gained further importance. In the mid-1900s, measurements of light by scientific techniques started to take over the strategies based on visual evaluation, since by that time, the scientists had recognized that the physiological conditions strongly affect the brightness perceived by the human eye. In this period, though, quantitative measurements had become quite established in many areas of science, yet the quantification of light was still in its infancy (Johnston 2001). As the light measurements entered the commercial arena, the requirement for proper and reproducible measurements lead to the interest of the academic community in the measurements which manifested in the form of responsibilities for measurement

reliability on national laboratories, which then started establishing standards for light intensity (Johnston 2001). This period also witnessed the shaping up of professional bodies like CIE (Commission Internationale de l'éclairage or International Commission on Illumination), which in turn initiated the standardization of light measurement (ABOUT THE CIE 2022). As industrialization further progressed, both the aspects of light measurements, viz., radiometry (physical aspect of light measurement) and photometry (physiological aspect of light measurement) started to gradually rise to prominence (Johnston 2001).

In the present time, the necessity of quantification of light has not simply been confined to the illumination business. Imperative sectors like defense, pharmaceutical, aeronautics, aviation, etc. are dependent on reliable measurements of light, both radiometry and photometry. Additionally, in the era of atmospheric extremities, excessive use of energy has turned into a serious concern. The assurance of good energy efficacy of lighting is sure to assume a big role, which in turn requires reliable and accurate light measurement. Along these lines, over around 150 years, estimation of light has progressed from basic visual scrutiny to a shift toward the quantum regime (Zwinkels et al. 2010; International metrology in the field of Photometry and Radiometry BIPM 2020; Johnston 2001).

Apart from geometric consideration, as light is perceived through vision, photometric measurements gained more initial attention as compared to the radiometric measurements leading to a photometric parameter holding the position of being one of the seven base units.

Realization of Candela

Candela being the unit of luminous intensity is defined as lumen per steradian. The radiant power of electromagnetic radiation when weighed by the average human eye response gives the luminous power. Thus, candela quantifies the luminous power from a light source within a solid angle of one steradian (Roychoudhury 2014).

Fig. 2 shows a geometrical representation of derivation of luminous intensity. The ray lines in the figure depict luminous power (dP_v) being emitted by the source along a direction and within a solid angle of $d\Omega$. In this case, luminous intensity would be

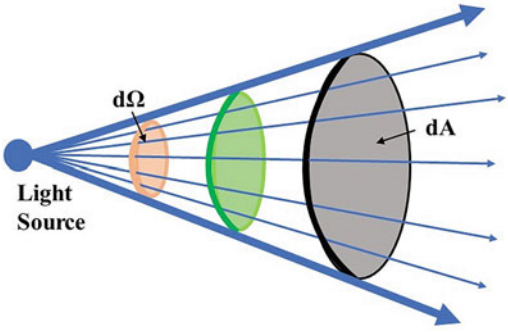

Fig. 2 Geometrical representation of derivation of luminous intensity

defined as $dP_v/d\Omega$, i.e., luminous power per unit solid angle. It can be observed from the figure that the number of lines representing luminous power remains invariant within the solid angle $d\Omega$, even on moving away from source. Thus, the geometrical aspect of the unit candela ensures its invariance over source-observer distance. Realizing this merit, metrologists are in a constant pursuit for realization of luminous intensity with lowest possible uncertainty.

The Path to Candela: Initial Realizations

During the period of late nineteenth century, luminous intensity was termed as candle power and was evaluated through comparison with candle flames. Luminous intensity was realized independently by various nations on the basis of their own respective artifacts which were the prototype candles. The British characterized their unit of candlepower, in 1860, on the basis of a prototype whale-oil candle (Treese 2018). During this period, the French too demarcated Carcel as the unit for measurement of light intensity which was based on the Carcel burner (Treese 2018). This was closely followed by the Germans in the late part of the nineteenth century as the Hefner lamp was established for defining their own unit of Hefner (Treese 2018). As compared to the present-day definition of candela, the British and the Germans were almost near to one candela; however, the French unit of Carcel was close to ten candela (Treese 2018). Nonetheless, the early day principles compromised on reliability owing to the reliance on composition of fuel and the testing conditions. Though this had provided some initial standardization, measurement uniformity across the nations could not have been seamlessly possible. In the standards based on combustion, the composition of fuel played a major role apart from the laboratory environment. Hence, its reproducibility would have been an issue that was to be addressed.

The drawback of the standards based on fuel combustion would later be overcome by a physical method. The physical standardization was proposed on the basis of emission of light from a one square centimeter area of molten platinum maintained at its melting point. The corresponding standard unit was named as Violle and in 1889, one Violle was characterized to be equivalent to twenty standard candle. However, owing to the issue of dealing with high temperature, the physical standardization based on molten platinum was dropped, though a similar principle was to be later accepted internationally. In this time, new candle was characterized equivalent to one Hefner unit (Treese 2018).

The international candle was adopted in 1909 as a mutually agreed standard by England, the USA, and Germany (Treese 2018; A Proposed International Unit of Light 1909). However, the standardization was now based on incandescent carbon-filament lamp, a transition toward an international physical standard. Nonetheless, standards based on respective lamps based on combustible fuels were still maintained by France, England, and Germany. However, adoption of the international candle based on incandescent carbon-filament lamp by the CIE, in 1921, lead to its wider international acceptance (Treese 2018).

The standardization based on filament lamp was a giant leap in the direction of defining an internationally acceptable unit of light measurement. Till this time of the early twentieth century, though a mutual international unit existed, it was still far away from the present-day definition of candela in terms of physical realization. Though the application of the incandescent lamps was a major step toward physical standardization as compared to the fuel-based combustion lamps, it had its own set of difficulties. The filament-based lamps' light output does not remain stable as the lamps age over time (Treese 2018; Greene 2003). Further, the fluctuation in the electrical inputs also contributed to the instability of the light output of the lamps. These were enough a reason for the metrologist to look for a standard that would remain stable as well as easily reproducible internationally.

Candela Based on Blackbody Radiation

The search for a source of light that would qualify to be an internationally accepted standard took the metrologist back to the point of standardizing the illumination from a body at high temperature. A perfect blackbody is such a source whose emission is the property of its temperature. For a blackbody radiator, the total power emitted and the spectra of the emission are a function of its temperature. Thus, the emission from the blackbody, being based on the physical quantity of temperature, could be standardized easily across laboratories. Planck's law establishes mathematically the spectral radiance produced by a blackbody radiator over a wide spectral range, in vacuum at a particular temperature. According to Planck's law (Zwinkels et al. 2010):

$$L_{e,\lambda}(\lambda, T) = \frac{2hc^2}{\lambda^5}\left(\frac{1}{e^{hc/\lambda kT} - 1}\right) \tag{1}$$

where the symbols represent physical quantities as follows:

h: Planck's constant.
k: Boltzmann's constant.
c: speed of light in free space.
λ: wavelength of radiation in vacuum.
T: temperature of blackbody radiator in Kelvin.

The blackbody now proved to be a more consistent candidate for defining luminous intensity as compared to its predecessors, combustion lamps, and incandescent lamps.

In this scenario by 1937, the CIE and CIPM (*Comité International des Poids et Mesures* or International Committee of Weights and Measures) prepared a new definition for luminous intensity based on the physical phenomenon of blackbody radiation. CIPM promulgated this definition, in 1946, through its resolution and thus a unit then known as "new candle" got an international acceptance which would read as (Taylor and Thompson 2008):

> "The value of the new candle is such that the brightness of the full radiator at the temperature of solidification of platinum is 60 new candles per square centimetre."

As apparent, the definition now standardized that a one square centimeter area of a blackbody radiator held at a temperature as that of freezing platinum would radiate a brightness of sixty new candles (Taylor and Thompson 2008; Parr 2000). This was a giant leap for photometry; as of now, its basis would change to be relying on a physical phenomenon instead of being based on artifacts. This paved the path for universal realization of a photometric unit leading to international acceptance. Being reliant on a physical principle, the unit took a step further on international reliability and lowering of uncertainty in measurements, in contrast to the artifact-based standardizations. In the year 1948, the apex intergovernmental body of measurement approved the definition in the ninth CGPM and adopted the present-day internationally accepted name for the unit of luminous intensity, the candela (cd) (Taylor and Thompson 2008; Resolution 7 of the 9th CGPM 1948).

Till now, though photometric unit had matured to be defined on the basis of a physical phenomenon, it was not recognized as one of the base units. In 1954, the tenth CGPM recognized candela as one of the base units in the system of units (Resolution 6 of the 10th CGPM 1954; Taylor and Thompson 2008). This system of units would later (in 1960) be coined as presently known *Syst'eme international d'unit'es* abbreviated as SI (Resolution 12 of the 11th CGPM 1960). The recognition of candela as one of the seven base units was followed by its amendment in 1967, by the thirteenth CGPM. Now the candela was defined as (Resolution 5 of the 13th CGPM 1967):

> "The Candela is the luminous intensity, in the perpendicular direction, of a surface of 1/600,000 square metre of a blackbody at the temperature of freezing platinum under a pressure of 101,325 newtons per square metre."

This definition had two main changes. The new candle (later renamed as candela) was established in terms of sixty new candle which was defined to be the radiation from one square centimeter of a blackbody held at a particular temperature. The new definition now defined one candela instead of sixty new candle, and hence, instead of defining the radiation from a one square centimeter area, the radiation from an area of one part of six hundred thousand square meters was defined. Thus, there was no change in realization of the level of brightness but the change brought the definition close to the base units. The other important change, worth a mention, was the introduction of a definite pressure. The freezing point of a substance does change with pressure. Hence, by defining the pressure, a more accurate temperature point was established by the new definition. With this, an internationally accepted candela was established which was defined in terms of law of physics.

The blackbody radiator thus overcame the predicament of instability of the filament-based lamps, and on the other hand, it also addressed the issues with reproducibility of the fuels for the combustion lamps. By now, as the spectrum of the blackbody radiation was well established; hence the concerns of visual physiological assessment of the Candela, prevailing during the photometric measurements, could be dealt with (Sperling and Kuck 2019).

The spectrum of the radiation emitted by a blackbody radiator at a particular temperature is defined by Eq. 1. The radiance of the radiation being emitted by the blackbody (in vacuum) at a particular temperature T can be calculated by integrating the LHS of Eq. 1 over wavelength and can be represented as:

$$L_e(T) = \int_0^\infty \frac{2hc^2}{\lambda^5}\left(\frac{1}{e^{hc/\lambda kT} - 1}\right)d\lambda \tag{2}$$

Eq. 2 gives the radiance of the blackbody radiator which is basically the radiant power being emitted by a unit area of the radiator into a solid angle of one steradian. If the area of the radiator is known, then the radiant intensity can be calculated by multiplying Eq. 2 with the said area. As the definition of candela defines the area of the radiator to be one part in six hundred thousand square meter and the temperature to be the freezing point of platinum at a pressure of 101,325 pascal (T_{FP}), hence the radiant intensity which would be perceived as one candela by an average human eye would be given as:

$$I_e(T = T_{FP}) = \frac{1}{600000}\int_0^\infty \frac{2hc^2}{\lambda^5}\left(\frac{1}{e^{\frac{hc}{\lambda kT_{FP}}} - 1}\right)d\lambda \tag{3}$$

and the corresponding spectral radiant intensity would be given as:

$$I_{e,\lambda}(\lambda, T_{FP}) = \frac{1}{600000}\frac{2hc^2}{\lambda^5}\left(\frac{1}{e^{\frac{hc}{\lambda kT_{FP}}} - 1}\right) \tag{4}$$

Since the blackbody radiator was accepted internationally as the standard for radiometric and photometric measurements, leading National Measurement Institutes (NMIs) like NPL, UK; NIST, USA; PTB, Germany; and VNIIOFI, Russia, established their radiation scale toward the end of the last century (Johnson et al. 1994; Harrison et al. 1998; White et al. 1995; Sperfeld et al. 1995; Sapritsky et al. 2003). In 2007, India established its variable temperature blackbody radiator (BB3200pg, 1800–3200 K, emissivity ~0.999) as the primary standard for optical radiation at CSIR-National Physical Laboratory (CSIR-NPL), India.

Defining the Response Function of Human Eye

In nature, two basic aspects of light are emission and detection. The light sources, natural or artificial, emit light, and on the other hand, detectors, natural (eye) or man-made, detect the light. On similar lines, the optical radiation standards can be source based (blackbody radiators discussed in previous section) and detector based. The detector-based standard, in contrast to blackbody radiators, should detect and quantify the radiation. However, all detectors have their own characteristic spectral responsivity. Hence, when the detectors are being used for measurement of

photometric parameter like candela, then it becomes important for the detectors to imitate the human eye response, as they detect and quantify radiation. Not only the unit of candela but all photometric parameters are quantified on the basis of the power of radiation as per the perception of the human eye.

Thus, for reliable measurement of photometric parameters, it becomes important to closely correlate measurements with vision characteristic of an average human observer. This correlation can be established only by knowing the human eye characteristics. Though we cannot perceive this in our day-to-day life, our eye does not behave in the same way to all the wavelengths of radiation even within the visible range.

With the passage of time, requirement for the standardization of the physiological aspect of vision was felt. A number of experiments involving a plethora of human subjects were carried out to reach a point where a function of the responsivity of the eye with respect to wavelength could be deduced (Schanda 2007). The response function of an average human eye is now well known and is in the form of a bell-shaped curve peaking at a wavelength of 555 nm and going to zero on either side around 380 nm and 780 nm (Fig. 3). The curve truly depicts the natural marvel of the eye, which responds best to green light (555 nm) and its vision dies down in the UV and IR regions for the human. This function came to be known as the luminous efficacy function of an average human eye, and it was adopted by the CIE in 1924 in its initial form (Rovamo et al. 1996). In other words, luminous efficacy function for the human eye depicts the inverse of the relative radiant power that would be required to produce the identical visual perception of brightness over all the wavelengths in the visible region of the electromagnetic spectrum (Schanda 2007). Luminous efficacy is basically the luminous value perceived per unit radiant power. It is important to point out that the luminous efficacy function being currently discussed represents the human eye response in well-lit conditions and hence also

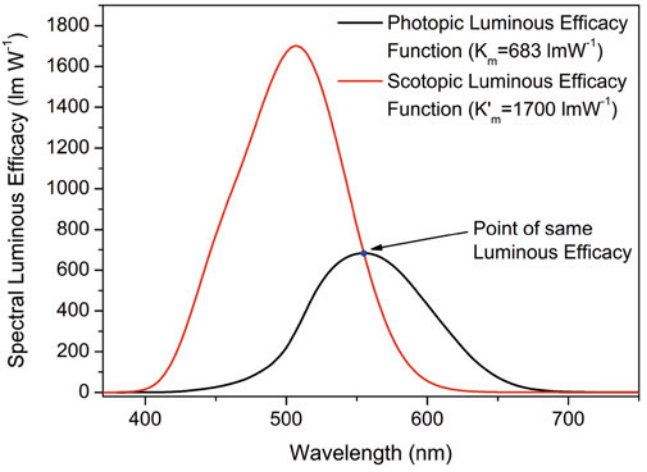

Fig. 3 Response function of a light-adapted (photopic) and dark-adapted (scotopic) average human eye (Goodman et al. 2016; Ohno et al. 2020)

known as the photopic response. This photopic luminous efficacy function, V(λ), represents the relative spectral responsivity of an average human eye in a light-adapted vision for a 2° field of view (cone vision) (Rovamo et al. 1996; International Commission on Illumination (CIE) 1983).

The luminous efficacy function being adopted by the CIE in the year 1924 was a major step for scientific measurement of light as per the perception of the average human vision. It will not be an exaggeration in stating that this laid the base of human-observer independent photometric measurements and in true sense provided a physical base to the physiological phenomenon of vision. This function was important not only for detector-based standards but also for the source-based standardization of light measurements. Without this luminous efficacy function, the photometric measurements would have always relied on human observers, which can never provide an absolute measurement with good reproducibility and would have lacked a scientific base. Thus, the V(λ) function gave a precise methodical basis of characterizing the visual impression of the electromagnetic radiation, closely correlating it with the vision of an average human eye stimulated by the radiation. This adoption played a crucial role not only in the field of photometric measurements, but also laid the foundation for the standardization of the CIE color matching functions.

As mentioned earlier, the luminous efficacy curve being discussed is for an average human eye in a bright field of vision. However, the illumination condition strongly affects the responsivity of the human eye. As we all have experienced, when we encounter sudden darkness, our eye requires some time before it can start perceiving the surroundings in the dark. This is the time when our eye adapts to the darkness or technically the responsivity of the natural detector changes. Now, since it is giving stimulus in lower radiant power, the value of luminous efficacy should increase. To provide a physical base for the human vision in the dark (rod-only vision), an efficacy function was defined which is known as the scotopic luminous efficacy function, V'(λ), that gives relative spectral responsivity of dark-adapted vision of an average human eye (International Commission on Illumination (CIE) 1983). The photopic as well as the scotopic luminous efficacy curves are characterized by a similar bell shape (Error! Reference source not found (Goodman et al. 2016; Ohno et al. 2020).). However, the peak of the scotopic luminous efficacy function can be observed to be at a shorter wavelength as compared to that for the photopic luminous efficacy curve. The disparity between the two functions is coined as Purkinje effect (Wyszecki and Stiles 1982) and the level of illumination between brightness (cone dominated vision) and complete darkness (rod dominated vision) is called the Purkinje illumination range or mesopic range (Zwinkels et al. 2010; Goodman et al. 2016).

The following equation is used to derive the photometric parameters by weighing the corresponding radiometric functions with this luminous efficacy function (Ohno et al. 2020):

$$X_V = K_m \int_0^\infty V(\lambda) X_e(\lambda) d\lambda \tag{5}$$

where the symbols represent physical quantities as follows:

X_V: Photometric quantity.

K_m: Maximum luminous efficacy in the photopic regime which is defined at a wavelength of 555 nm and has a value of 683 lmW^{-1}.

$X_e(\lambda)$: Spectral density of the corresponding radiometric parameter.

λ: Radiation wavelength in air.

Toward the end of the twentieth century, the growth of semiconductors made the use of solid-state detectors viable. The responsivity of the detectors now could be tuned by manipulating the material of the detectors and by using various filters. As the responsivity of the detector is tuned to the photopic luminous efficacy function ($V(\lambda)$) for a 2°-Observer, the detector would now become a photometer. A detector without $V(\lambda)$ correction measures radiometric quantity while the one having the $V(\lambda)$ correction can measure photometric quantities.

With this, one candela, now, could be derived from the radiant intensity given in Eq. 3 (obtained using Planck's formula), by weighing it with the luminous efficacy function as discussed in Eq. 5. Thus, using Eqs. 3 and 5, one candela can be defined as:

$$1cd = \frac{K_m}{600000}\int_0^\infty V(\lambda)\frac{2hc^2}{\lambda^5}\left(\frac{1}{e^{hc/\lambda kT_{FP}}-1}\right)d\lambda \tag{6}$$

On the other hand, the maximum photopic luminous efficacy (K_m) may also be determined as the ratio of the photometric quantity of one candela to the integral involving the radiometric quantity. Thus K_m can be mathematically derived using Eq. 6 and expressed as:

$$K_m = \frac{1}{\frac{1}{600000}\cdot\int_0^\infty V(\lambda)\frac{2hc^2}{\lambda^5}\left(\frac{1}{e^{hc/\lambda kT_{FP-1}}}\right)d\lambda}\left[\frac{cd}{m^2\cdot\frac{W}{sr.m^2}}\right] \tag{7}$$

The equation also leads to the value of K_m for photopic vision to be nearly 683 lm/W (Sperling and Kuck 2019). This maximum luminous efficacy is achieved at the wavelength of 555 nm corresponding to the peak of the bell-shaped curve (Fig. 3). In 1977, CIE adopted the value of the peak luminous efficacy of monochromatic radiation of a frequency of 540×10^{12} Hz (wavelength of 555 nm) as 683 lm/W (Resolution 3 of the 16th CGPM 1979).

Candela Based on Human Vision

The mid-1940s witnessed the transition of defining candela using Planckian radiators at a particular temperature and phasing out of the artifact lamps. Till the late 1970s, owing to the definition being based on the Planckian radiator, candela remained to be dependent on the International Practical Temperature Scale (IPTS)

as the realization of the photometric quantity was defined at the freezing point of platinum. However, as time progressed, the IPTS also witnessed its upgradation. During the decade of the 1960s, the freezing point of platinum changed based on the IPTS-68 (Saha et al. 2020). Thus the dependence of candela on the IPTS led to a direct variation in the scale of photometry also. On the other hand, the other disadvantages of the blackbody radiator (as the standard for photometric quantity), especially its challenging and expensive maintenance, were starting to be highlighted (Taylor and Thompson 2008; Parr 2000). Further, excessive divergence in the results for the realization of primary standards initiated a search for an alternate standard, apart from the blackbody (Resolution 3 of the 16th CGPM 1979).

The adoption of the standard value of the peak photopic luminous efficacy by the CIE in 1977 paved the way for a new definition of candela which would be independent of temperature. In 1979, the 16th CGPM adopted the new definition of candela as (Resolution 3 of the 16th CGPM 1979):

> The Candela is the luminous intensity, in a given direction, of a source that emits monochromatic radiation of frequency 540×10^{12} hertz and that has a radiant intensity in that direction of 1/683 watt per steradian.

The new definition did not depend on any temperature. Further, instead of relying on a broadband source like a blackbody, the new realization of candela was based on a monochromatic radiation. The 540 THz frequency is not only the frequency where the photopic luminous efficacy shows a peak, but a close observation of Fig. 3 would reveal that both the photopic and scotopic luminous efficacy curves intersect at that wavelength of 555 nm (corresponding to 540 THz) and both have the same luminous efficacy there (Fig. 3). It would be interesting to point out here that apart from the photopic and scotopic luminous efficacy curves, all those in the Purkinje illumination range also pass through the same 540 THz frequency and has the same luminous efficacy at the said point (Saha et al. 2020). This shows that the response of an average human eye remains independent of illumination for a radiation of frequency 540 THz. This choice of frequency for defining candela proved to be beneficial as the originally defined K_m value of 683 lm/W could be preserved, which was adopted after a large number of measurements on the blackbody radiator through tedious international measurement campaign (Sperling and Kuck 2019).

The choice of the wavelength and the value of K_m acted as a bridge between the new and old definitions. Hence, on the one hand, it paved the avenue for establishment of candela through the detector-based measurement of optical power, and on the other hand, even with the introduction of the new definition, the standards based on blackbody radiators were not absolutely obsolete. Further, the luminous intensity at different wavelengths could now also be relatively established as the luminous efficacy curve was adopted by the CIE (Parr 2000).

The definition of 1979 was a point wherein the detector-based realization of candela entered the arena and proved to be advantageous over the source-based standard. Though the solid-state detectors now seem to be an easy way for measurement of candela, its realization based on a physical quantity was enabled by the development and advancement of the electrical substitution radiometers. Since light

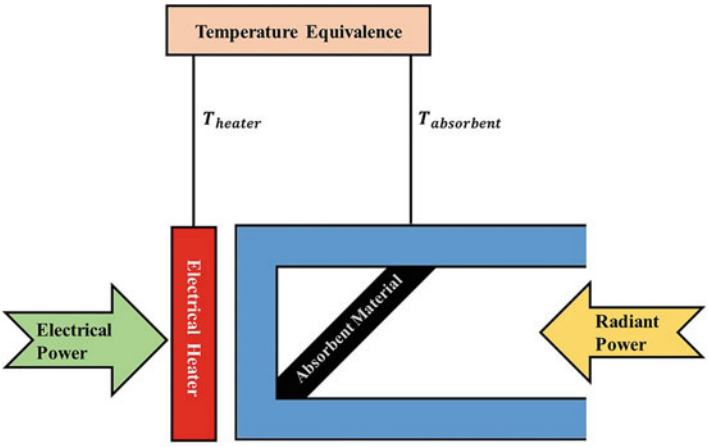

Fig. 4 Schematic of absolute radiometer

is a form of energy, hence an incident radiation, when absorbed by a body, leads to a rise in temperature of the absorbent. The electrical substitution radiometer exploits this principle and quantifies the incident radiant power by comparing it with an electrical power that could lead to the same rise in temperature (Fig. 4). The equivalence of the optical power and electrical power is established through the heating caused by the respective powers. This principle provides an edge to the electrical substitution radiometers, which relies on equivalence of a physical parameter, over the solid-state detectors as they rely on the radiometric response of the material. The electrical substitution radiometers are also known as absolute radiometers as the radiant power is derived in reference to physical laws and the measurements are completely independent of any comparison with other radiant power. As the measurements now depend on the principle of equivalence of temperature caused by two different types of power (radiant and electrical), a low background temperature proves to be helpful in increasing the sensitivity and hence reducing measurement uncertainties. For this reason, the absolute radiometry is done in the cryogenic temperature region which gives the name cryogenic radiometer.

NMIs across the globe initiated and worked to develop and establish cryogenic radiometer for establishing candela during the 1970s to 1980s (Parr 2000; Ohno et al. 2013). The absolute radiometer at room temperature was established at CSIR-NPL, India, in 1988, which was followed by the establishment of the cryogenic radiometer scale in the year 2012 (Zwinkels et al. 2010).

Candela and Defining Constant, *Kcd*

As science and technology progressed into the twenty-first century, the demand of higher accuracy and lower uncertainty emerged. During this time, the scientific measurements and their scope of improvement were strongly pursued at BIPM (*Bureau International des Poids et Mesures* or International Bureau of Weights

and Measures) and NMIs around the world. As the values of fundamental constants used in physics (e.g., speed of light, Planck's constant, etc.) had achieved fairly accurate values, the stage had arrived when the SI base units could be established in terms of the defining constants (Resolution 1 of the 24th CGPM 2011a). Bearing in mind the necessity of stable and self-consistent SI base units that would be practically realizable and be based on fundamental physical constants, CGPM initiated the formal transition of SI units from the beginning of the second decade of twenty-first century (Resolution 1 of the 24th CGPM 2011b). The metrology community agreed to give exact values to the defining constants (without any uncertainty) which would lead to further reduction of measurement uncertainty (Resolution 1 of the 24th CGPM 2011a). Now the definitions of SI units could be independent of processes and artifacts. Thus, in 2018 the SI units were redefined wherein the definitions of all the seven base units were established on the basis of seven defining constants, which were being used as fundamental constants of nature (Resolution 1 of the 26th CGPM 2018). As the new SI units based on fundamental constants of physics were adopted by the 26th CGPM, the luminous efficacy of 540 THz monochromatic radiation was coined as K_{cd}, having a value of 683 lm/W. The definition adopted by the 26th CGPM and implemented from 20th May 2019 may be read as (Resolution 1 of the 26th CGPM 2018; The International System of Units (SI) 2022):

> "The Candela, symbol cd, is the SI unit of luminous intensity in a given direction. It is defined by taking the fixed numerical value of the luminous efficacy of monochromatic radiation of frequency 540×10^{12} Hz, K_{cd}, to be 683 when expressed in the unit lm W^{-1}, which is equal to cd sr W^{-1}, or cd sr $kg^{-1}m^{-2}$ s^3, where the kilogram, metre and second are defined in terms of h, c and $\Delta\nu_{Cs}$."

The new definition establishes Candela on the basis of the constant, K_{cd}, with the unit lm/W, wherein the value of the constant K_{cd} is equal to 683 lmW^{-1} or 683 cd sr $kg^{-1}m^{-2}$ s^3 for 540 THz monochromatic radiation. Thus.

$$1\ cd = \frac{K_{cd}}{683} kg\ m^2 s^{-3} sr^{-1} \tag{8}$$

As the candela depends on other base units, it follows (Saha et al. 2020):

$$1cd = \frac{1}{\left(6.626\ 070\ 15 \times 10^{-34}\right)\left(9\ 192\ 631\ 770\right)^2 683} \left(\Delta v_{Cs}\right)^2 h K_{cd} \tag{9}$$

From Eq. 9, the dependence of the definition of candela on Planck's constant and the ground state hyperfine splitting frequency of the cesium 133 atoms, apart from the K_{cd}, can be inferred (The International System of Units (SI) 2022). Moreover, the measurements in electrical substitution radiometry are based on electrical measurements. Hence, the definition of candela would be prone to variations in the definition of electrical units. However, in the present scenario when the accuracy in the realization of candela is limited by uncertainties of the order of 100 ppm, any practical change in the strategies for realization of photometric and radiometric quantities is hardly expected (Sperling and Kuck 2019; Yadav and Aswal 2020).

Toward Quantum Candela

Photometry has progressed immensely especially through the twentieth century. It has seen the evolution of standards from very basic combustion candles to sophisticated systems like blackbody radiator and cryogenic radiometer. The dawn of the twenty-first century saw the redefinition of candela to be based on K_{cd}. However, the scientific community is still active to reach a more fundamental definition of candela, which would be based on photon. The "quantum candela" is expected to be the future of photometry and would establish its traceability to SI through the quantum regime of the electromagnetic radiation (Mise en pratique for the definition of the candela and associated derived units for photometric and radiometric quantities in the SI 2019). The quantum candela would be established in terms of photon number of known wavelength, thus depending on the particle nature of electromagnetic wave. This has led toward evaluation of the photometric, radiometric, and spectral radiant quantities with reference to photon counts (Sharma et al. 2009; Cheung et al. 2007).

The counts of photon can be related to the radiation power as

$$P = nh\nu = n\frac{hc}{\lambda} \tag{10}$$

where the symbols denote as follows:

P: radiometric power.

n: Number of photons per second.

h: Planck's constant.

ν: Frequency of radiation.

Thus, the dependence of the new quantum candela on the value of Planck's constant, speed of light, and wavelength (or frequency) of radiation leads to its uncertainty to be governed by the uncertainties in the wavelength (or frequency) determination and hence, in turn on the realization of the scale of time, i.e., one second (Cheung et al. 2007). However, since the values of h and c being defining constants have been accepted to be without any uncertainty, the measurement of radiometric power would be having uncertainties that would be contributed by the measurement strategies of the number of photon and on the other hand the wavelength (or frequency) of the light. It is important to point out that the accuracy of frequency measurements limits the precision with which the rate of photon can be measured. At present, the uncertainty in frequency (rate of photon) measurement is 10^{-16}, while the uncertainty in wavelength measurement is 10^{-12} (Sperling and Kuck 2019). Hence, in the current scenario, the measurement of radiometric power cannot be expected to have an uncertainty better than 10^{-12}.

The spectral radiometric parameters would be required to be derived from the corresponding quantities based on photon counts. The equation thus would be (Sperling and Kuck 2019; Zwinkels et al. 2016):

$$X_{e,\lambda}(\lambda) = \frac{hc}{\lambda} n_a(\lambda) X_{p,\lambda}(\lambda) \tag{11}$$

where the symbols represent physical quantities as follows:

$X_{e,\lambda}(\lambda)$: Spectral radiometric parameter.

$X_{p,\lambda}(\lambda)$: Corresponding parameter based on photon number.

$n_a(\lambda)$: Refractive index of air.

The photometric parameters were derived from the radiometric parameter through Eq. 5. In similar lines, for light-adapted vision, a photometric parameter (X_V) would be related to its corresponding quantity defined on the basis of photon number (X_p) by the equation (Sperling and Kuck 2019):

$$X_V = K_m hc \int_0^\infty X_{p,\lambda}(\lambda) \frac{n_a(\lambda)}{\lambda} V(\lambda) d\lambda \tag{12}$$

At present, one candela is defined as that luminous intensity which is perceived by an average human eye when exposed to a radiation of 540×10^{12} Hz frequency having a radiant intensity of 1/683 watt per steradian. Hence, putting the values of the constants in the Eq. 12 gives the number of photons which would be required, from a 540 THz source, every second in a solid angle of one steradian for establishing one candela and it comes out to be 4.092×10^{15} photons/second/steradian (Cheung et al. 2007). Thus defining one candela on the basis of photon number would require a source of photon and a device to count it for emitting and counting photons at a rate of the order of 10^{15} photons per second. Thus, for radiometry and photometry deriving traceability from photon counting, emitters with precise control on the number of photons would be required and simultaneously, devices for accurate counting of the rate of incoming photon would also be necessary. A source, of definite wavelength, which is capable of emitting a definite number of photons which are temporally well separated would be replacing the blackbody and absolute cryogenic radiometer as the standard source when the transition to photon number-based radiometry and photometry becomes a reality. Parallelly, counting system would be necessitated for establishing the single-photon sources as primary standards (CCPR Strategy Document for Rolling Development Programme 2017).

A single-photon source, which emits photons at symmetrically spaced temporal intervals, is usually excited for emission through its triggering by a periodic optical pulse or electronic signal (Vaigu et al. 2017). Fundamental research for absolute source of photon has already gained momentum. Some of the popular methods for achieving definite photon flux are by using spontaneous parametric down conversion (SPDC), quantum dots, attenuated laser beam, and surface acoustic wave (SAW) (Cheung et al. 2007). The method of laser attenuation, which would use multiple attenuators that are calibrated over a multi-order magnitude, can prove to be useful for obtaining multi-photon emission and hence establish a steady photon flux required (Eisaman et al. 2011; Gerrits et al. 2020). In principle, a two-level system that is capable of producing a spontaneous emission can act as a single photon emitter. Such emissions over a wide temperature range (initially cryogenic temperatures to even ambient conditions) have been demonstrated in the solid-state regime using quantum structures (Senellart et al. 2017). The report of acousto-electronic single photon source in the year 2000 (Foden et al. 2000) has been followed by

reports wherein photon emission based on the principle of SAW has been reported, as a lone electron transports in SAW potential through a n-i-p junction and then sees recombination with holes (Hsiao et al. 2020). Apart from the solid-state techniques, single photon source is also realized by the phenomenon of SPDC where photons are produced in pair and are of probabilistic nature (Eisaman et al. 2011). SPDC produces photons, that are correlated and of lower frequency as compared to the parent beam, through interaction of high energy photon with a non-linear medium.

Sourcing of single photon is not, the only important aspect of photon-based realization of photometric and radiometric parameter. Photon counting by single-photon detectors would also be required to be mature before candela can be redefined based on photon counting. In principle, a single-photon detector or a photon counter is based on its ability to detect faint optical pulse (Chunnilall et al. 2014). This incoming optical pulse is efficiently converted into a suitable electronic signal for detection. The "efficiently converted" phrase is of paramount importance in the preceding line. Detection of short-lived faint optical pulse is required for an efficient photon counter. Largely a photon counter has two parts: conversion of the incoming photon to electronic signal and the electronics for using the signal. Currently, both of these aspects enjoy the attention of the scientific community as efforts are being made to achieve a reliable photon counter that may support the definition of quantum candela (Eisaman et al. 2011; Kapri et al. 2021; Kapri et al. 2020).

The realization of photon-based candela may not be that lucid as it sounds from the text and current literature. The practical establishment of quantum candela requires emission and detection of photons in the order of 10^{15} photons per second. Efforts are still required to achieve a reliable system for emitting and detecting photons at this rate (Zwinkels et al. 2010; Sperling and Kuck 2019). Apart from a solution of emission and detection, another important aspect is the determination of pulse frequency as photon counts are deduced from counting of excitation pulses (Sperling and Kuck 2019). However, the NMIs, academia, and the scientific community are relentlessly working toward the realization of quantum candela (Chunnilall et al. 2014; O'Brien et al. 2009; Rodiek et al. 2017).

Candela at Various NMIs

Many NMIs across the globe have realized candela in different ways which is available on the BIPM Key Comparison Database (BIPM-KCDB) website (few are tabulated in Table 1) (BIPM 2022). The measurement capability span of the photometric quantity is currently quite broad among the NMIs, wherein the list of Calibration and Measurement Capabilities (CMCs) at the KCDB database shows that the candela is being realized from as low as 1 m cd to as high as 1000 k cd. Though the traceability is taken from a primary standard but for establishing measurement capabilities for dissemination, NMIs can be seen to prefer standard lamps and photodetectors for establishing candela. There can be a number of ways for establishing traceability to any photometric or radiometric unit from a primary standard. Here we present a typical way of establishing candela with traceability from source- and detector-based primary standards (Parr 2000; Ohno et al. 2013).

Table 1 "Candela" across various NMIs (sorted w.r.t. country code from selected data, downloaded from BIPM-KCDB website) (BIPM 2022)

NMI/institute	Quantity	Instrument/ artifact	Instrument type/method	Measurand		Expanded uncertainty at coverage factor of $k = 2$	
				Min value	Max value	Min. value	Max value
INTI, Argentina	Luminous intensity	Tungsten lamp	Reference photometer and photometer bench	50	1500	0.8	0.8
BEV, Austria	Luminous intensity	Tungsten lamp	Substitution on photometric bench	1	10,000	1.4	1.4
NMI, Australia	Luminous intensity	Tungsten lamp	Reference photometer	1	1500	0.6	0.6
BIM, Bulgaria	Luminous intensity	Tungsten lamp	Standard lamps, substitution on photometric bench	10	1000	1.5	1.5
INMETRO, Brazil	Luminous intensity	Tungsten lamp	Photometric bench and reference lamp/photometer	10	500	2	2
BelGIM, Belarus	Luminous intensity	Tungsten lamp	Photometric bench and reference lamps/photometer	5	35	1.5	0.8
BelGIM, Belarus	Luminous intensity	Tungsten lamp	Photometric bench and reference lamps/photometer	35	500	0.8	0.8
BelGIM, Belarus	Luminous intensity	Tungsten lamp	Photometric bench and reference lamps/photometer	500	1500	0.8	1.5
NRC, Canada	Luminous intensity	Tungsten lamp	Photometric bench and standard lamps/photometer	15	500	1.1	1.1
METAS, Switzerland	Averaged luminous intensity	LED	Measurement with reference photometer	0.1	100	2	2

NIM, China	Luminous intensity	Tungsten lamp	Standard lamps and photometer, equal distance method	1	5000	0.5	0.5
NIM, China	Averaged luminous intensity	LED	Standard lamps and photometer, photometric bench	0.1	100	1.8	1.8
CMI, Czechia	Luminous intensity	Tungsten lamp	Photometric bench and reference lamps or photometer	1	10,000	0.9	0.8
CMI, Czechia	Luminous intensity	Tungsten lamp	Photometric bench and reference lamps or photometer	1	10,000	0.8	0.9
PTB, Germany	Luminous intensity	Tungsten lamp	Network of lamps and photometers, photometric bench	0.001	1	1.5	0.4
PTB, Germany	Luminous intensity	Tungsten lamp	Network of lamps and photometers, photometric bench	1	1000	0.4	0.4
PTB, Germany	Luminous intensity	Tungsten lamp	Network of lamps and photometers, photometric bench	1000	100,000	0.4	1.5
PTB, Germany	Averaged luminous intensity	LED	Photometric bench	0.01	1000	1.9	1.2
IO-CSIC, Spain	Luminous intensity	Tungsten lamp	Photometric bench and reference photometer	1	10,000	0.8	0.8
MIKES-Aalto, Finland	Luminous intensity	Tungsten lamp	Reference photometer	1	10,000	0.5	0.5
LNE-LCM/Cnam, France	Luminous intensity	Tungsten lamp	Reference photometer	1	10,000	1	1
NPL, UK	Luminous intensity	Tungsten lamp	Photometric bench and reference lamps/photometer	1	10,000	0.4	0.5
BFKH, Hungary	Luminous intensity	Tungsten lamp	Reference photometer	1	3000	0.9	0.9

(continued)

Table 1 (continued)

NMI/institute	Quantity	Instrument/ artifact	Instrument type/method	Measurand		Expanded uncertainty at coverage factor of $k = 2$	
				Min value	Max value	Min. value	Max value
NPLI, India	Luminous intensity	Tungsten lamp	Reference lamps and photometers	1.00	100	1.6	1.4
NPLI, India	Luminous intensity	Tungsten lamp	Reference lamps and photometers	100.00	1000	1.4	1.4
INRIM, Italy	Luminous intensity	Tungsten lamp	Reference lamps and photometer	1	10,000	0.9	0.9
NMIJ AIST, Japan	Luminous intensity	Tungsten lamp	Network of lamps, photometer and photometric bench	10	3000	0.64	0.64
KRISS, republic of Korea	Averaged luminous intensity	LED	Illuminance meter	0.1	100	1.4	1.4
KRISS, republic of Korea	Luminous intensity	Tungsten lamp	Illuminance meter	1	10,000	1.1	1.1
KazStandard, Kazakhstan	Luminous intensity	Tungsten lamp	Photometric bench and reference lamps/photometer	5	35	1.6	0.8
KazStandard, Kazakhstan	Luminous intensity	Tungsten lamp	Photometric bench and reference lamps/photometer	35	100	0.8	0.8
KazStandard, Kazakhstan	Luminous intensity	Tungsten lamp	Photometric bench and reference lamps/photometer	100	1000	0.8	1.6

CENAM, Mexico	Luminous intensity	Tungsten lamp	Photometric bench and reference photometer	1	10,000	1	1
NMIM, Malaysia	Luminous intensity	Tungsten lamp	Photometric bench and reference lamps/photometer	100	1000	0.7	0.7
VSL, Netherlands	Luminous intensity	Tungsten lamp	Distance and illuminance measurement	20	5000	2	2
VSL, Netherlands	Averaged luminous intensity	LED	Distance and illuminance measurement	0.05	1000	1.7	1.7
MSL, New Zealand	Luminous intensity	Tungsten lamp	Reference photometer	10	5000	0.8	0.8
GUM, Poland	Luminous intensity	Tungsten lamp	Photometric bench, photometer, and reference lamps	5	1000	1.5	1.5
IPQ, Portugal	Luminous intensity	Tungsten lamp	Reference photometer, reference lamps	50	1000	1.6	1.6
INM, Romania	Luminous intensity	Tungsten lamp	Photometric bench, reference photometer	10	1000	1.5	1.5
DMDM, Serbia	Luminous intensity	Tungsten lamp	Photometric bench, reference lamps/photometer	1	10,000	2	2
VNIIOFI, Russian Federation	Averaged luminous intensity	LED	Photometric bench and reference LED and photometer	0.01	10	1	1
VNIIOFI, Russian Federation	Averaged luminous intensity	LED	Photometric bench and reference LED and photometer	10	5000	1.3	1.3
VNIIOFI, Russian Federation	Averaged luminous intensity	LED	Photometric bench and reference LED and photometer	0.01	10	3	3
VNIIOFI, Russian Federation	Luminous intensity	Tungsten lamp	Photometric bench and reference lamps/photometer	1	100	0.86	0.86

(continued)

Table 1 (continued)

NMI/institute	Quantity	Instrument/ artifact	Instrument type/method	Measurand		Expanded uncertainty at coverage factor of $k = 2$	
				Min value	Max value	Min. value	Max value
VNIIOFI, Russian Federation	Luminous intensity	Tungsten lamp	Photometric bench and reference lamps/photometer	100	1000	0.86	0.86
VNIIOFI, Russian Federation	Luminous intensity	Tungsten lamp	Photometric bench and reference lamps/photometer	1000	10,000	0.86	0.86
VNIIOFI, Russian Federation	Luminous intensity	Tungsten lamp	Photometric bench and reference lamps/photometer	0.001	1	1.5	0.86
RISE	Luminous intensity	Tungsten lamp	Substitution on optical bench	1	10,000	1.2	1.2
RISE, Sweden	Luminous intensity	Tungsten lamp	Measurement of illuminance on optical bench and inverse square law	0.001	0.01	3	3
RISE, Sweden	Luminous intensity	Tungsten lamp	Measurement of illuminance on optical bench and inverse square law	0.01	1	2	2
NMC, A*STAR, Singapore	Luminous intensity	Tungsten lamp	Reference lamps/photometers	10	2000	1.1	1.1
NMC, A*STAR, Singapore	Averaged luminous intensity	LED	Spectroradiometer with irradiance mode input optics/ reference illuminance meter	0.1	100	1.8	2.5
SMU, Slovakia	Luminous intensity	Tungsten lamp	Comparison to standard lamps of luminous intensity	8	3000	3	3

UME, Türkiye	Luminous intensity	Tungsten lamp	Photometer	1	10,000	1.5	1.5
CMS, Chinese Taipei	Luminous intensity	Tungsten lamp	Realization according to definition of candela	70	10,000	0.8	0.8
CMS, Chinese Taipei	Luminous intensity	Tungsten lamp	Detector-based	25	1500	1.2	1.2
CMS, Chinese Taipei	Averaged luminous intensity	LED	Photometric bench	0.01	10	1.8	1.8
NSC IM, Ukraine	Luminous intensity	Tungsten lamp	Photometric bench and reference lamps/photometer	1	1000	1.2	1.2
NIST, USA	Luminous intensity	Tungsten lamp	Photometric bench	0.01	1	1.5	0.5
NIST, USA	Luminous intensity	Tungsten lamp	Photometric bench	1	10,000	0.5	0.5
NIST, USA	Averaged luminous intensity	LED	Photometric bench	0.01	1000	3.1	3.1
NMISA, South Africa	Luminous intensity	Tungsten lamp	Realization according to definition of candela	1	10,000	1	1
NMISA, South Africa	Luminous intensity	Tungsten lamp	Inverse square law	1	10,000	2	2

(a) Traceability of Candela from Source-Based Standard (Blackbody).

The spectral radiance of the blackbody standard source can be established if the temperature of the Planckian radiator is known (using Eq. 3). The spectral radiance scale can be transferred to a spectral irradiance lamp through a spectroradiometer. If the distance (d) between the radiator source and the spectroradiometer is specifically known, and the radiating area (A_1) can be accurately determined using a precision aperture, then the spectral irradiance of the blackbody source can be deduced as:

$$E(\lambda) = \frac{L(\lambda).A_1}{D^2} \tag{13}$$

where D is a parameter having contributions from distance, area of precision aperture at the source, and the detector aperture and considers the finite size of the apertures (Parr 2000). The standard spectral irradiance lamp can be used to calibrate an illuminance meter. The calibrated illuminance meter, in turn, can be employed to determine luminous intensity of a source if the distance (d) between the two is precisely known.

$$I_v = E_v.d^2 \tag{14}$$

Thus, the scale of luminous intensity is established using a source-based standard, i.e., blackbody.

(b) Traceability of Candela from Detector-Based Standard (Cryogenic Radiometer).

The primary standard detector or the absolute cryogenic radiometer establishes the radiant power of a source. This calibrated radiant power is used to characterize a standard detector, like trap detector, from which the absolute spectral responsivity of a photometer ($s(\lambda)$), having unit of A/W, is deduced. The absolute spectral responsivity scale can then be used to deduce the responsivity of a standard photometer (R_{vf}) by using the equation:

$$R_{vf} = \frac{\int_{\lambda} P(\lambda)\, s(\lambda) d\lambda}{K_m \int_{\lambda} P(\lambda)\, V(\lambda) d\lambda} \tag{15}$$

where, P(λ) denotes the spectral power distribution of the test source. The unit of the R_{vf} is hence derived in A/lm. If the area of detector aperture (A_1) is precisely known, then the responsivity of the photometer for illuminance (R_{vE}), having unit A/lx, can be established as:

$$R_{vE} = A_1.R_{vf} \tag{16}$$

The photometer calibrated for R_{vE} can be used to measure illuminance, from which luminous intensity can be derived if the source-detector distance is known, and the distance is sufficient to assume the source as a point source. If a current (I) is generated by the photometer on being exposed by a point source at a distance d, then the luminous intensity (I_v) of the source can be given by:

$$I_v = r^2 . \left(\frac{I}{R_{vE}} \right) \tag{17}$$

Thus, the scale of luminous intensity can be established deriving traceability from an absolute cryogenic radiometer.

The principles for realization of candela though seems to be simple, but their effective realization may take years, especially in evaluation and making efforts for reduction of uncertainties in the measurements. Apart from the radiometric and photometric quantities, the dimensional parameters like distance and area contribute toward uncertainty at each step. Current supplied to the lamps as well as the measurement of current from a photometer or a detector also plays a role in the uncertainty calculations. Further, the color correction factor, alignment, and parameters of optical components like spectrometer and filters play a key role (Ikonen et al. 1995). The range of uncertainties in candela realization may be anticipated from the data of KCDB database. It can be seen that the uncertainties in realization of candela currently stand in the range 0.4–8.5% for various types of lamps. Presently NPL, UK, and PTB, Germany, have the lowest uncertainty in realizing which stands at 0.4%. This is closely followed by 0.41% for NMI, Australia, and 0.5% for NIST, USA, and NIM, China. The current capability for realization of candela at CSIR-NPL, India, is in the range 1–1000 cd with an uncertainty of 1.4% (100–1000 cd) and 1.5% (1–100 cd), through a set of reference lamps and photometers.

Role of CSIR-NPL, India, in Disseminating Optical Radiation Standards

India, as a country, and the world, as a whole, are looking toward reducing energy consumption and hence its carbon footprint, through a number of green initiatives. With government initiatives, our country is adapting an energy-efficient lifestyle through various programs like Star Labelling Programme and UJALA scheme. These have popularized the energy-efficient LED lights among the masses. Further, the Indian economy being in a growing trajectory and supported by initiatives like "Atma Nirbhar Bharat" and "Make in India" has resulted in India becoming a region of budding industries. These factors have led to the establishment and progress of the LED light manufacturing industries. As demands increased, the industry started to work toward designing new products for domestic as well the international market. Further, the increase in production of LED lamps and luminaires has resulted in imports of raw materials and LED chips and panels. Hence, the beginning and the end of production lines of the lighting industry required reliable photometric measurements for the benefit of the consumers as well as the producers. A reliable testing

of the LED chips is required for deterring poor-quality imports, and at the same time there remains a requirement for qualifying the worthiness of the final product for safeguarding the consumer interest. CSIR-NPL is in the process of establishment of an apex level calibration and testing facility for solid-state light (SSL) sources, particularly for LED-based lighting. Recently, C-type Goniophotometer (LMT GODS-2000) has been installed in the newly created LED Photometry Laboratory. The system primarily measures total luminous flux of a lamp in an absolute manner. The luminous flux is one of the important photometric parameters characterizing a light source. It provides the numeric quantification of the total power radiated in all directions as would be perceived by the human eye. The luminous flux is an important parameter with respect to characterization of light sources for their energy efficiency, as it is used to calculate the luminous efficacy of a lamp. The luminous efficacy, in turn, is used for star labelling of the lighting product. The installed C-type Goniophotometer is a mirror-based Goniophotometer that can measure light sources up to two meters and weighing up to 50 kg. A schematic diagram of the Goniophotometer system is shown in Fig. 5.

The other system that has been installed in the project is the Optical Radiation Test System. It would be used for the measurement of spectral power distribution, radiance, irradiance, radiation exposure, specific effective radiant ultraviolet power, illuminance, and other parameters of LED, LED modules, LED lamps, LED luminaires, fluorescent lamps, HID lamps, halogen lamps for both general lighting services, and special light source. The measurements of the system would facilitate characterization of light source in reference to photobiological safety.

Apart from the LED light industry, illumination industry in general which includes likes of auto lamp manufacturers, operation theater light industry, etc. also require reliable photometric assessment of their products. Moreover, the current pandemic situation has opened the market for home-made and imported UV germicidal irradiation (UVGI) systems, which require proper measurement and characterization for UVGI properties to ensure proper disinfection. Radiometric measurements are required for ensuring the reliability of such UVGI products (Sharma et al. 2022). The industries are supported with these calibrations and testing services through a number of public, private, and internal laboratories. For reliability of these measurements and international acceptance, it is vital to ensure that these

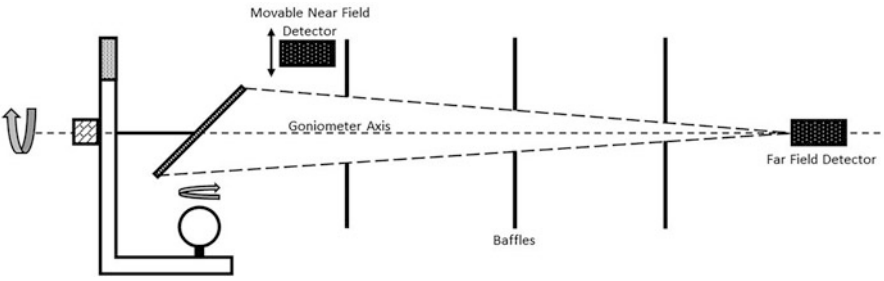

Fig. 5 Schematic diagram of C-type Goniophotometer system

Table 2 Current CMCs of CSIR-NPL, India, in the field of photometry, colorimetry, and radiometry (BIPM 2022)

Sl. No.	Parameters (bumber of CMCs)	Range	Expanded uncertainty at coverage factor of $k = 2$
1	Luminous intensity (2)	1 cd to 10^3 cd	±1.4–1.6%
2	Luminous flux (2)	1 lm to 2×10^4 lm	±1.8–2.0%
3	Illuminance (2)	1 lx to 5×10^3 lx	±1.6–2.0%
4	Luminance (3)	1 cd/m^2 to 10^4 cd/m^2	±1.6–2.0%
5	Correlated color temperature (1)	2000 K to 3000 K	±20 K
6	Spectral irradiance (3)	(wavelength range) 280 nm to 400 nm 400 nm to 800 nm 800 nm to 2500 nm	±1.9–6.0% ±1.9% ±3.3–5.2%
7	Illuminance responsivity (1)	In illuminance range 1 lx to 5×10^3 lx	±1.4%

measurements, from the testing laboratories, have their traceability to the SI units. CSIR-NPL, being mandated to be the NMI of India, plays the role of providing the traceability of SI units for the measurements being carried out by the testing and calibration laboratories.

CSIR-NPL, India, realizes, maintains, upgrades through R&D, and disseminates the standards for photometry and radiometry. The current list of CMCs of CSIR-NPL, India, in the field of photometry, colorimetry, and radiometry is summarized in Table 2. Apart from the parameters mentioned in Table 2, measurement facilities for spectral transmittance and spectral reflectance (diffuse and specular) in the wavelength range 380–780 nm with an expanded uncertainty of ±1.5% at a coverage factor of $k = 2$ are also available (Optical Radiation Standards 2022; Sharma et al. 2010).

Conclusion

Among the seven SI base units, candela is the unit of luminous intensity and is the one used for measurement of light as perceived by the human eye. This is the only SI base unit linked with a physiological perspective of human vision. Over a period spanning over almost two centuries, the measurement of light has evolved from being based on observation to now being derived from the laws of physics. As the society accepted the need for light measurement, its unit travelled also a long way, initially being defined separately by different countries as candlepower, Carcel, and Hefner, which was ultimately unified and became internationally accepted as Candela. Along with the unit, the standards were also evolving. Initially the combustion lamps were being used as standards which later gave way to carbon-filament incandescent lamps to be used as standards for light measurement. As the

technologies developed, the definitions and hence the standards were taken to be based on law of physics. The initial definition of candela to be established from the laws of physics was based on blackbody radiation, and it laid the base of source-based photometry. As the technology for detectors advanced, with the growth of the semiconductors, detector-based photometry gained importance which was quickly followed by the definition of candela being based on monochromatic radiation and the emergence of absolute cryogenic radiometer as the standard. The most recent definition of candela came with the redefinition of the seven SI base units in terms of seven defining constants. Without disturbing the essence of previous description of candela, the new definition defined candela in terms of a defining constant K_{cd}. The time is not far when the current definition would further give way for the quantum realization of candela which has shown a hope of reducing the current level of uncertainties.

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The Mole and the New System of Units (SI) 13

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Abstract

The mole (symbol: mol) – the unit of the amount of substance – was first realized and redefined with the Avogadro constant in the context of the revision of the base units of the International System of Units (SI) in 2019. The number of entities in 1 mol is equal to the numerical value of the fixed Avogadro constant: 1 mol = $6.02214076 \times 10^{23}$ specified elementary entities. Starting with a historical survey, the realization and dissemination after the revision of the SI is explained with the X-ray crystal density (XRCD) method of the Avogadro project: "counting" silicon atoms in a 1 kg Si sphere. As a key experiment, the determination of the molar mass of silicon – for this purpose highly enriched in the isotope ^{28}Si – with its successive improvements is reported. The aim of this chapter is to provide a state-of-the art and compact description of the mole, its history, changes, and the new definition. Links of the amount-of-substance to

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other relevant amount- and mass-related quantities are outlined, and examples of use for the practical scientist are given.

Keywords

International System of Units SI · Revision · Mole · Amount of substance · XRCD method · Avogadro project · Silicon

Introduction

The base unit of the amount of substance – the mole (symbol: mol) – is the latest of the seven base units, implemented in the SI, the International System of Units, in 1971, just five decades ago. However, the members of the Meter Convention agreed at that time on a definition linked to another SI unit – the kilogram. The mole was prior to its revision in 2019 linked to the mass of 12 g of the ^{12}C isotope – a certain mass of a certain entity. The term "mole" and its underlying concept have been of extreme importance for chemists and scientists as well as students, teachers, scholars, and others when dealing with chemicals and/or chemical reactions which have to be quantified either for analysis or synthesis or even just for teaching purposes. The principle of the mole enables the chemist to work with a basically infinitely large number of entities (e.g., atoms, molecules, ion, electrons) in a manageable way via the relation between a huge number of microscopic entities and the respective macroscopic portion – accessible via its mass and molar mass. The motivation of this chapter was based on the possible changes resulting in theory and practice from the new definition of the mole after the revision of the SI, which has been released on The General Conference on Weights and Measures (CGPM) at its 26th meeting (BIPM 2018; BIPM) in 2018. In this meeting, the CGPM considered the SI units should be self-consistent, provide long-term stability, and should be described on state-of-the art level physical experiments and respective laws. The SI should be defined based on seven fixed (invariable) defining constants (e.g., the Planck and the Avogadro constant) without associated uncertainties by definition. The new SI should be uniform and should support scientific as well as topics of trade, health, and others. In the new SI, effective from 20 May 2019, additionally, several defining fundamental constants are fixed. The Avogadro constant (symbol N_A) is $N_A = 6.02214076 \times 10^{23}$ mol^{-1}. This numerical value of N_A – among others – has been released in 2017 by the Task Group on Fundamental Constants (TGFC) of the Committee on Data for Science and Technology (CODATA) after a special least-squares adjustment (LSA) of the input values (Newell et al. 2018). The most prominent base unit – the kilogram – mainly triggered the revision of the SI, due to the inadequateness of congruence after comparing the International Prototype of the Kilogram (IPK) with its official copies on a regular basis over the last 100 years showing an increasing mismatch. To remove this last artifact – as an evident driving force as well as the historical attempts to harmonize the definitions of the SI units by introducing a set of invariable and fixed defining constants which

enable the realization and dissemination of the SI units by suitable laboratory experiments – was another important motivation for the revision. One of the initial ideas was given by *Max Planck*, addressing a universal system of constants, which will be discussed later in the context of the historical background and evolution of the mole (Liebisch et al. 2019). Prior to the revision of the SI, numerous institutes, especially national metrology institutes (NMIs) and the International Bureau of Weights and Measures (BIPM), were strongly involved in research and discussions concerning the new definition of this very SI base unit. One outcome was the Ad-Hoc Working Group on the Mole of the Consultative Committee for Amount of Substance: Metrology in Chemistry and Biology (CCQM), paving the way for its redefinition. The key references of the mole in the new SI are given in the SI Brochure and the *Mise en pratique* for the definition of the mole in the SI which were released by the BIPM (BIPM, CCQM; BIPM 2019a, b). After the revision of the SI which entered into force as of 19 May 2021, the revised definition of the mole is:

"The mole, symbol mol, is the SI unit of amount of substance. One mole contains exactly $6.02214076 \times 10^{23}$ elementary entities. This number is the fixed numerical value of the Avogadro constant, N_A, when expressed in the unit mol^{-1} and is called the Avogadro number.

The amount of substance, symbol n, of a system is a measure of the number of specified elementary entities. An elementary entity may be an atom, a molecule, an ion, an electron, any other particle or specified group of particles."

For the understanding of the basic concept of the mole – in the previous and in the new SI – it is necessary to give an overview on the historical background and development. This is done in the following. The ideas and pros and cons of the revision of the mole are addressed briefly, too. Since the mole is now defined via the Avogadro constant N_A, solely

$$1\,\mathrm{mol} = \left(\frac{6.02214076 \times 10^{23}}{N_A}\right) \tag{1}$$

it is now a completely independent and self-consistent SI base unit. Figure 1 shows a photograph of approximately 1 mol of pure silicon as a sphere with natural isotopic composition to visualize this SI unit in the macroscopic world. The sphere, which is used for demonstration purposes only, yields a balance reading of approximately 28 g and has a diameter of approximately 28 mm.

It is one concern of this chapter to explain the current state-of-the art procedures to realize and disseminate the mole. In this context, it must be noted that the planned revision of the SI unit kilogram gave a kind of priming for the revision of the SI in general in 2019. Finally, it was agreed that the artifact-based kilogram was defined via the fixed Planck constant h (Wood and Bettin 2019). Two experimental routes have been developed and asserted themselves: in one of them, the watt balance approach – now called Kibble balance experiment – the Planck constant h can be measured via a comparison of mechanical and electrical power with lowest associated uncertainty u, yielding finally $u_{rel}(h) < 1 \times 10^{-8}$ prior to the revision (Wood and

Fig. 1 A "mole sphere" made of 1 mol silicon with $m \approx 28$ g and a diameter $d \approx 28$ mm on a laboratory balance for the demonstration of the macroscopic meaning of "1 mol"

Bettin 2019; Petley et al. 1987; Gupta 2019). With this primary method at hand, the kilogram can be disseminated using the expression

$$1\,\mathrm{kg} = \left(\frac{h}{6.62607015 \times 10^{-34}}\right)\mathrm{m}^{-2} \times \mathrm{s} \tag{2}$$

The other primary method developed in parallel to realize the kilogram is the x-ray crystal density (XRCD) method. This experimental approach is used to determine the Avogadro constant N_A via "counting" the silicon atoms in a macroscopic single crystalline silicon sphere with a mass of 1 kg (Fujii et al. 2016). This approach has been applied for the first time by the former National Bureau of Standards (NBS, today: National Institute of Standards and Technology, NIST) in 1974, and since the 1980s, the XRCD method has been further developed by PTB and other institutes (Fujii et al. 2016; Deslattes et al. 1974). Initially, the XRCD method aimed at the definition of the kilogram via the Avogadro constant

$$1\,\mathrm{kg} = 10^3 \times \{N_\mathrm{A}\} \times u \tag{3}$$

with the numerical value of the Avogadro constant $\{N_A\}$ and the atomic mass unit u (with u = 1/12 $m(^{12}C)$ prior to the SI revision) (Becker 2003; Bettin et al. 2013). Equation (3) shows that the kilogram can be simply expressed via N_A, too. However, after the SI revision, the kilogram is defined via h. Fortunately, h and N_A can be converted into each other; thus two complementary primary methods were available to realize and disseminate the kilogram. The redefinition was introduced when both methods yielded results for the relevant constants in agreement with each other within a relative measurement uncertainty of 2×10^{-8}, which was one prerequisite of the redefinition of the kilogram and thus of the revision of the SI in 2019 (Stenger and Göbel 2012; Stock et al. 2019).

Consequently, the XRCD method is a primary method for the realization and dissemination of the mole, symbol mol, the SI base unit of the amount-of substance, symbol n, because it yields smallest uncertainties associated with n, it is completely understood, a complete uncertainty budget can be prepared, and an absolute value without the need of any reference standard is needed. Other names for the XRCD method which can be found in the literature are "Avogadro Project" and "Silicon Route" (Becker and Bettin 2011; Andreas et al. 2011; Becker et al. 2003). An overview of the principles of the XRCD method with special focus on the determination of the isotopic composition of silicon is given later. In the "mise en pratique" for the definition of the mole, it is stated that after the revision, any experimental method can be used to realize and disseminate the mole as long as it is traceable to the fixed defining constant (BIPM 2019b). Beside the XRCD method, some more practical methods are presented as well, e.g., the gravimetric approach which is still necessary in the daily life of laboratories using the relation between the macroscopic mass of an entity, its molar mass M, which has to be corrected for by the molar mass constant M_u, and the resulting amount of substance n. Other areas like electrochemistry derive n via the ratio of charge over charge number in electrolysis. The amount of substance is inversely proportional to the fixed Faraday constant times that ratio. Also, equations of state for gases can yield n having information about the pressure/volume and temperature relations. In all methods, the knowledge of the purity of the samples (entities) is one of the main limiting uncertainty contributions. In this chapter, also some typical relations between the amount of substance, the Avogadro constant, and other quantities and constants are reported with special emphasis to any changes prior and after the revision of the SI. One outcome of the revision of the SI is a clear distinction between the quantities amount of substance n and mass m establishing a respective unique feature, because prior to the revision n was dependent from m per definition. Also, after the revision in 2019, the practical measurement results of n are numerically the same as prior to the revision. In the chapter, the need of the redefinition and related changes are highlighted.

Additionally, a reference of the use of the mole and implications after the SI revision also for practical scientists may be used as an amendment for students' textbooks. Meanwhile, several scientific articles and reviews have been published concerning the revision of the mole. Here, only a short selection is cited and references therein are referred to Milton and Quinn (2001), Milton and Mills

(2009), Milton (2011), Marquardt et al. (2017), Güttler et al. (2017, 2019, 2019), Meija (2017), Brown et al. (2021), and Mills et al. (2006).

Brief Historical Survey of the Mole

The historical development of the term "mole" and its later introduction as the seventh SI base unit as well as its underlying quantity "amount of substance" is described in detail in several recent articles and reviews (Milton and Mills 2009; Güttler et al. 2019, 2019). The interested reader is also referred to the article of L. Cerrutti, who gave a comprehensive historical overview on the development and relations of the mole, the Avogadro constant, and the persons who discovered the respective quantities and constants with a variety of original cited research works (Cerrutti 1994). In this context, another comprehensive review article should be mentioned: A detailed review concerning the history, its development, and tools for the determination of the Avogadro constant has been published by P. Becker, initially demonstrating the purpose of the determination of N_A in the interrelation of the revision of the SI (Becker 2001).

Briefly, the concept of the mole started at least with the considerations of the British chemist J. Dalton (1808), who stated that atoms of one element are equal and have a special atomic mass and volume (Dalton 1808). This is in accordance with his "law of multiple proportions" (1803) which states that the combination of elements occurs in a ratio of (small) integers. At that time, there was no possibility for an exact experimental verification of these pioneering ideas which gave them a more axiomatic character. These findings were supported by the Italian scientist A. Avogadro, who published 1811 his hypothesis that presuming constant temperature and pressure, all gases of the same volume exhibit the same number of particles or molecules (Smith 1911). In the first half of the nineteenth century, these pathbreaking ideas almost felt into oblivion. Using the concepts of the atomic weight, in 1858 the Italian chemist S. Cannizzaro set up a list of chemical formulas and relative atomic masses (atomic weights) for all elements known at that time, which establishes the ideas and terms related to atoms, molecules, and atomic weights and later on pushed the development of the periodic table of the elements (Becker 2001). The basic concepts of atomic weights introduced by Cannizzaro gained a widespread attention and later on acceptance at least on the occasion of an international Chemistry meeting in Karlsruhe in 1860, mainly concerned with atomic and molecular theoretical concepts (Cerrutti 1994). The use and application of the atomic and molecular theory accelerates in the late nineteenth century. The concept of the "mole" has been probably first mentioned and applied by the chemist W. Ostwald. In 1893, he published a "Hand- und Hilfsbuch zur Ausführung Physiko-Chemischer Messungen." There, it is written "... let's generally call the weight in grams of a substance that is numerically identical to the molecular weight of that substance, one mole..." (Ostwald 1893). At that time, especially in Germany terms like "gram-molecule," "g-molecule," or "g-mole" circulated in the communities of chemistry and physics. In a publication of 1898, W. Nernst used explicitly the German term "Mol" and

"g-Molekel" as a "chemical mass unit" and in the context of the use of the equation of gases (Cerrutti 1994; Nernst 1893).

In the early twentieth century, the term "mole" already developed a twofold meaning: a description of a number and/or of a mass. A milestone was one of the first experimental determinations of the Avogadro constant by J. Perrin (1909), establishing the number of entities in 1 mol to be referred to as the Avogadro constant (Perrin 1909). Perrin used the expression "gram-molecule" at that time by the postulation that "any 2 g-molecules contain the same number of molecules (which is the number of the Avogadro constant)." Even A. Einstein used the term "gram-molecule" already in 1905 (Einstein 1905). During the next decades, these terms were used in ambiguous meanings. In 1955 U. Stille published a textbook of metrology where the use of the term "mole" was classified in a threefold manner (Stille 1955): The first use of the unit "mole" is that of a chemical mass unit of a special compound or atom (e.g., 1 mol = 28 g silicon). This is the relative atomic mass given in g. The second meaning is a number (of moles) n_{mol} according to the relation $n_{\mathrm{mol}} = N_{\mathrm{ent}}/\{N_{\mathrm{A}}\}$ with N_{ent} as the number of entities (which might be rather large) and $\{N_{\mathrm{A}}\}$ denoting the numerical value of the Avogadro constant (Avogadro number). The third application of the "mole" is the "amount of substance" which means that 1 mol is the amount of substance that contains as many entities as the relative atomic weight of, e.g., an element. Stille has introduced the term "amount of substance" for the first time, which is now commonly used as a key quantity in chemistry with the symbol n.

After progress in mass spectrometric techniques, notably the discovery of the isotopes ^{17}O and ^{18}O by W. F. Giauque and H. L. Johnston, the atomic weights were still based on a historic scale using the isotope ^{16}O as an anchor point with $A_{\mathrm{r}}(^{16}\mathrm{O}) = 16$ (Giauque and Johnston 1929). This was changed in 1959 to $A_{\mathrm{r}}(^{12}\mathrm{C}) = 12$ as a result of long and intense discussions between chemists and physicists (Guggenheim 1961). During the 1950s and 1960s, the concept of the "amount of substance" was then further immersed in the scientific communities, and its special need for chemistry was recognized. In 1971 the mole has been introduced as the seventh SI base unit at the 14th General Conference of the Meter Convention (CGPM) using the amount of substance as the underlying base quantity (CGPM, Resolution 3 of the 14th CGPM 1971; BIPM 1972; Terrien 1972). This step has been prepared by a joint recommendation of several international societies like the International Union of Pure and Applied Chemistry (IUPAC), the International Union of Pure and Applied Physics (IUPAP), and the International Organization for Standardization (ISO). The isotope ^{12}C of carbon was selected as the reference point of the scale of relative atomic weights at that time with $A_{\mathrm{r}}(^{12}\mathrm{C}) = 12$. This led to the old definition of the base unit mole prior to the revision of the SI:

"The mole is the amount of substance of a system that contains as many elementary entities as there are atoms in 0.012 kg of carbon 12. When the mole is used, the elementary entities must be specified and may be atoms, molecules, ions, electrons, or other particles, or specified groups of such particles."

As one outcome of the introduction of the mole as a base unit, in 1993 an international committee – the Consultative Committee for Amount of Substance

(CCQM) – was founded, working on the issues related to this base quantity "amount of substance" (BIPM).

Motivation for and Impact of the Redefinition of the Mole

As outlined, there existed increasing – almost accelerating – efforts for a revision of the SI during the last decades motivated and triggered mainly by the SI base unit of the mass, the kilogram, in its form of a (last) artifact, the International Prototype of the Kilogram (IPK), showing increasing discrepancies with its copies during the last 100 years. The kilogram would preferably be linked to an invariant constant of nature; thus already in 2006, ambitious efforts were undertaken to revise the SI although the prerequisites were not completely fulfilled (Mills et al. 2005). Another – not neglectable – advantage of a definition by an invariant constant of nature is the simple fact that the link to an artifact gets worse due to a stepwise sub-division of each, increasing and accumulating the measurement uncertainty. One main reason for the redefinition of the mole was to establish this SI base unit independently from any other unit as being previously from the unit of the mass – the kilogram. With the new definition of the mole, it is defined by one fixed constant of nature, a fundamental constant: the Avogadro constant N_A with no uncertainty any longer associated with. This fits the new definition of the mole into a set of invariant fundamental constants, independent from any artifact or experiment. For this reason, there is no standard or specification written in the definition about how to realize this base unit using a special experimental route. This can be chosen according to any available state-of-the art experiments when exhibiting the character of a primary method. From the point of view of the involved public – usually non-specialists in fields of metrology – especially teachers, students, and industrial scientists, who rely on the definitions and rules given by, to some extents, outdated textbooks, a more practical definition than the old one relating to 12 g of the carbon ^{12}C isotope, with the 1980 added specification that it referred to atoms in their ground state and at rest would have been desirable. In fact, in the comprehensive IUPAC Technical Report released by Marquardt et al. concerning the redefinition of the mole and related chemical quantities, several surveys were reported concerning the understanding of the definition of the mole of these user groups (Marquardt et al. 2017). Interestingly, the term *mole* is understood by most of them as a number of entities equal to the Avogadro number $\{N_A\}$. This circumstance legitimates the new definition of the mole via the Avogadro constant being closer to the common perception. A central benefit and legitimation for the introduction of the mole as an SI base unit in 1971 and more clearly after its new definition in 2019 is the fact that it presents an independent unit that is not based on artifacts such as the kilogram prototype or on any relation to specific elements or isotopes or any advice on how to realize the units. This unit is directly proportional to a number of entities which have to be defined when discussed. Usually, these elementary entities are "particles" used in chemistry like atoms, molecules, ions, and electrons. With the aid of this special quantity – the mole – scientists, especially chemists, can handle and calculate stoichiometric

relationships and interactions, independently from the mass. Extremely large numbers of these microscopic particles are present in the laboratory scale which usually has a macroscopic dimension. Thus, the Avogadro constant links the microscopic with the macroscopic range, and the mole is the SI base unit needed for this linkage. The mole-principle supported thus the understanding of chemical reactions and the use of stoichiometric relations for many, possibly most, chemists, which also justifies this change introduced by the new definition. The mole is no longer dependent on a material property as its relation to exactly 12 g of the ^{12}C isotope prior to the revision of the SI and even no longer dependent on an artifact, the kilogram via its relation to the mass. The new definition of the mole in terms of a fundamental constant has now a universal character as all the other six SI base units. It must be noted that it was important to take a few requirements into account prior to a new definition. If SI base units are defined via fundamental constants of nature, the set necessary to define them must be as small and exact as possible. It should not be overdetermined and the set of constants must not over-constrain the system. One SI base unit should be defined only by a certain numerically fixed fundamental constant without uncertainty: e.g., the second (hyperfine transition frequency $\Delta\upsilon_{cs}$ of ^{133}Cs), the metre (speed of light in vacuum c), the kilogram (Planck constant h), the ampere (elementary charge e), the kelvin (Boltzmann constant k), the candela (luminous efficacy K_{cd}), and the mole (Avogadro constant N_A). The system of units should be consistent and coherent prior and after the revision. Another central key requirement of a new definition is that the former relationships among the related quantities are untouched. However, some of them will have associated uncertainties after the new definition and have to be determined experimentally (e.g., the molar mass constant M_u, the mass of the ^{12}C isotope $m(^{12}C)$ or the molar mass of the ^{12}C isotope $M(^{12}C)$) and vice versa. As an example, the Avogadro constant N_A has a fixed number without an associated uncertainty after the revision of the SI. The following relation illustrates this change of numbers (and uncertainties) without changing the relation itself:

$$N_A = \frac{M\left(^{12}C\right)}{m\left(^{12}C\right)} \tag{4}$$

Since N_A has now a fixed value ($N_A = 6.02214076 \times 10^{23}$ mol^{-1}) without an associated uncertainty, $M(^{12}C)$ can no longer have a fixed value after the new definition of the mole. Prior to it, the exact relation $M(^{12}C) = 0.012$ kg mol^{-1} without an uncertainty was valid. This implies that the molar mass constant M_u

$$M_u = m_u \times N_A \tag{5}$$

(with the atomic mass constant m_u) which had prior to the new definition the exact value $M_u = 0.001$ kg mol^{-1} (without an associated uncertainty) must have an associated uncertainty after the new definition and therefore a variable numerical value which has to be determined experimentally, too. At present, M_u has the following value: $M_u = 0.99999999965(30) \times 10^{-3}$ kg mol^{-1} with a relative standard uncertainty of $u_{rel}(M_u) = 3 \times 10^{-10}$ (Tiesinga et al. 2021). The numbers

in parenthesis are the standard uncertainties of the last two digits. The atomic mass constant is $m_\mathrm{u} = 1.66053906660(50) \times 10^{-27}$ kg with the relation (Tiesinga et al. 2021)

$$m_\mathrm{u} = 1\,\mathrm{u} = \frac{1}{12} m\left({}^{12}\mathrm{C}\right) \tag{6}$$

Here, u is the unified atomic mass unit and $m({}^{12}\mathrm{C})$ the mass of the ${}^{12}\mathrm{C}$ isotope. Another relation is noteworthy, introducing the dalton (symbol Da with the unit kg)

$$m_\mathrm{u} = 1\,\mathrm{u} = 1\,\mathrm{Da} \tag{7}$$

These considerations yield $M({}^{12}\mathrm{C}) = 11.9999999958(36) \times 10^{-3}$ kg mol^{-1} after the revision of the SI (Tiesinga et al. 2021). Comparing these numbers with those prior to the revision of the SI, it is obvious that the numerical changes are tiny, and in most practical cases, there are no changes in the results noticeable within the limits of the obtained uncertainty. The relation

$$m_\mathrm{u} = \frac{m_e}{A_\mathrm{r}(\mathrm{e})} = \frac{2hR\infty}{A_r(\mathrm{e})\alpha^2 c} = \frac{1}{12} m\left({}^{12}\mathrm{C}\right) \tag{8}$$

shows in more detail which quantities will have an impact on the determination of $m({}^{12}\mathrm{C})$ and thus m_u: m_e is the mass of the electron, $A_\mathrm{r}(\mathrm{e})$ is the relative atomic mass of the electron, R_∞ is the Rydberg constant, α is the fine-structure constant, and c is the speed of light in vacuum. The largest impact on the uncertainty associated with m_u stems from $u_\mathrm{rel}(\alpha) = 1.5 \times 10^{-10}$ (Tiesinga et al. 2021). For comparison, $u_\mathrm{rel}(R_\infty) = 1.9 \times 10^{-12}$, and $u_\mathrm{rel}(A_\mathrm{r}(\mathrm{e})) = 2.9 \times 10^{-11}$ yield additional, but extremely small uncertainty contributions (Tiesinga et al. 2021). The "new" experimentally determined molar mass $M({}^{12}\mathrm{C})$ is obtained combining Eqs. (4) and (8)

$$M\left({}^{12}\mathrm{C}\right) = m\left({}^{12}\mathrm{C}\right) N_A = 12 \frac{2hR\infty}{A_\mathrm{r}(\mathrm{e})\alpha^2 c} N_A \tag{9}$$

Another important fact is that analogue to Eq. (6) prior to and after the revision of the SI also the relation

$$A_\mathrm{r}\left({}^{12}\mathrm{C}\right) = 12 \tag{10}$$

and

$$M\left({}^{12}\mathrm{C}\right) = A_\mathrm{r}\left({}^{12}\mathrm{C}\right) M_\mathrm{u} = 12\, M_\mathrm{u} \tag{11}$$

with the relative atomic mass A_r remain valid and exact. At this point, it is necessary that for the calculation of a molar mass $M(\mathrm{X})$ of an atom or molecule X, the general relation is applied after the redefinition

$$M(\mathrm{X}) = M_{\mathrm{x}} = A_{\mathrm{r}}(\mathrm{X})\, M_{\mathrm{u}} \tag{12}$$

For the mass $m(\mathrm{X})$ of an atom (molecule) X the analogue expression holds

$$m(\mathrm{X}) = m_{\mathrm{x}} = A_{\mathrm{r}}(\mathrm{X})\, m_{\mathrm{u}} \tag{13}$$

For practical purposes, the relative atomic masses A_{r} of the respective isotopes are released and updated from the latest atomic mass evaluation (Wang et al. 2021).

The link between the microscopic and macroscopic world (see Fig. 2) is then given by

$$\frac{h}{m_{\mathrm{u}}} = N_{\mathrm{A}}\, \frac{h}{M_{\mathrm{u}}} \tag{14}$$

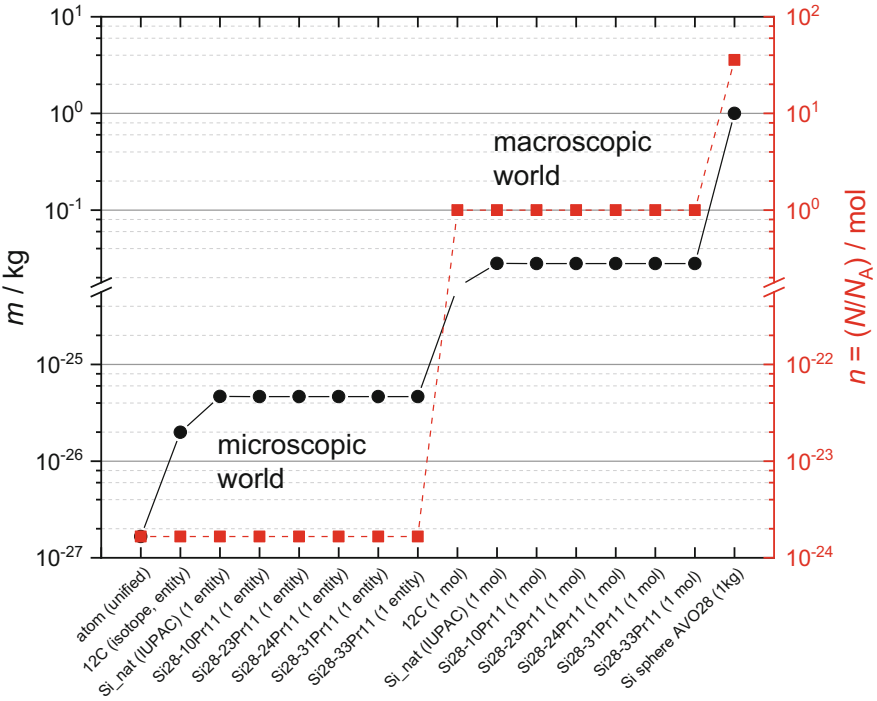

Fig. 2 Comparison of the ranges of microscopic entities and macroscopic bulk material. The mass m (in kg) is visualized vs. the amount of substance n (in mol). The first entity represents the unified atomic mass unit. The subsequent entries relate to the respective bulk single crystalline silicon material (Si_nat: silicon with natural isotopic composition; Si28-Pr11 species denote the respective silicon crystal material highly enriched in ^{28}Si). The plotted masses in the microscopic regime (left hand side) refer to a downscaled mass of the respective material according to one artificial entity of the corresponding macroscopic material. On the right-hand side (macroscopic regime), the respective species of the bulk material are shown (normalized to $n = 1$ mol). Additionally, the amount of substance of a silicon sphere of a hypothetical mass of 1 kg is shown

This relation summarizes clearly, that fixing h and N_A, the third quantity M_u or m_u must be determined experimentally, yielding a value almost equal to that prior to the revision with an associated (most likely very small) uncertainty.

Quantities and Constants Related to the Amount of Substance and Mole

Due to the revision of the SI, the mole is an SI base unit independent from any other base unit. Currently, the XRCD method provides the most exact method for the realization and dissemination of the mole. The amount of substance n of a (pure) substance X is related to its mass m via

$$n(\mathrm{X}) = \frac{m(\mathrm{X})}{A_r(\mathrm{X})\,M_u} = \frac{m(\mathrm{X})}{M(\mathrm{X})} \tag{15}$$

with the general relation

$$n(\mathrm{X}) = \frac{N(\mathrm{X})}{N_A} \tag{16}$$

The relative atomic masses A_r of an isotope of an atom (element) or a respectively composed molecule is available from the regularly updated atomic mass evaluation (AME) tables with associated relative uncertainties usually $<1 \times 10^{-10}$ (Wang et al. 2021). For A_r, the equivalent terms "molecular weight" or "atomic weight" are often used for historical reasons (IUPAC 2007). The A_r are experimentally determined mass ratios, and therefore, they are not depending on the changes of the SI. Every few years, the A_r are updated and completed due to improvements in their experimental realizations. To calculate M(X), it is, however important, to correct M(X) with the valid molar mass constant M_u (Tiesinga et al. 2021). Equations (15) and (16), respectively, compare in a simple manner the practical and theoretical access to the amount of substance n. N(X) denotes the number of elementary entities of a pure substance X in a sample; thus N can be classified as a particle number. However, counting the number of entities and yielding N is not possible using a standard experimental setup. Thus, in practice, in chemistry laboratories, Eq. (15) is applied using a balance, determining n gravimetrically – or less precise volumetrically, if the densities are available.

A practicable relation to determine the amount of substance n(X) in a chemical laboratory is

$$n(\mathrm{X}) = \frac{m(\mathrm{X})}{A_r(\mathrm{X})\,M_u} = \frac{m(\mathrm{X})}{M(\mathrm{X})} = \frac{w(\mathrm{X})m}{M(\mathrm{X})} \tag{17}$$

Having a sample with a macroscopic mass m, the mass $m(\mathrm{X})$ of the substance of interest X in this sample is yielded by the knowledge of its mass fraction $w(\mathrm{X})$. Another common and implemented problem is that a substance treated in a laboratory is never 100% pure. If the purity of a substance is known, Eq. (17) must be multiplied with a "purity coefficient" f with $f < 1$. In metrology, the quantities used must be traceable to the SI. According to Eq. (17) the amount of substance n is still traceable to the SI base unit mass. Commonly used and derived quantities of the amount of substance n are: the amount-of-substance fraction x (unit: mol/mol), the amount of substance concentration c (unit: $\mathrm{mol/m^3}$, in the older literature called "molarity"), and the amount of substance content (unit: mol/kg) (Milton and Quinn 2001; IUPAC 2007). It must be mentioned that this practical method of the gravimetrical determination of $n(\mathrm{X})$ in case of high purity substances suffers from the uncertainty of the mass determination (the balance and the weighing process). Then, $u_{\mathrm{rel}}(n(\mathrm{X}))$ is often $>1 \times 10^{-5}$. In summary, the mole as an SI base unit links chemistry with the SI. This means, the traceable and comparable quantifications are ensured globally and over time. Table 1 contains several quantities related to the amount of substance and molar mass: constants, numerical values, uncertainties, and their relations used in chemistry.

XRCD Method

Currently, the x-ray-crystal-density (XRCD) method serves as a primary method in metrology as the state-of-the art experiment for the realization – and after the SI revision – for the dissemination of the SI base units mole and kilogram (BIPM 2019a, b). The XRCD method, also known as "silicon route" or "Avogadro experiment," is a set of several combined experiments aiming at the measurement of the Avogadro constant N_{A} by "counting" the silicon atoms in a chemically pure silicon single crystalline sphere with a mass of approximately 1 kg (Gupta 2019; Fujii et al. 2016). This "Avogadro Project" has been operated by the International Avogadro Consortium (IAC), consisting of several national metrology institutes (NMIs) and research facilities from all over the world. Initially, the XRCD method has been developed to provide a primary method for the realization of the SI base unit of the mass, the kilogram, complementary to the development of the Kibble balance (Becker 2003). The origin of the XRCD method is based on the pioneering work of *Bonse* and *Hart* in the mid-1960s, who developed the first X-ray interferometer used for the measurement of the lattice parameter a in a silicon crystal (Bonse and Hart 1965). In 1974, Deslattes et al. published the first application of the XRCD method for the determination of N_{A} using single crystalline silicon with natural isotopic composition yielding a relative uncertainty of $u_{\mathrm{rel}}(N_{\mathrm{A}}) = 1.05 \times 10^{-6}$ (Deslattes et al. 1974, 1976). In 1981, Becker et al. presented a new method to determine the lattice parameter in silicon, resulting in an Avogadro number increased by $\Delta_{\mathrm{rel}}(N_{\mathrm{A}}) = +5.4 \times 10^{-6}$ (Becker et al. 1981). This large change in N_{A} initiated the "Avogadro Project" aiming initially at the redefinition of the kilogram on the basis of N_{A} four decades ago. For the determination of N_{A} via "counting silicon atoms" – the

Table 1 Quantities and constants related to the amount of substance and molar mass

			Former SI		New SI			
Quantity or constant (symbol)	Description	Unit	Numerical value	Uncertainty	Numerical value	Uncertainty	Relation	Remark[a,b]
N	Number of entities (particles)	1		Yes		Yes	$N = n \times N_A$	[c]
n	Amount of substance	mol		Yes		Yes	$n = N \times N_A^{-1}$	
N_A	Avogadro constant	mol^{-1}	$6.022140\,857 \times 10^{23}$	74×10^{-9}	$6.022140\,76 \times 10^{23}$	No		[d,e]
x	Amount of substance fraction	$mol \times mol^{-1}$		Yes		Yes	$x_i = n_i \times n^{-1}$	
m	Mass	kg		Yes		Yes		
M	Molar mass	$kg \times mol^{-1}$		Yes		Yes	$A_r \times M_u$	[f]
M_u	Molar mass constant	$kg \times mol^{-1}$	1	No	$0.99999999965 \times 10^{-3}$	$0.00000000030 \times 10^{-3}$	$M_u = m_u \times N_A$ $M_u = M(^{12}C)/12$	[e]
m_u	Atomic mass constant	kg	$1.6605387820 \times 10^{-27}$	$0.0000000024 \times 10^{-27}$	$1.66053906660 \times 10^{-27}$	$0.00000000050 \times 10^{-27}$	$m_u = 1\ u = 1\ Da$	[f,g]

u	Unified atomic mass unit	kg	$1.6605387820 \times 10^{-27}$	$0.0000000024 \times 10^{-27}$	$1.66053906660 \times 10^{-27}$	$0.00000000050 \times 10^{-27}$	$m_u = 1\ u = 1\ Da$	[f,g]
Da	Dalton	kg	$1.6605387820 \times 10^{-27}$	$0.0000000024 \times 10^{-27}$	$1.66053906660 \times 10^{-27}$	$0.00000000050 \times 10^{-27}$	$m_u = 1\ u = 1\ Da$	[f,g]
$M(^{12}C)$	Molar mass of ^{12}C	kg × mol^{-1}	12×10^{-3}	No	$11.9999999958 \times 10^{-3}$	$0.0000000036 \times 10^{-3}$	$M(^{12}C) = 12 \times M_u$	[f]
$A_r(X)$	Relative atomic mass of an entity X (isotope...)	1	See Wang et al. (2021)	Yes	See Wang et al. (2021)	Yes	$A_r(x) = 12 \times m(x)/m(^{12}C)$	

[a]Value, definition, uncertainty prior to May 20, 2019 (prior to revision of SI)
[b]Value, definition, uncertainty after May 20, 2019 (after revision of SI)
[c]For conversion into amount of substance, the entities must be specified. An elementary entity may be an atom, a molecule, an ion, an electron, any other particle, or specified group of particles
[d]Mohr et al. (2016)
[e]Newell et al. (2018)
[f]Tiesinga et al. (2021) experimentally determined and updated, updated on a regular basis
[g]Taylor (2009)

number N – in a single crystalline sphere, the lattice parameter a, the volume V of the sphere, its mass m, density ρ, and molar mass M have to be determined with lowest measurement uncertainties possible. The number of silicon atoms in a unit cell is 8:

$$N_{\mathrm{A}} = \frac{8\,V\,M}{a^3\,m} = \frac{8\,M}{\rho\,a^3} = \frac{N}{n} \tag{18}$$

The mass of the electron m_{e} can be calculated using

$$m_{\mathrm{e}} = \frac{2\,R_\infty\,h}{c\,\alpha^2} \tag{19}$$

with the fixed Planck constant h, the Rydberg constant R_∞, the speed of light in vacuum c, and the fine structure constant α. This way, N_{A} can be expressed via

$$N_{\mathrm{A}} = \frac{M_{\mathrm{u}}\,A_{\mathrm{r}}(\mathrm{e})}{m_{\mathrm{e}}} = \frac{c\,\alpha^2}{2\,R_\infty\,h}\,A_{\mathrm{r}}(\mathrm{e})\,M_{\mathrm{u}} \tag{20}$$

with the relative atomic mass of the electron $A_{\mathrm{r}}(\mathrm{e})$, and the molar mass constant M_{u} (Fujii et al. 2016).

After the revision of the SI with a fixed value of N_{A}, the amount of substance n can be obtained from this relation via

$$n = \frac{8\,V}{a^3}\,\frac{2\,R_\infty\,h}{c\,\alpha^2}\,\frac{1}{A_{\mathrm{r}}(\mathrm{e})\,M_{\mathrm{u}}} = \frac{8\,V}{a^3}\,\frac{1}{N_{\mathrm{A}}} \tag{21}$$

Equation (18) is a somewhat simplified expression for N_{A}, because several changing material properties of the single crystalline silicon have to be corrected for. Impurities (mainly carbon, oxygen, and nitrogen) and vacancies in the crystal lattice have to be quantified. Surface contaminations (oxide layers, etc.) have to be detected, quantified using x-ray fluorescence (XRF), x-ray photoelectron spectroscopy (XPS), and ellipsometry. Thus, the exact macroscopic mass of a silicon sphere is given by

$$m = \frac{2\,h\,R_\infty}{c\,\alpha^2}\,\frac{M}{A_{\mathrm{r}}(\mathrm{e})}\,\frac{8\,V}{a^3} - m_{\mathrm{deficit}} + m_{\mathrm{SL}} \tag{22}$$

The term m_{deficit} includes the point defects (impurities, vacancies) in the crystal lattice, and m_{SL} is the mass of the surface layer.

Although the realization of n according to Eq. (21) is based on the measurement of the sphere volume V and the lattice parameter a only, the respective silicon sphere must be characterized completely beforehand with regard to its isotopic composition (molar mass M) and mass or density. However, since a and M are considered as long-term stable quantities, it will be sufficient to re-measure the surface conditions and the diameter (and thus the volume) for the dissemination of n using a silicon sphere. Briefly, the main experiments are summarized. The lattice parameter a (with the

dimension of a length) describes the distance of lattice planes in a "perfect" single crystal. In the x-ray interferometer (XINT), three lamellas with {220} orientation were cut from the initial silicon single crystal ingot, thus representing the material property itself. The first lamella is necessary to split the X-ray bunch into two beams. The second lamella acts like a mirror, reflecting the two beams, which are recombined in the third analyzer lamella. The latter is moveable perpendicular to the X-ray beam. This induces a periodic change of the intensity of the transmitted x-rays correlated with the diffracting-plane spacing period. An optical interferometer measures the movement of the third lamella yielding the period of the X-ray interference fringes representing the lattice spacings. The measurement of the fringe period is finally related to a wavelength standard. The lattice parameter a must be measured at the same ambient conditions (temperature and pressure) as the volume of the sphere. The volume V in Eq. (22) of a "perfect" sphere can be determined by measuring the mean diameter of the sphere. This is one reason why a sphere was used as a macroscopic artifact in the XRCD method. Moreover, a spherical shape is less sensitive towards potential damage. Currently, two similar but different experimental setups are used to determine V using an optical sphere interferometer. In the PTB setup, the sphere is located in the center of two spherical etalons (reference surfaces). The sphere can be moved up and down and no rotation of the sphere is required. Thus, the diameter can be measured from numerous positions ($>$10,000) around the sphere simultaneously. However, the determination via the mean diameter requires an almost perfectly round shape of the sphere, with a roundness deviating only a few nanometers. The distance between the two etalons is measured with and without the sphere in the laser beam of the interferometer. When tuning the optical frequencies, the fractional interference orders of the beam are measured using the concept of phase shifting interferometry (Fujii et al. 2016). The other setup used at the National Metrology Institute of Japan (NMIJ) operates a sphere interferometer with flat reference surfaces, where the sphere is rotated for the determination of the volume. In these setups, the sphere is located in a temperature-controlled vacuum chamber to avoid influences of the refractive index of ambient air. Parallel to the volume measurement, the surface conditions of the sphere have to be monitored, because changing surface layers cause a phase retardation of the reflected beam in the interferometer. Therefore, the surface conditions of the sphere – for mass and optical corrections – are determined with complementary surface analytical techniques: spectroscopic ellipsometry (SE), X-ray reflectometry (XRR), X-ray fluorescence analysis (XRF), and x-ray photoelectron spectroscopy (XPS). For the realization and dissemination of N_A, the macroscopic mass m of the sphere is determined with lowest associated uncertainty using high-resolution mass comparator balances with a nominal load of 1 kg. Measurements in vacuum do not require additional air buoyancy and sorption effect corrections. The mass resolution is in the range of 100 ng which is 1×10^{-10} on a relative scale in case of a 1 kg sphere. Relative uncertainties of mass determinations are in the 10^{-9} range. For comparison purposes, the density ρ is measured using hydrostatic weighing techniques. However, the improved surface quality of the spheres enabled a density determination via diameter and mass measurements with a reduced uncertainty compared to direct

density measurements (Fujii et al. 2016). "Counting" silicon atoms in a sphere assumed a perfect crystal in both geometry and composition. However, at least due to the production process, tiny amounts of point defects (chemical impurities and vacancies) cannot be excluded and have to be determined experimentally for subsequent correction. For this purpose, Fourier transform infrared absorption spectroscopy (FTIR) is applied. For the enriched Si sphere (example: Si28kg01a) the main impurities are carbon ($N/V = 0.89(14) \times 10^{15}$ cm^{-3}; $x = 17.8(2.8) \times 10^{-9}$ mol/mol); oxygen ($N/V = 0.132(21) \times 10^{15}$ cm^{-3}; $x = 2.64(42) \times 10^{-9}$ mol/mol); vacancies ($N/V = 0.33(11) \times 10^{15}$ cm^{-3}; $x = 6.6(2.2) \times 10^{-9}$ mol/mol) (Güttler et al. 2019). The determination of the molar mass M is outlined in a separate section, due to its particular initial uncertainty contribution which has been reduced by several orders of magnitude using new theoretical and experimental mass spectrometric techniques. Prior to the revision of the SI, three main results of N_A with successively reduced uncertainties have been reported by the IAC using the XRCD method: $u_{rel}(N_A) = 3 \times 10^{-8}$ (2011) (Andreas et al. 2011), $u_{rel}(N_A) = 2.0 \times 10^{-8}$ (2015) (Azuma et al. 2015), and $u_{rel}(N_A) = 1.2 \times 10^{-8}$ (2017) (Bartl et al. 2017).

Silicon Crystal Production

The realization and new definition of the mole required the successful application of the XRCD method using a pool of macroscopic silicon spheres highly enriched in ^{28}Si with high chemical purity and an almost perfect single crystalline structure. Prior to 2007, N_A has been determined using Si spheres with a natural isotopic composition ending with $u_{rel}(N_A) = 3 \times 10^{-7}$, which was at least one order of magnitude too large for a planned redefinition of the kilogram and the mole (Becker et al. 2003; Pramann et al. 2020). Simulations predicted that artificially enriched silicon consisting of the main isotope ^{28}Si exclusively will enable a reduced uncertainty $u_{rel}(N_A) < 2 \times 10^{-8}$. The main challenge at that time was the production of a silicon sphere highly enriched in ^{28}Si. A crystal with a mass larger than 1 kg was needed also providing samples for molar mass and purity measurements as well as parts for the construction of the lamellae of the x-ray interferometer and others. Under the custody of the IAC, a project was started in 2004 to produce a large silicon crystal (mass of the crystal ingot approximately 5 kg) with an amount-of-substance fraction of $x(^{28}\text{Si}) > 0.999\ 9$ mol mol^{-1} yielding two spheres with a mass of 1 kg each (Becker et al. 2006). The first silicon crystal enriched in ^{28}Si has been produced in 2007, ready for the characterization with the XRCD method. For comparison purposes and the planned subsequent dissemination of the kilogram and the mole, additional silicon spheres highly enriched in ^{28}Si were necessary. Thus, in two side projects starting in 2012, a cooperation between PTB and Russian companies and research institutes and the German Leibniz Institute for Crystal Growth (IKZ) was aimed at the production of ten even higher enriched silicon spheres (Abrosimov et al. 2017). The crystal production is illustrated in Fig. 3.

Briefly, silicon with natural isotopic composition was shipped to Russia for the production of high purity silicon tetrafluoride (SiF_4). In cascades of gas centrifuges,

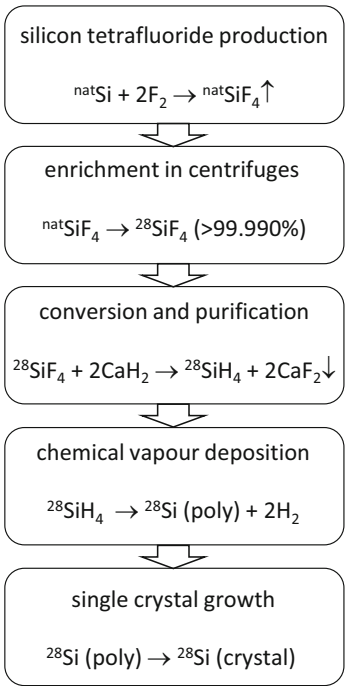

Fig. 3 Steps of the production of silicon crystals highly enriched in ^{28}Si. ^{nat}Si denotes silicon with natural isotopic composition

^{28}Si in SiF_4 was enriched up to $x > 0.999\,99$ mol mol^{-1}. After the conversion of SiF_4 into silane SiH_4, including several rectification and purification steps, SiH_4 was deposited in a pyrolytic (chemical) vapor deposition process yielding polycrystalline silicon, highly enriched in ^{28}Si. The polycrystalline rod was then processed into a single crystalline ingot using a float zone technique at the IKZ in Germany. Finally, the spheres and additional parts and samples were cut from the ingot at PTB. In an improved manufacturing chain, the spheres have a surface roughness smaller than (0.20 ± 0.02) nm and a deviation from perfect spherical shape smaller than (50 ± 10) nm (Fujii et al. 2016). Figure 4 shows the main steps of the crystal and sphere production.

Molar Mass of Silicon Determined Using the Virtual-Element-IDMS Method

Although the first large silicon crystal highly enriched in ^{28}Si has been successfully produced in 2007, a new unexpected problem has been imported in this context: The molar mass M(x) is usually best determined via isotope ratio mass spectrometry by the measurement of isotope ratios of the material x. In case of silicon, M can be expressed by the general relation

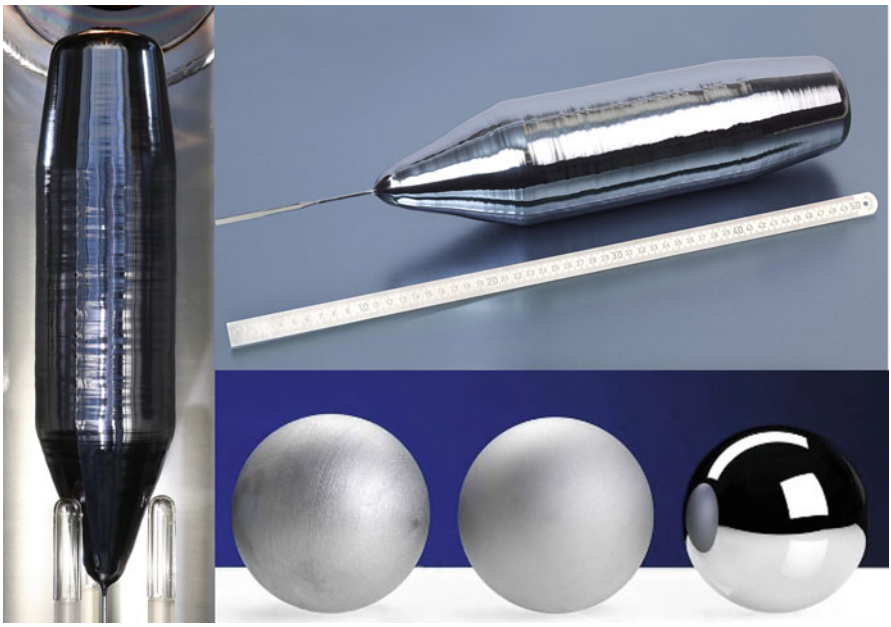

Fig. 4 Steps of the production of silicon crystals and spheres highly enriched in ^{28}Si. (Left) A silicon single crystal ingot (length ≈ 50 cm, mass ≈ 5 kg) produced in a float zone process. (Top) Si crystal prior to cutting. (Bottom) Sphere after cutting towards polishing

$$M = \sum_{i=28}^{30} \left[x\left({}^{i}\mathrm{Si}\right) \times M\left({}^{i}\mathrm{Si}\right) \right] \tag{23}$$

with the amount-of-substance fractions $x({}^{i}\mathrm{Si})$ of the silicon isotopes ^{28}Si, ^{29}Si, and ^{30}Si and the molar masses $M({}^{i}\mathrm{Si})$ of the respective isotopes. The $x({}^{i}\mathrm{Si})$ are accessible by the measured and mass bias corrected isotope ratios R

$$x\left({}^{i}\mathrm{Si}\right) = \frac{R_i}{\sum_{j=28}^{30} R_j} \tag{24}$$

Isotope ratios are usually determined using the most abundant isotope as the denominator. In fact, the new silicon material was enriched in ^{28}Si by almost 100% ($x(^{28}\mathrm{Si}) > 0.999\ 95$ mol mol^{-1}). This would yield ratios $x(^{29}\mathrm{Si})/x(^{28}\mathrm{Si}) \approx 4 \times 10^{-5}$ mol/mol and $x(^{30}\mathrm{Si})/x(^{28}\mathrm{Si}) \approx 1 \times 10^{-6}$ mol/mol in case of the first enriched silicon crystal. The large deviation of these ratios from unity results in uncertainties associated with the molar mass, orders of magnitudes too large to be applicable for the realization of N_A with an associated uncertainty $< 2 \times 10^{-8}$. To overcome this problem, at PTB *Rienitz* et al. have developed and improved a new method to determine the molar mass M of enriched silicon using a principle called Virtual-

Element Isotope-Dilution-Mass Spectrometry (VE-IDMS) (Rienitz et al. 2010; Mana et al. 2010). This approach was adopted to and modified from the classical isotope dilution mass spectrometry (IDMS) technique – a primary method in metrology in chemistry (De Bièvre 1990; Heumann 1992). In this context, it must be noted that the artificial and high enrichment of the silicon results in large efforts to be made to avoid contamination with silicon having a natural isotopic composition which is almost ubiquitous. Care must be taken to avoid this kind of contamination using silicon-free materials as good as possible. The advantage of the inductively coupled plasma mass spectrometry (ICP-MS) is the use of liquid samples which simplifies the preparation of blends and also the measurement of blank solutions in the same way as the sample solutions enabling the subsequent correction of contaminants.

The VE-IDMS method treats the silicon material highly enriched in ^{28}Si as theoretically consisting of ^{29}Si and ^{30}Si only (configuring a "new" virtual element VE) in an excess matrix of ^{28}Si. The ^{29}Si and ^{30}Si isotopes in the new crystal material were initially not desired, but due to technical reasons in the enrichment process, a tiny portion of these "impurities" were always abundant in the crystal and thus had to be determined as accurate as possible. In the VE-IDMS method, the ^{28}Si isotopes are first completely excluded, and mainly the ratios $x(^{30}\mathrm{Si})/x(^{29}\mathrm{Si})$ are measured in the sample x and in a blend bx prepared from the sample and a "spike" material y (which is the same element (silicon) with an almost inverted isotopic pattern). In the spike material the ^{30}Si isotope is highly enriched; however, the knowledge of the exact enrichment is not necessary. Additionally, the masses m_x and m_{yx} (mass of solid sample and spike components in the blend bx) have to be known exactly. The molar mass M is obtained via the VE-IDMS equation

$$M = \frac{M\left(^{28}\mathrm{Si}\right)}{1 + \frac{m_{yx}}{m_x} \cdot \frac{M\left(^{28}\mathrm{Si}\right)\cdot(1+R_{x,2}) - M\left(^{29}\mathrm{Si}\right) - R_{x,2}\cdot M\left(^{30}\mathrm{Si}\right)}{R_{y,3}\cdot M\left(^{28}\mathrm{Si}\right) + M\left(^{29}\mathrm{Si}\right) + R_{y,2}\cdot M\left(^{30}\mathrm{Si}\right)} \cdot \frac{\left(R_{y,2} - R_{bx,2}\right)}{\left(R_{bx,2} - R_{x,2}\right)}} \tag{25}$$

Additionally, the masses m_x and m_{yx} (mass of solid sample and spike components in the blend bx) have to be known. The isotope ratios in the respective materials and blends in Eq. (25) are $R_{j,2} = x_j(^{30}\mathrm{Si})/x_j(^{29}\mathrm{Si})$ and $R_{j,3} = x_j(^{28}\mathrm{Si})/x_j(^{29}\mathrm{Si})$ (Pramann et al. 2011a, 2014, 2015). Figure 5 shows the principle of the VE-IDMS method for the determination of the molar mass of enriched silicon.

The experimental details applied to several different silicon crystals highly enriched in ^{28}Si are given elsewhere (Pramann et al. 2011a, b, c, 2014, 2015, 2017; Pramann and Rienitz 2019; Rienitz and Pramann 2020). In the meantime, the VE-IDMS method has been successfully applied by several other national metrology institutes (NMIs), and an interlaboratory comparison (CCQM-P160) has been conducted (Yang et al. 2012; Narukawa et al. 2014; Vocke et al. 2014; Ren et al. 2015; Rienitz et al. 2020).

Parallel to the challenge of the determination of M with smallest measurement uncertainty, one experimental problem always accompanied with isotope ratio (R) measurements is the occurrence of biased isotope ratios, usually called "mass bias." The measured isotope ratios have to be corrected for using calibration K factors.

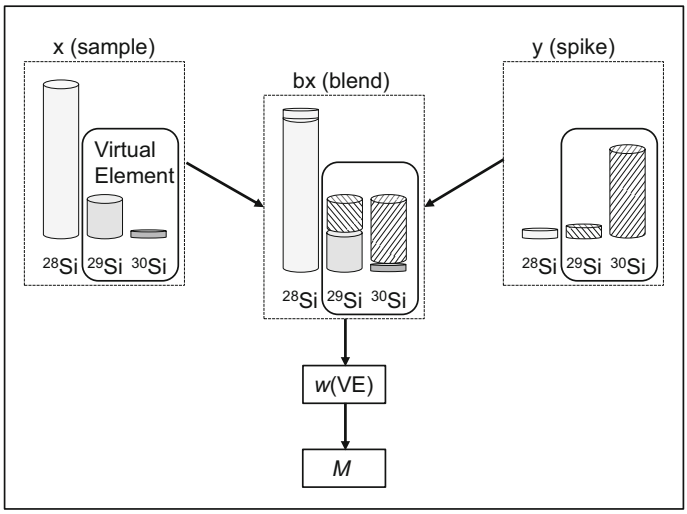

Fig. 5 Principle of the VE-IDMS method. The ^{28}Si signal is mainly ignored in the isotope ratio measurements. The sample x of the enriched silicon material is blended with another silicon material highly enriched in ^{30}Si (spike material y) yielding a blend bx with an adjusted ratio $x(^{30}\mathrm{Si})/x(^{29}\mathrm{Si}) \approx 1$. The mass fraction of the sum of ^{29}Si and ^{30}Si (the virtual element VE or "impurity") and subsequently the molar mass M of the silicon with all three isotopes can be calculated. According to Eq. (25) it is not necessary to determine the respective amount-of-substance fractions $x(^{i}\mathrm{Si})$; however they can be retraced, if necessary

Since there exists no reference material for enriched silicon, an analytical closed-form procedure has been developed in a cooperation of the Istituto Nazionale di Ricerca Metrologica (INRIM, Italy) and PTB (Mana and Rienitz 2010). In this method, gravimetrically prepared blends consisting of the parent materials are used. This enables the experimental determination of K factors without the need of iterative processes yielding an associated measurement uncertainty. Briefly, in case of the silicon isotope ratios $R_j(^{30}\mathrm{Si}/^{29}\mathrm{Si})$ and $R_j(^{28}\mathrm{Si}/^{29}\mathrm{Si})$, the relation between the "true" (corrected) R_j^{true} and measured isotope ratios R_j^{meas} is given by

$$R_{j,2}^{\mathrm{true}} = K_2 \cdot R_{j,2}^{\mathrm{meas}} \quad \text{with} \quad R_{j,2}^{\mathrm{meas}} = \frac{I_j\left(^{30}Si\right)}{I_j\left(^{29}Si\right)} \quad \text{and} \quad j \in \{x, y, \mathrm{bx}\} \tag{26}$$

$$R_{y,3}^{\mathrm{true}} = K_3 \cdot R_{y,3}^{\mathrm{meas}} \quad \text{with} \quad R_{y,3}^{\mathrm{meas}} = \frac{I_y\left(^{28}Si\right)}{I_y\left(^{29}Si\right)} \tag{27}$$

with the measured ion signals I_j (voltages at the respective detectors). Two blends b1 and b2 are necessary and sufficient for the determination of K_2 and K_3. In b1, the parent material z highly enriched in ^{29}Si together with material y (highly enriched in ^{30}Si) are mixed, whereas in b2 silicon with natural isotopic composition (material w) is blended with material z. The application and improvement of the combination of

the VE-IDMS method and the closed-form gravimetric K factor determination has established a primary method for the determination of the molar mass of silicon highly enriched in ^{28}Si. During the last decade, the uncertainty $u_{\mathrm{rel}}(M)$ associated with the molar mass M has been reduced by almost three orders of magnitude from 3×10^{-7} down to 8×10^{-10} as a function of both the high enrichment and the new methods (Güttler et al. 2019; Rienitz and Pramann 2020). This strong reduction in measurement uncertainty yields to the best of the authors' knowledge the most precise measurement results in chemistry so far, necessary for the revision of the SI in 2019.

Dissemination of the Mole

The ways for the practical realization of the SI base unit mole are described in the *Mise en Pratique*, an appendix of the SI brochure acting as an official guide which is released by the Consultative Committee for Amount of Substance – Metrology in Chemistry and Biology (CCQM) of the International Committee for Weights and Measures (CIPM) (BIPM 2019a, b). As both a condition and a consequence of the revision of the mole and the kilogram, the fundamental constants N_{A} and h have been assigned to fixed numerical values without any associated uncertainties. The most accurate and precise way to disseminate the mole is via the XRCD method using a characterized silicon sphere highly enriched in ^{28}Si. This method serves as a primary method considering the enriched spheres as primary standards. Equation (21) summarizes the quantities which have to be determined with lowest associated uncertainties for the current dissemination and realization of the SI base unit mole.

The determination of the lattice parameter a and the volume V (via diameter measurements) as well as surface analysis have to be performed, once a sphere has been characterized with regard to the main quantities like the molar mass, crystal purity, mass, and density according to Eq. (18). Equation (21) demonstrates the "island position" of the mole as a discrete independent SI base unit, which is defined by a certain number of a considered entity (atoms, ions. . .). In practice however, it is still common to realize the amount of substance via the mass of a primary sphere and subsequent comparison with the respective entity using Eq. (22).

For practical purposes some more routine procedures are necessary which can be also adopted from industrial laboratories. PTB has developed a concept of three different categories of silicon spheres which can be distinguished via their purity, surface quality, and isotopic composition which causes three ranges of uncertainties associated with the respective masses (Knopf et al. 2019). This concept is a consequence of the few available ^{28}Si spheres which are extremely expensive due to a time-consuming high level production. Thus, silicon spheres in part commercially manufactured with natural isotopic composition can be used as transfer standards for dissemination purposes. Three categories of silicon spheres will be used for the dissemination: first, the primary standard "^{28}Si" spheres, which are highly enriched in ^{28}Si, enabling the smallest relative uncertainty of $u_{\mathrm{rel}}(m) = 2 \times 10^{-8}$ associated with m (roughness ≤ 0.3 nm). These special spheres were manufactured at PTB. The

next level of uncertainty is represented by the so-called "quasi-primary" spheres ("$^{nat}Si_{qp}$") with $u_{rel}(m) = 3 \times 10^{-8}$ (average roughness ≤ 0.5 nm) which are also manufactured at PTB. Finally, the third category of silicon spheres is called "secondary" silicon spheres ("$^{nat}Si_{sc}$"). These industrially produced spheres can be used for practical (industrial) applications with $u_{rel}(m) = 3 \times 10^{-8}$ (average roughness ≤ 1 nm). The XRCD method thus describes the top-level method for the realization and dissemination of the SI base unit mole when expressing the amount of substance n with lowest associated uncertainty. In common practical (laboratory) applications, more convenient and common experimental methods are used with the drawback of larger uncertainties.

The typical and most common method to determine the amount of substance n in chemical laboratories is using a balance. This is called the "gravimetric-approach" (instead of the volumetric treatment of substances). Simply, a gravimetric determination of n is achieved by Eqs. (15) and (16), and n(x) can be easily accessed using the molar mass M(X) and the mass m(x) of the sample X. The product of the relative atomic mass A_r(X) and the molar mass constant M_u yields M(X). Often, the mass fraction w(X) of the substance X must be accessible, if X is part of a mixture and not a pure substance.

The amount of substance n can be accessed in gases in the simplest case using the ideal gas equation

$$n = \frac{pV}{RT} = \frac{pV}{N_A kT} \tag{28}$$

p is the gas pressure, V is the volume, T is the temperature, and R is the molar gas constant $R = k\,N_A$ (with the fixed Boltzmann constant k). Generally, the uncertainty associated with n is also a few orders of magnitude larger than $u_{rel}(n)$ obtained via the primary realization method using the XRCD method which is a consequence of the uncertainties of the respective input parameters p, V, and T.

In electrochemistry, the amount of substance n is accessible by charge transfer processes at electrodes. The amount of substance n can be visualized directly by the charge Q according to

$$n = \frac{Q}{zeN_A} = \frac{Q}{zF} \tag{29}$$

with the charge number z, the elementary charge e, and the fixed Faraday constant $F = e\,N_A$. Another example for the determination of n is provided by coulometry where a current I is measured in a time interval

$$n(\mathrm{A}) = \frac{1}{z \times F} \int I \mathrm{d}t \tag{30}$$

Besides these formal and defining methods for the realization and dissemination of the SI unit mole and the underlying constantly used quantity "amount of substance" n, the practical methods used in research laboratories as well as in industry

should be highlighted. The practical realization and dissemination of the mole, traceable to the International Systems of Units – the SI – is done using primary standards with a certain value (e.g., mass fraction w, amount concentration c) and an associated uncertainty (which in case of primary standards should be as small as possible). For practical purposes, a dissemination chain from the "highest metrological level" primary standard (usually provided by a national metrology institute NMI) down to a standard released to the respective end user is established. The "primary standards" provided by an NMI have to be characterized by primary measurement procedures (Milton and Quinn 2001). In case of the amount of substance n, not only the value of n but also the kind of entity X have to be specified. The amount of substance is a key quantity in metrology in chemistry, because its practical dissemination must be facilitated for every entity (substance) X which is required. After a careful characterization with regard to its purity, usually a liquid solution of a dissolved element is gravimetrically prepared and characterized by a primary measurement method. Here, primary ratio measurements like isotope dilution mass spectrometry (IDMS) represent the state-of-the art methods with lowest associated uncertainties in analytical chemistry (De Bièvre 1990; Heumann 1992; Sargent et al. 2002). Taking copper (X) as an example, the practical way for the dissemination of the amount of substance n(X) is as follows: A national standard (a "reference point") – usually a mono-elemental solution – is gravimetrically prepared from a high purity copper salt or copper metal sample (with precedingly determined purity) dissolved in a high purity acid (compare Fig. 6). For elements,

Fig. 6 Visualization of the traceability chain of the amount of substance n with the SI base unit mole (1 mol represented here by a sphere made of 28 g silicon) of a certain element (here, copper). High purity solid Cu granules (in the storage bottle), sometimes available as a certified reference material, are converted by gravimetric preparation into liquid primary solutions (ampule in the background)

these mono-elemental solutions represent the link to the mole. The uncertainties associated to the amount of substance include the mass, the purity, and the molar mass of the respective element in the solution as well as the contributions due to the preparation. Whereas the uncertainty of the purity of the respective material expressed as a mass fraction w is in a range $\leq 5 \times 10^{-5}$ g/g, the relative expanded uncertainty ($k = 2$) associated with the mass fraction w of an element is $\leq 10^{-4}$ g/g in case of a primary solution (Matschat et al. 2006). These primary solutions are used in primary measurements for the characterization of secondary solutions (also known as transfer solutions) establishing a traceability chain towards calibration laboratories. The relative uncertainties associated with the transfer solutions are in the range of $\approx 2 \times 10^{-3}$ g/g. Generally, the uncertainties associated with the content (mass fraction) of these national standards (primary solutions) are unaffected by the revision of the SI because the impact of the uncertainties associated with the molar masses needed for the gravimetric preparation is in the range of 10^{-10}, relatively, and can thus be neglected in most cases. A schematic of the traceability chain is shown in Fig. 7. In order to maintain the quality of the dissemination chain, the international comparability, and to prove the ability of the realization on the primary level, international comparisons are carried out with focus on important public areas like clinical chemistry, environmental protection, climate control, and related topics. The worldwide umbrella organization in metrology, the "Metre Convention" has set-up 1993 the "Consultative Committee for Amount of Substance: Metrology in Chemistry and Biology" (CCQM) which acts as an official board for metrology in chemistry. The main task of the CCQM is the establishment and maintenance of international traceability structures in chemistry. These tasks are further fostered by regionally operating metrology organizations like EURAMET (European Association of National Metrology Institutes) or SIM (America), APMP (Asia), COOMET (Euroasia), AFRIMETS (Africa), and GULFMET (Golf region) and also other organizations that support a metrology-based infrastructure for measurements in

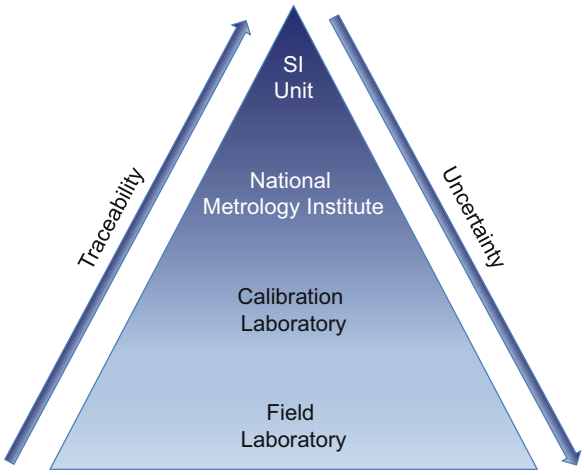

Fig. 7 Schematic traceability chain: Amount of substance n with the SI base unit mole is provided by a National Metrology Institute (NMI) and is disseminated towards accredited calibration laboratories and subsequently to laboratories in the field to ensure the traceability of routine measurements

chemistry such as Eurachem (a network of European organizations of metrology in chemistry), and on an international level, the CITAC (Cooperation on International Traceability in Analytical Chemistry).

Conclusion

After the admission of the SI base unit mole in 1971 which was then related to the SI base unit mass via a "certain" amount (12 g) of a "certain" entity (namely, the ^{12}C isotope), the revision of the SI in 2019 results in a complete independent definition: the mole as the base unit of the amount of substance n is now defined via the Avogadro constant N_A, a fixed fundamental physical constant, solely: "The mole, symbol mol, is the SI unit of amount of substance. One mole contains exactly $6.02214076 \times 10^{23}$ elementary entities. This number is the fixed numerical value of the Avogadro constant, N_A, when expressed in the unit mol^{-1} and is called the Avogadro number. The amount of substance, symbol n, of a system is a measure of the number of specified elementary entities. An elementary entity may be an atom, a molecule, an ion, an electron, any other particle or specified group of particles." This chapter provides the reader with a concise overview about the history, the development and need, as well as the realization, dissemination, and impacts on practical scientific life of the "youngest" SI base unit. For more specific information, a number of reviews are given. A focus in this presentation is the application of the mole in chemistry, and the consequences are shown after the revision of the SI. To understand the turning away from a definition via the relation to an artifact – the former International Prototype of the Kilogram – the XRCD method was presented in more detail, especially the chemical part which is the determination of the molar mass and isotopic composition of the silicon material used. Here, the Virtual-Element IDMS method and the respective closed-form calibration factor method are emphasized as two key approaches enabling the determination of the respective quantities with so far smallest measurement uncertainties which were a prerequisite for the revision of the mole. Both the advantage and the challenge of the new definition of the mole will be that in the future this SI unit might be realized and disseminated via any appropriate experiment with measurable quantities traceable to the SI and considerable low associated uncertainties. The need for related experimental results with lowest possible uncertainties requires thus the maintenance and improvement of existing and developing of future experiments.

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Progress of Quantum Hall Research for Disseminating the Redefined SI

14

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Abstract

As the global implementation of new technologies continues to progress, one should hope that a more universal accessibility to the quantum SI is established. This chapter intends to give historical context for the role of the quantum Hall

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effect in metrology, including a basic overview of the effect, supporting metrology technologies, and how the research in this field has expanded the world's general capabilities. The present-day era of resistance metrology, heavily dominated by the transition from using gallium-arsenide-based devices to graphene-based ones, will be summarized in terms of how the new 2D material performs, how the world has started to implement it as a resistance standard, and how the corresponding measurement infrastructure is currently adapting to the new standard. In the third section, emerging technologies based on graphene will be introduced to give a brief overview of the possible expansion device capabilities. These ideas and research avenues include p-n junction devices, quantum Hall array devices, and experimental components of ac metrology and the quantum ampere. The chapter then concludes by discussing the possible limitations of graphene-based technology for resistance metrology and looks to explore topological insulators as one potential candidate to, at the very least, supplement graphene-based QHR devices for resistance and electrical current metrology.

Historical Context

Introduction

To fully appreciate the impacts that the discovery of the quantum Hall effect (QHE) had on electrical metrology, it may benefit the reader to cultivate a general understanding of the phenomenon (Von Klitzing and Ebert 1985; Von Klitzing et al. 1980). For the purposes of this handbook, a basic overview will be given. The QHE may be exhibited by a two-dimensional (2D) electron system when placed under a strong magnetic field perpendicular to the plane of the system. These conditions lead to quantization, or discrete energy states for charged particles in the magnetic field. These energy values, determined by solving the Schrödinger equation, are known as Landau levels. In precise measurements, one defines the quantized Hall resistance R_{xy} as the measured voltage, perpendicular to the direction of the applied current, divided by that same current. The characteristic longitudinal resistivity ρ_{xx} goes to zero as R_{xy} approaches a quantized value over a range of magnetic field strength (nominally, a QHE plateau).

Since the discovery of the QHE at the start of the 1980s, the electrical metrology community has sought to implement a resistance standard based on the theoretical relation: $R_{\mathrm{H}}(\nu) = R_{xy}(\nu) = h/\nu e^2$, where ν is an integer related to the Landau energy level, and h and e are the Planck constant and elementary charge, respectively. In 1990 the QHE was assigned a conventional value as an empirical standard based on the determination of fundamental constants through experiments such as the calculable capacitor, which uses the Thompson-Lampard theorem (Thompson and Lampard 1956) to experimentally realize the unit of impedance, the farad, as defined by the International System of Units (SI). When the redefinition of the SI occurred in 2019, the constants h and e were assigned globally agreed-upon exact SI values such that the electrical units of the ohm, volt, and ampere are defined by relation to these fundamental constants. Interestingly, the electrical quantum standards of resistance and voltage now realize the SI unit of mass, the kilogram, through the exact value of

h and an experiment called the Kibble Balance (Kibble 1976; Schlamminger and Haddad 2019). Within the redefined SI, many other fundamental constants are exact or have greatly reduced uncertainties (Tiesinga et al. 2021). For instance, the von Klitzing constant is now exact, calculated from h/e^2 ($R_K = 25\,812.807\,45\ldots\,\Omega$).

Before the QHE, resistance metrologists would employ standard resistors made from copper-manganese-nickel and similar stable alloys. These standards behaved differently in locations around the world due to variation of their nominal resistance (Witt 1998), and this robust, time-independent definition based on the ratio of fundamental constants has developed to allow expansion of the focus of the field.

Predecessors for Quantum Hall Standards

Quantum Hall standards began their journey shortly after the QHE was discovered. Some of the earliest devices were based on silicon metal–oxide–semiconductor field-effect transistors (MOSFETs). For Si systems, the electric field needed to define the energy state of electrons in two dimensions was generated by a planar metallic voltage gate separated from the surface of the semiconductor by an insulating oxide layer. In work performed by Hartland et al., a cryogenic current-comparator (CCC) bridge with a precise 1:1- or 2:1-ratio was used to compare two quantized Hall resistances, one from the MOSFET and the other from a gallium arsenide (GaAs) heterostructure, which employs layers of semiconductors to create a 2D quantum well (Hartland et al. 1991). Both QHE devices were operated at the same temperature and magnetic field, but on different Landau levels. The critical components of the CCC were immersed in liquid helium at 4.2 K and were shielded from external magnetic fields using superconducting lead foil and low-temperature ferromagnetic alloy (Hartland et al. 1991).

This work focused on comparing the quantized Hall resistance (QHR) of the GaAs/AlGaAs heterostructure (measured at the $\nu = 2$ plateau, or $h/2e^2$, which is approximately 12.9 kΩ) to that of the silicon MOSFET device (measured at the $\nu = 4$ plateau, or $h/4e^2$, about 6.45 kΩ). The deviations from the expected ratio of 2:1, summarized in Fig. 1, are represented as Δ_{24} and the ratio of the winding ratio allows two resistances to be compared while canceling out the multiplicative factor of 2 between them. These results suggest that the two QHR values agree within 2 $[1-0.22(3.5)\times 10^{-10}]$, where the uncertainty is in parentheses. Given the novelty of the QHE at the time, this type of experiment was effective in supporting the notion of representing the plateaus as based on the fundamental constants.

One of the disadvantages of MOSFETs was the high magnetic field required during an experiment. Laboratory magnetic fields of the time typically yielded a resistance plateau that was only quantized over a narrow range of magnetic field. As seen in (Hartland et al. 1991), the Si MOSFET device required 13 T to access its $\nu = 4$ plateau, and currents above 10 μA would cause the QHE to break down, or lose its precise quantized value. These devices did not stay long in favor since GaAs-based devices quickly demonstrated more optimal behavior (Cage et al. 1985; Hartland 1992). In GaAs-based devices, a 2D layer of electrons forms when an electric field forces electrons to the interface between two semiconductor layers (the other, in this case, being AlGaAs). For many devices with this type of interface, the layers were

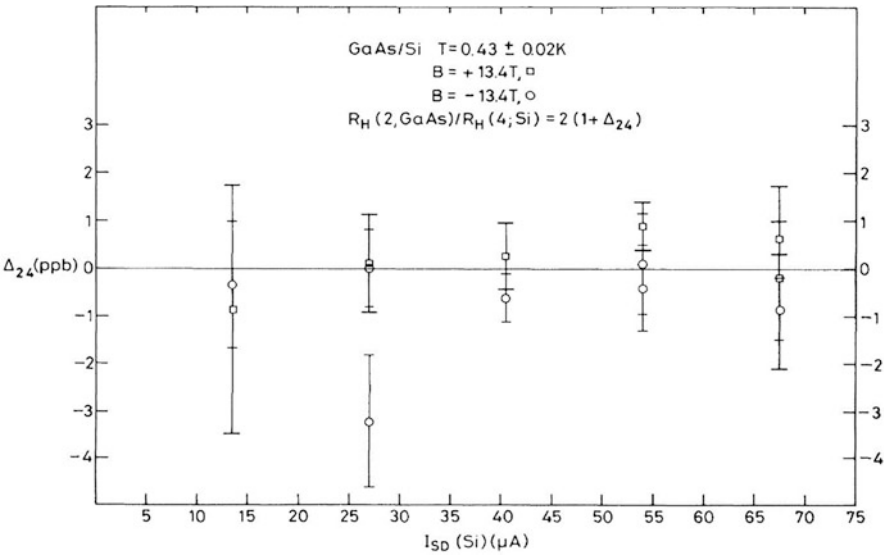

Fig. 1 Measurements of Δ_{24} expressed in parts per billion (ppb, or nΩ/Ω) for the direct comparison of the GaAs-based device (at the $\nu = 2$ plateau) and the Si MOSFET device (at the $\nu = 4$ plateau). These data are shown as a function of the applied current through the Si MOSFET device, and the error bars represent a 1σ random error of the average of Δ_{24}. (Reprinted figure with permission from (Hartland et al. 1991). Copyright 1990 by the American Physical Society)

grown via molecular beam epitaxy (Tsui and Gossard 1981; Hartland et al. 1985). Similar heterostructures were developed in InGaAs/InP, which were obtained via metal-organic chemical vapor deposition (Delahaye et al. 1986).

These early GaAs-GaAlAs heterostructure devices were grown with excellent homogeneity and exhibited high mobilities (on the order of 200 000 $cm^2V^{-1}s^{-1}$) at 4 K. The success of this material in providing highly precise quantized resistances led in 1990 to the definition of the $R_{K\text{-}90}$ representation of the ohm as recommended by the Consultative Committee for Electricity (CCE) (Taylor 1990). For another 20 years, technologies continued to improve, allowing both GaAs-based resistance standards and cryogenic measurement methods to offer increased sensitivity and precision (Jeckelmann et al. 1997; Williams 2011).

During the 1980s, metrologists at national metrology institutes (NMIs) utilized these high-quality devices to confirm the universality of the von Klitzing constant R_K over the course of many experiments. An example of the use of GaAs-based devices for metrology can be seen in Cage et al. (1985), where the authors looked to adopt GaAs as a standard used to maintain a laboratory unit of resistance. The work again demonstrated the universality of the QHE and showed the viability of the device as a means to calibrate artifact standards. The devices were grown by molecular beam epitaxy at an AT&T Bell Laboratory facility in New Jersey (Cage et al. 1985) and had dimensions as shown in the inset of Fig. 2. The magnetic field sweep data for the Hall and longitudinal voltages are also shown in Fig. 2a. The second part of the

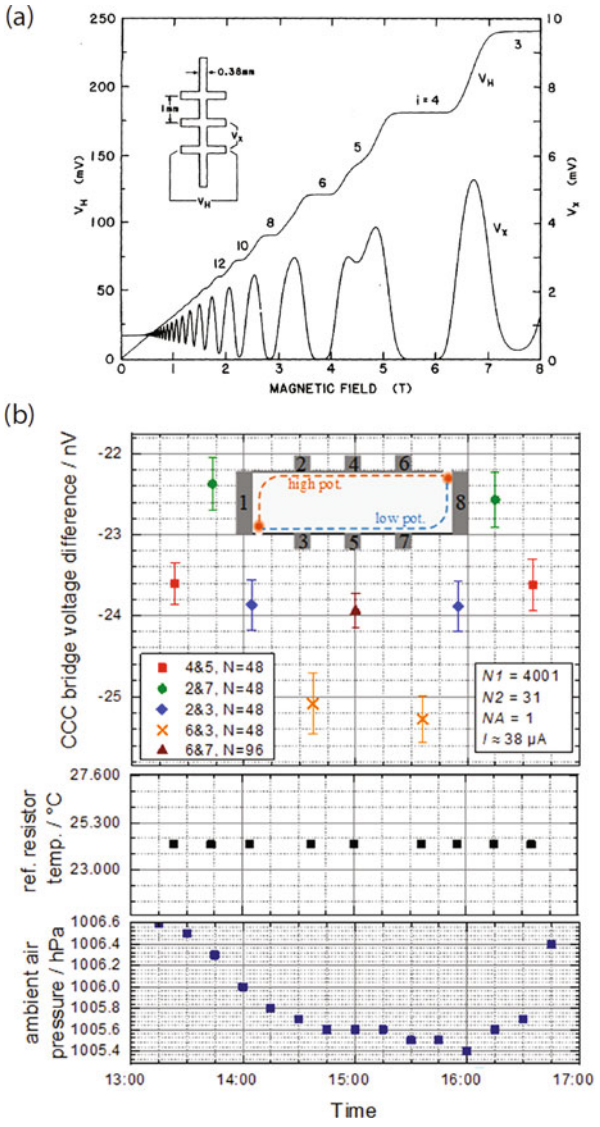

Fig. 2 (**a**) Hall and longitudinal voltage measurements from a magnetic field sweep of a GaAs-based device. The temperature was 1.2 K and the applied current was 25.5 µA. © 1985 IEEE. Reprinted, with permission, from (Cage et al. 1985). (**b**) A practical example of the final high accuracy device characterization procedure prior to calibration at a fixed *B*-field where the deviation from $R_K/2$ is expected to be below 1 nΩ/Ω. The characterization procedure of the device (CCC bridge voltage difference) involves a series of CCC measurements at diagonally and orthogonally aligned Hall contact pairs with the primary purpose of identifying instabilities over time as well as asymmetries in the device properties. The quantity *N* describes the number of measurement cycles in the CCC measurement. The error bars represent type A expanded measurement uncertainties ($k = 2$). To identify instabilities related to the room-temperature reference resistor, the ambient air pressure and the reference resistor temperature are recorded simultaneously as one characterizes the QHR device

experiment involved transferring the QHE value through 1:1 comparisons to a set of 6453.2 Ω resistors, and from there to the US ohm, maintained with a group of resistors at the 1 Ω level (Hamon 1954). Comparisons were done using a direct current comparator (DCC) resistance bridge and reconfigurable series-parallel resistor networks at several resistance levels (Cage et al. 1985).

Before a QHR device can be utilized for calibration, it needs to pass a characterization procedure. The main purpose here is to identify instabilities over time as well as asymmetries in the device properties for different combinations of contacts. The systematic series of CCC measurements shown in Fig. 2b involves various combinations of orthogonally and diagonally aligned pairs of Hall contacts inside the device. It is performed at the same fixed *B*-field intended to be later used for the calibration procedure in which the QHR will be the reference. The suitable *B*-field, where the deviation from $R_K/2$ is expected to be on the level of 1 nΩ/Ω, is typically identified prior to this procedure by characterizing the longitudinal resistivity ρ_{xx} at several points within the resistance plateau.

A series of eight symmetrically arranged Hall measurements at five pairs of Hall contacts (see inset in Fig. 2b) are applied in the following order: (1) contacts 4 and 5, (2) contacts 2 and 7, (3) contacts 2 and 3, (4) contacts 6 and 3, (5) contacts 6 and 7, (6) contacts 2 and 3, (7) contacts 2 and 7, (8) contacts 4 and 5. Whereas the contact pairs 2 and 7 (and 6 and 3) are diagonally aligned and thus have a longitudinal resistance component, the other contact pairs 4 and 5, 2 and 3, and 6 and 7 are orthogonally aligned Hall contacts. Therefore, in the case where a device exhibits equal quantization in all regions, the Hall resistances and corresponding bridge voltages at the Hall contact pairs 4 and 5, 2 and 3, and 6 and 7 should be the same within the expanded uncertainties. Since the remaining contact pairs 2 and 7 (and 6 and 3) have a longitudinal component across the full accessible length of the device, they should deviate from the previous three pairs according to their longitudinal resistance component. The results of the practical example, plotted in Fig. 2b, represent a typical pattern of such a measurement series. The difference voltage in nV represents average data derived using a CCC in a series of current-reversed measurements. The measured bridge voltage difference can be used to calculate the value of the unknown resistor if the reference resistor value, winding ratio, and compensation network configuration are known quantities (Götz et al. 2009).

In the case of instabilities in the QHR device or the reference resistor, the measurement results can be asymmetrically distributed or not be reproducible over time within the expanded uncertainties. Additionally, the noise and uncertainty figures of the individual measurements should be similar for all pairs of contacts. A typical expanded ($k = 2$) type A uncertainty of a CCC bridge voltage in a QHR versus a 100 Ω measurement is below 0.5 nV after 48 measurement cycles. To be able to identify instabilities caused by the reference resistor, it is recommended to simultaneously record the ambient pressure and resistor temperature during the measurement as shown in the bottom panels of Fig. 2b. The final dc calibration procedure in which the QHR is used as a reference is typically realized by using the center Hall contact pair 4 and 5.

Expansion of QHR Device Capabilities

QHR devices soon became the norm in the electrical metrology community, with many of the NMI efforts implementing the new standards based on GaAs devices (Small et al. 1989; Cage et al. 1989; Delahaye and Jeckelmann 2003; Jeckelmann and Jeanneret 2001). Compared to the part-per-million or larger changes that occurred over the years before 1990 in many NMI ohm representations based on standard resistors, better agreement was obtained (by an order of magnitude) between the various worldwide NMIs in later resistance intercomparisons. However, to calibrate standards at resistance levels far removed from the QHR value, several stages of resistance ratio scaling were required and the uncertainty increased with each stage. The next natural step for metrologists was to examine whether or not these QHR devices could accommodate other values of resistance so that the calibration chain could be shortened. More advanced experiments and models were developed by studies of QHE behavior and showed how this may be accomplished by constructing quantum Hall array resistance standards (QHARS) (Oe et al. 2013; Poirier et al. 2004; Ortolano et al. 2014; Konemann et al. 2011).

For instance, in Oe et al., a 10 kΩ QHARS device was designed and consisted of 16 Hall bars, providing a more easily accessible decade resistance value compared to previous work (Oe et al. 2013). The nominal value of the device was about 34 nΩ/Ω from 10 kΩ (based on R_{K-90}). The design and final device can be seen in Fig. 3. The device was measured using a CCC to compare the device against an artifact calibrated using another well-characterized QHR device. In this case, a 100 Ω

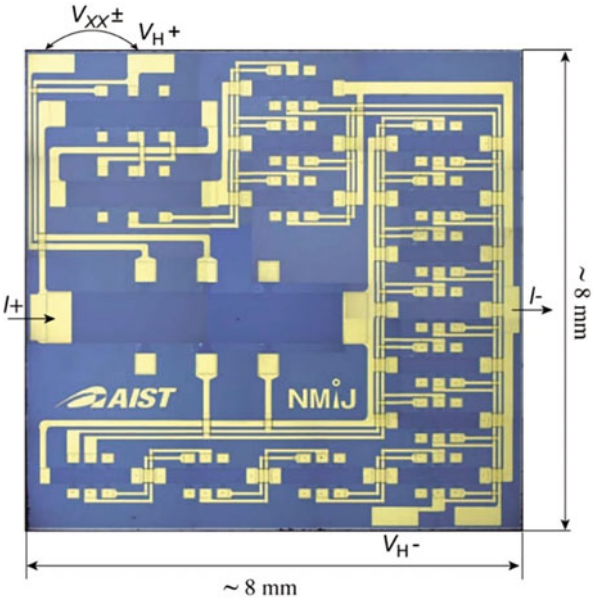

Fig. 3 A 10 kΩ QHARS device is shown with 16 Hall bars. The chip has lateral dimensions of 8 mm per side and was fabricated to be mounted on a standard transistor outline (TO)-8 package. The contact pads are labeled based on whether they are used for current injection or voltage measurements. (© 2013 IEEE. Reprinted, with permission, from (Oe et al. 2013))

standard resistor was used to verify that the array device agreed with its nominal value to within approximately one part in 10^8. The work also proposed new combinations of Hall bars such that the array output could be customized for any of the decade values between 100 Ω and 1 MΩ (Oe et al. 2013).

In addition to the benefits gained from expanding GaAs-based devices further into the world of metrology using direct current (DC), expansion was also explored in the realm of alternating current (AC). Various NMIs began standardizing impedance by using the QHE in an effort to replace the calculable capacitor, which is a difficult apparatus to construct and time-consuming to operate (Ahlers et al. 2009; Cabiati et al. 1999; Bykov et al. 2005; Hartland et al. 1995; Wood et al. 1997). The QHE still exhibited plateaus even at very high excitation frequencies, as shown in Bykov et al., which described the behavior of the 2D electron system in GaAs-based devices for frequencies between 10 KHz and 20 GHz (Bykov et al. 2005). However, when the applied current was AC, a new set of oscillations dependent on the magnetic field was found. This is relevant for impedance metrology because having a longitudinal resistance very close to zero is one mark of a well-quantized device, and having AC frequency-dependent oscillations would undoubtedly increase the uncertainty associated with the QHR. Challenges in adopting QHR technology still exist today in this branch of electrical metrology.

The Graphene Era Begins

Comparing Graphene to GaAs

At the end of the 2000s, research in 2D materials like graphene became prevalent (Zhang et al. 2005; Novoselov et al. 2005, 2007; De Heer et al. 2007). It was evident that QHR measurements could be graphene-based, with fabrications performed by chemical vapor deposition (CVD) (Jabakhanji et al. 2014), epitaxial growth (De Heer et al. 2007; Janssen et al. 2012) (Fig. 4), and the exfoliation of graphite (Giesbers et al. 2008). Given the many methods of available graphene synthesis, efforts to find an optimal synthesis method for metrological purposes were underway. Exfoliated graphene was widely known to exhibit the highest mobilities due to its pristine crystallinity. It was a primary initial candidate as far as metrological testing was concerned. It was found in Giesbers et al. that devices constructed from flakes of graphene had low breakdown currents relative to GaAs-based counterparts, with currents on the order of 1 μA being the maximum one could apply before observing a breakdown in the QHE (Giesbers et al. 2008).

Among the other forms of synthesis, epitaxial graphene (EG) proved to be the most promising for metrological applications (Tzalenchuk et al. 2010). In their work, Tzalenchuk et al. fabricated EG devices and performed precision measurements, achieving quantization uncertainties of about 3 nΩ/Ω. An example of their measurements is shown in Fig. 5, where (a) is a demonstration of how viable the graphene-based device was to be a suitable QHR standard. It was the start of a new chapter for standards, but additional work was required to exceed the stringent temperature requirement of 300 mK and low current capability of 12 μA, relative to modern day

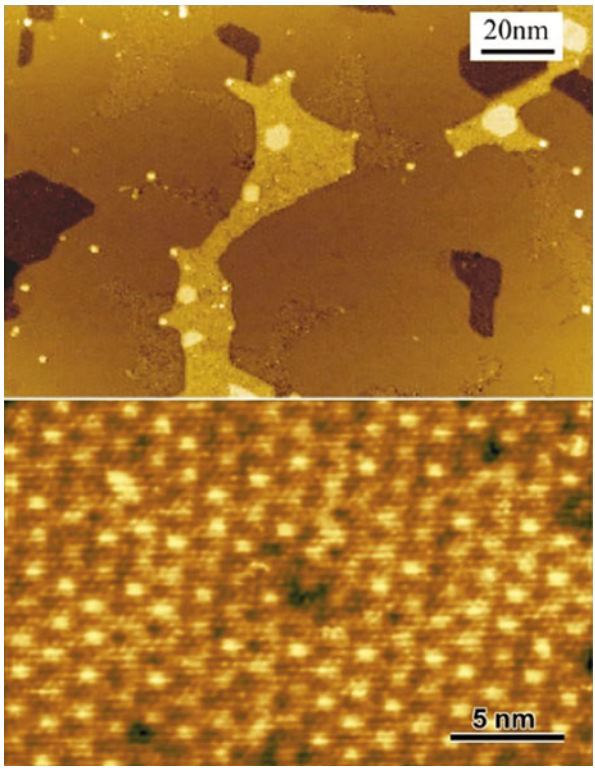

Fig. 4 Scanning tunneling microscopy images (taken at 0.8 V sample bias, 100 pA) show a single layer of epitaxially grown graphene on SiC(0001). The top and bottom panels show two different magnifications of a neighborhood of the grown crystal. (Reprinted from (De Heer et al. 2007))

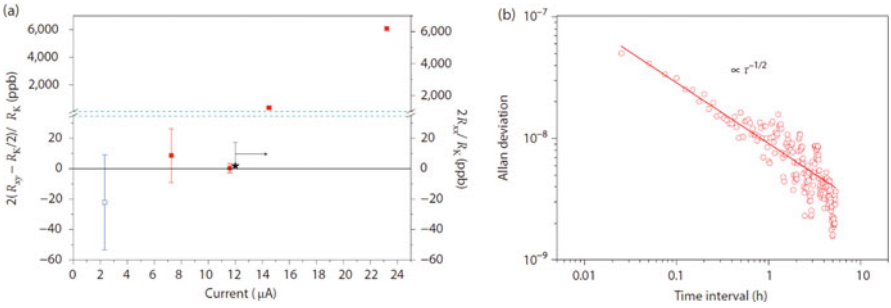

Fig. 5 The Hall resistance quantization accuracy was determined in these measurements. (**a**) The mean relative deviation of R_{xy} from $R_K/2$ is shown at different currents (ppb, or nΩ/Ω) on the right vertical axis. Recall that R_{xy} is nominally measuring the $\nu = 2$ plateau of the QHE. The value of the deviation at the smallest current was measured at 4.2 K (blue squares), and all other measurements were performed at 300 mK (red squares). To achieve the highest accuracies, using an 11.6 μA source–drain current (14 T, 300 mK), at least one of the measurements was performed over 11 h. The right vertical axis shows R_{xx}/R_K, which is represented as the black star along with its measurement uncertainty. (**b**) The Allan deviation of R_{xy} from $R_K/2$ is plotted as a function of time (of the measurement). The square root dependence (as fit by the solid red curve) is indicative of white noise. (Reprinted by permission from Springer Nature Customer Service Centre GmbH: (Tzalenchuk et al. 2010))

capabilities (Tzalenchuk et al. 2010). EG graphene as QHR standards soon provided increased current capability and showed impressive plateau width for high and low magnetic field and higher temperatures. EG was also synthesized on SiC via CVD in 2015, showing that graphene could provide standards-quality resistance at temperatures up to 10 K (Lafont et al. 2015). This CVD method relied on the SiC substrate to provide a template for the growth of EG from a gas containing a hydrocarbon precursor.

Establishing Graphene as a Global Resistance Standard

EG has been used as part of the traceability chain of electrical resistance dissemination in the United States since early 2017 (Janssen et al. 2011; Oe et al. 2019; Woszczyna et al. 2012; Satrapinski et al. 2013; Rigosi et al. 2019a). The preceding years from 2010 onward were dedicated to the study of basic fabrication processes and measurement techniques so that EG-based QHR devices could be accepted as the replacement for GaAs-based QHR devices. With optimized EG devices, performance far surpassed that of the best GaAs-based devices. A compilation of these efforts can be linked to multiple institutes (Lara-Avila et al. 2011; Rigosi et al. 2017, 2018, 2019b; Riedl et al. 2010; Janssen et al. 2015; Kruskopf et al. 2016; He et al. 2018). EG-based devices suitable for metrology required high-quality EG to the point of substantial scalability (centimeter-scale) as well as stabilization of electrical properties to have a long shelf life and end-user ease-of-use.

In one example of EG-based QHR development, implemented by two separate groups, devices had become compatible with a 5 T table-top cryocooler system (Rigosi et al. 2019a; Janssen et al. 2015). Janssen et al. first demonstrated this type of measurement with a table-top system in 2015, pushing the bounds of operability to lower fields and higher temperatures. The advantage of such a system also includes the removal of the need for liquid helium, enabling continuous, year-round operation of the QHR without the use of liquid cryogens. The $\nu = 2$ plateau in EG devices is the primary level used to disseminate the ohm, much like for GaAs-based QHRs. In fact, EG provides this plateau in a very robust way and for a large range of magnetic fields because of Fermi level pinning from the covalently bonded, insulating carbon layer directly beneath the conducting graphene layer (Lara-Avila et al. 2011).

For the 5 T table-top system at the National Institute of Standards and Technology (NIST), the EG QHR device plateau value was scaled to a 1 kΩ standard using a binary CCC and a DCC (Rigosi et al. 2019a). The uncertainties that were achieved with this equipment matched those obtained in GaAs-based QHR systems (i.e., on the unit order of nΩ/Ω). The results of some of these measurements are shown in Fig. 6 at three different currents. The limit of nearly 100 μA gives EG-based QHRs the edge over GaAs, especially since these measurements were performed at approximately 3 K. Another example of graphene being established as the new QHR standard comes from Lafont et al., whose work hardened the global question of when graphene would surpass GaAs in various respects (Lafont et al. 2015). An example of this investigation is shown in Fig. 7, where the sample of choice was

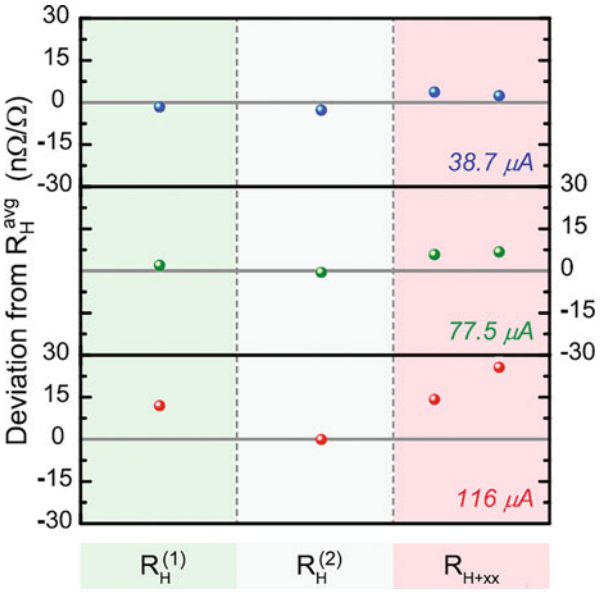

Fig. 6 Binary winding CCC measurements were performed with an EG-based QHR device in a cryogen-free table-top system at 5 T and 3 K. The CCC measurement data are displayed in three rows, with the upper, middle, and lower data corresponding to the source-drain currents of 38.7 μA, 77.5 μA, and 116 μA, respectively. Each column shows a comparison between the resistance of the $\nu = 2$ plateau and a 1 kΩ resistor, with the left two columns representing orthogonal contact pairs ($R_H^{(1)}$ and $R_H^{(2)}$). The third column represents the two diagonal pairs formed by the same contacts as the two orthogonal pairs. It is labelled as R_{H+xx} and indicates the impact of the residual longitudinal resistance. The type A measurement uncertainties are smaller than the data points, and the type B uncertainties are under 2 nΩ/Ω. (© 2019 IEEE. Reprinted, with permission, from (Rigosi et al. 2019a))

CVD-grown epitaxial graphene on SiC. Overall, with the success of the national institute's QHR efforts, the evidence for EG-based devices surpassing GaAs-based QHRs has been widely accepted.

Improvements in Measurement Infrastructure

Given the establishment of better QHR devices for resistance standards, the corresponding equipment and infrastructure with which one can disseminate the ohm had to also improve or at least be shown to maintain its compatibility with EG-based QHRs. The measurement of the ratio between a QHR device and a standard resistor must achieve uncertainties that are comparable to or better than the stability of those resistors. This criterion would support the notion of reliable traceability in that system. Measurements with a longer integration time produce better results because the ratios are well-maintained. The obvious benefits from room-temperature bridge systems like the DCC include an ability to deliver

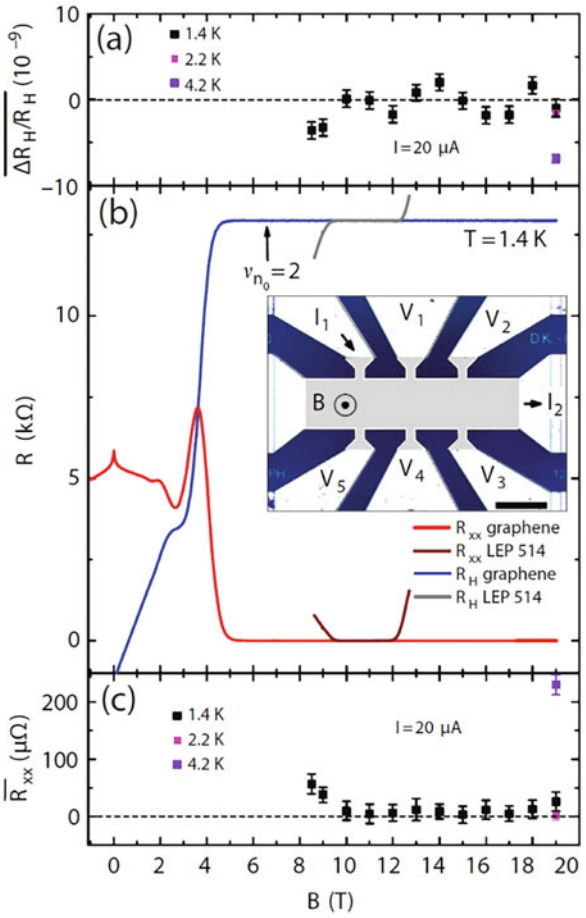

Fig. 7 The magnetic field dependence for Hall resistance in CVD-grown epitaxial graphene is shown. (**a**) The Hall resistance (R_H) deviation is measured on the $\nu = 2$ plateau at 1.4 K (black), 2.2 K (magenta), and 4.2 K (violet). (**b**) R_{xx} and R_H were measured with a source-drain current of 20 μA, with both types of voltages measured using the (V_1, V_2) and (V_2, V_3) contact pads, respectively. The magnetic field was energized to values between 1 T and 19 T for the graphene device (shown by the red and blue curves) and from 8 T to 13 T for the GaAs device (maroon and gray curves). The onset of the Landau level was calculated by using the carrier density at low magnetic fields, which is related to the slope of R_H near zero field. The inset shows an optical image of the device (scale bar is 100 mm). (**c**) Data for precision measurements of R_{xx} are shown at 1.4 K (black), 2.2 K (magenta), and 4.2 K (violet). The error bars represent combined standard uncertainties (1σ). (Lafont et al. (2015) is an open-access article distributed under the terms of the Creative Commons CC BY license, which permits unrestricted use, distribution, and reproduction in any medium)

measurements to laboratories in both NMI and non-NMI settings with greater frequently (year-round) due to their operation not requiring cryogens (MacMartin and Kusters 1966). Additionally, both DCCs and CCCs have been improved so that they are more user-friendly and automated, removing, at least in the case of the DCC,

most of the dependence on specialized knowledge that one normally expects for the more complex cryogenic counterparts (Williams 2011; Drung et al. 2009). Some NMIs have shown that the automated binary-ratio CCCs have type B uncertainties below 0.001 μΩ/Ω, a useful feature when dealing with precise QHR comparisons (Sullivan and Dziuba 1974; Grohmann et al. 1974; Williams et al. 2010). CCC scaling methods can achieve resistance ratios of 4000 or more, enabling measurements for sub-nanoampere current levels, whereas DCCs achieve similar current sensitivity only with larger currents.

The DCC is a room-temperature current comparator, and this simplifies the operation and cost compared to SQUID-based CCC systems. The work by MacMartin and Kusters shows the development of a DCC for comparing four-terminal resistances (MacMartin and Kusters 1966). It works by measuring a current ratio, represented as a turns ratio, that is balanced by detecting magnetic hysteresis in magnetic cores using a modulator. Manual dials allowed the ratio to be adjustable in part-per-million steps. The two current sources in the DCC are isolated so that there is no current in the galvanometer circuit when fully balanced. Their DCC was able to measure and compare the ratio of two isolated direct currents and adjust the ratio to balance the galvanometer. A detailed description provided a discussion of its accuracy limitations (MacMartin and Kusters 1966). In the optimum operating range, the authors achieved accuracies that were better than one part per million. The bridge was designed to accommodate the scaling of resistance standards from 100 Ω to a less than 1 mΩ and can be used for any ratio from 1 to 1000, thus permitting one to calibrate low value resistors and current shunts whose accuracies would be limited only by the level of noise and the stability of the resistor (MacMartin and Kusters 1966). Modern DCC systems are fully automated to control current levels, ratios, and resistance values, and allow scaling of resistance standards from 100 kΩ to below 10 μΩ.

From the previously mentioned work at NIST (Rigosi et al. 2019a), the two bridges demonstrate the applicability of using a modern room-temperature DCC with an EG-based QHR device to obtain uncertainty near 0.01 mΩ/Ω. As shown in Fig. 8a, the DCC measurements extend the range of the applied source-drain currents when scaled from the EG-based QHR device to a 1 kΩ resistor. The inset of Fig. 8a shows how sample heating from a high applied current affects the overall sample temperature. This information was relevant because high currents applied to EG-based QHR devices improves the resolution of the DCC measurements, but will degrade the accuracy of the QHE due to too high a temperature. Ultimately, the work found that table-top systems, in which the QHR is cooled indirectly by conduction and is in vacuum, may limit how much current can be applied (Rigosi et al. 2019a). Larger cryomechanical chillers are designed to immerse the device in a small volume of liquid helium and provide better temperature stabilization by contact with the liquid bath.

As of the present day, CCC bridges outperform DCCs in terms of achievable uncertainties. However, in order to more easily disseminate QHR technology globally, room temperature DCCs are preferred in terms of ease-of-use and fewer resource needs as well as year-round operability. For this reason, a comparison between the two methods was examined and is shown in Fig. 8b, where the number

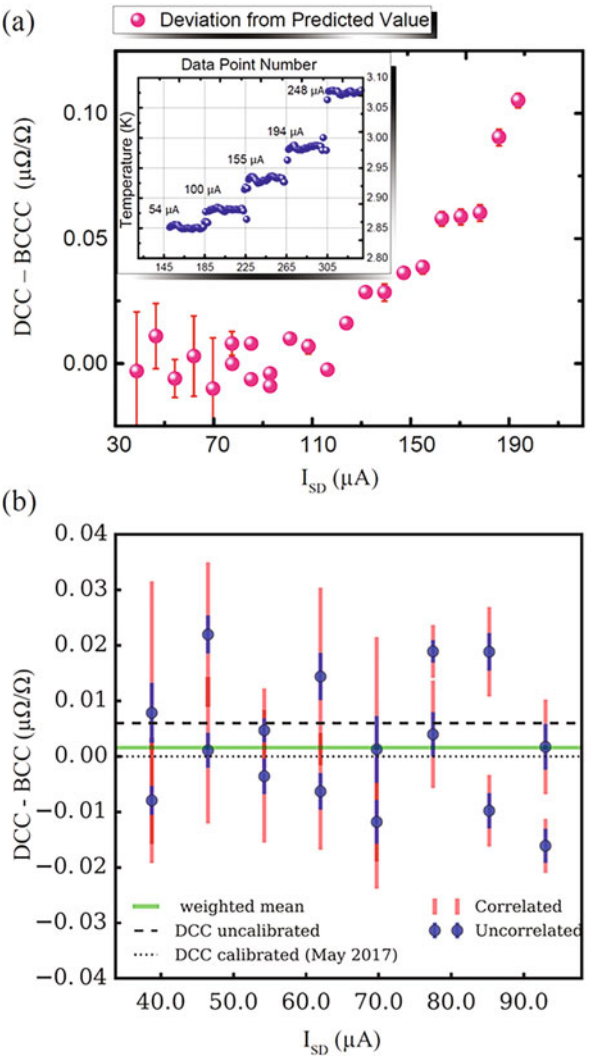

Fig. 8 (**a**) Current dependence data of DCC ratios for a 1 kΩ resistor are shown in magenta, based on the same EG QHR device used with a binary winding CCC (BCCC). The DCC results confirm that the QHR device remains quantized to within 0.01 μΩ/Ω up to approximately 116 μA. Data for higher source-drain currents shows the effect of heating of the sample caused by power dissipation, with temperature data for extended current ranges shown in the inset with blue data points. Some type A measurement uncertainties (in red) are smaller than the data points. (**b**) DCC data for the ratio of the QHR to 1 kΩ are shown for increasing source-drain currents from 38 μA to 94 μA and are normalized to the average results of BCCC scaling (using 38.7 μA and 77.5 μA). The red error bars show the expanded ($k = 2$) with uncorrelated uncertainties whereas the blue error bars show the expanded ($k = 2$) uncertainty reported by the measurement device. (© 2019 IEEE. Reprinted, with permission, from (Rigosi et al. 2019a))

of DCC data points averaged was varied inversely with the square of the applied voltage $V \approx I_{SD} \times R_H$ to obtain a similar type A uncertainty for all values of the source-drain current, with measurement durations ranging from 110 min for the 0.5 V measurements to 24 min for the 1.2 V measurements. Further, in Fig. 8b, the uncertainties indicated in blue were produced by the DCC software. The larger, red error bars are uncertainties that take into account known statistical correlations in the data (Zhang 2006). Potential noise from the cryogen-free mechanical refrigeration system may also interfere to some degree with the balancing algorithm. A similar issue exists for CCC systems, where the additional noise from vibration may

increase noise and may cause the SQUID to lose the delicate balance required to maintain its set point through feedback.

Even though DCCs would make global dissemination easier for some research institutes and smaller NMIs, CCC technology is still important for international metrology at the highest levels. Williams et al. demonstrated a design for an automated CCC in order to perform routine NMI measurements (Williams et al. 2010). Their system uses a CCC in a low loss liquid helium storage vessel and may be continuously operated with isolated supplies coming from the mains power. All parameters were shown to be digitally controlled, and the noise sources present in the system were analyzed using the standard Allan deviation, leading to the conclusion that one may eliminate non-white noise sources simply by choosing the appropriate current reversal rate. New generations of CCC systems with integrated fast ramping current sources, nano-voltmeter, and a compensation network for improved bridge balancing have become widely used at worldwide NMIs (Drung et al. 2009).

As progress continues, NMIs have both anticipated and in many cases achieved these goals for EG-based QHR devices. Lastly, to begin expanding on the functionality of EG-based devices, it would need to be shown that precise resistance scaling could be done to reference resistors from the QHR by using voltages larger than those available in standard DCCs and CCCs. These efforts are underway and hope to eventually accelerate dissemination efforts.

Expanding the Use of the Quantum Hall Effect in Graphene

Assembly of p-n Junctions in Graphene-Based Devices

With EG-based QHR devices established as national standards at several NMIs, as well as the metrology community in agreement that comparisons against GaAs-based QHR devices had been accomplished, the next steps became clearer concerning how the EG-based QHR could be further developed. Unlike the QHR in GaAs, only the $\nu = 2$ plateau is well quantized, and although other Landau levels exist in graphene, they simply do not offer the same level of precision as the $\nu = 2$ plateau (Zhang et al. 2005; Hu et al. 2021). Since the early 1990s, it has been of interest that QHR devices have a means of interconnecting several single Hall bar elements without loss of precision, and this has since been of great research interest. As discussed earlier, this work allowed QHR devices to output more than the single value at the $\nu = 2$ plateau (about 12.9 kΩ) through arrays of devices, using what are described as multi-series interconnections (Delahaye 1993). With graphene, which supports both electron conduction and hole conduction (with positive carrier charge) the first natural question is whether one may use the plateau at $\nu = -2$, in addition to the plateau at $\nu = 2$. This array would avoid the need for manufactured interconnections entirely, using an interface of electronic states internal to the 2D conductor.

This alternative for outputting new quantized values utilizes *p-n* junctions (*pn*Js) to provide useful and controllable values of resistance quantization (Hu et al. 2018a, b; Woszczyna et al. 2011; Rigosi et al. 2019c; Momtaz et al. 2020). Woszczyna et al. approached this issue a decade ago (Woszczyna et al. 2011), noting that since metallic

leads had to cross paths in traditional GaAs-based devices, typical fabrication methods would be overly complicated, having to include some form of multilayer interconnect technology. Any leakage between leads would likely generate an additional Hall voltage, thus becoming a detriment to the achievable uncertainty. Graphene is more accessible for the fabrication of small structures than the GaAs heterostructure interface, where contacts must be alloyed through the insulating surface layer. Charge carrier control by gating offered the opportunity to combine two regions of opposite charge carrier polarity. Rather than requiring interconnections, such a device only needs a set of uniform top gates to modulate regions between the $\nu = -2$ and $\nu = 2$ plateaus. This work demonstrated that *pn*Js were possible and worth exploring as a viable extension for QHR standards.

The fabrication of *pn*J devices larger than the order 100 μm is presently limited by the difficulty of producing of high-quality, exfoliated single-crystal hexagonal boron nitrate (*h*-BN) as an insulating material for top gating. As demonstrated by Hu et al., numerous pnJ's can be fabricated in a single device of size 100 μm or less, rendering it possible to implement reliable top gates for adjusting the carrier density in gated EG-based devices (Hu et al. 2018a). The underlying physics of these *pn*Js makes it possible to construct devices that can access quantized resistance values that are fractional or integer multiples of R_K. Hu et al. made an assessment for one such *pn*J device at the $\nu = -2$ and $\nu = 2$ plateaus (summing to about 25.8 kΩ), with data shown in Fig. 9. A DCC was used to provide turn-key resistance traceability and

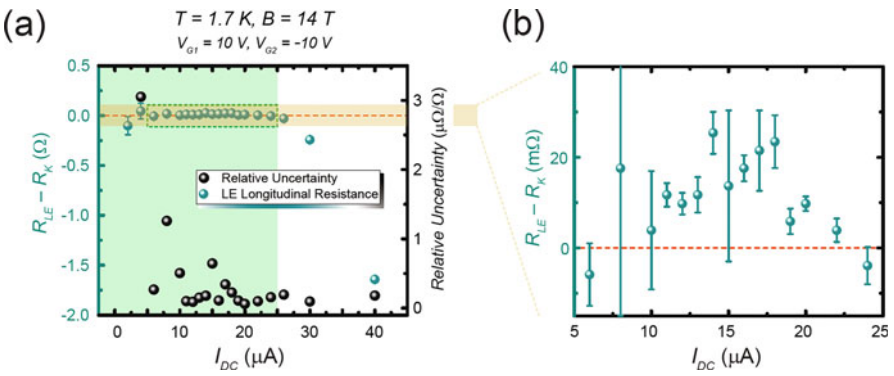

Fig. 9 (**a**) The lower edge (LE) of the device in Hu et al. was compared against a 10 kΩ standard resistor with a DCC to give an assessment of the quality of quantization exhibited by the device. The resistor was selected based on its traceability to a quantum resistance standard at the National Institute of Standards and Technology. The turquoise points show the DCC measurements as deviations from R_K on the left axis and the relative uncertainties of those deviations with DC current. The relative uncertainties improve with increasing current, but the device loses its optimal quantization after the critical current of 24 μA. The shaded green area indicates the well-quantized region. (**b**) The beige shaded area in (**a**) is magnified to show the deviations' error bars as well as the reference to zero deviation, marked as an orange dashed line. The error bars represent a 1σ deviation from the mean, where each data point represents an average of a set of data taken at each value of current. (Hu et al. (2018a) is an open access article distributed under the terms of the Creative Commons CC BY license, which permits unrestricted use, distribution, and reproduction in any medium)

compare against a 10 kΩ standard resistor (see Fig. 9a). The measurement time for each data point with the DCC was 15 min, with an orange shaded region representing Fig. 9b, which clarifies the deviation of the DCC measurements with respect to zero. In Fig. 9a, the right axis is represented by black points and gives the relative uncertainty of each measurement as a function of source-drain current. For this *pn*J device, a precision of about 2×10^7 was achieved as shown in Fig. 9b. Although this may be one or two orders of magnitude below what is possible for a conventional Hall bar device, one must recall that, based on this work, a programmable resistance standard may be built using the demonstrated techniques. Such flexibility and expansion of accessible parameter space could justify its further technical exploration within resistance metrology.

These types of devices are not as well-studied as other conventionally prepared devices, but given their interesting and rather unique properties, *pn*J devices could offer a second, more fundamental path to avoid any resistance from interconnecting metallic contacts and multiple device connections. This possibility is illustrated in a recent experiment by Momtaz et al. and demonstrates how a programmable quantum Hall circuit could implement an iterative voltage bisection scheme, thus permitting the user to access any binary fraction of the $\nu = 2$ plateau resistance (Momtaz et al. 2020). Their proposed circuit designs offer potential advantages for resistance metrology, as summarized here: first, their circuit contains no internal Ohmic contacts, a recurring problem in interconnected QHR circuits. Second, there is a logarithmic scaling of the complexity of the design as a function of the required fractional resolution. This scaling feature is a major advantage compared with a standard QHARS device. The latter might use hundreds of distinct multi-contact Hall bars, whereas a bisection circuit can output a similar number of values while only needing a small number of elements. The approach is thought to match, or become comparable to, the present limits of QHR standards.

The design does have some limitations, as pointed out by Momtaz et al. They noted that, even though the last bisection stages of the device were controlling a finer portion of the output value, each stage still relied on QHE states equilibrating across a junction barrier, emphasizing the importance of the quality of the junction itself. Their preliminary numerical estimates suggest that imperfections in the device would be partially fixable since any absolute errors caused by imperfect mixing would not be amplified through the remaining sections of the device. Nonetheless, this design warrants further study as one way to expand on available quantized resistance outputs.

Using Arrays to Expand the Parameter Space

Recent developments have utilized superconducting materials like NbN and NbTiN to create interconnections compatible with EG-based QHR devices (Kruskopf et al. 2019a, b; He et al. 2021). These metals have high critical temperature (≈10 K) and critical field (≈20 T) so that they may be applied to QHR devices during measurement, as shown by Kruskopf et al. (2019b). They argue that array technology based

on superconducting metals is preferred and can eliminate accumulated resistance and voltage errors at contacts and interconnections. They demonstrated that the application of NbTiN, along with superconducting split contacts, enabled both four-terminal and two-terminal precision measurements without the need for insulated crossovers. The split contacts are inspired by the multiple series approach described by Delahaye (1993), with reduced separation between the interconnections. Since the resulting contact resistances become much smaller than R_K, it becomes straightforward to fabricate series and parallel connections as fundamental device elements. The limits of this technology have not yet been determined, nonetheless, the merits of this technique seem to point to such structures as the next generation of QHR devices.

Another example of array technology that expands on the parameter space comes from Park et al. (2020), where they successfully construct 10 single Hall bars in series with EG on SiC. They operated this device at the $\nu = 2$ plateau near 129 kΩ with precision measurements made using a CCC. While measuring the device at a magnetic field of 6 T and a temperature of 4 K, they were able to achieve a relative uncertainty of approximately 4×10^{-8}, approaching the state-of-the-art for this resistance level. Despite only being able to inject a low double-digit electrical current (in μA), their efforts added support to the notion of expanding QHR values with QHARS devices.

One difficulty that could arise from making these arrays, especially as they increase in lateral size, is how to make their carrier densities uniform. For that, there are two prime examples of accomplishing this task, with both methods being user-friendly and attaining a long shelf life for the device. The first method involves functionalizing the EG surface with $Cr(CO)_3$ (Rigosi et al. 2019b), and what this process enables the user to do is apply heat for a specified time to obtain a predictable carrier density. The advantage is that this process is reversible and may be cycled without damaging the device. One of the features of this method is that storing the device in air for long periods of time simply resets the Fermi level to an energy close to the Dirac point. The same anneal may be applied to reacquire the value of the carrier density. Once in an inert environment or at colder temperatures, the carrier density of the device remains stable. The second method involves a polymer-assisted doping process (He et al. 2018), which allows the user to adjust the carrier density by also using an anneal. This method retains some reversibility as long as the polymer will retain its matrix. Both methods are highly scalable and give extended shelf life to EG-based standards.

With the ability to make very large QHARS devices having controllable and uniform carrier densities, researchers were able to construct a 1 kΩ array based on 13 single Hall bar elements in parallel (Hu et al. 2021). A CCC was used to measure two 13-element arrays. At 1.6 K, the array devices achieved useful quantization above 5 T. One such array measurement is shown in Fig. 10. In this case, the DCC measurements verified that, in the high-field limit, the resistance for both B-field directions approached the value $R_K/26$ (about 992.8 Ω) to better than one part in 10^8 for currents up to 700 μA. As is typical, the CCC ratio uncertainty was below one part in 10^9, and most of the uncertainty had originated from the 100 Ω artifact resistance standards.

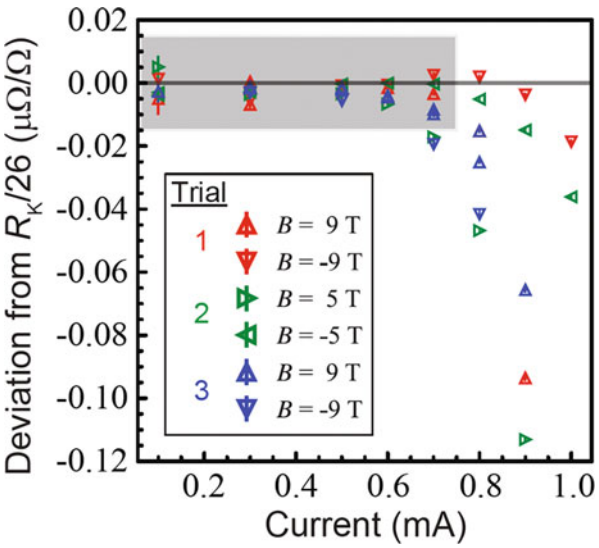

Fig. 10 The $\nu = 2$ plateau was measured at selected values of the magnetic field (B) for device 1 at 1.6 K in a conventional cryostat using a room-temperature DCC. DCC measurements verify that, in the high-field limit, the resistance for both B-field directions approaches the value $R_K/26$ (about 992.8 Ω) to better than one part in 10^8 for currents up to 700 μA, confirming that this array device utilizes highly homogeneous graphene. All expanded uncertainty values given here are for a 2σ confidence interval. (Hu et al. (2021) is an open access article distributed under the terms of the Creative Commons CC BY license, which permits unrestricted use, distribution, and reproduction in any medium)

Recent work by He et al. shows quantum Hall measurements performed on a large QHARS device containing 236 individual EG Hall bars (He et al. 2021). Given the difficulty of verifying the longitudinal resistance as zero for every element, they utilized a direct comparison between two similar EG-based QHARS devices to verify the accuracy of quantization. The design of this array is such that two subarrays with 118 parallel elements are connected in series, having a nominal resistance of $h/236e^2$ (about 109.4 Ω) at the $\nu = 2$ plateau (Fig. 11). The Hall bars were designed to be circular in order to both have a symmetrical design and to be able to fit many devices into a small area (He et al. 2021). The contact pads and interconnections were fabricated from NbN and support currents on the order of 10 mA at 2 K and 5 T. Additionally, a split contact design was implemented to minimize the contact resistance, much like other previously reported work (Kruskopf et al. 2019a, b). The two QHARS devices were compared using high precision measurements, showing no significant deviation of their output resistance within 0.2 nΩ/Ω. Within the next few years, given the increasing complexity of EG-based QHARS devices, even larger and more versatile arrays are expected to make available an abundance of new quantized resistance values.

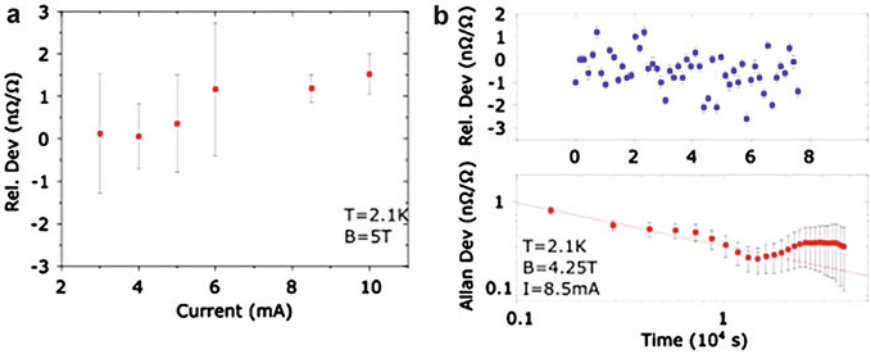

Fig. 11 High bias current measurements on arrays. (**a**) CCC measurements are shown of a direct comparison between subarrays to show that no significant deviation occurs until 8.5 mA. The data consist of the mean of 5–10 CCC readings, each of which is 20 min long. The top graph shows the relative deviation, with error bars of one standard deviation. The bottom graph shows the corresponding Allan deviation. The standard error is limited to 0.25 nΩ/Ω. (He et al. (2021) is an open access article distributed under the terms of the Creative Commons CC BY license, which permits unrestricted use, distribution, and reproduction in any medium)

From DC to AC to the Quantum Ampere

As the *quantum* SI continues to expand in its applications, new research directions are plentiful and of interest to the electrical metrology community. The QHE will continue to be our foundation for disseminating the ohm in the DC realm. The vast improvements in EG-based QHR technology are beginning to inspire efforts to develop more sophisticated devices suitable for AC resistance standards. This subfield of electrical metrology focuses on the calibration of impedance and is conventionally obtained from systems like a calculable capacitor (Thompson and Lampard 1956; Clothier 1965; Cutkosky 1974). Such a system has allowed for the calibration of capacitors, inductors, or AC resistors, which is essentially a measurement of complex ratios of impedance, where the signal phase depends on the type of standard. Historically, the design and operation of this kind of system have been challenging because of unavoidable fringe electric fields, imperfections in capacitor electrode construction, and a long chain of required bridges and standards.

The next step for improving AC standards may be to introduce EG-based QHR devices, as Kruskopf et al. have done recently (Kruskopf et al. 2021). They used a conventional EG-based device to analyze the frequency dependence of losses and to determine the characteristic internal capacitances. The environment of the device included a double shield used as an active electrode, to compensate for capacitive effects, as shown in Fig. 12a. Figure 12b displays the set of magnetocapacitance measurements corresponding to a configuration of the capacitance (Cx) between the active electrode (left side of (a)) and the EG Hall bar between points A and B in Fig. 12a. Cx was compared with a variable precision reference capacitor using a simple configuration used for other traditional measurements (Kruskopf et al. 2021).

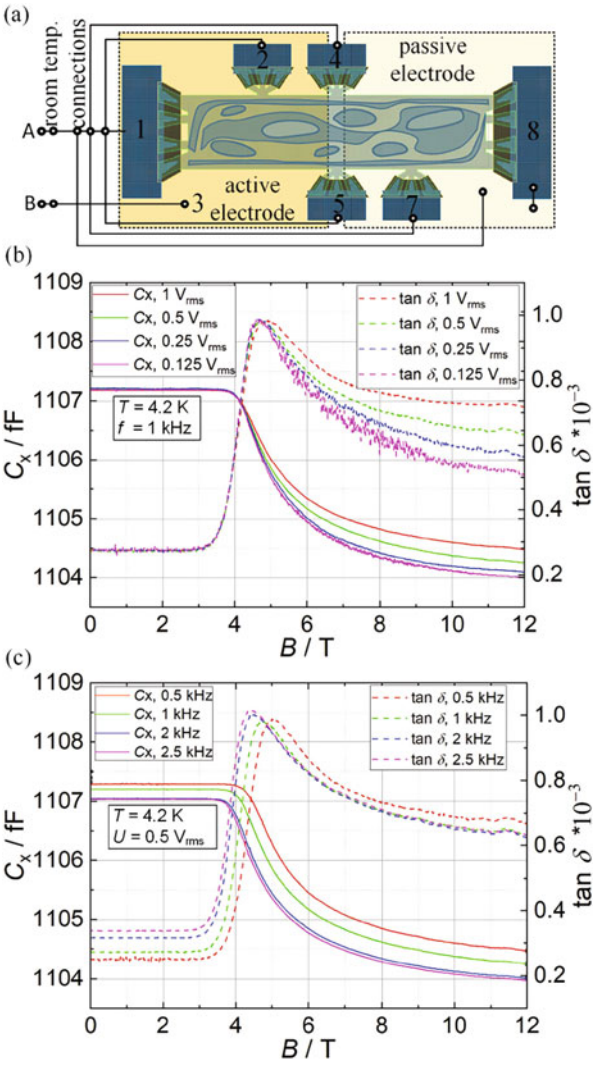

Fig. 12 Magnetocapacitance measurement data of an example device are shown (taken at 4.2 K). (**a**) An illustration captures the various elements of the used magnetocapacitance measurement configuration. As one electrode gets shorted to the labeled passive electrode (pin 8), the other electrode is then used to characterize C and tan(δ). Compressible and incompressible states of the 2D electron system are represented as different colors within the EG Hall bar. (**b**) The capacitance is plotted as a function of magnetic field for a set of different voltages. (**c**) A similar plot to (**b**) is shown for a set of different frequencies. The dissipation factor (tan(δ)) is also plotted representing the losses between the active electrode and the EG Hall bar device. (Kruskopf et al. (2021) is an open access article distributed under the terms of the Creative Commons CC BY license, which permits unrestricted use, distribution, and reproduction in any medium)

Figure 12a shows the passive electrode on right side, shorted to the EG contact, and therefore not contributing to the measurement of *Cx*.

The AC QHR may directly access the units of capacitance and inductance with high precision (Kruskopf et al. 2021; Lüönd et al. 2017). Kruskopf et al. show the voltage and frequency dependencies of the magnetocapacitance in Fig. 12b, c, respectively, along with the associated dissipation factor for an example device. By appropriately modeling the compressible and incompressible states, the observed dissipation factors may be explained rather well. At low magnetic field values, the 2D electron system in the EG is not quantized and is thus accurately representable as

a semi-metal with dominant compressible states. However, when the magnetic field is increased, *Cx* starts to decrease around 4 T and does so by nearly 3 fF in both cases as 12 T is approached. This phenomenon may be due to the increase in incompressible regions that form, which are themselves transparent to electric fields. Furthermore, the dissipation factor is observed to first increase, peaking during the transition region as the resistance plateau in EG forms. As the quantization accuracy improves at higher magnetic fields, the observed dissipative losses decrease again to about tan $(\delta) = 0.0003$. Overall, these are but a few magnetocapacitance measurements that demonstrate the viability of efforts to better understand the physical phenomena driving the observations made in the QHE regime while under AC conditions. This pursuit of AC QHR standards aims to advance how units such as the farad and henry are realized by using fundamental constants instead of dimensional measurements.

In addition to expanding the influence of the QHE to AC electrical metrology, one may also expand into the realms of electrical current and mass metrology by utilizing EG-based QHARS devices. The realization of the quantum ampere lacks accurate traceability to within 10^{-8} despite the various efforts that exist for developing a current source using single-electron tunneling devices (Giblin et al. 2012; Pekola et al. 2013; Koppinen et al. 2012). The alternative route one may take to realizing the quantum ampere is by building a circuit that effectively combines the QHE and a programmable Josephson junction voltage standard. This combination has been recently assembled to attain a programmable quantum current generator, which may disseminate the ampere at the milliampere range and above (Brun-Picard et al. 2016). The work reported by Brun-Picard et al. demonstrates this construction with a superconducting cryogenic amplifier and to measurement uncertainties of 10^{-8}. Their quantum current source, which is housed as two separate cryogenic systems, can deliver accurate currents down to the microampere range. Considering the orders of magnitude involved, this work renders the programmable quantum current generator a pronounced supplement to electron-pumping mechanisms.

Future Improvements and the Quantum Anomalous Hall Effect

Limitations to the Modern QHR Technology

In order to make an accurate assessment of future needs for quantum metrological applications, it would benefit us to know the limits of the current technology, at least to some extent. For instance, it is not known with substantial certainty how high a current could be applied to a single EG Hall bar, or the upper bound on operational temperature for EG-based devices.

There is much research to be conducted on newer technologies that build on the single Hall bar design despite the noted benefits from their development. For instance, the *pn*J devices have conceptual access to different fractional or integer multiples of R_H (or $R_K/2$) as possible resistances, especially if source and drain currents are allowed for more than two total terminals. If one defines q as a coefficient of R_H, then the following relation may be used (Rigosi et al. 2020):

$$q_{M-1}(n_{M-1}) = \frac{q_{M-2}(n_{M-1}+1)}{n_{M-1} + \frac{q_{M-2}}{q^{(0)}_{M-1}}} \tag{1}$$

In Eq. (1), M is the number of terminals in the *pn*J device, n is the number of junctions between the outermost terminal and its nearest neighbor, and $q^{(0)}_{M-1}$ refers to a default value the device outputs when the configuration in question is modified such that its outermost terminal moves to share the same region as its nearest neighbor (meaning that one of the two outermost regions containing any source or drain terminals has both a source and a drain connected to it) (Rigosi et al. 2020). The key takeaway with this algorithm (Eq. 1) is that an incredibly vast set of available resistances becomes hypothetically possible by simple reverse engineering. The algorithm assumes that the Hall bar is of conventional linearity. That is, each *p* region is adjacent to two other *n* regions unless it is an endpoint. The same would hold true when *p* and *n* are swapped. The equation breaks down when the *pn*J device geometry changes to that of a checkerboard grid or Corbino-type geometry. In all of these cases, the available resistances in this parameter space are vastly abundant and will obviously not be a limiting factor for this species of device. Instead, limitations may stem from imperfections in the device fabrication, an almost inevitable manifestation as the device complexity increases (Rigosi et al. 2019c).

In the limit of the purely hypothetical, should the resistance metrology community wish to scale to decade values only, as per the existing infrastructure, then a programmable resistance standard may be able to provide many decades of quantized resistance output by following the designs proposed by Hu et al. (2018a). The proposed programmable QHR device is illustrated in Fig. 13, with each subfigure defining a small component of a total device. When programmed in a particular way, this single device can output all decades between 100 Ω and 100 MΩ, as summarized by Table 1 (where voltage probes are labeled in reference to Fig. 13c).

With all *pn*J devices, better gating techniques are warranted. In the case of top gating, which is the basis for forming the device measured by Hu et al. (2018a), fabrication is limited by the size of the exfoliated *h*-BN flake typically used as a high-quality dielectric spacer. For bottom gating of EG on SiC, as seen with ion implantation (Waldmann et al. 2011), there has not yet been a demonstration of metrologically viable devices, though there may still be potential for perfecting this technique. Since gating is likely to be the largest limiting factor for *pn*Js, one must instead turn to array technology. Although QHARS devices can theoretically replicate *pn*Js, the necessity of an interconnection will ultimately result in a smaller available parameter space. Nonetheless, the output quantized resistance values offered by this technology may still be sufficiently plentiful for future applications in electrical metrology.

When it comes to arrays, there are several types that can take shape for a potential QHR device. There are the conventional parallel or series devices, which are the subject of recent works (Hu et al. 2021; Kruskopf et al. 2019a, b; He et al. 2021; Park et al. 2020). As one departs from this simpler design, the number of potential quantized resistances that become available rapidly increases. Designs with varying

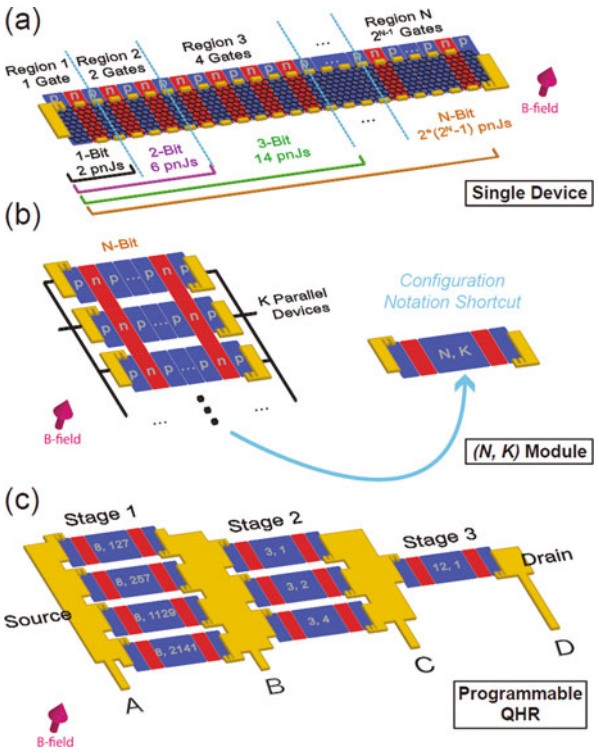

Fig. 13 The proposed device illustrated represents a programmable QHR device for scalable standards. (**a**) An *N-bit* device is illustrated showing how each region is defined and the maximum number of *pn*Js that can be used. (**b**) This device, when connected in parallel with *K* copies of itself, becomes the foundation of the (*N*, *K*) *module*. Each region has a set of gates that extend to all *K* parallel branches. (**c**) The proposed device is illustrated and composed of eight (*N*, *K*) *modules*, four of which are in parallel in stage 1, three of which are parallel in stage 2, and a single *module* in stage 3. All three stages are connected in series and all connections and contacts are proposed to be superconducting metal to eliminate the contact voltage differences to the greatest possible extent. The *modules* in stage 1 are *8-bit* devices with more than 100 parallel copies per *module*, whereas the *modules* in stage 2 are *3-bit* devices with four or fewer parallel copies per *module*. Stage 3 is a single *12-bit* device with no additional parallel branches. These numbers for (*N*, *K*) are required should one wish to reproduce the values in Table 1. (Hu et al. (2018a) is an open access article distributed under the terms of the Creative Commons CC BY license, which permits unrestricted use, distribution, and reproduction in any medium)

topological genera, as shown in Fig. 14, along with the predictive power of simulations like LTspice, Kwant, or traditional tight-binding Hamiltonians, allows the designer to customize devices accordingly. Depending on the genus and final layout, QHARS devices can still be metrologically verified by means of measuring a specific configuration and its mirror symmetric counterparts. In that sense, future QHARS devices whose longitudinal resistances cannot be checked must either be compared with a duplicate of itself, or must have appropriate symmetries such that a

Table 1 This shows all possible resistance decades achievable with the proposed programmable QHR device in Fig. 13

Resistance	Stage 1	Stage 2	Stage 3	Voltage probes used	Deviation from decade value
100 Ω	00010010 00011001 00001001 00010001	None	None	*A, B*	0.714 μΩ/Ω
1 kΩ	10001001 10001011 10000111 10010001	None	None	*A, B*	0.108 μΩ/Ω
10 kΩ	00110001 00100111 00110011 00111100	011 010 010	None	*A, C*	14.8 nΩ/Ω
100 kΩ	00111101 00111101 00111100 00111110	010 010 010	000000000011	*A, D*	0.043 nΩ/Ω
1 MΩ	00101011 00100001 00110101 00110000	010 001 001	000000100110	*A, D*	0.0243 nΩ/Ω
10 MΩ	01011110 01011001 01010011 01001100	110 101 110	000110000010	*A, D*	0.346 pΩ/Ω
100 MΩ	01110111 10001001 01101101 01100111	100 011 011	111100100001	*A, D*	1.21 nΩ/Ω

The values listed in this table are described in more detail in Hu et al. (2018a) and can achieve values of seven decades of resistance. Each module in each stage is assigned a binary string. As long as the exact configuration is used, the measured voltage between the two probes will measure a near-exact decade value, to within a deviation defined in the rightmost column. Hu et al. (2018a) is an open access article distributed under the terms of the Creative Commons CC BY license, which permits unrestricted use, distribution, and reproduction in any medium

same-device comparison can be made. For instance, a square array could be the subject of a two-terminal measurement using same-sided corners. This configuration would have a four-fold symmetry which can be measured and compared to verify device functionality. All types of arrays would be thus limited by the EG growth area, which at present has been optimized by the use of a polymer-assisted sublimation growth technique (Kruskopf et al. 2016). The total growth area may be the most demanding limiting factor for this species of device. Nonetheless, one can hope that the latter technique, and any similar technique to be developed, will enable homogeneous growth on the wafer scale such that the whole EG area retains metrological quality.

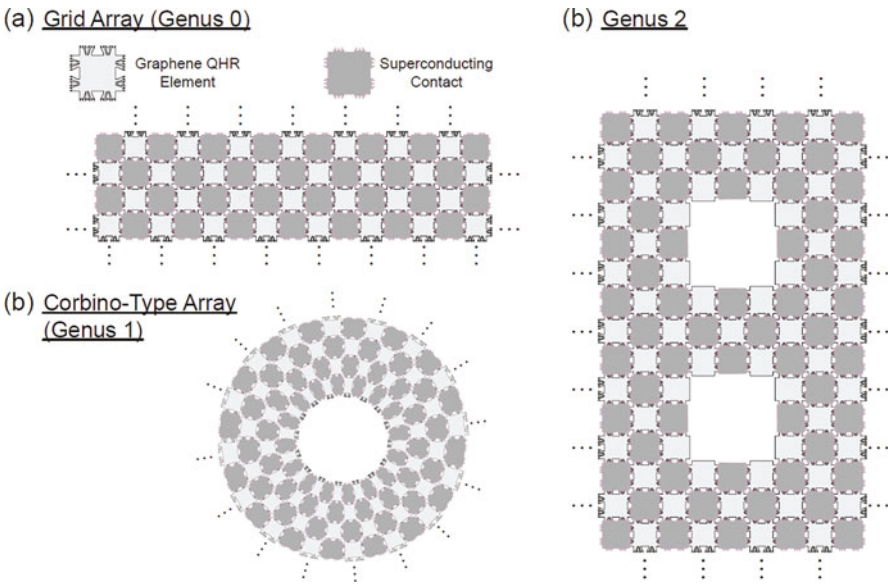

Fig. 14 Hypothetical array designs of varying topological genera. Determining the predicted value of output quantized resistance between any pair of contacts can be done with various modeling techniques done for similar systems. (**a**) The use of superconducting contacts enables the design of larger and more complex arrays, provided the EG QHR elements are small enough. For genus 0, grid arrays can take on user-defined dimensions. (**b**) Corbino-type geometries could also be implemented, and these are just examples of genus 1 topologies. (**c**) A final example of array type comes from those that have a genus 2 topology. Even more customized designs are possible and not well-explored

The Quantum Anomalous Hall Effect

Regardless of EG device size, the magnetic field requirement will always be a limit. This is inherently tied to the band structure of graphene. In addition to this limitation, there are at least two others that prevent SI-traceable quantum electrical standards beyond the ohm from being user-friendly and more widely disseminated, which at present, confines their global accessibility to mostly NMIs. These other limitations include the sub-nA currents obtained from single electron transistors (in the case of the quantum ampere) and the Josephson voltage standard's aversion to magnetism, complicating the use in a single cryostat with a QHR device to create a compact current standard. Ongoing research on magnetic topological materials has the potential to avoid compatibility problems.

The physical phenomenon underpinning this research is the quantum anomalous Hall effect (QAHE). This effect yields a quantized conductance in magnetically

ordered materials at zero applied magnetic field. Certain types of material display a quantized resistance plateau at zero-field suitable for metrology measurements, as has been shown in some recent work (Fox et al. 2018; Götz et al. 2018). The QAHE is a manifestation of a material's topologically nontrivial electronic structure. The QAHE, along with the Josephson effect and QHE, is a rare example of a macroscopic quantum phenomenon. There are several types of materials that exhibit the QAHE, with many being classified within the following categories: magnetically doped topological insulators (TIs), intrinsic magnetic TIs, and twisted van der Waals layered systems.

Fox et al. explored the potential of the QAHE in a magnetic TI thin film for metrological applications. Using a CCC system, they measured the quantization of the Hall resistance to within one part per million and, at lower current bias, measured the longitudinal resistivity under 10 mΩ at zero magnetic field (Fox et al. 2018). An example of the data they acquired is in Fig. 15a, b. When the current density was increased past a critical value, a breakdown of the quantized state was induced, and this effect was attributed to electron heating in parallel bulk current flow. Their work furthered the understanding of TIs by gaining a comprehension of the mechanism during the prebreakdown regime, including evidence for bulk dissipation, thermal activation and possible variable-range hopping. A concurrently reported work by Götz et al. also looked to present a metrologically comprehensive measurement of a TI system (V-doped $(Bi,Sb)_2Te_3$) in zero magnetic field (Götz et al. 2018). When they measured the deviation of the quantized anomalous Hall resistance from R_K, they determined a value of 0.176 μΩ/Ω ± 0.25 μΩ/Ω. An example of their data is shown in Fig. 15c. The steps both works made are vital to our eventual realization of a zero-field quantum resistance standard.

One of the remaining major limitations, besides finding a TI material system with a large band gap, will be to lift the stringent temperature requirements, which are currently in the 10 mK to 100 mK range. Fijalkowski et al. show, through a careful analysis of non-local voltages in devices having a Corbino geometry, that the chiral edge channels closely tied to the observation of the QAHE continue to exist without applied magnetic field up to the Curie temperature (≈20 K) of bulk ferromagnetism in their TI system. Furthermore, it was found that thermally activated bulk conductance was responsible for quantization breakdown (Fijalkowski et al. 2021). The results give hope that one may utilize the topological protection of TI edge channels for developing a standard, as has been demonstrated most recently by Okazaki et al. (2021). They demonstrate a precision of 10 nΩ/Ω of the Hall resistance quantization in the QAHE. They directly compared both the QAHE and QHE from a conventional device to confirm their observations. Given this very recent development, more efforts are expected to follow to verify the viability of TIs as a primary standard for resistance. In the ideal case scenario, TI-based QHR devices will make disseminating the ohm more economical and portable, and will, more importantly, serve as a basis for a compact quantum ampere.

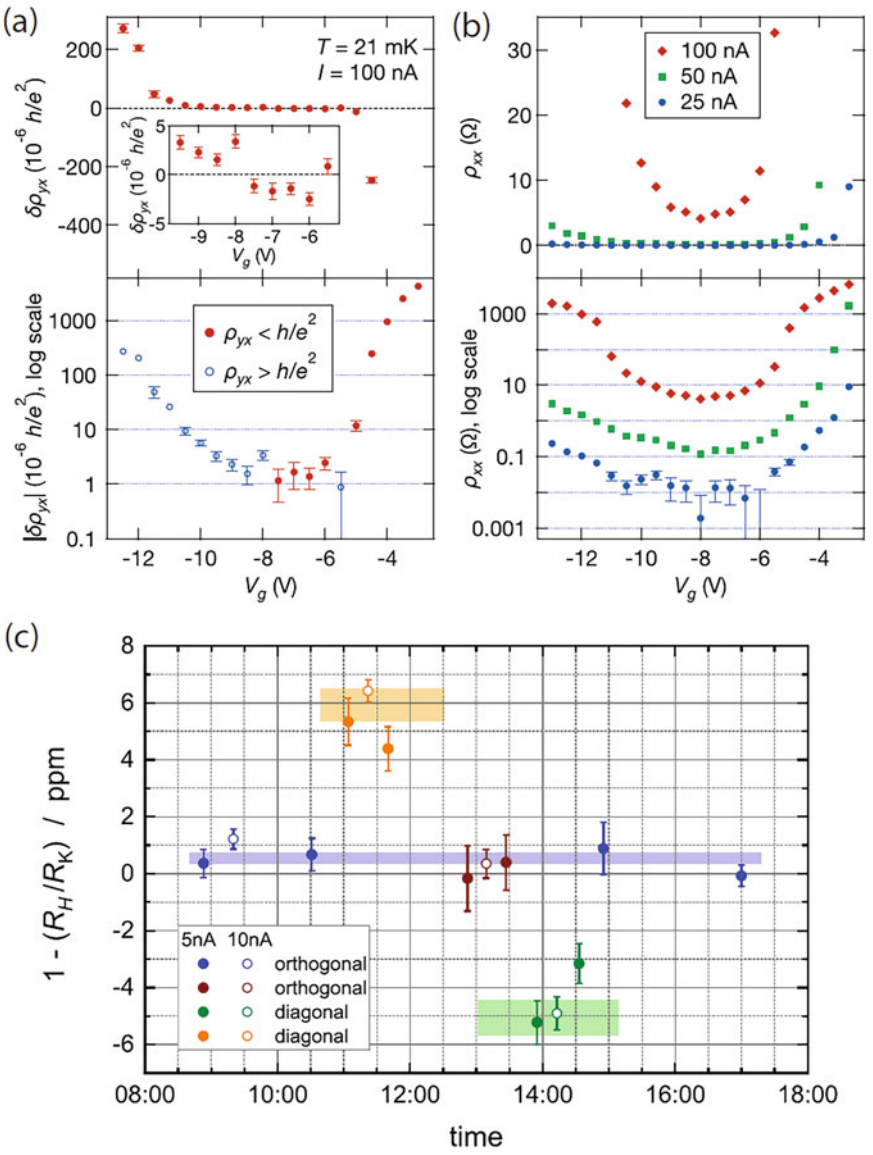

Fig. 15 (**a**) CCC data are shown for measurements of ρ_{yx} in an example device using a 100 nA current at 21 mK. The plateau in the upper panel shows the deviation from R_K for a range of gate voltages. The inset shows a magnified view of the Hall resistance deviations in the center of the plateau ($\nu = 1$). The bottom panel shows the logarithmic behavior of the deviations as one departs from the optimal gate voltage. (**b**) ρ_{xx} of the same device was measured as a function of gate voltage for three different bias currents. The data show strong current and gate voltage dependence, as displayed on a linear and log scale in the top and bottom panels, respectively. At 25 nA, with the gate voltage near the center of the plateau, the resistivity nearly vanishes and the measurements approach the noise floor. All error bars show the standard uncertainty and are omitted when they are

Conclusion

As the global implementation of new technologies continues to progress, one should hope to see a more universal accessibility to the quantum SI. This chapter has given historical context for the role of the QHE in metrology, including a basic overview of the QHE, supporting technologies, and how metrology research has expanded these capabilities. The present-day graphene era was summarized in terms of how the new 2D material performed compared with GaAs-based QHR devices, how the world began to implement it as a resistance standard, and how the corresponding measurement infrastructure has adapted to the new standard. In the third section, emerging technologies based on graphene were introduced to give a brief overview of the possible expansion of QHR device capabilities. These ideas and research avenues include *pn*J devices, QHARS devices, and experimental components of AC metrology and the quantum ampere. The chapter then concludes by discussing the possible limitations of graphene-based technology for resistance metrology and looks to explore TIs as one potential candidate to, at the very least, supplement graphene-based QHR devices for resistance and electrical current metrology.

It has become evident throughout the last few decades that the quantum Hall effect, as exhibited by our modern 2D systems both with and without magnetic fields, has the marvelous potential to unify the components of Ohm's $V = IR$ relation. That is, to bring together all three electrical quantities allowing several traceability capabilities will undoubtedly improve electrical metrology worldwide. Throughout all the coming advancements, it will be important to remember that these milestones should keep us motivated to continue learning how to better enrich society with the quantum Hall effect:

> It is characteristic of fundamental discoveries, of great achievements of intellect, that they retain an undiminished power upon the imagination of the thinker. – Nikola Tesla, 1891, New York City, New York

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Fig. 15 (continued) smaller than the data point. Reprinted figure with permission from (Fox et al. 2018). Copyright 2018 by the American Physical Society. (**c**) Another series of measurements on a topological insulator at the $\nu = 1$ plateau. Measurement currents of 5 nA and 10 nA were used in both orthogonal and diagonal configurations. The colored rectangles represent the weighted average and standard deviation of the data from those configurations. (Reprinted from (Götz et al. 2018), with the permission of AIP Publishing)

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Quantum Pascal Realization from Refractometry

15

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Abstract

After the redefinition of the International System of Units (SI), it has been realized that the pascal, the SI unit of pressure, can be calculated more accurately because, according to ideal gas law, the pressure depends on temperature, gas number density, and few fundamental constants like Boltzmann's constant, the universal gas constant, and Avogadro's number. Among those, the temperature is redefined by evaluating the thermal energy. The fundamental constants are already calculated using quantum electrodynamics (QED) with the utmost accuracy. Therefore, the precise measurement of pascal depends majorly on gas number density, which

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can be computed using optical techniques. Quantum calculations involve calculating pressure using optical methods, also known as the quantum realization of pascal. This chapter reviewed the optical techniques for measuring pressure using refractometry proposed by various national metrology institutes.

Keywords

Pascal · Traceability · Calibration · Metrology · Refractometry

Introduction

One of the most often observed physical phenomena in daily life, pressure is essential to many physical functions, such as human physiology, weather forecasting, civil aviation and space travel, earth core exploration, deep ocean research, heavy engineering, the power plant industry, etc. The development of mercury manometers by Italian scientist and mathematician Evangelista Torricelli in 1643 is the tale of barometric pressure and the first vacuum measurements. Some of the earliest barometric measurements were performed by carrying a liquid-filled mercury manometer to a mountain and periodically measuring the height of the mercury column as they moved toward the peak of the mountains. It was observed that the lower the mercury column height, the higher the altitude, and the higher the mercury column height, the lower the barometric pressure. One of the pressure units is torr, which is equal to 133.322 Pa and is still widely used on many vacuum gauges; it was named in honor of Torricelli and is ostensibly equal to 1 mm of mercury column height. Since Torricelli's time, mercury manometers have gradually improved based on a more accurate and precise resolution of column heights. The National Institute of Standard and Technology (NIST), USA, invented a device called the ultrasonic interferometer manometer (UIM) in 1975 (Heydemann et al. 1977) that uses pulsed ultrasound to precisely measure the column heights in a 3 m tall mercury liquid-column manometer within the resolution of 10 nm. These UIM pressure uncertainties are at par or better than any existing primary pressure standards. The National Physical Laboratory of India (NPLI) developed a comparable UIM system in 1982 (Kumar et al. 2019; Thakur et al. 2021). Various NMIs have phased out the UIM due to Mercury's neurotoxic nature and the instruments' huge, heavy, and complicated design. Even though digital pressure transducers, which rely on variations in the resistance, capacitance, or frequency of mechanically strained diaphragms, have replaced mainly mercury manometers used by various research and industry, they still require a UIM to complete the traceability chain. Different regulatory bodies prohibit the purchase and use of mercury-containing items due to mercury's position as neurotoxic and environmentally unfriendly. The National Metrology Institutes (NMIs) and the international metrology community are now working toward pascal's quantum realization, and other NMIs are joining the hunt for the most effective optical approaches for realizing pressure. In addition to making all quantum-based pressure realizations and measurements immediately traceable to the SI, further advancement will put an end to mercury's four-century hegemony in

pressure metrology. In quest of a more accurate, mercury-free, and physics-based method of realizing pressure than using a physical item. An alternate measuring technique is made possible by the relationship between pressure p and energy density. The relationship (1) is true for an ideal gas (the ideal-gas law),

$$p = \left(\frac{N}{V}\right) k_B T \tag{1}$$

where N is the number of particles filling volume V, k_B represents the Boltzmann constant, and T represents the ambient temperature.

By using gas refractivity, one may determine a gas's number density, $\rho_N = N/V$; in this case, the equation for pressure is $p \propto (\eta - 1)k_B T$, where η and $(\eta - 1)$ stand for the gas's refractive index and refractometry, respectively. The refractive index of a medium, which alters an electromagnetic (EM) wave's frequency in the presence of gas as opposed to a vacuum, is where the name "refractometry" originates (Pendrill 2004). The pressure is computed based on fundamental constants such as the Boltzmann constant, the Avogadro number, and frequencies.

The revised definition of Kelvin based on fundamental constants states that it deals with the quantity of thermal energy $k_B T$ (Fischer 2015). Since thermal energy can now be computed with the highest degree of precision thanks to Eq. (1) and Kelvin's redefinition, the only important variable left to determine pressure or temperature is the gas number density (ρ_N). Due to the interaction of gas molecules with laser light, or EM radiation, the most accurate way to calculate gas number density is by optical methods. Depending on the range of the measuring pressure, it may be obtained via dispersion, absorption, fluorescence, or any other light-matter interactions. The EM wave resonates with one of the gas molecules' quantum states in the event of the absorption condition.

NIST started an entirely new optical pressure standard that connects the pascal to quantum calculations of helium's refractive index. A gas's density, which relies on temperature and pressure, determines its refractive index. Calculations for atomic helium will enable us to connect pressure, refractive index, and temperature with an accuracy greater than 1 ppm. Quantum mechanics explains the precise link between these variables. Based on the atomic characteristics of helium, we can accurately measure temperature and refractive index to estimate the pressure and convert the instrument into the main pressure standard. On the other hand, if other methods can precisely measure pressure, the thermodynamic temperature may be calculated by measuring the refractive index. With the use of laser interferometry, it is possible to measure the refractive index with precision far greater than previously conceivable. Photons interact at the quantum level, causing light to move through gases more slowly than it does in a vacuum, which makes it possible to estimate the pressure of a gas. When a length is measured by laser interferometry, the presence of a gas, such as helium, makes the measurement appear somewhat longer than it would in a vacuum. The apparent difference would consequently enable us to calculate the refractive index if we could compare two equal lengths in gas and vacuum.

Fixed Length Optical Cavity

A fixed-length optical cavity (FLOC) was devised, constructed, and tested by the NIST team in order to accurately determine the refractive index. The FLOC comprises two Fabry–Perot (FP) cavities, each composed of a set of mirrors atop a spacer built of ultralow-expansion (ULE) glass (Avdiaj et al. 2018) to prevent variations in interferometer length with temperature. The most common FP cavities have lengths ranging from a few tens to hundreds of millimeters. In a vertical FP cavity, the lower chamber is kept at vacuum, while the upper cavity is filled with gas. The length between the two cavities changes based on the gas's refractive index, density, and atomic or molecular characteristics. First-principles calculations were used to determine the atomic characteristics of helium; therefore, the measurement of refractivity results in a determination of density, which then yields a determination of pressure.

The reference cavity is kept vacuum-sealed to reduce noise and other systematic errors, which helps to increase precision. The reference cavity, which can be seen in the schematic of FLOC Fig. 1a, is a hole bored through the glass block and sealed at each end with mirrors. The top cavity, by contrast, has a slit to readily enable gas to flow in and out. In addition, a vertical tube enables us to connect the reference cavity with the vacuum-pump (Fig. 1a). To increase temperature stability and guarantee that the gas species is known and, thus, has a known refractivity, the glass cavity is housed inside a leak-proof metal chamber. Quantum mechanics may be used to determine the refractivity and virial density coefficients for gases with simple electron structures and few isotopes, such as helium (Puchalski et al. 2016). This calculation can produce refractivity with better than 1 ppm of error. The refractivity values for additional gases such as N_2, Ar, and O_2 can be measured or estimated, but with far greater uncertainty. FLOC was designated as the principal realization of pressure, since pressure computation only requires temperature and refractivity. They were able to calculate nitrogen's refractivity $\eta - 1 = (26485.28 \pm 0.3) \times 10^{-8}$ at $p = 100.0000$ kPa, $T = 302.9190$ K, and $\lambda_{vac} = 632.9908$ nm (Egan et al. 2015), which outperforming mercury manometers (Egan et al. 2016). The NIST team (Egan et al. 2016) also independently measured the Boltzmann constant using helium's predicted refractive index. At low pressures, the FLOC has less uncertainty

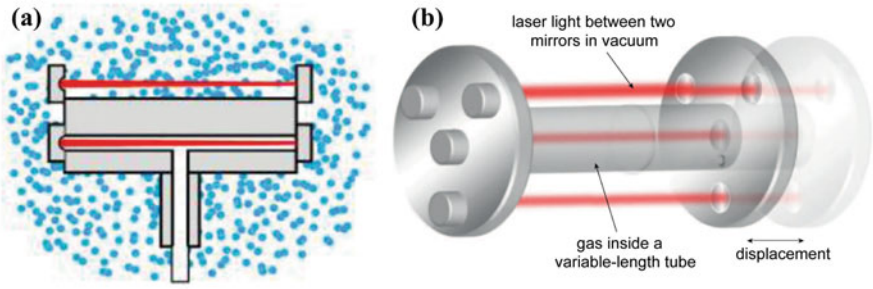

Fig. 1 Schematic of (**a**) FLOC (dots represent gas molecules) and (**b**) VLOC (Ricker et al. 2018)

than a mercury manometer, and at pressures higher above the operating range of manometers, the FLOC will reach parts-per-million uncertainties.

Although creating such a major pressure standard at NIST was a wonderful thing, it was also necessary to figure out how to transfer its better precision to sites outside the standards lab. As a result, NIST also created a FLOC that is less complicated and will provide more sensitivity than the existing pressure standards in a much more compact form factor. Because it is more refractive than helium and 100 times less susceptible to impurities, nitrogen (N_2) is used as the working medium in the transportable FLOC standard. Fixed-length cavities are desirable as pressure standards because they may be constructed from materials with little temporal instabilities and will provide a threefold reduction in measurement errors over current commercial technologies for vacuum pressures as low as 1 Pa. Although the FLOC is the primary, there are two significant flaws that must be taken into consideration when doing a high precision measurement. The first is the deformation of the glass as a result of pressure. As you exert force on the exterior surfaces, the glass is compressed in volume. The reference cavity is at a different pressure than the glass, which results in non-uniform forces that cause the glass to bend non-uniformly in addition to the bulk compression. Each glass cavity has a unique set of distortions, but the correction may be estimated empirically and rectified with a degree of error. The second problem is that glass may bulge as a result of helium absorption. The absorption can be tracked through time and extrapolated back to zero with adequate interferometer data, although there is some uncertainty involved. Overall, a FLOC standard may reach a nitrogen uncertainty of 9 ppm (Egan et al. 2016), although this might be further decreased with improved index determination. Additionally, using various gases with known refractive indices at the same pressure would be the ideal way to evaluate pressure distortions. Gas species won't affect the distortions, which may be solved to estimate the error's size. Better refractivity measurements result in better pressure standards because if the refractivity is not established properly, this might add ambiguity.

Variable Length Optical Cavity

It is obvious that the FLOC's aberrations would restrict its effectiveness for doing a high-accuracy measurement of refraction. Similar in operation to the FLOC, a different idea known as the Variable Length Optical Cavity (VLOC) has important distinctions that make it possible to quantify refractivity. First, as illustrated in Fig. 1b, the VLOC contains four chambers, three of which are spaced equally apart from the center cavity. While the surrounding cavities will be vacuum-sealed, the middle cavity, also known as the measuring cavity, will be exposed to a gas at a constant pressure. The exceptionally robust monolithic platforms on which the mirrors at either end of the device are constructed ensure that the movement of one mirrored end-piece causes equals displacements in the inner and outer interferometers. Three interferometers are placed in a vacuum to measure and regulate angular tilts when the end-piece is moved, which is required to prevent Abbe errors.

With the help of the VLOC, it is possible to test the refractivity at one cavity length, modify the cavity's length, and then measure the refractivity at a second length. Both lengths will exhibit the same systematic errors, such as distortions, helium absorption or bending. In order to get the refractive index, we must subtract the two measurements as follows:

$$\eta = \frac{OP_2 - OP_1}{L_2 - L_1}$$

where numerator and denominator on the right side of the above equation are given by displacement and optical path, respectively, and L represents the cavity length.

The refractive index of helium may be determined using the VLOC, and the value can then be converted to N_2 using a piston gauge. The piston gauge had been employed to create a steady pressure, starting from helium and subsequently in N_2. As a result, the helium's quantum traceability may be utilized to determine N_2's refractivity, so creating quantum traceability through the refractometry. This experimentation only has to be carried out once, and then any refractometer may be used to determine pressure and obtain traceability via the quantum fundamental quantities of the gas because the refractivity is an unchanging attribute of the gas. The VLOC should be able to estimate refractometry of helium with 1 ppm of uncertainty or less (Stone et al. 2014), which will significantly lower the uncertainties for nitrogen-based pressure determination using FLOC. In order to employ other inert gasses in a FLOC, refractivity data for those gases will also be accessible.

Several challenging conditions must be addressed to construct a VLOC for measurement to 1 ppm. The gas must, first and foremost, be highly pure. For gases like helium, whose refractive properties range greatly, the purity must be more than 50 ppb. The second crucial criterion is that the cavity must be built to alter lengths. Cavities that stretch 15 cm are ideal, since the longer the movement, the better resolution. The mirrors cannot be impacted by the movement, which is the last significant restriction. It is not possible to utilize alternative mirrors or switch out components since you want the mirrors to be the same at both places. Without adding faults exceeding the system's dimensional tolerance of 3 picometers (pm), it is almost hard to add something to change the path length. By 3 pm, the movement cannot transmit any pressures that would change the dimensions.

The VLOC built at NIST is the first of its type in the world and was built in accordance with the above description to achieve the uncertainty objective of 1 part per million (ppm). To avoid stress on mirrors, the design has a neutral axis mount and can compress from 30 cm to 15 cm. The complete apparatus will be evacuated to a vacuum of 10^{-5} Pa within a vacuum chamber. The linear stage is moved by a hand crank, which compresses a bellows with a 15 mm diameter to contain the measuring gas. The bellows' purity is guaranteed by employing a steady stream of pure helium that has undergone cryostat purification, preserving all metal seals, and using ultrahigh purity tubing. It will be possible to switch between helium and other gases using a commercial pressure balance that can deliver stability of better than

0.5 ppm. Various NMIs have been working on creating the optical interferometer manometer (OIM) for pascal realization utilizing laser refractometer alongside NIST.

Development of Pressure Standard at NMIs

National Institute of Standard and Technology (NIST)

The foundation of method, which uses a dual FP cavity refractometer and gas refractivity to measure pressure, is described in Egan et al. (2015). By silicate-bonding, ends of FP cavity in ULE spacer, mirrors were being attached. In order to determine the optical paths in both cavities, a He-Ne laser must be dither locked to both the cavities at resonance. The optical length is given by the formula ηL, where L is the physical distance between the mirrors. Initially both the cavities kept at vacuum but in one of them gas is filled such as that the cavity kept at vacuum and pressure are known as reference and measurement cavities, respectively. A gas refractivity is determined by computing the optical path between both the cavities and using the Lorentz-Lorenz equation and gas laws relate the pressure to gas refractivity in a linear fashion. As demonstrated in Fig. 2 with the actual image of the FP cavity inset, the laser refractometer created by NIST provides one of the most precise realizations of the pascal available anywhere in the world. Each refractometer system's FP cavities, fabricated of glass from various sets and have marginally dissimilar designs or dimensions (different compressive distortion terms), have a few minor differences. Additionally, the mirrors in each system are made from various coating runs and have a reflectivity of 99.8% and 99.7%.

The relation between pressure, density, and temperature for non-ideal gas is given by Eq. (2) (Moldover 1998),

$$P = k_B T\left(\rho_N + B_\rho {\rho_N}^2 + C_\rho {\rho_N}^3 + - - - - -\right) \tag{2}$$

where B_ρ and C_ρ are virial coefficients.

The Pascal is expected to switch the current apex level pressure standard UIM (Heydemann et al. 1977; Kumar et al. 2019) for pressure measurements less than 180 kPa since the thermal energy $k_B T$ can be estimated with an error of 1 ppm (Fischer 2016). Refractometry should be monitored with the least uncertainty, and the associated cavity's distortion throughout the experiment should be taken into account, which is the sole challenge in achieving pressure from OIM.

The technique of an OIM depends on fundamental physical quantities, i.e., frequency change of EM waves, which is steady with the Pascal's quantum realization, to calculate the pressure. This can be done by first calculating the refractometry, which can be done by calculating the polarizability. In order to capture pressure with an uncertainty below 1 ppm, theoretical calculations involving atomic characteristics, such as polarizability, were done at the relativistic and QED levels (Puchalski et al. 2016).

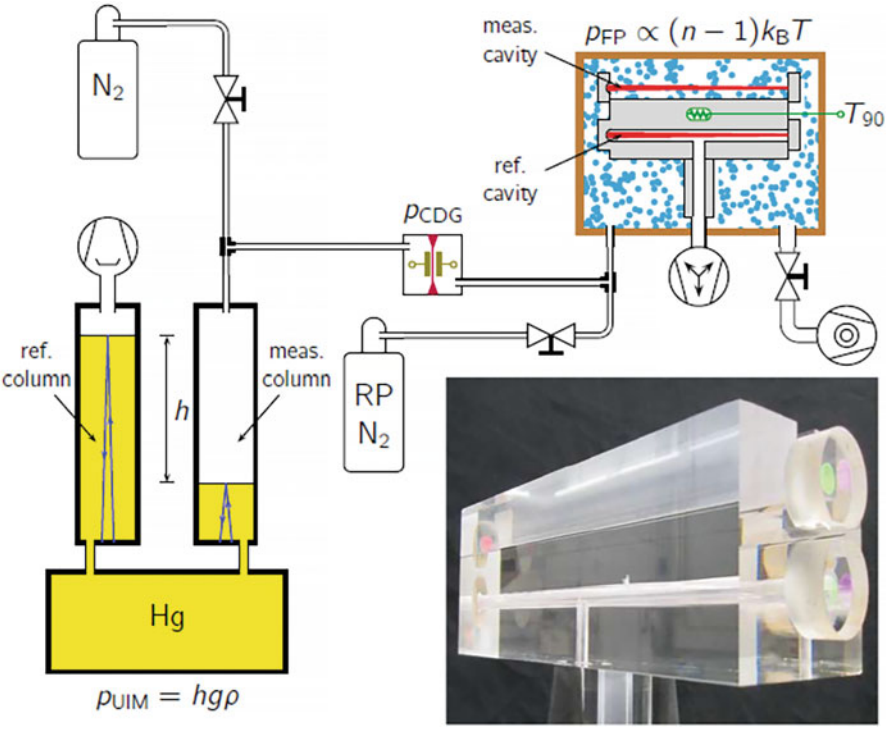

Fig. 2 Schematic of comparison setup of OIM and UIM with the inset of real picture of FLOC (Egan et al. 2016; Ricker et al. 2018)

Any inert gas that is appropriate for measuring the refractive index, such as He, N_2, Ar, or any other, may be selected. Due to the fact that it is absorbed by the FP cavity formed of ultralow expansion glass (ULE), helium gas exhibits one drawback when used as the testing gas for pressure computation utilizing OIM (Avdiaj et al. 2018). The ULE chamber served as a refractometer to determine the gas' refractometry. Figure 1 depicts the actual OIM chamber manufactured by NIST in the USA. The chamber contains two FLOCs, one at the top exposed to the gas input and pertinent vacuum system, known as the measurement cavity, and the other at the bottom linked to the vacuum pump.

A laser whose wavelength is fixed to an FP cavity at resonance. The servo modifies the laser's frequency f to preserve the cavity's resonance if the gas density (i.e., pressure) varies. The refractive index is determined by a change in frequency, as shown by the equation below,

$$n - 1 = \frac{\Delta f}{f} + n d_m p + d_r p \tag{3}$$

The effective fractional frequency can be calculated using Eq. (4),

$$\frac{\Delta f}{f} = \Delta\nu_{FSR}(1+\epsilon_\alpha).\frac{\frac{f_i - f_f}{\Delta\nu_{FSR}} + \Delta m}{f_f + \nu_f} \tag{4}$$

where f laser's frequency in vacuum and Δf is the difference in frequency between the measurement and reference cavities. One can calculate the free spectral range (FSR) by computing the frequency difference between measurement and reference cavities.

The effective fractional frequency is dependent on FSR of the measurement cavity $\Delta\nu_{FSR}$ with mirror phase shifts correction $(1 + \epsilon_\alpha)$, beat frequencies when both the cavities are kept at vacuum (f_i) and pressure (f_f); and resonant mode number change (Δm) of the measurement cavity.

Starting with f_i, one gets f_f, by filling the measuring cavity with gas. When the bulk modulus compresses, the cavity length changes from the initial (L_i) to final (L_f), resulting in the first distortion term, $d_m p = (L_i - L_f)/L_f$. The reference cavity's mirror bending and bulk modulus compression result in the second distortion factor, $d_r p = (\nu_i - \nu_f)(1 + \epsilon_\alpha)/(f_i + \nu_f)$, whose resonance frequency shifts from initial ν_i to final ν_f.

Under applied pressure, the vacuum and reference cavities deform. The deformation occurred in the cavities because of the existence of bulk modulus . Thus, the consequences of the distortion are neutralized.

The Lorentz-Lorenz equation (Born and Wolf 1980; Pendrill 1988, 2014), which is provided by Eq. (5) for an ideal gas, is used to determine refractometry,

$$\frac{\eta^2 - 1}{\eta^2 + 2} = \frac{4\pi}{3}(\alpha + \chi)\rho_N = A_R\rho_V \tag{5}$$

where χ and α are the magnetic susceptibility and dynamic polarizability, respectively, A_R is the molar dynamic polarizability or virial coefficient or molar refractivity (Egan and Stone 2011), and ρ_V is the mass density.

For a non-ideal or real gas, Eq. (5) can be rephrased as given below,

$$\frac{\eta^2 - 1}{\eta^2 + 2} = A_R\rho_V + B_R{\rho_V}^2 + C_R{\rho_V}^3 + - - - - - \tag{6}$$

B_R *and* C_R are temperature dependent virial coefficients.

The term $\frac{\eta^2-1}{\eta^2+2}$ reduces to $\frac{2}{3}(\eta - 1)$ at lower pressure.

Equation (6) can be rewritten as,

$$\eta - 1 = \frac{3}{2}A_R\rho_V + \frac{3}{2}B_R{\rho_V}^2 + \frac{3}{2}C_R{\rho_V}^3 + - - - \tag{7}$$

Since by perceiving Eqs. (3) and (7), refractometry $(n - 1)$ can be inscribed in the terms of pressure (p),

$$\eta - 1 = c_1 p + c_2 p^2 + c_3 p^3 + - - - - \tag{8}$$

Where

$$c_1 = \frac{3}{2k_BT}A_R, c_2 = \frac{3}{8(k_BT)^2}\left(A_R^2 - 4A_RB_P + 4B_R\right),$$

$$c_3 = \frac{3}{16(k_BT)^3}\left(5A_R^3 - 4A_R^2B_P + 4A_RB_R + 16A_RB_P^2 - 16B_PB_R - 8A_RC_P + 8C_R\right)$$

For suitability, disregarding higher orders in Eq. (8),

$$\eta - 1 \approx c_1 p$$

A dual FP chamber refractometer may now detect pressure generated by the FLOC thanks to a final series reversion (Egan et al. 2016),

$$p_{FP} = \frac{1}{c_1 - d_m - d_r}\left(\frac{\Delta f}{f}\right) - \frac{c_2 - c_1 d_m}{(c_1 - d_m - d_r)^3}\left(\frac{\Delta f}{f}\right)^2 + \frac{2(c_2 - c_1 d_m)^2 - c_3(c_1 - d_m - d_r)}{(c_1 - d_m - d_r)^5}\left(\frac{\Delta f}{f}\right)^3 - - - - \quad (9)$$

Equation (9) is used by the NIST's fixed-length OIM to measure pressure, and it produced an expanded uncertainty of $(0.002\ Pa)^2 + (8.8 \times 10^{-6}p)^2)^{0.5}$. The atomic or molecular characteristics of the gases employed in the FP cavities are the focus of the OIM. The Boltzmann constant, the Avogadro number, and frequencies serve as its foundations. OIM may be used to realize Kelvin as well.

One of the drawbacks of the NIST-made OIM is that the mode order m varied drastically per 0.9 kPa of pressure range (Egan et al. 2016).

Istituto Nazionale di Ricerca Metrologica (INRIM)

A refractometry-based OIM (Mari et al. 2019) in the 1 kPa to 120 kPa range was also created by INRIM, an Italian NMI. The optical system's response has first been assessed by assessing its sensitivity for N_2 and air, in the form of relationship between a change in pressure and the quantity of interferometric fringes recorded. The interferometer has then undergone geometrical characterization to provide an absolute measurement of the gas's refractive index, enabling an optical measurement of the gas pressure. Two capacitance diaphragm gauges (CDGs) that determine pressure values in the range of 1 kPa – 120 kPa and the standard have been compared.

Method and System

The suggested approach is an alternative Pascal realization method that makes use of FP cavities and is established on the evaluation of a gas's refractive index using a homodyne Michelson interferometer (Egan et al. 2015). He-Ne laser having wavelength (≈633 *nm*) is frequency-stabilized and used as the light source. It is connected

to a fiber that ends with collimeter and maintains polarization and incidents the light into the cavity. A homodyne Michelson interferometer having fixed arms is depicted schematically in Mari et al. (2018). The measurement arm basically comprises a dual mirror assembly which is evident of the reflection of bean between A and B (Pisani 2009; Mari et al. 2014), whereas the reference arm incidents the laser light to the mirror M_R. Depending on the angle between B and A and incidence angle on A, the double mirror multiplication setup enables the collection of an N-count of reflections (Pisani 2009). The beam splitter "BS" is used to combine the interference-producing reference and measurement beams. A high-speed camera's CMOS sensor "D" picks up the interference fringes, which appear as a dark or bright pattern after passing through a polarizer "P." The program administers the data collection and analysis for the experiment. Mirrors A and B are produced of ULE ceramic glass (Manufacturer: Clearceram) and attached to a circular plate made of the same material with a nominal diameter of 160 mm using the "hydroxide bonding" process, effectively acting as a single glass block, in order to make the interferometer as temperature stable as feasible. The mirrors had the following nominal measurements: width, height, and thickness given by 70 mm, 20 mm, and 10 mm, respectively. They adhered at an angle of around 0.4° (Mari et al. 2018). The plate has a circular pumping hole and is mounted on proper kinematic spherical supports to lessen mechanical strains and vibrations during the pumping or gas input operation. The plate is placed within the chamber V_1 made of stainless steel having height and internal diameter of 62.5 mm and 180 mm, respectively. This chamber is where the standard pressure p is produced, and it is where all the interferometer's other optical components are bonded. According to (Mari et al. 2014), an in-depth description of interferometer functioning is given. A double-knife CF40 flange-encased exchangeable conductance that connects V_1 to a pumping system is sandwiched between the conductance and the pumping system and a KF40 butterfly valve.

Three ports of the V_1 have a barometer placed on them that is calibrated against the INRIM HG5 main standard (Alasia et al. 1999) and two CDGs. V_1 is injected at an initial absolute pressure p_0 of around 0.1 Pa at the start of operation. The reference refractive index at p_0 is given by $\eta_{vac} \approx 1$. The gas is then inserted into the calibration compartment till the chosen nominal value is obtained, and wait for adequately enough time for stabilizing the temperature, then the standard pressure p is observed in V_1. One can calculate the final refractive index at pressure p using Eq. (10),

$$\eta = \eta_{vac} + \frac{\phi \frac{\lambda}{2}}{L} \approx 1 + \frac{\phi \frac{\lambda}{2}}{L} \tag{10}$$

where L is the interferometer's vacuum imbalance, or the difference in optical paths between the reference and measurement arms of the interferometer, and ϕ is the fringe counts observed between reference pressure p_0 and measured pressure p. The relationship between the molar density ρ_N and refractive index (Pendrill 1988, 2014) provided by the equation is established using the Lorentz-Lorenz equation (5).

Equation (11) yields the standard pressure p while excluding terms of higher order,

$$p = \rho_N RT\left(1 + B_\rho \rho_N\right) \tag{11}$$

T is the measured temperature at p in V_1.

Interferometer's Unbalance Measurement

Through a focused experiment shown in Fig. 3, an amplitude modulated fiber coupled diode laser having wavelength ≈ 640 nm, frequency f_M using a bias-T on the bias current to produce a synthetic wave with wavelength Λ and group velocity v_g, the interferometer's unbalance was determined using (12).

$$\Lambda = \frac{v_g}{f_M} \tag{12}$$

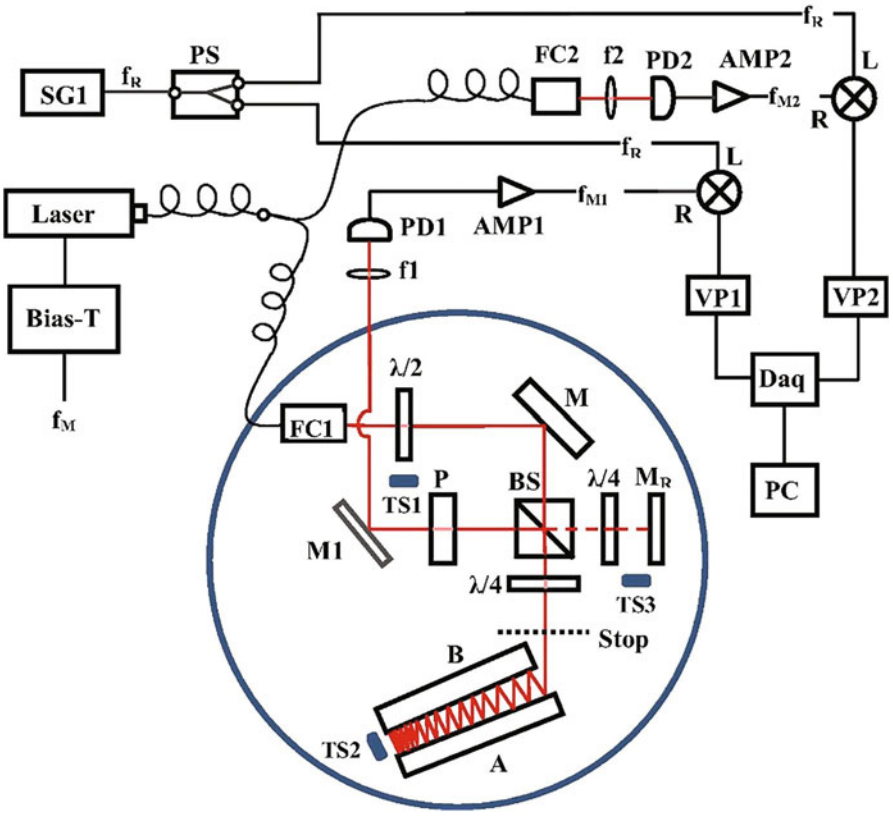

Fig. 3 Labeled diagram of the experimental assembly for path difference of the interferometer; M1: deflection mirror; voltage preamplifier: VP; power splitter: PS; signal generator: SG1; lens: f; fiber coupler: FC; amplifier: AMP; photo-detector: PD; AMP; mixer channels: R, L (Mari et al. 2019)

The laser emits two rays of radiation; one beam is incident to the photodetector PD2, which produces the reference ray at f_{M2}, and the other ray is thrown into the interferometer's input port. The interferometer's output port's photodetector PD1 produces the measurement signal at f_{M1}, which is the first measurement point. It is often easy to determine the optical path in terms of Λ obtained from the phase difference between the two signals f_M from the two photodetectors.

By adjusting the synthetic frequency f_M at the appropriate time, the uncertainty is removed because the measured phase only provides information about the excess fraction of wavelength. The superposition of the interferometer's two arms' beams, which PD1 was able to detect. To effectively detect the interferometer's imbalance or the relative differences between the two arms, they assessed the phase difference between the two signals (PD1 and PD2) by alternately blocking either of the two arms of the interferometer.

With frequencies of around GHz, which correspond to synthetic wavelengths of about tens of mm, we may reach a precision of fractions of mm. The resolution of the measurement falls as the modulation frequency f_M is increased. The signals from PD1 and PD2 are then combined with a signal at $f_R = f_M + 100$ kHz to perform a super-heterodyne down conversion at $f_{sh} = 100$ kHz. The down-conversion of the signal to the super-heterodyne frequency at 100 kHz preserves the signal's phase information at the order of GHz. The data at 100 kHz are recorded in Daq card and phase difference is evaluated using a computational method.

The imbalance L for several synthetic frequencies was measured between 0.7 and 1.3 GHz (Mari et al. 2019). Every point is the consequence of the variation in route lengths during an interval of a few minutes. After accounting for group velocity, the interferometer's resultant imbalance length in vacuum is $L = (1.4717 \pm 0.0002)$ m, with repeatability – measured as the standard deviation of the mean – contributing the majority of the uncertainty. The repeatability has been assessed using a number of normality tests. For instance, the Shapiro-Francia and Kolmogorov-Smirnov tests both produced p-values larger than 0.90, confirming the null hypothesis of normal distribution at the 0.05 level of significance. The incomplete elimination of common mode effects, for instance, because of fiber length changes during the minutes of measurements, may be the source of the data's dispersion.

Pressure Measurement by Optical Technique

This optical system's two potential methods for measuring pressure are outlined. First, the "sensitivity approach" which requires frequent calibration using a pressure reference standard, permitted the use of the device as an optical pressure sensor. Through the geometrical calibration of the interferometer arms, the second approach, known as the "absolute refractive index approach," permitted the use of the instrument as a pressure standard.

Sensitivity Approach

The initial step to detect and compute the sensitivity (S) using optical system's response (Mari et al. 2018), which is given by the ratio $\Delta p/\phi \approx p_s/\phi$, where Δp is the difference between the pressure p_s (usually from 120 to 130 kPa) and p_0 is the

reference pressure, and ϕ is the total number of fringes counted in the pressure difference Δp; p_s was obtained using barometer which is calibrated against INRIM primary standard HG5 (Alasia et al. 1999). An adjustable valve to control the leak was installed in series with the exchangeable conductance and adjusted to achieve an input time of V_1 of ≈ 5 minute. This allowed the pressure to change from p_0 to p_s while minimizing mechanical stress on the system and negating the opening change in temperature that would have happened in the situation of a rapid pressure deviation (Jousten et al. 2014; Jousten 1994), at standard temperature $T_{st} = 20$ °C, the findings of the S values for N_2 gas and ambient. The S value and associated standard uncertainty were determined to be (78.718 ± 0.014) Pa/fringe for N_2 and (80.322 ± 0.030) Pa/fringe for air, respectively. The uncertainty was calculated while also taking into consideration the pressure transducer's long-term stability. The findings of the sensitivity tests for N_2 and air are consistent with those of Table 1's data in Pendrill (2004). The last column shows the ratio percentage from each variance to total variance.

It is important to note that the established structure might be used as a pressure "sensor" in the sensitivity approach as early results, using the appropriate normalized sensitivity of each gas S_{gas} at T_{st} to acquire its associated pressure value p_j by counting the corresponding number of fringes ϕ_j arose between the p_0 and p_j at the temperature T_j:

$$p_j = \phi_j S_{gas} \frac{T_j}{T_{st}} \tag{13}$$

It may be assumed that ϕ_j has no effect of thermal expansion of ULE glass. The acquired findings further demonstrate the inherent stability for long span of the entire apparatus when used as a sensor because the interferometer was kept in the same settings.

Absolute Refractive Index Approach

To create and quantify standard pressure p, the second part of the investigation outlined above was then put into action. The interferometer's L-unbalance must be

Table 1 Uncertainty budget for N_2 gas at p= 100176 Pa (sensitivity approach) (Mari et al. 2019)

Input x_j	Value	Uncertainty source	$u(x_j)$	$c_i\, u(x_j)/p_j$	% Variance
S (Pa/fringes)	78.718	Stability, repeatability calibration uncertainty	0.014	1.82×10^{-4}	74.6
ϕ_j (fringes)	1271.76	Systematic effects, repeatability	0.05	3.93×10^{-5}	3.5
T_j (K)	293.342	Temperature gradients in chamber V_1, repeatability, calibration uncertainty	0.029	9.88×10^{-5}	21.9
Combined relative standard uncertainty ($u_c(p_i)/p_i$)				2.1×10^{-4}	100

determined, which calls for the specialized experimental setup seen in Fig. 3 and is a necessary prerequisite for this approach to be used rigorously. A sequence of data were recorded at a notional pressure of 120 kPa for 24 weeks after the imbalance L was determined so as to describe the OIM. The measured ϕ_j and T at optical pressure p and barometer pressure p_b and hence η and standard pressure p by Eqs. (10) and (11), respectively, with B_ρ acquired from (Egan et al. 2016; O'Connell 1981).

The outcomes p were contrasted with the calibrated barometer's pressure measurement p_b (standard uncertainty ≈ 5 Pa). The relative differences values obtained from the barometer and optical system at 120 kPa lie from -7.0 x 10^{-5} to +9.4 x 10^{-5} which is below the uncertainty bar limit or relative standard uncertainty of 1.7×10^{-4}.

Uncertainty Budget

Sensitivity Approach

Uncertainty was evaluated using model equation given by (13) in the sensitivity method example; Table 1 provides a summary of the corresponding uncertainty budget. The technique described in Mari et al. (2014) is used to count the interference fringes ϕ_j that occurred between the pressure at which the reference pressure was first set p_0 under vacuum and the pressure at which the interference occurred (p_j). Presently, there is 0.05 fringes of standard uncertainty. PT100 sensors were used to evaluate the temperature T_j. Calibrating, repeating, and accounting for temperature gradients within volume V1 result in a standard uncertainty of 0.029 K. However, as a first approach, the "sensitivity method" enables coverage of the pressure varies from 1 to 120 kPa with a relative standard uncertainty of 2.1×10^{-4} at standard barometric pressure. Table 1 makes it abundantly evident that the component owing to gas sensitivity determination has a significant impact on the combined standard uncertainty.

Absolute Refractive Index Approach

Equations (5, 10, 11) may be used to create the model Eq. (14) to calculate the uncertainty for the "absolute refractive index approach."

$$p = \frac{\left(1+\frac{\phi\lambda}{2}\right)^2 - 1}{\left(1+\frac{\phi\lambda}{2}\right)^2 + 2} \frac{RT}{A_R} \left(1 + B_i \frac{\left(1+\frac{\phi\lambda}{2}\right)^2 - 1}{\left(1+\frac{\phi\lambda}{2}\right)^2 + 2} \frac{1}{A_R} \right) \tag{14}$$

Table 2 summarizes the uncertainty budget associated with the model Eq. (13) for $p = 100164$ Pa. This discovery must be regarded as the first introductory step in the direction of a pascal realization optically because the obtained pressure standard has a relative standard uncertainty of 170 ppm at standard barometric pressure. It is simple to infer from Table 2 the key advancements required to achieve the desired uncertainty of 5 ppm: The measurement of the interferometer's imbalance L and the

Table 2 Uncertainty budget at $p = 100164$ Pa (Mari et al. 2019)

Input x_i	Value	Uncertainty source	$u(x_i)$	$c_i\, u(x_i)/p_i$	% Variance
T_i (K)	293.124	Temperature gradients in chamber V_1, calibration uncertainty, resolution, repeatability	0.029	9.89×10^{-5}	32.5
A_R (m^3/mol)	4.44585×10^{-6}	Mole refractivity (Egan and Stone 2011)	6 x 10^{-11}	-1.35×10^{-5}	0.6
ϕ (fringes)	1274.82	Systematic effects, repeatability (Mari et al. 2014)	0.05	3.92×10^{-5}	5.1
λ (nm)	632.9908	Laser wavelength (Jousten et al. 2014) (Stone et al. 2009)	9.5×10^{-13}	1.50×10^{-6}	0.0
L (m)	1.4717	Repeatability, calibration uncertainty, stability, resolution	0.0002	-1.36×10^{-4}	61.5
B_ρ (m^3/mol)	-5.95×10^{-6}	(Egan et al. 2016; O'Connell 1981)	2.4×10^{-7}	9.87×10^{-6}	0.3
Combined relative standard uncertainty ($u_c(p_i)/p_i$)				1.7×10^{-4}	100

temperature measurement within the calibration container V_1 are the two factors that contribute most to the combined standard uncertainty $u_c(p)$ in the current experimental setup.

The biggest challenge may be obtaining an accurate measurement of temperature because it necessitates further research to modify the system's current plan and design a thermal management and experimental setup that can achieve an uncertainty of the order of 0.3 mK while also considering the inhomogeneity of the temperature inside V_1. As per the computing the imbalanced L of the interferometer with an accompanying uncertainty of roughly 3 m, a different method might be used (van den Berg et al. 2015). Additionally, this aim will necessitate alterations to the optical setup in order to increase the thermal and mechanical stability of the numerous constituents. It will be essential to increase the signal-to-noise ratio, lessen the impact of the interferometer's non-linearities, build a more complicated compensating algorithm, and minimize the measurement of fringes ϕ (Gregorčič et al. 2009). Then, helium could be used because its refraction index can be determined by first principles with least uncertainty and its molar refractivity can be evaluated with higher accuracy, making it possible to compare the results of "first principles calculations" with the measurement made by the OIM to assess the effect of pressure on thermomechanical deformations of the dual mirror multiplication setup (Stone and Stejskal 2004; Schmidt et al. 2007; Moldover et al. 2014).

In order to account for the non-ideality of gas, the mathematical model represented in Eq. (13) must be updated. To do this, Eqs. (5) and (11) must be amended to include larger terms of refractivity and density virial coefficients, as detailed in (Pendrill 2004; O'Connell 1981; Achtermann et al. 1991).

National Metrology Institute Japan (NMIJ)

The OIM, which NMIJ created, can constantly evaluate an 18 kPa pressure range (Takei et al. 2020). The light source was a Littrow external cavity diode laser (ECDL) having a large adjustable frequency range, as opposed to a He-Ne laser. Because instead of a cavity with two optical routes, one optical path cavity was used for this investigation. One flat and one convex mirror, or two convex mirrors, can be employed in optical pressure measuring devices. The below equation provides the vacuum resonance condition of FP cavities.

$$\nu_{0,m} = \frac{c}{2L}(m + \gamma) \tag{15}$$

where $\gamma = \frac{1}{\pi}\cos^{-1}\sqrt{\left(1 - \frac{L}{R_1}\right)\left(1 - \frac{L}{R_2}\right)}$

Here R_1 and R_2 are the curvature radii of each mirror, c is the light speed, and γ is the ratio of the spatial to longitudinal modes of the cavity. The resonance criteria for a gas-filled at p is

$$\nu_{p,m} = \frac{c}{2\eta_p L}(m + \gamma) \tag{16}$$

If m does not change for gas and vacuum modes, then the refractive index at pressure p (η_p) is given by Eq. (17),

$$\eta_p = \frac{\nu_{0,m}}{\nu_{p,m}} \tag{17}$$

When m changes between vacuum and gas mode with the quantity Δm, it is essential to evaluate the FSR, $\nu_{FSR} = c/2L$, separately in a vacuum. While calculating η_p, the Δm is compensated and Eq. (17) is rewritten as Eq. (18),

$$\eta_p = \frac{\nu_{0,m} + \Delta m \times \nu_{FSR}}{\nu_{p,m+\Delta m}} \tag{18}$$

Thus Eq. (18) can be used to calculate η_p for the FP cavity to further evaluate the pressure.

Experimental Setup

Figure 4 displays a picture of the FP cavity that was employed in this work. Two mirrors with a 99.7% reflectivity were optically connected to a spacer constructed of ULE glass ceramic (Manufacturer: CLEARCERAM-Z) (http://www.ohara-inc.co.jp/en/product/electronics/clearceram.html). A finesse of approximately 1000 indicates that the full width at half maximum of the resonant peak is one thousandth of the full scale range. One mirror had a flat surface, and the other had a concave surface with a curvature radius of 75 cm. The length of cavity was around 5 cm, making the

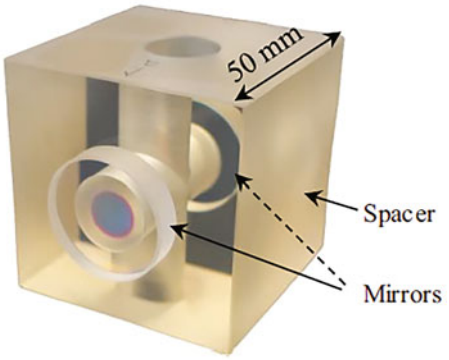

Fig. 4 Real picture of FP cavity having length of 5 cm (Takei et al. 2020)

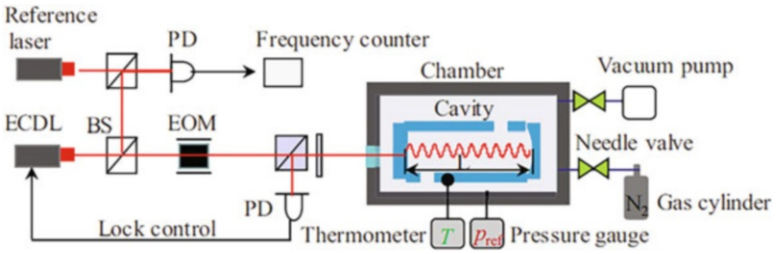

Fig. 5 Labeled diagram of OIM setup by NMIJ (Takei et al. 2020)

ν_{FSR} around 3 GHz. The hole was positioned within the vacuum compartment (Fig. 5). Sufficiently high thermal conductive metal aluminum was used to make the chamber in order to keep a consistent temperature within. The conduits that were joined to the chamber's exterior wall were also circulating water that was kept at a consistent temperature. According to the standard deviation, the thermal stability on the hollow surface was around 1 mK over a day. The cavity's temperature was considered to be the gas temperature. A calibrated pressure gauge was used to measure the pressure after the insertion of N_2 gas simultaneously to compare the readings obtained using the optical pressure measuring device (PACE1000, GE) in a pressure range of 130 kPa. This pressure gauge was calibrated with against a reference gauge with in the pressure range of 10–130 kPa with the standard uncertainty falls between 1.5 and 1.7 Pa.

An ECDL (manufacturer: New Focus 6005 Vortex, Newport) having a wavelength ≈ 632.99 nm in the variable range used as the light source for the optical pressure monitoring device. Using the Pound-Drever-Hall method, it was possible to determine the difference between the resonance and laser frequencies (Drever et al. 1983). A photodetector (PD) detected the laser beam that was reflected from the FP cavity after electro-optic modulator (EOM) modulates it at 15 MHz. Demodulating the signal from the PD yielded the frequency difference (i.e., the error signal). A proportional-integral controller transmitted the noise signal return to both ports to

adjust the frequency of the laser beam. One port was used to adjust the semiconductor chip's current, which has a quick reaction time but a limited range of adjustment. The other port was used to tune the piezo voltage, which has a sluggish reaction but a broad adjustable range, to change the diffraction grating's angle. In order to determine n_p, the laser's locked frequency was recorded. Beam splitters were used to combine a portion of the unmodified ECDL and I_2 stabilized He-Ne laser beams. A PD picked up the merged laser beam. Then this signal from PD resembles the difference between the two laser frequencies (i.e., beat frequency), which was further evaluated by a frequency counter having a gate time of 0.5 s. The ECDL's absolute laser frequency, which served as the reference frequency for the I_2 stabilized He-Ne laser, was calculated from the beat frequency. This optical pressure monitoring structure's constantly computable range is its major characteristic. The ECDL has an adjustable frequency range of more than 120 GHz. The detector and frequency counter frequency ranges are DC to 12 GHz and 0.25 GHz to 14 GHz, respectively. The continuous measuring range in the optical pressure evaluation setup with constant m is around 18 kPa since a change in pressure of 1 Pa resembles to a shift in frequency of roughly 1.3 MHz for N_2.

Discrete Pressure Measurements

Pressure was applied to both the internal and external side of the FP cavity during gas filling, which caused a little shortening of the cavity length. Even though this article merely provides a straightforward Eq. (18) for calculating refractive index η_p, the degree of deformation must be taken into account in reality. A calibrated gauge was used to compare the pressure points obtained from NIML's OIM for whole pressure range starts from vacuum to barometric pressure in order to quantify the deformation. Without altering the laser's frequency, m was varied throughout the trials at pressures of 10, 30, 50, 80, and 100 kPa.

At vacuum of 1 mPa, the resonant frequency $\nu_{0,\ m}$ was obtained. Resonant frequency $\nu_{p,\ m\ +\ \Delta m}$ was recorded when N_2 gas was filled to the specified pressure p by switching the resonance order from m to $m + \Delta m$. The resonant frequency was then measured once again when the pressure was changed to the subsequent set values. The temperature, frequency, and pressure were taken for every set of pressure values for 10 min at consecutive gaps of 8 s, and the obtained data points were utilized for the computations that followed. Five times this sequence of studies was carried out.

Using Eqs. (4), (8), and (12), pressure p was optically calculated from the observed temperature and frequencies. The recorded pressure values obtained by the calibrated gauge and optical pressure measuring equipment. The graph's slope coefficient was 0.997757. The cavity material's bulk modulus determines how the pressure variation causes the cavity to deform. The bulk modulus is equal to the nominal value of the glass ceramic employed in the spacer, which is 6.0×10^{-12}/Pa, supposing that the slope error is solely due to deformation (Silander et al. 2018). In reality, the slope value includes the impacts of both the bias and the distortion of the temperature sensor's calibration value. By dividing slope coefficient, the optically recorded pressure was adjusted for. The pressure computed using a calibrated gauge

and the pressure measured optically differ from one another. Several measurements at each pressure point exhibited repeatability of 0.1 Pa, while the range's non-linearity having a standard deviation of 0.5 Pa.

Continuous Pressure Measurements

The laser frequency was fixed to the cavity's resonance frequency while pressure was continually recorded over a wide range utilizing the optical pressure measurement method. The optical pressure measuring system can continually compute the range of ±9 kPa due to the optical system's detectable range of ±12 GHz, by fixing the resonant order m. (100 ± 9) kPa was chosen as the target pressure range. The needle valve was used to admit nitrogen gas at a pressure rise rate of around 0.5 Pa/s. The transition from 91 to 109 kPa in pressure took roughly 10 h. The beat frequency was evaluated by the frequency counter and the pressure as recorded by the calibrated gauge. The frequency range was precisely ±11.4 GHz. However, because the frequency counter employed can evaluate frequencies in the range of 0.25–12 GHz, the beat frequency curve was broken in the vicinity of about ±0.35 GHz. By dividing the slope coefficient, the pressure that was optically measured was taken into account. This experiment's frequency ranged greatly; therefore, the impact of the refractive index's dispersion (Peck and Khanna 1966) was mitigated. The readings from the pressure gauge from 91 to 109 kPa were linearly adjusted using the two calibrated points at 100 kPa and 80 kPa because the pressure gauge was calibrated at distinct pressure values. The discrepancy between the pressure values obtained using the optical pressure measuring device and the pressure gauge (Takei et al. 2020). The standard deviation of the range's non-linearity was 0.15 Pa (Takei et al. 2020).

The calibrated gauge's uncertainty was not exceeded by the slope of about 36 mPa/kPa. Because the frequency counter employed in the experiment is not capable to compute low frequencies, the beat frequency graph is broken in the middle. Beat frequencies in practically all ranges, with the exception of roughly 0 Hz, could be monitored with the addition of a frequency counter for low frequencies. The frequencies of the totally continuous entire range may be evaluated if the reference laser frequency was altered during the trials or if a second reference laser was employed. The graph's line width had an 88 mPa standard deviation. The gauge's pressure measurements exhibited a dispersion with a standard deviation of 64 mPa.

While the optical pressure measuring setup's temperature scattering data had a standard deviation of 0.08 mK, the scattering of its pressure measurements at 100 kPa had a standard deviation of 27 mPa. The optical pressure measuring setup's frequency scattering data in vacuum had a standard deviation of 0.01 MHz, which is equivalent to a scattering in the pressure measurements with a standard deviation of 7.7 mPa. This number would be equal to the dispersion of frequency measurements in gas-filled environments. Because of the temperature data, the line width was wider than the sum of the scatterings. The gas temperature along the light channel should be recorded in the optical pressure measuring system. However, the surface temperature of the cavity was recorded since it is technically challenging to detect the gas's

temperature along the path of the light due to the sensor's self-heating and spatial laser blockage. Continuous measurements may contain errors due to minute variations in temperature data and variations between the gas and cavity surface.

In this investigation, a continuous range of 91–109 kPa was used to compare the proposed optical pressure measuring system with a pressure gauge. Pressure gauges are calibrated at certain pressure values using pressure standards like pressure balances and mercury manometers. The sensitivities of the gauges' whole measurement range are adjusted using the calibration points at the discrete pressure values and the assumption that the sensitivity between the pressure points is roughly linear. Although a pressure range around atmospheric pressure was continually recorded for this investigation, it would be ideal to be able to measure a larger range of pressures, from vacuum to atmospheric pressure.

The measurement of beat frequencies at frequencies greater than 12 GHz presents a problem in extending the measurable range; however, this problem might be overcome by utilizing an optical comb with a variety of reference frequencies. Changing the gas species is a further strategy for extending the measured range. At atmospheric pressure, nitrogen's refractivity is 270×10^{-6} whereas helium's is 32×10^{-6}. The optical pressure measuring system's range might be extended to around ± 72 kPa by using helium gas with a lower refractive index, although measurement scattering would be greater.

Korea Research Institute of Standards and Science (KRISS)

The laser resonance frequency in a vacuum state, which served as a reference pressure, and that in an arbitrary pressure state were compared in a two-channel FP cavity structure also created by the KRISS team (Song et al. 2021). The cavity was constructed out of Zerodur. Taking into account the He-Ne laser's wavelength (633 nm), the pressure range to be employed, and the range of a single mode, the cavity was created as a rectangular parallelepiped with dimensions of 50 mm in width and 150 mm in length. Instead of using a readily accessible commercial laser, they independently created a single-mode (SM) He-Ne laser that allowed them to arbitrarily vary the frequency and obtain a single polarization regardless of the quantum number. With the help of our method, the internal pressure of a cavity served as a gauge for the beat frequency and pressure for about 1 Pa was measured.

Fabrication of a Cavity

They created the FP cavity structure in this work utilizing an optical approach with two canals to compute a random pressure by comparing the laser resonance frequency in a vacuum state, which serves as the reference pressure, and that in an arbitrary pressure state (Song et al. 2021). The FP cavity manufactured by the KRISS team is comparable to that of the NIST (Egan et al. 2015). Zerodur, a Schott-produced material with a thermal expansion coefficient of 1×10^{-8}/K, was chosen as the cavity's material. The FP cavity was a rectangular parallelepiped measuring 50 mm in width and 150 mm in length; it contains a rectangle channel measuring

7 mm in width and 11 mm in height at the upper section, and a circular channel measuring 7 mm in diameter at the bottom. The He-Ne laser was used as the light source, and mirrors were mounted to both extremities of the FP cavity for reflection and resonance. The two mirrors were of different shapes: one was flat, the other concave. In order to reflect more than 99.7% of the He-Ne laser utilized, the mirrors attached to the FP cavity's two ends were coated.

The mirror's curvature radius was determined using a Gaussian beam chart (Kogelnik and Li 1966), which is necessary for a precise resonance in the cavity. The lens, which has a 250 mm focal length and a 370 mm radius concave mirror, was created using a Gaussian chart. Additionally, the reflectance of each mirror was taken into account while calculating the finesse appropriate for the mirror (Eq. 19). The optical performance improves with increasing finesse. However, it is better to design the finesse to have an optimum value taking reflectance into account since it turns out to be more sensitive to the atmosphere as the finesse increases. The finesse in this study was created to have a value of 1500.

$$F = \frac{\pi (R_1 R_2)^{\frac{1}{4}}}{1 - (R_1 R_2)^{\frac{1}{2}}} \tag{19}$$

where R_1 and R_2 are mirrors's reflectivities attached to the two ports of the FP cavity.

The designed cavity length of 150 mm seems to be appropriate for the cavity to work as a stable resonator (Weichel and Perdrotti 1976). FP cavity and mirror were joined using an optical contact technique without epoxy. The vacuum level in the constructed cavity was monitored because one of the two channels of the FP cavity must retain a vacuum condition (a pressure $\leq 10^{-2}$ Pa). Epoxy is commonly employed in a variety of chemical bonding techniques, however in this situation, the outgassing from the epoxy during vacuum extraction might provide a concern. Despite using an epoxy-free optical contact approach, the finished cavity showed a minimum pressure of roughly 3 mPa after being continuously evacuated for 12 h. This demonstrated the usefulness of the optical contact approach for connecting the mirrors to the cavity.

Fabrication of a SM Polarized Laser

Instead of utilizing a commercially available He-Ne laser, they independently created an SM He-Ne laser, since it is difficult to freely modulate the frequency and to produce a single polarization regardless of the quantum number when using a commercial He-Ne laser. The optical length will vary even though the mechanical length of the resonator length L does not change when the polarization is altered in accordance with the quantum number due to the anisotropy of the cavity mirror, which is an issue. They therefore altered the Brewster's angle in a commercial laser tube for an iodine (I_2) stabilized He-Ne laser in order to give it a single polarization independent of the quantum number.

Zerodur was specifically employed to reduce the space between the two mirrors. The plane mirror jig was also equipped with a lead zirconate titanate (PZT) module

for frequency modulation (made by PI Co.), which may alter the length of the cavity. To concentrate the light between the two mirrors, the angle adjustment jig was fitted on the laser tube and the plane mirror. They measured the beating frequency with the reference I_2 stabilized He-Ne laser using a spectrum analyzer in order to assess the properties of the manufactured SM He-Ne laser. The FSR was around 890 MHz, which was close to the design frequency of 903 MHz. In comparison to the I_2 stabilized He-Ne laser, the laser displayed a beating frequency with a $\pm$ 2 MHz stability. This appears to be caused by changes in the ambient temperature.

Setup of the Laser Interferometer

The created SM He-Ne laser was locked into an FP cavity using a laser interferometer (Song et al. 2021). The laser's light beam travels via the isolator, which suppresses light produced by reflection, and the EOM, which generates a side band. It then moves successively via the focus lens, $\lambda/4$ wave plate, and polarization-beam-splitter (PBS) before entering the resonator. In order to alter the polarization of the light produced by the cavity, it is reflected and then sent via the focus lens and $\lambda/4$ wave plate once more. This light is reflected by the PBS and is then fed to the lock-in amplifier through a photodiode. Additionally, a function generator is used to provide the phase-delayed signal to the lock-in amplifier. The proportional-integral (PI) controller then uses a PZT control based on the error signal acquired from the lock-in amplifier to maintain the resonance.

With this setup, the laser frequency was locked to the upper cavity's resonant frequency while the internal pressure of the cavity was monitored using a CDG over a pressure range from 360 to 480 Pa. Without altering the mode number, the system was capable of measuring a range of 120 Pa. (m). The pressure was recorded by the CDG and the beat frequency determined by the spectrum analyzer (Song et al. 2021). It was discovered that 1 MHz beat frequency is about equivalent to 1 Pa of pressure, which is quite comparable to the findings reported by NIST (Egan et al. 2015, 2016) and NMIJ (Takei et al. 2020).

They are aiming to increase temperature stability by employing a double-chamber arrangement with an outer aluminum chamber and evaluating the cavity's arbitrary pressure using a measurement of its resonance frequency.

National Institute of Metrology (NIM)

The National Institute of Metrology (NIM), China has created an OIM on optical refractometry utilizing nitrogen gas (Yang et al. 2021). A dual FP cavity constructed of ULE glass, a copper compartment to house it, a thermal management setup, and an optical setup are further components of the NIM's OIM. By frequency-locking two tunable diode lasers to the reference and measurement cavities at a wavelength of 633 nm, the pressure within the copper compartment was recorded. The FP cavity's performance was examined, along with the impacts of creeping and the coefficient of thermal expansion. Within 1 mK, the copper chamber's temperature homogeneity was measured. All measurements were made with the leakage and

outgassing taken into account, and the so-called heat-island effect was explained and evaded. For the deformation of the reference cavity, non-linear and hysteretic behavior was seen. Direct comparison of the major piston gauges from NIM and OIM for the pressures assessed between 20 and 100 kPa indicated no deviations more than 10 ppm. The OIM's expanded uncertainty was calculated as $[(0.13\ \text{Pa})^2 + (23 \times 10^{-6} p)^2]^{0.5}$ (Yang et al. 2021).

Experimental Setup

The dual FP cavity employed in this study is comparable to the NIST design (Egan et al. 2016), except it is shorter (100 mm), with a 1.5 GHz FSR. While the thermal homogeneity of the shorter cavity length is advantageous, greater tuning ranges for the laser frequencies and a wider measuring range for the frequency counter are required. The cavity was created by optically connecting vacuum port, mirrors, and the spacer made of ULE glass (Corning Inc.) having a coefficient of thermal expansion (CTE) $\approx 3 \times 10^{-8}$/K. One mirror in each cavity is concave, with a radius of curvature of 500 mm, while the other is flat. A cavity finesse of $\approx$ 1000 corresponds to a reflectivity of 99.8% at 633 nm for all mirrors.

A copper vacuum compartment with impacted windows for coupling laser was used to seal the dual FP cavity. Two bellows located at the bottom of the copper chamber allow for gas filling and evacuation. A longitudinal slit having 2 mm width on the spacer's top connects the measuring cavity to the copper compartment's volume. About 63.5 mL of the chamber's total interior capacity may be filled with gas. The reference cavity is kept at vacuum, which is enclosed in a connecting tube made of polytetrafluoroethylene, and is not visible from the copper chamber (PTFE). The heat sharing between the copper compartment and the turbo pump-connected bellows is reduced by the PTFE tube. The bellows are submerged in a substantial sandbox to reduce vibration from the pump. The copper chamber was roasted for a number of weeks at a temperature of around 100 °C after the system had been put together and pumped down to vacuum. An ion gauge mounted in the manifold showed that the offset vacuum in the copper compartment was obtained to be less than 1 mPa.

The pressures determined by the OIM are directly influenced by the gas temperature. A temperature consistency of under millikelvin (mK) in the measuring cavity is necessary for the OIM to exceed cutting-edge pressure standards like a piston gauge and mercury manometer. Consequently, a unique thermal control mechanism was created. The next parts will provide an explanation of the optical setup and temperature control system.

Thermal Control System

Four poles were used to hang the copper chamber from an optical table, and a cylindrical temperature-controlled enclosure encased it. According to Fig. 6, the enclosure is made up of three coatings: outermost layer is filled with water which is maintained at 24.000 (3) °C, two aluminum shells with heating foils which are stable at 28 and 30 °C using thermal controllers (Eurotherm, model 2704). The thermal control setup was planned by considering following points:

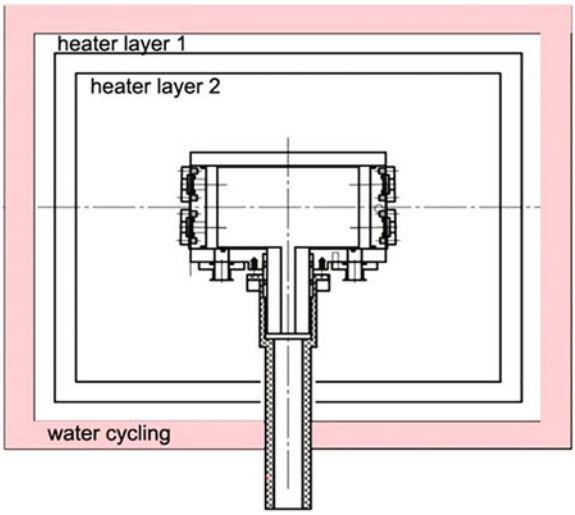

Fig. 6 Thermal controlled setup (Yang et al. 2021)

(a) The copper chamber's placement in the local area was made possible by the water-cycling shell, which also served to form a steady and constant thermal atmosphere. This is essential because the temperature of the lab is only stabilized by an air conditioner, resulting in a $\pm$ 0.5 °C difference.
(b) It was distant from the optical table to minimize vibrations from the water bath. The water tubing is about 8 m long, and the thermal steadiness of the water bath relies on ambient temperature. In order to increase the thermal constancy in the compartment, supplementary thermal-controlled heating layers were added. Dual heating layer setup's advantage over single heating layer setup is to minimize the thermal fluctuations and increase the uniformity in temperature inside the complete enclosure.
(c) The thermal enclosure's center included the location of the copper compartment, which should have a reasonable degree of temperature homogeneity which causes due to its strong thermal conductivity. Despite having many centimeters of gap between them, the copper compartment was thermally linked to inner heater layer through air. The copper chamber's temperature, which was established as stated below, is extremely stable since the aluminum shell's temperature changes are muted.

Four standard platinum resistance thermometers (SPRTs), of the capsule type, were put into the compartment at various locations to measure the temperature and its uniformity. The four SPRTs were able to measure temperatures with a maximum variation of roughly 1 mK, and the mean temperature fluctuated within 1.5 mK throughout the course of 9 h.

The degree to which the gas temperature deviates from the copper chamber's temperature is a different problem with temperature determination. This problem

was addressed by Ricker et al. (2021) in relation to a structure that had a copper compartment and the spacer made of ULE glass of equivalent size. As per the finite element analysis (FEA) simulations, it was assured that the temperature in the spacer was about 1.5 mK higher than that of inside the copper compartment after sufficient duration of applying the N_2 gas.

In seconds at the specified dimensions, the nitrogen attains the temperature of the walls, as will be demonstrated in a subsequent paper by Rubin et al. (2022). It is reasonable to presume that the temperature of the gas is somewhere between that of the copper compartment and that of the ULE glass. It should be noted that in order to prevent any potential heat-island effects, the copper chamber must be kept at low pressure (∼1 Pa) for a long enough period of time (∼15 min) during the evacuation process. Without performing this step, the temperature difference between copper compartment and the spacer is equal and greater than 10 mK which was given in more detail in section "Heat-Island Effect."

Optical Setup

The major component of the optical setup was two tunable diode lasers (Toptica, DL pro), which were frequency-locked to the reference and measurement cavities, respectively (Fig. 7). Both lasers had tuning ranges of around 20 GHz and operated at a 633 nm center wavelength. In free running mode, both the lasers have the linewidth of 200 kHz; by locking the lasers' frequencies to the FP cavities, this linewidth was reduced even further. When the diode current is changed, diode lasers operating at 633 nm frequently enter multimode. It is crucial to maintain the laser operating in single-mode in order to lock the laser to a cavity. The mode state of the diode lasers was observed using an etalon that was a commercial FP interferometer. The laser frequency was locked using dither technology to the FP cavity's resonance peak, which corresponds to the cavity's maximum light transmission. As a result of providing an oscillating signal to the diode current, the diode laser was briefly frequency and amplitude modulated at 10 kHz, causing the light transmission from the FP cavity while approaching resonance at 20 kHz. A lock-in amplifier was used to measure the photodiode's (PD) 10 kHz oscillating signal, which becomes zero when it reaches resonance. To maintain the laser frequency locked to the cavity, a proportional-integral servo was used to regulate the diode laser's piezo voltage. A PD having a 1.5 GHz bandwidth and to evaluate the beat frequency of the two lasers after they had been locked to the measurement and reference cavities, a frequency counter with gate time of 1 s was used.

Principle for Pressure Measurement

The beat frequency varies in accordance with the gas filled in the copper compartment with great N_2 sensitivity of 1.2 kHz/mPa by continually locking the lasers to the FP cavities. On the basis of fundamental physics rules, the relevant mathematical model was developed to count the pressure evaluation and assess the uncertainty.

When the FP cavities are kept at vacuum, then the locked frequencies help to solve the Eqs. (20a, 20b),

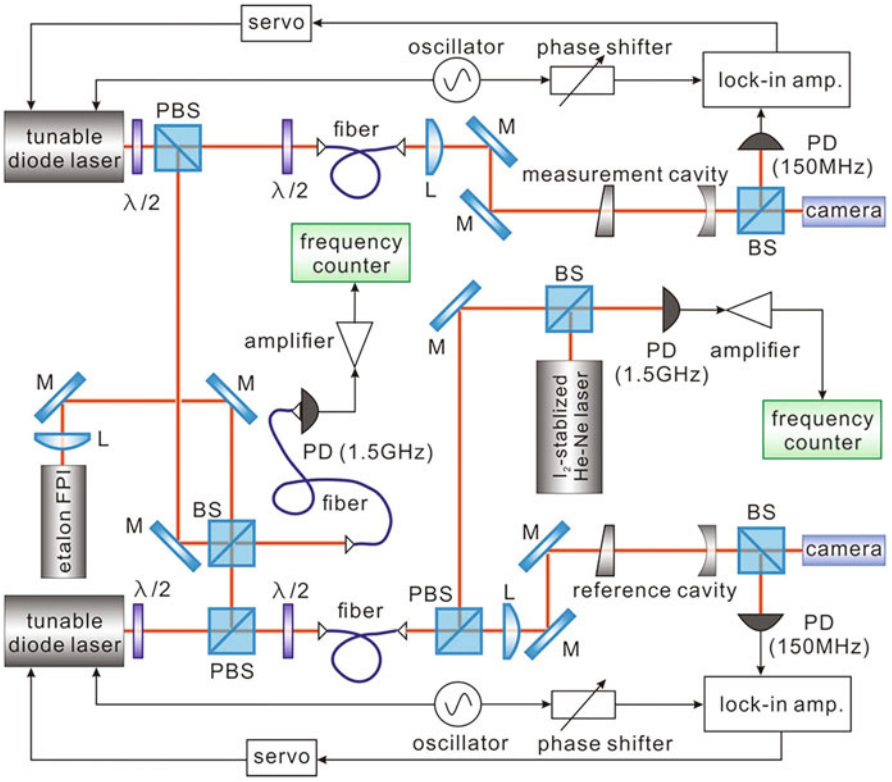

Fig. 7 Schematic drawing of the optical setup. Mirror: M, Lens: L, Half-wave plate: λ/2, Photodiode: PD, Beam splitter: BS, Polarizing beam splitter: PBS (Yang et al. 2021)

$$\nu_0^M = \left(m_0^M - \frac{\phi_0^M}{\pi}\right)\nu_{FSR}^M \tag{20a}$$

$$\nu_0^R = \left(m_0^R - \frac{\phi_0^R}{\pi}\right)\nu_{FSR}^R \tag{20b}$$

where superscripts R and M symbolize the reference and measurement cavity, respectively, ν_0 is the locking frequency, ϕ_0 is the mirror phase shift, m_0 is the mode number, ν_{FSR} is the free spectral range. Their heart rate was immediately determined to be $f_0^{MR} = \nu_0^M - \nu_0^R$. The frequency ν_0^Rf was calculated by measuring the beat frequency $f_0^R = \nu_0^R - \nu_{\mathrm{He-Ne}}$ with respect to an I_2 stabilized He-Ne laser. The locked laser frequencies vary in accordance with η and the cavity length's compression when the reference and measurement cavities are kept at vacuum and pressure p, respectively. Eqs. (20a, 20b) become:

$$\eta(1-d_m p)\nu_p^M = \left(m_0^M + \Delta m - \frac{\phi_p^M}{\pi}\right)\nu_{FSR}^M \tag{21a}$$

$$(1-d_r p)\nu_p^R = \left(m_0^R - \frac{\phi_p^R}{\pi}\right)\nu_{FSR}^R \tag{21b}$$

where ν_p^R and ν_p^M are the locking frequencies for the reference and measurement at pressure p, Δm is the change in the mode number and d_r and d_m are the distortion coefficients for the reference and measurement cavity, respectively. Mode hopping is also required if the beat frequency tends to zero, which cannot be measured with the frequency counter, in order to maintain the beat frequency below 1.5 GHz and permit measurement. Subtracting Eqs. (20a) and (21a) gives:

$$\eta - 1 - nd_m p = \left[\left(\nu_0^M - \nu_p^M\right)\left(1 + \frac{k\nu_{FSR}^M}{\pi}\right) + \Delta m\nu_{FSR}^M\right]/\nu_p^M \tag{22}$$

where k is given by $\left(\phi_p^M - \phi_0^M\right)/\left(\nu_p^M - \nu_0^M\right)$ and called as the mirror dispersion coefficient. With the beat frequency measurements $f_p^{MR} = \nu_p^M - \nu_0^M$, Eq. (22) can be written as follows:

$$\eta - 1 - nd_m p + d_r p = \frac{\left(f_0^{MR} - f_p^{MR}\right)\left(1 + \frac{k\nu_{FSR}^M}{\pi}\right) + \Delta m\nu_{FSR}^M}{\nu_0^M(1 + d_r p) + f_p^{MR}} \equiv \frac{\Delta f}{f} \tag{23}$$

The pressure may be calculated from $\frac{\Delta f}{f}$ using the Lorentz-Lorenz model and the gas state equation, as was previously mentioned (Egan et al. 2016). The beat frequency between reference and measurement (f^{MR}) is used to determine the term $\frac{\Delta f}{f}$ in Eq. (23). If just one cavity is to be used with the OIM, d_r is set to zero. The definition of $\frac{\Delta f}{f}$ switches to the right side of the equation in this situation (22). In this scenario, the difference in frequency obtained from standard (I_2 stabilized He-Ne laser) and measurement cavity is presented by adding their beat frequencies f^R and f^{MR}. In any instance, using equation, the pressure is then calculated from this (9). The high order components in Eq. (9) at a temperature of around 300 K were approximated for N_2 and He based on the published estimates of the virial coefficients (Puchalski et al. 2016; Egan et al. 2016, 2017).

Performance of the NIM's OIM

Measurements of the CTE, FSR, creeping effects, and heat island mechanism of the FP cavity were all done to study the OIM's performance at NIM. Pressure data of OIM was compared with the piston gauge (PG). An extrapolation technique was used to investigate and adjust the outgassing impact on the pressure determined by the OIM. Non-linear and hysteretic pattern for the d_r was observed and described, and the OIM's uncertainty was assessed.

FSR of the Measurement Cavity

A crucial factor for determining pressure is the ν_{FSR}^{M} of the measuring cavity, as illustrated in Eq. (23). The measurement and reference cavities, which were both evacuated to background vacuum and were in thermally stable circumstances, were locked to by the lasers in order to determine ν_{FSR}^{M}. $\Delta m = 1$ modified the locked TEM00 mode for the measuring cavity. Measurements were made of the two beat frequencies, f_0^R and f_0^{MR}. It was determined that the laser's absolute frequency was $\nu_{\mathrm{He-Ne}} + f_0^R + f_0^{MR}$, where $\nu_{\mathrm{He\ -\ Ne}}$ is defined as 473612514.643(11) MHz. Following the instructions in Axner et al. (2021), each of the beat frequency determinations for the two neighboring modes mentioned was finished in few minutes to reduce the drifting effects of the cavity length.

The experiments were performed several times, and it was discovered that the neighboring modes' frequency difference Δf was 1480294.2 ± 0.9 kHz. The FSR should be adjusted by $\frac{k\Delta f}{\pi} = -6.2 \times 10^{-6}$, where k is equal to $-1.3129 \times 10^{-8}\ \mathrm{MHz}^{-1}$ supplied by the mirror manufacturer, to account for the mirror dispersion effect. As a result, it was discovered that the measuring cavity's FSR was 1480285.0 kHz with a standard variation of 0.9 kHz. The He-Ne laser's short-term stability and the adjustment for mirror dispersion are both part of the FSR's type B uncertainty, which is expected to be 1.5 kHz.

Coefficient of Thermal Expansion

According to the manufacturer, the FP cavity's material has a CTE, or fractional change in length per K, of $0 \pm 3 \times 10^{-8}$/K. The FP cavities locking frequencies when it was kept at vacuum were continually tracked while the thermal management setup was operating in order to confirm the declared CTE value. By measuring the beat frequencies f_0^R and f_0^{MR}, respectively, it was possible to determine the cavities's locked frequencies. Even though the heat-island effect caused the FP cavities' temperature to differ from the copper chamber's under vacuum conditions, the temperature change was considered to be the same.

The CTE was derived from the frequency's rate of temperature variation (Yang et al. 2021). The effects of temperature increases and decreases on two thermal equilibrium processes were investigated. The "rapid" heating and cooling cycles that required less than 9 h were used to capture the two temperature ranges. The range of temperature near the thermal equilibrium value, which required several hours to attain.

The temperature relationship was obscured during this time because the frequency drift began to outweigh the temperature-induced drift, hence the data are not displayed. The reference cavity's CTE spans from -1.2×10^{-8}/K to 7.5×10^{-9}/K, whereas the measurement cavity's CTE falls between -2.9×10^{-8}/K and -7.5×10^{-9}/K (Yang et al. 2021). Despite the fact that the ULE spacer in the two cavities is the same, the non-uniformity in the temperature gradient and spacer's material may be to blame for the variance in CTE (or a combination of the two). In any event, these numbers match those that the manufacturer claims.

Equation (23), for example, demonstrates that a key measurand is the beat frequency between the reference and measurement cavities, f_0^{MR} and f_0^R. From its thermal dependency behavior, it is demonstrated that the frequency change was about 7 MHz/K per degree change in temperature. This suggests that the beat frequency will change by 70 kHz for a temperature change of 10 mK.

Creeping Effect of the FP Cavity

The creeping effect was noticed few 100 days by analyzing the baselines of both cavities kept at vacuum. The baselines for both the cavities were rose by 5–7 MHz during a 500-day period, showing a reduction in the spacer's length with the drift rate of about 1×10^{-8} per year. The copper compartment's temperature was maintained within a maximum difference of 0.07 °C during the experiments. The frequency shift brought on by a change in temperature will be below 1 MHz, as per the CTE value stated in section "Coefficient of Thermal Expansion." As a result, an aging procedure having spontaneous relaxation was given credit for the frequency creeping (Kobayashi et al. 2016). The reference cavity originally drifted more quickly when the cavity was repeatedly inflated and released. After pressurization at high pressures, it was discovered that the reference cavity's baseline had greatly risen (e.g., 100 kPa). If the hollow had been drained for a while, this tendency would become more obvious. For instance, the cavity was vacuum-stored from 90 to 500 days. After applying the 100 kPa, reference cavity's baseline rose by 1.3 MHz, whereas measurement cavity's baseline reduced by 0.77 MHz. This might be attributed to the tension created by the optical contact used to link the ULE vacuum port and reference cavity. Since the ULE vacuum port is cylindrical in shape, it shrank more slowly than the cavity. Over time, stress will accumulate and may be released during pressurization by relaxing, which may result in the baseline changes shown. The data before and after applying pressure was plotted for initial phase, whereas for long-term phase, only the data after applying pressure was plotted. Because of this, the initial rapid drift rates outside the linearly fitted curves were not the actual creeping rates of the cavity.

Heat-Island Effect

The gas temperature drops which further reduces the temperature of the cavity surface if the compartment is evacuated rapidly. The temperature gradient between the copper chamber and the cavity will be preserved if the background vacuum (< 1 mPa) is maintained, as a result of the poor thermal contact between them (known as the heat-island effect). The copper compartment is thermalized, but because of the ULE spacer's poor thermal conductivity at background vacuum, the spacer will continue to be colder than the chamber for a considerable period of time. When gas is added to the chamber, the PV work will increase the temperature of the gas while also transferring heat from the hollow walls to the chamber. Such an effect did occur, which shows the temperature of compartment as recorded by the SPRTs. The compartment's temperature decreased by 10 mK in 30 min after being filled with N_2 gas at 100 kPa, despite the gas's increased temperature as a result of its PV action,

showing that there existed previously a thermal gradient between the cavity surfaces and the compartment of at least 10 mK. The measurements collected after 5 h, when the gas was pushed out, also show that the thermal impact of the PV operation was around 2 mK. This PV work's scope is comparable to that of reference (Ricker et al. 2021).

The pressure readings will be impacted by a baseline offset brought on by the temperature gradient. After the gas has been evacuated, the copper chamber shouldn't be immediately fetched at vacuum; instead, it should be occupied with a nominal pressure of ≈100 Pa for a significant length of time to prevent this offset. Therefore, for baseline measurements, the copper compartment was merely evacuated to the ambient vacuum. A subsequent study will demonstrate that the cooling period after pumping out the cavity near vacuum has a time constant of a few hours. As a result, the baseline measurements were finished in 10 min for all the pressure evaluations indicated in the following sections. The temperature gradients between the copper compartment, the cavity, and the gas were calculated to be within 2 mK by applying this rule.

Pressure Measurements

A complete setup was built to link the OIM, resonant silicon gauge (RSG; Manufacturer: Yokogawa) and the PG for pressure comparison measurements. For this investigation, nitrogen 6.0 was employed, which has a purity rating of higher than 99.9999%. No data were recorded between 90 and 500 days and all the findings in this section are based on pressure readings taken after day 500 (Yang et al. 2021). As stated in section "Creeping Effect of the FP Cavity," if the aging cavity is evacuated for a lengthy period of time, the stress effect between the reference cavity and ULE vacuum port will almost certainly always take place. The stress effect theory is supported by first pressure readings from day 500. The nitrogen gas was pumped into the manifold, and during the course of 11 h, the pressures from 100 Pa to 100 kPa were reached. A Python script was used to record the OIM's frequency outputs, copper chamber temperatures, and RSG data all at once.

First, for each pressure level, d_r was determined as:

$$d_r = \frac{1}{p} \frac{f_p^R - f_0^R}{\nu_{\mathrm{He-Ne}} + f_p^R} \left(1 + \frac{k\nu_{FSR}^M}{\pi}\right) \tag{24}$$

where f_0^R is the baseline and f_p^R is the beat frequency for reference cavity with respect to the I_2 stabilized He–Ne laser at a pressure of p. It was anticipated that the cavity would deform linearly with pressure, yielding a uniform d_r value. Unexpectedly, however, showed notable variability, ranging from $1.54 \times 10^{-11}\mathrm{Pa}^{-1}$ to $1.66 \times 10^{-11}\mathrm{Pa}^{-1}$, and was steadily rising over time. The previously mentioned stress effect between ULE vacuum port and cavity is one reason for this phenomenon. As previously mentioned, pressurization caused the associated baseline to change, but the d_r calculated using Eq. (24) employed a constant baseline taken before the compartment was occupied by gas.

The baseline rose by around 1.3 MHz following the initial pressure computations (Yang et al. 2021). If d_r is assumed to be constant for all pressures, it is possible to compute the baseline change during pressurization using the d_r data. The baseline changes are computed under the constant d_r assumption of 1.60×10^{-11} Pa^{-1}. The greatest alteration took place when the highest pressure of 100 kPa was applied, and it recovered when the pressure was steadily reduced to 5 kPa. The assumption that had been formed was validated by the baseline change computed at the final 5 kPa being rather near to the baseline change seen following the initial observations.

The stress effect between the ULE vacuum port and cavity, which influence becomes considerable after creeping for a protracted period of many months, may be responsible for the variations at high pressure in the bonding state. Due to the fact that this bonding is accomplished without glue through an optical contact, it could not be strong enough to endure the load. High pressure pressurization may have relieved the tension in this situation. The baseline change with and without applying pressure was obtained to be less than 0.2 MHz, as evidenced by further observations following the initial measurements, as will be explained below. Therefore, in order to establish a stable baseline when a cavity is held for such a long duration, a pre-pressurization of up to 100 kPa is required. Eq. (9) and the virial coefficients for N_2 taken from Egan et al. (2016) may be used to determine the pressure as recorded by the OIM using the copper compartment's temperature and frequency values. The OIM pressure functioned as a single cavity utilizing $d_r = 0$ as previously described, taking into account the unpredictable d_r value affected by the stress effect. The FEA simulation outcome (Yang and Rubin 2018), which was $9.936 \times 10^{-12}Pa^{-1}$, was used to calculate d_m. The data obtained from RSG which was calibrated with respect to the PG with an expanded uncertainty of 2 Pa were compared to the OIM pressure data. The initial measurements' pressure discrepancies between the RSG and the OIM fall within a wide range of uncertainty of the RSG (Yang et al. 2021).

The difference in pressure rose when the compartment was occupied by gas, going from 100 Pa to 8 kPa, and the PG was not exposed to the components of the calibration setup like tubes, bellows, valves, etc. (Yang et al. 2021). This rise was brought on by the buildup of gas that was mostly water (water's A_R is 17% lower than nitrogen's). By shutting the valves, the leak rate of the calibration setup components, including the RSG and OIM, was evaluated. About 2 Pa/h of pressure rise was occurring. Since the OIM pressure was computed using pure nitrogen, it would have decreased if the leakage gas had simply been water vapor and increased the pressure differential relative to the RSG by roughly 0.34 Pa/h. The pressure differential growth rate is somewhat slower than the calculated rate of 0.34 Pa/h (Yang et al. 2021). This makes sense because air leakage, whose A_R is only 2% lower than that of nitrogen, may also contribute to the 2 Pa/h pressure rise rate in addition to water outgassing.

It was unable to clearly analyze the outgassing impact for the data collected after 5 h (Yang et al. 2021). The PG was left open to the manifold at that period, and once the manifold was evacuated to lower the setting pressure, gas that had collected due to outgassing or leaking was also pushed out. As a result, there was sporadic

reduction in the leakage effect on the pressure differential. The outgassing impact when the OIM and PG were compared is examined in considerably more detail in the next section.

The best PG utilized at NIM (China) was compared to the OIM's performance in the pressure range of 20 to 100 kPa in order to show the OIM's maximum potential metrological performance. The effective area (A_p) of piston-cylinder assembly (PCA) in PG (Fluke, model 7607) was determined at reference temperature (t_0) using dimensions data calibrated at NIM and rarefied gas dynamics (Yang and Yue 2011; Sharipov et al. 2016). The pressure (p(PG)) determined by PG at temperature (t) using effective area is given by,

$$p(\mathrm{PG}) = \frac{mg}{A_p[1 + \alpha(t - t_0)]} + p_{vac} + p_{head} \tag{25}$$

where g is the acceleration due to gravity, m mass of the load applied, α is the CTE of PCA, p_{vac} is the residual vacuum in the bell jar, and p_{head} is the head correction between reference levels between OIM and PG. The expanded uncertainty for absolute pressure when OIM was calibrated to PG evaluated to be ($0.05\ Pa + 1.4 \times 10^{-5}p$) (Yang et al. 2021).

To consider the leakage effects, comparison experiments were carried out in a time-controlled way. Each pressure experiment tracked a procedure known as 0-P-0: in this procedure, the baseline was initially logged for less than 10 min, then the valves are shut, and the time was documented. This procedure is similar to that used in the gas modulation refractometry (GAMOR) procedure (Silander et al. 2018). The measurements then began once the chamber was filled with nitrogen gas to float the PG at the desired pressure p. In order to record the baseline, again for less than 10 min, the manifold was pushed out until background vacuum was reached. Using a Python script, the OIM pressure, PG pressure, temperature of copper's compartment, and time stamp were logged as the measurements were being made. When the valves were shut, the time plotted against the pressure differential between the PG and OIM was 0 (Yang et al. 2021). The pressure differential trend was consistent with expected behavior for air and water leaks, both of which had values lower than nitrogen outgassing. A formula that combines two exponential decay functions was used to predict the consequences of air and water leaks. The two exponential decay functions have temporal constants of 6.35 h and 0.319 h correspond to air leakage and water outgassing, respectively (Yang et al. 2021). The genuine difference between the PG and OIM was found by fitting the data with the model and projecting to time 0. Here, using a linear fit for the initial brief period is an easy, quick, and accurate enough method (Yang et al. 2021). By utilizing the linear fit, the difference between the PG and OIM is calculated to be 0.18 Pa, whereas the exponential fit results in a difference of 0.15 Pa. The fitting method's inherent uncertainty was calculated to be within 0.05 Pa.

The relative differences between the OIM and the PG at pressures ranging from 20 kPa to 100 kPa were evaluated after the outgassing/leakage effect was removed using the fitting approach (Yang et al. 2021). There were four measurement runs

performed. All of the values fell between −5 and 10 parts per million. As a result, despite the PG's hesitation, the OIM and PG were in good agreement. As previously said, the PG should be rather stable at 1 ppm. The OIM was responsible for the scatter of roughly 10 ppm, which might also be attributed to the temperature gradient in the copper compartment, the variability of the baseline, or the unpredictability of the fitting procedure (Yang et al. 2021).

Consider the scatter to be the OIM's type "A" uncertainty. When the average of the differences across all pressures was calculated, the outcome showed a systematic inaccuracy between the OIM and the PG of 3.3 ppm. The systematic mistake is rather tiny, however it can be caused by the OIM's d_m error or the PG's effective area error. Prior to and following each pressure measurement, the baselines acquired for both cavities were noted. The history of the baseline changes illustrates the pressure measurement cycle, which includes upwards and downwards measurement. The variations in the baseline for pressure readings under 80 kPa were within ±50 kHz. The baseline changed more when the pressure was measured at 100 kPa, while the changes remained within ±100 kHz. In comparison to what was seen during the initial measurements, the baseline's stability was far better. Compared to what was seen in the earlier measurements, the variance of d_r was substantially less (Yang et al. 2021). The deformation of the reference cavity, however, was shown in non-linear and hysteretic behavior types in the history of d_r. It is unknown why these behavioral patterns exist, and more research is needed. One theory is that as tension built up on the bonding surface, the reference cavity underwent different deformation than the ULE vacuum port did. In this study, the OIM pressure was computed as a single cavity using just d_m due to the considerable uncertainty of d_r resulting from the non-linear and hysteresis behavior. The findings of the comparison between the OIM and the PG showed no non-linear or hysteretic behavior, proving that the d_m was constant and stable.

Uncertainty Evaluation

Equation (9) was used to determine the uncertainty of the OIM. Each parameter involved in calculating pressure was taken into consideration, and the involvement in the uncertainty in pressure was computed by altering the parameter's value by its uncertainty while maintaining the values of the other parameters constant. The outgassing/leakage impact was also evaluated using the fitting errors and included additional variables that contributed to the uncertainty. In Table 3, each parameter's uncertainty contribution is indicated.

The virial coefficients (A_R, B_R, C_R, B_p, C_p) values and uncertainties were taken from reference (Egan et al. 2016). Due to the small temperature difference of only 0.4 °C between this study and reference (Egan et al. 2016), greater uncertainties were chosen for B_p and C_p. Because their contributions were proportional to or quadratic in relation to the pressure, the pressure uncertainties supplied by B_R, C_R, B_p, and C_p indicated in Table 3 are at 100 kPa and were considerably decreased at lower pressures.

The history of the copper compartment temperature throughout a pressure measurement period was displayed in order to assess the uncertainty brought on by the

Table 3 Uncertainty budget of the OIM for N_2 when calibrated against PG (Yang et al. 2021)

Parameters	Value	Standard uncertainty	Uncertainty
A_R (cm^3/mol)	4.446139	1.5×10^{-5}	$3.4 \times 10^{-6}\, p$
B_R (cm^6/mol^2)	0.81	0.1	$< 0.89 \times 10^{-6}\, p$
C_R (cm^9/mol^3)	−89	5	$< 0.0018 \times 10^{-6}\, p$
B_p (cm^3/mol)	−3.973	0.12	$< 4.8 \times 10^{-6}\, p$
C_p (cm^6/mol^2)	1434	102	$<0.16 \times 10^{-6} p$
T (K)	303.28	0.00195	$6.4 \times 10^{-6} p$
d_m(Pa^{-1})	9.927×10^{-12}	1.9×10^{-14}	$7.1 \times 10^{-6} p$
Baseline (kHz)	Stability	50	40 mPa
ν_{FSR}^{M} (kHz)	1480285.0	1.8	$1.2 \times 10^{-6} p$
Outgasing/leakage	Fitting error	0.05 Pa	50 mPa
Expanded uncertainty of pressure at $k = 2$ ($U_c(p)$)		$[(0.13\ Pa)^2 + (23 \times 10^{-6} p)^2]^{0.5}$	

temperature measurement (Yang et al. 2021). The temperature changed with the fluctuation of 3 mK. It was anticipated that the temperature of the gas would match that of the copper chamber, as was previously mentioned. With the typical temperature 1 h after a pressure shift of 100 kPa, there is a maximum offset of 1.5 mK. The expanded uncertainty in the temperature measured was calculated to be 3.9 mK after accounting for the uncertainty components of SPRT (1 mK), the temperature uniformity (1 mK) and the potential temperature gradient in the copper's compartment (2 mK), and the temperature's fluctuation as computed (3 mK).

Since the OIM was used as a single cavity for the pressure computation, the uncertainty of d_r had no bearing on the OIM pressure. Due to the uncertainty data around Poisson's ratio and Young's modulus, it was challenging to get a reliable estimate for d_m from the FEA simulation. The data compared between the pressure obtained from OIM and the PG, however, may be used to calculate the d_m value and uncertainty. The PG's relative standard uncertainty was 7 ppm, while the mean difference between the OIM and the PG was 3.3 ppm. The OIM's uncertainty budget, which serves as a secondary pressure standard, is displayed in Table 3. By combining all the components of uncertainty, the uncertainty in pressure obtained from OIM was obtained to be $[(0.13\ Pa)^2 + (23 \times 10^{-6})^2]^{0.5}$.

Conclusion

The NMIs have established an OIM based on FP refractometry for vacuum pressure measurements in the maximum range of 100 Pa to 180 kPa (Jousten et al. 2017; Thakur et al. 2020). The main constituents of the OIM are an FP cavity fabricated from ULE which is enclosed by a copper compartment, a thermal management setup with a temperature homogeneity of 1 mK and the optical setup used two tunable diode lasers operating at 633 nm wavelength for the determination of beat frequency and locking frequencies in reference and measurement cavities. In addition to the

NMIs mentioned above, other NMIs and other institutes or universities (Silander et al. 2021; Beica et al. 2019; Booth et al. 2019), have also made attempts for developing OIM but still required further experimental data to be as competent as UIM (Heydemann et al. 1977); such as (1) low pressure (< 100 Pa) measurement unlike that of in UIM, 1 Pa can be measured, (2) uncertainty (8.8 ppm) which significantly higher than that of in UIM (i.e., 5.2 ppm) (Heydemann et al. 1977), and (3) pressure measurement range (i.e., 100 Pa – 180 kPa) which is far lower than UIM (i.e., 1 Pa – 360 kPa) (https://www.nist.gov/laboratories/tools-instruments/low-pressure-gauges). Despite these shortcomings, OIM still has a significant advantage over UIM, i.e., the mechanism behind measuring pressure based on the kinetic theory of gases, including fundamental constants. It can widen the pressure range using more dense inert gas that cannot be dissolved into the FP cavity glass and lower the uncertainty by adapting corrections while experimenting, like minimizing the errors in distortion coefficients.

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Part III

Applications of Time and Frequency Metrology

Time and Frequency Metrology

16

An Introduction

Poonam Arora and Amitava Sen Gupta

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Abstract

Time plays a central role in metrology and in the definitions of other SI units. Time interval or frequency is the quantity that can be determined with the highest accuracy. Nowadays, using optical frequency standards, it is possible to measure frequency with accuracy better than 1 part in 10^{17}. If such a clock runs continuously, it will not lose or gain 1 second over the age of the universe. Such precise frequency standards have become important resources for pushing the frontiers of science and technology for the benefit of the society. Ever-reducing uncertainties in frequency measurement have been steadily driving the innovations in the field of lasers, low-noise electronics, high-resolution imaging, statistical tools, and many others. Before we start to understand various concepts on time and frequency covered in this section, it is better to have a brief review of commonly used terms in time and frequency domain followed by a concise overview of the forthcoming chapters on timescales, atomic frequency standards, and time transfer techniques.

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Basics of Time and Frequency Measurements

Time-of-the-day, time interval, and frequency are the three basic information of interest in time and frequency metrology. Time-of-the-day, defined in terms of hours, minutes, and seconds, sometimes clubbed with date and year, is what is displayed or recorded by a clock. Setting two clocks to display the same time is called synchronization. Time interval is the duration between two events, measured in seconds. Time is one of the seven base units of the SI system and is used to define many other units. Frequency is the rate at which an event repeats. It is reciprocal of time period or time interval. It is measured in events/cycles per second and the standard unit is hertz (Hz). A device that is source of frequency is called an oscillator. The process of setting different oscillators to same frequency is called syntonization. An oscillator's quality is often defined in terms of accuracy, stability, and precision. Accuracy is the extent of closeness of a measurement, or the average of a set of measurements, to the definition of the quantity being measured. Accuracy is related to the offset from an ideal value. Stability indicates how well an oscillator can produce the same frequency offset over a given time interval. It doesn't indicate the correctness of a quantity being measured but only if it stays the same. It is basically a statistical estimate of time or frequency fluctuations of a signal over a given time interval. Fluctuations over intervals less than 100 s usually define short-term stability, while those over longer intervals (usually a day or longer) define long-term stability of an oscillator. Stability can be measured in either the frequency domain or time domain. Due to the non-static or dynamic data, the stability of an oscillator in time domain is defined with Allan deviation rather than standard deviation.

The stability of an oscillator improves as the averaging period gets longer, as some noise types can be removed by averaging, until it hits the noise floor. Noise floor is the point beyond which longer averaging does not improve the performance. The slope of the Allan deviation vs. averaging time graph (frequency instability graph) helps in the identification of the noise type prevalent in the measurement. Five noise types (with different slopes on the frequency instability graph) are white phase, flicker phase, white frequency, flicker frequency, and random walk frequency.

In the frequency domain, the standard measure for characterizing phase and frequency instabilities is one half of the double-sideband spectral density of phase fluctuations. The amplitude instability in the frequency domain is one half of the double-sideband spectral density of the fractional amplitude fluctuations. The unit of such measurements is decibels or dBc/Hz.

Precision is related to reproducible consistent measurements. It refers to the extent with which a set of measurements of a quantity stays close to the mean value of the set. For more details on various terminologies and basics on time and frequency metrology, the reader is referred to (Jespersen and Fitz-Randolph 1999; Riehle 2004; Lavine 1999; Arias et al. 2018).

The signal for time measurements is usually a pulse that occurs once per second (1 pps), while the signal for frequency measurements is usually a frequency of 5, 10,

and 100 MHz. Any clock or oscillator can be characterized and calibrated with its frequency or 1 pps signals.

Timekeeping and Timescales

Since the redefinition of SI second in terms of cesium transition frequency, the timekeeping is being done with atomic clocks worldwide (Panfilo and Arias 2019; Petit et al. 2015). For critical applications, an ensemble of clocks is used instead of relying on a single clock in order to avoid any single-point failure. In an ensemble of clocks, the time outputs of individual clocks in a group are combined, using a statistical averaging or timescale algorithm, to form a timescale. National Metrology Institutes (NMI) of several countries operate national timescales (based on ensemble of atomic clocks) and contribute the data to BIPM on regular basis for generating the international reference timescale known as Coordinated Universal Time (UTC). The International Bureau of Weights and Measures (BIPM), an international organization, established by the Metre Convention, is responsible for global timekeeping as it calculates "the time" for the world by processing the data collected from atomic clocks kept in national institutes around the globe. Since January 1, 1988, the maintenance of International Atomic Time (TAI) and Coordinated Universal Time (UTC) has been the responsibility of BIPM. TAI is a uniform and stable scale which is not related to the slightly irregular rotation of the Earth. The timescale with astronomical definition based on rotation and orbital motion of the Earth is known as UT1. UTC is derived from TAI by the adding or removing an integral number of seconds so that UTC tracks the irregular rotation of the Earth. UTC is identical with TAI except that from time to time "one second" (known as "leap second") is added or removed to ensure that the difference between UTC and UT1 remains within 0.9 s. The dates of application of the leap second are decided and announced by the International Earth Rotation and Reference Systems Service (IERS). The local realization of UTC at laboratory "k" is named UTC(k). On a monthly basis, BIPM issues "Circular T," a scientific bulletin, with deviation and uncertainties of UTC – UTC(k). More details on timescales are given in the chapter on "Atomic Timescales" in this section.

Standards of Time and Frequency

The quest for better timekeeping has been about finding better and stable oscillators. In the preatomic era, timekeeping and standard frequency measurements were mostly based on mechanical and electrical oscillators (Bize 2019). There were mechanical clocks (based on balanced wheel, pendulum, etc. as oscillators) and later on quartz clocks (based on periodically vibrating crystal as oscillator). With the advent of atomic frequency standards, there has been a rapid improvement of

timekeeping accuracies, which improved by nine orders of magnitude in the last 100 years alone. There can be several standards, but the primary standard is the one which realizes the SI unit of a quantity with highest accuracy. Primary frequency standard as present is cesium fountain clock, where Cs atoms are laser-cooled before they are tossed through the microwave cavity and interrogated for change in state. However, there is a possibility of redefinition of SI second in terms of an optical transition in the near future. Each type of atomic frequency standard is discussed and described in the chapter on "Atomic Frequency Standards" in this section.

Time and Frequency Transfer Techniques

Various modern applications require clocks or oscillators at different locations to be synchronized or syntonized. Different kinds of time and frequency transfer techniques are used to compare and adjust clocks and oscillators at different locations. It can be done through signals broadcast via many different media, including coaxial cables, optical fiber, radio signals, telephone lines, and the Internet (Bauch 2015; Matsakis et al. 2014; Lavine 2008). For synchronization, one requires an on-time pulse as well as a time code broadcasted or obtained from the reference or transmitting station. For syntonization, one requires extracting a stable frequency from the broadcasted or transmitted information. The frequency can be extracted from the carrier or time code or other information modulated onto the carrier.

Precision techniques for time and frequency transfer are based on satellites. The most common satellite time and frequency transfer technique in based on global navigation satellite system (GNSS), GPS being the most widely used satellite system for precise navigation and timing (PNT) services. Chapter on "Time Transfer via GNSS" in this section elaborates more on GNSS-related technical details.

The most precise satellite time and frequency transfer technique at present is two-way satellite time and frequency transfer (TWSTFT) which allows time transfer with ± 1 ns accuracy. TWSTFT is mostly used by NMIs and mission-critical strategic sectors for maintaining the traceability with respect to a better known reference. Basics, present, and future of TWSTFT technique is presented in chapter on "Time and Frequency Transfer via TWSTFT" in this section.

To complete the section, chapter on "Time and Frequency Transfer via Telephone, Radio Waves, and the Internet," has been included. These are lower accuracy transfer methods and used as common methods of time and frequency dissemination to the masses.

Summary

This section covers all important aspects of time and frequency metrology starting from the realization of the SI unit of time to the time and frequency transfer techniques while discussing the applications of precision time and frequency metrology.

After this brief introduction chapter, chapter 2 is on "Atomic Timescales," chapter 3 is on "Atomic Frequency Standards," chapter 4 is on "Precise Time Transfer via Telephone, LWR, and Network," chapter 5 is on "Precise Time and Frequency Transfer via TWSTFT," and finally, chapter 6 is on "Precise Time and Frequency Transfer via GNSS."

After going through the section, readers will realize that time interval and frequency are the most precisely measured physical quantities. Several critical applications, which have become integral part of modern lifestyle such as navigation and high-speed optical communication, rely completely on atomic clocks and allied technologies. There has been phenomenal improvement in the field of timekeeping in the last several hundred years especially in the last 100 years which saw nine orders of magnitude in performance. The realization and rapid development of optical frequency standard since the beginning of twenty-first century is enabling newer applications and opening new frontiers in the understanding of the universe. Ever-reducing uncertainties in frequency measurement are still steadily driving the innovations in the field of lasers, low-noise electronics, high-resolution imaging, statistical tools, and many others. Not alone the atomic frequency standards, there has been an equivalent amount of emphasis on developing better and more accurate time and frequency transfer techniques. However, there is still a lag in the intercontinental comparison of optical clocks. Redefinition of SI second on the basis of optical clocks will not be done until remote comparison of optical clocks over different continents becomes feasible. It is not possible to predict how this dynamically growing field will shape up, but there will surely be more accurate oscillators with further reduced uncertainties and newer unexplored applications will continue to emerge in time and frequency metrology.

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Atomic Timescales

17

J. M. López-Romero and C. A. Ortiz-Cardona

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Abstract

In this chapter general aspects of atomic timescales (AT) are discussed. As an introduction, the history and importance of AT and their applications is presented. Then, the construction of AT from an ensemble of atomic clocks is exposed. Different types of AT exist depending on their applications. There are AT defined by a physical signal, typically an electrical signal, usually distributed in real time by different means. An example of this type of AT is the UTC(k) timescale. For most developed countries, the UTC(k) atomic timescale is the reference for the

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national official time. Meanwhile, there exist also AT that are not defined by a physical signal, alternatively defined by a mathematical algorithm. An example of this type of AT is the Coordinated Universal Time, UTC, timescale which is produced by the *International Bureau of Weights and Measures* (BIPM). The UTC has a remarkable role worldwide because, mainly, it is the international reference for the local UTC(k) timescales. This chapter discusses how the UTC atomic timescale is produced by the BIPM and how it and UTC(k) timescales provide traceability to the time unit of the International System of Units (SI), the second. Finally, the most important technical requirements in setting up an AT laboratory are listed.

Introduction: A Brief History of Time Scales

When the history of timescales began? With the aim to give an answer to this interesting question, we certainly can say that timekeeping is as old as the civilization itself. The current sexagesimal structure to measure time (minutes of 60 s, hours of 60 min) is a reminiscence of the ancient cultures civilizations and their methods in measuring time. It is well known that these ancient cultures used mathematics in base 60. The reader can remember here that our nowadays common mathematics are built in base 10 and computers work in mathematics in base 2. As a curious fact, we would like to draw the reader's attention on the fact that the "average" human heartbeat rate is about one beat per second. Here we can ask ourselves if this is just a casual coincidence or, by the contrary, the Sumerians used the human heartbeat as unit of time to build timescales (minutes, hours, days, and so on)? A positive answer to this question about Sumerians will certainly explain many facts about time measurement through the different epochs. Any way, we will not go further over this question here. Soon in the history appears a practical way to measure time using the position of the Sun. It was known that the most prominent celestial body, the Sun, lasts 24 h to repeat its position in the sky so that measuring the position of the Sun provides a practical way to measure the time of the day. Sundials were artifacts broadly used through ancient times to measure the time of the day. With minor importance, other astronomical objects were also used to measure time, and so build calendars. One of those astronomical objects was the Moon. An example of a very old civilization that used the Moon to measure time, and build calendars, was the Hebrew civilization.

Astronomical methods to measure time remained until 1967 as the best way (due to the redefinition of the second in the SI) to determine the time of the day and build calendars as well. That situation changed around the Renaissance when mechanical artifacts started to be used as clocks. Galileo Galilei (1564–1642) in Florence, now Italy, first noticed that the period of a pendulum remained nearly constant for small amplitude oscillations. That fact was used to develop the first clocks based on pendulums. Slowly, mechanical clocks got more importance as good artifacts to measure time. In the sixteenth century already existed very accurate pendulum clocks. In some point astronomical methods and mechanical clocks were in a kind of competition to be the most precise method to keep and measure time. Mechanical

clocks were an excellent alternative for timekeeping in quiet environmental conditions (no mechanical vibrations, no big changes in humidity and temperature). However, mechanical clocks showed significant limitations in harsh environments (considerable humidity and temperature variations, and important mechanical oscillations and vibrations like those in a ship on sea). One of the facts that marks the beginning of the Renaissance was the discovery of the American continent. The secure navigation of ships on their transatlantic voyages was a very important issue that involved precise time measurements. The competition between astronomical methods and mechanical ones as the best way to measure time was evident, when the British Government offered, through the longitude act in 1714 (by the parliament), a prize of 20,000 £ (an immense fortune in those days) for a solution which could provide a method to measure the longitude to within half a degree (2 min of time) (Burton 2017; Seidelmann and Hohenkerk 2021). The methods would be tested on a ship sailing from Great Britain to any such port in the West Indies as those commissioners choose without losing their longitude beyond the limits before mentioned and should prove to be tried and found practicable and useful at the sea. Many of the most brilliant and educated people worked on an astronomical method to solve the longitude problem. However, it was solved by John Harrison, a working-class joiner with little formal education. Harrison concentrated on the rigorous scientific establishment of his time and won the longitude prize through extraordinary mechanical insight, talent, and determination. The era of secure navigation on seas by using accurate mechanical clocks was born. One of the most precious elements on a ship was precisely its clock.

The discovery of piezoelectric effect (Uchino 2017), and its application to develop quartz oscillators, was an important milestone in timekeeping. The first crystal-controlled oscillator was built in 1917 by Alexander M. Nicholson at Bell Telephone Laboratories in the United States (Nicolson 1940; Barrett-Gonzalez and Wolf 2022). Quartz crystal oscillators and their application to build clocks were a remarkable technological revolution for timekeeping.

Once quantum physics was developed with the contribution of a few dozens of brilliant scientists during the first half of the twentieth century, the astronomic era was ending and the atomic era began. Quantum physics predicted different energy states for the hydrogen, and hydrogen-like, atom (Davies 1992; Jaeger 2022). By the 1930s the scientific community was curious if predictions of quantum mechanics for hydrogen-like atoms were true or not. Around that date, Isidor Isaac Rabi, at the University of Columbia, conducted an extraordinary experiment, among other objectives, to test if magnetic resonance produces results compatible with predictions of quantum physics (Chodos et al. 2009). Time after, in 1945, Norman Ramsey, at Harvard University, developed the separated oscillatory magnetic fields method used to test the hyperfine structure of hydrogen-like atoms (Ramsey 2000). Both, Isaac Rabi and Norman Ramsey, received by their contributions the Nobel Prize in physics in 1944 and 1989, respectively (Ramsey 1990; Rigden 2000). By 1955 Louis Essen, at the National Physical Laboratory (NPL) in England, using the separated oscillatory magnetic fields method developed by Norman Ramsey, built the first atomic clock (Essen and Parry 1957). Microwaves at 9.192 GHz from a

quartz oscillator were used to produce the transition between the two hyperfine levels of the ground state of a hydrogen-like atom, the ^{133}Cs atom. Time measurement using this atomic clock was found to be more accurate than the astronomical time based on the Earth rotation. In 1967, during the 13th General Conference on Weights and Measures (CGPM), the unit of time of the SI was defined in terms of the frequency of the electromagnetic radiation associated with the hyperfine transition of the ^{133}Cs ground state, as the duration of 9,192,631,770 periods of such radiation (Barrell 1969; Newell et al. 2019). The atomic timekeeping era was born. Nowadays precision on timekeeping is such that the best atomic clock will accumulate an error of a second after nearly a million years of operation.

As a continuous accumulation of seconds produces a timescale, the continuous accumulation of atomic seconds produces an atomic timescale. The current UTC timescale was implemented in 1972 (Smith 1972; Nelson et al. 2001), and it is produced through the participation of timekeeping laboratories which operate atomic clocks and share data with the BIPM. In section "Atomic Timescale," several atomic timescales that the BIPM produces as a preliminary step before getting the UTC are introduced.

The importance of timekeeping in a uniform, consistent, and broadly recognized fashion has long been recognized. Since centuries ago, it was well known that accurate timekeeping is needed for navigation because longitude (Sobel 2004), the geographical coordinate, can be determined by time measurements along with astronomical observations (Tavella and Petit 2020). Local timescales and timekeeping conventions originated centuries ago to support commerce, navigation, and later transportation systems (Dohrn-van Rossum 1996). More recently, timescales that were used internationally date back to Greenwich Mean Time (GMT), an astronomical timescale based on the prime meridian at the Royal Observatory Greenwich in London, England. In 1880, GMT was legally adopted as the official time of Great Britain, and numerous other nations legally recognized GMT as official time during the next few decades. At the twentieth century, in 1928, based on a recommendation of the International Astronomical Union (IAU), the GMT became known as Universal Time (UT). Several improved versions of UT, including UT1 and UT2, were in use before the advent of atomic timekeeping (Howse 1997; Bartky 2007) in 1967.

The development of quantum physics at the beginning of the twentieth century set the bases for the invention of atomic clocks, which first appeared in the late 1940s (Essen and Parry 1955). The accumulation of seconds produced by atomic clocks led to atomic timescales that were far more accurate and stable than the astronomical timescales. Coordinated Universal Time (UTC) was a name first used in the early 1960s to refer to an international timescale based on the use of atomic clocks. However, neither atomic time nor UTC was internationally accepted at that time. In 1967, at the 13 CGPM (*Conférence Générale des Poids et Mesures*), the International System of Unit (SI) second was redefined not in terms of astronomical observations but in terms of quantum properties of the cesium atom. The SI second was established as the duration of 9,192,631,770 periods of the electromagnetic radiation associated with the transition between the two hyperfine levels of the ground state of the ^{133}Cs atom. In 1972 was when the world's current system for

keeping official time went into practice; the use of UTC was corrected periodically with integer leap seconds to keep it in phase with the Earth's rotation angle and so with astronomical timescales (Nelson et al. 2001). The *Bureau International des Poids et Mesures* (BIPM) is the international organization responsible for maintaining and disseminating the UTC timescale (Quinn 2011). The definition of the SI second based on the ^{133}Cs atom formally marked the beginning of the atomic timekeeping era, after many centuries of referencing time interval measurements to astronomical observations.

Currently, the measurement of time is of the utmost importance for many applications, including global satellite navigation systems (Riehle 2006; Giorgi et al. 2019; Tavella and Petit 2020), communication networks (Li et al. 2017; Mahmood et al. 2019), electric power transportation (Lombardi et al. 2016), astronomy (Bhonsle et al. 1981), electronic transactions (Lombardi et al. 2016), and national defense and security.

Atomic Timescale

The continuous accumulation of atomic seconds produces an atomic timescale. There are different types of atomic timescales. Among the most important timescales, it can be mentioned the following (Gérard et al. 2015): the free atomic timescale (EAL), the International Atomic Timescale (TAI), the Coordinated Universal Time (UTC), the physical realizations of the UTC (the UTC(k) atomic timescales), and the Terrestrial Time (TT). EAL, TAI, and UTC timescales are produced by the Time Section of the BIPM with data contribution from about 80 timing laboratories geographically distributed worldwide. The UTC(k) timescales are produced by different timing laboratories, most of them national metrology institutes (NMIs), equipped with atomic clocks that are maintained within a continuous local and/or international comparison.

Clock Ensemble

Timekeeping for critical applications requires demanding characteristics on reliability, stability, and accuracy. Because of the susceptibility to failure of any single clock, critical applications that need time measurement, like national official timescale, or timescales for navigation purposes, among other applications, require the simultaneous operation of more than one atomic clock. A group of N ($N \geq 2$) atomic clocks, which are maintained under a continuous comparison among them, can be used to produce a timescale with superior metrological characteristics to those obtainable from any single member of the clock set. The set of such atomic clocks is referred to as a clock ensemble. The most efficient use of a clock ensemble implies the generation of a timescale which is produced in terms of time difference measurements among the participant clocks. Such timescale is called an averaged timescale.

Averaged Timescale

If a set of N atomic clocks are under comparison among each other, then an averaged atomic timescale TA can be defined in the simplest way as an average of the readings of the clock ensemble:

$$TA(t) = \frac{1}{\sum_{j=1}^{N} \omega_j(t)} \sum_{i=1}^{N} \omega_i(t) h_i(t), \tag{1}$$

where $h_i(t)$ and $\omega_i(t)$ are the reading and weight of the clock i at time t, respectively. Weights must satisfy the normalization condition $\sum_{i=1}^{N} \omega_i(t) = 1$. As a first approach, it is possible to define weights $\omega_i(t)$ as constant in time. However, the best scenario is given when a dynamic weighting approach is implemented, which increases or decreases the weight given to a clock according to its frequency stability characteristics, which, in turns, can change over time for different reasons. Unfortunately, the eq. (1) cannot be implemented experimentally as it is written, because the readings from clocks $h_i(t)$ are not physically observable. Experimentally it is possible to measure time differences between two clocks. The equation can be written in terms of time differences between pairs of participating clocks by subtracting from each side of equation the reading $h_k(t)$ of clock k at time t:

$$TA(t) - h_k(t) = \frac{\sum_{i=1}^{N} \omega_i(t) |h_i(t) - h_k(t)|}{\sum_{j=1}^{N} \omega_j(t)}. \tag{2}$$

If we define $x_k(t)$ as the time difference between clock k and the TA at time t, and define $x_{ki}(t)$ as the difference between clock k and clock i at time t, it is $x_{ki}(t) = h_i(t) - h_k(t)$, then

$$x_k(t) = \sum_{i=1}^{N} \omega_i(t) x_{ki}(t). \tag{3}$$

This equation provides a way to access TA values at the time t through the reading of clock k of the ensemble and the time differences x_{ki} between clock k and the rest of clocks at time t. Notice that, by using the previous equation, once the set of time differences $x_{ki}(t)$ are measured, it is then possible to compute the TA. However, this way of computing the TA corresponds to a post-processing scheme. In cases where a computation of the timescale values are needed not in a post-processed regime but in real time, it is necessary to introduce a method of prediction for the time differences of clocks with respect to the TA. For some applications, like the formation of UTC and UTC(k) timescales, it is important to predict future values of time differences between clocks of the ensemble.

One of the simplest models to predict the time difference $x_{ki}(t+\tau)$ of a clock k with respect to the *TA* once the values $x_{ki}(t)$ are known is (Weiss and Weissert 1991)

$$\widehat{x}_k(t+\tau) = x_k(t) + \left[y_k(t) + \frac{D_k\tau}{2}\right]\tau + \ldots, \tag{4}$$

where $\widehat{x}_k(t+\tau)$ is the prediction of the time difference of clock k respect with *TA* for the future time $t+\tau$; $x_k(t)$ is the time difference between clock k and the *TA* at time t; $y_k(t)$ is the fractional frequency difference of clock k respect to the *TA* frequency at time t; and, finally, D_k is a constant accounting for linear changes in $y_k(t)$.

Once the future time $t+\tau$ is reached, it is possible to measure the time differences among participating clocks, and then, it is also possible to compute the value of the *TA* for that time $t+\tau$. Of course, the predicted value of the *TA* computed at time t for the time $t+\tau$ will not be equal to the computation of *TA* at time $t+\tau$. Once the time difference measurements $x_{ki}(t+\tau)$ at time $t+\tau$ are known, the prediction $\widehat{x}_k(t+\tau)$ for $t+\tau$ can be corrected according to the following equation:

$$x_k(t+\tau) = \sum_{i=1}^{N} \omega_i[\widehat{x}_k(t+\tau) - x_{ki}(t+\tau)]. \tag{5}$$

On the other hand, the prediction $\widehat{y}_k(t+\tau)$ of the fractional frequency deviation of clock k with respect to the *TA* can be made according to

$$\widehat{y}_k(t+\tau) = \frac{\widehat{x}_k(t+\tau) - x_k(t+\tau)}{\tau}. \tag{6}$$

Finally, when the future time $t+\tau$ is reached, the fractional frequency deviation prediction $\widehat{y}_k(t+\tau)$ can be corrected by

$$y_k(t+\tau) = \frac{1}{1+m_k}[\widehat{y}_k(t+\tau) + m_k(\tau)y_k(t)], \tag{7}$$

where m_k is given by

$$m_k(\tau) = \frac{1}{2}\left[\sqrt{\frac{1}{3} + \frac{4}{3}\frac{\tau_{min,k}^2}{\tau^2}} - 1\right], \tag{8}$$

and $\tau_{(min,k)}$ is the integration time at which the noise floor (Tavella and Thomas 1991; López-Romero and Díaz-Muñoz 2008) of clock k is reached.

Paper Clock (Virtual Clock)

The iteration of the equations presented in the previous section allows the formation of an *TA*. It is important to notice that, because such a *TA* is a result of a mathematical

process, there is not a physical signal associated with *TA*. In that sense, *TA* behaves as a "paper clock" or "virtual clock." It is also important to mention that very important atomic timescales, like EAL, TAI, UTC, and TT(BIPM), are all paper clocks.

In many applications, not only accurate and stable paper clock but also accurate and stable physical signal that can be distributed by different means is needed. Once a paper clock *TA* is produced, it is then possible to implement an experimental setup to "convert" *TA* in a physical signal TA_0. A way to achieve that goal is selecting one clock of the ensemble, typically the most stable and reliable, to correct its output frequency signal in order to "copy" the frequency of the paper clock *TA*. Let us refer to a such selected clock as Master Clock (MC). The corrected frequency of the MC is added as a "new clock" into the ensemble. Now, let us refer to the corrected frequency of the MS as y_0. In order to avoid any undesirable perturbation on *TA* due to the manipulation of the frequency $y_0(t)$, its corresponding weight $\omega_0(t)$ must be set to zero; it is $\omega_0(t) = 0$ for all time t. The frequency $y_0(t)$ is then adjusted to produce time differences with respect to *TA* as small as possible; it is $x_0(t) \approx 0$ for all t. In other words, in the conversion process of the paper clock *TA* into a real clock TA_0, it is necessary to correct the frequency of the MC in order to copy the frequency of the *TA*. Eventually time adjustments could be also needed (Fig. 1).

Coordinated Universal Time (UTC)

The BIPM, located in Sevres, France, is responsible to produce the international reference timescale, the Coordinated Universal Time (UTC). The UTC is the

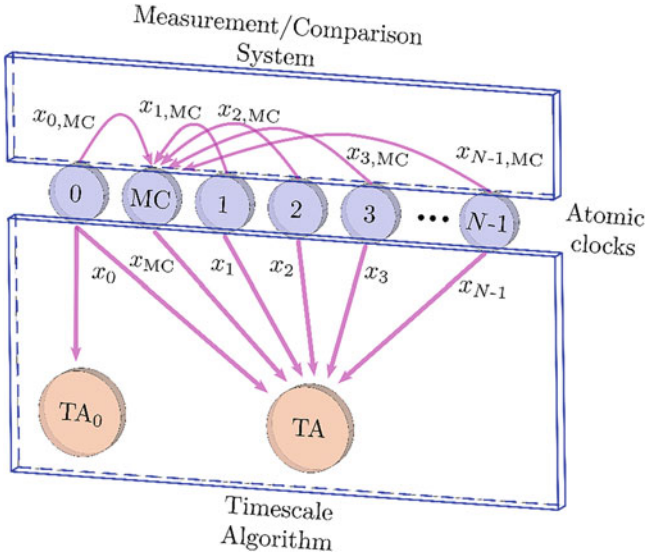

Fig. 1 Clock ensemble and timescale algorithm producing a paper clock *TA* and its physical realization TA_0

international reference timescale that forms the basis for the coordinated dissemination of standard frequencies and time signals; UTC is obtained from International Atomic Timescale (TAI) by the insertion of leap seconds according to the advice of the International Earth Rotation and Reference Systems Service (IERS) to ensure approximate agreement with the time derived from the rotation of the Earth, the UT1 timescale. The UTC timescale is defined by the next equations (Beard and The role of the ITU-R in time scale definition 2011; McCarthy 2011):

$$UTC = TAI + n, \tag{9}$$

where n is an integer number of seconds that made

$$|UTC - UTC1| \leq 1\ s. \tag{10}$$

UT1 in turns is an astronomical timescale which is proportional to the rotation angle of the Earth (Arias and Petit 2019; Wang et al. 2020). The IERS is the international organization in charge of *UT1*.

The TAI timescale is produced through a comparison of the fundamental time interval of the free atomic timescale (EAL) with respect to primary realizations of the SI second maintained in about a dozen NMIs. The EAL timescale is produced by clock data contribution of about 80 laboratories that operate atomic clocks. The main characteristic of the EAL is to keep a robust and continuous timescale with very high frequency stability. The UTC is an averaged timescale which is produced by a similar algorithm to that described in the previous section. Figure 2 shows the process at the BIPM to produce the UTC timescale.

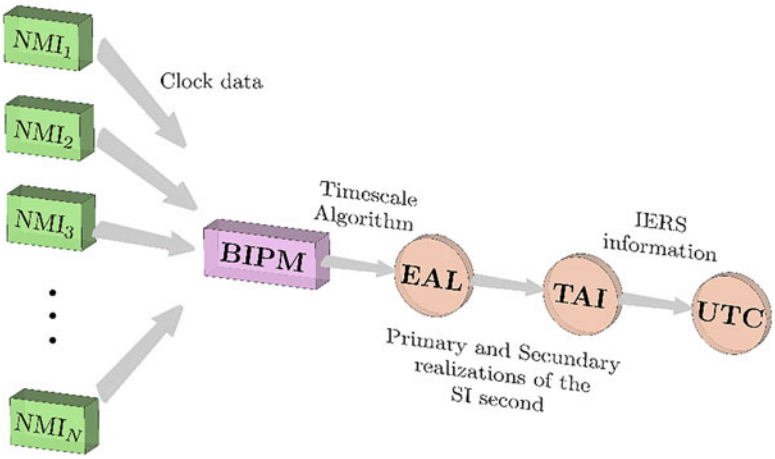

Fig. 2 Production of UTC timescale at BIPM through data clock contribution from NMIs. There are about 80 NMIs contributing with clock data to the BIPM ($N \geq 80$), around 420 atomic clocks contributing to the formation process of the UTC and 17 primary and secondary realizations of the SI second

The UTC(*k*) Timescales

With the aim to achieve enough level of reliability, stability, and accuracy, national measurement institutes (NMIs) usually keep in operation, and under a permanent comparison among them, more than one atomic clock to produce its local physical realization of the UTC timescale. The physical realization of the UTC, made by a laboratory k, where k is an identifier of such laboratory, is referred to as UTC(k). Physical realizations of UTC, named UTC(k), are maintained mainly in NMIs and astronomical observatories, which contribute to UTC computation by sending periodically their clock measurements to the BIPM. The UTC(k) atomic timescales can be seen as predictions of UTC. Values of [*UTC-UTC*(k)] at 5-day intervals are published in the monthly BIPM Circular T document (BIPM 2022a). For those institutes participating in the CIPM MRA (Di Giuseppe et al. 2013; MRA-D, CIPM 2016), their results are also published in the form of a key comparison, CCTF-K001. UTC (BIPM 2022b), resulting in a monthly publication of the degrees of equivalence with respect to the worldwide reference timescale UTC. Currently, there are about 80 laboratories (Panfilo and Arias 2019; Bauch et al. 2020), distributed worldwide, which maintain their own physical realization, UTC(k), of the UTC. The BIPM publishes the complete list of those laboratories on the Circular T. Some physical realizations of the UTC with high frequency stability and excellent approximation to UTC are UTC(USNO), UTC(NIST), UTC(PTB), UTC(OP), UTC(ROA), and UTC(NPL), among others. The identifiers of laboratories are USNO for US Naval Observatory, NIST for National Institute of Standards and Technology of the United States of America, PTB for *Physikalish Technische Bundesanstalt* of Germany, OP for Observatory of Paris of France, ROA for *Real Observatorio de la Armada* of Spain, and NPL for National Physical Laboratory of England.

In Fig. 3(a) and (b), it is shown the time differences between UTC and UTC (USNO) and UTC(PTB), respectively, from 50,800 MJD (December 18, 1997) to 59,699 MJD (April 30, 2022), as an example of performance of UTC(k) atomic timescales.

Physical realizations of the UTC are very important for economy and demanding real-time applications that need accurate time. Such is the case of telecommunications industry, navigation systems, defense, and electrical power distribution systems, among others. Signals of UTC(k) are distributed by different techniques: GNSS common-view and all-in-view methods (Weiss et al. 2005), two-way time and frequency transfer methods (Kirchner 1991), radio signals (Lichtenecker 1997), time transfer protocols by the Internet like precise time transfer protocol (PTP) (Marc et al. 2016), network time transfer protocol (NTP), (Mills and others 1985) and telephone.

The Circular T of the BIPM

The BIPM Circular T document is a monthly publication of the BIPM Time Department. Circular T is the source of traceability to UTC for the physical

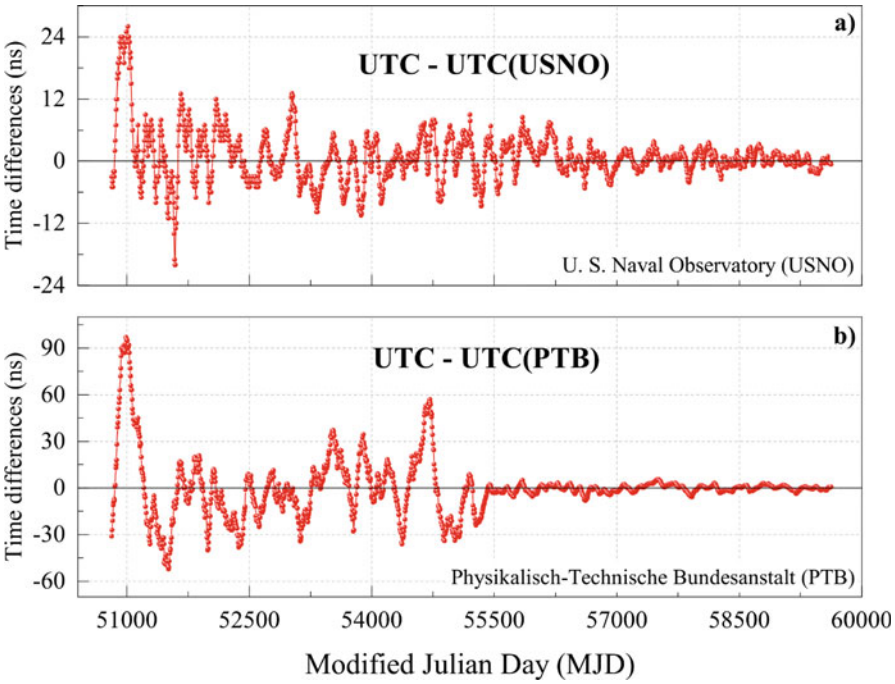

Fig. 3 Time differences between (**a**) UTC and UTC(USNO), (**b**) UTC and UTC(PTB) atomic timescales from 50,800 MJD (December 18, 1997) to 59,699 MJD (April 30, 2022)

realizations UTC(k). Circular T provides one numerical value of the time difference [*UTC-UTC*(k)] for every 5 days and corresponding uncertainties, for about 80 institutes that contribute regularly to the BIPM with their clock comparison data. It includes also other relevant information like time links used for the computation of TAI, calibration information of the time transfer equipment, and the corresponding uncertainties. Figure 4 shows a portion of the first page of the BIPM Circular T 412 (May 2022).

Other Atomic Timescales

The Rapid UTC, UTCr

Because UTC values are published monthly providing one value for every 5 days, UTC is not well adapted for real and almost-real-time applications. With the aim to serve the international community that needs high accuracy time measurements in real and almost-real-time regime, the BIPM produces the rapid UTC, denoted as UTCr (Petit et al. 2013; López-Romero et al. 2019). The UTCr is a preliminary computation of the UTC, with the purpose to know, with several weeks in advance, an estimation of UTC values that will be published in the next Circular T. Values of UTCr published by the BIPM weekly provide daily values of *UTCr-UTC*(k) time differences.

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The contents of the sections of BIPM Circular T are fully described in the document "Explanatory supplement to BIPM Circular T" available at https://webtai.bipm.org/ftp/pub/tai/other-products/notes/explanatory_supplement_v0.6.pdf

1 - Difference between UTC and its local realizations UTC(k) and corresponding uncertainties.
From 2017 January 1, 0h UTC, TAI-UTC = 37 s.

Date 2022 0h UTC	MAR 31	APR 5	APR 10	APR 15	APR 20	APR 25	APR 30	Uncertainty/ns		Notes
MJD	59669	59674	59679	59684	59689	59694	59699	uA	uB	u
Laboratory k				[UTC-UTC(k)]/ns						
AGGO (La Plata)	545.4	554.5	549.5	559.6	597.3	606.7	612.7	1.0	20.0	20.0
AOS (Borowiec)	-3.0	-2.2	-1.4	-0.8	-0.5	0.2	0.4	0.3	3.1	3.1
APL (Laurel)	-0.2	-1.2	-1.4	-1.2	-1.4	-1.9	-2.5	0.3	19.7	19.7
AUS (Sydney)	-539.4	-547.1	-538.9	-534.2	-525.3	-523.7	-514.9	0.3	11.2	11.2
BEV (Wien)	-14.1	-18.1	-24.9	-38.7	-45.1	-47.6	-47.6	0.3	2.6	2.6
BFKH (Budapest)	4569.6	4603.9	4638.6	4666.4	4699.0	4737.9	4768.8	1.5	20.0	20.1
BIM (Sofiya)	16011.5	16022.9	16040.0	16057.5	16064.5	16094.8	16119.3	0.3	7.1	7.1
BIRM (Beijing)	0.0	-1.4	-3.8	-7.0	-7.0	-0.5	9.5	0.3	3.0	3.0
BOM (Skopje)	-	-	-	-	-	-	-			
BY (Minsk)	0.5	1.3	0.9	0.7	0.3	0.5	-0.1	1.5	2.8	3.2
CAO (Cagliari)	-	-34953.1	-35068.3	-35185.6	-35298.0	-35412.7	-35526.8	1.5	20.0	20.1
CH (Bern-Wabern)	-1.4	-1.0	-0.8	-0.7	0.4	1.3	1.7	0.3	1.5	1.6
CNES (Toulouse)	-11.1	-13.8	-14.6	-14.2	-9.1	-2.3	2.1	0.3	2.6	2.6
CNM (Queretaro)	0.7	3.4	0.8	-4.0	3.4	-2.4	6.6	1.5	4.0	4.2
CNMP (Panama)	17.5	18.7	6.0	14.6	15.9	6.7	12.0	0.7	5.2	5.3
DFNT (Tunis)	1793.2	1899.4	1991.2	2086.8	2177.8	2271.9	2375.0	0.7	20.0	20.0
DLR (Oberpfaffenhofen)	-7.8	-7.5	-7.2	-7.2	-6.7	-7.0	-7.6	0.7	2.6	2.7
DMDM (Belgrade)	25.7	18.5	-	-	-	-	-	0.3	3.5	3.5
DTAG (Frankfurt/M)	-47.9	-44.6	-31.9	-25.4	-16.1	-15.1	-15.0	0.3	3.0	3.0
EIM (Thessaloniki)	7.1	7.5	6.8	6.1	13.2	10.0	7.4	4.0	11.2	11.9

Fig. 4 BIPM Circular T 412 (May 2022) with time difference UTC-UTC(*k*) data from March 31 to April 30, 2022

Terrestrial Time, TT

The Terrestrial Time, denoted as TT, is defined by the International Astronomical Union (IAU) as $\mathrm{TT} = \mathrm{TAI}^{*} + 32.184$ s, where TAI^{*} is a recalculation of the TAI (United States Naval Observatory 1992; Panfilo 2022). TT is defined to be exactly 32.184 seconds offset from TAI at 00:00 of January 1, 1977. Before 1955, TT must be extrapolated backward using ephemeris time (ET), a timescale defined by the motions of solar system bodies (Petit 2003). ET was originally based upon the length of the tropical year 1900. Different realizations of TT are denoted by $\mathrm{TT}_{(xxx)}$, where *xxx* is an identifier of the laboratory/organization in charge of such realization (Matsakis 2007). One of the most important realizations of the TT is that of the BIPM, denoted as TT(BIPM), computed in deferred time, each January. TT(BIPM) is based on a weighted average of evaluations of the frequency of TAI by the primary and secondary frequency standards. It is mostly used in scientific applications requiring long-term frequency stability and high-frequency accuracy. Values of the TT(BIPM) can be found at the BIPM website (BIPM 2021).

Hybrid Atomic Timescales and the Redefinition of the SI Second

Since around 2000 year, secondary representations of the SI units based on optical transitions became with equivalent or better uncertainty that the primary realizations based on the ^{133}Cs atom. Figure 5 shows progress on the fractional frequency uncertainty of optical clocks. Since then, optical clocks have improved their uncertainty by about two orders of magnitude with respect to cesium clocks. Due to this fact, the international time and frequency metrology community, the BIPM, the

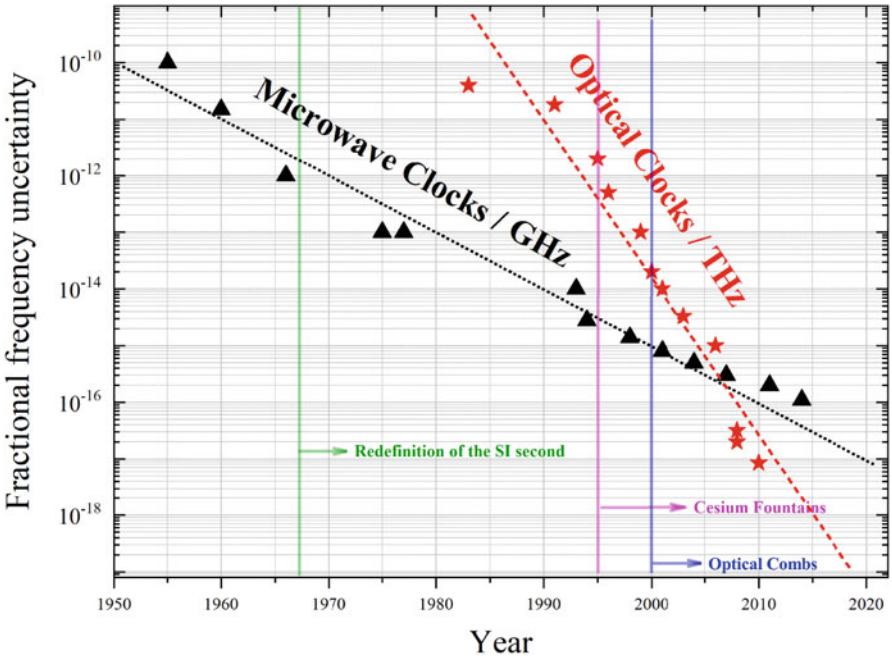

Fig. 5 Progress of microwave and optical clock uncertainties (Margolis 2010; Ludlow et al. 2015; McCarthy and Seidelmann 2018)

CCTCF, and CIPM are preparing the road to adopt a new definition of the SI second based on one, or even several, optical transitions.

One technical requirement to be in place to adopt such a new definition is the one related with capabilities to produce atomic timescales based on an optical transition, or in other words, the capability to produce atomic timescales based on the operation of optical clocks. Currently some research groups are developing the knowledge and technology to produce timescales based on the operation of both types of atomic clocks, radiofrequency (RF) atomic clocks (Cs, Rb, H), and optical atomic clocks (Yb, Sr, Al, Hg, etc.) (Margolis 2010; Ludlow et al. 2015; McCarthy and Seidelmann 2018). A timescale produced by the contribution of RF and optical clocks is called a hybrid atomic timescale, or, simply, hybrid timescale. Methods based on machine learning (ML) algorithms, particularly neural networks, are among the investigations in progress to produce hybrid atomic timescales (Jian et al. 2019).

Technical Requirements for Setting up a UTC(*k*) Laboratory

Figure 6 shows the basic infrastructure for a UTC(*k*) laboratory. Of course, the atomic clocks are the main equipment of the whole laboratory system. It is recommended to operate, at least, three atomic clocks to be redundant producing

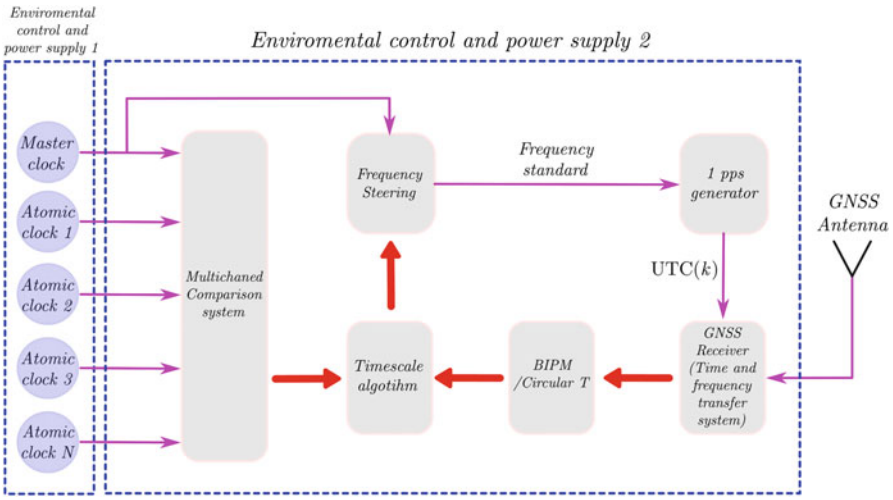

Fig. 6 Basic infrastructure for a timekeeping laboratory/UTC(k)

atomic seconds but also to be capable to determine, at any time, if a particular clock has or not an adequate metrological behavior. If a unique clock is in operation, then it is not possible to know if that clock is working properly. If two clocks are in operation, and time differences or frequency differences exhibit an anomaly, there is no way to determine which clock is not working properly. With at least three clocks in operation, it is possible to determine which clock is misbehaving (if there is any anomalous condition). Also, an ensemble with a least three atomic clocks allows a better clock performance characterization, among other reasons, because it is possible to use the three-cornered hat method to estimate the absolute, not relative, frequency stability of any clock of the ensemble (Gioia et al. 2019; Parker et al. 2022).

Additionally, it is necessary to have a comparison system to measure continuously time or frequency differences between atomic clocks. Ideally, it is necessary to compare any clock with the rest of the ensemble; however, this technique requires a robust and powerful comparison system. For instance, if N atomic clocks are in operation, then, the number of channels needed to support these comparisons is $N^2(N-1)$. This number increases rapidly as N^3 when N increases. However, an affordable way to compare the N atomic clocks is comparing a "master clock" with the rest of clocks of the ensemble. Let say that $x_{0,i}(t) = T_i(t) - T_0(t)$ is the time difference between the master clock 0 and the i^{th}-clock at time t. Here, $T_i(t)$ and $T_0(t)$ represent the reading of clock i^{th} and master clock at time t, respectively. Now, suppose you are interested to find $x_{(k,j)}(t)$, the time difference between an arbitrary pair of clocks j^{th} and $k^{\text{th}}x_{(k,j)}(t)$ at any time t, then you can find it by using the relation $x_{(k,j)}(t) = x_{(0,j)}(t) - x_{(0,k)}(t)$. Using this strategy to compare N atomic clocks, a comparison system with *N-1* channels and a well-selected "pivot clock" (here named also as a master clock) is enough. Actually, this comparison philosophy is

used by the BIPM to produce the UTC with the participation of around a hundred timekeeping laboratories. In the case of the BIPM, the PTB (*Physiskalish-Technische Bundesanstalt*) is selected as the "pivot laboratory."

The most efficient use of an ensemble of N atomic clocks is through a timescale algorithm, so that a system containing such algorithm feed by the measurements data is required to steer the frequency output of the master clock. The frequency steered will be the frequency standard of the laboratory. This timescale algorithm must also consider the information on the Circular T of the BIPM to maintain the frequency standard of the laboratory as close as possible to the definition of the SI second. Now, the UTC(k) is defined by a pulse per second (PPS) electrical signal, which is typically a square pulse with a rise time of around 1 ns a duration of around 1 μ s and an amplitude of about 5 V (Fig. 7). Therefore, a 1 PPS generator is also needed as part of the fundamental infrastructure of a timekeeping laboratory.

Finally, the 1 PPS signal that defines the UTC(k) must be compared to the UTC. To do that, a time and frequency transfer system is required. Typically, a global navigation satellite system (GNSS) receiver is used. The GNSS receiver measures the phase difference between the local 1 PPS signal and the 1 PPS signal of a GNSS constellation (GPS or GLONASS). That information is used along with a technique named common view (Allan and Weiss 1980; DangLi et al. 2011) to determine the time difference between the UTC(k) timescale and the UTC(k') produced by another timekeeping laboratory k'. The UTC(k)-UTC(k') time differences are used by the BIPM to produce, first, the free atomic timescale (EAL) and then with the participation of primary frequency standards produce the International Atomic Timescale (TAI), and, finally, with information from the International Earth Rotation and Reference Systems Service (IERSS), the BIPM produces the UTC. Information published by the BIPM about the time differences UTC-UTC(k), available in a monthly fashion on the Circular T document (BIPM Circular T 2022), must be used by a k laboratory to keep the UTC(k) as close as possible to the UTC. In other

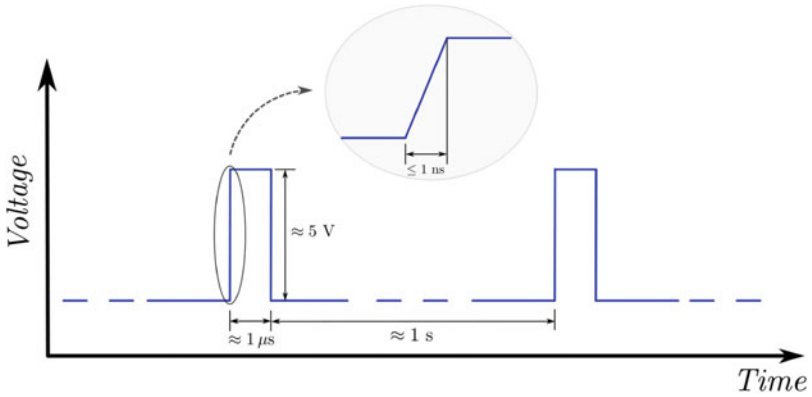

Fig. 7 A typical 1 PPS signal used to define the UTC(k) timescale.

words, the UTC(k) timescales are the prediction a month ahead, and physical realization as well, of the UTC.

The Most Used Atomic Clocks in UTC(k) Laboratories

There are several types of commercial atomic clocks which are used by UTC(k) laboratories, designated institutes (DIs), or NMIs worldwide to generate an AT and create an official and/or reference time with superior metrological performance for different purposes.

Every atomic clock, independent of the atomic specie, has, in general, a quartz oscillator which is the flywheel that can be adjusted using information of atomic transitions like rubidium (Rb), cesium (Cs), or hydrogen (H) atoms. Also, an option of a rubidium or quartz oscillator adjusted using GNSS is commercially available. The type of oscillator used depends on the metrological requirements of a particular application. Specifically for UTC(k) labs, the BIPM reports the number of clocks participating in the UTC calculation since 2004. There are two kinds of clocks reported, hydrogen masers and cesium clocks (data are presented in Fig. 8), which general aspects are presented below.

Cesium Beam Atomic Clock

It is the most used atomic clock worldwide by UTC(k) laboratories, telecommunications companies, and different sectors which require a very precise and reliable

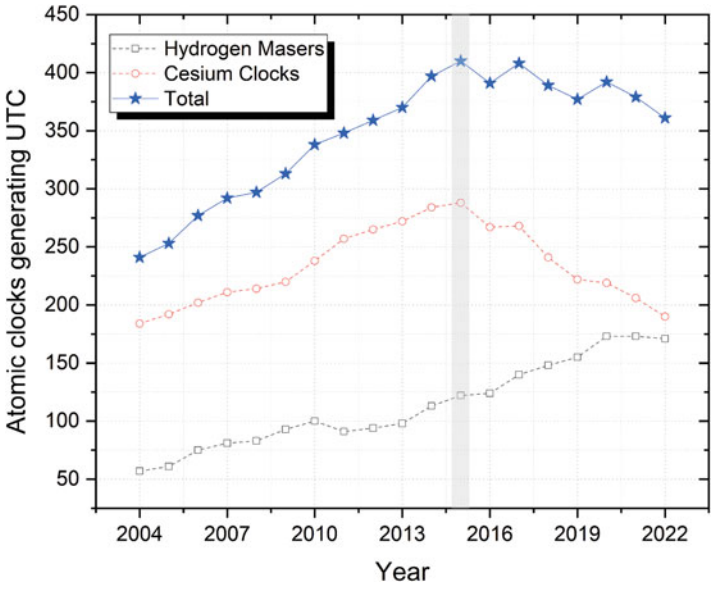

Fig. 8 Number of atomic clocks (hydrogen masers and cesium) generating UTC, reported by BIPM since 2004

source of time. The market prices are round some tens of thousands of US dollars. Cesium atomic clock uses the intrinsic properties of the cesium atom to operate, particularly, the alignment of the nuclear and electron spins. The minimum energy state of this system corresponds to both spins pointing in opposite directions. The required energy to align the spins is 38 μeV which according to Planck's equation corresponds to a frequency of 9,192,631,770 Hz.

In this kind of clock (Fig. 9), cesium is evaporated at the atomic source to create a thermal beam of atoms which travel at less than 300 ms^{-1}, through a low-pressure chamber. First, atoms in the minimum energy state are selected using a magnetic selector. Then, selected atoms pass twice through a very precisely controlled microwave field in a very uniform magnetic field, tuned to the resonance frequency which produces that cesium atoms evolve to a higher energy level. The cavity is isolated from external magnetic fields using a magnetic shield to avoid perturbations from external magnetic fields. Then, a second magnetic selector deviates the atoms which did not complete the transition to be absorbed by a carbon getter and deflects toward a detector those which evolves to the excited state. Using this information, the quartz oscillator is adjusted automatically by a servo control to maximize the number of cesium atoms detected, keeping the microwaves tuned to the correct frequency. After the small remaining biases are determined and eliminated, the short-term stability (observation time of 1 second) is less than 1×10^{-11} and reach the noise base at 1×10^{-14}.

Hydrogen Maser

It is the most precise, elaborate, and expensive frequency standard commercially available. The market prices are round some hundreds of thousands of US dollars.

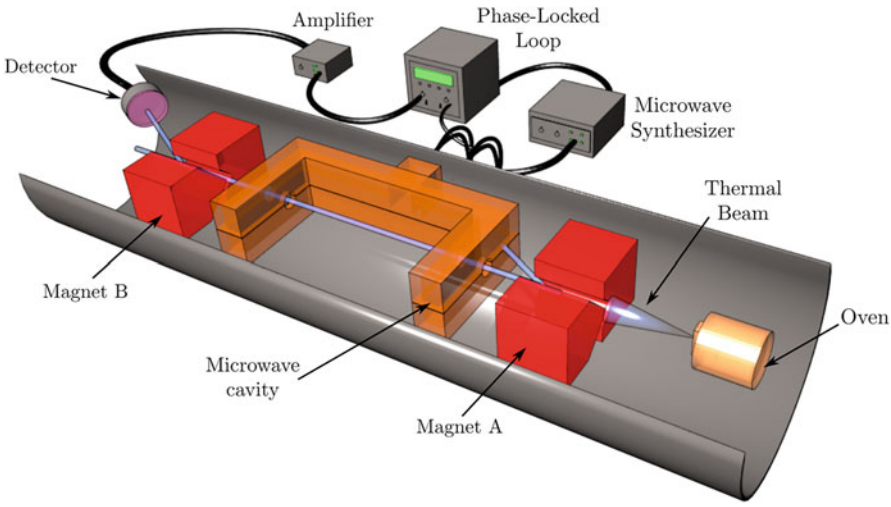

Fig. 9 Schematical diagram of a cesium-beam frequency standard using magnetic state selection and detection

The word maser means microwave amplification by stimulated emission of radiation (Townes 1965; Kekez 2018). The maser effect was the antecessor of the laser effect (Schawlow and Townes 1958; Kronenberg and Traxer 2019) in which the photon frequency is higher. In a maser, microwave photons are "copied" using a seed photon as reference to obtain an adequate signal for practical applications.

A hydrogen maser uses the intrinsic properties of the hydrogen atom to operate. Particularly, the alignment of the proton and electron spins. The minimum energy state of this system corresponds to both particle spins pointing in opposite directions. The required energy to align the spins is 5.87 μeV which according to Planck's equation corresponds to a frequency of 1,420,405,751.786 Hz.

In this artifact (Fig. 10), a controlled amount of molecular hydrogen gas, H_2, is fed into a discharge lamp to dissociate the molecules using an electric arc. Then, hydrogen atoms pass through a collimator and state selection magnets to be stored in a quartz cavity covered with polytetrafluoroethylene (PTFE) to reduce perturbations, chemical interactions, and absorption. This cavity is inside another microwave cylindrical resonant cavity made of precisely machined copper or ceramic covered with silver, tuned to the atom's resonant frequency to align the spins of the electron and the proton (1.420 GHz). Eventually some atoms decay emitting microwave photons which stimulate other photons to drop to a lower energy

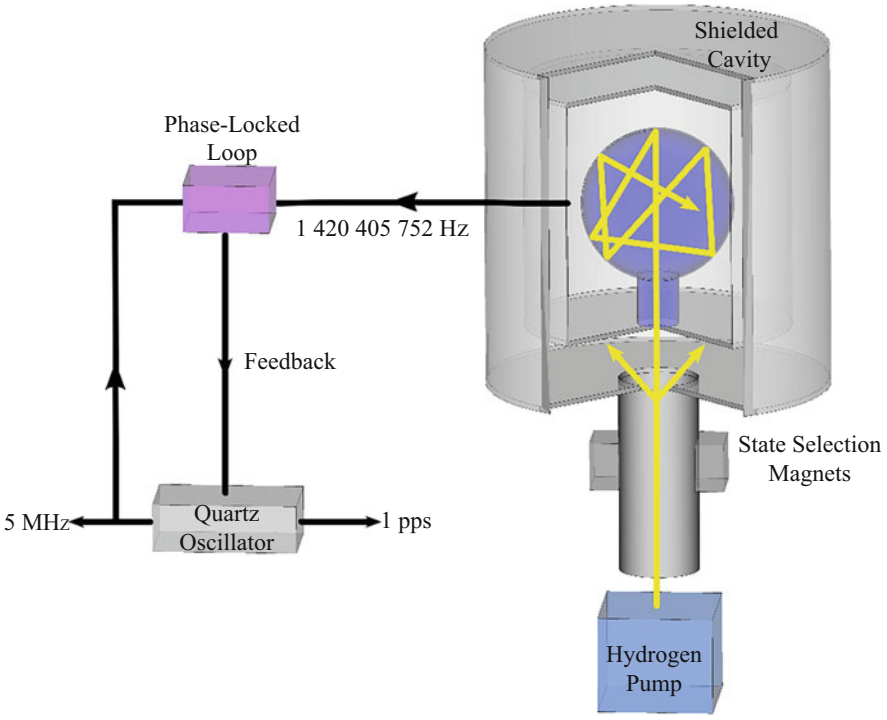

Fig. 10 Schematical diagram of a hydrogen maser frequency standard

level and, therefore, emit additional photons, and it creates a self-sustained microwave field. The cylindrical cavity redirects photons back to the system to maintain oscillations. Also, an axial low-intensity magnetic field is applied to break the degeneracy of the magnetic sublevels (Zeeman sublevels). Additionally, the cavity is surrounded by magnetic shields to minimize perturbations produced by external magnetic fields. Finally, a high-spectral purity signal tuned to the hydrogen resonance is produced, and it is continuously emitted as new atoms are fed into the system. This signal is used as reference to adjust the frequency of a quartz oscillator.

The hydrogen resonance frequency is lower than that of cesium (9,192,631,770 Hz); however, the width of the hydrogen resonance is around some Hz. Therefore, the quality factor, Q, is about 10^9, or at least an order of magnitude better than a commercial high-performance cesium clock. The short-term stability (observation time of 1 second) is less than 1×10^{-12} and reaches the noise base at 1×10^{-15}. Due to the quadratic variation of the resonance frequency as time function, the long-term stability is worse in comparison with cesium atomic clocks. However, the most recent models include the cavity tuning feature to compensate the frequency drift and achieve better long-term instabilities than cesium beam clocks.

There are two kinds of hydrogen masers: actives and passives. In the former case, the microwave cavity is the oscillator itself which requires a high atomic density and a high Q (Bai et al. 2018; Tiezhong et al. 2020). Meanwhile, in a passive hydrogen maser, the cavity is fed by an external frequency reference, and it is tuned to maximize the cavity response. The active hydrogen masers are typically four times more stable than the passive ones and one hundred times more stable than high-performance cesium beam clocks.

Traceability to the SI Second

Participation of primary frequency standards on the process to steer the free atomic timescale (EAL) implies that the fundamental time interval of the UTC timescale is a realization of the definition of the SI second (Panfilo and Arias 2019), so that the UTC provides traceability to the unit of time. It is important to noticed that, because UTC is a virtual timescale (no physical signal exists to define UTC), practical access to UTC is through the UTC(k) timescales. UTC(k) is the physical realization of the UTC established by laboratory k. Time dependence of the UTC-UTC(k) differences gives information about the frequency difference between those two timescales. In other words, Circular T contains information about traceability of UTC(k) to UTC, so that a calibration process of an oscillator with respect to the UTC(k) timescale can be taken as part of a chain of comparisons with respect to the international realization of the SI second. An adequate estimation of uncertainties on the calibration process of a local oscillator with respect to the UTC(k), along with the information published on the Circular T, establishes the traceability to the SI second.

Conclusions

Time measurements are strategic in contemporary economy. Different systems for critical applications, like telecommunications, power distribution, global navigation, deep-space vehicle navigation, defense, precise astronomical observations, and stock markets, among others, strongly depend on atomic clocks and AT. The AT results from a coherent accumulation of seconds. Therefore, it is important to understand well this topic and ensure that time requirements are attended. UTC and UTC(k) AT can provide traceability to the SI second. The next generation of AT will provide traceability to the upcoming definition of the SI second, which will be in terms of an optical atomic transition (Gill 2016; Bize 2019). The SI second, the AT, and time measurements are the SI core and the basis of ultra-high accurate measurements.

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Atomic Frequency Standards

18

Poonam Arora and Amitava Sen Gupta

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Abstract

From tracking the movement of celestial bodies to mechanical clocks and now prevalent atomic clocks, timekeeping has been one of the greatest quests for finding stable and better oscillators. Measurement of transition frequency between two designated levels of an atom forms the basis of atomic clocks. Time is derived from the measurement of frequency and that is the reason that atomic clocks are more appropriately called atomic frequency standards. Since the realization of first atomic frequency standard based on cesium in 1955, the field of precision time and frequency metrology has been revolutionized. Along with the evolution of timekeeping, a detailed description of various types of atomic frequency standards, applications, and the roadmap toward redefinition of SI second is presented in this chapter.

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Keywords

Timekeeping · Atomic Clocks · Cesium fountain · Optical clocks

Introduction

Four coordinates that are required to precisely define when and where something takes place consist of three spatial ones and one temporal one, i.e., time. Time may or may not be fundamental and often is illusive for researchers. Even Einstein said that time is relative for each and every observer and there is nothing like "absolute time." But time is real and it does exist as we are able to observe measure and quantify its passage. Measuring time has been a fundamental quest for humans and various ways have been developed over centuries for tracking it. The quest for precision in timekeeping has made it possible to measure time with nanosecond accuracy routinely. The modern technology of atomic clocks has been a game changer in pushing the frontiers of precision timekeeping. This chapter begins with section "Brief History and Evolution of Clocks" on brief history of timekeeping and evolution of clocks. The classification of atomic frequency standards is described in section "Classes of Atomic Frequency Standards" along with description, principle of operation, and other significant details on each type of atomic frequency standards. Some of the major applications which critically depend on atomic clocks are discussed in section "Applications."

Brief History and Evolution of Clocks

Evolution of time tracking technologies is related to the human evolution. In the early ages, people needed to measure just years and months. This led to track the time based on a calendar to define interval of time. Tracking the position of sun in the sky was good enough to have a perception of time, and further precise measurement of time was simply not needed then. Egyptians invented sundials in 1500 BC which divide day into smaller parts based on duodecimal system. The time was decided on the basis of lunar cycles (12 or 13 lunar months) in a year and numbers of stars were used to count the passage of night. However, the Babylonians created sexagesimal system for their mathematics and astronomy which was based on Sumerians counting (counting in 60s). In the year 46, Julian calendar was formed by Julius Caesar during Roman Empire and was further modified by Pope Gregory XIII in 1582 which was adopted worldwide.

For tracking the passage of time, various other methods were adopted as sundials had limited use depending on the presence of sun. Water clocks, candle clocks, hour glass, etc. were used before the mechanical clocks were invented as reliable means of timekeeping. With the rise in industrialization, the need was felt to have timekeeping device which could measure passage of time in seconds, based on physical mechanism. In the year 1656, a Dutch scientist, Christiaan Huygens, developed first

pendulum clock-based period of oscillation. The clock developed had shown error less than 1 minute in 1 day. Further Huygens in 1675 developed clock having balance wheel and spring assembly, which was able to achieve time with even better accuracy. In 1761, John Harrison from England developed clock based on chronometer which could be used to calculate longitude while at sea and was able to measure time with one fifth of a second a day accuracy. By then, it was realized that every clock works on some version of pendulum (or oscillator). The progress of evolutions of clocks has been presented in Table 1. Mechanical clocks worked well but the gears and springs wore down with time, thus limiting the time accuracy gradually. For details of timekeeping, evolution of clocks and basics of time and frequency standards, readers are referred to books (Vanier and Audoin 1989; Riehle 2004; Major 1998; Jespersen and Fitz-Randolph 1999; Sullivan et al. 1990).

Another breakthrough in timekeeping was the discovery of quartz crystal. It was now possible to develop a clock based on oscillations produced in crystalline quartz on applying electric field (due to piezoelectric effect). However, quartz crystal exhibits disadvantages as resonance frequency depends on thickness of the resonator and also quartz oscillation frequency alters with temperature and humidity variations.

Advances in science, technology, and many industrial processes and machines required to have further precise measurements of time. Most of the modern digital

Table 1 Progress of clocks

Standard	Oscillator	Counter	Date of origin	Time uncertainty (24 hrs)	Frequency uncertainty (24 hrs)
Obelisk	Rotating earth	Sundial	3500 BC	N.A.	N.A.
Verge escapement	Verge and crown wheel	Foliot	Fourteenth century	15 min	1×10^{-2}
Pendulum	Pendulum swing	Gears and hands	1656	10 s	7×10^{-4}
Harrison chronometer	Springs and balances	Gear train	1759	300 ms	3×10^{-6}
Quartz crystal	Quartz mechanical oscillations	Electronic counter	1927	1 s	1×10^{-10}
Cesium beam clock	Cs resonance	Microwave counter	1955	1 ns	1×10^{-14}
Rubidium clock	Rb resonance	Microwave counter	1958	1 ns	1×10^{-12}
Hydrogen maser	Hydrogen resonance	Microwave counter	1960	1 ns	1×10^{-14}
Cesium fountain	Cs resonance	Microwave counter	1991	100 ps	1×10^{-16}
Optical clocks	Narrowband optical transition in ion or ensemble of neutral atoms	Frequency comb	2001–2005	100 fs	1×10^{-18}

technologies require clocks that run faster than the processes and machines in order to control and manipulate them for their proper usage. These technologies require time to be measures in milliseconds, microseconds, and even nanoseconds with astounding accuracy. Quartz-based oscillators continued to serve as laboratory frequency standards, but at the same time, quest began for a universal time standard whose pendulum (or oscillator) will never wear down.

As predicted by James Clerk Maxwell (Maxwell 1870), each atom was found to be a perfect oscillator for measuring the time. The transition frequency between the designated states of atoms formed the basis of atomic clocks. Since frequency is inverse of time, an accurate measurement of frequency in atoms transcribes into precise measurement of time. Due to this reason, atomic clocks are better known as atomic frequency standards. First experiments in this direction led to the measurement of microwave transition frequencies in ammonia molecules. Cesium atom was later on found to be better and preferred for measuring time. The first working cesium clock was built in 1955 by Essen and Perry at NPL UK (Essen and Parry 1955). Finally, at the 13th General Conference on Weights and Measures (CGPM) held in 1967, second of atomic time in SI unit was defined as "the duration of 9,192,631,770 periods of the radiation corresponding to the transition between the two hyperfine levels of the ground state of the cesium-133 atom." Since then, research in this area has led to astonishing improvements in timekeeping accuracies. Today, atomic clocks operate at different frequencies including optical frequencies. Cesium fountain clocks remain the most accurate primary standards, but the optical clocks have been demonstrating increasingly competitive performance than their microwave counterparts. Based on the consistent measurement of fractional uncertainties lower than their microwave counterparts, one can expect that optical clocks will replace the cesium hyperfine transition in the definition of the second in the International System of Units. Gearing up for a possible redefinition of the SI second on the basis of one of the many candidates being investigated, the International Committee of Weights and Measures (CIPM) has approved a few candidates as "secondary representations" of the second.

Classes of Atomic Frequency Standards

Standards are often classified as primary and secondary. A primary standard is a device that is used to directly realize the SI unit with highest precision. In the case of atomic frequency standards, a primary frequency standard is the one that realizes the definition of the SI second based on the ground-state hyperfine transition in the cesium 133 atom. Moreover, all the sources of frequency shift in a primary frequency standard can be identified and determined quantitatively as well as reproducibly. As the cesium fountain clocks provide the best realizations of SI second with respect to accuracy and frequency instability, they are the primary frequency standards. A

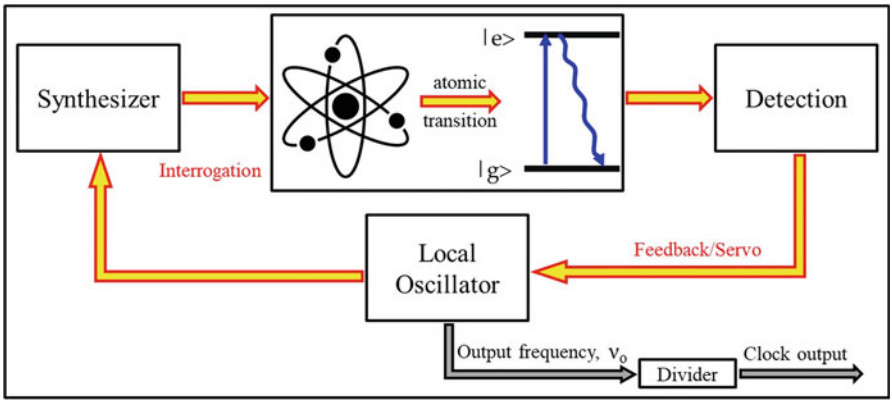

Fig. 1 Block diagram of an atomic frequency standard

secondary frequency standard realizes secondary representation of SI second based on an approved list of secondary representations. In standards, such as Rb clock or H maser, there are sources of frequency shifts that cannot be quantified. Thus, these categories of atomic standards are known as secondary frequency standards.

In any atomic frequency standard, one needs a mechanism to probe the transition frequency between the designated levels of the atomic species. In most of the cases, a local oscillator (LO) is used as reference to lock its frequency onto the atomic transition as shown in Fig. 1.

At present, there are two broad classes of AFSs: microwave frequency standards and optical frequency standards. In a microwave frequency standard, the atomic transition frequency to be probed lies in the microwave region of the electromagnetic spectrum and therefore, the LO is generally a quartz crystal oscillator. In an optical frequency standard, the atomic transition frequency lies in the optical region and therefore, the LO is a laser system.

The quartz oscillator typically has a 5 MHz frequency output. A microwave synthesizer takes this 5 MHz as reference and produces the appropriate microwave frequency for interrogating the atomic transition. An error signal is generated based on the difference between the synthesized microwave frequency and the atomic transition frequency. Based on this error signal, a feedback servo loop generates a control voltage to tweak the LO until the microwaves are tuned to the atomic transition frequency. The same concept is used in the optical frequency standards with a difference that the LO is generally a laser and an optical frequency comb is used as the counting mechanism which converts the output signal from the optical domain into the RF domain. Atomic frequency standards have been reviewed in several articles (Lewis 1991; Hellwig 1975; Beehler 1967; Ramsey and Res 1983; Bauch and Telle 2002; Audoin et al. 1976; McCoubrey 1967; Levine 1999; Schmittberger and Scherer 2021).

Microwave Frequency Standards

Microwave frequency standards are mostly based on alkali atoms (in particular H, Rb, and Cs). These atoms have a single valence electron, relatively simple energy spectrum, and long-lived ground states, thus resulting in slow relaxation. Hyperfine levels of these ground states, differing in relative orientation of the nuclear and electron magnetic moments, have narrow transition linewidths centered at frequencies in the range of 1–10 GHz.

Four types of microwave frequency standards are being widely used. Cesium fountain clocks, the most accurate atomic clocks at present (based on present definition of SI second), are the laboratory standards. There are about 15 of such primary frequency standards which have been indigenously built in only ten countries so far. Cs beam clocks are the workhorses, contributing to more than two-thirds of atomic clocks being used for timekeeping the world over. Hydrogen masers, with excellent short-term stability (active masers), are widely used in timekeeping laboratories worldwide. Low-performance, lightweight vapor cell atomic clocks based on rubidium atoms confined in a cell with a buffer gas are used for satellites, low-accuracy timekeeping in telecommunications, military, etc. and as a time base for many test and measurement devices. Apart from these four variants, chip-scale atomic clocks have also been developed, and a lot of work is going on to improve the performance and to reduce the power requirements of portable microwave frequency standards.

Microwave frequency standards are further classified into active and passive devices based on the interrogation method of the atomic transition. In an active device, the atoms are confined to a storage volume which is enclosed by a very high Q microwave cavity, tuned exactly to the transition frequency. Initially, a small thermal noise at the resonance frequency present in the cavity is able to amplify itself by inducing transitions in the confined atoms. Low loss cavity allows high-intensity atomic vapor to interact with microwaves in a way to generate self-sustained oscillations. A small amount of this energy is tapped out as a usable signal, which is then used to phase lock a quartz oscillator at 5 MHz. Active hydrogen masers fall in this category.

In a passive device, on the other hand, the atoms are allowed to interact with a microwave signal, externally synthesized from a quartz oscillator at 5 MHz. If the synthesized signal frequency matches the atomic transition frequency, many atoms make transition to the other state. The number of atoms detected after state-change is recorded, and a frequency control servo adjusts the quartz oscillator frequency so as to maximize the detected signal. Under this condition, the accuracy and stability of the atomic standard are transferred to the quartz oscillator. Cs clocks, Rb clocks, and passive hydrogen masers fall in this category.

Cesium Beam Frequency Standards

From the first cesium atomic clock to almost all commercially available Cs clocks are beam frequency standards (Bauch 2005). In these, hot Cs atoms leave the oven in

the form of an atomic beam that travels through a vacuum tube toward a pair of state-selection magnets which produces a strong magnetic field gradient resulting into two physically separated beams of the Cs atoms (one with atoms in the F = 4 and another in F = 3 state; F is the quantum number corresponding to the hyperfine levels). One of the state-selected beams of Cs atoms interacts with microwaves in a microwave cavity that is excited with a 9.192 GHz synthesized from a quartz oscillator. When the microwave frequency matches that of the Cs hyperfine transition frequency, many atoms are excited to the other hyperfine level. After passing through the interaction region, the Cs beam passes through another pair of magnets (similar to state-selection magnets) which physically separates the atoms in two different states. The magnets are followed by a hot wire ionizer detector in such a position as to detect the atoms that have made transitions during interaction (Fig. 2). Those atoms which did not change their state during microwave interaction are deflected away from the detector. The detector drives a servo feedback loop that continually tunes the frequency of the quartz oscillator till the detector output is maximized. This way, the quartz oscillator is kept locked to the Cs hyperfine line and produces the usable output frequency of the standard, usually at 5 MHz. This frequency is further divided to generate standard 1 PPS (pulse per second) signal to drive the clock output. As a matter of detail, the geometry of the microwave cavity used is U-shaped, also known as the "Ramsey cavity," named after Sir Norman Ramsey who discovered the novel interaction method of separated oscillatory fields (Ramsey 1949; Markowitz et al.

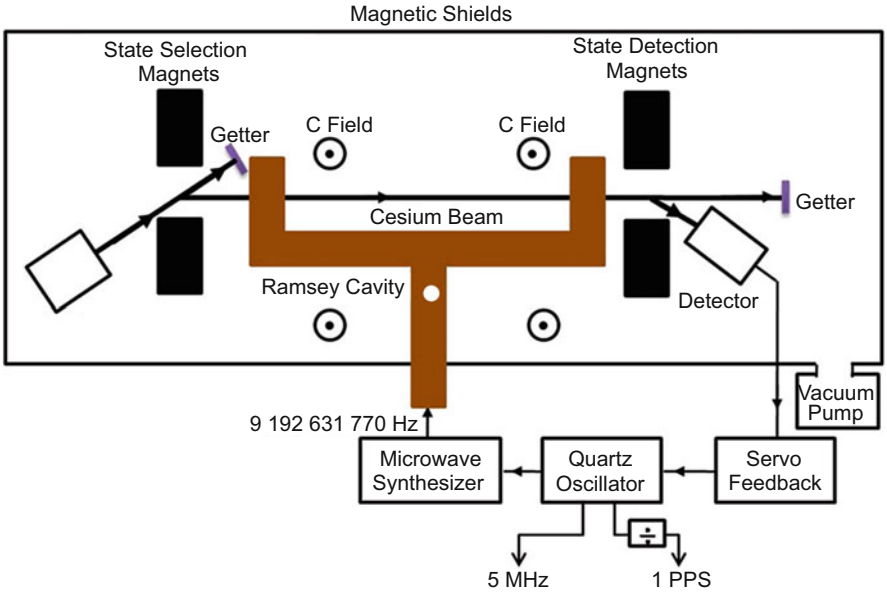

Fig. 2 Diagram of a cesium beam frequency standard using magnetic state selection and detection. The form of Ramsey interrogation involves a U-shaped microwave cavity, called a Ramsey cavity, where the oscillatory fields are spatially separated

1958). Cs beam interacts with the same phase of the microwave excitation in the two spatially separated regions. This results in an interference effect (in temporal domain) that produces a narrower atomic resonance linewidth than would occur if the atoms had interacted with the microwaves for the entire time. A uniform small magnetic field, known as the "C-Field," is produced over the whole interaction zone by a solenoid to define the quantization direction and to lift the degeneracy of the hyperfine levels. In order to avoid any perturbation from external magnetic fields, the whole interaction region is covered with layers of magnetic shields. There are few systematic effects which cause the frequency shifts in the Cs standard, but these can be evaluated quantitatively (within small measurement uncertainties). Typical values for stability of commercial Cs clocks are about few parts in 10^{12} for 1 s averaging time. The length of the Ramsey cavity is typically less than half a meter, thus limiting the interaction time of hot Cs atoms with microwaves to a few milliseconds, and the transition linewidth to a few hundred hertz. Since the instability of an atomic clock is directly proportional to the linewidth of atomic transition, laboratory models of Cs beam clocks have been developed with Ramsey cavity lengths of 1 m or more (to increase the interaction time). Recently, in some models of Cs clocks, optical pumping technique (using a carefully tuned and stabilized laser) has been used for state selection of atoms rather than magnetic field gradient. Optically pumped Cs clocks are more accurate with larger number of state-selected atoms leading to better signal-to-noise ratio and significant reduction in the frequency shifts caused by inhomogeneous magnetic fields due to the absence of strong state-selection magnets (Makdissi and de Clercq 2001; Shirley et al. 2001; Hasegawa et al. 2004).

Cesium Fountain Frequency Standards

As discussed in the previous section, the primary limitation of the frequency stability of a Cs beam clocks arises due to the transition linewidth caused by the short interrogation time. This time is essentially the interaction time of the Cs atoms as they pass through the Ramsey cavity. Laboratory models with longer cavities were built to increase the interaction time but these could not be translated into sustainable commercialization. Another way to increase the interrogation time by choosing slower atoms at room temperature did not work as the slow atomic beam is relatively weak and slowly sags down along its horizontal movement due to gravity. These limitations were practically overcome in two ways. Firstly, with implementation of laser cooling (Chu 1992; Cohen-Tannoudji 1998; Phillips 1998), the atoms are cooled down to a few micro Kelvin temperatures, thereby reducing the atomic velocities to few cm/s. Secondly, the problem of sagging of a horizontal atomic beam due to gravity is taken care of by using a vertically moving atomic beam. The basic idea of Cs fountain was proposed for the first time in the 1950s by J. Zacharias (Zacharias 1954); however, the first successful practical realization of the fountain principle with laser-cooled Cs atoms was done at BNM-SYRTE (presently LNE-SYRTE) at Observatoire de Paris in the year 1991 (Clairon et al. 1991). For laser cooling of Cs atoms, the system typically comprises a set of six mutually orthogonal counter-propagating beams of a laser whose frequency is tuned slightly to the red of a strong transition of the Cs atoms at 852 nm. The atoms, moving in the

direction opposite to a laser beam, see the laser frequency upshifted and closer to the resonance due to the Doppler effect. Hence, the absorption and reradiation rate of photons is higher than for the case atoms that are moving along the direction of the laser beam. At the intersection of the six orthogonal counter-propagating laser beams, no matter which way an atom moves, it encounters a laser beam opposing it. Due to the momentum exchange with photons, the atoms keep losing their energy and thus slow down rapidly. This mechanism is known as Doppler cooling and system is often termed an optical molasses.

A Cs fountain works on the same basic principle as a Cs beam frequency standard discussed in the previous section. Schematic of a typical Cs fountain is illustrated in Fig. 3. The difference primarily is that here the Cs beam consists of a bunch of laser-cooled atoms, which are launched vertically upward with a velocity of few m/sby slightly offsetting the frequencies of the two vertical lasers and then turning all six lasers off and which, after attaining a height of a few tens of cm, fall back downward due to gravity. Just after launching, the atoms in the bunch are subjected to a process of state selection, so that the upgoing atoms are mainly in one of the two ground-level hyperfine states. During flight, the atoms pass through a microwave cavity twice – on their way up and down. The microwave cavity is excited with the hyperfine frequency of 9.192 GHz, synthesized from a quartz oscillator. On their way down, atoms are detected using laser-induced fluorescence. The fluorescence intensity is directly proportional to the number of atoms and thus transition probability is estimated. As the microwave interrogation frequency is slowly tuned about the nominal value, the transition probability goes through successive maxima and minima exhibiting the typical Ramsey pattern. The top of the central fringe in the Ramsey pattern signifies the maximum transition probability and corresponds to the microwave frequency that alters the states of most of the cesium atoms and maximizes their fluorescence. This frequency is the natural resonance frequency of the cesium atom (9,192,631,770 Hz), or the frequency used to define the second. Due to the vertically moving slow atomic beam, the total interaction time (from upward till downward transit of atoms through the Ramsey cavity) is fairly long. Typically, it is possible to have a very narrow central Ramsey linewidth of the order of 1 Hz or even smaller – almost two orders of magnitude smaller than in the conventional Cs beam device. This implies frequency instability of a typical fountain approaches 10^{-15}(may approach 10^{-16} for few best ones) at 1-day averaging time. Since the development of first fountain at BNM-SYRTE (France), similar clocks have been developed in national metrology institutes (NMIs) in only 11 countries around the world. The countries which developed Cs fountain clocks which were approved as primary frequency standards (PFSs) and contribute/contributed data for the calibration of International Atomic Time (TAI) are France, Germany, the USA, the UK, Japan, Italy, India, Russia, China, Switzerland, and Canada (Clairon et al. 1995; Gerginov et al. 2010; Guéna et al. 2012; Jefferts et al. 2002; Szymaniec et al. 2005; Kurosu et al. 2004; Kumagai et al. 2008; Domnin et al. 2013; Heavner et al. 2014; Levi et al. 2014; Acharya et al. 2017; Fang et al. 2015; Jallageas et al. 2018; Weyers et al. 2018; Beattie et al. 2020). Some of the fountains are running almost continuously, and typically reports from about six to eight fountains are being generated

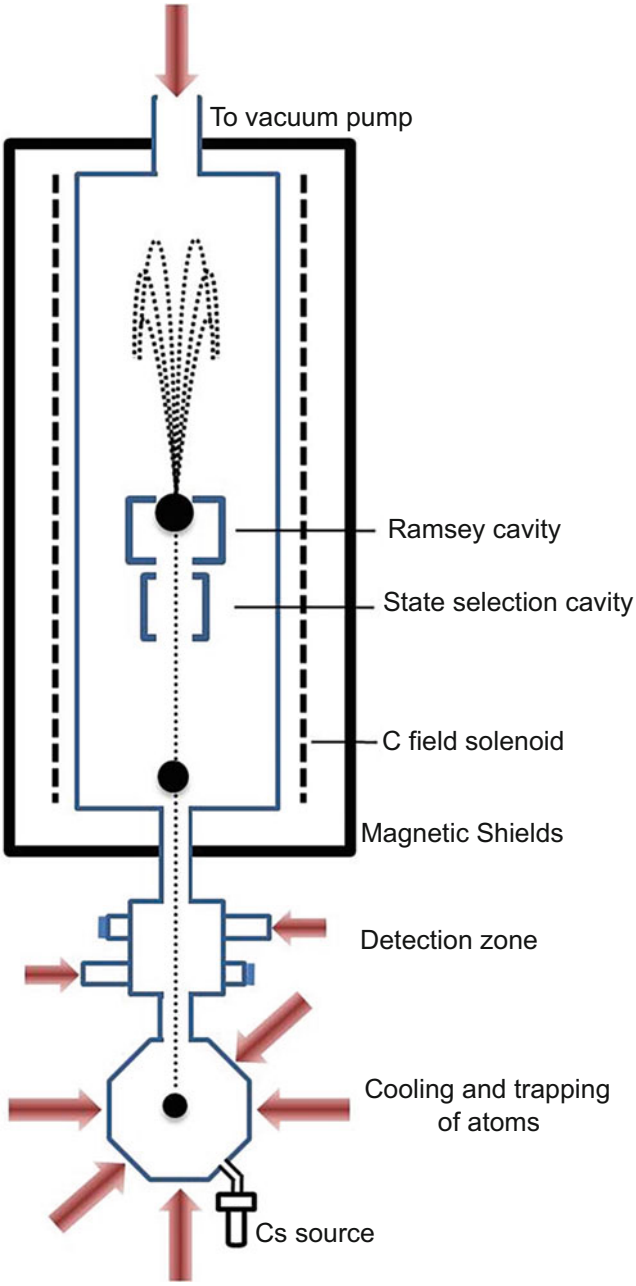

Fig. 3 Schematic of a Cs fountain

monthly data to calibrate TAI. Laboratories like PTB (Germany), LNE-SYRTE (France), USNO (USA), and INRIM (Italy) are steering their timescales with fountain data, thus greatly improving stability and performance of their timescales (Bauch et al. 2012; Peil et al. 2014; Rovera et al. 2016). Even with a possible redefinition of SI second on the basis of an optical clock, Cs fountains will continue to play a significant role in T&F metrology just the way Cs beam clocks still continue to be the field workhorses for precision timekeeping. Even today, some of the laboratories like NPL (India), KRISS (Korea), and NMIJ (Japan) are developing new Cs fountain frequency standards (Bharath et al. 2019; Heo et al. 2017; Takamizawa et al. 2022).

Rubidium Frequency Standards

Just like cesium, rubidium atoms also have a ground-state hyperfine transition in the microwave domain which is widely used for making atomic clocks for field deployment (Happer 1972; Camparo et al. 2005; Vanier and Mandache 2007; Kitching et al. 2001). The clock transition $F = 1$, $m_F = 0$ to $F = 2$, $m_F = 0$ in the $^2S_{1/2}$ ground state of ^{87}Rb with corresponding frequency of 6,834,682,608 Hz is also recommended as a secondary representation of second (SRS). LNE-SYRTE (France) and USNO (USA) have developed and regularly operate Rb fountains with the former contributing the Rb fountain data regularly for the calibration of TAI as a SRS. However, most commercial versions of Rb frequency standards are based on vapor cell or glass cell filled with low-pressure ^{87}Rb vapor with a buffer gas (neon, helium, argon, etc.). Optical pumping is used for state selection of atoms which then interact with microwaves and are detected optically using a photodiode. Initially, the two states of the clock transition are equally populated in thermal equilibrium. Atoms are then optically pumped into the $F = 2$ state typically by using a discharge lamp containing ^{87}Rb after passing its light through an absorption/filter cell containing ^{85}Rb. The lamp light can induce optical frequency transitions from both the $F = 1$ and $F = 2$ states to higher excited levels. Natural rubidium consists of a mixture of two isotopes, ^{85}Rb and ^{87}Rb, with an overlap in their atomic energy levels. Due to this reason, the lamp light component which could induce transitions from the $F = 2$ state is absorbed by the^{85}Rb filter cell, thus letting the transmitted lamp light with the component which can excite only the $F = 1$ state of the clock transition. Atoms in $F = 1$ state get optically excited and then decay into either one of the ground states, while those in $F = 2$ state cannot be excited by the filtered lamp light anymore. $F = 2$ state becomes a "dark state" and after a few absorption-emission cycles, almost all ^{87}Rb atoms are optically pumped into the $F = 2$ state, and the absorption cell becomes transparent to the filtered light, whose power is monitored with a photodetector. A resonant microwave signal at 6.834 GHz, derived from a quartz oscillator, is then applied to induce transitions from $F = 2$ to $F = 1$. This restores some population to the light-absorbing state with more atoms in the absorption cell to absorb the filtered light, thus resulting in less light arriving at the photodetector. A servo loop adjusts the microwave frequency such that the detector signal is minimized. The quartz oscillator is thus kept on atomic resonance frequency and provides a usable output of the Rb frequency standard (Fig. 4).

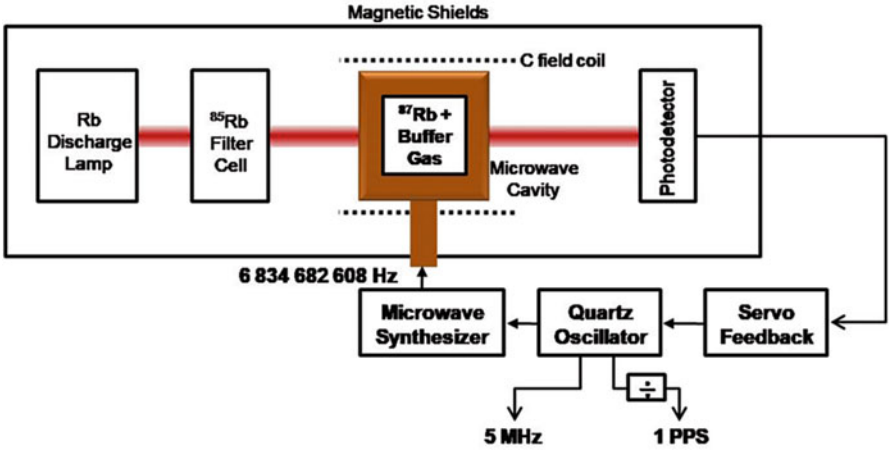

Fig. 4 Block diagram of Rb vapor cell clock

Typical Q-value of Rb oscillators is several 10^7 and typical stability is in the range of 1×10^{-11} at 1 s. Atomic collisions as well as the simultaneous action of the light and the microwave signals on the same atom cause frequency shifts which depend strongly on the composition, temperature, and pressure of the buffer gas and on the intensity of the light. Just like quartz oscillators, Rb frequency standards need frequent calibration to compensate for the frequency drift and aging.

In attempt toward miniaturization and reduction in power consumption, microwave excitation was replaced with all-optical excitation using lasers. Chip-scale atomic clocks are based on MEMS micro fabrication, laser pumping, and the phenomenon of coherent population trapping (CPT) (Arimondo 1996; Knappe et al. 2004; Wynands and Nagel 1999; Vanier 2005). CPT can be understood with a three-level quantum system. When the two hyperfine ground states are simultaneously resonantly excited by two coherent lasers, atoms are pumped into a so-called dark state, and there is an increase in optical transmission through the vapor cell. The interaction of the two fields creates coherence in the ground states by CPT. Practically, instead of using two phase-locked lasers, a single laser source is used with its current modulated at exactly half the hyperfine frequency, thus generating two phase-coherent side bands. Fast modulation of the laser current provides microwaves, thus completely eliminating the need for a tuned microwave cavity. Rb or Cs compact cell standards based on this interrogation scheme have been developed. The typical stability of a CPT-based rubidium standard has been reported in the range of 1×10^{-12} at 1 s.

Hydrogen Masers

The acronym maser stands for microwave amplification by stimulated emission of radiation. Masers have been built with various atomic and molecular species for the purpose of microwave spectroscopic investigations. A hydrogen maser, also known

as hydrogen frequency standard, is a specific type of maser that uses the intrinsic properties of the hydrogen atom to serve as a precision frequency reference (Goldenberg et al. 1960; Vessot 2005). Hydrogen masers are extensively used for building stable timescales. There are two variants of hydrogen masers: active and passive. Active H-masers are relatively bulky and have excellent short-term stability, while the passive hydrogen masers are smaller and lower-performance variants. The ground-state hyperfine transition $F = 0$, $m_F = 0$ to $F = 1$, $m_F = 1$ of atomic hydrogen, with a transition frequency near 1420 MHz, is the clock transition. A beam of atomic hydrogen is generated from H_2 gas in a radio frequency discharge. The atoms in the upper ground states are focused by a pair of state selecting magnet into a storage bulb inside a microwave cavity resonant with the 1.4 GHz frequency of the hyperfine transition, while the atoms in the other ground states are defocused and do not reach the storage bulb (Fig. 5). Hence, the storage cell, which is placed in a microwave resonator, contains more atoms in the upper state which can emit radiation by stimulated emission. When the number of incoming atoms and their storage time in the bulb is large enough, self-sustained maser oscillations are generated. These oscillations are detected by an antenna and used to tune the

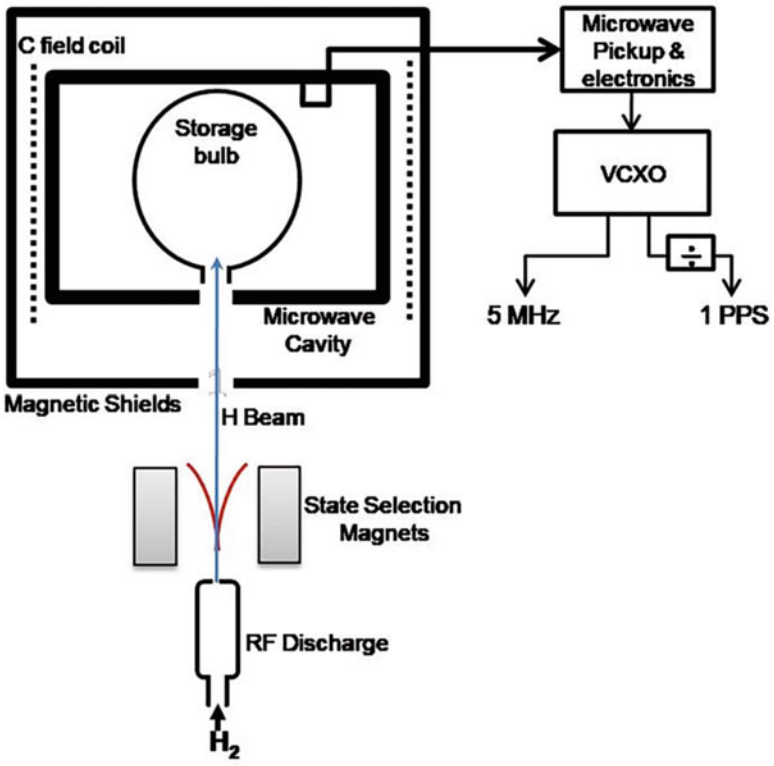

Fig. 5 Block diagram of an active hydrogen maser

frequency of a voltage-controlled crystal oscillator (VCXO) from which the output signal of the frequency standard is derived. The atoms typically spend about 1 second in the storage bulb, and due to this long interaction time, a narrow linewidth atomic transition is detected in a maser.

In an active maser, the losses of the cavity are small enough so that the device oscillates at the hyperfine transition frequency. The frequency of oscillation of an active maser is a function of both the cavity resonance frequency and the atomic transition frequency, so that some means of stabilizing the cavity is usually required. Collisions with the walls of the bulb also affect the frequency of the device. Various strategies are used to minimize these effects, including applying special "nonstick" coatings to the inside of the bulb and actively tuning the resonance frequency of the cavity so that it is exactly on atomic line center. In spite of these techniques, hydrogen masers usually have frequency offsets that are large compared to the stability of the device. The fractional frequency offset of a hydrogen maser is typically of order 5×10^{-11}, while the fractional frequency fluctuations at 1 day can reach 2×10^{-16}. In any case, the short-term stability of an active hydrogen maser is much better than the best commercial cesium beam clock. Active hydrogen masers are therefore selected as flywheel for generating stable timescales.

The construction of a passive hydrogen maser is similar, except that a microwave signal at 1.4 GHz is fed into the cavity and is amplified by the hydrogen atoms inside the cavity. The maser acts as an amplifier with narrow linewidth limited by linewidth of atomic transition. The atomic transition is probed by measuring the amplified signal and the external oscillator frequency is tuned until output signal is maximized. The passive maser is made with a smaller and compactor microwave cavity and therefore is generally smaller in size as compared to active maser. However, smaller size comes at the expense of larger short-term instability.

One of the key components of a hydrogen maser is the storage bulb, which is coated with Teflon or similar materials so that a hydrogen atom survives thousands of wall collisions before it gets spin depolarized. Storage times of several seconds are thus possible for the hydrogen atoms. The main advantage of a hydrogen maser is its low relative short-term instability. However, its long-term stability suffers from drifts mainly due to chemical changes and aging of its wall coating, which in turn changes the frequency shift due to the wall collisions.

Optical Frequency Standards

Any clock is characterized in terms of stability and accuracy. Accuracy is a measure of closeness to the true value (duration of SI second in case of clocks), and stability is the measure of ability to produce the same value over extended time (clock ticks produced at regular intervals in case of clocks). Stability as well as accuracy of atomic clocks is limited due to several factors. Stability of atomic frequency standards is assessed through Allan deviation. Fractional frequency instability (σ) of AFS, as a function of averaging time (τ), is expressed as

$$\sigma(\tau) = \frac{1}{\pi}\frac{\Delta\nu}{\nu_0}\frac{1}{S/N}\sqrt{\frac{T_{cycle}}{\tau}} \quad (1)$$

Here $\Delta\nu$ is the width of the atomic transition with center frequency ν_0, S/N is the signal-to-noise ratio of the detection process, and T_{cycle} is the time required for a single measurement cycle.

From this equation, it can be immediately understood that the instability is reduced for narrower lines ($\Delta\nu$), shorter measurement cycles, and higher signal-to-noise ratio.

Optical atomic clocks operate at frequencies around five orders of magnitude higher than the frequency of the cesium primary frequency standard, while linewidth-limiting processes in the two spectral regions are almost similar (Gill 2005). For a given linewidth $\Delta\nu$ of the absorption feature, the line quality factor $Q = \nu_0/\Delta\nu$ increases on operating at optical frequencies instead of microwave frequencies. An enormous reduction in instability can thus in principle be gained by basing a frequency standard on a transition in the optical region of the spectrum. This gain is, however, not as large as the 10^5 increase in frequency, because of various other fundamental and technical factors. The exact improvement factor depends on the signal-to-noise with which the atomic absorption signal is observed. It can also be seen from Eq. (1) that atomic frequency standards follow the $\tau^{-1/2}$ white noise behavior until hitting a noise floor at a certain time which differs for different standards. Cs fountains typically reach $\sim 10^{-15}$–10^{-16} stability levels at averaging times approaching a day. Optical frequency standards reach the similar stability levels at few seconds averaging times with many optical clocks reaching the noise floor of 10^{-17}–10^{-18} within a day.

While considering frequency uncertainty, optical frequency standards also suffer from a number of systematic shifts due to various physical effects, e.g., Doppler, Zeeman, Stark, collisional, gravitational, etc. In most of the vases, the magnitude of these shifts is inversely proportional to the frequency, thus making optical frequency standards less susceptible to environmental, electromagnetic perturbations. In addition, evaluation and suppression of systematic frequency shifts can be accelerated and measured with higher precision with optical standards possessing much smaller instabilities.

Thus, optical frequency standards offer potential advantage, over their microwave counterparts, both in terms of stability and accuracy.

Optical frequency standards are based on weak forbidden optical transition in a certain species of laser-cooled ions or neutral atoms. There are three key elements, which are necessary for the realization of an optical frequency standard (OFS) as shown in Fig. 6. These are stable local oscillator, optical transition in laser-cooled atomic species (atoms/ions), and a frequency counting mechanism. For optical clocks, the local oscillator is a highly stable laser, generally servo-locked onto a narrow resonance of a high-finesse, environmentally isolated, ultralow expansion (ULE) cavity. The atomic reference or clock transition is a weak, forbidden optical transition in a single cold ion or cold atomic cloud. Using laser-cooled atomic

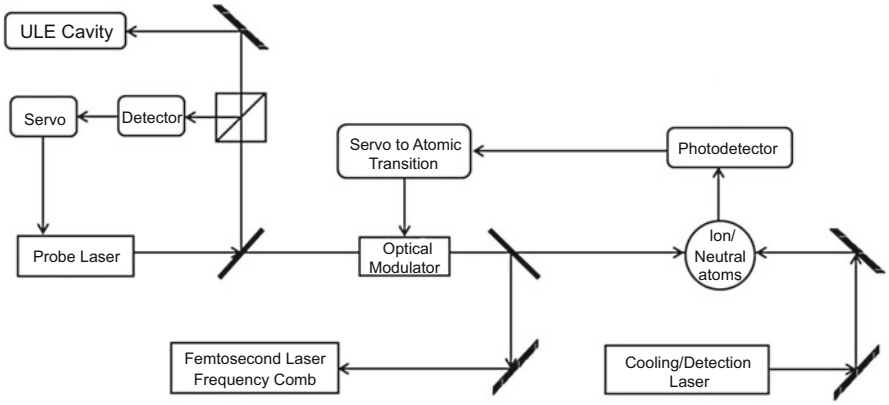

Fig. 6 Schematic of a generic optical frequency standard

samples enables the extended interaction times required to observe a narrow transition linewidth in the sub-1 Hz to a few hertz range, depending on the precise element. The frequency counting mechanism or clockwork, which is capable of relating the frequency of the standard to other standards, is a femtosecond laser frequency comb, which generates a large number of modes across the optical near-infrared region. In the year 2000, with the development of optical frequency combs (OFC) (Hansch 2006; Hollberg et al. 2005), realization of optical frequency standards became feasible.

Initially, atoms or ions are laser-cooled using a cooling laser, tuned slightly below the center frequency of a strong (allowed) transition. A second (repumping) laser is used to pump the atoms/ions back to the cooling cycle. To avoid broadening of the transition via the ac Stark shift, another (probe) laser is used to interact with the cold atomic species after switching off the cooling laser using an optical modulator. After a brief interaction time, the probe laser is turned off and the cooling laser is switched back on to interrogate the atoms/ions and the corresponding transition probability is measured. The probe laser is detuned in steps to measure the transition probability at different clock frequencies and generate a line profile of the clock transition. A servo loop is closed by locking the probe laser on the center of the clock transition. The frequency of the stabilized probe laser is measured using a frequency comb, which down converts the frequency from the optical to the microwave domain and serves to produce the RF/microwave output of the frequency standard.

For developing state-of-the-art OFS, the choice of the atomic reference transition is one of the most critical aspects. The required narrow linewidth transitions (clock transitions), almost insensitive to external perturbations such as frequency shifts caused by magnetic and electric fields, exist in both ions and neutral atoms. Therefore, there are two choices for realizing an OFS based on working with few ions or a large number of neutral atoms. Depending on the priority for low instability or low uncertainty, one can choose a single ion or a cloud of cold atoms. With many atoms, one obtains a much higher signal and thus higher stability, but on the other hand,

those atoms could potentially interact with each other to produce larger shifts. Single trapped ion-based clocks are more accurate due to long observation time with no dependence on inhomogeneity of external magnetic or laser light fields.

Optical clocks have made significant progress in recent years (Chou et al. 2010; Huntemann et al. 2016; Katori et al. 2003; Rosenband et al. 2007; Ludlow et al. 2015; Lemke et al. 2014; Peik et al. 2006; Falke et al. 2011; Margolis et al. 2004; King et al. 2012; Le Targat et al. 2013; Yi et al. 2011; Takamoto et al. 2011; Lin et al. 2015; Huang et al. 2012; Oskay et al. 2006). Out of the 11 atomic transition frequencies recommended by the consultant committee of time and frequency (CCTF) as the secondary representations of the second (SRS), ten transitions are in the optical frequency region besides Rb transition (https://www.bipm.org/en/publications/mises-en-pratique/standard-frequencies-second n.d.).

Neutral Atom Optical Frequency Standards

In neutral atom-based optical atomic clocks, large numbers of neutral atoms are cooled, trapped, and probed simultaneously, thus enabling an extremely high S/N ratio and potentially very high stability. As the fractional instability depends inversely on the square root of the number of atoms in the system, neutral atom optical frequency standards are expected to have unprecedented stability. However, absolute frequency uncertainty estimation is complicated and varies from one atomic system to another.

Methods of confining neutral atoms, as done in Cs fountain frequency standards, are efficient but, during the probe cycle using a laser beam, lead to shifts in the transition frequency due to Doppler and scattering effects. As a solution to the confinement problem, optical/laser lattice is used to trap the neutral atoms. In this method, counter-propagating laser beams interfere to create spatially localized potential wells which attract sufficiently cool neutral atoms to its minima. Such a periodic potential well structure is called optical lattice which spatially confines laser-cooled neutral atoms in so-called Lamb-Dicke regime where a carefully chosen wavelength of the lattice laser beams leads to minimized shifts of the clock transition (Dicke 1953). Such a confinement arrangement results in long interaction times and small Doppler shifts, along with the good S/N ratio due to large numbers of neutral atoms.

With prudent choice of atomic transition and lattice wavelength, perturbations to the clock frequency induced by the lattice beams are easily suppressed better than sub-10^{-17} level. Even with the additional experimental efforts required, optical lattice clocks are now dominating the field of high-precision neutral atom clocks. Such optical lattice clocks are being developed at several NMIs around the world. Most common choice of neutral atom systems currently under development for optical clocks includes Sr, Yb, and Hg atoms (Lodewyck et al. 2010; McFerran et al. 2012a; Falke et al. 2014; Wang et al. 2009; Blatt et al. 2008; Akatsuka et al. 2015; Yamaguchi et al. 2012; Sherman et al. 2012; Levi et al. 2016; Yasuda et al. 2012; Park et al. 2013; McFerran et al. 2012b). Many of these optical lattice clocks are being continuously operated and contribute to TAI as a secondary representation of second (SRS). Work is under progress in developing miniaturized and

transportable versions of optical lattice clocks. With rapid advancements in this area, the best laboratory optical lattice frequency standards have reached record levels of instability (ADEV of 4.7×10^{-17} at 1 s) and fractional uncertainty (5.8×10^{-19} after 1 h of averaging).

Trapped Ion Optical Frequency Standards

The idea of using single trapped ion as optical frequency reference was first proposed by H. G. Dehmelt in 1973 (Dehmelt 1973; Dehmelt and Walther 1975). A single ion confined in an electromagnetic trap, such as the Paul trap, is the simplest way of investigating weak optical transitions in the trapped ion. The environmental perturbations can be effectively controlled with relatively simpler experimental arrangements. In effectively cooled and trapped ion, the thermal motion-induced Doppler effects which shift the resonance frequency are minimized. With sufficiently deep trapping potentials, cold ions-based systems can be operated in the Lamb-Dicke regime. Moreover, an ion can be probed while being confined in the trap; thus, long interaction times can be achieved. The only limiting factor for trapped ion systems is the poor S/N ratio and thus increased instability. Despite having lower S/N, high-performance optical frequency standards have been developed by observing a weak optical (clock) transition in an ion virtually at rest in near-unperturbed conditions over extended periods (days). A number of laboratories worldwide are investigating the stability and accuracy of different trapped ion optical frequency standards. Till date, several kinds of ion optical clocks have been realized, which are $^{40}Ca^{+}$ion clock, $^{88}Sr^{+}$ ion clock, $^{199}Hg^{+}$ ion clock, $^{171}Yb^{+}$ ion clock, and Al^{+} ion clock (Matsubara et al. 2012; Harlander et al. 2012; Barwood et al. 2001; Dube et al. 2009; Tamm et al. 2009; Godun et al. 2014; Hannig and Pelzer 2019; Che et al. 2017). State-of-the-art ion optical frequency standards have achieved short-term fractional frequency instabilities nearing $10^{-15}/\sqrt{\tau}$ and fractional uncertainties of 9.4×10^{-19}.

Applications

Measurement of time is part of our daily lives with time-bound human activities which require timing accuracies of minutes and seconds, at best. However, accurate and precise time and frequency measurements done with atomic frequency standards (AFSs) have enabled many modern applications which have revolutionized our daily lives. SI unit of time, i.e., second, is the most accurately measured quantity which is used for measurement of other SI units as well. Apart from atomic timekeeping and metrology, there are applications such as navigation, telecommunication, power grids, banking and finance, cybersecurity, etc. which rely on accurate and precise atomic frequency standards.

Navigation is all about locating and tracking the position of an object/person. Global satellite navigation systems (GNSS) like GPS are based on precise measurement of time taken by signal radiated by satellites to reach a particular location. Each satellite has an atomic clock onboard, and atomic clocks on all the satellite systems

should be synchronized to about one billionth of second per day. A difference of just one microsecond between clocks can lead to a positioning error of 300 meters. Nowadays, miniaturized GNSS receivers integrated in mobile phones have enabled easy access to accurate navigation within few meters, thus making navigation an application of atomic clocks touching daily lives.

Another everyday application of AFSs is telecommunications. High-speed mobile networks need time synchronization (better than few microseconds) between base stations of different operators; otherwise, the information may be lost or corrupted in these packet-switched networks. Modern high-speed telecommunication systems require atomic clocks for reliable, uninterrupted services.

Electricity distribution networks, based on smart power grids, need to link sources with identical frequency and synchronized phase. Precise timing synchronization of the order of one microsecond to hundreds of nanoseconds between the power substations is required for the seamless operation of smart grids.

Banking systems and financial markets also need very accurate clocks to guarantee the time and date stamps of high-frequency transactions with microsecond accuracy. Cybersecurity also relies on accurate time stamps and synchronized systems; otherwise, cyber frauds can never be chronologically correlated and solved.

Among the high-end scientific exploration applications of atomic clocks are astronomy, geodesy, test of fundamental constants, etc. (Will 2014; Safronova et al. 2018; Hees et al. 2016; Takano et al. 2016; Mehlstaeubler et al. 2018). With the use of atomic clocks, arrays of distant radio telescopes can be synthesized together in very-long-baseline interferometry (VLBI) to give angular resolutions of 250 micro arc sec which is 250 times better than the best optical telescopes including the improved Hubble telescope. VLBI also provides the most accurate measurements of distances along the surface of the Earth and thereby aids in the study of the Earth's crustal dynamics. Atomic clocks are needed to measure the stability of pulsars, of the Earth's rotation, and of other periodic phenomena. The constancy of the fundamental constants can be tested by comparing different kinds of clocks at different times. With the advancement in atomic clocks technology in terms of accuracy, size, and affordability, many newer applications of atomic clocks will emerge in evolving quantum systems.

Future and Beyond: Towards Redefinition of SI Second

From tracking the movement of celestial bodies to mechanical clocks and now prevalent atomic clocks, timekeeping has been one of the greatest quests for finding stable and better oscillators, having least effect of changing surroundings. A quartz crystal is better than a pendulum, and an atomic transition is much better than a man-made crystal. Continuing with the similar objective, transitions of neutrons and protons in a nucleus are better isolated than electronic transitions in atoms (Peik and Tamm 2003; Beck et al. 2007). Nuclear transition is not affected by stray electric and magnetic fields like electronic transitions and thus could make ideal candidate for realizing more accurate and stable clocks. Typical nuclear transitions require

much higher energy than the normal lasers can provide with the exception of ^{229}Th, where a nuclear transition between a pair of closely spaced energy levels can be excited using ultraviolet lasers. However, a lot of work needs to be done before a single ^{229}Th ion trapped in a Paul trap can provide a new standard of time.

Since 2000, several clocks based on optical transitions in ions and neutral atoms have been developed and have been consistently showing superior performance with around two orders of magnitude lower relative uncertainty (at 10^{-18} level). A 10^{-18} level uncertainty means an error of less than a second in a billion years, the age of our universe. Based on better and consistent performance, optical clocks have been pushing the case for a possible redefinition of SI second in terms of an optical transition (McGrew et al. 2018). However, there is no consensus on the candidate/element and the confinement method for the eventual redefinition. An optical clock can be realized either with single trapped ion or neutral atoms trapped in a lattice. Earlier realization has the advantage of long coherence time and thus lower uncertainty, while the latter one offers large S/N ratio, thus providing better stability. At the same time, there are various species of ions (Yb+, Sr+, Al+, Ca+, Hg+, etc.) and atoms (Sr, Yb, Ca, Hg, etc.) which are being investigated but no candidate has a clear and obvious advantage for redefining the time. Moreover, high-accuracy time transfer/comparison links are required for intercomparing the optical clocks being operated at different countries and continents around the world. It is clear that a significant amount of work still needs to be done before the redefinition of SI second. An envisioned possible timeframe for the redefinition is as early as 2026. However, in order to ensure a smooth transition to an optical second, the International Committee of Weights and Measures (CIPM), in 2018, published a detailed framework with fixed milestones which have to be achieved in the domain of optical clocks before implementation of the new definition (Riehle et al. 2018; Lodewyck 2019). The set of criteria include demonstration of better accuracy and stability, reproducibility of results, continuity with the current SI definition, and even by intercomparison of results from different laboratories with suitable time transfer methods. A number of optical transitions, however, have been accepted as secondary representation of second (SRS) and are already contributing to the maintenance of International Atomic Time (TAI).

In 1967, SI second was redefined from astronomical definition to the atomic transition in cesium atoms, and few years ahead in time, it may be replaced by a transition in the optical frequency region. Another redefinition might happen once nuclear clocks are realized and tested for better stability and accuracy over optical clocks. Until the next redefinition, cesium fountain clocks will remain the primary standards of time and frequency.

Conclusion

Most of the modern world applications including navigation, telecommunications and the Internet, electrical power grids, banking, smart transport, finance, defense, cyber physical systems and upcoming technologies of 5G, artificial intelligence and

Internet of things, etc. rely heavily on precision time measurement and time synchronization. Additionally, precise time with ns accuracy is also required for high-end research in science, namely, measurement of fundamental physical constants, detection of the gravitational waves, geodesy, deep space navigation, radio telescopes, etc.

To meet the growing demand, there has been a remarkable progress in the field of timekeeping with atomic clocks, especially after the developments of the laser cooling and trapping atom techniques. Clock instability, reproducibility, and accuracy have been improved significantly in last half-century. Primary Cs fountain frequency standards, more accurate than Cs beam clocks, achieved a total uncertainty of a few parts in 10^{16}. Since 2000, trapped ion and neutral atom optical frequency standards have reached an uncertainty in a range of 10^{-18}–10^{-19}, thus exceeding the performance of best Cs fountain clocks. Due to the consistent performance of optical clocks, ten optical transitions have been recommended as secondary representations of the second. However, high-accuracy comparison between optical clocks located in different countries or continents is still a challenging issue.

From the present definition of SI second, best realized with laser-cooled cesium fountain clocks, scientific endeavor in the field of optical clocks is pushing for a future redefinition of SI second on the basis of an optical transition. With a number of viable candidates and confinement techniques, a framework has been set up by CIPM to ensure smooth transition to a new definition of SI second in future.

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Precise Time Transfer Techniques: Part I

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Telephone, LWR, and Network

Pranalee Premdas Thorat, Ravinder Agarwal, and Dinesh K. Aswal

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Abstract

Time dissemination from authentic common reference is a key for synchronized operations. This chapter discusses various techniques available for precise time transfer. Depending on the dissemination techniques used, the time synchronization accuracy varies. This chapter provides a brief overview of various dissemination services. Telephone time dissemination technique was establishes owing to its widespread availability in early days. With the development of Internet services and its reachability to masses, time dissemination via NTP is currently the most common methodology with millisecond/microsecond accuracy. Satellite time dissemination is for very precise users whose accuracy demands are of the order of nanoseconds. This chapter elaborates details about all such time transfer techniques with protocols and methodologies adapted and discusses their corresponding results.

Keywords

ACTS · Teleclock · CVGNSS · LWR · PTB · Network Time Protocol (NTP) · Time dissemination · Internet time dissemination · PTP · WR-PTP · Round-trip time · Offset

Introduction

Precise time synchronization and timestamping are critical for the country's overall growth and security. Time is an extremely important factor, but it is frequently overlooked. Time is only seen as a broader aspect of everyday life, matching time with some events. However, time is incredibly critical and demanding for many

applications and sectors. This chapter discusses some key aspects of the strategic sectors, such as navigation, electronic transactions and cybercrime, digital archiving, international trade, national security, weather prediction, and disaster management, transportation, application in the power grid, identifying underground resources, and so on.

Strategic sectors such as air, naval, and army, defense research laboratories, space sectors, and astronomical observatories, among others, necessitate extremely precise time measurement and synchronization. Surveillance, precise targeting, satellite launch and placement, space applications, and astronomical research all rely on time synchronization. Navigation is used for precise positioning. Time synchronization with nanosecond precision is required. Satellite navigation relies heavily on synchronization. With increasing demands for positioning accuracy, synchronizing the onboard satellite clocks with the base receivers and transmitters is critical. While progressing toward digitalization, digitization of economic involvements plays a crucial role. But unless these e-business dealings are synchronized to a common reference of time, they can turn down the economy instead of elevating it. Avoiding the chances of cybercrime by having traceable time is the key here. Digital archiving is linked to the orderly storage of electronic data. This is critical as otherwise, it may affect the smooth functioning of a multibrach business/sector, as well as their lawful accountability, which can also be in danger.

To avoid any disaster in international trading, accurate time synchronization requires international trade involving all other countries. Then, for national security monitoring of any violence or national calamity, as well as an immediate action against it, time synchronization among security agencies and alarming systems is required. Transfer techniques, weather prediction, and disaster management are all essential for any precise time. The dynamic mapping and evolution of any natural disaster, such as a cyclone, flood, or tsunami, is dependent on precise time synchronization among the devices involved. Furthermore, time synchronization of network devices and associated systems is required for public awareness of natural disasters, civil alerts, traffic information, and early warning on health or weather. Transportation is another critical factor in the efficiency of an automatic signaling system, air traffic control, and navigation of all transport systems, including road traffic, railways, and aviation, relying on the associated devices' accurate time synchronization.

The power grid application demonstrates that significant amounts of electric power can be saved if grids are time-synchronized, as power can be rerouted and grid failures can be detected without wasting time. Finally, identifying underground resources, and tomography of the ocean floor, and determining underground resources are accomplished by sending a sound wave and receiving the reflected signal, followed by geologists post-processing the data. This technique relies on the precise time synchronization of the associated transmitters and receivers. In many fields of technology and science, precise time and frequency are required. Directly or indirectly, the networked synchronization in time or frequency plays a critical role in these industries. A time or frequency signal is broadcast from a standard source to synchronize remote locations. The received time and frequency information is used to synchronize local time scales at the users' locations. However, the accuracy of

synchronization will be determined by the method or technique used to distribute the reference time and frequency signal, the network topology used, and the precision of the users' system. Different time synchronization techniques provide varying degrees of synchronization accuracy. However, depending on whether time information or frequency is transmitted, the transfer techniques must meet different requirements. As the delay is comprised due to many factors (cable delay, path delay), it must be properly considered or corrected for time transfer. As a result, the accuracy with which synchronization of transmitter and receiver is achieved influences source or transmitter capability, linking connecting source and transmitter and receiver capability. The uncertainty of the time transfer is summed up by all of these contributions. However, if frequencies are to be compared as a result of the signal's periodicity, the delay must be constant.

There are three types of time dissemination methods: unidirectional, bidirectional, and common view. These methods are commonly used to send time and frequency data. In the unidirectional method, path delays are modeled and adjusted using auxiliary data. However, path symmetry is critical in this model because the bidirectional method relies on cancelling the online path delay experienced during dissemination. Furthermore, several stations receive data from a common source via paths with roughly equal delays.

Unidirectional Methods

When data flows from transmitter to receiver in only one direction, the communication is referred to as unidirectional communication. This technique is primarily used in functional reactive programming. In this method, only the transmitter can send signals to the receivers, and the receivers have no way of communicating back to the transmitter. This set is useful in many control system applications where better data flow control is required. However, this is not appropriate for many applications where improved data flow control is not the goal.

Using supplementary data, the transmitter-receiver delay is computed in numerous systems. GNSS signal transmission in which path delays are estimated using various parameters that are also sent in the routing information, radio broadcast of time and frequency signals, and some telephone time dissemination services all fall under the unidirectional time dissemination method (refer to Fig. 1). In terms of precision, these methods have limitations. Nonetheless, these services are widely used because they provide adequate accuracy while remaining simple to use.

Bidirectional Methods

The majority of the time dissemination methods are bidirectional. The transmitter and receiving stations are online in this method, and they actively transmit and receive time data (refer to Fig. 2). Because the transmission and reception are bidirectional, the delay can be measured and compensated for. The best example

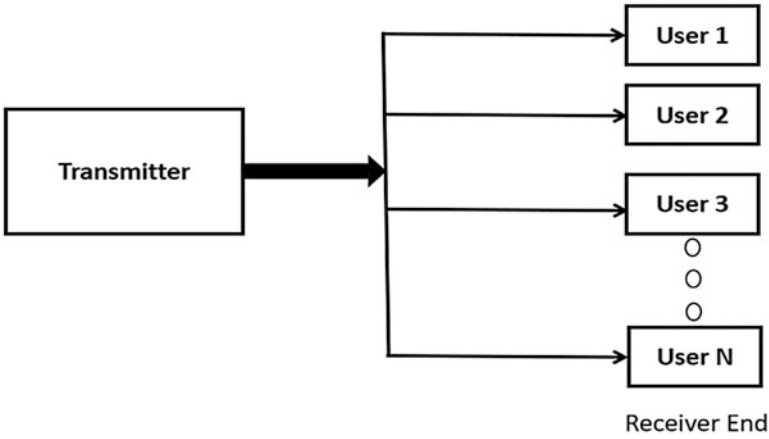


Fig. 1 Unidirectional time transfer

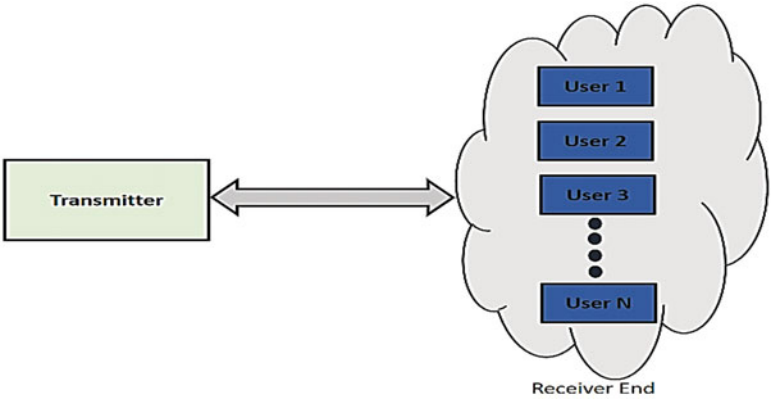


Fig. 2 Bidirectional time transfer method

of bidirectional communication is the two-way satellite time and frequency transfer system (TWSTFTS). This technique compares clocks located in different locations using geostationary satellites.

A modem used in the TWSTFT system generates a code that is synced with local clocks' 1 Hz signal. An uplink carrier frequency (14 GHz) is then modulated by pseudorandom code generated by TWSTFT modem. This signal is uplinked to a geostationary satellite. Downlink is done at 12 GHz frequency for modulated signal retransmitted by satellite. The receiver receives the signal, demodulates it, extracts the code, converts it into a synchronization signal, and verifies it with the available local replica of the same signal. It's a real-time bidirectional, full-duplex communication system with accuracy greater than 1 nanosecond.

Similarly, a commonly used and popular method of time dissemination is Network Time Protocol (NTP) method. In this client-server model, a request to the server is originated by a client which is fulfilled by the server with appropriate information within a standard format. The client extracts information from this data format while adjusting the network delay to half of the total path delay and synchronizing its clock. PTB Germany's telephone time dissemination service, Telefon, is another example of a bidirectional time dissemination method. This service is available via the public telephone network at +49 (0)531 512038. This service operates at a baud rate of 1200 and in full-duplex mode.

Common-View Methods

GPS/GNSS time transmission is a very good example of bidirectional network time transfer. The information generated by one source (onboard clock) is received by the devices in a network comprising several locations which could be termed as the easiest demonstration of the common-view method. Each station keeps track of the time it takes for a specific signal to arrive at its station. After that, the measurements are verified at their end. If there is symmetrical delay observed in the paths followed, then two stations in the same network can be compared using the network source with counting for it. Also, the effects of jitter involved in these paths get cancelled. This category includes CV- GNSS and AIV-GNSS methods.

Typically, the CVGNSS (Levine 2002) method employs atmospheric paths with microwave signals (Fig. 3). Short-wavelength signals are well suited to the requirement that the transmission path is well defined and easy to characterize because they travel in nearly straight lines and can be modeled using geometrical optics approximations. It is impossible to achieve global coverage with a single transmitter at short wavelengths. This can be addressed by the all-in-view method.

One method of achieving global coverage is to use multiple physical transmitters with synchronized signals to a common reference, which is the origin of the AIV and melting pot method. Because there is no actual transmitter whose signal dissipates to such long-distance locations at the same time, this method proves to be very useful. Long distances like India and Canada can use the AIV method.

Measurements in global navigation satellite system (GNSS) are dependent on two main factors:

- The time interval between the satellite
- Reception of pseudo-range noise (PRN) codes

The pseudo-range can be computed using the following equation:

$$P = c(t_r(\text{rec}) - t_e(\text{sat}))$$

where c represents the velocity of light. In applications, where sub-millisecond accuracy is needed, GNSS can be the best method for time synchronization.

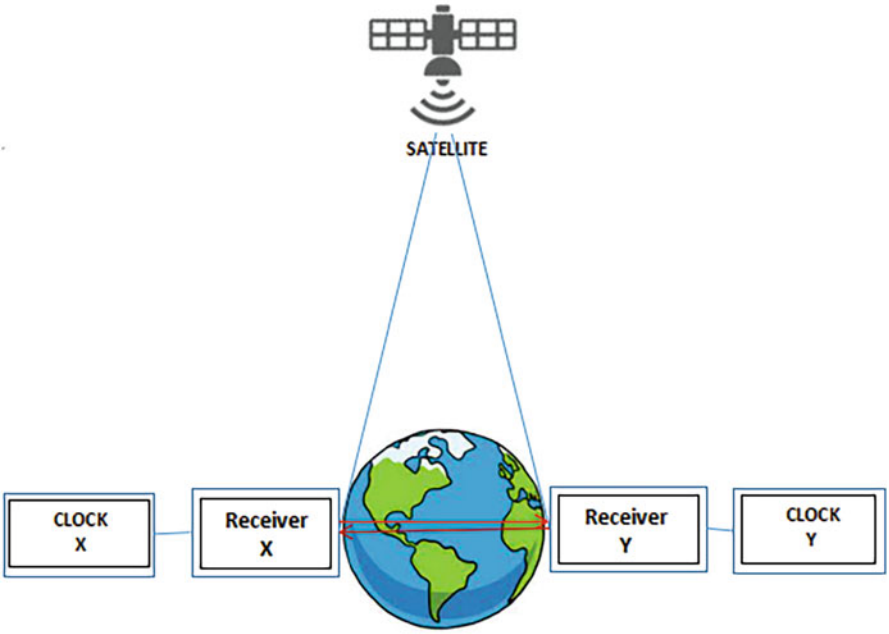


Fig. 3 Time transfer using common-view GNSS

Banking, telecommunication networks, energy transport and distribution network, etc. are some areas of applications where GNSS is utilized for time synchronization.

Telephone Time Dissemination Technique

A telephone is a device that allows for the simultaneous transmission and reception of voice. Telephones are inexpensive and simple to use. Telephone lines can also be used to transmit audible data. Some techniques have also been developed to transmit accurate time over telephone lines. Time transfer over a telephone line can serve a wide range of users who require millisecond precision. This service only requires the existing telephone exchange architecture to transmit the timestamp. Telephone time dissemination has been regarded as a valuable supplement to radio time dissemination services. It avoids the real and perceived problems of antenna placement and signal availability.

The synchronization of digital system clocks using a public switched telephone network (PSTN) is a simple way by which a remote clock can be set accurately. Additionally, it is the least expensive way to synchronize a remote clock. A total of 20 NMIs around the world have implemented telephone time dissemination. As shown in Fig. 4, the basic architecture of this technology includes a time server at the NMI laboratory end and receiver modules at the user end. The time reference for these servers is the time scale of the corresponding NMI. The following section

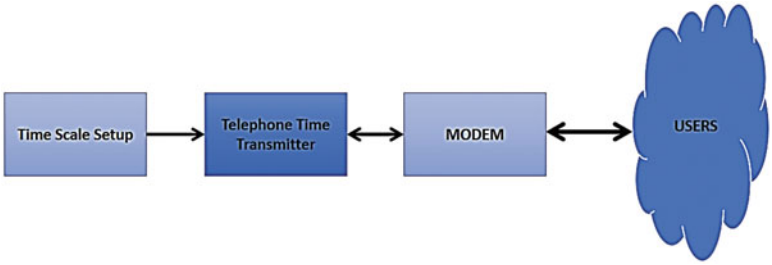
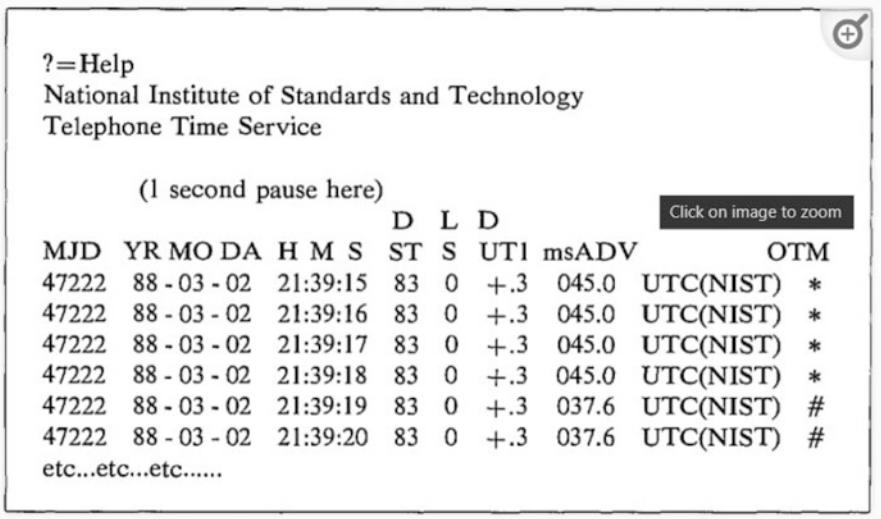


Fig. 4 Generalized block diagram of telephone time transfer

explains various key features of such telephone time synchronization services which are operated by NIST, USA; NPL, UK; NICT, Japan; PTB, Germany; NRC, Canada; and INRIM, Italy. NPL, PTB, and INRIM all use the same data format and transmission scheme. At CSIR-NPL, India is operating a service named Teleclock since 1998.

The NIST in the USA offers a time synchronization service known as Automated Computer Time Service (ACTS). It has been in operation since 1988 (Set Your Computer's Clock via the Telephone: NIST Automated Computer Time Service (ACTS)). ACTS uses analogue modems that connect to telephone lines. The time server can be reached by dialing 303 494-4774 or 808 335-4721. ACTS (Lombardi 2002) can operate at a maximum speed of 9600 baud. This technology uses 8 data bits, 1 stop bit, and no parity bit. For accurate and effective reception of full-time code, the receiver must connect at a speed of 1200 baud. At every second, ACTS transmits full-time code and the code carries more information than the 300 baud time code, which is sent every 2 s. Using this service (ACTS), any clock can synchronized itself with an accuracy of a few milliseconds. This technology has a millisecond delay. This delay is determined by the oath's physical length and the signal's group velocity. The ACTS Transmission Formats are explained and illustrated in Fig. 5:

- The modified Julian day number is represented by the MJD column; the MJD advances by one at 0000 Coordinated Universal Time (UTC) every day. On January 1, 1989, the MJD was 47,527.
- The next six digits represent UTC in years since 1900. This displays the month, day, hour, minute, and second.
- DST is a column that specifies whether a daylight saving time correction is required now or will be required soon. This flag is legal in most of the USA. DST of 00 indicates that standard time is in effect. The value 50 in the DST column indicates that daylight saving time is under effect. If the DST column has a value ranging between 99 and 51, it indicates that a transition to daylight saving time may occur. If the value of DST is between 01 and 49, it means that the transition from daylight saving time to standard time may occur in the near future.
- The leap second flag, LS, is generally not set, i.e., it is zero. If it is set to 1, the leap second will be added on the last day of the current month. The leap second will be



?=Help
National Institute of Standards and Technology
Telephone Time Service

(1 second pause here)

Click on image to zoom

MJD	YR MO DA	H M S	D ST	L S	D UT1	msADV		OTM
47222	88 - 03 - 02	21:39:15	83	0	+.3	045.0	UTC(NIST)	*
47222	88 - 03 - 02	21:39:16	83	0	+.3	045.0	UTC(NIST)	*
47222	88 - 03 - 02	21:39:17	83	0	+.3	045.0	UTC(NIST)	*
47222	88 - 03 - 02	21:39:18	83	0	+.3	045.0	UTC(NIST)	*
47222	88 - 03 - 02	21:39:19	83	0	+.3	037.6	UTC(NIST)	#
47222	88 - 03 - 02	21:39:20	83	0	+.3	037.6	UTC(NIST)	#

etc...etc...etc.....

Fig. 5 Data format of ACTS

labeled 23:59:60 UTC, and the second after 23:59:60 will be labeled 00:00:00 the following day. If the leap second flag is set to 2, it indicates that a second will be dropped at the end of the current month's last day.

- The column DUT1 represents the difference between a time scale and UTC. It measures the difference using the rotation of earth. Mathematically,

$$\text{DUT1} = \text{UT1} - \text{ms ADC}$$

- The on-time marker's advance in milliseconds is represented by msADV. To arrive at the user on time, the center of the on-time marker's stop bit leaves the transmitter by this amount of time earlier. The method used to estimate this parameter, as well as its uncertainty, is discussed below.
- The on-time marker is denoted by the symbol OTM. It is either "*" or "#," as discussed further below. The center of this character's stop bit is intended to arrive at the user at the time specified by the previous characters on the same line (Levine et al. 1989).

PTB offered a telephone time service named Telefon via the public telephone network under the phone number +49 (0)531 512038. The methodology follows the protocol CCITT V.22 and the connection is established using telephone modems. It can prove an accuracy ranging from 1 s to a few milliseconds. The time code transmitted by Telefon follows the rules defined by the European telephone time code.

TEL-JJY, which is used by NICT to distribute time over the PSTN, can be reached by dialing +81 45 411 7223. With a standard v22 analogue modem, the connection offers baud rates of 300, 1200, or 2400. During the synchronization

process, the path delay between the user and the transmitter is computed. It provides synchronization accuracy of a few milliseconds between the transmitter and the special type of receiver.

NPL, UK, also offers a time synchronization methodology over a telephone network named Phoneclock. This service can be accessed by dialing 020 8943 6333. The synchronization accuracy of the Phoneclock system is around 30 ms (Laverty 1996).

A clock system is developed by NRC, Canada, and a company named Leitch Video Limited which can set the digital clocks automatically. This system uses telephone lines to communicate with the time server. The protocol sends the date, time, and telephone loop delay, and the information is sent in the form of ASCII code formats. The service can be availed by dialing (613) 745-3900. Synchronization accuracy is in the order of a few milliseconds. This system uses a simple protocol to disseminate time information. The protocol utilizes ASCII characters transmitted by frequency shift keying techniques. The UTC is transferred from master to slave clock in this protocol. In this methodology, the slave clock makes a call to the master clock. The call puts the master into a prompt echo mode. The receiver then measures the delay time and applies half of the delay time as a correction to the time it receives from the master clock. This technique of time transfer provides an accuracy of a millisecond. This accuracy is possible over even a relatively low-grade telephone link (Jackson and Douglas 1986).

The CSIR NPL's Teleclock service (Banerjee et al. 2012a, b) provides synchronization accuracy of the order of 1 s. The current service makes use of a CCITT V.22 analogue modem with a baud rate of 1200. The Teleclock transmitters are connected to two accessible telephone lines from different service providers. Users can get the time by dialing 01145608687 and 01145608688. During time broadcast, information is transmitted in hours, minutes, and seconds. This service broadcasts time data in the form of hours, minutes, and seconds. It is a television broadcasting service. When a receiver communicates with the Teleclock transmitter and a successful connection is established, the transmitter begins broadcasting time information five times per second. There is no need for any special hardware to synchronize the computer clock with the Teleclock. The only requirement is to connect the PSTN line to the computer's built-in modem. Like India, few other countries, such as Nepal and Saudi Arabia, used the Teleclock service for disseminating time information. This service is serving many customers in India including Parliament House, CSIR headquarters, airports, railway stations, police control rooms, CSIR laboratories, etc.

Figure 6 depicts the basic unit for gaining access to the Teleclock. It is composed of three major components: a modem, a data identifier, and a clock system (Banerjee et al. n.d.-c, 2009; Thorat and Banerjee 2010, 2011). The modem is important because it allows you to access data from the PSTN line. Data deciphering unit is derived using the output of the modem, which retrieves the data, verifies its accuracy, and stores it for delivery to the clock. The clock system receives data from the modem and corrects the time displayed by the computer or client system's local clock. Digital time information is transmitted over the PSTN as an audio frequency range signal (300–3300 Hz). To send digital data over a telephone network, the digital data must be modulated so that it can be sent as an encoded audio frequency

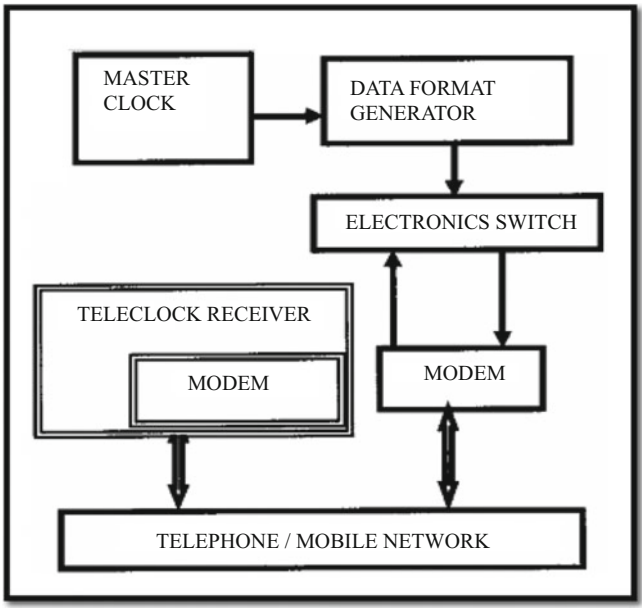


Fig. 6 Scheme of Teleclock service

over PSTN. A demodulator is also required at the receiver end to extract digital information from the modulated signal. To achieve greater accuracy, full-duplex communication is required. As a result, both ends of the system should have a transmitter and receiver, as well as a modem (modulator-demodulator). In the present work, the Conexant standard v.92 56 k USB modem was used.

Improved Teleclock Receiver/Mobile Teleclock

With the advent of telecommunication technology, telephone time dissemination via landline receivers was becoming a limitation. Hence, improved version of Teleclock receivers in the name of mobile Teleclock was developed at NPLI. These receivers were using a SIM card to access Teleclock transmitters. Mobile Teleclock receivers used GPRS modems. The limitation of availability of landline connecting at remote locations was overcome by mobile Teleclock service. The accuracy of this service was less than 1 s. However, this service was shut down in 2016 (Fig. 7).

FonOclock is the upgraded version of Mobile Teleclock (Arora et al. 2017) which is expected to give accuracy of the order of 10 ms. The system includes a transmitter server, a receiver, and a transmission channel. The connectivity over PSTN is used as the transmission channel. The receiver communicates with the time server by dialing the transmitter number on the ver PSTN network. After receiving the data from the transmitter, the receiver corrects its time for path delay as well as the actual date and time. There is a provision to correct the incurred path delay which makes the system better than that of earlier ones with better accuracy (Agarwal et al. 2019). FonOclock

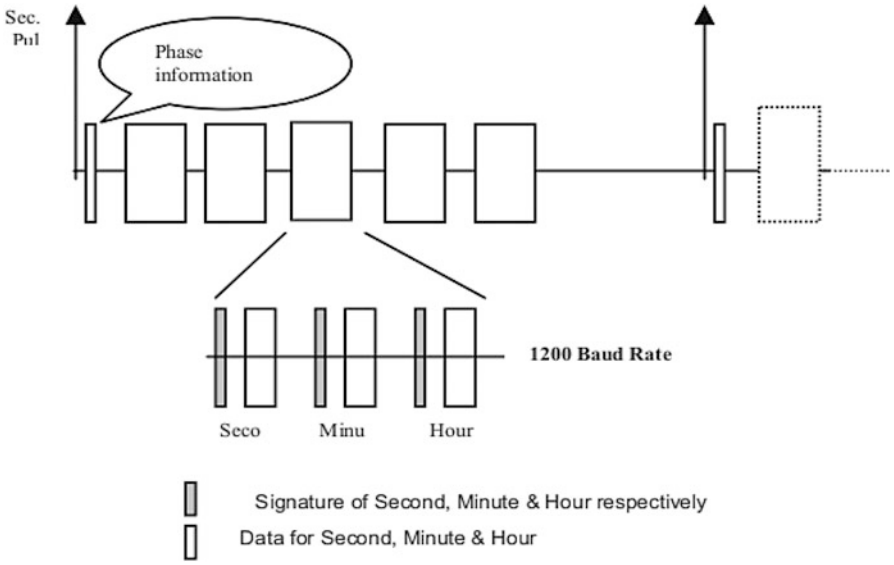


Fig. 7 Teleclock service data Format

transmitter, like Telethelclock, is connected to the public switched telephone network (PSTN) via a dedicated telephone line (Poudel et al. 2019). The receiver contacts the transmitter and requests timing information during the synchronization process. The link delay between the transmitter and receiver is calibrated after the authentication and link establishment processes are completed successfully. The delay compensated digital time information is then transmitted over the PSTN as an audio frequency signal. When a time code is successfully received, the receiver sets its clock and acknowledges the time server (Fig. 8).

The primary difference between transmitter and receiver is that no external 1 PPS is required to run the clock in the receiver. Using a software counter, the duration of a second is directly extracted from the oscillator used in it. The system employs a standard 10–20 ppm crystal oscillator. However, if greater stability or performance is required, better oscillators such as temperature-compensated oscillators, oven-controlled oscillators, and so on can be used. However, the system's cost will inevitably rise.

Network Time Dissemination

Daytime Protocol

This protocol is commonly used by small computer systems running MS-DOS, Windows, and other similar operating systems. The server listens on port 13 and responds to the client in either tcp/ip or udp/ip format in this protocol. Daytime Protocol is mentioned as a service in RFC 867 (1983), which defines Internet

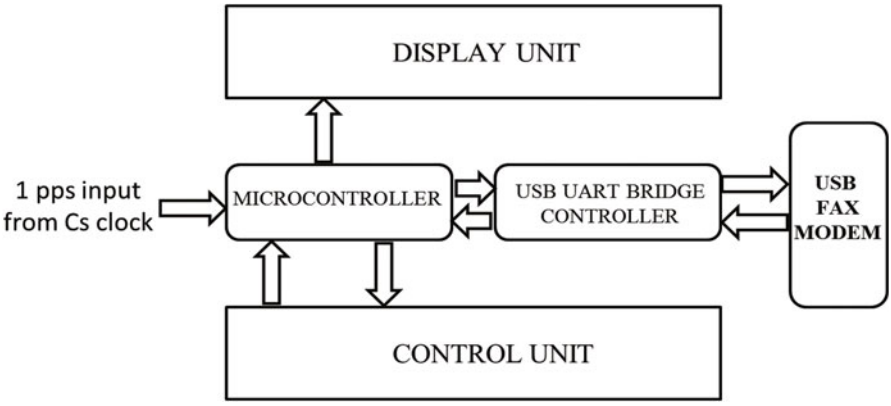


Fig. 8 Block diagram of transmitter/receiver hardware of FonOclock at CSIR_NPL

protocol suite (IPS) (Postel 1983). It is possible to form and measure computer networks. On TCP or UDP port 13, a host can communicate with a Daytime Protocol server. In an unspecified format, the server returns an ASCII character string containing the current date and time.

Time Protocol

The Time Protocol is a network protocol that was defined in 1983 in RFC 868 (Postel and Harrenstien 1983) by Jon Postel and K. Harrenstein as part of the IPS. It is aimed at providing a date and time which could be understood by machine and also is not position reliant. The Time Protocol can be implemented using either the TCP or the UDP protocols on port 37. The time is transmitted as 32-bit unsigned integer corresponding to the number of seconds elapsed since 00:00 (midnight) January 1, 1900, GMT. The connection of the server is terminated after this. Because there is no UDP connection setup, operation over UDP necessitates sending any datagram to the server port. Because of the fixed 32-bit data format, the timestamp resets every 136 years, with the first occurrence on February 7, 2036. Programs that use the Time Protocol must be carefully designed to distinguish these dates from those in 1900 by utilizing context-dependent information. The Time Protocol was used by many Unix-like operating systems for clock timestamping using the date utility. But NTP provides a more advanced solution with accuracy in milliseconds.

This straightforward protocol yields a 32-bit unformatted binary number representing UTC time in seconds since January 1, 1900. It's no longer commonly heard. The server accepts Time Protocol requests on port 37 and responds in tcp/ip or udp/ip format. The time is converted to local time by the client program (if necessary). The 32-bit binary format, with a resolution of 1 s, can represent times ranging from 136 years to 136 years. There are no provisions for extending the resolution or the time limit. The strength of the Time Protocol lies in its simplicity. Because many computers internally store time as the number of seconds since January 1, 1970 (or another date), converting received time to the required format

is often as simple as binary arithmetic. However, no additional information, such as advanced notification of leap seconds or daylight saving time, or information about the server's health, can be sent using this format.

Time Dissemination Using the Internet: NTP

Network Time Protocol

Network Time Protocol is abbreviated as NTP. It's a protocol for synchronizing the time on the Internet. It uses the Internet to synchronize computer clocks to a time reference. Prof. David L. Mills of the University of Delaware devised this protocol. NTP can sync all connected computers within a few milliseconds of Coordinated Universal Time (UTC) (Riehle 2004). It uses a modified version of Marzullo's algorithm to select the correct time server. It is intended to reduce the impact of network latency on time synchronization. Asymmetric routes and network congestion can result in errors of 100 ms or more (Lavineavine 2008; Blair 1974).

NTP provides a mechanism for synchronizing the local machine time with the Coordinated Universal Time. It employs a returnable-time design in which a distributed subnet of time servers self-organizes. NTP accuracy is determined by the precision of local-clock hardware as well as transaction latency. Even if the primary reference source is lost, the mechanism can maintain millisecond accuracy. NTP timestamps are represented as an unsigned fixed-point number of 64 bits (Mills 1989).

NTP is commonly referred to as a client-server architecture, but peer-to-peer architecture is also associated with NTP. Figure 9 depicts (Riehle 2004):20. Implementations use the User Datagram Protocol (UDP) on port 123 to send and receive timestamps (Mills 1981; Su 1981). Other modes for configuring NTP server are broadcasting or multicasting (Eliezer et al. 2007). The current version of the NTP, according to RFC 590,5, is NTPv4. It was, as documented in RFC 5905, the proposed standard.

Development of NTP

The application of network time synchronization finds its root in 1979. However, satellite-based Internet time synchronization technology was exhibited in New York's Conference on National Computer, which was mentioned in the 1981 published Internet Engineering Note (IEN) 173. This is how a public protocol got evolved as RFC 778 (Mills 1981). Realized using the Fuzzball router, this technology was used in LAN with preexisting "Hello" protocol.

The Daytime and Time protocols are two network tools for time synchronization. They are used to record the time of events and the timestamp of ICMP messages. They are also used to timestamp Internet protocol (IP) (RFC781) (Su 1981). The

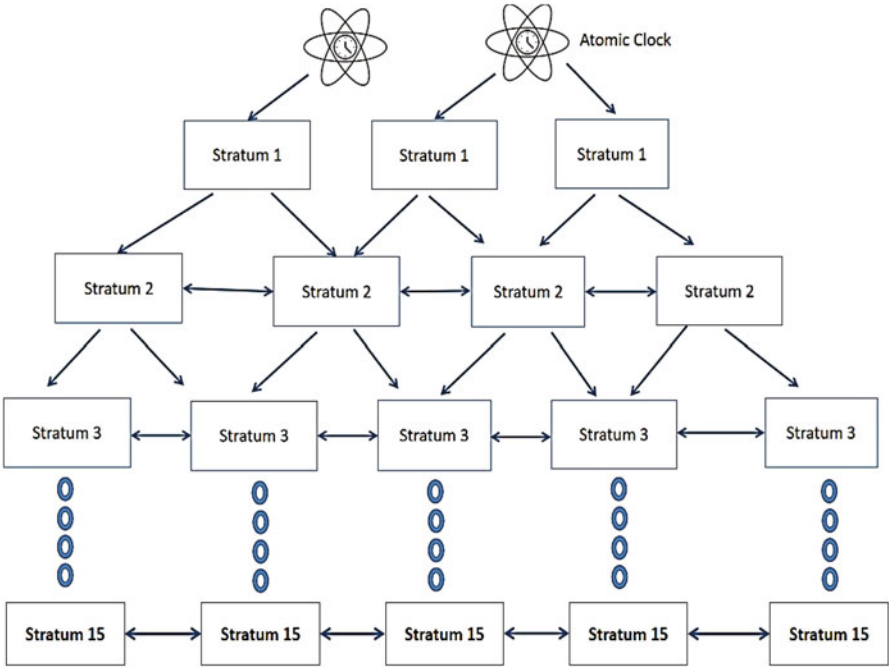


Fig. 9 Stratum structure of NTP architecture

Unix timed daemon provided better synchronization than the Daytime and Time protocols. An election algorithm was used in this protocol to identify the master server for client synchronization. However, this protocol did not perform any data analysis and did not use a clock disciplined algorithm like NTP. The Digital Time Synchronization Service (DTSS), like the NTP architecture, had a hierarchical structure of synchronization servers.

In 1985, Fuzzball- and Unix-based implementation of the initial version of NTP was achieved in the name of NTPv0. The RFC 958 elaborates on the important computations involved in the latest NTP form NTPv4 like offset and network delay (Mills 1985a). It is the specialty of this protocol that the best performance (surpassing 100 ms) was achieved even with contained devices available then.

RFC 1059 (Mills 1988) published an NTPv1 protocol with more specifications and associated algorithms in 1988. NTPv1 pioneered and explained client-server and peer-peer architecture which is existing to date and is considered as the best methodology for network time synchronization. RFC 956 (Mills 1985b) details the algorithm for clock filter with results as was published in 1991 IEEE Transactions on Communications by David L. Mills.

NTPv2 was published in 1989, using a state machine and pseudo-code to describe its operation, which was documented in RFC 1113. Management protocol with cryptographic authentication scheme was first time implemented in this version of

NTP. The DTSS penalized NTPv2 for a lack of formal correctness. As a result, from NTPv3 onward, Marzullo's algorithm was adapted for clock selection (Eliezer et al. 2007). NTPv3 was implemented in 1992 and documented in RFC 1305 (Mills 1992). The RFC contained an analysis of all potential sources of error, allowing calculation of a method for best server election. The broadcast mode was added to this version. NTPv3 was in use for quite a long time till 2010 when RFC 5905 proposed NTPv4 provisions (Mills et al. 2010).

Basic Features of NTP

NTP makes use of a reference clock, like a GPS, calibrated cesium clock, calibrated rubidium clock, NIST telephone time service, PTB telephone time service, or radio navigation service like LORAN-C. These clocks determine the correct time of the system. The NTP architecture's stratum-0 is the reference clock. For greater accuracy, NTP servers synchronize with the reference clock. UTC is typically used as the reference time for NTP time services. NTP is a fault-tolerant protocol. NTP is both accurate and scalable. NTP provides up to tens of milliseconds of accuracy over a wide area network (WAN) and less than 1-millisecond accuracy over a local area network (LAN). NTP provides good accuracy even in the absence of a network (temporarily). In the absence of network connectivity, NTP uses measurements from previous values to synchronize the local clock.

Architecture of NTP

NTP employs a hierarchical, semi-layered time source system. It's a top-down synchronization approach, where each step/layer is termed as stratum. The stratum starts with a reference clock itself and is identified the d as 0th stratum. The next layer, i.e., stratum-1, is directly connected to the previous stratum and gives synchronization to the following stratums in the architecture. It should be noted that the stratum numbers only specify the distance of the NTP server from the reference atomic clock (i.e., stratum-0) and do not necessarily depict the performance of the NTP server. It shows that the server is in synchronization in the hierarchy. It must be noted that clock strata are defined differently in telecommunications systems. The basics of various strata are explained in brief below:

Level: Stratum-O

Reference source races themselves are termed as 0th stratum Reference time source can be anything as calibrated cesium source, calibrated rubidium clock, NIST,

USNO telephone time service, or radio navigation services like LORAN-C or satellite constellations like GPS, GLONASS, or NAVIC. Precise time and frequency signal generated by this source is used to synchronize the connected NTP server, which is then termed stratum-1.

Level: Stratum-1

The servers which are first in layer and are directly connected with reference sources, i.e., 0th stratum, are termed as stratum-1. Due to this, microsecond-level synchronization to that of reference clocks is possible in this level. For sanity checks and backup, stratum-1 NTP servers may peer with other stratum-1 NTP servers (Mills et al. 2010). They are primary time servers (Lavineavine 2008; Blair 1974).

Level: Stratum-2

These are the systems that are network-synchronized with stratum-1 systems and provide synchronization to the next level in stratum (i.e., stratum-3). Besides being an interface layer between stratum-1 and stratum-3, it works on peer synchronization within this layer as well. Hence, NTP becomes an extremely robust architecture with consistent timing.

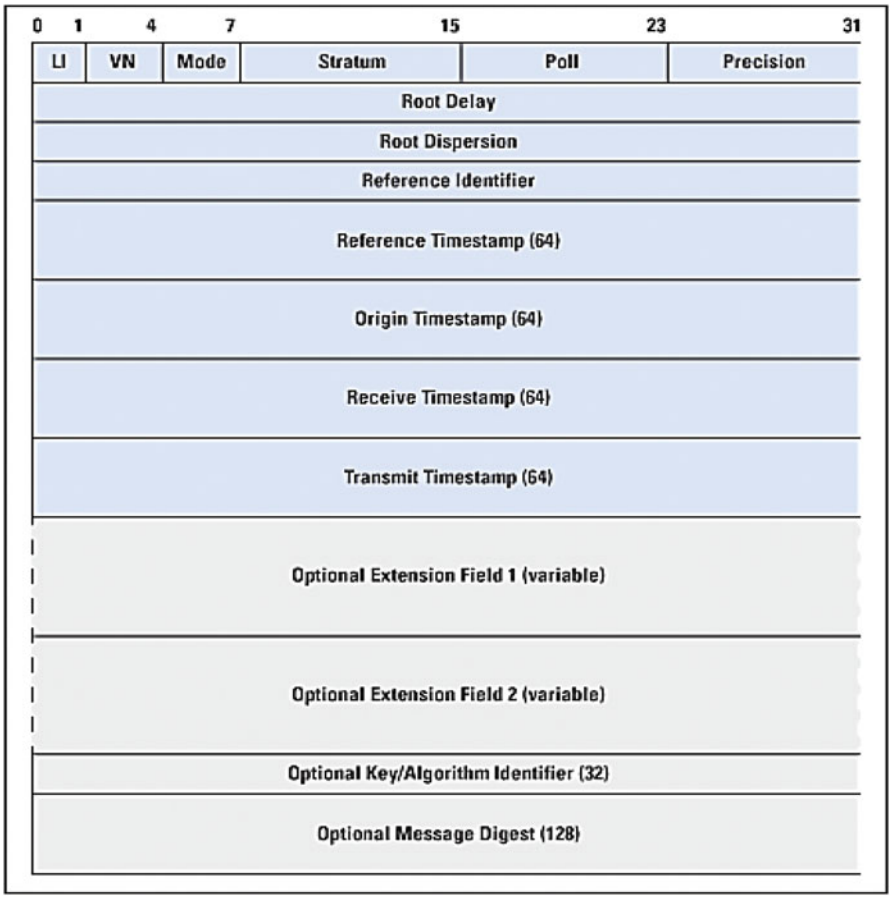
Level: Stratum-3

As mentioned in stratum-2, this is an intermediate stratum between stratum-2 and stratum-4. The functionality of this stratum is almost similar to that of earlier stratum-2 by taking synchronization from stratum-2 and providing synchronization to stratum-4 with peering capabilities. This continues till stratum-15.

The last stratum in the architecture is stratum-15; the next level, stratum-16, indicates that a device's clock is out of sync. Each computer's NTP algorithm collaborates to construct a Bellman-Ford shortest-path spanning tree in order to reduce the total round-trip delay to stratum-1 servers for all clients (Riehle 2004).

NTP employs a 64-bit timestamp. Seconds are represented by 32 bits, and 32 bits are used for displaying the fractional value of the second. NTP employs the January 1, 1900, epoch. The protocol is expected to be extended to 128 bits in future NTP versions. But Dr. Mills believes that the 64-bit value is more than sufficient as it can signify the second's value to the utmost resolution and any ambiguous depiction of time can be avoided using these timestamps (Banerjee et al. 2012b).

NTP Message Format



Algorithm

For details of each field of the message format, refer to RFC 4330 (Fig. 10).

A typical NTP client will poll three or more servers on different networks on a regular basis. To synchronize its clock, the client must compute its time offset and round-trip delay using Eqs. (1) and (2). The time offset and the round-trip time are denoted by Eqs. 1 and 2, respectively. T_1 is the client's request packet transmission timestamp, T_2 is the server's request packet reception timestamp, T_3 is the server's response packet transmission timestamp, and T_4 is the client's response packet reception timestamp.

Figure 2: Reference Identifier Codes (from RFC 4330)

Code	External Reference Source
LOCL	uncalibrated local clock
CESM	calibrated Cesium clock
RBDM	calibrated Rubidium clock
PPS	calibrated quartz clock or other pulse-per-second source
IRIG	Inter-Range Instrumentation Group
ACTS	NIST telephone modem service
USNO	USNO telephone modem service
PTB	PTB (Germany) telephone modem service
TDF	Allouis (France) Radio 164 kHz
DCF	Mainflingen (Germany) Radio 77.5 kHz
MSF	Rugby (UK) Radio 60 kHz
WWV	Ft. Collins (US) Radio 2.5, 5, 10, 15, 20 MHz
WWVB	Boulder (US) Radio 60 kHz
WWVH	Kauai Hawaii (US) Radio 2.5, 5, 10, 15 MHz
CHU	Ottawa (Canada) Radio 3330, 7335, 14670 kHz
LORC	LORAN-C radionavigation system
OMEG	OMEGA radionavigation system
GPS	Global Positioning Service

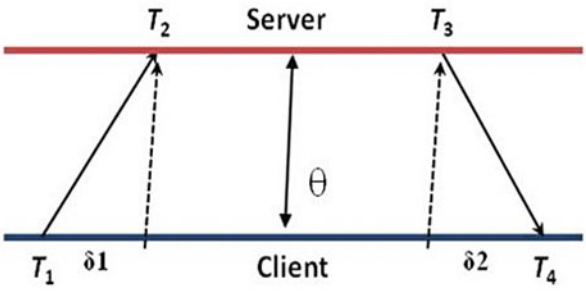
Fig. 10 Reference Identifier Codes (from RFC 4330)

The synchronization is correct when both the paths (i.e., from client to server and vice versa) followed between client and server are the same or encounter a delay of a similar value (Fig. 11).

But this is not the case in practical networking conditions. Hence, a systematic bias of half the difference between the forward travel time and backward travel times always exists (Matsakis et al. 2014).

NTP Security

NTP is linked to a slew of security issues. NTP servers are prone to man-in-the-middle attacks. As a result, cryptographic signing of NTP packets is required for authentication. Security of transport layer T-L-S, security extension of DNS, caching methods like DNS, gateway protocol like B-G-P, and other permanent login systems are some of the concerns with the NTP service. A Network Time Protocol (NTP) server can be damaged or degraded by a variety of NTP server misuse and abuse behaviors. NTP has been utilized in DDoS assaults. In this NTP server receives a small request spoofing the return address. This makes the NTP server send heavier data packets in response, which allows the hacker to significantly upsurge the number of packets transmitted to the destination. This issue is being targeted and it is expected that in upcoming versions, i.e., NTPv4, it will be taken care of.



$$\theta = \frac{1}{2}[(T_2 - T_1) + (T_3 - T_4)] \quad (1)$$

$$\delta = (T_4 - T_1) - (T_3 - T_2) \quad (2)$$

Fig. 11 Methodology of NTP

Modes of Operation

Based on the peers involved in the network, there are three classes in which the NTP time servers may operate: multicast, procedure-call, and symmetric.

Multicast class is used in high-speed LANs with a large number of workstations. The highest level of precision isn't sought in this case. The servers running in multicast mode regularly send NTP timestamp packets. Based on an expected delay of a few milliseconds, peers in client mode calculate the accurate time. Clients receive synchronization from the server in multicast mode, but the server does not receive any messages from the clients.

The procedure-call class is intended for use with file servers where great precision is required. The client sends an NTP message to the corresponding NTP server in the procedure-call. The server then takes the following steps:

- Swaps the addresses.
- Necessary timestamps are inserted.
- Performs the recalculations of checksum and optional authenticator, and sends the message (return back) promptly.

Symmetric active mode is intended for servers that operate at or near the top of the stratum. The servers that communicate with a huge number of peers regularly but on an irregular basis use the passive symmetry mode. These servers operate at low stratum levels.

A loosely linked connection is established when two symmetrically functioning servers exchange messages for the first time. Each server instantiates the NTP machine with persistent state variables; nonetheless, the protocol machine's primary goal is to preserve timestamps and related data, not to guarantee delivery. As each message is received in synchronic mode, the server updates the reachability status

every time, and if this status has not been updated for a long time, the server dissolves the association and restores state storage (Mills and Member, IEEE 1991).

NTP v4 Features

NTPv4 features quick synchronization at startup and after a network failure, which reduces network delay in time synchronization. The clock discipline algorithm in this version has been redesigned. It also allows clients to use the automatic server discovery feature, which aids in the discovery of live servers. It also includes updated and new reference clock drivers. It also supports new platforms and operating systems, allowing it to work with a wide range of new platforms. Public key cryptography is used. It works with the nano-kernel.

Advantages of NTP Synchronization

Event synchronization and server log file forensics both benefit from accurate timekeeping. The time on the Internet must be precise and consistent to support both of the responsibilities mentioned above. The Coordinated Universal Time is used to synchronize the clock on a server. A list of NTP's numerous benefits are as follows:

Accurate Time

The delay introduced due to communication over Internet is improved (for time) by the time server. This inaccuracy is generated due to the time taken by the synchronized signal to reach the user. If this inaccuracy is not removed, it may lead to bad results.

Reliable Time

To ensure accurate time delivery to all network clients, the time server uses multiple sources and a high-speed NTP processor. A local time server also ensures network reliability.

Security and Verifiable Audit Trail

For securing the operations over the network, secure timestamps play a major role. NTP servers provide security features like keypad lockout which gives controlled access to the server. Client authentication and independent network ports are the features considered for network deployment and are available with NTP servers with associated software.

Ease of Use

Network time servers are simple to set up and maintain. Their upkeep is simple. IPv4 and IPv6 are both supported by time servers (Amtest-TM Kft 2021).

Time Dissemination Using the Internet: Precision Time Protocol (PTP)

In 2002 PTP was established as IEEE 1588 standard. This is the standard established to provide precise (better than milliseconds as was provided in NTP) synchronization of networked clocks which is also required for control synchronization applications (Mills and Member, IEEE 1991). Ethernet is a competent technology for transporting data over a LAN and WAN. However, putting it into practice in mobile backhaul is a huge difficulty. Nanosecond accuracy is required for synchronizing base stations in mobile backhaul. Due to the high precision, the radio spectrum is not distributed into neighboring cells. As a result, if phase synchronization is not strictly enforced, most of the signals will get cancelled themselves in LTE-MIMO design. The IEEE 1588v2 standard was developed specifically for this purpose.

A computer network's clock is synchronized via the Precision Time Protocol. On a LAN, it achieves sub-microsecond clock accuracy, making it useful for measurement and control systems. Financial transactions, mobile phone tower transmission, and networks that are not connected to satellites but require accurate time can all benefit from PTP. For time distribution and clock synchronization, the architecture supported by PTP is a hierarchical master-slave one. Many slave/boundary clocks with single or multiple communication networks comprise a dissemination system of time in hierarchical architecture. To ensure master and slave device synchronization, timestamps of incoming and outgoing packets must be recorded (Fig. 12).

An ordinary clock has only one networked connection and serves as either a master or a slave for the synchronization reference. One section of a network is precisely synchronized with another section with the help of a boundary clock. It has

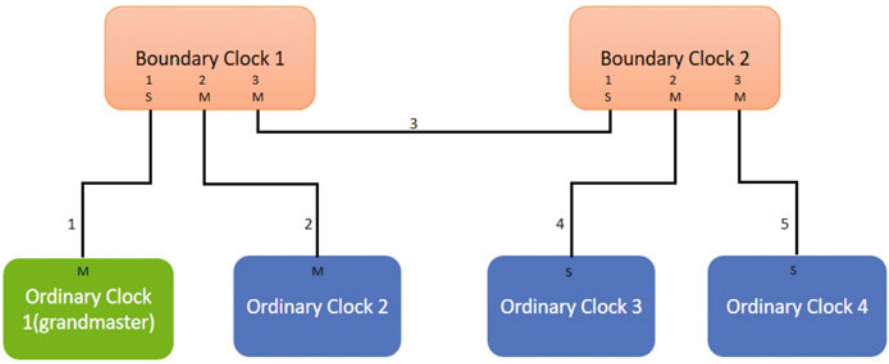


Fig. 12 Hierarchical structure of PTP time synchronization

many connections. A simple PTP network comprises ordinary clocks connected in the same network, and there are no boundary clocks. It's a networked clock system with a master-slave topology. An ordinary clock can only be a source or a receiver and has one network. There are five kinds of PTP clocks: transparent clocks include conventional clocks, boundary clocks, end-to-end transparent clocks, peer-to-peer transparent clocks, and management nodes (Laird 2012).

Ordinary clock is any equipment with single port which is capable of functioning as a master as well as slave clock.

A boundary clock is a multi-port device that can function as both a master and slave clock. A boundary clock is typically composed of a built-in slave clock that recovers a clock. This clock is then used to power the built-in master, which in turn supplies the clock to the next node.

End-to-end transparent clock is a multi-port device that is neither a master nor a slave clock, but rather serves as a bridge between the two. Its role is to forward and add correction. The bridge residence time is corrected by adding it to a correction field in the message's header.

Transparent peer-to-peer clock is a multi-port device that is neither a master nor a slave clock. It serves as a link between the slave and master clocks. Sync and follow-up messages are adjusted and forwarded by this device. Header of the message is updated with the delay incurred and this is how adjustments for delay are accomplished.

Management Node: It is a device that performs two main functions: configuration and monitoring of clocks. The round trip of PTP is shown in Fig. 13.

PTP selects the master clock/time source by employing the BMC algorithm. An ordinary clock computes the master-ordinary clock offset based on the relationship between the ordinary clock and the master clock:

$$\mathrm{del(t)} = \mathrm{Ord(t)} - \mathrm{Mas(t)} \tag{3}$$

Here, del(t) is the offset between the time of the slave clock and the time of the master clock, Ord(t) is the ordinary/slave clock time, and Mas is the time of the master clock (t). The master clock broadcasts the correct time to the slave clocks.

At time $t1$, the master clock sends a sync message to all slave clocks, kicking off the broadcast. When the slave clock receives the sync message, it records the internal time $t1'$. The master sends a follow-up message with the timestamp t1. To be accurately synchronized, the clocks must calculate the transit time of the sync message independently. Calculating the round-trip delay between each clock and its master can be used to determine the transit time. The slave clocks initiate a conversation with their master to determine the transit time k. Let us assume at an instance t2 ordinary/slave clock queries Delay_Req message to the master. This query is received at the master clock at instance $t2'$. In response to the slave clocks query, the master clock replies with a response message as "Delay_Resp." The corresponding timestamps generated are t1, $t1'$, t2, $t2'$. Let us assume that if the time taken by sync message is k and the difference of time between the master clock and slave clocks "del′" is uniform, in that case

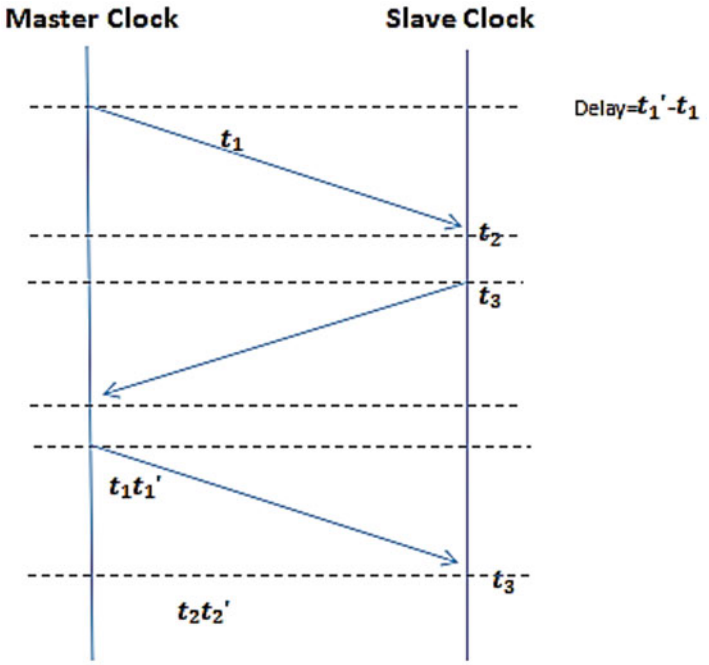


Fig. 13 Round trip of PTP

$$t1' - t1 = del' + k \tag{4}$$

$$t2' - t2 = -f' + k \tag{5}$$

By combining Eqs. (4) and (5), the findings are

$$f' = \frac{1}{2}(t1' - t1 - t2' + t2) \tag{6}$$

The clock can now synchronize itself with the master clock by calculating the offset. It has a grandmaster clock as its primary timing reference. Each of the system's Internet segments has its own synchronization master. The timestamp is transmitted via the network by this grandmaster clock. The timestamp provided by the grandmaster clock is then compared to all of the slave clocks. Then, based on the obtained timestamp, adjust their time.

PTP Messages: The PTP messages are used to synchronize the slave clock with the master clock. There are two kinds of messages in the PTP: event messages and general messages. Event messages are timed messages that generate an exact timestamp during transmission and reception. A timestamp is not required for a general message, but it may include timestamps for the event messages with which it is associated. The messages used in PTP are shown in Table 1.

Table 1 Communication massages used in PTP synchronization

Event messages	General messages
Sync message	Announce message
Delay_Req message	Follow_Up message
Pdelay_Req message	Delay_Resp message
Pdelay_Resp message	Pdelay_Resp_Follow_Up message
	Management
	Signaling

Best Master Clock Algorithm: The best master clock (BMC) algorithm chooses the best candidate clock by considering the clock properties.

Identifier – All devices have a specific identifier for the clock. This is usually derived from the MAC address of a device.

Quality – In PTP data field stratum computations are mentioned in its version1. However, both versions define the quality of the clock depending upon the methodology adopted for clock configuration, a stratum of the clock, and many other features. The two parameters, clock accuracy and clock class, specify clock's quality as per PTPv2. PTPv1 corresponds to IEEE1588:2002 and PTPv2 corresponds to IEEE1588:2008.

Precedence – Grandmaster precedence is assigned by using BMC. A Boolean inconstant is used to mention priority in PTPv1; however, two 8-bit fields are used in PTPv2 for the same.

Stability – In PTP, stability/variance is computed in comparison with the reference clock.

IEEE 1588–2008 utilizes hierarchical selection algorithm, as per the features (IEEE 2008) listed as:

First priority definition – Proactive precedence definition of the user is done with the help of prioritizing every clock in the PTP network. The lower the value, the better the priority.

Class – Every clock belongs to a particular class, with each class having its priority.

Accuracy – Expressed in nanoseconds, the offset value of clock and UTC is defined.

Variance – It defines the jitter value.

Second priority definition – In some situations when normal priority definition doesn't help, a backup order also termed as final priority is defined. Again as mentioned in first priority definition, smaller priority has higher weightage.

Unique identifier - There can be many clocks that are equally capable and match in every respect with another clock, but time selection of the clock is very difficult. That time as a selection criteria, MAC address is considered as the basis. When all other properties are similar, the selection based on MAC address is used as a tiebreaker.

PTP Message Header Format

Figure 14 depicts the PTP message's header, which is 34 bytes long. The messageType field indicates which type of message is contained in the message body, such as Sync, Delay Req, Delay Resp, and so on. The message length field indicates the total length of the PTP message, including the header, body, and any suffix (but excluding any padding) (Calnex Solutions Ltd 2009). The domain number field identifies the domain to which the PTP message belongs. Domains are logical groups of clocks that use the protocol to synchronize with one another but may or may not be synchronized with clocks in another domain. The flags field contains various flags that indicate status.

A correction field includes a correction value in nanoseconds for the residence time of a transparent clock as well as a path delay for peer-to-peer transparent clocks. The source or identity field specifies the port from which this message is originated. The sequence field contains a sequence number for each message type (with a few exceptions). The control field is historical, and its value is determined by the message type; it complies with the standard's version 1 specification. The messageType field is identical, but it has fewer options. The type of message influences the logMessageInterval field.

Announcement Message – When two clocks are in the same domain, the announcement message is used to show the capabilities of one clock to the other

Bits								Octets	Offset
7	6	5	4	3	2	1	0		
transportSpecific				messageType				1	0
Reserved				versionPTP				1	1
messageLength								2	2
domainNumber								1	4
Reserved								1	5
Flags								2	6
correctionField								8	8
Reserved								4	16
sorucePortIdentity								10	20
sequenceID								2	30
controlField								1	32
logMessageInterval								1	33

Fig. 14 PTP massage header format

Bits								Octets	Offset
7	6	5	4	3	2	1	0		
header (12,3)								34	0
originTimestamp								10	34
currentUtcOffset								2	44
Reserved								1	46
grandmasterPriority1								1	47
grandmasterClockQuality								4	48
grandmasterPriority2								1	52
grandmasterIdentity								8	53
stepsRemoved								2	61
timeSource								1	63

Fig. 15 PTP announces massage format

clocks. This enables the establishment of the master-slave hierarchy. Figure 15 depicts the format.

Sync Message

When the sync message was sent, a master clock sent a sync request message containing the master time. If the master clock is a two-step clock, the timestamp in the sync message is set to zero, and the actual sending timestamp is sent later in the associated follow-up message. Both types of PTP delay measurement mechanisms redirect synchronization messages (Fig. 16).

Delay_Req Message

The Delay Req message uses the same format as the sync message. A Delay Req message is sent by the slave clock, which included the slave time at the time the Delay Req message was sent. Delay Req messages are only sent in the delay request-response mechanism (Fig. 17).

Follow_Up Message

A master clock may send a follow-up message that contains the master time at the time the sync message was sent. When the master clock is a two-step clock, sync and

Bits								Octets	Offset
7	6	5	4	3	2	1	0		
header (13.3)								34	0
originTimestamp								10	34

Fig. 16 PTP sync message Format

Bits								Octets	Offset
7	6	5	4	3	2	1	0		
header (13.3)								34	0
originTimestamp								10	34

Fig. 17 PTP delay request message format

Bits								Octets	Offset
7	6	5	4	3	2	1	0		
header (13.3)								34	0
preciseOriginTimestamp								10	34

Fig. 18 PTP follow-up message format

follow-up messages are used. Both types of PTP delay measurement mechanisms send follow-up messages (Fig. 18).

Delay_Resp Message

When the master clock receives a Delay Req message, it sends a Delay Resp message containing the master time. Only Delay Resp messages are sent during the delay request-response process (Fig. 19).

The requestingPortIdentity field contains the source or identity field of the slave that requested this Delay Resp message.

Pdelay_Req Message

A "delay requester" peer-to-peer clock transmits the Pdelay_Req message and includes the time of "delay requester" peer-to-peer clock when the Pdelay_Req

Bits								Octets	Offset
7	6	5	4	3	2	1	0		
header (13.3)								34	0
receiveTimestamp								10	34
requestingPortIdentity								10	44

Fig. 19 PTP delay response message Format

Bits								Octets	Offset
7	6	5	4	3	2	1	0		
header (13.3)								34	0
originTimestamp								10	34
reserved								10	44

Fig. 20 PTP peer delay requester massage format

message was transmitted. Pdelay_Req messages are only sent in the peer delay mechanism. The bridge message length may vary due to asymmetry errors caused by networks with different transmit times. Therefore, the reserved field is used to make the message length the same as the Pdelayrespmessage (Fig. 20).

Pdelay_Resp Message

A "delay responder" peer-to-peer clock sends the PdelayResp message. It keeps track of the "delay responder" peer-to-peer clock time when the PdelayReq message was received. Only the peer delay mechanism sends PdelayResp messages. The source or identity field of the "delay requester" peer-to-peer clock that sent the Pdelay Request message is contained in the RequestingPortIdentity field in the PdelayReq message header (Fig. 21).

Pdelay_Resp_Follow_Up Message

A "delay responder" peer-to-peer clock can send the Play Resp Follow Up message, which includes the time of "delay responder" peer-to-peer clock when the PdelayResp message was delivered. It is used when the "delay responder" is a two-step clock with two stages – PdelayResp and Play Resp Follow Up message. Instead of a sending timestamp, this message can send a turnaround time: the time it takes from receiving a PdelayReq to receiving a Pdelay Resp. Only in the peer delay mechanism are PdelayResp Follow Up messages sent (Fig. 22).

Bits								Octets	Offset
7	6	5	4	3	2	1	0		
header (13.3)								34	0
receiveReceiptTimestamp								10	34
requestingPortIdentity								10	44

Fig. 21 PTP peer delay response message format

Bits								Octets	Offset
7	6	5	4	3	2	1	0		
header (13.3)								34	0
responseOriginTimestamp								10	34
requestingPortIdentity								10	44

Fig. 22 PTP peer delay responder follow-up massage format

Bits								Octets	Offset
7	6	5	4	3	2	1	0		
header (13.3)								34	0
targetPortIdentity								10	34
One or more TLVS								N	44

Fig. 23 PTP signaling message format

Signaling Message (Fig. 23)

The address of the message's target port/ports is stored in the targetPortIdentityfield.

TLV = Type Length Value Identifier

Management Message

PTP management messages are used to send information from a clock to a node manager and from a node manager to one or more clocks.

The targetPortIdentity field contains the address of this message's target port/the number of boundary clocks by which this message can be retransmitted, as specified in the starting message. Parameter BoundaryHops.

The boundaryHops field indicates how many boundary clock retransmissions are still available for this management message request or response. This field has the

Bits								Octets	Offset
7	6	5	4	3	2	1	0		
header (13.3)								34	0
targetPortIdentity								10	34
startingBoundaryHops								1	44
boundaryHops								1	45
Reserved				actionField				1	46
Reserved								1	47
managementTLV								M	48

Fig. 24 PTP management massage format1

Bits								Octets	TVL Offset
7	6	5	4	3	2	1	0		
tlvType								2	0
lengthField								2	2
managementID								2	4
dataField								N	6

Fig. 25 PTP management massage format2

same value as the startingBoundaryHops field for the first transmission from the issuing clock/node. The action field of this management message specifies the type of action that must be taken. The various types of commands are Get, Set, Response, Command, and Acknowledge. Figure 24 depicts the management fields.

Figure 25 shows how to set the type field to MANAGEMENT (0x0001). The length field specifies the length of the TLV. The format is 2 + N, where N is an even number. The management field specifies the class of management message. Initialize, Enable Port, and Disable Port are some examples.

Applications of PTP Time Transfer

There is a wide range of PTP applications in the digital and cyber world. PTP is a protocol for effective communication between devices utilized in various digital IT applications. It synchronizes PCs and digital devices such as ATMs, IoT devices, routers, and switches by providing timestamping. PTP is found in a variety of microprocessors, network interface cards, and protocol stacks. It's also utilized in

radio frequency instrumentation, aircraft monitoring instruments, and GPS-linked clocks (https://www.geeksforgeeks.org/precision-time-protocol-ptp/. Accessed 20 Apr 2022).

Advantages of PTP

PTP provides several advantages that are very beneficial for digital networks and devices (https://www.artel.com/blog/the-benefits-of-ptp/. Accessed 21 Apr 2022).

Multiple Media Formats – The PTP synchronization method can calculate the precise current phase of any media signal, even if the signals have different time bases. There is no need to have one flavor of PTP for US media networks and another for European media networks; both systems can operate on the same precision timing signal.

No Sync Overlay Network – Ethernet connections are used to disseminate the PTP clock signals, and they are also used for media signals, so an overlay network is not necessary to distribute synchronization. The complexity of cabling systems that connects the cameras to the control units can be greatly reduced.

Suitable for Any IP-Connected Device – There are various interface cards and modules are present in the market that provide timestamping over PTP. These cards are suitable for almost all devices and provide timestamping over PTP.

Highly Accurate for Media Applications – As PTP has an accuracy of better than 1 microsecond, it provides very precise time synchronization for video and audio applications.

Synchronization of Multiple Locations - The PTP grandmasters utilize GPS signals to get their clock synchronized; therefore, they can deliver accurate clock signals to multiple locations.

Time and Frequency Transfer: White Rabbit

CERN created White Rabbit, a time distribution system that can provide nanosecond precision. White Rabbit can distribute time with greater than 1-nanosecond accuracy and frequency with greater than 10-picosecond stability (Diaz et al. 2018). Optical filters and regular Ethernet connections are used to reach this level of accuracy. White Rabbit standards are achieved by enhancing many technologies, such as the Precision Time Protocol, which is utilized for clock synchronization, and the use of the L1 synchronization mechanism (inspired by the SyncE protocol).

PTP services were created to provide an alternate method of synchronization with great precision. However, PTP has challenges with precision over long distances, so the farther a network node is from a master clock, the less confident you may be that high accuracy will be maintained. The White Rabbit was created to address the issue of precision, allowing an Ethernet network to attain sub-nanosecond accuracy across a vast region. Using White Rabbit, networks can run tens of thousands of nodes all

on the same master clock. Telecommunications, financial services, smart grids, and air traffic control are among the commercial sectors that employ it for time synchronization because of its sub-nanosecond accuracy. The White Rabbit is flexible enough that it is compatible with many time transfer application that needs to work over some distance. It is increasingly used in time labs and commercial sectors to ensure sub-nanosecond accuracy (https://www.netnod.se/blog/what-white-rabbit-and-why-does-it-matter-high-accuracy-time-services, Accessed 21 Apr 2022).

The White Rabbit differs from other existing technology in several ways. Satellite-based time transfer systems such as GNSS and GPS all give accurate time at a reasonable cost. They are, nevertheless, quite sensitive. They are easily spoof-able and do not work indoors. PTP, on the other hand, offers a higher level of accuracy that is suitable for practically all current digital devices and applications. The chromatic dispersion of the fiber is compensated with the White Rabbit. Long-distance fiber connections between a master atomic clock and network nodes can be affected by this dispersion. The White Rabbit is designed to provide sub-nanosecond precision over vast distances. It is an open standard. It has open-source software and hardware designs.

NIST disseminated time using White Rabbit. NIST employs WR-LEN modules manufactured by Seven Solutions. A WR time link consists of a grandmaster module with a local 10 MHz and a pulse-per-second (PPS) reference and a slave module with a remote 10 MHz and a PPS, both connected by a single (bidirectional) fiber. The NIST connection's master and grandmaster modules communicate at 1310 nm, while the slave modules communicate at 1550 nm. The NIST time scale room is approximately 100 m away from the time transfer room, which houses the time transfer equipment. A WR grandmaster module in the time scale room receives data directly from the UTC (NIST) signal plane using a 10 MHz and a PPS signal. Through Fiber Link 1, the grandmaster module is connected to a slave module in the time transfer room that reproduces the UTC (NIST) PPS and 10 MHz signals (FL1). These signals will be used to power a TWSTFT Earth station and GPS (Global Positioning System) receiver for UTC (NIST) distribution. By connecting Fiber Link 2 (FL2) to a local slave module in the time scale room, the second fiber port on the remote WR module becomes the master. The outputs of this module are used to monitor and measure WR link stability. Figure 26 shows the setup at NIST.

An optical fiber network was successfully deployed to send precise time signals. A time signal with an 8-ns error was transmitted over an optical fiber link between Delft and Amsterdam. NIKHEF (National Institute for Nuclear and High-Energy Physics) in Amsterdam receives a timing signal from the VSL Time laboratory's atomic clock via an optical connection. A 137-kilometer optical line transports the NIKHEF timing signal back to VSL. In VSL, the received signal and the original signal are compared. After accounting for instrument delays on the Delft-Amsterdam-Amsterdam-Delft link, the received time signal reproduces the broadcast signal with an 8-nanosecond error, as shown in Fig. 27. White Rabbit Precision Time Protocol (WR-PTP) is used to send the time signal. This protocol employs the use of Ethernet. It is possible to measure and compensate for delay (caused by the glass fiber's limited light speeds) by routing Ethernet packets back and forth through the same fiber. By passing the Ethernet

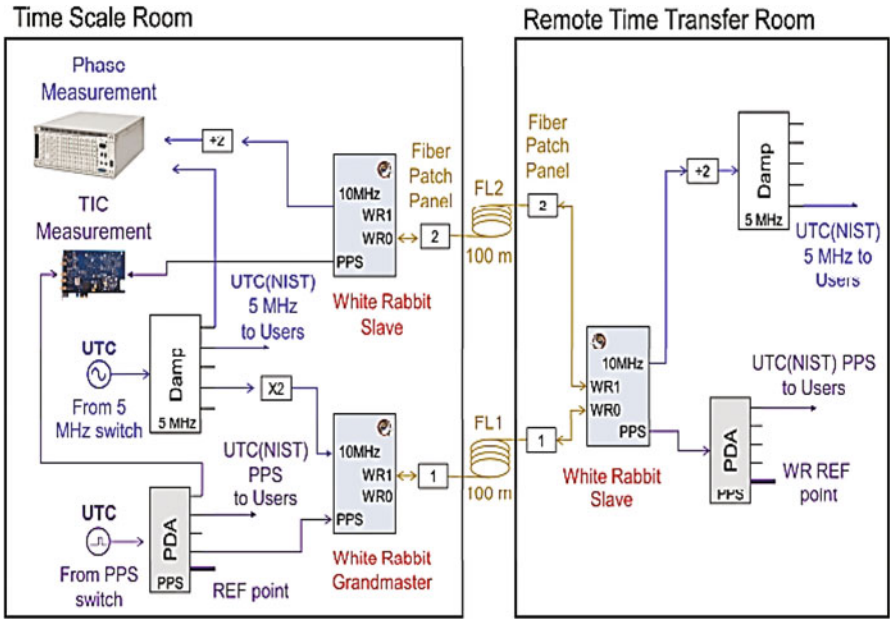


Fig. 26 Layout of the White Rabbit time transfer system used to disseminate time and frequency from the NIST time scale room to the remote location

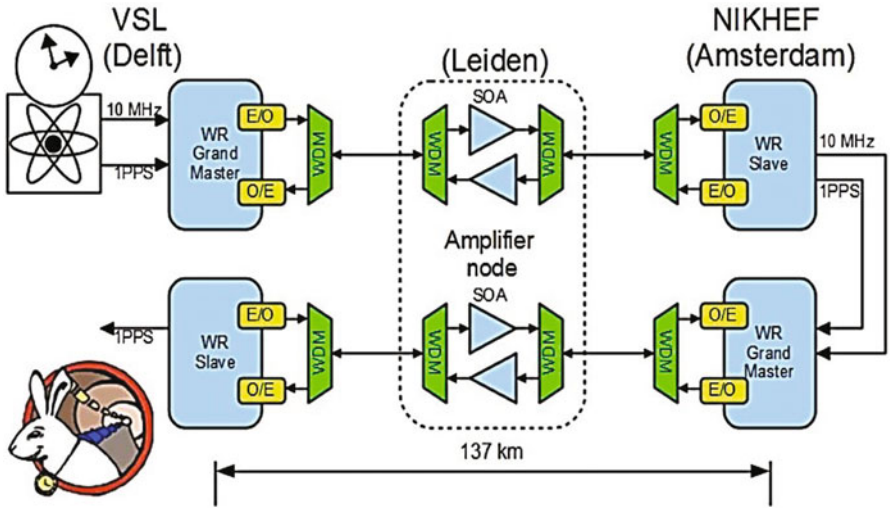


Fig. 27 WR-PTP Delft-Amsterdam-Amsterdam-Delft experiment link

packages back and forth across the same cable, the delay (caused by the glass fiber's limited light speed) can be monitored and adjusted. Because data in fiber typically flows in only one direction, delivering Ethernet packages in both directions over the same fiber is significant in standard telecommunications networks. The Eindhoven University of Technology created a special quasi-bidirectional amplifier to make the WR-PTP possible. They are now focusing on improving the connection calibration approach to further reduce the uncertainty of the time transfer. WR-PTP should be able to achieve sub-nanosecond uncertainty. This means that a timing signal can be disseminated with high precision (i.e., precision of an atomic clock) using the optical fiber, even in case of large distances. In the Netherlands, investigations have begun to roll out a larger network based on WR-PTP for precise time distribution (https://www.vsl.nl/en/about-vsl/news/accurate-time-dissemination-white-rabbit-precision-time-protocol, Accessed 21 Apr 2022).

Time Dissemination over Long-Wave Radio Waves

Long waves of the radio spectrum have a longer wavelength than the medium wave broadcasting band in the case of radio. Radio wavelengths greater than 1000 m are used in this band (frequencies up to 300 kHz). The highest limit is sometimes thought to be higher than 300 kHz, but not higher than 520 kHz. AM broadcasting is carried out across Asia, Africa, and Europe on frequencies spanning from 148.5 to 283.5 kHz. Long wave is a word used to describe a low-frequency broadcasting band that is part of the radio spectrum's frequency in the range of 30–300 kHz.

The radio waves can go beyond the horizon due to their enormous wavelength. Typically, they follow the earth's contour. Long waves propagate mostly via the ground (Seybold 2005). At lower frequencies, the ground absorption losses encountered are smaller compared to higher-frequency signals. The attenuation decreases as frequency lowers. The transmitter can send out low-frequency radio waves as farthest as 2000 km away.

The military employed these waves to communicate with submerged submarines since their penetration power allows them to go deep through saltwater for multitudes of depths. Low-frequency waves are reflected from the ionosphere, which is why they travel such long distances (the real mechanism is refraction), though this method, known as sky wave or "skip" propagation, is not as common as at higher frequencies. The ionospheric E or F layers reflect light. Sky wave signals can be detected up to 300 km (190 miles) away from the transmitting antenna (Melia and G3NYK 2009).

Long-wave receivers fitted inside radio-controlled clocks receive the time calibration signals. The delay encountered and corrected from transmitter to receiver is the same for any location in long-wave time dissemination as in this the signal always travels along the earth's surface. Long waves, unlike medium and short waves, are ground waves that travel along the earth's surface. Higher-frequency signals do not go beyond a few kilometers above the earth's surface, but they can travel as sky waves, bouncing off different levels of the ionosphere at different times

of the day. The time lag for each signal received can vary due to the various propagation pathways. The terrestrial distance of the receiver and transmitter along with the speed of the light through air defines the offset between transmitted and received long-wave signal.

As the time lag in signals received from the same signal station is roughly similar, a single constant shift forward from the time code in the signal may adjust for all long-wave signals received at any one location.

Long-wave transmitters are used to spread timing signals in numerous areas, including the USA, Japan, and Germany. For example, the German Telekom operates the DCF77 long-wave transmitter near Frankfurt (Mainflingen), Germany, at 50 01′ north and 09 00′ east, and the *Physikalisch-Technische Bundesanstalt* (PTB) supervises it. The time signals of the transmitter are guided by commercial Cs atomic clocks at the plant's location. A 77.5 kHz carrier frequency transmits time and frequency, with the amplitude modified by second markers.

The success of the radio-controlled clock is largely due to the features of long-wave emission. Long-wave signals have a significant advantage over time transmission through satellites: they penetrate structures and are unaffected by impediments such as trees or tower block buildings. They can be received without the use of external antennas, particularly with small ferrite antennas included in radio-controlled clocks. Compact radio-controlled clocks powered by batteries or solar cells can be created and operated without the need for a cable connection to an external antenna, thanks to the capability of the long wave. In contrast, receiving signals from the GPS navigation system, as well as the European equivalent Galileo in the future, necessitates the use of an external antenna with a clear view of the sky. If this antenna is deployed, time transmission will have undeniably lower uncertainty. As a result, satellite time transmission and long-wave time propagation will complement rather than replace one other.

By receiving a time signal from radio stations all across the world, an RCC may synchronize its local time to an atomic clock. The majority of radio-controlled clocks pick up signals from the low-frequency band of the radio spectrum (Depari et al. 2015), i.e., 40 to 80 kHz. Long-wave transmitters for time dissemination will continue to be useful in the future, despite the growing relevance of GPS. Prices are lower, power consumption is lower, and the receivers can be used indoors. The LF signal has a coverage larger than that of a radio frequency signal amidst low power, and it is also capable of performing in case of unavailability of line 0f sight with the following advantages:

Reliability on Every Front: Long-wave transmitter can transmit their signals in the kilohertz range. Long-wave radio is used not only for receiving international radio programs but also for transmitting national information.

Superior Performance: With various modes of communication channels available, it is quite difficult to choose the fastest and cheapest mode of communication. Long wave is the robust and reliable channel of communication to disseminate time worldwide.

Mechanical Robustness: The systems have a long-running life and have a low maintenance cost.

Most Economical: Radio transmission is a less expensive way to communicate with a large number of receivers without the need for costly infrastructure such as cables (IFR India 2021). Synchronizing phases in power plants, controlling traffic lights and guiding air traffic, and synchronization of computer and telecommunication networks are among the applications. Furthermore, the time transmitted over long wave is used for accurate phone billing, stock trading, and radio-controlled wristwatches.

Long radio waves are used by NIST to broadcast time and frequency as shown in Fig. 28. Millions of people across North America watch WWVB broadcasts. It transmits the time to digital clocks, clock radios, and wristwatches, among other devices. On a carrier frequency of 60 kHz, WWVB continually broadcasts digital time codes. The time codes are synced with the carrier frequency of 60 kHz and are continually disseminated in two forms at a rate of 1 bit/second utilizing pulse width modulation and phase modulation. The north and south antennas used by WWVB are identical. The north antenna was designed for WWVL broadcasting, whereas the south antenna was designed for WWVB 60 KHz broadcasting. The transmitted signal has a frequency uncertainty of less than one part in 10,000. If the route delay is taken into account, the WWVB may produce UTC with a 100-μs uncertainty. If correct receiving and averaging procedures are applied, the uncertainty can be lowered to the same level as the broadcast signal's uncertainty (https://www.nist.gov/pml/time-and-frequency-division/time-distribution/radio-station-wwvb, Accessed 20 Apr 2022).

Receiving the WWVB signal in ordinary household and official settings has some difficulties. Some regions receive high levels of on-frequency interference from the

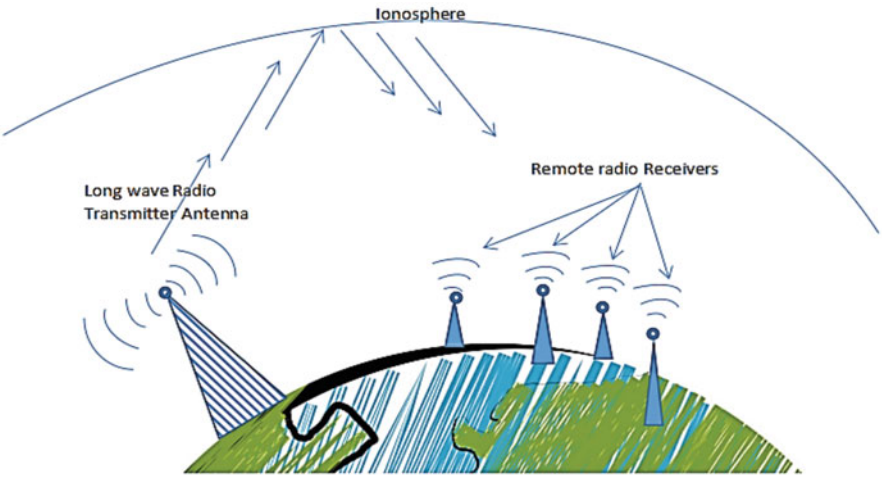


Fig. 28 The long-wave radio broadcast

MSF station in the UK. To address these issues, NIST developed a new protocol and modulation scheme (Eliezer et al. 2007). The enhancements made to the WWVB broadcast increased reception without harming the existing devices. Phase modulation adds several new capabilities while keeping the AM characteristics, allowing for backward compatibility. The new technique enhances timing resolution, but it was meant primarily to improve system robustness by lowering the signal to interference and noise ratio (SINR). The National Institute of Standards and Technology (NIST) began broadcasting a 30-second test signal every 8 min starting November 15, 2021, and every 48 min starting November 16, 2021, on WWV. The Ham Radio Citizen Science Investigation (HamSCI) and NIST have created this signal to assist in ionospheric research.

For weather forecasting, Japan employs the long-wave radio broadcast method. Radio wave broadcast forecasts, which give current and future conditions of radio wave propagation for telecommunications, are the source of Japanese space weather forecasts. In the year 1940, this service began. Because of advancements in rocket and satellite technologies, the significance of radio wave communications has diminished throughout time. NICT's space weather and environment informatics laboratory now provides Japan's operational weather forecast (Nagatsuma 2013).

In LORAN-C, demonstrated in the year 1957, permanent radio beacons transmitted signals to help in identifying the receiver's location. These are low-frequency radio broadcasts. It has a hyperbolic coverage area. It benefited by providing wider coverage with great accuracy which was previously incompatible qualities (Frank 1982). The cost of the equipment required to decipher the signals was a disadvantage. Hence, it was used only for defense purposes initially. The original LORAN was proposed by Alfred Lee Loomis in 1940.

Each LORAN station has a set of specialized equipment that produces precise time signals. The modulation/driving of the transmitting equipment is accomplished using these signals. To create 5 MHz and pulse per second signals, up to three commercial cesium atomic clocks are used. The timing equipment uses these signals to generate GRI-dependent transmitting instrument drive signals. LORAN-transmitter Cs has a max power of 100–4000 kW. The mast radiators in LORAN-C are typically 190–220 m long. The ground is kept away from these radiators. The output power of some LORAN-C (Frank 1983) transmitters reaches 1000 KW and greater. The mast radiators on these transmitters are 412 m tall.

The omnidirectional pattern is radiated by all LORAN-C antennas. As precise positioning is the base for the navigational calculations, using backup antennas is conceptually not possible in the stations of LORAN-C. A backup antenna's slightly altered physical location would result in lines of position that differed from the primary antenna's. LORAN-C has some limitations, such as electronic weather impacts and sunrise-sunset triggered ionospheric impacts. A ground wave following across the surface of the earth proves to be the utmost precise signal (Samaddar 1979). Due to the ionosphere, the ancillary radio wave is reflected on the earth's surface at night. This issue arises because several signals may arrive via various routes.

Results

PTB's telephone time dissemination service operates with three systems (TDS1, TDS2, and TDS3). AS TDS2 works as primary maximum calls are routed to TDS2, e.g., in January 2016, almost 1400 calls were received by TDS2, and TDS which is serving as backup has received approximately 400 calls.

Various results are shown in figures 29, 30, and 31.

Figure 32 shows the results displayed in the number of calls per day averaged over 1 month and their division to TDS1 and TDS2 for the period January 2016–December 2017. The service is mainly used in Germany by utility providers (gas transport sector), to get secure legal time information without the use of the Internet. The number of calls received by TDS2 and TDS1 from 2016 to 2018 is nearly about the same.

Almost 1800 calls per day were received almost every year from 2013 to 2017. The number of calls was averaged over a year. Figure 33 shows that there is a slight decrease in the calls in 2017.

NIST has received 2000 calls per day in telephone time-of-day service in 2019 and 2000 calls in 2020; however, it is reduced to half, i.e., 1000 calls per day in 2021. The service outage duration has also increased in these years as shown in the results below (Fig. 34):

The users can hear the audio portions of the WWV and WWVH broadcasts service using their phones. The delays in time dissemination vary when different mediums are used in communication. The time announcements are delayed by less than 30 ms when landlines are used. It gives stability (delay variation) of less than

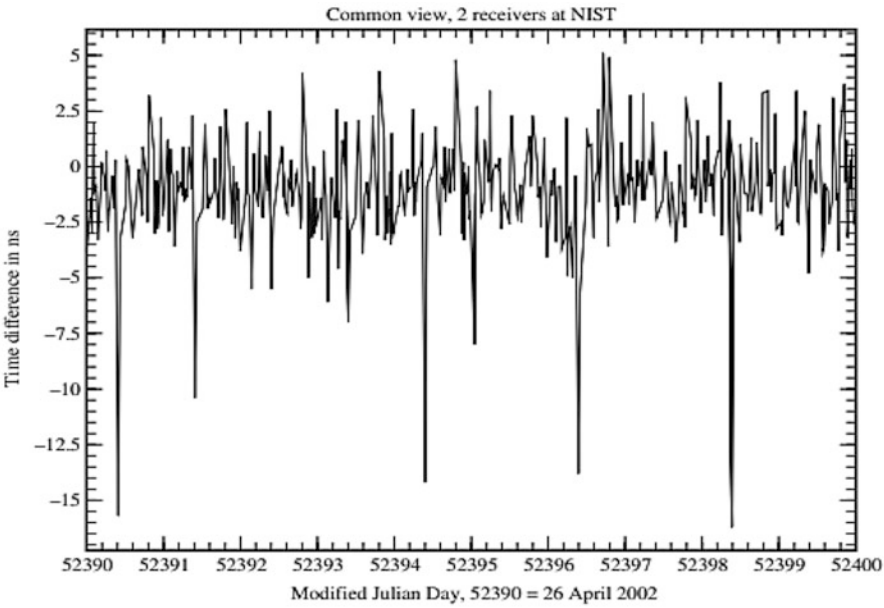


Fig. 29 Bidirectional communication-common view of GNSS receiver at NIST

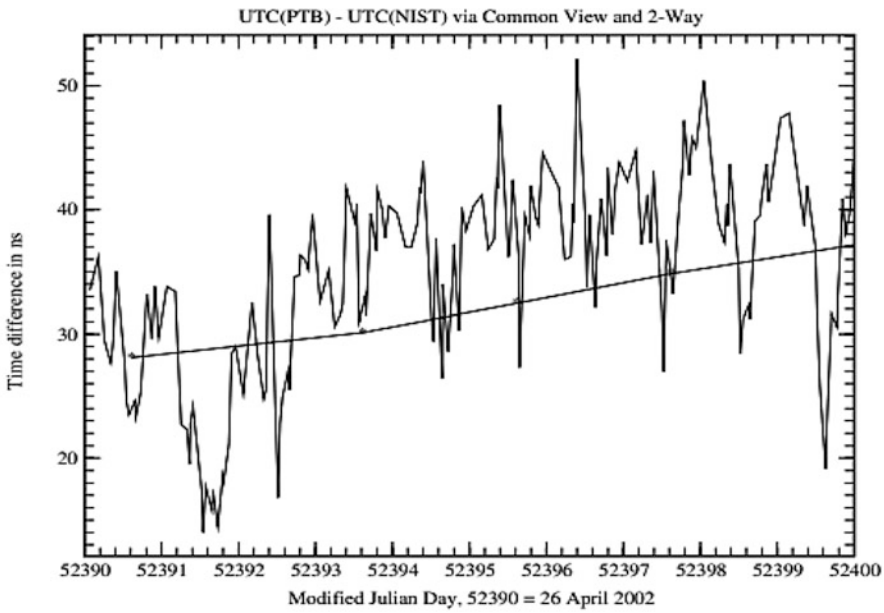


Fig. 30 Results of bidirectional time dissemination: common view and two-way communication offset between UTC (PTB) and UTC (NIST)

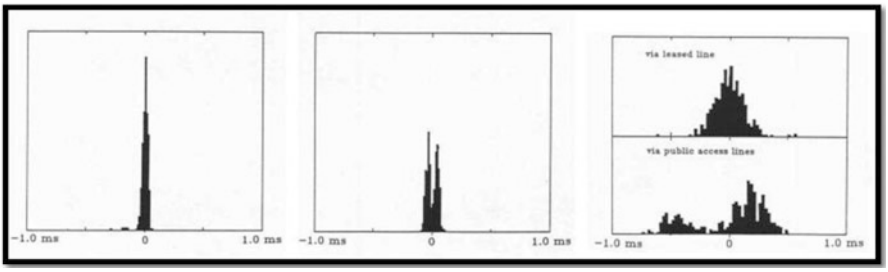


Fig. 31 Histogram depicting precision of NRC telephone time service over 500 calls

1 ms. The delays can be as much as 150 milliseconds in the case of mobile phones or voice over IP networks. The time is delayed by more than 250 ms if the connection is made via satellite phones.

In terms of data reliability received by the receiver, the NPLI Teleclock service has produced extremely good results. Figure 35 shows that by checking and comparing at least three sets of received data, more than 99% confidence in data reliability can be obtained.

Similarly, Teleclock service was able to synchronize the computer clock in real time using GPRS modem. In Fig. 36 results displayed show IST synchronization of computer clock using NPLI's Teleclock service.

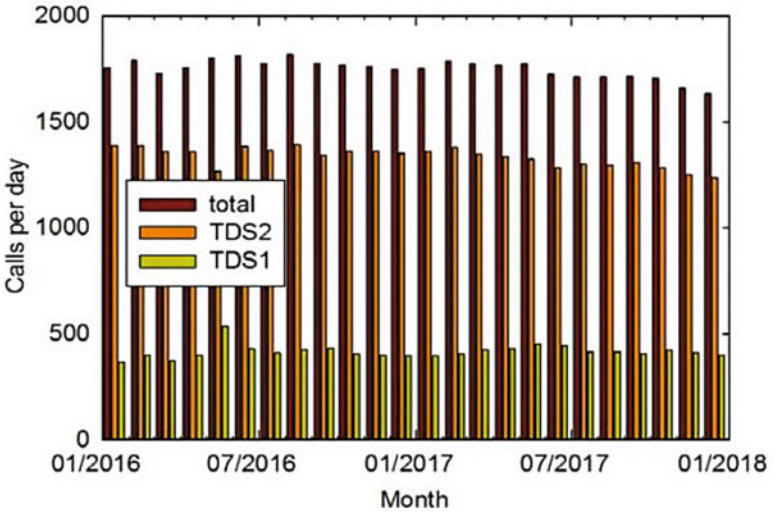


Fig. 32 Histogram demonstrating the usage of PTB's telephone time dissemination service over 2 years

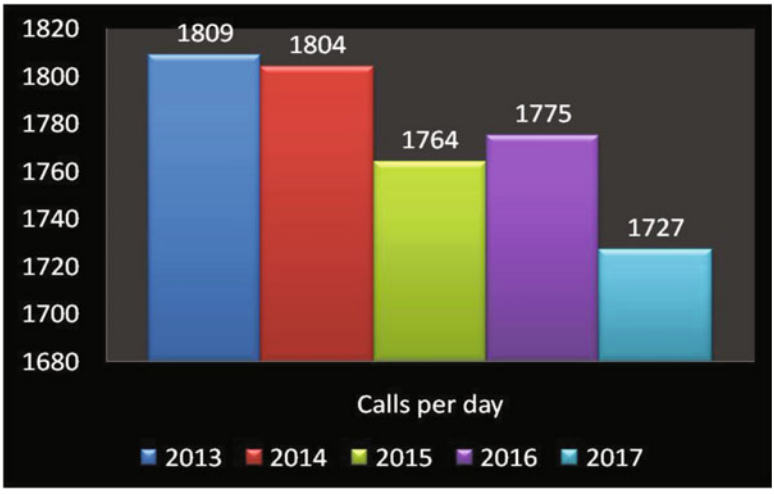


Fig. 33 Histogram depicting usage of telephone time service of PTB from 2013 to 2017

NTP services can be easily accessed by an inbuilt NTP client in Windows systems. The same is displayed in the results shown in Fig. 37.

NTP time dissemination service at NPLI is very popular, as the number of requests received every day is ranging from 12,000 (May 26, 2017) to 142,000 (June 10, 2017) as depicted in the results shown in Fig. 38.

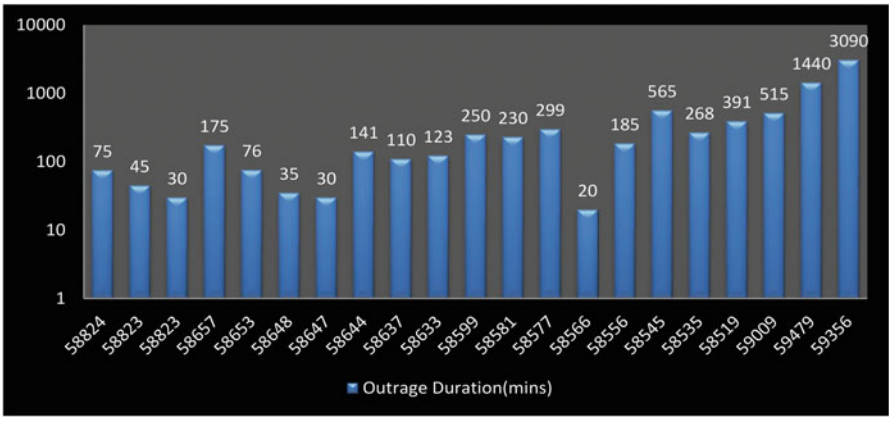


Fig. 34 Results displaying service in minutes of telephone time-of-days service of NIST, USA, from 2019 to 2021

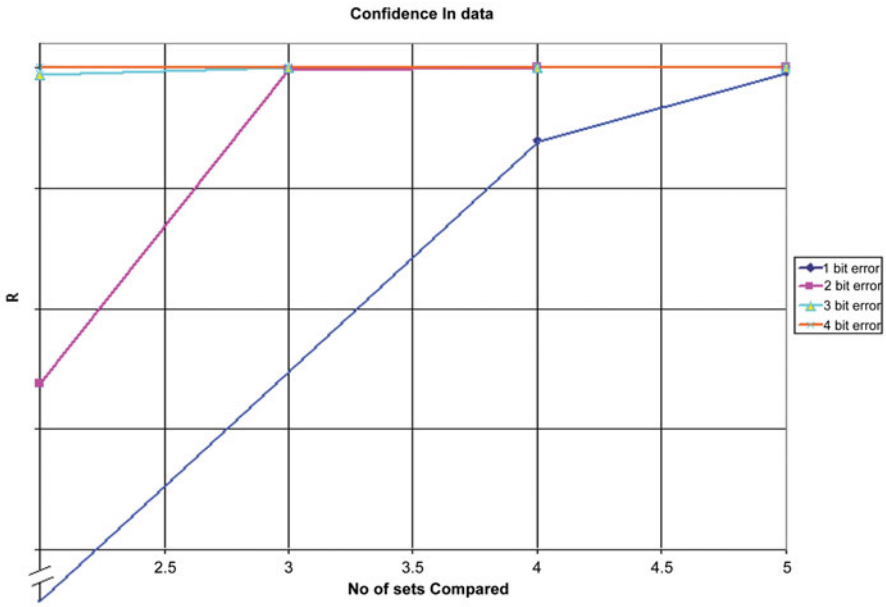


Fig. 35 Results depicting confidence in Teleclock-telephone time dissemination service by NPLI data

NTP services by NPLI are used extensively by many users and some of them are the major ones. For example, IP 110.172.171.194 who have accessed NTP service 249,813 times over a month and IP 203.101.52.55 used this service 3019 times in 1 month as shown in results displayed in Fig. 39. Various stratum-1 NTP servers are

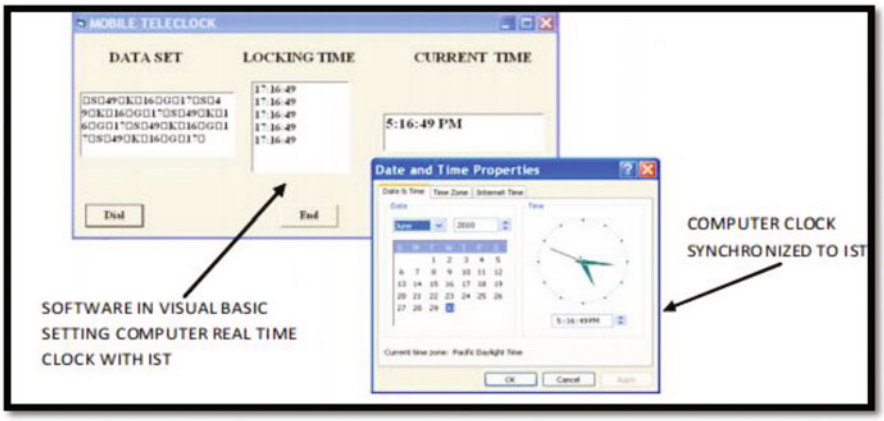


Fig. 36 Results of real-time synchronization of computer time using Teleclock service of NPLI

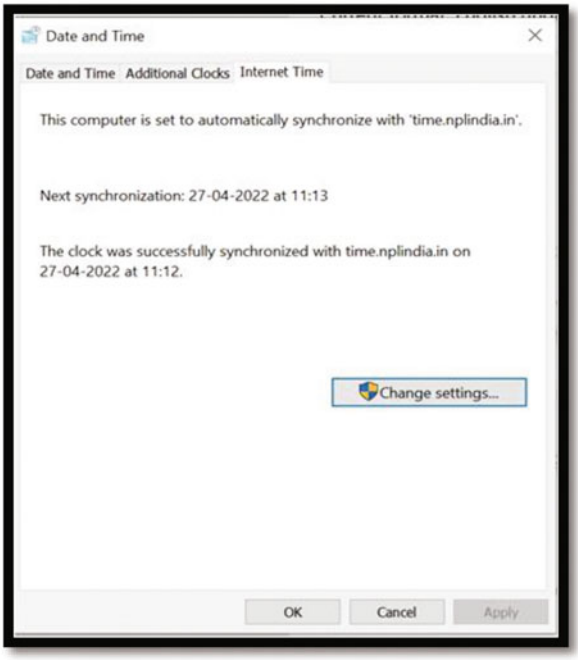


Fig. 37 Results of Windows-based PC synchronization using NTP time dissemination of NPLI

available worldwide now. The performance of geographically diversified NTP services over WAN in terms of RTT is shown in Fig. 40.

Figure 41 shows the NTP service of NPLI used worldwide. However, major portion, i.e., 81.8%, of this service is used by Indian users, 15.5% is used by the USA, some 3.3% are reserved users, and 1.1% are other users.

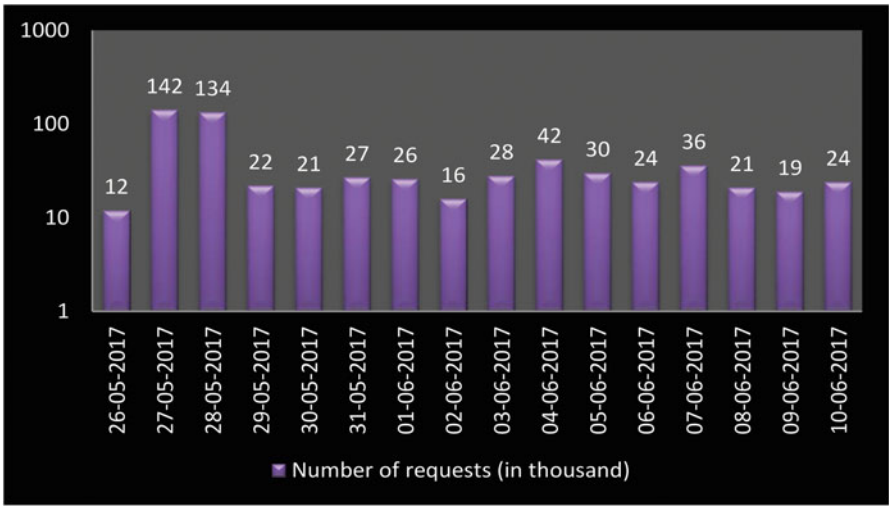


Fig. 38 Histogram depicting the popularity/usage of NTP service of NPLI

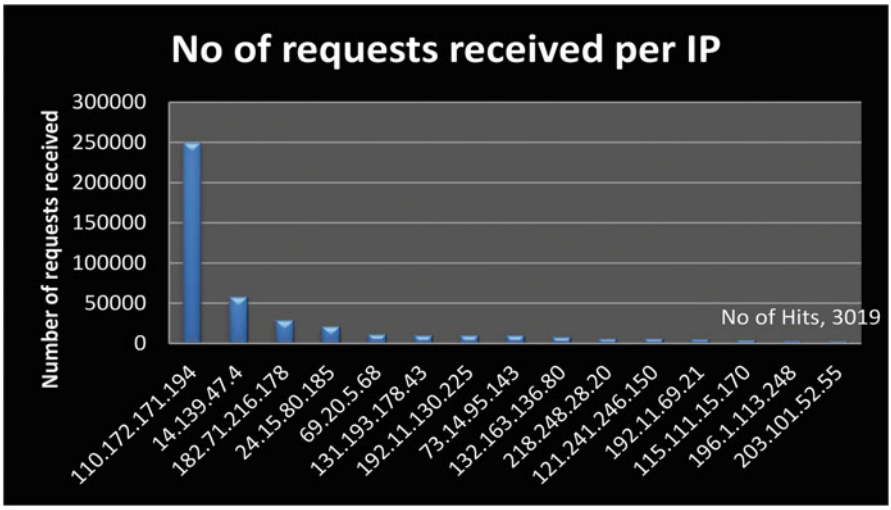


Fig. 39 Histogram depicting the usage of NTP time dissemination service of NPLI by various public IPs over a month

In the experiments, the following configuration is used (refer to Fig. 42). The laboratory experiments' purpose is to provide a better understanding of the field synchronization error behavior and to investigate the PTP clock and/or network parameter settings that can result in an improvement in overall synchronization performance. The experiments in this section are carried out using PTP slave and

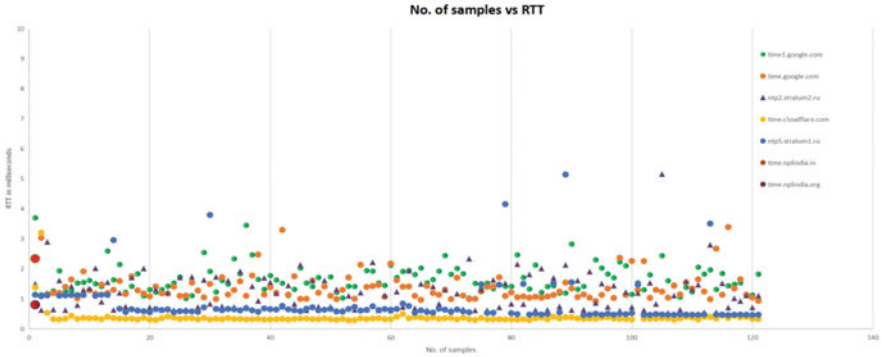


Fig. 40 Results of no. of samples versus RTT of global NTP servers

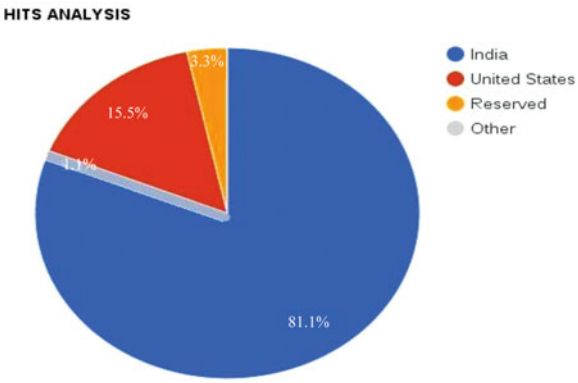


Fig. 41 Countrywide usage of NTP service of NPLI over 1 month

one-step grandmaster prototypes produced by the same manufacturer and referred to as SP and GMP, respectively.

Figure 43a and b show the WAN experimental results (b). The GMP and SP clocks are deployed in remote sites of Hydro Québec's WAN, and the SP clock is synchronized to the GMP via PTP messaging. The local reference clock is a GPS receiver that is synchronized to UTC and is installed in the same room as the SP. The offset between the GPS receiver's 1-PPS output signals and the SP clock corresponds to the synchronization error and is collected every second using an oscilloscope. We conduct two field trials, with the SP clock in Site H and Site R, respectively. For both trials, the rate of the PTP Sync (and Delay_Req/Delay_Resp) messages is set to 64 packets/second (Kassouf et al. 2013).

The low-frequency transmitter DCF77 by PTB, Germany, is by far the best-known service to disseminate legal time for Germany and can also be received in most of Central Europe. A standard frequency of 77.5 kHz is transmitted, as well as

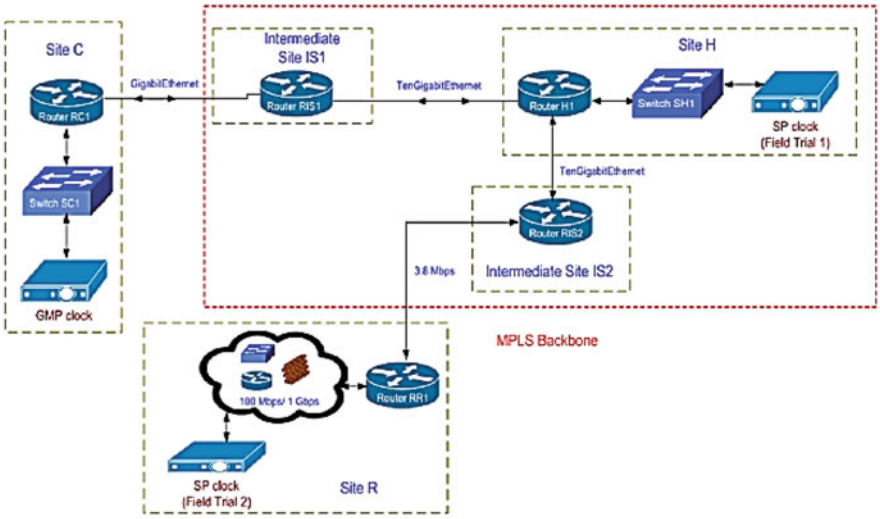


Fig. 42 Hydro Québec's WAN configurations for field trials

coded time information. The Allan deviation was used to calculate the output clock frequency stability, which is depicted in Fig. 44. On the ADC clock signal (ADC RATE), the Allan deviation estimation has been performed. The results indicate how much averaging is required to remove phase noise and achieve the desired stability. Comparable results can be obtained by connecting the DCF77 IN signal to a counter with a gate time that is compatible with the IIR LPF time constant (i.e., 100 ms). The stability of the reference (Levine 2002)-PPS clock signal has been assessed and is depicted in Fig. 45.

Lightning causes powerful noise in low-frequency time dissemination in the LF band or lower frequencies. Figure 46 depicts the effect of this in disseminating a 60 kHz signal. At 60 kHz, atmospheric noise levels are typically 100 dB higher than thermal noise if a theoretical antenna with a gain of 0 dB is assumed. The electrically and physically small receiving antennas used in commercial RCC products are very lossy, and the level of atmospheric noise received in them is less than the thermal noise in the receiver's front end. Lightning from nearby locations can still act as an impulsive interference source, corrupting part of a reception.

Similarly, electromagnetic interference is caused by noises produced by electronic devices and appliances. Figure 47 depicts the effect of this EMI on a WWVB signal at 60 kHz. The effects of EMI caused by office equipment such as computers are also depicted in Fig. 47.

The performance of the LORAN-C signal is evaluated with 100 kHz signal, 100 kW radiated power, sky wave delay of 100 kHz signal, and secondary and additional-phase-factor, smooth earth model for 100 kHz signal, as shown in Fig. 48.

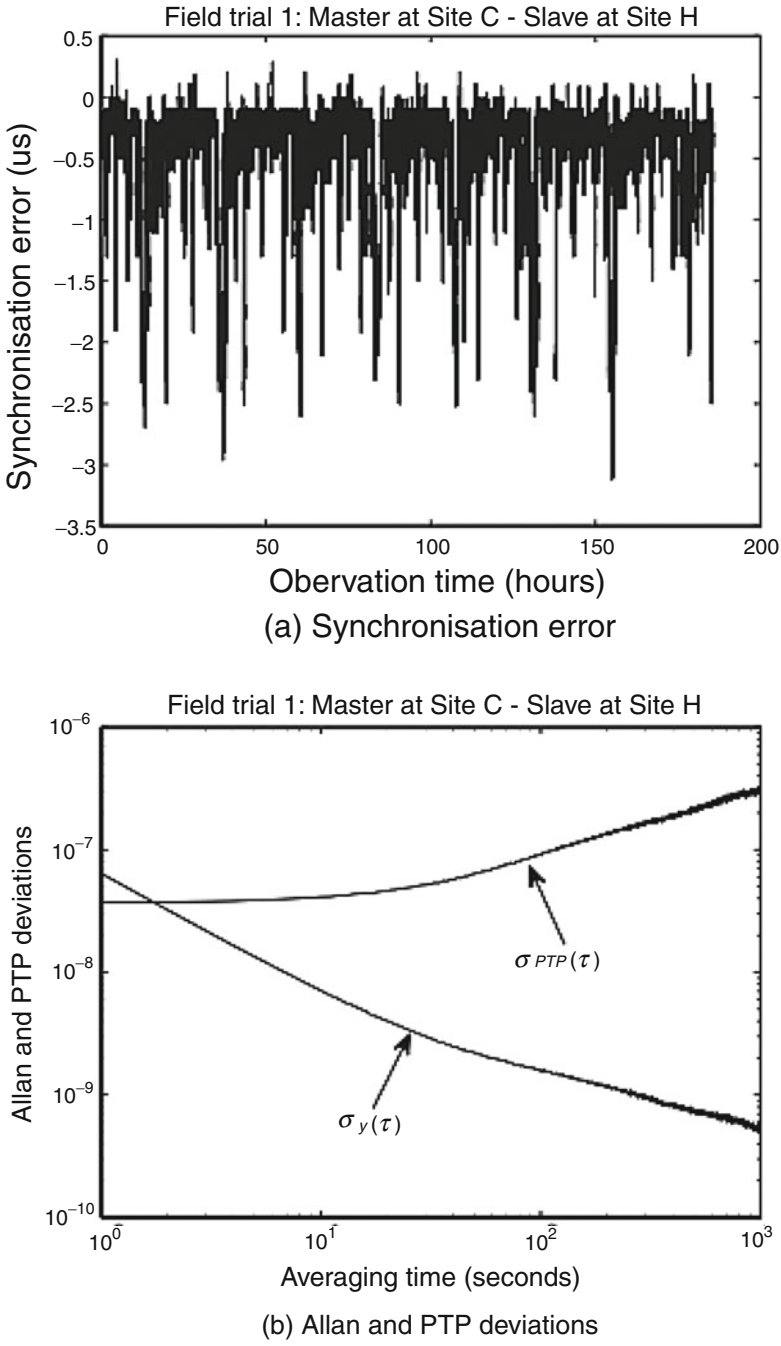


Fig. 43 Field trial 1 using a synchronization messages rate of 64 packets/sec

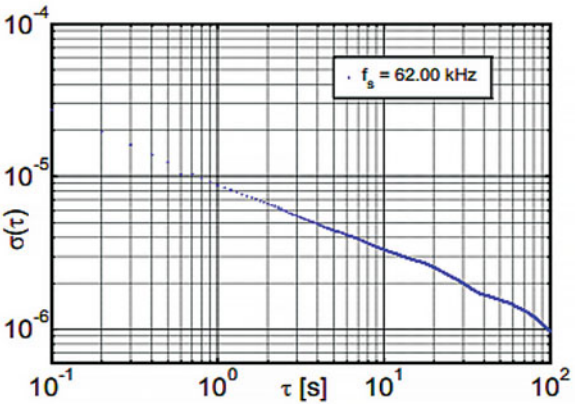


Fig. 44 Output clock frequency stability in LWR time dissemination

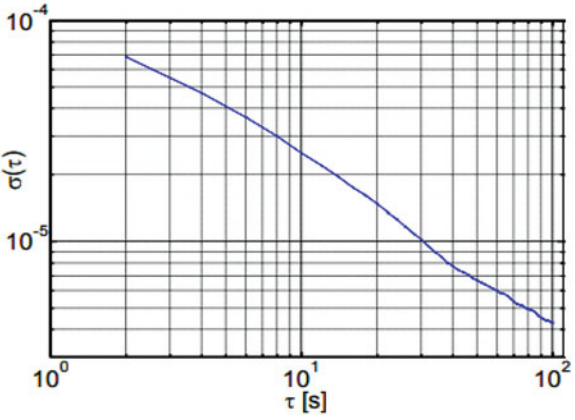


Fig. 45 Reference clock signal (1PPS) stability for LWR of NPL

Various time dissemination services run by various NMIs as per BIPM circular are given in Table 2. This depicts that NTP time dissemination is the most widely used technique.

Conclusion

Over the centuries, precise time dissemination techniques have grown in popularity. Regardless of their nature, all dissemination techniques were equally important. The telephone time dissemination technique was widely used at the time and

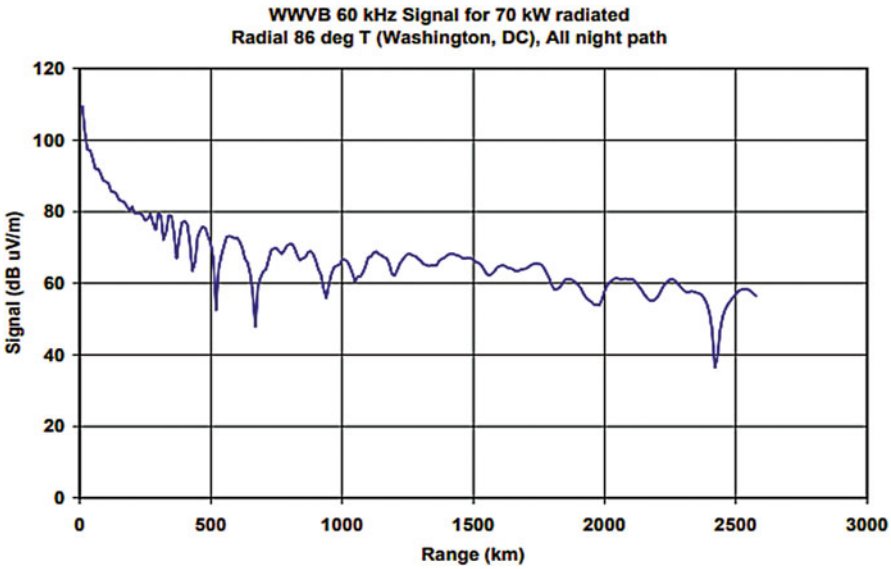


Fig. 46 Distance versus signal strength of WWVB

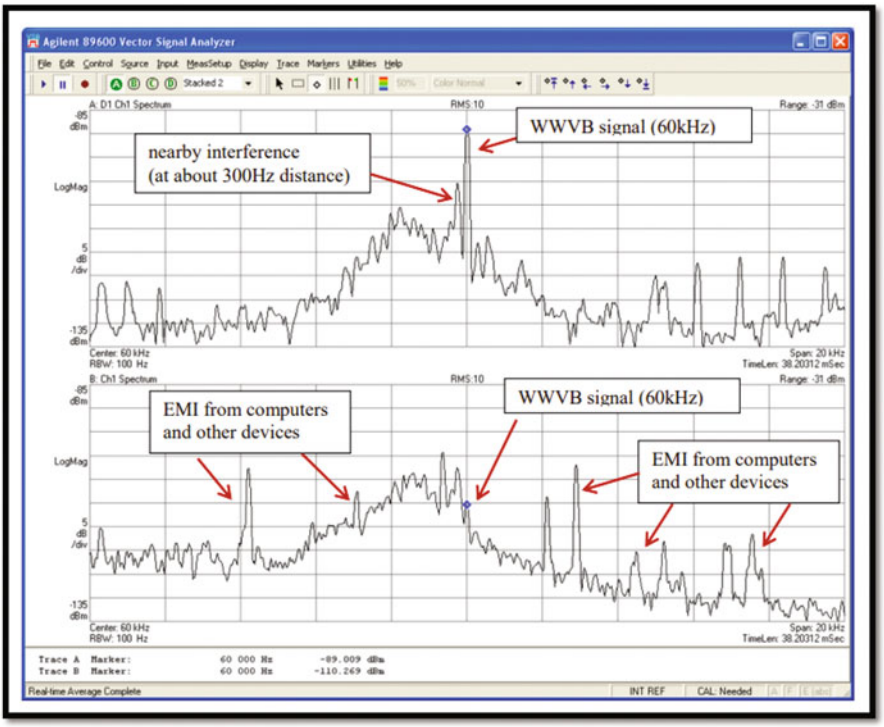


Fig. 47 Signal analysis of WWVB signal

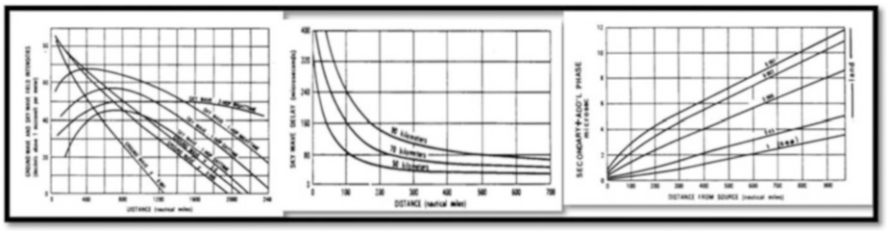


Fig. 48 Performance of LORAN-C 100 kHz signal strength, sky wave delay, and phase factor plotted against distance

made use of the existing telecommunications infrastructure. It was enormously popular because it served the common man's dissemination needs in society to a large extent. Some countries, such as the USA (NIST), Japan (NICT), and Germany (PTB), have upgraded their systems for telephone time dissemination from PSTN to packet-switched networks with infrastructure upgrades. Long-wave radio time dissemination has been critical and continues to be extremely useful for applications such as radio clocks, disaster warnings, etc. The accuracy of this dissemination technique is up to 1 s, but only five NMIs are running this service, including Japan, Thailand, Canada, Brazil, Germany, and the USA, because the installation cost for this service is enormous, and there are now many alternative dissemination techniques available that provide the same benefits. The Internet dominated the nineteenth and twentieth centuries, as did the time dissemination technique. Internet time dissemination via NTP is the most popular dissemination technique because it provides unrivaled simplicity and ease of operation with millisecond accuracy over WAN. Almost all NMIs support this method of dissemination. In some controlled environments, NTP could also support microsecond accuracy. NTP, on the other hand, is incapable of dealing with network asymmetry issues. This is handled very well in the Precision Time Protocol for time dissemination over the network. PTP has evolved from a control system protocol to a network time synchronization protocol over the years and has proven to be very efficient in the process. PTP provided synchronization accuracy on the order of a few nanoseconds. This accuracy is also improved in White Rabbit PTP to sub-nanosecond levels with more than 10-picosecond stability. GNSS time dissemination and two-way time dissemination methods are typically only useful for high-end applications, limiting their use. Even though most countries have not adopted optical cable-based infrastructure, the optical time transfer technique has made tremendous progress. One can conclude that time distribution over the network has been updated regularly to meet the most recent techniques and synchronization requirements. This field has been constantly expanding and improving, and it is expected to do so in the future.

Table 2 Summary of time dissemination services provided by various NMIs with their accuracy as per the latest Cir T

	Common dissemination services for masses *Accuracy: from a few seconds to milliseconds*			**Specialized time services** *Accuracy: from microseconds to picoseconds*		
Time and frequency service (Accuracy)	**Radio wave time transfer** (<1 s)	**Time transfer using NTP/web clock** (0.2–0 ms)	**Telephone time transfer/ speaking clock** (<1 s)	**Satellite time transfer/ TWSTFT** (10 ns)	**Time transfer over PTP** (0.2–10 μs)	**Time transfer over optical links** (6–7 ps)
NMIs	NIST, NIMT, NRC, ONRJ, PTB	NIST, NICT, NPL, NPLI, NMIJ, NIM, PTB, BIPM, INPL, INM, AOS, AUS, BelGIM, BEV, BoM, CENAM, CENAMEP, DMDM, EIM, GUM, HKO, ICE, ILNAS, IMBH, INACAL, INRIM, INTI, JV, KRISS, KZ, LNE-SYRTE, LRTE, LT, MASM, METAS, MIKES, MSL, NAO, NIMB, NIMT, NMISA, NMLS, NRC, NSC IM, NTSC, ONBA, ONRJ, ORB, RISE, ROA, SG, SIQ, SL, SMD, SNSU-BSN, TL, TP, UME, USNO, VMI, VNIIFTRI, VSL, NMIA	NIST, NICT, NPL, NPLI, NIM, PTB, BIPM, INM, AUS, BEV, CENAM, CENAMEP, GUM, HKO, INRIM, KRISS, MSL, NRC, ONRJ, ORB, RISE, ROA, TL, USNO, NMIA	NIST, NICT, NPL, NPLI, NMIJ, BIPM, INPL, AUS, IGNA, IMBH, USNO, NMIA, PTB, NIM	NICT, JV, MIKES, NSC IM,	NIST, NICT, NPL, NMIJ, BIPM, INRIM,NIM

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Time Transfer via GNSS

20

P. Banerjee

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Abstract

With the advent of satellite, accuracy of time transfer has seen significant improvements to nanoseconds and sub-nanoseconds. The major event in the direction is the implementation of GNSS being initiated by the USA and followed by Russia, European Union, China, Japan, and India. This chapter describes the various aspects of GNSS configuration as fielded by the respective countries. One may access time from a normal GNSS receiver with an accuracy of around 30 nanoseconds. But for timing applications, a special GNSS receiver is also available in the market. With these receivers, time accuracy may be improved. In order to do so, various techniques have been evolved. They are mainly common view, all-in-view, carrier-phase technique, P3, and precise point positioning (PPP) methods. These techniques basically identify sources of error and take due measures to eliminate or mitigate them leading to an accuracy of a nanosecond or better. All these aspects have been elaborated in this chapter.

Introduction

Time transfer verbally means giving access of correct time to users. Since 1335 time telling to the public started through chiming/tolling of church bell every hour in Europe. This was followed by the dropping of "time ball" at specified time by NPL, UK, since 1800 and by USNO, USA, since 1845. There was a real breakthrough of time transfer during the early twentieth century when synchronized time pulses were transmitted via electromagnetic wave. So, this technique of time transfer via electromagnetic wave was explored. Soon, the time dissemination via HF, VLF, OMEGA, and Loran was initiated in different parts of the world. Through these methods (few of them has been covered in Chap. 22), an accuracy of milliseconds, microsecond, and even to sub-microseconds could be achieved with some special effort.

After the advent of Earth artificial satellites in the mid-twentieth century, supported by rapid technological advancement and innovation, the technique of the navigation through satellites ushered in a new era of positioning, navigation, and time (PNT) service providing a scope of positioning and time transfer with unprecedented accuracy. Over the years GPS has matured to be fully effective in its utilization. Thus, the last three decades (prior to 2021) have seen explosive improvement in accuracy of PNT. The first operational satellite system for PNT service was ensued by the USA in the name of Global Positioning System very widely known as GPS with the launch of its first satellite on February 22, 1978. Though originally GPS was planned for the USA's armed forces around the globe, this service was soon globally released for a wide range of civil applications. The GPS ultimately became a blueprint for a whole family of global navigation satellite systems (generically called GNSS) subsequently established. It is beyond the scope of this chapter to elaborate on the full technical aspects of GNSS (Hofmann-Wellenhof et al. 2001; El-Rabbany 2002). But the brief description of the working function of GNSS is a necessary prelude for comprehensive discussion of time transfer via GNSS.

GNSS system

The past decade has seen a dramatic change in the landscape of satellite navigation. Economic, political, and strategic considerations have led to the buildup of independent satellite navigation systems by four nations or federations for global service. They are GPS by the USA, GLONASS by Russia, BeiDou by China, and Galileo by the European Union. Of course, there are other systems like NavIC by India and QZSS by Japan. But they are basically meant for regional application.

To explain the basic operation of GNSS briefly, let us concentrate on the description of configuration of each constellation, signal characteristics, types of clocks on board the satellite, and reference system for geodetic and time. This would be followed by operational principle of the GNSS scheme. Whatever has been discussed in this section has been summed up in Table 1 for a quick overview.

Configuration of Constellations

For designing the constellation, it was kept in mind that a minimum of four (to be explained later) satellites of every constellation should be visible everywhere all the time. To achieve this, all constellations have been configured almost in a similar way but necessarily with minor changes for obvious technical reasons. All the constellations for global operation satellites are designed in MEO orbit with zero eccentricity (circular).

The *GPS constellation* has been planned for 32 satellites in three orbital planes. Each is inclined to the equatorial plane by 56°, and the planes are offset from each other by l20° in longitude. Eight satellites are in a circular prograde l2-sidereal-hour orbit in each orbital plane.

GLONASS constellation has 6 orbital planes with a plan of 24 satellites, each plane inclined to equatorial plane by 64.8°. The orbital period is 11 h16 m.

The *Galileo satellite constellation* has circular orbits (with altitude of 23,222 km) and orbital inclination of 56°. It has three equally spaced orbital planes, each with eight operational satellites.

The global *BeiDou-3* system comprises 24 satellites in medium Earth orbit (MEO) distributed over 3 orbital planes. These MEO satellites take 12 h53 m to complete one revolution. The MEO satellites are, furthermore, complemented by three satellites in inclined geosynchronous orbit (IGSO) and three satellites in geostationary orbit (GEO) for regional service.

Clocks

At the heart of the GNSS, there are more than one (for the sake of redundancy) atomic clocks (see Chap. 21) onboard each satellite. GPS and GLONASS constellations have normally cesium and rubidium clocks, whereas Galileo and BeiDou constellations have opted for rubidium and hydrogen masers.

Table 1 Comparative description of GNSS

Parameters		GPS	GLONASS	Galileo	BeiDou
Launch of first satellite		February 22, 1978	12 October 1982	21 October 2011	30 March 2015
Constellation specifications (see Fig. 1.)	Orbit	MEO (circular)	MEO (circular)	MEO (circular)	MEO (circular), GEO, IGSO
	Altitude	20,180 km	19,130 km	23,222 km	21,150 km
	Inclination	56°	64.8°	56°	55°
	Eccentricity	0°	0°	0°	0°
	Orbit planes	6	3	3	3
	Period (h)	11h58m	11h16m	14h05m	12h53m
	Period (rev/sid)[a]	2/1	17/8	17/10	13/7
Number of satellites	Planned	24	24	30	27, 5, 3
	Current status	31 (sept. 01, 2021)	23 (sept. 19, 2021)	21(Sept.04. 2021)	26 (Sept.11,2021)
Signal characteristics	Frequency (MHz) (refer Fig. 2)	L1 1575.42 L2 1227.60 L5 1176.45	L1 1602.00 L2 1246.00 L3 1202.025	E1 1575.42 E5a 1176.45 E5b 1207.14 E6 1278.75	B1 1561.098 B2 1207.14 B3 1268.52
	Spread spectrum	CDMA	FDMA (L1/L2) CDMA(L3)	CDMA	CDMA
Clock on board	Clock type $	Cs, Rb	Cs, Rb	Rb, H	Rb, H
Reference	Geodetic reference	WGS-84	PE-90	GTRF	CGS2000
	Time reference	Steered to UTC (USNO)	Steered to UTC (SU)	Steered to a set of UTC (k) of the European Union	Steered to a set of Chinese UTC (k)

[a]A period of n/m revolutions (rev) per sidereal day (d/sid) results in a ground-track repeat track after m rotations of the Earth (approximately m ×23 h 56 m)
$ *Cs* cesium clock, *Rb* rubidium clock, and *H* hydrogen maser

Each constellation follows a different independent timescale.

GPS time is steered to UTC (USNO). It is continuous timescale implying that GPS time does not introduce leap seconds. GPS time was set to match UTC in January l980 but has since diverged. Thus [TAI – GPS time] = 19 s and is maintained.

Galileo time is steered to a set of UTC (k) of the European Union. Like GPS time, Galileo time follows the same relation with TAI as [TAI – Galileo time] = 19 s.

BeiDou time is steered to set of Chinese UTC (k). BeiDou time maintains constant relation with TAI as [TAI –BeiDou time] = 33 s.

Unlike these three constellations, *GLONASS time* is steered to UTC (SU) with leap seconds.

Unlike the year, month, and day format of the Gregorian calendar, the GNSS date is expressed as a week number and a second-of-week number.

For example, the week number in GPS data is transmitted as a ten-bit field in the navigation messages, and so it becomes zero again every l,024 weeks (l9.6 years). GPS week zero started at 00:00:00 UTC on January 6, l980, and the week number became zero again for the first time at 23:59:47 UTC on August 2 l, l999. To address this concern, the modernized GPS navigation messages use a l3-bit field, which only repeats every 8 l92 weeks (i.e., l57 years), and thus, the week number will not return to zero until near the year 2 l37.

Signal Characteristics

Each satellite system has specific signal characteristics, but each system attempts to be compatible with the others in order to prevent the interferences and attenuation between the signals.

Frequency bands used by GNSS according to ITU frequency allocations are confined to transmitting signals in only two frequency bands like the upper L-band (1559–1610 MHz) and the lower L-band (1164–1300 MHz) which have been established for worldwide satellite navigation use. S-band is also used in few cases of navigation system. Details of frequencies used by four constellations have been given in the Fig. 1 and also in Table 1.

Each satellite of GPS, BeiDou, and Galileo constellations broadcasts spread-spectrum-coded information in *code division multiple access (CDMA)* mode in which each satellite uses a distinctly different ranging code. But GLONASS uses a *frequency division multiple access (FDMA)* modulation, in which the individual satellites transmit at slightly different frequencies. However, similar to other GNSS, the L3 signal of GLONASS will make use of CDMA modulation which will be available with the next generation of K2 satellites expected by 2022.

Binary offset carrier (BOC) signals became a kind of standard in GNSS signal design besides using BPSK and QPSK. BOC signals are applied in GNSS in order to

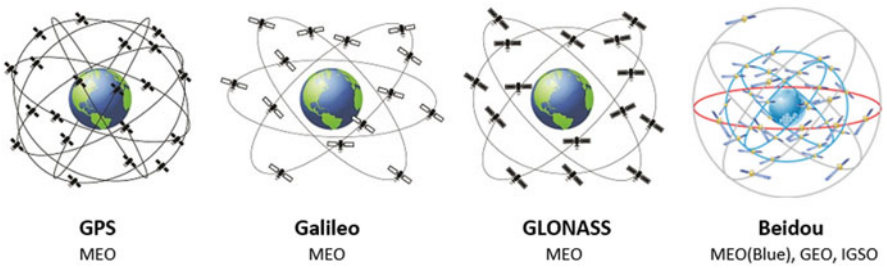

Fig. 1 Orbit configuration of four GNSS constellations

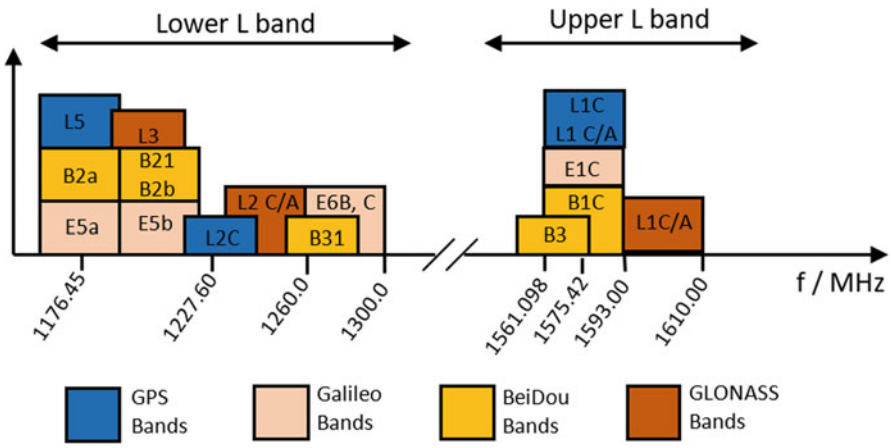

Fig. 2 Frequency bands used by global satellite navigation systems according to ITU frequency allocations

fulfil spectral separation requirements between different noninteroperable signals of different GNSS and to enhance synchronization performance. Galileo uses a composite binary offset carrier (CBOC) modulation for reduced ranging noise and improved multipath resistance.

Signals of all constellations are necessarily superimposed with navigation data, including satellite ephemeris data, an atmospheric and propagation message, and satellite clock bias information. These data are refreshed by the respective ground control station at a regular interval of time.

Relativistic Effects on Clocks in Orbit

While the satellites are in the orbit, the movements and height change relative to the Earth-centered inertial reference frame. According to the special and the general theories of relativity, the clocks on the satellites are thus affected by their speed, as well as by their gravitational potential, respectively. The fractional frequency offset ($\Delta f/f_0$) in GNSS clocks caused by these effects is covered by the following relation:

$$\frac{\Delta f}{f_0} = -\frac{1}{2}\left(\frac{v}{c}\right)^2 + \frac{\Delta U}{c^2} \tag{1}$$

where ΔU is the change in the gravitational potential at the satellite, v is the velocity of the satellite, and c is the velocity of light. Thus, the atomic clocks at the GNSS orbital altitude will tick more rapidly because they are in a weaker gravitational field than are atomic clocks on the Earth's surface. Atomic clocks moving at the orbital speeds of GNSS satellites will tick more slowly than stationary ground clocks. These combined effects lead to a frequency offset in clock.

Let us take the case of GPS constellation. The frequency of the base oscillator in a GPS satellite is l0.23 MHz, from which all other signals coherently are derived. Note that for GPS at the height of ~26,562 km and (v/c = 2.7×10^{-5}), the combined effect of offset comes out to be of 4.46×10^{-10}. Thus, to account for the relativity effect, the frequency of the base standard on board each satellite is set at l0.22999999543 MHz, instead of l0.23 MHz, prior to launch. Similar corrections in clock frequency are applied in the case of all other three constellations before launch.

Working Principle of GNSS

The basic principle of GNSS is based on measuring the range of the satellites at the receiving end. In fact, the receiver measures the travelling time of the signal with respect to its internal clock. By multiplying this time by the light velocity (c), we get the range between the satellite and the receiver. Code correlation technique is used to measure the time difference between the received and generated replica code. The measured range is different from the true range and so is called the pseudo-range (Pi) of the ith GNSS satellite which may be formulated as follows:

$$P_i = \rho_i - c\left(t_{rec} - t_{ref}\right) + c\left(t_i^{sat} - t_{ref}\right) + \left[d_{ion} + d_{trop} + d_{multi} + d_{rec}\right] \quad (2)$$

where:

ρ_i is the true range of ith satellite.
t_{ref} is the GNSS time.
t_{rec} is the time of receiver internal clock.
t^{sat}_i is clock time of ith satellite.
d_{ion} is delay error caused by ionosphere.
d_{trop} is delay error caused by troposphere.
d_{multi} is delay error caused by multipath effect.
d_{rec} is delay error caused by receiver noise.

Terms in parenthesis [] of RHS of (2) are error terms that equivalently become responsible for the deviation of the signal from the line-of-sight mode.

Please note the ($t^{sat}_i - t_{ref}$) is modelled as follows:

$$\left(t_i^{sat} - t_{ref}\right) = a0 + a1 \times \Delta t + a2 \times \Delta t^2 \quad (3)$$

The coefficient a0, a1, and a2 are transmitted in the navigation message. These coefficients are uploaded to the satellite and refreshed at a regular interval of time by the control center at the ground. Δt represents the time elapsed from the time of last update.

The true range may be expressed in cartesian coordinates as follows:

$$\rho_i = \sqrt{(x_{rec} - x_i)^2 + (y_{rec} - y)^2 (z_{rec} - z_i)^2} \quad (4)$$

Where:

x_{rec}, y_{rec}, and z_{rec} are the coordinates of the receiver.

x_i, y_i, and z_i are instantaneous coordinates of the ith satellite.

Please note that instantaneous satellite coordinates are determined from the satellite orbit parameters transmitted through navigation message.

So, if we carefully look at the Eq. (2), there are four unknowns to be determined (ignoring the error terms). This implies that for getting solution for four unknowns, four simultaneous measurements are required. *This necessitates the availability of minimum four satellites at a time.*

Limitation of GPS Accuracy

There is always a propagation delay when the time signal travels from the transmitting end to the receiver. The accuracy of time transfer largely depends on the degree of correctness of this delay assessment. GNSS signals are transmitted in L-band. At this frequency, signals travel from satellites to the receiver in a line-of-sight mode. But in practice, signals do deviate from the line of sight to some extent. So, it is important to identify the sources of the error of this deviation. Six major factors (Langley 1997) are normally identified as the major sources of error such as ephemeris error, satellite clock error, ionosphere error, tropospheric error, and errors caused by multipath and by receiver noise. They are explained in the following subsections.

Ephemeris and Satellite Clock Error

GNSS satellites broadcast navigation message that provides satellite orbit and clock parameters. Receivers use these messages in order to solve the equations. Through the navigation message, the coefficients (a0, a1, and a2) of clock prediction model are transmitted to the receiver. These are used to estimate the satellite clock offset with respect to the GPS system time. However, predictions may not be perfect causing some error in the estimate of the clock offset.

Similarly, parameters to define the current orbit status are sent to the receiver though navigation message. These are used to estimate the instantaneous position coordinates of the satellite. It may not be possible to correctly estimate the instantaneous position of the satellite due imperfection in prediction of ephemeris data.

Further, the parameters (clock coefficient and orbit parameters) that are sent through navigation message are continuously estimated at the control segment but only periodically uploaded to individual satellites. Once uploaded, the satellite position and clock prediction errors increase until the next upload refreshes them.

Ionospheric Error (d_{ion})

The ionosphere is the layer of the atmosphere between around 80 km and 600 km above the Earth. Electromagnetic wave signals are perturbed when travelling through the ionosphere.

The GPS navigation message includes values of the parameters of a simple ionospheric model known as the Klobuchar model. NeQuick-G, the ionospheric model is adopted by the Galileo system. A Klobuchar-like model is also used by the BeiDou system.

In GLONASS, there is no ionospheric model applied, and thus, there is no broadcast of any parameter. However, any of the GPS or Galileo ionospheric models could be used for GLONASS signals, by applying a correction factor given by their relative squared frequencies ratio.

Nevertheless, these models can give rise to some error.

Tropospheric Error (d_{trop})

The troposphere is the lowest layer of the Earth's atmosphere (10–15 km). Refraction in the troposphere has a dry component and a wet component. The only way to mitigate tropospheric effect is to use models. Tropospheric models depend on empirical models by considering all values of temperature, pressure, relative humidity, and mapping function. Examples of such models are the Hopfield and Saastamoinen models.

Multipath Error (d_{multi})

Multipath describes what happens when reflected GNSS signals arrive at a receiver's antenna along with one coming directly from the satellite. When a circularly polarized wave is reflected from a surface such as a wall or the ground, the sense of polarization is changed. The multipath affects the pseudo-range measurement. This effect can be mitigated by special antenna design, by processing the data, or by weighing the measurements according to the elevation angle.

Receiver Noise (d_{rec})

GNSS signals are very weak. Typically, less than 100 watts is transmitted from a distance of 20,000 km to 25,000 km. When received at the surface of the Earth, the signal strength may be as low as -160 dB W (1×10^{-16}) watts. Inherent noise within the receiver causes jitter in the signal.

Table 2 The standard error model: L1 C/A (the values are very rough estimates as indicative, so they vary slightly from one literature to another)

Source	Position error (1σ)	Time error (1σ)
Ephemeris	2.5 m	7.5 ns
Satellite clock	1.5 m	4.5 ns
Ionosphere dion	7 m	21 ns
Troposphere dtrop	0.7 m	2.1 ns
Multipath dmulti	1.2 m	3.6 ns
Receiver noise drec	1.5 m	4.5 ns
Total	7.85 m	24.0 ns

Combined Error

The dilution of precision (DOP) or geometric dilution of precision (GDOP) is a term used to describe the effect of the geometric strength of a satellite configuration on accuracy. The degradation of time and position solutions due to a suboptimal geometric arrangement of observable satellites is termed dilution of precision (DOP). The higher is the value of DOP, the lower is the accuracy.

Table 2 summarizes the values of the error contributed by individual sources. Time error has been calculated from the estimate of position error by multiplying it with the velocity of light. Let us note that 1 m range error corresponds to 3 ns time transfer error. Assuming that each component of errors is uncorrelated, the total error may be found by root of square of sum (RSS) of all these components. Of course, this total value needs to be multiplied by the DOP factor to take care of the effect of satellite geometry with respect to the receiver location. Refer to Table 2, where if we assume the TDOP to be 2, the time error amounts to about 48 ns.

However, one may achieve an accuracy time transfer *much better than this* by averaging over reasonably good number of observations.

Direct Access to Time from GNSS Receiver

All four GNSS constellations are by now fully operational. So, GNSS receivers capable of receiving signals from all constellations and generating solutions for PNT are already available in the market. All GNSS receivers not only show position coordinates but also have a hardware output of one pulse per second. With the help of these GNSS receivers, one can get direct access of time but with a limited accuracy to about 30 ns. An absolute calibration of the GNSS receiver delays is required for this technique.

A GNSS disciplined oscillator (GNSSDO) (Banerjee et al. 2007, 2008) can generate time and frequency signals by controlling its own oscillator with the precise time provided by the GNSS receiver. GNSSDO is already available in the market at a reasonable cost. For many applications requiring precise and stable frequencies, the expensive purchase of an atomic clock can be substituted by the use of GNSSDO.

Improving Accuracy of Time Transfer

It has been seen that the user can always find a time and position solution if a minimum of four satellites are simultaneously visible. But if one needs to know the time at a static location whose precise coordinates are known, then one can get the time by receiving the signal from a single satellite only. Special timing receiver that are designed to work on this principle are already available in the market. A few models for multichannel dual-frequency timing receiver are Septentrio PolaRx3eTR, MESIT GTR55, TTSx-EMDE Electronics, PikTime, and SPECTRACOM SecureSync (see Fig. 3.).

For the useful operation, such receivers should be configured in the way as shown in Fig. 4. The receiver should be fed with a second pulse (1PPS) and frequency of the receiver which is to be compared with the time of GPS of the receiver. It is important to determine precise coordinates of the antenna location in advance. Irrespective of the type of receiver, it is important to feed the values of coordinates to the receiver. These receivers have a time interval counter with high resolution (~20 ps) built into them. The output of this receiver is available in one common GNSS generic time transfer standard (CGGTTS) format as defined by the Consultative Committee for Time and Frequency (CCTF) (Defraigne and Petit 2015; Allan and Thomas 1994). The common output format has been devised for the convenience of exchange of data between the comparing clocks.

Fig. 3 Few special timing receivers that are commonly used by timing laboratories

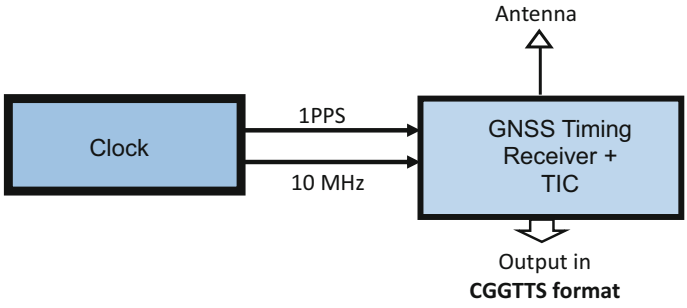

Fig. 4 Working configuration of multichannel dual-frequency riming receiver. Predetermined precise coordinates of the antenna location are to be fed to the receiver in advance

```
CGGTTS GPS/GLONASS DATA FORMAT VERSION = 02              +Eph+Ion
REV DATE = 1996-10-20
RCVR = SRC AOS TTS-4, SN:0131, 2009, HW:133.32, SW:2.39 2016/01/14
CH = 216 (GPS-L1C/L2C/L1P/L2P/L5P Galileo-E1/E5a Glonass-L1C/L2C/L1P/L2P SBAS)
IMS = IGS estimates
LAB = NPLI
X = +1243910.11 m (GPS,GLONASS)
Y = +5462559.47 m (GPS,GLONASS)
Z = +3038748.42 m (GPS,GLONASS)
FRAME = ITRF, PZ-90->ITRF Dx = 0.0 m, Dy = 0.0 m, Dz = 2.0 m, ds = 0.0, Rx = 0.0, Ry = 0.0, Rz = -0.000002
COMMENTS = NO COMMENTS           CREATED BY cggtts_cat.pl from header gmlit458.172
INT DLY = [ns] GPS: L1C:-14.96 L2C:-14.96 L1P:-14.96 L2P:-14.96 L5P:-14.96, GLO: L1C:-218.32 L2C:-218.32 L1P:-218.32 L2P:-218.32
CAB DLY = 142.20 ns (GPS),  142.20 ns (GLONASS)
REF DLY = 115.50 ns (1PPS DLY: 115.50 ns, phase corr: 0.00 ns)
CKSUM = 8B
SAT CL  MJD  STTIME TRKL ELV AZTH   REFSV      SRSV     REFSYS    SRSYS  DSG IOE MDTR SMDT MDIO SMDI MSIO SMSI ISG FR HC FRC CK
             hhmmss  s  .1dg .1dg    .1ns     .1ps/s     .1ns    .1ps/s .1ns     .1ns.1ps/s.1ns.1ps/s.1ns.1ps/s.1ns
 10 FF 58172  010 0  780 456 3096   -1631177    -38        -79    -83    5 901  112   -6   67   -3   42 9999 999  0  0 L1C BC
 32 FF 58172  010 0  780 396 2229    5313985     -7        -82    -21    9 901  126  -18   74   -9   33 9999 999  0  0 L1C BF
 14 FF 58172  010 0  780 169 2180     942954     17        -112    21   18 901  274 -102  116  -17   47 9999 999  0  0 L1C 0F
 20 FF 58172  010 0  780 226  707   -5015862    -15        -109   -24   19 901  207   36  103    9   55 9999 999  0  0 L1C E3
 15 FF 58172  010 0  780 215  433    3568339    -16        -143   -18    9 901  218   62  105   15   66 9999 999  0  0 L1C D8
 10 FF 58172  026 0  780 484 3194   -1631223    -47        -83    -92    5 901  107   -5   64   -3   39 9999 999  0  0 L1C D8
 32 FF 58172  026 0  780 461 2290    5313980    -14        -75    -28    8 901  111  -12   66   -7   30 9999 999  0  0 L1C A4
 14 FF 58172  026 0  780 232 2216     942982      2        -90      6   12 901  203  -54  102  -14   40 9999 999  0  0 L1C C0
```

Fig. 5 Sample of CGGTTS format of NPLI dated 02-23-2018 (MJD 58172)

In the sample of CGGTTS format shown in Fig. 5, tenth column "REFSYS" contains the offset "reference user clock – system timescale," i.e., the offset between the user clock and the GNSS time.

With the help of the special timing receiver, it has been possible to compare two distant clocks with much better accuracy. By using this receiver, it has been possible to improve the time transfer accuracy by eliminating or reducing the effect of the sources of error. Many techniques have been conceived to achieve this.

Common View (CV)

Common view technique started since the early 1980s. In this method, two comparing stations have to observe only those satellites that are in common view. Measurements with common view satellites are also to be made simultaneously. Let us discuss the common view between station A and B as comparing sites (see Fig. 6). We may write the following equation:

$$TA - TB = \frac{1}{N} \sum w_i \left[(G - t_A)_i - (G - t_B)_i \right] \tag{5}$$

TA and TB are time difference between the two clocks; $(G\text{-}t_n)_i$ is the observed time difference between the GNSS time (G) and the reference clock at that site (t_n, i.e., t_A or t_B), as determined from GNSS satellite i measurements at site n (e.g., $(G\text{-}t_n)$ shown in tenth column of Fig. 5).

The Bureau International des Poids et Mesures (BIPM) used to publish tracking schedules (see Table 3) for single-channel common view every 6 months so that observations between pairs of stations could be made simultaneously on the same satellite.

The GPS common view technique has been used for many years by the BIPM as one of its main techniques for international time comparisons.

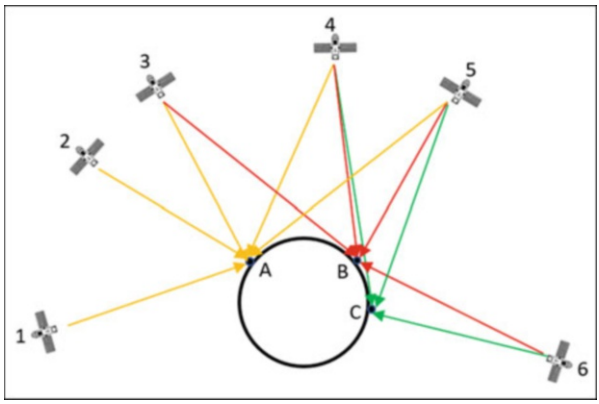

Fig. 6 Common view technique: satellites 3, 4, and 5 are in common view between station A and B, though satellite 1 and 2 are also visible from A and satellite 6 is also visible from B. No satellite is in common view between A and C

Table 3 Typical sample of GPS Schedule No. 49

India				
		Start		
Class	PRN	h	m	Connects
38	31	0	10	H
D4	31	0	26	H
A0	23	0	42	ME, EA, E
A4	20	0	58	SAF
AC	03	1	14	A
AD	03	1	30	A
AE	03	1	46	A, EA
AC	19	2	2	A
A0	13	2	18	ME, EA, E
F1	03	2	34	I
AD	19	2	50	A, EA
A0	25	3	6	ME, EA, E

The accuracy of common view GPS is better than for one way since the stability of common view is better, and also only a differential calibration is required between the two receiving stations. A differential calibration is made by carrying a portable GPS receiver between the two stations. This provides information on the difference between the delays of the receiving equipment at the two stations. An absolute calibration of the equipment delays at each station is not required. The accuracy of common view time transfers is typically in the 1 to 10 ns range.

Satellite clock errors are nearly equal in common view, so they are nearly cancelled out.

Orbit error is also attenuated but with increasing distances, the number of simultaneously observed satellites decreases, and the number of high-elevation satellites becomes low. But ionosphere and troposphere are different for different sites. Ionosphere error may be cancelled over a short baseline. So, the improvement

in ionosphere and troposphere errors depends on the baseline length. Ionosphere errors do not cancel over long baseline.

Local effects like multipath reflections and temperature sensitivity of receiver are not common in common view-mode and thus are not cancelled.

GPS CV had been used for the calculation of UTC at the BIPM until 2006. It is still in use, particularly for short baselines, as it is simple to use (Matsakis et al. 2014; Defraigne et al. 2007).

One main disadvantage of common view is that few of all visible satellites at the respective ends may be in common view. For example, though satellite 1 and 2 are also visible from A and satellite 6 is also visible from B in Fig. 6, they are not in common view between A and B. Further, the number of satellites in common visibility reduces with the increase of baseline distance. Thus, all visible satellites are not fully utilized. Common view cannot be applied for two widely separated stations like station A and station C as shown in Fig. 6.

Concept of All in View (AV)

Considering that CV has the constraint of using all visible satellites, All in view (AV) (Petit and Jiang 2008) has been proposed where all satellites in view are used to compare the local clock to a reference timescale through the satellite clocks at the two comparing stations. Difference between averages of the timing solutions of two station is subtracted to get the final result as shown in the following equation.

At each station using all visible satellites, one gets the difference between the combined results of two sites:

$$TA - TB = \frac{1}{N}\sum w_i(G - t_A)_i - \frac{1}{M}\sum w_i(G - t_B)_i \tag{6}$$

N is the number of observed satellites at station A and M is number of satellites at station B.

AV eliminates the constraint of simultaneous observations, so that two observing stations can be linked whatever their distance.

Improving Accuracy of AV through IGS Network

The errors from satellite clock estimate and the ephemeris estimate do not cancel in AV as they do at least partially in CV technique. To improve the accuracy of AV method, some of the IGS products are utilized.

Conceived in the early 1990s, the International GNSS Service (IGS) is a self-governed, voluntary federation of more than 300 contributing organizations from more than 100 countries around the world (see Fig. 7.). IGS provides satellite orbit and clock products (see Table 4) and other resources with the highest possible precision and accuracy. The IGS products are provided openly for the benefit of all scientific, educational, and commercial users. Though the service was started for GPS, in course of time, all GNSS constellations are covered under this umbrella.

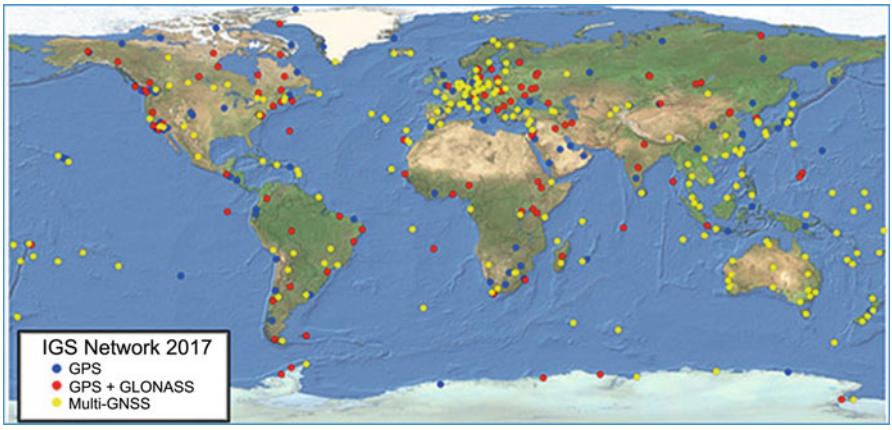

Fig. 7 IGS network. (Source: https://www.gpsworld.com/the-international-gnss-service-25-years-on-the-path-to-multi-gnss/)

Table 4 Specifications of IGS products

Type	Parameter	Accuracy	Latency
Broadcast (navigation message)	Orbits	~100 cm	Real time
	Sat. clocks	~5 ns RMS ~ 2.5 ns S dev	
Ultra-rapid	Orbits	~5 cm	Real time
	Sat. clocks	~3 ns RMS ~ 1.5 ns S dev	
Rapid	Orbits	~2.5 cm	17–41 hours
	Sat. clocks	~75 ps RMS ~ 25 ps S dev	
Final	Orbits	~2.5 cm	12–18 days

Using IGS rapid products, the uncertainties due to satellite orbits and clock average get significantly reduced.

The availability of IGS products made it possible to use the AV method instead of the CV method, resulting in improved performance of the time links independent of the distance. With wide availability of multichannel receivers, AV was implemented in place of CV since Sept. 2006 for all UTC links.

Concept of P3

The traditional GPS common view technique uses single-channel C/A code for computation in which the ionospheric delay can be reduced by an ionosphere model (as discussed in section on "Common View"). Over the years, most GPS receivers have become dual frequency. Keeping this in mind, better ionosphere removal can be achieved in modern receivers by using the fact of the frequency dependence of the ionosphere delay.

P1 and P2 are the pseudo-range measurements on two frequencies (say, L1 and L2 for GPS). These measurements are available in modern geodetic receiver.

Noting that ionospheric error $I_i = \frac{40.3}{f_i^2}$ (ignoring higher-order terms), one may prove that:

$$P_3 = \frac{f_{L1}^2}{f_{L1}^2 - f_{L2}^2} \times P_1 - \frac{f_{L2}^2}{f_{L1}^2 - f_{L2}^2} \times P_2$$

For GPS, where L1 is 1575.42 MHz and L2 is 1227.6 MHz:

$$P_3 \approx 2.55 \times P_1 - 1.55 \times P_2$$

P3 is the ionosphere-free code measurements.

Instead of using standard model, this method can be used to minimize the ionospheric error significantly.

Figure 8 clearly shows the improvement in P3 technique. In fact, P3 code removes 99.9% of the ionosphere delays, and Klobuchar models remove only 60%. By linear combination of L1 and L2 frequency, corrected range is more noisy by a factor of 3 as shown below:

$$Factor \approx \sqrt{2.55^2 + 1.55^2} \approx 3$$

With the help of GPS P3, sub-nanosecond statistical uncertainty for the time transfer could be achieved.

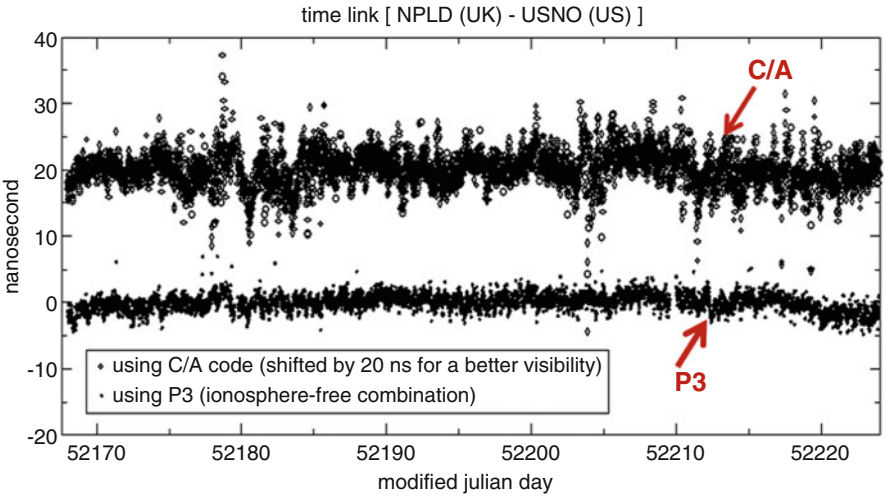

Fig. 8 The improvement in jitter in P3 over C/A code. (Source: Defraigne et al. 2007)

Concept of Carrier Phase

GNSS signals offer two types of observables. The pseudo-range code observable is one type, and the carrier phase that it modulates is the second. The pseudo-range's code length is about 1 ms, implying a wavelength of 300 meter; the carrier has a wavelength of around 20 cm (for GPS L1 carrier of 1.575 GHz has wavelength of 19 cm). The wavelength of carrier of GNSS signal is much shorter than code length. The resolution available from these carrier signals depends on the signal to noise and is typically around 2 mm, approximately 1% of its wavelength. So, the carrier-phase measurement noise and multipath error are about 100 times lower than the noise of the code measurements.

For this reason, the potentiality of GNSS carrier phases for time and frequency transfer is studied and demonstrated by different researchers (Ray and Senior 2003). The modern geodetic GNSS receiver measures the carrier phase and also code range.

So, the phase difference between two receivers p and q is as follows:

$$\varphi_p^i - \varphi_q^i = -f\left(\tau_p^i - \tau_q^i + \Delta t_{pq}\right)$$

$$+iono(p) - iono(q) + tropo(p) - tropo(q) + N_p^i - N_q^i$$

Δt_{pq} is the clock offset between two receivers p and q. Due to the cycle ambiguities, it is not possible to determine the absolute value of Δt_{pq}.The carrier-phase observable can thus be used for very precise determination frequency transfer by the determination of the rate of change of Δtpq (Fig. 9).

A combined analysis using both carrier phase and code observations is necessary to get the absolute value of the clock offset, with a precision limited by the code noise level (depending on the receiver type, but smaller than 1 nanosecond).

Concept of PPP

Code-only techniques like CV and AV have the limitations in performance due to multipath effects and errors in the tropospheric model and code noise. The effects of these error sources can be fairly minimized by phase measurements from geodetic-type receivers (Ray and Senior 2003; Kouba and Heroux 2001).

Precise point positioning (PPP) for time transfer is the technique that uses GNSS dual-frequency carrier phase and code measurements to get a better accuracy. The satellite orbit and clock data from IGS network solutions are used in addition to find the solutions for timing and positioning.

To facilitate the functioning of PPP time transfer, several online software services, using packages like Atomium, NRCAN, JPL, and Bernese, have been developed and are available for use. For performing PPP, one requires a dual-

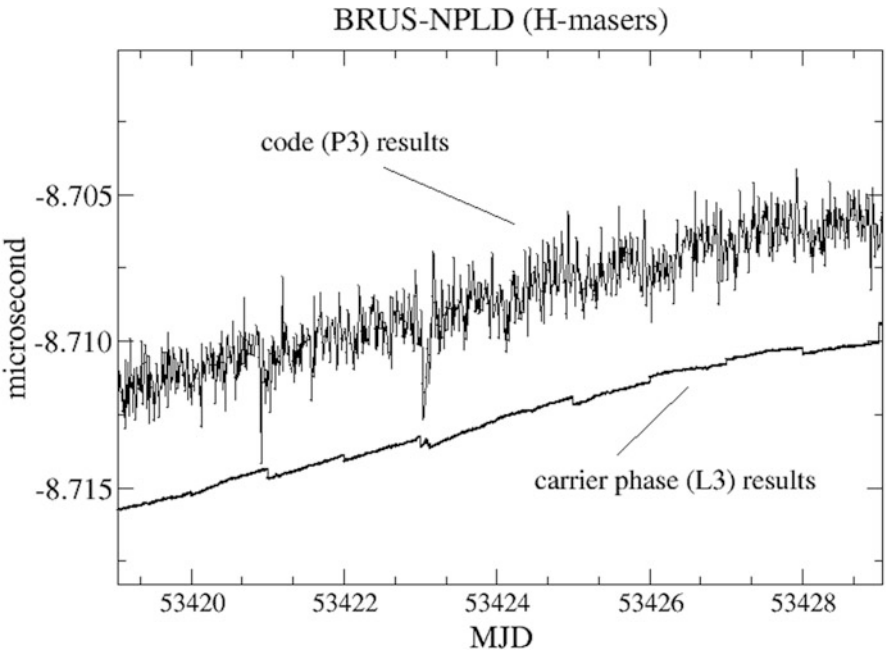

Fig. 9 Comparison of P3 and carrier phase through Δt_{pq} between labs. (Source: Defraigne et al. 2007)

frequency geodetic receiver which can generate both C/A code pseudo-range and carrier-phase observations, typically in RINEX format. These data are to be fed to any of the available software packages that subsequently generate the final output. This includes the time difference between a local clock and the timescale maintained by IGS with the precision of the carrier phase and the accuracy of the code. PPP technique has the potentiality of achieving time comparison with an accuracy of 500 ns. BIPM regularly publishes data of IGRT-UTC (k) through using PPP. NPLI is one of the laboratories that contribute to PPP link.

Comparison of GNSS Techniques

Table 5 includes the conservative estimate of the performance of different time transfer methods through GNNS system. It has been shown that GNSS can easily give 30–20 ns accuracy of time transfer, but by the improved method like PPP technique, GNSS promises to push accuracy to as good as 500 ps. There is a Two Way Satellite Time Transfer (TWSTFT) (see Chap. 24) technique which has capability a shade better that by GNSS techniques.

Table 5 Tabulation of GNSS time transfer methods describing the relative performance

Method	Uncertainty	Remarks
Direct access GNSS	50 ns–20 ns	Ephemeris and ionosphere error dominate
CA code CV/AV	1–1.5 ns	Ionosphere and troposphere do not cancel for long baseline. IGS product should be accessed for AV
GNSS P3	700 ps	Ionosphere error minimized
GNSS PPP	500 ps	Better uncertainty but requires significant post-processing, IGS data for orbits

NavIC on Timing Applications

Navigation with Indian Constellation (NavIC) or Indian Regional Navigation Satellite System (IRNSS) is a constellation of seven geosynchronous (GSO) satellites with coverage over the Indian landmass and in the region extending to 1500 km beyond the Indian geopolitical boundary. Three of the NavIC satellites are in GEO (32.5°E, 83°E, and 129.5°E), while four of the satellites are in two inclined geosynchronous orbits (IGSO) having inclination of 29 degrees each. While one of the IGSO planes crosses the equator at 55°E longitude, the other crosses at 111.75° E longitude. NavIC system has been realized by the Indian Space Research Organization (ISRO) and has been operational since 2018. It broadcasts NavIC system time which is realized at IRNSS Network Timing facility (IRNWT) and is steered to UTC (NPLI). NavIC, thus, provides reliable and precise dissemination of UTC (NPLI) and UTC.

Already many GNSS receivers that are available in the market have the provision to receive signals from NavIC constellation in addition to other GNSS constellations. Potentiality of NavIC service for precise timing applications through these receivers is being demonstrated through experiments and publications of the outcome of these efforts (Bhardwajan et al. 2021; Santra et al. 2022).

Concluding Remark

Combining multi-constellation GNSS can increase the number of available satellites and thus can improve the redundancy of the receiver clock parameters. Many laboratories contributing to UTC operate multi-GNSS receivers, allowing the computation of time links using not only GPS but also the other GNSS such as GLONASS, Galileo, and BeiDou, which have all reached operational status.

Various methods to combine the data from all four GNSS constellations have been studied (Verhasselt and Defraigne 2019) to improve the time transfer performance with respect to any single-GNSS solution. The optimal use of satellites of all GNSS constellations supplemented by the regional constellation is still being explored so that maximum benefit can be accrued out of such a good number of available satellites.

Cross-References

- ▶ Atomic Frequency Standards
- ▶ Precise Time and Frequency Transfer: Techniques
- ▶ Precise Time Transfer Techniques: Part I

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Precise Time and Frequency Transfer: Techniques

21

Two Way Satellite Time and Frequency Transfer

Huang-Tien Lin

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Abstract

TWSTFT (Two Way Satellite Time and Frequency Transfer) technology was developed over decades and has become one of the most precise and accurate techniques for comparison of the frequency standards located at remote sites. Since 1999, TWSTFT has been used in TAI (International Atomic Time) generation. More than two-thirds of TAI clocks and almost all the primary frequency standards are transferred using TWSTFT. The advantages of TWSTFT are its small calibration uncertainty and its long-term link stability. To further improve the time transfer precision and stability, several calibration methods have been

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developed and the possible instability-inducing sources were investigated. Although many instability sources of TWSTFT were studied, we still have not identified all sources since the observed instabilities in real TWSTFT links are larger than expected and the causes of diurnal effect (Diurnal effect is the daily variations, with peak-to-peak amplitude could be up to 2 ns in extreme cases, which periodically appear in the TWSTFT measurement results) still cannot be explained well. Further investigations are necessary. For the future development, the SDR (software-defined receiver)-based TWSTFT methods show competitive performance with the GPS PPP and much less diurnal effect than the conventional TWSTFT. Some of them are very promising methods for the next generation of TWSTFT. This chapter gives a review and brief introductions of the above topics.

Introduction

The concept of Two Way Satellite Time and Frequency Transfer (TWSTFT or TW) was proposed about 50 years ago, and several TWSTFT experiments, using small on-site earth stations and pseudonoise sequence, have been performed around 1970 (Jespersen et al. 1968; Chi and Byron 1975). In the beginning, only two or three measurements a week were performed, and now many single-links with more frequent measurements per day have constructed a worldwide network. Since 2003, TWSTFT has been used as a primary time transfer technique for the realization of coordinated universal time (UTC) and is currently one of the most precise and accurate techniques for comparison of the frequency standards located between remote sites. More than two-thirds of TAI clocks and almost all the primary frequency standards are transferred using TWSTFT.

By using TWSTFT method, frequencies can be compared with an uncertainty in the 10^{-15} range at an averaging time of 1 day, and time scale differences can be compared at the nanosecond level (Bauch et al. 2006; Piester et al. 2008). The achieved performances are based on the reciprocity of the propagation path that many delay variations of the transmitted signals cancel out in TWSTFT. However, the reciprocity is not completely fulfilled in practical situation. This is mainly due to: (1) the different signal paths through different electronic components in the ground stations and generally in the satellite; and (2) the different frequencies used for uplink and downlink to and from the satellite. Especially for the cases of long baselines, phase variations with amplitudes of 1 ns and more seem to limit the time and frequency transfer precision. Besides, as more and more timekeeping institutes use very stable maser clocks as their references, the diurnal fluctuations of about several hundred picoseconds appear in the continuous or hourly two-way time transfer results (Parker et al. 2003; Parker and Zhang 2005). In order to increase the accuracy and stability of TWSTFT, there is a keen desire to develop new technologies for time and frequency transfer. Moreover, the TWSTFT system should be calibrated and the sources of instabilities also need to be investigated.

Conventional TWSTFT is based on pseudorandom noise (PRN) codes, which are generated by commercial modem, to perform the measurements. The PRN occupying 2.5 MHz of up/down satellite link in usual case provides its resolution of 1 nanosecond. Increasing the bandwidth may get better time-transfer resolutions but will also increase the operational cost. Therefore, it becomes an important issue to develop some novel technique to improve the resolution without increasing the operational cost. Considering this, some new SDR (software-defined receiver)-based methods were proposed. National Institute of Information and Communications Technology (NICT) has developed software-based two-way time transfer modems (Amagai 2006; Amagai and Gotoh 2008; Gotoh and Amagai 2008), and their ability in improving the time transfer precision by using a Dual Pseudo-random Noise (DPN) signal has been demonstrated in the experiments between NICT(Japan) and TL(Taiwan) (Gotoh et al. 2011; Huang et al. 2011a; Tseng et al. 2012). In comparison with the conventional TWSTFT data, the DPN time transfer results of NICT-TL experiment have higher precision and less diurnal effect**.** Afterward, TL proposed the SDR transfer technology with the open-loop method. This technology has been verified in the pilot study organized by the CCTF WG on TWSTFT, and the comparison results have been utilized in the UTC generation. Recently, NICT has further devoted to develop the SRS (software ranging system) modem which allows the code and carrier phase measurements, with the goals of carrier-phase frequency transfer with instability below 1.0E-16 and time transfer by the code phase with lower than 1 ns uncertainty. These SDR-based TWSTFT technologies will be introduced in section "SDR-Based TWSTFT."

The precision of the TWSTFT highly depends on the residual nonreciprocity delays. Kirchner (Kirchner 1991) has discussed many causes of nonreciprocity. Several studies have also tried to evaluate all the potential sources of time delay instability and eliminate them (Parker and Zhang 2005; Piester et al. 2007). Nevertheless, we still have not identified all instability sources, since the observed instabilities in real TWSTFT links are larger than expected and the causes of the obvious diurnal effect still cannot be well explained. To obtain more precise and stable time transfer results, further investigations for evaluating all potential sources of time transfer instability are necessary.

In this chapter, we will review the basic principle of TWSTFT first, then introduce the conventional method and the SDR based technologies, the current adopted calibration methods, and finally explain the possible instable sources of TWSTFT technique.

Basic Principle of Conventional TWSTFT

The schematic of basic setup and signals propagation of TWSTFT is illustrated in Fig. 1. The equipment used for a TWSTFT Earth station includes the following:

1. A satellite terminal called VSAT (Very Small Aperture Terminal) type which usually consists of a dish antenna of 1.8 m ~ 3.6 m, a power amplifier, a low noise

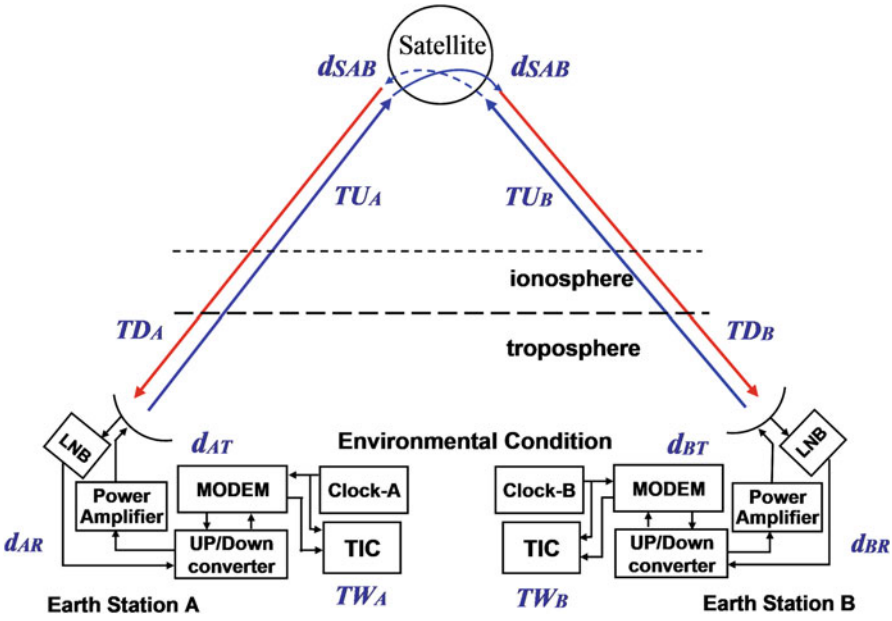

Fig. 1 The schematic of basic setup and signals propagation of TWSTFT

amplifier, and up/down converters. Most of the present TWSTFT links use Ku-band signal of the commercial communication satellite, and the up and down link frequencies are around 14 GHz and 12 GHz, respectively.

2. The other important equipment is the special designed time transfer "modem," which usually adopts the pseudo random noise (PRN) code modulation technique using the same frequency band of the satellite transponder to minimize the delay variation in the satellite. Currently, SATRE (Satellite Time Transfer and Ranging Equipment) modems, developed by a German company, are commonly used in Europe, Asia, and USA.

In Asian TWSTFT links, a different type of modem has been developed by the NICT in Japan (Imae et al. 2001; Imae et al. 2003), and used for conventional TWSTFT comparisons. This type of modem can perform the TWSTFT operation up to 7 stations simultaneously. Afterward, it was replaced by the newly developed DPN and SRS modems, which will be introduced in a later section of this chapter.

To perform the TWSTFT experiment, a pair of Earth stations transmit and receive pseudorandom noise (PRN)-coded signals via communication satellite. In each Earth station, the clock signal (typically 1 pulse per second, 1PPS) is modulated at an IF carrier frequency, up-converted to the up-link frequency, amplified by a high-power amplifier (HPA), and then radiated to the satellite by the antenna. The signal from the counterpart station clock is received at the satellite down-link frequency by the same antenna, amplified by a low-noise amplifier (LNA), and then down-converted to the

IF frequency and demodulated to the received clock signal. At the same time as the local clock pulse is starting the TIC, the pulse is also transmitted to the counterpart station. So the time interval counters (TICs) are started by a pulse from the local clock and stopped by the received pulse from the clock of counterpart. The same process goes on at both stations. The time interval information that is recorded at each station contains the clock differences as well as various delays as shown in the equations below.

$$TW_A = T_A - T_B + d_{BT} + TU_B + d_{SBA} + TD_A + d_{AR} + \Delta\tau_{BA} \quad (1)$$

$$TW_B = T_B - T_A + d_{AT} + TU_A + d_{SAB} + TD_B + d_{BR} + \Delta\tau_{AB} \quad (2)$$

where $\boldsymbol{TW_A}$ and $\boldsymbol{TW_B}$ are the time interval counter readings, $\boldsymbol{T_A}$ and $\boldsymbol{T_B}$ are the respective clock times, and the $\boldsymbol{d_{XY}}$s are the, respective, delays of the transmit and receive parts (equipment delays) of earth stations A and B. ***TUx*** and ***TDx*** are the up and down propagation delays illustrated in Fig. 1. $\boldsymbol{\Delta\tau_{BA}}$ and $\boldsymbol{\Delta\tau_{BA}}$ are Sagnac corrections, and $\boldsymbol{\Delta\tau_{BA} = -\Delta\tau_{BA}}$.

Since the time up to and back from the satellite is on the order of a quarter of a second, TW_A and TW_B are always positive for reasonably well synchronized clocks.

Assuming that the transmission and reception paths are almost the same, the atmospheric and geometrical delays between the Earth station and the satellite are canceled. Then calculation for the timescale difference between station A and station B by the TWSTFT can be simplified. However, the reciprocity is not fulfilled, so the equation for calculating the timescale difference contains all the delays terms. From Eqs. (1) and (2), the time difference between clocks at A and B sites can be determined by:

$$\begin{aligned} TA - TB = {} & (TWA - TWB)/2 \text{ (TIC readings)} \\ & +[(dAT - dAR) - (dBT - dBR)\,]/2 \text{ (Earth station equipment)} \\ & +[(TUA - TDA) - (TUB - TDB)]/2 \text{ (Propagation delay)} \\ & +(dSAB - dSBA)/2 \text{ (Delay in satellite)} \\ & +2\Delta\tau\text{RAB (Sagnac effect)} \end{aligned} \quad (3)$$

Figure 1 shows a model of the paths of the 1pps signals propagating between stations A and B and also gives a concept for analyzing the possible reasons for the time transfer instabilities of the TWSTFT technique. The sources of instability may include the following:

1. Atmospheric effects: troposphere, ionosphere.
2. Satellite motion effects: Sagnac effect, path delay difference.
3. Environmental conditions:

Ground station's internal delays affected by temperature, humidity, pressure.
Satellite transponder delays affected by heating due to solar radiation.

To further increase the precision of TWSTFT and reduce the related uncertainties, it is very important to evaluate all potential sources of time transfer instability, and then reduce or eliminate them.

SDR-Based TWSTFT

By using TWSTFT method, two clocks at two earth stations, with their distance very far away but within a common satellite's footprint, can compare to each other to get their time difference with its uncertainty up to nanoseconds or better within high levels of confidence. Convential TWSTFT relies on the hardware "commercial Modem" to perform sigal modulation and de-modulation in T/F transfer. PRN codes with code rate of 2.5 Mcps were usually used to achieve a measurement precision of 1 ns.

However, there has been a rapid development of lab-type atomic clocks and optical clocks in recent years, and their stabilities have greatly improved. That is why there is a keen interest to improve the precision of time and frequency transfer. The usage of broadband PRN codes could achieve more precise time transfer, but it requires a wider bandwidth of the satellite transponder and will lead to the increase of connection fees of commercial communication satellites. In order to improve the time transfer precision without increasing the operational cost, the development of novel techniques becomes an important issue.

In this section, some SDR-based methods are introduced. The SDR was originally designed for implementing the dual pseudorandom noise (DPN)-based or carrier-phase (CP)-based TWSTFT measurements for improving their short-term stabilities (Tseng et al. 2012; Piester et al. 2007; Fujieda et al. 2014).

DPN Method

The National Institute of Information and Communications Technology (NICT) has developed the Dual PRN (DPN) signal method, which is based on the Binary Offset Carrier (BOC) signal that will be further used on Galileo and GPS civil navigation systems. The DPN is realized by modulating a PRN with two different subcarriers. Measuring the DPN group delay is equivalent to measuring the propagation delay. Because the frequency period between DPN's separated subcarriers can be regarded as a wider operational bandwidth, it was expected that higher resolution could be achieved. Piecewise bandwidth is the most notable advantage that the DPN signals do not really occupy wider bandwidth of satellite transponder, so the operation costs will not increase.

The DPN test was first proposed and evaluated around 2006 (Gotoh and Amagai 2008). In August 2008, NICT conducted the common clock test (Gotoh et al. 2011).

The test was successful, and the TDEV reached 20 picoseconds at 1-second averaging, as expected. In Oct. 2008, NICT compared national standard with the free-running Hydrogen Maser (Huang et al. 2011a) by using DPN method with 20 MHz bandwidth. The TDEV reached 18 picoseconds (MDEV = 5.2E-14) at 10-mimute averaging, and the results agreed with the GPS PPP ones.

The first international DPN experiment has been successfully performed between NICT and TL and lasted for several months in 2010 (Gotoh and Amagai 2008; Gotoh et al. 2011; Huang et al. 2011a) and 2011. In summary, the numerical results show that DPN results of NICT-TL experiment have higher precision and less diurnal compared with the conventional TWSTFT results. It is also comparable to GPS PPP in the short term. Moreover, the DPN technology can adopt real-time processing and will enable the cost-effective utilization of the limited satellite bandwidth in TWSTFT systems.

SDR Transfer with the Open-Loop Method

The precision of TWSTFT was limited by the so called diurnal effect, which is the daily variation of about ±0.5 ns periodically appearing in the measurement results. To investigate the diurnal effect, several articles have indicated that the daily variations may be caused by atmospheric, environmental condition, and so on. However, the diurnals cannot be completely explained by anyone of these factors, so the diurnal effects are regarded as the combination of these non-reciprocities in TWSTFT.

Based on the cooperation experience on TWSTFT DPN technlogy between NICT and TL, TL further developed the SDR receiver with the open-loop method trying to mitigate the diurnal effect.

A. The Concept of The Open-Loop Method (Huang and Tsao 2015)

The principle of two-way time difference measurement can be simply expressed as Fig. 2. The earth station A sends its on-time tick to station B and vice versa based

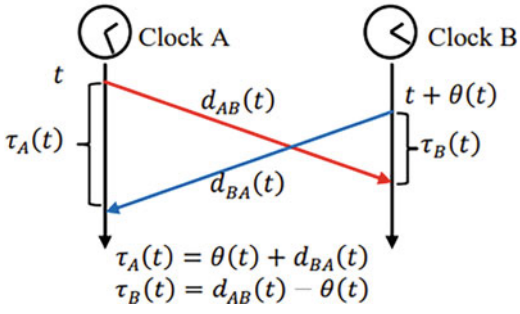

Fig. 2 The principle of two-way time difference measurement method

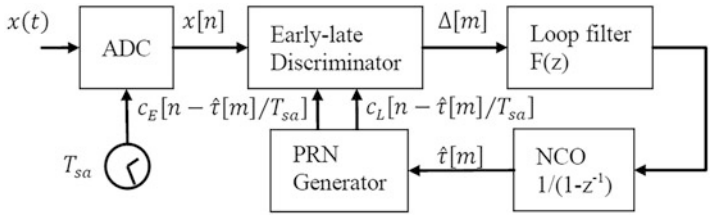

Fig. 3 Conceptual representation of the DLL with an E-L discriminator (Huang and Tsao 2015)

on their clocks, where we assume A leads B by (t) seconds at time t. Then the TOAs, $\tau_A(t)$ and $\tau_B(t)$, are measured and exchanged. If the forward and opposite paths between A and B are reciprocal (i.e., $d_{AB} = d_{BA}$), the time difference can be obtained by $\theta(t) = (\tau_A(t) - \tau_B(t))/2$.

For conventional TWSTFT modem, the most common method used for time-of-arrival (TOA) measurement was the delay-locked loop (DLL). As shown in Fig. 3, the error signal Δ[m], which is introduced to identify the code phase mismatch between the measured and true values, is fed into a loop filter F(z), to remove high-frequency components. Then by accumulating the filtered error signal using a numerical controlled oscillator (NCO), the TOA measurement is obtained. According to the TOA measurement, PRN generator updates the early and late PRN waveforms for the next cycle. It was found (Huang and Tsao 2015; Huang et al. 2016) that adopting the DLL method may cause error accumulation, and the TOA measurement will also be affected by input signal power.

In previous study (Huang and Tsao 2015), a first-order DLL was emulated and the common clock experiment was evaluated. The results showed that the first-order DLL with different loop gains will result in different TOA measurements, so that the reciprocity may depend on some parameters, such as loop order or bandwidth, used in the DLL.

For conventional receivers, a second-order DLL was also widely implemented to continuously track the PRN signal. The DLL following the correlator output will filter out the high-frequency terms of TOA measurement and reduce the measurement noise. However, it was found that the transfer behavior of the second-order DLL depends on parameters such as C/N_0 value, code-phase velocity, and code-phase drift (Holmes 2007). In the situation of multiple transmitting stations in a TWSTFT network, the C/N_0 values of all the participating Earth stations may be different due to satellite coverage or antenna pattern. Different transfer behaviors caused by DLLs would lead to different biases in TOA measurements. The biases cannot be canceled out and thus will cause an error in time-difference results. Besides, the errors caused by low quality of receiving signal will be accumulated and propagated by the closed-loop behavior of DLL.

To avoid the effects of DLL character and error accumulation, a parameter-free measurement scheme, the so-called open-loop (OL) method, was proposed to remove the loop filter and the feedback mechanism of DLL.

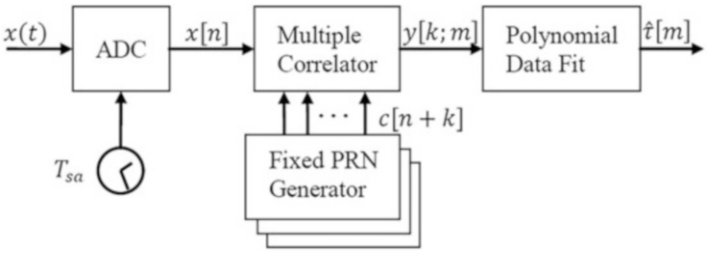

Fig. 4 Proposed open-loop TOA measurement method (Huang and Tsao 2015)

The concept of the Open-Loop(OL) method is shown in Fig. 4 that the local codes with predetermined code phase differences were generated, and they were cross-correlated with the input PRN code to get multiple correlation results. Then, these results were fitted to a second-order polynomial. The input code phase was then determined by finding the global maximum of the polynomial.

The results of previous common clock experiment, performed between two OL-based SDR and two SATRE modems, showed that the OL-based SDR could have higher reciprocity than SATRE modem. Therefore, receivers with OL method may reduce the TWSTFT uncertainties including the diurnals (Huang and Tsao 2015).

B. Implementation of SDR Receiver.

The composition of the SDR receiver and the procedure to obtain TOA will be introduced in this section.

The hardware of the SDR receiver includes an Analog-to-Digital Converter (ADC), some RF components, and a computer. To sample the receiving signal, the VSSP32 (Nitsuki 2016) and N210 (Ettus n.d.) were adopted as the ADC. The sampling frequency of ADC should set to be higher than twice of the PRN code-chipping rate, so that the PRN signal could be completely described and have better time resolution. Therefore, VSSP32 was set with a maximum sampling frequency of 64 MHz and a voltage resolution of 4 bits, and N210 was set with a maximum sampling frequency of 50 MHz and a voltage resolution of 8 bits. To avoid quantization error, an amplifier and an attenuator were used to make the signal level fulfill the dynamic range of the ADC input. Then the digital signal processing (DSP) algorithms were run in a high-performance personal computer (PC) to measure the TOA from the samples.

Besides, some previous works indicate that the effects of multipath interference and multiple access interference (MAI) would lead to diurnals (Zhang et al. 2009; Zhang and Parker 2013), so the high-resolution correlator (HRC) (Van Dierendonck et al. 1992; McGraw and Braasch 1999) and successive interference cancellation (SIC) algorithms (Pateland and Holzman 1994) were implemented on the SDR to reduce these effects.

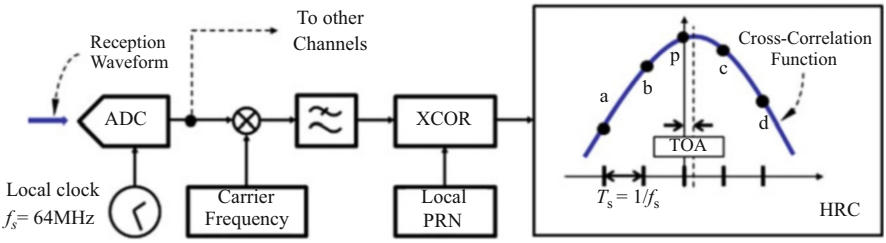

Fig. 5 The TOA measurement procedure using proposed SDR, where fs and Ts are sampling frequency and sampling period, respectively (Huang et al. 2016)

As mentioned in the previous section that the SDR is designed with the open-loop concept, so the possible errors caused by DLL could be avoided. Furthermore, the samples used for determining adjacent TOAs do not overlap. For example, using N210 with 50 MHz sampling rate, the PRN code length of 4 ms would be equal to 200,000 samples. The first 200,000 samples are used to derive the first TOA, and the following 200,000 samples will determine the next TOA. This procedure makes all TOA measurements independent, so that anomalies do not cause error propagation. Therefore, some anomalies due to low signal quality would merely affect time-difference performance, since the unusual results can be easily detected and removed.

The TOA measurement procedure by the SDR is shown in Fig. 5 (Huang et al. 2016). Conventional algorithms (Holmes 2007) are still used for the implementation of some functions, such as PRN acquisition, frequency tracking, and carrier phase locking. The received signal is first sampled and quantized by an ADC, e.g., VSSP32, and then the samples are down-converted to the baseband by mixing the carrier frequency with them and filtering out the high-frequency components. The baseband samples and desired PRN codes with different time delays are then correlated to yield the cross-correlation function (CCF). This cross-correlation procedure (XCOR) is performed once per code period, and the peak location of CCF is defined as the TOA. However, the location of the peak cannot be determined exactly due to signal processing in the discrete-time domain. Therefore, for precisely estimating the peak location, five largest CCF samples around the peak, denoted in order as p, c, b, d, a, are used on the SDR. By using HRC algorithm, the TOA measurement can be obtained with Eq. (4) from these CCF samples.

$$\mathrm{TOA} = \left(\frac{b-c}{b-2p+c} - \frac{a-d}{a-2p+d}\right) Ts \tag{4}$$

It was found that the experment results using SDR receiver showed similar long-term stability and better short-term performance in comparison to those results using conventional SATRE modem. Adopting insensitive earth station could further reduce the Diurnal effect. In 2016, the International Bureau of Weights and Measures (BIPM) and the Consultative Committee for Time and Frequency (CCTF) working

group (WG) on TWSTFT launched a pilot study on the application of SDR receivers in the TWSTFT network for UTC computation.

The first results of the pilot study, demonstrating that SDR TWSTFT shows superior performance compared to that of SATRE TWSTFT for practically all links between participating stations, were reported to the CCTF WG on TWSTFT annual meeting held in May 2017. In particular, the use of SDR TWSTFT results in a significant suppression of the diurnals. For the very long inter-continental links, e.g., the Europe-to-USA links where the diurnals are less pronounced, SDR TWSTFT achieved a smaller but still significant gain of 30%. These findings were also supported by the evaluation of some of the links with an alternate technique based on GPS signals (GPS IPPP). Stimulated by these results, the CCTF WG on TWSTFT prepared a recommendation for the 21st CCTF meeting for proposing the introduction of SDR TWSTFT in UTC generation.

After the calibration of the OP-PTB link, and an observation period which ensured the reliability of the data transmission and processing chain, SDR data was used in UTC calculation for the first time in March 2020 (https://www.bipm.org/en/-/2020-twstft-sdr).

The SDR is an open platform for the operators of the TWSTFT ground stations. The interested researchers may make deeper studies, e.g., by taking out the carrier phase information for various applications (Jiang et al. 2018).

TWSTFT Carrier Phase (TWCP) for Frequency Transfer

The precision of conventional TWSTFT is limited to signal bandwidth and is inversely proportional to the code chip rate. It was demonstracted in previous studies that, in comparison with conventional TWSTFT that adopted the code modulation technique, using carrier phase information (TWCP) could further improve the precision of TWSTFT. Sch¨afer et al. introduced the use of the carrier phase to TWSTFT (Schäfer et al. 1999), and Fonville et al. demonstrated an intercontinental test result for a few hours (Fonville et al. 2004).

The principle of TWCP is briefly introduced as the following: (Fujieda et al. 2014).

To conduct the TWCP experiment, two Earth stations A and B work as a pair. In addition to the time difference between A and B, there are three more unknown values that need to be solved. These unknown values are the phase fluctuations induced by the translation of the received to the transmitted signal at the satellite and two geometrical distances between the satellite and the two stations.

The TWCP technique requires that each site observes both its own satellite-transfered signal and the signal of its counterpart site, so that the phase information from A to B, from B to A, from A to A, and from B to B could be expressed by four equations. The Doppler frequency shift due to the satellite motion and the phase noise of the onboard oscillator could be canceled out by using the delay measurement information of earth station's round-trip signal.

Let us denote the uplink and downlink signal angular frequencies as w_u and w_d, respectively, and angular frequency of the translation onboard oscillator could be expressed as $w_s = w_u - w_d$.

Due to the Doppler effect, a signal $\sin(w_u t + w_u \tau_a(t))$ transmitting from station A can become $\sin(w'_u t + w_u \tau_a(t))$ when it arrives at the satellite, where $w'_u = w_u\left(1 - \frac{v_a(t)}{C}\right)$ and c is the speed of light.

With the onboard signal $\sin(w_s t + w_s \tau_s(t))$, the transmitting signal is further down-converted to $\sin[(w'_u - w_s)t + w_u \tau_a(t) - w_s \tau_s\,(t)]$. Due to the Doppler effect, this signal shifts to $\sin(w''_d t + w_u \tau_a(t) - w_s \tau_s(t))$ when it arrives at station B. Then with the local signal $\sin(w_d t + w_d \tau_b(t))$ at station B, the received signal is finally down-converted to

$$\sin\left((w''_d - w_d)t + w_u \tau_a(t) - w_s \tau_s(t) - w_d \tau_b(t)\right). \tag{5}$$

With the radial velocity from station B to the satellite being denoted as $v_b(t)$, the angular frequency received at station B with the Doppler shift is expressed as:

$$w''_d = (w'_u - w_s)\left(1 - \frac{v_b(t)}{c}\right) = w_d - w_d \frac{v_b(t)}{c} - w_u \frac{v_a(t)}{c} + w_u \frac{v_a(t)}{c}\,\frac{v_b(t)}{c} \tag{6}$$

The fourth term on the right hand is negligible, as it is less than 10^{-16}.

According to the above discussion, the phase information $\varphi_{ab}(t)$ in radian of the signal transmitted from station A and received at station B could be expressed as:

$$\begin{aligned} \varphi_{ab}(t) &= w_u \tau_a(t) - w_s \tau_s(t) - w_d \tau_b(t) - (w_u \nu_a(t) t + w_d \nu_b(t)\, t)/c \\ &\quad + w'_u I_{ua}(t) + w''_d I_{db}(t) + w'_u T_a(t) + w''_d T_b(t) \\ &\quad + w_u \frac{v_a(t)}{c}\,\frac{v_b(t)}{c} t \\ &\approx w_u \tau_a(t) - w_s \tau_s(t) - w_d \tau_b(t) - (w_u \rho_{as}(t) t + w_d \rho_{bs}(t))/c + w_u I_{ua}(t) + w_d I_{db}(t) \\ &\quad + w_u T_a(t) + w_d T_b(t) \end{aligned} \tag{7}$$

where T_a and T_b are tropospheric delays in seconds which are frequency independent over stations A and B, respectively. ρ_{as} and ρ_{bs} are the geometrical distances between the satellite and stations A and B. I_{ij} is the ionospheric delay correction term in seconds with frequency f_i (i = u or d) at position j which can be represented as follows:

$$I_{ij}(t) = \frac{40.3 \cdot TEC_j(t)}{c \cdot {f_i}^2} \tag{8}$$

here TEC_j (t) is the total electron content, i.e., the total number of free electrons along the signal path at position j, which is conventionally measured in TEC units. f_{u} and f_{d} are the uplink and downlink frequencies, respectively.

The phase information from B to A, from A to A, and from B to B, can be similarily represented by:

$$\begin{aligned}\varphi_{ba}(t) =& w_u\tau_b(t) - w_s\tau_s(t) - w_d\tau_a(t) - (w_u\rho_{bs}(t)t + w_d\rho_{as}(t))/c \\ &+ w_uI_{ub}(t) + w_dI_{da}(t) + w_uT_b(t) + w_dT_a(t)\end{aligned} \tag{9}$$

$$\begin{aligned}\varphi_{aa}(t) =& w_u\tau_a(t) - w_s\tau_s(t) - w_d\tau_a(t) - (w_u\rho_{as}(t)t + w_d\rho_{as}(t))/c + w_uI_{ua}(t) \\ &+ w_dI_{da}(t) + w_uT_a(t) + w_dT_a(t)\end{aligned} \tag{10}$$

$$\begin{aligned}\varphi_{bb}(t) =& w_u\tau_b(t) - w_s\tau_s(t) - w_d\tau_b(t) - (w_u\rho_{bs}(t)t + w_d\rho_{bs}(t))/c + w_uI_{ub}(t) \\ &+ w_dI_{db}(t) + w_uT_b(t) + w_dT_b(t)\end{aligned} \tag{11}$$

Since the geometrical distance from station A to the satellite and the corresponding distance from station B to the satellite are not symmetrical, there is a difference of about a few milliseconds in the arrival times of signals from stations A and B to the satellite. Let us assume that the onboard translation oscillator signal is stable during this time difference, and thus the induced phase jitter is negligible.

By subtracting Eq. (7) from Eqs. (9), and (10) and (11), one can obtain that:

$$\begin{aligned}&\varphi_{ab}(t) - \varphi_{ba}(t) = w_+(\tau_a(t) - \tau_b(t)) - w_-(\rho_{as}(t) - \rho_{bs}(t))/c \\ &+w_-(T_a(t) - T_b(t)) + w_u(I_{ua}(t) - I_{ub}(t)) - w_d(I_{da}(t) - I_{db}(t))\end{aligned} \tag{12}$$

and

$$\begin{aligned}&\varphi_{aa}(t) - \varphi_{bb}(t) = w_-(\tau_a(t) - \tau_b(t)) - w_+(\rho_{as}(t) - \rho_{bs}(t))/c \\ &+w_+(T_a(t) - T_b(t)) + w_u(I_{ua}(t) - I_{ub}(t)) + w_d(\,I_{da}(t) - I_{db}(t))\end{aligned} \tag{13}$$

Here $w_+ \equiv w_u + w_d$ and $w_- \equiv w_u - w_d$, and the terms related to the onboard signal are canceled.

In TWCP, the geometrical distances $\rho_{as}(t)$ and $\rho_{bs}(t)$ in Eqs. (12) and (13) are unknown, and we can further subtract the above intermediate results to cancel them to obtain:

$$\begin{aligned}&w_+(\varphi_{ab}(t) - \varphi_{ba}(t)) - w_-(\varphi_{aa}(t) - \varphi_{bb}(t)) \\ &= \left(w_+{}^2 - w_-{}^2\right)(\tau_a(t) - \tau_b(t)) \\ &+2w_uw_d[(I_{ua}(t) - I_{ub}(t)) - (I_{da}(t) - I_{db}(t))]\end{aligned} \tag{14}$$

The time difference between stations A and B could finally be obtained by using the following equation:

$$\tau_a(t) - \tau_b(t) = \frac{w_+\alpha(t) - w_-\beta(t)}{{w_+}^2 - {w_-}^2} + \frac{2w_u w_d}{{w_+}^2 - {w_-}^2} \times [(I_{da}(t) - I_{ua}(t)) - (I_{db}(t) - I_{ub}(t))] \quad (15)$$

where

$$\alpha(t) \equiv \varphi_{ab}(t) - \varphi_{ba}(t)$$

$$\beta(t) \equiv \varphi_{aa}(t) - \varphi_{bb}(t)$$

On the right hand of Eq. (15), the first term is derived from the four observed phases, the second one is for ionospheric delay correction, and the tropospheric delays are canceled. Since the uplink and downlink frequencies of TWSTFT are around 14 GHz and 12 GHz, the coefficients $w_+/({w_+}^2 - {w_-}^2)$ and $w_-/({w_+}^2 - {w_-}^2)$ are about 7.0E−12 and 8.0E−13, respectively. Thus, a measurement precision of 1.0E-13 seconds could be expected when the phase detection has a resolution of less than 0.1 radians.

Besides, the ionospheric correction term can be calculated, when $TEC_j(t)$ is obtained by theoretical model or actual measurement. The ionospheric delay needs to be considered carefully because of the frequency difference between uplink and downlink and the TEC difference between the ionospheres at stations A and B. Since the TWCP frequencies are higher than those used in GPS, the effect due to the ionosphere is smaller than that encountered in GPS time transfer. However, the amplitude of the ionosphere effect could be up to a few hundred picoseconds, so it cannot be neglected in TWCP, even though it is usually assumed to be negligible in conventional TWSTFT.

Some TWSTFT experiments using TWCP technique in recent year have showed encouraging results. One of these TWCP experiments was performed between NICT and PTB (Fujieda et al. 2014), with it ranging up to about 9000 km, and it confirmed that TWCP is applicable over very long baselines. The experimental results showed a good agreement with those measured by TWCode and GPSCP and clearly demonstrated that the measurement precision of TWCP is superior to that of TWCode and GPSCP.

Later study confirmed again that TWCP can provide higher stability than that obtained by conventional TWSTFT (Fujieda et al. 2016). The advanced TWCP technology had also been applied to optical clock comparisons between NICT and KRISS. It was demonstrated that the satellite-based techniques have the potential to realize uncertainty at the 1.0E−17 level, when optical clocks are continuously operated for a week or more (Fujieda et al. 2018).

"Software Ranging System (SRS)" Modem Developed by NICT (Fujieda et al. 2019, 2020)

To extend the application of the TWCP, the National Institute of Information and Communications Technology (NICT), Japan, is further devoted to develop the SRS

modem which allows both the code and the carrier phase measurements, with the goals of carrier-phase frequency transfer with instability below 1.0E-16 and time transfer by the code phase with lower than 1 ns uncertainty.

The newly developed SRS modem consists of FPGA (Field Programmable Gate Array), DAC (Digital to Analog Converter), ADC (Analog to Digital Converter), 1-GHz VCO (Voltage Controlled Oscillator), and PLL (Phase Locked Loop). A design consisting of mostly digital components makes the modem cost-effective and reconfigurable.

The performance of time and frequency transfer in a common clock and non-common clock measurements were evaluated, and encouraging results were obtained. Frequency stability of low 10^{-17} level has been achieved in the initial carrier phase measurements performed over domestic distances with a common-clock configuration. The observed delay repeatability also promises a calibration uncertainty lower than 1 ns (Fujieda et al. 2020).

Hopefully, the new SRS modem could be further improved by combining code-phase and carrier-phase measurements to achieve better time and frequency transfer stabilities, and then applied to more TWSTFT links.

Delay Calibration of Twstft Links

Based on the path reciprocity of signals, the TWSTFT technique cancels out the first-order effects. To increase the precision of time transfer, especially when the data of TWSTFT link will be applied to TAI computation, the parameters of nonreciprocal residual effects need to be determined. The possible delay differences due to nonreciprocal residual effects may happen in: (1) the propagation in the atmosphere for up and down links, (2) the transponder path in the geostationary satellite, (3) the earth rotation (Sagnac effect), and (4) inside each earth station including transmission and reception paths. The last parameter seems to be the most important source of error.

Most of the TWSTFT time transfer links were calibrated by using GPS common-view measurement results before 2006, and thus the attainable uncertainty was limited to several nanoseconds. Considering on this, the CCTF Working Group on TWSTFT stimulated to perform calibrations of TWSTFT links using more adequate technique.

According to the guideline (Jiang et al. 2016) proposed by BIPM, there are two major methods as follows:

1. Using a portable earth station (PS), which is in the relative calibration mode, either as an independent time transfer technique or as a reference to determine the relative delays of the earth stations to be calibrated with respect to the PS.
2. Using GPS traveling calibration equipment.

Besides, some other calibration methods have also been developed for the TWSTFT calibration, as follows:

3. Using a satellite simulator in the absolute calibration mode.
4. Using triangle-closure condition of TWSTFT links.

These methods for TWSTFT delay calibration will be briefly introduced in the following sections. Moreover, since the SDR method has been adopted for UTC calibration, the calibration rule proposed for SDR links will also be mentioned.

TWSTFT Calibration Using Portable TWSTFT Station

The differential delays of TWSTFT-participant earth stations can be determined by means of a portable TWSTFT station which is operated in consecutive experiments side-by-side with the participating earth stations in a common clock setup. The use of portable TWSTFT equipment has allowed time transfer with nanosecond accuracy, and the calibration uncertainty is usually uB $\leq$ 1 ns (Galindo et al. 2016; Piester et al. 2014; JCGM 2012; Bauch et al. 2011). All the details about the organization, data processing, uncertainty evaluation to be included in the final report, and the CALR (link calibration result) implementation of the calibration campaign can be found in references (Jiang et al. 2016; Galindo et al. 2016).

In the calibration campaign, TWSTFT will sequentially operate at participant laboratories during the period of the calibration trip. To verify the stability of the portable station during the trip, the measurements at the first site can be repeated again after visiting all other participating institutes in each campaign (Fig. 6) (Piester et al. 2005, 2006).

NICT and TL have performed bilateral comparison campaigns in 2006. An uncertainty of about 1 ns in the differential calibration was achieved. Although calibrating the TWSTFT earth stations by portable station is commonly used

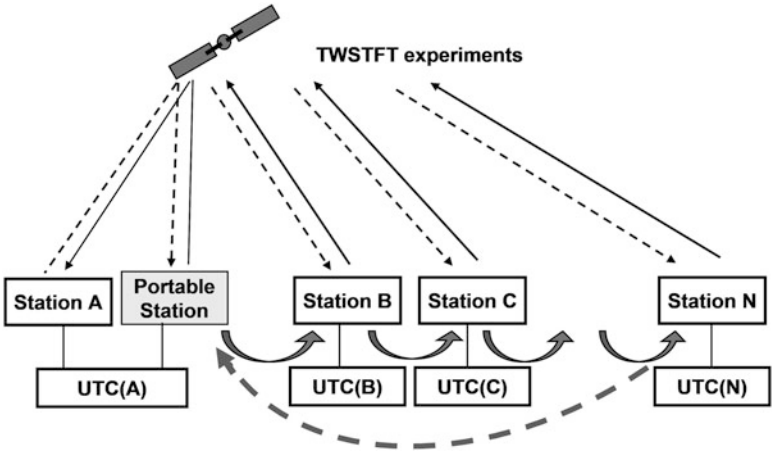

Fig. 6 Schematic set-up of TWSTFT calibration trip by using portable station (PS)

among European labs, it is not convenient for laboratories located in different continents. For example, in Asia-Pacific Rim region, TWSTFT calibration using portable TWSTFT station needs to transport the portable station overseas and thus make it very difficult. Besides, the calibration scheme of circulating a portable TWSTFT reference station is applicable, only in the case that the single satellite transponder is used. This condition is not always fulfilled in the intercontinental links, e.g., between Asia and Europe. In such case, calibration of the TWSTFT link can be achieved by adjusting the TWSTFT results to those obtained with a calibrated GPS link. PTB has proposed a method exchanging two portable TWSTFT stations and two GPS receivers between the two earth stations to be calibrated. The overall uncertainty for the calibration was estimated to be at a level of 2 ns (Fujieda et al. 2007).

TWSTFT Calibration Using GPS Traveling Calibration Equipment (TWSTFT 2012)

When calibration via the portable TWSTFT station is not applicable, the GPS time link calibration can be used as an alternative one, with which the uncertainty of 1.5 ns is attainable (Jiang et al. 2015). The GPS link calibration differs from the GPS receiver calibration. The latter is based upon a common clock difference (CCD) of the two GPS receivers (the traveling and the local receivers). The GPS link calibration measures the double clock difference (DCD) directly, which is calculated by subtracting the GPS link between the traveling calibrator and the remote site master receiver from the TWSTFT link. A link calibration of either TWSTFT or GPS is referred to the UTC pivot point (PTB at present).

Using GPS to calibrate a TWSTFT link is a simple alternative way. However, reporting the results to BIPM is essential and all key points, as what are requested for the calibration using a portable TWSTFT station, should be addressed in an adequate way.

Earth Station Delay Measurement Using Satellite Simulator (SATSIM)

In the past decades, several laboratories and companies have devoted time to the development of satellite simulator for the absolute calibration of TWSTFT station. This method was first proposed by De Jong (de Jong et al. 1989). By using the method and instrumentation developed at NMI Van Swinden Laboratorium (VSL, the Netherlands), six European Earth stations (participating in one TWSTFT experiment) were calibrated in an absolute way in 1994 (de Jong et al. 1995). These six stations have also been calibrated in a relative way by a portable station. Most results of these two methods showed an agreement to the level of 10 ns, although few of the data showed significant deviations of multiples of 100 ns. LNE-SYRTE (France) also had developed a satellite simulator and implemented it into the earth station. Afterward, VSL designed and assembled a dual-stage quad-mixer satellite simulator

(Mubarak and Dierikx 2011). With this satellite simulator, all delays in the TWSTFT station can be determined in absolute terms. The satellite simulator was implemented in the TWSTFT station and can automatically perform the delay calibration measurements. As for most of the newly developed equipment, the optimum transmission power level for each measurement path is underdetermined to achieve an optimal signal-to-noise (S/N) ratio, and then its long-term behavior can be determined.

The SATSIM equipment was mainly used for the Earth station delay measurement (absolute delay difference calibration and monitoring), and the measurement accuracy could be at a level of few hundred picoseconds. SATSIM can also be used to measure the delay fluctuations in earth stations. These fluctuations are the main source of instability that degrades the accuracy of TWSTFT. TL had installed a satellite simulator system, which was developed by a German company, on an earth station for the investigation of signal path delay variations in the TWSTFT earth station (Tseng et al. 2007). The results showed that the long-term and short-term temperature variations have different contributions to the measured delay value of the earth station. Besides, a very high correlation between the delay of earth station and the outdoor temperature data was observed.

Calibration of TWSTFT Links by Triangle-Closure Condition

Although the high accuracy of TWSTFT has a potential subnanosecond uncertainty, it is based on a metrological calibration. The calibration campaign is labor- and cost-intensive, and also limited by the complexity of the organization and the availability of equipment. Therefore, it is not practical applying either of the above calibration methods to all TWSTFT links.

Actually, three TWSTFT links, among three time-keeping laboratories, can make their contribution through the triangle constraint. The sum of three TWSTFT links will form a closed triangle, the so-called the triangle closure. In ideal case, a closure is expected to be zero if there are no measurement errors, namely, the triangle closure condition (TCC). BIPM (Jiang et al. 2008) has proposed a strategy to transfer the existing TW calibrations of the uncalibrated links using the TCC in the UTC TW network. The Europe-America TW network, comprised of 10 national time-keeping laboratories, was calibrated in this way. The related UTC data within the same period was used for the computation. The uncertainties of the TCC are 2 ns or 6 ns, depending on the original calibration uncertainties of TWSTFT or GPS, respectively. This is satisfactory for most of the scientific and metrological applications. By using this strategy, most of the time transfer links in a TWSTFT network can be easily calibrated.

The Calibration Issue of SDR TWSTFT Method (Jiang et al. 2018)

Since SDR TWSTFT shows better performance in comparison to the conventional SATRE TWSTFT, it was considered to apply the SDR TWSTFT results in the

generation of UTC, which required the signal delays along the full propagation path to be fully determined.

The SDR link calibration can be achieved by adopting the following rules:

1. For UTC links:
   (a) If the SATRE link is calibrated, the SDR link should be aligned to it.
   (b) If the SATRE link is not calibrated, both links – the SATRE and the corresponding SDR link – should be calibrated together by using a portable TWSTFT station or GPS calibrator with the "link" method (TWSTFT 2016).
2. The non-UTC link delay should be calibrated by the triangle closure calibration (TCC) method (TWSTFT 2016).

The conventional uncertainties of the UTC SATRE links, with uA and uB being 0.5 ns and 1.0 ns (using TWSTFT portable station calibration) or 1.5 ns (with GPS Calibrator), respectively, are assigned by BIPM. Usually, higher uncertainty values than the estimated ones will be assigned, e.g., the estimated uB for conventional SATRE links is 0.6 ns, while the actual values of 1.0 ns were assigned. These values are derived from a detailed report on the used link data in Section 5 of Circular T and are assigned to time differences UTC-UTC(k) as reported in Section 1 of Circular T at a 5-day grid.

Regarding the SDR TWSTFT links, its corresponding values of uA and uB are 0.2 ns and 1.0 to 1.5 ns, respectively (Jiang et al. 2018). It has been demonstrated that for short baselines, e.g., OP-PTB, a gain factor of two or three is obtained in the stabilities of SDR links in comparison with those of SATRE links. For long baselines, e.g., NIST-PTB, with much smaller diurnal comparing to most of other links, the stability of the SATRE link is already below to 0.2 ns. Therefore, assigning the uA value of 0.2 ns to a SDR link is reasonable.

As for uB, the conventional value is applicable for both the SATRE and the SDR links because the conventional uB of the SATRE link is often higher than the measured one. Therefore, the combined uB of the SDR link could be covered by the conventional uB of the SATRE link.

The conventional value of the uB for the SDR TCC calibration is 2 ns, the same as that when using SATRE data (Jiang et al. 2008). The uB of SDR links may be further improved if a direct SDR link calibration is performed.

Analysis of the Instabilities in TWSTFT

Instability Sources of TWSTFT

Potential causes of nonreciprocity and evaluating the sources of delay instability have been discussed by many researchers . Recently, reference clocks of many TWSTFT stations have changed to hydrogen masers (which provide better short-term stability) and the TWSTFT measurements are performed more frequently (even 24 h a day for Asia-Pacific TWSTFT network comparisons). Therefore, the

influences of instabilities in TWSTFT have emerged from the clock noise and become an important issue. The major instability sources of TWSTFT include (Recommendation ITU-R 2015): the Sagnac effect, environmental temperature, ionosphere, ground station delay (out-door and in-door), and path delay difference due to satellite motion.

Sagnac effect is associated with the Earth's rotation, which causes the relative position of a geostationary satellite to vary slightly over the course of a 24-h period. The variation of Sagnac can be computed from the ephemeris data of the satellite (Parker and Zhang 2005; Piester et al. 2007; Tseng et al. 2011).

The environmental temperature is also an important instability source. Previous studies demonstrated the TWSTFT measurements may show high correlation with local temperature of station sites. The NIST have measured temperature coefficients of group delay for several different TWSTFT sets of equipment (Parker and Zhang 2005). Their data showed that the temperature coefficients of TWSTFT system could be as large as several 10 ps/°C. TL employed a satellite simulator system (SATSIM) to measure the differential path delays (i.e., (TX-RX)/2) in 2005 (Mubarak and Dierikx 2011; Tseng and Lin 2005). By using a high chip rate of 20 Mcps (chip per second), a very high correlation between the delay of earth station and the outdoor temperature data was observed. During TWCP measurements, it is also observed that the transmission power decreased by about 10 dB when the outdoor temperature increased to more than 30 °C. In the case of common clock TWCP measurement, there was a daily variation of the measurement results with an amplitude of about 20 ps, and it had a correlation with the variation of the outdoor temperature (Fujieda et al. 2014). Nevertheless, if the whole TWSTFT equipment could be placed in a well-controlled environment, the influence of temperature could be regarded as a controllable factor.

Since TWSTFT use different frequencies for up- and down- links, the transmitting and receiving signals go through the ionosphere with different delays, which cannot be completely canceled in the two-way difference, and will thus cause instability. The ionospheric delay varies with respect to some factors, such as the change of total electron content (TEC) and the elevation angle of a TWSTFT station. Because the TEC varies with solar activity, the TEC shows both diurnal and annual variations. Besides, for TWSTFT link with long-baseline, its elevation angle will be lower, so longer ionospheric delay will be included and will cause larger ionospheric delay difference in the TWSTFT measurements. For a short-baseline time transfer, the difference for the ionospheric delay is usually less than 100 ps. While for an intercontinental measurement, this delay term could be larger from 150 to 300 ps. (e.g., 300 ps for NIST-PTB link (Parker and Zhang 2005), and 150 ps for NICT-PTB link (Piester et al. 2007)).

There are two other nonreciprocity terms related to the satellite motion. One is the path delay difference with its impact of less than 15 ps. The other term is frequency shift due to the Doppler effect, which is proportional to the relative velocity of satellite motion. Due to the error possibly caused by the imperfection of the TWSTFT receiver, the Doppler shift will cause the additive variation into the measurements (Tseng et al. 2011). A study on multiple access interference between

multiple pseudorandom noise (PRN) coded signals has tried to explain this effect (Huang et al. 2011b). The amplitude of the "measurement error" was estimated as large as 1.5 ns for a chip rate of 2.5 Mcps (chip per second). This term is likely to be eliminated by reducing the interference between PRN codes, and worth further studies.

Regarding the ionospheric delay, since it is inversely proportional to the square of the signal frequency, it cannot be canceled out in TWSTFT because of the frequency difference between the uplink and downlink. In TW Code, the impact of the ionospheric delay is assumed to be negligible (JCGM 2012). However, it should be considerable in the TWCP link taking the high performance into account. Related investigation on the ionospheric delay, along with some other error sources in TWSTFT carrier phase over very long baseline, was obtained in 2014 (Fujieda et al. 2014).

Although previous research investigated many instability sources of TWSTFT, it is clear that we have not identified all sources, since the observed instabilities in real TWSTFT links are larger than expected. Moreover, the causes of the obvious diurnal effect can still not be explained well. Anyhow, the main instabilities summarized from past studies are compiled in Table 1.

The Issue of Diurnals (Jiang et al. 2019)

The major source of statistical uncertainty in current TWSTFT is the daily variations, so-called diurnals, observed in the TWSTFT link data. Although the diurnals have been investigated extensively in the past (Parker and Zhang 2005; Piester et al. 2007; Zhang and Parker 2013; Mubarak and Dierikx 2011), the real causes of this effect are not fully understood and only some strategies for suppressing it have been developed up to now. Based on previous studies, the possible sources that lead to diurnals could be separated into two categories as follows.

The first category is related to the variations of the path delay difference (PDD) of the two signals traveling, in opposite directions, between the pair of TWSTFT ground stations A and B. For example, the Tx signal of station A transmits through the ground station equipment, free space, satellite, and down to station B. The Tx signal of station B also transmits in opposite direction to station A. The PDD of these two signals could be affected by a couple of factors, e.g., those mentioned as the following, typically with a daily pattern and thus cause a diurnal pattern in the TWSTFT results:

(a) The geometrical path delay difference: due to the gravity of the sun and maneuvers for position keeping, the geostationary satellite will swing around its nominal position. In combination with the time offset at which the signals from the ground stations arrive at the satellite, a geometrical path delay difference may be caused. The satellite daily movement data are not available to us, but the geostationary satellite is about 35,786 km above the earth surface, and

Table 1 Summary of instabilities in TWSTFT system

Instable terms and its potential sources	Amplitude of variation (ps)	Comments
Ionospheric delay difference (caused by TEC variation)	<100 (area) 150 ~ 300 (transoceanic length)	TEC varies with solar activity Elevation angle Larger difference for longer baseline
Sagnac effect (caused by satellite motion)	50	NICT-TL link via satellite IS-8
	10 ~ 150	NICT-PTB very long TWCP link (Fujieda et al. 2014)
Path delay difference (caused by satellite motion)	15	Satellite velocity Difference of arriving times of two signals
Doppler shift, and interference between multiple PRN codes (caused by satellite motion)	30 ~ 1500 (for 2.5 Mcps)	May be enlarged in the code-tracking process Depends on chip rates
Ground station delay (caused by outdoor temperature)	120 ~ 350 (variation range of outside temperature: 5 ~ 16 °C	Temperature coefficient: −22 ps/°C for TL's ground station (outdoor up/down converter) Measured by SASTIM in 2006 using a chip rate of 20 Mcps
Indoor equipment (caused by indoor temperature)	40 ~ 150	The variation of the room temperature may be up to 1.5 °C at TL
Troposphere delay difference	3	NICT-PTB very long TWCP link (Fujieda et al. 2014)
Phase jitter in satellite transponder	Unknown	
Phase jitter in instruments	0.2 (frequency converter)	
Phase variation in instruments	100	
2nd-order Doppler shift	2	

thus the related daily change of PDD was estimated to be approximately 200 ps (Zhang and Parker 2013).

The satellite movement also changes the magnitude of the relativistic Sagnac effect, while the nominal Sagnac effect of a baseline has been included in the link calibration.

(b) Temperature changes of satellite's transponders: Different surface parts of the satellite get exposure to radiation from the sun during the day and may cause temperature changes to the transponders and lead to delay changes. The effect could be canceled, if only one transponder is used in the TWSTFT link (such as the Europe-to-Europe links). The PDD change could be significant if two transponders are used (such as the Europe-to-USA links) for long-baseline comparison.

(c) Different frequencies adopted for the up- and down-link signals transmission: different frequencies used for the up- and down-link signals transmission: since the ionosphere is a dispersive medium, the path delays can be slightly different for signals transmitting to and receiving at the satellite. The troposphere is also dispersive but with a much lower level of influences at the adopted frequencies (Hobiger et al. 2013); by using Ku-band frequency, the change of ionospheric delay has been estimated to be less than 150 ps (Zhang and Parker 2013) and the tropospheric delay would be even smaller (Piester et al. 2007).
(d) Environmental condition of TWSTFT stations: Mainly temperature, but potentially also humidity and pressure, may cause variations of the delay between TX and RX signal paths in the equipment of ground station. Dependent on the equipment quality of the earth station, the environmental impact on the PDD could be as large as 1 to 2 ns.

The second category is related to signal interferences and multipath effects. For example, the signal interferences under the situation of multiple transmitting stations (such as Europe-to-Europe TWSTFT links) in a TWSTFT network, and multipath effects that may be caused by impedance mismatch in the equipment. These effects may, depending on the characters of the modem, lead to hardly quantifiable shifts in the time-of-arrival measurements of the RX module at the ground station. The diurnal pattern of the TWSTFT results look similar with those from the rather systematic effects discussed above, while it is difficult to distinguish them from each other.

Based on recent studies, adopting some newly developed TWSTFT technology, e.g., SDR method or SRS modem (carrier phase), could suppress the diurnal effect of many TWSTFT links. For intracontinental TWSTFT results, the improvement by adopting SDR receivers in reducing diurnals is significant, although some residual diurnals still exist. As for the TWSTFT SDR results of the inter-continental links, the diurnals are only slightly suppressed. Therefore, further studies are needed for full understanding of the entire causes of the diurnals.

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Part IV

Certified Reference Materials/Bharatiya Nirdeshak Dravya (BND): Need and Boon for Atma Nirbhar Bharat

Certified Reference Materials (CRMs)

22

An Introduction

Nahar Singh

Certified reference materials are required to ensure the quality and metrological traceability of products, validate the test methods/procedures and calibration of analytical instruments (Zakaria and Rezali 2014). Indian institutes/industries/R&D Sectors/accredited labs, etc., are procuring several reference materials for quality assurance. Due to the non-availability of indigenous reference materials, it is not easy to establish the quality of the procured product. The developed nations have their regulations and programs to procure quality products (Trienekens and Zuurbier 2008). The use of certified reference material can establish the quality of the procured product through validation of procedures and calibration of measuring equipment (Guide 2009). The stability, homogeneity and traceability of measurement ensure the quality of the reference material (Bunk 2007). The shelf-life of reference material is also an essential factor, as sometimes it expires even before usage because of low shelf-life, thereby creating an extra burden on users (Pauwels et al. 1998). Procuring from abroad is tedious; extra resources are involved, and the expiry time starts from the time it is dispatched. The transit time is also counted as a lifetime. To cater to the need of institutes/industries/R&D labs/accredited laboratories and avoid import, CSIR-NPL has taken the initiative to develop the certified reference materials (Bharatiya Nirdeshak Dravya: BND®) in-house and in collaboration with Reference Material Producers who are accredited as per ISO 17034: 2016.

Although, CSIR-NPLI has produced more than a hundred BNDs, in-house and in collaboration with RMPs, many more are yet to be developed (Pant et al. 2020). Still, there is significantly less awareness about the BNDs among institutes/industries/R&D labs/accredited laboratories and even the assessors. There is a need to create awareness about BNDs among the stakeholders. In this direction, BIS has taken the step to publish Indian standard specifications equivalent to ISO 17034:2016 (Aswal

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2020). Recently, the Bureau of Indian Standard (BIS) and the National Accreditation Board for Testing and Calibration Laboratories (NABL) has approved and encouraged industries to use Indian reference Materials (BNDs) instead of expensive imported CRMs (Pant et al. 2020). Another way of popularizing the BNDs is bringing out technical publications and publishing handbooks, book chapters, etc. However, these should not be very scientific in content but written in a manner that the accredited laboratory personnel, assessors, and even the common public can understand their utility. Therefore, these chapters are not scientific but presented so the reader can understand the reference material, why it is necessary to develop it, how it can be created, and where it is available. Also, the primary and secondary standards, their needs, and the distinction between them are discussed in the book. Thus, the book chapters will be a guide for the researcher working on reference materials which are required in almost all fields.

BNDs are the primary reference standards that ensure the reliability and comparability of the measurements as a bench-mark for quality. For global visibility, the BNDs produced by CSIR-NPL in different sectors are uploaded to the international database, i.e., COded'Indexation des Matériaux de Référence (COMAR) (Aswal 2020). In addition, Indian Reference Materials (BND) acts as an import substitute and is economically viable for many academic institutions and industries. Some of the major existing industries in India are taking the traceability or reference materials from abroad at a considerable cost. These reference materials have to be procured at a very high cost and need to be procured repeatedly after the validity of the reference material has expired. Thus, there is a substantial recurring loss of national resources. Also, we are procuring many materials, products, and sophisticated instruments and many times, we get substandard products and need to develop an adequately certified calibration facility to check them. These substandard materials are not only causing loss of national resources but also causing various kinds of pollution, as these materials, when thrown, degrade, and most are unusable.

Indian Reference Materials (Bharatiya Nirdeshak Dravya: BND®) division of CSIR-NPL is actively involved in the preparation of various BNDs in different sectors such as petroleum, cement, water, hardness, breath alcohol, pH, minerals, food etc. It may be noted that India faces a massive problem due to poor quality imports on the one hand and rejection of our export at foreign borders, on the other hand, causing a substantial economic loss to the country (Athukorala and Jayasuriya 2003). BNDs are poised to bring a rapid transition of India from a developing state to a developed state by strengthening its quality infrastructure and manufacturing. The CRMs are being procured at a very high cost. India is importing CRMs from other countries to the tune of several hundred crore rupees/annum. It indicates that the availability of indigenous CRMs (BNDs) will save huge import costs, enhance the export and thereby increase the country's GDP by many folds, reinforcing the "Make in India" and "self-reliance" pledge of the Government of India. The reference materials will boost the overall quality and safety of life in the country and international trade. For clarification, the definition of reference materials, certified reference materials, and Bharatiya Nirdeshaka Dravya is presented below:

Reference Material (RM): Material sufficiently homogeneous and stable concerning one or more specified properties, which have been established to fit its intended use in a measurement process (Ulberth-Buchgraber et al. 2011).

Certified Reference Material (CRM): The reference material is characterized by a metrologically valid procedure for one or more specified properties, accompanied by a certificate that provides the value of the specified property, its associated uncertainty, and a statement of metrological traceability (Beauchamp et al. 2020).

Bharatiya Nirdeshak Dravya (BND): A CRM issued by CSIR-NPL characterized by a metrologically valid procedure for one or more specified properties with associated uncertainty and accompanied by a certificate or certificate of analysis, providing a statement of metrological traceability (Pant et al. 2020).

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CSIR-NPL provides measurement traceability by calibrating equipment using its national primary standards and BNDs. BNDs are well characterized using state-of-the-art measurements to determine the chemical composition and/or physical properties. BND ensures the accuracy and reliability of measurement required for controlling the quality of products developed by the industry for discoveries in science, innovative technologies, pharmaceuticals, and monitoring the environment for better health and safety of the people of India.

The section is divided into nine chapters. One of the chapters describes the usage of BNDs for the calibration of the measuring instruments. CSIR-NPL has developed some BNDs related to the calibration of sophisticated instruments such as X-ray diffractometer, FTIR, and scanning electron microscopy, and the list is increasing rapidly. These BNDs are traceable to national standards. The need, how to produce, the market size, etc., have been described in this chapter. CSIR-NPL has also developed 18 nos. BNDs of the primary gas mixture, which are needed for calibrating various environment monitoring equipment. The primary reference gas mixtures of CO, CO_2, CH_4, and N_2 are presently available with CSIR-NPL. Another chapter discusses the role played by RMs and CRMs in standardization, QC and QA and their application for various purposes by the respective bodies – in the manufacturing and services sector – and international trade. The discussion reveals that ISO/IEC 17025:2017 embodies the spirit of the Plan-Do-Check-Act (PDCA) cycle. Taking the case of the Indian cement industry, the requirement of calibrations and RMs at various production and quality control stages is also discussed.

One of the chapters is related to cement and building material BNDs. CSIR-NPL started developing the cement-related BNDs in 2018 in collaboration with the National Council for Cement and Building Materials (NCCBM). It has contributed to the growth and development of the cement and construction industry, quality control, testing, method validation, technology transfer, productivity, and cost-effectiveness. India is the second-largest cement producer globally and has attracted

huge investments from Indian and foreign investors. The cement sector has to play a pivotal role in the Government of India's initiative of developing hundred smart cities, housing for all, dedicated freight corridor, water transport, concrete road projects, etc. In 2020, cement production had reached about 337 million tonnes and was projected to reach 450 million tonnes by 2022 (Karkun 2017). Due to the increasing demand in various sectors such as housing, commercial construction, and industrial construction, the cement industry is expected to reach 550–600 million tonnes per annum by 2025, contributing to a market size of approximately 10 billion USD (Nandamuri et al. 2017). India could become the leading clinker and grey cement exporter to the Middle East, Africa, and other developing nations in the next 10 years (Pareek and Pincha 2015). Therefore, there is a need for more stringent regulations on product quality where BNDs have to play a significant role. The current CRM market in India is approximately INR 5 to 6 crore per annum (Aswal 2020). Twelve BNDs have been developed and released in the market, and the list is multiplying. These BNDs have been sold in India, Nepal, Bhutan, Bangladesh, Africa, Hong Kong, and the Middle East. The details of production, the method of certification, etc., are described in detail in the chapter.

Another area where CSIR-NPL produces BNDs is food, chemical, and biologics. Food and water contamination is a significant concern globally. Unsafe and contaminated food consumption leads to food-borne infections, allergic reactions, and food intoxication (Gallo et al. 2020). Approximately 1.7 billion cases of childhood diarrhea diseases occur every year globally due to unsafe food and water (according to World Health Organization 2020) (Lee et al. 2020). The quality of the laboratory test is a deciding factor and an essential prerequisite that impacts health, the environment, and the economy. The BNDs are pivotal in providing uniform, traceable and reliable measurements. Food, water, and environmental testing laboratories extensively use mass spectrometry, chromatography, and other analytical techniques to check for adulteration, ensure health safety, and require periodic calibration using CRMs.

CSIR-NPL produces BNDs for food, water, and environment sectors with NABL accredited Aashvi Technology LLP, Ahmedabad, Gujarat and Fare Labs, Gurugram, Haryana, for direct benefit to the testing laboratory, educational institutes, R&D centers, industries, in-house labs. CSIR-NPL with Aashvi Tech LLP has developed 27 BNDs for food, water, and environment, and the list is growing fast to minimize the import of CRMs. Currently, the focus is on toxins, contaminants, vitamins, pesticide residue, and some specialized BNDs. BNDs for copper, nickel, lead, arsenic, mercury, cadmium, zinc, iron, and TDS standard solution, potassium hydrogen phthalate, potassium dichromate, sodium chloride, sodium carbonate, calcium carbonate, and pH buffer solutions 4.00, 7.00, and 10.00 are available with CSIR-NPL.

One chapter in the book is related to Petroleum BNDs. India is the second-largest refiner in Asia and the fourth largest globally (Nichols 2014). Its crude oil import was US dollars 101.4 billion in 2019–20. The economic progress of India is closely linked to its energy demand. Therefore, the need for oil and gas is expected to grow more, making the oil sector more attractive for investment. It makes it indispensable

to have a stringent check on the quality of imports and exports of petroleum products. The results from lab tests play a significant role in ascertaining the quality of petroleum products. Using BNDs ensures that laboratory instruments perform to the standard of predefined value, assuring laboratory tests' accuracy. With the non-availability of indigenous certified reference materials, domestic refiners have been importing the materials from other economies. CSIR-NPL has partnered with HPCL and BPCL to produce 29 BNDs, and the list is growing. The petroleum BNDs are expected to save foreign exchange and time, avoiding delays in receiving imported reference materials. Also, with the availability of BNDs to the Middle East, the African continent will love to expand business with us. The petroleum BNDs include BNDs for the various concentrations of sulfur, flash point (PMCC and Able), smoke point, distillation standards, pour point, color saybolt, color ASTM, total base number, total acid number, FIA-aromatics, olefins, saturates, density, kinematic viscosity at different temperatures, freezing point, etc.

Yet another chapter deals with the precious metal BNDs. The Global precious metal market was valued at USD 182.1 billion in 2019 and is expected to grow at a compound annual growth rate of 9% in revenue from 2020 to 2027. Asia-pacific is the primary producer and consumer of gold. The metal is significantly consumed in China and India. As the value of the precious metal is determined from a measurement of fineness, most countries have established laws regulating the manufacture and sale of precious metal items to protect the market and consumer. Different countries have varying standards and regulations that govern the manufacture and sale of good jewelry (Kannabiran and Bhaumik 2005). They differ in the marking systems, allowable tolerances, and fineness standards, among other things. With so much variation, it is easy to see why international trade in gold jewelry can be hampered by the standards and regulations of individual countries.

BND 4201 is India's first high purity (99.99%) gold reference standard manufactured by CSIR-NPL and Government Mint, Mumbai, in collaboration with Bhabha Atomic Research Centre, BARC, Mumbai (Pant et al. 2020). It will benefit the consumer and public to ensure the purity of gold. The BND 4201standard gold bars are 25% cheaper than the imported version. BNDs of other purity percentage gold will be prepared in collaboration with GoI Mint and Jalan and Company. These BNDs are available in the market, allowing jewelers to use instrumental methods rather than the conventional fire assay to ensure purity. The availability of precious metal BNDs at a low cost will enhance the export of precious metals.

One of the chapters deals with the mechanical properties of materials. Metals and the automobile sector are the major contributors to the Indian economy in international trade (Sahoo and Ashwani 2020). The foremost requirement for barrier-free metal and auto parts trade is to meet international quality standards. Material properties, such as strength, ductility, hardness, and wear resistance, are essential parameters for designing a product that needs SI traceable service measurements (Canale et al. 2008). CSIR-NPL with the Global PT provider has started producing BNDs for hardness, strength, and other parameters. Three BNDs of hardness have been released, and others are coming soon. The availability of the BNDs will enhance the capabilities of Indian laboratories and help the automobile, metallurgical

testing lab, steel manufacturing, railway, defense, and aerospace industry in a big way and thereby enhance the export in these sectors.

One of the chapters deals with pesticides. According to a source, 1.2 billion people died worldwide in 2019 due to bacterial resistance (Dadgostar 2019). It was reported that pesticides poison 385 million people and cost 11,000 death every year in India (Boedeker et al. 2020). India also has an unclear picture of the exact burden of all food-borne illnesses arising from pesticide contamination and excessive antibiotic use. Their management is mainly unregulated as unsuspecting consumers freely use their residues. Numbers of pesticides are used in agricultural production to control pests and weeds (Pimentel 2009). Despite having several benefits from pesticides and antibiotics, it is highly essential to monitor and control their amount in the food chains as they are the main culprit for potential toxicity to humans and animals (Okocha et al. 2018). Comparability and the accuracy of the measurement results depend mainly on certified reference materials appropriate for a particular application. To support measurement laboratories, it is urgent to develop cheaper CRMs for pesticide analysis in foodstuffs, serving as a representative material for vegetables or fruits with a high-water content.

One chapter deals with metals and alloys. Alloys are a mixture of pure metals and are mixed to improve their performance and properties (Zhang et al. 2012). A summarized detail of numerous alloy systems is discussed in this chapter. Metal alloys are classified into two categories, ferrous and non-ferrous. Different steels thus fall into the first category, and popular alloys like brass, bronze, invar, duralumin, hindalium, etc., fall into the second category. However, any alloy or material must be checked for its quality, production and performance repeatability; thus, standardization is necessary. Any such standardization involves checking and benchmarking the acceptable composition range of that material. When mixed, many properties depend on the individual metals' crystalline phase, so these also deserve to be bench-marked. In addition, their properties depend on the thermal cycling processes, heat treatment procedures and associated microstructure that result in, i.e., on the thermodynamics of their preparation process. These preparation processes have to be repeatable and, therefore, need to be standardized, i.e., measured, quantified and catalogued. This process will produce standard alloys or certified reference materials (CRMs). They may be compared with any subsequent batch-to-batch product to check the expected properties' consistency. Different National Metrology Institutes (NMIs) worldwide have been thus working on standardizing different alloys of interest to grow their economies (Tzalenchuk et al. 2022). The chapter summarizes the different standard alloy samples available with the major NMI's of the developed world like NIST-USA, PTB-Germany, NMIJ-Japan and NPL-UK or their surrogates. The American Society for Testing of Materials (ASTM), American Iron and Steel Institute (AISI), and Society of Automobile Engineers (SAE) have independently developed a lot of standard data for several alloys, irrespective of whether similar data have been published or not by any NMI (Lingaiah 2003).

Consequently, several alloy makers follow the ASTM standard data for trade purposes when NMIs do not have specific standards of interest. Based on

commercial prospects and demand, they are the ones that are waiting to be standardized in future by any of the global NMI. In the case of emerging economies like India, irrespective of what other NMI have or have not worked on, this write-up discusses how any alloy as a case study can be successfully standardized well for all its parameters. It shall be worth the effort as that knowledge may be commercially exploited for the local and global market and extrapolated to make other alloy standards and quality alloys in due time.

Medical devices can be instruments, apparatus, machines, appliances, implants, and reagents for in-vitro or in-vivo use that are intended to be used, alone or in combination, to address humans' medical requirements (Sharples et al. 2012). When a device or material performs with an appropriate host response in its specific application, it is said to have biocompatibility (Williams 2008). In medical device development, appropriate physicochemical and biological evaluations are devised to establish biocompatibility (Bernard et al. 2018).

Qualified biomaterial with proven safety and efficacy is most important to its functionality in the human physiological environment (Sivakumar 1999). Although critical performance analysis in pre-clinical evaluations can control the risk elements of such materials and devices, the databases of regulatory authorities report a considerable number of complications and adverse events annually (Tarricone et al. 2014). The main factors that challenge the process of biomaterials evaluation for qualifying them for medical use are non-uniformity in regulations, the difference in material development and methods of their evaluation, preconceived post-marketing data, the cost involved, etc. (Herrera Morales 2015). Established biocompatibility characteristics through documented material evaluation matrix within a risk-based quality management platform are essential for clinical acceptance (Vinodkumar et al. 2022). Selecting strategies for evaluating biomaterials could be performed wisely to address the regulatory requirements. Qualifying biomaterials to meet regulatory requirements insist on traceability to international standards in physicochemical characterization and biological evaluations. Appropriate Certified Reference material (CRM) is identified as one of the critical elements to address this requirement.

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Bharatiya Nirdeshak Dravya for Antibiotics and Pesticide

23

Reference Materials for Food Analysis

Arvind Gautam and Nahar Singh

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Abstract

Antibiotics and pesticides have a great impact on human life. However, antibiotics fight against the bacteria and prevent as well as treat infections; on the other hand, pesticides are reasonably essential as they not only help protecting crops from pests, diseases, and weeds but also increase the production to almost double. Without using antibiotics and pesticides, there will be a huge loss in terms of lifesaving and food production, respectively. The excessive uses of both create many acute health problems when disposed of into the environment through sewage and other sources. The worst part of antibiotics and pesticides is that they do not degrade over the period of time and remains in soil, water, and other environmental components. Finally, their residues reach to the food supply chains. The misuse of antibiotics build a resistance against the particular drugs made to help for fighting against the bacteria. In the same way, pesticide excessive uses have resulted in contaminations in food items, soil, and water bodies, leading to serious health implications. To cope with these environmental issues, certified reference material of both can play a very crucial role as it provides reliable measurements and benchmark for the quality assurance. CSIR-NPL has started Bharatiya Nirdeshak Dravya (BND®) program for the development of the certified reference material to mitigate the needs of the country in terms of quality assurance as per ISO 17034.

This chapter describes the importance, national need, and various steps involved in the development of certified reference material (BNDs) of antibiotics and pesticides. The metrological traceability of BNDs established linked to SI units is explained. Further, the metrological perspective, such as testing, calibration, method of validation, repeatability, reproducibility studies, control chart preparation, interlaboratory comparison, quality control, and quality assurance, has been discussed in detail. Further detailed discussion on BND process starting from planning to intended parameter with matrix, homogeneity, stability, assignment of property value with its associated uncertainty, and certification process are highlighted. The synergy among accredited bodies, stack holders, and regulatory bodies with NMIs is emphasized. The impact of BNDs in the formulation of regulations in trade is narrated.

Introduction

Due to rapid industrialization in the sector of agriculture and pharmaceutical, it has put an immense chemical burden on the environment. Pesticide contamination alone causes ≈ 3 million cases of poisoning worldwide, leading to death around 220,000 annually (Organization WH 2009). In the same manner, antibiotics put global burden

and staggering death toll of drug resistance bacteria. According to a source, 1.2 billion people died worldwide in 2019 due to bacteria resistance; it was reported that 385 million peoples are poisoned by pesticides, which causes 11,000 death every year in India (Walker et al. 2013). This is all because of limited understanding of estimates of diseases and deaths from all food safety-related issues. India also have unclear picture on the exact burden of the rise of all foodborne illnesses due to contamination of pesticide and antibiotic excessive uses, and their management is largely unregulated as their residues are freely used by unsuspecting consumers (Fafchamps et al. 2008). A numbers of pesticides are used in agricultural production with the main purpose of controlling pests and weeds (Meissle et al. 2010). Despite having several benefits from pesticides and antibiotics, it is highly essential to monitor and control their amount in the food chain as they are the main culprit of a potential toxicity to humans and animals (El-Shahawi et al. 2010). On account of this, individual maximum residue levels (MRL) for pesticides are regulated in food and feeds (Regulation (EC) No. 396/2005) to ensure that food products are safe and appropriate for consumption (Muhammad 2019). In the same way, MRL limit for antibiotics should also be regulated as per Food Safety and Standards (Contaminants, Toxins and Residues) Regulations, 2011. In a fast-growing country like India, in order to achieve the green growth policies and sustainable development goals, there is an urgent need for the adoption of safe production practices in agriculture. This can be achieved through scientific intervention and management of supply chain systems (Malisch 2017). The literature reveals that more than 2 million tons of pesticides is consumed worldwide; however, we are unable to decrease the net loss around 30% production of total agricultural produce. India has also lost approximately annually Rs 6000 crores due to the impact of pests, weed, insects, and rodents on agriculture produces (Bunch et al. 2003). The negative impact of various microbes on crops is continuously increasing due to resistance developed against several antimicrobial agents (Noyes et al. 2009). The excess use of synthetic pesticides and antibiotics causes detrimental effects not only on the environment but also affecting living organisms through the consumption of contaminated (Mahmood et al. 2016). The prolonged consumption of food and water contaminated by pesticides and antibiotics may cause several diseases like cancer, endocrine disruption, birth defects etc. long-term health effects and also like abdominal pain, loss of appetite, and cancer (Mostafalou and Abdollahi 2013). The Centre for Science and Environment (CSE), India, has reported the presence of several antibiotic and pesticide residues in human tissues of Indian people. Apart from raw agriculture food stuffs, pesticide and antibiotic residues have also been found in packed food items like soft drinks, bottled water, etc. The types of pesticides used in India as per the year 2019 is depicted in Fig. 1.

Food safety and globalization is attracting the attention of the researchers as trade is becoming complicated due to food safety norms across the globe (Johnson and Kumar 2003; Henson et al. 2000). To mitigate these issues, there is an urgent requirement for reliable measurements which will provide quality check for trade barrier and internal consumption (Umali-Deininger and Sur 2007).

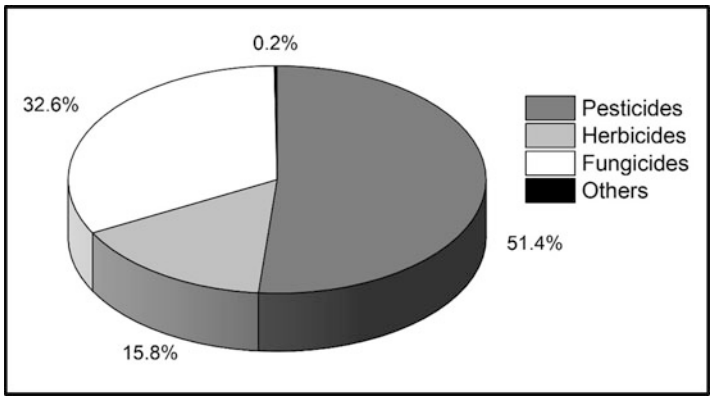

Fig. 1 Composition of pesticide uses in India 2019

Also, the certification of food products and continuous watch of the market are part of the policy. There is a consistent need for the testing of agro-products to evaluate the presence of contaminants like pesticides, antibiotics residues, etc. to ensure the quality of the food products. As a matter of fact, various inspection and testing laboratories associated to check the residues and other toxins persevere to increase rapid growth globally (Feigenbaum 1991). Worldwide, several countries like the USA, Australia, Japan, etc. have developed their own mechanism related to legal and regulatory control on food stuff being procured locally or from other countries (Maskus 2000). Recently Turkey rejects Indian durum wheat, i.e., shipping 56,877 tons consignment, due to phytosanitary concerns. Referring to this, certified reference materials that show their importance in terms of calibration of measuring instrument by providing measurement traceability are the most effective tool to minimize such disputes (Vesper and Thienpont 2009). The CRM plays a crucial role as quality control tools for measurement results (Campana 2001). However, it is important that usage of such matrix should match the reference material worldwide for comparability, accuracy, and traceability and to resolve the dispute during acceptance of the products (Tahoun et al. 2019). CRMs serve also as a lead material for testing and calibration of analytical tools which require to analyze residues of pesticides and antibiotics in vegetables or fruits with a high-water content (Obadina and Olotu 2017). To boost the national quality infrastructure and backing measurement laboratories, it is an urgent need to develop cheaper CRMs for pesticide and antibiotic analysis in agricultural food stuffs and animal food stuffs.

Pesticide's Effect on Human Health

Exposure to pesticides can cause various health effects; the most prominent effect is on the nervous system. General health effect may vary from short term to long term, called acute effects and chronic effects that can happen depending upon the duration

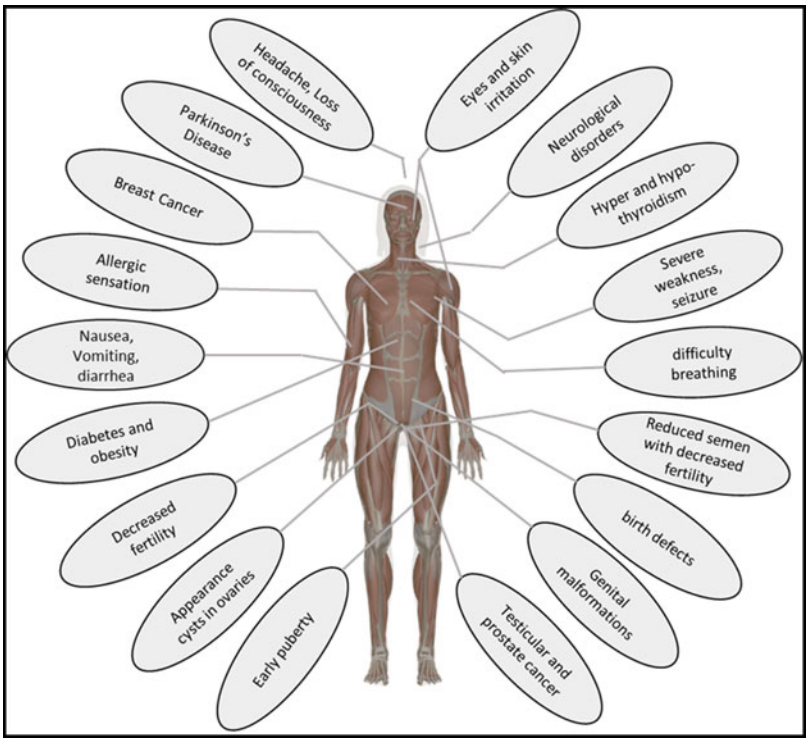

Fig. 2 Pesticides and human health

of the exposure (Damalas and Koutroubas 2016). Several pesticides are banned worldwide, but they exist in the environment and known as persistent organic pollutants (Leino et al. 2013). There are several pathways by which pesticides enter in the human body. Some of the examples of acute health effects are diarrhea, rashes, blisters, stinging eyes, blindness, nausea, dizziness, and death (Bhatia et al. 2014). The short-term and long-term effect of pesticides on human health are shown in Fig. 2. Human being can be exposed to pesticides into two ways: (i) through occupational and (ii) passively through non-occupational (Vera-Herrera et al. 2021). Generally, there are three ways through which pesticides can enter into our body, i.e., through skin, by inhalation, by mouth during mixing, and application. Occupational exposure may be carried out during industrial production, carriage, marketing, and application. The dermal absorption impact depends on the concentration, duration of exposure of pesticides, temperature, and humidity (Moonen et al. 2020). The absorption rate is very high through the skin, groin areas, eyes, ear canal, etc. (Shah 2020). Sometimes, pesticides are kept in food and drink containers, so accidental oral exposure to pesticides may be consumption of water from unlabeled containers. Pesticides may be found throughout the human body through the circulatory system and may be expelled from the body through urine, skin, exhale breath, etc. (Overholt and Castleton 1989). Respiratory exposure in human beings also

occurs due to inhalation of pesticide-contaminated aerosol generated during fumigations. The pesticides change into toxic gas when get contacted with moist air (Dowling and Seiber 2002). Worldwide, several tragic accidents have been reported; one example among them is Pakistan, where around 2800 workers got affected by malathion during spray to control pests (Tariq et al. 2007). The non-occupational exposure has also been reported in human beings by consumption and usage of pesticide-contaminated food, water, air, clothing, and tools because of precautionary fumigation or residual indoor sprays to control pest (Damalas and Eleftherohorinos 2011). The other common sources through which pesticides may enter in human skin are squish, tumble, and during cleaning of contaminated instruments (Organization WH 2020). Besides these, some other example of non-occupational exposure has also been reported through treatment of ectoparasite in pets (Larry et al. 2008).

(a) Types of Pesticides

Pesticides are chemicals and are grouped according to the types of pests which they kill including insecticides (insects), herbicides (weeds), larvicides (larvae), bactericides (bacteria), fungicides (fungi), nematicides (nematodes), and rodenticides (rodent and mice). The literature suggests that pesticides can be classified based on functional groups, chemical classes, mode of action, and their toxicity (Abubakar et al. 2020). Pesticides exhibit potential applications in the area of agriculture, medical, veterinary, etc. to control microbial infection. Pesticides may be organic and inorganic in nature. Organic pesticides further can be divided into natural or synthetic form (Lau et al. 2015). Organic pesticides contain carbon chain and having more complex structure than inorganic pesticides (Weed and Weber 1974), whereas inorganic pesticides may be in metallic and mineral form like copper, sulfur, and sulfates of copper, iron, etc. Pesticides may be further divided on the basis of application like insecticides and miticides which are utilized to control pest in agriculture, forestry, medical, etc., fungicides to control fungal infection, herbicides to control weeds, algaecides to inhibit growth of algae, rodenticides to kill mice and rats, etc.

On the basis of cited literature, it is essential to know the concentration of pesticides in various matrixes. There are several food matrixes, and they may contain numbers of contaminants like organic species (pesticide, antibiotics, fatty acids, dioxins, nutritional constitute, etc.), inorganic species (metal and non-metal element), and microbial (algae, bacteria, fungi, etc.). The reference material is required for the validation of method, calibration of instrument, and traceability compliance. The matrixes matching the reference material is a prerequisite requirement for precise and accurate determination of targeted species. The determination of pesticides in various matrixes can be carried out by using reference material, which may be in the form of high purity or matrix match.

Presently, it is difficult to get matrix match reference material. In such cases, calibration of instrument can be done by using high purity reference material or by adding known quantity of reference material to the test sample of nearly same matrix

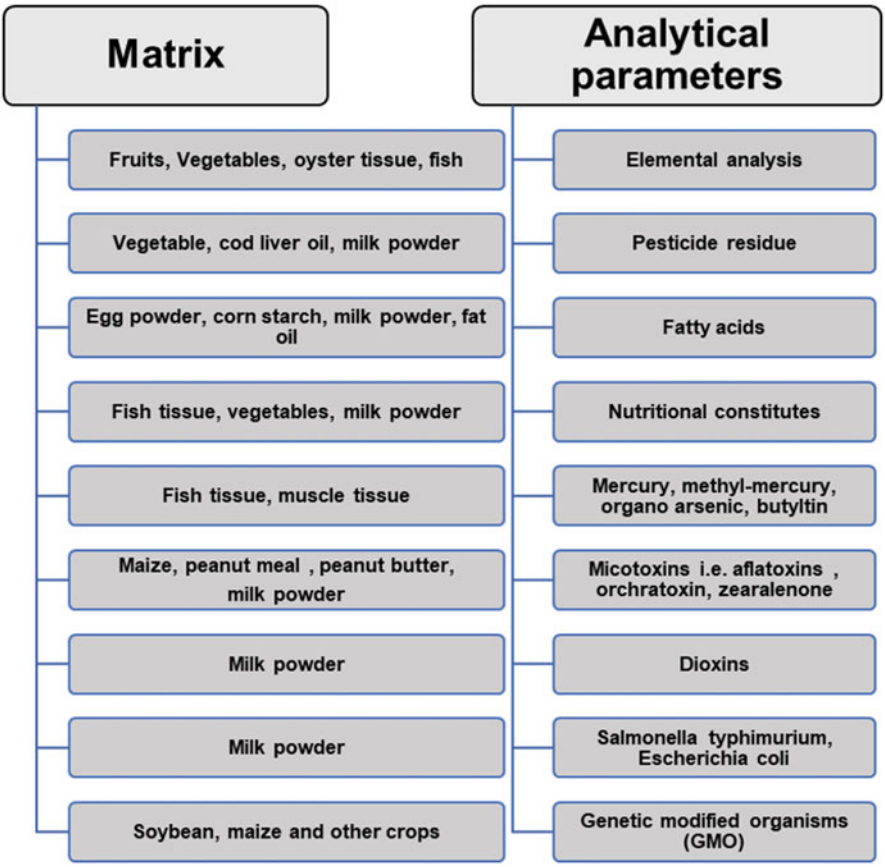

Fig. 3 Matrix and analytical parameters

along with the process blank. Figure 3 depicts some matrixes of food items along with important respective parameters.

(b) Types of Antibiotics

An antibiotic is an antibacterial material which fights against bacterial infections. The antibiotics are utilized as precautionary measure or treatment against bacterial and some protozoan infections. Antibiotics may kill or inhibits the growth of bacteria; however, they are not effective against viruses. The selection of antibiotics to patents is based on empiric or definitive therapy. So, on the basis of mechanism of action, the antibiotics can broadly be divided into two groups, the first one which kills bacterial infection and the second one which inhibits bacterial growth. Antibiotics can also be divided into several categories based on their chemical structure

spectrum of activity. For example, the antibiotics have main groups including β-lactams (penicillins, cephalosporin etc.), macrolides (azithromycin, erythromycin etc.), fluoroquinolones (ciprofloxacin, levofloxacin, etc.), tetracyclines (oxytetracycline, doxytetracyclin, etc.), and aminoglycosides (streptomycin, gentamycin, etc.). These are the subclasses of β-lactams, macrolides, fluoroquinolones, tetracyclines, and aminoglycosides, respectively.

Pesticides and Antibiotics in Our Path

Pesticides are usually released into the environment by several means in order to abolish these pests, but residues from these toxic chemicals also turn out in the air, water, and even in food stuffs (Carvalho 2017). Generally, pesticide is any substance used to kill, repel, or control certain forms of plant or animal life that are considered to be pests. Pesticides may be biological, chemical substances, or a mixture of substances in nature (Saravi and Shokrzadeh 2011).

Some targeted pests include insects, gnawers, weed, plant pathogens, mollusks, nematodes (roundworms), molds, fungus, and microbes that destroy assets, cause nuisance, spread disease, or are major cause of vectors for diseases. Pesticides and antibiotics are both continuously released in three essential commodities, i.e., foods, water, and soil. As mentioned in Fig. 4, the residual toxic chemicals are transmitted to raw material and finally go in to consumer products, causing so many diseases and

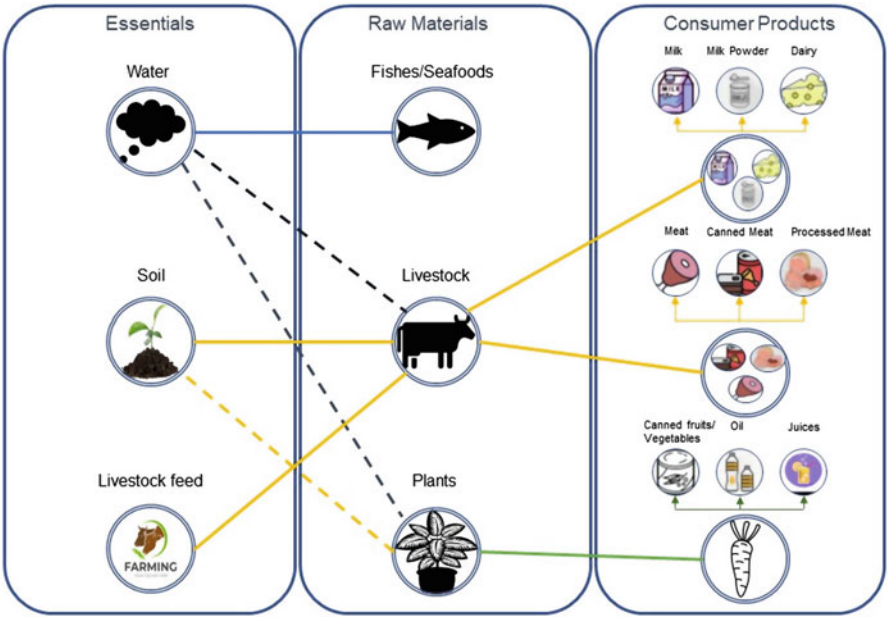

Fig. 4 Pesticides and antibiotics and their path in living things

drug resistances. Pesticides and antibiotics standards help us to precisely screen soil, water, and food quality in agreement with national and international regulations.

Pesticides and Antibiotics CRMs and Their Importance

The global population is growing very fast and expected to be reached ≈ 9.8 bn by 2050, which will lead greatly to increase in food demands. Therefore, combining of growing global demand and India's strong legacy points will come out as an opportunity for intensifying both commodity and value-added exports (Hertel 2011). To fulfill the demand of the growing population, it is important to enhance the production capacity. The enhancement in production yield can be achieved by using a precautionary measure like using appropriate pesticides to control pest. The literature revealed that thousands of varieties of insects, pathogen, and weed reduces more than 30% crop production worldwide (Reddy 2015). The major reduction has been observed in the production of cereals (32%), vegetables (54%), and fruits (78%) without using pesticides. So it is evident from the literature that pesticides play a crucial role in the enhancement of crop production, which is also important for future food security (Hashemi et al. 2014).

In the same way, standard reference materials for antibiotics, namely, International Standards for Antibiotics (ISA), are very much necessary for establishment of standardization and quality control of antibiotic and pharmaceutical products. It is an urgent need of the country to protect not only from counterfeit products but also provide quality assurance in pharmaceutical and food sectors. There are a huge market of the antibiotics and pesticides CRMs as we have more than 10,000 pharmaceuticals and about 150 industrial units manufacturing pesticides and about 500 industrial units engaged in formulations of pesticides. The effective evaluation of the quality of analytical results is very essential and can be achieved by matrix certified reference materials (CRMs) and proficiency testing (PTs). The CRMs come with documentation that mentioning metrological traceability, property values, and related their uncertainty, batch number, lot number with expiring date, etc.

Certified Reference Materials (CRM) and Quality Control

To support measurement laboratories, CSIR-NPL has mandate to produce BNDs for pesticides and antibiotics for residue analysis in crops, vegetables, or fruits with a high water content. The identification of certified pesticides is carefully selected based on different criteria such as their general use (insecticide, fungicide, and nematicide), being typical of different chemical classes (neonicotinoid, organochlorine, organophosphorus, azole, benzimidazole, dicarboximide, strobin, pyrethroid, and N-methyl carbamate), or having an authorized or banned use in India. Consideration is also given to different degrees of difficulty for their analysis when present in a food stuff. The mass fractions of the following pesticides with their associated uncertainties need to be analyzed by using the certified reference materials:

chlorpyrifos, carbendazim, acetamiprid, cypermethrin, diazinon, alpha-endosulfan, fenitrothion, azoxystrobin, imazalil, etc. (Fantke et al. 2014). Certified reference materials (CRMs) are very essential and play a key role in quality control by establishing the metrological traceability for measurements; they are also very important to confirm measurement results reliability (De Bièvre et al. 2011). In total, CRMs are very much essential to support international trade and national legal metrological control of the country. Highlighting the main purposes of CRMs is extensively used in calibration of equipment and assessment of measurement methods/procedure. In accredited laboratories, it is mandatory to use the CRMs for various purposes like internal measurement activities, in proficiency testing (PT) activities, and in legal metrological activities (Kaarls 2018).

It is observed that a country needs consistent and trusted measurement system and establishment of quality infrastructure for easy certification of food items for import, food product registration, market observation, export rejection, etc. Various examples are reported in newspapers regarding the rejection of consignment for foods, tea, and other essential commodity due to high level of residues of pesticides and antibiotics in agro-products. There is need to bridge the gap between measurement and calibration and standard reference materials. It is reported that majority of CRM (10,000 crores) are generally imported from abroad every year. To check the MRL of pesticide and antibiotic residues, mycotoxins and other toxic contaminants with indigenous CRMs need to be developed to make country self-sustain in this broad area (Kovač et al. 2021). To align with national mission, the number of RMPs (in association with NMI) and testing laboratories (related to quality assurance) is continuously increasing in the country. To justify the needs of CRMs, countries like the USA, the UK, Japan, and China have stringent legal and regulatory control and regulatory policies on food products. Therefore before exporting the foods and other agro-products it is need be monitoring the all the parameter later may cause for the rejection. All we can say, CRMs are much needed for the calibration of measuring instrument and regulatory control (imposed by legal entity) of measurement results when it is subjected to establish metrological traceability. The precise and exact measurement in food is imperious to minimize pointless disagreements (De Bievre 2008).

Pesticide Metabolite Standards

It so happened that active compounds of some pesticides may be either converted by the target insects and plants or may be degraded in the environment. In short, the monitoring of pesticide metabolites is required. A wide range of pesticide metabolite neat standards and solutions in various solvents are available in the market. Testing laboratories and food industries can use metabolites standards for the validation of certain analytical methods and can take effect the control of their measuring system. To demonstrate the metrological traceability of the certified values of the CRMs,

NMI needs to establish the degree of equivalence for the analytical methodology which is used with other NMIs through the participation of inter-comparison programs organized by CCQM. For the effective metrological control of the quality of food stuffs, the government has taken the initiatives for production of these CRMs in compliance with the relevant international standards such as ISO Guide 34 and ISO/IEC 17025.

Pesticides and Antibiotics Reference Material (RM)

Reference materials are sufficiently homogeneous material and stable in terms of specified properties; these properties have been established to be fit for its intended use in measurement of nominal properties. Although RM do not provide the same level of accuracy as certified reference materials, their advantage is that RMs are significantly easier to produce. RM is also compliant with ISO requirements and for less exacting requirements on traceability; RM can offer an excellent cost-effective alternative to CRMs. It stated that reference materials are:

- High purified.
- Qualitative and quantitative analysis.
- Detailed description about its properties.
- Certified through National Standardization Technical Committee.

Certified Reference Material (CRM)

The reference material when, represented by its documentation issued by an authoritative body and demonstrating one or more specified property values with associated uncertainties and traceabilities, using valid procedures called certified reference material. Certified Reference Materials (CRMs) occupy the second highest rank in the hierarchy of reference materials. In other terms, Certified Reference Materials are considered to enables the highest level of accuracy, uncertainty and traceability to an SI unit of measurement.

Metrological Traceability

Traceability refers to the value of a standard where it can be related to stated references (national or international standards) through an unbroken chain of comparisons, all having stated uncertainties (ISO). The term “metrological traceability” is described as “Property of a measurement result whereby the result can be linked to a reference through a documented unbroken chain of calibrations, each contributing to its measurement uncertainty.” It is well known that the measurement uncertainty of the property value of a reference material employed in a measurement process will contribute to the uncertainty of the final measurement but should contribute less than

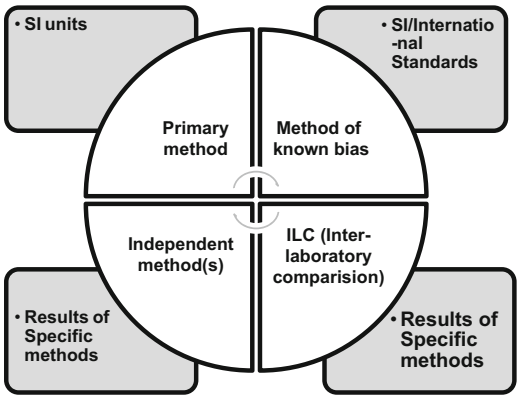

Fig. 5 Measurement methods and associated traceability

one third of the overall measurement uncertainty. In universal hierarchical system, usually, used for assigning property values to product/materials and even if not stated, their traceability can be described in Fig. 5.

Legal Metrological Control

The Central Government of India has enforced Legal Metrology Act 2009 to maintain the units for measurement. The act fully implemented on first April, 2011, in India. The main objective of Legal Metrology is to ensure public guarantee from the point of view of security and accuracy of the weighments and measurements. Figure 5 describing the overall legal metrology flowchart with associated activities which directly contribute to metrological assurance (Fig. 6). The legal metrological control includes:

(i) Measuring instruments and their legal control.
(ii) Metrological management.
(iii) Metrological proficiency.

Selection of CRMs

The ISO accreditation of a CRM guarantees accuracy, identity, and quality the material without additional contamination or error. A certified reference material provides not only the instruction to maintain homogeneity and stability but also provides traceability required under ISO guidelines. Laboratories have many choices for reference and calibration standards but the selection of the correct reference material can be difficult in some cases. However, some of them may be bestowed priority over others; the other parameters are as follows:

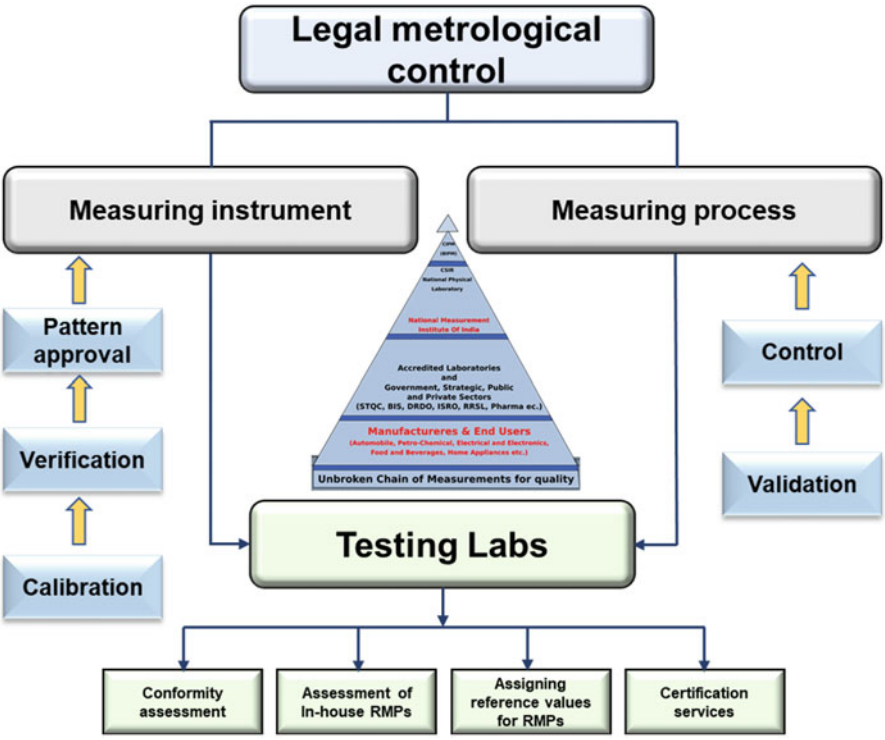

Fig. 6 Legal metrological control that requires the use of pesticides and antibiotics CRMs

Fit for Purposes

The CRMs have a matrix in a physical state and form that is as close as possible to those of the material subjected to the measurement process. Fit for purpose decisions in selection of reference materials can depend on several factors ranging from regulatory requirements, level of accuracy, availability, and type of testing applications. Apart from this, it is also important to consider the other factors like what types of tests, i.e., equipment calibration, method validation, identity, stability assessment, and internal quality control. Certified reference material should have properties at the appropriate level so that it can fulfill the purpose for specific requirement.

Characteristics

A fundamental term in metrological traceability is the measurement of SI units. The CRM needs to be meaningfully compared, across difference places, at different times, by different people, using different equipment. The measurement result must be related to a reference through a documented and unbroken chain of

calibrations, tracing back to the SI unit of measurement. The main important factor is the uncertainty associated with the certified value which is stated as quality parameter and reliability. The uncertainty budgets are an additional aid for specifying the extended measurement uncertainty. For example, the certification data should state accuracy, consistency, homogeneity, purity, and stability along with the expanded uncertainty (U) using a coverage factor $k = 2$ which specify the level of confidence nearly 95%. The precision and trueness requirements must be well matched with associated uncertainty of the certified value (Fig. 7).

Legal Requirements

The CRM should be able to fulfill the legal requirements on the precision and trueness for some special analyses. In some of the countries, there may be required special requirements on the packaging and label specially to mitigate the climatic conditions for CRMs to be imported from aboard.

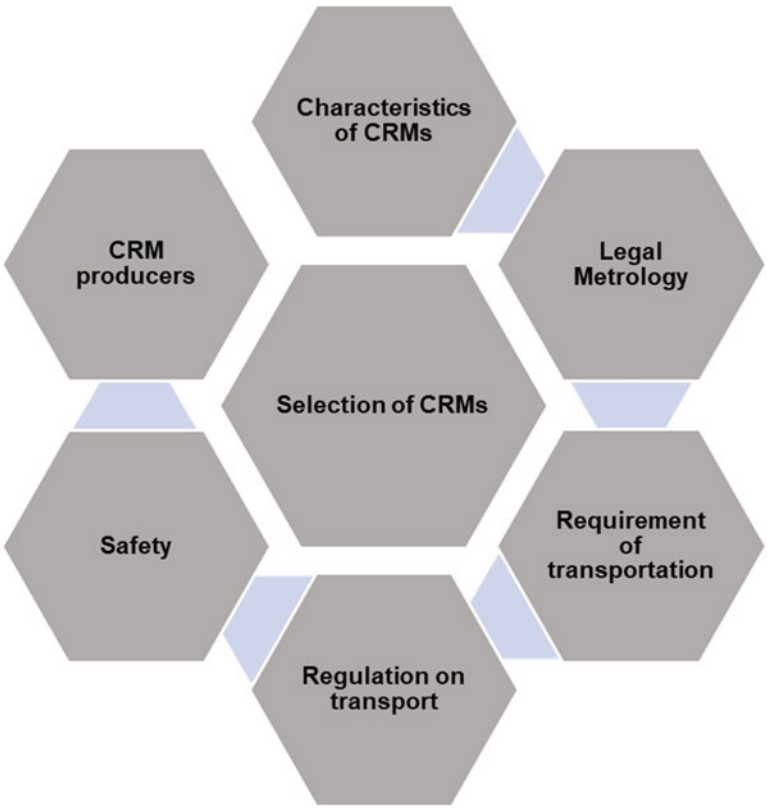

Fig. 7 Aspects considered in selecting CRMs

Regulation on Transportation

International regulation implies for the transportation of CRM which falls under stringent legal policies and norms. Several restrictions have been imposed on importing of certified reference material as they may be biological materials. For import of the CRMs, it is requiring a legal permit as they belong to restricted class categories.

Safety of CRMs

Certified reference material of pesticides and antibiotics requires a lot of attention in terms of safety and stability. Utmost care should be taken to ensure the safety and stability of materials especially for CRMs which are biological in nature, e.g., the materials need to be properly irradiated before packaging.

CRMs Producer

The curiosity attracts the attention of the users who can make the certified reference material. The answer is majority of the certified reference material are developed by the national metrological institutes (NMI) and thereby offered in the market. CSIR-NPL as an NMI of India and as per the mandate has taken the lead role for the developed in-house CRMs as well in association with reference material produces. The RMPs must have the ISO:34 accreditation is the general requirement for the competence to produce the reference material. For any National Measurement Institute (NMI), the CMC (calibration and measurement capability) the most important recognition among other NMI and can be recorded on Key Comparison Database (KCDB) set up by BIPM (http://kcdb.bipm.org) (Bock 2014). All we can say, the essential requirement for the CRM producer is ISO 17025 and 17,034. The accreditation criteria are set out in ISO 17034. The core competence of an RMP in producing reference materials includes the following:

- Functioning of a management system.
- Technical competence of staff.
- Suitability of the production plan.
- The RMs produced are sufficiently uniform and stable.
- The certification values are metrologically traceable to a reference.
- Fitness of the RM distribution process.

It is always advisable to CRM users that they should get the overall information of the CMC through KCBD. It will be help in preferring the which institute's products should be chosen.

Development of Pesticide and Antibiotics Reference Material

CSIR-NPL has started Bharatiya Nirdeshak Dravya (BND®) program for the development of various Indian reference material with trade name BNDs at NPLI. CSIR-NPL developed in-house BNDs as well as in association with reference material producers. The process of developing reference material is very cumbersome and it includes preparation of samples to analysis, which are the most common processes with PTs. The major process and steps involved are as follows:

Homogeneity Testing

As per the ISO Guide 35, homogeneity needs to be assessed by quantifying target pesticides and antibiotics prepared and chosen for CRMs and/or PTs. Homogeneity includes both within and between bottles. By using one-way analysis of variance (ANOVA), well-known statistical approach is applied to evaluate within-bottle and between-bottle differences (Kim et al. 2018). The basic method for evaluation of homogeneity is F-test. Any sample is considered to be homogeneous if the F-test values are smaller than those of the critical values at the 95% confidence interval. Standard deviations between bottles (S_{bb}) and within bottles (S_{wb}) are calculated using Eqs. (1) and (2) in accordance with ISO Guide 35 (Ellison and Botha 2018) as follows:

$$S_{bb} = \sqrt{\frac{MS_{among} - MS_{within}}{n}} \tag{1}$$

$$S_{wb} = \sqrt{MS_{within}} \tag{2}$$

where MS_{among} = squares between groups.
MS_{within} = within a group.
n = the number of measurements per bottle.

According to ISO Guide 13,528 by using Eqs. (3)–(5), the values of S_w and S_s are determined as follows:

$$S_x = \sqrt{\sum \left(X_{t\ldots} - \overline{X}\right)^2/(g-1)} \tag{3}$$

$$S_w = \sqrt{\sum w_t^2/(2g)} \tag{4}$$

$$S_b = \sqrt{S_x^2 - \left(S_w^2/2g\right)} \tag{5}$$

where S_x = the standard deviation of the average sample analyzed.
t = sample (t = 1, 2, 3, g),
X = data analyzed.
$\bar{X}$= average data.
S_w within-samples standard deviation.
S_b between-samples standard deviation.

An agreement between value of S_b less than 0.3r indicates acceptable homogeneity, where r is the target standard deviation representing a relative standard deviation (RSD) of 25% multiplied by the analytical values.

Finally, EURL-PT suggests using Eqs. (6) and (7) to determine and the constant c as follows S_b^2 (Ellison and Botha 2018):

$$S_b^2 = S_x^2 - \left(\frac{S_w^2}{2}\right) \tag{6}$$

$$c = F_1(0.3\sigma)^2 + F_2 S_w^2 \tag{7}$$

Determination of Assigned Value and Uncertainty

The assigned value can be gotten from the average value calculated from numbers of samples. The uncertainty caused by possible inhomogeneity can be evaluated as per ISO Guide 35 (Ellison and Botha 2018) that indicates by using Eq. (8) we can determine the value of uncertainty (u_{bb}) as follows:

$$u_{bb} = \sqrt{\frac{MS_{within}}{n}} 4 \sqrt{\frac{2}{\nu MS_{within}}} \tag{8}$$

where n = the number of measurements per bottle.
νMS_{within} = the number of degrees of freedom of MS_{within}.

If MS_{among} is > MS_{within}, the S_{bb} of Eq. (1) and u_{bb} of Eq. (8) will be employed; in addition Eqs. (1) and (8) can be applied. The uncertainty arises because inhomogeneity may be treated using the larger of the values for either S_{bb} or u_{bb} to prevent overestimation. If the $MS_{among} < MS_{within}$, only Eq. (8) can be applied.

Stability Testing

The result obtained from stability testing specifies the shelf life or expiry date of the CRMs. It is reported in the literature that there are two types of stabilities, i.e.,

degradation at elevated temperature for short duration, and second stability test must be performed at storage temperature to obtain information about the stability during storage called long-term stability.

The samples preparation and other environmental conditions like temperature and humidity for analysis is very important for stability testing. The samples are periodically withdrawn for the testing in stipulated time and the statistical data obtained for the evaluation for the stability.

In the pharma sector, generally two types of stability testing are practiced: (i) real-time stability testing and (ii) accelerated stability testing. Pharmaceutical products are kept under accelerated conditions. The changes under extreme conditions like temperature, light intensity, and humidity are monitored. The test results are compared with the specified standards. After assessing the result, the shelf life of the product is established. Stability testing tests the drug stability and monitors changes in physical, chemical, and microbial characteristics. If changes are detected in the physical conditions of target pesticides and antibiotics, the uncertainties for long-term stability is encompassed in the uncertainties of certified values in accordance with ISO Guide 35 (Ellison 2015). As per ISO guidelines, the stability is assessed on the basis of the analysis of variance (ANOVA) results of the linear regression of respective residual in the stability data (Grimalt et al. 2015). Temperature is an important factor and need to be monitored and evaluated in accordance with ISO Guide 35. There are two approaches for ANOVA: fixed batch effects model and random effect model. According ISO Guide 35, an analyte is considered to be stable provided absolute value of the slope, |b|, is smaller than the product of student's t factor ($t0.95, n-2$) by the uncertainty associated with the slope, $s(b)$, as presented in Eqs. (9), (10), (11), and (12). The slope (b_1) and intercept (b_0) of the linear regression is calculated using Eqs. (9) and (10) as follows (Monteiro et al. 2013):

$$b_1 = \frac{\sum_{i=1}^{n}\left(X_i - \overline{X}\right)\left(Y_i - \overline{Y}\right)}{\sum_{i=1}^{n}\left(X_i - \overline{X}\right)^2} \tag{9}$$

$$b_0 = \overline{Y} - b_1\overline{X} \tag{10}$$

where X_i and Y_i denote the time and the concentration at time I, whereas $\overline{X}$ $\overline{Y}$ denotes the average of X_i and Y_i, respectively. The $s(b_1)$ value can be calculated by using Eq. (11) and s value can be calculated using Eq. (12):

$$s(b_1) = \frac{s}{\sqrt{\sum_{i=2}^{n}\left(X_i - \overline{X}\right)}} \tag{11}$$

$$s^2 = \frac{\sum_{i=1}^{n}\left(Y_i - b_0 - b_1X^i\right)^2}{n-2} \tag{12}$$

Moisture Content

Advanced techniques such as Karl Fischer titrator, dynamic vapor sorption (DVS), and thermal gravimetry analysis also play an important role in the determination of moisture content. As per standard guides, all the samples must be sealed in bottles at 25 °C and 50% relative humidity and must be placed in electronic desiccators. At the time of bottling, the mean % weight difference between before and after the experiment should be around 5.0% (RSD of 0.85%). The moisture content can be estimated by taking 1−g sample from 10 random bottles ($n = 10$) and put the samples in a desiccator (anhydrous calcium sulfate or silica gel) at room temperature for 7 days (Wong et al. 2008). After 7 days, the samples were weighed to get the weight difference. Moisture content can also be calculated by advance techniques DSC-TGA and KFT techniques for structural water present in the samples.

Process of Assigning the Certified Values

The process of getting certified values for CRM is quite comprehensive. As per ISO 35 guidelines to obtained reliable analytical data, it is highly recommended to participate in CCQM a Consultative Committee on the Quantity of Material. The CCQM is a committee whose primary function is to document, develop, and improve the equipollence of national standards (CRMs/RMs). Apart from that, CCQM is held responsible for international benchmarking and comparison to evaluate analytical capabilities of national metrology institutions (Sturgeon et al. 2003). There are so many technically valid procedures available and can be used for the characterization of material to get the property values. To achieve the certified values, a collaborative study needs to be conducted with a PT provider. Consequently, results obtained from above studies showed the accuracy of analytical data and establish global comparability of measurement results, where traceability to the SI units is not yet feasible. PT is necessary for the analysis of various pesticides and antibiotics to confirmed international conformity. The analysis performed are based on a modified method for providing certified values (for CRMs) and reference values (for PT samples) (Raposo et al. 2008). For example, to avoid any predisposition associated with certain analytical methods like various extraction and clean-up procedures, LCMS and GS injection techniques and and LCMS and GC columns are applied to provide the certified values (for CRMs) and reference values (for PT samples) (Hajšlová and Zrostlıkova 2003). Pesticide and antibiotic quantification can be performed by using LCMS with tandem, Q-NMR, GC-MS, HPLC, TG-DTA, Karl Fischer, and other latest techniques (Yarita et al. 2015). In cases if reference values are available by an official method, the validation is conducted by some of the methods mentioned in the IS standards.

Certification of CRMs

All reference material certificates specify its associated uncertainty and a statement of metrological traceability characterized by a metrologically valid procedure. It is clearly distinguished between non-certified and certified values that can be used for calibration, performance checks, and trueness control. All CRMs must be accompanied by detailed certification report including material processing, homogeneity, stability and characterization studies, and value assignment. The certified values can be achieved through comparison of the analytical results obtained by two or more analytical methods used for each pesticide and antibiotics. Uncertainty analysis is not envisioned and should not be considered to account for accidents, mistakes, improper use, improper transport, etc. of the CRM (Ellison 2015; Otake and Yarita 2021). The uncertainty calculated from the mathematical operation and the mathematical representation of this model is a collective sum of property value obtained from the characterization, correction of property value due to the between bottled homogeneity, and correction of the property value of the batch due to long-term and short-term stability (Iso and Oiml 1995) (Fig. 8).

For example, NPLI certified values of the uncertainties are estimated from standard uncertainty due to analytical results (as mentioned above) (Fig. 9).

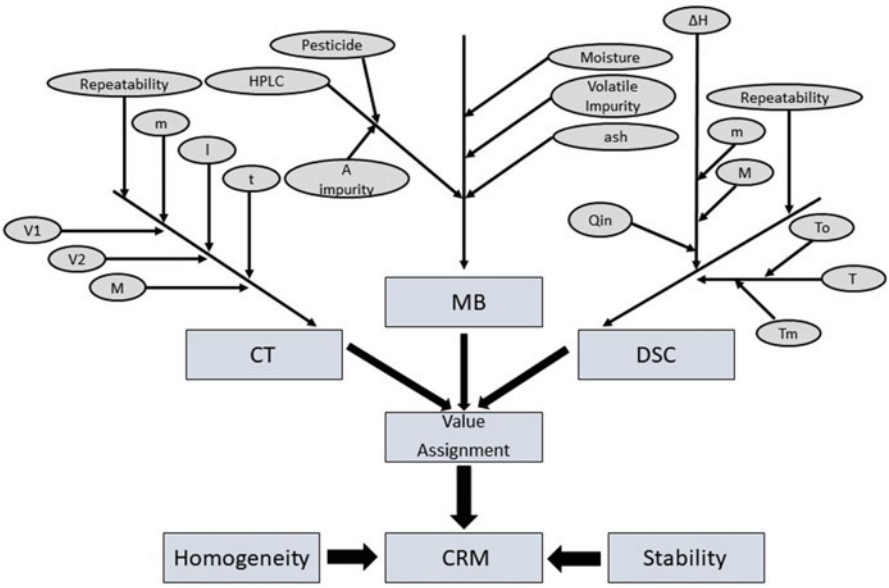

Fig. 8 Alternative model for pesticides and antibiotics CRMs

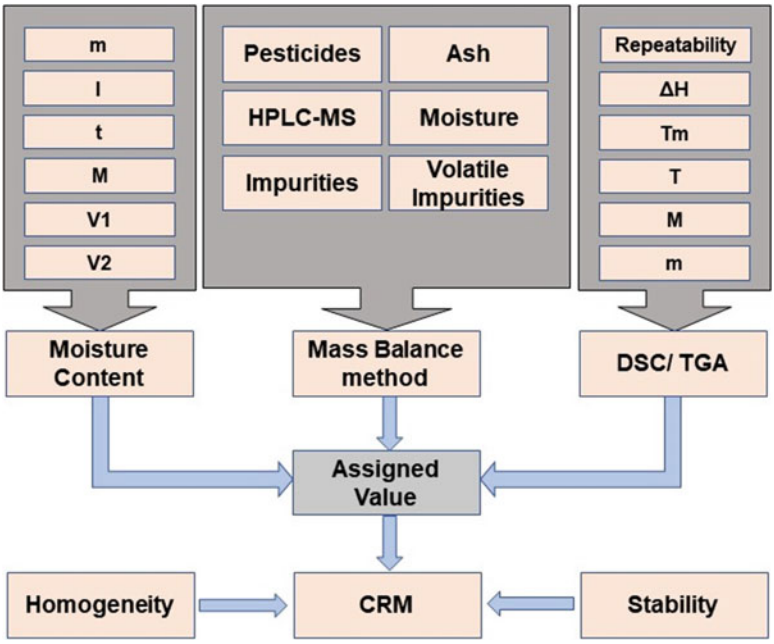

Fig. 9 Process of making pesticides and antibiotics CRMs

Assessment of Participant's Analytical Results

The performance can be assessed and the median value and normalized interquartile range (NIQR) evaluated instead of arithmetic average or standard deviation. PT scheme statistical analysis is applied to each population of analyte results to calculate robust z-scores which are used to fairly assess the testing performances of participants. Robust z-scores are calculated by replacing the mean and standard deviation (SD) in the "classical" z-score equation with the median and normalized interquartile range (NIQR), respectively. The NIQR is determined from the trimmed population of results by first calculating the interquartile range (IQR) of distribution across the middle 50% of results when arranged in ascending order. Z-scores may be two types as its standard deviation from proficiency testing can be calculated from the analytical results reported by participants (z) and another can be calculated from assigned values reported by NMI-NPLI (Z_{NPLI}). The z-scores (z) can be calculated using Eq. (13) (Shiffler 1988):

$$z = \frac{x_i - X}{\text{NIQR}} \tag{13}$$

where x_i denotes the results by each participating member, X denotes the agreement value calculated from the participant's labs results, and NIQR denotes a normalized interquartile range.

$$(\text{IQ} = \text{UPPER QUARTILE} - \text{LOWER QUARTILE})$$

The normalized IQ range is 0.7413× (quartile 3–quartile 1) (Otake et al. 2016).

The z-score is a numerical measurement that describes a value's relationship to the mean of a group of values and can be calculated by using Eq. (14):

$$z_{\text{NPLI}} = \frac{x^i - X_{\text{NPLI}}}{\text{SD}_R} \tag{14}$$

where X_{NPLI} denotes the NPLI reference value and SD_R represents the precision repeatability measure between-laboratory. SD_R can be calculated by using modified Horwitz equation used for quality benchmark and X_{NPLI}. The Horwitz equation is suitable for exponential relationship of the variability of chemical measurements in the interlaboratory trials. Horwitz equation is one of the first empirical parameters applied in the evaluation of the laboratories quality control and proficiency testing (PT) by using reference values. Each z-score interpretation is as follows:

$z \leq |2|$: it is being satisfactory
$|2| < z < |3|$: It is being questionable
$|3| \leq z$: it is being outlier.

Conclusion

Pesticides have been gaining importance since the last 60 years, as they marked fast, easy to use, cheaper substitute for eradicating weeds and pests (Nayak and Solanki 2021). Similarly, antibiotics also play a key role in the management of various diseases and improving human health and life expectancy significantly. However, human health risk is associated with antibiotic residue in the environment. To fulfill the requirement of analysts, industries, and academia, CSIR-National Physical Laboratory New Delhi has started the BND program for developing most promising pesticides and antibiotics standard reference material with trade name Bharatiya Nirdeshak Dravya (BNDs) which will be used in the quantification of pesticides and antibiotics residues. BNDs are high-quality analytical standards for many different application areas and market segments at par with international standards (Pant et al. 2020). These standards are raging from the food and beverage industry to environmental analysis and also to those used in the cosmetic, petrochemistry, and pharmaceutical industries. This chapter covers all the aspects related to the pesticides and antibiotics standards, their needs, and development and more stretches upon glimpse of the overall solutions for quality assurance. Country like India in terms of pesticides and antibiotics is more export-oriented in nature and overwhelming economy of India. Pesticide use has generally been implemented in terms of risk avoidance approach, but sometimes the

possibilities of pest incidence/critical pest inhabitants are regularly perceived in a wrong way to be on the higher side. Therefore, the main concerns on the negative outwardness of chemical pesticides across the globe have led to increasing awareness on the pesticide use, especially in the socioeconomically advanced humanities. The analysis has pointed out to the need for a detailed look on the pesticide certified reference material development indigenously and policy for the legal enforcement for quality assurance and regulatory mechanism at a micro level. It is well established that pesticides have a crucial role to play by increasing production of crops and subsequently raising the farmers' earning also. India has become self-sufficient in production of pesticides and also an important exporter of pesticides. The big challenge for India is to maintain a balance between production and quality. Although India is one of the largest producers of a number of agriculture commodities, on the other hand, EU is one of the largest export markets for India but faces several severe issues like microorganisms exposed to this increasing load of antibiotics in soil and water which can develop resistance (Gruère and Rao 2007). There is a possibility that traces of antibiotics and pesticides remain in edible parts of the plant longer after they are sprayed, affecting human health or making them resistant to antibiotics. As we know, safe food and health are linked together and unsafe foods caused more than 200 diseases. Pesticides use and its management are highly unregulated in India. India registered excellent growth in agriculture, and in 2022, India's export of agricultural commodities crossed $50 bn which is much higher than EU's export to India (Mukherjee et al. 2019). Despite this surge in production, existing literature reported that India faces trade issues as developed nation imposed legal policies and regulatory which in turn affect the export adversely. India's stated aspiration of doubling farmer income by 2022–2023 requires a commensurate increase in the output of the agriculture sector (Rodrik 2001). To mark a "paradigm shift" from residual export after meeting the national demand to targeted export in accordance to the preferences of foreign market and to realize this vision, the governments will have major role to play in strengthening the quality assurance of pesticides and antibiotics in agriculture, as also in attracting the private sector that has the capacity to build demand-driven CRMs and value-added exports. Globally India marked 12th position and in Asia stands on third position in pesticide use. Approximately 292 pesticides are registered in India and more than 50% of the pesticides used are of insecticides. The total of pesticide consumption in India was somewhere around 69,282 tons during 2017–2018. Maharashtra alone consumed the most chemical pesticides followed by Uttar Pradesh, and both the states cumulatively used 40% of total pesticides in India. Biopescticides is an alternative of synthetic pesticides and can helps farmer by replacing it by biopesticides which can minimize the environmental issues. The sustainable goal can be achieved through judicious and minimal uses of pesticides, using alternative chemical pesticides, legal framework, and by awareness to farmers. There is an urgent need to joining hands with all the government institutions, RMPs and public sectors to tackle these issues by anticipating BNDs development.

Cross-References

- ▶ Certified Reference Materials (CRMs)
- ▶ Legal Metrology and Global Trade
- ▶ Role of Certified Reference Materials (CRMs) in Standardization, Quality Control, and Quality Assurance

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Bharatiya Nirdeshak Dravya for Assessing Mechanical Properties

24

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Abstract

Materials for making engineering goods must have the appropriate mechanical strength to work under various loads. These properties of materials characterize the resistance to deformation in the presence of external forces. In this chapter, strength, hardness, toughness, hardenability, fatigue, etc. of the materials are described in detail. CSIR-National Physical Laboratory, in collaboration with Global PT Provider, has developed hardness blocks BND 2201, BND 2202, and BND 2204 for measuring Rockwell and Vickers hardness. These BNDs are useful to test and verify the HRC/HRV indentation hardness Testing Machine. Also, other mechanical properties for which BNDs are under development are discussed. This chapter will be useful for beginners to understand the mechanical properties and certified reference materials.

Introduction

The mechanical properties of material influence the materials' strength and service life. The mechanical properties of materials characterize the resistance to deformation in the presence of external forces (Pelleg 2012; Soboyejo 2002). Materials for making engineering goods must have the appropriate mechanical strength to work under various loads. Strength, hardness, toughness, hardenability, fatigue, etc. are some significant mechanical properties that decide the quality of materials (Wiederhorn et al. 2006). These different mechanical properties of materials are generally evaluated as per IS standards (Forster 2015). Standards are documents published to ensure the credibility of the materials, products, processes, and services. These establish material requirements, specifications, guidelines, characteristics, and procedures designed (Aswal 2020a). Organizations can promote their products and service consistency, security, and efficiency by adhering to these standards. In today's scenario, where products are assembled from parts manufactured in different countries and sold worldwide, proper standards have become more critical than ever. Measures are very much crucial for the preparation of certified reference material (CRMs) (Aswal 2020a).

CRMs are essential for generating accurate, precise, and reliable measurement data of major and minor constituents of different materials (Aswal 2020b). These have to be used to calibrate analytical instruments and validate test methods to successfully implement the quality assurance program of several governments and nongovernment agencies such as pollution control boards, public health laboratories, industries, R&D organizations, etc. The data generated using CRMs must be traceable to the national/international measurement system for global acceptability. International Organization for Standardization defines the reference material (RM) and CRM and is universally accepted (Aswal 2020a).

As the National Metrology Institute (NMI) of India, National Physical Laboratory (NPL), collaborates with public and private industries to prepare and disseminate CRMs (Aswal 2020a). These are launched as Bharatiya Nirdeshak Dravya (BND) or

Indian reference materials. Several BNDs are developed at NPL, including cement, petroleum, mineral oil, gasoline, diesel, water, pH, and hardness block (Yadav et al. 2021). The availability of BNDs will enhance the capability of Indian laboratories and help the automobile, material testing lab, railways, defense, and aerospace industries in a significant way and thereby enhance the export in these sectors.

CSIR-NPL has developed hardness blocks BND 2201, BND 2202, and BND 2204, respectively, in partnership with Global PT Provider Pvt. Ltd., for measuring Rockwell and Vickers hardness (Pant et al. 2020). Hardness is the most common parameter in the acceptance criteria of the steel industry (Senuma 2002). It provides significant info and insight into the toughness, strength, flexibility, and competence of materials. It is a commonly used test method in industries to validate the quality of components. Hardness block BNDs are produced for the standardization of hardness measurement. Presently, the Rockwell and Vickers hardness BNDs are in practice. Customers of these BNDs include the industries that currently require traceability to an NMI, reference material producers, instrument manufacturers, and end users. BND development for measuring other mechanical properties, such as strength, fatigue, hardenability, etc., is also in progress as per IS standards. In this chapter, the present hardness block BNDs available in India and other future BNDs for measuring mechanical properties are discussed in detail.

Hardness Tests

Hardness is the ability of the material to resist permanent deformation. The main types of hardness measurements are (1) scratch, (2) indentation, and (3) rebound, respectively (B. TEST n.d.). The most widely used in the engineering sector is indentation hardness, mainly for measuring metals. Indentation hardness testing is the most suitable method to investigate lesser volume materials. In this technique, indentations are made on the surface of the materials by indenting a rigid indenter of specific geometry (Khodabakhshi and Gerlich 2020). It is a nondestructive technique compared to other techniques which involve destructive action such as compression, extension, or bending of bulk samples (Broitman 2017). Indentation hardness testing is also crucial to other areas such as microelectronics, micromachines, and the magnetic recording industry (Sangwal et al. 2003). Indentation hardness is measured in different testing methods, including the Brinell, Rockwell, Vickers, and micro-hardness test, based on the type of sample materials (Broitman 2017; Walley 2012).

Brinell Hardness Test Method as per ASTM E10 and ISO 6506

The Brinell hardness test is the most widely accepted test method for coarse or rough surfaces, e.g., castings and forgings, and it is tested as per ASTM E10 and ISO 6506 (Gabauer 2000). In this testing method, the metal surface is indented for a standard time of about 30 sec with a 10 mm diameter steel or carbide ball with a high load of a maximum of 3000 kgf. The resulting indent impression is measured with a specially

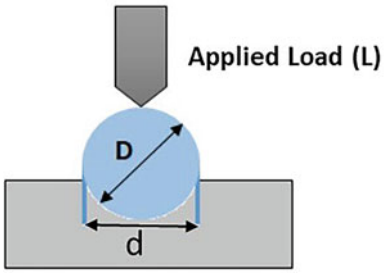

Fig. 1 Schematic representation of Brinell hardness test method (Kim et al. 2005)

designed Brinell microscope system across two diameters. Figure 1 shows the Brinell hardness test method (Kim et al. 2005).

The Brinell hardness (BHN) is calculated as a load (L) by the surface area of indentation (Barua et al. 2021; Muangma 2018) and can be expressed as follows:

$$\mathrm{BHN} = \mathrm{L}/(\pi \mathrm{D}/2)\left(\mathrm{D} - \sqrt{(\mathrm{D}_2 - \mathrm{d}_2)}\right)$$

where: L = applied load in kilogram,
D = diameter of the ball in mm.
d = indentation diameter in mm.

Rockwell Test as per ASTM E18 and ISO 6508

The hardness is obtained by setting the diamond point of the indenter on the material surface and recording the penetration depth under arbitrarily fixed conditions (Asgharzadeh et al. 2020). It is tested as per ASTM E18 and ISO 6508 (Kumar 2019).

Initially, a minor load is first applied to set the indenter on the material, which causes an initial penetration and holds it in position. Then, depending upon the scale being used, a major load is applied to increase the indentation depth. Firstly, the major load is removed, leaving a minor load still acting on the surface. The difference in penetration between the major and minor loads gives the Rockwell number, as shown in Fig. 2 (Chandler 1999).

Vickers Hardness Test as per ASTM 92 and ISO 6507

Vickers hardness test is generally used for harder metal surfaces, including heat-treated steel. The metal surface is exposed to a normal pressure for a standard time using a pyramidal-shaped diamond indenter with a vertex angle of 136° and a load that ranges from 1 kg to 120 kg. It is frequently used to determine thin film hardness and identify specific phases in metallography. Vicker hardness test is carried out as

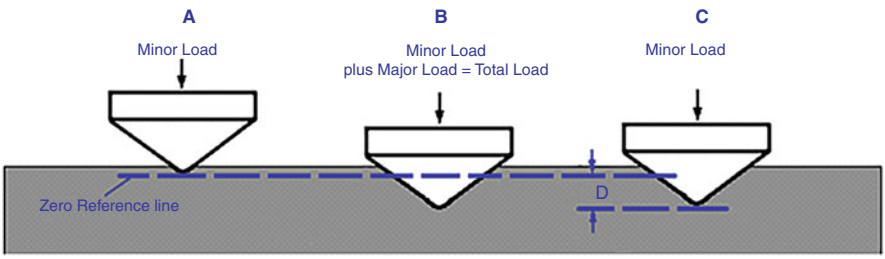

Fig. 2 Schematic representation of Rockwell hardness test method (Chandler 1999)

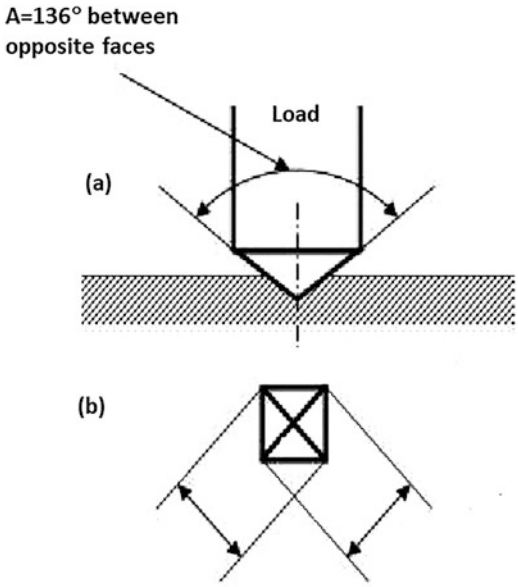

Fig. 3 Shape of Vickers indenter: (**a**) side view and (**b**) top view (Lim and Chaudhri 2006; Gall et al. 2001)

per ASTM 92 and ISO 6507 (Ellis 2003). In the Vickers hardness test, hardness is independent of the applied load since indentations with different loads are geometrically identical (Petrík and Palfy 2014). The shape of the Vickers indenter is shown in Fig. 3 (Lim and Chaudhri 2006; Gall et al. 2001).

The Vickers hardness is calculated from the following expression:

$$HV = 1.854P/L^2$$

Knoop Hardness Test

The Knoop hardness test is performed to measure the microhardness using the universal or microhardness testing machine (Clinton and Morrell 1987). Tests are

carried out as per ASTM E384 (Hetzner 2003). Generally, microhardness testing is done on thin films and brittle materials to avoid cracking. The Knoop indenter is a pyramidal diamond indenter with long and short diagonals at approximately 7:1. The indentation depth is about one-thirteenth of the length of the longer diagonal indenter. The Knoop hardness number (KHN) is obtained from the measurement. It is the number obtained by dividing the load applied (L in Kg) to a pyramidal diamond indenter of the specific size to the projected area of the impression (A in mm^2) (Revilla-León et al. 2021):

$$KHN = L/A$$

Knoop hardness is derived from the following formula:

$$HK = 14.229\ (F/D^2)$$

F is the applied load (kgf), and D^2 is the indentation area (mm^2).

Metallographic polishing is usually required to obtain a smooth surface. In the Knoop hardness test, the long diagonal impression is not affected for about 300 g load. In contrast, a small indentation is formed for lighter loads, resulting in a small elastic recovery. These factors generally give a high hardness reading. Therefore usually Knoop hardness number increases as the load is decreased below about 300 g. Schematic representation of the Knoop hardness test is shown in Fig. 4 (Ben Ghorbal et al. 2017; Sumiya and Ishida 2019).

Scope of Hardness BNDs

Rockwell and Vickers hardness BNDs, developed at CSIR-National Physical Laboratory, help the Indian industry suffering from traceability issues. All measurements have been carried out to establish traceability through an unbroken measurement chain with a stated uncertainty. Calibrated test blocks are used to confirm the instrument's

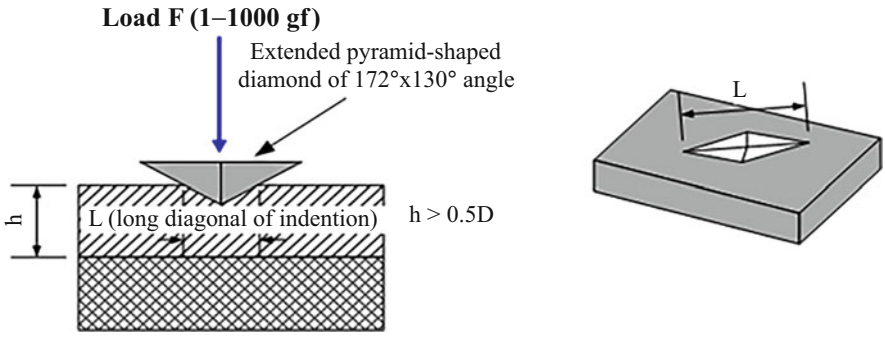

Fig. 4 Schematic representation of Knoop hardness test (Ben Ghorbal et al. 2017; Sumiya and Ishida 2019)

performance and calibration, ensuring the accuracy and traceability of hardness measurements. The hardness BND number and certified values are given in Table 1.

The Rockwell hardness block BND was prepared from steel and calibrated using Rockwell hardness tester ZHR4045/4150 according to IS 1586 (P-1), ISO 6508, and ASTM E18. A cone-shaped indenter was used for hardness measurement. The preload of about 10 kilograms and test load of about 150 kilograms were applied. Rockwell harness blocks (two sets) exhibited the certified value with the expanded uncertainty which was 44.33 ± 0.83 (HRC) and 61.04 ± 0.77 (HRC), respectively. For Vickers hardness for the load of 5 kg/49.03 N, the obtained value is around 735.6 ± 13.9 HV. The coverage factor for the reported values is $k = 1.96$, which agrees to a coverage probability of about 95% for a normal distribution (Fig. 5).

Table 1 Hardness block BND number and the certified values

Hardness BND number	Certified value
BND no. 2201	44.33 ± 0.83 HRC
BND no. 2202	61.04 ± 0.77 HRC
BND no:2204	735.6 ± 13.9 HV-5

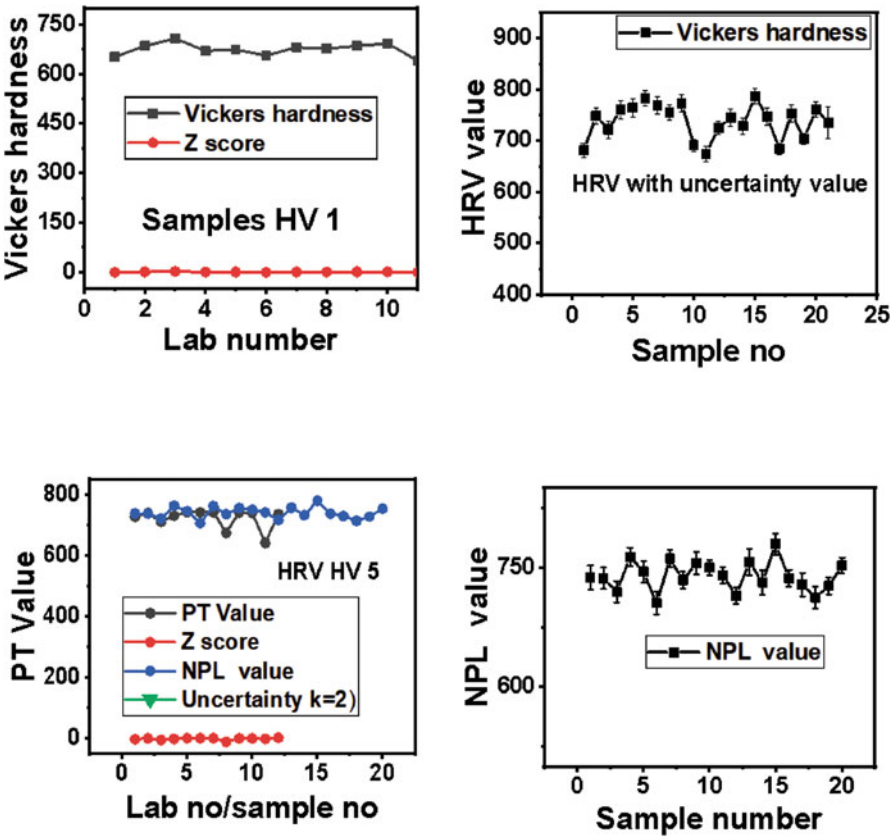

Fig. 5 PT results for Vickers hardness

During the hardness measurement, the selection of test location on the specimen is most important and must follow several precautions: If the indenter is placed closer to the edge of the specimen, the edge will bulge, and the hardness value will decrease accordingly (Kongkavitool et al. 2007). Distance from the edge of the material to the center of indentation should be at least a minimum of 21/2 times the indentation diameter. It ensures that any new indentation is not influenced by the work hardening of the material around a previous indentation.

Experimental Procedures Followed for the Development of Hardness Block BNDs at CSIR-NPL

Procedure for Rockwell Hardness Measurement

Rockwell hardness was tested using Model ZHR4045/4150 (Fig. 6), and the measurements were made as per ISO 6508 and ASTM E18 with preload of 10 kg and test loads of 60, 100, and 150 kg, respectively. The instrument's resolution was 0.1 digits, and the preload was used to cancel out errors. The use of a minor load significantly increases the accuracy of the test since it removes the effects of backlash in the measuring system that causes the indenter to break due to slight surface roughness. It is a rapid and nondestructive technique. The entire procedure

Fig. 6 Rockwell hardness tester

needs 5 to 10 seconds. The block used as a test sample is shown in Fig. 7. The Rockwell number represents the difference in depth from zero datum position due to the application of major load.

Procedure for Vickers Hardness Test

Figure 8 shows the Vickers hardness test equipment used for measuring the Vickers hardness as per IS 1501 (P-1)/ISO 6507-1/ASTM E92. Generally, indention is done

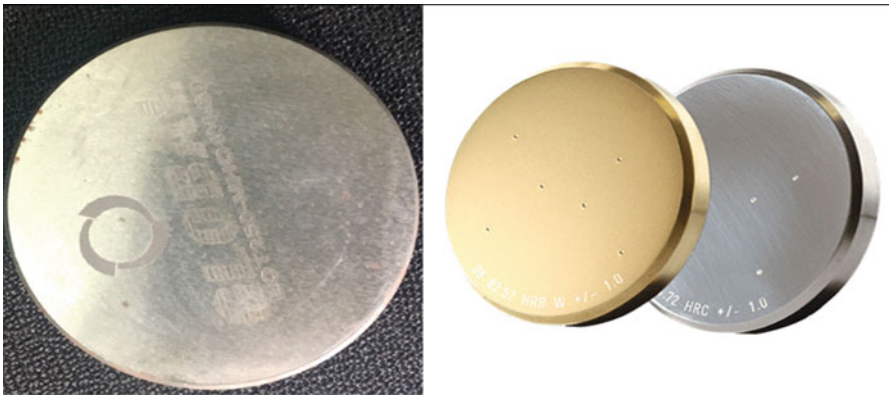

Fig. 7 Test piece used for hardness testing

Fig. 8 Vickers hardness tester

with the help of a pyramid-shaped diamond indenter with a square base. The angles between opposite faces are 136 degrees (see Fig. 2). A test force of 100 kgf is applied for 10 to 15 s. The mark is measured using a microscope.

Instructions for Storage, Commutability, Precautions, and Use for Handling

The BND should be stored in the box supplied with the reference material at a temperature range of 25 ± 10 °C with a relative humidity of 60% or less and should be protected from scratches, cavities, and other mechanical damage, mainly on the bottom surfaces. It is prone to corrosion due to moisture and skin oils. The user should wear protective gloves such as lint-free cotton or powder-free latex gloves while handling the BND. After using, the gloves should be placed back in a container and stored under defined storage conditions (25 ± 10 °C with a relative humidity of 60% or less) to minimize deterioration.

The sample's surface should be cleaned gently, and the top and bottom surfaces should be washed entirely with lint-free paper wetted with ethyl alcohol. The surface should be immediately dried using a soft lint-free paper before the alcohol evaporates, and the surface(s) should be blow-dried using filtered air.

Tensile Test

The tensile test is the standard test used to evaluate the strength of materials and is also considered an acceptance test for the specification of materials. The tensile test is carried out as per ASTM D638 and ISO 527 (Bergonzi et al. 2018). In this test, a specimen is deformed to fracture, with constantly increasing uniaxial tensile load, and simultaneously, elongation of the specimen is observed (Hart 1967). A test sample is steadily pulled or stretched to its breaking point to find the material's ultimate tensile strength. The standard tensile test specimens are shown in Fig. 9.

Fig. 9 Tensile test specimens

Generally, circular-shaped samples are used as the tensile test specimen, but sometimes rectangular-shaped specimens can also be used. For the tensile test, the "dog bone"-shaped specimen is used because during the stretching of the sample, deformation is restricted to the narrow center region to reduce the likelihood of fracture at the ends of the sample. The end of the sample is mounted into the holding grip of the testing apparatus. The tensile machine is designed in such a way to continuously and simultaneously measure the instantaneous load applied and resulting elongations during the testing process. The output result of the tensile test is recorded as a force vs. elongation plot (Sharifishourabi et al. 2014).

The tensile test result is displayed as force (F) applied to the sample vs. elongation (ΔL). The strength of the material is expressed in stress (σ) and strain (**ε**). Stress is defined as the force F divided by the sample cross-sectional area A ($\sigma = F/A$). Strain is the change in length, ΔL, by the initial length of the sample (L) ($\boldsymbol{\varepsilon} = \Delta L/L$). These values are then presented on an X and Y plot called a stress-strain curve. The parameters like modulus of elasticity, yield strength, ultimate tensile strength, strain hardening characteristics, ductility, Young modulus, and the Poisson ratio of material can be calculated from the stress-strain curve (Park et al. 2017). Several parameters such as the composition of metal or alloy, heat treatment condition, plastic deformation, temperature, strain rate, and state of stress applied during the test determine the curve's shape and magnitude. Figure 10 shows a classical stress-strain curve for engineering materials (Tu et al. 2020).

The slope region of the stress-strain curve is referred to as the modulus of elasticity (Young's modulus). The modulus of elasticity measures the stiffness of the materials. It is a factor used to calculate how materials resist deflection under

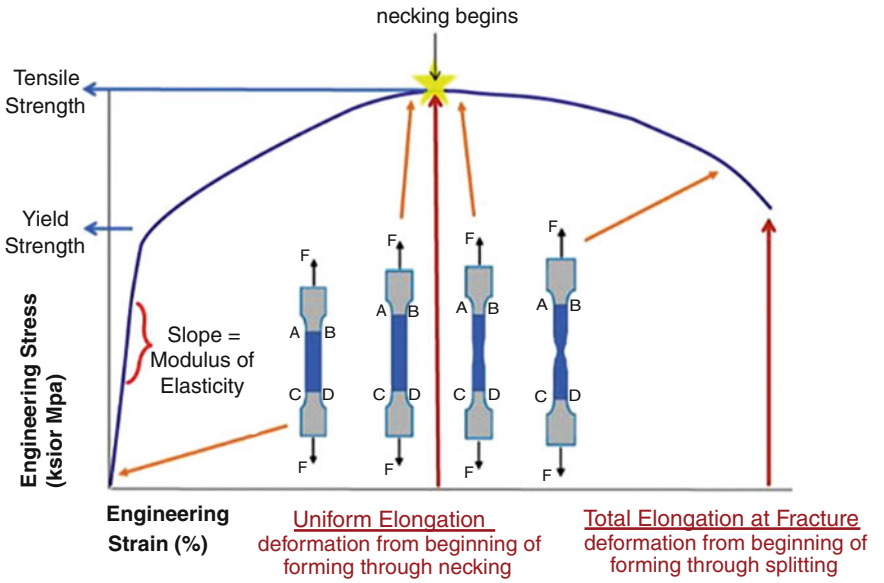

Fig. 10 Stress-strain curves of engineering materials (Tu et al. 2020)

applied load. This property reflects the elastic behavior of the materials. The maximum limit of the elastic behavior is noted as yield strength. It is the point where maximum stress can be developed on the material without causing plastic deformation.

Metals become stronger due to strain hardening as the plastic deformation increases. The load required increases with further straining until it reaches the maximum tensile strength. It is also known as the ultimate tensile strength (UTS) (Zhao and Takeda 2000). UTS is the maximum applied load at failure (P max) divided by the original area (A_0):

$$\text{Ultimate tensile strength (UTS)}\ \sigma\ \text{max}\ = \text{P}\ \text{max}\ /\text{A}_0$$

The diameter of the sample reduces rapidly beyond maximum load due to necking. The load needed for continuing the deformation is stopped as the diameter of the specimen decreases with maximum load until the specimen fractures. The tensile testing unit is shown in Fig.11.

Fig. 11 Tensile testing unit (Sharifishourabi et al. 2014)

Impact of Tensile BND

CSIR-NPL is in the process of developing tensile BND standards for metals and alloys. Steel and aluminum alloy standards are proposed to develop as the tensile sample. It is intended to be used to directly verify tensile testing machines, which can conduct tests following the ASTM A370/ASTM E8/IS 1608–1/ISO 6892-1. Thousand tensile specimens in a year were catered for automobile, aerospace industry, commercial metallurgical laboratories, etc. Presently only indirect verification is being done in India, and direct verification is not available. So these reference materials will effectively help the Indian market with minimum cost.

Fatigue

Fatigue is an essential criterion in the design of engineering materials. Fatigue is a failure in structures such as bridges, machine components, and aircraft subjected to dynamic and fluctuating stresses (Suresh 1998; Sharpe and Wilks 2002). It is determined by the endurance limit or the S-N curve. S-N curve is the plot of cyclic stress amplitude S on the *y* axis and the number of cycles for failure N on the *x*-axis (Lotsberg and Sigurdsson 2006; Ünal et al. 2016).

Fatigue life can be determined as the number of stress cycles that the materials can resist before the failure. Based on the properties of materials, the failure occurs in a few cycles, or sometimes it takes many cycles. The fatigue life cycle is divided into different types such as (1) low cycle fatigue (LCF), in which the cycles range from ~10^4 to 10^5, (2) high cycle fatigue (HCF) (~10^5 to 10^7 cycles), and very high cycle fatigue (VHCF) where the cycle number is greater than 10^7 (Wang et al. 2002a). Fatigue failure generally happens in three different stages and includes (1) crack initiation, (2) crack propagation, and (3) crack failure, respectively (Schütz 1979; Abd Allah et al. 1997).

Fatigue cracking initiates due to the materials being subjected to many cyclic stresses below the ultimate tensile stress or yield stress. Fatigue causes localized damage to the structure due to cyclic loading. The material experiences continuous and repeated loads at various points. When loads are high enough, they cause crack initiation, crack growth, and fracture.

Crack initiation is also known as a crack formation. A microcrack that grows to 0.5 or 1 mm in length for the defined cycles causes fatigue crack propagation. The sum of fatigue crack initiation and propagation is fatigue life. Fatigue failures frequently cause catastrophic failure in engineering materials. For most metals, polymers, and ceramics, the crack propagation takes place slowly, and in the later stages, they are also susceptible to sudden fatigue failures. Fatigue failure depends on cyclic loads, temperature, strain rate, oxidizing environment, structural deformation, embrittlement, structural deformation, frequency of the applied load, etc.

Figure 12 (Zalnezhad et al. 2014) shows the schematic representation of an apparatus used for fatigue testing. Applied stresses may be axial, flexural, or tensional. For simultaneous bend and rotation, the tension-compression stresses are applied on the specimen, and these are frequently conducted by using uniaxial

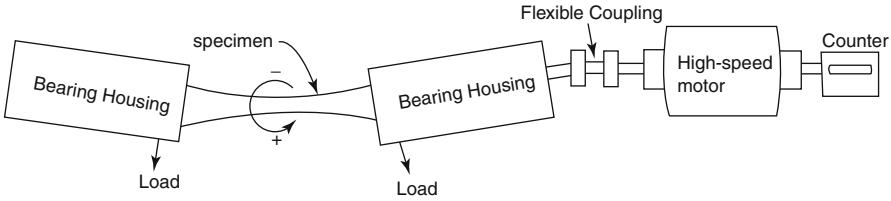

Fig. 12 Schematic representation of a rotating-bending test apparatus used for fatigue testing (Zalnezhad et al. 2014)

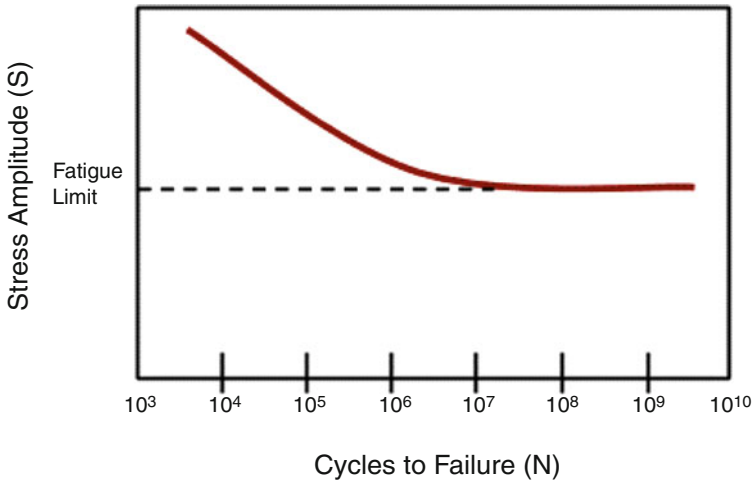

Fig. 13 S-N curve (Callister and Rethwisch 2018)

tension and compression stress cycle alternatively. The test sequences are started by exposing the sample to a stress cycle with relatively large maximum voltage amplitude, usually in sequence 2/3 of the static tensile strength and the number of cycles required for failure. The obtained data are plotted as stress S vs. the log. of the number N of cycles to failure, and it is generally called as S-N diagram (Fig. 13) (Govindaraju et al. 2021; Callister and Rethwisch 2018). It provides the relation between applied stress amplitude, S, to a material specimen and the number of loading cycles, N, the specimen can withstand before eventually failing. The S-N curve becomes horizontal for some materials at low stresses, indicating that the material will not fail at stresses below this limit (Wang et al. 2002b). The fatigue or endurance limit is the stress level below which there is no fatigue failure. The endurance limit is the stress below which a material can withstand an infinite number of repeated load cycles without exhibiting failure (Shen and Carpenter 1929).

Metals such as Fe and Ti alloys have a specific limit, as seen in Fig. 14 (Campbell 2008). In contrast, other materials such as Al and Cu do not have any specific limit, and they finally fail even if small stress amplitudes are applied. The fatigue strength test unit is shown in Fig. 15 (Il Seo et al. 2005).

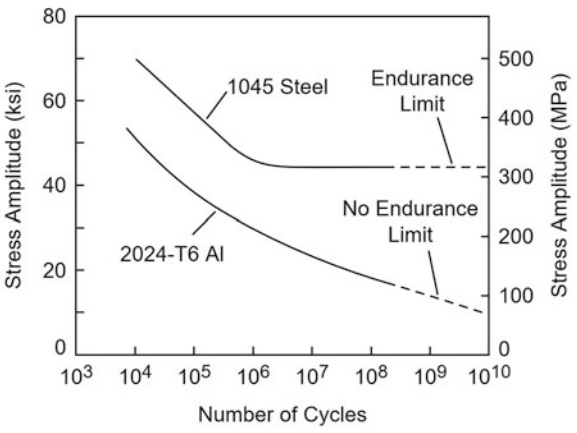

Fig. 14 S-N curve of aluminum and low carbon steel (Campbell 2008)

Fig. 15 Fatigue strength test unit (Il Seo et al. 2005)

Toughness

Toughness is the capacity of materials to absorb energy and withstand shock up to fracturing (Hahn et al. 1972). It is also defined as the amount of energy per unit volume that a material can absorb before fracture. The stress vs. strain curve representing the area of toughness measurement is shown in Fig. 16 (Nataraja et al. 1999).

Toughness can be measured by integrating the area under the stress-strain curve. Tensile toughness is measured in units of Jm^3 in the SI system. In general, tough

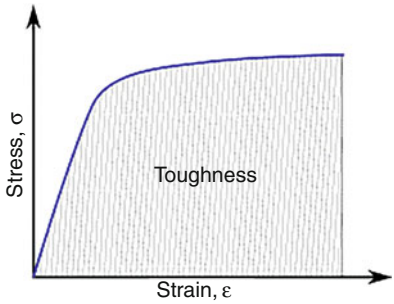

Fig. 16 Stress vs. strain curve representing the area of toughness (Nataraja et al. 1999)

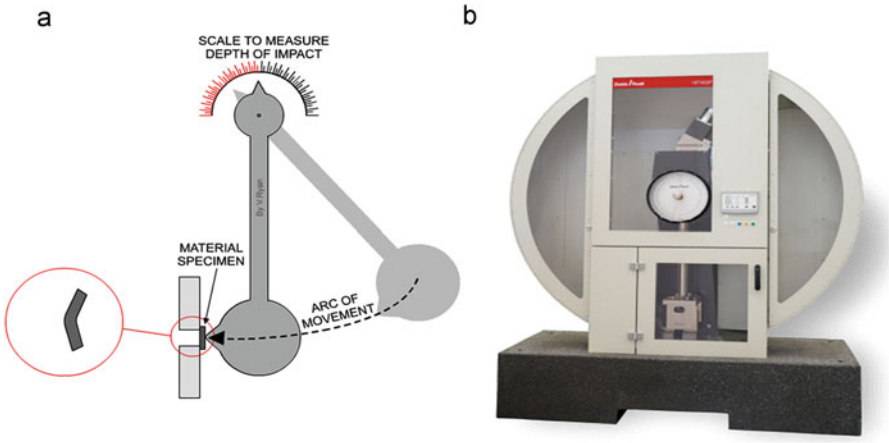

Fig. 17 (**a**). Schematic representation of pendulum Charpy test (Hayden et al. 1967), (**b**) Pendulum impact tester HIT450P (Brezinová et al. 2020)

material must be strong and ductile, and also it should have the capacity to withstand both high stresses and strains.

Ceramics are strong, brittle, and not tough. These also have less degree of ductility. However, very ductile materials with low strengths, such as aluminum, are also not tough. Strength specifies how much force a material can withstand, while toughness represents how much energy a material can absorb before the rapture. The Charpy impact test measures the toughness of the materials (Toth et al. 2002). The schematic representation of the pendulum Charpy test and modern pendulum impact tester HIT450P is shown in Fig. 17a and b (Hayden et al. 1967; Brezinová et al. 2020).

Impact BND standards for metals and alloys are also among the BNDs developed at NPL. To develop a sample, low, mid, and high (CVN) standards are proposed. This reference material is intended to verify the Charpy V-notch machines as per ASTM E23. The Charpy impact test standard is shown in Fig. 18 (Leis 2013; Bokuchava and Petrov 2020). One thousand sets, each of low-, middle-, and high-impact (CVN) RMs, in a year catering to automobile, forging, aerospace industry,

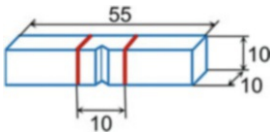

Fig. 18 Charpy impact standard (Bokuchava and Petrov 2020)

commercial metallurgical laboratories, forging, defense, and aerospace industry. NPL is committed to helping the whole Indian industry, which has suffered from traceability issues for a long time, and launching more Indian reference materials, effectively reducing the cost of Indian laboratories.

Hardenability

Hardenability is the alloy's ability to be hardened by the heat treatment process (Grange 1973). It is a measure of depth to which ferrous materials are hardened after quenching. It is also referred to as the depth of hardening. It is one of the essential parameters for the welding process and is inversely proportional to weldability, i.e., joinability of materials (Andersson 2011).

The Jominy end quench test (Newkirk and MacKenzie 2000) measures the hardenability of the ferrous alloys as per ASTM A255. When a hot steel workpiece is quenched in cool water, the area of the steel in touch with the water cools immediately. The workpiece temperature attains equilibrium with the quenching medium like water, oil, or air. However, the inner depths of the workpiece do not cool so rapidly. In larger steel workpieces, the cooling rate is slow enough to fully transform austenite into a structure other than martensite or bainite phases. Therefore, the workpiece is prevented from having the same crystal structure throughout its entire depth, creating a softer core and more rigid outer shell. The inner soft core is the combination of ferrite and cementite.

The hardening property of steel materials depends on the amount of carbon, alloying elements, and grain size of the austenite phases. The relative amount of the different elements present in the steel is calculated by determining the material's carbon equivalent (CE) (Kang 2011).

The cooling rate of the quenching medium depends upon the thermal conductivity and specific heat capacity. Generally, brine solution and water cool the steel faster than air or oil. Cooling occurs more rapidly if the fluid is agitated. The specimen's geometry also influences the cooling rate. For example, considering the two samples of the same volume, the higher surface area sample will cool faster than the lesser one.

Figure 19a represents Jominy end quench test. In this test, the cylinder-shaped sample is used with a diameter of 25.4 mm and a length of 102 mm, respectively (Peng et al. 2020). Generally, the steel sample is normalized to remove the differences in microstructure due to forging and austenitizing at 800 to 925 °C to transform the steel microstructure into austenite. The test steel sample is transferred

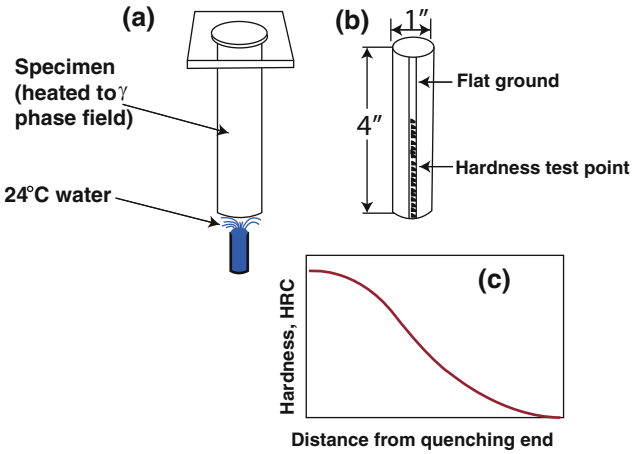

Fig. 19 (**a**) Jominy end quench test to check the hardenability, (**b**) specimen after test, and (**c**) hardness vs. distance from quench end (Callister and Rethwisch 2018)

rapidly to the test fixture, where one steel end is quenched with the controlled water flow. The cooling rate differs from one quenched end to another end of the sample. The cooling rate is very fast at the quenched end and slower cooling rates at the other end, similar to air cooling. The round sample is made flat by grinding along its length on opposite sides to a depth of at least 0.38 mm to eliminate decarburized materials formed during the forging process (Fig. 19b). The sample should not heat up during grinding as it can cause tempering and softening of the microstructure. The hardness is measured at regular intervals away from the quenched end of the specimen. The hardness is typically measured at about 1.5 mm intervals for alloy steels and 0.75 mm for carbon steels as close to the quenched end. The hardness decreases as the distance increases from the quenched end (Peng et al. 2020). High hardness was observed in the region with high-volume martensite fractions (Kandpal et al. 2021). Lower hardness occurs in the region where bainite or ferrite transforms into pearlite microstructures. Hardness measurement is commonly done using a Rockwell or Vickers method. A hardness curve describes the hardenability of the steel (Fig. 19c). It refers to the plot of hardness value measured at a certain length away from the quenched end of the sample. From the plot, the hardness decreases with an increase in quenching distance.

Conclusions

Strength, hardness, hardenability, toughness, and fatigue are some of the material's most important mechanical properties, which decide their suitability for engineering applications. In laboratories/industries, the mechanical properties are tested as per IS/ASTM standards for selecting materials as per the requirements. Several CRMs are developed to calibrate the instruments for accurate measurements. In India, these

CRMs are sold under the trade name BND. Recently CSIR-NPL has developed hardness blocks BND 2201, BND 2202, and BND 2204, respectively, in partnership with Global PT Provider Pvt. Ltd., for measuring Rockwell and Vickers hardness. Several BNDs are under development to cater to the nation's future needs.

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Role of Certified Reference Materials (CRMs) in Standardization, Quality Control, and Quality Assurance

25

S. K. Breja

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Abstract

In the globally connected world, measurement unit and metrological traceability are the two essential requirements for reporting measurement results. While use of System International (SI) ensures international parity in reporting calibration and testing results in similar unit of measurements, laboratories must ensure metrological traceability also. Without traceability and knowledge of uncertainty of measurements and generated data, at each stage of calibrations, technical validity of results remains questionable. Further, wherever direct calibrations of the equipment are not possible, reference materials (RMs) and certified reference materials (CRMs) must be used. Thus CRMs – reference materials with certified values – accompanied with measurement uncertainty play an important role in confidence building in measurements conducted by the laboratories. Use of RMs and CRMs is mandated in respective product standards and ISO/IEC 17025:2017/ ISO17043:2010/ISO 9001:2015 system standards, for quality control (QC) and quality assurance (QA). Decisions on acceptance of production lots, based on associated measurement results, are critical to the survival of businesses, and, therefore, it is essential for the manufacturers and the laboratories to ensure that reported measurement data and results are technically valid. It is needless to mention that today it is all about data, analytics, artificial intelligence (AI) robotics, and automation – the hallmark of Industry 4.0. Data are the basis of all decisions.

This chapter discusses the role played by RMs and CRMs in standardization, QC, and QA and their application for various purposes by the respective bodies – in manufacturing and services sector – and international trade. Discussion reveals that ISO/IEC 17025:2017 embodies the spirit of plan-do-check-act (PDCA) cycle. Taking the case of Indian cement industry, the requirement of calibrations and RMs at various stages of production and quality control is also discussed.

Keywords

Measurements · Metrology · International System of Units (SI) · Traceability · Measurement uncertainty · Testing · Calibration · Reference material (RM) · Certified reference material (CRM) · Standardization · Quality control · Quality assurance · International trade

Introduction

Data and information management play a vital role in the world economy. The world economy is expanding. World GDP is expected to be of $100 trillion size, even before 2024 (Centre for Economics and Business Research (CEBR) 2021). In the globally connected world of today, quality of products and services determines a nation's competitiveness. All measurements have to be technically valid and acceptable across the world. Thus, quality infrastructure, system of accreditation, generation of results by competent laboratories, and traceable measurements and calibrations – using standard equipment and/or reference materials (RMs) – play an important part in equipment standardization, quality control (QC), and quality assurance (QA) in the industry.

RMs with certified values are required for equipment calibration and maintaining metrological traceability. RMs with certified values, called certified reference materials (CRMs), are measurement standards. These are essential for quality control and quality assurance in the industry. CRMs ensure traceability and accuracy of measurements with associated value of uncertainty – at the stated confidence level – and accuracy, or inaccuracy for that matter, decides the fate of a production lot or export consignment. Thus, RMs or CRMs are needed primarily for making accurate and technically valid measurements and a variety of purposes, like standardization, calibration, method validation, quality control and quality assurance, etc. The world today exchanges goods and services at an unprecedented speed and scale. We have global supply chains, and slightest disruption – least of all of technical nature – in any part of the chain has cascading effect. Data accuracy and precision help break technical barriers to trade and assure better quality of life (Kumari et al. 2022). In the global economy, world CRM market, valued US$1190.32 million in 2019, is expected to grow at a CAGR of 6.8% (Global CRM market growth (2020(2027) 2021).

Measurement Units and International System

Reference materials play an important role in a nation's economy. Competitive countries have technical capabilities to produce and use RMs and CRMs across industry sectors. In the industrial world, product quality and acceptability decisions require making measurements. Decisions have to be made on the basis of data and facts. One can't imagine working without data, large data, analytics and automation in Industry 4.0 – characterised by artificial intelligence, machine learning, Internet of Things (IOT), cloud computing and analytics – which is changing the ways of manufacturing and distribution of products and services. Manufacturers are integrating the new technologies into their production facilities and throughout their operations, and using the embedded software for data collection, analysis and better decision making (What is Industry 4.0? 2022)

Scientific, R&D, and engineering organizations have to make measurements and express results in universally used units. In this context, standard unit system helps us understand the measurements in one set of unit system, and this facilitates global trade (Understand the International System of Units and its seven basic SI Units n.d.). System of units is the set of base units and derived units, together with their multiples and submultiples, for a given system of quantities, and the **International System of Units** (SI) is the system of units which is based on the International System of Quantities, adopted by the 11th General Conference on Weights and Measures (CGPM) in 1960 (ISO/IEC Guide 99 2007).

Global Trade relies heavily on monetary standard (gold or currency) and internationally accepted product, process, system and measurement standards. Here lies the importance of international institutions, like, International Bureau of Weights and Measures (BIPM), General Conference on Weights and Measures (CGPM), International Organisation for Standardisation (ISO), **ISO/REMCO** Committee on Reference Materials, Eurachem- A network of organisations in Europe, EUROMET- The European Association of National Metrology Institutes, National Metrology Institutes (NMIs) of various countries, International Laboratory Accreditation Cooperation (ILAC), International Accreditation Forum (IAF), National accreditation bodies of various countries, quality infrastructure in the countries world over, which have together contributed to the development of knowledge on reference standards and materials. In India, we have authoritative organisations, like, CSIR-NPL, i.e. National Physical Laboratory, India (NPLI), Bureau of Indian Standards (BIS), Quality Council of India (QCI) which – apart from National Accreditation Board for Testing and Calibration Laboratories (NABL) – has other boards also under it, and various other R&D organisations in different sectors (Aswal 2020)

Measurement Standards and Reference Materials

Reference material is basically a measurement standard. Measurement standard is defined as "realization of the definition of a given quantity with stated quantity value and associated measurement uncertainty, used as a reference" and, national measurement standard – standard for a country – is a measurement standard which is recognized by a national authority for assignment of quantity values to other measurement standards, with the purpose to serve in a particular state/economy (ISO/IEC Guide 99 2007). Cross-country use of national measurement standards also is often made for comparison or whenever needed, as traceability through these standards can be disseminated worldwide. Definitions of these standards are given in ISO/IEC Guide 99:2007 (ISO/IEC Guide 99 2007). Hierarchy of measurement standards can be depicted as in Fig. 1. The figure gives a brief description also of these standards.

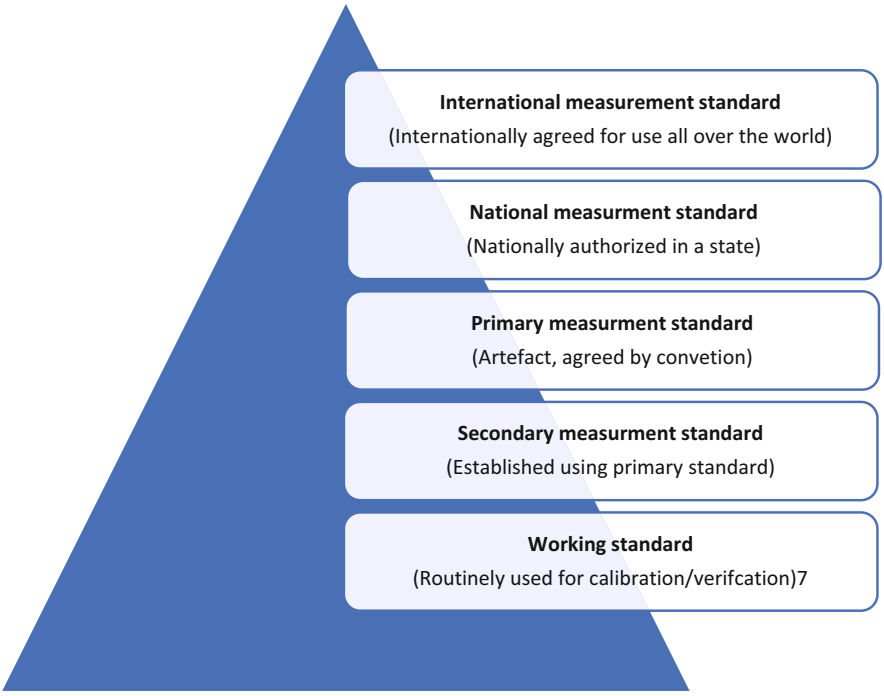

Fig. 1 Hierarchy of measurement standards (ISO/IEC Guide 99:2007)

Classification, Selection, and Use of Reference Materials

Reference materials are classified into three classes: primary, secondary and in-house or working standards (ILAC-G9 2005; EA-4/14 INF 2003; EEE/RM/062rev3 2002), with uncertainty increasing down the level from primary to working standard. ISO/IEC Guide 99:2007 (ISO/IEC Guide 99 2007) mentions two classes: Certified Reference Materials (CRMs) and Reference Materials (RMs)

RMs are materials whose property values are well established. These can be used for instrument/equipment calibration, assessment of a measurement method, and assigning value. CRMs are accompanied with additional documented information and have greater utility in the industry. Measurands of CRMs are traceable to SI units, and uncertainty – at a stated level of confidence – is reported in the certificate.

A measurement operation may serve more than one quality purpose, and there may be functional overlap also (EEE/RM/062rev3 2002). Thus, we need different types of RMs for different functions or levels of control. For example, a CRM, also called standard sample, is desirable for equipment standardization and calibration (standardization of Blaine apparatus for determining cement fineness), but a working-level reference material is adequate for checking hourly samples (for

Table 1 Reference materials – types (ILAC-G9:2005)

Type	Description
Pure substances	These are characterized for chemical purity and/or impurities present in minor/trace quantity
Standard solutions and gas mixtures	These are prepared using gravimetric procedures, mainly from pure substances, and are used for calibration
Matrix reference materials	These are prepared from matrices which contain components of select nature
Physicochemical reference materials	These are characterized for a variety of properties, like melting point, viscosity, etc.
Reference objects or artefacts	These are characterized for a variety of functional properties, like flash point, hardness, etc. microscopy specimens and microbiological specimens are also included in this category

determining fineness) and quality control in cement industry, as per STI/269/7, 2016 (STI/269/7 2016), and IS3535:1986 (IS 3535 1986).

According to ILAC-G9:2005 (ILAC-G9 2005), international guidelines for selecting and using reference materials, and EA-4/14 INF:2003 (EA-4/14 INF 2003), guidelines for selecting reference materials, the RMs (presented in Table 1) are of five types.

Reference materials play an important role in method validation, equipment calibration, estimation of measurement uncertainty, internal quality control (QC), external quality assurance (QA), training of personnel, and proficiency testing (ILAC-G9 2005)

Further, as per ILAC-G9:2005 (ILAC-G9 2005), validity of measurements can be assured through use of validated methods and appropriate equipment, deployment of competent personnel, establishing metrological traceability and estimating measurement uncertainty, participating in interlaboratory comparisons (ILC) and proficiency testing (PT), and laboratory accreditation.

ISO Documents (Guides and Standards) on Reference Materials and Important Definitions

ISO Guides and Standards

ISO has published a series of guidance documents on reference materials. The list of these documents is presented in Table 2.

RM and CRM Definition

ISO Guide 30:2015 (ISO Guide 30 2015) defines reference material (RM) as "Material, sufficiently homogeneous and stable with respect to one or more specified properties, which has been established to be fit for its intended use in a measurement

Table 2 ISO documents on reference materials

Document type and number	Document name
ISO Guide 30:2015	Guidance document for terms and definitions (ISO Guide 30 2015)
ISO Guide 31:2015	Guidance document for certificate contents, labels and related documentation (ISO Guide 31 2015)
ISO Guide 32:1997[a]	Guidance document for calibration in analytical chemistry and use of CRMs (ISO Guide 32 1997)
ISO Guide 33:2015	Guidance document for good practice in using RMs (ISO GUIDE 33 2015)
ISO Guide 34:2009[b]	General requirements for the competence of RMPs (ISO Guide 34 2009)
ISO 17034:2016	Standard for general requirements for the competence of producers of RMs (ISO/IEC 17034 2016)
ISO Guide 35:2017	Guidance document for checking homogeneity and stability (ISO Guide 35 2017)

[a]Withdrawn and revised by ISO Guide 33:2015
[b]Published as standard

process." CRM is an RM with certified property (SI traceable) results, with associated measurement uncertainty reported in the certificate. The standard defines CRM as "Reference Material characterized by a metrologically valid procedure for one or more specified properties, accompanied by a certificate that provides the value of the specified property, its associated uncertainty, and a statement of metrological traceability" (ISO Guide 30 2015).

Role of Reference Materials in Standardization, QC, and QA (ISO Requirements)

ISO/IEC 17025:2017 Requirements: PDCA Approach

ISO/IEC 17025:2017 (ISO/IEC 17025 2017) deals with the subject of reference materials in Sect. 6 (resources) and Sect. 7 (process requirements) under three aspects: metrological traceability (Subsect. 6.5); selection, verification, and validation of methods (Clause 7.2.2); and ensuring the validity of results (Subsect. 7.7). Subsection 8.6 deals with management system improvement.

Metrological Traceability and Measurement Uncertainty

Metrological traceability in ISO/IEC 17025:2017 (ISO/IEC 17025 2017) is considered a "resource." Clause 6.5.1 of the standard requires the laboratories to establish and maintain metrological traceability of measurement results. Traceability can be assured by direct realization of the SI units, calibrations, or RMs. Traceability has to

be documented through unbroken chain of calibrations and linked to an appropriate reference.

Measurement uncertainty. Measurement uncertainty is defined as follows: *Uncertainty of measurement is a non-negative parameter; it characterizes the dispersion of the quantity values which are attributed to a parameter*, according to ISO/IEC Guide 99:2007 (ISO/IEC Guide 99 2007), referred in ISO/IEC 17025:2017 (ISO/IEC 17025 2017).

According to Clause 6.5.2, the laboratories are required to ensure that measurement results are traceable to the International System of Units (SI) through using the services of competent (ISO/IEC 17025 accredited) agency – for calibration or procuring CRMs – which ensures traceability to SI.

Clearly, reference materials with certified values, reporting measurement uncertainty, and assured traceability play all important role in QC and QA by providing evidence of technical validity and thereby building customer confidence. For claiming competence, reference material producers have to fulfil the requirements of ISO 17034:2016 (ISO/IEC 17034 2016).

Method Selection, Verification, and Validation

Sect. 7 is about process requirements in the ISO/IEC 17025:2017 (ISO/IEC 17025 2017) Scheme. A note in Subclause 7.2.2.1 of Clause 7.2.2 (validation of methods) under Subsect. 7.2 (selection, verification, and validation of methods) in the standard recommends for calibration or bias evaluation using RMs, systematic assessment of influencing factors, testing robustness of method, comparison with other methods, interlaboratory comparison (ILC), and evaluation of measurement uncertainty. All these techniques need the application of reference materials for method validation.

Validity of Results

Clause 7.7.1 of Subsect. 7.7 (ensuring the validity of results) under process requirements (Sect. 7) of ISO/IEC 17025:2017 (ISO/IEC 17025 2017) requires planned monitoring of the laboratory quality control activities and, as a minimum, recommends to:

(a) Use RMs or QC material.
(b) Use calibrated alternative instruments.
(c) Conduct functional check(s) on laboratory equipment;
(d) Use working standards for routine calibration.
(e) Conduct intermediate checks on equipment.
(f) Conduct replicate tests or calibrations on test/calibration items.
(g) Retest or recalibrate samples/items retained by the laboratory.
(h) Correlate results for different characteristics of a sample.
(i) Review test/calibration results.

(j) Participate in interlaboratory comparisons (ILC).
(k) Test blind samples, the samples whose parameter values are not known to the analyst.

Monitoring and Review

Further, Clause 7.7.2 of Subsect. 7.7 of the standard requires monitoring laboratory performance by participating in proficiency testing (PT) conducted by ISO/IEC 17043 (ISO/IEC 17043 2010)-accredited competent PT providers and/or interlaboratory comparisons. Thus, the issue of nonavailability of accredited PT providers is addressed through allowing participation in ILC schemes.

Finally, Clause 7.7.3 of Subsect. 7.7 is about review and action for improvement. It requires that data from monitoring activities should be analyzed, and used, for control and improvement of the laboratory's activities; appropriate action is required to be taken in case of deviations (ISO/IEC 17025 2017). Statistical techniques should be used for defining and analyzing problems and planning actions for improvement.

The requirements of reference materials, prescribed in ISO/IEC 17025:2017 (ISO/IEC 17025 2017) and discussed above, are summarized in Table 3.

It is very clear from the above that the technical validity of results cannot be assured without discussion on metrological traceability and implementing the recommended measures in all quality control activities: method selection, verification and validation, and monitoring and review, i.e., the activities spanning planning, implementation, checking, and action, all stages of plan-do-check-act (PDCA) cycle, which is also called PDSA (plan-do-study-act) cycle, quality control cycle, or management cycle or Deming cycle, and is the basis for continual improvement.

Japanese organizations, after the Second World War, learnt from the teachings of Dr. Juran and Dr. Deming and implemented PDCA and quality circle approach and total quality control (TQC), now known as total quality management (TQM), and established international competitiveness of the Japanese industry. The main requirements of ISO/IEC 17025:2017 (ISO/IEC 17025 2017) regarding ruse of reference

Table 3 Important sections of ISO/IEC 17025:2017 on reference materials

Section number	Section name	Statement
6.5.1	Metrological traceability	Laboratories have to establish and maintain metrological traceability of their measurement results, linking them to an appropriate reference
6.5.2	Metrological traceability	Measurement results have to be traceable to the SI units through certified values of CRMs
7.2.2	Method validation	The techniques mentioned for method validation, particularly evaluation of measurement uncertainty, need application of CRMs
7.7.1	Validity of results	Laboratories need to use reference materials as one of the most important tool for ensuring technical validity of results

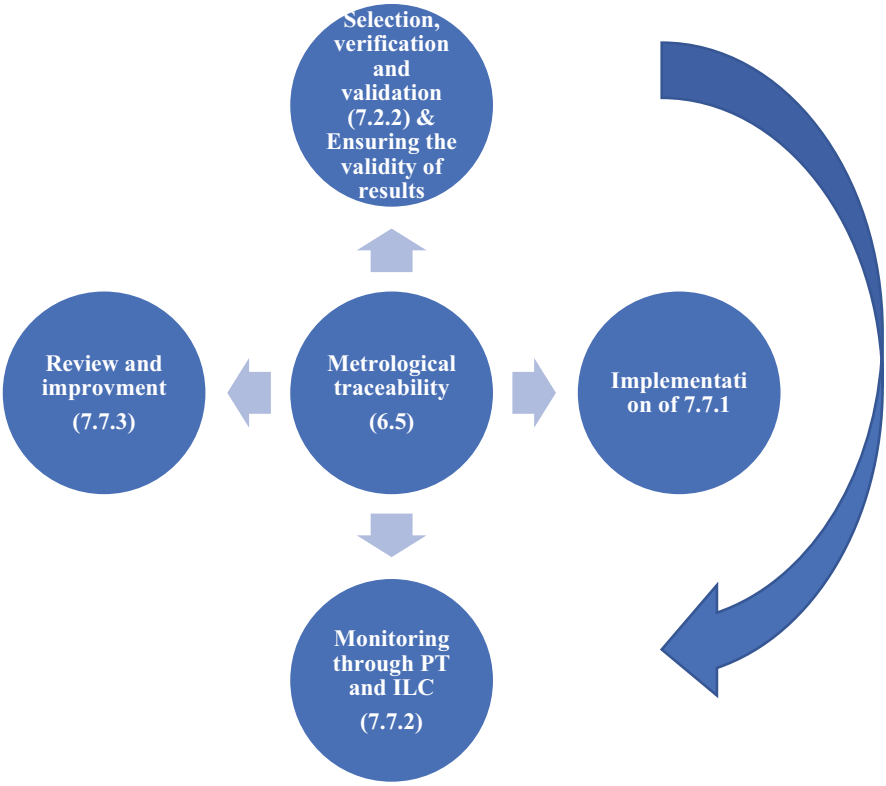

Fig. 2 QC and QA with application of ISO/IEC 17025:2017 requirements

materials in quality control (QC) and quality assurance (QA) are presented in Fig. 2. In the context of the laboratory competence, these measures help control the quality of measurements and also give assurance about quality, internally and externally, to interested parties.

System Improvement

Along with monitoring of process elements, monitoring and review of the management system are also required. Laboratories are required to improve the activities and results in a system's framework. Improvement is facilitated by adopting plan-do-check-act (PDCA) model for problem-solving and conducting periodic internal audits and management review. ISO/IEC 17025:2017 (ISO/IEC 17025 2017) in Subsect. 8.6 stresses on these aspects, along with risk management, use of policies and procedures, data analysis, and PT results. The structure of the management and technical systems, in fact, is based on PDCA framework (Fig. 3) in all 17,000 series

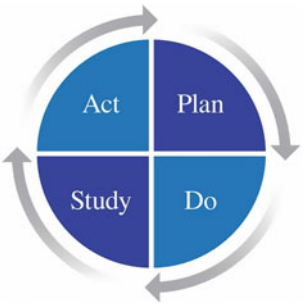

Fig. 3 PDCA cycle of continual improvement

standards, like ISO/IEC 17025 (ISO/IEC 17025 2017), ISO 17034 (ISO/IEC 17034 2016), ISO/IEC 17043 (ISO/IEC 17043 2010), and ISO 9001 (ISO 9001 2015).

ISO 17034:2016 on Production of Reference Materials

About ISO 17034:2016

ILAC-G12:2000 (ILAC-G12 2000) guidelines for RMPs contained 13 management system requirements and 19 technical requirements. ISO Guide 34: 2009 (ISO Guide 34 2009) which laid down requirements for the competence of RMPs contained 15 management requirements and 18 technical requirements.

ISO 17034:2016 (ISO/IEC 17034 2016), earlier Guide 34 (ISO Guide 34 2009), lays down general requirements for the competence of reference material producers (RMPs). Like ISO 17025:2017 (ISO/IEC 17025 2017), ISO 17034:2016 (ISO/IEC 17034 2016) includes structural, resource, and management system requirements. To address unique production aspects of reference materials, process requirements are detailed as technical and production requirements in the standard. ISO 17034:2016 (ISO/IEC 17034 2016) has been harmonized with the recent editions of ISO Guides on reference materials. The term quality manager has been removed in the standard and clause on risk management included. The production of multiple batches has been addressed explicitly, and homogeneity must be assessed for each property of interest (Trapmann et al. 2017). Figure 4 presents the family of reference material guides which support ISO 17034:2016 (ISO/IEC 17034 2016).

ISO 17034:2016 (ISO/IEC 17034 2016) shows the following major changes when it is read with ISO Guide 34:2009 (ISO Guide 34 2009):

- The standard is restructured as are the other ISO 17000 standards on conformity assessment.
- The standard includes requirements for all types of RMs and specifies additional requirements for CRMs.
- The standard is harmonized with revised versions of ISO Guide 31 (ISO Guide 31 2015) and ISO Guide 35 (ISO Guide 35 2017).

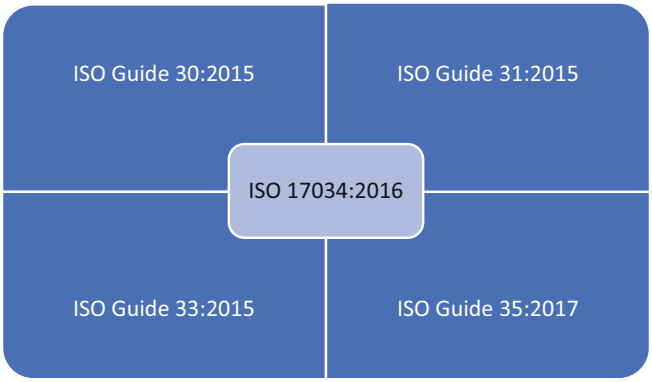

Fig. 4 Family of ISO Guides on reference materials supporting ISO 17034:2016

- The standard, like other 17,000 documents, requires addressing risks and opportunities.

ISO 17034:2016 Criteria for Competence of RMPs

ISO 17034:2016 (ISO/IEC 17034 2016) lists the following principal requirements for RMPs: general requirements, structural requirements, resource requirements, production and technical requirements, and management system requirements.

Sect. 7 on production and technical requirements in ISO 17034:2016 (ISO/IEC 17034 2016) contains the following subsections: 7.1 general requirements, 7.2 production planning, 7.3 production control, 7.4 material handling and storage, 7.5 material processing, 7.6 measurement procedures, 7.7 measuring equipment, 7.8 data integrity and evaluation, 7.9 metrological traceability and certified values, 7.10 assessment of homogeneity, 7.11 assessment and monitoring of stability, 7.12 characterization, 7.13 assignment of property values and their uncertainties, 7.14 RM documents and labels, 7.15 distribution service, 7.16 control of quality and technical records, 7.17 management of nonconforming work, and 7.18 complaints.

The process of production of RMs begins with need identification. Basically, production and delivery have to be planned for the following activities:

- Customer need identification.
- Material selection.
- Measurement method selection and verification.
- Material processing.
- Material testing and evaluation.
- Labeling and packaging.
- Homogeneity assessment.
- Stability checking.
- Value assignment.

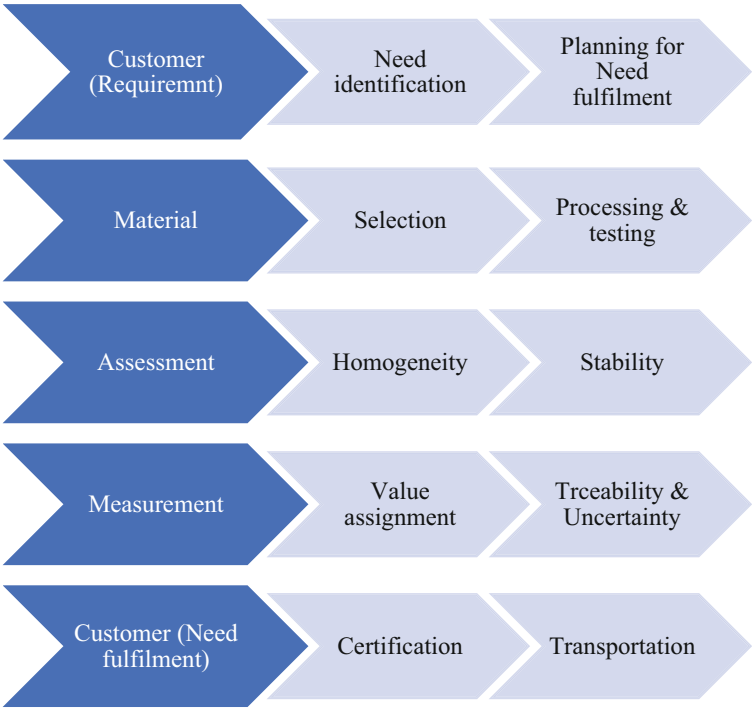

Fig. 5 Five-stage CMAMC approach to reference material production and delivery

- Certification.
- Transportation and delivery.
- Customer feedback and complaints analysis.

The steps of the process are presented as five stage process (CMAMC) relating to customer (requirement), material, assessment, measurement, and customer (need fulfilment), in Fig. 5. Establishment of metrological traceability and statement of measurement uncertainty is mandatory. The process begins with customer and ends with customer.

To avoid risk of technical invalidity of results, due care has to be taken for maintenance of storage and environmental conditions. This needs discussion on storage requirements for RMs as also their homogeneity and stability.

Facilities, Storage, and Environmental Conditions for Production and Distribution of RMs

Provision of facilities and environmental conditions and control of material handling and storage are important requirements for maintaining the technical validity of

reference materials. ISO 17034:2016 (ISO/IEC 17034 2016), firstly in Subsect. 6.4, requires that all facilities for production and distribution of RMs: laboratory facilities, calibration areas, testing areas, material handling, and processing and packaging areas, including energy sources, temperature and humidity conditions, lighting, and ventilation, should be such which ensure proper material handling, processing and packaging, and performance of various activities. Monitoring and strict control of these facilities are required. Further, ISO 17034:2016 (ISO/IEC 17034 2016) in Subsect. 7.4 lays down the requirements for material handling and storage. Personnel handling the materials should have undergone training on material handling. Adequate safety measures for environmental protection need to be taken. Precautions have to be taken to protect against adverse environmental conditions and any unforeseen or unintended contamination. Candidate RMs and RMs have to be identified and preserved, and integrity of these materials has to be ensured throughout the production process. Packaging and labeling process also have to be controlled and condition (stability) of RMs assessed at appropriate intervals.

Homogeneity, Stability, and Value Assignment

Homogeneity of RM candidates and stability of RMs are important requirements, a prerequisite, during their production and use. ISO 17034:2016 (ISO/IEC 17034 2016) requires establishing documented procedures for assessing homogeneity and stability on packed items. Homogeneity assessment is done by performing repeatability (within sample/bottle standard deviation) and reproducibility (between sample/bottle standard deviation). For stability check (checking condition of the materials during their utilization period) – which is conducted at periodic intervals, same laboratory – same test method and same parameters are checked. ISO Guide 35:2017 (ISO Guide 35 2017) provides approaches for assessment of homogeneity and stability, management of the risks associated with possible stability issues, characterization and value assignment of properties, evaluation of uncertainty for certified values, and establishment of the metrological traceability of certified property values. ISO 13528:2015 (ISO 13528 2015) also is a useful standard for the purpose.

Indian Reference Materials

Gross domestic product (GDP) in India is expected "to grow at 12-12.5 per cent (8 per cent real GDP and 4-4.5 per cent inflation) in the current financial year, and the economy will attain a size of USD 3,350-3,400 billion in 2022-23 and touch USD 5 trillion by 2026-27" (PhD Chamber 2022). In this environment, there is a growing market for RMs and CRMs. The reference materials produced in India by the premier organizations are of the highest quality and meet traceability requirements. Certificates issued for CRMs contain uncertainty statement as required. Due care is taken in homogeneity assessment, stability checking, storing, stocking, and

transporting. Indian organizations are getting ISO 17034:2016 (ISO/IEC 17034 2016) accreditation also.

Bharatiya Nirdeshak Dravyas (BNDs)

National Physical Laboratory of India (NPLI) has established a scheme of producing Bharatiya Nirdeshak Dravyas (BNDs). Now RMs produced by competent organizations are elaborately checked in coordination with reference material producer (RMP) by CSIR-NPL and certified for traceability. These RMs are then marked as BND by CSIR-NPL. BND is the trademark, and this marking allows dissemination of traceability throughout the world for test measurement of certified parameters; BNDs in different sectors are uploaded on the international COMAR database (Pant 2018).

Collaboration between CSIR-NPL and the National Council for Cement and Building Materials (NCCBM)

Apart from CSIR laboratories, the National Council for Cement and Building Materials (NCCBM), a premier governmental autonomous organization, has been making RMs for over four decades and delivering to cement and construction material organizations, R&D bodies, and third-party laboratories in the country and abroad (Breja et al. 2000). The CRMs produced by NCCBM are cost-effective, and quite a few were listed in ISO/REMCO directory (ISO/REMCO Document N 330 List of producers of certified reference materials, Information by Task Group 3 n.d.) and COMAR (Code d'Indexation des Matériaux de Reference) database also since early times. Over the years, it made 78 types of CRMs, including the ones for meeting quality control needs of cement industry, like CRMs of Blaine fineness and chemical composition for cement of different types and pozzolanic materials and CRM of pure phases of cement clinker, coal, etc., and the efforts support government of India's "Make in India" campaign (Breja et al. 2017).

Indian cement industry and overseas cement manufacturing plants routinely use CRMs produced by NCCBM, for equipment standardization and calibration and product quality control and quality assurance. Some of these CRMs, like the one for Blaine fineness, have been explicitly mandated by Indian standard codes (IS standards on various types of cement) and Bureau of Indian Standards Scheme of Testing and Inspection (BIS-STI), for use in quality control, maintaining traceability of measurements, and ensuring parity in results produced by different cement testing laboratories. These CRMs are used for ensuring technical validity of measurements also and meeting the requirements of Clause 7.7 of ISO/IEC 17025:2017 (ISO/IEC 17025 2017).

Now going one step further, NCCBM has collaborated with CSIR-NPL and developed Bharatiya Nirdeshak Dravyas (BNDs). "The National Measurement Institute (NMI) of respective country is to maintain the measurement standard

which is metrologically traceable to the international standards at International Bureau of Weights and Measures (BIPM) through unbroken chain of measurements with stated uncertainties. NPLI is developing IRMs (Indian Reference Materials) in association with National Council for Cement and Building Materials" (Kumari et al. 2022). NCCBM is the first organization, outside CSIR, to collaborate and jointly develop and issue these CRMs. The first five BNDs of cement and pozzolanas, produced by NCCBM in collaboration with NPLI, were released by Dr. Harsha Vardhan, the then minister of science and technology, Govt. of India, on 16 August 2018. Among these, the BND 5001 of Ordinary Portland Cement (OPC) for Blaine fineness and BND 5051 for chemical parameters of OPC, BND 5004 for fly ash for Blaine fineness are most sought for by the cement industry. More BNDs of NCCBM were dedicated to the nation by Prime Minster Mr. Narendra Modi on 4 January 2021, foundation day of NPLI (Yadav 2021). NCCBM has received ISO 17034:2016 (ISO/IEC 17034 2016) accreditation also from the National Accreditation Board for Testing and Calibration Laboratories, New Delhi.

Honourable Minister of Science and Technology, Govt. of India (fourth from right), releasing the BNDs at CSIR-NPL. Also present are Director CSIR-NPL (third from right) and the author (second from right)

Role of Reference Materials in Production, Quality Control, and Quality Assurance in Manufacturing Industry

Manufacturing processes operate under strict control. The essential equipment and instruments used in controlling input, process, and output quality parameters have to be checked and calibrated wherever measurements impact process and product

quality requirements. Quality control and quality assurance laboratories of the manufacturing units must be competent and use calibrated equipment and instruments. Third-party testing and calibration laboratories also must use calibrated equipment. RMs and CRMs wherever required have to be used. Case of quality control and quality assurance and the role of reference materials in cement industry are discussed in this section.

Quality Control in Cement Industry

India is the second-largest producer of cement in the world with 210 large cement plants and around 350 mini cement plants. Indian cement industry produced 334.48 million tons of cement in 2019–2020 and 232.57 million tons in 2020–2021 up to January 2021 (Cement Units, Lok Sabha Unstarred Question No. 4617 2021). Cement manufacturing is under mandatory licensing, and the Bureau of Indian Standards (BIS) is the standards publishing and licensing organization in the country.

Cement manufacturing is a complex process, involving mining of raw materials – mainly limestone and clay – crushing to −80 mm size, grinding to a fine −90 micron size, sintering at a temperature of 1450 °C, cooling the output clinker, and grinding with gypsum in a cement mill. In India, 90% of cement is packaged in bags and distributed. Clinker and cement quality are governed by raw material composition and control of process parameters. Indian cement industry has world-class cement plants and testing laboratories. Quite a few laboratories have received ISO/IEC 17025:2017 (ISO/IEC 17025 2017) accreditation also.

Sampling and Inspection (IS 3535:1986 Requirements) in Cement Manufacture

IS 3535:1986 (IS 3535 1986) prescribes the methods for sampling and the criteria for conformity of hydraulic cements. The manufacturer is required to draw representative samples in accordance with the standard requirements (presented in Table 4) for various cement types. This standard was first issued in 1966 and then revised in 1986. It lays down the requirements for sampling and inspection to meet the quality control and quality assurance needs of the cement plants at all stages of cement production: clinker manufacturing, clinker grinding to make cement, and final inspection of cement.

BIS Scheme of Testing and Inspection (BIS-STI): Requirement of Calibration and RMs

All mechanical and chemical test equipment, including balance, sieves, Blaine apparatus, vibrating machine, humidity chamber, compression testing machine, thermometer, data logger, sophisticated chemical testing instruments, glassware, etc., in the laboratory must be calibrated. Casting molds and other appliances are

Table 4 Frequency of inspection and tests for production of hydraulic cement (IS 3535:1986)

Sl. no.	Test	Clinker/pozzolana	Cement grinding	Cement packing
1	Chemical composition	Every day: one composite sample for clinker Every week: one composite sample for pozzolana	Every week: one composite sample	Every week: one composite sample
2	Fineness	Every day: one composite sample (prepared from clinker sample taken hourly)	Every alternate hour: one sample Every day: one composite sample	Every day: one composite sample
3	Setting time	-do-	a. Every hour: one sample b. Every day: one composite sample	-do- –
4	Soundness	-do-	-do-	-do-
5	Compressive strength	-do-	-do-	-do-

Table 5 BIS scheme of testing and inspection for certification of ordinary Portland cement according to IS 269: 2015 (sixth revision)

Sl. no.	Test equipment	Frequency of calibration
1	Blaine apparatus for fineness determination	Every day: with producer's own standard cement sample (working standard) Every month: using NCCBM-supplied standard cement sample (CRM)
2	Compressive strength testing machine for compressive strength determination	Every month: with producer's own proving ring and the ring from ISO/IEC 17025-accredited laboratory or NPL or NPL-accredited proving ring manufacturer (once in 2 years)
3	Autoclave pressure gauge used for volume expansion test	Every month: with producer's own dead weight pressure gauge tester or Every 6 months: from ISO/IEC 17025-accredited laboratory or NPL/NABL-accredited manufacturer of such gauges
4	Vibration machine used for cube casting	Every month: with producer's own tachometer. The tachometer has to be calibrated from ISO 17025-accredited laboratory (once in a year)

required to be periodically checked. As a minimum, BIS Scheme of Testing and Inspection (STI/269/7 2016) for ordinary Portland cement (IS 269 2015) lays down the requirements (Table 5) for equipment calibration. Similar guidelines have been issued for other varieties of cement also. Use of NCCBM standard cement sample is explicitly mentioned for calibration of Blaine air permeability apparatus.

CTM Calibration in Cement Plant Lab

Blaine Apparatus Calibration in Plant Lab

National Council for Cement and Building Materials (NCCBM) was the first organization outside CSIR-NPL to sign a memorandum of understanding (MOU) with NPLI for BND marking of its CRMs, for use in cement and construction sector. BNDs are the primary standards; the metrological traceability of BND is linked to SI units (Kumari et al. 2022). The point is that in the area of reference materials, BND – being approved by NMI – is the first link in the traceability chain and occupies the top spot in the traceability pyramid in India.

Conclusion

World of business works on commonly accepted standards and codes. Any deviation or violation of the accepted standards and codes impacts the quality of measurements and leads to problems at consumer's end and return of consignments. This also results in delays/disruptions and shortages in various parts of the world, and disruptions have a huge economic cost. Therefore, it is important for the countries to work toward self-reliance, upgrade their systems to international levels, and compete. Technically, all countries have to raise the standard of measurements, undertake all test and measurements as per standards and guidelines, and maintain accuracy of measurement results. Results of all calibrations must have evidence of metrological traceability and provide statement of measurement uncertainty, and test results should be reported after ascertaining measurement uncertainty at a given value. Here lies the importance of RMs and CRMs. Self-reliance in terms of production of CRMs is a measure of a nation's technical capabilities and reduces dependence on costly imports. India has a sound accreditation system and national standards authority (BIS), national metrology institute (NMI), and a scheme of BNDs to serve the country's needs. NCCBM, a governmental autonomous body, is playing a pivotal role by contributing in the development of BNDs for cement and construction sector and ensuring the traceability of reference material values. Premier organizations in other sectors of the economy also should come forward and support

the mission of BND development for meeting the country's requirements. Such initiatives should be taken by other countries also for meeting the requirements of CRMs for their respective product standards and competence requirements for RMPs.

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Precious Metals: Bharatiya Nirdeshak Dravya for Pure and Sure Jewelry

26

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Abstract

India and China have a demand for lots of gold and silver due to their age-old customs. Also, purchasing gold and silver is considered an investment. There is a very large market for these metals and due to the unavailability of the proper knowledge and regulation, there is a possibility of rigging in this. Therefore, there is a need for government regulations that these precious metals should be sold only after testing. For testing, one needs standard equipment to calibrate the equipment reference materials. Generally, these reference materials are costly in the international market, and therefore for ease, each nation should produce its own reference materials, which should be comparable to international standards. India has taken an initiative in this direction and registered BND (Bharatiya Nirdeshaka Dravya) as Indian reference materials. In the case of Gold, the first BND was released in collaboration with the Mint, Mumbai, and, currently, 7 BNDs have been released in association with Jalan and Company, New Delhi. This chapter describes the methods and procedures in detail.

Introduction

Precious metals are rare and have a high economic value due to various factors, including their scarcity, use in industrial processes, and role throughout history as a store of value. The most popular precious metals with investors are gold, platinum, and silver (Schoenberger 2011; Grimwade 2009; Barefoot and Van Loon 1999). On average, the Earth's crust contains 11 parts per billion of gold, 70 parts per billion of silver, and 1 part per billion of platinum instead of the abundantly found iron (23,000,000 parts per billion) (ter Haar 1949).

India is the largest consumer of gold, silver, and platinum jewelry, as precious metals are ancient Indian culture(Narang and Singh 2012). The business of jewelry in India is nearly a trillion dollars. Precious metals are easy to handle and craft due to their mechanical properties like malleability, which means they can be deformed, pressed, or rolled into sheets. They possess ductility – the property of being bent, because of which they can quickly be drawn into long wires without fracture (Lee et al. 2003; Yu et al. 2022). A gram of gold can be drawn into a wire 2300 m long. These metals have high optical reflectivity, which is responsible for their attractive luster and sheen. Gold reflects 95%, silver 97%, and platinum 73% of the light incident.

They withstand corrosion and tarnishing by not chemically reacting with their surroundings. They can last for years without diminishing quantity or quality, thus retaining their monetary value (Washbourne 2021). However, metals with similar mechanical properties (but are much more abundant), like copper and the alloy brass (mainly copper and zinc), are not used for jewelry manufacturing.

Global Market

The Global precious metal market was valued at USD 182.1 billion in 2019 and is expected to grow at a compound annual growth rate CAGR of 9.0% in revenue from 2020 to 2027 (Size 2020). Asia-pacific is the primary producer as well as consumer of gold. The metal is significantly consumed in China and India as the precious metal value is determined by finance measurement. Countries have established laws regulating the manufacture and sale of precious metal items to protect the market and consumers.

Therefore, through the Board of Indian Standards, the Government of India has a proposition that gold with a certain purity, 9 karats, 14 karats, 18 karats, 20 karats, 22 karats, and 23-karat, will be available with the jewelers. Another gold purity will not be allowed in the market. Regulation authorities have made the rule mandatory to have a Hallmark on Gold items (Goyal et al. n.d.).

Possible Impurities

The purity of gold is measured in karats. Thus, 14 k gold will be 58.5%, and 10-carat gold will be 41.7% pure. In simple terms, the higher the carat, the higher the gold's purity, and the greater its value (Kinneberg et al. 1998). Sometimes, it becomes confusing as gold comes in various colors, like, rose gold, white gold, and green gold. The variation in color of gold is made by mixing some percentage of other metals like copper (rose gold), platinum (white gold), etc. Similarly, silver has antimony and lead in the majority. Physical examinations are insufficient to determine the item's purity; the supplier provides a purity certificate.

Role of CRMs and Need for BNDs

For a typical person, differentiation in a metal can be made by the corresponding color like gold is yellow, silver and platinum is white. It is challenging to identify the mixing of other low-cost metals/alloys by physical examination. It led to cheating and forgeries in the market, and the customer ended up having a low purity item even paid for higher.

Many standard test procedures were used to define the purity of the specific metals, including sophisticated instruments like X-ray fluorescence (XRF) calibrated with certified reference materials (CRM) (Stankiewicz et al. 1998). India is currently procuring CRMs for various metals at a considerable cost due to additional load freight charges. Suppose the low-cost CRMs at par with the international level can be produced locally. In that case, CRMs can be made available to all the jewelers and mandatory to calibrate their equipment for certification. Therefore, there is a need to make different grades of CRMs for metals like gold, silver, platinum, etc.

Methods/Techniques of Testing Purity and Concentration of Metal

BAM Federal Institute for Materials Research and Testing certified three gold-based reference materials (CRMs) suitable for X-ray fluorescence spectrometry (rose gold, white gold, and yellow gold) (Venkateswaran 2013). The alloying elements' mass fractions, gold, silver, partially copper, nickel, and zinc, were certified. Characterization of the CRMs was performed using wet chemical methods and fire assay.

Gold Purity Testing

There are two methods to determine the purity of gold: destructive and non-destructive. The destructive techniques are fire assays and inductively coupled plasma mass spectrometer (ICP MS). Other methods are not destructive. Among the nondestructive techniques, touchstone and energy dispersive X-ray analysis can only determine the purity of the top surface. The nondestructive method like weighing balance, density meter, or ultrasound can measure bulk properties but cannot tell about the element present as an impurity.

Fire Assay Method

Initially, sampling is done. If the sample is in the ingot or dore bar, then sampling is done by drilling diagonally from the upper and lower side of the sample.

The fire assay method (Singh and Rugged 2012; Hoffman et al. 1998) is carried out per the BIS standard IS 1418:2009. It is a destructive method used for purity determination in gold samples, jewelry, etc. This method has been used for a long time. The cupellation process is used in this method. In the cupellation process, the ore of precious metal, pure silver, and copper are treated at high temperatures in an oxygen environment. At high temperatures, precious metal does not get oxidized. Still, other base metals present in the ore and lead are used for wrapping the precious metal from their metal oxides. These metal oxides, especially lead oxide, are absorbed by the cupel of magnesium oxide/calcium phosphate.

In this method, the known amount of the sample is weighed in triplicate/duplicate. A known amount of pure silver is inquarted in the gold sample to facilitate the parting of the gold by nitric acid. A known amount of pure copper is added to the gold sample to stop spurting during cupellation. To minimize possible systematic errors in the procedure, assaying of standard proof sample is also carried out parallelly.

After weighing all these materials, these are wrapped in known weighted lead foil, and a small ball-type shape is made mechanically with the help of balling pliers. This ball is carefully kept in the cupels and held in a preheated cupellation furnace at 850 °C. The temperature of the cupellation furnace is increased to 1100 °C, and the cupellation

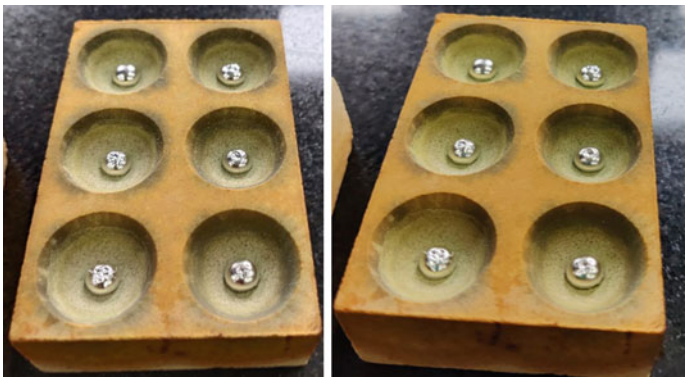


Fig. 1 Gold-silver alloy button in cupel after cupellation process

Fig. 2 Elongated gold-silver alloy fillets after annealing at 800 °C in an annealing tray

at 1100 °C for 30 min is done. In the cupellation process, all the lead gets absorbed by the cupel. After it gets cooled down, a button type shape of a gold-silver alloy is obtained (as shown in Fig. 1).

That button-shaped alloy is cleaned with a soft brush and flattened with the help of a polished hammer. This flattened button is annealed at 700 °C for 5 min in an annealing tray to facilitate the softening of the flattened button. These flattened buttons are again flattened with the help of a rolling mill and made in elongated fillets of approximately 0.2 mm thickness (Fig. 2).

Then this fillet is annealed at 800 °C to facilitate the softening of the fillet. A cornet is made from that soft elongated fillet, and the cornet is kept in the parting tray (Fig. 3). The parting tray is immersed in.

1.2 sp. gr. nitric acid for separating silver from gold by selectively dissolving the silver in nitric acid at 90–95 °C for approximately 30 min or until no more nitrous

Fig. 3 Cornet in small thimbles, which are kept in a parting tray

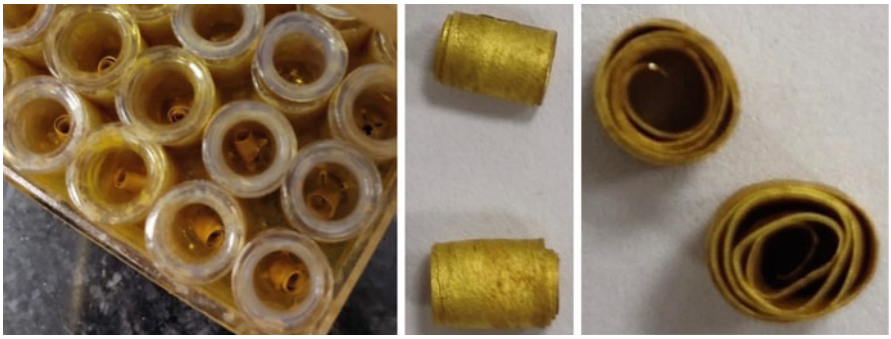


Fig. 4 Cornet in the thimbles after the parting and annealing process

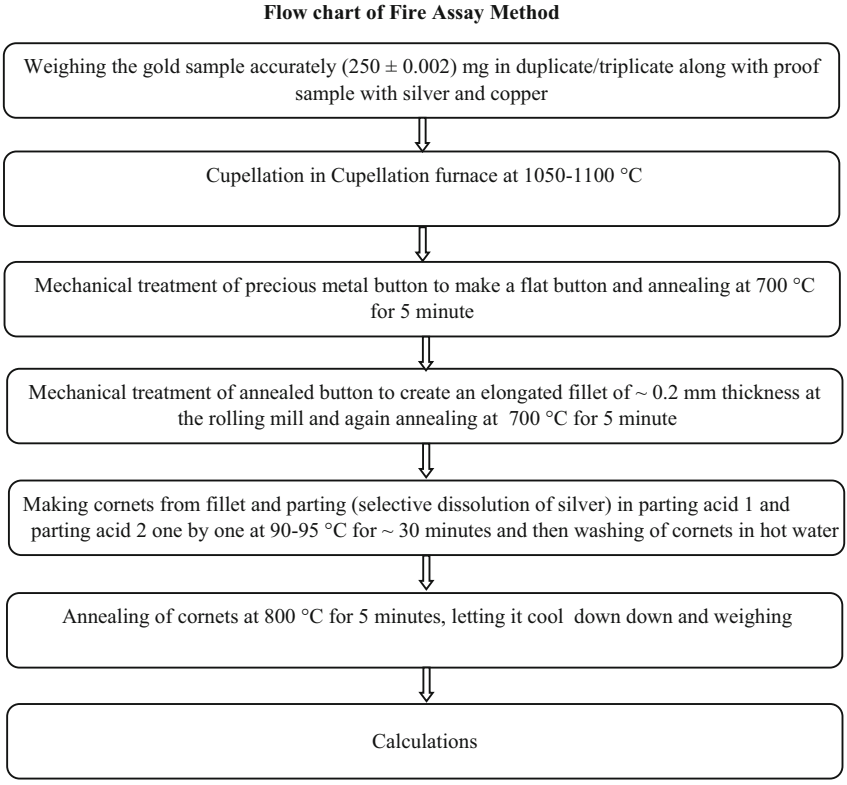
fumes are observed. Then it is washed by immersing the parting tray in hot distilled water. It is dipped in 1.3 specific gravity nitric acid at 90–95 °C for approximately 30 min to refine gold further. Furthermore, it is washed thoroughly to remove all the nitrates. Then the cornet is dried by heating and annealed at 800 °C for 5 min (Fig. 4). Finally, the weight of the cornet is taken.

$$\mathrm{w} = \frac{m_2 + \Delta}{m_1} * 1000 \tag{1}$$

w is the purity of the precious metal in part per thousand (PPT). m_1 and m_2 are the initial and final weights of the test sample

$$\Delta = \frac{\Delta_1 + \Delta_2}{2} \tag{2}$$

Δ_1 = Initial weight of first proof sample – Final weight of the first proof sample
Δ_2 = Initial weight of second proof sample – Final weight of the second proof sample



Gravimetric Method

As described in method (Gee 1877), the gold metal's surface is initially cleaned. Five replicates are taken in separate beakers after washing and drying the known amount of gold (0.2 g). These samples are dissolved in 20 ml aqua regia (3:1 ratio of hydrochloric acid and nitric acid) in 10 ml of de-ionized water. Then these samples are heated on a hot plate to digest samples to make them in syrup-like conditions. Then 5 ml of hydrochloric acid and 50 mL of de-ionized water are added and heated up to boiling. Approximately 10–12 g of hydroxylamine hydrochloride/hydroxyl ammonium chloride was added to digested samples under hot conditions and boiled once again. The golden yellow/ brown color precipitate is formed. After cooling down the precipitate at room temperature, it is filtered through a cleaned, dried, and pre-weighed G-4 crucible. Then the precipitate is dried at 105 °C for 1 h. It is allowed to cool down and be weighed. The weighing of G-4 crucibles before and

after precipitate filtration is carried out at room temperature. The initial and final weight difference of the G-4 crucible gives total gold.

Silver Purity Testing

Gravimetric Method

The method used for silver assaying/testing in silver and silver alloys is carried out per the BIS standard IS 2113:2014 (Nadin et al. 2001). In this method, precious metal or ore of the precious metal is dissolved in nitric acid (Tripathy et al. 2018). Nitrate salt of silver is formed. Silver nitrate remains soluble in acid, and silver is precipitated in silver chloride using hydrochloric acid. Silver chloride precipitate is filtered in a previously weighed sintered glass crucible. Then it is dried at 140 °C for 1 h. Weight of crucible along with AgCl ppt is taken. Then Ag is estimated in the form of AgCl.

Briefly, in this method, a 1.00 g ± 0.01 mg silver alloy sample is weighed in triplicate. These are transferred to a 250 ml beaker. Then 15 ml of nitric acid in a 1:1 ratio is added. The beakers are kept on a hot plate at approximately 100 °C till the samples are completely dissolved. The nitrous fumes are expelled (Fig. 5). Then the beaker is removed from the hot plate, and the solution is diluted with 100 ml of distilled water. Now 20 ml of hydrochloric acid is added with the help of a glass rod continuously stirring to facilitate the precipitation of silver as silver chloride. Again, the beaker is covered with a watch glass, kept on a hot plate, and heated. After 1 h, a few drops of dilute hydrochloric acid (1:9) are added to check the completeness of the precipitation process of silver. Then the beaker is kept on a hot plate for at least 2 h.

The beakers are removed from the hot plate and allowed to cool down (Fig. 6). Then all the content in the beaker is carefully filtered through a previously dried and weighed sintered glass crucible. All the walls of the beaker are washed with distilled water to avoid losses.

Now sintered glass crucible is heated at 140 °C for 1 h in the oven to dry the precipitate. After the crucibles are cooled down, the weight of the crucible is taken.

Fig. 5 Beakers on the hot plate during nitrous fumes expelled

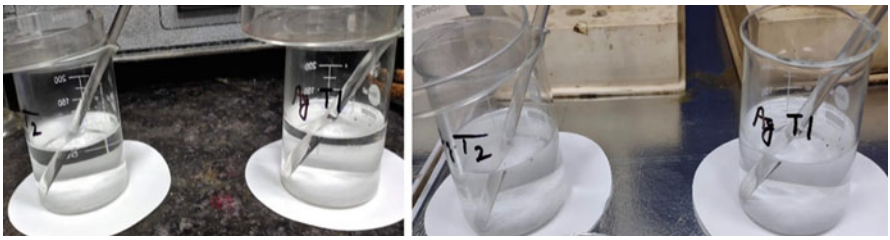
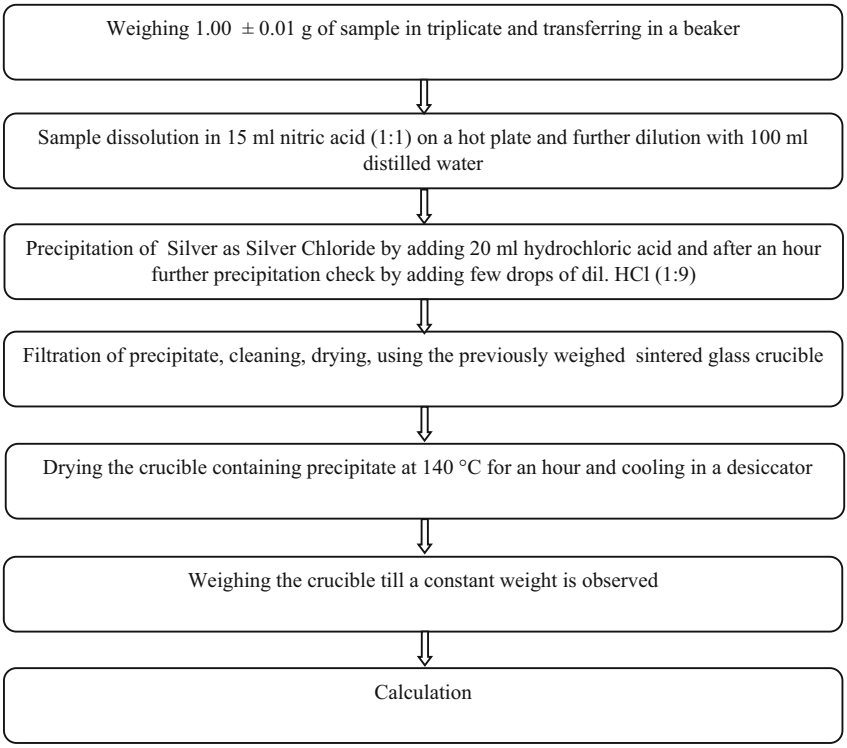


Fig. 6 Precipitation of silver as silver chloride

$$\text{Silver (in part per thousand)} = \frac{\text{A} * \text{molar mass of silver} * 1000}{\text{B} * \text{molar mass of silver chloride}} = \frac{\text{A} * 0.7526 * 1000}{\text{B}} \quad (3)$$

A is the weight of silver chloride in g, and B is the weight of the sample in g.

Flow chart of gravimetric method (Silver assay)



Volumetric Method

In this method (Easterbrook 1957), a known amount of the sample is dissolved in nitric acid. This solution is titrated with standard sodium chloride or potassium chloride solution using the potentiometric indication of the endpoint. The silver content is calculated based on consumed standard sodium chloride or standard potassium chloride solution.

Supporting/Secondary Test Methods

X-Ray Fluorescence (XRF)

X-ray is a nondestructive technique for determining the material's qualitative and quantitative composition (Nayak and Singh 2007). X-ray is considered to be the fingerprint of a material. Each element has characteristics of X-ray energy, and depending on the energy, the element can be determined. A quantitative analysis can be done depending on the intensity of the peak. In X-ray fluorescence spectroscopy, a high-energy beam falls on the sample, and X-rays are emitted. X-ray characteristics tell about the element and the intensity of the peak of the concentration of the component. Thus, XRF can give both qualitative and quantitative information about a sample.

Using XRF, samples can be characterized into two modes: wavelength dispersive X-ray fluorescence (WDXRF) and energy dispersive X-ray fluorescence (EDXRF) (Brouwer 2006). There is separate instrumentation for these two. WDXRF has better accuracy than energy-dispersive XRF. However, for practical purposes, EDXRF is considered enough to measure gold purity within the 0.1% range. On the other hand, WDXRF is expensive and requires many operational, installation, and maintenance costs. It also has limitations of chamber volume. Therefore, for general purposes, EDXRF is considered suitable for work related to gold. However, for research purposes, WDXRF can be very good where the purity has to be determined to be very high, like in certified reference material. For certified reference material, the uncertainty in the measurement should be as low as possible; therefore, WDXRF will be more suited.

EDXRF uses a less powerful X-ray source, so the requirement for a water chiller, etc., is not a requirement (Yusoff et al. 2002). Thus, the instrument is cheaper and smaller than the WDXRF, and a reasonable accuracy is achieved. However, the device needs to be calibrated against the known standards.

Recently, using XRF has become common as it is a handheld instrument. It serves several purposes, like (i) checking the concentration of gold, (ii) finding the presence of other elements, and (iii) checking the authenticity of the sample. XRF is used for forensic and pharma purposes also. For the forensic department, it is challenging to know whether a yellow metal is a bar of gold or not! The instrumental technique provides quick and accurate results. The test can be done at several points of a sample. Thus, the composition can be detected in several places. Depending upon the metal and sample, the composition of the top surface of 10–50 micron thickness can be analyzed.

Trace Element Analysis

Platinum group metals in the samples (insoluble in nitric acid) can lead to defects if they are conveniently removed without detection. Other techniques, such as touchstone, partitioning, atomic absorption spectroscopy (AAS), and induction-integrated plasma spectrometry (ICP), are suitable for gold analysis. While some require minute samples, they are destructive techniques, and most require prototyping. Furthermore, except for ICP, these methods do not have sufficient accuracy (Amais et al. 2020).

The most common official method of determining silver is potentiometric titration. It is an extinction technique based on potentiometric monitoring of precipitation of silver chloride (ISO Standard 11,427). Spectroscopic techniques can determine silver, but its solutions are unstable, and the results are not accurate enough.

Flame Atomic Absorption (AA)

The sample is dissolved in aqua regia and absorbed in an acetylene flame. The beam of light travels through the flame at the wavelength of gold. The gold in the sample is the ratio of light to the concentration of the element in the solution. Absorption is compared with standard solutions to determine the concentration of gold in the sample (Tuzen et al. 2008).

Inductively Coupled Plasma-Atomic Emission Spectrometry (ICP-AES)

The aqueous sample is absorbed into the plasma, and the gold in the sample emits light at distinct frequencies (Su et al. 1995). The intensity of light emitted is measured by software with standard gold-bearing solutions. The software then calculates the gold concentration of the sample.

Inductively Coupled Plasma-Mass Spectrometry (ICP-MS)

Gold in aqueous form is ionized. The intensity of the signal at mass 186 is measured, and the signals of any potentially interfering isotopes are measured. Intensity is measured in 186 μ to determine the concentration of gold in the sample and compared to solutions containing standard gold by software (Zhao et al. 2011).

Instrumental Neutron Activation Analysis (INAA)

The residue is extracted from the filter paper and radiated into the center of the nuclear reactor (Miransky et al. 1993). Most elements are radioactive and emit gamma-ray radiation of specific wavelengths. Gold wavelengths are measured and compared to gold standards to determine the concentration of gold in the sample.

Indian Reference Material (Bharatiya Nirdeshak Dravya-BND®)

Various countries are wearing standards and regulations governing gold jewelry manufacture and sale. The difference in the marking system's allowable tolerances and Standards of Finance, among other things with so much variation, makes it easy

to see why international trade in gold jewelry can be hampered by the standards and regulations of individual countries.

India's First 4N pure Gold BND: BND registered 4201 is India's first high purity 99.99% gold reference standard manufactured by CSIR and government mint Mumbai in collaboration with Bhabha Atomic Research Centre BARC Mumbai (Fig. 7a). BND 4201 standard gold bars are 25% cheaper than the imported version. Further, recently high purity gold and gold-alloys BNDs have also been produced by Jalan and company (Fig. 7b). India's high purity silver and silver alloys BNDs have been produced by Jalan and Company in collaboration with CSIR-NPL.

Fig. 7 (**a**) BND 4201: India's first high purity gold reference standard, (**b**)

Gold is very highly respected in Indian and Chinese culture. It is considered very auspicious and an excellent investment for the future. Therefore, it is rare to find how to know the purity of the material necessary.

Future Plans for High Purity Gold and Silver Standard

BND of other purity gold will be prepared in collaboration with the Government of India MINT and Jalan & company. Figure 8 shows a photograph of the new development in this direction. Gold and silver BNDs made in collaboration with Jalan was launched on 4 January 2022 and 20th May 2022.

Certification of Reference Materials and Testing is done based on the relevant ISO guidelines ISO-Guide 34 and ISO-Guide 35 [7.8] and the Guidelines for the Production of BAM Reference Materials (Venelinov and Sahuquillo 2006).

Uncertainty in the measurement was evaluated per the GUM guide, a guide to the expression of uncertainty measurement – JCGM 100: 2008. Uncertainty is a non-negative parameter characterizing the dispersion of the quantity values being attributed to a measurand based on the information used. Measurement uncertainty includes components from the systematic effect, such as components associated with corrections and assigned quantity values.

In general, measurement uncertainty comprises many components. Some of these may be evaluated by Type-A evaluation of measurement uncertainty. It is evaluated

Fig. 8 Photograph depicting the release of gold and silver BNDs made in collaboration with Jalan and Company

based on a statistical analysis of a series of measurements of the quantity value and based on standard deviation. At the same time, Type B evaluation of measurement uncertainty can also be characterized by standard deviations, evaluated from probability density functions based on experience or other information.

Traceability: Property of the result of a measurement or the value of a standard whereby it can be related to stated references, usually national or international standards, through an unbroken chain of comparisons, all having stated uncertainties.

In collaboration with the Jalan and company, CSIR-NPL has produced the following BNDs.

BND 4201A: pure gold 9999 (high purity gold)
BND 4202: gold-alloy 916 (22 karats)
BND 4203: gold-alloy 833 (20 karats)
BND 4204: gold-alloy 750 (18 karats)
BND 4101: pure silver 999
BND 4102: silver-alloy 925

These BNDs will soon be available in the market, allowing jewelers to use instrumental methods rather than the conventional fire essay to ensure the purity of previous precious metal at a low cost compared to the imported.

Conclusions

This chapter described the need for reference materials, and initiatives taken by India. Till now a total of six BNDs have been made in collaboration with the reference material producer, Jalan and company and many more are in the pipeline. These BNDs shall help in reducing the cost of the imported reference materials for calibration of the XRF instrument used for checking the purity of gold and silver.

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Indian Reference Materials for Calibration of Sophisticated Instruments

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Abstract

Reference materials play a significant role in the calibration of sophisticated instruments. The rapid advancement in science and technology and globalization of economies poses a strong need for more stringent metrological regulations in trade and commerce. For achieving greater heights in application-oriented science, accurate measurements of the requisite quantities are the basic essentials nowadays. It can only be possible by the calibration of corresponding instrument with suitable reference materials. Large numbers of reference materials are produced by various National Metrology Institutes (NMIs) throughout the world. CSIR – National Physical Laboratory (NPL) (NMI of India) is also making Indian reference materials (Bharatiya Nirdeshak Dravyas: BND®) for calibration of numerous sophisticated instruments. The present chapter focuses on the principle and applications of sophisticated instruments like powder X-ray diffraction (PXRD), FTIR, and UV-Vis spectroscopy. We have briefly mentioned about its role for measuring various physical quantities and the importance of calibration by using the Indian Reference Materials (BNDs).

Introduction

Accurate and precise measurement plays an important role in the development of science and technology. The measurement system has always stood as the backbone of a country, which is unnoticed by most of its population. Starting from daily shop to international trading and from laboratories to industries, everywhere proper measurement is one of the key factors for human beings to get better life. The branch of science that deals with the study of measurement is known as metrology. The major work of this branch is to reduce the error or uncertainty of measurements from the true values (BIPM 2017). Measurements are carried out by various scientific instruments to obtain appropriate results for which calibration of those specific instruments are most desirable. Calibration is a process of comparison between true or standard value and measured value obtained by the instrument. In order to acquire accurate measured value, calibration of an instrument with reference

materials is a basic and integral objective for most of the measurements (Koeber et al. 2010; Fanton 2019; de Medeiros Albano and ten Caten 2016; Pradhan and Tarafder 2016). The materials which fulfill the required criterion for calibration of various instruments are known as reference materials (RMs). In simple language RMs can be defined by specific types of materials whose property values are sufficiently homogeneous, stable, and can be used for establishing property value of material. When those reference materials are characterized by a metrological valid procedure for one or more specified properties, accompanied by a certificate that provides the value of the specified property, its associated uncertainty, and a statement of metrological traceability, then they are known as certified reference material (CRM) (de Albano and ten Caten 2014; Rutkowska et al. 2020). Several countries are producing various RMs and CRMs by their National Measurement Institutes (NMIs) for calibration of different instruments. Among them, NIST, USA, is one of the lead producers of RMs whose products are known as standard reference materials (SRMs). The production of RMs and CRMs boosts a countries' economic growth in many ways. Firstly, uses of RMs and CRMs will increase the confidence level of measurements whereas their production will create jobs for the future generations. As the preparation and production of RMs and CRMs are related to the branch of metrology, it will also introduce a new chapter to the education system. It will also helps in boosting the exporting capabilities of a country. From the last few decades, India stood amongst the top countries while importing but fails as one of the top exporter. The main reason of less export is rejection of our products by the other countries due to their quality. The use of RMs and CRMs will increase the quality of products and help in improving exports as well as getting good products while importing. This will lead to exponential growth of the countries' economy and automatically the socioeconomic life of the people will get better (Pauwels et al. 1998; Ihnat and Stoeppler 1990; Linsinger et al. 2001; Kumari et al. 2022). Now India is also producing Indian reference materials (IRMs) for sophisticated instruments calibration and various products by its NMI known as National Physical Laboratory, India (NPLI). These RMs produced by NPLI are known as Indian Reference Materials (IRMs) or Bharatiya Nirdeshak Dravya (*BND*®). The major purpose of producing BND® is to calibrate and increase the quality of products in the country. It will also help in developing the socioeconomic level of India. The information regarding the IRMs for calibration of various sophisticated instruments like PXRD, FTIR, and UV-Visible spectroscopy are discussed in details in forthcoming sections.

Why Calibration?

We are surrounded by measurements at every step of our life. But it is difficult to answer whether the measurements we performed are accurate and precise. In order to overcome such a situation, calibration of instruments is a basic requirement. Calibration helps to reveal the accuracy and confirm the reliability of equipment (Menditto et al. 2007; Stallings and Gillmore 1971; Shehata et al. 2014). There are measurement errors or large deviation of outcomes, so that it is no longer within a specific range. These types of errors in measurements display that there is no assurance of quality. By calibrating the equipment, the deviation in results can be

identified and documented. If the outcomes are out of permissible range, the instrument requires to be regulated. In this procedure, measuring device is calibrated, so that errors of results are reduced and deviations are within the equipment specifications. Hence calibration computes and regulates errors and uncertainties within measurement procedure to a tolerable level. In short, calibration of instruments is equally essential as measurements. It allows customers and trades to have self-assurance in outcomes and quality control of products. It enhances the self-reliance of accurate and precise measurements essential in technology, scientific research, invention, development, manufacture of millions of products and facilities worldwide. This can be understood by taking the following example. In case of any medicinal product, testing results may be wrong if measured through non-calibrated instruments. Consumers taking those medicines may feel sick or get into risk. These types of problems can be avoided by using calibrated instruments with CRMs or RMs. It assures the user and ultimately the consumers that the outcomes are accurate and precise if the instruments are calibrated and hence develops confidence in the consumer or person operating the instruments in getting the correct measurements.

Calibration is very significant in our daily life starting from trade, pharmaceutical, medical, transportation, aerospace, scientific laboratory to academia, etc. One can think what would happen in the trade if measurements are not accurate and precise. Both the buyer and seller will be in confusion regarding the quality of a product and their reasonable share in market. Similarly in the field of pharmaceutical how comfortably will one take medicine if equipment for measurements is not often calibrated. Calibration permits billions of individuals on the earth to take medicine with assertions of safety and effectiveness. The advantages of calibration involve enhanced profitability, revenue, innovation, safety, and cutting cost by decreasing production errors. Calibration of instrument helps to conserve money by avoiding untruthful acceptance and refusal of goods and by extending the lifetime of fabricating devices. Calibration is imperative as it supports to keep the world proficiently running, progressive, and secure.

Role of Reference Materials

The science of measurements plays a key character for growth engine and constant evolution of industries. The correctness of measurements during performance of analysis and control laboratories is crucial for operation of current society. Now academia, research laboratories, and industries are focusing on quality assurance, quality control, and traceability of outcomes. Reference materials (RMs) play a significant role in these consequences. RMs can be defined as those substances that are key factors in maintaining a healthy measurement system and to get very precise value (White 2011; Shehata et al. 2019). These stable substances with steady homogeneous property value, which can be used for calibration of an instrument, are known as reference materials (RMs). RMs can be in pure form or a mixture of gas, solid, or liquid. RMs are useful to validate and control the analytical procedure. Recently, RMs have become more crucial as quality desires have been turned sterner with customer requirements for suitably verified and controlled the outcomes.

In our daily life, we deal with measurements, which are also essential for modern science and technology. Precise measurements are the most important to have business with foreign countries. The comparable measurements are tested, measured, internationally accepted, and inevitable to improve the quality life of users and the social-economic growth of society through unobstructed trade. If the observed measurements are inaccurate, it will lead to critical consequences. So it is important to develop reference standards for satisfactory quality in research and development activities for various instruments. In order to calibrate these high-tech instruments, the certified reference materials (CRMs) are required. CRMs are commonly used for calibration of measuring instruments and validation of outcomes from these calibrated instruments. It also helps to evaluate measurement procedure and finally provide a quantity value with a corresponding uncertainty value of measurement results. The use of CRMs in measurements is an essential need for worldwide approval of products and test statements in the current situation of globalization of economy (Rutkowska et al. 2020). Now assurance of high quality in measurements and traceability to the analytical measurements with international measurement system (SI unit) can be confirmed using these CRMs.

Why Need of Indian Reference Materials?

There are some trade barriers due to differences attached with measurement data. These mismatches arise due to instruments without calibration. As mentioned above, the calibration of the instrument is done by some reference materials. If a nation does not have reference materials, it will be difficult to assure the quality of products for import/export. Then it will import these standards. It will affect the socioeconomic fiber of the nation. In this regard, NPLI is producing Indian reference materials (IRMs) with association of some specified reference materials producers (RMPs) by following the stern guidelines of ISO17025 and ISO 17034 & 35. Indian Reference Material or Bharatiya Nirdeshak Dravya (BND®) is the trademark of certified reference material (CRM) developed by NPLI. IRMs play crucial role in the industrial and scientific development by providing consistency and comparability of the measurements for the quality assurance (Trapmann et al. 2017; Ashwal 2020; Aswal 2020; Velichko 2010). IRMs are employed for calibration of equipment, allocation of property value of other CRMs, and validation of analytical procedure.

The production of IRMs will support the government for making India a developed country. The aim of the BND® program is upgradation of the socioeconomic growth of the nation through quality assurance of products for import, export, and by providing resourceful job opportunity. BNDs/IRMs are the core of scientific, academia, research, and industrial development by ascertaining quality assurance and controls in measurements.

National Physical Laboratory, India (NPLI) is devotedly involved in preparation of BNDs/IRMs with association of medium and large industries. NPLI has already worked on BNDs for drinking water, petroleum products, building materials, gold, powder X-ray diffractometer, etc. As every field is associated with complex

instruments, calibration of instrument is possible using RMs. So now the most important task is to develop IRMs. If IRMs are not available for any imported goods for long time then we will not be able to check the quality of goods or it may expire. So keeping in mind the significant role and requirement of IRMs for the nation, NPLI has initiated its mission mode program on development of IRMs in numerous fields. Now importance of BND® activity at national level will definitely bring a vast change in the socio-economic status through assurance and quality control for import/export and domestic costumer products in all the fields.

Powder X-Ray Diffractometer (PXRD) and Its Applications

There are numerous instruments available for various material characterizations. Most of the instruments need to be calibrated before being used in order to get a precise result. Among all the available characterization technique, a very well-known technique, i.e., powder X-ray diffraction technique is vastly used in research laboratories, academia, and industries across the world. In the following section, the importance of powder X-ray diffractometer (PXRD) and the effect of calibration on this instrument are explained in detail (Das et al. 2014; Mo and Zhang 1995; Polizzi et al. 1991). Also the various RMs used in PXRD for calibration of different parameters is discussed in the forthcoming section.

PXRD is one of the oldest and powerful techniques for characterizing various materials. The identification of phase, quantitative analysis, lattice dimensions, structural imperfections, and particle size of sample from various disciplines like polymeric, environmental, geology, pharmaceutical, and forensic sciences are measured using this tool (Upadhyayula et al. 2009; Sharma et al. 2012; Das et al. 2014). It is employed for identification and characterization of unknown minerals, inorganic compound, etc. The basic assumption of powder diffraction is that for every set of planes, there are statistically relevant numbers of crystallites that are properly oriented to diffract the incident beam. The recorded diffraction pattern is the characteristic of any crystalline phase. The intensity, position, shapes, and width of diffraction lines provide information about the specimen. Powder diffraction data comprises a record of photon intensity versus detector angle 2θ. The following information can be determined using XRD patterns as given in Table 1:

Working Principal of PXRD Instrument

The basic principle on which a PXRD instrument works is Bragg's law. This law suggests that atoms present in the crystal structure form an imaginary plane. When X-ray falls on the atoms, it diffracts, but from Bragg's law it is interpreted as "When X-ray is diffracted from atoms, it appears such as reflecting from an imaginary plane." The constructive interference between two X-ray beams occurs if

Table 1 Information about different parameters by PXRD

Information from PXRD	Detailed parameters
Unit cell lattice dimensions and Bravais lattice symmetry	1. Index peak positions 2. Lattice parameters may vary and so giving information about alloying, strains, solid solutions, doping, etc.
Phase composition of a specimen	1. Quantitative phase analysis can govern the relative quantity of phases in a mixture by referencing relative peak intensities
Residual strain (macro-strain) and Crystallite size and micro-strain	1. Indicated by peak broadening 2. Other defects (stacking faults, etc.) can be evaluated by investigation of peak shapes and peak width

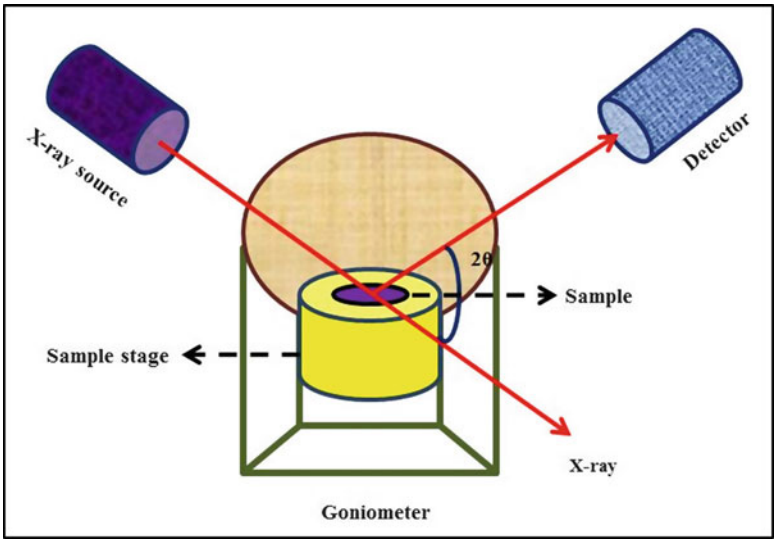

Fig. 1 Schematic diagram of powder X-ray diffractometer

$$2d\sin\theta = n\lambda \tag{1}$$

Where "d" is lattice spacing between two planes, "θ" is angle of incidence of X-ray, "n" is order of diffraction and "λ" is wavelength of X-ray (Mendenhall et al. 2015; Chauhan and Chauhan 2014). This law is valid for most of the X-ray diffraction-based instruments like Single crystal X-ray diffraction, X-ray Fluorescence, etc.

Basically, PXRD consists of three major sections, i.e., source, goniometer, and detector. The schematic diagram for a PXRD is shown in Fig. 1. Mainly a source is an X-ray tube with copper as source material. A goniometer is a sample stage where sample is spun or fixed as per requirement. The detector portion is primarily a scintillation counter detector to detect the diffracted X-ray photons. When the

sample is fixed on sample stage of a goniometer and X-ray is allowed to fall on it with making an angle θ, it gets diffracted from atoms, ions, or molecules present in sample in every possible directions. Now the detector is rotated through an angle "2θ" by Bragg-Brentano geometry and detects the constructive diffracted ray satisfying Bragg's law in Eq. (1). The fingerprint of the material is obtained on a computer screen as a graph between 2θ and Intensity (No. of photons counts) in X and Y axis, respectively.

As we are living in an instrumental world, there might be possibilities of getting error while measurements are carried out. This can be explained by taking a simple demonstration. Let us consider a case, where some instrumental defects in PXRD are giving some random errors in XRD pattern of a sample. It will lead to produce unnecessary confusion in material characterizations. Specifically if we want to detect one specific material say "X," the result might give the sign of other material say "Y" due to that instrumental defect. Similarly there might be several other possible errors caused by the observer while carrying out some experiments. These issues in an instrument can be avoided if the XRD fingerprint of sample is already known. It will allow us to identify whether the XRD is working properly or not. In order to fulfill such requirements, the instrument should be calibrated with some CRMs. As all the property values of CRMs are already known, the instrument should give exactly same known results for proper working condition. There are several advantages of calibrating an instrument. One of the major advantages for research and development department of country is that "the experimental data produced by the instrument will be internationally accepted." This will increase the impact of R and D and scientific research of the country.

Indian Reference Materials (IRMs) for PXRD

X-ray diffractometer has various applications in the field of academia and research, material sciences, pharmaceutical, forensic sciences, microelectronics, polymer glass industries, etc. It is a significant method for material characterization. (Bunaciu et al. 2015; National Institute of Standards and Technology Certificate Standard Reference Material® 640f Line Position and Line Shape Standard for Powder Diffraction (Silicon Powder)). PXRD has seen an enlarged relevance in the area of mineral science and mineral process optimization. With this, powder methods have advantages to determine new polymorphic forms. The XRD technique can be employed to measure lattice dimensions, identify impurities, detect phases, and analyze mixtures of polymorphs. Every single application shows a particular challenge with regard to the development of reference materials. Accuracy is crucial for d spacing (2θ line position), line shape, intensity, and full width half maxima. But it is affected by many instrumental factors like axial divergence, sample-displacement error, step size, deviation in X-ray wavelength, and due to some specimen factors. A conventional method to acquire accurate and precise measurement is calibration of instrument. Calibration can be performed by certified reference materials. In recent decades, enormous numbers of certified reference materials are being developed for

different parameters like line position (2θ), line shape, intensity, quantitative phase analysis, full width half maxima, etc. CSIR-NPL is working on the development of such Indian reference materials, which is described in the following sections.

Reference Materials for Calibration of Various Parameters in PXRD

Line Position Calibration with "Si" Powder

The fingerprint of PXRD pattern of a sample is identified by a graph between 2θ and intensity. Now the line position can be calibrated with the help of SRM "Si (640f)," which is stable cubic structured material (National Institute of Standards and Technology Certificate Standard Reference Material® 640f Line Position and Line Shape Standard for Powder Diffraction (Silicon Powder)). In order to calibrate PXRD instrument, experiment is performed on the CRMs and the certified values are compared. If observed results are not matching with the certified results then experimental conditions will be modified till obtaining the certified one. CSIR-NPL is also working on "Si" to develop it as its own certified IRM, which will be cost effective. This will help in boosting the economy of our country.

Line Shape Calibration with LaB_6 Powder

The peak profile analysis (PPA) in X-ray diffraction is a usual calculation procedure in powder diffraction. The presence of diffraction peaks broadening in a PXRD is primarily due to contribution of instrumental and sample broadening. The instrumental broadening produces unnecessary misinformation about the sample. Thus instrumental broadening in PPA for a PXRD instrument should be identified. This issue can be resolved by the use of SRM 660c, i.e., LaB_6. SRM 660c holds a peculiar property, i.e., absence of size and strain-induced broadening. LaB_6 also preserves regularly spaced, non-overlapping, and high intense diffraction peaks whose FWHM is a quadratic function of 2θ. It shows distinct diffraction peaks in this material. These features make the material prepared for calibration of line shape. Hence, values of lattice parameters are given in the certificate for calibration of instrument (National Institute of Standards and Technology Certificate Standard Reference Material® 660a Lanthanum Hexaboride Powder Line Position and Line Shape Standard for Powder Diffraction; Cline 1994). CSIR-NPL is also working on this material to develop it as a cost effective IRM.

Quantitative Analysis with α-Alumina

Quantitative analysis in PXRD instrument determines the crystalline phase purity in the sample, i.e., the amount of amorphous and crystalline content in the sample can be identified by this analysis. SRMs 676a α-alumina is used as an internal standard for quantitative analysis calibration in PXRD instrument (National Institute of Standards and Technology Certificate Standard Reference Material® 676a Alumina Powder (Quantitative Analysis Powder Diffraction Standard)). NPLI has already developed its own cost-effective certified reference material α-alumina for quantitative analysis.

Calibration of Infrared (FT-IR) Spectrophotometer Using BND®2004

Fourier Transform Infrared (FT-IR) Spectrophotometer and Its Applications

Infrared (IR) spectrophotometer has become an important analytical technique finding applications in each and every area of materials characterization and scientific research. Fourier transform infrared (FT-IR) spectrophotometer provides the analysis of molecules in electromagnetic spectrum falling between visible and microwave regions. The most useful range of analysis falls in the range of 4000–400 cm^{-1}. This technique reveals various fundamental facets of materials in many areas such as chemical, pharmaceutical, analytical, biomedical, food industries, etc. Indeed, FT-IR measurement is very fast, eco-friendly, simple, user-friendly, repeatable, and requires very small number of materials. The molecular FT-IR is based on Michaelson interferometer and also known as interferometric infrared spectroscopy. Except for enantiomers, no two molecules can give the same IR spectrum. The technique offers both qualitative and quantitative measurements of materials by simple inspections and using standards, respectively, and therefore is widely used for quality control and ensuring the quality of the final products. FT-IR spectroscopy is used in various industries to provide quality control, like in pharmaceutical field (for medicines), biomedical area (for protein, vitamin, lipids, starch, uric acid, etc.), and food industry (water, spirit, honey, oil, nut, flours, flakes, meat, rice, drinks, etc.).

The Working Principle

FT-IR spectroscopy measures the interaction between matter and electromagnetic fields in IR region at a molecular and atomic level. IR radiation absorbed by the molecules, especially organic molecules, is converted into energy of molecular vibrations, which itself is accompanied by several rotational energy changes. These vibrational-rotational energy changes occur between 4000 cm^{-1} and 400 cm^{-1} and is captured and presented by FT-IR spectrophotometer. The frequency or wavelength of absorption depends on the relative mass of atoms, force constant of the bonds, and geometry of the atoms. The position of the bands in IR spectrum is presented in wavenumber (ν) with a unit in cm^{-1} while the band intensity can be presented in transmittance (T) or absorbance (A). Only the vibrations that result in rhythmic change of molecule's dipole moment are observed in IR spectrum. The assignments of the bands can be approximated by application of Hook's law mentioned in the equation given below-

$$\nu = \frac{1}{2\pi c}\left[\frac{f}{(M_x M_y)/(M_x + M_y)}\right]^{\frac{1}{2}} \tag{2}$$

FT-IR spectrophotometer can be used for characterization of gases, liquids, and solids. Gases are characterized by expansion in evacuated cells. Liquids may be tested in neat or in solution transparent in region of interest. Solids are characterized as mull, pressed disks, or *as*-deposited glassy films.

Calibration of FT-IR Instrument and Reference Standard

The FT-IR instrument is one of the main tools for characterization, identification, inspection, counterfeit detection, and quality control, and therefore, a reference standard for calibration of FT-IR spectrophotometer is required (Lowry et al. 2000). Indian Pharmacopoeia commission (IPC) and other country's Pharmacopoeia like British Pharmacopoeia and the US Pharmacopoeia, etc., have set standards for all the drugs that are manufactured, sold, and consumed in that particular country. The resolution standards are defined for particular wavenumbers. The validation of FT-IR instrument performance is based on five tests, i.e., power spectrum, resolution, wavenumber accuracy, wavenumber repeatability, and absorbance reproducibility. The reference standard is used for the calibration of FT-IR instrument and validation of its performance. There are different reference standards for calibration of wavenumber accuracy, FT-IR instrument's photometric linearity, diffuse reflectance in NIR, Mid IR and UV regions, reflectance, NIR wavelength standard, ATR reference standard, etc.

For preparation of reference standard, researchers have realized the requirement of stable material with characteristic peaks covering the wide range of IR region. In this direction, a number of molecules, polymers, and materials were tested to be used as reference materials. Among all, polystyrene was the most suitable in terms of stability and suitable peaks position for calibration. Generally, polystyrene and its films are used in different applications such as in rigid food service containers, CD cases, appliance housings, envelope windows, automobile body panels, photoluminescence, and many other products. Polystyrene is the most employed aromatic thermoplastic polymer synthesized from the polymerization of styrene monomer and has been found to be used as reference standard on preparation of standard films as per ISO Guide 35 with high accuracy.

Material Preparation

Polystyrene (PS) films are used as reference standard for calibration of wavenumber/wavelength in IR range. Styrene monomer of high quality is used for polymerization either by thermal or catalyzed reaction to produce amorphous polystyrene polymer (Fig. 2). The produced polystyrene may be of different grades, like rubber modified high impact polystyrene (HIPS), Medium impact polystyrene (MIPS), general purpose polystyrene (GPPS), and expandable polystyrene (EPS). GPPS is highly used due to its clear appearance, rigidity, and multipurpose applications (Wackerly and Dunne 2017). MIPS is used for high flexibility and HIPS is applied where high

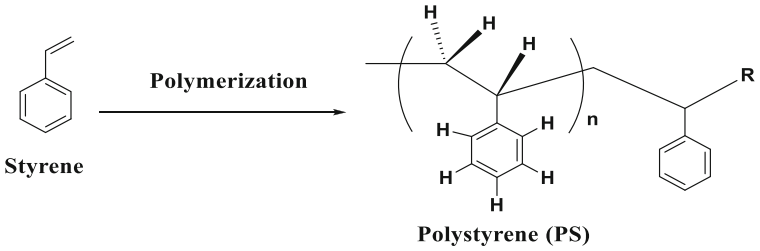

Fig. 2 Polymerization of styrene for the synthesis of polystyrene

impact resistance is required. EPS shows the insulating behavior. The modification of polystyrene is performed by addition of plasticizers, release agents, and stabilizers to achieve the desired properties. For specific applications, other additives like flame retardants, colorants, UV stabilizers, and impact modifiers are added to the formulation. The relative stereochemical relationship of consecutive phenyl groups determines the tacticity, which has an effect on various physical properties of the material.

Polystyrene is transformed into different products. One of the most important applications of polystyrene in science field is the preparation of polystyrene film for calibration of FT-IR spectrophotometer. In this regard, the National Institute of Standards and Technology (NIST), USA developed a reference material for calibration of wavelength/wavenumber scale of spectrophotometer in IR spectral region with the trade name "SRM 1921b." This reference standard and derived certified reference material (CRM) are most widely used across the world and therefore also imported in India for calibrating the wavenumber scale of spectrophotometers on the IR region. There is urgency for India to get self-reliant in calibration reference standards so that dependency on a foreign country is completely removed. CSIR-NPL, which is also the NMI of India maintains the primary standard for the International System (SI) units and provides traceability of many parameters for calibration. CSIR-NPL developed polystyrene-based reference standard for calibration of wavelength values of FT-IR spectrophotometers in the region from 4000 cm^{-1} to 400 cm^{-1} with the trade name BND® 2004. The developed BND® 2004 is comparable to other reference standards being used for peak position, absorption/transmittance, stability, thickness, uncertainty range, etc. The indigenous development of FT-IR standard (BND®2004) saves a huge amount of foreign exchange by replacing the purchase of SRM1921b from NIST and other imported IR standards.

Indian Reference Material BND®2004

FT-IR spectrophotometer operates on different measurement methods, such as diffuse reflection method and attenuated total reflectance. In diffuse reflection method, a sample holder is used to place the sample between IR source and detector. The beam passes through the sample and the transmitted energy of the IR radiation is

measured and presented in the form of either percentage transmittance or absorbance spectrum. For calibration of FT-IR instrument, BND®2004 is used, which is a matte-finished polystyrene film with a thickness of 38 (±2) μm placed in the center of a cardholder. CSIR-NPL has developed the reference standard by a cost-effective in-house method. During the development of the films, special care is taken to avoid any kind of impurity or dust getting added. A batch of ~100 films is prepared and characterized by calibrated FT-IR spectrophotometer by randomly picking 25–30 films. The films should exhibit the clear peaks of standard polystyrene with good resolution. The homogeneity of the films has been checked thoroughly by statistical method measuring at five different positions of a 25 mm diameter to detect any impurity or interferences due to undetected problems during preparation. Ideally there is a difference in peak position, the intensity from one point to other should be negligible, and the film is called highly homogeneous with respect to given characteristics. The stability of the film is also measured as per ISO Guide 35. There are two types of stability to be considered in reference materials, one is long-term stability of the material and the other is short-term stability. Long-term stability is estimated by repeated measurements as time elapses determining the lifetime of reference material. Short-term stability is recorded at different temperatures providing the information for the appropriate storage conditions and transport conditions.

The process developed by CSIR-NPL is simple, fast, and cost-effective as there is no requirement for costly and sophisticated equipment. The precursor materials are required with a specific molecular weight range, polydispersity, and high purity. The process used is hot press method at a specific temperature with pressure using special metal plates to obtain matte-finish roughness. The developed films are of uniform thickness and stable under appropriate storage conditions. This is an eco-friendly process and low capital cost is required to set up the facility. The advantages of the process are requirement of a small amount of precursor, no wastage of materials, no side product formation, and no disposal are required. The product film is a matte finished polystyrene film (38 ± 2) μm thick with a ~ 32 mm diameter and > 80% transmittance (Fig. 3). The thickness of the films is tested by stylus profiler.

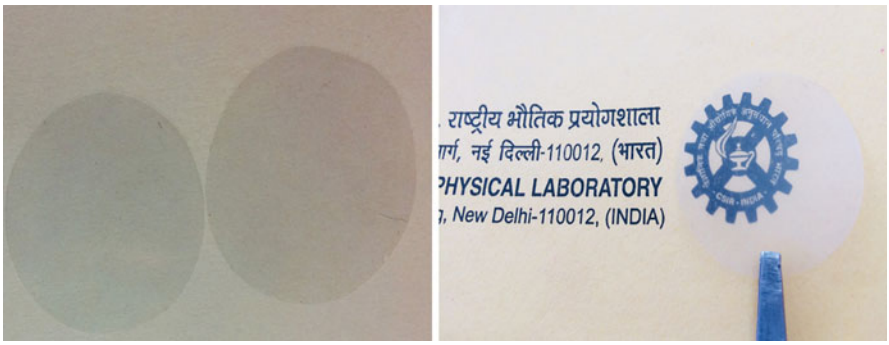

Fig. 3 Photograph of polystyrene films and transparency

The characteristic FT-IR peaks of BND®2004 is shown in percentage transmittance (%T) (Fig. 4) and absorbance (Fig. 5). The specific 14 peaks positions considered for calibration are summarized in Table 2. The peaks cover the full range and peak positions are used for calibration. Pharmacopoeia defines the differences between the absorbance at the absorption minimum of 2870 cm^{-1} and the absorption maximum of 2849.5 cm^{-1} peak is to be >0.33 or > 18 in %T. The difference between the absorbance at the absorption minimum of 1589 cm^{-1} and the absorption maximum at 1583 cm^{-1} is to be >0.08 or > 10 in %T (Figs. 5 and 6). BND®2004 complies with all the requirements and is being sold in India for calibration of FT-IR equipment.

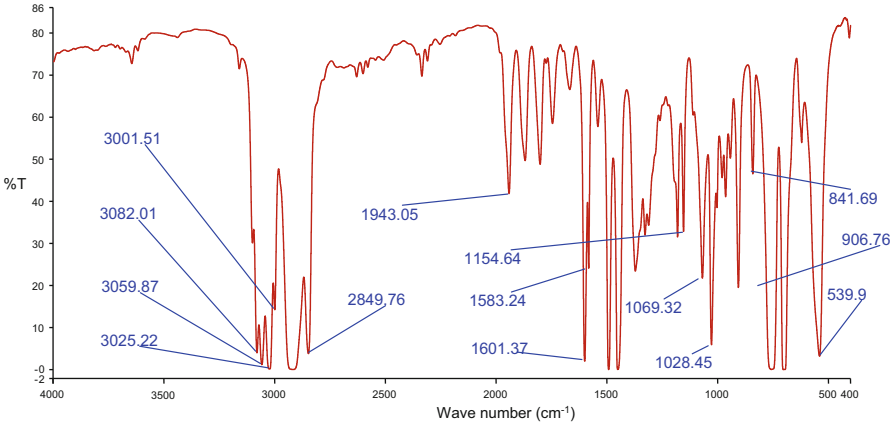

Fig. 4 FT-IR spectrum of polystyrene film

Fig. 5 Absorption spectra of polystyrene film

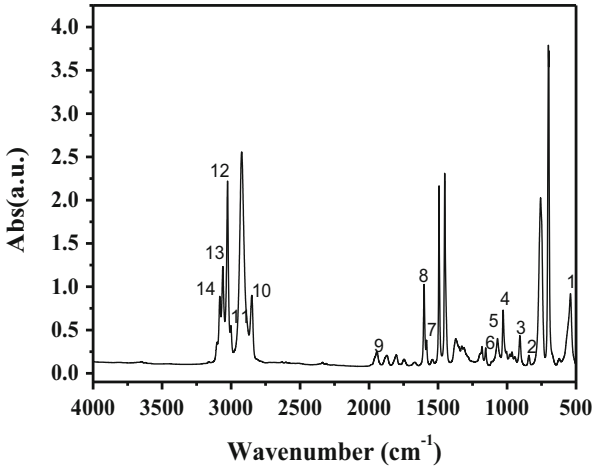

Table 2 The FT-IR peak positions of polystyrene film

Measurand	Peak positions (cm^{-1})	Peak No.
Wavenumber	539	1
	841	2
	906	3
	1028	4
	1069	5
	1154	6
	1583	7
	1601	8
	1943	9
	2849	10
	3001	11
	3025	12
	3059	13
	3081	14

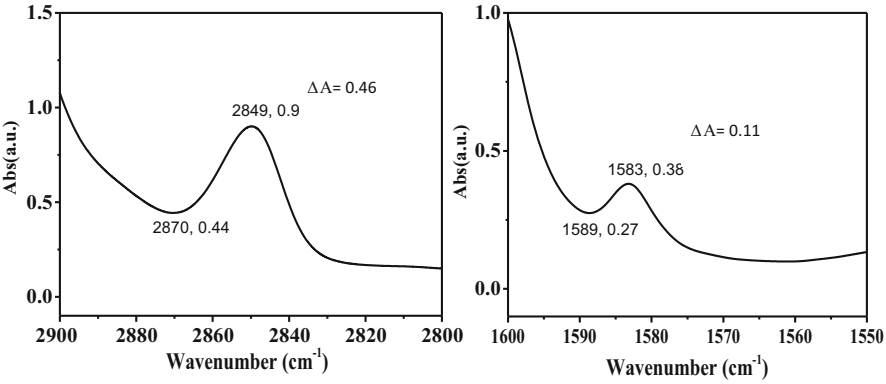

Fig. 6 Peak to peak resolution of the peaks as per Pharmacopoeia

Ultraviolet-Visible (UV-Vis) Spectroscopy and Its Applications

Spectroscopy is a general methodology that can be adapted in many ways to extract the information we need (energies of electronic, vibrational, rotational states; structure and symmetry of molecules; dynamic information) and to understand how light interacts with matter and how one can use this to quantitatively understand the sample.

Ultraviolet-Visible (UV-Vis) spectrophotometry is based on the absorption of the electromagnetic spectrum in the ultraviolet and visible region (Skoog et al.). It measures the amount of discrete UV/visible light wavelengths absorbed/transmitted through a sample and provides potential information about the composition and

concentration of the sample. UV-Vis spectroscopy technique has broad aspects in science, ranging from identification and quantification of organic compounds, bacterial culture, and quality control in the beverage industry, drug identification, and chemical research.

It works on the principle of Beer-Lambert Law, which establishes a relationship between attenuation of light through the sample. It states that "*the quantity of light absorbed by a substance dissolved in a fully transmitting solvent is directly proportional to the concentration of the substance and the path length of the light through the solution*."

Mathematically,

$$A = \log\left(\frac{I_0}{I_t}\right) = \varepsilon cl \tag{3}$$

Also,

$$T = \frac{I_t}{I_0} \tag{4}$$

where,

A = Absorbance
c = Concentration
l = Path length of sample cell
ε = Extinction coefficient
T = Transmittance
I_t = Intensity of transmitted light
I_0 = Intensity of incident light

The instrument comprises light source (deuterium or tungsten lamp), monochromator, sample holder, and detector. There are typically three basic designs for UV-Vis spectrophotometer – single beam, double beam, and simultaneous type instrument. The single-beam instrument (Fig. 7a) analyses one wavelength at a time using a filter or a monochromator between the source and the sample. The double-beam instrument (Fig. 7b) comprises a light source, a monochromator, a splitter, and a series of mirrors. The arrangement of splitter and mirrors split the beam between the reference sample and the sample to be analyzed, allowing more accurate measurements. On the other hand, the simultaneous instrument (Fig. 7c) does not have any monochromator; instead, a diode array detector is used, which allows the instrument to detect the absorbance at all wavelengths simultaneously. The simultaneous instrument is usually much faster and more efficient.

UV-Vis spectroscopy relies on the absorption of UV/Visible light of the electromagnetic spectrum by the analyte. Specific energy is required for electron transition

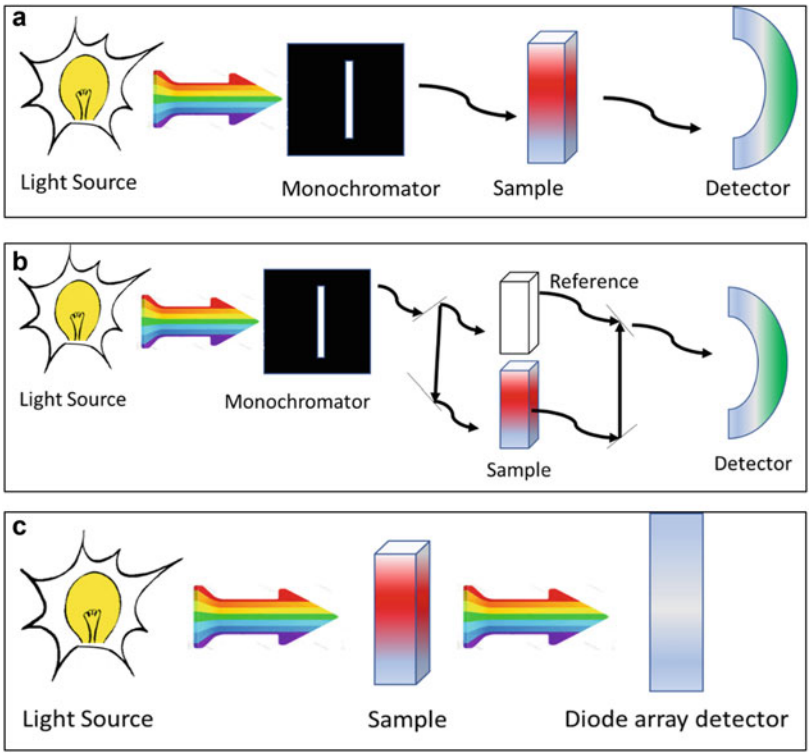

Fig. 7 (**a**) Schematic of single beam UV-Vis instrument. (**b**) Schematic of a double-beam UV-Vis instrument. (**c**) Schematic of a simultaneous UV-Vis instrument

from lower to higher energy levels and results in absorption spectra. Electrons in different bonding environments require different energies for transition to higher states; therefore, absorption occurs at different wavelengths for different materials. The UV-Vis region corresponding to the wavelength range of 800–200 nm of electromagnetic spectrum covers energy in the range of 1.5–6.2 eV.

Electronic transitions resulting from the absorption of electromagnetic radiations in the ultraviolet and visible ranges form the basis for quantitative analysis of compounds. These electronic transitions mainly occur from the highest occupied molecular orbital (HOMO) to the lowest unoccupied molecular orbital (LUMO) on radiation absorption with energy equivalent to the energy difference between the excited and the ground states. Valence electrons in most of the molecules can be found in one of the following orbitals listed in ascending order of their energy levels: σ (bonding) orbital $<$ π (bonding) orbital $<$ n (non-bonding) orbitals. The unoccupied high-energy orbitals are π^* (anti-bonding) orbital and σ^* (anti-bonding). The possible transitions (Figs. 8 and 9) that occur in most of the organic molecules are

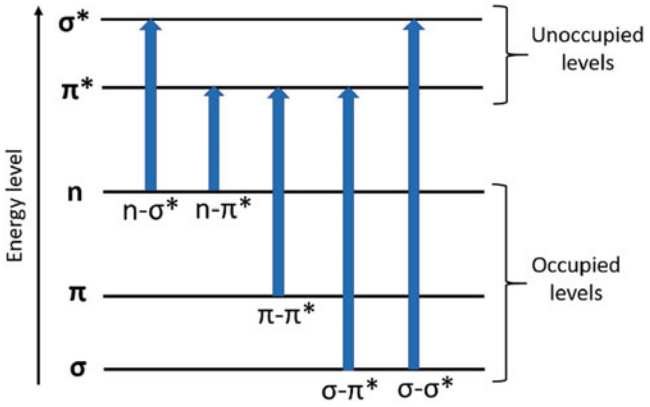

Fig. 8 Electronic transitions and their energy levels

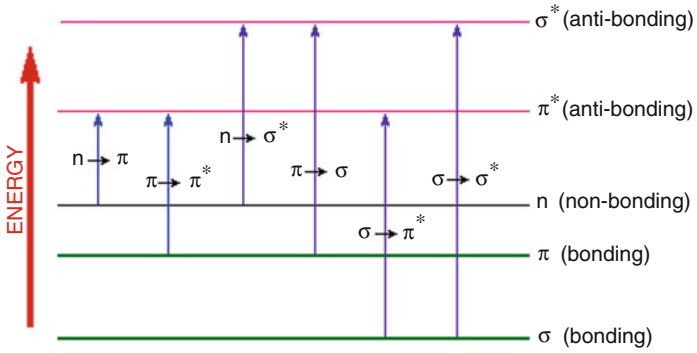

Fig. 9 Schematic of possible transition

σ−σ*, σ−π*, π−π*, n−σ*, n−π*. The σ−σ* transition that occurs only in alkanes requires very high energy, which does not fall in the UV-Vis region. The energy for the rest of the transitions falls in the UV-Vis region and gives rise to a band spectrum due to superimposed vibrational and rotational levels in molecules (Pavia et al. 2001). The electronic transitions that precisely lie in the visible region belong to the d-d transitions or the charge transfer phenomenon occurring in transition metal complexes and are responsible for their characteristic color (Lee 1999).

σ → σ* transition: This type of transition is observed with saturated compounds, e.g., methane (CH_4), which has a C-H bond only, undergoes σ → σ* transition having absorbance maxima at 125 nm.

π → π* transition: Such transitions are observed in compounds containing multiple bonds like alkenes, alkynes, nitriles, carbonyl, aromatic compounds, and show absorbance in the region 170–205 nm.

n → σ* transition: Such transitions are observed in saturated compounds containing atoms with lone pair of electrons like O, N, S, and Halogens and show absorbance in the 150–250 nm region.

n → π* transition: Compounds containing double bonds involving heteroatoms (C=O, C ≡ N, N=O) show that such transitions have absorption at a longer wavelength around 300 nm.

σ → π* transition and π → σ* transition: Such electronic transitions are forbidden transitions and are only theoretically possible.

The UV visible spectrometer different components present in the UV-Vis spectrophotometer are as follows:

Source of Light

UV-Vis spectrophotometer is a light-based technique that essentially requires a steady and wide range of wavelengths source. A high-intensity single Xenon lamp is most commonly used as a light source for both UV and visible ranges. However, such lamps are comparatively costlier and are less stable compared to tungsten and halogen lamps. Some instruments are based on two lamps, tungsten or halogen lamp, commonly used for visible light and a deuterium lamp as a UV light source. For the whole range scan, i.e., UV to the visible region, there is a switchover between the light sources, which typically occurs during the scan between 300 and 350 nm. Here the light emission is similar from both light sources, which results in a smooth transition.

Wavelength Selection

A suitable wavelength of light must be selected from the broad wavelength light source for the sample analysis. There are certain available methods for the selection of wavelength and are discussed below:

- **Monochromator**: The desired wavelength of light is achieved by rotating diffraction grating by tuning the incoming and reflected angles of light. The groove frequency of diffraction grating defines the optical resolution. The higher the groove frequency, the better will be the optical resolution and the narrower the usable wavelength range. Typically, 300–2000 grooves per mm are being used in gratings for UV Vis spectroscopy. On the other hand, a prism disperses the radiation into different wavelengths according to the respective refractive index. It provides the different wavelengths at different angles allowing for the selection of specific wavelengths. The dispersion is linear in the

case of grating, whereas it is nonlinear for prism, making it advantageous to use grating over a prism.

- **Absorption filters**: Absorption filters are generally colored glasses or plastics designed to absorb specific wavelengths of light.
- **Interference filters**: These are also known as dichroic filters, made of many dielectric material layers, which cause light interference between thin layers of materials. Such filters eliminate undesirable wavelengths by destructive interference and act as a wavelength selector.
- **Cut-off filters**: Such filters allow light either below (short pass) or above (long pass) specific wavelength to pass through.
- **Band pass filters**: Band pass filters allow a range of wavelengths to pass through for the analysis of the sample
  The filters are often used with monochromators to narrow down the wavelengths of light for more accurate and precise measurements and improve the signal-to-noise ratio.

Sample Holder

The sample is placed between the light source and the detector. In double-beam spectrophotometers, a reference sample holder is used for baseline correction and considered as the "blank sample." The liquid samples are placed in cuvettes of different path lengths depending on the sample's characteristics. Most plastic and glass cuvettes are inappropriate for UV absorption studies as plastics generally absorb UV light, and glass can act as a filter. Therefore, a quartz sample holder is generally preferred to identify sample signatures in the UV region.

Detectors

In UV-Vis spectrophotometers, photoelectric coatings or semiconductor-based detectors are generally used, which convert the light into a readable electronic signal as light passes through the sample.

On exposure to light, a photoelectric coating ejects negatively charged electrons, which generate electric current proportional to the light intensity. A photomultiplier tube (PMT) is the most commonly used detector in UV-Vis spectroscopy. A PMT can generate a larger electric current by sequential multiplication of the ejected electrons; therefore, it is useful for detecting very low light signal levels (Amelio 1974).

On the other hand, an electric current proportional to the light intensity is generated in semiconductor detectors when exposed to light. Photodiodes and charge-coupled devices (CCDs) (Tauc and Grigorovici 1966) are two of the most common detectors based on semiconductor technology.

Applications

Pharmaceutical Analysis

UV-Vis spectroscopy provides spatially and temporally resolved absorbance measurements, which are highly useful in pharmaceutical analysis. It provides real-time monitoring of drugs and drug release kinetics of sustained-release drug formulations, along with quantitative and qualitative analysis. Many drug molecules contain chromophores that absorb specific wavelengths of ultraviolet or visible light; also, the concentration of constituents in drug molecules can be estimated using the Beer-Lambert law, where the absorption of spectra generated from these samples at given wavelengths can be related directly to the concentration of the sample.

Food Analysis

Food is mainly comprised of water, fat, proteins, and carbohydrates together with numerous other components. The functional properties of such components, governed by their molecular structure and their intra−/intermolecular interactions, define the characteristics of food products. The demand for high quality and safety in food production requires high quality and process control standards, which requires appropriate analytical tools to investigate the food. UV-Vis spectroscopy is an important, flexible, widespread analytical method primarily used in the food industry for quality control of food and beverages. It allows the identification of the origin and the detection of adulteration, among other vital characteristics in food products that plays a crucial role in examining the quality of edible oils.

Identification of Unknown Sample

The absorption spectra of a particular compound are unique. A data bank of UV spectra can be used to determine an unknown compound.

Qualitative and Quantitative Analysis

UV-Vis spectroscopy is commonly used in analytical chemistry for the quantitative and qualitative analysis of various analytes, such as transition metal ions, conjugated organic compounds, and biological macromolecules. Mainly absorption in organic compounds is observed due to chromophores, which usually contain π (unsaturated) bonds. When such chromophores are inserted into a saturated hydrocarbon (which exhibits no UV-visible absorbance spectrum), it produces a compound with absorption between 185 nm and 1000 nm. Some of the most common chromophores and its corresponding maximum absorption are given in Table 3.

Table 3 List of common chromophores and their absorbance maxima

Selected chromophores and their absorbance maxima			
Chromophore	Formula	Example	Imax (nm)
Carbonyl (ketone)	RR'C=O	Acetone	271
Carbonyl (aldehyde)	RHC=O	Acetaldehyde	293
Carboxyl	RCOOH	Acetic acid	204
Amide	RCONH2	Acetamide	208
Nitro	RNO2	Nitromethane	271

Concentration Determination

UV-Vis is a fast and reliable method to determine the concentration of an analyte in a solution. Beer-Lambert law establishes a linear relationship between absorbance, the concentration of absorbing species in the solution, and the path length as given in Eq. (3). UV-Vis spectroscopy is an effective technique for determining the concentration of the absorbing species for a fixed path length.

Bandgap Estimation

The optical band gap estimation in semiconducting materials is crucial for many optoelectronic applications. To this end, UV-Vis spectroscopy offers an easy and appropriate method of estimating the optical band gap as it explores electronic transitions between the valence band and the conduction band.

The optical band gap can be estimated using the following two methods

Method 1:

Bandgap energy,

$$E_g = \frac{hc}{\lambda} \tag{5}$$

where $h = 6.626 \times 10^{-34}$ Js

and $c = 3 \times 10^8$ m/s

$$E = \frac{1.24 \times 10^6}{\lambda} \text{eV}, (\lambda \text{ in m}) \tag{6}$$

Method 2:

The energy band gap of a given material can be more precisely calculated using the Tauc (Amendola 2017) plot method as given below.

$$\alpha = \frac{A\left(h\nu - E_g\right)^n}{h\nu} \tag{7}$$

Rearranging the above equation,

$$(\alpha h\nu)^{1/n} = A^{1/n} h\nu - A^{1/n} E_g \tag{8}$$

where

$$\alpha = \frac{\ln(1/T)}{x}$$

α = Absorption coefficient
T = Transmittance
x = Thickness of the sample
E_g = Bandgap of the material
n = 2, 1/2, for direct allowed and indirect allowed transitions, respectively.

By plotting the graph of $(\alpha\, h\nu)^{1/n}$ vs. $h\nu$, we will get slope as $A^{1/n}$ and y-intercept as $A^{1/n}E_g$. Dividing y-intercept by A^n, the band gap can be estimated.

Size Determination in Metal and Metal Oxide

Among metal nanoparticles, gold nanoparticles (AuNP) have been studied in detail because of their extraordinary optical properties due to surface plasmon resonance (SPR). The size of concentration of AuNP determines the optical, electrical, and chemical properties of AuNPs. UV-Visible spectroscopy has proved to be an excellent tool in identifying the concentration and size of the AuNPs.

Mie (Haiss et al. 2007), in 1908, presented a solution for the calculation of extinction and scattering efficiencies of small metal particles. Later, Heiss et al. (Chauhan and Chauhan 2014), using Mie theory, have developed methods to calculate the particle diameter (d) of gold nanoparticles ranging 3–120 nm using UV-Vis spectra. They derived the analytical relations between the extinction efficiency (Qext) and particle diameter (d), which further determines the particle concentration (c). The particle diameter is given by:

$$d = \frac{\ln\left(\frac{\lambda_{spr} - \lambda_0}{L_1}\right)}{L_2} \tag{9}$$

where d is diameter, λ_{spr} is peak value, L_1 and L_2 are fit parameters, and λ_0 is the experimental value. The average of the absolute error in calculating the experimentally observed particle diameters was observed only 3% for the fit parameters determined from the theoretical values for d $>$ 25 nm ($\lambda_0 = 512$; $L_1 = 6.53$; $L_2 = 0.0216$); therefore, Eq. 9 allows a precise determination of d in the range of 35–110 nm (Brus 1987).

Size Determination in Semiconductor Nanoparticles

Many theoretical models were proposed to give a quantitative agreement of dependencies of energy bandgap (E_g) on the crystal size (R) with the experimental data for a wide range of semiconductors. Brus Model (Bunaciu et al. 2015) (i.e., the effective mass approximation) is the most used theoretical model for the experimental data analysis of energy bandgap E_g (R) dependencies. According to this model, the bandgap of semiconductor nanoparticles (considered as a sphere with radius R) is given by:

$$E_g(R) = E_y(\text{Bulk}) + \frac{h^2}{8m_0R^2}\left(\frac{1}{m_e^*} + \frac{1}{m_h^*}\right) - \frac{1.8e^2}{\varepsilon_0\varepsilon_r R} - 0.248\frac{4\Pi^2 e^2 m_0}{2(4\pi\varepsilon_0\varepsilon_r)\left(\frac{1}{m_e^*} + \frac{1}{m_h^*}\right)} \tag{10}$$

where,

E_g = bulk band gap value
h = plank's constant
m_0 = electron mass
m_e^* and m_h^* = electron and hole relative effective mass, respectively
e = electron charge
ε_0 = permittivity of vacuum
ε_r = relative dielectric constant of the semiconductor

The bandgap shift with respect to the bulk value is given as

$$\Delta E_g = \frac{h^2}{8m_0R^2}\left(\frac{1}{m_e^*} + \frac{1}{m_h^*}\right) - \frac{1.8e^2}{\varepsilon_0\varepsilon_r R} - 0.248\frac{4\Pi^2 e^2 m_0}{2(4\pi\varepsilon_0\varepsilon_r)\left(\frac{1}{m_e^*} + \frac{1}{m_h^*}\right)} \tag{11}$$

The first term is referred to as the quantum localization term or the kinetic energy term, which shifts the $E_g(R)$ to higher energies proportionally to $1/R^2$.

The second term results from the screened Coulomb interaction between the electron and hole, which shifts the $E_g(R)$ to lower energy as $1/R$.

The third is a size-independent term related to the salvation energy loss. This term is usually small and can be ignored.

Instrument Calibration

Ultraviolet-Visible (UV-Vis) spectrophotometry is routinely used in academics and industrial settings like the food and pharmaceutical industry to analyze compounds for quality assurance and testing. Any form of critical evaluation, whether in clinical

Table 4 List of RM and CRM for calibration of UV-Visible spectroscopy (source www.starna.com)

Parameter	Available reference material	Range (nm)
Absorbance	Potassium dichromate solution	235–430
	Neutral density filter	440–635
	Nicotinic acid	210–260
	Metal-on- quartz-filters	250–2850
	Starna green	250–650
Wavelength	Deep UV (DUV)	190–230
	Didymium solution	290–870
	Didymium filter	430–890
	Holmium solution	240–650
	Holmium filter	270–640
	Holmium/didymium solution	240–795
	Rare earth solution	200–300
	Samarium perchlorate	230–650
	Starna SRM-2065 equivalent	335–1945
	NIR reference	930–2550
Stray light	Cut-off filters	200–390
Resolution	Toluene in hexane	265–270
	Benzene vapor	230–270
Resolution (derivative)	Toluene in methanol	265–270

biochemistry, environmental research, food, and beverages, etc., essentially demands the accuracy of results depending on the instrument's performance. A regular performance testing of UV-Vis spectrophotometers under standard operating procedures is a prerequisite to ensure the proper functioning of the equipment and meet the required benchmarks of standardization. The UV-Vis spectrophotometer must be routinely calibrated for absorbance, wavelength accuracy, stray light, and resolution using RMs or CRMs as listed in Table 4.

CSIR-National Physical Laboratory, India (NPLI) has recently started indigenous development certified reference materials (CRMs) under the registered trade name of Bharatiya Nirdeshak Dravya (BND®) for the calibration of UV-Vis spectrophotometer as most of the CRMS are being imported. The absorbance calibration CRM BND®2021 is already available, and wavelength calibration standard BND®2022 is under preparation.

Conclusions

This book chapter explains about the importance of measurements, calibration, RMs and CRMs in metrology. Especially, the vital role of Indian Reference Materials in uplifting the socioeconomic life is elaborately discussed. Further, the chapter highlights the role of CSIR-NPL, India to enrich the economy of the country by

producing IRMs. The calibration and its importance on a few of the sophisticated instruments like PXRD, FTIR, and UV-visible spectrometer by using various RMs, CRMs, and especially IRMs are elaborately discussed. It emphasizes that calibrated instrument can produce more accurate results with minimum uncertainty than non-calibrated one. In case of PXRD there are several measuring parameters, which are calibrated using different RMs. Si, LaB_6, and α-alumina are few of them used to calibrate line position, line shape, and quantitative analyses in PXRD. Similarly, polystyrene is used to calibrate FTIR spectrometer. The details of all absorption peaks of polystyrene present in IR region are also given in this chapter. The process of calibration of FTIR using polystyrene is also described, which suggests calibration increase the confidence level in the measurement results of FTIR. Similarly, the working principle, applications, and its calibration using various RMs and CRMs are explained to demonstrate the significance of reference materials and calibration in sophisticated instruments. Also, the importance of the role of RMs, CRMs, and IRMs for calibration of instruments, the role of IRMs in uplifting India's economy and its effects are deliberated in this chapter. Till date, all the certified reference materials are procured from other countries with very high cost. Not only the cost and they need to procure it again and again after the validity of the reference material is expired. Thus, there is huge recurring loss of natural resource. Also we are procuring so many materials, products, and sophisticated instruments, and many times we get substandard products and we have certified calibration facility to check it. When the developed countries procure products from other countries, they send their representative and check the quality before importing it. Even some of the major existing industries in India are taking the traceability or reference materials from abroad at huge cost. Developed certified reference materials traceable to SI unit (BNDs) at CSIR-NPL and any RMPs will bring a paradigm shift in the socioeconomic fabric of the country through quality control assurance for export, import, and domestic consumer products in every sector.

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Certified Reference Material for Qualifying Biomaterials in Biological Evaluations

28

N. S. Remya and Leena Joseph

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Abstract

The importance of using certified reference material (CRM) for qualifying materials is identified at various levels in biological evaluations as prescribed by international standards like ISO, ASTM, USP, etc. The initiatives under quality management systems and regulatory controls in employing CRMs increase the reliability and acceptance of biocompatibility evaluations leading to safe clinical

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use of medical devices. This chapter explores the possibilities and challenges for the routine use of CRMs in the biocompatibility evaluation of materials and medical devices.

Keywords

Reference material · Negative control · Positive control · Biological evaluation · Cytotoxicity · Implantation · Certified Reference Materials

Introduction

Medical devices can be any instrument, apparatus, machine, appliance, implant, and reagent for in vitro use or in vivo use that are intended to be used, alone or in combination, for addressing medical requirements of humans. When a device or material performs with an appropriate host response in its specific application, it is said to have biocompatibility. In medical device development, a set of appropriate physicochemical and biological evaluations are devised to establish biocompatibility.

Devices with proper design and development phases within a regulatory framework ascertain their success in clinical use. A work platform contained within international standard ISO 13485: Medical devices – Quality management systems – Requirements for regulatory purposes definitely paves guidelines for getting regulatory approvals of a developing device. ISO 14971: Medical devices – Application of risk management to medical devices is applicable in identifying hazards and estimating risk at different device development stages from the design phase onwards. Special care in the selection of raw materials and the use of material additives are having much relevance in medical device development concerning biocompatibility. Impurities in raw material or residues of trace elements from the manufacturing process or residues of disinfection agents, etc. may cause undesirable effects in the clinical performance of the final product.

By definition "any substance or combination of substances, other than drugs, synthetic or natural in origin, which can be used for any period of time, which augments or replaces partially or fully any tissue, organ or function of the body, to maintain or improve the quality of life of the individual" is identified as a biomaterial (Bergmann and S. A. 2013). In biocompatibility evaluations, the performance of the biomaterials including chemical, toxicological, physical, electrical, morphological, and mechanical properties will get characterized and established (ISO10993-1; 2018). The use of qualified biomaterial with proven safety and efficacy is of at most importance with respect to its functionality in the human physiological environment. Although critical performance analysis in preclinical evaluations can control the risk elements of such materials and devices, the databases of regulatory authorities report a considerable number of complications and adverse events annually. The main factors that challenge the process of biomaterial evaluation for qualifying them for medical use are non-uniformity in regulations, the difference in material development, and methods of their evaluation, preconceived post-marketing data, the cost

involved, etc. For clinical acceptance, established biocompatibility characteristics through documented material evaluations matrix within risk-based quality management platform is very essential. The selection of strategies for evaluating biomaterials could be performed wisely to address the regulatory requirements (Masaeli and Tayebi 2019). Qualifying biomaterials to meet regulatory requirements insist traceability to international standards in physicochemical characterization and biological evaluations. The use of appropriate certified reference material (CRM) is identified as one of the key elements to address this requirement.

International Standards for Biological Evaluations: ISO 10993

The international standard ISO 10993 is formulated for the safe use of medical devices or biomaterials considering the potential biological risks arising from their clinical use. These standards can guide in biocompatibility studies or biological evaluations which are meant for assessing the fitness without any harmful physiological effects (Ratner et al. 2013).

In risk-based thinking for medical device development, it is important to conduct appropriate research and experiments on each component material prior to use. According to ISO 10993, a reference material (RM) is defined as material with one or more property values that are sufficiently reproducible and well established to enable the use of the material or substance for the calibration of an apparatus, the assessment of a measurement method, or for the assignment of values to materials (ISO10993-5 2009.). The use of an RM will facilitate the comparability of the response between laboratories and help assess the reproducibility of the test performance within individual laboratories. Properly characterized control materials designated as RMs are recognized as the integral part of any biocompatibility evaluations by optimizing quantity and cost for in vitro tests, in vivo tests, and animal experiments. Materials used as RMs shall be characterized specifically for the biological evaluation for which the use of the material is desired (Joseph et al. 2009) .

Biomaterial Classification and Selection of RMs

Selection of the material is the principal criteria for the proper functioning of any medical device. Any corrections in device design cannot improve device performance if the material is not compatible. Anticipated anatomical position and physiological conditions of in vivo or in vitro use determine what material should be used in the development of a device. Lightweight polymer or ceramic materials may be selected for cochlear and dental prostheses, whereas for total hip replacements and total knee replacements, mechanical properties of metallic devices are considered beneficial. Devices like artificial heart valves used in blood circulation may be composed of different components made of metals, ceramics, and/or polymers to achieve the required functionality (Navarro et al. 2008).

According to the site of application, requirements on material characteristic properties may vary. A smooth surface finish of a metallic heart valve may not be

appropriate for the osseointegration of metallic screws used in orthopedic applications (Vayalappil et al. 2016). In such cases, the engineering design of devices is governed by the compliance of material properties with the end-user requirements. Apart from the physicochemical properties, biological response or reactivity of the material of choice will be the deciding factor in designing medical devices. These properties have to be characterized by evaluating responses to appropriate test methods like cell culture studies, animal implantation studies, etc.

The characterization process should address the identification of hazards in the usage of medical devices, arising from variations with the manufacturing process, or risks caused by various factors like insufficient control of the manufacturing process, variation in concentration of additives, and impurities in raw materials. Such screening evaluations in accordance with ISO 14971 can be performed with test samples of device materials.

The ISO 10993 standard allows the selection of characterization matrix based on end- use of materials. By following the matrix, biomaterials may get qualified through a series of suitable biological evaluations. Cytotoxicity evaluation is the preliminary screening test widely employed in medical device development. The individual biomaterial components are exposed independently to a monolayer of cells and the morphological changes to the cells if any will be microscopically scored (Fig. 1). The results will be compared with positive and negative reference material.

Experimental controls are used in biological evaluations to either validate evaluations or compare the results between materials, thereby qualifying the material for end-use. As per the specifications of evaluation methodology employed, experimental controls negative and/or positive controls shall be used. RMs when used as experimental controls in biomaterial evaluations should be of the same class as that of the test material, i.e., polymer, ceramic, metal, etc.

Metals

Mechanical properties of metals make them preferable biomaterials for load-bearing applications and are used in supporting or replacing strenuous activities that need

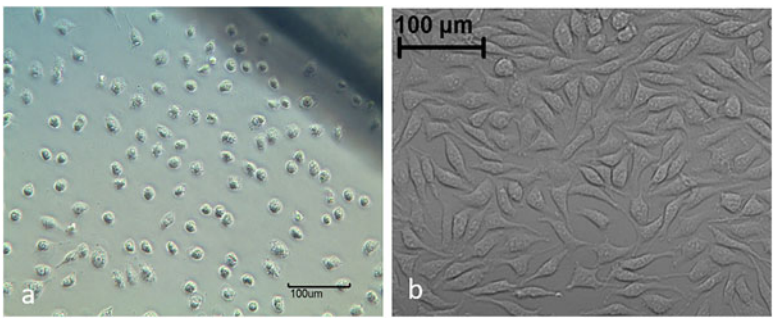

Fig. 1 (**a**) L929 cells around PVC-based RM material with positive reactivity. (**b**) With negative control; viable cells in proximity of the sample

sufficient fatigue strength. When used for orthopedic applications, they must have corrosion resistance, have excellent wear resistance, and need to support osseointegration. When used in vascular systems, they should not cause hemolysis. They have to be nontoxic and should not cause any inflammatory or allergic reactions in the human body. Improvement in the biocompatibility of materials is achieved by alloying pure metals, modifying material surfaces using different treatment methods, or providing coating, etc. Stainless steel 316 L, cobalt-chromium alloys (CoCrMo), titanium and its alloys (Ti-6Al-4 V), dental amalgams, gold, etc. were generally used in medical device development in the early days of biomaterial applications (Gad 2017). With time, biomedical researchers preferred more biocompatible metal alloys and corrosion and wear resistant than pure metals. In the early 1970s, titanium alloys become popular in medical device development due to their excellent specific strength, lower modulus, superior tissue compatibility, and higher corrosion resistance (Dowson 1992). CP Ti-commercially pure titanium-ASTM F67 was the first of its kind, with its natural oxide layer formation which had excellent osseointegration properties; that is, human bone cells bonded and grew on the titanium oxide layer quite effectively. But with low mechanical strength, its usage was limited in specific applications like hip prosthesis, dental implants, heart valves components, etc. Even though with a better performance of Ti-6Al-4 V, its usage is restricted by the presence of vanadium that causes cytotoxicity and other adverse tissue reactions in clinical use. Thus only a few of the alloying elements are identified as not to cause any harmful reactions when implanted inside the human body and having sufficient mechanical strength and machinability. Examples of such alloys are titanium, molybdenum, niobium, tantalum, zirconium, iron, and tin (Nag and Banerjee 2012).

In the recent past, Nitinol, shape memory and super elastic NiTi alloy, is being extensively used in vascular stents and various other applications such as orthopedic fixation devices. They are discovered by William J Buehler and his co-workers in 1963 in the Naval Ordnance Lab (NOL), and hence more commonly known as "Nitinol" where "Niti" stands for nickel-titanium and "Nol" stands for Naval Ordnance Lab (NOL), USA. It is a titanium-based inter-metallic material. Nitinol contains an equal amount of nickel and titanium and shows shape memory with super elastic properties due to thermoplastic martensitic transformation (Shabalovskaya and Van Humbeeck 2008).

Ideally, all biomaterials, whether stainless steel, Co-Cr alloys, titanium alloys, or Nitinol, should remain biocompatible under varying mechanical and electrochemical human physiological conditions. Studies report that a certain degree of toxicity resulting from the release of nickel ions into the human body is associated with the use of Nitinol (Oshida et al. 1990). This concern on Nitinol use has stimulated serious research on its surface modifications and coatings (Wadood 2016). Advancements in metallurgy and manufacturing technologies spur innovations in design and diversification in implantation material. Three-dimensional (3-D) printing technologies are being widely utilized to meet customized patient care needs in shapes and geometries using a variety of metals.

Fig. 2 An example of metallic reference material

Metallic Reference Materials

To satisfy regulatory criteria in a risk management platform, materials have to go through multiple biocompatibility evaluation matrices like cytotoxicity; sensitization; irritation or intracutaneous reactivity; acute, subchronic, and chronic systemic toxicity; material-mediated pyrogenicity; genotoxicity; implantation; hemocompatibility (hemolysis, complement activation, thrombosis); carcinogenicity; reproductive or developmental toxicity; and biodegradation (for absorbable materials). In such evaluations, the probable adverse effects at the local site of biomaterial or distant target tissues were validated with the use of appropriate RMs as positive or negative controls (USFDA 2019). RMs can be developed, qualified, and certified from appropriate commercially available materials according to the biological characterization procedure for which the CRM is to be produced.

ISO 10993-Part 12 suggests negative controls RMs of metallic materials made of stainless steel, cobalt/chromium alloys, and commercially pure (CP) titanium alloys (Fig. 2). In preclinical animal implantation studies, considering animal welfare and ethics, the use of positive controls are not recommended.

Polymers

Polymers, based on their origin, are classified as natural polymers, modified natural polymers, and synthetic polymers and are used widely in medical devices by altering the composition or physical properties such as molecular weight, polydispersity, crystalline, and thermal transitions. Most of the natural polymers are present in the tissues of living organisms; and they seem to be more biocompatible compared to synthetic polymers. But with less mechanical strength and rapid degradation characteristics, their use in modified forms is preferred for many applications. Unlike natural polymers, custom-made synthetic polymers can offer reproducibility of medical devices without any concern on disease transmission. Possibilities of designing and modification of biocompatibility parameters according to end-use make them more attractive. Synthetic polymers are the preferred choice of biomaterial because of their tailored properties and tuneable synthesis procedures. Natural polymers, including collagen, gelatin, chitosan, and synthetic polymers like poly(lactic acid) (PLA), poly(glycolic acid) (PGA), and polycaprolactone(PCL), are used in tissue regeneration applications (Ouchi and Ohya

2004; Puskas et al. 2004). Thus biocompatible polymers can directly be used in device development or used as a coating to minimize the chances of rejection when it comes in contact with the body. Advancements in technologies with shape memory materials, biodegradable polymers, tissue engineering, and coronary stents are some of the areas in which polymers find application (Isabel Cristina Celerino de Moraes Porto. 2012).

In tissue engineering (TE) applications, polymeric biomaterials are widely used for preparing scaffolds that provide the structural support for cell attachment and subsequent tissue development and growth. Additive manufacturing (AM) otherwise known as 3-D (three-dimensional) printing is the general method of scaffold preparation. In additive manufacturing, precise control of scaffold characteristics like porosity, pore size, pore connectivity, and internal flow channels that are vital for good scaffold function is possible with polymeric materials (Leong and Chua 2014). It is confirmed that tissue engineering requires artificial scaffolds with physicochemical, mechanical, and biocompatibility properties mimicking natural extracellular matrix (ECM) without causing immunological reactions and should degrade in a controlled manner with non-toxic products that can be excreted through metabolism. Specific additives to promote the formation of new tissues may be incorporated in the scaffold preparation. Since polymers own highly flexible design capability, their properties can be easily tailored to meet specific requirements by controlling their chemical compositions and structures (Shi et al. 2016).

Polymeric Reference Materials

Polymeric materials may contain traces of low-molecular-weight chemical substances such as catalysts, polymerization agents, or other additives, monomers, or oligomers.

Such toxicity leachable from polymer can migrate to the human tissues during use. In this scenario, any residue from the manufacturing processes, intentional or unintentional additives or contaminants, that may get integrated into biomaterials may have undesired performance of medical devices. The suitability of material can be assessed at different grades of biological reactivity like none, slight, mild, moderate, severe, etc. CRMs with certified properties can be used to compare and qualify biomaterials using (1) negative control RMs for reactivity "none" and (2) positive control RMs for reactivity's such as "slight," "mild," "moderate," or "severe." Polyethylene and silicon are the commonly used negative control materials, whereas PVC (polyvinylchloride), PU (polyurethane), natural rubber latex, Genapol X-080, etc. are used as positive controls (Fig. 3). Possible hazards derived from materials could be screened out when it is being evaluated against control RMs whose characteristic properties are well established.

Ceramics

Ceramic biomaterials otherwise called bioceramics are suitable for orthopedic and dental applications to support hard or soft tissue. Their mechanical and osseointegration characteristics and microstructure like grain size and porosity

Fig. 3 An example of polymer reference material. (Positive control material made of PVC)

make them suitable for medical device development. Generally, bioceramics can be grouped into bio-inert or bioactive materials. The bio-inert materials with their corrosion resistance, high wear resistance, and high mechanical resistance do not elicit adverse reactions with the surrounding tissue. Such properties are appropriate for orthopedic applications involving articulation and implantation sites subjected to loads and friction (Brunette et al. 2001).

Bioactive ceramics are materials whose surface characteristics contribute to bone-bonding ability or enhance new tissue formation. The bioactive ceramics are again categorized as resorbable or non-resorbable materials. At the interface of resorbable type implant material and the tissues, time-dependent changes are expected to support or restore the physiological functionality (Ducheyne and Q. 1999). In bone tissue engineering, defects are filled with appropriate mechanically strong bioceramic materials that could initiate bone growth. Such materials could get degraded in the growth rate of newly forming bone tissues (Gao et al. 2017). Physical forms of ceramics may be varied either in porous or dense bulk forms according to their molecular arrangement. Based on end- use, materials in the form of granules or coatings can be used (Huang 2007).

Aluminum oxides, calcium aluminates, titanium oxides, calcium phosphates, carbon, bioglass, etc. are the commonly used bioceramics. Alumina and zirconia belonging to the group of bio-inert ceramics and calcium phosphates, bioactive glasses, glass-ceramics, etc. are examples of bioactive ceramics. The presence of a wide range of oxides is common in bioactive ceramics. Crystalline hydroxyapatite is a bioceramic that has similar mineral properties to human hard tissues (Saenz and Rivera-Muñoz 1999).

Ceramic Reference Materials

The standard procedure for qualifying biocompatibility evaluations involves in vitro cell culture experiments, as a low-cost first step before going for in vivo use. But for ensuring the appropriateness of the biomaterials; researchers have to investigate and confirm the capability of biomaterials to stimulate or enhance tissue regeneration in preclinical in vivo conditions. The selection of RMs both negative and positive control RMs, to compare the cellular response, is therefore critical and essential (Wu 2009). The importance of using validated positive controls with well-known cytotoxicity is provided in relevant international standards for medical device evaluations. Commercially available latex is one example of a positive control

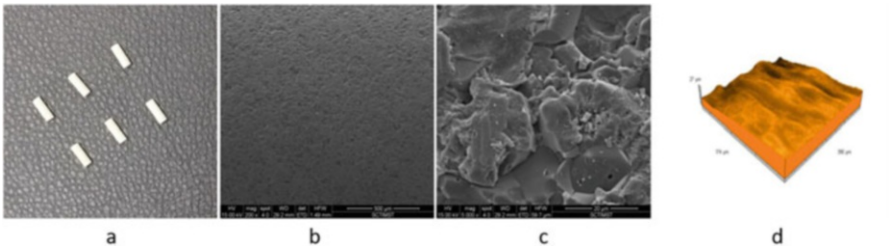

Fig. 4 (**a**) Ceramic reference materials (alumina), (**b**) and (**c**) SEM images of alumina RM at 200× and 5000× magnification, (**d**) surface profile of ceramic RM (alumina) – (imaged with Talysurf 1000)

used in in vitro testing (Lourenço et al. 2015). Even though biological evaluations and toxicity concerns are well understood, the selection of control materials for bioceramics is not defined with clarity (Thrivikraman and Basu 2014). According to research publications available, polymeric RM materials are widely used as control materials in the case of bioceramics also. Addressing the RM needs for ceramic biomaterials is an area that needs to be defined and established further (Fig. 4).

Evaluation Matrices and Methods of Evaluation

The products/materials intended for human application have to be thoroughly checked for their safety and efficacy before clinical translation. Devices have to get regulatory approval in both safety and efficacy terms with human safety being the primary concern for the regulators. Safety aspects can be considered only in comparative terms. Every device carries a certain amount of risk and could initiate complications in specific situations. The current approach to device safety is to estimate the potential of a device becoming a hazard that could result in safety problems and assessment of this harm or risk versus benefit.

The International Organization for Standardization (ISO) has produced a document (ISO 14971: 2019) providing the medical device manufacturers with a framework including risk analysis, risk evaluation, and risk control for risk management in medical device design, development, manufacturing, as well as monitoring the safety and performance of the device after marketing (ISO14971 2019). The standards specify that the evaluation strategies should follow a risk assessment-based approach for the safety clearance of biomaterials and medical devices. Device developers should identify all the possible risks associated with the usage of the intended medical device. There are a lot of processes involved in the manufacturing of medical devices (Fig. 5). At each stage, the potential factors compromising biocompatibility have to be identified. The extent of risk has to be estimated, and appropriate management strategies have to be adopted to mitigate or eliminate risk to the least extent possible to get regulatory approval. In this context, the selection of evaluation methodologies requires much attention.

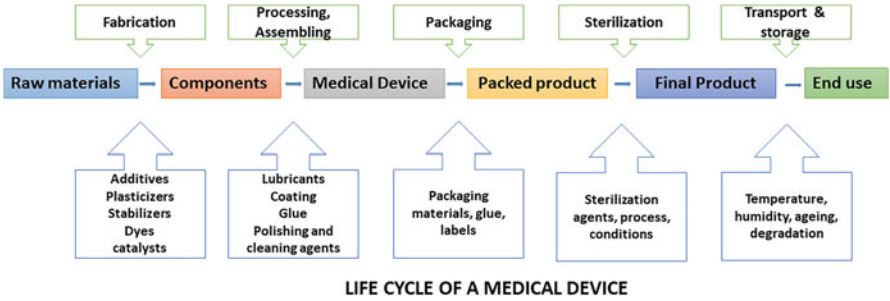

Fig. 5 Factors influencing the biocompatibility of a medical device

Table 1 Potential biological hazards encountered in a medical device development

Short-term effects	Long-term/specific effects
Acute toxicity	Sub chronic toxicity
Skin irritation	Chronic toxicity
Eye irritation	Sensitization
Mucosal irritation	Genotoxicity
Sensitization	Carcinogenicity/tumorogenicity
Hemolysis	Teratogenicity
Thrombosis	

ISO 10993, Part 1, is the primary horizontal standard for biological safety evaluation. It provides a framework to plan the biological evaluation of medical devices, selection of appropriate in vivo tests along with in vitro assays, and material/chemical characterization data. The standard specifically emphasizes that the evaluation should address all the relevant risks that could impact the biological safety of the final product (ISO10993-1 2018).

The potential biological hazards that are commonly encountered in a medical device development are given in Table 1.

Accordingly, the evaluation process should follow a tiered approach. Material characterization is the crucial first step. The extent of material characterization depends on the nature and duration of patient exposure, knowledge of raw materials, and available toxicological data. Possible residual processing aids, as well as the degradation profile of the component materials, should be evaluated before proceeding to biological evaluation.

Table 2 is an excerpt from the standard ISO10993, Part 1, depicting the evaluation methodologies to address the common potential factors encountered in medical device development compromising biocompatibility/safety of medical device

Reference Materials in Biological Evaluation

Reference material as per the definition is a particular form of measurement standards that are employed to check the quality and metrological traceability of products, to validate the methodologies, or to calibrate the instrument. Here the

Table 2 Evaluation methodologies for risk-based endpoint assessment

		Medical device category								
		Surface contacting device			Externally communicating device			Implant device		
Endpoints		Limited	Prolonged	Permanent	Limited	Prolonged	Permanent	Limited	Prolonged	Permanent
Physicochemical characterization		x	x	x	x	x	x	x	x	x
Cytotoxicity		x	x	x	x	x	x	x	x	x
Sensitization		x	x	x	x	x	x	x	x	x
Intracutaneous reactivity		x	x	x	x	x	x	x	x	x
Material mediated pyrogenicity				x	x	x	x	x	x	x
Systemic toxicity	Acute			x	x	x	x	x	x	x
	Subacute			x		x	x		x	x
	Chronic			x			x			x
Implantation			x	x		x	x	x	x	x
Hemocompatibility						x	x	x	x	x
Genotoxicity				x		x	x	x	x	x
Carcinogenicity							x			x

(Adopted from ISO10993 part 1)[2]

biological evaluation methodologies suggested by ISO standards also advocate the usage of certified reference material wherever possible. Reference materials are incorporated as 3 categories in the evaluation methodologies (a) Solvent control/vehicle control, (b) Experimental control, and (c) Controls for comparison

Solvent/Vehicle Control

The basis of biological application in medical device development is to identify all the potential factors that could affect the biocompatibility response of the biomaterials intended for the fabrication of particular medical devices.

Some of the factors include the following:

- Residual monomers.
- Residual solvents.
- Degradation products.
- Sterilization products.
- Formulation additives.
- Bacterial endotoxins.

For addressing all these, the device components/biomaterials are extracted in extraction media at a specific surface area/volume or mass/volume ratio. The sample preparation methodologies are detailed in ASTM (American Society for Testing and Materials) 619 and ISO 10993-12 Standards (ASTMF619–14 2014; ISO10993-12 2021). The prepared sample is used for subsequent in vivo analysis/assays. Generally, physiological saline and cottonseed oil are used for mimicking the hydrophilic (polar) and lipophilic (non-polar) physiological compartment of the human body. Other solvents that are used for extraction are alcohol saline, polyethylene glycol, cell culture medium, etc. for various biological assays. The solvent/extraction media are selected according to the different classes of polymers/plastics that are generally used for the fabrication of medical devices (Table 3).

Accordingly, in the evaluation of medical devices, the following in vivo biological assays use vehicle control/blank for the interpretation of test results.

- Systemic toxicity test.
- Intracutaneous reactivity test.

Table 3 Classification of plastics and extraction medium as per USP (USP31 <88>)

	Medical grade plastics/polymers classification as per USP					
Extraction media	I	II	III	IV	V	VI
Physiological saline	x	x	x	x	x	x
Alcohol-saline		x	x	x	x	x
Polyethylene glycol 400			x		x	x
Cotton seed oil/sesame oil			x	x	x	

- Animal skin irritation test.
- Special irritation tests.
- Sensitization tests.

In the above tests, the blank/reference control consists of the same quantity of extraction medium maintained in the same experimental conditions as that of test material extract. Extraction conditions as per standard are 37 °C for 72 hours/ 24 hours, 50 °C for 72 hours, 70 °C for 24 hours, or 121 °C for 1 hours. The control extract is administered to separate control animals or at different sites on the same animal as per the test procedure. Clinical observations and visual scoring will be done and for test materials and controls, and the test results will be interpreted accordingly. If there is an equivocal response on both control and test sites, the particular test will be repeated.

Experimental Controls

Incorporating concurrent controls (positive and negative control) improves the validation and acceptance of the assay performed. A positive control is a reference material that is known to produce a positive response under the conditions of the test. And a negative control is a reference material that particularly gives no reaction under the conditions of the test. The performance of positive and negative control determines the integrity of test results of the test procedure. They may further help in generating a historical control database that would further demonstrate the proficiency of the particular assay, specifically in a testing lab that works under a quality platform. Some of the biological assays that include positive and negative reference materials are cytotoxicity assays, genotoxicity assays, hemolysis tests, etc. A list of positive controls used in various assays is detailed below (Table 4). Generally, for in vivo biological assays, incorporation of positive control is not advised for every single test because of the animal welfare requirements. Nevertheless, the standard prescribes using a positive reference material (material known to produce some response) occasionally to ensure the proficiency of the particular in vivo assay.

Comparing the test results with that of positive control helps in estimating the extent of risk or probability of the material causing harm to the biological system. On the other hand, comparing the result with that of negative control would ensure the extent of safety associated with the exposure of the material to the physiological system. Employing statistical methods for the comparison of test results with both the experimental control results (positive and negative reference material) thereby significantly enhances the integrity of results of the biological evaluation.

Controls for Comparison

In biocompatibility evaluation, especially when the local tissue response of the material is evaluated, the test material is compared with a similar kind of reference

Table 4 List of positive controls used for biological evaluation

Assays	Chemicals/positive controls
Cytotoxicity assays	Sodium lauryl sulfate Zinc diethyldithiocarbamate Phenol Doxorubicin Triton X
Genotoxicity assays	Methyl methane sulfonate Mitomycin C 4-Nitroquinoline N oxide Cytosine arabinoside Benzo(a)pyrene Cyclophosphamide monohydrate Congo red 2-Aminoanthracene
Hemolysis	Washed Buna N rubber Vinyl plastisol
Sensitization/irritation assays	Sodium lauryl sulfate Mercaptobenzothiazole Hexyl cinnamic aldehyde Benzocaine 2,4-Dinitrochloro benzene

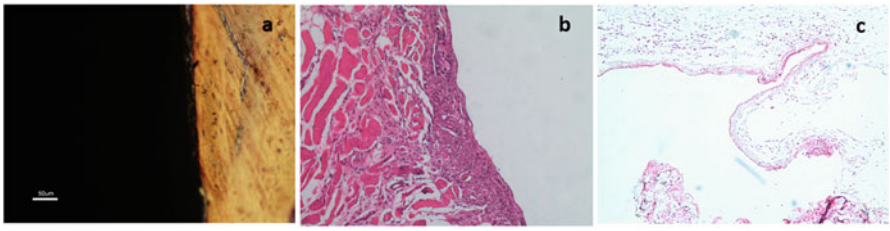

Fig. 6 Microphotographs of tissue surrounding the implant, (**a**) bone tissue, (**b**) muscle tissue, (**c**) subcutaneous tissue

material. In such a scenario, the reference material is of particular importance as the interpretation of test results depends on the control response. The local tissue response is evaluated by implanting the test materials in subcutaneous tissue or muscle tissue or bone tissue or any other clinically relevant sites (brain or vascular system) depending on the intended application. Reference material of the same class fabricated in similar dimensions as that of test material needs to be implanted at the control site in the same animals. At desired time intervals, both the samples are retrieved along with the surrounding tissue and are processed for histopathological examination. The cellular and inflammatory response around the material is microscopically scored and the test score is compared with that of the control score to report its response as non-irritant, mild, moderate, or severe irritant (Fig. 6).

Selection and Preparation of RM

Development of RMs involves the identification of the requirements and setting specifications required for reference material based on testing or evaluations. Relevant international standards draw guidelines in the selection of material type and their preparation. Commercially available industrial-grade materials after proper characterization are also allowed to use as RMs. It is a mandate that, appropriateness of relevant property values in line with the test or evaluation must be established before their routine use of proposed RMs. This confirmation shall be done through a series of physicochemical and biological evaluation series as directed by relevant international standards and ISO 10993 series. Standardization of procedures in RM preparation can then be followed to establish a system for development and production. ISO 10993 part 12 provides guidelines for the selection and preparation of RMs.

Physicochemical properties, biological response in a physiological environment, and the aging behavior of RM material are characterized and documented to generate material safety data. In routine production of RMs, the homogeneity of the samples in each batch of RMs with respect to the selected characteristic property is to be verified. ILC (interlaboratory comparisons) with the different participants can be employed to increase the acceptability of homogeneity test results. Based on the homogeneity and characterization studies, the material is assigned with a reference value. This could be qualitative or quantitative depending on its application. RMs with quantitative reference values could be assigned metrological traceability. But traceability establishment in RMs with qualitative reference values is not easy. In such cases, participation in interlaboratory comparisons (ILC) with multiple accredited laboratories may help to certify the assigned reference value. In general, steps in RM production can be summarized as follows:

- Material identification.
- Material preparation.
- assessment of homogeneity and stability,
- Material characterization.
- Assigning property values.
- Certification of RMs.

Maintaining RM Facility and Accreditation for RM Production

Reference material producers (RMP) shall define their scope of activities in terms of the types of reference materials, the properties to be certified, and the ranges of Assigned value, uncertainty, and best reference value capability of the reference materials they produce. Laid down procedures shall be followed for routine production of RMs. All metrological procedures of testing, calibration, and measurements in relation to homogeneity, stability and characterization assessments, etc. shall be available, and recorded results of measurement must be kept with RMP.

Accreditation for RM Facility

Since 2005, the availability of ISO Guide 34 and ISO 17025 in combination was the requirement for recommending accreditation of reference material manufacturers. ISO guides 30, 31, 35, etc. were the normative references at that period. With the publication of ISO 17034: General requirements for the competence of reference material producers in 2016, ISO guides became its informative references (TEC1–008 2019).

In the process of accreditation, the assessment body (AB) confirms that all aspects of ISO 17034 are complied by the RMP. ISO 17034 advocates the measurement procedures and metrological traceability in specifying accuracy of the RM property values should meet the requirements of ISO/IEC 17025. Concerning the methods and traceability, the RM producer shall pay special attention to the following aspects:

- Standard methods used are suitable for the intended use and in line with the latest published version.
- Non-standard methods used are developed by qualified personnel with adequate resources and are validated.
- Measuring equipment used in RM production must comply with relevant requirements of ISO/IEC 17025.

Market Availability and Use

With the increasing demand for satisfying regulatory criteria, evaluation phases of device development are expected to be employing international standards in a platform of risk-based thinking. Biological evaluation stages demand the use of RMs in relevant biomaterial classes. For addressing the requirement for a wider range of materials, it is not easy for the reference material producers to establish production facilities dedicated to RMs as control materials. One reason for this may be due to the greater risk of cross-contamination during the processing of positive control materials.

Joint Research Centre (JRC), one of the Directorate General of the European Commission, is an independent agency that provides scientific and technical support to community policy-making. They advised new dedicated facilities for individual reference material programs as batch production, for achieving targeted measurement benchmarks in response to changing requirements and environments. Hence a scientific and technical facility with dedicated laboratories and a multipurpose pilot plant for material processing is a requirement for large-scale RM production.

Very few research laboratories at national levels are available as RM sources for biological evaluations. IRMM (EU), National Institute of Standards and Technology (USA), United States Pharmacopeia (USA), and Hatano Research Institute(Japan) are the current major suppliers in the field.

In India, National Physical Laboratory (NPL), New Delhi, is the apex metrological body for facilitating national RM requirements in the area of physicochemical

characterization. Currently, Indian Reference Material or Bharatiya Nirdeshak Dravya (BND) available from NPL does not include RMs for biological evaluation. With respect to medical devices development and their evaluation, Sree Chitra Tirunal Institute for Medical Sciences, Kerala, an Institute of National Importance has an established system in a quality management platform of ISO 13485 and ISO 17025. Initiatives for the availability of RMs in metallic, polymeric, and ceramic classes are established for cytotoxicity evaluations, hemolysis studies, muscle and bone implantation studies, etc. at SCTIMST as an in-house RM facility. Both positive and negative control materials were validated through an interlaboratory comparison at this facility.

With the booming initiatives for medical device development across nations, indigenous RM facility is always the need of the hour in many countries. It is expected that India can become self-reliant in the BND availability along with expanding medical device industry.

Conclusion and Future Perspectives

Qualifying a device for safe clinical use depends on the physical and chemical nature of its materials in addition to the nature of the device's exposure to the body. There are chances for a biomaterial with good physical properties and cost effectiveness in a particular application may cause adverse toxicity effects in a different physiological environment due to its chemical properties. Properly designed biological evaluations help in such cases for determining potentially adverse or toxic effects of medical devices.

For regulatory approvals and global acceptance, such qualitative evaluations can be made traceable only through comparisons against certified reference material. An unlimited healthcare need around the globe compels the medical devices industries to operate in national or international regulatory framework. RM use explicitly stated in ISO 10993 standard series will then become an essential element in the evaluation of biomaterials. The non-availability of national-level RM sources in each country drives the need for establishing national facilities for the production and certification of RMs for biological evaluations.

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Alloys as Certified Reference Materials (CRMs)

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Ferrous and Non-Ferrous in Global Perspectives – A Review

Nirmalya Karar and Vipin Jain

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Abstract

Alloys are a mixture of pure metals and are mixed to improve their performance and properties. A summarized detail of numerous alloy systems is discussed in this chapter. Metal alloys are classified in two categories, ferrous and nonferrous. Different steels thus fall into the first category, and other popular alloys like brass, bronze, invar, duralumin, hindalium, etc. fall into the second category. However, any alloy or any material for that matter has to be checked for its quality, production, and performance repeatability, and thus a standardization protocol is necessary. Any such standardization involves checking and benchmarking the acceptable composition range of that material. Many of the properties of such alloys are further dependent on different crystalline phases of the individual metals when they are mixed – so these also deserve to be benchmarked. Additionally, the properties of alloys are also dependent on the thermal cycling

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processes, heat treatment procedures, and associated microstructure that they result in, i.e., on the thermodynamics of their preparation process. In some cases, it can also depend on the alloys' resistivity and magnetic properties. In order to achieve repeatability in their preparation process, all of these need to be standardized, i.e., measured, quantified, and cataloged. Such a process will result in production of standard alloys or certified reference materials (CRMs). They may be compared with any subsequent batch to batch product for checking consistency of expected properties. Thus, different national metrology institutes (NMIs) all over the world have been working on standardizing different alloys of their interest for growth of their own economies. Here we shall summarize the different standard alloy samples available with the major NMIs of the developed world like NIST-USA, PTB-Germany, NMIJ-Japan, and NPL-UK or their surrogates. The American Society for Testing and Materials (ASTM) or American Iron and Steel Institute (AISI) or Society of Automotive Engineers (SAE) has independently developed an extensive database for a large number of alloys, irrespective of whether similar data has been published or not by any NMI. Consequently, for trade purposes, several alloy makers from all over the world follow these standard data in situations where NMIs do not have specific standards of interest. A summarized discussion on all of them shall be made along with their consequences on their economies in this chapter. In the case of emerging economies like NMI-India, irrespective of what other NMIs have or have not worked on, this write up discusses how any alloy as a case study can be successfully standardized. It shall be worth the effort as that knowledge may be commercially exploited for the local and global market and extrapolated for making of other alloy standards and quality alloys in due time.

Keywords

Alloys · Certified reference material · NMI · NIST-USA · PTB-Germany · NMIJ-Japan · NPL-UK · Bharatiya Nirdeshak Dravya (BND®)

Introduction: Why Metal CRMs?

In this chapter, alloy systems (ferrous and nonferrous) are briefly summarized with major compositions tabulated. Ferrous alloy systems like cast irons, low-alloy steels, stainless steels, tool steels, and heat-resistant grades are briefly discussed. In the nonferrous category, aluminum (Al)-, copper (Cu)-, and titanium (Ti)-based alloys are summarized. We shall also discuss why all these alloys need better standardization parameters and how we could possibly implement them.

In case of any physical measurement or derived measurements involving primary SI units, its accuracy, precision, and repeatability are checked using established measurement methods that are acceptable to all national metrology institutes (NMIs) the world over. This is called validation of test results. These are traceable to the same primary SI units that are maintained by different NMIs globally.

However, for most materials sold in the market, such primary or secondary units do not exist. So, their quality is to be verified using a specific quality benchmark of a similar standard material prepared earlier that performs to a predefined performance level in a similar manner. As an example, let us take an example of brass. In order to specify its desirable quality, one may define its required expected density, composition, melting temperature, electrical resistivity, indentation hardness, a certain surface roughness in terms of reflectivity, surface stability in terms of discoloration upon keeping unused for 12 months in noncorrosive sulfur dioxide-less ambience, crack proofness upon hitting/exposing with a pressure of a certain pascal, etc. Or all these parameters can be compared in short using a known brass material that is known to conform to all of these parameters. Such an alloy material is called a reference alloy material, in this case a reference brass material. As another example, we could consider the case of setting quality standards for the widely used representative A4 paper characteristics as a reference material. It could be setting its acceptable discoloration rate in a decade, its weight uniformity in gms/cm^2, its minimum and maximum lignin content acceptance levels, its tear resistance uniformity in pascals/cm^2, its chemical and physical uniformity standards setting in terms of light intensity decrease on transmission, and its acceptable ink diffusion coefficient setting, i.e., smudging properties; else all of these can be compared in short with a reference A4 sheet that conforms to all of the above for brevity and simplicity and ease of measurement. So we can have scope and prospects for a reference A4 paper – it is compared with any batch of A4 sheet for declaring that batch as passed or failed the test in short. Zinc coating can be another example. Zinc coating is used for preventing oxidation of iron surfaces. But the issue that is always present is what should be the quality benchmark for this coating? This zinc can be melt-dip coated or electro-plated. The latter is much more expensive. The parameters under scrutiny will be what should be the minimum thickness of the zinc coating so that the material retains its noncorroded properties even 50 years after production, e.g., the telegraph and telephone poles and electric pylons and street lighting masts seen all over the country or steel rebars for construction of engineering structures or cheaper nails and screws for daily use. The more sturdy nails and screws are of stainless steel for durability. What should be its maximum and minimum acceptable surface roughness and porosity as well in SI units? A simplified comparison methodology for any routine batch could be to use a standard zinc-coated surface which meets all the set standards.

In this manner, it is seen that there is always a lot of scope for preparation and practical usage of standard alloy and other reference material samples for setting of better-quality benchmarks. Better quality benchmarks lead to better industrial products in the related segment and thus lead to better commercial success under ideal conditions. However, such better-quality products have a certain price tag that needs to be paid due to the procedure and protocols and checks it necessarily has gone through; it will not necessarily be the lowest quoted product (L1).

During sample analysis for elemental content using nondestructive analytical techniques, like energy dispersive X-ray analysis (EDAX) or X-ray fluorescence (XRF) or inductively coupled plasma optical emission spectrometry (ICP-OES),

accuracy of the measurements is improved if results obtained for an unknown sample can be compared with results for a sample of similar matrix and known composition. If such a reference sample is widely used due to the nature of usage, it may make sense to commercially characterize it and market them, after extensive analysis of its composition, phase, etc. It then leads to a standard reference material (SRM). So, in the case of analysis of an unknown gold sample, by EDAX or XRF would require a comparison with a gold sample of known purity, as analyzed by chemical and fire assay. This reference gold sample is thus an SRM.

A reference material (RM) is defined as "any material, which is satisfactorily stable and homogeneous with respect to one or more specific properties and which has been recognized for its intended use in a metrological activity." Certified reference material (CRM) or standard reference material (SRM) is defined as an RM described by a metrologically valid procedure for one or more specified parameters or properties and which is accompanied by a certificate of analysis that provides a specific value of the specified property and its associated uncertainty along with a statement of metrological traceability, i.e., how directly or indirectly it is attributable to one or more primary or secondary SI units (ISO Guide 30 2015). Several national metrological institutes (NMIs) or their associating agencies across the world produce alloy-based CRMs/SRMs for the requirements over the globe. Among them, alloy CRMs produced by the developed economies/NMIs, such as National Institute of Standards and Technology (NIST)-USA, Physikalisch-Technische Bundesanstalt (PTB)/BAM-Germany, National Metrology Institute of Japan (NMIJ)-Japan, and National Physical Laboratory (NPL)/LGC Standards-UK, are discussed. Later in this chapter, the status of alloy-based CRMs with Council of Scientific and Industrial Research-National Physical Laboratory (CSIR-NPL), India, is also discussed, which is currently one of the developing economies/NMIs in the world.

It needs to be borne in mind that there are several metallurgical methodologies for preparation of different alloys. The preparation method to be used depends on what type of alloys needs to be prepared. The methods used for different steels are not the same as that for different aluminum-, gold-, copper-, or titanium-based alloys. Details on aluminum-based alloy preparation are discussed by Persson (Persson 2011). Iron-based alloy production methodologies are discussed by Neveille and Cain (Neville and Cain 2013). Gold and such precious metals are prepared by a combination of chemical melt and electrochemical refining. Due to their high price, such methodologies are suitable only for such precious metals. Information on different copper-based alloy preparation are discussed by Zahner (Zahner 2020). Details on preparation methodologies of titanium alloys are to be found in the book by Sanchez (Sanchewz 2010). As will be realized, each metal is unique in its preparation process as much as in its properties. Alloys are mostly about special properties related to their special performance, their performance stability, and repeatability over thousands of times over several years. Since these have a distinct commercial connotation, in each case, their exact preparation process, related parameters and repeatability issues, related production data, and mostly commercial secrets of each of the companies that prepare and market them are often proprietary

items. Since alloys are mostly about performance, it is not always possible quantify all the desirable parameters, though mathematically definable properties in terms of primary units can be measured and are enforceable. But parameters like corrosion resistance-related performance for a few years or a knife's cutting edge's ability to cut certain items at a certain speed repeatedly for a certain number of months without sharpening or say the metallic properties of a gun barrel to fire certain number of rounds without cooling with a certain accuracy and then maintaining this ability for a certain number of years may or may not be totally quantifiable in terms of primary SI units. The ability of a specific composition-based alloy to maintain its shape upon repeated cyclic bending in a certain format up to a limit may or may not be totally quantifiable. The ability of a metal plate of a certain composition to maintain its overall physical shape and composition upon random shock or projectile impregnation and related fires may or may not be totally quantifiable. But it is for these "no totally quantifiable properties and parameters" that specific alloys are preferred and used in special applications. It is not always thus possible to have reference materials for these as all of their properties are not always totally quantifiable. Most known alloys fall into this category. Now, it may be noted that a reference material is a material that is used for standardizing or checking the quality in terms of a quantifiable specific predefined property of a similar material in terms of expected properties. It is not necessary to discuss about how it was prepared. It is not a patented item that its preparation details have to be disclosed in detail. After several years or several decades, such information may come into the common man's domain. So discussing preparation methodology in general for alloys is not practically feasible. Only those few metal alloys that are very well-defined, including its composition, with known unique preparation methodology and numerically definable performance parameters, can be classified into the category of reference materials. Otherwise, different alloy manufacturers often use their own preparation standards and secretive parameters to obtain repeatability in performance. No NMI can fully replicate these due to absence of all information. As an example, let us consider the case of stainless steel of certain grades. In terms of ASTM standards discussed here, it will be seen that it discusses the expected alloy composition profile alone with ranges to their possible values but discusses nothing on the preparation methodology, the acceptable metallic crystalline profiles, or different possible acceptable crystalline profiles. So it is seen that commercial stainless steel of many producers the world over has different crystalline profiles, even though they may have the same ASTM grade in terms of composition (Karar et al. 2022).

In this way, most current standards of alloys in general, as discussed below, talk of possible-acceptable element-wise composition ranges, with no information on acceptable production processes or density. Their microhardness or general hardness profiles, resistivity profile, and/or magnetization profile, perhaps their Young's modulus at times, and at best some representative information on their microstructure in terms of some representative etched micrographs are described only in cases where specific alloys are patented. As a representative case, the Hitachi coarse-grained oxide (CRGO) alloy for transformer core material-related patent can be cited [Metglas] (World Intellectual Property Organization 2016). There is no standard

reference material for this sort of alloy for use as transformer core material. The only parameter of concern is its magnetic property and nothing else – though all of those stated above, i.e., composition, resistivity, density, and crystalline properties, can be of practical significance in terms of checking its repeatability and quantification. They contain no information on the acceptable crystalline phases in terms of X-ray diffraction profile (XRD) and their full width at half maximum (FWHM) values or their correlatable Raman spectral data, which are both quantitative in nature and can in principle substitute the qualitative micrographs, in terms of repeatability. If not under patent, all alloys of need can in principle be quantified by NMIs using all possible parameters as stated above, much more than what current practice states. It then becomes a reference material. If there is a certificate issued with details of the quantitative analysis along with the related uncertainty value in the measurement parameters studied and certified and how these are traceable to the primary or secondary SI units, one gets the certified reference material (CRM). The SI units are maintained by the major NMIs of most major developed countries. These reference materials are internationally recognized because irrespective of which national NMI does a recheck of the measurement by a particular internationally acceptable method, the results obtained are expected to be the same within the uncertainty limits. As another example, we can state the case of hardness blocks. These are metal alloys of a particular shape and dimension that have to withstand hardness-related repeated indentations and still give the same expected value year after year, within a certain error level. It does not matter what is its resistivity, microstructure, magnetic property, composition, preparation methodology, or heat tempering cycle. The only parameter of concern here, in this context, is its repeatable hardness value. In case of precious metals like gold, current standards and norms are concerned only about how much purity a certain gold item is of, and density, not what is its microstructure or what is its resistivity or its magnetic property. So the only parameters of concern in a reference material are those parameters that give it the desired repeatability and the market value to the materials with which they are compared.

Ferrous Alloy Systems

Ferrous alloy systems include different alloying elements including iron (Fe) usually as the principal component. Some examples of alloying additions are C, Si, Mn, Cr, Ni, Mo, V, Nb, etc. Typical properties of ferrous alloys are durability, high yield and tensile strength, poor corrosion resistance, recyclable, good thermal conductivity, good conductor of electricity, etc. Various types of ferrous alloy systems broadly include different grades of cast irons, low-alloy steels, stainless steels, tool steels, heat-resistant steels, etc. Selected nominal compositions of different types of cast irons, case hardening (low alloy) steels, different grades of stainless steels, different tool steels, and heat-resistant casting alloys have been listed in Table 1 (Radzikowska 2004; Guide to Engineered Materials 2001), Table 2 (Vander Voort 2004a), Table 3 (Vander Voort 2004b), Table 4 (Vander Voort et al. 2004), and

Table 1 Range of composition for typical non-alloyed and low-alloyed cast irons (Radzikowska 2004; Guide to Engineered Materials 2001)

	Composition (wt%)				
Type of iron	C	Si	Mn	S	P
Gray (FG)	2.5–4.0	1.0–3.0	0.2–1.0	0.02–0.025	0.002–1.0
Compacted graphite (CG)	2.5–4.0	1.0–3.0	0.2–1.0	0.01–0.03	0.01–0.1
Ductile (SG)	3.0–4.0	1.8–2.8	0.1–1.0	0.01–0.03	0.01–0.1
White	1.8–3.6	0.5–1.9	0.25–0.8	0.06–0.2	0.06–0.2
Malleable (TG)	2.2–2.9	0.9–1.9	0.15–1.2	0.02–0.2	0.02–0.2

Table 5 (Vander Voort et al. 2004; Nickel Development Institute, Canada 2009), respectively.

"**Cast iron** is an iron-carbon (Fe-C)-based alloy also containing other elements of the periodic table and is made by remelting of pig iron, iron scrap, etc." In order to differentiate these from steel and cast steel, these may be described as a cast alloy having carbon content more than 2.03%. Its composition leads to an eutectic solidification. Alloying or non-alloying of cast iron is possible and depends upon the chemical composition. Table 1 (Radzikowska 2004; Guide to Engineered Materials 2001) illustrates the list of compositions of non-alloyed cast irons. The possible range of alloys that can be made in cast iron format is much wider, and many of these may consist of higher percentages of other elements, e.g., Si, Mn, or other special elemental additions.

"**Steel** may be defined as an alloy containing a minimum of 50% of iron and, in addition, one or more other alloying elements. These possible alloying elements may include C, Si, Mn, Ni, Mo, Cr, P, Cu, V, Ti, B, Nb, or Al." "Alloys designated as low- and very low-carbon steels are those that contain carbon less than 0.25%." "Carbon steels, usually called plain carbon steels, have their carbon content from approximately 0.10% to below 2.0% and are generally classified into three categories, i.e., low-carbon steels, medium-carbon steels, and high-carbon steels." In the USA, examples of plain carbon steels are AISI/SAE 1020, 1040, 1080, etc. "The American Iron and Steel Institute (AISI) along with the Society of Automotive Engineers (SAE)" have developed a coding system for different plain carbon steels, low-alloy steels, and stainless steel grades. It is a four-digit number designation for different types of steel. In the given three examples above, first two digits, i.e., "10," specify the steel of the class plain carbon steel, and the last two digits, i.e., "20," "40," and "80," represent the nominal carbon content and are stated in 0.01% increments in carbon. Hence, AISI/SAE 1020 steel can be described as the plain carbon steel having carbon content of 0.20%. Similarly, AISI/SAE 1040 steel can be described as the plain carbon steel having carbon content of 0.40% and so on. For surface hardening of steels, different processes are developed to allow diffusion of alloying elements through surface layers. "Such thermochemical processes include carburizing, nitriding, carbonitriding, nitrocarburizing, etc." These processes are usually found suitable for the low-carbon steels (containing carbon below 0.25%) and allow development of a case of high hardness. The advantage of this methodology

Table 2 Nominal composition of typical case hardening steels (Vander Voort 2004a)

	Composition (wt%)							
Steel	C	Mn	S	P	Si	Ni	Cr	Mo
Carbon steels								
1010	0.08–0.13	0.30–0.60	0.050 max	0.040 max	–	–	–	–
1020	0.17–0.23	0.30–0.60	0.050 max	0.040 max	–	–	–	–
1039	0.37–0.44	0.70–1.00	0.050 max	0.040 max	–	–	–	–
Alloy steels								
3310	0.08–0.13	0.45–0.60	0.025 max	0.025 max	0.20–0.35	3.25–3.75	1.40–1.75	–
4320	0.17–0.22	0.45–0.65	0.040 max	0.035 max	0.20–0.35	1.65–2.00	0.40–0.60	0.20–0.30
8620	0.18–0.23	0.70–0.90	0.040 max	0.035 max	0.20–0.35	0.40–0.70	0.40–0.60	0.15–0.25
8822	0.20–0.25	0.75–1.00	0.040 max	0.035 max	0.20–0.35	0.40–0.70	0.40–0.60	0.30–0.40
9310	0.08–0.13	0.45–0.65	0.025 max	0.025 max	0.20–0.35	3.00–3.50	1.00–1.40	0.08–0.15
Nitralloy 125[a]	0.20–0.30	0.40–0.70	–	–	0.20–0.40	–	0.90–1.40	0.15–0.25
Nitralloy N[a]	0.20–0.27	0.40–0.70	–	–	0.20–0.40	3.25–3.75	1.00–1.50	0.20–0.30
H13[b]	0.35	–	–	–	–	–	5.0	1.50

[a]These steels also contain 0.85 to 1.2% Al
[b]HI3 steel also contains 1.0%V

Table 3 Nominal composition of typical tool steel grades (Vander Voort 2004b)

	Composition (wt%)									
AISI type	C	Mn	Si	Cr	Ni	V	W	Mo	Ti	Co
W1	0.6–1.4	–	–	–	–	0.25	–	–	–	–
S1	0.5	–	0.75	1.5	–	0.2	2.5	–	–	–
O1	0.9	1.0	–	0.5	–	0.2	0.5	–	–	–
A2	1.0	0.7	–	5.25	–	0.2	–	1.1	–	–
A10	1.25–1.5	1.6–2.1	1.0–1.5	–	1.5–2.05	–	–	1.25–1.7	–	–
D2	1.5	0.5	–	12.0	–	0.2–0.9	–	0.8	–	–
H11	0.35	–	0.9	5.0	–	0.4	–	1.5	–	–
H13	0.35	–	1.0	5.25	–	1.0	–	1.3	–	–
T1	0.7	–	–	4.0	–	1.0	18.0	–	–	–
M1	0.S	–	–	4.0	–	1.1	1.5	8.5	–	–
M4	1.3	–	–	4.5	–	4.0	5.5	4.5	–	–
M42	1.1	–	–	3.75	–	1.15	1.5	9.5	–	8.0
L1	1.0	–	–	1.4	–	–	–	–	–	–
P5	0.1	–	–	2.25	–	–	–	–	–	–
AHT	1.0	–	–	3.0	–	0.25	1.05	1.1	1.0	–

Table 4 Nominal composition of standard wrought stainless steels (Vander Voort et al. 2004)

Type	UNS designation	Composition (wt%)							
		C	Mn	Si	Cr	Ni	S	P	Other
Austenitic type									
201	S20100	0.15	5.5–7.5	1.00	16.0–18.0	3.5–5.5	0.03	0.06	0.25 N
202	S20200	0.15	7.5–10.0	1.00	17.0–19.0	4.0–6.0	0.03	0.06	0.25 N
301	S30100	0.15	2.00	1.00	16.0–18.0	6.0–8.0	0.03	0.045	–
304	S30400	0.08	2.00	1.00	18.0–20.0	8.0–10.5	0.03	0.045	–
304L	S30403	0.03	2.00	1.00	18.0–20.0	8.0–12.0	0.03	0.045	–
310	S31000	0.25	2.00	1.50	24.0–26.0	19.0–22.0	0.03	0.045	–
316	S31600	0.08	2.00	1.00	16.0–18.0	10.0–14.0	0.03	0.045	2.0–3.0 Mo
316L	S31603	0.03	2.00	1.00	16.0–18.0	10.0–14.0	0.03	0.045	2.0–3.0 Mo
330	N08330	0.08	2.00	0.75–1.5	17.0–20.0	34.0–37.0	0.03	0.04	–
384	S38400	0.08	2.00	1.00	15.0–17.0	17.0–19.0	0.03	0.045	–

Ferritic type									
405	S40500	0.08	1.00	1.00	11.5–14.5	–	0.03	0.04	0.10–0.3 Al
430	S43000	0.12	1.00	1.00	16.0–18.0	–	0.03	0.04	–
434	S43400	0.12	1.00	1.00	16.0–18.0	–	0.03	0.04	0.75–1.2 Mo
442	S44200	0.20	1.00	1.00	18.0–23.0	–	0.03	0.04	–
Martensitic type									
403	S40300	0.15	1.00	0.50	11.5–13.0	–	0.03	0.04	–
410	S41000	0.15	1.00	1.00	11.5–13.5	–	0.03	0.04	–
416	S41600	0.15	1.25	1.00	12.0–14.0	–	0.15 min	0.06	0.60 Mo
420	S42000	0.15 min	1.00	1.00	12.0–14.0	–	0.03	0.04	–
440A	S44002	0.60–0.75	1.00	1.00	16.0–18.0	–	0.03	0.04	0.75 Mo
440C	S44004	0.95–1.20	1.00	1.00	16.0–18.0	–	0.03	0.04	0.75 Mo
Precipitation hardening type									
15-5 PH	S15500	0.07	1.00	1.00	14.4–15.5	3.5–5.5	0.03	0.04	2.5–4.5 Cu, 0.15–0.4 Nb
17-4 PH	S17400	0.07	1.00	1.00	15.5–17.5	3.0–5.0	0.03	0.04	3.0–5.0 Cu, 0.15–0.4 Nb
17-7 PH	S17700	0.09	1.00	1.00	16.0–18.0	6.5–7.75	0.04	0.04	0.75–1.5 Al

Table 5 Nominal composition of Alloy Casting Institute (ACI) heat-resistant casting alloys (Vander Voort et al. 2004; Nickel Development Institute, Canada 2009)

ACI designation	UNS number	ASTM specifications	Composition[a] (wt%)			
			C	Cr	Ni	Si (max)
HA	–	A217	0.20 max	8–10	–	1.00
HC	J92605	A 297, A 608	0.50 max	26–30	4 max	2.00
HD	J93005	A 297, A 608	0.50 max	26–30	4–7	2.00
HE	J93403	A 297, A 608	0.20–0.50	26–30	8–11	2.00
HF	J92603	A 297, A 608	0.20–0.40	19–23	9–12	2.00
HH	J93503	A 297, A 608, A 447	0.20–0.50	24–28	11–14	2.00
HI	J94003	A 297, A 567, A 608	0.20–0.50	26–30	14–18	2.00
HK	J94224	A 297, A 351, A 567, A 608	0.20–0.60	24–28	18–22	2.00
HL	J94604	A 297, A 608	0.20–0.60	28–32	18–22	2.00
HN	J94213	A 297, A 608	0.20–0.50	19–23	23–27	2.00
HP	–	A 297	0.35–0.75	24–28	33–37	2.00
HT	J94605	A 297, A 351, A 567, A 608	0.35–0.75	13–17	33–37	2.50
HU	–	A 297, A 608	0.35–0.75	17–21	37–41	2.50
HW	–	A 297, A 608	0.35–0.75	10–14	58–62	2.5
HX	–	A 297, A 608	0.35–0.75	15–19	64–68	2.50

[a]Bal. Fe for all compositions. Mn content: 0.35–0.65% for HA, 1% for HC, 1.5% for HD, and 2% for the other alloys. P and S contents: 0.04% (max) for all. Mo is intentionally added only to HA, which has 0.90 to 1.20% Mo; maximum for other alloys is set at 0.5% Mo. HH also contains 0.2% N (max)

is achieving the case of high strength and wear resistance that cannot be attained by the core metal. "Nitriding is a thermochemical treatment that is usually suitable for medium-carbon low-alloy steels (containing carbon in the range of 0.25–0.6%), although stainless steels and tool steels can also be taken through the nitriding process." Table 2 (Vander Voort 2004a) depicts composition of some of the more common case-hardening steels. Most steels that may contain an initial carbon content of 0.10–0.25% are subjected to carburizing or carbonitriding process to achieve desirable properties. Medium-carbon steels (containing 0.25–0.6% C) are occasionally carburized so that they may satisfy specific property requirements.

"**Tool steels** are high-alloy steels, and they are usually heat treated in different ways in order to achieve much higher hardness values than what most carbon or alloy steels can offer." Table 3 (Vander Voort 2004b) summarizes typical composition for various grades of tool steels. "The different tool steel grading methodologies include the most common water hardening (W-grade) or air hardening (A-grade, cold working), shock resisting (S-grade), D-type (cold working), oil hardening (O-grade), and hot working (H-grade)." "W-grade tool steels are basically high-carbon steels and usually water quenched. These steels can attain high hardness but are relatively brittle as compared to other tool steels. A-grade tool steels are

characterized by higher content of chromium (Cr) that provides better response to heat treatment. The A-grade tool steels can depict high machinability. Additionally, they also possess great wear resistance along with toughness. D-type tool steels combine the general characteristics of W-grade and A-grade. The D-grade tool steels contain a higher amount of carbon as compared to W-grade; however, they have the properties described above that are typical to the air hardening type (A-grade)." "Due to the higher chromium content, the D series tool steels may also be categorized as stainless, but their corrosion protection is pretty limited. The abrasion resistance properties are high for the D-type tool steels. The O-grade tool steels are general purpose oil hardening-type tool steels. They are characterized by good abrasion resistance and toughness for a comprehensive range of applications. S-grade tool steels are usually designed to counter shock at low or high temperatures. Their low carbon content is essential to achieve the required toughness. This grade of tool steels possesses high impact toughness but low abrasion resistance. Finally, the H-grade tool steels are suitable to cut materials at high temperatures. The H-grade tool steels are designed to have sustainable extra strength along with hardness at elevated temperatures. They are low in carbon content and moderately high in alloying additions."

"**Stainless steels** are multielement alloys containing at least 11% Cr along with other elements and have different grades like austenitic, ferritic, duplex, martensitic, and/or precipitation-hardenable grades." The nominal composition of typical standard wrought stainless steel grades has been given in Table 4 (Vander Voort et al. 2004). Stainless steels are commercially important due to their exceptional corrosion resistance properties. Stainless steel is normally 200 times more corrosion resilient than any form of mild steel or low-carbon steel grades. Stainless steels (SS) are divided into five groups as mentioned above. "Austenitic SS is the most weldable grade among all stainless steels and can be loosely divided into three subgroups: the more common chromium-nickel (300 series), the manganese-chromium-nickel nitrogen based (200 series), and the more specialty alloys." All these grades are generally nonmagnetic and are non-heat-treatable. "The ferritic SS are composed of trace amounts of Ni, 12–17% Cr, and less than 0.1% C and have other alloying additions like Mo, Al, Ti, etc." These alloy grades have good ductility and formability; however, their strength at high temperatures is poor as compared to similar austenitic grades. They are magnetic but cannot be heat treated. They are strengthened using cold working processes. "Martensitic SS contains at least 11–17% Cr, less than 0.4% Ni, and up to 1.2% of C. Such carbon contents of these steels make them hardenable but affect their forming and welding characteristics." In order to achieve beneficial properties and prevent any cracking, compulsory preheating and post-welding heat treatment are necessary. "Martensitic SS, e.g., 403, 410, 420, etc., are magnetic and heat treatable as well." "Duplex SS typically contain, along with Fe, 5% Ni and 22–25% Cr, along with traces of Mo and N." These steels have higher yield strength and better stress corrosion cracking resistance against chloride as compared to austenitic stainless steels. Finally, "precipitation hardening SS are Cr-Ni grades that also contain other alloying elements, such as Al, Cu, or Ti."

These alloys are hardened by a combination heat treatment of solution and aging. After aging, these grades could be either austenitic or martensitic.

The Alloy Casting Institute (ACI) has given nominal compositions for cast stainless steels, in keeping with the designations as above, in Table 5 (Vander Voort et al. 2004; Nickel Development Institute, Canada 2009) for the **heat-resistant SS grades**. These heat-resistant grade steels have relatively higher carbon contents for strengthening as compared to steel castings in corrosion-resistant grades. "The heat-resistant casting alloys comprise of those compositions that contain more than 12% Cr which are capable of sustaining adequately when used for applications at temperatures more than 650 °C." "Heat-resistant compositions, as a group, are higher in alloy content as compared to the corrosion-resistant-type steels." These alloys are composed primarily of iron (Fe), chromium (Cr), and nickel (Ni), along with a small percentage of other elements. Castings developed using these alloys are expected to satisfy two basic requirements: (i) a better surface film stability (in terms of oxidation and corrosion resistance) under different atmospheric conditions and also at the temperatures to which they are exposed and (ii) they should have sufficient mechanical ductility and strength in order to attain high-temperature usage requirements. "The ACI designations 'H' indicates alloys used in conditions where the metal temperature may exceed 650 °C." "The second alphabet denotes nominal nickel content, increasing from alphabet A to alphabet X. These alloys can be classified mainly in three groups (Vander Voort et al. 2004): i) Cr-Fe alloys: HA, HC, HD; ii) Cr-Ni-Fe alloys: HE, HF, HH, HI, HK, HL; and iii) Ni-Cr-Fe alloys: HN, HP, HT, HU, HW, HX. The Cr-Fe alloy group is composed of alloys having predominantly Cr with maximum 30% Cr and 7% Ni." These alloys are ferritic and possess relatively low high-temperature strength. The second type of Cr-Ni-Fe alloys have better high-temperature strength, hot and cold ductility, and also resistance to oxidizing and reducing conditions. They contain 8–22% Ni and 18–32% Cr and may have either a partial or completely austenitic microstructure. The third type, Ni-Cr-Fe alloys, is fully austenitic and contains 25–70% Ni and 10–26% Cr. These alloys can be satisfactorily used up to the temperature of 1150 °C, as there is no brittle-phase formation. They have good weldability along with high machinability.

Nonferrous Alloy Systems

"Nonferrous alloy systems could be any metal matrix system other than iron-based alloys, such as copper alloys, zinc alloys, aluminum alloys, magnesium alloys, titanium alloys, nickel alloys, lead alloys, etc." However, in this chapter, three more common alloy systems, i.e., based on aluminum, copper, and titanium, will be discussed. A**luminum alloys** primarily cover a variety of range of chemical compositions and product forms that could be fabricated by numerous metalworking processes and standard casting routes. Pure aluminum (Al) and Al-based alloys could be manufactured in various standard forms, e.g., sheet, plate, rod, bar, tube, wire, pipe, foil, etc. These alloys can also be shaped in numerous engineering forms for specific applications by various metal-forming processes, like extrusion, forging,

stamping, machining, and powder metallurgy. Aluminum alloys may encompass more than 300 commonly documented alloy compositions and several additional variations optimized as a result of supplier-consumer chain relationships. Usually, the commercial Al alloys contain some Fe and Si as well as two or more other elements purposefully added to achieve improved characteristics. The major groups of commercial Al alloys have been classified in Table 6 (Warmuzek 2004; Hatch 1984). The primary objectives of Al alloying are strengthening using components like Si, Cu, Mn, Zn, Mg, Mg + Si, etc. in combination with strain hardening, heat treatment, or both.

"Aluminum alloys are principally divided in two major categories: **castings** and **wrought compositions**." Further, subdivision of these alloys is based on "**heat-treatable** and **non-heat-treatable alloys**." "Cast and wrought aluminum alloy-related nomenclature as is currently used globally in this trade was developed by the Aluminum Association System in the USA." In **wrought aluminum alloys,** again "a four-digit system has been commonly used to segregate the following groups of wrought compositions":

- 1*xxx* Controlled unalloyed (pure) composition.
- 2*xxx* Alloys comprised of Cu as the main alloying constituent.
- 3*xxx* Alloys wherein Mn is the main alloying constituent.
- 4*xxx* Alloys having Si as the major alloying element.
- 5*xxx* Alloys wherein Mg is the major alloying constituent.
- 6*xxx* Alloys where Mg and Si are the primary alloy additions.
- 7*xxx* Alloys composed of Zn as the major alloying element.
- 8*xxx* Alloys which also contain Sn and some Li – they are characterized as miscellaneous compositions.

In the same manner, "**casting compositions** are described in this trade using a three-digit system followed by a decimal value. The decimal (i.e., 0.0) in all such cases concerns with the limits of cast alloying. Decimals 0.1 and 0.2 represent those ingot compositions, which are obtained after melting and processing. For casting compositions, the alloys are grouped in the following manner":

- 1*xxx* represents controlled but unalloyed, i.e., pure compositions.
- 2*xxx* alloys have Cu as the primary alloying element but may include additional alloying elements that may be specified.
- 3*xxx* are those alloys wherein Si is the primary alloying element; however, other elements, e.g., Cu and Mg, may also be specified.
- 4*xxx* are those alloys that have Si as the other primary element.
- 5*xxx* are those alloys in which Mg is the major alloying constituent.
- 6*xxx* designation is currently unused.
- 7*xxx* designation denotes alloys where Zn is the main alloying addition, but other constituents, e.g., Cu and Mg, may be specified.
- 8*xxx denotes* alloys in which Sn is the primary alloying constituent.

Table 6 Nominal alloy composition range for various groupings of commercial aluminum alloys (Warmuzek 2004; Hatch 1984)

Alloy group	Composition[a] (wt%)										
	Mg	Si	Ti	Cr	Mn	Fe	Ni	Cu	Zn	Zr	Other
WROUGHT ALLOYS											
lxxx (Al>99.00%)	0.006–0.25	0.006–0.7	0.002–0.06	0.01–0.03	0.002–0.05	0.006–0.6	–	0.006–0.35	0.006–0.05	–	–
2xxx (Cu)	0.02–0.08	0.10–1.3	0.02–0.3	0.05–0.2	0.05–1.3	0.12–1.3	0.05–2.3	0.8–6.8	0.10–0.8	0.05–0.5	–
3xxx (Mn)	0.05–1.3	0.3–1.8	0.05–0.1	0.05–0.4	0.05–1.8	0.1–1.0	0.05	0.05–0.5	0.05–1.0	0.1–0.5	–
4xxx (Si)	0.05–2.0	0.8–13.5	0.04–0.30	0.05–0.25	0.03–1.5	0.20–1.0	0.15–1.3	0.05–1.5	0.05–0.25	–	–
5xxx (Mg)	0.2–5.6	0.08–0.7	0.05–0.2	0.05–0.35	0.03–1.4	0.10–0.7	0.03–0.05	0.03–0.35	0.05–2.8	–	–
6xxx (Mg+Si)	0.05–1.5	0.2–1.8	0.08–0.2	0.03–0.035	0.03–1.0	0.08–1.0	0.2	0.10–1.2	0.05–2.4	0.05–0.20	–
7xxx (Zn)	0.10–3.7	0.10–0.50	0.03–0.15	0.04–0.35	0.02–1.5	0.10–0.70	0.10	0.05–2.6	0.8–8.7	0.05–0.18	–
8xxx (other element)	0.02–1.4	0.10–1.0	0.08–0.2	0.01–0.2	0.02–1.0	0.10–2.0	0.2–1.3	0.03–2.2	0.03–1.8	0.04–0.16	Li, B, Sn, Ga
CAST ALLOYS											
lxxx (Al>99.00%)	–	0.10–0.15	0.15–0.35	–	–	0.25–0.8	–	0.05–0.10	0.05	–	–
2xxx (Cu)	0.03–2.3	0.05–3.5	.06–0.35	0.15–0.4	0.05–0.7	0.04–1.5	0.03–2.3	3.5–10.7	0.05–2.5	–	–
3xxx (Si+Cu/Mg)	0.03–1.5	4.5–23.0	0.04–0.25	0.05–0.35	0.03–0.8	0.06–1.5	0.10–3.0	0.03–5.0	0.03–4.5	–	–
4xxx (Si)	0.05–0.10	3.3–1.3	0.2–0.25	0.25	0.05–0.5	0.12–1.3	0.05–0.5	0.05–1.0	0.05–0.5	–	–
5xxx (Mg)	1.4–10.6	0.10–2.2	0.10–0.25	0.25	0.05–0.6	0.10–1.3	0.05–0.4	0.05–0.30	0.05–0.20	–	–
7xxx (Zn)	0.2–2.4	0.1–0.3	0.1–0.25	0.06–0.6	0.05–0.6	0.10–1.4	0.15	0.1–1.0	2.0–7.8	–	–
8xxx (Sn)	0.1–0.9	0.4–6.5	0.2	–	0.1–0.5	0.5–0.7	0.3–1.5	0.7–4.0	Sn, 5.5–7.0	–	–

[a]Balance: aluminum

Heat-treatable Al alloys are those whose strength can be increased by a controlled cyclic heating and cooling. Al alloys that are usually denoted as above in the 2*xxx*, 6*xxx*, and 7*xxx* series are solution heat treatable and are strengthened by such cyclic quenching. Additional strengthening may be achieved by controlled deformation at room temperature. Another example of heat treatment is fully annealed O-temper Al alloy 2024. It has yield strength of approximately 186 MPa. After such cyclic treatment, followed by natural aging (T-3 temper), the yield strength can get enhanced up to 483 MPa.

There are also some **non-heat-treatable Al alloy groups** which are hardenable only by cold working and not using any heat treatment methods. The preliminary strength of such alloys, usually in the 1*xxx*, 3*xxx*, 4*xxx*, and 5*xxx* series, come from the alloying-related hardening effects alone. Their strength can be increased further by applying cold working denoted by H-temper. Cold working can significantly enhance strength properties of the aluminum alloys. As an example, the ultimate tensile strength (UTS) in Al alloy 3004 could be increased to up to 283 MPa after H-38 temper as compared to 179 MPa in its O-temper condition.

Copper and copper alloys have performed an important role in the human life and human evolution since ages. The combination of exceptional thermal and electrical conductivity, strength, workability, corrosion resistance, and their abundance makes these copper-based alloys essential to all societies and industries all through the ages. "Copper alloys have conventionally been classified by composition and as wrought or cast in the groups depicted in Table 7 (Caron et al. 2004; Joseph 1999). These alloys are listed by the UNS designation which is managed by the Copper-Development Association, USA." Chosen alloys for each group are classified in the table. The alloys designated as coppers may contain a minimum of 99.3% copper. These alloys possess highest thermal and electrical conductivity. Impurities, such as phosphorous (P), tin (Sn), selenium (Se), tellurium (Te), and arsenic (As), are detrimental to their electrical conductivity and recrystallization temperature. However, if intentionally alloyed, then these additions can improve other specific desirable characteristics. High-copper alloys contain about 96 to 99.3% copper in the wrought products. In the case of cast alloys, copper content is more than 94%. The major alloying elements are cadmium (Cd), beryllium (Be), and chromium (Cr). Copper-zinc-based alloys (i.e., different grades of brasses) contain Zn as the other major alloying constituent. Such alloys in wrought form "are further categorized as copper-zinc alloys, copper-zinc-lead (leaded brass), and copper-zinc-tin (tin brasses)." Cast brasses have four subdivisions: "(i) copper-zinc-tin- and copper-zinc-tin-lead-based alloys (they are colloquially denoted as red and leaded red, semi-red and leaded semi-red, and leaded yellow brass), (ii) copper-manganese-zinc and copper-manganese-zinc-lead (high-strength and leaded high-strength brass), (iii) copper-silicon (i.e., silicon-based brasses and bronzes), and (iv) copper-bismuth and copper-bismuth-selenium (i.e., copper-bismuth- and copper-bismuth-selenium-based alloys)." Bronzes may include copper-based alloys that do not contain Zn or Ni as the principal alloying constituent. The four subgroups of wrought alloys are "(i) copper-tin-phosphorous (phosphor bronze), (ii) copper-tin-phosphorous-lead (leaded phosphor bronze), (iii) copper-aluminum (i.e., aluminum

Table 7 Nominal alloy composition for typical copper and copper alloys (Caron et al. 2004; Joseph 1999)

UNS No.	Name	Composition (wt%)
Wrought copper		
C10100	Oxygen free electronic copper (OFE)	99.99 (min) Cu
CI1000	Electrolytic tough pitch copper (ETP)	99.90 (min) Cu
C12500	Fire-refined tough pitch copper (FRTP)	99.88 (min) Cu
Wrought high copper alloys		
C17200	Beryllium copper	Bal Cu, 1.90 Be, 0.40 Co
C18200	Chromium copper	Bal Cu, 0.9 Cr
C18700	Leaded copper	Bal Cu, 1 Pb, 0.05 P
Wrought brasses		
C26000	Cartridge brass, 70%	70 Cu, 30 Zn
C26800	Yellow brass, 66%	66 Cu, 34 Zn
C28000	Muntz metal, 60%	60 Cu, 40 Zn
C36000	Free-cutting brass	62 Cu, 3 Pb, 35 Zn
C44300	Admiralty, arsenical	71.5 Cu. 27.5 Zn, 1 Sn (0.04 As)
C46400	Uninhibited naval brass	61 Cu, 38 Zn, 1 Sn
Wrought bronzes		
C51000	Phosphor bronze, 5% A	Bal Cu, 5.0 Sn, 0.2 P
C63000	Aluminum bronze, 10%	82.2 Cu, 10A1, 3 Fe, 4.8 Ni
C64700	Silicon-nickel bronze	Bal Cu, 1.9 Ni, 0.6 Si
C68700	Arsenical aluminum brass	77.5 Cu, 20.3 Zn, 2.2 Al, 0.04 As
Cast high-copper alloy		
C81500	Chromium copper	98 (min) Cu, 1.0 Cr
Cast brasses, bronzes, and Ni silver		
C83600	Leaded red brass	85 Cu, 5 Sn, 5 Zn, 5 Pb
C86200	Manganese bronze	64 Cu, 26 Zn, 4 Al, 3 Fe, 3 Mn
C90300	Tin bronze	88 Cu, 8 Sn, 4 Zn
C92600	Leaded tin bronze	87 Cu. 10 Sn, 2 Zn, 1 Pb
C95300	Aluminum bronze	89 Cu, 10 Al, 1 Fe
C95500	Nickel aluminum bronze	81 Cu, 11 Al, 4 fe, 4 Ni
C95600	Silicon-aluminum bronze	91 Cu, 7 Al, 2 Si
C97800	Nickel silver	66 Cu, 25 Ni, 5 Sn, 2 Zn, 2 Pb

bronze), and (iv) copper-silicon (i.e., silicon bronze). The cast bronzes are colloquially known as copper-tin (i.e., tin bronze), copper-tin-lead (i.e., leaded and high-leaded tin bronze), copper-tin-nickel (i.e., nickel-tin bronze), and copper-aluminum-iron and

Table 8 Nominal alloy composition for typical commercial titanium alloys (Donachie 2000; Stefanescu and Ruxanda 2004)

Designation	Composition (wt%)				
	Al	Sn	Mo	Zr	Others
α alloys					
Ti-0.3Mo-0.8Ni	–	–	0.3	–	0.8Ni
Ti-5Al-2.5Sn	5	2.5	–	–	–
Ti-6Al-2Sn-4Zr-2Mo	6	2	2	4	0.08Si
Ti-6Al-2Nb-1Ta-0.8Mo	6	–	1	–	2Nb, 1Ta
Ti-2.25Al-11Sn-5Zr-1Mo	2.25	11	1	5	0.2Si
α-β alloys					
Ti-6Al-4V	6	–	–	–	4V
Ti-6Al-6V-2Sn	6	2	–	–	0.75Cu, 6V
Ti-7Al-4Mo	7	–	4	–	–
Ti-6Al-2Sn-4Zr-6Mo	6	2	6	4	–
Ti-6Al-2Sn-2Zr-2Mo-2Cr	5.7	2	2	2	2Cr, 0.25Si
Ti-4Al-4Mo-2Sn-0.5Si	4	2	4	–	0.5Si
β alloys					
Ti-10V-2Fe-3Al	3	–	–	–	10V
Ti-13V-11Cr-3Al	3	–	–	–	11Cr, 13V
Ti-3Al-8V-6Cr-4Mo-4Zr	3	–	4	4	6Cr, 8V
Ti-11.5Mo-6Zr-4.5Sn	–	4.5	11.5	6	–
Ti-15V-3Cr-3Al-3Sn	3	3	–	–	15V, 3Cr

copper-aluminum-iron-nickel (i.e., aluminum bronze)." Copper-nickels are obtainable as wrought and cast alloy forms. Copper-nickel-zinc-based alloys (in such wrought or cast format) are colloquially called nickel silvers. Other copper alloys comprise groups of specialty alloys, copper leads, and brazing alloys. Oxygen and hydrogen cause interference with the conductivity of copper and/or copper alloys. However, very small and controlled amounts of oxygen are in fact advantageous for the conductivity improvement since oxygen can combine with and remove from solution other impurities, such as iron that is far more detrimental.

A brief summary of selected commercial **titanium-based alloys** is presented in Table 8 (Donachie 2000; Stefanescu and Ruxanda 2004). Titanium alloys are a mixture of titanium and other elements of the periodic table. They show extremely high toughness and tensile strength (even at elevated temperature conditions) as compared to the above-discussed aluminum and copper alloys in the nonferrous category. Titanium alloys are light, display extraordinary corrosion resistance, and can withstand extreme temperatures. They are commonly classified into three major alloying phase groups, "designated as α, α + β, and β. The commercially used fully α alloys have several grades of pure Ti or ternary Ti-Al-Sn. In the α+β alloys, additional elements are included to stabilize and strengthen the α phase together with 4–6% of β-phase stabilizing additions. These alloys may contain 10–50% β-phase at ambient temperature. The α+β alloys have the maximum commercial

viability as per the current world scenario. Ti-6Al-4V alloy is the single composition which has up to 50% 'commercial market share' among all α+β Ti-based alloys, both in Europe and the USA." Consequently, wrought Ti-6Al-4 V alloy is considered the standard alloy when shortlisting any titanium-based alloy for any specialist requirement. The high-temperature stability of Ti-6Al-4 V alloy is only up to 400 °C. Therefore, for more elevated-temperature applications, Ti-6Al-2Sn-4Zr-2Mo (+Si)-based alloys are preferred. During the last few decades, numerous other titanium alloys have also been developed. However, none of them could match the price and performance of Ti-6Al-4 V alloy. Other Ti alloys for high-performance applications include Ti-10 V-2Fe-3Al, Ti-6Al-6 V-2Sn, and Ti-13 V-11Cr-3Al. The other titanium alloys mentioned in Table 8 are presently in various stages of development.

Standardization in Measurements: Pathway to CRMs/SRMs

In the world of metrology, the result of any measurement for an unknown value is obtainable only after it can be tallied with a known value. A standard is a benchmark in terms of dimensions or performance of known quantity in terms of measurement units that can be compared. So a standard is always used for comparison or as a reference point for the evaluation of an unknown thing. Global quality control lexicon mandates use of acceptable standards for all measurements. According to ISO 9001:2015 (ISO 9001 2015) guidelines, "Quality assurance is based on a methodology that ensures that a product, process, or service will fulfill the expected quality requirements," and quality control is "a way to follow the operational flow chart required to meet due Quality requirements." Certified reference materials (CRMs) ensure two issues to be taken care (Pant 2018). The first is preparing these materials and second is their characterization. Hence, CRMs need to be produced or at least supervised carefully by specialized agencies, such as different national metrology institutes (NMIs). In the Indian context, a metal or alloy-based CRM can be prepared by CSIR-National Metallurgical Laboratory (CSIR-NML), Jamshedpur, whereas a cement-specific CRM can be produced by the cement research institute in association with CSIR-NPL, New Delhi. The National Council for Cement and Building Materials (NCCBM), Ballabgarh, has an association with CSIR-NPL for the development of CRMs in the field of cement and allied materials, which are marketed in India and abroad with the trade name Bharatiya Nirdeshak Dravyas (BNDs). The International Organization for Standardization (ISO) has laid down specifications through ISO Guide 35:2017 (ISO Guide 35 2017) for such preparation and characterization activities. As the custodian of the national standards for physical measurements, CSIR-NPL, New Delhi, characterizes and shares all data for reference materials. It issues characterization certificates, listing their specific measurand properties (as mean value) along with related uncertainty data.

The world over, in all capitalist economies, CRMs have multiple valuable utilities. They are routinely used globally for engineering and industrial process

quality control by all quality control and quality conscious organizations. They are useful for the calibration of instruments and methods and for round-robin intercomparison between different similar instruments in participant laboratories. This activity ensures compatibility between the measurement laboratories and leads to metrological traceability of measurements to national standards. The developed economies/NMIs, such as NIST-USA, BAM-Germany, NMIJ-Japan, NPL-UK/LGC Standards, etc., provide SI traceable CRMs to the global market. NIST-USA has developed more than 1300 standard reference materials (SRMs) in the areas of industrial materials, for environmental monitoring and for health safety control (Montgomery and Crivellone 2021). Bundesanstalt für Materialforschung und–prüfung (BAM), Germany, produces more than 400 CRMs for analysis of chemical composition and control of industrial and engineering products (Recknagel 2021). The European Union as a whole has around 800 CRMs in diverse areas of food hygiene and safety analysis, environmental impact control, engineering, medical applications, etc. (Certified Reference Materials 2022). NMIJ-Japan has developed different CRMs for process control of their industrial materials, high-purity inorganics, polymer materials, different gases, etc. (NMIJ CRM Catalog 2021–2022 2021; NMIJ Certified Reference Materials (NMIJ CRMs) n.d.). On the other hand, the developing economies/NMIs, such as CSIR-NPL (India), have developed till date about 120 CRMs (with the tradename BND®) in various fields, such as chemicals, cement, petroleum products, hardness, ores, minerals, precious metals, etc. (https://www.nplindia.org/wp-content/uploads/2021/11/Available-Bharatiya-Nirdeshak-Dravya-list-251220.pdf; Singh et al. n.d.).

Most of the time, there is little overlap in the category and types of CRMs developed by the NMIs of different countries. There is no upper limit to the number of different metal and alloy-based reference materials that can be produced and marketed. Essentially in the practical sense, it all depends on what is the marketability of any reference material that is produced. If the market demands it, then it can be produced and sold. Else, in our opinion, it is only an academic exercise. In general, once any patent-protected proprietary alloys are mass manufactured by several other manufacturers, within the same country or globally, then for the sake of standardization of quality, reliability, and benchmarking, it may be necessary to produce a related reference material. Again, CRGO steel, Metglas, different military-grade alloys, or different shape memory alloys can be taken as an example in this context. In any country, as and when manufacturing picks up in any industrial segment, related reference material preparation and production also pick up, and they need to compete globally in terms of quality and value for money. In the context of many developing countries, so long as their trading community has the political power, it is only material trading that happens, rather than local manufacturing. As and when the local educated class evolves and learns to also enjoy the political power in such developing country, the local manufacturing picks up pace. Only then can the related reference material preparation and commercialization pick up pace in general.

"NIST"-Traceable Ferrous and Nonferrous SRMs/CRMs

"The National Institute of Standards and Technology (NIST) certifies and provides more than 1300 standard reference materials (SRMs) or CRMs through their precise and harmonious measurements (Montgomery and Crivellone 2021)." These CRMs are accompanied with scientific data on well-characterized properties and compositions. Their CRMs are globally used for instrument calibrations in quality monitoring programs. Industry, academia, and government agencies make use of NIST CRMs to expedite commerce and trade and also to accelerate R&D activities. In the USA, NIST CRMs also facilitate the mechanism for supporting measurement traceability. Each NIST CRM/SRM comes with a "certificate of analysis" and a "material safety data sheet (MSDS)." The NIST CRMs are commercial reference materials with well-documented metrological traceability relationship to different existing NIST standards. This traceability relationship is recognized through norms and protocols maintained by NIST.

Table 9 (Montgomery and Crivellone 2021) illustrates typical ferrous-based CRMs/SRMs supplied by the NIST-USA. This table comprises a short description of the CRMs in different categories, such as low-alloy steels, plain carbon steels, stainless steels, high-alloy steels, and different grades of tool steel. The available form of CRMs, i.e., chips (small metal turnings), rod, disk, or any other form, is mentioned. The SRM/CRM number is also mentioned in the table. These CRMs are used for the calibration of optical emission and X-ray fluorescence (XRF) spectrometers and different other possible analytical instruments for chemical analysis. They have used different steel alloys for getting a wide range in values of elemental content. Similarly, Table 10 (Montgomery and Crivellone 2021) illustrates selected nonferrous (Al, Cu, and Ti alloy)-based CRMs/SRMs supplied by the NIST-USA. They include titanium-based, copper-based, and aluminum-based alloys among others. The available form of the CRM (i.e., chips, disk, or block form) is also mentioned in the table along with the CRM number. The aluminum-based CRMs are usable in chemical and/or instrumental methods for analysis of aluminum-based alloys. Similarly, copper-based and titanium-based CRMs are used for the analysis of copper alloys and titanium alloys, respectively, by chemical and instrumental methods. The certificate of analysis usually includes the following details: i) certified mass fraction values of elements in the ferrous/nonferrous CRM, ii) date of expiry of the certificate, iii) instructions for the maintenance of the CRM certification, iv) instructions for handling storage and use of the CRM, v) method/s employed for the preparation and analysis of the CRM, etc. The material safety data sheet (MSDS) of each CRM usually provides the following information: a) substance and source identification, b) its extent of hazardousness, c) composition and information of hazardous constituents, d) recommended first aid measures, e) possible firefighting methodologies, f) handling of accidental release situations, g) all storage and handling issues, h) requirements on exposure control and personnel protection (PPE), i) physical and chemical properties of the CRM, and j) stability and reactivity of the CRM in different ambients.

Table 9 Typical ferrous CRMs available with NIST-USA (Montgomery and Crivellone 2021)

Category and form	NIST CRM no. and description
Plain carbon steels (chip form)	**8k**: Bessemer steel, 0.1% C; **12h**: basic open-hearth steel, 0.4%C; **13g**: 0.6% C steel; **14g**: carbon steel (AISI 1078); **19h**: basic electric steel, 0.2%C; **20g**: AISI 1045 steel; **178**: 0.4C basic oxygen furnace steel; **368**: carbon steel (AISI 1211)
Low-alloy steel (chip form)	**30f**: Cr-V steel (SAE 6150); **32e**: SAE 3140; **33e**: nickel steel; **72g**: low-alloy steel (AISI 4130); **100b**: Mn steel; **125b**: LA steel, high Si; **129c**: LA steel, high sulfur; **139b**: Cr-Ni-Mo steel; **155**: Cr-W steel; **163**: Cr steel; **291**: Cr-Mo steel (ASTM A-213); **2171**: LA steel (HSLA 100)
Low-alloy steels (disk form)	**1134**: low-alloy high-Si steel; **1135**: LA steel, high Si; **1224**: LA steel, carbon (AISI 1078); **1225**: LA steel (AISI 4130); **1226**: LA steel; **1228**: LA steel, 0.1% C; **1264a**: LA steel, high C; **1265a**: electrolytic Fe; **1269**: line pipe (AISI 1526 modified); **1271**: LA steel (HSLA-100); **1286**: low-alloy steel (HY 80); **1761a**: low-alloy steel; **1762a**: low-alloy steel; **1763b:** low-alloy steel; **1764a**: low-alloy steel; **1765**: low-alloy steel
High-alloy steels (chip form)	**126c**: high-Ni steel; **344**: 15 Cr-7 Ni steel; **345b**: Fe-Cr-Ni alloy UNS J92180; **346a**: valve steel; **862**: high-temperature alloy L605; **868**: high-temperature alloy (Fe-Ni-Co)
High-temperature alloys (disk form)	**1230**: high-temperature alloy A286; **1246**: Incoloy 800; **1247**: Ni-Fe-Cr alloy UNS N08825; **1250**: high-temperature alloy Fe-Ni-Co; **C2400**: Fe-Cr-Ni alloy UNS J92180
Stainless steel (disk form)	**C1151a**: SS, 23Cr-7Ni; **C1152a**: SS, 18Cr-11Ni; **C1153a**: SS, 17Cr-9Ni; **C1154a**: SS, 19Cr-13Ni; **1155a**: SS, Cr 18 Ni 12 Mo 2 (AISI 316); **1171**: SS, Cr 17-Ni 11-Ti 0.3 (AISI 321); **1219**: SS, Cr16-Ni2 (AISI 431); **1223**: Cr Steel; **1295**: SS (SAE 405); **1297**: SS (SAE 201)
Tool steels (chip form)	**50c**: W-Cr-V steel; **132b**: tool steel (AISI M2); **134a**: Mo-W-Cr-V steel
Specialty steels (disk form)	**1157**: tool steel (AISI M2); **1158**: high-Ni steel; **1772**: tool steel (S-7)

"BAM" Traceable Ferrous and Nonferrous CRMs

"The Federal Institute for Materials Research and Testing (German: Bundesanstalt für Materialforschung und–Prüfung or BAM) is a scientific and technical federal organization in Berlin, Germany. It is under their Federal Ministry for Economic Affairs and Energy. It performs the duties of testing, conducting research, and advising their government for protecting people, the environment, and their material goods." The BAM provides its clients around the world with high-quality reference materials targeted to their needs. Globally BAM is one of the oldest certified reference materials' producers. Started in 1912 with a "normal steel" for the determination of carbon, their development of CRMs has increased progressively. Since 2016, Deutsche Akkreditierungsstelle GmbH (DakkS) has accredited BAM as a producer of certified reference materials within ISO 17034:2016 (ISO 17034 2016).

Table 10 Typical nonferrous CRMs available with NIST-USA (Montgomery and Crivellone 2021)

Category and form	NIST CRM no. and description
Aluminum base alloys (chip form)	**87a:** Si-Al alloy; **853a**: Al alloy 3004; **854a**: Al alloy 5182; **855a**: Al casting alloy 356; **856a**: Al casting alloy 380; **858**: Al alloy 6011
Aluminum base alloys (disk form)	**1240c**: Al alloy 3004; **1241c**: Al alloy 5182; **1255b**: Al alloy 356; **1256b**: Al alloy 380; **1258-I**: Al alloy 6011 (modified); **1259**: Al alloy 7075
Copper base alloys (chip form)	**158a**: Si bronze; **458:** Be-Cu (17510); **459**: Be-Cu (17200); **460**: Be-Cu alloy; **871**: phosphor bronze (CDA521); **872**: phosphor bronze (CDA 544); **874**: cupronickel, 10% (CDA 706) "high purity"; **875**: cupronickel, 10% (CDA 706) "doped"
Copper base alloys (disk form)	**1107**: naval brass UNS 46400; **1110**: red brass B; **1111**: red brass standard; **1112**: gliding metal; **1113**: gliding metal; **1114**: gliding metal; **1115**: commercial bronze standard; **1124**: free cutting brass (UNS C36000); **1276a**: cupronickel (CDA 715)
Copper base alloys (block form)	**C1115:** commercial bronze A; **C1117**: commercial bronze C; **C1251a**: phosphorus deoxidized copper-Cu VIII; **C1252a**: phosphorus deoxidized copper-Cu IX; **C1253a**: phosphorus deoxidized copper-Cu X
Titanium base alloys (chip form)	**173c**: Ti alloy UNS R564000; **647**: Ti alloy, Al-Mo-Sn-Zr; **648**: Ti base alloy 5Al-2Sn-2Zr-4Cr-4Mo; **649**: Ti base alloy (15V-3Al-3Cr-3Sn); **2431**: Ti base alloy (6Al-2Sn-4Zr-6Mo); **2432**: Ti base alloy (10V-2Fe-3Al); **2433**: Ti base alloy (8Al-1Mo-1V); **2452**: hydrogen in Ti alloy
Titanium base alloys (disk form)	**641**: spectroscopic Ti base standard Ti alloy, 8Mn (A); **643**: spectroscopic Ti base standard Ti alloy, 8Mn (C); **654b**: Ti alloy, Al-V; **1128**: Ti base alloy (15V-3Al-3Cr-3Sn)

Their scope of accreditation comprises CRMs in numerous categories, i.e., nonferrous metals and alloys, glass and ceramics, soils, lubricants, fuel, solutions of ethanol/water, stable isotopes in aqueous solutions, packaging of food, etc. The BAM-Webshop currently offers online access to more than 400 CRMs (EURONORM CRMs) in various diverse areas, such as iron and steel, polymers, environment, food, etc. (https://www.webshop.bam.de/default.php?cPath=2321_3161&sort=1a&page=1&language=en). If the matrix and analyte levels of a CRM compare closely with those of an unknown sample, the analyst is capable to assure that the measurements are conducted adequately up to the required level of precision. "EURONORM-certified CRMs are prepared under the umbrellas of the European Committee for Iron and Steel Standardization (ECISS) in collaboration between the different national producing organizations including BAM." The following types of CRMs are presently available as EURONORM CRM: unalloyed steels {0}, alloyed steels {1}, highly alloyed steels {2}, special alloys {3}, cast iron {4}, ferro-alloys {5}, ores {6}, ceramics {7}, and slags {8}. Their above system of sample numbering permits an easy idea about the type of material. The "first digit of the sample number designates the type of material (0 represents unalloyed steel, 1 represents low-alloyed steel, 2 represents highly alloyed steel, 3 represents special alloys, etc.).

Table 11 Typical ferrous CRMs supplied by BAM-Germany (Recknagel 2021)

Category and form	BAM CRM no. and description
Unalloyed steels (chip form)	**D 030-4**; **D 031-3**; **D 032-2**; **D 035-2**; **D 036-1**; **D 042-1**; **D 077-3**; **D 079-2**; **D 082-1**; **D 083-1**; **D-083-2**
Low-alloy steels (chip form)	**D 126-1**; **D-128-1**; **D-129-3**; **D 130-1**; **D 179-2**; **D 180-1**; **D 181-1**; **D 182-1; D 183-1**; **D 187-1**; **D 187-2**; **D 191-2**; **D 192-1**; **D 193-1**; **D 194-1**; **D-194-2**
High-alloy steels (chip form)	**D 226-1**; **D 227-1**; **D 231-2**; **D 235-1**; **D 237-1**; **D 271-1**; **D 278-1**; **D 283-1**; **D 284-2**; **D 284-3**; **D 286-1**; **D 288-1**; **D 289-1**; **D 290-1**; **D 291-1**; **D 294-1**; **D 297-1**; **D 299-1**
High-alloy steels (disk form)	**D 271-1**; **D 284-3**; **D 288-1**; **D 289-1**; **D 290-1**; **D 291-1**; **D 294-1**; **D 297-1**; **D 299-1**
Special alloys (chip form)	**D 326-1**; **D 327-2**; **D 328-1**

Table 12 Typical nonferrous CRMs supplied by BAM-Germany (Recknagel 2021)

Category and form	BAM CRM no. and description
Aluminum base alloys (chip form)	**201**: GAlSi12; **300**: AlMg3; **301**: Al99.8; **BAM-M319**: AlMgSc
Aluminum base alloys (disk form)	**ERM-EB307a**: AlMg4.5Mn; **BAM-M308a**: AlZnMgCu1.5; **BAM-310**: Al99.85Mg1; **BAM-311**: AlCuMg2; **ERM-EB312a**: AlMgSi0.5; **BAM-M313a**: AlMg3; **ERM-EB314a**: AlSi11Cu2Fe; **ERM-EB315a**: AlSi9Cu3; **ERM-EB316**: AlSi12; **ERM-EB317**: AlZn6CuMgZr; **BAM-M318**: AlSi1.2Mg0.4; **BAM-M320**: AlMgSc; **BAM-M321**: AlCu4Mg1
Copper base alloys (chip form)	**223**: CuZn39Pb2; **224**: CuZn40MnPb; **227**: Rg7; **228**: Rg10; **BAM-229**: CuZn37; **BAM-M365a**: pure copper
Copper base alloys (disk form)	**BAM-368**: CuZn20Al2; **BAM-369**: OF-Cu; **BAM-370**: OF-Cu; **BAM-371**: OF-Cu; **BAM-372**: OF-Cu; **BAM-374**: CuSn8; **ERM-EB375 (BAM-375)**: CuZn39Pb3; **BAM-M376a**: pure copper; **ERM-EB377 (BAM-377)**: CuSn6; **ERM-EB378 (BAM-378)**: CuSn6; **BAM-M381**: pure copper; **BAM-M383b**: pure copper; **BAM-M384a**: pure copper; **BAM-M384b**: pure copper; **BAM-M385a**: pure copper; **ERM-EB387 (BAM-M387)**: CuZn20Ni5; **ERM-EB388 (BAM-M388)**: CuAl5Zn5Sn; **ERM-EB389**: CuNi25; **ERM-EB393a**: CuZn21Si3P; **BAM-M394**: CuZn40Pb2; **BAM-M394a**: CuZn40Pb2; **BAM-M396**: CuZn33Pb1AlSiAs; **BAM-M397**: CuSn4Zn2PS; **BAM-M397a**: CuSn4Zn2PS

Their second and third digit designate the specific sample. Another digit, if any, separated by a hyphen suggests the number of editions the material has passed through." Table 11 enlists typical ferrous CRMs, whereas Table 12 enlists typical nonferrous CRMs supplied by the BAM in various forms, such as chips and disks forms (Recknagel 2021). Many of these CRMs are also available in other forms like powder, pins or balls, etc.

Content of the CRM certificate. Each EURONORM CRM is provided with a certificate of analysis. The EURONORM number and the type of material of the CRM are given on the top. The mean values of the laboratories associated in the

certification are provided in a table together with the indicative values. The mean values of the parameter data sets, their overall standard deviations, and the standard deviations of the each of the laboratories are also reported in the table. "The certified reference values are given in a second table along with the associated uncertainties. The additional information made available includes a) the specific European owner lab, b) characterization methodology to be used for, c) laboratories involved in the certification process, d) the methodologies used in the elemental determination, etc."

The aluminum and copper alloy CRMs shown in Table 12 are produced and certified by BAM. This is carried out in association with the working groups of the Committee of Chemists of the Society of Metallurgists and Miners (GDMB Gesellschest der Metallurgen und Bergleute e.V.). The analyses are carried out in BAM and in other sister research laboratories. They are available in glass bottles with 100 g each. Cylindrical sample blocks are developed for a) arc/spark optical emission spectrometers (OES) and b) X-ray fluorescence (XRF) spectrometers. In interlaboratory comparisons, they use aluminum disks having height of 2.5 to 5 cm and diameter of 4 to 6 cm. Their copper blocks in cylindrical shape have a height of approximately 3 cm and 4 cm diameter. Each CRM is disseminated along with a certificate that shows the certified values together with their uncertainties and the indicative values. Other details in their certificate include a) the mean values of the accepted parameter data sets, b) their overall standard deviations, and c) the standard deviations from individual participating laboratories.

"NMIJ"-Traceable Ferrous and Nonferrous CRMs

The National Metrology Institute of Japan (NMIJ) has four types of research institutes under its umbrella, i.e., engineering measurements, physical measurements, materials-chemical measurements, and standardization of analytical instruments. These four types of research institutes take charge of supplying measurement standards in the activities primarily related to all measurement standards. The Center for Quality Management of Metrology takes care of processing administrative tasks, whereas the Research Promotion Division of NMIJ is responsible for overall planning and coordination. NMIJ produces CRMs to estimate measurement values in chemical measurements, such as calibration of analytical instruments and evaluation of different analytical procedures. In various aspects of Japanese and Asian social lives, an indispensable role is played by the NMIJ CRMs. The NMIJ CRMs are essential for industrial technology and R&D, commercial activities, environmental protection, and human health and well-being. NMIJ makes available these CRMs not only in Japan but also for the global users. A representative example can be the successful Japanese automobiles and analytical instruments and their related production methodology. As compared to NIST-USA and BAM-Germany, NMIJ-Japan has limited number of CRMs in the areas of ferrous and nonferrous alloys. Table 13 (NMIJ CRM Catalog 2021–2022 2021; NMIJ Certified Reference Materials (NMIJ CRMs) n.d.) illustrates various types of ferrous and nonferrous CRMs developed

Table 13 Ferrous and nonferrous CRMs supplied by NMIJ-Japan (NMIJ CRM Catalog 2021–2022 2021; NMIJ Certified Reference Materials (NMIJ CRMs) n.d.)

Category and form	NMIJ CRM no. and description
Industrial material CRMs for EPMA[a] (block form)	**1001-a to 1005-a**: Fe-Cr alloy; **1006-a to 1010-a**: Fe-Ni alloy; **1017-a**: stainless steel; **1018-a**: Ni(36%)-Fe alloy; **1019-a**: Ni(42%)-Fe alloy; **1020-a**: high-Ni alloy
Industrial material CRMs	**1016-a**: Fe-Cr alloy (Cr 40%)
Industrial material CRMs for positron defect measurements (size: 15mm×15mm×3mm)	**5607-a**: stainless steel
CRMs for thermophysical properties (size: 10mm×10mm×30mm)	**5805-a**: high-purity copper

[a]Electron probe microanalysis (EPMA)

and supplied by the NMIJ. The table shows category of the CRM and form, CRM no., and short description of the alloy system.

The certificate of each NMIJ CRM includes its scope, the measurand certified value, sample form, the preparation methodology used, analytical method used, instructions for use, etc. In some instances, indicative values are also provided. Most NMIJ CRMs have decent homogeneity and long-term stability. However, their stability can vary and some of these CRMs may have a short shelf file. The expiry date and storage conditions are specified in the certificate and attention should be paid to them. NMIJ frequently participates in the international interlaboratory comparison. In addition, like other NMIs, NMIJ is also regularly assessed for technical capability by other overseas NMIs. "The calibration and measurement capabilities (CMCs) and the range of determining the certified value for NMIJ CRMs and in fact most other NMIs are available in the key comparison database [KCDB: CIPM MRA Appendix-C (CIPM-MRA Key Comparison Data Base (KCDB) n.d.)]. The 'KCDB' is a comprehensive database managed by the International Bureau of Weights and Measures (BIPM)-France and is published on the website of BIPM. Most of the NMIJ CRMs are registered in this database and are internationally recognized."

"UK-Alloy Standards-LGC Standards": Ferrous and Nonferrous CRMs

NPL-UK as an NMI does not directly seem to have its own or outsourced metal or metal alloy CRMs. However, LGC is a British reference material producer and markets a lot of CRMs whose details are given herein, possibly taking their standard references from global NMIs including NPL-UK and NIST-USA. LGC Standards is a leading global producer and distributor of reference materials. LGC Standards is part of the LGC Group and is headquartered in Teddington, Middlesex, UK. LGC is the UK's designated national measurement institute (NMI) for chemical and bioanalytical measurements. They have expertise to produce as per highest expected

standards, by following norms set in ISO/IEC 17025:2017 (ISO/IEC 17025 2017) and ISO 17034:2016 (ISO 17034 2016). Their wide range of reference materials and proficiency testing schemes and skills is acknowledged by their experience in outsourcing and customized products. On the other hand, US-based Analytical Reference Materials International (ARMI), now part of LGC, manufactures an extensive range of high-quality metal alloy certified reference materials. In 2018, LGC, ARMI, and MBH Analytical merged into one company, creating the largest portfolio of metal alloy reference materials in the metal and alloy industry. The LCG-ARMI-MBH currently provides a wide range of CRMs including metal alloy CRMs, CRMs for nonmetallic standards, and reference materials for other industrial and for geological usage. Their CRMs are available in disks, chips, and other forms to suit various analytical techniques including optical emission spectrometry (OES), X-ray fluorescence (XRF) spectrometry, inductively coupled plasma optical emission spectrometry (ICP-OES), etc. As of today combined LGC group catalog advertises more than 600 different CRMs that include low-alloy steels, carbon steels, high-temperature and stainless steels, copper, nickel, cobalt, titanium, aluminum, and organic materials for different possible global requirements.

ARMI's typical ferrous and nonferrous CRMs are enlisted in Tables 14 and 15, respectively (Certified Reference Materials: Products and Services Catalogue 2015;

Table 14 Typical ferrous CRMs supplied by ARMI (part of LGC Standards UK) (Certified Reference Materials: Products and Services Catalogue 2015; gsometal.ru/Catalogues%202011/ARMI.pdf)

Category	ARMI CRM no./(UNS#) and description
25-piece low-alloy steel set	**K00095**: pure Fe; **G10180**: AISI 1018; **G10450**: AISI 1045; **G11170**: AISI 1117; **G11440**: AISI 1144; **G12150**: AISI 1215 B1; **G12144**: AISI 12L14; **G12150**: AISI 1215; **G41400**: AISI 4100; **G43400**: AISI 4340; **G46200**: AISI 4620; **G48200**: AISI 4820; **G52986**: AISI E52100; **G61506**: AISI E6150; **G86200**: AISI 8620; **N/A**: AISI 86L20; **G87400**: AISI 8740; **G93106**: AISI E9310; **K11572**: 1 ¼ Cr, ½ Mo; **K12822**: C ½ Mo-F1; **K21590**: 2 ¼ Cr-1Mo; **K24065**: NIT135M; **K42544**: 5Cr- ½ Mo; **K90901**: F91; **K90941**: 9Cr-1Mo
6-piece low-alloy steel set	**CLA1**; **CLA3**; **CLA5**; **CLA7**; **CLA9**; **CLA11**
13-piece austenitic SS set	**S30100**: AISI 301; **S30200**: AISI 302; **S30300**: AISI 303; **S30323**: AISI 303Se; **S30400**: AISI 304; **S30430**: AISI 302 HQ; **S30900**: AISI 309; **S31000**: AISI 310; **S31254**: Alloy 254SMO; **S31600**: AISI 316; **S31703**: AISI 317L; **S32100**: AISI 321; **S34700**: AISI 347
6-piece duplex SS set	**S32101**: alloy 2101; **S32205**: alloy 2205; **S32304**: alloy 2304; **S32550**: alloy 255; **S32750**: alloy 2507; **S32760**: Zeron 100
11-piece martensitic SS set	**S41000:** AISI 410; **S41600**: AISI 416; **S41800**: Greek Ascoloy; **S42000**: AISI 420; **S42200**: AISI 422; **S43000**: AISI 430; **S43100**: AISI 431; **S44004**: AISI 440C; **S44600**: AISI 446; **S45000**: custom 450; **S45500**: custom 455
15-piece tool steel set	**T11301:** AISI M-1; **T11302**: AISI M-2; **T11304**: AISI M-4; **T11350**: AISI M-50; **T12001**: AISI T-1; **T20811**: AISI H-11; **T20813**: AISI H-13; **T30102**: AISI A-2; **T30106:** AISI A-6; **T30402**: AISI D-2; **T31506**: AISI O-6; **T41901**: AISI S-1; **T41905**: AISI S-5; **T41907**: AISI S-7; **T61206**: AISI L-6

Table 15 Typical nonferrous CRMs supplied by ARMI (part of LGC Standards UK) (Certified Reference Materials: Products and Services Catalogue 2015; gsometal.ru/Catalogues%202011/ARMI.pdf)

Category	ARMI CRM no./(UNS#) and description
9-piece cast aluminum alloy set	**A02082**: 208.2; **A02952**: 295.2; **A03190**: 319; **A13330**: 333; **A13562**: 356.2; **A03831**: 383.1; **A13900**: 390; **A14132**: 413.2; **A08500**: 850
18-piece wrought aluminum alloy set	**A91050**: 1050; **A91100**: 1100; **A92014**: 2014; **A92018**: 2018; **A92024**: 2024; **A93003**: 3003; **A93004**: 3004; **A94043**: 4043; **A94145**: 4145; **A94343**: 4343; **A94643**: 4643; **A95052**: 5052; **A95083**: 5083; **A95182**: 5182; **A96061**: 6061; **A96063**: 6063; **A97050**: 7050; **A97075**: 7075
21-piece cast copper base alloy set	**C81500:** CDA815; **C83600**: CDA 836; **C84400**: CDA 844; **C85700**: CDA 857; **C86300**: CDA 863; **C87200**: CDA 872; **C87500**: CDA 875; **C89320**: Magnolla B; **C89510**: SeBiLoy 1; **C89520**: SeBiLoy 2; **C90300**: CDA 903; **C90700**: CDA 907; **C92200**: CDA 922; **C93200**: CDA 932; **C93700**: CDA 937; **C95400**: CDA 954; **C95410**: CDA 954 Mod.; **C95500**: CDA 955; **C95800**: CDA 958; **C96400**: CDA 964; **C97600**: CDA 976
18-piece wrought copper base alloy set	**C11000**: CDA 110; **C14500**: CDA 145; **C17200**: CDA 314; **C31400**: CDA 314; **C36000**: CDA 360; **C46400**: CDA 464; **C48200**: CDA 482; **C48500**: CDA 485; **C51000**: CDA 510; **C54400**: CDA 544; **C62300**: CDA 623; **C63000**: CDA 630; **C64200**: CDA 642; **C65500**: CDA 655; **C67500**: CDA 675; **C69300**: CDA 693; **C70600**: CDA 706; **C71500**: CDA 715
15-piece titanium base alloy set	**R50700**: Ti CP grade 4; **R52400**: Ti CP grade 7; **BT1-0**: Ti CP 0.5Al; **R54620**: Ti 6Al-2Sn-4Zr-2Mo; **R54520**: Ti 5Al-2.5Sn; **R54810**: Ti 8Al-1Mo-1V; **R56201**: Ti 6Al-2Nb-1Ta-1Mo; **R56320**: Ti 3Al-2.5V; **R56400**: Ti 6Al-4V; **R56410**: Ti 10V-2Fe-3Al; **R56620**: Ti 6Al-6V-2Sn; **R56700**: Ti 6Al-7Nb; **IMI 550**: Ti 4Al-4Mo-2Sn-0.5Si; **R56740**: Ti 7Al-4Mo; **R58640**: Ti 3Al-8V-6Cr-4Mo-4Zr

gsometal.ru/Catalogues%202011/ARMI.pdf). A key feature to these CRMs is the metrological traceability of each CRM to values from the NIST and other international certifying bodies (NMI). "These CRMs are available in the form of thin disks and chips as well depending upon the requirement. For example, thin disks may be used for X-ray fluorescence (XRF) analysis, whereas chips may be used for inductively coupled plasma optical emission spectrometry (ICP-OES) analysis." All LGC/ARMI CRMs are accompanied with a comprehensive certificate of analysis. These certificates provide a full description of the material to which they relate and summarize the analysis undertaken during the characterization process. Their certificates of analysis are prepared in compliance with ISO Guide 31:2015 (ISO Guide 31 2015), which specifies content, describing reference materials and required specifications in the certificates, and clarifies different labeling issues.

"CSIR-NPL, India": The Developing Economy/NMI and Prospects of Ferrous and Nonferrous CRMs

As discussed above, the certified reference materials (CRMs) are purposed for precise calibration of testing equipment. These CRMs are SI traceable and produced under rigorous conditions following the international standards. This is a generic requirement for achieving quality assurance in laboratories and industries. They are also used in checking calibration, testing, quality control, method validation, assigning a value to different test parameters, and establishing of traceability. In India, on behalf of the Indian government, by an Act of Parliament, maintenance of all primary and secondary standards linked to SI units and their methodologies that are acceptable globally is the mantle of CSIR-National Physical Laboratory (NMI of India). Reference materials have been developed and marketed on a limited scale by CSIR-NPL since 1984 onward. Initially these involved most essential daily life-related standard materials like water quality, calibration of alcohol meters, etc. Later, with progressive development of national industrial production base, these CRMs which were initially developed in-house have also been developed in collaboration with other national testing and certification organizations or reference material producers (RMPs). Subsequently, reference materials developed by the CSIR-NPL have started to be registered with the trademark (TM No. – 3669341) of Bharatiya Nirdeshak Dravyas (BND®). The Indian government, through its different ministries, has started to more actively support and encourage development, production, and use of such locally produced reference materials (BNDs). In the form of collaborative development of reference materials by different vendors with support from the Indian government through CSIR-NPL, over the last few years, there has been a gradual competency and capability development in the private sector. It also leads to growth of analytical laboratories, better quality control of local products, and better employment prospects for local educated youth. This is not a 1-day or a one-off process. It needs continuous support, monitoring, and often long-term handholding. In addition, these BNDs are used by government sectors and institutions, academia, private industries, etc. They are being marketed to other countries worldwide as well. If used properly, these BNDs ensure prime accuracy and reliability of their published data, with minimum uncertainties in measurements.

In the Indian context, the primary responsibility of alloy-related research and development lies with CSIR-NML. They also have their portfolio of reference materials within the alloys and related raw material items. Military usage-related niche sector alloys and their quality issues are handled by a few specific laboratories of Defense Research and Development Organization (DRDO). It is expected that CSIR-NML will slowly improve their portfolio of reference materials within the BND® brand. Large-scale steel production in India has been traditionally handled by Steel Authority of India Ltd. (SAIL). However, in the recent past, a lot of private players have started slowly entering this field. So there needs to be a standardization and quality control of all their products. In addition, there are also a lot of cheap/economic imports of different grades of alloys but of questionable quality (Karar et al. 2022; Paul 2009). All of these need uniformity of quality so that the Indian

population is not cheated of their hard-earned money. Steel Authority of India Ltd. (SAIL) has been traditionally getting its quality control done by its in-house Research and Development Center for Iron and Steel (RDCIS). It is thus possibly imperative that in the long term, RDCIS in association within CSIR-NML also slowly comes within the BND® umbrella for all steel-related reference materials for better control of all local and imported steel items. Currently, there is no Indian-made reference material for aluminum- or titanium-related alloys, even though India has significant deposit of its ores and is currently extracting both these metals and producing related alloys. In the medium and long term, CSIR-NML may, in due course, release related BNDs for better quality control of these too.

CSIR-NPL has recently initiated a mission mode program for greater dissemination of different BNDs through different reference material producers (RMPs) in the country primarily accredited for their ISO 17034:2016 scope by the National Accreditation Board for Testing and Calibration Laboratories (NABL), Gurugram (India). Under the aegis of this national-level initiative, CSIR-NPL shall establish the measurement traceability of all such BNDs, in various possible commodities, like chemicals, cement, petroleum products, food, textile, precious metals, instruments, ores and minerals, etc. Related agreements have been signed by CSIR-NPL with numerous RMPs, viz., Aashvi Technologies LLP, Ahmedabad; NCCBM, Ballabgarh; HPCL, Vizag; FARE Labs Pvt. Ltd., Gurugram; Global PT Provider Pvt. Ltd., New Delhi; MINT, Mumbai; BPCL, Mumbai; Jalan and Co., New Delhi; CSIR-IITR, Lucknow; etc., in various sectors like chemicals, cement, petroleum-based products, food safety, hardness blocks, precious metals, pharmaceuticals, etc. As of today, close to 120 BNDs were established by CSIR-NPL in the above-described sectors in collaboration with the different RMPs and also under the in-house BND® development program (https://www.nplindia.org/wp-content/uploads/2021/11/Available-Bharatiya-Nirdeshak-Dravya-list-251220.pdf; Singh et al. n.d.; Pant et al. 2020).

Each BND® certificate supplied by CSIR-NPL contains, on the top, the BND® no. and BND® name, parameter certified, and certificate number. The BND® supplier name, its intended use, packing details, method of certification, and details of standards followed are given. The certificate also contains details, such as certified value of the parameter and expanded uncertainty, traceability, instructions for storage, usage, and precautions for handling. Additionally, the date of certification and expiration period, conditions for maintenance of certification, etc. are also given in the certificate. A disclaimer is also given in the certificate incorporating zero liability for damages due to any misuse of information, material, apparatus, process, or method disclosed in the certificate or any warranties for material safety. In spite of tremendous efforts made by CSIR-NPL on numerous fields of BND® development, the development of ferrous- and nonferrous-based CRMs/BNDs is yet in the initial stages. In this line, CSIR-NPL has recently developed precious metal BNDs (such as high-purity gold and silver and their alloy BNDs) in collaboration with MINT, Mumbai, and Jalan and Co., New Delhi. Although a long way to go, the huge initiatives of CSIR-NPL for CRMs, under the "Make in India" program of the government of India in the long run is expected to bring an enhancement in quality

of products in the country for local consumption, export, and import. Production and certification of BNDs will accelerate economic growth and employment in the country and enhance global viability of Indian products.

Summary and Conclusion

Certified reference materials (CRMs), as defined above, have a vital significance in the economic growth of any country. Numerical values measured using a comparison with different CRMs are considered accurate and reliable Standard Reference Material (SRM). In this manner, all CRMs are used in the calibration of different analytical equipment. They ensure reliability and traceability to the international measurement system (SI units). CRMs available with developed economies/NMIs, such as NIST-USA, BAM-Germany, NMIJ-Japan, and ARMI (part of LGC Standards, UK), have been discussed in this chapter. In earlier sections, a summarized discussion is made on different types of cast irons, steels, tool steels and stainless steels Ferrous Alloy Systems and aluminum alloys, copper alloys and titanium-based alloys Non Ferrous Alloy Systems developed internationally. In the later sections, the ferrous- and nonferrous-based CRMs are paid attention during the discussions. While NIST-USA on an overall supplies more than 1300 CRMs, BAM-Germany produces more than 400 CRMs; the LCG-ARMI-MBH group has catalogued more than 600 different CRMs in a variety of fields; therefore, their impact on the global economy is enormous. The NIST-USA, BAM-Germany, and LGC-UK (with ARMI) produce a significantly large number of ferrous and nonferrous CRMs (see Table 9, Table 10, Table 11, Table 12, Table 14 and Table 15). In contrast, NMIJ-Japan produces only limited number of these CRMs (see Table 13). CSIR-NPL, India, is a developing NMI and produces close to 120 CRMs as of today with the tradename BND® in the areas of chemicals, cement, fuels and petroleum products, food, textile, hardness blocks, ores and minerals, precious metals, etc. The CSIR-NPL's CRM/BND development initiatives fall under the "Make in India" program of the government of India. However, CSIR-NPL is in the initial stages of development of ferrous- and nonferrous-based CRMs/BNDs. Finally, it may be concluded that CRMs play an important role in the human life today and they are essentially needed for the economic growth of the developed as well as developing countries/NMIs. Further, it is concluded that the developed economies/NMIs contribute to the major share of ferrous and non-ferrous based CRMs supplied to the international market today and the developing economies/NMIs have yet to make a visible impact in this frontier area of Science and Technology.

Cross-References

- ▶ Bharatiya Nirdeshak Dravya for Assessing Mechanical Properties
- ▶ Certified Reference Materials (CRMs)
- ▶ Mechanical and Thermo-physical Properties of Rare-Earth Materials

▶ Petroleum-Based Indian Reference Materials (BND)
▶ Role of Certified Reference Materials (CRMs) in Standardization, Quality Control, and Quality Assurance

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CRMs

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Ensuring the Quality of Cement and Building Materials for Civil Infrastructure

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Abstract

Cement industry plays a vital role in the growth and economic development of the country because of its strong linkage to other sectors such as infrastructure, construction, housing, transportation, coal, power, etc. To ensure the conformity of products,

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cement manufacturers are calibrating the equipment with Certified Reference Materials (CRMs) from time to time to improve the quality of the products and fulfill the norms set by BIS, the standardization body of GoI. These CRMs help to maintain the quality control and quality assurance of the products with accurate and precise measurements, which will improve the country's quality infrastructure.

Introduction

India has emerged as one of the fastest-growing major economies in the world and is expected to be one of the world's top five to six economic powers shortly (https://www.ibef.org/economy/indian-economy-overview). Among the eight core industries of India, the cement industry plays a vital role in the growth and economic development of the country because of its strong linkage to other sectors such as infrastructure, construction, housing, transportation, coal, power, etc. The recent focuses of the Government of India (GoI) on Housing for all by 2022, Make in India, World Class Cement Concrete Highways, Creation of 100 Smart cities, Dedicated Freight Corridors, Clean India Mission, Ultra Mega Power Projects, Connectivity improvement including Water Transport are bringing a big boost in infrastructure developments and housing sectors. The cement industry has to play a pivotal role in the success of these initiatives of GoI.

In the global cement market, India is the second largest cement producer in the world after China and it has evolved to become one of the best in terms of energy efficiency, quality control, and environmental improvements. Nonetheless, the cement industry has strongly contributed to employment, fiscal revenue, and community development while achieving manufacturing and technological advancement/excellence. The installed capacity of the cement industry in India as per a survey (NCCBM) in 2018–2019 has shown to be 537 million tonnes, with cement production of around 337 million tonnes, which comprises 145 integrated large cement plants, 107 grinding units, 62 mini cement plants, and 5 clinkerization units (NCCBM 2019). Cement consumption in India is still around 235 kg/capita against the global average of 520 kg/capita, which shows significant potential for the growth of the industry (https://www.business-standard.com/article/news-cm/domestic-cement-consumption-around-235-kg-per-capita-against-global-average-of-520-kg-per-capita-119020600509_1.html).

To ensure the conformity of products, cement manufacturers are calibrating the equipment with Certified Reference Materials (CRMs) from time to time to improve the quality of the products and fulfill the norms set by BIS, the standardization body of GoI. These CRMs help to maintain the quality control and quality assurance of the products with accurate and precise measurements, which will improve the country's quality infrastructure.

About CRMs

In the present scenario of globalization of economy, globally acceptable measurements are required where the availability of Reference Materials becomes crucial for universal acceptance of products and test reports. Reference material is characterized

by a metrologically valid procedure for one or more specified properties, accompanied by a reference material certificate that provides the value of the specified property, its associated uncertainty, and a statement of metrological traceability. These materials ensure high quality in measurement and provide SI traceability to analytical measurements. Certified Reference Materials (CRMs) of cement and cementitious products are regularly used in the cement and concrete industry. CRMs are an important tool in realizing several aspects of measurement quality and are used for calibration of online and offline X-ray analyzers, validation of test methods, establishing purity of materials, and studying microstructural & phase compositions.

In the broader sense, CRMs ensure:

- Clearly defined customer requirement
- Validated methods and equipment calibration
- Global comparability of measurements
- Independent evidence of performance through proficiency testing
- Well-defined QC and QA procedures

Reference materials of various commodities such as cement, ores, rocks, semi-processed ores, food, drugs, and many more are being produced by several organizations. The top contributors in the area of reference materials are NIST (National Institute of Standards and Technology, USA), Certipur Reference Materials, PTB, Germany, China Reference Materials, NIM, China, European Reference Materials, JRMM, Belgium, Thailand Reference Materials, NIM, Thailand, etc. Besides, many ISO 17034 accredited laboratories are also producing CRMs in the areas of Food, water, cement, petroleum products, metal, etc.

Recently, National Physical Laboratory (CSIR-NPL), India, has taken a step forward to develop CRMs named BND in various commodities like Cement, Water, Petroleum, food, etc. CSIR-National Physical Laboratory (NPL-India) is mandated to be India's "National Measurement Institute" (NMI) by the act of Parliament. It is the custodian of "National Standards" with responsibility for the dissemination of measurements to the needs of the country. It brings a boost to harmonize the quality infrastructure of the country and improve the economy of the Nation.

Cement-Related CRMS

Reference Materials (RM) in the area of cement and cementitious materials around the world (Standard Reference) are Portland Cement Fineness, Portland Cement (Blended with Slag), Portland Cement (Blended with Fly Ash), Calcium Aluminate Cement, White Portland Cement, Portland Cement (Blended with Limestone), Calcined Clay Cement, Portland Cement Clinker, Coal Fly Ash, Bituminous Coal, Silica Fume, Bauxite, Calcined Clay, Hydrated Lime, Petroleum Coke, Feldspar, Iron Ore, Laterite, Limestone, etc.

Requirement/Consumption of CRMS in Indian Cement Industry

In India, there are 145 integrated large cement plants, 107 grinding units, 62 mini cement plants, and 5 clinkerization units and 3400+ accredited testing laboratories as per ISO 17025, out of which >1350 laboratories are (Directory of Accredited) serving the construction industry wherein they require CRMs for their regular quality checks. On average, with a minimum of 10 units of CRM consumption in cement plants/laboratories, the annual requirement of CRM units would be around 17,000 units. As the construction industry and cement plants are growing consistently, the industry's requirement for CRMs is increasing yearly.

Currently, in India, there are 17 reference material producers in various fields as per ISO 17034 and NCB is the leading organization among them which develops CRMs in cementitious materials. NCB has been facilitating the cement and construction laboratories by supplying reference materials of cement and Cementitious materials since 1990. Besides, NCB is accredited with ISO 17034:2016 as a Reference Material Producer (RMP) in India and follows all the relevant standards while developing the CRMs to meet the specific requirement. Annually approx. 12,000 units of CRMs are being supplied to the cement and construction laboratories, saving approximately the import of CRM's worth INR 250 lakhs/annum (source: NCB databank).

Production Process of CRM

The process for developing Certified Reference Materials (CRMs) is very important and a systematic approach is retained to establish the reference material's property value.

The process is covers the following (ISO17034:2016):

- Material selection
- Verification of the identity of the material
- Maintaining suitable environments for all the aspects of production
- Material processing
- Choice of measurement procedures
- Validation of measurement procedures
- Verification and calibration of measuring equipment
- Specification of acceptance criteria for, and assessment of homogeneity, including sampling
- Specification of acceptance criteria for, and assessment and monitoring of, stability, including sampling
- Designing and organizing appropriate characterization, including sampling
- Assigning property values
- Establishing uncertainty budgets and estimating uncertainties of certified values
- Defining acceptance criteria for measurand levels and their uncertainties
- Establishing metrological traceability of measurement results and certified values

An example of chemical parameters and physical parameters of cement and cementitious materials are illustrated hereunder.

Chemical Parameters

The raw materials used to manufacture cement consist of CaO, SiO_2, Al_2O_3, Fe_2O_3, MgO, Na_2O, K_2O, and SO_3. Cement consists of lime (CaO) at about 60–67%, and it is cement's important and major ingredient. Lime is the main material for the manufacture of cement. Lime imparts strength and soundness to the cement; if it presents in excess quantity, it makes it expand and disintegrate. It adds strength and usability to cement. If the amount is too much, it makes cement expand and disintegrate. Silica is the next major ingredient in cement. The composition of silica in cement is about 14–25%. Adding a sufficient quantity of silica imparts strength to the cement. However, excess silica in cement will increase cement's strength, but cement's setting time also increases. Alumina is about 3–8% in cement, giving the cement an instantaneous setting property. It acts as a flux and reduces the clinkering temperature. An excessive amount of alumina reduces cement fixed time but, simultaneously, reduces cement strength. Iron oxide is around 0.1–5.0% in cement. It imparts color to cement and acts as a flux. It enters into a chemical reaction with calcium and aluminum at very high temperatures to form tricalcium alumino-ferrite. Tricalcium aluminum ferrite gives hardness and resistance to cement. Also it contains MgO (0.1–4.0%), Na_2O and K_2O (0.1–1.3%), and SO_3 (1–3%) (Taylor 1997).

NCB develops several raw materials and cement BNDs such as coal, fly ash, raw meal, limestone, clinkers, OPC, PPC, and composite cement. The developed BNDs are validated and certified for their chemical parameters per Indian Standards. These standard methods are essential adjuncts to the cement specifications because defective test methods may lead to wrong conclusions about cement quality. Parameter such as Loss of Ignition (LOI) is estimated as per IS 1760 (Part 1):1991, SiO_2 as per IS 1760 (Part 2):1991, and Fe_2O_3, Al_2O_3, CaO, MgO as per IS 1760 (Part 3):1992. The amount of SO_3 is evaluated as per ASTM – C-25:2017 and TiO_2 as per IS 12423:1988. The analysis of Mn_2O_3 and P_2O_5 has been carried out as per the NCB monograph (MS-5-79). Estimating Cl is as per IS 1760: (Part 5):1991. However, in the case of clinkers and cement, including OPC and PPC, LOI, CaO, SiO_2, Fe_2O_3, Al_2O_3, MgO, SO_3, IR, and Cl have been estimated carried out as per IS 4032:1985. The Alkalis such as Na_2O and K_2O are analyzed as per NCB standard procedure, MS-13-2010. The associated measurement uncertainty was calculated at a 95% confidence level with coverage factor $k = 2$, considering significant sources of uncertainty, including measurement replication, instrument background, the mass taken, possible heterogeneity, and stability factors according to ISO GUM Guide (JCGM 100.2008) and ISO Guide 35 as given in Table 1. Figure 1 shows the limestone, raw meal, clinkers, and composite cement BNDs. Figure 2 shows the instrument used for making and characterizing the cement BNDs.

Loss of ignition is the process of measuring the weight change of a sample after it has been heated to a high temperature causing some of its content to burn

Table 1 Certified values and associated measurement uncertainties of chemical parameters of limestone, raw meal, clinkers, and composite cement, respectively

	Lime stone		Raw meal		Clinkers		Composite cement		Portland Pozzolana Cement	
Chemical constituents	Certified values (Wt. %)	Expanded uncertainty, $k = 2$	Certified values (Wt. %)	Expanded uncertainty, $k = 2$	Certified values (Wt. %)	Expanded uncertainty, $k = 2$	Certified values (Wt. %)	Expanded uncertainty, $k = 2$	Certified values (Wt. %)	Expanded uncertainty, $k = 2$
LOI	36.79	±0.02	35.58	±0.05	0.60	±0.01	1.73	±0.02	1.25	±0.02
SiO_2	11.14	±0.05	13.14	±0.09	20.74	±0.04	32.88	±0.08	–	–
Fe_2O_3	0.92	±0.04	2.45	±0.05	4.01	±0.02	4.84	±0.11	–	–
Al_2O_3	1.65	±0.03	3.59	±0.06	5.31	±0.05	12.37	±0.23	–	–
CaO	46.98	±0.15	40.95	±0.13	61.88	±0.16	41.53	±0.11	–	–
MgO	0.66	±0.05	2.66	±0.06	4.21	±0.10	1.64	±0.02	2.27	±0.05
IR	–	–	–	–			27.12	±0.19	24.5	±0.38
SO_3	0.63	±0.03	0.20	±0.02	1.59	±0.01	2.84	±0.03	1.91	±0.02
Na_2O	0.17	±0.016	0.21	±0.02	0.25	±0.02	0.26	±0.03	0.49	±0.04
K_2O	0.32	±0.005	0.48	±0.02	0.85	±0.013	0.79	±0.01	0.77	±0.03
Cl	0.026	±0.002	0.014	±0.003	0.025	±0.002	0.012	±0.002	0.012	±0.002
TiO_2	0.129	±0.009	0.22	±0.05	0.43	±0.02	–	–	–	–
Mn_2O_3	0.081	±0.005	0.19	±0.07	0.07	±0.01	–	–	–	–
P_2O_5	0.108	±0.014	–	–	–	–	–	–	–	–

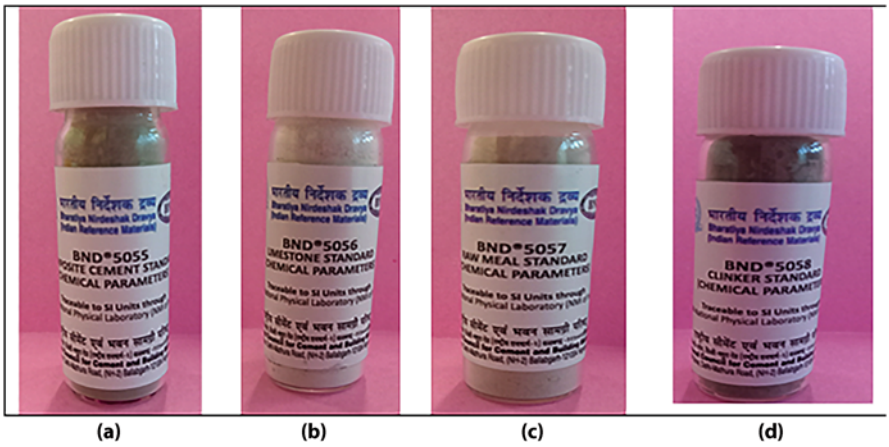

Fig. 1 (**a**) Composite cement, (**b**) limestone, (**c**) raw meal, and (**d**) clinkers BNDs

or to volatilize. The cement industry utilizes the LOI method to determine the cement's water content and/or carbonation as these reduce the quality. These tests are carried out at high temperatures (900–1000 °C) using the muffle furnace. The EDTA titration method determines parameters such as CaO, MgO, Al_2O_3, and Fe_2O_3. Alkalies are estimated using flame photometry or by Atomic Absorption spectrophotometer. TiO_2, MnO_2, and P_2O_5 are determined using UV-Vis spectroscopy.

Physical Parameters

The development of composite cement is illustrated (BND 5006) on physical parameters such as Blaine fineness as an example. The development of said certified reference material is followed as per the guideline of ISO 17034 and guide 35. The process covers material selection and verification, material processing (blending/grinding), Homogeneity, characterization, stability, packaging and certification, etc.

A systematic approach is presented below:

Material Selection and Verification

A similar matrix of a new lot of the material (Composite Cement Brand) for preparing CRM is sourced from the industry. The material is checked for compliance with product specifications laid down by BIS as per IS 16415:2015.

Material Processing

The material is sieved to eliminate any foreign particles and further blended and ground in a laboratory ball mill to ensure the proper sample homogeneity.

Fig. 2 Instruments used for making and characterizing the cement BNDs

Packaging

After proper homogenization, the sample is packed in an airtight bag to avoid the hydration of cement during storage prior to preparation and packaging. Then each bag is packed into different vials and capped, then packed in a box. Approx. 600 vials were produced, each containing approximately 10 g of cement. The vials are randomly selected (randomly block design) and sent to the subcontracted laboratories for measurements: homogeneity, characterization, stability, etc. After analyzing the results, the vials are packaged in small boxes containing four vials each.

Homogeneity

After the material is packed, the material is tested for homogeneity check to determine whether the material in different vials is the same. Minimum number of samples (ten nos) as per ISO GUIDE 35:2017 is drawn from the homogenized sample (randomize block design) for statistical check of homogeneity. The criterion for acceptance is decided before the assignment of property value.

Minimum number of units Nmin for a homogeneity study of materials characterized for a quantitative property given by:

$$\text{Nmin} = \max\left(10, 3\sqrt{\text{NProd}}\right) \tag{1}$$

where max (. . . , . . .) indicates the maximum of the terms within the parentheses. The maximum number of units for assessment of homogeneity is limited to not more than 30 units.

The laboratory is asked to measure triplicate samples from each vial. The received data is analyzed and evaluated through statistical analysis through the ANOVA method (ANOVA two factor without replication) to validate the homogeneous nature of the sample.

Characterization

To process the characterization of the material, an operationally defined measurand using a network of competent laboratories method is followed as per ISO GUIDE 35:2017. The analytical/testing work for characterization is done in sub-contracted laboratories accredited as per ISO 17025 and following IS test method. Composite Cement vials (total 24 nos) is sent to the subcontracted laboratories for fineness determination. The laboratories are asked to measure duplicate samples from each vial. The received data is analyzed and evaluated through statistical analysis. For determination of the property value of composite cement, a robust statistic calculation along with a Guide to the Uncertainty of Measurement (GUM) is considered for evaluation of the results.

Description of Method for Determining Blaine Fineness

The Blaine measurement is described in IS 4031(Part 2):1999. The principle of operation is that the permeability of a bed of fine particles is proportional to the fineness of the particles. Therefore, the test measures the flow rate of air through a bed of cement particles. Figure 3 shows the air permeability apparatus.

In brief, the test is carried out by packing the cement to be measured in a cell of known volume and placing it on top of a U-tube manometer that contains a non-hygroscopic liquid of low viscosity and density, e.g., dibutyl phthalate. The cell is placed on the U-tube so that a tight seal is created and a vacuum is created under the cement cell so that the liquid in the manometer is higher toward the cell. Then, the air is allowed to flow back only through the cement sample. The time for the liquid in the manometer to descend a set distance is measured. This time is used to calculate the fineness quantified by the surface area (S) of the cement defined using the following formula:

Fig. 3 Air permeability apparatus

The volume of cell = 1.6778
Bed weight (W) = Volume of cell × specific gravity × porosity (1 − e)

$$S = \frac{\rho_0}{\rho} \times \frac{(1 - e_0)}{(1 - e)} \times \frac{\sqrt{e^3}}{\sqrt{e_0^3}} \times \frac{\sqrt{t}}{\sqrt{t_0}} S_0 \tag{2}$$

S_0 is the specific surface of the reference cement (cm^2/g).
e is the porosity of the bed of cement under test.
e_0 is the porosity of the bed of reference cement.
t is the measured time for the cement under tests(s).
t_0 is the mean of the three times measured on the reference cement(s).
ρ is the density of the cement under test (g/cm^3).
ρ_o is the density of the reference cement (g/cm^3).

Stability

Minimum number of Samples as per ISO GUIDE 35:2017 is drawn from the composite cement lot for statistical stability check.

$$|XCRM - Xmon| \leq k\sqrt{(u2CRM + u2mon)} \tag{3}$$

XCRM = Certified value
Xmon = Monitoring value
uCRM = Standard uncertainties of the certified value
umon = Standard uncertainties of monitoring value
k = coverage factor at confidence level of 95%

Table 2 Certified values and associated measurement uncertainties of physical parameters of composite cement

Measurand	Estimation method	Certified value and expanded uncertainty
Specific surface area (Blaine)	IS 4031 (Part 2):1999	360 ± 8.2 m^2/kg

The certified value has been standardized with a specific gravity of 2.85

Fig. 4 Composite cement (Blaine fineness)

Certification

As the CRM of composite cement is developed, the CSIR-NPL team witnessed the test results to ensure the results validation and further certification for Bharatiya Nirdeshak Dravya (BND) through NMI of India. The certified value of the CRM is mentioned in Table 2 and the CRM specimen of composite cement is shown in Fig. 4.

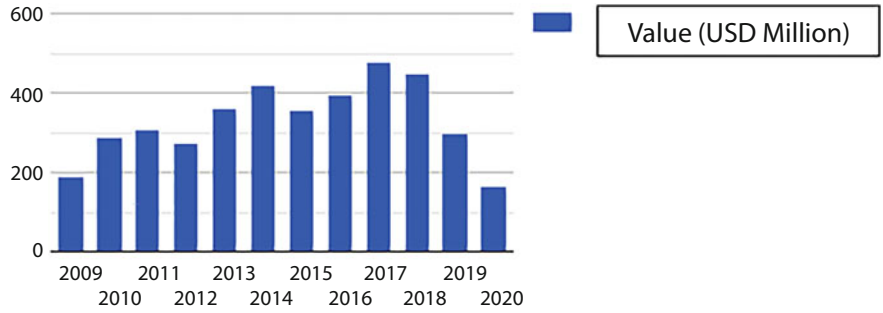

Fig. 5 Historical export analysis of cement from India (https://connect2india.com/global/export-Cement-from-india)

Impact on Import/Export

Cement manufactured in India is traded all over the world. Almost 176 countries and territories like the USA, United Arab Emirates, Sri Lanka, Nepal, Bangladesh, etc., actively import cement from India. The combined value of total export is 164.01 USD million for the year 2020. Historical export analysis of cement from India is given in Fig. 5 (https://connect2india.com/global/export-Cement-from-india).

Historical export analysis of cement from India shows that the cement export is increasing consistently from 2009 onwards except for 2019 and 2020 due to the Covid-19 pandemic. It shows that the acceptability of cement manufactured in India further increased among the countries due to precise measurement contribution after the development of BNDs in India.

CRMs are being produced at a very high cost; India is importing CRMs worth INR 1000 crores/annum (BND Coffee Table Book, NPL). It emphasizes a need for indigenous development of CRMs which will contribute to saving the import cost and also support the "Make in India" and "Self-reliance" pledge of Govt. of India.

In view, the Government of India decided to develop its standard material as Bharatiya Nirdeshak Dravya (BNDs), i.e., Indian Certified Reference Materials, through National Physical Laboratory (NPL), India, in the year 2018. CSIR-NPL and NCB have been working together to provide traceability for the reference materials in cementitious products for making such CRMs that are traceable to SI units. So far, 16 BNDs have been developed in Cement and Building materials and more number of BNDs are in the development phase. The list of available BNDs and their application is given in Table 3.

Requirement of New CRMS

Sustainable development has led to the necessity to implement changes toward balancing all the areas of human activity, including the construction sector. As the cement and construction industry is moving toward sustainability, there is a demand

Table 3 BNDs and their application

Sl. no.	Name of BND	Purpose/intended use
1.	Ordinary Portland Cement (OPC)-Blaine fineness **(BND 5001)** – range: 250–300 m^2/kg	Calibration of Blaine air permeability apparatus
2.	Ordinary Portland Cement (OPC)-Blaine fineness **(BND 5021)** – range: 320–360 m^2/kg	Calibration of Blaine air permeability apparatus
3.	Ordinary Portland Cement (OPC)-Blaine fineness (**BND 5011)** – range: 400–450 m^2/kg	Calibration of Blaine air permeability apparatus
4.	Fly ash-Blaine fineness **(BND 5004)**	Calibration of Blaine air permeability apparatus
5.	White Portland Cement (WPC)-Blaine fineness **(BND 5007)**	Calibration of Blaine air permeability apparatus
6.	Portland Pozzolana Cement (PPC)-Blaine fineness **(BND 5002)**	Calibration of Blaine air permeability apparatus
7.	Portland Slag Cement (PSC)-Blaine fineness **(BND 5003)**	Calibration of Blaine air permeability apparatus
8.	Composite Cement-Blaine fineness **(BND 5006)**	Calibration of Blaine air permeability apparatus
9.	Limestone-Chemical parameter **(BND 5056)**	Calibration of instrument (UV-Vis spectroscopy, flame photometer) and validation of method for characterization of the measurand for analysis of limestone
10.	Ordinary Portland Cement (OPC)-Chemical parameter **(BND 5051)**	Calibration of instrument (UV-Vis spectroscopy, flame photometer) and validation of method for characterization of the measurand for analysis of OPC
11.	Clinker-Chemical parameter **(BND 5058)**	Calibration of instrument (UV-Vis spectroscopy, flame photometer) and validation of method for characterization of the measurand for analysis of clinker
12.	Portland Pozzolana Cement (PPC)-Chemical parameter **(BND 5052)**	Calibration of instrument (UV-Vis spectroscopy, flame photometer) and validation of method for characterization of the measurand for analysis of PPC
13.	Raw Meal-Chemical parameter **(BND 5057)**	Calibration of instrument (UV-Vis spectroscopy, flame photometer) and validation of method for characterization of the measurand for analysis of raw meal
14.	Fly ash-Chemical parameter **(BND 5054)**	Calibration of instrument (UV-Vis spectroscopy, flame photometer) and validation of method for characterization of the measurand for analysis of Fly ash
15.	Coal-Chemical parameter **(BND 5091)**	Calibration of instrument (Bomb Calorimeter) and validation of method for characterization of the measurand for analysis of coal

(continued)

Table 3 (continued)

Sl. no.	Name of BND	Purpose/intended use
16.	Composite Cement-Chemical parameter **(BND 5055)**	Calibration of instrument (UV-Vis spectroscopy, flame photometer) and validation of method for characterization of the measurand for analysis of composite cement

for more CRMs in cement and building materials. NCB has produced 18 BNDs, including solid fuel (Coal). In addition, NCB is also producing other Reference Materials(RMs) viz. Granulated Glass Blast Furnace Slag- Blaine fineness, Calcined clay, pet-coke, silica fume, Iron ore, Red ochre, sand, Bauxite, White Portland Cement- chemical, Oil well cement-chemical, etc. Nevertheless, other CRMs available in the global market like Flue Gas Desulfurization Gypsum, phospho gypsum, Refractories (powder form), clinker phases for XRD calibration, etc.

In cement manufacturing, XRF/XRD is a state-of-the-art instrument that can provide complete quality control of clinker and cement. The X-ray fluorescence technique (XRF) is used to perform chemical elemental analysis on cement-making materials and X-ray diffractometry (XRD) equipment is typically required to determine the phase content in clinker or cement. On the other hand, the (XRD) system has the capability of measuring *quartz in raw meal*, *free lime* (CaO), and clinker phases, as well as calcite ($CaCO_3$) in cement. NCB has been developing secondary standard materials for XRF/XRD calibration for the specific cement plant product in the last two decades to improve its quality system.

In view of the above, the existing RMs is needed to be transformed into BNDs shortly and further development of the new BNDs like XRD for clinker phases, FGD gypsum, etc. meet the global gradient.

Future Goals

Research and Innovation play a pivotal role for any organization to remain competitive and keep pace with the demand and growth of society and the country. National Council for Cement and Building Materials (NCB), a premier Research, Development and Innovation institution, is equipped with multi-disciplinary expertise and state-of-the-art testing and evaluation facilities. Keeping in view the growing importance of CRMs/BNDs of cement and cementitious materials in India and Global, NCB and CSIR-NPL are aiming to develop more no. of Indian Certified Reference Materials in the areas of Cement and Building products to meet the National and Global demand. Our mission is always to provide our stakeholders with the best innovative solutions and quality products based on top-level advanced techniques.

Role in Indian Economy/GDP

Honorable Prime Minister of India, Narendra Modi Ji, dreams of making India a USD 5 trillion economy (https://pib.gov.in/) and a vision of "Atmanirbhar Bharat" or **self-reliant India**. CSIR-NPL has started a great initiative toward Atmanirbhar Bharat and Make in India through the BND program. These BND products will substitute imports and create a market capitalization for overseas industries and laboratories. The increasing supply of BNDs to domestic and international markets will contribute to the country's GDP and boost India's economy.

Conclusion

Certified Reference Materials (CRMs) of cement and 60 cementitious products are regularly used in the cement and concrete industry. CRMs are an important tool in realizing several aspects of measurement quality and are used for calibration of online and offline X-ray analyzers, validation of test methods, establishing purity of materials, and studying microstructural and phase compositions. The development of said certified reference material is followed as per the guideline of ISO 17034 and guide 35. The process covers material selection and verification, material processing (blending/ grinding), homogeneity, characterization, stability, packaging, and certification. As the CRM of composite cement is developed, the CSIR-NPL team witnessed the test results to ensure the results validation and further certification for Bharatiya Nirdeshak Dravya (BND) through NMI of India.

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Petroleum-Based Indian Reference Materials (BND)

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Production and Dissemination

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Abstract

India is the second-largest refiner in Asia and the fourth largest globally. Its crude oil import was US dollars 101.4 billion in 2019–2020. The economic progress of India is closely linked to its energy demand. Therefore, the need for oil and gas is expected to grow more, making the oil sector more attractive for investment. It makes it indispensable to have a stringent check on the quality of imports and exports of petroleum products. The results from lab tests play a significant role in ascertaining the quality of petroleum products. Using BNDs ensures that laboratory instruments perform to the standard of predefined value, thus assuring laboratory tests' accuracy. With the nonavailability of indigenous certified reference materials, domestic refiners have been importing the materials from other economies. CSIR-NPL has partnered with HPCL and BPCL to produce 28 BNDs, and the list is growing continuously. The petroleum BNDs are expected to save foreign exchange and time, avoiding delays in receiving imported reference materials. Also, with the availability of BNDs to the Middle East, the African continent will love to grow our business. The petroleum BNDs include BNDs for the various concentration of sulfur, flash point (PMCC and Abel), smoke point, distillation standards, pour point, color saybolt, color ASTM, total base number, total acid number, FIA aromatics, olefins, saturates, density, kinematic viscosity at different temperatures, freezing point, etc.

Introduction

Certified reference materials (CRMs) are essential for ensuring the quality infrastructure of any economy through testing and calibration with accurate and precise measurements traceable to standard units (Aswal 2020). Reference materials are needed due to rapid industrialization in various industries, including the environment, food and agriculture, petroleum products, etc. (Clapp 1998). The primary standard that guarantees the accuracy and comparability of the measurements as a baseline for the quality assurance attained through global networking is Bharatiya Nirdeshak Dravya (BND) (Neyezhmakov and Prokopov 2014). Many academic institutions and enterprises are made economically feasible by the BNDs produced at CSIR-NPL, which acts as an import substitute.

Accurate analytical results are the foundation for many essential conclusions on trade, business, legal matters, and resolving disputes (Gundlach and Murphy 1993). Accurate analytical results entail using certified reference materials, validated procedures, trained personnel, and CRMs (Maier 1991). Numerous international, national, and professional organizations have formed and strongly advocated using CRMs in analytical chemistry for quality control objectives (Charantimath 2017). It is essential to characterize samples together with CRMs to sustain the measurement procedure's long-term reliability (Kestens and Koeber 2022). Introducing petroleum CRMs facilitates quick access to petroleum-based laboratories and aid in the production of high-quality petroleum data (Pant et al. 2020).

CSIR-National Physical Laboratory (NMI of India) produced 10 BND®s in 2018 for chemical properties in petroleum products as per ISO 17034:2016 and ISO Guide 35 accreditation and legal metrology alignment with NMI. To produce BNDs on a broad scale, coherence between these bodies and NMIs must be planned. Now more than ever, NMIs must expand their skills to create various CRMs and BNDs in response to market needs. CSIR-NPL has taken the lead in this area by establishing MoUs with numerous RMPs to meet national and international demand.

Significance of Reference Standard and International Network

From the Bharat Stage (BS) VI emission standards and air quality perspective, the most important parameter defined in the fuel quality specifications is the maximum sulfur content of gasoline and diesel fuels in both cases; sulfur content is limited to a maximum of 10 ppm in the proposed BS VI regulation, which matches global best practices (Bharj et al. 2019). In BS-VI fuels, sulfur content is one of the critical parameters; it helps to meet BS VI emission norms and to decrease environmental pollution, low carbon footprint, and significant improvement in human health (Borkhade et al. 2022). The quality of many petroleum products is related to the mass fraction of sulfur present (Orr and Sinninghe Damsté 1990). There are also regulations promulgated in international, national, state, and local agencies that restrict the amount of sulfur present in some fuels, which causes a release of harmful sulfur oxides into the atmosphere (Engdahl 1973).

To date, internationally known sulfur standards are available, viz., SRMs were produced by NIST (National institute of standards and technology, USA) and a few other global ISO 17034:2016 accredited RMPs. All users had to procure these imported standards at very high prices and with long lead times for receipt (Martin et al. 2000).

There were no established reference material producers in India to cater to requirements. HPCL (ISO 17034:2016 accredited reference material producer) and CSIR-National Physical Laboratory (NMI of India) came forward to address the need in the year 2018; a total of 29 BND®s were launched jointly, and 10 sulfur BND®s are also part of the same.

Significance of Indian Reference Standard

The use of petroleum-based BNDs/IRMs for metrological perspectives such as testing, calibration, method validation, repeatability, reproducibility studies, control chart preparation, inter-laboratory comparison, quality control, and quality assurance has been discussed (Vempatapu and Kanaujia 2017). A detailed production/development process starting from planning the intention to produce the parameter with matrix to preliminary studies of homogeneity, stability, validation, assignment of property value with its associated uncertainty, and complete certification process of petroleum-based BNDs/IRMs is described. It is also discussed how the BND/IRM Division supports the reference material producers (RMPs) and various commodities

to produce BNDs, in which the traceability is established by NPL(I). Through dissemination of BNDs and rendering services to the Accreditation Body and Bureau of Indian Standard (BIS), the quality infrastructure of India can be strengthened. The impact of BNDs in the formulation in trade is narrated along with the conclusions and future perspectives. Further, it is planned to produce a series of petroleum-based BNDs starting from quantum standard, thermoelectric metrology, nanometrology, etc., which will be the future avenue of metrological activities. BNDs are Indian reference standards developed by Indian NMI CSIR – National Physical Laboratory and other Indian reference material producers.

Significance of NMI of India in Petroleum-Based BNDs

To establish national standards and disseminate traceability throughout the nation, the National Metrology Institute (NMI) plays a significant and vital role (Aswal 2020). NMI is responsible for maintaining national standards and approved reference materials (Kumari et al. 2021). As the NMI of India, the CSIR-NPL not only took part in and contributed to the BIPM and APMP but also achieved many CMCs for the nation. Various county stakeholders are informed of the metrological traceability, directly or through legal metrology, and NABL-approved laboratories.

Petroleum Products (BNDs)

Petroleum BNDs are helpful in:

1. Bringing traceability of measurement to Indian National Standard.
2. Reduce dependence on imported reference materials.
3. Calibration and performance evaluation of measuring equipment.
4. Method (measurement procedure) validation.
5. Comparison of different measurement procedures, etc.

Stakeholders:
1. Govt. of India “Maharatna” CPSE, Forbes 2000 Company.
2. Refining and marketing of petroleum products.
3. Listed on BSE and NSE.
4. Owns two refineries (Mumbai and Visakhapatnam).
5. Stake in HMEL and MRPL.
6. Retail outlets – 20,025, LPG Dealerships – 6243.
7. Locations: 140, ASF-47, pipeline length: 3775 KM.

CSIR–NPL achieved a significant milestone by releasing 28 petroleum BNDs in association with HPCL (Pant et al. 2020). The 28 petroleum BNDs jointly developed by HPCL and CSIR-NPL will provide traceability for all vital parameters of petroleum products testing and certification, comprising 12 physical properties,

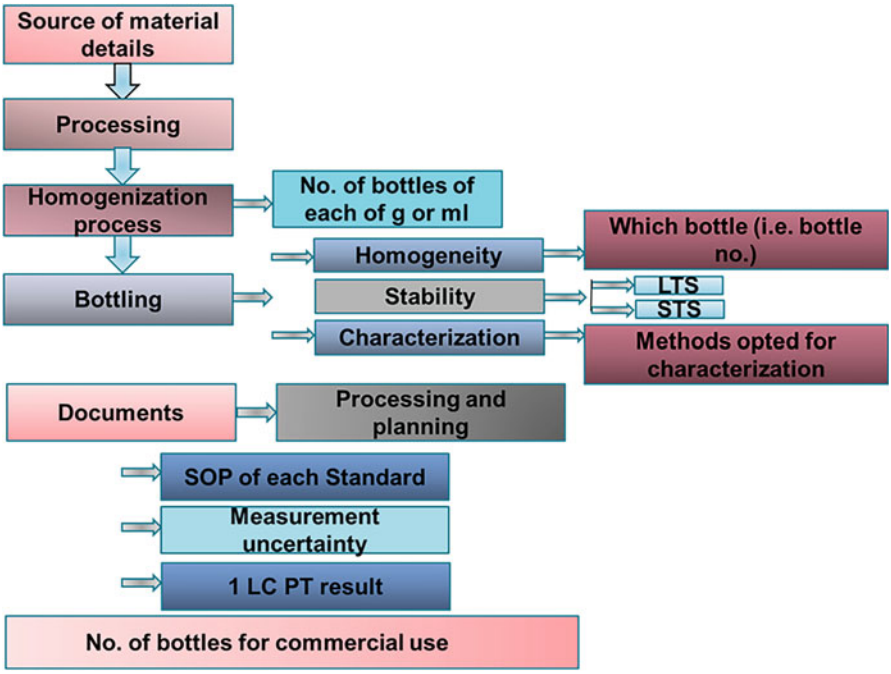

Fig. 1 Process of development of petroleum BNDs

3 physicochemical properties, and 13 chemical properties, including BND for sulfur content measurement at lower concentrations which will be of immense use for BS-VI fuels (Pant et al. 2020). The initiative will save vital foreign exchange through import substitution for certified reference materials (CRMs: BNDs). This unique initiative will be a milestone in ensuring our country's quality assurance of petroleum products. Figure 1 shows the protocols described as per ISO guidelines used for developing petroleum-based BNDs. Figure 2a and b shows honorable PSA to Govt. of India, R Chidambaram, DG-CSIR Dr. Shekhar C Mande, and other dignitaries releasing petroleum BNDs and petroleum BNDs by BPCL.

Physical, Chemical, and Physicochemical Parameter-Based BNDs

The physical properties-based petroleum BNDs have been developed and traceability for all vital parameters of petroleum products has been provided for testing and certification comprising 12 physical, 13 chemical, and 3 physicochemical properties. The flowchart shows the chemical and physical-chemical parameters of petroleum BNDs. The reference material producers are working in true spirit towards making India **"Aatmanirbhar Bharat"** as well as producing indigenous reference materials addressing in-line with **"Vocal for Local"** concept and thus contributing in response to the call of our Hon'ble Prime Minister Shri Narendra Modi towards making a

Fig. 2 (**a**) Releasing of petroleum BNDs by honorable PSA to Govt. of India, R. Chidambaram, DG-CSIR, Dr. Shekhar C. Mande, and other dignitaries and (**b**) petroleum BNDs by BPCL

"self-reliant India" (Paroda et al. 2020). In a step forward, HPCL started the development of BNDs in 2018–2019 in a tie-up with CSIR–NPL. Currently, 28 petroleum BNDs have been jointly developed by HPCL and CSIR–NPL. A few other petroleum-related BNDs are under development.

HPCL collaborated with CSIR-National Physical Laboratory (NMI of India) and produced 16 BND®s in 2018 for physical properties in petroleum products as per ISO 17034:2016 and ISO Guide 35, as represented in Tables 1 and 2. Lubricating oils are used in almost all industries for the maintenance of plant machinery, vehicles, engines, etc. (Holmberg et al. 2017). Total acid number (TAN) and total base number (TBN) standards help manufacturers to accurately dope the correct amount of additives in fuels/lubricants which will help prevent corrosion and related damage to engines and machinery components (Shahabuddin et al. 2012), which arises due to the formation of acidic/basic components (SOx and NOx) during the combustion process or the use of machinery. TAN/TBN is used as a guide in the quality control of lubricating oil formulations (Chamberlin et al. 1996). It is also sometimes used as a measure of lubricant degradation in service. TAN/TBN measured values can give information about the fitness of oil and balance useful life as shown in Table 3.

Table 1 List of chemical parameters of petroleum BNDs including diesel, gasoline, and kerosene

Sr. No.	BND	BND Details
1	BND® 7001	Sulfur Reference Standard in Diesel (mass fraction 2.5 ± 0.75 mg/kg)
2	BND® 7002	Sulfur Reference Standard in Diesel (mass fraction 5.0 ± 1.5 mg/kg)
3	BND® 7003	Sulfur Reference Standard in Diesel (mass fraction 10.0 ± 1.5 mg/kg)
4	BND® 7004	Sulfur Reference Standard in Gasoline (mass fraction 25.0 ± 2.5 mg/kg)
5	BND® 7005	Sulfur Reference Standard in Diesel (mass fraction 50.0 ± 2.5 mg/kg)
6	BND® 7006	Sulfur Reference Standard in Gasoline (mass fraction 50.0 ± 2.5 mg/kg)
7	BND® 7007	Sulfur Reference Standard in Kerosene (mass fraction 1000.0 ± 20.0 mg/kg)
8	BND® 7026	Sulfur Reference Standard in Gasoline (mass fraction 2.5 ± 0.75 mg/kg)
9	BND® 7027	Sulfur Reference Standard in Gasoline (mass fraction 5.0 ± 1.5 mg/kg)
10	BND® 7028	Sulfur Reference Standard in Gasoline (mass fraction 10.0 ± 1.5 mg/kg)

Table 2 List of physical parameters of petroleum BNDs including petroleum oils and mineral oils

Sr. No.	BND	BND details
1	BND® 7008	Density Reference Standard in Petroleum Oils
2	BND® 7009	Kinematic Viscosity at 40.0 °C Reference Standard in Petroleum Oils (Heavy Distillates)
3	BND® 7010	Kinematic Viscosity at 50.0 °C Reference Standard in Mineral Oils
4	BND® 7011	Kinematic Viscosity at 100.0 °C Reference Standard in Mineral Oils
5	BND® 7012	Kinematic Viscosity at (−)20.0 °C Reference Standard in Petroleum Oils/(Jet A1)
6	BND® 7013	Freezing Point Reference Standard in Petroleum Oils/Middle Distillates
7	BND® 7014	Flash Point-Abel Reference in Petroleum Oils/Middle Distillates
8	BND® 7015	Flash Point-PMCC Reference Standard In Mineral Oils
9	BND® 7016	Flash Point-COC (Cleveland Open Cup) Reference Standard in Mineral Oils
10	BND® 7017	Smoke Point Reference Standard for Petroleum Oils
11	BND® 7018	Distillation Properties Reference Standard of Petroleum Oils
12	BND® 7019	Distillation Properties Reference Standard of petroleum Oils
13	BND® 7020	Pour Point Reference Standard of Petroleum Oils, Lube, or Mineral Oils
14	BND® 7021	Color Saybolt Reference Standard of Petroleum Oils, Mineral Oils
15	BND® 7022	ASTM Color Reference Standard of Petroleum Oils, Lube, or Mineral Oils
16	BND® 7025	Aromatic, Olefins and Saturates content Reference Standard in Petroleum Oils

The quality of petroleum products depends on many physical properties measured values such as density, kinematic viscosity, distillation, pour point, color, and flashpoint (Al-Besharah 1989). Measurement of these physical properties plays a vital role in international trade, crude refining process, storage and transportation of

Table 3 BND details of TAN/TBN lubricant mineral oils

Sr. No.	BND	BND details
1	BND® 7023	Total Base Number – Reference Standard of Lube or Mineral Oils
2	BND® 7024	Total Acid Number – Reference Standard of Petroleum Oils, Lube, or Mineral Oils

Table 4 Chemical and physicochemical parameters of petroleum BND

S. No.	Measurand	Certified value and uncertainty
1	Color ASTM	4 ± 1 color ASTM unit
2	Color saybolt	24 ± 1 color saybolt unit
3	Flash point	40.5 ± 1.5 °C
4	Viscosity at 100 °C	10.64 ± 0.04 cSt
5	Density (kg/m^3)	799.8 ± 3.2
6	Viscosity at 40 °C	2.917 ± 0.02 cSt
7	Viscosity at 50 °C	170.2 ± 0.6 cSt
8	Viscosity at −20 °C	3.171 ± 0.022 cSt
9	Freezing point	−57.50° ± 1.5° C
10	Flash point PMCC	74.5 ± 3.5° C
11	Flash point COC	212 ± 15° C
12	Pour point	−6.00 ± 3.0° C

various petroleum products, and calibration and a performance check for measuring equipment (Wauquier 1995). There are also regulations and specifications promulgated by international, national, state, and local agencies that restrict the limits of these physical properties to address basic requirements, viz., environmental aspects, fuel economy, and technological advancement in the automobile sector. In recent, internationally known standards were available viz. SRMs were produced in NIST (National Institute of Standards and Technology, USA) and a few other global ISO 17034:2016 accredited RMPs. All users had to procure these imported standards at very high prices and with long lead times for receipt. Table 4 represents the chemical and physicochemical parameters of petroleum BNDs.

Density

Density is crucial to the performance of both machines and lubricants (Gropper et al. 2016). The mass per unit volume of a substance determines its density. For example, an object constructed of steel or iron will have less volume than an identical object made of a variety of less dense materials (e.g., water, petroleum ether, etc.). A substance's density varies with pressure and temperature. For solids and liquids, this variation is small; however, for gases, it is significantly more. Since most systems are made to pump a fluid with a given density, their efficiency changes when the fluid's density changes (Mwesigye and Yılmaz 2020). The main technique for establishing the traceability of reference hydrometers is the hydrostatic weighing

Table 5 Industry/market demand of various fluid

Fluids	Industry	Manufacturer (India)
Air compressor Lubricants	Rotary screw compressors	Indian Oil Corp. Ltd. (IOCL) Bharat Petroleum Corp. Ltd. (BPCL) Raaj Unocal Lubricants Ltd. Masterline Lubricants Pvt. Ltd. MOSIL Lubricants Pvt. Ltd. Castrol India Ltd. Rizol Petro Product Pvt. Ltd. ELF India, Gulf Lubricants Shell India Pvt. Ltd. GS Caltex India Pvt. Ltd. Valvoline Cummins Ltd. Amul Ltd. Asian Paints Ltd. Etc.
	Reciprocating compressors	
	Air compressor cleaners	
	Breathing air compressor	
	Blower lubricants	
Refrigeration Lubricants	Ammonia and CO_2 lubricants	
	HFC and HCFC lubricants	
Heat transfer fluids	Different temperature ranges	
Industrial lubricants	Chain lubricants	
	Gas compressor lubricants	
	Gear lubricants	
	Hydraulic fluids	
Concrete	Cement industry	
Ketchup, cream, etc.	Food industry	
Paint	Paint industry	

method (also known as the Cuckow's method) (Kumar et al. 2013). This method uses a balance and the Archimedes principle to measure the actual masses of a hydrometer in air and liquid at a specified stem point scale. When using hydrometers in the lower range, a sinker with known mass and volume is employed for Hydrometer calibration by hydrostatic weighing with automated liquid surface positioning (Aguilera et al. 2007). The primary solid density standard, the silicon sphere, can be used to calculate density (mass/volume) and is traceable to mass and length standards (for volume) (Gupta 2002). Due to silicon's low density and low thermal expansion coefficient, the need for air buoyancy correction is limited (Mizushima et al. 2004). Liquid is chosen according to the required stem scale point, and its density is evaluated using a silicon sphere (Mizushima et al. 2004).

Density Certified Reference Materials are used to verify laboratory equipment used to test petroleum and derivative products for density in accordance with methodologies including, but not limited to, ASTM D4052, IP 365, and other comparable international methodologies (Zanão et al. 2018). Kilogram per cubic meter (kg/m^3) is the SI unit of density (Table 5).

Viscosity

A fluid's viscosity is a gauge of how difficult it is for it to flow. It refers to the internal friction of a fluid in motion. Because of the high internal friction caused by its molecular structure, a fluid with high viscosity opposes flow (Thorpe et al. 1894). A fluid with lower viscosity, on the other hand, flows smoothly because of its molecular structure, which produces very little friction when it is in motion. The step-up approach is used to establish standard viscometers and standard viscosity oils using distilled water as the primary standard and water's kinematic viscosity (Fakhrabadi 2021). First, two master viscometers are calibrated using water at 20 °C and

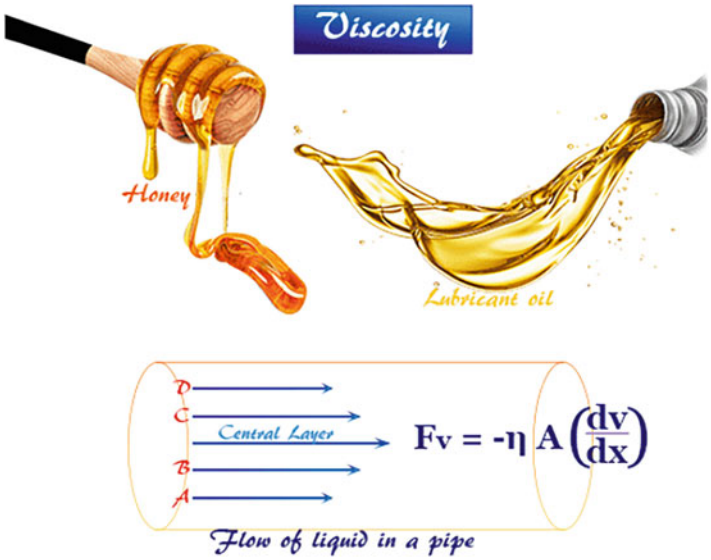

Fig. 3 Viscosity explanation with the help of fluid flowing through the pipe (https://www.priyamstudycentre.com/2020/11/viscosity.html)

calibration constants between (0.001–0.003) mm^2/s^2. These two master viscometers evaluate the kinematic viscosity of two standard oils at 40 °C. The appropriate corrections are performed for surface tension, temperature, and buoyancy.

In Fig. 3, the fluid is flowing in a pipe. The fluid does not move at the same speed throughout the pipe as it is flowing. Almost stationary is a thin layer that is in direct touch with the pipe wall. The core layer's velocity increases as you move away from the pipe's wall and stays at its highest level. As a result, the velocity of the flowing liquid has a gradient (dv/dx). The internal friction of the two layers causes each layer of the liquid surface that is flowing at a higher velocity to have a retarding effect. Viscosity is the amount of resistance a fluid offers to another fluid part moving at a different speed than it does to a portion moving at its own speed (Vogel 2020).

Flash Point

The lowest temperature at which a liquid (often a petroleum product) will produce a vapor in the atmosphere at its surface that will "flash," when in contact with an open flame (Stachowiak and Batchelor 2013). The flash point is a broad indicator of a liquid's combustibility (Glassman and Dryer 1981). There is not enough vapor present below the flash point to support combustion. Flash points are determined by carefully controlling the heat of a liquid to a range of temperatures, then applying a flame to the vapor rising to the liquid's surface. Either a "open cup" or a "closed cup" device is used for the test (Lance et al. 1979). The sample is placed into a test cup that is totally open at the top for the open cup test. Before the sample is heated, a thermometer is placed inside it. Every time the sample temperature rises by 2 °C, the test flame is passed over the cup. The flash point is attained when the sample vapors temporarily catch fire. The ASTM D92 Cleveland Open Cup (COC) test is the most widely utilized

Fig. 4 Picture of flash point process (https://www.bcl.co.za/news/q8oils-and-blue-chip-sign-lubricant-agreement-for-s-a-market/)

test procedure (Phillips and Totten 2003). In the closed cup testing, the sample is put into a test cup with a lid that is sealed and opened when a flame is applied as the source of ignition. As opposed to the open cup approach, when the vapors are exposed to the atmosphere, the closed cup technique captures all of the vapors produced during the heating of the sample. The fact that the closed cup test produces fewer flash points than the open cup test is therefore not surprising. Closed cup flash points are often calculated using the ASTM D93 Pensky-Martens Closed Cup (PMCC) test (Catoire and Naudet 2004). Flash point process picture is shown in Fig. 4.

Distillation

The term "refining of petroleum" refers to the process of separating the different components of petroleum from one other (Meyers 2016). This is accomplished by a procedure known as **fractional distillation**, which is premised on the observation that the various petroleum constituents have distinctively different boiling points (Shishkin 2006). Crude petroleum is heated in a furnace to a temperature of around 400 °C or somewhat higher during fractional distillation. The three most significant petroleum fractions that result from refining are as follows: gasoline, kerosene, and petroleum gas. Apart from these, diesel, lubricating oil, paraffin wax, and bitumen are the constituents of petroleum as shown in Fig. 5.

Pour Point

The pour point, which determines the temperature below which crude oil turns plastic and ceases to flow, is crucial for recovery and transportation (Ilyin and Strelets 2018). The range of pour points is from 32 °C to below −57 °C. Standard Test Procedure for Pour Point of Crude Oils is ASTM D97 (Venkatesan et al. 2002). For the purpose of promoting the growth of paraffin wax crystals, the specimen is cooled inside a cooling bath. The test jar is removed and inverted to look for surface movement at around 9 °C above the anticipated pour point and then every 3 °C after that. The jar is maintained horizontally for 5 s when the specimen does not flow when tilted. The pour point temperature is calculated by adding 3 °C to the

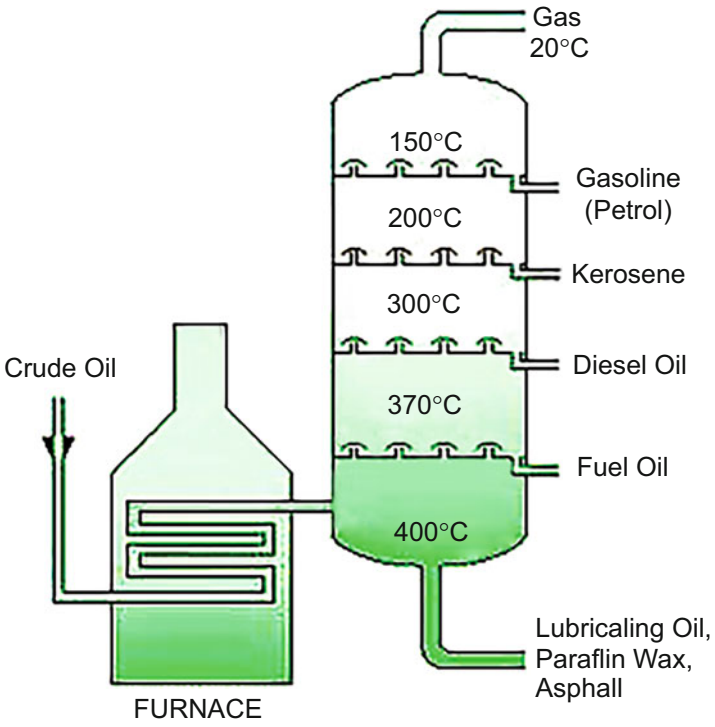

Fig. 5 Schematic of refining process and constituents of petroleum (https://www.geeksforgeeks.org/fractional-distillation-of-petroleum/)

equivalent temperature if it does not flow. It is also important to keep in mind that failure to flow at the pour point could potentially be brought on by viscosity or the specimen's previous thermal history (Donaldson et al. 1985). As a result, the pour point could provide a misleading impression of the oil's handling properties. The conduct of additional fluidity tests is also possible. The upper and lower pour points of the specimen can be used to determine an approximate range of pour points. An alternative to the manual test process is ASTM D5949, Standard Test Method for Pour Point of Petroleum Products (Automatic Pressure Pulsing Method) (Nadkarni and Nadkarni 2007). When reporting at a 3 °C temperature, it employs automatic equipment and produces pour point values in a format identical to the manual method (ASTM D97) (Standard 2001).

Dynamic Viscosity BND™ and BND® of Non-Newtonian Fluids Using Magneto-Rheometer and their Instrumentation

The CSIR-National Physical Laboratory, India (NPLI) develops Indian Certified Reference Materials (BND™) for research into accurate measurement methods, calibrating instruments, and providing quality control. NPLI certifies the BND for

rheological properties of non-Newtonian fluids, which show shear rate dependence of viscosity. Accurate measurements of these material properties are critical for predicting the behavior of these fluids in the complex flows they experience during processing and use. The development of this new BND is based on a multiphase approach. A round robin test of the BNDs is also planned, involving instrument manufacturers and users in industry and at universities. The round robin will assess the lab-to-lab variability in these measurements.

This BND (IRM) is intended for use in the calibration of the rheometer for the measurement of the rheological properties of non-Newtonian fluids. This consists of a nanofluid (10 ml $\pm$ 0.01 ml) with the dispersion of Fe_3O_4 nanoparticles (with a particle size distribution of 8–10 nm) in kerosene.

Absolute viscosity (also known as dynamic viscosity) is the measurement of the fluid's internal resistance to flow (Špeťuch et al. 2015). It is the ratio between the applied shear stress and the rate of shear of a liquid.

NPLI develops the BND for rheological properties of non-Newtonian fluids, which show shear-rate dependence of viscosity. Accurate measurements of these material properties are critical for predicting the behavior of these fluids in the complex flows they experience during processing and use. Figure 6 shows parallel plate geometry for dynamic viscosity measurements.

The measurement of non-Newtonian liquids is only possible with the rotational viscometer (ISO 3218.2) (Yu et al. 2022). In this type of rheometer, the tested fluid is sheared between two parallel surfaces, one of which is rotating. Each determination of viscosity must be accompanied by the temperature (with a constancy of $\pm$ 0.2 °C) at which the measurement was made. Usually, the angular velocity is imposed and the response of the material is monitored by the measurement of the torque. The range of this viscometer is 10 mPa-s to 10^9 mPa-s. The manufacturers recommend the use of standard oil of known viscosity to verify that the instrument is operating correctly. For non-Newtonian fluids, the results obtained are preferred in the form of flow curves which must be interpreted, assuming the validity of various laws of flow (Coleman et al. 2012). One very important aspect of providing certified data for a BND is to include estimates of uncertainties in the data as well as in any quantified

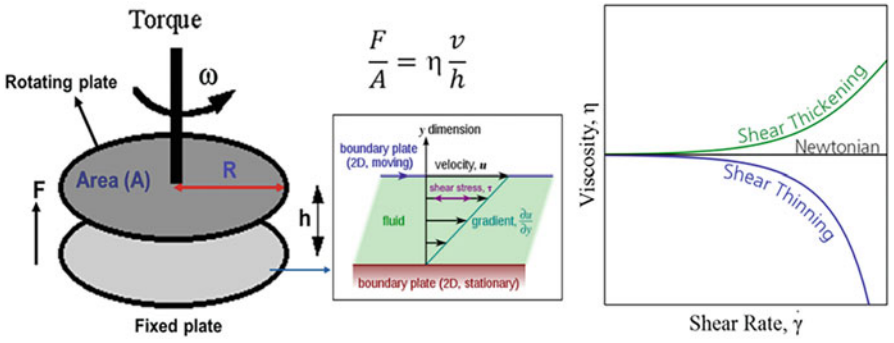

Fig. 6 Parallel plate geometry for dynamic viscosity measurements

parameters. The test report is to be drafted in accordance with the specifications in the standards. It includes individual and mean values at each temperature.

Dynamic Viscosity BND® of Newtonian Fluids (Silicon Oils)

Almost every industry that deals with liquids or semiliquids relies heavily on Newtonian fluids, particularly silicone oils, because of their unique properties and behavior (Lansdown 2004). The primary criterion for determining a fluid's viscosity to ensure high-quality goods is dynamic viscosity (η) (Chhabra 2010).

A fluid's dynamic viscosity, which measures the flow-resisting force a fluid exerts, is defined for a Newtonian fluid as

$$\eta = \tau/(dv/dn)$$

where τ is the shear stress, v is velocity of fluid in the shear-stress direction, and dv/dn is gradient for v in the direction perpendicular to flow direction. Concentric (or coaxial) cylinder systems (DIN EN ISO 3219 and DIN 53019) are an effective instrument for assessing dynamic viscosity, which measure the dynamic viscosity. With the help of rheometer, shear rate and shear stress are calculated with precision using bob and cup geometry to calculate the dynamic viscosity as shown in Fig. 7a–c.

From concentric cylinder geometry, the shear rate is (Pant et al. 2020)

$$\gamma = 2 \times \omega RC/(RC)2 - (Rb)2,$$

and shear stress is (Pant et al. 2020)

$$\tau = M/2\pi(Rb)2$$

These equations describe how the measured torque, the bob's radius, and its length affect the shear stress on the surface of the bob. Newton's law uses the shear

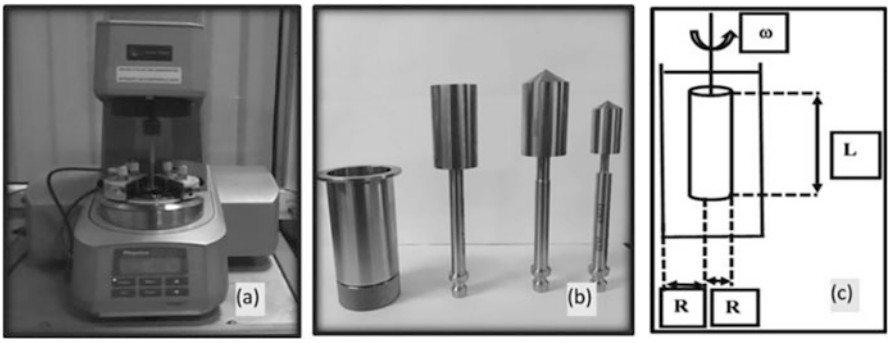

Fig. 7 (**a**) Magneto-rheometer; (**b**–**c**) cup and bobs of concentric cylinder geometry

rate (γ) and shear stress (τ) factors to calculate a Newtonian fluid's dynamic viscosity (η). For silicon oil, measurements are performed in the temperature range of −20 to 100 °C and a dynamic viscosity range of 1 Cp to 105 Cp.

Multi-Way Principal Components Analysis (MPCA)

Decreasing the number of dimensions of a complex data can be done by marking the greatest variances' direction. This method is called principal components analysis (PCA) (Rao 1964). One of the popular methods for algorithms is to decompose a matrix that is being observed, say X, into a set of scores T and loadings W while maintaining the form T = XW; this is singular value decomposition (Höskuldsson 1995). Each constituent of the loading's matrix W will be able to define one component of the low-dimensional subspace which will be interpreted, e.g., a chromatogram. The dimension is decreased by maintaining only the L constituents of W that define more than a threshold variance in the data, where L < the rank of X. So, this requires the complex data to be arranged in form of a matrix, a two-way array to be precise. We can use this method for higher orders also, then it becomes a multi-way array. In the case of a three-way array, the matrix X must have some I × J × K dimensions, which should be arranged into a two-way array X' of dimensions I × JK. One very good example of a multi-way array is the arrangement of GC-MS, because this three-way array is in such a way that each way represents different profiling, mass spectra, and elution times. Now the two-array equivalent will have different profiling and the mass spectra along with the chromatograms (de Carvalho Rocha et al. 2017).

The use of the MPCA approach in Elektrociepłownia Białystok is made to find leaks in a steam boiler's piping system before they cause any leak (Indrawan et al. 2021). When a leak first appears in a pipeline or when it enlarges to the point where the operational staff can clearly identify failure signs, it can be quite difficult to pinpoint exactly when it occurs in an industrial steam boiler. The length of a boiler's pipeline section is several dozens of meters, and cracks might occur anywhere along it. As a result, the development of these faults often varies over time and has a very diverse impact on process variables. In many cases, a single leak results in cracks in nearby tubes, a process known as failure propagation and multiplication (Davoodi and Mostafapour 2014). Measurements of a number of process variables must be used in the tube leak detection method, and they must be correlated with common defect patterns (Swiercz and Mroczkowska 2020).

Parallel Factor Analysis (PARAFAC)

Now we can develop our solution into a multidimensional one, in other words, a multidimensional equivalent of principal component analysis, which is named as parallel factor analysis (PARAFAC) (Harshman and Lundy 1994). In this method, the observational matrix X will be divided into scores matrices A, B, and C such that its dimensions are I × F; J × F; K × F, respectively, here F represents a number of factors, an analogue of number of components (L) in the MPCA method. This

analogue makes it clear that the value of F is always less than the rank of the matrix X. If we can make it so that all the A, B, and C matrices are positive at all the time, then we can solve PARAFAC using a nonlinear optimizer like alternating least squares (Paatero 1997).

Kohonen's Self-Organizing Map (SOM)

Kohonen's self-organizing map is one method in which the data is projected in a 2D space as close as possible to the interpretation (Su and Chang 2001). We can say it is a neural network. It takes the manipulation of the data required to map the 2D space and location of the data space to project, corresponding to a particular node. As each iteration runs, the data in each node comes closer and closer to the original sample data. There should be more nodes in the set than the number of sample data so that a particular data can be associated to its neighboring nodes in the neural map. The more similar the sample data, the closer they are in the neural map; so, this can analyze the nonlinear relationships within the sample database. For example, a straight line means a convolute and a nonlinear path. The nodes in the neural map can be mapped in terms of a L × N grid. We can describe it into a three-way array M with dimensions L × N × S. There are two types of distances between nodes. The distance between two nodes M_{ln} and M_{op} in the data set is the Hellinger distance D_h. The distance between the same two nodes in the map is the Euclidian distance D_E. This is given by the equations below.

$$D_H^2(M_{LN}, M_{OP}) = 1 - \Sigma_s(M_{LNS}M_{OPS})^{\frac{1}{2}}$$

$$D_E^2(M_{LN}, M_{OP}) = (L - O)^2 + (N - P)^2$$

Conclusion

CSIR-NPL has launched the mission mode program on the development of BNDs in diverse sectors with consideration for the increasing importance of reference materials in the worldwide market. This chapter focuses on the production of petroleum-based BNDs. CSIR-NPL has produced 29 BNDs in collaboration with HPCL and BPCL, and the list is continually rising. Numerous physical properties, including density, viscosity, distillation, pour point, and flashpoint, are assessed to determine the quality of petroleum products.

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Part V

Industrial Metrology: Opportunities and Challenges

Industrial Metrology

32

Introduction

Sanjay Yadav, Shanay Rab, S. K. Jaiswal, Ashok Kumar, and Dinesh K. Aswal

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Abstract

Metrology, in a broad term, is defined as the science of measurement or the study of measurement. The term metrology is the combination of Greek terminologies "metron" and "logos," which refers to the study of measurement, including both experimental and theoretical. In the present era of globalization, it is an imperative element in many fields, including manufacturing, engineering, science, and technology. It is used in diversified applications, including characterization and structural analysis in engineering; construction of different products in aerospace; process and quality controls in manufacturing; energy generation, transmission, and distribution in energy; and the development of medical products, devices, safety standards, diagnostic, monitoring, control, and therapeutic purpose in healthcare. The list of applications is endless, and only a few are mentioned. Metrology is applied to corroborate and confirm already defined specifications and standards in these applications. The metrological traceability of the standards is often established by calibrating, testing, and verifying them against better standard(s), supported through a well-recognized quality system implemented in the certified, accredited laboratories authorized by certification bodies. A fleeting investigation of the metrological practices being followed is presented in this chapter, enabling an understanding emphasizing the evolution of industrial revolutions, some of the application-oriented trends in industrial metrology, the contribution of metrology in achieving Sustainable Development Goals (SDGs) set by United Nations Organisation (UNO), and the role of industrial metrological measurements in the national quality infrastructure. It is hoped that the chapter will provide a good resource material in a single window for the readers following and working in the field of industrial metrology.

Keywords

Metrology · Standards · Accreditation · Traceability · Industries

Introduction

Life began in this universe more than a million years ago. Humans started living as a part of nature, finding food articles and water sources for survival and searching dwellings and shelters in forests, plains, rivers, caves, mountains, etc. Survival mainly depended upon the convenience of available resources based on plants, creatures, and animals within their reach in nearby places with a workable distance and their ability to collect and catch them. No one might have thought of any need for measurement in these ancient communities. However, they might be using some measurement assessment like the food is heavy or light, good or bad, far or near, adequate or inadequate, etc. However, in all walks of human life, humanity tends to explore and evaluate qualitatively and quantitatively various things around, say objects, elements, phenomena, and processes. The only way to acquire objective evidence and data is to perform measurements. Lord Kelvin has rightly said that "When you can measure what you are speaking about and express it in numbers, you know something about it; but when you

cannot express it in numbers, your knowledge is of a meager and unsatisfactory kind; it may be the beginning of knowledge, but you have scarcely, in your thoughts, advanced to the stage of science, whatever the matter may be." Of course, our knowledge and expertise are highly dependent upon the measurements. Measurement is thus the heart of all our activities in the material world, in industry, academia, legal, trade and commerce, space, defense, etc. Measurements are characterized as scientific and metrological measurements (Aswal 2020a, b; Rab et al. 2020; Peiser et al. 1979; Fanton 2019; Yadav et al. 2020a, b; Chakrabarti et al. 2008; Gupta 2009; Javaid et al. 2021; Garner and Vogel 2006; Kumar and Albashrawi 2022; Zafer et al. 2019).

Scientific measurements are performed to provide exact and comprehensive information about the phenomena extracted from interfering convolutions in terms of characteristic principles and theories of nature. The metrologist is the trained expert to perform such metrological experiments and compare them quantitatively with an established base for validation because only measurements are not adequate to fulfill the objectives of science. It is the duty and responsibility of metrologists to make descriptive, comprehensive, and valid interpretations of the measurements made, compared, and validated in his/her evolutionary ability to forecast and correlate with unexplored natural phenomena. Such investigations, results, and insights are well recognized and receive appreciation from the scientific community. With continuously improved precision and accuracy, metrologists are now able to design experiments and perform measurements to meet societal needs (Peiser et al. 1979; Fanton 2019; Yadav et al. 2020a; Bailey 2017; Lean 2015; Rab et al. 2021a; Yadav and Aswal 2020; Mari and Wilson 2014).

Industrial metrology basically deals with the application of measurement science in industrial manufacturing, processes, quality controls, monitoring, recording, and their use in life, health, trade, and society for ensuring the appropriateness of measuring instruments/devices, their traceability, and quality control. Metrology is categorized into three subareas (Fig. 1) as (i) scientific or fundamental metrology, (ii) applied or industrial metrology, and (iii) legal metrology. Fundamental metrology deals with the realization of SI units, unit, and quantity systems, establishing primary and secondary standards, and developing need-based new measurement methods and techniques. Applied metrology deals with disseminating measurement standards to users calibrating, testing, and verifying their instruments, maintaining secondary and working standards, and quality control of measuring instruments used in the industry. Legal metrology deals with the regulatory framework and regulatory aspects of measurement and measuring devices (Aswal 2020a; Rab et al. 2020; Peiser et al. 1979; Fanton 2019; Yadav et al. 2020a).

Industrial Revolution

Historically, the transformation of the economy from ancient handmade goods and handicrafts to machine-made or industrial processing manufacturing started in the eighteenth century in Europe and especially Britain with the introduction of innovative methods, techniques, and ways of doing things. Accordingly, society's living status also changed enormously, later spreading globally. Arnold Toynbee (1852–1883), the

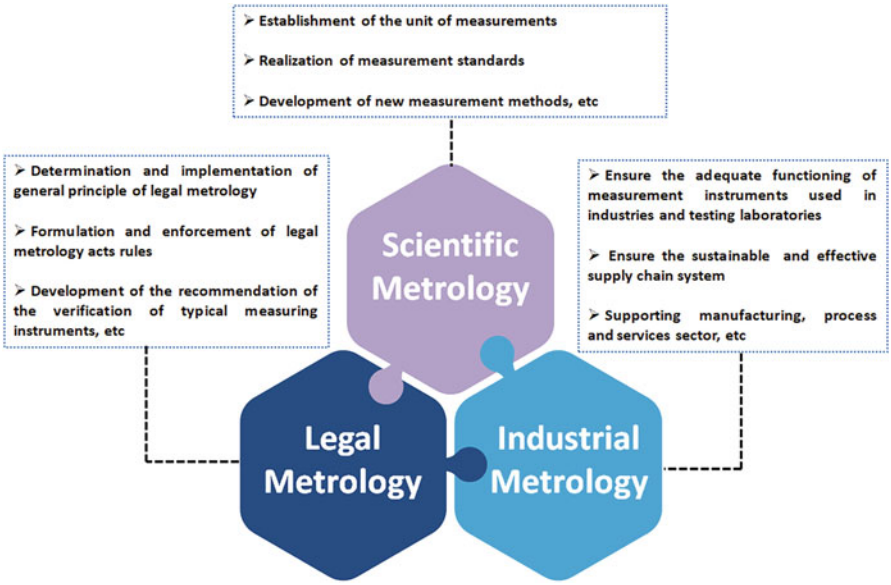

Fig. 1 Subareas of metrology

famous historian and writer, introduced the terminology for changes during the industrial revolution. However, such changes of the industrial revolution could not enter India and China until the twentieth century, while it entered the USA and Western Europe by the nineteenth century. The industrial revolution certainly revolutionized all walks of human life. This visibly transformed the social, economic, technological, and cultural status. The people started using iron, steel, new sources of energy like fuels and mechanical power, coal, steam, petroproducts, electric power, thermal energy, spinning wheels, power loom, manufacturing units, factories, various transport, and communication means like the steam engine and ships, automobiles, aviation, telephone, telegraph, radio and later on TVs, frequent and increased application of science to industry utilizing natural resources, and the large-scale production of manufacturing articles and goods (Popkova et al. 2019; Pozdnyakova et al. 2019; Koc and Teker 2019).

Further emphasis was simultaneously placed on the development of nonindustrial sectors to cater the raw material demands, mainly improvement for the agricultural and farming sectors, improving wider distribution network; cooperation for international trade, regulations, and policies to help and assist in fulfilling needs of the industrialized society; and edifying renovations placing focused attention on imparting training and enhancing skills of the workers to enable them to utilize machines, tools, devices, instruments in factories, plants, and manufacturing hubs (Varshney et al. 2021; Rab et al. 2022a).

As mentioned above, the ***First Industrial Revolution*** started in the eighteenth century with the introduction and use of steam power and machine for manufacturing.

For example, the threads prepared using hand-driven spinning wheels were transformed into power-driven mechanized looms, increasing the production of thread manifolds and saving time. The introduction of the use of steam power in industrial units and transportation was the breakthrough of the era employing steam engines and steamships for many applications. The ***Second Industrial Revolution*** started with the invention of electricity during the nineteenth century. With the use of electricity, industries could exploit several natural resources that were not hitherto utilized, such as metals, rare earth materials, new compositions and alloys, and synthetic products, new energy sources. Newer machines, tools, and devices were developed and utilized for a number of industrial applications, which also gave rise to automatic factories. Although fewer factories were partly made automatic, complete automation could only begin in the early twentieth century with the introduction of computers. The third quarter of the twentieth century may be characterized as the beginning of the ***Third Industrial Revolution*** with the advancement in automation with the use of programmable memory chips, controls, and computers. This could result in the automation of the whole production line with minimal human interventions, for example, robot-driven assembly lines.

The ***Fourth Industrial Revolution,*** also known as ***Industry 4.0***, is said to be started at the beginning of the twenty-first century with the technological application of information and communication in the industry. The computer-assisted production lines are being expanded, modernizing high-speed network connections, using cloud computing and digital twins. Potentially strong, Industry 4.0 is now able to deliver some implausible innovations and advancements in the industrial manufacturing sector. Now, smart machines, as part of smart factories, can assess, identify, detect, and predict failures and initiate, trigger, and alarm maintenance processes automatically for the probable plan, act, carry out maintenance, and rectify errors amicably. The digitalization of the manufacturing process enables flexibility for acquiring and passing correct information to the appropriate staff at the right time. In a nutshell, Industry 4.0 has proven to be a game changer for industrial units. Digital manufacturing would probably change how we think and work and finally result in how products are manufactured, repaired, maintained, serviced, refined, and improved.

After passing less than a decade since the introduction of Industry 4.0 and the interaction of machines with humans, the next industrial revolution is already in sight called the Fifth ***Industrial Revolution 5.0 or Industry 5.0***. In this revolution, humans and machines interact extensively and work together to find and improve the ways and means for increasing production efficiency. It is very interesting to mention that Industry 4.0 and Industry 5.0 are both in the same era. It is a very complicated situation for those industries tuned and adopting Industry 4.0 and simultaneously feeling the need for Industry 5.0. Sometimes, it is funny and confusing for them to go for well-established Industry 4.0 or choose Industry 5.0, which is just at the introductory stage. The financial viability of the newer products is also an issue to be tackled as it is determined using costing software for the manufacturing industry. However, it could easily be inferred that Industry 4.0 is for today and Industry 5.0 is for tomorrow.

Until the introduction of the First Industrial Revolution, metrology was not recognized as an important activity because all the available products prior to the industrial revolution were handmade. The need for measurement was probably felt necessary and important when post handicraft era, the produced goods flowing out of the production line were similar/identical in safe, size, dimension, weight, volumes, etc. Such requirements in industrial production could give rise to the concepts of quality control, quality assurance, etc. These developments had an impact on measurement science and technology. Accordingly, for the first time, the man felt the need to have an international system of units for "all people all the time," and indeed it is now called the SI system of units. Figure 2 shows the different industrial revolution timelines and their major theme.

The recent COVID-19 outbreak and the use of digital and online technologies have changed the ways we used to work and created/opened new lines of research and means of utilization for effective, timely, customer-oriented, driven, and satisfying products and services. The visionaries have now visualized the concept of ***Industrial Revolution 6.0***. Virtual reality and artificial intelligence (VR/AI), the Internet of Things (IoT), and cybersecurity are the emerging technologies pushing for the Sixth Industrial Revolution. The main focus of the Sixth Industrial Revolution is on medical technology, infusing quick and accurate fully automatic medical diagnosis, adaptive manufacturing/printed controlled medicines, machine, and body parts, and assisting medical practitioners and critical administrative workers absolving their additional work burden to enable them to focus on many trivial and critical issues. Search and use of alternative energy sources, robotic cleaning, task-specific

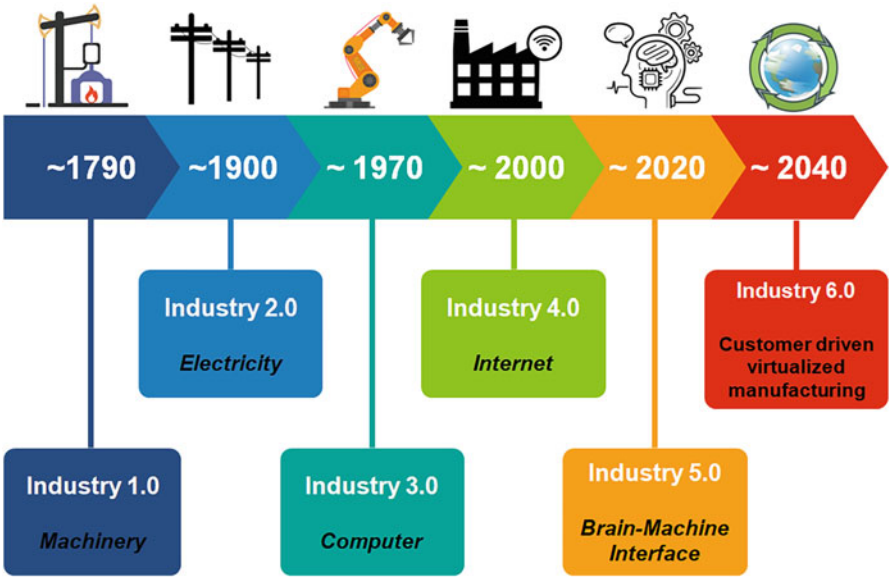

Fig. 2 Industrial revolution timeline

deployment, and fusion of novel and innovative ideas would lead the future in Industry 6.0 (Chourasia et al. 2022).

Some of the Application-Oriented Trends in Industrial Metrology

For the sake of the simplest example in dimensional quantities, i.e., the values of length and width of a component part, the exact measurement uncertainty in manufacturing such component parts is extremely important in the industrial sector. The noncompliance cost often surpasses the price of the manufactured product. Sometimes, even it results in the recalls of finished and marketable manufactured goods from the market. This not only tarnishes the image of the manufacturers but also results in heavy losses to customers as well as the company. Most of the time, outcome is legal proceedings and even incurs penalties, mental harassment, physical injuries, and violation of set regulations. From the customer's point of view, damage to a brand's name is seriously endangered due to the amount of noncompliance, besides keeping safety and health at risk. Further, noncompliance breaches confidence, a focal point and significant for establishing the brand name today. This is why most companies working in a globally competitive market have an effective quality control system backed by well-established metrological infrastructure (Wang et al. 2018; Bauer et al. 2015; Schmitt et al. 2016; Rab et al. 2021b).

Similarly, metrological processes and quality control are essential strides in the manufacturing industry. Actually, the quality control process is an investment in the long run. Few industries still avoid it as much as possible, bearing in mind that it is an additional cost to consumers. Even nowadays, in many developing nations leading to produce in bulk quantities, it is implemented to a bare minimum for avoiding slowing down the production line. This mindset has to be changed to keep pace with the brand establishment and recognition.

Nowadays, the core issue is not producing/manufacturing more but producing/manufacturing better and as per demand. Here lies the role of industrial metrology, which should help and support the production process via data collection to correct the process. If it is impossible to correct the process, at least it should be used to halt the process before it starts producing squandered morsel. The choice of appropriate measurement standard is mainly based on situation to situation, considering several issues like the measurement uncertainty needed, type and pace of measurement performed, and operational easiness and use.

Global metrological and technological advancements are approaching semi- or full automation, rapid, quick, and flexible measurement alternatives. Financial aids are easily available for viable projects and measuring solutions, which were not accessible due to high cost, are now available. It is very difficult to judge which is the top-class measuring instrument or the best measurement process control device. It all depends upon the precision, accuracy, and wholeness of the operational functions. For example, a measuring device should have process requirements of faster results, no environmental hazards, and compliance with safety standards. Normally in the automation process, ease of operation and mounting are highly important.

The applications related to measurements in manufacturing, manufacturing control, monitoring, and recording processes for use in society, for judging the operational integrity and ensuring the suitability of measuring instruments/devices, their calibration, testing, inspection, verification, and quality control fall under the category on industrial metrology. The last two decades have witnessed enormous technological advancements in industrial metrology. In order to complete a metrological or process task, quickly, precisely, reliably, and accurately, a single setup or a combination of multiple systems is required. One such beautiful example is the use of computer tomography (CT). CT was first invented and used in health science during the year 1972. Since then, many advancements have been carried out in CT scanning to examine optically, measure, and control the different component parts and methodologies. It is a wonderful technology to describe the number of metrological steps in a complete process, i.e., multiple inspection systems and techniques: (i) electromagnetic testing involving a measure of electric current or magnetic field; (ii) liquid penetrant testing having low viscosity fluids for the material under test, which dribbles through defects/cracks/pores and then controlled using a developer to bring it upward to assess the proper flaw; (iii) destructive testing, allowing damage of physical part for the inspection of internal parts; and (iv) visual inspection involving the operator to continuously monitor/gauge the test part using magnifier or computer (De Chiffre et al. 2014; Carmignato et al. 2012; Rab et al. 2022b).

Industrial metrological requirements like precision and accuracy are extremely important in health science, especially medical parts. The speed of manufacturing, testing, and supply at the proper locations of the essential items is another contemporary example, citing the special case of scientists, medical experts, and other staff working toward the COVID-19 vaccine to ensure its safe rollout across the globe. This could only be possible through metrology. A CT can efficiently scan multiple images of the same part or multiple parts (sometimes in hundreds) and reject the defective parts if not as per specification. The ultrasound-based detection, analytical, therapeutical, and characterization techniques, tools, and data are widely used and utilized in a number of industrial metrological applications. The data and results about ultrasonic velocity in samples, attenuation, refraction, reflection, bulk tissue properties, quantitative distance, Doppler measurements, and tissue elasticity of the organs thus generated are complex but become reliable and appropriate if the techniques used are properly made traceable through calibration and standardization. The imaging methods providing information about the present nature and amount of matter can be considered biomarkers and are conceptually similar to laboratory assays. Quantitative imaging includes developing, standardizing, and optimizing anatomical, functional, and molecular imaging acquisition protocols, data analyses, display methods, and reporting structures. Thus, such metrological measurements save a lot of time, money, and manpower (Sullivan et al. 2015; Rab and Yadav 2022; Javaid et al. 2022).

As per an estimate by CAGR, the industrial metrology market is expected to grow at 6.0% per annum from 2021 to 2026. A huge demand from the manufacturing industry for inspection, verification, testing, calibrations services, tools, instruments, quality control, etc. and big data analytics are the major sources for the growth in the industrial metrology sector. The major industry tycoons like Reliance, BHEL, GAIL,

Faro, Carl Zeiss, Nikon, Mitutoyo, etc., are developing and supplying several data analytics, control, process, management, security software tools, applications, and metrological solutions for the industrial metrology stakeholders (Aswal 2020a; Global Newswire 2022; Kuster 2020; Meškuotienė et al. 2022).

The demand for the development, installation, commissioning, use, and applications of energy-saving and efficient devices and lightweight materials in all walks of human life, especially automation, health, aviation, space, automobile, etc., is increasing exponentially. This has created a huge demand for metrological tools/solutions. Automotive, electronics, and power generation industries need precisely designed and fabricated components/parts with improved tolerances. The 3D measurement and manufacturing, portable CMMs, robotics-dependent CMM scanners, cloud computing platforms, and several forms of geometrical metrology solutions have fueled the growth in industrial metrology. As mentioned above, the utilization of cloud-based metrology solutions, form and geometrical dimensioning and tolerances, robotic-assisted metrology, laser scanning, automation, and installation of multisensory metrology software are contributing to increase the growth further (Chua et al. 2017; Emmer et al. 2017; Tsukahara 2007).

The humongous outbreak of COVID-19 situation recently has probably changed the perspective of working behaviors and environments. The special arrangements were made/being made involving minimum human resources in less time in a collaborative and cohesive way, keeping physical-distancing norms in mind. The global impact of COVID-19 has pushed all to move to alternative arrangements of virtual and online platforms, which is again fueling the growth in digital metrological tools and solutions. Measurement standards and metrological activities provide scientific and metrological infrastructure to the country. For industrial and scientific developments in any country, the role of national metrology institutes (NMIs) is significant and irreplaceable in the advancement of the field and its related technologies. It is the responsibility of all the stakeholders involved in metrology, be it manufacturers, standard and conformity bodies, NMIs and accredited laboratories, regulators, and users, to ensure that the instruments, devices, tools, and software solutions used are functioning properly, as per regulatory framework, and traceable to metrological standards (Rab et al. 2022a; Tzalenchuk et al. 2022; Davis 2018; Mustapää et al. 2020; Thiel 2018).

Considering the demands for futuristic metrological requirements, industries, administration, policymakers, and regulators are also swapping to automated and paperless digital systems as far as possible. As an example, there are more than 100 million electricity, gas, water, or heat legal meters in use in Germany. Additionally, a lot of metrological and other measuring devices are in use in laboratories, petrol pumps, weighing balances, and scales in the food sector. If all these instruments and devices are automated and digitally traced, the operator or the service users would not require the accuracy check. Automation of the inspection task and traceability would help industries, manufacturers, and laboratories to get the benefits of frequent inspection and rapid processing.

It is also extremely important for government agencies to work toward automated legal metrology solutions and reliable and accurate measuring systems to ensure

users get the correct commodities timely and quantitatively. The safeguarding of consumer rights is also highly critical ensuring the publication of mandatory requirements on packaged goods. Priority is also required from government agencies to formulate and execute legal regulations for medical devices, automotive, and software industries providing automated metrological solutions for consumer safety. Government, industrial functionaries, and all other stakeholders have to keep pace with the digital transformation, metrological solutions, emergence, and introduction of a number of emerging technologies. The number of such measuring instruments and technologies is countless, and they are being used or intended to be used for generating huge data banks. These instruments are required to be inspected, tested, verified, and calibrated at various stages for the grant of model/pattern approval, surveillance audits, and manufacturers' services. Big data analytics are required to maintain and manage this huge data bank. Another interesting and key area is 3D metrology, which utilizes 3D optical scanners capable of providing useful data on malfunctioning products.

In the global industrial metrology growth, the share of Asia-Pacific is comparatively higher than the other regions, and it is further expected to grow in the coming decade. The growing demand from automotive, aerospace, health, and manufacturing would lead the growth in the region. Moreover, Asia-Pacific nations are contributing aggressively to R&D and funding extensively, including India, China, Thailand, South Korea, Vietnam, and Japan, in the industrial metrology market in the region. Process assurance and product inspection are becoming much more pronounced areas of metrology in companies. The safeguarding of processes takes place on the line; here, the requirement for speed predominates; the inspection of dimensions, shape, and position remains in the measuring rooms; here, the requirement for accuracy predominates.

Emphasis is placed on making metrology simpler for manufacturing users, especially those unaware of and who are experts in metrology. Weight is being placed behind on operational easiness and lucidity of the results. Newer interfaces, conceptually simpler approaches, fast and quick execution and results, avoiding unnecessary repetitions, and concentrating on essentials are the special features being focused on for users in manufacturing. Another feature being added is the direct feedback and alert system to the processing machine to address and rectify the issue, amicably an error/problem if noticed/detected. Artificial intelligence is being used/applied to make extrapolation solutions through projections and trends to minimize production runouts.

Contribution of Metrology for Achieving Sustainable Development Goals (SDGs)

Metrology is a focused key area that contributed immensely to achieving the SDGs because it is crucial and indispensable to shield the planet, to ensure the exalted and improved lives for all individuals, and to accomplish inclusive economic growth and prosperity. It is extremely crucial and vital for trade, scientific comparison,

innovative and incipient technologies, mutual scientific cooperation, or even simple exchange of ideas and information. In the fast-changing, technologically advanced world, measurements are also getting more and more precise, accurate, and sophisticated. Consequently, measuring methods, devices and instruments are also becoming technologically advanced rapidly. Therefore, there are demands for improved measurement standards, adoption, and introduction of new and novel metrological concepts in newer fields, viz., nanotechnology, biotechnology and healthcare, food, clean water, clean and green environment, and energy sources, which are going even at a much faster speed. These are some of the important SDGs for the country to achieve. The role of national metrology institutes (NMIs) and the CSIR-National Physical Laboratory (CSIR-NPL), in the case of India, is paramount, not only for scientific and metrological developments in the country but also achieving SDGs (Yadav et al. 2020b; Fisher Jr et al. 2019; da Cruz et al. 2019; Grek 2020; Galimullina et al. 2020; Lazzari et al. 2017). Figure 3 shows the SDGs set by the UN.

The accurate and precise measurements through strong metrological infrastructure are essentially required for/help for achieving the different SDGs due to the following objectives:

(i) To ensure that all stakeholders (producers and manufacturers, farmers, etc.) get the correct payments for their products and that all consumers get the correct amount of goods.
(ii) To design, develop, fabricate, manufacture, and deploy suitable metrological techniques, methodologies, instruments, devices for industry, and manufacturers for helping product innovation, process improvement, and quality assurance.
(iii) To make sure that the component parts and finished goods comply with the regulatory requirements, documentary standards, and product specifications.

Fig. 3 The United Nations' 17 Sustainable Development Goals (courtesy of UN/SDG)

(iv) To ensure that quality expectations like product cost, durability, and reliability of the stakeholders (consumers) are met.
(v) To ensure that the control of prepacked goods/commodities is actually supporting/reducing market fraud.
(vi) To ensure the correct measurements of raw materials exported in bulk, which enable them to pay the correct price. It also helps governments to collect correct taxes on exports.
(vii) To also help improve the economic conditions for all concerned and assist in poverty reduction through metrological controls.
(viii) To remove technical barriers to trade through global acceptability of measurements and test results.
(ix) To facilitate international trade and contributes to inclusive economic growth, access to opportunities for small- and medium-sized enterprises (SMEs), and a level playing field for developing economies.

The cohesive metrological and documentary standards, varying calibration, testing, and certification systems help improve product quality, remove trade barriers, and reduce costs, time, and market uncertainties. Though metrology provides the impetus for growth in each of the SDGs, but major contributions can be listed as follows.

Metrology and the SDGs for Good Health and Well-Being for People (Goal 3): "Ensure Healthy Lives and Promote Well-Being for All at All Ages"

The innovations in medical electronics have strengthened the existing healthcare infrastructure in diverse dimensions, such as digitizing medical tests and diagnostic and therapeutic procedures. Recently, the central government made regulations for medical devices, under Section 33 of the Drugs and Cosmetics Act, 1940, called the "Medical Device Rules 2017" (https://cdsco.gov.in/opencms/export/sites/CDSCO_WEB/Pdf-documents/medical-device/Classificationg1.pdf. Accessed 31 Oct 2022), which came into force on January 1, 2018. CSIR-NPL has also initiated a major program on biomedical metrology to achieve the following objectives as part of SDG (Goal 3), reproduced from (https://www.bipm.org/documents/20126/43974911/Presentation-CGPM26-Sarmiento-SDG.pdf/c2dacbcf-8e92-ea6f-56e0-7253ff777769?):

- Medical measurements are fundamental to preventing, diagnosing, and treating diseases and other medical conditions. Getting measurements right improves patient outcomes, saves time, and reduces costs.
- Internationally recognized and accepted equivalence of measurements in laboratory medicine and traceability to appropriate measurement standards will lead to:
  - Improvement in the quality of healthcare of patients
  - Reduction of false-positive and false-negative test results
  - Reduction in costs for government and healthcare insurers

- Improvement of efficiency of healthcare
- Global acceptability of measurements and tests, which removes technical barriers to trade

Most of the NMIs are already providing calibration, testing, and consultancy services and conducting training programs for users and industries in various medical devices, i.e., blood pressure measuring instruments, clinical thermometers, medical weighing balances, infusion pumps, ultrasonic power measurements, and defibrillator analyzers, apart from various process parameters. The services for more medical devices are being introduced as per requirement. The CSIR-NPL, being the NMI of India, has also taken the noticeable initiative to "Ensure healthy lives and promote well-being for all at all ages."

Metrology and the SDGs for Ensuring Access to Affordable, Reliable, Sustainable, and Modern Energy for All (Goal 7)

As per the target set by 2030, using renewable energy and alternative energy sources would certainly reduce carbon dioxide emissions (Mission 500 GW by 2030). Such initiatives supported by directives and regulations would address all the aspects of the energy supply chain and measures to reduce energy consumption at the point of use. This strategy focuses efforts on sustainable and secure energy supplies (generation, transmission, and consumption), methods to reduce greenhouse gas emissions, and increasing competitiveness. CSIR-NPL is also working toward establishing primary standards for the calibration and testing of solar cells with comparable measurement uncertainty across the globe (Aswal 2020a; Yadav et al. 2021). This would be a unique facility as this is a gray area in the country, and no existing infrastructure is available for the calibration of solar photovoltaic devices. After establishing these measurement standards and facilities, CSIR-NPL is committed to dedicating them to national services and providing metrological traceability to users, academic institutes, and PV industries/testing agencies. It would finally help to achieve the national mission of self-sustainable India by providing quality measurements in the country. It would also help in a clean and green environment. Such efforts and initiatives would certainly play a major role in achieving SDG (Goal 7) for ensuring access to affordable, reliable, sustainable, and modern energy for all and SDG (Goal 13) for climate actions.

Metrology and the SDGs for Industry, Innovation, and Infrastructure (Goal 9): "Build Resilient Infrastructure, Promote Inclusive and Sustainable Industrialization, and Foster Innovation"

The main contribution of metrology in achieving SDGs is for industry, innovation, and infrastructure (Goal 9). The following important objectives are the key to any country's industrial revolution and growth:

- To manufacture and trade precisely made and tested products and components that trading partners accept for the economic success of a country.
- To control manufacturing processes and guarantee the quality of products, there is a need to align instruments to reference standards and ensure measurement traceability.
- To develop and deploy appropriate metrological methods being key for the industry and supporting product innovation, process improvement, and quality assurance.
- To ensure that components and finished products meet regulatory requirements, documentary standards, and specifications.
- To ensure that consumer and industrial quality expectations are met, including product value/price and reliability.

Right from its inception in 1950, the CSIR-NPL is fulfilling its primary charter of being the custodian of national measurement standards and providing various services to all the stakeholders to outshine in their efforts of quality products and suitably compete in international trade (Aswal 2020a; https://www.bipm.org/documents/20126/43974911/Presentation-CGPM26-Sarmiento-SDG.pdf/c2dacbcf-8e92-ea6f-56e0-7253ff777769?; Mission 500 GW by 2030; Yadav et al. 2021; Sanjay et al. 2020a, b, c, d; Yadav et al. 2018). It is also responsible for formulating and advising the country's policies, guidelines, and recommendations on evolving advanced metrology programs for government organizations, institutions, regulators, industries, and other stakeholders. The industrial system based on process, electrical and electronic parameters, sensors, and transducers with high-precision metrology is the backbone of Industry 4.0. After the historic decision taken by CGPM to redefine SI base units based on quantum standards/fundamental constants on November 16, 2018, and implemented from May 20, 2019, the role of CSIR-NPL is further increased manifold, not only to link and establish all SI units to fundamental constants but also educate all the stakeholders, regulators, administrators, policymakers, academic institutions, industries and users about the new changes and their implications in industry, innovation, and national metrological infrastructure.

As a continued and sustained effort, CSIR-NPL provides traceability services to almost 2500 industries and user organizations, generating more than 3000 calibration and test reports annually. The major customers are from national- and state-level R&D institutions; calibration, testing, and R&D laboratories; various sectors like oil, petro-products, refineries, automobile, and lighting industry; electronic, electrical, power, civil supplies and aviation, defense, space, healthcare, medicine, and safety and it's standards, pollution controls, monitoring, process, and manufacturing industries; and SAARC NMIs (Sanjay et al. 2020a, b). To strengthen the ties between NMI and industries, government organizations, and other stakeholders, almost 20–30 awareness and training programs are organized per annum for R&D laboratory, NABL labs, legal metrology officers, industry participants, and SAARC NMIs. Need-based consultancy services are also extended to industries in most of the measurement parameters.

The recent COVID-19 pandemic has generated new insights among researchers to work toward developing new technologies for testing PPE and medical devices.

Rapid industrialization, increasing demand for more precise measurements, updated technologies having a combination of IoT, artificial intelligence, soft metrology, and highly skilled manpower have increased the strong role of CSIR-NPL in the national measurement system, quality infrastructure, as well as the need for more stringent metrological regulations and certifications in trade and commerce. Its pioneering works have made a significant contribution in several sectors, such as infrastructure, industries, healthcare, defense, energy, environment, space programs, academia, etc.

Therefore, the contribution of CSIR-NPL for achieving SDGs is not only important and crucial for industry, innovation, and infrastructure (Goal 9) but also for socioeconomic growth (Goal 8) for promoting sustained, inclusive, and sustainable economic growth, full and productive employment, and decent work for all. This would not only help foster innovation, quality, and competitiveness in product manufacturing and improve export, development of import substitutes, and industrial growth, which in turn translates to improve our GDP, but also generate expertise in metrology, translating finally into employment generation.

Metrology and the SDGs for Climate Action (Goal 13) to Take Urgent Action to Combat Climate Change and Its Impacts by Regulating Emissions and Promoting Developments in Renewable Energy

The role of metrology is crucial in achieving SDG (Goal 13) wherein the following listed points need to be considered before achieving objectives:

- Accurate measurement is central to understanding climatic changes by identifying long-term trends of small magnitude from data that can vary enormously over very short timescales.
- Millions of measurements are made every day covering several climatic parameters using different techniques all around the world.
- The accurate and high-quality air pollution data are helpful in proper policy formulation and devising better mitigation techniques.
- The data has to be consistent so that it is meaningful and can be combined, by making measurements that are fully traceable to SI units ensuring the stability of measurement over time.
- The emergence of emissions monitoring, carbon trading, and other technologies such as carbon capture and storage all bring down their own measurement challenges.

CSIR-NPL is serving the nation through environmental metrology services, which is a key requirement for the generation of precise and high-quality air pollution data. CSIR-NPL has taken a massive program and is working to develop a national certification program in this area to help various stakeholders like industries, researchers, and policymakers (Sharma et al. 2020). Recently, CSIR-NPL has been authorized by the act of parliament as the "Certification Agency for Air Pollution Monitoring Equipment" by the Ministry of Environment, Forest and

Climate Change (MoEFCC), Govt. of India. Now, CSIR-NPL has developed a certification scheme (NPL-ICS) for certification of air pollution monitoring equipment, including Continuous Ambient Air Quality Monitoring Systems (CAAQMS), Online Continuous Emission Monitoring Systems (OCEMS), and particulate matter samplers in the country, and has already been notified by the MoEFCC for adoption under the National Clean Air Programme (NCAP). These efforts themselves speak volumes of the contribution of CSIR-NPL in achieving SDGs for climate action (Goal 13) to take urgent action to combat climate change and its impacts by regulating emissions and promoting developments in renewable energy.

Metrological Industrial Measurements and Their Significance for Indian Infrastructure

Metrological measurements are essential and important in our daily life for preventing the underuse, overuse, and misuse of product quantity and services and to identify disparities in the delivery and outcomes. Measurements are used for quality improvement, benchmarking, and accountability. Scientifically accurate and valid measurements have the potential to significantly improve quality and efficiency. Measurements play a key role as an engine of the industrial revolution and the growth of any country. Measurement standards and related activities provide scientific and metrological infrastructure to the country. With the advent of better-quality products through metrological advancement, our industries can compete internationally and overcome trade barriers/constraints and finally achieve export targets. This finally translates into the growth of industries through rapid industrialization, economic growth, and societal upliftment. Therefore, it is fundamentally important in industry and trade, not only for the consumers but also for the manufacturers, because accuracy, quality, and reliability of the products and measurements are equally important for both (consumers and manufacturers). Metrological activities ensure that the measurements are stable, comparable, and coherent. Suppose such metrology is not part of the manufacturing; in that case, we will end up with substandard products, which will be very damaging to the Indian economy, especially in view of the thrust we are giving to local industries (Aswal 2020a; Rab et al. 2021a; Lazzari et al. 2017; https://cdsco.gov.in/opencms/export/sites/CDSCO_WEB/Pdf-documents/medical-device/Classificationg1.pdf. Accessed 31 Oct 2022; https://www.bipm.org/documents/20126/43974911/Presentation-CGPM26-Sarmiento-SDG.pdf/c2dacbcf-8e92-ea6f-56e0-7253ff777769?; Mission 500 GW by 2030; Yadav et al. 2021; Sanjay et al. 2020a, b; Ačko et al. 2020; Bhardwajan et al. 2021; Garg et al. 2021).

What Needs to Be Done to Ensure Good Quality Measurement

Metrological infrastructure is the foundation stone for the practical realization of the Hon'ble Prime Minister's call for AtmaNirbhar Bharat. For both Make in India and

AtmaNirbhar Bharat programs, a robust national QI (metrology, standards, and accreditation: all of them harmonized with respective international counterparts) is essential to ensure that all products are in compliance with regulatory requirements. Metrology is absolutely essential for making India, trade, innovation, and new technology developments, which in the national context are the pillars on which the "AtmaNirbhar Bharat" would grow. The five pillars of development, as put forward by our honorable Prime Minister, viz., economy, infrastructure, system, demography, and demand, can be achievable with precise measurements and their certification or product approval. Thus, the honorable Prime Minister's call for "Vocal for Local" leading to "AtmaNirbhar Bharat" is conceivable, and metrological traceability and reliability in measurements would play a big role culminating in indigenous products with global acceptance and quality, and it would finally result in a quantum jump in economic growth. Much coordinated and effective synergy between all the stakeholders is always required. The country needs to enhance its metrological capabilities at par with international metrological institutes, developing more primary standards traceable to seven base SI units, enhancing calibration and measurement capabilities (existing 236, in comparison to leading NMIs of the world having more than 1000), participation in international inter-comparisons of standards for compatibility and global visibility, and finally the stronger network system of unbroken chain to disseminate the measurement standards to users and clients to get their maximum benefits. As per an estimate, there are more than 400,000 testing, verification, and calibration laboratories in the country in organized and unorganized sectors. Most of these laboratories would require traceability in coming years as the awareness of AtmaNirbhar Bharat is rising to ensure quality in their measurements, process, products, and services.

How This Is Critical in the International Market Given the Standards Applicable

Documentary standards are used to facilitate measurement, manufacturing, communication, and commerce. Standards play an important role in the economy, enabling companies to comply with relevant laws and regulations; providing interoperability between new and existing products, services, and processes; speeding up the introduction of innovative products to market; and facilitating business interaction for mutual understanding. In the case of metrological standards, as science evolves, advanced technologies are developed that demand even more precise and accurate measurements. As a result, the need is felt to focus research efforts on metrology, realize all the seven SI base units in terms of fundamental constants at CSIR-NPL, and establish quantum-based primary standards, BNDs, and primary and secondary standards of allied parameters. To keep pace with global advancement in metrological technologies and also as a responsibility of custodian of measurement standards, CSIR-NPL requires support from all stakeholders to establish and develop new measurement standards based on quantum metrology; to maintain the existing measurement standards and calibration facilities to cater to the rapid industrial

demand; to participate in international comparisons to be at par with international NMIs and to upgrade, extend, and intend measurement ranges and improve upon measurement uncertainties of existing measurement standards and measurement facilities; to participate in BIPM and APMP meetings to maintain and upgrade existing CMCs. In a nutshell, standards help organizations in reducing costs by minimizing errors and redundancies; increasing productivity and efficiency; mitigating risks; maintaining consistency, customer confidence, and uniformity; and eliminating trade borders.

What Can Be Done by Department Heads/District Collectors in This Regard

Traceable measurements are the prerequisite for quality assurance. All economies have their own mechanism imposed through government organizations that are responsible for monitoring the measurement accuracy, demonstrating the importance of the integrity of measurement to a nation's economic growth and prosperity. In the Indian existing system and mechanism of monitoring, CSIR-NPL provides traceability to the Regional Reference Standard Laboratories (RRSLs) of weights and measures and the controllers and inspectors are responsible for implementation at the state levels and at the district levels, respectively. This mechanism needs to be strengthened further by local authorities.

Every department should make it a target to ensure that the manufacturing and services are done with a quality framework that can be traced to the reference standards provided by the National Physical Laboratory. Such calibration and referencing could be suitably incentivized and monitored to ensure quality. In districts, district collectors also can sensitize entrepreneurs about the necessity of products and services to match up to the standards internationally recognized.

Conclusion

An introduction to the fundamentals of industrial metrology is described in this chapter. The study is divided into four parts. The first section described how measurements fit into the industrial revolution. The introduction of important ideas such as application-oriented trends in industrial metrology follows. The contribution of metrology to achieving sustainable development goals is discussed. Finally, a detailed description of the industrial metrological measurements and their importance for Indian infrastructure is given. Since metrological applications impact all aspects of our daily lives, measuring, practicing, and following well-established measurement techniques, methods, and procedures are paramount and critically important. As per the applications and use, metrology is divided into three categories: scientific or fundamental metrology, industrial or applied metrology, and legal metrology, which are interconnected. Industrial metrology serves two purposes. First, it is crucial to specify potential measurement-based acceptance/rejection of goods and services and carry out functions for measuring,

monitoring, controlling, and recording operations to ensure that they adhere to the required standards. Second, it also enhances the likelihood that the measuring apparatus complies with any applicable technical or legal specifications for its intended usage, which is often application specific. Modern industries and all the advanced production systems are centered and dependent upon the ever-increasing intelligent, adaptable, and quick-learning metrology systems. These metrology solutions will accelerate the manufacturing more inventive, well-designed, and high-quality goods while significantly reducing downtime in networked factories. The metrological solutions currently available in the market can be utilized to improve the manufacturing efficiency at all levels, even though a lot of advances in this field will continue to enter the market.

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Importance of Ultrasonic Testing and Its Metrology Through Emerging Applications

33

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Abstract

Ultrasonics or ultrasound is the sonic waves with frequencies higher than the upper audible limit of humans and is used in numerous studies and applications. Ultrasonic waves are applied in a variety of industries, including the manufacturing and process industries, medicine, domestic, civil, marine communications, wind turbines, and many more. In the medical field, ultrasound is used for both diagnostic and therapeutic applications. In industries, it is used during manufacturing, flaws, or voids for quality assessment and chemical processing. Due to its numerous advantages, ultrasonic testing is one of the most extensively

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utilized nondestructive testing (NDT) methods for the examination of structures. The present chapter is mainly focused on the significance of ultrasonic testing and related metrology. Emerging ultrasonic measurement techniques are also briefly discussed, as they would lead to greater measurement capabilities in the future. It is expected that the comprehensive overview of the progress in ultrasound in recent years would prove to be a useful information and knowledge bank for the student, researchers, engineers, scientists, and metrologists working or involved in the field.

Keywords

Nondestructive testing · Ultrasound · Inspection · Medical · Defects · Applications · Metrology

Introduction and Historical Background

The ultrasonic field is defined as the study and utilization of mechanical vibrational waves having frequencies above the human audible threshold frequency of 20 kHz. The ultrasonic waves pertaining to the characteristics of the sound travel in a similar fashion as light waves. Ultrasonic waves require a medium such as liquid, solid, or gas to propagate. The wavelength of the ultrasound in the medium depends upon the elastic characteristics of the medium and particle motions (Pandey and Pandey 2010). There are three ways in which ultrasound can be used, applying directly to the product, coupling with the device, and submerging the object in the sampler or ultrasonic immersion tank. Before World War II, ultrasound pioneers were inspired by sonar, the practice of generating sound waves in water and analyzing the back echoes that may identify the location of the submerged objects. Sokolov investigated the use of ultrasonic waves to identify objects from 1929 to 1935. For the first time, in 1931, Mulhauser was granted a patent for employing ultrasonic waves to discover the faults in solids using two transducer methods. Later, Firestone (1940) and Simons (1945) developed ultrasonic pulse-echo testing using a single transducer approach. Just after World War II researchers in Japan began investigating the medical diagnostic possibilities using ultrasound (Hellier 2013; Thabet et al. 2018). In the medical, ultrasound is used for health and safety concerns. Worldwide medical experts have then presented the use of ultrasound to detect gallstones, breast lumps, and cancer. Although the use of a high, 15 MHz frequency for detection of histologic structure of the living intact human breast was proposed as early as 1952, it took over five decades for commercial imaging probes to operate in the range of this frequency. The early ultrasonic devices displayed data in A-mode on an oscilloscope screen with blips. After that, a B-mode presentation with a two-dimensional grayscale image was discovered. The ultrasonic principle is based on the wave characteristics wherein ultrasonic waves are traveled into the specimen and then received back and analyzed for the information of the material under test. Many modern defect detectors with a mathematical function and ultrasonic approaches are

already available on the market for precise and fast location detection of flaws (Roshan et al. 2019; Kumar and Mahto 2013). The ultrasound has numerous applications in different domains like medical, industrial, manufacturing, underwater communication, adulteration detection, material quality assessment, concrete testing, and material characterization (Samokrutov et al. 2006; Ermolov 2005; Cawley 2001; Drewry and Georgiou 2007; Muthumari and Singh 2011). Ultrasound sensors technology is being used in automotive parking systems, flow measurement, and industrial metrology operations in ongoing development advancements. Ultrasonic has become a multidimensional field of study because of its various useful industrial and medical applications. There are various reasons why ultrasound is preferred over other techniques: relatively safe (no radiation like an X-ray), eco-friendly, the ability to maintain the original properties of the material, and cost-effective. It does not even require any pre-processing or post-processing of the samples. The ultrasonic technique is sensitive to both surface and subsurface discontinuities and consumes less energy compared to any other process. Its adaptability allows it to switch from one setup to another in a matter of minutes; devices have a long lifespan and require little maintenance. In some cases, particularly when using the pulse-echo approach, only one-side access to the sample is required (Lavender 1976; Cherfaoui 2012; Dwivedi et al. 2018).

Sources for the Generation of Ultrasonic Waves

There are numerous methods for the generation of ultrasound like mechanical, electrostatic, electrodynamic, electromagnetic, magnetostrictive, piezoelectric, and laser (Krautkrämer and Krautkrämer 2013). The mechanical method, sometimes known as the Galton Whistle method, is an early method for generating ultrasonic waves. For the generation of ultrasound in the frequency range of 100 kHz to 1 MHz, mechanical shock or friction is used. The electrostatic approach can produce ultrasonic waves in the range of 10–200 MHz. In the electrodynamic approach, ultrasound is produced using the magneto-inductive effect. Magnetostriction is the mechanical deformation of ferromagnetic materials in a magnetic field. Through direct interaction between the optical electromagnetic field and the material acoustic field, crossed laser pulse excitation generates high amplitude, counter-propagating ultrasonic waves (acoustic phonons of specified wave vector). The method allows for the optical creation of ultrasonic waves that can be tuned to at least 20 GHz (Nelson et al. 1981). Ultrasonic waves are generated directly into the sample by RF pulses applied in a coil positioned near a metal sample in the presence of a magnetic field (Dobbs and Llewellyn 1971). This is commonly known as an electromagnetic acoustic transducer (EMAT). The Piezoelectric effect is the most prevalent mechanism for producing ultrasound for various applications. The inverse piezoelectric action is used to generate ultrasound in response to the electrical signal applied to the transducer (Pandey and Pandey 2010).

Ultrasonic Metrology

Metrology is the science of measurement and its application. It is not about simply performing the routine measurements; it is about the infrastructure that ensures our confidence in the measurement's correctness. Scientific discoveries and innovation, industrial manufacturing, and worldwide trade enhancing the quality of life and protecting the global environment all rely on metrology. It is critical as a catalyst for any nation's industrial revolution, growth, and overall development (Aswal 2020). Metrological standards and activities contribute to the scientific and economic sustainability of the country. The protocols for the evaluation of performance characteristics of the developed device, testing methodology, manufacturing practices, product standards, scientific protocols, compliance criteria, labeling, or other technical policy criteria should be followed as per standard guidelines for metrological assessment (ISO 17025). Ultrasonic metrology plays a crucial role in providing traceability to medical and industrial equipment (Segreto et al. 2016; Zeqiri 2007). The calibration standard/parameters and their measurements associated with ultrasonic metrology are shown in Fig. 1.

Fig. 1 The layout diagram presenting the calibration of parameters/standards in ultrasonic metrology

In medical, total ultrasonic power/intensity is the critical parameter to be considered for both diagnostic and therapeutic applications. Ultrasound is progressively being used in variety of therapeutic applications due to its ability to cause irreversible changes in tissue when applied with suitable frequency, and intensity. Modern ultrasound diagnostic devices employ harmonic imaging on a routine basis and operate at center frequencies of about 15 MHz. The development of high-quality ultrasonic hydrophones based on membrane design has been the most significant breakthrough in ultrasound metrology (Harris 1996). Existing acoustic standards and guidelines require the acoustic output of imaging transducers to be characterized using calibrated hydrophones (Lewin 2010). International Standard ISO 14708-1: 2014 provides criteria for active implantable medical devices that are generally applicable and tests that must be performed on samples of an active implantable medical device to demonstrate conformity (ISO 14708-1:2014; BS EN 45502-1: 2015). In industries, ultrasonic metrology deals with the calibration of NDT equipment such as ultrasonic flaw detectors (UFD), ultrasonic transducers, and ultrasonic standard blocks (IIW-V1, IIW-V2, reference rods, and step wedge) for the various properties measurement (Ushakov and Davydov 2006; Rawding 1963; Dubey et al. 2015). To calibrate the ultrasonic reference block, both ultrasonic contact and immersion techniques are employed (Kalpana et al. 2022). Ultrasonic metrology has a wide range of applications, comprising material assessment, condition monitoring, physiotherapy, cleaning, navigation, and crack identification. For industrial application standards, IEC TS 63081:2019 is followed for key quantities relevant to ultrasonic materials characterization, specifies methods for direct measurement of many key ultrasonic materials parameters. Also, there are Indian and International standards followed by laboratories such as IS 4904:2006and ISO 2400:2012 for V1 block characterization. In concrete testing IS 12211 is followed for the quality assessment. Figure 2 depicts a few ultrasonic standard blocks used in industries.

Techniques and Systems for Ultrasonic Testing

In industries, ultrasonic testing is considered the most suitable method for quality inspection and assessment. The ultrasonic non-destructive testing approaches used are described below.

Concrete Testing

Concrete is a composite construction material consisting of sand, gravel, crushed rock, or other coarse and fine aggregates bound together in a stone-like mass by a binder like cement and water. Concrete is the oldest construction material worldwide. Some cracking is expected if substantial tensile stress is created in the concrete structure. The compressive strength of the confined concrete is required to measure. The quality of concrete is measured by ultrasonic testing. The ultrasonic concrete testing method is a contact method, and the proper contact between transducer and

Fig. 2 Ultrasonic standard blocks used for the calibration

Table 1 Grading of concrete structure based on ultrasonic pulse velocity measured through the concrete

Longitudinal ultrasonic velocity (km/s)	Concrete quality
Above 4.5	Excellent
3.5–4.5	Good
3.0–3.5	Medium
Below 3	Doubtful

material is made by applying acoustical couplant (grease, ultrasound gel or water, etc.). The testing of concrete needs to be accessible from both sides of the material as for the inspection purpose through-transmission approach is used. The ultrasonic pulse-echo method can also be used in concrete testing, but it required maximum optimization and efficiency in the method with increased resolution (Wollbold and Neisecke 1995). By both approaches, ultrasonic pulse propagation velocity in concrete structures provides useful information on the structure's strength and quality. The quality assessment in the concrete is evaluated by measuring ultrasonic pulse velocity in the material being inspected as shown in Table 1 (IS 13311-1).

Ultrasonic pulse velocity (UPV) has been designated as an official testing method. Because the high-frequency wave propagation through the concrete has no negative effects on the integrity of the concrete structure under test, the UPV approach is a true NDT technology. The transmitting transducer pulse applied to the concrete surface is allowed to transmit and generate ultrasonic waves into the material. At the other end, the receiving signal from the separate transducer gives information on pulse transit time in the concrete material (Helal et al. 2015). The velocity of the ultrasonic pulse provides detailed information about the sample under examination. Standard criteria for ultrasonic pulse velocity testing can be found in the standards (1) ASTM C 597: Standard Test Method for Pulse Velocity Through

Concrete (2) BS EN 12504-4:2004 Testing Concrete (ASTM International; BS EN 12504-4:2004). The UPV method is a great way to investigate the uniformity and durability of concrete quickly and easily. Literature also shows study has also been done on determining the morphology of concrete using NDT, which describes the UPV method as an ideal tool for measuring the homogeneity of the concrete. The study found that UPV testing can be used to discover relatively small faults in concrete structures that could cause structural concerns (Tushar et al. 2015; Lorenzi et al. 2015).

Electromagnetic Acoustic Transducer Testing (EMAT)

Electromagnetic acoustic transducer (EMAT) is a non-contact method for the generation and detection of ultrasonic waves in electrically conducting materials. So this method interacts with the material without making any physical contact with the test material. EMAT generation and detection process involve three factors, a high-frequency coil, a strong magnet, and the test object. The schematic of EMAT is shown in Fig. 3. EMAT generates ultrasonic waves using electromagnetic coupling with the material. The testing is based on the principle of generation of Lorentz force within the skin depth or magnetostriction method or both together. The induced eddy current in the presence of a static magnetic field causes the creation of time-varying magnetic flux. Time-varying magnetic flux induced by receiving coil generates the voltage across the coil. EMAT, which consists of a magnet and an electrical coil, employs electromagnetic forces to inject sound energy into the test object via Lorentz force and magnetostriction combination. EMAT testing provides various advantages in the field of ultrasonic inspection. One of the main advantages of this technique is ultrasonic waves which can be detected in the conducting material by EMAT without making any physical contact; thus hot or moving objects can be used for measurement purposes. Another major advantage of this technique is that various ultrasonic wave modes can be generated in the material by using a coil of different patterns and magnet arrangements. There are some disadvantages too of EMAT. This technique is extremely sensitive to noise so in the detection of the defects, the signal

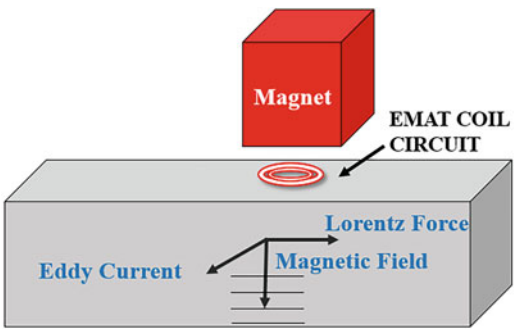

Fig. 3 The schematic of the EMAT method

received also contains noise. EMAT technique plays an important role in the inspection of the material in rail, pipeline inspection, and industrial applications. It allows for the detection of coating disbandment as well as the identification of various coating types (Zhai et al. 2014).

Ultrasonic Pulse-Echo and Through Transmission Testing

The ultrasonic pulse-echo method is most widely used in ultrasonic testing. The convenience and affordability of contact ultrasonic testing make it a popular choice for on-site inspections. In pulse-echo testing same transducer emits and receives ultrasonic energy. The signal received is in the form of echoes generally originated due to acoustic impedance mismatch at an interface, crack, or flaw. The strong signal is reflected from the back of an object. In this method, the probe is made in direct contact with the specimen, and no need to access both sides of the sample. In contact method couplant like gel, grease, oil, or water is used. On the other hand, ultrasonic through-transmission testing utilizes a transducer to transmit ultrasonic waves to the object, and a separate receiving transducer to receive the ultrasonic energy. Both methods can be used for the measurements and evaluation of dimension, material characterization, and identification of flaws in the surface or interior part of the material. For the detection of flaws, one has to measure the transit time of the ultrasonic waves and monitor ultrasonic velocity (Raišutis et al. 2008; Yadav et al. 2021; Rajagopalan et al. 2007).

The longitudinal ultrasonic velocity (c) is determined by the formula

$$c = \frac{2d}{t}$$

where d is the thickness of the sample in meters and t is the time of flight of ultrasonic waves in seconds.

Air Coupled Ultrasonic Testing

Ultrasonic inspections necessitate the use of an acoustic coupling medium. To achieve excellent results, uniform coupling between the sample and probe is required. However, due to air bubbles and line scale, it is difficult to provide a constant coupling of large components. An acoustical mismatch between transducer and test sample limits the measurement. The major advantage of using air-coupled testing (ACT) is that no coupling medium is required as shown in Fig. 4.

The typical air-coupled testing system consists of transducers, a transmitter for intense pulse excitation, an ultra-low preamplifier, a receiver amplifier, and a digital to analog converter, as well as a computer for configuration and assessment of the results (Stoessel et al. 2002; Gaal et al. 2019). The sensitivity of transducers for air-coupling is the most significant characteristic and is given by

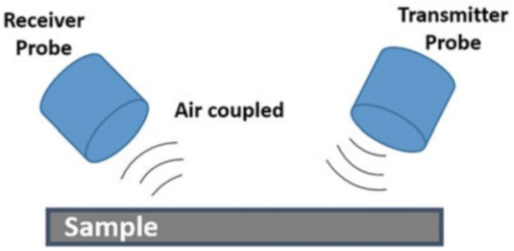

Fig. 4 The schematic of the ultrasonic air-coupled testing

$$S = 20 \log (\mathrm{VR}/\mathrm{VT})$$

where VT is the excitation in volts and VR is the receiver signal amplitude in volts.

Despite the acoustic mismatch, air-coupled testing can provide a much clearer indication of faults than water-coupled testing.

The challenge in the ACT is to maintain a large acoustic impedance difference between Air and Solid (test objects); also the material surface reflects 99% of the ultrasonic signal (Kays et al. 2007; Yilmaz et al. 2020).

Ultrasonic Interferometers for Liquid Testing

Ultrasonic propagation velocity is generally determined using one of two methods: continuous wave or pulse-echo method. The ultrasonic interferometer approach which utilizes continuous excitation is widely used for measuring ultrasonic propagation velocity in liquids. The thermo-acoustic, physical, and chemical properties of liquid can be determined using ultrasonic velocity. The principle of operation is based on the precise determination of wavelength in the medium. A quartz crystal fixed at the bottom of the liquid cell produces ultrasonic waves at a particular frequency. A moveable metallic plate positioned parallel to the quartz crystal reflects these waves. Standing waves are generated in the medium and current maxima or minima are produced when the distance between the transducer and the plate is exactly multiple of the ultrasonic half wavelength. In an ultrasonic interferometer, radiofrequency voltage is applied to excite the piezoelectric transducer (Sharma et al. 2019, 2020).

For ultrasonic velocity measurement, the excitation frequency and the distance between successive maxima and minima are considered. The following equation is used to calculate ultrasonic velocity (v):

$$v = \lambda \times f$$

where λ is the wavelength of ultrasound propagating in liquid and f is the excitation frequency.

Fig. 5 Ultrasonic interferometer apparatus developed for the measurement of ultrasonic velocity and attenuation in liquids

CSIR-NPL developed an ultrasonic interferometer shown in Fig. 5 which is capable to measure ultrasonic velocity and ultrasonic attenuation in liquid samples (Sharma et al. 2019). The system is now commercially available in the market.

The parameters that are derived from ultrasonic velocity are compressibility, effective Debye temperature, excess enthalpy, hydrogen bonding, intermolecular free length, solvation number/hydration number, Vander wall's constant, Rao's constant, Wada constant, and the study of molecular interaction.

Laser Ultrasonic Testing

Laser ultrasonic testing (LUT) is a type of ultrasonic NDT in which ultrasound is generated and detected by using lasers. The generation mechanism generally does not allow for wave type selection, and longitudinal, shear, and surface waves (or plate waves) are all released at the same time, which might be advantageous or disadvantageous depending on the application. Like traditional ultrasonics, laser-ultrasonics has similar applications, such as thickness measurement, flaws and crack identification, and material characterization. LUT testing is used because of its various advantages including couplant-free operation, ease of automation, relatively long distance to the object under test, harsh industrial environment feasible, difficult measuring positions possible, and high ultrasound bandwidth. With LUT ultrasound can be generated up to GHz frequency. Laser ultrasound is generated by two mechanisms: thermoelastic and ablation or vaporization method. The thermoelastic mechanism is the NDT approach while vaporization and ablation are invasive and based on the ablation of the sample. In LUT a short laser pulse irradiated on the

material surface causes localized thermal expansion that generates thermoelastic waves in the materials and can be detected by the optical interferometers. The signals obtained can be utilized to determine the material's elastic and density properties as well as discontinuities such as cracks or delamination (Cheng et al. 2013; Monchalin 2007).

Applications of Ultrasound

Ultrasound is extremely useful in various sectors. Table 2 provides a summary of some of the ultrasound applications.

Recent Technologies

For many decades, ultrasonic testing (UT) has been used, and ultrasound technology has been improved significantly. Workflow improvements in current generation ultrasound machines include automation/semi-automation of measurement, auto-image optimization, and reducing repetitive user tasks. The machinery used in NDT like ultrasonic flaw detectors have become portable devices having speedy analysis and accurate mathematical measurements. All the operator has to do is attach the transducer, and the instrument will automatically adjust variables like frequency and probe drive. EMAT technologies are being preferred due to their couplant-free operation. Now laser EMAT ultrasonic systems are also used for defects and time of flight measurement. However, the typical ultrasonic testing method requires acoustic interaction between piezoelectric transducers and test structures. In some cases, it is impossible to implement such as heated or complex-shaped, or fast-moving components. So in that particular case laser EMAT, ultrasonic testing is preferred (Boonsang and Dewhurst 2005; Lévesque et al. 2002; Senthilkumar et al. 2021).

In medical ultrasonic applications, elastography distinguishes liver tumors from healthy tissue, as well as determines the presence of fibrosis in the liver. Histotripsy, a high-intensity ultrasound therapy, is being studied at the University of Michigan for the non-invasive treatment of blood clots. Miniature ultrasonography at a low cost is used these days, and medical ultrasound imagers, like computers, have been getting smaller and smaller. Vscan Extend, handheld, pocket-sized ultrasound, portable healthcare devices, Sonosite, and Lumify portable systems are some of the latest point-of-care ultrasound systems in the market. When it comes to 3-D ultrasound, the technology is always improving. Scientific studies have progressed beyond basic 2-D and 3-D imaging to offer innovative approaches to assemble images to speed up and simplify analyses. Recent studies show that artificial intelligence (AI) has huge potential to assist repetitive ultrasound tasks, such as automatically identifying good-quality acquisitions and providing instant quality assurance. The AI methodology in the medical is used to evaluate adnexal masses, the risk of lymph node metastases in endometrial cancer, pelvic organ function, and

Table 2 Application of ultrasound in specific sector with example

Field	Trade/sectors	Descriptions
Industrial	Oil and gas	Due to the advantage of low power ultrasound, it does not alter the inherent property of the medium, low cost, and high efficiency; ultrasound has significant potential applications in the petroleum industry. Power ultrasound penetrates oil/water medium-well also creates and transmits high energy-specific density (10–1000 W/cm^2) (Luo et al. 2021; Qiu et al. 2020). The ultrasonic impact is used in oil transportation and storage to lower oil viscosity, reduce pumping costs, and clear oil storage tanks of asphalt and paraffin deposits that reduce tank volume and enhance corrosive activity (Palaev et al. 2019a, b)
	Defense	Ultrasonic testing is used once the rocket motor structure's resin structure and layered rubber insulation have been completed and gives a precursor to the crucial data needed to determine the system's structural integrity, defects, and micro inspection (Nudurupati 2021)
	Manufacturing	In manufacturing automated ultrasonic testing enables the viewing of the geometry of the tested element, allowing for the determination of its correctness and the magnitude and position of faults in welding for joints assessment. In industries, ultrasound is used in a wide range of processes, such as cleaning, welding plastics and metals, cutting, forming, testing materials, separating, mixing, de-gassing, atomizing, localizing, measuring, and many others
	Rail industry	In rail, rail tracks are subjected to extreme stresses, which can affect their structural integrity during usage and result in rail breakage. To monitor these types of defects, ultrasonic testing is useful (Bombarda et al. 2021)
	Food industry	It is important to harvest crops at their peak ripening stage. These days studies have extensively focused on developing nondestructive, noninvasive, and noncontact methods for determining food quality (Yildiz et al. 2019). It is determined that fruit characteristics are correlated with ultrasonic parameters (velocity and attenuation) (Yildiz et al. 2018; Natarajan and Ponnusamy 2020)
Research and development	Metrology	The key application of ultrasonic testing in metrology is the radiation force balance, which measures total output power, and the piezoelectric hydrophone, which measures acoustic pressure level. Ultrasonic metrology ensures users can make traceable measurements it maintains and disseminate primary standards for the measurement of key acoustical quantities (Zeqiri 2007; Mineo et al. 2017). Rail industries widely use ultrasonic testing for the inspection of rails. Both standard and angle beam techniques, as well as pulse-echo and pitch-catch techniques, are employed
	Cleaning	Ultrasonic cleaners are used to clean substrate surfaces in a variety of applications because of their ability to remove contaminants. By ultrasonic cleaning blind holes, thread roots, parts with complex geometry, minute surface

(continued)

Table 2 (continued)

Field	Trade/sectors	Descriptions
		contours, and several otherwise impossible cleaning tasks can be easily accomplished (Azar 2009). In ultrasonic cleaning, the high frequency (25–50 kHz) and high-intensity (up to 250 W/cm^2) ultrasound are used
	Spectroscopy	High-resolution ultrasonic spectroscopy is an analytical technique for monitoring molecular and microstructural alterations in liquids and semi-solid materials in a direct and non-destructive manner. High-frequency (MHz range) compression and decompression waves (longitudinal deformations) are used in ultrasonic spectroscopy to investigate the elastic properties of materials determined by intermolecular interactions and microstructural studies (Buckin et al. 2002; Buckin 2018)
Medical	Diagnostic	Ultrasound can image internal body organs non-invasively. Ultrasound is used to diagnose and accordingly treat medical conditions. Ultrasound in contrast to X-rays does not emit any radiation and has minimal risk. It is widely regarded as having a very low hazard, mostly as a result of the fact that it forms the image using non-ionizing radiation
	Therapeutic	The therapeutic ultrasound goal is to interact with body tissues in such a way that they are altered or eliminated. Ultrasounds are used to cure and treat medical conditions. In therapeutic ultrasound waves with frequencies of 1–2 MHz are generally preferred to drive the transducer (Pei et al. 2012)

breast lesions, assess aneuploidy risk, predict fetal lung maturity, perinatal outcome, shoulder dystocia, and brain damage, also estimate gestational age in late pregnancy, and classify standard fetal brain images as normal or abnormal (Drukker et al. 2020).

Ultrasonic Metrological Activities and Technologies at CSIR-NPL

The CSIR-National Physical Laboratory, India (NPLI), is responsible to maintain the national standards in the fields of ultrasonic and non-destructive testing. The activity has indigenously developed primary ultrasonic power measurement facility used for the measurement of total, time-averaged ultrasonic power generated by transducers. The frequency range is from 0.5 to 20 MHz and uses the radiation force balance (RFB) approach for ultrasonic power measurement in the range from 10 mW to 20 W. The system's functionality was validated by participating in the BIPM international key comparison (CCAUV.U-K3.1). The international key comparison is in the field of power measurement aimed to compare various National Measurement Institute's (NMIs) ultrasonic radiation conductance measurement capabilities (Haller et al. 2016). Other calibration facilities include calibration of ultrasonic standard reference blocks such as the IIW-V1, IIW-V2 (according to IS: 4904),

step wedge, ultrasonic pulse velocity (UPV) rod, cylindrical FBH blocks (according to ASTM E-127), ultrasonic flaw detector (as per BS: 4331-II/ASTM E-317 and IS: 12666), and ultrasonic probe (beam profile, bandwidth as per ASTM E-1065). Testing ultrasonic facilities are available for the measurement of propagation velocity, impedance, and attenuation for solids and liquids. The ultrasonic department has various facilities, and researchers are working in the direction of the development of technologies.

Conclusion

Ultrasound is significant since it opens up new options in a variety of fields such as industry, medicine, aerospace, food processing, undersea applications, manufacturing, and research and development. This chapter presents a concise report on various measurement techniques in ultrasonic, and ultrasonic metrology, and a review of metrological and technical requirements of documentary standards is presented. Over the last few years, numerous research efforts have been focused on the dependability of ultrasound, and many of them are detailed in this chapter. The present study discusses the various aspects of ultrasonic testing applications. This field has become more demanding, and applications are increasing day by day as we are surrounded by technologies. This chapter shows that some of the ultrasonic NDT techniques are having some limitations and unique advantages as well. The study obtained the most recent research on current testing and examined it to identify potential application areas.

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Mechanical and Thermo-physical Properties of Rare-Earth Materials

34

Vyoma Bhalla and Devraj Singh

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Abstract

The ultrasonic nondestructive technique (US-NDT) is the mostly preferred technique for material characterization. It has gained much attention in the area of materials science as there is no impairment of properties and does not affect the future usefulness of the sample under testing. The ultrasonic techniques are the versatile tools to investigate the material properties such as microstructure, elastic modulus, and grain size and the mechanical properties, velocity, attenuation, etc. This field has evolved mainly in the last few decades and is involved in developing new materials and modifying available materials by gaining a better understanding of characteristics under different physical conditions. These materials are thus referred to as condensed materials which have proven their potentials in many applications. Out of emerging materials, the exceptional structural, elastic, electrical, magnetic, phonon, and thermal characteristics of rare-earth compounds have attracted attention, as they are of technological significance. These materials crystallize into a basic NaCl structure, making them interesting samples for experimental and theoretical study. One of the most important reasons for the theoretical investigation of rare-earth compounds was the existence of a single crystal of chosen materials. After it was demonstrated that some rare-earth compounds can be grown epitaxially on III-V semiconductors, the interest in these materials has expanded even more recently. It has paved the path for their use in the manufacturing industry, infrared detectors, optoelectronic devices, spintronics, and several research domains.

Keywords

Condensed materials · Elastic constants · Ultrasonic velocity · Thermal properties · Ultrasonic attenuation

Introduction

Ultrasonic NDT is gaining societal importance in medical, industrial processing and defense, robotics, and materials characterization (Raj 2000). Due to rising competition and the demand for improved material productivity for use in electrochemical energy and renewable energy systems, the material-producing industries are focusing more on severe process and quality control requirements. This necessitates the use of two crucial methods:

(i) Materials characterization
(ii) Industrial metrology principles for quality control

The examination of numerous characteristics, such as elastic behavior, associated mechanical properties, material microstructure and morphological features, etc., is part of the materials characterization process. The following approaches are available for complete material characterization:

(i) Destructive testing (DT)
(ii) Semi-destructive testing (SDT)
(iii) Non-destructive testing (NDT)

Ultrasonic NDT: Material Characterization Technique

Ultrasonic echolocation is the use of sound waves and echoes to determine where the object is positioned (Chandrasekaran 2020). It is applied in NDT to fetch information regarding the wised variety of material analysis applications without affecting the samples' properties. Ultrasonic testing can be used to find holes, cracks, or corrosion in materials, inspection of weld, and imperfections in solids. Acoustic emission, a category of NDT study, uses ultrasonics for detecting the structural flaws in materials (Grosse 2013; Vary 1980; Nanekar and Shah 2003). High-frequency sound waves can also be used to compute and identify some fundamental mechanical, structural, or compositional characteristics of solids and liquids (Yamanaka et al. 1999; Xiang et al. 2014; Kumar et al. 2003). The ultrasonic material analysis is based on a fundamental physics principle: the mobility of any wave is altered by the medium through which it passes. Thus, changes in any one or more than one of four easily measurable parameters associated with the passage of a high frequency sound wave through a material-transit time, attenuation, scattering, and frequency content can often be correlated with changes in physical properties such as elastic modulus, density, hardness, and homogeneity or grain structure. The piezoelectric, electrostriction, and magnetostriction effects are primarily used to create and detect ultrasonic waves (Singh 2014).

Ultrasonic nondestructive testing (US-NDT) is preferred over other basic tools for characterization as the samples being tested are left undamaged by the process. It is also a very precise kind of inspection, making it ideal for material quality control and assurance (or products).

The NDT approach may be used to characterize the microstructures of a range of materials, as well as to analyze defects and determine physical attributes including density, thermal conductivity, and electrical resistivity. Ultrasonic measurements recorded during material manufacture and heat treatment can be used to ensure that the desired microstructure is obtained and to prevent the formation of faults, such as flaws in welds between different materials. Understanding how ultrasound interacts with microstructure is also crucial for tackling a variety of material issues. Defect detection is hampered by attenuation and backscattering, notably in rare-earth

monopnictides with coarse grains or complicated microstructures. As a result, minimizing attenuation is important in order to optimize the utility of ultrasonic testing. Material description investigations, such as nondestructive grain size measurement, can also benefit from microstructure information. Another important measure in ultrasonic characterization is wave propagation velocity, which can reveal crystallographic texture. The elastic constants and density of rare-earth monopnictides are closely linked to ultrasonic velocity. The elastic constants, in particular, are useful for determining material stability and stiffness (Bala et al. 2022). The goal of this chapter is to go over the methods for calculating ultrasonic characteristics of materials. Ultrasonic measurements, which rely on elastic waves propagating through a material, are one of the most widely used non-destructive evaluation (NDE) techniques (both experimentally and conceptually) and have played a vital role in illuminating the materials' global characteristics.

Ultrasonic Metrology

Ultrasonic metrology is one of the key metrological fields for industrial research because of its wide applications in industrial processing, manufacturing, and testing. The theoretical and experimental studies in wave mechanics provide a better understanding of the sound fields radiating from ultrasonic transducers and from reflectors or acoustic emitters. The results of these studies are used to develop measurement methods and reference standards for acoustic emission and pulse-echo ultrasonic system which are becoming increasingly important for quality control. The approach includes analyzing transient wave propagation in different directions based on numerical approach/theoretical computations. The use of ultrasonics is to measure the intrinsic and extrinsic parameters which include material elasticity, stress, surface hardness strength, etc.

Today's most sophisticated metrology systems drive quality assurance (QA) which has become a fundamental digital task which in and of itself facilitates efficient and cost-effective production processes. In a very real sense, QA data today controls how products are made and drives the bottom line and the repeatability in production which is vital in all pan-industrial manufacturing scenarios.

Rare-Earth Materials

Presently, rare-earth (RE) compounds are the materials of particular interest due to their power functionality and wide applications in metallurgical machinery, petrochemical, ceramic glass, agricultural, industrial, and textile industries (Massari and Ruberti 2013). Recently, RE materials have gained importance in electrochemical energy storage.

These are the key materials that cannot be ignored by both new energy, energy conservation, and efficiency enhancement routes and are often known as "the gold of world industrial manufacturing." These reputations basically reflect the three

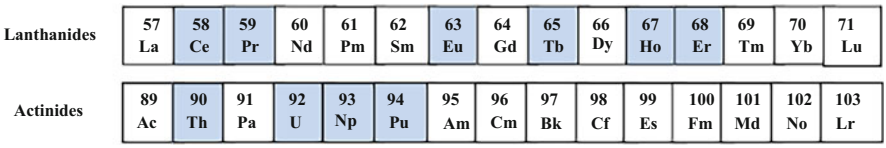

Fig. 1 Series of lanthanides and actinides as listed in Periodic Table

characteristics of rare earths: one is rarity, the other is powerful functionality, and the third is wide application. Rare earths can play a vital role in a variety of conventional industrial domains, notably in industries. In addition to these fields, the application of rare earths in new materials also has a broad space, such as permanent magnet materials, catalysts, high temperature superconductors, laser materials, precision ceramics, giant magnetostrictive materials, and electrothermal materials (Zhang et al. 2020).

The Modern Periodic Table consists of two rows of Ce-Lu and Th-Lr collectively known as f block which are divided into lanthanides (Ce-Lu) and actinides (Th-Lr). These are the "inner transition elements" involving the filling of inner-subshells such as 4f or 5f. Figure 1 shows the series of lanthanides (La) and actinides (An) (Moeller 1970).

The general properties of transition elements are as follows:

(i) They possess several oxidation states.
(ii) They are usually high melting point metals.
(iii) They are usually paramagnetic in nature.

The study is focused on two different material groups, XPn and XCh, where X = La and An; pnictides, Pn = N, P. As, Sb, and Bi and chalcogenides, Ch = S, Se, and Te.

Our main aim in this chapter is to do the analysis of past and current data by associating quantifiable uncertainty to measurements taking into the fact that multiple variables can influence readings. By this we are providing fit-for-purpose data.

This will eliminate the need to apply complicated mathematical formula to offset data discrepancies and thus improving accuracy.

Monopnictides (Pn) and Monochalcogenides (Ch)

In recent years, both the rare-earth monopnictides and monochalcogenides, REPn and RECh, are of great interest because most of them are characteristically low carrier sturdily correlated systems with a simple fcc (face-centered cubic) rock-salt type crystal structure. In such materials each RE atom coordinates with six neighboring atoms. The pnictides (N, P, As, Sb, and Bi) with electronic configuration ns2np3 ($n = 3, 4, 5$) and the chalcogenides (S, Se, and Te) with ns2np4 ($n = 4, 5$) configurations are the anions in the REPn and RECh type of compounds. Anomalous

physical phenomena such as the dense Kondo behavior, the metal–insulator transitions, the valence fluctuation behavior, the complicated magnetic states, etc. have been observed in these compounds. Most of them behave as semimetals. The rare earth metals nitride compounds in particular are the attention-seeking compounds. This group consists of intrinsic ferromagnetic semiconductors to half metals and thus has prospective to become spintronic materials.

The RE monochalcogenides are interesting ionic solids as these have attracted scientific community because of their electrical and magnetics properties. The semiconducting behavior is observed in case of divalent state and metallic when the rare earth ion is in trivalent state. The bonds have ionic character. The RE monopnictides and monochalcogenides form an interesting set of materials. To understand several physical phenomena in these material groups, the systematic study has been performed and is of significant attention for pure and applied research (Duan et al. 2007; Natali et al. 2013; Petit et al. 2010).

Applications of Ultrasonics in Research

Ultrasonic waves have a variety of uses in a variety of areas, some of which are listed below:

- Cavitation
- SONAR (Sound Navigation and Ranging)
- Biological ultrasonics
- Industrial ultrasonics
- Processing
- Ultrasonics in electronics
- Chemical uses
- Ultrasonic thermometry

Measurement Techniques for Metrology Solutions

Picosecond Ultrasonics (PULSETM Technology)

It has been widely utilized in thin metal film metrology because of its quick, non-contact, non-destructive nature and ability to measure many layers simultaneously. The simultaneous assessment of velocity and thickness for transparent and semi-transparent films has a lot of potential for not just process monitoring but also insight into device performance. Picosecond ultrasonics (Stoner et al. 2001) provides a full metrology solution in advanced radio frequency (RF) applications, including the measurement of diverse thin metal films for a wide range of thicknesses with exceptionally high repeatability. This could meet stringent process control requirements, simultaneous multilayer measurement capability, and simultaneous

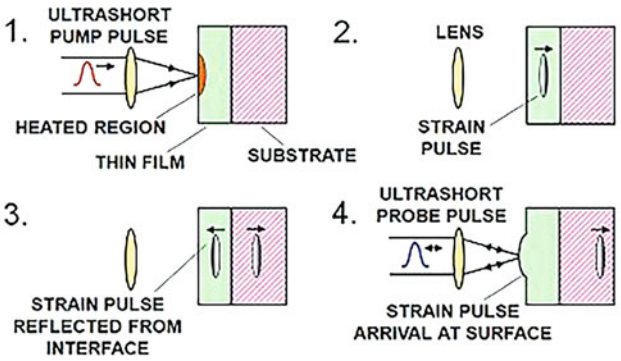

Fig. 2 Picosecond ultrasonic technology (PUT). (© Wikipedia)

measurement of sound velocity and thickness for piezoelectric films which play a vital role in the performance of RF devices.

Picosecond ultrasonic technology (PUT) is a nondestructive, non-contact laser acoustic technique for measuring film thickness, sound velocity, Young's modulus, density, and roughness. It has become the standard instrument for metal film thickness measuring in semiconductor fabs all over the world. A 100 fs laser pulse (pump) focused onto the film surface in Fig. 2 launches an acoustic wave in the film. The acoustic wave travels at the speed of sound through the film away from the surface. A component of the acoustic wave is reflected and returns to the surface at the interface with another material, while the remainder is transmitted. As the reflected acoustic wave reaches the wafer surface, the probe pulse detects it. The strain of an acoustic wave causes a change in optical reflectivity, which may be detected.

The thickness of a material may be easily derived using first principles utilizing standard sound velocity in the material. Film density, sound velocity, Young's modulus, and surface roughness can all be quantified in addition to thickness, depending on the application and the stacking of films. For single layer and whole stack thickness measurements, PUT delivers great reproducibility and stability. Because of the narrow beam spot and quick measurement time, direct measurements on real device structures are possible, as well as numerous dye measurements. The PUT's capacity to simultaneously measure thickness and sound velocity distinguishes it. The PUT has a unique technological advantage in that it can measure both thickness and sound velocity at the same time.

Ultrasonic Testing (UT)

The measurements and examinations are carried out using high-frequency sound energy (Balageas et al. 2016). Ultrasonic inspection is used to discover and evaluate flaws, take dimensional measurements, characterize materials, and more. As shown

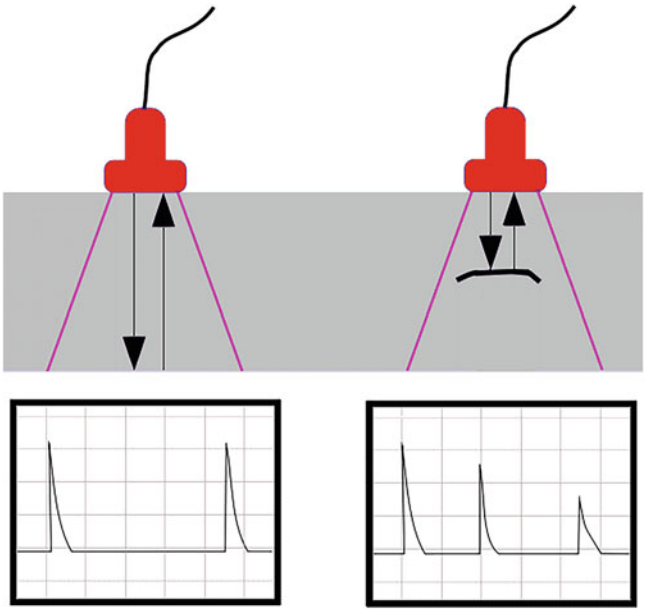

Fig. 3 UT inspection system. (© Wikipedia)

in Fig. 3, a standard pulse/echo inspection arrangement is utilized. The pulser/receiver, transducer, and display devices are among the components of the UT inspection system.

Pulser/receiver: It's a high-voltage electrical pulse-producing electronic device. The pulser drives the transducer, which produces high-frequency ultrasonic radiation. Sound energy is injected and propagates in the form of waves through the materials. When there is a break in the wave path (such as a fracture), some of the energy is reflected back off the defect surface. The transducer converts the reflected wave signal into an electrical signal, which is then shown on a screen.

The reflected signal intensity is shown against the time from signal production to when an echo is received in the applet below. The signal travel time can be directly related to the distance that the signal travelled. Information regarding the reflector's location, size, direction, and other characteristics may occasionally be gained from the signal.

Transducer: The pulser, which creates high frequency ultrasonic energy, drives the transducer. Waves propagate across the materials as the supplied sound energy travels through them. When a discontinuity in the wave path, such as a fracture, is detected, some of the energy is reflected back off the fault surface. The transducer converts the reflected wave signal into an electrical signal, which is then shown on a screen.

Display Devices: The reflected signal intensity is shown against the time between when the signal was generated and when an echo was received. The distance travelled by the signal may be directly attributed to the signal travel time.

Information regarding the reflector's location, size, direction, and other characteristics may occasionally be gained from the signal.

Theoretical Evaluation of Elastic Constants and Related Parameters

Elastic Constants

All materials are made up of atoms or molecules which are connected to each other by interatomic forces. The magnitude of these interatomic forces acts as a criterion for determining the vibration properties of the atoms and molecules of the medium. The solids with different structures have variable nature of forces, and hence the wave propagation characteristics also differ here (Decius and Hexter 1977; Stone 2013).

When an external force is applied to the crystal, the deformation takes place. When the external force is removed, internal forces oppose the deformation and return the sample to its original condition. This property is defined by "elasticity." The elastic constants of crystals link the structure of the lattice with its macroscopic behavior. The theoretical formulation for the evaluation of elastic constants is based on two common methods.

The first is the ab initio modeling of materials from their known crystal structures (Ching et al. 2009; Güler and Güler 2014). This method is based on the study of the total energy of appropriately strained states of the materials (volume conserving technique), whereas the other way (stress-strain method) is based on the examination of changes in calculated stress values resulting from changes in strain (Hiki 1981). In the present investigation, we used the stress-strain method to obtain higher-order elastic constants (HOECs) such as second-, third-, and fourth-order elastic constants (SOECs, TOECs, and FOECs) developed by Mori and Hiki (1978) under the assumption that stress is proportional to strain within the elastic deformation limit expressed by Hooke's law. For anisotropic materials, Hooke's law is as follows:

$$\sigma_{ij} = C_{ijkl}\eta_{kl} \tag{1}$$

and its inverse

$$\eta_{kl} = S_{ijkl}\sigma_{ij} \tag{2}$$

Here, $i, j, k, l = 1, 2, 3$, C_{ijkl} is the fourth rank tensor of elastic stiffness or elasticity tensor and contains $6 \times 6 = 36$ independent components; σ_{ij} and η_{kl} are the second-rank stress and strain tensors for the anisotropic (aeolotropic/anisotropic) materials, and S_{ijkl} is the elastic compliance tensor. The elastic stiffness of the material determines the propagation properties of elastic waves in anisotropic media. The components of Lagrangian strain tensor η_{kl} under finite deformations are given by

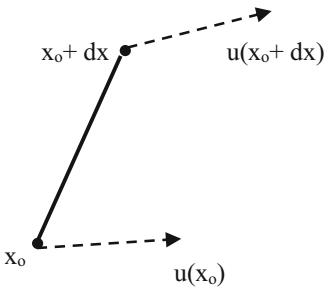

Fig. 4 The generalized displacement of the atoms in a crystal lattice

$$\eta_{kl} = \left(\frac{1}{2}\right)\left[\left(\frac{\partial u_k}{\partial x_l}\right)\left(\frac{\partial u_l}{\partial x_k}\right) - \delta_{ij}\right] \quad (k, l = 1, 2, 3) \tag{3}$$

where the displacement vector components **x** and **u** are the initial and final position of material points shown in Fig. 4. The vector **x** is the Cartesian coordinates of a point of the continuum in the unstrained state, and **u** is of the deformed state at time t. Thus, $u_i = u_i(x_1, x_2, x_3, t)$.

Under the circumstances of high stress and strain, the deviation from Hooke's law will take place. The complexity in the nonlinear theory of elasticity increases which leads to the introduction of HOECs. The determination of HOECs is important for identifying anharmonic characteristics of materials. Atomic displacements impact anharmonic properties such as thermal conductivity, nonlinear elasticity, and thermal expansion, as well as static and dynamic properties of lattices (Cowley 1963).

Now, the elastic potential energy density of the continuum is expressed by free energy expanded through Taylor series expansion in terms of the elastic strain components.

$$\phi = \frac{1}{2!}C_{ijkl}\eta_{ij}\eta_{kl} + \frac{1}{3!}C_{ijklmn}\eta_{ij}\eta_{kl}\eta_{mn} + \frac{1}{4!}C_{ijklmnpq}\eta_{ij}\eta_{kl}\eta_{mn}\eta_{pq} \tag{4}$$

Here, $C_{ijkl,}$ C_{ijklmn}, and $C_{ijklmnpq}$ are the usual SOECs, TOECs, and FOECs. The nth-order ($n \geq 2$) elastic constants defined by Brugger (1964) as

$$C_{ijklmn\ldots} = C_{IJK\ldots} = \left.\frac{\partial^n \phi}{\partial\eta_{ij}\partial\eta_{kl}\partial\eta_{mn}\cdots}\right|_{\eta=0} \tag{5}$$

The shorthand notation of elastic tensor C_{ijkl} (Voigt 1966; Love 1927) which is given by Voigt has been used for the indices ranging from 0 to 6:

$$11 \rightarrow 1,\ 22 \rightarrow 2,\ 33 \rightarrow 1,\ 23 \rightarrow 4,\ 31 \rightarrow 5,\ 12 \rightarrow 6$$

Thus, the elasticity tensor is now expressed as C_{IJ}. Also, the explicit forms of quadratic, cubic, and quartic terms of free energy are

$$
\begin{aligned}
F_2 &= \frac{1}{2!} C_{ijkl}\eta_{ij}\eta_{kl} \\
&= \frac{1}{2} C_{11}\left(\eta_{11}^2 + \eta_{22}^2 + \eta_{33}^2\right) + C_{12}(\eta_{11}\eta_{22} + \eta_{22}\eta_{33} + \eta_{33}\eta_{11}) \\
&\quad + 2C_{44}\left(\eta_{12}^2 + \eta_{23}^2 + \eta_{31}^2\right)
\end{aligned}
$$

$$
\begin{aligned}
F_3 &= \frac{1}{3!} C_{ijklmn}\eta_{ij}\eta_{kl}\eta_{mn} \\
&= \frac{1}{6} C_{111}\left(\eta_{11}^3 + \eta_{22}^3 + \eta_{33}^3\right) \\
&\quad + \frac{1}{2} C_{112}\left\{\eta_{11}^2(\eta_{22} + \eta_{33}) + \eta_{22}^2(\eta_{33} + \eta_{11}) + \eta_{33}^2(\eta_{11} + \eta_{22})\right\} + C_{123}\eta_{11}\eta_{22}\eta_{33} \\
&\quad + 2C_{144}\left(\eta_{11}\eta_{23}^2 + \eta_{22}\eta_{31}^2 + \eta_{33}\eta_{12}^2\right) \\
&\quad + 2C_{166}\left\{\eta_{12}^2(\eta_{11} + \eta_{22}) + \eta_{23}^2(\eta_{22} + \eta_{23}) + \eta_{31}^2(\eta_{33} + \eta_{11})\right\} \\
&\quad + 8C_{456}\eta_{12}\eta_{23}\eta_{31}
\end{aligned}
$$

and

$$
\begin{aligned}
F_4 = \frac{1}{4!} C_{ijklmnpq}\eta_{ij}\eta_{kl}\eta_{mn}\eta_{pq} &= \frac{1}{24} C_{1111}\left(\eta_{11}^4 + \eta_{22}^4 + \eta_{33}^4\right) \\
&\quad + \frac{1}{6} C_{1112}\left\{\eta_{11}^3(\eta_{22} + \eta_{33}) + \eta_{22}^3(\eta_{33} + \eta_{11}) + \eta_{33}^3(\eta_{11} + \eta_{22})\right\} \\
&\quad + \frac{1}{4} C_{1122}\left(\eta_{11}^2\eta_{22}^2 + \eta_{22}^2\eta_{33}^2 + \eta_{33}^2\eta_{11}^2\right) \\
&\quad + \frac{1}{2} C_{1123}\eta_{11}\eta_{22}\eta_{33}(\eta_{11} + \eta_{22} + \eta_{33}) + C_{1144}\left(\eta_{11}^2\eta_{23}^2 + \eta_{22}^2\eta_{31}^2 + \eta_{33}^2\eta_{12}^2\right) \\
&\quad + C_{1155}\left\{\eta_{11}^2\left(\eta_{31}^2 + \eta_{12}^2\right) + \eta_{22}^2\left(\eta_{12}^2 + \eta_{23}^2\right) + \eta_{33}^2\left(\eta_{23}^2 + \eta_{31}^2\right)\right\} \\
&\quad + 2C_{1255}\left\{\eta_{11}\eta_{22}\left(\eta_{23}^2 + \eta_{31}^2\right) + \eta_{22}\eta_{33}\left(\eta_{31}^2 + \eta_{12}^2\right) + \eta_{33}\eta_{11}\left(\eta_{12}^2 + \eta_{23}^2\right)\right\} \\
&\quad + 2C_{1266}\left(\eta_{11}\eta_{22}\eta_{12}^2 + \eta_{22}\eta_{33}\eta_{23}^2 + \eta_{33}\eta_{11}\eta_{31}^2\right) + 8C_{1456}\eta_{12}\eta_{23}\eta_{31}(\eta_{11} + \eta_{22} + \eta_{33}) \\
&\quad + \frac{2}{3} C_{4444}\left(\eta_{23}^4 + \eta_{31}^4 + \eta_{12}^4\right) + 4C_{4455}\left(\eta_{23}^2\eta_{31}^2 + \eta_{31}^2\eta_{12}^2 + \eta_{12}^2\eta_{23}^2\right)
\end{aligned} \tag{6}
$$

The total free energy density at a finite temperature T is given by

$$
F^{\text{Total}} = F^0 + F^{\text{vib}} = \left(F^2 + F^3 + F^4\right) + F^{\text{vib}} \tag{7}
$$

F^0 is the crystal's internal energy per unit volume, with all ions at rest at their lattice sites, and F^{vib} is the vibrational free energy given by

$$
F^{\text{vib}} = \frac{k_B T}{N V_c} \sum_{i=1}^{3sN} \ln 2 \sinh(\hbar\omega_i / 2k_B T) \tag{8}
$$

where V_c represents the volume of an elementary cell, s represents the number of ions in the cell, N represents the number of cells in the crystal, and k_B represents the Boltzmann constant.

Higher-Order Elastic Constants

SOECs, TOECs, and FOECs at a given temperature provide the strain derivatives of F^0 and F^{vib}, which are determined by adding the vibrational contribution to the static elastic constants given by Brugger (1964):

$$C_{IJ}(T) = C^0_{IJ} + C^{\text{vib}}_{IJ}; C_{IJK}(T) = C^0_{IJK} + C^{\text{vib}}_{IJK}; C_{IJKL}(T) = C^0_{IJKL} + C^{\text{vib}}_{IJKL} \tag{9}$$

where the superscript "0" indicates the static elastic constant at 0 K and the superscript "vib" indicates the vibrational portion of the elastic constant at a certain temperature.

The classic rigid-ion model applied on NaCl structure (Borg and Dienes 1992), in which the interaction energy between the pair of atoms arises from changes in the wave functions as the atom approaches, has been examined. The resulting total energy, ϕ, for the interaction as a function of atom separation, r, is the sum of two components: a long range interaction (lr), i.e., Coulomb potential, $\phi(r_o)$, and short-range interaction (sr), i.e., Born-Mayer short-range repulsive potential, $\phi\left(\sqrt{2}r_o\right)$, which is given in general by

$$\begin{aligned}\phi(r) &= \phi_{lr}(r) + \phi_{sr}(r) = \phi(r) = \phi(r_o) + \phi\left(\sqrt{2}r_o\right) = \phi(r) \\ &= \pm\left(\frac{e^2}{r_o}\right) + A\exp\left(\frac{-r_o}{b}\right)\end{aligned} \tag{10}$$

where e denotes the electronic charge, $\pm$ sign for like and unlike ions, r_o is the nearest-neighbor distance, b denotes the hardness parameter, and A stands for strength parameter given by

$$A = -3b\frac{e^2}{r_o}S_3{}^{(I)}\frac{1}{6\exp(-\rho_o) + 12\sqrt{2}\exp\left(-\sqrt{2}\rho_o\right)}; \rho_o = r_o/b \tag{11}$$

Tables 1 and 2 (Mori and Hiki 1978) provide detailed formulations for the static and vibrational components of SOECs, TOECs, and FOECs.

Here,

$$\begin{aligned}x = h\omega_o/2k_BT, \omega_o^2 &= \left(\frac{1}{M_+} + \frac{1}{M_-}\right)\left(\frac{1}{Hbr_o}\right), \quad H \\ &= \left\{(\rho_o - 2)\phi(r_o) + 2\left(\rho_o - \sqrt{2}\right)\phi\left(\sqrt{2}r_o\right)\right\}^{-1}\end{aligned} \tag{12}$$

k_B is the Boltzmann's constant; h is the Planck's constant. The values of lattice sums are obtained taking into consideration the Coulomb interactions between all the ions in the crystal up to the second nearest neighbor.

Table 1 Static and vibrational terms associated with SOECs, TOECs, and FOECs

Static	Vibrational
Second-order elastic constants (SOECs)	
$C_{11}^{0} = \frac{3}{2}\frac{e^2}{r_o^4}S_5^{(2)} + \frac{1}{br_o}\left(\frac{1}{r_o}+\frac{1}{b}\right)\phi(r_0) + \frac{2}{br_o}\left(\frac{\sqrt{2}}{2r_o}+\frac{1}{b}\right)\phi\left(\sqrt{2}r_0\right)$ $C_{12}^{0} = C_{44}^{0} = \frac{3}{2}\frac{e^2}{r_o^4}S_5^{(1,1)} + \frac{1}{br_o}\left(\frac{\sqrt{2}}{2r_o}+\frac{1}{b}\right)\phi\left(\sqrt{2}r_o\right)$	$C_{11}^{vib.} = f^{(1,1)}G_1^2 + f^{(2)}G_2$ $C_{12}^{vib.} = f^{(1,1)}G_1^2 + f^{(2)}G_{1,1}$ $C_{44}^{vib.} = f^{(2)}G_{1,1}$
Third-order elastic constants (TOECs)	
$C_{111}^{0} = -\frac{15}{2}\frac{e^2}{r_o^4}S_7^{(3)} - \frac{1}{b}\left(\frac{3}{r_o^2}+\frac{3}{br_o}+\frac{1}{b^2}\right)\phi(r_o) - \frac{1}{2b}\left(\frac{3\sqrt{2}}{r_o^2}+\frac{6}{br_o}+\frac{2\sqrt{2}}{b^2}\right)\phi\left(\sqrt{2}r_o\right)$ $C_{112}^{0} = C_{166}^{0} = -\frac{15}{2}\frac{e^2}{r_o^4}S_7^{(2,1)} - \frac{1}{4b}\left(\frac{3\sqrt{2}}{r_o^2}+\frac{6}{br_o}+\frac{2\sqrt{2}}{b^2}\right)\phi\left(\sqrt{2}r_o\right)$ $C_{123}^{0} = C_{144}^{0} = C_{456}^{0} = -\frac{15}{2}\frac{e^2}{r_o^4}S_7^{(1,1,1)}$	$C_{111}^{vib.} = f^{(1,1,1)}G_1^3 + 3f^{(2,1)}G_1G_2 + f^{(3)}G_3$ $C_{112}^{vib.} = f^{(1,1,1)}G_1^3 + f^{(2,1)}G_1(2G_{1,1}+G_2) + f^{(3)}G_{2,1}$ $C_{166}^{vib.} = f^{(2,1)}G_1G_{1,1} + f^{(3)}G_{2,1}$ $C_{123}^{vib.} = f^{(1,1,1)}G_1^3 + 3f^{(2,1)}G_1G_{1,1} + f^{(3)}G_{1,1,1}$ $C_{144}^{vib.} = f^{(2,1)}G_1G_{1,1} + f^{(3)}G_{1,1,1}$ $C_{456}^{vib.} = f^{(3)}G_{1,1,1}$
Fourth-order elastic constants (FOECs)	
$C_{1111}^{0} = \frac{105}{2}\frac{e^2}{r_o^4}S_9^{(3)} + \frac{r_o}{b}\left(\frac{15}{r_o^3}+\frac{15}{br_o^2}+\frac{6}{b^2r_o}+\frac{1}{b^3}\right)\phi(r_o) + \frac{r_o}{b}\left(\frac{15\sqrt{2}}{4r_o^3}+\frac{15}{2br_o^2}+\frac{3\sqrt{2}}{b^2r_o}+\frac{1}{b^3}\right)\phi\left(\sqrt{2}r_o\right)$ $C_{1112}^{0} = C_{1155}^{0} = \frac{105}{2}\frac{e^2}{r_o^4}S_9^{(3,1)} + \frac{r_o}{2b}\left(\frac{15\sqrt{2}}{4r_o^3}+\frac{15}{2br_o^2}+\frac{3\sqrt{2}}{b^2r_o}+\frac{1}{b^3}\right)\phi\left(\sqrt{2}r_o\right)$ $C_{1122}^{0} = C_{1266}^{0} = C_{4444}^{0} = \frac{105}{2}\frac{e^2}{r_o^4}S_9^{(2,2)} +$ $\frac{r_o}{2b}\left(\frac{15\sqrt{2}}{4r_o^3}+\frac{15}{2br_o^2}+\frac{3\sqrt{2}}{b^2r_o}+\frac{1}{b^3}\right)\phi\left(\sqrt{2}r_o\right)$ $C_{1123}^{0} = C_{1144}^{0} = C_{1255}^{0} = C_{1456}^{0} = C_{4455}^{0} = \frac{105}{2}\frac{e^2}{r_o^4}S_9^{(2,1,1)}$	$C_{1111}^{vib.} = f^{(1,1,1,1)}G_1^4 + 6f^{(2,1,1)}G_1^2G_2) +$ $3f^{(2,2)}G_2^2 + 4f^{(3,1)}G_1G_3 + f^{(4)}G_4$ $C_{1112}^{vib.} = f^{(1,1,1,1)}G_1^4 + 3f^{(2,1,1)}G_1^2(G_{1,1}+G_2) +$ $3f^{(2,2)}G_{1,1}G_2 + f^{(3,1)}G_1(3G_{2,1}+G_3) + f^{(4)}G_{3,1}$ $C_{1155}^{vib.} = f^{(2,1,1)}G_1^2G_{1,1} + f^{(2,2)}G_{1,1}G_2 +$ $2f^{(3,1)}G_1G_{2,1} + f^{(4)}G_{3,1}$ $C_{1122}^{vib.} = f^{(1,1,1,1)}G_1^4 + 2f^{(2,1,1)}G_1^2(2G_{1,1}+G_2) +$ $f^{(2,2)}\left(2G_{1,1}^2+G_2^2\right) + 4f^{(3,1)}G_1G_{2,1} + f^{(4)}G_{2,2}$

(continued)

Table 1 (continued)

Static	Vibrational
	$C^{vib.}_{1266} = f^{(2,1,1)}G_1^2G_{1,1} + f^{(2,2)}G_{1,1}^2 +$ $2f^{(3,1)}G_1G_{2,1} + f^{(4)}G_{2,2}$ $C^{vib.}_{4444} = 3f^{(2,2)}G_{1,1}^2 + f^{(4)}G_{2,2}$ $C^{vib.}_{1123} = f^{(1,1,1,1)}G_1^4 + f^{(2,1,1)}G_1^2(5G_{1,1} + G_2) +$ $f^{(2,2)}G_{1,1}(2G_{1,1} + G_2) + 2f^{(3,1)}(G_{1,1,1} + G_{2,1}) + f^{(4)}G_{2,1,1}$ $C^{vib.}_{1144} = f^{(2,1,1)}G_1^2G_{1,1} + f^{(2,2)}G_{1,1}G_2 + 2$ $f^{(3,1)}G_1G_{1,1,1} + f^{(4)}G_{2,1,1}$ $C^{vib.}_{1255} = f^{(2,1,1)}G_1^2G_{1,1} + f^{(2,2)}G_{1,1}^2 +$ $f^{(3,1)}G_1(G_{2,1} + G_{1,1,1}) + f^{(4)}G_{2,1,1}$ $C^{vib.}_{1456} = f^{(3,1)}G_1G_{1,1,1} + f^{(4)}G_{2,1,1}$ $C^{vib.}_{4455} = f^{(2,2)}G_{1,1}^2 + f^{(4)}G_{2,1,1}$

Table 2 Values of $f^{(n)}$ and G_n

$f^{(n)}$	G_n
$f^{(2)} = f^{(3)} = \frac{\hbar\omega_o}{8r_o^3}\coth x$ $f^{(1,1)} = f^{(2,1)} = f^{(2,2)} = f^{(3,1)} =$ $-\frac{\hbar\omega_o}{96r_o^3}\left(\frac{\hbar\omega_o}{2k_BT\ \sinh^2 x} + \coth x\right)$ $f^{(1,1,1)} = \frac{\hbar\omega_o}{384r_o^3}\left(\frac{(\hbar\omega_o)^2\coth x}{6(k_BT)^2\sinh^2 x} + \frac{\hbar\omega_o}{2k_BT\ \sinh^2 x} + \coth x\right)$	$G_1 = 2\{(2 + 2\rho_o - \rho_o^2)\phi(r_o) + 2(\sqrt{2} + 2\rho_o - 2\rho_o^2)\phi(\sqrt{2}r_o)\}H$ $G_2 = 2\left\{(-6 - 6\rho_o - \rho_o^2 + \rho_o^3)\phi(r_o) + \left(-3\sqrt{2} - 6\rho_o - \sqrt{2}\rho_o^2 + 2\rho_o^3\right)\phi\left(\sqrt{2}r_o\right)\right\}H$ $G_3 = 2\left[\left\{(30 + 30\rho_o + 9\rho_o^2 - \rho_o^3 - \rho_o^4)\phi(r_o) + \left\{(15/2)\sqrt{2} + 15\rho_o + (9/2)\sqrt{2}\rho_o^2 - \rho_o^3 - \sqrt{2}\rho_o^4)\phi\left(\sqrt{2}r_o\right)\right\}\right]H$ $G_{1,1} = \left\{\left(-3\sqrt{2} - 6\rho_o - \sqrt{2}\rho_o^2 + 2\rho_o^3\right)\phi\left(\sqrt{2}r_o\right)\right\}H$ $G_{2,1} = (15/2)\sqrt{2} + 15\rho_o +$ $(9/2)\sqrt{2}\rho_o^2 - \rho_o^3 - \sqrt{2}\rho_o^4)\phi\left(\sqrt{2}r_o\right)\}H$; $G_{1,\ 1,\ 1} = 0$

$$S_3^{(1)} = -0.58252; \quad S_5^{(2)} = -1.04622; \quad S_5^{(1,1)} = 0.23185$$
$$S_7^{(3)} = -1.36852; \quad S_7^{(2,1)} = 0.16115; \quad S_7^{(1,1,1)} = -0.09045$$

Using the above formulation, the SOECs, TOECs, and FOECs are computed in further chapters.

Pressure Derivatives

When a cubic crystal is subjected to higher hydrostatic pressure, the arrangement of the component of the crystal's specific structure is preserved. The idea of effective elastic constants is used to characterize nonlinear elastic characteristics, C_{IJ} (P). The Taylor series expansion explored by Birch (1947) yields the equations for the effective second-order elastic constants, C_{IJ}(P), as a function of pressure P. The first and second-order pressure derivative expressions are calculated.

$$C_{IJ..}(P) \approx C_{IJ..} + \frac{dC_{IJ..}(P)}{dP} P = C_{IJ..} + C'_{IJ..} P \tag{13}$$

The values of pressure derivatives, $C'_{IJ..}$, are computed using SOECs and TOECs as expressed in Table 3. It also lists the partial contractions of the FOECs calculated using these pressure derivatives.

Mechanical Properties

Elastic moduli give a macroscopic and global perspective of a material's stiffness. It reflects the interatomic bonding energies as well as the interatomic connections. Many of a crystalline material's mechanical and physical qualities are related to its elastic properties. The bulk modulus (B), shear modulus (G), Young's modulus (Y), and Poisson's ratio are all calculated elastic constants that are used to determine the crystal's reaction to external forces (v). These properties aid in the determination of material strength and are required for material characterization. The applied forces, which might be in the form of pressure or temperature, provide information regarding material structural changes.

The three independent elastic constants are linked to elastic moduli in Table 4 for single crystals that are elastically anisotropic. Single crystal elastic constants values are further used to compute polycrystalline properties under Voigt (V), Reuss (R), and Hill (H) approximation (Voigt 1966; Reuss 1929; Hill 1952).

Table 3 Pressure derivatives expressed as a function of SOECs and TOECs

First-order pressure derivatives	Second-order pressure derivatives
$\frac{dC_{11}}{dP} = (2C_{11} + 2C_{12} + C_{111} + 2C_{112})/C_A$ $\frac{dC_{12}}{dP} = (-C_{11} - C_{12} + 2C_{112} + C_{123})/C_A$ $\frac{dC_{44}}{dP} = -(C_{11} + 2C_{12} + C_{44} + C_{144} + 2C_{166})/C_A$	$\frac{d^2C_{11}}{dP^2} = \left[(1+3C_B)C_{11} + (4+3C_B)(C_{111} + 2C_{112}) + C_{1111} + 4C_{1112} + 2C_{1122} + 2C_{1123}\right]/(C_A)^2$ $\frac{d^2C_{12}}{dP^2} = [(1+3C_B)C_{12} + (4+3C_B)(2C_{112} + C_{123}) + 2C_{1122} + 5C_{1123}]/(C_A)^2$ $\frac{d^2C_{44}}{dP^2} = \left[(1+3C_B)C_{44} + (4+3C_B)(C_{144} + 2C_{166}) + C_{1144} + 2C_{1155} + 4C_{1255} + 2C_{1266}\right]/(C_A)^2$
$\frac{dC_{111}}{dP} = -(-3C_{11} - 6C_{12} + 3C_{111} + C_{1111} + 2C_{1112})/C_A$ $\frac{dC_{112}}{dP} = -(C_{11} + 2C_{12} + 3C_{112} + C_{1112} + C_{1122} + C_{1123})/C_A$ $\frac{dC_{123}}{dP} = -(-C_{11} - 2C_{12} + 3C_{123} + 3C_{1123})/C_A$ $\frac{dC_{144}}{dP} = -(C_{11} + 2C_{12} + 3C_{144} + C_{1144} + 2C_{1255})/C_A$ $\frac{dC_{166}}{dP} = -(-C_{11} - 2C_{12} + 3C_{166} + C_{1166} + 2C_{1255})/C_A$ $\frac{dC_{456}}{dP} = -(-C_{11} - 2C_{12} + 3C_{456} + 3C_{1456})/C_A$	Where $C_A = \frac{1}{(C_{11}+2C_{12})}$ $C_B = (4C_{11} + C_{111} + 6C_{112} + 2C_{123})/C_A$ $Y_{11} = C_{1111} + 4C_{1112} + 2C_{1122} + 2C_{1123}$ $Y_{44} = C_{1144} + 2C_{1155} + 4C_{1255} + 2C_{1266}$ $Y_{12} = 2C_{1112} + 2C_{1122} + 5C_{1123}$

Table 4 Elastic moduli for the symmetrical cubic crystal

Elastic properties	Formulae	Definition
Isotropic shear modulus	$G = (G_V + G_R)/2$ where, $G_V = (C_{11} - C_{12} + 3C_{44})/5$ $G_R = 5[(C_{11} - C_{12})C_{44}]/[4C_{44} + 3(C_{11} - C_{12})]$	It shows the relation between shear stress and shear strain. G_V is Voigt's shear modulus corresponding to the upper bound of G values, and G_R is Reuss's shear modulus corresponding to the lower bound of G values
Bulk modulus	$B = (C_{11} + 2C_{12})/3$	It provides a good link between the macroscopic elasticity theory and the atomistic viewpoints such as lattice dynamics
Young's modulus	$Y = \frac{9GB}{G+3B}$	It relates a unidirectional stress to the resultant strain and defines the stiffness of the materials
Poisson's ratio	$\nu = (3B - 2G)/(6B + 2G)$	It is defined as the ratio of lateral contraction to the longitudinal extension
Anisotropic ratio	$A = 2C_{44}/(C_{11} - C_{12})$	It represents the ratio of two elastic-shear coefficients extremes
Tetragonal shear modulus	$C_s = C_{44}$ for shear on <100> $C_S = \left(\frac{C_{11}-C_{12}}{2}\right)$ for shear on <110>	

Born Stability Criteria

For an unstrained cubic crystal to be mechanically stable, the elastic stability criteria given by Born (1940) must be satisfied. It is necessary that the following inequalities must be fulfilled for the representation of free energy by positive defined quadratic form.

$$C_{11} + 2C_{12} > 0, C_{44} > 0, C_{11} - C_{12} > 0 \tag{14}$$

Cauchy's Relations

The Cauchy's relation for elastic constants in cubic crystals has been confirmed (Cousins 1971) and is true only if:

1. The forces of contact between the atoms or molecules of a crystal are central forces, as in rock salt.
2. Each atom or molecule is the point of symmetry.
3. A harmonic potential can be used to represent the interaction forces between the building pieces of a lattice vibration in a crystal.

These relations for SOECs and TOECs of a cubic crystal are defined by Cousins (1967)

$$C_{12} = C_{44}; C_{112} = C_{166}; C_{123} = C_{456} = C_{144} \tag{15}$$

Generally, Cauchy's relations hold good at 0 K.

Elastic Waves in Anisotropic Materials

Elastic waves produced by mechanical vibrations of material media under applied stress are the consequence of collective vibrations of the medium's atoms and molecules. The magnitude of interatomic forces helps in determining the vibration properties of the medium's atoms and molecules. For solids with varied structures, the nature of these forces and, as a result, their wave propagation properties are variable. The materials chosen in our case are of rock-salt type (face-centered cubic). The computed elastic constants are related to the velocities of acoustic plane waves using the approach as suggested by investigators (Blackman 1955; Thurston and Brugger 1964) which are either pure longitudinal or pure transverse and are only propagated in the principal symmetry directions <100>, <110>, and <111> for cubic materials. The ultrasonic velocities in different crystallographic directions are computed using the classical linear theory of elasticity; the solution of equation of motion given by Eq. (16) for a freely vibrating body (Bedford and Drumheller 1994) for elastic wave propagation in a homogeneous anisotropic solid is obtained and listed in Table 5.

Table 5 Ultrasonic velocities along three crystallographic directions

Direction of wave propagation	Polarization direction	Wave mode	Expression
<100>	<100>	Longitudinal	$V_L = \sqrt{\frac{C_{11}}{\rho}}$
	<010>	Shear	$V_{S1} = \sqrt{\frac{C_{44}}{\rho}}$
	<001>	Shear	$V_{S2} = \sqrt{\frac{C_{44}}{\rho}}$
<110>	<110>	Longitudinal	$V_L = \sqrt{\frac{C_{11}+C_{12}+2C_{44}}{2\rho}}$
	<001>	Shear	$V_{S1} = \sqrt{\frac{C_{44}}{\rho}}$
	< 1−10 >	Shear	$V_{S2} = \sqrt{\frac{C_{11}-C_{12}}{2\rho}}$
<111>	<111>	Longitudinal	$V_L = \sqrt{\frac{C_{11}+2C_{12}+4C_{44}}{3\rho}}$
	< −110 >	Shear	$V_{S1} = V_{S2} = \sqrt{\frac{C_{11}-C_{12}+C_{44}}{3\rho}}$

$$\sigma_{ij,j} = \rho \ddot{u}_i \tag{16}$$

where $\ddot{u}_i$ indicates the partial derivative of displacement with respect to time and $\sigma_{ij,\,j}$ is the partial derivative of stress with respect to coordinate index j and ρ is the density of medium given by

$$\rho = \frac{nM}{N_a a^3} \tag{16a}$$

where, n, number of atoms per unit cell with $n = 2$ for body-centered cubic (bcc); $n = 4$ for fcc and $n = 8$ for diamond lattices; N_a, Avogadro's number; M, molecular mass, respectively.

Anisotropic materials can propagate both shear and longitudinal waves. The propagation velocities through such media are determined by the orientations of the stresses in relation to the medium's directions. It's worth noting that (a) the two shear waves degenerate into the same velocity in some directions, and (b) the shear wave in some wave guides is actually torsional in nature, with only one torsional wave velocity because the torsional mode automatically averages the stress in two perpendicular directions.

This also aids in the calculation of the Debye average velocity, Vm, which is a crucial parameter in determining thermal characteristics such as Debye temperature and thermal relaxation time of materials (Singh et al. 2011; Mason and Rosenberg 1966). The Debye average acoustic velocity is given along <100> and <111> directions by

$$V_m = \left[\frac{1}{3}\left\{\frac{1}{V_L^3} + \frac{2}{V_{S1}^3}\right\}\right]^{-1/3}; \; V_{S1} = V_{S2}$$

and along <110> direction

$$V_m = \left[\frac{1}{3}\left\{\frac{1}{V_L^3} + \frac{2}{V_{S1}^3} + \frac{1}{V_{S2}^3}\right\}\right]^{-1/3}; \; V_{S1} \neq V_{S2} \tag{17}$$

The Debye temperature (θ_D) is used to explain several lattice thermal processes and to define phonon excitation. θ_D correlates with several physical characteristics of solids, such as specific heat, elastic constants, and melting temperature, as an important parameter. Acoustic vibrations are the only source of vibrational excitation at low temperatures. As a result, the Debye temperature derived using elastic constants is the same as that predicted from specific heat measurements at low temperatures (almost at room temperature).

$$\theta_D = \frac{\hbar}{k_B}\left[\frac{3n}{4\pi}\left(\frac{N_a \rho}{M}\right)\right]^{1/3} V_m \tag{18}$$

where, n, number of atoms in the molecule and $\hbar = h/2n$.

Breazeale's Nonlinearity Parameter, β

The waveform distortion caused by ultrasonic waves travelling through a solid substance is induced by the medium's microstructural characteristics. The existence of three types of acoustic modes for a given direction of wave propagation in an anisotropic material is known from section "Elastic Waves in Anisotropic Materials." The harmonic production may be distinguished using the numerous interaction mechanisms of these acoustic waves. The nonlinearity of the component's material, which is characterized by a nonlinear stress–strain relationship, causes harmonic production. A nonlinearity parameter, β, can be used to quantify the degree of nonlinearity present in a material. Theoretically, it constitutes a separate material characteristic and is characterized in terms of HOECs from the nonlinear stress–strain relationship.

It characterizes the simple harmonic generation of longitudinal waves given by the negative ratio of non-linear term to the linear term in the nonlinear wave given by Philip and Breazeale (1983):

$$\beta = -(3K_2 + K_3)/K_2 \tag{19}$$

where K_2 and K_3 are linear combination of SOECs and TOECs, respectively, given in Appendix A.

Grüneisen Parameters

Grüneisen parameters (GP), which describe the relative strain dependence of elastic wave velocity and shed additional light on the thermal and mechanical characteristics of materials, are used to investigate the anharmonicity of a crystal lattice. It aids in the description of characteristics such as lattice-specific heats at high temperatures, thermal conductivity, thermal expansion, temperature fluctuation of elastic constants, and so on. The Grüneisen parameters define the phonon contribution to a variety of anharmonic solid characteristics such as lattice specific heat, thermal expansion, thermal conductivity, and so on. These parameters are represented as different weighted averages of the first-order Grüneisen tensor (Nava and Romero 1978).

$$\gamma^j_{\alpha\beta} = -\omega_i^{-1}\partial\omega_i(q)/\partial\eta_{\alpha\beta} \tag{20}$$

where ω, angular frequency, and $\eta_{\alpha\beta}$ are the six strain components ($\alpha, \beta = 1, 2, 3$). For instance, the thermal expansivity is relative to the specific heat weighted,

$$< \gamma_{\alpha\beta} >= \sum_{q,j} C_{q,i}\gamma^j_{\alpha\beta} / \sum_{q,i} C_{q,i} \tag{21}$$

Thermal conductivity weighted averages of the product may be used to represent the thermal Grüneisen parameters, γ, and the ultrasonic attenuation's (shear) Grüneisen parameter. Brugger (1965) calculated the components of the Grüneisen tensor in terms of SOECs and TOECs for anisotropic elastic continuum. Appendix B contains formulae for Grüneisen parameters (Klemens and Mason 1965) in various crystallographic orientations.

Ultrasonic Attenuation

The energy is dissipated by numerous mechanisms such as absorption, diffraction, and scattering as the ultrasonic wave passes through the object. The reasons for attenuation may be divided into two categories.

First is the scattering caused by discontinuities at grain boundaries, phonon-phonon (p–p) scattering, electron-phonon (e–p) scattering, inclusions, etc. Secondly, the absorption of ultrasound may be explained on the basis of thermoelastic effects, dislocation damping due to screw and edge dislocations, ferromagnetic effects, relaxation, and the effects of neutrons and electrons upon the sound beam. Scattering effects are generally much greater than the absorption effects.

The study of ultrasonic attenuation in crystals as a function of temperature is a useful tool for understanding the process of ultrasonic wave interaction with crystal lattice and intrinsic characteristics of all sorts of substances, including metallic, covalent, semiconducting, and dielectric crystals. The knowledge regarding the interaction of ultrasonic waves with individual phonons in the region $\omega\tau_{th} \ll 1$ (Mason 1967), where ω is the angular frequency and τ_{th} is the thermal relaxation time, may be deduced from variations in ultrasonic attenuation as a function of temperature.

Attenuation Due to e–p Interaction

The exchange of energy happens in metals free electrons and interacting lattice, according to Debye's theory of specific heat at low temperatures. At high temperatures, the electron mean free path is not equivalent to phonon wavelength; hence there will be no attenuation owing to electron-phonon interaction. The e–p interaction is very important for studying the optical and electronic properties. It is also responsible for many phenomena such as Kohn effect, temperature-dependent electrical resistivity, conventional superconductivity, etc. The ultrasonic attenuation of the wave becomes important for study below 100 K. Further, at extreme lower temperature along with other factors, dislocation losses vanish. Only e–p interaction remains effectual. In present case all the investigations have been done at 100 K. So, there is no consideration of e–p interaction in the chosen materials.

Attenuation Due to p–p Interaction

Akhiezer (1939) proposed this type of theory for the first time in the region ($\omega\tau_{th} \ll 1$) which was later simplified by Bömmel and Dransfeld (1960), and more complete version for the case $\omega\tau_{th} \gg 1$ was considered by Woodruff and Ehrenreich (1961). Finally, Mason (1967) enunciated the simplest form of this theory. He introduced the usage of Grüneisen parameters, $< \gamma_i^j >$, which has been already explained in section "Grüneisen Parameters." When the stress is applied, the non-equilibrium separation of phonon modes takes place. Thus, the change in elastic constants occurs and the ultrasonic attenuation, α, is given by

$$\alpha = \frac{\Delta C \omega^2 \tau_{th}}{2\rho V^3 (1 + \omega^2 \tau_{th})} \tag{22}$$

When the strain, S_j, is applied, the changes occur in the frequency of propagation mode given by

$$\omega_i = \omega_{io} = \left(1 - \sum\nolimits_{j=1} \gamma_i^j S_j\right) \tag{23}$$

We get, on differentiation

$$\gamma_i^j = -\frac{\partial \omega_i / \partial S_j}{\omega_{io}} \tag{24}$$

The thermal energy associated with the modes considering Debye approximation theory is

$$U_{th} = 3\hbar \sum\nolimits_i \frac{N_i}{\omega_{gi}^3} \int_0^{\omega_{gi}} \frac{\omega^3 d\omega}{e^{\hbar\omega/k_B T} - 1} \tag{25}$$

On performing differentiation on the total elastic energy and sum of thermal energy of all modes, we get

$$T_j = \left[C_{ij}^0 + 3 \sum\nolimits_l E_i \left(\gamma_i^j\right)\right] S_j + 3 \sum\nolimits_l E_i \gamma_i^j \tag{26}$$

where T_j represents the stress corresponding to the strain, S_j. E_i is the thermal energy associated with the mode propagating along a particular direction. The increase in elastic moduli is observed given by

$$\begin{aligned} \Delta C &= 3 \sum E_i \left(\gamma_i^j\right)^2 \quad \text{for shear modes} \\ \Delta C &= \left[3 \sum E_i \left(\gamma_i^j\right)^2 - \gamma^2 CT\right] \quad \text{for longitudinal modes} \end{aligned} \tag{27}$$

Using this value of ΔC in Eq. (22)

$$\alpha = \frac{\omega^2 \tau_{th} E_o (D/3)}{2\rho V^3 \left(1 + \omega^2 \tau_{th}^2\right)} \tag{28}$$

The expressions for calculating ultrasonic attenuation due to p–p interaction [$(\alpha/f^2)_L$ and $(\alpha/f^2)_S$] using Eq. (28) are

$$\left(\alpha/f^2\right)_L = \frac{2\pi^2 \tau_{th} E_0 D_L}{3\rho V_L^3}; \quad \left(\alpha/f^2\right)_s = \frac{2\pi^2 \tau_{th} E_0 D_S}{3\rho V_S^3} \tag{29}$$

where α is the ultrasonic attenuation constant, f denotes frequency of the ultrasonic wave, ρ is the density, V is the sound wave velocity, V_L and V_S are the ultrasonic velocities for longitudinal and shear waves, respectively, E_0 is the thermal energy density, κ is the thermal conductivity, and γ_i^j denotes Grüneisen number (i, j are the mode and direction of propagation, respectively). D, the acoustic coupling constant, is the change in elastic modulus caused by strain given by

$$D = 9\left\langle \left(\gamma_i^j\right)^2 \right\rangle - \frac{3\left\langle \gamma_i^j \right\rangle^2 C_v T}{E_0} \tag{30}$$

where C_v is the specific heat per unit volume and τ_{th}, the thermal relaxation time for the exchange of acoustic and thermal energy given by Herring (1954) which is

$$\frac{1}{2}\tau_L = \tau_S = \tau_{th} = \frac{3\kappa}{C_v V_m^2} \tag{31}$$

where κ is the thermal conductivity. Both C_v and E_0 are function of θ_D/T, which differ by a numerical factor and are obtained by AIP Handbook (Gray 1972). As a result, the major mechanisms that will raise the attenuation considerably above 50 K and higher include p–p interaction, dislocation damping, and thermoelastic loss. At room temperature, the p–p interaction is the dominant source of ultrasonic attenuation in all types of materials, including metallic, semiconducting, and dielectrics (Barrett and Holland 1970; Jyoti et al. 2021).

Loss Due to Thermoelastic Mechanism

When compared to total attenuation, thermoelastic loss causes very little attenuation. The p–p interaction causes the energy loss of ultrasonic waves, which may be calculated using Mason's method (1967). Hysteresis absorption is caused by the p–p interaction. Determining the thermal phonon relaxation time is therefore of importance. The propagation of an acoustic phonon and the re-establishment of the equilibrium as the relaxation phenomena have been postulated to yield the equilibrium distribution of phonons in a crystal.

The thermal phonons may be thought of as a fluid with a viscosity that results in f^2 absorption when the sound wavelength is substantially longer than the mean free path of the thermal phonons in the material. The requirement for this conclusion to hold owing to thermoelastic relaxation $(\alpha/f^2)_{th}$ process may also be expressed as

$$\left(\alpha/f^2\right)_{th} = \frac{4\pi^2 < \gamma_i^j >^2 \kappa T}{2\rho V_L^5} \quad \omega\tau_{th} \ll 1 \tag{32}$$

The total attenuation is given by

$$\left(\alpha/f^2\right)_{\text{Total}} = \left(\alpha/f^2\right)_{\text{Akhiezer}} + \left(\alpha/f^2\right)_{th} \tag{33}$$

Attenuation Due to Dislocation Damping

Dislocation damping, a process that has been proven to be beneficial in metals and rocks, is the last of the potential mechanisms to be addressed here. Absorption may be caused by dislocations in four different ways:

- The dislocation length's resonance absorption. This loss is dependent on f^2.
- A dislocation relaxation process (Bordoni loss), which is f dependent as well.
- A dislocation breaking away from the impurity pinning locations causes a strain-dependent hysteresis loss.
- The mobility of kinks in the dislocation length causes a strain-independent hysteresis loss.

Due to the thermal mechanism in the crystal, dislocations are subjected to phonon viscosity damping effects (Mason 1960). The viscosity of phonons contributes directly to acoustic attenuation and indirectly to dislocation motion damping given by

$$\eta = \frac{E_0 \kappa}{C_v V_m^2} \tag{34}$$

The dislocation, also known as the linear imperfection in a crystal, is made up of a combination of screw and edge dislocations. In the literature (Mason and Rosenberg 1966), the dislocation damping due to screw and edge dislocations has been computed.

$$B_{\text{screw}} = 0.071\eta \text{ and } B_{\text{edge}} = \left(0.0532\eta + 0.0079(\mu/B)^2\chi\right)/(1-\nu)^2 \tag{35}$$

where ν, μ, B, χ and η are Poisson's ratio, shear modulus, bulk modulus, and compressional viscosity.

Thermal Conductivity

Understanding the thermal conductivity of crystal lattice is critical for improving thermoelectric and optoelectronic device performance. Thermal conduction in a material refers to the rate of heat flow between opposing sides of an infinite slab with a 1 K temperature differential and is defined as the rate at which heat flows through phonon transport in a temperature gradient. It is reliant on the interatomic potential's anharmonicity (Ashcroft and Mermin 1976).

It is important to have a better knowledge of the thermal characteristics of rare-earth materials. Low thermal conductivity materials are required in thermal barrier coatings (for gas turbines) and thermoelectrics, for example. Thermal anisotropy is also required for heat spreading applications such as electronic packaging and magnets. To determine the magnitude and temperature dependence of lattice thermal conductivity, a simple model of lattice heat conduction proposed by Slack (1973, 1979) and Berman (1976) was used to estimate the magnitude and temperature dependence of thermal conductivity, which takes the form

$$\kappa = A.\frac{\overline{M_a}\theta_D^3 \delta n^{1/3}}{\gamma^2 T} \tag{36}$$

where δ (in Å) is the volume per atom, $A = 3.04 \times 10^{-8}$, n is the number of atoms in a molecule, γ is the Grüneisen constant, $\overline{M_a}$ (in amu) is the average atomic mass, and T is the temperature (in K).

Furthermore, the thermal conductivity of materials is related to the ultrasonic wave velocity associated with lattice vibration. As a result, the lattice's lowest thermal conductivity along with various crystal orientations (Cahill et al. 1992) is given by

$$\kappa_{\min} = \frac{k_B}{2.48} q^{2/3}(V_L + V_{S1} + V_{S2}) \tag{37}$$

where "q" is the atomic number per unit volume. This property is important when the material is subjected to thermal stresses and optical distortions. The thermal conductivity helps in obtaining the values of thermal relaxation time (τ_{th}) which is directly proportional to ultrasonic attenuation. It also provides base for calculation of dislocation damping coefficient due to screw and edge dislocations.

Computational Analysis

All computed data have an inherent uncertainty or error associated. Adapting to the good scientific practices assists in recognizing these limitations and, thus, providing a clear indication of the precision and reliability involved in any procedure. The correct interpretation of the measured or computed results requires knowledge about its uncertainty. The size of the uncertainty from each source with random error in the data is estimated and has been listed in Tables 6 and 7.

Table 6 Elastic constants for PrN and NpAs at different temperature range (in GPa)

Material	Temp (K)	Elastic constants					
		C_{11}		C_{12}		C_{44}	
		Uncertainty		Uncertainty		Uncertainty	
PrN	0	55.08	**58.585 ± 3.5**	22.65	**21.815 ± 0.835**	22.65	**22.81 ± 0.16**
	300	62.09		20.98		22.97	
NpAs	0	50.91	**50.135 ± 1.0**	12.88	**11.02 ± 0.57**	12.88	**13.01 ± 0.04**
	100	46.62		12.13		12.93	
	200	48.25		11.39		12.98	
	300	49.95		10.65		13.04	
	400	51.67		9.90		13.09	
	500	53.41		9.16		13.15	

Table 7 Uncertainty in the values of elastic constants for BP and InP with respect to experimental values (in GPa) at 300 K

	Elastic constants					
	C_{11}		C_{12}		C_{44}	
Material	Uncertainty		Uncertainty		Uncertainty	
BP	330.64	**352.48 ± 30.0**	81.19	**82.43 ± 9.80**	85.81	**133.34 ± 23.82**
Wang and Ye (2003)	411.8		66.11		154.2	
Meradji et al. (2004)	315		100		160	
InP	121.64	**123.15 ± 4.22**	49.04	**50.41 ± 0.70**	31.41	**30.17 ± 1.60**
Wang and Ye (2003)	131.1		51.3		32.1	
Hajlaoui et al. (2015)	116.7		50.9		27.0	

Elastic Constants

The variation of SOECs with temperature has been examined, and it has been discovered that C_{11} and C_{44} increase while C_{12} decreases as temperature rises. The variability in atomic interactions with temperature causes the elastic constant to increase or decrease, similar to what is observed in B_1 structured materials (Kumar et al. 2012).

SOECs of Group IIIrd Phosphides

The SOECs are calculated using a method that has been described in literature (Bhalla et al. 2013). The deviations between the estimated values of elastic constants using the current approach and previously published values are listed in Table 7. This discrepancy is due to changes in the methods used to estimate elastic properties; nonetheless, the similarity in order validates the viability of our method for calculating elastic constants using the basic interaction potential model approach.

Ultrasonic Velocity

Table 8 shows the computed longitudinal and shear elastic wave velocities in the temperature range 100–500 K. The ultrasonic velocity is directly proportional to SOECs and density (Singh et al. 2011).

Grüneisen Parameters

The ultrasonic Grüneisen parameters (UGPs) are determined using SOECs and TOECs for various modes and vary depending on the combined effect of the elastic constants and the material's molecular weight. It is important in the study of materials with anharmonic characteristics. Physical characteristics of materials such as thermal expansion, thermal conductivity, and temperature fluctuation of elastic constants are described. The resulting values of average UGPs are observed to vary very little with increasing molecular weight, although the average squares of the Grüneisen parameters decrease as molecular weight increases (Table 9).

Table 8 Ultrasonic velocities V_L and V_S of TbP (all in the units of 10^3 m/s)

	<100>	
Temp (K)	V_L	$V_{S1}=V_{S2}$[a]
100	2.73	1.43
200	2.79	1.44
300	2.87	1.45
400	2.94	1.46
500	3.01	1.47
Uncertainty	**2.868 ± 0.05**	**1.45 ± 0.007**

[a]Shear wave polarized along <100> direction

Table 9 Grüneisen parameters for CePn along different directions in temperature range 100–500 K

Material	Grüneisen parameters	100 K	200 K	300 K	400 K	500 K	Uncertainty
Ultrasonic longitudinal wave propagates along <100>							
CeN	$<\gamma^j_i>$	0.4624	0.4486	0.4343	0.4206	0.4078	**0.4347 ± 0.01**
	$<(\gamma^j_i)^2>$	1.7199	1.6368	1.5526	1.4751	1.4051	**1.5579 ± 0.06**
CeP	$<\gamma^j_i>$	0.4565	0.4399	0.4238	0.4089	0.3951	**0.4248 ± 0.01**
	$<(\gamma^j_i)^2>$	2.0588	1.9535	1.8560	1.7690	1.6918	**1.8658 ± 0.07**
CeAs	$<\gamma^j_i>$	0.4587	0.4408	0.4239	0.4084	0.3941	**0.4252 ± 0.01**
	$<(\gamma^j_i)^2>$	2.1473	2.0325	1.9296	1.8390	1.7592	**1.9415 ± 0.07**

Current Applications

Currently, the rare-earth elements and compounds are generating lot of interest among scientists and technology developers because of their very important role in green applications. There are numerous areas where rare-earth is being used for increasing the efficiency and enhancing the properties of the samples.

Area	Application
Electronics and technology	Optoelectronic devices, semiconductors, silicon chips, high-temperature superconductors, computers, and cell phones
Renewable and thermal energy	Electrothermal energy storage (ETEs), hybrid automobiles, rechargeable batteries

Conclusion

The data obtained in the research work provides a better insight to the properties of RE materials. It enables for further research into prospective industrial applications such as nondestructive testing (NDT), thermal conductors and insulators, spintronics, device fabrication, high-temperature materials, electronic industry, scientific research, and experimentation. The preliminary results of this study can be

used for further experimental investigation using the pulse echo overlap (PEO) technique for ultrasonic measurements, as well as traditional analytical techniques such as polarizing microscopy, surface tension analysis, solid-state nuclear magnetic resonance (NMR), scanning electron microscopy (SEM), X-ray diffraction (XRD), and transmission electron microscopy (TEM) add a whole new dimension to the study of the material.

In the future, the fundamental properties of these monopnictides and monochalcogenides will be widely used in the construction of hybrid devices, such as high-energy spectroscopy, hybrid automotive engines, laser sources, television screens, and electronic industries, among other things.

Cross-References

► Industrial Metrology

Appendix A

Table A1 Non-linearity parameters along different crystallographic directions

Direction	<100>	<110>	<111>
K_2	C_{11}	$\frac{1}{2}(C_{11}+C_{12}+2C_{44})$	$\frac{1}{3}(C_{11}+2C_{12}+4C_{44})$
K_3	C_{111}	$\frac{1}{4}(C_{111}+3C_{112}+12C_{166})$	$\frac{1}{9}(C_{111}+6C_{112}+12C_{144}+24C_{166}+2C_{123}+16C_{456})$

Appendix B

Table B1 Equations for Grüneisen numbers along <100> for longitudinal waves

Type of wave	No. of modes	Equations for $<\gamma_i^j>$
Longitudinal	1	$\frac{3C_{11}+C_{111}}{2C_{11}}$
Shear	2	$\frac{C_{11}+C_{166}}{2C_{44}}$
Longitudinal	2	$\frac{C_{11}+C_{112}}{2C_{11}}$
Shear	2	$\frac{2C_{44}+C_{12}+C_{166}}{2C_{44}}$
Shear	2	$\frac{C_{12}+C_{144}}{2C_{44}}$
Longitudinal	2	$\frac{2C_{12}+C_{112}+2C_{144}+C_{123}}{2(C_{11}+C_{12}+2C_{44})}$
Shear	2	$\frac{2C_{12}+C_{112}-C_{123}}{2(C_{11}-C_{12})}$
Shear	2	$\frac{C_{12}+2C_{44}+C_{166}}{2C_{44}}$
Longitudinal	4	$\frac{2(C_{11}+C_{12}+C_{44})+C_{111}/2+3C_{112}/2+2C_{166}}{2(C_{11}+C_{12}+2C_{44})}$

(continued)

Type of wave	No. of modes	Equations for $<\gamma_i^j>$
Shear	4	$\frac{2C_{11}+(C_{111}-C_{112})/2}{2(C_{11}-C_{12})}$
Shear	4	$\frac{C_{11}+C_{12}+C_{144}+C_{166}}{4C_{44}}$
		Longitudinal
4		$\frac{5C_{11}+10C_{12}+8C_{44}+C_{111}+6C_{112}+2C_{123}+4C_{144}+8C_{166}}{6(C_{11}+2C_{12}+4C_{44})}$
Shear	4	$\frac{4C_{11}+5C_{12}+C_{44}+(C_{111}+3C_{112})/2+(C_{144}+5C_{166})/2-2C_{123}}{6(C_{11}-C_{12}+C_{44})}$
Shear	4	$\frac{2C_{11}+C_{12}+C_{44}+(C_{111}-C_{112})/2+(C_{144}+C_{166})/2}{2(C_{11}-C_{12}+C_{44})}$

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Hardness Metrology

35

Riham Hegazy

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Abstract

This chapter presents different types of hardness test method, and calibration of testing machines (both direct and indirect method). Conventional static hardness methods (Brinell, Rockwell and Vickers) and dynamic Leeb rebound hardness with corresponding uncertainty calculation is discussed in this chapter. The recent advance of instrumented hardness is also presented and its important m and field of application.

Keywords

Hardness · Dynamic hardness · Uncertainty · Metrological chain

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Introduction

Hardness is the resistance of the material to plastic deformation induced by indentation. Hardness test is considered the most rapid test and the most popular test used for mechanical properties of materials. Types of hardness are classified as shown in Fig. 1. The conventional indentation hardness methods (Brinell, Rockwell, and Vicker) were introduced in the early part of the twentieth century. These hardness methods became the most useful tool for quality control for products in industry and testing property of materials (Mohamed 2014; Mohamed 2008). The testing method for the three scales of hardness test is fundamentally similar. An indenter with specific shape for each method is brought into contact with the material to be tested; then apply a specific test force to the indenter until the indenter makes a permanent deformation in the tested material. The resulting indentation is measured by a specific method for each scale. Brinell and Vickers hardness tests are based on measuring the area of area of the indentation after the removal of the test force. The Rockwell hardness test is based on measuring the depth of the indentation. The classification of the hardness methods is shown in Fig. 1.

To determine dynamic hardness, the indenter is forced under high loading rate which leads to impact or chock the test material. Thus, the indenter could be shot the tested material like a projectile onto the target surface (Low 2006; El-Ezz 2007). The

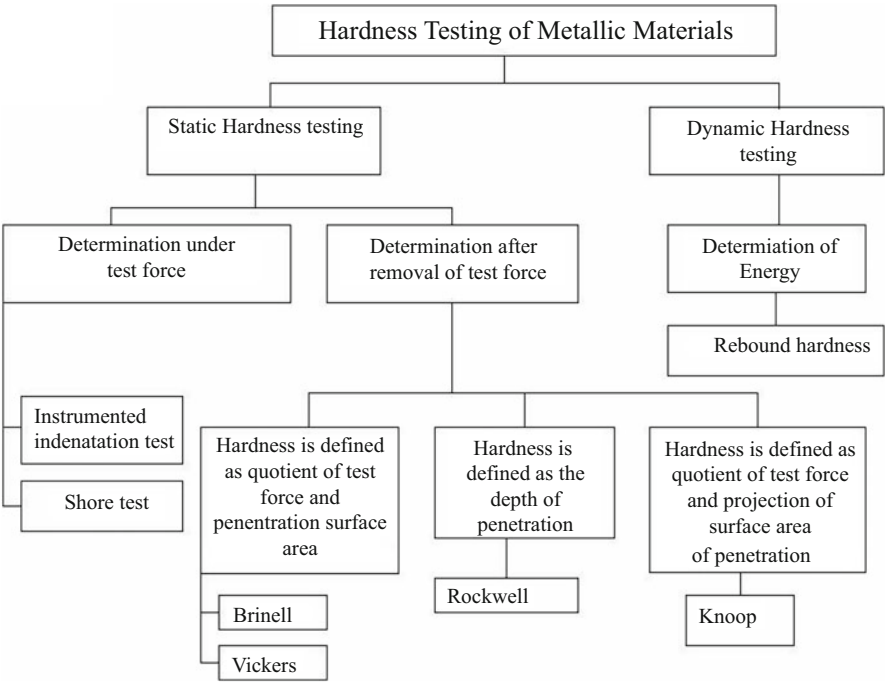

Fig. 1 Classifications of hardness

test material must be permanently deformed, so the kinetic energy should be chosen to assure plastic deformation of tested material. The dynamic hardness value depends on the measured quantities like impact energy, impact velocity, impact momentum, time of collision, deformation energy, indentation size, rebound energy, etc.

Metrological Chain for Hardness Test

There are four levels in the metrological chain as shown in Fig. 2.

International Level

The first and highest level which starts the metrological chain. International comparisons for different hardness scales are carried out.

National Levels

Primary hardness reference blocks are calibrated by the primary hardness machines in the national level, at the National Metrology institutes. The primary hardness

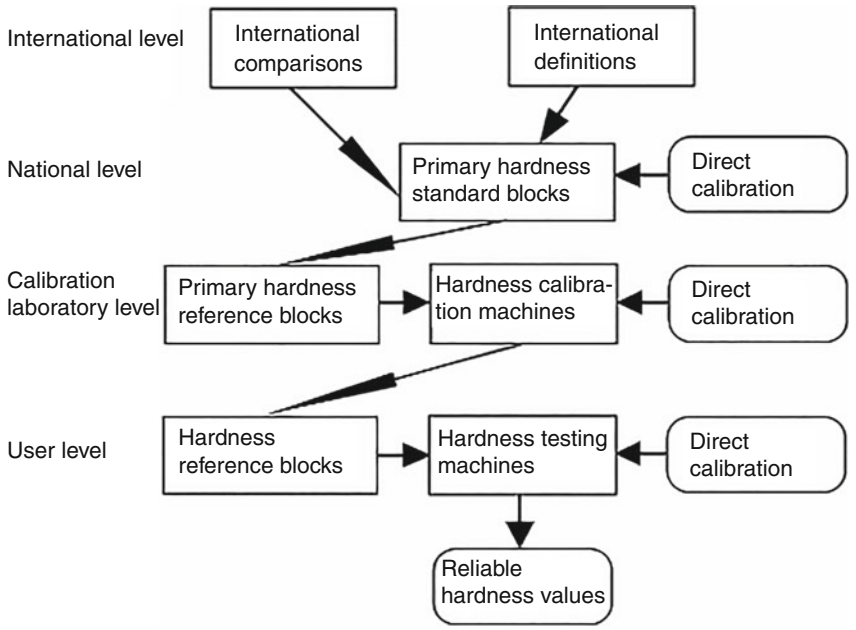

Fig. 2 Metrological chain for the definition and transmission of the hardness scales (Herman 2011)

machines are calibrated by direct method, and international comparisons are used to check and verify the behavior and accuracy of the primary machines.

Calibration laboratory level: The hardness reference blocks are calibrated at the calibration laboratories by calibration machines calibrated by direct and indirect methods to achieve traceability to the national standards.

User level: The hardness reference blocks which calibrated at calibration laboratory level are used in the user level to check the performance of the hardness testing machines used in the industry and other fields at user level (Germak et al. 2010; Herrmann et al. 2004a).

The Main Structure of Primary Hardness Testing Machine

Basically, a dead weight hardness testing machine consists of the following:

1. Machine structure
   (a) Test load values (dead weights). The load generation system (dead weights) should be traceable to the primary standard of mass in kilogram and local value of acceleration due to gravity in meter/sec2 within the range of uncertainty of ± 100 ppm.
   (b) Supporting frame (fixed frame). The supporting frame should be rigid enough to resist buckling due to the compressive load.
   (c) Load selection system. It is a simple system used to select the desired load values.
2. Indentation measuring system. It measures the produced impression due to applied load.
3. Control system. It aims to control test cycle through controlling the test time and speed of indentation, etc. (Properties of engineering materials 2004; Shigley and Mischke 1996; Thomas and Brown 2005; Lingaiah 2002; Herrmann et al. 2004b; ISO/DIS 7500–1 2016).

Conventional Static Hardness Methods

Direct Calibration

In direct calibration of the three conventional hardness machines (Brinell, Vickers, and Rockwell), the main influencing parameters of the hardness testing machines shall be calibrated (ISO 6508-1 2016; ISO 6506-1 2014; ISO 6507-1 2018; ASTM 2017a, b, c); these influencing parameters are as follows:

- Force calibration.
- Measuring system.
- Verification of the indenter geometry.
- Verification of the testing cycle.

Direct verification should be carried out under controlled environmental conditions at a temperature of (23 ± 5) °C (ISO 6507-2 2018; ISO 6506-2 2014; ISO 6508-2 2016; ES-5178-1 2016; ES-5179-1 2016; ES-5180-1 2016).

Force Calibration

The calibration of force shall be carried out by force proving instrument with class 1 or better according to ISO 376 (ISO 376 2011), and the instrument used for force calibration shall be traceable to national standards. Each force in the Brinell, Vickers, and Rockwell scale shall be calibrated by three measurements for each force at least at three positions of the indenter movement (ASTM 2017c; ISO 6507-2 2018; ISO 6506-2 2014; ISO 6508-2 2016).

Each measurement shall not deviate by more than ±1% of the nominal test force for Brinell hardness (ISO 6506-1 2014), and each force measurement in the Vickers hardness shall not exceed the tolerance limit given in Table 1 (ISO 6506-2 2014).

For Rockwell hardness scale, each preliminary force F_0 shall be calibrated, and the tolerance of measurement shall be ±2%, and the tolerance of the total force F shall be ±1% (ISO 6506-2 2014).

Calibration of Measuring System for Brinell Hardness

The measuring system of the hardness machines is based on two measuring criteria: the first depends on measuring the length of diameter indentation and the other depends on calculating the diameter from the projected area.

For diameter measurement system, each objective lens shall be calibrated by using the standard scale. The measuring system of diameter shall be graduated to allow the measuring diameter to be within ±0.5%. Minimum of four intervals shall be measured in each working range. The difference between each measurement and the reference value is not greater than 0.5% using eq. 1

$$\Delta L_{rel} = \frac{\overline{L} - L_{RS}}{L_{RS}} \tag{1}$$

where ΔL_{rel} is the relative deviation.

$\overline{L}$ is the measured mean value.
L_{RS} is the reference value.

For systems which measure the projected area, every objective length shall be calibrated. At least four standard circular reference images covering the range of areas are measured. The maximum error shall not exceed 1%.

Table 1 Tolerances of Vickers scale test force

Ranges of test force F, N	Tolerance %
F ≥ 1.961	±1
0.09807 ≤ F >1.961	±1.5

Table 2 Resolution and the maximum permissible error for Vickers measuring system

Diagonal length mm	Resolution of measuring system mm	Maximum permissible error mm
$d \leq 0.040$	0.0002 mm	0.0004 mm
$0.040 > d \leq 0.200$	0.5% of d	1% of d
$d < 0.200$	0.001 mm	0.002 mm

Table 3 The tolerance of the ball indenters

Ball diameter mm	Tolerance mm
10	0.005
5	0.004
2.5	0.003
1	0.003

Calibration of the Measuring System for Vickers Hardness

The resolution of the measuring system must be suitable for measuring the diagonal of the smallest indentation. Table 2 shows the minimum resolution and the maximum permissible error, and for measuring the diagonals of the indentation, each objective lens shall be calibrated by using the standard scale. A minimum of five intervals on the working range shall be verified.

Calibration of Depth Measuring System for Rockwell Hardness

The depth measuring system shall be verified using gauge block and shall have a maximum expanded uncertainty of 0.0003 mm. A minimum of four increments through the full range of the depth shall be measured. The working depth is 0.25 mm for regular Rockwell scales (A, C, D, B, E, F, G, H, K) and 0.1 mm for superficial Rockwell scales (N, T). The depth measuring system must indicate within ± 0.001 mm for the scales A to K and within ± 0.0005 mm for scales N and T, i.e., within ± 0.5 of a scale unit, over each range.

Brinell Ball Indenter Verification

To verify the size of the ball indenter, the minimum average measurement of three readings must be within the specified values given in Table 3.

Hardness: The ball shall be tested in accordance with ISO 6507-1. The hardness shall be not less than 1500 HV.

The tungsten carbide composite ball can be tested directly on its spherical surface or by sectioning the ball and testing on the ball interior.

b) Density: $\rho = (14{,}8 \pm 0{,}2)$ g/cm3.

Verification of the Diamond Vickers Indenter Geometry

The shape and dimension of the indenter geometry can be verified by direct measurement or by using profile projector and measure these dimension on its projection on the screen.

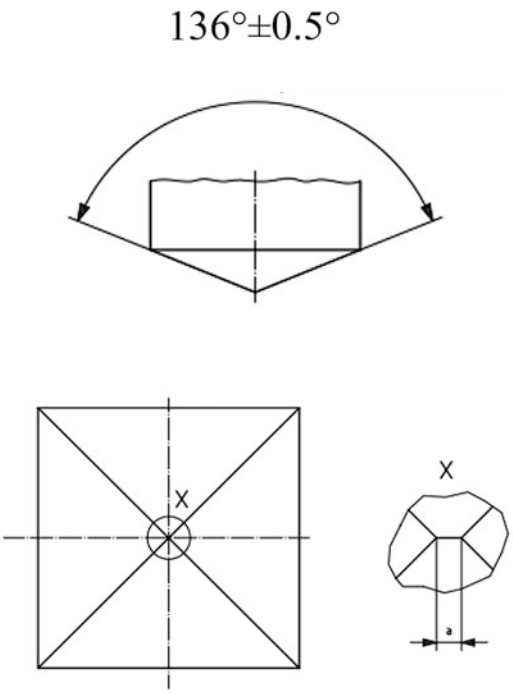

Fig. 3 Vickers indenter geometry

Table 4 Maximum length of line of conjunction

Range of the test force	Maximum permissible length of the line of conjunction a mm
F ≥ 49.03	0.002
1961 ≤ F <49,03	0.001
0,098 07 ≤ F <1961	0.0005

The angle between the opposite faces of the diamond pyramid shall be 136° ± 0.5 as shown in Fig. 3.

The angle between the axis of the indenter and the indenter-holder (normal to the seating surface) shall be less than 0.5 degree. The four faces shall meet at a line with maximum permissible length (a) shown in Table 4.

Direct Calibration and Verification of the Rockwell Diamond Indenter

The verification shall be done at a minimum of four unique equally spaced axial planes.

The instruments used for verification of the diamond indenter geometry shall have expanded uncertainty of 0.1° for the angle and 0.005 mm for the radius. The diamond cone shall have an included angle of (120 ± 0.35)°. The tip end of the indenter must be in a spherical shape. The radius of the spherical shape for each

single value shall be within (0.2 ± 0.015) mm. The mean value shall be within (0.2 ± 0.01) mm. Local deviations from a true radius shall not exceed 0.002 mm.

Verification of Testing Cycle

The testing cycle of the hardness machine shall be verified at the time of the manufacturing, at the failure of the indirect calibration, when the indirect calibration exceeds 13 months, and when any maintenance or repairs are done on the machine which may affect the testing cycle.

The testing cycle shall not differ from the reference value by more than 1 sec.

Uncertainty of Measurements

Uncertainty of the Test Force The combined uncertainty of force measurement is calculated from eq. 2

$$u_F = \sqrt{u_{FRS} + u_{FHTM}} \tag{2}$$

where:

$\mathbf{u_{FRS}}$ is the relative standard uncertainty of measurement of the force transducer (from calibration certificate).

$\mathbf{u_{FHTM}}$ is the relative standard uncertainty of the test force generated by the hardness testing.

machine.

F_{RS} (Nominal Test fFrce).

The calculation of force measurements is shown in Table 5.

where SF_i is the standard deviation of the force measurement at each position (European Co-operation for accreditation 2002; Herrmann 2016; Brice et al. 2003a, b).

Max u_{FHTM} for the three positions in Table 5 is used in the calculation of the uncertainty in Table 6.

Expanded uncertainty

$$U_F = K \times \sqrt{u_{FRS} + u_{FHTM}} \tag{3}$$

Table 5 Measurement of force calibration

Test force calibration at different position	Reading 1 F1 N	Reading 2 F2 N	Reading 3 F3 N	Mean value $\overline{F}$ N	Relative deviation ΔF_{rel} %	Relative standard uncertainty u_{FHTM}
1	F11	F12	F13	$\overline{F1}$	$\frac{\overline{F1}-F_{RS}}{F_{RS}} \times 100$	$\frac{SF_1}{\overline{F1}} \times 100 \times t$
2	F21	F22	F23	$\overline{F2}$	$\frac{\overline{F2}-F_{RS}}{F_{RS}} \times 100$	$\frac{SF_2}{\overline{F2}} \times 100 \times t$
3	F31	F32	F33	$\overline{F3}$	$\frac{\overline{F3}-F_{RS}}{F_{RS}} \times 100$	$\frac{SF_3}{\overline{F3}} \times 100 \times t$

Table 6 Calculation of the uncertainty of the test force

Quantity Xi	Value	unit	Sensitivity coefficient C_i	Distribution	Devisor	Relative uncertainty contribution
Force transducer	U_{FRS}	%	1	Normal	K	$u_{FRS} \times ci/K$
Force measurement	Max u_{FHTM}	%	1	Normal	1	$u_{FHTM} \times ci$

Calculation of the maximum relative deviation of the test force must include the uncertainty of calibration of the test force

$$\Delta F_{max} = |\Delta F_{rel}| + U_F \tag{4}$$

Uncertainty of the Measuring System The calculation of the uncertainty of the indentation measuring system is calculated from the equation

$$u_L = \sqrt{u^2_{LRS} + u^2_{ms} + u^2_{LHTM}} \tag{5}$$

Where:

u_{LRS} is the relative uncertainty of measurement of (reference standard) from the calibration certificate.
u_{ms} is the relative uncertainty of measurement due to the resolution of the measuring system.
u_{LHTM} is the relative standard uncertainty of length measurement of the hardness testing machine.

The calibration of measuring system is shown in Table 7.

Max u_{LHTM} for the four intervals in Table 4 is used in the calculation of the uncertainty in Table 8.

Calculation of the maximum relative deviation of the measuring system must include the uncertainty of calibration of the measuring length.

$$\Delta \mathrm{Lmax} = |\Delta \mathrm{Lrel}| + \mathrm{UL} \tag{6}$$

Indirect Calibration for Brinell Hardness

Indirect calibration shall be done by using calibrated reference block in the temperature of (23 ± 5)°C. Each force/ball size scale shall be calibrated by the suitable

Table 7 The calibration of measuring system

Reference value of length calibration L_{RS}	Reading 1 L1 mm	Reading 2 L2 mm	Reading 3 L3 mm	Mean value $\overline{L}$ mm	Relative deviation ΔL_{rel} %	Relative standard uncertainty u_{LHTM}
1	L11	L12	L13	$\overline{L1}$	$\frac{\overline{L1}-L1_{RS}}{L1_{RS}} \times 100$	$\frac{SL_1}{\overline{L1}} \times 100 \times t$
2	L21	L22	L23	$\overline{L2}$	$\frac{\overline{L2}-L2_{RS}}{L2_{RS}} \times 100$	$\frac{SL_2}{\overline{L2}} \times 100 \times t$
3	L31	L32	L33	$\overline{L3}$	$\frac{\overline{L3}-L3_{RS}}{L3_{RS}} \times 100$	$\frac{SL_3}{\overline{L3}} \times 100 \times t$
4	L41	L42	L43	$\overline{L4}$	$\frac{\overline{L4}-L4_{RS}}{L4_{RS}} \times 100$	$\frac{SL_4}{\overline{L4}} \times 100 \times t$

where SL_1 is the standard deviation of the length indication values for the i-th indication value of the reference.

Table 8 Calculation of the uncertainty of the measuring length

Quantity Xi	Value	unit	Sensitivity coefficient C_i	Distribution	Devisor	Relative uncertainty contribution
Reference standard length	U_{LRS}	Mm	1	Normal	K	$U_{LRS} \times$ ci/K
Length measurement	Max u_{LHTM}	Mm	1	Normal	1	$U_{LHTM} \times$ ci
Measurement system resolution	δ_{ms}	Mm	1	Rectangular	$\sqrt{3}$	$\delta_{ms} \times Ci/\sqrt{3}$

Table 9 Ranges of the hardness reference blocks

Hardness range	$0.12 \times F/D^2$			Permissible repeatability of hardness machine %	Permissible error of the testing machine %
	10	5	≤ 2.5		
1	$\leq$100HBW	70> HBW	70 > HBW	±3	±3
2	100 HBW to 200HBW	70HBW to 100 HBW	N/A	±3	±3
3	200 < HBW	10 < HBW	N/A	±3	±3

reference block. A minimum of two reference blocks shall be selected from two different ranges specified in Table 9 (ISO 6506-3 2014).

For each reference block, five measurements must be distributed uniformly over the surface of the reference block; for each reference block, let d1, d2, d3, d4, and d5 be the average of the measured diameter for each indentation arranged in increasing order and H1, H2, H3, H4, and H5 the corresponding hardness values; the average of the measured diameter for each indentation arranged in increasing order is calculated from Eq. 7 as follows:

$$\overline{d} = \frac{d1 + d2 + d3 + d4 + d5}{5} \tag{7}$$

The repeatability of the hardness testing machine r = d5−d1.
The error of the machine E is calculated from Eq. 8 as (ISO 6506-3 2014)

$$E = \overline{H} - H_C \tag{8}$$

where H_C is the certified value of the hardness block:

$$\overline{H} = \frac{H1 + H2 + H3 + H4 + H5}{5}$$

Uncertainty of the Brinell Indirect Calibration

The uncertainty of measurement of the indirect calibration of the hardness testing machine is calculated from Eq. 9 as follows:

$$u_{HTM} = \sqrt{u^2{}_{CRM} + u^2{}_{CRM-D} + u^2{}_H + u^2{}_{ms}} \tag{9}$$

where:

u_{CRM} is the calibration uncertainty of the hardness reference block according to the calibration certificate.
u_{CRM-D} is the drift of the reference block since its last calibration.
u_H is the standard uncertainty of hardness testing machine when measuring CRM.
u_{ms} is the standard uncertainty due to the resolution of the hardness testing machine.

$$u_H = \frac{t \times s_H}{\sqrt{n}} \tag{10}$$

where:
t = 1.14 for n = 5.

s_H is the standard deviation of the hardness measurement.

Budget of the Uncertainty of Indirect Measurement

Quantity	Distribution	Unit	Sensitivity coefficient Ci	Divisor	Uncertainty contribution
u_{CRM}	Normal	HBW	1	K	u_{CRM}/K
u_{CRM-D}	Rectangular	HBW	1	$\sqrt{3}$	$u_{CRM-D}/\sqrt{3}$
u_H	Normal	HBW	1	1	u_H
u_{ms}	Rectangular	Mm	$\frac{\partial H}{\partial d} = -\frac{H}{d} \times \frac{D+\sqrt{D^2-d^2}}{\sqrt{D^2-d^2}}$	$\sqrt{3}$	$u_{ms} \times ci/\sqrt{3}$

Maximum deviation of the hardness testing machine must include the uncertainty of measurement

$$\boldsymbol{\Delta H} = U_{HTM} + |\overline{b}| \tag{11}$$

Indirect Calibration for Vickers Hardness

Indirect calibration shall be done by using calibrated reference block in the temperature of (23 ± 5) °C. Each force shall be calibrated by the suitable reference block. A minimum of two reference blocks shall be selected from two different ranges specified as follows (ISO 6507-3 2018):

- >225 HV.
- 400 HV to 600 HV.
- >700 HV.

For each reference block, five measurements must be distributed uniformly over the surface of the reference block; for each reference block, let d1, d2, d3, d4, and d5 be the average of the measured diagonals for each indentation arranged in increasing order and H1, H2, H3, H4, and H5 the corresponding hardness values; the average of the measured diameter for each indentation arranged in increasing order is calculated from Eq. 7 as follows:

$$\overline{d} = \frac{d1 + d2 + d3 + d4 + d5}{5} \tag{12}$$

The repeatability of the hardness testing machine r = d5−d1.

The error of the machine E is calculated from Eq. 8 as

$$E = \overline{H} - H_C \tag{13}$$

where H_C is the certified value of the hardness block

$$\overline{H} = \frac{H1 + H2 + H3 + H4 + H5}{5}$$

The repeatability r of the hardness testing machine shall not exceed the following limits in Table 10, and the error of machine shall not exceed the tolerances shown in Table 11.

Table 10 Limits of the repeatability of the Vickers testing machine

	Repeatability of the testing machine max						
	r_{rel} %			r HV			
				HV 5 to HV 100		HV 0.2 to < HV 5	
Hardness of the reference block	HV 5 to HV 100	HV 0.2 to < HV 5	< HV 0.2	Hardness of the reference block	HV	Hardness of the reference block	HV
≤ 225 HV	3	6	9	100 200	6 12	100 200	12 24
< 225 HV	2	4	5	250 350 600 750	10 14 24 30	250 350 600 750	20 28 48 60

Table 11 Limits of the repeatability of the Vickers testing machine

	Maximum permissible percentage error, Erel, of the hardness testing machine HV															
Hardness symbol	50	100	150	200	250	300	350	400	450	500	600	700	800	900	1000	1500
HV 0.01																
HV0.015	10															
HV0.02	8															
HV0.025	8	10														
HV0.05	6	8	9	10												
HV0.1	5	6	7	8	8	9	10	10	11							
HV 0.2		4		6		8		9		10	11	11	12	12		
HV0.3		4		5		6		7		8	9	10	10	11	11	
HV0.5		3		5		5		6		6	7	7	8	8	9	11
HV1		3		4		4		4		5	5	5	6	6	6	8
HV2		3		3		3		4		4	4	4	4	5	5	6
HV3		3		3		3		3		3	4	4	4	4	4	5
HV 5		3		3		3		3		3	3	3	3	3	4	4
HV10		3		3		3		3		3	3	3	3	3	3	3
HV20		3		3		3		3		3	3	3	3	3	3	3
HV30		3		3		2		2		2	2	2	2	2	2	2
HV50		3		3		2		2		2	2	2	2	2	2	2
HV100				3		2		2		2	2	2	2	2	2	

Uncertainty of the Vickers Indirect Calibration

The uncertainty of measurement of the indirect calibration of the hardness testing machine is calculated from Eq. 9.

Budget of the Uncertainty of Indirect Measurement

Quantity	Distribution	Unit	Sensitivity coefficient Ci	Divisor	Uncertainty contribution
u_{CRM}	Normal	HV	1	K	u_{CRM}/K
$u_{CRM\text{-}D}$	Rectangular	HV	1	$\sqrt{3}$	$u_{CRM\text{-}D}/\sqrt{3}$
u_H	Normal	HV	1	1	u_H
u_{ms}	Rectangular	Mm	$\frac{\partial H}{\partial d} = 2\,H/d$	$\sqrt{3}$	$u_{ms} \times ci/\sqrt{3}$

$$\boldsymbol{\Delta H} = U_{HTM} + |\overline{\boldsymbol{b}}|$$

Uncertainty of the Rockwell Indirect Calibration

The uncertainty of measurement of the indirect calibration of the hardness testing machine is calculated from Eq. 9 (ISO 6508-3 2016).

Budget of the Uncertainty of Indirect Measurement

Quantity	Distribution	Unit	Sensitivity coefficient Ci	Divisor	Uncertainty contribution
u_{CRM}	Normal	HR	1	K	u_{CRM}/K
$u_{CRM\text{-}D}$	Rectangular	HR	1	$\sqrt{3}$	$u_{CRM\text{-}D}/\sqrt{3}$
u_H	Normal	HR	1	1	u_H
u_{ms}	Rectangular	HR	1	$\sqrt{3}$	$u_{ms} \times ci/\sqrt{3}$

Advanced Instrumented Indentation Testing

Instrumented indentation testing (IIT) is the recent and most important method to measure the mechanical properties of materials, also called depth sensing indentation. It is the same in the principle of the conventional hardness tests such as Brinell, Rockwell, Vickers, and Knoop. The principle is a hard indenter; usually the indenter of diamond shape is brought into contact with the tested specimen (Mark 2017), and the test force, F, and the displacement, h, of the indenter are measured simultaneously. However, the difference between the traditional methods and IIT method is that the traditional method measures only the deformation for one applied force and for the IIT method measures the force and displacement continuously during the contact of the indenter with the material. This continuous measurement makes the advance for the ITT method. The indentation depth hc, which is assigned to the contact of the indenter with the material, characterizes the plastic deformation under

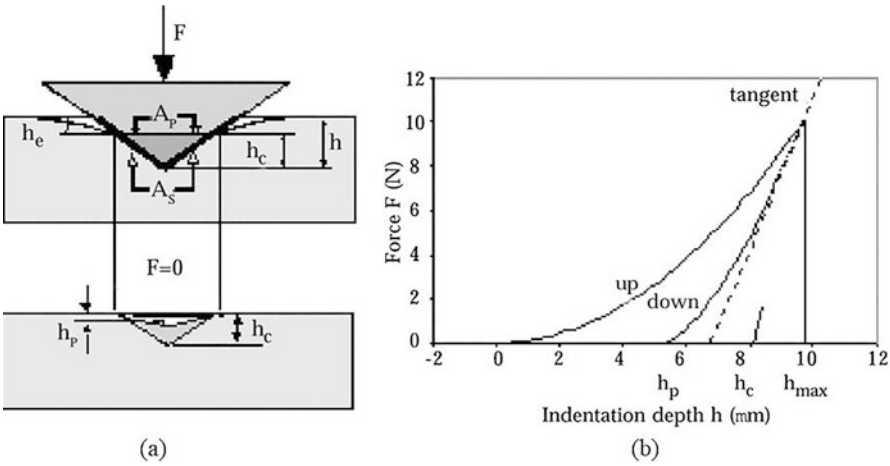

Fig. 4 Simplified representation of indentation process (Herrmann et al. 2004a)

the indenter. In addition, an elastic deformation of the surface occurs that is described by h_e. Figure 4 represents the ratios under the assumptions already summarized: Neither elastic resiliencies of the indentation inside the surface nor changing effects (pile-up) occur (Burik et al. 2015). A typical force-displacement dependence for an application of force and a removal of force (referred to as indentation curve) are shown in Fig. 4(b). If the indentation were purely elastic, there would be no difference between curves of increasing and decreasing forces. The area enclosed by the indentation curve characterizes the plastic indentation. The symbols hp., hr., h_c, and h_{max} occurring on the abscissa are explained in closer detail in the previous section and partly assist in analysis in an upcoming section. Part 1 of ISO 14577 (Brice et al. 2003b) deals with the fields of application (nano, micro, and macro range), the indenter types (Vickers pyramid, Berkovich pyramid, hard metal sphere, spherical diamond indenter), and the requirements for a reproducible measurement of the indentation curve (ISO 14577-1 2015; ISO 14577-2 2015). The definition of the hardness and other material parameters is given in the annex of Part 1 of this standard for pyramidal indenters.

IIT is very suitable for low thickness material such as thin films, particles, or other small features. The important property is the Young's modulus (E) which is the relationship between stress and strain in the elastic zone. This property is the most important property for design, which can know from this property the strain from stress and vice versa. In metals, the indentation causes the stress and strain on the specimen, which the hardness depends on. The hardness is the simplest method to measure the yield stress within a class of metals; the higher hardness gives higher strength of metal. In addition to Young's modulus and hardness, IIT has also been used to measure complex modulus in polymers and biomaterials yield stress and creep in metals and fracture toughness in glasses and ceramics (ISO 14577-3 2015).

The indenter contacts, and sense the surface. The sensing is depending on the stiffness of the mechanism of supporting the indenter columns. The user input the rate , and the limits of the test.

1. The test may be force controlled or distance controlled so the indenter stops pressing when the max force or maximum displacement is reached.
2. The user inputs the dwell time which the indenter stops with constant maximum force.
3. The test piece is then unloaded, and the indenter is moving upward at the rate suitable to the loading rate until 10% of maximum force.
4. The force is held constant at this small percentage. The purpose of this test segment is to acquire enough data to determine how much of the measured displacement should be attributed to thermal expansion and contraction of the equipment and/or test material, called "thermal drift." If thermal drift is expected to be small relative to the overall penetration of the test, this segment may be omitted.
5. The sample is fully unloaded (Fig. 5).

Fields of Application

The standard (Brice et al. 2003b) has a high universality, as both indentation depths of less than 200 nm and test forces up to 30 kN as well as different geometric of the indenters are included. Three fields of applications have been defined:

Nano range h $<$200 nm.
Micro range h $\geq$ 200 nm and F $\leq$ 2 N.
Macro range F $>$ 2 N.

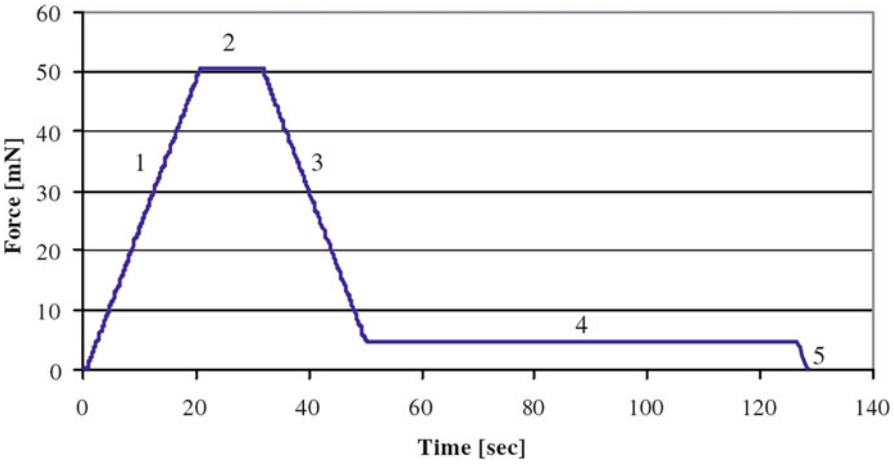

Fig. 5 Force–time diagram for instrumented indentation test (Brice et al. 2003b)

As a consequence, the formulations in the standard must take all characteristics of the nano range, the micro range, and the macro range into account. In addition, the four geometries of the indenters (Vickers pyramid, Berkovich pyramid, sphere, and cube edge) must be determined (ISO 6506-3 2014). For this reason, the standard is comprehensive and contains many explanatory notes. The required large measurement ranges of test force, indentation depth, and initial unloading slope are demonstrated in Table 12 on four materials. To investigate polymer layers 3 µm in thickness, the test force must, for example, amount to less than 100 µN. Adhesion forces to the surface as well as the condition of the surface can obviously influence the force-displacement curve before the contact takes place. The test forces up to 30 kN included in the standard, however, generate indentation depths of approximately 1 mm in metals. If the range of the expected elastic modulus is taken into account, it is also necessary to measure the initial unloading slope of 0.003 to 1400 N/µm. Whereas knowledge of the real geometry of the indenter is important for the results in the nano range (h <200 nm), the type of displacement measurement is important in the macro range (F > 2 N) with regard to the compliance of the machine and of the indenter. The field of application of the standard (Herrmann et al. 2004a) relates to all material types and composite materials, including metallic and nonmetallic layers. Because of their great importance, the particularities for determination of the mechanical properties of thin layers are dealt with separately in Part 4 of the standard.

Test Cycle

It is the main objective of each standard to obtain comparable results by the specification of test conditions, independent of test machines and test laboratories. This is why some influential factors and the specified tolerances are presented because most of the material parameters are time-dependent; the development of force and displacement with time plays an important role. The standard (Herrmann et al. 2004a) requires that all test cycle phases must be completely described in the test report. Figure 6 shows an example of a test cycle in force control, with linear progressive application of the test force up to a phase of constant force at F. This

Table 12 Martens hardness and indentation modulus for selected materials (Properties of engineering materials 2004)

Material	Martens hardness N/mm2	Indentation modulus N/mm2	Test force 100 µN		Test force 2 N		Test force 30 kN	
			h mm	$\frac{dF}{dh}$ N/µm	h mm	$\frac{dF}{dh}$ N/µm	h mm	$\frac{dF}{dh}$ N/µm
	50	1100	0.28	0.003	39	0.5	4770	60
	60	70.000	0.08	0.06	11	8	1380	980
	2000	200,000	0.04	0.08	6	11	750	1370
	15,000	400,000	0.02	0.05	2	7	280	860

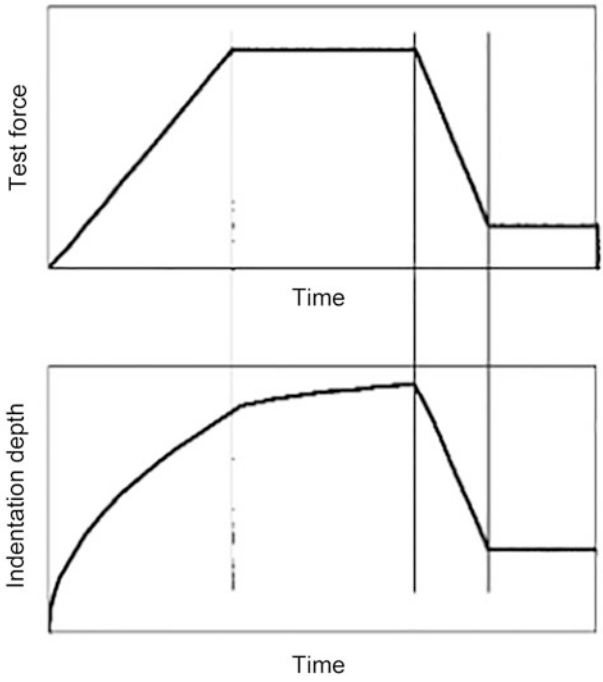

Fig. 6 Test cycle in force control (ISO 6506-3 2014)

increase is followed by a linear force drop with increased force rate up to 0.1 F_{max} followed by another phase of constant force. The test cycle ends with the rapid removal of the force (ISO 6506-3 2014).

An example of a test cycle in indentation control is shown in Fig. 7. Depending on the case, a change from force control to depth control and vice versa between the phases of the test cycle can be advantageous. When material testing machines are used in the macro range, test cycle phases with controlled traverse displacement may occur. Owing to the compliance of the machine, such phases can realize neither a linear increase in force nor a linear increase in indentation. What is decisive is the complete description in the test report. Although the comparability of the results is ensured by such complete information, the determination of the indentation modulus as closely as possible to the elastic modulus imposes an additional requirement regarding the duration of the holding period and the velocity of force removal.

During the beginning of the removal of force, creep of the material may lead to an additional increase in the indentation depth. To limit the potential deviation during the determination of the initial unloading slope, S, it is prescribed in Part 4 of the standard that the velocity of force removal (in N/s) must be at least ten times the product from initial unloading slope (in N/µm) and creep rate of the 30 last data points of the holding period (in µm/s).

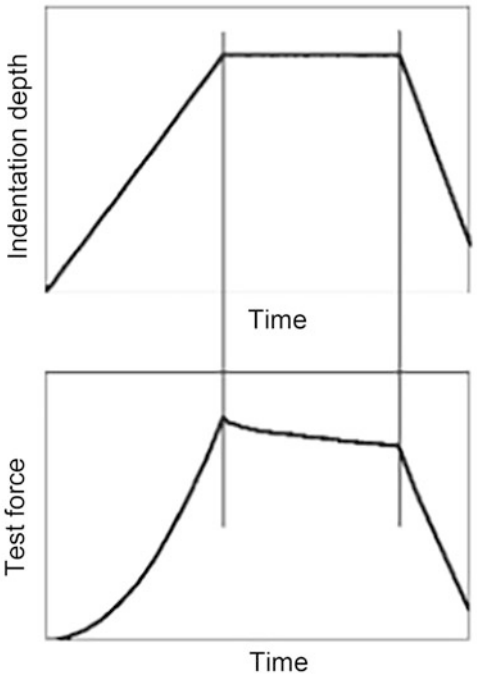

Fig. 7 Test cycle in indentation depth control (ISO 6506-3 2014)

Measurement Uncertainty

Calculating Measurement Uncertainty of the Material Parameters from Results Obtained on Reference Samples The current version of the standard on the instrumented indentation test (Brice et al. 2003b) does not contain any detailed instructions for determining the measurement uncertainty the next time the standard is revised; a procedure analogous to those for conventional hardness tests will presumably be introduced. According to these standards, the results of the indirect test carried out on the reference samples calibrated for this particular materials parameter are used in test laboratories to determine the measurement uncertainty of that parameter. Contrary to conventional hardness tests, not only does the test result consist of only a single hardness value, but several material parameters can require that the associated measurement uncertainty be explicitly stated. For this reason, differently calibrated reference samples, RM_x, must be available. The combined measurement uncertainty, U_x, is calculated as (ISO 6506-2 2014; ISO 6508-2 2016; ES-5178-1 2016; ES-5179-1 2016).

$$U_x = k.\sqrt{U^2{}_{RMx} + U^2{}_{PMx} + U^2{}_{Lmsx} + U^2{}_{bx}} \tag{14}$$

where:

K is the coverage factor of the expanded measurement uncertainty (k = 2).
URM_x is the standard uncertainty of the materials parameter, x, for which the reference sample has been calibrated.
UPM_x is the standard uncertainty of the testing machine when measuring the materials parameter, x, on the reference sample.
U_x is the standard uncertainty when measuring a test sample.
U_{msx} is the standard uncertainty due to the resolution of the measuring systems.
U_{bx} is the standard uncertainty when determining the correction bx.

In the GUM, it is recommended that the correction bx be used in order to compensate for systematic effects. The measurement result of the materials parameter, x, is then given by

$$\overline{x}_{corr} = \left(\overline{x} + \overline{b}_x\right) \pm U_x \tag{15}$$

Dynamic Hardness

An impact device as shown in Fig. 8 released an impact body with a permanent magnet and the very hard indenter sphere itself to the surface of the tested material. The velocity of the impact body is recorded in three main test phases (Maier-Kiener and Schuh 2017; ISO 16859-1 2015; ISO 16859-2 2015; ISO 16859-3 2015; ASTM A956 / A956M 2017):

1. The pre-impact phase: the impact body is accelerated by spring force to the surface of the tested material.
2. The impact phase: where the impact body is in contact with the tested material. The tested material deforms elastically and plastically. After the impact body is fully stopped, elastic recovery of the test material and the impact body takes place and causes the rebound of the impact body.
3. Rebound phase: where the impact body leaves the test piece with residual energy, not consumed during the impact phase.

Leeb-number or Leeb hardness (HL) is determined by the ratio of the rebound velocity (vr) and impact velocity (vi), multiplied by 1000:

$$\frac{V_r}{V_i} \times 1000$$

When the rebound (vr = vi) would cause L = 1000, all energy is elastically recovered and there is no plastic or permanent deformation. The L is decreased with decreasing material hardness.

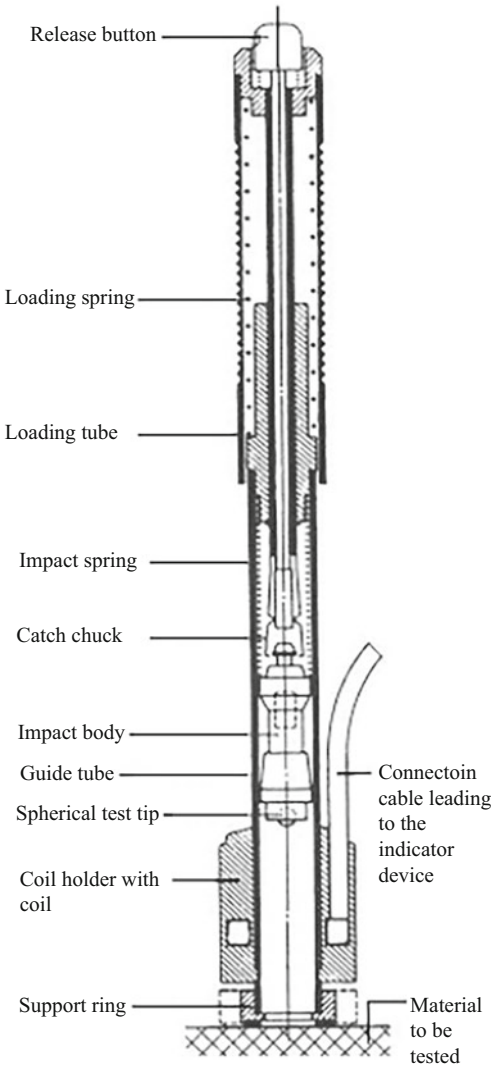

Fig. 8 Schematic diagram for impact device

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Torque Metrology

36

Koji Ogushi and Atsuhiro Nishino

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Abstract

Torque is one of the most important physical quantities which are used for the development of various industries. Torque is defined as the moment of force, also called the rotational force or the turning effect of force. The authors here introduce precise torque realizing method, torque standard machines, torque calibration machines, torque measuring devices, and so on. We also describe the present status of inter-laboratories comparisons of torque. Moreover, the chapter also introduces the SI traceability system of torque, including the description of each traceability component, such as torque testing machines, torque

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wrench testers, and hand torque tools. Finally, the authors mention the recent activities concerned with the multi-component force/torque measurement and the dynamic torque measurement techniques.

Keywords

Torque · Torque standard machine · Torque calibration machine · Torque measuring device · Reference torque wrench · Torque testing machine · Hand torque tool · Multicomponent force · Dynamic torque

Introduction

"Torque" is one of the most important physical quantities used in various industries. Torque is measured in multiple scenes for rotational driving equipment such as engines in automobiles (Judge 2012), electrical motors (Bakshi and Bakshi 2020), machining tools (Lacalle and Mentxaka 2008), as well as fastening control tools such as power-driven and hand torque tools (HTT) (Bickford 1998). It's easy to imagine that a truck couldn't climb an upslope without the required torque and that a wheel with loosened screws would cause tires to fall off, leading to a severe accident, for example.

The authors would like to introduce how the precise value of torque confirmed its reliability and how people use the reliable torque in this chapter. First, we define the torque value and then explain how the precise torque is realized (or generated). Second, we introduce how to measure the torque precisely realized, confirming the traceability of measurement in the industrial field from a national torque standard in a country/economy. Next, some typical inter-laboratories comparisons of torque standards are introduced, which contribute to how to confirm the equality of national torque standards. Finally, we also describe the present situation of research activities such as multicomponent force/torque measurement and dynamic torque measurement techniques.

Definition of Torque

Torque is defined as the moment of force, also called the rotational force or the turning effect of force. It's expressed by the cross-product of distance and force. The definition of torque is shown in Fig. 1.

Torque $\boldsymbol{T}$ is defined with the following equations:

$$\boldsymbol{T} = \boldsymbol{r} \times \boldsymbol{F}, \tag{1}$$

$$|\boldsymbol{T}| = |\boldsymbol{r} \times \boldsymbol{F}| = |\boldsymbol{r}|\,|\boldsymbol{F}| \sin\theta, \tag{2}$$

where $\boldsymbol{r}$ is the distance vector from the axis of rotation to the loading point, $\boldsymbol{F}$ is the force vector, and θ is the angle between $\boldsymbol{r}$ and $\boldsymbol{F}$ vectors.

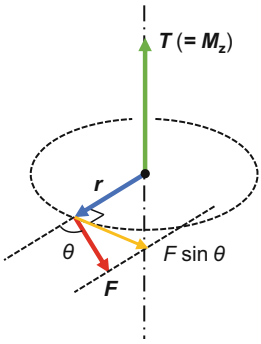

Fig. 1 Definition of torque

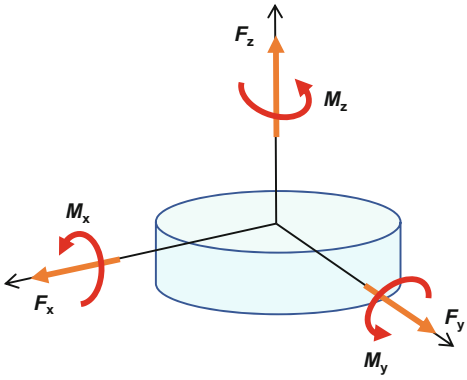

Fig. 2 Six components of force and moment on the Cartesian coordinate system

On the other hand, torque T can be expressed by the following equation when a rigid body having the mass moment of inertia I rotates with the angular acceleration α:

$$T = I\alpha, \tag{3}$$

where T is a scholar value in the case. Equation (3) is utilized for the rotational dynamics.

The unit of torque shall be expressed by "N·m" or "N m." A middle-dot point or a space is required between the unit of force, Newton, and the unit of length, meter, according to the International System of Units, known as the international abbreviation "SI" (BIPM 2019). Torque is distinguished from the unit of energy, Joule, expressed by "J." The expressions such as "Nm," "N-m," or "N.m" are all wrong. Incidentally, space between a numerical value and a unit is also necessary, like "20 N·m." The expression of "20N·m" is false too.

There exist six components of force, including torque, in the natural world. The forces acted on a body are exactly expressed by the forces $\boldsymbol{F}$ and moments $\boldsymbol{M}$ acting on x, y, and z directions on the Cartesian coordinate system as shown in Fig. 2.

Twisting moment $\boldsymbol{M_z}$ generally expresses torque $\boldsymbol{T}$. In contrast, $\boldsymbol{M_x}$ and $\boldsymbol{M_y}$ typically represent the bending moments.

Traceability of Torque Measurement

Generally, the end users use low accuracy but usable measuring devices in the industry. The torque values indicated on such measuring devices ensure the traceability of measurement, finally making the calibration chain to the national torque standards. Such intrinsic torque standard is typically realized by the combination of weights with a known mass, local gravity, and arms with a known length.

Figure 3 shows an example of a typical SI traceability system of torque developed in Japan. The authors believe other countries/economies have established or are establishing similar traceability systems.

First, the National Metrology Institute (NMI) has developed and maintained torque standard machines (TSMs), which realize precise torque (typically, 0.002 % to 0.010 % of relative expanded uncertainty). NMIs have assembled the torque values by themselves from the basic units of mass, length, and time, as explained later.

There are two main flows in torque traceability. One is the "pure torque flow," as shown on the left side of Fig. 3. We try to realize and measure only the M_z

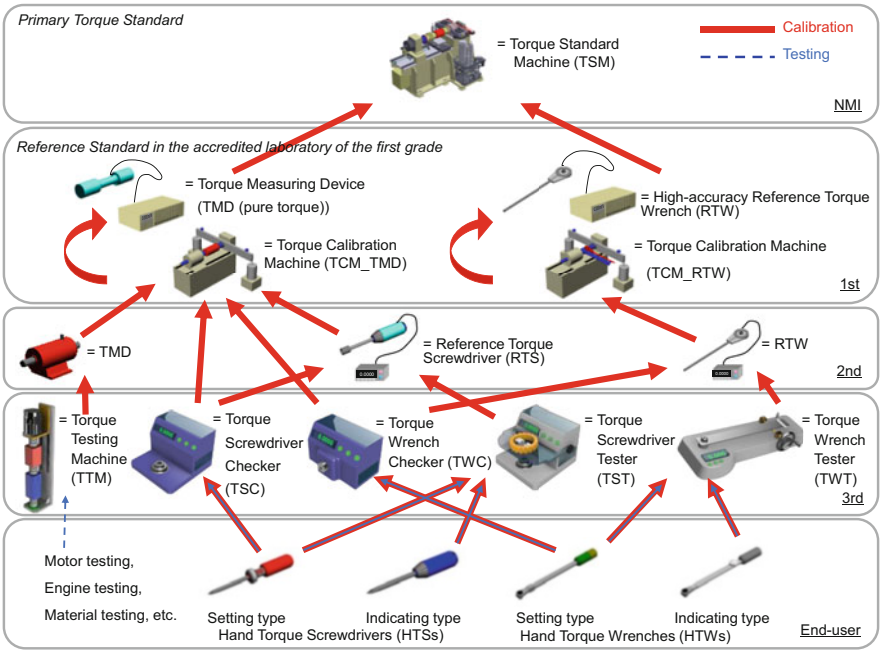

Fig. 3 SI torque traceability system in Japan

component without other components in the pure torque loading. The other is the "torque wrench flow," as shown on the right side of Fig. 3. In the torque wrench loading, bending moments (M_x, M_y) and transverse forces (F_x, F_y) are necessarily and parasitically accompanied when generating torque.

In the pure torque flow, the accredited laboratories calibrate the torque testing machines (TTMs), such as the electric motor testing machines and twisting fatigue testing machines, using torque measuring devices (for pure torque) (TMDs). TMDs are calibrated using torque calibration machines (TCMs) as well. Various products are tested in the industry, such as motors, engines, power trains, and structural materials, using TTMs. As mentioned later, hand torque screwdrivers (HTSs), which measure pure torque, are also included in the pure torque flow.

In the torque wrench flow, the accredited laboratories calibrate the torque wrench testers (TWTs) using reference torque wrenches (RTWs). As mentioned later, hand torque wrenches (HTWs), which measure torque generated by the torque wrench loading, are undoubtedly included in the torque wrench flow.

Here, somebody says that it would be doubtful whether the calibration result in the case of the torque wrench loading becomes so different from the case of pure torque loading. The National Metrology Institute of Japan (NMIJ) in the National Institute of Advanced Industrial Science and Technology (AIST) had continued successful research and found that the torque wrench loading was necessary, apart from the pure torque loading when the relative expanded uncertainty of less than 0.1 % to 0.2 % is required (Ogushi 2016, 2017, 2018, 2020a).

The following sections explain each traceability component in establishing the SI torque traceability system above.

Torque Realization

A torque standard machine (TSM) is a primary torque calibration equipment that realizes precise torque by itself, constructed from quantities of the basic unit such as mass, length, and time. On the other hand, a torque calibration machine (TCM) is a secondary standard. TCM realizes the torque with a precise TMD as a transfer torque standard calibrated by the TSM. This section introduces typical TSMs and TCMs.

Deadweight Type Torque Standard Machine

The machine of this type generates the precise torque loading deadweights at the tip of the moment-arm. Figure 4 shows a schematic of the deadweight type torque standard machine (DWTSM). The following equation expresses the torque T realized by DWTSM:

$$T = mgL\left(1 - \frac{\rho_a}{\rho_W}\right), \tag{4}$$

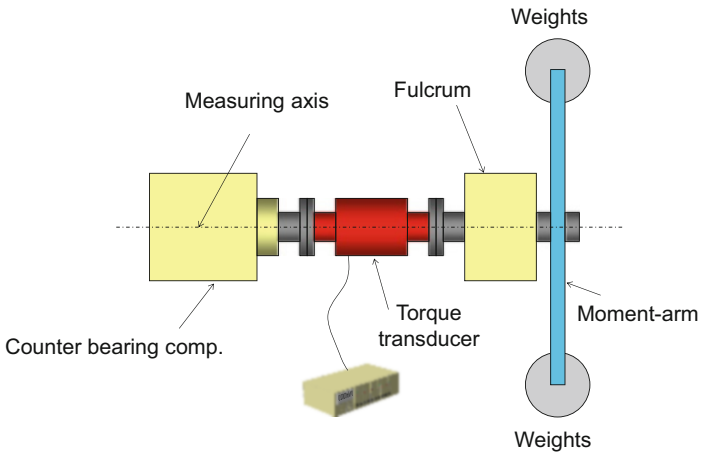

Fig. 4 Schematic of the deadweight type torque standard machine (top view)

where m is the mass of deadweights, g is the local acceleration due to gravity, L is a moment-arm length, ρ_a is the air density, and ρ_W is the density of deadweight material. That is, the influence of air buoyancy is considered.

Many NMIs adopted this type as the national torque standard machine (Adolf et al. 1995; Peschel 1997; Röske 2014a; Ohgushi et al. 2002a, b; Nishino et al. 2010; Gassmann et al. 2000; Carbonell et al. 2006; Park et al. 2007; Averlant and Duflon 2018; Arksonnarong et al. 2019). In order to maintain the torque standard in the smaller uncertainty level, it makes sure not only the traceability for the mass of deadweight, length of the moment-arm, and the acceleration due to gravity to each national standard but also the evaluations of the deformation of the moment-arm (Röske 1997; Ohgushi et al. 2003, 2007), the influence of temperature (Ohgushi et al. 2007), the friction at the fulcrum (Nishino et al. 2010, 2014), and so on.

Although DWTSM is able to realize the most precise torque at present, it takes a lot of cost to maintain its performance. This type's relative expanded uncertainty range for torque is approximately 0.002 % to 0.010 %.

For example, a DWTSM with a rated capacity of 1 kN·m is shown in Fig. 5 (Ohgushi et al. 2002b). The deadweight series consists of five stacks of 1 N with 11 pieces, 2 N with 22 pieces, 10 N with 11 pieces, 20 N with 22 pieces, and 100 N with 22 pieces. They are one inner linkage weight type. The double arm type was adopted for the moment-arm part. The nominal length of one side is 500 mm. Therefore, this DWTSM can realize the torque from 500 mN·m to 1.1 kN·m for both directions of clockwise (CW) and counterclockwise (CCW) torques.

Force Transducer Type Torque Standard Machine

The machine of this type generates the torque using hydraulic cylinders and a moment-arm, or an electric motor. The torque is measured by the force transducers and another moment-arm at the opposite side of the calibrated torque transducer.

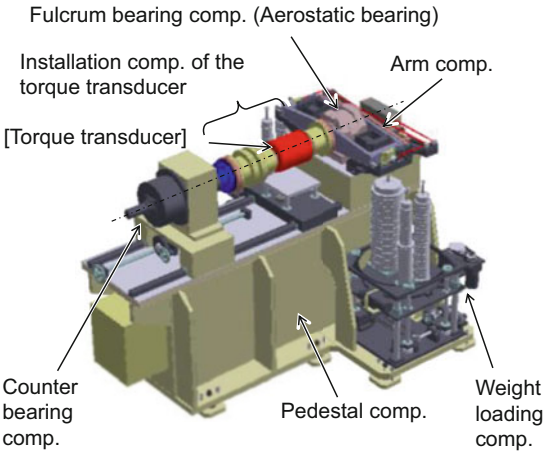

Fig. 5 An example of the deadweight type torque standard machine (1-kN·m-DWTSM at NMIJ)

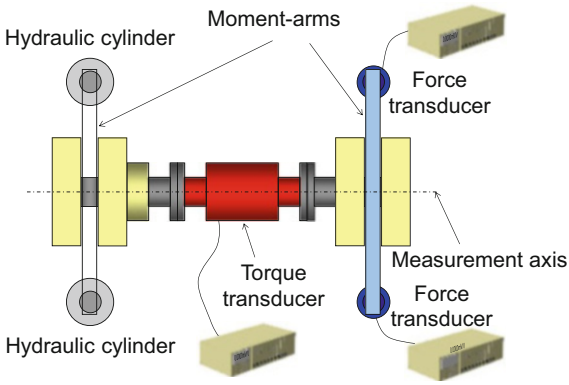

Fig. 6 Schematic of the force transducer type torque standard machine (top view)

Figure 6 shows a schematic of the force transducer type torque standard machine (FTTSM).

The following equation expresses the torque T realized by FTTSM:

$$T = F\,L, \tag{5}$$

where F is the value measured by the force transducers and L is the opposite-side moment-arm length.

The alignment adjustment at the tip of the moment-arm and finding the actual loading point of the force transducer are quite difficult in this type. On the other hand, it's relatively easy to realize the huge torque such as mega-Newton-meter (Kenzler et al. 2002; Peschel et al. 2005). This type's relative expanded uncertainty range for torque is approximately from 0.02 % to 0.10 %.

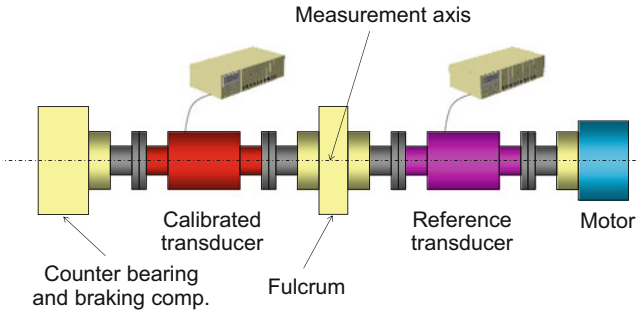

Fig. 7 Schematic of the reference type torque calibration machine

Reference (Comparison) Type Torque Calibration Machine

In a machine of this type, a reference torque transducer and a calibrated one are connected in series. Figure 7 shows a schematic of the reference type torque calibration machine (RTCM). It's also called a comparison type torque calibration machine.

An electric motor generates the torque with the breaking system put on the opposite side. The output from the calibrated transducer is compared with one from the reference transducer. This equipment is easily constructed, although it cannot be a primary torque standard, which means TSM (Peschel 1996; Brüge 2015; Kiuchi et al. 2020). The reference torque transducer is to be calibrated by a superior TSM. It's also advantageous to shorten the total calibration time by executing the continuous calibration, but not the step-by-step one (Saenkhum and Sanponpute 2017).

Figure 8 expresses an example of RTCM at NMIJ (Kiuchi et al. 2020). The calibration range can be set from 100 mN·m to 10 N·m by exchanging the rated capacity of the reference torque transducer.

This type's relative expanded uncertainty range for torque is approximately from 0.040 % to 0.20 %. The calibration and measurement capability (CMC) mainly depends on the long-term instability of the reference torque transducer inside.

Electromagnetic Force Type Torque Standard Machine

In the machine of this type, the torque is realized by electrical quantities applying the principle of Kibble balance, which was invented by Kibble et al. (1990) for the new definition of the kilogram. Nishino et al. utilized the principle for generating the precise small torque and developed the first electromagnetic force type torque standard machine (EMTSM) (Nishino et al. 2016; Nishino and Fujii 2019). Figure 9 shows a schematic of the EMTSM.

The principle of torque realization can be explained using Figs. 10a and 10b. A rectangular coil is placed in a homogeneous magnetic field. A Lorentz force F is

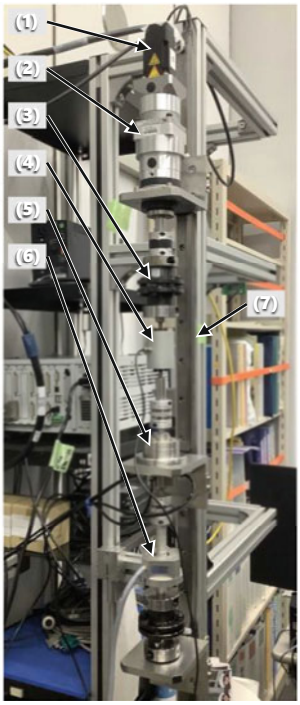

Fig. 8 An example of the reference type torque calibration machine (10-N·m-RTCM at NMIJ), with (1) a servo motor, (2) a reduction gear, (3) couplings, (4) a reference torque transducer, (5) an aerostatic bearing, (6) a calibrated torque transducer, (7) a linear motion guide

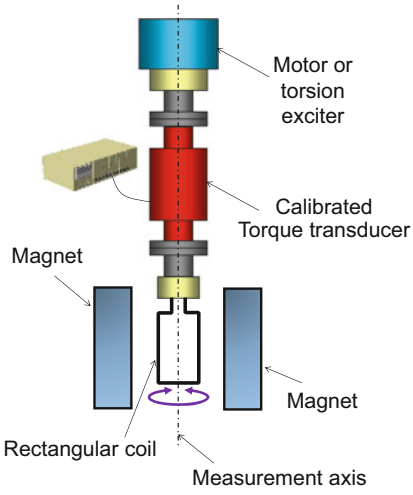

Fig. 9 Schematic of the electromagnetic force type torque standard machine

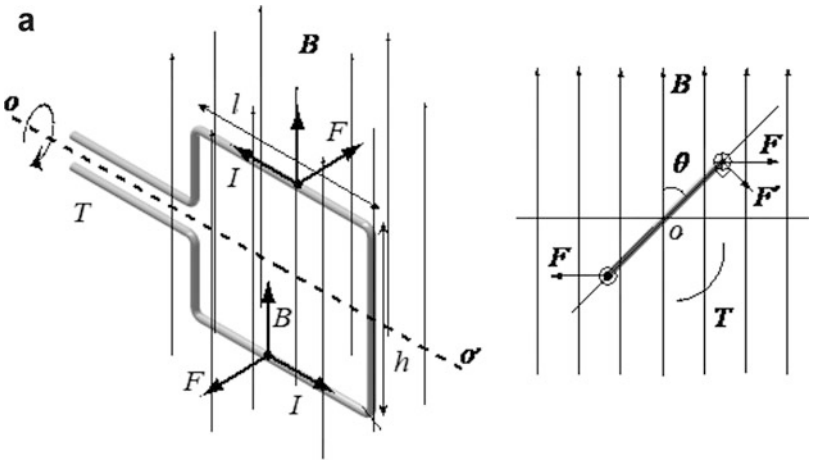

Fig. 10a The principle of torque generation with EMTSM (torque generation mode)

generated when the electric current I flows into the coil, as shown in Fig. 10a, by the following equation:

$$F = IBl \text{ or } F = -IBl, \tag{6}$$

where l is the length of the rectangular coil parallel to the rotation axis O–O′. These two forces are a couple, so the torque T is expressed by

$$T = NABI \cos\theta. \tag{7}$$

where A (= hl) is the cross-sectional area of the rectangular coil, N is the number of turns of the rectangular coil, B is the magnetic flux density, and their product NAB is a kind of eigenvalue in the apparatus.

Next, the rectangular coil is rotated at a constant angular velocity ω, as shown in Fig. 10b. An induced electromotive force (voltage) V is generated as expressed by the following equation:

$$V = NAB\omega \sin\omega t. \tag{8}$$

V reaches its maximum value $V_{\max}$ when the value of sin ωt equals unity like as

$$V_{\max} = NAB\omega. \tag{9}$$

T reaches its maximum value $T_{\max}$ when cosθ equals unity like as

$$T_{\max} = NABI. \tag{10}$$

Therefore, $T_{\max}$ is calculated from Eqs. (9) and (10) as the following equation:

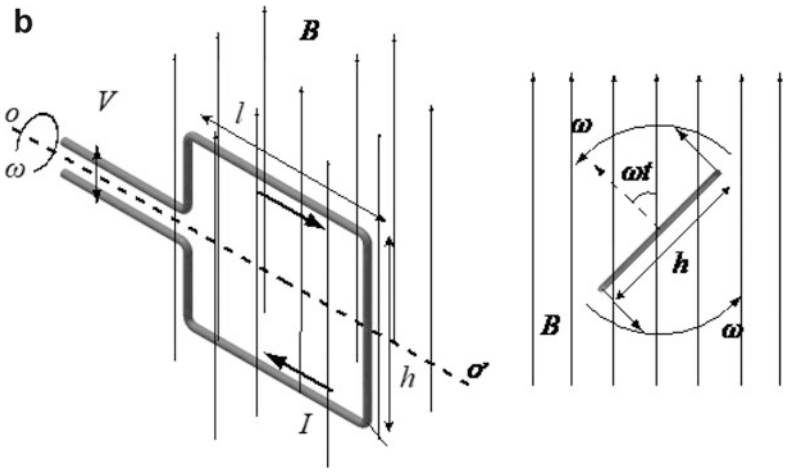

Fig. 10b The principle of torque generation with EMTSM (induced electromotive force generation mode)

$$T_{\max} = \frac{V_{\max}}{\omega} I. \tag{11}$$

By measuring V, ω, and I in each mode, $T_{\max}$ can be obtained without evaluating N, A, and B, respectively. The EMTSM can realize the torque by electric quantities without using mass and gravity.

The realization of precise torque with this method is still tricky mainly because of the nonuniformity of the magnetic field and the unstableness of the angular velocity. The influence of the heat generation also pretends the precise and large torque measurement. This method's range of torque realization is approximately from nano- to tens of milli-Newton-meter. The relative expanded uncertainty level is approximately from 0.1 % to 0.5 %. Figure 11 shows the prototype of the EMTSM developed at NMIJ (Nishino et al. 2016). The torque-generating machine using electromagnetic force is also being developed at the Korea Research Institute of Standards and Science (KRISS) (Kim 2021).

Torque Measuring Devices

The torque standard can be effectively disseminated to the industry when the triple combination of "equipment" realizing precise torque, "a device" measuring the torque precisely, and "a method" calibrating the torque correctly, is prepared. We have already discussed the torque realization in section "Torque Realization." We describe the torque measuring devices (TMDs) in this section.

A torque measuring device is defined as a complete set of a torque transducer, a cable, and an indicating device (with amplification), where the torque transducer

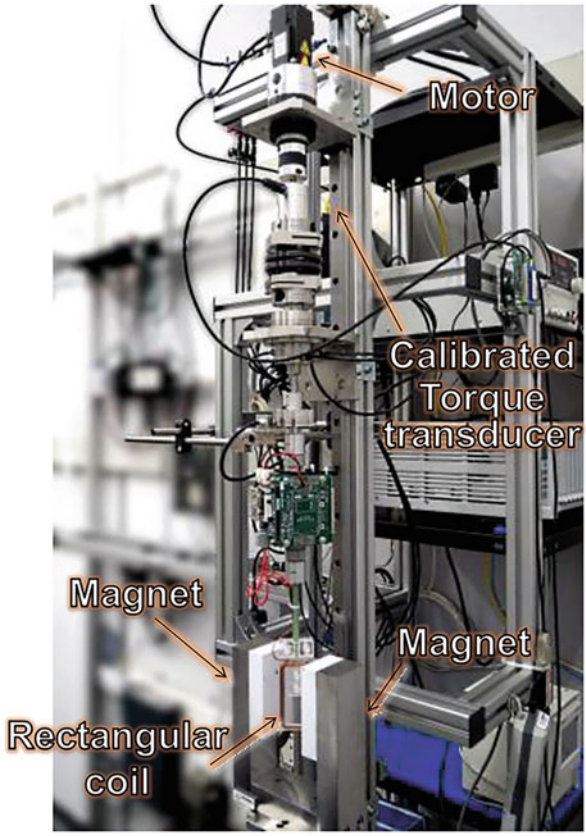

Fig. 11 The EMTSM developed at NMIJ

exchanges the change of torque value into the ones of other quantities such as displacement, voltage, and electrostatic capacity.

The TMD is the device that generally measures the torque with the pure torque loading (DIN 2005; EURAMET 2011; JMIF 2004). On the other hand, the reference torque wrench (RTW) is the device that measures the torque entailing the bending moments and transverse forces, as mentioned in section "Traceability of Torque Measurement."

The TMDs are categorized in strain gauge type, magnetostrictive type, phase difference type, electrostatic capacity type, piezoelectric type, and so on according to their transducing principles.

The strain gauge type measures the torque with an electroresistive strain gauge that is adhered to a spring element or a torque transmission shaft when it's twisted, and the strain occurs. This type is used mainly for torque measurement in the industry (Hannah and Reed 1992). The type has a small drift and is suitable for high-precision measurement. Still, its relatively low natural frequency limits itself to

the measurement with high-speed rotating bodies and high-frequency torsional vibration.

The magnetostrictive type measures the torque with a magnetic bridge circuit. It utilizes a phenomenon that the magnetization state changes (permeability changes) when a ferromagnet such as iron or nickel deforms. When the initial magnetization strength is sufficiently large, the change in magnetic flux caused by torque is measured, and then the torque can be calculated. This type has the feature that torque can be detected without contact in principle (Du 2014).

In the phase difference type, two soft iron disks with the same number of teeth are attached to the transmission shaft with the twisted shaft in between. When a pickup coil is installed with a slight gap between the teeth of each disk, a magnetic flux change occurs when the teeth of the disk pass through the magnetic field generated by this pickup coil, and an alternating voltage is induced. Since the output voltage of both pickup coils has a phase difference due to the twist of the shaft, the torque can be obtained from this phase difference (Sydenham 1985). This type can make high sensitivity, whereas the measurement system becomes expensive.

In the electrostatic capacity type, there are two types, one that uses the change in the gap between the opposing plates and the other that uses the change in the area of the plates. The measurement can be performed without contact with the shaft. When two cylinders dug a groove at regular intervals along the generatrix direction face each other so that their central axes coincide, the capacitance formed between the two electrodes becomes the relative angular displacement between the two cylinders. This change in capacitance is measured to obtain the torque (Beeby 2004). In this method, the key to improving accuracy is how to keep the variation in electric capacity small.

In the piezoelectric type, piezoelectric materials are substances that generate an electric charge when subjected to mechanical stress. The charge obtained is proportional to the mechanical stress applied, and the charge amplifier converts this charge into a measurable, for example, 0 V–10 V signal. The output voltage is proportional to the mechanical stress. By an appropriate arrangement of several piezoelectric force sensors sensitive to shear forces, it would be possible to measure torques. Piezoelectric sensors have a fast response, but have a large drift. Therefore, they are suitable for measuring quasi-static and dynamic torque (Rupitsch 2018).

The torque transducer has rotary and non-rotary styles as a structure. In the rotary type, it's necessary to convert the amount of torsional strain in any of the above types and then transmit a signal from the rotor to the stator. The method includes slip ring/brush, rotary transformer, optical transmission, telemeter types, etc.

There are shaft, flange, and disk shape torque transducers. Most torque transducers that measure pure torque M_z are axisymmetric.

Figure 12 shows a typical TMD consisting of a nonrotating style, a shaft shape, and a strain gauge type torque transducer with an amplifier/indicator.

As a traveling device for an international comparison between NMIs, a TMD that can be expected to have relative stability and repeatability of about 10^{-6} is required. On the other hand, as an industrial application, a system for mass production of

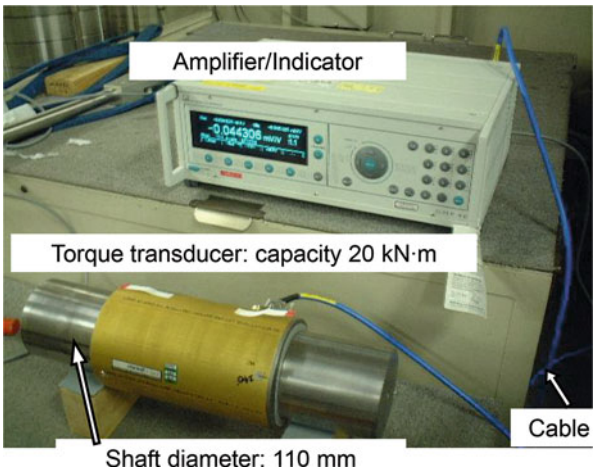

Fig. 12 Typical TMD

torque measuring equipment that can measure with a relative expanded uncertainty of equal to or less than 0.1 % is desired in a specific severe condition. Still, it has not reached that level yet.

Intercomparisons of Torque

Each NMI insists on its calibration and measurement capabilities (CMCs), the smallest uncertainty for the calibration results when the NMI calibrates the almost ideal TMD through the daily calibration procedure. International comparisons are usually carried out to verify the validity of those CMCs.

The Working Group of Force and Torque (WGFT), in the Consultative Committee for Mass and Related Quantities (CCM), in the International Committee of Weights and Measures (CIPM), is conducting related international key comparisons of force and torque for the Mutual Recognition Arrangement (CIPM_MRA). In the field of torque, CCM.T-K1 was completed in 2004 and CCM.T-K2 in 2008, respectively.

In the scheme of CCM.T-K1, two torque transducers with a rated capacity of 1 kN·m were used as traveling devices. The torque steps of 500 N·m and 1 kN·m were compared then. Eight NMIs participated in this comparison. The final report said that all measurement results matched within their CMCs range. Therefore, the international equivalence of measurement standards was confirmed (Röske 2009).

In the scheme of CCM.T-K2, two torque transducers with a rated capacity of 20 kN·m were used as traveling devices. The torque steps of 10 kN·m and 20 kN·m were compared then. Four NMIs participated in this comparison. The final report said that all measurement results matched within CMCs. Consequently, the international equivalence of measurement standards was confirmed too (Röske and Ogushi 2016).

After the above key comparisons, CCM.T-K1.1 through CCM.T-K1.4, additional comparisons of the above key comparisons were also performed. In addition, each Regional Metrology Organization (RMO) has conducted many key comparisons and supplementary comparisons, ensuring the international equivalence of the measurement standard for torque worldwide. The website of the International Bureau of Weights and Measures (BIPM) can be visited for more detailed information on the Key Comparison Database (KCDB) (BIPM 2022).

Reference Torque Wrenches

A reference torque wrench (RTW) is defined as a complete set of a torque transducer with a form of hand torque wrench, a cable, and an indicating device (with amplification). RTW is also called a torque transfer wrench (TTW). Currently, only the strain gauge type transducer for RTW is on the market.

The measuring side of the transducer is a square drive form, whereas the reacting side is a lever form. The lever receives a reaction force within a predetermined lever length range at the point of effort. The RTW would be the best reference standard for calibrating the torque wrench tester (TWT) or torque wrench calibration device (TWCD). Figure 13 shows a typical RTW, in which the square drive size is 10 mm, and the effective lever length range is from 300 mm to 500 mm. The rated capacity is 100 N·m.

Generally, the relative expanded uncertainty for RTWs is from 0.01 % to 0.2 %, depending on the performances of both the RTW itself and the superior calibration equipment.

Torque Testing Machines

Most torque testing machines (TTMs) have built-in torque transducers as the reference standards. In the TTM, torque is generated by a drive assembly such as a motor, and torque is applied to the test object.

Figure 14 shows an example of a material torsion testing machine, which is one of TTM. A motor generates torque. Torque and angle are measured by a built-in torque transducer and a rotary encoder. The shear strength and other mechanical characteristics of the material are evaluated.

Other examples included in TTMs are an engine test stand, a motor testing machine, a driveshaft testing machine, etc. The relative expanded uncertainty of the calibration required for TTMs is approximately 0.1 % to 1 %.

Hand Torque Tools

The hand torque tool (HTT), defined in ISO 6789-1 (ISO 2017), is an essential tool required in the assembly process in the industry as well as a precision measuring instrument.

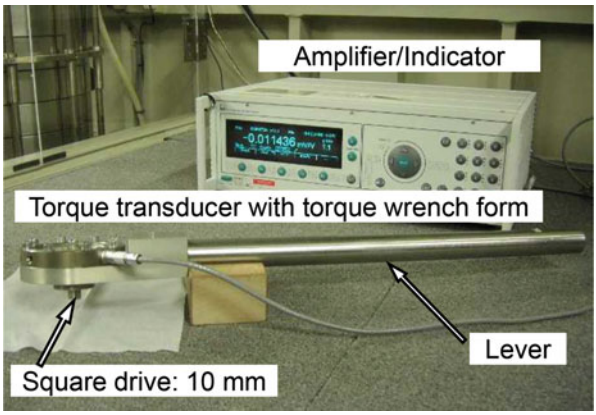

Fig. 13 Typical RTW

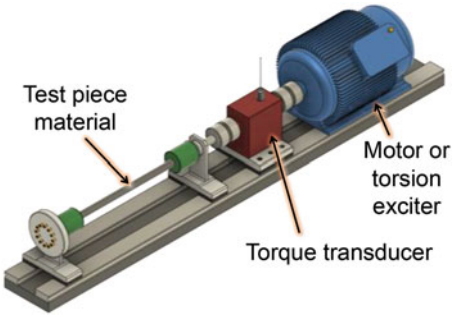

Fig. 14 Typical TTM

HTTs are classified in detail as follows, as defined by ISO 6789-1.

(a) Indicating torque tools (Type I):
   1. Class A: HTW, torsion or flexion bar.
   2. Class B: HTW, rigid housing, with scale or dial or display.
   3. Class C: HTW, rigid housing, and electronic measurement.
   4. Class D: HTS, with scale or dial or display.
   5. Class E: HTS, with electronic measurement.

(b) Setting torque tools (Type II):
   1. Class A: HTW, adjustable, graduated, or with display.
   2. Class B: HTW, fixed adjustment.
   3. Class C: HTW, adjustable, non-graduated.
   4. Class D: HTS, adjustable, graduated, or with display.
   5. Class E: HTS, fixed adjustment.
   6. Class F: HTS, adjustable, non-graduated.
   7. Class G: HTW, flexion bar, adjustable, graduated.

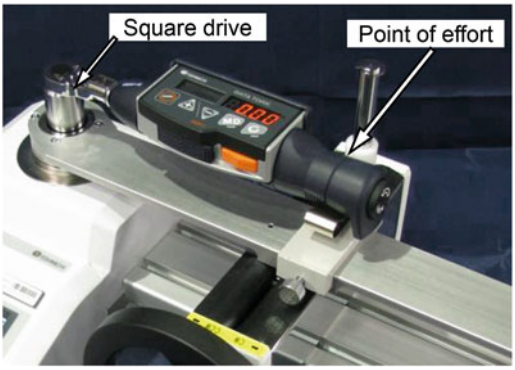

Fig. 15 An example of an HTT (digital torque wrench) installed on a TWT

Figure 15 expresses a typical HTT, a Type I, Class C, indicating torque wrench with an electric measurement (so-called digital torque wrench).

The hand torque wrench (HTW) is calibrated (or tested) by the torque wrench tester (TWT) or the torque wrench checker (TWC). In contrast, the hand torque screwdriver (HTS) is calibrated (or tested) by the torque screwdriver tester (TST) or the torque screwdriver checker (TSC), as shown in Fig. 3. Here, TWT and TST have a loading mechanism then can keep loading stably. They can calibrate both the indicating and setting types of HTTs. On the other hand, TWC and TSC have no loading mechanism, so that they can accept only hand-loading by a human. They can calibrate only setting types of HTTs. The TST is also included in a kind of the TTM.

ISO 6789-1 stipulates 4 % or 6 % as the maximum permissible deviation of HTTs, which varies depending on the type and rated capacity.

Various research has been conducted to improve the calibration or test method for HTWs. Röske proposed the evaluation method of measurement uncertainty for the HTWs calibrated according to ISO 6789:2003 (Röske 2014b). Brüge investigated the measurement uncertainty of setting type HTWs when they were used as transfer devices for comparisons or proficiency tests (Brüge 2014). Bangi et al. showed a simple uncertainty evaluation method for the calibration results of HTWs in action (Bangi et al. 2014). Khaled et al. proposed a new calibration procedure for setting type HTWs to reduce the measurement uncertainty differentiating from ISO 6789-1 (Khaled and Osman 2017). Ogushi researched test results of various digital torque wrenches using the same TWT (Ogushi 2020b).

The research concerning the reliability of torque values indicated or pre-set by HTS might be less discussed. Ogushi et al. have investigated the influence of reference standards on the calibration results of TST (Ogushi et al. 2012), the perfect traceability of HTS to the national torque standard (Ogushi 2016), and the testing method of setting type (Ogushi 2019) and digital HTSs (Ogushi 2021).

Compared with HTTs, there are also hydraulic, pneumatic, and electric driving torque tools. However, there are no international standards and no international agreement on the traceability of measurement. It would be considered difficult to

uniformly determine the permissible deviation, etc., due to the dynamic control of the fastening torque.

Multicomponent Measurement

As shown in Fig. 2, natural forces are represented by six components of force and torque. A six-component force balance is required to measure all three forces and moments that determine an aircraft's motion in a wind tunnel (Smith et al. 2001). Multicomponent measurement is also necessary for six-component force control at a joint of a robot (Siciliano and Villani 1999). There is much research on developing multicomponent force sensors (e.g., Joo et al. 2002; Park et al. 2008).

On the other hand, the development of multicomponent calibration systems has been less reported because of the system complexity. In addition, the rated capacity of each component would be limited, and then multicomponent transducers, which a specific calibration system can calibrate, would also be limited. Merlo and Röske proposed a transducer-based six-component calibration system where six reference force transducers were put in position, and six components could be measured (Merlo and Röske 2000). Röske developed a system to generate and measure the arbitrarily directed forces and moments (Röske 2003). Both driving and measuring units were realized as hexapod structures with the same geometry but mirrored arrangement.

Baumgarten et al. developed a precise F_z-M_z two-component calibration system based on a 1 MN deadweight type force standard machine at Physikalisch-Technische Bundesanstalt (PTB) (Baumgarten et al. 2016). It can calibrate a two-component transducer which is used for the calibration of the screw tightening evaluation system.

The multicomponent calibration systems are generally quite costable. Therefore, a more reasonable and universal calibration method is required in this field.

Dynamic Torque Measurement

In the above sections, we have described a static torque calibration method that takes sufficient time to stabilize torque. However, there is a need in the industry to measure the torque that changes with time accurately.

For example, measuring the fluctuation torque in automobile engines and motors is essential for energy saving (e.g., Liu et al. 2016). The torsional fatigue test of the drive shaft measures the amplitude and phase of sinusoidal torque with high accuracy (e.g., Mayer et al. 2015).

As shown in Eq. (3), the dynamic torque is expressed by the product of the mass moment of inertia and the angular acceleration. It becomes extremely costly and technically challenging to apply a dynamic torque loading to a torque transducer to be evaluated, with a larger moment of inertia at a higher frequency.

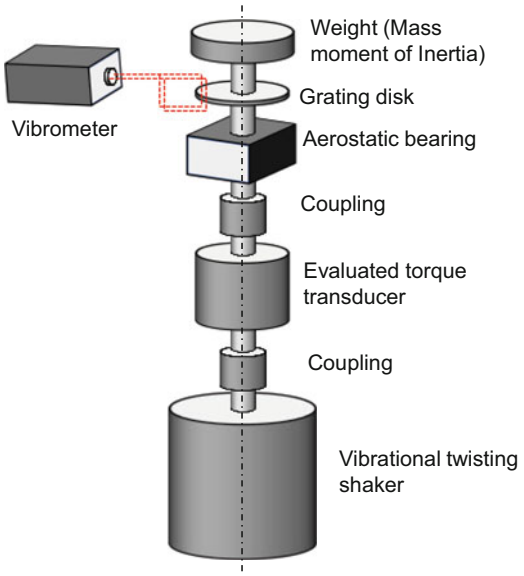

Fig. 16 Dynamic torque standard machine at PTB

Bruns and Klaus et al. are developing a dynamic torque standard machine (Bruns 2003; Klaus et al. 2015). Figure 16 shows a schematic diagram of the machine developed by them. In this machine, a torque transducer is mounted coaxially with the torsional shaker, and the change in angular velocity when torsionally vibrated by the torsional shaker is measured by a laser Doppler vibrometer through a grating disk. The torque can be calculated together with the model parameters (moment of inertia, spring constant, and damping viscosity coefficient) of all the machine elements obtained experimentally. Calibration is performed by comparing the torque calculated by Eq. (3) (but more complicated) with the output torque from the torque transducer.

Oliveira et al. tried to obtain the dynamic torque from an apparatus in rotation, consisting of a motor, a torque transducer, an encoder, two bearings, and different configurations of mass moment of inertia. They made the acceleration pulse between two angular speed steps. They also tried to evaluate the uncertainty of dynamic torque measurement (Oliveira et al. 2019).

Hamaji et al. tried to evaluate a dynamic response of a torque transducer based on the principle of Kibble balance, where the torque was generated with an electromagnetic force, as shown in Fig. 11 (Hamaji et al. 2022). Figure 17 shows a photograph of the electromagnetic force type dynamic torque evaluation equipment developed at NMIJ. Dynamic torque with an amplitude of 0.5 N·m and a frequency of up to 10 Hz has been realized.

We expect a high-reliable evaluation method for the dynamic torque is established and accepted worldwide in the future.

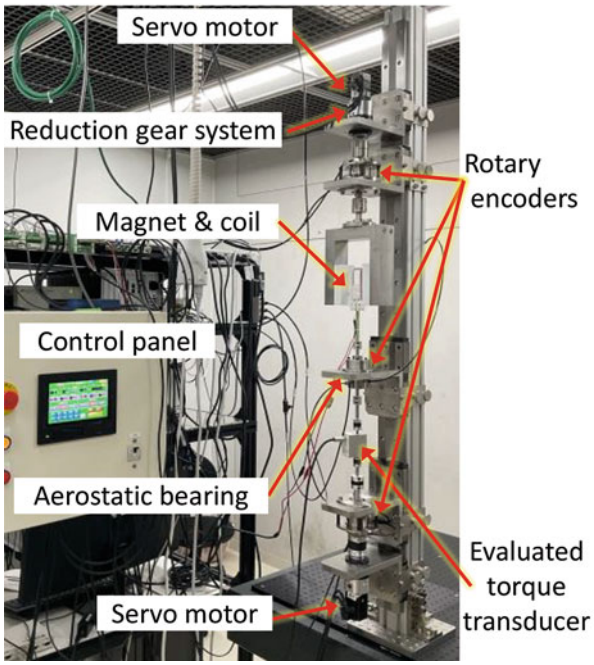

Fig. 17 Electromagnetic force type dynamic torque evaluation equipment at NMIJ

Summary

In this chapter, the authors briefly introduced torque metrology, including how to realize, disseminate, and maintain the national torque standards in each country/economy. First, we define the torque value and then explain how the precise torque is realized (or generated). Second, we also described how to confirm the traceability of measurement in the industrial field. Some inter-laboratories comparisons of torque standards were mentioned too. Finally, we described the present situation of research activities such as multicomponent force/torque measurement and dynamic torque measurement techniques.

There is also a simple method of directly calibrating a torque wrench tester or a torque screwdriver tester using a weight and bar system, although the authors didn't mention it here. We would like to mainly explain the calibration methods that directly ensure the traceability of the torque by torque in the SI unit.

We hope this review paper will be helpful as an introductory text for students or young engineers to learn about torque metrology.

Acknowledgments The author appreciates Ms. Misaki HAMAJI, a researcher in NMIJ, for her contribution to the drawing figures; Mr. Michio NOTE, technical staff in NMIJ; and Mr. Hiroshi MAEJIMA, senior staff in NMIJ, for their assistance in data acquiring, photographing, and processing.

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Recent Trends and Diversity in Ultrasonics 37

Deepa Joshi and D. S. Mehta

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Abstract

Ultrasonic technology is growing rapidly with a promise of having tremendous potential. A range of diverse applications using ultrasonic sensors have been witnessed recently in several areas ranging from healthcare, food sector, non-destructive testing, level measurement, etc. In industry, it is used for processes including cutting, forming, cleaning, and welding of metals and plastics. New hybrid forms of ultrasonics are also finding latest interest and attraction owing to the improved resolution and penetration depth for medical imaging. The present chapter explores the basics of ultrasonics along with a wide range of applications. Improvements in ultrasonic sensors have now resulted in low-cost, user-friendly, and compact devices. Being a radiation-free alternative, ultrasound has become ubiquitous attracting a wide range of interest resulting in the growing customer market. In retrospect, it should not have been so astonishing that ultrasound is soon to be covering the day-to-day field of importance areas widely.

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Ultrasonics: An Introduction

The last few decades have witnessed phenomenal growth in the field of ultrasonics and associated areas. At present, it is hard to find any area where ultrasonic application has not entered. Some of the state-of-the-art ultrasonic-based applications are given as follows:

(i) Industrial cleaning (removing dirt even from inaccessible portions and from delicate components), cutting and drilling brittle materials, plastic welding, sintering, non-destructive testing (metal, concrete, polymer), surveying.
(ii) Chemistry and biotechnology emulsification (to disperse one liquid phase into small droplets in a second phase), homogenizer (to reduce small particles in a liquid to improve uniformity and stability), degassing, sonication, trans-esterification (increases the yield of the trans-esterification of oil into biodiesel), nanoparticle preparation.
(iii) Oceanography sonar (submarine detection, surveillance, sea-bed mapping), fish tracking, acoustic release, tomography.
(iv) Medical diagnostics, blind aid, therapy (pain reliever), nebulizer (asthma management), phacoemulsion (cataract removal), lithotripter (calculi disintegrator), cancer hyperthermia.
(v) Material characterization, elastic moduli, residual stress, intermolecular interaction, grain size, dimension analysis, concentration (quality check). Among all these areas, ultrasonic metrology (non-destructive testing, surveying, characterization, etc.) has made its own niche as one of the important branches of scientific research and development.

This chapter specifically deals with the development of new measurement methods based on ultrasonic principles that are useful for scientific laboratories as well as for industries. The unique capability of ultrasonic techniques in comparison to other techniques is its versatility, as it can be applied to systems even if they are optically non-transparent. Besides, rapid, precise, and non-destructive characterization along with the scope of fully automated operations places ultrasonics at the elevated place. The NDE embraces various analytical techniques to determine several properties associated with materials, components, or systems without causing significant change to its physical and chemical properties.

Sound waves above 20 kHz are termed as ultrasonic waves. The physical principles of classical acoustics, however, hold equally good for the audible as well as ultrasonics regime (White 1963; Birnbaum and White 1984; Hutchins 1988; Edwards et al. 1990; Cedrone and Curran 1954). As mentioned in an excellent

compilation by M S Srinivasan in his book entitled, *Physics for Engineers* (Srinivasan 1998), mechanical vibrations above human audible range were achieved more than a century ago by Rudolph Koenig. But ultrasonics as a technology was born during the First World War (1918). It began with the construction of a high power ultrasonic generator by Paul Langevin using quartz crystal. Next important development was in 1927, when Wood and Loomis generated ultrasonic waves in the frequency range 100–700 kHz by exciting quartz disks at about 5 kV. Application of ultrasonics for flaw detection in various substances was first suggested by Sokolov in 1929. This was the beginning of non-destructive testing (NDT) by ultrasonics. During 1945–1946, Firestone in the USA and Sprout in the UK independently developed pulse echo test unit. Developments in the field of transducer technology have played a significant role in the development of ultrasonic test machines. The introduction of polarized ceramic (barium titanate—a piezoelectric material) in 1947 has revolutionized industrial applications of ultrasonics. This has led to generation of higher acoustic power and lowered the cost of ultrasonic machines. The use of computers in ultrasonics in recent years is another important advancement in this area.

Physical Properties of Ultrasonic Waves

Ultrasonic waves are capable of propagating in all the media except vacuum. The ultrasonic waves propagate through the material as stress or strain waves. Elastic properties of the medium govern the propagation of ultrasonic waves. These waves obey laws of reflection, refraction, and diffraction as in the case of optics. The ultrasonic waves are classified as longitudinal waves, transverse waves, Rayleigh waves, and Lamb waves depending on the particle displacement of the media. The longitudinal waves are also known as compression waves. They are commonly used in the ultrasonic inspection of materials. The waves travel through the material media as a series of alternate compression and rarefaction, in which particles vibrate back and forth in the direction of the wave travel. Fluids support only longitudinal wave propagation.

When the particles of the medium vibrate up and down in a plane perpendicular to the direction of motion, the wave propagation is termed as shear or transverse waves. These waves are more often related for solids. The surface waves or Rayleigh waves propagate in elliptical orbits along flat or curved surface of thick solids without influencing the bulk of the medium below the surface. The depth to these waves propagating below the medium surface with effective intensity is of the order of a wavelength only. These waves are abundantly used to detect flaws or cracks on or near the surface of test objects. Lamb waves or flexural waves are also known as plate waves and are produced in thin metal whose thickness is comparable to wavelength.

The ultrasonic waves obey laws of reflection, refraction, diffraction, interference, and dispersion like light waves. Another important parameter for material characterization is the acoustic impedance, which is extensively used for intermolecular

interaction studies in liquids. For plane harmonic waves, acoustic impedance is determined as ρc where ρ is the density of the medium and c is the velocity of the ultrasonic waves. Acoustic impedance decides the fraction of ultrasonic waves reflected or transmitted at an interface. The ultrasonic attenuation is the loss of energy in an ultrasonic wave propagating through a material and can be attributed to scattering and absorption due to heat conduction, viscosity, and elastic hysteresis (Myers et al. 1959; Papadakis 1967, 1976).

The ultrasonic waves travel in liquid with a velocity dependent on compressibility and density. In a lossless medium, it does not show any change with frequency. However, in some medium, the ultrasonic velocity changes with frequency. This is called dispersion, like that in light. In a dispersive medium, velocity is categorized in two types: (i) group velocity and (ii) phase velocity. The group velocity is the velocity of the wave packet. In a non-dispersive medium, the ultrasonic velocity is the group velocity. The phase velocity is the velocity of individual frequency component in a medium, more significant for dispersive medium.

Generation and Reception of Ultrasonic Waves

Piezoelectric Method

The piezoelectric effect was discovered by Curie brothers in 1880 (Aindow and Chivers 1982), who found that certain crystals like quartz, tourmaline, and Rochelle salt develop electric charges across their faces when a mechanical pressure is applied. The converse piezoelectric effect is also observed, i.e., when the electric field is applied across the crystal, a change in the crystal dimension is observed. The direct piezoelectric effect is used to detect the ultrasound while its reciprocal effect is used to generate ultrasound.

Laser Method

Compared to contact piezoelectric transducers, a laser can generate orders of magnitude higher ultrasonic signals. Fast laser pulses generate high frequency broadband (wide frequency range in one pulse) ultrasonic signals. The frequency of generated waves depends upon the choice of laser and material properties. Pulsed lasers with rise times of the range of nanoseconds up to picoseconds have also been used to generate ultrasound in a range of samples. It is generally described in terms of two extreme regimes, the ablative regime and the thermoelastic regime (Rogez and Bader 1984; Ernst et al. 1992; Horváth-Szabó et al. 1994). In the ablative regime, the laser energy is focused onto the surface of the sample which will form plasma at sufficiently high energy densities. On the other hand, in the thermoelastic regime, the laser energy having a sufficiently low energy density is directed onto the surface of the sample to avoid ablation. The laser energy is absorbed within the skin depth of the sample and rapidly heats the exposed volume of the sample. While there may be no superficial damage to the sample, there may be subsurface damage due to the absorption of laser energy, being a major drawback.

Electromagnetic Acoustic Transducer (EMAT)

The electromagnetic acoustic transducers (EMATs) are electrical devices that can transmit and receive ultrasonic waves in an electrically conducting material, without contacting the material being inspected and emerging as a mainstream non-destructive evaluation (NDE) technique. The EMATs offer several distinct advantages over traditional piezoelectric transducers. These advantages include contactless, no need of a coupling fluid, high temperature operation, and the ability to utilize shear horizontal waves. The EMATs are also ideally suited for launching and receiving Rayleigh waves, Lamb waves, and shear horizontal plate waves.

The principle of EMAT is given as follows:

> When, near the surface of an electrically conducting object, a coil of wire is placed and driven by an alternating current at arbitrary ultrasonic frequency, it produces a time varying magnetic field. This field induces eddy currents in the material under test. If a static magnetic field is present, the interaction of these eddy currents with the static magnetic field results in a magnetic volume force whose direction and intensity is determined by the vector eq. $F = J \times B$, where F is Lorentz force, B is magnetic field induction, H is magnetic field, and J is induced eddy current. This magnetic volume force is called the Lorentz force, and this force results in the generation of a wave that propagates within the specimen.

Ultrasonic Non-destructive Testing (NDT) Methods

Basically, two methods of ultrasonic testing, (i) contact method and (ii) immersion method, are commonly used (Fig. 1).

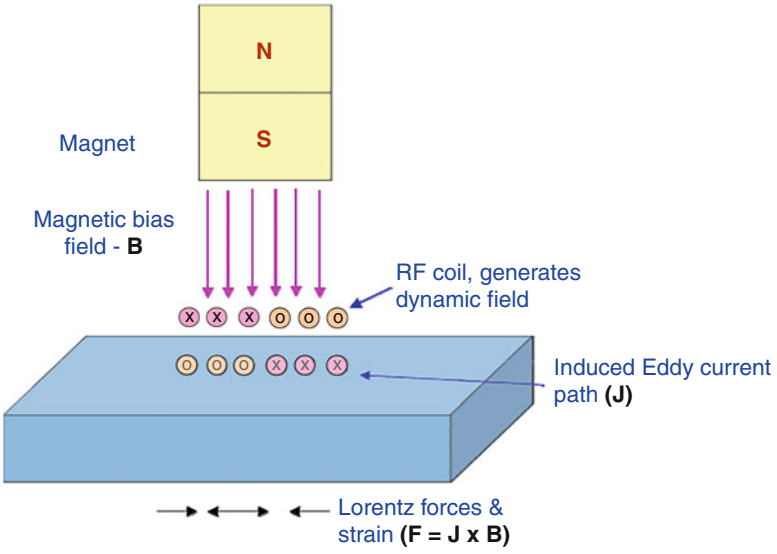

Fig. 1 Working principle of EMATs

Contact Method

In this method, the transducer is brought in contact with the test specimen using a suitable couplant like water, oil, grease, etc. In the absence of couplant, air bubbles are formed between the material under test and the transducer. The amplitude of the reflected pulse is affected by the type of couplant, its thickness, and uniformity of the couplant thickness. For this reason, contact method may not be the right choice where amplitude of echo is important mainly for attenuation measurement as it is often difficult to maintain the uniform pressure. However, this method allows quick testing of surfaces.

Immersion Method

In this method, the test specimen and the special-type immersion transducer are immersed in a liquid, usually water. The liquid acts as a couplant in the transfer of sound energy from the transducer to the test specimen. In this method transducer can be moved under water to introduce the sound beam at any desired angle, thus provides testing flexibility. It also provides constant path length of couplant (water) between the transducer and the material under test. This ensures uniform pulse height throughout the experimental measurements.

Techniques of Ultrasonic NDT

There are three basic ultrasonic test systems that are commonly used in industries. They are pulse echo, through transmission, and resonance system.

Pulse Echo System

Here the ultrasonic waves are not transmitted into the material continuously but in pulses. The pulse is reflected back from the boundaries of the material or discontinuity. The reflected pulse or echo is analyzed for evaluation of material properties.

(a) **Spike Excitation**

In this method, the short duration high amplitude pulses coming from the transmitter are given to the transducer. The transducer converts the spike into a broad band short duration pulse which is radiated into the test piece via a coupling medium. If it encounters a flaw in its path, the flaw reflects an echo wave, a portion of which depending on its form and size reaches the receiver. Echo of flaw indicates transit time from transmitter to flaw and to receiver which gives distance of flaw. A transducer acts both as a transmitter and a receiver. Both the transit time and intensity of ultrasound are measured in this method.

(b) **Tone Burst Excitation**

In high power ultrasonic applications, tone burst generators are often used. In this method, a single frequency pulse, usually matching with the transducer's resonant frequency, is generated. Low-voltage signals are taken and then converted into high-

power pulse trains for the most power-demanding applications. Tone burst capability is mostly useful to work with difficult, highly attenuative materials.

Through Transmission System

In this method two transducers are required; one is used as a transmitter and the other one as a receiver. Short duration ultrasonic pulses are transmitted into the material. The receiver is aligned with the transmitter on the opposite side to pick up the sound waves passing through the material. The quality of the material is determined by change in the pulse shape as it transverses the material. A marked reduction in the amplitude of the received energy indicates a discontinuity.

Resonance System

This system makes use of the resonance phenomenon to detect flaws. Here continuous longitudinal waves are transmitted into the material. The frequency of the waves is varied till the standing waves are setup within the specimen causing the specimen to vibrate or resonate at greater amplitude. Resonance is then sensed by a device. A change in the resonant frequency cannot be traced to a change in the material thickness, but it is usually an indication of a discontinuity in ultrasonic non-destructive testing (NDT).

Techniques for Velocity Measurement (Review of Literature)

Ultrasonic velocity has become an important tool for the study of physical properties of matter. These measurements offer a rapid and non-destructive method for the characterization of materials. During the process control of manufacturing procedures and also in routine monitoring of quality changes of viscoelastic products during and after manufacturing processes, they are often employed. Sound velocity is a favorable parameter for such applications because of its remarkable sensitivity to liquid states.

Basic Pulse Echo Methods

An ultrasonic transducer converts a pulsed radio frequency (r.f.) signal of given frequency into pulsed ultrasonic wave of the same frequency. Pulse echo methods are the most simple and widely used techniques for measuring sound velocities in liquids. Extensive literature survey reveals velocity methods utilizing plane wave sound propagation (Hosoda et al. 2005; Crawford 1968; Carstensen 1954; Cerf et al. 1970; Kaatze et al. 1988; Kaatze et al. 1993; Tong and Povey 2002; Bakkali et al. 2001; Taifi et al. 2006; Sachse and Pao 1978; Kline 1984). This group of pulse measurements is referred as "narrow-band pulse techniques." A sinusoidal signal is not a single frequency. In addition to the carrier frequency, there are side bands, resulting in an effective bandwidth (Goodenough et al. 2005). This bandwidth has to be taken into account for phase velocity measurement. Methods of determining directly the difference between the sound velocities in the sample and in a reference liquid (Cedrone and Curran 1954; Greenspan and Tschiegg 1957) use nearly single-frequency signals. The output of the receiver transducer is added to a reference signal

from the harmonic signal generator, to measure the phase of the signal transmitted through the specimen cell. The sound velocities can also be obtained as a derivative from pulse-modulated attenuation measurements (Myers et al. 1959; Aindow and Chivers 1982).

The other group employs sharp pulses (Tardajos et al. 1994; Hosoda et al. 2005; Crawford 1968; Ernst et al. 1992; Forgacs 1960; Van Venrooij 1971; Papadakis 1973; Bilaniuk and Wong 1993; Høgseth et al. 2000) referred here as wideband techniques having wider frequency spectrum with sharp pulses required to make broadband transducer systems. A single sharp pulse is applied to the transmitting transducer, containing a wide range of frequency components. Pulse propagates through a known distance in the liquid medium and reflects back as received signal. The frequency components of the received signal are extracted using a Fourier transformation of the received signal. Time of flight measurements are crucial for sound velocity determination. The velocity is determined from distance divided by time of flight. One more method for velocity measurements employs pulses propagating through the sample and echoes reflected back and forth (Carstensen 1954; Benedetto et al. 2005). Pulse echo methods are employed for narrow-band signals as well as for sharp pulses. The broad frequency spectrum of sharp pulses allows to determine group and phase velocities and the attenuation coefficient in a considerable frequency range (Kaatze et al. 1988; Meier and Kabelac 2006; Letang et al. 2001; Papadakis 1967; Papadakis 1976; Horváth-Szabó et al. 1994). The broadband technique is based on a Fourier analysis of a primarily sharp pulse. The principle of three main techniques used for ultrasonic velocity measurement is presented briefly in the following section.

(a) **Sing Around Method**

In this method a transducer placed at the end of the sample opposite the transmitting transducer acts as a receiver. Trigger is acquired by the received signal for the pulse generator thereby generating a succession of pulses. Since pulse repetition rate of sequence depends on time of flight in the sample, the velocity may be determined by measuring the repetition rate (Kaatze et al. 1988; Tong and Povey 2002; Papadakis 1976; Horváth-Szabó et al. 1994; Tardajos et al. 1994).

(b) **Pulse Echo Superposition Method**

A series of radio frequency pulses from a pulse generator is transmitted into the sample, and their repetition rate, controlled by frequency of continuous wave oscillator, is adjusted to correlate approximate multiple of an acoustic round trip transit time in the sample. The time delay between superimposed phase pulses is the reciprocal of the continuous wave oscillator frequency.

(c) **Pulse Echo Overlap Method**

As described with the above method, series of r.f. pulse are introduced in the sample with controlled pulse repetition rate. During the acoustic time measurement

oscilloscope is switched to an X-Y mode in which continuous wave oscillator provides sweep. CRT intensity is reduced and two amplified echoes are visible. The echo can be made to overlap cycle for cycle. This method is useful to measure group velocity as well as phase velocity. Ultrasonic velocity can be measured both in liquids and solids, thus making this method versatile and accurate.

Continuous Wave Method

Two transducers, one operated as a transmitter and the other one as a receiver, are used for velocity measurement (McSkimin 1961; Mitaku and Sakanishi 1997; McClements and Fairly 1991; Elias and Garcia-Moliner 1968; Pearson et al. 2002; Nolting 1999; Hubbard 1931; Eggers 1967/68; Pethrick 1972; Dev et al. 1973; Sarvazyan 1982). The transmitting transducer is excited at constant amplitude and at fixed frequency resulting continuous wave propagation in the medium. The ultrasonic wave after interaction with the medium is received by the other transducer. From values of frequency and wavelength, the velocity is calculated. Current techniques employ interferometers, in which the path length of interactions of the ultrasonic waves with the sample is increased by multiple reflections giving high sensitivity of measurements.

Optical Method

When a sound wave passes through a transparent material (solid and liquids), periodic variation in refractive index occurs. A diffraction grating is formed due to maxima at compression and minima at rarefaction. The diffraction grating has spacing equal to half the ultrasound wavelength. A monochromatic light beam is passed through a slit, and further a convex lens is used to make them as parallel rays through the medium. The velocity is determined using Bragg relation. Bragg scattering (Kaatze et al. 1987) and Brillouin scattering (Eggers 1992) are challenging for required applications. Improved signal-to-noise ratios, however, are reached in Bragg reflection as well as in stimulated (Nakajima and Arakawa 1993; Eggers 1994) and forced (Eggers 1997) Brillouin scattering techniques.

All the techniques discussed so far have their own merits and demerits. Most of the techniques are laboratory methods and hence cannot be used for online purposes. Temperature stability is also one additional crucial parameter. New techniques have been designed and developed by the author to overcome the limitations of existing methods (Joshi et al. 2009, 2014). The present chapter manifests current state-of-the-art regarding recent trends and diversity in emerging areas of ultrasonics.

Motivation

It is apparent from the aforesaid sections that during the last few decades, ultrasonic technology has emerged as one of the important branches of scientific research and product development. We also see that one of the established applications of ultrasonics is in non-destructive evaluation (NDE), which embraces various analytical techniques to determine several properties associated with materials,

components, or systems without causing significant change to its physical and chemical properties. The present chapter specifically deals with the development of new measurement methods based on ultrasonic studies which are useful for laboratories and industry.

Ultrasonic non-destructive testing (NDT) provides one of the larger markets for industrial equipments. However, it is only one of many areas in which low-intensity ultrasonic energy is applied. The development of metrology applications of ultrasound has occurred in parallel with the development of medical applications using high as well as low intensity ultrasound. Ultrasound equipment over the last few decades has gotten physically smaller, generates less heat, and has become more power efficient. These upgrades, along with vast enhancements in image quality, have pushed point-of-care ultrasound to widely popularize in emergency rooms and obstetric clinical practices. Availability of instant real-time diagnostic data has consequently helped reduce overall healthcare costs by replacing more expensive diagnostic exams comparatively.

Recent years have witnessed promising growth within the multidisciplinary field that encompasses ultrasonics in diverse applications from medical imaging to several industries. The speed, efficacy, low cost, and non-invasive nature of ultrasound imaging are some of the key attributes that have given this technology an edge over the other medical imaging modalities. In addition, ultrasound equipment is economical compared to other systems; even the most advanced ultrasound systems cost only about one-fifth of the price for a low-end magnetic resonance imaging (MRI) system. Similar technological evolution has been experienced by the ultrasound imaging. Device miniaturization and Windows PC-based architecture have made it possible to pack increasing amounts of processing units into smaller and smaller medical devices.

Diversity in Ultrasonics and Emerging Trends

To assess the performance and safety of medical ultrasonic equipment, ultrasonic power measurement is essential. The power generated from medical transducer can be used to bring about irreversible changes in tissue, being several watts for physiotherapy applications. This characteristic of ultrasound to whether be used for diagnostic or for therapy applications prompts a grave need for accurate knowledge of the intensities of what the patient is being exposed for safety. During the design and optimization of an ultrasonic equipment, acoustic output needs to be determined essentially, as it plays an important role in the performance level optimally. Ultrasonic sensors reliably detect transparent and other demanding objects where optical technologies do not suffice.

The acoustic impedance is basically the opposition exerted by the medium to the displacement of medium's particle by sound energy. Ultrasonic waves on passing through an interface of two media are reflected and transmitted, based on the acoustic impedance of the two media. The acoustic impedance of any medium is governed by inertial and elastic properties of the medium; hence its determination serves as an

important parameter to study the material characteristics of different substances. The greater difference in acoustic impedance will lead to greater reflection of sound energy. The acoustic impedance directly using piezoelectric resonator has also been measured (Bhatnagar et al. 2010). Joshi et al. (Joshi et al. 2013) also measured the acoustic impedance of solid salol directly by measuring successive echo amplitudes. Later Kumar et al. (Kumar et al. 1997) introduced a unique method of evaluation of the acoustic impedance measurement in thin samples like a membrane which was considered to be an unattainable task. In general, relation of acoustic properties with equilibrium constants is a major advantage of sound velocity technique in addition to being non-destructive and cost-effective. More often measurement of ultrasonic velocity and density is carried out by researchers to study the undergoing interactions. It was observed that non-linear variation of the ultrasonic velocity with concentration is the signature of complex formation between unlike molecules. For a particular value of concentration, the maximum echo height also signifies the maximum interaction at that particular value of concentration. In addition to this, density is also one of the basic parameters to act as a probe for intermolecular interaction. The accurate determination of density is rather very critical and generally carried out in a long procedure using a pyknometer. The ultrasonic velocity measurement is usually performed by interferometer method where the accuracy is mainly a function of degree of parallelism between the quartz disc and reflector. All these analyses essentially require measurement of ultrasonic velocity and density, separately, at different time using individual set-up. This usually enhances the chances of error in the measurements because in two different experiments of velocity and density, there is always a possibility of different temperatures in the vicinity of two experimental set-ups. Such temperature difference can further lead to remarkable change in the derived parameters. Also employment of two different set-ups for velocity and density measurements may change concentration due to volatile property of organic liquids, further causing error in the measurements. The relatively small impedance mismatch between soft tissues allows for the ultrasonic waves to propagate across several interfaces providing imaging capability up to large depths. The ultrasonic transducer design plays an important role in determining the quality of images. For an efficient transfer of acoustic energy from the body to the transducer and electrical energy between the system and transducer, the acoustic and electrical impedance matching has to be optimized. Significant technological advances employed in commercial systems have amplified scope and the range of applications of ultrasound in the last few decades. This chapter includes discussion of many of these advances and improvements. Many of the advances that took place in the field of ultrasonics recently are directly related to developments in electronics, data processing techniques, and market demands.

Applications of Ultrasonic Imaging

The ultrasonic NDT and medical diagnosis are two main technical areas utilizing ultrasonic waves and their interaction to material discontinuities. The two important

quantities for which traceable measurements can be made are the acoustic pressure and the acoustic power. To enable measurements undertaken by different parties to be compared on a meaningful basis, the calibration of the devices must be traceable to a national primary standard. From the perspective of modern automization in road safety, a variety of sensors assist self-driving vehicles in navigation and general operation, helping with blind-spot detection and self-parking. The unique advantages of the ultrasound technologies still do not make it a one-stop shop for every need, leading the emergence of new hybrid form of ultrasound-based technologies.

Emerging Technology

Newer technologies are set to revolutionize ultrasound practice effectively. Some recent industrial applications are packaging industry, electronics industry, and food technology offering wide variety of object geometries and surfaces however with narrow space requirements for sensors providing high safety.

The ultrasonic technology and applications are continuously expanding, and these sensors are steadily gaining recognition as an industry standard.

Current Ultrasound Applications in Health Care

Ultrasound technology is an inseparable part of the health care and is a promising technology with several recent advancements. It is utilized for a variety of diagnostic purposes. However, so far, all an ultrasound could be used for was viewing the inside of the human body. Ultrasounds are typically and traditionally used for imaging, usually in internal medicine and prenatal applications. Several newer applications for ultrasound technology include dental descaling and shock wave lithotripsy. This process is very popular now as an alternative to surgery and utilizes targeted ultrasound waves to break up kidney stones and calcium-based growths in the body that are too large to pass naturally.

Futuristic Promising Technology

To look inside the human body, invasive techniques such as surgery can potentially damage the body. Therefore, non-invasive biomedical imaging modalities, such as projection radiography, x-ray computed tomography (CT), positron emission tomography (PET), single-photon emission computed tomography (SPECT), magnetic resonance imaging (MRI), and ultrasound (US), are widely used in clinics today for the diagnosis and treatment of various diseases. Some of these modalities (projection radiography, x-ray CT, PET, and SPECT) use ionizing radiation, which exposes the human body to harmful radiation. On the other hand, much safer MRI is time consuming, expensive, and cannot be used for patients with metallic implants as it uses high-strength magnetic fields. US imaging offers a cheap, real-time, easier to operate, and safe alternative. However, poor soft tissue contrast and difficulty in imaging through skull/bone limit its use in many body parts. Nonetheless, all these

imaging modalities are capable of imaging deeper (∼20 cm inside soft tissues in the case of US imaging), which is a major advantage for clinical applications.

Conventional optical microscopy provides images with high spatial resolution and rich optical contrast, making it possible to visualize cellular and subcellular structures. However, due to light scattering in biological tissue, imaging depth is limited to not more than ∼1 mm. Therefore, even though optical microscopy is a gold standard in pathology, it is not suitable for in vivo human imaging. Another high-resolution purely optical imaging modality, known as optical coherence tomography (OCT), can image tissue with high resolution (∼1 μm) and at an imaging depth of up to ∼2 mm. However, OCT also has limited application to eye retinal imaging to date. Therefore, researchers are trying to come up with various techniques to overcome these depth limitations of optical imaging. Diffuse optical tomography provides imaging depth up to a few centimeters deep inside tissue, but the spatial resolution is relatively low—one-third of the imaging depth. Therefore, high-spatial-resolution images at deeper imaging depths are always a challenge for purely optical imaging techniques.

Ultrasound Mediated Optical Tomograpghy (UOT)

Noninvasive diagnostic imaging technologies to detect diseases such as cancer and others come in a wide variety. Most of these technologies include sophisticated techniques such as MRI and X-ray, but they lack optical contrast which is beneficial for detection of small lesions without using the contrast agents or ionizing radiation. Famous techniques such as optical coherence tomography (OCT) and diffuse optical tomography (DOT), which are optical ion nature show good contrast but the resultant depth and resolution, respectively, are limited by the strong light scattering from the tissue. It is quite challenging to measure with high resolution at penetration depths larger than millimeter order due to the high optical scattering which results in a strong dampening of ballistic photons. In contrast to X-ray these optical techniques are non-invasive and use non-ionizing radiation. The absorption coefficient of light is wavelength dependent making spectroscopy possible, which can be used to determine blood oxygenation levels. On the other hand, the ultrasound in the few MHz range has a scattering coefficient less than light, which allows for superior penetration depth while retaining spatial resolution. There is however a drawback in which ultrasound lacks the benefits of optical contrast. Recently emerging imaging techniques which interplay sound and light are hybrid forms as acousto-optics imaging (AOI) or ultrasound modulated optical tomography (UOT) and photo-acoustics imaging (PAI). These two techniques combine the high resolution of ultrasound with strong contrast obtained with optical techniques. Development of an ultrasonic modulated optical tomography system (Fig. 2) couples ultrasounds and light in order to reveal the local optical contrast of absorbing and/or scattering objects embedded within thick and highly scattering media such as biological tissue and human breast tissue. Thus, ultrasound would be focused into the tissue/sample under test causing periodic compression and rarefaction. In parallel, if the tissue would be illuminated with a laser light source of sufficiently long coherence length, then the intensity distribution at the output will be laser speckle pattern consisting of

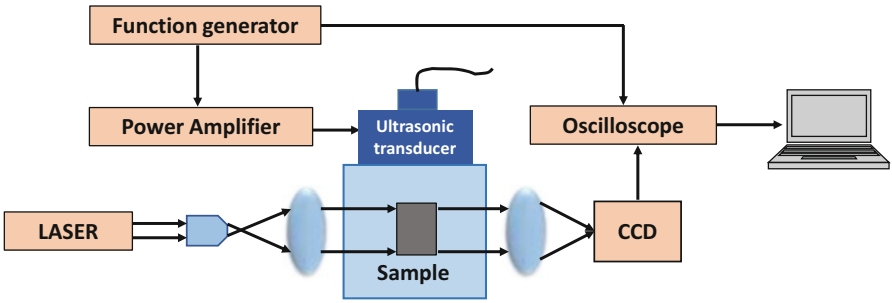

Fig. 2 Schematic setup for ultrasonic mediated optical tomography

distribution of bright and dark speckle grains. The optical effect of ultrasound would result in modification of local refractive index and modification of the position of the optical scattering sites. The combination of light and ultrasound to measure local optical properties through thick and highly scattering media is a tantalizing approach for breast cancer detection. Speckle reduction minimizes the scattering noise and thus increases both the resolution and optical contrast in the biomedical imaging.

Photoacoustic Tomography (PAT)

Hybrid imaging techniques utilizing couplings of physical modalities have attracted considerable attention in recent years. Optical imaging of biological tissue is able to provide structural and functional information, which potentially enables early detection of abnormalities such as tumors and detection of tissue oxygenation. However, one disadvantage of such technique is the difficulty to image objects at depth. By comparison, ultrasound imaging is able to provide good image resolution at depth and is an established clinical imaging modality. However, ultrasound imaging is based on the detection of the mechanical properties in tissues which is not as sensitive as optical techniques in, e.g., detecting tumors. Several efforts have been made in the recent years to develop new imaging modalities which lie visible and near infrared regions and are based on the optical properties of soft biological tissues. At these wavelengths, radiation is nonionizing, and the optical properties of biological tissues are related to the molecular structure, offering potential for the detection of functions and abnormalities. With the growing application of photoacoustic imaging in medical fields, there is a need to make them more compact, portable, and affordable. Besides, medical imaging is a challenging and most needful issue currently. Depending on the organ to be inspected, different waves (X-ray, ultrasound) are used. Tissues are highly scattering media. When using light only, like in diffuse optical tomography, the resolution of a breast tissue image is usually around 10 mm so that doctors can hardly detect an emerging tumor. Use of radiography also could not be much efficient. The ultrasonic waves scatter much less than light waves and thus provide better localization; however it suffers from contrast problem.

It is the need of the hour to get the millimeter resolution, which compelled researchers to associate ultrasonics with optics and thus named as photoacoustic

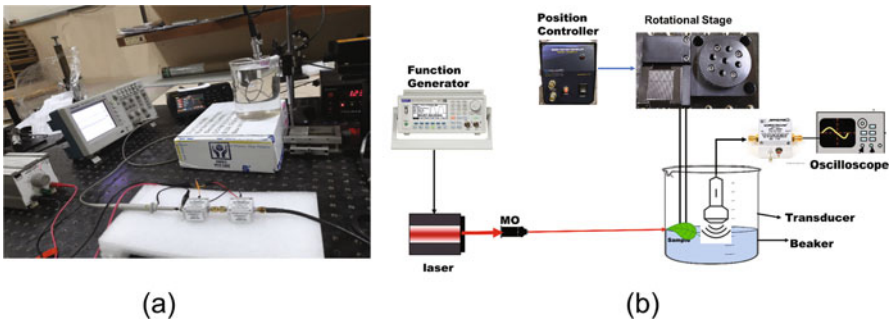

Fig. 3 (**a**) Laboratory set-up of the laser-based PAT system to study biological tissues. (**b**) Modified photoacoustic instrumentation with 360-degree rotational stage

tomography (PAT). In PAT, an image is formed through detection ultrasound waves induced by laser beam. Pulsed lasers with nanosecond time duration having relatively high energies (milli joules)—including Q-switched Ti: Sapphire laser, optical parametric oscillator (OPO), Nd: YAG laser, and dye laser—which are most commonly used excitation sources in PAT systems. However, their high cost, bulky sizes, and strict maintenance requirements significantly limit their practical use in the biomedical applications. Pulsed laser diodes (PLDs) were proposed to address some of these drawbacks because PLDs are relatively simple, compact, inexpensive, and highly power efficient. Design of a very low-cost PAT system is shown in Fig. 3, by replacing the expensive and sophisticated laser with a low-energy pulsed laser diode. It has been shown by several researchers that using PLD, the biological tissue can be imaged with adequate signal-to-noise ratio (SNR) and spatial resolution. Furthermore, since PLDs can operate at a high repetition frequency (up to megahertz), it provides significant advantage for real-time imaging applications. To have a complete information of the sample, 3D images can be built up by moving the focused ultrasonic beam. Interplay of strong optical contrast with high ultrasonic spatial resolution can thus be advantageous for medical imaging. In PAT, due to the conversion of absorbed optical energy into a temperature rise, a corresponding pressure change generates an acoustic signal that can be detected by the ultrasonic receiver, and a high resolution image of optical absorption can be reconstructed revealing the internal structure and functioning. With the growing application of photoacoustic tomography in medical fields, there is a need to make them more compact, portable, and affordable.

Conclusion

Several recent developments in the field of ultrasonics have proved it as a promising tool for medical imaging and non-destructive testing with a promising future full of opportunities. One of its prime attributes is the ability to obtain real-time imaging compared to MRI at a comparably low cost, and ease of operation has also made

impact in daily crucial fields like food safety, medical emergency, and a lot of applications. The past few years have witnessed unprecedented growth in ultrasonic devices in terms of portable size and cost-effectiveness resulting in increased popularity. The financial benefit alone is compelling. In the near future, ultrasonic-based sensors are believed to have a significant role in much wider range of applications.

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Calibration of Snow and Met Sensors for Avalanche Forecasting

38

Neeraj Sharma

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Abstract

Accurate avalanche forecasting is imperative for the safety of man-machine and habitations and operations including snow removal, road clearance, etc. in higher Himalayas and later use of the road for safe passage. In common parlance, avalanche forecasting means the daily assessment of the avalanche hazard for a given region. Real-time snow and meteorological data are important parameters for the study of snow and avalanche forecasting. The timing and size of an avalanche depend on the amount of snow in the formation zone area, the stability of the snow as well as the triggering factors. Therefore, snow and meteorological data from the avalanche formation zone are the most useful input in avalanche forecasting. The data is measured using the snow and meteorological sensors, and it is of paramount importance as this is the basic input for providing avalanche forecasts at various strategic locations in the snow-bound region of the Indian Himalayas. Therefore, periodical calibration of these sensors/instruments is foremost important for ensuring good quality of snow and meteorological data. All electronic devices are sensitive to changes in their working environment, the same applies to sensors. Changes in the working environment and aging of the sensors may result in erroneous and undesired output due to malfunctioning of the internal elements of the sensor. Therefore, sensor calibration is a very crucial aspect that needs to be addressed to maintain the accepted performance level and reliability of sensor measurement. It becomes evident from the above facts that calibration proves instrumental in improving the performance of the sensors and accuracy of measurement which in turn can help to improve the forecast precision.

Keywords

Snow · Avalanche · Sensor · Calibration · AWS · Traceability · Uncertainty

Introduction

This chapter will cover the need for snow and meteorological sensors in snow-bound regions of the Indian Himalayas. What are the different kinds of sensors used for the study of snow and avalanche forecasting? What is the need for sensor calibration from a snow and avalanche forecasting perspective? Along with a brief insight into the workings of the avalanche forecasting system. It may be noted that wet snow avalanches are a complex process that has not been completely understood and are quite difficult to forecast; water is a prime ingredient in the snow cover, to forecast wet snow avalanches, it is extremely important to get a fair idea of the liquid water content of the snow cover. Evaluation of wet snow instability by field measurements

and experiments is quite difficult, and generally, physical models are used to estimate the liquid water content in the snow cover utilizing metrological input. It may thereby be understood the paramount importance of calibration for the sensors which read these data, for example, the liquid water content in the snow cover. Any minuscule error can cause inaccurate measurement with disastrous consequences.

Avalanche Forecasting

Avalanches are set in motion due to certain peculiar topographical and snowpack conditions, in most cases, the formation zones are generally having slope angles of 30–45°, accelerated by barren hills with scanty forest/vegetation cover.

The main aim of avalanche forecasting is to provide valuable and crucial information for enhancing the mobility of man and machine in snowbound avalanche-prone areas by evaluating snow and avalanche hazards. The information disseminated is a result of analysis of snow and meteorological data using highly complex mathematical algorithms running on high computation machines/supercomputers. The snow and meteorological data are mostly gathered from Automatic Weather Stations (AWS), and also manually from various observatories installed at specific places of interest in the snow-bound regions.

Avalanche forecasting had been defined by McClung (2000) as "The prediction of current and future snow instability in space and time relative to a given triggering level." The temporal and spatial variability of snow cover is unstable, along with snow and weather condition changes, variability in human perception, and estimation which make forecasting a difficult task and can be seen with skepticism.

The validation of avalanche forecasting is closely associated with the spatial scale that forecasting is designed for; McClung and Schaerer (1993) categorized the scales into three categories, i.e., synoptic, meso, and micro. Synoptic scales are in the order of 10,000 km^2 (e.g., mountain ranges); meso scales are in the order of 100 km^2 and used for terrain features of independent actions like ski runs.

Some institutes in the world are working in the field of avalanche forecast and mitigation strategies. Few of the most renowned institutes in the field of avalanche forecasting are:

- American Avalanche Association & US Forest Service National Avalanche Center in collaboration with US Avalanche Centres, United States (Bridgeport Avalanche Center, Chugach NF Avalanche Center, Colorado Avalanche Information Center, etc.).
- Avalanche Canada, is a nongovernment, non-profit organization dedicated to public avalanche rescue and safety. Avalanche Canada issues daily avalanche forecasts all-round the winter for major avalanche-prone mountainous regions of Canada.
- WSL Institute for Snow and Avalanche Research SLF, Switzerland.

- Defence Geo-Informatics Research Establishment (DGRE), DRDO India. Erstwhile Snow & Avalanche Study Establishment (SASE).

This list is not limited to the above entries but is exhaustive as there are many regional players which have developed methods and algorithms for providing avalanche forecasts for specific areas, especially for the skiers who are at high risk in avalanche-prone areas.

As elaborated above there are several institutes and centers which are operationally committed to the safety of human lives and infrastructure deployed in various snow-bound areas of the world. In the Indian context Jammu, Kashmir, Ladakh, Himachal Pradesh, Uttarakhand, and the higher reaches of Sikkim and Arunachal Pradesh are among the most vulnerable regions from the avalanche threat perspective.

As detailed by LaChapelle, conventional level entry forecasting consists of the following characteristics:

- A well-built element of determinism formed by the interaction of physical process in the snow cover with the weather, coupled with a non-determinate still decision concerning the actual forecast.
- Reliance on redundant & diverse sources of information about weather, snow, terrain, and avalanche occurrence.
- A high level of human skill which is developed through experience & field learning.

Needs and Types of Snow and Meteorological Sensors

Avalanche forecasting is the science of prediction of avalanche occurrence based on results obtained from complex mathematical algorithms and expert opinions. To have both algorithms and expert opinion work on snow and meteorological data from the snow-bound regions is of utmost importance. The accuracy of algorithms will depend upon the accuracy and reliability of data from the snow-bound locations. To measure and disseminate snow and meteorological data from a network of observatories (automatic/manned) is deployed over the area of interest. The snow and meteorological data from these observatories is passed over to a central location twice a day or on an hourly basis where it is fed into various algorithms and provided to experts for opinions which finally provides the avalanche forecast.

Sensors are imperative to the chain of avalanche forecasting since they form the first block in the chain (Fig. 1).

Various weather variables are used to express meteorological and snow, e.g., temperature, wind speed, precipitation, wind direction, pressure, solar radiation, snow depth, and snow surface temperature. In the case of sensors, special attention has to be given to their accuracy and long service life. Especially in the case of sensors for snow data where they have to be exposed to sub-zero temperature conditions with high thermal changes and extreme weather conditions.

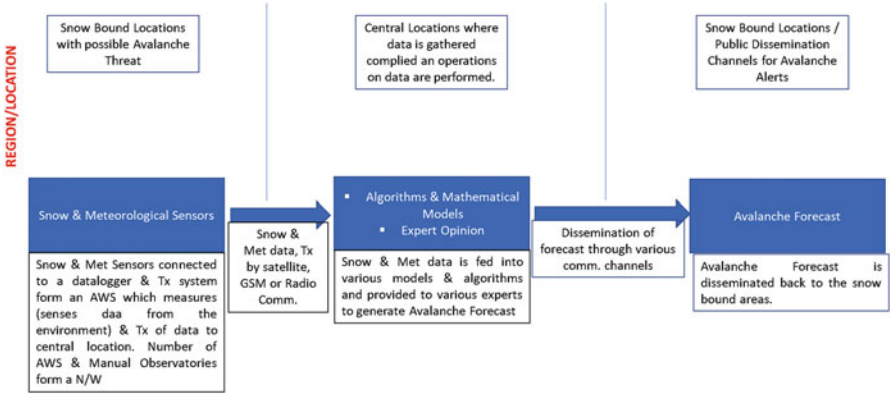

Fig. 1 Avalanche forecasting chain

Automatic Weather Station

Conventionally, data had been collected with manual efforts which suffer from various inherent drawbacks, viz. delay in data reception, exposure of humans to severe weather conditions, measurement errors, and sparse intervals of recording. To obviate these problems and eliminate the risk to human life, automatic data collection with the help of the Automatic Weather Station (AWS) has been adopted worldwide. Sensors form the ears and eyes of the AWS which sense and measure various snow and meteorological parameters for recording these parameters at fixed intervals and disseminating.

The AWS system will primarily consist of Snow and meteorological Sensors, a Datalogger, Data Transmitter with Antenna, a Power supply (Battery, Charger, and Solar Panel), and a Mounting tower. The Datalogger, Data transmitter, and Battery with charger will be housed inside the NEMA-4X enclosure. The lightning arrestor, Antenna, Solar Panel, Sensors, and NEMA-4X enclosure will be mounted on the AWS tower. The advantage with the AWS is that they can be installed even in the formation zones of an avalanche site before the onset of winter. Since the data observation is completely software-guided, it is free from any human error (Fig. 2).

Prolonged, unattended operations of the Automatic Weather Stations are achieved by utilizing low power consumption electronic modules, batteries and solar panels, wireless data transmission, and substantial onboard data storage capacity. AWS reports nonstop, i.e., all meteorological and snow conditions are reported 24 h a day, 365 days a year. They are fitted with custom snow condition sensors or snow depth sensors. Barometric Pressure sensor, Snow Surface Temperature, Ambient Temperature, Water Temperature, Precipitation, Solar Radiation, Wind Speed & Direction, Snow Depth, Relative Humidity, Soil Temperature, etc. are the most common sensors attached to the AWS.

Fig. 2 AWS in snow-bound area

In a typical satellite transmission-based, AWS, the snow and meteorological data is received at Earth Receiving Station (ERS) from the remote AWS for further utilization in the study of snow and avalanche forecasting. The ERS consists of a Dish antenna, Low Noise Amplifier (LNA), frequency Down Converter, Demodulator, Data Management Computer, and Server (Fig. 3).

Typically, the sensors utilized in the AWS for measuring the snow and meteorological data specific to avalanche forecasting are:

(i) Ambient Temperature Sensor
(ii) Relative Humidity Sensor
(iii) Atmospheric Pressure Sensor
(iv) Wind speed and Wind Direction Sensor
(v) Radiation Sensors (Albedometer and Pyrgeometer)
(vi) Precipitation Gauge
(vii) Snow Surface Temperature Sensor
(viii) Snow Depth Sensor

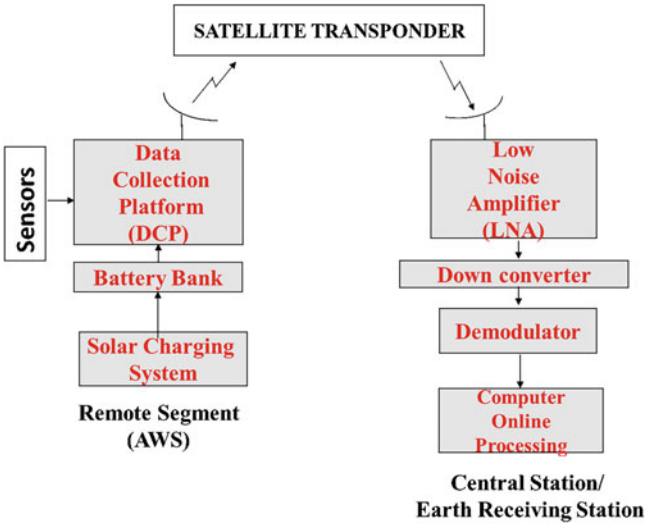

Fig. 3 Block diagram AWS and ERS setup

The parameters and specifications of sensors are given in Table 1 below. The specification of sensors is defined as per the typical installations that exist worldwide for avalanche forecasting purposes.

Grímsdóttir and Ingólfsson (2019) worked on data usage from automatic snow sensors in forecasting avalanches in Iceland. They had relied on snow depth data from different types of snow depth sensors installed in the avalanche starting zone for over 20 years. They measured the output temperature profile based on which an algorithm was developed to calculate snow depth in real-time. The temperature profile also shows the value for avalanche forecasting, since the metamorphism of snow crystals depends on the temperature gradient.

Similarly, Herwijnen and Schweizer (2011) used a seismic sensor to monitor avalanche activity. Rockfalls, landslides, avalanches, etc. are considered hazardous mass movement and are detected remotely by seismic sensors. Avalanche detection through seismic sensors was shown first time by St Lawrence and Williams (1976). Biescas et al. (2003) used runout distance to compare signals between wet snow and dry snow avalanches. They showed that the former generates larger signals than the latter, which can be attributed to the higher mass density of the former.

While avalanche forecasting is being done by both meteorological and snow data, however, it is not simple to predict the influence of meteorological conditions on avalanching. Heightened minimum temperature and change in snow affects the above-average avalanche activity mostly. A combination of prolonged snowfall and mean air temperature near freezing points generally increases the avalanche activity. Furthermore, days with high main wind speed frequently coincide with avalanching. Thereby, it is clear that the period of increased avalanche is related to abovementioned two different processes (Schweizer and Jamieson, 2003).

Table 1 Make and models of various sensors

S. no.	Sensor name	Make	Model no.(s)
1.	Ambient temperature and relative humidity sensor	Rotronic	Hygro Clip HC2A-S3, HC2A-SM, and MPA 101A
		Lufft	WS300
2.	Barometric pressure sensor	Sutron	5600-0120-1S
		Vaisala	PTB210
		RM Young	61,032 V
3.	Wind speed and direction sensor	RM Young	05103 V, and 05108-45
4.	Albedometer	Kipp and Zonen	SMP6
		Deltaohm	LP PYRA 06
5.	Pyrgeometer	Kipp and Zonen	CGR3
6.	Snow precipitation gauge	OTT HydroMet	Pluvio^2
		Belfort Instrument	5915 series
7.	Standard field view infra-red radiometer	Apogee	SI-111-SS, and SI-411-SS
		Everest Interscience, Inc.	Snow-Therm
8.	Ultrasonic snow depth sensor	Campbell Sci	SR-50A-L, and SR-50
		Max Botix, Inc.	MB7374
		Judd Communication	–

A schematic of the operational avalanche forecasting for Western Himalayas is shown in Fig. 4 (Ganju and Singh, 2004).

The following tables give the details of various make and models of the sensor, which are used in AWS specifically deployed for data collection for avalanche forecasting.

Examples of a Few Sensors with the Source of Error

Snow Precipitation Gauge

The density of new snow coupled with the amount and rate of loading of the snowpack provides the precipitation data. It may also be noted that the variation in these quantities is affected by the interaction of wind with the topography. To accurately determine the snow load to trigger an avalanche needs data from both the sensor and the avalanche-prone site. Thereby, it's imperative to measure data from a sufficiently protected site and snow data independent of wind speed or direction (Marriott 1984). For water equivalent measurements, the most used sensors are gauge sensors; however, these sensors also suffer several errors, including missed catch, capping, and evaporation.

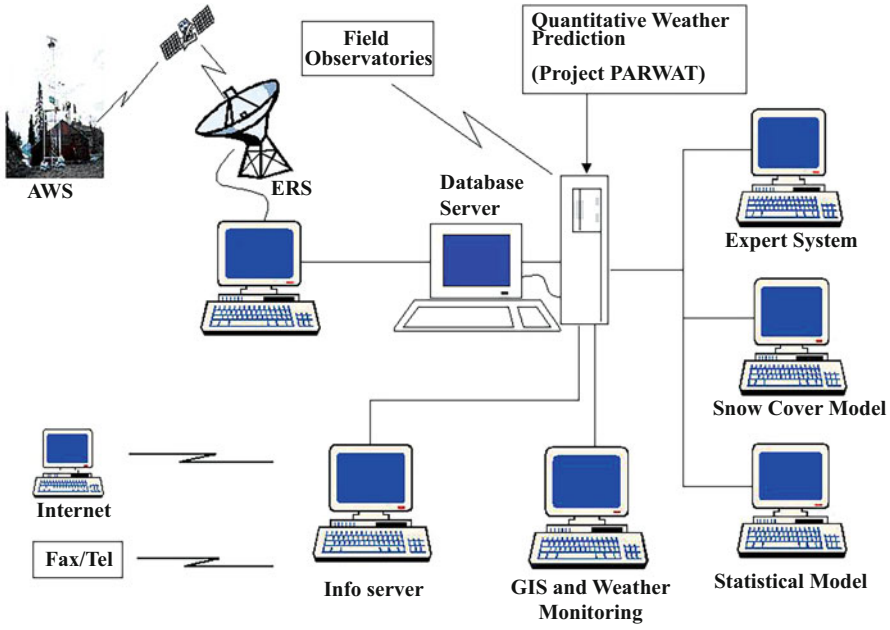

Fig. 4 Schematic of forecasting

Missed catch highlights the wind effect on measuring sites, which may introduce serious errors. Worth by Larson and Peck (1974) shows that wind effects can introduce substantial error engage measurement. The magnitude is related to three factors windspeed, vein speed, and vertical profile, and the obstruction to the airflow presented by the gauge. However, it is possible to develop, a calibration-based correction factor for the lost catch.

Gauge capping indicates the accumulation of snow along the rim of the collection cylinder of in during moderate to heavy snowfalls. This accumulation leads to a decrease in the effective orifice size, thereby reducing the measured data of precipitation. Also, sometimes the eventual melting of the accumulated snow result in an overestimate of the current precipitation.

Evaporation occurs inside a heated gauge, and it appears in the zone of dry snow at a low snowfall rate.

Wind Sensor

Snow stability forecasting involves the estimation of the degree and nature of snow deposits over the topography; both these factors are in turn related to wind information. Variation in local winds is created by both terrain orientation and the synoptic scale veins interacting with the mountain topography. This problem is often further

complicated by veins driven by Mesa scale pressure difference across a mountain range, small-scale channeling effects, or drainage wind.

For wind sensors, site selection for wind measurements is of supreme importance, which requires evaluation of all the potential effects of all the Lanes on the Avalanche starting zones of interest. Choosing a location for measuring veins involves identifying a site that can give sufficient wind information that can be used to infer the starting zone winds. The main error with wind sensors occurred during periods of rime formation. Taylor and Kuyatt (1994) described the process to minimize riming of the instrument by locating the anemometer downwind over the Terrain feature; this process removes most of the supercooled water droplets causing riming.

Air Temperature Sensor

Forecasting of snow stability includes many factors such as crystal types, surface melt, rainfall occurrence, the density of snowfall, snowpack metamorphism, etc., and all these factors are intricately related to the air temperature and subsequently to the snowpack strength. Temperature fluctuations in large mountain areas are dependent on air freezing levels or diurnal variations or the free air circulation. Sometimes minor effects like localized terrain-induced factors can also cause temperature fluctuation.

Topographic effects may cause variation in temperature from the scale of a few meters, to up to a scale of hundreds of kilometers. The biggest contributor to an error in air temperature measurement is the lack of shielding from heat and radiation sources. Locating sensors close to heated buildings or direct sunlight without proper radiation shielding may produce errors in measurement, making observation difficult to interpret and correct.

All these errors categorically signify the need for precise functioning of the sensor by proper maintenance and periodic calibration, to circumvent the errors and produce realistic and reliable data. In those instances where sensor data alone must be relied upon, the best solution appears to be several measuring sites with several types of sensors at each site. This type of input often helps to resolve potential ambiguities produced by individual sites or sensors. Similarly, an increase in the number of measuring sites can overcome deficiencies at individual sites and provide information on mesoscale weather features.

Need for Sensor Calibration from a Snow Avalanche Perspective

The classic technique for assessing avalanche hazards is as described by Lachapelle (1980). The technique uses daily weather, snow, and snow cover data observed and measured at selected locations for a given area. The data are acquired through sensors, which are then evaluated by human expertise, which in turn is strictly dependent on the accuracy of the sensor for that specific region (Fig. 5).

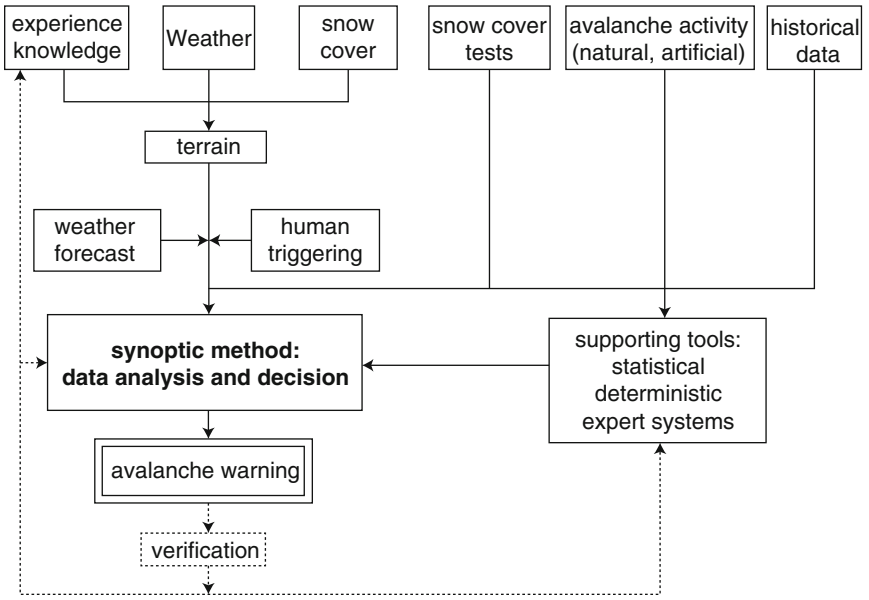

Fig. 5 The classical conventional method to forecast avalanches on a regional scale (Schweizer and Fohn, 1996)

The accuracy of avalanche forecasting heavily depends on the quality of data received from various sensors installed on AWS and various other portable scientific instruments located in observatories/field locations. The data recorded by the sensors is of paramount importance as this is the basic input for providing avalanche forecasts for high-altitude locations in snow-bound regions. Therefore, for ensuring a good quality of data from these sensors/instruments, periodical calibration of these sensors is foremost important. The calibrated sensors will provide more accurate and reliable data, which in turn will help in providing better avalanche warning and weather forecasting to the inhabitants and personnel deployed in snow-bound areas. This will enhance the safety of mankind and infrastructure in snow-bound areas which is of paramount importance.

Snow and weather conditions over a large area can be inferred from the snow and weather parameters, both these parameters aren't necessary for avalanche forecasting, and for this purpose, the point measurement of the factors is needed to analyze snow stability. The human experience is the main tool for inferring and interpreting these data in the mountain environment, as the process is a very complex or qualitative correlation between the measured point letter and the variation in the conditions of the two basic parameters needed generally.

There can be two major errors in the avalanche forecasting system. The first error is associated with the site chosen for measurement, i.e., the topography and environmental conditions, which may vary over short distances and are often hard to predict. For example, the temperature may vary by complications of mesoscale or

synoptic scale weather conditions. This makes it undetectable from a valley site. Thereby, multiple measurements need to be taken to remove ambiguity.

There are a wide variety of sensors available for measurement of each parameter, and generally, these cause the second source of error. All sensors have the individual type of responses for the same environmental conditions, thereby there can be very different readings at the same location but from different instruments. An accurate and systematic calibration of the sensors to remove anomalies/data ambiguities are of paramount importance.

Calibration Methods

Specialist knowledge is required to determine the frequency at which instruments should be calibrated. For example, an instrument can have a margin of 2% inaccuracy, in that case, certain performance degradation can be allowed if after recalibration the inaccuracy becomes 1%. The pattern of performance degradation is quantifiable, thereby before reducing accuracy to the defined limit, the instrument can be recalibrated.

There are three standard calibration methods used for sensors as follows:

1. One-point calibration
2. Two-point calibration
3. Multipoint curve fitting

1. This type is for linear sensors and correcting sensor errors of a single level measurement accuracy. This is the fastest calibration method out of the three and uses a zero-point adjustment. One-point calibration is typically done in the lower 20% of the sensor range. The single point-based calculation is done to differentiate the instrument reading and the reference value, and finally, this produces an offset correction factor.

   In the case of gauge transducers, this method is as simple as venting the device's both input and reference ports to the atmosphere. An absolute transducer may require the instrument to be brought down in the vacuum range for zeroing. This depends on the full span pressure of the DUT.

   This method is generally ideal for transducers that possess a constant offset value since the adjustment applies to all the points across the complete range. For example, temperature sensors are usually one-point calibrated.
2. This type is for correcting both offset errors and the slope, the usual process being zero and span adjustment. This type of calibration works on instruments with linear drift throughout the range and having a zero error. This method is different from the one-point method as it requires the instrument to be pressurized to the top 20% of the range for getting the second point reading or the span. The span adjustment is used to create a multiplier that is factored in at every point within

the measured pressure. Additionally in this type, sensors output has to be reasonably linear over arrange and with two reference values, i.e., high and low.

3. This type is for nonlinear sensors, which also require some adjustment to get an accurate measurement. Transducers generally have a zero shift or span drift and at times it will be inconsistent linearity throughout the range. Sometimes the zero or span point doesn't produce offset values, but it may still produce errors throughout the range. For instruments with this type of behavior, it will need a multipoint adjustment.

   In this method, the calibrator can use a minimum of three to a maximum of 11 reference points for adjustment, where each point set can be treated as two-point calibration. Though it takes more time, the multipoint method gets the best result. This type of calibration is typically referred to as performing a "linearization" of the device. For example, this is usually done for thermocouples, which are used in extremely hot or extremely cold conditions.

   In the case of snow and meteorological sensors specifically required for avalanche forecasting purposes, the calibration has to be done in the sub-zero range as the sensor is exposed to such temperatures during most of the period which is of interest to the forecaster. The calibration systems designed and in use for sensors utilized in avalanche forecasting should have at least two points in 0 °C and below ranges preferably the lowest measuring point of the sensor or the lowest temperature which the sensor is expected to encounter in such environments.

Principles of Calibration

When the output of any instrument or sensor getting tested for the first time, it is always compared against the output of an instrument that has an accuracy level known to the testers this method is called calibration; however, both the instruments should receive the same input for the calibration process the input can be out of a various range which covers the whole measurement range of the sensor and the calibration process ensures that the accuracy of all the instruments in questions for measuring purpose is known over the whole measurement range but one factor has to be considered that is the environmental condition should be same for all the instruments and sensors that are being calibrated.

It is a well-known factor that over time characteristics or specifications of any instrument or sensors can change, thereby the calibration process has to always be repeated mostly at previously predefined intervals to set the sensors updated. The sensors have changes in their specification or output or characteristics do too many external factors which even include dirt dust chemicals and even temperature changes in the environment and additionally the mechanical wear.

In the process of calibration, both the frequency of calibration and the system used for calibration are to be established properly, and thereby it is very important that the system always remains efficient and up to date by periodical review. However, sometimes cost efficiency is a matter because repeated calibration can

be expensive, and thereby alternative system can be adopted which may be less expensive but there shouldn't be any compromise for the effectiveness of the calibration. When a review of calibration is done primary basis becomes the previous history of the calibration of the sensor, thereby our instrument can be calibrated frequently because either it is an old instrument that the aging factors are taking into consideration or because the operating environment has changed. The environmental changes and the repeated use or frequent use of any instrument sensors can affect the sensor both adversely or positively, though there is always a possibility that the calibration interval can be decreased or can be increased as per the situation in need.

The international standard ISO 900 is the quality control function used for any calibration process to any measurement; previously the British quality standard BS5750 was used. Taking consideration of the human factor of human error or manual error 9000 plus always made it very specific that the person dealing with the calibration equipment needs to be properly trained. They are not various national standard organizations that can monitor both the mechanical testing laboratories and the calibration of instruments. Minerally different countries have their separate mechanisms for the maintenance of standards but always the frameworks have to be equivalent in the effect of ensuring the requirement of ISO/IEC 17025 standards. This helps to retain a global standard for the goods and services that are cross boundaries and also properly calibrated.

Whenever the instrument gets calibrated, it is termed as the device under test (DUT). To be calibrated the duty configuration should have known shell input stimuli for the sensors, and this process helps to determine actual errors in the measurement.

The calibration process helps us to determine the following results:

- No error noted on the DUT.
- An error is noted and no adjustment is made.
- Adjustment meant is made to remove the error and the error is corrected to the desired level.

Control processes are adjusted by monitoring the control system and for that sensor calibration is necessary. Automatic systems also apply sensor calibration to get error-free results.

Traceability

Calibration acts like a chain reaction whereby each successive instrument is calibrated for accuracy concerning this previous or higher-up model in the chain. Thereby to trace the measurement standard for the instrument which lies at the bottom of the chains, all the aspects of all elements of the calibration chains must be known beforehand. Traceability is thereby known as the knowledge of the full complete chain of instruments involved in the total calibration procedure, it is also mandatory for having an ISO 9000 standard.

An unbroken chain of calibration always contributes to the measurement of uncertainty, and when this gets documented, the result can be referred to and it becomes a property of the measurement and the property is called metrological traceability. Data validation needs both the measurement data and the traceability of the test.

Traceability is characterized by six essential elements.

i. The fixed reference point can be either a national or international standard, and to reach this point an unbroken chain of comparisons is needed.
ii. Uncertainty of measurement: This needs to be calculated for each step in the chain, and this then culminates in the whole chain's overall uncertainty.
iii. Documentation: All steps must be performed properly as per well documented procedures and the results must be recorded.
iv. Competence: There should always be recorded evidence of technical computer competence of the laboratory performing the steps.
v. Difference to SI units: The chain of comparisons must, where possible, end at primary standards for the realization of the SI units.
vi. Calibration intervals: There should be repeat calibrations with appropriate intervals; however, the intervals can depend on several variables, for example, uncertainty, frequency of use, way of use, etc.

ISO/IEC 17025 Standards always stress the accuracy or validity of the results from all equipment used for tests and or calibrations, including equipment for subsidiary measurements. Traceability to SI units is maintained by calibrating all the equipment. However, standards include other mechanisms as well, other than traceability.

ISO15189 standards use either natural constant or SI units or any stated reference, both the trueness verification and calibration have to be traceable to any of those three standards. Most of the time, laboratories need to design specific programs for this purpose.

All the calibration and medical laboratories have to get checked for the requirements and acceptable traceability while testing. This task is being overseen by the national accreditation board for testing and calibration laboratories (NABL, 2012). All laboratories, therefore, need to have a set calibration program and traceability to SI units (national or international standards). Calibrations of equipment are performed from:

i. National Physical Laboratory, India, or a national metrology institute that is a signatory to the CIPM Mutual recognition agreement (MRA).
ii. Accredited calibration laboratories: When the MRA covers calibration activities. Calibration, certificate bearing accreditation, body symbol, or a special reference to the accreditation status by a recognized accreditation body shall only be considered for valid traceability.

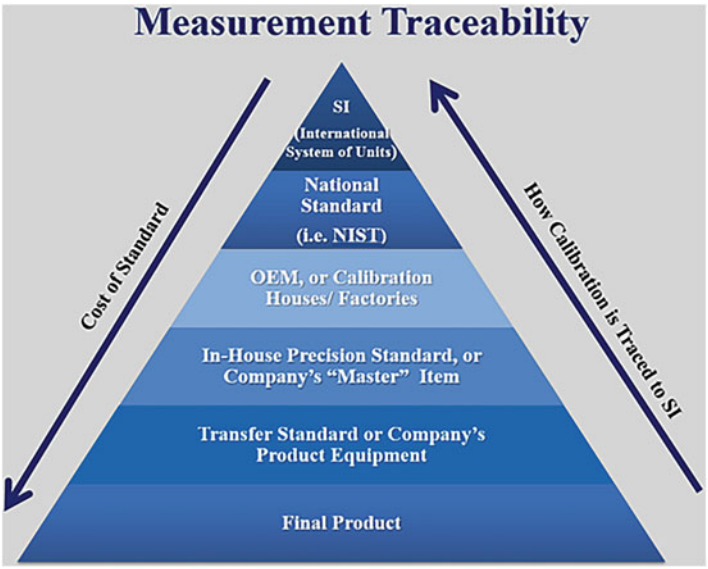

Fig. 6 The concept of measurement traceability

Sometimes there are cases when such traceability as indicated above is not possible and reasonable. In that case, the following options can be taken:

i. Certified reference material is needed for a reliable chemical or physical characterization, and traceability is a demonstration of such performance.
ii. Specific methods and or consensus standards are also recognized by NABL, which are multipartite and acceptable to all. The laboratory involved must participate in proficiency and performance testing, then only it can provide satisfactory evidence of correlation of results.

With all the procedures detailed above, NABL thus maintains a policy on traceability of measurement results (Fig. 6).

Uncertainties in Calibration and/or Measurement

Measurement Uncertainty

Higher accuracy is the most demanded component in the manufacturing process; however, in any measurement or calibration, such factors are always uncertain pump and thereby measurement uncertainty is the natural parameter which defines all these uncertain factors in the measurement. Various industries produce more or less similar measurement equipment, and often the uncertainties can be estimated by the better equipment. The intent of the industries depends on the competitive advantage of the

fact that uncertainty is inversely proportional to the quality, and thereby the equipment's effectiveness and accuracy can be predicted within the range of its performance. However, uncertainty is a complex procedure and there may be a situation where actual accuracy is lower than the predicted performance level accuracy.

Generally, an instrument is evaluated by the errors found during the operation however error can also occur because of improper uses of the instrument and also incomplete knowledge of the test artifact. This instrument's unrelated contributors to the errors give rise to the test uncertainty.

A measurement system only becomes acceptable when it produces accurate measurement results and also the verification of the expected quality output. Any measurement result can only be complete when it is added with a statement of its uncertainty which considers all possible measurement errors at each level of handling.

The quantity subject to measurement is called measurand (ISO, 1993), and measurement uncertainty is thereby defined as the parameter, associated with the result of a measurement that characterizes the dispersion of the values that could reasonably be attributed to the measurand.

During measurement, there can be two factors that influence the process: one is the expected accuracy of the output and the other the factors influencing the measurement process, measure, and the specific quantity subject to the measurement (Fig. 7). A few examples are as follows:

i. Temperature is a prime factor while dealing with metal items, thereby temperature actors are better when micro-level accuracy is needed in measuring metal tools.
ii. The length of a row acts as a measurement during measurement as the length is always affected by the tension in the road.

A Fishbone diagram can be used to depict the contribution of various sources of measurement errors and uncertainty in the final measurement output. It is also called

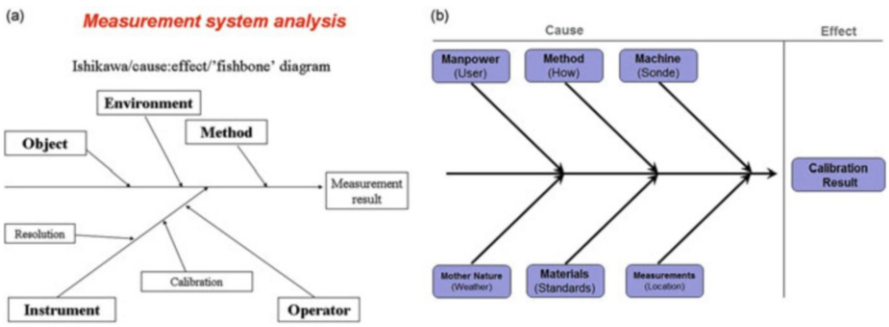

Fig. 7 (**a**) Fishbone diagram of measurement uncertainty. (**b**) Fishbone diagram of measurement uncertainty

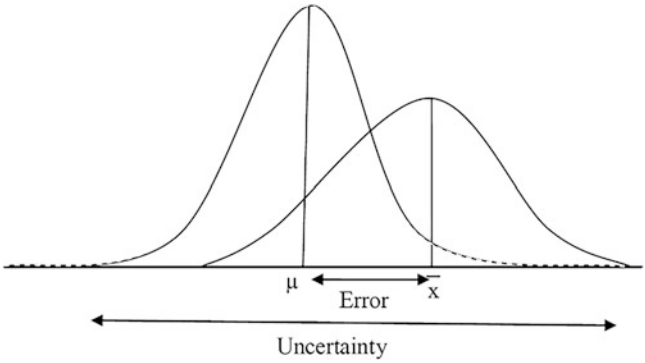

Fig. 8 Difference between error and uncertainty

as Ishikawa diagram, simply it's a hierarchical representation of cause and effects as stated above.

There is the general depiction of 6 Ms (manpower, mother nature, material, measurement, method, and machine), which are the different categories of causes. These 6 Ms can in turn impact calibration uncertainty by influencing the sources of uncertainty, i.e., instrument, standards, location, weather, how, and the user (Figs. 8 and 9).

Uncertainty Budget

Uncertainty has many sources that require identification, quantification, and characterization, a list of which items all components contribute to the uncertainty is called an uncertainty budget.

This tool is used by engineers, physicists, and meteorologists globally for uncertainty analysis while doing the quality measurement. This budget provides the laboratory with a formal record of the uncertainty analysis process which can be shared with other professionals who can validate the results.

How Uncertainty Budgets Improve Measurement Quality

There are several ways uncertainty budgets can help improve measurement quality. Here is a list of common benefits associated with these budgets:

1. Have ISO 17025 standards
2. Obtain or maintain ISO 17025 accreditation
3. Estimate the CMC uncertainty expressed in the Scope of Accreditation
4. Less time and effort are needed for measurement uncertainty calculation
5. Provide objective evidence that an uncertainty analysis was performed

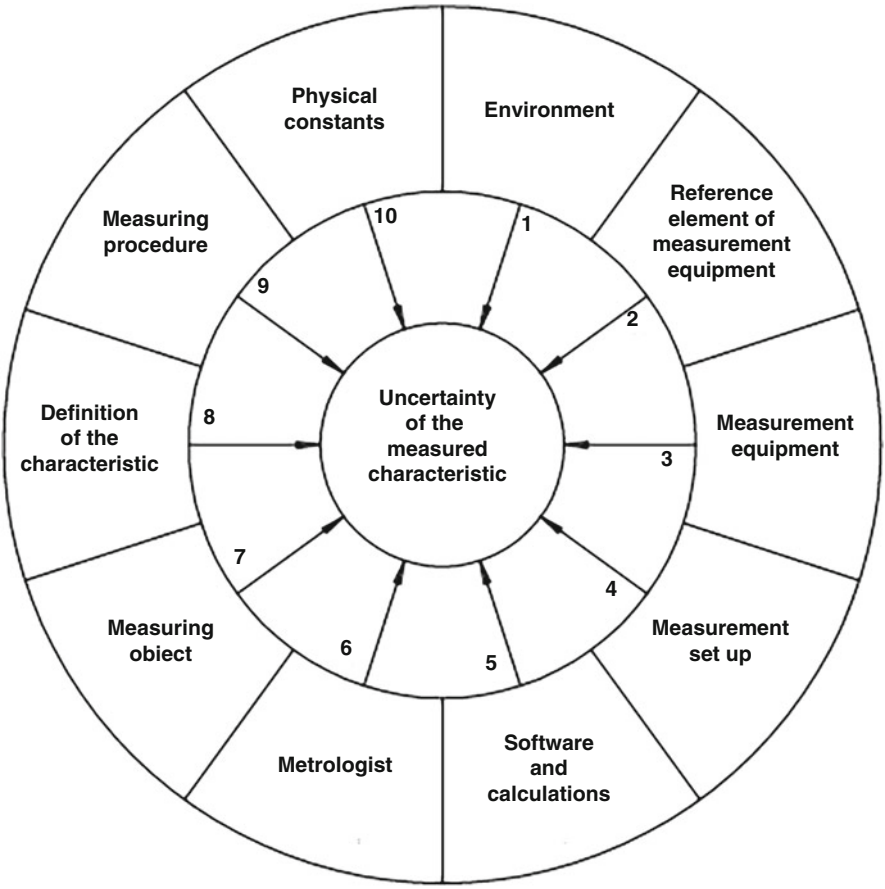

Fig. 9 Uncertainty contributor to measurement

6. Quality improves as uncertainty contributors are evaluated (i.e., find the greatest contributors and identify where reductions can be made)
7. Decision makers have more confidence
8. Errors in measurement results are reduced or prevented completely
9. Less measurement risk (e.g., Statements of Conformity and Decision Rules)
10. Increase customer satisfaction and improve customer service

Example of the Uncertainty Budget

An example is shown below, which also clarifies the several principles of uncertainty analysis. The reported value for a test item is the average of N short-term measurements where the temporal components of uncertainty were estimated from a three-level nested design with J short-term repetitions over K days.

Table 2 Example of error budget for type A and type B uncertainties

Type A components	Sensitivity coefficient	Standard deviation	Degrees freedom
1. Repeatability	$a_1 = 0$	s_1	$J - 1$
2. Reproducibility	$a_2 = \sqrt{(K-1)/K}$	s_2	$K - 1$
3. Stability	$a_3 = 1$	s_3	$L - 1$
4. Instrument bias	$a_4 = 1$	s_4	$Q - 1$

The number of measurements made on the test item is the same as the number of short-term measurements in the design, i.e., N = J. Because there were no repetitions over days or runs on the test item, M = 1; P = 1. The sensitivity coefficients for this design are shown on the foregoing page.

This example also shows the effective use of bias correction, when the sample instrument is biased with regard to other instruments. The sensitivity coefficient, given that the bias correction is based on measurements of Q artifacts, is defined as $a4 = 1$, and the standard deviation, s4, is the standard deviation of the correction (Table 2).

a. *Standard Uncertainty (UI)*
   It is the representation of each component of uncertainty that contributes to the uncertainty of measurement, by an estimated standard deviation.
b. *Combined Standard Uncertainty (UC)*
   It is the combination of all the standard uncertainties, which represents the standard deviation of the result. It is usually the square root of the sum of the squares of the individual standard uncertainties.
c. *Expanded Uncertainty (U)*
   It is the combined standard uncertainty times the coverage factor. The expanded uncertainty forms a boundary about the measurement result y within which the measure Y lies.

$$\mathrm{y} - \mathrm{U} \leq \mathrm{Y} \leq \mathrm{y} + \mathrm{U} \,\mathrm{or}\, \mathrm{Y} = \mathrm{y} \pm \mathrm{U}$$

d. *Coverage Factor (k)*
   A number larger than one by which a combined standard measurement uncertainty is multiplied to obtain an expanded measurement uncertainty.
e. *Accuracy*
   Accuracy is the closeness of agreement between a measured quantity value and the true quantity value of a measurand. A measurement is said to be more accurate when it offers a smaller measurement error.
f. *Bias*
   Bias is the difference between the average value of all the measurements (μ) and the true value (μo). Bias is a measure of the amount by which a tool is consistently off target from the true value, and bias can be positive or negative.

$$\mathrm{Bias} = \mu - \mu\mathrm{o}$$

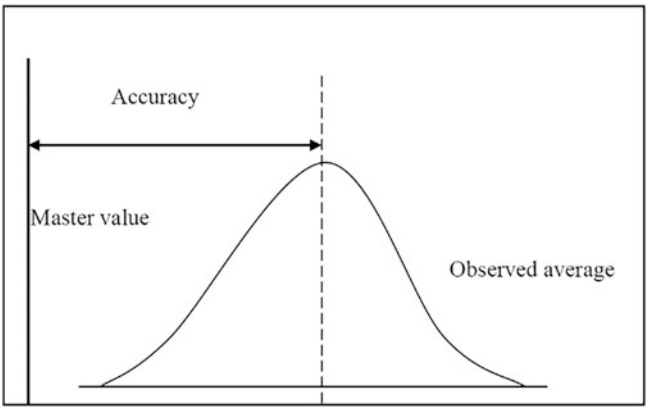

Fig. 10 Accuracy

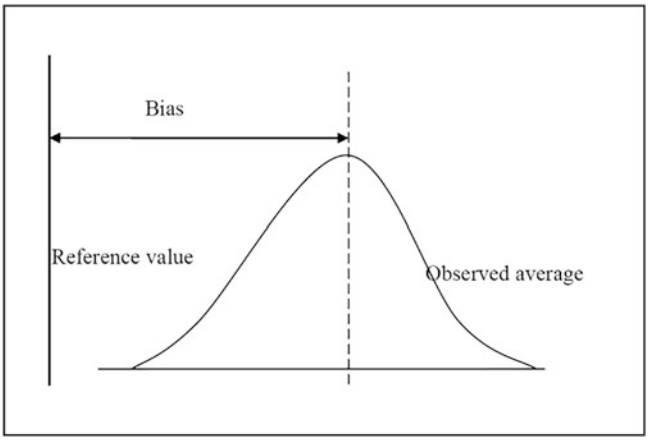

Fig. 11 Bias

g. *Precision*

Precision is the measurement of the extent of variation between different measurements. The standard deviation of the measurement distributes quantify the total variation in the measurement that is the precision. Precision is inversely proportionate to standard deviation, and it is also the degree of closeness of the measured value for each other (Figs. 10 and 11).

Test Uncertainty Ratio (TUR)

Various phases of production process certify the measurement equipment performance. Generally the measurement capability of the equipment has to be perfect and

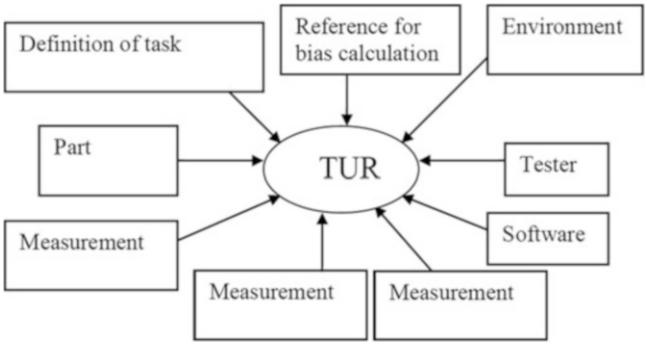

Fig. 12 Contributors to TUR

also the acceptance of the manufactured product has to be reliable. The method to obtain these two is called the test uncertainty ratio. The ratio between the uncertainty in determining the measured value and the tolerance for a specific measurand is the TUR. In the numerator lies the specified tolerance and in the denominator lies the uncertainty former and the higher value of the TUR always indicates a better performance of the test (Fig. 12).

Test Uncertainty

The outcome of any test is always uncertain and this factor is called the uncertainty whenever a new piece of equipment gets tested. During calibration, some permanent sources of calibration errors can be found in the measurement equipment itself. The errors generate due to manual inputs provided by the person handling the equipment, and thereby from all the instruments which produces the reference value. However, during calibration, the instrumental errors are not included in the test uncertainty. The uncertainty is only reflected in the ability of the test to evaluate properly the instrument, thereby it is different from measurement uncertainty and the value is smaller as well. In simple language calibration process verifies the performance of the instrument whereas the measuring instruments are to be calibrated by using the reference to determine the uncertainty.

Calibration Hierarchy

Celebrations can be performed in sequence starting from a reference and ending with the final measuring system, the hierarchy became the outcome of calibration where each of the previous celebrations in turn influences the next calibration. Measurement procedures are operated by measuring systems, and the measurement standards form the elements of the calibration hierarchy. However, in general, a simple comparison between two different standards can also be called calibration, only if

the comparison is useful for correcting the quantity value, and also the uncertainty can be attributed to any one of the standards used.

Sensor-Specific Calibration Systems

The parameters and specifications of various calibration systems for calibration of snow and meteorological sensors are given in Table 3 below.

Wind Speed Sensor Calibration System

To have a calibration system for the wind speed sensor which is to be utilized in sub-zero ranges of temperature, a specially designed wind tunnel is proposed. Brief design characteristics and standard requirements of the tunnel will be discussed in this section.

The wind tunnel design shall be based on a wind tunnel calibration facility that employs all sensors traceable to national/international references and as per guidelines mentioned in ASTM D 5096-02, ISO 17713-1, and IEC 61499-12-1 standards.

The wind sensor calibration system, design envisages having a full-fledged, standalone, and self-sufficient calibration facility for calibrating wind speed and direction sensors/anemometers/wind vanes at different temperatures and wind speeds.

The design will be such that the system would be able to calibrate and test the wind sensors and anemometers with the design proposed for the wind tunnel calibration facility.

Functional Requirements and Specifications for the Wind Speed Sensor Calibration System

- The system must be capable of calibrating all types of phase-shifts (ultrasonic, laser Doppler), Pressure Types (pressure tube, pressure plate, sphere anemometers) and thermoelectric (hot wire and hot plate anemometers), electro-mechanical wind speed sensors, Standard Cup Counter Anemometer within the size of the existing test section as mentioned in the preceding paras.
- Max Wind Speed: 60 m/s with Max Uncertainty 0.1 m/s.
- Closed Circuit – Closed Wall test Section type wind tunnel.
- Min Test Section Dim: The test section should be able to accommodate the sensors intended to be calibrated. The width and height of the test section will be calculated based on the blockage ratio and size of the anemometers to be calibrated. The min test section dimensions should be following the ASTM D 5096-02, ISO 17713-1, and IEC 61499-12-1 standards which define that the Blockage Ratio should be less than 5%. Blockage Ratio calculation will include a summation of Frontal Area of Sensor Under Test (Approximate Calculation of Projected Area/Area Perpendicular to Wind Direction Provided below), Sensor Mounting, Reference Wind Speed Sensor/Pitot Tube, Pressure Sensor,

Table 3 Parameter and specification of various calibration systems

Sl. no.	Calibration systems and parameters	Specification
1.	**Temperature calibration system**	It consists of a dry block calibrator, liquid temperature bath calibrator (for glass thermometers), and platinum resistance thermometer with temperature indicator and data acquisition system for both calibrators
	a) Platinum resistance thermometer (PRT)	
	(i) Temperature range	40 °C to +420 °C
	(ii) Leads	4 Wire
	(iii) RTPW	25.5 ± 0.5 Ω or 100 ± 0.5 Ω,
	(iv) W(Ga)	≥1.11807
	(v) Measurement uncertainty	±0.010 °C or better(−40 °C to 420 °C)
	(vi) Sheath	Inconel/Quartz Glass
	(vii) Diameter	6 ± 1 mm
	(viii) Length	450 ± 50 mm
	(ix) Connection cable	4-wire Measurement Leads compatible with Temperature
	(x) Indicator	Specified below
	(xi) Calibration	NMI Traceable Calibration as per ISO/IEC 17025: 2017
	(xii) Carrying case	To be supplied along with PRT
	b) Temperature indicator with data acquisition system	
	(i) Temperature range	40 °C to +420 °C
	(ii) Temperature resolution	0.001 °C or better, 0.0001 Ω or better
	(iii) Temperature accuracy	±0.008 °C to ±0.020 °C or better in the full range
	(iv) Sensing current	1 mA
	(v) Inputs	Two or more channels (for 4 wire RTD)
	(vi) Display units	°C, Ω
	(vii) Temperature conversation equation	ITS-90, Callendar-Van Dusen, with provision to set new calibration coefficients
	(viii) Communications and data logging	Compatible Communication port to interface with external computer and software for data logging
	(ix) AC power	230 V AC (±10%), 50/60 Hz
	(x) Calibration	NMI Traceable Calibration as per ISO/IEC 17025: 2017
	(xi) Carrying case	Supplied along with indicator
	c) Dry block calibrator	
	(i) Temperature range	40 °C to +140 °C
	(ii) Temperature resolution	0.1 °C or better
	(iii) Immersion depth	150 mm or higher
	(iv) Temperature stability	±0.030 °C or better in the above range
	(v) Temperature axial uniformity	±0.070 °C or better (40 mm) in the above range

(continued)

Table 3 (continued)

Sl. no.	Calibration systems and parameters	Specification
	(vi) Temperature radial uniformity	±0.050 °C or better in the above range
	(vii) Block insert	(i) One – suitable for max. 16 mm diameter sensor probe (ii) One – suitable for various RTDs
	(viii) AC power	230 V AC (±10%), 50/60 Hz
	(ix) Calibration	NMI Traceable Calibration as per ISO/IEC 17025: 2017
	(x) Carrying case	Supplied along with dry block calibrator suitable for field application
	d) Liquid temperature bath calibrator	
	(i) Temperature range	40 °C to +100 °C with single fluid silicone oil
	(ii) Temperature stability	±0.010 °C or better in the above range
	(iii) Temperature uniformity	±0.020 °C or better in the above range
	(iv) Set point resolution	≤0.01 °C
	(v) Display resolution	≤0.01 °C
	(vi) Bath volume	≤45 l
	(vii) Bath access opening	Wide opening
	(viii) Bath immersion depth	≥170 mm
	(ix) Control probe	4 Wire PRT
	(x) AC power	230 V AC (±10%), 50/60 Hz
	(xi) Calibration	NMI Traceable Calibration as per ISO/IEC 17025: 2017
	(xii) Bath liquid	50 l **silicone oil** suitable for above temperature range
2.	**Humidity calibration system**	The proposed humidity generator/calibrator is based on a two-pressure method along with an air supply source to achieve a better accuracy level for calibration of DRDO-SASE humidity probes: 1 Unit
	(i) Relative humidity range	10–95% RH or extended range (at test chamber temperature range)
	(ii) Relative humidity resolution	0.01% RH or better
	(iii) Humidity generation principle	Based on the two-pressure method
	(iv) Flow gas type	Nitrogen or air
	(v) Relative humidity uncertainty	±0.5% or better with above RH range
	(vi) Chamber temperature range	10–70 °C or an extended range
	(vii) Chamber temperature stability	±0.05 °C or better
	(viii) Chamber temperature uniformity	±0.1 °C or better
	(ix) Temperature resolution	0.01 °C or better
	(x) Test chamber pressure	Ambient

(continued)

Table 3 (continued)

Sl. no.	Calibration systems and parameters	Specification
	(xi) Gas flow rate	Adjustable with 0.1 resolution
	(xii) Inlet gas pressure	>150 psi along with air supply source and accessories
	(xiii) Test chamber size	25 × 25 × 25 cm or bigger
	(xiv) AC power	230 V AC (±10%), 50/60 Hz, single phase
	(xv) Data acquisition	Compatible software, data cables, and accessories
	(xvi) Interface port	RS-232/USB/IEEE
	(xvii) Calibration	NMI Traceable Calibration as per ISO/IEC 17025: 2017
3.	**Pressure calibration system**	Proposed barometric pressure calibration is setup which consists pressure/vacuum source, pressure controller, vacuum chamber, secondary standard, and data acquisition system, a completely automatic system
	a) Barometer pressure controller	
	(i) Range	0–2000 mbar (abs)
	(ii) Accuracy	Better than ±0.008% full-scale deflection (FSD)
	(iii) Precision	±0.004% FSD
	(iv) Resolution	Four to six digits
	(v) Panel	Touch Screen Panel for control
	(vi) AC power	230 V AC (±10%), 50/60 Hz, single phase
	b) Vacuum pump	
	(i) Type	Rotary vane type
	(ii) No. of stage	Double stage
	(iii) Pumping speed	16 m^3/h
	(iv) Ultimate vacuum	$\leq 5 \times 10^{-3}$ mbar
	c) Air compressor/nitrogen cylinder	
	(i) Cylinder	Air compressor/nitrogen cylinder
	(ii) Regulator	ISI mark regulator for cylinder
	(iii) Accessories	Accessories to generate pressure less than 2000 mbar with proper safety valve for overpressure and regulated output
	d) Vacuum chamber	
	(i) Dimension	Vacuum chamber dimension with minimum working space inside the chamber 1.5 × 1.5 × 1.5 ft
	(ii) Material	Chamber made of SS 304/metal with a frontside full optical window/vacuum glass door which can easily open to keep the device under test (DUT) inside the chamber
	(iii) Illumination	Provision for illumination for good vision
	(iv) Connection	Suitable connecting vacuum port, and feed-through for input power supply and output voltage/current of at

(continued)

Table 3 (continued)

Sl. no.	Calibration systems and parameters	Specification
		least three devices under test. The vacuum chamber should be leak tested for high vacuum compatibility
	e) Test bench/working table	Calibration test bench with at least three lockable drawers and powder-coated CRC frame, focusing lamp, perforated backplate with one wheelchair
	f) General	The primary requirement is a complete remote-controlled, automated pressure calibration bench in the barometric pressure range from 0 to 2000 mbar. The accuracy of the measuring devices is at least 0.008% of full scale. All the above parts mentioned must be fully compatible with each other. Any hardware, connectors, plugs, software, and any items required to ensure complete compatibility will be provided by the bidder. The pressure and vacuum pump provided should be able to maintain the pressure inside the vacuum chamber from 0 to 2000 mbar. However, if the bidder has an integrated solution as mentioned above, the same will be accepted if it meets the technical specifications
4.	**Infra-red temperature calibration system**	
	a) IR radiation thermometer	
	(i) Temperature range	40 °C to +80 °C
	(ii) Emissivity	≥0.95
	(iii) Spectral response	8–13 μm
	(iv) Resolution	≤0.1 °C or better
	(v) Response time	≤5 s
	(vi) Uncertainty	≤0.5 °C
	(vii) Laser aiming sight	Laser marker or marker through the eyepiece
	(viii) Control interface	RS232/IEEE ports for serial data transfer to an external computer for online data evaluation (USB adapter), software for online data evaluation (Windows version), cables, and necessary accessories
	(ix) Operating temperature	23 ± 3 °C or wider
	(x) Mounting accessories	Suitable mounting accessories (if any)
	(xi) AC power	230 V AC (±10%), 50/60 Hz
	(xii) Calibration certificate	NMI or NMI Traceable Calibration certificate as per ISO/IEC 17025:2017
	b) **Blackbody source with accessories**	
	(i) Temperature range	40 °C to +100 °C
	(ii) Temperature stability	±0.2 °C or better
	(iii) Temperature uniformity	±0.2 °C or better
	(iv) Spectral range	8–14 μm
	(v) Accuracy	±0.5 °C or better

(continued)

Table 3 (continued)

Sl. no.	Calibration systems and parameters	Specification
	(vi) Resolution	0.1 °C
	(vii) Aperture diameter	50 mm diameter or more
	(viii) Emissivity	≥0.95
	(ix) Control probe	4 Wire PRT
	(x) Control interface	Interface with control software, cable, and accessories, real-time data logging software
	(xi) Calibration	Confirmation certificate for stability and uniformity
	(xii) Cooling arrangement	To be provided to achieve 40 °C (coolant to be provided by the supplier)
	(xiii) Anti-frost system	To avoid frosting at low temperature (optional as per requirement)

Temperature Sensor, Relative Humidity Sensor in the test section, and Support Apparatus in the test section (if any).

- Blockage Ratio: Less than 5%.
- Flow Uniformity: Uniform Velocity Profile within 1% variation inside the test section.
- Horizontal Wind Gradient: Less than 2% (Within Test Section).
- Turbulence Intensity: Less than 1% (Within Test Section).
- Air Density Uniformity: Better than 3% (Within Test Section).
- Reference Wind Speed Sensor: Pitot Tube (max uncertainty: 0.1 m/s; traceable to national/international reference standards)
- Parameters to be monitored inside the test section with the following sensors:
  - Temperature Sensor with max uncertainty ±0.5 °C
  - Relative Humidity Sensor with max uncertainty of ±2%
  - Pressure Sensor with max uncertainty ±1 mbar

  All these sensors must be traceable to some reference national/international standards.
- Temp Range Control: –30 °C to +30 °C (in steps of 5 °C). The system must have the capability to maintain the set temperature within a tolerance of ±0.5 °C for a period of 10 min. The max time taken to attain a temp of –40 °C from 30 °C should not exceed 4 h.
- Data Acquisition System Sample Frequency 10 Hz: The system should be able to acquire data from Reference Pitot Tube, Temp Sensor, Relative Humidity Sensor, Pressure Sensor, and anemometer under test inside the Test Section at min 10 Hz sampling frequency. Min resolution of acquiring wind speed should be 0.02 m/s, Temp 0.1 °C, RH 1%, and Pressure 0.5 mbar. The DAQ should have the capability to store the data for at least 72 h and generate a calibration summary for the sensor under test as per the standards mentioned above.

The design criteria should cater to excellent flow quality in space and time, ease of anemometer mounting, and good accessibility to the test section.

Sample Calculation for Approximate Frontal Area of Sensor Under Test and Blockage Ratio

As an example, two sensors have been considered for calculating the frontal area and blockage ratio:

1. RM Young Mechanical Wind Speed Sensor Model No. 05103, 05106, 05305, 09101, 09106, 05108-45. The technical details and specifications can be accessed from http://www.youngusa.com/products/7/. Max Size: 65 × 20 × 40 cm (L × W × H)
   a. Propeller Swept Area: πr^2 (r = 0.2/2) = 0.032 m^2
   b. Approximate Area of Sensor Body Perpendicular to the Flow = 0.02 m^2
      Total Frontal Area: 0.052 m^2
2. Standard Cup Counter Anemometer as per Indian Standards IS 5912:1997. The technical details and specifications can be accessed from https://archive.org/details/gov.in.is.5912.1997.
   Max Size: 39 × 39 × 30 cm (L × W × H)
   a. Cup Swept Area: 0.0488 m^2
   b. Approximate Area of Sensor Body Perpendicular to the Flow = 0.01776 m^2
      Total Frontal Area: 0.06656 m^2
      Format for Calculation of Blockage Ratio:
      Total Projected Area Perpendicular to the Direction of Wind Flow
      i. Frontal Area of Sensor Under Test (Standard Cup Counter Anemometer as per Indian Standards IS 5912:1997): 0.067 m^2
      ii. Frontal Area for Ref Sensor/Pitot Tube: an m^2
      iii. Frontal Area of Mounting for Sensor Under Test: b m^2
      iv. Frontal Area of Temperature Sensor: c m^2
      v. Frontal Area of Relative Humidity Sensor: d m^2
      vi. Frontal Area of Pressure Sensor: e m^2
      vii. Frontal Area of Support Apparatus (if any): f m^2
      Total Blockage Area (x) = (0.067 + a + b + c + d + e + f) m^2
      Height of Test Section: y m
      Width of Test Section: z m
      Area of Test Section (q) = y × z m^2
      Blockage Ratio = $\frac{x}{q}$ (Blockage Ration should not be more than 5%)

Automatic Calibration of Temperature Sensors

Orzylowski et al. (2000) in IEEE Instrumentation and Measurement technology conference in 2000 discussed the essential problems associated with the automatic calibration of industrial thermometric sensors in the temperature range of 200–1300 ° C carried out utilizing the comparison method.

The distance of the resistive sensors, the measurement junctions, and the reference computers are to be located in a uniform temperature field to perform a proper

calibration. For this purpose special furnaces are used in high-temperature ranges and liquid thermostats are used for the low-temperature region, which is up to 200 ° C.

During the calibration procedure, there may be several temperatures at which calibration needs to be done. For this purpose, a uniform and stable temperature failed, and possibly a short setting time is required, for which measurement inserts temperature control is used. Sometimes algorithm is used for accuracy requirements when the automatic calibration is being done at several temperature fields.

The paper discussed that for the calibration of standard thermocouples in the target temperature range, the requirements for the temperature field surrounding the sensor junctions can be defined as below.

i. There cannot be a difference of more than 10 K between the insert temperature mean value and the nominal calibration temperature.
ii. 0.2 K is the optimum value for the differences in temperature between calibrated thermocouple junction and the reference function. However, the value should be less than zero point when the temperature is 2000 ° C.
iii. 0.5 K/cm is the threshold for temperature gradient along a sensor, additionally at least 5 cm of distances are required.
iv. During the calibration measurement, 0.05 K should be the minimum temperature mean value stability.

During automatic calibration, the supervisory PC checks and monitors the stable temperature for several program levels according to the requirements. The accessory to the computer is a heater controller through a serially galvanically insulated interface and also a multi-channel temperature measuring system. The reference insert and the calibrated sensors produce data and the computer uses the data to calculate the signals for the temperature controller, and finally the record values of temperature are obtained in the measurement insert for each level of calibration (Fig. 13).

Digital data processing was used to reduce the measurement noise and to obtain the desired resolution. The supervising microprocessor performed the following functions:

i. Drives the auto-zero switches of the input amplifiers
ii. Controls the switching of gain in the input amplifiers
iii. Provides the timing signals for the A/D converter
iv. Compute the temperature concerning the characteristics of the sensors used
v. Controls the communication with a supervisory computer

The basic problems concerning the design of a system for the calibration of industrial thermometric sensors can be summarized as follows:

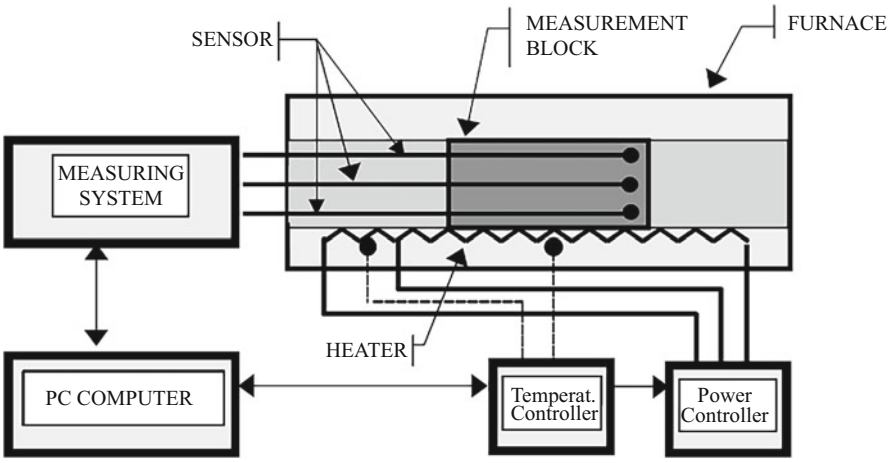

Fig. 13 The scheme of calibrating system

i. A tubular furnace with a multisection heater creates a uniform temperature field in which particular attention was paid to the phenomenon of radiation inside the furnace chamber.
ii. A measurement inserts temperature predictive control, where precise temperature control was obtained within the measurement insert and much-reduced setting time for the necessary conditions of the calibration measurements.
iii. Finding an algorithm for the automatic detection of the fulfillment of the accuracy requirement for the measurement.
iv. Analysis of the measurement error sources and the workout of an accurate multichannel measurement system with insulated channels for measurement of temperature sensor signals.

All the factors together helped in designing a perfect system that fulfills all the requirements of efficient calibration.

Temperature and Relative Humidity Calibration System

Ambient temperature and relative humidity mostly impact the calibration process, thereby both factors are needed to be maintained within a tolerable limit. Uncertainty is caused due to deviation from this limit, thereby disqualifying the complete calibration work. The ambient temperature and relative humidity are always reported on calibration certificates, establishing their importance.

Maintenance of both these factors requires not only an air handling system but also monitoring devices that ensure that the system is operating properly and parameters are within acceptable limits.

Rick Walker and Alan Cordner (2006) have detailed the procedure for temperature and relative humidity calibration system accordingly. Presently there are digital recorders available that are more accurate with remote access functions, which are hardier and can record data for a much longer period. They described a specific digital temperature/relative humidity recorder (Fluke Hart), which can store more than a year of data for two sensors and is accurate to ±0.125 ° C and ±1.5% relative humidity. The recorder of interest has a display unit and dual sensors, the sensors are used for both temperature and relative humidity sensing. The sensors are connected directly to the display unit, thereby simultaneous monitoring of two different locations can be done for both the data.

The sensor units contain analog to digital converters, and digital representation of the measurement is passed from the sensor units to the display unit. The sensor units need not always match the display unit, as the final accuracy of the sensor unit is the requirement, therefore, and only sensors are calibrated. Each sensor unit is calibrated independently as each has its unique characteristics. This uniqueness is stored in the nonvolatile memory of the sensor, thereby the digital parameters are also calibrated along with the sensor (Fig. 14).

A sensor always has non-volatile memory which produce date of the direction and the slope with reference to the judgment parameters; by setting both ways factors measurement errors of the sensors can be corrected. The opposite and slope adjustment parameters are also related to the relative humidity and temperature measurements. Calibration should include points at the extreme of the ranges and the point near the middle. All the relative humidity generally does not affect the temperature measurement but the temperature fluctuation does affect the measurement. Normally at the midpoint of the temperature range, the relative humidity is calibrated, a

Fig. 14 Fluke Hart scientific temperature/relative humidity recorder

specification of selective humidity then also allows for operation in other temperatures but within the specific temperature range.

Once the sensors are installed in the chamber it is expected that the calibration will be automatic. Normally the computer turns all the works while operating the humidity generator and collecting the sensor data, but that computer needs at least one temperature/relative humidity recorded display unit, with which the computer would also be connected. To preserve self-heating conditions, all sensors would have to be powered continuously.

Calibrating one or two temperature/relative humidity sensors at a time in an environmental chamber is common practice. The uncommon issue is the large number of sensors that needed to be calibrated at a time, and the low uncertainty required of the calibration.

Finally, the performance of the system is presented in terms of calibration and certainties, with contributing components as detailed during the procedures.

Icing Cloud Calibration of Icing Wind Tunnel

The icing cloud calibration process needs to be revised because the features of snow clouds are different from small supercooled droplets in terms of size, shape, and density. Then, relevant parameters in the acceptance criteria need to be updated. The basic procedure remains unchanged, including size distribution, water content, and cloud uniformity measurements. But the instrumentation and techniques to be used for the calibration of snow conditions change. As an alternative to tunnel calibration, the tunnel can operate with calibrated instrumentation that is used to measure the actual test conditions being obtained at the test point. This provides the actual conditions for each test point, including the effect of the test article. A wind check test is needed as the main acceptance criteria by testing a model to demonstrate the ability of the test facilities to reproduce snow accretion phenomena. This in turn allows an intercomparison of the test facilities, and assess the impact of any change in the test facility configuration.

Snow Conditions

Snowflakes, also called snow crystals, are aggregates of many single ice crystals. In nature, ice crystals often form in mixed clouds, where ice crystals grow via water vapor deposition at the expense of evaporating supercooled liquid water droplets, once the environment becomes saturated with water. This effect is called the Bergeron Findeisen effect, and it corresponds to the air transport of water vapor from the liquid to the ice phase, where the water vapor transforms directly into solid. Furukawa and Wettaufer (2007) had classified the snow crystal shapes into,

a. Plates and dendrites from zero to –3 °C.
b. Needles, columns, and prisms from –3 °C to –10 °C.
c. Solid, thin, and sectored plates and dendrites from –10 °C to –22 °C.
d. Solid plates and columns below –22 °C.

Finally, snowflake aggregation mostly appears at air temperatures near 0 °C. And this is predominantly affected by the air temperature and the shape of the aggregating ice crystals. Snowflake diameters are mainly between 2 and 5 mm and inversely proportional to the snowflake density, i.e., the larger the flakes, the lower the density (Table 4). This constant of proportionality between snowflake diameter and the density of snowflakes is almost four times larger in wet flakes than in dry flakes (Fig. 15).

Icing/Snow Cloud Parameter

- Total water content (TWC) or ice water content (IWC) are more relevant than liquid water content (LWC) and solid phase content (snow).

Table 4 Performance target and acceptance criteria

Parameters	**Measurement Instrument Maximum Uncertainty**	**Tunnel centerline Temporal stability**	**Spatial Uniformity**	**Limit value**
Aerodynamic parameters				
Airspeed	±1%	±2%	±2%	N/A
Static air temperature <–15 °C	±2 °C	±2 °C	±2 °C	N/A
Static air temperature between –15 °C and + 2 °C	± 0.5 °C	± 0.5 °C	± 1 °C	N/A
Flow angularity	±0.25	N/A	±2°	±3°
Flow turbulence				
Atomization pressure/snow generation system off	±0.25%	±2%	<2%	N/A
Atomization pressure/snow generation system on (maximum)	±0.25%	±2%	<2%	N/A
Pressure altitude	±50 m	±50 m	N/A	N/A
Cloud and cloud uniformity parameters				
Total water content/ice water content/ liquid water content	±15%	±20%	±20%	N/A
Liquid water ratio/wetness of the snow	±20%	±25%	±25%	N/A
Median mass diameter	±10%	±10%	N/A	N/A
Absolute humidity	±3%	N/A	N/A	N/A
Snow particle bulk density/snow bulk density	±15%	±20%	±20%	N/A
Parameters	**Measurement Instrument Maximum Uncertainty**	**Accuracy**		
Real atmospheric conditions representativeness and continuity check				
Snow/no snow accretion	N/A			
Growth rate (mm s^{-1})				

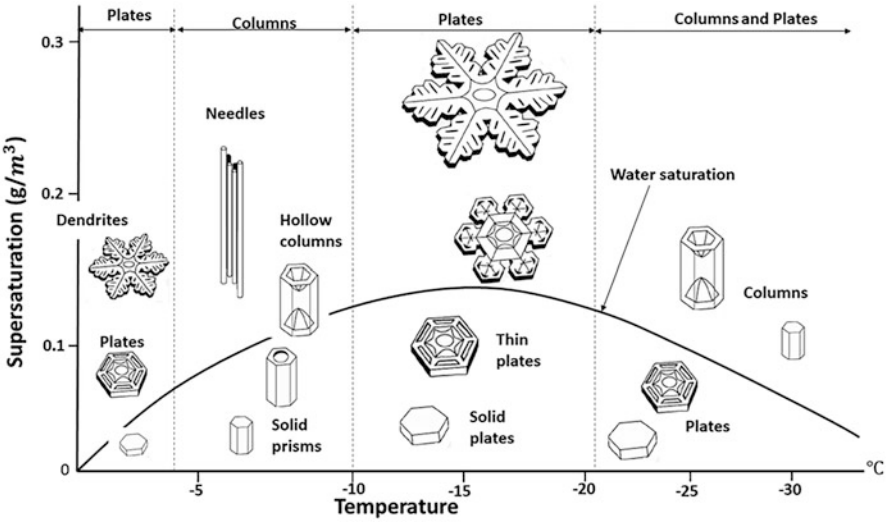

Fig. 15 Snow crystal morphology diagram (Tan et al., 2022)

- Median much diameter (MMD) is used instead of median volume diameter (MVD) for snow because of the variability of the density of the snowflakes.
- Absolute humidity is chosen instead of relative humidity to avoid ambiguity when liquid and solid phases are simultaneously present. Snow, particularly bold density, and snow density are added since they constitute a key parameter to assess the quality and the type of simulated snow and represent it with regard to real snowflakes.
- Liquid water issues, since it is expected to be one of the most important parameters of snow accretion.

The Main Measurement Objectives to Be Made to Ensure Calibration with the Selected Instrumental Payload Are as Follows

- Measurement reliability of large ice particle properties like size-dependent crystal number and mass. Up to 10 mm and beyond if possible.
- Ensure the measurement capabilities of bulk snow water content containing large snow crystals.
- Reasonably good morphological analysis and indicator for dry and wet snow of the snow particles. Along with the use of a high-resolution grayscale imager.
- Ensure cloud homogeneity in the test section.

Few additional adjustments for the most adequate instrument payload are as follows:

- The more the ice particle is at the upper size limit of a corresponding probe, the higher the probability of truncated images that need to be reconstructed.

However, with some uncertainties, in general, the retrieved reconstructed image underestimates the real size and mass when image truncation occurs.

- There is a potential that snow crystal fragmentation in a wind tunnel may produce non-negligible submillimeter crystal mass. That needs to be documented. Thereby 2 T stereo probe is recommended in such cases.
- High-resolution imaging probes are required for the entire snow particle size range and disposing of extremely large sample volumes for good statistics of snow microphysical properties and concentration. High-resolution imaging probes can improve the knowledge of cloud particle morphological information with the potential to create a proxy or an indicator of the presence of liquid water. This new particle thus distinguishes wet and dry snow.
- Bulk ID measurement devices for implementation of reliable snow water content measurement in test facilities.
- Liquid cloud droplets are injected into the test section in addition to the snow-type particles, and there is a clear need for phase discrimination of particles (solid/liquid). In such cases, data from high-resolution imaging probes or cloud droplets spectrometers may be used.
- The possible small ice crystals, contamination, and loss of last snow particles due to ice particle shattering should be avoided. Thereby, artifact minimization is needed during snow particle sampling.

Humidity

Humidity is an important parameter for icing that can be characterized using different values like relative humidity, absolute humidity, or dew point temperature. It is not possible to measure correctly the humidity in the working section of the facilities in icing conditions because the measurement of the sensors is biased by the presence of super cool droplets or/and snow. Therefore, humidity measurement is generally done upstream. The cloud generation system in the still chamber (Table 5).

Particle Size Distribution (PSD)

Snowflake size varies between 2 and 5 mm, ranging up to 15 mm. PSD measurement requires at least the use of two probes to cover the whole size range. The test facility should provide a detailed description of processing methods to account for shattering or out-of-focus verticals. If the instrument is also used for evaluating water content, and mass size relationship, it should be substantiated. In such a case it has to be monitored. Such instrumentation shouldn't be used as a sole basis for water content calibration (Table 6).

Particle Morphology

In addition to imaging probes, spanning the entire snow particle size range and disposing of extremely large sampling volumes for good statistics of snow microphysical properties. High-resolution imaging probes have been effective in considerably improving the knowledge of cloud vertical morphological information. These have the potential to create a proxy or an indicator of the presence of liquid water in the snow particle, thus distinguishing at least dry and wet snow. However, with

Table 5 Acceptable instrumentation for humidity measurement

Instrumentation	Principle
Chilled mirror hygrometer	This device measures the dew-point temperature by controlling the temperature of the mirror to condensate the water vapor of the sampling flow. This technology can measure very low dew-point temperature (–70 °C) but response time and stability could be affected by the sampling flow rate. Sometimes, an error on dew-point temperature is possible due to an uncertainty on the state of water, supercooled water, or ice, over the mirror for negative temperature
Optical sensors	Water vapor has an interesting absorption spectrum with some peaks in the IR and UV wavelengths. Based on this property, some devices measure the extinction of a laser beam crossing a sample of air and calculate the vapor concentration from Beer-Lambert's law. The absorption coefficient of the medium depends also on its pressure and temperature and must be corrected to ensure good accuracy
Capacity hygrometer	This device measures relative humidity and the local temperature. The principle of measurement is based on the change of the permittivity of material with relative humidity. The local temperature is used to calculate the local vapor pressure. This technology cannot operate accurately in very dry or wet conditions

extremely limited sample volume that does not allow quantifying particle statistics with high frequency (Table 7).

The Snow Cloud Uniformity Assessment

The cloud coverage area and concentration uniformity in the test section is another important wind tunnel measurement to support the calibration activity. For liquid clouds, the icing calibration grid is the recommended method, which is based on the rime ice accreted. For the snow clouds, cloud uniformity is measured through the measurement of total water content at a matrix of points across the test section.

Modes of Calibration

- Baseline calibration: It is the full calibration of the facility and includes an error, thermal calibration, and icing in snow cloud calibration. Full calibrations are required on initial commissioning or following any major facility modification that changes the icing/snow cloud.
- Interim calibration: It should be performed on an annual basis for the first 2 years following the baseline calibration, resulting from initial commissioning or a major change to the facility. These should include either confirming that icing/snow cloud uniformity has not changed from that established during the baseline calibration, or confirmation of tunnel axial centerline LWC/IWC measurements. The interim calibration should also include confirmation of the MVD/MMD calibration using a representative sampling from the baseline calibration.
- Check calibration: It should be performed every 6 months, except when it is superseded by a baseline or interim calibration. This consists of measurement of icing/snow cloud uniformity and centerline LWC/IWC measurements, over a representative sampling of uniformity.

Table 6 Instrumentation for PSD measurement

Instrumentation	Principle
Two-dimensional stereo probe (2D-S)	Optical array probe developed by SPEC. A laser beam deflects the cloud particle entering the field of vision which in turn fails in collecting optical and counted at a rate such that the pixel resolution in time is the same as that in the across-array dimension. This forms a digitized shadow image of the particle
Cloud imaging probe (CIP)	Cloud Imaging Probe developed by DMT. A collimated Ledger beam reflects the shadow images of the particles passing through, which had been projected on a 64 photodetectors linear array. A change in the light level of the diode represents each particle. The changes in the photodetectors should be compatible with the pro velocity and the instrument size resolution. Particle images are reconstructed from individual "slices," where a slice is the state of the 64-element linear array at a given moment in time A slice must be stored at each time interval that the particle advances through the beam at a distance equal to the resolution of the probe. Optional grayscale imaging gives three levels of shadow recording on each photodetector, allowing more detailed information on the particles The CIP also contains a Hotwire LWC sensor. This sensor estimates liquid water content using a heated sensing coil. The system maintains the coil at a constant temperature, usually 125 °C, and measures the power necessary to maintain this temperature. More power is needed to maintain the temperature as droplets evaporate on the coil surface and cool the surface and surrounding air. Hence, this power reading can be used to estimate LWC The CIP measures cloud particles in the 25–1600 μm range with a resolution of 25 μm (64 photodiodes)
Precipitation imaging probe (PIP)	Optical array probe developed by DMT. Cloud particles enter the field of view passing through a laser beam. Particle shadow falls on a collecting optical array (normally a photodiode array). This is sampled at a rate such that the pixel resolution in time is the same as that in the across-array dimension. This forms a digitized shadow image of the particle The Precipitation Imaging Probe (PIP) is a state-of-the-art probe that measures particles in the 100–6200 μm range. It is an ideal choice for measuring rain, snow, graupel, and hail
High volume precipitation spectrometer (HVPS)	The HVPS-3 (Version 3), developed by SPEC, combines 2D-S optoelectronics with HVPS optics and probe tips designed to minimize shattering Probe images precipitation sized particles using the 2D-S 128-photodiode array with 150-micron pixel resolution The HVPS probe measures particles in the 150–19,200 μm range. It is an ideal choice for measuring rain, snow, graupel, and hail

Table 7 Acceptable instrumentation for ice particle morphology measurement

Instrumentation	Principle
High speed imager (HSI)	Developed by ARTIUM, the HSI records high-resolution (5 μm) digital images of cloud particles as they pass through the measurement volume of the instrument. It collects the images of the particles on a 2000 by 900 CMOS array with sufficient resolution for obtaining the drop size and shapes over the size range 3–4000 μm, with the images having a resolution of 3 μm. The method of data processing leads to the particle size distribution and calculates the usual microphysical parameters: ice particle concentration, mean particle size, and ice water content. The classification of the shape of the ice particles can also be derived
Cloud particle imager (CPI)	The CPI is a major improvement on existing imaging/shadowing probes such as the PMS classical OAPS. The CPI can image and count particles in the size range of 2.3–2300 μm, with the images having a nominal 2.3 μm resolution. Image analysis and data processing software provided with the probe takes particle size information (including area and volume) and ice crystal habit classification from the images to produce histograms, etc.
Multi-angle snow camera (MASC)	The Multi-Angle Snowflake Camera, or MASC, was developed to address the need for high-resolution multi-angle imaging of hydrometeors in free fall, while simultaneously measuring their fall speed The MASC consists of three cameras, separated by 36 and each pointing at an identical focal point approximately 10 cm away. The focal point itself lies within a ring through which hydrometeors fall. The ring houses a system of near-infrared emitter-detector pairs, arranged in two arrays that are separated vertically by 32 mm. Hydrometeors passing through the lower array simultaneously trigger each of the three cameras as well as a bank of lights aimed at the center of the camera depth of field. Fall speed is calculated from the time it takes

- Continuity check: A model should be tested during or before and after the above calibration test to assess changes or stability of the tunnel's ice/snow accretion characteristics.

Aerothermal Calibration of Icing Wind Tunnel

A dryer air thermal calibration should be conducted to determine the basic airflow qualities of the facility. The flow properties should be as below:

- Airspeed distribution
- Temperature distribution
- Turbulence intensity distribution
- Centerline airspeed correction
- Clean tunnel flow angularity distribution

- Humidity time history
- Altitude time history

References cum Literature Review

Avalanche Hazard forecasting through sensor-acquired data is a regular activity being carried out by all mountainous countries. For example, USGS Scientists from the northern Rocky Mountain Science Centre had been studying snowpack stability and shifting in snowmelt, and runoff for the last 26 years. They particularly focus on wet snow avalanches. Which occurred in the spring when snowmelt weakens the strength of the snowpack? This action may prove crucial to future generations, as wet snow and avalanches are likely to become more common with the prevailing climate change scenario (Clark et. al, 2017).

Eric Trenbeath, an avalanche forecaster for the USFS Utah, had spoken about reading patterns in the snow and how that fits in the Avalanche forecast. In his blog, The Art of Avalanche Forecasting, special emphasis was given to the accuracy of sensors.

Similarly, the importance of calibration for multi-sensor imaging has been described in detail by K. Thome in 2017, at the International Geoscience and Remote Sensing Symposium.

Lachapelle (1980) made the first study on the fundamental process of conventional avalanche forecasting. The forecasting was said to be cumulative integration of a widely diverse body of information, gathered over a specified period. Information about snow conditions is highly diverse. The data readily fall into three categories, viz. meteorology, snow cover structure, and mechanical state of snow on a slope. Every available item, no matter how trivial, uncertain, or seemingly irrelevant, enters the integration.

The Shoda diagram illustrating the chain of causation in an avalanche formation is shown in Fig. 16 (De Quervain et. al, 1973).

The path of data flow in avalanche forecasting is shown in Fig. 17 (De Quervain et. al, 1973).

The forecasting data sequence, meteorologist snow structure mechanics, follows the chain of causation from distant approximate, and also follows a sequence of decreasing number of event classes and especially are decreasing population of elementary events in each class. The supreme importance of removing the uncertainty by identifying the errors in data thereby becomes crucial. And the sensors providing the data to the Algorithm need to be calibrated for perfection.

At World Congress on natural disaster mitigation, a new approach to avalanche forecasting for the western Himalayas was presented (Ganju and Singh 2004). The paper discussed a fresh approach by integrating state-of-the-art technology with conventional tools, thereby paving a way for the future development of sensor-dependent forecasting. It established the fact that communication of field data through modern telemetry systems and remote sensing satellites can overcome the shortcomings of the non-availability of timely data. Thereby, leading to frequent

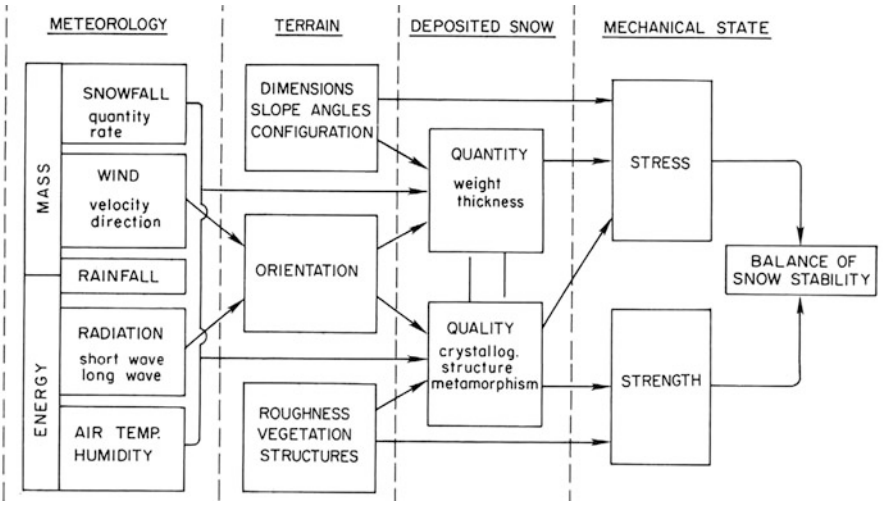

Fig. 16 The Shoda diagram

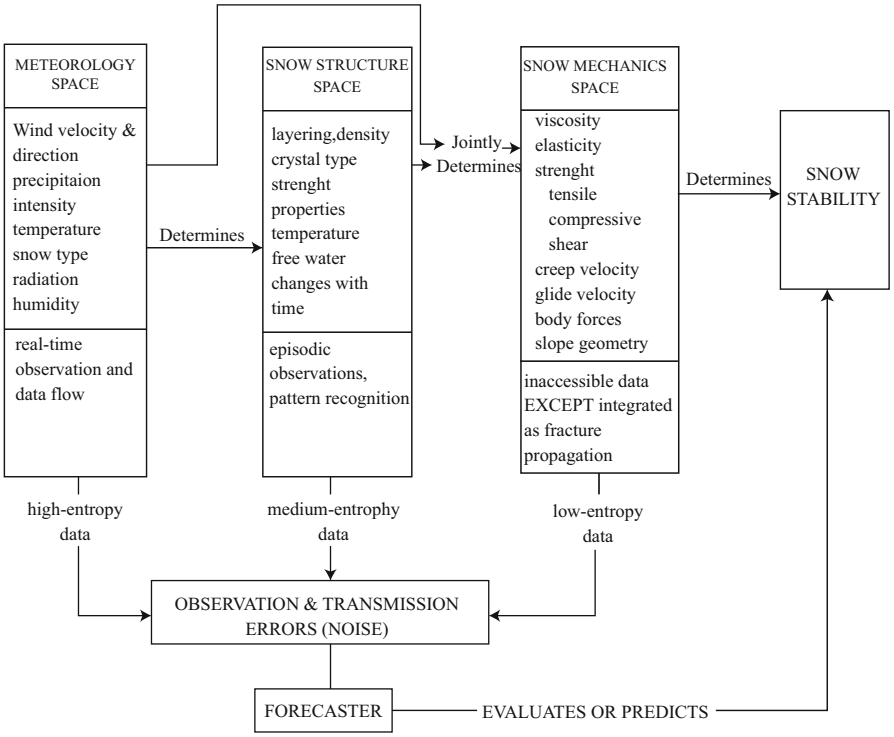

Fig. 17 Avalanche forecasting data flow

disasters. Presumably accurate functioning of the sensors was of paramount importance for the smooth operation of all the AWS, and appropriate functioning of the sensors can only be ensured by accurate calibration.

Studies made by Vilajosana et al. do not show the importance of the sensor location relative to the avalanches. However, to analyze the signals in great detail, sophisticated equipment in a single avalanche path is required. This also signifies the crucial need for proper calibration. Depending on the size of the event, avalanches that occur up to several kilometers away from the sensors can be detected (Leprettre et al. 1998). However, automatic avalanche detection is based on even detection using an amplitude threshold and event classification; single interpretation is relatively limited, and due to the amplitude threshold only larger avalanches are registered in the sensor. Thereby, in the recent past, automatic avalanche monitoring systems were not much widely used, as identification of avalanches remains problematic (Navarre et al. 2009).

In situ calibration of meteorological sensors has been discussed in detail by Merlone et al. It was elaborated that checking the measurement by comparison against reference high-quality instruments is a non-metrological approach. And this can only be overcome by the introduction of systemic periodic calibration that guarantees expressed traceability. The prototype of the Portable Climatic chamber coded EDIEO (Earth Dynamic Investigation Experiment O) was designed in 2010, aiming to promote this modus operandi by allowing in situ calibration of temperature, pressure, and humidity sensor, while evaluating their mutual influences. The relevance of the work lies in providing traceability and metrological quality to mountain atmospheric data, and a complete analysis of any uncertainty source in the calibration process was reported along with the total uncertainty results satisfactory for the application.

In another study by Grykałowska et al. (2015), the basic principles for calibration of meteorological temperature sensors were discussed in detail. The availability of reliable data defines the success of any meteorological and climate observation. The study also detailed the establishment of active cooperation between the World Meteorological Organisation (WMO) and the International Bureau of Weights and Measures (BIPM), to combine the requirement of monitoring climatic conditions, along with the responsible assurance of precise measurements, in short paving the way for accurate calibration. In the study, the calibration of temperature sensors was divided into the following phases.

- Characterization of the parameters of the measurement system used in the experiment (The measurements were carried out in liquid air equipment and a climatic chamber).
- Calibration was carried out using the equipment.
- Study of humidity impact on the readings.
- Calibration in the climatic chamber.
- Comparative in-situ measurements.

It may be noted that the importance of calibration in sensors especially related to snow and avalanche forecasting is not a young field of research. Almost four decades ago Marriott and Moore has explored the errors in measurement and their interpretation, thereby highlighting the requirement of calibration, in one of the first reported studies. They have shown that measurement of weather and snow parameter for forecasting may frequently contain false or misleading information, such errors can be either attributed to poor site selection or inconsistent response of the sensor due to changing weather conditions. It's a combination of sensors and direct human observation to provide a correct forecast. However, proper functioning of the sensors to provide useful information optimal for stability analysis and thereby leading to avalanche forecasting is the most crucial factor. Erroneous data of even a Milli scale of measurement may lead to false predictions associated with a disastrous consequence, thereby once more necessitating the requirement of proper calibration. Calibration is an important process to ensure that the data products meet the requirement of scientific utilization (Freeman 1992). Calibration and data quality evaluation are important aspects of performance of a sensor; the consistency and quality of any sensor data depends on the calibration and quality and worked as parameters to evaluate and monitor the sensor for a long-term regular output.

The calibration methodology for snow test facility has been worked out with funding by European Union. The work has proposed a calibration methodology for snow and mixed face condition. Not even parameters for calibration with their acceptance criteria for measurement accuracy, temporal stability, spatial uniformity, and target values have been defined. The possible instrumentation for particle size distribution and water content measurements are also described.

The National Institute of Standards and Technology (NIST) has also described procedures for calibrating several different types of industrial thermometers. One of the main types they described is a liquid in glass thermometer, which includes both mercury and organic field models. These thermometers covered a temperature range from 20 ° C to 400 ° C (Vaughn and Strouse 2001)

Conclusion

The accuracy and reliability of snow avalanche forecasting heavily depend on the quality of snow and meteorological data received from various sensors. Therefore, for ensuring a good quality of data from these sensors/instruments, periodical calibration of these sensors is foremost important. A summary of various sensors and corresponding calibration systems is given in Table 8 below.

On the Khumbu valley on the Nepal side of Mount Everest, a dedicated facility for calibration of AWS has been established; this is called the Everest pyramid laboratory cum observatory. Such an establishment itself makes it very clear that calibration is having paramount importance in the field of forecasting. This facility represents a fruitful collaboration between environmental science and metrological activities also; this is the highest metrology laboratory in the world. In this time with the climate change issues, climate science needs an intricate collaboration with

Table 8 Summary of various sensors, and corresponding calibration systems

Sr. no.	Sensor name	Range	Accuracy	Supply voltage	Calibration system
1.	Ambient temperature and relative humidity sensor	40 °C to +50 °C	≤±0.2 °C	5 or 12 V DC (typical)	Temperature calibration system, and humidity calibration system
		0–100% RH	≤±2%		
2.	Barometric pressure sensor	500–1100 mbar	≤±0.5 mbar	5 or 12 V DC (typical)	Pressure calibration system
3.	Wind speed and direction sensor	0–60 m/s	≤±1% of reading	5 V or 12 V DC (typical)	Temperature controlled wind tunnel
		0–360° (mechanical)	≤±3°		
		0–355° (electrical)			
4.	Albedometer	0.3–2.8 μm (shortwave)	<2% (nonlinearity)	5 or 12 V DC (typical)	Radiation sensor calibration system
5.	Pyrgeometer	4.5–42 μm (longwave)	<2% (nonlinearity)	5 or 12 V DC (typical)	
6.	Snow precipitation gauge	≥1500 mm	≤±1 mm	5 or 12 V DC (typical)	Snow precipitation gauge calibration system
7.	Standard field view infra-red radiometer	50 °C to +55 °C	≤±0.5 °C	5 or 12 V DC (typical)	Infra-red temperature calibration system
8.	Ultrasonic snow depth sensor	0.5–6 m	≤±0.5% of the distance to the target	5 or 12 V DC (typical)	Snow depth sensor calibration system

metrological activities, which can improve data quality and forecasting activities, for robust incomparable measurement. The same has been achieved in 2010 when World Meteorological Organization (WMO) signed the Mutual Recognition Arrangement (MRA) with International Committee for Weights and Measures (CIPM). The field of climate change also depends on the possibility of measurements, and there is a regular need to test and evaluate the behavior of a calibration system to assess the proper activity and maintenance of sensors and even reporting failures which are otherwise difficult to identify. Evaluation of proper calibration and the associated uncertainty always needs proper calibration. There is an actual objective and need of any project to deliver long-lasting quality data on climate and global warming; all these activities may help to formulate policy based on reliable assessments and deliver an applicable and feasible mitigation strategy (Merlone et al. 2015).

Tell oppression being mostly manual tasks there is always an inherent relationship with the field of psychology and neurophysiology as the human mental attribute dealing with uncertainty needs to be countered.

Garner (1974) demonstrated the role of redundancy in surmounting basic limitations, in perceiving and assimilating information. Normally numerous measurement data are being collected as there are always natural variations or observational errors. Example of forecasting needs some multiple measurements with multiple variables, and the soul importance of calibration can also be highlighted once more.

Cross-References

► The Quantifying of Uncertainty in Measurement

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Pressure and Its Measurement

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An Introduction

Vinay Kumar

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Abstract

Measurement is the most fundamental idea in science. Measurement of pressure is the next-most measured process parameter after temperature. It is required almost in all process and manufacture industry. A large number of different types of pressure-measuring devices are on hand to fulfil the requirement of industry. In this chapter a glimpse on pressure, pressure units and its conversion, various types of pressure, commonly used pressure-measuring instruments, and pressure sensor's metrological characteristics is described. The basic definitions related to metrology are described. The measuring properties of different pressure gauges are also tabulated. The approved pressure calibration and measuring capabilities (CMCs) of different countries' national metrology institutes (NMIs) are compiled in this chapter. The NMIs have CMCs of different ranges starting from 0 pascal to 1400 MPa.

Keywords

Pressure · Pressure instruments · Pressure standards · Calibration

Introduction

Measurement is maybe one of the most fundamental ideas in science. The word "measurement" is taken from the Greek word "metron," which means "limited proportion." Measurement is a procedure in which properties of an object are determined by comparing them to a standard. Measurement has a great significance not only now but from the start of human civilization. A speedy advancement of new instruments and measurement techniques started in the nineteenth century during the Industrial Revolution to fulfill the requirements of commercial production. Since then, there has been a significant and fast development in measurement technique and instruments. This has been especially visible during the last decades of the twentieth century.

The science of pressure measurement started in the middle of the seventeenth century. The credit of measuring atmospheric pressure goes to Evangelista Torricelli. His discovery was immediately started to be used in a lot of scientific research. After World War II, there was rapid development of pressure measurement devices, vacuum to high pressures. The accuracy of these instruments has major impact on the quality of product manufactured by industries using these instruments.

The accurate measurement of pressure plays an important role in manufacturing and processing industries such as automobiles, aerospace, steel, synthesis of high-strength materials, research, etc. In the industry accurate pressure measurement is important not only due to quality considerations but also to safety requirements. No single instrument can measure the pressure in the entire range of interest with the same accuracy and reproducibility. The instruments may also not be sensitive enough to small changes in pressure and insensitive over a working temperature range. The pressure, velocity, and temperature are the most important properties of fluids. Pressure and temperatures play a major role in the fields of "pipeline constructions," "water supply," "hydraulic lifts," "hydraulic presses," "hydraulic diggers," "hydraulic for irrigation," "thermal power," "hydraulic power," "aviation of flights," human blood, etc. The pressurized fluids can be used to perform work as they exert forces. The pneumatic and hydraulic systems use fluids for their working. These systems are used in most of the industries such as "automobile manufacturing" (brakes), "hoists" found in service stations, "dentist's chair," "aeronautics," "submarines," the "shipping industry," etc. A small pressure applied on a very large area can make a very large force. When any fluid flows over or through an object, it delivers small force pushes on the surface of the object. These forces, over the entire surface, are defined as pressure and can be calculated as force per unit area. The study of physical quantities like pressures, velocity, and temperature is very important to understand the fluid flow. In this chapter, focus will be on describing pressure, its unit, its type, and various pressure-measuring instruments. The basic definitions related to metrology and metrological characteristics of a pressure sensor are also described. Lastly CMCs of various country NMIs including uncertainty and type are also tabulated.

Pressure

In mechanical concept, the pressure on any object can be generated by applying normal force on the surface of the object. The physical quantity pressure is defined as the force applied perpendicular to the surface per unit area. It is a scalar physical quantity. The fundamental mathematical expression for pressure is given in Eq. 1:

$$\text{Pressure (p)} = \frac{\text{Force (F)}}{\text{Area (A)}} = \frac{\text{m g}}{\text{A}} \tag{1}$$

where m is applied mass to generate pressure force and g is acceleration due to gravity.

Any physical state of matter that possesses weight, whether at rest or in motion, exerts a pressure upon the surface in contact.

Units of Pressure

Many units are in use to measure the pressure (https://blog.beamex.com/pressure-units-and-pressure-unit-conversion) depending on the perspective of use. It will not make any sense if we say maintain air pressure of my car's tire at 206843 Pa, rather 30 psi.

SI Unit of Pressure

The unit of pressure in International System of Units (SI system) is pascal (Pa); 1 pascal is the pressure exerted by 1 newton force on area of 1 m^2. It is same as newton per square meter (N/m^2). Pascal is a small unit of measurement, for example, the standard atmospheric pressure is 1.013×10^5 Pa.

Imperial Units

In some countries like the United States and United Kingdom, the mass is being measured commonly in pounds or ounces and area and distance with inches or feet. So, some other pressure units derived from these are pounds per square feet (lbf/ft^2) and pounds per square inch (psi). In the United States, the most common pressure unit is psi. Nearly in all country, we measure the tire pressure in psi.

Liquid Column Units

The older pressure-measuring devices often use liquid in a transparent tube or U-shaped tube. Therefore, pressure is also expressed in terms of height of liquid column in the tube or difference of height of liquid in the two arms of a U-shaped tube. This unit of pressure is still in use. We measure blood pressure in millimeter of mercury (mmHg). We get a long list of pressure units in terms of height, just to mention a few: "centimeter," "inch," "millimeter" (torr) and "micrometer" (mtorr) of mercury, "millimeter," "centimeter," "meter," "inch," and "foot" of water.

Atmospheric Units

The "standard atmosphere" (atm) unit is used for measurement of the atmosphere's absolute pressure. It is approximately equal to typical air pressure at Earth mean sea level and is defined as 101325 Pa. There is another unit to measure the atmosphere's absolute pressure which is a "technical atmosphere" (at) which is close to "atm." The "at" is equal to pressure generated by 1 kilogram force at 1 square centimeter. The torr is another unit to express the atmospheric pressure. It is approximately 1/760 of atm. Torr is not included in the SI system.

Non-SI Unit of Pressure

In the "centimeter-gram-second" (cgs) system, the unit of pressure is "barye" (Ba), which equals to pressure exerted by 1 dyne force on area of 1 square centimeter. The "bar" is a metric unit of pressure, equal to pressure exerted by 1000000 dyne force on area of 1 square centimeter, but not part of the SI system. Units derived from the bar

include the "megabar" (symbol, Mbar), "kilobar" (symbol, kbar), "decibar" (symbol, dbar), "centibar" (symbol, cbar), and "millibar" (symbol, mbar).

A few other units are kilogram force per square meter (symbol, kgf/m^2), gram force per square meter (symbol, gf/m^2), and kilogram force per square centimeter (symbol, kgf/cm^2).

Pressure Unit Conversion Standards

The pressure is measured in various units; some of them have been discussed above. Sometimes there is need to convert pressure expressed in one unit to another pressure unit. Many conversion tables, even online conversion calculators, are available. The one such conversion factor (Rab et al. 2022) is shown in Table 1.

Types of Pressure

In this section of the chapter, various types (Gaßmann 2010) of pressure will be discussed.

Absolute Pressure

The pressure whose reference point of the measurement is zero pressure or an absolute or perfect vacuum is "absolute (abs) pressure." It is denoted by "P_{abs}." The pressure used in perfect gas law is an example of absolute pressure.

Gauge Pressure

The pressure whose reference point of the measurement is ambient pressure is "gauge pressure." Mostly pressure instruments measure gauge pressure since these instruments indicate the pressure relative to atmospheric pressure.

Differential Pressure

Differential pressure reads the difference in pressure of two different points of measurement.

Note that the same instrument can be used for all three types of absolute, gauge, and differential pressure; only the reference is different. Differential pressures are commonly used in industrial process systems. The relation between absolute, gauge, and vacuum pressures is shown in Fig. 1.

Table 1 Conversion factors for pressure units, table taken from reference (Rab et al. 2022)

Unit To ↓ From	psi	mbar	bar	Pa	kPa	MPa	mmH_2O	$in.H_2O$	mmHg (Torr)	in.Hg
psi	1	68.95	0.0689	6895	6.895	0.006895	703.8	27.71	51.715	2.036
mbar	0.0145	1	0.001	100	0.100	0.0001	10.21	0.402	0.75	0.0295
bar	14.504	1000	1	100000	100	0.1	10210	401.9	750.1	29.53
Pa	0.000145	0.01	0.00001	1	0.001	0.000001	0.102	0.00402	0.0075	0.000295
kPa	0.14504	10.0	0.01	1000	1	0.001	102.07	4.019	7.5	0.295
MPa	145.04	10000	10	1000000	1000	1	101971.6	4014.6	7500.6	295.3
mmH_2O	0.001421	0.098	0.000098	9.8	0.0098	0.0000098	1	0.0394	0.0735	0.00289
$in.H_2O$	0.0361	2.488	0.002488	248.8	0.2488	0.00025	25.4	1	1.866	0.0735
mmHg	0.01934	1.33	0.001333	133.3	0.1333	0.00013	13.61	0.536	1	0.0394
in.Hg	0.4912	33.86	0.03386	3386	3.386	0.00386	345.7	13.61	25.4	1

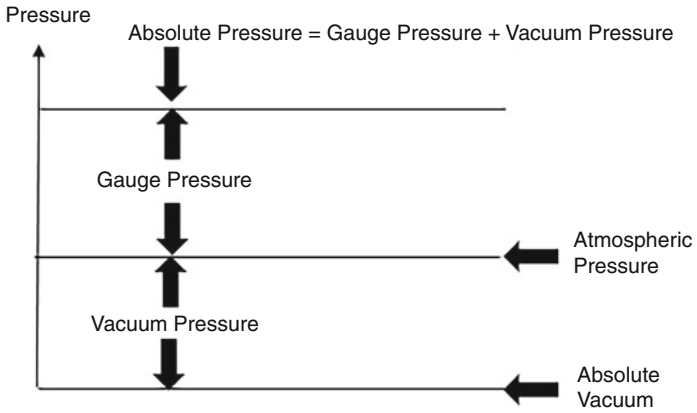

Fig. 1 Relation between absolute, gauge, and vacuum pressures

Hydrostatic Pressure

The pressure exerted by a fluid at any given point is called the "hydrostatic pressure." Pressure exerted by nonmoving (static) fluid on the body of a container is called "static," when the fluid is moving, it is called "dynamic." The principles of fluid dynamics can be used to describe the pressure in closed conditions. Bernoulli's equation given in Eq. 2 can be used to determine the pressure at any point in a fluid:

$$p + \frac{1}{2}\rho v^2 + \rho gh = \text{constant} \tag{2}$$

where p is "pressure"; ρ, "density"; v, "velocity"; g, "acceleration due to gravity"; and h, "elevation."

The applications of hydrostatic pressure are in "hydraulic brakes," "artesian well," "blood pressure," "hydraulic head," "plant cell turgidity," "Pythagorean cup," etc.

Static Pressure

The pressure uniform in all directions is called "static" pressure. If pressures exerted by a fluid at different direction are independent of direction, it will be a static pressure.

Dynamic Pressure

When fluid flows, it applies extra pressure on surfaces which are perpendicular to the fluid flow direction and not much effect on surfaces which are parallel to the flow

direction. This additional pressure in a moving fluid is called "dynamic pressure." Dynamic pressure is referenced to static pressure; therefore, it is a differential pressure.

Stagnation Pressure

The sum of the static and dynamic pressures is called the total pressure or stagnation pressure.

Static pressure and stagnation pressure are related to each other as shown in Eq. 3:

$$p_o = \frac{1}{2}\rho v^2 + p \tag{3}$$

where p_o is "stagnation pressure"; ρ, "density"; v, "flow velocity"; and p, "static pressure."

Explosion Pressure

Pressure generated during ignition of explosive gases, mists, and dust/air suspensions, in free or enclosed spaces, is known as "explosion" or "deflagration pressure."

Negative Pressure

Negative pressure is the situation whereby the pressure of one place is lower than in comparison to another.

Surface Pressure

The pressure in two-dimensional area is known as surface pressure. It is calculated by the lateral perpendicular force applied on a unit length of a line. Surface pressure is denoted by π and can be calculated using Eq. 4:

$$\pi = \frac{\mathrm{F}}{\mathrm{l}} \tag{4}$$

where F is lateral perpendicular force and l, length of line.

Pressure of an Ideal Gas

The pressure exerted by the ideal gas molecules confined within a rigid container is determined using "ideal gas law" given in Eq. 5:

$$\text{Pressure (p)} = \frac{nRT}{V} \tag{5}$$

where n is the "number of gas molecules"; R, "ideal gas constant" (8.314 J mol^{-1} K^{-1}); T, "temperature" of the gas; and V, volume of the container.

Vapor Pressure

When vapor is in a "thermodynamic equilibrium" with its condensed phase (liquid or solid) in a closed system, the pressure exerted by the vapor above the liquid or solid surface is known as "vapor pressure" or "equilibrium vapor pressure." The evaporation rate of a liquid depends upon equilibrium vapor pressure. As temperature of liquid increases, vapor pressure also increases. The vapor pressure is nonlinearly proportional to temperature according to "Clausius-Clapeyron" relation. The total pressure in the system is called "partial vapor" pressure if only vapor pressure of a mixture contributes to total pressure.

Fluid Pressure

The fluid exerts pressure. The pressure exerted by a fluid depends on its depth and density. The pressure of a static fluid can be easily calculated if the density of the fluid is known. P is the pressure on top of fluid as shown in Fig. 2. The absolute pressure at a depth H in a fluid is calculated as given by Eq. 6:

$$P_{abs} = P + (\rho \times g \times H) \tag{6}$$

where ρ is the density of the fluid and g acceleration due to gravity at that place.

Pressure Ranges

Some arbitrary defined pressure ranges (https://www.nist.gov/document/pmc-2pdf) are given in Table 2.

Pressure-Measuring Instruments

The pressure-measuring instruments are called pressure gauges (used to measure pressure above atmospheric pressure) or vacuum gauges (used to measure pressure below atmospheric pressure). Pressure gauges are used in automobiles, aeronautics, oil production, food industry, chemical industry, hydraulics, etc. Many instruments (Morris and Langari 2016; Soloukhin et al. 1991; Bourdon Tube Pressure And Vacuum Gauges 1971; https://www.budenberg.co.uk/store/gauges/pressure;

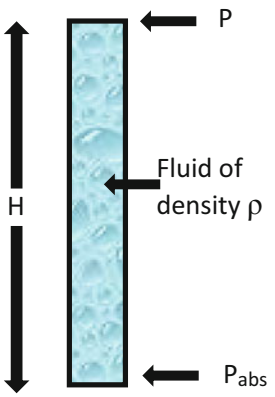

Fig. 2 Pressures exerted by liquid column

Table 2 Pressure ranges, table taken from reference (https://www.nist.gov/document/pmc-2pdf)

Range	Pressure range (Pa)	Pressure (conventional unit)
"Very high pressure"	10^{8}–10^{11}	1–1000 kbar
"High pressure"	10^{5}–10^{8}	15–15000 psi
"Atmospheric pressure"	10^{2}–10^{5}	1 torr–1 atm
"Low vacuum"	10^{-1}–10^{2}	10^{-3}–1 torr
"High vacuum"	10^{-4}–10^{-1}	10^{-6}–10^{-3} torr
"Very high vacuum"	10^{-7}–10^{-4}	10^{-9}–10^{-6} torr
"Ultrahigh vacuum"	10^{-10}–10^{-7}	10^{-12}–10^{-9} torr

Jones n.d.; Brombacher 1970; Cross 1963) have been invented to measure the pressure from 1×10^{-9} Pa to 1×10^{9} Pa till now, having different advantages and disadvantages. The cost of a pressure-measuring instrument depends upon pressure range, sensitivity, and dynamic response of that instrument. The oldest type is the "liquid column manometer" invented by "Evangelista Torricelli" in 1643. The "U-tube manometer" was invented by "Christian Huygens" in 1661. The different types of measuring instruments are shown in Fig. 3.

A "vented gauge pressure transmitter" reads pressure with respect to ambient atmosphere pressure. The main advantage of this gauge is that it reads zero when pressure is not applied, and its pressure measurement is not affected due to change in atmospheric temperature and pressure. At one end of the pressure sensing diaphragm, pressure to be measured is applied, and the other end is exposed to ambient atmospheric pressure via a vented cable or porous surface or a hole on the other side of the device.

A "sealed gauge reference" is very similar to vented gauge except that atmospheric pressure is sealed at another end of the diaphragm. The "sealed gauge" never reads zero as atmospheric pressure changes continuously. The sealed gauge is generally used to measure high pressure. At high pressure, accuracy of the reading is not affected due to change in atmospheric pressure.

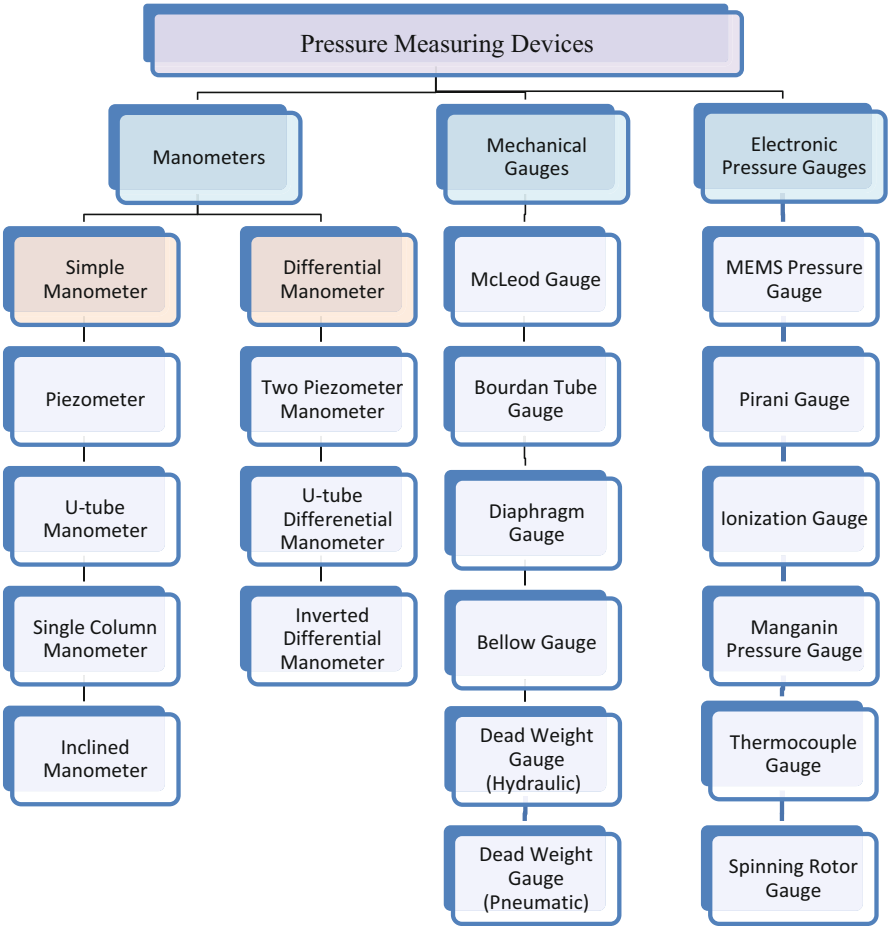

Fig. 3 Type of pressure-measuring instruments, content of figure adopted, and modified from Fig. 3 in reference (Rab et al. 2022)

An "absolute pressure" is pressure measurement whose reference point is zero pressure or absolute vacuum. The barometer reading the absolute pressure is the best example of this pressure.

Liquid Column Gauges

A manometer (https://solutionpharmacy.in/differential-u-tube-manometer/) is an instrument which measures pressure near to atmospheric pressure. These instruments are the cheapest, easy to use, easy to construct, and accurate pressure measurement devices. These instruments are widely used primary standards. It consists of a tube, half filled with the liquid. In a manometer, the weight of liquid

columns is balanced by pressures. It is used for measuring the "static pressure." Manometers inherently measure differential pressure. To determine absolute pressure, the reference side pressure must be known or negligible. Manometer gauges comprise of a vertical column of liquid either in a tube or in a U-shaped tube.

The different types of manometers are as follows:

1. "Simple manometer"
2. "Differential U-tube manometer"
3. "Inverted U-tube manometer"
4. "Inclined manometer"
5. "Small manometer"

The U-tube differential manometer is shown in Fig. 4. The ends A and B of this tube are subjected to different pressures P_1 and P_2. The liquid in columns will rise or fall in accordance with pressures at both ends until the weight of liquid becomes equal to the pressure to be measured. The end A is connected to the region where pressure needs to be measured, while the other end B to the "reference pressure." If we want to measure "absolute pressure" at end "A," the end "B" has to be connected to zero pressure or "absolute vacuum," and for "gauge pressure" measurement, end "B" has to be exposed to the atmosphere. The liquid height difference is directly proportional to the applied pressure. The difference (h) between two levels of liquid is measured by vertical ruler. For precise length measurement, high precision of about 10^{-3} mm micrometeor can be used; this can enhance overall accuracy of the instrument. The acceleration due to gravity (g) at the place of the instrument also required to be measured precisely. If we are taking reference density of fluid, the fluid must be of high purity. The most commonly used purification technique is vacuum distillation. The pressure can be calculated by using Equation "6" excluding the term P. Any fluid can be used in the manometer, but for high accuracy, mercury is preferred due to its high density (13.534 g/cm^3), chemical stability, small temperature coefficient, low vapor pressure, and does not stick to the walls of the tube. Since

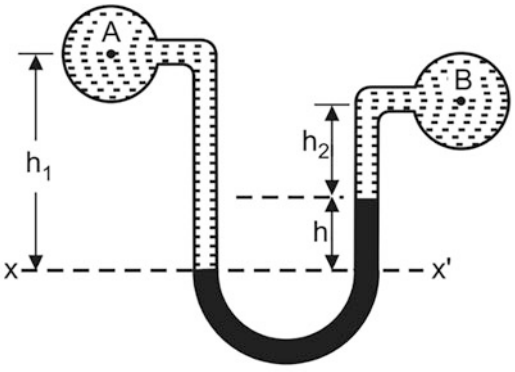

Fig. 4 U-tube differential manometer. Picture has been taken from reference (Song et al. 2020)

mercury expands with temperature and is compressible, its density depends upon temperature and pressure; therefore reference density values are to be specified for a particular temperature and pressure. For low-pressure measurement and to obtain higher resolution, lesser-density oil compared to mercury density can be used. In the "U-tube manometer," there is no fixed reference point as liquid height changes in both limbs of the tube in accordance with pressure. This makes the height difference measurement more difficult in comparison if one surface could be kept at some fixed level. The performance of manometers depends on length measurement, temperature measurement, operator skill, and environment conditions. The magnitude of random errors such as properly locating the liquid surface, measuring the height difference, variations in capillarity effect, and variations in temperature may be evaluated experimentally by repeated observations. The manometers have some drawbacks like small operational range of order of 1 MPa and fragile. The errors in measurement occur due to change in temperature as density of fluids used in manometers changes with temperature. Good commercial manometer using mercury can attain as low as 20–30 ppm of the measured pressure near atmosphere. The manometers are used in gas systems, to construct swimming pools, climate forecasting, for blood pressure measuring, etc.

McLeod Gauge

"McLeod" (https://en.wikipedia.org/wiki/McLeod_gauge) gauges are used to measure absolute pressure at very low pressure range between 10^{-4} torr and 10^{-6} torr. It was invented by Herbert McLeod in the year 1874. McLeod gauge is shown in Fig. 5. It consists of mercury reservoir, piston, reference column, bulb, scale, and measuring capillary. This gauge works on the principle of Boyle's law. It takes some volume of gas from a vacuum chamber whose pressure needs to be measured. After taking the small volume, the instrument is tilted. When the instrument is tilted, the infilling mercury compresses the small volume of taken gas. A mercury manometer is used to measure the pressure of compressed small volume of gas. By knowing the compression ratio, the pressure of the original vacuum can be determined. In this gauge there is linear relation between applied pressure and height. This gauge can be used as reference standard to calibrate other pressure gauges. McLeod gauges cannot give continuous output; therefore it measures only on sampling basis. Nowadays, these gauges are replaced largely by electronic vacuum gauges.

Aneroid Gauges

In "aneroid gauges," the pressure sensing element is made up of elastic metal. The working principle of "aneroid gauges" is the metallic sensing element flexes elastically when the pressure difference is applied across it. Aneroid gauges do not use any fluid for its working. "Aneroid gauges" can be used to measure the pressure of both liquid and gas. Working these gauges does not depend on the type of gas or liquid

Fig. 5 McLeod gauge. Picture has been taken from reference (https://en.wikipedia.org/wiki/McLeod_gauge)

whose pressure is being measured. The pressure sensing element may be a "Bourdon tube," a "diaphragm," a "capsule," or a set of "bellows."

Bourdon Tube Gauge

The "Bourdon tube gauge" is one of the most broadly utilized devices. It was invented in about 1849 by "Eugene Bourdon"; since then its use is increasing day by day. He patented "Bourdon tube pressure gauge" in the year 1849 in France. This gauge can be used for measuring the pressure of all types of liquids and gases including steam, water, and air. It can operate from 60 kPa to nearly 700 MPa. This gauge is useful for measuring static or slow-changing pressures. A Bourdon gauge (https://en.wikipedia.org/wiki/File:WPGaugeFace.jpg) indicator side is shown in Fig. 6, and inner construction is shown in Fig. 7. It uses a coiled tube as an elastic transducer. One end of this tube is connected to pressure to be measured and another end to the pointer through mechanical linkage. When pressure increases in the tube, it uncoiled due to expansion. When tube expands

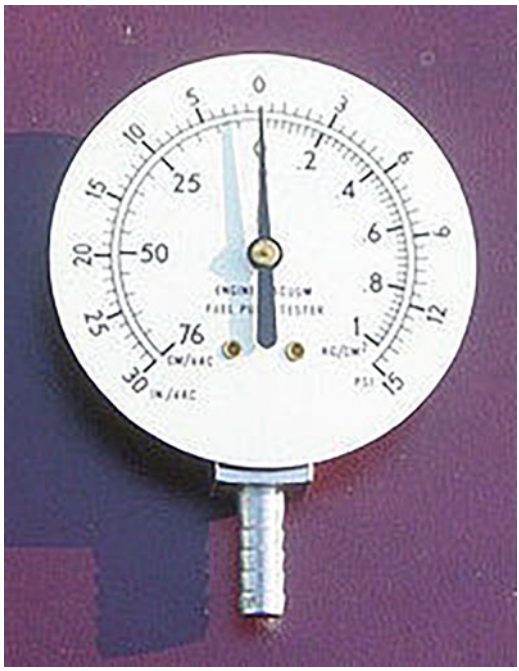

Fig. 6 Indicator side with card and dial Bourdon gauge. Picture has been taken from reference (https://en.wikipedia.org/wiki/File:WPGaugeFace.jpg)

it rotates the pointer connected with the tube via adjustment link, segment leaver, and pinion. When pressure decreases the tube contracts and tends to coil more tightly. This motion is transferred to an indicating needle. The pressure can be read directly with the help of calibrated scale. The performance of these gauges is limited to 0.1%. The precision of measurement is also limited by visual detection. The repeatability of gauge is also limited due to inelastic behavior of the tube material. Temperature changes also affect the reading because of thermal expansion of components. To reduce these problems, electronics can be blended with mechanical components. The mechanical linkage, arm, and pointer can be replaced by optical lever, and deflection of tube can be detected electronically and output displayed digitally.

Diaphragm

Diaphragm pressure gauges (https://en.wikipedia.org/wiki/File:Barograph_03.jpg) have a simple working mechanism. It consists of a thin circular membrane of a corrugated metal sheet called a diaphragm as shown in Fig. 8. The diaphragm can be made using a variety of materials like Inconel, nickel, titanium alloys, stainless steel, fused quartz, silicon, etc. (https://www.nist.gov/document/pmc-2pdf). In diaphragm gauge the flexible membrane defects if there is pressure difference in both sides of the membrane. The deformation of diaphragm is proportional to the pressure difference on two faces of thin flexible membrane. The one face of the membrane is kept at

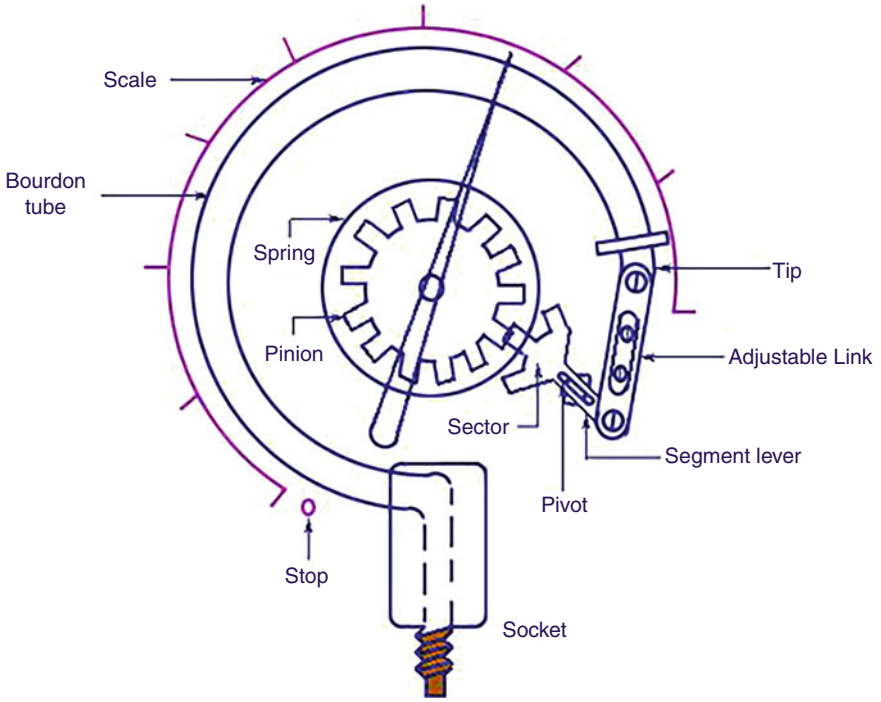

Fig. 7 Inner construction Bourdon gauge. Picture has been taken from reference (Maitra n.d.)

Fig. 8 A pile of pressure capsules with corrugated diaphragms in an aneroid barograph. Picture has been taken from reference (https://en.wikipedia.org/wiki/File:Barograph_03.jpg)

pressure to be measured, and the reference face can be exposed to the atmosphere, at any pressure, or can be sealed according to the type of measurement like gauge pressure, differential pressure, and absolute pressure, respectively. The deformation can be measured using mechanical, optical, or capacitive techniques. This gauge is used for low-pressure measurement from -100 kPa to 40000 kPa (https://tameson.com/pressure-gauge-diaphragm.html) in food and beverage, pharmaceutical, petrochemical, mining industries, etc.

Bellow Gauge

The pressure sensing element of bellow gauge is a bellow (https://instrumentationtools.com/bellows/). These gauges are more compliant than Bourdon gauges and the diaphragm gauges. The bellow is a thin-walled metallic tube which can contract and expand according to pressure variation as shown in Fig. 9. Generally, bellows contain 12 to 24 folds and its diameter ranges from 1.2 to 30 cm. When pressure is applied to the inlet (one end) of the bellow, it will expand or contract according to the applied system or line pressure. Another end of the bellow is connected to a spring and a pointer. As the bellow expands or contracts, either pressure can be read directly on scale, or an electrical signal is generated which can be processed to measure pressure. Within the elastic limit of the material of bellow, the relation between deflection and pressure is linear. The bellow gauge can measure pressure from 0.6 kPa to 100 kPa. The bellow gauge is more sensitive than the Bourdon tube gauge, thus useful for low-pressure applications.

Differential Pressure-Measuring Devices

The "differential pressure-measuring devices" measure the difference of two pressures. The devices which are used for differential pressure measurement have two input ports. The pressures whose difference has to be measured are applied to ports. The instrument output gives the differential pressure. Any pressure-measuring devices described earlier can be used to measure differential pressure in special forms. These devices are suitable for flow measurement, pumping plants, etc.

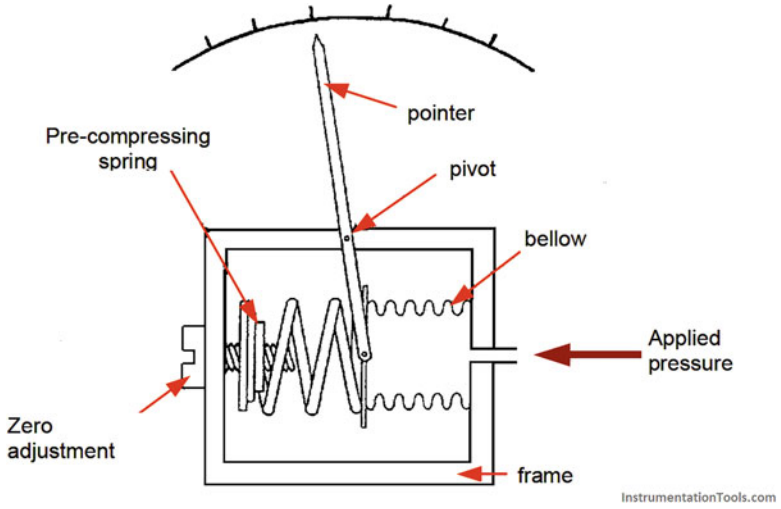

Fig. 9 The bellow gauge. Picture has been taken from reference (https://instrumentationtools.com/bellows/)

Deadweight Piston Gauge

The deadweight piston gauge (de Malherbe and Firth 1970; http://www.calref.it/clrf_eng/manual/Manual%20PDWT-10.pdf) measures the pressure using the fundamental units, mass and length; therefore they are used as a primary standard. Many other names of piston gauges are "deadweight gauges," "deadweight tester" (DWT), "gauge tester," "piston manometer," and "pressure balance." The DWTs are proven primary standards and the most accurate instruments available in the market for the calibration of secondary pressure gauges and sensors. It has therefore been used for years in factory, calibration laboratories, industry, national metrological institutes, and research laboratories. The basic construction of deadweight piston gauge is as illustrated in Fig. 10. The main element to construct these gauges is piston and cylinder assembly. The piston is implanted into a close-fitting cylinder. At one end of the piston, fluid pressure is applied, and weights are loaded at the other end to generate pressure to balance the fluid pressure. The different types of piston gauges have been constructed to accommodate the method of loading and method of rotating the piston and controlling the clearance to reduce friction between piston and cylinder resulting to increase the accuracy of the gauge. Three commonly used designs of cylinder are the simple, the reentrant cylinder, and the controlled clearance cylinder (Rab et al. 2022; Jain et al. 1992). Controlled clearance DWTs contain an external jacket through which hydraulic pressure can be applied to vary the clearance between the piston and the cylinder. Three different design configurations,

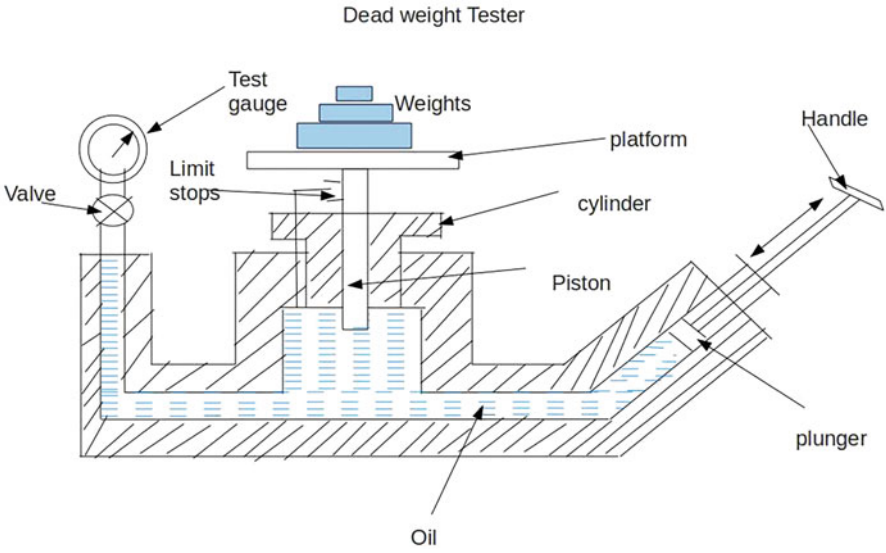

Fig. 10 Component of dead wight tester. Picture has been taken from reference (http://4.bp.blogspot.com/_SEJ_mMmfxvU/TKhjEXQ_mnI/AAAAAAAAAHw/h5qGU3EqnyQ/s1600/dead+weight+tester.jpg)

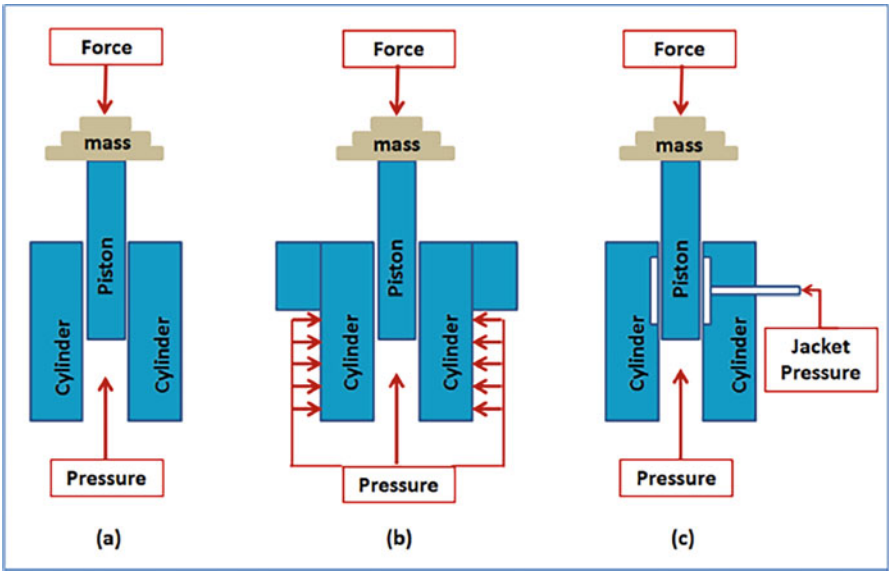

Fig. 11 Different design configurations of piston-cylinder assembly, (**a**) simple-type DWT, (**b**) reentrant-type DWT, and (**c**) controlled clearance-type DWT. Picture has been taken from reference (Rab et al. 2022)

simple, reentrant, and controlled clearance of piston-cylinder assembly (Rab et al. 2022), are shown in Fig. 11. The DWTs can use oil or gas for its operation. Gas can be used for pressure near to atmospheric pressure and lower part of high-pressure range in both gauge and absolute mode. Oil-operated gauges can be used for high and very high pressure in gauge mode. To achieve high accuracy by reducing the friction between piston and cylinder, piston and weight stack are rotated with respect to cylinder. The clearance between piston and cylinder must be small. The uncertainty of the DWTs is calculated by uncertainties of the acceleration due to gravity, mass of the weights, and effective area. In most cases, effective area of piston limits the accuracy of these gauges. The distortions of piston and cylinder introduce additional uncertainties. The piston gauges are very sensitive and stable. Commercial oil-operated gauges are available up to 1.3 GPa, and special purpose gauge have been built for operation up to 2.6 GPa (Thomson and Heydemann 1967; Eremeev 1974; Nishibata et al. 1980).

Pressure can be calculated by knowing the mass applied to piston, area of piston, and acceleration due to gravity at that location using Eq. 1. When pressure is applied to the piston and cylinder assembly, the area of piston gets charges. For accurate pressure calculation, the effective area of piston must be determined, and some other factor must be included. The effective area is determined by cross-floating with a known reference primary standard DWT. With a formula given in Eq. 7 (3rd International Congress on Mechanical Metrology (CIMMEC2014) 2015), the most accurate pressure can be calculated:

$$p = \frac{\left[m_p\left(1-\frac{\rho_a}{\rho_{mp}}\right)+m_r\left(1-\frac{\rho_a}{\rho_{mr}}\right)\right]g_l+\sigma C}{A_{0,T}\left[1+\left(\alpha_c+\alpha_p\right)(\theta-20)\right](1+\lambda p_n)} \pm \rho_{fluid} g_l \Delta h \tag{7}$$

where:

p is "the measured pressure, given in Pa"
m_p is "the mass of the piston, in kg"
ρ_a is "the specific volumetric mass of the air, in kg/m^3"
ρ_{mp} is "the volumetric mass of the piston material, in kg/m^3"
m_r is "the sum of the remaining masses operating on the system, in kg"
ρ_{mr} is "the specific volumetric mass of the remaining masses, in kg/m^3"
g_l is "the acceleration due to the local gravity, in m/s^2"
σ is "the surface tension of the fluid, in N/m"
C is "the length of the circumference of the piston, in m"
$A_{0,\ T}$ is "the area of the piston-cylinder assembly, in m^2"
T is "the reference temperature of the piston, in °C"
$\alpha_c + \alpha_p$ is "the coefficient of the linear thermal expansion of the piston-cylinder assembly, in °C^{-1}"
θ is "the temperature of the piston at the time of the measurement, in °C"
λ is "the distortion coefficient of the piston-cylinder assembly, generally in (10^{-6}) MPa^{-1}"
p_n is "the rated pressure of the measurement given in MPa"
ρ_{fluid} is "the specific volumetric mass of the fluid used, in kg/m^3"
Δh is "the gap between the piston and the cylinder at the location where the pressure is measured and is given in m"

DWT is simple in construction and easy to use. Desired range of pressure measurement can be easily obtained by adding weights or by changing the piston-cylinder arrangement.

Electronic Pressure Gauges

Electronic pressure gauges (EPG) are special forms of the pressure gauges used for automated devices like washing machines. The output of these pressure gauges is in millivolts (mV). The output varies according to the pressure applied. In these gauges the electronic techniques are applied to improve performance.

EPG Types by Sensing Principle

Different types of sensors are available to measure pressure (https://www.processindustryinformer.com/a-world-under-pressure-pressure-measurements-for-process-industries/). Piezo-resistive sensors are the most used sensors, due to high accuracy and robust construction. Most piezo-resistive sensors are based on a Wheatstone bridge principle. When strain/pressure is applied, the value of each

resistor in the bridge changes, and then this change can be used into a variety of electrical outputs.

Capacitive sensors come also in a big variety of shapes and forms, with generally very simple shapes. One plate of capacitor is a thin diaphragm. The pressure applied on the diaphragm will cause the elastic deformation/movement of the diaphragm and hence a change in the electrical capacitance. By measuring the change in capacitance, applied pressure can be measured. Capacitive sensors have high sensitivity.

Inductive sensors have a similar approach to capacitive sensors. A sensing (elastic) capsule moves a core element inside of a linear variable differential transformer. The applied pressure can be measured in terms of the induced electromotive force (emf).

Resonant pressure transducers are some of the most accurate sensors. When stress is applied to resonant sensor, frequency of a resonator changes in accordance with the applied stress. Applied stress is measured in terms of change in frequency.

EPG Types by Output

There are two main types of pressure sensors based on output signal. One is continuous output (mV, V, mA, and Hz) and another discrete or digital output. An mV output is desirable if the signal has to be modified in a customized application and display is at short distance, while a V output or mA output can be sent over relatively long distances. Nowadays digital output pressure sensors are becoming more popular as they can be easy integrated into digital systems (computer, wireless, Bluetooth, etc.).

Pressure Sensor Metrological Characteristics

It is extremely significant to understand metrological characteristics of pressure sensors to ensure a suitable sensor is chosen for the intended application.

The main metrological parameters are as follows.

Signal Offset

The output of the sensor at zero applied pressure is known as signal offset. The ability of "re-zero" is desirable in a sensor. Measuring and resetting such offsets require a calibration and maintenance plan to ensure accuracy in the measurement. Offset is shown in Fig. 12.

Sensitivity

The sensitivity of a sensor is the ratio of the output signal change and the pressure change. Let the output of any sensor change be 3 V, while the pressure changes by 1000 Pa; then the sensitivity will be 3 mV/Pa. This is an important parameter for the way we would use the signal in the application.

Precision

Precision is the term used to describe sensors' behavior in terms of its repeatability, linearity error, and signal hysteresis. Precision tells the behavior of the sensors. For example, is it repeatable? Is it linear? Is there hysteresis? Repeatability is the

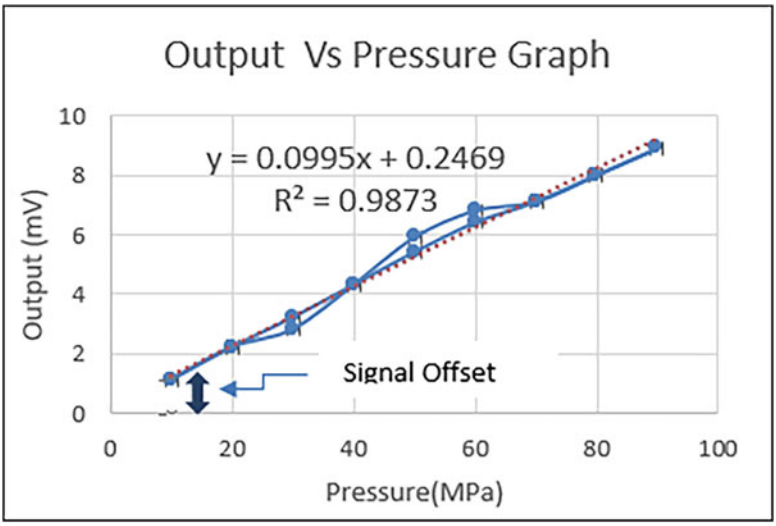

Fig. 12 Output vs. **pressure** graph for a sensor

closeness of the agreement between the successive measurement results of the same pressure carried out under the same conditions of measurement in a relatively short period of time. Often the repeatability is determined as the standard deviation of the repeated measurements. Linearity error is calculated taking the difference between the measured value and the theoretical line which assumes linear behavior of the sensor. R-squared (R^2) indicates the percentage of variance. If R^2 is close to "1," this means linear error is less. Linearity error is shown in Fig. 12. Solid line (-) represent the measured data and dotted line (. . .. . .) represent theoretical.

Hysteresis

Pressure hysteresis error is the difference between two separate measurements taken at the same point but one where the pressure is increasing and another where pressure is decreasing. The hysteresis graph is shown in Fig. 11.

Long-Term Stability (Drift)

Stability of a measuring instrument is the property, whereby its metrological properties remain constant in time. The drift is expressed as a tolerance range, D = +/- 10 Pa; hence it should be included in the overall accuracy of the instrument. Generally, the pressure sensor exhibits some form of drift over time; hence it is important that systems are designed with the potential to be adjusted for both: offset and span drift. This problem requires a calibration and maintenance program.

Accuracy and Uncertainty

Accuracy measures the product's performance quality. Generally, uncertainty is the measure of statistical dispersion of the values measured, including the impact of systematic error, random error, and drift.

The accuracy of a pressure sensor or of a measurement is obtained as part of the measurement uncertainty. Most of the time, the accuracy is evaluated as expanded uncertainty. Type A, type B, standard, and expanded uncertainty can be calculated using equations described elsewhere (Sawla 2000). Without accuracy evaluation, any measurement has no meaning. If accuracy of pressure measurement is not in desired limit, it can be dangerous specially in aircraft, chemical, and process industry. Accuracy estimation is a difficult task. Several common terms (Compilation of ASTM Standard Definitions 1986) for accuracy measurement are defined below. These definitions are commonly used in metrology.

Accuracy. This indicates how close a measurement to its true value. Since we cannot know true value exactly, accuracy can be estimated only. Most of the users use the error for accuracy. For example, measured value is 98.5 % of accepted value, it is said to have an accuracy of 1.5%.

Confidence level. We define confidence level in terms of standard deviation (sigma, σ) (https://news.mit.edu/2012/explained-sigma-0209). For accurate measurements, we take a large number of observation, and plot of results follows normal distribution. The deviation is how far the measurement point is from the average value and the standard deviation is square root of average of all squared deviations. One standard deviation (1σ) is the region that includes 68% of all the points in normal distribution curve. Two standard deviation (2σ) includes 95% and three standard deviation (3σ) includes 99.7%. If the errors are assumed to be distributed normally and uncertainty is specified 0.5% at the 2σ level, it means that there is 95% probability that the measured value is within 0.5%.

Error. It is the difference between measured value and true or expected value. Error can be estimated only. Errors can be represented as absolute error, relative error, and percentage errors, but generally the error is represented in percentage.

Linearity. Performance quality of any instrument is described in terms of linearity. It tells us how perfectly the instrument measures correspond to reality. It is a misleading indicator.

Precision. It is referring to closeness of two or more observations at the same point to each other.

Random error. Random errors are erratic differences in repeated measurement of a quantity. It is often characterized by standard deviation.

Repeatability. The closeness of agreement of measurement made under the same conditions is characterized by repeatability.

Reproducibility. The closeness of agreement of measurement made under the different conditions is characterized by repeatability. Reproducibility may also be on the measurement that is repeated over a long period of time.

Sensitivity. The ratio of change of output to change in input is known as sensitivity.

Traceability. The documented relationship between uncertainty of the instrument and the recognized standard. For example, traceable to CSIR-NPL means the instrument is calibrated relative to the standard maintained by the CSIR-NPL.

Head correction. If position of pressure gauge under calibration and pressure standard is at different elevation, the hydrostatic head contributes to error in calibration. The hydrostatic head correction is independent of pressure; it changes with fluid density and height difference between calibrator and gauge under calibration. For a fluid with a density of 1 g/cm^3 in acceleration due to gravity of 980 m/s^2, the correction will be -98 Pa for every cm increase of height of gauge under calibration.

Influence Quantities

Influence quantities (temperature, humidity, atmospheric pressure, electromagnetic fields, vibrations, noise, etc.) are any external (i.e., not included in the input/output) quantities, which can influence the performance of a pressure sensor. Most of the time, the pressure sensor manufacturers provide the range for the influence quantity and its effect on the metrological characteristics of the sensor. The influence quantities and their effect should be supplied by manufacturers through their datasheets.

Calibration of Pressure Sensors

Calibration is a process of configuring an instrument to provide an accurate measurement. The calibration process generally involved a calibrator, an instrument which is more accurate (4 times more) than the sensor to be calibrated. In past years, pressure sensors would be taken out of the system into which they are incorporated and calibrated in a metrological laboratory. In modern times, most of the sensors are calibrated "on the spot" using pressure calibrators. Also, many of the calibrators can evaluate the calibration uncertainty (many times referenced as "accuracy").

MEMS Pressure Sensors

"MEMS" (microelectromechanical systems) pressure sensors (Song et al. 2020) can be used to measure the pressure of gases including air. Their basic construction is shown in Fig. 13. The pressure is sensed by a thin silicon diaphragm of typical thickness 100 mm on a silicon substrate. When the pressure to be measured is applied to the diaphragm, it bends under the action of the applied pressure.

The deflection of the diaphragm can be measured by the following methods:

1. By mounting piezo-resistors on the diaphragm surface
2. By measuring the change in capacitance between the diaphragm and a silicon beam above the diaphragm

The relation between output and input of a piezo resistor is linear. In type 1 the output is considerably affected by temperature changes. The size of devices based on

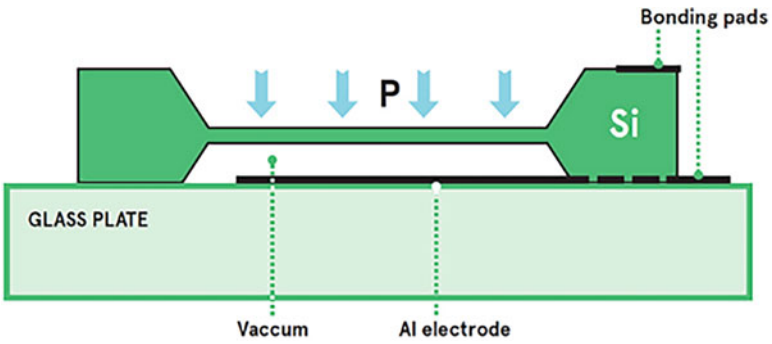

Fig. 13 Structure of the MEMS pressure gauge. Picture has been taken from reference (https://www.avnet.com/wps/portal/abacus/solutions/technologies/sensors/pressure-sensors/core-technologies/mems/)

the capacitive method are much larger, but output is less impacted by temperature changes. They have a nonlinear input-output relation. MEMS gauges based on capacitance are cheaper than piezo-resistive ones. Different versions of MEMS pressure sensors can measure gauge and differential pressure. The accuracy of pressure measurement by MEMS gauge is typically ± 3% of full-scale reading. They are used in large numbers of industry including automotive and medical.

Thermal Conductivity Pressure Gauges

The thermal conductivity pressure gauge is based on change of thermal conductivity of gases according to pressure change. Generally, when pressure of a real gas increases, its density also increases, and hence the ability of gas to conduct heat increases. These gauges consist of a wire filament, a thermocouple, or resistance temperature detector (RTD). The filament is suspended in a chamber whose pressure has to be measured. This filament is heated by flowing current in it. RTD or thermocouple can be used to measure the temperature of the filament. The temperature of filament is dependent on the thermal conductivity of surrounding gas and hence on pressure. Therefore, by measuring the temperature of filament, we can measure the pressure of chamber. These gauges can be used from 0.5 torr to 10^{-3} torr.

Pirani Gauge

The "Pirani gauge" was invented by "Marcello Pirani." The construction of Pirani gauge (Morris and Langari 2016) is shown in Fig. 14. The main component of a Pirani gauge is a metal wire made up of gold-plated tungsten or platinum. This wire is exposed to the pressure being measured. The wire is heated by passing the current through it and cooled by the surrounding gas. The cooling effect of the wire is directly proportional to the thermal conductivity of gas and hence on pressure. The resistance of the wire varies with temperature. Resistance of wire can be calculated

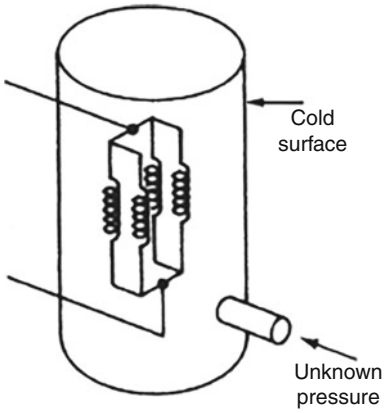

Fig. 14 Pirani gauge. Picture has been taken from reference (Morris and Langari 2016)

by noting supply voltage and current which heats the filament. By knowing the resistance, the gas pressure can be determined. Useful range is 10^{-5} mbar to 1 mbar.

Ionization Gauge

Ionization gauges (Soloukhin et al. 1991) are used to measure very low pressure as they are the most sensitive gauges for very low pressures. The structure is shown in Fig. 15. They sense pressure indirectly by measuring the electrical ions produced when the gas is bombarded with electrons. Fewer ions will be produced by lower-density gases. The calibration of an ion gauge is unstable and dependent on the nature of the gases being measured, which is not always known. Thermionic emission generates electrons, which collide with gas atoms and generate positive ions. The ions are attracted to a suitably biased electrode known as the collector. The current in the collector is proportional to the rate of ionization, which is a function of the pressure in the system. Hence, measuring the collector current gives the gas pressure. There are several subtypes of ionization gauge. Useful range is 10^{-10}–10^{-3} torr (roughly 10^{-8}–10^{-1} Pa).

Capacitive Pressure Gauges Using Dielectric Material

The solution for adverse effect of temperature and inelastic behavior on the gauge performance is given by many researchers using capacitive pressure gauge (Andeen et al. 1971; Collwel 1979; Kumar et al. 1999; Kumar et al. 2003; Kumar and Yadav 2017; Kumar 2021). Andeen and co-worker (Andeen et al. 1971) used a variation of capacitance of parallel plate capacitor of CaF_2 solid dielectric with applied pressure to measure the pressure. This gauge demonstrated good sensitivity and repeatability of the pressure measurement. Colwell (Collwel 1979) made significant improvement

Fig. 15 Ionization pressure gauge. Picture has been taken from reference (https://www.directindustry.com/prod/stanford-research-systems/product-7218-1865284.html)

in the design of the gauge by using two capacitors of different materials. Vinay et al. (Kumar et al. 1999; Kumar et al. 2003; Kumar and Yadav 2017; Kumar 2021) demonstrate the use of $Pb(Mg_{1/3}Nb_{2/3})O_3$ [PMN – lead magnesium niobate] and $Pb(Fe_{2/3}W_{1/3})O_3$ [PFW – lead iron tungstate] relaxor material either single or in combination with other to fabricate dielectric pressure sensor for high-pressure measurement up to 450 MPa.

High-Pressure Measurement (Greater than 7000 Bar)

Pressure range above 10^9 Pa is usually measured with "manganin pressure gauge" (Morris and Langari 2016). These gauge works on the principle of change in resistance of special material with pressure. Manganin is the preferred material for these gauges since its variation of resistance with pressure is more linear in comparison of other material and it is most sensitive. The pressure equation can be obtained by studying metrological properties using piston manometer. A manganin pressure sensor consists of a coil of manganin wire enclosed in sealed, kerosene-filled, flexible bellows, as shown in Fig. 16. The two ends of the wire are soldered to electrical lead. The unknown pressure is applied to one end of the bellows, which transmits the pressure to the coil. The coil resistance changes according to the applied pressure. By measuring the resistance of the coil, pressure can be calculated.

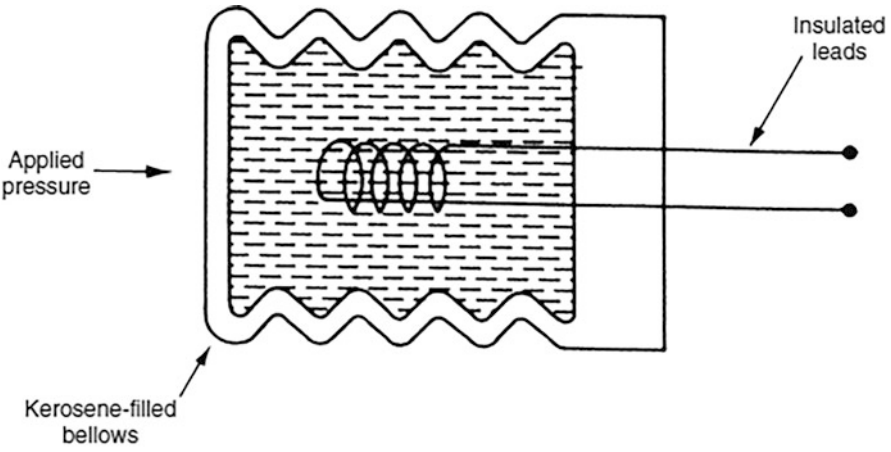

Fig. 16 Structure of manganin pressure sensor, picture has been taken from reference (Morris and Langari 2016)

Pressure up to 3000 MPa can be measured by a "manganin wire pressure sensor," with a typical inaccuracy of ± 0.5%.

Measurement Standards and Facilities

The measurement standards can be classified in to two types: primary standard and secondary. The primary standards are not calibrated against other standards, but for accuracy check, their measurement result must compare to other standards (intercomparison), although parts of primary standard may be calibrated to standards of other units. The design of primary standards must be based on physics laws relating quantity (pressure) to be measured to the measuring components. For example, in case of pressure standard, the relationship between applied pressure and measuring quantities like mass, length, temperature, time, etc. must be known. The uncertainty analysis of primary standards requires thorough knowledge of operation of the instrument and effect of environment conditions. Examples of primary pressure standards are "McLeod gauges," "liquid column manometers," and deadweight testers.

The secondary or transfer pressure standards are calibrated against primary standards. Any instrument with well-established stability can serve as secondary standards. The uncertainty of instrument is always greater than the primary standards. Repeated calibrations of transfer standard are required. If there is much difference in measurement value, either secondary or primary standard is malfunctioning.

The approved CMCs of different countries' NMIs for the different pressure range and with uncertainty are given in Table 3 (Rab et al. 2022; Calibration and measurement capabilities 2022).

Table 3 Approved CMCs of different countries' NMIs, data is taken from reference (Rab et al. 2022; Calibration and measurement capabilities 2022)

Countries, NMI	Pressure ranges and approved CMCs	Reported measurement uncertainties through equation (pressure p, p_l, p_{line}, and D_p are in Pa, if not specified)	Mode and medium of pressure
Argentina	1500 Pa to 5000 Pa	2 Pa	Gauge, gas
	1E8 Pa to 4E8 Pa	4E-05p + 2.1E-13p^2	Gauge, oil
	5000 Pa to 7E6 Pa	1 + 4E-05p	Gauge, gas
	−1500 Pa to 1500 Pa	0.3 Pa	Gauge, gas
	7E6 Pa to 12E6 Pa	5E-05p	Gauge, gas
	1E8 Pa to 4E8 Pa	4E-05p + 2.1E-13p^2	Absolute, oil
	−5000 Pa to −1500 Pa	2 Pa	Gauge, gas
	2E5 Pa to 1E8 Pa	30 + 5E-05p	Absolute, oil
	1E5 Pa to 1E8 Pa	30 + 5E-05p	Gauge, oil
	1500 Pa to 4.00E4 Pa	4Pa	Absolute, gas
	−1E5 Pa to −5000 Pa	1 + 4E-05p	Gauge, gas
	4E4 Pa to 7E6 Pa	2.0 + 5E-05p	Absolute, gas
	7.1E6 to 1.21E7	6E-05p	Absolute, gas
Australia	2 kPa to 7000 kPa	1 to 1.7E2 Pa	Gauge, gas
	0 kPa to 105 kPa	1 Pa	Absolute, gas
	105 kPa to 7000 kPa	3.5 to 2.1E2 Pa	Absolute, gas
	600 kPa to 500 MPa	1.0E-5 to 4.5E-2 MPa	Gauge, oil
Austria	4E7 to 4E8	8E-05p + 2.5E-13p^2	Gauge, oil
	5000 Pa to 1.20E5 Pa	0.5 + 4E-05p	Absolute, gas
	3500Pa to 1.7E5 Pa	0.5 + 3E-05p	Gauge, gas
	500 Pa to 5000 Pa	0.5 + 4E-05*p*	Absolute, gas
	1.7E5 Pa to 4E6 Pa	0.5 + 4E-05p	Gauge, gas
	5E5 Pa to 4E7 Pa	6 + 4E-05p + 1.2E-13p^2	Gauge, oil
	4E6 Pa to 1.2E7 Pa	1E-04p	Gauge, gas
Belarus	1E7 Pa to 1E8 Pa	0.02%	Gauge, oil
Bosnia and Herzegovina	1E4 Pa to 3.5E5	0.3 Pa + 3.0E-05p	Absolute, gas
	4000 Pa to 1.2E7 Pa	0.3 Pa + 3.0E-05p	Gauge, gas
Brazil	0.1MPa to 20 MPa	6.00E-05	Gauge, gas
	12 kPa to 700 kPa	5.00E-05	Gauge, gas
	5 kPa to 110 kPa	10 Pa	Absolute, gas
	0.4 MPa to 80 MPa	6.0E-05	Gauge, liquid
	1.4 kPa to 172 kPa	4.5E-05	Absolute, gas
	12 kPa to 700 kPa	5.00E-05	Absolute, gas
	1.4 kPa to 172 kPa	4.50E-05	Absolute, gas
	1 MPa to 250 MPa	6.00E-05	Gauge, liquid
	0.014 MPa to 7 MPa	5.00E-05	Gauge, gas
	0.1 MPa to 20 MPa	6.00E-05	Gauge, liquid
	0.014 MPa to 7 MPa	5.00E-05	Absolute, gas

(continued)

Table 3 (continued)

Countries, NMI	Pressure ranges and approved CMCs	Reported measurement uncertainties through equation (pressure p, p_l, p_{line}, and D_p are in Pa, if not specified)	Mode and medium of pressure
Bulgaria	1E4 Pa to 1E5 Pa	5E-04*p*	Negative gauge, gas
	2E4 Pa to 7E6 Pa	1E-04p	Absolute, gas
	2E4 Pa to 2E5 Pa	6E-05p	Gauge, gas
	5000 Pa to 1E4 Pa	1E-03*p*	Absolute, gas
	100 Pa to 4000 Pa	0.2 + 1E-04*p*	Gauge, gas
	8E6 Pa to 4E7 Pa	8E-05p	Gauge, oil
	8E7 Pa to 2.5E8 Pa	5E-04p	Gauge, oil
	2E5 Pa to 2E6 Pa	5.5E-05p	Gauge, gas
	4E7 Pa to 8E7 Pa	8E-05p	Gauge, oil
	2E6 Pa to 7E6 Pa	6.5E-05p	Gauge, gas
	5E4 Pa to 7E4 Pa	2.5E-04*p*	Differential, gas
	13 Pa to 130 Pa	5E-03*p*	Absolute, gas
	7E4 Pa to 7E5 Pa	1E-04p, p_{line} 1E + 05 Pa	Differential, gas
	8E5 Pa to 8E6 Pa	8E-05p	Gauge, oil
	2.60E4 Pa to 1.30E5 Pa	13 Pa	Absolute, gas
Canada	5 kPa to 350 kPa	2.00E-05	Absolute, gas
	1 Pa to 200 Pa	0.034 + 4.2E-06*p*	Absolute, gas
	5 kPa to 350 kPa	2.00E-05	Gauge, gas
	25 kPa to 1750 kPa	3.60E-05	Gauge, gas
	0.1 MPa to 7 MPa	5.00E-05	Gauge, gas
	25 kPa to 1750 kPa	3.60E-05	Absolute, gas
	0.1 MPa to 7 MPa	5.00E-05	Absolute, gas
China	10 MPa to 100 MPa	3.2E-05*p*+8.9E-14p^2	Gauge, oil
	5 kPa to 175 kPa	0.2+1.3E-05*p*	Gauge, gas
	600 kPa to 7000 kPa	2.0E-3 %	Gauge, gas
	100 MPa to 200 MPa	3.3E-05*p*+9.0E-14p^2	Gauge, oil
	0 Pa to 2500 Pa	0.2 Pa	Gauge, gas
	175 kPa to 7000 kPa	3.0E-05*p*	Absolute, gas
	0.1 Pa to 1 Pa	2.1%	Absolute, vacuum
	200 MPa to 500 MPa	2.4E-05*p*+1.4E-13p^2	Gauge, oil
	175 kPa to 600 kPa	1.5E-3 %	Gauge, gas
	1E-4 Pa to 1E-3 Pa	1.2 %	Absolute, vacuum
	5 kPa to 175 kPa	0.3+1.4E-05*p*	Absolute, gas
	1 MPa to 10 MPa	2.4E-3 %	Gauge, oil
	1E-3 Pa to 1 Pa	0.32 %	Absolute, vacuum
Chinese Taipei	6E-4 Pa to 1 Pa	2.9E-2	Absolute, gas
	172 kPa to 700 kPa	2.60E-05	Gauge, gas
	2.8 MPa to 28 MPa	3.30E-05	Gauge, oil
	700 kPa to 7000 kPa	4.20E-05	Absolute, gas

(continued)

Table 3 (continued)

Countries, NMI	Pressure ranges and approved CMCs	Reported measurement uncertainties through equation (pressure p, p_l, p_{line}, and D_p are in Pa, if not specified)	Mode and medium of pressure
	700 kPa to 7000 kPa	4.20E-05	Gauge, gas
	1 kPa to 372 kPa	0.65, 2.6E-05p,	Absolute, gas
	20 Pa to 5700 Pa	0.76 + 4.16E-04*p*	Gauge, gas
	5E-6 Pa to 6E-4 Pa	7.8E-2 to 7.4E-2	Absolute, gas
	372 kPa to 700 kPa	3.50E-05	Absolute, gas
	28 MPa to 273 MPa	7.40E-05	Gauge, oil
	17 kPa to 172 kPa	3.40E-05	Gauge, gas
	1 Pa to 1000 Pa	1.8E-2	Absolute, gas
Colombia	50 MPa to 200 MPa	3.0E-05p + 40	Gauge, oil
	2 MPa to 50 MPa	2.0E-05p + 35	Gauge, oil
	1.75 MPa to 7 MPa	4.1E-05p+5.2	Gauge, gas
	200 MPa to 500 MPa	6.0E-05p + 50	Gauge, oil
	0.1 MPa to 2 MPa	2.7E-05p + 25	Gauge, oil
	0.35 MPa to 1.75 MPa	2.8E-05p + 1.0	Gauge, gas
	1.5 kPa to 5 kPa	7.0E-05p+0.70	Gauge, gas
	5 kPa to 350 kPa	2.5E-05p + 0.55	Gauge, gas
Costa Rica	10 kPa to 7000 kPa	3.9E-05p	Gauge, gas
Croatia	1500 Pa to 1.40E5 Pa	6.1 + 4E-05p	Absolute, gas
	1.40E5 Pa to 1.40E6 Pa	4 + 5E-05p	Absolute, gas
	1.50E5 Pa to 7.00E6 Pa	25 + 5E-05p	Gauge, oil
	1.40E6 Pa to 1.70E7 Pa	7E-05p	Gauge, gas
	0 Pa to 1.70E7 Pa	1E-04p + 10	Differential, gas
	1.40E5 Pa to 1.40E6 Pa	4 + 5E-05p	Gauge, gas
	−1000 Pa to 3000 Pa	0.6 + 1.3E-04*p*	Gauge, gas
	0 Pa to 1.00E7 Pa	2E-04p + 50	Differential, oil
	7.00E6 Pa to 1.40E8 Pa	280 + 1E-04p	Gauge, oil
	3000 Pa to 1.40E5 Pa	1.8 + 7E-05p	Gauge, gas
Czechia	1 Pa to 1.50E4 Pa	0.02 + 2.80E-05*p*	Absolute, vacuum
	1.50E4 Pa to 7.60E6 Pa	0.3 + 1.10E-05p +2.00E-13p^2	Absolute, gas
	5000 Pa to 1.00E7 Pa	0.2 + 1.00E-05p +2.00E-13p^2	Gauge, gas
	100 Pa to 1.00E6 Pa	1 + 5.00E-05p	Differential, gas
	100 Pa to 1.00E6 Pa	1.0 + 1.20E-05p +2.00E-13p^2	Absolute, gas
	7.60E6 Pa to 1.00E7 Pa	1.0 + 1.20E-05*p* + 2.00E-13p^2	Absolute, gas
	1 Pa to 1.50E4 Pa	0.02 + 2.80E-05*p*	Gauge, gas
	5000 Pa to 1.00E5 Pa	1 + 1.20E-05*p*	Negative gauge, gas
	1 Pa to 1.50E4 Pa	0.02 + 2.80E-*05p*	Negative gauge, gas
	1E-4 Pa to 1 Pa	1.00E-05 + 1.5E-02*p*	Absolute, vacuum

(continued)

Table 3 (continued)

Countries, NMI	Pressure ranges and approved CMCs	Reported measurement uncertainties through equation (pressure p, p_l, p_{line}, and D_p are in Pa, if not specified)	Mode and medium of pressure
Denmark	−1000 Pa to 0 Pa	0.3 Pa	Negative gauge, gas
	1.00E5 Pa to 8.00E7 Pa	36 + 1E-04*p*	Gauge, oil
	1.00E4 Pa to 2.00E7 Pa	3 + 1E-04*p*	Gauge, gas
	1500 Pa to 1.50E4 Pa	6 Pa	Absolute, gas
	1.50E4 Pa to 1.50E5 Pa	9 Pa	Absolute, gas
	0 Pa to 1000 Pa	0.3 Pa	Gauge, gas
	8.00E7 Pa to 5.00E8 Pa	5E-04p	Gauge, oil
	5 Pa to 20 Pa	6E-02 + 1E-02*p*	Absolute, vacuum
	2.00E7 Pa to 8.00E7 Pa	18 + 1E-04p	Gauge, gas
	3500 Pa to 1.00E4 Pa	2 + 5E-04*p*	Gauge, gas
	1000 Pa to 1500 Pa	3E-04*p*	Gauge, gas
	5.00E5 Pa to 2.00E7 Pa	1E-04*p*	Differential, gas
	20 Pa to 1500 Pa	1E-02*p*	Absolute, vacuum
Egypt	2.00E5 Pa to 5.00E6 Pa	32 + 2.5E-05p	Gauge, gas
	1.00E4 Pa to 4.00E7 Pa	70 + 3.3E-05p	Gauge, oil
	5.00E6 Pa to 1.00E8 Pa	100 + 3.2E-05p	Gauge, oil
	1.00E4 Pa to 4.00E5 Pa	1.2 + 1.1E-04p	Absolute, gas
	1.00E8 Pa to 5.00E8 Pa	500 + 5E-05p + 3E-13p^2	Gauge, oil
	4.00E5 Pa to 4.00E6 Pa	1.2 + 1.1E-04p	Absolute, gas
Finland	1.50E4 Pa to 3.50E5 Pa	1 + 2.5E-05p	Absolute, gas
	5.00E5 Pa to 5.00E7 Pa	12 + 2.5E-05p	Gauge, oil
	2.00E8 Pa to 5.00E8 Pa	1.2E-04p	Gauge, oil
	0 Pa to 1.50E4 Pa	0.02 + 4E-05*p*	Negative gauge, gas
	1.50E4 Pa to 1.00E5 Pa	0.5 + 2.2E-05p	Gauge, gas
	5.00E-4 Pa to 0.5 Pa	3E-05 + 3E-02*p*	Absolute, gas
France	0 Pa to 2.00E7 Pa	10 +1.5E-06p_{line} + 2E-05p	Differential, gas
	2.00E8 Pa to 1.00E9 Pa	120 + 1.5E-05p + 1E-13p^2	Gauge, oil
	1.00E-3 Pa to 1 Pa	7E-06 + 5E-03*p*	Absolute, gas
	3500 Pa to 1.00E5 Pa	0.2 + 1.9E-05p	Absolute, gas
	1.00E6 Pa to 1.00E7 Pa	2.9 + 8E-06p + 9E-14p^2	Gauge, gas
	1.00E7 Pa to 2.00E7 Pa	10 + 2.1E-05p	Gauge, gas
	1 Pa to 1000 Pa	4E-03 + 1.3E-03*p*	Absolute, gas
	1000 Pa to 3500 Pa	1 + 1.3E-03*p*	Absolute, gas
	1000 Pa to 1.00E4 Pa	1 + 1.3E-03p	Gauge, gas
	0 Pa to 1000 Pa	4E-03 + 1.3E-03p	Gauge, gas
	0 Pa to 4.00E6 Pa	6 + 1.5E-06p_{line} + 2.5E-05p	Gauge, gas
	2.00E7 Pa to 4.00E7 Pa	60 + 3.6E-05p	Gauge, gas
	1.00E4 Pa to 1.00E6 Pa	0.1 + 7E-06p	Gauge, gas
	1.00E-6 Pa to 1.00E-3 Pa	1E-07 + 5E-02*p*	Absolute, gas

(continued)

Table 3 (continued)

Countries, NMI	Pressure ranges and approved CMCs	Reported measurement uncertainties through equation (pressure p, p_l, p_{line}, and D_p are in Pa, if not specified)	Mode and medium of pressure
	4.00E7 Pa to 8.00E7 Pa	100 + 3.1E-05p	Gauge, gas
	2.00E5 Pa to 1.00E7 Pa	5.7 + 8E-06p + 9E-14p^2	Gauge, oil
	1.00E5 Pa to 1.00E6 Pa	0.2 + 7E-06p	Absolute, gas
	0 Pa to 5.00E5 Pa	1.1 + (1E-11p_{line} + 4E-05)p	Differential, gas
	1.00E7 Pa to 2.00E8 Pa	50 + 1E-05p + 1E-13p^2	Gauge, oil
Georgia	1.00E5 Pa to 7.00E6 Pa	1.8E3 Pa	Absolute, gas
	0 Pa to 2.00E6 Pa	6.3E2 Pa	Gauge, gas
Germany	1.00E4 Pa to 1.00E5 Pa	4.2E-04*p*	Absolute, gas
	1.00E-7 Pa to 0.02 Pa	0.030*p* to 0.0066*p*	Absolute, gas
	4.00E8 Pa to 1.00E9 Pa	2.6E-05p + 1E-13p^2	Gauge, oil
	2.00E6 Pa to 1.00E7 Pa	1.6E-05p + 2E-13p^2	Gauge, gas
	1.00E-9 Pa to 1.00E-7 Pa	0.10*p* to 0.030*p*	Absolute, gas
	0 Pa to 4.00E7 Pa	10 + 4E-06p_{line} + 2.4E-05p	Differential, gas
	0 Pa to 1.00E4 Pa	0.6 + 3E-08p_{line} + 3E-04*p*	Gauge, gas
	1.00E7 Pa to 4.00E8 Pa	2E-05p + 1E-13p^2	Gauge, oil
	1.00E9 Pa to 1.40E9 Pa	4E + 06,	Gauge, oil
	1.00E7 Pa to 1.00E8 Pa	2.2E-05p + 2E-13p^2	Gauge, gas
	1.80E5 Pa to 1.00E6 Pa	0.2 + 8E-06p	Gauge, gas
	1.00E6 Pa to 2.00E6 Pa	1E-05p	Gauge, gas
	30 Pa to 1.00E4 Pa	(2.8E-03 + 2.85E-09p^2)$^{0.5}$	Absolute, gas
	9.00E-4 Pa to 1 Pa	0.0087*p* to 0.0026*p*	Absolute, gas
	3.50E5 Pa to 7.00E6 Pa	1.6E-05p + 2E-13p^2	Absolute, gas
	-4000 Pa to 4000 Pa	0.02 + 5E-05*p*	Gauge, gas
	100 Pa to 1.80E5 Pa	0.06 + 6E-06p	Absolute, gas
	1 Pa to 30 Pa	0.0024*p* to 0.0016*p*	Absolute, gas
	−1.00E5 Pa to −4000 Pa	0.16 + 1E-05*p*	Gauge, gas
	1.80E5 Pa to 3.50E5 Pa	0.2 + 1.2E-05p	Absolute, gas
	0 Pa to 2.00E7 Pa	10 + 4E-06pline +2.2E-05p,	Differential, gas
	4000 Pa to 1.80E5 Pa	0.16 + 6E-06p	Gauge, gas
	1.00E5 Pa to 1.00E7 Pa	5 + 1E-05p	Gauge, oil
Hong Kong, China	1 kPa to 3.5 kPa	1E-04P, P in kPa	Absolute, gas
	3.5 kPa to 7000 kPa	3.5E-05P, P in kPa	Gauge, gas
	0.5 MPa to 2.3 MPa	2E-04P, P in MPa	Gauge, oil
	2.3 MPa to 84 MPa	4E-05P, P in MPa	Gauge, oil
	84 MPa to 140 MPa	6E-05P, P in MPa	Gauge, oil
	3.5 kPa to 7000 kPa	3.5E-05P, P in kPa	Absolute, gas
Hungary	3.50E4 Pa to 7.00E5 Pa	2.30E-05p	Absolute, gas
	1 Pa to 3500 Pa	1.10E-01 + 9.50E-05*p*	Gauge, gas
	3500 Pa to 2.00E5 Pa	3.00E-01 + 2.50E-05p	Gauge, gas

(continued)

Table 3 (continued)

Countries, NMI	Pressure ranges and approved CMCs	Reported measurement uncertainties through equation (pressure p, p_l, p_{line}, and D_p are in Pa, if not specified)	Mode and medium of pressure
	7.00E5 Pa to 7.00E6 Pa	3.60E-05p	Absolute, gas
	10 Pa to 3.50E4 Pa	2.1 + 4.00E-05*p*	Absolute, gas
	1 Pa to 10 Pa	8.00E-02*p*	Absolute, vacuum
	7.00E5 Pa to 1.00E7 Pa	3.90E-05p	Gauge, gas
	1.20E8 Pa to 4.00E8 Pa	4.50E-04p	Gauge, oil
	2.00E5 Pa to 7.00E5 Pa	2.60E-05p	Gauge, gas
	3500 Pa to 1.00E5 Pa	3.00E-01 + 2.50E-05p	Negative gauge, gas
	6.00E7 Pa to 1.20E8 Pa	1.00E-04p	Gauge, oil
	1 Pa to 3500 Pa	1.10E-01 + 9.50E-05*p*	Negative gauge, gas
	1000 Pa to 9.00E6 Pa	1.00E-04p + 1.60E-06p_{line}	Differential, gas
	3.00E5 Pa to 6.00E6 Pa	7.20E-05p	Gauge, oil
	6.00E6 Pa to 6.00E7 Pa	7.20E-05p	Gauge, oil
	1.00E-3 Pa to 1 Pa	4.00E-02*p*	Absolute, vacuum
India	6.5 kPa to 360 kPa	$[(1.4E\text{-}01)^2 + (12E\text{-}06p)^2]^{1/2}$	Absolute, gas
	0.036 MPa to 4 MPa	22E-06p	Gauge, gas
	30 kPa to 150 kPa	$(2.5E\text{-}06p_l)^2 + (2.9E\text{-}05D_p)^2]^{1/2}$	Differential, gas
	12 MPa to 20 MPa	33E-06p	Gauge, gas
	4 MPa to 8 MPa	26E-06p	Gauge, gas
	200 MPa to 500 MPa	135E-06p	Gauge, oil
	100 MPa to 200 MPa	50E-06p	Gauge, oil
	8 MPa to 12 MPa	32E-06p	Gauge, gas
	1.00E-3 kPa to 130 kPa	$[(9.2E\text{-}03)^2 + (7.2E\text{-}06p)^2]^{1/2}$	Gauge, gas
	0.1 MPa to 50 MPa	45E-06p	Gauge, oil
	50 MPa to 100 MPa	48E-06p	Gauge, oil
	1.00E-3 kPa to 130 kPa	$[(9.2E\text{-}03)^2 + (7.2E\text{-}06p)^2]^{1/2}$	Absolute, gas
	0.05 Pa to 10 Pa	4E-03*p*	Absolute, gas,
	500 MPa to 1000 MPa	250E-06p	Gauge, oil
	20 MPa to 40 MPa	36E-06p	Gauge, gas
	20 kPa to 360 kPa	12E-06p	Gauge, gas
	3.00E-6 Pa to 0.1 Pa	2E-02*p*	Absolute, gas,
Indonesia	0.5 MPa to 5 MPa	1E-04p	Gauge, oil
	5 MPa to 50 MPa	2E-04p	Gauge, oil
	50 MPa to 100 MPa	2.8E-04p	Gauge, oil
		p pressure in MPa	
Italy	6400 Pa to 7.00E6 Pa	0.5 + 2.4E-05p	Gauge, gas
	6400 Pa to 7.00E6 Pa	0.5 + 2.4E-05p	Absolute, gas
	7.00E6 Pa to 2.00E7 Pa	0.5 + 2.8E-05p	Gauge, gas
	2.00E6 Pa to 1.00E8 Pa	30 + 2.8E-05p	Gauge, oil
	5.00E7 Pa to 5.00E8 Pa	$1.4E\text{-}13p^2$ + 3.0E-05p + 720	Gauge, oil

(continued)

Table 3 (continued)

Countries, NMI	Pressure ranges and approved CMCs	Reported measurement uncertainties through equation (pressure p, p_l, p_{line}, and D_p are in Pa, if not specified)	Mode and medium of pressure
	3 Pa to 1.50E4 Pa	3E-05p + 0.02	Gauge, gas
	3 Pa to 1.50E4 Pa	3E-05p + 0.02	Negative gauge, gas
	3 Pa to 1.50E4 Pa	3E-05p + 0.02	Absolute, gas
Japan	5 kPa to 175 kPa	400 + 13p, *p in kPa*	Absolute, gas
	5 kPa to 175 kPa	100 + 14p, p in kPa	Gauge, gas
	175 kPa to 7000 kPa	400 + 20p, *p in kPa*	Absolute, gas
	175 kPa to 7000 kPa	20p, *p in kPa*	Gauge, gas
	7 MPa to 70 MPa	40p, *p in kPa*	Absolute, gas
	1 MPa to 500 MPa	100 + 20p + $0.2p^2$, *p in MPa*	Gauge, oil
	10 Pa to 150 Pa	0.18 %	Absolute, vacuum
	0.1 Pa to 1 Pa	2.8 %	Absolute, vacuum
	1.00E-6 Pa to 1.00E-4 Pa	4.3 % to 3 %	Absolute, vacuum
	1.00E-3 Pa to 10 Pa	0.38 % to 0.32 %	Absolute, vacuum
	1 Pa to 10 Pa	0.36 %	Absolute, vacuum
	1 Pa to 1.00E4 Pa	11 + 14*p, p in kPa*	Differential, gas
	1.00E-4 Pa to 1.00E-3 Pa	0.91 %	Absolute, vacuum
Malaysia	175 kPa to 700 kPa	2.50E-05	Absolute, gas
	175 kPa to 700 kPa	2.50E-05	Gauge, gas
	0.7 MPa to 7 MPa	4.50E-05	Absolute, gas
	0.7 MPa to 7 MPa	4.50E-05	Gauge, gas
	0.345 MPa to 50 MPa	5.00E-05	Gauge, oil
	50 MPa to 138 MPa	6.00E-05	Gauge, oil
	10 kPa to 175 kPa	2.5E-5	Absolute, gas
	10 kPa to 175 kPa	2.5E-5	Gauge, gas
Mexico	1 kPa to 120 kPa	0.5–2.5 Pa	Absolute, gas
	5.00E-3 kPa to 150 kPa	15E-06 p_l + 15E-06 p_s	Differential, gas
	10 kPa to 175 kPa	1.50E-05	Gauge, gas
	10 kPa to 175 kPa	1.50E-05	Absolute, gas
	175 kPa to 350 kPa	1.80E-05	Gauge, gas
	175 kPa to 350 kPa	1.80E-05	Absolute, gas
	0.35 MPa to 1.75 MPa	2.60E-05	Gauge, gas
	0.35 MPa to 1.75 MPa	2.60E-05	Absolute, gas
	7 MPa to 80 MPa	5.00E-05	Gauge, oil
	0.2 MPa to 100 MPa	4.50E-05	Gauge, oil
	100 MPa to 200 MPa	5.00E-05	Gauge, oil
	200 MPa to 500 MPa	2.00E-04	Gauge, oil
	1.00E-4 Pa to 1 Pa	9.0E-3	Absolute, gas
	10 Pa to 100 Pa	2.5E-3	Absolute, gas
	1.75 MPa to 7 MPa	3.8E-5	Gauge, gas

(continued)

Table 3 (continued)

Countries, NMI	Pressure ranges and approved CMCs	Reported measurement uncertainties through equation (pressure p, p_l, p_{line}, and D_p are in Pa, if not specified)	Mode and medium of pressure
	1.75 MPa to 7 MPa	3.8E-5	Absolute, gas
	1 Pa to 10 Pa	3.5E-3	Absolute, gas
Netherlands	5000 Pa to 3.50E5 Pa	0.2 + 3E-05p	Absolute, gas
	5000 Pa to 5.00E5 Pa	0.5 + 3E-05p	Gauge, gas
	5.00E5 Pa to 2.00E7 Pa	1 + 4E-05p	Gauge, gas
	1.00E6 Pa to 8.00E7 Pa	5E-05p	Gauge, oil
	8.00E7 Pa to 5.00E8 Pa	1E-04p	Gauge, oil
New Zealand	35 kPa to 350 kPa	2.00E-05	Absolute, gas
	2 kPa to 350 kPa	2.00E-05	Gauge, gas
	350 kPa to 3500 kPa	6.00E-05	Absolute, gas
	350 kPa to 1.10E4 kPa	6.00E-05	Gauge, gas
	0.2 MPa to 17 MPa	1E-04 + 6.6E- 05p	Gauge, oil
	17 MPa to 280 MPa	$6.6E\text{-}05p + 7E\text{-}07p^2$	Gauge, oil
	2 kPa to 35 kPa	1.3E-5	Absolute, gas
	1 Pa to 9000 Pa	6E-03 + 4.5E-05*p*	Differential, gas
Paraguay	2.00E5 Pa to 1.00E8 Pa	1.98E-04p + 2.14E +2	Gauge, oil
Peru	0.3 MPa to 1 MPa	6.00E-05	Gauge, oil
	1 MPa to 84.5 MPa	5.00E-05	Gauge, oil
Philippines	0.2 MPa to 20 MPa	7.1E-05p	Gauge, gas
Poland	3500 Pa to 1.70E5 Pa	0.2 + 2.5E-05p	Gauge, gas
	1.70E5 Pa to 7.00E6 Pa	3.0E-05p	Gauge, gas
	1.70E5 Pa to 7.00E6 Pa	3.0E-05p	Absolute, gas
	2.00E5 Pa to 1.00E7 Pa	6 + 3.0E-05p	Gauge, oil
	1.00E7 Pa to 1.00E8 Pa	$3.0E\text{-}05p + 2.1E\text{-}13p^2$	Gauge, oil
	100 Pa to 3500 Pa	0.14 Pa to 0.31 Pa	Gauge, gas
	3500 Pa to 1.70E5 Pa	0.2 + 2.5E-05p	Absolute, gas
	1.00E8 Pa to 2.50E8 Pa	2.0E-04p	Gauge, oil
Portugal	3500 Pa to 1.70E5 Pa	0.4 + 3.5E-05p	Absolute, gas
	3500 Pa to 1.70E5 Pa	3.0E-05p	Gauge, gas
	1.70E5 Pa to 2.00E6 Pa	3.0E-05p	Gauge, gas
	2.00E5 Pa to 1.00E7 Pa	$[(27 + (5.8E\text{-}05p)2]^{1/2}$	Gauge, oil
	1.00E7 Pa to 1.00E8 Pa	5.5E-05p	Gauge, oil
	-1.00E5 Pa to -500 Pa	3.3E1 Pa	Gauge, gas
	1.00E8 Pa to 1.00E9 Pa	1.5E-04p	Gauge, oil
Republic of Korea	0 MPa to 20 MPa	$10 + 3.5E\text{-}05p_{line} + 3.5E\text{-}05p_{diff}$	Differential, oil
	20 kPa to 350 kPa	0.2 + 2.0E-05p	Absolute, gas
	20 kPa to 350 kPa	2.0E-05p	Gauge, gas
	350 kPa to 7000 kPa	3.00E-05p	Absolute, gas
	0.35 MPa to 20 MPa	3.5E-05p	Gauge, gas

(continued)

Table 3 (continued)

Countries, NMI	Pressure ranges and approved CMCs	Reported measurement uncertainties through equation (pressure p, p_l, p_{line}, and D_p are in Pa, if not specified)	Mode and medium of pressure
	0.05 MPa to 5 MPa	3.5E-05p	Gauge, oil
	5 MPa to 100 MPa	4.0E-05p	Gauge, oil
	100 MPa to 500 MPa	9.0E-05p	Gauge, oil
	5.00E-7 Pa to 1.00E-3 Pa	0.01p to 0.06p	Absolute, vacuum
	0.01 kPa to 20 kPa	4.0E-2 Pa to 4 Pa	Absolute, vacuum
	1.00E-3 Pa to 0.01 Pa	0.01p	Absolute, vacuum
Romania	1.00E4 Pa to 3.50E5 Pa	6.00E-05	Absolute, gas
	1500 Pa to 1.00E6 Pa	6.00E-05	Gauge, gas
	1.00E6 Pa to 3.50E6 Pa	6.00E-05	Gauge, gas
	2.00E5 Pa to 5.50E7 Pa	6.00E-05	Gauge, oil
	5.50E7 Pa to 1.10E8 Pa	8.00E-05	Gauge, oil
	–1.00E5 Pa to –1500 Pa	6.0E-5	Gauge, gas
	10 Pa to 1500 Pa	0.1 Pa	Gauge, gas
Russian Federation	270 Pa to 1.30E5 Pa	6.7 Pa	Absolute, gas
	4.00E4 Pa to 6.30E5 Pa	2.00E-04	Gauge, gas
	1.30E5 Pa to 7.20E5 Pa	5–36 Pa	Absolute, gas
	5.00E4 Pa to 1.00E7 Pa	8E-06 to 2E-05	Gauge, oil
	1.00E7 Pa to 6.00E7 Pa	5.00E-05	Gauge, oil
	6.00E7 Pa to 2.50E8 Pa	5.00E-04	Gauge, oil
	2.50E8 Pa to 1.60E9 Pa	3.00E-04	Gauge, oil
	5000 Pa to 4.00E4 Pa	1.1 Pa	Differential, gas
	–9.50E4 Pa to 4.00E4 Pa	2.0E1 Pa	Gauge, gas
	0.1 Pa to 100 Pa	0.1 Pa	Differential, oil
	0.1 Pa to 1000 Pa	7.2E-03 + 1.0E-04*p*	Absolute, gas
	1.00E-3 Pa to 1000 Pa	1.0E-4 Pa to 6.6 Pa	Absolute, gas
	100 Pa to 5000 Pa	0.34 Pa	Differential, oil
	1000 Pa to 1.00E4 Pa	8.0E-01 + 1.0E-05*p*	Absolute, gas
	1.00E-7 Pa to 1.00E-3 Pa	7.0E-9 Pa to 4.0E-5 Pa	Absolute, gas
Saudi Arabia	350 kPa to 7000 kPa	45 +1.7E-5p	Gauge, gas
Singapore	1.00E4 Pa to 8.00E6 Pa	1.00E-04	Absolute, gas
	2.00E4 Pa to 6.00E6 Pa	6.50E-05	Gauge, gas
	6.00E6 Pa to 8.00E6 Pa	1.00E-04	Gauge, gas
	2.00E6 Pa to 1.00E7 Pa	1.00E-04	Gauge, oil
	1.00E7 Pa to 1.00E8 Pa	5.00E-05	Gauge, oil
	1.00E8 Pa to 2.00E8 Pa	8.00E-05	Gauge, oil
	2.00E8 Pa to 4.00E8 Pa	1.30E-04	Gauge, oil
	5000 Pa to 1.00E4 Pa	2.0E-4	Absolute, gas
	6.70E-4 Pa to 0.13 Pa	5.4E-2	Absolute, gas
	5000 Pa to 2.00E4 Pa	1.0E-4	Gauge, gas

(continued)

Table 3 (continued)

Countries, NMI	Pressure ranges and approved CMCs	Reported measurement uncertainties through equation (pressure p, p_l, p_{line}, and D_p are in Pa, if not specified)	Mode and medium of pressure
	1 Pa to 1.30E4 Pa	1 Pa	Differential, gas
	0.13 Pa to 1300 Pa	3.3E-2	Absolute, gas
Slovakia	3000 Pa to 3.00E6 Pa	2.8E-05p	Absolute, gas
	3000 Pa to 7.00E6 Pa	2.8E-05p	Gauge, gas
	2.50E5 Pa to 6.00E6 Pa	2.2E-05p to 3.2E05p	Gauge, oil
	6.00E6 Pa to 2.00E7 Pa	3.2E-05p	Gauge, oil
	2.00E7 Pa to 1.00E8 Pa	4.8E-05p	Gauge, oil
Slovenia	3500 Pa to 2.20E5 Pa	3.2E-05p + 0.3	Absolute, gas
	3500 Pa to 2.20E5 Pa	3.2E-05p + 0.2	Gauge, gas
	2.20E5 Pa to 7.00E6 Pa	3.2E-05p + 3	Absolute, gas
	2.20E5 Pa to 7.00E6 Pa	3.2E-05p + 3	Gauge, gas
	0 Pa to 7.00E6 Pa	1E-04p + 0.6	Differential, gas
	5.00E6 Pa to 2.00E8 Pa	$3.8E\text{-}05p+1.8E\text{-}13p^2 + 100$	Gauge, oil
	1.00E-5 Pa to 0.5 Pa	$0.013p + 2.4E\text{-}06$	Absolute, vacuum
	0.5 Pa to 1300 Pa	$6E\text{-}03p + 3E\text{-}03$	Absolute, vacuum
	1300 Pa to 3500 Pa	5 Pa	Absolute, vacuum
	3.00E5 Pa to 6.00E6 Pa	$50+7E\text{-}05p$	Gauge
South Africa	1.4 kPa to 180 kPa	0.1 + 2.5E-05p	Gauge, gas
	180 kPa to 700 kPa	1 + 2.8E-05p	Absolute, gas
	0.7 MPa to 7 MPa	1 + 5E-05p	Absolute, gas
	1 MPa to 120 MPa	5.4E-05 + 5.8E-05p	Gauge, oil
	0.1 MPa to 1 MPa	1.4E-04 + 4.4E-05p, p in MPa	Gauge, oil
	120 MPa to 500 MPa	$2.3E\text{-}03–3.8E\text{-}07p +5E\text{-}07p^2$, p in MPa	Gauge, oil
	700 kPa to 1000 kPa	2.0 + 4.5E-05p	Gauge, gas
	180 kPa to 700 kPa	2.9E-05p	Gauge, gas
	1.4 kPa to 180 kPa	$0.1 + 2.5E\text{-}05p$	Gauge, gas
Spain	3500 Pa to 1.70E5 Pa	3.1E-05p	Absolute, gas
	1.70E5 Pa to 7.00E5 Pa	2.7E-05p	Absolute, gas
	5.00E4 Pa to 2.00E6 Pa	1.6E-05p	Gauge, gas
	2.00E6 Pa to 2.00E7 Pa	2.4E-05p	Gauge, gas
	2.00E7 Pa to 5.00E7 Pa	4.0E-05p	Gauge, gas
	5.00E7 Pa to 7.00E7 Pa	5.0E-05p	Gauge, gas
	1.00E5 Pa to 5.00E5 Pa	8.0E-05p + 10	Gauge, oil
	5.00E5 Pa to 2.00E6 Pa	3.5E-05p + 10	Gauge, gas
	7.00E5 Pa to 7.00E6 Pa	3.0E-05p	Absolute, gas
	7.00E6 Pa to 2.00E7 Pa	3.0E-05p	Absolute, gas
	2.00E6 Pa to 1.00E8 Pa	3.5E-05p	Gauge, oil
	1.00E8 Pa to 2.00E8 Pa	4.0E-05p	Gauge, oil
	1.00E5 Pa to 5.00E8 Pa	1.0E-04p + 10	Absolute, oil

(continued)

Table 3 (continued)

Countries, NMI	Pressure ranges and approved CMCs	Reported measurement uncertainties through equation (pressure p, p_l, p_{line}, and D_p are in Pa, if not specified)	Mode and medium of pressure
	2.00E8 Pa to 5.00E8 Pa	9.5E-05p	Gauge, oil
	5.00E8 Pa to 1.00E9 Pa	2.3E-04p	Gauge, oil
	100 Pa to 5.00E6 Pa	1 + 5.00E-05p	Differential, gas
	1.00E-4 Pa to 0.05 Pa	3.5E-02*p* + 9.1E-06	Absolute, vacuum
	0.05 Pa to 0.1 Pa	3.6E-02*p*	Absolute, vacuum
	20 Pa to 3500 Pa	2.0E-04*p* + 9.0E-02	Gauge, gas
	0.1 Pa to 23 Pa	8.2E-03*p* + 1.8E-03	Absolute, vacuum
	23 Pa to 1000 Pa	5.4E-03*p* + 7.0E-02	Absolute, vacuum
	20 Pa to 3500 Pa	2.0E-04*p* + 9.0E-02	Negative gauge, gas
	3500 Pa to 5.00E4 Pa	3.0E-05*p*	Gauge, gas
	3500 Pa to 1.00E5 Pa	3.1E-05*p*	Negative gauge, gas
	1000 Pa to 3500 Pa	5 Pa	Absolute, vacuum
Sweden	3500 Pa to 1.70E5 Pa	0.2 + 2.5E-05p	Gauge, gas
	3500 Pa to 1.70E5 Pa	0.5 + 3E-05p	Absolute, gas
	1.70E4 Pa to 7.00E5 Pa	1.0 + 3.0E-05p	Gauge, gas
	1.70E4 Pa to 7.00E5 Pa	1.2 + 3.0E-05p	Absolute, gas
	7.00E5 Pa to 7.40E6 Pa	4.6 + 3.0E-05p	Gauge, gas
	7.00E5 Pa to 7.40E6 Pa	4.6 + 3.0E-05p	Absolute, gas
	7.40E6 Pa to 4.00E7 Pa	2.7 + 5.0E-05p	Gauge, gas
	5.00E5 Pa to 5.00E7 Pa	3.7E-05p + 1.7E-13p^2	Gauge, oil
	5.00E7 Pa to 5.00E8 Pa	5.2E-05p + 2E-13p^2	Gauge, oil
	5.00E-6 Pa to 5.00E-4 Pa	12E-02*p*	Absolute, gas
	1 Pa to 1.50E4 Pa	0.08 + 7E-05*p*	Differential, gas
	3 Pa to 133 Pa	0.12 + 6.0E-03*p*	Absolute, gas
	5.00E-4 Pa to 3 Pa	3.0E-05 + 2.7E-02*p*	Absolute, gas
	133 Pa to 3500 Pa	1.2 + 9E-04*p*	Absolute, gas
Switzerland	7000 Pa to 3.80E5 Pa	1.50E-05p + 0.5	Absolute, gas
	7000 Pa to 3.80E5 Pa	1.50E-05p + 0.5	Gauge, gas
	1.00E5 Pa to 4.00E6 Pa	5.00E-05p + 20	Gauge, gas
	1.00E5 Pa to 4.00E6 Pa	5.00E-05p + 20	Differential, gas
	4.00E6 Pa to 5.00E7 Pa	1.00E-04p	Gauge, gas
	5.00E7 Pa to 5.00E8 Pa	5.00E-05p + 3.0E- 13p^2	Gauge, oil
	5.00E8 Pa to 1.00E9 Pa	5.00E-04p	Gauge, oil
	1.00E-4 Pa to 1 Pa	0.020*p* to 0.0036*p*	Absolute, gas
	1 Pa to 7000 Pa	7.00E-05*p* + 0.4	Absolute, gas
Thailand	3.50E5 Pa to 1.75E6 Pa	2.6E-05p	Absolute, gas
	1.75E6 Pa to 7.00E6 Pa	2E-05p	Gauge, gas
	1.00E5 Pa to 7.10E6 Pa	3E-05p	Absolute, oil
	7.10E6 Pa to 7.00E7 Pa	4E-05p	Absolute, oil

(continued)

Table 3 (continued)

Countries, NMI	Pressure ranges and approved CMCs	Reported measurement uncertainties through equation (pressure p, p_l, p_{line}, and D_p are in Pa, if not specified)	Mode and medium of pressure
	70.1 MPa to 140.1 MPa	5E-05p, p in MPa	Absolute, oil
	3.50E5 Pa to 1.75E6 Pa	1.5E-05p	Gauge, gas
	3.50E5 Pa to 1.75E6 Pa	2.2E-05p	Gauge, gas
	1.75E6 Pa to 7.00E6 Pa	2.5E-05p	Gauge, gas
	1.75E6 Pa to 7.00E6 Pa	3.0E-05p	Absolute, gas
	5.70E5 Pa to 7.00E6 Pa	3E-05p	Gauge, oil
	7.00E6 Pa to 7.00E7 Pa	4E-05p	Gauge, oil
	7.00E7 Pa to 1.40E8 Pa	5E-05p	Gauge, oil
	1.40E8 Pa to 2.00E8 Pa	6.5E-05p	Gauge, oil
	2.00E8 Pa to 5.00E8 Pa	1.23E-04p	Gauge, oil
	3500 Pa to 1.00E4 Pa	0.8 Pa	Gauge, gas
	3500 Pa to 1.00E4 Pa	0.8 Pa	Absolute, gas
	1.00E4 Pa to 3.50E5 Pa	1.3E-05*p*	Gauge, gas
	1.00E4 Pa to 3.50E5 Pa	2.4E-05*p*	Absolute, gas
	1.00E4 Pa to 3.50E5 Pa	2.0E-05*p*	Gauge, gas
	−1000 Pa to 3000 Pa	3.0E-04*p*	Gauge, gas
United Kingdom	1.40E8 Pa to 2.80E8 Pa	$3.0E\text{-}05p + 1.8E\text{-}13p^2$	Gauge, oil
	2.80E8 Pa to 5.00E8 Pa	$3.2E\text{-}05p + 2E\text{-}13p^2$	Gauge, oil
	7.00E6 Pa to 2.00E7 Pa	$2.8E\text{-}05p + 2.6E\text{-}13p^2$	Gauge, gas
	5.00E5 Pa to 7.00E6 Pa	$3.1E\text{-}05p + 1.9E\text{-}13p^2$	Gauge, oil
	3500 Pa to 1.60E4 Pa	2.3E-05*p*	Absolute, gas
	3500 Pa to 1.60E4 Pa	2.2E-05*p*	Gauge, gas
	7.00E5 Pa to 7.00E6 Pa	$2.4E\text{-}05p + 2.4E\text{-}13p^2$	Gauge, gas
	1.60E4 Pa to 7.00E5 Pa	2.0E-05p	Gauge, gas
	7.20E5 Pa to 7.00E6 Pa	$2.5E\text{-}05p + 2.4E\text{-}13p^2$	Absolute, gas
	1.60E4 Pa to 7.20E5 Pa	2.1E-05p	Absolute, gas
United States	1 Pa to 30 Pa	0.003 + 6.00E-05*p*	Absolute, gas
	0.1 Pa to 4 Pa	$[(2E\text{-}03)^2 + (1.4E\text{-}03p)^2]^{1/2}$	Differential, gas
	3.00E-6 Pa to 9.00E-5 Pa	0.0048 + 2.00E-08/*p*	Absolute, gas
	0.3 MPa to 1.4 MPa	$[(1.03E\text{-}05)^2 + (3.8E\text{-}12p)^2]^{1/2}$	Absolute, gas
	1.4 MPa to 7 MPa	$[(1.28E\text{-}05)^2 + (2.0E\text{-}12p)^2]^{1/2}$	Absolute, gas
	1.4 MPa to 7 MPa	$[(1.24E\text{-}05)^2 + (2.0E\text{-}12p)^2]^{1/2}$	Gauge, gas
	1.00E-4 Pa to 1 Pa	3.0E-3	Absolute, gas
	1.00E-7 Pa to 3.00E-6 Pa	0.012 + 5.00E-10/*p*	Absolute, gas
	26 MPa to 272 MPa	3.20E-05	Gauge, oil
	27 kPa to 300 kPa	$[(0.15/p)^2 + (1.04E\text{-}05)^2]^{1/2}$	Absolute, gas
	3 Pa to 140 Pa	$(4E\text{-}03)^2 + (36E\text{-}06p)^2]^{1/}$	Absolute, gas
	0.1 Pa to 3 Pa	$[(8E\text{-}04)^2 + (1E\text{-}03p)^2]^{1/2}$	Absolute, gas
	4.00E-3 kPa to 10 kPa	$[(6E\text{-}03)^2 + (6.4E\text{-}06p)^2]^{1/2}$	Differential, gas

(continued)

Table 3 (continued)

Countries, NMI	Pressure ranges and approved CMCs	Reported measurement uncertainties through equation (pressure p, p_l, p_{line}, and D_p are in Pa, if not specified)	Mode and medium of pressure
	1 MPa to 26 MPa	2.20E-05	Gauge, oil
	9.00E-5 Pa to 3.00E-3 Pa	0.0042 + 7.00E-08/p	Absolute, gas
	0.1 kPa to 360 kPa	$[(6E\text{-}03)^2 + (5.2E\text{-}06p)^2]^{1/2}$	Gauge, gas
	3.00E-3 Pa to 0.1 Pa	4.2E-3	Absolute, gas
	27 kPa to 300 kPa	$[(0.15/p)^2 + (9.8E\text{-}06)^2]^{1/2}$	Gauge, gas
	0.14 kPa to 160 kPa	$[(6E\text{-}03)^2 + (5.2E\text{-}06p)^2]^{1/2}$	Absolute, gas
	7 MPa to 17 MPa	$(1.32E\text{-}05)^2 + (2.24E\text{-}12p)^2]^{1/2}$	Gauge, gas
	0.1 kPa to 360 kPa	$[(6E\text{-}03)^2 + (5.2E\text{-}06p)^2]^{1/2}$	Absolute, gas
	0.3 MPa to 1.4 MPa	$[(9.7E\text{-}06)^2 + (3.8E\text{-}12p)^2]^{1/2}$	Gauge, gas
Uruguay	1.00E5 Pa to 7.00E6 Pa	120 + 4 E-5p	Gauge, gas

Conclusion

Nowadays, physical quantity pressure is a very important quantity with which we encounter in daily life like tire pressure, hydraulic breaks, hydraulic power in different machines, etc. Many instruments are available to measure pressure. The accurate measurement of pressure plays an important role in many industries such as automobiles, aerospace, steel, chemical, medical, and manufacturing. The accurate pressure measurement is important for quality and safety requirements. No single instrument can measure the pressure in the entire range of interest with the same accuracy and reproducibility. "Pressure" and "temperatures" play a major role in the fields of "pipeline constructions," "water supply," "hydraulics for irrigation," "thermal power," "hydraulic power," "aviation of flights," "human blood," etc. In this chapter, I dealt with study of pressure and its various pressure-measuring instruments. Reader can understand basics of pressure, pressure units and its conversion, various types of pressure, commonly used pressure-measuring instruments, and pressure sensor's metrological characteristics. Also, reader can find the basic definitions related to metrology in section "Pressure-Measuring Instruments." The approved pressure CMCs of different countries' NMIs are compiled in this chapter. The NMIs have CMCs of different ranges starting from 0 Pa to 1400 MPa. Some countries have enough CMCs but some have very less. This chapter is highly useful for students, researchers, and stakeholders as they find most of the information related to pressure at one place. Although large numbers of equipment are available to measure the pressure in different ranges, still more research is required to develop instruments with improved accuracy, portability, and cost-effectiveness.

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Traceable Synchrophasor Data for Smart Grid Metrology and Its Application in Protection and Control

40

Avni Khatkar and Saood Ahmad

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Abstract

Recently, as the power industry is revolutionizing, the smart grid concept is gaining momentum continuously around the world. For real-time supervision, protection, and control of the behavior of the power grid, phasor measurement units (PMUs) are being installed at a large scale. This chapter presents in detail the electrical power system, the structure of the Indian power grid, synchrophasor technology, wide area monitoring system (WAMS), PMUs in monitoring, and control of the power grid. Also, the Unified Real Time Dynamic State Measurement (URTDSM) project, which is being implemented in the Indian Power System, is elucidated, along with the details of PMU and the role of

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synchronization techniques in PMU. The IEEE synchrophasor standards associated with the measurements, compliance, and testing and the PMU calibration system of the CSIR-NPL, along with its traceability to the country's national standards, are dealt with in detail. The chapter finally ends by highlighting the implementation issues and challenges faced by the power grid and their solutions, such as load management and automation of the grid, which are essential to transform and augment the power sector into a responsive, stable, and secure system.

Keywords

Indian power sector · Synchrophasor · PMU · Calibration · IEC/IEEE 60255-118-1:2018 · Stability

Introduction

The power grid is currently evolving at a very swift rate. The modernization of the grid is full of challenges as the power grid network is very dynamic and heterogeneous in nature. The power industries across the globe are facing new cutthroat difficulties in planning, generation, operation, and development of this sophisticated infrastructure, which is capable of capturing the grid dynamics and can also support diverse requirements of varied applications. Also, as globally the power demand is rising exponentially which is putting a lot of pressure on the existing infrastructure. To increase the generation capacity of power sector and focusing on the green energy to reduce carbon dioxide emission, new renewable sources are being introduced into the grid. With the integration of these substantial generating resources in the grid, the complexity of the power grid is also intensifying. The power systems, being very sophisticated, are susceptible to blackouts and outages (Khatkar et al. 2022a), and when the instances of outages and blackouts increase in the power network, the control and protection of the grid becomes very difficult. Conventionally, supervisory control and data acquisition (SCADA) systems were used to keep track of the electrical power network (HyungSeon 2017), but these systems are protracted, nonsynchronous, and steady-state measurement systems with the inability to monitor the dynamics of the system. Hence, to observe and check the dynamics of the power system networks, WAMS is overtaking SCADA system.

WAMS is now becoming a widely accepted monitoring system, which renders addendum to the existing SCADA system. In WAMS, PMU plays a major role as it is the device that records the time-stamped synchrophasor data at a very high speed. In the twenty-first century, the PMU is attaining high importance in order to monitor, supervise, and correct the power grid (Nicorescu et al. 2019).

Here, in this chapter, complete details of the electrical power system, Indian perspective of power system, synchrophasor technology and its applications, and WAMS and its architecture, advantages, and applications along with the presence of WAMS in the Indian power grid are described in detail. Furthermore, a brief

description of PMU, its architecture, the role of different synchronization techniques in PMU, IEEE standards, CSIR-NPL traceable PMU calibration system, and the performance evaluation of PMU as per the IEEE standard are elucidated in detail in this chapter. The chapter ends with the protection systems that are possible due to synchrophasor inputs and power grid automation and load management, which are the need of the hour in order to transform and augment the power sector.

Electrical Power System

The electrical power system is described as the network of electrical components that are used to generate, transmit, distribute, and consume electric power (Agüero et al. 2017). Traditionally, electric power is delivered to the end users from the generators by using an infrastructure that consists of AC power systems. The voltage and current waveforms in the normal operating conditions of the power system are modeled as sinusoidal signals, as shown in Eq. (1):

$$x(t) = A_o.\cos(2\pi f_o t + \varphi_o) \tag{1}$$

where A_o is the nominal peak amplitude, φ_o is the initial phase at $t = 0$, and f_o is the nominal power system frequency.

The power system exhibits dynamic behavior even in normal operating conditions and is never in a steady-state operation (Tan and Novosel 2017). Hence, the parameters, as shown in Eq. 1, are hardly ever time-invariant because of the changes related to generation imbalances, large generator inertia, automatic speed control of generators, and frequency fluctuations, which are among the most evident phenomena. During faults or events of switchover, these frequency fluctuations are even more pronounced. So, based on these dynamic events, a generalized signal model incorporating all these phenomena can be expressed, as shown in Eq. (2):

$$\begin{aligned} x(t) = a(t)\cos(2\pi f(t)t + \psi_o) + \sum\nolimits_{h=1}^{H} A_h(t)\cos(2\pi f_h(t)t + \psi_h) \\ + A_{DC}(t)e^{-\frac{1}{\tau}} + \varepsilon(t) \end{aligned} \tag{2}$$

where the main tone is represented by the first term, the instantaneous frequency $f(t)$ is close to the nominal frequency f_o(50 Hz or 60 Hz), $A(t)$ represents the instantaneous peak amplitude, and the instantaneous phase is represented by $\Psi(t) = 2\pi f(t)t + \phi_o$.

It is assumed that the initial phase ϕ_o is constant, which implies that the instantaneous variations in phase are attributable only to the frequency variations of the generator, thereby removing the possibility of multiple couples of frequency that give the same $\Psi(t)$. The superposed sinusoidal tone and the harmonic effects, interharmonics, and subharmonics are modeled by the second term of Eq. (2). In the second term, $f_h(t)$ represents the instantaneous frequency of each tone, $A_h(t)$

represents the peak amplitude, and the instantaneous phase is described by $\Psi_h = 2\pi f_h(t)t + \phi_h$, where ϕ_h is the initial phase.

The peak amplitude in the case of normal operating condition is one order lower than the main tone $A(t)$. In Eq. (2), the third term represents the decaying DC component, A_{DC} being the initial amplitude and τ the arbitrary time constant. The last term $\varepsilon(t)$ expresses the wideband noise.

In electrical engineering, the phasor transformation is adopted to simplify electrical signal analysis. Eq. (1) can be rewritten as:

$$x(t) = A\ cos(2\pi ft + \Psi) \tag{3}$$

$$= R_e\left\{A.e^{j(2\pi ft+\Psi)}\right\} \tag{4}$$

$$= R_e\left\{A.e^{j\Psi(t)}\right\} \tag{5}$$

In Eqs. (3), (4), and (5), as $x(t)$ is stationary, dependency of $\psi(t)$ on time is disregarded. And then, the sinusoidal term $x(t)$ can be associated with complex number X, which is called a phasor, and can also be represented in polar or rectangular coordinate as shown in Eqs. (6) and (7):

$$x(t) \leftrightarrow X \approx A.e^{jw} \tag{6}$$

$$= X_r + jX_i \tag{7}$$

In Eq. (7), r represents the real part, and i represents the imaginary part.

Hence, phasor is a complex number that is used to represent both magnitude and phase angle of sinusoidal wave or time-varying electrical quantity. The phasors are used to describe the behavior of the power system when it is dynamic in nature. Therefore, the phasors function as basic tools for analyzing AC circuits and represent the steady-state sinusoidal waveform of fundamental power frequency.

Indian Power Grid

The power grid is an amalgamation of generation of electrical power, its transmission, and consumption (Gyanendra Singh and Tiwari 2017). The Indian power grid stands third based on power generation globally (Rao 2019). India produced 1669 TWh of energy as opposed to 8537 TWh by China and 4381 TWh by the United States in 2021. The total installed capacity of power stations in India in February 2022 was around 395.6 GW. The contribution of thermal, nuclear, and renewable sources was 235.93 GW, 6.78 GW, and 152.89 GW, respectively (Installed Capacity Report 2022).

In early sixties, the management of grid on regional basis was initiated, and the grids of the states were integrated to form a regional grid. By the integration of grids, all the regions of India were divided into five regions, namely, Northern (NR),

Eastern (ER), Western (WR), Northeastern (NER), and Southern region (SR). In 1991, taking the concept of integration further, the ER and NER grids were also connected. Moving further in March 2003, the WR and ER-NER were also synchronized. In August 2006, the Northern grid and Eastern grid were interconnected. Hence, a central grid was formed with four regional grids, i.e., NR, ER, WR, and NER, synchronously operating at one frequency. On December 13, 2013, the SR was attached to the central grid in synchronous mode. This was done with the commissioning of 765 KV Raichur-Solapur transmission line. By doing so, the grid of India achieved the target of "One Nation-One Grid-One Frequency" (Sarkar et al. 2015).

Synchrophasors

The phasor concept is a stationary hypothesis and hence cannot be applied directly for the real AC power system analysis as it does not hold good in the real world. During electromechanical oscillations or power swings, that is, when the power system is not stationary and is in dynamic state, the current and voltage variations can be treated as quasi-steady approximation, that is, the stationary hypothesis is applied, and, hence, phasor analysis of waveform can be done. Without following the sinusoidal steady-state hypothesis, it becomes impossible to approximate phasor measurements at various positions of the grid without having a common time reference because the phase of a sinusoidal waveform is dependent on time scale t, and, hence, it cannot be precisely expressed without a common time reference. The phasor concept is extended to synchrophasor, owing to this reason.

The phasor representation was represented by Steinmetz in 1893 (You et al. 2021), and, through the development of the PMU by Phadke and Thorp in 1988, the Steinmetz's technology was used for real-time phasor measurements, which are synchronized by a common time reference (Vallakati and Ranganathan 2017). These synchronized phasor measurements are referred to a synchrophasor.

The IEEE 1344-1995 (1995) first defined the synchrophasor concept as a phasor that is calculated from samples of data, which are referenced to a standard time reference. Again, the concept of synchrophasor was reviewed and defined by IEEE C37.118.1-2011 (2011) standard as instantaneous phase angle, which is relative to the cos function at nominal frequency [50 Hz or 60 Hz] and is referenced to Coordinated Universal Time. However, the definition of synchrophasor is confining and gives rise to ambiguity. Hence, an inclusive definition of synchrophasor is one which refers to both phase estimations and the amplitude referred to the common time.

So, for the signal $x(t)$ represented by Eq. (2), the correlated synchrophasor is expressed as:

$$x(t) \leftrightarrow X(t) \triangleq A(t).e^{j\Psi(t)} \tag{8}$$

$$= A(t).e^{j(2\pi f(t)t+\Psi)} \tag{9}$$

Table 1 SCADA system and synchrophasor technology (PMU) comparison

Attribute	SCADA	PMU
Observation	Steady state	Dynamic
Resolution per second (sample)	1 in 2–4 seconds	10–60
Measurements	I, |V|	I, |V|, δ, frequency, rate of change of frequency (ROCOF)
Synchronization	No	Yes
Accuracy	Low	High
Focus area	Low area	Wide area

Table 2 Synchrophasor technology applications

Offline applications	Online applications
Power system model validation	Visualization and monitoring of wide area of power system
	State estimation
Post disturbance analysis	Monitoring of voltage, instability, oscillations, frequency instability
	Fault location identification and protective relaying

In Eq. (9), $A(t)$ represents the instantaneous peak amplitude, and $\Psi(t)$ represents the instantaneous phase of $x(t)$.

Traditionally, SCADA was used to observe the power system, but these measurements were protracted and were based on steady-state analysis of the system and hence were unable to observe the dynamics of the power system (Domijan and Jain 2011). The comparison between SCADA and synchrophasor technology or PMU measurements is given in Table 1.

As, PMUs bring forth the best knowledge and precise information of the current or dynamic situation of the grid, hence are being deployed at very high rate in the grid across India as well as the whole world. In 2000, there were only 15–20 PMUs deployed across the Indian National Grid, whereas, currently, around 1700 PMUs are deployed and are providing real-time information of the Indian National Grid for wide area situational awareness.

Table 2 sheds light on the application of synchrophasor technology. The applications are varied, such as monitoring of voltage stability, angle and frequency, real-time control of power system, thermal overload monitoring, steady-state and dynamic estimations, state measurement, wide area stabilization, congestion management, restoration of system, post-event analysis, adaptive protection, and planned power system separation.

Wide Area Monitoring System (WAMS)

As the power sector is proliferating in India, the Indian power grid is forced to operate very close to its operational limits to efficiently and effectively exploit all available resources. Hence, it becomes essential for the grid operators to remain

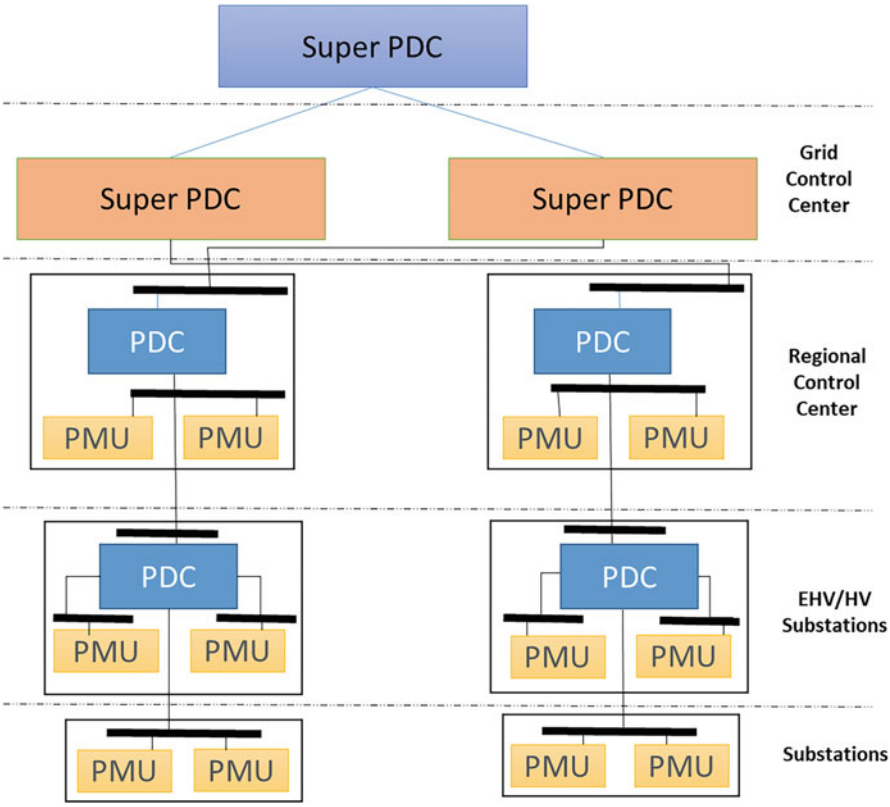

Fig. 1 Architecture of wide area monitoring system

extra vigilant and alert in observing and monitoring the health of the grid so that it functions smoothly. The situational awareness of the grid can be enhanced by deploying PMU-based WAMS (Abdullah Sufyan et al. 2018). The PMUs that are deployed over a large area and are interconnected together to monitor and control the regions are generally known as WAMS.

The WAMS network has the capability to transfer data at a very high rate generally 10–60 samples per second as opposed to one sample per 2–4 second in the existing SCADA system. Synchronized measurements are an indispensable part of the WAMS network to get the panoramic view of the complete grid. The WAMS architecture is depicted in Fig. 1. Typically, WAMS networks consist of PMUs, phasor data concentrator (PDC), and a communication network. The PMUs are optimally placed throughout the grid to get a comprehensive grid view (Saha et al. 2015).

WAMS: Importance and Application

The deployment of PMUs and the successful implementation of wide area management system are playing crucial roles in proper visualization, monitoring, and

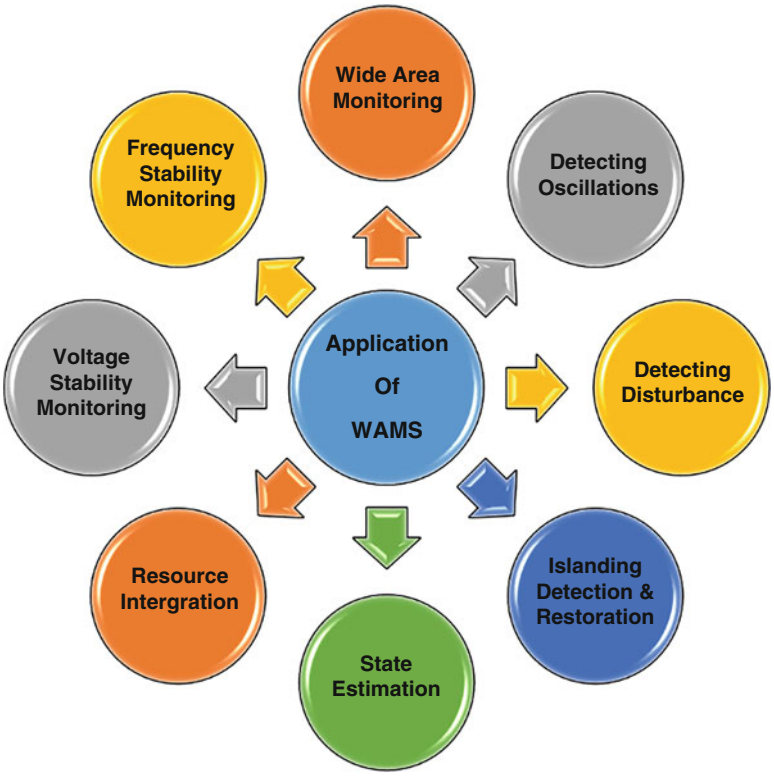

Fig. 2 Applications of wide area monitoring system

control of the power grid. The application offered by successfully implemented WAMS is summarized in Fig. 2 (Rihan et al. 2013). The main benefit offered by WAMS is that it works as a tool to provide better understanding and control of various events occurring in the grid. Because of the grid operator's real-time situational awareness, WAMS plays a critical role in preventing grid collapse. WAMS also provides better visualization of imbalances in power grid in the form of low-frequency oscillations and helps in preventing power outages and blackouts. WAMS facilitates the grid operators by being an analytical tool providing accurate and precise measurements to help secure the power grid in an effective and unerring manner. Also, WAMS helps in postmortem analysis of the grid by providing data in the offline mode (Kanabar et al. 2013).

WAMS in the Indian Power Grid

In India, the synchrophasor initiative was pioneered in the form of a pilot project in May 2010 in the Northern region of the grid. In a similar fashion, many projects were conceived in different parts of the Indian power grid, which is summarized in Table 3

Table 3 Summary of PMU and PDC installed in different pilot projects

Sr. No.	Region	No. of PMUs	No. of PDCs
1	ER	8	5
2	WR	11	1
3	NR	8	1
4	SR	6	1
5	NER	6	1
6	NLDC	18	1

(Saha et al. 2015). After the successful implementation of these pilot projects at regional levels, these projects were integrated at the national level and these synchrophasor projects communicated through PDCs at regional level to the PDC that was located at National Load Despatch Centre (NLDC), New Delhi. This PDC in New Delhi receives data from around 604 PMUs for real-time event analysis and monitoring. Also, some vendors have installed PMUs apart from the ongoing pilot projects, but these PMUs also report to the NLDC (Agrawal et al. 2011a).

By using the data from these pilot projects, monitoring classification, analysis and detection of fault, mitigation of low-frequency oscillations, synchronization, dynamic model validation, coherence group detection, islanding detection, and PSS tuning are accomplished. Gleaned from the positive insights from these WAMS pilot projects, a synchrophasor project, known as URTDSM, was envisaged and implemented in the Indian Power System in 2012 (Agarwal 2012; Agrawal et al. 2010, 2011b; Unified Real Time Dynamic Measurement System 2012; Zuhaib and Rihan 2017; Soonee et al. 2015). In URTDSM project, installation and commissioning of around 1669 PMUs across pan-India has been done. This project is put into effect in a phased manner. The PMUs that were installed earlier in the pilot projects are also integrated into URTDSM project. The PDCs that lie in the state control centers are integrated to the regional PDCs, and these are integrated to the national PDC, also known as super PDC. The URTDSM (WAMS project) covers 34 state, 5 regional and 1 national control center along with backup. The URTDSM project is providing the required wide area situational awareness to the operators. However, in order to improve the system and utilize the data properly, some improvements in the system are sought. Based on the operational experience of the existing system, Power System Operation Corporation Limited (POSOCO) has suggested some improvements for proper functioning of URTDSM project in 2021 (Power System Operation Corporation Limited 2021). This includes the improvement in the graphical user interface (GUI) or visualization of the data in real time. The phase voltage should be displayed in real time in order to identify the fault in the phase and understand the issue in real time.

Along with this, a single user interface should be used in which the operator can visualize the real-time data and can compare it with the historical data. Also, the window size to picture the real-time data should be increased. In some locations the need is felt to increase the range of monitored frequencies for which high sampling rate of PMUs is required. Also, the PDC's capability should be increased to store data of high sample rates. In the current scenario, the PDCs can store data of 25 Hz

only. In addition to this other improvements which are sought in the existing system are challenges related to interoperability aspect of PMUs of different manufacturers, development of logic based analytical tools to enhance situational awareness, non-generation of alarms failing to detect oscillations which are present in the system, high noise in PMU, which results in bad data analytics, loss of confidence in system and poor performance, frequent issues related to time synchronization and low of pulse per second (PPS) signal. Besides these challenges, some future requirements of the PMU project are analytics for this big data system to full potential of the system, techniques to store and compress historian data with minimal loss, secure and reliable communication infrastructure, event detection tool, development of real-time analytical applications, online inertia assessment, voltage stability monitoring, and automatic islanding detection (Power System Operation Corporation Limited 2021).

Phasor Measurement Unit

The PMU is a device that monitors and captures information at high rate and time-stamps each input to a common time reference (Khatkar et al. 2022a). The principal role of PMU is elucidated in Fig. 3 (Khatkar et al. 2022a). The main components of PMU are:

- Analog inputs: The current or potential transformers that are installed at sub-stations deliver the measured current and voltage values. The secondary windings of these transformers are connected as the analog input to the PMU.
- Anti-aliasing filters: These are low-pass analog filters that filter out the components having greater frequencies from the input signal to ease out phasor estimation.
- Global Positioning System (GPS): GPS offers very accurate time synchronization signal. It utilizes high-accuracy clock from the satellite to deliver synchronized time at diverse positions of the grid.

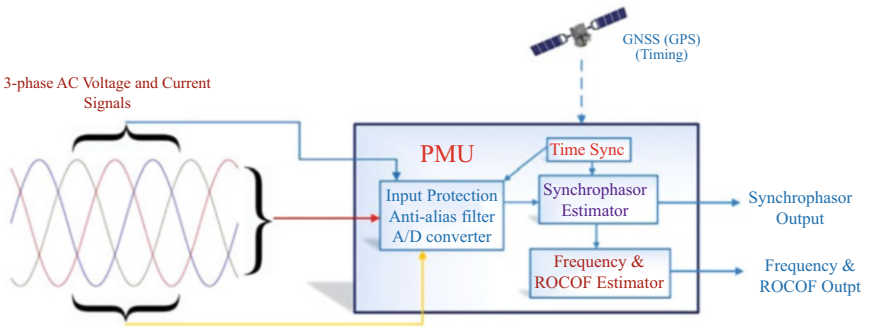

Fig. 3 Phasor measurement unit

- Phase-locked oscillator: This oscillator and GPS reference source delivers a high-speed synchronized sampling.
- Analog to digital converter (ADC): The ADC converts a continuous time and amplitude analog signal into a discrete time and amplitude signal. The sampling rate of converter is the rate at which new digital values are sampled from the analog signal.
- Microprocessor: The microprocessor calculates positive frequency values of all the current and voltage signals using the discrete Fourier transform (DFT) techniques.
- Modem (modulator-demodulator): It is the device that is used to modulate the analog signal in order to encode the digital information and again demodulate this signal to decode the transmitted information from signal.

PMUs are gaining importance in the twenty-first century, as these are the synchronized devices used for magnitude and phase angle estimation (Adamiak et al. 2005). These provide situational awareness of real-time measurements to the grid operator. Thus, PMUs have a key role in the expansion of the grid along with keeping a track of the health of the grid. A large number of PMUs are being optimally placed in the power grid to monitor real-time conditions. For real-time measurements from the PMU, the time synchronization of these devices is crucial. This is done through the global positioning system (GPS), and all the devices are synchronized to the Coordinated Universal Time (UTC) (Stenbakken and Zhou 2007; Stenbakken and Nelson 2007; Tang and Stenbakken 2012).

Time Synchronization in PMU

PMUs of different vendors have electronic clocks that sometimes provide inaccurate time. The various reasons that may be accounted for this inaccuracy are the manufacturing defects, oscillator aging, interference of electric and magnetic fields, or temperature changes. These inaccuracies are at times due to random variations in clock or can also be systematic due to uncalibrated clock. In timekeeping, even the smallest fraction of errors are critical as they significantly add up in due course. So, accurate time reference plays an important role in correct functioning of PMU. The reference of the time is from the instant when the samples of voltages and currents at nodes are taken. The samples are either synchronized directly to the timing reference, i.e., 1 PPS signal, or through software by post-processing. Accurate and precise synchronization of the sampling pulses is essential to accomplish a common timing reference and is achieved either by internal or external synchronization source deployed with the PMU. In the case of internal, the synchronization source is built-in, i.e., integrated with PMUs, while, in external, the timing signals are provided by external source (local or global) to the PMU. In both cases the timing signal is referenced to UTC throughout the grid without any interruption. The various synchronizing sources, which are used to synchronize PMUs, are IRIG-B, 1 PPS, and high-precision format time code. The IRIG-B format has a total of 100 bits per second, and it repeats every second. This time code is described in IRIG Standard 200-04 (Inter-range instrumentation group time codes n.d.). These bits contain

framing (sync) bits, time assignment bits, and control function bits. IRIG-B code is used in unmodulated or amplitude-modulated format with 1 kHz carrier (Dickerson 2007). The accuracy of unmodulated format is limited by slew rate of the digital signal and is more than 1 microsecond, whereas the accuracy of modulated IRIG signal exceeds 1 millisecond (one period of 1 kHz) but is generally not more than 10 microseconds. The 1 PPS synchronization signal is a one pulse per second positive pulse. The rising edge of the pulse is on time, and the second change of the pulse is responsible for precise time synchronization (IEEE Standard C37.118-2005 2006). However, as all the pulses are identical and it is not possible to ascertain which second a pulse is associated with, 1 PPS signal always has a 1-second uncertainty, which is resolved by making use of a data channel simultaneously. The IEEE Standard 1588 (IEEE 2020) describes the high-precision time code format. By using a standard connection, it delivers hardware-level accuracy of time. By using this time format and using a dedicated timing hardware for each port, timing accuracy comparable to IRIG-B or 1 PPS signal is achieved.

The IEEE Synchrophasor Standard

With the development of PMUs, a need for a standard was felt, which can be followed throughout the globe for accurate and comparable measurement results. An IEEE working group (WG) was formed in 1992 to articulate a standard to ensure compatible results of PMUs globally, and the first standard IEEE 1344-1995 (1995) for synchrophasor was developed. This standard has details only about the signal samples, communication protocol, and timing signal, but accuracy requirements were not clearly indicated. The WG in 1999 started revising this standard and published the revision as IEEE C37.118-2005 (2006). This standard transformed the communication protocol and included time stamp to UTC, and in order to assess the phasor measurement requirements, the concept of total vector error (TVE) was included with levels 0 and 1 as compliance levels to assess the measurement accuracy. The main flaws of the standard was that only steady-state operating conditions were included and dynamic operating conditions were not incorporated. In 2008, drafting of new standard was initiated, which contained dynamic conditions, and other imperfections of the existing standard were rectified in the revision. The International Electrotechnical Commission (IEC) also considered to develop a synchrophasor standard. So, in 2011, the IEEE C37.118-2005 standard was bifurcated into two standards, one for measurements, i.e., IEEE C37.118.1 (2011), and one for communication aspects, i.e., IEEE C37.118.2 (2011) standard. In measurement standard, measurement (M class) and performance classes (P class) were added for compliance testing, and it updated the synchrophasor definitions and formulas. In this, the steady-state performance compliance requirements were polished, and for dynamic performance tests further specifications were added. The IEEE C37.118.1a2014, a revision to the 2011 standard (IEEE Standard C37.118.1a-2014 2014), was published in 2014. The currently existing standard IEC/IEEE 60255-118-

1 (2018) jointly prepared by IEC/IEEE working group (WG) replaced the C37.118.1 and its amendment and was available in December 2018.

PMU Calibration System

In today's rapidly developing world, as we move to the smart grid concept, it is important to calibrate PMUs for phase and amplitude and seriously assert that the multi-vendor PMUs in the Indian grid comply with IEEE standards. The PMU calibration system (PMU-CAL) is an important tool to help PMU manufacturers verify that their PMUs comply with IEEE standards (Svensson et al. 2018). The calibration system helps to ensure the reliability of the PMU by validating the outputs of the PMU (Acanski et al. 2016). The system provides the ability to validate the performance of PMUs before deploying them in the field and periodically calibrate the PMUs, thus ensuring quality and reliability (Tang et al. 2013; Pirret 2012). Conventional PMU calibrators are complex and time-consuming and require skilled professional operators to operate them manually. Performance depends on operator expertise handling complex measurement setups, which ultimately makes the process costly and time-consuming. Calibrating the PMU in manual mode usually takes 2–6 weeks. In addition, to ensure accuracy and reliability, the calibrator must be traceable to the primary standards. The recommended uncertainty for the PMU calibrator is less than 10% of the limits provided in the IEEE standard. PMU manual calibration systems sometimes fail to meet these stringent accuracy requirements due to the potential for human error and also require high test processing times. The automatic PMU calibration system overcomes these limitations of manual methods. Therefore, for consistent, fast, authentic, and traceable PMU calibration, these manual techniques are replaced by automated PMU calibration systems (Ahmad et al. 2021). This calibration system can be used easily, even by someone who is not an expert in the field, producing complete and accurate test results in hours instead of weeks in the case of manual tests.

CSIR-NPL PMU-CAL System

The PMU-CAL system illustrated in Fig. 4 is located at CSIR-NPL, New Delhi, India. The built-in PMU calibration system conforms to IEC/IEEE 60255-118-1: 2018 and C37.242-2021 (2021) to perform testing as well as the certification process. The CSIR-NPL PMU calibrator is an integrated and fully automated system; therefore, even amateur users can operate the system and perform tests on the PMU. The system can perform more than 300–500 automated tests (including steady-state, dynamic, and latency tests) over a period of 6–7 hours for different frame rates and measurement and protection class conforming to IEC/IEEE 60255-118-1: 2018 standard. The calibrator's power supply system consists of three power supply standards, one for each phase and are connected together in a server rack. The power supply system is connected to the timer controller and the host PC. The GPS

Fig. 4 CSIR-NPL PMU-CAL system

receiver provides the time source for the system and the PMU that is being tested (6135A/PMU Operators Manual 2020). This time is used by the PMU to mark the time for each report made by the Device Under Calibration (DUT). The client personal computer is connected to the server and operates the calibration system. The server controls the calibration system by receiving information from the client's PC. The PMU calibrator can simulate static and dynamic conditions, close to what the PMU experiences in the power grid, and test the PMU against the performance standards written in IEEE C37.118.1a-2014, IEEE C37.118.2, and IEC 60255-118-1:2018. The PMU-CAL system is more than ten times accurate than the PMU test limits described in the standard. The PMU's calibration system can generate the required voltage and current signals under conditions similar to those of the electrical

network in the grid. Figure 3 illustrates the calibration setup to evaluate the PMU (Khatkar et al. 2022a).

Traceability of PMU-CAL System

CSIR-NPL, as a National Metrology Institute (NMI), is responsible for verifying, maintaining, and improving the country's primary standards. Figure 5 shows the traceability of the PMU-CAL system that has been established from CSIR-NPL national standards. The PMU-CAL system is traceable to national standards (programmable Josephson voltage standard (PJVS) for DC voltage, quantum Hall

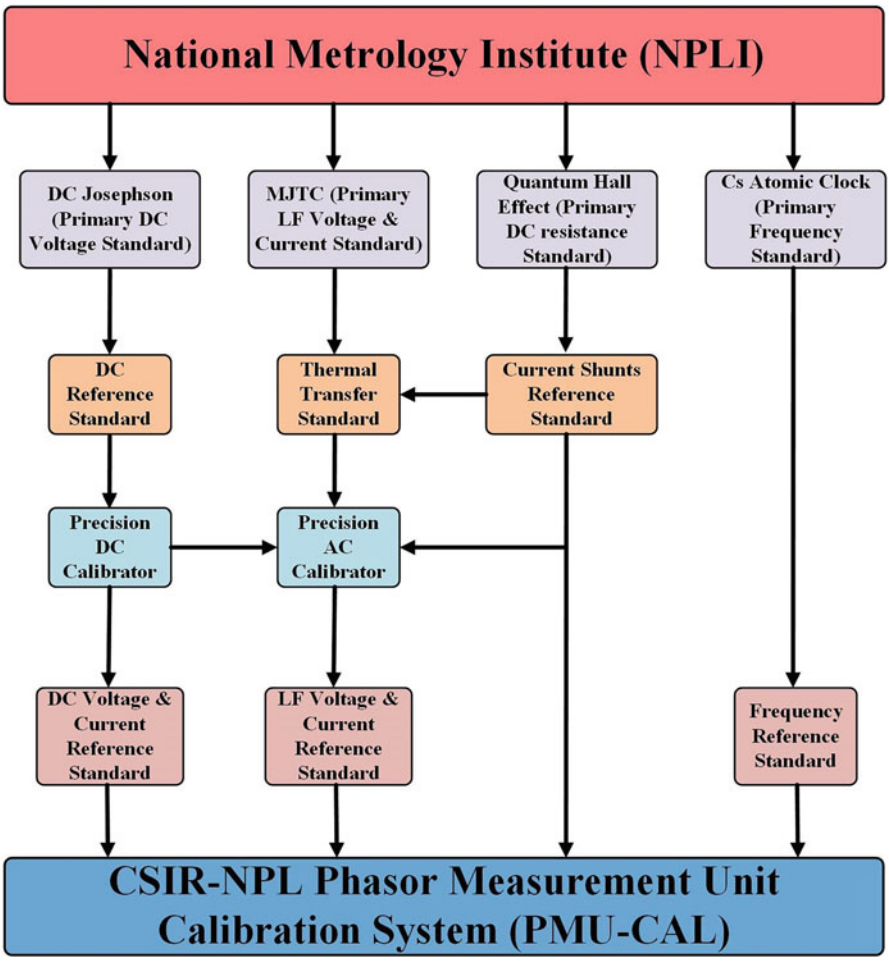

Fig. 5 Traceability of the PMU-CAL system

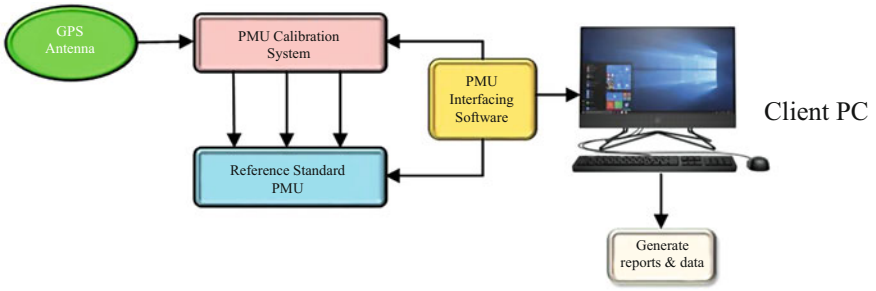

Fig. 6 Calibration setup of PMU-CAL system

resistance (QHR) for DC resistance, multi-junction thermal converter (MJTC) for low-frequency voltage and current, and cesium atomic clocks for frequency) (Ahmad et al. 2021). These primary standards are used periodically to properly calibrate the CSIR-NPL PMU calibration with measurement uncertainties ranging from ±50 ppm to ±100 ppm (Khatkar et al. 2022a). The PMU-CAL system traceability for SI units ensures that the PMU measurements are consistently stable and compliant with IEEE synchronization standards, thus avoiding ambiguous, potentially misleading data, which can impede the security of the national electricity grid and lead to financial loss for the country (Fig. 6).

Synchrophasor Calibration

During calibration, the PMU can be configured for a nominal frequency of 50 Hz or 60 Hz, and for the PMU to generate synchrophasors, it must have a built-in or external device for time synchronization with UTC. Alternatively, the PMU can be tested as a measurement (M class) or protection (P class) device (Khatkar et al. 2022b; Qiu et al. 2022; Barros et al. 2017) as per the latest IEC/IEEE standard 60255-118-1 (2018). Compliance testing (steady-state, dynamic, and latency testing) is performed according to the IEEE Guide for Synchronization, Calibration, Testing, and Installation of PMUs (IEEE C37.242-2021 2021). The performance evaluation criteria as per IEC/IEEE 60255-118-1 is used to perform reliability studies presented in the chapter. Figure 7 depicts test results after configuring the PMU as a measuring device for offline monitoring and post-event analysis. The M class test results for 63 V, 1 A, 50 Hz, and 50 frames per second are shown in Table 4. All the reported values of total vector error (TVE), magnitude error (ME), phase error (PE), frequency error (FE), and rate of change of frequency error (RFE) are highly stable and are well within the range as per IEC/IEEE 60255-118-1:2018 as shown in Table 4. The maximum measurement reporting latency reported is 109.651 ms, while the limit as per the standard is 140 ms. The minimum reporting latency is 108.359 ms, and the standard deviation is 0.090145 ms.

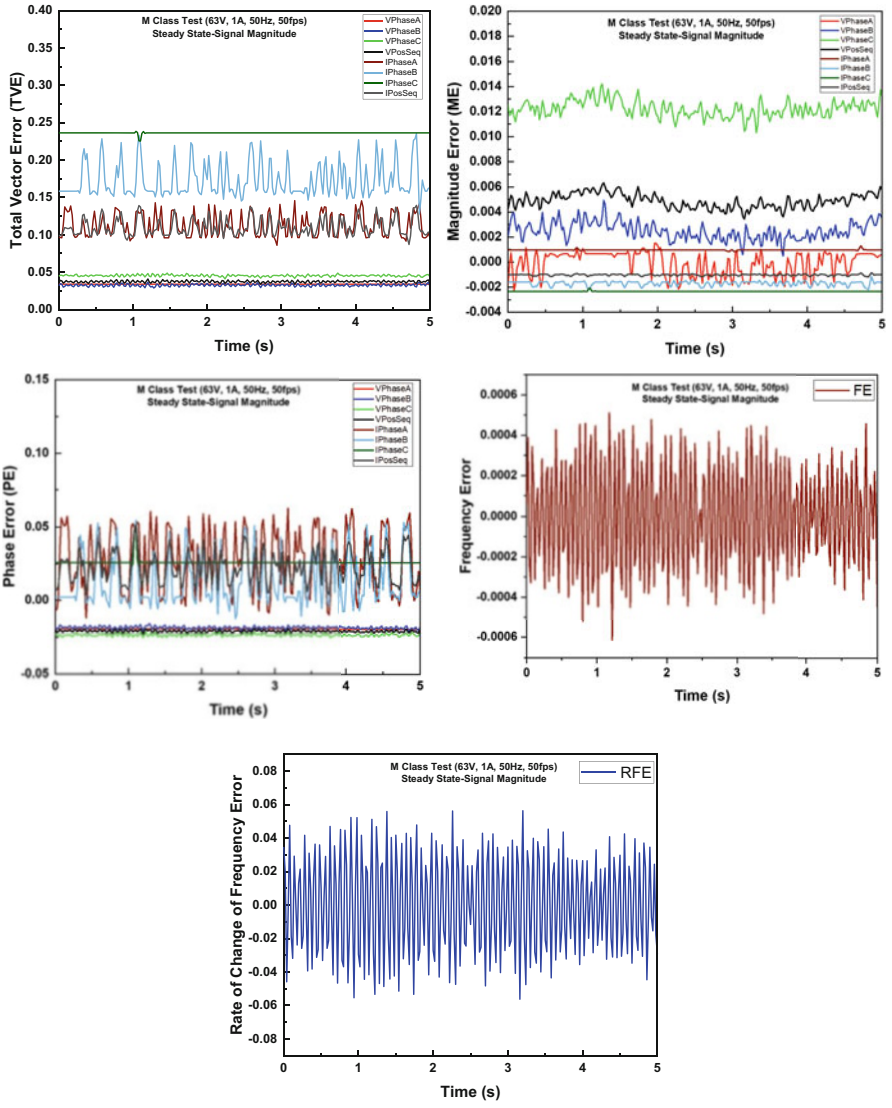

Fig. 7 Results of M class test at 63 V, 1A, and 50 frames per second

Automation and Load Management of Power Grid

Automation and load management of the power grid is a subset of WAMS to proactively identify and isolate stressed areas of the grid in real time, automate load management and load shedding functions, and integrate and implement tele-protection functions in order to maintain the efficiency, health, and stability of the

Table 4 The M class test results

Test	Parameter	Voltage phase A	Voltage phase B	Voltage phase C	Voltage positive sequence	Current phase A	Current phase B	Current phase C	Current positive sequence	Limit
Steady state: Frequency response	TVE	0.2076	0.2047	0.1877	0.1353	0.398	0.4078	0.4635	0.2533	1
	Fe	0.0007	0.0007	0.0007	0.0007	0.0007	0.0007	0.0007	0.0007	0.005
	RFe	0.0697	0.0697	0.0697	0.0697	0.0697	0.0697	0.0697	0.0697	0.1
Steady state: Harmonic distortion	TVE	0.04482	0.04293	0.05701	0.04712	0.2241	0.2412	0.2803	0.1403	1
	Fe	0.00066	0.00066	0.00066	0.00066	0.00066	0.00066	0.00066	0.00066	0.025
	RFe	0.06576	0.06576	0.06576	0.06576	0.06576	0.06576	0.06576	0.06576	Inf
Steady state: Out-of-band interference	TVE	0.1122	0.108	0.1019	0.08577	0.2898	0.3228	0.3877	0.2078	1.3
	Fe	0.00333	0.00333	0.00333	0.00333	0.00333	0.00333	0.00333	0.00333	0.01
	RFe	0.4697	0.4697	0.4697	0.4697	0.4697	0.4697	0.4697	0.4697	Inf
Steady state: Signal magnitude	TVE	0.1049	0.1005	0.1012	0.09555	0.5745	0.83	0.8498	0.5124	1
	Fe	0.00343	0.00343	0.00343	0.00343	0.00343	0.00343	0.00343	0.00343	Inf
	RFe	0.3043	0.3043	0.3043	0.3043	0.3043	0.3043	0.3043	0.3043	Inf
Dynamic: Ramp of system frequency	TVE	0.1902	0.1881	0.1802	0.1231	0.3606	0.3669	0.4273	0.2216	1
	Fe	0.00063	0.00063	0.00063	0.00063	0.00063	0.00063	0.00063	0.00063	0.01
	RFe	0.06635	0.06635	0.06635	0.06635	0.06635	0.06635	0.06635	0.06635	0.2
Dynamic: Phase modulation	TVE	0.06537	0.06367	0.07339	0.0659	0.2235	0.2558	0.3255	0.1435	3
	Fe	0.06308	0.06308	0.06308	0.06308	0.06308	0.06308	0.06308	0.06308	0.3
	RFe	5.45	5.45	5.45	5.45	5.45	5.45	5.45	5.45	14
Dynamic: Amplitude modulation	TVE	0.1184	0.123	0.1453	0.127	0.2909	0.3322	0.3697	0.2242	3
	Fe	0.00123	0.00123	0.00123	0.00123	0.00123	0.00123	0.00123	0.00123	0.3
	RFe	0.07988	0.07988	0.07988	0.07988	0.07988	0.07988	0.07988	0.07988	14

Dynamic: Input step change (phase)	Phasor response time	0.07	0.074	0.076	0.072	0.072	0.074	0.076	0.072	0.14
	Phasor delay time	0.002	0	0.00096	0.002	0.002	0	0.00097	0.002	0.005
	Phase overshoot	0.4867	0.4965	0.5007	0.4569	0.4863	0.4981	0.5007	0.4571	10
	Frequency response time	0.192	0.192	0.192	0.192	0.192	0.192	0.192	0.192	0.28
	ROCOF response time	0.232	0.232	0.232	0.232	0.232	0.232	0.232	0.232	0.28
	Frequency overshoot	1.055	1.055	1.055	1.055	1.055	1.055	1.055	1.055	Inf
	ROCOF overshoot	68.94	68.94	68.94	68.94	68.94	68.94	68.94	68.94	Inf
	Amplitude overshoot	0	0	0	0	0	0	0	0	Inf
Dynamic: Input step change (amplitude)	Phasor response time	0.03	0.03	0.028	0.03	0.032	0.03	0.03	0.03	0.14
	Phasor delay time	0.00033	0.00146	−0.0013	0.00017	0.00045	0.00145	−0.0012	0.00024	0.005
	Phase overshoot	0	0	0	0	0	0	0	0	Inf
	Frequency response time	0	0	0	0	0	0	0	0	0.28

(continued)

Table 4 (continued)

Test	Parameter	Voltage phase A	Voltage phase B	Voltage phase C	Voltage positive sequence	Current phase A	Current phase B	Current phase C	Current positive sequence	Limit
	ROCOF response time	0.082	0.082	0.082	0.082	0.082	0.082	0.082	0.082	0.28
	Frequency overshoot	0.00153	0.00153	0.00153	0.00153	0.00153	0.00153	0.00153	0.00153	Inf
	ROCOF overshoot	0.1645	0.1645	0.1645	0.1645	0.1645	0.1645	0.1645	0.1645	Inf
	Amplitude overshoot	5.928	6.31	6.291	6.12	0.1019	0.1042	0.1061	0.09973	10

national power grid at all times (Ivankovic et al. 2017). Each PMU provides phasor status, frequency and frequency offset, frequency change rate, phase angles and relative phase shifts, and configuration information at very high data rates according to the IEEE standard. This data can be effectively used for automation and load management of the power grid to implement grid stabilization, grid automation, and automated load management to realize a (intelligent) national power grid. This would automatically identify and isolate the problem areas that are overloading the grid so that the problem in any one sector of the unified national grid does not threaten or collapse the complete national power grid. This would comprise PMUs operating in M class mode providing data at 50 frames per second, user configurable interface, and implementation and integration of teleprotection functions to initiate automated trip commands. To achieve this, the IEEE 37.118 PMU data from the multiple PMUs shall be fed to the server to monitor the phase shifts and phase angles at various points to automatically identify, isolate, and island the problem areas by disconnecting the offending areas from the national power grid, thereby ensuring the stability of the national power grid. In order to realize reliable functioning, it is extremely essential that the IEEE C37.118 data from all PMUs be time-stamped to a 1PPS source that is calibrated to a national reference and with an accuracy of better than ± 1 microsecond to accurately account for any phase shifts and phase angle deviations occurring within the power grid and take remedial actions to protect it. Accurate monitoring of phase angles and their deviations/phase shifts across the entire power grid using the IEEE 37.118 PMU data can be effectively used with modern IEEE 61850 protection protocols, such as GOOSE, to effectively implement

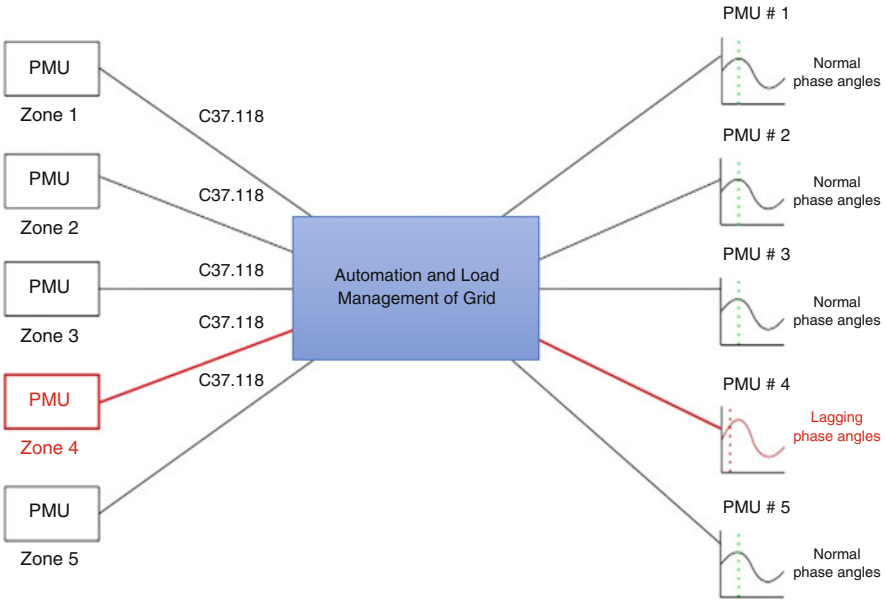

Fig. 8 Typical application of automation and load management of grid

and automate load management functions. GOOSE was primarily designed for layer 2 networks and designed to work within the substations. But GOOSE messages, using teleprotection protocols, can be extended beyond the boundary and the confines of the substation network over encapsulated IP packets, C37.94 optical links, and E1 channels to trip protection relays in remote locations to effectively manage the stability of the power grid and to isolate the problem areas that are excessively loading the grid, which were resulting in phase angle changes/deviation in the phase angles/phase shifts that are beyond the grid operator's mandated limits. A block diagram, illustrating a typical automation and load management system, is provided in Fig. 8. In this, IEEE C37.118 data is received from multiple PMUs across various zones. The system analyzes time-stamped data it receives from various PMUs, and if the phase angle variation of any one zone exceeds the user prescribed threshold, it automatically generates warning messages, alarms, and trip commands to the protection relays to disconnect and isolate the offending zone from the main power grid to ensure that it remains healthy and stable.

Conclusion

PMUs are widely deployed in the power grid for real-time monitoring of the static and dynamic performance of the smart grid. The operators at power grid rely exclusively on PMU data for power grid protection and analysis, so it becomes essential to evaluate the performance of the PMU before deploying it in the network. PMUs deployed without any operational and viable software that can be effectively used for an automated load management and grid stabilization application will not serve the purpose of transforming the national grid into a smart grid. All power grid PMUs must be integrated with automation and load management system to automatically identify hotspots and isolate stressed areas of the line using real-time analysis and actively help maintain the efficiency, health, and stability of the national electricity grid. The load management system will automatically manage the load of distribution line and load shedding functions and will actively assist in maintaining the efficiency, health, and stability of the national power grid, which are essential to transform and augment the power sector into a responsive, stable, and secure system.

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Part VI

Industry 4.0 and Engineering Metrology: A Perspective

Using AI in Dimensional Metrology

41

Arif Sanjid Mahammad

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Abstract

The use of artificial intelligence in metrology became inevitable after declaring the theme of Metrology in the Digital Era on the World Metrology Day 2022. Consequently, the 27th meeting of the General Conference on Weights and Measures (CGPM) has considered the digital transformation to collect, aggregate, analyze, and interpret the networked sensor data that would be acquired during metrology applications (Draft Resolutions of the General Conference on Weights and Measures 2022). It emphasizes the secured data sharing at the international apex down to a local level at various stakeholders (i.e., governments, industry,

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academia, and civil society). Digitizing the International System of Units (SI) provides confidence in the accuracy and global comparability of measurements. CGPM also drafted a resolution to adopt the FAIR (Findable, Accessible, Interoperable, and Reusable) principle to develop and promote an SI Digital Framework. The framework covers (1) digital representation of the SI – dissemination of SI standards – exchange of metadata; (2) digital data exchange standards and protocols, digital certificates; and (3) digital twins to simulate the measurement process (Proceedings of the 26th meeting of the General Conference on Weights and Measures (CGPM) 2018; International Committee for Weights and Measures (CIPM) 2021). Turing's legacy in computing is far from leaping from the equations to the phenomena, from calculation to modeling (Machinery, Computing 1950). Turing is the first to think of competing humans with computer intelligence.

Introduction

Traditionally, researchers fit the data points extracted from a surface to a known geometrical shape. Digital representations of the surface of geometric object used coordinate measuring machines. Computerization is important to estimate the accuracy of fitting the measured coordinate cloud. The tolerances required in the design process and the accuracy of the fitted model parameters are both needed to accept the built model. Often, the geometric fitting problem is a nonlinear minimization problem to be solved through iteration and has been widely recognized as an analytically and computationally difficult problem.

In contrast, present-day metrology applications demand next level of intervention of the computer simulation (Arif Sanjid 2021). Recent arena of the additive manufacturing, X-ray tomography, and super-resolution microscopy adopt the framework of simulation techniques from bioinformatics (Shu et al. 2021). Especially, dimensional metrology that determines the shape of objects with nanometer accuracy is quickly emerging into computer-aided simulations of experimentation (Zhao et al. 2020). Unlike earlier, metrologist aims to demarcate the computational methods to simulate measurements, propagation of uncertainty, and optimization of experimentations (Boudjemaa et al. 2003). Simulation-aided measurement originates from the matured field of metrology, statistics, probability, and evolving computer applications.

Dimensional Metrology

Engineers design new products with improvements to meet the functional requirements of consumers. The art of creating the novel designs and new routes of production needs critical understanding and the knowledge of the dimensional metrology. Dimensional metrology stretches from defining the geometric product

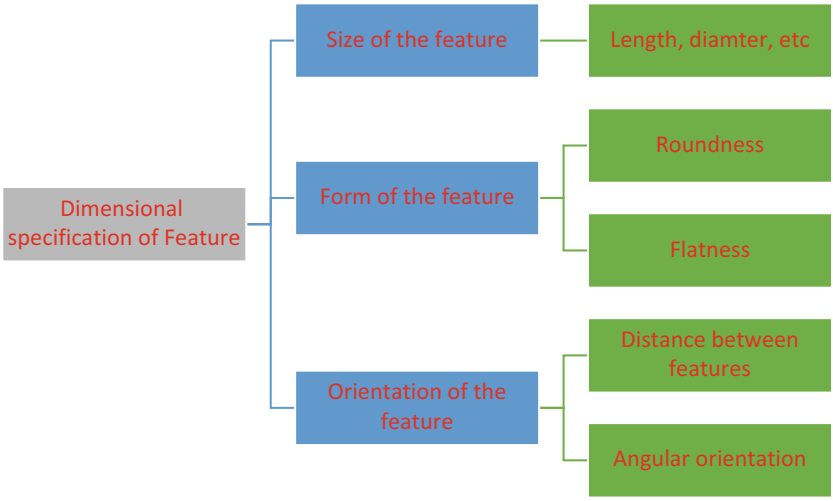

Fig. 1 Dimensional parameters of products

specifications, producing the parts according to design tolerances and the quality assurance of processes and products. Therefore, the digital transformation of dimensional metrology encompasses aspects of measurement of dimensional parameters, their standard measuring methods, and machines to supplement these requirements (Leach et al. 2020). Figure 1 depicts a simple classification of the geometric features referring to the dimensional measurements.

The basic dimensioning starts by identifying various features (e.g., line, point, surface, circle and cylinder, etc.), specifying their size (i.e., linear parameters – radius, length, and pitch). Moreover, the location, the orientation, and the form of features are also essential to achieve the desired functionality of mechanical parts. The geometric parameters are broadly into six groups. They are (1) radiations of Mise en Pratique – Matter of Practice, (2) linear dimensions, (3) orientation or angular parameters, (4) form of the surfaces, (5) complex measurands, and (6) basic instruments. Figure 2 shows the chart of linear dimension artifacts. The remaining five categories can also be branched as depicted in Fig. 2. The artifacts realize the length in desirable form according to the application. For example, the geodesic lengths are measured using a woollen tape or displacement measuring electronic devices. Over the above, the quantum phenomena based standard optical radiations realizes the SI definition of length. Therefrom, the optical radiation shines the end standards (i.e., gauge block) in an interferometer to disseminate the traceability down to basic working instruments, viz., micrometer, vernier caliper, dial gauge, etc. (Flack and Hannaford 2006).

It is apparent from the chart of linear measurements that the artificial intelligence is likely useful. Also, the AI algorithms are based on heuristic strategy; hence they tend to be less accurate. On the contrary, the traditional and standardized dimensional measuring methods are sufficiently suitable for the accurate mathematic

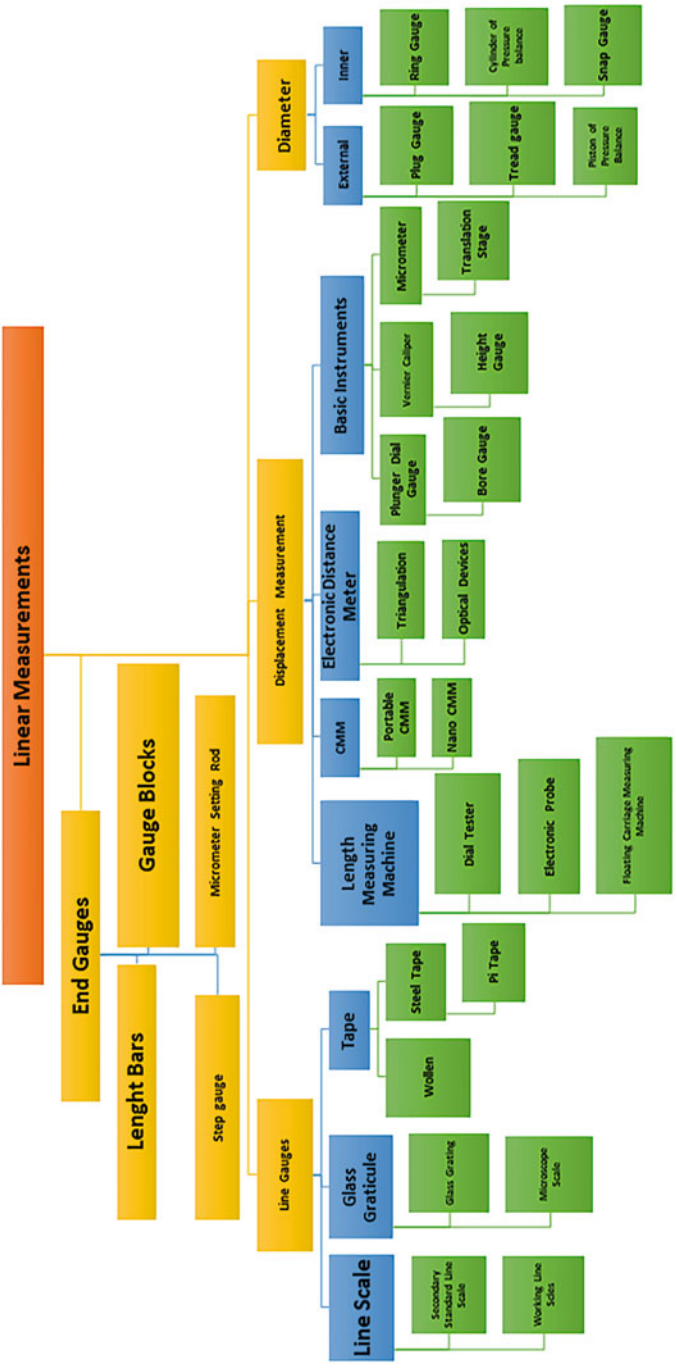

Fig. 2 Linear measurement and associated standard artifact standards

analysis of the observation/ readings. Nevertheless, the diameter measurements insist the optimization of the peak detection of cylindrical curvature. Recently, simulation methods are employed to rectify the misalignment and peak detection for the measurement of the inner as well as the external diameter measurements (Sanjid et al. 2018). Some worker used genetic algorithm to reduce the eccentricity during roundness measurement of flick standard and the algorithm is registered under a copyright (Arif Sanjid and Chaudhary 2013). Another AI algorithm that is the swarm particle optimization is workout to estimate flatness and straightness measurement (Wang et al. 2012).

Present scenario of digitizing the International System of Units (SI) and the FAIR Framework will need the new computational algorithm to generate metadata and simulate the digital twins. Consequent automation of measurement method invariantly seeks the AI methods. A brief background of AI is discussed in the next section. Subsequently, the prospects of using AI in dimensional metrology are envisaged.

Artificial Intelligence

AI techniques enable computers to mimic human behavior; more specifically, they are inspired by the biological intelligence. Some of the well-established AI techniques such as (1) neural networks, (2) self-organizing maps, (3) growing cell structures, (4) evolutionary algorithms, (5) cellular automata, (6) expert systems, (7) fuzzy logic, (8) simulated annealing, etc. are discussed here. Traditionally, machine learning algorithms define a set of features in their data. These features are handcrafted. Machine learning algorithms process data in order to achieve the tasks as shown in Fig. 2. Different algorithms, viz., support vector machine, naïve Bayes classifier, linear regression, K-mean, hierarchical clustering, Apriori, the principal component analysis, Q-Learning, etc., are the example. The list is growing with the advances in research (Fig. 3).

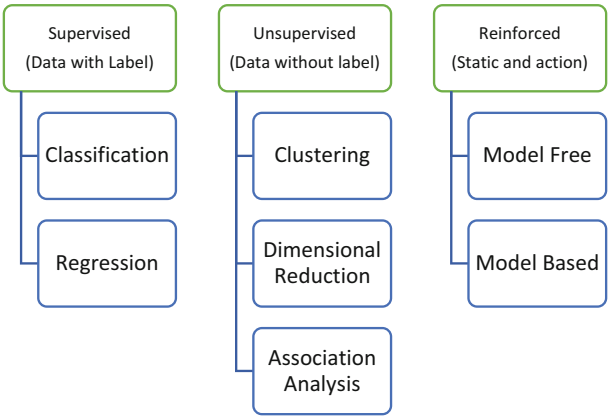

Fig. 3 A task-based classification of machine learning algorithms

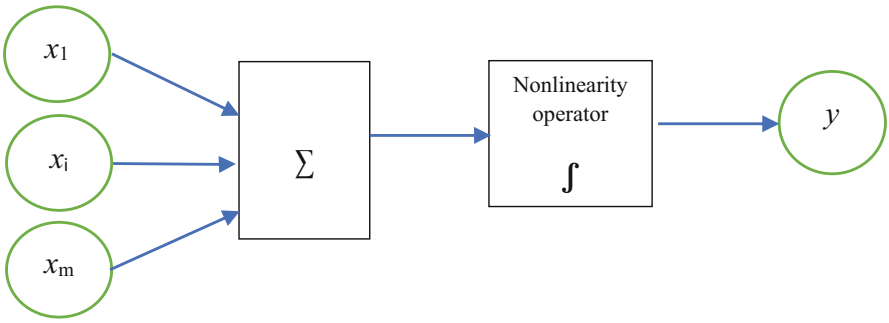

Fig. 4 An AI neuron

Neural Networks

The preceptor propagates the weighted stimuli after summing through a nonlinear threshold to result a response. Thus, the preceptor adopts the behavior of biological neuron. The weights are adjusted (mathematically) according to training data set using back propagation algorithm. The training data set is a map of input to responds of the given physical system, which has to be simulated. Figure 4 depicts a basis neuron cell.

$$y = g\left(\sum_{i=1}^{m} w_i x_i\right) + b_0 \tag{1}$$

y: Output of AI neuron
$g()$: Nonlinear function/threshold
w_i: Weight of the input x_i
x_i: The ith inputs up to m
b_0: Bias if any

Iterative tuning of weights finds the local minimum of the objective function of given application. Setting the learning rate dictate the accuracy vs. speed of handles large data. Adoptive learning rate optimize between the small learning rate – slow convergence – and large rate, where overshoot occurs. The application of neural network (NN) is effective, when there is no need of feasible/reasonable to assign a mathematical model (Chandrasekaran et al. 2021). However, the NN enables to establish a new model for perfectly known or using measured data with a sufficiently less uncertain.

Self-Organizing Maps

When metrologists attempt to compare a given type measuring apparatus, it might need to arrange them into a group by distinguishing their differences as well as

correlating their functionalities. The self-organizing map (SOM) identifies similar objects set and cluster them together so that the similarities appear. Thus, the measuring machines or their results can be visualized according to their multiple attributes. The SOM performs clustering by estimating how far the points are, and it reveals them by projecting onto two-dimensional space (i.e., reduce the high dimensions into lower dimensional scale). SOM is more democratic – every node occupies a site of equal importance in regular lattice of dimensions. Unlike NN, the links in SOM are inactive, and the vector of weights at each node provides the memory. The learning of SOM is unsupervised and predicts the pattern of characteristics of objects. A numerical measure of the quality of the clustering use the Euclidian distance between the vectors of characteristics. A neighborhood is chosen by assigning a cut-off distance. The drawback of SOM is that the learning time increases with the quantum of data and the overestimation issue arises during the splitting of clustering. The SOP is useful for classification, associating, and projection – reducing dimensionality and modelling (Cartwright 2008).

Genetic Algorithm

Genetic algorithms (GAs) are heuristic search and stochastic optimization that imitate the natural selection and biological evolutionary process. GA combines the notion of survival of the fittest, random, and yet structured search. In such situation, relatively high fitness structures have a larger chance to survive and to produce higher fitness offspring. This evolutionary process can be applied to the real-world problems where heuristic solutions generally lead to unsatisfactory results (Mikó et al. 2021).

GA starts with an initial set of random solutions called a *population*. Each individual in the population is called a *chromosome*, representing a solution to the solved problem. A chromosome is usually a binary bit string. The chromosomes evolve through successive iteration, called *generations*. During each generation, the chromosomes are *evaluated*, using some measures of *fitness*. To create the next generation, new chromosomes, called *offspring*, are formed by either (1) mating two chromosomes from the current generation or (2) modifying a chromosome using a *mutation* operator. A new generation is formed by (1) selecting best parents and offspring based on fitness values and (2) rejecting others so as to keep the population size constant. Fitter chromosomes have higher probabilities of being selected. Figure 5 depicts the algorithm of generic algorithm.

There are three basic genetic operators including selection, crossover, and mutation, used to generate new chromosomes (solutions) from generation to generation. Selection is a biased reproduction process to determine which chromosomes will be reproduced in the new population. Chromosomes with higher fitness values (better solutions) have larger chances of being selected for reproduction. The crossover operator is the mating operator that allows production of new chromosomes through partial information exchange between pairs of chromosomes. The mutation operator significantly changes the structure of a chromosome and hence produces albinos.

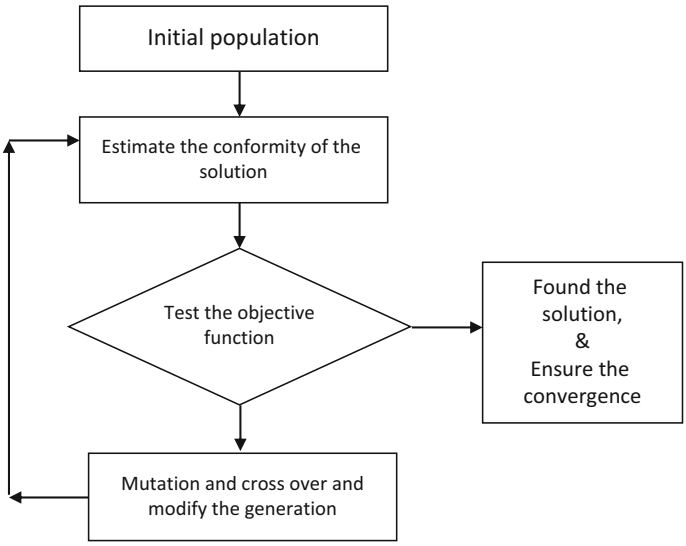

Fig. 5 Flowchart of basic genetic algorithm

Mutation serves as a guaranteed procedure to prevent solutions from being trapped in local optima. GAs are useful where the search space is large, nonlinear, and noisy.

Several characteristics of GAs suggest that they may have advantages relative to alternative heuristic solution algorithms for optimization problems. They are (1) dealing with coded parameter set, not the parameters themselves; (2) simultaneous search from a population; (3) use of objective information; and (4) use of stochastic rather than deterministic transition rule in the search process.

The fitness function is of importance for the GA to optimize a complicated problem, and it is a bridge between the optimization algorithm and the problem to be solved. The fitness function is expressed in terms of the factors affecting the optimization object.

Cellular Automata

In many dimensional metrology applications, the tactile probing system acquires temporal or spatial data. The process of moving the probe to the desirable location in coordinate measuring machine is solved using a set of differential equations. When attempted to assess the re-entrant type of surfaces, a simple analytical solution cannot be found. Then, it is necessary to apply numerical integration. In cellular automata, discrete approximation is used in many small cells to circumvent the use of differential equations. A limited number of variables are assigned to each cell change according to a set of rules. CA do not learn but offer to evolve the solution.

The cells of CA follow transition rule to update their states in each recursion/iteration. Transition rules define how the state of cell is renewed by using the current state of cell and its neighbor cell (Cartwright 2008).

Expert Systems

Expert system focuses on a single specific field unlike general-purpose problem-solving machines. It complements the human knowledge with its reasoning ability to deduce the solution from the given data. ES poses two components (1) a database relevant to the field of application and (2) the rules that set the context of the problem and method of manipulating the supplied data to it (Cartwright 2008).

Bee Colony Algorithm

CMM is suitable for flatness deviation measurement, when surfaces are of small diameters. Work is carried out using particle swarm optimization (PSO) algorithm to serve the nano scale micro/nano surface flatness error evaluation by a minimum zone method (Lei et al. 2014). There are many types of swarms, and the kind of intelligence they use varies from swarm to swarm. Their self-organization feature comes from the collective behavior by means of intuitive interactions among the simple agents. The self-organization in swarms has four characteristics:

(i) *Positive feedback*: promoting the formation of opportunity structures by recruitment and reinforcement.
(ii) *Negative feedback*: the act of countering the positive feedback and helping to stabilize the collective pattern. A negative feedback mechanism avoids the saturation.
(iii) *Fluctuations*: Randomness is significant for emerging the structures and the discovery of new solutions.
(iv) *Multiple interactions*: Agents in the swarm use the information among them to spread trigger throughout the network.

Bee colony algorithm is developed by Karaboga in 2005 that mimics the bees searching for honey (Karaboga et al. 2014). The optimization algorithms are usually divided into three kinds of bees: employment, observation, and reconnaissance. For the traditional artificial bee colony algorithm, in the last stage of iteration calculation, the employed bees will search the nectar, thereby increasing the post calculation workload. It affects the speed of search for the optimal solution with tabu strategy. Thus, bees do not search the local optimal solution again. Tabu strategies refer to form tabu list to search the local optimal solution and, at the same time, mark them so that you can avoid to search this local optimal solution late again (Ge et al. 2015).

Fuzzy Logic

The "fuzzy" word means the things that are not clear or are vague. This concept provides the possibilities which are not given by computers but are similar to the range of possibilities generated by humans. Rule base is a component used for storing the set of rules, and the if-then conditions given by the experts are used for controlling the decision-making systems (Cartwright 2008).

Let there be five measurements of a given geometric parameter of a product. The horizontal line represents the tolerance limit/reference value/conventional true value. The vertical error bars correspond to expanded measurement uncertainty. Figure 6 graphically represent the different cases of measurements.

The conformity decision would be taken as per the decision rule stipulated by the laboratory. A formal agreement with a customer or authentic approval from the regulator may be considered for each test to decide upon the decision rule. It is all about how do you pass or reject the product under test after accounting for the measurement uncertainty. Table 1 gives an example of the compliance decision taken by a lab and the associated risk. While zero (unity) represents the false (true), respectively, the fuzzy logic deals with the real number between zero and unity. In other words, it can be employed where it handles the concept of partial truth to be

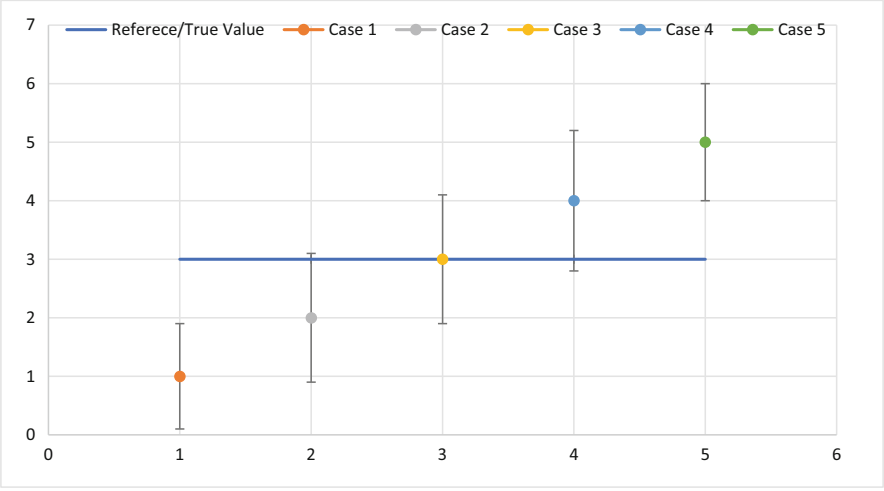

Fig. 6 Plot of measurement result with the uncertainty of measurement

Table 1 The decision taken and associated risk

	Compliance decision	Risk of decision
Case 1	Pass	No risk
Case 2	Pass	Low
Case 3	Pass/fail	High
Case 4	Reject	Moderate
Case 5	Reject	Low

dealt with. The truth value may range between completely true and completely false. In Table 1, case 3 is the partial truth. A fuzzy interface system uses the uncertainty of measurement as a knowledge base (rule set) and nonlinear fuzzy logic to map the partial truth to convincing results. Moreover, the result of a tested product must be in conformity with the specification. The decision rule is a policy matter of the metrology laboratory that dictates the risk analysis.

ILAC-G8:09/2019 provides overall guidance on how to select appropriate decision rules and compile the required elements of a decision rule (ILAC-G8 2019). A decision rule and risk associated with the decision for five measurement cases are discussed.

Simulated Annealing

In metallurgy, annealing of metal involves a process of slowly cooling to reach a state of low energy. Simulated annealing (SA) is devised for optimization analogous to thermal annealing. The thermodynamics temperature corresponds to the randomness. The temperature is reduced slowly, starting from a random search at high temperature eventually converging to greedy descent as it approaches zero temperature (Nikolaev and Jacobson 2010).

In SA, the currently assigned values to variables are compared with randomly picked ones at each step. When there is an improvement, the algorithm updates current variable; else it accepts the assignment with some probability. The algorithm is shown as a flowchart in Fig. 7.

Some AI Applications

The ANSI Dimensioning and Tolerance StandardY14.5 (Dimensioning and Tolerancing Y14.5 2018) specifies the estimation of the form tolerances on a component concerning an ideal geometric feature. And, ISO Standards (ISO 1101 2017) advocates the form tolerance evaluation according to the concept of minimum zone. Flatness tolerance is defined as the least separation between two parallel planes that enclose most measurement points. The evaluation of the minimum zone association of flatness error is a non-convex problem. Intelligent computations, such as the genetic algorithm (GA) and particle swarm optimization technique, were used to solve the nonlinear problem of flatness evaluation (Meo et al. 2013; Huo et al. 2016; Liu et al. 2021; Chandrasekaran et al. 2020).

Some workers used a genetic algorithm for the evaluation of the roundness of circular profiles using the MZT method. The standard fitting criteria that establish the reference circle are the least squares method (LSQ), the maximum inscribed circle (MIC), the minimum circumscribed circle (MCC), and the minimum zone tolerance (MZT). The MZT useful for cylindrical mating parts involves two concentric circles that contain the measured points such that the difference in radii is the least possible value.

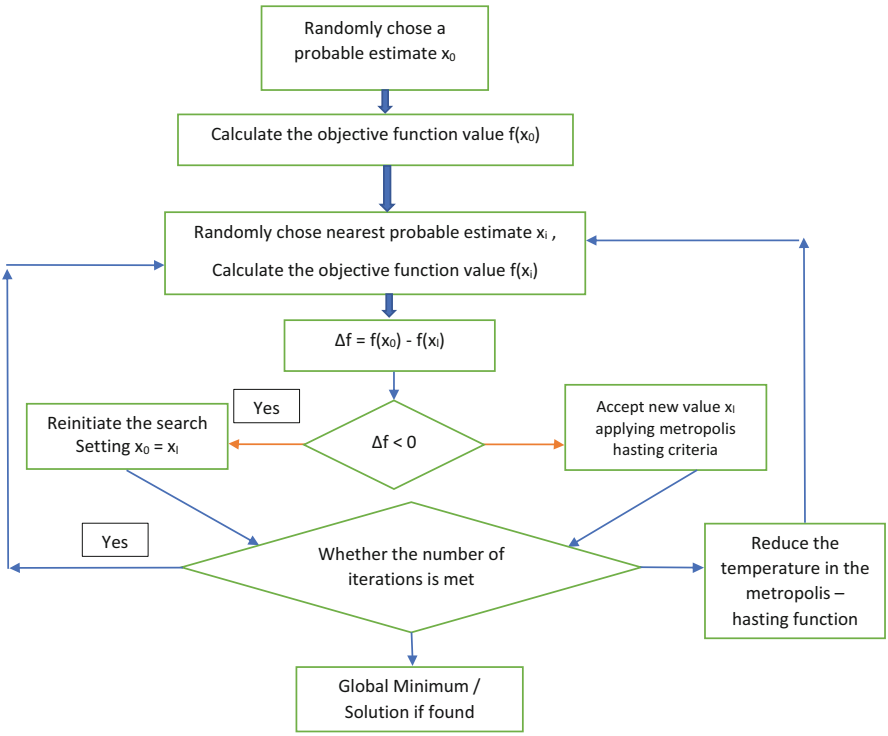

Fig. 7 Flowchart of SA algorithm

The popular characterization method of finding defects in thin film include a combination of optical defect detection with scanning electron microscopy (SEM), scatterometry or optical critical dimension (OCD) metrology, and X-ray scattering metrology. SEM register the deviations from the patterned template. An inverse algorithm is required to convert the scattering intensity data into a meaningful real space structure (Hannon et al. 2016). Despite the above discussed verity of applications, the following dimensional metrology applications are practiced.

Thermal Expansion Estimation

At length, dimension, and nanometrology division (NPLI), a study was conducted to determine central deviations and thermal expansion simultaneously. The measurement results are analyzed using simulated annealing algorithm (Zhou et al. 2018). A gauge block of size 100 mm is wrung on a platen in the automatic gauge block interferometer (GBI 300). The gauge block is measured at different temperature about 20 °C. The gauge block is thermally stabilized for at least 4 hours in the range of (20 ± 1) °C. At each temperature, readings of the gauge block are taken by entering different nominal thermal expansion coefficients values (TEC). The SA

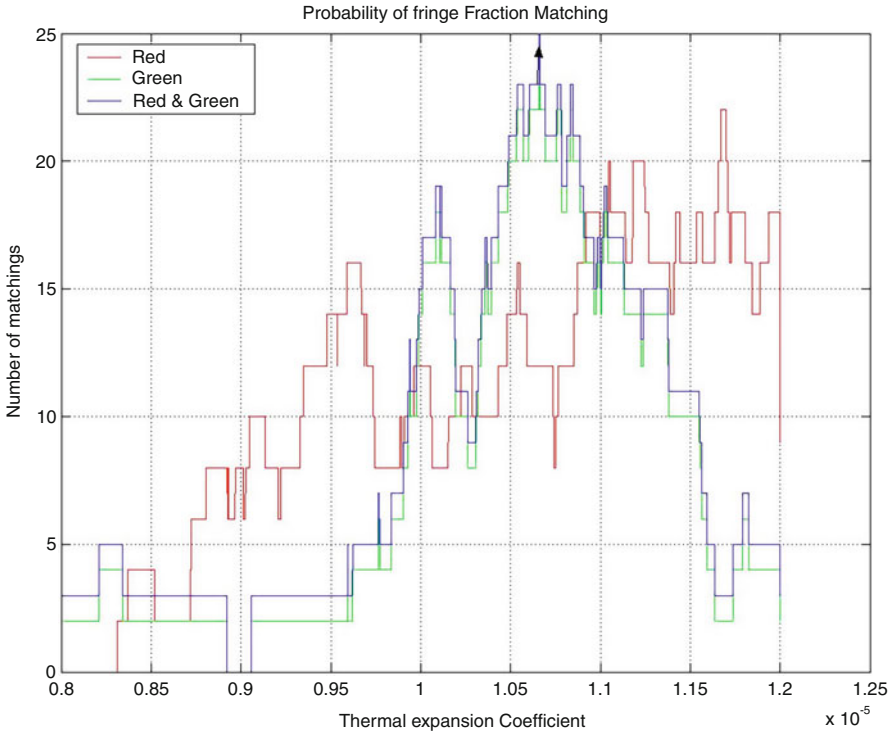

Fig. 8 Results of SA used for determining thermal expansion coefficient of gauge blocks

estimated the thermal expansion coefficient value as 10.11×10^{-6} /K. The probability distribution of measured fringe fraction deviation within the range of 19–21 °C is shown in Fig. 8. The convergence is shown as an arrow mark on the plot (Unpublished work 2012).

Flick Standard

Flick standard is a material artifact standard used to calibrate the roundness measuring machines. Flick standards have cylindrical seggetta with a discontinuity. Generally, a flick of 0.01 mm amplitude is sufficient to verify the linearity of LVDT probes and set the calibration (amplification) factor of most of the form measuring machines. The graphical parameters of the seggetta and amplitude of flick indicate filtering characteristics. Usually, the uncertainty associated with flick standard calibrated on form testing machine is the order of 0.1–0.2 micrometer. And the traceability is established through reference standard artifact – reference measuring machine; obviously their associated uncertainty is superior to that under test. A method to calibrate the flick standard using laser interferometer with rotary table on a X-Y stage is developed employing a genetic algorithm. An LVDT probe is used as

Fig. 9 Flick artifact centering using GA

null detector, and heterodyne displacement laser interferometer is used as reference measuring system. Henceforth the traceability hierarchy is established to the primary laser – heterodyne displacement laser interferometer. Further, the measurement uncertainty is improved. We have observed a main uncertainty component in this method is from type A, which is due to cantering. The associated uncertainty of measurement down to 40 nm is achievable. Software is also prepared for roundness filter, and roundness attributed by the flick standard is determined (Fig. 9).

The flick standard under test is aligned over Moore's indexing table. A laser interferometer fitted with probe is used to touch and feel the peripheral surface of flick standard. Moore's indexing table is rotated to obtain the surface profile coordinates of the flick standard at laser interferometer (Unpublished work 2014). The readings of the laser interferometer corresponding to the rotation of the index table are fed to a genetic algorithm-based program. The software program is developed using PERL (**an open-source language**) into two parts. The program accepts the probe displacement and angular position of index table readings to predict **eccentricity error of** alignment of flick standard (Sanjid and Chaudhary n.d.). The values of the eccentricity and flick depth are binary coded to implement the genetic operations (i.e., mutation, cross-breeding). The program simulated the coordinates of the flick standard with an arbitrary eccentricity error and flick depth values. The deviation of the simulated values and the measured coordinates are estimated to calculate the cost function value of the genetic algorithm. The generic algorithm

iterates the genetic operations to reduce the difference between the simulated coordinates and the measured coordinates.

Prospects of AI

Dimensional metrology involves the definition of geometric product specifications of components to be manufactured considering the feasibility of their assessment through a globally acceptable verification along with the traceable chain that snaps to SI unit of length. The partition operator is suggested in ISO 17450 to cluster the coordinate data points on the specific location on the part. After filtering the extracted points, the operation advocated in 17,450 associates them to given shape through optimum fitting (ISO 17450-1 2011). Therefore, the clustering, optimization for finding global minimum, and association operations demanded in dimensional metrology can be solved using SOM, GA, bee swarm, SA, etc. The complex structures with multiple characteristics require other than traditional methods. But the detection methods used for the complex structural components can be handled by collecting more data points. SOM is useful to classify and remove the outliers from the collected set of data.

Spirit Vials Calibration

Spirit levels are constructed by installing spirit vials of different shapes and lengths in a prismatic metallic case. Machinist's precision level is patented as US2316777A in 1940 by William B Fell. The main claims are (1) the graduated scale on a glass container of the liquid with an air bubble and (2) a metallic holder for the vial unit having pad portions. The specification and testing method are given in Indian standard 5706 (Indian Standard 5706 2006) and ISO 17123-2 (ISO 17123-2 2001). The test method recommends estimating the accuracy of the spirit level by determining the sensitivity of the bubble. A known tilt is applied to float the bubble between the graticules. Figure 10 shows the calibration setup and a spirit level. A micrometer fitted on one end of a metallic arm is used to tilt the platform. The spirit level to be tested would be placed on the platform. The operator carefully notes the stable location of bubble after tilting the spirit level. The micrometer must be adjusted precisely to shift the bubble exactly by a division. This exercise needs to be repeated 50 times to calibrate a spirit level. Its automation definitely reduces the time consumption and averts the influence of manual fatigue. Neural network is best suitable for the automation of calibration of spirit level.

The rotation of micrometer can be controlled through automation. A camera captures the location of bubble between the graduations on the vial. Tilt the spirit level, and the corresponding micrometer reading will be feed to the controller. Repeat this exercise to train the neural network. Therefore, the sensitivity of spirit level can be estimated.

Fig. 10 Calibration setup for sensitivity of spirit level

Digital Twins

Other than optimization, automation simulation of metrology experiments is in high demand with the advent of digital SI. The digital twin is a computer program that represents the practical measurement setup. Data acquired in the measuring experiments is used to train the virtual model. Further, the virtual model is evolved by reducing uncertainty components. Artificial intelligence algorithms are essential to design and develop the virtual infrastructure akin to the dimensional measuring machine and its application. The concepts of the digital twins are still progressive. Therefore, authorized/harmonized/standard methods for developing virtual dimensional metrology infrastructure instigate new field of technology (Flamigni et al. 2021; Harper et al. 2019).

The last but not the least, the advanced version of evolutionary method is genetic programming (GP). It is not limited to evolve the solution to a problem, but a computer program will evolve such that when executed will yield the desired solutions. The implementation of the GP follows the basic concept of evolutionary algorithm, i.e., genetic algorithm. The strings of command/code of the GP are randomly mutated and crossover for each cycle of generation. The crossover (mutation) operations mimic the biological cross breeding (mutation), respectively.

This chapter covered the AI and their metrology application. It is emphasized that AI methods prospect toward the implementation of digital twins and data processing applications.

Conclusion

This chapter covered the different applications of dimensional metrology supplemented by different AI techniques. GA and Swarm particle optimization are popular in analysing flatness and straightness. The chapter explained some recent AI applications to develop a digital transformation framework.

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Artificial Intelligence

42

Implementation and Obstacles in Industry 4.0

Kirti Soni, Nishant Kumar, Anjali S. Nair, Parag Chourey, Nirbhow Jap Singh, and Ravinder Agarwal

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Abstract

In the fourth industrial revolution (Industry 4.0), Artificial Intelligence (AI) had been a cognitive science that produces actual value needed for relevant data with processing capabilities and algorithms. Manufacturing in the Internet-of-Thing (IoT) era will be more efficient, better quality, easier to manage, and more transparent through integration of physical and cyber technologies in Industry 4.0-based smart factories. Factory automation relies heavily on sensors and AI to make the system intelligent. Sensor technology advancements and developments linked to Industry 4.0 serve as the backbone for the inclusive expansion of industry and the economic success of any country. It is imperative that

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manufacturing organizations and supply chains have access to the latest low-cost sensor technology for collecting data and putting it to good use. Standard sensor types include position sensors, flow, temperature, flow rate, pressure, and force. A wide range of fields, including motorsport, health care, manufacturing, the armed forces, and agriculture all make use of them on a day-to-day basis. Increasing efficiency through automation is the goal of Industry 4.0. The purpose of this chapter is to provide a brief overview and viewpoint on the most recent advancements in AI and the associated problems.

Attempts to define the main ideas and tools behind this new era of manufacturing in the early years of the so-called fourth industrial revolution (Industry 4.0) always ended up referring to the concept of smart machines that could communicate with each other and with the environment. When it comes to the new industry 4.0, it's the defined cyber physical systems connected by the IoT that get all the attention. Nonetheless, several tools and applications will benefit the new industrial environment, complementing the actual formation of a smart, embedded system capable of performing autonomous tasks. And the majority of these revolutionary ideas are based on the same background theory as artificial intelligence, in which the analysis and filtration of massive amounts of incoming data from various types of sensors aids in the interpretation and recommendation of the best course of action. As a result, artificial intelligence science is well suited to the challenges that arise during the fourth industrial revolution's consolidation. The purpose of this chapter is to provide a brief overview and viewpoint on the most recent advancements in AI and the associated problems.

Keywords

Industry 4.0 · Smart sensor · Artificial Intelligence (AI) · Machine Learning (ML) · Digital metrology

Introduction

As Information and Communication Technologies (ICT) are increasingly integrated into industrial and manufacturing units, the concept of Industry 4.0 is becoming increasingly important in the digital transformation. Industry 4.0 includes the integration of cyber-physical systems, cloud computing, data exchange, retrieval, and storage, as well as a security framework, into industrial and manufacturing units in recent years (Hsu et al. 2018). In the twenty-first century, Industry 4.0 instantly became the focus of a global world that promoted the computerization of manufacturing (Kalsoom et al. 2020). It has revolutionized the manufacturing process, leading to intelligent manufacturing promising self-sufficient manufacturing processes by using machines and devices that communicate with each other through digital connectivity. Industry 4.0 incorporates numerous types of smart sensors that are used to measure and process quality metrics (Stăncioiu 2017; Varshney et al. 2021). These sensors process/transfer data by means of electronic

medium or wireless signal. Industry 4.0, the pioneer of smart factories, has access to a wide range of advanced technologies, including big data analytics, Artificial Intelligence (AI), advanced robotics, 3D printing, and cloud computing, all of which are critical to the modern manufacturing industry's success (Kalsoom et al. 2020).

AI is a branch of computer science that aims to create intelligent machines. It has become an essential part of the technology industry (Charniak 1985). AI creates intelligent machines and computer programs to perform various tasks that require human intelligence. Until recently, AI was confined to science fiction and academic arguments bringing about major impact of technology on the modern world (Dautenhahn 1998). But today, AI is an integral part of our daily lives, serving as a critical function in a wide range of industries. Industries including manufacturing, healthcare, and supply chains all feel the effects of this change. AI's ability to execute tasks that humans are unable to opens up a plethora of new possibilities for enhancing efficiency and productivity (Makridakis 2017). AI is a highly technical and specialized field of study. Programming computers to have particular abilities, such as knowledge, reasoning, problem-solving, perception, learning, planning, ability to operate and move objects, etc., is at the heart of AI.

Smart and integrated networks are essential in Industry 4.0 for intelligent manufacturing development (Kalsoom et al. 2020). Sensors and AI can collect and analyze data to assist in decision-making, and self-optimization in production line automation may be beneficial. Production lines can be made more efficient by automating the production process. Firms must work with experts to develop bespoke solutions due to the complexity of applying AI in industrial automation. The cost of developing the necessary technology is prohibitive for most manufacturers, who also lack the necessary qualifications and experience. Because of the widespread use of Computer Numerical Control (CNC) machines and industrial robots, production processes have become more adaptable, while Computer-Aided Design (CAD) and Computer-Aided Processing Planning (CAPP) have made computer integrated manufacturing a reality (Culler and Burd 2007). It is possible for manufacturers to implement digital transformations from several perspectives because of the IoT's practical usage, such as increased productivity, automation, a focus on the customer, and other competitive benefits. Any digital economy is seeing an increase in the collection, measurement, and transmission of precise data that can be traced back to international standards. Although Industry 4.0 was announced several years ago, digital transformation is still in the early stages of growth. Data storage and processing speed, smart sensor affordability and availability, effective data management, cyber physical system development, and other factors all play important roles in the digitalization process. However, for maintaining the highest levels of accuracy, calibration of smart sensors with respect to some digital or electrical standards is essential. The definitions and consequences of SI units also give inspiration for the development of sophisticated reference standards with lowest levels of measurement uncertainty for calibrating this smart equipment (Garg et al. 2021; Varshney et al. 2021). The IoT is more frequently employed in metrology for calibration and testing purpose.

Smart Sensor

An Industry 4.0 system consists of many elements/phases that must be designed to meet the demands of the manufacturer or industry (Vaneker et al. 2020):

- Historical data collection.
- Live data capturing via sensors.
- Data aggregation.
- Connectivity via communication protocols, routing, and gateway devices.
- Integration with Programmable Logic Controllers (PLCs).
- Dashboards for monitoring and analysis.
- AI applications: Machine Learning (ML) and other techniques.

A sensor is a device that detects and responds to a measurable digital signal to the input stimuli, which may be any quantity, property, or condition from the physical world. As an example, the input stimulus might be anything from pressure to force to flow to light to heat to motion to wetness (Schütze et al. 2018). For reading or other distribution, the response output is typically a digital signal in the form of an electric signal such as voltage or current, capacitance or resistance, frequency, or other unit of measurement. The industrial IoTs, which incorporates various types of sensors for regulating and monitoring process parameters, is a fundamental component of Industry 4.0. Depending on the configuration, these sensors can communicate with the process control equipment through radio or by placing them nearby. Varshney et al. (2021) summarize the importance of smart sensors in Industry 4.0. Process automation begins with smart sensors and actuators. Figure 1 summarizes the classification of sensors in brief of their industrial/manufacturing uses (Bentley 1984; Eaton and Smith 1997; Fan et al. 2002; Chong and Kumar 2003; Chen and Lu 2005; Rai 2007; Hu et al. 2009; Wang et al. 2010; Lebosse et al. 2011; Gómez-Robledo et al. 2013; Lee et al. 2016; Clark 2018; Leal-Junior et al. 2018; Lewis et al. 2018; Schütze et al. 2018; Sattar et al. 2019; Benitez et al. 2019; Jeon et al. 2020; Landaluce et al. 2020; Moheimani et al. 2020; Gupta et al. 2021; Lee et al. 2021; Sakai et al. 2021).

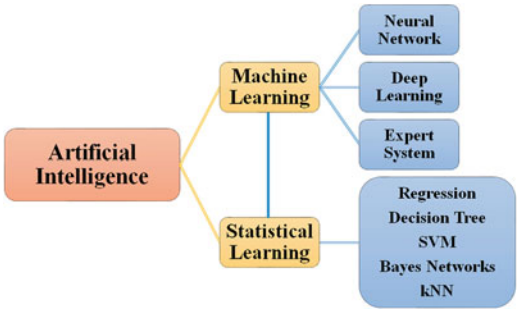
Artificial Intelligence Relevance and Implications

Artificial intelligence (Fig. 2) is a set of technologies that enable software and computers to perceive, comprehend, act, and learn on their own or in conjunction with humans. In comparison to human labor, industrial manufacturing may attain improved efficiency using AI (Lee et al. 2018; Peres et al. 2020). Furthermore, AI can allow robots to execute activities that a human would be unable to accomplish, such as managing hazardous raw materials or tiny components (Peres et al. 2020). It's critical to understand that many of these industrial robots aren't very sophisticated right now. In many circumstances, they are capable of doing a wide range of activities, although they are only programmed to a limited extent. You'll have to

Fig. 1 Classification of smart sensor

reconfigure it if you need more. The fact that this technology is improving with Industry 4.0 is driving ongoing expansion in the field.

Fig. 2 Artificial Intelligence



In 1955, John McCarthy established the notion of AI, claiming that a computer could replicate all aspects of learning and essential domains of intelligence (Andresen 2002). "While it is difficult to pinpoint the roots of AI," Haenlein and Kaplan (2019) wrote, "it is plausible to trace back to the 1940s, particularly 1942, when the American Science Fiction writer Isaac Asimov produced his short book Run about." The plot of Run-On centers on a robot developed by an engineer. Marvin Minsky and John McCarthy, a Stanford computer scientist, coined the phrase "artificial intelligence" in 1956. "The study of intelligent problem-solving behaviour and the creation of intelligent computer systems" is how AI is defined. In other words, AI describes how computers do tasks that would need intelligence if performed by humans" (Wisskirchen et al. 2017).

AI is split into two categories, according to Wisskirchen et al. (2017): weak AI and strong AI. When AI is poor, "the computer is essentially a tool for evaluating cognitive processes, and the machine mimics intelligence." Strong AI, on the other hand, refers to "processes in which computers are cognitive, self-learning processes," in which "computers can grasp and optimise their behaviour based on their prior behaviour and experience with the suitable software / programming" (Wirtz et al. 2019). Strong AI has the ability to automatically network with other computers, which has a large scalability effect (Osuizugbo and Alabi 2021). Deep learning, robotization, dematerialization, the gig economy, and self-driving cars are just a few of AI's most important economic fields (Wisskirchen et al. 2017; Osuizugbo and Alabi 2021).

Despite the fact that AI has been studied for decades, Benko and Sik Lányi (2009) believe that the discipline is wide and vague, making it one of computer science's most perplexing topics. From reasoning robots to search algorithms used in board games, AI is claimed to include everything. To put it another way, AI is expected to have applications in almost every aspect of how humans use computers in society. AI is also used in subtler ways, such as analyzing purchase histories and influencing marketing decisions (Simon 1995). According to Buchanan (Buchanan 2019), robots have always been a part of the public's perception of intelligent computers, despite early robotics efforts focusing more on mechanical engineering than intelligent control. Because of the potential of AI, robots, on the other hand, are becoming

more powerful vehicles for testing hypotheses about intelligent behavior all across the world. However, it has been argued that AI is not just concerned with robots. It's also about using computers as test beds to learn more about the nature of intelligent cognition and behavior (Simon 1995; Benko and Sik Lányi 2009; Buchanan 2019; Osuizugbo and Alabi 2021).

AI gets improved each time, and its costs falls. Implementing more complicated AI algorithms has also helped businesses to assess the purchase of new technologies that enable them to solve issues and make more complex and secure decisions (Dopico et al. 2016). AI has resulted in a shift in the way industry operates, owing to a new kind of interaction between man and machine. Intelligent factories, which have given birth to Industry 4.0, are characterized by cloud-based interactions between people and cyber-physical systems. Intelligent factories absorb automated structures and feature digital enablers that enable machines to connect with one another and with the factory's overall systems through an IoT setup. These abilities are in high demand by industries across the board, as they attempt to maintain the competitiveness of their manufacturing units in an increasingly technology environment. When used correctly, AI may provide various benefits to the industrial business (Dopico et al. 2016; Rouhiainen 2018; Azizi 2019). A list of three of the most important benefits for the manufacturing industry today is compiled and stated below:

Error Reduction – Intelligent algorithms, if taught, may perform very effectively in jobs that are prone to human mistake. Algorithms should be immune to the effects of external stimuli since they are not vulnerable to them.

Cost Reduction – Robots are being used by a number of e-commerce sites and banks to commence client assistance. If the situation is more complicated, the human attendant is summoned. Companies may cut personnel expenditures or reassign people to more important responsibilities, allowing them to enhance profits and concentrate on their core business.

Revenue Growth – Decision-makers will have more time to think about the core business and leave other AI chores to personnel who make fewer mistakes and concentrate on more vital operations.

Implication of Industry 4.0 and Industrial AI's Impact on Manufacturing

To achieve the aforesaid efficiencies, intelligent manufacturing incorporates real-time data analysis, AI, and ML into the production process. According to Trend Force, this industry is expected to develop significantly (Fig. 3). According to the International Federation of Robotics (IFR) report, the worldwide smart manufacturing industry is expected to expand to more than $ 320 billion by 2020, with more than 2.6 million industrial robots deployed in companies.

The influence of AI on manufacturing may be divided into five categories (Waltersmann et al. 2021; Kinkel et al. 2022):

i. **Predictive quality and yield**: For manufacturers of all stripes, reducing production losses and minimizing production process inefficiencies has always been an ongoing challenge. This is truer now than ever before, as rising demand meets greater competition. On the one hand, consumer expectations are greater than they have ever been, and global consumer habits are increasingly "westernizing" as the population boom continues. According to recent polls, the world population will increase by 25% by 2050, resulting in an extra 200,000 mouths to feed every day. Consumers, on the other hand, have never had access to so many options in practically every product conceivable. According to recent research, shoppers are more inclined to permanently abandon even their favorite companies if, for example, a product isn't accessible on the shelf.

   Manufacturers can no longer afford to ignore process inefficiencies and the resulting losses in the face of such a background. Every loss in waste, yield, quality, or throughput eats into their bottom line and gives the competitors an inch – providing their manufacturing methods are more efficient. Many industries, especially those with complicated processes, face a glass ceiling when it comes to process optimization. Some inefficiencies have no clear reason, and process specialists are stumped as to why they exist.

   Predictive Quality and Yield use Industrial Artificial Intelligence to uncover the underlying causes of many of the day-to-day production losses that businesses suffer. This is accomplished by continuous, multivariate analysis, which employs ML algorithms that have been specifically taught to comprehend each distinct manufacturing process. "Supervised learning" is the AI/ML approach utilized here, in which the algorithm is taught to discover trends and patterns in the data. Automated suggestions and warnings may then be created to notify production teams and process engineers of an impending issue, as well as offer critical information about how to avoid losses before they occur.

ii. **Predictive maintenance**: One of the most fundamental and well-known uses of Industrial AI is predictive maintenance. Predictive maintenance, rather than performing maintenance on a predetermined schedule, uses algorithms to predict

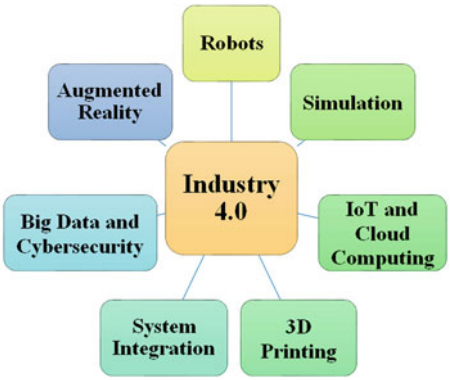


Fig. 3 Component of Industry 4.0

the next failure of a component, machine, or system, and then alerts personnel to perform focused maintenance procedures to prevent the failure, but not too early to waste unnecessarily downtime. Predictive maintenance systems, too, use ML methods to make predictions (although in a different category – unsupervised learning rather than supervised learning). The benefits are vast, and in many circumstances, they may dramatically cut expenses while obviating the need for scheduled downtime.

Systems may continue to work without interruptions if a failure is predicted using a ML technique. When maintenance is required, specialists are notified about the components that need inspection, repair, or replacement, as well as the tools and procedures to utilize. Predictive maintenance also extends the Remaining Useful Life (RUL) of machinery and equipment by preventing secondary damage and using fewer labor resources to complete maintenance operations.

iii. **Human-robot collaboration**: According to the IFR, there will be 2.67 million industrial robots in use globally by 2020. Workers will be provided training for higher-level employment in programming, design, and maintenance when occupations are taken over by robots, according to the general approach. Meanwhile, as industrial robots are certified for work alongside people, the efficiency of human-robot collaborative work is improving. As the use of robotics in manufacturing grows, AI will play an increasingly important role in protecting the safety of human workers while also empowering robots to make judgments that may improve processes based on real-time data acquired from the factory floor.

iv. **Generative design**: AI may also be used in the design process by manufacturers. Designers and engineers may utilize an AI algorithm, often known as generative design software, to explore all the potential configurations of a solution using a precisely stated design brief as input. Restrictions and definitions for material kinds, manufacturing processes, time limits, and budget restraints may be included in the brief. ML may then be used to test the set of solutions created by the algorithm. The testing step adds to the knowledge of which ideas/design choices succeeded and which did not. Additional improvements may be made in this manner until an ideal solution is identified.

v. **Market adaptation/supply chain***:* AI is pervasive across the Industry 4.0 ecosystem, not only on the factory floor. One example is the employment of AI algorithms to improve industrial operations' supply chains and enable them better adapt to and predict market changes. An algorithm may use demand patterns classified by date, location, socioeconomic features, macroeconomic behavior, political state, weather patterns, and other factors to provide market demand estimates. This is groundbreaking information for manufacturers, who may use it to improve inventory management, personnel, energy usage, raw materials, and financial choices about the company's strategy.

Metrology and Impact of Industry 4.0

Metrology, or measurement science, is an important part of production that is often disregarded (Aswal 2020a, b). It is critical to undertake this examination to verify that all component pieces fit together and function correctly. As a manufacturer, you'll want to think about more than simply the cost of a project; you'll want to think about the precision, consistency, and speed of production as well. It's crucial to think about when the measurement should take place in this procedure. Depending on your field, this will differ.

The growth of the Industry 4.0 idea brings with its new problems and a stronger metrological foundation (Varshney et al. 2021). As a result, it is necessary to reconsider measuring phenomena in order to bring Industry 4.0 and Metrology 4.0 together. For prospective industrial needs, there is a great need for online metrology. Industry 4.0's major objective is to improve throughput quality while increasing efficiency (Castelo-Branco et al. 2019). The measurement of multi-dimensional quantities in real-time mode while maintaining the accuracy of measurement findings is one of metrology's primary problems. The creation of sophisticated national and international standards governing the industry's present and future requirements is the most important aspect. Furthermore, Industry 4.0 (Wisskirchen et al. 2017) requires the adoption of proper measuring techniques and procedures that comply with applicable national and international standards, as well as the selection of appropriate equipment.

AI/ML is seeping into a variety of processes in the fabrication and packaging industries, but not always for the intended purpose (Dal et al. 2019; Bécue et al. 2021). The semiconductor business is just now starting to figure out where AI fits in and where it doesn't. AI works best as a tool in the hands of someone who has extensive subject knowledge. Certain tasks, such as pattern matching across large data sets, are ideally suited to AI. It is not as effective as engineers with years of expertise in fields like metrology, test, and inspection. However, when AI and experienced humans work together, they often deliver greater outcomes than any of them could do alone.

In the sphere of science and technology, metrology is the experimental and theoretical assessment of any measurement uncertainty. Scientific metrology, which deals with measurement standards, industrial metrology, which deals with measurements in an industry, and legal metrology, which deals with the legislation and protection of users of measuring devices (Aswal 2020a), are the three forms of metrology. The traceability of calibration and the analysis of uncertainty using international standards guidelines are two of the most important aspects of metrological infrastructure. With the use of smart sensors linked with ML algorithms for automated decision-making, the elements of measurement are changing as a result of digital transformation of industries and data analytics. The goal is to strengthen the metrological infrastructure necessary for better quality infrastructure in smart factories by implementing traceable calibrations in compliance with applicable national and international standards set in response to future industrial needs (Andonov and Marija 2018). Under the aegis of Industry 4.0, establishing a virtual link between the

calibration laboratory and client locations will be critical in calibrating various sensors in far less time without dismantling them in the system, thereby improving the accuracy, precision, and standardization of products and services (Castelo-Branco et al. 2019). Calibration may be done online/remotely via data transmission rather than instrument carriage in this scenario, and data can be supplied electronically from even the highest-level bodies such as National Metrology Institutes (NMIs) to customer sites. This will save time and money while maintaining the instrument's accuracy and specification. Sensors have an impact on smart cities and smart mobility, which is not restricted to industry. These sensors are self-calibrated and tested, and their status is recorded based on the kind and method of calibration. Calibration certificates are used to store this information. Smart sensors are utilized in Industry 4.0, which can connect through wireless communication protocols, allowing calibration results to be transferred to remote locations without requiring a lot of human work. Because today's sensors are mostly IoT devices, this measurement may be coupled to cloud services to build an Internet of Measurement Things (IoMT). The physical calibration system, the Metrological Information Infrastructure Cloud (MIIC), and an application software that may be utilized in metrology are all incorporated into the IoMT architecture provided in reference (Dopico et al. 2016; Rouhiainen 2018; Waltersmann et al. 2021).

Challenges in Industry 4.0 and AI

I defy anybody in the manufacturing business to spend a day without hearing the term "Industry 4.0" (Castelo-Branco et al. 2019; Compare et al. 2019). And, at an era when the majority of businesses are pursuing digital transformation, it's natural for manufacturing technology to evolve to cover both automation information and data interchange (Taymanov and Sapozhnikova 2018). With cyber-physical systems, augmented reality, IoT, and cloud computing all operating in unison, the possibilities for Industry 4.0 in manufacturing and beyond are almost limitless. When investing wisely in the factory of the future, keep an eye out for the following possible roadblocks (Fig. 4) (Vempati 2016; Andonov and Marija 2018; Taymanov and Sapozhnikova 2018; Horváth and Szabó 2019; Aswal 2020a; Ghadge et al. 2020; Vuksanović et al. 2020; Varshney et al. 2021):

i. **A Gap in Technical Skills**: The labor requirements are always changing. Are your personnel up to the task? When recruiting for unfilled roles, seek for candidates that have "digital dexterity," or an understanding of both industrial processes and the digital tools that support them. Only with the appropriate staff will company models be able to properly integrate and sustain new technologies.
ii. **Computing Power**: The quantity of power used by these power-hungry algorithms is a deterrent for the majority of developers. ML is the building blocks of modern AI, and their effective operation requires an ever-increasing number of cores and Graphics Processing Units (GPUs). We have concepts and skills to apply ML frameworks in a variety of disciplines, including asteroid monitoring,

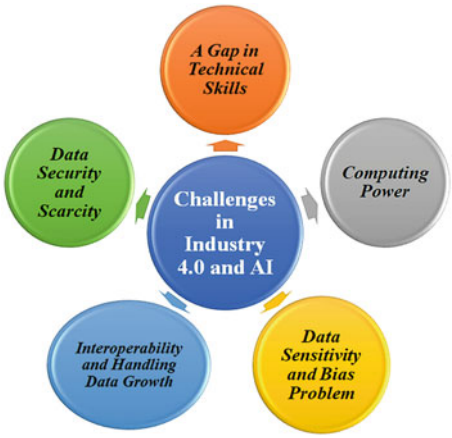


Fig. 4 Challenges in Industry 4.0 and Artificial Intelligence

healthcare deployment, and cosmic body tracing. They need the computational capacity of a supercomputer, and certainly, supercomputers are not cheap. Although the availability of cloud computing and parallel processing systems enables engineers to work more efficiently on AI systems, these benefits come at a cost. Not everyone can afford it, especially as the influx of enormous amounts of data and the complexity of algorithms continues to grow.

iii. **Data Sensitivity and Bias Problem**: The advancement of technology has also increased worries about the privacy, ownership, and management of data and intellectual property. A frequent occurrence? To develop an AI system properly, data is necessary to train and test it. To do this, data must be shared. Many businesses, on the other hand, are hesitant to share their data with third-party solution developers. Additionally, our present data governance standards for intra-organizational usage are insufficient to facilitate inter-organizational data exchange. Data is a valuable asset – safeguard it!

The quality of an AI system is entirely dependent on the quantity of data on which it is trained. As a result, the capacity to acquire high-quality data is the key to developing future-proof AI systems. However, in fact, the data that businesses acquire on a daily basis is insufficient and has no intrinsic value. They are prejudiced and serve solely to identify the nature and characteristics of a small group of individuals who have similar interests on the basis of religion, ethnic origin, gender, community, and other racial prejudices. The only way to effect significant change is to define some algorithms capable of tracking these issues effectively.

iv. **Interoperability and Handling Data Growth**: Another key problem is that protocols, components, products, and systems are not isolated. Regrettably, interoperability limits businesses' capacity to innovate. Additionally, since users cannot readily "switch out" one vendor for another or one component of the system for another, interoperability restricts system component upgrades.

As more businesses grow reliant on AI, they will be confronted with an increasing amount of data created at a quicker rate and presented in numerous forms. To sift through these massive volumes of data, AI systems must be more understandable. Additionally, these algorithms must be capable of combining data of different sorts and time periods.

v. **Data Security and Scarcity**: Another key problem is the threat posed by present and potential industrial vulnerabilities. Real-time interoperability is enabled by the physical and digital technologies that comprise smart factories – but at the cost of expanding the attack surface. When a smart factory's various machines and gadgets are linked to a single or several networks, vulnerabilities in any one of those pieces of equipment might expose the system to attack. To overcome this problem, businesses must anticipate both enterprise system and operational weaknesses at the machine level. Businesses are ill-equipped to deal with these security concerns, with many depending on technology and solution providers to identify weaknesses.

With big firms such as Google, Facebook, and Apple facing criminal accusations for unethical use of user data, many nations such as India are enforcing rigorous IT regulations to control the flow. As a consequence, these organizations now confront the dilemma of leveraging local data to construct global applications, which would result in bias. AI is heavily reliant on data, and labelled data is used to teach robots to learn and predict. Certain businesses are attempting to invent new approaches and are concentrating their efforts on developing AI models that can provide correct results in the face of data shortage. When information is skewed, the whole system may become defective.

Artificial Intelligence Application

In health sector currently, AI is being utilized in managing the medical data and records. Main application of AI is data management. AI-based robots store, collect, re-format, and trace data to provide quicker more constant access. In this section different applications of AI in health sector is discussed. Currently, AI is being utilized in managing the medical data and records. Main application of AI is data management. AI-based robots store, collect, re-format, and trace data to provide quicker more constant access. AI is playing important role in doing repetitive jobs accurately and faster of health sector such as CT scans, X-Rays, analyzing tests, data entry, etc. AI-based collected data is helping in treatment design by analyzing the notes and reports of patients.

As a result of our scientists' efforts, a low-cost and speedy diagnostic of COVID-19 patients is being developed in the healthcare industry (Mhlanga 2021; Srivastava et al. 2021). At least some people will be saved, and they will have access to an enormous quantity of information that they may use as soon as possible to protect themselves and their loved ones. AI has been used extensively in the search for a cure for COVID-19 in order to discover new drugs. The application of artificial intelligence (Fig. 5) by many researchers is aimed at discovering new treatments and

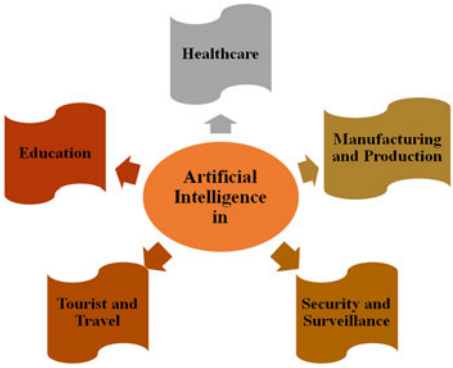


Fig. 5 Application of Artificial Intelligence

therapies for cures, with some computer scientists concentrating on the detection of infectious individuals using medical image processing, such as X-rays and CT scans (Kalyanakrishnan et al. 2018; Dal et al. 2019).

Manufacturing in the current era comprises logistic systems with ubiquitous and dominant computing networks. These systems are powered by millions and trillions of bits of data generated by systems, people, sensors, smart devices, and machines (Khemani 2012). With the fast expansion of data, the system's calculation skills must be analyzed more thoroughly and comprehensively than in the past. All of this has raised the relevance and usefulness of AI, ushering in a new era of technology known as Industry 4.0, also known as Smart Factory. Progressive cognitive computing and deep learning techniques have begun to be employed in various industrial processes and systems, such as production line inspection, problem identification, and keeping up with essential maintenance. Active efforts are made to implement and strengthen the various ways for handling the needed materials in the plants and production lines. The various methods available based on AI technology, IOT, research approaches, and cyber-physical systems are being used.

Smart manufacturing envisions systems that give us with detailed information on each stage of a product's lifespan. Deep Learning is used in descriptive analytics to evaluate product quality, diagnostic analytics to identify faults, and predictive analytics to anticipate defect occurrence (Wang et al. 2018). The first facet of implementing Deep Learning in manufacturing is product quality control. Machine vision combined with Deep Learning has been demonstrated to be beneficial for surface quality inspection. It is used to identify abnormalities on textured or non-textured surfaces such as scratches, wears, dirt, discoloration, and burrs. Furthermore, deep convolutional networks enable us to do object identification, which may be used in robot-assist tasks (e.g., a robot arm mounting a part on the CNC machine, robot object manipulation or sorting) (Hernavs et al. 2018). The second group includes diagnostic analytics and defect assessment tasks, when Deep Learning approaches are required. CNNs were utilized in equipment to analyze problems in bearings, gearboxes, rotors, and wind turbines. Under varied operating circumstances, the planetary gearbox was diagnosed using AEs and a five-layer deep neural

network (Jia et al. 2016). These examples demonstrate that Deep Learning models outperform classic Machine Learning approaches using designed features, such as Support Vector Machine and backpropagation neural networks. The last field of Deep Learning applications is predictive analytics for fault diagnosis, which includes maintenance and service prediction. RNN was used in (Malhi et al. 2011) to predict the long-term health of rolling bearings. Prediction of machining tool wear was made possible by combining CNN and bidirectional LSTM. Vanilla LSTM predicted the remaining usable life of an aircraft turbofan engine with high accuracy. These and comparable applications are critical for increasing production productivity and lowering maintenance costs.

Additive Industrial is a sort of AI used in manufacturing operations (AM). AM is commonly employed in the manufacturing process (Luckin et al. 2016). It has developed fast and massively, but its resilience and competency still need to be enhanced. As is generally known, AI is capable of jobs that humans are unable of performing with remarkable results when compared to human competence. AI-based technologies have the potential to reduce the workforce while increasing AM productivity and utilization by a factor of ten. Existing intelligent agents are used in problem statements including manufacturing, product design, and process design. The research proposes that by incorporating cloud edge computing into the already existing AM framework, the system may be greatly enhanced. As a result, as a general reflection, response time would be improved.

The various distributed AI in industrial systems is described more below. The Distributed AI (DAI) framework is utilized here to leverage the various structure and setup used for process planning in different plants and assembly facilities (Kubassova et al. 2021). The technique described above is used to divide whole production lines and jobs into smaller or subtasks. These smaller activities are carried out with the help of fundamental components known as intelligent agents. The solution is derived from all of the subsystems cooperating to solve the problem at hand – this is accomplished through the use of a method known as fuzzy management of the agents to derive the best suited solution to the problem as well as the most efficient and appropriate for implementation on the shopfloor. The foregoing is realized with the assistance of inputs from CAD, design data, short-term plan, DAI system, shopfloor observation, and so on. This offers a production plan that is possible for the actions conducted on the assembly line.

Agriculture is playing the main role in economic sector in India, and the automation of agriculture is the main concern nowadays. Traditional methods of farming are time consuming and are not sufficient to fulfill the requirement of our country. Application of AI-based sensors in various agriculture activities such as irrigation, plant disease detection, spraying, harvesting, weeding, etc. has brought a revolution. By using AI-based technologies, fertility of soil can be maintained, excess use of pesticides, herbicides, and water can be controlled, and also productivity and quality of crop can be improved. In automatic weeding manual labor is replaced by robot that uses lasers to kill weeds. In this technology robot drives through crop fields and cameras beneath display crops and weeds. AI-based software detects weeds and gives a command to remove them by attached high-energy lasers without disrupting

the soil. Similarly, in automatic harvesting systems, robots having camera, AI software, and mechanical arms are used for harvesting, to increase the speed of work and business, which is more efficient than humans and can work 24/7. AI also plays a key role in plant disease detection; around 20–40% of global food production is destroyed by pests and plant diseases. Nowadays AI-powered image recognition technique is being used for detection of more than 60 diseases.

AI plays a critical role in ensuring security in such a manner that only authorized users are permitted to access the material in order to prevent data leakage, which is referred to as digital leakage (Haenlein and Kaplan 2019). Security is the most concerned field of all time, and it should be included in all technological tools and apps, indicating the protection of each user's data and its respective information by avoiding the danger to be confronted. Several government entities are attempting to use these in order to ensure secure communication among authorized users (Ai et al. 2003). Several studies are also being conducted in order to cooperate with AI and IOT. Several Bots have been built in order to enable secure order delivery, and secure data transfer is also conceivable with this technology (Khanzode and Sarode 2020). For secure data transmissions, accurate results are produced. Data transmission occurs in a sequential way, i.e., first in, first out, in order to avoid data overlapping, and authentication and encryption of data are necessary at all times in order to enable safe data transfers. Technologies used for secure data transfer are very secret, and every time a system check is necessary to ensure that it is only accessible to authorized users. Deep learning is a branch of AI that has a copy of human learning and also analyzes multimedia, which has been the most challenging portion in the last decade.

Wireless communication appears to be the most efficient mode of communication, but it is still more prone to data leakage. There is a danger of data leaking during transmission, but it is the most efficient mode of information transfer. To prevent data leakage, the concept of artificial intelligence enters the discussion because the main concept behind artificial intelligence is completely based on human intelligence, and human intelligence includes the complete study and analysis of data encryption and decryption, and in order to successfully transmit the data, the complete study of artificial intelligence is required. Deep learning is a branch of AI that has a copy of human learning and also analyzes multimedia, which has been the most challenging portion in the last 20 years. Wireless technology is relatively loose in a certain region, and, for example, every mobile network has a data network.

The Intelligent Surveillance System (ISS) is a brand-new type of surveillance system with a focus on data analysis. The intelligent surveillance system's features include the ability to evaluate data that is used to detect or follow the behavior of forthcoming occurrences without interruption from humans. Using AI, sensors, and signal processing, this new technology can be used for a variety of purposes, including traffic monitoring, home security, and more. Sensory information from cameras is collected in enormous numbers by practical surveillance systems. A new technology in the sphere of national security, AI, is gaining a lot of traction in

technical advancement. Robotics and AI are being promoted by the US military's Department of Defense (DOD), which is expanding internationally at a remarkable rate for military objectives (Board 2019).

Travel and hospitality service companies have begun implementing AI into their operations, too, in the tourism industry (Ivanov and Webster 2017). Digital travel sales are expected to take off in the near future, and the business is expected to produce a sizable amount of revenue. These studies suggest that Internet and self-service technology are preferred by customers. Researchers believe that interactive self-service technology, based on AI, has the potential to boost customer satisfaction. These research findings indicate more than just clients' technological preferences. AI used to discover financial risk measures and monitors, which can be problematic if there is a lack of data or a lack of data protection, resulting in calculation errors. Manufacturing models, environments, and processes have been transformed by the incorporation of information technology, communication technology, product expertise, and AI. We are entering an era in which AI technology will be integrated with the Internet of Things (IoT), current generation information technologies, new renewable energy technologies, biotechnology materials, and other cutting-edge technologies. AI and AM are rising at a great rate in the manufacturing industry simultaneously, which allows them to scale the company, in the current scenario.

ML and AI analytics can be used to improve teaching quality and learning effectiveness by identifying at-risk students and providing early interventions, as part of the shift from a one-size-fits-all approach to a more personalized and personalized approach. The use of AI and other related technologies in assessing students' learning challenges may allow teachers to intervene in real-time to improve student learning outcomes. It is possible that AI can help humans become more productive. With disappointing results, computers were previously used in education. This has changed, and current AI research in education is providing extremely clear and firm outcomes (Zhang and Aslan 2021). Education resources like digital libraries and student data databases are maintained by computers. Programming abilities are being taught in schools using these computers/microcomputers, which are being used in a variety of industries. The educational sector has given a lot of thought to using computers with AI in the classroom.

Initially, AI took the shape of computer-based software and technology, eventually evolving into web-based intelligent education systems (Lee et al. 2020). Web-based chatbots and robots may now fulfill responsibilities and activities of instructors autonomously, thanks to recent technical advancements such as embedded computer systems and other technologies. Using these cutting-edge online platforms, teachers may now execute many administrative activities more efficiently and rapidly, such as evaluating and grading student performance and assignments, resulting in greater teaching quality. Many people confront obstacles while going to interviews in this competitive environment, such as introversion, insecurity, and a lack of hard or social skills. Training is required to enhance interview performance. Researchers recently created an intriguing application that simulates interviews by

merging virtual reality and artificial intelligence. They designed an interactive app for interview preparation by integrating the benefits of both technologies. To provide an appropriate answer, the system also examines the person's emotional intelligence and facial expressions.

AI is also applied in the automation of marksheet assessments and robotic teaching techniques (Malik et al. 2019). These varied approaches benefit both students and teachers, making their lives simpler. People nowadays employ advanced Natural Language Processing (NLP) to create sophisticated intelligent teaching systems. It also improves students' learning and comprehension rates by tailoring the course to each student's unique traits. The use of AI can help students acquire a variety of skills, such as the capacity to come up with innovative solutions and resolve contradictory assertions, among others.

The primary purpose of artificial intelligence in education is to give students with a personalized learning environment that helps them based on their preferences and personal attributes (Mogali 2014). Precision education requires learners to be provided with preventative and intervention methods by analyzing their learning patterns and pace, allowing the software to work as an intelligent tutor by incorporating knowledge and critical analysis of learner behaviors.

AI in India

An important part of India's strong IT infrastructure is based on the potential of various sectors, including health, education, energy, agriculture, and the environment, which have been considered by the National Science and Technology Anchor for smart applications, smart power generation and distribution, and smart management (Vempati 2016; Aswal 2020a, b; Garg et al. 2021). There are many plans for AI development in India. For this reason, the government has decided to invest their money in AI and transform India into a society that is completely automated. AI models have been pursued in order to estimate the timeline of India's oilseed output by taking into account increasing demand, policy environment, slow development in domestic production, and import charges (Marda 2018; Ghadge et al. 2020). Oilseed cultivation is an essential part of Indian agriculture because of the country's vast resources. AI is being applied in a few domains in Indian organizations, such as product creation, retail, internet, and conversational administrations (Garg et al. 2021), to discover concerns. A growing number of companies are turning to AI as a way to tailor the experience for customers. More and more businesses are adopting AI for a wide range of purposes as the technology improves and becomes smarter. There are many ways in which AI is being used in the banking business; this is just one example of how it is being used (Malali and Gopalakrishnan 2020). AI has been implemented by Indian banks to improve operational efficiencies and customer satisfaction. Its primary goal in the banking industry is to anticipate customer behavior and make operational decisions based on that information.

Conclusion

To support the Industry 4.0, a smart sensor and AI-enabled industrial infrastructure must be in place that is traceable to digital metrology, traceability, and certification. This chapter discussed the many parts of this infrastructure. This chapter covers the different issues that arise when incorporating AI into Industry 4.0 with the use of smart sensors. Manufacturing processes are made more intelligent through the use of AI and digital quality infrastructure, which is demonstrated in this chapter. The purpose of this study is to provide a succinct overview of the most recent technological advances and obstacles in this subject. AI has helped us better grasp what intelligence is and how it might be put to use in a variety of contexts. AI is a rapidly developing field that is allowing society to live as simple a life as possible. From online shopping to watching TV, it has slowly but surely permeated every facet of our life. In other words, it is the study of computers and their components. Currently, AI is a trendy topic in every industry. There are both advantages and disadvantages to the technology being developed at such a rapid pace. It has a wide range of applications, according to the research I've done. These include everything from health care to farming to businesses to transport to education. As a result of the use of high-end processors, these applications virtually eliminate the possibility of human error and human engagement throughout the process. Because AI makes education more accessible, it can help alleviate some of the unemployment that has resulted as a result of the technology's widespread adoption. As a result, AI is a huge asset to our society, which necessitates further advancement in the form of AI. There are a number of strong pillars for building a national digital quality infrastructure in the metrology area, such as the planned metrology cloud, digital calibration certificate, and development of the Measurement Information Infrastructure (MII) as that focused. A solid foundation for digitalization is provided by the integration of Indian industry with India's IT infrastructure, the latter of which is the largest IT exporter in the world. For IT as well as other fields, this is a win–win situation for expanding the field of R&D. IoT and various technological advancements in conjunction will lead to research and development in smart sensors, AI, wireless protocols, IoT devices, and cloud computing for Industry 4.0. This will serve as the backbone of inclusive growth in the economy of a country, as well as a means of increasing productivity.

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Fusion of Smart Meteorological Sensors, Remote Sensing Techniques, and IoT in Context of Industry 4.0

43

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Abstract

In general, the Internet of Things (IoT) concept allows any type of electrical equipment to connect the Internet (thing). As both an outcome, any "thing" will be assigned an Internet Protocol (IP) identifier and will be reachable using normal Web protocols such as Domain Name System (DNS). The IoT will be a technology or functionality associated with conventional ISA-95 automation systems in an automated environment. IoT automation elements include sensor, actuators,

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Programmable Logic Controllers (PLCs), control loops, data analytics, control models, simulators, and optimizers.

The IoT, Industry 4.0, and intelligent systems might all be combined to produce a strong mixture that foreshadows an amazing future with enhanced efficiency. In many areas, especially the industrial sector, long-term growth is crucial. With the help of an intelligent system, we may be able to detect and address significant issues in Industry 4.0. As a consequence, the concept may be thought of as a physical device or a feature computed in a software system that runs on any device with sufficient computing power. An IoT device presently lacks a standardized technology. Several projects to standardize are now in the works. For example, the International Telecommunication Union has a standardization working group SG20 on "IoT and its applications, including smart cities and communities," and Institute of Electrical and Electronics Engineers (IEEE) has an ongoing IoT Ecosystem study as well as the IEEE P2413 draught standard for an architectural framework for the IoT.

This chapter provides a review of the literature on revolutionary industrial growth and the difficulties and advantages of Industry 4.0. One of the most pressing issues in the current scenario is achieving long-term growth. Air quality has become a major problem on a global scale. This chapter also includes a case study on the merging of meteorological sensors with remote sensing devices. This case study demonstrates the relationship between meteorological sensors, such as temperature, wind, and relative humidity, and Atmospheric Boundary Layer (ABL) as well as its importance in regional air quality monitoring. The case study also highlights the necessity for advanced technologies such as the IoT and Artificial Intelligence (AI) to be integrated into the field of environmental monitoring.

Introduction

The Industrial Revolution changed society in acquiring new knowledge, analyzing events, and innovations. Many structural changes have occurred in cities and their populations over the course of the industry's long history, as well as in product demand, supply chain processes, production process development research, and workplace culture. The emergence of an industry can be categorized into three fundamental pillars: knowledge, experimentation, and entrepreneurial innovation as shown in Fig. 1.

These three pillars are the root of evolution and growth of an Industry. Industry evolution has successfully acted as a medium for creating new employment opportunities for various products to satisfy people's demands (Vaidya et al. 2018). Consequently, the industry's development is spread out over a number of stages and is of equal importance. There are four stages of industrialization: the First, Second, Third, and Fourth revolutions. Industry 4.0 has utilized the previous industrial revolutions with much faster pace and incorporates a wider range of new

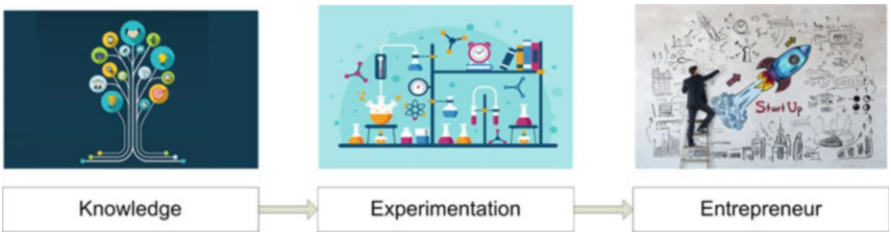

Fig. 1 Three pillars of Industry evolution

technology. Industry 4.0 has changed the internal and external aspects such as product development, distribution techniques, consumer needs, and job culture. Industry 4.0's capabilities and potential must be understood in order to maximize its potential. It is critical to be aware of the new opportunities presented by Industry 4.0 in order to collaborate with stakeholders and contribute to its growth (Prisecaru 2016).

Industrial revolutions can be planned based on the combination of the three main characteristics of industry's creation, namely, knowledge, experimentation, and entrepreneurial innovation as shown in Fig. 1. In the past, a scholar or an inventor improved their ability to absorb and integrate information, knowledge, and their research for the faster growing industry. The experimentation and collaborative works enhance the production and development process. The third component of entrepreneurial innovation is employers' belief in new discoveries, work practices, and ways of manufacturing and developing their company on unimaginable scales in accordance with the opinion in and reinvest in these breakthroughs. The development of new technologies, such as energy sources, equipment, materials, and processes alone cannot account for industrial revolutions. To be sure, economic growth over the last two and a half centuries relied heavily on these forces. Hydrostatic mills, for example, were in use many years before the invention of the computer. As a result, this era is marked by the widespread use of machines, which we refer to as machinery (Shariatzadeh et al. 2016). There have already been four major industrial revolutions, as seen in Fig. 2 (Jeschke et al. 2017).

Water, animal energy, and wood were the primary construction and combustion materials at the beginning of the first industrial revolution. Iron was commonly employed in construction throughout the Industrial Revolution, coal was the dominant source of energy, and steam was the major source of energy (Zhou et al. 2015). There were three distinct stages of industrialization documented. The steam engine was created during first of them, the so-called First Industrial Revolution, and Adam Smith was its greatest theorist. He is also credited with inventing economic liberalism (Chen 2017).

The Industry 1.0 came to a conclusion in the nineteenth century, and significant developments occurred during this time. The industrial production and consumer goods divisions have extended to nations other than England throughout this time period. Railroads, telegraphs, and steel-hulled transoceanic steamboat sailing

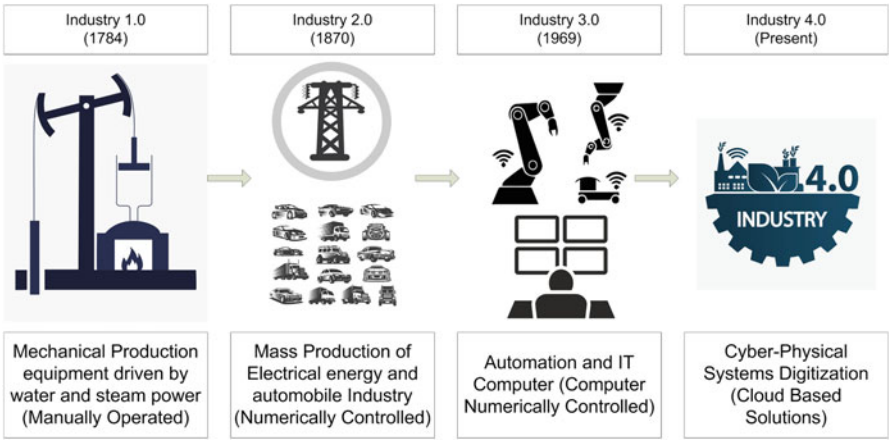

Fig. 2 Historical timelines of Industrial Revolutions

changed transportation and communication in the mid-nineteenth century. Expansion of a company's activities beyond national borders, which was formerly seen to be a combination of business and adventure, is now achievable and financially encouraged (Zhong et al. 2017).

Following Industry 1.0, the so-called Industrial 2.0 began in the late nineteenth century. It included breakthroughs in the chemical, electrical, oil, and steel industries, as well as other advancements like water steel ships, improved aeroplanes, food canning, bulk consumer product manufacturing, mechanical cooling systems, and the invention of the electromagnetic mobile phone. Throughout this time, a modern generation of industry of great potential began in Germany, the United States, France, and the United Kingdom. Taylor's modern management program (Patel et al. 2017) is how the Industrial 2.0 is known in the United States. New resource and energy sources, like electricity and, progressively oil, have aided in the changeover of Industry 2.0.

Around the turn of the nineteenth century, the world economy started to show indications of cyclicality. In the 1920s and 1930s, the growth of automotive and aviation sectors, as well as telephone and radio broadcasts, intensified the transformation in communication and transportation technology (Zhong et al. 2017). According to the author (Shariatzadeh et al. 2016), an Invention of the steam engine occurred in the latter part of the nineteenth century. While the former depended on iron and coal-fired steam power, the later relied on electricity and steel, resulting in substantial advances in communications, chemistry, and petroleum use. Because they had not yet totally replaced the previous ones, these improvements did not begin to stand out until the twentieth century.

As part of the third Industrialization (Industry 3.0), which began in the 1970s, the economic system offered collaboration as a new paradigm, stating that an organization can only be realized collectively (Domingo Galindo 2016). As a consequence, a large number of businesses were employed, and their development became more

focused on developing strong couplings with the same level of collaboration. Third-generation revolutions have ushered in a new approach to economic management, stressing more horizontal and flexible bargaining across enterprises in order to accomplish more precise and integrated economic activity (Chen and Limchimchol 2006). The development of financial economy instruments, often known as the "market economy," enabled the Third Industrial Revolution to speed media and transportation growth, as well as globalization (Patel et al. 2017). Biotechnology, computer science, Computer Numerical Control (CNC), microelectronics, and telematics integrated systems are just a few of the technologies that were widely used in Japan in the mid-1970s as part of this revolution.

"Industry 4.0" refers to a new industrial revolution that emphasizes and incorporates the most recent technical advancements, allowing for more rapid and customizable manufacturing. The phrase was coined in 2011 in Germany to represent closely related improvements in automation and information technology (Pandiyan et al. 2018). The integration of the Internet and production processes with the assistance of smaller sensors and Artificial Intelligence (AI) on machines is envisaged as part of the Fourth Industrial Revolution's objective of enhancing productivity through "smart factories" (Zheng et al. 2017). For industry 4.0, there are many different definitions and nomenclatures. However, as defined in this chapter, Industry 4.0 is a modern manufacturing model that includes a varied range of techniques that is not essentially unreported, but merged with each other and that of entire industry, and is distinguished by high immersive, digital, and scientific performance.

Although this model is still in its early stages, it already has chaotic characteristics in terms of environmental performance, as it is considered a sustainable manufacturing model, and human work, as it provides a set of potential technologies that aid the work performed by people in the industry, eliminating aspects such as intense physical effort and market positioning, as Industry 4.0 has an excellent response time to internal and external stimuli. As early as 2014, enterprises in Germany began assessing their readiness for Industry 4.0 implementation. At the time, at least 41% of German businesses had heard about "industry 4.0" and had launched some concrete efforts. The road ahead, though, is long, and many companies are still unfamiliar with the term. This is especially true for small businesses, as 44% have no idea what Industry 4.0 is.

The modern manufacturing model, on the other hand, is well-known among larger businesses, with just 17% unaware of its existence. There is also a time lag between big industries and small- and medium-sized firms in implementing Industry 4.0 strategy (SMEs). Only 17% of SMEs in Germany had implementation strategies, despite the fact that nearly 20% of original equipment manufacturers had effective implementation plans (Jiang 2018). Industry 4.0, according to Xu, (2012), is not only a technological challenge, but a reality that will fundamentally transform firm organizational structures. The authors provide five disruptive technology views, which are not meant to be exhaustive. As early as 2014, enterprises in Germany began analyzing their readiness for Industry 4.0 implementation. At the time, at least 41% of German enterprises had heard of the term "Industry 4.0" and had started the process. There is still a long way to go, and certain industries are still catching

up. According to a new poll, 44% of small-scale businesses have no understanding what Industry 4.0 is. Industry 4.0 is a paradigm shift in the following aspects:

- Improved social – technical interaction: Autonomous and self-organized production resources carry out planning activities in value chains between enterprises.
- Innovative goods: Both the products and the acceptable operating parameter of a certain manufacturing process are well-known. These goods can be bundled together to increase production.
- Personalized manufacturing: During the design, planning, production, and recycling processes, companies can take into consideration the individual qualities of client demand and product.
- Automation: Employees are in charge of configuring and controlling intelligent production resources based on context-oriented targets.
- Data Science: Managing a product's life cycle is made easier by the use of product-related data.

Significant Things of Industry 4.0

According to Tao et al. 2014, three general features of Industry 4.0 are smart plants, smart manufacturing, and smart logistics. For example, the "smart plant" is an important component of future smart infrastructure, concentrating on massive production structures and procedures, and also network distribution in production facilities and establishing the factory's designed to be completed (Bhardwaj et al. 2010). Figure 3 depicts the relationship between the three elements listed above, as shown in reference (Tao et al. 2014).

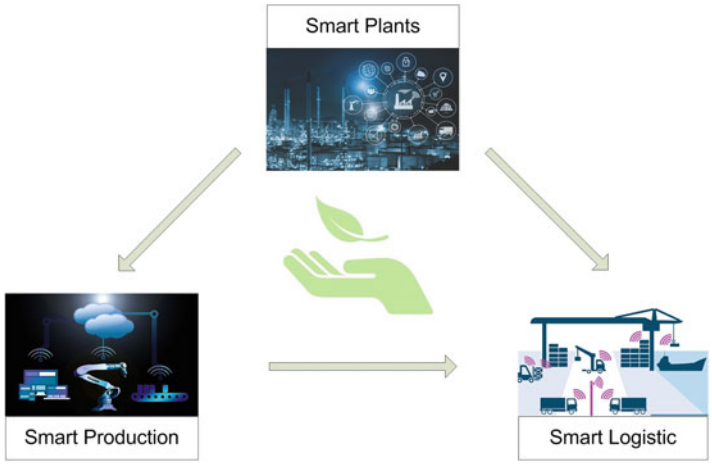

Fig. 3 The most significant things of Industry 4.0

Because of human-to-machine (H2M) contact and the use of 3D/4D technology in industrial processes, "intelligent production" is a high-capacity industrial chain capable of being flexible, customized, and actively maintained in a network. On the supply and demand sides, "intelligent logistics" works in combination with high-performance logistics resource efficiency, particularly via the Internet and logistics resource integration networks, to provide optimal logistical assistance. The production system in Industry 4.0 is made up of three unique components that work together to perform each other's jobs. Smart manufacturing, which attempts to manufacture tailored goods for consumers (Peng et al. 2009), is the foundation of this new sector.

Industry 4.0 Standards and Guidelines

Industrial Revolution 4.0 is a production industry trend or idea that stresses automation process and data transmission. The IoT, computer systems, and cloud computing are all used in this project. It is a German-based organization that pushes for total industrial process automation.

Interoperability, Virtualization, Decentralization, Real-time capability, Service orientation, and Modularity are the crucial six design principles as shown in Fig. 4 for optimum utilization of Industry 4.0.

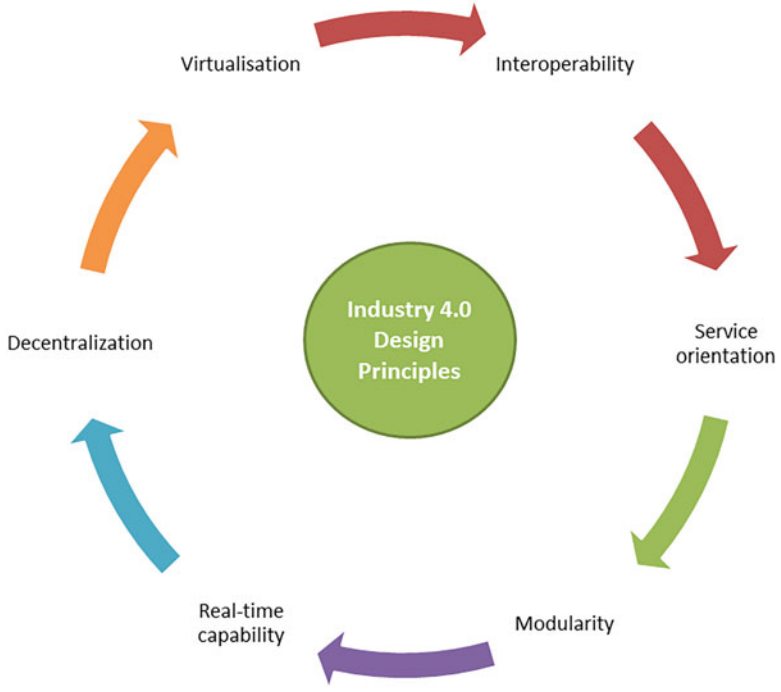

Fig. 4 Industry 4.0 standards and guidelines

Interoperability

A system has interoperability when people, machines, and objects all work together in an organization. The ability to link everything in a company, everywhere, and with everyone is critical for utilizing the insights provided by data in order to improve efficiency and procedures. In order to achieve interoperability, integrating things with cloud computing and digitization are the crucial aspects. Smart integration using open-source platforms and software such as Linux, Android, Python, Apache, and so many others are very important. Businesses can save money and time by allowing information to flow freely between systems, eliminating unnecessary duplication and allowing third-party applications to be used when necessary.

Virtualization

The term "virtualization" is used to describe two distinct scenarios, first one is where multiple physical resources are combined into a single virtual resource and second one is where many virtual resources are derived from a physical resource or a combination of resources. To begin, a business's hardware environment is replicated by building digital "twins" of actual assets derived from remotely sensed data. In the second case, software divides a single hardware resource into many virtual servers.

Using virtualization, hardware costs and physical resources can be reduced. Improved reliability and greater flexibility are two of the benefits of virtualization. With virtualization, all of an organization's computing resources are no longer tied to a single physical device. It's best to begin virtualizing for less critical assets such as desktops and servers before moving on to more significant ones.

Decentralization

In order to achieve virtually limitless scalability and adaptability, businesses have long embraced decentralization by moving systems away from a centralized computer. Using the cloud to store and transfer data, as well as automating repetitive tasks, is an example of decentralization. With both, the focus is shifted away from a single location. Industry 4.0 allows for the creation of decentralized systems across all industries on a global scale as all technology is decentralized.

Any company that adopts new technology will inevitably begin to see decentralization in their operations. To help this process, managers can disperse decision-making among a larger number of workers on the ground and delegate responsibility for day-to-day operations to smaller groups.

Real-Time Capability

The phrase "real-time capacity" refers to the ability to gather and analyze information in real time in order to make judgments anywhere at given time.

This enables a user to observe the business, and make real-time decisions to improve productivity. It also gives you the ability to look at micro trends, respond quickly to production line failures, and be proactive in addressing anomalies or inefficiencies. Data from sensors and internet-connected devices is now readily

available to the vast majority of businesses. Processing and analyzing data is a difficult but necessary task if business operations are to be continuously improved.

To help small- and medium-sized businesses (SMEs) build an inexpensive and integrated reporting platform for insights across multiple data sources, open-source tools are critical. Many services now offer embedded analytics that can be accessed at any time and from any location via cloud computing, allowing for real-time management of this data without the need to wait for reports.

Service Orientation

As a result of big data and interoperable systems, businesses now have the ability to respond in real time to their customers' needs. This enables businesses to respond to customers' evolving needs and expectations as they arise, allowing for a more personalized service. A consequence of this is a shift away from mass production and toward customized services for individual customers across all industries.

A growing number of farmers are using data insights to improve their equipment and supply chains and distribution models in agriculture. Using cloud computing and IoT devices, healthcare providers can streamline medication regimens, health reports, exercise regimens, and more for their patients.

Predictive modeling is at the heart of these services, which rely on data collection and analysis. If you want to stand out from the competition in a service-oriented model, you need to use data insights to create unique offerings for your customers. This includes identifying non-core activities that can be reduced or eliminated using technology so that employees can focus on the most important aspects of the company.

Modularity

Modularity, like service orientation, refers to a company's ability to respond quickly to shifting market conditions. Modularity can be viewed in two different ways. Small, well-defined teams that focus on specific aspects of the business operation in the first instance. In the second place, businesses that use modular systems, such as accounting or HR software platforms, are able to buy only what they need now and add more in the future as their needs change. In contrast to outsourcing, modular systems are designed to interact and connect to other parts of the business.

Both allow companies to concentrate on their core competencies. As a result, employees become experts in their core activities, providing value to customers in a variety of ways. Core activities can then be expanded as the business grows, increasing the company's agility and scalability, and allowing it to adapt more quickly to changing market conditions.

Industry 4.0 for Sustainable Growth and Smart Environment Monitoring

Industry 4.0 is changing the approach of manufacturers and regulatory authorities in the direction of sustainable growth. Sustainable growth is the need of the moment. Sustainable growth aims to develop a peaceful and prosperous society for people and

the planet. The primary focus of Industry 4.0 is upgrading the existing technologies with the latest techniques and preserving the environment, oceans, and forests. The global economy is growing at a very fast rate, which has a lot of negative effects on the environment and ecology. In order to capture the growth of the Industry while maintaining a healthy and clean environment, an effective environment monitoring system is needed (Ghobakhloo 2020).

The study of concepts and computer networks that can do activities that would ordinarily require human intellect, such as sense input and decision-making, is known as AI. "The ability of a system to accurately perceive external information, learn from it, and use what it has learned to achieve specific goals and adjusts its behavior accordingly," according to Kaplan and Haenlein (2019). To drive systems toward stated objectives, AI acts on external information supplied from IoT and other big data sources, employs knowledge-based rules (given by developers), or uses Machine Learning (ML) to identify the underlying rules and patterns. A really intelligent system can learn, generalize (if that's possible), acquire information, create goals and priorities, and reduce risk in decision-making processes.

The series of crises, such as COVID-19, overpopulation of the world, and lack of natural resources, demand an alternative approach for the world economy to grow and develop toward sustainable growth. Industry 4.0 is a revolutionary outlook for the world. It is equipped with advanced sensors, embedded systems, and data acquisition systems. The integration of the IoT makes data collection and data analysis much easier. It allows regulatory authorities to make better decisions in real time. The volume of the data has also increased multiple folds.

The cloud storage techniques solved the problem of managing a large amount of data from any world location. The ample amount of data opens the opportunity to implement the AI and ML solution on top of conventional approaches and provides an intelligent solution to the consumer. These advanced technologies lead to predictive maintenance, optimization of the process, and automation improvements to achieve high efficiencies and responsiveness to customers, which was not previously possible (Deepa and Prabadevi 2020).

"Expert systems" built on rules may be used to bring AI into industry, and this method is known as "rule-based" AI (Nilsson 1982). All of the rules that govern the dynamic behavior of the system are based on either physical principles or human expert knowledge gained via experience. In this strategy, IoT applications or sensors are used to continuously monitor the system or process. Individual system descriptors are extracted from the IoT sensors and examined using the rules. To display patterns, raw data is occasionally filtered and saved in a database. The participation of numerous unknown factors in atmosphere, the variability of meteorological parameters, and atmospheric pollutants all contribute to the high complexity of air-quality systems. This makes it difficult to translate scientists', meteorologists', and industry professionals' knowledge into well-articulated, well-defined procedures (computer programs) that may be used in AI-based expert systems (Goyache et al. 2001). Nonetheless, rule-based AI is appropriate in situations where real-time choices and control are required and regular training is not possible.

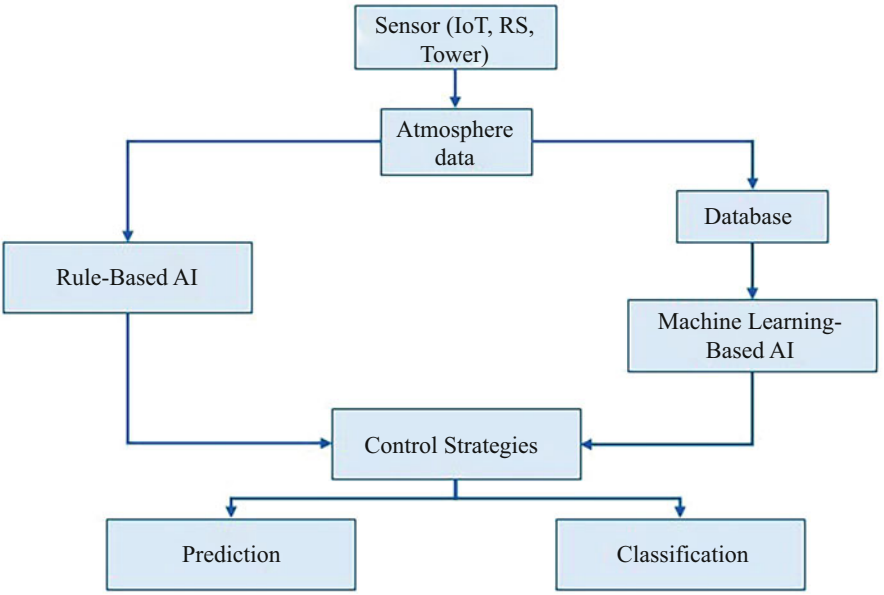

Fig. 5 Simple flowchart from sensor to data preparation and the role of AI

Because of the drawbacks of rule-based AI, ML-based AI has gained traction in recent years (Michalski et al. 2013). There could be a lag in data collection (sensing) and developing predictions or judgments in ML-based AI (for example, via IoT sensors) since the system is designed to look for data patterns (Fig. 5). ML-based AI systems currently do not employ human intelligence-based computer rules (i.e., obedience to any form of science, physical reality, or expert experiences); instead, they are completely data driven. ML-based AI is well suited to systems where frequent training is not a need and better precision is sought, as is the case with atmospheric systems.

Because the capacity to learn from the environment is frequently associated with intelligence, ML is one of the most vital aspects in AI. ML is an AI development approach that allows computers to learn without being explicitly programmed (Goyache et al. 2001; Michalski et al. 2013). Simply said, learning algorithms extract and agglomerate information from unstructured data into computer programs that may perform helpful activities like alerting users or triggering essential processes (Nilsson 1982). It delves into the research and architecture of algorithms that can learn from and forecast data; these algorithms go beyond merely static programming instructions by generating data-driven predictions and judgments (Solé 2007). The three primary categories of ML algorithms are shown in Figs. 6 and 7 shows the basic steps and process of ML.

Figure 8 depicts the numerous models that fall into these categories, as well as their technical applications. There are further in-depth descriptions of ML

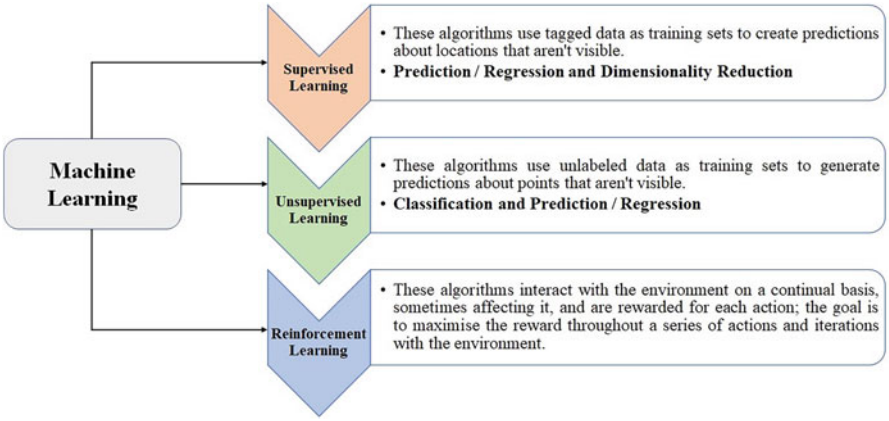

Fig. 6 Different types of ML

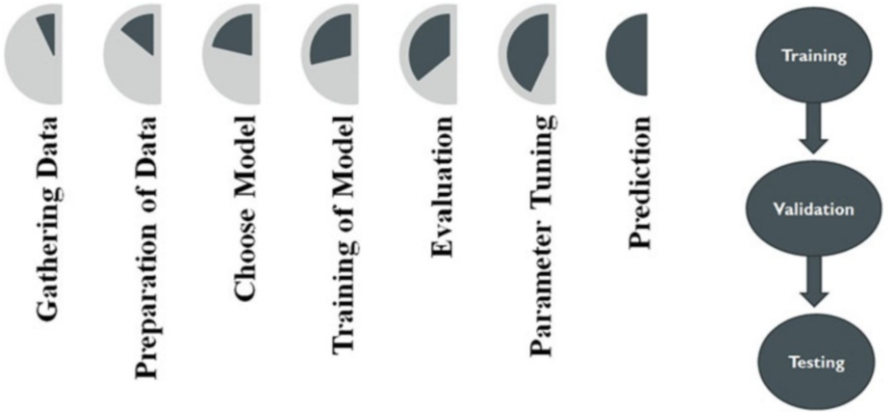

Fig. 7 Basic steps and process of ML

approaches and their atmospheric applications in (Bochenek and Ustrnul 2022; Moosavi et al. 2021).

Monitoring climate change in a time of substantial growth in the industry is also a challenging task. India is the fastest growing economy in the world. Therefore, an effective regional level monitoring of critical environmental parameters is pivotal (Chourey et al. 2022a).

The end objective of AI or big data processing in the industrial context is usually production control or automation. However, the ultimate deployment of all IoT, big data analyses, and AI systems is still up for debate, and cloud also involves collecting insights, making forecasts, and sending out alerts, in addition to automation (fault detection). The learning approaches' final nodes in Fig. 8 show some frequent instances of end applications for which data insights are developed using ML technologies.

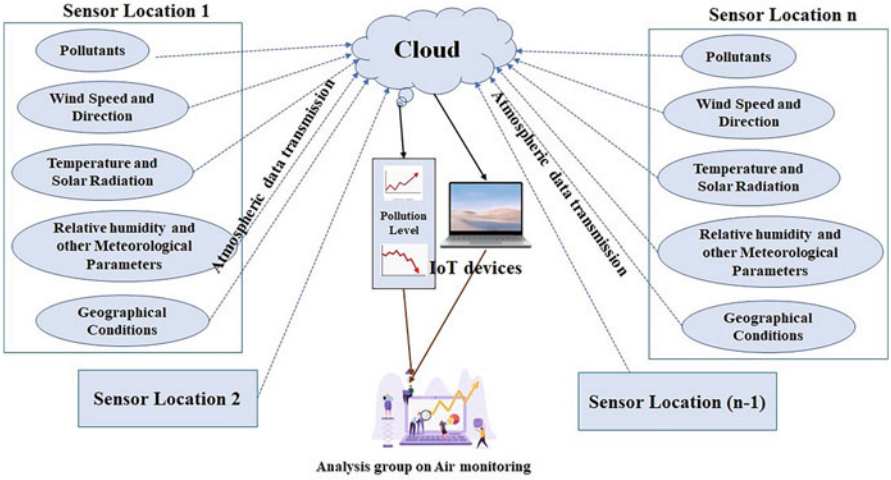

Fig. 8 Basic diagram of IoT framework in the context of air quality analysis

IoT-Enabled Meteorological Sensors and Remote Sensing

Meteorological and remote sensing sensors are critical in collecting observations from the environment. Including IoT and soft computing in the monitoring system offered insights for deeper environmental knowledge and analysis. Sensors are essential for a wide range of industrial applications (Chourey et al. 2022a). The chapter Smart Sensors for the Industrial IoT compiles the most recent research in intelligent sensor technology. It introduces the reader to the numerous industrial uses made possible by this technology. The chapter provide a broader perspective on the application of intelligent sensor devices in a variety of multidisciplinary fields such as Intelligent Transportation Systems, Healthcare, Agriculture, Drone Communications, and Security. This chapter guides the reader through the utility and improvements in intelligent sensors and their applications in a variety of study domains by providing an insight into intelligent sensors for industrial IoT.

Creating smart factories offers a fantastic chance for the manufacturing industry to reach the fourth industrial revolution. The analysis of massive volumes of big data generated from factory floor sensors provides real-time visibility of industrial assets. It can give tools for undertaking predictive maintenance in order to reduce downtime on equipment (Kumar et al. 2021a).

Using increased IoT devices in smart factories accelerates output and improves quality. Using Intelligence visual analytics to replace traditional business evaluation models reduces manufacturing errors while also saving money and time. Quality control personnel may monitor manufacturing activities from practically anywhere using a cellphone connected to the cloud. ML algorithms can help manufacturers discover flaws relatively soon, when fixes are more difficult (Talavera et al. 2017).

The principles and technologies of Industry 4.0 may be used in a variety of industries, including continuous and process production, and also oil and gas, mining, and other sectors.

The IoT promises to bring the same benefits to everyday objects as its widespread application has brought countless benefits to rural citizens around the world in the last two decades. The most adorable benefit was the ability to consume and produce data and services in real-time, which the IoT promises to bring the same benefits to everyday objects. Environmental monitoring is a broad use of the IoT that includes everything from monitoring Ozone levels to protecting national forests from smoking. With the use of Bluetooth, data from each node is transferred to a central node. The characteristics of environment monitoring sensors are shown in Table 1 for each center, which consists of five sensors with properties given in Table 1. The sensors used in environmental monitoring are listed in Table 2.

Table 1 Common details of SODAR-based meteorological sensor network (Chourey et al. 2022b)

Parameter	Sensor/Model	Sensitivity	Range
Atmospheric Boundary Layer Height (ABLH)	Monostatic SOnic Detection And Ranging (SODAR)	10 m	50–3000 m
Wind speed	Barani-MeteoWind-2	0.1 m/s	0–89 m/s
Wind direction	Barani-MeteoWind-2	1 degree	0–359 degree
Temperature	Barani-Meteotemp RH + T	0.3 °C	0–80 °C
Relative humidity	Barani-Meteotemp RH + T	1%	0–100%

Table 2 ML-based research on IoT-enabled meteorological sensors

Research	Purpose	Data and technique
Prediction of Air Pollution (Ayele and Mehta 2018)	Air pollution control	RFID, gas sensor, and IoT
ABL height and Air pollution (Chourey et al. 2022b)	Monitoring of air quality	SODAR, meteorological parameter, and IoT
ABL structure (Kumar et al. 2021a)	Atmospheric condition monitoring	Monostatic SODAR and classification model
Air Pollution (Shetty et al. 2020)	Monitoring of air quality	Gas sensors from mobile vehicle data, IoT, and ML
ABL height Prediction (Kumar et al. 2021b)	ABL height	Monostatic SODAR and LSTM Model
Indoor air quality (Yang et al. 2019)	Monitoring air quality in a confined space	IoT, Volatile Organic Compound (VOC), Long range WAN
Intelligent air quality analysis (Thu et al. 2018)	Air quality	Temperature, humidity, dust and carbon dioxide sensor, long range WAN
Characterization of air quality (Kumar and Chakramani 2020)	Monitoring of air quality	Heterogeneous sensors, ML-based predictive model
Wearable sensors (McNeill et al. 2017)	Temperature and humidity monitoring	Wireless and wearable sensor technology

Analysis of the Web and Social Media

The exponential expansion in data quantities across the Internet has stemmed from the tremendous increase in the number of computer and mobile devices in use. In 2016, 3.3 million Facebook posts, 448,800 tweets, 65,972 images, and 3.8 million Google queries are predicted to be generated each minute of Internet activity (Allen 2017). Social media data includes blogs, chats, or remarks on internet sites (e.g., news pages), blogging (Twitter), community projects (e.g., Wikipedia), social media platforms (Facebook and LinkedIn), and internet forums (e.g., YouTube and Instagram) (Olsen and Christensen 2015). Followers, shares (or retweets), likes, comments, mentions, and metadata linked with the material are all examples of metrics that may be derived from social media content provided by users. Depending on what the user wanted to reveal, the metadata connected with their activity may contain information like age, nationality, education, career, geolocation, and more.

In a nutshell, social media data is information acquired from social networks that indicates how people share, view, or interact with a user's or organization's content or profiles. As the most prominent example, people who "like" Facebook pages of corporations see brand page updates and material in their news feed. This allows companies to communicate with individuals who like or follow their pages, as well as the other way around. Almost every food company has a social presence, with a large number of individuals who follow their Facebook pages and so participate in the social interaction and data production process (Misra et al. 2020). The goal of a typical social big data analysis is to extract important information chunks from a massive quantity of complicated user-generated content (typically expressed verbally or visually) and assist government, industry, or stakeholders in making informed decisions (Fig. 9).

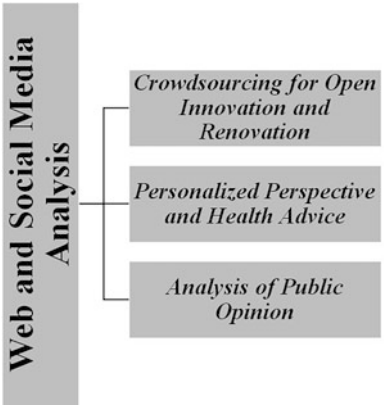

Fig. 9 Types of web and social media analysis

Case Study

Over the last century, meteorological measurements have grown in importance for aviation safety, especially with the start of the aviation age and the introduction of numerical weather prediction models. In the early decades of the twentieth century, humans gave these observations, but as technology evolved, sensors that could take many of the same types of observations autonomously were developed. As these sensors became more common, human weather watchers began to rely on them for the more basic measurements such as temperature, relative humidity, wind, etc., while concentrating on the more subjective ones.

A case study on Delhi to enumerate the fusion of remote sensing technique, meteorological sensors, and IOT for Industry 4.0 is elucidated. The revolutionization of industries imposed greater demand on ensuring stringent regulation on pollution load. Vivid parameters are taken into consideration while understanding the role of Atmospheric Boundary Layer (ABL) had been relevant. The manifestation of ground-based acoustic remote sensor, the Sound Detection and Ranging (SODAR) deployed for probing the lower layer of atmosphere is deeply understood. The fumigation period, stability layer, and the convection period are inevitable while considering the Environmental Impact Assessment (EIA) of Industrial set up. It is internationally recognized and recommended by the Environmental Protection Agency (EPA) for air quality dispersion modeling in EIA (Iyer and Raj 2013). SODAR is a cost-effective instrument for real-time monitoring of ABL height up 1–2 km. ABLH-derived ventilation coefficient (VC) is imperative to comprehend the dispersion and assimilative capacity of pollutants by the atmosphere over a locus (Gera et al. 1996). It is the need of the hour of Industry 4.0 era to contemplate the pollution aspect for ensuring environmental safeguard and sustainable enhancement of the resources.

CSIR-National Physical Laboratory (NPL) New Delhi had successfully initiated the task of development of SODAR for effective monitoring of ABL structures. Numerous modifications were undertaken and thereby the growth of industrialization demanded the incorporation of the study of ABLH for the pollution abatement plans and strategies. Wherein CSIR-NPL has been successful in installing SODARs at seven locations over India, which include Central Pollution Control Board (CPCB) and NPL at New Delhi; Indian Institute of Technology (IIT), Roorkee; Aligarh Muslim University (AMU), Aligarh (Uttar Pradesh); Indian Meteorological Department (IMD) Hisar (Haryana), Rajasthan State Pollution Control Board (RSPCB) Alwar (Rajasthan); Punjab Pollution Control Board (PPCB) Sangrur (Punjab). Here, an analysis of meteorological parameters along with ABLH using meteorological sensors and SODAR to study characteristics of ABL over seven different sites is provided. Basic variables such as relative humidity, wind speed, and temperature may be utilized to assess the boundary layer processes and thermodynamic structure. Lower tropospheric phenomena, air pollution modeling, weather prediction, and environmental monitoring might all benefit from the network of meteorological sensors and SODAR. The next part discusses a comprehensive comparative study of different climatic factors and ABLH among the selected sites.

Atmospheric Boundary Layer Height (ABLH)

The ABLH during the period from March 2020 to February 2021 is analyzed for this case study. Fig. 10 shows the ABLH for the selected seven locations. The above two graphs are from the CPCB and CSIR-NPL. The bottom five graphs are from the Aligarh, Alwar, Roorkee, Sangrur, and Hisar. The data is not consistent throughout the year and therefore month-wise seasonal analysis is also presented in the upcoming sections. In Fig. 10, daily average variation with maximum and minimum values during each day is plotted.

The data set has a temporal value (1 h = one value) for 12 months (March 2020 to February 2021). The dataset has been classified into four seasons based on North Indian meteorology, namely, summer or pre-monsoon (March to June), monsoon (July to Sept), post-monsoon (Oct to Nov), and winter (December to February) for the analysis of seasonal ABLH. All seasons ABLH is presented in the Fig. 11 (a, b, c, and d).

The maximum value of ABLH is observed during summer season while the minimum height is found during monsoon. Summer in general has large growth rate compared to other seasons. The growth is found to significantly diminish during onset of monsoon season and winter season. The ABL achieved height ~2 km during summer season. Slightly higher value tends to occur during post-monsoon than during monsoon. The governing components of the atmosphere were analyzed to characterize the fluctuation in various seasons (temperature, relative humidity, and wind speed). The vigorous mixing caused by the strong solar radiation causes the ABLH to be high in the summer. High RH means that there is more moisture in the air, which promotes intense convection and the formation of ABLH. The high ABLH throughout the summer was due to significant insolation generated by solar heating and convection.

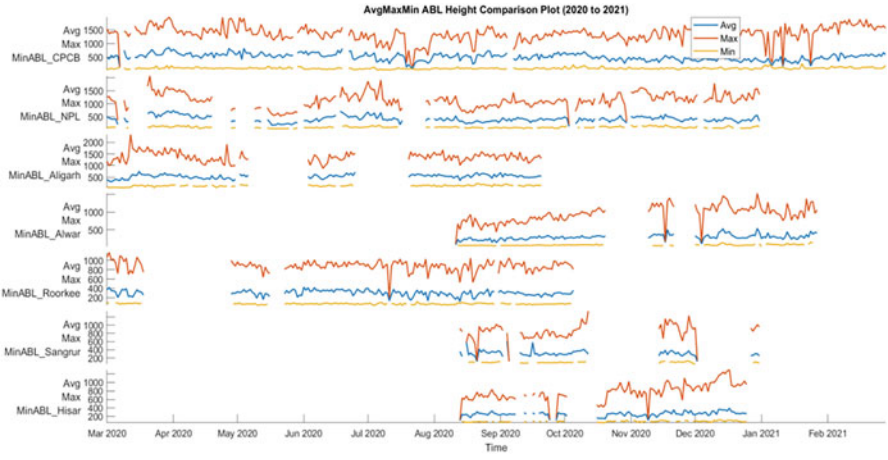

Fig. 10 ABLH Comparison of ABLH for Delhi (two locations), Aligarh, Alwar, Roorkee, Sangrur, and Hisar

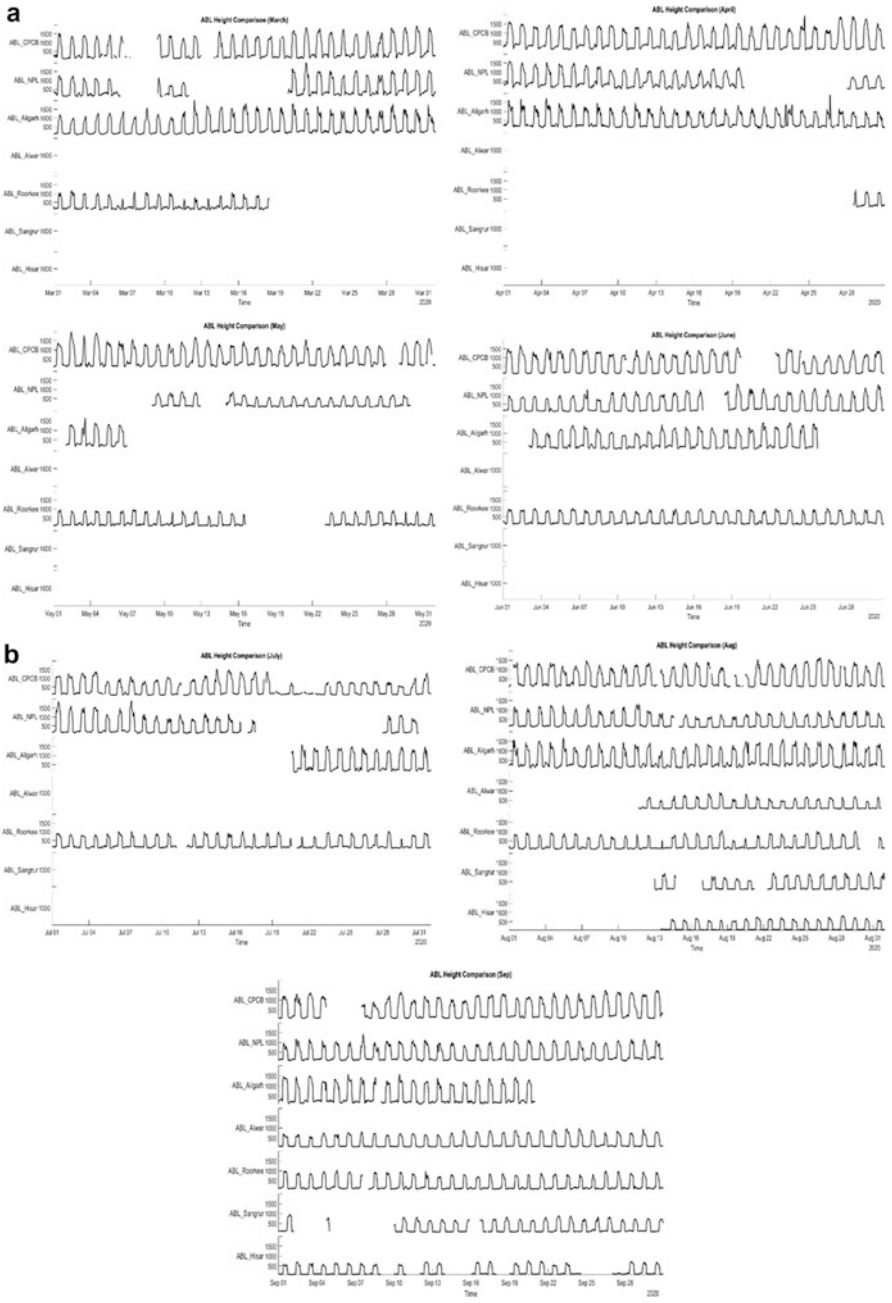

Fig. 11 (**a**) Pre-monsoon season ABLH. (**b**) Monsoon season ABLH. (**c**) Post-monsoon season ABLH. (**d**) Winter season ABLH

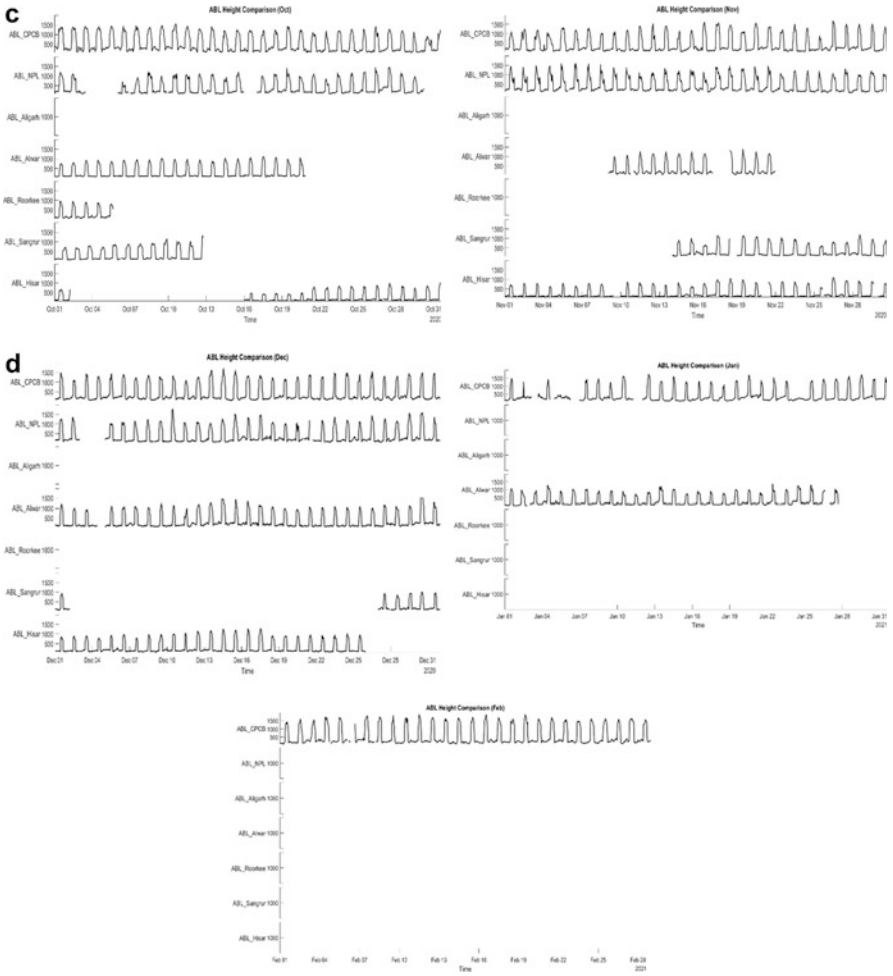

Fig. 11 (continued)

Overall, Aligarh had the highest ABLH at 2340 m; this was followed by NPL (2180 m), CPCB (1970 m), and Alwar (1860 m) (1573 m). Roorkee, Hisar, and Sangrur were to be the locations with ABLH maximums less than 1500 m. Roorkee (1390 m), Sangroor (1340 m), and Hisar (1360 m) had the highest ABLs (1290 m).

For all the seven sites, it was found that there are 2 months (August and September), which fall in commonality and thereby a comprehensive comparative study was performed for these 2 months. Variability in maximum and minimum ABLH during 2 months was stimulated through the following observatory analysis. It was found that the ABLH for Aligarh site (1680 m) was found to be maximum followed by CPCB (1650 m) and NPL (1440 m). Alwar and Hisar had maximum

ABLH of 990 m and 830 m, which was found to be the lowest maximum ABLH in comparison to other sites. Minimum ABLH fluctuated between 40 m and 90 m among these sites. Where Aligarh, CPCB, NPL, Roorkee, and Sangrur witness maximum increase of more than 1 km in their ABLH, Alwar and Hisar showed a data with maximum ABLH reaching just below 1 km. The minimum value of ABLH in all sites were witness to be less than 100 m with lowest shown by NPL having minimum ABLH of just 40 m on 12.09.2020.

Having an observatory analysis on the diurnal variation in these 2 months, it was noted that in Aligarh and CPCB the convective period during the selective months started even early at 7.00 IST and lasted till 17.00 IST. Whereas in the other sites, the convective period started at 9.00/10.00 IST, which is 2–-3 h late from the Aligarh. However, it was found that the duration of convective period over Sangrur was less in comparison to other sites. On a whole, it was figured that the rate of convection period was found to be of shorter duration for Roorkee and Sangrur. Due to the data inconsistency, the comparative meteorological parameter analysis is performed for the August and September months only.

Meteorological Parameters Analysis

It was found that the temperature in these sites ranged from ~19 °C to ~38 °C. The lowest temperature was found for Roorkee and Sangrur on different durations of time period for number of days. Since temperature has a positive correlation with the ABLH, low temperature induces maximum of inversion and low ABLH, clarifying the fact that Sangrur exhibiting minimum duration of convection period. The temperature range of Roorkee (24–36 °C), Sangrur (22–35 °C), Hisar (19–38 °C), Alwar (23–38 °C), CPCB (24–37 °C), Aligarh (26–37 °C), and NPL (24–37 °C). The temperature variation over these sites is depicted in Fig. 12.

On average, Aligarh's ABLH was found to be the highest, followed by CPCB and NPL. After already being warmed by solar radiation, the surface quickly passes the heat into the air via the ABL's turbulent motion, changing wind, temperature, and humidity, as well as modifying the ABL's thickness. The evolution of ABLH is influenced by the temperature. The next session provides a quick overview of the impact of temperature on ABLH accomplishment.

Sunlight is visible throughout the day but not at night. The lack of convection at night causes vertical stratification, which is eroded by daytime convection. During the day, the efficacy of erosion is influenced by the quantity of cloud cover. A dense layer of clouds may block much of the Sun's rays, reducing the amount of heat that reaches the Earth. Convection often engulfs the lowest 1–2 km of the atmosphere due to mixing processes. The night usually ends with a shallow nocturnal boundary layer (NBL), which is created by wind friction and causes mixing. The thickness of this layer varies depending on the prevailing wind velocity and the surface roughness, although it seldom surpasses 300 meters. Due to heat loss to space throughout the night, the air above the NBL is mildly stratified. Convective movements swiftly exceed wind shear turbulence as the sun rises, heating the atmosphere from below. Penetrative convection

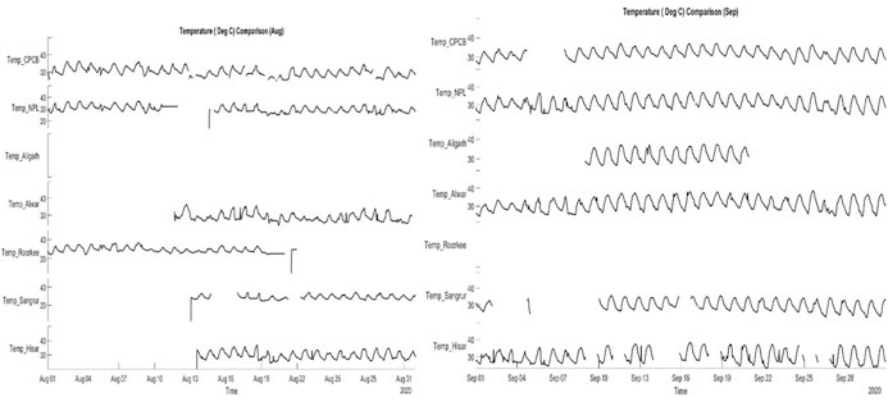

Fig. 12 Comparative analysis of daily variation in temp. (°C) for 2 months across seven sites

then takes over, gradually dissolving the stratified layer that formed throughout the night and replacing it with the convective boundary layer (CBL). Penetrative convection may be rather quick depending on how thin the stratification was above the NBL. Except near the base of the CBL, where both have comparable strength, wind-induced turbulence is substantially less than convection-induced turbulence. The surface layer is the lower layer, while the CBL is the remainder of the convective zone. The developing convective zone finally reaches the same height as the previous day as time passes (assuming perfect repeatability from 1 day to the next). Convection is halted when heat input from the surface is reduced after sunset, and the CBL transforms into an un-agitated fossil mixed layer known as the residual layer. Wind builds a fresh NBL near the ground by causing friction.

The thickness of the ABL is determined by the amount of water evaporated into the air and the degree of surface heating. In terms of humidity, NPL had 32–93%, CPCB had 36–98%, Alwar had 24–98%, Aligarh had 49–89%, Hisar had 32–92%, Roorkee had 58–98.4%, and Sangrur had 45–85%. Humidity and ABLH have an anti-correlation. Figure 13 depicts the difference in RH (%) among seven distinct sites.

Taking wind speed into account, the wind speed of NPL was found to be in the range of 0.1–4.1 m/s, for CPCB it was 0.47–7.31 m/s, for Alwar it was in range of 0.1–5.3 m/s, for Aligarh it was 0.1–2.3 m/s, for Hisar it was 0.1–1.2 m/s, for Roorkee it was 0.1–3.5 m/s, and for Sangrur it was 0.1–5.1 m/s. The wind direction or wind speed shear was able to accelerate turbulent and molecule motion from the ground up, causing heat convection to rapidly diffuse to high altitude, which was favorable for the formation of a deep boundary layer. Strong wind speed or wind direction shear reduced the strength of the temperature inversion at the top of the CBL by boosting the diffusion and entrainment effect, which carried sensible heat flow from the ground to the upper air and enhanced convection efficiency. This would greatly increase CBL and provide a dynamic environment that would encourage the production of ultra-thick CBL. Figure 14 depicts the variance in wind speed at seven locations.

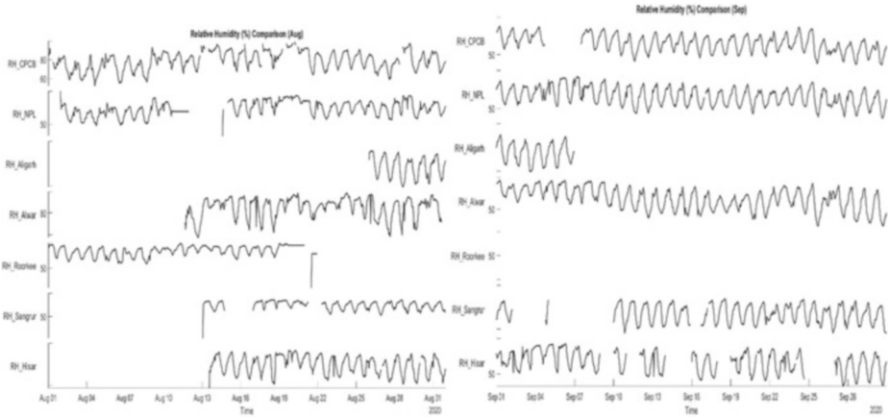

Fig. 13 Comparative analysis of daily variation in RH (%) for 2 months across seven sites

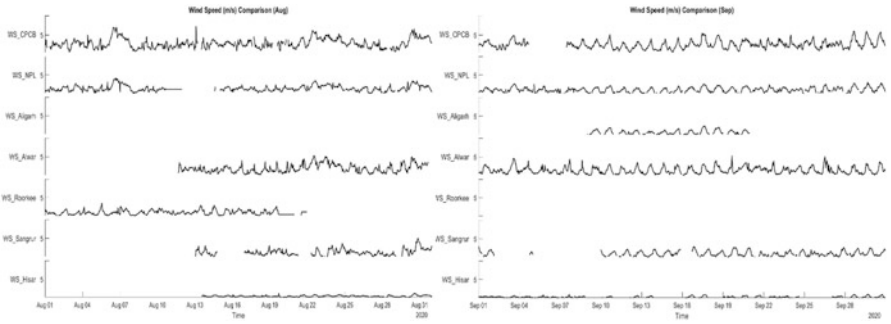

Fig. 14 Comparative analysis of daily variation in wind speed (m/s) for 2 months across seven sites

The range of wind speed showed highest wind speed for CPCB. Wind induces turbulence and causes increase in ABLH, thus sharing a positive correlation with wind speed. Thereby high wind speed can be known as a predominant factor for increased ABLH during the course of selected time interval of 2 months.

Correlation Analysis

The ABL is a turbulent, dynamic system in which horizontal and vertical air mixing is largely determined by weather and geomorphology (Bendl and Hovorka 2017). ABL shows diurnal variation with convective and stable boundary layer formation due to the influence of surface fluxes (net radiation and sensible heat) and turbulence (Mehta et al. 2017). Meteorological parameters are vital in the development of ABL. It is inferred that the ABL is more positively correlated with wind speed and temperature and more negatively correlated with RH. As the temperature increases,

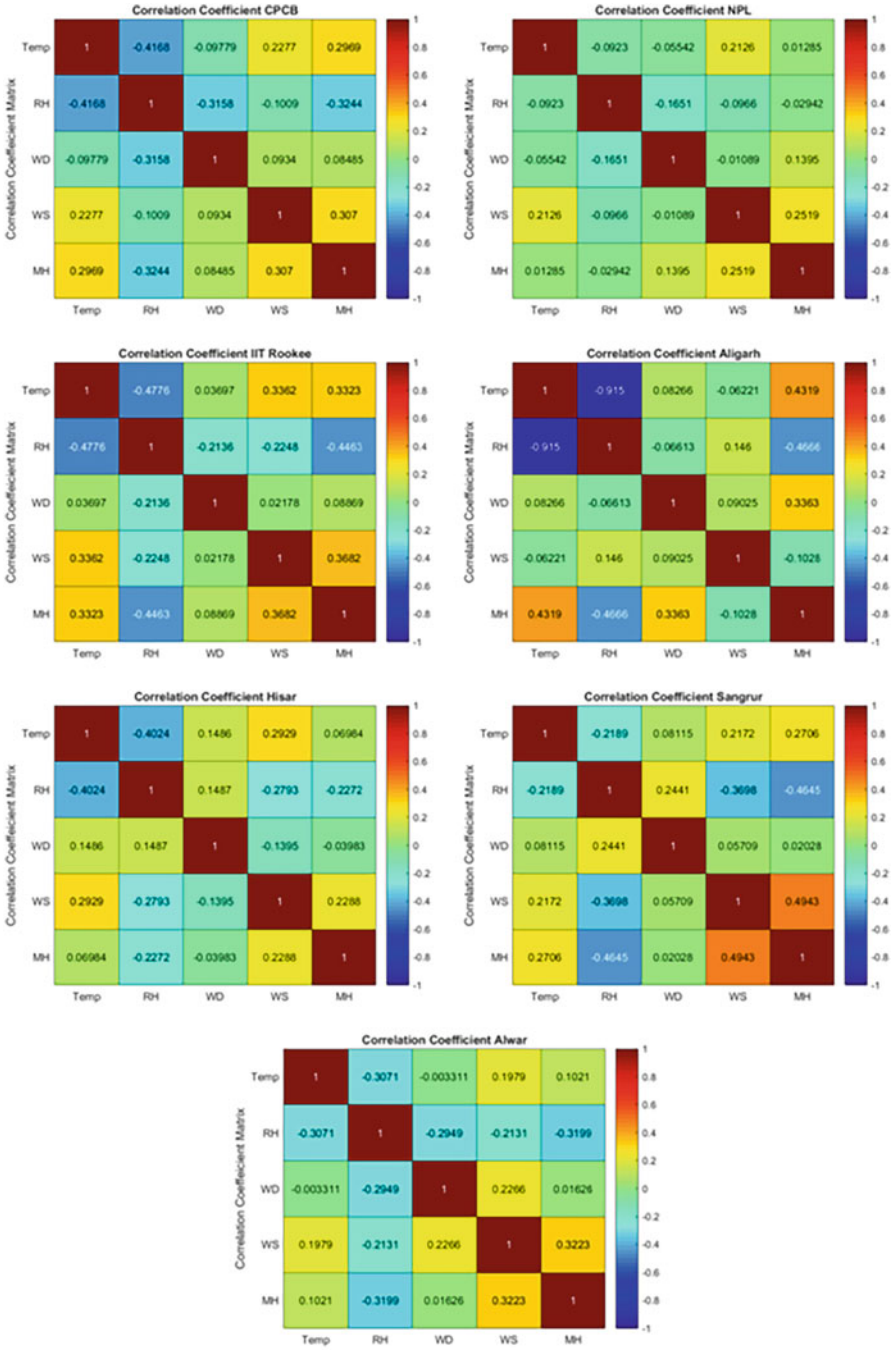

Fig. 15 Correlation analysis between the ABLH, temperature, relative humidity, and wind for seven locations

the sensible heat flux eventually increases resulting in increased boundary layer height. Similar increased turbulence due to high wind speed induces elevation in ABLH.

Strong negative correlation states the anti-correlation of RH with ABLH. The correlation between the meteorological parameters shown in Fig. 15 clearly indicates the influence of these parameters on ABL.

The case study clearly presented in this chapter is to give brief understanding about the importance of having a network of meteorological sensors and remote sensing for monitoring of air quality in a specific region. The integration of advance technology such as IoT and AI makes the monitoring and controlling more effective.

Conclusion

Together with environmental monitoring, smart sensor products may undoubtedly build an internal or even worldwide information network, combining IoT with industrialization. As a result of this tendency, the goal of Industry 4.0, i.e., a sustainable growth with a fully context-sensitive, controllable, describable, manageable, and self-regulating production system, is becoming a reality. It may be concluded that a smart and novel environmental monitoring system is required for the fast-growing Industry 4.0. The IoT-enabled sensor network is a new way to establishing "a Comprehensive Broadband Infrastructure" that focuses on the field of "Computer Networks and Communications." This chapter provides a comprehensive overview of the Industrial Revolution, encompassing invention and application, technology, protocol, and standards. Furthermore, this chapter demonstrates how smart gadgets are evolving across the world, with a substantial trend that has already spread from people's daily lives to the global industry. The case study given in this chapter highlights the need of a linked monitoring network for regional air quality measurement as a key aspect of Industry 4.0 for attaining sustainable growth. Chips, terminals, base stations, networks, software tools, testing devices, operating systems, and APPs will all be affected by Industry 4.0, as connected smart devices all over the world will provide secure, real-time, and reliable sensing, communicating, and computing services via the Internet, affecting the following aspects: chips, terminals, base stations, networks, software tools, testing devices, operating systems, and APPs. The regulatory authorities' decision-making is greatly improved by the AI and IoT-enabled environment monitoring system.

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Machine Learning in Neuromuscular Disease Classification

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Niveen Farid

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Abstract

This work reviews the recent techniques used to analyze the EMG signals to extract and classify features. The discussed techniques are used in medical applications for the diagnosing of neuromuscular disorders. Techniques such as wavelet transform (WT), principal component analysis (PCA), empirical mode decomposition (EMD), artificial neural network (ANN), support vector machine (SVM), extreme learning machine (ELM), k-nearest neighbors (kNN), and deep learning are among the powerful techniques used for feature extraction and disease classification. The many improvements made to the algorithms could increase the performance and accuracy of the models discussed in this area. Recently, the researchers focused their attention on the classification of

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neuromuscular illnesses using artificial intelligence to acquire and classify myoelectric signals via electromyography (EMG). Amyotrophic lateral sclerosis (ALS) is one of the neuromuscular illnesses that drives the attention of the researchers in this field. The current work focuses on the most recent methods for extracting and categorizing characteristics from EMG data to diagnose neuromuscular disorders in medical applications. Feature extraction approaches presented here include the wavelet transform (WT), principal component analysis (PCA), and empirical mode decomposition (EMD). Artificial neural networks include artificial neural networks (ANNs), support vector machines (SVM), extreme learning machines (ELM), k-nearest neighbors (kNN), and deep learning. An example on the uncertainty evaluation in machine learning classification is introduced.

Keywords

EMG signal · Disorder classification · Machine learning · Uncertainty

Introduction

Machine learning is a successful soft metrology application for extracting data and developing models from a certain database collection. The current work gives substantial information on measurement error in neuromuscular classification models.

Machine learning is based on recording signal data and loading into a machine to be automatically assessed. This work demonstrates the distinction between normal and abnormal signals, allowing the abnormalities in the human body to be identified. In biomedical engineering, machine learning is a potential tool for disease classification. It is concerned with extracting information from raw data signals using mathematical operations to be developed and analyzed (Fig. 1). If such techniques are developed to manage massive data connected to medical operations, they can be valuable for biomedical engineering. Since machine learning approaches have developed in many fields addressing categorization, biological signals can now be evaluated in multiple dimensions (Szkoła et al. 2011; Mannini et al. 2016; Balli and Palaniappan 2009; Ene 2008; Mierswa et al. 2006; Kordylewski et al. 2001; Lanyi and Adler 2004; Raez 2006; McCool et al. 2014; Berger et al. 2006). Supervised learning tools are examples of machine learning approaches applied in diagnostic and disorder classification (Ene 2008; Mierswa et al. 2006). Machine learning techniques are trained on a set of data and then used to reduce massive time series signals to optimally discriminated features (Kordylewski et al. 2001).

EMG is a useful diagnostic tool for studying the electrical signals generated in a muscle during contraction (Lanyi and Adler 2004). It has been widely employed by physicians and researchers for the diagnosis of neuromuscular problems in recent decades. The term “neuromuscular disorder” refers to a variety of disorders affecting

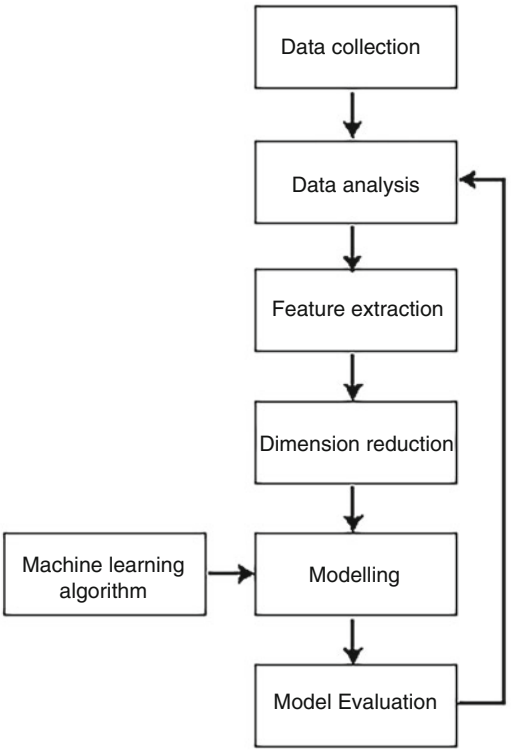

Fig. 1 Machine learning framework

the neurological system or muscle fibers. ALS is a neurodegenerative illness that affects the spinal cord and is classified as a form of neuromuscular disorder known as motor neuron disease. Another important category of neuromuscular condition is myopathy, which refers to a set of disorders that affect skeletal muscle tissue (Raez 2006). It's necessary to have an accurate diagnosis of these disorders in the early time in order to manage and treat them. In this chapter, a variety of classification and feature extraction strategies are discussed for the diagnosis of neuromuscular illnesses.

Detection and Processing of EMG Signal

The electromyographic (EMG) signal is a biological signal that detects the electrical currents generated in a muscle during contraction, resulting in neuromuscular activity (McCool et al. 2014). The smallest functional unit that can describe brain control during muscle contraction is the motor unit (MU). The motor unit action potential (MUAP) is a collection of action potentials generated by all muscle fibers in a motor unit. Noninvasive and invasive procedures can both be used to measure

MUAPs. The noninvasive technique uses EMG electrodes placed directly on the skin surface, whereas the invasive technique uses needle electrodes that penetrate muscle tissue to acquire EMG data.

The noninvasive technique, on the other hand, appears to be more flexible, but it only detects the total of actions from various motor units. Invasive techniques can detect the potentials of a single motor unit, providing additional information regarding muscle activity.

Many researchers use the EMG signal as a source of data, while developing diagnostic tools for neuromuscular illnesses. They preprocess the raw information acquired by the EMG electrodes, identify relevant features, and then classify the input signal into distinct categories using a classification technique (i.e., healthy, myopathy, and ALS patients). The major procedures for diagnosing neuromuscular diseases using EMG signal analysis are EMG signal acquisition, signal processing, features extraction, and signal classification.

Key features are extracted from the acquired EMG signal and used as input to the classification stage. The extraction and selection of features are critical for improving classification performance in the diagnosis of neuromuscular illnesses. Frequency domain (FD) features, time domain (TD) features, and time-frequency domain (TFD) features are the three types of features that can be retrieved from a non-stationary signal like EMG.

The TD aspects are analyzed by focusing on signal amplitude, which is a function of time and is affected by muscle types and cases at the time of observation. This kind requires no further signal manipulation and has the simplest analysis. The parametric approaches are applied to the power spectrum density of the signals to compute the FD characteristics. Combining TD and FD yields TFD features, which characterize frequency variation at different temporal locations on the signals (Berger et al. 2006).

To extract TD, FD, or TFD characteristics from EMG signals, several researches used wavelet transform, empirical mode decomposition, principal component analysis, and cepstral analysis.

Principal Component Analysis (PCA)

PCA is a statistical technique that has been widely employed in the analysis of biomedical signals. It's a nonparametric method for extracting important data from large datasets (Kehri et al. 2017). Data patterns can be compared and contrasted using PCA. The data dimensionality can be decreased without losing much information by finding patterns (Shaw and Bagha 2012). PCA is the most prevalent technique for dimensionality reduction for these reasons, and it may be used effectively to analyze big datasets of EMG signals (Fig. 2). The PCA algorithm is shown in Fig. 3 according to Garrone (2020).

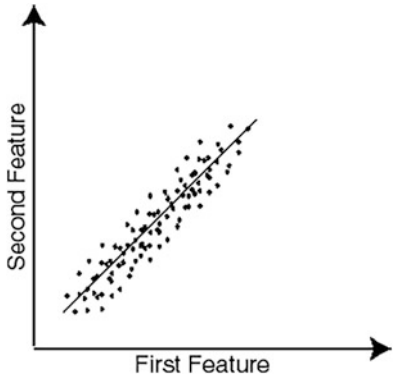

Fig. 2 Principal component analysis for two features

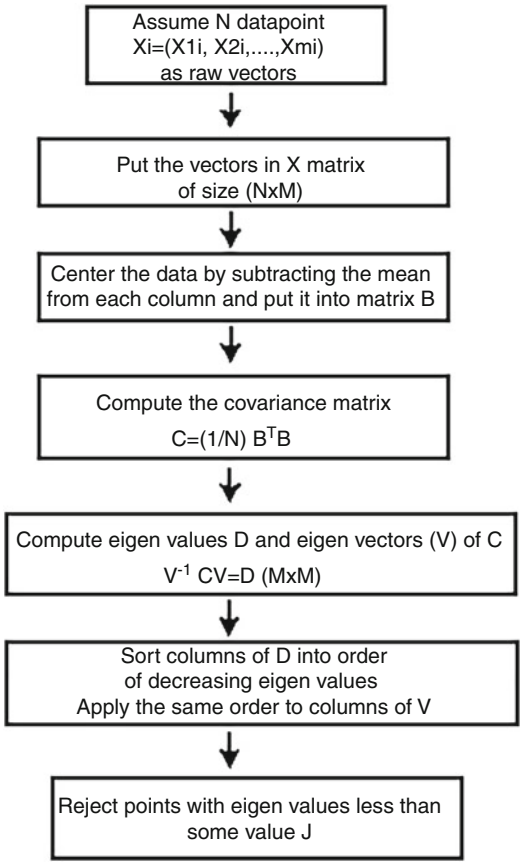

Fig. 3 The flow chart of PCA algorithm

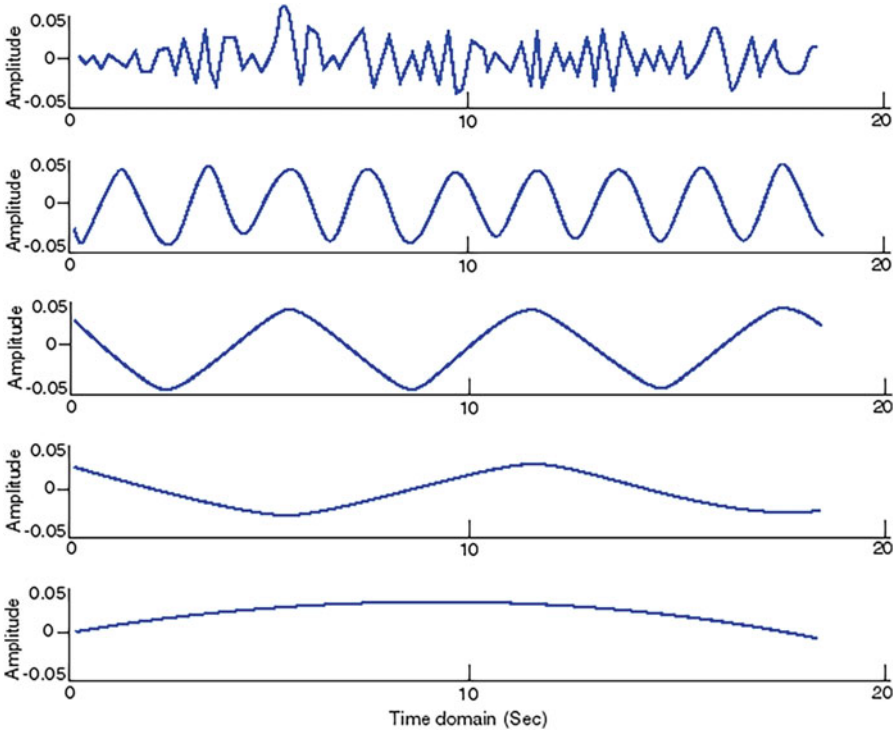

Fig. 4 Reduction of a simulation signal by EMD

Empirical Mode Decomposition (EMD)

The EMD methodology does not rely on an external decomposition function. Instead, it is a self-adaptive time-space analytic method that decomposes nonlinear and nonstationary signals like the EMG signal. It can keep the signal's temporal domain while decomposing it into many intrinsic mode functions (IMFs). IMFs must meet two main requirements, that is, only one difference between the number of zero crossings and the number of extrema should exist and the local average between the upper and lower envelopes should be zero. The EMD can be used in a variety of audio, biomedical, and image processing applications (McCool et al. 2014). EMD is very beneficial for processing complex nonlinear and nonstationary signals (Fig. 4). Figure 5 shows the algorithm of EMD method according to Cura et al. (2020).

Wavelet Transform (WT)

To handle nonstationary signal issues, the WT describes the time function in simple block form. Such blocks are a collection of functions formed from a wavelet using translation and dilation. In all frequency ranges, this approach delivers the best time-

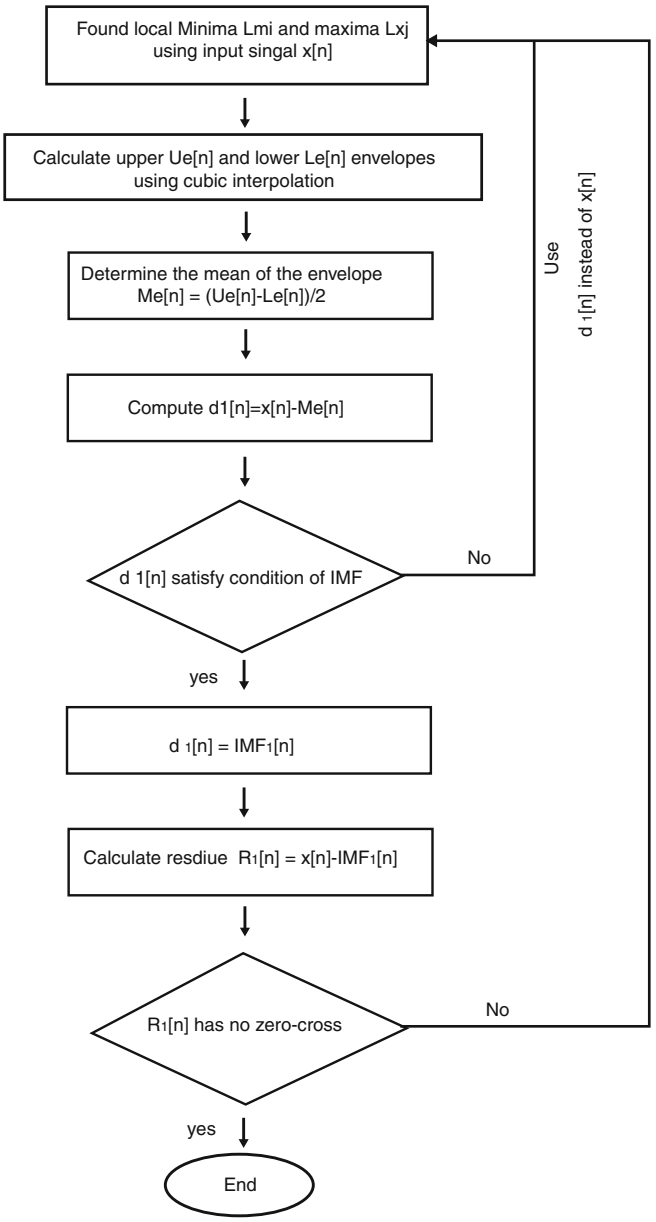

Fig. 5 The flow chart of EMD algorithm

frequency resolution (Kehri et al. 2017) (Fig. 6). Since it gives both time and frequency information at the same time, the discrete wavelet transform (DWT) can be used in a variety of biomedical engineering signal processing applications, such as EMG signal processing. The discrete Fourier transform (DFT), on the other hand,

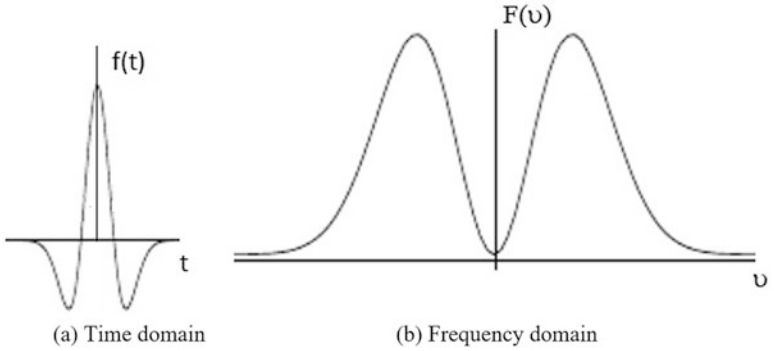

Fig. 6 (**a**) Wavelet in time domain and (**b**) its transform in the frequency domain

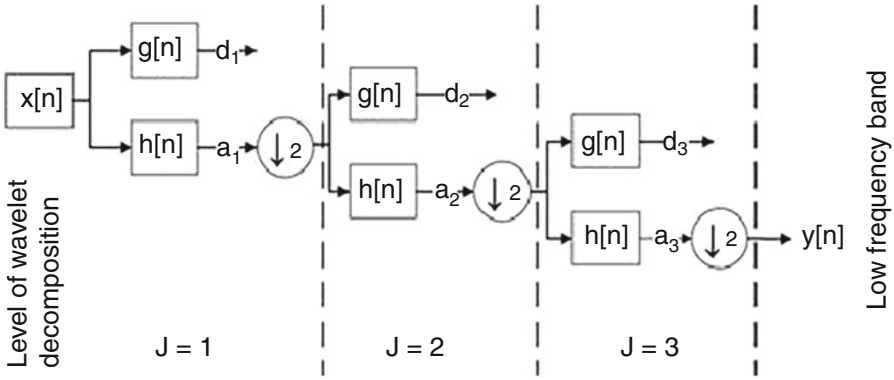

Fig. 7 Flowchart of multi-resolution wavelet decomposition (Alzaq and Üstündağ 2018)

is exclusively confined in the frequency domain (Khan and Hasan 2015). Many researchers have utilized DWT to extract features from EMG signals in order to classify neuromuscular diseases (Kehri et al. 2017; Khan and Hasan 2015; AbdelMaboud et al. 2015; Khan et al. 2016; Meena and Bansal 2016; Subasi et al. 2018). Figure 7 shows a diagram for wavelet decomposition, the block diagram of the two-channel four-level discrete wavelet transform decomposition (J = 3) that decomposes a discrete signal into two parts.

Classification Approaches

Classification approaches such as k-nearest neighbors-kNN, artificial neural network-ANN, support vector machine-SVM, and deep learning have been employed by many researchers. This section gives a quick overview of each of these methods.

K-Nearest Neighbors (kNN)

A distance function, such as the Euclidean distance function, is evaluated between a test set and k-neighboring points from normal and diseased participants in the kNN classifier. In classifying the test record, the labels of the k-nearest pattern are employed with the k values offering the best classification accuracy.

The kNN can calculate the distances between the test record and the signal records and display the k records with the least distance in ascending order. In pattern recognition, on the other hand, it can be used to classify objects based on the closest training examples defined in the feature space (Bhuvaneswari and Kumar 2016; Fattah et al. 2013). This method takes a long time and is based on learning by relationship using the Euclidean distance, with the unknown record being chosen as the most well-known class among its k-nearest neighbors (Witten et al. 2011; Saxena 2016) (Fig. 8).

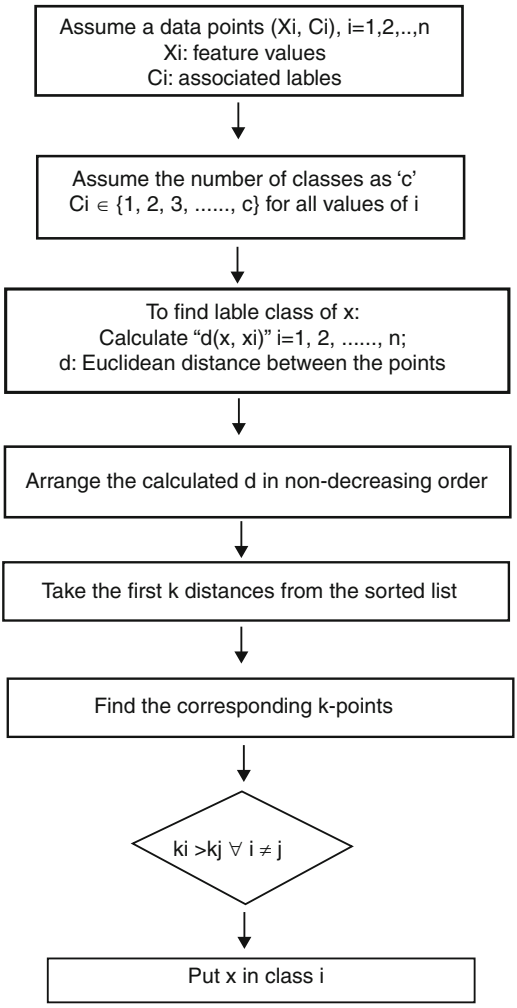

Fig. 8 Flow chart of kNN algorithm

Artificial Neural Networks (ANN)

ANNs are commonly employed in biomedical engineering to analyze data because of its effectiveness, quick recognition, and real-time diagnosis. The ANNs are trained by examples rather than by own emotional states, exhaustion, or environmental circumstances (Sharma et al. 2010).

The key feature here is the capacity to accurately simulate system complexity in input and output data using either linear or nonlinear analogies. Using transfer functions and connection settings, data is taken from an external source and transported to subsequent layers.

In pattern recognition and signal processing, the back propagation (BP)-based neural network offers a wide range of applications (Priyadharsini et al. 2015; Elamvazuthi et al. 2015). It's a multilayered structure. As a result, the data is passed to the input layer, which is connected to hidden layers with weights and then to the output layer, which produces the required output (Fig. 9). If the expected and actual outputs are not identical, the error is evaluated to update the weights, and the procedure is repeated until the error is removed (Singh et al. 2017).

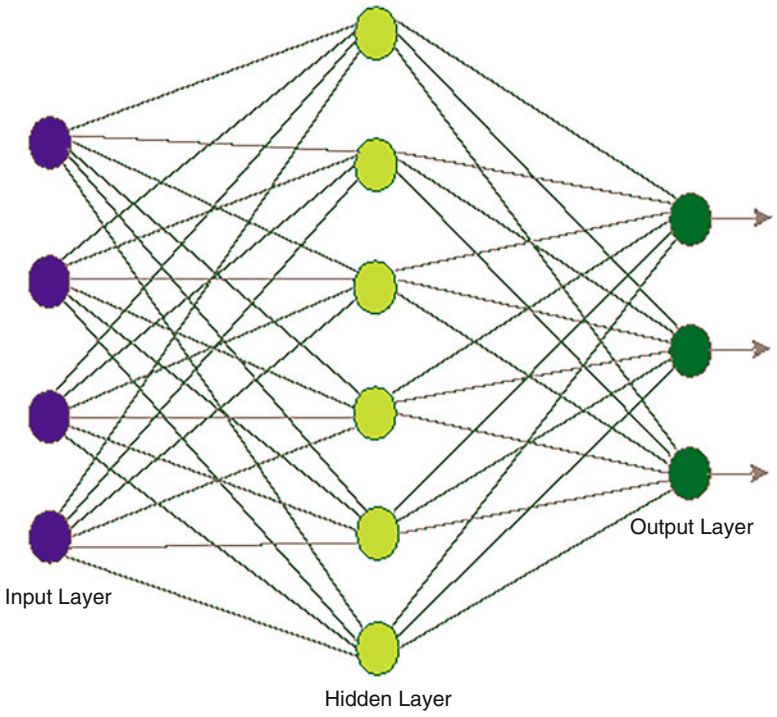

Fig. 9 Artificial neural network

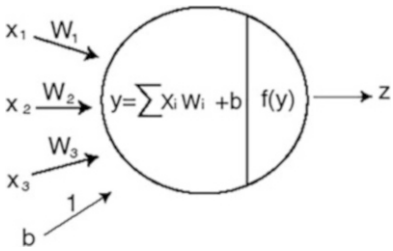

Fig. 10 Schematic diagram for a neuron function

The neuron is the main constituent in biological networks that performs computation. The connection weights manage the input signals, which are then aggregated and compared to a threshold value, which is the neuron's bias. An activation function controls the output signal (Fig. 10).

In Fig. 7, x_1,x_2,x_3 denote the inputs, while w_1,w_2,w_3 are the weights, and b is a bias. The neuron then performs a function to compute the output z.

Hidden layers occur between the input and output layers in a multilayer feed-forward network, and the neurons are linked layer by layer (Zhou 2012). Figure 11 shows a flow chart of ANN according to Illias et al. (2015).

Support Vector Machine (SVM)

It is the most accurate binary classifier for pattern recognition, and it uses a hyperplane to characterize the margin to distinguish distinct classes of data (Fig. 12). SVM models can be linear or nonlinear. The data points in the linear one are separated by a gap, and straight hyperplanes can be utilized to divide classes. Nonlinear classifiers are recommended in some circumstances where linear SVMs are ineffective. Multiclass classifications (Fig. 13) utilize one-against-all (OAA) and one-against-one (OAO) relationships in nonlinear SVM classification (Witten et al. 2011). Figure 14 shows a flow chart of evolutionary SVM according to Subasi (2015).

Deep Learning

Deep learning can deal with vast amounts of high-dimensional data with a large number of features. Auto-encoders, deep belief networks, deep Boltzmann machines, convolutional neural networks, and recurrent neural networks are examples of deep learning (Alaskar 2018). The most successful deep learning model is convolutional neural networks (CNNs) (Fig. 15). They are self-learned and self-organized networks that have lately been applied in the classification of biological data.

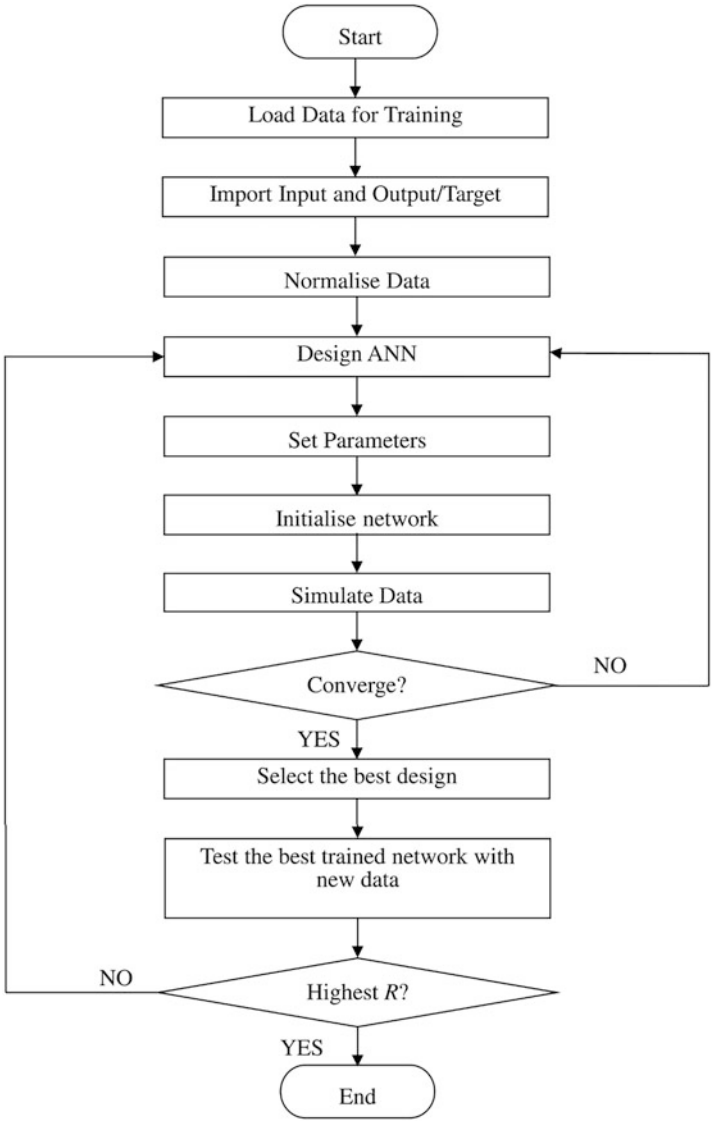

Fig. 11 Flowchart of ANN (Illias et al. 2015)

Diagnosing Neuromuscular Disorders by Machine Learning

The steps of diagnosing neuromuscular disease are shown in Fig. 16 including feature extraction and classification.

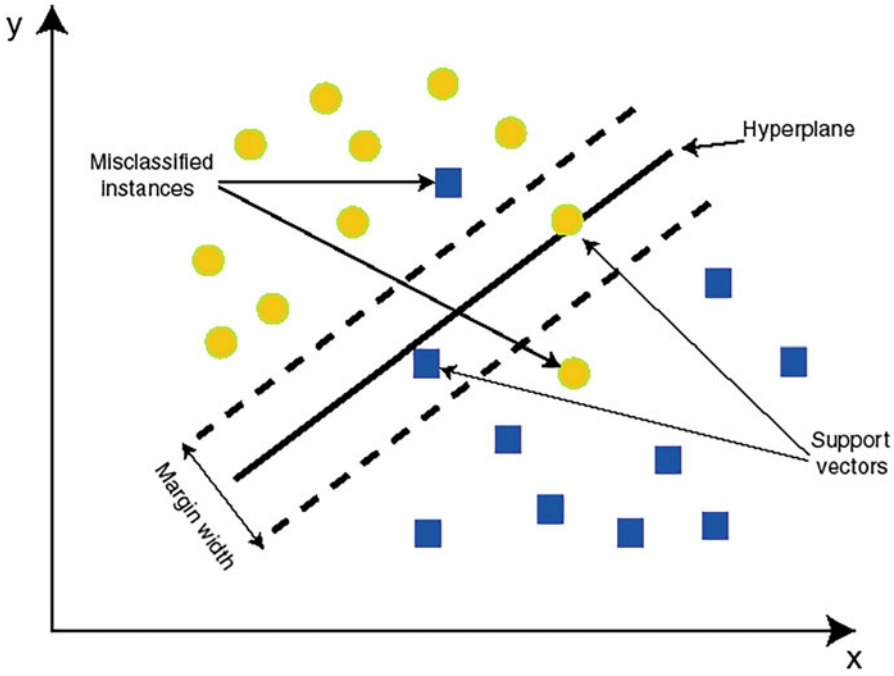

Fig. 12 A structure of support vector machine used in generating the input-output mapping from training dataset

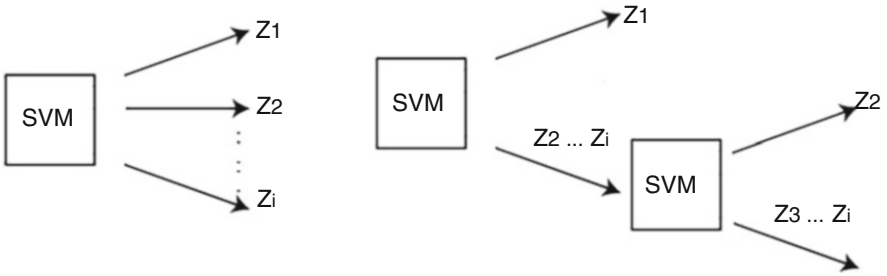

Fig. 13 Single multiclass and multistage SVM

The probabilistic neural network (PNN) was employed in Shaw and Bagha (2012), to classify the surface EMG output and distinguish between normal, myopathy, and neuropathy patients. The study was capable of extracting useful clinical information from simultaneously acquired EMG signals. They could provide a clustering technique to classify the signals with sufficient accuracy and speed and gave online information on patients' status. PCA was used to extract features from the signal. With respect to pattern length, the accuracy of the PNN obtained varies between 85 and 98%, yielding an average classification rate of 91.72%.

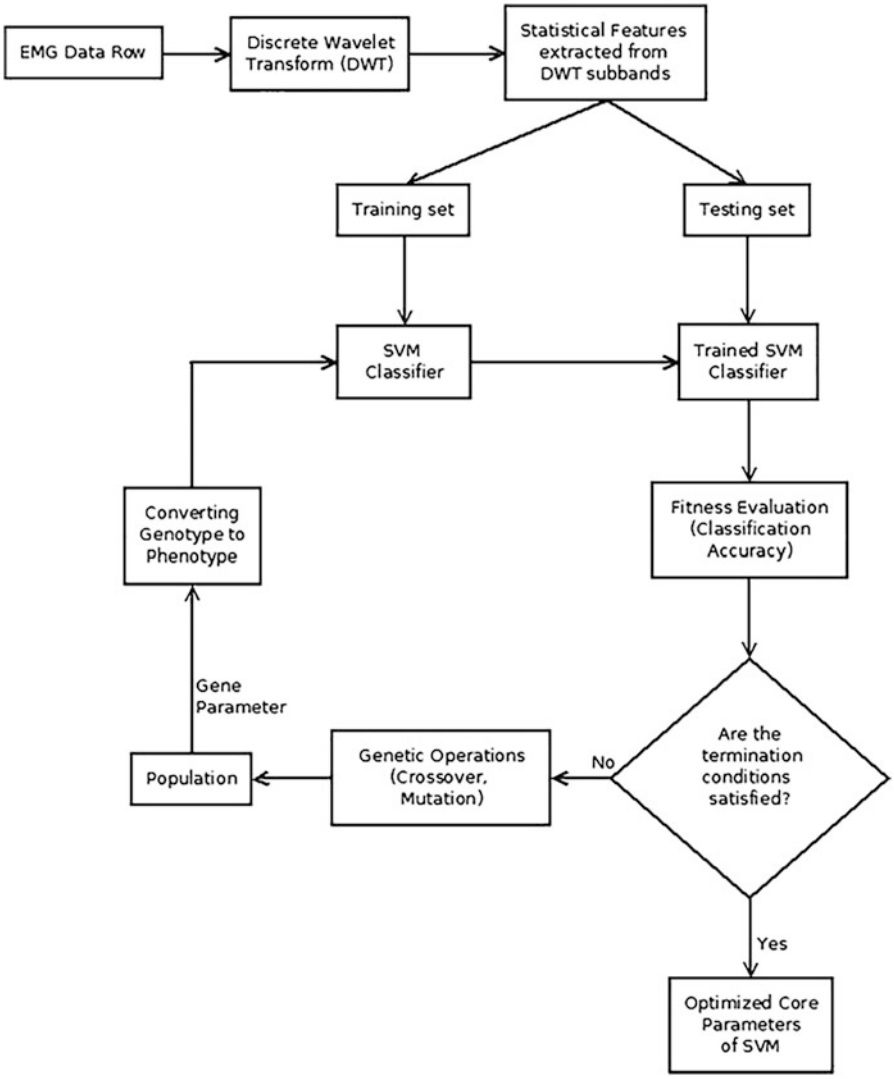

Fig. 14 Flowchart of evolutionary SVM showing feature extraction, training, and classification (Subasi 2015)

A multilayer perceptron neural network with a backpropagation algorithm was used in Elamvazuthi et al. (2015), to analyze EMG signals from 15 patients' biceps muscles (5 healthy, 5 myopathy, and 5 neuropathy). One hundred samples were randomly selected from the signal for each class (60 for training, 20 for validation, and 20 for testing). Five distinct feature extraction algorithms were used, including root mean square (RMS), zero crossing (ZC), AR, waveform length (WL), and mean absolute value (MAV). Five separate datasets (healthy/unhealthy (abnormal),

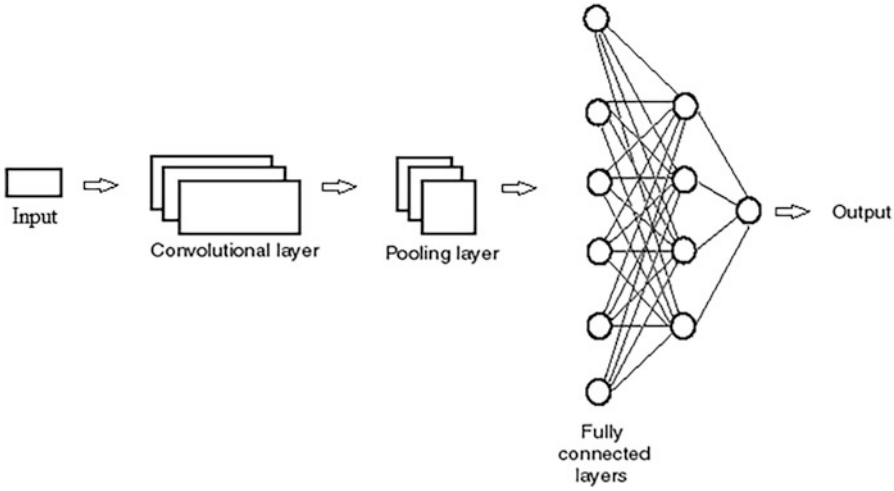

Fig. 15 CNN for deep learning

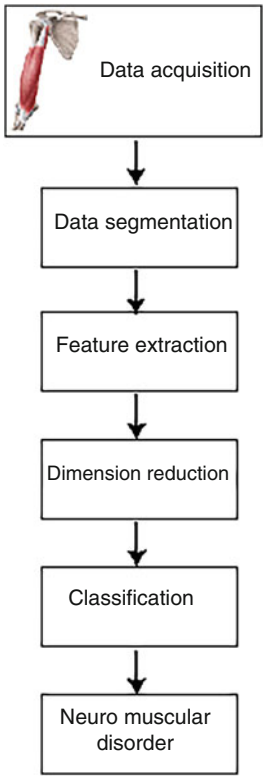

Fig. 16 Neuromuscular disorder classification framework

healthy/myopathy, healthy/neuropathy, myopathy/neuropathy, and healthy/myopathy/neuropathy) were created using the output data from the feature extraction procedure.

Using RMS, ZC, AR, WL, and MAV characteristics, it is possible to produce five distinct classification accuracies for each group. Using RMS and WL, the group of healthy and unwell subjects had the maximum classification accuracy of 82.5%. AR has the best results in the group of healthy and myopathy participants, with an 83% success rate. AR and RMS received the highest score of 83.5% in the third group. The AR approach generated the best results for the fourth and fifth groups, with accuracies of 82.5% for myopathy and neuropathy participants and 86.3% for healthy, myopathy, and neuropathy subjects. By achieving the best accuracy for four groups of five, the AR showed to be a good feature extraction technique.

The study in Priyadharsini et al. (2015), made a comparison between DWT and wavelet packet transform (WPT) as feature extraction techniques to detect neuromuscular disorders from EMG signal. The ELM was used for the classification process. They found that the proposed techniques ELM-DWT and ELM-WPT were better than the existing techniques in terms of classification accuracy and number of feature vectors. They proved that the learning speed of ELM was extremely fast, and more suitable in the applications that required fast prediction and response capability. They could achieve classification accuracy of 100% for any type of activation function.

ELM is a learning algorithm proposed by Huang et al. (2006) for single hidden layer feed-forward neural networks (SLFNs). It can choose hidden nodes randomly and determines the output weights of the SLFN analytically. The ELM technique has an extremely fast learning speed and yields to a generalization performance that is better than the imitative gradient-based learning algorithms. The ELM differs from traditional gradient-based learning algorithms in that it may be used to a variety of non-differentiable activation functions. An EMG dataset of 15 healthy participants, 1 myopathy subject, and 11 neuropathy subjects was used to test the suggested approach. Three electrodes were inserted on the biceps to record the EMG signal. With a sample rate of 1 KHz and 16-bit resolution, the signal was acquired using LabVIEW software and a data collection card. A total of three participants (1 healthy, 1 myopathy, and 1 neuropathy) were collected from a different dataset in addition to the prior one. The features were extracted individually using DWT and WPT approaches, and the ELM classifier was used to classify them. The WPT reduced the number of characteristics to a minimum and took less time to compute than the DWT. The ELM demonstrated to be a successful classification method for EMG signal categorization and diagnosis of neuromuscular diseases, with classification accuracy of 100% for both feature extraction techniques.

To classify EMG signals into healthy, myopathy, or neurogenic groups, researchers in Bozkurt et al. (2016), employed two types of neural networks: mixed neural networks and feed-forward error backpropagation artificial neural network (FEBANN). The suggested approach was tested on 1200 EMG signal patterns collected from 27 people (7 healthy, 7 myopathy, and 13 neurogenic), with 400 patterns separated into training and testing for each class. The EMG signal

was recorded from the biceps brachii muscle during a 5-second voluntary contraction and sampled at 20 kHz using a 12-bit acquisition data card resolution. To extract features from MUPs of EMG signals, researchers used AR parametric techniques such as Yule-Walker, covariance, modified covariance, and Burg procedures, as well as subspace-based methods like multiple signal classification and eigenvector techniques. The eigenvector performed the best out of all of these strategies. The FEBANN with eigenvector features had the highest classification accuracy of 93.3%, while CNN with the same feature extraction method had a higher accuracy of 94%.

ANN was employed in Singh et al. (2017), to classify EMG signals using parameters derived from the empirical mode decomposition procedure, such as mean, standard deviation, variance, and entropy of the IMFs. This method was used on 51 healthy, 111 myopathy, and 148 neuropathy subsets in a dataset. For signal categorization, 1000 samples were used for each subgroup. A 25 mm needle electrode was used to collect signals from the tibialis anterior muscle at a sampling rate of 50 kHz. When compared to other feature extraction techniques such as multiscale entropy, WT, AR coefficients, and statistical features, IMF statistical features attained a classification accuracy of 94%.

kNN was utilized in Bhuvaneswari and Kumar (2016), to differentiate healthy and neuropathy people using EMG patterns. They employed a dataset of 25 records from healthy people with no history of neuromuscular disease, people who have had polymyositis for a long time, and people who have chronic low back pain and neuropathy due to right L5 radiculopathy. A Medelec Synergy N2 EMG monitoring system with a sampling rate range of 20 Hz to 50 KHz was used to collect the data. The cepstral analysis was utilized to extract the features. Cepstral analysis is a nonlinear signal processing approach that deals with the real part of the signal and is calculated using the inverse of the discrete Fourier transform signal's log. The signal was first converted from the time domain to the frequency domain. The signal's spectrum representation was then decomposed using wavelet decomposition, which employed a low-pass filter for approximation coefficients and a high-pass filter for detail coefficients. Then, using the cepstral analysis procedure, characteristics were retrieved. After that, the dimensions of the features were decreased using minimum redundancy maximum relevance (MRMR), and the results were sent into the kNN classifier as inputs. The accuracy of categorization varies depending on the features chosen and the dimensionality reduction approach utilized.

The discussion in Fattah et al. (2013), included the employment of the kNN to classify ALS and normal signals. The EMG frame was used to extract the DWT features, which were then classified using the kNN classifier. The suggested approach was tested using EMG signals from five normal participants and five ALS patients, with 15 datasets in each class. The higher-valued DWT coefficients, as well as their mean and maxima, were employed to lower the feature dimension of DWT for ALS EMG signal detection. The classification accuracy of these three features was 100%, with the zero crossing (ZC) and mean frequency (MF) features achieving 72.2% and 69.5%, respectively.

In Kiran et al. (2018), kNN was used to distinguish between healthy people and ALS patients. The tunable Q-factor wavelet transform (TQWT) was utilized to extract characteristics from an EMG signal in order to distinguish between healthy and ALS signals. The TQWT could divide an EMG signal into sub-bands, from which statistical parameters like mean absolute deviation, interquartile range, kurtosis, mode, and entropy can be obtained and fed into the kNN classifier. In addition to the kNN classifier, the least squares support vector machine classifier was employed to categorize the same features, and the results of both classifiers were compared. The dataset utilized to evaluate the proposed approach contains 89 ALS and 133 healthy EMG signals that were amplified and sampled at a frequency of 10 kHz using a frequency range of 5 Hz–5 kHz filter settings. The TQWT sub-bands affected the classification accuracy of each classifier. The classification accuracy of the LS-SVM classifier was 74.55–90.45%. The kNN, on the other hand, outperformed the others, with a classification rate of 90.45–95%.

Through EMG signal analysis, Goen (2014), used binary SVM to distinguish between normal, myopathy, and neuropathy patients. An EMG dataset including 11 myopathy, 11 neuropathy, and 12 healthy patients was used to test the suggested approach. The EMG signal was first dissected into MUAPs. The statistical features from the MUAPs were then computed and supplied to the classifier. For two-class (normal/diseased) and three-class (normal/myopathy/neuropathy) recognition, binary SVM, SVM ensemble, and RBFNN were utilized. With classification rates of 90.1 and 91.2, the SVM ensemble offered the greatest performance for both two-class and three-class classification.

In identifying healthy, myopathy, and ALS EMG signals, the discussion in Khan and Hasan (2015), compared kNN and SVM classifiers. To extract wavelet characteristics from the data, the scientists employed DWT. The collected features were classified with an accuracy of 82% using the kNN classifier, while SVM classified them with an accuracy of 92.33%. When compared to the kNN technique, SVM demonstrated to be more effective at classifying EMG signals in this study.

The research in Artameeyanant et al. (2016), suggested a method for extracting features from EMG signals for myopathy and ALS classification based on a normalized weight vertical visibility algorithm (NWVVA). This method focuses on generating features from relationships between vertical visibility nodes and their amplitude differences as weights, as well as extracting sampling nodes based on sampling theory. The statistical features were calculated using selective statistical mechanics and measurements. These characteristics were then fed into the classifier as input. In this work, kNN, MLPNN, and SVM classifiers were employed to categorize the data, and their classification accuracies were compared. The accuracy of the kNN, MLPNN, and SVM classifiers was 96, 97, and 98%, respectively. Among the three classifiers used in this study, the SVM classifier had the highest accuracy.

The efficacy of different classifiers and feature extraction strategies in classifying EMG signals into normal and pathological patients was compared in Kehri et al. (2017). EMG data from the brachial biceps muscles were obtained using a conventional needle electrode. The signal's sample rate is 10 kHz. There are five normal and

five aberrant signals in the dataset. DWT+ANN, PCA+ANN, DWT+PNN, and DWT+SVM were used as classification approaches, with classification accuracies of 90.4%, 92.4%, 93.05%, and 94.28%, respectively. The best accuracy rate was achieved when DWT characteristics were classified using an SVM classifier.

Single and bagging ensemble classifiers were compared with different classification approaches in Subasi et al. (2018). EMG signals from the biceps brachii muscle were recorded with a concentric needle electrode and sampled at 20 kHz for 5 seconds for this comparison. The signals were obtained from 7 healthy people, 7 people with myopathy, and 13 people with neuropathy. First, the signals were decomposed into wavelet coefficients using DWT. Then, statistical features such as MAV, SD, average power, ratio of absolute mean values, skewness, and kurtosis were calculated to lower the features dimension. SVM, ANN, kNN, classification and regression trees (CART), C4.5, reduced error pruning (REP) tree, logical analysis of data (LAD) tree, and random tree were among the classifiers employed. SVM has the greatest classification accuracies of 98.96% and 99% for single classifier and bagging ensemble classifier, respectively, when compared to other approaches.

The study in Sengur et al. (2017), suggested a deep learning-based approach for classifying ALS and normal EMG signals in 2D temporal frequency pictures. Each input image is classified using the CNN based on the spatial information of the pixels incorporated into the network nodes. The CNN is made up of convolution and pooling layers, which create the characteristics that the fully connected layer uses to classify the data. During the training step, which is controlled by the convolution backpropagation technique, the parameters are fine-tuned. To increase the efficiency and speed up the training process, a new learning strategy called reinforcement sample learning strategy was used. For translating EMG signals into their time-frequency representation, spectrogram, continuous wavelet transform, and smoothed pseudo-Wigner-Ville distribution (SPWVD) were used in this study. The proposed approach was put to the test on a dataset that included 89 ALS and 133 normal EMG signals with a sampling rate of 24 kHz. The CNN had a classification accuracy of 96.80%.

For the diagnosis of neuromuscular illnesses, a hybrid classifier was developed in AbdelMaboud et al. (2015). This classifier was tested on a dataset of EMG signals from five healthy people, five people with myopathy, and five people with ALS. For EMG signal analysis, the DWT was used. The statistical values RMS, MAV, ZC, SSC, and STD were then computed. The hybrid classifier combined SVM and ANN, two different types of classifiers. To categorize one class against the other, each of these classifiers was applied to all of the individual features separately. Using the SVM classifier with the RMS feature as input, the highest classification rate of 98% was obtained for ALS. The ANN, on the other hand, demonstrated its capacity to accurately diagnose myopathy cases with an accuracy rate of 86.6%. This hybrid classifier has an overall accuracy of 85.5%.

In Shijiya and Thomas (2016), fuzzy logic and artificial neural networks were combined to increase the classification accuracy of EMG signals from normal, ALS, and myopathy. The spectral features like MF, mean power (MP), total power (TP),

peak frequency (PF), wavelet energy (WE), wavelet mean (WM), wavelet average power (WAP), wavelet standard deviation (WSD), and temporal features like MAV, variance, SI, and integrated EMG (IEMG) were used. The dataset for this study includes 75 signals from each of the 3 groups, as well as 15 signals from Parkinson's disease patients and 20 signals from Huntington's disease patients. ANN has a classification accuracy of 93.46%, and ANN with fuzzy logic has a classification accuracy of 94.61%. When compared to an ANN classifier, fuzzy logic with ANN significantly improved accuracy.

The researchers in Khan et al. (2016), also presented a multi-classifier strategy to improve EMG signal classification accuracy for the diagnosis of neuromuscular diseases. On MUAPs derived from an EMG signal, they used both time domain and time-frequency domain characteristics. A dataset of 150 EMG signals from normal, neuropathic, and myopathic patients was used to test the proposed approach, including 50 samples from each class. Different SVM and kNN single and multiple classifiers were examined. Using the time-frequency feature, the multi-classifier approach described in this work achieved average accuracy of 97%, while weighted kNN offered accuracy of 95%.

The SVM-kNN classifier has also been shown to be effective in other studies, such as Meena and Bansal (2016), where the Coiflet wavelet transform was used to evaluate the EMG signal and generated statistical features such as energy, mean, and standard deviation. These characteristics were then utilized to classify normal, myopathy, and neuropathy signals. This study employed a dataset of 500 signals from each class (350 for training and 150 for testing). To classify the collected features, a variety of classifiers, including SVM and SVM-kNN, were used to determine which classifier had the maximum accuracy. SVM had a 92% accuracy rate, whereas SVM-kNN had a 95% accuracy rate.

The article Abdel-Maboud et al. (2022), compared several bagging and boosting ensemble learning algorithms for classifying EMG data for neuromuscular disease diagnosis. The wavelet-packed coefficients (WPC) were initially estimated from the signals. Then, from the WPC of the signals, six statistical features were generated and utilized as inputs to the classifier: MAV, average power, SD, ratio of absolute mean values, skewness, and kurtosis. This study included 7 healthy people, 7 people with myopathy, and 13 people with neuropathy. With 99.08% accuracy, the AdaBoost learning algorithm in combination with the random forest ensemble classifier produced the best result.

Hence, it is noticed that ANNs are commonly employed in EMG signal categorization and have achieved great accuracy rates in identifying neuromuscular abnormalities. It's also worth noting that SVM has proven to be effective in the same field, particularly for multiclass classification.

Table 1 shows the employment of the machine learning techniques by some recent studies for classifying neuromuscular disorders. The table includes the classification techniques, datasets, feature extraction methods, and the corresponding accuracies.

Table 1 Neuromuscular disorder classification in machine learning

Study	Feature extraction	Classifier	Dataset	Aim	Accuracy (%)
Singh et al. (2017)	Functions (IMFs) generated by (EMD) process	FFBP-NN	PhysioNet (Goldberger et al. 2000)	Myopathy, neuropathy	94
Sengur et al. (2017)	C-WT and smoothed pseudo-Wigner-Ville	CNN	EMG lab (Nikolic 2001)	ALS	96.80
Khan and Hasan (2015)	DWT	kNN SVM	EMG lab (Nikolic 2001)	Healthy, myopathy, ALS	82 (kNN) 92.33 (SVM)
Subasi et al. (2018)	Features extracted from DWT	Bagging ensemble with SVM	Neurology Department of Gaziantep University	Healthy, myopathy, neuropathy	99

Uncertainty Analysis

The uncertainty is a quantity that evaluates the dispersion of the values attributed to a specific measure. The evaluation of the uncertainty is based on the systematic factors associated with the corrections. Due to the complexity of some measurement models, both stochastic and non-stochastic methods are combined to simplify the computations.

Among the publications that addressed the uncertainty in soft metrology is Lachish and Murray (2018), in which the uncertainty was evaluated based on the error factors such as the conversion and calculation errors. They started by identifying the uncertainty sources and the model by which the uncertainty was propagated to evaluate the combined uncertainty. In a different study (Pathmanathan et al. 2020), the uncertainty was evaluated based on a data-driven approach in which both the measurement and inference of variables contribute to the uncertainty.

In machine learning, the following parameters are used in estimating the uncertainty (Vallejo et al. 2020):

- Model output including error and covariance which are relevant to the deviation in the model output from the true value
- The calibration dataset that is influenced by measurement point uncertainties and used in training the model
- The input variables that are used in the inference model and have uncertainties
- The output uncertainties outside the calibration dataset

Many machine learning applications, such as decision-making, forecasting, automating scientific modeling, data compression, experiment design, and discovery, rely on accurate uncertainty estimate. Probability and statistics can help in quantifying the anticipated value and variability in the observations from the domain in case the system exhibits noise. In an incomplete system domain, probability can also be used to quantify and predict both the distribution and density of data to evaluate the capability and performance of a predictive model.

Uncertainty in SVM

In the classification stage in SVM, it is necessary to use dataset as inputs for training the classifier. But the training dataset may contain inaccurate classes leading to uncertainty probabilistic labels.

Considering the widely used support vector machines (SVMs), there is two-class classification problem, and it is necessary to find the separating hyperplane maximizing the margin between the examples of both classes. An approach in Niaf et al. (2011), proposed using both labels and probabilities as input.

The approach assumed a learning dataset (x_i, l_i) of input vectors (x_i) and labels (l_i) such that:

- In the classification stage, the class labels $l_i = y_i \in \{-1, +1\}$ for $i = 1, 2, \ldots, n$
- In regression stage, the real labels $l_i = p_i \in [0, 1]$ for $i = n + 1, \ldots, m$

So, p_i associated to (x_i) refers to the uncertainties about x_i 's class.

In order to handle the uncertain labels in SVM classification, they simulated different datasets and used RBF kernel $\mathrm{k(u, v)} = \mathrm{e}^{-\|\mathrm{u}-\mathrm{v}\|^2/2\sigma^2}$ and $\mathrm{C} = \mathrm{C}^{\sim} = 100$ to evaluate the classification performance and the probabilistic predictions by estimating the probabilities using Platt's scaling algorithm.

The performance was evaluated by computing the accuracy (Acc) of predicting the test set and Kullback-Leibler distance (KL) as follows:

$$\mathrm{D_{KL}(P\|Q)} = \sum_{\mathrm{i}=1}^{n} \mathrm{P(y_i = 1|}x_i) \log\left(\frac{\mathrm{P(y_i = 1|}x_i)}{\mathrm{Q(y_i = 1|}x_i)}\right)$$

for probability distributions P and Q.

The generated datasets were unidimensional and labeled +1 and −1 based on a normal distributions of variances $\sigma_{-1} = \sigma_1^2 = 0.3$ and means $\mu_{-1} = -0.5$ and $\mu_1 = +0.5$.

The dataset $\left(x_i^{\mathrm{l}}\right)_{\mathrm{i}=1,..,n^{\mathrm{l}}}$ denoted the learning data for $n^{\mathrm{l}} = 200$,while $\left(x_i^{\mathrm{t}}\right)_{\mathrm{i}=1,..,}$ n^{t} denoted the test set for $n^{\mathrm{l}} = 1000$. The true probability $\mathrm{p(y_i = +1|}x_i)$ was computed for each x_i and belonged to class + 1.

For $i = 1, \ldots, n^l$, the regular SVM dataset was obtained by using a probability of 0.5 as the threshold for assigning class labels y_i associated to point x_i.

If $p(y_i^l = 1 | x_i^l) > 0.5$, then $y_i^l = 1$, while in case $p(y_i^l = 1 | x_i^l) \leq 0.5$, then $y_i^l = -1$

The dataset $\left(x_i^l, y_i^l\right)_{i=1,..,n^l}$ is used to train the C-SVM classifier.

Another dataset $\left(x_i^l, \widehat{y_i^l}\right)_{i=1,..,n^l}$ was defined such that for $i = 1, \ldots, n$:

- If $p(y_i^l = 1 | x_i^l) > 1 - \eta$, then $\widehat{y_i^l} = 1$
- If $p(y_i^l = 1 | x_i^l) < \eta$, then $\widehat{y_i^l} = -1$, otherwise, $\widehat{y_i^l} = p\left(y_i^l = 1 | x_i^l\right)$

If the probability values are sufficiently close to 0 or 1 through precision and confidence, then they belong, respectively, to class −1 or 1.

The test set $(x_i^t)_{i=1,..,n^t}$ was used to find the true probabilities $p(y_i^t = 1 | x_i^t)_{i=1,..n^t}$ that could lead to the probability prediction error (KL). Figure 17 shows the probability prediction performances by the classical SVM.

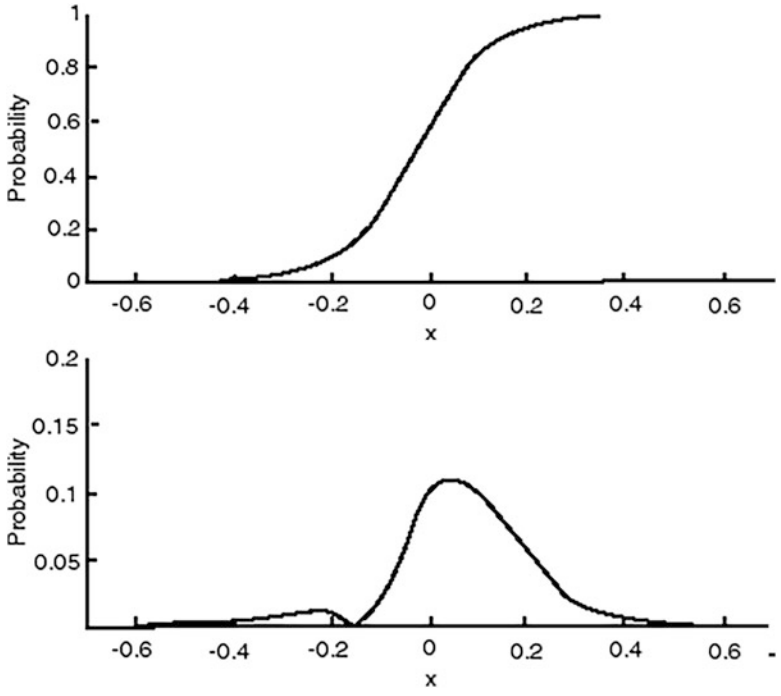

Fig. 17 Probability estimation. Top plot shows the true posterior probability with SVM estimations. Lower plot shows the distance between true probability and estimations (Niaf et al. 2011)

Cross-References

- ▶ Artificial Intelligence for Iris-Based Diagnosis in Healthcare
- ▶ Metrological Aspects of SEMG Signal Acquisition, Processing, and Application Design
- ▶ Precision Measurements in Healthcare Systems and Devices

Conclusion

In this work, the performance of different classifiers of machine learning was evaluated. The scope was to compare between the different techniques for the classification of neuromuscular diseases with special interest to the algorithm, EMG signal analysis, and accuracy. Several challenges which face researchers in EMG signal analysis were due to signal's complexion and nonstationary nature. A comprehensive discussion on advances in feature extraction and classification was presented. The involved techniques proved significant efficiency in this field, such as DWT for feature extraction and SVM for classification. A case study on the uncertainty evaluation for the SVM classification was introduced. For future work, efficient EMG classification method with high accuracy rate can be used in real-time neuromuscular disorder diagnostic systems. The chapter provided a valuable tool that can support researchers in making decisions about which classifier could help in the diagnoses of specific diseases. Based on the comparison, the SVM model is the best-performing algorithm for classifying medical datasets.

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Part VII

Metrology in Additive Manufacturing: Present Scenario and Challenges

Role of Metrology in the Advanced Manufacturing Processes

45

Meena Pant, Girija Moona, Leeladhar Nagdeve, and Harish Kumar

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Abstract

Metrology is the science of measurement used to measure various features (majorly dimensional and geometrical parameters) of components to improve the quality and reliability of parts for scientific and engineering applications. It needs to keep up with developments in manufacturing (e.g., automation) and current products with their associated technology, such as nanotechnology, microtechnology, and additive manufacturing to support quality management and process control effectively. When dimensions are decreased to micrometer or nanoscale, advanced measuring instruments are required to measure a

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manufactured component's dimensional and geometrical characteristics. This chapter discussed the role of metrology in the advanced manufacturing process. The in-line metrology role also is discussed for real-time data accessing during manufacturing. The economic benefits of metrology are highlighted in different domains. Precise metrology is required to regulate the measuring instruments to enhance consumer protection and minimize instrument uncertainty.

Keywords

Metrology · Microtechnology · Nanotechnology · Additive Manufacturing · In-line metrology · Quality

Introduction

Measurement is helpful in science, technology, and the economy because it establishes a foundation for fair trade, product optimization, and customer and corporate confidence in products. It is a process of collecting quantitative information of products in terms of units. According to Lord Kelvin in 1883, "whatever the subject may be, you have barely advanced to the science stage unless you can measure what you are communicating about and convey it in numbers. If you cannot measure it and cannot express it in numbers, your knowledge is of the same vague and unsatisfactory kind. It is the beginning of knowledge, but you have barely advanced to the science stage. This statement is applied to all research and development domains in science and engineering technologies. A measurement is an act of measuring quantities that everyone performs in our day-to-day life. All ordinary people unknowingly practice the aspect of measurement daily, but most of them are unaware of its importance. It includes units expressed with quantified numbers" (Kunzmann et al. 2005; Moona et al. 2019). The formation of units, development of measurement protocols, and a result of artifacts serve as measurement standards. Geometrical and dimensional metrology has evolved significantly in precision engineering due to increased demands generated by new manufacturing technologies and requirements for part functioning. Dimensional and geometrical measurements are the decisional basis for controlling the design, process, function, tolerances, and quality control (as shown in Fig. 1). Also, these are the primary basis with which other physical, electrical, mechanical, chemical, and thermal properties are linked. Precision in measurement becomes crucial to measure components for fitting and functional assemblies, measure required dimensions, measure an average number of pieces produced, and quality inspection to match with standard sizes. Firstly, it is done to establish units, then implement units, and finally applied for traceability (Carmignato et al. 2020; Weckenmann et al. 2007). Precision measurement achieves the following objectives.

1. To confirm items that are within acceptable tolerances.
2. To describe different manufacturing processes.

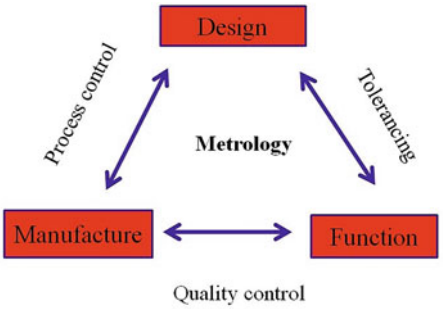

Fig. 1 Metrology in manufacturing process (De Chiffre 2007)

3. To develop standard techniques that help to decrease inspection costs and maximize measurement accuracy.
4. To maintain size variances under control and within established ranges.
5. To control the effectiveness of measurement devices.

However, it is notable that measurement science is evolved with the modern manufacturing processes. Presently, manufacturing of complex parts requires high precision to accomplish technical requirements, design parameters, and norm guidelines. Accuracy and traceable dimensional metrology support a wide variety of industry requirements, from macro-engineering to nanotechnology, and help to achieve the purpose of doing it correctly the first time, making it an unavoidable aspect of advanced manufacturing (De Chiffre 2007). However, the advancement in measuring instruments enables measuring and controlling the dimensioning up to the nanoscale range. Precision dimensional metrology, combining multiple measuring techniques into a single device with excellent resolution and accuracy, opens up various possibilities. The role of dimensional metrology becomes critical in micro and nanomanufacturing by utilizing unique techniques of measurement data fusion. This study describes the part of metrology in microtechnology, nanotechnology, and additive manufacturing. In-line metrology was discussed during the manufacturing process. The economic benefits of metrology are also explored in the manufacturing domain.

Historical Evolution of Metrology

The value of metrology was first recognized by Ancient Egyptians and applied to the construction of pyramids with incredible precision. The royal Egyptian cubit was based on the sections of a pharaoh's or king's body used as measurement standard ruling in that era. However, human-referenced measuring systems were no longer quite acceptable due to industrialization and the evolution of mass production at the end of the eighteenth century. The advancement in the manufacturing industries recognized the need for a metrology concept. Artisans had to secure the functionality of individually created parts by adapting the parts to each other. In contrast, the

industry had to provide component interchangeability to reap the economic benefits of mass manufacturing and supply interchangeable spare parts. Consequently, it wasn't enough to compare and rework two parts to make them fit together. The final dimensions of parts had to be created and measured precisely enough to ensure the assembly functions independently. Therefore, dimensional metrology had to maintain synchronization with the industry's ever-increasing demands for advanced technology. The meter agreement was accepted and internationally recognized as a length unit in 1875. It is always a comparison of the length to be measured against a length standard, the length unit. In 1983, the speed of light redefined the meter unit as "the distance traveled by light in a vacuum for 1/299792458 seconds due to the invention of laser" (Weckenmann et al. 2009, 2012; Quinn and Kovalevsky 2005). The time unit second and the time measurement define the length of a unit, which is calculated with good precision by many magnitudes. Berndt established the measurement uncertainty should not surpass the tolerance's one-tenth (or one-fifth in extreme instances) from 1924 to 1930. Over time, more precise measuring tools were created, such as length measuring machines, microscopes, profile projectors, etc. However, they could only make one- or two-dimensional measurements. The evolution of two-dimensional measuring devices such as fringe projectors and gauge blocks occurred from 1950 to 2000. The development of the coordinate measuring machine (CMM) became a boon for manufacturing. It is used to quantify various geometric features such as lines, planes, circles, cylinders, and spheres by measuring numerous points on the workpiece surface. However, due to the limited technological development, measurements were performed between two points in one dimension. Advanced measuring instruments like X-ray tomography and scanning electron microscopes are used to detect a dimensionally poor quality product. 3D noncontact metrology is contributing significantly to advance manufacturing applications. With increasing demands for accuracy, it was found that two-point measures could be confusing. For example, when measuring the diameter of a shaft with significant form deviation, it was addressed by developing improved metrology devices, such as measuring microscopes and profile projectors, which allowed two-dimensional measurements in a plane section on the one hand. However, the manufacturing industry confronted new challenges with smaller components, complicated geometry, higher tolerances, and stringent quality control. A reliable and accurate dimensional metrology environment was required to overcome these difficulties, underlining the significance of advanced dimensional metrology solutions in production. Coordinate metrology has proven to be a revolution for manufacturing, assessing deviations by taking dimensional measurements on numerous places on the work surface, and displacing basic equipment like calipers, gauge blocks, and micrometers. CMMs can measure the dimensions of a wide range of objects. Oversized items, such as aircraft, can be measured with CMMs and laser trackers, whereas micro-CMMs can perform measurements in the nanometer range. Metrology is developed for in-process adaptation for real-time monitoring of process parameters utilizing embedded sensors. The components are meticulously checked during or after production to preserve product consistency and conformity to dimensional standards (Quinn and Kovalevsky 2005). Figure 2 shows the dimensional and

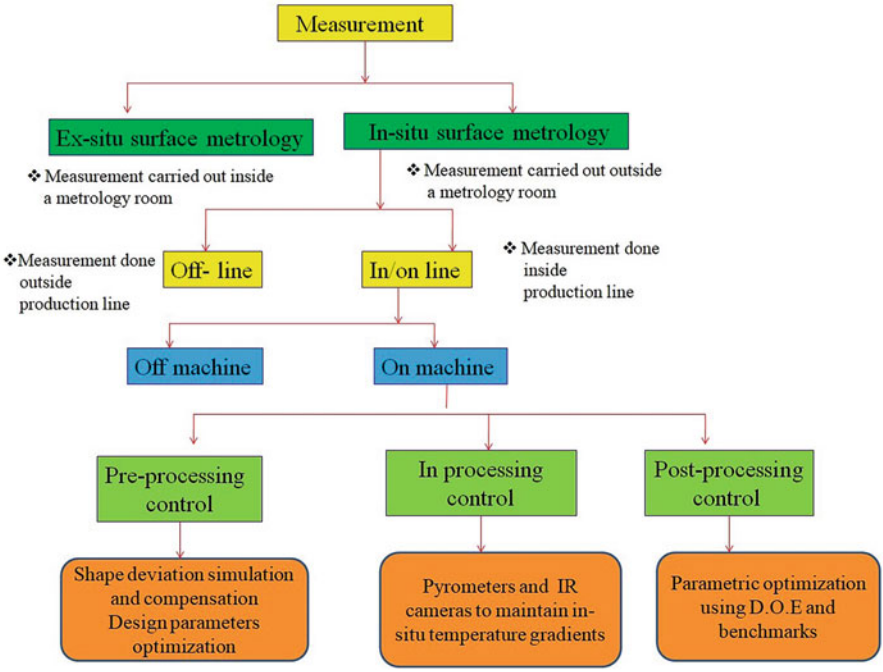

Fig. 2 Different measurement methods for precision manufacturing (Everton et al. 2016)

geometrical measurement control methods that can be performed before, during, and after production. Table 1 briefly discussed some common measurement methods in industries.

State of the Art of Metrology in Advanced Manufacturing

As the manufacturing processes became increasingly advanced to manufacture complex components that fulfill various functional and technical requirements, measuring science progressed simultaneously. The present metrological systems are more flexible, self-adaptable, and intelligent, based on cyber-physical systems, the Internet of things, and cloud technology. In this manner, the quantity of information and quality assurance of an object can be improved at the same time. Manufacturing metrology demands are set by production, so it must follow production engineering trends. Presently, additive manufacturing, Industry 4.0, microtechnology, robotics and automation, and nanotechnology are emerging fields in the manufacturing industries. Additive manufacturing has transformed the manufacturing scenario by opposing the conventional manufacturing principle. Industry 4.0 refers to a Fourth Industrial Revolution (following water/steam power, mass manufacturing, and automation through IT and robotics) and introduces the concept

Table 1 Measurement methods of metrology

Method	Role	References
Dimensional metrology	It is concerned with measuring geometric features of an object such as size, distance, angle, form, or coordinates to meet design requirements. It is required as a "toolbox" to aid manufacturing and reduce total loss. The toolbox must have a variety of instrument modules such as rulers, scales, Vernier caliper, and micrometer for measuring the dimensions of objects	Carmignato et al. (2020) and Musso et al. (2016)
Surface metrology	The measurement of a surface's characteristics (uniform patterns, abnormalities, roughness, curvatures, and critical dimensions) is known as surface metrology. It is significant for the new materials utilized in modern processes such as bearings, thermal, advanced coatings, optical, and semiconducting/electronic devices. It can be done using tactile or optical techniques such as Taylor-Hobson, atomic force microscopy, stylus profilometers, optical profilometers, photogrammetric, and interferometer	Jiang and Whitehouse (2012) and Senin and Leach (2018)
Coordinate metrology	In industry, coordinate measuring machines (CMMs) are the most popular instruments used for 3D monitoring physical elements. CMMs are mechanical systems that use the movement of a measuring probe to identify the coordinates of points on the surface of a component instead of directly measuring the attributes. Numerous CMMs are available, and all can accurately measure tiny details of components from micro- to nano-level. CMM is available with different configurations, including probe (contact or contactless, single or multiple), mechanical structure (cantilever, moving bridge, fixed bridge, horizontal arm, gantry, column, etc.), and drive controller (computer-assisted data processing, direct computer control, post-processing software, and so on). Today, noncontact probe technologies such as optical sensors, X-ray computed tomography (CT), optical interferometry, and magnetic resonance imaging are utilized to quickly increase measuring speed and the number of measurement points	Mian and Al-Ahmari (2014) and Berry and Barari (2018)

(continued)

Table 1 (continued)

Method	Role	References
Geometirc dimensioning and tolerancing (GD&T)	It plays a significant role in product functionality and quality assurance. To assure geometric conformity and standards, the product must fulfill the limit, fit, and tolerance specifications. The two primary methods, experimental methods based on geometric benchmark test artifacts and analytical methods like finite element analysis, are preferred for GD&T manufactured products. In such situations, prior assessment of geometric tolerances, i.e., GD&T features, is critical to avoid part rejection and material waste	Rupal et al. (2020a) and Ameta et al. (2015)
Ex situ/in situ metrology	Based on the environmental conditions, measuring instruments are divided into two groups: Those installed in the metrology room and those installed outside the metrology room. When the measuring environment is different from the manufacturing environment, surface metrology performed in the metrology room is categorized as ex situ metrology. The component dimensions are measured on the same workspace without isolating the component from the production environment, referred to as in situ metrology. In situ metrology ensures component quality and detects errors before final fabrication. As the component design becomes complex, the calibration and traceability procedures for advanced in situ operations become difficult. It is further divided into two groups; outside the metrology room based on their proximity to the manufacturing line. Some inspections are performed outside of the production process. When needed, the component must be moved out from the production line for surface measurement, checked by one or more equipment, and returned to the line for continuing manufacture. This type of measurement is known as "off-line" surface metrology	Gao et al. (2019) and Everton et al. (2016)

of cyber-physical systems to differentiate this new evolutionary phase from the electronic automation that has gone before. Industrial robots are frequently used to boost production, provide flexibility, and assist workers with physically demanding and risky activities. Quality control is necessary to operate intelligent automation solutions in the advanced manufacturing sector. This innovation allows for an even

greater connection, enabling manufacturers to keep their competitive advantage in a constantly changing world and to react quickly and flexibly to customer demands. Modern manufacturing metrology must keep up with these changes and broaden its horizons. It is an important task that must be performed essentially because a part's size, thickness, and shape complexity significantly impact the selected process. It becomes even more complicated when absolute dimensions are reduced to microscale or nanoscale levels. Precision is a critical function widely needed in microtechnology and nanotechnology for process control and quality assurance (Durakbasa et al. 2015).

Microtechnology

Microtechnology is a crucial technology with a rapidly expanding market in various industries. In recent years, it has advanced significantly in different application domains. Now it plays a vital part in everyday life without being widely recognized. Small electronic, mechanical, and optical components are found in computers, cell phones, and automobiles daily, and their quality must be guaranteed (Schmitt and Pavim 2008). It is a rapidly developing discipline that takes advantage of the unique features of ultrasmall-scale materials. The micro components are used in various industrial applications such as micro gears or micromotors, medical items such as hearing aids or ear implants, chemical analysis and processing (passive microfluid components), small freeform lenses in mobile phones, optical devices (optical switch component), and radio frequency technology components. It is often hard to achieve all the micro-level inspections with a single inspection technique. Different measurement concepts or configurations need to perform to inspect the various shapes, dimensions, and geometric specifications. It indicates that the measurement system relies on the object's attributes. This reliance is problematic for monitoring flexible manufacturing lines, such as small-series manufacturing printed circuit boards (PCBs) or micro-lasers. Multiple sensors technology is mandated during the production to assess the individual features of different parts to expand a microproduction unit's inspection range and adaptability (Nicholson and Wallace 2007; Tosello et al. 2009). As a result, advanced measuring devices have been required to measure dimensional and geometrical features of micro components in the submillimeter to micrometer range. Many micro-metrology procedures are already used in the industrial sector. Some are based on optical or nondestructive test (NDT) principles due to their micro-world benefits (touchless, noninvasive, fast, and accurate measurement results). The measurement system's simple interaction with the (micro-) items under scrutiny could harm or damage them if the surface is not smooth. Microelectromechanical system (MEMS) is a specialist field that refers to technologies that allow existing sensors, actuators, or system products to be miniaturized (De Chiffre et al. 2003).

Nanotechnology

The science of measurement at the nanoscale is known as nanometrology. Today, nanometrology is dependent on semiconductor technology advancements to a significant extent. Nanometrology is imperative in producing nanomaterials and devices with high accuracy and reliability in nanotechnology. The demands for new nanostructured sample characteristics exceed current measurement science capabilities. Due to the advancement of the requisite metrology techniques, a comprehensive range of measuring devices has been designed to evaluate surface structures down to the 0.1 nm level. Dimensional nanometrology is a branch of dimensional metrology that refers to a set and verification of dimensional values on components having at least one important dimension or functional feature smaller than 100 nm and whose uncertainty is expressed in nanometers (Tosello et al. 2009; Ukraintsev and Banke Jr 2012). Understanding the importance of manufacturing precision in industrial metrology in the nanometric range is essential for success on a global scale. When modern computer-aided production metrology is heavily utilized to regulate manufacturing to meet this requirement, product quality based on geometrical specifications improves as dimensional product tolerances tighten. Advanced laboratory research must ensure that the product meets its design criteria at nano-level parameters for leading semiconductor and mechanical engineering industries. Optimizing dimensional tolerances and selecting appropriate machining and measuring procedures are vital to the proper functioning of any nanocomponent (Hong and Chang 2002; Lazos-Martınez and Gonzalez-Rojano 2013). In recent years, AFM with laser interferometers has been created and employed to measure nanoscale standards such as step height, one- and two-dimensional gratings, and line width. Other measurement techniques, including computed tomography, X-ray reflectometry, optical CMM, transmission electron microscopy (TEM), scanning electron microscopy (SEM), interferometry microscopy, 3D optical profilometer, and optical scatterometry, have been used to inspect nanodimensional measurements for surface characterization and layer thickness measurements. Practically, all microscopic methods, such as AFM, TEM, SEM, and optical interference microscopy, measure from the top; they cannot acquire enough data at sharp corners of structures to estimate metrics like line edge and width roughness. There are many other challenging tasks in this domain, such as sidewall measurements and accurate 3D measurements of nano- and microstructures. Some measuring instruments are listed in Table 2 to measure the micro and nanocomponents (Dai et al. 2008; Koumoulos et al. 2018).

Additive Manufacturing

Additive manufacturing (AM) is a new paradigm that opposed the subtraction principle in the manufacturing industry. Usually, a complex part had to undergo various processes to produce a net shape in the conventional manufacturing process. The inspection is required to check the dimensional stability and quality at each step.

Table 2 Methods for measuring capability (Bauer et al. 2015)

Design limit	Measuring instruments
>10 μm	Coordinate measuring machine (CMM), optical systems, pneumatic and mechanical comparators
10 μm – 1 μm	Micromechanical comparators, CMM, optical and electric comparators spin resonance
1 μm – 100 nm	CMM, optical interferometer, dark-field microscopes, phase microscopes, electrostatic and magnetic comparator
100 nm – 10 nm	Roughness measurement devices, laser interferometers, and fluorescence microscope
10 nm – 1 nm, 1 nm	Laser confocal microscope, X-ray microanalyzer, scanning electron microscope (SEM), transmission electron microscopy (TEM), scanning tunneling microscopy (STM), atomic force microscope (AFM), electron and X-ray diffraction system

In contrast, AM evolved with a new revolution where a complex part was made from a single machine source with less assembly operation. The wire, powder, and sheet form materials are joined into a layer-by-layer pattern. It is an early phase of progress; it still provides enormous design freedom opportunities in lightweight components for aerospace, intricate design features, cooling channels for the automobile industry, porous structures for medical implants, etc. (Behera et al. 2021). Metrology is vital in engineering applications, including aerospace, automotive, biomedical, marine, and medical, where accuracy is crucial, and quality control depends on precise product attributes and process control information. The most visible unfavorable exterior consequences on additive-produced components are shape variation and texture. Many diverse factors contribute to these effects, but each layer's height and thermal stresses' easing play a significant influence. Therefore, a metrological assessment of the components is required for quality assurance and post-process planning. Focusing on the metrological aspects of AM components is necessary because the part is fabricated with a less tight tolerance scheme (Vora and Sanyal 2020; Huang et al. 2015; Leach 2016; Shulunov 2016). However, no particular provision of standardization and tolerance has been established yet.

However, the National Institute of Standards and Technology (NIST) has attempted to narrow the AM competence and qualification gap. NIST hosted a session on "Measurement Standard Issues in Rapid Prototyping" in 1997. NIST issued rapid prototyping industry standards in 1999. The ASTM F-42 group was in charge of setting guidelines for categorizing the AM method, which resulted in seven different classes. The ASTM E-28 committee developed the tensile test requirements for AM parts (Jurrens 1999; Jadayel and Khameneifar 2020). Then, the American Society for Testing and Materials (ASTM) and the International Organization for Standardization (ISO) analyzed the potential of AM and developed a roadmap of standardization for AM technologies. These two organizations jointly worked on standardizing AM process chains, quality parameters, test procedures, supply agreements, and fundamentals to avoid the progress of several competing and potentially inconsistent standards (Rupal et al. 2020b; Leach et al. 2019; Li et al. 2018; Moroni

Table 3 Standardization activities for AM ((ISO) IOfS 2020; Standardization IOf 2020; Villarraga-Gómez et al. 2018)

Organization	Role
ISO/ASTM 52910	Highlights the effect of triangle sizing on the surface texture, effects of build direction on tolerances, the effect of overhang elements requiring support structures, etc.
ASTM F42.01	Test methods
ASTM F42.91	Terminology
ISO17296-3	It describes additively manufactured products' texture, size, dimensions, and geometric tolerances
ISO1302 and ISO 4288	Mention specification and measurement of surface texture
ASTM F42.04	Design
ISO 1101 and ISO 2768-2	Refers geometrical tolerancing
American Society of Mechanical Engineers (ASME) Y 14.46	Defines the product definition requirements particular to AM
ASTM F42.90	Executive
ASME Y 14.41	Within a digital product description data set, create a three-dimensional (3D) model
ASTM F42.94	Strategic planning
ASME B46	Defines surface topography of AM parts
ASTM F42.05	Materials and processes

et al. 2017). Table 3 shows the role of different organizations in the standardization of AM. However, the nondestructive, noncontact, real-time, in situ measurements methods are helpful for the AM process to obtain accurate, less time, and cost-consuming reliable processes and improve process control.

In-Line Metrology

With advancements in manufacturing processes, it is analyzed that the product feature should be measured simultaneously to save energy and time during production. Now, metrology can be incorporated to collect real-time data about any product or service outside the production (ex situ or off-line) and inside the production (in situ or in-line/on-line) in modern manufacturing processes. Presently, in-line metrology significantly affects accessing the real-time information of any product or process. An in-line measuring technique would be the most excellent solution for scaling up production while maintaining regularity. Traditional CMM technology provides the best metrological feedback. However, due to its time and effort, it cannot be used for in-line measurements, especially in an AM chain where component measurement is required for post-processing planning. It is based on obtaining measurement data during the production cycle and using it for immediate process control and quality assurance. The requisite changes in production processes are easily detectable using this method. Quick reactions can improve operational

stability and product quality early in the manufacturing process. It is incredibly significant for monitoring, regulating, and optimizing product fabrication and manufacturing processes to save cost by better utilizing raw material, troubleshooting, defect minimization, and improving production yield and energy efficiency (Moenning 2006; Kiraci et al. 2017). It benefits only when in-line metrology is included in the early stages of production planning and setup.

Additionally, there are two ways to carry it out: on-machine (while the workpiece is mounted on the machine) and off-machine (when the workpiece is moved out of the machine). Usually, noncontact methods are preferred for in-line metrology because sometimes contact methods can be sluggish and cannot measure hidden features or abnormalities. Metrology instruments can become a motivating factor for in situ metrology in advanced manufacturing involving additive manufacturing, microtechnology, and nanotechnology. Efficient imaging techniques such as optical, X-ray, infrared red, e-beam, computer tomography, optical CMM, ultrasound, or scanning probes can be utilized for advanced manufacturing applications (Tofail et al. 2018; Bordron et al. 2019).

On-Machine Metrology

On-machine inspection, also known as in-process inspection, can be done simultaneously during the machining operation. The inspection does not take additional time; hence, on-machine metrology can maximize efficiency. The sensor head must withstand the potentially harsh environment, including machining fluid, generated chips, vibration, and high temperatures. The associated machine tool defects, such as spindle/slide motion errors, thermal deformations, and vibration, are a substantial difference between on-machine surface metrology and off-machine measurement utilizing a stand-alone surface measuring instrument. An additional error cause is the sensor head alignment problem on the machine tool. It is also vital to look at reducing and separating these errors from on-machine surface measurement results and the problems with calibration and traceability they cause. Full-field and probe-scan systems are two measurement systems for on-machine and in-process surface metrology. A light beam is focused onto a specific work surface area to collect three-dimensional (3D) information about the surface over a region, often referred to as the system's field of view. A 1D or 2D probe, contact or contactless, is rotated by the machine tool axes to scan the work surface in a point-by-point or continuous-path mode in an on-machine probe-scan system. Table 4 shows the advanced on-machine measurement techniques (Shimizu et al. 2021; Imkamp et al. 2016).

Metrology for Economic Growth

Generally, metrological development refers to increasing the amount of information acquired about an object or decreasing the cost per unit of measured information. Some scientists, practitioners, and industrialists have viewed it as a cost factor rather

Table 4 Noncontact measuring and inspection methods

Technique	Features	Advantages	Disadvantages	References
Interferometry	It is a noncontact measuring process where the known surface shape is correlated with the unknown surface. The interference fringe is depicted as a series of differences between the surfaces	High resolution Easily measure the smooth surface Simple	High cost Error created when measuring rough surface	Abdelsalam and Yao (2017)
Fringe projection	It is a three-dimensional noncontact measurement process where a projector is used to produce different light-dark fringe patterns on measuring components	High accuracy Complex features can be easily measured Fast data processing	Expensive Uneven illumination distribution	Pineda et al. (2019) and Liu et al. (2021)
X-ray computed tomography	It is a nondestructive technique used to inspect various materials such as polymer, metal, ceramic, light metal, and fibers. It can generate geometric data for the characterization of material structure	Used for analyzing the internal and external structures High resolution Used for dimensional accuracy measurement	High cost Operational maintenance required	Villarraga-Gómez (2016) and Zanini et al. (2018)
Optical CMM	Optical CMM systems automatically use image processing technologies to provide fast and accurate measurement results. The rapid development of these measurement tools has paralleled the steady advancement of factory automation at production facilities	Portable Use for small features High resolution and speed	Required post-processing Only provide data on a few points Costlier	Sładek et al. (2011) and Gapinski et al. (2014)

(continued)

Table 4 (continued)

Technique	Features	Advantages	Disadvantages	References
High-speed camera	A photodiode and a high-speed CMOS (complementary metal-oxide-semiconductor) camera can examine the quality of three alternative geometries: Nearby scan vectors, hanging regions, and sharp edges	Fast speed Low-power consumption	Inaccuracy due to lens dislocation and misalignment	Chua et al. (2017)
Infrared camera	An infrared camera detects and measures the infrared energy of objects (also known as a thermal imager). The camera turns the infrared data into an electronic image that depicts the object's apparent surface temperature. An optical system in an infrared camera concentrate's infrared energy onto a particular detector chip (sensor array) with thousands of detection pixels stacked in a pattern	Provide thermal details of an element	The accuracy depends on the camera resolution; a blurred quality image is obtained if the resolution is poor	Rockett et al. (2021)
Radiography testing	The volume of a specimen is examined via radiographic testing (RT, X-ray, or gamma-ray). It is a good NDT technique to obtain suitable quality components. Radiography combines X-rays and gamma-rays to create a radiograph of a specimen, revealing any variations in	Inspection of assembled components Surface preparation isn't necessary Crack growth is determined Applicable to all types of conducting and nonconducting materials	Health risks associated with radiation Expensive	Panas et al. (2021)

(continued)

Table 4 (continued)

Technique	Features	Advantages	Disadvantages	References
	thickness, abnormalities (internal and exterior), and assembly features.			
Optical surface topography	Optical surface height sensitivity at the nm level is required for optical surface topography measurement. Vertical resolution is the most frequently mentioned metric concerning surface height measurement sensitivity	3D measurements, fast measurement speed, simple to use, automated	Surface gradient limitation, material surface demand	Foreman et al. (2013) and Feng et al. (2019)

than a value-adding activity in the manufacturing process. Measurement is the main cost element throughout manufacturing because faulty measurement devices can lead to erroneous and uncertain outgrowths. However, if precise measurement is not considered during manufacturing, incorrect decisions may be made, resulting in rougher sampling and increased measurement uncertainties, making it costly. Hence, there is always a compromise between the amount of reliable information available and the expenses associated with gathering it through measures. Cost-benefit assessments must be carried out to determine which measuring technique qualifies best for a specific application. Suppose metrologists, engineers, and management personnel have enough knowledge about the measurement process. In that case, they can estimate what information about a process or product is needed to check its functionality, tolerances, and quality. When metrology is applied at each step of the production process, it can be considered a value-adding process. It may provide significant cost-benefits (as shown in Fig. 3) when employed concurrently during or soon after the production; hence, it can improve the quality and productivity of the overall system (Rodrigues Filho and Gonçalves 2015; Savio et al. 2016). The economic impact of metrology can quickly be evaluated by checking the cost involved in measurements and identifying the flaws in the value chain. Metrology exists from the design phase when a new product's functionality and specifications are defined and continue to the product development phase when all specifications and tolerances are set, followed by post-manufacturing characterizations. According to the studies, around 75% of all manufacturing errors are investigated in the early stages; hence, any wrong decision and insufficient or ineffective feedback regarding the process and product results in delays in the whole product life cycle, which affects the servicing (Link 2021). The following points recognize the economic benefits of metrology in manufacturing:

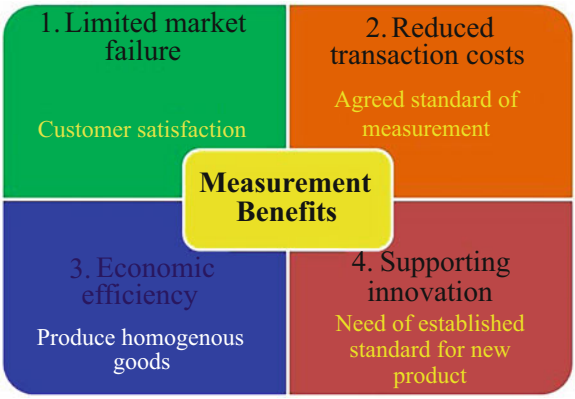

Fig. 3 Economic benefits of measurement (Robertson and Swanepoel 2015)

- Measurement acquired data is used to develop and plan the product generation and design evaluation to diminish failure modes.
- If proper inspection of the product has not been done in earlier stages, the cost of eliminating errors increases by a factor of 10 for each subsequent step of the product life cycle. Approximately 90% of a product's future costs are caused by difficulties already implemented in the earlier phases, such as fixing product attributes and designing components to determine their tolerances.
- Initially, enhanced dimensional measurement procedures may result in a slight decline in production, but better measurement capabilities open the door to improving productivity in the long run.
- Measurement helps decide the acceptance and rejection of a manufactured component. For example, in the aerospace industry, measuring in-service degradation of turbine blades concerning multiple blade dimensional parameters saves the time and cost of manufacturing.
- Telecommunications and navigation, both becoming more important in today's world, require the most consistent time and spatial measurement standards.
- Consistent measurements are essential to human health and safety in diagnosis and medical treatment.
- Only accurate and reliable measurements can protect the environment from industrial activity's short- and long-term detrimental consequences.
- Global environmental studies rely on accurate and consistent data from various disciplines over long periods, which can only be secured by utilizing measurement standards based on fundamental and atomic constants.

Conclusion

Metrology is a vital tool for gathering data at every stage of the manufacturing process. The present chapter describes the role of metrology in advanced manufacturing systems. Precision dimensional metrology is essential to ensure that

the finished product fulfills design criteria and works as expected. However, it should be noted that any false interpretation about the process and product makes metrology mere a cost factor. Metrology in manufacturing should be implemented from the start of the design process to increase product quality and reliability. In-built measuring instruments should be attached to the production line to find faults in dimensional and geometrical features. Progress of measurement science is the guidance for controlling the production process, improving quality, and increasing customer satisfaction.

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Additive Manufacturing: A Brief Introduction

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Abstract

In recent years, both academically and industrially, additive manufacturing (AM) has experienced rapid expansion because of its capability of producing customized products of intricate shapes. 3D printing has evolved as a promising technology for the creation of engineering parts or applications, as it is an additive technique, unlike traditional manufacturing techniques. The advantages of 3D printing include reduced material waste, easiness in manufacturing, minimal human participation and post-processing, and efficiency in energy, which make it a viable option for industrial application. The chapter provides a brief introduction to additive manufacturing and also addresses a variety of 3D printing techniques, as well as their benefits and drawbacks. A detailed discussion of the various materials that are suitable for each type of 3D printing procedure is discussed along with the application of AM and the challenges associated with it.

Keywords

Addictive manufacturing · Applications · 3D printing · Materials

Introduction

As the years progressed, the process of manufacturing has been changed as per the requirement of better flexibility, efficacy, and customization, and because of these reasons, additive manufacturing comes into the picture. Additive manufacturing (AM) also known by other names as well such as three-dimensional (3D) printing, layered manufacturing (LM), rapid prototyping (RP), and solid freeform fabrication (SFF) is a novel method of manufacturing that builds three-dimensional parts layer upon layer using computer-aided designing (CAD) data. The ISO/ASTM 52900 standard (ISO/ASTM 52900 2015) defines additive manufacturing (AM) as "the technique of combining materials to build parts from the data of 3D model, generally layer by layer, in opposition to subtractive and formative manufacturing methodologies of production." Unlike traditional subtractive manufacturing methods like conventional machining which involves removing material until the desired part is obtained, AM creates the product by adding material. The utilization of AM is viewed as another Industrial Revolution (Lipson 2014; The Economist 2012), subsequent to steam engines, computers, and the Internet. Its ability to create customized prototypes or parts of even complicated geometrical configurations with reduced production time and cost has outperformed the traditional manufacturing settings. Individuals from various fields like aerospace (Uz Zaman et al. 2019; Gradl et al. 2022), dental (Chen et al. 2016), biomedical (Ruban et al. 2022; Whenish

et al. 2022; Sontakke and Jaju 2022), construction, and defense industries (Davies et al. 2022), etc. have benefited from this. The steady increase in market competitiveness to fulfill customer requirements and satisfaction in the quickest feasible period is one of the major causes behind the fast rise of AM in diverse fields. It allows for more efficient and faster manufacturing of parts as there is no need for fixtures, tooling, and coolants in 3D printing; it also reduces wastage of material (Williams 2016), gives freedom for design, and can print complicated structures with a higher resolution and remarkable flexibility (Ngoa et al. 2018).

AM includes a variety of techniques, materials, and machines that have grown through time and have the potential to revolutionize the manufacturing and logistics sector. In 1986, Charles Hull introduced this technology by a process called stereolithography (SLA), which was accompanied by advancements like selective laser sintering (SLS), fused deposition modeling (FDM), laminated object manufacturing (LOM), binder jetting (BJ), and selective laser melting (SLM) (Liu et al. 2021) as shown in Fig. 1.

AM's ability to 3D print small batches of customized parts or products for a reasonable price is especially advantageous in the field of biomedicine, where one-of-a-kind patient-specific products are frequently required. Products that are custom-made have become the new trend in 3D printing (Berman 2012). Metals, polymeric materials, ceramics, and concrete are some of the most often utilized materials in 3D printing. The most common polymers employed in the 3D printing of composites are polylactic acid (PLA) and acrylonitrile butadiene styrene (ABS). However, the material chosen for 3D printing largely depends on the type of application it will be used (MacDonald and Wicker 2016).

Earlier 3D printing was used as just a rapid prototyping method. Furthermore, as technology advances, 3D printing is now utilized for full-fledged production with reduced time in production, lower cost, and greater flexibility; thus, it is also known as rapid manufacturing (Mohamed et al. 2015). The various incentives for pursuing AM are depicted briefly in Fig. 2. The major purposes include the production of prototypes, development of the product, fabrication of new products that are not possible to produce using traditional techniques of manufacturing, improved efficiency in production, lower production cost, fabrication of products that are customized/personalized, source of revenue, expansion of new production lines, improvement in customer satisfaction, improvement in the management of supply chain and its logistics, and others.

There are numerous variations in additive and conventional manufacturing as conventional manufacturing offers the manufacturing of a variety of parts for mass production, whereas additive manufacturing is a system that is material and design specific. Also, AM can fabricate parts in a variety of sizes ranging from the microlevel to the macrolevel. But the precision or quality of the printed parts, on the other hand, is determined by the accuracy of the printing procedure used and the printing scale.

In 3D printing, the first step is, to begin with, the basic design of the part which is to be fabricated, and it is created by utilizing CAD software. CAD then creates the file and translates it into standard tessellation language (STL) format and sends it to

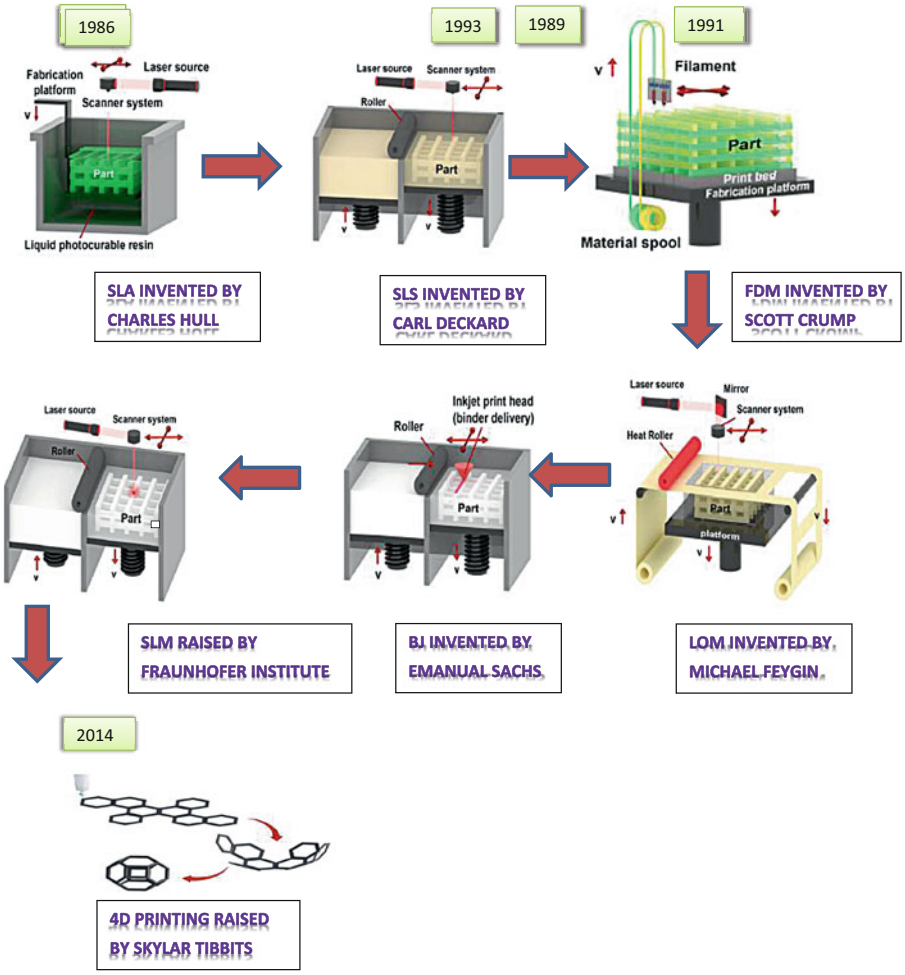

Fig. 1 Evolution of additive manufacturing (Jandyal et al. 2022)

the printer where it reads that file and builds the part by stacking layers on top of each other (Raina et al. 2021). Instead of reading the part as a whole, 3D printers read it as a single two-dimensional layer at a time. The basic procedure for creating 3D parts utilizing 3D printers is shown in Fig. 3.

The following are some of AM's distinct capabilities:

1. **Flexibility in design**. AM methods are distinguished by their layer-by-layer manufacturing methodology, which allows practically any geometrically complicated shape to be created. This contrasts with subtractive methods, which limit design flexibility because of the necessity for fixtures and a wide range of tooling as well as the risk of crashes and the complexity of maneuvering the cutter's

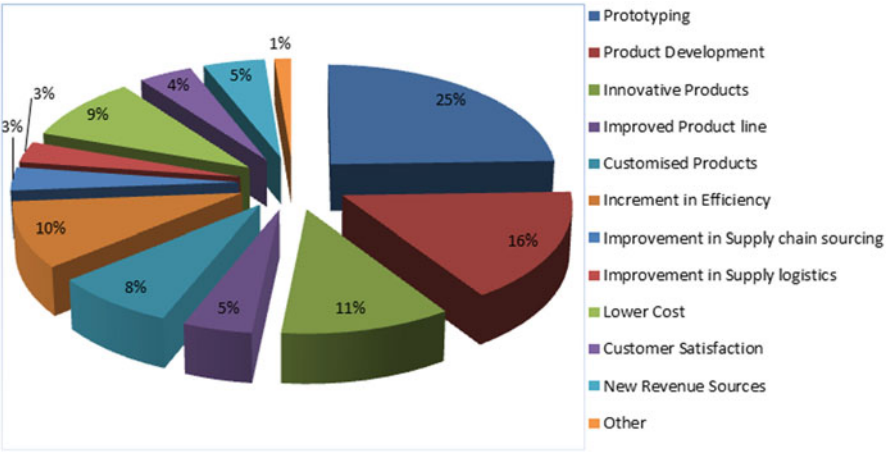

Fig. 2 Major reasons for employing AM (Dizon et al. 2018; Gartner 2014)

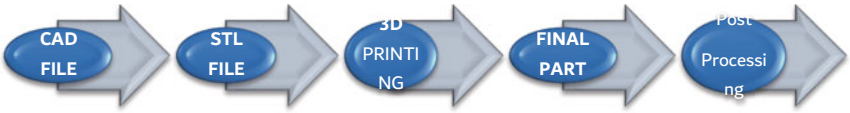

Fig. 3 Basic process in 3D printing

ability to reach deeper and parts that are not visible when it comes to creating complicated geometry.

2. **Cost related to geometrical intricacy**. In conventional manufacturing, there is a clear link between the complexity of the part to be produced and the cost associated with it, while employing AM there is no additional expense since there is no such necessity for extra tooling, refixturing, and more specialization of the operator or even the time it takes to construct the part.
3. **The need for assembly**. AM methods allow the creation of geometric designs that might otherwise need the assembly of many components if manufactured traditionally. Additionally, AM may be used to create "single-part assemblages" with integrated components. The joints and components are printed in a position and held by support material or sintered powder that should be ejected during posttreatment.
4. **Reduced overall production cost**. AM has the ability to cut inventory expenses as well as supply chain and delivery costs. When producing components with AM, there is often relatively little wastage of material, while there is some material trash in powder bed fusion technologies owing to support structures and powder recycling. Also, the ratio of the quantity of material bought to the quantity of material present in the finished component for AM techniques is extremely less (Brice and Taminger 2011). Improved designs combined with lower supply chain and delivery costs make AM technologies fit for a wide range of uses.

This chapter gives a brief introduction to additive manufacturing. The remaining sections of the chapter are arranged in the following way. Sections "Classification of AM Techniques" and "Materials Used in Different AM Techniques" describe the classification and materials used in different AM techniques, respectively, and section "Applications of AM" discusses its application. Section "Challenges and Future Scope in AM" gives an overview of the challenges and future scope of AM. Eventually, section "Conclusion" will present the conclusion.

Classification of AM Techniques

A wide range of 3D printing technologies has been developed, each with its own set of capabilities. As per the American Society for Testing and Materials (ASTM), AM is primarily divided into seven classes, viz., material jetting (MJ), binding jetting (BJ), powder bed fusion (PBF), material extrusion (ME), sheet lamination (SL), vat photopolymerization (VP), and direct energy deposition (DED) (Guo and Leu 2013) as shown in Fig. 4.

Fused deposition modeling (FDM), selective laser sintering (SLS), laminated object manufacture (LOM), selective laser melting (SLM), and stereolithography (SLA) are among the most often employed technologies.

These techniques of 3D printing have been discussed briefly including their merits and shortcomings in the subsequent subsections.

Fig. 4 Types of AM techniques

Vat Photopolymerization

Charles Hull from 3D Systems Corporation introduced the first commercialized AM technology in 1984, employing the stereolithography process (SLA), in which an ultraviolet laser was used to precisely polymerize ultraviolet curable resins to form a solidified material layer (Cooper 2001; Noorani 2006). After that, the layers are cured one by one until the component is completed. SLA builds parts at a relatively faster rate, and the minimum layer thickness is determined by the curing depth (Pham and Ji 2000). The disadvantages of this approach are process defects caused by over curing, scanning line shape, and expensive supply and material prices (Wong and Hernandez 2012). This technology is fundamentally confined to photopolymers because it relies on photopolymerization.

Material Extrusion

Scott Crump, a co-founder of Stratasys Limited, devised an AM technique in 1988 that produced layers by physically extruding molten thermoplastic material such as polylactic acid (PLA) or acrylonitrile butadiene styrene (ABS) upon a substrate (Williams et al. 2011). Fused deposition modeling (FDM) is a technology that needs generally high working temperatures and produces prints with extreme porosities. However, among the DIY crowds, affordable and flexible extrusion technologies are gaining appeal.

Binder Jetting

Binder jetting technology was marketed by Z Corporation and ExOne after it was investigated at the Massachusetts Institute of Technology (MIT) (Li et al. 2020). It is a type of inkjet printing that has been modified (Gibson et al. 2020). This procedure employs an inkjet to bind the components rather than lasers. It uses inkjet 2D printing technology and advances in layers to build 3D products. The liquid binder is accurately sprayed with the aid of a printhead that moves on two axes. This method also starts with the creation of a three-dimensional model and then transferring it to the printer software, just like any other 3D printing process (Chen and Zhao 2016). Because printing necessitates constant supply, a dispenser provides such supply by storing powder that will be utilized. The printing head adheres the binder according to the requirements after applying a powder layer of varied thicknesses. The binder-containing solvent is dried using fluorescent or electric lamps before proceeding to the next layer. The powder bed is then dropped, and a new powder sheet is deposited.

Powder Bed Fusion

The powder bed fusion process employs a beam of energy, namely, a laser beam or an electron beam so that the powder bed can be melted sequentially. After the

scanning of one layer, the very next powder layer is disseminated using a rolling device. The following layer is scanned and merged with the one before it. Polyamides and polymer composites are commonly processed using polymer powder bed fusion, which was first devised by (Deckers et al. 2012) in the mid-1980s. The most common metal powder bed fusion processes are electron beam melting (EBM), direct metal laser sintering (DMLS), and selective laser melting (SLM). EOS GmbH (Germany) and Arcam AB (Sweden) created them in 1995 and have been commercially accessible since 2005. DMLS and SLM use a concentrated laser beam to heat and melt consecutive pre-alloyed, atomized powders (Yang et al. 2010; Kruth et al. 2005; Buchbinder et al. 2011; Murr et al. 2012) like 17-4 PH stainless steel, whereas EBM uses a scanning electron beam (Shellabear and Nyrhila 2004).

Direct Energy Deposition

This approach, unlike other 3D printing procedures, is used for repairing and maintenance rather than generating components. Direct energy deposition methods facilitate material creation by melting the material as it is deposited (Gibson et al. 2015). The deposition head, which is a combination of a source of energy and two power feed nozzles, is the apparatus utilized in this process. This procedure can use either metal powder or a thin wire. The to-be-fabricated part is placed on a platform, and in certain circumstances, tubing of inert gas is also provided. The deposition head that deposits the powder beam and the laser beam simultaneously is a four- or five-axis machine. The DED technique employs a centered heat source either an electron or laser beam, and the material is adhered layer by layer as it solidifies, resulting in fixing and creating new material objects on existing products.

Sheet Lamination

In 1986, Helisys Incorporation which is now known by the name of Cubic Technologies 10 created laminated object manufacturing (LOM) technologies, which were patented in 1987.

The components produced have lower internal tension and fragility, excellent surface finish, and reduced process, material, and machine costs which are all benefits of LOM (Mueller and Kochan 1999). It is a method that uses plastic, paper, or metal laminates that have been successfully glued together and then a laser cutter to produce the required shape of the object or model. The procedure begins with a heated roller adhering a sheet to a substrate, next, using a laser or mechanical cutter to precisely cut the subsequent layers and bond in order. The finished layer platform moves lower, and the current sheet metal is rolled into place, while the platform returns to its original position to collect the next layer. The procedure should be followed until the prototype is completed.

Material Jetting

Material jetting AM techniques use drop-on-demand inkjetting to deposit wax or photopolymer droplets directly onto a substrate, which is similar to the process of inkjet printing where the ink droplets are transferred from a fluid channel to a paper's substrate in a drop-by-drop way (Le 1998; Calvert 2001; deGans et al. 2004; Sheoran and Kumar 2020).

Printing Characteristics of AM Techniques

The characteristics of printing a 3D part that are important to consider when choosing an AM technique and a construction pattern have been described in this division. These characteristics as well as other aspects might affect the printed part's quality.

Time of Build

The build time for a specific object or assembly is generally determined by the printer's speed of printing an object, the size of the object to be printed, the thickness of the layer, as well as the orientation of the build. The bigger the size of the part to be produced, the more amount of time it will take to print irrespective of the process adopted for the printing.

Resolution of Feature

The pattern of energy or material is principally responsible for feature resolution in AM systems.

Quality of Surface

The printed layer thickness determines the surface quality of a printed component.

Support and Anchors

Every AM method must have some way of sustaining the printed characteristics of the following layers to construct complicated geometries like undercuts, assemblies of printed components having moving parts, and overhangs (Mumtaz et al. 2011).

Post-Processing

Post-processing techniques that separate printed products with built-in support material are required. The procedures and easiness of uninstallation differ depending on the printing methods and building material used.

Materials Used in Different AM Techniques

The different methods of 3D printing have been discussed in the earlier section. After classification, the next thing that comes is the materials that are utilized in these methods, what sort of qualities they give, and for what processes and applications they are used. As a result, material features relevant to the 3D printing techniques have been explored in the following sections.

Figure 5 shows the material usage by 3D printer service providers in the industrial sector.

Polymers

Polymers are the most extensively utilized materials in the three-dimensional printing sectors because of their versatility and simplicity of application to varied 3D printing techniques. Reactive monomers, thermoplastic filaments, powder, and resin are all examples of polymers used in additive manufacturing. Many industries, including medical, architectural, toy manufacturing, and aerospace, have been exploring the capacity of using 3D printing of polymers and composites for some

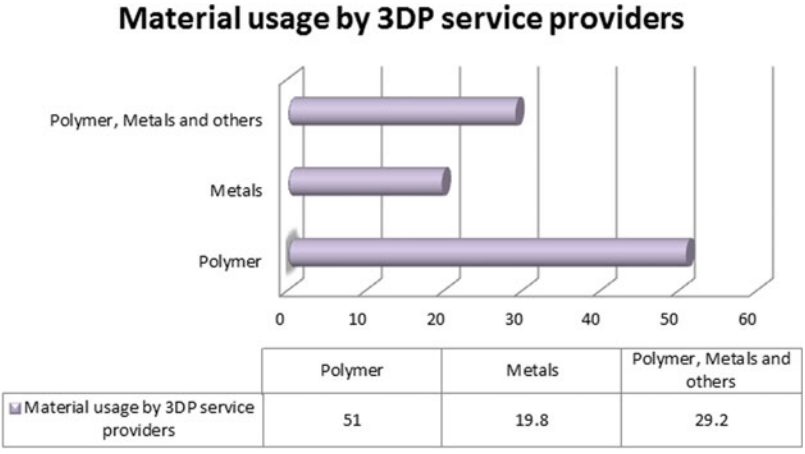

Fig. 5 Material usage by 3D printer service providers (Dizon et al. 2018; Wohlers 2017a)

years. The ability to tailor geometry with great accuracy is one of the advantages of producing composites with 3D printing. In addition, for tailored items, this procedure may be more economical than other classic formative processes like molding and extrusion (Wang et al. 2017a; Takezawa and Kobashi 2017).

In stereolithography (SLA) printing technology, photopolymer resins typically polymerize when exposed to UV light. As per Wohlers Associates' annual industry study, prototypes created utilizing photopolymers account for over half of the 3D printing market in manufacturing industries. The second leading class for 3D printing is plastic for selective laser sintering (SLS). Polyamides, thermoplastic, and elastomer polystyrene are just a few of the SLS polymers available (Ligon et al. 2017). A variety of 3D printing technologies can be utilized to process thermoplastic polymers, such as polylactic acid (PLA) (Postiglione et al. 2015), acrylonitrile butadiene styrene copolymers (ABS) (Zhuang et al. 2017), etc. The creation of 3D-printed scaffolds for tissue engineering is presently being done with PLA-based composites.

Stereolithography, selective laser sintering, fused deposition modeling, 3D bioprinting, and inkjet printing are some of the 3D printing processes available for manufacturing polymers and composites. The manufacturing of polymer composites and thermoplastics with low melting points is most typically done with FDM (Postiglione et al. 2015).

Over time, 3D printing of polymeric composites has evolved, allowing for the investigation and exploration of novel materials. The combination of 3D printing with a polymer matrix composite has a lot of potential in terms of industrial production with higher efficacy and mechanical performance.

Metals

Metal 3D printing/additive manufacturing has a lot of potential for expansion. Among the 49% of enterprises dealing with metal AM, the number of firms selling AM systems increased from 49 in 2014 to 97 in 2016 (Wohlers 2017b). This technique has mostly been employed in aerospace applications for research, prototyping, or sophisticated applications. It's also employed in medicinal, defense, and automotive applications (Wohlers 2017b). In comparison to traditional production processes, metal AM allows for a great deal of flexibility when it comes to creating complicated geometries with unique interconnections.

Melting metal feedstock which is in the form of wire/powder with a source of energy like an electron or laser beam is often the first step in 3D printing metals. In a pattern of layer, the molten metal is turned into a solid portion. The most widely used methods for printing a metal include direct energy deposition (DED) and powder bed fusion (PBF) although there are other approaches, such as direct metal writing (Chen et al. 2017), cold spraying (Sova et al. 2013), binder jetting (Bai and Williams 2015) friction stir welding (Sharma et al. 2017), and diode-based procedures that have recently been developed. These methods can provide greater precision or speed.

PBF AM methods may be used to create a variety of metallic materials, including titanium and its alloys, stainless and tool steels, nickel-based alloys, and certain aluminum alloys. Titanium and its alloys, particularly, are high-performance materials that are widely employed across a wide range of sectors (Sheydaeian and Toyserkani 2018). Based on traditional production processes, they are associated with the high cost of machining and a longer lead time. Reduction in cost of tooling, the ability to design and manufacture complex and lighter-weight parts, and multiple component unification that can remove the part assembly are among the advantages of metal additive manufacturing.

Concrete

The construction sector has embraced additive manufacturing technologies. Contour crafting, a process comparable to inkjet printing, has evolved as the primary approach for the additive production of architectural structures (Khoshnevis 2004). To extrude the paste of concrete, this technology requires bigger nozzles and higher pressure. Trowel-shaped equipment attached to the printer has been devised to achieve a smooth finish rather than a layer-by-layer look. In comparison to standard fiber-reinforced concrete, fiber reinforcement of concrete composites by 3D printing has the advantage of regulating the orientation of fiber in a printed structure. Flexural strength was greatly improved by allowing carbon fibers to be oriented freely along multiple printing pathways (Hambach and Volkmer 2017).

Ceramics

AM is now a popular technology for producing sophisticated ceramics in biomaterials and tissue engineering industry, such as bone and tooth scaffolds (Wen et al. 2017). Despite the precision of the printing, the key problems for 3D printing of ceramics are the layer-wise look and a restricted number of materials. Sintered ceramic components' post-processing to get the required shape is a time-consuming and expensive operation. Various approaches and materials have been explored in order to improve the mechanical properties of ceramic lattices produced by 3D printing when compared to traditional methods.

By adding CaSO4 with dextrin, Li et al. (2016) created a porous alumina ceramic whose flexural strength came out to be higher. Maurath and Willenbacher (2017) revealed that by enhancing the rheology and uniformity of the ink-printing process and optimized sintering, a honeycomb structure with higher specific strength and greater dimensional stability may be created. Powder bed fusion (PBF), paste extrusion, inkjet, and stereolithography (SLA) are the most common methods for 3D printing ceramics. Inkjet is considered to be the most common technology for producing solid ceramic samples that don't require post-processing (Travitzky et al. 2014). Ceramic powders may also be 3D printed using selective laser sintering (SLS). Another method called selective laser gelation (SLG), which combines SLS with the solgel method, also has been developed for printing ceramics (Liu et al. 2011).

Applications of AM

In recent years, the development of novel, sophisticated AM processes has accelerated, resulting in a growing number of commercial applications. AM, as opposed to subtractive manufacturing, is particularly well suited to making low-volume items, particularly components with complicated geometry.

In Construction

In recent years, automated building construction using 3D printing technology has gotten a lot of attention. It allows for significant time and manpower savings during construction (Wu et al. 2016). It has the potential to transform the building sector and make development on the moon easier for astronauts (Labeaga-Martínez et al. 2017). In the sector of construction, 3D printing can be used in areas with limits, such as geometric complications and hollow constructions. In 2014, the FDM process was used to create the first 3D-printed residential construction in Amsterdam (shown in Fig. 1) (3D Print Canal House – DUS Architects). The project was initiated by DUS Architects, who intended to demonstrate the printer's mobility with low material waste and transportation expenses, opening the way for it to enter the construction industry (Fig. 6).

Lim et al. represented three large-scale 3D printing procedures for the construction sector, each of which differs in terms of materials used and application process. The production procedure for D-shape and concrete printings is normally done

Fig. 6 First 3D-printed house made by DUS Architects (3D Print Canal House – DUS Architects)

off-site and is frame-mounted and gantry-based. On the other hand, the robot or crane-mounted contour crafting technique can be used for on-site applications (Lim et al. 2012). Because the raw materials and manufacturing technique used in 3D printing differ from traditional building methods, competent individuals with the capacity to combine robotic and civil work remain a challenge for AM in the construction industry. Although additive manufacturing (AM) is still in its early stages in the construction business, it is clear that it has the potential to improve traditional construction methods. A new study into and development of this technology is predicted, to result in more opportunities and problems in the construction sector.

Automotive Industry

The automobile industry has adopted AM technology as a vital tool in the development and design of automobile parts because it has shortened the development process and reduced manufacturing and product costs. Small amounts of structural and functional pieces for luxury, low-volume autos, such as driveshafts, gearbox parts, engine exhausts, and brake systems, have also been created utilizing AM technologies. Unlike passenger automobiles, racing vehicles are often made of lightweight metals (such as titanium) with very complicated construction and limited manufacturing numbers. AM methods have also been successfully used to create functioning components for racing automobiles by companies and research organizations. CRP Technology (Italy) has successfully developed and produced many components for motorsports using AM processes such as SLM, SLS, and EBM. F1 gears (titanium), MotoGP 250R air boxes, motorcycle dashboards and supports, F1 suspension systems, camshaft covers for MotoGP engines, reed valves, and other parts are among the items manufactured. Vilaro et al. created a water pump for racing vehicles by using SLM utilizing aluminum alloy (AlSi10Mg). Their tests revealed that the components generated had mechanical qualities that are comparable to heat-treated AlSi10Mg (Vilaro et al. 2008). Using LENS, Optomec created Ti6Al4V components for the Red Bull racing vehicle, including suspension mounting brackets and driveshaft spiders, which results in a $>$ 90% material reduction, as well as considerable time and cost-savings (Optomec). Concept Laser used SLM to make a variety of steel and aluminum automotive parts, including wheel suspensions, engine blocks, oil pump housings, exhaust manifolds, and valve blocks. Many studies have shown that AM can build complex components for the automotive sector utilizing casting patterns, molds, and cores created by SLA, SLS, 3DP, FDM, and LOM.

Aerospace Industry

Complex geometries are common in aerospace components, which are typically comprised of sophisticated materials such as nickel superalloys, titanium alloys,

special steels, ultrahigh-strength steels, and high-temperature ceramics, which are difficult, expensive, and time-consuming. For small quantities, AM is more cost-effective than traditional approaches since it doesn't need costly types of equipment like molds or dies. Concept Laser used SLM to create an engine housing. In the aircraft sector, AM-built plastic components, like vents and ducts, also have been employed. In the meantime, flame-resistant polymers, like PEEK, were developed for AM techniques to suit aircraft standards. Aeroengine components, turbine blades, and heat exchangers are examples of metallic and nonmetallic items that may be made or repaired using AM. For quick prototyping of components and manufacturing fixtures and interiors composed of ceramics, plastics, and composite materials, nonmetal AM technologies such as stereolithography, multi-jet modeling, and FDM are employed. In conjunction with several aerospace firms such as Piper Aircraft, Bell Helicopter, and NASA, Stratasys embraced FDM for fast prototyping, manufacturing tooling, and part production. NASA, for example, used Stratasys FDM technology to print 70 components of the Mars rover, resulting in a lightweight and sturdy construction (Stratasys|3D Printing Solutions for Aerospace). Another aerospace use of AM is the creation of wind tunnel testing models for airplanes, missiles, airfoils, and other designs to evaluate their aerodynamic properties. The use of additive manufacturing technology reduces the time and cost of producing these models, which typically have complex geometry. Daneshmand et al., for example, used SLS to construct a wing-body-tail launch vehicle configuration model out of glass-reinforced nylon (Daneshmand et al. 2006). In addition to directly producing functional components for aerospace applications, AM methods are also utilized to repair aircraft engine parts like compressors, housing parts, turbine and combustor castings, and blades to minimize the cost and prolong the lifetime of such parts. LENS has been shown to effectively repair elements used in gas turbine engines such as vanes, stators, seals, and rotors, as well as geometrically difficult parts including airfoils, ducts, and diffusers, according to Optomec. As a result, AM technology is particularly well suited to aircraft applications.

In the Medical Field

Since every individual is unique in the medical arena, AM offers a lot of promise for tailored and customized medicinal applications. There are several fields such as bio-fabrication, dentistry, surgery, and pharmaceutics in which there are a huge number of applications of AM.

Bio-Fabrication

The production of vascularized organs is a major problem in bio-fabrication (Murphy and Atala 2014). For large organs to maintain their metabolism, they require blood vessels. Therefore, strategies for creating complicated vascular networks and innervation will be required. 3D bioprinting of blood arteries and efficient vascularized tissue had made significant development recently (Zhang et al. 2017). The existing limitations of printing at fine resolutions at a microscale, along with the

absence of mechanical qualities for vascularized tissues, indicate that further study and progress are needed in this field to manufacture 3D-printed tissues. However, breakthroughs in adaptable bio-inks highlight additive manufacturing's promise in tissue engineering of delicate human components (Lee et al. 2010). Bioreactors are being investigated in conjunction with substances that stimulate angiogenesis and innervation to produce mature bioprinted components with the required properties. The use of a mix of bio-inks and advanced manufacturing methods to create human-scale tissue structures is yielding promising results (Kang et al. 2016). In situ bioprinting for tissue regeneration is also a cutting-edge concept that will transform the biomedical business.

In Surgery

In maxillofacial and oral surgery, AM is a widely used process. It is the procedure of connecting materials to items constructed from virtual model data (Levi et al. 2012). Oral and maxillofacial surgeons face a difficult task in managing missing craniofacial tissues due to congenital defects, trauma, or cancer therapy. Many approaches to treating such deformities have been developed, but autogenic bone transplants endure the gold standard in rehabilitative bone surgery (Petrovic et al. 2012). However, cell-based therapies combining adipose stem cells with osteoconductive polymers or scaffolds have emerged as a viable alternative to autogenous bone transplants. Customized 3D scaffolds that meet functional and cosmetic needs, provide appropriate blood supply, and meet the load-bearing necessities of the head are frequently required in such treatment procedures. Presently, such personalized 3D frameworks are being manufactured using 3D printing technology.

Pharmaceutics/Drug Delivery

The pharmaceutical industry's future is shifting away from mass manufacturing and toward tailored dosing for distinct groups of patients based on metabolic uniformity, except for direct energy deposition and sheet lamination deposition. Fused deposition modeling and inkjet printing are the most used advance manufacturing technologies in medicine delivery and pharmaceutics. Although these technologies give more precision at cheap cost, fast drug release, patient-personalized medications, and higher drug dose loading, poor productivity when compared to traditional medicine manufacturing processes. Furthermore, because 4D printing is a novel idea in the pharmaceutical business, only restricted technologies have been applied, such as direct ink writing for intravenous drug distribution systems (Durga Prasad Reddy and Sharma 2020). Similar 3D printing processes may find their way into the pharmaceutical business in the future. Another issue with AM in the pharmaceutical business is the scarcity of biomaterials. Sterilization procedures for polymer-based pharmaceuticals, for example, can't be strict or employ a temperature greater than the temperature of biopolymer's glass. Shape recovery hydrogels, on the other hand, showed significant promise for future medicinal uses. Finally, the most significant issue that

prevents AM technology from providing mass manufacturing of personalized medications is the inability to scale up innovations in this sector.

Dentistry

Patients in dentistry have a variety of needs, such as crowns, implants, and bridges. Dentistry can benefit from a variety of AM technologies. Dental prostheses are made using binding jet technology. The dried particle level, binder quantity, drying period, and powder spreading speed are all taken into account. The outcome demonstrates that it gives a precise implant at a reasonable cost with increased strength (Miyanaji et al. 2016). Patient-specific eruption guiding appliances are made via additive manufacturing, which improves patient comfort while also lowering the model's overall cost (Barone et al. 2018). Jet binding metallic powder is used to create partial dentures using 3D printing technology. To produce a 3D-printed model, micro-computed tomography is employed to scan the current framework. The results demonstrate that it can reach a density of more than 99% while controlling shrinking at a lower cost. It is extremely suitable for the creation of complex-shaped dental implants with precise dimensions (Kunrath 2020). The precision of dental restorations made utilizing additive manufacturing versus subtractive manufacturing methods such as wax or zirconia milling has yet to be determined. The results reveal that dental restorations made using additive manufacturing techniques are more accurate than those made using subtractive methods. AM is also a digital way of storing inventory in digital form, which has the added benefit of lowering inventory costs (Mostafaei et al. 2018). For teeth manufactured using the FDM process, an experimental investigation for dimensional correctness is performed. Three distinct layer thicknesses are employed to build a virtual 3D model and constructed model using the intraoral 3D scanner. GOM Inspect program evaluates the model's dimensional correctness, revealing that the lowering layer thickness readily reaches a high degree of precision (Bae et al. 2017). AM allows dentists to meet their unique needs in less time and at a lower cost (Milde et al. 2017).

Challenges and Future Scope in AM

Although 3D printing's benefits, like design freedom, customization, and the capacity to create complicated structures, some disadvantages would need more study and technical advancement. High prices, restricted uses in big constructions and mass manufacturing, poorer and anisotropic mechanical qualities, material limitations, and flaws are only a few of the disadvantages. In comparison to conventional processes like casting, fabrication, extrusion, or injection molding, AM of a part often takes longer. The long time taken and the greater cost of 3D printing are the key barriers to large-scale production of any repetitive components that can be readily produced using other traditional technologies in a fraction of the time and price. However, AM can be more economical when it comes to a personalized product with a complicated structure, such as a 3D-fabricated

scaffold for bone tissue creation (Duta and Caraiane 2017). One of the most significant disadvantages of 3D printing is the creation of voids between successive layers of materials. Because of the reduced interfacial adhesion between printed layers, the extra porosity induced by AM can be rather considerable, lowering mechanical performance (Hollister 2005). The production of voids is more prevalent in processes that employ filaments of materials, like FDM or contour crafting, and is regarded as one of the primary faults that result in weaker and anisotropic mechanical characteristics. After printing, this gap creation might cause delamination between layers (Wang et al. 2017b). Due to complete contact between succeeding layers, Paul et al. found that rectangular nozzles produce fewer voids than cylindrical nozzles. Rectangular nozzles, however, make 3D printing of complicated objects, particularly at joints, more challenging (Gibbons et al. 2010). One of the most difficult aspects of AM is anisotropic behavior. The microstructure of the elements within each layer differs from that at the borders between layers due to the nature of layer-wise printing. The addition of successive layers reheats the borders of preceding layers in metals and alloys 3D printed by heat fusion (SLS or SLM), resulting in distinct grain morphology and anisotropic behavior owing to thermal gradients (Paul et al. 2018). The laser beam's heat penetration into each layer is critical for managing the sintering process as well as minimizing anisotropic behavior. The main tool for designing an item that may be 3D printed is computer-aided design (CAD) software. Because of the constraints of additive manufacturing, the printed item may have a few faults that were not anticipated in the planned element. Solid geometry and boundaries are combined in the CAD system. To approximate the model, it usually uses tessellation ideas. Transferring CAD to a 3D-printed object, on the other hand, frequently results in mistakes and faults, especially on curved surfaces (Carroll et al. 2015). Although a very fine tessellation may be able to alleviate this problem to some extent, the computational processing and printing would be time-consuming and complex. As a result, post-processing (heating, laser, chemicals, or sanding) to remove these flaws is occasionally contemplated. Due to the nature of additive manufacturing, layer-by-layer appearance is also an issue. If the 3D-printed object is buried in the end use, such as scaffolds for tissue engineering, the look may not be significant. A flat surface, as opposed to a layer-by-layer look, is desirable in various applications such as structures, toys, and aircraft. Sintering, for example, is a physical or chemical post-processing procedure that can eliminate this fault, but it adds to the manufacturing time and expense (Oropallo and Piegl 2016).

Future Scope

Even while AM has advanced significantly in recent years, it is still not universally recognized by most sectors. The main goals for the next 5–10 years are to improve technology to the point where it can change people's thinking and acquire industry acceptance, as well as widen, develop, and find industrial uses that are only conceivable with AM methods. Novel AM methods, such as those for bioapplications

employing cells, biologics, or biomaterials as building blocks, as well as for micro/nano-engineering, must be explored and developed to widen and create new applications. AM technology and its applications will require a substantial amount of research and development in terms of materials, designs, new procedures as well as machines, process modeling and controlling, bio-additive manufacturing, and energy sustainability applications to reach these goals.

Conclusion

The key advantages of 3D printing include design flexibility, product customization, and the capacity to build complicated structures with minimal waste. A detailed assessment of 3D printing technologies, materials, and the present status of trending applications in a varied range of fields was conducted. Also, the primary challenges associated with 3D printing were explored. 3D printing has the ability to replace traditional technologies since it is a sustainable technology. In addition to being cost-effective, AM is also environmentally benign and hence can assist to minimize the negative environmental impacts of industrial growth. Based on studies, it can be concluded that a variety of 3D printing technologies have emerged, each of which is compatible with a particular material. Different advantages and cons are linked with each 3D printing method. Aside from being able to handle complicated and elaborate designs, 3D-printed items need relatively little posttreatment. FDM is the most prevalent 3D printing process; however, it works best with polymeric materials. SLS and other powder-based methods have several drawbacks, including difficulties in manufacturing. Powder-based technologies like SLS encounter several challenges, including powder transport and storage.

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Additive Manufacturing Metrology

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Challenges

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Abstract

The potential of additive manufacturing (AM) to build products in a layer over layer basis from a digital three-dimensional model with tremendous variability in accordance with the criticality of design has propelled it to the forefront in the manufacturing technology sector. Furthermore, AM requires no extra tooling equipment and can make parts with little to almost no material wastage. Even after having these technological benefits, AM has still yet to realize its full potentiality, owing to a dearth of proper knowledge of all AM procedures and coordinated efforts in the area of standardizing, metrology (measurement science), qualification, as well as certification. Consequently, AM manufactures products with greater complexity and characteristics but with lower dimensional accuracy, precision, requisite tolerances, and desired material characteristics. Method-specific standardized metrology and inspection procedures for AM parts, in particular, play a critical role in achieving the requisite quality and, as a result, simplifying the AM product certification process. This chapter offers a brief introduction to the metrology of AM, its need, and the challenges associated with it along with a detailed evaluation of general metrology methods as well as for AM-specific metrology methods.

Keywords

Additive manufacturing · Challenges · Inspection and qualification · Metrology

Introduction

Additive manufacturing (AM) also known as 3D printing, unlike various manufacturing techniques, has developed as a feasible technology for the creation of engineering components, as it is an additive process. Despite traditional manufacturing subtractive methods, where a tool has been used to remove undesirable material from a workpiece to form a product, AM is a method of fabricating components from a 3D model information by joining materials layer-by-layer (Standard 2012). When low manufacturing numbers, significant design complexity, and regular design changes are necessary, AM proves to be useful. Overcoming the design limits of traditional manufacturing processes, AM allows for the production of complicated parts. As AM does not require fixtures, tools, or coolants, it enables more efficient and quicker component manufacture. It also eliminates material waste, provides design freedom, and can print difficult structures with a greater resolution and surprising flexibility (Ngoa et al. 2018). In the aerospace, defense, automotive, biomedical, and construction industries, AM methods are widely employed (Pham and Dimov 2003; Campbell et al. 2018). AM processes have been mainly classified into seven categories – powder bed fusion (PBF), material extrusion, binder jetting, vat photopolymerization, sheet lamination, material jetting, and directed energy deposition (DED) (Astm 2015). Most commonly used 3D materials are polymers, metals, ceramics, and concrete. Although AM has numerous benefits, still AM

technology has not yet reached the point where it can be used in applications in the real world. There have been downsides and obstacles, along with advanced technological developments, that need to be investigated. The limitations of part size, and the construction of overhang surfaces, as well as high costs, anisotropic mechanical properties, low manufacturing efficiency, scalability limitations, top layers gaps, misalignment of layers, poor accuracy, over-extrusion, under-extrusion, elephant foot, warping, mass production, and material limitations, are all challenges that have been faced in AM (De Jong and De Bruijn 2013; Chen et al. 2017). Despite its enormous popularity, the lack of common metrology, inspection, and standards criteria has hampered AM's implementation (NIST 2012). Until the generated things were confirmed as appropriate and acceptable, AM's great potential would be intangible (Everton et al. 2016; Leach et al. 2019). This is one of the most significant obstacles to overcome until AM can be used extensively in industrial and military applications. Metal powder AM components are frequently characterized by high surface roughness values and might have undesirable material qualities. Furthermore, while a component cannot be made using subtractive methods without even a dimensional consistency system, it is currently unknown how to use tolerance concepts for AM parts. AM isn't any different than subtractive manufacturing in terms of metrology. However, existing AM equipment and procedures lack integrated metrology, which makes the commercialization of the produced items difficult (Berglund et al. 2018; Mani et al. 2017; Seifi et al. 2016). To supplement present AM technologies for manufacturing components with desired properties and accurate dimensions, new measurement techniques should be developed (Slotwinski et al. 2012; Stavroulakis and Leach 2016a). In this regard, metrology is a critical tool for analyzing and optimizing AM performance. Metrology in AM is crucial for identifying and then executing mitigation strategies to obtain dimensionally accurate products with the appropriate surface smoothness and material properties.

Background: Metrology – Measurement and Inspection Methods

Metrology is far more common than one may assume; practically everyone uses it unintentionally in their daily lives. Any conclusion that may be expressed numerically and in terms has been included. Metrology includes the establishment of units, the development of measurement techniques, the creation of artifacts that act as measuring norms to allow measurement traceability, and the analysis of uncertainties and accuracies (Raghavendra and Krishnamurthy 2013). Metrology and inspection are important and economical ways to improve AM quality. Testing would have to combine both developing and current methodologies for AM capabilities to keep growing. Different classifications for measuring and inspection methods include contact/contactless, destructive/nondestructive, in situ/ex situ, and real-time/off-time depending on the type of the method. Contactless, nondestructive, real-time, in situ measurements, as well as precise and time- and cost-effective technologies that are reliable and assist process control, are more suitable for AM. Different state-of-the-art metrology and inspection techniques are as follows:

Dimensional Metrology

Dimensional metrology deals with geometric characteristics, such as size, length, angle, shape, or coordinates. It is extremely crucial for controlling and measuring production processes when mechanical equipment interactions cause geometry deviations. Physical measuring equipment range from simple scales and rulers to advanced optical measuring and interferometry devices.

Linear Measurement

Linear measurement is performed with a variety of measuring equipment developed for industrial use. The majority of linear measuring tools are advanced versions of simple rulers and scales. Depending on the measurement needs, these are either precision or non-precision and graduated or non-graduated. Verniers, calipers, and micrometers are some of the most common linear measurement devices.

Angular Measurement

For alignment reasons, angular measurements are required not just to measure the angle, but also to determine straightness, flatness, and parallelism. The most commonly used instruments for this type of measurement are Vernier, micrometer, and protractor.

Comparators

Comparators use relative measuring principles instead of absolute measurement, with the sole distinction that seems to be that the sizes are assessed and compared to established measurements or standards. Electrical, mechanical, pneumatic, and mechanical-optical are the four major categories they fall under. Electrical, mechanical, pneumatic, or optical principles drive its fundamental working mechanism, as the name indicates.

Surface Metrology

Surface metrology tends to refer to the measurement of variation inside a surface or between two places on the same area. Surface qualities (such as topography, surface polish, or roughness) are crucial in the manufacturing industry. That's because the qualities of the mating surfaces have a considerable influence on the successful production of the entire system in terms of friction, corrosion, stress, dependability, aesthetic appearance, and so on when the components are put together. A careful examination of any surface shows surface abnormalities like waviness and roughness, which are usually related to the manufacturing process. Surface roughness measurements are mainly of two types: contact and contactless.

Coordinate Metrology

Coordinate metrology is the most sophisticated approach for accurately measuring three-dimensional (3D) coordinate values (Vora and Sanyal 2020). The information

regarding the coordinates of the specific position is critical for 3D measurements. Only coordinate metrology tools provide for the present capacity to produce things with the utmost accuracy (micro to nanoscale). Electronics, mechatronics, optics, mechanics, and computer science developments have all aided in the creation of coordinate metrology systems. These measurements give not only 3D surface data but also GD&T data, allowing for the rapid and precise detection of exterior (surface finish, etc.) and interior (porosity, etc.) faults in the 3D arena. Overall, coordinate metrology delivers great measurement precision and accuracy, although the equipment is more costly and measurements take longer.

Coordinate Measuring Machine

A coordinate measuring machine (CMM) is a device that comprises the contact probes that contact the test object's surface, a mechanical system that allows the probe to travel in three directions (X, Y, and Z), and a control system, either manual or automatic that keeps track of data of each axis' three-dimensional coordinates. Nowadays, different types of CMMs are available which have variations in mechanical structures, drive controllers, and configurations in the probe (Sładek 2016).

X-Ray Computed Tomography

X-ray computed tomography (CT) is mostly employed as a medical diagnostic instrument in the health industry. It has become increasingly popular for measuring the dimensions of engineering components because this process can measure the outer and inner geometry of a part without destroying it. It may be utilized to collect information on product internal structures for wall thickness examination, dimensional measurement, and component size and voids (Villarraga et al. 2015).

Magnetic Resonance Imaging

Paul C. Lauterbur devised magnetic resonance imaging (MRI) (Lauterbur 1973). Noncontact coordinate measuring and scanning systems are MRI systems. MRI systems are commonly employed in medical diagnostic procedures. The concept of nuclear magnetic resonance underpins MRI. A strong magnetic field region gradient is formed by the operation of large magnets, which impacts hydrogen atoms. A computer captures these changes to build a cross-sectional picture (Gao et al. 2015).

Geometrical Dimensioning and Tolerancing

The theory and measuring procedures of geometrical dimensioning and tolerancing (GD&T) are briefly discussed in this section. It is generally recognized that just supplying the tolerancing (plus/minus) and dimensioning of the design drawing is insufficient in production. To assess components and process capabilities, data on GD&T factors such as flatness, parallelism, straightness, roundness, squareness, cylindricity, and runout are required.

Measurement of Material Properties

Material qualities must be measured to evaluate part performance. Material characteristics can be evaluated through in situ metrology and mechanical testing. Mechanical testing is an important aspect of the examination process for determining the parts' functional and mechanical characteristics. Manufacturing process advancements provide several metrological difficulties, which may be efficiently addressed by employing in situ metrological technologies. Furthermore, the utilization of these approaches for real-time monitoring systems is one of the key advantages of in situ metrology.

In Situ Metrology

In measurements, in situ refers to taking a measurement using the system while maintaining the original test circumstances. In situ measurement is critical for ensuring the quality of produced parts. As machines become more complicated, new obstacles to standardizing procedures for modern in situ operations arise. In-process control and monitoring algorithms for the operation of the machine are always being improved.

Role of Metrology in AM

Metrology, or measurement science, comprises the equipment used to measure the dimensions, or even the measurements themselves and data processing (Musgraves et al. 2018). Some of the main reasons we invest so much effort into measuring what we produce are:

1. To determine if a part is suitable for its intended use; for example, would a shaft fit through a hole while still providing sufficient space for the passage of lubricating fluids?
2. To allow the gathering of complicated parts; without acknowledging the sizes of objects as well as their corresponding tolerances, fitting one component to the other becomes nearly impossible – this is especially true when assembling components produced by different companies or various parts of the same company.
3. Metrology is important for quality control because it allows us to try things like net-shape production and get it correctly the first time.
4. To provide control of a production process; there seems to be little control without measurement. For example, we may wish to adjust the speed of a cutting tool based on the surface texture it produces – we must measure the texture during the machining operation.
5. To reduce energy consumption; the fewer repetitive manufacturing processes necessary, the less energy is needed to make a product.

6. Customers' trust in a product ("customers" in this sense may be another manufacturing concern that has to utilize your components) – without tolerances and quality assurance, there will be a loss of faith in assembly procedures down the track.

Metrology is a very important and low-cost way to improve AM quality. Some of the applications are (1) validating whether components are within necessary tolerances, (2) characterizing various AM methods, and (3) developing standard techniques that assist decrease inspection costs and maximizing measurement accuracy. When different new materials and component geometries are continually introduced, form metrology, or the measurement and characterization of a component shape, is crucial for quality standards of AM components and AM machine makers to efficiently characterize and optimize their AM methods. Metrology for AM is critical in discovering and then implementing mitigation methods to achieve dimensionally correct products with the desired surface finish and material attributes. The present AM literature clearly states that metrology is required for many AM technologies, yet there are currently just a few solutions and guidelines available (Monzón et al. 2015; Seifi et al. 2016; Lane et al. 2016).

Challenges in Metrology of AM

As AM is comparable to conventional machining in terms of metrology. Therefore, the existing AM equipment and techniques lack integrated metrology, which makes commercializing the resulting parts very difficult. To provide the customer's faith in a product, the customer could be other manufacturing firms that have to utilize the product so without providing the quality control and tolerances, assembly operations will lack confidence. For instance, an aircraft company would not use a turbine blade created with additive manufacturing unless its metrology can provide a high level of assurance.

Advancements in metal AM equipment and procedures resulted in direct manufacturing of functional parts. This direct manufacturing of parts, particularly parts having complex structures, necessitated shared understanding between the one who designs, manufactures, and the end users concerning (a) the intent of the design, (b) specifications of material, (c) part testing and inspection essentials, and (d) procedures and part qualification and certification concerns. Numerous technical roadmaps have been developed in the past decade to highlight the major technological challenges to the mainstream adoption of AM (Bourell et al. 2009; NCMS Report 0199RE98 1998; NIST 2013).

All such roadmaps agree that AM-specific specification standards are needed to bring uniform and consistent methodologies for establishing requirements and testing the final components, by lowering the risk of implementing this technique. The first organization for developing standards to recognize the necessity and possibilities of AM standards was the American Society for Testing and Materials (ASTM International). The standards are defined at three levels according to the framework of ASTM and ISO: first is the basic standards (e.g., shared concepts and

requirements across all applications); second is standards for materials, processes, types of equipment, and finished items in general; and the last is standards for a particular material, procedure, or application. Based on their range, norms for geometrical specifications and measurements of AM components may correspond to all three levels of this system. However, there are currently just a handful of developed standards in this area. This shortfall is primarily due to the more immediate requirement to select materials and procedures for this developing technology. The most significant ISO and ASTM recommendations for geometric requirements and metrology standards produced so far are linked to designing for AM (ISO/-ASTM 52910 2017) and some basic concepts concerning quality attributes of AM components, with accompanying suitable test techniques (ISO 17296-3 2014). Part geometry and properties are mentioned specifically in the AM design standards (ISO/ASTM 52910 2017). The dimensions of the component, the surface texture, the maximum aspect ratio, the minimum spacing of feature, the minimum feature size, the maximum part size, and the maximum unsupported feature characteristics are all detailed. Part dimensions, surface texture, minimum feature size, maximum aspect ratio, minimum feature spacing, maximum part size, and maximum unsupported feature attributes are all stated in detail.

When it comes to the quality of AM products, a variety of measuring and inspection obstacles must be addressed. Among them, some are mentioned below.

Shape Complexity

Metal AM products have unique properties, such as complex shape and surface roughness, which make coordinate measurements of surface form and dimensions difficult. However, shape complexity is critical, particularly for interior holes as well as channels, porosity, and undercuts. They all can be printed readily by utilizing AM, but they cannot be measured or inspected easily. Consequently, product complexity has an inverse relationship with quality inspectability and measurement.

Effect of Post-Processing on Metrology and Inspection

The post-processing methods have an adverse effect on metrology and inspection. The post-processing of components in AM is unavoidable and preferred to fulfill the application's demands for metals, polymers, and composites. Shrinkage and cracks can also occur in AM components (Shahzad et al. 2013). Support structures are an important component of AM sections with overhanging components. Support structures must be eliminated before final use to avoid warpage in the AM component (Dimitrov et al. 2006). Removing the support structures leaves the component with a poor surface quality, which can be addressed with post-processing techniques (Svensson et al. 2010). Sintering-based AM techniques are an example of this approach. AM component shrinkage during solidification is a problem that may be mitigated by allowing for shrinking allowances. As a result, these post-processing procedures must be integrated into the component design to guarantee proper GD&T

(Islam et al. 2013; Seepersad et al. 2012). Before the slicing procedure, these size compensations are integrated into the CAD model (Limaye and Rosen 2006). This stresses the need for metrology in additive manufacturing.

Surface Texture Metrology

Metal AM surfaces are frequently complex, and uneven, due to which the texture measurements can reveal considerable variations in measuring technology. In the case of surface texture metrology, particularly the majority of measurement problems are connected to the topographies of the produced surfaces rather than material attributes. PBF procedure produces rougher surfaces than grounded surfaces, although not being as abrasive as other AM procedures like DED and FDM (Thompson et al. 2017). Because of vertical walls, many discontinuities, and reentrant characteristics, PBF surfaces pose considerable measuring issues (Townsend et al. 2016).

Powder Measurement

Besides, it is well recognized that environmental conditions such as relative humidity can affect powder measurements, particularly bulk density and flow measurements.

Internal Feature Metrology

As AM reaches the application arena of high-quality functional parts. The metrology issues associated with internal and hidden details are also rising (De Chiffre et al. 2014). Internal characteristics comprise internal cooling channels, which can already be expressed using old GD&T symbology, but not infill patterns and lattice structures, which are presently being developed using new GD&T symbology. Furthermore, though AM's process variability isn't as high as subtractive methods, inspection for internal material defects like cracks and pores is still required (Thompson et al. 2016).

Metrology and Inspection Methods for AM

AM can manufacture items with the utmost complex geometry and a wide range of materials, posing the same challenging metrology methods for measuring the performance of AM (Brown et al. 2013; Russell and Fielding 2014). Recent research has focused on not just the design and manufacturing of the component, but also on the requirement for metrology to assess the part's dimensions and functional quality.

Nondestructive Testing and Evaluation

The ubiquitous requirement for nondestructive testing (NDT) and evaluations (NDE) for AM is generally established based on the outcomes of numerous

conferences, meetings, and journal articles supplied by academia, industry, and government (Waller 2018). According to studies, AM technology is capable of creating a large variety of parts when especially in comparison to the component quantity, and it is recommended to NDT and NDE techniques to reuse the tried components. To develop real-time process monitoring, measurement, and control of AM technology, the National Aeronautics and Space Administration (NASA) and National Institute of Standards and Technology (NIST) have advised employing NDT/NDE techniques for AM parts and artifacts using in situ, contactless, and real-time metrological instruments for dimensional and materials characteristic measurement (Waller et al. 2014). Owing to the difficulty of the dynamic of AM techniques and the absence of formal predictive methods required for process control, some commercial AM systems are not provided with the in situ process and attribute evaluation with closed-loop process control systems. Internal flaws and surface texture monitoring in AM-specific products are currently the subjects of less investigation (Stavroulakis and Leach 2016b). It is suggested to check the industry criteria for component dimensions and measurement concerns before deciding on the optimal inspection technique for AM. Most of the inspection methods used for AM are nondestructive methods. The most commonly used NDTs for inspection purposes of AM are as follows:

Computed Tomography

The usefulness and feasibility of employing computed tomography (CT) (Ferrucci et al. 2015; Carmignato et al. 2014; Ontiveros et al. 2012) to both subjectively and statistically analyze AM-generated components has recently aroused attention. Using CT for the metrological assessment of AM components and artifacts has been a focus of study in recent years. For dimensional metrology, CT may be utilized to give information on the interior structures of products (Saint John et al. 2014). As a nondestructive technology for obtaining structural characterization of both internal and exterior geometric shapes of AM components, CT offers various advantages, and it is sometimes the sole feasible choice for extracting component dimensions of internal or hidden characteristics (De Chiffre et al. 2014; Villarraga-Gómez et al. 2014). As a result, CT technology is a partner in the "3D printing revolution" for AM product research and inspection. CT seems to have been beneficial in real-time process monitoring, in situ in AM.

Coordinate Measuring Machine

Coordinate measuring machines (CMMs) are commonly used to gather and assess three-dimensional metrological geometry data on a product. With sub-micrometer precision, they can determine the dimensions of spatial locations on the sample surface (Park et al. 2006). CMMs are commonly utilized as semiautomated to fully automated inspection techniques that are best suited for the production environment. CMMs are increasingly being utilized to help in AM inspection. Controlling both geometry and characteristic fluctuation is difficult with complicated components produced by AM machines. Tactile CMMs have been utilized in a variety of industrial applications for decades and can determine form and measures with

more precision than noncontact CMSs. Tactile CMM is contact equipment that takes many points with a probe to measure the surface of an element. An operator can manage the probe's position manually or automatically using a computer (Kupriyanov 2018). They are, however, slower and frequently sample a smaller number of spots on the measured area. Principles and features of tactile probing methods are presented elsewhere (Weckenmann et al. 2004), and interest in this area has dwindled since then. Metal AM methods and associated error factors are frequently studied using tactile CMMs as reference measurements. An optical CMM is a noncontact equipment that uses pictures to do measurements, relatively similar to measuring microscopes and optical comparators. Depending on whether or not they employ their means of illumination in contrast to ambient light, optical coordinate measurement techniques may be divided into two categories: active and passive (Carmignato et al. 2010). Active sensors, on the other hand, give superior measurement uncertainty and performance (Se and Pears 2012). Passive sensors are frequently less expensive, lighter, and more compact. As a result, active optical sensors are often the most accurate optical sensors for monitoring AM metal components. Both give high precision, but only optical CMM is used for smaller characteristics. Both give data of few points, are expensive, are slow, and also require post-processing.

Structured Light Testing

Controlling both geometric and characteristic variation is difficult with complicated products manufactured by AM methods. Laser scanning and structured light are optics-based methods for evaluating the surface of an AM component, along with dimensional analysis and the identification of residual stresses caused by warping (Mandache 2019). Structure light testing (ST) approaches enable real-time imaging performance and are widely employed in a variety of 3D imaging applications (Nguyen et al. 2015).

Ultrasonic Testing

Ultrasonic testing (UT) is an NDE method for identifying and analyzing surface and internal issues. UT can detect flaws as deep as several meters from most metals, although the component thickness is a severe constraint for other NDT techniques (Cerniglia et al. 2015). In AM, ultrasonic testing (UT) can identify voids or inadequate deposition layers. It is one of the most capable NDT technologies for industrial component testing (Fathi-Haftshejani and Honarvar 2019; Shakibi et al. 2012). Many new sophisticated ultrasonic methods have been introduced and deployed in a variety of industrial applications in recent years. It is a nonhazardous process and can also detect deep flaws. But it requires experienced technicians, and the surface must be smooth and accessible.

Interferometry

Interferometry is an NDT, contactless method which has been accomplished for surface topology measurements. Surface shape interferometric measurements are comparative measurements in which the form of a defined surface is compared to

that of an unknown surface. The difference is shown as a sequence of interference fringes on the surface (Abdelsalam and Yao 2017). Interferometry can measure smooth surfaces and provide high contrast and resolution. It has disadvantages like high cost and causes errors while measuring rough surfaces.

Scanning Force Microscopy

Scanning force microscopy (SFM), also called atomic force microscopy (AFM), is a kind of scanning probe microscopy (SPM) that has attained nanoscale resolution, which is 1000 times better than the optical diffraction limit. It makes a 3D surface profile and also works in both liquid and ambient air environments. It has a higher resolution than scanning electron microscopy (SEM).

Optical Profilometry

Optical profilometry is a surface metrology technology that is fast, nondestructive, and contactless (Fainman et al. 1982). An optical profiler is a microscope that divides light from a bulb into two paths using a beam splitter. In one direction, the beam is guided at the region under test, while in the other, it is directed toward a reference mirror. The two surface reflections are merged and then projected upon an array detector. When the path difference between the recombined beams is smaller than a few light waves, interference can occur. This interference provides information on the test surface's surface contours. It is an automated process and easy to use but has a limited surface slope.

Neutron Diffraction

Neutron diffraction is the application of neutron scattering to determine a material's atomic and magnetic structure. A specimen is immersed in a stream of hot or cold neutrons, which produces a diffracted that gives structural information about the material. It provides atomic structural information with great precision. Because of the irregularities in the neutron beam, the measurement accuracy may suffer.

Laser-Ultrasonic Testing

Laser ultrasonics is a cutting-edge NDT approach in which ultrasonic waves are generated and then measured in a material using lasers. When a laser beam is directed onto a sample surface, a thermal shock occurs, enabling the sample surface to quickly expand. As a result of the rapid expansion, the test piece generates ultrasonic waves. Generally, LU is performed in one of two ways: ablation or thermoelastic (Scruby and Drain 2019). Laser-ultrasonic testing can create ultrasound in objects without contacting them, has a tiny footprint that allows it to be deployed to asymmetrical shapes, and uses fiber optics to get access to inaccessible regions. It is a contactless method and remote examination of samples at high temperatures, such as during welding with limited access, is possible. The footprint is small and customizable. Inspection of tiny and complicated geometries is possible. It can detect flaws that are undetectable by ultrasonic testing. It has few disadvantages as efficiency is a function of material absorption qualities, which is somewhat costly and necessitates preventative precautions against probable laser risks.

X-Ray Computed Tomography

When it comes to interior geometry, X-ray computed tomography (XCT) is the sole viable option for product inspection in AM (Hiller and Hornberger 2016; Villarraga-Gómez et al. 2018). It employs X-rays to generate the virtual 3D representation of the inspected object without damaging it. Due to various benefits of tactile and optical measuring technologies, XCT is widely being employed to coordinate dimensions of AM components (Thompson et al. 2017). Furthermore, XCT may give surface texture measures, internal feature geometry assessment, and material quality inspection all at the same time. Different research areas are presently available in metrology to evaluate XCT as a referencing device for quality assurance of AM components. It provides information on interior flaws that other AM approaches cannot correctly deliver. If a part has complicated free-form geometries, high-roughness surfaces, difficult-to-access areas, or components with varied optical and surface qualities, XCT can benefit (Sbettega et al. 2018). A vast variety of influencing variables increases the uncertainty in XCT coordinate determinations. When scanning metal components, the metal and its penetrating thickness, which affects X-ray attenuation directly, have a significant impact on dimensions and form measurements (Carmignato et al. 2018). It is somewhat costly, and operational safeguards are necessary.

Conclusion

In real-life applications, AM technology shows great promise and has the potential to transform design, fabrication, logistics, maintenance, and acquisitions. Nevertheless, there are still a number of obstacles to conquer before AM becomes a viable tool in the industry's tool kit. "If you can't measure it, you can't enhance it," as stated by Lord Kelvin, and this simple phrase encompasses a tremendous discourse on why measuring is important and serves as a necessary component of the manufacturing system. That is why as AM progresses, the only strategy to assure that this new technique emerges as a viable production capability is to prioritize the development of suitable measurement methods and calibration regimens.

The present state of metrology for AM has been examined in the chapter, as well as its requirements. Present AM specifying standards and the need for wide usage of AM procedure have considerable gaps. Due to a variety of factors, including the existence of internal deformities and complicated surface topography, additively made parts are characterized by metrological problems and complications. Specific properties of metal AM products, such as complicated shapes and surface characteristics, with typically significant roughness, make coordinate measurements of surface form and dimensions challenging.

A broad analysis of general metrology and in situ real-time inspection techniques utilized in traditional production methods has been discussed. In order to use the stated approaches for components generated by AM methods, a full description of metrology and in situ real-time inspection methods is offered.

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Metrological Assessments in Additive Manufacturing

48

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Abstract

Today, additive manufacturing (AM) is used in many industries, including aircraft, transportation, medicine, architecture, toys, arts & design, and construction. The 3D model is created in a computer-aided design (CAD) software, and this model is converted into a standard translating language (STL) and then transferred to a 3D printing machine, creating a physical model via a material joining process into a layer-by-layer pattern. AM is expected to have a tremendous impact on the industry and, eventually, every aspect of our life. To make the manufacturing revolution a reality, though, there needs to be an ongoing research effort into AM metrology. AM processes require the implementation of tolerance and quality control techniques, starting with offline metrology and moving toward closed-loop control using inline metrology. The several significant capabilities of AM are covered in this chapter, along with their metrological implications. Additionally, several significant challenges of AM for measurement systems are briefly explored.

Keywords

Additive Manufacturing · Metrology · Standardization · Measurement · Quality

Introduction

Additive manufacturing (AM) has proliferated over the past decades. The American Society for Testing and Materials F-42 classified the AM process into seven different processes based on the powder, liquid, and solid forms of material (as shown in Fig. 1). In terms of general operation, all AM technologies use the same base. The procedure begins with the 3D modeling, converting the CAD model into a standard

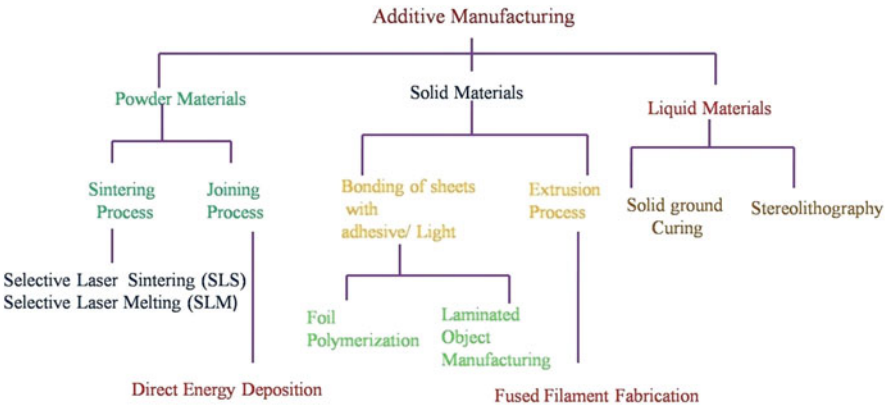

Fig. 1 Classification of AM processes (Gardan 2017)

translating language (STL) or AMF file format. Usually, STL file format is preferred as compared to AMF because it is heavier and requires more storage. This file was transferred to a 3D printing machine where slicing of the model was done based on printer slicing software; then ultimately final part was fabricated. It has the unique feasibility of design freedom, efficiently fabricating complex and customized parts without any constraints for mold, dies, or prerequisite tool preparation. The manufacturing industry has been attracted toward incorporating AM into the production system entirely or in hybrid mode. Despite many benefits, it is still not feasible to see AM-produced parts as finished, ready-to-use parts and for mass production, especially when it comes to metal additive manufactured parts. Metal AM part still has many discrepancies such as dimensional inaccuracy, surface roughness, warpage, porosity, and defects. Although industries have partially adopted the AM industry, however, it cannot replace the conventional manufacturing process entirely due to a lack of standardization, metrology (measurement science), dimensional inaccuracy, and qualification and certification issues in AM (Pant et al. 2021, 2022a; Tofail et al. 2018; Paul et al. 2018). Since the inception of manufacturing, metrology has been indispensable for the manufacturing industry because it provides valuable input for process control and post-process troubleshooting. It is strenuous to build up production processes and maintain production tolerances to reduce the number of scrap components without quick and precise metrology. In the field of AM, where many machines are employed on the same production floor to scale up output, metrology is important in every phase. To obtain precise tolerances on the products being created, each machine could be considered an individual production process or manufacturing line that requires process feedback. Although various authors have noted hurdles to the widespread adoption of AM in the industry, the following may be highlighted as examples of these challenges:

1. The link between material qualities and powder properties is not precise, and this must be recognized.
2. Machine-to-machine or day-to-day variability is caused by underlying variables that are not fully understood.
3. Traditional methods for material qualification and certification of AM materials are largely impracticable in terms of time, effort, and expense. Standardized procedures for qualifying and certifying AM components materials are absent or insufficient.
4. There is a lack of AM material data, including both generic and high-quality, pedigreed, reproducible data required for design-allowable databases.
5. Limited build volumes/part sizes, poor part accuracy, and surface finishing.
6. There are no established, uniform techniques for analyzing AM materials across laboratories.

AM is a multidisciplinary field that necessitates tight collaboration between information and communication technologies (ICT), design, material, and technology. The adoption of AM principles will make it possible to move from mass production to the future specialized, demand-driven, and environmentally friendly

manufacturing needs. This chapter discusses the metrology role of AM. The different methods needed during the preprocessing, in-process, and post-processing are discussed. It also discussed the present organization working on the standardization and developing a road map of measurement science for AM industry.

Metrology Need for Different Phases of AM

AM has been widely used to produce parts with internal cavities in free-form geometries, typically impractical or prohibitively expensive with traditional machining techniques. The use of modern technologies, which model, simulate, and store every step of the manufacturing process (i.e., the whole life cycle of a product, from its design to production, assembly, testing, and maintenance), increases productivity and quality. The function of metrology, and its application, has altered substantially due to countless advancement. The inconsistent geometric dimensioning and tolerancing (GD&T) of 3D printed objects is one issue preventing the widespread usage of AM industry. Traditional inspection techniques, such as tactile coordinate measuring machines (CMM) and contact measuring devices used for castings or forged components, are also utilized to measure system and material properties on a specific combination of additive machine and alloy. The limitations on its wider acceptability are variances in part quality concerning material qualities, dimensional tolerances, surface roughness, and defects. Today's process control methods, which rely on heuristics and experimental data, only slightly increase part quality. The measurement standard was required in the following areas (Javaid et al. 2022; Pant et al. 2022b).

Preprocess Measurements

Powder bed fusion (PBF) is the most common process in the metal AM process, where the material is used in powder form. Hence, it is critical to quantify and measure the powder properties to predict the final quality of components. The literature reported that if the two powder particles show the same behavior, their dynamic behavior is still a variation. It demonstrated how various batches of AM steel powders can operate differently in a metal AM system despite having similar particle size distributions and dramatically varying flow characteristics. Undoubtedly, the technologies employed to analyze AM powders are not sensitive enough or appropriate to do so accurately, given the importance of particle properties to AM process. The shear, dynamic, and bulk properties measurements should be included in best practices to understand powder flow behavior in AM systems fully. Hall flowmeter measurements frequently used to characterize metal powders also have limits. The final quality outcomes can be influenced by particle characteristics such as morphology, surface roughness, surface chemistry, size distribution, and environmental factors (Forster 2020; Slotwinski and Garboczi 2015; ASTM International 2010).

Process and Equipment Measurements

3D design is the foundation for ensuring that parts are accurate, fit correctly into a functioning assembly, and line up with other essential features and components. Specialized software has been created for design, simulation, feasibility, and costing solutions. Materials with various pricing ranges are required for fabrication, including composites, carbon, stainless steel, aluminum, brass, and copper. The 3D model is examined for scale and manufacturability before any material is created, which increases the process efficiency. The digital thread that connects computer-aided engineering (CAE) manufacturing and metrology solutions is centered on the CAD model. The activities of CAD-based inspection software are based on a 3D model. A metrology tool, such as a portable arm and integrated scanner, collects 3D point data or scanned point clouds from the manufactured part connected to the digital model. The scan set and primary component are compared in the inspection program to determine whether the part passes or fails after the data has been aligned to the CAD model using key reference points. The temperature and melt-pool shape were measured using the charged-coupled-device (CCD) camera. It resulted in mapped photos of the entire build area with more precise and localized melt-pool signals. Scientists could identify part deformation and overheat close to overhanging structures by measuring variations in the photo detector output (Dunbar et al. 2016).

Modeling and Simulation

Modeling and simulation are crucial for evaluating the characteristics of AM products before the start of actual production. It aids in the optimization of input variables to produce products with the desired features and attributes. Real-world metrology results assist in validating the outcomes of modeling and simulation. Usually, a range of physical factors controls the metal AM process, including powder layer creation, laser-powder particle interaction, and heat transfer, fluid dynamics of the melt pool, phase transitions, and microstructure growth. Any deviation in these systems could impact the final component's quality. Inconsistent product quality is a significant obstacle to the metal AM technology's broad adoption. Investigating the sources of variability, quantifying the uncertainties that arise at various stages of the process, and figuring out how much they have impact on crucial output values will all help to improve the quality of the final result. Computational models and simulations are potential techniques to comprehend the dynamics, complicated phenomena, and variability present in the process because undertaking uncertainty quantification via physical tests is frequently expensive (Seif et al. 2016; Moges et al. 2018).

Post-process Measurement

The post-process measurements are often focused on the part quality and based on dimensional accuracy, surface texture, and mechanical properties. The parts made with AM can potentially crack and shrink. The support structures are an essential

component of the overhanging structure AM components. Support structures must be taken down to prevent the warpage of the AM component before final use. When these support structures are removed, the component has a poor surface finish, which is improved by post-processing processes. Coordinate metrology of the exterior form is essential for metal AM product quality assurance and for providing input for AM process optimization. Metal AM items eventually exhibit form defects due to intricate interactions between the material and the energy source, warpage brought on by thermal gradients, and residual stress inside the part. Tactile CMMs are commonly used as reference measurements for the assessment of metal AM operations and related error sources. There is so much literature that has been reported on various materials to check the tensile, fatigue, density, porosity, and residual stress properties to correlate with AM machine parameters and compare with conventionally manufactured parts (Leach et al. 2019; Chen et al. 2022).

Qualification and Certification

An industrial insight analysis of the AM market estimated that the AM market would increase at an average annual rate of above 20% over the next 5 years. Due to the superior overall performance of toughness, strength, hardness, and wear and heat resistance of metallic products as compared to those of polymeric and ceramic counterparts, metal AM, which includes directed energy deposition (DED), powder bed fusion (PBF), binder jetting, and sheet lamination techniques, plays an increasingly important role in the AM industrial market. The primary purpose of certification is to satisfy a certifying authority or organization. Most of the time, certification and qualification are interchangeable because they attempt to fulfill predetermined requirements. However, certification emphasizes whether the authority meets deliverables (such as products). In contrast, qualification focuses more on whether a product is created or manufactured following industry criteria. According to the technical requirements and the specification of the product manufacturing process, the operation, technique, and process parameters should be evaluated. A series of production qualification for sample testing, consistency, and reliable production performance validation could be required throughout metal AM operations and post-processing. Qualification and certification organizations play a crucial role in driving the commercialization into different sectors by offering a range of services, including feasibility assessments, training, consulting, accreditation of quality management systems, testing, and audit, as well as surveys (Seifi et al. 2017; Chen et al. 2022) (Table 1).

Advanced Measurement Methods

The function of metrology within the manufacturing system has evolved dramatically due to advancement like intelligent multisensor systems, virtual metrology, and metrology-driven operations. In the past, measuring solutions were typically used for

Table 1 Several advanced techniques for quality control and inspection (Vora and Sanyal 2020)

Factor	Property	Measurement approach	Measurement technique
Material characterization	• Powder, wire filament liquid feed morphology	Off-line/at-line	X-ray photoelectron spectroscopy, energy-dispersive X-ray analysis, atomic force microscopy
In-process monitoring	• Dimensional inspection • Surface chemistry, macrostructure, and porosity microstructure • Internal surface inspection • Thermal management	Off-line/in-line On-line/in-line On-line/in-line On-line/in-line	CMM, gauges, light scanners, Raman spectroscopy, scanning electron microscope, transmission electron microscope Ultrasonic, radiography, tomography IR: infrared
Post-build processing	• External surface inspection • Internal stress • Grains, grain boundaries, cracks, and defects including dislocations • Surface finish	Off-line Off-line Off-line Off-line	Visual, penetrate, E-magnetic X-ray diffraction visible light microscopy Atomic force microscopy, Profilometry

quality checks at the end of verifying a product's compliance or as post-processing activities. The most recent industrial revolution has resulted in the availability in real time of a wide range of measurable data needed for management, monitoring, and diagnostic activities that can be used to speed up inspections and analysis. Feeler gauges, micrometers, slide calipers, checking fixtures, and a coordinate measuring machine (CMM) equipment are used for quality control (QC) and inspection. Measurement science has made considerable strides in the previous ten years and now provides the technology and software required for inspection. With an emphasis on accessibility and software modification, it has enabled the broad use of high-precision measurement for vehicles, aircraft, consumer goods, electronic enclosures, and other production applications. As mechanics, technicians, tool makers, and other nonprofessional metrologists employ portable metrology devices today, the person who 10 years ago used feeler gauges to check a part does so without hesitation. Modern manufacturers are discovering the value of feedback from metrology data during production, so they may change the procedure and stop spending time inspecting the items. Robots, laser trackers, sensors, software, and hardware, all of the technology needed to make these notions possible, are all readily available now. Manufacturers are placing metrology procedures on the shop floor directly adjacent to the production of parts at the OEM level (McCann et al. 2021; Santos et al. 2020). Robotic and metrology-based automated cell systems can take apart, load it into a scan cell, start the cycle, and produce a report at the conclusion. Many nondestructive (NDT) methods have been reported to evaluate the quality of additive manufactured parts, which are discussed below.

X-Ray Computed Tomography (XCT)

AM can fabricate complex and intricate parts; however, these parts require a delicate measurement method to measure the dimensional accuracy, surface texture, and internal and external features. Sometimes, it is difficult to measure such delicate components using conventional measurement methods. XCT is an optical measurement tool used to measure porosity, volume density, dimensional accuracy, and internal structures. In the XCT method, an X-ray cone beam scans an AM part repeatedly from various angles, enabling a 3D reconstruction of the part to expose the 3D complex volume without causing any damage. This method offers fantastic advantages over conventional inspection methods. It can observe the internal and exterior properties simultaneously. By quantitatively sampling and reconstructing the intricate lattice structures, flaws and porosity can be found, and their distribution can be determined (Matsushima et al. 2006; Garboczi and Bullard 2017).

In-Line Metrology

Modern in-line/on-line machine measuring facilities should adhere to all the features and observe the geometric related flaws and other internal features of the additively created components during the production. In-line metrology involves taking measurements while a process is in progress and has several benefits over conventional approaches, which treat inspection as a specific activity for manufacturing operations. Using in-line metrology concept, it becomes easier to take quick action and detect the flaws immediately. Previously, the additional setup and transport activities needed to move the items to an inspection department can cause delays. In-line inspection speeds up the process and enables the elimination of some processes. Because early problem detection of measuring items in-process can help to reduce rework and scrap (Everton et al. 2016; Fieber et al. 2020).

Robots

In today's workplaces, robots assist in creating and calibrating complex measuring devices; nevertheless, robots frequently rely on these same sophisticated measuring devices to complete their tasks. The requirements of robots have occasionally served as an inspiration for the creation of sensors, although robots perform poorly in poses, which restricts their usage for high-precision tasks, particularly for measurement applications. They provide six degrees of freedom (DoFs) of movement, a fast rate of execution, and a considerable deal of versatility. These qualities continue to be very desirable and may enable the 3D digitization of complicated parts produced by AM while utilizing the ongoing reorientation offered by robots. The surface quality of the AM parts is still challenging because it does not meet the requirements for final use. To resolve this problem, measuring the form and dimension variation is vital to thoughtfully plan the post-processing processes. Additionally, the advanced measurement phase should be wholly integrated into the production line near the AM

process. Robots enable quick access to the majority of orientations and positions required to digitize complicated parts (Tankova and da Silva 2020; Bhatt et al. 2020).

Laser Trackers

The development of novel optical measurement techniques, including laser trackers, structural light sensors, and optical sensors, is a result of the advancement of production processes. These techniques are utilized in manufacturing, metrology, and reverse engineering. Since laser sensors enable the generation of more than 100,000 points in a matter of seconds, they are distinguished by having a quick measuring time. When the proper scanning approach is used, they can perform measurements accurately. Therefore, the proper scanning strategy configuration might improve accuracy while utilizing measurement speed and the dense point cloud. In general, quality indicators often assessed by artefact measurements are used to indicate the measurement quality of a laser sensor. Most often, noise and trueness are used to describe the caliber of measurement data. A parameter that assesses the dispersion of the measured points concerning a related theoretical feature is referred to as noise. In contrast, trueness is defined as the difference between a reference distance and a measured distance between two measured characteristics (Macy 2015; Chesser et al. 2022).

Present Organization in AM

The rapid prototyping community first became interested in standards for AM in the 1990s, even though these technologies were not frequently used to create actual parts for direct production. At the time, rapid prototyping technology was still primarily focused on formal/functional prototypes. The National Institute of Standards and Technology (NIST) in the United States held a workshop for the industry on "Measurements and Standards Issues in Rapid Prototyping" on October 16–17, 1997. Numerous organizations and specialists have urged the establishment of particular standards for additive manufacturing over the past decades. However, the most significant advancements have been made in the last few years, mainly via the efforts of international organizations like the International Organization for Standardization (ISO) and the American Society for Testing and Materials (ASTM), with the assistance of technical groups and projects aimed at the standardization (Jurrens 1999; Musgraves et al. 2018). Standards are instruments that eventually increase the user confidence in a product. Different measurement techniques, standards, procedures, and terminology are applied correctly and consistently. The Society of Automotive Engineers (SAE), Aerospace Material Specifications (AMS), American Society of Mechanical Engineers (ASME), and American Welding Society (AWS) are examples of organizations working on standards for metal AM that cover materials, processes, post-processes, and inspections. The corresponding AM committees are ISO Technical Committee (TC) 261: Additive Manufacturing and ASTM F42: AM Technologies. Currently,

ASTM F42 is working on about 20 work items for proposed new standards, with ten of those being approved standards. Additionally, ASTM F42 has developed a strategic plan for standards creation that specifies the fields in which high-level, particular, and technical standards are required in the future (Mani et al. 2017; GovTrack HR 1988; U.S. Congress 1988). The National Science Foundation (NSF) and NIST both contribute to defining the technical difficulties preventing the more general deployment of AM technology. The academic community, business, and governmental groups all participated in both organizations. The NIST project proposed broad industry strategies for tackling those technological problems, particularly on demands related to metal-based AM systems.

Challenges

The widespread use of AM is hampered by problems with part quality, including dimensional inaccuracy and form flaws, undesirable porosity, layer delamination, and unclear or inadequate material characteristics. The setup of the AM process parameters, which is currently commonly done via a trial-and-error manner for each material, might be attributed to part quality difficulties. With AM, many new challenges have emerged from a metrological point of view:

1. However, there are not enough adequate and pertinent metrological instruments and methods. Due to that, AM has not yet established a sustainable quality assurance system.
2. The lack of material data still necessitates a deeper analysis of the materials' issues relating to AM-built items.
3. Internal faults can significantly impact the quality, mechanical properties, and safety of finished products such as powder agglomeration, balling, porosity, internal cracks, and thermal/internal stress. Defect inspection techniques are therefore essential for removing manufactured flaws and improving the surface quality and mechanical properties of AM components. To convey information more clearly and from different angles, three-dimensional multicamera modeling objects need to be investigated. It will increase the accuracy of defect identification.
4. AM processes must be controlled for tolerance and quality, first with offline metrology and then with closed-loop control using inline metrology. The advancement of AM also requires concurrent research activities and global standardization.

Conclusion

This chapter summarized and identified the measurement science requirements essential to real-time AM process control. The study is structured to show the measurement role during the different phases of AM to manufacture better quality

products. Metrological characterization is required for AM not just from a technological aspect but also due to the market's requirement for products with consistent and dependable performance. The future manufacturing industry is built on more innovative, faster, actionable metrology. The automated control systems used in factories with industrial robots and automated material handling are based on rapid feedback and verification. Manufacturers need to adapt their inspection processes to overcome physical barriers and close the gaps in the metrology information between different value chain stages to support autonomous digital production.

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Advances in Additive Manufacturing and Its Numerical Modelling

49

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Abstract

Additive manufacturing (AM), commonly known as 3D printing, is gaining popularity in academia and business due to its distinct benefits over old-fashioned subtractive manufacturing. However, its processing parameters are challenging to control since they can significantly influence the printed parts, microstructure, and subsequent product performance. Building a process-structure-property-performance (PSPP) connection for AM is a complex undertaking. Understanding the combined effects of strain, strain rate, and temperature is essential for understanding the crashworthiness of the AM components. This can be done using numerical-analytical models and topology optimization. This chapter reviews the progress made in using topological optimization and numerical-analytical models for several parts of the am whole chain, including

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model creation, in situ monitoring, and quality assessment. The current difficulties in using conventional numerical and analytical models to analyze AM are then discussed, along with possible remedies.

Keywords

Additive manufacturing · Modelling · Metrology · Stress · Strain · Topology

Introduction

In contrast to conventional manufacturing, which comprises subtractive manufacturing technologies and formative manufacturing procedures, additive manufacturing (AM) is the "process of combining materials to build things from 3D model data, usually layer upon layer" (ISO/ASTM 52900-15) (A Short History of 3D Printing). AM offers significant advantages over conventional manufacturing processes, such as the capacity to create intricate structures, drastically reduced resource usage, and quicker product development cycles. Since freeform manufacturing enables the construction of novel structures with greater strength and less weight than conventionally manufactured components, metal additive manufacturing (AM) has gained appeal. Powder-based metal additive manufacturing (AM) has advanced from quick prototyping to quick production during the past 20 years. The two main process types are powder bed fusion (PBF) and directed energy deposition (DED). Some powder bed AM processes go by historical or branded names like direct metal laser sintering (DMLS), selective laser melting (SLM), electron beam melting(EBM), selective laser sintering (SLS), etc. Directed energy deposition techniques are also referred to as laser-engineered net shaping (LENS), direct metal deposition (DMD), and laser-melted deposition (LMD). Some of the significant milestones for the development of AM are listed in Table 1.

The Additive Manufacturing File Format

The 3D file format known as AMF, or additive manufacturing file Format, is used to store and describe things that will be manufactured via additive manufacturing. The stereolithography (STL) file standard is what AM machines utilize to create the slices needed to make the part. As seen in Fig. 1, the STL format approximates the part surfaces using planar triangles. The component representation is erroneous due to this piecewise planar approximation, especially for strongly curved surfaces (Zhao and Luc 2000; Zhao et al. 2009; Pandey et al. 2003; Jamieson and Hacker 1995). The AMF format, which is based on XML, was created to include native support for file specifications including scale, geometry, lattices, color, material, and orientation.

The main body of the part can be colored in accordance with a function from the original design by using nested color assignments. Vertices, triangles, volumes, objects, or materials can be colored in red, green, or blue and have a certain amount

Table 1 Milestones in the development of additive manufacturing/3D printing

Additive manufacturing milestones	Year	Reference
The Battelle Memorial Institute made the first attempts to polymerize resin with two laser beams of different wavelengths in order to make solid objects	1960s	A Short History of 3D Printing; Wohlers and Gornet 2014
The application of stereolithography, a method that uses a laser to solidify tiny layers of UV light-sensitive liquid polymer, in commercial settings	1987	Mukhtarkhanov et al. 2020; Zhang et al. 2021
Together, 3D systems and Ciba-Geigy developed stereolithography materials and released the first-generation acrylate resins for commercialization	1988	Krishna et al. 2021
Materials for stereolithography under the Somos trademark were developed	1988	Mouzakis 2018
Technologies of stereolithography were commercialized by NTT data CMET in Japan	1988	Touri et al. 2019
Technologies of stereolithography were commercialized by Sony/D-MEC	1989	Wohlers and Gornet 2014; Mouzakis 2018; Min et al. 2018
The German company electro optical systems (EOS) sold its first stereolithography system	1990	Bose et al. 2019
The Mark 1000 SL system, developed by Quadrax, used visible light resin	1990	Butt and Shirvani 2018
Beginning to fused deposition modelling (FDM)	1991	Rezayat et al. 2015
SLS, or selective laser sintering, was developed by DTM, which is now a part of 3D systems	1992	Wang et al. 2020
Denken introduced a solid-state laser-based SL system	1993	Kruth et al. 1998
The first XB 5170 epoxy resin product was commercialized by 3D Systems and Ciba	1993	Waterman and Dickens 1994
A laser-sintering-based machine named EOSINT was commercialized by the German company EOS	1994	Shellabear and Nyrhilä 2004
First stereolithography machine to be commercialized	1995	Hopkinson and Dickens 2006
With the help of a technology that deposits wax material layer by layer, 3D systems sold its first 3D printer	1996	Vanderploeg et al. 2017
A method known as laser additive manufacturing (LAM) was developed by AeroMet	1997	Kobryn et al. 2006
Direct metal deposition was announced by precision optical manufacturing	2000	Smurov and Yakovlev 2004
Prodigy, developed by Stratasys, develops parts using ABS plastic and FDM technology	2000	Smurov and Yakovlev 2004
The EOSINT M 270 direct metal laser sintering system from EuroMold was introduced	2003	Esteve et al. 2017; Strauss 2013
Spectrum Z510, a color 3D printing machine released from Z Corp.	2005	Gao 2016
First SLS machine achieves viability in the marketplace. On-demand 3D-printed part industrialization is starting to make sense	2006	Lipson and Kurman 2013

(continued)

Table 1 (continued)

Additive manufacturing milestones	Year	Reference
With the introduction of the first prosthetic limb in medicine, 3D printing acquires more recognition	2008	Manero et al. 2019
The patents protecting the fused deposition modelling (FDM) printing method expired in 2009. This made it possible for new companies to begin making commercial FDM 3D printers	2009	Monclou Chaparro 2017
The development of the first printed aircraft and prototype car demonstrates the scalability and potential	2011	Molitch-Hou 2018
The specification for its additive manufacturing file formats was made available by the ASTM international committee F42 on additive manufacturing technologies	2011	Lee et al. 2019; Krueger 2017
SLS patent of Carl Deckard expires. The high-resolution, low-cost technology offers new potential for 3D printing businesses to grow. The first 3D printer with zero-G technology is sent to orbit by NASA	2014	Bayraktar 2022; Werkheiser et al. 2014
Introduction of 4D printing	2014	Tibbits 2014
The Swedish company Cellink releases the first commercial bio-ink that can be used to create cartilage tissue. This development benefits researchers worldwide as 3D bioprinting becomes more accessible	2015	Jose et al. 2016
Additive manufacturing for high-strength alloys	2016	Mukherjee et al. 2016
The first household to reside in a 3D-printed house. The house took 2 days to print and is now fully habitable	2018	Hossain et al. 2020
Self-healing elastomer materials with free-form designs produced using additive manufacturing	2019	Yu et al. 2019
World's first human cells-based 3D-printed heart created	2019	Noor et al. 2019
Introduction and implementation of aerial additive manufacturing with multiple autonomous robots	2022	Zhang et al. 2022

of transparency added to them. Because clear components can be produced by various AM methods, including vat photopolymerization, the transparency value may not be useful. Color values could be used in conjunction with other material-based factors to control an AM process in a variety of ways.

Regarding the picture data assigned to objects, the color assignment is inapplicable. The texture operator, however, can be used to accomplish this. First, a texture is assigned geometrically by scaling it to the feature, which ensures that each pixel applies to the object uniformly. These pixels will have intensity, and that intensity will be given a color. It should be noted that this is not a physical texture, such as ridges or dimples, but rather an image texturing process like computer graphics. It is possible to allocate different volumes to be made from various materials. Multiple

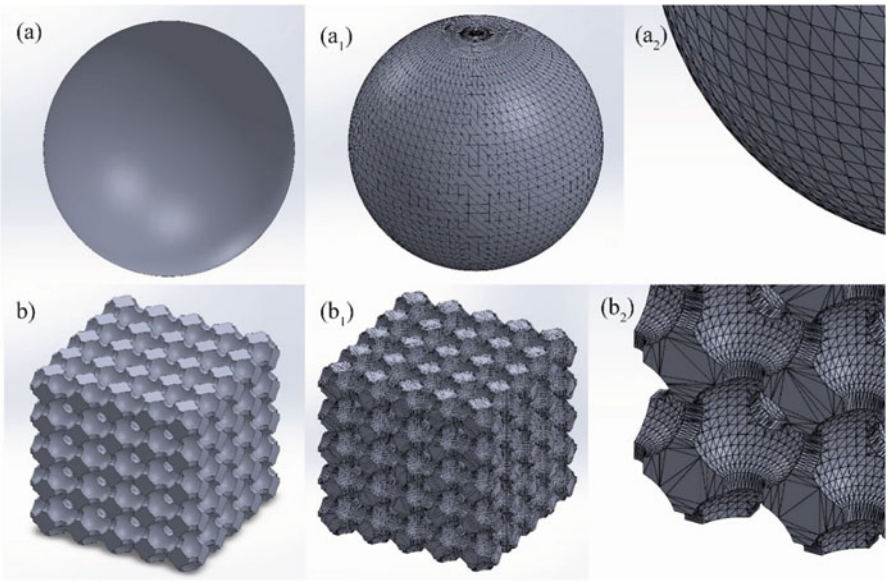

Fig. 1 CAD part (**a**, **b**) and STL file (**a1**, **a2**, **b1**, **b2**)

material parts can now be produced using the Connex machines from Stratasys and a few other extrusion-based systems. The operating system of the machine must now go through a laborious redefining process before designing parts. It is feasible to continue this from the design stage onward by having a material specification within AMF. The fundamental structure of the component to be manufactured can be altered using AMF operators.

The parts produced by AM machines using the STL file representation as a guide are inaccurate as a result of this approximation (Navangul 2011). By using smaller triangles and increasing the tessellation in the STL file, the approximation can be lessened. However, the number of triangles in the file grows exponentially as a result, leading to huge file sizes. Vertex translation algorithm (VTA)-based selective modification of the STL file depending on the error threshold of various surfaces has been suggested by Navangul et al. (2011). The amended file still employs planar facets even though this technique has been shown to be effective in decreasing the approximation error while regulating the growth in the number of facets. Consequently, there is a limit to how much better the tessellation may be made using this method.

The STL format has a lot of issues, despite being effective, as has already been mentioned. An alternative format will probably be needed as AM technologies advance to accommodate various materials, lattice structures, and textured surfaces. In May 2011, the ASTM Committee F42 on Additive Manufacturing Technologies published ASTM 2915–12 AMF Standard Specification for AMF Format 1.1, and ASTM also proposed a whole new additive manufacturing file (AMF) format.

(de Normalización 2013). Second-degree Hermite curves are used in this AMF format to depict curved triangles that split back into planar triangles by the AM machines as they read the AMF file to produce the slices.

Although it is still in its early stages of development, some commercial and beta-stage software has already adopted this file format. AMF, which is much more complicated than STL, intends to incorporate a wide range of new part descriptions that have impeded the advancement of current AM technology. These have the following characteristics: triangles that curve: in STL, the surface normal and the triangle vertices to which it is connected are on the same plane. The normal vector's starting point does not need to be in the same plane in AMF, though. If so, a curved triangle must be used in its place. This method of describing the triangles allows for far less triangle usage in a normal CAD model. This solves the issue of big STL files for high-resolution systems due to complicated geometry models. Since there is a limit to how much curvature may be applied, the curved triangle method is still only an approximation. However, there has been a significant improvement in overall accuracy in terms of cusp height deviation.

Santosh et al. (Allavarapu 2013) have developed a curved format for AM machines that piecewise approximate the CAD surfaces using modified rectangular biquadratic Bezier surfaces. Although the Bezier format can considerably reduce mistakes, cutting the Bezier patches to generate the contours cannot be done analytically, necessitating the use of numerical or optimization methods, which introduces flaws in the solution.

A new file format that uses curved Steiner patches rather than flat triangles for both approximating the part surfaces and creating the slices is introduced by Paul and Anand et al. in a research paper (Paul and Anand 2015). Bounded Roman surfaces called Steiner patches can be parametrically expressed by rational Bezier equations. This new Steiner file format will be more accurate than the conventional STL and AMF formats due to Steiner surfaces' greater order, which will also result in fewer geometric dimensioning and tolerancing (GD&T) mistakes in components produced by AM procedures. The slicing of the Steiner format can be done with relatively low processing complexity because the intersection of a plane and the Steiner patch is a closed-form mathematical solution. The Steiner representation is also generated using an error-based adaptive tessellation approach that minimizes the number of curved facets while enhancing the Steiner format's correctness. The adaptive Steiner, STL, and AMF format representations are used to digitally construct the test components, and the GD&T errors of the fabricated parts are calculated and compared. The findings show that, in comparison to the STL and AMF formats, the modified Steiner format can greatly reduce the chordal and profile defects. The aim of this chapter is to provide an overview of modelling and simulations. The modelling is organized as follows:

(a) *Heat source modelling*
(b) *Residual stresses and distortion*
(c) *Melt pool characteristics*

(d) *Surface characteristics modelling*
(e) *Microstructure modelling approaches*
(f) *Porosity modelling*
(g) *Material properties modelling*
(h) *Topology optimization*

Modelling Considerations in AM

To evaluate the build quality, process modelling offers first-hand data (such as temperature evolution and stress evolution). A strong computational method should contain both microscale and macroscale models due to the nature of layer-by-layer production. The microscale modelling objective is to investigate the basic operations of an AM process at short lengths and time scales, such as the interaction of an energy beam with a material in a microsecond, transient melting pool dynamics in a single scan, temperature evolution in a few scans, or the building of multiple layers. Macroscale modelling adopts the outcomes of microscale modelling to anticipate the surface integrity of the manufactured component, such as residual stresses, part deformation, and surface roughness. Utilizing multiscale modelling methods at the process level not only provides essential information for future microstructure prediction but also makes it possible to optimize the process and product design for higher component quality. This section describes the advancements and modern computing techniques in macroscale and microscale modelling. Metal AM processes have been facing various challenges in terms of quality-wise component production (Rezayat et al. 2015). These challenges primarily include the framing of defect remedies and their in situ/post-process minimization/mitigation.

Hit-and-trial experiments and numerical modelling are the two methodologies needed to address the physical phenomena involved in the occurrence, correction, and prevention of these problems. Therefore, to prevent this from happening, an analytical and numerical modelling technique is required. This approach offers a proper set of process parameters for component manufacture with the fewest possible flaws. Finite element, boundary element, and finite difference methods (FEMs, BEMs, and FDMs) are mesh-based numerical modelling techniques. Smoothed particle hydrodynamics (SPH) is a mesh-free numerical modelling technique (Dao and Lou 2021; Liu and Liu 2010).

The mesh-based finite element method (FEM) was developed by Turner et al. (1956). In every area of engineering, FEM-based numerical analysis has been a proven vital tool to tackle boundary difficulties, initial and eigenvalue problems (Hughes et al. 2014). To avoid wasting time and money on hit-and-trial testing, the usage of FEM has been expanded for modelling and parametric optimization (Mehboob et al. 2018). In light of this, FEM has emerged as the most popular modelling technique for predicting flaws, such as porosity, residual stress, and distortion. A few noteworthy evaluations in the areas of technologies, development, and long-term interests and qualities, defect, and problems in material processing and quality improvement.

Numerical Modelling in AM

The use of finite element analysis (FEA) techniques to forecast how various manufacturing processes will perform as process parameters, geometry, and/or material parameters change is becoming more and more common. To forecast the results of welding, forming, molding, casting, and other processes, manufacturers employ commercial software programs like SYSWELD, ANSYS, COMSOL, Mold Flow, and DEFORM (Vanderploeg et al. 2017). Predictive finite element methods have an especially difficult time simulating AM technology. For instance, utilizing physics-based FEA to accurately mimic the multiscale nature of metal powder bed fusion techniques like metal laser sintering and electron beam melting would take a very long time.

Due to the intrinsic multiscale nature of AM processes, it is necessary to use fine-scale finite element meshes with a size of 10 μm or less in order to fully represent the solidification physics surrounding the melt pool. Several academics are seeking for methods to establish assumptions that would allow them to copy and paste solutions from simplified geometries to construct a solution for big, complicated geometries using existing FEA tools. The advantage of this method is a quicker solution time, but for large, complex geometries, these predictions fall short of fully capturing the impacts of shifting scan patterns, intricate accumulations of residual stresses, and localized thermal properties.

Additionally, the simplified solutions may become invalid due to slight adjustments to the input conditions. Therefore, the objective of a predictive AM simulation tool is to create a simulation infrastructure that can swiftly build up a "new" response for every given shape, input condition, and scan pattern. In order to speed up FEA analysis for AM, researchers have recently started employing dynamic, multiscale moving meshes (Greco et al. 2015; Neugebauer et al. 2014; Luo and Zhao 2018). These multiscale simulations move along at speeds that are many times quicker than those of conventional FEA simulations. This section now moves on to overviews of modelling and simulation that take into account many aspects that affect the quality and use of parts made using additive manufacturing.

Heat Source Modelling

A number of process variables, including heat source power, deposition rate, thermal history, etc., affect how well AM components work. The numerical modelling reduces the failure rate of the designed structure, enhances the structure's quality, and aids in understanding the intricate phenomena that control the success of the AM. During metal AM procedures like selective laser melting (SLM) and directed energy deposition, the same high-energy heat source is used for material deposition. Around the melt pool, a significant temperature differential is produced by the high-energy concentric heat source. The manufactured structure of AM parts will

therefore always have residual stress and thermal distortion. Residual stresses can impair the designed structure's strength, fatigue life, and dimensional accuracy and are frequently linked to unexpected failures. This demonstrates that a significant component affecting the quality of the formed structure is the thermal history and temperature distribution during deposition.

The definition of the heat source is an essential component of thermal modelling. The melt pool's temperature distribution is determined by the thermal heat source. The volumetric heat source concept is frequently used in welding and additive manufacturing. The 1940s saw the beginning of research on the heat source model. A detailed characterization of the heat source is crucial to calculate the fusion zone (FZ), heat-affected zone (HAZ), temperature distribution along the deposition track, and peak temperature. For this model to precisely quantify the magnitude and distribution of the heat flux, it needs to know several crucial melt pool parameters.

Thermal Model

The melt pool generated by the laser heat source was defined using a transient heat input approach. The analysis is influenced by the activated parts at a particular spot in the model. For computations, the inactive elements are not considered. The controlling equation for the analysis of transient heat transfer is given by

$$\rho c \frac{\partial T}{\partial t}(x, y, z, t) = -\nabla . \vec{q}(x, y, z, t) + Q\,(x, y, z, t) \tag{1}$$

In the Eq. 1, ρ is the density, c specific heat capacity, T is the temperature, (x, y, z) are coordinates, t is the time, $\nabla$is special gradient, $\vec{q}$ is the heat flux, and Q is the heat generation.

Heat Source Models

The interaction between the arc and the metal material in a numerical simulation of the arc additive manufacturing process frequently corresponds to the heat source model. The numerical simulations in this research (Goldak et al. 1984) use the Goldak double ellipsoid model as the heat source model. Two semi-ellipsoids with the same minor axis but separate major axes are used to approximate this model, which accounts for the molten pool's actual geometry. Figure 2 displays the model's schematic diagram.

As per the Goldak double ellipsoidal model, the heat source model is expressed as:

$$Q = \frac{6\sqrt{3}}{\pi\sqrt{\pi}} \frac{\eta P f_i}{abc_i} \exp\left[-\left(\frac{3x^2}{a^2} + \frac{3y^2}{b^2} + \frac{3z^2}{c_i^2}\right), (i = 1, 2)\right] \tag{2}$$

where Q is the heat flux, P is the heat source power, η is the energy absorption rate, f_1 and f_2 are the energy distribution coefficients of the front and back ellipsoids, respectively, and a, b, c_1 and c_2 are the heat source shape parameters.

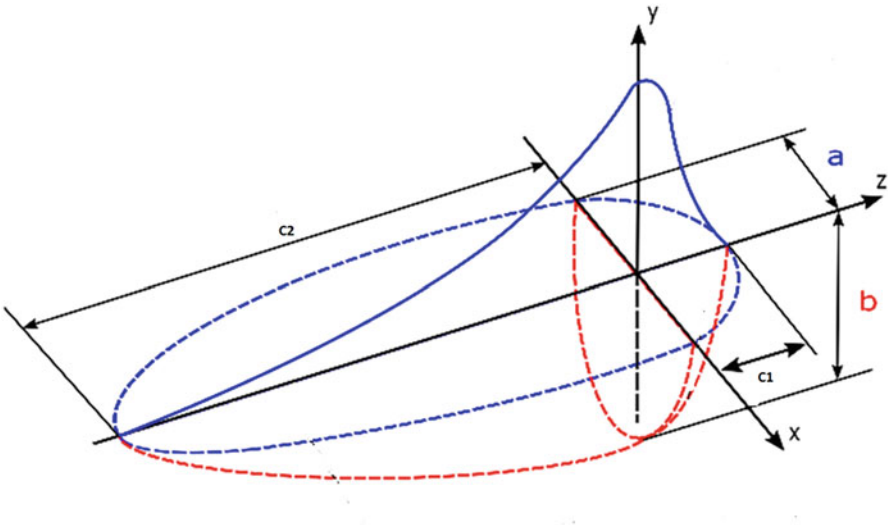

Fig. 2 Goldak double ellipsoidal model schematic diagram

Residual Stresses and Distortion

Distortion and residual stresses are two common and significant challenges that impact the manufactured part's performance and dimension accuracy when using primary AM technologies. Distortion refers to how the part created using additive manufacturing deviates from its intended dimensions or shape. It can happen during the additive process or after the created part has been detached from the substrate, resulting in a reduced precision that is difficult to recover. In the AM processes, significant variability of distortion and residual stress are caused by a long history of temperature cycles, a variety of processing factors, and complex structural designs. As illustrated in Fig. 3, the three aspects of structure, material, and processing comprise key influencing variables of residual stress and distortion.

Material shrinkage occurs often in different AM techniques. In the case of metal AM, the metal contracts and turns from liquid to solid. The resin also shrinks during the light curing procedure. In contrast, because the preheating temperature was so close to the glass transition temperature of the sintered material, the part produced by the selective laser sintering (SLS) technique was somewhat distorted. The SLS material experiences low contraction, which limits the built part's distortion. Thus, in various AM techniques, material shrinkage is a major contributor to distortion and residual stress.

Mode of Residual Stress in AM

The stresses that remain retained in the material after thermomechanical treatment are known as residual stresses. The three different types of residual stresses are

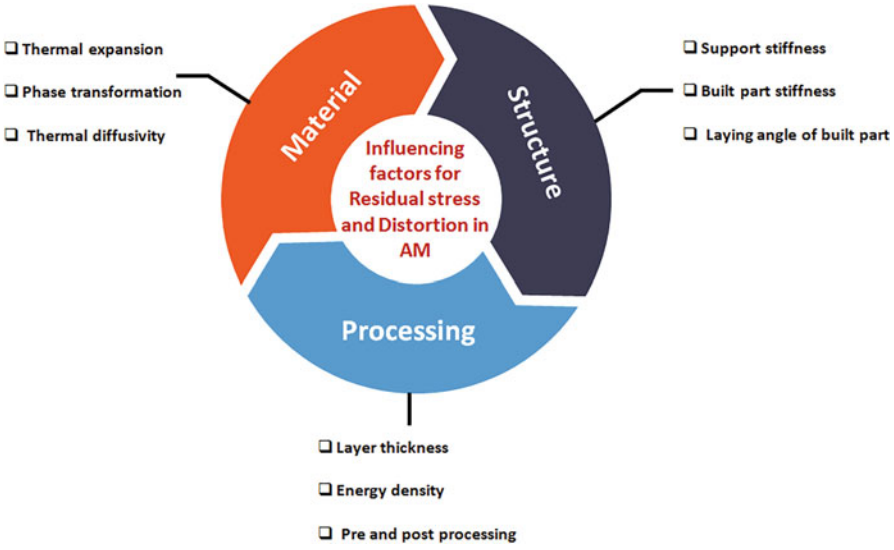

Fig. 3 Influencing factors for residual stress and distortion in AM

type I, type II, and type III. Long-range stresses of type I balance across macroscopic dimensions. Continuum models are used to calculate these types of stresses, which overlook a material's multiphase or polycrystalline nature. Type II residual stresses rely on the size of a material's grain, whereas type III dislocations exist spanning atomic dimensions. Residual stress is a type of long-range stress that continually varies over a length scale that is comparable to the macroscopic dimension of a component. On the other hand, thermal stress indicates the state of stress at high temperatures, making measurement procedures difficult. Thermal stress collected throughout the AM operations leads to residual stress, which is typically evaluated at ambient temperature. Cracks can emerge during the AM process and during the separation from the substrate when the heat stress is higher than the material's tensile strength.

Mode of Distortion in AM

In AM processes, when a small bit of material is heated to its melting point on an otherwise much cooler entity, distortion is an inevitable effect. While the top layer is still blazing hot, the material in the heat-affected zone (HAZ) naturally experiences thermal expansion, which causes the part to suddenly bow downward as the cooler material is pushed to bend to accommodate the expansion of the top. Thermal contraction occurs at the top of the part as the molten material solidifies during cooling, stretching the lower regions of the part and causing it to bend upward. Because of the high stresses experienced during expansion and contraction, the component is forced to give, which causes irreversible deformation.

By optimizing the model based on morphology and dimension, the stiffness of the deposited structure can be increased. Additionally, because it can change the slice of

the model and the related structural stiffness layer by layer, a part's laying angle in an additive process allows for adjustment in the stiffness of the deposited structure. Additionally, it is thought that the distortion is significantly influenced by how stiff the support system is. To lessen distortion, proper processing parameters, including heat input, are required. It is thought that excessive heat input will raise the constraining coefficient and the resulting distortion. Invar and other materials with low thermal expansion coefficients deform less than materials with high thermal expansion coefficients. Next, the part produced by additive manufacturing must be accurate and of high quality (particularly if it uses SLM or LPBF). It is claimed that distortion should take precedence over part quality because the latter can be enhanced through parameter optimization and post-processing techniques such as hot isostatic pressing, polishing, and remelting.

Melt Pool Characteristics

The processes that take place in the temporary melt pool that forms when a laser or electron beam contacts a metal powder bed are considered in models of the melt pool in powder-bed processes, as well as in the neighborhood (often within about 1 mm) of the melt pool. On a microsecond time frame, the beam melts the powder, which is typically made up of metal particles of 20–50 μm and generates a liquid melt pool that later cools and solidifies (Cook and Murphy 2020; Li et al. 2022).

The melt pool has been modelled using a variety of computational techniques, but regardless of the specifics of the formulations employed, the same fundamental set of physics equations must be solved. The formulation presented in the following sections is given from the viewpoint of computational fluid dynamics (CFD), the most popular technique for modelling fluid flow in general and melt pool modelling (Zhang and Zhang 2019).

The majority of CFD codes either employ a FEM (Zienkiewicz et al. 2005) or FVM (Ferziger et al. 2002) strategy. Both discretize the model domain into a mesh of tiny elements or control volumes, allowing the physics equations to be solved in more straightforward linearized forms before being assembled to reconstruct the entire domain. FEM has traditionally been regarded as having better handling of complex geometry and occasionally greater accuracy, whereas FVM has been regarded as being quicker and more user-friendly for developing novel physics. However, in contemporary codes, there is no obvious favorite, and these distinctions are highly hazy.

Melt pool simulation has been carried out using LBM and SPH (Liu and Liu 2010), two more computational techniques. In LBM, fictive fluid particles are produced at a lattice of sites rather than directly solving the Navier-Stokes equations. Realistic fluid behavior develops because of tracking their subsequent impacts and movement. With SPH, the material is represented by a sizable collection of imaginary Lagrangian particles, each of which is specified by a local kernel function. The common physics equations are used and then broken down into interactions between nearby "particles." Few LBM and SPH applications for melt pool simulation have

been made to date. Additionally, some crucial physics, such as temperature-dependent surface tension and intricate boundary conditions, are challenging to integrate in commercial software due to their lack of development. However, both approaches have potential in the long run because they implicitly handle complex interphase boundaries and can simulate the whole-body motion of solid metal particles.

The main requirements of the melt pool model are to predict the melt pool geometry and temperature distribution. Porosity, spattering, and other phenomena that are closely related to what happens in and around the melt pool can be predicted to arise. To compute the initial evolution of the microstructure surrounding the melt pool, temperature information is required. To improve the global model's approximations, mesoscale processes must be understood. Provide a global model of the thermal and residual stress in the part with an averaged melt pool temperature or a heat source.

A linked solution for heat transfer and fluid flow underlies the simulation and incorporates several physical processes, such as the melting and heating of powder particles. The coalescence of molten material has a wetness and surface tension effects. Flow occurs in the melt pool. Porosity development, expansion, and freezing, the motion of the liquid's free surface, heat transfer via convection and radiation, and the development of recoil pressure on the liquid's surface during evaporation are all factors. Metal that is changing from a liquid to a solid while also releasing or absorbing latent heat.

Surface Characteristics Modelling

As was previously discussed, AM is widely used in large applications due to its capacity to generate yet lightweight components. Despite its widespread use, AM has not completely supplanted conventional techniques, with poor surface quality being one of the major causes. Surface texture is vital to a component's operation and, if neglected, can lead to major issues with the manufactured parts. Therefore, to establish control over the surface quality, it is required to properly understand surface behavior and modelling regarding the factors affecting it. Furthermore, complex surface metrology issues are prevalent in all forms of additive manufacturing, but especially in metal AM. Nevertheless, given the process' enormous potential, these challenges must be overcome. Processes can therefore be comprehended, enhanced, and optimized with the correct surface modelling and advanced tools and techniques.

Surface measurement tools offer topographic data that can be utilized directly for qualitative characterization. The measured surfaces typically contain geometrical data regarding the surface's shape and surface texture relied on by the manufacturing process. Surface texture is recognized as the roughness features of the surface topography, and surface shape is frequently referred to as form and waviness. The primary surface is derived from the raw surface data in accordance with ISO 25178-2 (2012) following the use of an S-filter that eliminates all of the small-scale lateral components. The S-F surface is the surface that is generated from the primary surface after the form has been removed using the F-operator. The S-L surface is

the surface that is generated from the S-F surface by employing an L-filter to remove the large-scale components. Roughness or surface texture refers to the S-L surface, while waviness surface refers to the residual surface left over after applying the L-filter to the S-F surface.

The surface wavelength determines how the shape, waviness, and roughness of a surface vary. The waviness can be caused by manufacturing-related disruptions including vibrations, temperature effects, or in some cases, the nature of the manufacturing process itself. The form describes the low-frequency or long-wavelength component of the surface topography. The existence of the form may be caused by the sample's geometrical shape, tilt errors, or rotary form when measured on cylindrical, cone-shaped, or spherical samples. Wavelengths of the waviness components on the surface topography are often much shorter than those of form.

Microstructure Modelling Approaches

Only a small amount of research has incorporated microstructural characterization, despite the fact that it provides an understandable relationship between the method and the ensuing mechanical, physical, and even chemical properties. For their use in service, the mechanical property variability of metal parts produced via additive manufacturing is a major concern. The texture of the solidification, which is influenced by scan patterns and other process variables, is one of the parameters determining the feature. An avenue to regulate these features, which ultimately determine the final structural material qualities, can be provided by an understanding of how these textures emerge throughout the AM process. It is presented and qualitatively compared with reported literature by numerous researchers who have investigated grain evolution in multilayer depositions using different scan patterns in directed energy deposition (DED), metal laser sintering/selective laser melting (MLS/SLM), and electron beam melting (EBM). Metallic materials processed using SLM and EBM showed variation in grain size and orientation evolution, which was directly influenced by exposure to various cooling rates and temperature gradients. It is possible to anticipate continuum-level structural features at both the global and local length scales using CA-based modelling for forecasting grain orientation and size in metal AM processes. The general conclusion for the development of grain structure from various studies is as follows:

(a) Due to the heat source moving perpendicular to and parallel to the plane, cross-hatching scans include two different grain orientations.
(b) The grain morphology is unidirectional and bidirectional, when rotating an island scan pattern at 45°.
(c) Even when utilizing zigzag scan patterns (Fig. 4 shows different patterns), the morphology changed from zigzag to unidirectionally oriented grains by lowering the cooling rate and thermal gradient.
(d) In the vertical and horizontal parts, respectively, low cooling rates and temperature gradients are the main causes of large columnar and equiaxed grains.

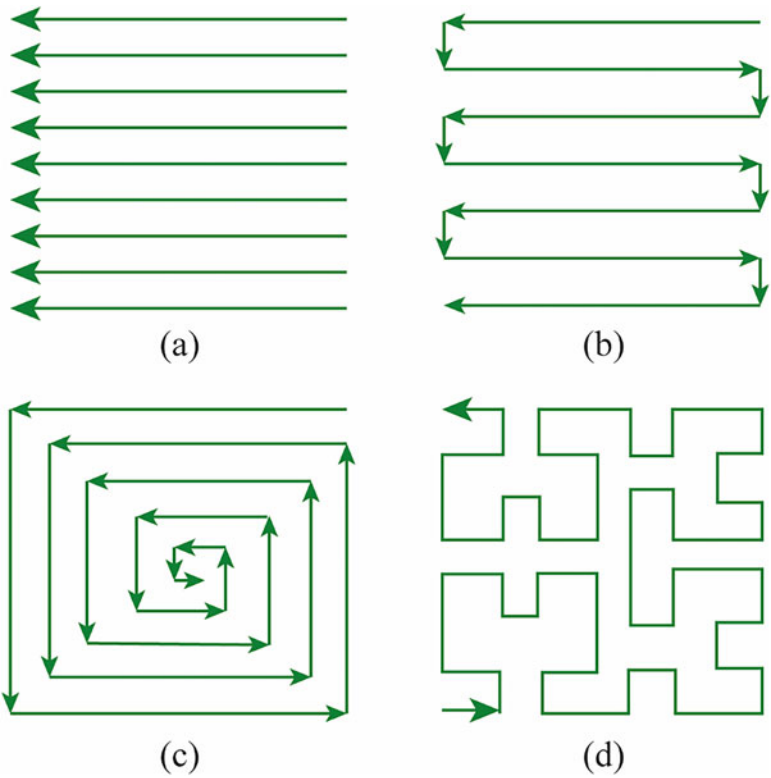

Fig. 4 Different deposition patterns: (**a**) raster, (**b**) bidirectional, (**c**) offset-out, and (**d**) fractal (Saboori et al. 2017)

(e) Curved columnar grains are seen in single-line scans at slower laser (energy source) speeds than straight columnar grains, which are seen at faster laser speeds. This finding explains why the grain morphologies produced by welding and additive manufacturing techniques differ.

If cooling rates and thermal gradients are known, it is possible to forecast grain evolution for many different processes (including SLM, DED, and EBM). It is possible to mimic the grain size of a material by entering the appropriate nucleation parameter for an alloy. Predicting the ultimate mechanical properties of components made via additive manufacturing will be much easier with the modelling, which includes grain size and orientation.

Porosity Modelling

The world is experiencing a rapidly increasing demand for mimics of natural geometry and physical structure, e.g., biomedical implants, biomedical scaffolds,

fractal-shaped microchannels in the electronic industry, filtration, and purification. In the same context, the application of porous parts has increased exponentially in recent years (Bayraktar 2022). This section aims to discuss only the development of porous structures in AM technologies and the related sciences. To argue in the favor of porous parts that can be developed easily by additive manufacturing, we may have a case of bone (implants due to an expanding and aging population, which increases both the rate and number of such incidents as trauma, bony tumors, and skeletal deformities) (Werkheiser et al. 2014). The AM process is used for various porous applications such as dental or orthopedic parts, batteries, electrodes, sensors, pharmaceuticals, fuel cells, etc. Figure 5 shows various porous applications.

The majority of AM research focuses on creating high-density parts with porosity values ranging from 0.1 to 5%, depending on the application. The AM approaches for porous parts were divided into two categories by Stoffregen et al. (2011) (i) geometrically defined lattice structure porosity (GDLSP) and (ii) geometrically undefined porosity (GUP). In contrast to GUP, where the required porosity and pore size are produced by optimizing the process parameters of AM technologies, in GDLSP, porosity and pore size are dictated by different lattice structures and strut thickness, respectively. GUP pores typically range in size from 1 to 100 nm, while GDLSP pores are between 100 nm and 1 mm. The porosity is at its highest point and the pore size is in the 10–30 m range when this technique is used (Abele et al. 2015).

There are different methods used to measure the porosity, as follows:

(a) Archimedes
(b) Bulk density measurement
(c) Mercury intrusion porosimetry
(d) Microscopy analysis
(e) X-ray computed tomography (XRCT)

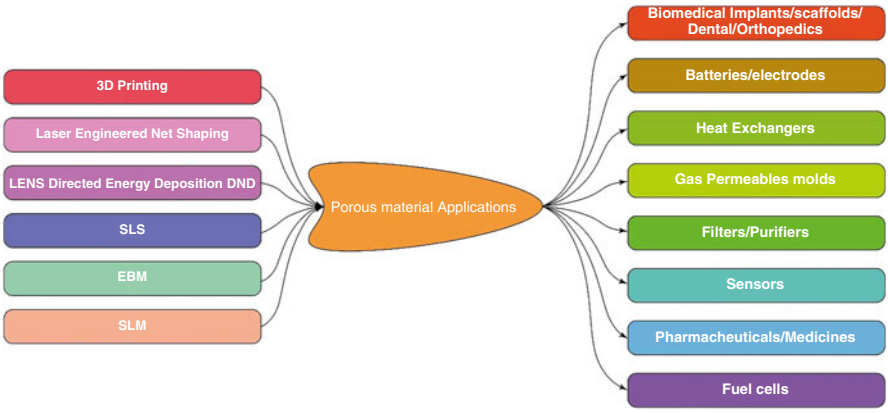

Fig. 5 AM process for various porous applications

(f) Mass and volume
(g) Ultrasonic wave speed measurement technique

Material Properties Modelling

Paper laminates, waxes, and polymeric materials were initially used to develop AM technology, some AM components may contain voids or anisotropy depending on the part orientation, process conditions, or how the design was fed to the machine (Gibson et al. 2015). With these questions, the study moved forward and observed that the prediction of deformation, residual stress, and thermal stress of a structure is crucial in determining its structural properties. This is particularly true when it comes to 3D printing concrete, which, due to its time-dependent rheological qualities, calls for accurate deformation forecasts and exact process parameters to ensure that the fabricated object stays within the tolerances given in the design. Furthermore, the accuracy of the process parameters guarantees that limitations on manufacturing, like the total amount of printing time, are not broken. Concrete additive manufacturing techniques using extrusion are highlighted by finite element analysis (FEA), which reveals the intricate relationships between process and material factors (Ekanayaka et al. 2022; Kruger and van Zijl 2021).

In a typical structural FEA, the deformations and stresses of a fixed input geometry are computed, but in AM simulations, the input geometry changes over time as new layers are built on top of older ones, adding another level of complexity to the simulated model. A variety of AM technologies can be used to simulate structures using current commercial FEA tools. Similar general procedures apply when utilizing commercial software to run typical FEM simulations, where the target CAD geometry is initially supplied together with the appropriate boundary conditions, material properties, and external forces. The governing physical equations are discretized and solved to determine nodal displacements after the structure has been constructed. Secondary factors like stress and tension can be subsequently determined.

Topology Optimization

In addition to the modelling discussed above, topology optimization is another numerical modelling technique used to develop parts for the metal 3D printing process. To identify the best material distribution in a part's design for a specific set of loading and boundary constraints, topology optimization is a mathematical technique (Fernández et al. 2021). FEA is the model's foundation. By eliminating the material from the area where it is not required to support the load or boundary condition, this procedure is carried out. The final topological structures produced by this procedure, which resemble tree branches and bones, are not uniform in cross-section (Orme et al. 2017). Making a rough 3D model and breaking down the specific load and boundary conditions into components are the first steps in the

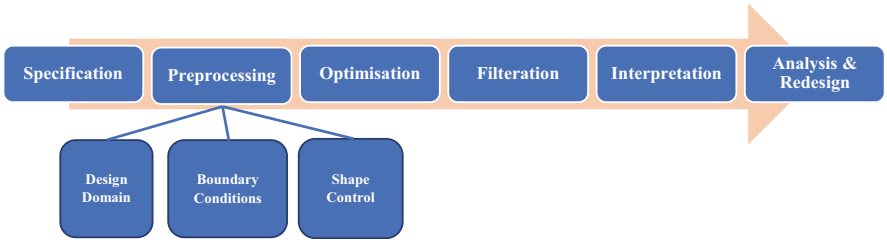

Fig. 6 Topology optimization process

topology optimization modelling process. Then, all the applied constraints are calculated by the optimization software. Then, the regions that are unnecessary will be removed. The final geometry will satisfy the mechanical and design criteria (Walton and Moztarzadeh 2017). The process stage of part modelling topology optimization is shown in Fig. 6.

Conclusion and Future Scopes

The modelling approaches that have been put forth in the literature primarily aim to represent the phenomena of heat, residual stress, distortion, microstructure modelling, and porosity creation in AM parts. These characteristics will determine whether the part can meet the design criteria. Insight into AM processes can be gained through computational modelling. This overview discusses and examines a number of numerical models used in AM. The modelling approach that predicts thermo-mechanical models most frequently is FEM. Similar to this, the PF modelling approach shows excellent promise for predicting microstructure during AM operations. Other researchers' modelling approaches have taken into account surrogate models, whose inputs either comprise a sizable amount of modelling experimental data or data produced by other modelling models.

It should be anticipated that mathematical advances will happen more frequently in the future as computational modelling sources become more widely available and mathematical techniques become more refined. This is especially true if the amount of experimental data also increases, creating a "big data" database. The main goal of AM modelling research is to develop a computational model for a certain phenomenon. Examples of this include modelling the Marangoni effect in the creation of melt pools or modelling aimed at understanding the mechanisms driving various occurrences. It is still necessary to fully incorporate modelling – modelling that takes into account the links between processes, structures, and properties – into the entire engineering design process. It is important to keep in mind that modelling requires a feedback-based strategy, where simulation results from modelling provide data that are processed to produce the next simulation until the modelling successfully fulfills all the necessary parameters (low distortion, desired microstructure, high cycle modelling fatigue, etc.).

Disclaimer The information on several companies in the area of 3D printing is only supplied for general informational reasons. The authors do not in any way recommend the companies or their products for a specific use or level of quality.

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Part VIII

Digital Transformations in Metrology: A Pathway for Future

Role of IoT in Smart Precision Agriculture 50

Kumar Gaurav Suman and Dilip Kumar

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Abstract

The growing population of the world has pressurized today's agriculture sector for more production with less available resources. The advancement in technology paved the path for the induction of new techniques replacing the age-old traditional methods in the agriculture sector. The internet of things (IoT) along with its components like wireless sensor network, big data analytics and cloud computing have contributed in transforming the traditional agriculture to smart precision agriculture. In smart precision agriculture methods, we are monitoring the field parameters like soil moisture, temperature, humidity, pH level etc. via different sensors available in real-time condition and supplying the optimum resource required to the field. This method avoids excessive use of resources and increases the production as well. This data generated can be stored in the cloud and future decision can be taken based on the analytic results. So, IoT can enable us to guide in smart farming in a way to save resource, optimum resource utilization, increase production, real-time monitoring, future decision and many ore

Keywords

Internet of things (IoT) · Smart precision agriculture · IoT agriculture architecture · Agriculture application

Introduction

The word "Agriculture" is a Latin word that means "Land Cultivation." In a wide sense, agriculture is the practice of cultivating plants and livestock. It is the most essential part of human civilization which leads to the production of food, and food supports the life system on the earth. The population of the world is growing at a very fast rate and is expected to touch about 8 billion by 2025 and 9.5 billion by 2050. So, to feed such a large population, agriculture production must be increased. However, in the twenty-first century, agriculture will face many challenges like the production of more food to feed such a large population with less available resources, smaller labor force, climate change sustainable crop, etc. (Accessed 2020; [1] Food and Agricultural Organisation of United Nations 2022).

In India, with a population of 1.27 billion, the agriculture sector is the largest source of livelihood. Here more than 70% of rural livelihood depends on agriculture. India still accounts for a quarter of the world's hungry people and is home to over 190 million undernourished people. Increasing stress on water resources of the

country would need a realignment and rethinking of policies. Desertification and land degradation also pose major threats to agriculture in the country. Water scarcity and falling water tables have been a key concern in recent years. Fortunately, the importance of judicious use of water is being increasingly recognized. India also needs to improve its management of agricultural practices on multiple fronts. So, one aspect is strengthening agriculture diversity and productivity (FAO 2022). At the same time, the percentage of food lost after harvest on-farm and at the transport, storage, and processing stages stands at 13.8 percent globally, amounting to over USD 400 billion a year (FAO 2021). To achieve the sustainable development goal of the United Nation along with the Government of India, information and communication technology can play a crucial role. It can be used to improve agricultural productivity to feed a large growing population and effective management of crop and natural resources.

The traditional methods of agriculture are slow and not sufficient to cope with the increasing demands of food. The application of the most advanced technology in agriculture is growing day by day. With this, the application of IoT in the agriculture sector cannot be left behind. As internet of things (IoT) itself implies interconnection of things that can communicate and exchange information. Therefore, things can be used to monitor agricultural parameters continuously. IoT can be useful in agriculture in some areas like precision agriculture, early disease detection, climate monitoring (Ma et al. 2020), crop management, data analysis, field monitoring, natural resource management, etc.

IoT is important in agriculture for the following reasons (Sharma et al. n.d.; Elijah et al. 2018):

(a) It can solve the problem persisting in the current scenario regarding water shortage, resource management, cost management, and productivity challenges to achieve the food security for all.
(b) The valuable resources can be utilized in a judicious manner with increase in the crop production or in other words well organized scheduling with available restricted resource.
(c) IoT can act as a predictive system for the control of pests and insects. It monitors the growth and development of weeds, pests, and other harmful insects in the crop via GPS-enabled devices. If left unmonitored, then the application of pesticides and fungicides will incur extra costs to the farmer and also pollute the environment.
(d) The traditional agriculture system is converted to data-driven services with decision tools and automation system.
(e) Due to declining in number of manpower, IoT can automate the agriculture field which may require a smaller number of labor and minimal chance of error.
(f) The field can be monitored from around the world in a real-time and predictive analysis can be done by smart decision tools to maximize the profit.
(g) Agriculture machinery connected via IoT with GPS can operate in auto-pilot mode at precise and accurate location. This can increase the efficiency of the system.

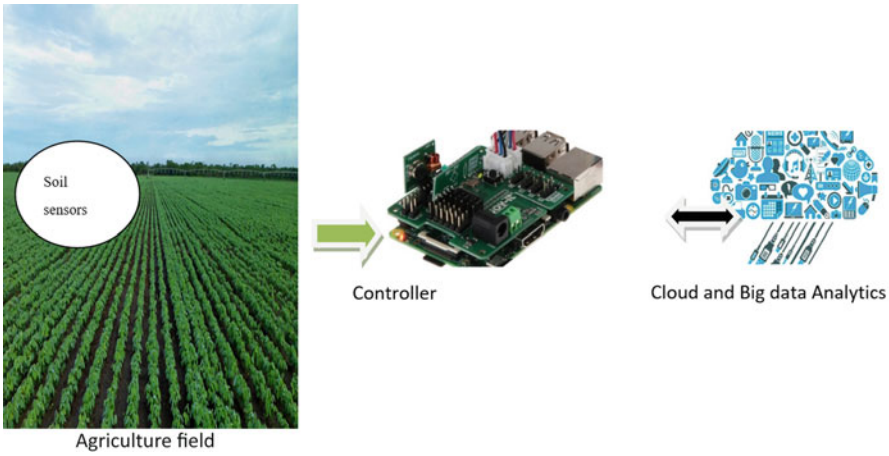


Fig. 1 Smart agriculture using IoT

(h) It can be useful in asset tracking in the field of supply chain and logistics for agriculture companies which may lead to reduction in loss after production stage.
(i) IoT can monitor the micro-environment in greenhouse by reducing human efforts, energy saving, and efficient farming and can provide customer connectivity.

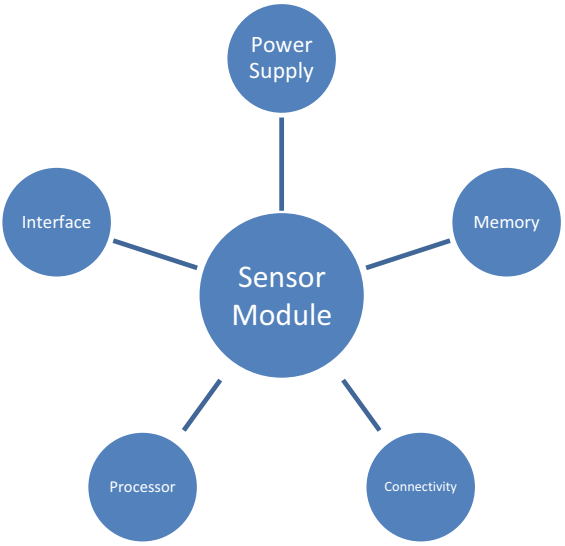
Besides these, a large number of the agricultural parameter can be monitored, sensed via different sensors, and the information from these sensors can be stored in the cloud server and analyzed via big data analysis, and action can be taken based on the outcomes. These data help us to understand the course and development of the farming system. The IoT has the capability to reduce human efforts and saves natural resources in a broader sense.

The general overview of the implementation of IoT in agriculture sector has been shown in the Fig. 1.

Sensors

Sensor is a device which measures physical parameters. In agriculture application, sensors can be used in a number of applications, viz., precision agriculture, disease detection, etc. A sensor transmits the output voltage proportional to sensed parameter in the form of binary information generally. The main components of a sensor module are power supply which powers the module, processor that acts like a brain of module, memory for data storage, connectivity for sending sensor data over wireless or wired medium to the Internet, and interface to connect nearby sensors or actuators. These can be shown in Fig. 2.

Fig. 2 Sensor components



In precision agriculture, sensors are used to monitor the agriculture parameters at different growth stage of the crop. Precision agriculture means supplying the optimum parameters at optimum time and at optimum amount for the growth of a healthy crop. Therefore, the actual growth stage and its required parameters are monitored by a variety of sensors. The sensors generate data from the parameters like soil moisture, temperature, weather condition, soil nutrients, humidity, air pressure, soil salinity level, CO_2 level, plant wetness, hydrogen level in plant, etc. The sensor information is sent to the controller which transmits it to the cloud for storage and analysis. Data analytics is used to give certain decision based on the analysis, and actuator is used to control the amount of water, pesticides, fertilizer, etc. in the field (Shafi et al. 2019). So, the correct information at the right time is provided which results in cost saving and increase in yields. In other words, the data received via sensors play a crucial role in smart crop irrigation (Table 1).

The various types of sensors are shown in Fig. 3.

Sensor Associated with Smart Agriculture

The various types of sensor with its application are illustrated in the following sections:

Table 1 List of sensor and parameters measured (Shafi et al. 2019)

S. No	Name of the sensor	Parameter measured
1	ECH2O soil moisture sensor	Soil temperature, soil moisture, conductivity
2	EC sensor (EC250)	Soil temperature, salinity level, soil moisture, conductivity
3	107-L temperature sensor (beta Therm 100K6A1B thermistor)	Plant temperature
4	237 leaf wetness sensor	Plant moisture, plant wetness, plant temperature
5	YSI 6025 chlorophyll sensor	Photosynthesis
6	SHT71, SHT75 (humidity and temperature sensor)	Humidity and temperature sensor
7	CM-100 compact weather sensor	Air temperature, air humidity, wind speed, air pressure
8	Cl-340 hand-held photosynthesis	Photosynthesis, plant moisture, plant wetness, CO_2, plant temperature, hydrogen level in plant
9	LW100, leaf wetness sensor	Plant moisture, plant wetness, plant temperature
10	Sense H_2 ™ hydrogen sensor	Hydrogen, plant wetness, CO_2, plant temperature

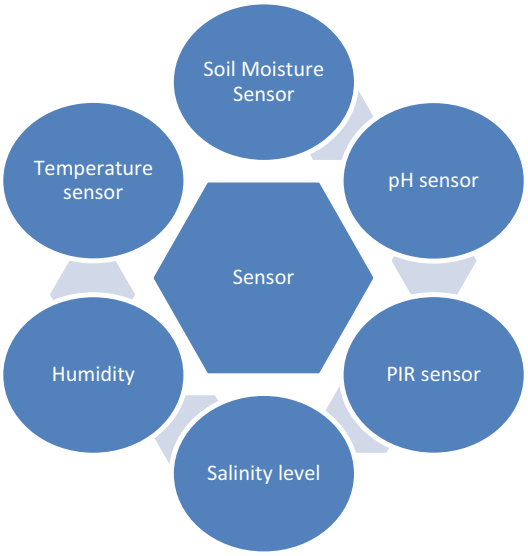


Fig. 3 Types of sensors used in agriculture field

Unmanned Aerial Vehicle

In agriculture, unmanned aerial vehicle (UAV) integrated with sensors can be used to detect disease and crop management. UAV can act as an eye of the farmer to view the entire field from a height very close to the surface. High-resolution images captured by the camera sensors can be analyzed to detect the presence of disease, pest, weeds, and growth stage of the crop from the field in a real-time situation. Images captured

by satellite are useful in identifying distributed shrublands and grasslands with an accuracy of 79% and 66%, respectively (Saha et al. 2018). Optical and multispectral techniques are being used via UAV drones equipped with RGB-D cameras or laser cameras to identify the presence of chlorophyll in the field. Based on the outcome of the image processing applied on the images, various measures can be taken like supplying the water to the soil; measures to protect against weeds, insects, and fungi; or any other countermeasures which seem fit depending on the situation. This early detection can protect the crop damage and farmer loss (Tripicchio et al. 2015). The pesticides can also be sprayed to exact location which have pests using UAV and thus save the crop and cost incurred by farmer.

Intrusion Detection

There is a significant loss in the production due to theft by human being or unwanted entry of animals in the field. Hence, IoT can be helpful in reducing this loss gap. The detection system employs PIR and ultrasonic sensor to record this event. The sensor, employed at the boundary of the field, can sense the motion of intruder and sends the information to the controller which in turn generates an alarm or informs the farmer in the form of text message. Thus, timely action can save the crop and increase the production efficiency. The sensors can communicate via single or multi-hoped connectivity in wireless sensor network which is a part of IoT (Roy et al. 2015).

Livestock Monitoring

Livestock is a part of agriculture system. Livestock monitoring helps the farmer to know about the real-time position, location, and health of the livestock. Farmers can track the body temperature, humidity level, and heartbeat rate of their cattle using different sensors. Body temperature sensor can inform about indigestion, milk infection, and disease like Influenza and anthrax in the cattle. Humidity level measurement can inform about the stress level faced by the cattle. A heartbeat sensor can predict the low and high blood pressure level found in the cattle. A high heartbeat rate signifies a pain with several disease. The sensed information is sent to the farmer via IoT using WSN who can monitor it even from outside the farm. An action can be taken in advance before the situation worsens for the cattle. Hence, the role played by the IoT in livestock monitoring cannot be undermined (Shah et al. 2019).

Greenhouse

The greenhouse or polyhouse is a polythene-covered house which protects the crops from outside uncontrolled environment parameters like harmful sun rays, extreme weather condition, excess rainfall, high speed wind, etc. Here, the farming is done in

a fully controlled environment to increase the production and quality of the crop. In the indoor greenhouse system, all the parameters can be monitored and controlled with IoT via different sensors. The sensor can check the variables like temperature, fan, soil pH, soil moisture, lighting, water, motion detector sensor, and openings for their timely supply and send the information to the server for appropriate action (Dagar et al. 2018).

Network Architecture

It plays an important role in the implementation of IoT in agriculture sector. It facilitates data transfer and reception between the devices. It sets some guidelines and rules on how data can be sent over the network, that is, path to be followed for lossless delivery to the intended receiver. Hence, it basically consists of network topologies, specification, and set of rules for IoT to be used in agriculture. IoT agriculture network consists of four layered architecture, namely, link layer, network layer, transport layer, and application layer. Each layer has its own protocols, and various technologies are available that work under particular layer.

Link Layer Protocols

This layer is responsible for implementing methods and process related to actuating and sensing different agricultural parameters. It decides on ways of the data transmission over physical medium like wireless or wired medium (Boursianis et al. 2020). As the field lies in the remote location, so wireless medium is most popular. IEEE 802.15.4 operates in ISM band having 2.4 GHz and is the most energy efficient physical layer protocol used in agriculture IoT. It is also low cost and a less complex protocol. Other standard wireless protocols available are Zigbee, Z-wave, and RFID. Cellular technology such as 2G/3G/4G/LTE is also used in some application due to its high data rate and Internet connectivity. The field agricultural parameters are transmitted to the nearest server storage or cloud via 4G LTE network. However, it consumes high power (Zhang et al. 2017). Some of the protocols are listed as follows:

IEEE 802.11

It is a bunch of wireless Local Area Network communication standards. It includes 802.11a having a bandwidth of 5GHz, 802.11b operates in 2.2 GHz, 802.11 g operates in 2.4/5 GHz, 802.11n operates in 60 GHz, and 802.11 ac operates in 5GHz. It has an overall frequency band of 5–60GHz with a data transfer rate varying from 1 Mbps to 7Gbps (Hiertz et al. 2010). It has high energy consumption with high cost. Its range varies from 20 to 100 m.

LoRAWAN

LoRAWAN stands for long range wide area networking, and it is a long range communication protocol. LoRAWAN has data rate of about 27 Kbps with a range of less than 30 KM. It is meant for battery-powered application as it has very low energy consumption (Adelantado et al. 2017) (Dias and Grilo 2020).

WIMAX 802.16

IEEE 802.16 is a bunch of wireless broadband standards. It has a frequency range of 2–66 GHz with a data rate of 1.5 Mbps to 1Gbps. It has a range of about 50 Km and medium energy consumption.

Zigbee

It is based on wireless 802.15.4 standard and developed by Zigbee alliance. It operates with the frequency of 2.4 GHz, and its range is low, i.e., 10–20 m. its data rate is 20–250 Kbps. It has low energy consumption and low cost. It has a characteristics of network resilience and interoperability. It is less secured than Wi Fi system and very commonly used in agriculture system (Ngangue Ndih and Cherkaoui 2016).

Bluetooth

It is based on IEEE 802.15.1 standard. It has a frequency band of 24 GHz with data rate of 1–24 Mbps. It has a very low energy consumption and low range personal area network. It has a range of 8–10 m. Many sensors used in agriculture are Bluetooth enabled.

Cellular 2G/3G/4G Mobile Network

Cellular network like 2G/3G/4G is available in most of the region of our country. Hence, IoT devices in agriculture can communicate over this cellular network area. It has a frequency of 865 MHz and 2.4 GHz. It has a data rate of 50–100 Kbps for 2G, 200 Kbps for 3G, and 0.1–1Gbps for 4G network. It has medium energy consumption and medium cost.

NB-IoT

Narrow band IoT (NB-IoT) is a cellular technology. It has long range, low data rate, and low energy consumption and is cost efficient. It consumes less power compared to cellular 2G/3G/4G cellular network. It promises long battery life for energy-constrained devices. It employs GSM frequency band and best suits for agricultural application as agriculture field lies in remote location in the rural area (Valecce et al. 2020).

Network Layer

This layer has the responsibility of sending data from source network to destination network. The packet contains both the senders and destination address, and this layer

performs packet routing to reach the intended network. The unique addressing is provided via IPv6. As the number of devices is increasing day by day, so IPv4 was not sufficient to provide unique addressing to all the connected devices over the network.

IPv6

It is an upgraded version of IPv4. With large number of devices being connected to the internet, it became difficult to uniquely identify in IPv4. Hence, IPv6 is being used. IPv6 uses 128 bit for addressing different devices. The total number of distinct devices that can be connected goes to 2^{128} devices.

6LoWPAN

When we talk about IoT in agriculture, we mean the things or devices which are generally energy constrained. Hence, 6LoWPAN, i.e., IPv6 over Low Power Personal Area Network is being used as Internet protocol in power-constrained devices. It operates in the frequency range of 2.4 GHz with a data rate of 250 Kbps. It is an IP-based communication protocol (Suryady et al. 2011).

Transport Layer

This layer supports end-to-end message delivery irrespective of underlying network. It is called host-to-host transport layer. The sensor data is obtained and encapsulated in this layer, i.e., data transferred from IP to IoT domain. It has two types of protocol, i.e., transmission control protocol (TCP) and User Datagram Protocol (UDP).

Transport Control Protocol (TCP)

TCP is a connection-oriented protocol. It is also a stateful protocol, i.e., each request from client to server does not contain all the information. It supervises ordered data packet transmission from sender to receiver in a reliable manner. It uses handshaking for reliable delivery. It also supports error detection and correction techniques in which lost data packet is resent and duplicity avoided. The flow control mechanism synchronizes the sender and receiver speed of data packets. It has higher overhead so consumes more power compared to UDP. TCP has lower packet sending speed.

User Datagram Protocol (UDP)

UDP is the connectionless protocol. Hence, there is no overhead for opening a connection, maintaining a connection, and terminating a connection. UDP is known as stateless protocol. It does not acknowledge the send packets been received or not. UDP is very fast compared to TCP. As it does have overhead, so it consumes very less power and an ideal choice for some application in IoT. UDP is also efficient for broadcasting and multicasting. It only has basic error checking mechanism.

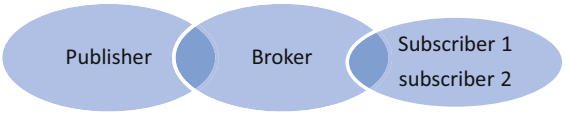


Fig. 4 Publish subscribe model

Application Layer Protocol

This layer involves the ways of how to interface with lower level in order to send data. The data files encoded using application layer and encapsulated using transport layer protocol are transmitted using network layer to reach the destination address. The different application layer protocols used in the IoT are MQTT, CoAP, AMQP, XMPP, and HTTP (Bansal and Kumar 2019). These protocols are light weight as it requires to work in energy-constrained environment. It works on different communication model like request-response or publish subscribe or web-socket (Al-Fuqaha et al. 2015).

Message Queue Telemetry Transport (MQTT)

MQTT is a communication model widely used in IoT. It is bandwidth efficient and consumes less power and is suitable for energy-constrained device. It works on publish-subscribe model over TCP/IP protocol and has one to many distribution. Publisher acts like a source of data which transfer the data to the subscriber subscribed to a topic managed by broker. Whenever a broker receives any information on a topic, it forwards to the subscriber. Neither is publisher aware about the subscriber nor vice versa. It is managed by a broker (Eurotech, IBM 2022) (Fig. 4).

Constrained Application Protocol (CoAP)

It is an application layer protocol which is best designed for constrained device. It is web transfer protocol based on User Datagram Protocol (UDP) and works on request-response model of communication. It is more suitable for IoT application as it reduces overhead of handshaking.

Extensible Messaging and Presence Protocol (XMPP)

This protocol is used for sending data in real-time environment. It uses client server architecture. It can deal with large and small file such as text, image, or video.

The overall architecture of IoT in agriculture sector is shown in Fig. 5.

Technologies in Agricultural IoT

Various technologies that are associated with IoT can be specifically used in agriculture IoT to modernize this sector. These technologies will give boost to age old traditional agricultural system that prevailed in our country. The various

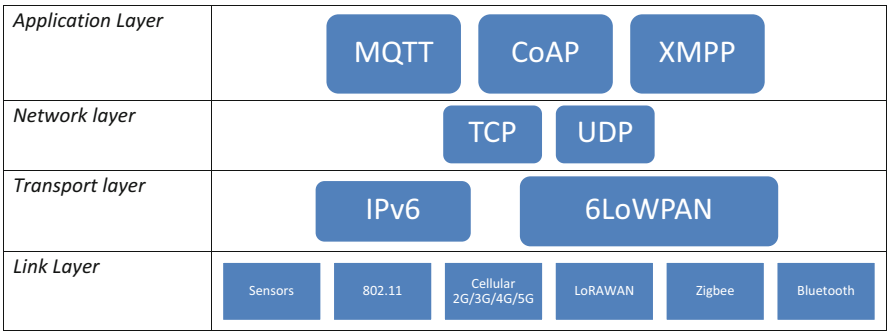


Fig. 5 IoT architecture in agriculture

technologies are cloud and edge computing, wireless sensor network, robotics, big data analysis, and communication protocols.

Cloud and Edge Computing

A cloud computing may be considered as an on-demand service like data storage, access, and cloud application available on the Internet. IoT can provide access to shared resource in agriculture. Application and architecture on the cloud can process and retrieve the agricultural information. A sensor data from the field can be sent to cloud for storage as it has large storage capability (Botta et al. 2016). Edge computing facilitates the computing at the source of data generation like sensors and actuators to reduce the burden at the cloud. These computing can used in smart farming as one of its features (Zamora-Izquierdo et al. 2019).

Big Data Analysis

Big Data refers to large volume of data. In agriculture, when the fields are connected with sensors via IoT, a wide variety of massive data is generated. This data are captured and stored in a huge storage space or in the cloud for further analysis and decision making. The decision can be predictive in nature or for real-time decision making or may be useful to design future business model (Wolfert et al. 2017).

Wireless Sensor Network (WSN)

A wireless sensor network is a network of large number of sensor nodes with each nodes having sensor to detect physical parameters. For monitoring large field size, WSN proved to be an optimal solution. WSN can be useful in different farming-related activities like irrigation scheduling, early stage detection and prevention of a

crop disease, real-time field monitoring, etc. It can also be used to control the microclimate within the greenhouse at different location. It has the capability to increase the farm production as well as optimum utilization of resources like water and pesticides. WSN can be useful to monitor the field parameters such as temperature, air humidity, soil pH level, soil humidity, wind speed and direction, etc. (Kalaivani et al. 2011; Khan et al. 2018). WSN has low power and very low data rate so it is an energy-efficient technology (https://www.cropin.com/iot-internet-of-things-applications-agriculture/ n.d.).

Robotics

Labor shortage in the agricultural sector causes a loss of 3.1 billion USD in the USA alone. So, focus is given to automation in this sector. The application of robotics like wedding robots, machine navigation, harvesting robots, material handling, and drones, etc. supports automation in this sector. Drone is a device which is equipped with camera and sensors for surveying, mapping, imaging, etc. There are two types of drones, viz., ground based and aerial. Ground based used robots for its work, whereas aerial drones use UAV for its operation. Drones can be remotely monitored to collect information like crop health, irrigation, spraying, planting, soil and field, etc. The collected data then can be analyzed to predict about events in agricultural sector (Sharma et al. 2021).

Machine Learning

AI and machine learning can be applied in the agricultural sector to analyze the operational data and improve decision making to enable smart farming. Based on the data available, AI and machine learning algorithm are useful in training the system so that they can take decision automatically and reduce human effort. This technology can be useful in predicting the pest breeding well in advance so that measures can be taken beforehand to save the crop (Al-Sarawi et al. 2017).

Communication Protocol

An IoT system relies on its network and communication protocol to transmit the information from sender to receiver. Various available wireless communication technologies and protocols that are useful in IoT are Internet Protocol version 6 (IPv6), Low Power Wireless Personal Area Network (6LoWPAN), Zigbee, Bluetooth Low Energy (BLE), Sigfox, Near Field Communication (NFC), etc. Some are low powered, and some are short range network. Based on the requirement and suitability, different protocols can be applied to transfer data over the network (Sreekantha and Kavya 2017).

Application of IoT in Agricultural Sector

IoT can be applied in agriculture sector in a number of ways. It can be used in smart precision farming, farm management, greenhouse monitoring, livestock monitoring, field fire detection system, etc. IoT is capable of revolutionizing the traditional agriculture system. Some of the applications are as follows:

Precision Farming

It is a method of farming in which agriculture parameters are utilized in a best suitable way so as to save the resources as well as to increase the yield. This can be achieved by monitoring the agriculture parameters via different sensors. The different parameters are soil quality, soil moisture level, weather condition, temperature, moisture, humidity, wind direction, etc. The information sent to the server are used for analysis, and decision taken is conveyed to the actuators via controller for appropriate actions.

Crop Monitoring

In this method, IoT can be used to monitor the growth stage and field crop condition in field crop so as to increase the productivity and cost reduction. Every growth stage is monitored, and water and other nutrients are supplied according to its need in appropriate quantity via controlled instruments. When the crop is ready to harvest, the sensors provide information. So this increases the productivity by timely harvest and reduces cost by supplying optimum required nutrients (Kour and Arora 2020).

Disease Detection

Pest or disease infection can destroy the standing crop in the field resulting in loss to the farmer. IoT can be beneficial in preventing this damage. Sensors and image processing play a crucial role in the early identification of the pest. Images of the crop can be taken via camera from the field directly, UAV drones, or satellites. The preventive measures are taken well in advance before the pest damages the entire crop by spraying pesticides. It prevents the crop from damage and loss to the farmers.

Greenhouse Monitoring

Greenhouse is a method in which we grow plants inside a covered transparent house. Greenhouse is generally used to provide healthy and necessary environment for the growth of protected cultivable plants like fruits and vegetables. Here plants are

Fig. 6 Greenhouse monitoring

grown in a controlled environment, so intense monitoring of parameters is required within it. IoT can be used to monitor the microclimate inside it via WSN with minimal human interference. A large number of sensors can be used in water management, plant monitoring, and microclimate monitoring inside greenhouse. Microclimate includes maintaining proper ventilation, balance of CO_2 and O_2 level, temperature, and humidity inside greenhouse. Different location has different parameter level. All the sensory modules are interfaced or connected to the controller to share the information for further processing. The controller is programmed to act based on the defined threshold value via actuators. For example, when the soil moisture falls below critical value, the controller turns on the actuator water pump and turns it off when the soil moisture level exceeds the limit. Sensor information are stored, analyzed, and used to take appropriate action in the future (Pahuja et al. 2013) (Fig. 6).

Fertigation Monitoring

The application of IoT in agriculture can be beneficial for the optimum use of fertilizers in the field. Excess application of fertilizers not only increases the salinity of the soil but also affects the fertility. The nutrients like nitrogen (N), potassium (K), phosphorus (P), and pH of the soil can be monitored via sensors with IoT for optimum use of fertilizers in the crop field to save the soil and production cost (Cambra et al. 2014).

Farm Machine Monitoring

Nowadays, due to scarcity of labor force, heavy automatic machines are used in the big as well as small farming. The sudden failure of these machinery can cause loss to the farmers. So, in order to have a smooth supply chain, all the equipment and machinery can be monitored via IoT to avoid sudden breakdown and real-time monitoring from remote location (Kaloxylos et al. 2012). Any problems can be diagnosed well in advance, and necessary action can be taken to failures.

Livestock Monitoring

Livestock are considered as a part of agriculture system in our country. The health status and position of the livestock in the field can be monitored via sensors using IoT. The heartbeat sensors can be useful in monitoring the health condition of the livestock. The unusual behavior can be tracked, and necessary action can be taken well in advance. This can be useful to prevent some communicable disease by isolating the infected livestock and taking proper care. In other ways, the position of some livestock can be monitored in real-time manner. In both these ways, sensors attached to livestock via wireless sensor network connected to IoT can transmit information to the server (Kumar and Hancke 2014).

Field Fire Detection

Sometimes during summer season naturally or accidentally, standing field crop catches fire. This can cause huge loss to the farmer. So, the temperature sensor and fire sensor connected to the Internet can sent alarm signal or alert to the farmer so that necessary action can be taken (Nesa et al. 2018).

Drip Irrigation

Drip irrigation is the best available technology for the efficient use of water for growing horticultural crops in large scale on sustainable basis. Drip irrigation is a low labor-intensive and highly efficient system of irrigation, which is also amenable to use in difficult situations and problematic soils, even with poor quality water. Irrigation water savings ranging from 36 to 79% can be affected by adopting a suitable drip irrigation system. Drip irrigation or low volume irrigation is designed to supply filtered water directly to the root zone of the plant to maintain the soil moisture near to field capacity level for most of the time, which is found to be ideal for efficient growing of horticultural crops. This is since at this level the plant gets ideal mixture of water and air for its development. The device that delivers the water to the plant is called dripper. Water is frequently applied to the soil through emitter placed along a water delivery lateral line placed near the plant row. The

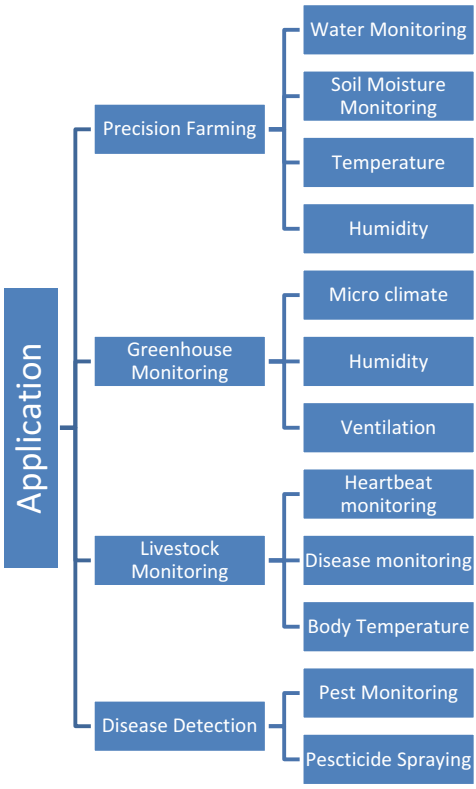


Fig. 7 IoT application in agriculture

principle of drip irrigation is to irrigate the root zone of the plant rather than the soil and getting minimal wetted soil surface. This is the reason for getting very high-water application efficiency (90–95%) through drip irrigation. The area between the crop rows is not irrigated; therefore, more area of land can be irrigated with the same amount of water. Thus, water saving and production per unit of water is very high in drip irrigation (Fig. 7).

Case Study

Smart precision agriculture uses different sensors and IoT devices for communication and control in the IoT network system. This system monitors different low-cost sensors which collect data from soil, air, water, pests, and other agricultural parameters. The controller makes the decision of irrigation, fertigation, pesticide spraying, and others automation based on the analysis of data.

The whole system has been divided into three layers. The first layer includes the sensors, hardware, and controller. Actuators. The second layer consists of database

Fig. 8 Smart irrigation layer architecture

operation, communication protocols, artificial intelligence, cloud services, other software related services, etc. The third layer comprises of web-based dashboard, mobile application, weather application, data analytics and decision-making commands, scheduled irrigation, and fertigation system decisions, etc. The different layers can be represented in Fig. 8.

Various sensors for parameter measurement like soil moisture sensor, pH measuring sensor, air pressure sensor, temperature and humidity sensor, and related actuators like water pumps, solenoidal valves, and fertigation system are spread out in the field which communicates with the controllers via wireless communication protocol IEEE 802.11. The controllers used are raspberry Pi which also acts as gateway and servers in this case. Raspberry pi can be programmed via different programming language like python to communicate and convert different sensors messages into common configuration. The server also acts as an interface and sends data to web-based dashboard or mobile application. The data are stored in the server or cloud space in case of large data for future analysis and decision making. The server can give command to actuators based on the sensor values. When connected to cloud or Internet, this sensor can be read, and actuators can be accessed from worldwide. This system performs smart precision irrigation using connected sensors and other data available about current weather. In the next phase, it irrigates field with appropriate water quality which can be checked with pH and turbidity sensor. If the water quality is not good, this system prevents it and sends and alerts message to the user. To conserve the precious water resource, this system employs drip irrigation technique. The drip pipe must be checked if there is any leakage or not. In case of leakage, the flow sensor will send an alert to the user with location and will cut the water supply through that route. This system also cleans the pipe and removes an unwanted sand particles through filtering technique. A variety of filters available to remove contaminants from irrigation water are gravity filter, wire screens, sand separator, screen filter, and gravel filter. This system also employs fertigation system as a system with drip irrigation; no additional pumps or any other device is required. The same tank is useful for mixing water and soluble fertilizers. Fertigation is done

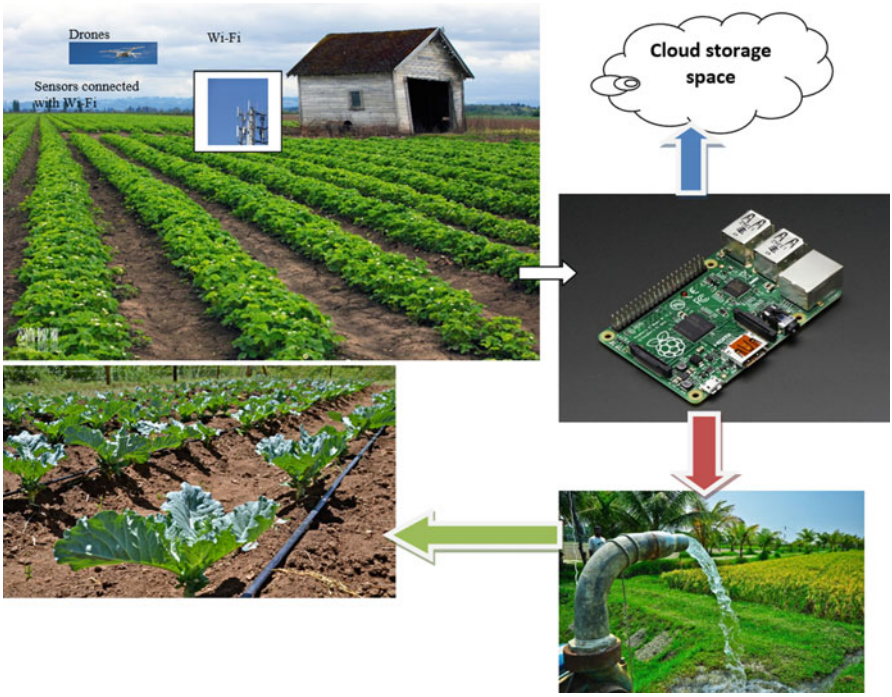


Fig. 9 Case study – smart precision agriculture

by the system after monitoring the soil nutrients, database related to the corresponding crops, and irrigation scheduling. The nitrogen-phosphorous-potassium (NPK) solution is supplied to the crop in various proportion based on the parameters obtained via sensors through drip irrigation (Jani and Chaubey 2021). In this system, a drone camera collects the image of the pests in the field with location. So, pesticide can be applied to accurate location with less cost and reduced health hazards. The whole smart irrigation system has been shown in the pictorial form in the diagram 9 as shown Fig. 9.

Challenges of IoT in Agriculture

Besides the benefits of IoT, it also posed some of the serious challenges in the country like India for its proper implementation (Elijah et al. 2018). Some of them can be listed as follows:

(a) In our country, 86.2% of the farmers are marginal farmers with land holding below 2 hectares. These are low-income farmers. They cannot afford costly automation system using IoT. So, our aim is to design and develop very low

cost IoT integrated system for the welfare of this large section of farmers and conservation of large natural resources.

(b) These marginalized farmers are not so educated. The farmers in the rural areas have lack of knowledge about IoT and its technology. So, we must provide training and awareness program to them for its efficient implementation.

(c) The policy of the government will play an active role in its successful implementation and creating an environment beneficial for the private players to invest on the development of low-cost products. More number of competitors will automatically lead to low-cost products.

(d) Besides these, the general challenges are security and privacy issues, cost factor, un-interoperable IoT devices, reliability of the hardware system, etc.

(e) There is need for resource optimization due to varying field size and available sensor types for profit margin.

(f) Ultralow-power devices and long-range connectivity for its sensors will give boost to its implementation in the agriculture field.

Conclusion

Thus, we can see that information and communication technologies with Internet of things (IoT) help to make our agricultural system smart, i.e., Agriculture 4.0. It is the future of our farming system which is the need of the hour to avoid wastage of agriculture resource and to tackle the climate change effect. The overall architecture and components required for smart precision agriculture has been discussed. The different possible applications are also explained in brief to give a wide view of its uses in agriculture sector. A case study about the application of IoT in the smart precision agriculture has been discussed which implements the smart irrigation, fertigation, pesticide spraying, and data analysis in the real-time in the field. IoT can be a game changer and offer several benefits in the agriculture sector if implemented with low cost devices for small and marginalized farmers. There are immense scope for the growth and implementation of IoT in agriculture in the future. IoT also has the potential to grow a new business model in the agriculture sector. Thus, IoT paved the way for resource utilization, production improvement, and future prediction for better decision making in the current scenario by utilizing the data for research and management. This technique will boost the food security requirements and conserve and utilize the available natural resource of the earth efficiently with less human labor requirement.

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Soft Metrology

51

Concept and Challenges from Uncertainty Estimation

Marcela Vallejo, Nelson Bahamón, Laura Rossi, and Edilson Delgado-Trejos

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Abstract

Nowadays, increasing demands in monitoring, control, and connectivity often require quantifying variables that are not easy to measure. Soft metrology allows

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the development of routines that objectively quantify magnitudes that are subjective, difficult, or expensive to measure. (As shown in this chapter, soft metrology measures quantities related to human perception or those derived from abstract representations, and this is a new conceptual paradigm for comprehending metrology in contrast to the well-known basic measurements directly associated with the International System of Units (SI).) However, despite the increase in the use of soft metrology systems, it is a developing area, and there are still many issues related to ensuring the validity of results, as there are no standardized uncertainty estimation procedures for these kinds of applications, and there still are open discussions about the uncertainty propagation through blocks of measurement processes made up by computer routines. This book chapter discusses the state of conceptual development and the proposed general structure for soft metrology systems, focusing on challenges involved in uncertainty estimation. Lastly, some discussions and final considerations are presented where epistemic and aleatory uncertainties are contrasted and associated with the representation quality and learning capability for inference, regression, and/or forecasting models in the framework of measurement processes and soft metrology.

Keywords

Soft metrology · Uncertainty · Machine learning · Deep learning · Epistemic uncertainty · Aleatory uncertainty · Representation quality · Representation space

Introduction

One of the main challenges of metrology in the context of Industry 4.0 is the development of measurement systems that allow more accurate and efficient manufacturing processes, maximizing quality and minimizing costs. Measurement is a key factor in engineering as it allows the quantification of important attributes in monitoring, analysis, and control of processes. This has led to the development of multiple sensors, designed to measure every imaginable magnitude in a wide variety of contexts. However, increasing complexity in industrial processes and in the requirements related to production efficiency, product specification compliance, and environmental constraints according to laws and regulations have resulted in increasingly more complex measurement systems.

One of the current challenges is the fact that there are many processes in which the relevant variables that need to be monitored to have a good comprehension of the system are difficult or expensive to measure or impossible to determine in-line. However, with current data analysis techniques, several strategies have been developed in which the available information of other easier, or less expensive to measure, variables in the process can be used to create models to infer the value of those that cannot be directly measured (Souza et al. 2016; Kullaa 2017). These techniques, developed since the 1970s, have been referred to as soft sensors, virtual sensors, inferential sensors, virtual metrology, and soft metrology.

Regardless of the term used to designate a soft metrology system, the main idea behind all these developments is that it is possible to find system variables that are more easily measured and that have a phenomenological relation with the difficult to measure variables, thus allowing the creation of a variety of indirect measurement models which can express the linkage between inputs and outputs corresponding to the measurand estimation. In Vallejo et al. (2020), all the terms used in the literature for indirect measurement systems and their application contexts are presented, and the term soft metrology is proposed as a global designation suitable to gather all other terms. Previous definitions were considered (Rossi 2016) and a wider one is proposed, which encompasses the concepts related to all the different cited terms, as follows: "A set of models and techniques that allows the objective quantification of magnitudes that are subjective, difficult, or expensive to measure, such as those related to human perception and/or to process dynamics. This quantification is intended as the process of experimentally obtaining one or more quantity values that can be attributed to the variable that needs to be measured by building a model that correlates it with other quantities, such as available measured variables from the same process, or physiological, operational, and psychological response in the case of human perception" (Vallejo et al. 2020).

Although there are several approaches to developing soft metrology systems, most of them rely on modeling tools based on machine learning algorithms to create the measurement model. In the literature, we find important challenges in the development of soft metrology systems (Vallejo et al. 2020), especially in terms of ensuring the validity of results, since there are no standardized methods to estimate the uncertainty in measurement models based on machine learning.

The remainder of this chapter is organized as follows: Section "Soft Metrology Systems: Approaches, Structure, and Applications" presents the structure and applications of current soft metrology systems, section "Challenges in Soft Metrology Applications" presents challenges for future developments, and section "Discussion and Final Considerations" discusses final considerations.

Soft Metrology Systems: Approaches, Structure, and Applications

There are two main approaches for developing a soft metrology system: model-driven and data-driven. The model-driven approach is based on the construction of a measurement model using the phenomenological knowledge of the studied system (Arpaia et al. 2015; Ferrari et al. 2017); this means that it is important to have a deep understanding of the process dynamics condensed in a series of equations that can be used to correlate the available measurements with the magnitude that needs to be inferred. On the other hand, the data-driven approach uses a structured database to generate a measurement model using machine learning techniques (Kadlec et al. 2009). The database can be obtained from an experimental design or from historical data. Given the complexity of most of the processes that require a soft metrology system, the data-driven approach has been widely researched and used in practical

applications (Kano and Ogawa 2010), and this chapter will focus mainly on this kind of soft metrology system.

Structure of a Soft Metrology System

The structure of a typical soft metrology system is depicted in Fig. 1, where physical sensors are used to acquire the magnitude of the system variables that are easy or unexpensive to measure, and that will be the input for the inference model, which consists of a machine learning algorithm that captures the relationship between inputs and the desired measurand.

Essentially, the idea of using machine learning is to create a measurement model without using phenomenological knowledge of the process. Instead, a database containing examples of the system behavior is used to make an initial model and adapt it until the system intrinsic dynamics is achieved, following the examples in the database. Thus, the learning algorithm is a program developed to help the dataset "learn" the behavior of the system. It means that the designer will define a certain initial structure for the measurement model and the algorithm will iteratively adapt the structure and/or its parameters until the model is able to express the dynamic represented in the database with a minimum error or bias.

The process for developing the machine learning algorithm depends on the technique chosen to model the process, and there are many machine learning routines that can be used in a soft metrology system. However, there is a common workflow for designing solutions, independent of the specific technique chosen, as illustrated in Fig. 2. The rest of this subsection will describe characteristics of this workflow and some approaches for each step in this design process.

The design process begins with database construction and preprocessing, where the goal is to collect the data that will be used in the following stages to build the measurement model. The structure of the database depends on the model choice. If a supervised method is chosen, the dataset must have an organized structure containing labeled samples, where the values of the inputs are registered for each value of the output (measurand); but with unsupervised methods, it is possible to

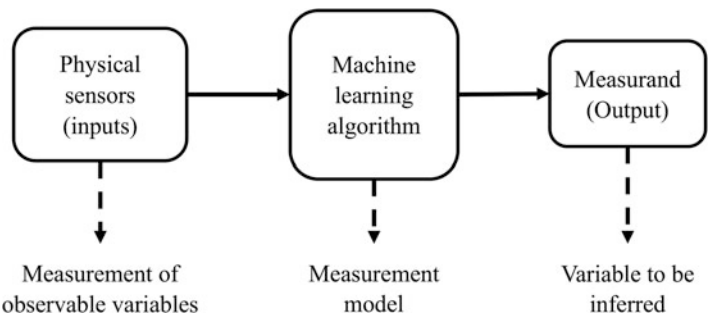

Fig. 1 Typical soft metrology system structure. (Source: Authors)

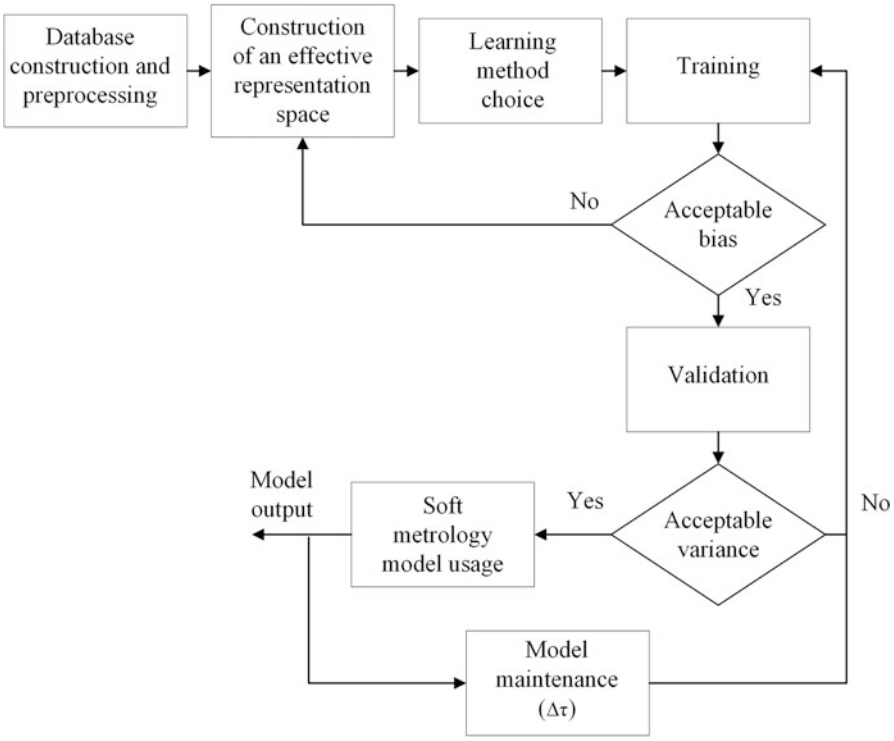

Fig. 2 Design of a data-driven soft metrology system. (Source: Authors)

have unlabeled samples, where only the inputs are stored. Later on, we will briefly discuss supervised and unsupervised learning methods.

To be able to train the models in the following stages, the database must be built keeping the following considerations in mind (Vallejo et al. 2020):

- It is important that the database includes values that cover the entire dynamic range that is to be modeled, as the system will not be able to model the process behavior in input or output values not included in the database.
- The representation quality of a soft metrology system depends on the reliability of the measurements embedded in the representation space for optimizing routines of prediction, regression, or inference of dynamic states. Additionally, the representation quality is deeply influenced by the amount of data in the database, i.e., the larger the database, the better the representation will be.
- The database must be preprocessed to find outliers and/or missing values. Also, normalization and noise filtering are usually performed.

However, building a database entails challenges (Susto et al. 2015) that sometimes make it unviable to achieve an optimal REsult. For starters, it is not always easy to obtain labeled samples, as the output is usually a difficult or expensive

variable to measure. It is also possible that some of the available input variables in the system may be irrelevant for the inference or several of them may be redundant. Likewise, some processes have delays between inputs and outputs that must be characterized in order to determine the concordance among the points in the database.

For these reasons, the following stage in the design process is the construction of an effective representation space. This stage is devoted to improving the quality of the representation, by mapping the points in the database to a proper representation space made up of features that capture the intrinsic information embedded in the measurements acquired with the physical sensors (Terzi et al. 2017). This is achieved by means of feature selection and/or feature extraction techniques, which do not only allow a better representation of the information but can also help reduce the dimensionality, thus improving the computational complexity in the model development. Feature selection techniques are essentially intended to eliminate irrelevant and/or redundant variables, which do not contain useful information for the inference process, while feature extraction techniques involve a functional transformation of the available variables to build a new space of features that better represents the information.

There are essentially four approaches to this stage (Chandrashekar and Sahin 2014):

1. Filter approaches, in which a relevance criterion is used to rank the input variables according to a correlation coefficient between inputs and outputs, such as Pearson correlation coefficient (Bidar et al. 2018), average mutual information (AMI) (Wang and Liu 2017), or variable importance in the projection (VIP) (Wang et al. 2015), among others. Filter approaches have the advantages of being simple and they avoid overfitting; however, they do not consider the performance of the following stages of the design process in the choice of the relevant variables.
2. Wrapper approaches, where subsets of the input variables are analyzed, and the inference error of each subset is used in an objective function to optimize the combination of variables. A common way of implementing a wrapper approach is the use of sequential selection algorithms, such as stepwise regression (SR), sequential forward selection (SFS), and sequential backward selection (SBS), where variables are added or removed according to the inference error (Kim et al. 2017). Another option for wrapper approaches is least absolute shrinkage selection operator (LASSO), which favors simple models by penalizing the complexity of the model reducing the coefficients of irrelevant variables to zero (Susto et al. 2015). Unlike filter approaches, wrapper approaches do consider the performance of the following stages, supporting an optimal representation. However, these methods are more complex and can suffer from overfitting.
3. Embedded approaches, in which the training of the learning method chosen to model the system and the obtainment of the effective representation space are performed at the same time. Therefore, there is no need for a separate stage involving feature selection/extraction procedures. Currently, the most commonly used method in this kind of approach is deep learning (Zeng and Spanos 2009).

These approaches also consider the performance of the following stages, but they are computationally less expensive than wrapper methods. Also, these methods can suffer from overfitting.

4. Transformation approaches, where the original variables are transformed to obtain new features (i.e., latent features), seeking a better representation of the underlying information. A widely known example of transformation approaches is principal component analysis (PCA) (Yuan et al. 2015; Maione et al. 2019). An important advantage of this technique is that it allows dimensionality reduction without the loss of information caused by eliminating features. Also, it helps avoiding overfitting. However, the new features can be difficult to interpret, and they are not chosen according to the performance of the following stages.

The effective representation space obtained from these procedures is then used as an input for a learning algorithm, which is used to model the system dynamics. In the following stage, a specific technique is chosen that will be used to create the model and that will essentially define the measurement model structure. However, there are several options in this step and the choice depends on many aspects. The most relevant elements that must be considered are the nature of the variable that needs to be estimated, the database structure, and the process linearity.

- *The nature of the variable that needs to be estimated is related to the objective of the model.* If the goal is to estimate a continuous numerical value (e.g., infer the value of a flow rate using information about the vibrations on the pipe (Vallejo et al. 2021)), the model can be created using a regression algorithm. The specific choice also depends on the linearity of the process. If the behavior is linear, it is possible to use techniques such as multiple linear regression (de Morais et al. 2019) or principal component regression (Slišković et al. 2011); but if the process is nonlinear, artificial neural networks (Hirai and Kano 2015), support vector regression (Popli et al. 2018), or extreme learning machines (Wang et al. 2019b) are more appropriate. However, if the objective is to obtain a discrete value that may correspond to a state, class, or type (e.g., inferring from electromyographic signals whether a patient suffers from myopathy or amyotrophic lateral sclerosis or is a healthy subject (Vallejo et al. 2018)), this requires a classification algorithm, such as linear discriminant analysis (LDA) (Wang et al. 2019b), K-nearest neighbors (K-NN) (Rajasekar and Sharmila 2019), support vector machines (SVM) (Haq and Djurdjanovic 2016), or artificial neural networks (ANN) (Ruiz-Gómez et al. 2018).
- *The database structure is also an important factor to keep in mind as it is the main element used to train the chosen method.* If the database consists of a collection of labeled samples, where every example of input values has a correspondent value for the output, supervised learning methods may be used. However, sometimes it is impossible to obtain labeled samples, and in these cases, it is necessary to choose a method that allows unsupervised learning. The difference between supervised and supervised learning will be discussed below.

Once the method has been chosen, it is necessary to train the model. The training process consists of an iterative transformation of the values and/or the structure of the model, until it has learned the system dynamics represented by the database. As mentioned before, there are two main approaches for this stage: supervised and unsupervised learning.

Figure 3 illustrates the supervised learning scheme, where the model starts with initial values for its parameters (this may be random), which are then updated according to the error between the model output and the expected value contained in the label. This updated procedure will be iteratively repeated until a desired minimum error has been reached.

On the other hand, if the database does not contain labels, it is necessary to use an unsupervised learning algorithm, where structures in the representation space are found according to some criteria (e.g., finding trends, patterns, or clusters). Figure 4 illustrates the scheme of an unsupervised method, were the model starts with initial values for its parameters, that may be random, which are then updated according to

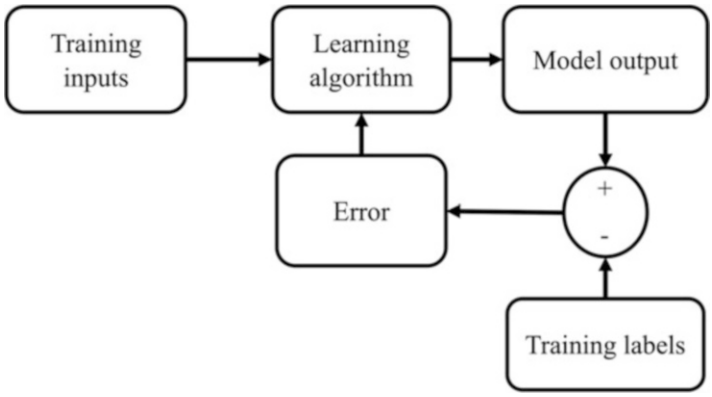

Fig. 3 Supervised learning scheme. (Source: Authors)

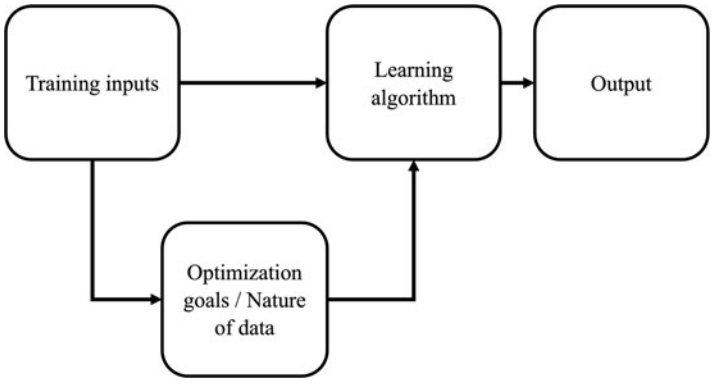

Fig. 4 Unsupervised learning scheme. (Source: Authors)

the nature of the data and the optimization goals, defined by the previous conceptualization that the designer has established for the system.

It is important to clarify that, before training, the database is separated in two portions (generally using a random split algorithm), and only one of them is used in the training procedure. The remaining portion is reserved for validating the model after training, as is explained below.

Training is successful if an acceptable model is obtained that represents the relationship between inputs and outputs with a relatively low bias, that is, with an acceptable error between the output given by the model and the target value contained in the database. If the training process does not yield a low bias, the structure of the representation space and/or the learning method choice must be reconsidered before attempting a new training procedure.

On the other hand, if training is successful, then the model must be validated. This means that the model is presented with the data contained in the remaining portion of the database. The error between the model output for this new data and the target value in the database will show if the model can generalize a good behavior in the presence of previously unseen data, with an acceptable variance in the results. If the system does not have generalization capabilities, it is necessary to go back to the training process.

Once the soft metrology model is successfully validated, it is possible to use it as a measurement model to infer the output value in the presence of new inputs. However, it is important to consider a periodical model maintenance, where the system is retrained with a certain periodicity ($\Delta\tau$), to ensure that the method adapts to changes in the process dynamics over time due to elements such as deterioration of pieces and changes in environmental conditions, among others (Lu and Chiang 2018; Souza et al. 2016).

Soft Metrology System Applications

As mentioned previously, soft metrology systems have been used in a variety of contexts and applications, although, sometimes, they are referred to with different terms. The first expression used for this type of indirect measurement strategy was "inferential sensor," which became popular in the 1970s with the development of inferential control for industrial applications, and it continues to be used in chemical and process industries (Mojto et al. 2021; Wang et al. 2021; Khatibisepehr et al. 2013).

However, in the 1990s, the term "soft sensor" was popularized in the literature, becoming a synonym of inferential sensor, and it was extensively used in many applications (Kano and Ogawa 2010). Some of the fields that have used the term soft sensor more frequently are in chemistry, process industries, and oil refineries, where it is possible to find works such as the one presented by Chen and Ngu (2023), who investigated the predictive performance of soft sensors based on locally weighted partial least squares (LW-PLS), illustrating the potential application in several processes related to chemistry such as the predictive control of a wastewater

treatment process and the polymerization of methyl methacrylate. Another example is reported in Beiroti et al. (2023), where a model was trained for estimating the biomass of *Pichia pastoris* yeast, which is used to express hepatitis B surface antigen for vaccine production. In an industrial context, there are many applications, such as the work by Vemulapalli and Venkata (2022), which describes a method to estimate the correct output of an orifice flow meter in the presence of disturbances such as plate contamination, flow density changes, pipe corrosion, or incorrect insertion of the plate, and the work by Amini et al. (2021), who used an artificial intelligence model to predict water flow. In oil refineries, a benchmark application is the measurement of the butane content on a debutanizer column (Guo et al. 2020; Yuan et al. 2018). However, the above are not the only application areas of soft sensors; for example, in the medical field, this type of technology has also been applied to the continuous monitoring of pressure and temperature, to minimize the potential harm of pressure injuries in hospitalized patients (Oh et al. 2021). In aquaculture, some studies have proposed water quality monitoring using soft sensors (Inderaja et al. 2022).

In some cases, the development of soft sensors can be assigned to different fields of application simultaneously. An example is reported in Jo et al. (2022), where a multifunctional soft sensor capable of simultaneously detecting six stimuli (pressure, bending stress, temperature, proximity, ultraviolet light, and humidity) was developed. Thus, there are many potential applications in the industrial, medical, and military fields.

Another designation frequently used is "virtual sensor," which is usually implemented in automotive and aircraft applications, such as the estimation of vehicle's unsprung mass (UM) vertical velocity in real time (Kojis et al. 2022), the estimation of vehicle roll angle (Sieberg and Schramm 2022), or the implementation of virtual angle of attack (AOA) sensors in aircrafts (Mersha and Ma 2022). Virtual sensors are also used in applications for intelligent buildings, for energy or ventilation systems (Koo et al. 2022; Torabi et al. 2021), as well as in air quality measurements for estimating the concentration of black carbon (Fung et al. 2020; Rovira et al. 2022).

There are also terms related to a very specific domain of applications, such as virtual metrology, which is used in semiconductor manufacturing (Chien et al. 2022; Choi et al. 2022; Shim and Kang 2022), and electronic tongues, used in the characterization of flavors in the food industry for many products like edible fungi (Yang et al. 2022), milk (Salvo-Comino et al. 2022), tangerine peel (Li et al. 2022b), honey (Lozano-Torres et al. 2022), peaches, and so on. Likewise, the food industry has used electronic noses for characterizing flavors and odors (Zhang et al. 2022; Li et al. 2022a), but this application can also be found in other contexts, such as breath test analysis for the detection of silicosis in miners (Xuan et al. 2022), the detection of pests and diseases in agriculture (Zheng and Zhang 2022), and the discrimination of pure biofuels (Mahmodi et al. 2022), among others.

As mentioned before, the term soft metrology has been proposed to encompass all designations, although initially it was specifically used to refer to measurement

systems that characterize human perception with the five senses (Bergman et al. 2020; Iacomussi et al. 2016; Fanton 2019; Rossi 2016; Pointer 2003).

In addition, many measurement applications use machine learning tools to model the corresponding process but do not use any of the aforementioned terms. An example is reported in X. Wang et al. (2022), where the system characterization in terms of product quality in an oil refinery is accurately estimated through a model and a parameter optimization process. In the same way, in Wen et al. (2022), deep learning algorithms are used to measure and differentiate movement features and joint positions in the human body.

Challenges in Soft Metrology Applications

Although nowadays soft metrology systems (soft sensors, virtual sensors, virtual metrology systems, etc.) are extensively used in many contexts, there are still many challenges in their development, especially in two aspects:

- The development of a soft metrology system for a specific application requires a careful assessment of the process under analysis and an experimental design to obtain the appropriate database. Each application may entail specific complexities and challenges. In subsection "Soft Metrology System Implementation: An Example of Flow Rate Measurement," an example is presented to illustrate this point.
- There is a lack of quality metrics associated with measurement for soft metrology systems. Particularly, there are no standardized procedures to determine the uncertainty associated with an indirect measurement system that has a machine learning-based inference process. This issue is further analyzed in section "Measurement Uncertainty in Soft Metrology Systems."

Soft Metrology System Implementation: An Example of Flow Rate Measurement

Flow rate measurement is an essential task in industrial processes and urban water supply systems (Dinardo et al. 2018), among others. For this reason, a wide variety of measuring instruments has been developed. However, these devices can be expensive (Campagna et al. 2015) and usually intrusive (Medeiros et al. 2015, 2016). These are critical disadvantages in cases where many sensors must be installed, when portable measurement systems are required, or when measurements are taken in adverse conditions. Hence, indirect measurement systems based on pipe vibration analysis have been developed with the aim of achieving less expensive, portable, and nonintrusive flow rate measurement devices (Vallejo et al. 2021).

Since the 1990s, the correlation between vibrations and flow rate in a pipeline has been studied. In 1992, the INEEL (Idaho National Environmental and Engineering Laboratory) led a series of analysis and found a direct correlation between the

standard deviation of the signal produced by an accelerometer attached to the pipe and the flow rate, which led to further studies intended to characterize this correlation with metrological purposes (Evans et al. 2004), so different techniques have been proposed for indirect flow rate measurement.

In the literature, the correlation between the fluid flow rate through a pipe ($\dot{Q}$) and the acceleration affecting the pipe wall in the radial direction is described by a series of linear relationships ($\propto$) expressed by Eq. (1) (Dinardo et al. 2018; Campagna et al. 2015):

$$\dot{Q} = A\overline{U} \propto u' \propto \tau_w \propto \frac{\partial^2 r_w}{\partial t^2} \tag{1}$$

where A is the pipe cross-sectional area, $\overline{U}$ is the average flow velocity, u' is the flow velocity fluctuation along the axis, τ_w is the shear stress in the pipe, and $\left(\frac{\partial^2 r_w}{\partial t^2}\right)$ expresses the acceleration in the radial direction.

However, there is no precise and unique phenomenological equation that can express this relationship, and several studies have shown that there is a wide variety in aspects such as the sensor used to measure vibrational features, the machine learning routines used to create the model, and the range of flow rates studied, among others.

In order to acquire signals that contain the vibratory characteristics of a pipe, three types of sensors have been used: accelerometers, which register the acceleration in several axes and are attached to the wall of the pipe (Medeiros et al. 2015, 2016; Evans et al. 2004; Kim et al. 2008; Venkata and Navada 2018; Pirow et al. 2018); laser Doppler vibrometers (LDV), which measure the amplitude and frequency of the vibration using a laser and an interferometer (Dinardo et al. 2013, 2018; Campagna et al. 2015; Fabbiano et al. 2020); and acoustic sensors, which focus on the vibrations associated with acoustic dynamics (Göksu 2018; Jacobs et al. 2015; Safari and Tavassoli 2011).

Each type of sensor exhibits specific advantages and disadvantages:

- Accelerometers measure vibrational characteristics in three axes, increasing sensor accuracy by improving the inference space (Mohd Ismail et al. 2019).
- LDVs do not require direct contact with the pipe surface, but this feature is problematic in cases with limited access or without line of sight (Rothberg et al. 2017).
- Acoustic sensors are contactless too, but they may be more susceptible to background noise.

The acquired signals must be filtered to reduce noise in the representation space (for any type of sensor), using several types of filters such as Butterworth filters, “notch comb” digital filters (Medeiros et al. 2015, 2016), median filters (Hu et al. 2014), and filters based on wavelet transform (Dinardo et al. 2018).

To improve the representation space, frequency analysis has been proposed, which relies on the fact that the natural frequency of a fluid moving through a pipe decreases when the flow rate increases (Evans et al. 2004). Frequently, the power spectrum is used as an input for the inference algorithm and the amplitude of the first harmonic (Dinardo et al. 2013, 2018; Fabbiano et al. 2020; Jacobs et al. 2015; Safari and Tavassoli 2011), or the center frequency of the first harmonic (Evans et al. 2004) as representation features. Time analysis has also been used, where flow rate is characterized in terms of the signal amplitude (Jacobs et al. 2015) or statistical parameters such as the statistical moments (Medeiros et al. 2015, 2016), the frequency domain average time series (Evans et al. 2004; Pirow et al. 2018), or the root mean square (RMS) value (Dinardo et al. 2018; Fabbiano et al. 2020; Jacobs et al. 2015). Lastly, some studies report approaches that use decomposition techniques that simultaneously analyze the signal time-frequency features, using wavelet transform (Göksu 2018).

As to the learning method choice, the simple linear model with least squares fitting is the most implemented (Dinardo et al. 2013, 2018; Campagna et al. 2015; Pirow et al. 2018), but nonlinear models, such as polynomial regression (Medeiros et al. 2015, 2016; Evans et al. 2004; Safari and Tavassoli 2011) and a third-order square root curve (Y. Kim et al. 2008), have also been proposed. Similarly, regression models by parts, which combine linear and polynomial regression, are also used (Medeiros et al. 2015, 2016). Finally, some studies have explored the use of ANN for flow rate estimation (Venkata and Navada 2018; Göksu 2018), since ANN performs well in highly nonlinear processes, and it is not necessary to have prior knowledge of the data.

Results reported in the literature of soft metrology systems applied to flow rate are diverse, and the comparison is complex due to differences in operational conditions related to magnitude, equipment characteristics, data acquisition, sampling, and installation parameters, among others. Some papers have reported the influence of certain operating parameters on each model, such as the pipe material and diameter (Campagna et al. 2015), the sensor location (Safari and Tavassoli 2011), the acquisition time (Medeiros et al. 2015, 2016), and the mechanism that drives the fluid (Dinardo et al. 2013). Also, the model accuracy varies for each specific work. Particularly, some authors report the results in terms of the R^2 parameter (Evans et al. 2004), the root mean square error (RMSE), or the mean absolute error (Kim et al. 2008; Göksu 2018; Jacobs et al. 2015). Other authors do not even report results in terms of accuracy, since their research focuses on trying to find a deterministic relationship between vibration characteristics and flow rate, and testing the influence of parameters, such as the pipeline characteristics (Campagna et al. 2015; Dinardo et al. 2013).

It is important to observe that soft metrology systems are usually developed for very specific contexts and applications, and it is necessary for the designer to carefully analyze the scope of the proposed system and the range of applications where it can potentially be used. Changes in aspects such as the measurement range, installation specifications, material, and disposition of the structures, among others, require that the system be trained again, with new databases that incorporate new

conditions or scenarios. According to this, several challenges rise around specific experimental designs for database construction, which must be oriented to a wide range of applications and operational or functional states.

Measurement Uncertainty in Soft Metrology Systems

In accordance with the International Vocabulary of Metrology (VIM), the measurement uncertainty is defined as a nonnegative parameter characterizing the dispersion of the quantity values being attributed to a measurand, based on the information used (Joint Committee for Guides in Metrology (JCGM) 2012). In this document, the word "quantity" is used as it is the one used in the translations of the VIM and the GUM into English. Note however that "magnitude" could also be used.

Measurement processes in engineering require the estimation of uncertainty associated with the value, which expresses the incomplete knowledge of the quantity and ensures the validity of results. The uncertainty estimation process is a key tool in measurement and is also a fundamental tool for achieving measurement compatibility. The importance of uncertainty can be translated to a broader sense, as it contributes to the achievement of the objectives of the organization of metrology worldwide. In particular, the International Committee of Weights and Measures (CIPM) expresses: "Mission: The main task of the CIPM is to promote worldwide uniformity in units of measurement. . .".

The main issue with the soft metrology systems is the lack of quality metrics that can be correlated with the measurement, specifically the absence of standardized procedures to determine the uncertainty of the estimated value. Likewise, it is difficult to find information in the literature that addresses the uncertainty estimation of the measured value in soft metrology. Only a few papers can be found that propose mathematical models to analyze uncertainty, and they are mostly focused on model-driven implementations, for instance, in Song et al. (2013), a virtual flow meter was developed including a procedure for uncertainty analysis. Additionally, in Cheung and Braun (2016), a general method for calculating the uncertainty of virtual sensors for packaged air conditioners is presented. These papers identify different uncertainty sources that can be categorized as:

- Database uncertainty included in the generation of the inference model.
- Physical sensor uncertainty related to the instruments used to measure the input variables in the inference model.
- Uncertainty due to environmental noise (i.e., external, not fully controllable, unpredictable disturbances).
- Uncertainty associated with the inference process.

Database uncertainty is associated with the instruments used to capture the database, with which the inference model is trained. Likewise, the physical sensor uncertainty corresponds to the input variable uncertainty in the model, but in this case, it is related to the online operational process after the model is trained.

The uncertainty due to environmental noise can also be considered as related to the measurement instruments, as the noise usually affects the circuits associated with the sensor conditioning and digitalization stages. The uncertainty associated with the inference process is directly proportional to the model training quality since this uncertainty is an expression of learning and representation capabilities that the inference routine has when capturing the system phenomenological dynamics.

Particularly, the sensor uncertainty corresponds to the same uncertainty estimated in conventional measurement systems which can be addressed with non-stochastic tools, such as the Guide to the Expression of Uncertainty in Measurement (GUM) (Joint Committee for Guides in Metrology (JCGM) 2008), and stochastic tools, such as Monte Carlo method (Joint Committee for Guides in Metrology (JCGM) 2008). It is important to note that the estimation of the sensor and database uncertainty, the GUM, is a worldwide success case as it has become the obligatory reference for uncertainty estimation. However, this guide has limitations, and some inconsistencies regarding the VIM or even the GUM supplements. From a technical point of view, the GUM presents a summary of steps for the uncertainty estimation:

1. Mathematically express the relationship between the measurand and the input quantities, given by

$$y = f(x_1, x_2, \ldots, x_N) \tag{2}$$

   where y is the measurand (output value), N is the number of input quantities, and x_i are input quantities.

2. Estimate the input quantities.
3. Evaluate the standard uncertainty of each estimated input.
4. Evaluate covariances associated with correlated inputs.
5. Calculate the estimated measurand value.
6. Determine the combined standard uncertainty of the measurement result.
7. If necessary, give an expanded uncertainty in order to have a greater probability of coverage.
8. Report and appropriately express the measurement result with the corresponding uncertainty.

In the case of a soft metrology system, it is not always possible to precisely apply the guide as the complexity of the system often makes it impossible to implement the first and the fifth steps, i.e., the relationship between the measurand and the input quantities is modeled using a machine learning algorithm that may be complex and does not always allow a classical mathematical function to be built or the evaluation of covariances. In any case, the GUM can be used to estimate the database and physical sensor uncertainty.

Another alternative is the Monte Carlo method, which simulates a large number of measurements using a random number generator; for this reason, this method can

be known as a "brute force" routine. To do this, algorithms for generating values associated with a particular probability distribution are implemented. In this regard, the Monte Carlo method requires a mathematical model for describing the measurand and the probability distributions associated with input quantities. Monte Carlo simulations have been used in the development of soft metrology systems as a tool to generate the measurement model, although not with the aim of estimating its uncertainty.

Commonly, GUM and Monte Carlo methods are compared, and pros and cons are analyzed. These methods are not totally independents as they use the same information of assumptions made for input probability distributions. The main advantage of the Monte Carlo method is that it does not require any assumptions about the output probability distribution. However, it has a high computational complexity, which can sometimes be unnecessary in scenarios where the GUM method describes the measurand well with its uncertainty. In general, these methods perform better when used in a complementary way.

On the other hand, the less studied source of uncertainty in measurement systems based on soft metrology approaches is related to learning models for inference, regression, or forecasting, which can be integrated within the framework of the whole expanded uncertainty. The rest of this document will focus on the source of uncertainty derived from computer applications, since the current literature on this subject shows analysis from the point of view of machine learning training efficiency, but not from the perspective of conventional metrology or from the implications that this has on an indirect measurement system. This means that in computational model uncertainty, there are important challenges that will need to be solved in order to reach a generalized use of soft metrology systems.

Computational Model Uncertainty

It is important to note that, depending on the field of study, different definitions of uncertainty may be found (Siddique et al. 2022), and, although the basic concept is the same in all cases, the terminology and specific methods related to uncertainty in metrology differ from those in machine learning or other computational applications.

The goal in a soft metrology system, as it is in any machine learning application, is to create an approximation model of the phenomenological relationship between the input variables and the magnitude of the value to be inferred. Indeed, a soft metrology system is affected by uncertainties inherent to any model that is an imperfect approximation of a phenomenological dynamic behavior, i.e., uncertainties associated with a process of inductive inference and from incorrect suppositions or approximations in the model structure, among others (Hüllermeier and Waegeman 2021). Traditionally, machine learning models have focused on training approaches that aim to optimize the model parameters to achieve the minimum prediction error, but do not consider uncertainty. However, there is a growing interest in uncertainty quantification, as well as in standardizing other quality metrics that

allow for model validation and evaluation of quality, reliability, and applicability (Vishwakarma et al. 2021).

Some authors propose that there are three dimensions of uncertainty in systems that model a behavior: the location, level, and nature of uncertainty (Walker et al. 2003).

- **Location of uncertainty:** This is where the uncertainty is manifested within the model structure. The specific list of uncertainty locations in a system may vary according to the system structure, but some general locations are representation boundaries and completeness, model structure, computational realizations, inputs, methods used to calibrate the model parameters, and, finally, the output accumulated uncertainty.
- **Level of uncertainty:** This refers to where the uncertainty can be placed between complete deterministic knowledge and total lack of knowledge of the study case system or process.
- **Nature of uncertainty:** This is an analysis of whether the uncertainty is due to a lack of knowledge or to the inherent randomness of the phenomenological dynamics.

This last dimension has been the most studied in machine learning literature and, although the source of uncertainty in machine learning is traditionally modeled in a statistical way, an increase in the use of machine learning in high-risk, optimal, or robust applications requires more demanding characteristics in quality and security, and for this reason, it is very important to identify uncertainty components that can be minimized and those which cannot be (der Kiureghian and Ditlevsen 2009). These uncertainty sources have been labeled epistemic and aleatory uncertainty, respectively, and this classification of uncertainty sources has become popular in machine learning. However, some authors point out that, although this categorization may appear evident, the determination of which uncertainty sources can be minimized is sometimes ambiguous as of the context may change the decision or representation conditions. Therefore, it is important to analyze the uncertainty sources in light of the context and the chosen model structure (Hüllermeier and Waegeman 2021; der Kiureghian and Ditlevsen 2009).

Epistemic Uncertainty

The concept of epistemic uncertainty is related to the lack of knowledge about the process dynamics, which can lead to a model that does not precisely represent the underlying function. Thus, epistemic uncertainty is related to improperly tuned model parameters, wrong assumptions about the model functional structure (Combalia et al. 2020; Nguyen et al. 2022), number of examples in the database, and data density in the range of study (der Kiureghian and Ditlevsen 2009; Osman and Shirmohammadi 2021; Posch and Pilz 2021). This uncertainty can be mitigated by taking actions that improve the model and data functionalities.

Epistemic uncertainty is usually formalized as a probability distribution over the model parameters or weights (Kendall and Gal 2017;

Lakshminarayanan et al. 2017). The main tool used in machine learning is Bayesian statistics, which provides a conceptual framework for expressing uncertainty toward algorithmic routines (Combalia et al. 2020). In Siddique et al. (2022), a categorization of approaches for uncertainty quantification in machine learning is mainly proposed as follows:

- *Frequentist inference*: This is the use of traditional machine learning techniques with a training procedure that is intended to give a point estimate for the model parameters w. In this case, the trained model will have a certain standard error, and therefore this approach does not actually provide information about uncertainty estimation.
- *Bayesian inference*: This is the use of Gaussian process regression (GPR) or Bayesian neural networks (BNN). This approach does not consider a point estimate for w, but rather considers a posterior probability distribution over the parameters $p(w)$, given the observed data in the training dataset.

To illustrate the use of Bayesian inference as a tool for modeling uncertainty in machine learning, we discuss BNN in contrast to conventional artificial neural networks (ANN).

Artificial neural networks (ANN) represent an arbitrarily complex function $y = f(x)$ using an interconnexion of relatively simple units called neurons. Each neuron is composed of a linear combination of weighted inputs and a nonlinear activation function, such that the output of each neuron is $y = \varphi\left(\sum_{i=1}^{N} x_i * w_i\right)$, where y is the neuron output, φ is the activation function, x_i is the *ith* input, w_i is a weight associated with the *ith* input, and N is the number of inputs to the neuron. Figure 5 illustrates the typical structure of a neuron, with N inputs.

An ANN is an arrangement of several neurons, structured in layers. Figure 6 shows an example of a three-layer ANN, although the number of hidden layers and neurons in each layer varies according to the designer's criteria. Each connection between neurons has a parameter w that represents the weight of that specific connection. The training algorithm for an ANN aims to find the optimal point

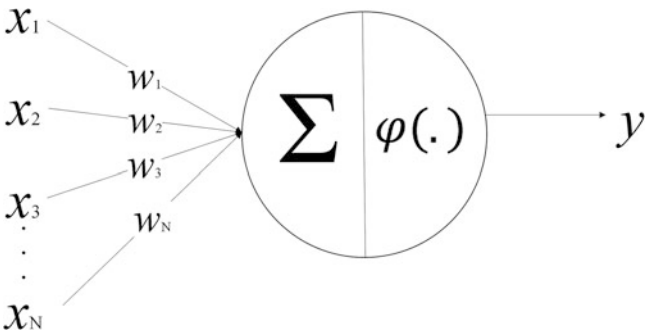

Fig. 5 Neuron structure. (Source: Authors)

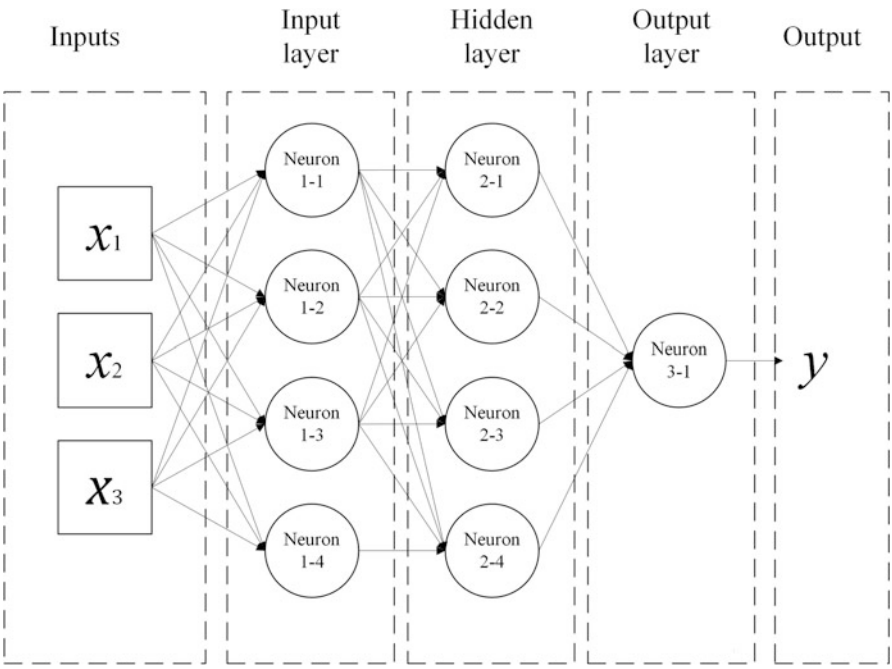

Fig. 6 ANN structure. (Source: Authors)

estimate for parameters w using strategies such as gradient descent, which are based on differentiation.

Bayesian neural networks are a type of neural network that use Bayesian inference to express the uncertainties related to the model. The main idea is to model the uncertainty in the weights w, so that the training data is used to determine a posterior distribution over the weights (instead of a single value) $p(W|X, Y)$, where $W = \{w_1, w_2, \ldots, w_N\}$ is the set of weights in the network and (X, Y) are the training examples, being $X = \{x_1, x_2, \ldots, x_N\}$ and $Y = \{y_1, y_2, \ldots, y_N\}$. From this posterior distribution, a predictive distribution of a test sample x is obtained, which is noted as $p(y|w, x)$, and is used to quantify the predictive uncertainty.

However, exact Bayesian inference is not computationally executable for neural networks because this approach establishes a high-dimensional and nonconvex problem where the needed integrals are too difficult to compute and directly sampling the posterior distribution is not viable. Several alternatives have been proposed to deal with these problems, such as:

- Markov chain Monte Carlo (MCMC) (Bardenet et al. 2017; Neal 1996), where the idea is to build a sequence where each sample depends only on the previous one, such that the samples are distributed following a desired distribution.
- Variational inference (Blundell et al. 2015; Louizos and Max 2016; Olivier et al. 2021), where this method scales better than MCMC, and a variational distribution

is proposed until it is as close as possible to the exact posterior, usually using as the measure of closeness as the Kullback-Leibler divergence (KL divergence).
- Probabilistic back-propagation (PBP) (Hernandez-Lobato and Adams 2015), where an algorithm sequentially approximates the distributions of the weights using one-dimensional Gaussians.

Additionally, epistemic uncertainty has been expressed using dropout training in deep neural networks (Gal and Ghahramani 2016), where an ANN is trained using aleatory dropout to estimate the uncertainty of the model. Dropout techniques can be configured on a regularization scheme to reduce overfitting. Particularly, Bayesian inference uses multi-class predictive entropy to analyze the epistemic uncertainty of the system (Kurmi et al. 2021).

Uncertainty estimation techniques for neural networks based on Monte Carlo sampling have been used in classification problems (Lakshminarayanan et al. 2017; Gal and Ghahramani 2016). Results show that uncertainty metrics can be used successfully to detect samples that are out of distribution or are difficult to integrate (Combalia et al. 2020). Also, Bayesian analysis has reported good results modeling uncertainty in conventional neural networks (Kendall and Gal 2017).

Aleatory Uncertainty

Aleatory uncertainty is due to nondeterministic elements that are inherently aleatory in the process dynamics and is mainly related to the noise in the observations, as well as to the influence of hidden variables or measurement errors (Kendall and Gal 2017; Tagasovska and Lopez-Paz 2019). Unlike epistemic uncertainty, aleatory uncertainty is not reduced by a deeper knowledge of the modeled process and is usually modeled by probability distribution in the output of the inference model (Kendall and Gal 2017). This type of uncertainty can be categorized as homoscedastic or heteroscedastic, depending on whether the noise is assumed to be identical or different in each of the process inputs (Wentzell et al. 2021; Ayhan 2018).

To model aleatory uncertainty, some studies, such as Ayhan (2018), propose data augmentation techniques, using simple random transformations on the data to capture heteroscedastic uncertainty in deep neural networks. A similar example is found in G. Wang et al. (2019a) where data augmentation techniques are used in medical imaging applications. In this approach, aleatory uncertainty is modeled by masking different geometries on images and analyzing prediction diversity according to the distribution variance and entropy. Another approach for estimating aleatory uncertainty consists of building a model that fits the information loss, guaranteeing that the model does not fit the noise in the data (Liu and Han 2021).

As mentioned earlier, the distinction between aleatory and epistemic uncertainty can be fuzzy or rough, and each one can be determined by a model choice (der Kiureghian and Ditlevsen 2009). In fact, some methods previously mentioned have also been used to estimate both aleatory and epistemic uncertainty. In this regard, in Junhwan et al. (2022), dropout Bayesian approaches with Monte Carlo simulation are proposed to predict and estimate uncertainty, where the aleatory uncertainty is

associated with noisy data location and outlier detection, while epistemic uncertainty is used to quantify the model performance.

Lastly, some studies make the distinction between epistemic and aleatory uncertainty, but the focus is not on the uncertainty estimation, but rather on identifying errors due to aleatory uncertainty so that they are not considered in the design process (Ali et al. 2021), or in data relabeling (Redekop and Chernyavskiy 2021).

Discussion and Final Considerations

Most applications of soft metrology are required in smart manufacturing systems, which digitally represent every aspect in a new technological platform, from the design stage to manufacture, using multi-objective control and monitoring tools, and multi-structural, multi-sensorial, and remote topologies of connectivity, in order to mobilize the dominant paradigm of internal combustion engines toward a culture of computer challenges and connectivity (Mesa-Cano and Delgado-Trejos 2021).

A new production ecosystem based on Industry 4.0 platforms requires a new perception and processing vision with new challenges and opportunities, where the concept of measurement has changed from a numeric value to representation data. In this sense, a measurement process must obtain enough information from a dataset to assess, understand, and make decisions on a phenomenological behavior. However, up to now, the uncertainty estimation in soft metrology applications is an open issue when computer routines integrate an important component in the whole measurement process. According to this, some new concepts of uncertainty estimation for computer routines have been presented in this chapter, with the aim of illustrating new analysis strategies for soft metrology approaches.

On the other hand, soft metrology structures can adapt representation spaces using measurement sets and selecting those with a high information level of the phenomenological dynamics, associated with a complex and/or expensive study variable. In this sense, decision-making systems based on measurements require prior knowledge about the nature of the data to optimize routines of prediction, regression, or inference of dynamic states, taking into account constraints in terms of the error and uncertainty mitigation within specific tolerance frameworks. Likewise, the construction of an effective representation space is problematic in terms of training sample size, as measurements or process observations can be easily obtained, but the categorization or labeling by technical experts is an expensive and complex procedure. It is also important to note that phenomenological comprehension, associated with the measurement sets, is not always achievable, which is a key factor for assuring reliability in results from the computational routines.

Mechanisms derived from soft metrology show promising research opportunities in terms of generating representation spaces with new indirect measurement methods inspired by human perception, multivariate statistical analysis, machine and deep learning, and decision support systems, among others, seeking out intrinsic structures of complex measurements for finally achieving inference results with a proper and rigorous uncertainty analysis.

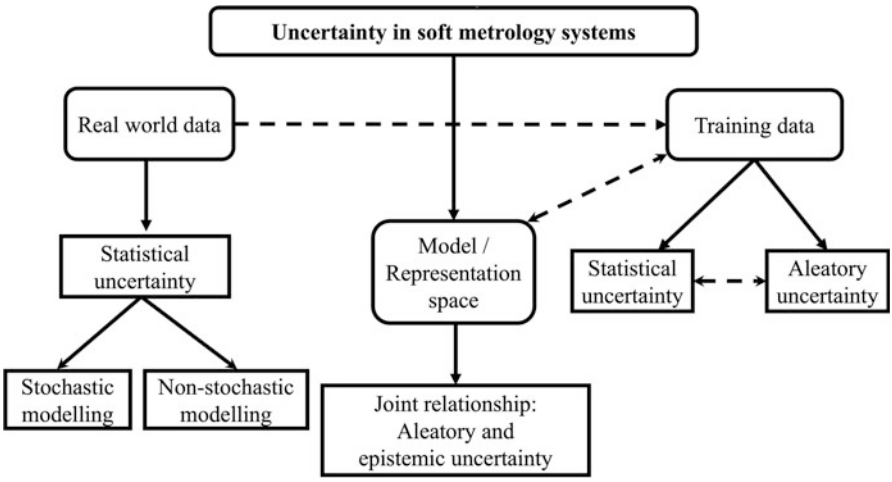

Fig. 7 Uncertainty in soft metrology systems. (Source: Authors)

Figure 7 depicts a structure that illustrates uncertainty components in a soft metrology system based on information from the literature. This structure considers two elements related to data: database uncertainty, associated with model training, and physical sensor uncertainty, associated with real-world data that constitutes the system input after training. The data-related uncertainty has an aleatory component, related to noise, and is modeled using either stochastic or non-stochastic methods. As the database is an essential input to the training stage, its uncertainty propagates throughout the model, which also has an epistemic uncertainty component, associated with the lack knowledge about the process features, i.e., the representation model/space does not have enough information about the intrinsic dynamics. As mentioned, the epistemic uncertainty can be reduced by increasing the amount of data, but this reduction is limited by the designer's phenomenological comprehension. Also, the epistemic uncertainty can be mitigated by estimating and selecting relevant features from a robust representation space, using relevance analysis techniques (Delgado-Trejos et al. 2009) .

In Vallejo et al. (2018), a conceptual scheme is presented, where training quality is directly related to representation quality, which is constrained within a certain range bound by the following factors: dataset size and phenomenological comprehension of the functional states of the variable of interest. From this conceptual analysis, for soft metrology systems, epistemic and aleatory uncertainty must be considered. In this regard, the conceptual scheme must be restructured, as depicted in Fig. 8 where the y-axis corresponds to three behaviors, which are the phenomenological comprehension (PhC), the epistemic uncertainty (EpU), and the aleatory uncertainty (AlU), as related to the dataset size and the learning capability in the x-axis. It is important to note that in this illustration the learning capability is directly proportional to the dataset size; therefore, the conceptual relationships between the y-axis variables and each x-axis variable are indistinctively assumed for all x.

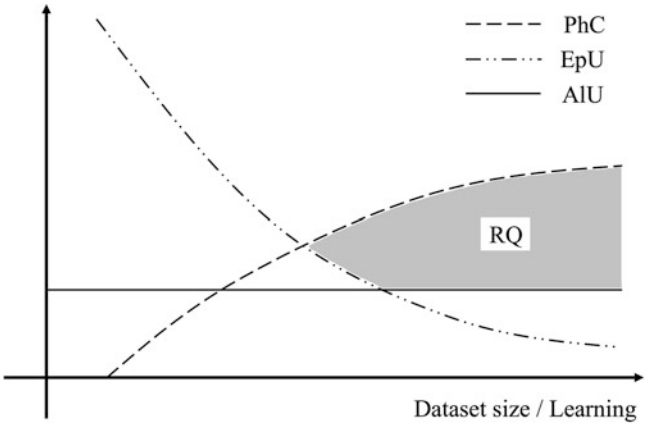

Fig. 8 Ideal region for representation quality (RQ) according to epistemic and aleatory uncertainty. (Source: Authors)

As observed in Fig. 8, the ideal region for representation quality (RQ) in soft metrology systems has an upper limit given by the phenomenological comprehension, which relates to how the computational models can be adjusted using the knowledge of technical experts, where factors that disturb the normal dynamics should preferably be included. Likewise, this region has a lower limit given by the aleatory uncertainty, as this uncertainty is linked to the internal randomness of the specific phenomenon, and with machine intelligence, it is not possible to figure out and establish decision boundaries below this lower limit. Also, as an additional constraint for some midpoints between the lower and upper limit, the epistemic uncertainty constitutes a boundary for the ideal representation quality, since this uncertainty is related to the lack of relevant representation features (knowledge derived from enough measurements regarding the specific phenomenon) that dictate how the system should behave, capturing its intrinsic dynamics.

Conclusions

In this book chapter, the concept of soft metrology and challenges related to two types of involved uncertainties are discussed. In general, the contribution of the epistemic and aleatory uncertainty in the global uncertainty of a measurement process for soft metrology must be focused on the uncertainty propagation derived from the computational effort in representation spaces, where the measure value is inferred from multivariate structures. Particularly, the intrinsic representation noise within data is associated with the aleatory uncertainty, since this natural noise comes from phenomenological disturbances produced by the interaction dynamics of the real world, and for this reason, it is considered independent of the model tuning of machine learning and the dataset size. In contrast, the epistemic uncertainty can be corrected by minimizing the model parameter tuning and assuring enough dataset

size, where the phenomenon understanding has an important role, considering that the upper limit of the representation quality in soft metrology is achieved based on how well the phenomenon is understood.

Finally, several soft metrology challenges are connected to the proper phenomenological comprehension for data acquisition where each representation state has enough number of samples, and this dataset allows the design of an appropriate indirect measurement method in terms of robustness and accuracy. Likewise, the uncertainty estimation, specifically, epistemic uncertainty, also requires phenomenological comprehension to lead the computational effort toward an effective representation space, where several data processing techniques, such as time-frequency analysis, nonlinear dynamics, and others, must be tested and analyzed in terms of representation quality and learning capability for inference, regression, and/or forecasting. Although there are many other challenges, it is important to note that the inclusion of strengths derived from computer intelligence routines in metrological processes must be further explored in the coming years.

Cross-References

- ▶ Advanced Techniques in Evaluation of Measurement Uncertainty
- ▶ Artificial Intelligence for Iris-Based Diagnosis in Healthcare
- ▶ Evaluation and Analysis of Measurement Uncertainty
- ▶ Machine Learning in Neuromuscular Disease Classification
- ▶ Measurement Uncertainty
- ▶ Modern Approaches to Statistical Estimation of Measurements in the Location Model and Regression
- ▶ The Quantifying of Uncertainty in Measurement
- ▶ Using AI in Dimensional Metrology

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Part IX

Optics in Metrology: Precision Measurement Beyond Expectations and Nano Metrology

Optical Dimensional Metrology

52

Arif Sanjid Mahammad and K. P. Chaudhary

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Abstract

Optical technologies continue to expand to ease engineering metrology and improve precision of measurements. Invariantly, the optical techniques are used for the continuous, high-speed inspections of items of all shapes, sizes, and materials in Industry 5.0. Combining optics with sensors and competitive digital procedure benefits the rapid, portable, and large measurement.

Though it is impossible to accommodate the vast knowledgebase of the optical metrology, an attempt is made to cover the glimpse of recent recommendations, well established, and accepted optical methods and dimensional metrology measurement that are covered in this chapter. The applications of different properties

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of light and their respect techniques are discussed. The implementation of optical concept for the dimensional metrology instrumentation is categorically described.

Finally, some of the new works and prospective research are given. Therefrom, new frontiers may be thought off.

Introduction

Optical technologies continue to expand and improve precision and ease in engineering metrology. Invariantly, optical techniques are used for the continuous, high-speed inspections of items of all shapes, sizes, and materials in industry 5.0. Combining optics with sensors and competitive digital procedure benefits the rapid, portability, and sensitivity to the large measurement volume.

Laser interferometry, confocal microscopy, and white light technique with the aid of computation-based digitizing technologies are extensively in practice for dimensional measurements (Harding 2008). On the contrary, X-ray tomography, and terahertz metrology are already in use for inspecting the internal geometry of the biological and engineering artifacts.

Metrology

Metrology can be precisely defined as the science of measurement concerning the application (OIML VIM 2000). Consumer requirements that inspire new products and their trade demand well-established regulations, authentic measurements, and reliable measuring instruments through legal metrology (ISO/IEC-17025 2017). Behind the scene, scientific metrology ensures timeline stability of reference standard value of quantities for the conformity of meeting the geometric specifications and the traceability of measurement to SI units (Quinn and Kovalevsky 2005; Pierre Fanton 2019; Foster 2010; Pendrill 2014). Nevertheless, the development of manufacturing, ensuring the suitability of measuring instruments and their calibration, is dealt with by applied metrology (Gonçalves and Peuckert 2011; ILAC 2007; CIPM MRA-D-05 2016). The optical method used in dimension metrology has its special place in all these categories. Because of obvious reasons and a large volume of literature, however, only unique concepts are emphasized here.

Measurement Standards

A physical quantity is measured by comparing it with a standard quantity of same kind (Ehrlich et al. 2007). The quantification will be expressed as a numerical value accompanied with by the unit. Since the meter convention, the most accepted unit of the standard of length (i.e., SI unit) is meter (https://www.bipm.org/en/publications/SI-brochure/ 2021). The standard quantity should qualify the following basic requirements (Aswal 2020; Phillips et al. 2001; Rumble et al. 2001):

(i) Easy to operate and feasible to disseminate.
(ii) Reasonable to scale to the measurand.
(iii) Quantitatively invariant – quantum-based standard.
(iv) Theoretically well defined – definition of SI units.
(v) Practically realizable – mise-en-practique.

Meter is defined by fixing a numerical value of the speed of light in a vacuum as 299,792,458 ms^{-1}. Here, the second is defined in terms of the Cesium frequency Δv_{Cs}. Numerically, the meter in terms of the defining constants c and Δv_{Cs} will be given by Eq. 1.

$$1\ m = \left(\frac{c}{299\ 792\ 458}\right)\mathrm{s} = 30,663\ 319\ \frac{c}{\Delta v_{Cs}} \tag{1}$$

The next requirement practical realization recognized by the BIPM is as follows:

(a) Direct measurement, i.e., time measurement of the transmission of light.
(b) Indirect measurement of the optical wavelength using optical interferometry.
(c) Using material measures.

The list of recommended optical frequencies for realizing the meter which are related to iodine isotope transitions is given at BIPM website (Darnedde et al. 1999; Lazar et al. 2009). On the contrary, gauge blocks, glass scales, and step gauges are the material artifacts (Doiron 2008). The laser interferometer realized the displacement as well as length of material measures. Haitjema narrated the comprehensive calibration of the displacement measuring heterodyne laser interferometer (Haitjema 2019).

Traceability to SI Standards

Traceability is a fundamental quality of measurements, which requires an unbroken chain of comparisons (through calibrations) with a defined measurement uncertainty (JCGM-100 2008). The measured is referenced to a standard that realizes the quantity as per the definition of its SI unit (Ehrlich and Rasberry 1997). Figure 1 depicts the hierarchy of traceability in practice.

Dimensional Metrology

Dimensional metrology is an applied science which deals with geometric measurements, methods (NPL, UK 2005). The invention of steam engines required a large piston that fits in the cylinder. Consequent industrial revolution demanded mass manufacturing the interchangeable parts at different places (Kacker et al. 2003; White 2011). The concepts of geometrical product specification (GPS) model, tolerancing, and the Taylor Principle of gauge design supplemented these

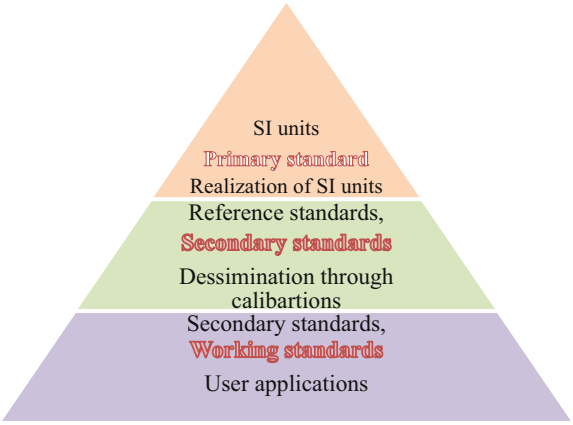

Fig. 1 Traceability of standards to SI units

requirements (ASME-Y 14.5 2018). Major milestones of the standard of dimensional measurements include the invention interferometer by Albert Abraham Michelson and the invention of the gauge block system by Carl Edvard Johannson, lapping practice by Joseph Whitworth. Dr. Richard K Leach 2003 advocated that "Metrology is not just a process of measurement that is applied to an end product," but rather the critical process requirement at every stage of product design to manufacturing (Richard Leach's Web Portal n.d.).

Conceptually, the dimensional parameters are measured in terms of length. Inserting a long cylindrical pin inside a deep cylindrical bore throughout its length is the basis of the first principle of dimensional metrology. This first principle by Taylor is based on the envelope principle (Raghavendra and Krishnamurthy 2013). The GO gauges are helpful to check and verify the violations and the conformance of the maximum material condition (MMC). The physical material distribution decides the boundaries of the separation (IS 8000 -1 2005).

Primarily, the material properties of a workpiece are important to serve its purpose (physical and chemical properties, internal discontinuities, and imperfections). Second, geometric shape and its surface condition directly influence the functional requirements. The geometrical tolerances of the workpiece are defined as deviations from geometrical ideal elements (i.e., line, surface, and curvature). Geometrical features (e.g., planes, cylinders, spheres, cones, etc.) have tolerance on (1) size deviations; (2) orientational deviations – bevels, squareness; (3) form deviations – roundness; and (4) surface conditions – waviness, roughness (ISO-14978 2018). One can identify the ratio between spacing and depth 1000:1 and 100:1 as waviness. On the contrary, roughness may be periodic or nonperiodic irregularities of a workpiece surface with the ratio between spacing and depth of the order of 150:1 and 5:1.

The ISO 17450 categorically describes the operators of the dimensional measurement process. The aim of the feature operators relevant here are as follows (ISO-17450 2011):

(a) Extraction operation acquires the coordinates from the surface of interest.
(b) Collecting the several functionally and desirable geometrical features associated with feature.
(c) Constructing virtual feature(s) from actual components within constraints (e.g., datum).

Anciently, the light is used for the measurements of distance and location. The study of the nature and properties of light is categorized as physical optics. On the contrary, the image-forming principles of lenses, mirrors, and other optical elements are covered under geometric optics. Hence, the essential concepts are reiterated.

Light and Its Interaction

The mathematical description of light propagation is given by parameters viz. the wavelength, phase, phase fronts, rays, etc. The speed of light is a quantum constant too. Figure 2 shows a snapshot of a harmonic wave that propagates in the z-direction.

$$\psi(z,t) = \mathrm{A}\ \cos\ \left[2\pi\left(\frac{z}{\lambda} - ft\right) + \gamma\right] \tag{2}$$

The field Eq. 2 of an electromagnetic wave represents the optical wave of amplitude A and offset phase γ. The wave propagates with a velocity $v = \lambda f$. Its wavelength is λ and its frequency is f. The concern of visible light where $\lambda = 400$ nm -700 nm (1 nm $= 10^{-9}$ m), and the f ranges from 400 THz to 750 THz (Gasvik 2002), is as follows:

The phase difference of the electromagnetic disturbance over a displacement in the z-axis direction is $\emptyset_1 - \emptyset_2 = 2\pi(z_1 - z_2)/\lambda$. For polychromatic radiation (say, it contains two angular velocities ω_1, ω_2), the rate of change of phase with time at a given point $|(\partial\ \emptyset\ /\partial t)z| = 2\pi f$ corresponds to the phase velocity $(\omega_1 + \omega_2)/\ (2\pi/\lambda_1 +$

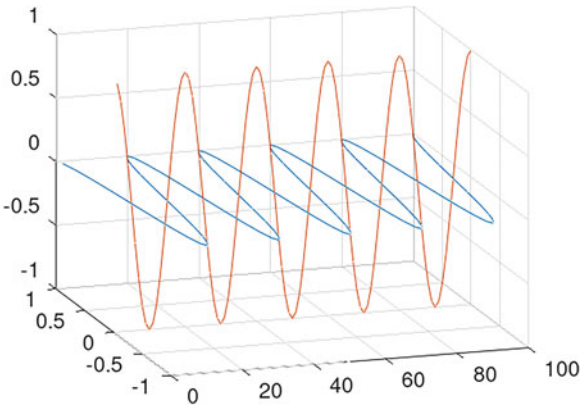

Fig. 2 Depiction of harmonic nature of light (units are arbitrary)

$2\pi/\lambda_2$). Further, the rate of change of phase with position at an instant is propagation constant. The group velocity corresponds to the slow-moving peak of envelope of polychromatic energy.

When the oscillating transverse electromagnetic radiation is propagating in z-axis direction having its electric component in x-axis direction, it is called polarized in x-axis. Consider two waves with their electric vector in x- axis and y-axis and having a phase difference ϕ. Equation 3 is their jones vector representation.

$$E = \begin{vmatrix} E_x \\ E_y \end{vmatrix} = \begin{vmatrix} E_{0x} \cos(kz - \omega t) \\ E_{0y} \cos(kz - \omega t + \phi) \end{vmatrix} \tag{3}$$

Here, k is propagation constant and E_{0x} (E_{0y}) is positional vector of electric field E_x (E_y), respectively. Alternatively, empirically measured stake's vector is used to represent the polarization in Mueller's calculus.

Primarily, the properties of light are described by the theory of classical electromagnetism. However, the quantum theory explores the interactions of light with atoms and molecules. The twentieth-century physicists adopted quantum electrodynamics (QED) as comprehensive theory of light despite the dual nature of light. QED combines the concepts of the classical electromagnetism, quantum mechanics, and the special theory of relativity.

Refraction

In 1621, Willebrand Snell discovered the mathematical relation (Snell's law) between the angles of incidence and transmission for a light ray at an interface between two media. The direction of propagation of light changes according to $n_1 \sin \theta_1 = n_2 \sin \theta_2$ at the interface of the media of refractive indices n_1, n_2. This is explained by Maupertuis with his principle of least action, that is, the path of light is of least length. Later, Fermat proposed that a ray takes the path of the least time. The optical path length of the transmission in a medium is equal to the geometrical path length multiplied by the refractive index of the medium. Accordingly, the phase difference $= k \times$ (optical path length).

The absolute index of refraction n of a medium is a ratio of the speed c of an electromagnetic wave in a vacuum to the speed v in that medium. When $\theta_2 = \pi/2$, the corresponding angle of incidence is called the critical angle given by $\theta_1 = \arcsin(n_2/n_1)$. This is called total internal reflection. This is useful to uniformly expand the plane wave from a laser beam.

Reflection

Ibn Al-Haytham, in his book "*Kitab al-manazir,*" correctly *proposed* that human vision is a passive reception of light rays reflected from objects. Following his studies, the metal-coated glass mirrors were discovered in modern-day Lebanon. However, due to the complexity of manufacture, vast majority of mirrors were simply polished disks of metal.

The mechanism of reflection in the case of dielectrics such as glass involves the influence of the electric field of the light on the electrons in the material. Consequently, the moving electrons generate fields and become new radiators. Thus, the light refraction from the glass is the reflected light which is the combination of the backward radiation of all of the electrons. On the contrary, the metal surface contains free electrons to oscillate with the incident light. The phase difference between their radiation field and the incident field is π (180°) at the interface. Here, the forward radiation cancels the incident light, and backward radiation would be the reflected light.

Diffraction

Diffraction effects studies by Jesuit mathematician Francesco Grimaldi were published in 1665. His experiments advocated the emerging of a cone of light from an aperture instead of a rectilinear ray. Diffraction refers to the bending of light around obstruct to encroach the shadow regions (Shimizu et al. 2019). In classical physic, the mechanism of diffraction of the optical wave is explained by the Huygens–Fresnel principle and the principle of superposition of waves. The propagation of light involves every particle of the transmitted medium on a wave front as a point source for a secondary spherical wave (Bhaduri et al. 2014). Encounter through an aperture gives interference bands depending on the wavelength of the light. Fraunhofer's model of the optical waves deals with far-field diffraction (i.e., at a long distance from the diffracting object), and it is focused at the focal plane using an imaging lens. On the contrary, Fresnel worked out the near-field diffraction using the approximation of the Kirchhoff–Fresnel diffraction equation. Diffraction limits the lateral resolution of detecting two adjacent light sources or illuminated objects. The angular resolution Θ due to diffraction-limited for a circular aperture diameter d at the first null of the Airy disk is $\Theta = 1.22\ \lambda/d$. Here, λ is the wavelength of illuminating light. Most often the microscope used in metrology has the diffraction limit, which is proposed by Abbe for a microscope as $d = 1.22\ \lambda/2n \sin \Theta$. Here, "n" is reflective index of medium through which light passes on to the object.

Interference

Optical waves are three directional, unlike, the two-dimensional as in the figure. Traditionally, the interference of light is explained using the classical wave model. The superposition principle states that at least two identical waves propagate to incident on a point; the results amplitude at that point are equal to the vector sum of the amplitudes of the individual waves (de Groot et al. 2021).

$2I \cos(\phi/2) \cos(kz - \omega t + \phi/2)$ is the mathematical representation of intensity due to the interference of two sinusoidal waves with phase difference ϕ. This indicates a constructive (bright fringe) interference for the phase difference of even multiples of π. The destructive interference occurs when the $\phi = \{\pi, 3\pi, 5\pi.\}$ between the monochromatic light waves (Dewhurst and Shan 1999). The generic equation of optical interference is $I_1(r) + I_2(r) + 2\sqrt{I_1(r)I_2(r)}\ \cos[\phi_1(r) - \phi_2(r)]$; here the displacement of wave is represented as exponential form $U(r) = A(r)e^{i(\Phi(r) - \omega t)}$ at point r. The

waves derived by the amplitude division and/or the wave front division supplement the essential requirement of interference that is same polarization (Ortlepp et al. 2020). Moreover, a wave of different polarization state occurs when the waves of different polarization are added. Michelson interferometer used for displacement measurement and the Mach–Zehnder interferometer are examples of amplitude-division systems (Michelson 1893). Young's double slit interferometer and Lloyd's mirror involve wave front-division.

Optical Techniques

Optical phenomena viz. reflection, refraction, diffraction, and polarization are used to device different instrument for different applications. The wave properties of light relate space and time too. Traditionally, the ray optics rely on the concept of rectilinear propagation of light. The ABCD matrix method deals with the ray optics (Saidane 2000). Nevertheless, the wave propagation describes diffraction, interference, with ease. Figure 3 depicts the broad categorization of dimensional metrology instruments (Castro 2008).

Ray Optics

Ray optics follow the three laws of geometrical optics:

1. Rectilinear propagation in a uniform, homogeneous medium.
2. Reflection – on reflection from a mirror, the angle of reflection is equal to the angle of incidence.
3. Refraction – total internal reflection that follows Snell's law.

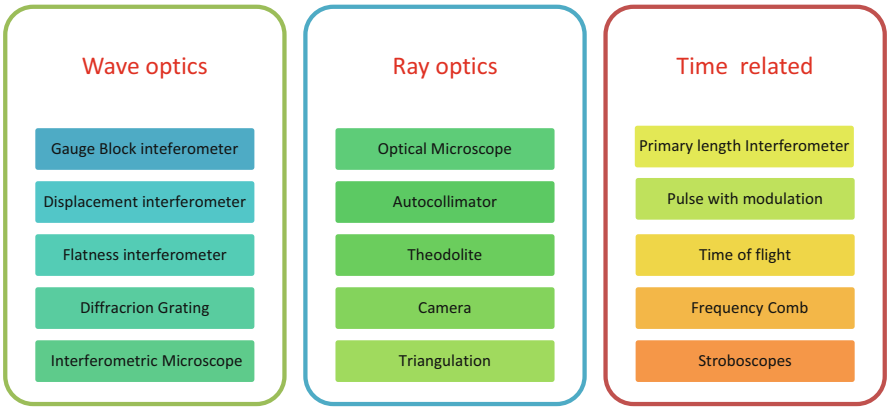

Fig. 3 Principles used for instrumentations

Telescope

The prime intention of telescope is a collect maximum optical flux. Larger the aperture size/objective of lenses, much denser will be the collected information. Thus, the collected light is focused to form image. The light-gathering power is equal to the product of the area of the collecting-lens and the solid angle subtended at the lens by the observed region of the source.

Image Forming

The imaging formation involves mapping of surface of a given object on to another surface known as image plane. Each illuminated point of the object corresponds to the projected image. The ratio of the size of the image to the size of the object is called the magnification of the image-forming optical system. The collected surface region of the image and the focal length of the lens determine the size of the field of view of the lens.

Aperture diffraction causes imperfect image despite of a perfect shape of mirror, or a lens. Practicable curvature of the optics induces image aberrations in addition to aperture diffraction (Fuerst et al. 2020). Monochromatic and polychromatic are main classes of aberrations. The first-order monochromatic aberrations are decomposed into five constituent aberrations by Philipp Ludwig von Seidel (Vrbik 2021). The five Seidel aberrations are (1) Spherical Aberration, (2) Coma, (3) Astigmatism, (4) Curvature of Field, and (5) Distortion.

Optical Coherence Tomography (OCT), a technique, views the subsurface images below the skin and through tissues. Ophthalmologists obtain details of the retina using OCT. Cardiologists take help of OCT to diagnose coronary artery disease. Researchers are involved in developing materials for preparing phantoms for the experiment of OCT.

Microscope

The optical microscope contains optics to produce an enlarged image of a sample placed in the focal plane. Often, microscopes have refractive or reflective optics to focus light on the eye or on to another light detector. The typical magnification of a visible light microscope is 1250X with a theoretical lateral resolution of around 0.25 μm.

Many modern, high-resolution microscopy techniques meticulously coupled the object illumination sections with detection optics to achieve optical sectioning (Wimmer 2017). This interface involves coinciding the illuminated point/plane and the collection point/plane (Schneider et al. 1999).

Wave Optics

The visible spectrum ranges from (4.3 to 7.5) $\times$ 10^{14} Hz. Most electronic detectors are unable to follow the electric variations of light. The interference phenomena average the intensity over time and space. Different types of instruments are developed based on the interference or diffraction and the spectrum of wave. In 1678, Huygens proposed that every point that a luminous disturbance meets turns

into a source of the spherical wave itself. The sum of the secondary waves, which are the result of the disturbance, determines what form the new wave will take.

Spectroscopy

Spectroscopy is the study of the interaction light with the matter. The electromagnetic energy is a function of the frequency, or the optical wavelength relates to the molecular spacing (Gaigalas et al. 2009). Various discovered spectroscopic effects are useful to investigate the response of different material and engineer novel materials. Absorption emission elastic scattering are some spectroscopy processes. Inelastic scattering phenomena involve a shift in the wavelength of the scattered radiation due to exchange of energy of radiation and the matter. These include Raman and Compton scattering.

Holography

Holography is a technique to exploit the superimposing of a second wave front (normally called the reference beam) on the wave front of interest. A wave front is recorded and later reconstructed to generate three-dimensional images. Holograms are also generated using computer to model the two wave fronts (Rehman et al. 2012).

Optical Dimensional Measurements

The dimensional measurements can be broadly divided into five categories. Different optical instruments are used to measure the dimensional parameters. Figure 4 shows the dimensional metrology categories.

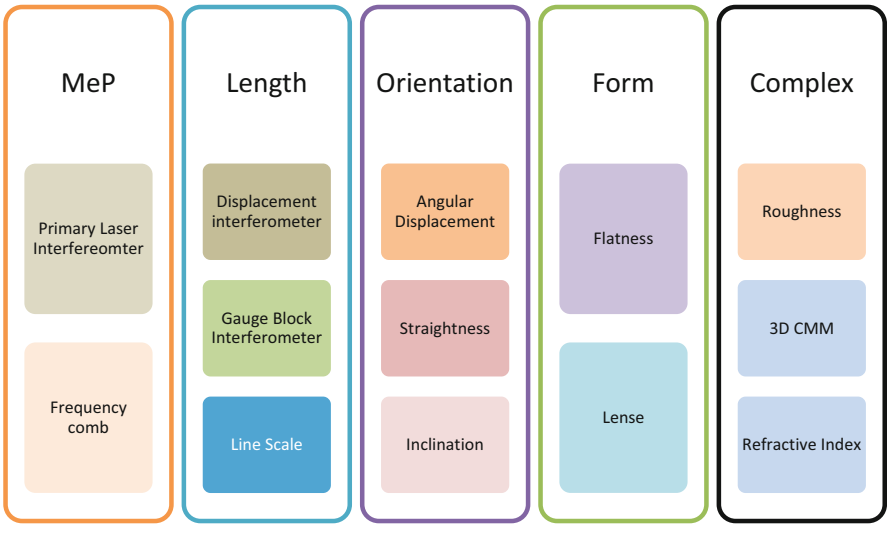

Fig. 4 Optical measuring methods of dimensional parameter

Optical Comb

Since, the frequency of optical radiations realizes a natural/quantum unit of length, the visible range of the spectrum is used for the measurement standard for the microdisplacements. Interferometry specifically solves the problem of inability of sensor to respond to the extremely high frequency of oscillation of the electromagnetic field (0.6 PHz). The optical path lengths of the reference and the "object" paths are compared to ascertain a difference of an integer multiple of the wavelength interference (Hall and Ye 2003). The length/displacement is determined by counting the interference fringes and applying the refractive index corrections of the medium of interest. On the contrary, the wavelength of the light that corresponds the interferometric fringes must be known precisely. Therefore, the calibration of the wavelength of recommended optical radiations is inevitable. Atomic transitions make ideal frequency references and hence are reproducible. Recently, optical frequency comb is adopted as a reference for calibrating optical frequencies (Araujo-Hauck et al. 2007). The optical field of the laser pulse train is described by a carrier frequency, $\nu_c = \omega_c/(2\pi)$, that is modulated by a periodic pulse envelope. Often, frequency comb is generated with a mode-locked laser (Fortier and Baumann 2019a). It produces a series of optical pulses separated in time by the round-trip time of the laser cavity. Typically, the time period of the optical pulses is (1–10) ns. The spectrum of such a pulse train approximates a series of Dirac delta functions separated by the repetition rate (the inverse of the round-trip time) of the laser. This series of sharp spectral lines is called a frequency comb or a frequency Dirac comb (Fortier and Baumann 2019b).

Popularly, iodine-stabilized He-Ne laser sources are used for dimensional metrology. The atomic energy level transition gives the laser radiation of multiple wavelengths; which corresponds to a wider bandwidth rather a sharp line of optical spectrum. The spectral line width is of Lorentzian form (Eickhoff and Hall 1995). Moreover, atomic vibration induces Doppler shift $\Delta v = v\sqrt{(8\pi kT \ln 2/mc^2)}$ between half power points. Here, m is mass of the optical frequency (v) emitting atom and k is Boltzmann constant. The output laser spectrum is convolution of the frequency-spectral response of resonator cavity and Doppler-broadened gain curve of the active medium (Rubin et al. 2022). Lamb dip, polarization, and saturation absorption stabilization are useful feedback control mechanisms to refine the separation of mirrors of the optical cavity. Allen variance serves as a measure of the stability of a laser source.

Linear Measurement

The measurement of a rectilinear displacement of movable structures, machine platforms, and objects, a distance between two designated marks on a surface, any end faces of solid block, and peaks or valleys are considered as linear measurements in dimensional metrology. Moreover, external and inner diameter of cylindrical objects, rings shapes, also belong to linear measurement (Li et al. 2014). Glass

graticules, steel scale, tape, and meter bar are popularly known as line reference standard artifacts (Swyt 2001). Similarly, gauge blocks, snap gauges, and diameter artifacts are called end standards (Stone et al. 2011).

Displacement Laser Interferometry

Displacement measuring interferometry (DMI) outperforms in position monitoring and measurement due to the high resolution, wide measurement range, and fast response. Moreover, the laser beam in the DMI can be used as a virtual axis of measurement to eliminate Abbe offset errors (Flügge 2014; Abduljabbar 2014; Haitjema 2008). Commercially, DMIs employed a homodyne or the heterodyne techniques to measure rectilinear displacement, plane angular – rotation, straightness, etc. (Kim et al. 2011). The reproducibility of measurement is supplemented by the wavelength-stable laser source, refractive index sensor electronics (Badami et al. 2016).

End Gauge Block Interferometer

The measurement of gauge block is based on the optical interference principle (Twyman-Green type) to estimate the fringe fraction corresponding to the length of the gauge block under calibration (Hu 2013). The length of the gauge in the ambient conditions of measurement is $L_A = (N + f)\lambda/2$. "N" is an unknown integer number of half wavelengths, and "λ" is the wavelength of the light (Jin et al. 2006). If the gauge length is known by mechanical measurement to be better than "$\lambda/2$," then the correct value for "N" can be calculated and the gauge length "L_a" calculated (Ali and Naeim 2014). The length measured under the ambient conditions must be corrected to the standard temperature of 20 °C: $L_{20} = L_A(1 + \alpha\Delta fT)$. Here, "α" is the thermal expansion coefficient of the gauge block and "Δt" is the difference between the temperature at which the gauge block has been measured and 20 °C. The lasers are frequency stabilized; the ambient conditions of the air are monitored in order to correct the wavelength for the refractive index of the air applying Elden's eq. (Schödel 2015).

Line Scale Measurement

The linear encoder is composed of a grating and a reading head system. The grating has the incremental grooves with an equal spacing structure. In the reading head, a laser beam from the laser diode (LD) is divided into two beams by a beam splitter (BS1). One beam passes through BS1 and propagates toward the displacement assembly, while the other is projected onto the scale grating after passing through the reference mask. The laser beam passing through a beam splitter 1 is then divided into another two beams by another beam splitter 2. The beams are projected onto the scale grating and a reference grating, respectively. The spacing of the reference grating grooves would be same as the incremental grooves of the scale grating. Two beams diffracted from the scale grating pass through BS2 and coincide with those from the reference gratings (Li et al. 2016). The reference and measured signals are analyzed to determine the line graduation spacings.

Angular Measurement

The geometric angle is essential dimensional parameter. Angle gauge blocks, regular polygon, and the angular indexed rotary stage realize the plane angle. The SI unit of plane angle (radian) is defined as the ratio of a unit circumference to a unit radius (Arif Sanjid 2013). Autocollimator, laser interferometer, spirit level, and theodolites are the typical instruments used to measure the angle. The inclination of the surface concerning the ground, the squareness between two planes, and the parallelism of end faces of prismatic objects are additional angular qualities of concern (Weichert et al. 2009). Often, the inclination is realized using a sine bar (sine arms) in terms of the length of the gauge blocks (a micrometer), and distance between the rollers (supports) of the sine bars (tilting table), respectively.

Autocollimator

An autocollimator is an optical instrument used for the measurement of deflection of the reflective plane (Vrhovec et al. 2007). The pitch (i.e., interior) angle of prismatic gauge blocks is calibrated using autocollimator on an angle-indexed rotary stage. An autocollimator projects a collimated optical beam of an image (say graticule) onto a target mirror. The return beam will be collected onto a graduated scale or an electronic detector (Igor et al. 2021). The least count of visual (electronic) autocollimator is of the order $0.2''$ ($0.01''$), respectively. Precision autocollimators can detect return beam with a measuring range up to $\pm 2000''$.

The calibration using two autocollimators is shown in Fig. 5. A precision index table (Moore Index Table), of 200 mm diameter and a least count of 15 min, is placed on a rigid, stable, leveled surface plate. A small table adjustable in three planes with leveling screws is placed on the top of the index table.

Two autocollimators with a range of ± 5 min and a least count of 0.2 s. of arc are placed on a rigid stand radially to the index table. The axis of the autocollimator objective lens is leveled and adjusted so that the working face of the angle gauge is in the field of view of the autocollimator and the face is square to the axis of the autocollimator. This can be checked by observing the position of reflected image by taking reflection on both the faces of the angle gauge in turn.

Laser Tracker

Laser tracker combines the principle of the theodolite and the state-of-the-art technology: a stabilized laser beam. The laser beam determines the displacement of the cube corner prism using time of flight method interferometry. The azimuth and elevation angles of the target direction are sensed using angular encoders. The embedded software use triangulation concepts to perform the volumatic analysis of large-scale metrology (Francis et al. 2010).

Form Measurement

The flatness refers to controlling the uniform local heights of a surface or a median plane. Technically, the flatness tolerance is defined as the separation of two parallel

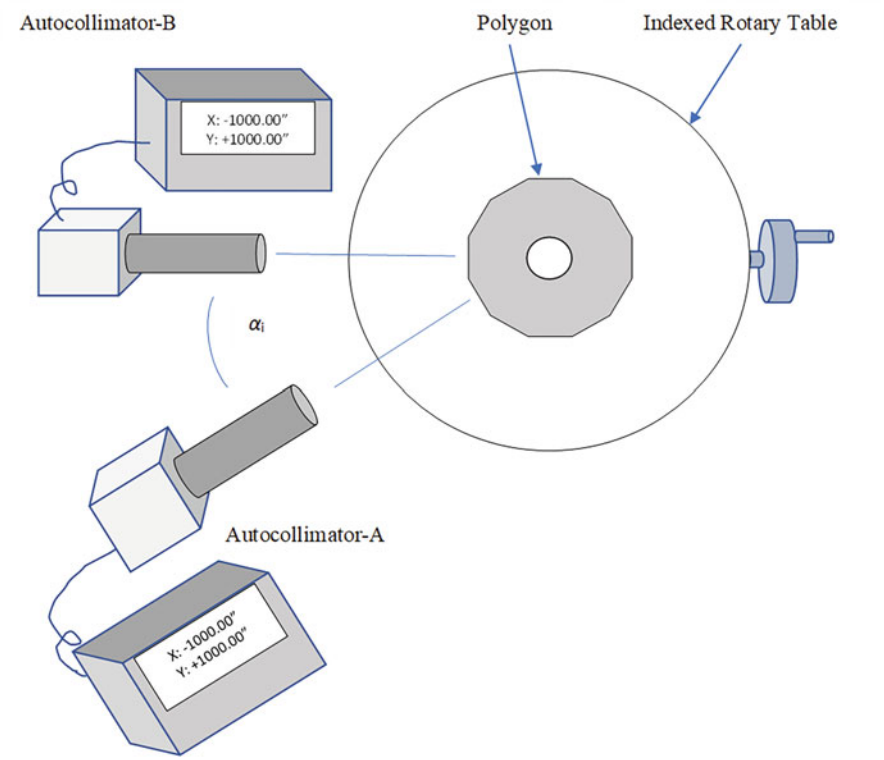

Fig. 5 Calibration of angular artifacts (regular polygon) using autocollimator

planes on either side of the assumed flat surface. The coordinates of the profile of the surface must lie between these two planes to pass the quality check of flatness tolerance. Primary flat surface is realized by a silver ambiguation liquid level. Despite of the influence of the curvature of earth, the liquid meniscus is perpendicular to the direction of gravity which is considered as flat as few nanoradians. At the National Metrology Institute of Germany, the concept of light propagation in straight-line is explored to establish secondary standard of flatness and straightness (Feng et al. 2004). These measurements are traceable to SI unit of length and angle. Diamond-lapping processes are ideal for producing reflective surfaces, which can be measured for flatness using this method directly after the lapping operation.

Usually, the flatness of a component like optical flats and grounded surface are measured using a monochromatic light source and a highly lapped reference optical flat. A sodium light emitting wavelength approximately 589 nm shines the interface of the reference flat and the surface under test. The resultant interferometric bands are made up of a bright and dark fringe and correspond to 294 nm. On the contrary, Fizeau interferometer uses a collimated laser beam of 633 nm and phase-shifting mechanism to measure optical flats with an accuracy of 10 nm (Nyakang et al. 2013).

Complex Measurement

The parameter viz. surface roughness, refractive index of lens, geometry of hardness indenter, and deformation of odd shaped objects due to the stress and thermal gradient are some of the complex measurements in dimensional metrology (de Groot 2015). They involve simultaneous measurement of multiple dimensions of length, angle, or unconventional optical methods of measurements. Spackle interferometry and fringe projection techniques are some of such unconventional practices.

Spackle Interferometry

Conventional interferometry deals with the optically smooth-polished surfaces and well defined the optical path. On the contrary, speckle interferometry deals with the ordinary diffusing objects, which roughness is comparable to the wavelength (Smith and Smith 2002). The objects' undertest is illuminated with coherent light, giving rise to speckle fields, i.e., strongly fluctuating distributions of light intensities and phases (Colonna De Lega 1997; Creath 1988; Moore et al. 1999).

Speckle interferometry (SI) aims to create and record a two-beam interference pattern involving at least one speckle wave. Speckle waves are spontaneously reflected or transmitted by diffusing surfaces exhibiting rapid spatial intensity and phase fluctuations. The phase shift between the spatial spackle waves is a function of the geometrical path difference $\delta(r)$ they traveled $\varphi(r) = 2\pi/\lambda\, n\, \delta(r)$. Here, λ is the vacuum wavelength of the light source and "n" the refractive index of the propagation medium.

SI operates in differential mode by taking an initial state most often related to a fixed configuration as a reference state and the other spackle state of the small perturbation. These specklegrams are recorded by a digital camera and correlated statistically to estimate the measurand (Tendela et al. 2015). SI is widely used for the complex dimensional measurement like surface roughness, deformation of structure/objects due to stress.

Fringe Projection

A shadow moiré pattern is a virtual interference pattern that follows geometric interference principle. Moiré fringe image is generated by covering the measurement area with a physical grating. Then the surface is illuminated and viewed at opposite directions. The fringes are projects on to the surface by which light passes through the grating at a tilted angle (Chasles et al. 2007). This shadow is observed at a tilted angle from the opposite direction to the light source. A moiré pattern can be seen as a pattern of fringes. The topology of the surface with the peak/valley rings represents the same height relative to the physical grating.

Summary and Future Prospective

In the dimensional metrology, optical methods are robust, reliable, and non-replaceable. Since the comprehensive exploration is vast enough to cover many volumes, the sustained and prevalent techniques are briefed in this chapter.

Moreover, only the gist and some critical issues are emphasized. Optical comb, laser trackers, and spackle photometry are emerging topics of research interest. The Michelson interferometer with Fourier transform can be dramatically enhanced with optical frequency combs (Badami et al. 2019; Picqué and Hänsch 2019). The optical medical imaging demands metrological scale for volume scattering of diffuse materials of the optical properties of biological tissue. Such materials are used as phantoms to evaluate and validate instruments that are useful for clinical trial which will be traceable to the scales for reflectance and transmittance (Lemaillet et al. 2016).

The picometer range displacement measurement length ruler demanded by the lithography of semiconductor chip fabrication is under experimentation by combining the X-ray diffraction and displacement interferometry (Köchert et al. 2011). Flatness interferometry for X-ray mirrors, establishing traceability to X-ray diffraction grating, are some more prospective areas of research (Alcock et al. 2016). Spherical and cylindrical interferometer designs are for measuring the volume of silicon spheres of the Avogadro project. NMI, Germany, has determined a precise diameter determination with an uncertainty of one nanometer or less. Optical methods of simulating curvatures and devising interferometric digital twin are frontiers in digitalization of metrology. Similarly, the NMI, Germany, developed a technique of assessing the tactile probe of a microcoordinate measuring machine using a microscope, and combination with contact trigger sensing of coordinates is still to be commercialized (Cui et al. 2012; Fan et al. 2012).

Conclusion

The concept of traceability of measurement to the SI unit of length, the internationally adopted recommendations and their practices for dimensional metrology are explored. The properties of the light and their dimensional application are explained. Thereafter, different optical techniques and instrumentation are discussed. The optical methods of dimensional metrology parameter are elaborated. Additionally, advance in dimensional metrology using optical techniques are emphasized.

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3D Imaging Systems for Optical Metrology

53

Marc-Antoine Drouin and Antoine Tahan

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Abstract

In recent years, new noncontact measurement devices known as 3D imaging systems have being introduced on the market. They combine one or more cameras, and most often, a controllable light source, which allows the acquisition of dense and accurate digital representations of the geometry of manufactured objects. Modern 3D imaging systems also require complex software stacks to

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extract useful information from the dense digital representation. Since such systems can be used to compute the deviation between as-built and as-designed objects, they tend to complement more traditional coordinate measurement machines. Due to their compactness, short acquisition time, and affordability, 3D imaging systems constitute an interesting tool for quality control operations, reverse engineering and metrology-assisted assembly, and/or manufacturing. These systems can thus act as important enabler for cyber-physical systems, which are critical to Industry 4.0.

Keywords

Active 3D imaging · Metrology · Optical metrology · Measurement uncertainty · Industry 4.0

Introduction

Many manufacturers are embracing the digital Fourth Industrial Revolution (comprising AI, robotics, IoT, 3D printing, etc.). The shift toward Industry 4.0 has led to ever closer interactions between the physical and the virtual (Alcacer and Cruz-Machado 2019; Uhlemann et al. 2017; Zezulka et al. 2016). Deploying cyber-physical systems requires a means to acquire dense and accurate digital representations of the geometry of manufactured objects, manufacturing equipment, and sites (Uhlemann et al. 2017). Three-dimensional imaging systems are tools that can enable such cyber-physical systems. The systems are affordable, small, and allow noncontact measurement with short acquisition times and high densities ($>10^5$ points/sec) (Catalucci et al. 2022; Drouin and Beraldin 2020).

Three-dimensional systems combine one or many cameras, and most often, a controllable light source (coherent or noncoherent), which allows to acquire dense and accurate digital representations of object geometries. Modern 3D imaging systems also require complex software stacks to extract useful information from the dense digital representation. They have short acquisition times, which make it possible to use them for quality control, digitization, metrology-assisted assembly, etc. Since 3D imaging systems can be used to compute the deviation between as-built and as-designed objects, they complement more traditional coordinate measuring machines (CMM). Typically, CMMs enable the acquisition of a few very accurate 3D points, while 3D imaging systems acquire millions of 3D points that are usually less accurate. The sheer volume of such lower accuracy data requires a robust software stack. In this chapter, we focus on 3D imaging systems that target the digitization of manufactured objects, with a special emphasis placed on this software stack.

Any discussion of 3D imaging systems characterization must be preceded by a rigorous definition of uncertainty, accuracy, precision, linearity, repeatability, and reproducibility (Gage R&R) in in situ conditions. Two authoritative texts on uncertainty management and terminology are the *Guide to the Expression of Uncertainty*

in Measurement (GUM) and the *International Vocabulary of Metrology* (VIM) (ISO 1995; Joint Committee for Guides in Metrology 2008). The ASTM's E2544 provides a definition and description of the terminology for 3D imaging systems (ASTM 2010). Due to space constraints, we refer the reader to the previous chapter of this book and to the above-mentioned authoritative texts for rigorous definition of those terms. The use of lasers is associated with eye safety hazards and, for more details, we refer the reader to the *American National Standard for Safe Use of Lasers* (ANSI Z136 2007) or to their particular organization's laser safety officer. Note that high-power low-coherence (and noncoherent) light sources can also pose eye safety issues (BS EN 2008).

The core section in this chapter covers the measuring principle of 3D imaging systems, as well as the sources of error associated with their use. We then discuss hardware and software aspects of 3D imaging systems. Toward the end of this chapter, we present some Industry 4.0 applications, and finally, we provide some concluding remarks.

Measurement Principle

Most of the 3D imaging technologies presented in this chapter are classified as optical noncontact 3D imaging systems because they emit light into the environment and use the reflected light to estimate the distance to a surface in the environment. The three most used measurement principles in commercially available systems are time-of-flight (ToF), interferometry, and triangulation. Other classifications of the principles have been proposed as well (Besl 1988; Blais 2004; Jahne et al. 1999; Nitzan 1988).

Time-of-flight is based on accurate clocks, interferometry requires accurate wavelengths, and triangulation systems are based on the intersection of light rays in 3D space. A triangle is defined when the values of two interior angles and the length of the included side are known. With these values, the lengths of the other two sides can be computed. The included side is known as the baseline of a triangulation system, while the two others are known as light rays. In the presence of a complex geometry, one of the light rays corresponding to a 3D point may be blocked by other parts of the surface being digitized. Obviously, when such occlusions occur, triangulations become impossible. Note that the two interior angles can be measured indirectly. The coordinates of a pixel in a camera can be transformed into interior angles when the camera's parameters are known (Hartley and Zisserman 2004). An active triangulation system is composed of one or many cameras and one projection system, while a passive one consists of a least two cameras. The remainder of this chapter will focus on 3D imaging systems based on triangulation, and which are well adapted to advanced manufacturing applications.

Typically, in interferometric systems, a laser beam is split into two paths. One of the paths is of known length, while that of the other is unknown (Jahne et al. 1999). The difference in path lengths creates a phase difference between the coherent light beams and this phase difference is used to compute the distance. Of interest are

systems based on optical coherent tomography (OCT), which can be designed with a very compact scanner head that allows accessing hard-to-reach areas of manufactured components (Zanardi et al. 2010). In ToF systems, the distance is computed from the round-trip time of light, which can be converted into distance (Drouin and Hamieh 2020). Obviously, interferometric and TOF systems are classified as active.

Some of the 3D imaging technologies presented in this chapter are classified as optical contact systems because they measure distance by emitting light into the environment and using the reflected optical energy on collaborative targets fixed on the environment's surfaces. Collaborative targets can be either corner reflectors, spheres, or 2D circular stickers. Corner reflectors for their part are typically used with interferometry or time-of-flight system, while spheres and stickers are used with triangulation systems. Optical contact systems can be used to extend the working volume of noncontact 3D imaging systems, and many vendors offer solutions based on this combined use. Here, we briefly discuss these contact technologies in the context of extending the working volume of noncontact 3D systems.

Sources of Uncertainty

Uncertainty quantification in the 3D imaging measurement of coordinates remains a complex endeavor. When trying to characterize the uncertainty associated with the measurement obtained by a 3D imaging system, many users concentrate their effort on the measurement device (i.e., the hardware). Manufacturers of 3D scanner usually provide one or many numerical values, which are used to characterize the uncertainty of the measurement taken by their systems (VDI 2005, 2008). It is crucial to realize that these values are obtained in very specific conditions, using specific methodologies and selected test objects. Such values, typically associated with a 3D scanner by the manufacturer, may not be representative of the uncertainty of the measurement taken by an end user. The uncertainty associated with a measurement (outcome) depends on the 3D scanning technology used, including all physical phenomena that define the point clouds acquisition process (i.e., hardware), the methodology (measurement strategy, sampling, etc.), the software (filtering, algorithms, hyper parameters, etc.), the ambient conditions in which the measurement is performed, the materiel properties of the sample being digitized, surface texture variations, the part size, geometric complexity, and finally, the people performing the measurement. These uncertainty sources are illustrated in Fig. 1, and are briefly described in the remainder of this section.

The impact of the people performing digitization is often underestimated. As an example, users' visual acuity and expertise can significantly impact the digitization process. Note that the methodologies and best practices followed by operators can have a significant impact on a measurement result. Use of a sound methodology and best practices can allow to reduce the errors induced by the operator and improve the system's repeatability and reproducibility. Because many 3D imaging systems are simple to operate, some users may fail to define and use a methodology adapted to

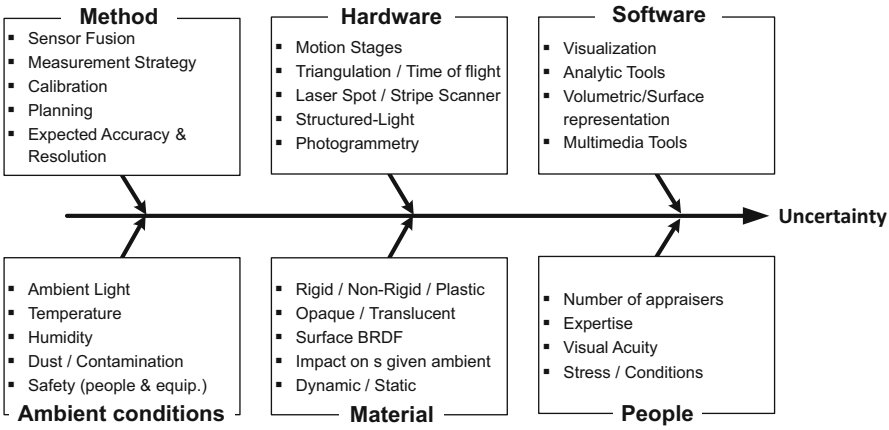

Fig. 1 Diagram showing typical sources of uncertainty for the most common 3D imaging technology use case. (Adapted from Drouin and Hamieh (2020). Figure courtesy of NRC)

the targeted application. For instance, the uncertainty associated with the measurement of many active 3D imaging systems varies within their specific measurement volumes in a nonintuitive manner (Drouin et al. 2017). This often results in methodological errors in which a system well adapted for a given task is positioned in a suboptimal manner.

Ambient conditions can significantly influence the outcome of a 3D acquisition. A case in point is where artifacts could be induced by a powerful ambient lighting source with a spectral distribution overlapping with that of the active 3D imaging system. Depending on the 3D acquisition system used, vibration can also introduce artifacts, while temperature variations can provoke changes to the relative position of the components within a 3D imaging system. For a triangulation system, this corresponds to a baseline change, and can result in a systematic scaling error of the digitized representation (Drouin and Beraldin 2020). In an interferometric system, it is critical to have no variation for the path of known length (Jahne et al. 1999). Any variation of the refraction index (and thus, on the speed of light) in the air between the imaging system and the artifact being digitized will introduce artifacts in the digital representation (Drouin and Hamieh 2020). The severity and type of artifact depend on both the measurement principle used and the implementation detail of a specific 3D imaging system. As an example, a temperature gradient in the air between the object being digitized and the 3D imaging system could introduce artifacts. When the gradient is perpendicular to the line of sight of the acquisition system, aberration will be introduced in the optical system, which can limit the performance of the imaging system. When the gradient is aligned with the line of sight, a change of effective focal length of the optical systems could be introduced. For triangulation system, this could induce scaling issues of the digital representation. In general, high-end 3D imaging systems will apply some form of compensation that takes any variation in the ambient environment into account. A simple and

popular approach for temperature is to simply include a warming period to bring the system to a given temperature prior to acquiring data. Another popular approach is to include sensors and to apply a correction function.

The properties of the object being digitized can greatly affect the overall uncertainty of a measurement. Other than the obvious fact that some objects are rigid, while others are not, the small-scale mechanical texture of an object (i.e., its surface roughness) can significantly impact the uncertainty of a measurement performed using a 3D imaging system. The characterization of the mechanical texture was discussed in Leach (2011) and Yoshizawa (2009). Because of this small-scale texture, the different coherent waves emitted by a 3D imaging system are reflected on the object and reach its detector with slightly different paths, resulting in different phases. This induces constructive and destructive interference, also known as speckle noise (Baribeau and Rioux 1991; Dorsch et al. 1994; Drouin and Beraldin 2020; Leach 2011). The presence of a semitransparent coatings and variations of the surface reflectance can also significantly affect the overall performance of a 3D imaging system. The geometric configuration of the surface of an object can generate specularity and inter-reflection, which can produce many artifacts. Descriptions of many of the artifacts introduced by the surface properties are included in Drouin and Beraldin (2020). Moreover, there may be some interactions between the object being digitized and the ambient condition at the time of digitization. The most obvious of such interactions involve the thermal expansion of the object being digitized.

The software used may also affect the uncertainty. While a coordinate machine measurement (CMM) will typically measure a few points on the surface of an object, a 3D imaging system generates millions of points, which must then be analyzed, and in some cases, visualized. The software/hardware divide is somewhat blurred as there may be processing going on directly on the acquisition hardware, while others are executed on a host computer. Processing can include filtering (smoothing), outlier removal, surface completion, fusion of multiple acquisitions, alignment (registration) of a digital representation with a reference one, etc. Each one of these processing steps may affect the uncertainty.

The uncertainty associated with a measurement taken by a 3D imaging system depends on the measurement principle used. Triangulation systems (in with the effect of out-of-focus blurring are neglected) have an uncertainty that increases with the distance, whereas with time-of-flight systems, the uncertainty should be independent of distance (Drouin and Beraldin 2020; Drouin and Hamieh 2020; Drouin and Seoud 2020). Note that for a subset of the active triangulation systems, an uncertainty model that takes into account out-of-focus blurring is presented in Drouin et al. (2017). The range uncertainty for practical scanner configurations has a slanted “U” or “W” shape (see Table 2 for an example). Typically, we expect that triangulation systems will operate at a shorter standing distance than time-of-flight ones. Also, the performance of a triangulation system is related to its physical size (i.e., the length of the baseline) and the uncertainty is expected to increase as their physical size is reduced. As a rule of thumb, noncontact 3D imaging systems that are designed to digitize objects with a working volume ranging from a few cubic centimeters to approximately a few cubic meters are expected to be based on

triangulation (Drouin and Beraldin 2020). Time-of-flight technologies are expected to be encountered for working volume starting at approximately a few cubic meters (Drouin and Hamieh 2020; Drouin and Seoud 2020). Optical contact system can be based on triangulation (when large baselines are used), or can be time-of-flight or interferometry systems.

Measurement Device

In this section, we review the taxonomy presented in Drouin and Beraldin (2020), which classifies active triangulation systems based on the method used for structured light. We conclude with a discussion on the contact and noncontact 3D imaging system combinations used. We refer the reader to Drouin and Beraldin (2020) and Leach (2011) for a more in-depth description of short-range 3D imaging technologies. We consider four distinct categories: spot scanners, stripe scanners, systems that use structured light patterns, and handheld scanners (see Fig. 2). While the first two categories typically use coherent light sources, the last two can use both coherent and noncoherent light sources. It is easier to mitigate the impact of ambient lighting when a system uses coherent or low-coherent light. However, coherent and low-coherent light source are more severely affected by speckles, which for their part, are a function of the mechanical texture of the component being digitized. Note that, the impact of speckle noise for a given surfaces can be reduced by shortening the wavelength and increasing the bandwidth of the light source (Baribeau and Rioux 1991; Dorsch et al. 1994; Leach 2011).

Spot Scanner

The first and simplest system is the spot scanner, in which a scanned laser beam illuminates a very small circular spot on the component being digitized. The simplest implementation of a spot scanner would use a linear detector, which consists of a camera with a single row of pixels. For each image acquired by the linear detector, a

Fig. 2 Left: Experimental spot-based scanner described in Blais (2004) and Drouin and Beraldin (2020). Center: Commercial stripe-based scanner mounted on a translation stage. Right: Commercial structured-light scanner. (Figure adapted from Drouin and Beraldin (2020). Figure courtesy of the NRC)

3D point is computed by triangulation using the position of the laser spot on the detector. Motion stages, galvanometer-mounted mirrors or a combination of both, are used to acquire a 3D scan of the object. This makes the acquisition slow and sensitive to vibration. Note that some high-end spot scanners allow to dynamically adjust the spatial sampling of the scanner, the optical power, and the focalization distance of the laser (Blais 2004; Drouin and Beraldin 2020).

Stripe Scanner

Typically, in a stripe scanner, a laser beam is passed through a cylindrical lens, or diffractive optical element in order to illuminate the object being digitized with a thin laser stripe. The simplest implementation of a stripe scanner would include a camera with each of its rows acting as a spot scanner. These systems are sometimes called profile scanners, and the term profilometry is frequently used in the literature in association with them. A complete digital representation of an object can be achieved by installing the scanner on top of a conveyor belt, or on a motion stage (Blais 2004; Drouin and Beraldin 2020).

Structured Light

The third type of active triangulation projects a structured-light pattern onto the scene (Jahne et al. 1999). These devices are sometimes referred to as area scanners (VDI 2002). Note that the use of the word *scanner* here is somewhat abusive since these systems do not scan a light beam over the object being digitized. Rather, they project structured patterns that simultaneously cover a large part of a component being digitized. This is illustrated in Fig. 2. The light patterns are structured either spatially, temporally, or both and allow building pairs of corresponding points between the projected and the captured pattern (Salvi et al. 2004). Those projector-camera pairs can be used to triangulate 3D points. Some systems use a single structured pattern, which minimizes the distortion due to vibration, while others use a small number of patterns that require a static environment. There are many types of patterns that can be used, and for more information, we refer the reader to Salvi et al. (2004).

Handheld

The last type of scanner is a handheld scanner, which can be used in dynamic environments where the operator moves the scanner in front of the object being digitized. Some such systems allow both the 3D scanner and the object to be moved simultaneously. Handheld scanners are composed of a positioning and a measurement system, and the latest devices are typically either stripe-based scanners or structured light systems. The positioning system is further described in §4.6. Note that handheld

scanner performs a significant amount of processing within the scanner as they are constantly merging newly acquired 3D points with existing ones. More details about their inner working can be found in Hebert (2001) and Tubic et al. (2003, 2004). The characterization of a handheld sensor is presented in Givi et al. (2019).

Multi-View Configuration

The commercial implementation of spot, stripe and structured-light scanners can use one or many cameras. Typically, a handheld scanner will contain more than one camera. In a popular two-camera design, the light source is located at the center of the system, with a camera located on each side. Such a configuration allows two triangulation strategies and some scanners can use both strategies simultaneously. The first strategy performs the triangulation independently for each camera. The system then selects the 3D points that seem to be the most reliable and discard the others. The second strategy is named active stereo. In this, the triangulation is performed between points in the two cameras. The projector is only used to find pairs of points in the cameras that correspond to the same 3D point. Some active-stereo systems use a single structured pattern and benefit from recent progress that has been made to the design of passive triangulation algorithms. In this context, the projector is used to add extra texture on the scene in order to improve the performance of passive triangulation algorithms (Drouin and Seoud 2020). Using two cameras allows to design systems that can reduce the impact of occlusion, specular reflection, speckle, and color variation of surfaces. These artifacts are described in detail in Drouin and Beraldin (2020).

Combining Pose and Surface Measurement

Contact and noncontact systems can be used together in order to extend the working volume of noncontact 3D imaging systems. This is a popular configuration used by many handheld scanners. Laser tracker and triangulation-based passive systems are two popular commercially available options. When using a passive triangulation system, 2D retroreflective target are positioned on the 3D scanners. When a laser tracker is used, spherically mounted retroreflectors are included in the mechanical design of the scanner. For each surface measurement, the position of the 3D scanner with respect to the positioning system is known, and all measurements can be transformed into a single coordinate system. Some handheld scanners can have an integrated positioning system. In that case, the 2D targets are installed on the object being digitized (Hebert 2001). The 3D imaging systems can also be installed on a robotic arm, and encoders of the arm can provide the positioning information required to bring all surface measurements into the same coordinate system. The reader is invited to explore the hand-to-eye calibration functionality included in the OpenCV library and review the work of Tsai and Lenz (1989). Descriptions of medium-range contact and noncontact 3D imaging systems can be found in Drouin and Hamieh (2020), Muralikrishnan et al. (2016), and

Remondino and Stoppa (2013). The fusion of multiple 3D scans can also be accomplished without a positioning system by using software that exploits the complex geometry of successive overlapping 3D scans (Salvi et al. 2007; Zhao et al. 2021). This fusion is known as a rigid registration. This registration can be used to align two point clouds (SCAN to SCAN) or a point cloud with a CAD (SCAN to CAD). Rigid registration methods are described in the next section.

Software

Essentially, a measurement error propagates through data processing pipelines including filtering, registration, datum fitting, and feature extraction (Forbes 2018; Senin et al. 2021). Computer-Aided Inspection (CAI) software provides various options to fit the measured points (source data) to a target *XYZ* coordinate system. Most CAI software applications propose different options for data filtering (strategy for removing outliers, simplification, down-sampling, Fig. 3 – step 0) and different fitting algorithms and options that optimally adjust the measured points to a nominal definition (CAD) in various methods (Gaussian best fit, Chebyshev best fit, etc.). As a result, the performance of a measurement system is an estimate of a combination of various measurement errors (random and systematic) that include hardware (equipment error), software (algorithmic error) and operators. Note that both the software and operator can still strongly correlate.

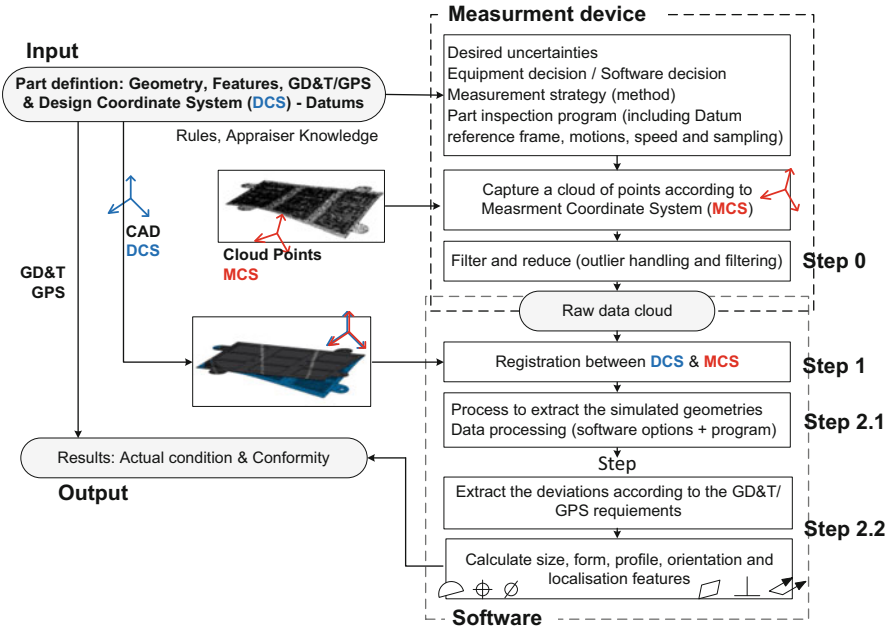

Fig. 3 Typical inspection process using the point cloud obtained with a 3D imaging system

The term *Algorithmic Error* is defined here as the inaccuracy resulting from a choice of the wrong algorithm, method, parameter, or hypothesis for achieving the intended result (i.e., size of a feature, orientation error, profile error, localization error, etc.). In the specific case of dimensional metrology, computational 3D metrology includes the implementation of algorithms to achieve computations on discrete data collected by measuring equipment. Standards such as the ASME Geometric Dimensioning & Tolerancing (GD&T) Y14.5 (ASME 2020) or the ISO Standards Geometrical Product Specifications (ISO 1101, ISO 5459, etc.) (ISO-GPS 2016) define concepts, symbols, and rules to communicate information on technical documents (included in 2D drawings or 3D annotation CAD files). The aim is to ensure a common understanding by all stakeholders or agents (i.e., design, manufacturing, quality control). Knowledge of the GD&T/ISO-GPS standards is essential to ensure that drawing information (requirements) is being interpreted properly. The manufactured part is inspected using a 3D imaging system and compared with the definition and requirements in order to verify dimensional and geometric feature specifications (i.e., actual size, form error, localization error, orientation error, etc.). Deviations are then computed and displayed as results. The comparison of these results with tolerances determines the conformity status of the manufactured part. Figure 3 presents the inspection process model step by step.

Presenting the contributions of each step in the inspection process in this manner is arbitrary. Other authors have presented the process somewhat differently. For example, with respect to coordinate metrology, Weckenmann et al. (2001) suggest that the main contributors to uncertainty can be subdivided into five groups: measuring devices, environmental conditions, work pieces, software (algorithms used, filtering, conditions for removal of outliers, stability of the algorithm, layout handling, etc.), operators (skills, certification, GD&T/ISO-GPS decoding and interpretation, etc.), and the measurement strategy (amount of data, sampling, number of measurements, etc.). An impressive volume of work has been produced by the community in terms of investigating the measuring device, environment, and workpiece components. Although no common understanding of software validation procedures currently exists, the reader is referred to Greif et al. (2006), as well as to the TraCIM EMPIR project (EURAMET 2020; Forbes et al. 2015; Müller 2014), for research that has been performed on software validation in the field of metrology.

Rigid Registration (Step 1)

In 3D optical scanning, the scanned point (SCAN) cloud is usually not in the same coordinate frame as the CAD model. Scan data lie within the measurement coordinate system (MCS) and the CAD model is located in the design coordinate system (DCS) (Fig. 3 – step 1). Therefore, the measured point cloud must be transformed (translation and rotation) into the DCS using a 3D transformation. This transformation can be rigid (Besl and McKay 1992), affine, flexible (Aidibe et al. 2020), or non-rigid (Myronenko and Song 2010). This operation is commonly called registration (Li and Gu 2005). The iterative closest point (ICP) (Besl and McKay 1992) and its

variants, are the most popular computational methods of the rigid registration. The ICP algorithm starts with coarse registration to find a good initial estimation of the rigid matrix transformation. This step can be carried out through principal component analysis (PCA) (Liu and Ramani 2009), features matching (Li et al. 2007), statistics methods (Attia et al. 2018), heuristic algorithms or Genetic Algorithms (Ji et al. 2017). A review of coarse registration methods can be found in Diez et al. (2015).

The fine registration process tries iteratively to minimize a sum of the squared Euclidian distances (or Hausdorff distance in some variants of the ICP (Ghorbani and Khameneifar 2021)) between the SCAN points and their closest estimated points on the CAD using a rigid transformation (translation vector T & rotation matrix R). The minimization can be obtained using a Singular Value Decomposition for point-to-point (Arun et al. 1987), point-to-plane (Chen and Medioni 1992), or point-to-surface errors (Mitra et al. 2004). The numerical solution of this optimization problem (minimization) can be obtained by various numerical methods (such as the quasi-Newton optimization algorithms, genetic algorithms, etc.) where the global optimum is not guaranteed and the solution will always depend on some hyper parameters that must be carefully chosen. The approach performs well when the point cloud matches the CAD. In the case of manufactured parts, the inherent variation in manufacturing processes induces deviations between the two sets (point cloud and CAD). Various studies have proposed to identify and eliminate outliers (attributed to the process-induced error) from data by the M-estimation (Ding et al. 2019) or signal-to-noise ratio metric (Zhu et al. 2007). To sum up, the ICP and its variants perform well with two identical geometric sets (SCAN & CAD) and the robustness is good even with a noise level of the measurement (Liu et al. 2018). However, when significant differences exist between SCAN & CAD (damage zones, manufacturing defects and nonconformity, scale and deformation) the registration process is affected by the unreliable correspondences (defined in damaged regions), and an optimum transformation cannot be achieved. The algorithm introduces averaging-out errors in the quality of the alignment. In addition, the registration algorithms are sensitive to the local density of the measured points. High-density areas will have significantly more weight, which may affect the alignment result.

Readers interested in hands-on registration and manipulation of 3D points obtained by 3D imaging systems can explore the Point Cloud Library (Rusu and Cousins 2011), the MeshLab open-source application (Cignoni et al. 2008), and the KinectFusion library (Newcombe et al. 2011). In addition, a comprehensive review of 3D image registration techniques, which includes an experimental comparison of a broad set of registration techniques to determine the rotation error, translation error, and root-mean-square error, is proposed in Salvi et al. (2007).

Features Extraction (Step 2)

The ISO-GPS (2016) and ASME Y14.5.1 (2020) standards define dimensional and geometrical specifications (tolerances) as a means of expressing the limits (threshold or statistical behavior) of surfaces and features variations with respect to nominal

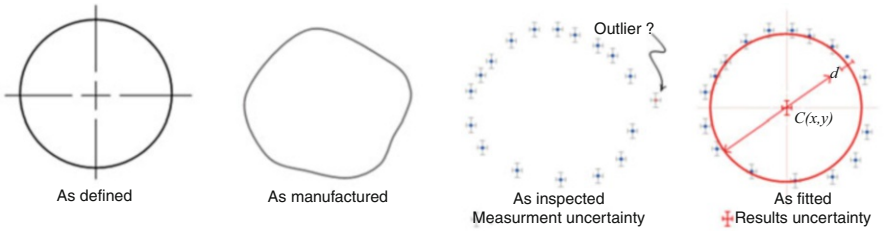

Fig. 4 Illustration of the feature extraction process

model definition (CAD). A quality control process uses a measuring device (including measurement equipment, the hardware, and CAI software) to verify the conformity of the parts to these tolerances. Figure 4 illustrates the feature extraction process. In addition, the ASME 14.5.1 (2020) standard presents a rigorous mathematical definition of GD&T consistent with the principles and practices of ASME Y14.5, which allow to determine actual values of each requirement (e.g., thickness, form error, localization, etc.). Mathematical equations in ASME Y15.4.1 define relationships between real numbers, 3D vectors, DCS associated with datum reference frames, and sets of these quantities. Indeed, once the registration step is completed, to convert the measurement points as captured, it is mandatory to select: (*i*) the appropriate algorithm (e.g., Gaussian least square, maximum inscribed size, etc.) and some related hyper-parameters; (*ii*) working hypotheses (e.g., treatment of outliers, noise filtering, missing data, etc.). It should be noted that this standard frames the mathematical definition but does not propose any specific algorithm to be used to find a solution, and even less so the set of hyper-parameters to be used (see Fig. 3 – step 2).

In GD&T/ISO-GPS, the extraction can be broken down into two steps. The first is the extraction of the feature size (such as a plan, circle, or free form profile, see Fig. 4). The second is the identification of its shape error and its conformity with some constraints (translations and rotations) related to datums defined by reference frames (see Fig. 4).

Feature Fitting (Step 2.1)

Generally, the fitting problem consists in selecting an appropriate algorithm to adjust data points collected from the inspection of a manufactured part to a geometric feature (e.g., plane, cylinder, circle, free form, etc.). The perfect form of estimation obtained by a fitting is called a reference feature or substitute feature (Forbes 2006a, b). ISO 14405-1:2016 posits size as a fundamental geometric descriptor. It also defines a new set of modifier tools for the size. Jbira et al. demonstrated that defect and noise measurement significantly affects results with ISO-14405-1 modifiers (Jbira et al. 2018).

Fundamentally, the fitting problem is an optimization problem defined in Eq. 1.

$$l_p = argmin\left(\sum_{i=1}^{n} |d_i|^p\right)^{\frac{1}{p}} \tag{1}$$

Where n is the number of measurement points and d_i is the shortest distance between measurement data and the reference form. If $p = 2$, the fitting problem is a Gaussian Least Square (LS) fitting. In that case, the solution is found by the pseudo-inverse operator. The solution is explicit and unique for a set of data. And if $p \to \infty$, l_∞ solution is the Chebyshev or *minimax* fitting. In this case, an iterative optimization algorithm is used to find an approximation of the solution. The global optimum is not guaranteed and the solution depends on several hyper parameters (convergence criteria, starting point, algorithms used, etc.).

Various methods have focused on the problem of fitting surfaces applied to many fields, but the estimation of form errors continues to be a challenge (Rhinithaa et al. 2018; Srinivasan et al. 2012; Xiuming et al. 2013). For example, in the specific case of circular, cylindrical, and spherical forms, one such method can be a Voronoi diagram used to determine the minimum circumscribed circle (MCC) or maximum inscribed circle (MIC). To obtain an MCC, the points of the diagram that lie farthest from its center are used. The center of the MIC lies on the Voronoi vertex or on the Voronoi edge. The distance between the circle center and the convex vertex represents the radius. The Delaunay Triangulation is usually used to calculate the convex and Voronoi diagram (Tran et al. 2015). In Nurunnabi et al. (2019), the authors compared different algorithms to facilitate the choice of the adequate execution method for MCC, MIC, and MZC in order to compute roundness errors. In addition, they used a new geometric concept based on the reflection of a mapping technique to assess roundness errors. They proposed a selected benchmark of algorithms in the literature in order to provide the optimal execution method. It was concluded that no single algorithm provides the best solution. Geometric primitive reconstruction is a significant problem in the field of computer-aided design (CAD). Goch and Lubke (2008) proposed a new algorithm to approximate geometry elements using Gauss and Chebyshev criteria. Nurunnabi et al. described the problem of circle fitting for complete and incomplete datasets with outliers (Nurunnabi et al. 2017, 2018). They proposed a robust approach for circle fitting, which had to merge the PCA and robust regression. Their experimental results confirmed the performances of the proposed approach with different percentages of tolerance of clustered outliers. Also, Guo and Yang (2019) proposed a new procedure for circle fitting. They used Taubin's approach to compute the center and radius, and then they identified and removed the outliers by calculating the Euclidian distance. Their experiments demonstrated that the iterative procedure could resist against the occurrence of outliers. More specifically, in the case of geometric elements of plane form, Deschaud and Goulette (2010) proposed an accurate algorithm to extract planes in noisy point clouds using filtered normal and voxel growing. In a first step, they estimate the better normal at the data points. The second step consists in computing a score of local planes, and then they apply the growing voxels. Finally, they evaluated the proposed algorithm on different numbers of points and compared it with existing algorithms. The method had a linear algorithmic complexity and was able to detect large and small planes in very large data sets. Nguyen et al. presented a comparative study of least square plane fitting algorithms with different segmentation methods (e.g., RANSAC, RGPL, Cabo, RDPCA) (Nguyen et al. 2017). They validated the study using two

real point clouds collected by a scan system. The results demonstrated that the RGPL method gives the best result for planar surface extraction. Marriott et al. (2018) presented an unsupervised extraction planar method. They proposed to adjust data with a piecewise-linear Gaussian mixture regression model whose components were skewed over planes. In Besl and McKay (1992) and Moroni et al. (2014), the problem of fitting full- and half-geometrical primitives (e.g., circular, spherical, and cylindrical primitives) was addressed. The authors used the Levenberg Marquardt (LM) method to approximate these geometries. They also proved that using the chaos optimization method improves the initial algorithm estimation. In fact, the chaos LM algorithm provides efficient results even when the input data points are incomplete and noisy. Chiboub et al. (2021) proposed a robust approach based on a non-vertex solution for generating reference soft gauges for complex shapes (free form) to estimate the form (profile) error based on the *minmax* fitting (Chebyshev fitting). The authors proposed two implemented fitting algorithms called the exponential penalty function (EPF) and the hybrid trust region (HTR). The two algorithms were tested on a number of generated reference pairs. Their performance was evaluated according to two metrics: degree of difficulty and performance measure.

GD&T/ISO-GPS Specification Extraction (Step 2.2)

An experimental investigation of three commercial inspection software packages to evaluate algorithmic errors was proposed by Jbira et al. (2019). The artifact used in the study is shown Fig. 5. The inspection software packages are typically used to analyze point clouds generated by 3D imaging systems. The benchmark and experimental case studies demonstrate the influence of the algorithm choice. Figure 6 contains an illustration of the testing artifact with ASME GD&T specifications. In this reproducibility study, only one measurement points cloud, one operator, the same CAD, and many fitting algorithms with different shapes (circle, plan, and oblong hole) and features (size, circularity, and flatness) were evaluated. The experience demonstrated the presence of a significant variation between the three CAI softwares. Using the example of a Gaussian least square (LS) algorithm (deterministic analytical solution is well known), results demonstrate a small variation between the CAI software packages. However, for the Chebyshev criteria (*minimax* algorithm), minimum circumscribed circle (MCC), or maximum inscribed circle (MIC), the results showed a significant variation between the CAI software packages. These differences can be due to filtering and smoothing operations, a reduction in the point density, the treatment of outliers, the registration algorithm, and optimization parameters (when the tolerances allow degrees of freedom in translation and rotation). All these sources have an effect on variations that which we call *Algorithmic Errors*.

According to Table 1, the LS algorithm provides the same diameter values for software #1 and software #2 (19.874 mm). In addition, software #*1* and #2 with an MCC algorithm provide the same diameter values (20.091 mm), but software #3 shows a somewhat different value (20.093 mm). The MIC algorithm shows the same values for software #1 and software #2 (19.760 mm), but software #3 provides a

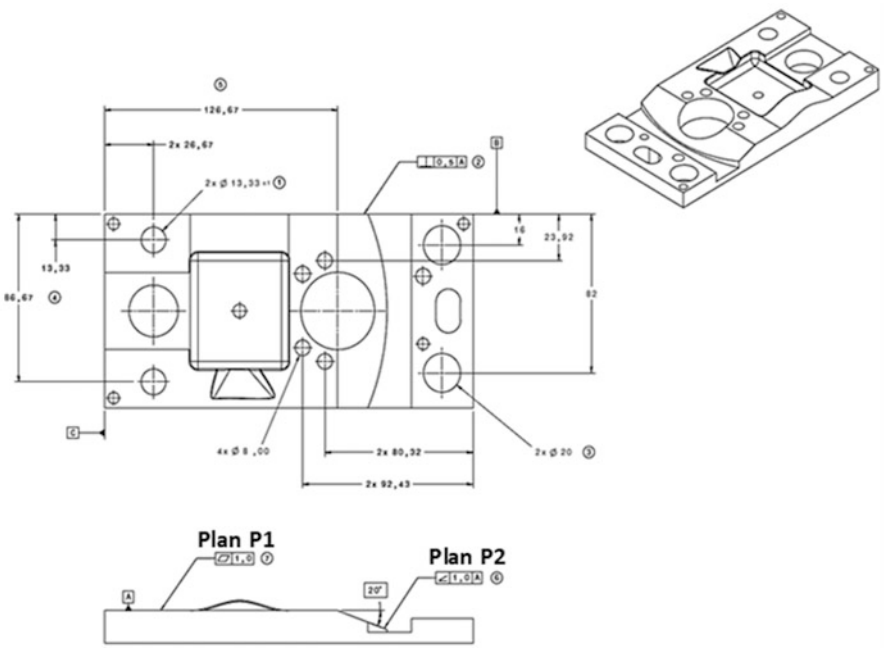

Fig. 5 Artifact used to evaluate algorithmic errors as proposed by Jbira et al., courtesy of (Jbira et al. 2019)

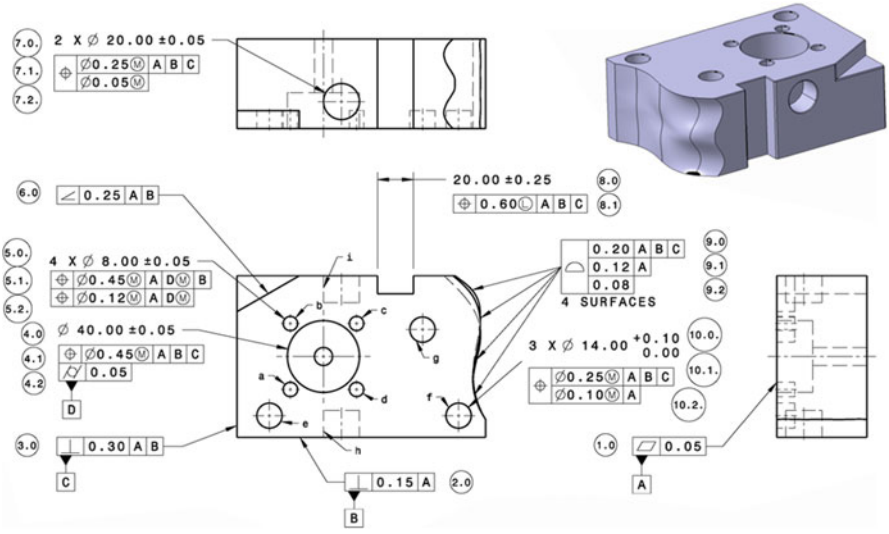

Fig. 6 SME GD&T-based benchmark with many DoF configurations. Note the complexity of the symbology used including modifiers and datum reference systems, courtesy of (Aidibe et al. 2020)

Table 1 Results for the inspection of size and surface features. The flatness and angularity shown are, respectively, for planes P1 and P2 of Fig. 5 [Dimensions in mm]

		Feature of size (circle)			Planar surface features	
Software	Algorithm	Diameter	Δ_{dia}	Cicularity	Flatness P1	Flatness P2
#1	LS	19.874	−0.126	0.186	0.107	0.070
	MIC(MIP)	19.760	−0.240	0.186	0.107	0.070
	MCC(MAP)	20.091	0.091	0.186	0.107	0.070
#2	LS	19.874	−0,126	0.205	0.119	0.031
	MIC(MIP)	19.760	−0.240	0.208	N/A	N/A
	MCC(MAP)	20.091	0.091	0.195	N/A	N/A
	Minmax	19.907	−0.093	0.185	N/A	N/A
#3	LS	19.873	−0.126	0.034	0.051	0.035
	MIC(MIP)	19.721	−0.279	0.034	0.051	0.035
	MCC(MAP)	20.093	0.093	0.034	0.051	0.073

slightly different value (19.721 mm). Finally, with the *minimax* algorithm, available only in software #2, the diameter is equal to 19.907 mm. According to Table 1, software #1 and software #3 provide the same circularity values for all methods: 0.186 mm and 0.034 mm (LS, MIC, MCC and *minmax*), while software #2 provides different results for each method.

In addition, in the same study, the authors computed the flatness of two planes (P1 (Datum A) and P2 with an angularity specification related to datum A) (Table 1). Results with the LS (l_2) option are shown in bold. According to Table 1, CAI software #1 and #3 offer three adjustment algorithms for P1 (Datum A), the LS, Minimum Plane (MIP), and Maximum Plane (MAP). CAI software #2 only offers one algorithm (LS) to fit a plane. Only the best fit is available. For P1, software #1 gives the same flatness value (0.107 mm) for the three algorithms (LS, MIP, and MAP). In the case of software #2, only the best fit is available (0.119 mm). From the Table 1, the results for P2 are similar to those for P1. Software #1 gives the same flatness value (0.070 mm) for the three algorithms. In the case of software #2, only the best fit is available (0.031 mm), but software #3 gives different values (0.035 and 0.073 mm).

The use of datums with GD&T/ISO-GPS features controls degrees of freedom (DoF). In 3D space and for a rigid part, three translations ($T = [T_x, T_y, T_z]$ or $[X, Y, Z]$) and three rotations ($R = [R_x, R_y, R_z]$ or $[U, V, W]$ define of DoF. Depending on the functional requirements (e.g., assembly, performance, aesthetics, etc.) rules of the GD&T/ISO-GPS can be applied by designers to block needed degrees of freedom. In some cases, designer can require all the DoF (Fig. 7, right side), or even leave all the DoF (e.g., flatness, profiles not related to a datum, etc.) as shown in Fig. 7 (left side).

Aidibe et al. propose a GD&T-based benchmark for an empirical evaluation of the performance of different laboratories and operators in a CAI context considering different criteria related to dimensional and geometrical features (Aidibe et al. 2020). In the study, an artifact was designed to include some basic geometries (such as the

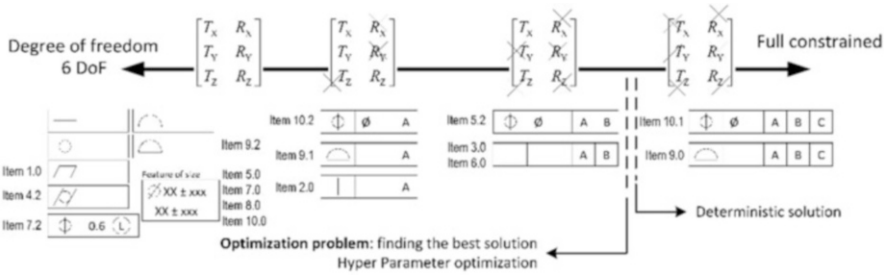

Fig. 7 Schematic presentation of the use of DoF with ASME GD&T (Item #x refers to GD&T requirements as defined in Fig. 6 (Aidibe et al. 2020)

cylinder and plane) and complex free-form surfaces with many configurations for DoF (Fig. 7). Five artifacts were manufactured including defects. All operators used a CMM (with, accuracy $\ll$ defect). The results from this inter-laboratory comparison study showed significant performance variability in three cases: (*i*) where the surfaces are complex such as a free form; (*ii*) in the case of a composite profile and localization GD&T/ISO-GPS; (*iii*) when the algorithm employed is not deterministic (e.g., when the tolerance is not fully related to the reference frame). This emphasized the importance of training and certification in order to ensure a uniform understanding among different operators, combined with a fully automated inspection code generator (including the use of the appropriate algorithm) for GD&T/ISO-GPS purposes.

A DoF with a GD&T/ISO-GPS implies that the tolerance zone can move (in translation T_i or in rotation R_i, or any combination) according to a specific degree of freedom. We refer readers to ASME (2020) and ISO (2011) for annotation rules and examples. Our interest in this chapter is to pay special attention to the algorithmic error that can be induced. Indeed, in the case of the complete blocking of the DoFs, there will be no optimization error, and the calculation of the conformity of a feature is deterministic. For example, the true position of a cylindrical element in 2D is calculated by the well-known formula: $2\sqrt{\delta_x^2 + \delta_y^2}$, and the profile error of a surface is calculated by:

$$2 \max \mid \overrightarrow{\delta_n} \mid .$$

When DoFs are involved, the estimation of deviations requires an optimization operation. Indeed, the tolerance zone can move in translation or in rotation (or both), and in some cases, its size is not constant. This is the case, for example, of circularity, cylindricity, and a dynamic profile (Y14.5-2018). Generally, the optimum is defined by the zone (1-D, 2-D or 3-D), which minimizes the maximum error (Chebyshev criterion). In these cases, there is no explicit or unique solution. The result is obtained by running an iterative optimization algorithm (e.g., Newton's method, gradient methods, evolutionary algorithms, genetic algorithms, etc.). Moreover, like any optimization algorithm, the identification of the global minimum is not guaranteed.

Applications

In this section, we briefly present two 3D imaging applications that demonstrate its use in the context of Industry 4.0. First, we present a 3D printing application where measurements taken by structured light scanners are used in a closed manufacturing loop that relies extensively on rigid registration. Then, we illustrate the use of a 3D imaging technology combined with augmented reality (AR) for jigless assembly.

Metallic Additive Manufacturing

Recent years have seen significant progress in the development of metallic additive manufacturing (MAD) (Frazier 2014). However, MAD is not perfect, with one of its drawbacks being the required labor intensive post-processing. It calls for the inclusion of support structures in printed components in order to prevent the deformation of overhanging geometries (Alfaify et al. 2020). The support structures need to be removed manually. Moreover, some components may require polishing and some un-sintered powder can be found on some surfaces. These manual tasks must be performed by skilled workers, which may introduce repeatability and reproducibility issues. A metalworking robotic cell produced by Rivelin Robotics allows to automate these labor-intensive tasks (see Fig. 8). The solution uses intensive 3D imaging

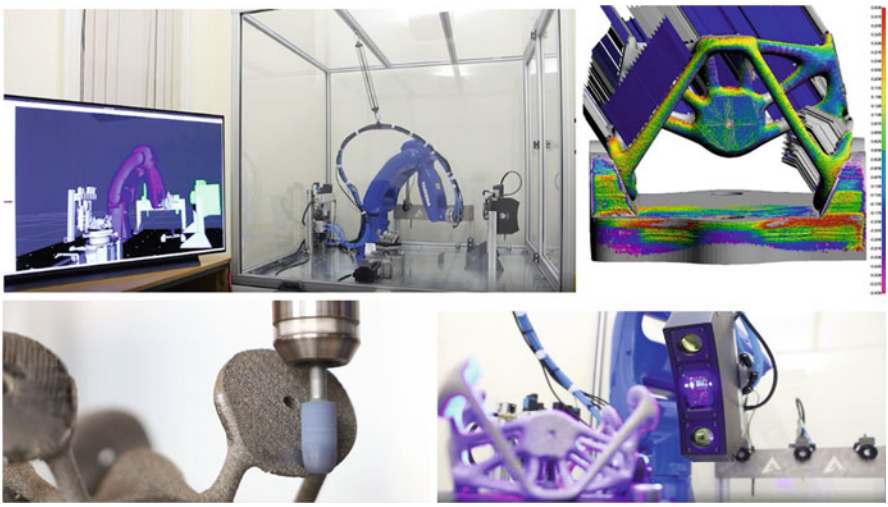

Fig. 8 Top left: Image of the metalworking robotic cell. The digital twin is visible on the left screen. Top right: A 3D scan of the printed component aligned with the CAD. The deviation between as-built and as-designed is color coded. Note that support structure is still attached to the printed component. Bottom left: The robot polishing a surface of the component is guided by the alignment between the acquired digital representation and the CAD. Bottom right: The robot is holding the 3D imaging system, which is acquiring a point cloud. The printed component is mounted on a rotation stage. Figure courtesy of Rivelin Robotics and NRC

and digital twining. The portion of the work presented here is a collaboration between Rivelin Robotics, Renishaw, the University of Sheffield, and the National Research Council of Canada. Note the results of the feasibility study we present here are not necessarily representative of the performance of the robotic cell currently being marketed by Rivelin Robotics. The remainder of this discussion focuses on the effort made, at the early phase of the project, to estimate the magnitude of the error related to the 3D imaging acquisition and processing. When defining a minimum viable product, the required functionalities and associated budgets are defined. Of interest here are the budgets related to the acquisition and processing time, the cost of imaging hardware and the uncertainty budget. For many new automating applications (including this one) that use in-process optical metrology, the available uncertainty budget, at a given manufacturing step, may be difficult to quantitatively define, or is simply unknown.

The manufacturer of the structured-light scanner selected uses a ball-bar test to associate an accuracy statement to their product (VDI 2005, 2008). We also performed our own validation using a well-characterized planar surface with surface properties similar to those of printed components. The artifact was installed on a fully characterized translation stage and was moved to different positions in front of the 3D scanner. At each position, an acquisition was performed. The acquisitions were repeated with different orientations of the planar surface. The root mean square (RMS) of the surfaces of all acquired range images can be computed. Moreover, since the translation is known, the error between the parallel surfaces (i.e., thickness error) can be computed. The results are shown in Table 2.

In a second round of experiments, a representative printed component was scanned at different distances from the scanner (see Fig. 8) for a representation of the printed component). Again, we used the characterized translation stage. It is possible to register two consecutive scans and compare the computed displacement

Table 2 A characterized planar surface was moved to different positions in front of the 3D scanner. The root mean square (RMS) error and distances to plan 1 are given for 11 different positions on the surface. In this experiment, the planar surface is perpendicular to the scanner

		Distance to plane 1	
Plane index	RMS (μm)	Nominal (mm)	Measured (mm)
1	26	0	0
2	20	10	9.997
3	17	20	19.993
4	15	30	29.993
5	13	40	39.996
6	13	50	49.996
7	16	60	59.990
8	20	70	69.987
9	26	80	79.976
10	32	90	89.976
11	39	100	99.975

Table 3 A representative printed component was moved to different positions in front of the 3D scanner. The SCAN-to-SCAN registration errors using two software packages were computed. We also evaluated the SCAN-to-CAD-to-SCAN error for one of the software packages

		Measured (mm)		
		SCAN to SCAN		SCAN to CAD to SCAN
Pose displacement	Nominal (mm)	Software #1	Software #2	Software #2
1–2	10	10.000	10.001	9.948
2–3	10	10.000	10.001	9.926
3–4	10	10.002	10.004	11.162
4–5	10	10.000	10.002	10.675
5–6	10	10.002	10.005	9.998
6–7	10	9.998	10.001	10.008
7–8	10	9.997	10.003	10.041
8–9	10	9.999	10.004	10.103

with the nominal one. While not shown here, this experiment was repeated using a rotation stage. The results are given in Table 3 for two different software packages that produced similar results. In both cases, the root mean square error between the registered models is approximately 33 μm for the entire registration models. We also computed the SCAN-to-CAD registration for each scan and then computed the SCAN-to-SCAN registration error using the SCAN-to-CAD-to-SCAN transformation chain. This was computed only using software #2, which has an option for SCAN-to-CAD alignment. This test was designed to quantify an upper bound for the scan-to-CAD registration error, which is a key algorithm used in this application.

Assembly Assisted by 3D Imaging Systems – Jigless Assembly

Traditionally, in many industrial fields, assembly is done with dedicated fixtures (jigs, templates). This is a method that is widely used to position different parts, subsystems, and systems in different industrial sectors, such as aerospace, rail transport, urban transport, and many others. The advantage of this traditional approach is its robustness and resource qualification requirements. However, it is based on obsolete vision: produce a single product at high volume. Its manufacturing cost and inadaptability to change are obstacles to productivity.

New functionalities of contact and contactless 3D measurement technologies (such as Indoor GPS, photogrammetry, laser trackers, and 3D imaging) allow to use 3D measurements, not in a classic role such as quality control, but as external encoders for positioning, tracking, and adjusting components during assembly operations (see Fig. 9). In other words, the assembly assisted by 3D metrology approach aims to extend the metrology capabilities over the entire space of a plant and reduces the use of templates, provides greater flexibility, and improves productivity.

Various scientific and technological challenges, inherent to this approach, need solution to its full industrial deployment. (*i*) The strategy to adopt for the fusion of

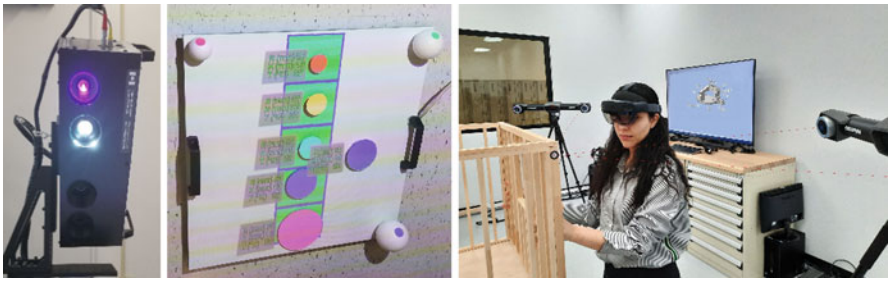

Fig. 9 Left: A structured-light system that incorporates a projector for augmented reality (see Boisvert et al. 2020). Middle: An example, used for public outreach, of a jigless assembly task guided by the structured-light system shown on the left. Right: Demonstration of a metrology-assisted assembly within a research laboratory context. Note the AR headset and the two noncontact 3D imaging systems used to guide the assembly

data from different 3D measurement technologies. (*ii*) Estimating the uncertainties of an assembly made by measurement. (*iii*) Ensuring metrological traceability. (*iv*) The strategies to adopt for setting monitoring targets. (*v*) The behavior of compliant (non-rigid) parts during assembly. Readers interested in the jigless assembly using augmented reality (AR) can explore (Chiboub et al. 2021; Gomes de Mello et al. 2020).

Concluding Remarks

In recent years, new families of affordable 3D imaging systems that target the entertainment industry were introduced (Drouin and Seoud 2020; Giancola et al. 2018; Zanuttigh et al. 2016). Ruggedized versions of some of these scanners are now available. This paves the way to low-cost Industry-4.0 applications that were not economically viable using conventional 3D imaging systems. Since these low-cost systems are targeted at acquiring 3D videos of humans, they are especially attractive for new robotic applications in which humans and machines interact closely (cobot).

As mentioned, earlier, the uncertainty associated with the measurements of optical metrology systems vary within their respective measurement volumes in a nonintuitive manner. This can lead to the use of problematic methodologies in robotic applications, which consist of many possible positioning of the 3D imaging system. Some of the software used by robotic experts can integrate the measuring volume of 3D imaging systems. However, to the best of our knowledge, they do not take into account the performance variation induced by the positioning of the 3D imaging system. Modeling this error when performing path planning would be a powerful enabler for Industry 4.0 applications and quality traceability.

Finally, the implementation of new advanced manufacturing solutions, which use in-process optical metrology, requires the collaboration of experts in the fields of metrology, mechanical engineering, and machine vision. Such a collaboration requires well-recognized terminology, standards, and characterization procedures for 3D imaging systems including hardware and software. Some researchers have

tried to unify the language of each field by developing a characterization procedure that is directly based on the GD&T (MacKinnon et al. 2013). The generalization of the use of such characterization methods would provide a well-defined common language centered on the final objective of the manufacturing process, which is to produce a good part in compliance with requirements. Many other challenges associated with the use of 3D imaging systems can be found in Catalucci et al. (2022).

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Speckle Metrology in Dimensional Measurement

54

Niveen Farid

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Abstract

Speckle can overcome the limitations of prior techniques in non-destructive testing of a product in industrial applications, allowing for a fast, reproducible, and automated description of deformation parameters such as strain, stress, displacement, surface roughness, expansion, and thickness. Speckle metrology has progressed from a research tool to a valuable tool for metrologists, having been proven in comparison to complex, time-consuming, and expensive approaches. The advancement of technology and image processing has attracted the attention of speckle metrology. The goal of this review is to cover the fundamentals of speckle methods, as well as some current improvements and applications.

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Keywords

Speckle pattern · Dimensional metrology · Digital processing · Correlation

List of Symbols

a_1	Varying amplitude for the reference mirror beams in speckle interferometer
a_2	Varying amplitude for the object beams in speckle interferometer
$a(x, y)$	Background noise
A_1	Complex amplitude of the reference wavefront in speckle interferometry
A_2	Complex amplitude of the object wavefront in speckle interferometry
A_r	Specular reflection of the object
A_T	Total amplitude
$A(x, y)$	Amplitude function at the object plane
$A(\theta_1, \theta_2)$	Scattered field amplitude in angular displacement
$A(u, v)$	Amplitude function at the image plane
$b(x, y)$	Modulation noise
$B\{A(q)\}$	Fourier Bessel transform function
$c(x, y)$	Complex term
C_n	Constant of the expansion
$C_{pp'}$	Cross-correlation function in speckle correlation
d	Distance between speckle and Young's fringes planes in angular displacement
Δd	Path difference (thin film thickness)
f_0	Carrier frequency
f_1, f_2	Spatial frequencies
$F(f_1, f_2)$	Fourier transform of the scattered wavefront
$g(x, y)$	Fringe pattern describing function in speckle interferometry
$g(w)$	Function of the spectral line profile
G_2	Geometrical factor
$G(f, y)$	Fourier transform of the fringe pattern describing function
I	Intensity
I_m	Speckle intensity for the substrate at the image plane in speckle interferometry
I'_m	Speckle intensity for the thin film at the image plane in speckle interferometry
I_{max}	Maximum intensity
I_{min}	Minimum intensity
I_T	Total intensity
$I(v_z, v_z)$	Power spectral density functions of the speckles before rotation

$I(v_z', v_z')$	Power spectral density functions of the speckles after rotation
$I(v_z, v_z')$	Correlation function between the speckles before and after rotation
$I(\omega)$	Intensity of the young's interference fringes in the image plane for angular displacement
k	Propagation wave vector
l	Average height
$2L$	Width of the illuminated area
m	Order of interference
M	Magnification factor
P	Point of observation
$p(x, y)$	Reference image in speckle correlation
$p'(x', y')$	Displaced image in speckle correlation
q	Radius coordinate in the object plane
r	Radius coordinate in the image plane
R_a	Average surface height variation
ΔR	Axial displacement
S	Area of integration
$S(x)$	Surface distribution function for a specific area in one dimension
$S(x, y)$	Surface distribution function for a specific area in two dimensions
$u(x, y)$	Displacement field in u-axis in speckle correlation
$v(x, y)$	Displacement field in v-axis in speckle correlation
V	Visibility of the speckle pattern
w_0	Frequency at the center of the spectral line
δ	Spectral half width
Δx	Displacement in the object plane in x-axis
Δy	Displacement in the object plane in y-axis
Z	Focal length of the transform lens
Δ^2	Squared Euclidean distance representing the displacement fields in speckle correlation
φ_1	Varying phase of reference mirror wavefront in speckle interferometry
φ_2	Varying phase of object wavefront in speckle interferometry
$\Delta\varphi$	Phase change between reference and object beams
$\varphi(x, y)$	Phase function
λ	Wavelength
λ_a	Average wavelength
$\Delta\lambda$	Shift in wavelength
σ_r	Standard deviation of a rough surface
θ	Angle of illumination
$\delta\theta$	Angle of object rotation
$\Im[c(x, y)]$	Imaginary component of $C(x, y)$
$\Re[c(x, y)]$	Real component of $C(x, y)$

Introduction

The need to assure the effectiveness of machines and tools used in industrial applications drive the attention to robust simple accurate techniques such as speckle. Speckle can test machines and tools capabilities to function under different conditions, including vibration and temperature. Traditional investigative methods require surface contact and may be limited to object's material and measuring time. Since it is critical to employ a simple nondestructive procedure in the investigation in order to retain the object's quality, and to make the testing service available at any time, and under any circumstance, speckle as a nondestructive technique can detect object's variation when subjected to various conditions (Pino et al. 2011; Bender et al. 2018; Patzelt et al. 2019). In this chapter we will explore different speckle techniques that have proved their reliability in dimensional metrology.

Digital Simulation of Speckle

Special attention has been paid to study the phenomenon of scattering of coherent laser beams forming "speckle" of random intensity and distribution in the light field if the beams are scattered from a rough surface. This phenomenon has become one of the most important research topics in the field of coherent optics. Since that time, the basic studies of the speckle pattern dealt with proposing many and varied applications in the field of scientific and industrial measurements. A large proportion of component failures are due to surface defects caused by either discontinuity of isolated manufacturing or gradual deterioration of surface quality. One of the important factors that ensure the safety of the surface is the surface roughness, which is referred to by the variation in the average heights of the surface points, especially since the manufacturing industry depends on the surface being within certain roughness limits. Hence, measuring surface roughness has become important for quality control work in many applications.

Assume the center line average height, which is the current standard assessment of the surface roughness (average surface height variation) defined by Farid (2008) and Stempin et al. (2021) as:

$$R_{\mathrm{a}} = \frac{1}{\mathrm{Area}} \iint |S(x,y) - l| \, \mathrm{d}x\mathrm{d}y \tag{1}$$

where $S(x, y)$ represents the height distribution function for a specific area of the surface, and the average height l is defined by:

$$l = \frac{1}{\mathrm{Area}} \iint S(x,y) \, \mathrm{d}x\mathrm{d}y \tag{2}$$

Assume $S(x, y)$ as multi sine functions of different amplitudes and phases as follows:

$$S(x,y) = k_1 \sin\left(\frac{2\pi}{\lambda_1}(x+y)\right) + k_2 \sin\left(\frac{2\pi}{\lambda_2}(x+y)\right) + k_3 \sin\left(\frac{2\pi}{\lambda_3}(x+y)\right) + \ldots \quad (3)$$

The laser light scattering from rough surface is determined by the diffraction phenomena based on the electromagnetic wave propagation theory. Then the amplitude function of the wavefront scattered from a surface of a height distribution function $S(x, y)$ is given by:

$$A(x,y) = \exp(i\varphi(x,y)) \quad (4)$$

where the phase function $\varphi(x, y)$ is defined as:

$$\varphi(x,y) = \frac{2\pi}{\lambda} \quad 2S(x,y)\cos\ \theta \quad (5)$$

λ is the laser wavelength that is incident on the rough surface at an angle θ. The Fourier transform of the wavefront is expressed in (Huynh et al. 1991) as:

$$F(f_1,f_2) = \int S(x,y)\ \mathrm{e}^{\left(-\frac{2\pi i}{\lambda}(f_1 x + f_2 y)\right)}\ \mathrm{d}x\mathrm{d}y \quad (6)$$

The spatial frequencies f_1 and f_2 are given by:

$$f_1 = \frac{\mathrm{XM}}{Z} \text{ and } f_2 = \frac{\mathrm{XM}}{Z} \quad (7)$$

Z is the focal length of the transform lens, M is the magnification factor, and X, Y are the coordinates on the image plane. The Fourier transform is reformulated using the expansions of the amplitude and the phase functions as:

$$F(f_1 + f_2) = \sum\nolimits_{n=1}^{\infty} C_n \iint_A \frac{(iS(x,y))^n}{n!}\ \mathrm{e}^{\left(-\left(\frac{2\pi i}{\lambda}\right)(f_1 x + f_2 y)\right)}\ \mathrm{d}x\mathrm{d}y \quad (8)$$

where the constant of the expansion C_n is:

$$C_n = \left(\frac{4\pi\cos\theta}{\lambda}\right)^n \quad (9)$$

In the case of smooth surface (i.e., very small $\varphi(x, y) < \pi/10$), the first term in Eq. (8) at $n = 1$ is predominant while other terms are negligible. As $\varphi(x, y)$ increases, the other terms become significant.

The half width of a spectral line profile of the light source affects the patterns visibility. The spectral line profile is described by the function $g(w)$ of Gaussian distribution:

$$g(w) = g(w_0)\mathrm{e}^{\left(-4\ln 2(w-w_0)^2\right)/\delta w^2} \quad (10)$$

$$g(w_0) = \frac{1}{\delta w}\left(\frac{4\ln 2}{\pi}\right)^{1/2} \tag{11}$$

such that w_0 is the frequency at the center of the spectral line, and δw is the spectral half width. Then the spatial intensity distribution of the speckle pattern is described by:

$$I = \int_{-\infty}^{\infty} F(f_1,f_2)\, F^*(f_1,f_2)\, g(w)\mathrm{d}w \tag{12}$$

Correspondingly, visibility of the speckle pattern is determined by:

$$V = \frac{\left\langle I^2_{\max}(x,y)\right\rangle^{1/2} - \left\langle I^2_{\min}(x,y)\right\rangle^{1/2}}{\left\langle I^2_{\max}(x,y)\right\rangle^{1/2} + \left\langle I^2_{\min}(x,y)\right\rangle^{1/2}} \tag{13}$$

such that $I_{\max}$ and $I_{\min}$ are the maximum and minimum intensities obtained by constructive and destructive interference, respectively. Figures 1, 2, and 3 show digital simulation for different assumptions of $S(x, y)$ using sinusoidal functions of

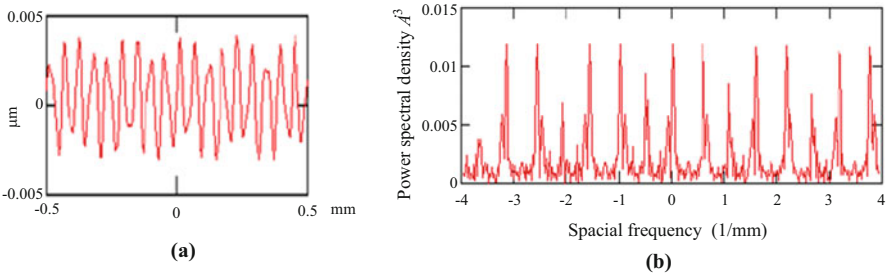

Fig. 1 Simulation of (**a**) the surface function and (**b**) the Fourier transformation of a surface of roughness 1.7 µm (Farid 2008)

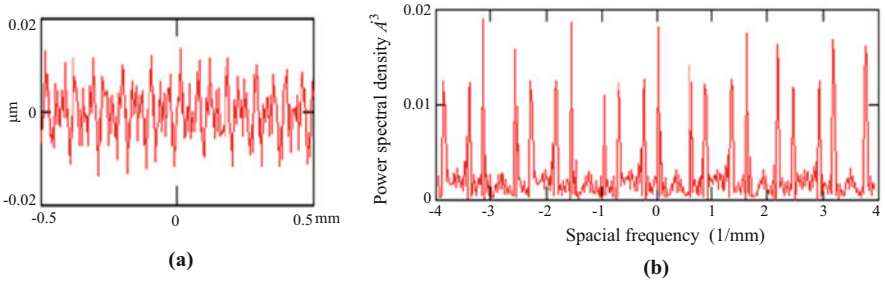

Fig. 2 Simulation of (**a**) the surface function and (**b**) the Fourier transformation of a surface of roughness 3 µm (Farid 2008)

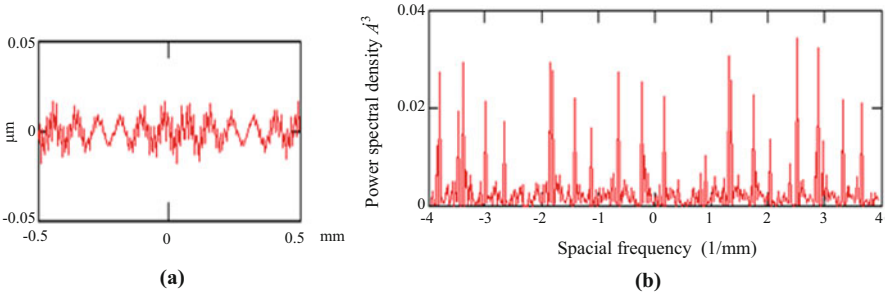

Fig. 3 Simulation of (**a**) the surface function and (**b**) the Fourier transformation of a surface of roughness 6.5 μm (Farid 2008)

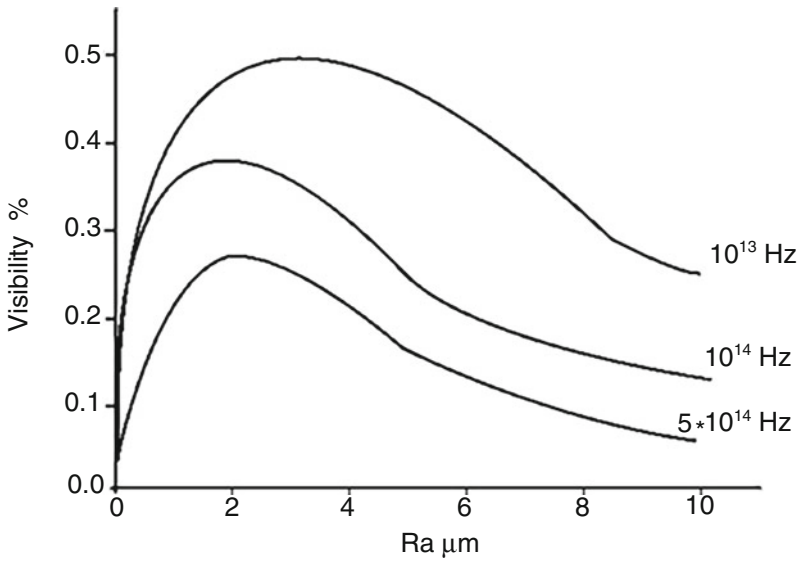

Fig. 4 Plot of the speckle visibility at different degrees of surface roughness, and different spectral lines (Farid 2008)

variant spatial frequencies and amplitudes, and the corresponding Fourier transformation FFT.

A simulation of the dependence of the speckle pattern visibility on the surface roughness for different spectral lines is shown in Fig. 4.

The wave reflected by a rough surface is the sum of many independent secondary waves from the scattering areas. Because the secondary waves arrive at different light paths at different points on the observation plane, the dephased secondary

waves interfere at these points. This interference varies from one point to another on the observation plane forming the speckle. This explains why the speckle appears as granular pattern. Secondary waves can interfere if their light path difference is less than the coherent length, and the coherent length that is associated with temporal coherence.

The simulation in Fig. 4 shows an inverse relationship between the speckle visibility and the spectral line half width, which is relevant to the fact that the temporal coherence length of the light source used in illumination is in inverse relation with the spectral half width.

The visibility of the speckle pattern increases with the increment of the surface roughness and the random interference until the visibility achieves its maximum limit. With further increment in the surface roughness, the scattered temporal incoherent beams increase leading to a reduction in the speckle visibility.

The pattern is an incoherent combination of the intensity, and the superposition of intensity will result in the reduction of speckle contrast since the speckle patterns are different for the lights of different bandwidths. So the spectral line of $\delta w = 5 \times 10^{14}$ Hz, which is of low temporal coherence and short coherence length reduces the speckle.

This is evident because of the fact that for larger spectral width, the speckle pattern becomes blurry and the contrast decreases. This also provides a quantitative way to optimize the image quality and also to determine the safety power level for different light sources when viewing the produced images by human eyes directly.

Speckle Photography

Speckle photography is a technique in which the object surface is illuminated with laser beams and the resulted random interference is recorded. It is a widely used nondestructive tool in metrology due to its simplicity. It is useful to analyze surface deformation that is induced by mechanical or thermal effects, and to evaluate various types of surface displacements and translations (Necklawi et al. 2007; Tausendfreund et al. 2021). In speckle photography, the surface of interest is illuminated with coherent light, and the scattered light is collected by a lens, and imaged by either photographic film or a CCD array (Fig. 5).

The scattered light will contain speckle due to the roughness of the surface, and the speckle pattern can be imaged and stored for the analysis. The speckle size can be controlled by the F-number of the imaging lens to be suitable to the size of the defect that the object encounters. Displacements, stress, or strain, etc., imparted to the surface can shift the speckle patterns proportionally to the direction of the local deformation, and the full deformation field can be determined either by the Fourier transform of the speckle double exposure or by the speckle cross-correlation.

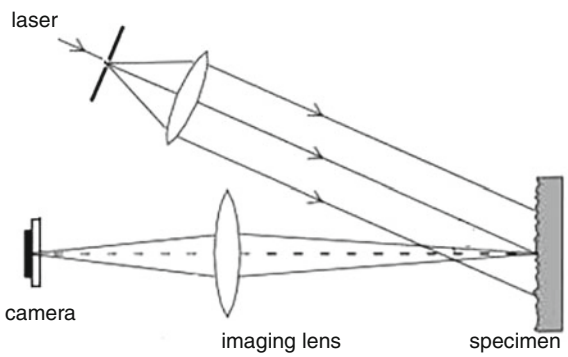

Fig. 5 Schematic diagram of the speckle photography setup

Fourier Transform of Speckle Double Exposure

Nondestructive testing can benefit greatly from optical approaches that offer sufficient information on displacement and deformation. The following items are necessary in an optical system to collect the data required for the testing procedure:

- The object's reference state.
- Optical wavefront.
- The altered object configuration.
- Imaging system.
- Detection process.
- Decoding process that analyzes the received data and converts it into a suitable format to a human or a machine.

A deformation/displacement applied to the surface creates a shift in the speckle pattern that is related to the magnitude and direction of the deformation/displacement.

In double speckle exposure, both initial and updated speckle patterns are recorded and then compared by image processing software. The difference between the speckle patterns of the two states produces speckle pairs representing a double exposure. A Fourier analysis of a double exposure speckle image determines the total surface deformation.

In the Fourier analysis method, the displacement field was determined by scanning a laser beam across the negative of the double speckle pattern in the early days of the speckle photography to form the Young's interference fringes whose orientation and spacing are relevant to the direction and the magnitude of the local deformation. Replacement of the photographic film by the CCD sensor could realize a substantial increase in the imaging speed, enable real time analysis using digital image processing, and make the speckle photography more suitable to industrial applications (Abdel Hady et al. 2012; Janák et al. 2020). Analyzing of speckle pattern using this method is effective in measuring in-plane and out-of-plane displacements.

In-Plane Displacement

In-plane displacement is a kind of displacement that occurs in the same plane of the object by mechanical or thermal effects. In order to measure the in-plane displacement of an object using the speckle photography, the speckle is recorded before and after the displacement, then the speckle images are added digitally yielding pairs of identical speckle patterns shifted by amounts Δx and Δy that are related to the displacements made in the object plane (Abdel Hady et al. 2012; Yan et al. 2019). The implementation of FFT to the speckle pairs produces Young's interference fringes, because each pair of the speckle acts as a pair of identical sources of coherent light, which interfere to form the patterns in Fig. 6. Visibility, separation, orientation, and number of the fringes are related to the amount and the direction of the displacement of interest.

Assume Fig. 7 displays the scattering system in the speckle photography. The scattering surface lies in the (x, y) plane and is illuminated by highly coherent light. Any point on the observation plane (u, v) placed at a distance R in front of this surface will receive contributions from all points on the surface. Assume that O is any point on the scattering surface plane, I is any point on the observation plane and R is the distance between O and I making an angle α with R.

The total complex amplitude at the image plane is (Necklawi et al. 2007):

$$(u,v) = \mathrm{e}^{-i\frac{2\pi}{\lambda}(R)} \int A(x,y)\, \mathrm{e}^{-i2\pi\left((x+\Delta x)\left(\frac{u}{\lambda R}\right)+(y+\Delta y)\left(\frac{v}{\lambda R}\right)\right)}\, \mathrm{d}x\mathrm{d}y \tag{14}$$

Correspondingly the intensity at the image plane is defined by:

$$I = 2\mathrm{FFT}(A(x,y))^2(1 + \cos 2\pi(f_1\Delta x + f_2\Delta y)) \tag{15}$$

where the spatial frequencies are represented by Eq. (16) showing linear relationships with object displacements.

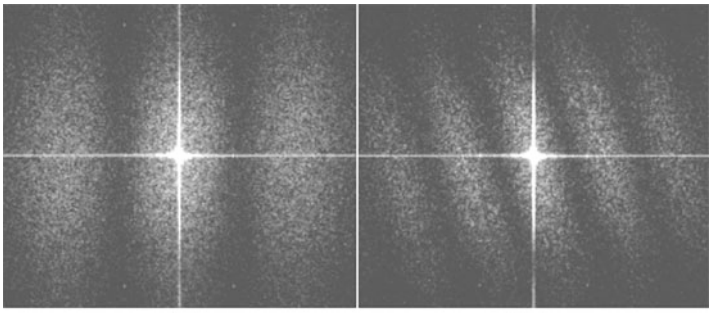

Fig. 6 Young's interference fringes obtained by applying FFT to the Speckle pairs indicate (**a**) horizontal displacement and (**b**) diagonal displacement (Abdel Hady et al. 2012)

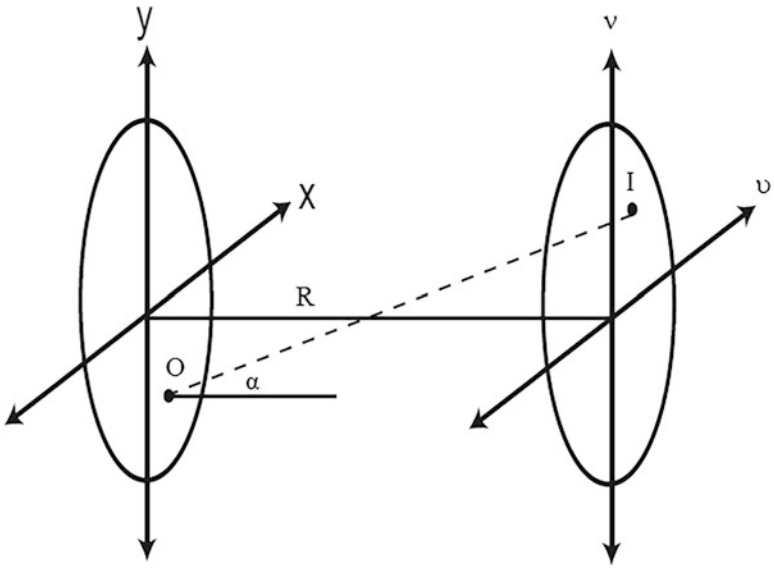

Fig. 7 The scattering system in which the scattering point *O* in the object plane is observed at the point *I* in the image plane (Abdel Hady et al. 2012)

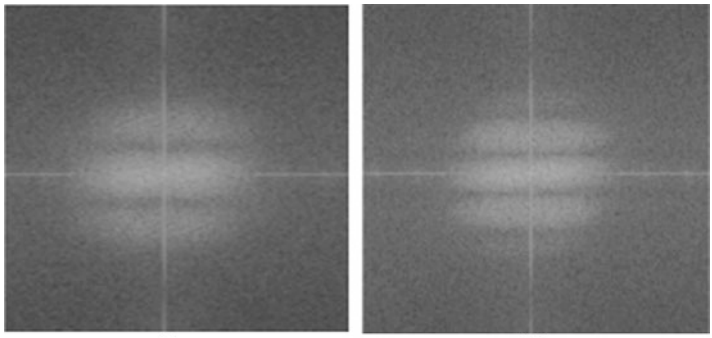

Fig. 8 Young's interference fringes provided by the Fourier transform of speckle pattern for an object under axial loads of (**a**) 0.5 kg and (**b**) 0.9 kg

$$f_1 = \frac{u}{\lambda R}, \qquad f_2 = \frac{v}{\lambda R} \tag{16}$$

Figure 8 shows Young's interference fringes produced by the implementation of Fourier transform to double exposure speckle that is resulted when placing different loads on the upper surface of an object. The interference fringes' separation, orientation, and number are relevant to the axial loads.

The results in Fig. 8 are in good agreement with the results given by automatic comparator balance of capacity 1000 g based on the three cycles method according

to OIML R 111–12,004 (OIML, 2004). The uncertainty in measurement was found to be ±2 μg.

Out-of-Plane Displacement

Out-of-plane displacement is the displacement in which the object moves out of its plane such as axial and angular displacements (Gu et al. 2018; Hu et al. 2021).

- **Axial Displacement**

In the axial displacement, the object (diffuser) is shifted axially forming an interferometer of two parallel planes in which the speckle patterns are modulated by a ring system. A double exposure of axially shifted speckle patterns forms interference circular rings when processed by Fourier transform. Each of the rings is localized in a particular plane and the number of the observable rings is limited by the speckle size (Fig. 9).

In this case the wave fields are represented in the polar coordinates and the intensity at the image plane is (Necklawi et al. 2007):

$$I = 2\left(\frac{S}{\lambda R}\right)^2 B^2\{A(q)\}\left(1 + \cos\ k\Delta R\left(\frac{r^2}{2R(R+\Delta R)} - 1\right)\right) \tag{17}$$

where S is the area of integration, $B\{A(q)\}$ is the Fourier Bessel transform function, $q = \sqrt{x^2 + y^2}$ is the radius coordinate in the object plane, and $r = \sqrt{u^2 + v^2}$ is the radius coordinate in the image plane.

At maximum intensity, the axial displacement is determined by:

$$\Delta R = \frac{2m\lambda R^2}{r^2} \tag{18}$$

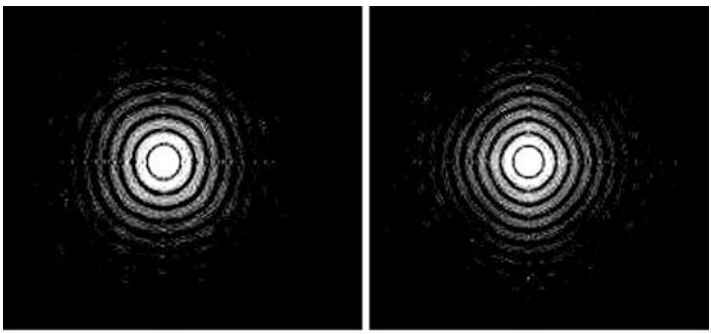

Fig. 9 Successive rings obtained by applying Fourier transform to a double exposure speckle pattern for (**a**) 20 μm, (**b**) 60 μm axial displacements (Necklawi et al. 2007)

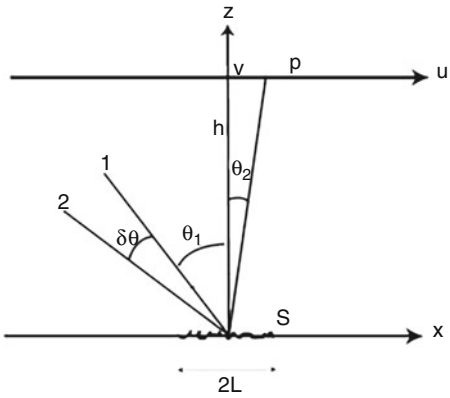

Fig. 10 Schematic diagram for the angular displacement of the object (Farid 2008)

- **Angular Displacement**

In the angular displacement, the object is rotated by an angle $\delta\theta$ (Fig. 10) such that the classical Young's fringes are formed by applying Fourier transform to the double exposure speckle patterns. As $\delta\theta$ increases, the correlation between the speckle patterns decreases.

Assuming P be the point of observation, and assuming one-dimensional rough surface represented by $S(x)$, then the scattered field $A(\theta_1, \theta_2)$ at P is given by Farid (2008) as follows:

$$A(\theta_1, \theta_2) = \frac{A_r G_2(\theta_1, \theta_2)}{2L} \int_{-L}^{L} e^{i(v_x + v_z S(x))}\, dx \tag{19}$$

where A_r is the specular reflection of the object, G_2 is a geometrical factor defined by Holzer (1978), and $2L$ is the width of the illuminated area.

The intensity of the young's interference fringes in the image plane is represented by Holzer (1978) as follows:

$$I(\omega) = \left(\frac{|A_r|^2 G_2^2(\theta_1, \theta_2)}{4L^2}\right)[I(v_z, v_z) + I(v_z', v_z') + 2I(v_z, v_z') \cos(2k \cos\theta_1 \delta\theta\omega)] \tag{20}$$

where $I(v_z, v_z)$ and $I(v_z', v_z')$ are the power spectral density functions of the speckles before and after rotation, respectively, $I(v_z, v_z')$ is the correlation function between the two speckles, $\omega/(\lambda d)$ is the frequency in the image plane, and d is the distance between the speckle plane and Young's interference fringes plane. The surface height distribution $S(x)$ is described by the standard deviation σ_r. For a very rough surface ($v_z \sigma_r > > 1$), the visibility of Young's interference fringes is described by:

$$V \approx \mathrm{e}^{-((2\pi/\lambda)\sigma \sin\theta_1 \delta\theta)^2} \tag{21}$$

which proves the inverse relationship between the visibility of Young's interference fringes and the angular displacement.

Speckle Cross-Correlation

Techniques that measure the displacement based on analyzing Young's interference fringes can only indicate the value of the displacement but not the field of the displacement. A technique such as the cross-correlation can identify the field of displacement by determining the cross-correlation function between the speckle patterns recorded before and after displacement or deformation (Hofer et al. 2018). The speckle patterns are recorded in a sequence of images as the object undergoes the deformation, and the speckle positions are traced, matched, and analyzed by iterative method (Sutton et al. 1983, 1986). This enables tracking the variation of the object's surface corresponding to the strength and the direction of the influencing factor that causes the deformation to the object (Fig. 11). The processed data can visualize the field of displacement with an accuracy that depends on the resolution of the recording sensor and the stability of the optical system. Both flexibility and simplicity of this technique increase its applications in surface engineering science.

The first image in the recorded sequence is considered as the reference case while the others show the gradual variation from the reference case. Each image is divided into sub-images in which the speckle movement is traced to extract the length and the direction of the displacement vectors by the maximum cross-correlation. The cross-correlation technique is more informative as it can provide complete data analysis for different kinds of materials and deformations. It compares the sequential speckle images to extract the relative motion of the speckles by calculating the cross-correlation function.

In Fig. 12, if $p(x, y)$ is the reference image, and $p'(x', y')$ is the displaced image, and $u(x, y)$, $v(x, y)$ are the displacement fields, then the relation between the two functions is represented by the squared Euclidean distance (Gonzalez and Woods 1992):

$$\Delta^2 = \sum\nolimits_{x,y} (P(x,y) - P'(x - u(x,y), y - v(x,y)))^2 \tag{22}$$

The expansion of this equation yields the cross-correlation function that finds the similarity between $p(x, y)$ and $p'(x', y')$ (Perie et al. 2002; Chu et al. 1985):

$$C_{\mathrm{pp}'} = P(x,y) * P'(x - u(x,y), y - v(x,y)) \tag{23}$$

$$P * P' = \mathrm{FFT}^{-1}(\mathrm{FFT}(P*)\ \mathrm{FFT}(P')) \tag{24}$$

The displacement fields $u(x, y)$ and $v(v, y)$ are determined by evaluating the normalized cross-correlation of the speckle patterns before and after the deformation.

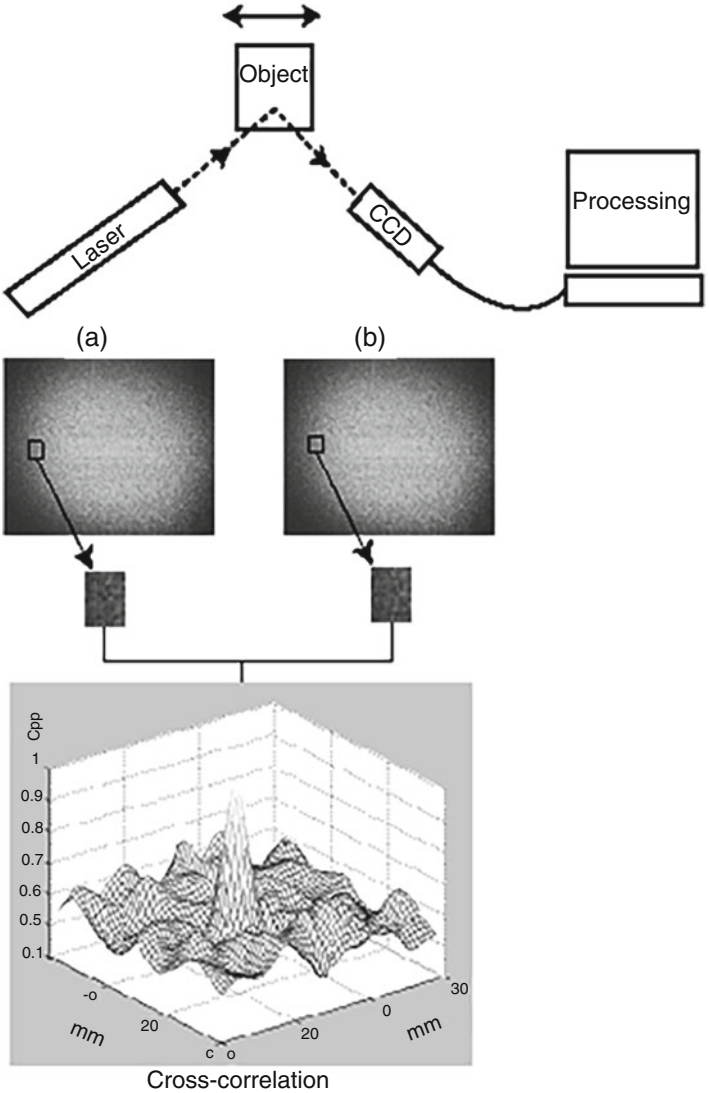

Fig. 11 Cross-correlation function that detects the field of displacement from the reference case in (**a**) to the shifted case in (**b**) (Farid et al. 2012)

Speckle photography could provide accuracy of ±4 μm for dynamic displacement of the object. The results in these measurements are affected by the cosine error influenced by the angle between the axis of object movement and the axis of measurement, while in case of static displacement of the object such as expansion and strain, the accuracy could achieve ±0.2 μm due to the absence of the cosine error.

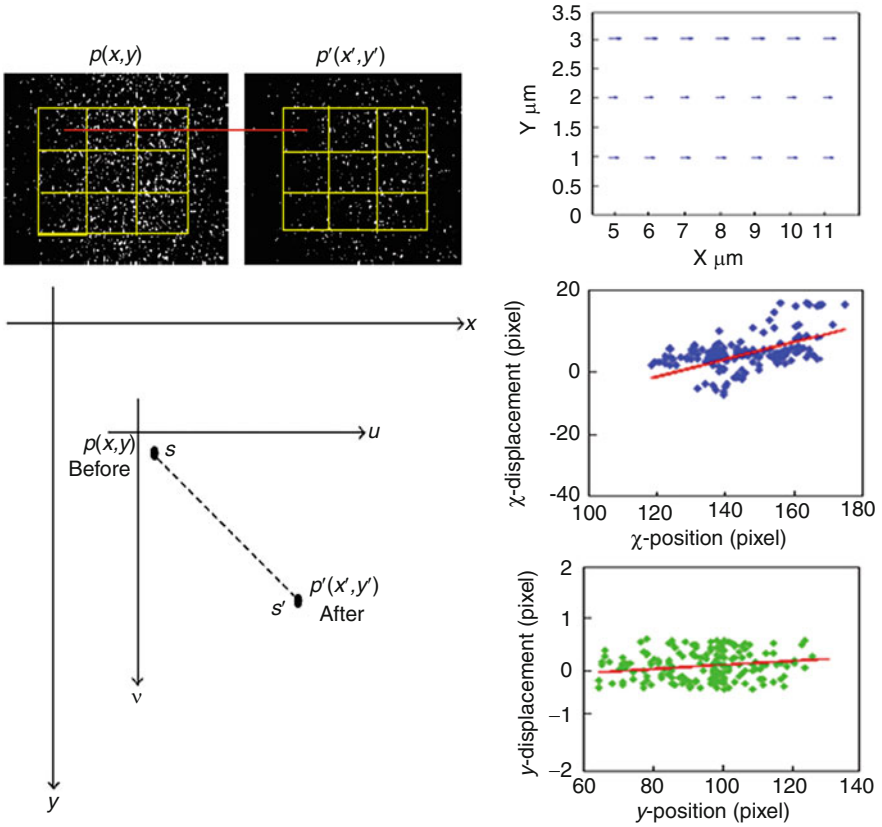

Fig. 12 The image $p'(x', y')$ is shifted from the object $p(x, y)$ by the displacement fields $u(x, y)$ and $v(x, y)$ (Farid et al. 2012)

Extending the Measurement Capabilities of Speckle Photography

The speckle patterns that characterize the scattering parameter of an object can be induced by coherent and incoherent sources. The scattered beams interfere coherently and induce speckle patterns with high contrast in case the scattering parameter is within the coherence length of the light source. As the scattering parameter increases to a value that is greater than the coherence length, the contrast decreases. A tunable source allows controlling the speckle contrast by wavelength sweeping, which is suitable to characterize very rough surfaces by synthesizing a comparable wavelength (McKinney et al. 1999).

However, incoherent sources can induce speckle patterns that are effective in measuring fine dimensions. Argon-ion lasers are of high advantage in illuminating large objects due to the high output powers, hundreds of milliwatts to several watts allowing them to be delivered by optical fibers with considerable intensities. Solid state laser diodes are small compact portable tunable sources, which provide variable

wavelength (Tatam 1998; Huang et al. 1997; Galannlis et al. 1995). This wavelength is shifted by the modulation of the cavity effective index via temperature control (low frequency modulation capability) or the injection current (high frequency modulation capability). Large wavelength shifts of 0.3–1.5 nm can be produced by the injection current inducing mode hops. Also the emission wavelength can be shifted by several nanometers at a rate of 0.3 nm/K by temperature tuning. Then tuning by temperature and by current is considered as coarse and fine tuning, respectively. Speckle photography uses diode lasers due to coherence, reliability, and low cost sources for surface measurement required in modern industry (Wei 2002). Surface roughness can be characterized by correlation methods in which the speckle images are taken either at different illumination angles (angular correlation) or at different wavelengths (spectral correlation), then the images are analyzed using point to point correlation or by Fourier transform in the far field (Jakobi 2000).

Some studies (Tchvialeva 2010; Strzelecki et al. 1998) could explore the effect of the wavelength on the speckle contrast, and the relation between the scattering surface and the speckle at arbitrary wavelength, to determine the surface roughness at arbitrary wavelengths. Moreover Shirley and Lo (1994); and Shirley and Hallerman (1996) discussed the dependence of speckle on the tunable laser emission. In Farid et al. (2015) the speckle photography used the diode laser to improve the visibility of the interference fringes to provide accurate measurement of roughness values at extended range. A suitable wavelength was selected for measuring a specific roughness based on the speckle visibility. As roughness increased, the speckle contrast decreased correspondingly due to the reduced temporal coherence associated with the coherence length. To avoid speckle reduction, the diode laser enabled scanning the wavelengths to produce a suitable synthetic wavelength of a larger coherence length. The large scattering parameter could then lie within the new coherence length, and consequently, the speckle contrast was improved.

The following formula presents the field amplitude of the wavefront scattered by a rough surface that is represented by a distribution function $S(x, y)$:

$$A(x,y) = A_o e^{i\varphi(x,y)} \tag{25}$$

such that $\varphi(x, y)$ is the phase function at wavelength λ and incidence angle θ, and described as:

$$\varphi(x,y) = \frac{2\pi}{\lambda} 2S(x,y)\ \cos\ \theta \tag{26}$$

The Fourier transform of the scattered wavefront is represented by:

$$F(f_1,f_2) = \int A(x,y)\ e^{\left(-\frac{2\pi i}{\lambda}(f_1 x + f_2 y)\right)}\ dxdy \tag{27}$$

Consequently, the intensity of the speckle pattern at single exposure becomes:

$$I = \int F(f_1,f_2)F^*(f_i,f_2)g(\upsilon)\mathrm{d}\upsilon \tag{28}$$

where $g(w)$ is the function that describes a symmetrical spectral line profile as follows:

$$g(w) = g(w_o)\mathrm{e}^{\left(\frac{-4\ln 2(w-w_o)^2}{\delta w^2}\right)} \tag{29}$$

where w_o is the central frequency, and δw is the spectral half width. The object is illuminated with different wavelengths λ_1, λ_2 for each exposure, and the amplitude at each wavelength becomes:

$$A_1(x,y) = A_{01}\,\mathrm{e}^{i\phi_1(x,y)} \text{ and } A_2(x,y) = A_{02}\,\mathrm{e}^{i\phi_2(x,y)} \tag{30}$$

$$\begin{aligned} &\text{and } \phi_1(x,y), \phi_2(x,y) \text{ are : } \phi_1(x,y) = \frac{2\pi}{\lambda_1}\,2S(x,y)\,\cos^{\circ}\theta \\ &\phi_2(x,y) = \frac{2\pi}{\lambda_2}\,2S(x,y)\,\cos^{\circ}\theta \end{aligned} \tag{31}$$

where $\lambda_2 = \lambda_1 + \Delta\lambda$.

The total amplitude after double exposure is:

$$A_T(x,y) = A_1(x,y) + A_2(x,y) \tag{32}$$

Assuming I' is the second exposure intensity, and then the total combined intensity at the double exposure $I_T = I + I'$ is:

$$I_I = |A_T|^2 = A_1^2 + A_2^2 + 2A_1A_2\,\cos\,\left(4\pi\left(\frac{1}{\lambda_1} - \frac{1}{\lambda_2}\right)S(x,y)\,\cos\,\theta\right) \tag{33}$$

$$I_I = I_1 + I_2 + 2\sqrt{I_1I_2}\,\cos\,\left(\frac{4\pi\,\Delta\lambda}{\lambda_a^2}S(x,y)\cos\,\theta\right) \tag{34}$$

where λ_a is the average wavelength.

The reconstruction of the double exposure speckle pattern in the image plane yields Young interference fringes of the following visibility (Farid et al. 2015):

$$V = \mathrm{e}^{-\left(\frac{4\pi\sigma\,\Delta\lambda}{\lambda_a^2}\,\cos\,\theta\right)^2} \tag{35}$$

$$\sqrt{-\ln(V)} = \frac{4\pi\,\sigma\,\Delta\lambda}{\lambda_a^2}\,\cos\,\theta \tag{36}$$

where σ is the standard deviation of the surface distribution function $S(x, y)$.

Farid et al. (2015) used transverse and longitudinal single mode where single frequency operation is maintained over several hundred GHz. Maximum Optical output power is 80 mW. The full lasing tuning range of the laser lies within 779–781 nm. Modulating the laser diode temperature rather than injection current could control the wavelength in order to keep the intensities of the speckle patterns unaffected by increasing the roughness degree from 4.5 to 18 μm. The object was illuminated by an arbitrary wavelength and the speckle pattern was recorded referring to the first exposure. The wavelength was then tuned by raising the temperature of laser cavity to 297.15 K, 303.15 K, 309.15 K, and 315.15 K.

and the second exposure was recorded. Selecting two different wavelengths induced a change in the speckle patterns. The construction of the combined speckle (double exposure) formed Young's interference fringes of number and visibility depend on $\Delta\lambda$ and σ of the object. Figure 13 shows the effect of tuning the wavelengths on improving the visibility of the interference fringes when measuring increased roughness. This proves the advantage of this method in extending the measurement capabilities of the speckle photography for high roughness degrees.

The curves in Fig. 14 explore the relation of the visibility of Young's interference fringes with the tuned wavelength for different roughness degrees (4.5, 9, 18 μm) based on Eq. (35).

From Fig. 14, it is clear that the visibility of Young's interference fringes decreases with roughness increment. By selecting an appropriate $\Delta\lambda$ the visibility can be improved to enable better analysis at the increased roughness.

Both experimental and theoretical results in Figs. 13 and 14, respectively, are similar, and show good agreement with the reference values of the standard roughness specimens used in the experiment. The speckle photographic technique using diode lasers as coherent sources could fit the very rough surfaces since the speckle contrast could be improved by scanning laser frequency to select the suitable wavelength. The accuracy of this technique was found to be ± 0.6 μm.

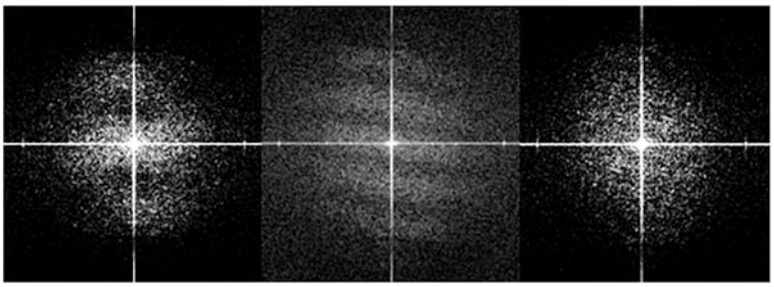

Fig. 13 Young's fringes' visibility for roughness $\sigma = 18$ μm at $\Delta\lambda = 0.4$ nm, 0.7 nm, and 1 nm

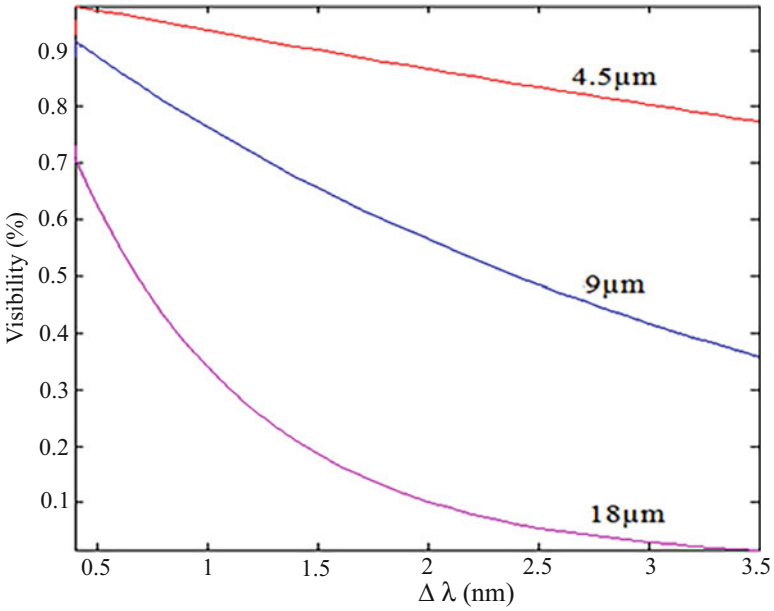

Fig. 14 Dependence of Young's interference fringes visibility on the tuned wavelength

Speckle Interferometry

A motion or deformation of an object surface causes a change in the intensity or phase of the individual speckles in a speckle interferometer. Interference fringes are a visual representation of this change that can be determined (Goodman 2007; Tokovinin et al. 2020).

The phase change caused by the thin film thickness is extracted by analyzing the intensities of the interference patterns in Fig. 15a, and the thickness can be calculated by converting the interference fringes into a wrapped phase map with a phase between 0 (black) and 2π (white).

An unwrapped phase map with a continuous grayscale can be obtained if 2π is correctly added or subtracted every time a discontinuous jump of 2π appears (Svanbro 2004). By this, both thickness and surface deformation of the film is obtained by one-dimensional FFT with phase unwrapping (Necklawi et al. 2006).

Assuming A_1 and A_2 the complex amplitudes of the wavefronts coming from the thin film and the reference mirror, respectively:

$$A_1 = a_1 e^{i\phi_1} \text{ and } A_2 = a_2 e^{i\phi_2} \tag{37}$$

where a_1, a_2 are varying amplitudes, and φ_1, φ_2 are varying phases.

The speckle intensity at the image plane on the substrate is:

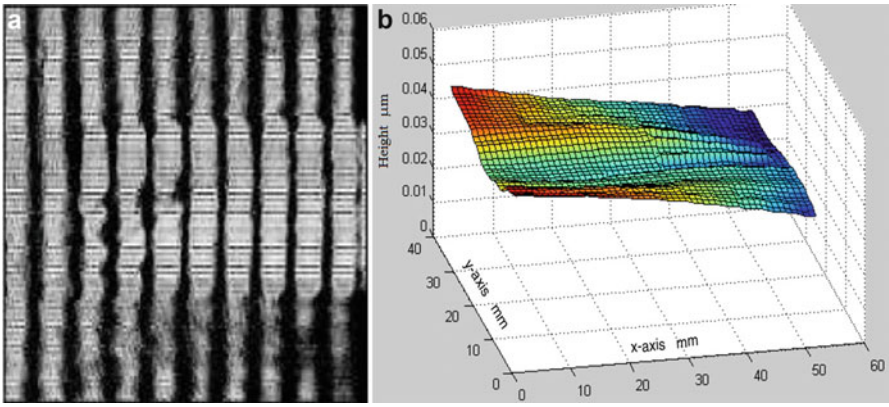

Fig. 15 (**a**) Thin film interference fringes captured from the speckle interferometry, and (**b**) Thin film profile mapping

$$I_m = I_1 + I_2 + 2\sqrt{I_1 I_2}\cos\ \varphi \tag{38}$$

such that

$$I_1 = A_1 A_1^*, \quad I_2 = A_2 A_2^*, \quad \phi = \phi_1 + \phi_2 \tag{39}$$

The speckle intensity on the thin film is:

$$I'_m = I_1 + I_2 + 2\sqrt{I_1 I_2}\cos\ (\varphi + \Delta\varphi) \tag{40}$$

Maximum correlation is reached at

$$\Delta\varphi = 2n\pi \tag{41}$$

While minimum correlation is at

$$\Delta\varphi = (2n+1)\pi \tag{42}$$

The variation in the correlation between I_m, I'_m represents the variation in the thin film thickness.

$\Delta\varphi$ depends on the path difference of the traveling waves resulted from the thickness of the thin film.

The path difference is given by:

$$\Delta d = \Delta\varphi \frac{\lambda}{4\pi} \tag{43}$$

where Δd is the thickness of the thin film, and λ is the wavelength of the light source.

Fourier Transform and Phase Unwrapping

When an object encounters a deformation, the uniform spacing between the fringes is lost as the fringes deform, resulting in localized areas of compressed and expanded fringe spacing. Scanning across this area would detect changes in fringe frequency that corresponded to fringe spacing changes. The essential phase information is separated by extracting this fringe frequency from the dominant carrier frequency to analyze surface deformation (Brayanston-Cross et al. 1994; Yun et al. 2017). This is achieved by applying the Fourier transform method to handle the fringe pattern described by the following equation:

$$g(x,y) = a(x,y) + b(x,y)\cos\ [2\pi f_0 x + \varphi(x,y)] \tag{44}$$

where $a(x, y)$ is the background noise, $b(x, y)$ is the modulation noise, f_0 is the carrier frequency in the x-direction, and $\varphi(x, y)$ is the phase information of interest.

Equation (4) is displayed in its complex form as follows:

$$\begin{aligned} g(x,y) = a(x,y) + \frac{1}{2}b(x,y)[&\cos(2\pi f_0 x + \varphi(x,y)) + j\sin(2\pi f_0 x + \varphi(x,y)) \\ &+ \cos(2\pi f_0 x + \varphi(x,y)) - j\sin(2\pi f_0 x + \varphi(x,y))] \end{aligned} \tag{45}$$

where $\cos(2\pi f_0 x + \varphi(x, y)) - j \sin(2\pi f_0 x + \varphi(x, y))$ is the complex conjugate.

Equation (45) can be rewritten in more convenient form thus:

$$g(x,y) = a(x,y) + \frac{1}{2}b(x,y)\left(e^{j\varphi(x,y)}e^{j2\pi f_0 x} + e^{-j\varphi(x,y)}e^{-j2\pi f_0 x}\right) \tag{46}$$

let $c(x, y) = (1/2)b(x, y)e^{j\varphi(x, y)}$ and $c^*(x, y)$ be the conjugate. Thus,

$$g(x,y) = a(x,y) + c(x,y)\ e^{j2\pi f_0 x} + c^*(x,y)\ e^{-j2\pi f_0 x} \tag{47}$$

The Fourier transform of the intensity distribution is given by

$$G(f,y) = A(f,y) + C(f - f_0, y) + c^*(f + f_0, y) \tag{48}$$

By choosing a carrier frequency that is suitable for the resolution of the image, and the cutoff frequencies, the phase information can be separated from the unwanted frequency modulation and the high frequency speckle. Then the phase map is calculated using the inverse transform of $C(x, y)$, and the phase distribution is given by

$$\varphi(x,y) = \arctan\left[\frac{\Im[c(x,y)]}{\Re[c(x,y)]}\right] \tag{49}$$

where $\Im[c(x,y)]$ and $\Re[c(x,y)]$ are the imaginary and real components of $C(x, y)$.

The solution of this equation is the wrapped phase in the form of sawtooth function with discontinuities, i.e., with edges as fraction $[\Im[c(x,y)]/\Re[c(x,y)]$ changes by 2π. Methods that eliminate the edges are known as phase-unwrapping methods, which can be realized by finding the edges and adding $\pm 2\pi$ to the wrapped phase when passing an edge (Robinson 1993; Stetson et al. 1997).

In order to obtain phase information from the interference fringes of the thin film, the Fourier transform technique requires only one fringe pattern, and the desired phase information can be extracted from the high frequency speckle and DC noise by carefully selecting cutoff frequencies. Despite the existence of discontinuities and pixel noise, a fringe pattern can be handled by combining the Fourier transform approach with phase unwrapping to display the phase information (Fig. 15b). This method requires one image, and as a result, the device is simple and noise-resistant. The speckle interferometer could measure the thickness of the thin film with accuracy of ± 0.02 μm.

Future Prospective in the Field of Speckle Metrology

The speckle techniques discussed in this chapter are very promising for future applications. For example Speckle can be used in measuring light wavelengths. This can develop compact wavemeters and spectrometers, although the resolution will be limited by the strong correlations between the speckle patterns produced by closely spaced wavelengths as in a diode laser. Analyzing the speckle images in a proper way can overcome this limitation.

Speckle can also be used as a tool to extract a unique code from the component's surface, and the pattern can then be used as a secure digital fingerprint. In this case, the intensity correlation can be used as a numerical identifier.

Furthermore, speckle photography is appropriate for in-process measurements. The limitation of this fast and robust measurement technique based on image correlation techniques is the inability to detect out-of-plane deformations in the measurement system's direction, which may increase the measurement error of in-plane deformations. Practical methods can be developed for inferring local out-of-plane motions of the object surface from speckle pattern decorrelation and reconstructing 3D deformation fields while maintaining in-process capability.

Moreover, compound speckle interferometer can be developed for measuring three-degree-of-freedom (3-DOF) displacement using a combination of heterodyne interferometry, speckle interferometry, and beam splitting techniques, while maintaining high resolution and simple configuration. This combination will enable high measurement capabilities to be applicable to real-world applications.

Traceability to SI Unit

To establish traceable measurement for speckle technique in a real-world environment, the optical system must be calibrated and a traceable reference source used.

Using a traceable "reference," such as a laser source with a wavelength suitable for the range of optical measurement, can disseminate length traceability to the measurand.

The measurands in the speckle techniques discussed here are represented ni metric dimensions. Because speckle analysis is dependent on image processing, the pixel must be converted to length unit, and this conversion is done according to pixel size in each magnification, which drives the attention to the importance of calibrating the optical system.

A preliminary test can be done to calibrate the speckle technique readability by moving the object laterally or axially in reference steps using calibrated motorized stage. The displacement represented by laser speckle is compared to the reference one to perform a traceable full-field measurement calibration. This can be used to assess a specific setup in a real-world environment related to a specific application. It can also help to understand the measurement system's performance by comparing measurements to a reliable, calibrated source, such as the motorized stage. The optical system calibration investigates the effect of measurement process variables on measurement accuracy. Understanding these influences increases confidence in the reliability of both singular and comparative measurements, while also assisting in the refinement of experimental design.

In some cases, calibration is accomplished by comparing optically measured quantities to theoretical predictions. In other cases, a variety of measurements are proposed for evaluating out-of-plane displacements.

Some approaches propose applying artificial deformation to a reference image in order to provide an accurate representation of a real speckle pattern. Or artificial speckle with well-known characteristics can be added to the image in a controlled manner to assess optical system calibration.

When evaluating the uncertainty, systematic and alignment errors are identified through the calibration of the optical system. In speckle technique that uses digital image correlation, it is necessary to consider the influence of variance errors on measurement uncertainty. This requires computing spatial and temporal standard deviations of the quantity of interest from static object images to quantify the variance errors.

Also, both standard deviation and spatial resolution are necessary in evaluating the uncertainty in dimensional measurement using speckle. The standard deviation shows the smallest value of the measured dimension that can be detected, while spatial resolution is the smallest distance between two independent measurement points. In addition, other factors including the environment, lighting, and speckle size contribute to the uncertainty in measuring displacements using speckle techniques.

Conclusion

Some interesting areas have been discussed including different applications of speckle metrology in nondestructive testing. Technological development in the speckle field provides fast and simple analysis of different types of deformation overcoming many limitations.

Cross-References

► Industrial Metrology
► Optical Dimensional Metrology

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Necessity of Anatomically Real Numerical Phantoms in Optical Metrology

55

A Study

Vineeta Kumari, Neelam Barak, and Gyanendra Sheoran

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Abstract

Optical phantoms with tissue-mimicking properties have long been recognized as an important intermediary stage for developing metrological techniques, and their use in preclinical studies is extensively growing. This chapter discusses the various phantoms currently being used in multiple biomedical applications using spectroscopy, optical tomography, and imaging. The chapter emphasizes on their material composition and optical characteristics. As a result, it is an

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endeavor to give the researchers well-informed range of potential applications of optical phantoms in optical metrology. This chapter also describes the current state of the art for measurement techniques used to define the key properties of phantoms used in medical diagnostic measurements. Also, measurement tools and methodologies are described in detail. It was analyzed that the maximum deviation of length measurements in the phantoms without calibration is -3.5 ± 0.18 *mm at* $k = 1$, *98% level* of confidence.

Keywords

Phantoms · Optical meteorology · Tissue mimicking materials · Scattering coefficient

Introduction

Optical metrology is the science of measurement of material properties with high precision and accuracy. The most common uses of optical metrology are in the manufacturing industry, where features can be measured with sub-nanometer accuracy. This has led to the creation of a wide variety of optical measurement devices – from telescopes and binoculars, to microscopes and cameras. However, optical metrology is also beginning to find applications in the life sciences, including in the field of oncology. Most commonly, optical metrology is used to determine the dimensions of tumor cells and other cellular phenotypes. Furthermore, it is being used to determine the size and shape of cells and tissues for cellular studies and determine the amount of fluid in the brain for neuroscience studies. Today, we are able to make images and data of human body tissues that would have been inconceivable just a few decades ago. Although the measurement of light to detect the presence of defects and abnormalities in human body is not a new idea, today, several techniques have been developed to measure light as it passes through the tissues in order to detect possible abnormalities, for example, using the backscattered light. The development of measurement and diagnostic imaging systems has necessitated the usage of multiple materials to mimic human tissues' properties in the testing of the instruments for its applicability, harmfulness, repeatability, or several other parameters (Cohen 1979; Seltzer et al. 1988; Olsen and Sager 1995; Pogue and Patterson 2006). These phantoms serve a variety of functions:

1. System design and development from the ground up
2. Improving current systems' signal-to-noise ratio
3. Controlling the devices' quality
4. To assess the systems' performance.

The optimization of the image quality and patient radiation exposure is one of the main requirements for clinical radiological imaging. Real patients can be used for the radiation measurements; however specific phantoms are needed for image quality

assessments. A number of phantoms are designed to accomplish both tasks simultaneously for this optimization. The sharpness and noise are the important parameters used to define the quality of the radiographic images. Their quantitative assessments are performed by physical measurements. For example, the sharpness and the noise of imaging systems can be described in the frequency domain by the modulation transfer function (MTF), and by the normalized noise power spectrum (NNPS), respectively. Detective quantum efficiency (DQE) is determined from these metrics and gives the efficiency of a detector when using input signal-to-noise ratio provided by a limited number of photons to form an image at a certain dose level. When the system fully establishes, and to be used in clinical trials with a proper ethical approval from the competent authority, there are standards or recommendations for imaging quality control phantoms for system performance and usage validation. The instrument's parameters can only be validated for clinical trials if the object is made in such a way that it retains the same properties as original tissue has, which are termed as phantoms (Vardaki and Kourkoumelis 2020). Moreover, medical physics organizations such as the Canadian Organization of Medical Physicists (COMP) and American Association of Physicists in Medicine (AAPM) gave recommendations on the necessity of phantoms for new technologies, systems, and to keep track of the existing systems (Pogue and Patterson 2006). Researchers have developed and demonstrated the phantom by various ways. This chapter provides a detailed study of the numerous tissues mimicking optical phantoms, their characteristics, and uses in metrological applications.

Optical phantoms usually approximate the optical features of human body tissues, which are often utilized to replicate interaction of light with biological tissues. The main purposes of using phantoms are: simulating light propagations with geometry of real tissues, calibrating optical devices, and performing a traceable measurement using instrument. Phantoms are used in a variety of imaging qualitative/quantitative measurement applications, such as radiology, dermatology, endoscopy, ophthalmology, and diagnostics to mimic the properties of human tissues for measurements. For example, a typical phantom is made of a material that simulates the soft tissue but also has optical properties in similitude with human tissue optical properties. Although, for precision measurements, phantom properties are carefully calibrated in order to closely match the optical properties of human tissues, and constructed using such tissue mimicking materials, generally, those materials are typically made of a plastic or other material that is typically white and has a shiny, gelatin-like appearance (Vardaki and Kourkoumelis 2020). Furthermore, they are also used to create images of human organs that would be difficult or impossible to image using other techniques, such as CT scans. However, properties for optical phantoms can be altered by adding specific materials to the medium that is used to form the phantom. This enables the optical properties of the medium also to be mimicked.

The necessity of the phantoms is shown in Fig. 1 (Vardaki and Kourkoumelis 2020).

In this study, we have evaluated advances in the design, development, and fabrication of phantoms for optical metrological applications that imitate the optical, geometrical, and mechanical characteristics, along with the intricate structures of

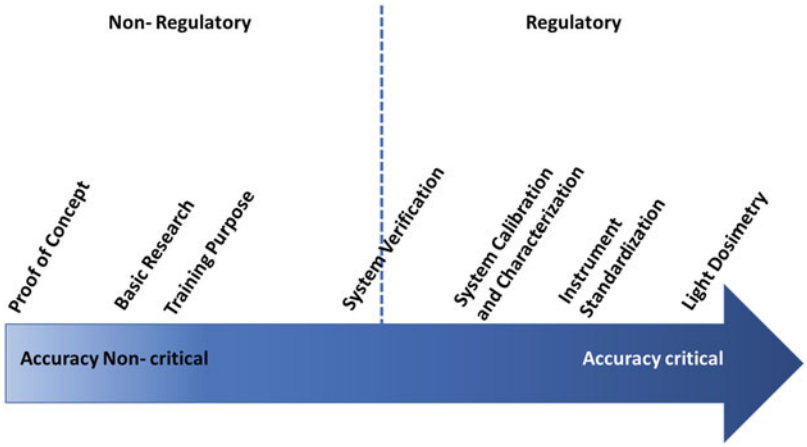

Fig. 1 Necessity of the optical phantoms in system development (Vardaki and Kourkoumelis 2020)

tissues, with a focus on materials with maximum endurance. Furthermore, this chapter attempts to analyze the advantages and disadvantages of each phantom kind, as well as topics such as system purpose, shape, and tissue type. In an effort to offer the most thorough information feasible at this level of development, the trade-offs between structural, biological, and chemical functionality are also mentioned.

Tissue Optical Properties

In the early 1980s, a boom in research interest for clinical studies in breast cancer imaging for transillumination in near-infrared, also known as diaphanography, prompted the creation of phantoms for near-infrared spectroscopy and imaging of tissues (Watmough 1982, 1983; Drexler et al. 1985). Later, multiple applications such as therapies using photodynamic and laser treatment drew interest in designing the phantoms because when it came to therapeutic effectiveness, understanding the optical fluence in tissues was crucial (Grossweiner 1986; Jacques and Prahl 1987; Flock et al. 1987). The development of measurement devices used for enhancing spatial and temporal resolution in the early 1990s prompted many researchers to examine imaging and spectroscopy of tissues, resulting in the creation of a variety of tissue phantoms (Grossweiner et al. 1987). Recently, the use of optics in medicine has grown substantially, with aesthetic laser surgery becoming the major commercial technique and diagnostics using fluorescence and reflectance as important commercialized challengers (Cerussi et al. 2012). In order to design and develop phantoms, it is the pre-requisite to measure the tissue optical properties to determine both the coefficients (i.e., scattering and absorbance). The details of the optical properties are discussed as follows.

Both diagnostic and therapeutic applications in optical metrology are influenced by the optical characteristics of human tissue. In diagnostic applications, the capacity of light to infiltrate a tissue and probe and then exit the tissue is critical to detect, whereas, in therapeutic applications, the capacity of light to permeate a tissue and deposition of energy via the tissue's absorption qualities is crucial (Jacques 2013). As a result, defining a tissue's optical characteristics is the first step in appropriately developing equipment, interpreting diagnostic results, and establishing therapy regimens. The cue to match and mimic properties of phantoms utilizing physical and biological aspects of tissues that govern its interaction with light (Arnfield et al. 1988). The optical properties always are specific to micro- and macroanalysis applications. It's critical to match the scattering coefficient (μ_s), absorption coefficient (μ_a), and anisotropic coefficient g (scattering angle cosine) in tissues for small-scale applications. Whereas, a scattering length is termed as the reciprocal of scattering coefficient across greater distances (i.e., greater than three times scattering lengths) $1/\mu_s$, it is also defined as the reduced scattering coefficient , i.e., $\mu_s{}' = \mu_s (1 - g)$. For transmission through thick tissues, it is possible to replicate the tissue's effective attenuation coefficient only, which is stated in the wavelength zone where the theory of diffusion is correct as $\mu_e = \left(3\mu_o\mu_s'\right)^{-1/2}$ (Salomatina et al. 2006; Bevilacqua et al. 1999). Furthermore, for spectroscopy applications, it is necessary to separate out absorption and scattering coefficients to facilitate spectral fitting, therefore the tissue must have both the values. The aforementioned properties are the prerequisites while choosing the useful phantom materials and design along with other aspects to consider such as thickness, heterogeneities, containers, and all geometrical designs (O'Sullivan et al. 2012). However, in real-time applications, just a few of these attributes are crucial, while the others can be disregarded or given a lesser priority. It was stated that the following features should be there while designing the phantoms (Cheong et al. 1990; Cheong 1995):

1. Depending on various tissues, the absorption and scattering characteristics might vary.
2. Multiple features have a wavelength dependency comparable to tissue.
3. Particular interest should be there for different molecules (e.g., NADH, FAH, collagen, tetrapyrroles, fluorophores, and actinometers).
4. Properties should be time independent and have higher stability under varying environmental circumstances (e.g., humidity, temperature).
5. Mechanical and surface characteristics are comparable to those of tissue.
6. Capability to incorporate optically diverse regions (e.g., inclusions of tumors and layered skin).
7. Brownian motion or flow can be included into the phantom.
8. Capability to integrate thermal characteristics resembling tissues.

For an accurate replication propagation of infrared or visible light in a particular tissue, its optical characteristics must be reproduced at the measurement wavelength. Furthermore, by reproducing optical qualities (scattering coefficient), the replication

of co-dependent variables such as the concentration of distinct chromophores (absorption coefficient) and the shape, size, and concentration of scattering components in the tissues (Strömblad 2015; Bashkatov et al. 2007) is done. Apart from that, fat emulsions like Intralipid and Liposyn, milk, TiO_2, latex, and polystyrene microspheres are all commonly utilized scatterers in the construction of tissue phantoms. Whereas, Inks (black ink, India ink, red ink), biological absorbers (hemoglobin, carotene, melanin) and dyes (naphthol green dye, blue dye, indocyanine green, nigrosin, indocyanine green) are all examples of absorbing medium. By combining the necessary amounts of absorbing and scattering agents in the volume of phantoms, the desired optical characteristics may be produced (Ebert et al. 2001; Bremer et al. 2003; Troy et al. 2004). Table 1 lists the scattering elements of several tissue imitating materials.

Furthermore, absorbers are also important in phantoms, which can be used with the three primary scatterers (lipid, TiO_2, microspheres) discussed above. Majorly, water is the main cause, due to which absorption of phantoms occurs at most of the near-infrared and visible wavelengths. However, the coefficient for absorption is so small, below the wavelength of 700 nm, that it may be ignored, and some other absorbers may be appended to adapt the absorption coefficient and spectrums to tissues (Bednov et al. 2004). Table 2 gives a quick overview of absorbers and how they're used.

Types of Phantoms

Solid phantoms**,** liquid phantoms, polymers, silicone phantoms, gelatin/agarose phantoms, and animal phantoms are some of the most often used optical phantoms (Pogue and Patterson 2006; Tuchin 2015; Esmonde-White et al. 2011b). Those are discussed in detail as follows (Fig. 2):

Aqueous/Liquid Phantoms

These are the simplest and most adaptable in terms of manufacture since volume and uniformity can be readily regulated and handled. Because of this, the optical characteristics of liquid phantoms may be changed simply by varying the relative amounts of scattering and absorbing components in the solution while making the phantom (Pogue and Patterson 2006; Tuchin 2015). In most cases, liquid phantoms are created by combining the emulsions (intralipid, liposyn), which is commercially available with an absorber (e.g., ink) (In et al. 2014). The mixing is usually done in a container and the material of the container may differ based on usage and the likelihood of propagation of light from the container walls. Also, because phantoms are liquid, they allow for substantial flexibility during measurements, such as the integration of scattering and fluorescence characteristics. Another significant benefit of liquid phantoms over semisolid/solid phantoms is their simple and quick synthesis technique (Viator et al. 2002).

Table 1 Scattering constituents of the mimicking materials

Scatterer material	Durability	Compatibility with biological constitute	Chemical (organic/ Inorganic)	Refractive index	Size of particle	Application	References
Lipids	NP	Y	O	1.45	10–500 nm	Multiple phantom contrast investigations, intralipid, milk, and combination testing	Merritt et al. (2003), Reble et al. (2009), van Staveren et al. (1991), Esmonde-White et al. (2011a) and Masson et al. (2018)
Polymer microspheres	P	Y	O	1.59	50 –100 μm	Most precise theoretical prediction of resin phantom characteristics	Ramella-Roman et al. (2003) and In et al. (2014)
TiO_2, Al_2O_3 powders	P	Y	O	2.4–2.9	20–70 nm	Used in conjunction with gelatin and resin phantoms	Kircher et al. (2012), Dong et al. (2015), Khan et al. (2014) and Tuchin et al. (2011)
Quartz glass microspheres	P	Y	O		250 nm	Used with resin phantoms	Dingari et al. (2011), Motz et al. (2004) and Maher et al. (2014)

Table 2 Brief summary of absorbers

Absorbers	Functionality	Shortcomings	Durability	References
Whole blood	Spectra of real tissues and oxygenated	Neither stability nor repeatability	Hours to few days	Srinivasan et al. (2004) and Dehghani et al. (2003)
Ink	Spectrum with flat absorption		If mixed then days	Madsen et al. (1992), Barman et al. (2009), Dingari et al. (2011) and Vardaki et al. (2015)
Molecular dyes	Spectrum with wavelength peaks		Weeks	Barman et al. (2009) and Dingari et al. (2011)
Fluorophores	High compatibility with liquid or aqueous solutions	Additional agents are needed to avoid aggregation	Few Days	Vardaki et al. (2015)
Absorption/ scattering/ Fluorescent	To check the feasibility of tomography and imaging	Due to light channeling, clear enclosures must be avoided, and changes in the refraction index can be severe for solid inclusions	Days	Dehghani et al. (2003)

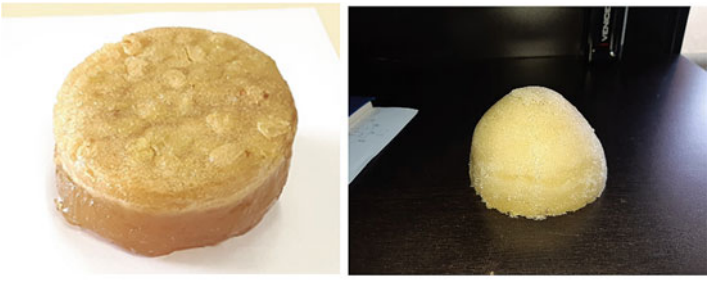

Fig. 2 Examples of tissue phantoms: (**a**) gelatin (**b**) agarose phantom (all two images are from the author's unpublished work)

Solid Phantoms

Solid phantoms are time-consuming and have low degrees of flexibility to make because their optical properties are difficult to control just by including or omitting elements from the bulk matrix (Pogue and Patterson 2006). Bulk matrices with various degrees of transparency, i.e., silicone, polymers, and wax, have been used to

create solid phantom samples. Despite the fact that polymer-based phantoms are less durable and instable in size during polymerization, they have better shelf life than animal or aqueous tissue phantoms.

Hydrogel Phantoms

Hydrogel-based phantoms, such as agarose and gelatin matrices, are a unique type of semisolid tissue phantom. Phantoms made by both of these materials have been extensively used for biocompatibility testing and imaging in metrological applications (Viator et al. 2003; Arteaga-Marrero et al. 2019). Because hydrogel phantoms are mostly composed of water, evaporation of the solvent can occur, causing the rapid change in phantom's size and optical characteristics (D'Souza et al. 2001). Also, these phantoms are susceptible to bacterial development; some of the preservatives may be employed to extend their life span.

It was stated that both types of tissue phantoms (i.e., liquid and solid) lack the genuine intricacy (Vardaki and Kourkoumelis 2020). Multiple layers of equal or different optical characteristics need to be added to increase the realistic complexity of such phantoms. Mineral inclusions to mimic tumors widely employed to increase the biological value of phantoms are used in solid phantoms. Also, to portray pathological calcifications, calcification powder (i.e., hydroxyapatite (HAP)) is frequently used into the phantom (Kerssens et al. 2010). Apart from this, if many inclusions with varying scattering and dispersion are required inside the simulated tissues some other materials are used to mimic the tissues, e.g., calcium carbonate, polymers, or trans-stilbene can be used. On one side usage of liquid tissue phantoms makes it simple to add or remove inclusions from the sample volume. Solid phantoms, on the other hand, cannot readily include inclusion characteristics after manufacturing, and also the surrounding tissue seems unrealistic, so the solution can be done using animal phantoms (Tuchin et al. 2011; Dingari et al. 2011; Roig et al. 2015).

Animal Phantoms

The most appropriate phantoms are animal phantoms because they effectively mimic the mechanical characteristics, heterogeneity, and uniformity of human tissue. Animal tissues are extensively employed in optical imaging not only for its inherent scattering and absorption capabilities, but also for its shape and chemical content (Ghita et al. 2016; Stone et al. 2010; Mosca et al. 2020; Asiala et al. 2017). It was stated in the literature that animal tissues excised and used ex vivo can serve as a realistic model due to its morphological complexity, which are also prevalent in human tissue. Despite their advantages, these are very challenging to manage for repeatable experiments due to their inability to modify optical characteristics exactly (Sharma et al. 2013). Furthermore, due to the reduction in blood volume during removal, lower absorption in tissues is predicted. Animal phantoms, on the other

hand, are valuable as a stopgap between human and in vivo investigations because they allow researchers to explore healthy and sick tissues. Such things would be impossible to do in a human patient owing to ethical concerns.

Material-Based Phantoms

Silicone Phantoms

Such phantoms were firstly employed by Oldenburg et al. (2005) in optical coherence tomography (OCT), demonstrating magnetomotive contrast using their mechanical characteristics. The most difficult aspect of making the phantom using silicone is that it is very difficult to achieve a uniform scattering and absorbing distribution. It should be free of sedimentation, aggregation, and formation of air bubbles. Also, to guarantee a homogeneous distribution of scatterers and absorbers across phantoms with silicone, several procedures such as sonication (Oldenburg et al. 2005), thinning of silicone, and evaporation (Kennedy et al. 2009; Bisaillon et al. 2008) can be utilized alone or in combination. Although, silicone phantoms have a disadvantage the most commonly used inorganic scatterers have refractive indices that are substantially bigger than biological structures, which is a disadvantage of silicone phantoms. Furthermore, the refractive indexes of alumina and titanium dioxide, for example, are 1.76 and 2.3, respectively. Whereas, other materials, such as silica microspheres, have a higher value of refractive index; therefore they have been employed less frequently in the creation of OCT phantoms (Bisaillon et al. 2011; de Bruin et al. 2010; Rehn et al. 2013).

Polarizing/Depolarizing Phantoms

Polarization has an impact on the scattering characteristics of tissue-mimicking materials. Many researchers have investigated the influence of particle size, density, and refraction index on scattered light polarization. The primary scatterers in biological tissues, as revealed by the results of these investigations, are organelles, nuclei, and tissue structures, which restrict photon penetration depth and depolarize light flowing through these media (Rehn et al. 2013). The nucleus and organelles are typically depicted as scattering particles having refractive indexes ranging from 1.34 to 1.46. Collagen and elastin are the materials with cylindrical or spherical structures in the ECM. Moreover, to make phantoms with scattering capabilities, microspheres and other microscopic particles are often suspended in liquid. Additionally, to affect the absorption qualities, India ink, hemoglobin, and colors are widely used. Furthermore, titanium dioxide (TiO_2) is another substance widely employed in optical phantoms to induce scattering. While utilizing TiO_2 particles in phantom fabrication, they are used in the mediums like polydimethylsiloxane or polyurethane to make polymers. It is possible to modify the amount of depolarization by changing the

concentration of TiO_2 particles. Zinc oxide (ZnO) particles are frequently incorporated with polymers to create such phantoms (Chue-Sang et al. 2019).

Fibrin Phantoms

Although silicone has a lot of benefits as a synthetic phantom material. But, its incompatibility with biological tissue components is a drawback. Fibrin phantoms get around this problem. Major advantages of the fibrin are its time-efficient construction, extended life span, stiffness at room temperature, and minimal scattering over routinely used physiologically compatible materials like agar and gelatin (Chue-Sang et al. 2019). Fibrin is a protein, which naturally occurs in humans to give structural support for blood clots (Lamouche et al. 2012). It is generated by proteolysis mediated by the enzyme thrombin from the protein fibrinogen. In surgical treatments and wound closures, it is used as an adhesive (Kennedy et al. 2010). To fabricate the fibrin, thrombin powder and fibrinogen are dissolved in saline separately and further combined to produce a gel. However, adding calcium chloride ($CaCl_2$), can enhance the gel's flexibility, the gel formation time can be sped up (Kaetsu et al. 2000).

It was discovered that most early experiments with phantoms were majorly focused on regular-shaped materials development with approximated scattering and absorption at specified wavelengths. However, the emphasis has switched to develop such phantoms that can duplicate the characteristics over a wider variety of wavelength ranges, mimic the tissue spectrum, absorption, and scattering values. There's also a lot of interest in creating highly compatible phantoms with biochemical and biological properties. Moreover, they can use biologically important molecules, e.g., melanin, hemoglobin, endogenous fluorophores like flavin adenine dinucleotide FAD, nicotinamide adenine dinucleotide NADH, and exogenous fluorophores like cyanine dyes, porphyrins (Bremer et al. 2003; Troy et al. 2004).

PVA-C Phantoms

Such phantoms were mostly used because of their mechanical properties: to create more realistic mechanical behavior of phantoms for OCT-based elastography (Lamouche et al. 2012; Kennedy et al. 2010). Poly (vinyl alcohol) is a thick, liquid hydrogel (cryogel (PVA-C)) made by dissolving a polymer (poly (vinyl alcohol), in dimethyl sulfoxide (DMSO) or water (Lamouche et al. 2012). When these solutions are frozen and thawed repeatedly, they cross-link to form cryogel. Here, the solvent (dimethyl sulfoxide, water), rate of thawing along with the additives all have an effect on the optical characteristics of PVA-C phantoms. When PVA is mixed with water, it generates a transparent film that becomes progressively opaque as the PVA concentration increases. Using alumina and block-printing ink, scatterers and absorbers may be introduced into the matrix to change the optical characteristics of PVA-C.

Tissue-Engineered Phantoms

Nowadays, methods that include tissue engineering have advanced while mirroring the structural features of tissues that may be generated or grown in culture. These tissues are particularly useful in cases when the intricate nuances of the tissue's thin-layered structure are not well described; hence inert tissue phantoms cannot properly replicate them (Pogue and Patterson 2006). This is particularly relevant in optics when anisotropic scattering occurs owing to features like collagen matrix or muscle fibers, or when the stacked sequence of tissues impacts light transit in and out of the tissue. While this discipline is still in its early stages, these models have a strong chance of becoming commonplace in molecular imaging research. This is becoming a realistic alternative as created tissues become more consistent across laboratories. Another justification for using these structures is to avoid using animals in research. In most laboratories, alternatives to animal models are accepted as long as the model is a genuine portrayal (Pogue and Patterson 2006).

Ex Vivo Tissues

Ex-vivo tissues aren't exactly phantoms, but its ubiquitous application in tissue imaging and spectroscopy is worth mentioning. Because of the biological intricacy of the tissue's absorption and fluorescence spectra, and the difficulty of effectively simulating layered structures, it's generally preferable to utilize excised tissue rather than phantoms. Also, in diffuse imaging applications, chicken or bovine muscle has been widely used as an ex vivo tissue to evaluate transmission parameters to see how the modeling or measuring system functions in actual tissue (Pogue and Patterson 2006; Vardaki and Kourkoumelis 2020). While phantoms constructed by tissue mimicking material are beneficial, there is always the risk that they do not accurately imitate tissue qualities. An ex vivo tissue might be defined as a valuable intermediary before beginning measurement experiments on humans. Apart from that, chicken breast tissues are frequently employed because they have extraordinarily low concentrations of blood resulting in a tissue with outstanding light penetration.

Although the scattering coefficient is unlikely to change considerably after removal of the tissues, even the absorption due to blood will decrease along with blood volume drops (Vardaki and Kourkoumelis 2020). Also, hemoglobin oxygenation will alter the properties and all the tissues will become ischemic after a few seconds of removal. Only pre- and post-cryogenic freezing can preserve the tissue's oxygen and energy level status. Moreover, excised tissue may retain thermal characteristics and provide a useful model of nonperfused organs for optical therapy experiments as well. However, ex vivo tissue loses the heat convection caused by blood flow, making it a poor model for long-term heat dispersion investigations in perfused tissues. Table 3 lists types of phantoms along with their usage in optical methodologies.

Table 3 Different types of phantoms that are used for multiple applications

Container/bulk matrix	Wavelength	Type of phantom	Absorbing element	Scattering element	Purpose of development	References
Agarose	785 nm	Liquid		Intralipid	To estimate depth of focus of the system	Chuchuen et al. (2013)
Cuvette	830 nm	Liquid	India Ink	Dairy element	To estimate offset of tissue properties	Bakhshaee et al. (2015)
Fused silica	808 nm	Liquid	India Ink	Intralipid	Evaluation of system	Barman et al. (2009)
Silica cuvette	830 nm	Liquid	India Ink	Intralipid	Calibration of blood system	Dingari et al. (2011)
Agarose	830 nm	Semisolid/ hydrogel		Liposyn	Calibration of bone CT data	Kircher et al. (2012)
Gelatin	785 nm	Semisolid/ hydrogel			Evaluation of bone studies	Esmonde-White et al. (2011a)
Gelatin	680 nm	Semisolid/ hydrogel			Assessment of cervix/water content	Masson et al. (2018)
Silicone	488 nm	Solid		TiO_2	Signal correction for skin	Reble et al. (2009)
PDMS	720 nm	Solid		TiO_2	Comparison of System	Roig et al. (2015)
Glass substrate	785 nm	Liquid/ animal		Intralipid/ chicken	Tissue assessment and performance analysis	Vardaki and Kourkoumelis (2020)
Petri dish	785 nm	Animal/ solid		Paraffin/ Acetaminophen	In-depth measurements	Khan et al. (2014)
Glycerbal	635 nm	Porous phantom			Assessment of agents and tissue evaluation	Pogue and Patterson (2006)
	488 nm, 535 nm, 630 nm, 775 nm	Inflamed phantoms	India Ink	TiO_2	To characterize and tune the properties	Oldenburg et al. (2005)
	930 nm	Solid		PVC plastisol Zinc oxide	Testing and assessment of NIRS system	Kaetsu et al. (2000)

(continued)

Table 3 (continued)

Container/bulk matrix	Wavelength	Type of phantom	Absorbing element	Scattering element	Purpose of development	References
Vacuum chamber	450 nm	Semisolid		Glycerine	Investigation of spectroscopy methodologies	Bisaillon et al. (2011)
	930 nm	Solid		PVC plastisol + ZnO	Skin tissue investigation	Roig et al. (2015)
Excised tissue	785 nm	Animal		Porcin tissue	Detection of calcifications in human breast	Madsen et al. (1992)
Excised tissue	770-830 nm	Animal		Porcin tissue	Optimization of the system for breast tissues	Esmonde-White et al. (2011b)
Quartz	785 nm	Aqueous	India Ink	Intralipid	Evaluation of system of nodes and lymph	Kircher et al. (2012)
Water	488 nm	Liquid	India Link	lipid	Spatial resolution analysis	Maher et al. (2014)
Isotopic Saline	830 nm			Blood plasma	Assessment of blood phantoms	Strömblad (2015)
Agarose	785 nm	Semisolid		Intralipid	Calibration of CT data for bone	Dingari et al. (2011)
Glass slides		Animal		Porcine tissue	System assessment for breast tissues	van Staveren et al. (1991)
Agar	785 nm			Intralipid	Calibration of CT data for bone	Esmonde-White et al. (2011b)
Silicone	488 nm	Solid	Red silopren	TiO_2	Signal correction	Dingari et al. (2011)
Agarose	785 nm	Semisolid	Nigrosin	Intralipid	Evaluation of depth measurement in skin	Chue-Sang et al. (2019)
Agarose	785 nm	Semisolid	India Ink	Liposyn	Penetration depth measurement and system evaluation	Grossweiner et al. (1987)
Polydimethylsiloxane	720 nm	Solid		TiO_2	Comparison of system	Merritt et al. (2003)
	785 nm	Animal		Ovine tissue	Lab prototype for bone	Dong et al. (2015)

Materials for Optical Phantoms and Their Durability

The material chosen in developing the phantom has the most influence on how the phantom may be employed (Pogue and Patterson 2006). These materials are also known as bulk matrix. The basic types of matrix materials are outlined in Table 4. Different materials are ideal for different uses while developing the phantoms. In the table below, the utilization of various phantom matrix compositions is also explained.

It was stated that to increase the durability of phantoms, some additives are added in the materials. Some of the most common additives are, e.g., formaldehyde in gelatin phantoms, which raise the melting temperature by enhancing fiber cross-linking (Pogue and Patterson 2006). This phantom may be used at room temperature without a need to be refrigerated. Apart from that, agar-based phantoms can also be used for this; however, they can grow frail and collapse when stressed. In addition, biochemical substances such as wood preservative, mild acid ethylenediamine tetraacetic acid EDTA, inclusion of penicillin, and sodium azide create a stable and durable phantom without any type of bacterial development. Usage of these additives will maintain good biological compatibility and stability for a longer period of time. Moreover, the usage of vegetable oil has also been defined in order to preserve water content. Furthermore, the addition of blood to hydrogel phantoms gives a good approximation of tissue spectra in the near-infrared, where hemoglobin and water are the prominent absorbers. These phantoms are suitable for therapeutic studies because they can have

Table 4 Bulk matrix and their composition

Phantom material	Durability (permanent/ temporary)	Solid/ liquid	Refractive index	Recommended use	References
Aqueous suspension	T	Liquid	1.34	Multiple phantom contrast investigations and initial usage	Jacques (2013)
Gelatin/agar matrix	T	Semisolid	1.35	Studies of phantoms using Bioabsorbers and fluorophores	Esmonde-White et al. (2011a)
Polyacylamide gel	T	Semisolid	1.35	Study of thermal therapies	Barman et al. (2009)
Epoxy resin	P	Solid	1.54	Validation and calibration	D'Souza et al. (2001)
Polyurethane resin	P	Solid	1.50	Inclusion of dye and comparisons of systems	Asiala et al. (2017)
RTV silicone	P	solid	1.4	Flexible and durable phantoms with complex structures	Pogue and Patterson (2006)

the same elastic characteristics as human tissue and similar thermal qualities as well. The designing and development of different optical phantoms, the research of interest, is very useful in order to assess and calibrate new measurement systems.

Measurement Modalities
This section describes the measurement modalities used for analysis of the different optical parameters while designing and developing the optical phantoms.

"A measurement result is complete only when accompanied by a quantitative statement of its uncertainty. The uncertainty is required in order to decide if the result is adequate for its intended purpose and to ascertain if it is consistent with other similar results." The Uncertainty of measurement provides a quantitative estimate of the quality of a test result, and therefore is a core element of a quality system for calibration and testing laboratories. To reflect this, various international metrological and standard bodies jointly developed a Guide to the Expression of Uncertainty in Measurement (GUM) to provide such laboratories with a framework of formal metrological terminology and methodology for expressing uncertainty of measurement. Subsequently, international standards ISO/IEC 17025 and ISO 15189 (17025 re-written for medical testing), have required complying laboratories to provide estimates of uncertainty for their test measurements, referring to GUM for the appropriate methodology (Bakhshaee et al. 2015). The uncertainty of measurement, traceability, and numerical significance are separate but closely related concepts that affect both the format and the information conveyed by a quantitative test result. A detailed error analysis of the measuring procedure is required for accurate measurement of optical characteristics. Developing a tight error budget for a specific procedure allows SI traceable measurements to be accomplished. Traceability to the international system of units is critical for guaranteeing that measurements made everywhere in space and time are comparable to measurements taken anywhere else in space and time. This is the most effective technique to ensure the long-term integrity of data generated by a measuring system. In addition, the use of SI units provides a consistent basis for the reporting of clinical laboratory data.

Uncertainty "A parameter associated with the result of a measurement, that characterises the dispersion of the values that could reasonably be attributed to the measurand" (VIM).

Traceability: "Property of the result of a measurement or the value of a standard, whereby it can be related to stated references, usually national or international standards, through an unbroken chain of comparisons all having stated uncertainties" (ISO 15189).

Numerical Significance The significant figures of a number are those that have some practical meaning. The significant figures of a number express its magnitude to a specified degree of accuracy.

Another method for ensuring long-term measurement repeatability and consistency is to compare measurements to a standard material manufactured using a

controlled process of fabrication and for which the user has assigned the conventionally true value to its physical properties of interest. This is often the only viable option. The creation of fluorescence intensity standards is a good illustration of this. Absolute estimation of fluorophore concentration from fluorescence intensity measurement necessitates knowledge of the molecule's quantum yield, among other factors. Measurements can be used to determine quantum yields, but they are problematic. To tackle the practical difficulty of fluorescence measurement standardization, researchers compare their readings to a solution of fluorophores of known concentration created in a controlled manner. A recipe-based technique might be used to standardize diffuse optical measurement. This technique has many flaws. The repeatability of the standard would be restricted by the variability of the optical characteristics of the stock intra-lipid solution or the variability in the size distribution of the TiO_2 powder. The long-term repeatability of the standard would also be jeopardized due to the unclear future availability of its ingredients. If the reproducibility of a phantom is based on a precise recipe rather than an accurate, stable, and recognized characterization setup, users of this specific phantom risk the manufacturer of one of the recipes' elements ceasing to make the product. The authors experienced this challenge when attempting to get a new batch of TiO_2 powder with a specific size distribution that was no longer accessible from any source. This issue is solvable if a characterization method that produces SI traceable measurements based on a rigorous metrological methodology with a narrow error margin is available. It will be feasible to identify a replacement and check that the new phantoms created with this product are equivalent to the prior one using this way. If phantoms are manufactured in the centralized manner described earlier by a small number of laboratories or firms, it may become feasible to measure the optical characteristics of each phantom generated and therefore reach the maximum degree of precision and quality control.

Uncertainty Parameters While Designing the Phantoms The majorly defined uncertainty parameters of optical phantoms are given as follows:

- Diffuse total reflectance (ToR)
- Reduced scattering coefficient, i.e., $\mu_s' = \mu_s\,(1 - g)\,\mu_s$ is a scattering coefficient and g is a scattering anisotropy factor
- The intrinsic absorption
- Scattering
- Fluorescence
- Refractive index of the matrix material
- Effective attenuation coefficient $\mu_e = \left(3\mu_o\mu_s'\right)^{-1/2}$
- Backscattering intensity
- Reduced scattering and absorption coefficients
- Aging-related alteration of optical properties, retainment, stability over a period of time, shelf life of the phantoms

- Poor repeatability between various instances of the phantom may not have a direct impact on the quality of the system itself, but may induce perceptions of inconsistency and poor repeatability with respect to the instrument itself when presented to the target user. Such scenarios are definitely not desirable from a commercialization point of view.

The sources of uncertainty in the imaging system, as above and apart from that magnification, pixel-to-millimeter unit conversion, and penumbra effect, are also considered, and the lengths of the phantom before and after imaging system calibration were compared. The maximum deviation of length measurements without calibration is -3.5 ± 0.18 mm at $k = 1$, 98% level of confidence. Furthermore, length correction values are expected to be useful for diagnosis and treatment planning, where precise length measurements are essential while designing and developing the phantoms.

Conclusion

In this chapter, a detailed overview of optical phantoms and materials is presented to highlight field of interest in an area with a wide range of applications and approaches. It is discussed that a significant variety of optical tissue phantoms are available, with different forms, geometries, and optical characteristics ranging from simple to complex. Optical tissue phantoms have become a typical feature of an experimental setup for investigations translating into a clinical setting due to their design flexibility, time efficiency, and low cost of production. However, due to lack of homogeneity and a "gold standard" for comparison, tissue phantoms have significant issues and problems. But, as imaging and spectroscopy systems utilized for human body tissues get regulatory certification for marketing and clinical use, standardized phantoms are needed to establish the accuracy and repeatability of newly created instruments and they should take on a new degree of importance in the scientific community. Unfortunately, commercial manufacturing of these phantoms is not readily available at this moment. Yet some researchers are developing anatomically accurate phantoms with realistic optical properties in order to bring the field closer to homogeneity. It is the need of hour to develop commercialized phantoms with well-controlled optical properties for pre-clinical trials. The detailed review presented in this chapter is the first stage in compiling a summary of the field progress. However, optical field is vast, so we have limited the scope of the chapter to near-infrared spectroscopy and optical imaging. It is concluded that optical phantoms developed using multimodality is an emerging area of interest. Therefore, major advancements are needed in the field of optical phantom designing for metrological applications in order to develop commercial pre-clinical imaging systems.

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Microscopy Using Liquid Lenses for Industrial and Biological Applications

56

Error Analysis and Uncertainty Evaluation

Neelam Barak, Vineeta Kumari, and Gyanendra Sheoran

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Abstract

Quantitative Microscopy refers to the surface metrology of microscopic structures. It deals with the interaction of light with material and quantifies the changes in terms of surface measurements. This chapter describes optical metrology techniques used in qualitative and quantitative microscopy and explains the different extended depth of focus techniques used in quantitative microscopy. The prime focus is on the usage of liquid lenses as a potential solution for extended depth of focus imaging in quantitative microscopy. This chapter summarizes a number of applications of liquid lenses in microscopy dealing with the scope in industry and biological sciences. In the last section, the concept of uncertainty of measurement along with the uncertainty budget has been explained taking into consideration the relevant factors responsible for uncertainty.

Keywords

Optical metrology · Optical microscopy · Quantitative microscopy · Extended depth of focus imaging · Digital holographic microscopy · Electrically tunable lens · Uncertainty

Introduction

Metrology is the science of measurement. Metrology finds many applications in the fields such as in the manufacturing industry for interchangeability, quality control, component matching, yield enhancement, and process development. Metrology and optics have always been intertwined. Optical Metrology deals with the qualitative and quantitative measurements that are based upon the interaction of light with matter. Optical techniques have been widely used for quantitative measurements, topographic measurements, roughness measurements, etc. With the beginning of new computing technologies and fast sensing and switching devices, a plethora of optical metrology technologies have developed to make the surface measurements easy.

Angle measurement equipment and telescopes enhance vision in estimating the sizes of faraway stars, while microscopes allow the eye to examine the surface grain structure and the internal features of living cells quantitatively. Small micro- and nanostructures are manufactured and utilized in a number of scientific and technological applications. Most applications of these structures are in semiconductor industry where the need for compact structures is increasing day by day. Microscopic methods are especially useful for surface characterization of micro- and nanostructures. This surface metrology is referred to as quantitative microscopy.

Microscopy is a branch of science that deals with the study and imaging of microscopic objects whose dimensions cannot be resolved with unaided eye (Croft 2011). Since its inception in the seventeenth century, optical microscopy has been utilized as a key tool for the imaging of micro objects and for surface metrology of micro- and nanostructures. The following are the main aspects that are required for a

precise and consistent structural metrology of microstructures with an uncertainty level below micrometers or less: careful control of the measurement conditions and measurement system parameter settings, and maintaining a high level of measurement reproducibility (Osten and Reingand 2012).

The widely recognized types of microscopy based upon the applicable frequency range in the Electromagnetic spectrum are optical scanning, electron scanning probe, and X-ray microscopy (Grubb 2012; Yamamoto and Fujimoto 2014; Takahashi et al. 1997; Zuo 2015). Optical microscopy is used in the visible region of light and provides moderate to high resolution images (up to submicrons). For nanometrology, electron and scanning probe microscopy (SPM) are used as they provide ultra-high resolution. An example of SPM is tunneling or atomic force microscopy. In X-ray microscopy, the soft X-ray band of light is used and it can achieve resolution up to 30 nm. All of these microscopic techniques have their own advantages and disadvantages, and they provide varied but usually additional information about the structures being measured. As a result, thorough and trustworthy measurements of complex structures with tough constraints (like measurement uncertainty), the necessity of using different microscopic methods arises. Visible region microscopic metrology has advantages over its contemporaries in the terms of cost, nondestructive behavior, and noninvasive property.

This chapter focuses on optical metrology techniques along with a brief of interference- and non-interference-based microscopic metrology techniques. The prime focus is on quantitative microscopy metrology and imaging of through-focus or extended depth of focus samples. A wide variety of extended depth of focus techniques that have been used in literature are discussed. These techniques range from systems that are over decades old to recently developed instrumentation that have broadened the new capabilities in quantitative microscopy. However, due to the promising achievements of liquid lenses in extended depth of focus, a detailed study is being done on the advent and applications of liquid lenses in microscopy, and numerous industrial and biological applications have been presented for use in quantitative and 3D microscopy. This chapter attempts to cover the majority of tunable liquid lens applications in the visible region; nevertheless, given the large number of techniques and articles reported, some exclusions are unavoidable.

Principle of Optical Microscopy

Optical microscopy is concerned with microscopic imaging in the visible area of the electromagnetic spectrum, i.e. (wavelength: 400–700 nm). It involves interaction of light with the specimen and the collection of scattered/transmitted light to generate an optical image (Grubb 2012). The different light phenomena associated with optical microscopy include diffraction, reflection, and refraction. Primarily, the principle of optical microscopy is to magnify the image of a microscopic specimen using an objective lens. The arrangement of the optical microscope has been described in Fig. 1. The optical microscope operates by collecting light from an optical source using a condenser lens, which focuses it on the condenser aperture. At

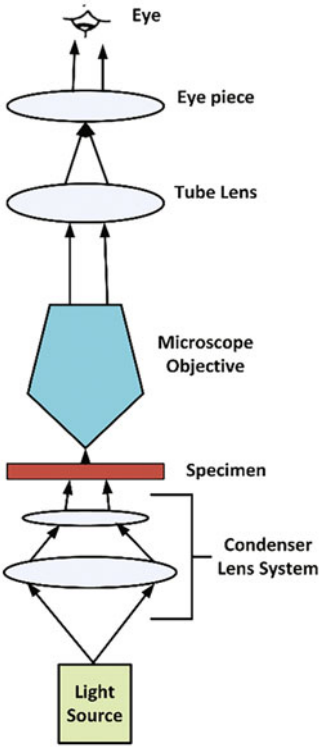

Fig. 1 Schematic of a conventional optical microscope

the focal plane of the objective, an image of the object present at the working distance is produced. The image acquired at the image plane is magnified using the eye piece lens. CCD/CMOS sensors are positioned at the image plane of the eyepiece to record a digital image of the specimen (Sluder and Wolf 2007).

Important Parameters in Optical Microscopy

1. **Magnification**: Magnification is referred to the measure of the degree of visual enlargement observed for a small entity. It is measured in multiples, for example, 10×, 12.5×, 25×, 40×, 60×, and 100×. The enlarged image is obtained at the stationary imaging plane of the objective.
2. **Resolution**: In microscopy, the term "resolution" is defined as the capability of a microscope to differentiate between two laterally placed distinct points of a specimen as separate entities. The resolution of a microscope is dependent on the objective's numerical aperture (NA) as well as the operating wavelength of light at which the specimen is examined. Resolution in diffraction-limited microscopic system is defined by Abbe's diffraction limit, $d = \lambda/(2 \times \mathrm{NA})$, where "$d$" is

the minimum resolvable distance between two distinct entities, "λ" is the operating wavelength, and "NA" is the objective's numerical aperture.

3. **Numerical aperture**: The numerical aperture is defined by the amount of light that enters the front aperture/pupil of the microscope objective. Numerical aperture gives a measure of resolving a fine specimen detail at a fixed working distance of the objective. The numerical aperture is given as $\mathrm{NA} = n \sin\theta$, where "$n$" is the refractive index of the medium and "θ" is the half angle of the light entering the microscope objective's front aperture.
4. **Field of view**: The field of view (FOV) number is the diameter of the field in millimeters, which is observed at the intermediate image plane. FOV number or Field number is used to represent the diameter of the focused object field in an optical microscope.
5. **Depth of focus (DOF)**: The thickness of the focal spot of the objective is referred to as its depth of focus. The terms depth of focus and depth of field are interchangeably used in image and object space, respectively. The focal depth of a microscope is the depth of the layer of the sample, which remains in sharp focus even if the distance between the objective lens and the specimen plane is changed (Grubb 2012).

Types of Optical Microscopy

Quantitative/Metrological Optical microscopy has been predominantly divided into further subcategories based upon its use in different application areas. The most commonly used optical microscopy techniques can be classified as:

1. Non-interferometry based
2. Interferometry based

Non-interference-based techniques include wide-field, bright-field, dark-field, light-sheet, oblique illumination, dispersion staining, fluorescence, confocal, and deconvolution microscopy. Interference-based microscopy techniques include differential interference contrast, interference reflection, phase contrast, and digital holographic microscopy. The interference-based microscopy techniques have an additional reference arm that is interfered with the object arm before imaging (Kim 2011). The interference allows the recording of phase information of the sample and the quantitative data about the sample can be retrieved. Figure 2 shows the different types of optical microscopy techniques and their categorization as interference and non-interference-based techniques.

Non-Interferometry-Based Microscopic Metrology Techniques

Bright Field Microscopy It is the most commonly used and simplest optical microscopy technique in which the sample is illuminated with an optical source

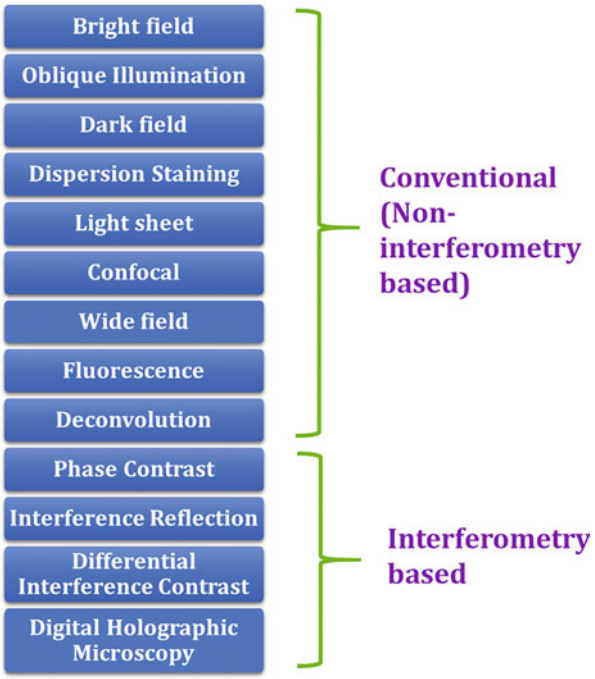

Fig. 2 Optical microscopy techniques

and a contrast is generated when the light is transmitted through the sample. The main advantage of bright field microscopy is the minimalism of the setup with only basic tools required for imaging. This technique is majorly used in the imaging of micro samples that have intrinsic color like chloroplast in plants, stained live cellular and tissue structures, peptide detection, and allows definitive identification, etc. (Michael Conn 2004). The major limitation of this technique is that contrast is very low for weakly absorbing samples, and it is not suitable for transparent specimen. One of the best methods of overcoming these limitations is the use of colored filters on light source and staining of samples in transparent cases. Moreover, bright field microscopes have magnifications limited up to 1300×.

Oblique Illumination Microscopy The light is projected obliquely onto the specimen in this approach to get a high contrast image. It is of great use for imaging unstained samples like living cells, crystals, diatoms, and other translucent or semi-transparent specimens. Because the orientation of the samples changes relative to the angle of incident light, oblique angle illumination might produce misleading pictures of the samples and lower resolution. Therefore, this microscopy is not suited for quantitative measurements or topographic measurements of the sample. However, if higher-order terms of the diffraction pattern of the sample are taken, the imaging resolution can be enhanced (Sanchez et al. 2018).

Dark Field Microscopy In this technique, the background light is eliminated from the optical image. It is most commonly used to image cell suspensions, pond water microbes like algae, planktons etc. It improves image contrast without the use of stain, and thus do not kill cells. The most common benefit is that no special sample preparation is required. The limitation of this technique is that large amount of light is required for imaging and also the sample should be thin enough to overcome the artifacts produced due to diffraction (Rijal n.d.).

Dispersion Staining It uses variations in an unknown material's refractive index dispersion curve compared to a reference material's dispersion curve to classify or describe that unknown material. It is an optical staining approach that produces colors without the use of a stain or dye. This approach is most commonly used to confirm the presence of asbestos in building materials. This technique is limited for use because both materials have similar refractive indices over some wavelength range in visible region but the dispersion curves for refractive indices must not be same for these materials. It can be utilized to differentiate them by constraining them to adopt separate colors (Mccrone 1974).

Light Sheet or Selective Plane Illumination Microscopy In light sheet microscopy, the illumination beam is incident at right angle to the imaging system to form a sheet of light through a 2D plane of the sample. It is applicable for 3D imaging in tissues, because of the capability of selective illumination of tissue layer. Furthermore, light sheet microscopy offers high speed, low light exposure, high penetration range, and low levels of photobleaching. However, the field of view (FOV) of light sheet microscopy is limited as the Gaussian beam has less focal depth. Another limitation is that the light-sheet images get deteriorated in quality due to scattering problems. Bessel light sheet microscopy, Structured Illumination Microscopy are the potential solutions for achieving confocal level resolution (Allahabadi 2016; Fahrbach et al. 2013).

Confocal In confocal microscopy, a spatial pinhole is utilized to blur the defocused information. The optical components used for both illumination and detection focus the light on a diffraction-limited spot and scan the specimen to make a complete image. The main advantage of confocal microscopy is the ability to control depth of field, and elimination of background noise from the focused plane. Its major application is in cell biology that relies on imaging both fixed and living cells and tissues. Furthermore, it has a resolution in the range of 1 nm, which is much better compared to conventional light microscopes. The main limitation of confocal microscopy is the narrow range of excitation wavelengths. This problem can be easily overcome by cost-shared microscope systems having different applications (Mccormick 2007; Shotton 1989).

Widefield Widefield microscopy refers to the imaging technique where the whole sample is illuminated with light. It is mostly used for capillary abnormality measurements in tissues and live cell imaging. The most common benefit of widefield

imaging is that field of view is enlarged as full aperture of the objective is used for imaging with same numerical aperture. The limitation of widefield imaging is its limited axial resolution and cannot image highly scattered thick samples. Usage of adaptive optics can improve axial resolution in widefield microscopy (Grubb 2012).

Fluorescence In this technique, the specimen is illuminated (excited) with a short wavelength light. It observes live cells in situ without the need for toxic and time-consuming staining processes. The high degree of specificity is possible in fluorescence microscopy. Despite the diverse range of probes available, fluorescence microscopy is limited by its dependence on probes. Also, prolonged exposure can bleach the sample and fluorescence intensity can be lost. To overcome this, epi-illumination is employed in which the light used for excitation is reflected onto the specimen through the objective (Mir et al. 2012).

Deconvolution Deconvolution microscopy is an analytical method in which the diffracted light is reallocated to its original place using calculations for the recorded image stacks. In comparison to traditional microscopic techniques, it has a better resolution, sensitivity, and dynamic range. It is mostly used in application areas of confocal and fluorescence microscopy along with 3D imaging of thick biological tissues and samples. The main limitation is that the computational complexity gets increased due to high processing and highly sensitive detectors are required for acquiring the images. Usage of high speed processors and automated scanning can increase the speed of measurements using deconvolution (Mir et al. 2012; Orth and Crozier 2012).

Interferometry-Based Microscopic Metrology Techniques (Quantitative Microscopy)

Background: Why Quantitative Microscopy?
Optical microscopes' resolution and magnification performance have significantly improved thanks to advancements in optical lens manufacturing methods. Ernst Abbe developed the scalar diffraction theory, which was widely used for many years to describe how optical microscopes operate in terms of imaging and resolution. However, in this theory, the effect of light polarization on microscopic images was not discussed and it was limited only to planar structures, i.e., two-dimensional structures, to be imaged. A conventional microscope is limited to providing only two-dimensional images of the sample at test. Therefore, conventional microscopes are not suitable while profiling 3D microscopic structures and high-end nanostructures used in semiconductor industries. This limitation necessitates the introduction of the third dimension, which is the height/depth of the microstructure. For the imaging of a third dimension, specific microscopy techniques like interference microscopy need to be used. The advanced diffraction theories like vectorial diffraction models consider the object's three-dimensionality. They calculate the light–matter interaction numerically, which is the fundamental for quantitative

microscopy. Primarily, the interference-based microscopic techniques are used in quantitative microscopy to extract the third dimension or phase of the object to provide the structural information. These are discussed as follows:

Phase Contrast (PCM) It is an optical microscopy technique that uses the changes in phase of the light transmitting through a transparent specimen and maps them to brightness changes in the captured image. It is highly used for imaging transparent thin samples and the most common benefit is that sample staining is not required. The capacity to produce visual contrast from materials that do not absorb light, such as cells and tissues in culture, is its major benefit. One drawback of phase contrast is that it does not work well with thick specimens because phase shifts from regions above and below the specimen defocus the resulting image. A phase plate can be used to reduce the contribution of the background light (Barty et al. 1998; Rodrigo and Alieva 2014).

Interference Reflection Interference reflection microscopy (IRM) is an optical microscopy technique that creates an image of an object on a glass surface using polarized light. It is mainly used to image cell adhesion on glass, cell mobility on glass coverslip. The main limitation is that a high power source needs to be used due to the presence of multiple optics (Verschueren 1985).

Differential Interference Contrast (DIC) It is also called Nomarski interference contrast (NIC) microscopy. It is used to improve the contrast in unstained, transparent samples by producing contrast image using interferometry. It is most used in the imaging of nanoparticles, surface slope measurements, etc. The main disadvantage is that the optical system becomes complex and specialized optics is needed (Mir et al. 2012).

Digital Holographic Microscopy (DHM) It works on the principle of digital holography applied to optical microscopy. The concept of holography was first introduced by Dennis Gabor (Gabor 1948). Digital holographic microscopy distinguishes itself from other microscopy methods as it does not capture the projected image of the object. It records the wavefront information originating from the object as a hologram and then uses numerical reconstruction algorithms to reconstruct the object amplitude and phase images (Kim 2011). DHM is a type of quantitative microscopic metrology and is mostly used for quantitative imaging of cell and tissue structures, in-flow imaging, particle tracking, imaging of engineered microsurfaces, etc. (Barak et al. 2018). The schematic of a basic digital holographic microscope is shown in Fig. 3. The key advantages of DHM are that it is a label-free, noninvasive, and nondestructive technique and its resolution ranges in the order of subnanometers (Kim 2011). Also, numerical refocusing can be performed to reconstruct out-of-focus information without mechanical scanning. One limitation of DHM is that numerical refocusing can be performed only within the recorded DOF of the objective. To image information outside the DOF and for gathering accurate

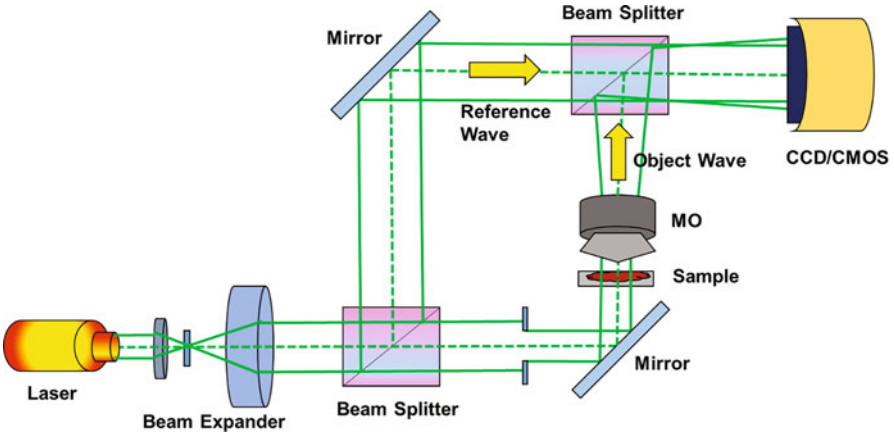

Fig. 3 Basic schematic of digital holographic microscopy

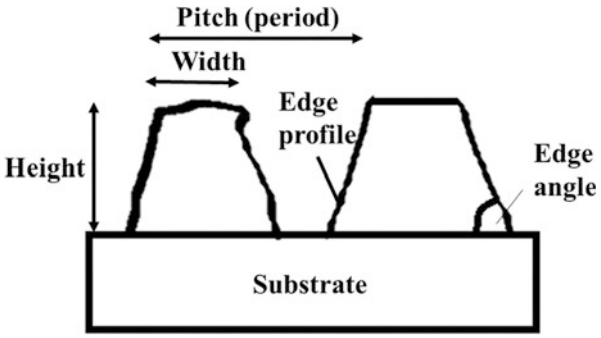

Fig. 4 Schematic representation of measurands in quantitative microscopy (Osten and Reingand 2012)

quantitative information of samples thicker than the DOF, extended depth of focus techniques need to be applied.

Application Areas of Quantitative Microscopy

Quantitative microscopy, as previously stated, measures the structural information of object having dimensions in the range of micro- and nanometers. The most important parameters of measurement in micro- and nanostructures are the widths and heights of the structure along with the distance or pitch between separate entities. Fine quantities like surface profile, edge angles/shapes or the length of a line, and shape/diameter of aperture are also some important measurands that can be of interest in quantitative microscopy. Figure 4 shows the schematic representation of important measurement entities in quantitative microscopy.

Depth of Focus and Field of View Challenges in Quantitative Microscopy

Microscopes have a limited depth of focus, therefore viewing a complete image of a three-dimensional (3D) object in a single view is unachievable. Because the magnification capability of microscopes is inversely proportional to the depth of field, this is the case. Because the specimen in a 3D scene is usually thicker than the microscope lens' allowable focal depth, obtaining a focused image of the whole sample is difficult. The numerical aperture, objective magnification, object distance from objective, objective working distance, and other parameters all influence the depth of field/focus of a microscope objective. The magnification of the objective is the most crucial limitation for expanding the depth of focus. In order to increase the depth of field of the objective, the magnification must be lowered down, thus lowering the resolution of the system. Therefore, it is a trade-off between the extended focal depth and the magnification and resolution of the microscopic system. To increase the axial range, a lot of researchers have worked on the development of extended depth of focus algorithms for obtaining complete structural information of optically thick microscopic samples (Pan 2013; Zalevsky 2010; Valdecasas et al. 2001; Dowski and Cathey 1995).

Techniques for Expanding the Depth of Focus in Quantitative Microscopy

Microscopic systems offer high lateral resolution in the axial range of the system, which called the depth of field or the depth of focus, depending on the object or image space measurements, respectively. While imaging outside this axial range, i.e., imaging optically thicker samples, the image becomes blurred because of the presence of additional phase in the pupil function of the system. This blur function introduces errors in the quantitative measurements due to alteration in phase. Various strategies have been proposed to circumvent this constraint and provide expanded focus depth. They are as follows:

Wavefront Coding

One of the earliest methods used in extending the focal depth of microscopic imaging systems was Wavefront coding (Dowski and Cathey 1995). In this method, the noncoherent optical system was altered by introducing an additional phase mask. This phase mask modulated the point spread function of the objective so that it became insensitive to blur. This alteration in phase allowed wider range of depths of the sample. The sampled intermediate image was reconstructed using digital signal processing by using phase deconvolution. Although this technique proved to be a fast technique and was sensitive to noise, it required modifying the optical hardware of the microscope, which made it a complex process.

Aperture Apodization

Aperture apodization is another technique used for extended focus imaging. The process is done by introducing an additional binary mask in the aperture plane, which performs blocking/transmitting (Zalevsky 2010). This technique provides large number of axial planes but it has the limitations of aperture size. While using small aperture, the system's resolution decreases as the light reaching the image plane gets decreased in intensity.

Image Fusion

Image fusion is another interesting approach utilized in extending the depth of focus. In this technique, different images of the specimen are obtained with different specimen portions in focus of the objective. To accomplish this, either the specimen stage, objective, or detector are scanned to obtain a focal stack. The focal stack contain images recorded at different focal depths of the objective corresponding to different axial planes. The stack is then merged into a single output image that includes all of the input image stack's in-focus sections. The information density and quality of the extended depth of focus image are improved by utilizing image fusion (Zalevsky 2010; Valdecasas et al. 2001). However, limitations associated with image fusion include high computational intensity and a large volume of data being required to form the input stack. Moreover, the scanning speed makes the process very slow, which limits its use in real-time scenarios.

Multiple Focal Plane Microscopy

In order to avoid mechanical scanning of the image plane, another option is to split the illumination beam into numerous fixed beams that are refocused concurrently on different imaging sensors placed corresponding to different axial planes in the extended focus. Another name for this method is Image plane sharing microscopy. Its key benefit is that it eliminates the need for physical scanning (of sample, objective, or imaging sensor). The extended focus images are captured all together in this mode, which allows for high-speed image collection. However, since the imaging beam is divided into many beams, the available optical intensity at each imaging sensor is diminished, thus, lowering the optical data signal-to-noise ratio (Tahmasbi et al. 2014). Another limitation is that the number of axial focal planes available is also limited to the number of times the illumination beam is split.

Multiple Focus Microscopy

For single shot acquisition of extended focus images, another solution is multiple focus microscopy (MFM). It allows rapid and concurrent acquisition of extended

depth of focus images. This is attained by the combination of a multifocus grating (MFG) (placed at the Fourier plane) along with a chromatic correction grating (CCG) and a prism (Pan 2013). This combination separates the principal image into several extended focus images and then projects them on the imaging plane altogether. In this technique, the imaging camera is segregated into different squares, which collect differently focused images (multiple focus images). This technique is limited by the sensor size as the imaging camera can collect a limited number of extended focus images in a single shot.

Wavelength Scanning Method

In the microscope objectives which are not achromatic, various wavelengths produce different focal spots due to chromatic aberration. The sample stage and the chromatic aberrated objective are both fixed in this technique. Light of various wavelengths is employed to illuminate the sample, allowing for prolonged focus image acquisition (Attota 2018). The wavelength scanning technique provides more mechanical stability than traditional stage-scanning optical microscopes because the position of the sample stage is not scanned. The expense of multi-wavelength/tunable wavelength sources is the primary constraint. Furthermore, the detector's wavelength range also restricts the range of extended focal depth.

Artificial Intelligence-Based Methods

Another option for extended focus imaging is to use learning methods to assess the activity function, such as artificial neural networks. To categorize the in-focus and out-of-focus portions of a picture, a decision map is trained. Using proper merging procedures, a mosaic of the in-focus areas is created. The processing time required to train the algorithm to differentiate between blur and in-focus regions is very high and limits the usage of this method. Furthermore, accurate projections need a vast amount of data for training the algorithm (Wei et al. 2021).

Numerical Refocusing Using Digital Holographic Microscopy

Digital holography microscopy is digital holography combined with microscopy. The numerical refocusing property of digital holography is utilized to numerically extract the refocused extended focal images from a single experimentally recorded hologram. This technique avoids the movement of sample physically to different focal regions. Extended focus images are obtained numerically refocusing the amplitude images to different z distances based upon the reconstruction technique used (Anand et al. 2014; Samsheerali et al. 2014). DHM improves the speed of microscopic image acquisition and provides single shot quantitative information

(Pandiyan et al. 2021). The reconstruction distance may be mathematically altered and made as minimal as required (Singh et al. 2014). However, digital refocusing in DHM is limited only within the DOF of the microscope objective. Therefore, it is possible to reconstruct images at positions lying within the DOF range. To extend imaging to further axial planes outside the microscope objective's DOF, other techniques in DHM have been implemented. These include:

(a) **Extended Focused Image (EFI)**: In digital holographic microscopy, the EFI concept has been further explored by P. Ferraro et al. (Ferraro et al. 2006). They emphasized the extended focus image as a combined image from variously focused subareas (EFI). The in-focus regions of each image stack were correctly selected using distance information given by the phase image. As a consequence, the final EFI was built with precision. The fact that this digital holographic EFI technology only works with a single object with an axial dimension bigger than the DOF is a common limitation of this methodology.
(b) **Optical Scanning Holography**: The use of optical scanning holography for EDOF has been provided in reference (Kim 2006). The solution of sectioning is found via Wiener filtering or iterative methods. However, because the complex phase information is not retrieved during processing, this technology can only offer amplitude information and cannot be used for only phase objects.
(c) **3-D Deconvolution**: In reference (Latychevskaia and Gehri 2010), the utility of 3-D deconvolution methods in DHM is presented. The accurate 3D distribution of microsamples is reconstructed by utilizing 3D Point spread function. The basic limitation of 3-D deconvolution is the requirement of large amount of memory. Also, data resampling is a necessity in this method; thus, spatial resolution is degraded.

Adaptive Optics for EDOF Imaging

Recently, the use of adaptive optical elements has been explored for EDOF quantitative microscopic imaging. Tip/tilt, deformable mirrors, spatial light modulators (SLM), are some examples of adaptive optics that can be utilized for EDOF image collection. These adaptive optics are used to change the phase of illumination and shift the focal spot. These types of optics have multiple applications in extended depth imaging (Ho et al. 2017; Salter et al. 2013), autofocusing (Gao et al. 2013), creating optical tweezers for 3D stacking of particles (Dasgupta et al. 2012; Xavier et al. 2012), etc. Another technique for EDOF using an axicon lens is stated by Ong et al. (2013). The use of axicon lens for cone shell imaging to achieve extended focal depth has been presented. A key advantage of these methods in EDOF imaging is high-speed image collection. Adaptive optics technologies and equipment are widely accessible, with biological applications being the most typical use. The

aberrations created in the wavefront owing to modulations are an often cited restriction with adaptive optics.

Extended Focus Quantitative Microscopy Using Liquid Lenses

With the advent of adaptive optics, the evolution of liquid lens also took place. The basic idea behind the liquid lens was to provide a lens with a variable focal length. At first, the idea of a variable focus lens was given by Berge and Peseux (2000), where the efficacy of a transparent drop was studied as a variable focus lens by applying electrowetting effect. A lens can be made from the meniscus of two immiscible liquids with different refractive indices. Electrostatic regulation of the solid/liquid interfacial tension causes a change in the meniscus' curvature, resulting in a change in focal distance.

The concept of variable focus liquid lens was introduced by Oku et al. (2004). They proposed a compact, high-speed variable-focus liquid lens using acoustic radiation. An annular piezoelectric ultrasonic transducer and an aluminum cell filled with degassed water and silicone oil constituted the lens. By applying sonic radiation force from the transducer, the profile of the oil–water interface may be rapidly changed, allowing the liquid lens to be used as a variable-focus lens. Electrically tunable lenses (ETLs) were introduced using similar techniques. These lenses have curvature-changing capabilities that allow for fast axial shifts of the focal plane, allowing for acquisition rates in the order of milliseconds. These liquid lenses are inspired by the human eye and can focus quickly. They have been used in a number of cell biology experiments (Note, Optotune Application 2013). The liquid lenses are made up of an optical liquid that can change its shape by application of external current/voltage or mechanical pressure. Based upon the type of actuation, the liquid lenses are broadly classified as: Mechanically actuated and electrically actuated liquid lenses. Different commercially available variants of tunable focus lenses are available in the market with different nomenclatures. These include electrically tunable lenses (Optotune), tunable acoustic gradient lens (Mitutoyo), etc. A photograph of electrically tunable liquid lens manufactured by Optotune is presented in Fig. 5.

The ETL provides major benefits like:

- High speed focus tuning
- Covers a wide focal range
- Fast control of Z-axis
- No mechanical movement required in Z-axis
- High Axial scanning range
- Compatible with finite and infinite objective lenses
- Vibration-free
- Broadband

Fig. 5 Electrically tunable lens (Optotune n.d.)

Liquid Lenses in Microscopy

Recently, the technology of adjustable focal length lenses or liquid lenses has advanced quickly. Liquid tunable lenses or variable focus lenses have been categorized as: Electrowetting (Berge and Peseux 2000; Hendriks et al. 2005), piezo-hydraulic actuation (Oku et al. 2004), and acoustic radiation force tunable lens (Koyama et al. 2011). Liquid lenses can be easily introduced into microscopic setups to serve a wide range of applications, including extended-depth-of-field imaging (Liu and Hua 2011). Developing high-speed adjustable lenses for quick focusing is another application (Oku et al. 2004), along with speeding up the procedure of axial focus changing in two-photon microscopy (Lee et al. 2010), and spherical aberration compensation caused by refractive index mismatch in tissues (Tsai et al. 2007).

As discussed earlier, extending the depth of focus of a microscope objective is a tedious procedure and requires different complex algorithms or mechanical movement of microscopic components to achieve larger depth of focus. Such procedures are time-consuming and cannot be implemented in real time. A liquid tunable lens or a commercially available electrically tunable lens provides a compact and fast solution for extended depth of field imaging. Recent researches have shown a huge interest of researchers for exploring the application of liquid lenses in microscopic imaging (Fahrbach et al. 2013; Jabbour et al. 2014; Haslehurst et al. 2018; Grewe et al. 2011; Jiang et al. 2015). Most researchers have focused on the use of tunable lenses to extend the depth of focus, depth of field, axial scanning range, and field of view of microscopic systems using liquid lenses. Rapid axial traversing,

autofocusing, and increased depth of field can all be done by incorporating an electronically tunable lens (ETL) in a microscope.

One of the earliest realizations of ETL in microscopic setup was implemented by (Grewe et al. 2011). They presented a combination of liquid tunable lens with a conventional microscope for in vivo two-photon imaging of neuronal populations. After that, ETL was introduced in confocal microscopy setup in reference (Jabbour et al. 2014; Jeong et al. 2016) for optical axial scanning and high speed quantitative measurements with a large field of view. The advantage of ETL over traditional z-scanning for fast axial measurements has been demonstrated. In another application, tunable lens has been employed in multifocal fluorescence microscopy for depth-sensitive epithelial cancer measurements (Zhu et al. 2014). Color imaging has been done at multiple depths by using a tunable lens with a microlens array and the benefit of vibration-free multiplane imaging has been explained. ETL has also been employed in light sheet microscopy in reference (Fahrbach et al. 2013; Haslehurst et al. 2018; Zhai et al. 2019; Hedde and Gratton 2018). The usage of ETL has been explored for improving axial resolution, and imaging in fluorescence light sheet microscopy. Also, fast selective plane light sheet microscopy has been presented where the uniformity of light sheet is maintained using ETL. The efficacy of ETL has been shown for rapid light sheet microscopy and no involvement of mechanical scanning has been implied upon. ETL has been employed in phase imaging in microscopy (Rodrigo and Alieva 2014; Zuo et al. 2013) for high speed and rapid quantitative phase measurements. Image stacks have been acquired by scanning using ETL and through-focus images have been acquired to extract phase information. Apart from that ETL has been employed in temporal focusing microscopy (Jiang et al. 2015) for fast 3D measurements and for in vivo multifocal photoacoustic microscopy (Li et al. 2014). Here, the speed of ETL has been utilized for rapid measurements and for fast focal point tuning. A comparison of extended depth of focus techniques in microscopy based upon their scanning speed, performance, axial range has been presented in Table 1.

The utility of liquid lenses in conventional microscopy has been described in the previous section. Apart from this, application of the electrically tunable liquid lenses has been explored in DHM for quantitative microscopic imaging. In (Schubert et al. 2014), the electrically tunable liquid lens has been used in a self-interference configuration in DHM where it has been employed to modify the amplitude and phase of the illumination light entering the objective. This is done to reduce the effects of coherence of illumination light on quantitative phase images. In another application, ETL has been used for modulating the phase of the reference wave to cope up with the phase changes in the object wavefront (Deng et al. 2017). Here, the phase of the reference wave is changed to match with the curvature of object wave so as to compensate any phase distortions. This is helpful in compensation of the curvature changes in the two arms. Furthermore, ETL has also been employed in digital holography to test aspheric lenses by producing deformed wavefronts (Wang et al. 2017). In this context, the ETL produces an adjustable deformed wavefront that

Table 1 Comparative analysis of various extended depth of focus techniques in quantitative microscopy

Extended DOF techniques in quantitative microscopy	Mechanical scanning	Quantitative information	Image extraction	Speed (in ms)
Wavefront coding (Dowski and Cathey 1995)	No	No	Computational	Hundred ms
Image fusion (Valdecasas et al. 2001)	Yes	No	Direct	Few hundred ms
Wavelength scanning (Attota 2018)	Yes	No	Direct	Few hundred ms
Multifocal plane microscopy (Tahmasbi et al. 2014)	No	No	Direct	High-speed (instantaneous)
Machine learning based (Wei et al. 2021)	No	No	Computational	Few ms
Adaptive optics (Salter et al. 2013)	No	Dependent	Direct	Few ms
DHM only (Colomb et al. 2010)	No	Yes	Direct	Few ms
Sectioning and merging (Colomb et al. 2010)	Yes	Yes	Computational	Few seconds
Optical scanning holography (Kim 2006)	Yes	Yes	Computational	Few seconds
3D deconvolution (Latychevskaia and Gehri 2010)	No	Yes	Computational	Few seconds
Tunable lens (Grewe et al. 2011)	No	Dependent	Direct	A few ms

helps in the reduction of aspheric lenses' high gradient. A double-exposure measurement can be used to estimate the absolute phase of an aspheric surface by decomposing it into two resolvable ones. Different ETL curvatures corresponding to ETL control currents can be used to test different aspheric lenses. In another application of ETL, it is employed to acquire out-of-focus holograms to retrieve the object's phase by employing synthetic aperture imaging in (Lee et al. 2015). Instead of using a mechanical translation stage, ETL is used to defocus images and perform synthetic aperture imaging. A calibration algorithm has been generated to register images, correct the image for different magnifications, and compute the axial locations of the image planes. Apart from these, ETL is used for autofocusing in DHM. Here, ETL focuses the out-of-focus plane information extracted using DHM reconstructions. For evaluation of focused image, a contrast-based autofocus criterion using sharpness metric is described (Kim and Lee 2019). However, the main limitation is that such autofocus algorithms lack efficiency when dynamic specimen is being imaged. A novel method of quantitative imaging of human red blood cells in a volumetric space has been presented in (Barak et al. 2020). Here, ETL has been used to image RBCs present at different focal depths in a volume and their quantitative profile has been obtained. Another method in DHM utilizing variable magnification zooming with ETL has been presented in (Sanz et al. 2020). Here, the

Table 2 Biological applications of liquid lenses in microscopy

Reference No.	Application area	Microscopic technique
Zuo (2015)	Phase imaging of the macrophage phagocytosis	Phase microscopy
Fahrbach et al. (2013)	Imaging beating zebrafish heart	Light sheet
Schubert et al. (2014)	Human fibro sarcoma cells imaging	Self-interference DHM
Jiang et al. (2015)	Ca2+ imaging of neurons in GaCamp6 labeled zebrafish	Two-photon excited fluorescence (TPEF) microscopy
Grewe et al. (2011)	Two-layer two-photon imaging of neuronal cell populations	Two-photo microscopy
Lee et al. (2015)	HEK-293 cell imaging	Synthetic aperture microscopy
Jabbour et al. (2014)	Epithelial tissue imaging	Confocal microscopy
Haslehurst et al. (2018)	Imaging a neuron's dendritic arbor in mammalian brain tissue	Light sheet microscopy
Li et al. (2014)	Imaging of in vivo microvasculature imaging of mouse ear and brain tissues of the mouse	Photoacoustic microscopy
Barak et al. (2020)	Imaging human red blood cells floating in water	Digital holographic microscopy

Table 3 Industrial applications of liquid lenses in microscopy

Reference	Application	Microscopic technique
Deng et al. (2017)	Phase compensation	DHM
Wang et al. (2017)	Measurement of aspheric lens	DHM
Kim and Lee (2019)	Autofocus tracking system	DHM
Barak et al. (2021b)	Zoom and telecentric system for imaging USAF chart	Widefield microscopy

application of ETL for variable zooming in holographic microscopy is explored and presented. ETL is used to reconstruct images at different z-planes using numerical refocusing with different magnifications. However, using a single ETL provides extended plane imaging and multiple ETLs need to be used for zooming into extended planes. An application of ETL with variable aperture controlled objective is presented in (Barak et al. 2021a; Barak et al. 2021b), where a single platform is used to provide telecentricity and zooming in microscopic domain. Tables 2 and 3 show the biological and industrial applications of ETL, respectively, in microscopic domain.

Evaluation of Uncertainty of Measurement

Measurement uncertainty is defined to be a "non-negative parameter characterizing the dispersion of the quantity values being attributed to measurands, based on the information used." Here, a quantity value is defined to be a "number and reference together expressing magnitude of a quantity," and a measurand is defined to be the "quantity intended to be measured" (Joint Committee for Guides in Metrology (JCGM/WG 2) 2006). The "*Guide to the Expression of Uncertainty in Measurement*," or GUM for short, is the generally accepted guide to all things measurement uncertainty related. All measurements are subject to measurement uncertainty. This is for the simple reason that one cannot be infinitely precise or accurate.

Hence, there are certain factors that always influence the measurement and these may be considered as Type B components. Before making any measurement, it has to be ensured that the system has the proper traceability and the record of environmental conditions is mandatory. To start with, first evaluate the Type A uncertainty, which is normally called the repeatability. For this, normally ten readings are taken at a single point of measurement and its mean is calculated dividing it by number of readings. Then standard deviation and standard uncertainty is evaluated as in Eq. (1).

$$S(x) = \sqrt{\frac{1}{n-1}\sum_{j=1}^{n}\left(x_j - \bar{x}\right)^2} \tag{1}$$

where, $S(x)$ is the Estimate of std. deviation, n *i*s the no. of elements, $\bar{x}$ is the mean value, and x_j is the value at *j*th reading.

After that, all of the unknowns should be combined together. To do so, however, one must first convert all previously observed uncertainties into standard deviations. Standard uncertainty is applied in the following phases. In a measurement uncertainty distribution, this is effectively one standard deviation. After then, using the equation of uncertainty propagation, one may aggregate all standard uncertainties (GUM-2008 2008). For this, Eq. (2) must be applied:

$$u(y) = \sqrt{\sum_{i=1}^{n} c_i^2 u(x_i)^2} \tag{2}$$

Here, $u(y)$ is the "combined standard uncertainty," c_i is a sensitivity coefficient, and x_i is an uncertainty component.

Expanded Uncertainty Although the combined standard uncertainty $u(y)$ is commonly used to express the uncertainty of many measurement results, for some commercial, industrial, and regulatory applications (for example, health and safety), a measure of uncertainty that defines an interval around the measurement result y within which the value of the measurand Y can be confidently declared to lie is often

required. Expanded uncertainty, recommended symbol U, is the measure of uncertainty used to achieve this criterion, and it is calculated by multiplying $u(y)$ by a coverage factor, indicated symbol k. This is given by Eq. (3)

$$U = k.u(y) \tag{3}$$

where U is the Expanded Uncertainty, $u(y)$ is the consined uncertainty, and k is the constant (Coverage Factor).

However, quantifying type B occurs in a somewhat different way compared to type A. In comparison to the normal distribution of measurements caused by type A uncertainities, a measurement distribution due to type B uncertainities often takes the form of a rectangular or triangular distribution, with the distribution width upper and lower limits being estimated by the experimenter. This is due to the fact that the type B distributions create errors that tend to remain constant throughout all measurements. A rectangular distribution is used if it is believed that it is equally probable for the uncertainty value to fall anywhere between the upper and lower limits. This is often used as the default distribution for type B uncertainties. A standard deviation can then be obtained from the type B distribution. In order to find the standard deviation of a rectangular distribution, Eq. (4) is used as follows:

$$s = a/\sqrt{3} \tag{4}$$

where "s" is the standard deviation, and "a" the limits of the rectangular distribution. A triangular distribution should be used if it is believed that the uncertainty value is more likely to fall in the middle of the upper and lower limits and less likely to fall near each limit.

The uncertainty parameters for extended depth of focus microscopy using ETL have been mentioned in the next section followed by the uncertainty budget. In the uncertainty budget, some of the parameters are not taken as their contribution is negligible.

Factors Affecting Measurement and Leading to Uncertainty of Measurement

There are several factors that affect the accuracy of measurements. These are called the uncertainity factors that affect the system parameters like repeatability, reproducibility, etc. The Type B uncertainty factors considered for the microscope system with a tunable lens are provided as follows:

(U1) **Resolution**: All optical microscopes are limited to a maximum resolution of around 200 nm. The microscope camera, for instance, has its own finite resolution. The resolution of a microscope is measured using the Rayleigh resolution criterion expressed as in Eq. (5):

$$d = \frac{0.61\lambda}{\text{N.A.}} \tag{5}$$

$$\text{N.A.} = n^{*}\text{Sin}(\mu) \tag{6}$$

where, "d" is the spatial resolution, λ is the imaging wavelength, N.A. is the numerical aperture of microscope given by Eq. (6), n^{*} is the Index of refraction, and μ is the 1/2 of angular aperture.

(U2) **Imperfect optics and sensor**: Imperfect optics may blur the edges of the object and an imperfect sensor might not pick up on the contrast that the measurand has compared with its surroundings.

(U3) **Image analyzer software**: It's also important to consider the level of uncertainty involved with any image software analysis. Perhaps the software has trouble differentiating the microsphere from its surroundings, or it isn't built to handle microspheres larger than a specific size. All of these issues must be considered.

(U4) **Contrast**: Only imaging dark objects with large index of refraction.

(U5) **Reproducibility**: When a microscope's experiment findings are repeated, the results gained through an experiment, a should be attained with a high degree of reliability yet again.

(U6) **Temperature effect**: The temperature variations due to heating of ETL and due to changes in external environmental conditions both lead to uncertainty in measurements.

(U7) **Coma error due to gravity**: When ETL is placed vertically with respect to the optical axis, the coma error gets introduced in the microscope due to displacement of lens liquid by the effect of gravity.

(U8) **Wavefront error**: The wavefront error produced by ETL is 0.5λ at 0 mA, 525 nm, which is another factor for uncertainty.

(U9) **Optical retardance**: The other uncertainty factor is optical retardance produced by ETL. It provides an optical retardance of 6.1 nm at 590 nm.

(U10) The microscope parameters like NA, FOV, magnification, DOF, working distance, resolution get distorted due to introduction of ETL in the microscopic arm. This is because ETL disrupts the conventional telecentricity of the microscope. This leads to type B uncertainty in microscopic measurements.

(U11) Stability

(U12) Bias

(U13) Drift

(U14) MFR/Cal.

(U15) MFR/Cal. Specification

(U16) Reference Standard

(U17) Reference Std. Stability

Uncertainity Budget: The table 4 provides the uncertainty budget for MO and ETL combination.

Table 4 Uncertainty budget for microscope and ETL combination

	Sensitivity coefficient	Value					Std. uncertainty	Degrees of freedom	Significance check
	(c_i)	(x_i)	unit	Type	Distribution	Divisor	($u(x_i)$)	v	
Repeatability	1	1000	ppm	A	Gaussian	2	0.5000	1	9.7%
Reproducibility	1	1000	ppm	A	Gaussian	2	0.5000	1	9.7%
Stability	1	1000	ppm	A	Gaussian	2	0.5000	1	9.7%
Bias	1	1000	ppm	A	Gaussian	2	0.5000	1	9.7%
Drift	1	1000	ppm	A	Gaussian	2	0.5000	1	9.7%
Resolution	1	1000	ppm	B	Rectangular	$\sqrt{3}$	0.5774	1E-200	11.2%
MFR/Cal. Specification	1	1000	ppm	B	Gaussian	2	0.5000	99	9.7%
Reference Standard	1	1000	ppm	B	Gaussian	2	0.5000	99	9.7%
Reference Std. Stability	1	1000	ppm	B	Gaussian	2	0.5000	99	9.7%
Temperature Effect	1	1000	ppm	B	Rectangular	$\sqrt{3}$	0.5774	1E-200	11.2%
Combined Uncertainty							$u_t(y)$	v_e	
Expansion Coefficient (k)							1.633	22.6	100.0%
Expanded Uncertainty [$ku_t(y)$]							2.074		
							3.39		

Conclusion

In summary, a survey of extended depth of focus algorithms techniques has been presented for extending the axial range in quantitative microscopy. Various computational and experimental methods have been discussed to extend the focal depth of a microscope ranging from decades old to recently developed new methodologies. The study concludes the advantage of using liquid lenses over other conventional techniques in microscopy for extending the axial depth range. It has been presented that liquid lenses prove to be a fast and vibrationless solution for imaging in the extended focus regions. Various applications of tunable liquid lenses in microscopy have been presented at length ranging from biological to industrial and dynamic scenarios. The discussions are provided in such a way that the reader might get familiar with the usage of liquid lenses in microscopic systems as well as can have a better understanding of what technique must be applied for a certain application area. We have discussed most of the utilities and benefits of liquid lenses in microscopy, however, there are certain limitations and precautions that need to be taken care of while introducing tunable lens in microscopic setups. These include the problem of telecentricity, image shift, poor resolution, coma aberrations, wavefront errors, gravity factors, etc. Most of the researches have proposed some potential solutions to these challenges; however, all the approaches that have been used in microscopy using liquid lenses are still exploring their utilization with better imaging quality. Further improvements in fabrication technologies of liquid lenses and advancements in extended depth of focus techniques is still a matter of research. In the final section of this chapter, an example has been quoted to evaluate the uncertainty of measurement by taking into account the various factors affecting the measurements as Type B components and the uncertainty budget is also presented.

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Part X

Metrology for Advanced Communication

Quantum Microwave Measurements

57

Yashika Aneja, Monika Thakran, Asheesh Kumar Sharma, Harish Singh Rawat, and Satya Kesh Dubey

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Abstract

The term "quantum microwave measurement" stands for the measurement of microwave parameters by exploiting the properties of atomic energy levels and related transitions in presence of optical/laser signal. In other words, it can be

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stated that atoms work like a transducer, converting microwave E-Field and H-Field strengths into observable changes in optical signals. A real-time application of microwave in this domain was first investigated by Christopher and their group, where they have sensed RF electric field strength by measuring the frequency split of electromagnetically induced transparency signal in the presence RF field applied between the Rydberg states (Holloway et al. 2014a). The beauty of this experiment was that the RF E-Field strength could be made directly traceable to Planck's constant thus reducing the uncertainties associated with other methods of measurement of E-Field using various classical probes. After that, numerous research was done by various researchers and still undergoing utilizing the optical and microwave field interaction with Rydberg atom-based systems, such as broadband RF E-Field sensing, frequency modulation, frequency up-conversion, frequency down-conversion, etc. The interaction of electromagnetic field with atomic system results in specific phenomena such as EIT and ATS (Boller et al. 1991; Marangos 1997; Anisimov et al. 2011), which also form the basis of all these applications. Therefore, in this chapter first we will discuss basic properties of Rydberg atoms, followed by the interaction of electromagnetic field with two-level atomic system. Later, EIT and ATS phenomena in a three-level system and applications will be discussed. Some of the devices based on these phenomena such as electric field sensors, microwave mixers as well as receiver for communication purposes will also be discussed in detail. As Rydberg atoms are the heart of the Quantum Microwave Measurements, different methods of fabrication of atomic vapor cells with their advantages and disadvantages will also be discussed in brief.

Introduction

The term quantum microwave measurement stands *for the measurement of microwave parameters by exploiting the properties of atomic energy levels and related transitions in presence of optical/laser signal. In other words, it can be stated* that atoms *work like a transducer, converting microwave E-Field and H-Field strengths into observable changes in optical signals. A real-time application of microwave in this domain was first* investigated *by Christopher and their group, where they have sensed RF electric field strength by measuring the frequency split of Electromagnetically Induced Transparency signal in the presence RF field applied between the Rydberg states* (Holloway et al. 2014a). *The beauty of this experiment was that the RF E-Field strength could be made directly traceable to Planck's constant thus reducing the uncertainties associated with other methods of measurement of E-Field using various classical probes. After that, numerous research was done by various researchers is still undergoing utilizing the optical and microwave field interaction with Rydberg atom-based systems, such as broadband RF E-Field sensing, frequency modulation, frequency up-conversion, frequency down-conversion,* etc. *The interaction of electromagnetic FIELD with atomic system results in specific phenomena such as EIT and ATS* (Boller et al. 1991; Marangos 1997; Anisimov et al.

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Rydberg Atoms

Rydberg atoms are named after Johannes Rydberg, who introduced an empirical formulation to determine energy levels of the highly excited atom (Scientists Make First Observation of Unique Rydberg Molecule 2022). Rydberg atoms are basically excited atoms having one or more electrons with a very high principal quantum number n (Gallagher 1988), where higher the value of n, larger the distance of electrons from the nucleus. This results in several distinct characteristics. First, higher value of n results in outermost electrons being weakly bound to the nucleus, making them highly sensitive to the environmental conditions of the atom. The sensitivity of atoms idealizes them to behave as ideal probe for detection of external electric (Holloway et al. 2014a, 2017) and magnetic fields (Sedlacek et al. 2012). Second, the strong interaction between the collectively excited Rydberg atoms induces dipole blockade effect (Comparat and Pillet 2010; Gallagher and Pillet 2008), which is further used in applications such as quantum information processing (Lukin et al. 2000; Saffman et al. 2009) and quantum computation (Lim et al. 2013). Moreover, the lifetime of the Rydberg state is of the order of 100 μs (Beterov et al. 2009) which is higher in comparison to low-lying excited states with the order of around 23 ns (Steck 2021). The other properties include low-ionization energy, larger transition dipole moment, level frequency spacing of the order of MHz and GHz, etc. which are proven to be quite useful for microwave and RF electrometry. These distinct characteristics of Rydberg atoms have made it a popular research topic in the field of quantum optics as well as for atom photon interaction. The dependency of the above discussed properties of Rydberg atoms on principal quantum number, n, is shown in Table 1. The size of the atom increases with the scaling of the factor of n^2

Table 1 Dependency of properties of Rydberg atoms on principal quantum number n

Property	Scaling as n
Size	n^2
Lifetime	n^3
Dipole moment	n^2
Ionizing field	n^{-4}
Polarizability	n^7

which implies as the principal quantum number of the atomic state increases, the size of the atom increases. The transition dipole moment also scales as n^2, leading to ease in coupling of two excited Rydberg states with the help of RF and microwave fields. The low ionization energy obeying the scaling law as n^{-4} suits perfectly well for sensing of external RF and microwave E-fields (Heckötter et al. 2017).

Atom-Photon Interaction

There are infinite number of energy states available for transitions in an atomic ensemble, each defined by a different energy level. Invention of lasers in 1960, by T. H. Maiman, led to the possibility of detailed study in this domain. The understanding of atom-photon interaction physics helps in manipulating the optical response of atom-based system. The interaction of excited atom and a photon can be understood in the easiest way for a single atom in two-level system. A closed two-level atomic system means that the relaxations from the other states in continuum are neglected and any population leaving the excited state relaxes to the ground state. Figure 1 shows a 2-level system coupled with a resonant laser light of frequency ω_L.The incident light excites the atom from ground state $|1\rangle$ to the excited state $|2\rangle$, and the excited state decays to ground state with decay rate γ_{21} with lifetime of $T = 1/\gamma_{21}$. Although an infinitely large number of atomic states are available for transitions in the atomic medium, the monochromatic light beam emitted by laser can couple only those two energy states which are resonant with the laser frequency, thus reducing the problem to a two-level system. A semiclassical approach can be taken to analyze this problem, where the light is treated classically and atom is treated quantum mechanically. Density matrix formulism (Blum 2012; Rand 2016) is used to treat this system mathematically. It is equivalent to using a wave function and representing the states of a quantum system, giving complete details regarding population and coherences in between the states.

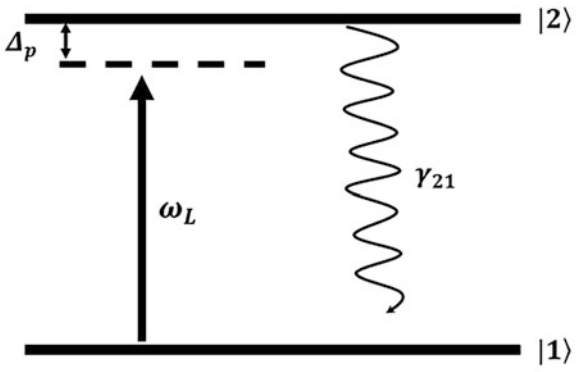

Fig. 1 Schematic representation of two-level system interacting with optical frequency ω_L

The wave function for a two-level system can be written as

$$|\psi >= C_1|1 > +C_2 \mid 2 > \tag{1}$$

Density matrix operator is expressed as follows:

$$\rho =| \psi > <\psi \mid \tag{2}$$

$$\rho = [C_1(t)|1 > +C_2(t)|2 >]\left[C_1^*(t)<1|+C_2^*(t)<2|\right] \tag{3}$$

$$= |C_1(t)|^2|1 > <1| + C_1C_2^*(t)|1 > <2| + C_1^*C_2(t)|2 > <1| + |C_2(t)|^2|2 > <2| \tag{4}$$

The density-matrix elements are

$$\rho_{11=}|C_1(t)|^2 \tag{5}$$

$$\rho_{12=}\ C_1C_2^*(t) \tag{6}$$

$$\rho_{21=}\ C_1^*C_2(t) \tag{7}$$

$$\rho_{22=}|C_2(t)|^2 \tag{8}$$

where, ρ_{11}- Probability of occupying the ground state
ρ_{22}- Probability of occupying the excited state

The diagonal matrix elements ρ_{11} and ρ_{22} give the probability of occupancy of population in level $|1>$ and $|2>$, respectively, while the other two off-diagonal elements give the coherences between the states.The same density matrix formulism can be further applied to systems having more than two atomic energy levels. In case of a multiple-level atomic system problem, the values of density matrix elements are averaged over states. In an ensemble of n systems in the state $|\psi_n\rangle$, the density matrix operator can be expressed as follows

$$\rho = \sum P_n \mid \psi_n > <\psi_n \mid \tag{9}$$

where P_n is the probability of finding a particular system in state $|\psi_n\rangle$.

This two-level system can serve as a unit for any atomic transition-based application, as it is easy to understand and mathematically analyze as well. This system can be coherently manipulated, leading to absorption of electromagnetic signal and other related phenomena.

However, the real-world scenarios involving atomic transitions in excited atom ensemble may be much more complex, varying from three-level atomic systems to six-level system or more. In case of three or higher-level system, these coherent manipulations lead to various constructive and destructive phenomena between the excitation pathways of atomic transition. The next section focuses on the phenomena of EIT and ATS in three-level atomic system.

Electromagnetically Induced Transparency (EIT)

The interplay of electromagnetic (EM) waves with a three-level atom system incites interferences in between the pathways of transition within the atom. Figure 2 shows the two, three-level models, which have been used to study EIT, by different groups (Agarwal 1997; Fulton et al. 1995; Anisimov and Kocharovskaya 2008), namely, cascade (Fig. 2a) and lambda (Fig. 2b) model. It demonstrates the two pathways of transitions between ground and higher energy states of atom due to interaction of weak (probe) and strong (pump) optical fields with the atom. When the frequency of weak coupling field (probe laser) is resonant with the frequency of transition from $|1>$ to $|2>$ atomic levels, this gives an absorption spectrum. When a stronger coupling field (pump laser) is applied in resonance with the $|2>$ to $|3>$ atomic transitions, these two states get further mixed to form pairs of dressed states, having almost equal energy. Destructive quantum interference is observed from each of these dressed states, due to opposite signs of the excitation amplitude (Holloway et al. 2014a), resulting in transparency in absorption spectra. This phenomenon of quantum interference in which transparency in absorption spectra is observed when a second stronger light of different frequency is incident on the system that has already absorbed a light of specific frequency is referred to as EIT.

Harris et al. reported EIT for the first time in 1989 (Imamoğlu and Harris 1989): Interference in between the dressed states led to zero in absorption profile. An experimental proof was reported in 1991 by Boller et al. (Boller et al. 1991) which exhibited an opaque atomic transition completely transparent at its resonance frequency.

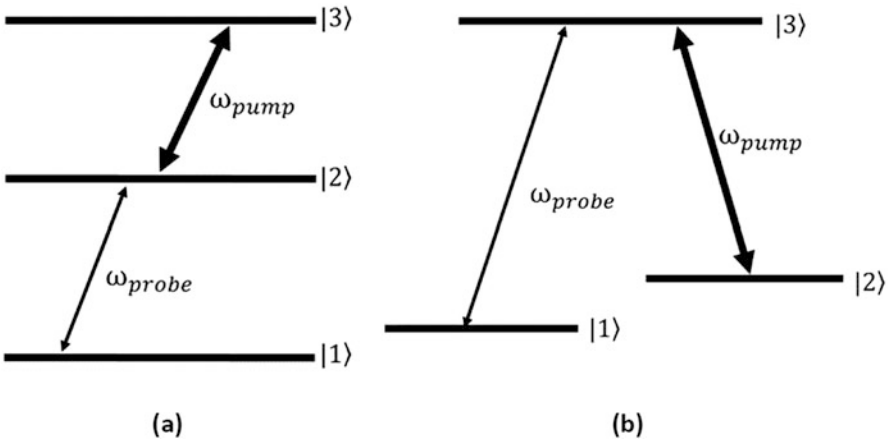

Fig. 2 Energy-level diagrams for three-level atom: (**a**) Ladder (cascade) model with weak probe light for coupling states $|1\rangle$ and $|2\rangle$, and strong pump beam for coupling states $|2\rangle$ and $|3\rangle$; (**b**) lambda model with weak probe light for coupling states $|1\rangle$ and $|3\rangle$, and strong pump beam for coupling states $|2\rangle$ and $|3\rangle$

Autler-Townes Splitting (ATS)

Autler-Townes splitting is a quite similar looking phenomenon to EIT, as both have a characteristic dip in the absorption spectra. It is very important to understand to identify the difference in between the two phenomena. The major difference lies in the parametric regime in which these two phenomena are observed. EIT is observed as a consequence of destructive interference (Marangos 1997; Fleischhauer et al. 2005) while on the contrary ATS is observed as a gap in between two resonances which is a consequence of the constructive interference (Davuluri et al. 2015). Autler and Townes (1955) reported this effect in 1995 on OCS molecule and depicted its microwave transition splitting in distinct components when a stronger microwave field is used to couple one of the resonant states with a third energy level. With the passage of time, lasers were developed and this effect had also been reported in optical regime (Hänsch and Toschek 1970). Availability of laser fields provided researchers with enormous opportunities to provide high-energy beams of fixed frequencies to study atom-photon interactions.

EIT and ATS in Three-Level Atomic System

As mentioned before, both EIT and ATS phenomena are quite similar in appearance and are also accompanied with cancellation in absorption of light, which is resultant of interferences between transition amplitudes (Anisimov et al. 2011; Anisimov and Kocharovskaya 2008; Tan and Huang 2014; Giner et al. 2013; Abi-Salloum 2010; Lu et al. 2015; Rawat and Dubey 2020); it becomes very important to realize whether reduction is observed due to EIT or ATS. Salloum (Abi-Salloum 2010) worked in this aspect and figured out the major difference between the two.

Figure 3a represents probe laser absorption curves for ladder model as shown in Fig. 2a. The Rabi frequencies of probe and pump lasers are represented as Ω_{pr} and

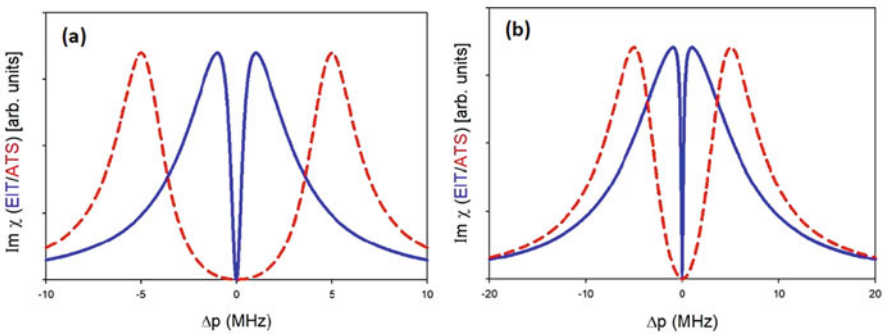

Fig. 3 The blue- and red-dashed curves represent the EIT and ATS, respectively, for (**a**) three-level ladder and (**b**) three-level lambda model. The probe laser's Rabi frequency for both is $\Omega_{pr} =$ 500 kHz. The pump laser Rabi frequency for EIT and ATS in both the models is $\Omega_{pu} = 2$ MHz and 10 MHz, respectively

Ω_{pu}, respectively. Both the curves are observed in a condition when $\Omega_{pr} \ll \Omega_{pu}$, i.e., weak probe regime, with $\Omega_{pr} = 500$ kHz and $\Omega_{\mathrm{pu}} = 2$ MHz. The blue curve, showing reduction in the value of absorption at $\Delta_{\mathrm{p}} = 0$ when Rabi frequency is less than the decay of the transition (i.e., $\Omega_{pu} \ll \gamma_{21}$), represents EIT, and given by:

$$\Delta_p = \omega_{21} - \omega_{pr} \tag{10}$$

Here, Ω_{pr} and Ω_{pu} are the Rabi frequencies for probe and pump lasers, respectively.

Δ_{p} is the detuning of the laser lights from the transitions
ω_{21} is the frequency for transition between level 1 and level 2
ω_{pr} is the frequency for probe laser
The decay rate from level $|2\rangle$ to level $|1>$; $\gamma_{21} = 6.06$ MHz
The decay rate from level $|3\rangle$ to level $|1>$or $|2>$; $\gamma_{31} = \gamma_{32} = 1$ kHz

The red-dashed curve showcases reduction in absorption, which represents the ATS effect when the pump Rabi frequency exceeds the decay of the transition, i.e., $\Omega_{pu} > \gamma_{21}$. The pump Rabi frequency for this case is 10 MHz, while keeping all other atomic parameters same. Figure 3b represents probe light absorption for lambda model in weak probe regime. The blue and red-dashed curves represent EIT and ATS phenomena for pump Rabi values 2 MHz and 10 MHz, respectively. It is quite observable that both of these effects occur in strong coupling field regime, but there lies a minimum threshold value for transition from EIT to ATS, and this threshold value is the decay rate (γ_{ji}) for the states on which probe light is tuned. In case when $\Omega_{pu} < \gamma_{21}$, the transparency in absorption is termed as EIT, whereas when the applied pump laser's Rabi frequency goes beyond this value, the transition from EIT to ATS is noticeable.

Till now, we have discussed the properties of Rydberg atom, interaction of two-level atomic system with optical field, followed by EIT and ATS phenomena in the three-level atomic system. Researchers have extended this work by exploring the various new phenomena and changes in the discussed phenomena by application of external RF/microwave fields on Rydberg atom-based system. Now, we will discuss the application of these quantum phenomena in microwave measurements.

Utilizing Advanced Laser-Atom-Interaction Phenomena in Microwave Applications

Rydberg Atom as E-Field Sensor

The conventional E-field measurement sensors are essentially antennas or probes, which operate in a designated frequency band, along with associated electronics and suitable microwave receiver or spectrum analyzer, etc. The antennas receive

the electromagnetic energy, the associated electronics converts it into received microwave power, and it can be read through the receiver or analyzer. The antennas and probe are made of metal or contain a metallic sensing element (dipole, monopole, patch, etc.) in them. The principles of EM wave reception on these antennas and probes are well established. However, the major issue with these is the fact that the metallic element of the antennas, while measuring the E-field, also perturbs the field. This results in inaccuracy to the measured field. Also, these antennas are very frequency selective and cannot cover the entire microwave spectrum and are hence needed to be designed for each different frequency band, as required. There are also size constrains, as the size of probe is quite large for lower frequencies (below 1 GHz) and very small for higher frequencies (above 30 GHz).

Rydberg atom-based sensors offer a unique solution to all above problems in the E-field sensing. As mentioned before, application of external RF field upon the atom already in higher state results in further coupling on the Rydberg states. The energy gap of the coupling Rydberg states is as per the frequency of external RF signal. As there are infinite number of Rydberg states available, the transitions/resonances can occur over a wide range of frequencies ranging from hundreds of megahertz (MHz) to terahertz (THz); this can serve as a single E-field sensor for entire frequency range in the spectra. Also, the atomic dipole moments for these frequencies range can be very large, thus making Rydberg atoms a perfect candidate for E-field sensing (Holloway et al. 2017).

Figure 4 shows a schematic of Rydberg atom-based E-field sensor. In this setup, first a weak probe beam is used to excite the alkali (Rubidium) atoms from ground state to first excited state, then a strong counterpropagating pump laser is applied, tuning the atoms from first excited state to a Rydberg state. This leads to the detection of EIT signal at the photodetector as shown in Fig. 5a.

When an external RF field is applied on the vapor cell, two Rydberg states are coupled depending upon the frequency of the RF signal applied. The EIT lines split

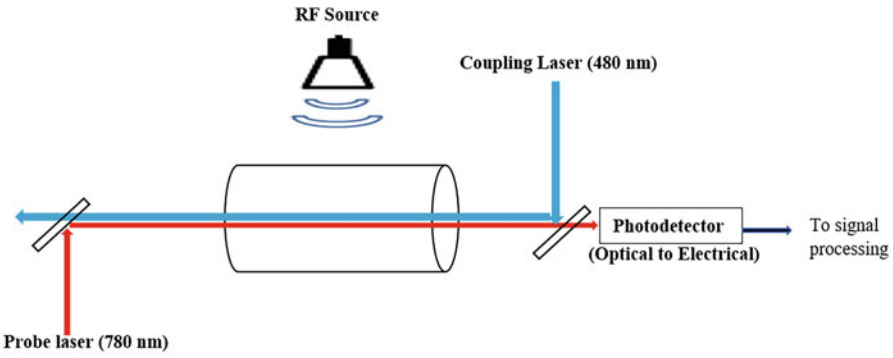

Fig. 4 Schematic of atomic electric field sensor for sensing RF fields

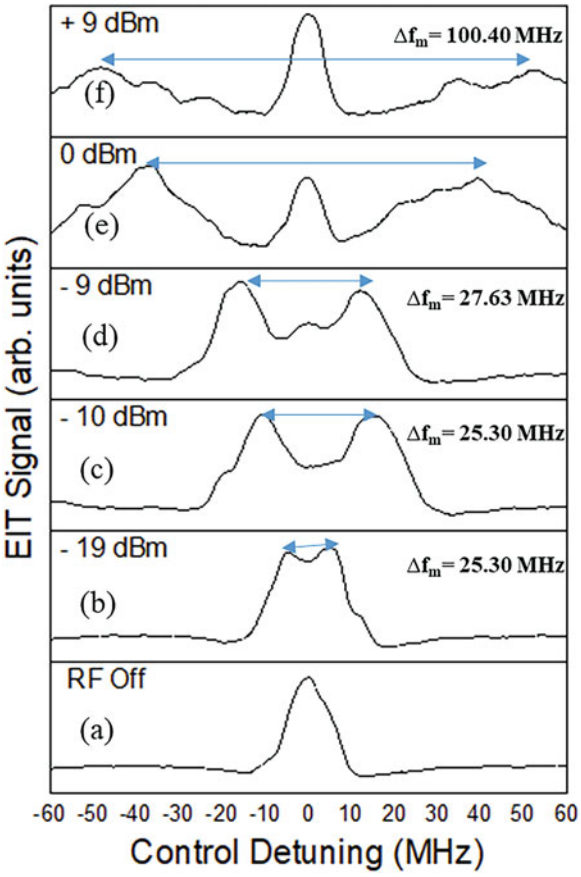

Fig. 5 Illustration of splitting of EIT peak with increasing RF power of signal generator at 15.09 GHz: (**a**) RF off; (**b**) – 19 dBm; (**c**) – 10 dBm; (**d**) – 9 dBm; (**e**) 0 dBm; and (**f**) 9 dBm. Splitting, indicated by double-sided arrow between the off-set peak, increases with increasing RF power. (Adapted from Rawat and Dubey (2020))

into a pair of two ATS lines. This splitting increases with increase in RF signal power. This is shown in Fig. 5b–e.

Rawat and Dubey (2020) have studied this for coupling $52D_{5/2}$ state to $53P_{3/2}$ using RF signal of frequency 15.09 GHz and obtained results in terms of Δf_m (frequency difference between two ATS peak) corresponding to the applied signal power (in dBm). The amplitude of electric field of RF can be calculated using the following relation (Holloway et al. 2014a):

$$| E | = \frac{h\Delta f_m}{d} \tag{11}$$

where Δf_m - is the frequency difference between two ATS peak,

d- is the electric dipole moment of the Rydberg transition,
h-Planck's constant

Table 2 Measured amplitude of RF E-field strength for various RF power levels

Power (dBm)	Δf_m (MHz)	$\|E\|$ (V/m)
19	9.61	0.44
−10	25.30	1.15
−9	27.63	1.26
0	77.12	3.52
9	9100.40	4.58

Adapted from Rawat and Dubey (2020)

These results with electric dipole moment of 1451.27×10^{-29} C-m (for $52D_{5/2}$ state to $53P_{3/2}$ transition) are compiled in Table 2.

However, this method has certain limitations. Since the Rydberg states' energy levels are quantized and not all can be coupled with each other, this phenomenon is only valid for limited range of frequencies, i.e., 500 MHz to 1 THz (Gordon et al. 2019). This problem can be resolved by exploiting the full quantum response of the Rydberg atom interaction with RF fields that even includes off-resonance AC Stark shift readout (Anderson et al. 2018), enabling direct ERMS measurement of continuous-frequency RF field frequencies over tens of gigahertz with a >60 dB dynamic range.

Also, this method becomes highly ineffective when E-field applied is too weak for ATS to occur (Gordon et al. 2019). This problem can be addressed by making use of Rydberg atom as a mixer, as discussed in the next section.

Rydberg Atom as Mixer

While using Rydberg atom as E-field sensor, it was observed that the ATS peaks in the absorption spectra split upon application of external RF field. However, this method is ineffective while dealing with weak RF signals, i.e., when the field is too weak for ATS peaks to split. The minimum detectable RF field capable of being detected with ATS and causing AT peak separation equivalent to EIT line width (Holloway et al. 2014b) is given by:

$$E_{\mathrm{AT}} = \frac{\lambda_p}{\lambda_c} \frac{2\pi\hbar\Gamma_{\mathrm{EIT}}}{\rho_{\mathrm{RF}}} \tag{12}$$

where λ_p is the wavelength of the probe laser,

λ_c is wavelength of the coupling laser,
Γ_{EIT} EIT line width,
ρ_{RF} is element of dipole matrix element of RF Rydberg transition field, and
$\hbar$ is Planck's constant

Several techniques were earlier explored for measurement of weak RF fields, such as optical cavities (Yang et al. 2018), narrow EIT line-width frequency-

modulated spectroscopy (Yang et al. 2018), homodyne detection with Mach Zehnder Interferometer (Kumar et al. 2017), etc. Simons and his group (Gordon et al. 2019; Simons et al. 2019) have suggested using Rydberg atoms as mixer for measuring phase and frequency of weak RF fields. The experimental setup is similar to E-field sensing, but an additional local oscillator is used to generate another RF field. The two signals interfere with each other in Rydberg states of the atom and generate a low- and high-frequency component. The intermediate frequency (IF) representing the difference between frequencies of two applied RF fields is detected by optically probing the Rydberg atoms (Simons et al. 2019). This IF signal modulates the probe laser intensity enabling the measurement of weak RF fields. Also, the phase associated with the IF signal corresponds to the measurement of phase of the RF field directly.

The setup for the experiment conducted is as shown in Fig. 6. The probe laser of wavelength $\lambda_p = 852$ nm (D_2 transition wavelength) excites the Cesium (Cs) atoms from ground state ($6S_{1/2}$) to first excited state ($6P_{3/2}$). These atoms are further excited to state $34D_{5/2}$ by counterpropagating coupling laser tuned to $\lambda_c = 511.148$ nm resulting in transparency in the absorption spectra for the probe laser.

In a conventional RF mixer, two RF signals are fed to the mixer. The resultant signals are sum and difference of the two input signals. RF mixers are used for up-conversion and for down-conversion of frequencies, usually in the case of transmitters. The Rydberg atom-based RF mixer setup also works in the similar way. The two different weak RF signals are incident on a vapor cell and can be represented as

$$E_1 = E_{\mathrm{LOC}} \cos(\omega t_{\mathrm{LOC}} + \varphi_{\mathrm{LOC}}); E_1 \approx E_{\mathrm{AT}} \tag{13}$$

$$E_2 = E_{\mathrm{SIG}} \cos(\omega t_{\mathrm{SIG}} + \varphi_{\mathrm{SIG}}); E_2 \leq E_{\mathrm{AT}} \tag{14}$$

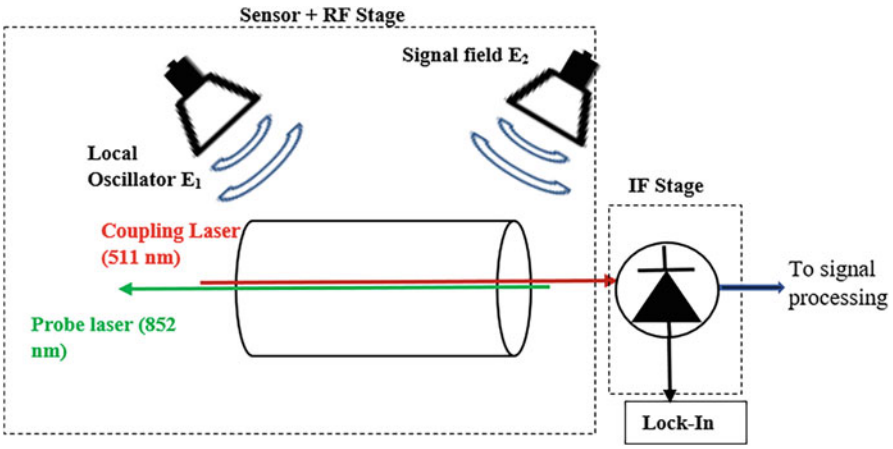

Fig. 6 Experimental setup for detection of weak electric field and measurement of phase of RF fields

Here, E_{AT} represents minimum detectable RF field (as per Eq. 12).

In experiment carried out by Gordon et al. (2019), the local oscillator is tuned to 19.626 GHz (in resonance with $34D_{5/2}$ to $35P_{3/2}$ Rydberg transition of Cs). The other RF signal to be sensed is tuned at 19.626090 GHz, so that it gets detuned by 90 KHz (intermediate frequency) from the other field.

These two RF signals interfere with each other and generate a high-frequency component and a low-frequency component. One of the electric components oscillating at higher frequency oscillates at ω_{LOC} resonant with Rydberg transition while the other component oscillates at a frequency $\Delta\omega = \omega_{LOC} - \omega_{SIG}$ resulting in modulation of probe laser intensity, which can be measured on photodiode and as EIT spectrum. Now the resulting frequency of the signal turns out to be 90 KHz which makes E_{SIG} detectable even when it is below $E_{AT.}$ The final output on photodiode is further passed to lock-in amplifier with reference same as the intermediate frequency. The lock-in output voltage is proportional to the applied weak field (Gordon et al. 2019).

The same setup can be used to detect the phase of the RF signal. The resultant second RF which oscillates at same frequency as ω_{LOC} creates a beat note which can be demodulated by the Rydberg atoms. The change in phase can now be measured as phase of demodulated signal and is directly associated with the incident RF signal (Simons et al. 2019).

Rydberg atom when used as a mixer overcomes limitations of conventional RF mixers (such as lineality, operating range, sensitivity, etc.) and also has other advantages such as tuning and narrow band frequency selection. The Rydberg mixer's ability to separate and distinguish between signals of various RF frequencies with a frequency resolution order of magnitude finer than the response bandwidth of the Rydberg transition makes it more advantageous. These characteristics indicate that the vapor cell can be used to characterize the weak RF E-field efficiently, enabling the development of quantum E-field sensors for broad frequency range and better sensitivity.

Rydberg Atom as a Receiver for Communication

Traditional antennas have been in use since a very long time for transmission and reception of signals. Different types of antennas are used for this purpose such as microstrip patch antenna, helical antenna, etc. It basically acts as a transducer which converts RF signals to electrical signals and vice versa. However, their uses are limited to the applications based on their size, frequency range, gain, and transmission requirements. Detection of time-varying and modulated RF fields using Rydberg atoms have offered new possibilities in the field of RF sensing and communication. Currently, researchers are focusing on Rydberg atom-based communication system because of its advantages over conventional antenna. A smaller single antenna can be used for a wider range of frequencies from few MHz to 100 GHz. It eliminates the necessity of demodulation circuitry as signal can be read out directly from the optical laser beam. The signal received is free from any electromagnetic interference, since the signal is received through optical laser beam.

Distinct EIT peaks can be used for locking, and message signal can be modulated over RF signals of different frequencies resulting in coupling of distinct Rydberg states. A vapor cell containing both Rb and Cs atoms can together allow multiband communication through a single atomic antenna. The communication becomes independent of the size limitation in the traditional antennas.

Different groups have worked and showed detection of various kinds of time-varying and modulated signals (Anderson et al. 2021; Meyer et al. 2018; Deb and Kjærgaard 2018; Holloway et al. 2019) using Rydberg atom-based system. Some of them include Rydberg atom-based transmission system for digital communications (Meyer et al. 2018), atom radio-over-fiber (Deb and Kjærgaard 2018), and a multiband atomic AM and FM receiver for radio communications (Anderson et al. 2021).

The basic principle for an atomic vapor cell to act as an antenna (receiver) lies in the fact that EIT observed explores a number of large differential dipole moments for different Rydberg states of an atom. H. S. Rawat, in his work (Rawat 2012), has demonstrated the use of Rydberg atom as a receiver for analog communication, and the experimental setup for the same is shown in Fig. 7.

In this setup (adapted from), counterpropagating probe and coupling laser beams are passed through vapor cell to excite the atoms to the Rydberg state. The absorption spectra is continuously monitored to observe the EIT. An audio signal is then AM modulated with a microwave source and incident on the vapor cell. The change in the quantum state of Rydberg atom due to this AM signal is observed by the detector. The detectors' output signal is directly connected to an output device (oscilloscope or speaker). Now, when both probe and coupling lasers are tuned at

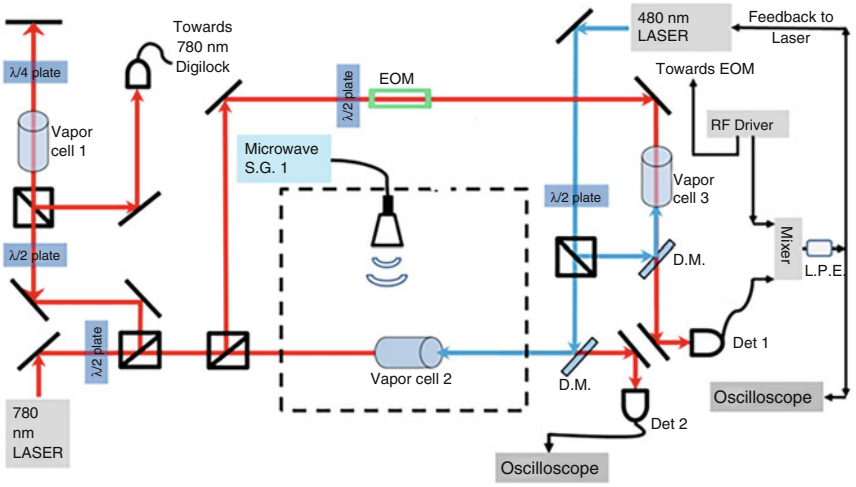

Fig. 7 Schematic of the experimental setup for atom-based receiver for analog communication. (Adapted from H. S. Rawat (2012))

specific transition frequencies, the output signal at detector will be same as the message signal in the AM-modulated signal.

In the schematic shown in Fig. 7, the probe laser (red) is locked at 780.24 nm by tuning the atomic states $5S_{1/2}$ and $5P_{3/2}$ via the frequency reference technique, achieved through "vapour cell 1." The coupling laser (blue) is tuned at 479.788 nm by coupling the states $5P_{3/2} - 61S_{1/2}$, locking at these transition levels by locking at the EIT peak detected at photodetector (Det 2), and feeding the signal to "vapour cell 3."

Afterward, ATS is observed in EIT for "vapour cell 2," by applying the AM-modulated microwave field. The microwave signal chosen for this experiment has frequency 15.175 GHz (for coupling the Rydberg states $61S_{1/2}$ with $61P_{3/2}$).

The amplitude-modulated message signal using signal generator is incident directly onto the vapor cell 2 (after the laser is locked at one of the inflection points of EIT peak obtained through vapor cell 3). The incident microwave field now interacts with the Rydberg atoms in vapor cell 2. This impacts the dc signal already observed for EIT and results in variation of the amplitude of E-field due to the presence of message signal. Since the message signal was an audio signal, the output from the detector (Det 2) is directly given to the speakers and the exact message signal was received without any delay or loss of the signal.

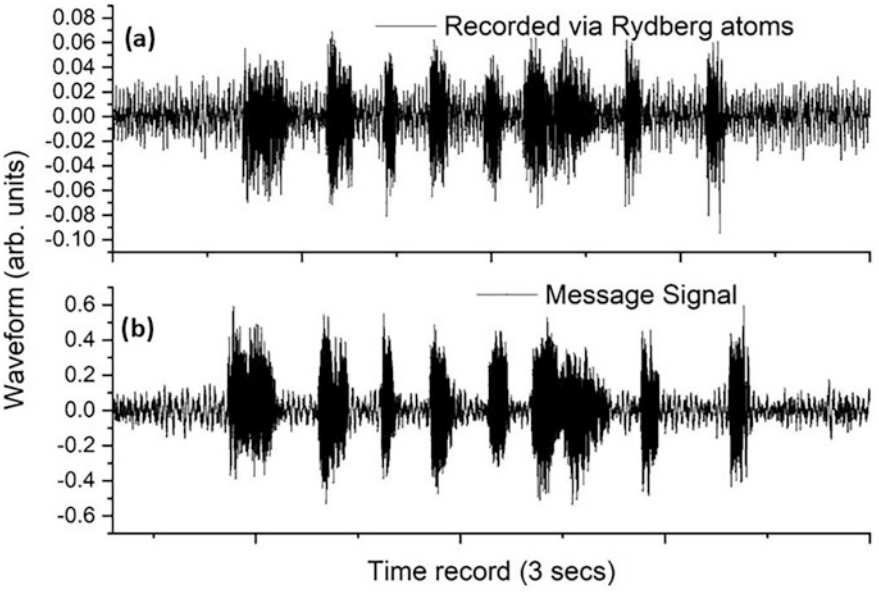

Fig. 8 The recorded waveforms of audio message signal amplitude modulated on the microwave. The message signal is "Hello mic testing alpha one check check": (**a**) The message signal recorded before modulation; (**b**) the message signal obtained via Rydberg atoms without any demodulation circuit. (Adapted from H. S. Rawat (2012))

Figure 8 shows the comparison of the output signal at detector (Det 2) (Fig. 8a) and the input audio signal (message signal) used for amplitude modulation of the 15.175 GHz microwave signal (Fig. 8b).

The important observation over here is that to receive the message signal, one does not require any demodulation and the message signal can be retrieved as it is when the atoms inside vapor cells are properly controlled.

It can be said that change in the quantum states of Rydberg atoms takes place, when a modulated signal is incident on it and that change in quantum states is heard/observed through speakers. These Rydberg atoms automatically demodulate the signal, without any extra demodulating circuitry, and one can get a direct readout of the message signal without any delay and with high fidelity.

This depicts that Rydberg atom as a receiver has major advantages over the existing conventional techniques, namely,

- It eliminates the necessity of demodulation circuitry as signal can be read out directly from the optical laser beam.
- A single receiver (Rydberg atom-based vapor cell) of such a smaller size can be used for RF signals ranging from 100 MHz to 500 GHz.

Fabrication Methods of MEMS Alkali Vapor Cells

As clear from the previous sections, alkali atoms inside the vapor cells containing the alkali atoms are at the heart of all the discussed applications and other various phenomena arising from laser-atom interactions of alkali metals. Fabrication of these vapor cells and miniaturizing them has been a task of major importance since the 1950s when absorption of microwave radiations by Cs/Rb vapors led to the discovery of atomic time and frequency standards (Lombardi et al. 2007; Sullivan 2001). Fabrication process of atomic vapor cell has always been a tedious task because

- The size of the cell has to be small.
- It should be sealed hermetically to protect vapors of alkali atoms and buffer gases from leaking.
- Moreover, the cell should be designed in a way that it allows light to pass through the atoms, and the material from which the cell has to be fabricated should neither be reactive with atomic vapors nor magnetic in nature.
- The size of the cell proved to be problematic, when the results showed instabilities of time parameters.

Several different methods are used in microfabrication of atomic vapor cell. These methods are based on three fundamental ideas:

1. Filling of pure liquid alkali metals
2. Chemical reaction of alkali compounds
3. Dispensing of alkali vapor from the solid-state dispensers

The following section discusses these in brief along with their limitations.

Filling of Pure Alkali Metals

Pipetting

Liew et al. (2004) proposed one of oldest known methods for fabrication of atomic vapor cell. The fabrication part included three main steps – first, fabricating a glass-silicon preform. Second, alkali metal is then directly introduced via pipetting in a high-vacuum chamber. Lastly, a buffer gas is then introduced at a pressure around 50 to 200 Torr, and then the second glass is anodically bonded to the silicon preform.

Pure alkali metal and appropriate mixture of buffer gases are obtained in this process, but manipulation in anaerobic condition is quite a difficult job.

Glassblowing

This process involves etching of two holes on a silicon wafer, which are connected through a channel. The second step includes anodically bonding a pyrex glass below the silicon wafer. The second glass wafer to be bonded above the silicon wafer is first drilled to have a hole, which has to be placed over one of the holes on the silicon wafer. This hole is further attached to a vacuum line for proper dispensing of alkali metal. After filling of the cell with buffer gas and alkali metal, the connectivity is cut off using a gas torch (The chip-scale atomic clock-recent development progress 2022).

Though pure cesium vapor is obtained from this method, the incompatibility of this method with MEMS technology has been a drawback in using this method (Knapkiewicz 2018a).

Chemical Reactions

Although the above discussed methods are useful, they have their limitations. The glass-blowing technique provided excellent results, but the size of the cell was not favorable for MEMS technology and pipetting method comes with negative impact of low-temperature anodic bonding (Liew et al. 2004). Hence, new methods for fabrication of atomic vapor cell were discovered by researchers and are discussed below:

On-Chip Chemical Reaction

In chemical reaction approach, silicon-glass preform is fabricated using anodic bonding method with holes etched on the surface of silicon wafer. The chemical reaction takes place inside the cavity in UHV chamber, which is as follows

$$B_aN_6 + C_sCl \rightarrow B_aCl + 3N_2 + C_s \quad (15)$$

Heating the preform at 120 °C induces decomposition of barium azide into Pure Ba and nitrogen gas. Then the silicon glass preform is covered with another layer of glass which is anodically bonded at voltage 1 kV and at a temperature about 200 °C.

Further, the final reaction takes place where elemental Cs is obtained along with barium chloride (Liew et al. 2004; Knapkiewicz 2018a).

This method offers advantage as the reactants are stable at room temperature. But, the by-product barium chloride, which is white in a color, can add difficulty in transmission of light through the cell. Although, the leftover nitrogen can act as buffer gas in this method.

Off-Chip Chemical Reaction

The same process of chemical reaction as described in the previous method was revised by Eklund et al. (2008). The major difference in two process is that, in this method, the same chemical reaction (Eq. 15) takes place inside an ampoule outside the glass-silicon preform at a temperature about 180 °C. In this way, pure Cs/Rb can be obtained and this can be filled in the form of a few drops in the preform cavity. Lastly, buffer gases are filled and another layer of glass wafer is hermetically sealed over silicon glass preform.

This method is quite advantageous since the problem produced by the presence of barium chloride is resolved. Moreover, desired mixture of buffer gases can be used and pure alkali metal can be introduced in the cell preform.

UV-Induced Chemical Reaction

Physical deposition process can also be used to deposit a thin layer of Cs/Rb azide. Researchers (Woetzel et al. 2013, 2014; Liew et al. 2007) have presented different ways to pippete the solution of Cs/Rb azide onto the inner surface of the cell. In this method first, the glass silicon preform is prepared using anodic bonding with holes already etched onto the silicon wafer. Then a thin film of cesium azide is deposited inside the preform. The second layer of glass is anodically bonded onto the silicon wafer. When the cell is exposed to UV radiation, Cs azide decomposes into pure Cs and nitrogen.

However, in this method, the layer of Cs azide lacks uniformity, which can result into variation in the amount of alkali metal produced in different cells. Besides this, it is not a very cost-effective method as well.

Off-Chip and On-Chip Dispensing

Apart from anodic bonding, other different types of bonding such as eutectic, thermo-compression as well as glass frit bonding can also be used to bond silicon with glass.

Off-Chip Dispensing Using Eutectic Bonding

The silicon wafer and glass are bonded through eutectic bonding, and then the preform along with another glass wafer, kept at a safe distance from it, is placed inside the vacuum chamber. Rb in liquid form is evaporated, so that it could condense in the cell preform resulting into deposition of droplets of Rb. Finally, the glass silicon preform is covered with glass wafer through thermo-compression bonding via a layer of indium (Petremand et al. 2010; Mileti et al. 2010; Vecchio et al. 2010).

However, eutectic bonding is not an ideal method to use because of its inability to attain proper vacuum levels and sustaining optical properties of the cell for longer periods (Knapkiewicz 2018a).

On-Chip Dispensing Using Anodic Bonding

One of the most favorable techniques for fabrication of MEMS cell is dispensing of alkali metal, when laser is irradiated inside the sealed cells. There are three major requirements for fabrication using this method, i.e., an optical chamber, a dispenser containing alkali metal (Saes Getters), and a connection channel. A cesium disperser is placed inside already bonded silicon glass preform. Irradiation of laser (near NIR regime) is used to dispense Cs out of the dispenser. With the rise in temperature to near about 720 °C, chemical reaction takes place between Cs and other reducing agents present in the dispenser. The quantity of the alkali metal produced is dependent on the laser power as well as on the time duration of radiation (Douahi et al. 2007; Knapkiewicz et al. 2010).

This method proved to be one of the best methods for fabrication of vapor cells because of the availability of micro dispensers (Knapkiewicz 2018b). The only requirement of this process is a laser in IR regime, irrespective of its wavelength. The amount produced can be directly controlled from power and time duration of irradiation. Although the size of dispenser can be one of the drawbacks, a recent research showed that this technique can be used for mass-scale production (Knapkiewicz 2018a).

Conclusion

Rydberg atom-based spectroscopic technique for the measurements of microwave parameters, driven by Electromagnetic Induced Transparency (EIT) and related phenomena, is a promising technology for microwave measurement and other applications. One of its major advantages lies in its ability to provide direct traceability to all existing microwave measurements against universal Planck constant. This technique also possesses other advantages like broadband nature, better sensitivity, no constrains related to size of device, and whether to be used as sensor, receiver, or other purpose. The tunability of the Rydberg states, based on probe and pump lasers and RF fields applied, allows for better selectivity and higher sensitivity simultaneously.

This chapter has described the role and impact of microwave in ongoing quantum physics study in term of hyperfine structure tunability which can lead to various new phenomena as well as provide platform to fine tune various experiments. The role of Rydberg atoms of alkali metals and various microwave devices such as sensors, mixers, receivers, and modulators are described in detail. Many more interesting phenomena can occur, if Rydberg series of ions, molecules, or excitons in semiconductors are used. Research involved with Rydberg atoms can have many more real applications, if cold atomic vapors are used instead of thermal or room-temperature vapors. In the recent years, cold atomic vapors have been used endlessly to produce unidentified types of novel matter, such as Rydberg molecules, cold neutral plasmas, and collective Rydberg matter. These cold Rydberg atoms have also found application in the creation of large-scale defect-free atomic arrays, the realization of high-fidelity quantum gates, the simulation of quantum spin models, and the proof of single-photon-level optical nonlinearity.

Research has also to find newer, faster ways to fabricate vapor cells. Now, even on-chip vapor cells are being fabricated to achieve portable Rydberg atom-based system. The emergence of MEMS-compatible vapor cell fabrication will lead to fast commercialization of such systems in many applications. These achievements also pave the way for continued success in Rydberg atom-based QIP and open up exciting possibilities for developing scalable quantum computation and simulation in the coming decades.

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Electromagnetic Metrology for Microwave Absorbing Materials

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Naina Narang, Anshika Verma, Jaydeep Singh, and Dharmendra Singh

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Abstract

In recent times, microwave absorbing materials have been playing crucial roles in various applications including communication, stealth, and shielding. The working of these materials largely depends on the careful characterizations that include measurement for morphological, electrical, and physical properties followed by microwave characterization. A significant amount of work has been carried out in exploring the physical, electrical, and magnetic properties of various organic and inorganic materials, polymers, nanocomposites, meta-surfaces, and meta-materials suitable for employing them as microwave absorbing material. Both intrinsic and extrinsic properties of materials play an important role in the

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development of microwave absorbing materials. The design of the functional materials largely depends on the optimization of the intrinsic properties of the raw material such that required extrinsic properties can be achieved. Depending on the application, a wide range of such parameters need to be characterized to test the performance of the material in a particular frequency range. An excellent microwave absorbing material is required to possess main properties like high reflection loss, wide bandwidth, low weight, lower coating thickness, chemically inactive, and cost-effectiveness. To measure these properties, the microwave characterizations can be difficult to implement in various scenarios such as when (1) operating in wideband frequency range; (2) size, shape, and thickness of the material are not uniform; (3) the sample of material is in difficult state for microwave characterization, i.e., semisolid or liquid; (4) there is an unavailability of the specific complex instrument or software; and (5) using novel materials with poor repeatability. All these factors contribute to the uncertainty in the measurement of performance parameters of microwave absorbing materials. In this chapter, the relationship of fundamental aspects of electromagnetic (EM) materials with the performance of the microwave absorption is presented. Indicative literature is presented showing the major sources of uncertainty and their contribution in the measurement. Various applications are discussed where characterization of this relationship becomes a key factor.

Keywords

Dielectrics · Microwave · Metrology · Measurement · Reflection · Shielding effectiveness · Transmission · Uncertainty

Introduction

Microwave materials (MMs) have received considerable interest over the years for their applications in stealth, communications, electronics, automotive, and avionics technologies. In defense applications, one of the main objectives is to detect threats as early as possible and be undetectable to increase survivability in the area monitored by advanced radar and microwave technology. In communication, these materials are used for various requirements such as miniaturization, EMI shielding, temperature stability, thermal conductivity, etc. Worldwide, multidimensional research is going on for the development of efficient absorbers using these microwave materials, generally referred to as microwave absorbing materials (MAMs) or radar absorbing materials (RAMs) or simply microwave absorbers (MAs). After initial widespread use of Salisbury screens and Jaumann absorbers, a significant amount of work has been carried out in studying the microwave absorption characteristics of organic and inorganic materials, polymers, nanocomposites, meta-surfaces, and metamaterials. The microwave absorption behavior of grading honeycomb composites (Zhou et al. 2015), frequency-selective surface (FSS) impregnated absorbers (Lee et al. 2015), graphene structures (Xu et al. 2015;

Olszewska-Placha et al. 2015; Zhang et al. 2015), different shapes of materials (Lv et al. 2015; Liu et al. 2015), and doping elements (Hong et al. 2016) has been studied.

Due to the unique properties of graphene and oxide, a large literature is available for the development of MAMs synthesized using graphene composites (Wu et al. 2015; Wang et al. 2015; Balci et al. 2015). As an excellent representative of a carbon material, reduced graphene oxides (R-GO), the thinnest and most lightweight 2D materials of the carbon family, have also been reported to exhibit excellent dielectric properties (Feng et al. 2016; Yi et al. 2017). Recently, molybdenum disulfide (MoS_2) has also been actively investigated for their microwave absorption properties due to its unique electrical, optical, and mechanical properties (Zhang et al. 2016; Xie et al. 2016). The composites of R-GO and MoS_2 have also shown promising absorption performance; however, wideband performance has not been achieved, and repeatability is a standing issue (Ding et al. 2016). For achieving wide bandwidth, the multilayering technique using metamaterials and FSS embedding is widely being adopted and has shown considerable improvement in the absorptivity of the materials (Chen et al. 2018; Yigit and Duysak 2019). A large literature on magnetic microwave absorbers is also available (Yao et al. 2016; Yuan et al. 2016; Wang et al. 2016), but the need for frequency-selective and low-density MAMs remains a challenge. A recent study of clustered $NiFe_2O_4$-rGO nanocomposite has been carried out showing highly tunable electromagnetic properties for selective-frequency microwave absorption (Zhang et al. 2018). These studies are crucial since control of dielectric and magnetic losses can be understood which directly helps in determining the microwave absorption mechanisms in MAMs.

Most recently, high-performance, tunable MAMs required for various applications are being actively researched upon using multidimensional approaches and materials discussed above (Hu et al. 2019; Tang et al. 2019). Tunable MAMs using tailored dielectric properties are also being explored and have shown significant improvement in the absorption properties (Quan et al. 2017; Cheng et al. 2016; Pan et al. 2017). A significant amount of effort has also been made in the field of development of lightweight, textile-based MAMs which have crucial applications in camouflaging and stealth applications (Song et al. 2016, 2017). Depending on the application, such as the absorbers required in EMI and EMC applications, characterization techniques differ, and thus a separate set of properties is studied (Khalid et al. 2017). For example, infrared emissivity is the parameter to quantify the heat transfer. MAMs, having a high emissivity in the microwave spectrum, are crucial in a wide array of applications, such as electromagnetic interference mitigation, stealth technology, and microwave remote sensing and radiometer calibration (Houtz and Gu 2017). Apart from absorption characteristics, the mechanical requirements, e.g., ultrathin, flexibility, lightweight, thermal stability, etc., are to be catered for application-based products and are widely stressed upon by the researchers (Song et al. 2016; Shanenkov et al. 2017; Jia et al. 2018; Machado et al. 2019; Aguirre et al. 2018).

In this chapter, the relationship of fundamental aspects of electromagnetic (EM) materials with the performance of the microwave absorption is presented.

The focus of the chapter is to demonstrate the metrological aspect in the characterization of these materials required for analyzing the absorption performance. The major sources of uncertainty in the measurement of the MAMs are discussed in detail. The chapter also covers a large number of new methods and applications being prevalent in the modern-day systems. A large set of references related to the topic is presented in the chapter for readers to get more details. In this chapter, the considerations to be made for the accurate measurement of microwave parameters, such as reflection, transmission, insertion loss, shielding effectiveness, electrical properties, and radar cross section, required for gauging the performance of microwave materials are discussed in detail. Depending upon the application in hand, the required measurements can be set up with the minimum uncertainty by considering and evaluating the different sources of error. For example, the absorption characterization is important for realizing microwave absorbing materials, and it depends upon the dielectric, magnetic, reflection, transmission properties of the microwave material. Hence, each parameter has to be carefully measured to estimate the absorption property of the said material.

Electrical Properties of Materials

How the materials interact, intercept, and absorb the incident wave, schematically shown in Fig. 1, has to be physically and analytically defined to help the researchers to avoid the trial-and-error method in the development of MMs and their use for any application. For example, various microwave techniques can be implemented to characterize the absorption at macroscopic level. These microwave absorbing materials are widely used in various applications such as electronics, stealth, and communication technologies. While analyzing the microwave absorbing absorption, the impedance matching is critical for maximum absorption, which can be given as

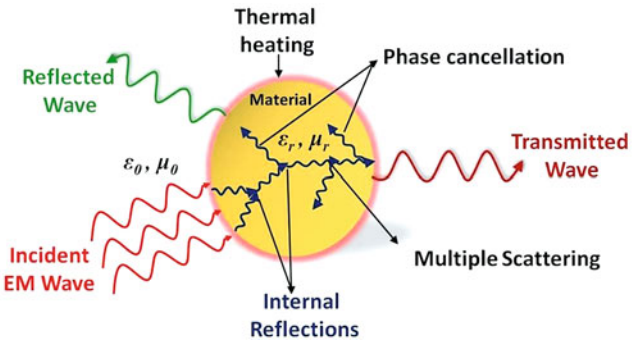

Fig. 1 Simple schematic representation of main mechanisms of EM wave and material interaction

$$\frac{Z_{in}}{Z_0} = \sqrt{\frac{\mu_r}{\varepsilon_r}} \times \tanh\left(j\frac{2\pi ft}{c}\sqrt{\mu_r\varepsilon_r}\right) \tag{1}$$

where μ_r is the relative permeability, ε_r relative permittivity, f is the frequency in Hz, c is velocity of light in m/s, and t is thickness of sample in m. The thickness of the material t is limited by the quarter wavelength cancellation law. For an ideal absorption, $Z_{in}/Z_0 = 1$, showing a perfect match between the free space and the sample. Using the characteristic impedance of the material, the reflection loss (RL, in dB) can be calculated as

$$RL(dB) = 20\,log\left|\frac{Z_{in} - Z_0}{Z_{in} + Z_0}\right|. \tag{2}$$

For 90% and 99% absorption, the return loss should be less −10 dB and −20 dB, respectively.

Depending upon how the material acts with the incident EM wave, the materials can be broadly classified as dielectric, semiconductor, or conductors. There may be more categories which again will largely depend upon the interaction with the incident wave. The major electromagnetic materials and their properties are given in Fig. 2. From the metrological point of view, the selection of characterization method, measurement setup, and corresponding uncertainty evaluation largely depends on the material under test (MUT). Therefore, once the class of the material is identified, further selections can be made accordingly. This point is further discussed in the subsequent sections, wherein the specific characterization will be employed for the specific MUT.

Moreover, the classification of the materials for microwave applications can also be based on the macroscopic properties exhibited by the material. Some examples are given in Fig. 3, which are grouped based on wave interaction, properties in different orientations, and number of constituents.

The classification of materials in Fig. 2 shows the broad categories of materials and their properties in general. The properties of any material can depend on the shape, size, geometry, propagation parameters, electrical transport properties, etc. These properties for electromagnetic materials are classified as intrinsic

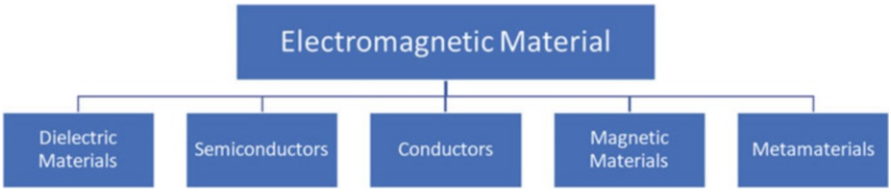

Fig. 2 General categories of electromagnetic materials based on their properties

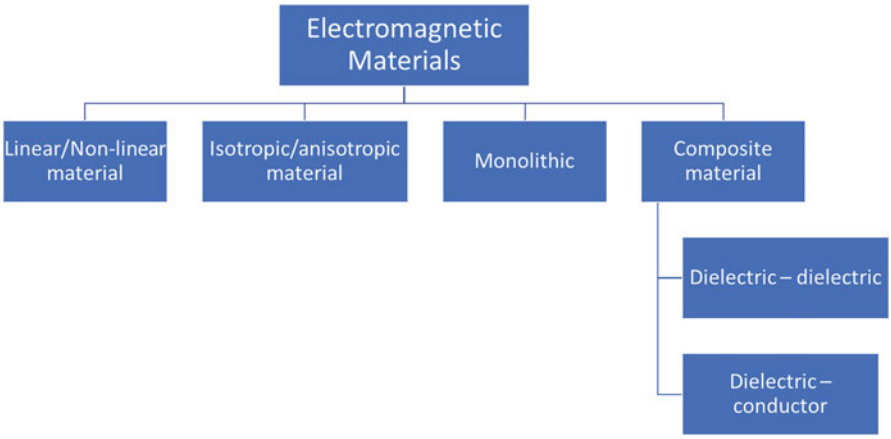

Fig. 3 General categories of electromagnetic materials based on their macroscopic properties

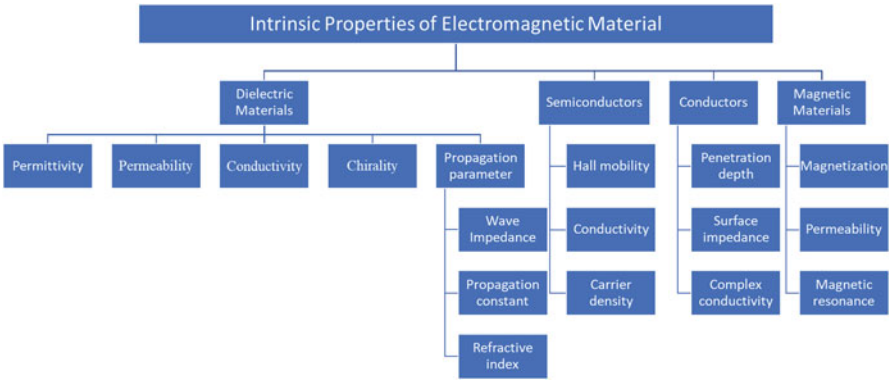

Fig. 4 Important intrinsic properties of different electromagnetic materials

(independent of size and geometry) and extrinsic (dependent on size and geometry) properties. Few of the important intrinsic properties of electromagnetic materials are given in Fig. 4. These properties play a crucial role in deciding the use of the material for a particular application and development of functional materials of desired extrinsic properties. For instance, the dielectric materials are most widely used as microwave absorbers due to their bulk parameters, and their structure and thickness can play a monumental role in achieving the desired extrinsic properties such as characteristic impedance, reflectivity, operating frequency, etc. In the next section, the use of the intrinsic properties to achieve the desired extrinsic properties is elucidated for microwave absorbing materials.

Microwave Absorption Principles

The essential principle of microwave absorption is to convert the microwave energy into thermal energy that dissipates into the environment through various absorption mechanisms. The absorption mechanisms of these materials are dependent on the intrinsic properties of the material, including:

- Heterogeneity of the composite and its effect on the wave propagation
- Electrical and magnetic properties (i.e., complex permittivity, permeability, conductivity, etc.)
- Morphological parameters such as shape, size, density, and cohesion of the composite materials
- Layered system of composites
- Thickness ratio in the multilayered system

The thickness of the material t is an important parameter while fabricating the MAM. It is limited by the quarter wavelength cancellation law and thereby can be taken as

$$t = \lambda_0/4 \times 1/\sqrt{\mu_r \varepsilon_r}. \tag{3}$$

The thickness parameter is also studied, and the lower bound for the thickness to bandwidth ratio is given (Rozanov 2000). The Rozanov limit given in the paper specifies that for the nonmagnetic MAM; the thickness to bandwidth ratio is limited to 1/17.2 at the 10-dB absorption level. Similar inequalities are given for narrow-bandwidth and magnetic materials. The Rozanov limit is computed from the inequality given as

$$|\ln \rho_0|(\lambda_{\max} - \lambda_{\min}) < 2\pi^2 \sum_i \mu_{s,i} d_i \tag{4}$$

where the operating band specified as $\lambda_{\min} \ldots \lambda_{\max}$ is the lowest possible reflectance, d_i is the thickness of *ith* layer (in case of multilayer system), and $\mu_{s,i}$ is the static permeability of *ith* layer.

Also, the absorption mechanisms in the materials are dependent on the intrinsic properties of the material such as dielectric and magnetic properties (complex permittivity and permeability) as well as the morphological parameters such as shape, size, density, and cohesion of the composite materials. The permittivity originated from electronic, ion, and intrinsic electronic and interfacial polarization can be described in terms of relaxation formula:

$$\varepsilon = \varepsilon_\infty + \frac{\varepsilon_0 - \varepsilon_\infty}{1 + (\omega\tau)^2} \tag{5}$$

where ε_0 and ε_∞ represent permittivity for angular frequency $\omega \to 0$ and $\omega \to \infty$, respectively, and τ is the relaxation time for polarization of composite. The microwave absorption can be enhanced by fabricating the effective absorbing material of favorable bulk parameters as well as by using advanced EM techniques such as multilayering, FSS imprinting, meta-surfaces, and metamaterials. However, in this chapter, the absorption enhancement techniques are not discussed, since the scope of the chapter is contained only to the metrological aspects.

Material Characterizations

The microwave material as an absorber is usually characterized using the recommendations given in IEEE Std. 1128 published in 1998 (IEEE 1998). However, the standard has limitation of the frequency range up to 5 GHz only which may not suffice for the current-day needs. Therefore, the revision to the standard is expected to address the frequency extension and may be including the modern test setups for the characterization of the absorbers. The IEEE Std 1128 focuses on the measurement methods for different levels, starting from the evaluation of (1) material bulk parameters, (2) reflectivity, and (3) practical use of the absorbers in the anechoic chambers/open area test site.

Characterization of Material Bulk Parameters

The most often used characterization techniques for measuring the bulk parameters of the materials include nonresonant and resonant methods. As the names suggest, the nonresonant methods are a characterization over a frequency range, and on the other hand, the resonant methods are specific to a particular frequency (or frequencies). The major methods under both the techniques are given in Fig. 5.

Considering the nonresonant methods, the measurement methods for the bulk parameters can be broadly categorized into frequency and time domain procedure, as shown in Fig. 6. The detailed explanation on the frequency domain method can be found in the literature and is currently the most widely adopted method. However, as discussed previously, the time domain method can be thought of as a suitable candidate to extend the frequency of operation up to millimeter wavelength. However, so far, the applicability of the free space characterization method is found to be suitable only for medium and high-loss materials (Liu et al. 2021), while free space techniques for electrical property measurements are preferred over cavity and waveguide methods for the following reasons:

- Materials such as ceramics, composites, etc. are inhomogeneous due to variations in manufacturing processes. Because of inhomogeneity, the unwanted higher-

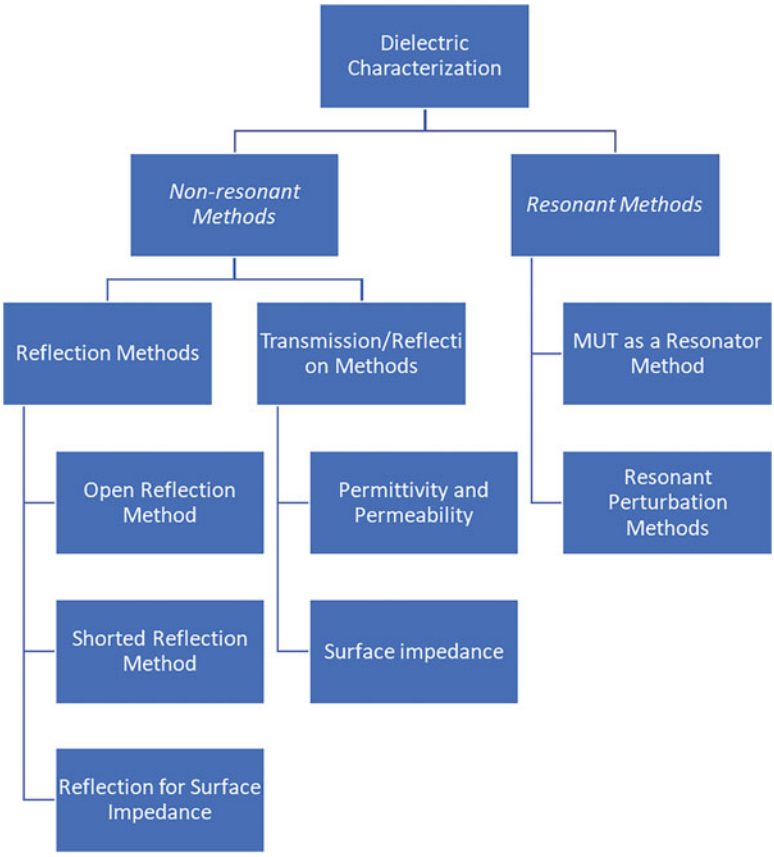

Fig. 5 Broad categories and subclassification of the prevalent dielectric characterization

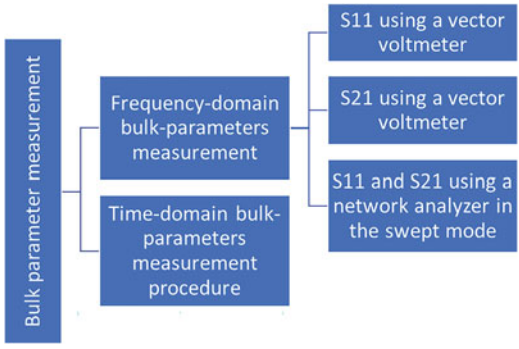

Fig. 6 The bulk parameter measurement methods

order modes can be excited at an air dielectric interface in waveguides and cavities.
- Dielectric measurements using free space techniques are nondestructive and contactless, particularly suitable for dielectric measurements at high temperature.
- In the cavity and waveguide methods, it is necessary to machine the sample so as to fit the waveguide cross section with negligible air gaps. This requirement will limit the accuracy of measurements for materials which cannot be machined precisely.

The inaccuracies in dielectric measurements using free space methods are mainly due to:

- Diffraction effects at the edges of the sample (minimized using lens antenna)
- Multiple reflections between the two horns via the surface of the sample (can be minimized using time-domain gating)

The free space method for measurement of dielectric constant ε and loss tangent (tan δ) of dielectric materials and the complex reflection coefficient $\mathbf{S_{11}}$ were measured by inserting a perfectly conducting plate behind the plate of unknown material. The metal-backed sample is placed at the focus of the transmit lens antenna. From transmission line theory, $\mathbf{S_{11}}$ for normal incidence plane wave is related to the complex relative permittivity ($\varepsilon^* = \varepsilon'(1 - j \tan \delta)$) by the following relationship:

$$S_{11} = \frac{jZ_{dn} tan\,(\beta_d d) - 1}{jZ_{dn} tan\,(\beta_d d) + 1} \tag{6}$$

where Z_{dn} is the normalized wave impedance in the unknown material.

For nonmagnetic materials,

$$Z_{dn} = \frac{1}{\sqrt{\varepsilon^*}}, \quad \beta_d = \frac{2\pi}{\lambda}\sqrt{\varepsilon^*} \tag{7}$$

where λ is the free space wavelength and d is the thickness of the sample permittivity which can be calculated with the help of iteration, and finding the zeros of error function,

$$E = |S_{11}^m - S_{11}^c| \tag{8}$$

where $S_{11}{}^m$ and $S_{11}{}^c$ are measured and calculated values of complex reflection coefficients, respectively.

Once the time domain transient response is recorded, then the bulk parameters can be extracted using the famous Nicholson-Ross and Weir algorithm, details of which can be found in various references (Nicolson and Ross 1970; Weir 1974; Smith et al. 2002).

Table 1 Few representative studies of free space method for dielectric characterization

Type of material	Application	Frequency band	Measurement technique	Calibration	Reference
Teflon, quartz, and boron nitride	High- and low-temperature dielectric characterization of materials	8–110 GHz	Free space measurement using spot-focused antenna	TRL	Hollinger et al. (2000)
Low-loss dielectric	To overcome the need of a priori information of the sample thickness in dielectric characterization	220–325 GHz	Bidirectional scattering (transmission and reflection) system using standard horns and lens	TRL with time domain gating	Kim et al. (2016)
Nanocarbon composite materials	Implementation of line-reflect (LR) error correction technique	30–50 GHz	Free space measurement using conical horn antenna	LR with time-gating	Hassan et al. (2016)
Highly conductive radar absorbing material	Iterative optimization model to extract the complex permittivity	18–27 GHz and 75–110	Free space measurement using conical lens horn antenna and frequency extenders	TRL	Vohra and El-Shenawee (2020)

It is evident from the review given in Table 1 that the through-reflect-line (TRL) calibration technique is important to establish the measurement setup. The complex electric permittivity and the magnetic permeability are calculated from the measured values of S_{11} and S_{21}. Diffraction effects at the edges of the sample are minimized by using spot-focusing lens antennas, and the errors due to multiple reflections between antennas via the surface of the sample are corrected by using a free space TRL calibration technique. The standards used in this calibration technique can be explained as follows:

- The *through* standard is realized by keeping the distance between the two antennas equal to twice the focal distance. For this standard, the common focal plane is placed at the front face sample location (reference plane).
- The *reflect* standards for port 1 (transmit horn) and port 2 (receive horn) are obtained by mounting a metal plate on the sample holder at the reference plane.
- The *line* standard is achieved by separating the focal planes of the two antennas. The distance between focal planes is approximately a quarter wavelength at mid-band.

Measurement Uncertainty

The major sources of uncertainties identified by the researchers are summarized from various references in Table 2. It is evident that the distortion in the cross section of the MUT, length of the MUT, and smoothness of the surface are some dimensional sources of uncertainties. Also, since the measurement is largely based on the S-parameters, therefore, the uncertainties are generated by the instrumentation, i.e., the VNA is another key source. The MUT orientation and placement are another factor contributing to the measurement uncertainty. The experimentalists need to overcome these sources of uncertainties. For example, for thin, flexible samples, the sample had to be sandwiched between two half-wavelength (at mid-band) quartz plates to eliminate the effect of sagging. Similar techniques need to be incorporated while executing these measurements.

Few other possible sources of uncertainty can be

- Drift: The drift in the physical system can contribute to the measurement; this can be included by taking the measurement over a period of time and evaluating the drift in the measurement results.
- Frequency: the measurands for calibration MAM are implicitly dependent on the frequency; this should be carefully incorporated when time-gating and other such software-based operations are applied during the measurement.
- Near field: Since the measurements are based on plane wave approximation, the effect of the near field should be incorporated in the uncertainty evaluation.

An example of uncertainty budget for a free space measurement of dielectric properties in the frequency range of 55–65 GHz is shown in Table 3.

Table 2 Major sources of uncertainties in the different characterization methods

Measurement method	Parameter	Sources of errors identified	±Error %	Reference
Loaded rectangular waveguide	Dielectric properties	Distortion in cross section Thickness of the material Sample preparation	>10%	Foudazi and Donnell (2016)
Free space method	Permittivity and conductivity	Instrumentation, impedance mismatch Distribution of sample thickness Interface alignment mismatch due to sagging Nonlinearity Environmental conditions	2%	Hassan et al. (2016)
Free space method	Permittivity and conductivity	Thickness Mismatch losses	5%	Singh et al. (2021)
Loaded rectangular waveguide	Permittivity	Cable stability Connector repeatability Drift, linearity Noise Dimensional uncertainty	–	Shoaib et al. (2016)
Coaxial line	Permittivity	S-parameter magnitude and phase Sample length	3%	Boughriet et al. (1997)

Table 3 Uncertainty budget of free space measurement (Singh et al. 2021)

	Elements of uncertainty				
Type	Sources	Uncertainty value (U_i)	Probability distribution and divisor	Standard uncertainty	Degree of Freedom
A	Independent repeated measurements	0.0548	Normal 1s $\sqrt{5}$	0.0245	4
B	Thickness of the specimen	0.0010	U-shaped $\sqrt{2}$	0.0007	∞
	Mismatch losses	0.0031	U-shaped $\sqrt{2}$	0.0022	∞
	Combined standard uncertainty, U_c	$U_c = \sqrt{\sum_{i=1}^{3} U_i^2}$		0.0246	
	Coverage factor, k	1.96			
	Expanded uncertainty, U_e	$U_e = k \times U_c$		0.0428	

Applications of Microwave Materials as Absorbers

Ultra-wideband (2–18 GHz) microwave absorbing materials in particular are extensively used for RADAR and applications such as camouflage nets and radar cross-section (RCS) reduction of planes for which stealth is of utmost importance. The MAMs are also required for civil applications such as reduction of electromagnetic (EM) pollution from mobile communication devices, electromagnetic interference shielding, radomes, etc. In the age of high electromagnetic (EM) pollution, the conventional MAMs, such as Salisbury screen; Jaumann absorber; coating of magnetic materials such as Fe_3O_4, $NiFe_2O_4$, $BaFe_{12}O_{19}$, $ZnFe_2O_4$, and $CoFe_2O_4$; and coating of dielectric materials such as ZnO and SiC, have limited practicability due to the narrow bandwidth, low dielectric or magnetic losses, thermal stability, high density, poor flexibility, and coating thickness. Significant efforts are being made by different research groups to overcome these limitations to serve various kinds of applications of MAMs. Till date, the majority of microwave absorbers prepared by the researchers for broadband application are thick sheets (>3 mm) and are made up of ferrite materials. The thickness of the layers is to be minimized, keeping the performance intact. Another technique of multilayering for enhanced microwave absorption is widely used, but this also suffers with thick coating and has low strength. With the incessant growth of the EM pollution and advanced engineering requirement in military stealth applications, the need of high-performing MAMs is indispensable. Considering the different applications of MAMs in the civil and defense sector, there is a vital need for cost-effective synthesis and fabrication of effective microwave absorbers using low-cost raw materials and less complex fabrication techniques. Thereby, in this section, few notable studies are discussed for various applications of MAMs.

EMI/EMC

The ever-increasing use of electronic equipment in our day-to-day life along with significant scientific advancements in the electronic area has led to increased EMI that is harmful to both the user and other electronic equipment. The basic EMI/EMC tests carried out for the aeronautics and space applications include the radiation and susceptibility tests. These tests are responsible to ensure that the radiation/conduction to and from any electronic unit will interfere neither with its operation nor with the other units. These tests can be further categorized as (Mallette and Adams 2011; Mathur and Raman 2020):

- Radiated emission test
- Conducted emission test
- Radiated susceptibility test
- Conducted susceptibility test
- Immunity test

Recently, a lot of promising approaches have been reported by several groups in EMI shielding with respect to different types of materials ranging from metallic, polymeric, nanomaterial-based, nanocomposite-based, porous systems, alloys, high-temperature shields, etc. Depending on the application, such as the absorbers required in EMI and EMC applications, characterization techniques differ, and thus a separate set of properties is studied. For example, infrared emissivity is the parameter to quantify the heat transfer. MAMs, having a high emissivity in the microwave spectrum, are crucial in a wide array of applications, such as electromagnetic interference mitigation, stealth technology, and microwave remote sensing and radiometer calibration. Apart from absorption characteristics, the mechanical requirements, e.g., ultrathin, flexibility, lightweight, thermal stability, etc., are to be catered for application-based products and are widely stressed upon by the researchers. Few applications where EMI/EMC studies are crucial and the use of microwave materials has been demonstrated are:

- Densely packed integrated circuits in telecommunication systems
- Aeronautical and space applications
- Printed circuit boards

EM Shielding

An EM disturbance or noise that degrades the performance of the equipment or even causes the failure of equipment is called electromagnetic interference (EMI). The problem of EMI can be eliminated by using EMI shielding materials. The shielding effectiveness (SE) is a number that measures the amount of dissipated energy or attenuated energy by a shielding material. SE is the ratio of the incident energy to the remaining energy. SE is expressed as:

$$SE(dB) = 20\,log\ \ \log\left(\frac{E_t}{E_i}\right) = 20\,log\ \ \log\left(\frac{H_t}{H_i}\right) = 10\,log\ \ \log\left(\frac{P_t}{P_i}\right) \tag{9}$$

where E, H, and P are the electric, magnetic field, and power intensity, respectively, and "i" and "t" represent the incident wave and transmitted wave, respectively.

Three mechanisms contribute to the SE of the material for any EMI. Some part of the incident wave is reflected by the shielding material, a part of it is absorbed, and the left part undergoes multiple internal reflections. Therefore, the total shielding effectiveness (SE_T), given in Eq. (10), of shielding material is equal to the sum of shielding effectiveness due to absorption (SE_A), shielding effectiveness due to reflection (SE_R), and shielding effectiveness due to multiple reflections (SE_M) in thin shielding material:

$$SE_T = SE_A + SE_R + SE_M \tag{10}$$

The study of the shielding effectiveness again depends on the MUT. A large database of materials is available in the literature where shielding effectiveness is used as the only parameter for a particular application (Mehdipour et al. 2011; Gupta and Tai 2019; Wu et al. 2021a, b). For example, the use of composites showing high shielding effectiveness in the high frequency is critical for aeronautic applications. Researchers have shown the material preparation and high shielding effectiveness of up to 90 dB using the highly conductive composites wherein the shielding effectiveness can be given as (Mehdipour et al. 2011)

$$SE(dB) = 20\,log\ \ \log\left|\frac{(\eta_0+\eta_m)^2}{4\eta_0\eta_m}\right| + 8.686\left(\frac{d}{\delta}\right) \tag{11}$$

where skin depth (δ) << thickness of the material (d) and η_0 and η_m are the intrinsic impedances for the TE10 mode for air and material, respectively. It is evident from Eq. (11) that the SE is dependent on the thickness of the material since there is a direct proportionality between SE and d.

Radar Absorbing Materials for RCS Reduction

Many researchers have suggested microwave absorbing materials as radar absorbing materials (RAMs) for RCS reduction in various configurations such as for stealth aircrafts (Jayalakshmi et al. 2019) and antenna (Genovesi et al. 2013; Wang et al. 2014) and most recently in automotive radar systems (Chipengo 2018). RCS quantifies the reflective strength of object terms of the normalized power flux density of the field scattered by the object in the direction of the receiving radar. The various types of RCS are pictorially shown in Fig. 7. The direction of the reflected and scattered field is used to categorize the RCS as monostatic (or backscatter), forward scatter, or bistatic. Mathematically, RCS σ is given as (IEEE Std 1502–2020):

$$\sigma = 4\pi \lim_{R\to\infty} R^2\left(\frac{|E_s|^2}{|E_i|^2}\right) \tag{12}$$

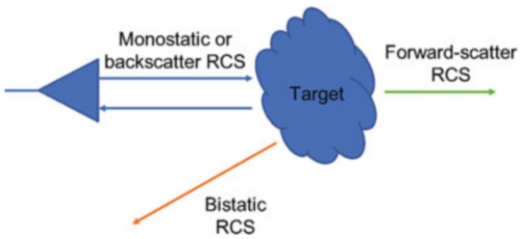

Fig. 7 Types of RCS based on the direction of reflected or scattered energy

where R is the range of the target, E_s is the electric field strength scattered by the target, and E_i is the incident electric field strength. The derivation of the equation and other details can be found in Richards (2014).

The basic technique employed for RCS reduction is coating a surface of any object with MAM of high-loss factor such that the incident wave can be reduced to heat or absorption, thereby minimizing the scattered fields and thus the RCS. There are MAMs that exhibit electric and magnetic losses and are used for RCS reduction. Few examples of such materials used in RCS reduction are summarized in Fig. 8.

Predominantly, RCS reduction was first desired in stealth applications where the invisibility of the aircraft or ships on the radar was envisaged as a countermeasure of radar detection. With the outset of new applications such as automotive radar systems, the use of RAMs is proposed in a few recent papers for cruise control and other autonomous driving requirements (Chipengo 2018; Kim and Kim 2019). The application of MAMs in automotive radar systems is an ambitious project which is still in its nascent stage. The most commonly researched MAMs for the automotive radar systems are given in Fig. 9 with their basic properties. The MAMs can play a crucial role in dictating the scene of the traffic for the efficient adaptive cruise control. For example, the RCS reduction of guardrails is an important task such that low RCS targets can be visible to the radar sensor on the autonomous cars (Chipengo 2018). However, employment of MAM on the guardrails may require coating of low cost, uniformity, and homogeneity of the material. Other studies have proposed the employment of MAMs for reducing the ghost targets produced due to the close proximity of the automotive radars (Kim and Kim 2019).

Conclusion

In this chapter, the microwave absorbing materials and their electrical properties are introduced. The most frequently used metrology techniques for the characterization of microwave absorbing materials used for various applications are presented. The recent progress in the field of MAM is driven by improvements in both materials,

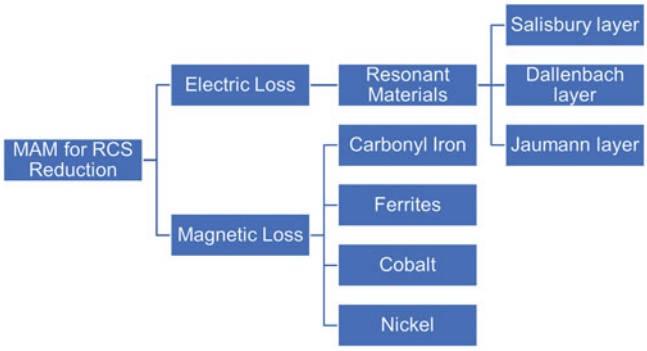

Fig. 8 Commonly used materials for RCS reduction application

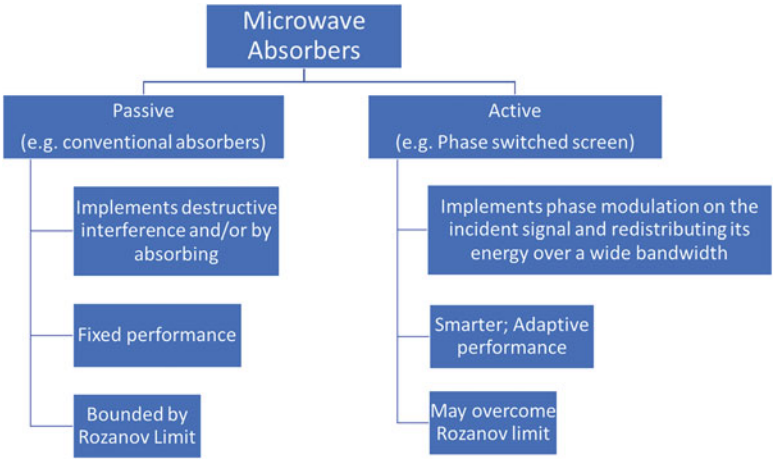

Fig. 9 Characteristics of the active and passive microwave absorbers used for RCS reduction

algorithms, and measurement techniques. Interestingly, methods and system designs vary substantially in terms of a number of antennas, measurement strategies, use of matching medium, algorithms, and so on. Therefore, it becomes essential to understand the applicability of a measurement system and its constraints and performance depending upon the material under test and its application. The major challenges and properties of the measurement setups are presented including the calibration techniques adopted by the researchers for a particular application. Majority of applications lie in the frequency range from 2 to 18 GHz. However, the important techniques of characterization of MAM bulk properties from 1 to 300 GHz is reviewed and presented from the metrological point of view. Major sources of uncertainty include the instrumentation, mismatch, sample dimensions, nonlinearity, and in some cases poor repeatability. In the end, the applications of MAMs are discussed and related with the characteristics of materials.

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Phased Array Antenna for Radar Application

59

Ashutosh Kedar

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Abstract

Phased array antenna (PAA) systems have emerged as the state-of-the art technology providing a solution to almost all of the modern-day communication requirements like 5G mobile system, upcoming 6G mobile system, Massive MIMOs, focal plane arrays, radars, SatCom, driverless vehicles, weather monitoring, air traffic control, etc. Phased array antennas are system of systems comprising of smaller subsystems being integrated to realize the whole system. Thus, the design and development of these requires a meticulous approach taking care of the needs of the end user, performance of the whole system, system life cycle, and environmental eventualities during the design cycle. PAA involves various subsystems such as antenna array, transmit receive modules, beamformer

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architecture, power supplies, cooling systems, etc. PAAs are complicated and costly systems due to involved active circuitry and fabrication expenditures; however, with the advent of the cheaper and more efficient semiconductor technology, the system cost is manageable. Henceforth, the use of these is extended for commercial applications as well. This class of antenna systems require specialized measurement techniques for assessment of their operational capabilities. These are called as array calibration techniques.

The chapter briefly discusses the phased array antennas, beamforming principles, system architecture for phased array antennas, and the various technologies involved and finally concludes by the measurement and calibration techniques specifically for radar applications. The chapter aims at providing the basic information to the readers about the phased array antennas with their applications focused toward radar applications.

Keywords

Phased array antennas · Radar · Transmit receive module · Beamformer · Antenna

Introduction

A Little Bit of History

The term Radio Detection and Ranging (Radar) was coined during World War II. In 1886, Heinrich Hertz demonstrated the reflection of radio waves for metallic objects. At the same time, similar investigations were reported by Sir J. C. Bose in India as well. Later in 1897, A. Popov developed an apparatus demonstrating a communication between the two ships in the Baltic Sea and noting an interference beat due to an interfering ship between the two ships. The German inventor, Christian Hulsmeyer, known as father of the radar, for the first time used radio waves for detecting the presence of a distant metallic object (see Fig. 1). He had shown that it is feasible to detect a ship in a dense fog and got a patent in 1904 and later got a patent on full radar system, called "telemobiloscope." In 1915, Robert Wattson-Watt also used radio direction for detection and worked further on shortwave transmission. *A. Hoyt Taylor* and *Leo C. Young*, US navy researchers, located a transmitter and receiver on the opposite sides of River Potomac and observed a fade in and out of the received signal due to interference by ships passing in between the beam. They suggested the phenomenon's usage for detecting the presence of ships in low visibility conditions. Later, *L.A. Hyland* at the *Naval Research Laboratory* (NRL) observed similar phenomenon, which led to a patent application leading to the further intensive research at NRL, where Taylor and Young were based at the time. The first radar systems developed in the USA were the SCR-268 (operating at 205 MHz) for antiaircraft gunfire control and the SCR-270 (operating at 100 MHz

Fig. 1 World war II radar produced by Germans. (Source: Internet)

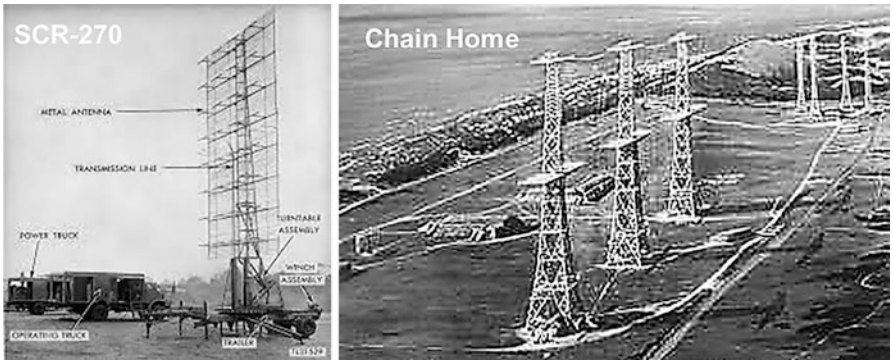


Fig. 2 First few radar systems. (Source: Internet)

frequency) for aircraft detection (see left-hand figure in Fig. 2). SCR-270 detected the approach of Japanese warplanes toward Pearl Harbor, near Honolulu, on December 7, 1941. In September 1938, the first British radar system, the *Chain Home* (see right hand figure in Fig. 2), was operated over a full day. It remained operational throughout the war, and it operated at about 30 MHz, i.e., the shortwave or HF and was invented by Sir Robert Watson-Watt. Thus, the title "father of radar" can be jointly shared by Hulsmeyer and Wattson. At the same time, similar research took place in Great Britain. The USA, Britain, Germany, France, the Soviet Union, Italy, the Netherlands, and Japan were the initial countries which started the work during WWII days on radar systems. A chronological detail of the various events is summarized in Fig. 3.

Year	Achievement
1886	Hertz tested maxwell's theories and demonstrated the similarities between light waves and electromagnetic waves and found that EM waves are reflected by mettallic and dielectric bodies
1904	Patent of obstacle detection and ship navigational aid by Hulsmeyer, a German Engineer
1922	Use of short waves for radio detection
1925	Principle of pulse ranging by US scientists for height measurement of ionosphere
1930	Pulsed radar systems into operation in US
1940	Modern radar development in the WAR years, development of high power magnetron in UK, MTI,MOPA etc.
1950	Long Range radars, Computer techniques
1960	Electronic scanning arrays concept
1970	Developed wide Bandwidth electronic scanning array Radars

Fig. 3 Chronological achievements

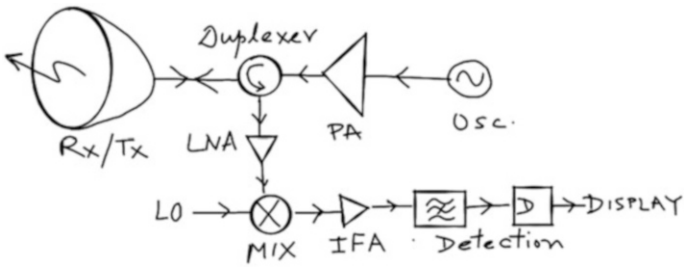


Fig. 4 Radar-functional block diagram

Radar Principle of Operation

A radar, in principle, radiates electromagnetic (em) energy in the space and receives the scattered and reflected echoes from the space, process it further to get the desired target information (Sarkar et al., 2006; Skolnik, 2008). Figure 4 shows the functional block diagram of the radar illustrating the various components comprising the overall system. The front end consists of the receive (Rx)/transmit (Tx) antenna which serves as "eyes and ears" of a radar, followed by a duplexer which isolates the Tx and Rx portions of the radar from each other. The Tx portion consists of a power amplifier (PA) which amplifies the signal from a signal generator "osc" to the necessary level required for transmission or radiation by antenna of a radar system

to achieve desired range. The Rx portion consists of a low noise amplifier (LNA), which removes the unwanted noise interference from the scattered/reflected signals received from space, followed by a mixer (MIX) which translates the radio frequency (rf) signal to intermediate frequency (IF) base band level, which is further amplified by an IF amplifier and detected through 2nd level detector and display on a video terminal. Thus, a radar may be considered as a "*system of systems"* requiring a team with vivid expertise ranging from electronics to electrical systems along with computer science experts to develop an efficient radar system.

Types of Radar

Radar can be classified broadly into three categories – *mechanically scanned radar (MSR)*, *passive electronically scanned radar (PESAR),* and *active electronically scanned radar (AESAR)* based on the *antenna type* used in the radar system (Kedar, 2020a) (see Fig. 5). The antenna type classifies the radar based on the beamforming and scanning strategy as adopted in it. MSR has a got a fixed antenna mounted on a pedestal fitted with a servo motor. The rotating mechanism rotates the antenna at fixed revolutions per minute (rpm) rate which in turn decides the scanning rate of the radar. *MSR* suffers from the problems such as slow reaction times, inertia in the servo motor mechanism, etc. *PESAR* overcomes this issue by providing the phase shifters at the element level leading to electronically controlling the phase shifts provided to the antenna elements, which in turn scan the antenna beam electronically. This increases the reaction time of the radar and eliminates the inertia factor. However, the power source or transmitter in MSA and PESAR is single source and may act as a source of complete failure of radar in case of any failure.

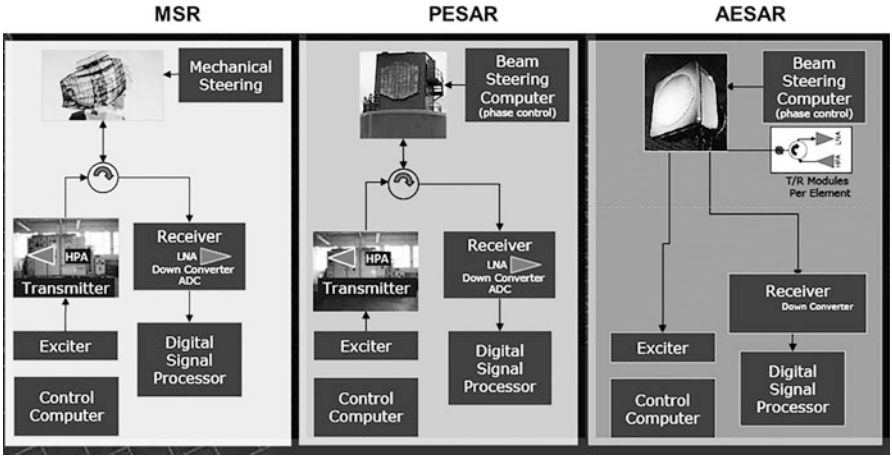


Fig. 5 Classification of radar systems

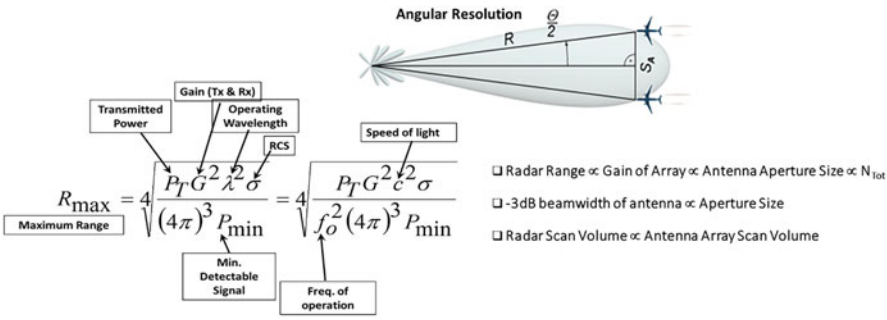


Fig. 6 Radar range equation and its characteristics

Henceforth, to take care of this issue the transmitter is distributed into small transmitters at the element level leading to the generation of *AESAR* class of antennas. Thus, the distributed transmitters provide a feature called as graceful degradation in AESAR, which means even in case of failure of these transmitters by 10–12% the AESAR will still operate with slightly degraded performance. Thus, antenna forms an important part of radar system controlling its overall performance and nomenclature of the radar system. Henceforth, *the antenna systems for radars form the theme of this chapter.*

A radar needs to illuminate the space for various surveillance and tracking applications. The important characterizing parameters of a *radar* are, namely, *radar range*, *angular resolution*, and clutter cancellation, which are totally dependent on the antenna system performance (Skolnik, 2008) (see Fig. 6). In addition, a radar needs to *cover or view* the complete hemisphere (360° of coverage) in a limited time which can be accomplished only by the movement of antenna beam throughout the hemisphere.

Figure 6 Illustrating the *radar range equation* and the radar parameters shows that radar range is mainly dependent on the *total power radiated* (P_T), *gain of antenna (G),* and *frequency of operation* ($\sim\lambda=c/f$). The *Gain, G,* of antenna depends upon the *number of elements*, N_{ToT} being collocated in an antenna array aperture in a definite geometric pattern decided by various design configurations, and the power radiated in turn depends upon the N_{ToT} *times the power fed to each of the antenna element*, P_{in}. The spatial coverage of radar depends upon the *grating lobe* (GL) free scan volume offered by the antenna array system (Skolnik, 2008; Kedar, 2020a; Balanis, 1997). Further, the radar spatial coverage requires *antenna beamforming*, which implies formation of customized antenna beam shapes like pencil beam, fan beam, multiple beams, cosecant squared beams, etc., as shown in Fig. 7 for surveillance and tracking applications (Skolnik, 2008; Kedar, 2020a; Brookner, 1991). Thus, the importance of antenna arrays is clear in design of radars and forms the theme of this chapter.

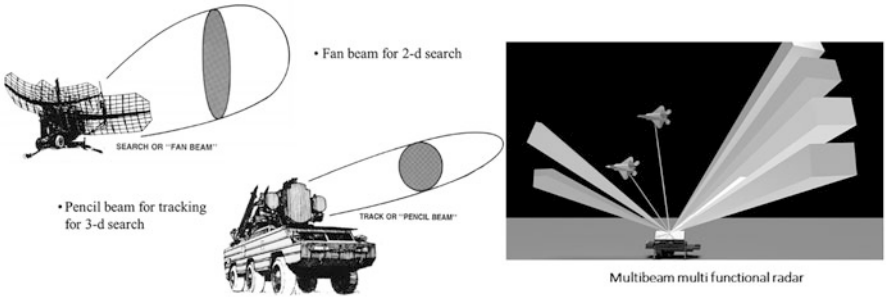


Fig. 7 Types of beamforming (source: Internet)

Antenna

Antenna in a radar does not contain a single element; however, it contains a group of elements, called as an *antenna array* as the *overall gain* of an antenna system is directly proportional to N_{ToT}, i.e., the number of elements in an ideal sense, which also decides the *range* of radar (Brookner, 1991; Kedar, 2022). Referring to Fig. 7, an antenna (actually an antenna array) covers the entire *area of interest* for radar either *mechanically* or *electronically* by the virtue of single or multiple beams (Jeffrey, 2008). The mechanical coverage is by the means of physical rotation of antenna system in a radar using servo-motor system, i.e., *MSR* or electronically by inputting *phase delay* at the element level through some *phase shifters* or *time delay units*, i.e., *PESAR* or *AESAR*. Based on this operational mechanism, the antenna is also characterized as *mechanically scanned antenna* (MSA) and *phased array* antenna (*PAA*) or electronically scanned antenna (*ESA*). The focus over here is on *PAAs* or *ESAs* which forms the integral part of the modern-day radar systems (Brookner, 1991; Jeffrey, 2008).

Phased Array Antenna

A *phased array antenna* (*PAA*) is a *group of radiating elements* in which *the relative phase shifts* of the respective signals fed to the individual antenna elements are varied in such a way that the *effective beam* (a planar wave front) formed by the array is reconstructed in a desired direction (at an angle, θ) and suppressed in undesirable directions (Kedar, 2022) as shown in Fig. 8. Figure 8 shows a phase lag $\alpha = (N-1)\phi$ being fed to n^{th} element in an N-element array. Thus, necessary phase delays are introduced in the antenna elements' input resulting in to the respective wave fronts emitted by individual radiator to interact constructively to form a primary planar wave front at an angle, θ. This phase delay can be provided either by using *phase shifter* or *true time delay* (TTD) lines (Sreenivasulu et al., 2020; Kedar et al., 2020a; Fourikis, 1997; Sturdivent et al., 2019).

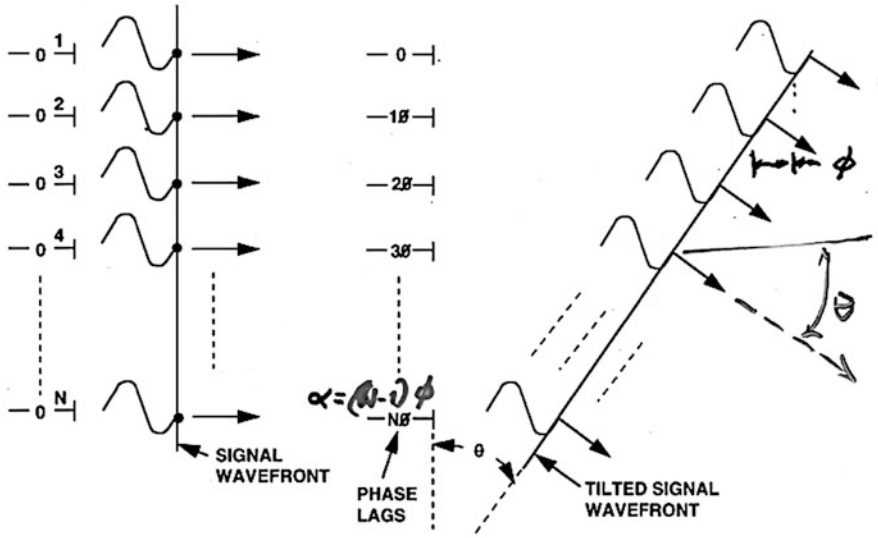


Fig. 8 Planar wave front formation in phased array antenna

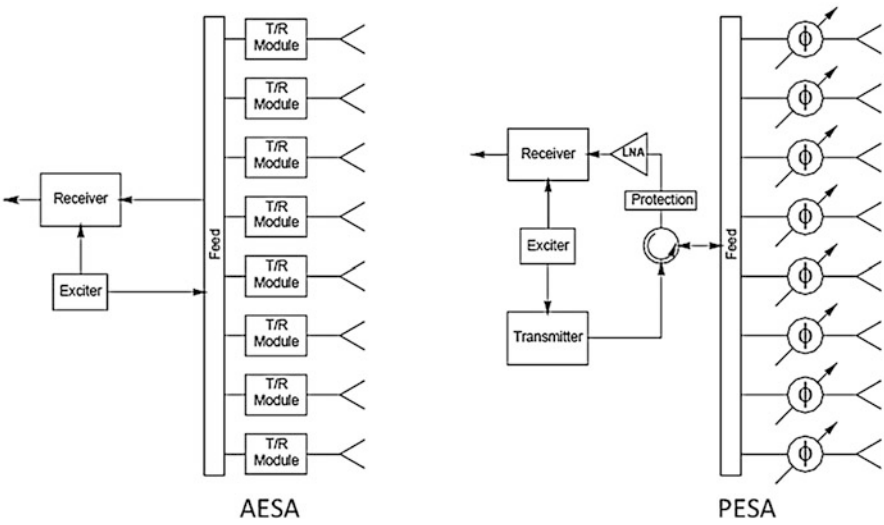


Fig. 9 AESA versus PESA

Figure 8 shows the comparison between active ESA (AESA) and passive ESA (PESA). In PESA, only the phase control is taken at the element level (see Fig. 9) whereas in AESA the amplitude as well as phase controls (together called as T/R module or TRM (Sreenivasulu et al., 2020; Kedar et al., 2020a; Fourikis, 1997; Sturdivent et al., 2019; Sturdivent & Harris, 2015)) is distributed at the element level

(see Fig. 9). Thus, AESA have better detection sensitivity and possess high reliability (i.e., *graceful degradation* (Fourikis, 1997), which implies continual operation of AESA even in the presence of few failures of the modules). In addition, the advent of digital analog to dc convertors and vice versa (i.e., ADCs or DACs) have made it possible to convert active electronics beyond power amplifier (in Tx chain of TRM) and low noise amplifier (in Rx chain of TRM) by the digital circuits (Sturdivent & Harris, 2015). These are called as *digital beamforming* (DBF)-based AESAs or *digital AESAs* (Sturdivent & Harris, 2015; Shang et al., 2017; Talisa et al., 2016). Thus, the present state-of-the-art technology and research trends for the radars are centered on AESAs.

An AESA has several advantages over MSAs and PESAs as given below:

- AESA provides an Inertia Free Beam Steering & Control (no physical movement).
- AESA facilitates multifunctionality, i.e., search, track, track while scan capabilities, etc.
- Adaptive Beam Pattern Control (dynamic beamforming, adaptive nulling, and multiple beam formation).
- Efficient use of radiated energy.
- Highly modular architecture (scalable architecture).
- High reliability and maintainability (good thermal management and repeatability in performance).
- Graceful degradation over time (operational even with few failed modules).
- Reduction of radar cross section.

Figure 10 shows the functional block schematic of active phased array antenna system or AESA showing prime constituent submodule comprising it. AESA may

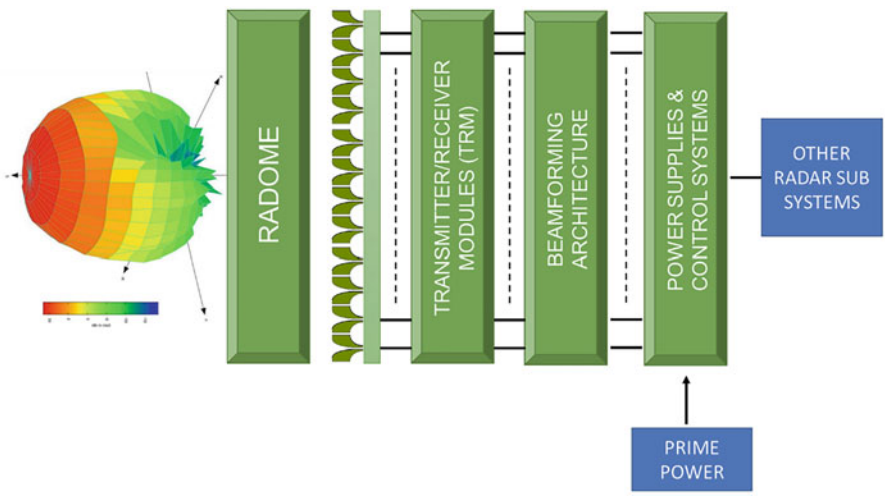


Fig. 10 Functional block schematic of active phased array antenna system

be considered as a system of systems which involves various technologies as listed below:

- Radome
- Antenna Element Technologies
- Transmit/Receive Modules
- Thermal Design, Reliability and Heat Transfer Technologies
- Array Beamforming
- Engineering Aspects (Mechanical Chassis and Integration)

Hence, an AESA designer or rather system designer shall be aware of these technologies to derive the overall specifications and guide the design and development team for AESA toward successful realization of submodules of an AESA, their integration, and evaluation (Fourikis 1997; Sturdivent et al. 2019). The next section discusses the most important technologies involved in the AESAs development.

Involved Technologies In AESAs

Radome

A Radome is a structure and weatherproof enclosure which protects an antenna from the effects of environmental exposure and conceals it from the public view, specifically in defense applications. The material used shall be RF transparent, having low dielectric constant and low-energy reduction. The technology spinoff for Radome occurred during World War II (Kozakoff 1997). The system bandwidth is limited by the *radome's bandwidth,* and the *Radome depolarization* limits the *dynamic range* in a frequency reuse application. *Radome transmission efficiency* (RTE) and *Insertion phased delay* (IPD) are the two main characterizing parameters of a radome.

A radome may have various designs based on its deploying platform like airborne, shipborne, or ground-based platforms which puts different specifications' stringency from the system perspective (see Fig. 11). Radome materials have continued to develop in two main areas: (a) ceramics, used primarily for hyper velocity missile radomes and (b) high-strength organic materials for sandwich composite radomes. The parameters that affect the radome's broadband performance are the following:

- Wall thickness (thinner radome is generally better)
- Wall design that includes the number of layers, thickness, and permittivity of each layer (more layers is better)
- Shape of the radome (flatness is desirable)
- Selection of low loss materials (small loss tangent is required)

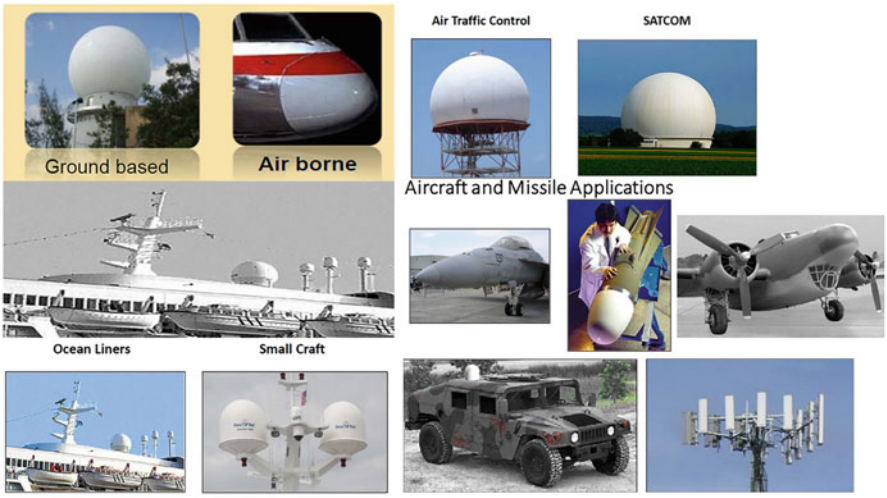


Fig. 11 Different platforms for the deployment of radome (source: Internet)

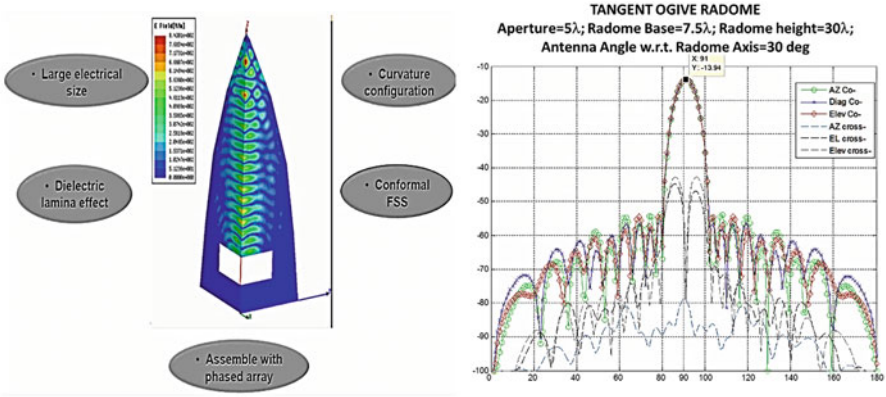


Fig. 12 Design challenges in radome

Figure 12 (left-hand side) illustrates the various constraints in design process of a radome. Figure 12 (right-hand side) shows the *em* simulation of a tangent ogive-shaped radome. A radome may affect the antenna far field radiation behavior in terms of 3 dB beam-width (HPBW), Side lobe level (SLL), and Axial ratio (AR). It may introduce Boresight error (BSE) and Boresight error slope (BSES) and may increase depolarization or the folding of energy from one polarization sense to the other. A radome may result into an increase in antenna VSWR and the insertion loss (IL) due to its presence. A radome design requires a proper choice of material, shape modeling as per curvature of antenna and mounting platform (conformability), and its modeling along with the phased array antenna (Kozakoff 1997; Kedar and

Revankar 2006; Kedar et al. 2020b). The performance of a radome affects the antenna performance as:

- Influences the antenna's far field radiation behavior in terms of HPBW, sidelobe level (SLL) of antenna array's radiation pattern, and axial ratio (AR). SLL generally increases inside radome due to the distortion and wall transmission effects as a wavefront propagates through a radome wall. This is a great hindrance in the design of low-sidelobe-level antennas.
- Boresight error (BSE), is the bending of the angle of arrival of a received signal relative to its actual angle of arrival due to distortions of the electromagnetic wavefront as it propagates through a dielectric radome wall and boresight error slope (BSES), is the rate of change of BSE as a function of the antenna scan angle within the radome. BSE depends upon the frequency, polarization, and antenna orientation. BSES can cause severe degradation for modern guidance systems. The radome manufacturing tolerance affects the BSES characteristics of the radome. The maximum allowable BSES is determined by the manufacturing tolerances of the radome.
- Power Transmission Coefficient
- Introduction of a registration error: Registration error is the difference in uplink frequency and downlink frequency, and it should be sufficiently small to assure that both the transmit beam and receive beam peaks are coaligned on the same satellite.
- Increased depolarization: It is the folding of energy, i.e., the change in the sense of the primary antenna polarization, for instance, the RHCP signal after reception transforms to LHCP signal. This is due to the radome wall curvature and difference in complex transmission coefficient between orthogonal polarized vectors.
- Increase in antenna VSWR: The antenna VSWR can greatly increase from RF power reflected from the inner radome wall surface. The VSWR increase from this radome reflected power will represent an additional gain/loss that can be critical to, for example, SATCOM communications systems.
- Insertion Loss due to the Presence of the Radome

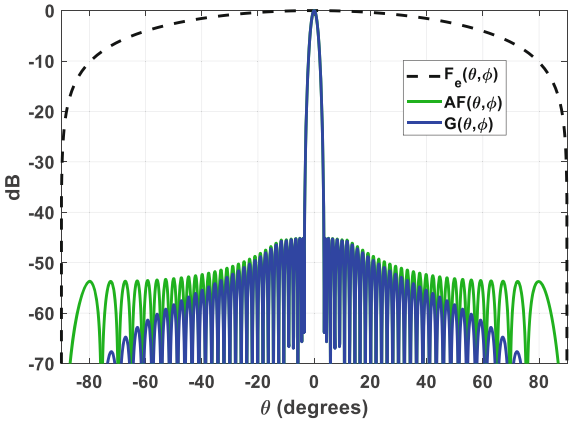
Figure 12 shows the performance of an antenna array in the presence of tangent ogive radome. The left-hand side shows the electromagnetic propagation inside radome as captured using electromagnetic simulations using Ansys HFSS. The right-hand side figure shows the antenna radiation pattern (co- and cross polarization field) in the presence of tangent ogive radome whose physical and electrical parameters are shown on the top of the plot. There are various classes and types of radome, and for the details of these one can refer to the ref (Kozakoff 1997). Thus, the choice of proper radome to ensure the minimally degraded performance of an antenna array is very important. The next important technology is antenna which is discussed in the next subsection.

Antenna Technology

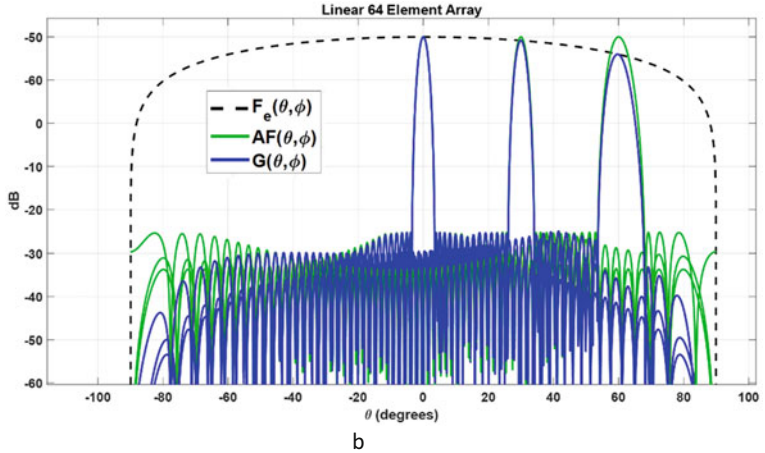
The radiation pattern, $G(\theta, \varphi)$, of an AESA's phased array antenna depends upon the design of the *antenna element* ($F_e(\theta, \varphi)$ is the element pattern) and the chosen *array configuration* (quantified by array factor, AF) (Kedar 2020a, 2022; Balanis 1997; Brookner 1991) given as in (1) (Kedar 2022).

$$G(\theta, \varphi) = F_e(\theta, \varphi) \times AF(\theta, \varphi) \tag{1}$$

In (1), F_e is the element pattern, AF is the array pattern, and (θ, φ) are the spatial direction of observation for sampling the respective radiation patterns. Figure 13 shows computed radiation pattern of a 64-element array with half a wavelength spacing between the elements in an array. Figure 13a shows the respective plots, F_e,



a



b

Fig. 13 Radiation pattern of a 64-elements antenna array (**a**) at boresight (**b**) scanned performance

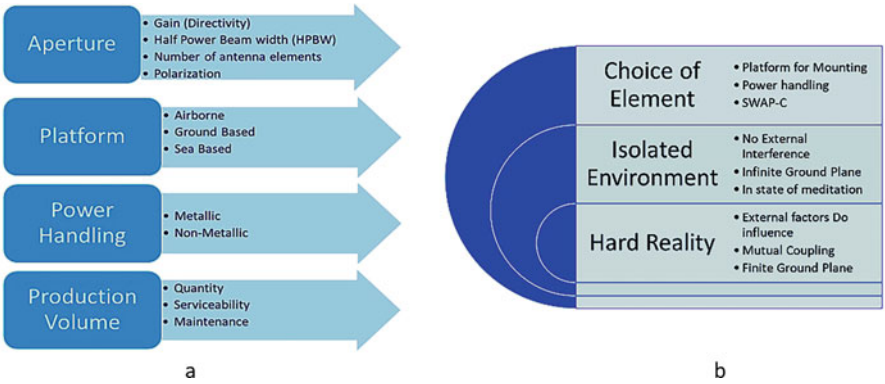


Fig. 14 Design challenges in antenna array

AF, and *G*. The effect of F_e is clearly visible in the far angles where the sidelobes of *AF* are suppressed in *G*. It is further clearly visible from the plot in Fig. 13b that the *AF* varies with scan angle in its pointing direction but with no change in its maximum gain and HPBW. However, the variation in gain and HPBW of overall radiation pattern is seen due to the variation in the element pattern with respect to the scan angle. Thus, the choice of element pattern is very critical to decide overall performance and the control over the variation in gain of the antenna array (Sreenivasulu et al. 2020). In the above example, $F_e = \cos^n(\theta, \varphi)$ is chosen which represents an isotropic antenna element's pattern (n = 1). However, in real life, n≠1.

Figure 14a shows the relationship between array's design parameters based on the design constraints. Figure 14b shows some real-life constraints which need to be accounted for while designing an antenna element for phased array applications (Brookner 1991; Kedar 2022; Fourikis 1997; Sturdivent et al. 2019; Sturdivent and Harris 2015). The practical design constraints while designing an antenna array are inclusion of the *mutual coupling* and *mounting platform effects* on the antenna performance (Kedar et al. 2020b; Singh and Jha 2013; Kedar 2021). The Eq. (2) describes mathematically the inclusion of these design constraints on the array pattern of the element chosen for the array realization. In (2), $\Gamma(\theta, \varphi)$ accounts for the mutual coupling effects from the neighboring elements in an array, A is the array's aperture area, λ is the operational wavelength, and (θ, φ) is the scan angle (Kedar 2020b, 2022).

$$F_{e,\,active}(\theta, \varphi) = \frac{4\pi A}{\lambda^2} \cos\theta \left[1 - |\Gamma(\theta, \quad \varphi)|^2\right] \tag{2}$$

The subscript "*active*" implies the value of element gain in the presence of the surrounding environment including other antenna elements in an array (Bhattacharya 2006).

The performance of an antenna elements is vividly different from its performance when it is surrounded by neighboring elements. Ideally, the design of an element starts in isolated environment assuming it is situated on an infinite ground plane with total isolation from the surrounding environment. The practical scenario is totally opposite where the element is affected by its surroundings due to the finite ground plane and the mutual coupling (Singh and Jha 2013; Kedar 2021; Bhattacharya 2006). Thus, to design an element for phased array antenna application the effect of mutual coupling needs to be taken care of (Kedar et al. 2020b). The various design challenges may be summarized as:

- Mutual coupling effects: effect of one antenna element on another in array environment. The gain, beam shape, and radiation impedance change due to coupling, and it may cause scan blindness.
- Antenna impedance matching over full scan volume.
- Active element pattern beam width optimization.

There are various ways to analyze an element's performance in array environment (Kedar 2022):

- Unit cell approach (Floquet mode approach) which considers:
  - It assumes an infinite array containing the element.
  - No array's finite edge affects.
  - Pairs of Perfect Es or Perfect Es and Perfect Hs surrounding the element.
  - Periodic boundaries – Linked Boundary Conditions (LBC).
  - It is a fast solver within few minutes.
- Finite array approach which considers:
  - Finite array.
  - Finite edge affects.
  - It is computationally expensive and time consuming.

Figure 15a shows a unit cell consisting of a dipole antenna over a metallic reflector on the left side. The right-hand side plots in Fig. 15a show the effect of mutual coupling on single element performance (SEP) embedded in an antenna array environment in E- and H- planes and also show the effect of scanning. It can be seen that in Fig. 15b SEP varied in a finite array (real-life scenario with fixed number of elements in an array) as compared to an infinite array (assuming infinite number of elements surrounding the element) environment. Thus, effect of the mutual coupling and finite edges of antenna array are apparent. Thus, a proper design of an antenna element accounting for all these considerations is crucial for the desired performance of an AESA.

Another example of antenna technology chosen for active phased array radars is shown in Fig. 16a. Usually, the printed patch antennas suffer from dielectric losses resulting into loss of antenna efficiency, which in turn reduces the overall array efficiency (Kedar 2019). A metallic antenna is completely devoid of substrates or any dielectric in turn resulting into removal of dielectric losses. Figure 16a shows the photograph of a metallic Vivaldi antenna (Kedar 2019) which is completely made

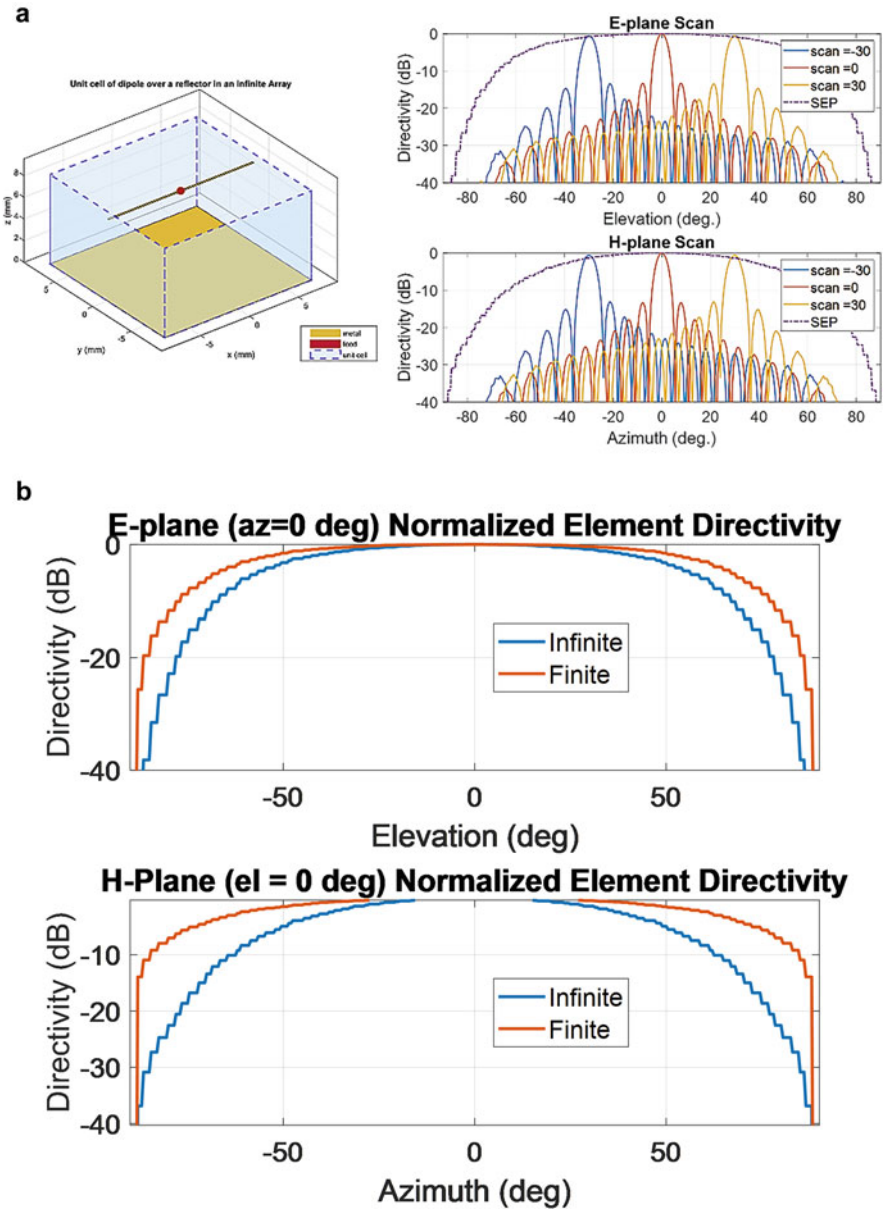


Fig. 15 Element performance in an array environment

from a metallic block by wire-cutting technology within precision of 50 μn. A 64-element (8 × 8) array is fabricated arranged in a triangular grid with Fig 16b showing the numbering scheme of the elements. Table 1 shows the comparative

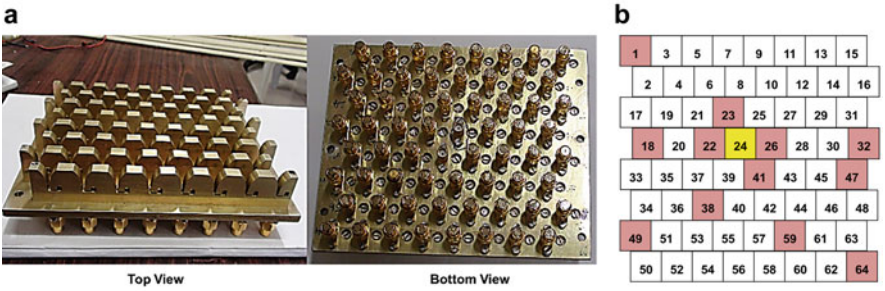


Fig. 16 Metallic Vivaldi antenna (MVA): (**a**) photographs of the top and bottom views; (**b**) numbering scheme for elements in an array

Table 1 Comparison of MVA pattern at 10 GHz

Case	Gain (dBi)	HPBW (deg)			x-pol (dB) over ±60° scan volume		
		E	D	H	E	D	H
Isolated	2.3	131	131.5	121	−19.3	−9.6	−22.8
Infinite array	4.7	141	125	93	−9.5	−32	−14.4
8 × 8 Array (measured)	3.2	118	–	95	−18	–	−22

summary of isolated element with infinite array and finite array (measured) in terms of gain, HPBW, and cross (x-) polarization levels. A substantial difference is observed in the HPBW and x-pol levels between isolated element and the element in the array environment. A variation is observed in the finite array and infinite array environment as expected because of the finite edge effects, as expected. Figure 17 shows the measured co- and cross-polarization radiation pattern plots over 9–12 GHz bandwidth in finite array environment.

The antenna element design is crucial, and once it is done, the next design aspect in PAA is the design of transmit receiver modules discussed next.

T/R Module Technology

The T/R module (TRM) (Sreenivasulu et al. 2020; Kedar et al. 2020a; Sturdivent and Harris 2015) forms a key element of an AESA which comprises of *distributed transmitter* and *receiver* providing necessary *amplitude* and *phase* coefficients to the array elements. It makes an extensive use of the semiconductor technology to provide necessary transmit power and low noise reception of *rf* signals to/from an antenna array in AESAs. It decides the desired power output and receiver sensitivity of the radar system. TRMs have necessarily four functions:

1. Provides gain and RF output power during transmission of PAA
2. Provides low noise reception during reception through PAA

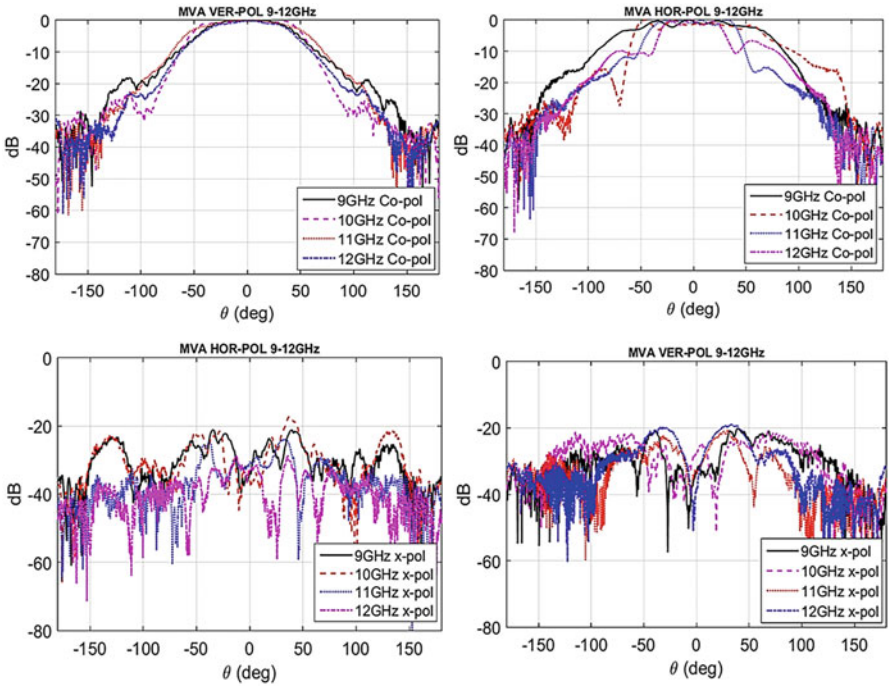


Fig. 17 Measured co- (top side) and cross- (bottom) radiation pattern plots of element in the finite array environment over 9–12 GHz

3. Provides dynamic switching between transmit and receive chains of PAA
4. Provides dynamically phase shifts and attenuation to the elements in PAA

The total power output, P_{out}, of radar is equal to the number of elements multiplied by individual TRM output. Figure 18 shows a typical design flow chart of a TRM with a functional block diagram of TRM shown as an inset. Figure 19a shows typical common leg functional bock diagram which is commonly employed in PAAs. This allows only half duplex operation, i.e., either TRM operates in Tx or in Rx mode only at an instant as shown in Fig. 19b.

The various semiconductor technologies available for the realization of monolithic microwave integrated circuits (MMICs) are based on Si, SiC, GaAs, LDMOS, or GaN processes (Sturdivent and Harris 2015). The choice of the particular technology depends upon the trade-off in the form factor of TRM, power output, module efficiency, operational bandwidth, and the cost. GaAs and GaN are the most suitable and utilized semiconductor technologies used for the realization of MMICs for wide band and high-power output TRMs in today's scenario. Figure 20 shows the comparison of these two technologies. GaN is the latest current semiconductor technology extensively used for high-power and high-gain applications in TRMs (Kedar et al. 2020a; Sturdivent and Harris 2015). Typically, considering a 4–6-inch

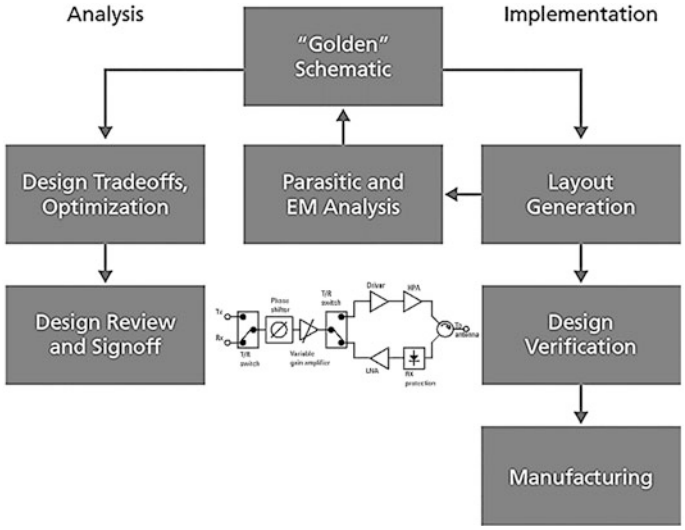


Fig. 18 TRM design realization flowchart

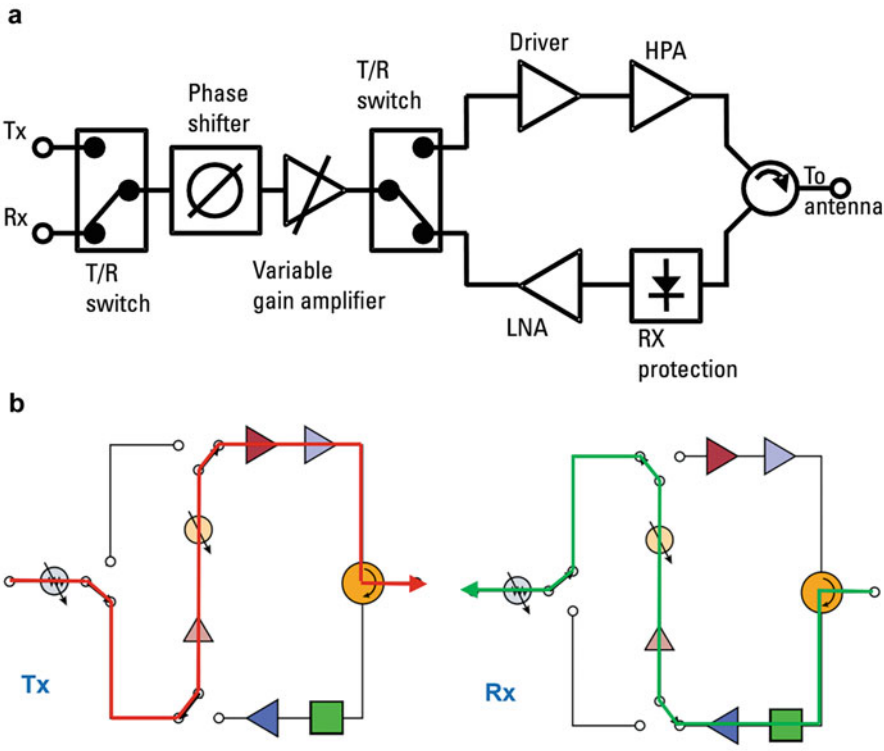


Fig. 19 (**a**) Typical common leg TRM design functional block diagram; (**b**) half duplex operation

Parameter	GaAs	GaN
Power Density	0.5 to 1.5W/ mm	>10W/mm
Operating Voltage	5-20V	25-50V
Max. Breakdown Voltage	~40V	>100V
Maximum Current	~0.5A/mm	~1A/mm
Thermal Conductivity (W/m-K)	47	390/490, SiC
Efficiency	>20-40%	>50%

Fig. 20 Comparison of GaAs and GaN technologies

wafer the power levels obtained from GaN technology are 2–3 times more than corresponding GaAs wafers (Sturdivent and Harris 2015).

The advantages offered by GaN over GaAs technology are the following:

- High-energy bandgap (2.5 times more)
- High breakdown field-strength (10 times more)
- High RF power densities (5–10 times more) on small chip sizes
- Broadband amplifier applications (multioctave design)
- High bias supply voltages
- High thermal conductivity ($\times 3$ (GaN), $\times 8$ (SiC))
- Radiation hardness

The specifications of TRM are derived from PAA-based radar specifications. Radar range determines power-aperture product of antenna (Kedar 2022), which in turn determines the output power radiated per element and henceforth, the power output of TRM. Total power-aperture product determines the number of TRMs and elements in PAA. For example, in C-band if 20 kW power is required to be radiated from an array, it can be served by choosing 1000 numbers of 20 W GaN-based TRMs, each feeding an element in PAA. However, if GaAs-based TRM is chosen, a maximum of 10 W is possible, and henceforth the number of TRMs will be doubled. Thus, the technology and the quantity of TRMs are determined from the radar requirements. Thus, size, power, noise figure, efficiency, phase and amplitude controls, and connectors for inter-connections in TRMs, all of these are derived from radar requirements. The size of TRM is determined by the inter-element separation in an array (Kedar 2022; Sturdivent and Harris 2015; Kedar et al. 2020b). The noise figure and receive chain gain of TRM determines the front-end noise figure of overall RX chain of PAA-based radar and thus is very critical and needs to be minimized (Kedar et al. 2020a; Sturdivent and Harris 2015; Kedar et al.

2020b). Thus, overall dynamic range of cascaded receiver chains is dependent on the noise figure and receive gain.

Phase shifter is another critical technology which determines the dynamic application of desired phase shifts for beam steering and forming single or multiple steered beams from PAA-based radar. It should provide 360° of phase control across the operating bandwidth while allowing minimum insertion loss, switching time, rms errors, etc. With the advent of digital phase shifters comprising of PIN diodes, MESFETs, etc. with phase increments dependent upon the least significant bit (LSB), it is possible. The typical 6bit phase shifter allows an LSB of 5.625° error. It will give a total of 2^6 states of phases across 360°. Thus, greater the number of bits, better is the performance but with the increase in cost and complexity due to intricacies of technologies involved (Sturdivent and Harris 2015). Similarly, attenuators control the amplitude of the signal fed to array elements and thus control the sidelobe level (SLL) of PAA. The number of bits of digital attenuator controls the minimum attenuation that is possibly fed to the element. Currently 6- and 7- bits attenuators and phase shifters are in use in radars arena. The power amplifier decides the power output as explained in the above text and again decides the semiconductor technology to be used, LDMOS, GaAs, GaN, etc. The circulator provides isolation between receive and transmit chains before antenna element, and limiter switch protects receive chain from high power leakage from transmit chain. Low noise amplifier (LNA) determines the noise figure and maximum rf signal that can be received by the TRM. The details of TRM constituents can be had from ref (Sturdivent and Harris 2015).

The concept of PAA is critically dependent on the availability of compact and minimum weight, low consumption, and high reliability T/R Modules. The large number of individual T/R Modules, integrated with the respective radiating elements, ensures a great degree of redundancy (Graceful Degradation). It decides the generation of Transmit Power, Low Noise Amplification of the Received Signals, Required Phase Shift in the Transmit and Receive Mode for Electronic Scanning, Variable Gain Setting for Aperture Weighting, and Proper Interface to the Array Beam Steering Controller as well as Array Beamformer and Radiating Elements. T/R microwave modules are thus the key elements of the active phased arrays.

Thermal Design, Reliability, and Heat Transfer Technologies

The involved MMICs in TRMS and transmission lines for carrying *rf* signals lead to practical issues such as thermal management and signal integrity. Thus, design of AESA is not complete without carrying out proper thermal management and signal purity analysis to mitigate any cross talk or heat buildup inside the TRMs. Further, TRMs involve multilayer printed circuit boards glued together involving interconnects binding multilayer boards' traces. Thus, design of TRMs involves careful consideration of these using *em* analysis based on the full wave techniques like method of moments (Sturdivent and Harris 2015).

Beamformer Technology

PAAs need to compute the progressive phase shifts, compute number of beam parameters, etc., digitally at a down level hierarchically in the PAA-cascaded chain. Usually these are relying on field programmable gate arrays (FPGAs) or application-specific integrated circuits (ASICs) technologies. Beamformer circuits are essentially digital PCBs which form the beams as per the radar requirements. Each beam input is split according to the number of elements, and amplitude and phase weights applied. The input to each element includes a power combiner which order is equal to the number of beams. The number of gain and phase control points is the product of the number of beams and number of elements. In order for the array to constructively beamform in the direction theta, the elemental paths must introduce a progressively increasing time delay of n*d*sin(theta) or an equivalent phase delay (only correct at one frequency). These converts rf received from TRM into digital signal using analog-to-digital convertors (ADCs) or converts digital to rf while transmitting the power to TRM using digital-to-analog convertors (DACs). The number of ADCs/DACs depends upon the number of channels in the receiver, which is again determined from radar requirements mainly beam switching time, cost, complexity, and the number of beams required by radar to be formed. There are various possible beamformer architectures available in literature (Sturdivent and Harris 2015) but not dealt with here due to limited scope of this chapter.

Phased Array Calibration and Assessment

The realization of AESAs is an engineering challenge involving assembly of different submodules, thermal management, power distribution, and compact framework housing these subunits. The integrated AESA needs to be calibrated and evaluated for the desired performance. It requires beam collimation and calibration inside an *anechoic chamber* isolated from surrounding noises (Balanis 1997; Brookner 1991; Rensburg et al. 2020) based on near field measurements. An optimal performance of an AESA is only achievable by employing suitable calibration techniques to mitigate the nonuniformities observed in the radiation pattern of the radiating elements due to the various factors such as mutual coupling, the amplitude and phase sensitivities observed in the associated active electronics (TRMs, beamformer hardware, etc.), and near-field scattering phenomenon.

Collimation Procedure for Active Array: The instrument calibration loop inside anechoic chamber is used to perform three distinct functions. First of all, it characterizes the instrument transfer function during the measurement modes, followed by the characterization of the individual TRMs and lastly the special external characterization mode.

Calibration: The amplitude and phase characteristics of the passive circuit viz., probe sampling the energy from AESA, Built In Test Evaluation (BITE) Feed Network inside the AESA, and cables in the RF injection path of each of the TRM are measured as *calibration ratio (CR)* in an antenna chamber, also called as *near*

field test range (NFTR) during the pattern measurement of an AESA. The CR measurement setup is shown in the Fig. 21.

$$\text{Calibration ratio} = \text{Probe value} - \text{Injection value} \tag{3}$$

In NFTR, CR measurement is carried out at a fixed temperature (room temperature) for multiple spot frequencies of operation for AESA (typically 11 or more) in all the operational modes of an AESA, i.e., *transmit (Tx)* as well as *receive (Rx)* modes.

The basic concept in calibration of TRMs is to Switch "ON" only one selected module out of "N" Modules with rest of N-1 Modules in OFF State in an AESA. Then, sample the amplitude and phase values of the "ON" Module through Sum/CAL Receive Channel of Central Unit by injecting the appropriate BITE/CAL RF Signal through BITE/CAL Feed Network.

Transmit path calibration: During Tx mode calibration as shown in Fig. 22, TX drive signal from exciter is fed as an input to TRMs through RF feed network. The selected TRM, i.e., its Tx chain, is "ON," amplifies the Tx drive signal, and radiates, while keeping other TRMs in "OFF" state (i.e., in Rx mode with proper isolation).

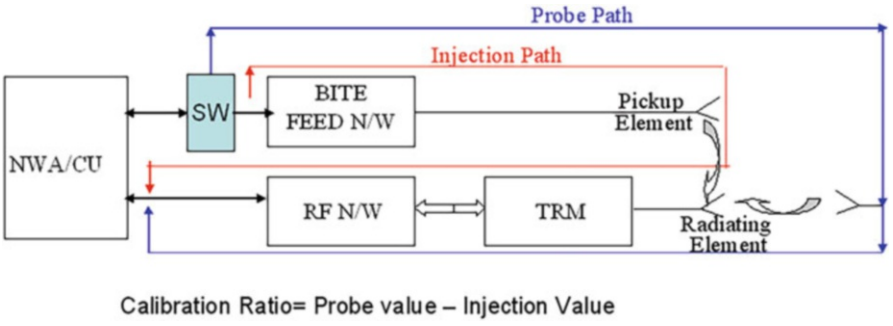


Fig. 21 Calibration ratio measurement setup

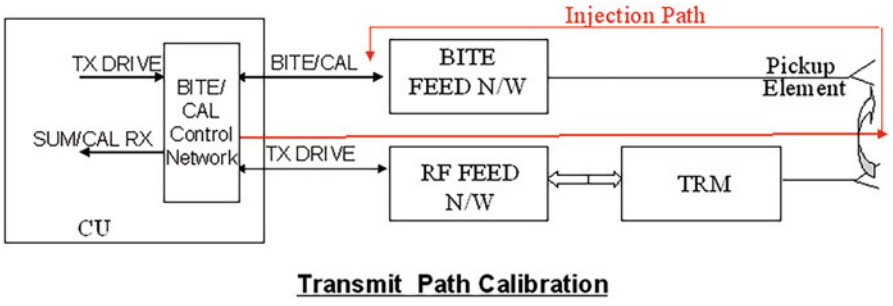


Fig. 22 Transmit path calibration

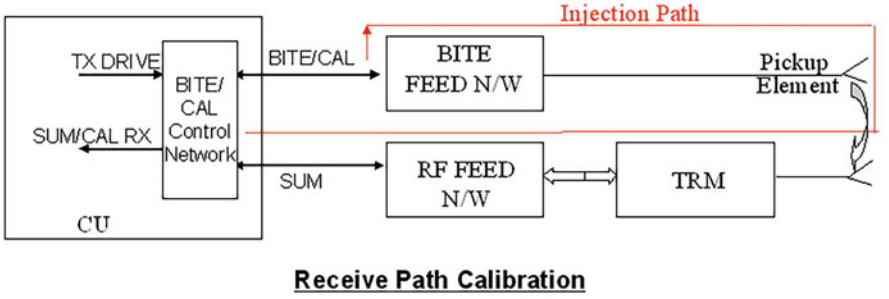


Fig. 23 Receive path calibration

This radiated signal is received by the pickup element of the radiating segment and passed through BITE feed network to the input of Rx channel in the receiver. Further, it is digitized as "I" and "Q" data and sent to the Radar Controller (RC) for computing the amplitude and phase values of the selected TRM. The module to be calibrated is set for 0 dB attenuation, 0° phase state during transmit mode at the first iteration. The amplitude and phase values are computed by adding this sampled (injection path) value to the respective CR value. As the Tx amplifiers are saturated, phase calibration only required to cophase all TRMs within 1 LSB of 0° phase value.

Receive Path Calibration: The Rx mode calibration is shown in Fig. 23 below, where Tx drive signal is fed to the pickup element of the AESA's radiating segment through BITE feed network. The selected TRM, whose receiver is "ON," amplifies the signal received through its antenna from the pickup element's radiation, while other TRMs are in "OFF" state (kept in Tx mode with proper isolation). This amplified signal passes through RF feed network, Sum channel of the beamformer unit, and is fed to the input of the receiver-exciter unit of radar. The sample is further sent to the RC for computing the amplitude and phase values of the selected TRM. The module to be calibrated is set for attenuation state and corresponds to distribution value, 0° phase state at the first iteration. The amplitude and phase values required for calibration are computed by adding this sampled (injection path) value to the respective calibration ratio value. As the receive amplifiers are linear, all TRMs need to be calibrated to their respective distribution values within 1 LSB and phase calibration is required to cophase all TRMs to within 1 LSB of 0° phase value. It is to be mentioned that the calibration network thus explained should conform to the dynamic range of the receiver exciter unit of the radar. Typically, the dynamic range is around −50 dBm to −70 dBm during in-system calibration for transmit and receive states. The noise level at the input of Sum/CAL receive channel is −90 dBm. Thus, *signal to noise ratio* (SNR) during calibration is >20 dB which is easily obtainable in AESAs.

The in-system calibration of TRMs is performed on request basis in the field either periodically over 100/200 h of system usage or after replacement of faulty

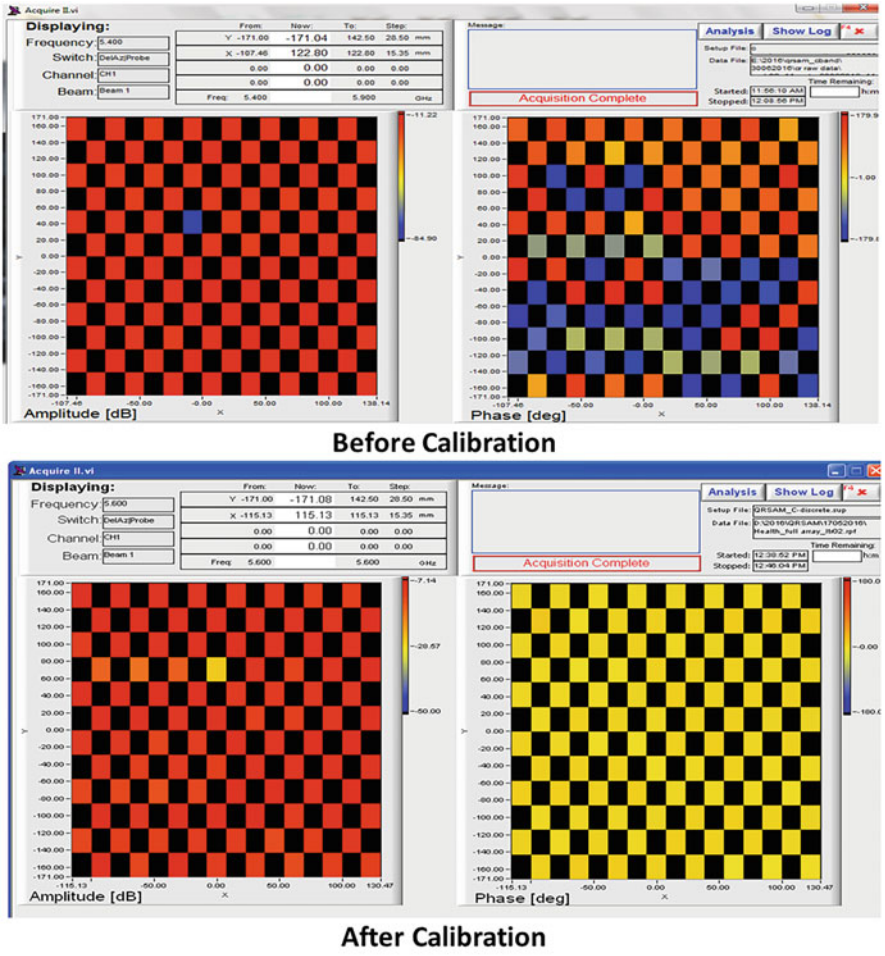


Fig. 24 Receive path calibration

modules in AESA. If there is a large variation in T/R Module parameters during Power on BITE or On-line BITE, array calibration can be performed for the randomly selected T/R Module or for all the T/R Modules based on the operator request. Figure 24 shows the amplitude and phase distribution in a 64-elements AESA before and after calibration process. It is clear that after calibration the amplitude and phase distribution over the radiating aperture of AESA is within 1 LSB of amplitude and phase distribution. Figure 25 shows the pattern performance of an AESA during the iterations involved in calibration process. A clear beam formation is visible after the completion of calibration process involving 2–5 iterations at a maximum.

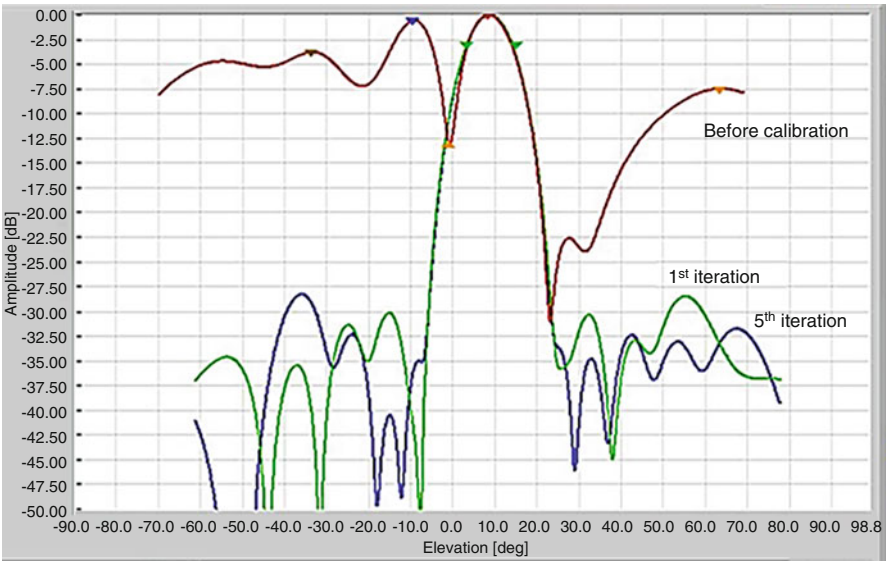


Fig. 25 Radiation pattern performance of AESA

Conclusion

The present chapter discusses the phased array antenna technology for radars and presents various involved technologies like radome, antenna, T/R modules, thermal integrity, calibration processes, etc. It presents a brief historical perspective of radar and the importance of an efficient design of phased array antenna to ensure desired performance of phased array radar. Phased array antenna is system of systems involving various subsystems involved in its buildup like radome for external environment protection without giving any hindrance to its performance, antenna array serving as eyes and ears of radar, T/R modules giving necessary power output and noise-free reception, and various technologies involved in developing these technologies. The various involved technologies are discussed briefly familiarizing the reader about them and a platform to start the detailed investigations on the topic of interest. A brief of operational constraints, characteristic parameters, and working mechanism of various technologies is presented. The chapter concludes with the calibration and collimation techniques for characterizing AESAs.

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Antennas for mm-wave MIMO RADAR

60

Design and Integration Challenges for Automotive Applications

Jogesh Chandra Dash and Debdeep Sarkar

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Abstract

Multiple-input Multiple-output (MIMO) antennas are one of the key enabling technologies for the modern (4G-LTE, 5G) cellular communication base stations/hand-sets/wireless access points (WAPs) and will continue to dominate the implementation of future generation wireless technologies (6G and Beyond). MIMO systems provide significant advantages over conventional SISO (Single-input Single-Output) systems, especially in combating multipath fading effects in NLOS (Non-Line of Sight) propagation scenario, leading to enhanced channel capacity (data-rate), lower bit-error probability, and better signal-to-interference-noise ratio (SINR). Transcending from cellular and WLAN (Wireless Local Area Network) domain, research on MIMO techniques are currently emerging significantly in the context of RADAR (Radio Detection and Ranging) systems as well. While the classical RADAR technology has been well-established in the context of defense applications (since World War II) and weather-monitoring systems, its integration with MIMO technology has opened new directions for RADAR usage in many industrial and civilian applications.

In this chapter, we first describe the standard operating principles of MIMO RADARs and elaborate upon their advantages over conventional phased array

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RADARs. Among the various application environments, MIMO RADARs have assumed a crucial role, especially in automotive applications like advanced driver assistance systems (ADAS). Subsequently, we describe the state-of-the-art mm-wave MIMO antennas available in the open literature for automotive systems. Besides highlighting the antenna design aspects, we also critically examine the different integration challenges when a RADAR antenna system is installed inside a vehicle.

Keywords

MIMO RADAR · Automotive · ADAS · Virtual array · Antenna

Introduction

Automotive RADAR systems, first introduced in the late 1990s, show a steady trend from safety-oriented vehicle functions (Active Safety: AS) to comfort (Advanced Driver Assistance System: ADAS). Nowadays autonomous vehicle is a reality and a very challenging research topic. As we all know, most of the high-tech, sophisticated, and luxurious automotive industries such as Daimler Mercedes-Benz, BMW, Tesla, etc. working proactively on future advanced driver assistance systems (ADAS), driver-less vehicles, or self-driven cars. The component or device, which is the basic building block to accomplish this concept, is the automotive MIMO RADAR sensor or collision avoidance RADAR. Automotive MIMO RADAR is the key component in autonomous driving and safety systems including a diverse range of applications shown in Fig. 1. RADAR-based systems tend to address scenarios of growing complexity, targeting urban or city traffic along with highway scenarios,

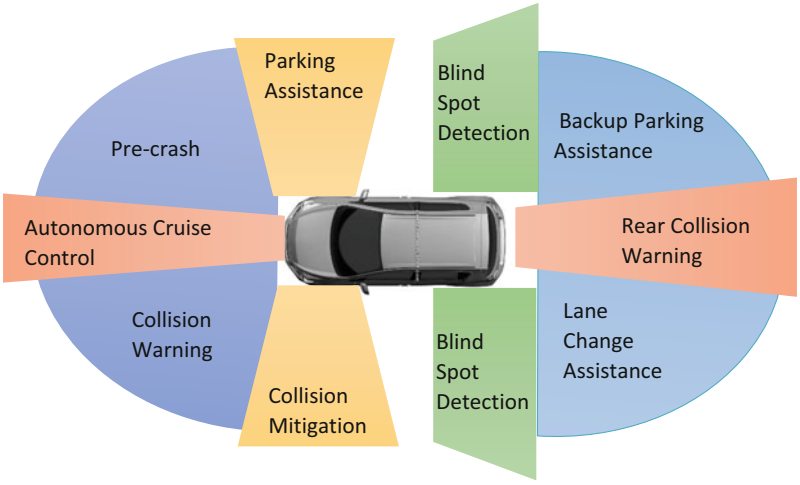

Fig. 1 Applications of automotive collision avoidance RADAR

which are the main application in recent years. In general, frequency-modulated continuous wave (FMCW) radar technology is used for this purpose. Resolution in range, azimuth angle, and target velocity are the key features to deal with demanding scenarios of high complexity and high dynamics (Rasshofer 2007). European Union has replaced the previously used 24 GHz ultrawideband RADAR sensors with a new frequency band from 77 to 81 GHz to enable future ADAS and AS systems. However, there are certain basic advantages of the 77 GHz band over 24 GHz as far as the radar antenna properties are concerned as listed in Table 1 (Viikari et al. 2009; Ramasubramanian et al. 2017).

The European-funded project "RADAR on Chip for Cars" (short RoCC) started in the fall of 2008 (RoCC press release 2009) in order to start the development of 79 GHz band technology. The main advantages of the 77 GHz to 81 GHz frequency range (79 GHz band) are that RADAR devices can be much smaller, and that a single technology can be used for all applications along with a greater capability for distinguishing between objects.

Further, as the degree of automation has reached levels 4 and 5 (Merlo and Nanzer 2022), situational sensing is an utmost priority in an autonomous vehicle, necessitating the rapid development of sensor technologies (Marti et al. 2019) with high-resolution mapping. Generally, this high-resolution sensing is achieved by using various sensors such as LIDAR (Light Detection and Ranging), camera, ultrasonic SONAR, and automotive RADAR, which have paved the path for the realization of fully autonomous vehicles or self-driving cars. In addition, it is supported by advanced monolithic microwave integrated circuits (MMICs), computer vision techniques, artificial intelligence, and improved signal processing (Engels et al. 2017). The optical sensors such as LIDAR and camera provide high angular resolution and visual representation of the target, respectively, but fail in low light conditions and face high attenuation in bad weather conditions such as rain, fog, dust, snow, etc. (Jokela et al. 2019). The SONAR sensor is suitable for low-speed and short-range target avoidance; however, its spatial resolution and maximum range are restricted for dynamic driving conditions and at the same time get attenuated in adverse weather conditions. On the other hand, RADAR is one of the most efficient sensors among others which is unaffected by weather conditions and independent of lighting conditions. Due to the development of mm-wave, MMICs automotive industries are now capable to realize the complete RADAR on a single chip (RoC) or system-on-chip (SoC). For example, a fully integrated 65-nm CMOS technology 77 GHz frequency-modulated continuous wave (FMCW) RADAR system for an automotive application is proposed in (Kucharski et al.

Table 1 Basic comparison between 24 GHz and 77 GHz automotive radar antenna parameters

Parameter	24 GHz	77 GHz
Frequency range (bandwidth)	24 GHz–24.5 GHz (250 MHz)	76 GHz–81 GHz (5 GHz)
Size	Higher	One-third the size of 24 GHz
Resolution	Lower detection range with resolution	Higher resolution and excellent detection range than 24 GHz

2019). In (Lee et al. 2010), a 79 GHz scalable RADAR based on a transceiver system having a single channel is proposed. Similarly, Usugi et al. (2020) propose another fully integrated 77 GHz LRR (Long-range RADAR) transceiver in 40-nm CMOS technology. Further, the MIMO (multiple-input multiple-output) concept in the RADAR system provides an additional degree of freedom to improve the angular resolution without increasing the physical antenna aperture. The MIMO technique increases the RADAR angular resolution by introducing the formation of a virtual array of receiver antenna that contains more number of antenna elements and a larger virtual aperture than the physical antenna through the use of waveform diversity (i.e., transmitting orthogonal waveform from transmitter antennas) and reduces the cost of LNAs and other active components present in the receive chain (Li and Stoica 2007). More information on angular resolution improvement concepts using MIMO RADAR can be found in (Dash et al. 2021a; Rao 2017; Vasanelli et al. 2017).

The design and analysis, present in the open literature on 77 GHz automotive RADARs, deals with the RADAR performance in free space. However, these automotive RADARs, in the practical application scenario, are integrated behind a bumper material having multilayered paint materials (Dash et al. 2019). The bumper produces additional path loss thereby affecting the sensitivity of the receiver and detection latency, consequently affecting ranging. In addition, the bumper distorts the antenna radiation pattern that causes ambiguity in the direction of arrival (DoA)/ departure (DoD). Therefore, a systematic study on the electromagnetic effects on the MIMO RADAR antenna performance due to the presence of bumper material property is essential for the automotive industry to choose a proper bumper material to achieve robust and uncompromised RADAR functionality.

MIMO RADAR Working Principle

MIMO RADAR system contains multiple transmitter antennas and multiple receiver antennas which generate multiple EM waveforms and provides the capability to access multiple degrees of freedom (Fishler et al. 2004). This feature has attracted automotive RADAR engineers. MIMO RADAR is of two kinds such as MIMO RADAR with collocated antennas (Li and Stoica 2007) and MIMO RADAR with widely separated antennas (Haimovich et al. 2008). The schematic representation of these two MIMO RADAR types is shown in Fig. 2. Automotive applications focus on the MIMO RADAR with the collocated antennas. This RADAR system contains multiple transmitter and receiver antennas on a single substrate placed at a few wavelengths apart. The signal transmitted by each antenna is different from each nearby antenna. Further, the reflected signal from the target belongs to each transmit antenna and can be separated easily due to signal orthogonality. This feature allows the formation of a virtual array that contains the signal information from each transmitter antenna to the receiver antenna. Figure 3 shows the schematic of the virtual array. Thus, the M transmit antennas and N receive antennas in the MIMO RADAR form the M.N independent transmit-receive pair, where each pair acts as a single monostatic RADAR. The M+N number of physical antennas form M.N signal pairs. This allows the advantage of wide FoV, high angular resolution, and compact

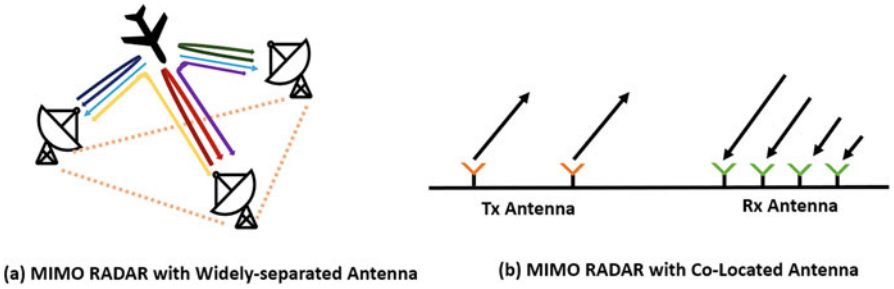

Fig. 2 Schematic representation of two different types of MIMO RADAR based on the antenna placement

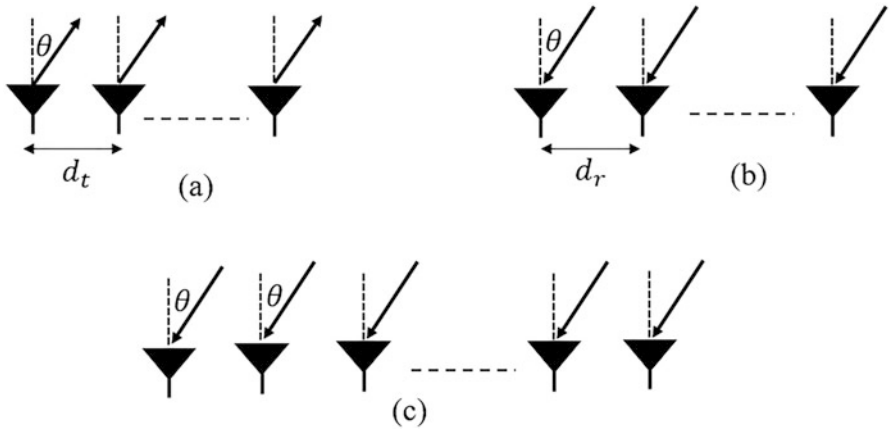

Fig. 3 Schematic to represent virtual array for MIMO RADAR with collocated antennas: (**a**) M transmit antennas; (**b**) N receive antennas; and (**c**) MN × 1 virtual array

size. Equations 1, 2, and 3 (Qu et al. 2008) represent the signal model for MIMO RADAR as shown in Fig. 3.

$$A_t(\theta) = \left[1, e^{-jkd_t \sin(\theta)}, e^{-j2kd_t \sin(\theta)}, \ldots\ldots\ldots, e^{-j(M-1)kd_t \sin(\theta)}\right]^T \quad (1)$$

$$B_r(\theta) = \left[1, e^{-jkd_r \sin(\theta)}, e^{-j2kd_r \sin(\theta)}, \ldots\ldots\ldots, e^{-j(N-1)kd_r \sin(\theta)}\right]^T \quad (2)$$

$$V(\theta) = A_t(\theta) \otimes B_r(\theta)$$

$$= \left[1, e^{-jkd_r \sin(\theta)}, \ldots\ldots, e^{-j(N-1)kd_r \sin(\theta)}, \ldots\ldots, e^{-jkd_t \sin(\theta)}, e^{-jk(d_t+d_r) \sin(\theta)}, \ldots\ldots,\right.$$
$$e^{-jk(d_t+(N-1)d_r) \sin(\theta)} \ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots, e^{-j(M-1)kd_t \sin(\theta)}, \quad (3)$$
$$\left. e^{-jk((M-1)d_t+d_r) \sin(\theta)} \ldots\ldots, e^{-jk((M-1)d_t+(N-1)d_r) \sin(\theta)}\right]^T$$

where $A_t(\theta)$, $B_r(\theta)$, and $V(\theta)$ are the steering vectors for the transmit, receive, and virtual arrays, respectively. d_t and d_r are the inter-element spacing in the transmit and receive array and $\otimes$ represent the Kronecker product.

Though it is quite possible to get high range and Doppler resolution from a stationary RADAR, it suffers from low angular resolution due to the smaller number of physical channels. Angular resolution is directly proportional to the antenna's physical dimension. The virtual array, produced by the MIMO RADAR, bears a larger effective dimension that increases the angular resolution within the small physical size. This unique feature of achieving higher angular resolution through the virtual array implementation has drawn the attention of automotive industries to perceive advanced driver assistance systems (ADAS) using MIMO RADAR for current and future autonomous vehicles. According to the Rayleigh criterion, angular resolution ($\Delta\varphi$) can be defined as $1.22\lambda/d_v$ (Hasch et al. 2012), where λ = free space wavelength and d_v represents the MIMO RADAR virtual array dimension. MIMO RADAR having a 3-transmitter and 5-receiver configuration showing the angular resolution of 5.82° is described in (Dash et al. 2021a).

Overview of Antenna System for Automotive RADAR Application

Automotive MIMO RADAR is classified into long-range RADAR (LRR), medium-range RADAR (MRR), and short-range RADAR (SRR) based on the maximum attainable range and field-of-view (FoV). Nowadays, automotive RADARs are commercially provided by several companies such as Autoliv, Continental ADC, Hitachi, Delphi, Bosch, Fujitsu Ten, InnoSent, Hella, Mitsubishi, TRW Autocruise, and Valeo (Menzel and Moebius 2012). These commercially available RADAR sensors are a complete transceiver module. The antenna system in this RADAR module is the main differentiating factor, because the standard antenna performance parameters such as gain, bandwidth, impedance matching, and the number of antennas used in a RADAR module decide the FoV and the angular separation between the target objects. A brief overview of four antenna system concepts for automotive RADAR application based on the beamforming properties is found in (Hasch et al. 2012) such as (a) quasi-optical beamforming, (b) digital beamforming, (c) analog beamforming, and (d) mechanically scanned array, though it does not address any antenna structure or type of antenna for an autonomous vehicle. However, the antenna type affects the system integration, performance, overall size, and cost of any RADAR module. Moreover, antenna design also decides the RADAR range and the coverage area, i.e., a mid or short-range RADAR requires a wide beam and low-gain antenna. There are certain general antenna types available in the literature (Menzel et al. 2002) for millimeter wave-automotive applications, broadly reflector antenna and planner antenna. Apart from that antenna-in-package and antenna-on-chip are a few state-of-the-art technologies to develop integrated antenna-on-chip, which requires advanced fabrication facility along with the selection of low loss substrate material at 77 GHz millimeter wave application.

Initially, Robert Bosch developed a 77 GHz RADAR lens antenna for automotive applications (Binzer et al. 2007). However, this antenna structure was quite bulky in size to integrate into a vehicle. Instead of complex lens structure planner, reflect array design using periodic or quasi-periodic structure is reported in (Menzel et al. 2002) to focus incident spherical wave and to produce a polarization conversion to the incident wave. An antenna concept for automotive RADAR sensor using planner end-fire Yagi-Uda radiators with cylindrical parabolic reflector is proposed in (Beer et al. 2009). This antenna is designed on an RF substrate while pointing the antenna toward the parabolic reflector. It reflects the EM beam in the desired direction perpendicular to the board in the elevation plane. Though the antenna configurations like reflector or lens are some of the good performance structures, the planner or microstrip structures outperform when the system integration and compatibility come into the picture. Simple structure, low profile, lightweight, low manufacturing cost, and easy to integrate behind the car bumper or fascia patch antenna arrays have been widely used in the automotive application at 76–81 GHz frequency band (Blöecher et al. 2012; Bloecher et al. 2009). A 18×8 element series-fed antenna with Taylor array pattern synthesis is proposed in (Shin et al. 2014) for short- and long-range automotive RADAR application. The sidelobe levels of this proposed antenna are 20.19 dB in the elevation plane and 25.08 dB in the azimuth plane, and a waveguide to microstrip transition is designed to feed the antenna. A transmit array in a two-layer printed circuit board (PCB) with Coplanar patch unit-cells etched on opposite sides of the PCB and connected by through-via is proposed in (Yeap et al. 2015) for automotive RADAR applications. To generate a high gain beam for automotive application, the proposed transmit array is combined with four-substrate-integrated waveguide (SIW) slot antennas as the primary feed. A patch array with a double coupled feeding structure is proposed in (Yoo et al. 2020). The antenna is designed to achieve wideband matching and gain bandwidths for automotive RADAR applications. The antenna consists of a microstrip line with a rounded rectangular radiator with a waveguide feeding. Two arrays having 1×18 and 4×18 elements are presented in (Lee et al. 2020) for millimeter wave application. These are capacitively coupled microstrip comb-line (CCMCA) array antennas for the millimeter-wave band having a low sidelobe level. An efficient leaky-wave antenna (LWA) based on a half-mode substrate-integrated waveguide is proposed in (Front RADAR sensor 2020) for 24 GHz-automotive RADAR application. The prototype exhibits 25% impedance bandwidth with 12.5 dBi gain and 85% efficiency at 24 GHz application frequency. The antenna has a half-power beamwidth of 12°. Apart from that, Table 2 provides more information on the various antenna design aspects at the 77 GHz operating band for automotive radar applications available in the open literature.

The antenna designs available in the literature for automotive RADAR applications are restricted to the design of individual antenna elements and lack discussion of RADAR implementation viewpoint. A modified binomial series-fed microstrip antenna with 3 Tx and 5 Rx MIMO configuration for 77 GHz-automotive RADAR application is provided in (Dash et al. 2021a). The antenna exhibits low sidelobe level ($<$16 dB), low cross-polar level ($<$30 dB), and 80° half-power beamwidths.

Table 2 Antenna types for 77 GHz automotive radar application

Reference	Antenna type	Substrate material	Dielectric constant (thickness: mm)	HPBW (degree) (E/H plane)	Cross polarization (dB)
(Yeap et al. 2015)	Dual-layer transmit array	RO4003C	3.38 (0.203)	15	–
(Hallbjorner et al. 2012)	Lens antenna	Taconic Taclam Plus	2.1 (0.1)	4/4.4	<-19
(Wang and Stelzer 2013)	LTCC patch array antenna	Low-temperature cofired ceramic	7.3 (0.27)	7	–
(Artemenko et al. 2013)	Integrated lens antenna	RO4003C	3.5(−)	18 (12.5 mm lens) and 27 (7.5 mm lens)	<-25
(Imbert et al. 2015)	Dielectric flat lens	Roger TMM6	6 (7)	20	<-20
(Arnieri et al. 2018)	Planar grid array	Roger 3003	3 (0.254)	5	<-22

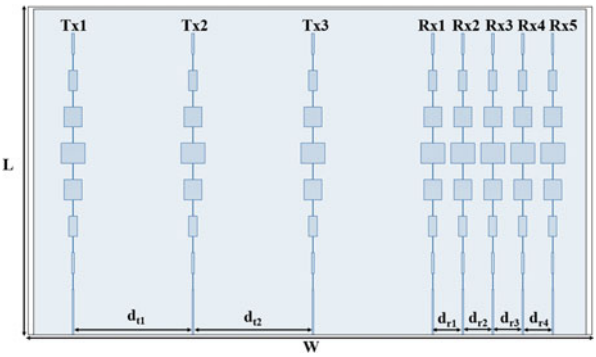

Fig. 4 Schematic of 3Tx-5Rx MIMO RADAR antenna. Dimensions in mm: L = 37.3, W = 22.65, $d_{t1} = d_{t2} = 8$, $d_{r1} = d_{r2} = d_{r3} = d_{r4} = 2$. (See Dash et al. (2021a) for more details). (© IEEE 2021)

Further, the proposed MIMO RADAR antenna configuration provided 5.82° angular resolution. The MIMO RADAR antenna prototype at 77 GHz operating is shown in Fig. 4. Fig. 5 shows the S-parameter curve and the 3D radiation pattern of the individual antenna element in Fig. 4.

More specifically, the antenna design shown in Fig. 4 is a one-dimensional (1D) MIMO RADAR antenna configuration. Note that the 1D MIMO RADAR antenna defines the nature of the formation of the virtual array (see Fig. 3) due to the antenna arrangement as shown in Fig. 4. This antenna arrangement is limited to azimuth angular resolution using 1D MIMO RADAR virtual array as shown in Fig. 3c. In order to support autonomous driving and better capability, angular resolution in both azimuth and elevation is required in addition to the range and

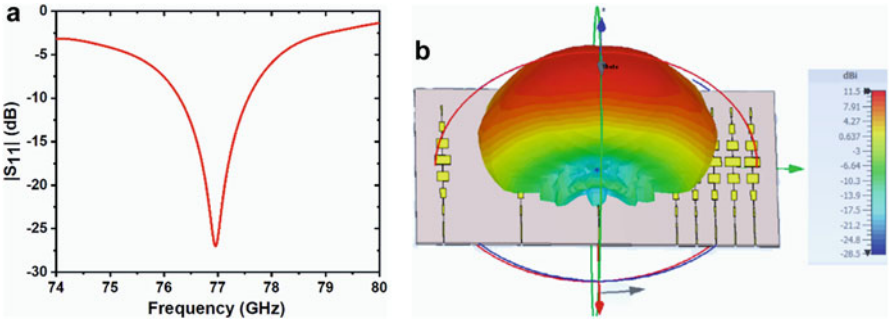

Fig. 5 (**a**) Simulated S_{11} magnitude versus frequency for the series-fed array Tx-1 in Fig. 4; (**b**) simulated 3D radiation pattern of Tx-1 in the MIMO RADAR antenna configuration at 77 GHz. (See Dash et al. (2021a) for more details). (© IEEE 2021)

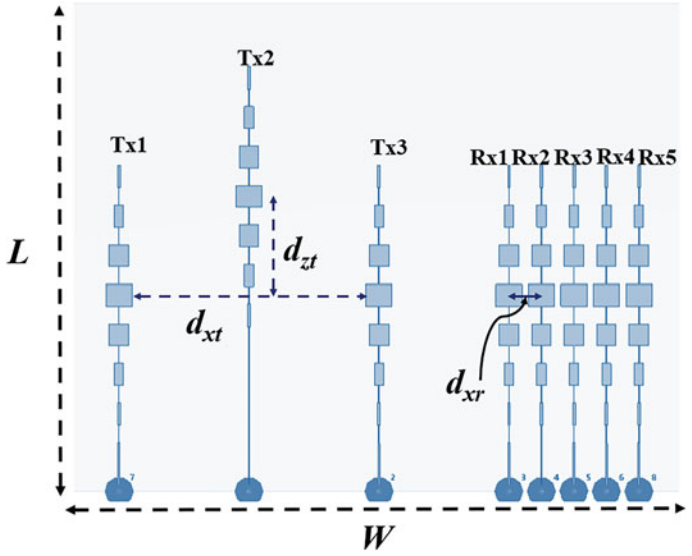

Fig. 6 Schematic of 3Tx-5Rx MIMO RADAR antenna. Individual antenna dimension in mm: L = 30.14, W = 37.3, d_{xt} = 16, d_{zt} = 6.1, d_{xr} = 2. (See Dash et al. (2022a) for more details). (© IEEE 2022)

velocity estimation. The angular resolution in both azimuth and elevation creates the 3D snapshot of the target\cite{cm}. 2D MIMO radar antenna supersedes the 1D MIMO radar antenna configuration by fulfilling the requirement of a 3D image capturing a target/object (Ma et al. 2011). However, the 2D array configuration needs more antenna elements which eventually increases the cost. A 3Tx-5Rx 2D MIMO RADAR antenna configuration is proposed in (Dash et al. 2022a). The schematic of the antenna is shown in Fig. 6. The individual antenna element in Fig. 6 has the identical property as in Fig. 4. Though the antenna type and arrangement for 1D and 2D MIMO RADAR configurations are looking similar, the offset design of Tx2 in

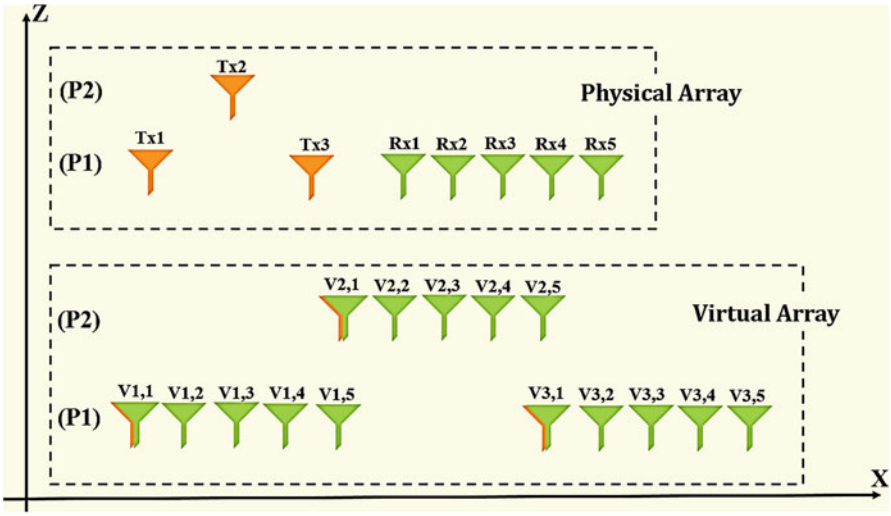

Fig. 7 Pictorial representation the virtual array formation for 2D antenna arrangement of Fig. 6. (See Dash et al. (2022a) for more details). (© IEEE 2022)

Fig. 6 creates the actual difference between 1D and 2D MIMO by providing an additional degree of freedom in elevation plane resolution. The 2D virtual array formation (as shown in Fig. 7) enables the elevation angular resolution of 44.8° and uninterrupted azimuth angular resolution of 5.82°.

Further, in commercial RADAR sensor, Bosch announced its third-generation RADAR sensor called MRR in 2011 (Hasch et al. 2012) and its 5th generation sensor in 2020 (Front RADAR sensor 2020). Compared to the earlier versions, it had different transmitter and receiver antennas on the same module. Instead of a combination of antenna patched and dielectric lens, planner antennas are used comprising four receivers and two transmit antennas. In (Waldschmidt et al. 2021), an MIMO sensor antenna studied by Toyota Laboratories is provided. The antenna consists of six subarrays, three for the transmitter and three for the receiver. Each antenna subarray consists of two series-fed arrays. All subarray provides a relatively wide azimuth beam, and a good angular resolution is achieved by beamforming. However, the issue of the RADAR integration inside a car is not addressed in these articles, which is one of the most critical segments of vehicle manufacturing that severely affect the RADAR performance along with other sensors.

Integration Challenges of Automotive MIMO RADAR Antenna in a Vehicle

Generally, automotive RADAR sensors are integrated behind the bumper, which contains multiple paint layers. Without disturbing the aerodynamic profile and outer body look of the vehicle, the RADAR sensors must be mounted behind the bumper,

which is a significant challenge. When the sensor antenna radiates the EM wave, the wave passes through the bumper painted with multiple paint layers. Hence, the wave must propagate through the bumper and then through the painted media in the transmitting mode and vice versa for receiving mode. The bumper with layers of paint can be considered as a stratified medium where each layer has its thickness, dielectric constant, and loss tangent. These painted bumpers affect the EM radiation of the RADAR and thereby affecting the RADAR range and velocity estimation.

In the simulation stage of the RADAR antenna and bumper-integrated system, these painted layers affect the overall simulation time and system memory. More specifically, it is well known that the vehicle-integrated system is electrically very large. On the other hand, at a 77 GHz millimeter wave band the EM simulator generates billions of mesh cells which require a very high simulation time as well as a large memory requirement. A high-power computing facility could be helpful to solve this kind of simulation requirement. However, to simplify this problem an alternative solution to this issue is provided in (Dash et al. 2019), which deals with the reduction of overall simulation time of the RADAR antenna and bumper-integrated system without affecting the accuracy. The technique uses the modeling of multilayer media to a single-layer equivalent media using the effective medium theory. The equivalent modeling of stratified media considers two different model approaches such as (a) parallel plate capacitor model, and (b) model based on the same transmission and reflection coefficient. Again, these models consider two simulation implementations. A simplified rectangular bumper with six paint layer model is considered for the validation of the proposed model, where the simulation approach (1) considers the equivalent model generation by bumper and paint layers to a single equivalent layer whereas simulation approach (2) considers the equivalent model generation by converting only paint layers to a single equivalent layer by keeping the bumper as a separate layer as shown in Fig. 8a, b, respectively. Detailed mathematical formulation and implementation using CST microwave studio EM simulation are described in (Dash et al. 2019). The proposed model reduces the simulation time by 88% and 57% in simulation approaches (1) and (2), respectively.

In a realistic autonomous vehicle application scenario, RADAR sensors are placed behind the car bumper because of which the sensors get affected by the bumper material as well as the high dielectric constant paint layers. In addition, the vehicle esthetic and the curvature of the bumper also affect the EM radiation from the RADAR. Moreover, the distortion in the antenna radiation pattern due to the bumper produces ambiguity in the DoA/DoD (direction of arrival/departure). There are certain earlier studies on the antenna with radome-integrated structures available in the literature. A physical optics-based tool is presented in (Schuster and Cassidian 2011) to study the effect of a spherical and ogival radome on an antenna. Similarly, an iterative physical optics-based technique is proposed in (Rim et al. 2016) to study the radiation properties at high operating frequency in the presence of an elliptical radome. However, it should be noted that the radome design considers electromagnetic transparency. On the other hand, bumpers consider the esthetic of the vehicle instead of electromagnetic effects. Therefore, the electromagnetic effect of bumper material property as well as the bumper curvature effect on the MIMO RADAR

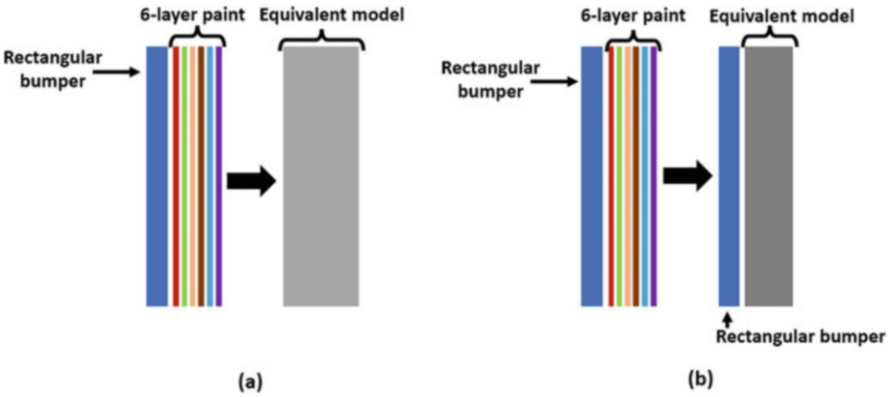

Fig. 8 Equivalent model of a rectangular bumper with multi-layered paints based on (**a**) simulation approach 1, and (**b**) simulation approach 2

antenna performance needs to be investigated since it will assist automotive industries to choose the proper bumper/paint material for robust and uncompromised functionality.

To study the RADAR antenna-bumper-integrated effect, a 2Tx-4Rx MIMO RADAR antenna configuration is designed as shown in Fig. 9 (accordingly to the design parameter provided in (Dash and Sarkar 2021)). The MIMO configuration is designed at 77 GHz automotive RADAR application frequency on a Roger 5880LZ substrate having dielectric permittivity and loss tangent 2 and 0.0021, respectively, with 0.25 mm substrate thickness. Other design parameters of the antenna are shown in Fig. 9. Similarly, the design of a simplified curved bumper configuration is shown in Fig. 10 (Dash and Sarkar 2021). The relative permittivity of the bumper is varied from 2 to 8 to study the effect of bumper dielectric property on the RADAR antenna radiation pattern. The 3D field pattern of Transmitter-1 for the integrated antenna-bumper system for the different values of bumper dielectric constant is shown in Fig. 11, which clearly shows the variation of the shape of the field pattern with respect to the change in the bumper dielectric constant. To further quantify the effect, two different analytical parameters such as (a) bidirectional loss model (BLM) (Dash et al. 2021b) and (b) antenna array ambiguity function (AAAF) (Dash and Sarkar 2021) are considered here.

(a) Bidirectional loss model (BLM): The BLM takes into account the two-way mismatch loss (ML) and dielectric loss (DL), for to and fro EM wave propagation of RADAR, as well as considers the effect of bumper shape (see Dash et al. (2021b) for a more detailed explanation).
(b) Antenna Array Ambiguity Function (AAAF): The AAAF considers the RADAR antenna pattern distortion when the RADAR is installed behind a bumper. It is evaluated using the MIMO antenna steering vector model as described in the "MIMO RADAR Working Principle" section. Mathematically, AAAF can be written as (Dash and Sarkar 2021)

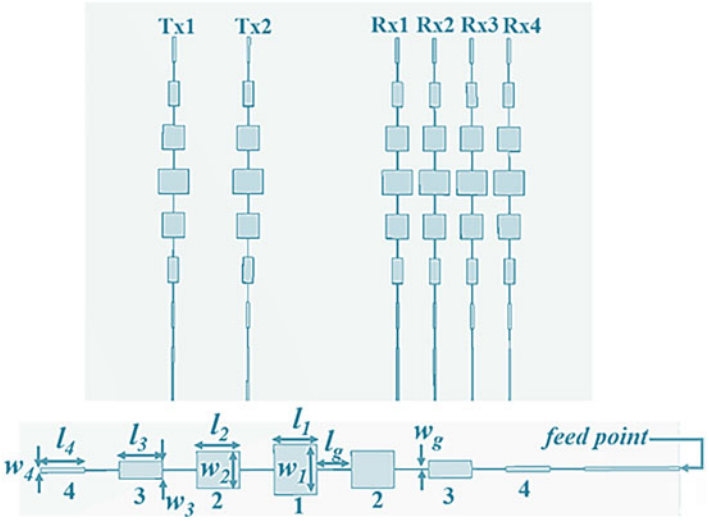

Fig. 9 Schematic of 2Tx-4Rx MIMO RADAR (individual antenna dimensions in mm: $l1 = 1.35$, $w1 = 1.6$, $l2 = 1.35$, $w2 = 1.185$, $l3 = 1.36$, $w3 = 0.553$, $l_4 = 1.39$, $w_4 = 0.185$, $l_g = 1.085$, and $w_g = 0.03$). © IEEE 2021

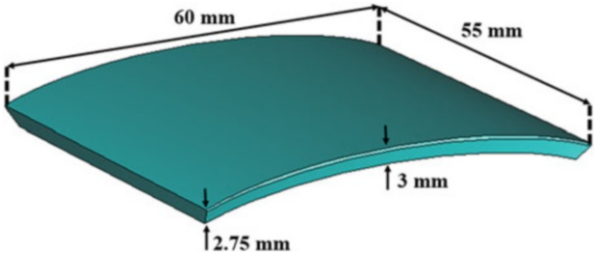

Fig. 10 Simplified curve-shaped 3D CAD model of the bumper used to integrate with the proposed MIMO RADAR as shown in Fig. 9 (Dash and Sarkar 2021). © IEEE 2021

$$\chi(\alpha_i, \alpha_j) = \frac{|\, Y^H(\alpha_i).Y(\alpha_j)\,|}{\|Y(\alpha_i)\|.\|Y(\alpha_j)\|} \tag{4}$$

where $(.)^H$ represents the Hermitian operator, $\|\,.\,\|$ represents the norm operator, $Y(\alpha)$ represents the antenna azimuth virtual array field pattern, and α represents the azimuth angle. Readers can refer (Dash and Sarkar 2021) for a more detailed explanation.

Figures 12 and 13 indicated the computed bidirectional loss (BL) and ambiguity function (AF) plot, respectively, for the integrated antenna-bumper system with the varied bumper material property. AF indicates that the bumper dielectric constants 5, 6, and 8 are creating more ambiguity in the field pattern than that of the other bumper permittivity under consideration. In addition, the bidirectional loss is

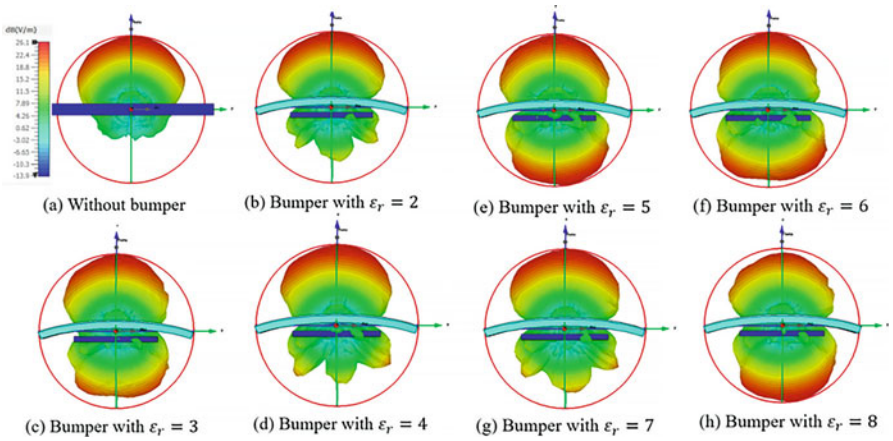

Fig. 11 The 3D E-field radiation patterns of transmitter-1 in the integrated antenna-bumper system shows pattern distortion caused by the permittivity variation of bumper. (See Fig. 10) (Dash and Sarkar 2021). (© IEEE 2021)

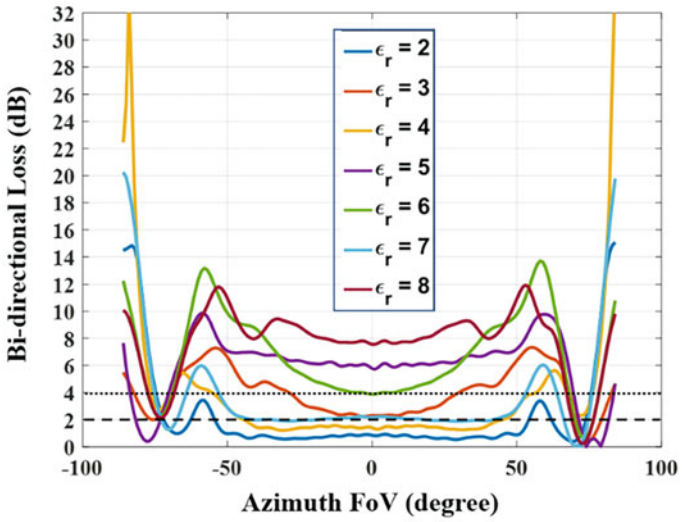

Fig. 12 Bi-directional loss plots for different material permittivity values of the bumper in the integrated antenna-bumper system (Dash and Sarkar 2021). © IEEE 2021

consistent with the ambiguity as it can be observed that the BL value is high for the same value of bumper permittivity as AF. This study provides the direction to study the antenna-bumper-integrated system which would be considered for other application scenarios where EM wave is attenuated due to the integration issue.

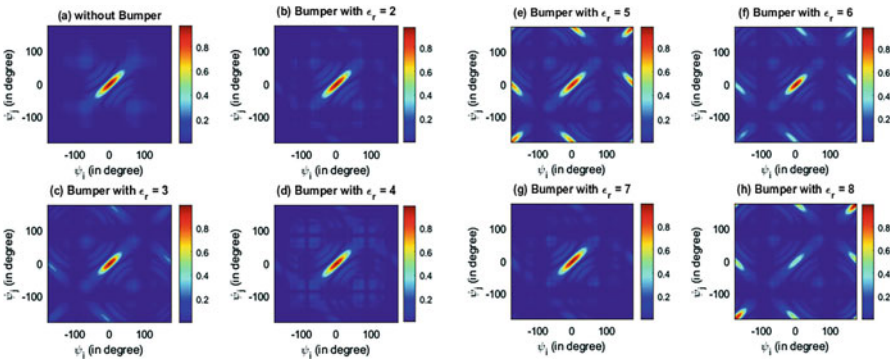

Fig. 13 Two-dimensional plots of AAAF for different bumper permittivity in the antenna-bumper-integrated system (Dash and Sarkar 2021). © IEEE 2021

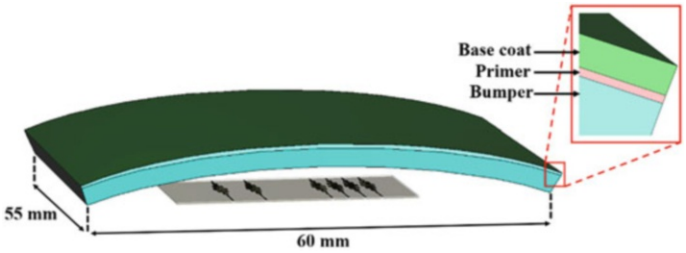

Fig. 14 Schematic of 2Tx-4Rx antenna-painted bumper-integrated system (Dash et al. 2022b). © IEEE 2022

Similar studies on the antenna-bumper integration effect, using BLM and AAAF methods, are conducted in (Dash et al. 2022b) to observe the effects when the bumper is painted. Here, to study the effect of paint on 2Tx-4Rx RADAR (as shown in Fig. 9), performance of two layers of paint material such as primer coat (ε_r =9, tan δ =0.06) and base coat (ε_r =9.99, tan δ =0.907) having thicknesses of 4 um and 15 um, respectively, are applied on the bumper (ε_r =2, tan δ =0.01) (see Fig. 14). In general, automotive industries put six-to-seven layers of paint on their vehicles with varied thicknesses (Dash et al. 2019). However, for the initial study purpose and to restrict the simulation time, the number of paint layers and their thicknesses are kept limited.

It can be seen from Fig. 15 that the bumper alone produces a loss of less than 2 dB in FoV(−40° to +40°). Gradually, the loss started fluctuating due to the addition of primer coat, and when the base coat is added the loss increases to near 4 dB within FoV though it is less in broadside. It can be seen from the AAAF plot in Fig. 16 that bumper alone has a negligible effect on RADAR FoV; however, the FoV reduces by the addition of primer coat (see Fig. 16c), and it reduces significantly due to the additional base coat layer (see Fig. 14d).

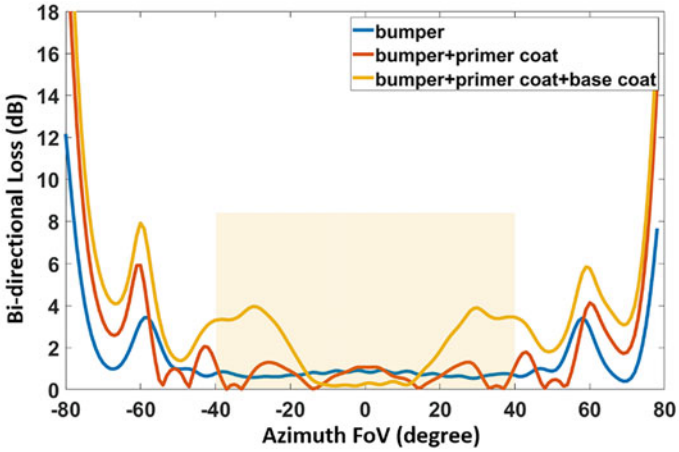

Fig. 15 Variation of bidirectional loss for the antenna-painted bumper-integrated system of Fig. 14 (Dash et al. 2022b). © IEEE 2022

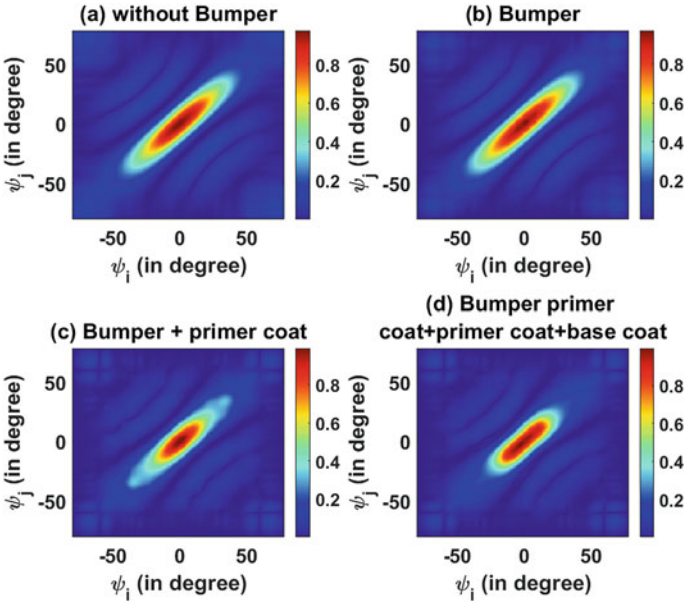

Fig. 16 Two-dimensional plots of AAAF for the antenna-bumper-integrated system of Fig. 14 (Dash et al. 2022b). © IEEE 2022

Conclusion

A comprehensive discussion on the application of MIMO RADAR in the advanced driver assistance system is discussed. The operating principle of MIMO RADAR with an explanation of the virtual array concept for 1D and 2D MIMO RADAR configurations is discussed. Further, a discussion on the state-of-the-art antenna design for the automotive RADAR application is provided based on the research literature as well as the commercially available design. Finally, the very practical issue of MIMO RADAR integration behind the multilayered vehicle bumper is explained as well as the challenges in a simulation environment, and the real environment is examined using the effective dielectric constant model and BLM and AAAF model, respectively. Such studies are potentially helpful to automotive industries to obtain more insights about integrated antenna performance and to provide better solutions with high system efficiency during real-time operation.

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Radar Cross-Section (RCS) Estimation and Reduction

61

Vineetha Joy and Hema Singh

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Abstract

The accurate prediction of radar cross-section (RCS) of various structures is crucial for controlling its visibility and hence plays a vital role in the development of stealth technologies. Several approaches like body geometry decomposition, low frequency techniques, high frequency techniques, and hybrid methods can be employed for RCS estimation based on the complexity of geometry and electrical size of the structure. Scattering entities especially electrically large ones give rise to a wide variety of electromagnetic phenomena like refraction, reflection, traveling waves, leaky waves, diffraction effects, etc. Although low frequency techniques provide accurate predictions for complex geometries, the computational complexity increases with electrical size of the structure. On the other hand, high frequency approaches, although computationally efficient for electrically large problems, give incorrect results when the problem under consideration has intricate geometrical features. Therefore, hybrid approaches incorporating both of them have been introduced to solve electromagnetic problems where neither class of formulations is sufficient by itself. This chapter presents an overview of these methods and certain case studies are also included for better clarity. Further, the popular approaches employed for reduction of RCS are also included with

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special emphasis on radar absorbing materials (RAM)/radar absorbing structure (RAS) and antenna array.

Keywords

Radar cross-section (RCS) · Radar absorbing structure · Radar cross-section reduction · Reflectivity · Absorber · Low RCS antenna

Introduction

The accurate prediction of radar cross-section (RCS) of various structures is crucial for controlling its visibility and hence plays a vital role in the development of stealth technologies. The challenge is to estimate the scattering from the objects of arbitrary shape, size, and material. Moreover, the RCS estimation and measurements depend on the frequency of the incident signal, the orientation of the object w.r.t. the radar, and the polarization (Knott et al. 2004; Knott 2006). In this direction, a concise description of few of the techniques used for estimation of RCS and the popular approaches employed for reduction of the same are presented. Some case studies and results are also included for better understanding. In order to reduce the target detectability, radar absorbing materials and structures (RAM/RAS) are employed over the target surface. In current trend, frequency selective surfaces (FSS), electromagnetic bandgap (EBG) materials, artificial magnetic conductors (AMC), etc. are incorporated in the RAS to achieve wideband electromagnetic (EM) absorption. However, the researchers face challenges in these design and development in view of payload constraints, bandwidth requirements, structural strength, RF power handling capability, and environmental conditions. In particular, RCS reduction in the antennas has added constraint because any modification in the design should not degrade the antenna radiation performance in terms of its VSWR, antenna gain, and beamwidth (Singh and Singh 2021).

Basic Principles of Radar Cross-Section (RCS)

When an electromagnetic (EM) wave is incident on a target, energy is reradiated back in different directions. This reradiation is called scattering and the target itself is called scatterer. RCS is the effective area contributing to the scattering of EM waves back toward the probing radar. It can be defined as

$$\sigma = \lim_{R \to \infty} 4\Pi R^2 \frac{|E_{\text{sca}}|^2}{|E_{\text{inc}}|^2} \tag{1}$$

where

σ: radar cross-section of the target (m^2)
E_{sca}: electric field intensity of scattered wave (V/m)
E_{inc}: electric field intensity of incident wave (V/m)
R: range from the radar to the target (m)

RCS is usually expressed in dB relative to one square meter (dBsm) as given below:

$$\sigma(\text{dBsm}) = 10 \log_{10} \sigma\left(\text{m}^2\right) \tag{2}$$

The significant factors which determine the magnitude of RCS are the following:

(a) Shape of the target and roughness of the surface
(b) Material composition of the target (EM properties)
(c) Frequency and polarization of the incident EM wave
(d) Aspect angle of the target w.r.t. incident wave

As an interesting fact, RCS of a stealth aircraft like Northrop Grumman B-2 Spirit, also known as the stealth bomber, is of the order of −40 dBsm which is comparable to that of a small insect.

Estimation of RCS

Scattering entities, especially electrically large ones, give rise to a wide variety of electromagnetic phenomena like refraction, reflection, traveling waves, leaky waves, diffraction effects, etc. In addition to the complex geometrical features of the primary reflecting surface, ground planes as well as other proximal structures contribute to these effects making the problem even more intractable. Earlier, such complex scenarios have been solved using body geometry decomposition. Here, the structure is considered as a combination of different canonical shapes. Using the known solutions for these generic shapes, the overall RCS is computed by adding together (coherently or incoherently) the individual contributions. The main advantage of this technique is that the RCS of electrically large structures can be estimated without any exhaustive computations. However, it considers only first-order/specular edge scattering and does not take into account the interaction between different components.

In order to overcome the limitations of body geometry decomposition technique, two main approaches have been proposed by researchers around the globe (Mitschang and Wang 1989). They are categorized as given below:

(i) High frequency techniques
(ii) Low frequency/numerical techniques

High frequency approaches precisely account for the diffraction phenomena occurring along curved surfaces/edges using physical theory of diffraction (PTD), geometrical theory of diffraction (GTD), etc. These optics-derived techniques provide comparatively accurate scattering predictions without resorting to extensive computations. Hence, they are the preferred mode of analysis for electrically large bodies composed of relatively simple generic shapes. However, the main disadvantage of these asymptotic techniques is that the diffraction coefficients are available only for a limited set of canonic geometries like cylinders, spheres, and wedges. Further, the simplifying assumptions made during calculations lead to comparatively approximate solutions rather than accurate predictions.

On the other hand, low frequency approaches like method of moments (MoM), finite element method (FEM), finite-difference time-domain (FDTD) method, etc. provide full wave solutions to scattering problems. These techniques provide accurate predictions for intricate geometries thoroughly incorporating the interaction effects between different parts of the structure. For instance, FDTD method is a differential time-domain numerical method. Here, Maxwell's equations are expressed in terms of finite difference approximations and then the space and time domains are discretized. The electric and magnetic fields are then determined with the field calculations interleaved in both time and space. FDTD requires the problem to be meshed with fine spatial discretization, and therefore the computational complexity increases with increase in electrical size of the structure (Teixeira and Chew 2000).

FEM is another numerical method for solving scattering problems in electromagnetics (Bondeson and Rylander 2013). In FEM, a problem is analyzed in four major steps as given below:

(i) Segmentation or meshing of the structure into a finite number of elements such that each element has a homogenous dielectric associated with it
(ii) Derivation of governing equations for a typical element
(iii) Assembly of all the elements of the structure to generate the system of equations
(iv) Evaluation of the system of equations to determine the unknowns

In FEM, we have to solve a linear system of equations in order to update the fields. Consequently, for the same number of cells used for both FEM and FDTD, FEM needs more computer resources, both in terms of CPU time and memory.

Integral equation (IE) technique is another popular numerical method for the analysis of metallic and dielectric structures of arbitrary geometry (Harrington 2001). Integral eqs. (IE) are inferred from Maxwell's equations through the use of equivalence principles. They are further discretized through MoM into a linear system of equations whose solutions are the aforementioned equivalent sources. In surface integral techniques (SIE), the dielectric-to-dielectric interfaces are removed by placing equivalent electric and magnetic surface currents. The prime concern of SIE is the evaluation of Green's function, which is the electromagnetic response of a point source in a given environment. Although mathematical derivation of SIE is

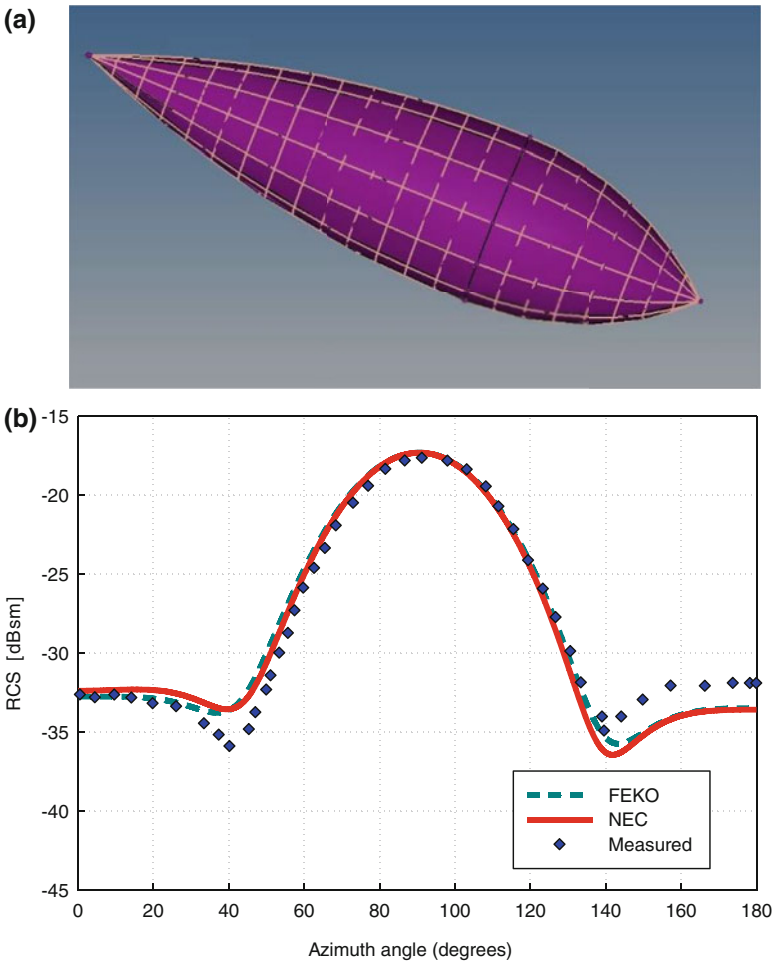

Fig. 1 Variation of monostatic RCS of metallic double ogive *w.r.t.* azimuth angle at 1.57 GHz. (**a**) Meshed model of metallic double ogive, (**b**) RCS characteristics (vertical polarization)

possible, Green's functions for arbitrary inhomogeneous medium is nontrivial and can be obtained only via numerical computation. This can be as tedious as solving the original problem.

Among all the low frequency techniques, MoM is exceptionally efficient for the scattering characterization of conducting structures. Currently, several commercial EM simulation software (like Altair FEKO) are available with MoM as the solver. Open-source codes like numerical electromagnetics code (NEC) are also widely used for the RCS analysis of metallic geometries. The monostatic RCS of a typical benchmark radar target like metallic double ogive computed using NEC and FEKO is shown in Fig. 1 (Joy et al. 2020). It is clear that the results are well matched to the measured data presented in Woo et al. (1993).

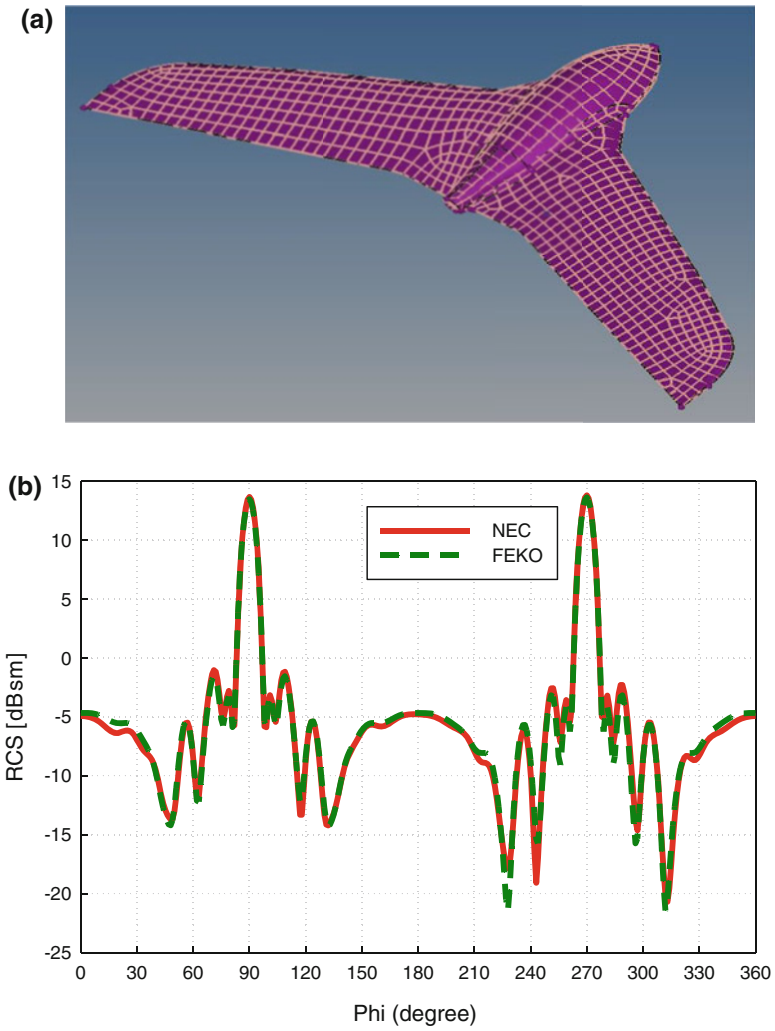

Fig. 2 Variation of monostatic RCS of metallic UAV *w.r.t.* azimuth angle at 600 MHz. (**a**) Meshed model of metallic UAV, (**b**) RCS characteristics

Further, the monostatic RCS of a complex aerospace structure like unmanned aerial vehicle (UAV) computed using NEC as well as FEKO is presented in Fig. 2. The plot clearly indicates very good agreement between both the results.

Next is the antenna RCS estimation. Here a single patch antenna is considered. Figure 3a shows the schematic of single patch antenna designed in full wave simulation software, FEKO and CST Microwave Studio, respectively. The patch element made of copper ($\sigma = 5.8 \times 10^7$ S/m) material has dimension of 6.5 mm $\times$ 9.2 mm. The antenna is fed by a microstrip feed line (length $=$ 5.75 mm, width $=$ 0.9 mm). The patch is backed by 1.588-mm-thick FR4 ($\varepsilon_r = 4.3$, tan $\delta = 0.025$)

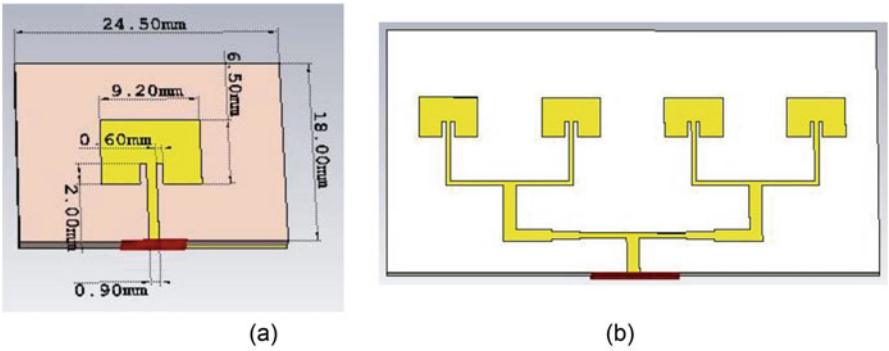

Fig. 3 Schematic showing the design of (**a**) single patch antenna, (**b**) four-element patch array with conventional PEC ground plane

substrate of dimension 18 mm × 24.5 mm. The ground plane is a copper layer with 0.018 mm thickness.

The radiation characteristics of the patch antenna have been analyzed using MoM-based FEKO and FEM-based CST. It can be noted from the return loss shown in Fig. 4a that the resonant frequency of the single patch antenna obtained through both the simulation software are approximately the same. The result obtained in CST shows that the antenna resonates at 9.93 GHz with a return loss of −18.23 dB. Using FEKO, minimum return loss (−20.79 dB) of the same patch is attained at 9.92 GHz. A slight variation in the gain of the antennas at their resonant frequencies is also evident. The maximum gain achieved is 4.85 dB (with CST), whereas it is 4.46 dB with FEKO (Fig. 4b). These discrepancies can be attributed to the different feeding techniques employed in the two cases – a wire feed in FEKO, whereas a waveguide port in CST.

The monostatic RCS of the single patch antenna is computed by impinging plane wave over the surface of the antenna. Figure 5 presents the structural RCS pattern obtained using both the full wave simulation software. The specular RCS of the antenna obtained with CST and FEKO is −25.79 dBsm and −25.59 dBsm, respectively.

A corporate-fed linear microstrip patch array with conventional metallic ground plane shown in Fig. 3b is designed using the same patch element (Fig. 3a) and FR4 substrate. The inter-element spacing between the patch elements has been kept more than 0.5λ in order to avoid mutual coupling effects. The array is placed on a FR4-based substrate of 40 mm × 77.2 mm × 1.588 mm dimensions. The bottom layer is a metallic ground plane.

The radiation and scattering characteristics of the four-element linear patch array are shown in Fig. 6. The results obtained are analyzed and compared using CST and FEKO. It can be noted that the patch array resonates at 10.56 GHz with a return loss of −46.3 dB in CST, while −31.68 dB at 10.48 GHz in FEKO (Fig. 6a). Further, it

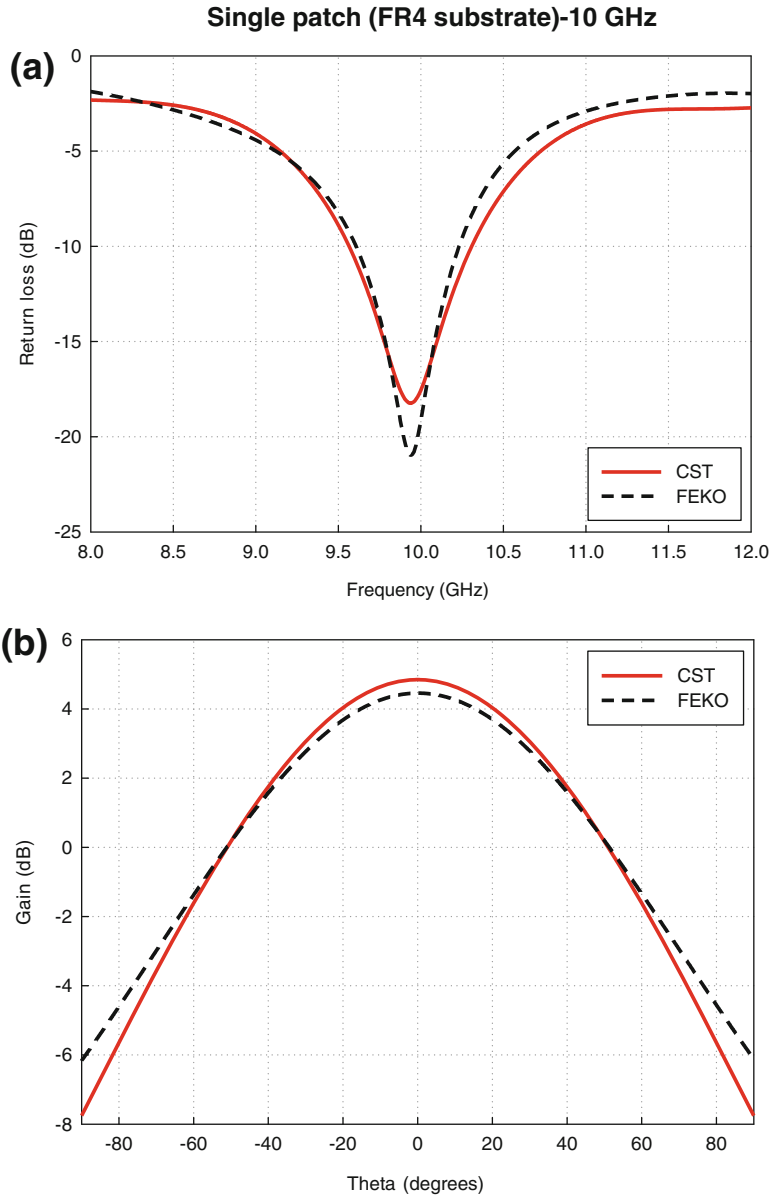

Fig. 4 Radiation characteristics of a single patch antenna with conventional ground plane. (**a**) Return loss, (**b**) gain

can be observed from Fig. 6b that there is a slight variation in the gain obtained in the two cases, i.e., 6.95 dB from CST, whereas 6.29 dB from FEKO. As already mentioned, the difference in the radiation characteristics obtained in the two cases

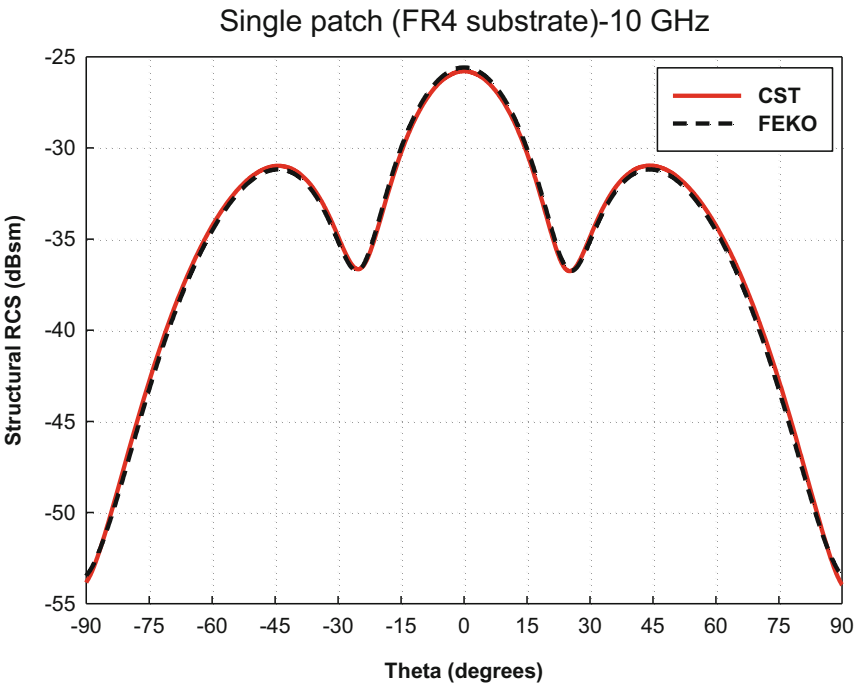

Fig. 5 Monostatic RCS of single patch antenna at 10 GHz

is due to the different feeding techniques involved in CST and FEKO. Figure 6c shows the structural RCS of the patch array for elevation angle ranging from −90° to 90° at 10 GHz. It can be noted that the structural RCS of the patch array obtained is similar using both CST and FEKO with a specular lobe of −8.83 dB at 10 GHz. The antenna RCS is estimated using plane wave source and the antenna structure only contributes to the array RCS.

However, when the antenna is radiating, i.e., switched on, then the antenna mode RCS comes into picture. This antenna mode (radiation mode) RCS is many folds higher than the structural RCS of the antenna (Singh and Jha 2015). This radiation mode is calculated by considering all the reflections and transmissions an impinging signal undergoes while traversing through the antenna and its feed network. This antenna RCS is also proportional to the antenna gain in addition to the frequency, scan angle, and antenna design parameters (Singh et al. 2020). All the impedance mismatches within the antenna and its feed network contributes to the antenna scattering. The RCS pattern of an antenna array is equivalent to the radiation pattern of the antenna array with double inter-element spacing. Essentially, the antenna RCS has twofold impact on the overall RCS of platform over which it is mounted. Thus, for stealth platforms, conformal antenna/array is employed to avoid any protruding structure giving rise to the scattering.

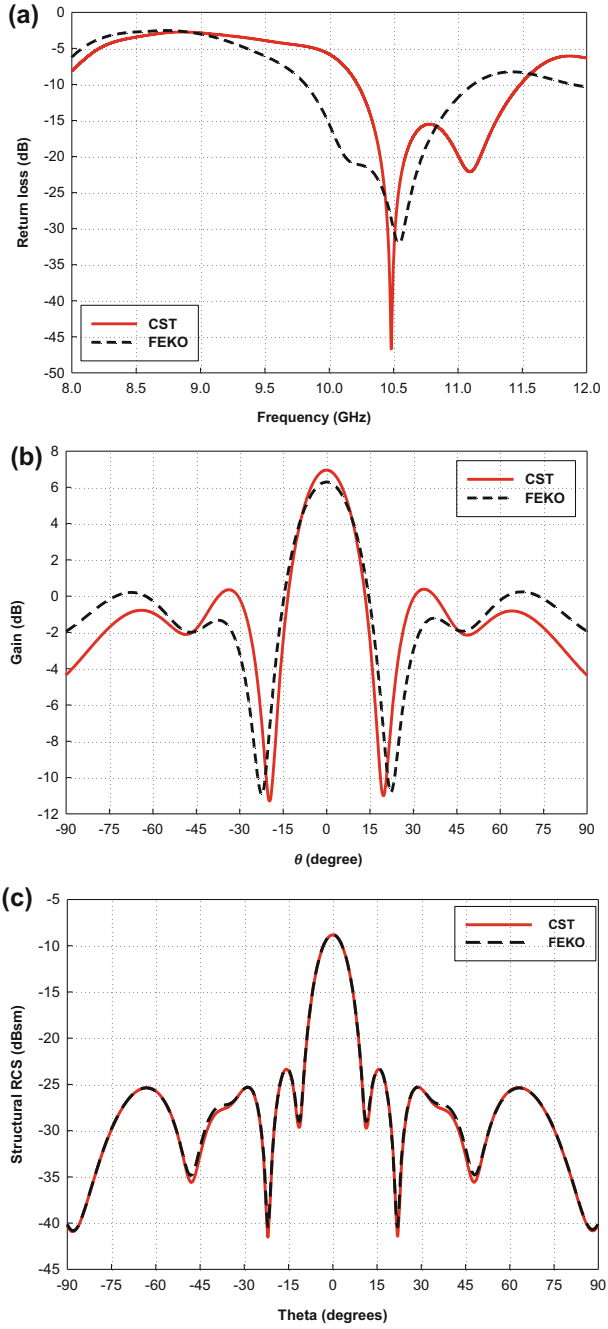

Fig. 6 Radiation and scattering characteristics of a four-element patch array with conventional ground plane. (**a**) Return loss, (**b**) gain, (**c**) antenna RCS at 10 GHz

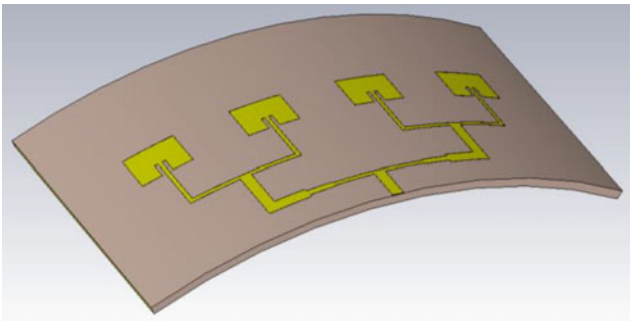

Fig. 7 Conformal patch array with conventional PEC ground plane

Thus, in order to demonstrate the RCS estimation, a patch array conformed to a cylindrical surface with radius of curvature 90 mm. Figure 7 shows the conformal patch array with conventional metallic ground plane. The patch elements are of the same dimension as mentioned for planar patch array, except for the substrate dimension being modified to 45 mm × 90 mm × 1.59 mm.

Figure 8 shows the radiation characteristics of the conformal patch array. It can be observed that the array resonates at 10.44 GHz, with a minimum return loss of 27.23 dB. The patch array has a maximum gain of 5.91 dB at the resonant frequency. Figure 9 presents the structural RCS at 10.44 GHz in the elevation plane ranging from $-90^\circ \leq \theta \leq 90^\circ$. It can be noted that the specular lobe has a null with a magnitude of −17.55 dBsm at the resonant frequency.

The variation of structural RCS of the conformal patch array for a frequency range of 8 GHz–30 GHz in azimuthal plane, $\theta = 0^\circ$ (specular RCS), is shown in Fig. 9b. The structural RCS in the specular direction for a planar and conformal four-element patch array are compared. It can be noted that a broadband reduction in specular RCS is obtained for the conformal patch array when compared to the planar patch array of the same design configuration. A maximum reduction of 12.12 dBsm is obtained in the X-band, i.e., at 11 GHz. In Ku-band maximum specular RCS reduction of 12.26 dBsm is obtained at 13GHz. In K-band a maximum reduction of 15.69 dBsm is achieved in specular RCS at 24 GHz, while in Ka-band a maximum reduction of 18.34 dBsm is obtained in specular RCS at 26 GHz. Thus, array RCS can be reduced by shaping the antenna array, i.e., conforming the antenna over the surface. In order to achieve both in-band and out-of-the-band RCS reduction in antenna array, other techniques (FSS/AMC, slotted/defected ground plane, meta-materials) are employed.

One needs to keep in mind that although low frequency approaches like MoM are suitable for a variety of scattering problems, computational intensity is a limiting factor especially in the case of electrically large structures. Here, the computation time and memory requirements increase with increase in the electrical size of the structure. Considering the inherent drawbacks of both low frequency and high frequency techniques, hybrid approaches incorporating both of them have been introduced to widen the range of EM scattering issues that can be precisely dealt

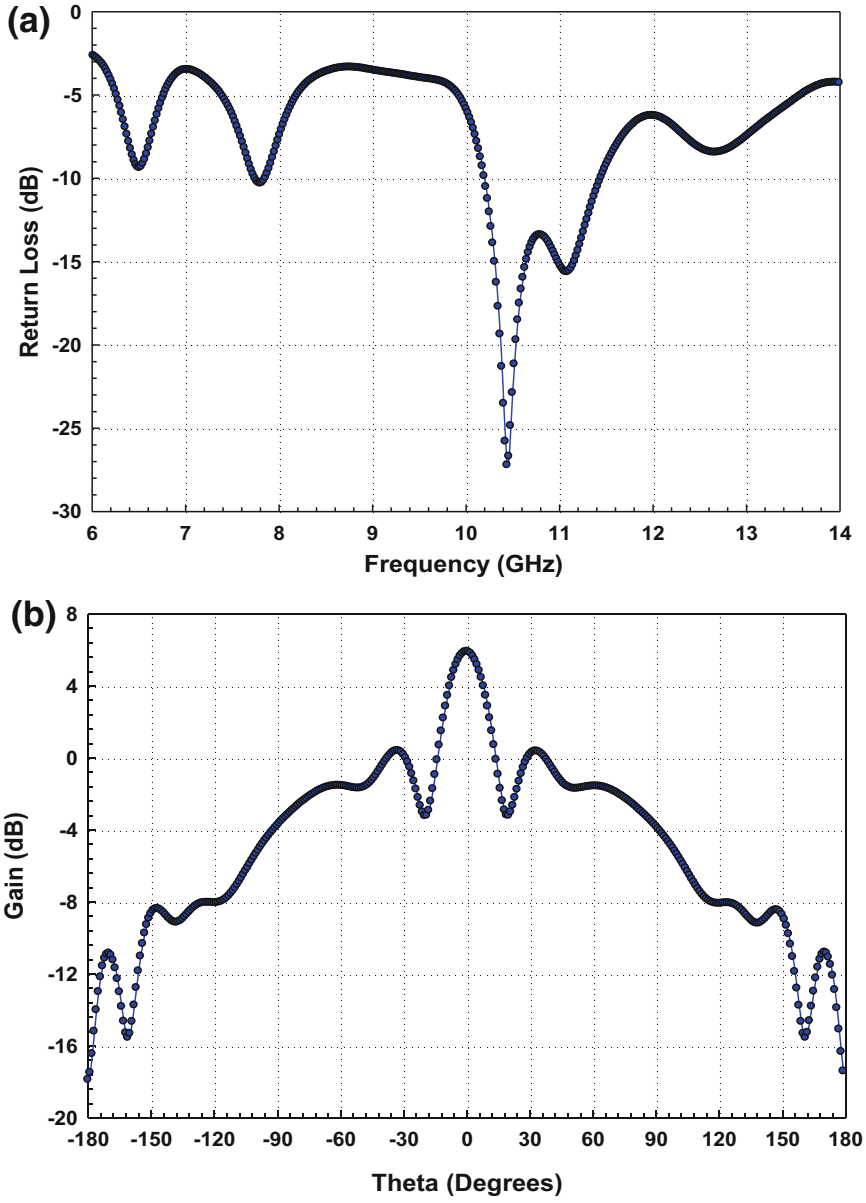

Fig. 8 Radiation characteristics of a four-element conformal patch array with PEC ground plane. (**a**) Return loss, (**b**) gain

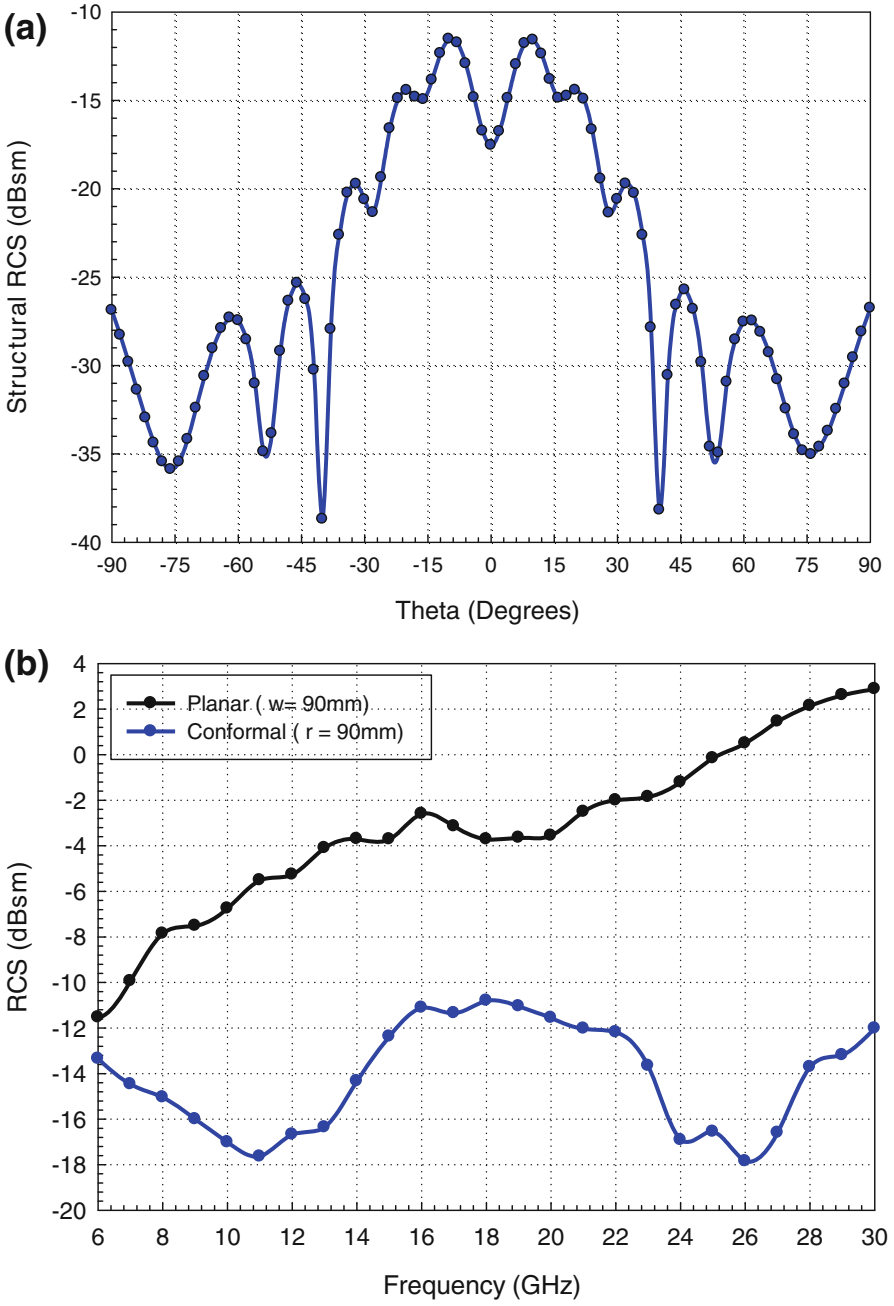

Fig. 9 Structural RCS of a four-element conformal patch array with PEC ground plane. (**a**) At frequency = 10.34 GHz, (**b**) specular RCS versus frequency

with (Bouche et al. 1993). Hybrid methods like physical optics (PO)-MoM technique are basically logical combinations of asymptotic and numerical methods for solving EM problems where neither class of formulations is sufficient by itself.

RCS Reduction

Majority of complex structures will have certain predominant hotspots which contributes significantly to the total RCS. For instance, a stealth aircraft has numerous hotspots like air intake cavity, antenna behind radome, dihedral corner reflector, leading wing edge, etc. These locations scatter majority of the electromagnetic energy incident on them, thereby increasing their visibility to probing radars. Therefore, in order to a make a structure stealthy, RCS of the major scatterers on it are to be reduced drastically.

RCS of a structure can be reduced in four ways as shown in Fig. 10. One approach used at the preliminary design stage is to modify the shape so as to deflect the incident waves in a direction different from that toward the probing radar. Another approach includes the application of radar absorbing materials (RAM) or radar absorbing structures (RAS) at pre-identified major scattering locations. The main objective here is to absorb or diffuse the incident signals so as to reduce the reflections toward rival radar systems. Passive cancellation is another technique where an additional scatterer is employed to cancel the reflections from the primary scatterer. An alternate approach called active cancellation uses active transmitters to generate modified waveforms that will destructively interfere with the original waves reflected from the structure (Cadirci 2009).

Among all these approaches, application of RAM/RAS is widely used due to the feasibility of varied range of designs catering to different locations and frequency ranges. Further, with rapid advancements in the field of material processing and fabrication technologies, the realization of RAM/RAS has become much simpler.

Methods to Minimize Reflectivity Using RAM/RAS

The primary goal while designing a RAM/RAS is to achieve superior power absorption over a wide range of frequencies for any angle of incidence and arbitrary polarization. Both reflection and transmission have to be reduced in order to enhance absorption. This section gives a theoretical background on the ways in which reflection can be reduced when an electromagnetic wave interacts with a homogenous media (Saville 2005). The reflection of an EM wave incident on a material (free

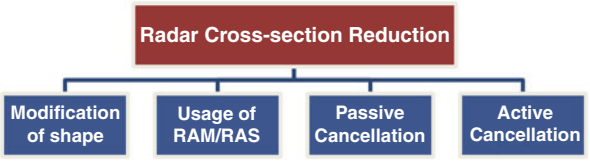

Fig. 10 Different approaches for reducing RCS of a structure

space is taken as the incident medium) can be expressed in terms of the reflection coefficient (Γ) at the interface as

$$\Gamma = \frac{Z_{\text{media}} - Z_{f_\text{space}}}{Z_{\text{media}} + Z_{f_\text{space}}} = \frac{Y_{\text{media}} - Y_{f_\text{space}}}{Y_{\text{media}} + Y_{f_\text{space}}} \tag{3}$$

where Z_{media} and Y_{media} are the impedance and admittance of the propagating medium and Z_{f_space} and Y_{f_space} are the impedance and admittance of free space, respectively. It can be observed from (3) that the reflection coefficient at the interface between two media depends on their respective impedance/admittance values. Based on (3), the following approaches can be employed for minimizing RCS:

(i) It is clear that for minimum reflection ($\Gamma = 0$), the impedance of the propagating medium must be perfectly matched to that of the incident medium, i.e., $Z_{\text{media}} = Z_{f_\text{space}}$. The intrinsic impedance of any medium is given by the ratio of the electric field intensity (E_x) to its transverse magnetic field intensity (H_y). The intrinsic impedance of free space can be expressed as

$$Z_{f_\text{space}} = \frac{E_x}{H_y} = \sqrt{\frac{\mu_o}{\varepsilon_o}} \approx 377\ \Omega \tag{4}$$

where μ_o and ε_o correspond to the permeability and permittivity of free space, respectively. Thus, in order to achieve zero reflection at the interface, the intrinsic impedance of the propagating medium should be equal to 377 Ω.

(ii) The reflection coefficient can also be expressed in terms of normalized impedance as

$$\Gamma = \frac{Z_n - 1}{Z_n + 1} \tag{5}$$

where

$$Z_n = \frac{Z_{\text{media}}}{Z_{f_\text{space}}} = \sqrt{\frac{\mu_r}{\varepsilon_r}} \tag{6}$$

where μ_r and ε_r corresponds to relative permeability and relative permittivity of the medium, respectively. When the normalized impedance is unity, i.e., $Z_n = 1$, which in turn imply μ_r being equal to ε_r, the reflection coefficient drops to zero. The relative permittivity and permeability can be expressed as $\varepsilon_r = \varepsilon_r' - j\varepsilon_r''$ and $\mu_r = \mu_r' - j\mu_r''$, respectively. The prime and double prime superscripts denote real and imaginary components, respectively. Therefore, it can be observed that the real and imaginary components of relative permittivity and relative permeability must be equal to reduce Γ to zero.

(iii) The general expression for an EM wave propagating through any medium along $+z$-direction can be written as

$$E(z, t) = E_o e^{-\alpha z} \cos(\omega t - \beta z)\widehat{a}_x \tag{7}$$

where E_o, α, and β correspond to amplitude of the electric field oriented along x-direction, attenuation constant, and phase constant, respectively. The amplitude of the EM wave gets attenuated by the factor $e^{-\alpha z}$ as it traverses through the medium. The term α can be expressed as

$$\alpha = \sqrt{\varepsilon_o \mu_o}\,\omega \sqrt[4]{a^2 + b^2} \sin\left(\frac{1}{2}\tan\left(-\frac{a}{b}\right)\right) \tag{8}$$

where

$$a = \left(\varepsilon_r' \mu_r' - \varepsilon_r'' \mu_r''\right)$$

$$b = \left(\varepsilon_r' \mu_r'' + \varepsilon_r'' \mu_r'\right)$$

For a medium with minimal thickness, α should be large for the attenuation to be high. Thus, high values of ε', ε'', μ', and μ'' are required to obtain high attenuation and henceforth minimum reflection. This approach must be applied keeping in mind the first condition, where large values of permittivity and permeability result in a large reflection coefficient. Therefore, this approach can be applied to inner layers for increasing the attenuation as the wave moves deeper.

Based on the location, frequency range of interest, and feasibility of material synthesis and application, any one of the abovementioned approaches can be used to design a RAM.

Further, artificially engineered electromagnetic entities like metamaterials, frequency selective surfaces (FSS), and metasurfaces can be used in combination with homogenous materials in order to enhance the absorption performance. Among these technologies, FSS and metasurfaces are widely used in the design of RAS as they are capable of improving the performance with reduced thickness/weight penalty. Here, superior absorption is achieved in two ways as explained below:

(i) *Impedance matching followed by absorption in lossy material:* In this case, the impedance of FSS or metasurface at the interface between free space and absorbing media is matched to that of free space, thereby reducing the reflection from first interface to zero. Once waves enter the lossy material, they get attenuated/dissipated as they propagate through it (Munk 2000).

(ii) *Cancellation of scattered waves*: Here, RCS reduction is obtained by carefully manipulating the phase difference/path difference between incident and reflected signals so that they interfere destructively. One way to accomplish this is to set the thickness of RAM/RAS equal to integral multiples of quarter

wavelength (λ/4). The category of absorbers working on this principle is called resonant absorbers.

Salisbury screen is one of the simplest forms of this type where a resistive sheet of surface resistivity equal to free space impedance (377 Ω/sq) is placed quarter wavelength away from a metallic ground plane. These layers are usually separated by a lossless dielectric spacer. At the frequency of resonance, the effective impedance of this structure becomes equal to 377 Ω resulting in matched condition. The impinging wave thus enters the dielectric material and comes out of it with a phase difference of 180° due to the reflection at the metallic ground plane. This results in destructive interference between the incident and reflected waves, thereby reducing effective RCS to zero. However, narrow bandwidth and relatively large thickness at lower frequencies are the major drawbacks of Salisbury screen.

Jaumann absorbers overcome the bandwidth constraint of Salisbury screen by stacking several layers of resistive sheets, with different surface resistivity, separated from each other by quarter wavelength at the frequency of interest. But wider bandwidth is achieved here with larger thickness. Dallenbach RAS is a different variant of resonant absorber where a homogenous lossy material of quarter wavelength thickness is placed on top of a metallic ground plane. In this case, the waves from first and second interfaces interfere destructively, thereby reducing overall reflection.

The resonant absorbers discussed so far used resistive sheets with pure resistive values. Circuit analog (CA) absorbers represent another class of resonant absorbers where the resistive sheets are configured as lossy FSS layers. This results in a complex sheet admittance, with both resistive and reactive components, which can be appropriately tailored to accomplish broader bandwidth at reduced thickness (Choi et al. 2014; Zhang et al. 2012). For the purpose of illustration, the unit cell of a three-layered CA absorber (total thickness = 6.228 mm) with square and trapezoidal pattern-based resistive layers is shown in Fig. 11. Commercially available substrates like AD255C and RO4835 have been used to configure the multilayered model (Kantikar et al. 2020). The absorber is placed on top of a metallic plane whose RCS is to be reduced. The absorption characteristics of the RAS *w.r.t.* frequency and angle of incidence for TM as well as TE polarization are presented in Figs. 12 and 13, respectively. It is clear from the plot that the power absorbed is greater than 90% for a wide range of incident angles (0° to 60°) for TM polarization. On the other hand, for TE polarization, the percentage of power absorbed is greater than 90% from 2.51 GHz to 24 GHz till 50°. The unit cell has been then used to form a 15 cm × 15 cm planar RAS as shown in Fig. 14. The monostatic RCS of the planar RAS is shown in Fig. 15 along with that corresponding to PEC plate and copper plate of identical dimensions. The plot clearly reveals that the CA absorber reduces the RCS of metallic plate by 8 dB over a broad frequency range from 2 GHz to 18 GHz.

In order to cater to conformal platforms, the planar RAS has been bent with a radius of curvature (ROC) of 280 mm (Fig. 16). The variation in monostatic RAS of the conformal RAS *w.r.t* frequency is shown in Fig. 17. It is evident that the performance of the conformal RAS model is similar to that of planar RAS without any degradation.

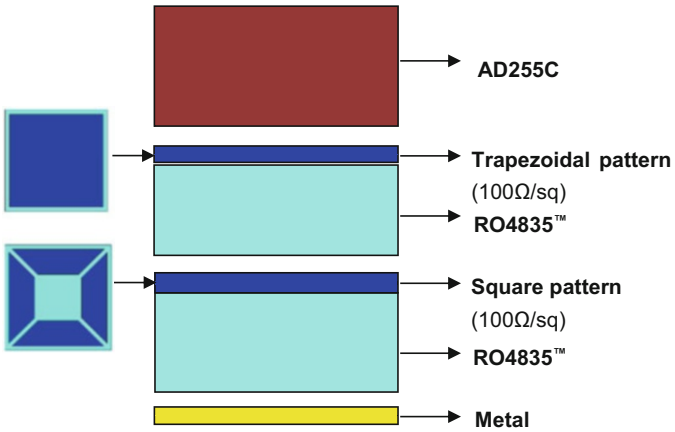

Fig. 11 Unit cell of a three-layered CA absorber

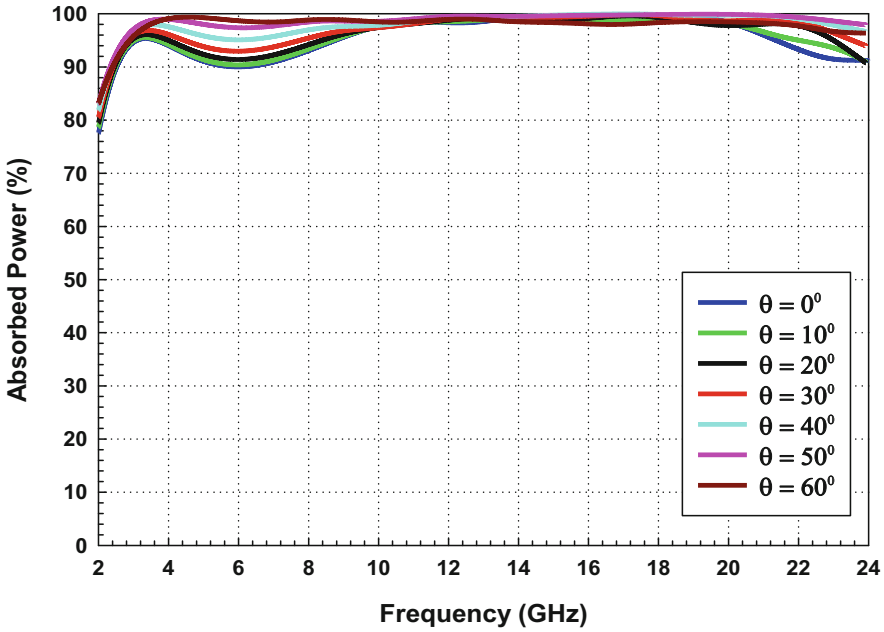

Fig. 12 Variation in the power absorption characteristics of CA absorber *w.r.t* frequency for different incident angles (**TM polarization**)

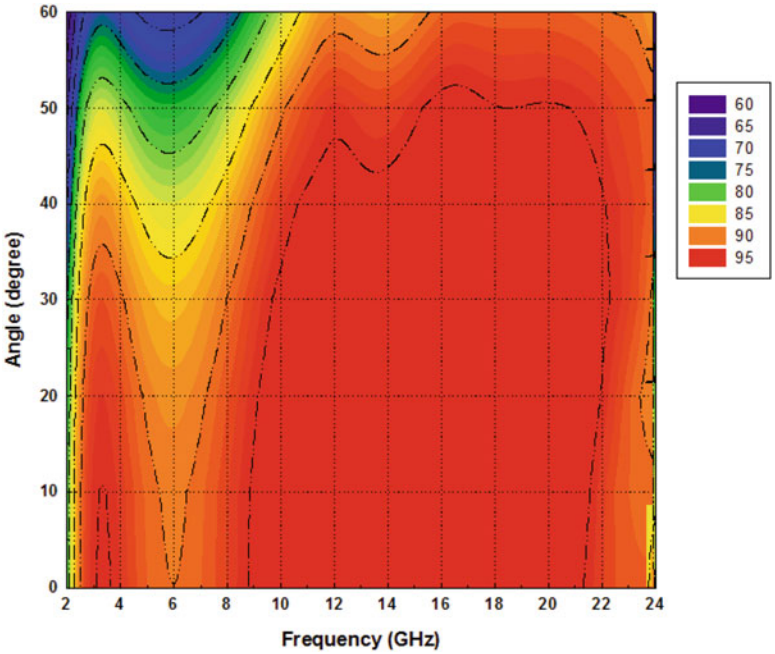

Fig. 13 Contour plot showing variation in the power absorption characteristics of CA absorber *w.r.t* frequency and angle of incidence (**TE polarization**)

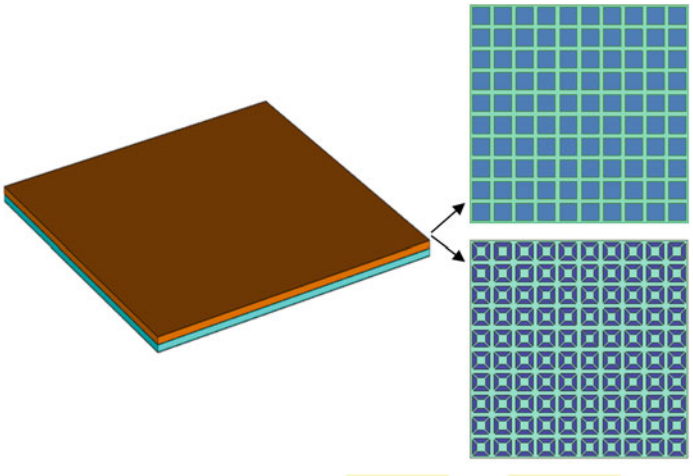

Fig. 14 Schematic of 15 cm × 15 cm planar RAS using CA absorber-based unit cell

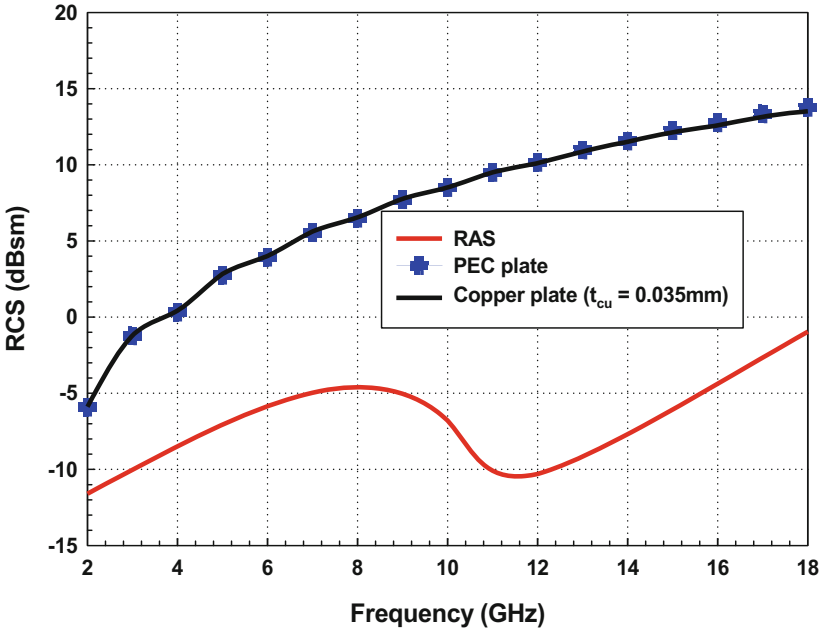

Fig. 15 Monostatic RCS characteristics of planar CA absorber (normal incidence)

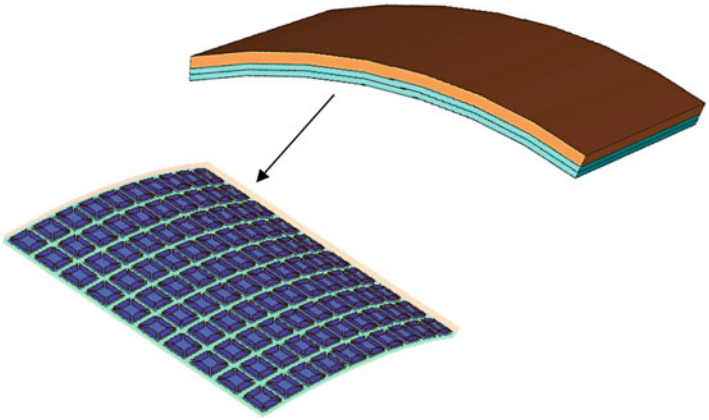

Fig. 16 Schematic of 15 cm × 15 cm conformal RAS using CA absorber-based unit cell

RCS Reduction in Antenna/Array

As mentioned already, antenna RCS is a significant parameter to be considered in a low observable platform. The antenna RCS is essentially a combination of both radiation mode and structural mode RCS. The total RCS of an antenna is given by

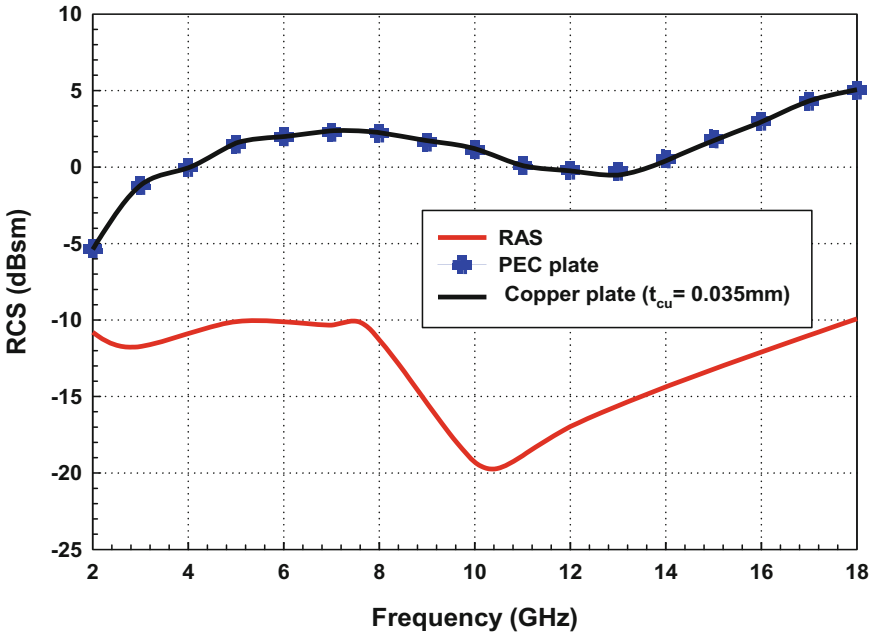

Fig. 17 Monostatic RCS characteristics of conformal CA absorber (normal incidence)

$$\sigma = |\sqrt{\sigma_{\text{struct}}} - (1 - \Gamma)\sqrt{\sigma_{\text{rad}}}e^{j\phi}|^2 \tag{9}$$

where σ_{struct} and σ_{rad} represent the structural mode and radiation mode RCS, respectively, Γ is the reflection coefficient of the antenna, and ϕ is the phase difference between σ_{struct} and σ_{rad}. One of the ways to reduce antenna RCS is to modify the ground plane of the antenna. It can be done by replacing PEC ground plane with a high-impedance surface (HIS) comprising of AMC and FSS elements. Such designs provide RCS reduction in out-of-band frequency range (Shater and Zarifi 2017). Another method is to incorporate a resistive FSS layer of single octagonal loop in the antenna design (Ramkumar et al. 2016). The implementation of embedded passive (EP) resistors without soldering facilitates RCS reduction and parasitic reactance, besides eliminating the soldering defects. Researchers have designed different combinations of AMC cells that can replace the conventional ground plane of microstrip patch array, so that the out-of-band RCS reduction can be achieved (Paquay et al. 2007; Iriarte et al. 2013). A low profile patch array with ground plane modified by hybrid FSS (Genovesi et al. 2012) shows out-of-band RCS reduction without any degradation in the radiation characteristics. An antenna array with nonuniform configuration of HIS elements can show broadband RCS reduction. However, the significant reduction in structural RCS of patch array over the desired frequency range requires extensive analysis.

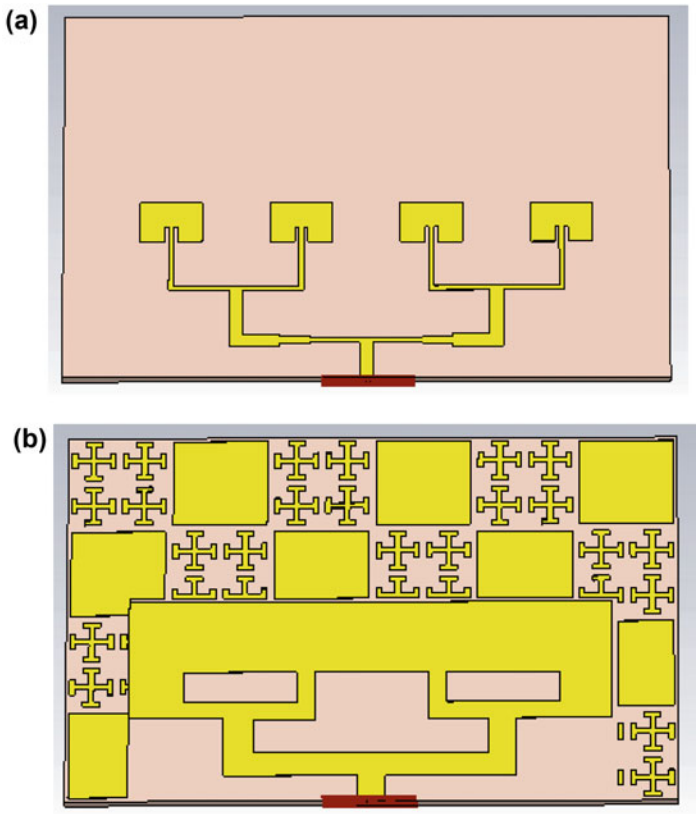

Fig. 18 A four-element linear patch array with hybrid HIS layer as ground plane at 10.5 GHz (substrate dimensions: 60 mm × 90 mm). (**a**) Top view, (**b**) bottom view.

Table 1 Radiation characteristics of a four-element patch array with hybrid HIS layer as ground plane at 10.5 GHz

Array performance	PEC ground plane	Hybrid HIS layer as ground plane
Return loss	−26.68 dB at 10.48 GHz	−22.85 dB at 10.37 GHz
VSWR	1.1	1.16
Gain	6.75 dB	7.3 dB

The main challenge is to achieve reduction in antenna RCS without any degradation in antenna radiation performance in terms of return loss, VSWR, and antenna gain.

Figure 18 shows a four-element patch array with HIS-based hybrid ground plane consisting of chessboard arrangement of patch and Jerusalem cross elements. The substrate is of 60 mm × 90 mm dimensions. It can be noted from Table 1 that the

resonant frequency of the patch array has shifted from 10.48 GHz to 10.37 GHz with a return loss of −29.24 dB, when the conventional PEC-based ground plane is replaced with HIS ground plane. The corresponding value of VSWR is 1.16. It can be observed from Fig. 19a that there is an improvement in the array gain. The gain of the patch array with HIS ground plane is obtained as 7.3 dB, in contrast to 6.75 dB with PEC-based ground plane.

Figure 19b shows the extent of structural RCS reduction over the frequency range of 6 GHz to 30 GHz. It can be noted that RCS reduction of 1.37 dBsm to 23 dBsm is achieved in the frequency range of 11 GHz to 30 GHz. Moreover, RCS reduction of 1.43 dBsm to 3.16 dBsm is obtained in the frequency range of 6 GHz to 9 GHz. A maximum RCS reduction of 23 dBsm is achieved at 25 GHz, while a reduction of 18.26 dBsm is obtained at 18 GHz. The stop-band characteristics of HIS layer are corroborated in the structural RCS plot of the patch array. The RCS reduction is not obtained at the resonant frequency of the patch array.

Figure 19c presents the variation of gain of four-element patch array with frequency. It is evident that the array gain is maximum at the resonant frequency. Moreover, the gain of the patch array with hybrid HIS layer as ground plane has increased in the frequency range of 8 GHz to 10.5 GHz, when compared to the patch array with conventional ground plane.

Conclusion

With rapid advancements in the field of stealth technologies, accurate estimation of RCS has become an inevitable component in the development of low observable structures. Different approaches used for RCS estimation such as body geometry decomposition, low frequency techniques, high frequency techniques, and hybrid methods have been briefly explained in this chapter. Special emphasis has been laid on enabling the reader to choose a suitable technique based on the electrical size of the structure. Low frequency techniques provide accurate predictions for complex geometries at the expense of increased computational cost, whereas high frequency approaches tackle relatively simple electrically large problems with less computational resources. Further, hybrid methods are logical combinations of asymptotic and numerical methods for solving EM problems where neither class of formulations is sufficient by itself. RCS of certain structures like metallic double ogive, UAV, patch antenna/array computed using commercial software, and open-source codes have been presented as well. Once RCS is estimated, the next major step is to reduce the scattering from predominant hotspots. In this regard, the popular approaches employed for reduction of RCS have also been included with special emphasis on minimizing reflectivity using radar absorbing materials (RAM), radar absorbing structure (RAS), FSS, metasurfaces, etc.

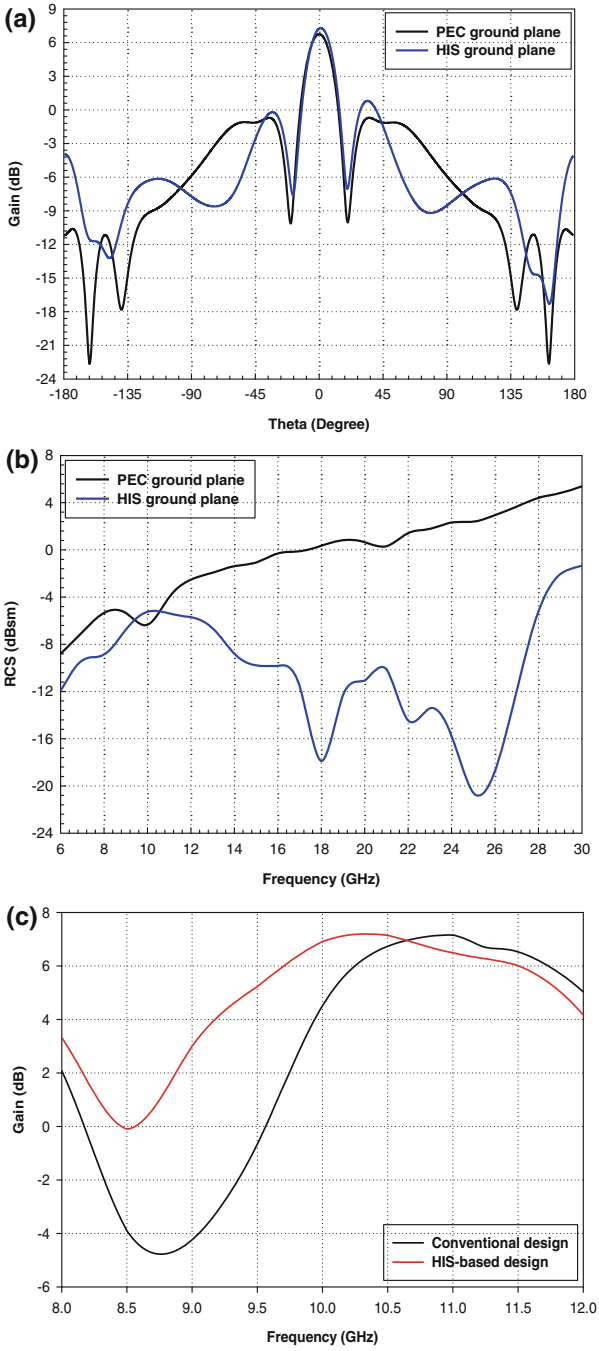

Fig. 19 Radiation and scattering characteristics of a four-element patch array with hybrid HIS layer as ground plane at 10.5 GHz. (**a**) Gain, (**d**) specular structural RCS, (**c**) variation of gain with frequency

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Part XI

Environmental Metrology

Environmental Metrology

62

An Introduction

Ravinder Agarwal and Susheel Mittal

Abstract

Meteorology is an important atmospheric science. It includes all aspects of the atmosphere, including atmospheric physics and chemistry. Meteorology is the short-term study of the atmosphere and its movements. Meteorology, also known as "weather," is the study of atmospheric elements that affect present or near-future circumstances. Temperature, humidity, precipitation amount and kind, wind direction and strength, atmospheric pressure, and cloud cover are just a few of the meteorological factors that characterize the atmosphere.

The "environment" refers to all natural materials that surround us, such as air, water, land, and noise, in which humans, animals, and plants coexist. Pollution wreaks havoc on the ecosystem. Our environment is poisoned in several ways. Smoke is produced by mills, factories, automobiles, burning of crop waste, and other similar sources. As a result, substantial air pollution develops, particularly in big cities and industrial areas. Pollution means contamination of this environment. The environment is polluted in many ways, smoke from the mills, factories, vehicles, crop residue burning, etc., resulting in serious air pollution especially occurring in big cities and big industrial areas.

Every day, 2 million tons of sewage and industrial and agricultural wastes are released into local water sources across the world. Human waste is the major source of water pollution; 2.5 billion people do not have access to basic sanitation facilities. Open defecation is practiced by 12% of the global population. Greenhouse gases,

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dangerous chemicals, smoke, industrial pollutants, pollen, and exhaust may all combine to generate smog layers, particularly over cities and metropolitan areas, raising health and environmental concerns.

This introductory chapter provides a brief overview of the international metrology program for ensuring metrology for monitoring air indoor and outdoor quality, pollution effect on health, traceability in analytical environmental measurements, and noise pollution monitoring in urban cities with a global measurement system traceable to SI units. This book presents many policymaking innovation approaches for the development of a knowledge economy and knowledge society. Pollutants have a negative influence on not just public health but also the global climate and ecosystems. Most programs to minimize air pollution provide a "win-win" solution for both health and the environment. Lower levels of air pollution improve the population's cardiovascular and respiratory health in the long and short term. Reducing ambient and home air pollution can also reduce emissions of carbon dioxide (CO_2) and short-lived climate pollutants such as black carbon particles and methane, so helping the near- and long-term mitigation of climate change.

In general, air pollution is one of our era's biggest scourges, not only because it influences climate change but also because of its impact on public and individual health owing to increased sickness and death. Several contaminants are substantial contributors to human illness. Particulate matter (PM), particles with varying but extremely tiny diameters, enter the respiratory system by inhalation and cause respiratory and cardiovascular illnesses, reproductive and central nervous system malfunction, and cancer.

Particulate matter is made up of a mixture of solid and liquid particles such as dust, pollen, ash, soot, dirt, and smoke. Some of these particles may be visible, as evidenced by the appearance of haze, while others are too tiny to be seen with the human eye. The accurate and exact measurement of ambient particulate matter (PM) is required for a better understanding of the possible environmental and human health implications of airborne particulate matter (PM). Several methods and devices are available, ranging from filter-based sample collection for offline laboratory analysis to online instruments that detect airborne particles and provide real-time information on size distribution, particle mass concentration, and chemical data (US EPA e-CFR40; US EPA 40 CFR 2020). Depending on the scientific aims of the investigation, different measuring methodologies, sample devices, and equipment based on different principles are utilized. Continuous methods' improved time resolution has proven useful in monitoring PM concentrations and characterizing ambient PM and is becoming increasingly important in allowing policymakers to make decisions and scientists to investigate sources of particulate pollution and to probe into the dynamics and mechanisms of aerosol formation in the atmosphere. It is critical for researchers working on the subject of air quality to have a thorough awareness of the principles and limits of various PM monitoring methodologies. In this regard, this chapter gives an overview of the measuring principles of PM measurement techniques that are accessible and utilized globally. Offline and online procedures are complementary in nature, with the online methodology improving the temporal resolution of PM characterization by delivering data in 1 hour or less and

the offline technique, being nondestructive, allowing the use of the same sample for analysis by other techniques. The employment of both online and offline techniques in tandem can give a better knowledge of the particulate matter by research needs.

Knowledge of atmospheric boundary layer (ABL) characteristics that impact pollutant concentration, diffusion, and transit is required for air pollution mitigation. The ABL is one of the most essential component zones of the atmosphere since it is vital to major human endeavors such as air pollution, aviation, and communication. It is also known as the biosphere's circulatory system. SODAR is a critical ground-based remote sensor capable of detecting ABL height in real time. SODAR is a well-known and well-acknowledged meteorological technology for providing real-time and essential ABL data in real time. SODAR contributes to city planning, industrial zoning, and other air quality management approaches (Kumar et al. 2021; Saha et al. 2019). The data obtained by SODAR echograms is processed, evaluated, and compared to the Modern-Era Retrospective Analysis for Research and Applications, Version 2 (MERRA-2) model sensor, to determine its efficacy and data accuracy. Following a careful examination, it was discovered that SODAR is effective ecologically friendly equipment that can be utilized for air pollution control and air quality measurements. To accomplish this purpose, the work addresses five aspects: SODAR as a monitoring technique, SODAR system calibration and validation, SODAR application in air pollution warning system, comparison of the ABL height of SODAR and MERRA-2, and SODAR application in determining the pollution loading capacity of the region. ▶ Chapter 65, "Development of Calibrated and Validated SODAR with Reference of Air Quality Management," is divided into nine sections. In a nutshell, a new SODAR monitoring system was developed employing a novel combination of a moving-coil transducer, parabolic dish, and acoustic ACP sheet baffle. To send sounds into the turbulent zone, an acoustical parabolic antenna was deployed. The system was utilized for long-term continuous data collecting with a minimal likelihood of electronic failure. SODAR data is used to determine the region's pollution loading capacity and air pollution alarms in various atmospheric circumstances (Parag et al. 2022).

Stubble burning has been identified as a major cause of air pollution, notably in South Asia. It greatly contributes to the formation of gaseous pollutants such as carbon monoxide (CO), nitrogen oxides (NO_x), methane (CH_4), and others. India's position is more austere because of the extensive rice-wheat rotation system, which creates a considerable amount of stubble. A huge amount of waste is generated after harvesting crops like wheat and rice, most of which is burned after the valuable constituents have been removed. Some of these wastes are used as animal feeds, for making cattle sheds and rural house roofs, or as fuel for domestic cooking.

Crop residue burning (CRB) is one of the top contributors to ambient air pollution ($PM_{2.5}$) in northwestern India. Despite the magnitude of the problems associated with high particulate levels, epidemiologic data on the associated health effects of CRB focusing on rural populations is limited in India. The present study was conducted to quantify the impact of ambient air quality ($PM_{2.5}$) from CRB on the respiratory health of rural/agricultural communities in Patiala. Particle size is closely related to its ability to cause health concerns (Sachin et al. 2020; Susheel et al. 2016).

Small particles fewer than 10 micrometers in diameter are the most dangerous because they can enter your lungs and, in some cases, your bloodstream. Such particles can have an impact on both your lungs and your heart. Numerous scientific studies have linked particle pollution exposure to a wide range of problems, including premature death in people with heart or lung disease, nonfatal heart attacks, irregular heartbeat, aggravated asthma, decreased lung function, and increased respiratory symptoms, such as airway irritation, coughing, or difficulty breathing. People with heart or lung illnesses, children, and the elderly are the most vulnerable to particle pollution. Among numerous sorts of environmental imbalances, air pollution is a global health concern with substantial public health consequences. There is a relationship between air pollution and a variety of health concerns, notably among children, pregnant women, the elderly, and those with comorbid conditions.

There have been reports of the negative impact of air pollution on neurological, cardiovascular, and respiratory diseases. Maximum deterioration in FVC and PEF values was seen in one of the respiratory system experiments during each crop season. When physiological markers were compared to their baseline levels, an unrecoverable impact was detected. The rice crop residue burning season had more impact on subjects than the wheat crop residue burning season due to the release of higher concentrations of PM in the ambient environment (Bai et al. 2018). This study revealed for the first time the role of BMI in the respiratory-related sickness of human subjects during exposure to ambient air particulate matter. In the future, if necessary steps are not taken to stop these burning activities, it may cause chronic diseases like cardiopulmonary disorders, irrecoverable lung capacity or asthma, and unrecoverable disorders in human subjects in the initial age of their growth. As a result, massive awareness campaigns are required to inform farmers about the availability of economically viable alternatives.

Indoor air quality (IAQ) is the quality of air in and around a building that directly impacts human health. Over the last few decades, the worldwide scientific community, environmental research, and governmental organizations have worked to enhance building inhabitants' comfort, health, and welfare. First and foremost, why is IAQ monitoring required? Because indoor air pollution (IAP) may have a substantial and harmful influence on both occupants' health and quality of life. Given that residents spend the bulk of their time indoors, the issue of IAQ represents a compelling strategy for mitigating air pollutant exposure, particularly amid the current health crisis caused by COVID-19. Various research organizations have employed cutting-edge technology to handle IAQ testing services that precisely assess a variety of characteristics and difficulties that cause discomfort at work. As a result, the purpose of this chapter is to answer the stated need and examine the existing understanding of IAQ measurement in terms of various elements, allied science, and applications to demonstrate the strategy of translational research studies on IAQ. This chapter also covers many elements of IAQ such as measuring methodologies, characterization methods, and source apportionment, as well as their implications on human health and the environment. Thermal comfort characteristics such as ambient temperature, relative humidity (RH), and wind speed are recognized to have a vital impact on indoor air quality due to continuously changing

technology. The chapter attempts to investigate how humans may live in and near good air quality environments – overall, creating incentives to enhance IAQ in ways that benefit building occupants, employers, and owners, as well as increasing respect for IAQ in green construction initiatives (Bai et al. 2018; Dubey et al. 2021). This chapter is a comprehensive scientific publication that covers all facets of the IAQ system, including measurement techniques, characterization methods, source apportionment tools, and ways of assessing health risks. This will benefit and be highly beneficial for researchers, policymakers, and government agencies in designing new strategies and policies for improving the IAQ system.

Because of the fast expansion of industrialization, motorized transportation, and urbanization, emissions from these activities are negatively altering air quality and threatening human health, agriculture, plants, forests, animals, and property. There is an understanding of the necessity of excellent air quality and the importance of comprehensive monitoring. Monitoring of industrial emissions in developed nations, notably in the United States, Japan, and Germany, has almost entirely transitioned to the use of automatic continuous monitoring systems based on specific physicochemical or electrochemical features of the pollutants. For assessment of impacts, devices giving time-weighted average values are most desirable, and for assessment of short-term hourly variations, real-time continuous monitoring systems are more desirable. Such equipment needs very particular sensors for each pollutant; calibration methods that necessitate calibration gases of precisely specified and known composition; a dust-free, air-conditioned atmosphere; and so on. Such systems become nearly unaffordable in terms of capital costs for industrial units in developing nations, particularly medium- and small-size units. Even more difficult to address are issues such as a lack of instrument parts and calibration gases in local markets, as well as skilled personnel for the effective operation and maintenance of these complicated systems. In India, experience has shown that wherever such systems have been built, they have been inoperative nearly from the outset.

The USEPA has established straightforward ideas for measuring both source emissions and ambient air quality. Although these have become outmoded in developed nations, they are still appropriate for adaptation in underdeveloped countries. Even in this country, instruments made in the United States, Japan, Germany, and other countries frequently tend to make the instrument difficult and costly, causing problems with operation, maintenance, replacements, and so on, in addition to the much higher prices. The instruments presented here are based on simple USEPA ideas and are easy to use and maintain in the field circumstances of developing nations' warm, tropical environments. These instruments are relatively inexpensive. Furthermore, a new sort of device is necessary to measure the exposure of people who move around while working. The movement of air in the local environment is critical for understanding the buildup or dispersion of contaminants. To safeguard those with hearing impairments, immediate and LEQ noise levels must be monitored (Pokhariyal et al. 2019).

In today's environment, the reliability of measurement data creates a solid foundation for successful policymaking, resource allocation, and other major

choices for a government. Accurate and reliable measurements serve as the foundation for the country's growth and development across all sectors (Yadav et al. 2020a). Every measured data has an uncertainty connected with it, which frequently calls into question its legitimacy in the realm of metrology. Thus, metrological traceability, which connects measured findings to a specified reference via an unbroken chain of comparisons while accounting for all acknowledged uncertainties, has become critical. Although estimating uncertainty is easier for physical measurements, this concept still has to be spread in the field of chemical/biological/environmental measurements. Environmental regulation is based on qualitative and quantitative data in environmental research, and even little inaccuracies/errors can be fatal. This increases the need for traceability in environmental measurements. Analytical procedures for environmental measures include multiple processes, including sample pretreatment (extraction), sample storage, and chemical/biological analysis. However, estimating errors in environmental measurements is a difficult task due to incomplete analyte recovery, inappropriate sample handling during extraction, and incorrect sample storage. Uncertainty connected with analytical environmental measurements and the techniques that influence the quality of the data provided must still be understood and investigated in depth (Pokhariyal et al. 2019). This chapter discusses how traceability using documented standards and certified reference materials can be perceived in analytical environmental measurements, with an attempt to link the associated uncertainties at each step of the measurement of atmospheric particulate matter (PM) concentration as well as during the analytical determination of its chemical components.

Accurate assessment of an analyte level in analytical environmental measurements is extremely complex and tough to perform realistically. The missing linkages in analytical environmental measures must be addressed for this aim. There is still a need for research into the references used in environmental studies so that precise measurements may be obtained without compromise. This ensures the traceability of measurement data to references and the precise assessment of environmental pollutants.

The current chapter provides a comprehensive review of noise monitoring studies, a comparison of prediction models including physical propagation models, and applications of artificial intelligence techniques, noise mapping, and noise pollution monitoring in the mining sector conducted by various researchers. The majority of noise pollution studies are concerned with the evaluation of traffic noise, whereas some are only concerned with noise monitoring for residential, educational, industrial, and commercial areas. The study demonstrates that early models were based on mathematical prediction models, while subsequent machine learning and deep learning approaches were commonly utilized for noise-level prediction and forecasting. The chapter provides a look back at noise mapping and control. Although certain studies have been recorded in several Indian cities, the approach, strategy, and methodology of noise mapping are still in their early stages in India. However, when more studies are completed, it will provide consultants, town planners, research institutes, labs, and pollution control authorities in India with greater information and expertise in performing noise mapping studies and implementing

noise control measures. Promoting good health and a high standard of living necessitates a comprehensive strategy that focuses primarily on periodic noise monitoring and assessment of noisy locations in cities, determining compliance with ambient noise standards, and developing appropriate noise action plans in conjunction with policy issues concerning environmental noise planning and control. Furthermore, noise pollution management and abatement techniques are noted as being critical for decreasing ambient noise levels in the country's urban centers.

Moreover, our everyday lives are intrinsically linked to sound. Noise, in particular, is frequently unpleasant, degrades our living environment, and may be harmful to our health. Many efforts have been made in the past to address noise-related problems. As a consequence, the techniques for achieving precise noise measurement are the subject of this chapter. First, for noise measurement, the frequency dependence of our audible sensation and the time variation of the noise are considered, and several noise indicators are proposed. The operational principles, actual implications, and calibration techniques of acoustic measuring instruments are then covered. And after that, quantitative standards which are needed for precise noise measurement are introduced. They are achieved as the sensitivities of special condenser microphones and ensure the reliability of acoustic measuring instruments.

Despite the challenge of eliminating anthropogenic environmental contamination, a good solution might be envisioned as a close collaboration of authorities, bodies, and doctors to normalize the condition. Governments must provide adequate information, educate citizens, and involve specialists in these concerns to properly control the rise of the problem. International collaboration in research, development, administration policy, monitoring, and politics is critical at this stage for successful pollution management (Sarangi et al. 2016). Air pollution legislation must be harmonized and updated, and policymakers should propose the development of a formidable environmental and health protection mechanism. Air pollution reduction technologies must be developed and implemented in all sectors. In the future, if necessary steps are not taken to stop these burning/pollution activities, it may cause chronic diseases like cardiopulmonary disorders, irrecoverable lung capacity or asthma, and unrecoverable disorders to human subjects in the initial age of their growth.

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Physical Quantities of Sound and Expanding Demands for Noise Measurement

63

Hironobu Takahashi, Keisuke Yamada, and Ryuzo Horiuchi

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Abstract

Sound is closely related to our daily life. Specifically, noise is frequently unpleasant, degrades our living environment, and may harm our health. So far, many efforts to deal with noise-related problems have been conducted. Therefore, this chapter focuses on the techniques for achieving precise noise measurement. First, the frequency dependence of our auditorial sensation and time variation of the noise are considered for the noise measurement, and several noise indicators have been proposed. Then, operational principles, practical use, and calibration techniques of the acoustic measuring instruments are discussed. Next, quantitative standards indispensable for precise noise measurement are introduced. They are achieved as the sensitivities of special condenser microphones and ensure the acoustic measuring instruments' reliability. Therefore, requirements for these measurement procedures and instruments are adequately provided in ISO and IEC standards. Finally, some recent advances in acoustic measurement are also introduced.

Keyword

Noise · Sound pressure · Frequency weighting · Time weighting · Sound power · Sound field · Sensitivity · Condenser microphone · Sound level meter · Calibration · Acoustic standard

Introduction

We are exposed to sound daily in a busy area, near a construction site or transit facilities, at the workplace, and even at home. Generally, noise is an unwanted sound often regarded as uncomfortable. Long-term noise exposure may cause severe health damage, such as hearing loss, in some cases. So far, precise and reliable measurement techniques for "acoustic" noise have been studied to ensure a safe environment. These techniques underpin the acceptable noise limit and are reflected in regulations in many countries. Furthermore, quantitative noise monitoring has been achieved in various situations.

Therefore, this chapter focuses on the topics associated with noise measurement, which covers technical terms, measuring instruments, measurement techniques, and instrument calibration. Additionally, current advances in precise acoustic measurement are also introduced. This chapter refers to a few well-known textbooks on acoustics such as Tachibana and Yano 2012, Kido 1990, and Garrett 2017, especially for explaining basic quantities of sound, indicators of noise, and sound measuring principles.

Basic Physical Quantities of Sound and Related Terminology

Sound Pressure and Frequency Weighting

Sound Pressure

The vibration of sound sources such as loudspeakers and musical instruments is transmitted to the surrounding air as a sound wave producing our auditorial sensation. The sound wave causes temporal and spatial air density distribution, resulting in pressure oscillation.

The difference between the time-varying instantaneous atmospheric pressure induced by the sound wave and the static pressure at a given measurement point is known as "instantaneous sound pressure" and is defined as

$$p_{\text{inst}}(t) = P(t) - P_{\text{stat}}, \tag{1}$$

where $p_{\text{inst}}(t)$, $P(t)$, and P_{stat} are the instantaneous sound pressure, instantaneous total pressure, and static pressure, respectively. The unit of quantities in Eq. (1) is Pa, named pascal, following the International System of Units (SI). However, the instantaneous sound pressure is time-dependent and takes both positive and negative values. Therefore, for convenience, the root mean square of the instantaneous sound pressure, such as the sound pressure p, which is mainly used in acoustics, is denoted as follows:

$$p = \sqrt{\frac{1}{T}\int_0^T p_{\text{inst}}^2(t)dt}, \tag{2}$$

where t is time and T is measurement time interval.

Acoustics deals with sound pressure in a wide range. The minimum audible sound pressure is 20×10^{-6} Pa for a sinusoidal sound wave of 1 kHz, while the sound pressure of noise around 20 Pa can be endured for a short time. Therefore, it is not convenient to express the sound pressure in the unit of Pa. Alternatively, the sound pressure is transformed into a logarithm known as "sound pressure level." The unit of the sound pressure level is dB and named decibel. The sound pressure level L_p is defined as follows:

$$L_p = 10\log_{10}\left(\frac{p^2}{p_{\text{ref}}^2}\right), \tag{3}$$

where p_{ref} is the reference sound pressure of 20×10^{-6} Pa, as abovementioned. Therefore, sound pressure from 20×10^{-6} Pa to 20 Pa is converted to sound pressure level from 0 dB to 120 dB. Another reason for the logarithmic expression of the sound pressure is that human sensation is proportional to the logarithm of the stimulus.

Frequency-Weighted Sound Pressure

The sound pressure level is incompatible with our auditory perception. In acoustics, the sensory magnitude of a sound is known as "loudness." Interestingly, loudness is frequency-dependent, and an equal-loudness-level contour shows the frequency dependency of the sound pressure level, which gives the equal loudness level. Furthermore, different loudness level causes a different curve, and for each contour, the loudness level is considered as the sound pressure level at the reference frequency of 1 kHz. Moreover, ISO 226 (ISO 226 2003) provides equal-loudness-level contours for the different loudness levels.

Frequency weighting is applied to sound pressure levels to reflect our auditorial characteristics approximately. Figure 1 shows the frequency dependency of the well-known frequency weightings A, C, and Z. Particularly, frequency weightings A and C approximate the inverse characteristics of the equal-loudness-level contours of 40 dB and 100 dB, respectively. However, frequency weighting Z is 0 dB for all the frequencies and implies that the sound pressure is not frequency-weighted. These frequency weightings are formulated in Annex E of IEC 61672-1 (IEC 61672-1 2013).

Frequency weighting A is commonly applied to noise and auditory measurements in practice, while that of C or Z is employed to measure physical quantities such as sound pressure level. In the frequency domain, the A- and C-weighted sound pressure levels, $L_{p\mathrm{A}}$ and $L_{p\mathrm{C}}$, are expressed as follows:

$$L_{p\mathrm{A}}(f) = L_p(f) + A(f), \tag{4}$$

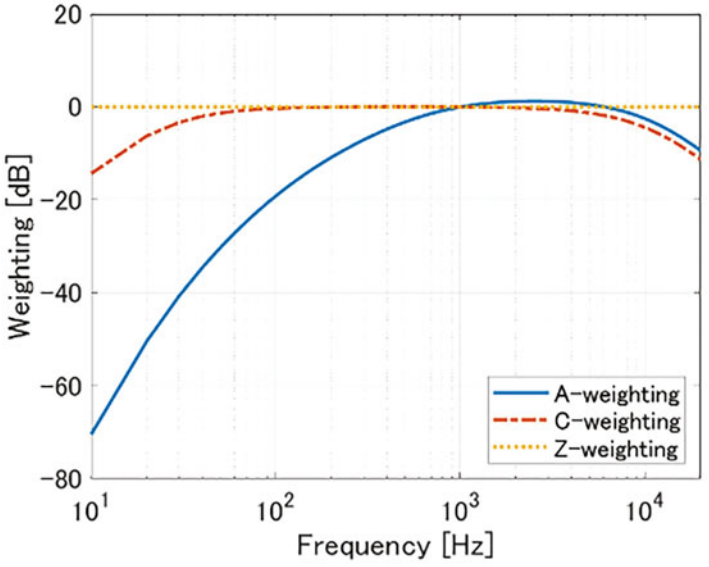

Fig. 1 Curves of frequency weightings A, C, and Z

$$L_{pC}(f) = L_p(f) + C(f), \tag{5}$$

where f is frequency, $L_p(f)$ is the sound pressure level, and $A(f)$ and $C(f)$ are the frequency weightings A and C, respectively. Figure 1 illustrates that our auditory system is more sensitive at mid frequencies than at lower and higher frequencies.

Particle Velocity

Particle velocity $\boldsymbol{u}$ is the air velocity within an infinitesimal volume caused by the propagation of sound waves. The equation of motion describes the relationship between particle velocity and sound pressure as follows:

$$\frac{\partial \boldsymbol{u}}{\partial t} = -\frac{1}{\rho}\nabla p, \tag{6}$$

where ρ is air density and ∇ is a gradient operator. Therefore, particle velocity is obtained by integrating Eq. (6) as follows:

$$\boldsymbol{u} = -\frac{1}{\rho}\int \nabla p dt. \tag{7}$$

Equations (6) and (7) deduce "instantaneous" particle velocity. Unless otherwise specified, particle velocity implies the root mean square value, similar to sound pressure. Additionally, particle velocity should not be confused with sound speed. Specifically, they are conceptually different, and sound speed in air is usually much faster than particle velocity.

Sound Intensity

The instantaneous sound intensity $\boldsymbol{I}_{inst}$ is the product of the instantaneous sound pressure p_{inst} and the instantaneous particle velocity $\boldsymbol{u}_{inst}$. This is denoted as follows:

$$\boldsymbol{I}_{inst}(t) = p_{inst}(t) \cdot \boldsymbol{u}_{inst}(t). \tag{8}$$

The sound intensity $\boldsymbol{I}$ is the time average of the instantaneous sound intensity and is expressed as the following:

$$\boldsymbol{I} = \frac{1}{T}\int_0^T I_{inst}(t)dt, \tag{9}$$

where T is the measurement time interval. As a physical quantity, sound intensity means sound power (see section "Sound Power") per unit area perpendicular to the direction of the propagating sound.

In the case of a plane progressive sound wave, the sound wave propagates unidirectionally. Accordingly, there is no phase difference between the instantaneous sound pressure and particle velocity, and the ratio is presented as follows:

$$\frac{p_{inst}(t)}{u_{inst}(t)} = \rho c, \tag{10}$$

where ρ is the density of air and c is the speed of sound in air. Therefore, the instantaneous sound intensity is described as the following:

$$I_{inst} = \frac{p_{inst}(t)^2}{\rho c}. \tag{11}$$

The sound intensity is also expressed in the logarithm and is known as the "sound intensity level." Furthermore, the unit of the sound intensity level is dB, known as a decibel. The sound intensity level L_I is defined as the following:

$$L_I = 10\log_{10}\frac{I}{I_{\text{ref}}}, \tag{12}$$

where I_{ref} is the reference sound intensity of 1.0×10^{-12} W/m^2. I_{ref} is calculated from p_{ref} of 20×10^{-6} Pa and ρc of 400 Pa·s/m for the plane progressive sound wave.

Sound Power

The sound power W is defined as sound energy emitted from a sound source per unit of time. The unit of sound power is watt, also known as joule per second. Let us consider the virtual space in Fig. 2. The sound power is formulated by using sound intensity $\boldsymbol{I}$ as follows:

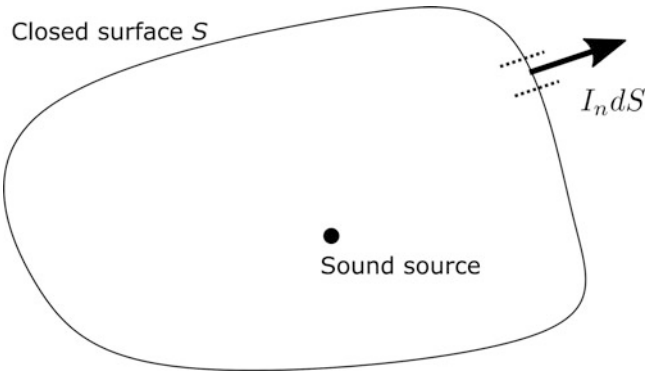

Fig. 2 Conceptual diagram of the relationship between sound power and sound intensity

$$W = \oint_S \boldsymbol{I} \cdot d\boldsymbol{S} = \oint_S \boldsymbol{I} \cdot \boldsymbol{n} dS = \oint_S I_n dS, \tag{13}$$

where S is the closed surface that surrounds the sound source, vector $\boldsymbol{n}$ is a unit normal vector, and I_n is a normal component of sound intensity.

For a plane progressive sound wave, the sound power is derived from Eqs. (9), (11), and (13) as follows:

$$W = \frac{1}{T} \oint_S \int_0^T \frac{p_{inst}(t)^2}{\rho c} dt \, dS. \tag{14}$$

Similarly, the sound power level L_W is defined as the following:

$$L_W = 10 \log_{10} \left(\frac{W}{W_{\text{ref}}} \right), \tag{15}$$

where W_{ref} is the reference sound power of 1.0×10^{-12} W and calculated in a similar approach as I_{ref} using the reference area of 1 m^2.

Sound Fields

Overview of Sound Fields

Testing and calibrating acoustic measuring instruments and noise measurement of sound sources require a special measurement environment. Therefore, this section describes the characteristics of a typical sound environment, such as pressure field, free sound field, and diffuse sound field.

Pressure Field

The pressure field is characterized by a uniform sound pressure in small, enclosed spaces. An acoustic coupler realizes the pressure field in its cavity, and the dimension of the cavity is much smaller than the wavelength of a sound wave. Therefore, the acoustic coupler is used for the calibration of microphone sensitivity under a pressure field (see section "Pressure Sensitivity Calibration by Reciprocity Method" and "Pressure Sensitivity Calibration by Laser Pistonphone Method").

Furthermore, the output voltage of earphones or headphones is measured by attaching it to the acoustic coupler, which simulates the acoustic characteristics of our auditory system.

Free Sound Field

The free sound field is also known as the "free field." Here, the sound wave travels without the influence of obstacles. Therefore, anechoic chambers have been designed to realize a free sound field.

The inner walls of the anechoic chamber, including a ceiling and floor, are frequently covered with wedged and porous sound-absorbing materials such as

glass wool. The length of the wedges determines the lower limit frequency of sound absorption. The total performance of the chamber is evaluated with the deviation from the inverse square law. This law means that the sound intensity should be inversely proportional to the square of the distance from the sound source. The procedures for qualifying the anechoic chamber are provided in Annex A of ISO 3745 (ISO 3745 2012). The space near the inner walls is practically unsuitable for the free sound field, because the sound reflection from the inner walls increases compared with the space at the center of the chamber.

The abovementioned anechoic chamber is sometimes known as a full-anechoic chamber, while a hemi-anechoic chamber has an acoustically reflective floor made of concrete, among others. The hemi-anechoic chamber is used for measuring the noise from heavy objects, such as cars and large machines, and the sound power level of sound sources.

Diffuse Sound Field

A diffuse sound field is also known as the "diffuse field." Here, sound energy density is uniform throughout the chamber, and sound energy is propagated from all directions with equal probability. Reverberation chambers have been designed to realize diffuse sound fields. The inner walls of the reverberation chamber are made of acoustically high reflective materials, such as hard stone and concrete. Additionally, the inner walls of the reverberation chamber are often placed unparalleled, allowing as many reflected sound waves to emerge from every direction as possible.

The acoustic performance of the reverberation chamber is evaluated with the reverberation time T_{60}. This parameter is defined as an interval to reduce the sound energy in a steady state by 60 dB after the sound source is suddenly stopped. The reverberation chamber is used for measuring sound absorption coefficient, acoustic transmittance of materials, and the sound power level of sound sources.

Sound Measuring Instruments and Related Equipment

Condenser Microphone

Mechanical Structure

An electroacoustic transducer, such as a microphone and loudspeaker, converts the acoustic signal into an electrical signal or vice versa. A microphone is a typical acoustic sensor and can be categorized by the conversion principle. A condenser microphone is also known as an electrostatic microphone and suitable for acoustic measurement because of a flat frequency response over a wide frequency range and sufficient short- and long-term stability.

The condenser microphone is classified into two types as follows: Type (a) requires external DC bias voltage to operate as a microphone, and it is known as the "externally polarized microphone." Figure 3a illustrates its typical structure. Specifically, a thin tensioned metal membrane and a metal backplate are placed approximately 10–30 μm apart and constitute the plates of a capacitor. The internal

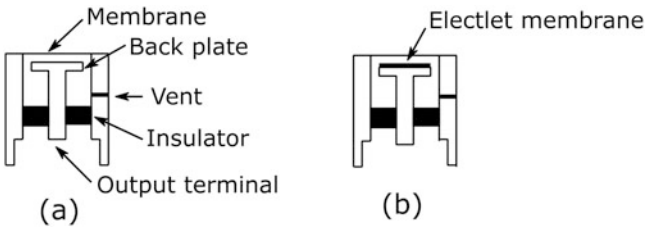

Fig. 3 Typical structure of condenser microphones: (**a**) externally polarized type (side of the housing is vented in this case) and (**b**) pre-polarized type (back plate is pre-polarized in this case)

cavity behind the back plate is exposed to atmospheric pressure through a small vent to equalize the pressure on both sides of the membrane. Furthermore, the vent passes through either the side of the microphone housing or the insulator.

Type (b) does not need DC bias voltage from an external power supply, and it is known as a "pre-polarized microphone" or "electret microphone." Figure 3b illustrates its typical structure. Morphologically, type (b) has almost the identical structure as type (a) except for pre-polarization. The microphone's membrane or back plate is partially made of ferroelectric material and stores electric charge semi-permanently. The convenient pre-polarized microphone is extensively used as a transducer of a sound level meter (see section "Sound Level Meter") and at the measurement site.

Microphone Sensitivity

The microphone sensitivity is defined as the ratio of open-circuit output voltage to sound pressure. As explained below, the sound pressure's physical meaning depends on the sound field.

Firstly, when the microphone membrane is exposed to a pressure field, as shown in Fig. 4a, the "pressure sensitivity," M_p, of the microphone is defined as follows:

$$M_p \equiv \frac{V_0}{p_p}, \tag{16}$$

where V_0 is the open-circuit output voltage of the microphone and p_p is sound pressure that is uniformly applied over the microphone membrane. The pressure sensitivity definition implies that the sound pressure p_p is "actually" acting on the membrane.

Next, let us consider the free sound field. If the measurement point is sufficiently distant from a sound source, the plane progressive sound wave is observed at that point, as shown in Fig. 4b. However, the sound field changes with the introduction of the microphone, because the sound wave is reflected and diffracted by the microphone housing, as shown in Fig. 4c. Therefore, "free-field sensitivity," M_f, of the microphone is defined to determine the sound pressure before positioning the microphone and formulated as the following:

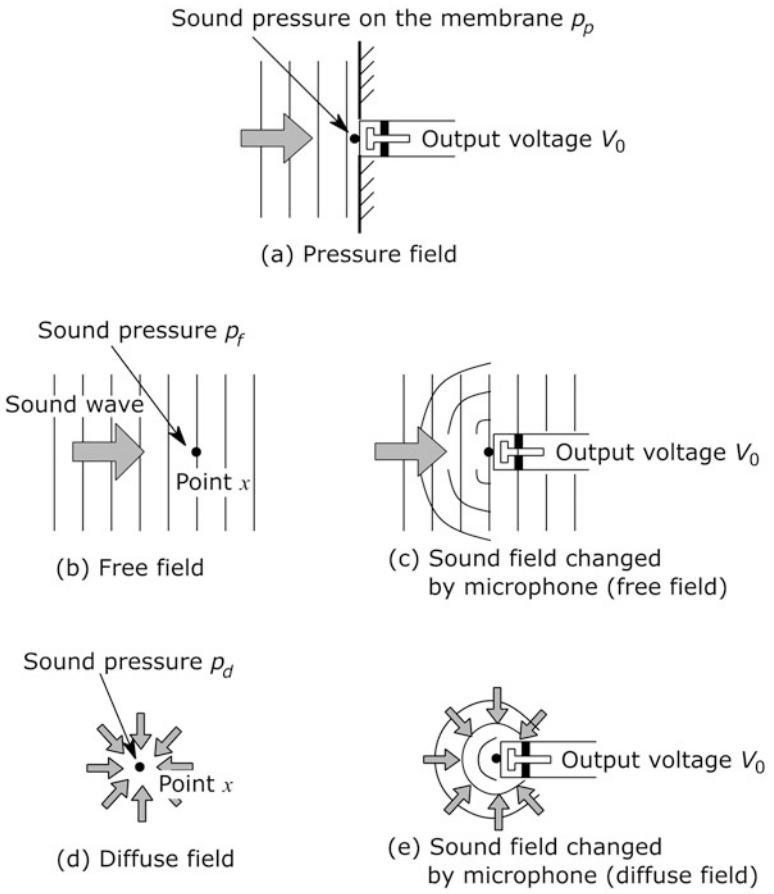

Fig. 4 Change of sound field by the introduction of a microphone

$$M_f \equiv \frac{V_0}{p_f}, \tag{17}$$

where V_0 is the open-circuit output voltage of the microphone, as shown in Fig. 4c, and p_f is sound pressure of the plane progressive sound wave that would exist at the point x without a microphone, as shown in Fig. 4b. The point x is the acoustic center (see section "Free-field Sensitivity Calibration by Reciprocity Method") of the microphone, where the microphone receives the sound pressure substantially. Free-field sensitivity depends on the sound wave's incident angle to the microphone; therefore, the incident angle should be clearly specified.

Finally, in a diffuse sound field, sound waves reach the point x from all directions with equal probability, as shown in Fig. 4d. "Diffuse-field sensitivity," M_d, of the microphone is defined in the same manner as free-field sensitivity and formulated as follows:

$$M_d \equiv \frac{V_0}{p_d}, \tag{18}$$

where V_0 is the open-circuit output voltage of the microphone, as shown in Fig. 4e, and p_d is sound pressure that would exist at the point x without a microphone, as shown in Fig. 4d.

Furthermore, random-incident sensitivity is defined as a root mean square of free-field sensitivities to sound waves coming from all the incident directions successively and with equal probability. The random-incident sensitivity has the same value as the diffuse-field sensitivity in principle and is experimentally determined instead of the diffuse-field sensitivity in practice.

The unit of the microphone sensitivity for each sound field is V/Pa and is usually expressed in logarithms. The unit of sensitivity level is dB, and the reference sensitivity is 1 V/Pa. Furthermore, microphone sensitivity slightly depends on environmental conditions, such as temperature and static pressure.

Classification of Condenser Microphones

Specifications of condenser microphones necessary for precise acoustic measurement are standardized in IEC 61094-4 (IEC 61094-4 1995), and such a microphone is known as the "working standard microphone."

Working standard microphones are categorized by grade, dimensions, and preferable sound field for use. Four characters represent the type name. The first two characters imply microphone grade. "WS" and "LS" mean "working standard microphone" and "laboratory standard microphone," respectively. A laboratory standard microphone (see section "Primary Calibration of Condenser Microphones") is a unique working standard microphone suitable for acoustic standards, and its specifications are prescribed in IEC 61094-1 (IEC 61094-1 2000).

The third character is a number from 1 to 3 accompanied by mechanical dimensions and the corresponding tolerance limits. The last character is "P," "F," or "D" and corresponds to pressure field, free field, or diffuse field, respectively. This character represents the sound field under which the microphone sensitivity is approximately flat in the broadest frequency range.

Table 1 presents typical characteristics of working and laboratory standard microphones. Figures 5 and 6 illustrate examples of these microphones.

Practically, working standard microphones that are commercially available have been classified by the housing diameter in inches. For example, the condenser microphone with a diameter of 3.175 mm is used for measuring sound pressure at frequencies up to 140 kHz; however, this is presently out of the scope in IEC 61094-4 (IEC 61094-4 1995), and there is no type name. Therefore, in this section, this type is named 1/8 inch microphone.

Furthermore, some working standard microphones are designed for special purposes, such as measuring infrasound at frequencies lower than 0.1 Hz, low sound pressure of less than 10 dB, and extremely high sound pressure of more than 180 dB.

Table 1 Typical characteristics of working and laboratory standard microphones

Type designation	WS1{P, F, D} LS1P	WS2{P, F, D} LS2aP, LS2bP	WS3{P, F, D}	1/8 inch type
Housing diameter	23.77 mm	12.7 mm	6.35 mm	3.175 mm
Frequency range	5 Hz to 10 kHz	5 Hz to 20 kHz	10 Hz to 100 kHz	10 Hz to 140 kHz
Sensitivity level	−20 dB to −30 dB	−25 dB to −40 dB	−40 dB to −60 dB	−60 dB
Self-generated noise	10 dB	20 dB	30 dB	40 dB
Maximum sound pressure level	140 dB	150 dB	160 dB	160 dB
Capacitance	50 pF	20 pF	5 pF	3 pF

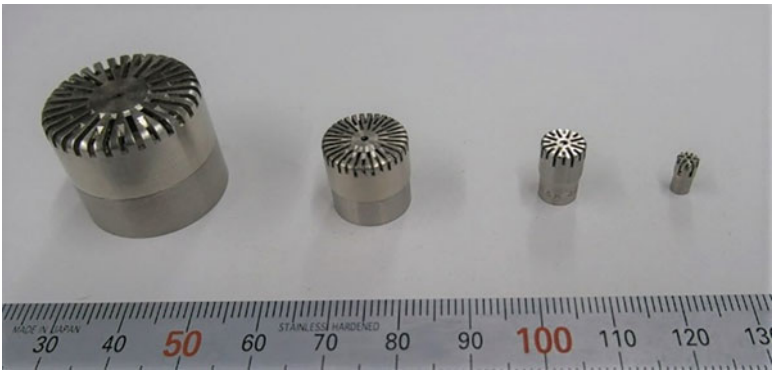

Fig. 5 Examples of working standard microphones: from left, WS1P microphone (Brüel & Kjær type 4144), WS2P microphone (Brüel & Kjær type 4134), WS3F microphone (Brüel & Kjær type 4939). The smallest microphone (Brüel & Kjær type 4138) is not currently a working standard microphone

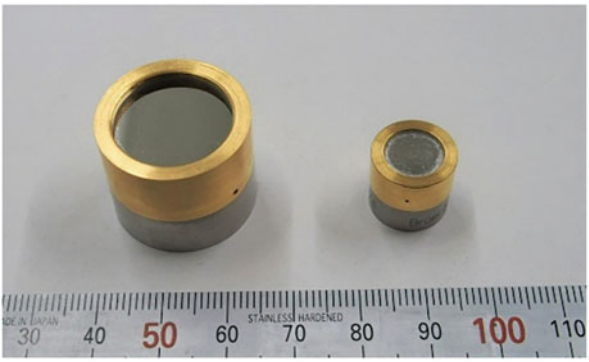

Fig. 6 Examples of laboratory standard microphones: LS1P microphone (left, Brüel & kjær type 4160) and LS2aP microphone (right, Brüel & Kjær type 4180)

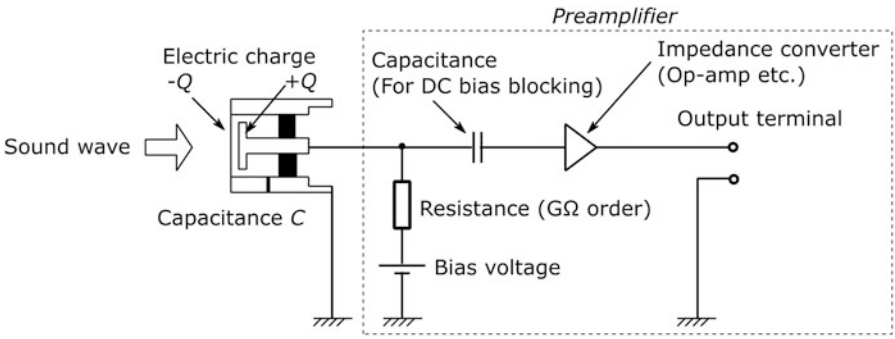

Fig. 7 Schematic diagram for obtaining electrical signal corresponding to sound pressure from an externally polarized condenser microphone

Usage of the Condenser Microphone

Figure 7 depicts the schematic for obtaining an electrical signal corresponding to sound pressure from an externally polarized condenser microphone. When sound pressure is applied to the microphone membrane, vibration of the membrane causes a capacitance change between the membrane and the back plate. Specifically, the capacitance of the condenser microphone is in the order of picofarad. This means that the microphone's output impedance is more than the order of megaohm and much larger than the input impedance of voltage measuring instruments, such as digital multimeters. Therefore, a microphone's output terminal is connected to the preamplifier as an impedance converter and then to the digital multimeter. The preamplifier gain is almost equal to 1; however, it can be obtained when necessary by applying the insert voltage technique (IEC 61094-2 2009), which applies a known electrical signal to the microphone instead of sound pressure.

Furthermore, a working standard microphone may be used without a protection grid. If necessary, a mechanical adaptor should be applied to the microphone to protect the membrane.

Sound Level Meter

A sound level meter is an instrument for measuring frequency-weighted and time-weighted sound pressure levels, which indicates the measured value on a display. The handheld type is used for on-site measurement, while the stationary type is used to monitor the noise continuously at some observation point. Figure 8 shows examples of the handheld type. Specifications for sound level meters are standardized in IEC 61672-1 (IEC 61672-1 2013). This standard provides class 1 and class 2 sound level meters. They have the same design goals, such as frequency weightings, but differ in tolerance limits and operational temperature range. Class 1 is superior to class 2.

Fig. 8 Examples of sound level meters: Rion type NL-27 (left) and Brüel & Kjær type 2250 (right)

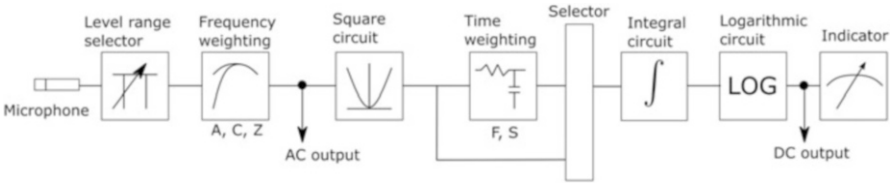

Fig. 9 Block diagram of sound level meter

Figure 9 illustrates the typical components of the sound level meter, such as microphone and preamplifier, level range selector, circuits for frequency weighting, square, time weighting, integral and conversion to logarithm, and indicator of the calculation results. Additionally, the level range is a measurement range of frequency-weighted sound pressure levels and should be selected appropriately to prevent overload and under-range. However, some advanced sound level meters do not need to change the level range because they can cover all the measurement ranges in a single level range. The formulation of the quantities measured and indicated by the sound level meter will be explained in section "Indicators of Environmental Noise Based on the Sound Pressure Level."

Time weighting is an exponential function of time with a time constant. The time constants of 125 ms and 1 s are specified in IEC 61672-1 (IEC 61672-1 2013) and are known as the "time weighting F" (fast) and "time weighting S" (slow), respectively.

The time constants are selected depending on the noise fluctuation characteristic. Time weighting F generates a time response like our auditory system, while time weighting S is used for easy reading by smoothing the fluctuating noise wave. A first-order low-pass filter realizes their time weighting circuits.

Most of the sound level meters have AC and DC output terminals, as shown in Fig. 9. DC output corresponds to the value indicated on the display of the sound level meter.

Furthermore, digital signal processing technology enabled analog circuits of sound level meters to be replaced by digital circuits, such as micro-processing unit (MPU) and digital signal processor (DSP). Some advanced sound level meters have an optional signal processing function, such as octave or FFT analysis for the noise.

Sound Calibrator

A sound calibrator is a portable sound source that produces a pressure field within its cavity when coupled to a microphone. The sound calibrator is used for checking and adjusting the sound level meter. It can also simply calibrate the pressure sensitivity of the working standard microphones. Typical sound calibrators generate a stable and sinusoidal sound pressure at 250 Hz or 1 kHz, ranging from 94 dB to 124 dB. Specifications for sound calibrators are provided in IEC 60942 (IEC 60942 2017). Figure 10 shows examples of sound calibrators.

Sound calibrators are categorized into two types by the mechanism of sound generation. The "pistonphone" creates sound pressure via the piston's vertical motion, converted from the rotating motion of a motor via a cam. Under the condition where the air in its cavity compresses and expands adiabatically, the resulting sound pressure p is given by

Fig. 10 Examples of sound calibrators: Rion NC-75 (left, loudspeaker type) and Brüel & Kjær type 4228 (right, pistonphone)

$$p = \frac{\gamma S_{SC} P_{\text{stat}} d}{V}, \tag{19}$$

where γ is the specific heat ratio of air, S_{SC} is a cross-section of the piston, P_{stat} is static pressure, d is the root mean square value of piston displacement, and V is the cavity volume formed by the pistonphone and microphone. Eq. (19) shows that the sound pressure depends on the static pressure. Therefore, when necessary, the sound pressure can be corrected to the value at the reference static pressure of 101.325 kPa. Furthermore, the cavity volume includes the microphone's acoustic characteristics. Correction values for the major microphone types are tabulated in the pistonphones' instruction manual.

Another sound calibrator type is often called "loudspeaker type" and generates sound pressure from a loudspeaker driven by a built-in oscillator. To compensate for variations in generated sound pressure by static pressure, some loudspeaker-type sound calibrators have an internal pressure sensor, and a feedback circuit automatically adjusts the sound pressure level. The more advanced type includes a feedback microphone to monitor the sound pressure level and adjust its change caused by the fluctuation of static pressure and by replacing the microphone under test.

Therefore, the loudspeaker-type sound calibrator is recommended for field use, while the pistonphone is suitable for indoor use in a well-controlled environment.

Sound Intensity Probe

A sound intensity probe is an instrument for measuring sound intensity by using two microphones that are positioned proximately. This measurement technique is known as the two-microphone method. Specifications for sound intensity instruments are provided in IEC 61043 (IEC 61043 1993). Figure 11 shows an example of the sound intensity probe.

Equation (8) shows that detecting sound intensity requires instantaneous sound pressure and particle velocity. However, direct measurement of particle velocity is difficult, and the particle velocity is approximately obtained from the difference in sound pressures observed at two proximate points. This situation is illustrated in Fig. 12.

$p_1(t)$ and $p_2(t)$ is the instantaneous sound pressure at two proximate measurement points, and Δr is the distance between these two points. The instantaneous sound pressure $p(t)$ and particle velocity $u(t)$ in the direction of sound propagation can be approximately obtained at the center point. Therefore, Eq. (6) is rewritten as follows:

$$\frac{\partial u}{\partial t} = -\frac{1}{\rho}\nabla p \approx -\frac{1}{\rho} \cdot \frac{p_1(t) - p_2(t)}{\Delta r}. \tag{20}$$

Equation (9) is similarly changed, and the sound intensity is obtained as follows:

Fig. 11 Example of sound intensity probe (Ono Sokki Co., Ltd. type MI-6410)

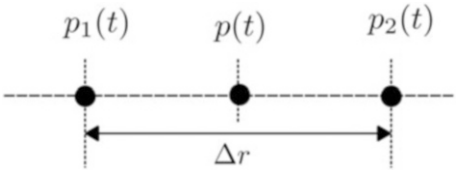

Fig. 12 Relationship between points and corresponding sound pressures

$$I(t) \approx -\frac{1}{\rho \Delta r} \overline{\frac{p_1(t)+p_2(t)}{2} \int_{-\infty}^{t} \{p_1(\tau)-p_2(\tau)\} d\tau}, \tag{21}$$

where the upper line denotes the time average.

When Eq. (21) is converted into the frequency domain, the intensity I in a frequency range from f_1 to f_2 is expressed as follows:

$$I \approx -\frac{1}{2\pi\rho\Delta r} \int_{f_1}^{f_2} \frac{\mathrm{Im}[G_{12}(f)]}{f} df, \tag{22}$$

where Im[·] denotes the imaginary part of a complex number and G_{12} is the cross-spectral density of $p_1(t)$ and $p_2(t)$. An FFT analyzer is extensively used to calculate the sound intensity based on Eq. (22).

The two-microphone method has an upper limit frequency determined by the relationship between the measurement point distance Δr and the wavelength of sound. Furthermore, the phase difference of the microphone sensitivities primarily causes the lower frequency limitation.

Reference Sound Source

A reference sound source is a stable, broadband noise source with a flat frequency response and omnidirectivity. Figure 13 shows an example of the reference sound source. The reference sound source usually generates turbulence noise from a

Fig. 13 Example of reference sound source (Brüel & Kjær type 4204)

sirocco fan. The fan rotates around a vertical axis by a motor. The fan's rotation speed depends on the voltage and frequency of the power supply and causes a similar dependency on the resulting sound power. Specifications and calibration procedures of the reference sound sources are provided in ISO 6926 (ISO 6926 2016).

The reference sound source is used as a reference for determining the sound power from a machine by using the comparison method. It is also used for measuring acoustic materials' sound absorption or insulation characteristics.

Environmental Noise Measurement Based on Sound Pressure

Overview of Environmental Noise Measurement

EU Directive defines environmental noise as "unwanted or harmful outdoor sound created by human activities, including noises emitted by means of transport, road traffic, rail traffic, air traffic, and from sites of our industrial activities" (EU 2002). ISO 1996-1 and 1996-2 (ISO 1996-1 2016 and ISO 1996-2 2017) provide indicators of environmental noise, which are determined from the sound pressure. Here, some indicators will be introduced after the classification of environmental noise. Note that the indicators' actual measurement and assessment procedures are not explained here because they should be reflected in national regulations or guidelines.

Classification of Environmental Noise by Time Fluctuation

As shown in Fig. 14, environmental noises are classified by the characteristics of time waveforms in ISO 12001 (ISO 12001 1996).

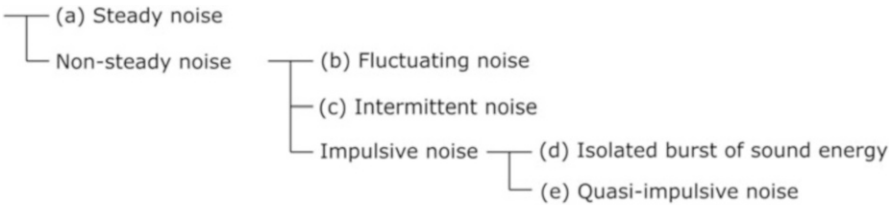

Fig. 14 Tree diagram of noises classified by time fluctuation

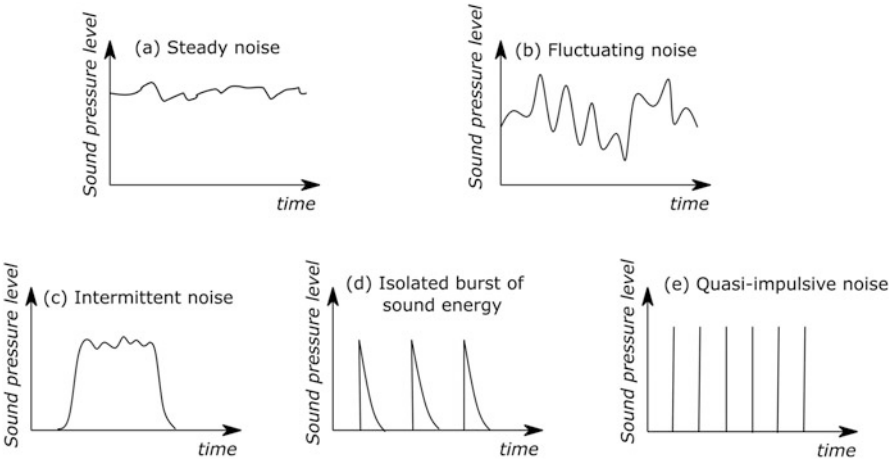

Fig. 15 Pictorial examples of noise time patterns

ISO 12001 (ISO 12001 1996) defines these noises. They are quoted below and illustrated in Fig. 15.

- Steady noise

A noise with negligibly small fluctuations of level within the period of observation (Fig. 15a).

- Nonsteady noise

A noise whose level shifts significantly during the period of observation

- Fluctuating noise

A noise whose level varies continuously and to an appreciable extent during the period of observation (Fig. 15b)

- Intermittent noise

A noise whose level abruptly drops to the level of the background noise several times during the period of observation. The time during which the level remains at a constant value different from that of the ambient is of the order of 1 s or more (Fig. 15c)

- Impulsive noise

A noise consisting of a series of bursts of sound energy, each burst having a duration of less than approximately 1 s

- Isolated burst of sound energy

A single burst of sound energy or a series of bursts with intervals larger than 0.2 s between the individual bursts (Fig. 15d)

- Quasi-impulsive noise

A series of noise bursts of comparable amplitude with intervals shorter than 0.2 s between the individual bursts (Fig. 15e).

Indicators of Environmental Noise Based on the Sound Pressure Level

Time-Weighted and Frequency-Weighted Sound Pressure Level

As introduced in section "Sound Level Meter," time-weighted and frequency-weighted sound pressure levels are the most fundamental quantity for measuring environmental noise.

ISO 1996-1 (ISO 1996-1 2016) refers to IEC 61672-1 (IEC 61672-1 2013) and mentions that time weightings F and S, and frequency weightings A and C are the standard weightings.

For example, F time-weighted and A-weighted sound pressure level $L_{\mathrm{AF}}(t)$ is defined as follows:

$$L_{\mathrm{AF}}(t) = 10\log_{10}\frac{\frac{1}{\tau_{\mathrm{F}}}\int_{-\infty}^{t} p_{\mathrm{A}}^2(\xi)\cdot \exp\left\{\frac{-(t-\xi)}{\tau_{\mathrm{F}}}\right\}d\xi}{p_0^2}, \tag{23}$$

where τ_F is the time constant for time weighting F, t is the time of observation, ξ is a dummy variable of time integration, $p_A(\xi)$ is A-weighted instantaneous sound pressure, and p_0 is the reference sound pressure of 20×10^{-6} Pa.

Maximum Time-Weighted and Frequency-Weighted Sound Pressure Level

The maximum time-weighted and frequency-weighted sound pressure level is used to evaluate single event noise with limited duration. Standard time weightings are F and S, and standard frequency weightings are A and C. For example, the symbol for the maximum F time-weighted and A-weighted sound pressure level is L_{AFmax}.

Sound Exposure Level

Sound exposure is the time integral of the square of frequency-weighted sound pressure over the interval T and gives a total sound energy of the nonsteady noise during T. If frequency weighting A is used, sound exposure level $L_{EA,T}$ is defined as

$$L_{EA,T} = 10\log_{10}\frac{\frac{1}{T_0}\int_{t_1}^{t_2} p_A^2(t)dt}{p_0^2}, \tag{24}$$

where T starts at t_1 and ends at t_2, $p_A(t)$ is A-weighted instantaneous sound pressure, T_0 is reference time of 1 s, and p_0 is the reference sound pressure of 20×10^{-6} Pa. Similar definition is applied to frequency weighting C.

The sound exposure level increases as the integration time T becomes longer. Therefore, the sound exposure level should be accompanied by a clear statement of the integral period T. Note that the sound exposure level for a single event is known as the "single event sound exposure level."

Equivalent Continuous Sound Pressure Level

The equivalent continuous sound pressure level is the time average of the square of frequency-weighted sound pressure over the time interval T and expressed in decibels. This quantity represents the energetic average of frequency-weighted sound pressure level for the nonsteady noise and is also known as the "time-averaged sound pressure level." A-weighted equivalent continuous sound pressure level $L_{Aeq,T}$ is defined as follows:

$$L_{Aeq,T} = 10\log_{10}\frac{\frac{1}{T}\int_{t_1}^{t_2} p_A^2(t)dt}{p_0^2}, \tag{25}$$

where T starts at t_1 and ends at t_2. $p_A(t)$ is A-weighted instantaneous sound pressure, and p_0 is the reference sound pressure of 20×10^{-6} Pa.

As illustrated in Fig. 16, the equivalent continuous sound pressure level of the nonsteady noise over a given period T is equal to the total sound energy of the steady

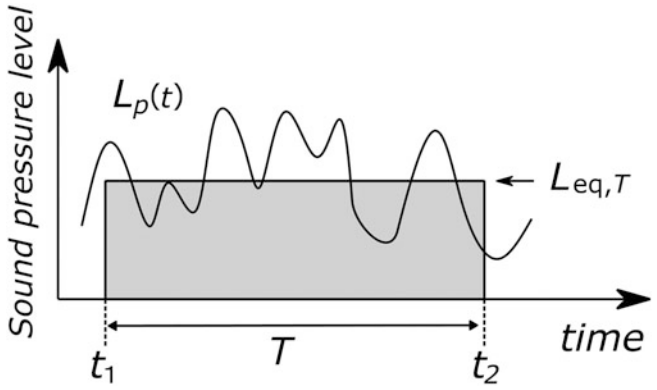

Fig. 16 Relationship between equivalent continuous sound pressure level and actual sound pressure level for nonsteady noise

sound pressure level. Consideration should be given to the measurement time interval of intermittent or impulsive noises, because this time interval may not include the whole noise event. In such a case, the start and end times of the measurement should be clearly stated.

N Percentage Exceedance Level

N percentage exceedance level represents a criterion of the sound pressure level that is exceeded for *N* % of the time interval *T*. The symbol $L_{\mathrm{AF}N,T}$ is given to F time-weighted and A-weighted sound pressure levels. The measurement procedure of $L_{\mathrm{AF}N,T}$ is as follows: Firstly, the sound pressure level is determined by the sound level meter with time interval T and sampling time ΔT. Then, the cumulative percentage distribution of the sound pressure level is obtained, as shown in Fig. 17. Finally, the sound pressure level conforming to $(100 - N)$ % of the cumulative distribution is considered as the N percentage exceedance level $L_{\mathrm{AF}N,T}$.

N percentage exceedance level is a statistic of the observed sound pressure, and the arithmetic or energetic mean obtained from a different time interval has no physical denotation.

Measurement of Noise Sources Based on the Sound Power

Overview of Sound Power Measurement

In the preceding section, several indicators of environmental noise have been introduced. They significantly contribute to monitor and assess the environmental noise. It is also important to precisely determine the sound energy emitted from the

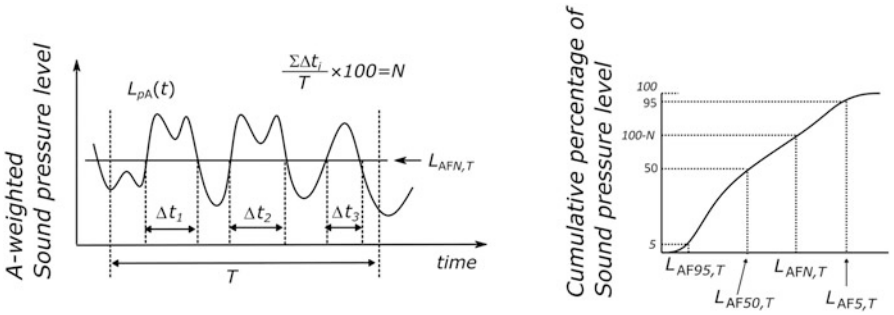

Fig. 17 Procedure to calculate *N* percentage exceedance level from acquired sound pressure level

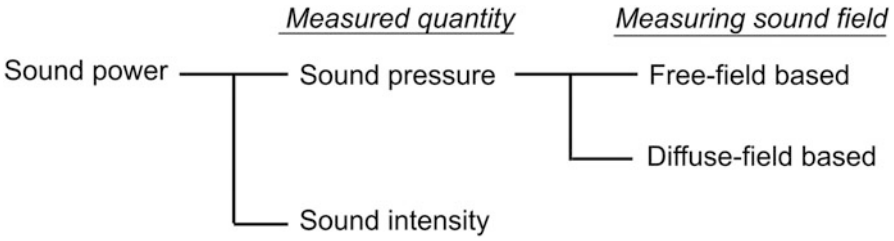

Fig. 18 Tree diagram of ISO standards for determining sound power level, categorized by measured quantity and environment

noise sources and consider their noise reduction. Sound power gives a unique value to each sound source; however, sound pressure depends on the measurement position relative to the sound source and the characteristics of the sound field. Accordingly, sound power is suitable for comparing the product noise. Furthermore, the sound power of the sound source can also be used to estimate the sound pressure at a given distance from the sound source.

Sound power is determined using Eq. (13) or (14) by taking the integral of sound intensity or sound pressure over the closed surface surrounding the sound source. Practically, the number of discretized measurement points for surface integral determines the measurement time and accuracy.

Various ISO standards have been published to obtain the sound power levels of noise sources. ISO 3740 (ISO 3740 2019) is a guidance document for the appropriate selection of the ISO standards considering the type of sound source, measurement principle, measurement environment, and required accuracy.

Here, these ISO standards will be comprehensively introduced. They are broadly classified by measured quantity and measurement environment, as shown in Fig. 18, and subdivided by the measurement method and accuracy. When applicable, the individual standards of a specific noise source should be preceded in measurement conditions.

Measurement Based on Sound Pressure in the Hemi-Free Field

Regarding the sound field for measuring sound power level, the hemi-anechoic chamber is preferable to the full-anechoic chamber for the following reasons. Firstly, the hemi-anechoic chamber has a hemispherical measurement surface; therefore, the number of measurement points is reduced by half compared with the full-anechoic chamber. Secondly, it is easy to place the sound source on a floor and at the center of the hemisphere because the hemi-anechoic chamber has a rigid floor. Finally, even heavy-weight objects can be transferred into hemi-anechoic chamber relatively easily.

Figure 19 shows the sound source placed on an acoustically reflective floor and at the center of the virtual hemisphere. The sound wave passing through the hemispherical surface is regarded as a plain progressive sound wave, because the hemisphere surface is sufficiently distant from the sound source. When the hemispherical measurement surface is divided into N segments, the sound power W in Eq. (14) is discretized as follows:

$$W = \sum_{i=1}^{N} \frac{\overline{p_i^2}}{\rho c} \Delta S_i, \tag{26}$$

where $\overline{p_i^2}$ is the time average of the square of sound pressure and ΔS_i is the area of the i-th segment, respectively. Eq. (26) is converted to logarithm as follows:

$$L_W = 10\log_{10} \sum_{i=1}^{N} \overline{L_{pi}} \cdot \Delta S_i, \tag{27}$$

where $\overline{L_{pi}}$ is the time-averaged sound pressure level. For convenience, the surface sound pressure level L_{pf} is introduced as follows:

$$L_{pf} = 10\log_{10}\left(\frac{1}{S}\right)\left[\sum_{i=1}^{N} \Delta S_i \cdot 10^{\frac{\overline{L_{pi}}}{10}}\right], \tag{28}$$

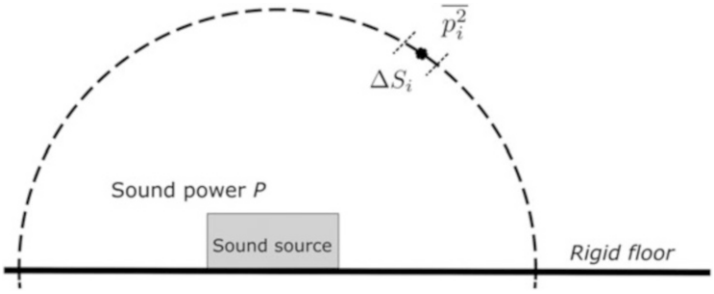

Fig. 19 Schematic diagram of sound power measurement based on sound pressure in a hemi-free field

where S is the total area of the hemispherical surface. L_{pf} is the energetic average of the time-averaged sound pressure level over the hemispherical surface. By using Eq. (28), Eq. (27) is rewritten as follows:

$$L_W = L_{pf} + 10\log_{10}\frac{S}{S_0} + C, \tag{29}$$

where S is equal to $2\pi r$ and denotes the surface area of a hemisphere with a radius r and S_0 is the reference area of 1 m^2. C is introduced to correct the temperature and pressure dependency of ρc and that of sound power of the sound source and to consider the absorption of a sound wave in the air.

There are three ISO standards for determining sound power level in a free field by using sound pressure level. They possess different accuracy, such as the "precision method" (ISO 3745 2012), "engineering method" (ISO 3744 2010), and "survey method" (ISO 3746 2010). Table 2 summarizes the differences between these ISO standards.

The accuracy depends on the degree of caution taken to determine the sound power level from Eq. (29), such as acoustic characteristics of the chamber, number of measurement points for obtaining L_{pf}, and the correction term C. The value of C is generally minute and considered only for the precision method. It is ignored in engineering and survey methods. Regarding the measurement surface, engineering and survey methods allow both hemisphere surface and parallelepiped surface.

While the precision method requires full- or hemi-anechoic chamber, engineering and survey methods allow the test chamber with higher sound absorption or outdoor. Such a practical acoustic environment will increase measurement error due to background noise and sound reflection. To compensate for this drawback, L_{pf} is replaced with L'_{pf} using the following equation for engineering and survey methods:

Table 2 Summary of ISO standards for determining sound power level from sound pressure level in free field

Standards	ISO 3744	ISO 3745	ISO 3746
Grade	Engineering	Precision	Survey
Test environment	Essentially free field over a reflecting plane	Anechoic or hemi-anechoic room	In situ over a reflecting plane
Volume of sound source	No limit; limited only by test environment	Less than 0.5% of test room volume	No limit; limited only by test environment
Number of measurement points	$\geq$9 Less than 9 is allowed in specific condition	$\geq$10	$\geq$4 Less than 4 is allowed in specific condition
Limitation for background noise	$K_1 \leq 1.3$ dB	$K_1 \leq 0.5$ dB	$K_1 \leq 3.0$ dB
Frequency range of interest	1/3 octave bands from 100 Hz to 10,000 Hz	1/3 octave bands from 100 Hz to 10,000 Hz	Octave bands from 125 Hz to 8000 Hz

$$L'_{pf} = L_{pf} - K_1 - K_2, \tag{30}$$

where K_1 and K_2 are corrections for background noise and sound reflection, respectively. Correction for background noise K_1 is obtained as follows:

$$K_1 = -10\log_{10}\left(1 - 10^{-\frac{L_{pf} - L''_{pf}}{10}}\right), \tag{31}$$

where L''_{pf} is the surface sound pressure level without the sound source operation.

Correction for sound reflection K_2 can be obtained by using the reference sound source as denoted below:

$$K_2 = L^*_W - L_{Wr}, \tag{32}$$

where L^*_W is the sound power level of the reference sound source that is obtained by engineering or survey method at the condition $K_2 = 0$ in Eq. (30). L_{Wr} is the sound power level of the reference sound source, precisely calibrated in advance.

From Eqs. (29), (30), and (32), the sound power level L_W is changed as

$$L_W = \left\{\left(L_{pf} + 10\log_{10}\frac{S}{S_0}\right) - K_1 - L^*_W\right\} + L_{Wr}. \tag{33}$$

The term in the bracket {} of Eq. (33) denotes the sound power level difference between the sound source under test and the reference sound source. Therefore, when the reference source is calibrated in advance, the sound power level of the sound source can be deduced by comparing it with the reference sound source and compensating for the sound reflection in the sound field.

Measurement Based on Sound Pressure in the Diffuse Field

As shown in Fig. 20, the sound power level can also be determined by measuring a sound pressure level in a diffuse field. In a steady state, the sound power W emitted from the sound source is equal to the sound power absorbed by the inner walls of the reverberation chamber, including the ceiling and floor, and it is formulated as follows:

$$W = \frac{c\overline{E}}{4}A, \tag{34}$$

where c is the speed of sound in air, $\overline{E}$ is the space-averaged acoustic energy density, and A is an equivalent sound absorption area of the reverberation chamber. Furthermore, $\overline{E}$ is derived as follows:

$$\overline{E} = \frac{\overline{p^2}}{\rho c^2}, \tag{35}$$

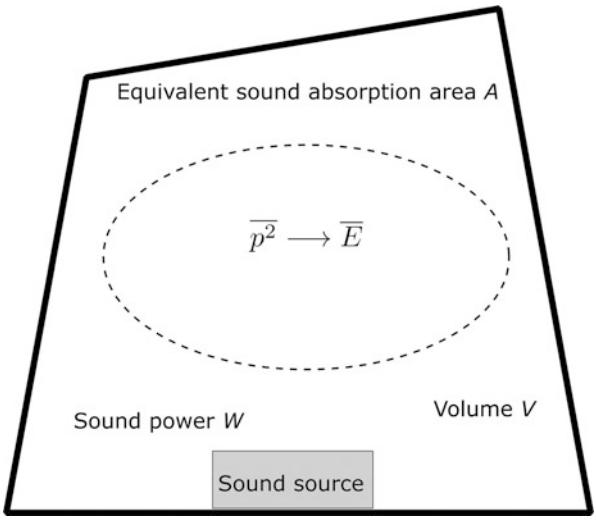

Fig. 20 Schematic diagram of sound power measurement based on sound pressure in a diffuse field

where ρ is the density of air, $\overline{p^2}$ is named "mean square sound pressure," and it means the time and space average of square sound pressure in the diffuse field. Substitution of Eq. (35) for Eq. (34) shows that

$$W = \frac{A\overline{p^2}}{4\rho c}, \tag{36}$$

where ρc is known as the "specific acoustic impedance" and is set to be 400 Pa·s/m for convenience, although it depends on temperature and static pressure.

By expressing Eq. (36) in logarithm, the sound power level L_W is expressed as follows:

$$L_W = L_p + 10\log_{10}\frac{A}{A_0} - 6, \tag{37}$$

where L_p is the mean sound pressure level in the diffuse field. A_0 is the reference of equivalent sound absorption area and is set as 1 m^2. Equivalent sound absorption area A is given by Sabine equation as follows:

$$A = \frac{55.26}{c}\left(\frac{V}{T_{rev}}\right), \tag{38}$$

where V and T_{rev} are the volume and reverberation time of the reverberation chamber, respectively.

In practice, the sound pressure is not spatially uniform in the diffuse field; therefore, the mean sound pressure level is obtained as follows:

$$\overline{L_p} = 10\log_{10}\left[\frac{1}{N}\sum_{i=1}^{N} 10^{\frac{L_{pi}}{10}}\right] - K_1, \tag{39}$$

where N is the number of measurement points, L_{pi} is the sound pressure level at the i-th measurement point, and K_1 is the correction for background noise in the diffuse field obtained in the same way as Eq. (31).

Equations (37) and (38) are the basic equations for calculating sound power levels in a diffuse field. As observed in the free-field measurement in section "Measurement Based on Sound Pressure in the Hemi-Free Field," methods for measuring sound power level using sound pressure level in the diffuse field are categorized by acoustic measurement environment and measurement accuracy. Table 3 summarizes the differences between the related ISO standards (ISO 3741 2010; ISO 3743-1 2010; ISO 3743-2 2018; ISO 3747 2010).

Actually, determining the equivalent sound absorption area A is difficult. Therefore, direct determination of the sound power level from Eqs. (37) and (38) are usually substituted with comparison measurement using the reference sound source calibrated in advance. The sound power level L_W of the sound source under test is given as follows:

$$L_W = \left(\overline{L_p} - \overline{L_{p_ref}}\right) + L_{W_{ref}}, \tag{40}$$

where $\overline{L_p}$ and $\overline{L_{p_ref}}$ are the mean square sound pressure levels of the object under test and the reference sound source, respectively, and both are obtained in the diffused field using Eq. (39). L_{Wref} is the sound power level of the reference sound source and calibrated in advance.

Table 3 Summary of ISO standards for determining sound power measurements from sound pressure level in a diffuse field

Standards	ISO 3741	ISO 3743-1	ISO 3743-2	ISO 3747
Grade	Precision	Engineering		Engineering and survey
Test environment	Reverberation test room	Hard-walled room	Special reverberation test room	Essentially reverberant field in situ
Volume of sound source	Preferably less than 2% of test room volume	Preferably less than 1% of test room volume		No limit; limited only by test environment
Limitation for background noise	$K_1 \leq 0.5$ dB	$K_1 \leq 1.3$ dB	$K_1 \leq 2.0$ dB	$K_1 \leq 1.3$ dB
Frequency range of interest	1/3 octave bands from 100 Hz to 10,000 Hz	Octave bands from 125 Hz to 8000 Hz		Octave bands from 125 Hz to 8000 Hz

Measurement Based on Sound Intensity

Equation (13) shows that sound power is principally obtained by measuring sound intensity. The acoustic environment such as a free field or diffuse field is not required for sound intensity measurement. The measurement surface of the sound intensity is divided into N segments, and Eq. (13) is discretized in the same way as section "Measurement Based on Sound Pressure in the Hemi-Free Field." The sound power level L_W is obtained as follows:

$$L_W = 10\log_{10}\frac{1}{S_0}\sum_{i=1}^{N} 10^{\frac{L_{ni}}{10}} \cdot \Delta S_i, \tag{41}$$

where S_0 is the reference area of 1 m^2. For the i-th segment, ΔS_i is its area, and L_{ni} is its sound intensity level that is normal to that segment.

Determination of the sound power level from the sound intensity level has the advantage that, in principle, the obtained sound power level is unaffected by sound sources or noises outside the measurement surface. This is because the sound intensity measurement considers only the sound waves propagating from inside to outside the measurement surface. However, measurement accuracy may deteriorate when sound-absorbing areas or objects are inside the measurement surface.

The two procedures can measure sound intensity at each segment. One is to measure the sound intensity at each discrete segment. The other is to move the sound intensity probe and scan at each segment continuously, either manually or automatically by a moving system. When a sound source has complex sound radiation characteristics, the measurement should be made at many segments. The measurement at many discrete segments will take a long time to achieve the necessary accuracy.

The scanning method considerably reduces the measurement time. However, manual scanning requires adequate operator training to achieve high measurement accuracy. In contrast, mechanical scanning does not need training, but the moving system's vibration sometimes causes a measurement accuracy problem.

Table 4 summarizes the differences between the related ISO standards (ISO 9614-1 1993; ISO 9614-2 1996; ISO 9614-3 2002) based on the measurement accuracy and method.

Table 4 Summary of ISO standards for determining sound power from the sound intensity level

Standards	ISO 9614-1	ISO 9614-2	ISO 9614-3
Grade	Precision, engineering, and survey	Engineering and survey	Precision
Test environment	No special test environment		
Volume of sound source	No restrictions		
Measurement method	Discrete points	Scanning	Scanning
Frequency range of interest	1/3 octave bands from 50 Hz to 6300 Hz		

Calibration of Acoustic Measuring Instruments and their Measurement Traceability

Traceability System of Acoustic Standards

Recently, various products and services have been required to conform to the ISO or IEC standards from the point of quality management on many occasions. For example, in acoustics, these international standards request that acoustic measuring instruments are traceable to national measurement standards to ensure their reliability.

Acoustic standards are the national measurement standards for acoustics. They have been realized as pressure sensitivities of laboratory standard microphones using more basic national measurement standards regarding voltage, length, pressure, and temperature, among others.

The National Metrology Institute (NMI) in each economy calibrates the acoustic standards. The NMI supplies the standards to calibration laboratories and manufacturers of acoustic measuring instruments, such as working standard microphones, sound level meters, and sound calibrators. They calibrate end-users acoustic measuring instruments compared with acoustic standards or other traceable acoustic measuring instruments. In some cases, the end-users instruments are calibrated by the NMI. In this way, the end-users can confirm that their instruments are traceable to the national measurement standards. Figure 21 illustrates an example of a traceability system. Other acoustic measuring instruments can also be calibrated by comparison.

Primary Calibration of Condenser Microphones

Pressure Sensitivity Calibration by Reciprocity Method

Technical requirements for the acoustic standards are as follows:

1. Acoustical characteristics are stable for a short and long time.
2. The frequency response is flat as possible over the entire frequency range.
3. The primary calibration method Is established and does not need other quantitative acoustic standards

In the audible frequencies from 20 Hz to 20 kHz, pressure sensitivities of the laboratory standard microphones satisfy the above requirements 1 and 2 and are absolutely determined using the reciprocity method. As explained in section "Classification of Condenser Microphones," laboratory standard microphones have been specifically designed for the acoustic standards (IEC 61094-1 2000). Furthermore, the reciprocity method is standardized in IEC 61094-2 (IEC 61094-2 2009).

A unique characteristic of condenser microphones is that the microphone converts an acoustic signal into an electrical signal and vice versa. The reciprocity method successfully uses this reciprocal characteristic. Firstly, two condenser

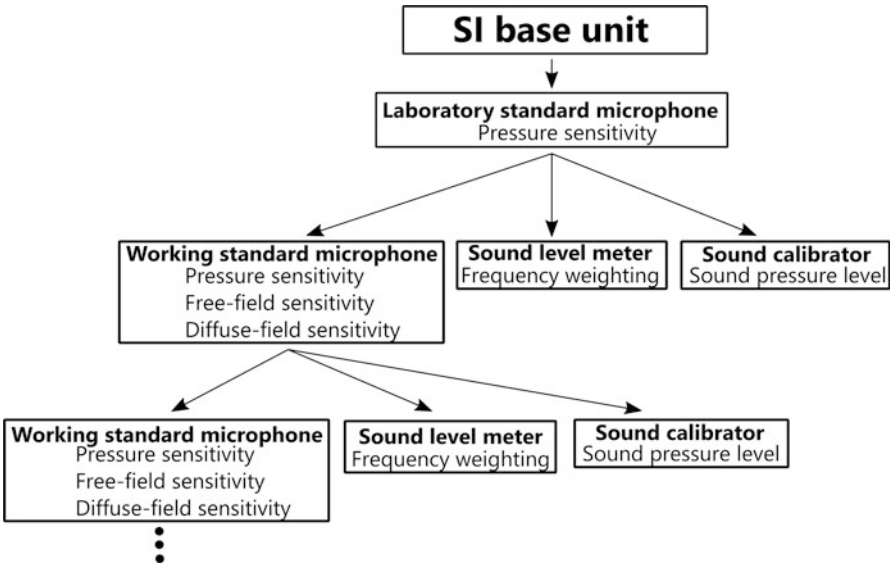

Fig. 21 Example of a traceability system for acoustic measuring instruments

microphones named x and y are placed apart in some sound field. The transmitter microphone x is supplied with the input current i_x and emits a sound wave. Next, the receiver microphone y receives the sound wave over its membrane and results in the output voltage v_y. This signal transmission is formulated as follows:

$$Z_{e,xy} \equiv \frac{v_y}{i_x} = M_x Z_{a,xy} M_y, \tag{42}$$

where $Z_{e,xy}$ is known as the "electrical transfer impedance" between microphones x and y. M_x and M_y are the sensitivities of microphones x and y, respectively. $Z_{a,xy}$ is the "acoustic transfer impedance" between microphones x and y, and it stands for acoustic coupling between the microphones. The acoustic transfer impedance is analytically derived for simple sound fields.

When pressure sensitivities of the laboratory standard microphones are determined, the microphones are acoustically coupled in an acoustic coupler through a cylindrical cavity, as shown in Fig. 22. Additionally, the acoustic coupler is usually fitted with a capillary tube to balance the static pressure inside and outside of the cavity.

When the sound pressure within the cavity is uniform and air in the cavity compresses and expands adiabatically, the acoustic transfer impedance is written as follows:

$$Z_{a,xy} = \frac{\gamma P_{\text{stat}}}{j\omega V}, \tag{43}$$

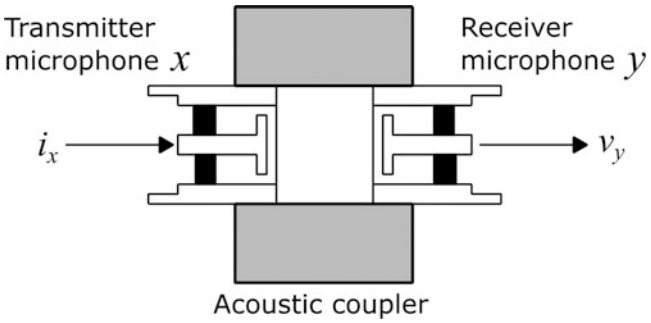

Fig. 22 Schematic diagram of microphones coupled acoustically through a cylindrical cavity

where γ is the specific heat ratio of air, P_{stat} is the static pressure, j is an imaginary unit of a complex number, ω is the angular frequency of the signal, and V is the inner volume of the acoustic coupler.

The acoustic transfer impedance should be corrected for heat conduction and viscous loss at the inner walls of the cavity and departures from uniform sound pressure distribution. Here, the corrected acoustic transfer impedance is denoted as $Z'_{a,xy}$.

Absolute determination of the pressure sensitivities requires three microphones of the same type, and they are labeled as "1," "2," and "3." Then, three pairs of transmitter and receiver microphones are sequentially made, namely (1, 2), (1, 3), and (2, 3). The following three equations are derived from Eq. (42) as follows:

$$\begin{cases} Z_{e,12}=M_1 Z'_{a,12} M_2 \\ Z_{e,13}=M_1 Z'_{a,13} M_3 \\ Z_{e,23}=M_2 Z'_{a,23} M_3 \end{cases}. \tag{44}$$

By solving Eq. (44), the pressure sensitivities of the three microphones are determined. For example, the sensitivity of microphone "3" is expressed as follows:

$$M_3^2 = \frac{Z_{e,13} \cdot Z_{e,23}}{Z_{e,12}} \cdot \frac{Z'_{a,12}}{Z'_{a,13} \cdot Z'_{a,12}}. \tag{45}$$

Precise calibration of the microphone sensitivity requires the following consideration: Firstly, as shown in Eq. (42), the electrical transfer impedance $Z_{e,xy}$ is the ratio of the output voltage of the receiver microphone to the input current of the transmitter microphone. Therefore, the insert voltage technique should be applied to the receiver microphone to precisely measure its open-circuit voltage (see section "Usage of the Condenser Microphone").

Secondly, as shown in Table 1, the capacitance of the condenser microphone is in the order of picofarad and causes a large electrical impedance. Therefore, electrical current passing through the transmitter microphone becomes smaller than the order

of microampere, and its precise measurement is difficult. When a capacitor or resistor with a known electrical impedance is connected to the transfer microphone in series, the voltage across this electrical element can be measured more efficiently and accurately. Otherwise, the electrical impedance of the transmitter microphone can be measured accurately without connecting any electrical elements.

Finally, the sensitivity of the condenser microphone slightly depends on temperature and static pressure. Therefore, the sensitivity should be corrected using its temperature and pressure coefficient if necessary.

Pressure Sensitivity Calibration by Laser Pistonphone Method

Sound waves with frequencies lower than 20 Hz are generally called infrasound. Although our sense of hearing cannot perceive the infrasound, it causes unpleasant feelings such as discomfort or oppression and makes windows or doors rattle. These sounds are generated by a large-scale blower or an air compressor installed mainly in factories. An interesting example is the occurrence of infrasound during a dam water release. Although various measures have been established to prevent infrasound generation, the number of infrasound complaints is increasing.

Therefore, precise measurement of the infrasound is required to investigate or evaluate it. Advanced sound level meters can measure the sound pressure level at frequencies down to approximately 1 Hz. However, the reciprocity method limits microphone calibration at frequencies lower than 20 Hz, because the signal-to-noise ratio deteriorates by the noise from external sound sources and internal electrical circuits.

Therefore, the laser pistonphone method (Rennie 1977) was developed and used to determine the pressure sensitivity of laboratory standard microphones. The IEC technical report covers this method (IEC TR 61094-10 2022). Figure 23 depicts the schematic of the microphone calibration system by the laser pistonphone method. A piston moves sinusoidally using a shaker and generates sound pressure in a cylindrical cavity. The piston displacement d is obtained by a laser interferometer, and the sound pressure p is given as follows:

$$p = \frac{\gamma P_{\text{stat}}}{\sqrt{2}V} S_{\text{pist}} d, \tag{46}$$

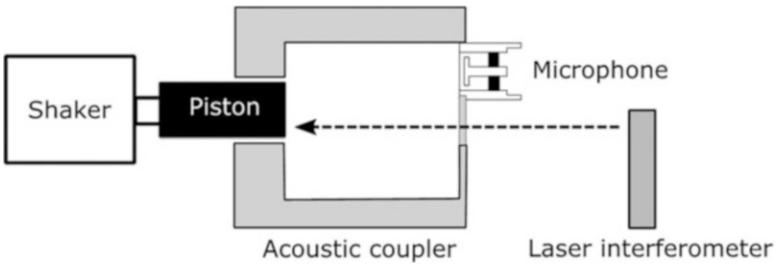

Fig. 23 Schematic diagram of laser pistonphone method

where V is the cavity volume and S_{pist} is the cross-sectional area of the piston. Piston displacement is usually determined using the interference fringe counting method and is written as follows:

$$d = \frac{n\lambda_l}{8}, \tag{47}$$

where n is the number of fringes and λ_l is the wavelength of the laser. Simultaneously measuring d and the open-circuit voltage e of the microphone under test results in the pressure sensitivity M as follows:

$$M = \frac{e}{p}. \tag{48}$$

Practically, the laser pistonphone method can generate sound pressure levels larger than 120 dB and does not have the issue of signal-to-noise ratio. The cavity's dimensions are much larger than in the reciprocity method due to the difference in measurement frequency range. A small gap between the piston and the cavity equalizes the static pressure inside and outside of the cavity.

Free-Field Sensitivity Calibration by Reciprocity Method

The reciprocity method can also determine the free-field sensitivities of the laboratory standard microphones. The calibration principle of free-field sensitivity is provided in IEC 61094-3 (IEC 61094-3 2016) and is the same as pressure sensitivity, except that the microphones are placed in a free field.

To calibrate free-field sensitivities, transmitter and receiver microphones are set in the anechoic chamber face to face and about 20–40 cm apart, as illustrated in Fig. 24.

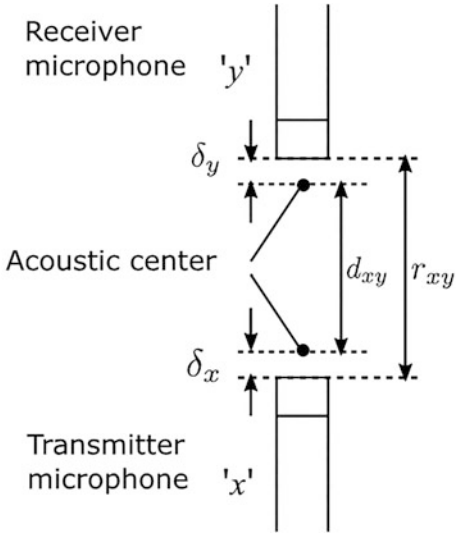

Fig. 24 Positioning of microphones mounted to rods in the free field for reciprocity calibration

In the free field, the acoustic transfer impedance, $Z_{a,xy}$, is expressed as

$$Z_{a,xy} \equiv j\frac{\rho_{xy} f}{2d_{xy}} e^{-\left(\alpha_{xy} r_{xy} + jk_{xy} d_{xy}\right)}, \tag{49}$$

where ρ_{xy} is the density of air, f is the signal frequency, α_{xy} is the air attenuation coefficient, and k_{xy} is the wave number. α_{xy} characterizes sound attenuation per unit length caused by sound travel through the air. k_{xy} is defined as $2\pi/\lambda$ using the wavelength λ of the sound wave. r_{xy} is the physical distance between the membranes of microphones x and y. d_{xy} is the virtual distance between the acoustic centers of the microphones. Theoretically, a sound source is represented by its acoustic center from which a spherical sound wave is emitted. When a reciprocal sound source is used as a sound receiver, the acoustic center of the sound receiver is defined as a point where sound pressure is effectively observed and equivalent to that of the sound source. In the case of the microphones x and y in the free field, the relationship between r_{xy} and d_{xy} is shown in Fig. 24 and formulated as follows:

$$d_{xy} = r_{xy} - \left(\delta_x + \delta_y\right), \tag{50}$$

where δ_x and δ_y are the distance between the membrane and acoustic center for microphone x and y, respectively. d_{xy} is experimentally obtained by changing r_{xy} with the least mean square method.

In practice, the reciprocity method in the free field takes much longer time than the pressure field due to the difference of signal-to-noise ratio. This is because the sound energy from the transmitter microphone is enclosed within the cavity in pressure field calibration while it spreads to all directions in free-field calibration. Therefore, the receiver microphone observes only a small portion of the sound energy. Furthermore, as shown in Eq. (49), the acoustic transfer impedance is proportional to frequency, and the signal-to-noise ratio deteriorates as frequency decreases. Consequently, performing the free-field calibration by the reciprocity is difficult when lower than 1 kHz.

Nevertheless, free-field sensitivities are frequently used by end users. The ratio of free-field sensitivity to pressure sensitivity is experimentally and analytically determined in advance and provided for the two types of laboratory standard microphones in IEC TS 61094-7 (IEC TS 61094-7 2006). Additionally, precisely determining this ratio requires pressure and free-field calibration techniques by the reciprocity method. The ratio depends on the shape and size of the microphone housing and the acoustic impedance of the microphone membrane.

So far, absolute calibration of pressure and free-field sensitivities by the reciprocity method has been performed in audible frequencies for laboratory standard microphones. Recently, noise measurement has expanded to the ultrasound range higher than 20 kHz, because high-power airborne ultrasound is used for applications such as parametric loudspeakers and haptic technology.

As explained above, the acoustic transfer impedance in the free field is proportional to the frequency. Therefore, from the signal-to-noise ratio, free-field

sensitivities of the microphones can be relatively easily obtained at airborne ultrasound range rather than audible range (Takahashi and Horiuchi 2018).

Conversely, precise calibration of pressure sensitivities by the reciprocity method becomes unrealistic at airborne ultrasound range. The dimensions of the cavity must be sufficiently small compared with the wavelength of the sound to realize uniform sound pressure distribution over the cavity. Specifically, the wavelength of the sound waves is at most 17 mm at 20 kHz and becomes smaller at higher frequencies. Therefore, preparing such a small cavity with high accuracy is difficult.

Secondary Calibration of Acoustic Measuring Instruments

Calibration of Microphone Pressure Sensitivity by Comparison Method

Calibrating the microphone sensitivity by comparison method requires two microphones: one whose sensitivity is unknown and the other calibrated in advance. They are known as the "test microphones" and "reference microphones," respectively, and their sensitivities are denoted as M_{test} and M_{ref}. Both microphones are exposed to the same sound field and the output voltage of the test microphone e_{test} and that of the reference microphone e_{ref} are measured. By assuming linearity of the microphone sensitivity and the calibration system, the ratio of the output voltages $e_{\text{test}}/e_{\text{ref}}$ is equal to that of the sensitivities $M_{\text{test}}/M_{\text{ref}}$. Therefore, the unknown sensitivity M_{test} is given as follows:

$$M_{\text{test}} = \frac{e_{\text{test}}}{e_{\text{ref}}} M_{\text{ref}}. \tag{51}$$

IEC 61094-5 (IEC 61094-5 2016) provides a comparison calibration procedure of pressure sensitivity. Both microphones are exposed to the same pressure field in an acoustic coupler simultaneously or sequentially. Additionally, the acoustic coupler has an integral sound source to produce sound pressure in the cavity. Multifrequency measurement can speed up by applying the signal, such as swept-sine or white noise, instead of the sinusoidal signal. Figure 25 shows a schematic of simultaneous and sequential calibration in an acoustic coupler using the comparison method.

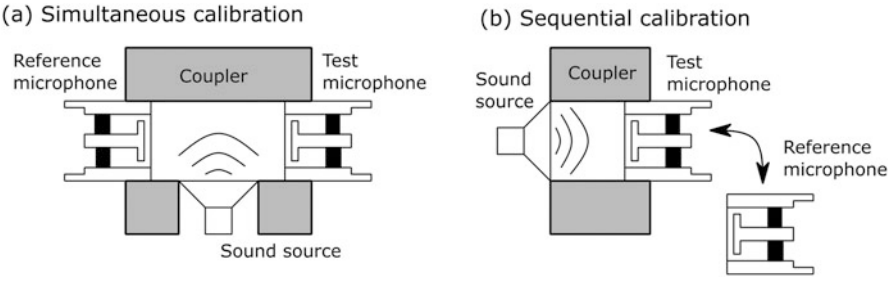

Fig. 25 Schematic diagram of pressure sensitivity calibration by comparison method

In both cases, the cavity of the acoustic coupler is designed to have sufficiently small dimensions to ensure uniform sound pressure distribution in the frequency range of interest.

Uniform sound pressure may be achieved in simultaneous calibration without needing an acoustic coupler. For example, when both microphones are mounted face to face with a separation of less than 1 mm and exposed to the sound field by an external sound source, uniform sound pressure can be applied over the membranes of the microphones at frequencies up to 20 kHz (IEC 61094-5 2016).

Furthermore, the same sound pressure should be applied to both microphones in sequential calibration. However, sound pressure may be changed in practice due to the switch of the two microphones or the fluctuating acoustic properties of a sound source. Therefore, the addition of a third microphone can monitor the change of sound pressure and correct for any changes that occur.

Calibration of Microphone Free-Field Sensitivity by Comparison Method

Calibration procedure of free-field sensitivity by comparison method is provided in IEC 61094-8 (IEC 61094-8 2012). As shown in Fig. 26, simultaneous and sequential comparisons are possible, similar to pressure sensitivity. The lower limit frequency of the free-field calibration is based on sound absorption in the sound-absorbing materials of the chamber. In a free field, microphones are positioned sufficiently far from the sound source, more than 0.5 m apart in usual cases, to realize plane progressive sound waves at the positions of microphones.

Practically, sequential calibration is preferable to simultaneous calibration. In the simultaneous calibration, it is difficult to set the two microphones at acoustically equivalent positions relative to the sound source, because the microphones and their rods for mounting interfere acoustically with each other and disturb the sound field. Sequential calibration does not have such a problem because microphones are substituted. Furthermore, in sequential calibration, the introduction of the

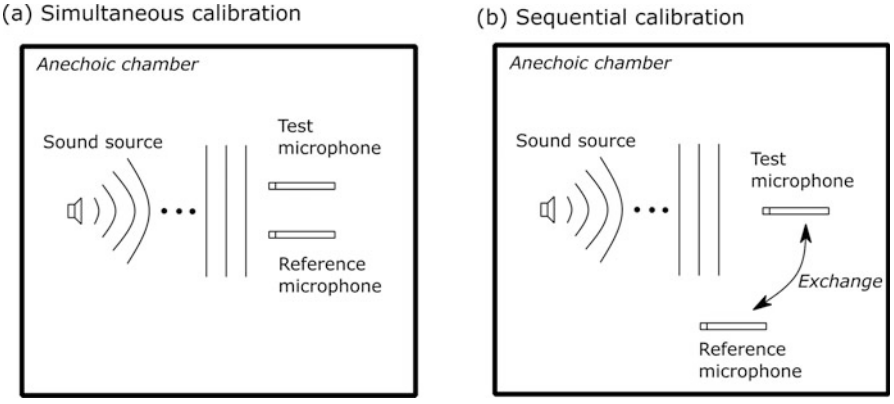

Fig. 26 Schematic diagram of free-field sensitivity calibration by the comparison method

monitoring microphone is recommended for precise measurement, because driving a loudspeaker for a long time may result in a drift of sound pressure.

Calibration of Sound Level Meter

As explained in section "Sound Level Meter," specifications for sound level meters are standardized in IEC 61672-1 (IEC 61672-1 2013). Test methods for verifying the conformity to these specifications are provided in IEC 61672-2 (IEC 61672-2 2013). A limited number of important tests are performed periodically following IEC 61672-3 (IEC 61672-3 2013). Frequency weighting is one of the essential functions of the sound level meter and is often calibrated referencing IEC 61672-2 (IEC 61672-2 2013), although this standard describes the test method to verify the conformity of the sound level meter under test. This calibration is performed for a specified frequency weighting by obtaining the difference between the sound pressure level measured with the laboratory standard microphone and the time-weighted and frequency-weighted sound pressure level indicated by the sound level meter. Therefore, this difference implies the experimental frequency weighting and is compared with the calculated design goal in the verification process.

The calibration procedure is nearly identical to the free-field sensitivity calibration of the microphone by comparison method in section "Calibration of Microphone Free-Field Sensitivity by Comparison Method," except that a device under test is changed from a working standard microphone to a sound level meter. Firstly, the sound source produces the sound wave for a given frequency, and the sound pressure level is determined at the reference point of the microphone of the sound level meter using the open-circuit voltage and free-field sensitivity of the reference microphone. This reference point of the microphone is located at the center of the membrane. Then, the sound level meter is placed instead of the reference microphone, with the microphone of the sound level meter at the same point as the reference microphone. The sound level meter's indication of time-weighted and frequency-weighted sound pressure levels is recorded visually or automatically. Sound level meters usually output the indication value from the DC output terminal or through interfaces such as RS-232C and USB. If DC output voltage is used, the quantitative relationship between the indication and the DC output voltage should be determined beforehand.

Practically, consideration is necessary for calibrating frequency weighting A at frequencies lower than 100 Hz. The design goal of the frequency weighting A is −50.5 dB at 20 Hz. However, even when the sound pressure level is set to 80 dB at the position of the sound level meter, the A-weighted sound pressure level becomes 30 dB. It is challenging to realize the stable measurement of the A-weighted sound pressure level by considering the self-generated noise of the sound level meter and the maximum sound pressure that can be produced from the sound source.

This problem can be resolved by considering the nature of frequency weighting. The following two factors cause deviation of the frequency weighting from its design goal: reflection and diffraction of the sound wave caused by the housing of the sound level meter and the influence of the internal electrical circuit of the sound level meter. Acoustical influence can be considered by applying the above

calibration procedure to frequency weighting C or Z in the free field, because the frequency weightings C and Z do not have the signal-to-noise ratio problem. A sensitivity $M_{\mathrm{SLM_}a}$ of the sound level meter is obtained as a ratio of AC output voltage $e_{\mathrm{AC_}a}$ to the sound pressure p_a, which is determined by the reference microphone. This sensitivity $M_{\mathrm{SLM_}a}$ includes the influence of sound reflection and diffraction.

On the other hand, electrical influence is obtained as follows: The microphone of the sound level meter is replaced with a dummy microphone with a capacitance similar to the real microphone. The sound level meter is set to the same frequency weighting as acoustic calibration. First, input voltage e is applied to the dummy microphone and the AC output voltage $e_{\mathrm{AC_}e}$ is recorded. $e_{\mathrm{AC_}e}$ includes only the influence of the electrical circuit of the sound level meter. The sound pressure corresponding to this AC output voltage $e_{\mathrm{AC_}e}$ is estimated as $p_{\mathrm{e}} = e_{\mathrm{AC_}e}/M_{\mathrm{SLM_}a}$ using the above sensitivity $M_{\mathrm{SLM_}a}$. Then, without changing the input voltage e, frequency weighting is changed to A, and the indication L_{AF} is recorded. Finally, the experimental A-weighing A_{exp} is obtained as follows:

$$A_{\mathrm{exp}} = L_{\mathrm{AF}} - 10\log_{10}\left(\frac{{p_{\mathrm{e}}}^2}{p_{ref}^2}\right). \tag{52}$$

The experimental frequency weighing can also be determined in the pressure field using the sound calibrator with a known sound pressure level. This method is frequently used to check the sound level meter or as a simplified testing method provided in IEC 61672-3 (IEC 61672-3 2013).

Calibration of Sound Calibrator

The sound calibrator's output sound pressure is typically calibrated according to Annexes A and B of IEC 60942 (IEC 60942 2017), which describes the pattern evaluation tests and periodic tests.

The sound pressure level of the sound calibrator is determined using the working standard microphone with a known pressure sensitivity level. When the microphone is coupled to the sound calibrator, the open-circuit voltage of the microphone is measured by the insert voltage technique (see section "Usage of the Condenser Microphone").

Furthermore, the sound pressure level can be simply calibrated with the reference sound calibrator having a known sound pressure level. Additionally, a common microphone is used to compare the output voltages for both sound calibrators. Therefore, the microphone does not have to be calibrated in this case.

Calibration of Reference Sound Source

ISO 6926 (ISO 6926 2016) standardizes two methods for calibrating the sound power level of reference sound sources as follows: One is the precision method in the hemi-anechoic chamber and refers to ISO 3745 (ISO 3745 2012). The other is also the precision method in the reverberation chamber and refers to ISO 3741 (ISO 3741

2010). Details of each precision method on sound power level measurement are explained in sections "Measurement Based on Sound Pressure in the Hemi-Free Field" and "Measurement Based on Sound Pressure in the Diffuse Field," respectively.

Recent Advances in Acoustic Measurement

New Noise Sources: Unmanned Aerial Vehicle Noise Measurement

Unmanned Aerial Vehicles (UAVs) have been widely used in many fields such as transportation, video, and agriculture. It may not be long before UAVs fly over our living space and become new noise sources. Recently, European Union has included the UAV noise in the regulation as one of the airframe specifications (Official Journal of European Union 2020).

However, current methods for evaluating the other noise sources cannot be applied to UAVs. For example, aircraft noise is evaluated by measuring the A-weighted sound pressure levels near the airport or under the predetermined flight path (ISO 20906 2009). On-site measurement of the sound pressure level is unsuitable for UAV noise, because the UAV does not necessarily move along the fixed route.

So far, various methods have been proposed for evaluating UAV noise based on sound pressure or sound power measurement. For example, outdoor measurement of the A-weighted sound pressure level is performed on the ground, which is assumed to be a hemi-free sound field (Kloet et al. 2017). However, proper measurement conditions, such as the flight altitude of the UAV and the microphone position relative to the UAV, should be discussed and provided to obtain comparable measurement results.

Alternatively, a sound power level of the UAV) gives a unique value independent of measurement conditions. However, conventional measurement methods are not intended for noise sources flying in the air. Therefore, the method specialized in measuring the sound power level of UAVs is being developed. Such a method is based on acoustic test rooms' sound power level measurement (Papa et al. 2016). However, flight operation of the UAV in a room is usually difficult because the satellite positioning system is unavailable and flight space is limited. Therefore, the hovering state is realized by fixing the UAV on a stand.

Alternative Use of Sound: Infrasound

The lower frequency limit for the infrasound complaints was around 1 Hz; however, the measurements at much lower frequency have recently been highlighted. Sounds below 1 Hz occur in natural events such as volcanic eruptions and landslides.

A well-known recent event is the 2022 Hunga Tonga–Hunga Ha'apai eruption and tsunami in the Tongan archipelago in the southern Pacific Ocean. Pressure

fluctuations, such as infrasound, caused by this eruption were observed on a global scale. As shown in this case, infrasound occasionally travels globally and long distances (Matoza et al. 2022).

To exploit this characteristic, some applications are already in operation. For example, one application is detecting or preventing natural disaster events. The basic idea of this application is to observe infrasound that occurs during the event at several points and then estimate the point where the event occurs from the time differences among the monitoring points by applying signal processing techniques.

Another application is for weather forecasts or climate change monitoring. For example, when infrasound travels long distances on a continental scale, atmospheric conditions, such as temperature and pressure, affect infrasound propagations. Therefore, observing the difference in infrasound propagation provides information on the atmospheric conditions, such as sudden stratospheric warmings. Furthermore, long-term observations on the scale of years also contribute to monitoring climate change (Pichon et al. 2019).

These applications require multiple observatories, and currently, various infrasound observation networks, from large-scale to small-scale, are in operation or under development by various organizations. The International Monitoring System (IMS), operated by the Comprehensive Nuclear-Test-Ban Treaty Organization (CTBTO), is a well-known network. This network monitors seismic, hydroacoustic, infrasonic, and radionuclidic phenomena at 321 monitoring stations and 16 laboratories built worldwide. Of these, 60 stations monitor infrasound (CTBTO 1997). The IMS network was designed and has been operated to detect nuclear explosions. The data collected by the IMS network is now used for disaster warnings and scientific studies, including climate change (CTBTO 2015).

In any infrasound observation network, the infrasound is measured by microphones, vibrometers, and microbolometers. However, the acoustic standard below 1 Hz has not been established nor have calibration methods for infrasound measuring instruments. Therefore, all infrasound observation networks are operating without calibration. Once calibration is accomplished, and the reliability of acquired data is improved, the various application performance is expected to improve. For instance, it can improve the location estimation where the natural disaster occurs. Accordingly, several national metrology institutes strive to realize the acoustic standard below 1 Hz and find calibration methods for various infrasound measuring instruments (EMPIR 2020).

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Strategies and Implications of Noise Pollution Monitoring, Modelling, and Mitigation in Urban Cities

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S. K. Tiwari, L. A. Kumaraswamidhas, and N. Garg

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Abstract

The present chapter provides an exhaustive review on the noise monitoring studies, comparison of the prediction models including physical propagation model, and applications of the artificial intelligence techniques, noise mapping, and noise pollution monitoring in mining sector carried out by various researchers. Most of the noise pollution studies deal with the assessment of traffic noise and some were focused exclusively on noise monitoring for the residential,

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educational, industrial, and commercial sites noise. The study reveals that early models were based on mathematical prediction models, later machine learning and deep learning methods were generally used for prediction and forecasting of noise levels. A retrospective view on noise mapping and control is presented in the chapter. Also, the noise pollution control and abatement measures are highlighted that shall be indispensable for reducing the ambient noise levels in metropolitan cities of the country.

Keywords

Noise assessment · Noise monitoring · Noise mapping · Noise prediction · Ambient noise standards

Introduction

Environmental noise pollution has become a serious concern since past decade across the globe. Noise assessment and modelling has been a subject of extensive research in larger parts of world as it poses serious risks to health and quality of life (Ising and Kruppa 2004; Stansfeld and Matheson 2003). Many studies have been conducted over last few decades on noise pollution, including noise sources (Zeng and Zhan 2005; Yang et al. 2011), noise pollution levels (Arana 2010), exposure levels (Brown and van Kamp 2009), strategic noise map (Ausejo et al. 2010; Makarewicz and Galuszka 2011), sampling strategies (Doygun and KuşatGurun 2008; Gozalo et al. 2013), and psychological and physiological effects (Morillas et al. 2005; Öhrström 2004). The European Union issued Directive 2002/49/EC (European Noise Directive 2002) relating to assessment and evaluation of noise pollution effects on community. This directive suggests the use of noise maps of the urban agglomerations in single noise parameters: L_{den} and L_{night} represented as day-evening-night and night equivalent levels over a year. The long-term noise monitoring strategy is expensive and cumbersome process. However, there are some notable example of cities which have installed extensive Noise Monitoring stations such as Lille in France (Chopard et al. 2007), Gdansk in Poland (Czyzewski et al. 2012) and Dublin in Ireland (Dublin City Council 2013).

There are three types of noise sources, i.e., Point source, linear source, and Surface source (Numan 2008). For point source, the waves propagates spherically in air. As a result, the intensity of the sound pressure level vary depending on the source to receiver distance. While, the linear source is a collection of points that travel in a straight line, separating one another at regular interval of equal distance, and surface source is a representation of two-dimensional wave emitted from the surface, such as wall window, machine room, or lab interface (Al-Taai 2021). The level of noise generated by a source are calculated using propagation models (De Coensel et al. 2005). Steele et al. (2001) thoroughly analyzed some of the significant noise prediction models. Some of the noise prediction models still in use are: Calculation of Road Traffic Noise model (CoRTN), Federal Highway Administration model (FHWA), Swiss Road Traffic Noise model (STL-86),

(MITHRA), Richtlinienfür den Lärmschutz an Strabenmodel (RLS-90), Nordic Prediction Method for Traffic Noise model (NORD-2000), French method for road traffic noise prediction model (NMPB-Routes-2008), Acoustical Society of Japan noise model (ASJ-2008), Common Noise Assessment Methods in Europe model (CNOSSOS-EU-2009), etc. The latest versions of some of these models are also available and used widely in some countries. In physical propagation model, the noise level at a receptor position is usually modelled as a function of traffic conditions (i.e., traffic speed, traffic volume, and traffic composition), absorbent ground cover percentage, distance between the source and receptor (Tiwari et al. 2022). Recently, artificial intelligence has become an emerging trend for prediction of noise data which requires past observational data. Artificial neural network (ANN) is one of the most used machine learning technique for prediction of noise levels. Cammarata et al. (1995) used neural network to predict noise levels in Italy. It is also compared with some traditional algorithms, where ANN shows better performance. Hamada et al. (2008) used regression based neural network in construction area based on BPNN to analyze the impact of noise. There are some limitations of neural network which is further addressed by using deep learning algorithm. Zhang et al. (2020) predicted environmental noise level using LSTM (long short term memory) at different interval of time. The proposed model was compared with stacked auto-encoder, random walk, and support vector machine. Navarro et al. (2020) predicted loudness level, and sound pressure level using LSTM (Long short term memory) from 1 to 60 min. The European Union has laid the foundation for creating noise maps of cities as a tool for environmental noise management and control (Štimac 2005; Kliučininkas and Šaliūnas 2006). Collecting data of the traffic flow for a specific time period, and area in the concerned city is a common strategy for conducting the street traffic flow analysis and obtaining a traffic noise forecast map (Popescu et al. 2011). The aim of the present study is to have a retrospective view on noise monitoring studies, and propagation models and prediction models. The objective of the work is summarized as:

- Ascertaining the noise scenario and methodology adopted for developing the various noise prediction models,
- Noise pollution monitoring, mapping and control studies carried out by various researchers and recommendation of harmonized noise mapping strategy in India,
- Ascertaining the significance and efficacy of the various control measures and their implications in Indian perspectives,
- Recommendations on policy measures for noise pollution control in the metropolitan cities.

Noise Monitoring Studies

Noise is typically used to refer to any annoying or intrusive sound, and noise pollution is among the general public environmental concerns. Noise monitoring strategies have a vital role in analyzing about the noise pollution scenario. Table 1

Table 1 Noise assessment studies of different cities

Author	Country	City	Error	Conclusions
Hunashal et al. (2012)	India	Kolhapur	L_{10}, L_{50}, L_{90}, NC, L_{eq}, NEI	The study indicated higher value of noise level in residential, commercial, educational, silence, industrial and recreational zones of the city.
Ravindra et al. (2016)	India	Chandigarh	L_{max}, L_{eq}	The sound pressure level exceed the permissible limit.
Yadav et al. (2021)	India	Gorakhpur	L_{eq}	The noise level was higher for all the three area considered for the study.
Masum et al. (2021)	Bangladesh	Dhaka	L_{eq}, NC, NPL	The study found that day equivalent noise level is 20–30 dB(A) higher than the night equivalent level.
Kuldeep et al. (2021)	India	Kota	L_{eq}	Noise level were measured at 10 locations of the city, and the study concludes that the noise levels were more than the limits suggested by the CPCB.
Esmeray et al. (2021)	Turkey	Safranbolu	L_{eq}	The noise level at 47 points of Safranbolu district is at medium level.
Sahu et al. (2020)	India	Burla	NC, NLP, TNI, NEI	The study indicated that the noise pollution reached to an alarming level in the city.
Balashanmugam et al. (2013)	India	Chidambaram	NLP, L_{10}, L_{50}, L_{90}, NC, L_{eq}	The noise level at 10 selected locations of Chidambaram town exceeds than the permissible limits suggested by the CPCB.
Nungate et al. (2022)	India	Imphal	L_{10}, L_{50}, L_{90}, L_{eq}	The noise level lies between 52.2–71.1 dB(A) for morning, noon, and evening hours at traffic junction of Imphal city.
Manojkumar et al. (2019)	India	Vellore	NC, NLP, TNI, L_{eq}	An assessment of noise level at 6 sites of Vellore city is done in afternoon, and evening hours and results showed that at 5 locations, noise level exceeds the permissible limit.

Note: L_{eq} – Equivalent noise level, L_{max} – Maximum equivalent noise level, L_{10}, L_{50}, L_{90} – Noise levels exceeds by 10%, 50%, 90%, *NC* Noise climate, *TNI* Traffic noise index, *NPL* Noise pollution level, *NEI* Noise exposure index

Table 2 Short-term and temporal strategies studies by various researchers

Author	Country	Year	Error	Conclusion
Utley et al. (1982)	UK	1982	Accuracy (in dB)	Sampling strategies can be a better substitute for long-term measurement for some applications.
Gaja et al. (2003)	Spain	2003	Mean, Standard Deviation, Probability [±1, ±2, ±3]	Random 9 days strategy is suggested for an error of ±1 dB.
Romeu et al. (2011)	Spain	2011	Correlation Coefficient, Skewness, R^2	15 min measurement is suggested for an error of ±2 dB(A).
Morillas and Gajardo (2014)	Spain	2014	Standard Deviation	Random 9 days strategy is recommended.
Garg et al. (2015)	India	2015	Standard Deviation, Prob. Dist.	Random 2 months strategy is suggested.
Geraghty et al. (2016)	Ireland	2016	ANOVA	Short-term noise monitoring cannot characterize long-term indicators accurately.
Tiwari et al. (2021a)	India	2021	Mean, Standard Deviation, Probability [±1, ±2, ±3]	Random 2 months strategy is suggested for all the four zones in India.

gives the detail of some studies on assessment of noise monitoring using various noise indices.

Continuous monitoring of noise level at each location of a city or country is a cumbersome and expensive process. Therefore, researchers have used short-term and temporal sampling strategies as a substitute which is described in Table 2.

Noise Prediction Models

Noise prediction models play a vital role for city planning for control and abatement of noise pollution. Therefore, a lot of mathematical/hybrid/artificial intelligence models had been developed in past years introducing sound propagation and sound specific emission empirical formulation. Thus, in order to achieve high performance, these models make use of high-speed processing computers. Therefore, it is crucial to conduct a scientific analysis and comparison of various models in order to determine their applicability generally and to identify which method is optimal for traffic noise modelling. The method of conceptually estimating the noise levels in a location of interest under a certain set of circumstances is the basic framework of noise modelling. A fixed representation of the relevant physical environment will make up the particular set of conditions for which the noise is being assessed. In reality, the physical world is typically not static but rather is characterized by continually changing circumstances. The actual sound environment

will then change in time and place as a result of these modifications in the real-world settings.

Noise Prediction Using Mathematical Models

The primary variables focused on developing a noise model are: (a) various noise sources, (b) an approximate representation of the physical variables through which noise will be transmitted from the source of the noise in the area such as built environment, terrain, and atmospheric conditions (temperature, wind, humidity), (c) path taken by sound from the noise source to the receiver position or spatial location. The mathematical noise prediction models developed due to various noise sources such as rail, industrial, road, or road traffic is given in the form of equation (Steele 2001):

$$L_{\mathrm{SPL}} = L_{\mathrm{w}} - A_{\mathrm{tot}} \tag{1}$$

where, L_{SPL} represent the sound pressure level at the receiver side, L_{w} denotes the sound power level, and $\mathrm{A_{tot}}$ represent the total attenuation factor between the source and receiver. The attenuation factor includes geometric divergence, ground absorption, diffraction dissipation, and air dissipation. Table 3 summarizes some of the Traffic noise models and their features.

Time-Series Prediction and Forecasting Approach

When a single variable is measured against time, or when time-series data are involved, time series analysis is a statistical technique that can be used to simulate real-world events. The development of intelligent models to anticipate noise levels prediction is a major area of research due to its rapid advances in computer technologies and the growing amount of traffic data. Computer systems learn to carry out tasks like pattern recognition, grouping, categorization, predictions, and other activities through the use of these models, which are closely related to artificial intelligence. Due to its great efficiency in identifying complex nonlinear features, outperforming statistical methods, and managing difficult noise level prediction forecasting problems, machine learning is becoming more popular for the task of predicting traffic flow. In this subsection, some machine learning algorithms such as Artificial neural network (ANN), Emotional Artificial neural network (EANN), Wilcoxon neural network (WNN), Extreme gradient boosting (EGB), Random forest with Artificial neural network (RF-ANN), Support vector machine (SVM) are highlighted. Deep learning approaches are now being used to improve noise levels predictions by utilizing the capabilities of artificial intelligence models. In order to learn high-level features, deep learning techniques are used, which is an advancement of machine-learning techniques, having significant amount of processing layers. In past years, it has been used as a prediction tool to predict and forecast ambient noise level, volume, and annoyance. In addition, the most recent deep

Table 3 Traffic noise models and their features

Mapping	Line to point	Multiple dual points to points	Line to point	Line to point	Line to point	Line to point	Line to point	Line to point	Multiple points to point	–
Prediction	No	No	No	No	No	No	No	No	No	No
I/P data	Traffic speed, flow, type, road data, and environmental data	Traffic speed, flow, type, environmental data, and emission data	Traffic flow, type, environmental data, and road	Traffic flow, type, environmental data, and road surface	Traffic flow, type, environmental data, and road	Traffic speed, flow, type, and barrier	Traffic flow, intensity, road surface, and temperature	Traffic flow, road gradient, vehicle category, and road surface	Traffic speed, flow, type, gradient, environmental data, and road surface	Traffic flow, type, temperature, and road surface
Vehicle type	Light, and heavy vehicle	Light, medium, and heavy vehicle	Light, & heavy vehicle, and urban trains	Light, and heavy vehicle	Light, medium, and heavy vehicle	Light, and heavy vehicle	Light vehicles (less than 3500 Kg), medium vehicle (between 3500 to 12,000 Kg), and heavy vehicle (more than 12,000 Kg)	Light vehicles (less than 3500 tonne), and heavy vehicle (more than 3.5 tonne)	Light, medium, and heavy vehicle	Light, medium, and heavy vehicle
Model type	Hybrid	Mathematical/ Hybrid	Hybrid	Hybrid	Hybrid	Hybrid	Mathematical	Mathematical	Mathematical	Mathematical/ Hybrid
Noise descriptors	L_{10} (1 to 18 h)	DNL, L_{Aeq}, and CNEL	L_{eq}	L_{10} (8 to 20 h)	L_m, L_{mE}, and L_{eq}	L_{eq}	L_{den},L_{night},$L_{Aeq,T}$ and LA_{eq}	$L_{Aeq,T}$ and LA_{eq}	L_{50}, and L_{eq}	L_{den},L_{night}, and L_{eq}
Traffic condition	Grades, and constant speed	Acceleration, grade, and constant speed	Grades, and constant speed	Grades, and constant speed	Interruption, grades, intersection, and constant speed	Constant speed	Urban road, motorway, and main road	Acceleration, deceleration, and constant speed	Acceleration, deceleration, roads, constant speed, and intersection	Grades, intersection, and constant speed
Source	Simple stream	Simple stream	Simple stream	Simple stream	Simple stream	Simple stream	Simple stream	Simple stream	Simple stream	Simple stream
Model	CoRTN	FHWA	STL-86	MITHRA	RLS-90	ERTC	NORD-2000	NMPB-Route-2008	ASJ-2008	CNOSSOS-EU

(continued)

Table 3 (continued)

Mapping	Line to point	Multiple dual points to points	Line to point	Line to point	Line to point	Line to point	Line to point	Line to point	Multiple points to point	–
Country users	UK	USA	Switzerland	France, and Belgium	Germany	Thailand	Sweden, Norway, Finland, and Denmark	France	Japan	Sweden, Norway, Finland, and Denmark
Proposed year	1975	1978	1986	1987	1990	1999	2000	2008	2008	2009
References	Givargis et al. (2008)	Hankard et al. (2006)	Balzari et al. (1988)	Patel et al. (2022)	Lee et al. (2009)	Patel et al. (2022)	Ögren et al. (2000)	Dutilleux et al. (2010)	Yamamoto et al. (2010)	Faulkner et al. (2022)

Table 4 Various artificial intelligence techniques utilized for noise prediction

Authors	Model	Input data	Performance matrix			
			RMSE	MAE	ACC%	R^2
Garg et al. (2016)	ANN	L_{day}, L_{night} (A & C weightings)	2.0 & 1.3 for A weighting & 1.2 and 1.5 for C weighting	0.6 & 0.4 for A weighting & 0.2 and 0.5 for C weighting	–	0.5 & 0.6 for A weighting & 0.3 and 0.6 for C weighting
Nanda et al. (2018)	WNN	L_{eq}	0.127	0.117	–	–
Zhang et al. (2020)	LSTM	Noise data (L_{eq}) for 1 min, 10 min, 30 min	1 min (1.77–3.20) 10 min (0.48–0.93) 30 min (1.14–2.38)	1 min (1.41–1.95) 10 min (0.31–0.69) 30 min (0.73–1.57)	–	–
Wen et al. (2020)	GBM	L_{eq}	<1.0	–	–	> 0.7
Yin et al. (2020)	EGB	L_{Aeq}	4.54	–	–	0.71
Ahmed et al. (2021)	RF-ANN	L_{eq}(15 min)	2.378	–	–	0.835
Zhang et al. (2021)	Bi- GRU	Traffic noise (L_{eq})	2.4 ± 0.7	2.0 ± 0.5	–	–
Tiwari et al. (2021b)	SVM	L_{day}, L_{night}	1.3, 1.6	1.2, 0.9	–	0.7, 0.6
Kuznetsov et al. (2022)	EANN	L_{Aeq}	–	–	95.6	–
Tiwari et al. (2022)	AE-LSTM	L_{day}, L_{night}	0.563	0.519	99.6	0.9

learning approach offers understanding of high-dimensional data mining. Some of the deep learning algorithms which has been used in the study were Long short term memory (LSTM), Gradient boosting model (GBM), Bi-directional gated recurrent unit (Bi- GRU), Auto-encoder-based Long short term memory (AE-LSTM) as shown in Table 4.

Noise Mapping: Strategy and Implications

Noise mapping is a method for representing estimated noise levels over a specific geographic area. The European noise directive defines noise mapping as "the representation of information on a current or estimated noise levels in terms of a noise indicator, specifying contraventions of any relevant maximum permissible in

force, the proportion of persons affected in a specific region, or the number of residences exposed to such values of a noise indicator (EC (European Commission) 2002).Numerous such studies on creating noise maps of different cities had been published in Europe and some other nations as well (Vogiatzis and Remy 2019; King et al. 2011; Escobar et al. 2012).The idea of noise mapping is a rigorous method that includes the following steps (de Kluijver and Stoter 2003):

- Identification of noise sources and collecting the noise indicators in that particular area;
- Noise level computation in a model;
- Presentation of noise level contour;
- Determination of noise impact and its effects.

Although the key phases of noise mapping are identifying hot spots, understanding the noise scenario, and determining the noise impact of various noise sources, the studies would be insufficient and ineffective without taking into account and implementing noise control measures.

Strategy and Implications

The noise maps may be created using established noise models incorporated with Geographical Interface Systems, or they may be produced using actual data taken at various locations using a specified sampling technique. Figure 1 displays the advantages of noise mapping, and Fig. 2a, b shows the framework of noise mapping. To offer an overall noise picture of the site, mapping can be done either strictly for particular noise sources, such as, railroads, road traffic, industries, aircraft, etc., or in combination with all the sources for the region under consideration (Akhtar et al. 2016). There are various considerations used in for mapping an area as follows:

- Sampling strategy
- Positioning of the microphone in accordance with the ISO 1996-2 (ISO 9613-2 1996) standard
- No of measurement points to be chosen in an area
- Control measures to be adopted

Noise Mapping Software

The noise map gives legislative authorities, local planners, researchers, and architects the baseline information they need to design and carry out new initiatives. Maps can be created either using data from noise monitoring or by using a recognized noise model. Many nations have created their own independently verified noise prediction models (Garg et al. 2021). Commonly used noise mapping software's are:

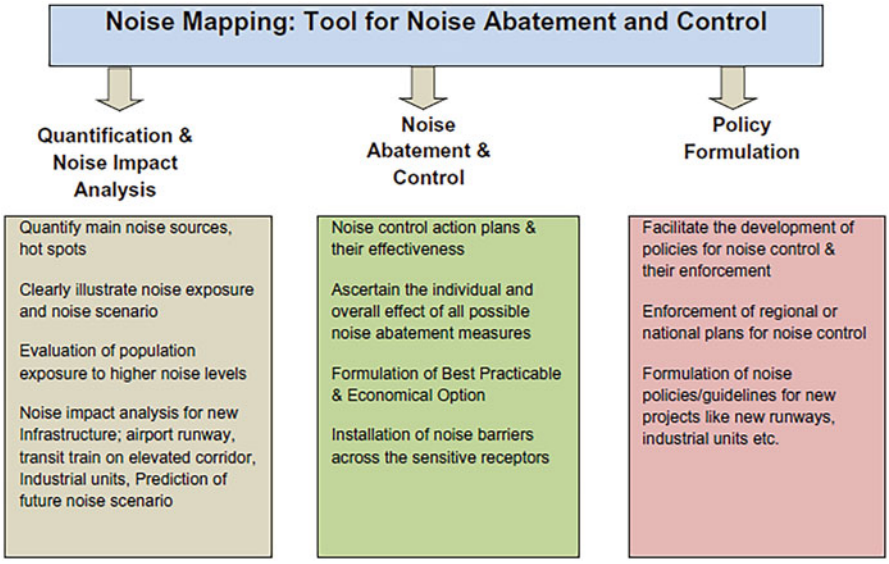

Fig. 1 Noise mapping tool. (Garg et al. 2021)

Arc GIS, Sound PLAN, Cadna A, and Predictor- LIMA. To create cross-sectional grid noise maps for various areas of the city, suitable locations should be chosen based on a clear sampling plan, in conjunction with long-term and short-term noise monitoring, and then an interpolation approach should be used. Additionally, comparing maps from various cities is made straightforward and easier by the use of a country's standardized color coding system.

Ambient Noise Standards

It is necessary to regulate and control noise producing and generating from different sources with the aim of maintaining the ambient noise levels in public areas from various sources, including generator sets, vehicular horns, industrial activity, construction activity, music systems, loud speakers, public address systems, and other mechanical devices, which have detrimental effects on human health and the psychological well-being of the people. The ambient noise standards recommended in India are tabulated in Table 5 (The Noise Pollution 2000):

- For the purpose of enforcing noise rules at various places, the State Government shall classify the areas as residential, silence, industrial, or commercial zones.
- The State Government must take action to reduce noise, including noise from moving vehicles, and must make sure that the current noise levels do not surpass given in these regulations.

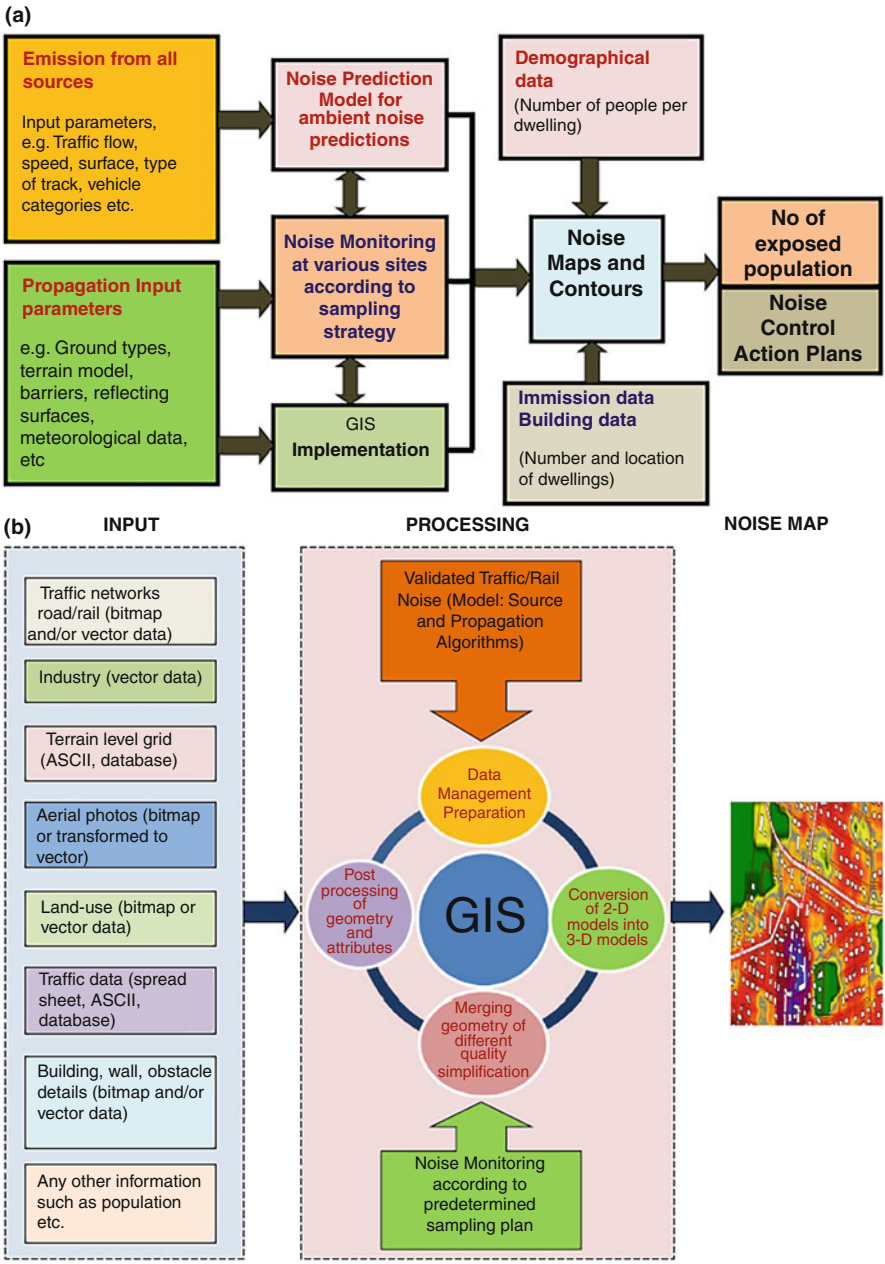

Fig. 2 Noise mapping framework. (Garg et al. 2021)

Table 5 Ambient noise standards recommended by Ministry of Environment and Forests and Climate Change

Zone code	Category of zones	L_{eq} in dB(A)	
		Day levels	Night levels
A	Residential	55	45
B	Silence	50	40
C	Industrial	75	70
D	Commercial	65	55

Garg et al. (2021); The Noise Pollution (Regulation and Control) Rules (2000)

Note:

1. Day equivalent noise levels will be measured from 06.00 a.m. to 10.00 p.m
2. Night equivalent noise levels will be measured from 10.00 p.m. to 06.00 a.m
3. L_{eq} is the equivalent sound pressure level for a certain period of time

- To avoid noise threat and to achieve the goal of maintaining the ambient noise standards, all local bodies, development authorities, and other concerned departments must include all aspects of noise pollution as an attribute of quality of life when planning developmental exercise or conducting out activities related to town and country planning.
- For the purposes of these rules, a silence area may be established in the vicinity of educational institutions, hospitals, and courts that is in range of at least 100 m.

Noise Impact Assessment

Noise impact assessment is concerned with the impact of a particular project or a new establishment on the ambient noise levels of an area or locality with respect to the noise standards recommended by the authorities. It is very helpful in the planning phase of many residential, silence zone or industrial areas in order to adopt appropriate protection for reducing the noise pollution and its impact on health and quality of life. The types of control methods involved can be: Avoidance; Reduction and Compensation. Avoidance can also be achieved by controlling or limiting the hours of operation; limiting the number of events; or limiting the number of different sources operating. Reduction involves the various measures like use of barriers, enclosures etc. Compensation may include measures like using the double glazing's in windows of the dwellings to protect from the outside traffic noise (Guidelines for Environmental Noise Impact Assessment 2014). The noise impact assessment should be exercised at the planning stage by the relevant authorities and all the control measures should be taken prior the new establishments or project so as to minimize the impact on community. Figure 3 summarizes the methodology of Noise Impact Assessment and Mitigation with some modifications to that proposed in report (Guidelines for Environmental Noise Impact Assessment 2014; Garg 2021).

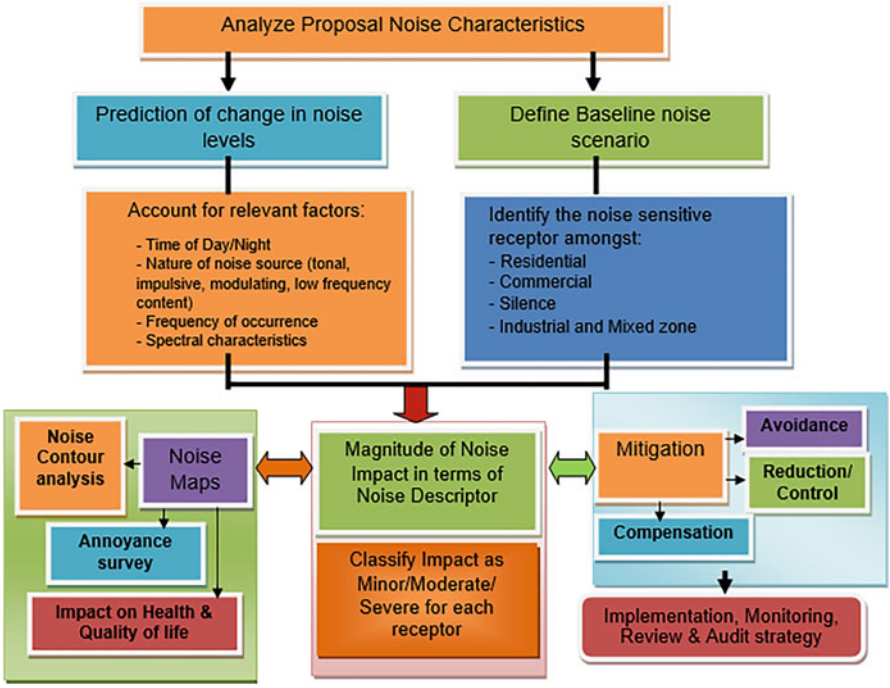

Fig. 3 Summary of Noise Impact Assessment and Mitigation. (Guidelines for Environmental Noise Impact Assessment 2014; Garg 2021)

Noise Pollution in Mining Sector and Construction Activities

Due to the introduction of modern and heavy machineries (such as bulldozers, continuous miners, mine trucks, and drilling machines) using advanced technology, natural mineral productivity has been massively increased over time to fulfil the exponentially growing needs from all over the world. Increased use of these types of machinery has resulted in increased road transportation and a significant increase in noise levels (Tripathy 2008). Mostly all mining activities generate noise from a variety of stationary, mobile, and impulsive sources, making it an inherent part of the mining area. It is described as sound having no pleasing musical quality or sound that is undesirable. Noise is a typical environmental concern in opencast mines, as it is generated by heavy earthmoving equipment. As the mining activity proceeds, the equipment and environmental circumstances alter. The overall mining noise generated from the mining equipment varies in quality and level depending on their placement. Most mining equipment's in opencast mines emit noise levels in the range of 90–115 dB(A), which can cause noise-induced hearing loss and other non-auditory health problems in miners over time (Tripathy and Patnaik 1994; Vardhan et al. 2003). Fig. 4 shows the mining operation in open cast mines.

Fig. 4 Mining operations

The mining operation includes:

- Blasting
- Drilling
- Excavation
- Loading and Transportation

Noise Assessment in Mines

Noise assessment in opencast or closed mines is done to determine:

1. Noise pollution condition inside and in nearby areas of mines.
2. The sound pressure level generated by heavy machineries used for various mining operations.
3. Octave band analysis of various equipment and noise induced hearing loss of workers working in mines.

Utley et al. (1980) described about the assessment, measurement, and prediction of noise levels generated in coal mines. The study also discussed about the control measures required for reduction noise pressure levels. Singh et al. (2004) considered various environmental pollution issues at mining locations due to use of heavy

machineries. Further, the study also analyzed about impact of noise on the residents and workers. Ghose et al. (1990) briefly discussed about the measurement, mapping and analysis of noise pollution in mines. Bhattacharya et al. (1994) measured noise levels generated from drill machine and crusher in Meghatuburu mines. The equivalent noise level was 101–106 dB(A), while maximum noise level was in between 106–110 dB(A). Pal et al. (1994) measured noise level in underground mines at a distance of 5, 15, 25 m from the noise source.

Noise Prediction in Mining Area

The effect of ambient noise levels in opencast mines is influenced by the sound power level of the noise generators, the current geo-mining environment, and the meteorological characteristics. The noise levels must be examined as a combined effect of the aforementioned variables. As the mining operation advances, the environment and equipment conditions are constantly changing. The common mining noise coming from the mines differs in quality and volume depending on where they are located. As a result, for environmental noise prediction models, the sound pressure level (SPL) at every receiver site must equal the sum of all the noise sources. It is generally known that it is important to precisely forecast the noise level produced by opencast mines. Some of the widely used noise prediction algorithms in Europe were Conservation of Clean Air and Water in Europe (CONCAWE), German Draft Standard VDI-2714 Outdoor Sound Propagation, and Environmental Noise Model (ENM) (Manning 1981; Tonin 1985; Nanda et al. 2009). The International Standard Organization (ISO) standard addresses various aspects in this regard (Cho et al. 2004; Nanda et al. 2012).

Noise Control

According to a US research study quantifying the traffic noise costs, $21 annually per housing unit per decibel increase is estimated (Perkins 1998). Therefore, it is unavoidable that every nation would have a robust noise control strategy with the basic goal of preventing and minimizing the detrimental impacts of noise on human wellbeing. As a result, it implies that regions with amplified ambient noise levels beyond the legal limits require noise action measures. In order to restrict exposure to environmental noise and consequently ameliorate its impacts, such as sleep disruption and discomfort, the European directives on environmental noise require that authorities to implement, establish, and carry out action plans against urban noise. Acoustic planning, which refers to reducing future noise by deliberate actions including traffic and land-use planning, noise reduction at the source, and sound insulation techniques, (Murphy and King 2010). Understanding the noise scenario, prioritizing control measures, and involving stakeholders and the public all contribute to the structure and prioritization of noise control measures in noise action plans. A traffic relocation strategy that involves building a bypass highway, for example, could be a helpful mitigation strategy (Lokhande et al. 2017). In regions with high

ambient noise levels, the potential decrease can be achieved by constructing noise barriers. The other appropriate strategic plans mentioned by Torija et al. (2012) include source control, which entails setting restrictive speed limits, reducing travel demand, prohibiting heavy vehicles in residential and sensitive areas, and testing vehicles' periodic noise emissions; mobility management, which encourages the use of public transportation systems urban planning, and reduces traffic jams; which entails land-use management and planning, such as pedestrianism, zoning, and other techniques. A harmonized noise policy at regional and national level for a sustainable environment and the integration of noise control planning into urban planning activities are essential. However, at the local and regional levels, the cost-effectiveness and applicability of the policy actions may vary.

A synergistic, uniform, and coordinated strategy to noise pollution control across the nation would be provided by the adoption of a harmonized national policy at both the regional and national levels. Figure 5 represents the cost-effectiveness score and analysis of different control techniques. Additionally, there should be coordination between various initiatives and plans, such as the air pollution control program and Swatch Bharat mission, with the cooperation of all related personnel. The Working Group on Health and Socio-Economic Aspects found that the current benefit of noise control in Europe be valued at 25 Euros per household/dB/year (Working Group on Health and Socioeconomic Aspects 2003). Clearly, this measurement can be interpreted from an Indian perspective as well as a coordinated strategy with a harmonized national policy for sustainable urban mobility strategy and noise pollution regulations at both national and regional levels; noise mapping of metropolitan areas and implementation of potential new initiatives would be indispensable in reducing the adverse health effects of noise exposure and prevalence of diseases linked to environmental noise in India.

Practical Implications and Recommendations

Whereas the idea behind noise mapping seems to be very intuitive and practical, there are still a number of open questions regarding noise mapping, public information disclosure, population exposure assessment, and noise strategic planning, which can sometimes make it challenging for urban planners and pollution control entities to adopt a standardised approach. Prior conducting the noise mapping of a city, it is essential to make decisions on the assessment of day and night noise levels in different seasons, sampling technique, impact of metrological parameter. Noise mapping procedure should be standardized, resulting in the lowest possible level of uncertainty in the noise forecasts. This would be achieved through the use of a validated noise model, an acceptable interpolation technique, and a harmonized representation with reference to color coding. The implementation of the same modelling technique and software that has been established from an Indian perspective would essentially constitute a unified strategy for full standardization of the noise mapping process. Additionally, the following recommendations are made (Garg et al. 2021):

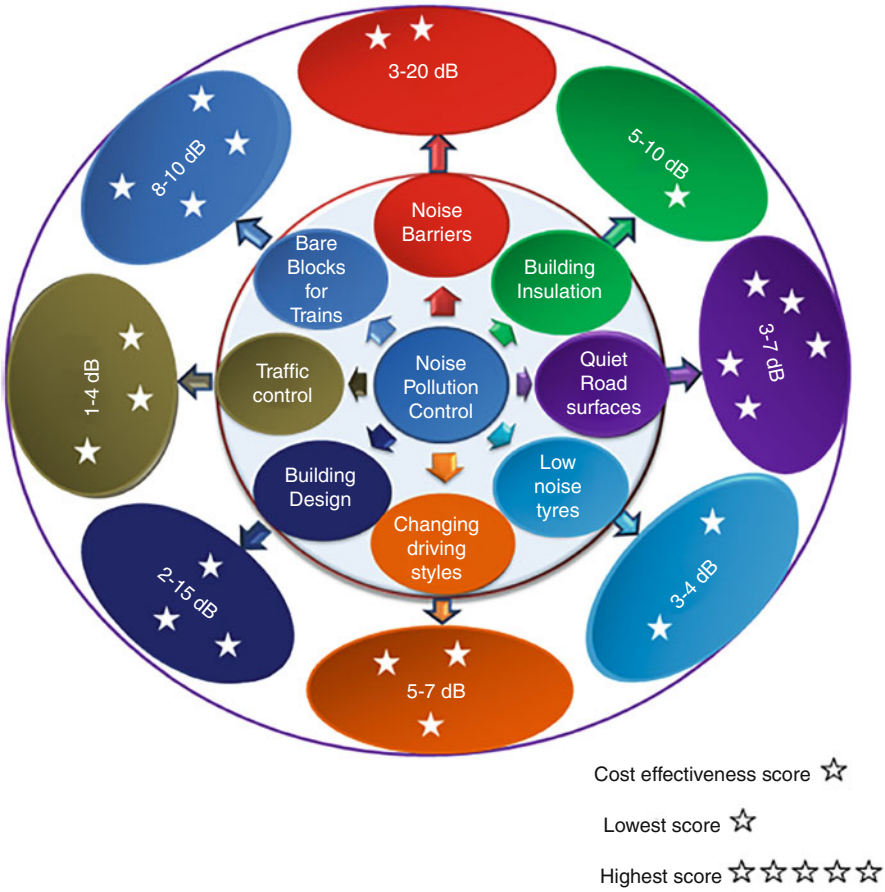

Fig. 5 Cost-effectiveness score and analysis of different control techniques. (Garg et al. 2021)

- A hybrid strategy that use interpolation algorithms and prediction models to anticipate the ambient noise levels at specific spots and the usage of noise monitoring stations for noise assessment at numerous locations may be used.
- Prior to conducting noise mapping studies, an optimal strategy has to be decided about the locations for noise measurements and accuracy. Increasing the grid size increases the map accuracy depicting the ambient noise levels, but at a higher expenses.
- The number of points to be sampled for the city's mapping may be determined based on a variety of factors, including the location of sensitive receptors and noise-generating sources, the division of the area into hot spots that are suitable for sampling, areas of high relevance, grid-based sampling, and the division of the area based on factors like traffic flow, road size, residential zones, and highway roads.

- For locations wherein noise assessment is not done, the predictive algorithms can cover the gap. However, reducing the uncertainty in noise forecasts would benefit from the model and software prior calibration.
- Short-term noise monitoring strategies (Tiwari et al. 2021a) can be utilized for efficient utilization of Noise Monitoring Terminals for nose monitoring of larger sites of the metropolitan cities.
- Noise Control measures utilizing the Multi Criteria Decision-Making (MCDM) techniques (Garg 2021) shall be indispensable for noise pollution control at the noisy hot spots. Also, policy measures and cooperation of all stakeholders, and awareness among the community shall be very helpful for reducing the ambient noise levels in cities.
- The noise barrier must physically fit in the available space, must completely break the line-of-sight between the noise source and the receptors. Noise barriers should be tall enough to provide noise reduction for the upper-most stories of nearby sensitive receptors. A barrier must achieve a Sound Transmission Class, STC (or weighted sound reduction index, R_w) of 30 or greater in accordance with ASTM Test Method E90 or ISO 717 in order to ensure adequate sound transmission loss characteristics and effective mitigation. Also, the source side of the barrier can be lined with acoustical absorptive materials in order to avoid reflections with building facades (Thalheimer 2000).

Conclusions

This chapter presents a detailed discussion of noise monitoring studies, noise prediction models, and noise mapping in different countries. It is evident that the artificial intelligence technique offers more reliable and accurate prediction capabilities as compared to the conventional mathematical models. The technique, strategy and methodology of noise mapping is at primitive stage sin India although some studies had been now reported in some cities of India. However, as an when more and more studies shall be conducted, it will give consultants, town planners, research institutions, laboratories, and pollution control bodies in India more knowledge and experience for conducting noise mapping studies and implementing the noise control measures. Promoting good health and a high standard of living requires a comprehensive strategy that focuses primarily on periodic noise monitoring and assessment of the noisy locations in cities, ascertaining their compliance with the ambient noise standards, and developing appropriate noise action plans in combination with policy issues on environmental noise planning and control.

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necessarily reflect the official policy or position of any agency of the Government of India. The content of the papers is solely to present a retrospective and prospective view of noise levels and may not be used or considered for disputes redressal in legal framework.

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Development of Calibrated and Validated SODAR with Reference of Air Quality Management

65

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Abstract

Air pollution control necessitates knowledge of Atmospheric Boundary Layer (ABL) parameters that influence pollutant concentration, diffusion, and transit. The ABL is one of the most important constituent zones of the atmosphere, as it plays a critical role in major areas of human endeavor such as air pollution, aviation, and communication. It can also be thought of as the biosphere's circulatory system. SOnic Detection And Ranging (SODAR) is an important ground-based remote sensor competent to detect ABL height in real time. It is utilized in air quality management as a diagnostic tool for monitoring air quality in a variety of hazardous circumstances that are directly related to human health issues. The present work focuses on the methods derived for probing the ABL using SODAR for effective air shed management. The data generated through SODAR echograms is processed, interpreted and compared with MERRA-2 Model sensor so as to determine its effectiveness and precision of data. After critical analysis it was observed that SODAR is an effectual environmentally friendly instrument that can be used for air pollution abatement and air quality measures.

Keywords

SODAR · Atmospheric boundary layer · Air pollution · Ventilation coefficient

Introduction

Ground-based SOnic Detection And Ranging (SODAR) is an active acoustic remote sensing device that is used to probe the Atmospheric Boundary Layer (ABL) structures and to determine the ABL height influenced by the convection and turbulence in the atmosphere. The SODAR system generates real-time and spatial echograms, as well as a visual representation of ABL dynamics. SODAR as a ground-based sensor provides appropriate vertical and temporal coverage with high resolution data that can be used for vivid applications like to study the thermal structure of the atmosphere, wind shear structure, as well as three-dimensional wind information in the lower part of the atmosphere (Crescenti 1997). SODAR echograms are used to infer ABL height influenced by meteorological conditions prevailing at the loci. This ABL height is detrimental for examining the pollution load of a place through ventilation coefficient, a factor determined by ABL height and wind speed. SODAR is acoustical instrument and works by sending and receiving directed acoustical signals. The two most critical needs in any SODAR system are a transmission system to illuminate the highly directed signal in the turbulent area and a sensitive reception signal to identify an air-scattered signal from the air. The present work describes the development and improvement of Monostatic SODAR and the adavancement underwent during the ongoing years. The calibration of the instrument has been putted it on the front with high accuracy and precision. The work also emphasizes the role of SODAR in air quality management.

Atmospheric Boundary Layer

The Atmospheric Boundary Layer (ABL) is the lowest part of the troposphere where the direct effect of surface heating and cooling has been seen (Stull 1988; Emeis 2010). All biological and human activities take place in this layer (Bradley 2007). The ABL extends to a height of convective mixing of buoyant parcels of air or the base of an elevated inversion layer during daytime unstable conditions, and the level of surface-based inversions determines the height of the ABL during nighttime stable conditions. Furthermore, turbulence is a necessary component of ABL (Stull 1988). It facilitates the movement of water vapor, heat, momentum, pollutants, and other tracer elements from one location to another. Wind speed normally increases with height in the ABL, while pressure decreases regularly with height, temperature decreases less regularly, and humidity decreases very irregularly (Stull 1988; Singal 2006). The derived parameter refractivity usually decreases exponentially, but sharp gradients can occur depending on weather conditions. ABL height is a critical parameter for understanding the ABL structure's characteristics such as turbulence, atmospheric stability, and diurnal pattern related to convection. The ABL structure varies from day to night and with synoptic and mesoscale weather conditions. The development and growth of ABL height is primarily influenced by various controlling factors such as surface turbulence flux, surface temperature, soil moisture entertainment of warm and dry air from the free atmosphere, and so on. Surface turbulent fluxes are related to the formation and development of ABL and are important feedback parameters in the coupled system.

The ABL is in convective equilibrium with the Sun-warmed Earth's surface (Emeis et al. 2008). The Sun is the primary energy source for the Earth's atmospheric system. The precise amount of this energy is determined by the time of year, the time of day, latitude, topography, and weather conditions. The nature of the soil and the type of surface determine the amount of energy reflected, absorbed, or transmitted from the Earth's surface (Stull 1988). The absorbed energy is converted to heat energy either directly or indirectly through evaporation or conversion to mechanical, electrical, or chemical energy. The direct heating of the surface of the Earth sets up convective motions in the atmosphere while the indirect heat energy gets stored in the Earth for long periods in the form of chemical energy (say as fossil fuels) before being converted back into heat energy.

Need for ABL Studies

Air quality management and environmental protection are global issues that require joint attention and action at the international, individual, national, and even grassroots levels of domestic human activities. The troposphere, particularly the bottom 2×10^3 m, plays an important role in important human endeavors such as communication, aviation, and air pollution (Seibert et al. 2000). These situations necessitate complete information on wind velocity, humidity, and thermal structure at a location in order to monitor potentially hazardous situations. As a result, it is critical to have a

thorough understanding of its microscale details as well as their distribution in real time and space.

Monitoring Techniques for ABL

Various monitoring techniques have been developed over time to monitor atmospheric parameters such as temperature, pressure, humidity, wind speed and direction, and their distribution in space, providing information about atmospheric structures and various dynamical processes active at a local scale (Stull 1988; Emeis et al. 2008). Table 1 (Seibert et al. 2000) depicts the schematic arrangement for using the various techniques to obtain information about a specific atmospheric volume. All the techniques have their relevance to monitoring any parameter.

These techniques are broadly classified as in situ techniques and remote sensing techniques based on their mode of operation. In situ techniques necessitate carrying measuring sensors or instruments aloft to the height range of interest. In situ techniques such as the radiosonde and microwave refractometer have the advantage of providing detailed height and space profiles, but they are not continuous, which is required for some applications (Singal 2006).

Different types of remote sensing techniques have been developed in recent years using acoustic and optical and radio waves are, SOnic Detection And Ranging (SODAR), Light Detection And Ranging (LiDAR), and RAdio Detection And Ranging (RADAR), respectively (Seibert et al. 2000). Radio And Sounding System (RASS) has also been developed using radio and acoustic wave technology.

Table 1 Different techniques for measurement of ABL or atmospheric volume

	Continuous data output	*Range covered well*			*Determination of turbulence parameters*
		10–100 m (Low SBL)	100–500 m (SBL/CBL)	0.5–3 km (CBL)	
In-situ *techniques*					
Aircraft	–	–	X	√	√
Radiosonde	–	–	√	√	.
Mast	√	√	–	–	√
Tethered balloon		√	√	–	X
Remote sensing techniques					
SODAR	√	–	√	√	√
RASS	X	–	X	X	X
RADAR	√	–	X	√	√
Mini-SODAR	√	√	–	–	√
LIDAR	√	–	X	√	√
Numerical models	X	X	–	√	√

Note: – √ means fulfilled, X partially fulfilled, and – not fulfilled

Microwave RADAR is similar to LiDAR or Laser RADAR. LiDAR may be used to derive air density profiles above 30 km and map aerosol distribution using intensity back scatter observations. For quantitative probing, resonant, Raman, and fluorescent scattering are being studied, with a focus on pollution monitoring. RADAR probing measurements of reflectivity and doppler shift also include attenuation measures of total water content in storms, as well as polarization effects indicating particle shape, Strom structure, rain water and snow distributions, and cloud, storm, convective cloud, and wind shear dimensions and heights, among other things (Stull 1988). The temperature of the Earth's surface, cloud tops, and other objects is measured using infrared and optical techniques, and vertical profiles of temperature, water vapor, and ozone are inferred using an inversion technique. Microwave radiometer probing is a passive technique that is confined in space and time to a line of sight to the sun. Acoustic techniques (Mono static, Mini, Doppler SODAR) have been used to detect and analyze volcanoes, earthquakes, ocean gravity waves, tornadoes, jet streams, and aurora, among other phenomena (Singal 2006). The following atmospheric information can be obtained using SODAR techniques: first, profiles of mean speed and direction, second, profiles of the three-dimensional spectrum of kinetic turbulence, i.e. intensity of mechanical turbulence as a function of height, wave number, and direction, third, profiles of the three-dimensional spectrum of temperature inhomogeneity, fourth, location and measurement of the intensity of temperature inversion, and finally, information about the intensity of temperature inversion. This chapter concentrated solely on acoustics waves and SODAR.

SODAR As a Monitoring Technique

Thomas (1968) recognized the potential of acoustic sounding for remote detection of thermal inversion layers in the atmosphere. He used an 18-meter-diameter parabolic dish with a transducer at its focus to work at a frequency of 190 Hz. Under low ground noise conditions, echoes from 600 m altitude were received. The received echoes were attributed to a temperature inversion, random atmospheric inhomogeneities, cloud layer boundaries, and temporary wind shear conditions. However, the echosounder/SODAR technique for remote sensing was used to study the index of refraction fluctuations within the atmosphere (McAllister 1968, 1971; McAllister et al. 1969). McAllister (1968) developed an array of loudspeakers and parabolic dish with a horn-loaded transducer mounted at its focus to be used as an acoustic antenna. The frequency used was *950 Hz* and therefore the echoes from an altitude of *1000 m* were detected. The increase in detection range compared to Thomas (1968) could be ascribed to McAllister (1971)'s use of a higher frequency, which increased the signal-to-noise ratio. The introduction of a facsimile recorder, which provided a 3-D display in terms of height, time, and intensity, improved the richness of the observations of atmospheric structure acquired by McAllister et al. (1969). The event of basic physical concepts of the influence of turbulent inhomogeneities on sound wave propagation and scattering within the atmosphere was surveyed by

Kallistratova (1963). However, the most theoretical and experimental results were obtained and summarized, the analysis of fundamental problems, requiring an answer for practical applications of sound waves within the atmospheric research, was given.

Little (1969) examined the possibility and limits of an acoustic RADAR and addressed several acoustic approaches for distant probing of the atmosphere. Fukushima et al. (1974) used a concrete paraboloid dish having an aperture of *16 m* and this antenna was used as the acoustic sounder for studying the lower atmosphere. The interrogating waves had a frequency of *850 Hz* within the sort of pulses of 60×10^{-3} *s* duration repeated every *3.6* or *12 s* depending upon the detection range. They had recorded the meteorological structures of the lower troposphere and realized their usefulness within the effective monitoring of the troposphere. Hall (1972) described an acoustic SOnic Navigation And Ranging (SONAR) for detecting the temperature and velocity structure of the Earth's physical phenomena. The specifics of the SODAR development and experiment work were documented by several experts (Brown and Hall 1978; Beyrich 1997; Ito 1998; Sgouros et al. 2011). Several studies (Stull 1988; Emeis 2010; Bradley 2007; Melas 1993; Antoniou et al. 2003; Kallistratova et al. 2018) provided outlines of the theory of SODAR research.

SODAR operates on the principle of active echo sounding, which employs acoustic waves to probe the lower atmosphere. The propagation of these waves through the medium is affected by meteorological factors such as temperature, relative humidity, and wind. These parameters exhibit their degree of variability through fluctuations in the refractive index of the medium, causing acoustic waves to be refractive, scattering, and attenuated as a result. Sound wave phase and amplitude fluctuations have been evaluated in terms of spatial and temporal fluctuations in atmospheric temperature, wind velocity, and humidity in a manner similar to that of electromagnetic waves. Table 2 shows that acoustic waves are considerably more sensitive to atmospheric fluctuations than are electromagnetic waves and further that wind and temperature fluctuations dominate for sound waves (Rossing 2014; Müller and Möser 2012; Lo and Lee 2013). The strong interaction of acoustic waves with fluctuations in temperature and wind velocity compared to electromagnetic and optical waves makes it possible to monitor the presence and behavior of various atmospheric structures more effectively using sound waves.

SODAR is one of the most effective remote sensing techniques for providing continuous real-time data on air quality, meteorological variables like mixing height/

Table 2 The sensitivity of remote-sensing equipment to atmospheric parameters

Parameter	Magnitude changes of the parameter	Corresponding change in refractive index		
		Acoustic	Radio	Optical
Temperature	1° K	1700	1	1
Humidity	1 mb	140	4	0.04
Wind	1 m s^{-1}	3000	2×10^{-6}	2×10^{-6}

inversion height, ABL stability, and wind profile (Stull 1988; Emeis 2010). It is well known on a global scale and has been shown to be cost-effective. It is internationally recognized and recommended by the Environmental Protection Agency (EPA) for air quality dispersion modeling in Environment Impact Assessment (EIA). All over the world, this technique is used in a variety of practical and ABL research applications. Thermal convection, inversion, elevated inversion, advection, subsiding air mass, sudden gust, waves, and other phenomena can be easily identified using SODAR data. SODAR data is useful for tracking the evolution of coastal boundary layer occurrences (Sea/Land breeze) (Kallistratova et al. 2018). Information has been used to evaluate and comprehend the real-time signal performance of communication links. In bright sunlight, the air warms significantly faster than air close to the Earth's surface. It becomes lighter and rises as a result of this. Thermal plumes have been observed on SODAR as a result of this vertical movement (dense inverted cones). SODAR is one of the most effective ABL approaches in real time and up to 3000 m in space (Seibert et al. 2000).

Mechanism of SODAR and Generation of Echogram

Small-scale refractive index fluctuations occur continually in any volume of atmospheric air, according to the literature (Stull 1988; Emeis 2010; Bradley 2007). These differences serve as tracers for the scattering of the acoustic beam. The variations are caused by turbulence inside an area of refractive index gradient associated to either temperature inversion or convective activity. Turbulent temperature changes half the length of the acoustic wavelength scattered the energy. When the wavelength was shortest, the dispersion was greatest. Without turbulence, temperature inversions were unlikely to have large enough temperature gradients to create substantial echoes. To induce scattering, sufficient turbulence fluctuations were necessary, as well as air stratification different than neutral (Stull 1988; Bradley 2007; Emeis 2010).

The mechanism of generation of turbulence due to temperature lapse rate other than dry adiabatic was discussed (Tatarskii 1961; Monin 1962; Antoniou et al. 2003; Emeis 2021). Tatarski (1961) obtained an expression for scattering cross-section $d\sigma$ as

$$d\sigma(\theta) = 2\pi k^4 v \left[\frac{1}{C^2} E_v(k) \cos^2\frac{\theta}{2} + \frac{1}{4T^2} \emptyset_T(k)\right] d\Omega \tag{1}$$

Kallistratova et al. (2018) and Monin (1962) used a different approach to effects of wind speed and temperature to obtain the expression for $d\sigma$

$$d\sigma(\theta) = 2\pi k^4 v \cos^2\theta \left[\frac{1}{C^2} E_v(k) \cos^2\frac{\theta}{2} + \frac{1}{4T^2} \emptyset_T(k)\right] d\Omega \tag{2}$$

This expression didn't consider the Doppler frequency shifts and scattering due to humidity fluctuations. However, such effects were normally quite small for sound

waves. Furthermore, in the development of the expression for scattering cross-section, they assumed that the field of fluctuations within the scattering volume didn't vary significantly during the passage of sound waves. The refractive effects on the scattering volume, scattering angle, and Doppler bordering of frequency were investigated in detail (Parry and Sanders 1972; Georges and Clifford 1974; Fortuna and Kozaczka 2014). They found that these effects were small for sounding up to a height of *1000 m*. The amplitude of the dispersed signal was proportional to the degree of turbulence, and the acoustic wave scattering occurred in all directions.

The estimation of received acoustic power in the case of a SODAR depends on the amount of atmospheric attenuation along its path in addition to various other parameters. In the case of hemispherical spreading of sound, the intensity I at a distance R from the source was expressed as

$$I = \frac{P}{2\pi R^2} e^{-\alpha R} \tag{3}$$

Harris (1963, 1966) published detailed results for sound absorption in the air as a function of humidity and temperature. However, this work was of significance for sea-level investigations and was somewhat limited for use in the free atmosphere because it didn't include the less well-defined effects and results from scattering and refraction in the free atmosphere.

After the development of the SODAR system, receiving power was determined to be a finite value. The receiving power was dependent on the number of parameters. These parameters were used to develop the system and affect the measuring of the ABL component.

$$\text{Receiving Power}, P_r = P_t \cdot \sigma \cdot \frac{c\tau}{2} \frac{A.L}{R^2} \tag{4}$$

$$C_T^2 = \frac{\sigma_{180} T^2}{4 \times 10^{-3} k^{-1/3}} \tag{5}$$

where σ = scattering cross-section; c = speed of sound; τ = pulse width; R = range; A = antenna area; L = factor containing the equipment efficiencies, antenna gain, and the atmospheric absorption; P_t = the transmitting power; C_T = temperature structure constant; k = angular wavenumber; T = temperature.

Backscattering was only caused by temperature inhomogeneities, according to the monostatic SODAR power Eq. (4) and temperature structure Eq. (5). These backscattered sound signals were examined by the monostatic SODAR system. As a result, dish curvature, antenna gain, and atmospheric absorption were critical to transmitting and receiving the acoustic wave. Wind and humidity fluctuations cause temperature fluctuations, which affect the temperature structure parameter. The received signal was processed to derive information on amplitude about the degree of prevalent turbulence, while the frequency contained information about the current wind speed.

Calibration and Validation of SODAR System

SODAR is an important piece of equipment that measures ABL height in near real time this has been employed as a diagnostic tool in the management of air quality for environmental monitoring in various hazardous conditions such as communication, aviation, wind energy sourcing evolution, meso-meteorology, and as a contemporary technique to other techniques. To achieve these aspirations and predictions, the most specialized ground-based distant sensing of the mean wind, temperature, and humidity within the troposphere is used to monitor and predict the weather. SODAR has been the preferred choice for a long time due to its combined capability to provide data on the dynamics of ABL thermal structures, the ability to disperse the environment, and real-time wind profile measurements in a variety of applications. SODAR is an active sensor that allows us to measure absolute values of the temperature structure parameter of turbulence (C_{2T}), a very useful parameter for studying the behavior of the optical refractive index in the atmosphere, among many other practical applications (Danilov et al. 1992). However, the primary interest of researchers has been in data validation to make sure the standard of knowledge and gain confidence before its direct use as an alternative to standard data availability from existing conventional direct measurement technologies like an instrumented tower, radiosonde, etc.

For the efficacy of the system, careful calibration of SODAR components is prerequisite. Evaluation methods for precision measurement and inter-verification interval measurement for measurement devices based on the concept of measurement uncertainty are required in order to determine the timing of the next scheduled or unscheduled verification of the correspondence of metrological characteristics to their standardized values (Vasilevskyi 2014). A number of possible sources of errors while measuring the absolute turbulence parameter-derived ABL Height are highlighted and are categorized into three categories, which includes geometrical error (errors in tilt angle, scattering angle), bias in ABL Height, wind estimation (uncertainties in calibration against which the calibration is done), and noise interferences (loss of transmitted and received signals). A few of these errors are indeed "operational problems," in the context that they can cause a change in calibration based on the location, conditions, or SODAR operating parameters (Bradley et al. 2005). Coefficients for converting electrical power to acoustic power and vice versa are commonly used to measure absolute values of structure parameters. Despite the development of several approaches for such calibration, most modern commercial SODARs and many research SODARs lack absolute calibration. Fukushima et al. (1974) used a corner reflector for outdoor calibration of the SODAR. Coulter and Kallistratova (2004) described a straightforward method for calibrating a parabolic-type transmitter receiver using a standard-size anechoic chamber and a theoretical antenna directivity pattern. Several works used calibration by comparing echo signal intensity to C_{2T} values measured at a nearby mast (Coulter and Kallistratova 2004).

Marshall et al. (1972) conducted a theoretical examination of horizontal and vertical wind profiles using SODAR, while Beran et al. (1974) and McAllister (1971) conducted practical evaluations. Little (1969), Beran et al. (1973), and Wesely (1976) investigated the viability of employing SODAR data to compute

the vertical humidity profile. The ABL was precisely measured using a SODAR, a parameter used by Greenfield et al. (1974) in the theoretical computation of temperature profiles inside the lower atmosphere under changing climatic conditions. Mahoney et al. (1973) conducted an experiment at Haswell including the use of 3-axis SODAR placed *240 m* apart at the corners of a triangle within which is a *150 m* instrumented tower capable of delivering information about temperature and wind at a rapid rate to correlate the information collected by the SODAR with meteorological factors. Although some of the SODAR data was understood in terms of the recorded turbulent heat flux and momentum, it was determined that the system required to be evolved into an independent quantitative instrument to unravel a large number of details seen. To ascertain the validity of SODAR records, dissimilar researchers had compared the SODAR observations with vertical temperature profiles obtained from radiosonde techniques, captive balloons, LiDAR, and RADAR (Lokoshchenko 2002; Neff et al. 2008; Richardson et al. 2013; Dang et al. 2019; Emeis 2021; Kumar et al. 2021a). They showed that echoes on the SODAR echograms correspond to regions of enhanced temperature discontinuities. Earlier the measurement of SODAR calibration parameters required either a tethered balloon, a high mast, or a very large anechoic chamber for getting precise results. However, a straightforward approach for assessing calibration parameters was described in 1992, using a standard size anechoic chamber and a theoretical antenna directivity pattern (Danilov et al. 1992).

Work at Council of Scientific and Industrial Research-National Physical Laboratory (CSIR-NPL), New Delhi began in 1972, with a preliminary experiment on SODARs. In 1973, after fabricating suitable directional acoustic antennas, a colocated SODAR was established, which was later converted to monostatic operation in 1974. The first calibration of SODAR was initiated in 1976 when another SODAR was set up at the Ayanagar Observatory of the India Meteorological Department in Delhi. The second SODAR was calibrated with the radiosonde data. However, during the calibration field measurement of SODAR antenna, stable windless weather condition along with the high mast or tethered balloon was required, which was found to be an inappropriate method for calibration as it lacked the precision due to the undefined variation in radiation patterns. If structure parameters are to be computed, accurate estimation of SODAR antenna parameters, back-scattering cross-section, and knowledge of acoustic absorption in atmospheric air are required. Thereafter, to delimit the limitations in calibration deployment of anechoic chamber for the acoustical calibration of SODAR transducer and horn with a theoretical estimate of Sant (area of SODAR antenna) was calculated (Danilov et al. 1992).

A method using a theoretical antenna directivity pattern, together with measurements of transduction characteristics of the acoustic transducer, was used for SODAR Antenna calibration (Danilov et al. 1992; Bradley et al. 2005). Theoretically (Eq. (6)) and experimentally determined directivities of the acoustic transducer, one from Russian Academy of Science, Moscow (LATAN-1) and other from CSIR-NPL, New Delhi have been compared (Danilov et al. 1992). The thermal structural values obtained for Sodar Latan-I observed in August 1990 at Mount Mayadank

(2600 m above sea level, Uzbekistan, West Parnero Alay) were calculated by means of the aforementioned approach and compared to those measured simultaneously using microthermal probes for calibration. The deviation between SODAR and in situ data was relatively small when compared to the large range of uncertainty in C2T values (around four times higher at given altitudes and eventually up to five times greater whenever the height dependence within boundary layer is taken into account) (Danilov et al. 1994). The height of ABL during the inversion period was computed using the bulk-Richardson-number approach, diagnostic equations for the equilibrium ABL height, and a relaxation-type prognostic equation, and was explained in terms of their physical foundation and applicability of experimental data (Stull 1988; Richardson et al. 2013). They concluded that the critical bulk Richardson number (R_{iBC}) of the ABL throughout the inversion period was not constant. They also stated that the Richardson number-based computation methodologies should only be used for rough ABL calculations. The output efficiency, input efficiency, and directivity distribution of the LATAN-1 and CSIR-NPL SODAR transducer-horn assemblies were measured in the anechoic chambers of the respective laboratories at operating frequencies of *2.0 KHz* and *2.2 KHz,* respectively. The theoretical directivity distribution behavior was also computed, and the two directivity distribution behaviors were displayed. The two theoretical and experimental curves match adequately within 20% of their values inside the angles at which the paraboloids of the two SODARs are observed from their respective foci, as can be seen in Fig. 1. With a view to illustrating the values of correction, the measured and corrected data were represented for the CSIR-NPL SODAR transducer in Table 3 (Danilov et al. 1992).

$$D_h(\theta_h) = \frac{1 + \cos(\theta_h)}{kR_h \sin(\theta_h)} J_1(kR_h \sin(\theta_h)) \tag{6}$$

where J_1 is the Bessel function, and R_h is horn aperture

Danilov et al. (1992) were computing the directivity behavior; for the SODAR LATAN-1, an exponential horn was used with $R_h = 0.085$ m, which at a working frequency of 2 kHz gives $kR_h = 3.14$**.** The horn used for the CSIR-NPL SODAR is also exponential and has $R_h = 0.075\ m$, which at a working frequency of *2.2 kHz* gives $kR_h = 3.04$. The angles at which the apertures of the reflecting paraboloids are seen from their respective foci are 61° for LATAN-1 and 67.5° for the CSIR-NPL SODAR.

It may be noted that the output efficiencies and the parameters, kR_h and b, for the two SODARS are more or less the same. It may also be noted that the numerical coefficient in Eq. (7):

$$K = 25\pi b^4 (kR)^2 (kR_b)^{-4} \tag{7}$$

is constant for a given device. For LATAN-I it has been computed and found to be $K = 10.7$.

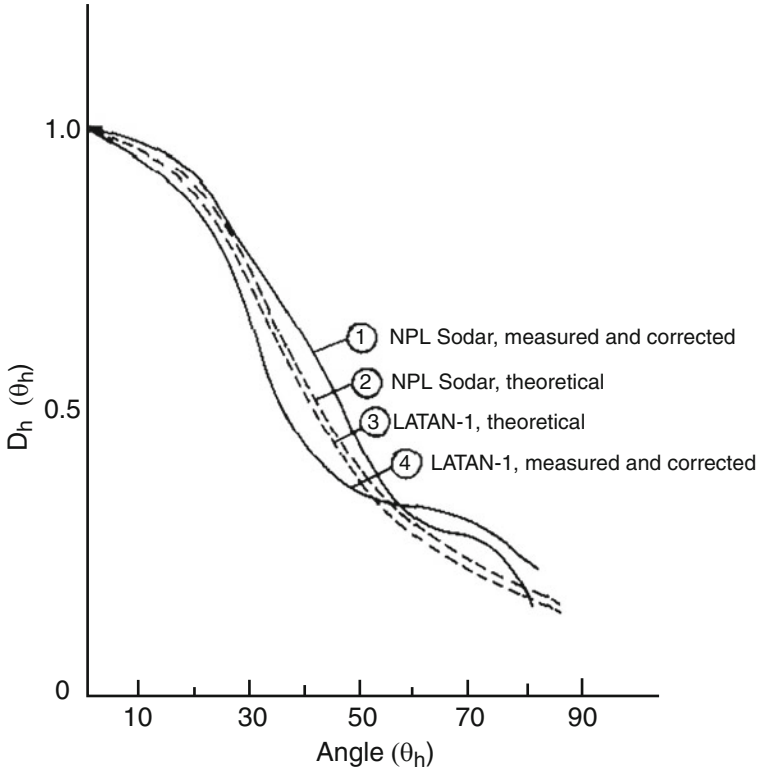

Fig. 1 The comparison of the theoretical and the real directivity diagrams of loudspeaker units with horns (Danilov et al. 1992)

Table 3 Measured and corrected data *for* directivity determination of the NPL SODAR source

Measured angle θ_m	0°	10°	20°	30°	40°	50°	60°	70°
Distance d from horn aperture to microphone (m)	0.80	0.80	0.81	0.83	0.86	0.88	0.92	0.95
Distance variation correction 20log(d/0.8)	0	0	0.1	0.32	0.65	0.87	1.30	1.65
Real angle θ_h	0°	12.5°	25°	37°	48°	60°	70°	78°
Measured level, dB	0	−0.3	−1.8	−4.0	−6.5	−10.5	−12.5	−16.0

The acoustic dish was created to investigate the lower atmosphere. For the acoustic antenna, a fiber parabolic dish with an aperture diameter of *1.2 m* and a dish depth of *0.22 m* was built. A commercial driver unit (60 watts RMS/90 watts maximum) was mounted on a tripod platform at the focus of the dish, *0.60 m* from its apex, with a fiber exponential horn. Figure 2 depicts the entire dish, horn, and driver unit. Figure 3 shows the directional response of this antenna arrangement at frequencies of *1000, 2000,* and *2500 Hz.* At *1000 Hz, 2000 Hz,* and *2500 Hz,* the 90°

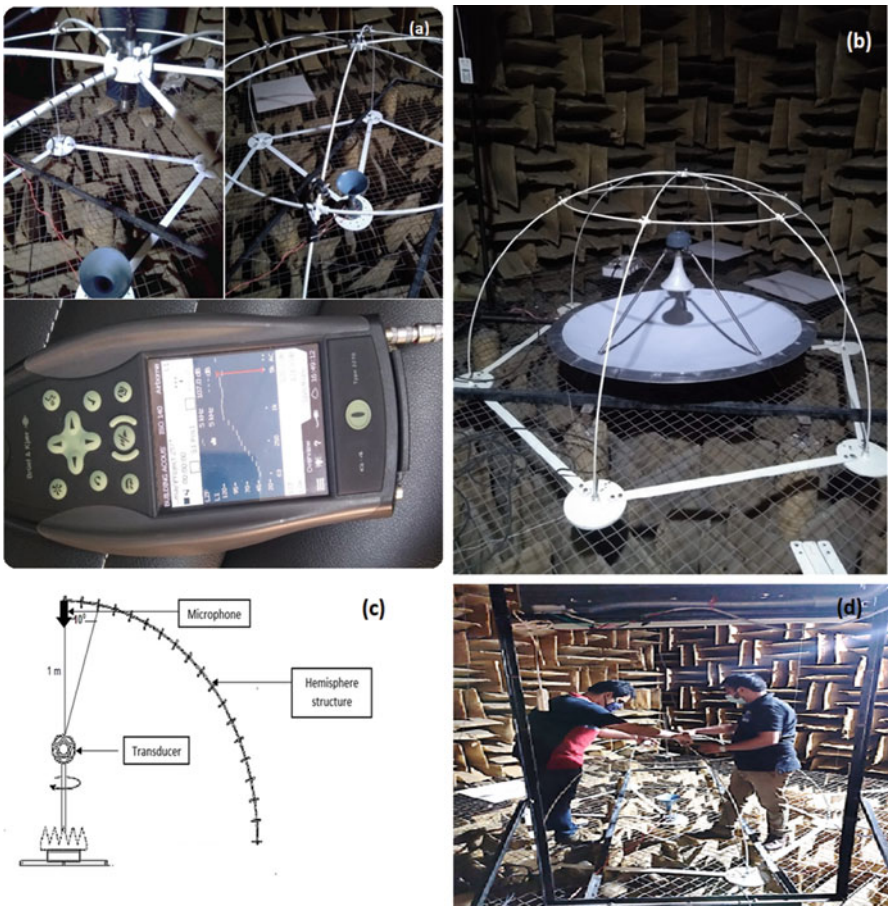

Fig. 2 Transducer testing in anechoic chamber

side lobe suppression is roughly *15 dB, 20 dB,* and *25 dB*, respectively. At *1000 Hz,* the primary lobe had a width of $\pm 12°$. Without the dish, the maximum impulse output was 110 dB for a 60 W input, and with the dish, it climbed to 136 dB at 1000 Hz utilizing a typical sound source. Under free field conditions, the system's receiving sensitivity was *12.5 mv/μbar*. The characteristics of a parabolic dish in a receiving transducer with a B&K pre-polarized free-field 1/2 microphone type 4189, set at its focus, with open-circuit sensitivity of 26.9 dB (1 V/Pa) and 0.2 dB uncertainty (95% confidence level). The boost in the received signal due to the paraboloid reflecting surface was found to be on the order of *25 dB*. (approximate). If the sounding source moved by an angle of $\pm 20°$ on either side of the axial line of the paraboloid surface, the received signal was reduced by *10 dB*, with *25 dB* side-lobe rejections at *90°*. To calculate the antenna transducer's receiving and

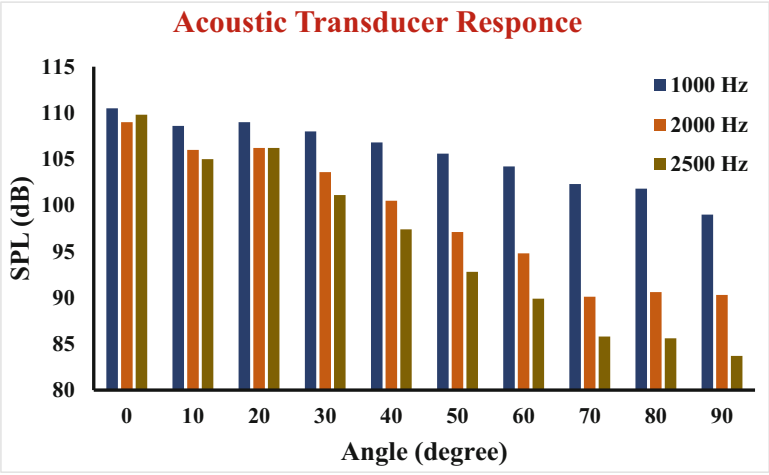

Fig. 3 Transducer response in transmitting mode

transmitting power efficiency, calculations were done (Bradley 2007). Receiving and transmitting efficiencies were calculated to be *21.3*% and *29*%, respectively.

SODAR Application in Air-Pollution Warning System

The use of concentration monitoring networks in air pollution investigations has some physical, technological, and economic restrictions. Air pollution dispersion models are commonly utilized for diagnostic purposes, forecasts, simulation studies, and climatological assessments as a result of this feature. Models are extremely important in air pollution control systems, where anticipated concentration levels are required to assess potential hazards, forecast violations of Air Quality Standards, and activate countermeasures.

The dispersion models are analytical expressions that describe the pollutant's concentration field as a function of emission, source characteristics, and meteorological conditions in their most generic form. The atmospheric stability category, mixing layer depth, and wind vector as a function of height are the meteorological data utilized in dispersion models, which are not dependent on their detailed structure. These data may be required in many formats depending on the purpose; for example, measurable data for diagnosis, projected values for air pollution forecasting, and values characteristic for a specific time period for climatological investigations. SODAR can provide useful information in all of these situations. SODAR recordings can be used to read real-time features of stability, inversion layers, and higher winds. The ability to connect Doppler SODAR to a modeling computer directly is particularly useful in this application. A series of regular SODAR observations is required for longer climatological investigations. The forecast is the most difficult scenario; it necessitates unique ways.

Furthermore, the SODAR technology has proven to be a useful diagnostic tool for line-of-sight communication lines and remedial actions. SODAR data provides detailed information on the dynamics of a variety of atmospheric events that influence communication quality in both analogue and digital systems. The stable conditions explored by the acoustic sounder are usually linked with severe fading decreasing the quality of reception in line-of-sight links. SODAR is the most practical instrument for diagnosing communication issues caused by extreme fading induced by reflected or refracted components from the ground and higher layers.

The SODAR system provides continuous data of ABL height (during convection and inversion condition) and its situations in air pollution, communication, and aviation. In this part, we show how SODAR data can be put to good use in a variety of different weather condition for air pollution.

Method of Determining ABL Height Using SODAR Echogram

The SODAR is a low-cost remote sensing system that employs sound waves. It enables round-the-clock monitoring. This is a straightforward operating principle. A SODAR sends out sound pulses at different levels of the atmosphere and receives back dispersed pulses in temperature inhomogeneities. SODAR echograms provide images of reduced climate turbulence as well as contaminant distribution responsibilities. The echogram's vertical profile of the acoustic refractive index divides atmospheric conditions into two categories: convective period and non-convective (stable) period (output of SODAR). Bradley (2007), Beyrich (1997), and Singal et al. (1997) have compiled methods and algorithms for calculating the ABL height using SODAR data. Being based on sonic principles, the range of SODARs is limited to a few hundred meters extending a maximum up to $3000\ m$ based on the power and frequency of the emitted pulse. The data must be divided into two categories by the observer: inversion time and convection time. Fog-layer, rising-layer, multilayer, and other ABL structure classifications exist, and each classification necessitates a distinct technique to ABL height estimate.

Aside from the convection period, the ABL height was directly extracted from the echogram using visualization. Because SODAR can record both thermal and shear echoes as a gauge of turbulence in the lower atmosphere, it can be used to estimate ABL height both during the day and at night. The most active ABL height can be seen during the day in the form of thermal plumes. The only factor that determines whether or not this is an accurate representation of the ABL height is SODAR sensitivity. A suitable empirical relationship, on the other hand, can always be used to approximate the theoretical limit stated by the Holzworth model. They were covered by a low-level elevated shear echo layer (Emeis 2010; Bradley 2007; Singal and Aggarwal 1979). So, the actual ABL height can be determined by using an empirical formula (Singal and Aggarwal 1979).

$$y = 4.24x + 95 \tag{8}$$

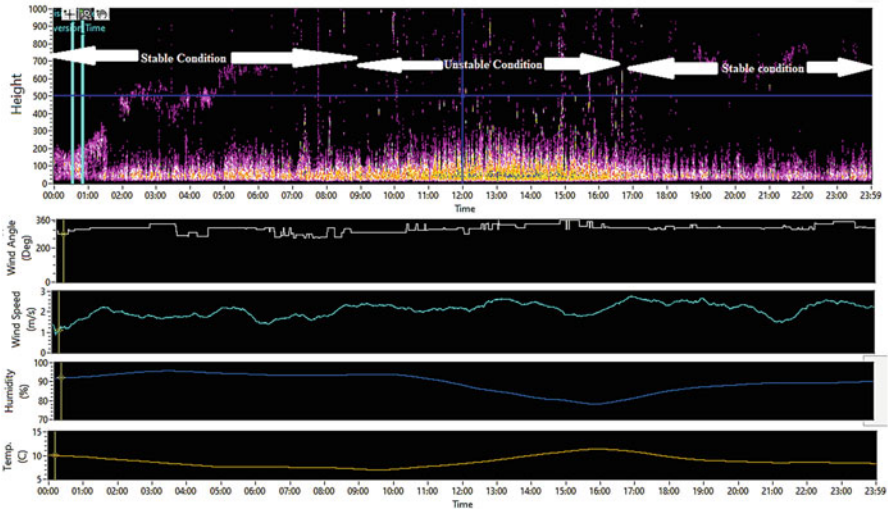

Fig. 4 Stability conditions observation at Delhi

where y is the calculated ABL height and x is the observed thermal plumes height in the echogram. This formula was determined by comparing SODAR data with Radiosonde data.

Thermally stable echo returns are visible at night. The ABL height of the atmosphere can be estimated by measuring the top of the acoustic echo. An advantage of this technique over more convectional direct measurement is that it records the presence and height of inversion layers in real time.

On the echogram, Fig. 4 depicts the stable and unstable situations for the atmosphere. These conditions were the result of many meteorological occurrences causing thermal inhomogeneities in the lower atmosphere.

As the sun sets, thermal plume activity decreases, and SODAR data shows that the characteristics of the echoes change. The structure depicts how nocturnal inversions occur on the ground surface as the air cools in contact with the cooler ground. Wind speed, cloud cover, and surface characteristics all have an effect on the thickness and lapse rate of the cooled layer, i.e., the depth and strength of the inversion layer. To demonstrate how the echo layer represents mechanically induced turbulent mixing and atmospheric stability, consider the following: an acoustical echo's base represents the depth of a mixed layer, its thickness represents a stable zone, and the echo layer's peak represents an inversion peak. SODAR recordings show a horizontally stratified layer or layers throughout the night. As soon as the sun comes up, the thermal plume activity begins to take over and replace the horizontally stratified structure. The diurnal pattern of the nocturnal inversion and the turbulent mixing depth can be determined from a typical SODAR night record. An echo layer that starts at the ground suggests that the stability begins at the surface and that the mixing height based on lapse rate is zero.

During the day, fog rises and persists near the surface at a height of 500 m, resulting in a significant increase in air pollution. The best time to observe these

phenomena is on a cloudy day following a night of intense radiation cooling. Aviation is also jeopardized as a result of vision deterioration. Such situations do occur on occasion as a result of inversion advection during a cold wave in Delhi. In reality, the current studies' primary goal is to try to determine the significance of SODAR records in providing micrometeorological information relevant for monitoring air pollution situations over a 3–5-year period.

SODAR Measurement at Delhi Region

India's Capital Region, i.e., Delhi (*28.70° N, 77.10° E*) in the homogeneous urban area is placed near the bank of the river Yamuna at *715 feet* above sea level (Kumar et al. 2021b). It is unique in that it serves as a gateway between the Thar desert to the southwest, the Aravalli range to the northeast, and the Himalayas to the north. The climate of Delhi is determined mostly by its remote inland location and the presence of continental air, as shown in Fig. 5. The climate of Delhi is known for its extreme dryness, as well as its hot summers and frigid winters. This is influenced by easterly or southerly air and is responsible for seasonal changes in temperature, humidity, cloudiness, and precipitation. Delhi's climate is influenced by the wind. The wind is moderate for most of the year. In June it is strongest and in November it is the lightest. During the summer months, hot and dust-laden winds can lead to storms and storms. During the period from September to May, the prevailing wind direction to the northwest is dominated by the easterly monsoon component. In summer, the temperature increases gradually to *46° C*, and in winter, to *2° C* (Kumar et al. 2021c).

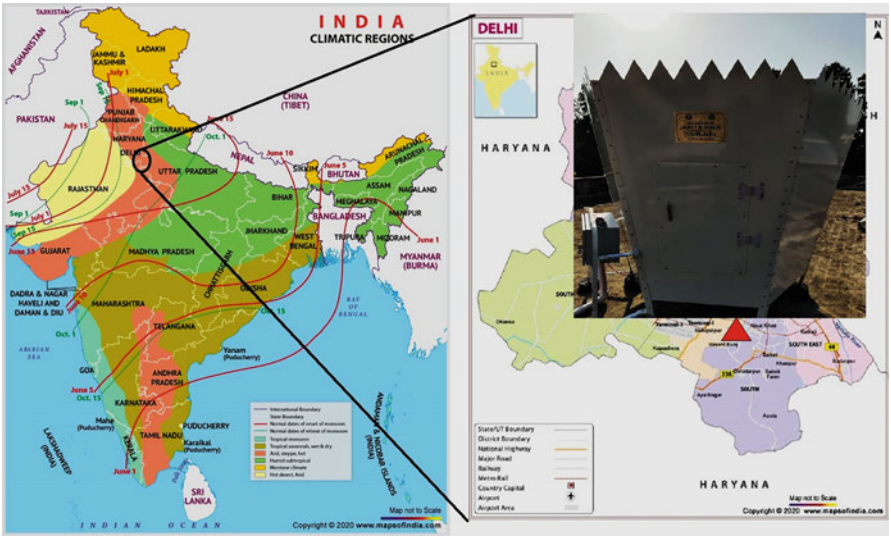

Fig. 5 Monsoon map of India and installed SODAR system at Delhi. (Courtesy: www.mapindia.com)

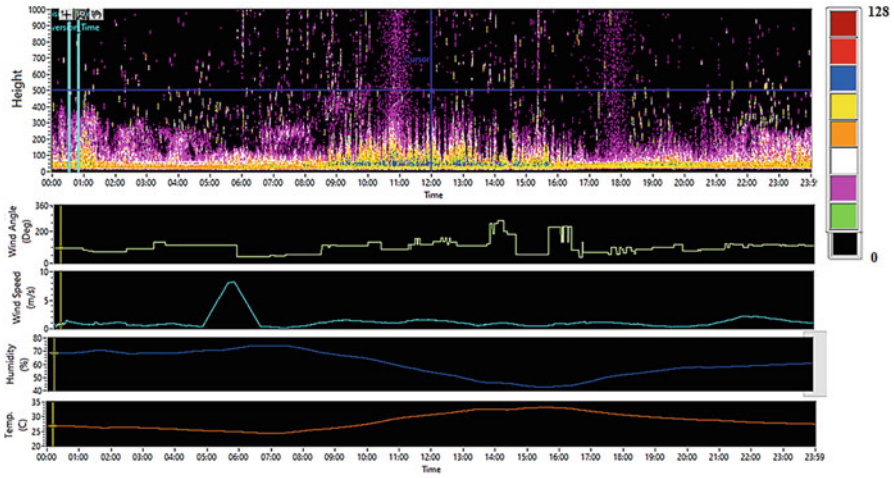

Fig. 6 SODAR echogram observation at Delhi

Monostatic SODAR has operated in Delhi for a few months, during the day and at night, under different atmospheric conditions. It is seen that stable and unstable conditions can be easily determined and classified on the echograms, together with various types of perturbations and Kelvin-Helmholtz blows. The echograms have studied details with the application of software resolution. These structures are caused by various meteorological phenomena and are the appearance of thermal inhomogeneities within the low atmosphere. The diurnal variation and varied architectures of the ABL with variations in wind direction, wind speed, relative humidity, and temperature are shown in Fig. 6. SODAR echogram for the 17^{h} *October 2019*, from *0000 to 2359 IST*. The principal regime is a "classical" 24-h pattern with two transition phases of around *0700–1000 IST* and *1600–1800 IST*, alternating between stable (inversion in the evening and night, *1900–0900 IST*) and unstable (convection in the daytime, *0900–1700 IST*) stratification. Local circulation is unclear in this case; the wind pattern is described as a land breeze with a direction of *150–250 degrees*. Temperature and relative humidity diurnal patterns are typical of fair-weather circumstances. Wind speed increases monotonously with elevation in the inversion layer reaching an extreme at the top of the inversion layer. Due to the disintegration of the stratified echo layer, the wind speed and direction change between *0430 and 0630 IST*. The ABL varied constantly throughout the day, as evidenced by these structures.

Comparison of the ABL Height of SODAR and MERRA-2

Data from 1980 are available in the Modern-Era Retrospective Analysis for Research and Applications, Version 2 (MERRA-2). Because of advancements in the assimilation technology that enable assimilation of recent hyperspectral radiance

and microwave observations, as well as GPS-Radio Occultation datasets, it was introduced to replace the original MERRA dataset (Gelaro et al. 2017; Bosilovich et al. 2015; Rienecker et al. 2011). MERRA-2 includes improvements to both the GEOS model and the GSI assimilation system. The spatial resolution is similar to MERRA (approximately 50 km in the latitudinal direction). Along with the meteorological assimilation improvements, MERRA-2 makes substantial progress toward GMAO's goal of an Earth System reanalysis (Bosilovich et al. 2015). The MERRA-2 reanalysis is the first long-term global reanalysis to incorporate space-based aerosol measurements and depict their interactions with other physical processes in the climate system (Rienecker et al. 2011). The ice sheets over Greenland and Antarctica are represented in MERRA-2.

Figure 7 shows that during the analysis, the MERRA-2-derived ABL height is monthly 0.5 × 0.625 (MERRA-2 model M2TMNXFLX v5.12.4) over January 2019 to December 2019, shape India, region 77.12E, 28.64 N. SODAR-derived ABL heights are compared to MERRA-2-derived ABL heights in the following. Scatter plots of both ABL heights SODAR and MERRA-2 for the Delhi region are shown in Fig. 8. There have been a significant number of MERRA-2 derived ABL heights that

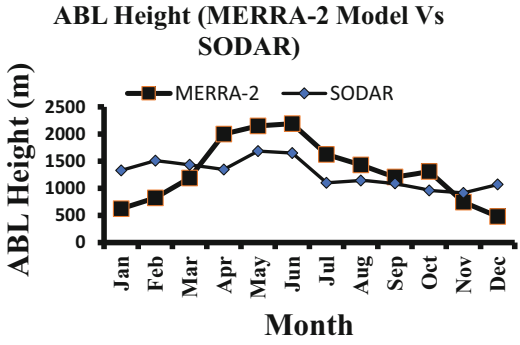

Fig. 7 Monthly variation of ABL height of Delhi

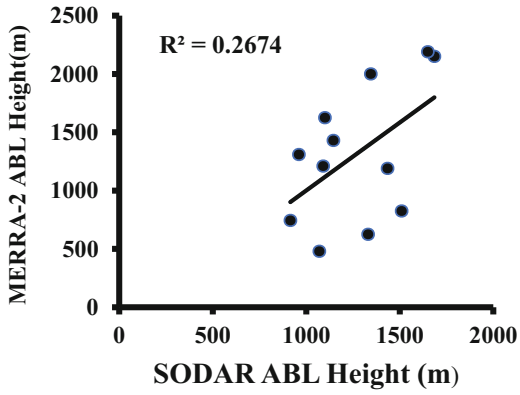

Fig. 8 Scatter plot between SODAR and MERRA-2 dataset

is larger than those derived from SODAR. The average deviation is approximately 450 m, and it tends to increase with increasing ABL heights.

SODAR Application in Determining the Pollution Loading Capacity of the Region

Because of its ability to provide continuous data about the thermal boundary layer and its variability in space and time, the SODAR system is used in dangerous situations such as air pollution, communication, and aviation. In this section, we will discuss how SODAR data can be used in pollution control.

Simmons et al. (1971) and Beran and Hall (1974) discovered that the concentration of pollutants in any location grows as the square root of the distance along the trajectory when the emission intensity is uniform. Also, the concentration rises parabolically with distance over the source area, and it falls off rapidly at first, then slowly as the air moves further away from the source. Upwind mixing and lateral diffusion can only reduce, not remove, pollutant levels no matter how far downwind they are. Solar heating and winds to an extent help to remove pollution but by far the most affective process to remove pollutants from the atmospheric air is the falling of rain. The rain removes the particles and gases present in the cloud drops and in the polluted air. The vertical change of temperature (low-level stability, inversion, topography and buoyancy, etc.) and the direction and strength of the wind are the primary determinants of pollutant concentration in the atmosphere (Kumar et al. 2017a; Saha et al. 2019). The rate at which contaminants spread upward is controlled by vertical temperature changes, and clear air from above mixes downward into the polluted air. The quantity of mixing and rate at which pollution spreads horizontally as it is carried downstream is determined by wind speed, while the wind field variance controls the atmosphere's stability. Aside from these two fundamental parameters, air quality in any given urban region is affected by local and surrounding terrain, landscape features, climatology, location and type of pollution source, political borders and legislation, demography, total effluent, types of effluent, time distribution of emission, and synoptic weather pattern, among other factors.

The ventilation coefficient is the product of the ABL (mixing height) measured by SODAR and the average wind speed; it is an atmospheric condition that indicates air quality and pollution potential, i.e., the ability of the atmosphere to dilute and disperse pollutants across a certain area (Kumar et al. 2015; Sujatha et al. 2016; Kumar et al. 2021d, 2017b). The higher the ventilation coefficient, the more effectively the atmosphere can dispose of contaminants, and therefore the air quality improves. Lower ventilation coefficients, on the other hand, result in poor pollutant dispersal, stagnation, and poor air quality, potentially resulting in pollution-related dangers. The ABL and the average wind speed across the mixing layer are directly connected to the ventilation coefficient. The ventilation coefficient varies in response to changes in the ABL and average wind speed. High pollution potential is indicated by a ventilation coefficient of less than 6000 m2s-1 during the afternoon hours, or by

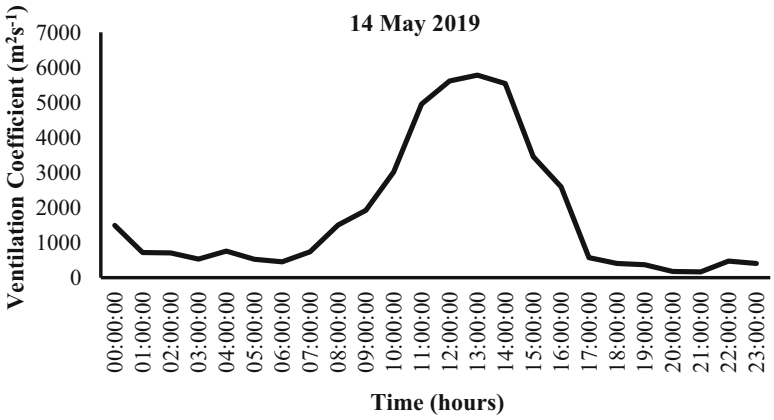

Fig. 9 Temporal variation of ventilation coefficient

an ABL height of less than 500 m during the morning hours. Using SODAR data, the ventilation coefficient over Delhi was calculated (Fig. 9), revealing severe pollution loading.

In varied atmospheric conditions, Fig. 10 depicted the decision algorithm's scheme for air pollution warnings. The synoptic analysis and actual ABL observations by SODAR were utilized to calculate various types of data and verify the stability of actual pollution conditions.

Conclusion

SODAR is a well-known and widely used meteorological tool that provides continuous real-time and critical ABL data. SODAR contributes to city planning, industrial zoning, and other air quality control methods. Data is essential for assessing environmental impact assessments and determining a city's carrying capacity for pollutants. For site-specific dispersions, SODAR has proven to be extremely useful in controlling air pollutants at a low cost. SODAR and other approaches were now being developed as part of these efforts. Every year, new advancements in devices and applications were reported, demonstrating technical versatility. More commercial enterprises have improved SODAR models with a new R&D concept and application potential.

Future Scope

A combination antenna baffle made of high-acoustics material and high-density foam may provide better atmospheric noise cancellation for low-strength backscatter signals from critical atmospheric structure information in the spatial domain. For better ABL structure identification and atmospheric noise reduction, a hardware-

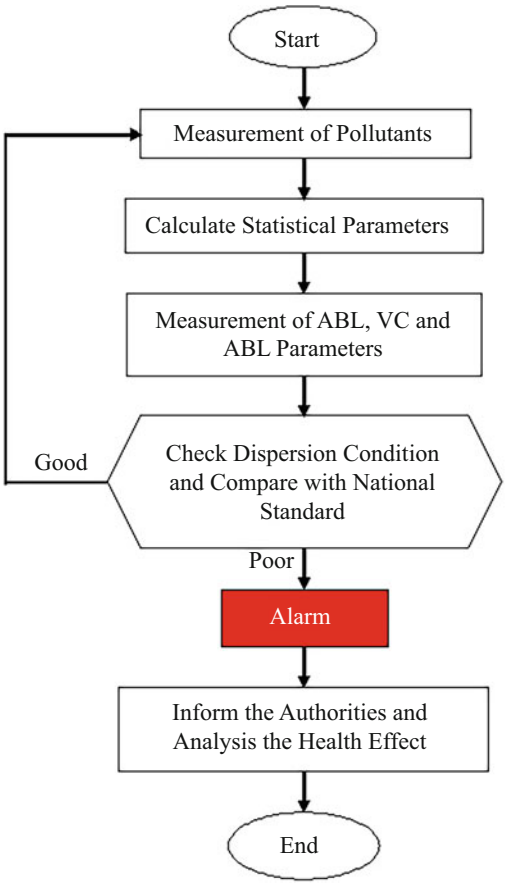

Fig. 10 Scheme of the decision algorithm for air pollution in different condition

based machine learning system composed of an electronic representation of the pre-amplifier output in real-time mode can be designed. The acoustic antenna structure can be created using 3D printing and lightweight fiber material.

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Measurements of Indoor Air Quality

66

Science and Applications

Atar Singh Pipal and Ajay Taneja

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Abstract

Air quality in indoor environments is defined as the eminence of air in and nearby the house that directly affects human health. In the last few decades, the international scientific forum, environmental research, and political institutions have strived upon the improvement of the comfort, health, and well-being of building occupants. First and foremost, why is monitoring of IAQ necessary? This is due to indoor air pollution (IAP) which can have a serious and negative impact on both health and quality of life of inhabitants. In view of majority of time spend indoors by occupants, the subject of IAQ beholds a compelling approach for

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mitigation of air pollutant exposure specifically during current health crisis owing to COVID-19. Various research groups have used the latest equipment to manage the IAQ testing services that precisely test a range of parameters and concerned issues causing discomfort at the workplace. Thus, the aim of this chapter is to address the necessity and review the current knowledge of IAQ measurement in terms of different aspects, allied science, and its applications in order to show the strategy of translational research studies on IAQ. In addition, the different aspects of IAQ such as measurement techniques, characterization methods, source apportionment, and their effects on environment and human health are also parts of this chapter. Due to the rapidly changing technology, thermal ease parameters such as temperature, relative humidity (RH), and wind speed are known to play an important role in indoor air quality. This chapter is an attempt to discover how persons can live in surroundings of the environment with a good air quality and comfortable lifestyle.

Keywords

IAQ · Measurement techniques · Characterization · Application and science

Introduction

Indoor air quality (IAQ) is basically defined as the circumstances within the house which usually comprise air quality admittance to views, daylight, lovely sound situations, resident activities, and thermal relief. IAQ is also demarcated by the illustration of air pollutant concentrations (gaseous compounds and particles such as non-biological and bioaerosols) and thermal circumstances which could be harmful and influence the fitness, coziness, and enactment of a building's inhabitants (Satsangi et al. 2014; USEPA 2020). IAQ definitions can be different and depends on perceptions of the human user, indoor air space, and source contribution profile to indoor air pollution (Bluyssen 2009).

IAP is considered within the five top risks to public health (USEPA 2020). The disclosure leads to health hazards including respiratory and cardiovascular syndromes and their carcinogenic effects (lung cancer and childhood leukemia) (Franklin 2007; Lin et al. 2013). The World Health Organization (WHO) recognizes an array of pollutants like particulates (PM), carbon dioxide (CO_2), nitrogen dioxide (NO_2), carbon monoxide (CO), radon (Rn), formaldehyde, polycyclic aromatic hydrocarbons (PAHs), and volatile organic compounds (VOCs) (benzene, naphthalene, trichloroethylene, and tetrachloroethylene) for assessing IAQ (WHO 2010). Around 4 lakh impulsive demises due to air pollution were reported in 2019 by the European Union (European Environment Agency 2019). Therefore, it is serious to afford declarative and conceptual consciousness about the survival of air pollutants and their sources, formation methods, limit values, and health effects along with their measurement methods and principles (Rohra et al. 2018, 2020; Pipal et al. 2021).

The IAP led to greater than four million premature deaths in offspring and adults as per the WHO report in 2012. It is also responsible for 1.5 million deaths and 2.7% of the worldwide burden of illness (WHO 2010). IAP is one of the nine utmost substantial global burdens of disease risk (Forouzanfar et al. 2015). In this connection, the climate variation, environmental ruin, quick growth in urban areas, and epidemics are the global challenges. In order to overcome these global issues, the review and modification are highly required in the current understanding/outdated methods of evaluating man-made effects on the environment, multidisciplinary tactics of integrated valuations, and sustainable environment development methods.

The Sustainable Development Goals (SDGs) and targets were sated by the United Nations which aims to close risky deficiency and generate a strong and workable world by the year 2030 (UN SDGs 2015). Air pollution is precisely stated in the diverse SDGs which are as follows: Goal 3 emphasizes on moral health and well-being, Goal 7 marks entree to fresh and reasonable energy, Goal 11 pays attention to maintainable cities and societies, and Goal 13 focuses on challenges for climate change. Apart from these goals, necessary action and strategies (like converting to cleaner energy, green economy, smart cities, cooking and transport solutions) should be taken for improvement of IAQ.

Consequently, these steps are to abstract what has been educated and relate it to the individual atmosphere which comprises the plans for keeping good IAQ. This has expanded significant importance in scientific community and public awareness during the COVID-19 pandemic period. IAQ has been debasing due to excessive emissions of air pollutants and directly results the extent of COVID-19. The $PM_{2.5}$ magnifies COVID-19 infectivity which permitting to the epidemiological study led in Italy (Borro et al. 2020). It was shown that growth (10–25 $\mu g/m^3$) in $PM_{2.5}$ concentration is responsible for increasing the infection rate at three times and enhance (twice) the mortality case. An epidemiological assessment was conducted by Cole et al. (2020) to find the association between air quality and COVID-19 losses in 355 municipalities of the Netherlands by a deleterious binomial regression model with chemical transport modeling (Cole et al. 2020).

Therefore, regular ventilation, proper cleaning, and other precautions in indoor environments are crucial to decrease the probable viral burden in the air which diminish the likelihood of infection with the COVID-19. In addition, the concentration of air pollutants (PM and VOCs) could be reduced which are responsible for various harmful symptoms (disorders, headaches, and dizziness). Metal oxide semiconductor (MOS) sensors used for direct measurement of VOCs and CO_2 measurement are also a sign for IAQ monitoring system (Höfner and Andreas 2021). The good air quality in classrooms and schools is also important to make ideal situations for learning and other scientific activities. In addition to avoid the high emission of air pollutants every day, regular ventilation system is the most actual mode to approve good IAQ. Thus, the environmental awareness and its consequences along with make approval for ventilation trials which are the basic requirement for maintaining a good IAQ. Moreover, agreeing to recent ideas of environmental knowledge and attitude in terms of different aspects and environmental behavior is a very important part. It has been displayed that the modest teaching of

environmental familiarity about the pollutants and principles of sensor technology is a basic requirement, and this type of information alone is not adequate to create a variation in environmental performance (Szagun et al. 1994; Milton et al. 1995). Thus, measurements, characterization, and source apportionment of IAQ are necessary in order to examine the entire aspects of IAQ system with respect to current science and its applications. However, IAQ is hard to measure and evaluate for several causes, prominent among them are:

(i) The absence of fixed metrics, standards, and consent on auspicious IAQ.
(ii) The diversity and difficulty of air pollutants which are found in indoor environments can distress public health and their comfort at very low levels.
(iii) The scarce understanding of relations among indoor pollutant levels, separate exposure, and combined air pollutants and its health effects (acute and chronic).
(iv) The variety of health issues which are related to IAP exposure also can affect humans in various ways.
(v) The probe of whether the pollutants being measured are the ones that really matter.
(vi) The lack of awareness of potential problems and remedies.

Moreover, climate variables such as temperature, rainfall, water vapor, WD, WS, and cloud properties affect air quality in direct and indirect ways. For instance, the higher temperature in outdoor environment leads to amplified O_3 level and subsequent reactions which generate air pollutants in indoor environment. Increased temperature and precipitation in indoor environment could increase chemical off gassing degrees from indoor resources and threat for flooding and water damage, respectively. The regional climate change is also responsible for affecting the flood arrays, droughts, wildfires, vegetation, and allied indoor exposures (mold, dust, bacteria, ignition products, chemical contaminants, pesticides, and pollen).

The different indoor air pollutants and their concentration level and sources along with health effects are shown in Table 1.

IAQ and Potential Pollutants

The research of IAQ includes concentration measurement of PM and primary (Rn, O_3, NOx, SO_2, CO, diatomic carbon, and VOCs) and secondary gaseous pollutants. Several studies have reported higher particle number concentration in ultrafine (<0.1 μm) than coarse range in urban indoor environment (Thomas and Morawska 2002; Gramotnev and Ristovski 2004). In indoor environments, the usage of chemical products has been increased brutally in the last decade. This can generate numerous hazardous chemicals in indoor under room temperature that are associated with several health issues. VOCs exposure is associated with different indications like respiratory symptoms, headache, vertigo, fatigue, nausea, and nose, eye, and throat annoyances (Yang et al. 2001; Gorai et al. 2014), whereas adverse effects of NO_2 toxicity are associated with fever, sneezing, coughing, diaphoresis, chest pain,

Table 1 Summary of different IAPs and its characteristics, sources, and health effects

Pollutants	Sources	Health effects	Levels	Steps to reduce exposure
Radon (Rn)	Rn sources are Earth crust, rocks beneath home, water, building construction materials	No immediate symptoms. Lung cancer deaths (7–30 k/year). Smokers are at greater risk for lung cancer	Average Rn level is 1.3 (pCi/L) and ~ 0.4 pCi/L in indoor and outdoor, respectively. It is based on a national residential survey done in 1991	Testing your home for Rn, fix your home if Rn level is >4 pCi/L and this Rn level still pose a risk
Environmental tobacco smoke (ETS)	Cigarette pipe and cigar smoking	Eye, nose, and throat annoyance, headaches, lung cancer, and heart disease. In addition, increased risk of lower respiratory tract infections and asthma	PM levels are lower as compared to outdoor concentration in indoors without smokers or other strong particle sources, while PM levels may higher many times in homes than outdoor smokers	Stop smoking inside the buildings. Rise ventilation and open windows and use exhaust fans in the smoking areas
Biologicals	Drizzly/humid walls, ceilings, carpets, and furniture strictly maintained humidifiers, dehumidifiers, and air conditioners	Eye, nose, and throat irritation, shortness of breath, and other infectious diseases	Indoor levels of pollen and fungi are lower than outdoor levels. The dust mites are higher in indoor as compared to outdoor levels	Fix and use fans vented to outside in kitchens and bathrooms. Daily clean cool mist and ultrasonic humidifiers, and refill with clean water. Clean and dry or remove water damaged carpets. Use dehumidifiers to preserve humidity (30–50%)
CO	Unvented kerosene and gas space heaters dripping and furnaces leaking	At low-level concentrations, fatigue and chest pain in healthy people and people with	Normal level (0.5–5 ppm) in homes without gas stoves while 5–15 ppm in near to well-	Keeping gas usages accurately adjusted. Acquiring a vented space

(continued)

Table 1 (continued)

Pollutants	Sources	Health effects	Levels	Steps to reduce exposure
	chimneys, gas water heaters, wood stoves and fireplaces gas stoves and automobile exhaust	heart disease, respectively. At higher concentrations: headaches, dizziness, and nausea	adjusted gas stoves. Poorly accustomed stoves may be 30 ppm	heater. Using good fuel in kerosene space heaters. Fitting and using an exhaust fan vented to outdoors over gas stoves. Properly sized woodstoves. Make sure that doors on all woodstoves fit tightly. Having a good inspect, clean and tune up central heating system
Nitrogen dioxide (NO_2)	Kerosene heaters, unvented gas stoves and heaters, ETS	Eye, nose, and throat annoyance could cause reduced lung function and amplified respiratory contagions in children	Average level in homes without burning usages is half of outdoors. With gas stoves, kerosene heaters, unvented gas space heaters, indoor levels than surpass outdoor	See steps under carbon monoxide
Organic gases	Household products (paints, strippers, and other solvents), wood protectors, aerosol sprays, cleansers, antiseptics and air fresheners, stored fuels, and automotive goods	Eye, nose, and throat anger, headaches, nausea, kidney and central nervous system diseases Several organics can cause cancer in animals; few are supposed to cause cancer in humans	The levels of organics found two to five times greater in indoors than outdoors. During certain activities (paint stripping), the levels increased to 1000 times of background (outdoor)	Household products should be used as per the maker's instructions. Confirm and provide amply of fresh air while using goods Keep out children and pets Never mixed household care products
Respirable particles	Firesides, woodstoves, and kerosene heaters	Eye, nose, and gorge frustration,	Particle levels in homes without smoking or	Vent furnaces to outdoors; keep other doors open

(continued)

Table 1 (continued)

Pollutants	Sources	Health effects	Levels	Steps to reduce exposure
		respiratory infections, and bronchitis lung cancer	other robust sources are similar or lower than outdoor levels	when using unvented space heaters. Select well-sized woodstoves; clean and tune up central heating system. Repair leaks quickly. Frequently change filters on central heating, cooling systems, and air cleaners
Formaldehyde	Wood products (hardwood, plywood wall paneling, and fiber board) and furnitures. Drapes and glues	Irascibility for the eye, nose, and throat, wheezing. Coughing, skin rash; allergic reactions cause cancer and other effects due to organic gases	Average concentrations in older homes are below 0.1 ppm. Homes with significant amounts of new wood products are greater than old homes	Use external rank pressed wood products. Use air conditioning and dehumidifiers to reserve adequate temperature and decrease humidity levels Increase ventilation to diminish formaldehyde sources
Lead (Pb)	Pb-based paints, polluted soil, dust, and drinking water	Pb affects all body systems. High levels of Pb (80 ug/dl) in blood can cause convulsions, coma, and death, while low levels can cause adverse health consequences on the nervous system, kidney, and blood cells. Pb in blood (10 ug/dl) can harm mental system and physical growth		Children playing areas should be kept clean and dust-free Do not disturbed Pb-based paint; do not sand or burn off paint. Do not bring Pb dust into the home Eat Ca- and Fe-rich balanced diet

headache, throat frustrations, and pulmonary edema (Hesterberg et al. 2009). Numerous studies expose that an adequate CO_2 level in indoor environments has been associated with fatigue and headaches and it leads to vomiting, dizziness, and nausea (Yu et al. 2009). IAP could be a reason for number of chemical reactions in the environment which generates secondary air pollutants. It was reported by various studies that secondary air pollutants are much dangerous as compared to primary pollutants (Wainman et al. 2000; Rohr et al. 2003). Indoor secondary pollutants (ISP) like O_3, NO_2, and SO_3 are commonly detected which cause major anxiety and injurious influences on human health. The measurement and prediction of ISP are more challenging due to their chemical complexities (Yu et al. 2009). Moreover, volatile, nonvolatile, and non-biological mediators cause a destructive effect on people living inside the environments. Biological creatures (pollen, fungi, bacteria, and insects), mycotoxins, and contagious chemicals also play a vital role in deteriorating the indoor air quality (Nolte et al. 2001).

Investigation of Indoor Air Quality: A Scientific Approach for Better Environment

IAQ is an important cause for comfort of public health and quality of air inside the home which is resolute via huge issues and, subsequently, diverse occupations that are intricate in allocating with solution of indoor air difficulties. The eminence of indoor environment has converted a rising problems and fear due to a range of undefined indicators (annoyance/aridity of mucous membranes, burning eyes, headache/fatigue) in the last eras. These symptoms was reported on the basis of survey and complained by occupants who are living in several indoor environments (Habil et al. 2013; Satsangi et al. 2014).

Moreover, these symptoms might be related to raise air pollutant concentrations in homes which were committed to evaluating climate situations and chemical components in the indoor environments. Therefore, the purpose of indoor air monitoring is to define the state of surroundings, evaluate the fears, analyze the influence of several discharge sources, and offer monitoring data for policymakers to make a conclusion. In addition, monitoring and characterization may contain relevant measurements for corporal atmosphere with the buildings' technical systems, measurement contact, studies of the occurrence of aggravation reactions, or signed reports due to IAP.

IAQ is affected by air exchange rate, RH, temperature, ventilation, air movement, etc. (Graudenz et al. 2005), as the buildings presently constructed are airtight and built by the use of sophisticated isolation ingredients which can aid to diminish the damage of energy. Nevertheless, the modern building envelope and the air conditioning systems cause a drop in motion of new air. Additionally, the chemical goods and man-made resources used in indoor atmospheres have contributed to diverse VOCs (Wang et al. 2004). The studies infer that people are still not innocuous from risks related with IAP.

Investigations of IAQ provide evaluation of air quality in community and profitable houses. The procedure for investigating IAQ is from gathering information to formation and testing of hypothesis. The following factors are known to influence IAQ:

- A building's inhabitants
- A heating, ventilation, and air condition (HVAC) system in buildings
- Probable contaminant ways
- Potential sources of impurity

Therefore, the investigation of IAQ requires several steps which are as follows:

Planning

- Gathering background information for building systems
- Interviewing precious people, understanding the grievances and indications, and checking for designs as to where and when it arises
- Setting purposes
- Approaching determination

Data Assembling

Air pollutant (CO_2, CO, PM, VOCs, chemicals, and bioaerosols) measurements and collection of other environmental conditions like temperature, relative RH, and ventilation rate in the building.

Analyzing the Data

Measurements to eradicate assured areas or alleged difficulties as well as irregularities that directly relate to zones which require further attention.

Reporting of Probable Outcomes

Entire results showing a need for helpful action which wants be conveyed.

Contribution Support

Preparing an IAQ executive strategy (conducting routine measurements and setting policies) for maintaining a good air quality.

Although many restrictions and snags arise in the calculation of indoor quality in different environments, still there is a robust necessity to bear scientific studies for monitoring of IAQ. Obtaining analytical data could be used for statistical analysis for getting complete aspects of indoor air quality in domestic zones. Figure 1 shows various aspects which are affecting the complexity of IAQ measure and estimation system.

The strategies for collection of scientific data and its analysis are very intricate tasks for the researchers due to the complexity in indoor environments. The objectives and interest of the study play an important role in examining the IAQ. Thus, the entire aspect of IAQ such as levels of different air pollutants, chemical and physical characteristics, sources, and their probable effects on human health along with

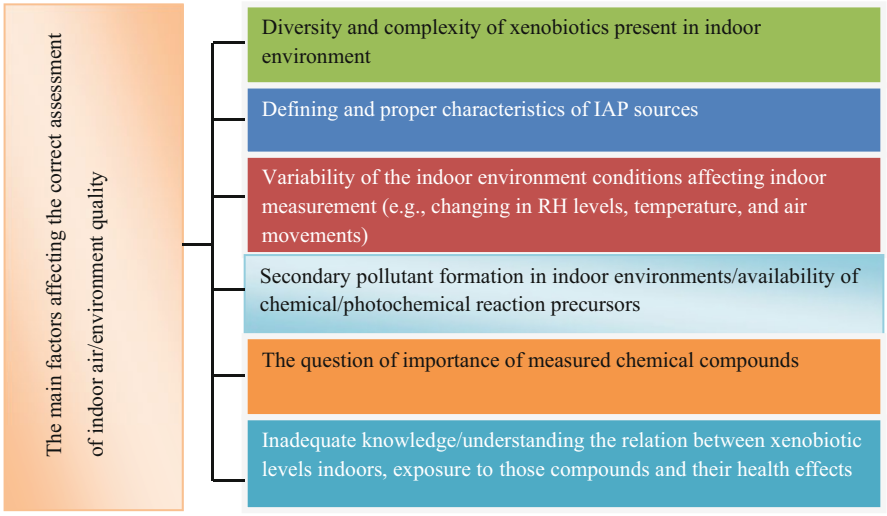

Fig. 1 Key features that disturb the effort to measure and guess the IAQ (Steinemann et al. 2017)

meteorological parameters should be considered as necessary task to obtaining the acceptable and comfort air quality (Pipal et al. 2021, Jan et al. 2017; Satsangi et al. 2014). In this regard, various researchers have been reported their finding in different microenvironments (schools, offices, and other environments) in terms of chemical composition (metals and ions) and morphological aspects (size, shape, and mixing states). A few of the studies mentioned here have reported different scientific techniques for data collection and statistical tools for data analysis and interpretation. Satsangi et al. (2014) have characterized fine ($PM_{2.5}$) and inhalable (PM_{10}) particles at urban and rural locations in terms of metal concentrations and their bioavailability, health risk assessment, and in silico study with nucleosomal protein. This study could deliver the standard data for health-associated contaminants of the urban and rural indoor environment. It is also useful for air pollution abatement programs and can encourage advanced studies on relative metal dispersal, source influence, and enrichment virtual to the fresh context. Pipal et al. (2021) used field emission scanning electron microscopy coupled with energy dispersive X-ray spectrometry (FESEMEDX) for morphological characterization (shape, size, mixing states, other morphology descriptor parameters) and surface elemental composition of individual particles in different microenvironments. They have used ImageJ software for individual particle image analysis and statistical tools for scientific data analysis and interpretation.

Habil et al. (2013) examined the acquaintance of children who are studying in schools of India to PM ranks and metal impurity along with their sources. Massey et al. (2012) characterized different air pollutant (PM_{10}, $PM_{5.0}$, $PM_{2.5}$, and $PM_{1.0}$) concentrations in roadside and urban households using Grimm aerosol spectrometer in Agra, India. Jan et al. (2017) measured coarse and fine particles and characterized

them in terms of anions, elements, morphology, and their threat calculation along with gaseous pollutants (CO_2, SO_2, NO_2, and O_3) in obviously ventilated schools in Pune, India. IAP surveys with outdoor air pollution offer a realistic data of personal air pollution revelation and for evaluating actions of health safety. Zhao et al. (2020) measured different sizes of PM concentrations and assessed individual exposure in 40 homes in the German cities of Leipzig and Berlin. Vardoulakis et al. (2020) reviewed air pollutant concentration and their sources in indoor environments linked with adverse health problems. They suggest the requirement for consistent IAQ measurement and systematic methods for long-time monitoring system over several locations.

The many indicators and sicknesses have been connected to poor IAQ in households in the last decades. Exposures to inorganic, organic, physical, and biological toxins in indoors at small ranks are mutual, pervasive, and constant. Building-related sickness is caused by various indoor ecological factors which are allocated into various groups according to the WHO such as sick building syndrome (SBS), building-associated sickness (BAS), and manifold chemical sensitivity (MCS)/Environmental Disorder (ED) (USEPA 2020).

- *Sick Building Syndrome (SBS)*: It is defined as group of symptoms (experienced by 30% of occupants) that are associated with the bodily atmosphere of definite erections (Marmot et al. 2006). These effects or symptoms are either confined in specific zones or extensive in a house (Redlich et al. 1997; Joshi 2008). These symptoms are inclining to exacerbate as a function of disclosure time in buildings and dissolve as people occupy major time away from the house (Guo et al. 2013). According to the WHO, SBS indications caused by IAP can be classified into different categories such as mucous membrane annoyance (the eye, throat, and nose), neurotoxic effects (headaches, irritability, and fatigue), asthma (chest tightness and wheezing), and skin irritation (dryness) (Kreiss 1989; USEPA 1991).
- *Building-Associated Sickness (BAS)*: BAS is a medical term used for diagnosable sickness linked to building habitation which defines sicknesses and signs with an known relevant chemical agent (formaldehyde, xylene, pesticides, and benzene along with biological agents which are much extensive). The sources of biological pollutants in houses are cooling towers, humidification systems, filter wet surfaces, and water-damaged building constituents (USEPA 2020). The BAS health symptoms are flu, fever, chills, chest tightness, muscle aches, and cough along with the grave lung and breathing complications (USEPA 2020). It has been mentioned that indoor air pollutants could be responsible for BRI symptoms in four main tools such as immunologic, infectious, toxic, and irritant (Seltzer 1994). Moreover, there are different factors like physical, chemical, biological, and psychosocial environments which are also linked to BRI.
- *Manifold Chemical Sensitivity (MCS)/Environmental Disorder (ED)*: IAQ can effect on production, individual coziness, house preservation charges, and health and care. It is an offensive situation where a distinct has or grows feeling to minor ranks of certain chemicals due to prolonged disclosure.

IAQ Guidelines and Standards

The deprivation of IAQ and its substantial hazards to human healthiness has been reported in various studies due to the long-term exposure of air pollutants and anthropogenic activities in indoor environments. Thus, the relevant organizations and the scientific community have developed indoor air standards and guidelines to assess the scenario of air quality and its problems which are established on an assimilated house tactic (Avgelis and Papadopoulos 2004). The part of IAQ strategies is to deliver a serious database that will be used as a reference for the avoidance of risky concerns of indoor pollutants and defense of community healthiness (WHO 2020). Table 2 reviews the IAQ rules for IAPs formulated by the WHO and USEPA (Al-Hemoud et al. 2018). However, it is essential to differentiate nonoccupational (such as inhabited houses, schools, and offices) plans from working (industrial) values (Abdul-Wahab et al. 2015). However, the IAQ guidelines are non-applicable to occupational sectors (Mandin et al. 2009) as these guidelines are applied for the control of air quality risks in different indoor environments such as households, schools, hospitals, public buildings, and offices (WHO 2010).

The IAQ is dynamic as people used a significant time in diverse indoor places and the presence of several pollution sources (traditional and building materials, furniture, cooking methods, and dusting causes). Therefore, the worldwide societies have set guidelines and inception standards to keep an ideal IAQ (Table 2). The guidelines of various indoor air pollutants and control policies in Indian perspective still are not available; therefore, policymakers, government bodies, and scientific societies should take action to formulate standard guidelines for indoor air pollutants. The crucial endorsements for logical exploration communities and the significance of indoor air quality program development are listed in Table 3.

Monitoring System for IAQ

Measurement and characterization of air quality are very important to examine its probable effects on human health and indoor environment. IAQ monitoring methods are very useful for the measurement of various types of air pollutants in indoor environments. The air quality measurement system which is based on various scientific methods and technologies is defined in different sections. These sections describe the required measurements to develop an IAQ profile with special emphasis on parameters of attentiveness in non-research studies and practically underlining.

Overview of Sampling Devices

There are different types of methods which can be used for measurement of indoor air pollutants as given below:

Table 2 Guidelines of indoor air quality for the different indoor air pollutants given by various scientific bodies

Toxins	Concentration levels (mg/m^3)	Coverage time	Organization
CO	100	15 m	WHO
	60	30 m	
	30	1 h	
	10	8 h	
	29	1 h	USEPA
	10	8 h	
CO_2	1800	1 h	WHO
NO_2	0.4	1 h	WHO
	0.15	24 h	
	0.1	1 year	USEPA
PM	0.15	25 h	USEPA
	0.05	1 year	
O_3	0.15–0.2	1 h	WHO
	0.1–0.12	8 h	
	0.235	1 h	USEPA
SO_2	0.5	10 m	WHO
	0.35	1 h	
	0.365	24 h	USEPA
	0.08	1 year	
Pb	0.0005–0.001	1 year	WHO
	0.0015	3 months	USEPA
Rn	100 Bq/m3	1 year	WHO
Xylene	8	24 h	WHO
Formaldehyde	0.1	30 m	WHO

Source: https://www.europeansources.info/corporate-author/europeancommission/

Selecting Measurement Devices

The increasing attention in IAQ interests the expansion of devices for examination purposes and house surveys. For evaluating the measurement device, the following criteria should be considered:

- **Simplicity of use**
  - Convenience
  - Direct interpretation vs. scrutiny mandatory
  - Sharpness
  - Real-time monitoring essential
- **Eminence pledge**
  - Accessibility of facilities and client sustenance
  - Preservation and standardization desires

Table 3 Recommendations for scientific research and the importance of indoor air quality policy management

Research theme	Scientific confidence in current knowledge and understanding	Precise resolution to meet vital slits in awareness	Prominence for strategy provision (passivity or well-being)
Indoor air pollution emissions	Characterization of IAPs (medium: further research is required)	Development of IAQ models with accurate description processes (imperatively essential to stop science gap)	Systematic quantification of IAP sources (high: highly vital for developments of new policies)
	Quantification of the IAP source fluxes (medium: further research is required)		Role of indoor sources in the entire experience to air contaminants (high: highly essential for developments of new policies)
	Chemical and physical processes affecting IAPs (short: growth is beneficial but might not need in the short term)		Interactions between outdoor and indoor pollutants

- **Productivity**
  - Time-averaged vs. prompt appraisals
  - Compassion
  - Compatibility with computer or data-logging decorations
- **Price**
  - Solo usage vs. refillable
  - Acquisition vs. rental

Strategies Used for IAQ Measurement and Monitoring

Material Development for IAQ Devices

The two-dimensional (2D) nanostructured material has exclusive somatic and chemical assets which fascinated excessive consideration for their usefulness in the plan and invention of gas-sensing devices in the last decade. In addition, this type of materials keeps unresolved features, like a huge surface-volume ratio, superb semiconducting goods, and great apparent compassion. The permutation of this kind of material with other dimensional materials is also planned as a capable method for developing high-performance IAQ monitoring sensors (Bag and Lee 2019). Recently, MOS gas sensors have been deliberated as a feasible stage for VOCs and CO finding in the indoor air atmosphere (Kim et al. 2016; Oosthuizen, et al. 2019). It is described that numerous metal oxides, such as SnO_2, ZnO, TiO_2, In_2O_3, Fe_2O_3, MoO_3, Co_3O_4, CuO, NiO, and CdO, have been used in IAQ sensing applications.

There are several categories of sensors which are currently available and used by various researchers in the field of environmental science. MOS (resistive sensor and semiconductor) respond on the sensor exterior in the air and amend its resistance. Some typical features of different types of sensors are:

(a) *Metal oxide sensors (NO_2, O_3, and CO)*:
- Low-cost sensor.
- Decent warmth (mg/m^3-$\mu g/m^3$).
- Due to variations in temperature and humidity, results may be affected.
- 5–50-minute-long response time.
- Output depends on input history.
- Instability could be observed.

(b) *Electrochemical sensors (NO_2, SO_2, O_3, NO, and CO)*: This type of sensors is formed on a chemical reaction between diverse gases:
- Average cost sensor
- Decent sensitivity (mg/m^3-$\mu g/m^3$)
- Dissolute answer time (30–200 s)
- Extremely delicate to temperature and humidity disparities which depend on electrolyte
- Discrimination: display cross-reactivity with comparable molecule

(c) *A photo ionization detector* ionizes volatile organic compounds and trials the electrical present.

(d) *Optical particle counters* perceive particulate matter contamination by computing the light scattered due to atmospheric particles.

(e) *Optical sensors* detect gases (CO and CO_2) by assessing the absorption of infrared light.

(f) *Photo ionization detector* (VOCs measurement):
- Reasonable cost sensor and handled device
- Respectable sensitivity (mg/m^3-$\mu g/m^3$)
- Partial temperature dependency and humidity effects
- Very fast response time
- Not selective: responds to VOCs that can be ionized by the UV lamp
- Important signal drift

(g) *Optical particulate counter ($PM_{2.5}$ and PM_{10})*:
- Moderate-cost sensor and handled device.
- 1-second response time.
- Compassion (1 $\mu g/m^3$).
- Capable to detect (PM_{10} and $PM_{2.5}$) particle size.
- Based on theoretical model, able to convert particle counts to PM mass.
- A diversity of parameters including particle shape, size, color and compactness, humidity, and refractive index are responsible for measurement of signal.

(h) *Optical sensors (CO and CO_2)*:
- Moderate-cost sensor and handheld device.
- Decent compassion for CO_2 (350–2000 ppm).

- Selectivity is good through characteristic of CO_2 IR spectra.
- Response time (20–120 s).
- Imperfect drift over time of the sensor calibration.
- Constraint for correction for the effects of temperature, humidity, and pressure.

Table 4 shows the different types of sensors and their measurement method.

Table 5 depicts the characteristics and various features of sensors which are used for monitoring of air pollutants.

The advanced rate and immense power ingesting of sensor nodes is one of the major alarms in the development of IAQ schemes. The sensor units are generally fitted in an industrial situation, inside homes, offices, and outdoor areas for considering the real-time applications of IAQ systems. However, the intention of the sensor is much emphasized on size, plan cost, power depletion, and message protocol along with presentation requirement on temperature and humidity disparities. In order to confirm the accurate real-time monitoring using LCS, the sensor standardization is the key trial for the researchers. The MOS is cheaper as associated with the optical and electromechanical sensors. However, numerous techniques such as wireless sensor network (WSN) and Internet of Things (IoT)-created models have been developed for real-time IAQ monitoring system.

Advanced IAQ Monitoring Technologies

There have been numerous techniques developed and publicized for real-time observing of IAQ (Saini et al. 2020). Among them, low-cost sensor (LCS), wireless sensor network (WSN), and Internet of Things (IoT)-based systems are the most current tools developed for IAQ monitoring due to their increasing possibility in the Industry 4.0 revolution.

Measuring IAQ with Low-Cost Sensors (LCS)

The LCS technology is attracting more attention to the measurement of air quality nowadays due to the minimum cost for air quality monitoring than conventional methods. This kind of sensor technology could be a game changer in measurement

Table 4 Different sensors and their measurement method

S. no.	Measurement process	Device	Abbreviation
1	Gravimetric	Gravimetric filter impactors	N/A
2	Optical	Photometers	N/A
		Optical particle counter	OPC
		Condensation particle counter	CPC
		Scanning mobility particle sizer	SMPS
		Fast mobility particle sizer	FMPS
		Aerodynamic particle sizer	APS
3	Electrical	Diffusion size classifier	DiSC
		Nanoparticle surface area monitor	NSAM

Table 5 Sampling approaches and key features of every sensor

Sensor for pollutants	Detection method of sensors	Sensor minimum resolution	Sensor correctness	Measurement range
CO	Electrochemical detection	$<1746\ \mu g\ m^{-3}$	<+/−10% of reading or 2% of full scale	0–58,200 $\mu g\ m^{-3}$
Formaldehyde	Electrochemical detection	$62.5\ \mu g\ m^{-3}$	<+/−10% of reading or 2% of complete measure	0–5000 $\mu g\ m^{-3}$
NO_2	Electrochemical detection	$41\ \mu g\ m^{-3}$	<+/−10% of reading or 2% of full scale	0–41,000 $\mu g\ m^{-3}$
O_3	Electrochemical detection	$2.14\ \mu g\ m^{-3}$	<+/−10% of reading or 2% of full scale	0–1070 $\mu g\ m^{-3}$
TVOC	Photoionization detection	$230\ \mu g\ m^{-3}$	<+/−10% of reading or 2% of full scale	0–115,385 $\mu g\ m^{-3}$

of air pollution, traffic organization, personal exposure and health inspection, citizen science, and assessment of air pollution in order to further improve the measurement quality in developing countries. The potential of LCS to abridge personalized sign about IAQ including citizen exposure to PM aims to tenacity-probable dichotomies between indorsing healthy IAQ and energy-effective houses. In order to measure spatial distributions of PM in indoor environments, the LCS networks are one of the capable scientific solutions in indoor air quality scenario. This kind of technological progress will positively change the picture of air quality measurements in the near future. Moreover, to understand uncertainties in LCS signals, it is necessary to assess and authenticate sensors with field and laboratory measurements.

The LCS technology has innovated rapidly with new research, expansions, and requests as published by numerous studies that have verified LCS technology in ambient air monitoring system (Mead et al. 2013; Miskell et al. 2017; Schneider et al. 2017; Spinelle et al. 2017; Fishbain et al. 2017; Lewis et al. 2016; Zheng et al. 2018; Liu et al. 2017). Due to the fast rising sensing technology, a wide variety of low-cost devices/sensor modules was made accessible in the market for indoor air measurement determinations. These devices are informal to use and deliver numerical evidence of air pollutant concentrations in diverse indoor environments (Chojer et al. 2020; Fanti et al. 2021; Kang et al. 2021; Howard et al. 2022). Rai et al. (2017) reviewed the scientific literature and reveals that there is a need to provide systematic guidance for selecting suitable LCS sensor and evaluated their performance which commercially existed for $PM_{2.5}$ and PM_{10} and gaseous air pollutant (CO, O_3, NO_2, and SO_2) measurement in outdoor environments. Several studies have stated that IAQ is exaggerated by outdoor air (Shrestha et al. 2019; Tofful et al. 2021; Kang et al. 2021) and noted the significant differences between outdoor and indoor air

qualities. Thus, there is concern and ambiguity for choosing the environment for suitable scientific monitoring system (Han et al. 2016).

There are several air pollutant sources which are spotted in indoor environments that depend on the carefulness of separate indoor space (Soreanu et al. 2013).

Furthermore, few review articles intensive on LCS are used for monitoring of IAP (Kumar et al. 2016a; Kumar et al. 2016b; Schieweck et al. 2018). Morawska et al. (2018) and Kang et al. (2022) showed the necessity for a scientific review of published studies with indoor and outdoor applications of LCS technology. Moreover, there is an urgent need to thoroughly review studies which compared LCS technology with other instruments used for IAQ monitoring in diverse indoor places (homes, offices, and schools). This technology is extremely significant due to the circumstances under which sensors calibrated in the laboratory do not overlay with the situations met in an indoor atmosphere (Morawska et al. 2018). In order to recognize the application of the low-cost sensing technology for indoor air observation, the current assessment attentive on studies concurrently used low-cost equipment/devices and other tools for contrast at indoor surroundings, with laboratory studies simulating indoor situations.

WSN-Based Systems

The effectiveness of air eminence observation and measurement was enhanced significantly by the WSN systems due to the huge number of nodes arranged to offer pertinent evidence of the IAQ spreading in diverse zones. The WSN is an automatic monitoring system with a self-directed, exact, and instantaneous real-time measurement of indoor air pollutants in dissimilar parts (Pitarma, et al. 2017). This method is generally useful for several VOC (benzene, ethyl benzene, toluene, and xylene) determination (Arroyo et al. 2019). It is also used for measurement of meteorological parameters (relative humidity, temperature), CO, CO_2, and luminosity (Pitarma et al. 2017). These structures regularly comprise a circulated sensor network associated with a cloud method (Arroyo et al. 2019). The improved cloud computing systems have been employed for monitoring, storing, processing, and visualizing achieved data from the sensor setup (Arroyo et al. 2018). In addition, the gained data is further analyzed and processed by artificial intelligence (AI) skills for optimization of air pollutant exposure.

Various researchers have functioned on WSN-built intentions with Zigbee in the last decade and found that WSN is the reliable communication protocol for the development of the monitoring system for IAQ.

In such situations, to improve the whole system performance, the scholars desired to effort on battery power executive structure. Nevertheless, it is noticed that few researchers successfully implemented energy and charge redeemable checking schemes using WSN (Yu and Lin 2015). Mostly, web servers are used as data access platforms in the WSN IAQ monitoring systems because it strains few surplus struggles to make real-time warnings on smart phone user to avoid dangerous circumstances.

IoT-Based Systems

The Internet of Things (IoT), big data, and machine learning (ML) technologies have been announced as trending tools with the expansion of mobile skills which provide excessive aptitude for IAQ real-time monitoring in recent years. The IoT built movable devices that can be used for real-time air quality monitoring and its control. These devices are proficiently transferring real-time data to a cloud accumulation constructed web server using an IoT sensor setup (Benammar et al. 2018; Chiesa et al. 2019).

Several IAQ watching schemes and IoT-based devices have been planned and familiarized; this includes open-source machineries for data processing and transmission (Pitarma et al. 2016). The "electronic nose" (E-nose) is one of the imperative applications of the IoT in IAQ observation method. These are biomimetic kind devices that mimic the functionalities of mammals' olfaction system (Persaud and Dodd 1982). E-nose systems contain four simple modules such as (i) a multisensor array, (ii) software with numerical pattern respect algorithms, (iii) an information processing unit (ANN), and (iv) reference library databases (Freund and Lewis 1995). E-noses can simply monitor various gas concentrations by using chemical sensor collections coupled with the grouping algorithms (Scott et al. 2007). Consequently, E-noses have worldwide concern over excessive curiosity for IAQ monitoring due to its low cost and portability (Chen et al. 2016). Tastan and Gökozan (2019) have proposed this technology system with a series of sensors for monitoring CO_2, CO, PM_{10}, and NO_2 and atmospheric meteorological constraints (temperature and RH) in real time.

Jo et al. (2020) presented a smart air method which collects consistent and precise data for IAQ. Figure 2 represents IoT-based summary for IAQ monitoring.

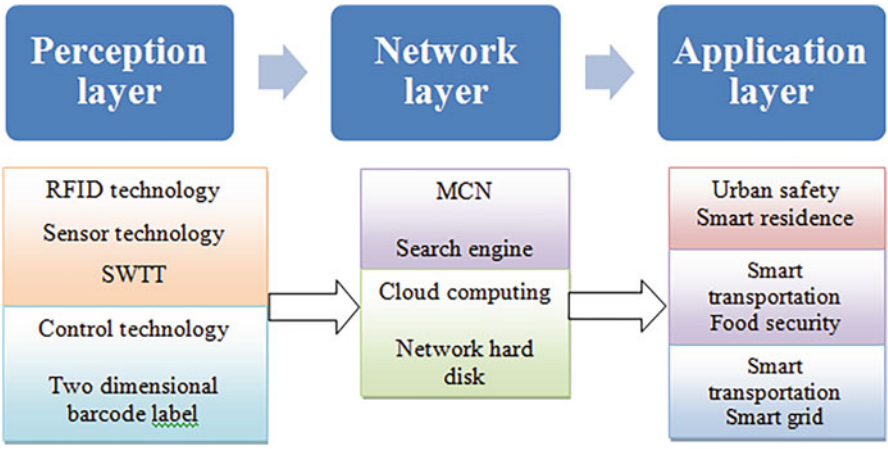

Fig. 2 IoT architecture for measurement of IAQ (Sung and Hsiao 2021). (*RFID* radio frequency identification, *SWTT* short-range wireless transmission technology, *MCN* mobile communication network)

The IoT smart monitoring methods are supposed to be the best trustworthy explanations for IAQ real-time measurement nowadays. Subsequently, researchers and industrial manufacturers are attracted toward this scientific technology system.

This IoT system can monitor a huge number of parameters, even without negotiating scheme enactment. Idrees et al. (2018) and Sivasankari et al. (2017) provided advantage monitoring systems for IAQ with impactful IoT architecture intention. However, few studies have been working on IAQ monitoring prediction systems in the past years. Therefore, in order to start a dependable forecast conclusion system, it is calmer to associate IoT checking methods with ML and profound learning linkages.

Other Technologies

Few researchers have been working on architecture-based technologies which are different from WSN and IoT and found a low performance as compared to IoT real-time monitoring systems. This type of newly developed monitoring system, i.e., C-Air platform, has been designed by Wu et al. (2017), and it is limited only to PM measurements. This is the major weakness of this kind of measurement system because IAQ system is exaggerated by different additional contaminants. Zampolli et al. (2004) have worked on several pollutant measurements, but the study was constrained to the replication atmosphere, and the practical process of such systems still is the real task. Cheng et al. (2014) implemented a forecast model with CAN edge which is again limited to PM. Moreno-Rangel et al. (2018) presented an appreciated study with Foobot monitors and deliberated for numerous real-time IAQ parameter analyses. But again the sensor adjustment was the applicable task to confirm desired performance. Therefore, it is necessary and highly required to do more scientific and technical research in order to overcome this type of issues to develop a smart IAQ measurement system.

Table 6 depicts the details of the air parameters and their working technology and other characteristics.

Indoor Air Quality and Modeling

The studies of chemical constituents and its modeling for diverse uses are extensively amplified due to the interest of scientific research. Air quality forecast (AQF) and evaluation method is help to conclusion creators for refining air quality with respect to community fitness via alleviating the event of severe air pollution periods in town zones and decreasing their connected influence on agriculture, ecosystems, and climate. The progressive tactics in AQF is association and collecting the state-of-the-art models, great resolution release portfolios, satellite explanations, and surface measurements of the applicable chemical components.

This system provides the study and forecast from global- to regional-level air pollution and downscaling for labeled nations, areas, and urban zones.

AQF was mostly executed using realistic methods and statistical models proficient with or fitted to antique air quality and weather data. These approaches have numerous normal problems for air quality forecast system which were reviewed by

Table 6 Air pollutant measurements and their technology

Parameters	Architecture	Communication interface	MCU	Data access	References	Remarks
VOC, PM_{10}, CO, temperature, and RH	IoT	Bluetooth, Wi-Fi, and RF communication module	TI MSP430	Wb server	Kang and Hwang (2016)	Real-time IAQ level indicator designed
RH, temperature, O_3, SO_2, NO_2, CO, $PM_{2.5}$, and PM_{10}	IoT	Zigbee and Wi-Fi ESP8266	ATmega328P	Web-based IBM Watson IoT platform, mobile app	Idrees et al. (2018)	Instinctive standardization method established for the sensor system. It executes power feeding and computer charge enquiry
RH, temperature, NO_2, CO, and smoke	IoT	UART	Raspberry Pi	IP address on the Web	Sivasankari et al. (2017)	Data can be monitored by logging into IP address

Zhang et al. (2012) and Baklanov and Zhang (2020). According to the recent data analyses and the literature reviewed, it was shown that there are various factors such as study gaps, challenges, applications, and advances which should be rectified to overcome the issues occurring in AQF approaches. In addition, the trends and research desires are also required in advance improvements of atmospheric constituent and air quality modeling and prediction.

IAQ models offer a mode to tie evidence about sources and home issues to guess concentration of indoor pollutants. The role of indoor air models in assessing the threat of indoor pollutants is displayed in Fig. 3. Numerous models have been developed to study IAQ which depends on the study's intended uses in the model. The most models used for f IAQ are based on the following points:

- Assessing the inhabitants' coverage to numerous indoor air pollutants
- Assessing the effect of separate sources on air pollutant concentration
- Guessing the influence of distinct sources and IAQ control options on personal exposure

The modeling data used for estimation of indoor exposure to $PM_{2.5}$ accounts for the huge popularity of the mortality load ascribed to whole exposure of $PM_{2.5}$ (Azimi Stephens 2018).

Risk analysis process and assessment for air pollutants are given in Fig. 3, and IAQ model categories are shown in Table 7.

The indoor environment is a very complex vibrant system where the heat and mass transfer and air flow phenomenon take place. In addition, there is a constant interface between the outdoor and the indoor atmosphere, and social involvement through building expertise is strengthened to control the relief situations. Figure 4 depicts the indoor house atmosphere resolute by several sources such as outdoor situation, occupants, and HVAC system. The different origin source over diverse

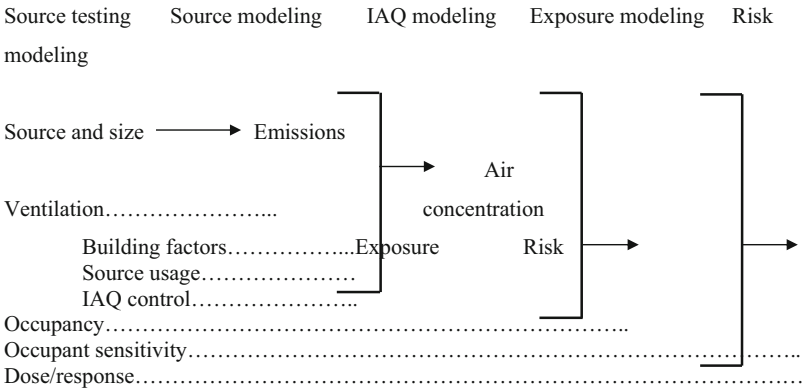

Fig. 3 Indoor air quality modeling system

Table 7 Indoor air model classes

Type of model	General purpose	Example
Statistical	Assessing population exposure	Ott (1984)
Mass balance	Source effect estimation	Sparks (1996), Axley (1988)
Computational fluid dynamics	Assessing close source individual exposure	Brohus (1997)

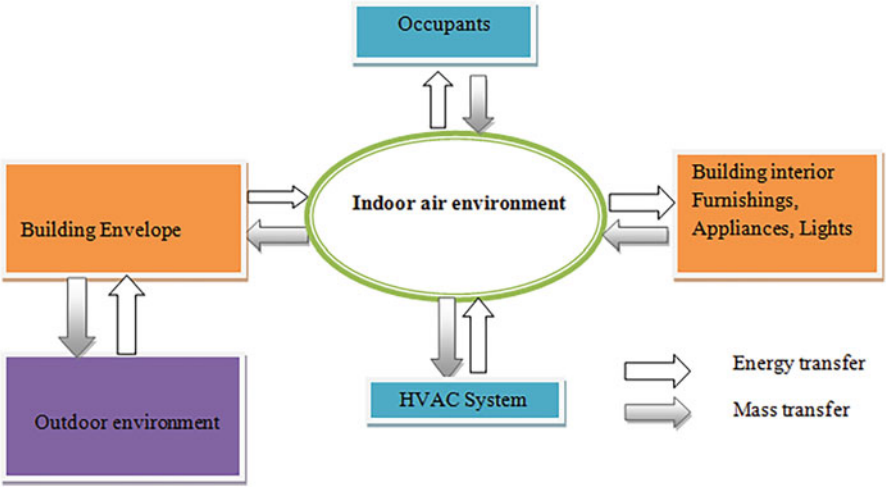

Fig. 4 Indoor air atmosphere as an addition of vibrant methods

heat and mass transfer paths and routes which are managed in the indoor climatic conditions is also presented in Fig. 4.

Various researchers have developed and modified numerous models which are generally used for prediction of air pollutants and their solutions.

Goyal and Khare (2011) have modified the solo partition mass balance-based model which was earlier established by Hayes (1989, 1991) for forecasting PM_{10}, $PM_{2.5}$, and $PM_{1.0}$ concentrations in classrooms of an urban school. The quantitative simulation was conducted by He et al. (2022) using the inclusive air quality model with additions (CAMx) on the vehicle influence of $PM_{2.5}$ between Zhongyuan Urban Agglomeration and adjacent cities in 2017.

The magnitude of IAP can be assessed with arithmetic models, and mass balance equations are used to forecast indoor air pollutant concentrations as the function of outdoor concentrations. This could be used to evaluate penetration rates, indoor source assets, pollutant decline amounts, and mixing causes.

The existence of IAQ has been examined constantly, and different researchers observed greater level of pollutant inside the buildings as compared to outside atmosphere (Lee 1997; Blondeau et al. 2005; Goyal and Khare 2011; Challoner and Gill 2014; Massey et al. 2012). Challoner and Gill (2014) observed that either

certainly or automatically ventilated houses are not active outfits to lessen IAP which derives from outdoor atmospheres. Goyal and Khare (2011) examined numerous limitations surviving in indoor location (PM and CO_2) and also establish that contaminated air imminent from the outdoor environments disturb the inhabitants' well-being and respiratory problems. IAQ control plans could be applied for reducing the contamination threat of COVID-19 through airborne communication (Zhang et al. 2020; Morawska et al. 2018). Refining indoor ventilation methods in homes using air scrubbing tools can develop the IAQ. Wearing a mask in indoors significantly cut the septicity threat of COVID-19. These approaches have been presented and conferred by various researchers (Zhang et al. 2020; Morawska et al. 2018; Dai and Zhao 2020) which was recommended by the WHO (2020). As per the literature, numerous models have been established, but only few have been certified against the observational data.

Statistical Models

In order to appreciate the acquaintance of entire inhabitant system or delicate subpopulations to indoor air pollutants, it is precisely and significantly required to study IAQ system for decreasing the human health risk caused by air pollution.

The charge and difficulty of defining resident's revelation practically have resulted in the improvement of numerous IAQ models. These models deliver the distribution of indoor pollutant exposure to complete inhabitants which depend on the nature of health hazard of disquiet. For instance, overall lifetime disclosure is typically allied with cancer threat, and chronic and annoyance hazards are openly linked with highest exposure.

Database of concentration as a function of time for various atmospheres, residents' movement, and breathing rates are essential for the numerical models. This model generates disclosure and probable quantity circulation using Mono Carlo techniques to associate the database. The basic algorithm is as follows:

1. Select an environment.
2. Sample the concentration database to produce a pollutant concentration.
3. Sample the activity design database to regulate time spent in the environment.
4. Calculate exposure in the environment.
5. Sample the breathing rate record to define breathing rate.
6. Calculate potential dose.
7. Repeat steps 2 through 6 to generate disclosure passage for particular atmosphere.
8. Replicate stage 1 over 7 till all atmosphere of concern are covered.

Mass Balance Models (MBM)

MBMs are designed to assess the source influence on IAQ control in indoor pollutant concentrations. These models are based on the principal of mass conservation which

have single or multiple compartments. The whole house is characterized as a solo planetary, and a building is signified by a sum of allied places for single and multi-compartment models, respectively. Some of these models are discussed in several studies for instances: Hayes (1989) has discussed the multichamber customer disclosure model (MCCEM), and Koontz et al. (1999) and Koontz and Nagda (1991) have estimated individual exposure.

Computation Fluid Dynamics Models

Computation fluid dynamics (CFD) models attempt to pretend the communication of fluids and gases where the surfaces are demarcated by edge circumstances. It provides useful information on the fundamental carriage phenomena in chemical and biochemical manners (heat, momentum, and mass transfer). The revision suggested by Furtaw et al. (1994) may be acceptable for the different conditions. Conversely, this alteration does not provide the desired information to the MBMs in different circumstances. These conditions encompass the necessity to estimate a native concentration instead of an average in room concentration. In such cases, models based on CFD offer the anticipated expectations.

CFD models are dissimilar from MBMs as (1) CFD models envisage air rate and pollutant concentration at separate points in a room instead of the average concentration expected by MBMs and (2) CFD models solve a set of partial differential equations instead of the ordinary equations solved by MBMs. It is revealed that mass balance models provide the necessary information on average air pollutant concentrations, while CFD models deliver the required evidence on air pollutant concentrations fact by fact. This means that CFD models are computationally more expensive than MBMs as are intended for individual computers, while CFD models are planned for small or bigger computers. The diffusion on native or town scales reflects hindrance effects which have been explored by Badeke et al. (2021) using CFD model.

In order to develop IAQ models with precise explanation of chemical and physical methods elaborated in outdoor-indoor air interface as well as treating and carriage of IAP in homes, it is desired to accurately report construction between the outdoor air quality and IAP sources. Further, innovative modeling is desirable for air-surface interfaces aiming releases and basins on diverse surfaces such as ventilation setup (Liu et al. 2013) and authentication of the indoor air models with measurements in environments.

Several researchers have reported comprehensive thorough reviews and observed the expansion and principles of worldwide and local 3D AQF models. They also acknowledged areas of enhancement in climatic predictions, chemical ideas, and model actions of atmospheric physical, dynamic, and chemical processes (Bai et al. 2018; Baklanov and Zhang 2020).

Improvement of Indoor Air Quality

There are different simple tactics to recover IAQ, and they provide significant information on the basis of recent systematic research and nominal appreciative of the indoor air quality matters at local as well as global level. There are different methods such as source control, ventilation improvement system, air cleaner techniques, and potted plants that have been used for improving the indoor air (Zhang 2020).

Zhang (2020) has applied these different scientific strategies in existing ventilation principles for refining and standard IAQ for comfortable lifestyle. These approaches had the possibility to advance IAQ by a factor of 200 outside the present standard, which would surge inhabitant fulfillment from ~80% to 100% (Fanger (2006).

Source Control Methods

The source controlling is the major and utmost imperative method among all IAQ approaches. It is the best effective method to eliminate the individual sources of air pollutants and reduce their emissions from the indoor environment. Few sources contain asbestos that could be bounded, and other sources such as gas ovens can be accustomed to decline the discharge amount of the pollutants. The source control is the price-proficient tactic for defending IAQ than growing ventilation. Native air dissipation is an alternative active source control plan which has the prospective to expand the air quality and cut the cross impurity by a factor of 1.4–10. This phenomenon depends on the local divider pattern, deplete screen, and stream amount (Dygert and Dang 2012).

Ventilation Improvement System

It is another alternative method to dropping the concentrations of air pollutants in homes by increasing the quantity of outdoor air to indoor environments. In maximum household warming and chilling methods containing forced air heating arrangements and do not automatically carry new air into the household. An advanced strategy in new home construction has started the mechanical systems that transport outdoor air into the indoor environments.

Few of these design homes contain energy-able heat retrieval ventilators (air-to-air heat exchangers). Ventilation and covering can help to control indoor temperatures and remove/dilute indoor contaminants imminent from diverse indoor sources. This kind of system decreases the level of impurities and recovers IAQ.

Maximum housing forced air heating, and air conditioning systems do not get air from the outside into the house by instinctively and penetration. Hence, the ventilation is used for bringing an adequate quantity of fresh outdoor air and delivering it

to the inhalation region and efficiently diluting the deteriorated air in indoor atmospheres. The automatically ventilated schoolrooms and workplaces in USA zones have ~20% of their air delivered from outdoors, and the respite air is recirculated (Zang et al. 2020). This was completed to except warming and freezing energy while preserving tolerable IAQ levels. To lessen the threat of the SARS CoV-2 virus septicity, the outdoor ventilation degree should be enlarged to the extreme operative volume of the house ventilation method (Zang et al. 2020). Movement ventilation is a well method for renewed air distribution and dipping the pollutants concentration in the occupied zone by a factor of 1.2–2 released from dwellers (Chen and Glicksman 2003).

Air Cleaner Techniques for Improving IAQ

Air cleaning approaches comprise air percolation/cleansing in the outdoor source, recirculated/assorted air duct, and at personal level. There are several kinds and sizes of air domestics in the market reaching from reasonable table top models to classy and posh full household methods. Out of them, some air cleaners are extremely active for particle elimination, while others (table top models) are very fewer capable. The success of an air cleaner depends on how well air cleaner gathers the air pollutants from indoor atmospheres and how much air these air cleaners pull through the sieving component. Zhao et al. (2020) conveyed a series of clean air carriage degree from 170 to 800 m^3/h with a median cost of $361 and electrical feeding of 56 W per unit. The results were consistent with an earlier laboratory study done by Chen et al. (2006) in which six movable air cleaners were tried for particulate and VOC deduction enactment.

Among the various interventions and tools, air purifiers are required to recover IAQ which is usually engaged in households, clinics, and workplaces. In order to control discharge sources of air pollutants in indoor environments, expansion of air refining tools and optimization of ventilation systems are the key slants to increase IAQ (Huang et al. 2016). Among them, the exploration of original skills to sieve and sanitize IAP has concerned an excessive consideration over the last era. Adsorption machineries (using carbon found in filter media) are the leading technologies for removing gaseous pollutants (Haghighat et al. 2008). There are several inhabited and profitable HVAC filters which have been established and used in several houses (domestic, shopping mall, schools, and workplace buildings) for removing numerous toxic contaminants (O_3, aerosols, VOCs, and PMs) (Zhao et al. 2007). The HVAC strainers can be intended in various shapes like filter cassettes made of flat sheet filter media and containers implanted by refined triggered carbon (Ligotski et al. 2019). Conversely, this expertise has showed a substantial restriction in terms of the adsorption ability due to its stability adsorption capability which ominously diminished the pollutants (Carter et al. 2011). Dubey et al.'s (2021) study has shown short-term use of air purifiers to remove the concentration levels of air pollutants from indoor environment of Agra, India, and found that some air purifiers have good efficiency for removing indoor air pollutants.

Potted Plants

Some studies have been suggested that potted plants have been reducing the ranks of chemicals in laboratory tests and indoor air pollutants (Pipal et al. 2012). However, a reasonable number of potted plants eliminate substantial amounts of air contaminants in indoor environments (households and workplaces).

Smart Home for IAQ Control

The term "smart home" is related with numerous home robotic methods (home utilizations, transportations, and entertaining electronics). In latest years, home robotics and the smart home mainly have fascinated excessive concern. It is governable via smartphone solicitations and IoT-based wearable devices which is an energetic part of smart cities. Its groundwork was the projected home energy management system (HEMS) (Zhou et al. 2016; Patil et al. 2019). In order to improve the energy usage effectiveness and elevate ventilation technology, the smart home was settled (Paetz et al. 2012; Missaoui et al. 2014; Chenari et al. 2016).

The future growth of smart home skill is probable to keep in two broad ways: (i) the control of ventilation, usages for a decrease of energy charges, and improvement of human fitness, care, and comfortability and (ii) support to old and spiked persons by the use of robotic machines and smart sensor tools (Schieweck et al. 2018). In addition, the smart home technology deals with chances to cut bad influences of IAP on human well-being. Kraakman et al. (2021) grant the latest developments in biological system for indoor air purifying methods as a "green" substitute to physicochemical methods, with prominence on present trials and occasions. This method can compromise for filtering IAQ, erection energy price reserves, and enlargements on indoor ease and well-being. This system has major challenges which are required to correctly discourse the bioavailability of air pollutant at low concentrations, assurance microbial safety, and integrate CO_2 removal.

Way Forward

The implementation of real-time IAQ monitoring system is preferred than an organized laboratory atmosphere or on imitation methods in real scenarios. With ease to control the adverse impact of IAP, the development of prediction systems remains a crucial alarm for forthcoming studies.

The core zones of research work for future researchers can be summarized as follows:

- An IAQ monitoring system should be developed which can work more efficiently in real-time measurements instead of replicated/laboratory built observations.
- Designing a cost-effective IAQ observing method to offer a harmless result for improved active locations in rural areas.

- Working on IAQ forecast methods so that suitable defensive trials can be followed on time.
- Designing a power-proficient and strong system for simple monitoring situations in urban and rural zones.
- Mounting novel effective method which can produce direct sirens to the users via email and SMS when IAP reached to threshold levels.
- The mobile app-based checking schemes should be developed which can be functioned by non-technology awareness persons as well.
- Increasing rapid aware systems with probable defensive actions such as switch on/off air conditioner, open/close frames, and check gas leakage. This will help to monitor people for strong control solutions of air pollutants in the active atmosphere.

Conclusion

This chapter entails about the IAQ scenario and their related effects of exposure to airborne and gas compounds which were considered as harmful products in different studies. The following lasting movements could help to endorse auspicious IAQ in indoor environments: (i) developing IAQ guidelines for significant air pollutants and giving credits for their decline and prevention; (ii) evolving an IAQ index or metrics; (iii) presenting acknowledgments for testing of releasing from goods used in indoor environments; (iv) conducting IAQ measurements inside the building environment; (v) accounting for cross-category effects of elucidations in other groups of IAQ system; (vi) steady recertification based on IAQ ranks, and (vii), usually, evolving incentives to recover IAQ that will profit to inhabitants, employers, and holders staying in indoor environments which create greater obligation of IAQ in greenhouse assignments. Overall, this chapter is an inclusive scientific document which covers the entire aspects of IAQ system such as measurement techniques, characterization methods, and source apportionment tools along with health risk assessment methods. In addition, this chapter is introducing the new scientific technology and its applications which are used to develop smart tools and devices for improving IAQ system. This will benefit and be very useful for researchers, policymakers, and government's agencies for developing a new strategy and policies for better IAQ system.

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Instruments for Monitoring Air Pollution and Air Quality

67

S. K. Gupta and Balbir Singh

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Abstract

Due to rapid increase in industrialization, motorized transport, and urbanization, the emissions from these activities are adversely impacting the air quality, and consequently human health, crops, vegetation, forests, wildlife, and properties are

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under serious threat. There is recognition of having good air quality and importance of its regular monitoring extensively. In industrialized countries particularly in the USA, Japan, and Germany, monitoring of industrial emissions has almost completely shifted to use of automatic continuous monitoring instruments based on some specific physicochemical or electrochemical properties of the pollutants. Such instruments require very specific sensors for each pollutant, calibration procedures needing calibration gases of exactly specified and known composition, and dust-free and air-conditioned environment. Such systems become almost unaffordable, from the capital cost point of view, for industrial units in developing countries, particularly for medium- and small-scale units. Even more difficult to tackle are problems of nonavailability of instrument spares and calibration gases in the local markets and also of competent manpower for proper operation and maintenance of these sophisticated systems. Experience in India shows that wherever such systems were installed, they remain inoperative almost from the very start.

The USEPA had developed simple concepts for monitoring of source emissions and also ambient air quality. These may have become obsolete in industrialized countries but still remain suitable for adaptation in developing countries. Even here, instruments manufactured in the USA, Japan, Germany, etc. often tend to make the instrument complex and costly, again creating problems of operation, maintenance, spares, etc., besides the significantly high costs. Instruments described here are based on simple USEPA concepts and are simple for use and maintain in field conditions of warm, tropical climate of developing countries. These instruments are also cost-effective.

Keywords

Air pollution · Air quality · Appropriate instruments · Manual samplers

Introduction

Growing air pollution is a matter of big concern now. Combustion from vehicles and industries, mining, and construction activities are major sources of pollution and chiefly responsible for deteriorating air quality which is adversely impacting human health, plants and vegetation, agriculture, wildlife, materials, and all other components of our environment. Impacts of some pollutants are short-term and of others are long-term. Recent studies have established that some pollutants are also responsible for climate change and reduced visibility. People working in mining areas are exposed to heavy concentrations of pollutants and suffer from respiratory disorders, bronchitis, asthma, and other deadly diseases like silicosis. Our residences and offices are increasingly becoming air-conditioned and poorly ventilated leading to buildup of carbon dioxide levels as the day progresses. Extensive use of paints and laminates adds VOCs in the indoor air. Instances of "sick building syndromes" are increasingly being reported in cities. Due to increased vehicular mobility and use of

loud musical devices, ambience is becoming noisy and our hearing is getting impaired. Local meteorology also plays a great role in either buildup or dispersion of pollutants in ambient air. Thus, it is evident that good air is essential for survival of living organisms. Regulatory bodies in all countries have laid down limits for emissions from various polluting sources and also standards for good air quality to protect the health of all human beings. Therefore, regular monitoring of emissions from various sources and ambient air quality as well as indoor air, noise, and local meteorology is essential and a necessity. There is a need of appropriate instruments for monitoring air pollutants and air quality. This chapter shall describe such instruments in detail.

Monitoring Air Pollution and Air Quality

To assess pollution loads, one often needs to monitor the emissions into the atmosphere from various sources that are likely to influence air quality in the area. The emissions could broadly be categorized into two groups: (a) formal emissions or discharges from a stack or duct which can be reliably sampled and quantified and (b) fugitive emissions. Formal emission could again be either from stationary sources existing at various heights above ground level like stacks, ducts, vents, etc. or from mobile sources, e.g., vehicular exhausts. Emissions from stationery sources are sampled as per the stack sampling technique and modeled as "point sources" of pollution. Emissions from mobile sources are monitored by various methods briefly described herein after and modeled as "line sources." Fugitive emissions cannot be very exactly quantified and are only approximately assessed. These are modeled as "area sources of pollution." Emissions from a mobile source like a vehicle vary not only from vehicle to vehicle; its state of loading also depends on a number of factors like the speed of movement, slope of the road, condition of the road surface, condition of the tires, and even the speed and direction of wind, etc. Also the pollution caused by a moving vehicle should not include only the emissions from exhaust or tailpipe but also the dust lifted up from the road surface and tire-wear and tear and also vapors escaped from fuel tank, piping, crankcases, etc. Government's transport departments usually mandate the Pollution Under Control (PUC) certificate for the vehicles. The PUC test assesses the emission of a vehicle and checks whether it follows the standard pollution regulations or not.

Air pollution and air quality measurements can broadly be described under the following categories:

(i) Source emissions
(ii) Ambient air quality
(iii) Personal exposure and work zone air quality
(iv) Indoor air quality
(v) Local meteorology
(vi) Noise

There are different techniques and instruments to measure any given pollutant. One should examine the options to determine which techniques and instruments are most appropriate, taking into account the main uses of the monitoring data, initial investment costs for equipment, operating costs, ease of operations, and reliability of systems. Considering this we would study simple methods and instruments to measure these categories.

Source Emission Monitoring

Generally, two types of techniques are followed for monitoring of emissions from stationary sources or stacks: extractive type manual sampling and continuous emission monitoring.

Extractive Type Manual Sampling

In this technique primary methods of sampling and analysis are followed wherein a stack gas sample is extracted from the stack/duct and dust particles present in the stack gas are collected on a filter media, their quantities are measured gravimetrically from the weight of dust collected on the filter media and results are calculated after dividing the weight of dust collected by the volume of gas sampled. Gases present in stack emissions are monitored by absorbing the filtered stack gas in suitable chemical reagents and subsequently analyzing them in the laboratory using standard wet chemical methods to know their concentrations.

The most common and widely used method for stack emission monitoring world over is extractive or grab sampling technique. In this technique, instrument used for sample collection is not restricted to a specific pollutant, but a variety of pollutants can be sampled with the same instrument. A representative sample is drawn from different points selected across the entire cross-sectional area of the stack or duct. These points are called traverse points (US EPA 1994). A typical view of selection of traverse points inside a stack or duct is shown in Figs. 1 and 2.

Instruments based on the primary techniques are designed to be portable to allow collection of sample of emissions from multiple stacks. In some cases they require installation of a monorail system on the chimney at the monitoring location. Instruments installed on monorails are usually heavy as the gas-filtering media or particulate collection media is placed outside the chimney in a heated box which makes the sampler probe heavy and, therefore, needs a monorail. In portable instruments filtering media is kept inside the chimney during monitoring; it makes the handling of sampling probe easy and convenient. The following photographs show a sampler mounted on a monorail as per USEPA Method 5(US EPA e-CFR40 2020) and a portable sampler with in-stack filtration as per USEPA Method 17(US EPA e-CFR40 2017). The schematic diagrams of sampling trains for both techniques are shown in Figs. 3 and 4.

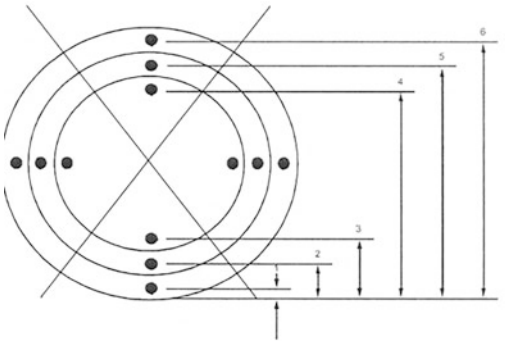

Fig. 1 Traverse points inside a stack

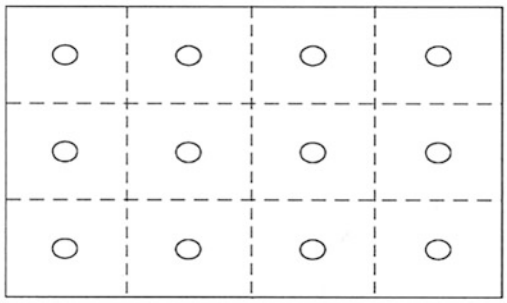

Fig. 2 Traverse points inside a duct

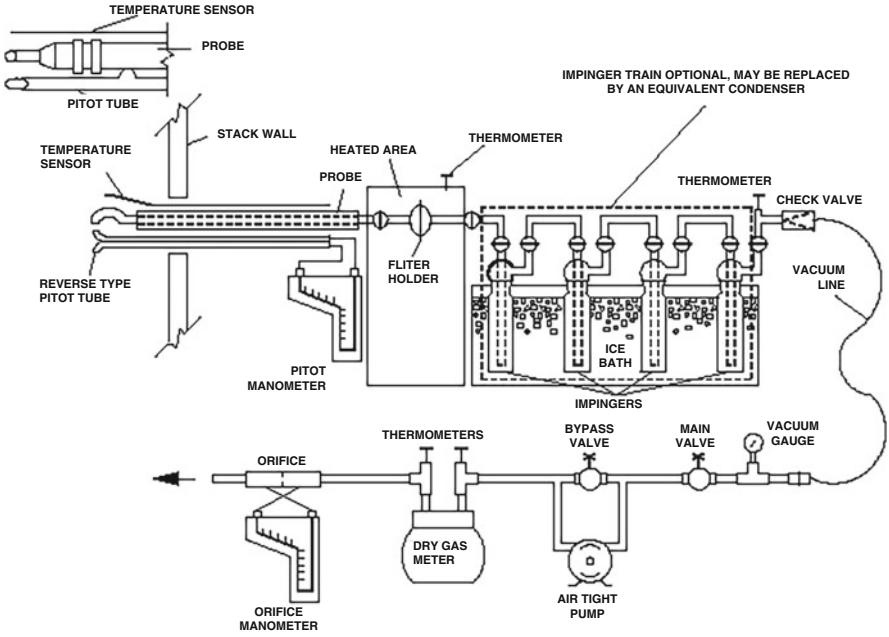

Fig. 3 Sampling train for EPA Method 5

Monitoring with a sampler installed on a monorail (as per USEPA Method 5)

Monitoring with a manual sampler as per USEPA Method 17 (in-stack filtration)

A conventional stack sampler consists of the following major components:

1. Pitot tube.
2. Thermocouple.
3. Probe set consisting of set of nozzles, thimble holder, probe pipes, and hoses.
4. Suction pump.
5. Instrument panel consisting of differential pressure meter, pyrometer, rotameters, stopwatch, and set of glass impingers mounted in a box with provision to keep ice.
6. Tripod stand and associated consumables like filtration thimbles, absorbing reagents, etc. A typical stack sampler is shown in Fig. 5.

Sampling Port Holes and Platform

Construction of a sampling port hole for drawing the stack gas sample and a platform for accommodating the stack sampler and the monitoring team on the stack are the

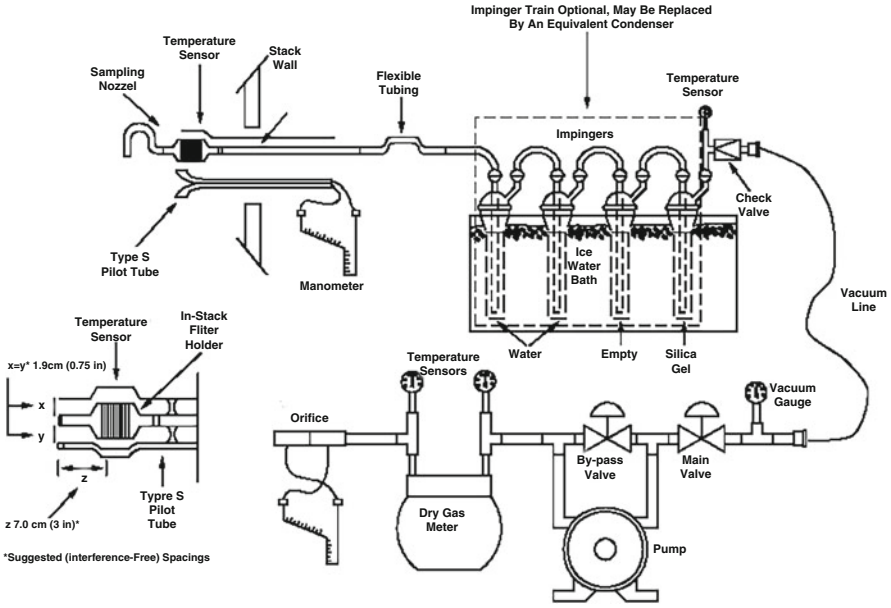

Fig. 4 Sampling train for EPA Method 17

Fig. 5 A stack sampler

prerequirements of the sampling. Sampling port hole of 100 mm internal diameter should be ideally located at more than 8 times of the stack diameter (8D) downstream from any merging point, bend or tapering and 2 times of stack diameter (2D) upstream from the exit point, bend, or any other flow disturbance

point (Guidelines on methodologies for source emission monitoring 2013–2014). Sampling platform should be about 1.5 m below the port hole, and a power supply point should be available for operation of monitoring instruments. A schematic diagram of the location of sampling port hole and the platform is shown in Fig. 6.

An important aspect of stack emission monitoring is that sampling for the particulate matter (PM) is carried out at isokinetic conditions, i.e., samples from the stacks or ducts are extracted at the same velocity which exists inside the stack or duct. And velocity across the entire cross-sectional area of a stack or duct may not be same; therefore, velocity is measured at various selected traverse points, and isokinetic flow rates are calculated at each traverse point. Samples are then drawn at each traverse point at the calculated flow rates. Stack temperature, stack pressure, stack gas moisture content, and the stack gas density are also measured and considered in calculating the isokinetic flow rates.

The first step in stack emission monitoring is to know the stack gas velocity. For this differential pressure is measured at selected traverse points across the cross-sectional area of the stack with the help of a pitot tube and a manometer available with the stack sampler. Stack gas temperature, stack pressure, stack gas density, and moisture content are also measured, and stack gas velocity is then calculated using the following equation (Emission Regulations 1985):

$$V = K\left(\frac{2 \times g \times \Delta P \times dm \times Ts}{da \times Ta}\right)^{1/2}$$

where

V = Stack gas velocity in m/s
K = Pitot calibration constant
g = Gravitational constant (9.81 m/s^2)
Δ P = Differential pressure across the stack in meters
dm = Density of water in Kg/m^3 (1000 kg/m^3)
da = Density of stack gas in Kg/m^3 (for air it is 1.25 kg/m^3 at 25 °C)
Ta = 273 + 25 in °K
Ts = Stack gas temperature in °K

During sampling, stack gas enters the sampler through the nozzle; therefore, the isokinetic flow rate through the nozzle needs to be set at the sampler which is calculated after multiplying the stack gas velocity with the cross-sectional area of the nozzle used. Usually a rotameter is used to set the flow rate on the stack sampler; therefore, isokinetic flow rate is derived after selection of a suitable nozzle so that the calculated flow rate falls within the measurement range of the rotameter of the sampler. Usually a set of 3–4 nozzles having different cross-sectional areas are available with the sampler. A schematic diagram of a nozzle, thimble holder, and sampling probe is shown in Fig. 7.

Temperature of the stack gas is measured with the help of a thermocouple and pyrometer available with the stack sampler. Moisture content in the stack gas is

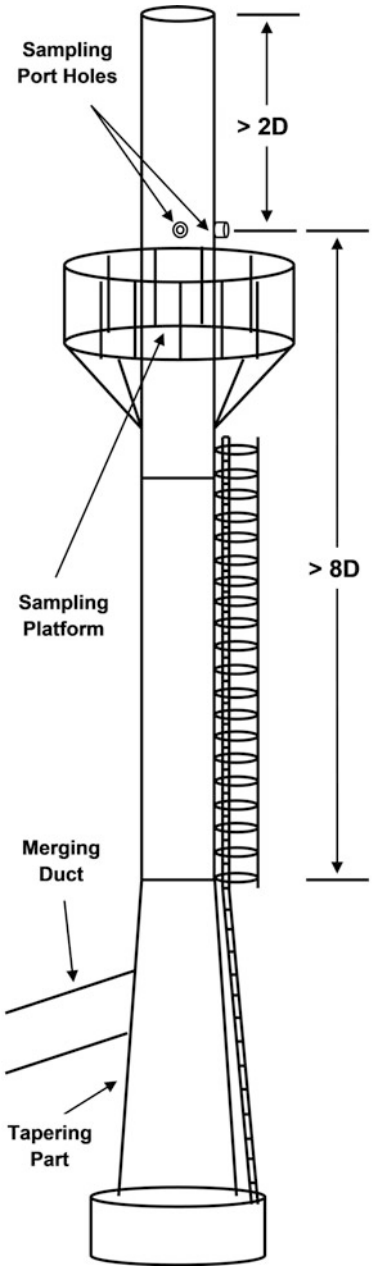

Fig. 6 Location of sampling port hole and platform

determined either by drawing a known volume of stack gas and condensing the water vapors present in the stack gas in an ice-cold water trap or by absorbing it in the activated silica gel bottle. To collect the moisture in ice-cold water trap, a known

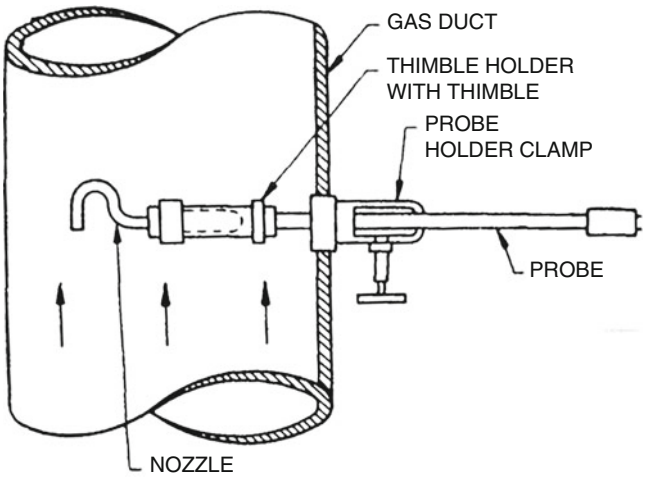

Fig. 7 Schematic diagram of a nozzle, thimble holder, and sampling probe

volume of ice-cold water is taken in a glass impinger maintained at low temperature by ice, and a known volume of gas is passed through it at a low flow rate. The water vapors present in the stack gas get condensed in the ice-cold water, and consequently, there is an increase in the volume of water from the initial volume of water taken in the impinger. Moisture content is then determined from the volume of water condensed and volume of gas passed through it. Another method for determination of moisture in stack gas is by absorbing stack gas moisture in activated silica gel. In this method, a known volume of stack gas is passed through the granular activated silica gel taken in a glass bottle. The amount of moisture absorbed in the silica gel is determined from the difference in weight of the silica gel bottle, after sampling and before sampling. Moisture content is then calculated from the weight of water absorbed by silica gel and volume of gas passed through it.

Stack gas density is determined by drawing a stack gas sample in a gas bottle or in a "Tedlar" bag and analyzing it on an Orsat apparatus or on a gas chromatograph (GC). Gas chromatographic analysis involves costly setup and highly skilled manpower, whereas an Orsat apparatus is a simple apparatus. It consists of a graduated measuring burette, an aspirating bottle, and three double reagent pipettes to absorb CO_2, O_2, and CO. Measuring burette is connected to aspirating bottle by means of rubber tube. The first reagent pipette next to measuring burette contains the saturated KOH solution to absorb CO_2. The second pipette contains alkaline solution of pyrogallic acid to absorb O_2, and the third pipette contains acidic solution of cuprous chloride to absorb CO. Each pipette is made of two pipettes joined together by a glass tube at the bottom as shown in Fig. 9. Each front pipette is provided with the number of glass tubes inside to increase the wetted surface area to accelerate the gas absorption. Initially, 100 ml of stack gas taken either in a gas bottle or in a Tedlar bag is transferred to the measuring burette, and it is brought in contact of KOH solution in the first pipette by raising the level of aspirating bottle and the process is repeated till absorption of CO_2 is completed. The same procedure is repeated with other two

pipettes for absorption of O_2 and CO to find the percentage analysis of O_2 and CO, respectively, in the flue gas. The remainder is N_2 gas. Thus, Orsat apparatus gives analysis of the dry gas (IS 13270:1992 1992). A typical Tedlar bag and an Orsat apparatus are shown in Figs. 8 and 9, respectively.

Continuous Emission Monitoring

In this technique, specific characteristics/properties of dust particles and gases are used for their detection and measurement. Since secondary properties are used for detection, these methods fall under secondary methods. Monitors or systems used here are known as continuous emission monitoring systems (CEMS) or online continuous emission monitoring systems (OCEMS). CEMS or OCEMS are usually permanently installed on the stacks, and in some cases stack gas is extracted through a pipeline and fed to the analyzer located at some nearby location under controlled environmental conditions.

Most monitors use one or more of the light interaction properties with the particles. When light falls on the surface of a particle, it may be absorbed, reflected, refracted, or diffracted by the particle fully or partly. Any one or more of the light interaction phenomena occurring are used to measure the concentrations of the particles. For this a suitable detector is used to measure the intensity of light received after their interaction with the particles. In the process detector also measures the intensity of reference beam. Change in intensity of the sample beam and reference beam is used for computation of the concentration of particles. In such monitors, usually, opacity or transmissivity through the cross section of the stack or duct is measured. A typical example of interaction of light with particle is shown in Fig. 10.

Continuous emission monitoring systems can be categorized into two types (Guidelines for Continuous Emission Monitoring Systems 2018):

1. Extractive type
2. In situ type

Fig. 8 A Tedler bag

Fig. 9 An Orsat apparatus

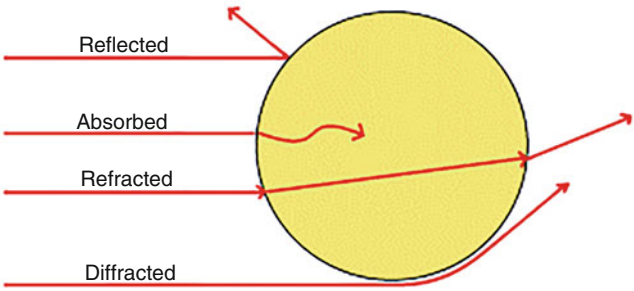

Fig. 10 Light interaction modes with particle

Extractive type systems draw flue gas from the stack and transport it continuously to the analyzer. These systems perform analysis at a single point, whereas in situ type systems carry out most of their operations in the stack itself. These can further be categorized as point in situ monitors and cross duct or path monitors. Point in situ systems carry out analysis at a single point in the stack, and path monitors carry out analysis over the entire stack or duct diameter. Again, there are two basic types of cross duct or path monitors:

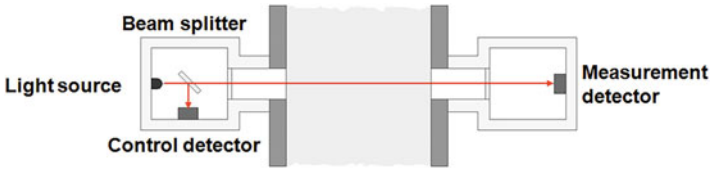

Fig. 11 Schematic diagram of a single-pass opacity monitor

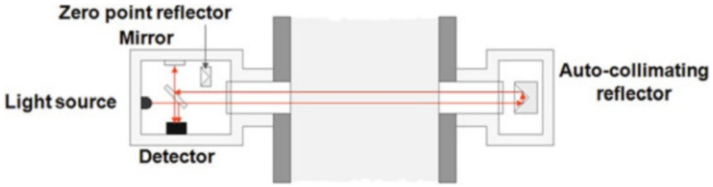

Fig. 12 Schematic diagram of a double-pass opacity monitor

1. Single-pass monitors
2. Double-pass monitors

In a single-pass monitor, one light beam from the source is passed to receiver or detector placed on the opposite side of the stack, whereas in a double-pass monitor, the beam is reflected back across the stack to the detector located on the side of light source. Double-pass monitors are more sensitive as compared to a single-pass monitor. Schematic diagram of a single-pass and double-pass opacity monitors is depicted in Figs. 11 and 12.

Again, all these monitors have two configurations: single-beam configuration and double-beam configuration. Single-beam configuration is simple and has a single beam of light passing across the stack, whereas in a double-beam configuration, the light emitted from the source is internally split into two beams, one becomes measurement beam and another becomes reference beam. Double-beam configurations are more sensitive as compared to the single beam.

Instruments with nonoptical technologies are also popular and used for measurement of particulate matter in stack emissions. Probe electrification technique is one of them. In this technique electrical current produced by particles interacting with a grounded rod protruding across the stack or duct is measured and correlated to dust concentration. Schematic diagram of probe electrification instrument is depicted in Fig. 13.

Presence of moisture in stack gases affects the monitoring response of all in situ technologies to an extent. In these instances, extractive systems give more reliable results. Beta attenuation is one of the widely used technologies for dust monitoring. In this technique moving gas stream is sampled and the particulate is collected onto a filter. The filter is advanced periodically into a measurement chamber, where radioactive beta particles are passed through the sample and the amount of beta particles

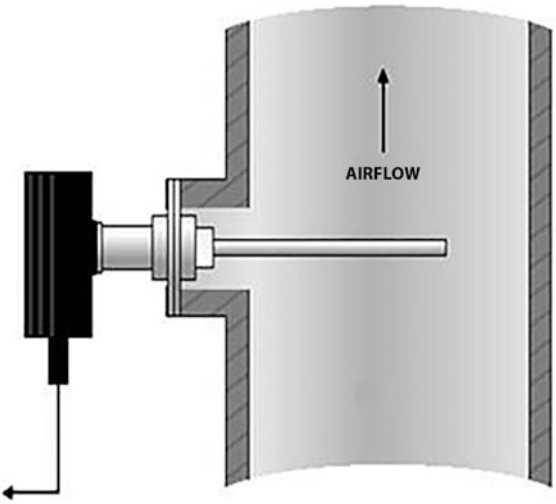

Fig. 13 Schematic diagram of a probe electrification instrument

transmitted through the sample is measured. The advantage of this technique is that the absorption of radioactivity is not significantly affected by the type of particle. Schematic diagram of a beta attenuation monitor is depicted in Fig. 14.

All the continuous particulate matter monitoring system (PM-CEMS) are calibrated against isokinetic sampling at different operational loads using a manual stack sampler, whereas gaseous analyzers are calibrated against standard span gases.

Another category of instruments used for obtaining information on stack gas constituents and in turn stack gas density are the flue gas analyzers. These analyzers are mostly sensor-based instruments and give quick information on the stack gas constituents. This data can be directly used for calculation of stack gas density required for computation of stack velocity and isokinetic flow rate. These instruments are microprocessor based and may also provide additional information like combustion efficiencies of the processes and also the presence of other pollutants in stack gases. Usually a specific type of sensor is used for detection of a particular constituent. A typical flue gas analyzer is shown in Fig. 15.

Ambient Air Quality Monitoring

Before planning air quality monitoring, it is necessary to have a clear idea of the monitoring objectives, since these determine the selection of appropriate pollutants and measuring methods, as well as the required level of accuracy and precision. Objectives may include the assessment of quality of air that a community or ecosystem is being exposed to and hence developing scientific understanding of

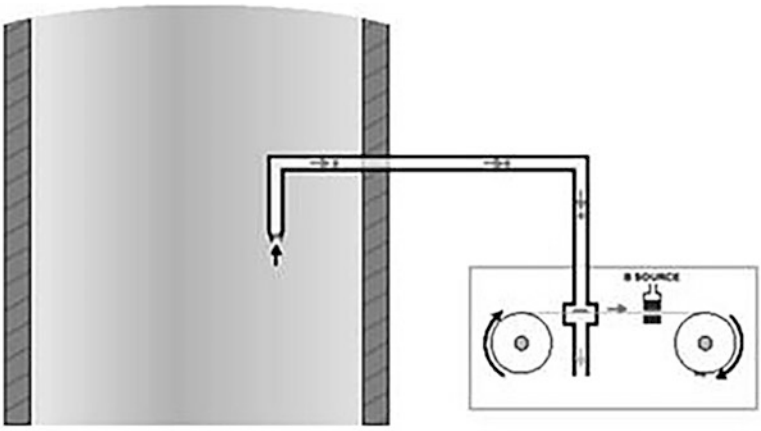

Fig. 14 Schematic diagram of a beta attenuation monitor

Fig. 15 A typical flue gas analyzer

causes; generation, impact, status, and trends; creating awareness; and for the purpose of setting standards/limits defining need for likely action or for evidence for punitive action.

More technical objectives may even include the identification and apportionment of pollution sources, land use planning, determination of spatial and temporal patterns of exposure, or identification of long-term trends. The long-term trends allow us to detect any deterioration in air quality arising from residential/industrial/economic development and/or the effectiveness of pollution control measures taken. In short, monitoring data are generally intended for use in developing strategies for management of air quality in a city, locality, region, or national level.

Air quality surveys are generally concerned with temporal and spatial variation of the concentrations of selected pollutants. The type and level of air pollution is a major factor in the quality of life of the community.

The data must also be of acceptable quality to satisfy the intended use, since proper quality assurance of data is the keystone to ensure realistic interpretation for correct decision-making. Obviously it would not be desirable to mix up networks with substantially different overall design, using different monitoring techniques or using varying levels of skilled manpower.

Air quality assessment is made using monitoring systems composed of carefully selected air quality monitoring techniques suitable for the application involved, and representative of the area. The classical pollutants measured at an ambient air monitoring station include suspended particulate matter (SPM), respirable suspended particulate matter (RSPM) or PM_{10}, fine particulate matter or $PM_{2.5}$, sulfur dioxide, nitrogen oxides, ozone, and carbon monoxide. However, these days many more other constituents are being considered as pollutants, and their measurement is also being carried out. Various methodical approaches, technical skills, and instruments to estimate the air pollutants have been developed and are still being developed. Broadly, monitoring systems available at present include manual samplers, online systems, and sensor-based instruments including remote sensing.

In the following sections, we have tried to briefly describe these techniques one by one. However, it may be stated that the choice of appropriate monitoring technique for any air quality assessment exercise will depend largely on its objectives, local resource constrains, and the availability of skilled manpower. There is a clear trade-off between instrument cost, complexity, reliability, and performance. More expensive systems, although they provide increasingly refined data, often tend to be less reliable and more difficult to use.

The simplest method that meets the specified monitoring objectives should be selected. Inappropriate, excessively complex, or failure-prone equipment can severely disrupt overall performance limiting the usefulness of data gathered. Automatic analyzers are required if short-term concentration peaks or the resulting acute health, amenity, or ecosystem effects are to be investigated, but manual samplers will often be adequate if long-term baseline levels or trends are of importance. Appropriate techniques and samplers for monitoring of particulates and gaseous pollutants in the ambient air are described here.

The dust found in the atmosphere is in the form of suspended dust particles and known as SPM. These particulates are small in size, usually below 100 microns, and remain suspended in the atmosphere for significant periods of time. Particles of size 10 micron and below present in the atmosphere are known as RSPM, and further

smaller particles of size 2.5 micron and below are termed as fine particulates. Fine particulates are usually more harmful than the bigger size of particles as these can penetrate deep into the lungs. Our respiratory tract retains particulates of varying sizes at different levels. Larger size particles are retained in the upper respiratory tract and only smaller ones can go down deeper.

Meteorological conditions such as wind speed, wind direction, temperature, relative humidity, rainfall, and atmospheric stability conditions play a very important role in the dispersion of the pollutants generated at any place, and therefore information on local meteorology need to be collected along with the monitoring of dust particles and gaseous pollutants for better understanding of the air quality. Broadly, four different techniques are used for monitoring ambient air quality:

(i) Manual samplers
(ii) Continuous ambient air quality monitoring systems
(iii) Sensor-based systems
(iv) Satellite-based systems

Manual Samplers

The basic technique for monitoring particulates in ambient air is filtration technique using manual samplers followed by gravimetric analysis. Among the particulates, coarser particles present in the atmosphere get settled quickly at the ground, whereas particulates of size 100 micron and below remain suspended in the air for appreciable time durations and are known as SPM or TSPM. SPM is measured using a high-volume air sampler wherein suspended dust particulates present in the air are directly collected on a filer paper, whereas particulates in other size ranges such as PM_{10}, $PM_{2.5}$, etc. are first separated in required size range and then collected onto a filter paper by passing a known volume of air through the filter paper using a particulate sampler, and the amount of dust collected on the filter paper is estimated from the difference in weight of the filter paper after the sampling and weight of filer paper before sampling. The common techniques for physical separation of dust particles are either an impactor or a cyclone. All such samplers have a particle separation device, an air flow meter for adjusting the required flow rate of air and a time totalizer for recording the total time of run. Volume of air sampled is then calculated by multiplying the average air flow rate with the total time of sampling. Concentrations of particulates present in the air are then calculated from the total weight of dust collected on the filter paper and dividing it by total volume of air sampled. Results are then reported in units of microgram per meter cube.

Instruments for Monitoring of SPM

High-volume air sampler (US EPA 40 CFR n.d.-a) is the most frequently used sampling device in air quality monitoring, so much so that they are often called

"Work horses of air quality management." The basic principle is sucking air at a constant prefixed rate for a known time through a filter (usually a glass microfiber filter) and determining the initial and post-sampling filter weights after appropriate conditioning of the filters. It is used for monitoring of SPM in air. The design of the air inlet of high-volume air sampler is unlikely to collect airborne particles with diameters greater than 100 micrometers (μm) in diameter. It is measured by passing air at a high flow rate of 0.8–1.2 cubic meters per minute through an $8'' \times 10''$ high efficiency glass fiber filter paper which retains the particles. Filter paper is installed over a wire mesh and tightened securely with the help of four screws. A detachable gable roof is provided to prevent the heavier settleable dust particles from reaching the filter. A water manometer and a time totalizer are provided for measurement of air flow rate and the duration of sampling. The volume of air sampled is determined from the average flow rate and duration of sampling, while the amount of particulates collected is determined by measuring the change in weight of the filter paper as a consequence of the sampling. A timer is also available in the instrument for programming the sampling schedule. If required, particulate matter retained on the filter can be analyzed to determine the concentration of metals such as lead or any other metals. A typical high-volume air sampler and an illustration of installation of a filter paper are shown in Figs. 16 and 17. These instruments are available with an attachment for sampling of gaseous pollutants which enables simultaneous sampling of gases like SO_2, NOx, etc.

Instruments for Monitoring of RSPM or PM_{10}

Respirable suspended particulates are the particles of size $\leq$10 μ. These particles are generated by combustion and non-combustion processes, including windblown dust, sea salt, industrial processes, motor vehicle engines, and fires. These particles can enter the human respiratory system and penetrate deeply into the lungs, causing adverse health effects. A respirable dust sampler (RDS) (Guidelines for Measurement of Ambient Air Pollutants 2012–2013) is used for monitoring of RSPM. It has a cyclone device to separate the dust particles in two fractions of size 10 micron and below and the other fraction of size coarser particles having size >10 microns. In such instruments particle cutoff of the cyclone close to 10 μ size is an important aspect and need to be properly designed. In a RDS air enters through the inlet cap and travels to the cyclone via the inlet pipe at a regulated flow rate in the range of 0.85–1.15 m^3 per min. In the cyclone dust particles get fractionated into two size ranges. Dust particles having size greater than 10 microns get separated in the cyclone and fall into the cup attached at the bottom of it, whereas finer particles which are of size 10 micron and below are carried to the filter paper and get deposited on the filter paper. The filtered air is then allowed to exhaust out of the instrument through the pump.

The instrument measures the flow rate with the help of a glass manometer directly calibrated in flow rate units of cubic meter per minute. A timer and a time totalizer are available for programming the sampling schedule and recording the sampling duration. Measuring the volume of air sampled and weight of dust collected on to a filter paper determines the concentration of PM_{10} particles in the air. The dust

Fig. 16 A high-volume air sampler

Fig. 17 Installation of filter paper on a high-volume air sampler

particles retained on the filter can be analyzed to determine the concentration of other pollutants. These instruments also have provision of attaching a gaseous sampling attachment for simultaneous sampling of gaseous pollutants. One can also assess the SPM by combining the dust collected in the cyclone cup with the dust collected on the filter paper. For very precise and accurate values, RDS or PM_{10} sampler has

electronically controlled system for maintaining constant flow rate. A typical respirable dust sampler and its schematic diagram are shown in Figs. 18 and 19.

Instruments for Monitoring of $PM_{2.5}$

The finer portion of the dust which enters deep into the lungs causes severe respiratory problems. This finer portion of the dust is usually referred to as $PM_{2.5}$ particles. A fine particulate sampler or $PM_{2.5}$ sampler is used for monitoring $PM_{2.5}$ in ambient air (US EPA 40 CFR n.d.-b).

In these instruments ambient air enters at a flow rate of about 16.7 liter per minute through an omnidirectional inlet designed to provide a clean aerodynamic cut point for particles of size greater than 10 μ. Particles in the airstream which are smaller than 10 μ proceed to a second impactor that has an aerodynamic cut point at 2.5 μ. The air containing fine particulates exiting from the $PM_{2.5}$ impactor is allowed to pass through a 47-mm-diameter Teflon filter membrane (PTFE filter) that retains the dust particles of size $\leq$2.5 μ. All these instruments are provided with a pump, a flow

Fig. 18 A respirable dust sampler

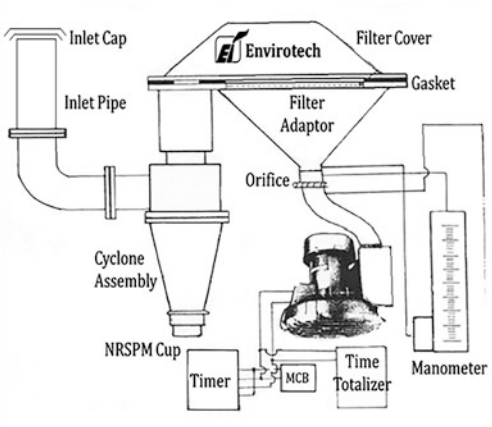

Fig. 19 Schematic diagram of a respirable dust sampler

rate-measuring device, a flow totalizer, timer, time totalizer, and a filter holder designed to accept any standard 47-mm-diameter filter media. A typical $PM_{2.5}$ sampler is shown in Fig. 20.

To ensure greater precision and accuracy, a mass flow sensor is also added to measure air flow rate. In these instruments sampling rate is held constant at 16.7 liter per minute by a microprocessor-based electronic flow controller which automatically adjusts the pump RPM with increasing pressure drop due to deposition of dust on the filter. An ambient air temperature sensor and barometric pressure sensor are provided for effective temperature and pressure corrections of volume of air. An additional filter pressure sensor is provided to record the filter pressure to ensure that the system terminates the sample collection if the pressure drop across the filter exceeds the safe limits. An electronic display of parameters in a $PM_{2.5}$ sampler is shown in Fig. 21.

Increase in temperature in the filter paper compartment may lead to loss of volatile matter present in the dust collected over the filter paper. Considering this, the suction pump is housed in a separate cabinet to minimize heat dissipation near the filter holder assembly. The two cabinets are then clamped together during operation.

Instruments for Monitoring Ultrafine Particulate Matter or PM_1

Recent studies sponsored by the USEPA have concluded that the submicron "ultrafine" particles penetrate further deep into the lungs as compared to PM_{10} and $PM_{2.5}$ particles and have serious adverse effects on the human health. These communities are now monitoring the submicron or PM_1 particulates in addition to the PM_{10} and $PM_{2.5}$. PM_1 monitoring instruments have similar features of a $PM_{2.5}$ sampler except that of the particulate separation device. Here an impactor to separate particles of size ≤ 1 μ is used. Sampling flow rate is maintained at about 10 liters per min. Ultrafine particulates leaving the PM_1 impactor are allowed to deposit on a PTFE filter paper, and the concentrations of PM_1 are determined from the amount of dust collected on the filter paper and the volume of air passed. A typical PM_1 sampler is shown in Fig. 22.

Fig. 20 A $PM_{2.5}$ sampler

Gaseous Sampling Attachments

In addition to particulates, a variety of gaseous pollutants like sulfur dioxide (SO_2), oxides of nitrogen (NOx), carbon monoxide (CO), ozone (O_3), ammonia (NH_3), and many more other gases also need to be monitored. These gases can be absorbed in specific chemical reagents/solutions. For this air is aspirated through these chemical reagents/solutions called absorbing solutions at a known flow rate for a specified time period. A rotameter of 0–3 lpm is available for setting the required flow rate. Usually, there is provision for placing up to four glass impingers of 35 ml capacity in an ice tray and a rotameter and a manifold is provided to set the required flow rates. For simultaneous sampling of one or more gases along with the particulate sampling, gaseous sampling attachments can be hooked with the particulate samplers.

Use of ice is common practice to maintain low temperature around impingers for better absorption of gases. But arranging ice all the time during sampling is often difficult and its availability in remote areas is not assured. To overcome this problem

Fig. 21 An electronic display of parameters in a $PM_{2.5}$ sampler

Fig. 22 A PM_1 sampler

gaseous attachments are now designed to deploy Peltier effect-based solid-state heat pumps to keep the temperature well below the ambient temperature. A built-in thermostat constantly monitors the cold box temperature and ensures that absorbing solution is maintained at set temperature. Silica gel filters are usually provided in the outflow path of each impinge to arrest the particulate that otherwise gets deposited in the needle valve which may result in drop of flow rate. A schematic of a typical gaseous sampling attachment having provision for cooling based on Peltier effect is shown in Figs. 23 and 24.

Instruments for Monitoring of Heavy Metals

Mining and mineral-based industries including crushers, cement plants, thermal power plants, steel and other metal refining and processing plants, coke ovens, brick/lime and pottery kilns, etc. all emit large quantities of pollutants in air, not just ordinary dust, smoke, CO_2, and CO but toxic, phytotoxic, and harmful gases and toxic particulates. Very often these include components of toxic heavy metals like As, Cd, Cr, Hg, Ni, Pb, etc. and also highly toxic organics. These cause serious human health problems, crop damage, and economic losses besides adverse ecological impacts. Since impacts of these on health are noticeable only over a long period, it is advisable to measure their annual averages.

A heavy metal sampler enables collection of dust samples at a flow rate of about 3 liters per minute for long durations. Samples of even 1-month duration can be collected over a single filter paper, and analysis of single sample gives the monthly average results of heavy metals, and one can obtain the annual averages from 12 such samples. A typical heavy metal sampler is shown in Fig. 25.

Continuous Ambient Air Quality Monitoring Devices

The exposure during peak concentrations of certain pollutants can be lethal, and it is therefore desirable to know the short-term, peak, and hour-to-hour concentration of

Fig. 23 A gaseous sampling attachment

Fig. 24 An independent gaseous sampler

pollutants. This helps to the communities to know about the critical periods of pollution levels during the day. It necessitated the requirement of real-time continuous recording type of instruments. Continuous recording type instruments use specific characteristics/properties of dust particles and gases for their detection and measurement. These devices have advantages such as ease of use, better accuracy, low cost per sample, robustness, and low time frequency up to only a few minutes. The principle used in these systems is some physical or chemical property of gas that can be continuously detected, usually by optical-electronic methods. Sampled air enters a reaction chamber where either optical property of the gas can be measured directly or a chemical reaction occurs and produces chemiluminescent or fluorescent light. A light detector produces an electrical signal which is proportional to the concentration of the pollutant being measured.

A sophisticated quality assurance is an integral part of the automatic analyzers. Continuous analyzers produce a large quantity of data which necessitates telemetry system for data acquisition and online compilers for subsequent processing and analysis.

However, these advanced capabilities of automatic analyzers are obtained at the expense of increased capital, maintenance, and support costs. In addition these instruments are also liable to be more susceptible to the technical problems than manual samplers, and require skilled manpower for routine operations.

All these analyzers and calibration systems are microcontroller-based instruments with large graphic LCD to provide continuous status updates including gas concentration, real-time zero/span calibration, sample flow, etc. employing a data acquisition system (DAS).

DAS is rack-mounted industrial PC running Windows-compatible software that has a client-server architecture. It is able to work unattended, i.e., captures analyzer values and sensor values automatically. If a user wishes to perform zero-span checks

Fig. 25 A typical heavy metal sampler

or simply wishes to view the data, he should be able to log in to DAS through a valid username-password. As such, the user can be located anywhere and not necessarily be close to DAS. Analyzers are interfaced to DAS and then get remotely connected to a user/scientist through modem and phone lines/Internet. It should be noted that man-machine interaction is achieved not only locally but remotely also. DAS is accessible by a remote computer for preparing reports or plotting graphs. Schematic diagram of a typical data acquisition system (DAS) is shown in Fig. 26.

For assessment of air quality of an area, district, or city, a typical continuous ambient air quality monitoring station consists of the following:

(a) Rack-mounted set of analyzers and calibration system
(b) Glass lined and heated inlet pipe with manifold
(c) Common external sample pumps with rack-mounted kit.

(d) Calibration system and span gases
(e) Data acquisition system (DAS)

A typical online air quality monitoring station produces thousands of data values every month. The report preparation software is therefore a vital aid to analyze the data recorded by the DAS. Good software can download data from different field stations to a central location for analysis, long-term storage, and dissemination of information. It supports communication via a variety of communication systems including PSTN land lines, TCP/IP, wireless or GSM modems, etc. The software is used to tabulate the data, export it to suitable formats such as MS Excel, or perform scientific analysis in the form of graphs/charts. A typical CAQMS is illustrated in Fig. 27. Common air pollutants and their measurement principles/techniques are given in Table 1.

Principles/methods used for measurement of these pollutants are described below.

Continuous Measurement of Sulfur Dioxide

The measurement of ambient SO_2 levels by the UV fluorescence SO_2 analyzer is based on the principle that SO_2 molecules absorb UV light at the wavelength of 214 nm. The absorbed UV energy excites the outer electrons to the next state. The excited electrons then return to the original state and emit photons at the wavelength of 390 nm. The first reaction step is:

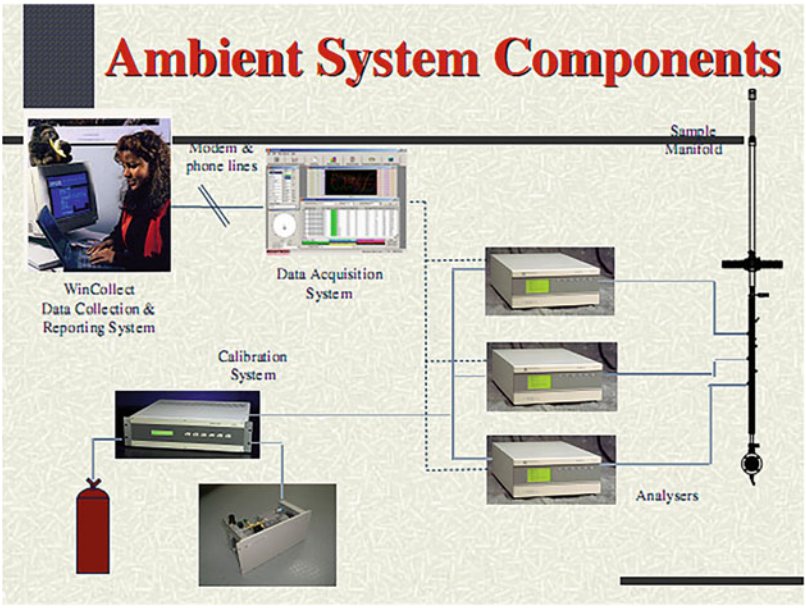

Fig. 26 Schematic diagram of a typical data acquisition system (DAS)

Fig. 27 A typical CAQMS

Table 1 List of common air pollutants and their measurement techniques

Air pollutants	Principle/technique
Sulfur dioxide (SO_2)	UV fluorescence
Nitrogen dioxide (NO_2)	Gas phase chemiluminescence
Ammonia (NH_3)	Chemiluminescence
Ozone (O_3)	UV photometry/chemiluminescence
Carbon monoxide (CO)	Nondispersive IR
PM_{10} & $PM_{2.5}$	Beta attenuation/TEOM
Benzene (C_6H_6)	Gas chromatography (GC)-based continuous analyzer/adsorption and desorption followed by GC analysis

$$SO_2 + h\upsilon_1 \rightarrow SO_2{}^*$$

Then in the second step, the excited SO_2* molecule returns to the original ground state, emitting an energy $h\upsilon_2$ according to the reaction:

$$SO_2{}^* \rightarrow SO_2 + h\upsilon_2$$

The intensity of the fluorescent radiation is proportional to the number of SO_2 molecules in the detection volume and is therefore proportional to the concentration of SO_2. A schematic diagram of a SO_2 analyzer is shown in Fig. 28.

Continuous Measurement of Oxides of Nitrogen

The measurement method is based upon the rapid chemiluminescence reaction of nitric oxide (NO) with excess ozone (O_3). The reaction is made to take place in a light-free chamber:

$$NO + O_3 \rightarrow NO_2^* + O_2$$

A portion of the resultant nitrogen dioxide is produced in a highly excited energy state (NO_2*) and subsequently decays to the ground level state emitting light in a broad frequency band with a peak at about 1200 nm:

$$NO_2^* \rightarrow NO_2 + h\upsilon$$

The intensity of the light emitted is linearly proportional to the nitric oxide concentration and is measured by a photomultiplier tube. This is done in two steps, in the first step NO present in the air sample is detected, and in the second step, NO_2 present in the air sample is first converted to NO in a catalytic converter and then total of NO and NO formed from the conversion of NO_2 is detected which gives NOx concentration. A schematic diagram of a NOx analyzer is shown in Fig. 29.

Continuous Measurement of Ammonia

The ammonia analyzer is a combination of NH_3 converter and an NO-NO_2-NOx analyzer. Ammonia in the air sample is oxidized to nitric oxide (NO) with molybdenum and carbon converters. The measurement method is based upon the rapid chemiluminescent reaction of nitric oxide (NO) with excess ozone (O_3). The reaction is made to take place in a light-free chamber:

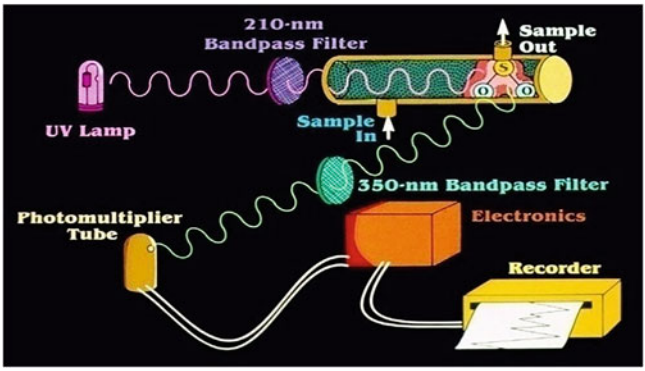

Fig. 28 Schematic diagram of a SO_2 analyzer

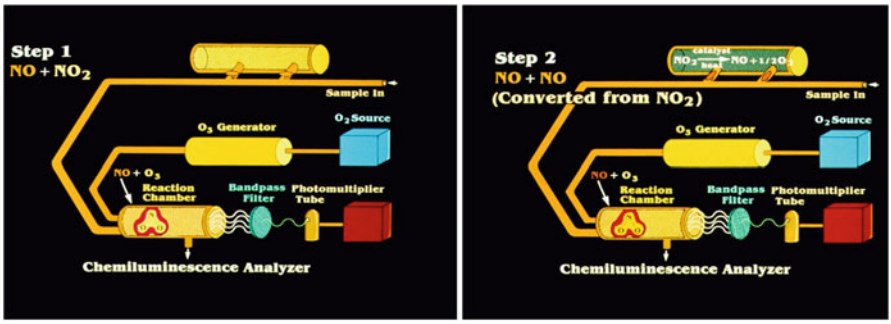

Fig. 29 Schematic diagram of a NOx analyzer

$$NO + O_3 \rightarrow NO_2^* + O_2$$

The instrument is designed for the measurement of total oxides of nitrogen (NOx) and nitric oxide (NO) and indirect determination of nitrogen dioxide (NO_2) and ammonia (NH_3). NO_2 and NH_3 are calculated by subtraction of NO from NOx and NOx from NOy as shown below:

$$\begin{aligned} NOx &= NO + NO_2 \\ NO_2 &= NOx - NO \end{aligned} \quad (1)$$

$$\begin{aligned} NOY &= NOx + NH_3 \\ NH_3 &= NOY - NOx \end{aligned} \quad (2)$$

Continuous Measurement of Ozone

The method is based on the photometric assay of absorption of ultraviolet light by ozone (O_3) molecules in a dynamic flow system. The concentration of O_3 is determined in an absorption cell from the measurement of the amount of light absorbed at a wavelength of 254 nm. The quantities are related by the Beer-Lambert law. A schematic diagram of an ozone analyzer is shown in Fig. 30.

Continuous Measurement of Carbon Monoxide

Carbon monoxide absorbs IR radiations at 4.7 μm. Instrument uses two parallel infrared beams one through a sample cell and the other through a reference cell and a selective detector. The detector signal is led to an amplifier, and the analyzer output is measured on a meter and recording system. A schematic diagram of a CO analyzer is shown in Fig. 31.

Continuous Measurement of $PM_{10}/PM_{2.5}$

A technique used for continuous measurement of PM_{10} and $PM_{2.5}$ particulates is beta-ray attenuation monitor (BAM). In this system dust-laden air is allowed to pass through a particle separation device usually a cyclone for separation of either PM_{10}

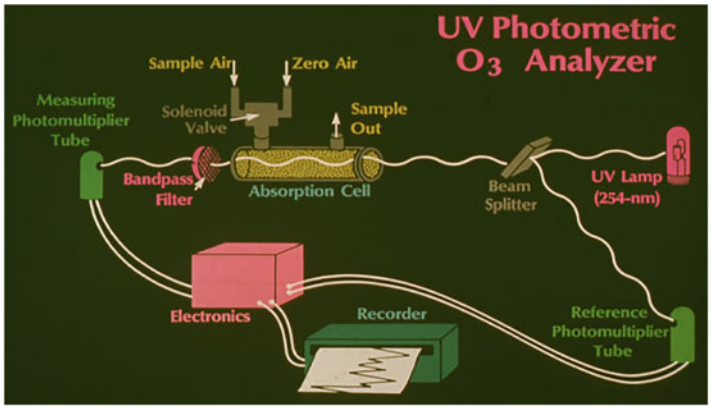

Fig. 30 Schematic diagram of an ozone analyzer

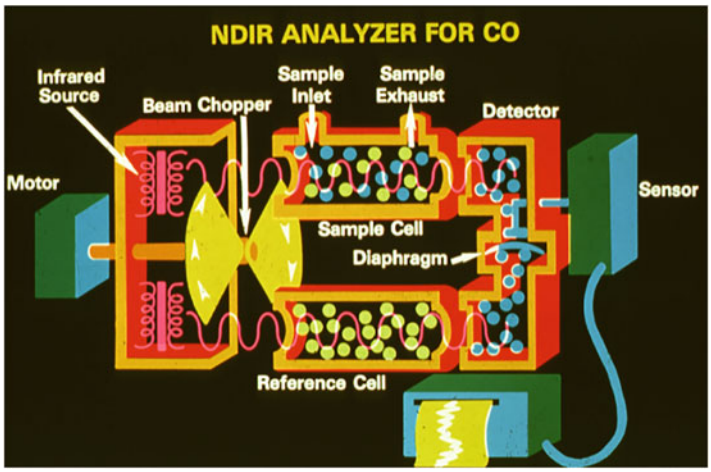

Fig. 31 Schematic diagram of a CO analyser

particles or $PM_{2.5}$ particles from the air. Instrument can measure either PM_{10} particulates or $PM_{2.5}$ particulates at a time. Required cyclone is placed in the line for measurement of either PM_{10} or $PM_{2.5}$ particulates. In this technique a small 14C (Carbon-14 or Krypton 85) element is a constant source of high-energy electrons known as beta rays through a spot of clean filter tape. These beta rays are detected and counted by a detector to determine a zero reading. The monitor automatically advances this spot of tape to the sample nozzle, where a vacuum pump then pulls a measured and controlled amount of dust-laden air through the filter tape, loading it with ambient dust. At the end of the hour, this dirty spot is placed back between the beta source and the detector, thereby causing an attenuation of the beta ray signal which is used to determine the mass of the particulate matter on the filter tape and the

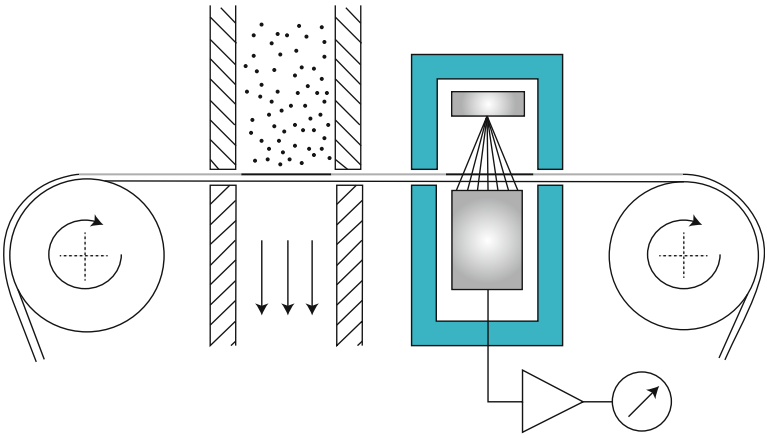

Fig. 32 Schematic diagram of a $PM_{10}/PM_{2.5}$ analyzer

volumetric concentration of particulate matter in ambient air. A schematic diagram of a BAM-based PM_{10}/ $PM_{2.5}$ analyzer is shown in Fig. 32.

Tapered Element Oscillating Microbalance (TEOM)

TEOM is an instrument used for real-time detection of aerosol particles by measuring their mass concentration. It makes use of a small vibrating glass tube whose oscillation frequency changes when aerosol particles are deposited on it increasing its inertia. TEOM-based devices have been approved by the US Environmental Protection Agency for environmental air quality monitoring.

The TEOM uses a hollow glass tube as a microbalance. Incoming particles are deposited on a filter at the tip of the tube, and the added mass causes a change in its oscillation frequency which is detected electronically. The element is periodically cycled to return it to its natural frequency. The inlet to the device only allows particles of the desired size range to enter. TEOM devices operate continuously and do not need frequent filter changes. A schematic diagram of a TEOM particulate monitor is shown in Fig. 33.

Sensor-Based Techniques for Monitoring Ambient Air

Sensor-based instruments provide high-resolution spatial and temporal air pollution data in real time. However, most of the particulate sensors used presently are based on light scattering in which the intensity fluctuations of the scattered light are detected in the presence of scattering particles and recorded as particle count. In this technique, recounting of same particle is a critical factor which often leads to errors in calculation of PM concentrations. In light scattering, reference index depends on composition of particles (carbon, silica, etc.), so the calibration factor developed on one kind of dust might not be relevant in another location. Besides, the calibration factor is strongly dependent on temperature and relative humidity which

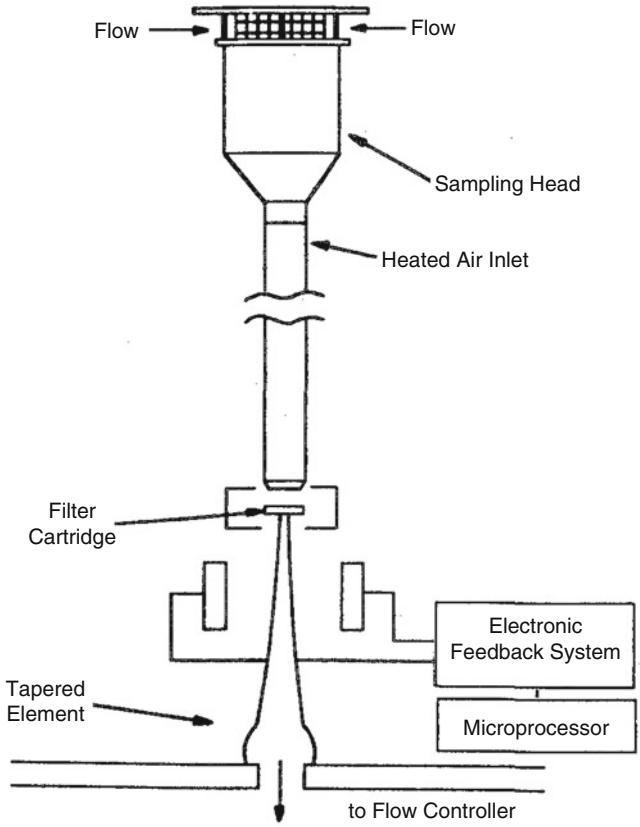

Fig. 33 Schematic diagram of a TEOM particulate monitor

are highly variable in Indian conditions, diurnally and seasonally. They convert number concentrations to mass using assumed density, shape, and composition. These instruments are not based on principles of gravimetric techniques or beta-ray attenuation monitors (BAM). That is why these are not presently used for regulatory purposes.

Air pollution sensors are still in an early stage of technology development, and have not yet been evaluated to determine the accuracy of their measurements. However, these low-cost, portable air quality sensors open an exciting opportunity for a wide range of applications beyond traditional regulatory monitoring. Some potential nonregulatory application areas for air sensors are summarized below:

1. Research aimed at discovering new air pollutants and identifying polluted areas/hot spots.
2. Assessment of personal exposure an individual is exposed to while carrying out his duty on his occupation.

3. Supplementing existing regulatory monitoring network.
4. Source identification and characterization of potential sources.
5. Education/information/awareness by using sensors for informal awareness on air quality.

Performance requirements of a sensor differ according to its application. The quality of a measurement is dictated by the basic performance of the sensor, the way the sensor is operated, and the way its measurements are analyzed. The quality of data collected with sensors varies greatly depending on sensor design and the deployment strategy. Understanding the strengths and limitations of an air sensor is important in making sure that the data collected is useful for the intended purpose. There are several things to think about before putting a sensor in use. First and foremost is to determine a target pollutant. This will guide each of the subsequent decisions, and ultimately lead to the most appropriate types of sensors for the intended application. Also important to consider are device specifications like detection range and detection limit, precision and bias, calibration procedures, etc. The major factors need to be considered in order to collect quality data from an air sensor are:

- Bias
- Precision
- Accuracy
- Drift
- Concentration range
- Detection limit
- Linearity of sensor response
- Life of sensor
- Calibration
- Interferences
- Climate susceptibility
- Data aggregation

Air Quality Monitoring Using Satellites

Monitoring air quality from satellites has played a key role in understanding the status of air pollution loadings and trends on the regional to global scale, by providing information on pollutant amounts, emission, and transport in a quantitative manner. The technology used to monitor air quality from space is complex. In general, satellite-based air quality monitoring operates by looking at how sunlight is scattered by ambient particulates, such as $PM_{2.5}$, PM_{10}, etc. However, this process can easily be disrupted by weather conditions, properties of the pollutant particles, atmospheric mixing, and other factors. Dependence on sunlight also means that this monitoring cannot operate during a cloudy day. In many applications, satellite air

quality monitoring data is fused with ground-based measurements to lend a "bigger picture" of air quality, for example, combining a variety of data points from the satellite monitor with cloud optical thickness or wind speed and direction and with ground-level air quality index (AQI). The commonly used satellites are polar orbiting satellites which generally have global coverage of once a day. Using such satellites, study of diurnal variations is not possible and spatial resolution is also about 10 km only. However, geostationary satellites which spin at the same rate of the earth and constantly focus on the same area enable the satellite to take a picture of the earth at the same location with less time duration.

Personal Exposure and Work Zone Air Quality Monitoring

The increasing general awareness of atmospheric pollution and its hazards on the health and well-being of industrial workers is resulting in greater stress on frequent assessment of work place pollution and worker's exposure. Work zone or workplace is the area where workers are engaged in various activities to complete their tasks. They usually work in shifts of 8 h and are exposed to the pollution during that period. Problem faced in monitoring worker's exposure is that most workers keep moving from one place to other on the shop floor as part of their normal duties. The instrument therefore must be worn by the worker. Instruments suitable for monitoring worker's exposure to dust and gaseous pollutants are described below.

Personal Samplers

Instruments used for measurement of workers' exposure are known as personal samplers. These instruments should be compact, lightweight, and battery operated and worn by the workers while working. These instruments usually operate at low flow rates of about 0–3 liters per minute. Collection of 8 hourly data makes it convenient to compare worker's dose to the specified TLV levels.

During sampling a personal sampler draws the air at low flow rates and allows it to pass through a filter paper of about 37 mm diameter. Dust deposited on the filter paper is measured gravimetrically, and volume of air sampled is obtained after multiplying the flow rate with the time of sampling. Concentration of total suspended particulate matter for which a worker is exposed is calculated in microgram per cubic meters and compared with the prescribed threshold limits. To measure workers' exposure to respirable dust fraction, instruments are equipped with a particle separation device such as a cyclone or an impactor which separates respirable fraction from the total dust and subsequently allows it to pass through a filter paper. In this case filter paper receives the respirable fraction of dust only, and concentration of respirable dust can be measured after gravimetric analysis of dust deposited on filter paper. A picture of the personal sampler is shown in Fig. 34.

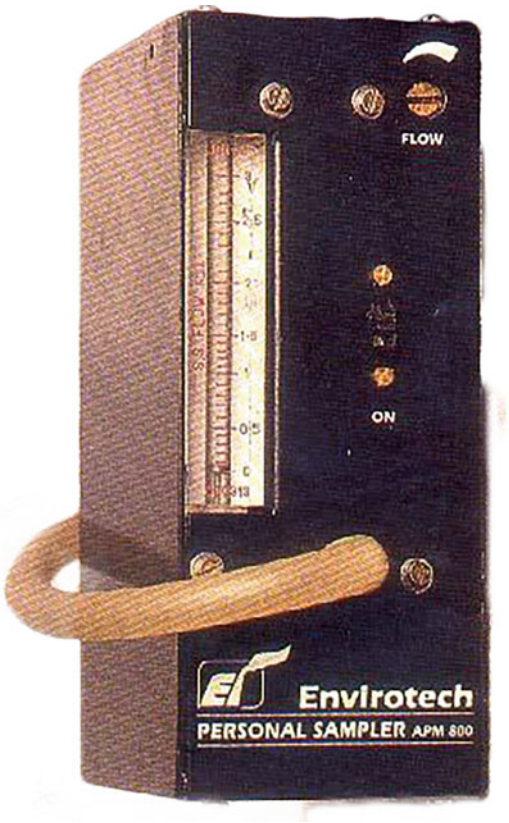

Fig. 34 A typical personal sampler

Handy Samplers

Workplace environment usually has higher concentrations of dust as well as gaseous pollutants. These are required to be monitored to assess the pollutant exposure on the shop floor. The instrument must be portable so that it can be kept at different locations on the shop floor and preferably battery operated. A typical handy sampler for such applications is shown in Fig. 35. Such instruments have provisions for placing a set of glass impingers in an ice tray, usually two impingers, for measurement of gaseous pollutants, where absorbing media is kept for the gases to be sampled. These instruments are compact, are lightweight, and have a flow rate-measuring device, usually a rotameter and a battery-operated pump to provide a flow rate in the range 0–3 liters per minute along with the battery backup of at least 8 h, so that samples for 8 h covering a full shift can be collected. A filter paper is put in the filter holder to arrest the particulates; filter holder is then attached to the impingers. The required sampling flow rate is set with the help of the rotameter, and instrument is allowed to run for the specified time duration. Instrument also has a timer to preset

Fig. 35 A typical handy sampler

the start and stop time of the sampling. Samples are subsequently analyzed in the laboratory using wet chemical methods. Dust concentrations can be assessed from the weight of dust deposited on the filter paper and the volume of air sampled.

VOC or Hydrocarbon Samplers

Over the years complex organic compounds are used in many diverse applications. While rural environments often have high levels of pesticides and other residues from sprays of weedicides and other chemicals used in modern farming, urban environments have high doses of a host of organics coming from paints and varnishes, lacquers, solvents, foam-blowing agents, sprays, etc. Majority of these compounds are highly toxic and many of them are reported to be carcinogenic. Replacement of lead by benzene and its derivatives in petrol as anti-knocking agent resulted their presence in atmosphere as part of vehicular pollution.

Unlike inorganic and some organic gases and vapors which react with and/or absorbed by solutions kept in impingers, most organic gases and vapors are too volatile and insoluble in aqueous reagents. The best option in most cases is adsorption on appropriate surfaces such as activated carbon. The flow rate has to be extremely low to allow such adsorption, generally lower than 200 ml per minute and often as low as 20 ml per minute, thus needing a low-flow pump and very accurate flow metering. Sampling flow rate can be set from 20 to 200 ml per minute. The hardware components include appropriate holders for the adsorbing media, i.e.,

activated charcoal tubes, flow control, and other auxiliaries. After allowing a known volume of sample to pass through the adsorption media, either it is desorbed through a thermal desorber or it is eluted with a solvent and analyzed in the laboratory. The method for measurement of VOCs/benzene requires gas chromatographic (GC) analysis with the following possible approaches:

(a) An air sample collected in the field in a Tedlar bag which can then be directly fed to a GC.
(b) VOCs present in ambient air are collected by their adsorption in activated charcoal tubes and desorbed later for analysis on a GC.

Both techniques require use of a battery-operated/portable instrument capable of drawing air sample at a low flow rate of a few milliliters per minute. A typical VOC sampler and activated charcoal tubes for sampling and adsorption of VOCs/HCs are shown in Figs. 36 and 37. Nowadays automated GC-based systems have also been developed for real-time monitoring of VOCs and their derivatives.

Fig. 36 A typical VOC/HC sampler

Fig. 37 Charcoal tubes for adsorption of VOCs/HCs

Indoor Air Quality Monitoring

Indoor air pollution is a relevant area of concern as it has a direct impact on mortality and morbidity. A large population throughout the world use coal and biomass (crop residues, wood, dung, and charcoal) as the primary source of domestic energy. Moreover, humans spend 80–90% of their routine time indoors, so indoor air quality (IAQ) leaves a direct impact on overall health and work efficiency. Our residential and office complexes are increasingly becoming air-conditioned and poorly ventilated leading to buildup of carbon dioxide levels as the day progresses. Also high levels of $PM_{10}/PM_{2.5}$ outdoors as well as high levels of noise prompt occupiers to keep windows shut in an attempt to prevent air pollution entering the inner spaces. Extensive use of air fresheners, paints, varnishes, and laminates is also adding VOCs in the air. Instances of "sick building syndrome" are increasingly being reported for our cities. The comfort parameters such as temperature, relative humidity, and airflow are important dimensions of indoor air quality. The essential parameters for IAQ measurement include CO_2, PM_{10}, $PM_{2.5}$, and PM_1, volatile organic carbons (VOCs), formaldehyde, ozone (O_3), carbon monoxide (CO), radon, bacteria, fungi, etc.

Fig. 38 Sensor-based indoor air quality monitor

Generally, low-cost sensor-based instruments are used for monitoring indoor air quality. Optical sensors are used for measurement of particulates (PM_1, $PM_{2.5}$, PM_{10}, TSPM, etc.), and electrochemical sensors, NDIR sensors, and PID sensors are used for measurement of various gaseous pollutants. Usually a small fan is provided to suck the air from the atmosphere and circulates it through the sensor. A typical low-cost sensor-based indoor air quality monitor is shown in Fig. 38.

Local Meteorology

Air pollutants disperse under the influence of local meteorological parameters like wind speed, wind direction, atmospheric temperature, atmospheric pressure, stability, and mixing height. No air quality or pollutant-dispersion modeling is possible without reliable information on the local wind trends and stability conditions at different times of the day and seasons of the year. A typical instrument to monitor local meteorological conditions is shown in Fig. 39.

These instruments usually have sensors for wind speed, wind direction, temperature, relative humidity, rainfall, barometric pressure, and solar radiation. Required numbers of sensors are attached to a processing unit and processed data is stored in a data logger.

Instrument can also display the real-time data on a LCD display. Data can be retrieved from the data logger through a laptop/PC or cloud server PC. A typical system configuration of a meteorological monitoring instrument is shown in Fig. 40.

An important use of these instruments is to plot wind rose diagrams. Wind roses are graphical charts that characterize the speed and direction of winds at a location. A wind rose gives a very concise but information-laden view of how wind speed and direction are typically distributed at a particular location. Presented in a circular format, the wind rose shows the frequency of winds blowing "from" particular directions. The length of each "spoke" around the circle is related to the frequency of time that the wind blows from a particular direction. Each concentric circle

Fig. 39 A typical meteorological monitoring station

represents a different frequency, emanating from zero at the center to increasing frequencies at the outer circles. The available software generates colored wind roses in the standard format as shown in Fig. 41.

Liquid in glass thermometers, bimetallic thermograph, resistance thermometer, or a thermocouple can be used for measurement of temperature. However, thermistors are preferred as air temperature sensors. It is because these semiconductor devices can make air temperature measurements from +100 °C to −100 °C with high resolution and accuracy and are available in a wide variety of shapes and sizes as bead, disc, and rods, and the response time can be manipulated better than a second.

For wind direction measurements, wind vane is the standard device. To get fairly fast response and avoid over−/undershooting, wind vanes are properly balanced and critically damped. Wind vanes are fixed on a stand that has markings for the 16 main directions (N, NNE, NE, ENE, E, ESE, SE, SSE, S, SSW, SW, WSW, W, WNW, NW, and NNW). Readings are taken every 15 min or as desired. For sensitivity one needs a low-friction bearing, but damping may also be needed to avoid too much fluctuation in readings.

Devices used for monitoring wind speed are of two types: those which employ mechanical sensors, e.g., cup anemometers, and propeller anemometers, and those which employ nonmechanical sensors, e.g., sonic anemometers and hot wire

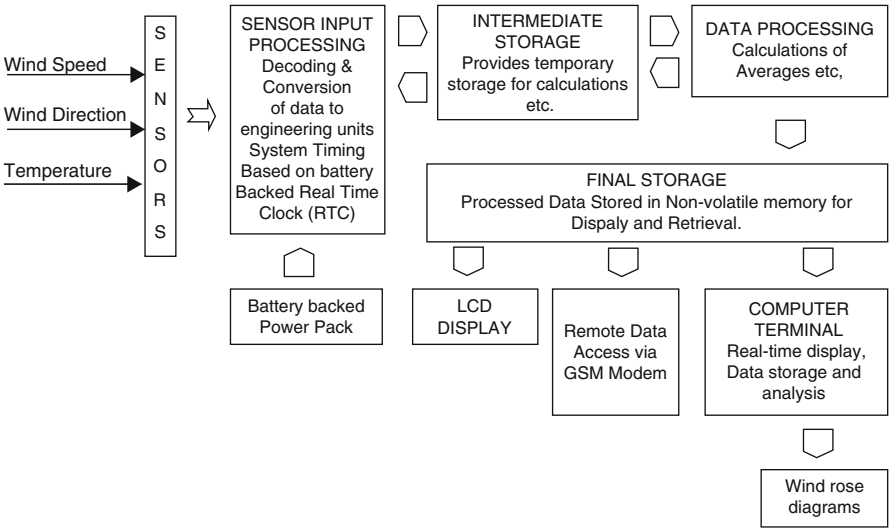

Fig. 40 System configuration

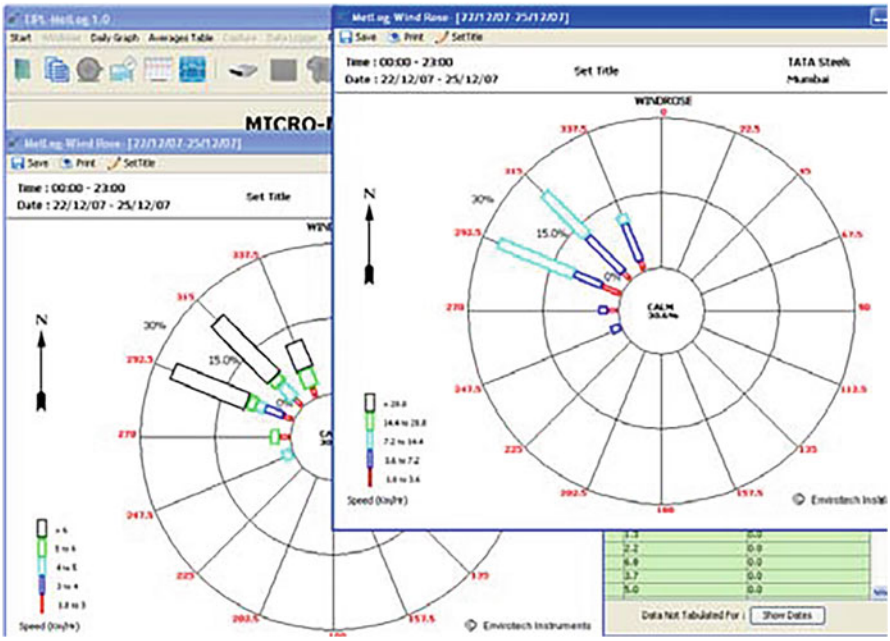

Fig. 41 A typical wind rose diagram

anemometers. In the mechanical devices, suitable transducer is coupled to convert rotations into electric signals which are recorded. Low inertial systems give fast response with a limited range. With proper design, fairly good response and measurement range can be obtained.

For relative humidity measurements, dry and wet-bulb psychometer, dew point hygrometer, and infrared humidometer are used. For continuous measurement of relative humidity, resistive sensors are most popularly used.

For barometric pressure measurements, mercury barometer, aneroid barometer, or a barograph can be used. As a sensor capacitance, aneroid is used. In this device the expansion of aneroid is translated into capacitive change by connecting one plate of the parallel plate condenser with the free end of the aneroid. With this arrangement continuous measurement of pressure is possible.

The solar radiation sensors measure the total energy or radiation coming from the sun. It is measured in watts per meter square (W/m^2). It has to be mounted in a vertical position without any obstructions from shadows and reflections. The sensor uses a semiconductor solar cell which converts solar energy to an electric voltage. The voltage is amplified and transmitted to the data logger via its cable. The data logger will convert the voltage back to W/m^2 Units or calories/cm^2/min (CCM).

The tipping bucket rain gauge is used for measuring the rainfall. It contains an 8-inch-diameter collecting funnel which directs the collected rainfall into a small precision tipping bucket. The bucket is calibrated to tip for every 0.5 mm of rainfall. When the bucket tips, a magnetic reed switch is triggered which sends a pulse signal to the data logger. The data loggers total up the received pulse and count the number of tips to calculate the total rainfall.

Noise Monitoring

Noise pollution is described as any disturbing or unwanted sound that interferes with or harms human health or wildlife. Cities are often considered to be very noisy. Exceptionally loud music, devotional songs, and prayers over loudspeakers are part of the daily life of our city dwellers. To this is added noise from honking of horns and other commercial and industrial activities. All limits are exceeded during periods of festivity and marriage season. However, excessive noise is known to be harmful and considered as a pollutant to be controlled.

Sound or noise is a form of energy produced by a vibrating object or by an aerodynamic disturbance propagated in the form of waves/pressure variations through a material medium. As these pressure variations reach our eardrums, they cause them to vibrate. The transmission of these vibrations to the inner ear and their interpretation by the brain result in our sensory perception of sound. Sound characteristics most important to human hearing include frequency and intensity. Humans can hear sounds in the frequency range of 20–20,000 Hz. All frequencies are not, however, heard equally well as humans hear best, those frequencies which correspond to human speech (500–4000 Hz). There are three internationally standardized characteristics, termed as “A,” “C,” and “Z,” weighting networks. A-weighted sound

level is useful for assessment of human response, as also for rank ordering of noises with respect to loudness level as long as these noises are of similar character. Weighting networks "C" and "Z" do not give good correlation. The reason being that equal loudness contours from where these networks have been chosen are based on experiments with pure tones while most common sounds are complex signals rather than pure tones. The loudness of sound is measured in decibels (dB).

A common instrument used for measuring noise is the Sound Level Meter (*SLM*). A simple *SLM* consists of a microphone, amplifier, attenuators, weighing networks, electronic circuits, and a readout display. The microphone detects the small air pressure variations associated with sound and changes them into electrical signals. These signals are then processed by the electronic circuitry of the instrument. The weighing networks are an integral part of the instrument. They take into account the subjective response of the human ear. The readout displays the sound level in decibels (dB). With most *SLMs*, the readings can be taken on either *SLOW* or *FAST* response. The response rate is the time period over which the instrument averages the sound level before displaying it on the readout. Workplace noise level measurements should be taken on *SLOW* response. Two types of sound level meters are available: Type 2 and Type 1. Type 2 *SLM* is sufficiently accurate for industrial field measurements, whereas Type 1 *SLMs* are primarily used in engineering, laboratory, and research work. These are more precise and accurate.

A standard *SLM* takes only instantaneous noise measurements which is adequate for workplaces with continuous noise levels. But in workplaces with impulse, intermittent, or variable noise levels, the *SLM* makes it difficult to determine a person's average exposure to noise over a work shift. An integrating sound level meter is an instrument that measures sound energy over a period of time and displays it in Leq values. An integrating sound level meter is more suitable for measuring the noise exposure at such places. Further, no data is valuable unless it is interpreted properly. With modern software tools, even a novice user can do the classical data analysis from the recorded data. Nevertheless, ambitious users can use high-end software tools to interpret their data and can get percentiles, scatterplots, pie charts, histograms, and exceedance curves etc. A typical sound level meter is shown in Fig. 42.

While making field measurements, the user will often be required to select/set up the following functions marked on the sound level meter:

1. Range: Typically the default range will be the middle range with an operating sound level from 30 to 140 dB.
2. Time weighting: This function alerts the response time of the instrument. Most sound level meters will have at least two modes – *SLOW* and *FAST*. Some instruments will also have an *IMPULSE* mode. The *SLOW* level has a response time of 1 Sec. and is the default mode in most instruments. Most field measurements will be made in this mode.

   Fast response has a time constant of 1/8 sec or 125 Milli-Sec and should be selected if rapidly changing sound levels are to be monitored, e.g., at a traffic junction.

Fig. 42 A typical integrating sound level meter

3. Averaging period: Besides instantaneous sound pressure level (*SPL*), it is often desired to monitor Leq and/or the sound exposure level (*SEL*). These measurements are integrated over time. In modern instruments a recording or data logging mode is available. In this mode Leq values integrated over 1 min are stored in the instrument's memory; these may be downloaded to a PC software to be used to compute Leq values over desired integrating/averaging period.

Application of various instruments and techniques described above for monitoring of air pollution and air quality is summarized below.

Type of instrument	Application
Stack sampler	Stack emission monitoring
Flue gas analyzer	Determination of stack gas composition
Probe electrification instrument	Continuous emission monitoring of PM
Beta-ray attenuation monitor (BAM)	Continuous monitoring of PM in emissions and $PM_{10}/PM_{2.5}$ in ambient air

(continued)

Type of instrument	Application
Tapered element oscillating microbalance (TEOM)	Continuous monitoring of PM_{10}/$PM_{2.5}$ in ambient air
High-volume air sampler	Monitoring of SPM in ambient air
Respirable dust sampler	Monitoring of PM_{10} in ambient air
$PM_{2.5}$ sampler	Monitoring of $PM_{2.5}$ in ambient air
PM_1 sampler	Monitoring of PM_1 in ambient air
Gaseous attachments to ambient air quality samplers	Monitoring of gaseous pollutants in ambient air
Heavy metal sampler	Monitoring of heavy metals in ambient air
Personal sampler	Monitoring of worker's exposure to dust in work environment
Handy sampler	Monitoring of gaseous pollutants in work environment
VOC/HC sampler	Monitoring of VOCs/HCs and other organic pollutants in ambient air
Sensor-based real-time monitors	Monitoring of TSPM/PM_{10}/$PM_{2.5}$/VOCs & other gaseous pollutants in indoor air and outdoor air.
Meteorological monitoring station	Monitoring of wind speed, wind direction, temperature, relative humidity, atmospheric pressure, rainfall, solar radiation, etc.
Sound level meter	Noise monitoring

Maintenance and Calibration of Air Quality Monitoring Instruments

To ensure smooth working of any instrument, it is essential to provide regular maintenance. Preventive maintenance helps to keep the instrument at its best performance and extends its life. Maintenance of manual samplers is easier as compared to real-time online continuous analyzers. Skills required for maintenance of manual samplers are also low and can be easily developed through a short training. Spare parts are also easily available for indigenous instruments, whereas maintenance of a real-time analyzer is difficult and requires highly skilled manpower. Availability of spares and consumables is a problem in case of analyzers, which often are imported. As a result of this, it becomes difficult to keep such systems operational.

Similarly, periodic calibration of instruments is very important to check the accuracy and to determine the traceability of the measurement. Several factors are considered to decide the calibration frequencies such as the environmental conditions in which instruments are used and the frequency of their use.

In addition to periodic calibration of instruments, time-to-time calibration checks or calibration verification by the users need to be performed at specified frequencies by using secondary calibrators. This is done to ensure that the results of calibration still meet the accuracy levels. Calibration checks are also performed after an instrument has gone some major repairs or the instrument is transported from one

location to another location or the instrument starts giving inaccurate results. Right calibration helps in overall quality assurance and quality control of the results. Required standards of performance can be achieved only through effective planning and execution of maintenance programs. Common issues related to maintenance and calibration of the air quality monitoring instruments are described below.

Preventive Maintenance of Manual Samplers

PM_{10} and $PM_{2.5}$ Samplers

1. Only distilled water need to be filled in the manometer.
2. Check the cracks and leaks in various rubber gasket and "O"-rings once every 6 months. Replace to avoid leakages.
3. Replace PVC pipes once in a year.
4. "O"-rings of cyclone where cup is fitted need to be checked and replaced once these are loose.
5. Inlet pipe gasket should be checked periodically and shall be glued if it has become loose.
6. Wash the rotameter with mild soap solution, when becomes dirty. Never use an organic solvent.
7. Cyclone and the inlet pipe should be cleaned thoroughly.
8. Inlet pipe bend should also be cleaned regularly.
9. Hardware needs to be tightened periodically.
10. Time totalizer checking must be done regularly.
11. Wire mesh should not be loose.
12. Electrical wires and connectors must be checked periodically.
13. Ensure stocking of some frequently required spares like PVC pipe, impinger, manometer tube, "O"-rings, gaskets, mains cord, and miscellaneous hardware.
14. Ensure regular calibration of rotameter, flowmeter, and time totalizer.
15. Ensure stabilized power supply to the instrument. It should be 220 $\pm$ 10 V output.
16. Ensure length of the mains cord should not be more than 10 meters and without joints.

Stack Samplers

1. Clean instrument after each monitoring.
2. Clean the brass fittings used to connect hoses once every 2 months.
3. Check "O"-rings of the filling/drain plug and replace them in case they are excessively flattened or damaged.
4. Clean the impingers After Every monitoring.
5. Replace the thermocol of impinge box after one year or earlier in case of damage.
6. Use detergent solution to clean hose pipes, if necessary allow the probes to dry, or otherwise flush the components with compressed air before use.

7. Check the oil level in the pump before every monitoring and top up with fresh recommended oil.
8. Check the mounting screws of the pump.
9. Check the wiring regularly.

Meteorological Instruments

The most important precaution while setting up a meteorological system is alignment of the direction sensors. This has been described below in brief. The installation of other sensors is easier, and the reader is recommended to refer respective manuals of different weather monitoring systems:

1. Position the tripod where you can get a clear and safe view of the sensor. The best position for mounting of sensors is usually to 10–30 meters from ground.
2. Level the compass by using the circular bubble level. The compass is correctly sighted on both arms: front side and the crossarm or orientation vane. Check the bubble in the circular level to confirm that it is still level. The north-seeking end of the needle should be pointing to 0 degree ± 0.5 degree. If the correct declination is used and the compass is properly sited, the crossarm or alignment vane should be aligned to true north. To confirm that the sensors are properly aligned, it is helpful to view the mast from the opposite side (180 degrees) of the orientation direction.
3. The use of an orientation pointer may make alignment of sensors to true north easier.
4. Alignment of sensors is best done by two persons, one person to orient the sensor and another person to read the compass.

Calibration

Periodic calibration of critical components of the instruments is required under the following situations:

1. When instrument is manufactured.
2. When instrument is transported to a new location and in different climatic conditions.
3. If instrument is operated after interruption of several months.
4. After repair or overhaul of key components.
5. After observing significant drift in zero or span values or in monitoring results.
6. On expiry of validity of calibration (usually 1 year).

Conclusions

Selection of an appropriate instrument for monitoring either emissions from sources or ambient air quality or indoor air is dependent on the objectives of monitoring. For assessment of impacts, devices giving time-weighted average values are most

desirable, and for assessment of short-term hourly variations, real-time continuous monitoring systems are more desirable. To assess exposure of workers who keep moving while working, a different type of instrument is required. Movement of air in the local environment is important to understand buildup or dispersion of pollutants. To protect people's hearing impairment, instant and Leq levels of noise are required to be monitored.

We hope authors have adequately described the various types of instruments to meet the objectives of monitoring.

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Traceability in Analytical Environmental Measurements

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Shweta Singh, Monika J. Kulshrestha, and Nisha Rani

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Abstract

Reliability of the measurement data in today's world lays a strong foundation for effective policy making, resource allocation, and many such significant decisions for a country. The world of metrology relies that every measured data has an uncertainty associated with it, the lack of which often questions its credibility. Thus, metrological traceability, which links the measured results to a stated reference through an unbroken chain of comparisons along with all the stated uncertainties, has become essential. However, estimation of uncertainty is quite simpler for physical measurements, but in the field of chemical/biological/environmental measurements, this idea still needs propagation. The qualitative and quantitative measurements in environmental science forms the basis for environmental legislations; even small inaccuracies/errors can prove to be lethal. This enhances the role of traceability in environmental measurements. The analytical methods of the environmental measurements consist of several steps pre-treatment (extraction) of the sample, sample storage, and chemical/biological analysis. However, the estimation of uncertainties is quite a challenging process in environmental measurements due to partial recovery of an analyte, improper handling of samples during extraction, and improper storage of the samples. The knowledge of uncertainty associated with analytical environmental measurements and its procedures which determine the quality of the data produced is still needed to be understood and explored in detail. This chapter discusses how traceability using documented standards and certified reference materials can be perceived in analytical environmental measurements with an attempt to link the associated uncertainties at each step of the measurement of atmospheric particulate matter (PM) concentration as well as during analytical determination of its chemical components.

Keywords

Traceability · Analytical environmental measurements · Uncertainty · Certified reference materials · Unbroken chain of comparison

Introduction

The accurate measurements in different sectors are the key to growth and development of a country. Be it cross-border trade, fashion industry, manufacturing industry, healthcare and medical sector, civil sector, environmental measurements, and other fields of science and technology, the fundamental component is accurate and precise measurements. The quality of goods relies on the quality of measurements carried out at different stages. All the measurement results have some uncertainty related with them. The uncertainty of measurements is an important parameter that is used to judge the quality of a measurement. Indeed, the results of measurement lack its credibility without the statement of associated uncertainty in measurement.

Consequently, the reliability of a measurement is an important part of product development and quality control which is established through traceability.

In the scientific world, numerous measurements and their comparisons are being made on a daily basis on local, regional, or global scale. The science of comparison usually examines the underlying similarities (or differences) in two or more things. Sometimes, achieving comparability becomes challenging when these measurements are made at different locations, at different times, and using different methods. Therefore, the need for traceability arises in the measurements which would be able to remove these biases. Nowadays, the metrological traceability has become crucial in the world of measurement sciences which is aimed to attain quality data. Traceability connotes that the result of a measurement, no matter where it is made, can be related to a national or international measurement standard, and that relationship is documented. In addition, the measuring instrument is calibrated by a measurement standard that is itself traceable. The idea of establishing traceability to national or international measurement standards has evolved considerably over the past few decades. However, there is still a necessity of its deep inculcation in every field of scientific measurement to compare these measurements in time and space. Metrological traceability implies that the measured results can be traced back to a common national/international standard through an unbroken chain of comparisons. The third edition of International Vocabulary of Basic and General Terms in Metrology (VIM) defines metrological traceability as "property of a measurement result whereby the result can be related to a reference through a documented unbroken chain of calibrations, each contributing to the measurement uncertainty" (JCGM 2008a). This definition of traceability includes three main components, i.e., stated references, unbroken chain of calibrations/comparisons, and measurement uncertainties. These components are required to be inferred in relation to the environmental measurements (Valcarcel and Rios 1999; Quevauviller 2000; Walsh 1999).

The stated references may be reference methods/reference materials/SI (Syste'me International) units as per definition of the ISO (International Standards Organization) (ISO 1993; King 1997). Theoretically, traceability indicates that all measurements should be traceable to SI units. The idea of traceability infers that measurement data are linked to stated references through an unbroken chain of comparisons, all with stated uncertainties (Yadav et al. 2020a). The traceability was initially limited to physical parameters like mass, time, length, etc. for more than 100 years. Now, it is has become important even for chemical, biological, and environmental measurements (Valcarcel et al. 1998; King et al. 1999; Quevauviller and Donard 2001; Quevauviller 2004). But in chemical/biological/environmental measurements, the theoretical aspect of traceability is impractical as there are several steps involved in such measurements with a possibility for missing links for unbroken chain of comparisons (King et al. 1999; Maroto et al. 1999). Nowadays, metrology in chemistry, biology, and environment is a matter of concern for reliable data quality (Valcarcel et al. 1998; King et al. 1999; Quevauviller 2001). Theoretical metrology is not directly applicable to chemical, biological, and environmental measurements as there is a considerable difference between these measurements and the physical measurement processes (Quevauviller and Donard 2001). However,

chemical/biological/environmental measurements are based on sampling methods, storage of samples, extraction of samples, comparisons of amounts, and instrumental response (sensitivity and selectivity) to an analyte. In the case of analytical chemical/biological/environmental measurements, traceability may be achieved in relation to either a reference material (RM) or certified reference material (CRM)) through a standardized method (King 1997). For example, analytical environmental measurements include many steps starting from its sampling to final determination of an analyte/component. Therefore, the application of concept of traceability and uncertainty in analytical environmental measurements is quite challenging due to complexities involved in measurement processes. In analytical environmental measurements, the measured results can even get influenced by the initial preliminary steps involved like sampling, storage of samples, extraction of samples, etc. Also, the quantitative estimation of environmental matrices is very difficult as the same chemical component can be analyzed through different analytical procedure in different matrices. The traceability concept is profound in the field of physical measurements; however, in chemical/biological/environmental measurements, this idea still needs propagation. This chapter discusses about how traceability to SI units through stated references, unbroken chain of comparisons, and measurement of uncertainties is perceived in analytical environmental measurements.

Traceability in Analytical Environmental Measurements

Environmental measurements include diversified range of several biotic and abiotic entities. Biotic entities comprise living organisms like flora and fauna, while abiotic entities include air, water, soil, etc. Any variation in the environment gets incorporated in these biotic and abiotic components which are well reflected in their physical and chemical properties. Thus, these entities have a great potential for environmental studies. Ever since the anthropogenic activities have started dominating and impacting the earth system, there is continuous need for qualitative and quantitative monitoring of these environmental entities which are analyzed at different concentration levels for a wide variety of chemical and biological components. Though environmental measurements are difficult due to its complex matrices, still there is a need for the quality data. Since the qualitative and quantitative estimation is quite sensitive and forms the basis for environmental regulations/legislations, even minute inaccuracies/errors can prove to be lethal. Thus, there is a need for a reliable data with all possible sources of uncertainty in environmental measurements. Thus, the role of traceability in environmental measurements is very crucial (Quevauviller 2004a). Any method which is traceable to a documented reference may not be the accurate one, but an accurate method is always traceable to the nearest true value (ISO 1993). Occasionally, traceability is misunderstood as accuracy and/or precision. Similarly, accuracy and precision are two different concepts and not to be confused with one another. Accuracy is the closeness of the measured result with the true value of a measurand, while precision is the closeness of a number of replicate measurements to each other or it's a set of dispersion of values.

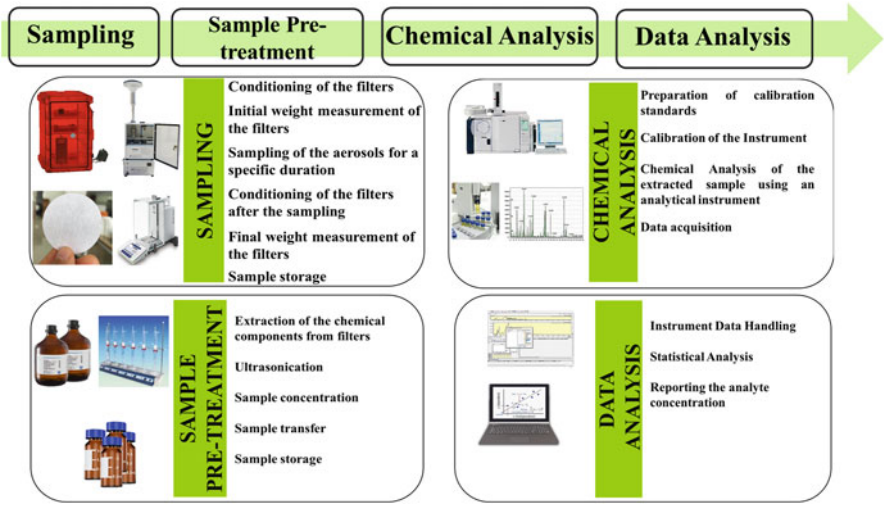

Fig. 1 Potential sources of uncertainty in aerosol monitoring and chemical analysis

Environmental measurements are far more complex in nature, so theoretical metrology may not be useful to that extent (Horwitz 1992; Griepink 1990). In the 1990s, many environmental chemists did not focus much on data quality and reliability of analytical results. The quality of data for environmental monitoring depends on the reliable analysis of abiotic and biotic entities. There are a number of carcinogenic and toxic constituents present in the environment in different concentration levels. During the last two decades, the focus on quality assurance (QA) of environmental analysis is increasing which has led to the development of a number of guidelines and documented standards, e.g., sampling, method validation, etc., and use of Certified Reference Materials (CRMs) in analytical procedures and interlaboratory comparison through proficiency testing (Gunzler 1996; Quevauviller 1995; Subramanian 1995; Barcelo 2000; Quevauviller and Maier 1999). Analytical environmental measurements mainly involve several steps, e.g., sampling for collection of samples, storage of samples, chemical extraction of collected samples for quantitation, calibration of instrument, analytical determination of chemical components of interest, data analysis, and results (Fig. 1). Considering these steps, let us understand the concept of traceability in the context of analytical environmental measurements.

Stated References/Documented Standards in Environmental Measurements

The documented standardized procedures include sampling protocols and analytical techniques with reference to specific measurements. In analytical environmental measurements, i.e., environmental monitoring and analytical procedures,

standardization is the key component. The documented standards contain appropriate steps of methods and/or techniques required for regulations of environmental measurements. It fulfills the minimum requirements of traceability which further helps in improving the comparability of analytical results. The implementation of a standardized method does not mean that an error will not occur; it only offers a background for a given measurement as the reference. In environmental monitoring, sample collection and storage are primary sources of error. These are the weakest links in traceability chain as it is very difficult to make a standard procedure for sample collection and storage (Quevauviller 1995; Barcelo 2000). Similarly, in analytical environmental procedure, it is difficult to get the 100% recovery of an analyte from the extraction of a sample (Ure and Davidson 1995). For example, in the determination of organic compounds in atmospheric particulate matter sample using a single or sequential extraction procedure, the measurements do not relate to specific organic compounds. If the documented extraction procedures are not properly followed, then the data comparability is not possible (Quevauviller 1998). Any change in the proper implementation of the documented procedure might lead to the infringement of traceability chain. Therefore, the documented extraction procedure would be considered as a reference for traceability of analytical results.

The Unbroken Chain of Comparisons in Environmental Measurements

The unbroken chain of comparison means that there is no missing link of information in analytical environmental measurements while following an environmental monitoring and analytical procedure. In environmental monitoring, the steps include sample collection and sample storage. The missing links in environmental monitoring may be improper sample collection and sample storage, which are crucial for unbroken chain of comparisons. The analytical procedure includes several steps like extraction of sample, storage of extracted sample, and analysis of sample using a suitable instrument. But the missing link may be the partial recovery of an analyte, contamination of a sample while doing extraction, improper storage of extracted sample, using an uncalibrated instrument, etc., which are crucial for unbroken chain of comparisons. The lengthier analytical procedure has more possibility of missing links which may disturb the unbroken chain of comparison. There are additional steps involved in an analytical procedure while using an analytical technique for detection of individual chemical components, e.g., lack of sensitivity and/or selectivity of the instrument which may break the chain of comparisons. Therefore, both environmental monitoring and analytical procedure for detection of chemical components in environmental measurements are essential parts of unbroken chain of comparisons for traceability. In analytical environmental measurements, the uncertainty component is inevitable for traceability, and it must be assessed at each stage of an environmental monitoring and analytical procedure. Since the unbroken chain of comparison and uncertainties are inter-linked, the smaller chain of comparison will provide better understanding of uncertainty of the results.

Uncertainty in Environmental Measurements

The word *uncertainty* generally relates to "not sure" or "doubtful." But scientifically, the uncertainty is related with the result of a measurement and tells the limits of dispersion of the values ascribed to the measurand. For a measurement, uncertainty does not indicate a distrust about the validity; rather it provides more confidence in the validity of a result. The term uncertainty is also misunderstood as precision. The term precision is closeness of agreement between the results obtained by applying the same experimental procedure several times under prescribed conditions. Both random and systematic errors are included in uncertainty while precision only includes the random errors (Valcarcel and Rios 1999). Further, error and uncertainty are two separate concepts. These two should not be confused for one another. The difference between the true value and the measured value of a parameter is called the error while a range of values within which a parameter is supposed to remain is called the uncertainty. Errors can be random or systematic. Random errors are always present. The sources of random errors may occur due to the errors made by the operator or changes in experimental conditions which are out of control of the operator. Random errors can be minimized by taking repeated measurements and calculating the average of the same. Systematic errors are typically present which may be predictable. The sources of systematic error are instrumental, physical, or human limitations, e.g., instrument is out of calibration. The systematic errors can be minimized by careful calibration of the instrument and by using best suitable analytical techniques.

As discussed above, one needs to consider three major components to achieve traceability in any measurements including analytical environmental measurements, i.e., stated references, unbroken chain of comparisons, and measurement uncertainties. Further, one needs to focus on quality assurance/quality control components to achieve three major components of traceability especially in analytical environmental measurements.

Quality Assurance/Quality Control in Environmental Measurements

To understand the concept of quality assurance and quality control (QA/QC), one should have clear concept of the term "quality." In general, the word quality implies to inherent characteristics of a product/device/service that guarantees to meet the needs of a customer. The quality includes design, conformance, and performance of the product/device/service. ISO 9000:2015 defines quality as "degree to which a set of inherent characteristics of an object fulfils requirements" (ISO 9000 2015). In analytical domain, the quality can be understood as furnishing the reliable information about a measurement data under specific conditions. The specific conditions include accuracy and precision of the measured data which is directly related to "fit-for-purpose." To achieve the quality of the desired result, method validation, measurement uncertainty, and metrological traceability are the key elements. Thus, to accomplish quality, a proper quality management system is required which should be

aimed at minimizing the frequency of quality failures along with their timely identification and eradication. ISO 9000:2015 provides the basic concept, principles, and vocabulary for quality management system (QMS). The ambit of QMS comprises two main components, quality assurance (QA) and quality control (QC).

EURACHEM/CITAC Guide, 2016 defines QA as "part of quality management focused on providing confidence that quality requirements will be fulfilled" and QC as "part of quality management focused on fulfilling quality requirements" (Eurachem/CITAC Guide 2016). QA includes all the activities undertaken by a laboratory to ensure the fulfillment of quality requirements, while QC includes the measures taken at an individual level to fulfill those requirements as shown in Fig. 2. QC is mainly intended to prevent or reduce the occurrence of errors.

However, various sources of errors can slide in when an analytical environmental procedure is performed, for example, sampling error, calibration error, loss of analytes from the sample, interferences of other components in sample matrix, incompetent operator, etc. The improper sampling or sample handling can produce significant amount of errors. Non-representative sampling and/or non-representativeness of sampling site must be avoided and taken care of to minimize errors. Loss of analytes during sample pre-treatment procedures (extraction) due to oxidation, volatilization, heat-degradation, etc. can also contribute significantly to errors. This type of error can be reduced by following a set of valid and documented procedures. Incompetent operator can also be the source of error and produce erroneous results. The errors can slide in while performing any step of analytical procedure like dilution, marking of samples, calculation mistakes, or biasness toward a method or result. Thus, the analyst should have proper educational qualification along with relevant hands-on training. The laboratory should maintain all the records from sampling till analysis and reporting of data.

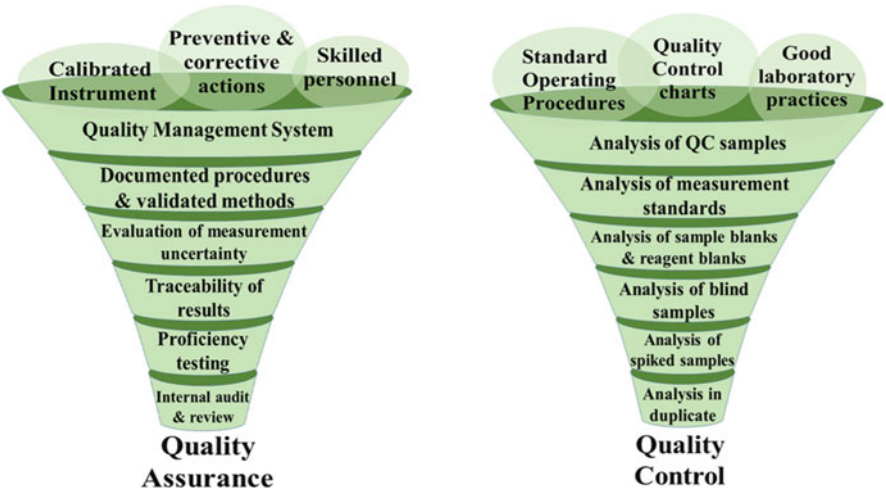

Fig. 2 Main elements of quality assurance and quality control

Method Validation in Environmental Measurements

The scientific laboratories around the world perform several experiments or test every day using various methods. The scientific measurements form the base of a society and its growth and development. These methods need to be checked several times before putting it into routine use so that their performance could be validated for a particular application. The method validation helps to perform check on such methods. The main aim of performing method validation is to validate that a method under consideration is "fit-for-purpose" (Eurachem Guide 2014). This means that the method is capable of producing similar results as per the requirement of the customer. The method validation also ensures that a result delivered by a laboratory is reliable or not (Stauffer 2018a). Method validation is needed when it is required to express that the performance characteristics of the method are suitable for a specific purpose.

As per ISO/IEC 17025 guidelines, a laboratory can validate non-standard methods, laboratory-developed methods, and standard methods used outside their intended scope or otherwise modified (ISO/IEC 17025 2017). Method validation is also required to state similarity of results acquired by two methods, e.g., a new method and an existing standard/regulatory method. The key approaches to determine the validity of a method involves linearity, selectivity, limit of detection (LOD)/limit of quantification (LOQ), ruggedness, trueness, precision, and measurement uncertainty as shown in Fig. 3 (Eurachem Guide 2014). Similar to accuracy in other fields, the method validation is very important in analytical environmental measurements. The different components of method validation have been explained as follows:

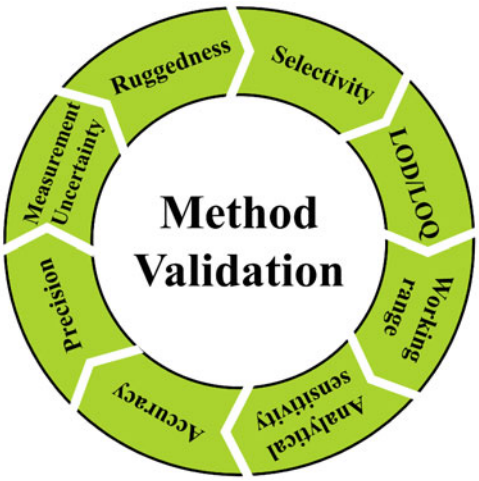

Fig. 3 Key components of a method validation

Selectivity

The selectivity of an analytical method means the ability of a method to determine a particular analyte in a mixture without interference from any other components having similar characteristics (Eurachem Guide 2014). The applicability of the method should be studied using response from a range of samples, i.e., from a measurement standard to mixture having complex matrices which is applicable in analytical environmental measurements.

Specificity The word selectivity and specificity are often used interchangeably; however, there is slight difference. Specificity implies that the analytical method determines or measures quantitatively the component of interest in the matrix without separation. The International Union of Pure and Applied Chemistry (IUPAC) defines specificity as the ultimate selectivity which means there is no expected interference (Bievre et al. 2011).

Analytical Senstivity Sensitivity of an instrument is a measure to detect or differentiate the small changes in the concentration (Eurachem Guide 2014). Sometimes sensitivity is also referred to as limit of detection (LOD). It can be measured by slope of the calibration curve (calibration sensitivity). It can be represented as:

$$S = mC + S_{Blank}$$

where, S = signal; m = slope of the calibration curve; C = concentration; S_{Blank} = signal for blank

Linearity (Working Range)

For an analytical method, the working range can be determined by inspecting the samples having different analyte concentrations and determining the range of concentration for which an acceptable uncertainty can be achieved. In other words, the interval between higher and lower levels can be determined with accuracy, precision, and linearity. Linearity of an analytical method can be determined by different levels of stock solution/standard dilution and plotting a calibration curve for more than three concentration levels which are very important in environmental measurements due to high variability in the concentration levels of chemical components in the samples (Zonen et al. 1998).

Calibration

Calibration is a process related to measured analytical signal to the concentration of analyte. All types of analytical methods require calibration for quantification. Calibration is defined as an "operation that, under specified conditions, in a first step, establishes a relation between the quantity values with measurement uncertainties provided by measurement standards and corresponding indications with associated measurement uncertainties, and in a second step, uses this information to establish a relation for obtaining a measurement result from an indication" (Eurachem/CITAC Guide 2016). Calibration curve is used to understand the response of an instrument

for a particular analyte and predict the analyte concentration in the unknown sample. Three calibration methods are commonly used for calibration purpose in analytical environmental measurements, i.e., standard addition, external standard, and internal standard.

Standard Addition

In analytical environmental measurements, the sample may contain multiple components called the matrix along with the analyte. The matrix might alter the analyte signal, and change its response, known as matrix effect. The method of standard addition is used to compensate for the matrix effect. In this method, known amounts of the standard analyte are spiked to the fixed amount of sample to establish response and create calibration curve. The concentration of the analyte is determined by extrapolating to the x-intercept (Taylor 1985; Danzer and Currie 1998).

External Standard

In analytical environmental measurements, the external standard method is most commonly used to determine the concentration of an analyte in an unknown sample. In this method, a number of standards including at least one blank are used. The standard solutions are prepared with known and increasing concentration of analyte, while the blanks contain no known amount of analyte. Same reagents are used in sample as well as standard preparation. The method is called external as the standard solutions of known concentrations and unknown sample are analyzed separately under similar conditions. The responses of the blank and standards are compared to remove the blank signal and get the response of the analyte only. The adjusted response is then used to get a calibration equation to find out the concentration of analyte in unknown sample (Stauffer 2018b; Harvey n.d.).

Internal Standard

In this method, a known amount of a chemical species known as internal standard different from the analyte of interest is added to the standard solutions, blanks, and the unknown sample to determine the concentration of analyte. Ratios of the signal of analyte to the internal standard and concentration of analyte to the internal standard are calculated for the blank and the standard solutions. The signal ratios are then plotted against the concentration ratios to get a calibration curve. The equation used for calculating relative response factor is as follows:

$$\frac{Area\ of\ analyte\ signal}{Concentration\ of\ the\ analyte} = F\left(\frac{Area\ of\ standard\ signal}{Concentration\ of\ the\ standard}\right)$$

where F is the response factor (also known as relative response factor) of the analyte which is further used in the calculation of the concentration level of an analyte in analytical environmental measurements (Stauffer 2018b; Harvey n.d.).

Limit of Detection (LOD)/Limit of Quantitation (LOQ)

LOD is the minimum concentration of an analyte that can be measured by the specific method at a stated level of confidence (Shrivastava and Gupta 2011). In other words, LOD can be defined as minimum concentration that an instrument can detect but cannot quantify. LOD is determined by signal-to-noise ratio (S/N) (magnitude of the signal)/(magnitude of the noise). Noise is an unwanted baseline fluctuation in the absence of analyte signal (standard deviation of the background). The amount of analyte necessary to yield a net signal equal to 2 or 3 times the standard deviation of the background (Harvey n.d.; Theodorsson 2008).

The term LOQ is defined as the lowest concentration at which the instrument is able to detect and quantify an analyte. LOQ is the lowest concentration of analyte in a sample that can be determined with acceptable level of accuracy and precision. Sometimes, LOQ is the concentration of lowest calibration point on the calibration graph (Fig. 4).

The determination of LOD and LOQ become significant when determining an undesired or forbidden component of a matrix, for example, pesticide in food and arsenic in drinking water and PAHs in environmental particulate matter samples.

The LOD and LOQ are calculated manually by multiplying standard deviation with suitable factors as represented in Fig. 5 (Eurachem Guide 2014).

Limit of Linearity (LOL): The concentration at which the linearity of the calibration curve departs.

Dynamic Range: The minimum to maximum range of concentration detectable by an instrument.

Accuracy/Trueness

Accuracy of an analytical method is the degree of closeness between the measured/observed value to the true value/reference value. Accuracy is also referred to as trueness of the measurement. The accuracy of the analytical procedure can be determined by "bias." Biasness of a method is the measure of inherent systematic

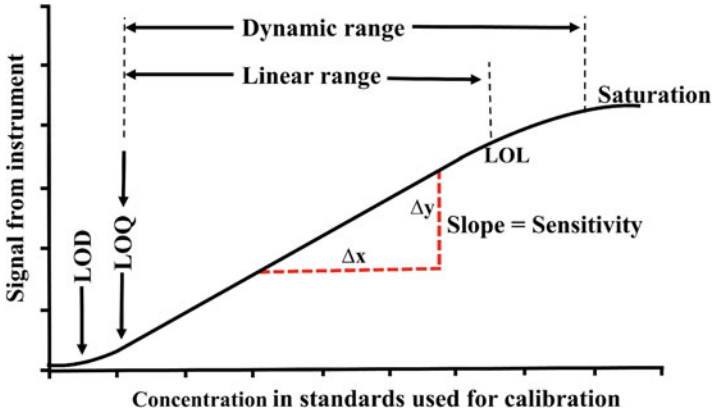

Fig. 4 Calibration curve representing limit of detection, limit of quantitation, and linear range

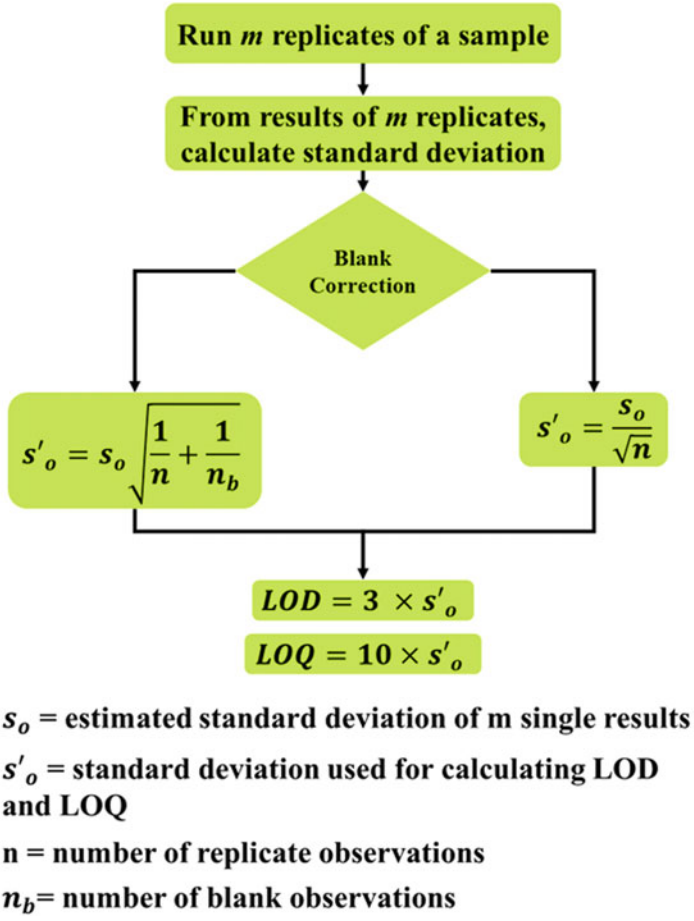

Fig. 5 Steps to determine limit of detection and limit of quantitation

errors, regardless of which laboratory uses it. Bias can be determined by analysis of RMs, by spike studies, or by comparing the obtained results to the results obtained by some other method (Eurachem Guide 2014).

Ruggedness (Robustness)

Ruggedness of an analytical procedure or an instrument is a measure of its capacity to remain unaffected by small yet deliberate variations in method parameters (Draghici and Manciulea 2016).

Precision

Precision is a measure of closeness of a series of measurement results determined under specific conditions. Repeatability and reproducibility are two types of precision. Repeatability measures the variability in values of observations when an

experiment is repeated by the similar person under similar environmental conditions. Reproducibility, whereas, measures the variation in values of observations when the measurements are performed in different laboratories. Reproducibility is not needed frequently in case of single-lab validation. In fact, it proves to be beneficial when many laboratories are going to use an analytical method or standardize it.

Tools to Validate Analytical Methods

To validate an analytical procedure in analytical environmental measurements, various tools that can be used are as follows:

Blanks

Blanks can help in knowing and evaluating the response of the instrument for a particular analyte. Reagent blanks and sample blanks are commonly used to check the measured signal of the instrument. Reagents that are used during analytical processes are analyzed to check their contribution to the measured signal. Sample blanks are sample matrices having no analyte. The measured signals are corrected from the signal obtained for these blanks to eliminate their contribution in measurement results.

Routine Test Samples

The routine analysis performed on the test samples provides useful information on precision, interferences, etc.

Spiked Material/Solution

A known amount of analyte of interest is deliberately added to the sample/standards resulting in spiked solution. Spiking can help in determining the response of the analyte to be measured and calculated in terms of the amount added, although the actual amount of the analyte in the sample was unknown before and after spiking of the sample.

Incurred Material

These are materials in which the analyte of interest may be essentially alien but has been introduced to the bulk at some point prior to the material being sampled. The analyte is thus more closely bound in the matrix than by spiking.

Reagents and Solutions

Proper labelling of reagents and solutions should be done. Deteriorated or outdated reagents and solutions should not be used. The date of opening a reagent or solution should be mentioned on the bottle. Care should be taken if the reagent or solution should be stored in ambient temperature or refrigerated. The expiry date of the reagents and solutions should be checked before use.

Certified Reference Materials (CRMs)/Reference Materials (RMs)

Certified Reference Materials (CRMs) are certified standards traceable to SI units. These standards are used to check the quality and metrological traceability of products, to validate analytical measurement methods, or for the calibration of instruments. Standards make a massive and positive influence to most aspects of human beings. Standards ensure desirable features of products and services such as quality, safety, reliability, environmental friendliness, efficiency, and interchangeability at an economical cost. When products and services meet the expectations, people tend to take this for granted and be unaware of the role of standards. However, when standards are absent, it is soon noticed. It is noticed when products turn out to be of poor quality, do not fit, are incompatible with equipment that one already has, or are unreliable or dangerous. When products, systems, machinery, and devices work well and safely, it is often because they meet standards. And the organization responsible for many thousands of the standards which benefit the world is the International Standards Organization (ISO). The ISO makes the development, manufacturing, and supply of products and services more efficient, safer, and cleaner; facilitates trade between countries and make it **fairer**; provides governments with a technical base for health, safety, and environmental legislation and conformity assessment; shares technological advances and good management practice; disseminates innovation; safeguards consumers, and users in general, of products and services; and makes life simpler by providing solutions to common problems.

The CRMs are used to achieve highest level of accurate measurements, uncertainty, and traceability; however, RMs do not provide the same level of accuracy in the measurements. One of the advantages of using RM is that they are cost-effective and easier to produce. RMs are also ISO compliant, without any certified uncertainty associated with it.

Proficiency Testing

Proficiency testing (PT) also known as external quality assessment is an external way of assuring quality control by comparing the performance of the laboratories. As defined by ISO/IEC 17043:2010, PT is the evaluation of participant performance against pre-established criteria by means of interlaboratory comparisons (ISO/IEC 17043 2017). In other words, PT is an interlaboratory comparison of a method or a standard to establish stated references under same conditions, i.e., equipment, analytical procedure, etc. PT enables to bring out the variations among the laboratories and also helps in identifying the underlying inaccuracies (Eurachem/CITAC Guide 2016). In the process of proficiency testing, the material/matrix is sent to different laboratories for the determination of value for a component. The material is tested by each laboratory, independently, and the results are submitted to the coordinating body. The results of all the participating laboratories are evaluated by the coordinating body, and a performance report is issued to each participating

laboratory. If there is drastic difference in the results of the different laboratories even by following the same method, this process helps in identifying the underlying sources of errors in the method. Proficiency testing for a single method establishes precision. The stated references must meet stability and homogeneity requirements. PT provides laboratories an opportunity to achieve comparability of their results to a reference, which is a very useful method for achieving comparability of measurements. PT is used to determine the reproducibility of a method, to develop reference standards, and to verify new method of analysis (Guide 2021).

The results obtained by different laboratories are often converted to a value/score so that the underlying biasness (if any) can be evaluated for taking proper remedial actions. Two methods that are widely used to assess the PT results are Z-score and E_n score.

Z score: Z-score is a statistical measure of how much the observed result deviates from the population mean. The evaluation identifies and excludes the outliers from the PT results. The Z-score is calculated as:

$$Z = \frac{\chi - X}{\sigma}$$

where, χ = measurement result of a participating lab; X = assigned value (indicates population mean after excluding the outlier values); σ = standard deviation of the population.

If |Z-score| $\leq$2, the result would be regarded as "satisfactory."

2 < |Z-score| <3, the result would be regarded as "questionable."

|Z-score| $\geq$ 3, the result would not be accepted and regarded as "unsatisfactory."

The assessment Z-score and acceptance of the results is based on the concept of normally distributed data. This implies that the normally distributed analytical results would lie within three standard deviations ($\pm 3\sigma$) with a probability of 99.7% and within two standard deviations ($\pm 2\sigma$) with a probability of 95%. The outliers or unsatisfactory results indicate possible inaccuracies in the method of analysis undertaken. The laboratories must examine the cause of underlying problems and eradicate that through proper remedial actions.

E_n score: When the participating laboratories submit their results with uncertainty estimate, the assessment of the results are done with the help of E_n score. The E_n score is calculated as:

$$E_n = \frac{\chi_{lab} - X_{ref}}{\sqrt{U^2{}_{lab} + U^2{}_{ref}}}$$

where χ_{lab} = measurement result of a participating lab; X_{ref} = measurement result of reference lab; U_{lab} = expanded uncertainty (at 95%) of participating lab; U_{ref} = expanded uncertainty (at 95%) of reference lab.

If | E_n| $\leq$1, the results are regarded as "satisfactory."

| E_n| >1, the results are regarded as "unsatisfactory."

Thus, PT also becomes a significant part of QA/QC program that enable to monitor the comparability of results at national and international levels.

Analytical Methods in Environmental Analysis

There are several spectroscopic techniques, i.e., atomic absorption spectrometer (AAS), inductively coupled plasma-atomic emission spectrometer (ICP-AES), inductively coupled plasma-mass spectrometer (ICP-MS), flow injection analyzer (FIA), UV-vis spectrophotometer, X-ray diffraction (XRD), and X-ray fluorescence (XRF), and chromatographic techniques, i.e., high-performance liquid chromatography (HPLC), liquid chromatography-mass spectrometer (LCMS), gas chromatography (GC), gas chromatography-mass spectrometer (GCMS), capillary electrophoresis (CE), capillary electrophoresis-mass spectrometer (CEMS), and ion chromatography (IC), for analytical environmental measurements (Fig. 6) (Draghici and Manciulea 2016). All these techniques require calibration for making measurements. Instrument validation is necessary for any analytical laboratory. Data produced by "faulty" instruments may give the appearance of valid data. The frequency for calibration, re-validation, and testing depends on the instrument and extent of its use in the laboratory. Whenever an instrument's performance is outside the "control limits," reports must be discontinued. Whenever an analysis is to be performed, important questions are needed to be answered for any analytical instrument, e.g., the purpose for instrument being used for, the standards used for the calibration of the instrument traceable to national standards, if the instrument is within specification and documentation to prove this, if the instrument is not within specifications, the deviation factor, and action to be taken to overcome the issue to

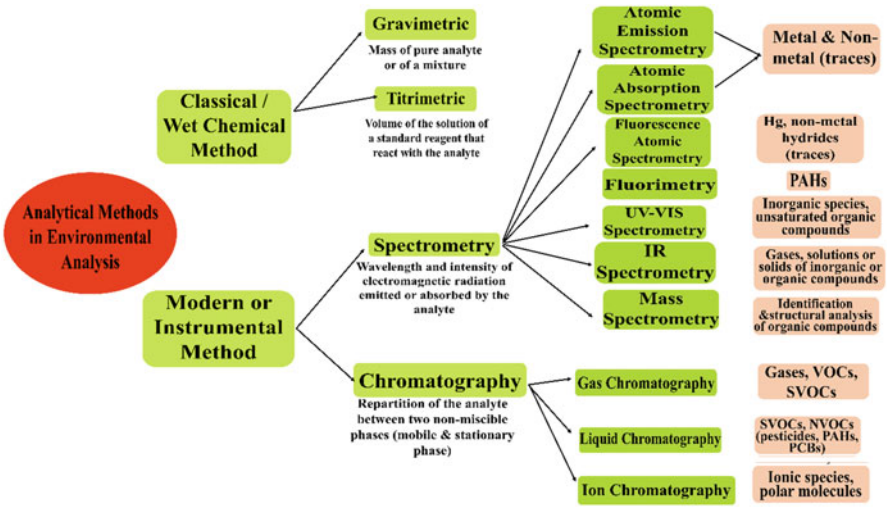

Fig. 6 Various analytical methods in environmental analysis

get the reliable analytical results. All types of analytical methods require calibration for quantification. In analytical environmental measurements, CRMs are available in matrix for a mixture of standard compounds. A minimum of three-point calibration curves are plotted in the range of the analyte to be measured. This helps to know the concentration of an unknown compound.

After considering all the abovementioned parameters during analytical environmental measurements, one may attempt to get the uncertainty for an analyte.

Evaluation of Uncertainty in Environmental Measurements

In 1980, the *Bureau International des Poids et Measures* published recommendation INC-1 on the statement of uncertainties, which dealt with expression of measurement uncertainty (JCGM 2008b). The guide suggests that the uncertainties of a measurement should be grouped and reported into two categories, Type A and Type B, depending on whether they were estimated statistically or based on scientific judgment. Type A uncertainties can be determined by using any valid statistical method like standard deviation in case of series of measurements, least squares to fit a curve to data in order to estimate the parameters of the curve, and their standard deviations one-way or two-way ANOVA (The NIST reference on Constants, Units, and Uncertainty 2022). Type B uncertainties which are determined on scientific judgment can include information from calibration certificate, manufacturer's specification, previous measurement data, etc. (The NIST reference on Constants, Units, and Uncertainty 2022). Further, in 1993, the document was revised as "Guide to the Expression of Uncertainty in Measurement" (GUM) which was later revised and updated accordingly with time. GUM provides a comprehensive outlook on how to deal with uncertainties in metrology across a broad spectrum of measurements. The current version of this guide is published as ISO/IEC Guide 98–3:2008 "Uncertainty of measurement –Part 3: Guide to the expression of uncertainty in measurement" (ISO/IEC 98–3 2008). EURACHEM/CITAC Guide titled "Quantifying Uncertainty in Analytical Measurement" is a document used for estimating the measurement uncertainty in the field of analytical chemistry (Eurachem/CITAC Guide CG4 2012). This guide demonstrates that the concepts and principles of physical measurements stated in ISO GUM guide can also be applied in chemical measurements. The methodology used in the EURACHEM/CITAC guide for determining uncertainty in chemical measurements is in close association with the requirements of ISO/IEC 17025:2017 (Damasceno and Couto 2018). As per EURACHEM/CITAC Guide, the following are the steps that can be followed to determine uncertainty in analytical measurements which is also useful in analytical environmental measurements:

Step 1: Specification of the Measurand In uncertainty estimation, the first and foremost step is to identify "what is to be evaluated and the mathematical expression associated." In other words, "specification of the measurand" which means a clear and explicit statement of what is to be measured is required along with the

mathematical expression that connects the value of the measurand to the various factors on which it depends. The factor may or may not directly influence the measurand.

Step 2: Identify and Characterize Uncertainty Sources The next step consists of collecting a comprehensive list of all the possible sources of uncertainty that may affect the factors of the identified measurand directly or indirectly. According to GUM guide, there are mainly two types of uncertainties, Type A and Type B. Type A uncertainty arises from statistical analysis, and it generally includes uncertainties arising from accuracy, linearity, repeatability, and reproducibility. Type B uncertainty is determined by scientific judgment or any other source of information. One of the most convenient ways to identify possible uncertainty sources is cause-effect diagram or fishbone diagram to establish the relationship among the measurand, variables affecting measurand, and sources affecting these variables. In the process of determining a measurement result, there may appear several uncertainty sources. Even if it is quite challenging to report all the possible uncertainty sources, a few of them are (Eurachem/CITAC Guide CG4 2012) as follows:

Sources of Uncertainty

(a) Ambiguous or incomplete definition of the measurand.
(b) Biased sampling procedures or random variations between different samples resulting in non-representative sampling.
(c) Personal bias in performing and reporting the measurement values of the experiment.
(d) Uncertainty in reference standard which is introduced from the purity of the reference material.
(e) Uncertainty arising due to stability of the reagents and reference material.
(f) Assumptions made during measurement procedures.
(g) Limitations of the instrument.
(h) Repeatability which is measurement of the precision under a set of repeatable conditions.
(i) Reproducibility that determines the variability in measurements under reproducible conditions, for example, operator vs operator, instrument vs instrument, method vs method, day vs day, and environmental conditions vs environmental conditions.
(j) Drift which determines the change in response of a measuring instrument over time.
(k) Environmental conditions at which measurements are made.

Step 3: Quantifying Uncertainty In the process of uncertainty measurement, the next step involves quantification of the estimate of uncertainty by considering all the identified sources in Step 2. In other words, the numerical value that is assigned to each possible source is calculated in this step. Further, all the uncertainties must be specified as a form of standard uncertainty, i.e., standard deviation or standard deviation of the mean. Fig. 7 shows the calculation of standard uncertainty from

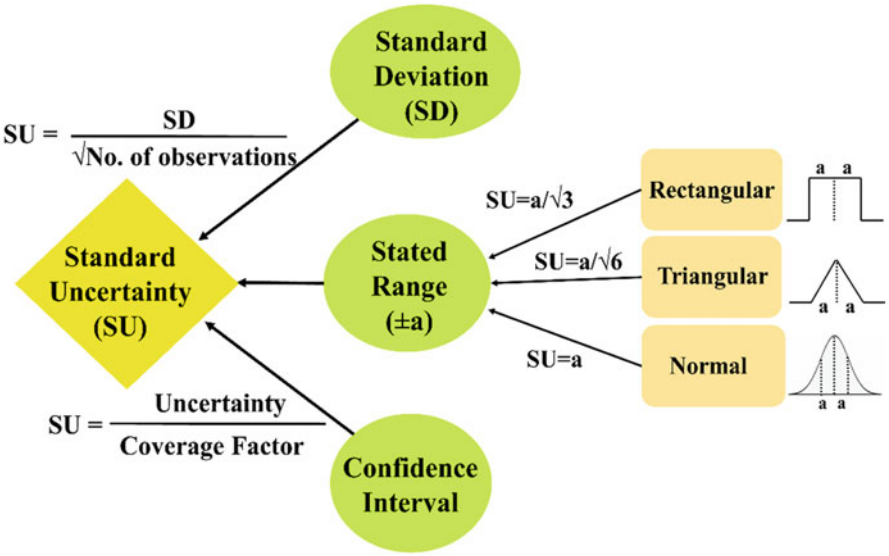

Fig. 7 Calculation of standard uncertainty from various forms of uncertainty

various forms of uncertainty. For this, the estimate of individual uncertainty component is divided by a probability distribution that characterizes the data set. Most common probability distributions are normal or Gaussian, rectangular or uniform, and triangular.

Step 4: Calculating the Combined Uncertainty The standard uncertainties calculated in Step 3 are combined and expressed as combined standard uncertainty which can be expressed as follows:

$CU = \sqrt{u_1}^2 + u_2^2 + \cdots + u_n^2$, where u_1, u_2, …, u_n are individual standard uncertainties.

In other words, root sum of squares of each standard uncertainty can give combined uncertainty.

Reporting Uncertainty As per ISO/IEC 17025:2017 requirements, uncertainty must be reported in the form of expanded uncertainty at the confidence interval of 95% in all situations unless specified.

$u = k \,.\, CU.$

(Result): (x ± U) (units)

where U = expanded uncertainty.

k = coverage factor.

CU = combined standard uncertainty (calculated in step 4).

x = the measurand, of which the uncertainty has to be estimated.

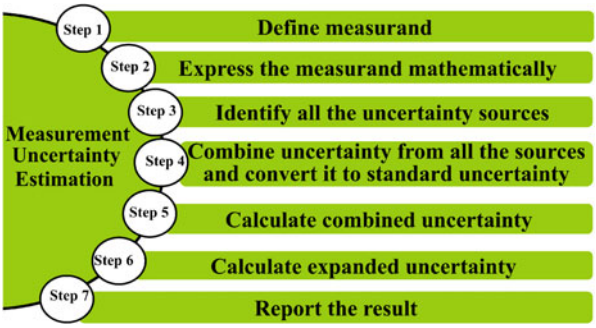

Fig. 8 Steps to evaluate measurement uncertainty

Figure 8 shows the simplified steps that can be followed while estimating the measurement uncertainty.

Environmental Sampling and Chemical Analysis

Environmental pollution especially air pollution is a topic of national and international importance along with transboundary pollution as a major problem in some countries. Can these data be actually compared? Surprisingly, no. Until, there exists a common documented measurement procedure, measured through an unbroken chain of comparisons, no measurements could be compared.

To better understand the concept of measurement traceability and uncertainty in environmental matrices, let's take example of air as matrix and discuss the involved steps from sampling to chemical analysis in detail.

Sampling of the Aerosols

Air consists of nitrogen, oxygen, and other trace gases. Apart from that, there exist numerous tiny suspended solid or liquid particles called aerosols, ranging from few nanometers to 100 μm. Additionally, some gaseous pollutants are also present in the air. These aerosols and gaseous pollutants are responsible for the contamination of the environment. The aerosols directly or indirectly modify the earth's radiation budget and have severe detrimental effects on plants and animals. Thus, aerosols form an important constituent whenever ambient air quality is considered. In most of the countries, aerosols are considered as crucial parameters in their ambient air quality guidelines. Mostly, aerosols having aerodynamic diameter $\leq$2.5 μm and $\leq$10 μm (i.e., $PM_{2.5}$ and PM_{10}) are considered for the regulatory purposes along with their measurement methods.

To monitor the physico-chemical properties of air, the samples of aerosols are collected. The aerosol samples are generally collected by two different methods offline and online measurements. Offline measurement (manual sampling) is the

most widely used method which determines the concentration of particulate matter in air. Offline measurement method further involves active and passive sampling. Active sampling implies that air is sucked at a constant rate using a pump while in passive sampling, the air is allowed to deposit over any collection media at natural rate of deposition. Active sampling can be used to collect total suspended particulate matter (TSPM), respirable suspended particulate matter (RSPM) or PM_{10}, and fine particulate matter ($PM_{2.5}$, PM_1, or finer fraction) on a collection media called "filters." There are various types of filters on which PM can be collected such as glass fiber, cellulose, quartz microfiber, nylon membrane, and Teflon (Watson et al. 2017). The selection of filters depends on the objectives of study due to their specific advantages and disadvantages. Since, passive sampling does not involve a fixed rate of air deposition, determining the volume of air deposited over the collection plate is quite difficult. Thus, the deposition flux (concentration over area) of various species is determined directly. In passive sampling, size selective sampling is not possible. In fact, it determines the average concentration of pollutants deposited over time. The documented steps that are included in active sampling of particulate matter are shown in Fig. 9 (40 CFR Appendix J 1987; 40 CFR Appendix L 1997; National Ambient Air Quality Standards for Particulate Matter 1997; CENTRAL POLLUTION CONTROL BOARD 2013; McMurry 2000).

The initial step involves the conditioning of the filter, which means the filter is to be kept in controlled environment (fixed temperature and humidity) for minimum of 24 hours. The recommended temperature is 25.0 ± 2.0 °C while humidity should be $45 \pm 5\%$. This is done to ensure that the moisture content of the filter should not affect the weight measurements. These standard conditions account for variations arising due to moisture during pre- and post-weighing period as the filter media have high affinity for changes in relative humidity. For the same reason, cellulose and cellulose ester membrane filters are not suggested for gravimetric assessment of particulate matter in ambient air. After conditioning of the filter, its weight is measured using analytical microbalance. During weight measurements, various sources of uncertainties can arise like uncertainty of analytical balance due to limited accuracy, variation arising from multiple measurements (run to run variation), and

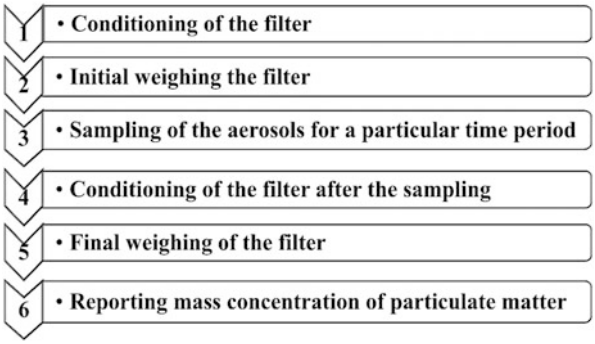

Fig. 9 Steps in sampling of particulate matter

the physical factors like temperature and humidity changes. The uncertainties arising from analytical balance can be determined by its calibration and considering the expanded uncertainty from calibration certificate. The uncertainties arising from the precision of the analytical microbalance can be determined from repeated measurements of the same sample. After weighing, the sampling is performed for a specific period of time to collect the particulate matter on a filter. During aerosol sampling, air is sucked through a size-selective inlet at a constant volumetric flow rate. The volume of the air collected is an important parameter which determines the mass of particulate matter in that particular volume of air. The volume of air collected depends on the flow rate of the instrument and duration for which the instrument is operated. The uncertainty arising from the flow rate and time measurements results in the uncertainty in volume measurements. Thus, calibration of flow rate and time are crucial parameters for traceability in environmental measurements. In these samplers, the individual units of flow measurement and time (time totalizer) are calibrated, which helps to find the uncertainty in the measurement of that parameter. There are various flow meters which are used to calibrate the flow rate of the instrument like manometer/rotameter. The inlet design depends on cut-off diameter which is designed on the basis of 50% particle collection efficiency. This means the inlet is collecting at least 50% of the particles in the size range similar to cut-off diameter. Still, it does not provide the uncertainty in size cut-off measurements. In India, there are no calibration facilities or reference method for calibration for air monitoring instruments. The CPCB uses Federal Reference Method/Federal Equivalent Method of the US EPA for calibration of their samplers (CENTRAL POLLUTION CONTROL BOARD 2013). The manufacturers get their instrument calibrated outside the country or sometimes a calibrated sampler is used as a reference sampler to calibrate other sampler by comparing mass of filters. Lately, CSIR-National Physical Laboratory (CSIR-NPL) has established first and unique National Calibration Facility that would test and calibrate $PM_{2.5}$ and PM_{10} samples for both their cut-off size and flow rate (CSIR-NPL n.d.). CSIR-NPL, India, has also been designated as the apex calibrating body for all such instruments getting manufactured in India. Further, there may be an uncertainty source arising from run to run variation which could be determined by repeated measurements. And finally, after sampling, the filters are removed, conditioned again for 24 hours, and weighed post sampling. Mathematically, the mass concentration of particulate matter is calculated as:

$$\text{Mass concentration} = \frac{\text{final weight of the filter} - \text{initial weight of the filter}}{\text{volume of the air sampled}}$$

After sampling, the sample (filter) is preserved and stored until samples are subjected to pre-treatment process. The main reason behind proper sample storage is to minimize physical and chemical losses in the sample due to volatilization, diffusion, chemical reactions, or photochemical reactions. In such case, the physical and chemical properties of the sample may vary to give erroneous results (Pokhariyal et al. 2019). To avoid this, the samples are sealed in baked aluminum foil and put in

locked polybags and kept inside airtight containers. These airtight containers are kept in refrigerator below 4 °C.

Monitoring Instrument

Figure 10 shows a manual low volume air sampler used to collect $PM_{2.5}$ samples. The air is drawn through inlet using a vacuum pump to pass through a quartz microfiber filter to collect particulate matter on the filter surface. The first impactor has cut-off diameter of 10 μm which allows particles having size $\leq$10 μm to continue to the stream in the sampler. Similarly, the second impactor has cut-off diameter of 2.5 μm which allows particles having size $\leq$2.5 μm to get deposited on the filter surface. The air is drawn at the rate of $1m^3$/hour which is held constant through a critical orifice. The total volume of the air sampled is measured directly with the help of a dry gas meter. The flow rate in this case can be determined by dividing the air volume sampled with total time for which the instrument is operated. The uncertainty in volume measurements can be estimated by calibration of the instrument (Type B) and run to run variation (Type A). The cause-effect diagram shows the various sources of uncertainty in calculating mass concentration of particulate matter (Fig. 11).

Steps to Determine Uncertainty Associated with Calculation of Mass Concentration of $PM_{2.5}$

To understand the uncertainty calculations that are involved in enumeration of mass concentration of $PM_{2.5}$, the $PM_{2.5}$ samples are collected over a 47 mm quartz microfiber filter using a low volume $PM_{2.5}$ sampler as shown in Fig. 10. The filter has to be conditioned for at least 24 hours prior to the sampling so that initial weight of the filter could be taken. The filter is then placed in the sampler operating at flow rate of $1m^3$/hour for a particular period of time. The duration of the sampling depends on the objectives of sampling. After that, the filter is removed, conditioned

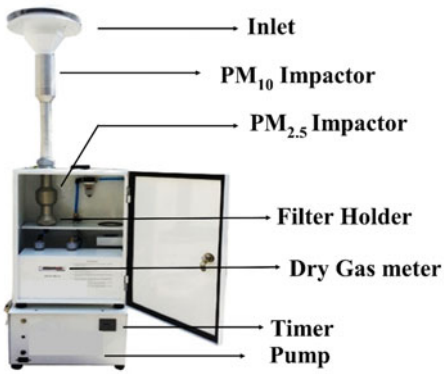

Fig. 10 Low volume $PM_{2.5}$ sampler

for 24 hours, and weighed again. For example, the initial and final weighings are repeated for 5 times, and the weights obtained are m_1, m_2, m_3, m_4, and m_5 mg for initial measurements and m_1', m_2', m_3', m_4', and m_5' mg for final measurements. The initial and final volume readings observed are V_i and V_f m^3, respectively.

As per, Eurachem/CITAC, QUAM 2012 the steps to determine uncertainty are:

1. **The First Step Is Defining the Measurand, i.e., What Is To Be Measured.**

So, the measurand here is mass concentration of $PM_{2.5}$ for which the mathematical equation is as follows:

$$PM_{2.5}\ mass\ concentration = \frac{m_f - m_i}{V}$$

where m_f = final weight of the filter.

m_i = initial weight of the filter.

V = total volume of the air sampled.

The measurand has two major variables which can affect its measurement, i.e., weight change and volume of the air sampled. The weight change of the filter can be calculated as the difference in averages of final and initial weights. Similarly, the volume of the air can be calculated as the difference in final and initial volume readings of the instrument.

Volume of the air sampled (m^3) = final volume –initial volume.

= ($V_f - V_i$) m^3.

$$= \Delta V\ m^3 \tag{1}$$

Thus, $PM_{2.5}$ mass concentration can be given as:

$$mass\ concentration, C_m = \frac{m_f - m_i}{V_f - V_i} = \frac{\Delta m}{\Delta V}\ mg/m^3$$

Since the mass concentration of particulate matter is generally reported in $\mu g/m^3$, the obtained mass concentration C_m can be written as:

$$C_m = \frac{\Delta m}{\Delta V} \times 10^6\ \mu g/m^3$$

$$C_m = X\ \mu g/m^3 \tag{2}$$

2. **Identify All the Uncertainty Sources Affecting the Variables of the Measurand.**

The uncertainties in weight change are due to calibration of the analytical balance (Type B) and repeatability of the measurements (Type A). Similarly, uncertainty in

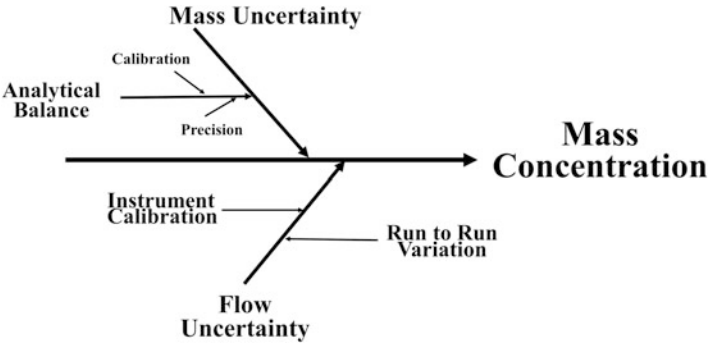

Fig. 11 Cause-effect diagram for determination of uncertainty in mass concentration of particulate matter

volume can be determined from the calibration certificate of the instrument (Type B). The cause-effect diagram is shown in Fig. 11.

3. **Combine All the Uncertainty Sources and Convert Them into Standard Uncertainties.**
   i. Uncertainty arising from repeated measurements of mass or precision of the measurement:
      In a single measurement, the standard deviation is used as standard uncertainty whereas for results subjected to averaging, the standard uncertainty is the standard deviation of the mean (Table 1).
   ii. Uncertainty arising from calibration of analytical balance:
      The calibration certificate of an electronic balance (N19040058/D1.01/C-011) states that at confidence interval of 95% (coverage factor of 2), the expanded uncertainty is ±0.32 mg for load weights up to 100 g while ±0.33 mg for load weights up to 200 g. In this case, the uncertainty would be ±0.32 mg, while to obtain standard uncertainty, the uncertainty would be divided by the coverage factor (2).
   iii. Uncertainty arising from calibration of the instrument:
      The uncertainty in volume readings of the manual low volume $PM_{2.5}$ sampler can be obtained from the calibration certificate of the sampler.

   Now, the uncertainty in mass, arising from two different components, can be combined together and expressed as:

$$u_{mass} = \sqrt{(uncertainty\ due\ to\ precision)^2 + (uncertainty\ due\ to\ calibration\ )^2}$$

where uncertainty due to precision is standard error of the repeated mass measurements = SE and uncertainty due to calibration of analytical balance = $(u_{cal})_{bal}$

Table 1 Repeated weight measurements of filter

Initial weight (mg)	Final weight (mg)
m_1	m_1'
m_2	m_2'
m_3	m_3'
m_4	m_4'
m_5	m_5'
Average initial weight = m_i	Average final weight = m_f

Weight difference = Δm; Standard deviation = SD; Standard error, SE = SD/√n, where n = no. of observations taken for weight measurement. Here n = 5

Thus,

$$(u_{mass})_{total} = \sqrt{(SE)^2 + (u_{cal})_{bal}^{\ 2}} \tag{3}$$

4. **Calculate Combined Uncertainty.**

The combined uncertainty is calculated using root sum of squares of all the relative standard uncertainties which is determined by dividing the standard uncertainty with the numerical value of the variable.

Combined uncertainty in mass concentration

$$\frac{u_{C_m}}{C_m} = \sqrt{\left(\frac{(u_{mass})_{total}}{\Delta m}\right)^2 + \left(\frac{u_{\Delta V}}{V}\right)^2} \tag{4}$$

Equation (4) can be re-written as:

$$u_{C_m} = C_m \times \sqrt{\left(\frac{(u_{mass})_{total}}{\Delta m}\right)^2 + \left(\frac{u_{\Delta V}}{V}\right)^2}$$

where u_{C_m} = uncertainty in mass concentration of $PM_{2.5}$.

C_m = mass concentration of $PM_{2.5}$, can be obtained from Eq. (2).

$(u_{mass})_{total}$ = uncertainty in mass measurement arising from both precision and calibration of analytical balance, can be obtained from Eq. (3).

Δm = difference in averages of final and initial weights (Table 1).

$u_{\Delta V}$ = uncertainty in volume of air sampled.

V = Volume of air sampled, obtained from Eq. (1).

Calculation of Expanded Uncertainty. The expanded uncertainty is determined at the coverage factor of 2 or 95% confidence interval.

Expanded uncertainty at k = 2.

$$U_{exp} = 2 \times u_{C_m}\ \mu g/m^3 \tag{5}$$

To report the measurement uncertainty in mass concentration of $PM_{2.5}$ for 24 hours sampling duration, the reported expanded uncertainty = (X ± U_{exp}) μg/m^3. The measurement result in this case would be traceable to the SI units as each step has its contribution to the total uncertainty. However, the uncertainty related to cut-off diameter is also needed to be considered in addition to volume in m^3.

Chemical Analysis

The atmospheric aerosols contain useful information about the chemistry of atmosphere, particles, and gases which is subject to proper chemical analysis. Thus, after sampling and gravimetric analysis, the filters can be used for qualitative and quantitative estimation of various inorganic and organic chemical components present in the atmosphere. Fig. 12 represents the steps involved in the chemical analysis of the particulate matter.

Inorganic compounds like trace metals, anions, cations, etc. can be detected by using different analytical instruments like atomic absorption spectroscope (AAS), inductively coupled plasma (ICP) with mass spectrometer or optical emission spectrometer as detector, ion chromatograph, etc., while organic compounds can be determined using liquid chromatograph (LC), liquid chromatograph-mass spectrometer (LC-MS), gas chromatograph, gas chromatograph-mass spectrometer (GC-MS), etc. But a common process involved in both inorganic and organic estimation is pre-treatment of the sample, i.e., extraction of the chemical components from the sample matrix (filters) using a suitable solvent. Deionized water or other aqueous solvents can be used as a solvent for inorganic determination (also known as aqueous extraction) while a suitable organic solvent can be used for organic determination based on the compounds of interest. Since the organic compounds possess more complex nature than inorganic compounds, the techniques involved in extraction are dependent on solubility of the analyte in polar or non-polar solvents and

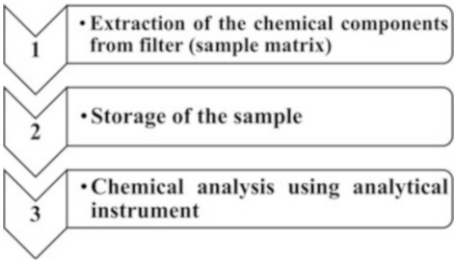

Fig. 12 Steps involved in the chemical analysis of the particulate matter

phase transformations. The procedures for extraction of some compounds are documented by the United States Environmental Protection Agency (USEPA) like Method TO-13A for PAHs, Method TO-4A for pesticides and polychlorinated biphenyls, MLD073 for levoglucosan and galactosan etc., which are usually followed among the scientific community to achieve high accuracy and precision (Centre for Environmental Research Information 1999a, b; California Environmental Protection Agency, California Air Resource Board 2014). The extraction of the chemical components from the sample matrix is not the only aspect involved in enumerating the levels of chemical species in atmospheric particulate matter. The other aspects which are considered for uncertainty in measurement results are sample storage, quality control, and quality assurance during chemical analysis of the samples. The sample storage involves both short-term storage that includes immediate analysis after extraction and long-term storage which includes reanalysis of the samples for precision and accuracy of the data. Thus, the sources of uncertainty during extraction and storage of the sample mainly arises due to accuracy of glasswares which can be determined through calibration and uncertainty values provided on the calibration certificate. In case of absence of calibration certificate of glasswares, repeated measurements for the samples can be done, and the uncertainty can be calculated by standard deviation of the repeated measurements. Another source that will produce uncertainty in the results would be purity of the chemicals provided by the manufacturer.

For analytical measurements, the primary step is to check for the calibration of the instrument to be used for chemical analysis. Accuracy of quantitative results depends on robustness of the analytical method and preparation of samples and standards. Traceability of the reference material and confidence in the property value (concentration) is critical for ensuring quality and accuracy of laboratory test results. ISO/IEC 17025 and ISO guide 34 require that laboratories maintain traceability and uncertainty of their reference materials as a key element of their quality system (ISO/IEC 17025 2017; ISO 17034 2017). The stated concentration of a solution standard is usually accompanied by an uncertainty statement expressed as an expanded uncertainty with a specified coverage factor. Understanding the uncertainty of the concentration of reference standards is critical for laboratories seeking to ensure compliance with ISO/IEC 17025.

Reference Material (RM) or Certified Reference Material (CRM) is needed for the calibration of an analytical instrument. ISO Guide 31:2015 Defines RM as "material sufficiently homogenous and stable with respect to one or more specified properties, which has been established to be fit for its intended use in a measurement process" (ISO Guide 31 2015). And CRM is defined as "reference material, characterized by metrologically valid procedure for one or more specified property, its associated uncertainty and a statement of metrological traceability" (ISO Guide 31 2015). CRMs are mostly produced by metrology institutes of any country like CSIR-National Physical Laboratory (CSIR-NPL), India; and National Institute of Standards and Technology (NIST), United States; and other such NMIs around the

world. The use of RMs and CRMs ensure the validity and accuracy of the analytical methods which help in comparability of the measurement results among various laboratories over time and space.

Different concentration levels of CRM or RM can be used to calibrate the analytical instrument. The slope of the calibration plot is used to enumerate the compounds of interest. The sources of uncertainties that can arise during quantitative estimation of a chemical analyte are uncertainty associated with CRM/RM obtained from the manufacturer's certificate, accuracy of glasswares, steps involved in making working standards from CRM/RM, instrumental accuracy and precision, and run to run variation.

Uncertainty Calculations in Analytical Determination of BaP in $PM_{2.5}$

The concept of measurement uncertainty estimation can also be applied in analytical environmental measurements. Let us understand the application of concept of uncertainty with an example of determination of particle-bound PAHs in the collected $PM_{2.5}$ sample. PAHs are group of non-polar organic compounds that naturally occur in coal and petroleum and emitted in the atmosphere by their ignition (EPA Factsheet for PAHs 2008). They are also emitted by burning of wood, trash, and tobacco. PAHs have proved to be carcinogenic, mutagenic, and teratogenic to various life forms. Thus, they have become a contaminant of significant concern.

Benzo(a)pyrene (BaP), a criteria pollutant in Indian National Ambient Air Quality Standards (NAAQS), is one of the PAHs, which is the most common carcinogen present in the ambient air with established effects on humans (Yadav et al. 2020b). To monitor and regulate the concentration of PAHs in atmosphere, CPCB considers BaP as representative of all PAHs and has set a maximum concentration of 1 ng/m^3 (24-h average) as NAAQS (Yadav et al. 2020b).

To understand the uncertainty calculations involved in determination of the concentration of PAHs (BaP as example) in $PM_{2.5}$ by using GC-MS as analytical instrument, the steps are as follows (Eurachem/CITAC Guide CG4 2012; Sungur et al. 2014).

Defining the Measurand

The measurand is concentration of particle-bound polycyclic aromatic hydrocarbons (PAHs). The concentration of PAHs (in ng/m^3) in aerosol samples is determined by the following equation (Elorduy et al. 2018):

$$Concentration\ \left(ngm^{-3}\right) = \frac{C\ \left(\mu g.mL^{-1}\right) \times V_s\ (mL)}{V_{air}(m^3)} \times 1000 \qquad (6)$$

where C = concentration of the target compound obtained from calibration ($\mu g.\ mL^{-1}$).

V_s = final volume of the sample (mL).

V_{air} = volume of the air sampled (m^3).

Identification of all the Uncertainty Sources Affecting the Variables of the Measurand and Converting Them into Standard Uncertainties

Here, the measurand depends on three main variables, i.e., the concentration of the targeted compound obtained from final volume of the sample, calibration curve, and volume of the air sampled. Thus, the various sources of uncertainty are shown in Fig. 13.

Uncertainty Derived from Final Volume of the Sample The sample matrix is extracted in 20 mL of dichloromethane and sonicated for 3 hours to extract the PAHs. In this case, the uncertainty arises from calibration of the measuring cylinder (Type B). The temperature variation in the laboratory also affects the uncertainty in glasswares (Type A). The extracted fraction is then transferred to Kuderna-Danish extraction apparatus to evaporate the solvent up to 2–3 mL using a heating mantle (~40 °C). Finally, the content is transferred to a vial and reduced to 1 mL using pure nitrogen gas stream. The uncertainty from final volume of sample depends on the uncertainties associated with measuring cylinder and vial which is calculated as follows:

$$\left(\frac{u_{volume\ of\ the\ sample}}{V_s}\right)^2 = \left(\frac{u_{measuring\ cylinder}}{V_{measuring\ cylinder}}\right)^2 + \left(\frac{u_{vial}}{V_{vial}}\right)^2 \tag{7}$$

where $u_{volume\ of\ the\ sample}$= uncertainty in the final volume of the sample.

V_s =volume of the final solution =1 mL.

$u_{measuring\ cylinder}$ = uncertainty in volume measurement of measuring cylinder.

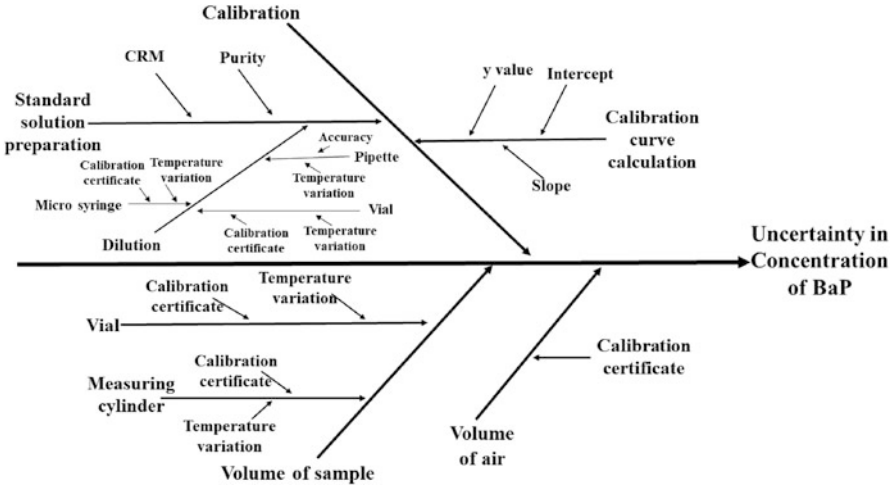

Fig. 13 Cause-effect diagram for determination of uncertainty in concentration of BaP in $PM_{2.5}$

$V_{measuring\ cylinder}$ = volume of measuring cylinder = 50 mL.
u_{vial} = uncertainty in volume measurement of vial.
V_{vial} = volume of vial = 1 mL.

The uncertainty in measuring cylinder due to calibration can be determined from calibration certificate (N19050149/D1.01/C-059) which states that the uncertainty for 50 mL volumetric flask is ±0.050 mL at 95% confidence interval. Thus, for standard uncertainty, the uncertainty given is divided by a factor of 2.

The temperature variation in the laboratory can affect the volume measurements of the glasswares. The EURACHEM/CITAC 2012 Guide mentions/states that a variation of ± 3 °C is allowed during the chemical measurements which can be calculated as follows (Yenisoy-Karakas 2012):

$$u_T = \frac{3 \times V \times Q}{1.73} \tag{8}$$

where V = measured volume and Q = coefficient of the volume expansion of the solvent used.

And for DCM, Q = 1.44x 10^{-3} $°C^{-1}$ (at 25 °C).

So the total uncertainty in volume measurement of measuring cylinder can be given as follows:

$$u_{measuring\ cylinder} = \sqrt{(u_{cal})^2 + (u_T)^2} \tag{9}$$

Similarly, volume of vial =1 mL.

Uncertainty in vial due to calibration can be obtained from calibration certificate (N19050149/D1.01/C-060) which states that the uncertainty for 1 mL vial is ±0.015 mL at 95% confidence interval. Thus, for standard uncertainty, the uncertainty value given is divided by a factor of 2.

Uncertainty in vial due to temperature variation, u_T, can be obtained from Eq. (8).

So, the total uncertainty in volume measurement of vial can be given as:

$$u_{vial} = \sqrt{(u_{cal})^2 + (u_T)^2} \tag{10}$$

And the relative uncertainty in volume of the sample can be calculated by putting Eqs. (9) and (10) in Eq. (7).

The account of loss or addition of the analyte (recovery) during pre-treatment procedures (extraction) is quite a challenging process but it should be considered while estimating the measurement uncertainty. In present case, the recovery of the sample is not considered for calculations due to some limitations. However, it can be estimated by adding a known amount of standard before the sample pre-treatment procedure and then performing the extraction. The concentration of the added standard can be determined from the analytical instrument and the percentage recovery could be estimated.

Uncertainty Derived from Calibration The uncertainty associated with the calibration can be determined by combining the individual uncertainties arising from preparation of calibration standards and fitting of calibration curve.

So, the uncertainty arising from calibration can be expressed as:

$$\left(\frac{u_{Cal}}{C}\right)^2 = \left(\frac{u_{std}}{C_{std}}\right)^2 + \left(\frac{u_{Cx}}{C_x}\right)^2 \tag{11}$$

where u_{std} is the uncertainty from preparation of the standard solution, u_{Cx} is the uncertainty from linear least square fitting, C_{std} is the concentration of prepared standard solution, and C_x is the predicted analyte concentration.

Uncertainty Derived from Preparation of Calibration Standards The preparation of working calibration standards from CRM involves an intermediate step of stock solution preparation. This means a stock solution has to be prepared from CRM by dilution first and then working calibration standards of different concentrations prepared by serial dilution of the stock solution formed. The sources of uncertainty contributing in preparation of working standards include uncertainty of CRM as per the certificate (Type B), stock solution preparation, and series of dilution. The stock solution and dilution series further consist of uncertainties arising from vial calibration (Type B), micro syringes calibration (Type B), accuracy of the pipettes (Type A), and purity of the solvent used (Type A). Thus, relative uncertainty in preparation of standards can be given by:

$$\left(\frac{u_{std}}{C_{std}}\right)^2 = \left(\frac{u_{CRM}}{C_{CRM}}\right)^2 + \left(\frac{u_{Dilution}}{Dilution\ Factor}\right)^2 + \left(\frac{u_{purity}}{Purity\ of\ the\ solvent}\right)^2 \tag{12}$$

where,

$$(u_{dilution})^2 = \left(\frac{u_{microsyringe}}{V_{microsyringe}}\right)^2 + \left(\frac{u_{pipette}}{V_{pipette}}\right)^2 + \left(\frac{u_{vial}}{V_{vial}}\right)^2 \tag{13}$$

where u_{CRM} is the uncertainty of CRM; $u_{Dilution}$ is the uncertainty due to dilution, u_{purity} is the uncertainty associated with the purity of solvent used, and $u_{microsyringe}$, $u_{pipette}$, and u_{vial} are the uncertainty due to micro syringe, pipette, and vial used, while C_{CRM} is the concentration of CRM and $V_{microsyringe}$, $V_{pipette}$, and V_{vial} are the volumes of micro syringe, pipette, and vial.

A CRM of mixture of 18 PAHs (CRM 47543, Sigma Aldrich) having concentration 2000 μg/mL is used for making the calibration standards. To prepare the stock solution of 20 μg/mL, the CRM having 2000 μg/mL concentration has been diluted two times. Further, the calibration standards of levels 0.1, 0.5, 1, 2, and 5 μg/mL can be prepared by serial dilution.

Initially a solution of 200 µg/mL is prepared by diluting 100 µL of the 2000 µg/mL CRM/standard with 900 µL of DCM. Further 20 µg/mL is prepared by diluting 100 µL of the 200 µg/mL CRM/standard with 900 µL of DCM. A series of working standards ranging from 0.1, 0.5, 1, 2, to 5 µg/mL are prepared by diluting the stock solution and the successive standards formed. For example, 1 mL working standard of 5 µg/mL can be prepared by diluting 250 µL of 20 µg/mL with 750 µL of DCM, 2 µg/mL can be prepared by diluting 400 µL of 5 µg/mL with 600 µL of DCM, 1 µg/mL can be prepared by diluting 500 µL of 2 µg/mL with 500 µL of DCM, 0.5 µg/mL can be prepared by diluting 500 µL of 1 µg/mL with 500 µL of DCM, and finally, 0.1 µg/mL can be prepared by diluting 200 µL of 0.5 µg/mL with 800 µL of DCM.

To prepare the standards, micro syringes (HAMILTON, 500 µL), micro-pipettes (Eppendorf, 100–1000 µL), and vials (Borosil) are used. Tables 2, 3, and 4 summarize the uncertainties obtained from micro-syringe, pipette, and vial for each dilution stage. Table 5 shows uncertainty arising from calibration standards of different levels.

Table 2 Uncertainty associated with microsyringe used during preparation of CRM solution for each dilution stage

Dilution stage	Concentration of the solution µg/mL	Volume of syringe used µL	Measured volume µL	$(u_{microsyringe})_{cal}$ µL	$(u_{microsyringe})_T$ µL	$(u_{microsyringe})$ total µL
1	200	500	100	0.075	0.25	0.79
2	20	500	100	0.075	0.25	0.79
3	5	500	250	0.075	0.62	0.97
4	2	500	400	0.075	0.99	1.24
5	1	500	500	0.075	1.25	1.46
6	0.5	500	500	0.075	1.25	1.46
7	0.1	500	200	0.075	0.5	0.90

Table 3 Uncertainty associated with autopipette used during preparation of CRM solution for each dilution stage

Dilution Stage	Concentration of the solution µg/mL	Volume of pipette used µL	Measured Volume µL	$(u_{pipette})$ µL	$(u_{pipette})_T$ µL	$(u_{pipette})$ total µL
1	200	1000	900	0.00035	2.25	2.25
2	20	1000	900	0.00035	2.25	2.25
3	5	1000	750	0.00035	1.87	1.87
4	2	1000	600	0.00035	1.50	1.5
5	1	1000	500	0.00035	1.25	1.25
6	0.5	1000	500	0.00035	1.25	1.25
7	0.1	1000	800	0.00035	2.00	2.00

Table 4 Uncertainty associated with vial used during preparation of CRM solution for each dilution stage

Dilution stage	Concentration of the solution µg/mL	Volume of vial mL	(u_{vial}) cal	$(u_{vial})_T$	(u_{vial}) final mL
1,2,3,4,5,6,7	200; 20; 5; 2; 1; 0.5; 0.1	1	0.075	0.0025	0.075

Table 5 Uncertainty associated with calibration standards

Concentration of the solution µg/mL	Dilution factor	Concentration of CRM µg/mL	u_{CRM} µg/mL	$\left(\frac{u_{std}}{C_{std}}\right)$
5	4	2000	29.5	0.026
2	2.5	2000	29.5	0.029
1	2	2000	29.5	0.031
0.5	2	2000	29.5	0.031
0.1	5	2000	29.5	0.025

Uncertainty in microsyringe due to calibration can be taken from calibration certificate (N19050149/D1.01/C-058) which states that the uncertainty value for 500 µL microsyringe is ±7.5 µL at 95% confidence interval. Therefore, for obtaining standard uncertainty, the value is divided by a factor 2.

Uncertainty in microsyringe due to temperature variation, u_T, can be calculated using Eq. (8). The uncertainty in microsyringe due to temperature variation is calculated for each dilution stage where measured volume is 100, 100, 250, 400, 500, 500, and 200, respectively, as shown in Table 2.

So, the total uncertainty in volume measurement of microsyringe can be given as follows:

$$u_{microsyringe} = \sqrt{(u_{cal})^2 + (u_T)^2}$$

Uncertainty in pipette due to accuracy can be estimated from manufacturer's specifications which states that the accuracy for 1000 µL volume is ±0.06%. To get the standard uncertainty, the accuracy value is divided by √3 by using rectangular probability (Fig. 7).

Uncertainty in pipette due to temperature variation, u_T, can be calculated using Eq. (8). The uncertainty in pipette due to temperature variation is calculated for each dilution stage where measured volume is 900, 900, 750, 600, 500, 500, and 800, respectively, as shown in Table 3.

So, the total uncertainty in volume measurement of pipette can be given as:

$$u_{pipette} = \sqrt{(u_{cal})^2 + (u_T)^2}$$

Since there are a total of seven stages of dilution, thus Eq. (13) can be re-written as:

$$(u_{dilution})^2 = \left\{ \left(\frac{u_{microsyringe}}{V_{microsyringe}}\right)^2 + \left(\frac{u_{pipette}}{V_{pipette}}\right)^2 + \left(\frac{u_{vial}}{V_{vial}}\right)^2 \right\}_1 + \left\{ \left(\frac{u_{microsyringe}}{V_{microsyringe}}\right)^2 + \left(\frac{u_{pipette}}{V_{pipette}}\right)^2 + \left(\frac{u_{vial}}{V_{vial}}\right)^2 \right\}_2 + \left\{ \left(\frac{u_{microsyringe}}{V_{microsyringe}}\right)^2 + \left(\frac{u_{pipette}}{V_{pipette}}\right)^2 + \left(\frac{u_{vial}}{V_{vial}}\right)^2 \right\}_3 + \left\{ \left(\frac{u_{microsyringe}}{V_{microsyringe}}\right)^2 + \left(\frac{u_{pipette}}{V_{pipette}}\right)^2 + \left(\frac{u_{vial}}{V_{vial}}\right)^2 \right\}_4 + \left\{ \left(\frac{u_{microsyringe}}{V_{microsyringe}}\right)^2 + \left(\frac{u_{pipette}}{V_{pipette}}\right)^2 + \left(\frac{u_{vial}}{V_{vial}}\right)^2 \right\}_5 + \left\{ \left(\frac{u_{microsyringe}}{V_{microsyringe}}\right)^2 + \left(\frac{u_{pipette}}{V_{pipette}}\right)^2 + \left(\frac{u_{vial}}{V_{vial}}\right)^2 \right\}_6 + \left\{ \left(\frac{u_{microsyringe}}{V_{microsyringe}}\right)^2 + \left(\frac{u_{pipette}}{V_{pipette}}\right)^2 + \left(\frac{u_{vial}}{V_{vial}}\right)^2 \right\}_7 \quad (13a)$$

The purity of the solvent, dichloromethane, used is ≥99.9% as per the manufacturer's specification that means the purity lies between 99.9 and 100%. And whenever a range is specified like a ± Δa, then, the uncertainty can be expressed as $\Delta a = \frac{max - min}{2}$. Thus, here the maximum purity can be 100% while minimum purity is assigned as 99.9%. Since there is no other information about the sources of uncertainties, rectangular distribution is applied.

$$\Delta a = \frac{100 - 99.2}{2} = 0.05$$

$$u_{purity} = \frac{0.05}{\sqrt{3}} = 0.03 \quad (14)$$

For uncertainty in CRM, a single compound, i.e., BaP, has been considered for calculation purpose. The uncertainty associated with BaP is ±59 μg/mL at coverage factor of 2 (CRM 47543 calibration certificate) for CRM having concentration of 2000 μg/mL.

$$u_{CRM} = \frac{59}{2} = 29.5\ \mu g/mL \quad (15)$$

$\left(\frac{u_{std}}{C_{std}}\right)$ is calculated by using Eqs. (13a), (14), and (15) in Eq. (12).

Uncertainty Derived from Linear Regression Fitting of the Calibration Curve The uncertainty derived from linear least square fit of calibration curve can be calculated as follows:

$$\left(\frac{u_{Cx}}{C_x}\right)^2 = \left(\frac{u_{y\ value}}{mean\ value\ of\ y}\right)^2 + \left(\frac{u_{slope}}{slope\ value}\right)^2 + \left(\frac{u_{intercept}}{intercept\ value}\right)^2 \quad (16)$$

where $u_{y\ value}$ is the uncertainty of linear regression curve and u_{slope} and $u_{intercept}$ are the uncertainty obtained for slope and intercept of calibration curve. Thus, the relative standard uncertainty from linear regression fitting of the calibration curve can be obtained using Eq. (16).

Determining Combined Uncertainty

The main sources of uncertainty in determining the concentration of particulate bound PAHs is calibration curve which is used to determine the concentration of the analyte, final volume of the sample, and volume of the air sampled. The combined uncertainty is calculated by combining the uncertainties from individual components using the following equation:

$$\frac{u_{combined}}{C} = \sqrt{\left(\frac{u_{cal}}{C}\right)^2 + \left(\frac{u_{volume\ of\ the\ sample}}{V_s}\right)^2 + \left(\frac{u_{volume\ of\ the\ air}}{V_{air}}\right)^2} \quad (17)$$

Equation (17) can be re-written as:

$$u_{combined} = C \times \sqrt{\left(\frac{u_{cal}}{C}\right)^2 + \left(\frac{u_{volume\ of\ the\ sample}}{V_s}\right)^2 + \left(\frac{u_{volume\ of\ the\ air}}{V_{air}}\right)^2}$$

where u_{cal} is the uncertainty from calibration (μg), C is the amount of analyte (ng/m^3), $u_{volume\ of\ the\ sample}$ is the uncertainty in final volume of the sample (μL), V_s is the final volume of sample (μL), $u_{volume\ of\ the\ air}$ is the uncertainty in air volume sampled (m^3), and V_{air} is the total air volume sampled (m^3).

The amount of analyte, C (ng/m^3), can be estimated using Eq. (6).

Determining Expanded Uncertainty

And expanded uncertainty at 95% confidence interval can be given by $U_{exp} = 2 \times u_{combined}\ ng/m^3$.

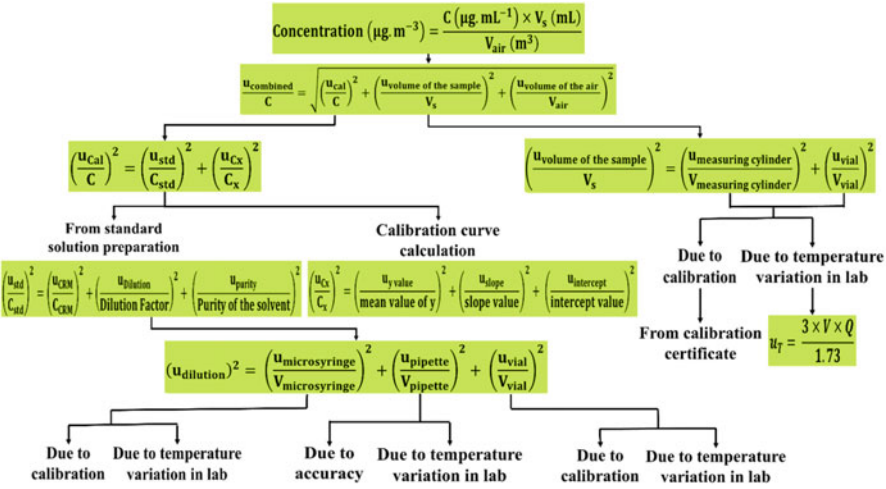

Fig. 14 Flowchart depicting the calculations involved in uncertainty estimation in analytical measurement

Reporting Result

The uncertainty in concentration of Benzo(a)pyrene in sample may be expressed as follows:

Reported Uncertainty $= (C \pm U_{exp})\ ng/m^3$.

Finally, the calibration curve, volume of the sample, and volume of the air sampled are identified as the sources of uncertainty in determination of particle bound PAHs in aerosol sample.

The above process of calculation of uncertainty at each stage is summarized as a flowchart in Fig. 14.

Conclusion

Accurate and reliable measurements form the basis of growth and development among various sectors of the country. Particularly, the fields of environmental protection, healthcare, and medical facilities along with food and nutrition require quantitative analyses. Metrological traceability and measurement uncertainty are becoming the latest buzzwords in the arena of measurement science to attain quality data. Both are required to remove the biases of measurements made in different time and space. The idea of traceability is well-rooted in the field of physical measurements; however, in chemical/biological/environmental measurements, it still needs propagation. Since, the qualitative and quantitative estimation in environmental measurements form the basis of environmental legislations, the uncertainty estimation of the measurement results become imperative. The uncertainty estimation in analytical environmental measurements requires traceability of the measurements

through documented references and unbroken chain of comparisons. However, the field of analytical environmental measurements is quite vast as it contains a variety of matrices which make sampling and sample handling completely different for different measurements. Thus, there cannot be a single standard operating procedure that could apply in all cases. Consequently, estimation of uncertainty becomes a challenging process in the field of environmental measurements.

An example of determination of mass concentration and particle-bound PAHs (BaP) in $PM_{2.5}$ with the calculation of uncertainties associated at each step has been discussed. The uncertainty measurements of mass concentration have two major variables, i.e., uncertainty in mass measurements of the filter (sampling substrate) and uncertainty in volume of air sampled. Uncertainty in mass measurements originates from calibration of the analytical balance and repeatability of the measurements, while uncertainty in volume measurement depends on calibration of instrument flow meter. At present, the manual air monitoring instruments generally do not have calibrated cut-off diameter of the impactor/sampler.

In the case of analytical determination of environmental measurements, the calibration curve, final volume of the sample maintained after extraction, and volume of the air sampled are key contributors in uncertainty in determination of particle bound PAHs (BaP) in aerosol samples. The identified missing link in the present study is the determination of uncertainty due to cut-off diameter of the instrument used for sampling of the particulate matter. Additionally, sample recovery during the extraction procedures are not considered for the uncertainty calculations.

Thus, in analytical environmental measurements, accurate determination of amount of an analyte is very challenging and difficult to be achieved practically. For this purpose, the missing links in the analytical environmental measurements are needed to be addressed. Still there exists a need of investigation regarding the references being used in the field of environmental studies so that accurate measurements could be achieved without any compromises. This will ensure the traceability of the measurement results to the references and accurate determination of the environmental contaminants.

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Particulate Matter Measurement Techniques

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Kritika Shukla and Shankar G. Aggarwal

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Abstract

For the better understanding of potential environmental and human health impacts of airborne particulate matter (PM), the accurate and precise measurement of ambient PM is prerequisite. Various methods and instruments are available, ranging from filter-based sample collection for off-line laboratory analysis to on-line instruments that detect the airborne particles and give information of

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size distribution, particle mass concentration, and chemical data in real time. Different measurement strategies, sampling devices, and instruments based on different principles are used depending on the scientific objectives of the study. The improvements in time resolution achieved by the continuous methods have proven to be useful in monitoring of PM concentration, characterizing ambient PM, and are becoming essential in allowing decision-makers to make policies and scientists to investigate sources of particulate pollution and to probe into the dynamics and mechanisms of aerosol formation in the atmosphere. It is important for the researchers working in air quality field to have a detailed understanding of the principle and limitations associated with different PM measurement techniques. In this view, here this chapter provides overview of the measurement principle of techniques available and used worldwide for PM measurement.

Keywords

Particulate matter · Aerosol measurement techniques · Manual method · Automated method · Chemical characterization

Introduction

Airborne particulate matter severely effects human health, and its control is still a challenging task. Many countries have categorized particulate matter (PM) as criteria pollutant under their National Ambient Air Quality Standards (NAAQS). Federal Reference Method (FRM) based on gravimetric measurement and Federal Equivalent Method (FEM) based on the beta attenuation and tapered element oscillating microbalance method for PM measurement are included in NAAQS of different countries (e.g., EN 12341 2014; US EPA 2017; Aggarwal et al. 2013). Apart from these, there are other measurement methods available based on optical property, electrical mobility, time-of-flight of particle, and many more. Therefore, it is important for the researchers making aerosol measurement to be able to access the details regarding measurement technique, size range, precision, accuracy, and appropriate application of each instrument available in the market to have a more refined choice of instrument according to the requirement of study.

Different techniques use different principle to determine particle size leading to difference in size reported for the same particle. The particle aerodynamic size obtained with impactors and aerodynamic particle sizers depend on particle shape and density while the electrical mobility size determined by electrostatic classification depends on particle shape and size. Optical sizes obtained by the amount of light scattered by individual particles depend on particle refractive index and shape. Also, converting sizes from one measurement technique to another is essential in utilizing aerosol measurements but it involves significant errors. Therefore, it is important to understand the available techniques used to measure sizes, PM concentration, and properties of aerosols including shape, density, and refractive index. The objective of this chapter is to provide readers with an up-to-date overview of the available PM

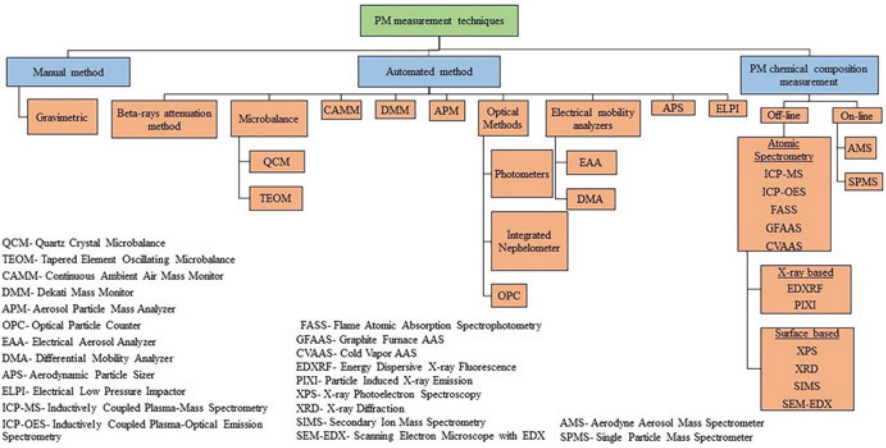

Fig. 1 Techniques and instruments for PM measurement

measuring techniques, substantiated with a technical discussion about the principle of measurement, features, and limitations intrinsic of each technique. Choosing the appropriate instrument for a particular application is of critical importance, and it is a function of a number of associated parameters such as the particle properties being studied, sampling and measurement time, access to available facilities, and the costs associated to equipment acquisition. Figure 1 classifies particle measuring instruments into manual method, automated methods, and methods for PM chemical composition measurement.

Manual PM Measurement Method

Gravimetric method, also known as manual method for the sampling of PM involves drawing of ambient air through filter material using a pump with subsequent quantification of particle mass with weighing balance. This method is considered as a primary or reference method and reliable in measurement as it is traceable to SI unit (Aggarwal et al. 2013). This manual method is used to measure the particulate matter concentration worldwide. The process involves weighing a blank filter on a laboratory balance, drawing ambient air through it at known flow rate and duration, re-weighing the filter, then the blank filter weight is subtracted, and the weight gain is divided by the sample volume to determine the PM concentrations over the sample period. The basic components of manual sampling system include a size-selective inlet, filter substrate, filter holder, pump, and flow controller (Chow 1995). The size-selective inlet defines the sampled size fraction of $PM_{2.5}$ and PM_{10} (particles with aerodynamic diameters $\leq$2.5 and $\leq$10 μm). They are designed to deliver all particles smaller than a specified diameter with 50% collection efficiency (i.e., D_{50}). Size-selective inlets use inertial, diffusion, or centrifugal classification. The most commonly used classifiers for selective $PM_{2.5}$ collections are Very Sharp Cut Cyclone

(VSCC) (Kenny et al. 2004) and Well Impactor Ninety-Six (WINS) (Peters et al. 2001).

Inertial Impaction

Inertial separators are widely used for the collection of fine and coarse particles. The general principle of inertial separation is that the particulate-laden air is forced to change the direction because of the obstacle. As the air changes direction, the inertia of the particles causes them to continue in the original direction and be separated from the air stream (Fig. 2). Impaction devices accelerate the aerosol through circular jet or slit toward an impaction substrate. Adhesion of particles is caused by the electrostatic attraction and molecular surface phenomena known as the van der Waals forces. The impactor separates the sampled particulate mass into the particles larger than the cutoff size (collected on the impaction plate) and the particles smaller than cutoff size (collected on the filter). The collection surface in many impaction devices is coated with a thin film of oil or light grease to increase particle retention or overcome the re-bouncing issue. The probability that a particle will diverge from the air streamlines can be expressed as a function of the Stokes number (Stk) defined by:

$$\mathrm{Stk} = \frac{\rho_\mathrm{p} C_\mathrm{c} D_\mathrm{p}^2 U}{9\eta d_\mathrm{n}} \tag{1}$$

where ρ_p and D_p are the particle density and diameter, U and η are the relative velocity and dynamic viscosity of the carrier air respectively, d_n is the nozzle diameter, and C_c is the slip correction factor. The higher the Stokes number, the more efficient the inertial transport is and the more likely particles will deposit by inertial impaction.

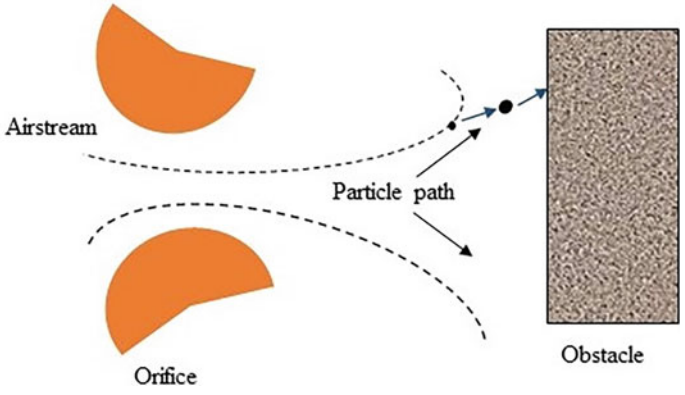

Fig. 2 Particle collection by impaction

The following section discusses various impactor-based devices:

Cascade Impactor

It is a multi-stage impactor containing stages with many, defined evenly spaced nozzles of different sizes in each stage. Impactors (stages) are arranged in the order of decreasing cutoff size by reducing the size of nozzle with the largest cutoff size first (Fig. 3). Filter at each stage can be removed for gravimetric and chemical measurement of the collected particles. This gives distribution of total mass fraction in each aerodynamic size range. In cascade impactors, particles are lost in the first two stages primarily due to inertial removal at bends in the flow path. These inter-stage losses can be reduced by modifying impactor in a way to minimize sharp bends in the inter-stage flow path of the initial stages by operating at a lower flow rate. Another problem with impactor is the re-entrainment, i.e., buildup of particle deposit

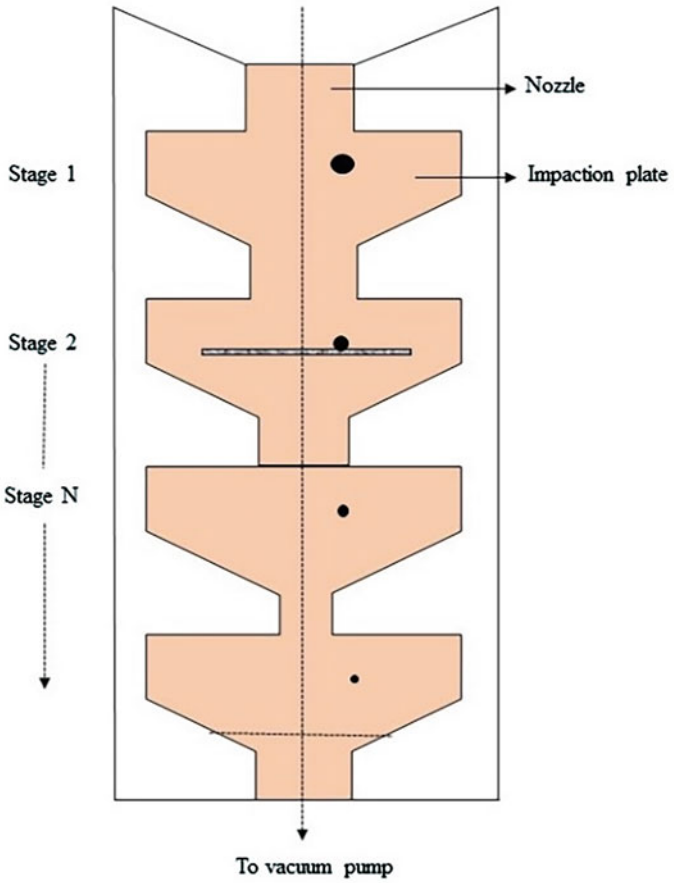

Fig. 3 Schematic of cascade impactor

which changes the flow geometry and the cutoff size. Andersen sampler is an eight-stage cascade impactor, which is used to measure the different particle size concentration ranging from the 0.4 μ m to 9.0 μ m. The stages of Andersen sampler are numbered 0 to 7, with different size range orifice divisions according to the particle aerodynamic diameter as stage 0 (>9.0 μm), stage 1 (5.8–9.0 μm), stage 2 (4.7–5.8 μm), stage 3 (3.3–4.7 μm), stage 4 (2.1–3.3 μm), stage 5 (1.1–2.1 μm), stage 6 (0.7–1.1 μm), stage 7(0.4–0.7 μm), and backup stage (<0.4 μm). Every stage of the impactor contains a series of nozzles through which the air is drawn, directing the particles toward the surface of the collection plate for that particular stage. Particles having enough inertia will impact on that particular stage collection plate, while smaller particles will remain entrained in the air stream and pass to the next stage. The stages are assembled in a stack or row in the order of declining particle size. The sampling flow rate of the sampler is 28.3 lpm. Another group of cascade impactors uses micro-orifice nozzles for jet acceleration, commonly known as Micro Orifice Uniform Deposit Impactor (MOUDI). This sampler is equipped with combinations of impactor stages, covering wide range of particle sizes and collecting particles smaller than 0.056 μm. It reduces particle bounce by rotating the impaction plate relative to the nozzles, thus resulting in uniform deposit and prevents heavy build-up of particles under the nozzles (Maenhaut et al. 1993).

Virtual Impactor

The problem of bounce, overloading, and non-uniform particle deposition can be overcome by virtual impactors. Here, instead of the larger particles impacting onto a solid impaction plate, they impact into a collection probe and are later collected on a filter downstream (Fig. 4). Large particles impact through a virtual surface and smaller particles are carried radially away from the jet axis by the major air flow and are collected on filter downstream. Most of the flow reverses in the collection probe carrying with it the particles that are smaller than the cutoff size. In virtual impactor, particles larger than cutoff size and those smaller than the cutoff size are collected on separate filters. Virtual impactors, beside an inertial classifier, can also be considered a particle concentrator for particles larger than cutoff size and this is useful when sampling particles that are of low concentration or sampling particles into an instrument that requires greater concentrations (Bergman et al. 2005; Keskinen et al. 1987; Liebhaber et al. 1991; Romay et al. 2002; Wu et al. 1989). Virtual impactors usually have only one or two stages due to complexity of controlling two flows for each stage. The first device to incorporate a virtual impactor was described by Conner (1966).

Cyclone

Cyclone is designed such that air drawn through it moves in either a circular or a helical path of decreasing radius, thereby increasing its collection efficiency for small particles. Cyclones are cone-shaped devices in which the sampled aerosol enters tangentially, rotates several times about the axis, and exits vertically through

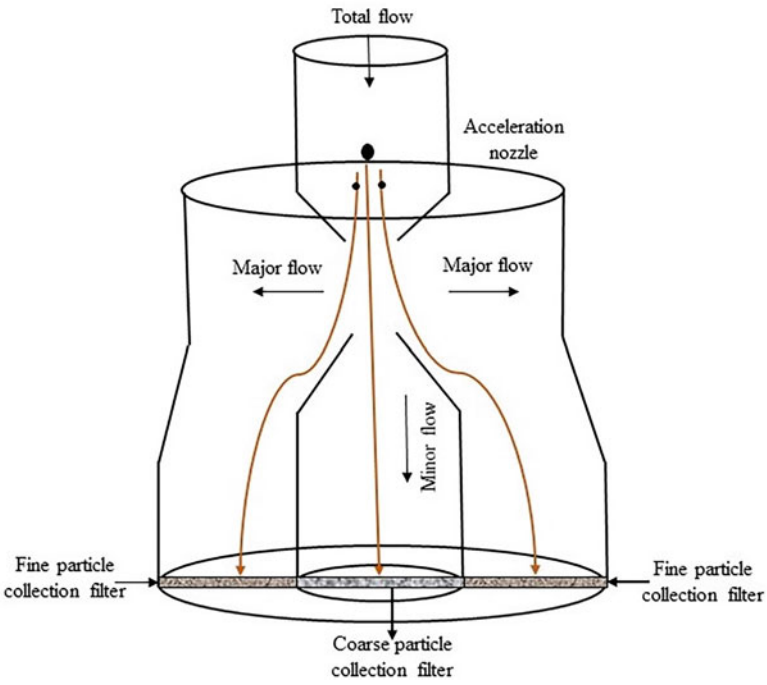

Fig. 4 Flow schematic of co-cylindrical virtual impactor

an opening located on the axis at the top. As the air stream surges through the cyclone, particles adhered to the cyclone's wall as the velocity approaches zero due to the centrifugal force. In most cyclone samplers, liquid particles adhere to the wall, while solid particles settle into a dust collection cavity (cup). The collection efficiency of a cyclone sampler is dependent on flow rate. VSCC $PM_{2.5}$ fractionator is a centrifugal separation device used to collect $PM_{2.5}$. In VSCC, there is no need for the application of oil as required in WINS impactor.

Diffusion

Diffusion technique is useful in the collection and size distribution of nanoparticles. Small particles do not settle down through sedimentation or impaction process, while due to Brownian diffusion, they coagulate easily. Diffusion samplers separate particles by differential diffusion mobilities. There are two types of diffusion samplers used in air sampling, a diffusion battery to measure the size distribution of sub-micrometer particles and a diffusion denuder to separate and collect gases or vapors from airborne particles. The rate of deposition of particles onto a unit area of surface is determined by applying Fick's first and second law:

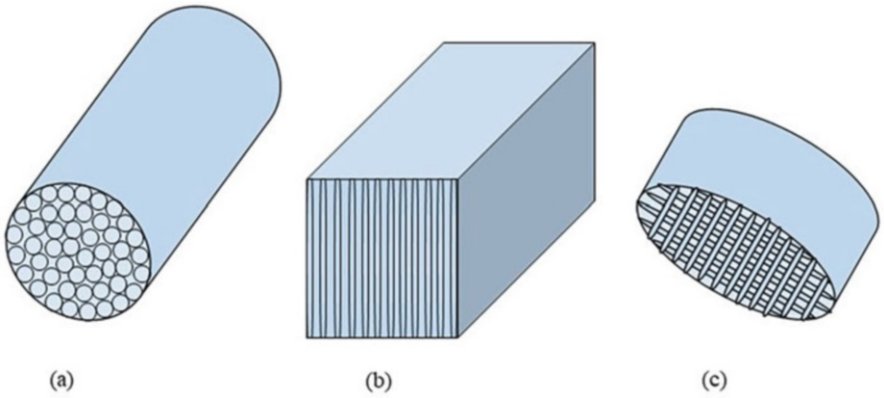

Fig. 5 Diffusion batteries: (**a**) tube bundle, (**b**) parallel plate, (**c**) screen batteries

$$J = n\frac{\sqrt{D}}{\pi t} \tag{2}$$

where J is the deposition flux, n is the initial number density, D is the diffusion coefficient, and t is the time. Equation (2) can be integrated to determine the concentration at each moment of time. Diffusion batteries remove small particles from a laminar-flow of aerosol. It is applicable to particle sizes from 0.002 to 0.2 μm. Several types of diffusion batteries have been designed including tube bundle, parallel plate, and screen batteries (Fig. 5).

Automated PM Measurement Methods

Manual method which is a filter-based sampling of particulate matter is more accurate and precise for the detailed study of the airborne particles (chemical analysis) during intensive experiments, but when used in routine monitoring, it has drawbacks associated with it. They do not provide information in real time. There is time lag between the sample collection and data availability which make it impossible for regulatory agencies to monitor peak concentration events. This time lag, long sampling time, and chances of sample contamination can be avoided by the use of real-time continuous instruments for PM measurement. Real-time instruments relatively need less attention; time resolution is in the order of minutes and can also be used for the study of PM in the areas where the concentration is generally low such as the North Pole. Available automated techniques and their principle of PM measurement are discussed in the following section.

Beta-Rays Attenuation Method

Beta gauge monitors based on beta-rays attenuation method are used to measure the concentration of PM (PM_{10} or $PM_{2.5}$) in ambient air. It is included in NAAQS. A schematic of beta gauge monitor is shown in Fig. 6. It consists of a size-selective PM_{10} inlet, a $PM_{2.5}$ classifier (VSCC), a movable filter tape for sample collection, a beta particle source, and a detector. It determines the mass concentration of collected PM by measuring the attenuation of beta particles when transmitting through the filter before and after sample collection. Continuous measurement is achieved by the automatic advancement of filter tape after specific time (1–6 hours). The relationship between the beta counts and sample mass was initially proposed in Jaklevic et al. (1981).

$$I = I_{0\ e^{-\mu x}} \tag{3}$$

where I and I_0 are the attenuated and un-attenuated beta counts, respectively, through mass density x (mg cm^{-2}) of collected sample and μ (cm^2 mg^{-1}) is the mass

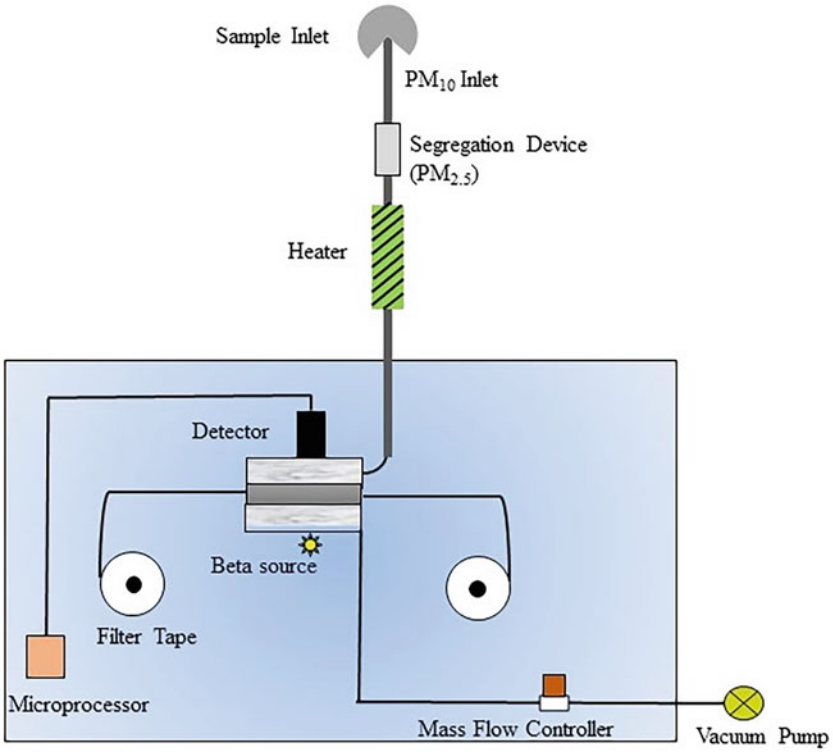

Fig. 6 Schematic of beta gauge monitor

absorption coefficient for beta particle absorption. The ambient concentration C (μg m^{-3}) of PM can be calculated as:

$$C = \frac{A \ln \frac{I_0}{I}}{\mu Q t} \times 10^6 \quad (4)$$

where A (cm^2) is the surface area of sample deposited, Q (L min^{-1}) is the sampling flow rate, and t is the time in minutes.

The choice of beta particle sources depends on the energy spectrum and half-life of radioisotope source. Beta particle sources which have been incorporated in beta gauge units in the past include ^{63}Ni ($t_{1/2} = 85$ y, $E_{max} = 67$ ke V), ^{14}C ($t_{1/2} = 5730$ y, $E_{max} = 156$ ke V), ^{147}Pm ($t_{1/2} = 2.62$ y, $E_{max} = 225$ ke V), and ^{85}Kr ($t_{1/2} = 10.76$ y, $E_{max} = 720$ ke V) (Wedding and Weigand 1993). However, current beta gauge monitors are usually equipped with ^{14}C source (Jaklevic et al. 1981). Detector should be sensitive to the beta particles, various detectors which have been used in the past include solid-state semiconductor, silicon surface barrier, ion-implanted silicon semiconductor, Geiger counter, and photomultiplier tube with scintillation device. Literature showed differences in PM concentration measured by gravimetric and beta gauge method (Arends et al. 2000; Chang et al. 2001). The major influence on measurement biases is due to high relative humidity. The inorganic hygroscopic particles which get collected on the filter absorb water as relative humidity rises above the deliquescence RH which result in growth in the size of particles, thus causing high concentration measurements by beta gauge monitors (Bagtasa et al. 2007; Chang et al. 2001; Khlystov et al. 2005; Salminen and Karlsson 2003; Shin et al. 2011; Tsai et al. 2006). Negative sampling artifact such as evaporation losses of semi-volatile species, such as NH_4Cl, NH_4NO_3, and organics (VOC) during sampling process, also contribute to the measurement difference between the methods (Le et al. 2020a; Liu et al. 2014). Biases in beta gauge measurement are also reported to vary with high particle mass loading and other ambient conditions such as low boundary layer height and ventilation coefficient along with high RH (Shukla and Aggarwal 2021). To overcome the effect of high RH, beta gauge monitors were modified to include an inlet heater to limit the relative humidity below 35%. Various correction approaches are applied to beta gauge measurements in order to obtain reference equivalent data from automated beta gauge monitors (Gehrig et al. 2005; Le et al. 2020b; Shukla and Aggarwal 2021).

Microbalance Methods

Quartz Crystal Microbalance (QCM)

The quartz crystal microbalance (QCM) is a well-established method for measuring mass (Mecea 2006), and it is commonly used to determine airborne PM mass concentration (Chen et al. 2016; Sarangi et al. 2016). It can measure nanogram to microgram level changes in mass per unit area. QCM uses piezoelectric crystal which undergoes mechanical deformations (vibration) when an electrical potential is

applied (Ward and Buttry 1990). Here, aerosol particles get deposited by inertial impaction or electrostatic precipitation onto the surface of an oscillating piezoelectric quartz crystal disk. Crystals have a resonant vibrational frequency that depends on the material thickness, and this frequency gets changed as the particle mass get deposited on vibrating surface. Then the change in sampling crystal frequency is electronically compared with that of a clean reference crystal, generating a signal that is proportional to the collected mass. Figure 7 shows the schematic of QCM. Frequency change (Δf) of oscillating quartz can be linearly related to its mass change (Δm) (Sauerbrey 1959).

$$\Delta f = -C_{\mathrm{f}} \times \Delta m \times A \tag{5}$$

where Δf is the change in the resonance frequency (Hz), Δm is the change in mass (μg), C_{f} is the sensitivity factor of the crystal (56.6 Hz ug^{-1} cm^2 for a 5 MHz AT-cut quartz crystal (cut made from y axis at 35° theta rotation) at room temperature), and A is the piezoelectrically active area (the area between the electrodes, cm^2).

Sensitivity of quartz crystal is around hundred hertz per microgram and measures mass concentration of about 10 μg m^{-3} in less than 1 min (Olin and Sem 1971). Collection efficiency can be increased by coating grease on the collection electrode to improve adhesion of particles to the quartz surface. This technique has some drawbacks, such as a low resonance frequency under high mass loading, sensitivity

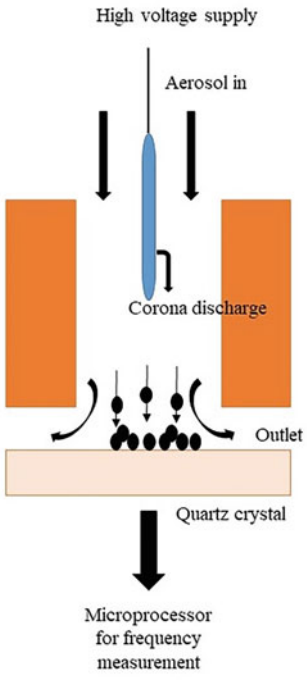

Fig. 7 Schematic of quartz crystal microbalance

to temperature and relative humidity, and a high frequency of the electrode surface, which disproves the relationship between frequency variation and mass estimation. The deposition surface needs to be cleaned and re-greased between each measurement to avoid particle bouncing (Tzou 1999). Some modifications can be incorporated in this instrument for more accurate and precise high concentration measurement such as lowering dead volume in the sensing chambers to reduce internal losses, improve response time, and include isokinetic sampling inlet.

Tapered Element Oscillating Microbalance (TEOM)

Tapered Element Oscillating Microbalance (TEOM) system is widely used to measure particle mass concentration in air quality networks. A schematic of TEOM is shown in Fig. 8. It draws ambient air through filter at a constant flow rate, continuously weighing the filter and calculating the mass concentration. The particles are

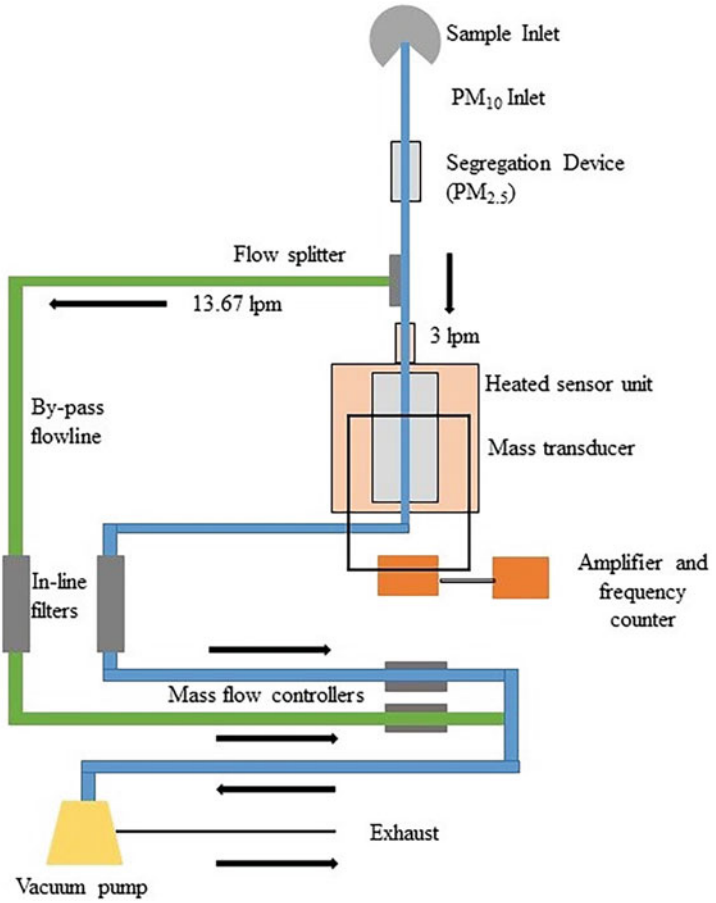

Fig. 8 Schematic of tapered element oscillating microbalance

collected on a replaceable filter, incorporated in a microbalance that is free to oscillate. Air is drawn through a size-selective inlet at the rate of 16.67 lpm. The sampled air is split into two channels, one with major flow (e.g., 13.67 lpm) which is discarded and the remaining with minor flow (e.g., 3 lpm) which is directed down a tube onto a sample filter resting on the end of a hollow tapered tube with the wider end of the tube being fixed. The air then passes through the filter with particles getting deposited on the filter and the filtered air then passes through the tapered tube to a flow controller. The tapered tube with the filter on its end is maintained in a fixed amplitude oscillating motion. The oscillation frequency of the tapered tube changes as the particles deposit on the filter and it depends upon physical characteristics of the tapered tube and the mass collected on its free end.

The resonant frequency of TEOM can be related to the mass of the sampled air by the equation of motion for simple harmonic oscillator:

$$\Delta m = K_{\mathrm{O}} \left(\frac{1}{f_{\mathrm{b}}^2} - \frac{1}{f_{\mathrm{a}}^2} \right) \tag{6}$$

where Δm is the mass of the sample deposit, f_{b} is the frequency of the oscillating element after sample deposit, f_{a} is the frequency before sample deposit, and K_{o} is a constant (spring constant) unique to each tapered element.

There are biases associated with the measurement of PM with TEOM which lead to discrepancy in data collected. The major problem associated with TEOM is the possibility of volatilization of collected material when heating the sample to 50 °C. This results PM concentration measured by the TEOM to be lower than measurements by the federal reference method due to loss of particle-bound semi-volatile compounds (Rizzo et al. 2003; Tortajada-Genaro and Borrás 2011; Zhu et al. 2007). Therefore, TEOM must be kept at a constant temperature to minimize thermal expansion. To overcome this issue, various modifications are incorporated in TEOM. Some modified versions are Real-Time Ambient Mass Sampler (RAMS) which eliminates gas-phase compounds to avoid losses of semi-volatile material (Eatough et al. 2001); Sample Equilibration System Tapered-Element Oscillating Microbalance (TEOM-SES) operates at a reduced temperature of 30 °C, and to avoid condensation of humidity, a diffusion dryer precedes the TEOM; differential TEOM which works with electrostatic precipitator to remove positive and negative filter artifacts (Jaques et al. 2004) and TEOM equipped with a Filter Dynamics Measurement System (TEOM FDMS) include a diffusion dryer and cold filter maintained at 4 °C to remove particles prior to passage to the TEOM.

Continuous Ambient Air Mass Monitor (CAMM)

Change in pressure drop across the filter due to particle deposit can be used to measure PM. The CAMM measures ambient PM by pressure drop difference between a reference airstream channel and particulate matter airstream channel. It

includes an impactor with a cut size of 2.5 μm, a diffusion dryer to remove particle-bound water, and a membrane filter tape which advances every hour. The detection limit of CAMM is ≈5 μg m^{-3} at a flow rate of 0.3 lpm and sampling time of 1 hour. The pressure drop across the filter depends on particle size. Chung et al. (2001) showed good correlation between the CAMM and reference sampler. Discrepancy of CAMM with semi-volatile material is reported by Lee et al. (2005).

Dekati Mass Monitor (DMM)

The Dekati Mass Monitor (DMM) is the on-line particle measurement instrument which is the combination of mobility analyzer and six-stage impactor. It consists of inlet with a separator of 1.3 μm cutoff, and particles get electrically charged in a corona charger, then pass through one-stage mobility analyzer where highly mobile small particles get precipitated to an electrode connected to an electrometer, and less mobile particles enter into six-stage electrical impactor with electrometer amplifier to determine the current of precipitated particles (Fig. 9). This instrument gives information about particle density and size distribution which are used to determine mass of particles. There are chances of error if the density is size dependent as mobility analysis is applied only to small particles and impaction to only large particles. Studies showed that DMM overestimate the mass (around 20–39%) in comparison to reference method (Lehmann et al. 2004; Mamakos et al. 2006).

Aerosol Particle Mass Analyzer (APM)

Measurement of aerosol particles in Aerosol Particle Mass Analyzer (APM) takes place according to particles mass-to-charge ratio (Ehara et al. 1996). Initially aerosol particles pass through a bipolar charger to achieve equal charge distribution and then enter into annular space between two coaxial rotating cylindrical electrodes having inner radius r_1 and outer radius r_2 (Fig. 10). The two electrodes rotate at same

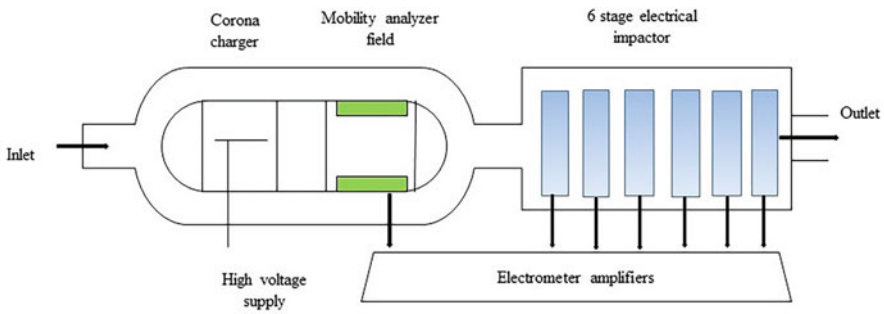

Fig. 9 Schematic of Dekati mass monitor

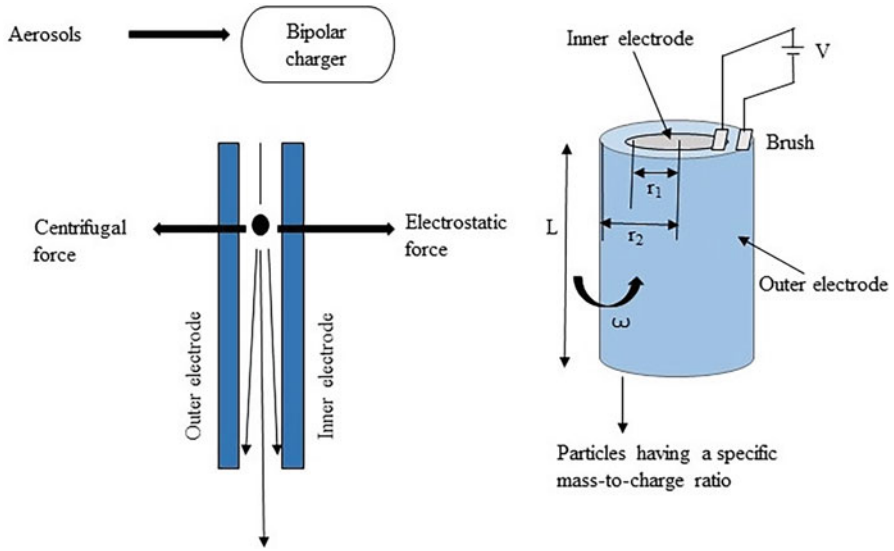

Fig. 10 Operating principle of aerosol particle mass analyzer

angular velocity ω, and q is the electrical charge of particles. The centroid mass, m_c, is derived from balance force as:

$$m_c = \frac{q\,V}{r_c^2\,\omega^2 \ln \frac{r_2}{r_1}} \tag{7}$$

where r_c is the center radius ($r_c = (r_1 + r_2)/2$) and V is the voltage applied. Two forces work on particle inside annular space, centrifugal and electrostatic force, and those particles for which these two balance each other, travel the entire length of electrode, and exit the annual space. The advantage of this instrument is its ability to measure particle mass without the need for particle collection for weighing.

Optical Methods

The optical property of aerosols and their interaction with light is also useful for measuring aerosol particle size and concentration. Optical methods are extremely sensitive and instantaneous with no requirement of particle collection on filter. Optical instruments, in real-time measurement of particle concentration, are based on principles of scattering, absorption, and light extinction. There are three instruments based on measurement of light scattered at fixed angles: photometers, integrated nephelometers, and optical particle counters. Light scattering instruments are sensitive to minute changes in scattering angle, particle size or shape, and refractive index of particles (Hinds 1999).

Photometers

Photometer measures PM concentration by measuring the intensity of light scattered from many particles in one or more angles reaching the detector. It measures the light scattered in the region of 90°, 45°, or less than 30°. This instrument is sensitive to aerosol composition and particle size distribution. To reduce the effect of particle size and refractive index on measurement by photometers, the precise choice of scattering angles and wavelengths are made. Some photometers studied include DataRam 4 (Costa et al. 2012), UCB-PATS (University of California Berkeley-Particle and Temperature Sensors) (Chowdhury et al. 2013), and DustTrak (Chowdhury et al. 2013; Prado et al. 2012). They showed good correlation with other instruments. UCB-PATS used smoke detector technology combining photoelectric and ionization sensors. It uses light-emitting diode (LED) and photodiode measuring light intensity scattered at an angle of 45°, and cyclone along with photoelectric sensor helps in measuring $PM_{2.5}$, whereas ionization sensor is more sensitive toward PM_1.

DustTrak sampler includes an impaction inlet, laser diode, optics, photodetector, vacuum pump, and electronic control panel. The continuous aerosol is drawn through the impaction inlet where particles with diameter greater than 10 μm, 1 μm, or 2.5 μm are removed depending on type of inlet. Then sample stream pass is intersected by light of 0.78 μm wavelength emitted from laser diode. The light scattered by particles is measured by the photodetector oriented at an angle of 90° to laser and sampler stream which is then converted to mass concentration by electronics. Proportionality constant used is determined by comparing DustTrak against gravimetric instrument with Arizona test dust.

Integrating Nephelometer

The integrating nephelometer measures particle concentration by the light scattered at several wavelengths in visible range. The light scattered at angles from 8 to 170° by particles was measured and used to determine particle scattering coefficient. This scattering coefficient is proportional to the mass of particulate matter and used to determine particle concentration. The nephelometer is used widely for the measurements of light scattering coefficient due to its high accuracy, portability, and moderate cost. Drawback associated with nephelometer are the biases in light scattering caused by different sizes and composition of particles, inlet loss of coarse particles, and inefficiency of optical measurement of light scattered in the near forward direction. Figure 11 shows schematic of nephelometer.

Optical Particle Counter (OPC)

Optical Particle Counter (OPC) measures the light scattered by each particle as the aerosol flow through a tightly focused beam of light. It uses laser diode as light source to excite sample of particle at a specific angle, and the light scattered by particles is collected by 90° lens and then is measured by photodetector as an electric pulse. Schematic of OPC is shown in Fig. 12. Particle size is determined from the magnitude of electric pulse by using a calibration curve obtained by measuring

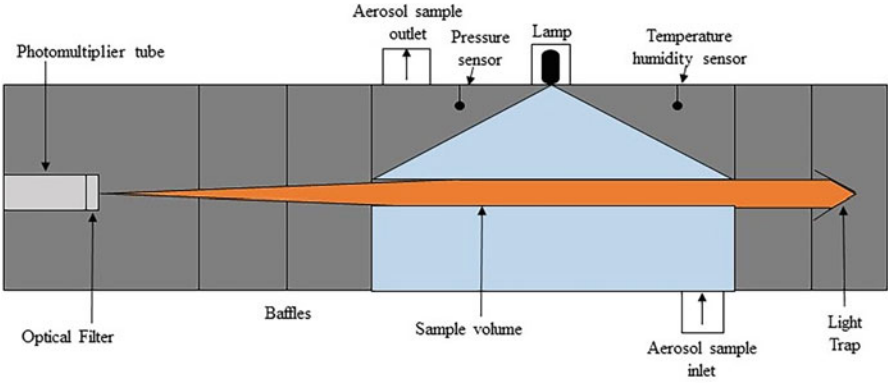

Fig. 11 Schematic of nephelometer

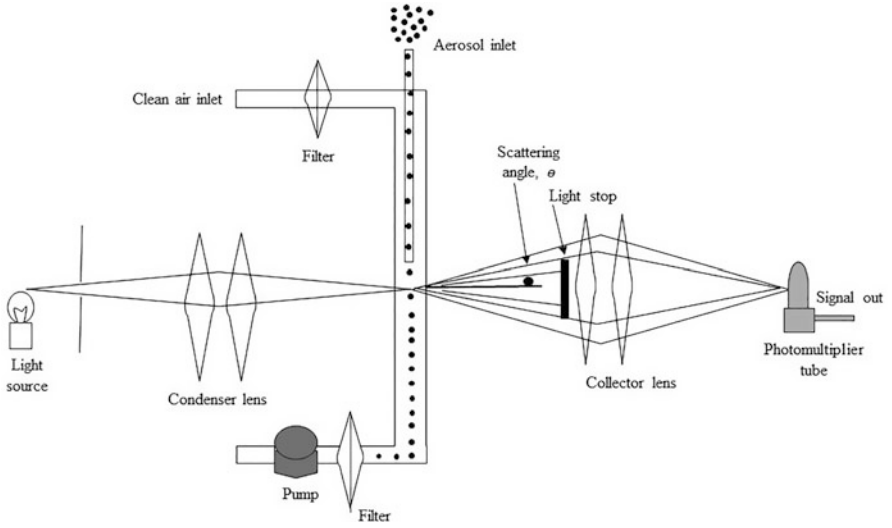

Fig. 12 Schematic of optical particle counter

known particle size and composition. Laser OPC are generally prefer than white light OPC as laser OPC have minimum detection limit of ~0.05 μm particle size and white light OPC cannot detect particle smaller than ~0.3 μm (Gebhart et al. 1976). Also, laser OPC have counting efficiency of 100% for particles greater than 0.1 μm. Its measurement depends on refractive index, shape, and morphology of particles. Varied particle chemical composition causes significant uncertainties in estimating refractive index. While measuring particle size distribution with OPC, it is necessary to dilute aerosol so that only one particle passes through the beam at a time for unbiased measurement.

Electrical Mobility Analyzers

Electrical mobility analyzers classify charged particles based on their mobility in an electric field. Generally, an electrical mobility analyzer consists of two oppositely charged plates parallel to each other having specific voltage with aerosol particles entering through the space between plates. Particles with mobility higher than specified voltage will get trapped to the plates, and particles with low mobility will pass through and get collected on filter (Fig. 13). The comparison of mass sampled in the absence of voltage on the plate to that for different voltages will give the particle distribution. Particles get classified based on their electrical mobility denoted by Z and for spherical particles based on Stokes drag, Z is given by (Hinds 1999):

$$Z = \frac{\mathrm{neC}}{3\pi\mu D_P} \tag{8}$$

where n is the number of charges carried by the particle, e is the magnitude of charge, C is the slip correction factor, μ is the viscosity of air, and D_p is the diameter of particle. This electrical mobility is based on Stokes drag, and if Reynolds number (Re) of particle is greater than 1, the true mobility will be greater. Mobility analyzer are able to measure much smaller particles and are not sensitive to differences in refractive index as optical technique but are sensitive to particle shape.

The following section includes commonly used electrical mobility analyzers for aerosol studies.

Electrical Aerosol Analyzer (EAA)

Electrical Aerosol Analyzer (EAA) is the first electrical mobility analyzer developed by Whitby and Clark (1966). In the EAA, aerosols flow at 5 lpm through a unipolar ion diffusion charger, where before entering into classifier, they get charged by small positive ions. The duration of particle stay in the charger and ion concentration are kept fixed to ensure the predictable charge number exposure to the specific size of particles. Particles having mobility greater than a cutoff mobility value (predetermined by flow rates and voltages) are precipitated in the concentric cylindrical classifier, while the particles with less mobility pass out of the analyzer and get collected on filter where the current produced by the capturing of charged particles is continuously monitored by an electrometer. The difference in current generated by the analyzer's two voltage settings is then related to the number of particles in the size range defined by the two voltage setting's set cutoff sizes. EAA measures size distribution in the 0.003–1 μm diameter range.

Differential Mobility Analyzer (DMA)

The differential mobility analyzer (DMA) is used as a monodisperse aerosol generator to produce sub-micrometer-sized aerosols for testing and calibration purpose and to measure particle size distribution in sub-micrometer size range. In DMA, aerosol first pass through bipolar ion diffusion charger, where it achieves Boltzmann charge equilibrium before entering into classifier. DMA consists of a cylinder and a

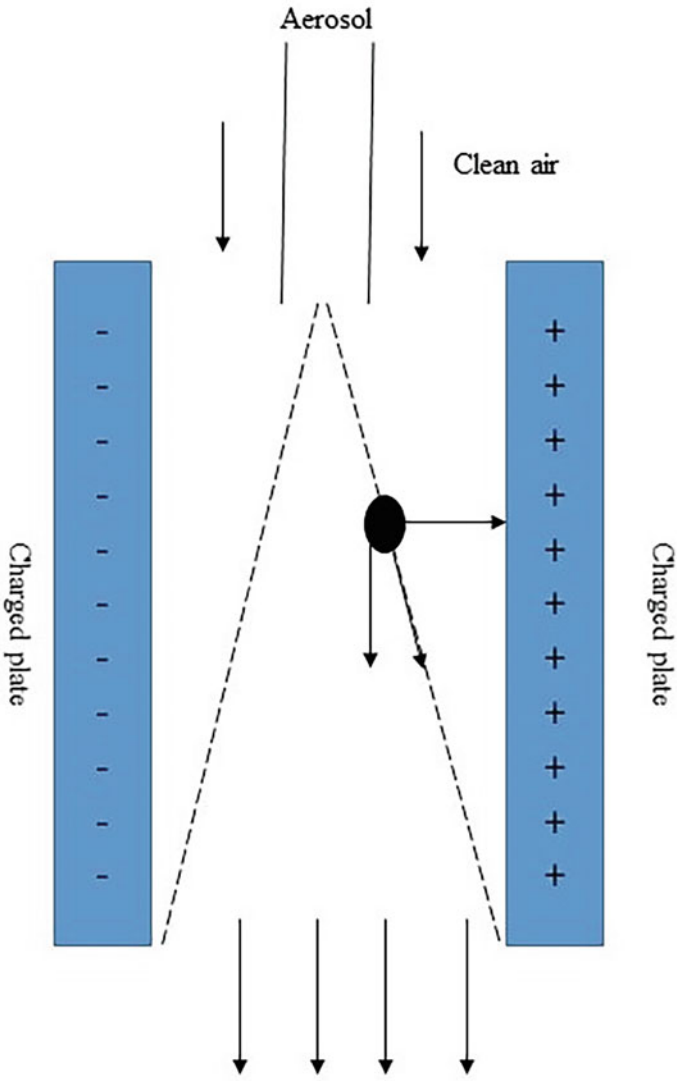

Fig. 13 Schematic of electrical mobility analyzer

charged central rod creating electric field in the space between the cylinder walls and the rod. Particle free clean air and particles through sample inlet move toward the central rod at a velocity determined by their electrical mobility. The particles then may exit through slit or through exhaust or get deposited on the central rod. The exiting aerosol are singly charged and nearly monodisperse whose size is controlled by adjusting the voltage on the central rod. Particles with narrow range of mobility get separated from the main flow by varied voltages and then delivered to detector or

counter for particle size and number concentration measurement. The complete particle size distribution can be determined by continuously scanning through voltage range. DMA measures size distribution in the 0.005 to 1.0 μm diameter range.

Aerodynamic Particle Sizer (APS)

Aerodynamic Particle Sizer (APS) determine particle size distribution ranging from 0.5 to 20 μm by measuring particle velocity relative to the air velocity within an accelerating airflow. It counts and sizes particle by an aerodynamic property time-of-flight (TOF) and an optical property (light scattering intensity). It is based on the principle of inertia. Aerosol accelerates through nozzle with particles lagging behind the carrier gas due to inertia (Wilson and Liu 1980). The difference between particle and gas speed tends to increase with increase in size and density of particles. APS operates at a flow rate of 5 lpm, with 4 lpm removed by passing through filter and re-introduced upstream of the acceleration nozzle as sheath flow and recombining remaining 1 lpm aerosol flow entering through nozzle. The combined aerosol and sheath flow accelerates through final outer nozzle and enters detector region (Fig. 14). The TOF of each particle is detected and converted to aerodynamic particle size using a pre-programmed calibration. Also, the scattered light intensity is measured by analyzing magnitude of pulses of scattered light from the laser. APS is widely used for the characterization of aerosols and evaluation of particle samplers due to its accurate and precise measurement of particle size and concentration. The gas velocity U_g in the nozzle can be determined from the Bernoulli equation of compressible flow:

$$U_g = \left\lfloor \frac{2RT}{M} \ln\left(\frac{P}{P-\Delta P}\right) \right\rfloor^{1/2} \tag{9}$$

where R is the universal gas constant, T is the absolute temperature, M is the gas molecular weight, P is the ambient pressure, and ΔP is the pressure drop across the nozzle. The velocity of particle V_p can be determined from:

$$V_P = \frac{U_g t_{min}}{t} \tag{10}$$

where t_{min} is the minimum transit time or time-of-flight (TOF) for particles and t is the transit time of particles. TOF is the time interval between two pulses of light scattered by particles as they pass through two beams of light, and it is proportional to particle velocity and thus aerodynamic diameter. As the particle size increase, the TOF increases because particle lags behind the air velocity. A calibration curve between aerodynamic diameter and TOF data stored in an accumulator in bins is

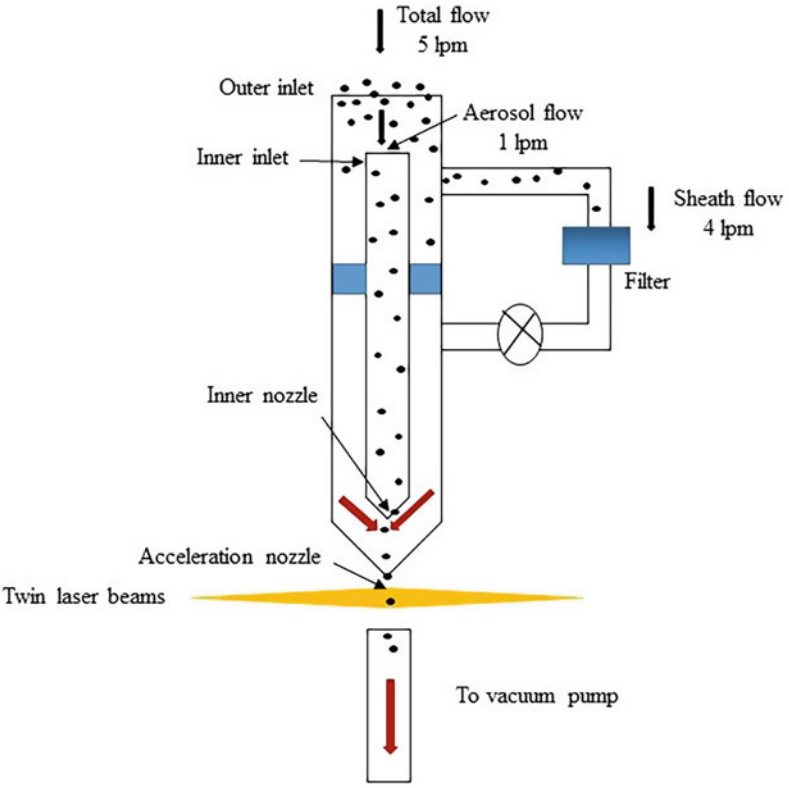

Fig. 14 Schematic of aerodynamic particle sizer

done for each instrument individually using monodisperse test particles with known density. This curve is based on the fact that for a given acceleration nozzle and light detection arrangement, the ratio of V_p/U_g as a function of Stokes number is unique. The counting efficiency η of the APS is the number of particles of a given size counted divided by the number sampled:

$$\eta = \frac{N_{\text{counted}}}{N_{\text{sampled}}} \tag{11}$$

Unlike optical counters, APS measurements are not compromised by Mie resonances. But if more than one particle is involved in the analysis of particle transit time, then APS give rise to coincidence effects resulting in phantom particle generation. It underestimates liquid droplets as they flatten and get greater velocities than spherical particles.

Electrical Low Pressure Impactor (ELPI)

Cascade impactors are widely used for PM measurements. Conventional impactors cover a size range from 0.3 to 10 μm. Impactor-based PM sampling requires long sampling duration to collect measurable particle mass on each stage and a sensitive microbalance. Therefore, to overcome limited number of sample collection in a day and to avoid errors associated with man-handling of filters during sampling and weighing, real-time automatic impactor-based Electrical Low Pressure Impactor (ELPI) were introduced (Keskinen et al. 1992). It determines particle size distributions by combining electrical detection of charged particles and the impactor collection technique. Its mass sensitivity for sub-micrometer range is better than QCM and comparable to EAA. Figure 15 depicts the operating principle and main components of ELPI. Briefly, ELPI generally consist of three components: a unipolar charger, a cascade impactor, and a multichannel electrometer. The aerosol is introduced into unipolar charger at flow rate of 10 lpm. The charged particles then pass through cascade impactor classifying the aerosol into 12 size fractions, and multichannel electrometer simultaneously measures the charge carried by the particles from each stage. The measured current signals are then converted to number concentration at different particle sizes. It also contains flush pump used to zero the electrometers by pumping clean air. ELPI measures size distribution in the range

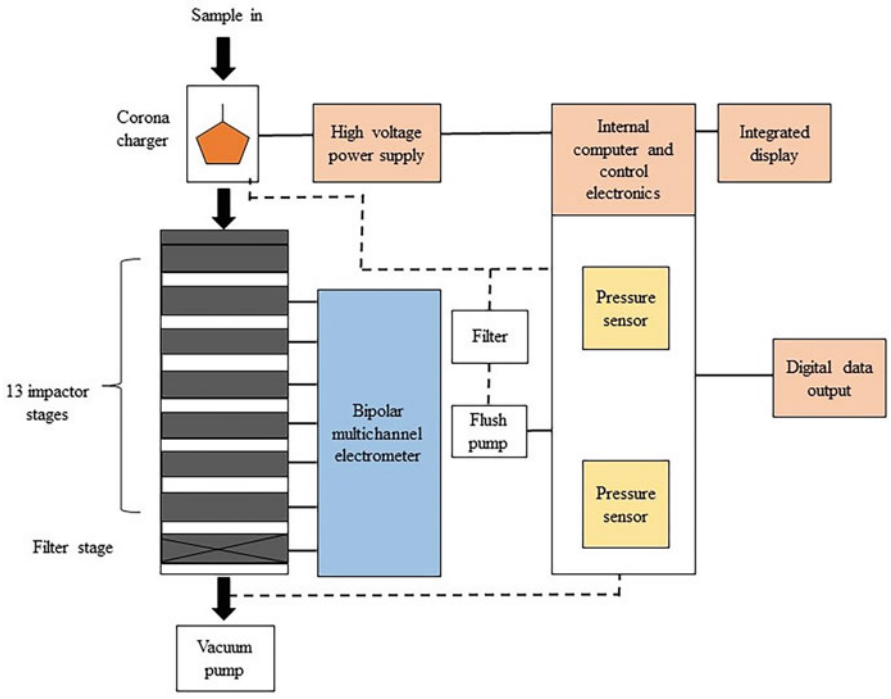

Fig. 15 Schematic of electrical low pressure impactor (ELPI+)

30 nm–10 μm. Dekati Ltd. in 2010 introduced conventional ELPI as ELPI+ with impactor stages increased to 15 stages (14 impactor stages + filter stage) allowing real-time particle size distribution measurement.

PM Chemical Composition Measurement Techniques

Aerosols interact with earth radiation budget and climate directly as well as indirectly. In a direct way, aerosols scatter or absorb solar radiation. Indirectly, aerosols modify the size of clouds affecting the cloud albedo. Size distribution and chemical composition of PM affects cloud condensation and nucleation activities. Therefore, it is important to know the chemical composition of PM using precise and accurate measurement techniques. The choice of measuring techniques depends on chemical compound that requires to be identified, the need to preserve the sample, if quantification is needed or information needed on morphology or size of particles. The major chemical components of PM which are analyzed include anions (sulfate, nitrates, sometimes chloride, and organic acids), cations (NH_4^+, water soluble potassium, sodium, calcium, and magnesium), carbons (OC and EC), and trace elements. A variety of methods and instruments are available ranging from filter-based sample collection for off-line analysis to on-line instruments which detects the chemical compositions of particles in real time (Aggarwal 2010), which are discussed in the following sections.

Off-Line PM Analysis Techniques

Off-line technique involves the prior collection of PM on filter, transportation procedures, and analysis to determine the chemical composition of PM. It provides detailed, specific chemical information about PM composition. However, there are positive and negative artifacts affecting off-line technique: evaporative losses during sampling, storage, and transport of samples, high particle loading causes some sample to bounce or be blown off the filter, and gases can be adsorbed onto filters and react with the surface. Off-line techniques can be divided into atomic spectrometry-based techniques, X-ray-based, and surface analysis techniques.

Atomic spectrometry is the widely used technique for elemental characterization of PM with Inductively Coupled Plasma (ICP) accounting for highest use in research. ICP is based on the ionization of sample under argon plasma atmosphere. ICP-Mass Spectrometry (MS) and ICP- Optical Emission Spectrometry (OES) provide quick and multi-elemental analysis with high sensitivity and low detection limit (ppb). Therefore, this technique is preferred for trace element analysis but the greatest disadvantage is that it is a destructive technique as it requires complete extraction and digestion of sample in acid medium. Another traditional technique for chemical analysis of aerosols is Atomic Absorption Spectrophotometry (AAS) based on the atomization or ionization of element at high temperature and detection at a specific wavelength by a monochromator. There are three variants of AAS: Flame

AAS (FAAS), Graphite Furnace (GFAAS), and Cold Vapor AAS (CVAAS). The difference among these variants lies in the ionization method. FAAS uses flame reaching 300 K, GFAAS uses atomization temperature over 3000 K, and CVAAS uses no vaporization system (Brown and Milton 2005; Wilson et al. 2002). AAS instruments are less costly in comparison to ICP but it is a single element analysis technique, thereby increasing the time of analysis.

X-ray-based techniques used a focused beam of charged particles which excite electrons in the samples with a wavelength characteristic of the element. Among X-ray-based techniques, the most applied ones are energy dispersive X-ray fluorescence (EDXRF) and particle-induced X-ray emission (PIXE). Both EDXRF and PIXE are non-destructive techniques. For EDXRF, there is minimal sample preparation required which reduces the risk of contamination, and it can possibly analyze liquid samples also. PIXE is an ion beam method based on detection of X-rays induced by a particle beam from electrostatic accelerators (Wilson et al. 2002). In contrary to EDXRF, PIXE has higher sensitivity for light elements such as Na, and it is less expensive.

Surface analysis techniques investigate chemical structure of a shallow layer of solid surfaces. Techniques commonly used for surface characterization of atmospheric particulates are X-ray photoelectron spectroscopy (XPS) (Gilham et al. 2008); X-ray diffraction (XRD) (Cienfuegos and Vaitsman 2000); secondary ion mass spectrometry (SIMS) (Janssens 2013); and scanning electron microscope equipped with EDX (SEM-EDX) (Janssens 2013; Sobanska et al. 2003).

On-Line PM Analysis Techniques

On-line techniques provide a real-time chemical characterization of the atmosphere with high temporal resolution of less than 1 hour. The mass spectrometers have been developed to determine the composition of particles in real time and a continuous manner. On-line mass spectrometry measures chemical changes in atmospheric particulate matter over a short period of time and can either determine the chemical composition of particulate assemblages, i.e., bulk analysis or single particles.

Aerodyne aerosol mass spectrometer (AMS) provide real-time information on mass distribution and chemical composition of volatile and semi-volatile component in PM combining standard vacuum and mass spectrometric technique. It detects only non-refractory compounds such as sulfate, nitrate, ammonium, chloride, and organics. It involves thermal ultra-high vacuum vaporization of PM inside tungsten vaporizer (400–600 °C) followed by vapor molecules ionization by a 70 eV electron impact ionization source. The AMS consists of three components: a sampling section, a section for particle sizing, and particle composition detection section. A schematic of AMS is presented in Fig. 16. Each section is pumped deferentially and separated by small apertures. The inlet samples a flow of 1.46 cm^3 s^{-1} focusing particles into narrow beam as they enter into vacuum with nearly 100% transmission efficiency to the detector for particle size ranging from 70 to 500 nm and substantial transmission for particles ranging from 30 to 70 nm and 500 nm to 2.5 μm. AMS

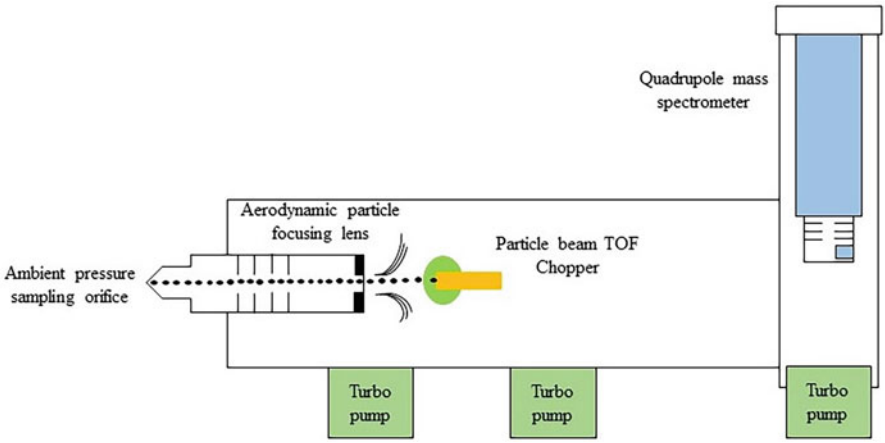

Fig. 16 Schematic of aerodyne aerosol mass spectrometer

alternates between two modes, mass spectrum and TOF with mass spectrum giving the average mass concentration of non-refractory species and TOF used to monitor mass distribution associated with a specific chemical compound as a function of aerodynamic diameter. Limitations associated with AMS are that it can only detect non-refractive particles; it cannot detect refractory particles such as soot, fly ash, metal oxides, and sea salt (Canagaratna et al. 2007); and it cannot provide complete mass spectra of single particle (Allan et al. 2003). Other versions of AMS available vary only based on the type of mass spectrometric detector: quadrupole (Q-AMS), time-of-flight (TOF-AMS), and high resolution TOF (HF-TOF-AMS).

In contrast to AMS that is based on thermal vaporization for characterizing the particles in bulk, aerosol mass spectrometry based on laser vaporization provides mass spectra for individual particles. Single-particle mass spectrometer (SPMS) involves the use of a high-power laser to vaporize and ionize the aerosol particles followed by mass spectrometric analysis. It provides qualitative or semi-quantitative information of chemical composition of individual particles. Several versions of SPMS instruments include aerosol time-of-flight mass spectrometer (ATOFMS) (Prather et al. 1994); laser mass analyzer for particles in the airborne state (LAMPAS) (Hinz et al. 1994); particle analysis by laser mass spectrometry (PALMS) (Murphy and Thomson 1995); laser-induced breakdown spectroscopy (LIBS) (Hahn and Lunden 2000); rapid single particle mass spectrometer (RSMS) (McKeown et al. 1991); nanoaerosol mass spectrometer (NAMS) (Wang et al. 2006); single particle laser ablation time-of-flight mass spectrometer (SPLAT) (Kamphus et al. 2008); and single particle aerosol mass spectrometer (SPAMS) (Li et al. 2011). The difference among the various versions of SPMS is in the laser wavelength used to perform the desorption and ionization and the method used to determine particle size (Sullivan and Prather 2005). Generally, SMPS uses specially designed inlets to transport particles from air into the vacuum and detected by optical light scattering.

Both off-line and on-line techniques are complementary in nature with on-line technique showing improvement in temporal resolution of PM characterization by providing data in 1 hour or less, and off-line technique being non-destructive allows the usage of same sample to analysis by other techniques. The combined use of both on-line and off-line technique can provide better understanding of particulate matter as per research requirement.

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Part XII

Metrology in Bio-medical Engineering

Precision Measurements in Healthcare Systems and Devices

70

Ravinder Agarwal, Amod Kumar, and Sanjay Yadav

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Abstract

Medical measurements are present in the everyday life of people and are fundamental processes in the prevention, diagnosis, and treatment of diseases. A growing interest lies in the role of metrological decisions and conformity assessment, where measurements are in support of health. Due to the inherent danger of life involved during the diagnosis and treatment of the patient, it is critical to

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measure amounts as precisely and reliably as possible. Such measures and measuring tools are common in people's daily lives and play a very important role in illness prevention, diagnosis, and treatment. Although medical measures (measurements that are important to medical diagnosis) are simply pieces in the larger puzzle of medical decision-making, they make a significant contribution.

Introduction

Over the last few decades, quality evaluation methodologies have become more sophisticated, allowing for meaningful comparisons between healthcare institutions or health plans. However, as they are now utilized, these approaches cannot be used to enhance healthcare quality to the maximum feasible degree since metrics that rely on data acquired for administrative purposes unavoidably contain measurement flaws at the individual patient level. Patients may be incorrectly declared fit or unfit for a wrong or incorrect measurement, or the data taken was not collected properly, or the measurement was not appropriate for them for other reasons, such as exclusion requirements that the measurement system missed. Precise measuring system technician assists with modeling validation and production by employing precision measurement instruments such as hand-help, bench, or vision system.

This chapter focuses on the use of medical devices to improve their metrological traceability, emphasizing the specific role of metrology in the field of health care and the impact of legal control within the framework of the medical device regulation with a measuring function.

Medical decisions are frequently based on statistical analysis and clinical study findings. These studies include medical measurements that are correlated with other medical findings. Assuming that medical measurements are related to International System of Units (SI) of quantities such as mass, temperature, length, and so on, it appears that the main concepts of industrial metrology can be adapted and used fairly easily in the medical field. Following this logic, it is obvious that the metrological traceability of measuring instruments must be improved for them to be usable (reproducibility and repeatability conditions). Metrological tools such as calibrations, legal metrological control, and reference measurement methods should be used to ensure measurement quality.

Measurement is essential in the medical, biomedical, and healthcare fields because it serves as the foundation for medical diagnosis, prognosis, and evaluation. Indeed, "measurement is the foundation of medical research and clinical practice." To make the best decision for their patient, medical professionals such as doctors and clinical laboratory scientists must have faith in the results reported by their instruments or measurement methods. In many industries, incorrect measurements would simply results in customer dissatisfaction or a loss of money; however, incorrect measurements in biomedical could be fatal and result in a loss of life. The proper instrumentation and measurement (I&M) is critical in the medical field. I&M is primarily concerned with the measurement, detection, monitoring, and recording of

a phenomenon known as the measurand, as well as the calibration, uncertainty, tools, and applications that go with it. The fundamentals of I&M apply to any type of instrument or measurement, including medical instruments and measurements.

In this chapter, an overview of the complete range of I&M aspects in the medical field: measurement accuracy, uncertainty, instrument reliability, calibration against the gold standard, and complying with various regional, national, or international medical regulations, has been covered. Measurements of physiological, biochemical, physical, and other patient-related variables are not only common in intensive care medicine and beyond, but the results of such measurements also provide critical information for clinical decision-making as well as research and technology development. Erroneous measurements can jeopardize patient safety and expose the most critically ill patients to serious risks.

In this handbook, chapters based on metrological aspects for blood pressure measurement, SEMG signal acquisition, cognition advances in EEG and fNIRS, tremor measurement during surgery, artificial intelligence techniques for disease predictions in soft tissue organs, and iris-based diagnosis have been discussed.

In recent years, the technological revolution has created opportunities for bringing medical services to everyone's door. This is accomplished not only through the numerous applications available for smartphones but also through specialized devices that are available in a portable format. For such devices, the I&M community faces two major open concerns, i.e., clinical quality measurements to be obtained in the home using low-cost measurement equipment and their operation by a layperson.

Improving signal quality is the most important key to answering these questions. Because the cost of specialized equipment is so high, low-cost equipment can be equipped with inexpensive advanced signal processing tools. Furthermore, specialized sensors can be replaced by less expensive sensors or measurement devices that can be operated by a layperson and have a dedicated mathematical model or data analysis technique that can compute the physiological parameter of interest from the acquired measurements.

Blood Pressure Measurement

The correct measurement of blood pressure (BP) is essential for the diagnosis and management of hypertension. An intra-arterial evaluation is required for direct BP measurement. This is impractical in clinical practice, as blood pressure is measured noninvasively. For clarity, we refer to noninvasive blood pressure measurements as BP measurements. BP determination for hypertension diagnosis has traditionally depended on readings obtained on the arm above the brachial artery during healthcare appointments, often referred to as office BP.

Before the twenty-first century, office blood pressure was measured by an observer listening to noises with a stethoscope while observing a sphygmomanometer (i.e., auscultation). Blood pressure (BP) is an important physiological measure utilized in the diagnosis of many life-threatening disorders such as heart attack,

ischemic stroke, hypertension, vascular disease, hypotension, and many others. As it is closely tied to human life, it plays a significant role in the diagnosis and monitoring of many health conditions. Reliable and exact BP measurement is always a priority for both researchers and medical practitioners (Kumar et al., 2020, 2021b). With recent technological advancements, the issue of precise measurements, standardization of methodologies, and efforts toward adopting certain universal methods remain disputed, particularly traceability in dynamic mode. There are several types of BP measurement equipment in the market. The accuracy of these instruments is an unresolved issue for academics, medical practitioners, and metrologists. Several national and international standards prescribe procedures for calibrating these devices; however, relatively few technologies/systems/instruments are available to calibrate these devices in SI traceable units convincingly. In this chapter, an exhaustive review is presented explaining various noninvasive BP measuring techniques and available national and international standards. ISO standard of simulator is also described, which is used for the calibration and testing of oscillometric-type BP devices. Insight on some of the recent advancements in BP measurement and challenges ahead in their calibration are also provided.

Metrological Aspects of SEMG Signal Acquisition, Processing, and Application Design

Surface electromyography (sEMG) is a complex interference pattern of electrical activity during muscle contraction. It is directly connected to muscular activity and fitness level. Its amplitude ranges from 0.01 mV to 10 mV, with the majority of its energy focused between the frequencies of 0 Hz and 500 Hz. The detection of sEMG signals using noninvasive technology is extremely useful in clinical diagnostics, rehabilitative medicine, and intelligent prosthesis control. EMG has been utilized in the identification of sign language gestures, gaming control, and wearable devices in recent years (Gupta & Agarwal 2021, Gupta et al., 2020). Over the last few decades, there has been significant growth in SEMG-based applications. It is mostly because of advancements in electronic manufacturing and computing capabilities. However, the difficulties related to metrological standards remain unaddressed and must be solved urgently. This chapter tries to put together suggestions, standards, and criteria for such systems, as well as a case study of a SEMG-operated lower limb prosthetic design. The SENIAM recommendations are extensively reviewed and included in the application design. To construct a biologically inspired control structure, high-level and low-level control modules are used. The high-level control module provides activity intention, whereas the low-level control module determines the required angular position of the joints. The SEMG signal is used differently by each control module. The high-level control module problem is written as a classification problem, whereas the low-level control module problem is written as a classification problem for knee angle prediction and a continuous value estimation problem for ankle angle estimation. SVM, LDA, and NN classifiers with time domain feature vectors were observed to be appropriate for the high-level control

module. SVMQ proved to be superior for knee angle prediction. The autoregressive exogenous neural network (NARX) model incorporating SEMG and knee angle information performs well across all locomotion modes for estimating ankle angle. The framework and techniques are presented here for the design and control of SEMG-based active lower limb prostheses, exoskeletons, and orthoses.

A Survey on Advancement in Measuring Cognition Using EEG and fNIRS

Mental workload (MWL) impacts persons who continuously interact with computers and other devices. The usage of technology in everyday life may place substantial cognitive demands on users as they traverse complicated interfaces. Mental overload can affect users' performance and sometimes their safety by raising mistake rates and causing weariness, loss of motivation, longer response, and disregard of essential information (cognitive tunneling). Taking into consideration the users' cognitive qualities and limits is therefore crucial in enhancing the design of human-machine interfaces (BMI) and running them efficiently by installing adaptive elements that can respond to changes in the MWL. It is simple to get an early assessment of a patient's cognitive functions by comparing the patient's score reference standards using appropriately validated cognitive impairment screening tests (Andrianopoulos et al., 2017, Uchitel et al., 2021). This provides for a more accurate diagnosis and treatment strategy. The three basic methods for testing everyday cognitive functions are self-report, informant report, and performance-based screening evaluations. Human cognition is the basic building block of human intelligence, and it is what defines us as individuals. Cognition is described as the ability to recognize and respond effectively to environmental events based on one's beliefs, actions, experiences, and senses. It is one of the most fundamental reasons for human life and one of the most significant features of the brain. Human cognitive processes are constantly growing and developing during childhood, adolescence, and maturity. Although some of these talents begin to decrease as one grows older and reaches elder maturity, others begin to weaken when neurons die and the mechanisms that replenish them become inadequate. Understanding cognition is essential for not only healthy cognitive development and survival but also for treating a wide range of neuropsychological problems, including Alzheimer's disease. Before understanding cognition, one must first investigate the cognitive functions of the human brain. Low-cost ways of monitoring and evaluating cognitive function include fNIRS and electroencephalography (EEG). The use of combined EEG-fNIRS is still in its early stages, but it may be beneficial soon in terms of improving the algorithms and physical system design of hybrid EEG-fNIRS, such as emphasizing spatial and temporal advantages from fNIRS and EEG appropriately by designing advanced feature extraction methodologies and splitting nodes in different targeted cortical functions. This chapter covers the principles of electroencephalography (EEG) and functional near-infrared spectroscopy (fNIRS), as well as numerous ways to

preprocess and analyze EEG and fNIRS data. This chapter also discusses the usage of simultaneous EEG-fNIRS.

Artificial intelligence (AI) is a computer program's capability to perform human intelligence-related functions such as thinking, learning, adaptability, sensory comprehension, and interaction. Artificial intelligence (AI) uses data and algorithms to reach judgments that are on par with, if not better than, those reached by people. AI is now a part of our everyday lives; it powers facial recognition technology, speech recognition in virtual assistants, self-driving cars, etc.

Use of Metrological Characteristics in Ultrasound Imaging and Artificial Intelligence Techniques for Disease Prediction in Soft Tissue Organs

Soft tissue organ diseases are causing havoc on the population as a result of lifestyle changes, infections, heredity, and other reasons. A systematic review is carried out in the related chapter to predict diseases of soft tissue organs based on their metrological characteristics associated with ultrasound images and the use of artificial intelligence algorithms. Different soft tissue organs such as the kidney, liver, thyroid, and breast have been considered, and the need for developing computer-aided diagnostic tools for disease prediction of these organs has been highlighted. This is followed by a detailed description of the different disorders affecting the soft tissue organs and their sonographic appearance. This chapter describes the current advancements in the design of CAD systems for the prediction of kidney, liver, thyroid, and breast disorders. The research was carried out on ultrasound images of the breast based on observations acquired from a detailed analysis of related studies. It has been established that some textural properties are better displayed in the transform domain than in the spatial domain for better differentiation of benign and malignant tumors (Kriti et al., 2019, 2020). The non-subsampled contourlet transform (NSCT) has been used to decompose input images into different resolutions to better capture characteristic information. Texture features were calculated from the decomposed image using the GLCM method, and the computed feature set was merged with morphological features to represent the metrological characteristics of the tumors as seen on ultrasound. The resulting feature set has been fed into the adaptive neuro-fuzzy classifier (ANFC) for tumor classification into benign and malignant classes.

Artificial intelligence (AI) has gained in popularity as a technology with enormous potential to enhance many facts of our life besides research in health care. Despite being a trendy issue, the world of artificial intelligence is robust and ever expanding. AI uses computer systems to mimic human intelligence processes. These processes include learning (acquiring knowledge and learning when to use it), reasoning (using rules to reach conclusions), and self-correction. AI works by integrating massive volumes of data with quick processing and sophisticated algorithms, allowing the program to learn automatically from patterns or features in the data. Artificial intelligence in medicine is a potential technology. Recent research

indicates that using artificial intelligence in medicine can lead to the development of excellent diagnostic and prediction systems for identifying various diseases. Because of advancements in the fields of medical imaging and information technology, computers now play a major role in the diagnostic process for diabetes patients. It is challenging to lower mortality rates to more normal levels by using early and precise detection methods. Data on the frequency and incidence of diabetes and prediabetes are important when trying to identify risk factors for complications, acute and long-term difficulties, mortality, and medical care costs. The International Diabetes Federation estimates diabetes prevalence analysis of the existing prevalence of diabetes globally. Furthermore, additional factors, such as physiological indicators, that may play a role in the development of diabetes are being examined. Early detection and proper treatment can prevent up to 95% of visual impairment.

Glucose Monitoring Techniques and Their Calibration

Over the last two decades, there has been an increase in demand for reliable and noninvasive blood glucose testing. Despite the fact that current blood glucose monitoring techniques, such as the widely available glucometer, are well established. Because of the invasive nature of glucometers, measurement results are frequently inaccurate. This review discusses various glucose testing techniques classified as invasive, minimally invasive, and invasive. The development of the enzyme-based invasive approach is also discussed, with an emphasis on the introduction of portable glucometers (Wei et al., 2022, Alsunaidi et al., 2021). The various minimally invasive and noninvasive glucose measuring methods that have been investigated over the last few decades, as well as their calibrations, are also discussed. Moreover, several techniques have been used for this purpose, but device with accuracies comparable to established glucometers remaining a device with accuracies comparable to established glucometers remains a pipe dream. Despite that various noninvasive devices have been developed, mass-scale commercialization has yet to happen. However, the economic benefits of developing a noninvasive device are significant, and commercialization of such a device may be possible soon.

Artificial Intelligence for Iris-Based Diagnosis in Healthcare

Complementary and alternative medicine, particularly iris-based diagnostics, has made tremendous progress in identifying diabetes and related disorders. Diabetes is the primary cause of renal failure worldwide. It increases the risk of heart and blood vessel disorders. Diabetic kidney disease is a long-term disease. The blood test is used to diagnose diabetic kidney disease, a difficult and stressful therapy for patients. This chapter combines iridology with modern computer-based techniques for diagnosing diabetic kidney disease. Participants were evaluated and classified depending on their health state, such as not having diabetes, having diabetes but not diabetic kidney disease, and having diabetic kidney disease (Atul at al., 2015; Piyush &

Ravinder 2019). Pre-image processing procedures are employed to extract the region of interest from the iris based on the iridology chart's specified regions. The region of interest was evaluated for first-order statistical and second-order textual features. Finally, numerous classification models were tested using various sets of attributes to find the best model. The highest accuracy of 94.4% was attained with the help of a fine Gaussian kernel of the SVM classifier. It has been empirically demonstrated that characteristics of particular parts of the iris are substantially connected with an individual's diabetic kidney disease condition.

Assessment and Elimination of Surgeon Hand Tremor during Robotic Surgery

Two major shortcomings of conventional surgeries are the large size of cuts made by the surgeon in the patient body and the physical discomfort of the surgeon. The large size of the cut leads to many aftereffects like more chances of infection, delayed curing of the wound, more blood loss, more scars (Chen et al., 2020, Patle at al., 2019), and more pain to the patient. A surgeon is physically not at ease because he is supposed to stand for many hours continuously. It is painful as, on average, three to four surgeries are generally performed by a surgeon per day.

Laparoscopic surgery, which revolutionized surgical procedures in the mid-1970s after endoscopy reached a satisfactory level of performance, overcame the first of the above shortcoming by using thin, long nose instruments to reach the site of surgery. The small camera of the endoscope, along with fiber-optic light source and surgical tools, only needed around a 1 cm diameter hole on the patient's body. The surgical procedure was aptly named minimally invasive surgery (MIS). The surgical tool was manipulated by the surgeon manually according to the image of the surgical site sent by the camera, which was displayed on a 2-D monitor in front of the surgeon. Till date, this is the most prominent method to teach surgical skills to the interns of surgery by expert senior surgeons in the hospitals. However, the drawback regarding physical fatigue of the surgeon persisted. Then came robotic surgery in the last decade of the last century.

Robotic surgery is that minimally invasive surgery that takes care of this very issue. Here, the surgeon sits comfortably on a chair and guides the surgical arms of the robot, which perform the surgery. Robotic surgery has many attractive features. It allows precision delicate surgeries that were till now thought to be impossible. The robot arms can reach those critical locations in the body where the surgeon's hands cannot reach. However, the real advantage of this technique lies in the fact that the tremor generated in the hands of the surgeon does not affect the precision of surgery as it can be filtered out before it reaches the surgical arms of the robot.

The related chapter of the book discusses the technique to detect and sense the surgeon's hand tremor during surgery and prevent it from reaching surgical tools. The hand tremor was simulated in the lab using an experimental setup and suitable signal processing of the acquired signal. High-order digital filters were designed and tested on the composite signal consisting of volunteer hand motion signal and

undesired tremor signal. The obtained results indicate that tremor was suppressed successfully to allow required precision in surgery.

Conclusion

Biomedical metrology is the application of electronics and measurement principles and techniques to develop devices used in diagnostic and therapeutic applications.

The various chapters of the book have been selected in such a manner that they not only describe an important device or procedure to measure a biological parameter but also lead to the solution of a medical problem in the most innovative manner. Chapters have been written by the authors in a manner that integrates the engineering principles and design concepts to medicine and biology for healthcare purposes. The chapters cover vast spectrum of biomedical measurement area – blood pressure, brain functioning, and medical imaging to surgery. All the metrology techniques have been applied with Artificial Intelligence at the center, which promises complete mitigation of human misery one day. At the same time, the described techniques indicate that biomedical devices of the future will have to conform more to applicable governmental regulations, which set standards for not only metrology but efficacy and safety of humans also.

Cross-References

- ▶ Advancements in Measuring Cognition Using EEG and fNIRS
- ▶ Artificial Intelligence for Iris-Based Diagnosis in Healthcare
- ▶ Assessment and Elimination of Surgeon Hand Tremor During Robotic Surgery
- ▶ Glucose Monitoring Techniques and Their Calibration
- ▶ Metrological Aspects of Blood Pressure Measurement
- ▶ Metrological Aspects of SEMG Signal Acquisition, Processing, and Application Design
- ▶ Use of Metrological Characteristics in Ultrasound Imaging and Artificial Intelligence Techniques for Disease Prediction in Soft Tissue Organs

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Assessment and Elimination of Surgeon Hand Tremor During Robotic Surgery

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Abstract

Tremor is the repetitive movement of some part of the body. It disturbs the daily routine activities of humans like eating, grasping objects, etc. Physiological tremor is a big concern during surgery as the tremor in surgeon's hand may

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lead to harmful results. Robotic surgery makes it possible to filter out the tremor before it is passed on to the surgical arms of the robot actually performing the surgery. In order to eliminate the effect of tremor, it is to be sensed and measured well before time. The research work reported in this chapter pertains to characterization and sensing of tremor by different methods in general and by electromyography (sEMG) in particular. SEMG signal was picked up from the tremor arm, processed, and applied to a classifier to detect the tremor and remove it. Total 14 subjects belonging to different age groups were included in the study. Seven features viz Mean, Standard Deviation, Skewness, Kurtosis, Mean Absolute Value, Zero-Crossing, and Slope Sign Change were extracted and fed to the classifier. Four classifier models viz Support Vector Machine, Logistic Regression, K-Nearest Neighbor, and Random Forest were compared. A decrease in median frequency and increase in power during tremor were observed. Classification accuracy was found as 99.5% with Random Forest model.

Keywords

Hand tremor · Surface Electromyogram (sEMG) · Features · Classification · Accuracy · Robotic surgery

Introduction

In the conventional surgery, a surgeon operating on the patient belly has to make a cut that is 6-to-12 inches long. It gives him the required space to reach the organ which he has to work on and to see what he is doing. Right from the beginning of twentieth century, efforts in laparoscopic surgery started changing this scene inside operation rooms in hospitals. For laparoscopic surgery, also called keyhole surgery, several small cuts are made by the surgeon (of average size half-inch). The name of the process has been derived from laparoscope, a thin and long tool that has a small video camera along with light at its tip. A tube is inserted through each port or opening (Fig. 1). The laparoscope and surgical tools enter into the body through this tube. Then the surgeon starts the operation. The camera image is displayed on a video monitor. Surgeon keeps looking at the monitor to see what is happening inside the body. Due to long nose instruments (Fig. 2), the surgeon does not need to reach into the patient's body. This is also the prevalent method to train surgery interns in the operation theater.

The laparoscopic evolution was initially slow largely because of the limitations of technology and partly by nonacceptance of the procedure by the medical fraternity. The early laparoscopic surgeons were subject of ridicule for conventional surgeons. During the mid-1960s and 1970s, gynecologists contributed greatly to laparoscopic technology and its safety aspects. Video technology was in the developing stages then; camera and monitor had poor resolution; and it was difficult to carry out operative laparoscopy. Moreover, the technology was thought to be used for teaching purpose. The video system adequate for surgery was developed by a surgeon (not an

Fig. 1 Incision in Laparoscopic surgery

Fig. 2 Surgical tool used in laparoscopy

engineer!) in the early 1980s, and it became possible for a group of surgeons to work as a team. Mr. John Wicker, an English urologist, named the procedure as "minimally invasive surgery." It proved to be a revolution in general surgery, specially in urology and gynecology, because of its numerous advantages: small cut size, early healing and consequent saving on hospital expenses, less infection possibility, etc. Patients were happy with this new method of surgery and instrument companies were quick to develop and manufacture long nose surgical tools required for this surgery. Very soon, laparoscopic technique was being used for virtually every abdominal operation (Fig. 3).

Laparoscopic surgery changed the face of surgery from the view point of both surgeon and patient; nonetheless, it has many shortcomings. The biggest difficulty is that there is a requirement of tear minimization at the incision point and, therefore, laparoscopic tools have inherent kinematic constraint. Available laparoscopic instruments have limited degrees of freedom – generally 4 – whereas in conventional surgery, the surgeon has the liberty to exercise 7 degrees of motion. Second point is that two-dimensional vision of the system is no substitute to the natural vision of surgeon. It demands a high hand-eye coordination and dexterity on the part of surgeon. Third, there being no haptic feedback, the surgeon does not have the psychological feeling of accomplishment of his task. The small sense of touch also

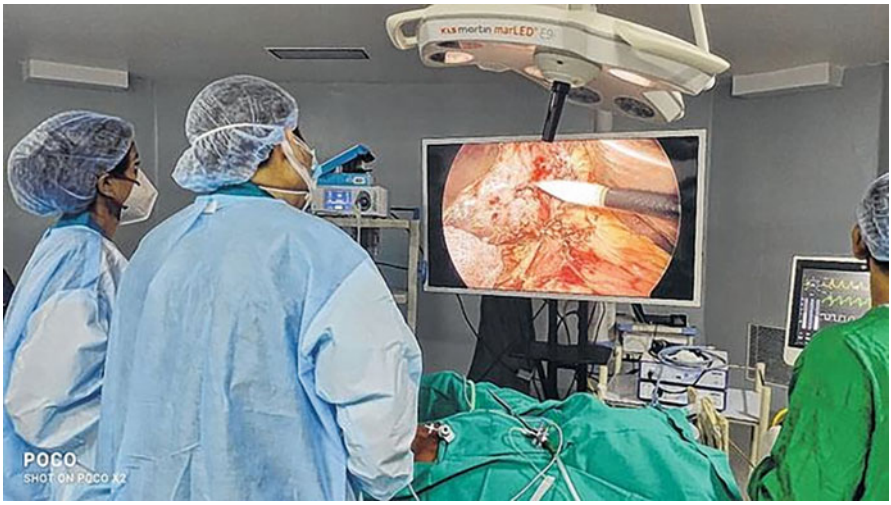

Fig. 3 A typical scene of laparoscopic surgery in an operation theater

makes surgeon's decisions more dependent on his monitor visualization. Finally, physiological tremors in the surgeon's hands due to fatigue cannot be prevented from reaching the surgical tools which can cause harm to the patient when it comes to precise surgeries. Due to these limitations, many complex procedures (e.g., pancreatectomy) have proved difficult to perform laparoscopically. For example, the surgeon needed to adopt awkward positions due to the 2-dimensional vision of the field (Sabiston 2004). Research continued in the field of surgery by both doctors and engineers as it seemed that only a new technology could lead to further development.

Robotic Surgical Systems

The dawn of the twenty-first century brought further advances in the area of minimal invasive surgery. These were the robotic surgery and telepresence surgery which, to a large extent, removed the limitations of laparoscopic procedures (Sabiston 2004; Ballantyne 2002).

The robotic instruments entered the operation theater of the hospital in late 1990s (Camarillo et al. 2004). Before that, PUMA 560 industrial robot arm, which was successfully being used for industrial applications like welding and spray painting, was deployed for a neurosurgical procedure in 1985. A voice-operated robotic system for endoscopic surgery, AESOP® Endoscope Positioner developed by Computer Motion Inc. Intuitive Surgery Inc. (earlier Integrated Surgical Systems), brought out the da Vinci Surgical System® which had two motion subsystems: One was called master handles manipulated by surgeon while the other was a pair of robot arms to execute surgical tasks. It was said to possess master-slave configuration. It was approved by FDA for abdominal surgery in the year 2000 and for

mitral valve repair in 2002. This led to an upsurge in the usage of robotic systems in different types of surgeries like urology, cardiac, pelvic, and neurological. While the robotic technology makes it possible to carry out precise surgery from the surgeon's point of view, it results in less blood loss, faster recovery, less chances of infection, and lesser complications on the part of patient. This surgery allows the surgeon to do the surgery while sitting comfortably while he can rest his hands for few seconds when he feels tired. The vision system of these robots is three-dimensional which shows the depth information related to the movement of the robotic arms and the surgical site. The motion controller of the robot is quite advanced. It performs the necessary motion scaling of the surgeon's hands who moves the master handles in inches while the motion transferred to the surgical arms may be in millimeters in a precision surgery. This allows the surgeon to be natural and look after other tasks related to the procedure. The motion controller also detects incorrect paths of surgical arms and allows their motion only through safe areas. Newer researches are adding advance features in the area of robotic surgery, making the systems more and more cost effective and ensuring more frequent usage of this technology in times to come (Hashizume and Tsugawa 2004).

The main drawback of manual or laparoscopic surgery is inducement of tremor in the hands of the surgeon with time. This tremor is an involuntary, undesirable repetitive movement generated by fatigue (Bhagwath 2001) which causes inaccuracy in microsurgery (Patkin 1977). The biggest advantage of robotic surgery – precision – may be lost due to tremor in the hands of surgeon. Suppression of fatigue-generated tremor is necessary to prevent in microsurgery. Consider the cannulation of retinal vein for injecting anticoagulants to treat occlusions. These occlusions lead to blindness. According to one estimate, about 1.6% of persons of 49 years or more get affected by this disease (Mitchell et al. 1996). The precision needed in this surgery can neither be achieved in conventional surgery nor in laparoscopic procedure (Weiss 1998) as the position accuracy of surgical tool required is of the order of 10 μm (Charles 1996). Another example is ophthalmologic arteriovenous sheathotomy, which is considered as a possible treatment for branch retinal vein occlusion (Opremcak and Bruce 1999), but nearly infeasible with precision available without the robotic surgery (Tang and Han 2000). Similar examples exist in otorhinolaryngological, neurological, and microvascular surgeries (Taylor et al. 1999).

Let us now look at the techniques to reduce the magnitude of physiological tremor during microsurgery. These include both technology- and nontechnology-based methods. At the lower end of the ladder, there are wrist rests and other mechanical supports. Controlling of sleep, the caffeine consumption by surgeon before the surgery also decreases tremor (Humayun et al. 1997). Similarly, beta-blockers have shown reduction of tremor amplitude (Elman et al. 1998). On the technology front, it is the robotic surgery which promises the kind of accuracy required in precise operations. Teleoperation in robotic procedure is the most effective provision to minimize tremor (Das et al. 1999; Jensen et al. 1997; Hunter et al. 1993). This is due to the physical distance between master handles and surgical arms that allows tremor filtering. Simultaneously, it is helpful in implementing

motion scaling. However, the initial cost of a teleoperating robot is very high. Also, there is hindrance to the free movement of persons within the operation theater which may be menacing if these persons have been working in the environment of conventional surgery and have not been trained for robotic surgery. Apart from a surgeon trained on surgical robot for years together, robotic surgery also requires stringent safety measures to be adopted. By combining robotic surgery with communication technology, its usefulness has increased manifold. First, telerobotic surgery was performed between politecnico di Milan, tele-robotics laboratory, Italy, and jet propulsion laboratory, Pasadena, on 7 July 1993 (Rovetta et al. 1993). Advancement was achieved for path planning with the combination of collision avoidance technique and geometric algorithms in the late 1990s (Patle et al. 2019). After some time, three-dimensional technique, planning, and motion flexibility was achieved with good dexterity. Today, success rate of robotic surgery is 94–100% (Robotic surgery). Common robotic surgeries are the following: cardiovascular, head and neck, gynecologic, knee replacement, abdominal colon, rectal, and urology. Some typical cases of oncology take up to 14 h for surgery. In India, presently, there are 8 hospitals where robotic surgery is being done on regular basis. In USA, about 1200 hospitals conduct robotic surgery.

The Surgical Robot Architectures

The ROBODOC® (Fig. 4) was introduced by Integrated Surgical Systems in the early 1990s.

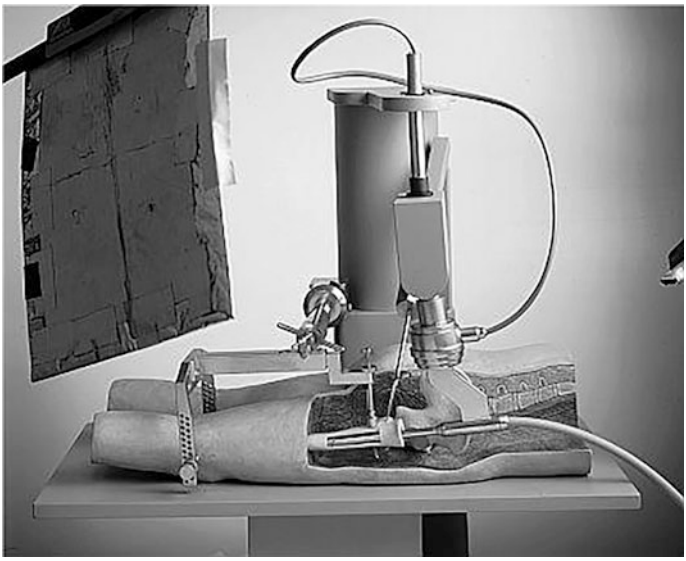

Fig. 4 ROBODOC performing surgery. (Source: www.avrasurgical.com)

For performing surgery by ROBODOC, the surgeon has to first make a choice of the required implant and its appropriate position in the patient's body by looking at the CT images of that area. The robot controller does the path planning, and the surgical tools move accordingly. Registration of the images is done before the surgery. Surgeon marks the surgical tie points so that CT images and patient are in synchronization.

Automated Endoscopic System for Optimal Positioning (AESOP) is an autonomous robot having the capability of imaging only. It is controlled by simple voice commands. It was developed by Computer Motion Inc. in 1994. Its prime function was to position an endoscope at the desired place during surgery. Endoscope is held by a 2-DOF wrist joint. The robot base is clamped to a rail on the operation table. The design of the robot is such that the endoscope may be placed very close to the organ under surgery. The robot is capable of storing three anatomical positions in its memory, and one voice command is enough to recall them (Nathan et al. 2006). The device became quite popular among surgeons because the endoscope supported by robot produced stable images. When the system was compared with manually supported endoscope, it was found that the average operative time was lesser with AESOP (213 min) than with a human assistant (235 min) (Berkelman and Ma 2009) (Fig. 5).

Zeus system has three robotic arms – two of them perform the surgery while the third is an endoscopic arm. The surgeon sits comfortably upright with a control console consisting of video monitor (Fig. 6). The robotic arms execute surgery by mimicking the movement of the surgeon. The endoscopic arm is voice-operated. Surgeon maneuvers the instrument handles which are positioned to maximize his comfort. The surgical arms have 7 degrees of freedom.

Fig. 5 AESOP robotic arm. (Source: http://www.spineuniverse.com)

Fig. 6 Zeus system setup. (Source: Ann Surg© 2004 Lippincott Williams & Wilkins)

Natural Orifice Transluminal Endoscopic Surgery (NOTES) has a master slave configuration where robotic arm enters the body through an orifice (Nakamura 2010). An articulating tube helps the movement of robotic arm in such a way that fulcrum effect is negligible. This makes the manipulations inside the body quit easily. Robotic arm is equipped with three end effectors – the first is a gripper, the second holds scissors, while the third is a clamp whose function is to place the tools at the point where surgery is to be performed. Depending on the surgical task, the electromechanical tool-changing mechanism of the system selects the appropriate tool. The surgical tool is controlled from remote by a joystick. The tool changer (Fig. 7) not only minimizes the operation time by optimizing the process of tool exchanges, but also reduces the tissue trauma during exchanges. Placing the actuators at site increases the effectiveness of robot as the end effectors can apply high force during operation. Kinematics of the robot does not pose restrictions in movement as the robot arm is quite far from the articulating tube.

The da Vinci system (Fig. 8) has 4 subassemblies: (1) a vision cart consisting of endoscopic arm having a pair of light source and a pair of cameras, (2) a 3-D visualization console where the operating surgeon sits, (3) a moveable patient cart consisting of two surgical robotic arms and one manually operated manipulating arm, and (4) surgical tools which are to be changed during surgery; these are fitted to the grippers of surgical arms depending on the surgical task to be done.

The endoscopic arm has a pair of cameras mounted at the tip. Each camera generates a 2D image from which 3D image is reconstructed. The 3D video console

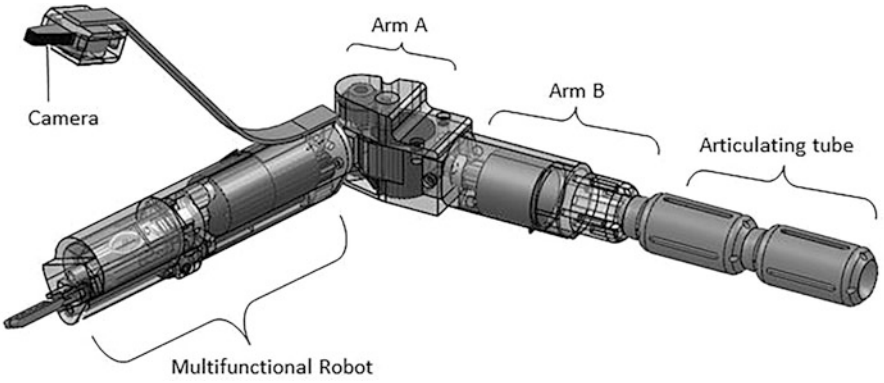

Fig. 7 NOTES multifunctional robot

Fig. 8 da Vinci system set up. (Source: http://www.intuitivesurgical.com)

shows a true three-dimensional image which has the depth information of the surgical site. It also has foot pedals by which endoscopic arm is controlled by the surgeon. The foot pedals also control electrocautery. The surgeon console is equipped with encoders or master handles whose motion is captured by the motion controller and transferred to the surgical robot arms after scaling down. The robotic arm grippers are operated through motion transfer by cables driven by motors at the other end. The gripper has a total of 7 degrees of freedom – three body motions, three wrist motions, and one open-close operation.

Advantages of Robotic Surgery

Robotic surgery overcomes many shortcomings of the conventional and laparoscopic surgery. Difficult surgeries become feasible because:

(i) In robotic surgery, surgeon has more flexibility in execution of surgical tasks.
(ii) Tremor in the surgeon's hand can be prevented from reaching the end-effecter by filtering the encoder signals using appropriate filters.
(iii) Large movements made by surgeon with encoders are scaled down into motions of mm size or less by motion controller.
(iv) The clarity of surgical site through 3D vision results in excellent hand-eye coordination.
(v) Surgeon is in the most comfortable position, and his body does not have to go through twists and turns to carry out surgical tasks.

Robotic surgery is most suitable for eye surgery, prostate surgery, and knee and hip replacement which demand high precision and accuracy. According to an estimation, by 2035, surgeons will be replaced by robots considering the dexterity in suturing and accuracy.

Challenges and Future of Robotic Surgery

The challenges to robotic surgery systems are as follows:

(i) Robotic surgery is a relatively new technique, and its propriety is yet to be established. Critics claim that the duration of robotic surgery is more as compared to corresponding conventional or laparoscopic surgery. It has been observed that on average, robotic surgery takes about 50 min more than laparoscopic surgery. Supporters of robotic surgery argue that this is mostly due to lack of training of persons associated with this surgery.
(ii) Initial cost of surgical robot is prohibitive. Only big hospitals can afford it.
(iii) Operation theaters are already congested having a large number of equipment. Robotic surgical system is voluminous and occupies substantial floor space. Accommodating the robot in the existing operating room along with the personnel is quite difficult. Future surgical rooms may be constructed bigger to overcome this disadvantage.
(iv) At present, nurse changes the surgical tool manually. This results in longer operating time. In future, this task may be made automatic so that tool changing time decreases substantially (Reuvekamp 2009).
(v) Robotic surgery requires massive training of surgeon and nurses. According to one estimate, minimum 15 complete practice surgeries are a must before the procedure is done on an actual patient.
(vi) A high-power back up supply is to be ensured because failure of electricity may put the patient's life in danger.

One requirement of successful robotic surgery is that patient and surgeon must not be at large distance away from each other so that surgeon's commands can be followed by surgical arms in real time. If it is required to keep this distance large, the commands to the arms may be communicated through optic fiber cables to maintain accuracy, but this results in two disadvantages. One, system's cost increases, and second, time lag between issuing of commands from master stage and their execution at slave stage may be so high that the patient is ultimately harmed. The development of wireless commands for robots may be the solution to this problem. With advances in communication technologies, robotic surgery of future is expected to possess wireless control (Electronic devices). Control of surgical arms using voice commands will result in lesser supporting staff in the operating room.

One point to be kept in mind is that due to expansion of robotic surgery, human beings will not be laid off but their skill set will undergo a change. They will be more competent than their present counterparts.

The main requirement of surgical robot is that it should possess good computational capabilities. The artificial intelligence (AI) is integrated in the design of this surgery. Surgeon takes the training, performs the procedures, and gets the experience even though the mistakes are committed. Main motive to implement AI in the field of medicine is to keep the surgeons alert while performing the surgery. The base for AI is heuristic knowledge by which the machine learns with real life experiences. Genetic algorithm (GA) and neuromorphic engineering are mostly used in AI. The meaning of neuromorphic engineering is to transform the microcircuitry in analog computing medium which can mimic the neural tissue. The structure requires the equivalent of neurons in hardware like resistors, capacitors, and silicon chip which can store the acquired signal. Latest research is to determine and copy the interconnections of neurons of brain. AI maintains the population of chromosomes and coding for control architecture. There are three requirements for robot to meet: learn, remember, and evolve. Surgeons are being trained on computer virtually for surgery based on preoperative or intraoperative imaging data of structures. With this process, the risk of mistakes during surgery can be avoided.

The integration between an unsolved problem and new approach to solve the problem is a challenge in robotic surgery. One solution is that the surgical tool should be designed in such a way that it may be used in some advanced way or the result should be better, i.e., mortality should be decreased. In surgical system, the imaging and manipulation should be enhanced. For development of surgery, the input and the guidance from the clinical area are required. The role of surgeons is there in two ways: first to collaborate with research team of technology developers and then to utilize the developed technology in their specialities. Institutional programs should be organized frequently on surgical innovations.

Ethical and Safety Considerations

Robotic surgery is technology-intensive electromechanical device, and therefore, any mishap may take place within the operation theater during surgery. It may even

put the life of patient in danger. Therefore, this surgery needs the presence of technically trained person in the operation room who can tackle any emergency related to the robot and who can help the surgeon during surgery. This surgery involves many legal and ethical issues also that require attention of medical, technical, and administrative authorities. Following are some of these issues:

(i) Is it ethically correct to charge the cost of robotic surgery from patient? He, in fact, never insisted on robotic surgery. It was the surgeon's preference and decision.
(ii) Who is responsible in case robotic surgery fails because of *technical problems* – the surgeon or the robot-manufacturing company?
(iii) If robotic surgery involves teleoperation and the surgeon and patient are located in different countries, how will the ethical and regulatory norms be followed?

Biological Tremor and Its Characteristics

The tremor in humans is a continuous and involuntary shaking of a body part characterized by its frequency, location, and pharmacological response (Smaga 2003; Mansur et al. 2007). The tremor movement may occur either due to neurological disorders or due to natural physiological reasons. Physiological tremor is always present in human beings (Borges and Ferraz 2006) to some extent. An example of neurological disorder tremor is the one during Parkinson disease (Almeida et al. 2010) which is caused by dysfunction of the central nervous system. There are few other, relatively rare, pathological tremors mentioned in the literature (Wyne 2005).

Physiological tremor can be categorized as resting tremor and action tremor (Borges and Ferraz 2006). As the name indicates, resting tremor occurs when the body is in rest position and suddenly the parts of body start shaking due to some disease. During this tremor, the muscles are relaxed, for example, when the hands are resting on the lap. Normally, this tremor affects only the hands or fingers. It happens due to some problem in that part of the brain which governs motor functions in the human body. It may occur due to neurological disorders such as stroke, brain injury, and neurodegenerative diseases, use of medicines especially for asthma, or drugs used for neurological disorders, alcohol withdrawal, thyroid problem, liver or kidney problem, anxiety, or panic (Rocon et al. 2007). Parkinson tremor is a type of resting tremor.

The action tremor comes to fore when a person contracts his muscles (Borges and Ferraz 2006). It occurs due to excessive force experienced by the muscles, mainly due to fatigue. Time domain behavior of tremor shows that the tremor increases with time.

Tremor can be classified into three categories on the basis of its frequency. Low-frequency tremor is less than 4 Hz, middle-frequency tremor is between 4 and 7 Hz, and high-frequency tremor has frequency components more than 7 Hz

(Borges and Ferraz 2006). To cover full range, frequency of physiological tremor may be taken as 3–15 Hz (Latt et al. 2011).

From the view point of location, tremor is most visible in the arms, legs, and the head. Most affected part of the body due to tremor is hand where it manifests itself by difficulty experienced in the tasks performed by hand such as writing or drawing or grasping of objects. It may sometimes be seen in periodic shaking of head or legs or as a shaky voice.

Tremor is a nonstationary phenomenon (Deuschl et al. 2001). Tremor is observed more when a person is in stress, physically exhausted, occupies certain posture or movements. Amplitude of tremor increases with fatigue or stress and disappears after taking some rest (Adhikari et al. 2016). Tremor increases with aging, and mostly it occurs after the age of 50 years, but people may suffer due to tremor at any age (Moghadam et al. 2018). Men and women are affected equally with tremor.

Tremor is not life-threatening but affects day-to-day routine work. During a typical surgical procedure, tremor is to be taken on priority. By holding the surgical tool for a long time with controlled movements, the muscles get tired and the resultant trembling of hand may erroneously lead to the cutting of some tissues other than the intended ones (Taylor et al. 1995). It has been observed that hand tremor increases continuously with the duration of surgery (Slack et al. 2008).

Techniques for Detection and Measurement of Tremor

Human tremor can be estimated by many methods. The most popular method in the clinic is by severity scales (Ramaker et al. 2002; Greffard et al. 2006). Here, the patients draw various drawing patterns which include geometrical figures, letters, and numerals (Fig. 9). Neurologists evaluate the deviation of drawn pattern from the reference and assign a numeric scale 0–5. Here, 0 means no visible tremor while 5 is severe disabling tremor. It is clear that this type of classification is subjective. Second, parameters characterizing the tremor – frequency, amplitude, and speed – cannot be determined from this method.

Clinical evaluation does not give much information about the characteristics of tremor or its genesis as it is not patient-specific. To overcome these shortcomings, electronic techniques have been developed for the assessment and analysis of tremor. These include digitization, accelerometry, position encoding using magnetic tracking system or active optical markers, spirography, and electromyography (EMG) (Almeida et al. 2010).

Digitizers promise the measurement of tremor quantitatively directly on the computer. The digitizing tablet is able to record the coordinates of the tip of the pen held by the patient on the digitizer surface. This device can detect the movement of a subject following standard drawing patterns placed on it (Elble 2003).

Spirography makes use of digitizing tablets (Almeida et al. 2010). In this method, patient is asked to reproduce the standard Archimedes' spiral. The tablet surface contains the reference image of this spiral, and the patient is required to trace the

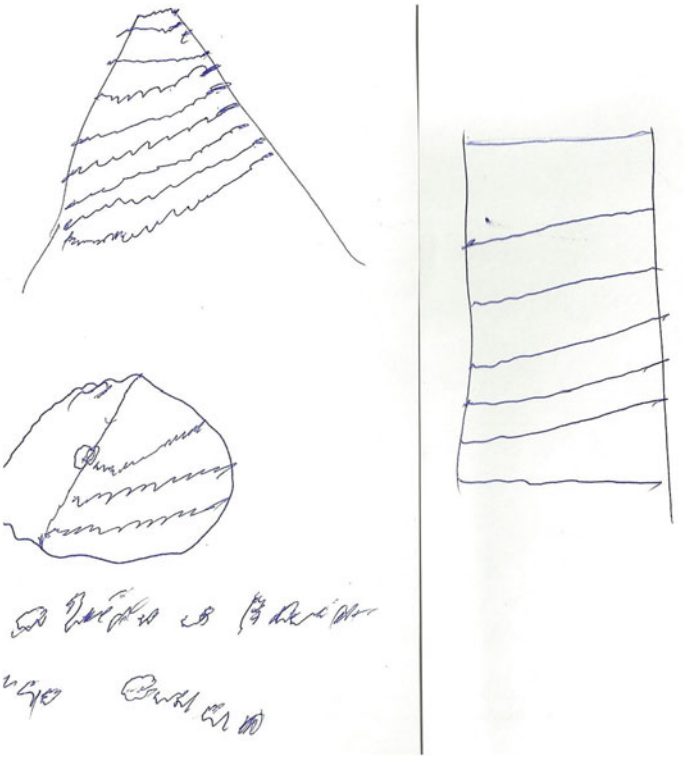

Fig. 9 Drawing patterns drawn by tremor patients

spiral image as accurately as possible. Archimedes spiral has several features which make it a popular method of tremor assessment:

- Its design is straightforward, and patients can easily understand it.
- There are no sudden changes in the shape, and it allows smooth movements.
- Digital spirography provides quantitative estimation of the movement control (Pullman 1998).
- The signals can be stored in a computer for later reference and analysis.

The analysis of patient signal can be done by image processing. Geometric transformations may be applied to the patient drawing image to extract relevant features (Saunders-Pullman et al. 2007). Statistical analysis can be done using advanced tools algorithms for data evaluation (Haubenberger et al. 2011; Bajaj et al. 2011).

Position encoders have also been used in the assessment of tremor (Gopal et al. 2013). Encoders provide a position feedback signal of an object. They are available with analog (0–5 V, 0–10 V, or 4–20 mA) or digital output. Both linear and rotary encoders may be used to measure tremor depending upon the attachment of sensor

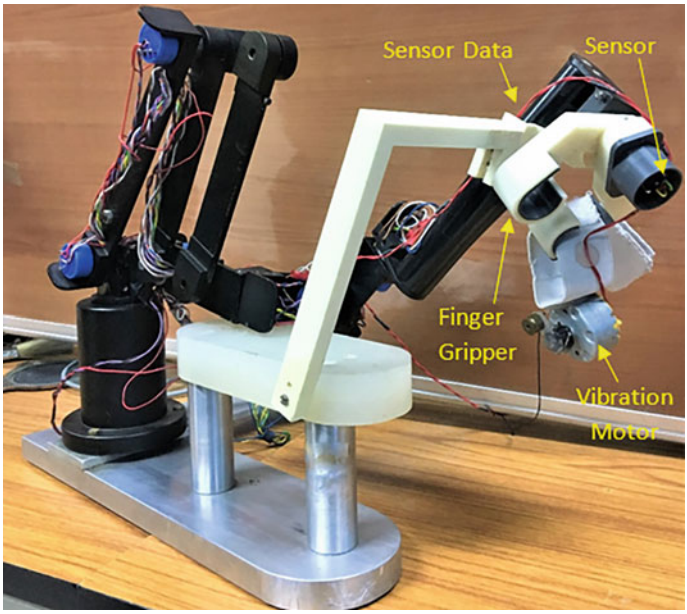

Fig. 10 Prototype of two Degree of Freedom (DoF) tremor generation and measurement arrangement

with the object under tremor. A prototype system of two degree of freedom (DoF) developed by authors of this chapter for generation and measurement of tremor by displacement sensor is shown in Fig. 10.

To simulate tremor, 12 V DC vibration motor was attached to a fabricated finger gripper. An analog rotary encoder with 12-bits resolution was mounted on the gripper joint to sense the tremor. The encoder output was a signal which was a total of surgeon's voluntary hand motion and involuntary tremor motion. The position was achieved with precise linearity by doing numerical integration of output signal of encoder. The calibration for sensor is not required as its rotational life (shaft revolutions) is 100 million. The advantages of using this sensor in over accelerometer are the following: noncontacting magnetic technology, highly resistant to vibrations, programmability at factory for zero position, highly repeatable, extremely long life, stability, linearity, programmable electrical angle, minimal sensitivity to environment parameters, and ability to withstand side load on arm. The output signal of sensor was amplified by a factor of 2. The output voltage was digitized by NI 9205, a 16-bit data acquisition system @ 1 kHz. Tremor was determined in both vertical and horizontal directions and its peak frequency displayed.

Artificial tremor in the range of 3–15 Hz was generated using vibration motor. Figure 11 shows the composite analog signal – high-frequency component tremor superimposed on low-frequency voluntary motion signal. The data was acquired and analyzed at various tremor frequencies. Figure 12 shows the Fourier Transform (FFT) of signal at 10 Hz tremor.

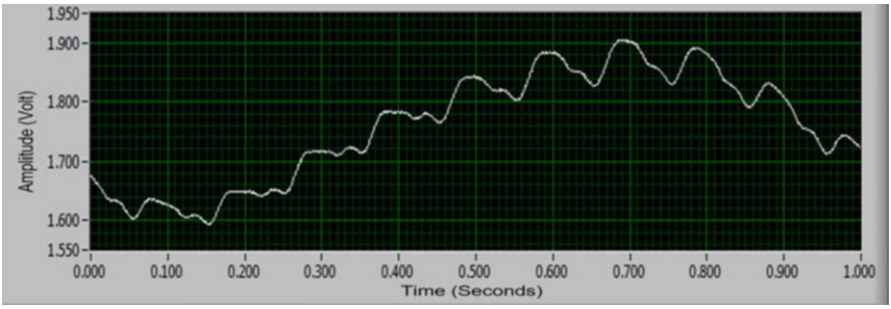

Fig. 11 Recorded raw signal having voluntary and involuntary motions

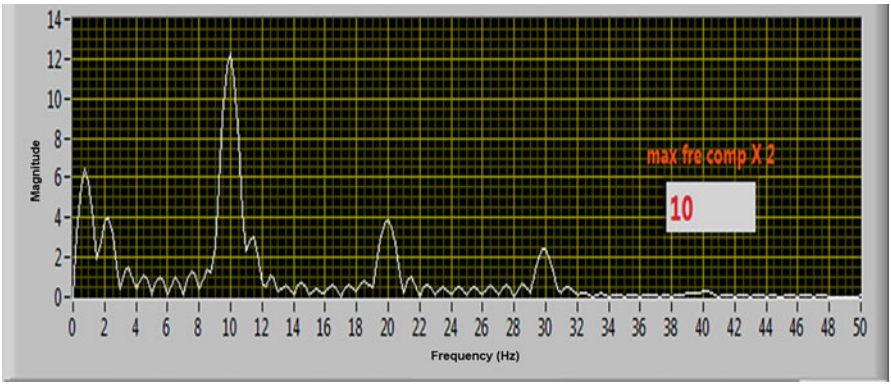

Fig. 12 FFT of raw signal having peaks of voluntary and involuntary motions

From the FFT diagram, it can be observed that peak power exists in the tremor signal of 10 Hz and then in the volunteer signal which is around 1 Hz. The other peaks exist at 20 Hz and 30 Hz which are the harmonic frequencies of the fundamental tremor frequency of 10 Hz. To keep only the volunteer frequency and tremor frequency and eliminate high frequencies, a low pass filter with cutoff frequency $f_c = 20$ Hz was designed. The composite signal was filtered by this filter. The output of filter now had only volunteer signal and fundamental tremor frequency.

To eliminate tremor, it was required to notch 10 Hz. A Butterworth filter having stopband from 8 to 12.5 Hz and passband frequencies of 4 and 18 Hz and of order 5 was realized which removed the tremor component. It was an adaptive filter which adjusted its notch frequency according to the change in the frequency of tremor signal. For this purpose, peak frequency of tremor signal is monitored continuously in real time throughout the duration of the experiment.

Accelerometry is another popular method of tremor determination. In this method, accelerometer monitors the vibrations produced by the tremors (Elble 2003; Raethjen et al. 2000). The principle of accelerometer is based on Newton's second law of motion. It measures linear acceleration along three orthogonal

directions. The sensor inside an accelerometer is an electromechanical device which makes use of either the piezoelectric phenomenon or the variation of capacitance caused due to vibrations. Accelerometer outputs the instantaneous value of the vibration of the body part to which it has been physically attached as a function of time. These values are logged in and can be analyzed later.

Tremor can be detected and measured quite easily by electromyography (EMG). The electromyogram signal is generated when a muscle undergoes contraction and relaxation. Neuromuscular disorders can easily be diagnosed by analyzing EMG of such patients. Two types of electrodes are used to acquire EMG. Needle or wire electrodes are inserted into the muscle if activity of a particular muscle is to be recorded (Elble 2003; Raethjen et al. 2000). If the signal is to be acquired from a group of muscles, it can be accomplished by placing electrodes on the skin surface. The parameters of EMG signal which characterize tremor include frequency (Hz), mean amplitude (mV), and pattern (synchronous or alternating) (Milanov 2001).

During acquisition of EMG signal, ECG signal generated by heart acts as an interfering signal. The effect of heartbeats on acquired EMG can be minimized if EMG is recorded with muscles carrying some weight. For tremor analysis using EMG also, some task is chosen which involves weights.

Yet another method to measure tremor is by gyroscope. While accelerometers measure linear acceleration, gyroscopes measure angular acceleration. Their output is expressed as degrees per second (°/s). Accelerometers measure inertial forces, and therefore, they provide a frame of reference in relation to gravity. Problem occurs when the accelerometer is also vibrating. In this situation, reference of accelerometer becomes difficult to maintain. An accelerometer can provide a false frame of reference if it is subjected to sustained static accelerations. Gyroscopes do not suffer from this drawback. They provide a stable, oriented, and accurate frame of reference. Thus, the gyroscope provides true course of motion. Gyroscopes are not usually used alone for vibration measurement, but when paired with accelerometers, they offer a much more complete picture of the vibrating environment. Currently, there are wireless devices that integrate accelerometers and gyroscopes for movement capture (Giuffrida et al. 2009).

There are some lesser used techniques that measure displacement and can be used to measure tremor. One such method is the magnetic tracker system which gives information about the linear movement along Cartesian coordinates (x, y, and z) and orientation in spherical coordinates (pitch, roll, and yaw) of an object (Ghassemi et al. 2006). Acceleration can also be measured using active optical markers, and the application of trigonometry makes it possible to describe the vector orientation and the estimation of posture (Albert and Kording 2011).

Measurement methods have been combined with advanced communication techniques to make tremor evaluation results available remotely. For example, tremor is done through videos (Uhríková et al. 2011). There are tools that use tremor sensor and transmit information through Internet and Bluetooth technology (Barroso Júnior et al. 2011).

After acquiring the tremor signal, it is to be processed to extract useful information carrying features. Usually, the signal processing is done in frequency domain for

which Fast Fourier Transform (FFT) of the signal is taken (Giuffrida et al. 2009; Barroso Júnior et al. 2011). There are some applied variants of FFT which are more effective when signal fulfills certain conditions. For example, its version Weighted Fourier Linear Combiner (WFLC) (Riviere et al. 1997) is more suitable for periodic or quasi-periodic estimation of motion with single dominant frequency. Similarly, Band-Limited Multiple Fourier Linear Combiner (BMFLC) (Veluvolu and Ang 2011a) is better when the tremor signal is band-limited signals and it contains multiple frequency components. Statistical tools may be used for extraction of parameters which may be used for classification of tremor. Other algorithms may also be applied depending upon the objective (Latt et al. 2011; Popovi et al. 2010). It is very difficult to pinpoint the best method of analysis as tremor is a random or quasi-random signal (Mansur et al. 2007).

It may be observed that any tremor sensor provides a composite signal in which hand tremor of surgeon is superimposed on voluntary motion corresponding to the surgical tool movement. Hence the signal processing is to be done first to separate the two motions before extracting the characterizing features of tremor to analyze it. Researchers have suggested some methods to accomplish this task and estimate the tremor. An Extended Kalman filter (EKF) was proposed to assess tremor in the presence of associated noise (Bó et al. 2009). An auto-regressive moving average (ARMA) model has been described in (Becker et al. 2008). Literature survey indicates that the solution to this problem has also been attempted using stochastic filters (Becker et al. 2008).

Summarily, the different methods of tremor estimation are based on sensing the displacement, velocity, acceleration, gyroscopic parameters, and electromyography (Bridgeman and Arsham 2017; Veluvolu and Ang 2011b; Teskey et al. 2012). Mostly, accelerometer is used for tremor measurement for the purpose of kinematic and dynamic studies of hand (Kim et al. 2009a; Zhang et al. 2011). However, accelerometer, being an inertial sensor, cannot predict the fatigue. To measure the muscle fatigue in time and frequency domain, surface electromyography (sEMG) has been suggested as the best technique (Galinsky et al. 1990; Chen and Chen 2020; Cifrek et al. 2009a).

Feature extraction and classification are main tasks to evaluate the tremor. Statistical measures of sEMG spectral parameters have been widely used for characterization of muscle fatigue (Basmajian and Luca 1985; Slack and Ma 2007a). The standard parameter of measurement of fatigue is median frequency (Chandra et al. 2018). Measurement of amplitude of sEMG, i.e., action potential, is also a widely accepted parameter (Bo et al. 2011; Ackmann et al. 1977). Previous researchers have reported attributes like root mean square (rms) (Shenoy et al. 2008), time domain parameters (Fougner et al. 2011), time frequency representation (Wang et al. 2006), auto regressive (AR) coefficients (Graupe and Cline 1975), multivariate AR model (Hu and Nenov 2004), sequential feedforward selection (Hargrove et al. 2009), etc. To reduce the vector dimensions of EMG features, various tools like discriminant analysis, prediction probability, and principal component analysis (PCA) (Englehart et al. 1999) have been used.

For suppression of hand tremor, the use of bandpass and high pass filters is a good approach as explained in (Zhang and Ang 2006). Some authors have used the fuzzy

logic proportional derivative controller to suppress the tremor (Bó et al. 2011). Tremor can be suppressed with proportional integral controller by means of functional electrical stimulation (FES) (Gallego et al. 2013). Tremor can also be suppressed with impedance-based approach in addition to FES by increasing the viscosity of muscles (Bó et al. 2014). With repetitive control (RC), the tremor is suppressed. With this approach, one need not modify the joint dynamics (Freeman et al. 2015). RC scheme is structurally efficient though it is not considered having a general form. Since distal joints are stronger than proximal joints, FES-based tremor suppression was found better for tremor in wrist (Bain 2002). Surface electromyography (sEMG) was used for measuring muscle fatigue and was found a faithful measure for assessment in both time and frequency domain (Cifrek et al. 2009b). Though accelerometer was used for dynamic and kinematic study of hand tremor, the tremor cannot be assessed with accelerometer alone as inertial sensors cannot capture the induced tremor (Kim et al. 2009b).

EMG-Based Tremor Sensing and Elimination

Spectral measure of sEMG provides the most used metric for physiological tremor (Slack and Ma 2007b). A decrement of the mean and median frequency of sEMG is certainly a standard measure of fatigue level; sometimes amplitude of the raw sEMG signal is also added in this list (Kallenberg et al. 2007). If proper feature extraction and classification techniques are used for sEMG signal, it can provide good results to predict the tremor. K-nearest neighbor and SVM are popular classifiers for EMG signal (Oskoei and Hu 2008). Combination of different neural and fuzzy techniques for classification also provides good results (Kim et al. 2004). Many researchers have used feed-forward backpropagation neural network and Fuzzy-based classifiers (Ajiboye and Weir 2005). Some researchers have used wavelet neural network to classify the sEMG (Subasi et al. 2006). The accuracy obtained was 90.7%. By decoding the ten individual finger flexion and extension with 32 channel surface sEMG signal for transradial amputees, the accuracy obtained was 90% (Tenore et al. 2008).

An EMG-based tremor sensing, signal processing, separation from volunteer motion, and elimination have been described in this section. Further, analysis of the separated tremor motion was done by extracting its characterizing features and subsequent classification. The remaining volunteer motion corresponding to surgical action may be transmitted to the surgical arms of the slave robot after necessary kinematics and inverse kinematics algorithms in the motion controller module of the surgical robot.

EMG Signal Acquisition

Total 14 subjects were recruited (eight males, six females, weight 58–74 kg, and height 154–180 cm). The demographic details of subjects are given in Table 1. Data was collected using in-house developed active electrodes (Fig. 13). Process of recording and handling of phantom was explained to subjects, and written consent

Table 1 Subject details

S. no.	Age (years)	Weight (kg.)	Height (cm.)
1.	26	58	154
2.	25	63	163
3.	24	59	163
4.	23	70	170
5.	29	70	166
6.	26	72	177
7.	26	58	165
8.	24	62	167
9.	19	50	170
10.	28	65	180
11.	44	64	162
12.	70	68	165
13.	68	72	169
14.	49	74	170

Fig. 13 Active electrode used for sEMG signal acquisition

was taken from all the subjects before the data collection. Ethical principles of medical research were followed during the whole process of signal acquisition (World Medical Association 2013).

During recording, the subjects were kept warm as cool limbs slow down the nerve conduction (Naik et al. 2016; Chang 2019). Electrode site was cleaned with mild soap and water. 4-channel raw EMG signal was acquired in two different trial conditions. During first trial, subjects were instructed to be in relaxed state as in Fig. 14 and the sEMG data was recorded by operating the phantom handle. The movement of phantom handle was kept similar to standard surgical hand movement. No fatigue was, thus, developed during this phase.

During the second set of trials of sEMG acquisition (Fig. 15), a weight of 2.7 Kg was placed on the subject's forearm in sagittal plane followed by hand exercise with hand gripper for about 4 min because of which sufficient fatigue and tremor was developed.

Four electrodes were placed on the muscles to acquire the sEMG signal (Haris et al. 2015) at the positions of Flexor carpi ulnaris and Palmaris longus and at Extensor carpi ulnaris and Extensor carpi radius. Between trials, sufficient time was given to subjects to get normalized. These electrodes were put as per the norms of

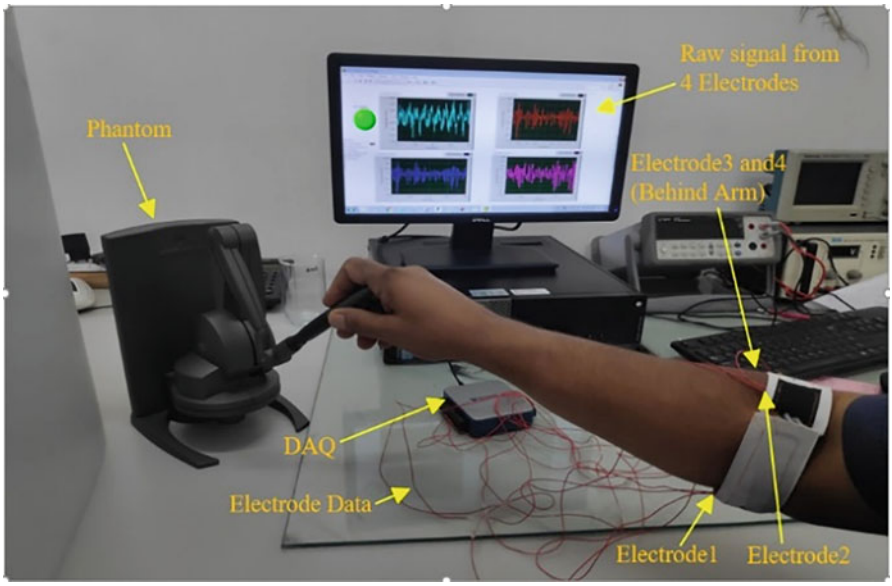

Fig. 14 Experimental setup for signal acquisition in relax mode

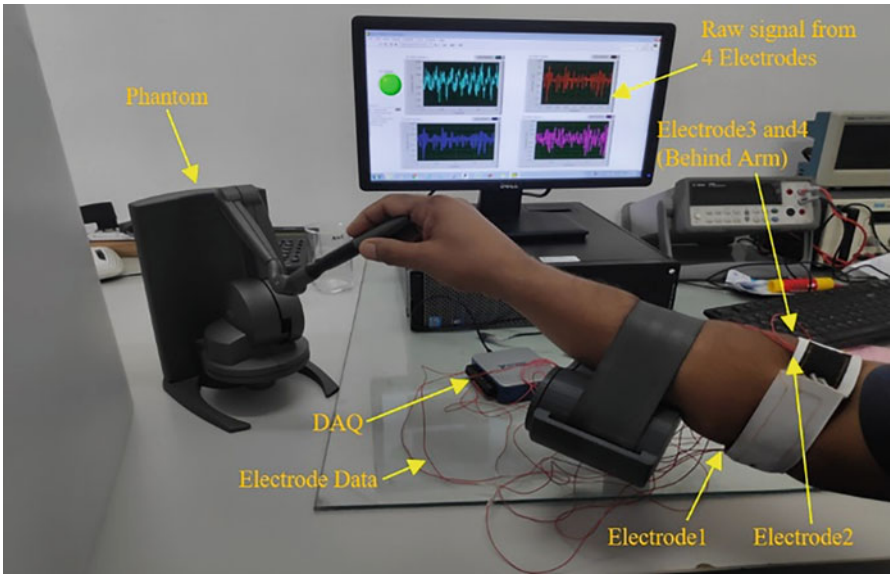

Fig. 15 Experimental setup for signal acquisition in tremor mode

Surface Electromyography for the Non-Invasive Assessment of Muscles (SENIAM) (Hermens et al. 2000). The sampling rate was set at 512 samples/sec and recording time 30 s. Amplification of the EMG signal was adjusted so that maximum output

voltage of active electrode was in the medium range of about 2.5 volts rms (approx.) during full contraction of muscles. sEMG signal was acquired with data acquisition card NI USB 6210 of National Instruments, UK.

EMG Signal Processing

Figure 16 shows the plot of 4-channel raw sEMG signal in relaxed state, and Fig. 17 is the raw signal obtained in tremor mode. In order to remove the power line interference of 50 Hz, the recorded raw signal was passed through a Butterworth bandstop filter of order 9. For this filter, passband ripple was selected as −2 dB and stop band attenuation as −50 dB. The edge frequencies were taken at 1, 3, 50, and 100 Hz. The transfer function of this filter was calculated as:

$$H_a(s) = \frac{1.31}{s^9 + 1.13s^8 + 4.34s^7 + 4.68s^6 + 7.07s^5 + 7.28s^4 + 5.1s^3 + 7.31s^2 + 3.74s + 1.3} \quad (1)$$

Another bandpass filter of 7th order with bandpass frequency of 3–15 Hz was used to remove the signals other than tremor. This Butterworth filter was designed for passband ripples of 1 dB and stop band attenuation of 40 dB. The transfer function of this was obtained as:

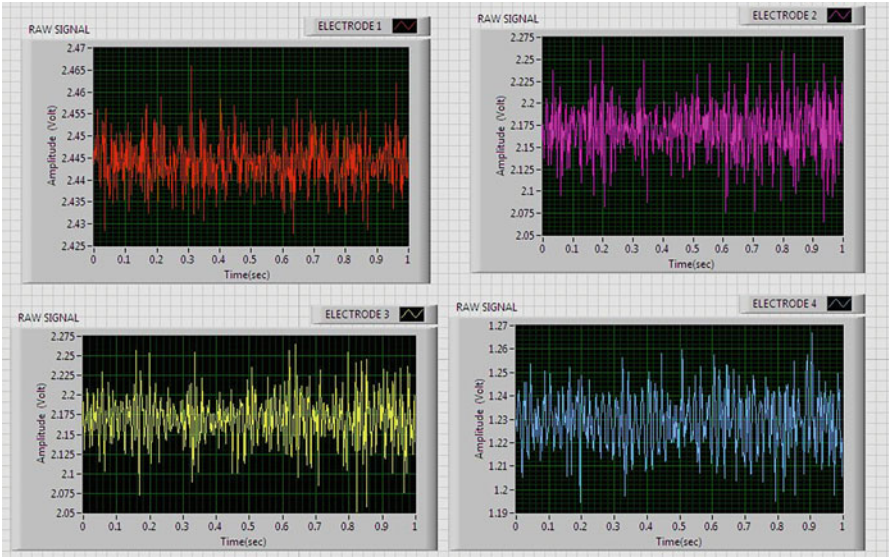

Fig. 16 Raw signal of electrodes in relax mode

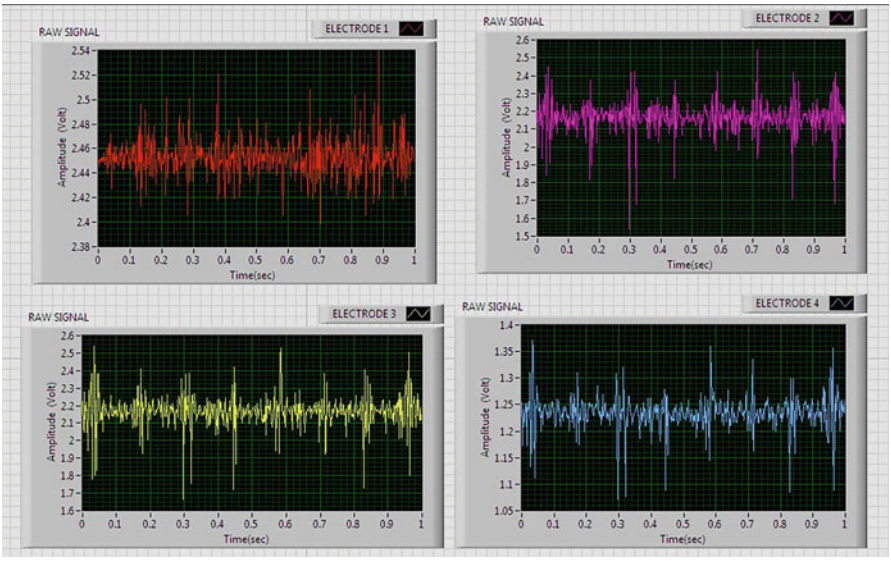

Fig. 17 Raw signal of electrodes in tremor mode

$$H_a(s) = \frac{1.32}{s^7 + 4.66s^6 + 11.53s^5 + 14.97s^4 + 14.87s^3 + 10.03s^2 + 4.09s + 1.31} \tag{2}$$

The filtered sEMG signal for both relax and tremor state is shown in Figs. 18 and 19. After filtering, the signal is ready for further analysis and tremor classification.

Feature Extraction

The 4-channel sEMG signals were analyzed with 1 s epoch without overlapping. Six linear features were extracted. These features are Mean, Standard Deviation, Skewness, Kurtosis, Zero-Crossing, and Slope Sign Change (Duan et al. 2016). The features were selected such that they have a significant relationship with tremor intensity (Das et al. 2001). To differentiate tremor from relaxed state, data was arranged in the tabular format required for classification. Since four electrodes were used, there would be 24 columns (6 × 4). One additional column for class, i.e., "tremor" or "relaxed," was kept. Data was acquired in four different time slots of 30 s each for two different hand movements (relaxed and tremor). Thus, 120 feature vectors of 30 s for relaxed and 120 feature vectors of 30 s for tremor state were generated. Hence, the dimension of the feature dataset was 6000 (240 rows × 25 columns). The model output determines the severity of tremor. The description of selected features is as below:

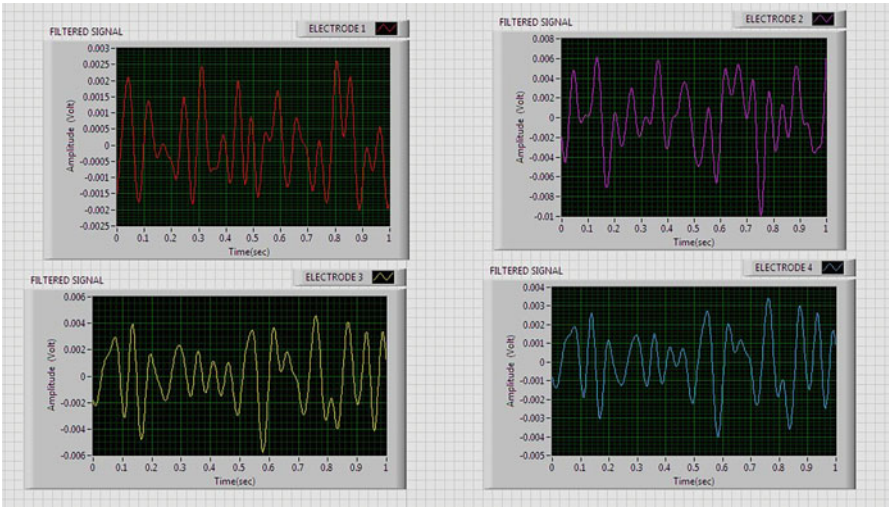

Fig. 18 Filtered signal of electrodes in relax mode

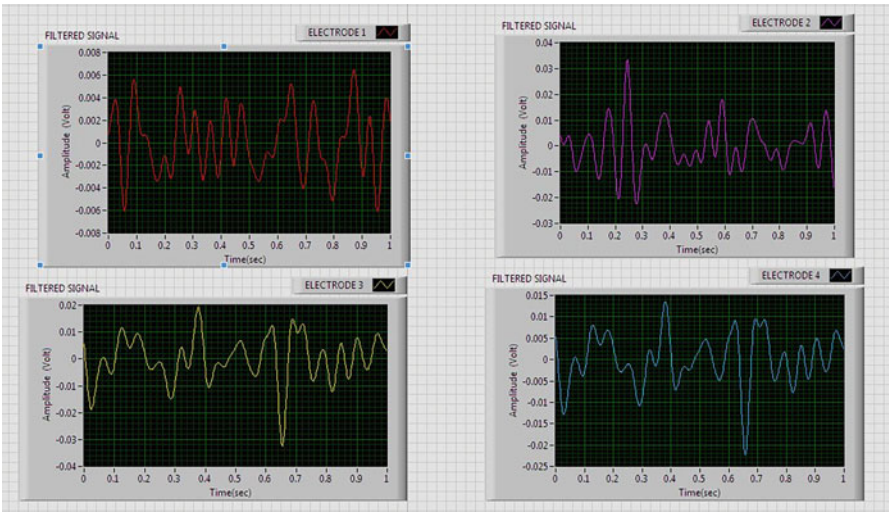

Fig. 19 Filtered signal of electrodes in tremor mode

Mean

Mean is a measure of central tendency which is calculated by aggregating all the values of the dataset by sum operation and then dividing it by the size of dataset. Mathematically, it is defined as:

$$\text{Mean of Dataset } \mu = \frac{x_1 + x_2 + \ldots + x_n}{n} = \frac{\sum_{i=1}^{N} x_i}{n} \tag{3}$$

Standard Deviation

It measures the amount of variation in the dataset from the mean value. If it is low, dataset values are close to one another and very close to the mean, and if it is large, it represents the values are far apart from one another and far from mean. Mathematically, it is represented as:

$$\sigma = \sqrt{\frac{\sum (x_i - \mu)^2}{N}} \tag{4}$$

where σ: population standard deviation; N: size of the population; x_i: each value from the population; μ: The population mean

Skewness

Skewness is a measure of the asymmetry of the probability distribution of a real-valued random variable about its mean. It may be +ve or –ve. In case of positive skew, tail of the probability distribution at the right side is longer as shown in Fig. 20a. It shows that maximum mass is concentrated toward the left of figure. In negative skew, the left tail is longer (Fig. 20b). Mathematically, Pearson's moment as coefficient of skewness is represented as:

$$\mu_3 = E\left[\left(\frac{X - \mu}{\sigma}\right)^3\right] = \frac{\mu_3}{\sigma_3} = \frac{E\left[(X - \mu)^3\right]}{\left(E\left[(X - \mu)^2\right]\right)^{3/2}} \tag{5}$$

Kurtosis

Kurtosis is a statistical measure that defines how heavily the tails of a distribution differ from the tails of a normal distribution. In other words, kurtosis identifies whether the tails of a given distribution contain extreme values. For univariate data, kurtosis is mathematically defined as:

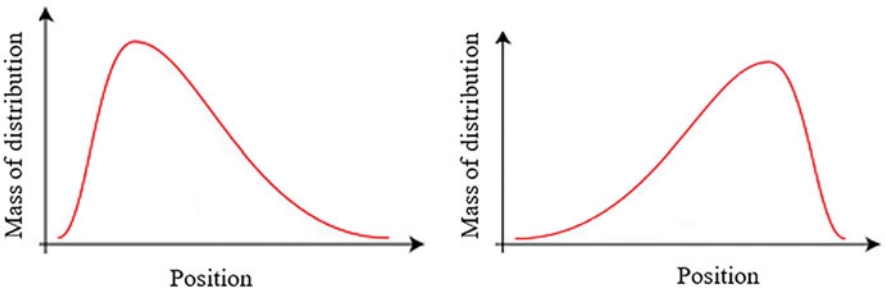

Fig. 20 (**a**) Positive skew. (**b**) Negative skew

$$\frac{\sum_{i=1}^{N}\left(Y_i-\overline{Y}\right)^4/N}{\sigma^4} \tag{6}$$

where Y_i's are the data points and σ is the standard deviation of the data. Since kurtosis is often compared with normal distribution and kurtosis value for normal distribution is 3, alternatively kurtosis is represented as:

$$\frac{\sum_{i=1}^{N}\left(Y_i-\overline{Y}\right)^4/N}{\sigma^4}-3 \tag{7}$$

Reason for calculating the kurtosis is to assess the sharpness of the input SEMG signal.

Zero-Crossing

A zero-crossing is a point where the sign of a mathematical function changes, represented by an intercept of the axis in the graph of the function. In image processing, zero-crossing represents the detection of an edge and is thus useful for finding contours.

Slope-Sign Change

This feature detects the change in mathematical sign of the derivative of the function. It is important because it processes the information whether a function is increasing or decreasing which helps to model the underlying data-generating function.

Principal Component Analysis

Principal Component Analysis (PCA) is used to reduce the dimensionality of the feature vector space. In this approach, the feature vector z (x_1, x_2, x_3. . .x_n), where x_n is the nth extracted feature for each subject, is decomposed into orthogonal basis vectors. First 3 principal components have been taken with maximum variance. Output of this stage was fed to the classification model.

Classification

Tremor is classified by acquiring the signal and then extracting features from it. Classification was done to differentiate between the physiological tremor in resting and contraction conditions of sEMG. Accuracy of classification was compared among four methods viz K-Nearest Neighbor (KNN), Logistic Regression, Support Vector Machine (SVM), and Random Forest.

K-Nearest Neighbor

K-Nearest Neighbor (KNN) is a supervised learning approach where the class of the respective feature vector is known for the training data set. Whole dataset was split into training and testing sets. The ratio of data was kept as 75:25. Selection of data for the two sets was random. The two output classes – "tremor" and "relax" – were considered. In KNN method, according to the number of nearest epochs selected for the system (value of K), output class is assigned to the test epoch by majority voting among the nearest epochs. Assigning the class to the respective testing data is done by calculating the distance of the testing feature vector with all training feature vectors. The various steps for KNN classification are as follows:

1. Choose the value of K.
2. Calculate distance of present testing dataset from all the training dataset.
3. Sort the calculated distance.
4. Find out the number of different classes for the first K nearest distance data.
5. Assign that class to the present testing data whose number is maximum.
6. Continue steps 2–5 for the complete test dataset.

Logistic Regression

It is a probabilistic supervised learning technique in which, primarily, maximum likelihood estimation is done given the parameter vectors. The parameter vector should be selected such that the probability of all the observed targets is maximized. Since the data points are identically and independently distributed, product of each predicted data point can be taken, and after taking logarithm of this product, the log sum can be maximized.

Support Vector Machine

It is a binary classification algorithm which assigns either 1 or -1 to the data points. Given the data points ($x_1, x_2 \ldots x_n$) and the corresponding label values ($y_1, y_2 \ldots y_n$), optimal margin classifier is trained to find a hyperplane which is at the farthest distance from the support vector(s). Support vector(s) are the data points which are at the minimum distance from the hyperplane, and our objective then becomes to select a hyperplane which is at maximum distance from the data points.

Random Forest

Random Forest is an aggregate predictor of decision trees. Decision trees greedily choose a feature split based on the impurity function score. Two impurity functions are used, viz Entropy and Gini Index. The feature which gives the maximum difference in impurity before and after applying the feature split is chosen for splitting. However, decision trees are prone to overfitting, and to deal with this problem, random forests are used as bagging bootstrap aggregator in which many decision trees are trained and each decision tree is supplied with a random subset of total features, and at the prediction time, voting is used for classification. This also deals with missing features problem in the dataset.

Results and Analysis

All the trials were conducted successfully, and subjects were healthy after trials. All the 14 subjects participated in the study for sample collection, and the analysis was carried out in three steps, i.e., time domain analysis, frequency domain analysis, and classification to know the total behavior of tremor signal.

Time Domain Analysis

Accuracy of voluntary signal depends upon the filtering of signal from raw signal. It was very much required to develop the optimized filter to get the tremor signal free from noise. The signals presented in Figs. 4 and 5 clearly demonstrate the change in amplitude from relaxed state to tremor state. In time domain, estimation of tremor becomes visible with the change in shape of signal. Figure 7 demonstrates the filtered tremor signal with different values of amplitude. Table 2 shows the increase in tremor values during fatigue.

Frequency Domain Analysis

Frequency domain analysis of tremor involves the calculation of median frequency and power contained in the sEMG signal. The sEMG median frequency was determined for all the subjects for different actions during surgery, and the results are presented in Fig. 21. These plots show decrease in median frequency for all four electrode signals during tremor. The decrement of median frequency indicates the existence of tremor due to fatigue. The power of sEMG signal is depicted in Fig. 22 for both tremor and relax states. It is clear that the power contained in the signal increases during tremor.

Classification

PCA was carried out, and the components of maximum variance were obtained. Two, three, four, and five components were evaluated one by one. All four classifier models were fed the PCA data to classify the tremor in relaxed and tremor states. Classification accuracy measures the ratio of correct predictions to the total number of cases evaluated. Table 3 shows the classification accuracy for PCA components 2, 3, 4, and 5 for all the classifiers. For KNN classifier, different values of K ranging from 4 to 7 have been considered. It can be seen from the obtained result that the accuracy increases for every classifier when the number of PCA components increases from 2 to 4. However, it does not change with further increase of PCA components to 5. The number of PCA components was finally selected as 4 (PC_1 to PC_4).

Table 2 Amplitude of filtered signal in relax and tremor mode of 4 electrodes

Mode	Electrode 1	Electrode 2	Electrode 3	Electrode 4
Relax	0.0038	0.008	0.008	0.006
Tremor	0.006	0.029	0.028	0.018
Increased amplitude (times)	**1.58**	**3.6**	**3.5**	**3**

Bold data are the increased amplitude (times) of tremor signal

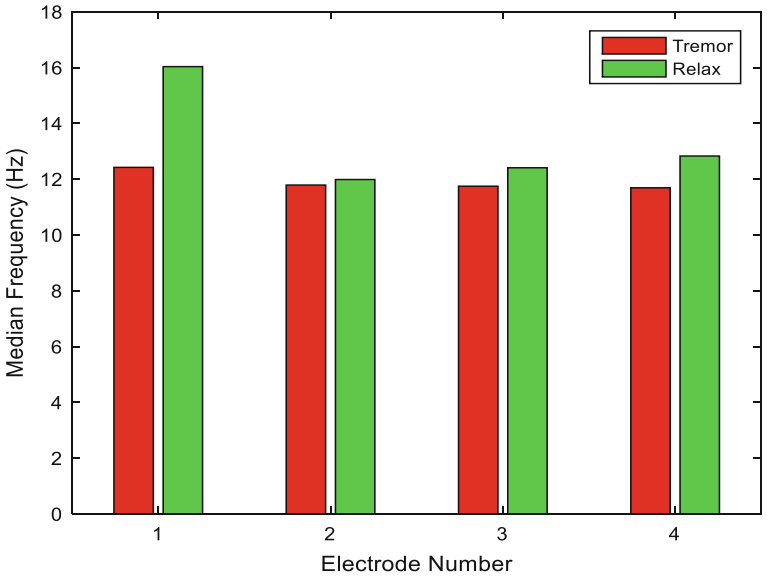

Fig. 21 Median frequency of sEMG signal in relaxed and tremor states

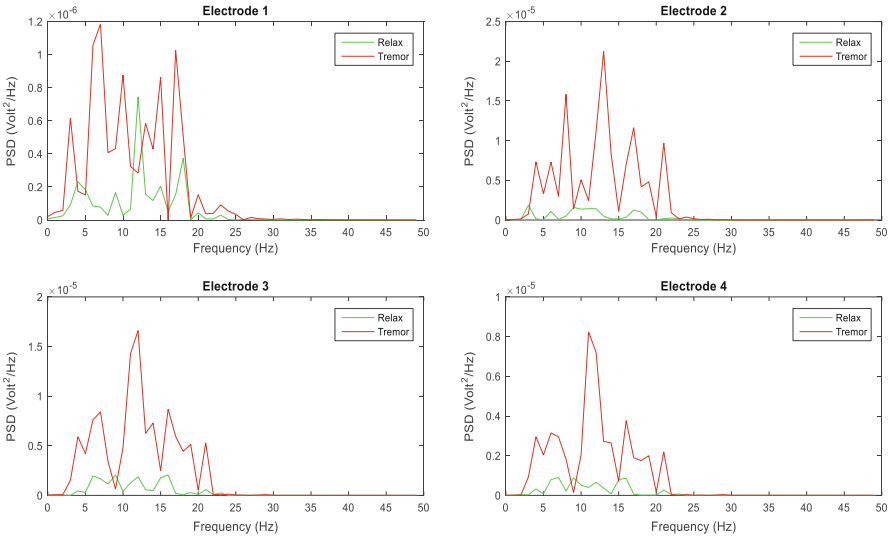

Fig. 22 Power of 4 electrodes in relax and tremor mode

Another conclusion that can be drawn is that for KNN classifier, the increase in size of neighborhood beyond 5 does not increase the accuracy. So, if KNN is to be used for classification, K may be taken as 5.

Table 3 Accuracy of different classifiers (in %) for various number of PCA components

No. of PCA components	Accuracy of KNN classifier with different values of K				Logistic regression	SVM	Random forest
	$K = 4$	$K = 5$	$K = 6$	$K = 7$			
2	96.5	97.2	97.3	97.3	98.4	99.2	100
3	97.7	98.0	98.0	98.0	99.5	99.3	100
4	98.2	98.4	98.4	98.4	99.6	99.5	100
5	98.2	98.4	98.4	98.4	99.6	99.5	100

The third observation is that different classifiers can be ranked from KNN to be the least accurate to Random Forest to be the most accurate. For KNN classifier, the maximum accuracy obtained was 98.4% for $K \geq 5$ and PCA components $\geq$4. For the second classifier – Logistic regression, the maximum accuracy was 99.6%, and for third classifier – Support Vector Machine, the maximum accuracy obtained was 99.5%. Fourth classifier (Random Forest) gave an accuracy of 100% for all components of PCA. From these values, it can be said that Random Forest classifier is the best for classification of tremor signal.

Conclusion

Researchers have worked since long to predict tremor in human arm so that it can be stopped. Present system has been designed, developed, and validated by using sEMG for measurement, analysis, and classification of tremor. Analysis was performed on relax and tremor data of subjects to identify the generated tremor. This method ensures that the tremor component is fully extracted without any damage or distortion to the rest of the sEMG signal. The sEMG signal, with muscle fatigue, was filtered with efficient Butterworth filters of 9th and 7th order which allow bare minimum ripples in the pass band and maximum attenuation in the stop band. Six features viz Mean, Standard Deviation, Skewness, Kurtosis, Zero-Crossing, and Slope Sign Change were extracted. Principal Component Analysis was done to prune the number of features. First, four PCA components were fed to the classifier. The classification of the fatigue was implemented on 14 subjects. Results show the decrease in median frequency of muscle signal and increase in sEMG signal power during tremor compared to relaxed state. Classification was performed with four classifiers viz K-Nearest Neighbor, Logistic Regression, Support Vector Machine, and Random Forest. Random Forest Classifier was found to be the best as the accuracy obtained with this was highest.

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Metrological Aspects of Blood Pressure Measurement

72

Rahul Kumar, P. K. Dubey, and Sanjay Yadav

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Abstract

Blood pressure (BP) is an important physiological parameter used in the diagnosis of many life-threatening diseases such as heart attack, ischemic stroke, hypertension, vascular disease, hypotension, and many others. It is directly connected to human life, it plays an important role in the diagnosis and monitoring of various health ailments. Accurate and precise BP measurement is always a priority for both researchers and medical practitioners. With recent technological advancements, the issue of accurate measurements, standardization of techniques, and efforts toward adopting some uniform methods remain debatable, particularly traceability in dynamic mode. There are numerous types of BP measuring devices in the market. The accuracy of these devices is always an unresolved issue for researchers, medical practitioners, and metrologists. Several national and international standards recommend methods for calibrating these devices, but very few devices/systems/instruments are available to calibrate these devices convincingly in SI traceable units. This chapter provides an in-depth explanation of various noninvasive BP measuring techniques as well as available national and international standards. Some recent advances in BP measurement, as well as challenges addressed and challenges ahead in their calibration, are also discussed.

Keywords

Blood pressure · Calibration · Testing · Accurate · Instruments · Techniques

Introduction

Regular blood pressure (BP) measurement is required for monitoring the health of the heart and its associated illnesses. BP measurement is commonly used to diagnose high blood pressure (HBP), hypertension, cardiovascular disease, heart attack, chronic renal disease, kidney failure, stroke, dementia, and peripheral vascular disease (Landry et al. 2022; Avolio et al. 2022; Kitagawa et al. 2022; Kovell et al. 2022; Kim et al. 2020; Liu et al. 2012; Chen et al. 2012; Schutte et al. 2022). HBP affected approximately 1.13 billion people in 2015, accounting for 19% of all fatalities (NCD Risk Factor Collaboration (NCD-RisC) 2017; Forouzanfar et al. 2017). Uncontrolled blood pressure affects up to 80% of people with HBP, putting them at a higher risk of morbidity and death (Egan et al. 2019). Thirty percent of people have HBP or have an increase in BMI and age. With an aging population and

rising obesity rates, it is becoming increasingly important to diagnose, treat, and control hypertension (Leenen et al. 2010; Franklin et al. 1997; Kumar et al. 2021a, b; Lubin et al. 2020). An international study of more than 1.5 million people discovered that more than 70% of those taking HBP medication had frenzied BP (>130/80 mmHg) (Beaney et al. 2020; Gakidou et al. 2017). According to the World Health Organization (WHO), over 17.90 million people died worldwide in 2016, with cardiovascular disease being the leading cause. The global prevalence of high BP in people over the age of 18 is defined as systolic and diastolic pressures greater than or equal to 140 and 90 mmHg, respectively. The number of adults with high blood pressure nearly doubled from 594 million in 1975 to 1.13 billion in 2015, with low- and middle-income countries (LMICs) accounting for the majority of the increase. Only LMICs account for more than 75% of cardiovascular disease deaths (https://www.who.int/en/news-room/fact-sheets/detail/cardiovascular-diseases-(cvds). Accessed 02 Apr 2022; https://www.who.int/health-topics/cardiovascular-diseases/#tab=tab_1. Accessed 02 Apr 2022; https://newsroom.heart.org/news/acc-aha-hfsa-issue-heart-failure-guideline. Accessed 02 Apr 2022). Statistics clearly reveals that BP measurements are extremely important in everyone's life.

Blood pressure measurement was made by the great scientist Stephan Hales, which is still regarded as a remarkable discovery of the eighteenth century (Booth 1977). The significance of BP measurements can also be understood by the famous statement of the great physiologist Johannes Mülle on the discovery of BP, which states that "the discovery of the blood pressure was more important than the discovery of the blood" (Booth 1977). Following the discovery of the BP, many researchers continued to focus their attention and efforts on BP, resulting in the development of various noninvasive blood pressure (NIBP) measurement techniques, which are summarizd in this chapter.

There are various NIBP measuring instruments and devices on the market that measure blood pressure using various principles (Kumar et al. 2021b). BP is measured using auscultatory techniques based on Mercury, but Mercury is a toxic element that poses health risks to humans. According to the Minamata Convention recommendations, mercury-based medical devices are either prohibited in medical practices or suggested to be phased out with alternative techniques by 2025 (see www.mercuryconvention.org for Minamata Convention on Mercury, United Nations Environment Programme, September 2017). In contrast, mercury-based sphygmomanometers are still used in some clinical settings as a Gold Standard for blood pressure measurements. The accuracy of blood pressure measurement with a mercury sphygmomanometer is also dependent on the medical practitioners' visibility and hearing ability (Babbs 2015; Mion and Pierin 1998).

NIBP Measuring Techniques

Various noninvasive BP measurement techniques thus developed, commercially available, and in use are depicted in Fig. 1.

Fig. 1 Various noninvasive blood pressure measuring techniques

Auscultatory Technique

One of the most widely used methods for measuring blood pressure is the auscultatory technique. It is a noninvasive method of measuring blood pressure. Despite being based on mercury and having been invented in 1896, it is still commonly referred to as the "Golden Standard." Because mercury is hazardous to human health, its use is prohibited in all public places, including medical facilities, research facilities, and laboratories. The use of mercury in sphygmomanometers is no longer as common as it once was in clinical practice (Penaz 1973). The auscultatory method of measuring blood pressure necessitates the use of skilled personnel. Moreover, the operator's capability of hearing and vision plays an important role in determining the BP readings. A typical traditional mercury sphygmomanometer is shown in Fig. 2.

The auscultatory method entails wrapping a cuff around the patient's upper arm of the hand and inflating it to a pressure of approximately 160–180 mmHg until the artery is not completely closed, resulting in the closure of blood flow through the artery. There is no audible sound in these circumstances. The sound produced when the cuff pressure is deflated is known as the Korotkoff sound. The frequency of the generated sound is so low that it cannot be heard by the human ear. A stethoscope is used by the medical practitioner or operator to listen to this Korotkoff sound. Korotkoff sound changes during the very short duration of cuff pressure deflation, which is classified into five different phases as shown in Fig. 3.

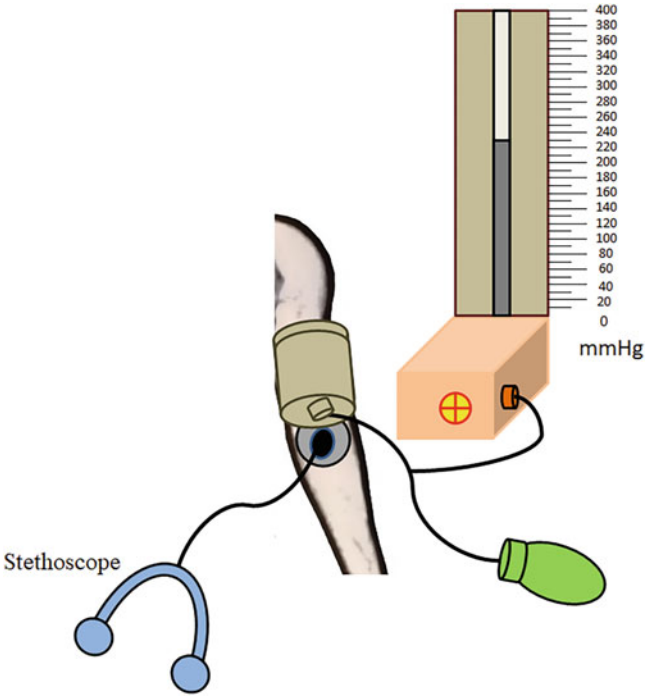

Fig. 2 Pictorial view of the traditional mercury sphygmomanometer

Phase	Korotkoff Sound	Pressure (in mmHg)
1	Thud	120 (Systolic)
2	A blowing noise	110
3	A softer thud	100
4	A disappearing noise	90
5	Nothing	80 (Diastolic)

Fig. 3 Different phases of the korotkoff sound generated during the deflation of the cuff pressure at the time of BP measurement using the auscultatory techniques

Phase 1 begins when the cuff pressure first begins to inflate to the point where blood can flow. A thud (clear tapping) sound is produced in Phase 1 to represent systolic blood pressure (SBP), and a blowing noise (softening of the tapping and a swishing element) is produced in Phase 2. A softer thud is associated with phase

3. Figure 3 shows the Phase 4, the Korotkoff sound begins to disappear (abrupt muffling), and in Phase 5, there is no sound available, which is known as diastolic blood pressure (DBP). When the artery just begins to open and the artery pressure rises above the cuff pressure, a phase Korotkoff sound is produced, which is referred to as SBP. When this happens, the blood in the artery has a turbulent flow as it travels through the partially compressed artery with the cuff pressure, and this sound does not stop even as the cuff pressure decreases. When the phase 5 Korotkoff sound (nothing or no sound) is produced, the pressure is considered to be in the DBP. In the century since its invention, the mercury sphygmomanometer has undergone no significant design changes. Mercury sphygmomanometers have a very simple design and basic operating principle, which means that there are very few points that could go wrong with them in comparison to other BP devices. This is the reason why they were/are so popular. One other glaring issue is the widespread utilization of mercury, which is inherently dangerous (see www.mercuryconvention.org for Minamata Convention on Mercury, United Nations Environment Programme, September 2017; Babbs 2015; Mion and Pierin 1998; Bailey and Bauer 1993).

Oscillometric Technique

In the year 1876, Marey was the first person to propose using the oscillometric approach (Marey 1876). Marey noticed the pressure oscillations in the cuff and recorded them for the period of the cuff pressure deflation, at the point of maximum oscillations corresponding to the arterial pressure. This was the first time that the oscillometric method had been used. When the pressure in the cuff rose above the SBP, the oscillations began. They continued even when the DBP dropped below the SBP.

One of the most common types of automatic NIBP approaches is known as the oscillometric technique. It is not necessary to wrap the cuff around the brachial artery when utilizing this method. In the oscillometric method, a microcontroller is utilized to perform the functions of controlling the motor, reading the analogue signal from the pressure sensor, converting the analogue signal to its digital equivalent, and estimating the SBP and DBP values on the logic that has been programmed. The amplitude of the pressure sensor's fluctuations is extremely minimal (approximate in the order of μV to mV). Therefore, it is necessary to increase the strength of the amplitude before transmitting the results to the microcontroller. Because the sensor's output is at a low level and is susceptible to noise from the environment, a filter is utilized to minimize the noise in the signal. Furthermore, the SBP and DBP are determined through the use of empirically derived algorithm. A pictorial view of the oscillometric method for BP measurement technique is shown in Fig. 4 (Mauck et al. 1980; Amoore et al. 2008; Zheng et al. 2011).

The most significant advantage of this method over the others is that the cuff position has no effect on the BP measurement. Another potential advantage of the method is that it is less susceptible to outside noises, with the exception of mechanical motion-induced low-frequency vibrations. The procedure is more convenient for patients, particularly in terms of making the cuff placement and removal easier.

Fig. 4 Photograph of the oscillometric method-based BP device during the BP measurement

Another advantage of this method over the auscultatory technique is that no stethoscope is required. The most significant disadvantage of using this method is that the algorithm used to estimate the SBP and DBP varies from device to device (Zheng et al. 2011; Jazbinsek et al. 2010).

Finger Penaz Technique

Continuous noninvasive arterial BP measurements are performed using the Penaz technique, which is a noninvasive (indirect) method for BP measurement. This technique was introduced in 1972 by Jan Penaz. In this technique the arterial pulse in finger was detected by photoplethysmograph (PPG). It is a simple and low-cost optical technique used to detect variations in the microvascular bed of tissue blood volume. It operates on the unloaded artery principle. In this method, a cuff is wrapped around the finger and inflated while the artery is not completely collapsed. An electro manometer attached to a cuff that measures cuff pressure after an artery has collapsed. The Penaz method keeps the finger inside the cuff. Because the close loop control system modulates the cuff pressure, the artery thickness remains constant. When cuff pressure is used, the artery thickness and intra-arterial pressure vary naturally. To maintain a constant transmural pressure, the modulating cuff pressure must be changed at the same rate as the intra-arterial pressure, which is inversely proportional. The closed loop control system is constantly working to maintain a consistent photoplethysmographic output at a constant nonzero transmural pressure. When the artery reached a zero transmural pressure state, the cuff pressure remained constant for arterial pressure and BP was recorded in a continuous and indirect manner. The plethysmograph output is used to operate a servo-loop, which quickly changes the cuff pressure to maintain a steady output while keeping the artery partially open. The oscillations of the cuff pressure are determined and found to resemble the arterial pressure in the majority of cases. This method measures SBP and DBP changes more accurately, but both SBP and DBP may be underestimated at times when compared to brachial artery BP measurements. Due to its inconvenient nature

and erroneous absolute BP measurements, this approach is currently underutilized in clinical settings. Figure 5 depicts a schematic diagram of the fundamental working principle of this technique (Penaz 1973; Van Egmond et al. 1985).

Ultrasound Technique

When the patient's pulse is difficult to detect with a stethoscope and the patient is hemodynamically unstable, the ultrasonic method is used to measure blood pressure. In this method, ultrasonic transducers are used to transmit and detect ultrasound around the artery while it is covered by the cuff, as shown in Fig. 6. Before compressing the artery, the cuff is inflated to obtain an accurate blood pressure reading. When the cuff pressure begins to fall, the reflected ultrasound undergoes a Doppler phase shift, causing the artery wall to begin moving at SBP. The DBP is the pressure measured after the arterial motion has either stopped or reached its end point. Utilizing

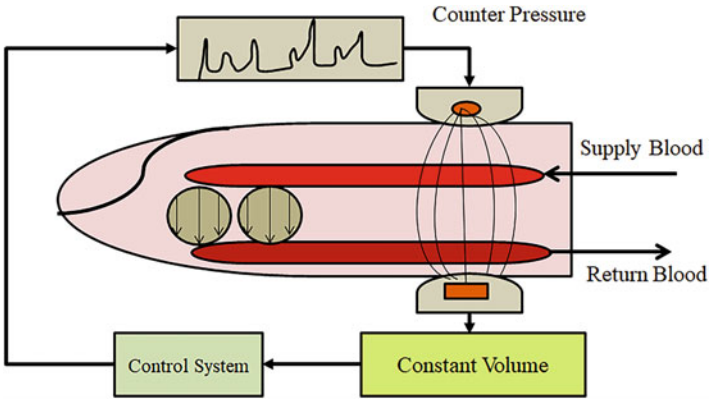

Fig. 5 Block diagram of the working principle of the Finger Penaz technique

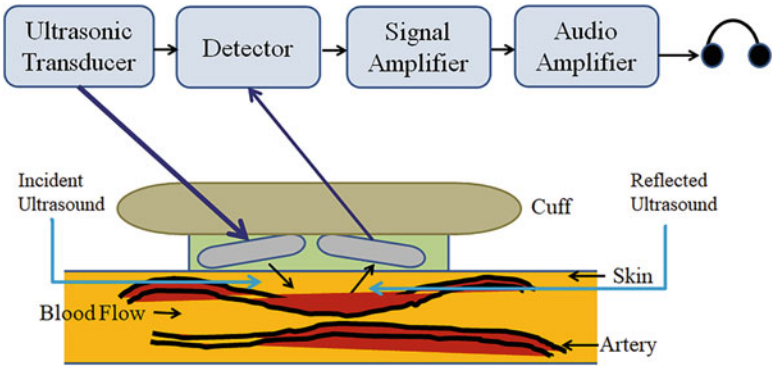

Fig. 6 Block diagram of the working principle of the ultrasonic technique

this technique, the user can also determine the ankle-arm index and blood flow in the infants and children using this method. Additionally, blood flow can also be determined (Elseed et al. 1973; Senador et al. 2022; Zakrzewski et al. 2018).

Tonometry Technique

Tonometry is essentially a noninvasive continuous BP measurement method. It uses a superficial artery waveform splinted (or partially squeezed) against a bone. In this technique, multiple or an array of pressure sensors, similar to cuff pressure, are used directly on the wrist to locate an artery slightly underneath and at the center of the sensor. Although pressure sensors are used to measure the beat-to-beat waveform of patients' wrists in this method, it is still considered the most accurate. Because calibration is required for each individual patient, the device cannot be deemed suitable for regular users. In the critical care unit (ICU), it is typically utilized during surgical procedures, particularly when patients are under the influence of anesthesia (Drzewiecki et al. 1983). The schematic diagram is shown in Fig. 7.

Wearable Sensor-Based Cuffless Technique

Continuous blood pressure measurement using this method is an invasive technique that is painful, uncomfortable, and carries the risk of infection. Given the variability of blood pressure, there is no viable solution for outpatient continuous BP measurement that has been shown to improve patient outcomes. Wearable sensors-based cuffless technique for continuous NIBP measurement is a new option for long-term monitoring and preventative care at home. Traditional cuff-based BP measuring instruments and devices cause discomfort. PTT and PPG are used in the cuffless technique. Commercially available PTT and PPG equipment does not meet the AAMI/ISO guidelines for blood pressure measurements (mean difference 5 mmHg and standard deviation 8 mmHg). Magnetic sensors have been used to measure PTT, vascular transit time (VTT), tonometry, and pulse wave velocity to address these constraints (PWV). The PAT is based on the heartbeat's journey from the heart to the finger's peripheral artery, as determined by the R-wave of the ECG and the peak of

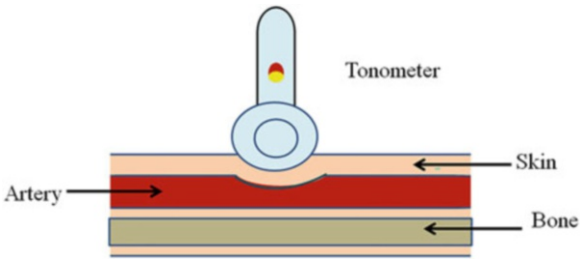

Fig. 7 Schematic diagram of the tonometry technique

the PPG signal. PTT is the interval between the ECG R-wave and the first PPG derivative. The VTT is used in phonocardiography to estimate the delay between the PPG fingertip and the chest. Several recent studies used a mobile phone camera to measure blood pressure, while others used the Windkessel model and neural system (Papadopoulos et al. 1993; Le et al. 2020; Elgendi et al. 2019; Khalid et al. 2020).

This section discusses various NIBP measurement techniques. Figure 8 depicts a comparative description of the benefits (in green) and drawbacks (in blue) associated with these NIBP techniques.

Location for BP Measurements

The location of the BP measurement is one of the important issues in BP measurement techniques. Several locations of the body being used for BP measurement are shown in Fig. 9.

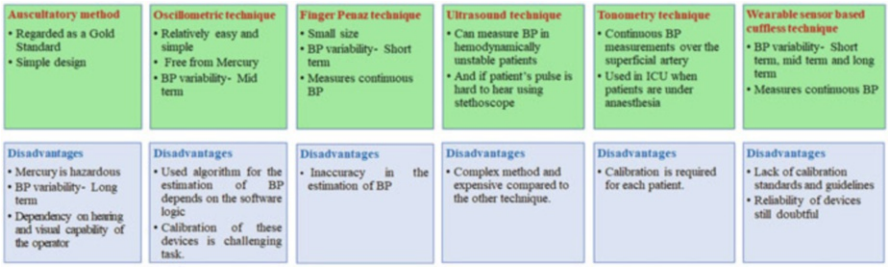

Fig. 8 Advantages and disadvantages of NIBP measuring techniques

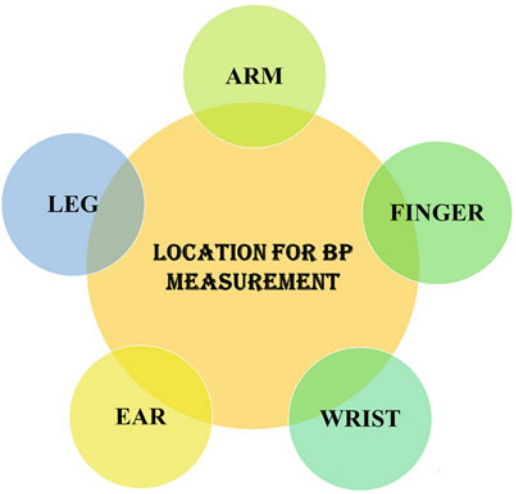

Fig. 9 Various sites for the blood pressure measurement

Arm

The majority of blood pressure measurements are taken on the upper left and right arms. To the best of our knowledge, there is no technique for determining which part of the body is best suited for accurate blood pressure measurement. However, an observational study was conducted in the neuroscience intensive care unit (NSICU), a comprehensive stroke academic center, to examine blood pressure measurements in extremely ill patients with nervous diagnoses. Data from 82 patients was collected and documented over the course of 4 months. All of the patients' blood pressure (BP) was measured in four different locations at the same time, including the left upper arm (LUA), right upper arm (RUA), right wrist, and left wrist, with specially designed wrist cuffs. Wrist sites are not commonly used in regular blood pressure measurements. When other measurement sites are inconveniently unavailable, the wrist is used, but it is included in the inter-site comparison. Forty-one of the 80 participants in this study were men. According to reports, the LA and RA are the most commonly used NIBP measurement sites. Based on the literature, the mean difference between the LA and RA in the total pressure range of 0–30 mmHg was determined to be 7.877.23 mmHg, which is very significant because a 5 mmHg variation in the SBP makes a difference in opinion when deciding whether to intervene or withhold treatment (Siaron et al. 2020; O'Rourke 2002).

Leg

The leg is an important site of the human body for blood pressure measurement. P. Sareen et al. studied 250 patients who required anesthesia for surgery and compared their blood pressure levels in the legs and arms as part of the study. The average SBP in the arm was found to be 127.72 ± 15.65 mmHg while the average SBP in the leg was found to be 142.97 ± 22.18 mmHg. The BP in the arm was 80.67 mmHg as compared to 75.66 mmHg in the leg. The MBP of 96.34 ± 11.71 mmHg was observed in the arm while it was 98.24 ± 13.94 mmHg in the leg. When it comes to SBP readings, the calf measurement was found to be 15.25 mmHg higher as compared to hand measurement for an individual. It varied from a 25 mmHg in the arm compared to an 80 mmHg in the leg. The upper limb DBP was 5 mmHg higher than the lower limb DBP on an average (Sareen et al. 2012).

Finger

The blood pressure is measured on the patients' fingertips. There are numerous devices on the market for measuring blood pressure with the finger, but the accuracy of these devices is questionable. Blood pressure measurement on the finger is often less reliable, but it is still used in clinical settings. The benefits of measuring blood pressure on the finger include the ability to continuously monitor blood pressure

without discomfort and the ability to simply detect the patient's hypertensive tendency after surgery in the intensive care unit due to continuous BP pulse measurement as well as long-term BP recording and monitoring. The reading of the finger BP measurement is less reliable than the brachial artery (Stergiou et al. 2018; Sesler et al. 1991; Molhoek et al. 1984).

Wrist

BP is sometimes measured at the wrist using specialized devices. Because of their smaller size than arm devices, wrist devices can be used by obese people. When using wrist monitors, a systematic inaccuracy is introduced into BP measurements due to the hydrostatic effect of wrist position changes with regard to the heart. To reduce the possibility of systematic errors, keep the wrist level with the heart when taking blood pressure readings. While a series of readings are examined, it is impossible to know whether this was done retroactively. Wearable blood pressure monitors have the potential to provide highly reliable blood pressure readings, but more research is required (Sareen et al. 2012; Mitchell et al. 1964).

Ear

In a recent study, Xing et al. used a PPG-based technique to measure blood pressure at the ear site. A multi-Gaussian decomposition PPG for the in-ear was performed to reduce noises caused by the extraction process. A four-element Windkessel model was applied to PPG data to generate BP estimation algorithms. To validate this new strategy, 53 patients were enrolled for a 1-month follow-up and 17 for a 2-month follow-up, and it was discovered that SBP estimation accuracy improved using inertance-based pre-classification, while DBP showed less improvement (Xing et al. 2021).

Sources of Errors in the BP Measurements

Various sources of errors identified in the BP measurements are depicted in Fig. 10 and described in detail in this section.

Body Position

The position of the arm is one of the most crucial elements to consider. As the arm moves from the up to the down position, there exist variation of 5–6 mmHg in the pressure. This shift in pressure is attributed due to increase in the hydrostatic pressure. During the BP measurements, it is recommended that patients take the measurements while seated in a chair that has a back support (Mitchell et al. 1964; Netea et al. 2002, 2003; Webster et al. 1984; Pickering 1993; Verberk and Mieke 2016).

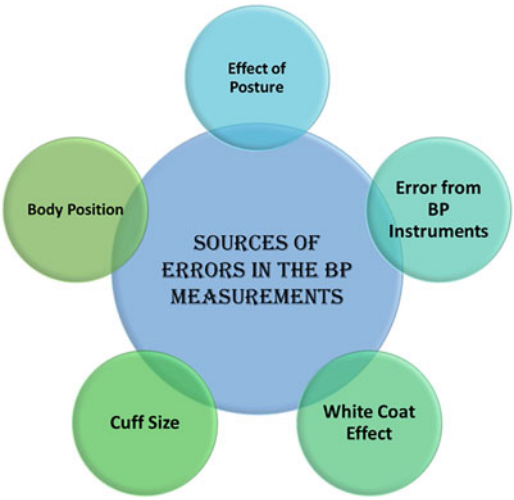

Fig. 10 Different types of sources of errors in blood pressure measurement

Cuff Size

The size of the cuff influences BP measurement and is the most common cause of BP overestimation and underestimation. The cuff should be chosen based on the patient's arm diameter. Except for the slimmest arms, properly sized cuffs reduce error. The British Hypertension Society (BHS) recommends a cuff size of 12.5–13 cm and a length of 35 cm for adults with an arm circumference greater than 33 cm (Pickering 1993; Maxwell et al. 1982; Van Montfrans et al. 1987).

Effect of Posture

The American Heart Association (AHA) recommends sitting as the best posture for measuring blood pressure (Pickering et al. 2005). According to Netea et al.'s study on 245 people of various ages, SBP was the same in both seated and supine postures, but there was a difference in DBP at the age of 30. The DBP in the seated position was approximately 10 mmHg higher than in the supine position; however, at age 70, the difference was only 2 mmHg. According to the research, BP measurements should be taken when the patient is calm and the arm is at heart level. Patients should thus choose the supine position for accurate blood pressure measurements (Netea et al. 1998).

White Coat Effect

When blood pressure is measured in a doctor's office, clinic, or hospital, it is found to be higher than in another location

shortly afterward. Recent research has established that when blood pressure is measured by operators wearing a white coat, this can cause anxiety, which is known

as the white coat effect (Jhalani et al. 2005). The white coat effect is observed to a greater or lesser extent in hypertension patients, but it is much less or absent in normotensive individuals. The distinction between clinics and ambulatory BP is reached. White coat hypertension refers to people whose clinic blood pressure is hypertensive but who are normotensive elsewhere (Verdecchia et al. 1995).

Error from BP Instruments

The key factor that determines the accuracy of the assessment of human BP is the measurement accuracy of the BP measuring instrument itself. The basic accuracy of the device is affected by a number of different phenomena, all of which are discussed above and contribute to the general decline in measurement accuracy. The measurement apparatus itself contains several errors, each of which is unique to a particular measurement technique. For instance, in a sphygmomanometer that uses mercury, the accuracy of the device is affected by factors such as the quality of the mercury and the diameter of the capillary. When it comes to aneroid sphygmomanometers, the precision of the pressure dial gauge that is utilized makes a significant contribution to the instrument's accuracy. The precision of the sensor, the accuracy of the bridge arrangement, and the quality of the excitation source are all potential causes of inaccuracy in oscillometric approaches. In addition to these major contributors, the software logic that is used to estimate SBP and DBP also contribute significantly (Cluett et al. 2021; Hwang et al. 2018).

International Standards and Recommendations for BP Devices

Indian Standards

The BP measuring devices for humans are covered by two Indian standards: IS 3390: 1988 and IS 7652: 1988. The first standard describes the mercurial sphygmomanometer's requirements, while the second specifies the aneroid sphygmomanometer's dimensions and other standards. According to IS 3390: 1988, all calibration must be completed against a standardized and validated reference mercury sphygmomanometer, starting with the highest pressure and then in decreasing orders of pressures at the rate of 2 mmHg per second without tapping. The IS 7652: 1988 stipulates that the three calibrations are required without tapping and with dropping pressures falling at a rate of about 2 mmHg per second. Calibrations are advised to be performed with the gauge in a vertical position and only after at least 24 h have been passed since the device was last used. It is recommended that observations be made sequentially, starting with the greatest pressure and working down in multiples of 30 mmHg. After the initial calibration has been completed for 6 h, a second calibration has to be done in the same manner, but with the gauge angled 45° toward the back (Indian Standard IS 3390 1988; Indian Standard IS 7652 1988).

World Health Organization (WHO) Guidelines

The WHO guidelines (2020) is the revised document of WHO guidelines (2005) for BP measurement devices. The revised documents amicably address the major concerns of the developing economies wherein access and availability of the accurate, precise, and high-quality BP devices are not sufficient and adequate. It stipulated the technical specifications for NIBP devices with a cuff, their specific characteristics, procedure for calibration, maintenance, and regulatory requirements. It further stipulates that the mercury-based appliances should be checked to confirm that the column of mercury is undamaged and upright, as well as the mercury meniscus is centered at zero. Calibration of aneroid sphygmomanometers is recommended to be performed by technical professionals half yearly or as suggested by the manufacturer to ensure accuracy within the acceptable limit. For digital devices, it is recommended to be calibrated against a reference manometer with an accuracy of 0.1 mmHg or compared with a well-maintained mercury sphygmomanometer with an accuracy ± 3 mmHg (WHO 2020).

OIML Recommendations

The technical documents on BP instruments recommended by the International Organization of Legal Metrology (OIML) are in two parts; one for noninvasive mechanical sphygmomanometers as OIML R16-1 and another for noninvasive automated sphygmomanometers as OIML R16-2.

According to OIML R16-1 recommendations, noninvasive mechanical sphygmomanometers and their related elements are utilized with an inflated cuff to measure arterial BP noninvasively. This recommendation covers sphygmomanometers with a mechanical pressure sensor and display used with a stethoscope or other manual techniques to detect *Korotkoff* sounds and inflate the cuff. *Korotkoff* sounds are a technique for measuring BP in the adults wherein sounds are heard above and closed artery when occluding pressure is gradually released, with the appearance of sounds that correspond to SBP, and the disappearance of sounds that correspond to DBP. To verify the first time, the maximum permissible error for measuring cuff pressure is 0.4 kPa (3 mmHg), and 0.5 kPa (4 mmHg) for using sphygmomanometers in the range of 15–25 °C and relative humidity of 20–85%. The sphygmomanometer must meet the maximum permissible error after 24 h at −20 °C and 70 °C with 85% relative humidity (International Recommendation OIML R 16-1 2002).

As per the scope of OIML R 16-2, the recommendations are specifically applicable to BP devices, which measures BP at the wrist, upper arm, and thigh. According to OIML R 16-2 recommendations, the NIBP system should have at least one cuff that is connected with the pneumatic system, one electromechanical transducer for measuring the cuff pressure, and a display to show the user the measured value of the pressure. Additionally, the system should have a pneumatic system that is connected with the cuff. During BP readings, the apparatus needs to have a deflation valve so that the air may be let out of the pneumatic system.

Kilopascals (kPa) or mmHg are the units of measurement that are suggested for use when referring to BP. The greatest amount of error that is acceptable in the cuff pressure indicator at any position along the scale range is ±0.4 kPa, which is equivalent to ±3 mmHg. It is advised that it be within ±4 mmHg in the event that it is being confirmed for the first time.

It is recommended that the observations conducted in the environmental conditions should be within 15–25 °C for ambient temperature and 20–85% RH for relative humidity. It is recommended to assess the performance of the BP measuring device by placing it in the climatic chamber at 5 °C for 24 h, followed by 24 h at 50 °C. Within the goal range of SBP and DBP, the air leakage must not exceed a pressure drop of 6 mmHg/min, and the deflation rate of the programmed deflation valve must be in the range 2–3 mmHg/s. The pressure should be released from the device at a rate of 260–15 mmHg within 10 s, and in the case of neonatal or new-born patients, the rate of pressure release should be less than 5 s at 20–0.7 kPa or 150–5 mmHg. There should be a provision for automated zero settings, and the device should have the capability to perform zero settings at the necessary interval. The device needs to be compatible with electromagnetic waves. After going through 10,000 simulated measurement cycles, the change in the cuff pressure indication should not be greater than 3 mmHg. This applies to the entire pressure range. On the display unit, the precision of the pressure value must be 1 mmHg (International Recommendation OIML R 16-1 2002).

Some Other Important Standards

The IEEE Standard 1708 defines wearable cuffless BP devices (WCBD) and evaluation processes. The authorization of the BP measuring instrument for performance evaluation, as recommended by the ESH standard (I.S. Association 2014; O'Brien et al. 2002), requires at least two trained observers. The procedure for calibrating WCBD is extremely difficult and varies from device to device. The observers must be familiar with the manufacturer's recommended operating instructions for each item; otherwise, an engineer from the manufacturer should be part of the evaluation team. The utilized sphygmomanometer must conform to ISO 81060-2:2013 or ANSI/AAMI SP10. Adult testing must include a minimum of 45 individuals, including at least 22 females and 22 men between the ages of 18 and 65, in accordance with ISO standards. Due to its complexity and length, the validation process is suggested by the ESH protocol to be divided into two parts. Twenty people are recruited in the first level, and any device that fails during this level is removed from the testing phase. After validating 20 participants, the data collected from various tests (e.g., static test, test with BP change from calibration point, test after a specific period of time after calibration) are individually analyzed and should be within 7 mmHg for both SBP and DBP measurements (I.S. Association 2014; O'Brien et al. 2002).

The above discussed various standards are summarized in concise form in Table 1.

Table 1 National and international standards for various BP devices

S.No.	National and international standard	Type of BP device	Some of the recommendations/ guidelines
1.	Indian standards: IS 3390: 1988 (63)	Mercury sphygmomanometer	1. All calibration shall be accomplished properly standardized and certified mercurial manometer. One calibration shall be made without tapping and with decreasing pressure only. Reading shall be taken at the highest point of scale and at all other points, which are multiples of 20 mm. The reading shall be taken with the pressure falling at a rate of 2 mm/s approximately **2. Leak test:** When the cuff of each completely assembled sphygmomanometer is tied around a rigid bar of appropriate diameter and each valve shall be subjected to an internal air pressure of 50, 150, and 300 mmHg. The instrument or the valve shall then be cut off from the air supply. The leakage rate of the completely assembled sphygmomanometer shall not exceed 10 mmHg per minute
2.	Indian standards: IS 7652: 1988 (64)	Aneroid sphygmomanometer	1. Three calibrations shall be made without tapping and with falling pressures only. The pressure shall be made to fall at a rate of approximately two millimeters per second. The first calibration shall be made only after at least 24 h have elapsed since the last application of the pressure to the instrument. The gauge shall be in vertical position. Readings shall be taken with decreasing pressure at the highest point of the scale and at all other points which are multiples of 30 millimeters. The pointer of the instrument shall be brought to an even division and the pressure read on the calibrating manometer and instrument under test are recorded. The second calibration shall be done not less than 6 h after the completion of the first, in the same manner as the first except that the gauge shall be inclined backward at an angle of 45” to the vertical

(continued)

Table 1 (continued)

S.No.	National and international standard	Type of BP device	Some of the recommendations/ guidelines
			2. Each completely assembled sphygmomanometer and its valve shall be subjected to internal air pressure of 50,150, and 300 mmHg. The leakage shall not exceed 10 mm of mercury per minute when the sphygmomanometer or its valve is cut off from the air supply
3.	WHO (65)	Automated non-invasive blood pressure measuring devices with cuff	**1. Efficacy measure:** Threshold for accepting accuracy of BP measurement at estimated probability of tolerable error ($\leq$ 10 mm Hg) $\geq$ 85% **2. Sample size:** $\geq$ 85 participants **3. Participating general population study:** Participants aged $\geq$12 years **4. Cuff sizes:** Minimum number of participants per cuff size, depending on number of test device cuffs. Requirement for arm circumference distribution according to range of use of test device **5. Reference BP:** Mercury sphygmomanometers or accurate non-mercury devices **6. Data collection:** Sequential BP measurements on the same arm are preferred **7. Pass criteria:** Average difference in BP and standard deviation criteria 1 and 2 of universal method. Absolute differences in BP $\leq$ 5, 10, and 15 mm Hg and scatterplots presented
4.	OIML R16–1 (66)	Noninvasive mechanical sphygmomanometer	**1. Environmental conditions** The maximum permitted error for measuring cuff pressure is 3 mmHg and 4 mmHg when utilizing sphygmomanometers at 15–25 °C and 20–85% relative humidity. After storage at −20 °C and 70 °C with 85% relative humidity for 24 h, the sphygmomanometer must meet the maximum permissible error 2. During rapid exhaust of the pneumatic system, with the valve fully open, the pressure reduction from 260 to 15 mmHg must not exceed 10s

(continued)

Table 1 (continued)

S.No.	National and international standard	Type of BP device	Some of the recommendations/ guidelines
			3. According to the supplier's declaration, mercury must be 99.99% pure 4. The mercury shall exhibit a clean meniscus and shall not contain air bubbles
5.	OIML R16–2 (67)	Noninvasive automated sphygmomanometer	1. Replace the cuff with the vessel. Connect the calibrated reference manometer by means of a T-piece connector to the pneumatic system. After disabling the electro-mechanical pump (if fitted), connect the additional pressure generator into the pneumatic system by means of another T-piece connector 2. For each of the following combinations of temperature and humidity, condition the device for at least 3 h in the climatic chamber to allow the device to reach steady conditions: 10 °C ambient temperature, 85% relative humidity (non-condensing) 20 °C ambient temperature, 85% relative humidity (non-condensing) 40 °C ambient temperature, 85% relative humidity (non-condensing) **3. Rapid exhaust:** During the rapid exhaust of the pneumatic system, with the valve fully opened, the time for the pressure reduction from 260 mmHg to 15 mmHg shall not exceed 10 s 4. For blood pressure measuring systems having the capability to measure in a neonatal/infant mode, the time for the pressure reduction from 150 mmHg to 5 mmHg during the rapid exhaust of the pneumatic system with the valve fully opened shall not exceed 5 s **5. Stability of the cuff pressure indication:** The change in the cuff pressure indication shall not be more than 0.4 kPa (3 mmHg) throughout the pressure range after 10,000 simulated measurement cycles

(continued)

Table 1 (continued)

S.No.	National and international standard	Type of BP device	Some of the recommendations/ guidelines
6.	IEEE Standard 1708 (68)	Wearable cuffless blood pressure devices	**1. Observer training and measurement:** Two observers are trained in accurate BP measurement and to get familiar with the data collecting procedure and device operation **2. Subject selection:** Twenty subjects are recruited at Phase 1of the assessment, and devices passing the criteria of this phase will proceed to Phase 2, where an additional 25 subjects are recruited. The characteristic of recruited subjects should follow the requirement stated in this standard **3. Main validation:** The procedure is broken down into three levels: static test, test with BP change from the calibration point, and test after a certain period of time from calibration. Practitioners should properly design their validation protocol to cover the validation from each of the three levels. Observer measurements and test device measurements are conducted with stated validation procedures **4. Data analysis:** The collected data are analyzed and compared to the stated accuracy criteria **5. Data reporting:** The results are presented in recommended format

Calibration of the Oscillometric BP Devices

Accurate blood pressure readings are critical for both diagnostic and therapeutic purposes. An incorrect blood pressure measurement could result in discrepancies between medical practice and clinical trial results. An important challenge for automated oscillometric instruments is assessing and determining the accuracy of the BP measuring device because the manufacturer does not disclose the algorithm used. As a result, various simulators are available on the market for testing and calibrating these devices. Furthermore, an international standard is available to ensure the simulator's quality and accuracy (ISO/TS 81060-5: 2020 standard for NIBP simulators).

There is a widespread usage of automated NIBP measurement devices in the hospitals and at homes. Consequently, a great deal of concentrated work is being expended on the development of calibration methodologies for the NIBP simulator. The International Organization for Standardization (ISO) has also developed a standard, ISO/TS 81060-5: 2020, which specifies the repeatability and reproducibility requirements for NIBP simulators. We have simply summarized a few needs; for more information, please refer to the complete standard mentioned here. It describes the acceptable repeatability and reproducibility limitations for the amplitude and shape of the oscillations produced. The standard specifies that the resolution must be at least 0.1 mmHg. The appropriate testing conditions, e.g., temperature and humidity are 15–25 °C and 15–85% RH, respectively. The variation in temperature over the entire test should be within 1 °C, and the variance in relative humidity should be within 5% RH of these values. The opening and closing of the doors and windows should be avoided during the test to avoid affecting the results (I.T.S. 2020; Kumar et al. 2020). Maximum error in pressure measurement at any location within the nominal measurement range must be less than or equal to ±1 mmHg. At any selectable pulse rate, the maximum error for measuring the pulse rate must be less than or equal to ±1.0/min.

The experimental standard deviation of oscillation amplitude over ten successive oscillations must be less than or equal to 0.05 mmHg in order to assess the repeatability of oscillation amplitude. For the purpose of the reproducibility test, the mean value will serve as the reference value. Calculating the mean value of the oscillation amplitude and comparing it to the reference value is one way to investigate whether or not the oscillation amplitude can be replicated accurately. The difference in each static pressure's two values must be less than or equal to 0.1 mmHg, and this difference must not exceed 0.1 mmHg. There is no additional criterion that must be met in order to test the repeatability and reproducibility of the envelope of the oscillations, as this is not required by the specifications. The measurement data was taken while conducting the reproducibility test and shapes of the oscillations captured and logged for periodic tests and re-tests of the NIBP simulator over the course of time (I.T.S. 2020; Kumar et al. 2020; Hung et al. 2007; Riedel et al. 2011; Amoore et al. 2007).

Future Directions and Confronts

Recently, considerable research attention has been focused on the development of novel, dependable, and portable BP measurement technologies. BP is measured during daily activities using nonlinear autoregressive models, feed forward neural network models, and pulse arrival time models. Multiple one-class support vector machine (OCSVM) models are used in this study to train the data in proportion of outliers. OCSVMs hyper planes assigned new BP estimations to clusters, and prediction intervals (PIs) were computed using BP error standard deviation linked with various clusters. When compared to individual model precision, covariance intersection fusion improved BP and PI predictions by 24%. The used model fusion

demonstrates ability for clinically assessing BP. The PI indicates that approximately 71%, 64%, and 29% of the data acquired from sitting, standing, and walking can yield accurate estimations of BP (Landry et al. 2022).

Holz et al. designed wearable glasses consisting of optical sensor. This technology can measure BP in real time and can run for a whole day after charging once. The PPT technique is used to calculate BP. The wearer's PPT on the angular artery, superficial temporal, and occipital is the primary signal. As a result, such gadgets would be particularly useful for continuous BP monitoring. With the help of developed device, the user can also monitor his BP at the time of drinking, moving, or doing any other activity, and record the BP readings (Holz and Wang 2017).

Wang et al. have developed a sophisticated BP measuring device that is based on smart phones. The vibrations caused by the beating of the heart are measured by the system using the accelerometer sensor available in the phone. The camera of the smart phone is used to monitor the pulse on the fingertip. Due to the fact that the system measures BP using the camera on a smart phone, the disadvantage is that it is difficult to use when engaging in physical activity (Wang et al. 2018; Liu et al. 2018; Miao et al. 2015).

Andrew Carek and Christian Holz designed and developed a wearable device that was sewn into a pair of shorts. The machine known as Naptics is able to measure blood pressure in patients without waking them up while they are sleeping. Naptics detects the optical reflections of the pulse as it travels down the leg of the wearer and records the micro vibrations of the patients that are caused by the heartbeat. This allows Naptics to determine where the beat wave is located in the wearer's leg (Carek and Holz 2018).

Automation based on the internet of things (IoT) and artificial intelligence (AI) is the primary emphasis of the current industry 4.0 age as well as also important component for the industrial revolution 5.0. Consequently, IoT-based BP measuring devices and remote BP measurement would be a significant improvement in the industry for convenient and continuous wireless monitoring and measurement. The recent literature survey on BP devices clearly reveals an uptrend in research, commercial, and clinical activity wherein increased use of wearable BP devices by the consumers and also adherence to certain standards and regulations for wearable BP devices. In-spite of well-known inherent difficulties with cuffless BP measuring devices, the discussions and research to use or not to use or some reliable replacement of the arterial cuff sphygmomanometer in the diagnosis and treatment of BP related ailments are still in focus (Wu et al. 2020; Avolio et al. 2022).

Conclusions

Because of the direct impact on the heart, blood pressure measurement, recording, and monitoring are vital for human health. It is a major cause of kidney failure, coronary heart disease, heart strokes, and the majority of other fatal diseases. We discussed various noninvasive blood pressure measurement techniques, as well as their benefits and drawbacks, in this chapter. The various types of errors caused by

posture, position, white coat effect, and devices, among other things, are thoroughly discussed. Various national and international BP standards are elaborated, and an ISO standard of simulator, which is used for calibrating and testing oscillometric type BP devices, is also described. We are confident that this comprehensive review, which includes information on BP measuring devices ranging from basic to advanced, will be very useful and will pave the way for researchers, students, metrologists, and academics who want to work in this challenging area of medical science.

Cross-References

- ▶ Precision Measurements in Healthcare Systems and Devices
- ▶ Pressure and Its Measurement

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Glucose Monitoring Techniques and Their Calibration

73

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Abstract

In the last two decades, there has been an increasing demand for accurate and non-invasive blood glucose tests. Despite that current blood glucose monitoring techniques, such as the widely available glucometer, are well established. Inaccuracies in measurement results are common, because of the invasive nature of glucometers. This review discusses various glucose testing techniques classified as invasive, minimally invasive, and invasive. The development of the enzyme-based invasive approach is also reviewed, with emphasis on the introduction of portable glucometers. The various minimally invasive and non-invasive glucose measuring methods that have been investigated over the last few decades and their calibrations are also discussed.

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Keywords

Glucose measurement · Invasive · Minimally invasive · Non-invasive · Calibration

Introduction

Diabetes is a chronic metabolic syndrome that is characterized by impaired functioning of the insulin hormone, resulting in higher glucose concentration in the blood (Asmat et al. 2016). According to the World Health Organization, it has been stated that there are about 420 million diabetes patients worldwide and this number is increasing alarmingly day by day (WHO 2022). The chances of complications like cardiovascular disease, retinopathy, neuropathy, etc. are higher in diabetic individuals. Therefore, it is advisable to monitor the levels of glucose at regular intervals. In addition, this is critically required for regulating the insulin dose (Lawand et al. 2015). The conventional method used for the monitoring of blood glucose levels uses electrochemical enzyme-based detection. However, sampling involves needle-based finger pricking and collection of blood on a strip immobilized with the glucose oxidase enzyme. This approach is painful and may also lead to infection (Paul et al. 2012). Other invasive glucose measurement method includes the hexokinase assay which is widely used in hospitals and laboratories. The history of glucose biosensors dates back to the 1950s, and since then different generations of glucose biosensors have come to the forefront (Nikhil et al. 2016). Even though all these methods are well established and accepted, the painful procedure of testing limits their frequency. Therefore, these days other minimally invasive and non-invasive methods are being extensively explored as an alternative that may have the potential to replace the conventional invasive approach. In this chapter, three major categories of blood glucose measurement, including the invasive, minimally invasive, and non-invasive methodologies, are discussed in detail. The emergence of different glucose measurement devices is also briefly explained. Glucose sensing using the electrochemical method is very popular, paving the way for the development of different generations of biosensors over the years. Even though invasive methods are widely used, measurement errors are common, which may include pre-analytical, analytical, and post-analytical errors. Therefore, certain established standards have been included and briefly discussed. In addition, the calibration approach for different glucose monitoring techniques is also explained.

Glucose Measurement Techniques

The timely management of diabetes is a critical parameter for preventing further diabetes-related complications. Regular monitoring of glucose levels helps in managing the disease. The popular glucose measurement devices are invasive in nature and may not be suitable for frequent monitoring. Therefore, a lot of research is being carried out toward the development of techniques that are non-invasive in nature. Glucose measurement devices can be divided into three categories:

(a) *Invasive glucose devices*: The conventional glucose measurement devices are based on the electrochemical method which involves puncturing the skin with the help of needles. This results in the release of blood which is sucked onto a measurement strip through capillary action. This technique provides instant glucose levels, and no technically skilled personnel is required for the measurement. The frequent puncturing can cause discomfort and may lead to infections; therefore, patient compliance becomes difficult (Tonyushkina and Nichols 2009).

(b) *Minimally invasive devices*: These devices allow the real-time continuous measurement of subcutaneous glucose levels. The working principle is based on the enzymatic reaction in the interstitial fluid using the glucose oxidase (GO_x) enzyme (Reddy et al. 2020). However, the prolonged usage of continuous glucose measurement (CGM) devices can cause tissue damage at the site of installation. The advantage of using the minimally invasive device is the availability of real-time variations in the glucose levels occurring throughout the day. This allows the patients to have a greater degree of metabolic control over their sugar levels through an optimized insulin therapy minimizing the risk of diabetes-associated complications (Koschinsky and Heinemann 2001).

(c) *Non-invasive devices*: The main disadvantage of the invasive or minimally invasive device is the probability of low patient compliance. Therefore, the development of a non-invasive device will bring about a paradigm shift due to vast market potential. Some techniques like Raman, NIR, MIR, and impedance spectroscopy and ultrasonics, etc. are being explored for the development of non-invasive devices. A few devices have been developed so far using optical

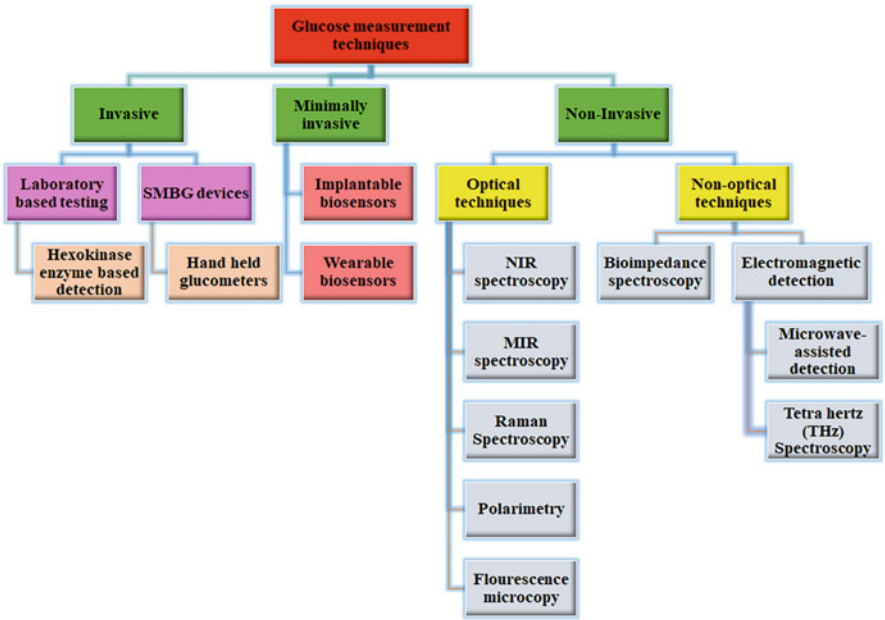

Fig. 1 Different types of glucose measurement techniques

techniques; however, they suffer from poor accuracy (Paul et al. 2012). Figure 1 summarizes the classification of glucose detection techniques (Kottmann et al. 2012).

Invasive Glucose Measurement Technique

Conventional glucose measurement devices allow the self-monitoring of blood glucose levels and are therefore also known as self-monitoring blood glucose (SMBG) devices. The pioneers of this field were Clark and Lyons from the Children's Hospital in Cincinnati. Their device was based on the determination of oxygen consumption, catalyzed by glucose oxidase enzyme. The working electrode consists of a thin layer of the enzyme entrapped over an oxygen electrode using a semipermeable membrane (Chen et al. 2013). This method proved to be a milestone in the measurement of diabetes (Wang 2001). The following set of reactions occurs in this process:

$$\text{Glucose} + O_2 \xrightarrow{GO_x} \text{Gluconic acid} + H_2O_2 \quad (1)$$

$$O_2 + 4H + +4e^- \rightarrow 2H_2O \quad (2)$$

It was only in the early 1970s that the SMBG devices became commercially available when the rights of this device were transferred to Yellow Spring Instrument Company which eventually launched it in 1975 as a 23 YSI analyzer. Since then, several variations in the basic methodology have occurred for the improvement of invasive glucose measurement devices, and several devices have flooded the market. However, the invasive method of detection suffers from multiple limitations like false negative/false positive, due to mishandling or exposure of the sensor to extreme ambient conditions, incorrect coding (leading to >30% measurement errors), interference from biomolecules like uric acid and triglycerides whose concentration is altered in certain diseased conditions like gout, and pancreatitis. Furthermore, the inconvenience of measurement which is not only painful but poses risks of infection

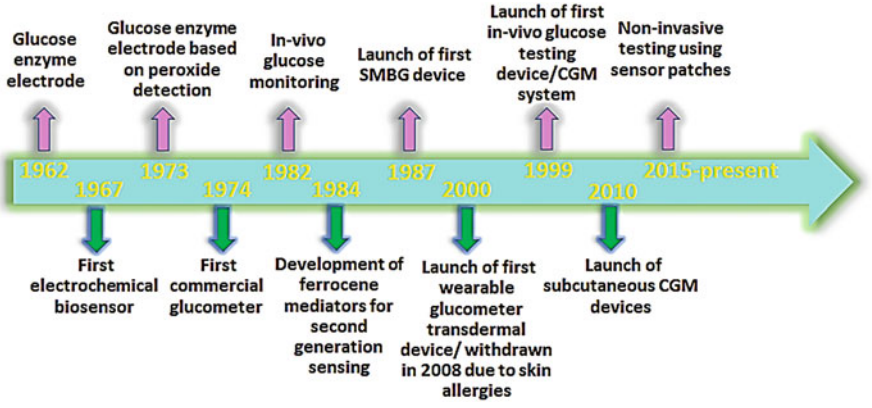

Fig. 2 Timeline of glucose measurement devices

due to frequent monitoring may also cause tissue abrasion (Ginsberg 2009; Erbach et al. 2016). The timeline of the advent of different glucometers is shown in Fig. 2 (Chen et al. 2013; Wang 2001).

The electrochemical method of detection is based on measurements mediated by any one of the three enzymes, namely, hexokinase, glucose oxidase (GO_x), or glucose-1-dehydrogenase (GDH) (Yoo and Lee 2010). The hexokinase assay is a common technique that is used for glucose detection in laboratories (Yoo and Lee 2010; Price 2003). The SMBG devices generally involve glucose oxidase or GDH enzyme for glucose detection. However, glucose oxidase-based biosensors are more frequently used because of their high selectivity for glucose (Yoo and Lee 2010). In addition, the glucose oxidase enzyme is less sensitive to variations in pH and temperature. The GDH-based glucose biosensors are also used, the GDH enzyme is divided into three major subcategories based on the type of cofactor involved, i.e., flavin adenine dinucleotide (FAD), pyrroloquinoline quinone (PQQ), and nicotinamide adenine dinucleotide (NAD), FAD and PQQ cofactors are tightly bound to their enzyme counterpart, whereas NAD is not. Even though GDH enzymes do not require oxygen substrate for their reaction, they still suffer from low specificity toward glucose limiting their applications in glucose biosensing (Teymourian et al. 2020; Lee et al. 2018).

The basic principle of detection in the electrochemical technique involves the oxidation of glucose in the presence of water and oxygen, leading to the formation of gluconic acid and hydrogen peroxide (H_2O_2). The H_2O_2 serves as the signal which is detected at the electrode as a result of its oxidation at the anode (Grieshaber et al. 2008). There are three ways in which glucose may be detected at the electrodes:

(i) Measurement of the oxygen consumption
(ii) Measurement of the total concentration of the H_2O_2 produced at the electrode
(iii) Employing a chemical mediator for the electrons transfer to the electrode interface

The first two approaches are considered the first generation of glucose biosensors, and the third approach is considered the second generation of glucose biosensors. The third generation of biosensors eliminates the need for any toxic mediators for electron transfer to the electrode and involves the direct transfer of electrons to the electrode surface (Dudkaitė and Bagdžiūnas 2022).

Basic Principle of Glucose Biosensing

The basic principle of a glucose biosensor involves the catalytic oxidation of glucose to gluconic acid in the presence of oxygen molecule. The cofactor FAD of GO_x is regenerated during the following reactions:

$$\text{Glucose} + GO_x - FAD+ \rightarrow \text{Gluconolactone} + GO_x - FADH_2 \quad (3)$$

$$GO_x - FADH_2 + O_2 \rightarrow GO_x - FAD + H_2O_2 \quad (4)$$

$$H_2O_2 \rightarrow 2H^+ + O_2 + 2e^- \quad (5)$$

The regeneration step results in the production of H_2O_2 which is electrochemically oxidized at the electrode interface to release the electrons which are directly proportional to the number of glucose molecules in the blood (Yoo and Lee 2010).

First Generation of Glucose Biosensors

In this generation of glucose biosensors, the glucose concentration has been estimated as a result of the measurement of the total amount of hydrogen peroxide released by the glucose oxidase enzyme after catalyzing the reaction leading to the formation of gluconic acid from glucose. The enzyme cofactors provide steric hindrance which limits the electrical interaction between the enzyme and the electrode surface. This interaction is, therefore, mediated by oxygen substrate which helps in shuttling the electrons to the electrode surface. The amount of H_2O_2 is equivalent to the concentration of glucose present. However, problems like deactivation of the enzyme as a result of H_2O_2 production and requirements of high working potentials are encountered which results in interference from other electroactive species like ascorbic acid, uric acid, etc. (Avari et al. 2020).

Second Generation of Glucose Biosensors

In this case, there is no need for a high operating potential; instead, electron acceptors which are commonly known as redox mediators are employed to mediate the electron transfer process. The mediator gets oxidized and reduced at the electrode interface resulting in an amperometric signal. Some examples of these redox mediators include ferrocene, ferricyanide, methylene blue, Os^{2+}/Os^{3+}-based complexes, etc. (Teymourian et al. 2020; Guven et al. 2021; Lin et al. 2019; Bollella et al. 2019).

Third Generation of Glucose Biosensors

In this case, the direct electron transfer takes place from the enzyme to the electrode surface mediated by nanostructured materials and their composites, without the use of any toxic mediators. Over the years, extensive research has been directed toward the synthesis of nanomaterials that possess high electron transfer capabilities for the development of glucose biosensors. This leads to an enhancement in the sensing performance of the bioelectrode because of the efficient coupling of the enzyme with the electrode. Various metal nanoparticles, metal oxide nanoparticles, and nanocomposites have been employed for this purpose (Rahman et al. 2010). Figure 3 shows a schematic representation of the different generations of glucose biosensors (Avari et al. 2020).

Table 1 summarizes multiple glucose sensors based on third-generation sensing.

Minimally Invasive Glucose Measurement Devices

CGM devices can be classified into three types:

(i) Invasive CGM devices: These devices are capable of measuring the fluctuating glucose levels in venous blood. However, they are preferably used in hospitals.

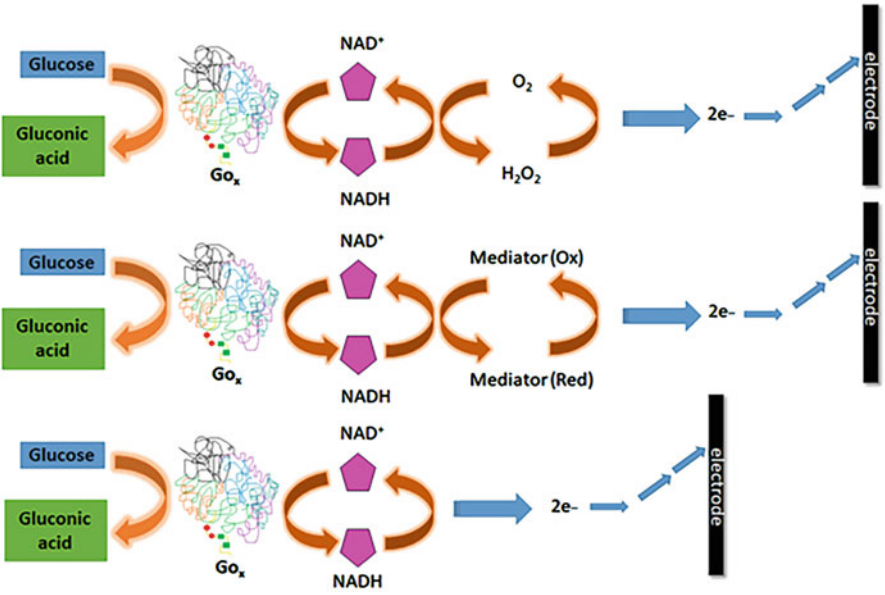

Fig. 3 Different generations of glucose biosensors

(ii) Minimally invasive CGM devices: They may include needlelike or implantable sensors that measure glucose levels in the subcutaneous tissue.
(iii) Non-invasive CGM devices: Such devices are yet to be marketed, and exhaustive research is being carried out in this direction to develop a device based on thermal, acoustical, optical, or electrochemical techniques. This is a popular area of research, and a lot of potential ideas have been introduced that may pave the way for non-invasive CGM devices (Freckmann et al. 2021).

Calibration of CGM Devices

The calibration of the commercially available CGM sensor is critical to ascertain the accuracy of the measurement. The devices are calibrated one 1–2 h after installation known as the warm-up period. It is recommended to calibrate the device every 12–24 h after the completion of the warm-up period. Linear regression techniques are quite popular for calibrating CGM devices (Acciaroli et al. 2018). Some of the commercially available CGM devices are Medtronic Guardian 3, Abbott Freestyle Libre, Dexcom G4 Platinum, Dexcom G5, and Dexcom G6. All of these devices measure the interstitial glucose levels based on the glucose oxidase-based chemical reaction, which is converted into an electrical signal which is proportional to the glucose concentrations. This conversion is mediated by a calibration function. This function must be updated from time to time due to variations that might occur as a result of instrumental drift, biocompatibility issues, etc. Furthermore, these devices may be factory calibrated, but still face challenges like inefficiency to detect sudden blood glucose changes and skin allergies due to the sensor glue (Heise et al. 2021).

Table 1 Third-generation glucose biosensors based on different nanomaterials and their nanocomposites

S. no	Sensor matrix	Detection technique	Linear dynamic range	Limit of detection	Ref
1.	Graphene nanoribbons	Cyclic voltammetry	50–2000 mg/L	20 mg/L	Mehmeti et al. (2017)
2.	ZnO nanorods	Amperometry	0.05–1 mM	0.06 mM	Ridhuan et al. (2018)
3.	PEC complex (chitosan/kappa-carrageenan) doped with gold nanoparticles (AuNPs)	Square wave voltammetry	10 μM–7 mM	5 μM	Rassas et al. (2019)
4.	Chitosan-supported mesoporous carbon nanocomposite	Amperometry	250 μM–3 mM	4.1 μM	Barathi et al. (2019)
5.	(3-Aminopropyl) triethoxysilane (APTES) stabilized titanium dioxide nanoparticles cross-linked to carboxylic graphene	Electrical impedance spectroscopy	50–1000 μM	24 μM	Ognjanović et al. (2020)
6.	Poly(vinyl alcohol) and poly (ethyleneimine) blend nanofibers activated by glutaraldehyde	Differential pulse voltammetry	2–8 and 10–30 mM/L	0.3 mM	Temoçin (2021)
7.	Mesoporous carbon nanostructure, lignin-based carbon nanofibers	Cyclic voltammetry	0.15–2.7 mM	89 μM	Beaucamp et al. (2021)
8.	Nanoporous copper oxides embedded into graphitic carbon layers (MOP-CO)		0–4000 μM	26 nM	Wei et al. (2022)
9.	Platinum nanoparticles electrodeposited on poly(azure A)	Amperometry	20 μM–2.3 mM	7.6 μM	Jiménez-Fiérrez et al. (2020)
10.	Single-walled carbon nanotubes (SWCNTs)	Amperometry	0.1–30 mM	0.05 mM	Tasca et al. (2011)

Metrological Issues Related to SMBG Measurements

All the laboratory tests including the lab-based glucose measurement technique are prone to erroneous results. The diagnostic efficacy of a test is largely influenced by abundantly large test errors irrespective of the origin of the error. Therefore, a

parameter known as the total error is important to the clinician as it includes all the different sources of errors that influence the overall measurement (Krouwer and Cembrowski 2010):

$$\text{Total error} = \text{Observed value} \sim \text{true value of glucose} \quad (6)$$

The various sources of errors affecting the overall glucose estimation are tabulated in Table 2.

Standards for SMBG Monitoring

The allowable limit for analytical errors associated with SMBG measurements is still a debatable issue. The American Diabetes Association (ADA) recommends that the total error of glucometers should be $<5\%$. Other organizations like the National Committee of Clinical Laboratory Standards (NCCLS) are a little lenient, with the allowable error limit set at $\pm 20\%$ for blood glucose >100 mg/dl and ± 10 mg/dl for glycemia ≤ 100 mg/dl (Solnica and Naskalski 2007). Furthermore, the International Organization for Standardization (ISO) recommends that ninety-five percent (95%) of the individual glucose results shall fall within ± 15 mg/dl of the results of the manufacturer's measurement procedure at glucose concentrations ≤ 75 mg/dl and within $\pm 20\%$ at glucose concentrations >75 mg/dl. Furthermore, ISO 15197 uses metrology terminology for accuracy "closeness of agreement between a test result and the accepted reference value" (Krouwer and Cembrowski 2010).

Non-invasive Glucose Sensing

Considering the importance of regular glucose monitoring, for the optimization of diabetes therapy, non-invasive glucose sensing is a popular research area. Given the need for measuring glucose levels at least five to six times a day, the invasive techniques are not fit for regular use. Different non-invasive techniques are being

Table 2 Various sources of error in SMBG measurement

Error type	Definition	Sources
Pre-analytical errors	Errors which may occur before testing	Lack of proper cleaning of test surface (finger) Blood sample from a hypotensive patient Dilution of blood specimen as a result of manipulation of puncture site (Krouwer and Cembrowski 2010)
Analytical errors	These include errors which may occur during analysis	Usage of expired test strips Untrained test personnel performing the test Temperature variations that may cause denaturation of bio-recognition elements (Nichols 2011)
Post-analytical errors	Errors which occur after the measurement	Delayed reporting Reporting of wrong information to the clinician (Krouwer and Cembrowski 2010)

explored, including optical techniques like Raman, fluorescence, NIR and MIR spectroscopy, polarimetry, etc., and other nonoptical techniques like bio-impedance spectroscopy, photoacoustic spectroscopy, microwave assisted detection, and electromagnetic detection. However, optical techniques are known to demonstrate the best results (Alsunaidi et al. 2021).

Raman Spectroscopy

In this technique a laser is used to induce molecular vibrations in the body fluids; the scattering of light which is influenced by the molecular vibrations is detected which gives an estimate of the concentration of glucose in the blood (So et al. 2012). The NIR laser is the preferred source of excitation because of its good penetration depth and less risk of photochemical and thermal damage (Pandey et al. 2017). Raman spectroscopy is highly sensitive to several molecular vibrations including the C–O, C–C, C–OH, C–O–H, and CH_2 stretching and C–O–H bending frequencies, which are specific to glucose (Wróbel 2016).

Raman spectroscopy is a popular technique in terms of the deep tissue analysis of the human body courtesy of the development of the spatially offset Raman spectroscopy (SORS) technique. This technique enabled the probing of the human tissues from several micrometers to several millimeters and in certain special cases up to several centimeters as well. Nowadays, it is possible to use SORS in association with surface-enhanced Raman spectroscopy (SERS) to exploit the advantages of both techniques. This allows for the in-depth analysis of diffusely scattering media especially tissues, enabling the detection of those biomolecules which exist in extremely low physiological concentrations like glucose. In a recent study, glucose levels were successfully monitored in the interstitial fluid, by SORS technique which detects the spectra generated using a SERS-based implant in vivo (Matousek and Stone 2013).

A Raman-based prototype GlucoBeam, RSP Systems A/S, Denmark, was tested in a study, on patients suffering from type I diabetes. The objective of the study was to develop calibration models and to test the efficacy of the device in real time. The study comprised 15 adults, and was conducted over a span of 25 days at home and in the clinic, where each subject underwent at least six measurements per day, two measurements were performed using capillary blood glucometers, another two using a CGM device, and another two using the GlucoBeam, RSP Systems A/S prototype being tested. Calibrations models were developed using data from 19 to 24 days, and the leftover 3–8 days were used for independent validation. The device demonstrated a good practical ability with around 94% of the values falling in zones A and B of the Clark error grid. These findings suggest the potential of the prototype in the development of a real-time Raman-based non-invasive glucose measurement device (Pleus et al. 2021).

In another study, the transcutaneous measurement of blood analyte, i.e., glucose was demonstrated, using Raman spectroscopy on a total of 17 human subjects. The calibration was performed using capillary blood glucose measurements. Calibration models were developed using partial least squares regression (PLSR) and were validated using leave-one-out cross-validation utilizing spectrally preprocessed

data using Savitzky-Golay filter using 13 window points smoothing to improve the signal-to-noise ratio. The raw data consisted of overlapping backgrounds, obscuring the Raman features. Two strategies were employed to preprocess the spectral data. In the first strategy, least square fitting was performed allowing the Raman features to be resolved, and in the second strategy, the background subtraction was not carried out and the spectra were analyzed in their raw form (Enejder et al. 2005). This report suggests that the spectral features of the glucose are an integral part of the calibrations.

Near-Infrared and Mid-Infrared Spectroscopy

This technique utilizes radiation in the NIR region of the electromagnetic spectrum. Owing to the high penetration depth of the NIR light, NIR spectroscopy has been extensively applied in the field of biology for deep tissue analysis. The NIR region is categorized into three main bands, the combination overtone band (2000–2500 nm), the first overtone band (1400–2000 nm), and the second or higher overtone band (750–1400 nm) (Zhang et al. 2016).

The overtones and combination bands represent the forbidden frequencies, resulting in weak signal intensities. This also results in overlapping spectral features, giving rise to the problem of multicollinearity in the NIR region which makes the specific detection of glucose a major concern. To solve this problem, chemometrics techniques are used for deriving maximal chemical information about the target analyte (Ozaki et al. 2018). Within the NIR region, two optical windows exist, one window from 650 to 950 nm and another from 950 to 2200 nm which also represents the short-wave NIR region (SW-NIR). The first window has been extensively explored; however, the exploratory analysis of the second window calls for specialized instrumentation. The SW-NIR region pertains to the electronic transitions which are responsible for the high permeability, making it particularly interesting to biologists.

For the non-invasive detection of glucose, both absorption and reflectance-based techniques have been utilized. The sites of measurement usually include fingers, lips, palms, and webbing between the fingers. However, obtaining glucose-specific information is quite challenging because of the interference from other NIR-active biological compounds like fats, hemoglobin, and proteins. Furthermore, the measurements are influenced by ambient factors like temporal variations, machine drift, time drift, etc. (Yang et al. 2018).

The SW-NIR region has been probed for non-invasive glucose detection. In a study by Jain et al., absorption and reflectance spectroscopy in the SW-NIR region were employed for the design of a non-invasive blood glucose monitoring system. The design of the system was such that it involved both absorption and reflectance measurements mediated by infrared emitters of 1300 and 940 nm in 25 subjects. The model has been validated against reference blood glucose measurements using an SD check invasive glucometer.

One of the common obstacles in glucose detection using NIR spectroscopy is the significant absorption by water. Therefore, identification of those bands in the NIR which are transparent to water is a prerequisite for glucose detection (Jain et al.

2019). The water-specific peaks in the NIR region fall between 1350 and 1520 nm and 1790 and 2000 nm. Therefore, the NIR band from 700 to 1100 nm is an important window for the non-invasive detection of glucose (Shokrekhodaei and Quinones 2020). Several glucose-specific peaks fall in this region, owing to the CH and OH vibrations. Golic et al. have pointed out the influence of H bonding and temporal variations on the CH and OH vibrations, respectively. Therefore, it is important to calibrate the glucose concentrations for the CH and OH features, to minimize the effect of H bonding and temperature. The major spectral features which correspond to the CH, CH_2 and OH stretching vibrations are located at 910 nm, 930 nm, and 970 nm respectively (Golic et al. 2003).

MIR spectroscopy technique follows the same principles of absorption as NIR spectroscopy. Here, the light in the MIR region from 2500 to 10,000 nm of the electromagnetic spectrum is employed for analysis. The absorption features corresponding to glucose in the MIR region are much sharper in comparison with those in the NIR region. However, the practical applications of this technique for non-invasive device development are challenging as a result of the poor penetration depth of the MIR light (Vashist 2012). Therefore, MIR-based technique can only be adopted in the reflective mode. The absorption feature of glucose in the MIR region occurs between 6250 and 11,110 nm. Water has a much more prominent absorption in the MIR range in comparison with the NIR range (Tang et al. 2020).

In a study, a combination of the quantum cascade radiation and photoacoustic detection of absorption was used. The in vitro data suggests that the usage of less number of IR wavelengths allows for better prediction; in addition, the accurate selection of glucose variant and invariant wavelengths is also critical. They have also shown that by using the MIR technique in conjunction with the photoacoustic method, the glucose-containing layer of the skin can be easily accessed, so resolving the issue of MIR light's shallow penetration depth. Also, by introducing a time delay of 10 min, it is possible to obtain a higher degree of correlation between blood glucose and the photoacoustic signal. All these results indicate the potential of MIR spectroscopy in the development of an in vivo blood glucose measurement system (von Lilienfeld-Toal et al. 2005).

Polarimetry

This technique has been adopted for non-invasive glucose monitoring for a long time. This is based on the principle that when a plane-polarized light is incident on a chiral molecule, the plane of the light gets rotated. Glucose is a chiral molecule that is optically active and can rotate the plane of polarized light. The polarization direction will form a deflection angle with the original incident direction, which is proportional to the glucose concentration. The technique enables direct estimation of glucose concentration, and is easy to operate and simple, but suffers from multiple limitations like interference from other optically active molecules, and scattering effects. The human body is a scattering medium and this may cause the polarized light to lose its polarization. Furthermore, the aqueous humor of the eye is the site of glucose measurement, but the measurement is largely affected by the motion artifacts, because of the variations in the corneal infringement (Tang et al. 2020). In a

study, a differential Mueller matrix polarimetry technique has been used. The estimation of glucose concentrations from aqueous glucose samples by measuring the optical rotation values (γ) and depolarization index (δ) in tissue phantom samples consisting of 0–500 mg/dl glucose and 2% Lipofundin with a limit of detection of 20 mg/dl was accomplished. The γ and δ values have also been obtained from the fingertips of healthy volunteers. The γ and δ values obtained from the fingertips of human volunteers were found to have a good correlation with those of the tissue phantom samples (Phan and Lo 2017).

Fluorescence Spectroscopy

When an electron absorbs a photon, it gets excited from its singlet ground state (S_o) to an excited state (S_n, $n > 1$). The electron then decays back to S_1, the lowest excited state, by dissipating a part of its energy in the environment. This process is known as internal conversion. When the electron returns to the ground state (S_o) with the emission of a photon having a radiative rate constant K_r, this phenomenon is known as fluorescence (Albani 2008). Glucose emits a fluorescent signal when exposed to visible or UV light. But the signal is weak and affected by other fluorescent interferants. For this purpose, glucose is usually coupled to other fluorescent probes known as receptors, which enhance the fluorescent intensity significantly. However, the measurement may be affected by several factors like scattering, skin pigmentation, the thickness of the epidermis, etc. (Alsunaidi et al. 2021). A variety of receptors have been employed for this purpose, including enzymes, glucose-binding proteins like concanavalin A and glucose galactose-binding proteins, nanostructured materials like carbon nanotubes (CNTs), quantum dots, etc. Fluorescent resonant energy transfer is a popular technique which is based on the interaction between a fluorophore (donor) molecule and the acceptor molecule (receptor). In the presence of glucose molecule, this link between the donor and the acceptor is broken which disrupts the electron sharing between the two and improves the fluorescent signal. This bond stays intact in the absence of glucose enabling electron transfer between the two and reducing the fluorescent intensity (Villena Gonzales et al. 2019; Klonoff 2012). Nowadays, CNTs are being extensively utilized for glucose detection. The technique involves the adsorption of glucose receptors onto carbon nanotubes. Here, the receptor molecules are absorbed by single-walled CNTs, and post the binding of glucose molecules to their receptors, the nanotubes' fluorescence gets altered because in the presence of glucose, this binding between the CNTs and the receptor is disrupted. The change in fluorescence is then relative to the glucose concentration (Klonoff 2012).

Bio-Impedance Spectroscopy

This technique comes under the broad term bioelectrical impedance analysis (BIA). BIA is a practical tool that is used to analyze the composition of the body in terms of fat content and water. In this technique, painless currents of low amplitude are passed through the human body using cables connected to electrodes or other conductive surfaces. This allows the measurement of resistance (R) and reactance (Xc). Both these parameters are used for estimating the impedance (Z), and the total water

content of the body, in addition to the extracellular and intracellular water. Based on this data, other parameters like fat-free mass (FFM), fat mass (FM), and body cell mass (BCM) are calculated (Mialich et al. 2014; Lukaski et al. 1985). This technique is also used for analyzing the compounds in the blood. The changes in the parameters discussed above reflect the changes in the dielectric/dimensions of the target (Losoya-Leal et al. 2012). In another version of this technique, known as dielectric spectroscopy, the variations in frequency are measured concerning the variations in the dielectric properties of the target matrix. A study has shown the potential of using bio-impedance spectroscopy, for the non-invasive detection of glucose. A wristwatch was used for the sensing purpose in healthy human subjects for clinical-experimental studies. The sensing system is composed of an open circuit that is coupled to the skin and another circuit that performs the impedance measurements. The sensor uses electromagnetic waves in the frequency range of 1–200 MHz. These waves then interact with the skin to monitor any changes in the electrical properties of the tissue underneath. The results showed a good correlation between the actual glucose levels and the sensor readings. However, environmental variations, sweat, and motion artifacts due to the sensor probe are bound to cause measurement errors which must be addressed (Caduff et al. 2003).

Electromagnetic Monitoring

Electromagnetic wave (EM) sensing has an edge over other techniques because of its capability to undergo strong interactions like absorption, scattering, and transmission, with various compounds inside the body. EM waves can be categorized into radio waves, microwaves, terahertz (THz) waves, visible/infrared (IR) light, ultraviolet, X-rays, and gamma rays particularly based on their frequencies or wavelengths. However, the EM waves may broadly be classified into ionizing (X-rays and γ-rays) and nonionizing radiations (THz waves, microwaves, IR waves), and the nonionizing radiation is more suitable for biological testing due to its nondestructive nature and minimal damage to the tissues and organs in the body. The IR-based techniques (NIR and MIR spectroscopy) have already been discussed in the sections above. THz waves have a frequency in the range of 0.3–3 THz, and they fall between microwaves and far-infrared waves. Terahertz time-domain spectroscopy (THz-TDs) is utilized for medical applications (Zhang et al. 2019). In a study, the optical properties of the human skin were characterized using optical THz time-domain spectroscopy. The variations in the attenuated total internal reflection (ATR) spectra were correlated to the changes in the glucose levels after the measurement of ATR spectra consecutively from 0 to 90 min in six volunteers after glucose intake. The ATR amplitude exhibited an increase with rise in glucose concentration (Cherkasova et al. 2016). In another study, THz waves in the frequency range from 0.2 to 0.9 THz were used for the quantification of blood glucose levels using a wave generated from an indium arsenide (InAs) wafer and an 800-nm femtosecond laser in transmission mode. The analysis was performed on 20 blood samples obtained from diabetic patients, and a good correlation with low error rates ($<15\%$) was found between the measurements and a commercial glucose meter (Chen et al. 2018). Furthermore, the principle of measurement through microwaves

involves the interaction between microwaves and matter which is under test. The incident signal may change in terms of velocity, attenuation, or reflection. Furthermore, this interaction manifests as a change in the permittivity of the sample. This is a nondestructive technique and doesn't require direct contact with the test object, assisted by penetrating waves. The nonionizing nature of the waves minimizes any health risks associated with the technique, making it a particularly attractive technology in the healthcare industry. However, temperature, moisture content, density, etc. may affect the measurement (Korostynska et al. 2014). The dielectric coefficient of blood is dependent on glucose levels. Based on this fact, for the non-invasive measurement of glucose levels, microwaves of a frequency most sensitive to glucose are incident on a tissue, and the reflected signal is then measured which is proportionate to the glucose concentration. In a study, a single spiral microchip-based sensor was developed, and the sensor was fabricated on a high-frequency substrate RO3000. The sensor could operate at two different frequency ranges, one from 0 to 2 GHz and another from 2 to 4 GHz to determine the dielectric coefficient of the blood. The variations in the permittivity of the tissue cause a shift in the resonance frequency of the sensor with a larger shift observed for a higher frequency (Buford et al. 2008). In another study, a novel microchip-based glucometer is developed which consists of a band-pass filter, which acts as the sensor, operates approximately at a frequency of 1.9 GHz, and possesses high sensitivity toward the dielectric coefficient of its target site (finger-superstrate). The variation in the blood glucose level leads to a change in the permittivity of the thumb, thereby affecting the response of the sensor (Baghbani et al. 2015). The calibration of these variations leads to the determination of blood glucose levels. The proposed sensor may allow the continuous self-monitoring of glucose levels. A few of the non-invasive devices which have been developed over the years are summarized in Table 3.

Calibration of Non-invasive Measurements

Multivariate calibration techniques are used for determining the analyte concentration in a complex mixture. Based on the non-invasive data, generally, the NIR spectra, the standard error of prediction (SEP), and correlation coefficients (R^2) are calculated. A higher R^2 value and a lower SEP are associated with a successful prediction. Successful predictions are dependent on numerous factors given below:

(i) Overfitting of the data must be avoided.
(ii) Chances of correlation with other time-dependent parameters must be avoided like instrumental drift, time drift, noise, etc. Such parameters affect the prediction performance of multivariate calibration techniques like partial least squares regression (PLS) and artificial neural networks (ANN). The NIR spectroscopy is an indirect method which increases the possibility of chance correlations, because PLS technique takes into account all the sources of variation, thereby increasing the chances of correlation of a nonspecific signal with the analyte (Liu et al. 2005).

Table 3 Non-invasive glucose measurement devices

S. no	Name of sensor	Sample type	Design	Detection limit	Response time	Wireless telemetry	Ref
1	Cavitas sensor	Saliva	Pt and Ag/AgCl electrodes are integrated with an enzyme membrane on a mouth guard	5–1000 μmol/L			Arakawa et al. (2016)
2	Smartphone-based sensor	Saliva	Glucose oxidase enzyme is immobilized in conjunction with a pH-responsive dye over a filter paper-based strip	50–540 mg/dL	20 s	Based on smartphone	Soni and Jha (2017)
3	Fast Fourier transform in near-infrared-based sensor		Sensor module/probe is printed from poly-lactic acid (PLA) which consists of a pair of one or two NIR LED and a photodiode		60 s	Transmission module is a conglomerate of modules used for monitor display and for transmission through Wi-Fi or Bluetooth	Jenie et al. (2017)
4	TensorTip Combo Glucometer	Tissue photography based (NO SAMPLE)	Users have to initially calibrate the device using blood for a few weeks following which they can painlessly measure their blood glucose levels by putting their finger in the space provided			Smartphone based	Segman (2018)
5	Grove's device	Fingertip or earlobe (NO SAMPLE)	Takes reading from blood on the capillary level instead of the skin level		20 s		Siddiqui et al. (2018)
6	Glucosense	Touch based (NO SAMPLE)	Silica gel that has ions having high sensitivity toward the infrared light the		30 s		NetScientific PLC (2015)

			reflected spectrum changes at the touch of a user. It uses a power efficient laser to acquire the readings				
7	glucoWISE	Area between the thumb and forefinger	Measures glucose level at the capillary level		10 s	Wireless data transmission using Bluetooth to generate alerts	glucoWISE®
8	GlucoTrack	Earlobe	It has two components, one is the ear clip that measures the blood glucose levels, and the other that displays the results by using electromagnetic, ultrasound, and thermal techniques				Lin et al. (2018)
9	iQuickIt saliva analyzer	Saliva	It employs disposable draw wick sticks for the samples which are transmitted to other smart devices			Transmission to smart devices	Siddiqui et al. (2018)
10	Contact lens	Tissue (eye)	It is composed of a wireless transmitter for transmitting the glucose readings and a static electrical charge for drawing power			Transmission to other devices	Keum et al. (2020)
11	Noviosense	Tears	Electrochemical			The device has a wireless module to transmit data and to power the device	Noviosense; Geelhoed-Duijvestijn et al. (2021)

(iii) The training and prediction sets must be taken at different time intervals.
(iv) The number of input data points must be larger than the parameters used in the fitting equations.

The calibration models may be generated by fitting the non-invasive data to the invasively determined values. The standard error of calibration (SEC) and calibration correlation coefficient are calculated based on this fitting (Khalil 2004).

Leave-One-Out Cross-Validation

The validation of the calibration model can be performed by evaluating the standard error of cross-validation prediction (SECV) and the cross-validation correlation coefficient (RCV) values. The process involves the prediction of a single data point by using the rest of the data points as the calibration set also known as the leave-one-out cross-validation (LOOCV). The prediction performance of the calibration model depends on a low SECV and a high RCV value (Khalil 2004).

The calibration of non-invasive devices depends on several invasive blood glucose measurements along with non-invasive measurements. Oral glucose tolerance tests (OGTT) are used to determine the changes in the glucose levels in vivo which are required for the calibration of non-invasive devices. The sequential collection of spectral data is preferred in OGTT, because randomized collection will add to the complexity of the calibration model (Delbeck et al. 2019). Furthermore, in the OGTT, a given concentration of glucose is administered to the subject, and the changes in the blood glucose levels are monitored over a fixed duration of time. However, time drift and the circadian rhythm may influence the calibrations due to variations in the OGTT. Therefore, a self-calibrating device is needed, which does not require frequent invasive measurements; for this purpose the spot test approach may be adopted, which is performed at random time slots over a long period, thereby minimizing the chance of correlations, and changes induced by variations in the circadian rhythm (Khalil 2004).

Conclusion

Various glucose measurement techniques with all their current prospects, technical aspects, calibration, and ISO standards for testing have been briefly reviewed. Furthermore, the different glucose measurement devices which have surfaced over the years have also been traced. In addition, the different non-invasive devices which have been developed over the past four decades are also highlighted. Over the years, a tremendous amount of research has been directed toward the development of glucose measurement devices. Given the rapid upsurge in the number of diabetic patients globally, the economic burden and healthcare costs have increased by several manifolds. This has led to the generation of several economic prospects and research opportunities. Even though the blood glucose monitors exhibit excellent accuracies, they suffer from several limitations like false negative/false positive, due to mishandling or exposure of the sensor to extreme ambient conditions,

incorrect coding (leading to >30% measurement errors), and interference from other biomolecules like uric acid, triglycerides, etc. Furthermore, the stability and shelf life of biomolecules are also questionable. Hence, extensive research is being carried out on the development of abiotic sensors which do not require need a bioreceptor (enzyme) for detection. The principle is based on the direct detection of an analyte using novel transducer materials exhibiting catalytic properties paving the way toward the fourth generation of glucose biosensors. In addition, minimally invasive devices are also being thoroughly explored for the detection of blood glucose levels. They may also cause several problems like skin irritations, allergies, generation of a vast amount of data, painful sensor insertions, etc. Therefore, over the past two decades, non-invasive glucose measurement is being employed as a possible alternative to invasive devices. Several techniques have been adopted for this purpose, but the development of a device that has accuracies comparable to the established glucometers is a farfetched dream. Even though several non-invasive devices have been developed, mass-scale commercialization is yet to happen. However, the economical fruits of the development of a non-invasive device are lucrative, and the commercialization of such a device may become possible soon.

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Advancements in Measuring Cognition Using EEG and fNIRS

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A Survey

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Abstract

Human cognition is the essential building block of human intelligence, and it is what makes us who we are. Cognition is defined as the capacity to recognize and respond appropriately to external stimuli based on one's beliefs, actions, experiences, and senses. It is one of the fundamental reasons for human existence and is one of the most important aspects of the brain. In childhood, adolescence, and maturity, the cognitive processes of humans are always evolving and developing. Although some of these abilities begin to diminish as one grows older and approaches older maturity, others begin to deteriorate when neurons die and the systems that replace them become insufficient. Understanding cognition is essential not just for healthy cognitive growth and survival but also for the treatment of

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a variety of neuropsychological conditions, such as Alzheimer's disease. It is necessary to examine the cognitive functions of the human brain before one can comprehend cognition. fNIRS and electroencephalography (EEG) are low-cost methods of assessing and evaluating cognitive function. The principles of electroencephalography (EEG) and functional near-infrared spectroscopy (fNIRS), as well as a number of methods for preprocessing and interpreting EEG and fNIRS data, are therefore covered in this chapter. Lastly, the use of simultaneous EEG-fNIRS is discussed along with its limitations and advantages.

Introduction

One of the fundamental characteristics that distinguish the human species from other animals is intelligent behavior. Humans are not one-trick ponies; they may participate in complex goal-oriented actions over time rather than just reacting impulsively to circumstances or stimuli in front of them (Rosati 2016). And one of the building blocks of human intelligence is human cognition, and intelligence and cognition are like the two sides of the same coin. The Latin verb *cognoscere*, which means "to know" or "to come to know," is where the term cognition originates (Bernhard 2007). It is the mental process through which humans learn and interpret things. At its most basic level, it is the capacity of a human to detect and respond appropriately to external stimuli based on its ideas, actions, experiences, and senses. The sensory information one gets is quite huge, extensive, and complicated, cognition helps distill all of this information down to its fundamentals, and humans interact appropriately with their surroundings. As a result, cognitive functioning is essential for daily functioning as well as in guiding human ideas and behaviors and is one of the building blocks of human intelligence. Cognition is comprised of a variety of cognitive processes, which may be further classified into six core neurophysiological domains, which include attention, learning, language, social cognition, motor function, and executive function (Andrianopoulos et al. 2017). Each of these domains deals with a specific function of human cognition starting from basic response to stimuli to higher-level capabilities that determine human intelligence.

Cognition continuously evolves and adapts to new knowledge and stimuli, controlling human behavior throughout their lives, and is heavily impacted by both environmental factors and heredity. The physical basis of cognition can be traced from the approximately 100 billion neurons found in the human brain. Each of these neurons has a significant number of connections with other neurons (Herculano-Houzel 2009). Therefore, cognition – which is directed by distinct brain circuits supported by a variety of neurotransmitters – is the primary regulator of human ideas and actions. Chemical messengers called neurotransmitters to pass messages from one neuron to another. Common neurotransmitters are dopamine, noradrenaline (norepinephrine), serotonin, acetylcholine, glutamate, and GABA (Jones 1986). Dopamine is a neurotransmitter that is produced by the brain's reward system. Hence, it is crucial to investigate cognition and the

underlying neurobiology that supports behaviors in both healthy and illness situations in order to better understand what drives them. Processes occurring inside various sections of our brain give rise to different cognitive functions, but only some of them make it into our conscious consciousness. Human cognitive processes are constantly improving during childhood, adolescence, and adulthood. However, as one ages and enters older maturity, some of these abilities begin to decrease as neurons die and the mechanisms that replace them become inadequate (Arain et al. 2013). Understanding cognition is vital not just for healthy cognitive development but also for the treatment of a variety of neuropsychological illnesses in which cognitive deficiencies emerge. Many of the world's most pressing health problems are linked to underlying cognitive deficiencies, which can be easily alleviated through early intervention. The real-time monitoring and assessment of cognition have the potential to alter human lives and prevent cognitive capacity from deteriorating at a quicker pace than it should as people naturally age.

Measuring Cognition

With suitably validated screening tests for cognitive impairment, it is simple to acquire an early estimation of a patient's cognitive functions by comparing the patient's score to reference standards (Andrianopoulos et al. 2017). This allows for a more accurate diagnosis and treatment plan. The three main methods for assessing daily cognitive function are self-report, informant report, and performance-based screening tests. Self-report has been demonstrated to have certain limits in cognitively impaired individuals as self-reports frequently deteriorate in these individuals (DeBettignies et al. 1990; Farias et al. 2008). In contrast, performance-based measures the cognitive functioning of patients using well-defined functional tasks, which are then compared with an appropriate comparison group (Gevins et al. 2011). These functional tasks are mostly used in clinical neuropsychological assessments to pinpoint the cognitive strengths and limitations of patients across a wide range of cognitive areas. A key part of performance-based tests is the cognitive test battery which was created to evaluate key cognitive abilities (Rapp et al. 2012). According to several researchers, this is the most valid and trustworthy approach to evaluating functional ability (Avitan et al. 2009; Giovannetti et al. 2006). The bulk of performance-based evaluations take place under controlled circumstances when the subject is encouraged to finish a job and is provided with the necessary tools. Although neuropsychological assessments like performance-based tests are the benchmarks for clinical evaluation of cognitive ability, the efficacy of these tests is highly subjective (Gevins et al. 2011). Multiple factors may affect the cognitive performance of the patient such as fatigue, sleepiness, affect, and mood, all of which may alter the results of the test (Gevins et al. 2011). However, in recent times, researchers have started using neuroimaging tools like positron emission tomography (PET), functional MRI (fMRI), electroencephalography (EEG), or functional near-infrared

spectroscopy (fNIRS) during cognitive testing to extract pertinent information regarding the cognitive performance as well as brain function during these tests (Herold et al. 2017; Smeha et al. 2022). When the various types of data available from direct brain function tests and neuropsychological testing are merged into a single index, it has the potential to deliver significant benefits for both clinicians and researchers (Gevins et al. 2011; Dahal et al. 2011). EEG and fNIRS are simple, portable, and cost-effective technologies that stand out as potential methods for assessing neural function in a variety of operating contexts, particularly in the field.

Electroencephalography (EEG) is a technique for recording the electrical activity of the brain caused by the firing of neurons within a brain. In order to perform it, numerous electrodes are placed on the scalp (Congedo and Sherlin 2011). EEG records the changes in electric waves and serves as an all-encompassing reflection of the electrical activity of nerve cells on the scalp or cerebral cortex surface (Klonowski 2009). It is a common practice to record EEG during certain moments of cognitive processing and then compares the results to those obtained by recording EEG during a resting or baseline time. The use of EEG has the benefit that cognitive events can last for a lengthy period of time, in the range of seconds or even minutes. Additionally, EEG has a high temporal resolution as it records electrophysiological responses at a millisecond scale. When different stimuli are presented to the brain, distinct parameters may be retrieved from the EEG and utilized as a metric in the study of cognition (Bell and Cuevas 2012).

Functional near-infrared spectroscopy (fNIRS) is a noninvasive neuroimaging method that is still in its infancy and is used to track cerebral blood flow (Farias et al. 2008). Compared to EEG, it has a higher spatial resolution, as well as its mobility, movement tolerance, and safety of usage; therefore, this technology is particularly well suited for researching brain activity in newborns and children, among other applications (Pinti et al. 2020). In the field of schizophrenia, fNIRS is being used extensively in research, with a focus on functional abnormalities in the prefrontal cortex (PFC) during verbal fluency tasks (Sun et al. 2020). With the use of fNIRS, it is possible to compute the temporal variations in cerebral blood flow by monitoring changes in the light absorption of hemoglobin. fNIRS is a technique that, in its most basic form, monitors variations in the concentration of oxygenated and deoxygenated hemoglobin in brain tissue by beaming near-infrared (NIR) light (650–950 nm) into the head (Pinti et al. 2020). Because of its low sensitivity to body motions and mobility, fNIRS is particularly well suited for monitoring cerebral hemodynamic responses during motor activities or walking tasks, among other applications. Because EEG signals are very susceptible to movement, functional near-infrared spectroscopy (fNIRS) offers an alternate method for monitoring everyday cognition in a noninvasive way (Liao et al. 2017).

In what follows, we first explain the functioning of the EEG and how it is used to measure cognition, as well as future challenges. Next, we elaborate on neurophysiological data acquisition using fNIRS and how it helps in understanding cognition. Lastly, we broach the topic of simultaneous EEG and fNIRS data collection and its impact on measuring cognition.

Measuring Cognition Using EEG: How EEG Works

How EEG Measures Cognition

Measuring cognition requires explaining cognitive processes and behaviors with regard to the functioning of the human brain and the underlying processes of the nervous system (Purves et al. 2008). Behavioral responses to cognitive activities are measured in order to get an understanding of the underlying processes when the nervous system creates appropriate responses to stimuli. When it comes to investigating cognition, neuroscientific methodologies may be classified into two major categories. In the first category, disruptions in brain function caused by degenerative changes or trauma have been utilized to investigate the role of the brain in certain cognitive functions, while in the second category measuring brain activity while a subject is engaged in a specific cognitive task has been utilized to understand the neural underpinnings of cognition. Brain network exploration and understanding are accomplished through the use of electroencephalography (EEG). Scientists have used EEG to map the brain areas that are active during a psychological process, in what is known as the hunt for the neural correlates of cognition (Purves et al. 2008). For example, researchers discovered that neurons in the prefrontal cortex increase their firing rate when information is retained in memory; this new understanding encouraged further investigation into the involvement of that brain area in a variety of executive functions. EEG was initially designed for the purpose of assessing individuals with epilepsy, and it is now utilized for a broad range of purposes, including exploring functional connectivity and functional brain networks, which allows for the evaluation of cognitive processes (Šneidere et al. 2020). Ongoing EEG signals measured over time are widely used in clinical settings to assess various aspects of brain function (Light et al. 2010). Mostly, EEG signals are segregated into multiple frequency bands and analyzed by looking into their respective band powers (Purves et al. 2008). It is possible to utilize the relative power in these bands, in conjunction with other features of the EEG, to assess the general condition of the brain, such as the degree of alertness and sleep phases, as well as to detect and monitor aberrant activity associated with epilepsy.

How Neurons Communicate

The human brain is made up of millions of neurons, each playing a significant part in managing the behavior of the human body in reaction to internal/external and motor/sensory impulses, among other things (Arai et al. 2019). These neurons will serve as information transfer agents between the human body and the brain (Kumar and Bhuvaneswari 2012). Understanding the cognitive behavior of the brain may be accomplished by the analysis of signals generated by the brain. It has been estimated that the typical human brain has around 86 billion neurons (Herculano-Houzel 2009) and that communication between neurons is the key to brain function. A cell body, dendrites (treelike structure), and a number of axons ending in synapses primarily

constitute a neuron. When neurons receive messages in the form of neurotransmitters (dopamine, epinephrine, acetylcholine), the receptors on their dendrites fire, resulting in electrical changes in the neuron, which is referred to as an action potential (Hodgkin and Huxley 1952). Membrane potential is defined as the differential in electrical charge between the interior and the outside of a cell membrane (Chrysafides et al. 2021). It is measured in microvolts. The formation of ion clusters on the outside and interior of the membranes results in a differential in electrical charge between the two membranes. The different ions that are present in the cell membranes are the positively charged sodium and potassium ions and the negatively charged chloride and organic anions. When the neuron is at rest, the sodium and potassium ions are prevalent outside the cell membrane, while the potassium and organic anions are prevalent inside the cell membrane. The average resting membrane potential for a neuron is approximately −70 millivolts. This is because the inside of the neuron has a greater negative charge than the exterior, which causes it to be more negatively charged. When the dendrites of a neuron receive neurotransmitters, it can lead to depolarization, i.e., causing it to move toward 0 millivolts. When neurotransmitters interacting with dendrite receptors cause repeated depolarization of the neuron, it forces the membrane potential to reach the threshold potential (−55 millivolts). When the membrane potential reaches the threshold, a large number of positively charged sodium ions are released into the inside of the cell leading to massive depolarization of the cell which results in the membrane potential being positively charged. This influx of positive ions creates the electrical signal known as action potential which then travels down the neuron (Chrysafides et al. 2021). Eventually, the action potential reaches its peak, which results in the closing of the sodium channels, and no more sodium ions are released inside the cell. In the same period, positive potassium ions are released outside the cell leading to repolarization. As a result, the neuron starts returning to the resting membrane potential; however, it overshoots it and becomes hyperpolarized. In the hyperpolarization stage (or refractory period, where the neurons try to return to resting membrane potential), it is very hard for the neuron to fire again. If the action potential is strong (excitatory), it is then sent to the axon ending at the synapses, resulting in the release of the neurotransmitters (Chrysafides et al. 2021). The axon consists of a myelin sheath that helps in protecting the signal from degradation during transmission. The neuron where the signal originates is called a presynaptic neuron. The neuron where the signal is received is called a postsynaptic neuron. Neurotransmitters interact with the receptors of the postsynaptic membrane resulting in an excitatory neuron or an inhibitory neuron. Excitatory neurons release neurotransmitters that fire an action potential in the postsynaptic neuron, whereas inhibitory neurons release neurotransmitters that inhibit the firing of an action potential.

The Source of EEG

An EEG signal is generated by a population of cortical neurons whose synaptic activity is synchronized. Because of the excitation of postsynaptic neurons, there is a

bigger negative extracellular voltage at the neural dendrites than there is everywhere else along the neuron; this leads to the formation of a dipole (Avitan et al. 2009). A dipole is an area separating a region of positive charge and a region of negative charge (Jackson and Bolger 2014). The electrodes in an EEG device detect the total number of positive and negative charges in their immediate environment. Electrodes can only detect dipoles if they are closer to either the positive or the negative end of the dipole. Since an electrode will measure neutral if it is equidistant from both the source and the sink of a dipole, this means that dipoles can only be detected by electrodes that are closer to either the positive or the negative end of the dipole (Jackson and Bolger 2014). This means that dipoles that are perpendicular to the surface (tangential dipoles) and parallel to the surface (radial dipoles) can both be measured in EEG (Jackson and Bolger 2014; Whittingstall et al. 2003). Because dipoles have a positive and negative side, they will create positive and negative deflections in various areas of the scalp. The dipole of a single neuron is very small, making it hard to be measured from the scalp. The dipoles from several neurons in a location will total up since electrodes sense the sum of charges in their surroundings. The amplitude of a single dipole is quantifiable as the sum of many individual dipoles in a region (Kandel et al. 2000). However, because electrodes in the brain detect the cumulative sum of both positive and negative ends of dipoles, neurons need to be active in both a parallel a and synchronous manner for there to be a quantifiable (nonzero) signal (Jackson and Bolger 2014). Therefore, if the neurons are all arranged in the same direction, their impulses can accumulate to make a bigger signal, and the parallel arrangement is required to produce a quantifiable dipole. In any other configuration, the positive and negative ends of the separate dipoles will accumulate and cancel each other out.

The transmission of the EEG signal across the brain is attained through volume conduction. The electrical signal generated by a collection of neurons (or neuronal activity) travels from the brain, through the dura layers, through the skull layers, through the scalp, and ultimately to the electrode in order to be measured with an EEG electrode placed outside the head on the scalp (Jackson and Bolger 2014). When the signal that is being carried by volume conduction reaches the boundary of the volume it is moving through, volume conduction can no longer take place because ions are unable to exit the volume. The signal's ability to travel from one volume to the next is due to a sort of capacitance that exists between the volumes. In its most fundamental form, a capacitor consists of two charge reservoirs that are physically separated by an insulating layer (a dielectric). If there were no insulating barrier, then the charges would be able to freely mix, and the end result would be a pool that was neutrally charged. However, the presence of the insulating layer prevents the ions from mixing. On the other hand, if there is an insulating layer between the membrane and the medium that surrounds it, there is a possibility that a charge difference will develop between the membrane and the medium that surrounds it. This will result in positive ions accumulating on one side of the membrane and negative ions pushing up on the other side of the membrane (Baars and Gage 2010; Jackson and Bolger 2014). The amount of charge that it accumulates on the other side is determined by the properties of the insulating materials (Keil et al.

2014). An array of layers starting from the brain to the dura layers forms a series of conductive volumes analogous to a stack of capacitors. The extracellular positivity will propagate via volume conduction to the near side of the closest insulating layer, at which point volume conduction cannot carry the signal further because the insulating layer is a physical barrier to ion flow. The positive ions that have collected on the near side of the insulating layer will resist the positive charges that have accumulated on the other side of the layer. Positive charges that are repelled and amass on the far side of the insulating layer will, in turn, repel and push other positive charges away from the insulating layer. This procedure will be repeated through each capacitor in the stack until it reaches the electrode at the end of the stack. It is thought that the capacitive arrangement described above accounts for the vast bulk of the EEG signal's content. The passage of electrons between the gel and the electrode (Usakli 2010) does result in the generation of half-cell (DC) potentials. EEG system users are instructed to allow the electrodes to "settle" for a few minutes before recording. This is done to allow the interaction between the electrode and the gel to reach a steady state, at which point the steady DC offset can be accounted for somewhere else in the system.

EEG Analysis

Event-Related Potentials (ERPs)

The continuous EEG signal is beneficial for analyzing the general condition of the brain; however, its value in researching individual cognitive functions is restricted due to the fact that it is not tied in time to any specific cognitive processes or events (Purves et al. 2008). It is more effective to use a time-locked average method such as event-related potentials (ERPs) to map scalp electrical activity to cognitive performance than to use a single point in time. Event-related potentials (ERPs) are very small voltages generated in the brain structures in response to specific events or stimuli (Sur and Sinha 2009). They reflect the summed electrical activity of neuronal populations specifically responding to these events. As such, they can provide high temporal resolution of neural processing underlying various cognitive functions (Blackwood and Muir 1990). ERPs are often smaller in size than the raw EEG signals in which they are contained; yet, in order to separate ERP signals from background noise, several trials need to be averaged over a period of time (Sur and Sinha 2009). These time-locked epochs of the EEG signal are averaged throughout the many occurrences of a particular cognitive event in order to extract ERPs. ERPs are then analyzed (Blackwood and Muir 1990). These random fluctuations in the EEG tend to average away, leaving only those voltage changes directly related to the processing of the event of interest. The amplitude of the EEG varies more or less randomly in relation to the timings of these events. The average ERP trace that can be acquired in this manner normally consists of a number of peaks, both negative and positive, that are labeled according to their electrical polarity (N or P for negative or positive) and their latency. These peaks are named according to their electrical polarity (Purves et al. 2008).

Recordings of ERPs are typically made when the participants being studied are engaged in a variety of cognitive tasks and are being exposed to a range of stimuli. The variability of ERP responses as a function of the type of stimulus and the task conditions is then utilized to infer something about the mechanisms behind the cognitive process that is under investigation (Purves et al. 2008). ERPS represents neural activity with a very high temporal precision since the signals arise from the electrical elements of neuronal activity. Because of this, ERPS is particularly effective for investigations in which the timing and sequencing of functional brain activity are very relevant. For instance, ERPs can indicate how early attention can exert an influence on the processing of sensory input, or they can indicate the time point in language processing at which the semantic analysis of a word begins. Both of these are examples of how early attention can have an effect on later processing.

Even though ERPs' primary advantage in the study of cognitive processes is their high temporal resolution, information may also be gleaned from the geographical distribution of the scalp recordings. This information can be used to get some insight into the potential origins of the activity, which can be a useful complement to ERPs' high temporal resolution. For instance, focused activation over the occipital scalp is likely to be originating from the visual cortex and, as a result, include visual processing (Sur and Sinha 2009). ERP recordings, because of their great temporal resolution, are able to provide a dynamic picture of the activity that occurs in the brain throughout time. When many electrodes are utilized, it is possible to estimate the locations of the neural generators producing scalp-recorded ERPs with the use of analysis algorithms (Rugg 2009).

EEG Spectral Bands

The frequency bands of the EEG waveform are commonly divided into five categories (alpha, beta, theta, delta, and gamma bands). In terms of the frequency range, delta oscillations have the smallest frequency range. Delta oscillations are neuronal oscillations arising from the thalamus or cortical region associated with global processing, mainly pertaining to evolutionary processes (Neske 2016). They span large cortical regions and have been hypothesized to be involved in integrating cerebral activities across diverse cortical sites to homeostatic processes (Knyazev 2012). Researchers have also found implications of delta oscillation in higher emotional involvement, behavioral inhibition, as well as synchronization of brain activity with autonomic functions (Neske 2016). When looking at ERP studies, it has been demonstrated that delta EEG power correlates favorably with the amplitude of the P300 component, which is often connected with activities in the dopamine reward system. Therefore, states associated with the necessity of meeting fundamental biological requirements can be related with greater delta activity, according to research. Theta oscillations (between 4 and 7 Hz) are frequently associated with memory and emotional control. According to a number of research, theta oscillations are important in the encoding of information during exploratory movements and spatial navigation. Theta band power variations have also been demonstrated to be connected with discriminating between pictures of positive and negative emotional valence, and people with low vs. high skin conductance responses exhibit different

power changes in theta band than those with high conductance responses. It appears that theta oscillations, like delta oscillations, are associated with the P300 ERP component and have a role in salience detection, demonstrating the strong relationship between motivational and affective processes. Theta predominates in the EEG of nonhuman animals, whereas delta predominates in the EEG of reptiles.

Alpha waves, also known as the alpha rhythm, are neural oscillations that occur in the frequency range of 8–12 Hz and are thought to be caused by the synchronous and coherent (in phase or constructive) electrical activity of thalamic cells in humans. According to prior research, the alpha frequency band is inversely connected to cortical activation, and more activity in the left prefrontal cortex is associated with increased approach desire (Harmon-Jones and Gable 2018). As a result, it has also been viewed as a predictor of emotionally valenced reactions that are positive in nature (Zhao et al. 2018). It can be inferred from the role of alpha oscillations in human EEG that the development of higher brain functions is heavily reliant on inhibitory mechanisms or other functions associated with alpha oscillations, such as working memory and mental representation of objects and events, which are all associated with alpha oscillations (Bazanova and Vernon 2014). It has been found that as youngsters grow older, the percentage and power of slow waves in their brains diminish but the power of alpha waves in their brains grows dramatically as they become older.

Beta oscillations (13–30 Hz) are frequently implicated in the sensory processing of the brain (Barone and Rossiter 2021). They are formed during stable postures and are reduced during active phases, such as movement planning and execution, to maintain a stable posture (Barone and Rossiter 2021). Just prior to and during movement execution, a decrease in the amplitude of beta oscillations throughout sensorimotor regions is observed. In contrast, a rise in beta amplitude above baseline values has been reported at the end of motor movement.

Finally, the building of object representation is related to the use of the gamma band (30–50 Hz or above). There are two main processes involved: the binding of distinct pieces of a single object through bottom-up processes and top-down processes that include the activation, retrieval, or rehearsal of an internal representation. When doing difficult and attention-demanding activities, the power of the gamma band rises, and this increase is generally considered to represent the neural substrate of cognitive processes (Tallon-Baudry and Bertrand 1999). It is well known that distinct aspects of things or events are encoded and processed in different areas of the brain, and it is possible that humans see coherent representations as a result of gamma oscillations. According to research, gamma activity is involved in the binding together of spatially diverse aspects in visual objects as well as the reflection of linkages between words and the meanings they imply. The integration of sensory and motor processes has also been linked to it in the context of human movement.

EEG Preprocessing

Among the most important steps in EEG data analysis is data preparation. This is important because data errors caused by noise in the data might lead to inconsistencies in the results. EEG data often contains a great deal of noise, which can make

it difficult to distinguish weaker EEG signals. Apart from that, people who take part in EEG experiments frequently blink and fidget, which results in the formation of ocular and muscle artifacts that might taint the data. When EEG data is preprocessed, it assists in the removal of undesired artifacts from the EEG signal and makes it appropriate for further processing. One of the most important aspects of EEG data processing is the detection/removal of defective channels because many analysis algorithms will break down if they are presented with egregiously bad signals (Bigdely-Shamlo et al. 2015). The following procedures are followed by the majority of EEG data processing pipelines: noise reduction, bad channel detection, bad channel removal, interpolation of the bad channels, artifact rejection, and referencing.

The following is a summary of EEG preprocessing:

1. Downsampling: It refers to the process of reducing the sampling rate of a signal. Downsampling is useful to compress the data size. It also helps independent component analysis (ICA) (Sun et al. 2005) to produce better decomposition by eliminating unnecessary high-frequency components.
2. Filtering:
   (a) Line noise removal: Usually, a notch filter at 60 Hz is used to eliminate line noise from EEG recordings. In practice, however, notch filters frequently have a notch width of 10 Hz or greater, resulting in severe signal distortion in the frequency range of 50–70 Hz. This distortion should not be a problem for analyses that apply a 40-Hz low-pass filter to the signal; however, it is problematic for studies that require high-frequency signals to be examined. Some investigations have also demonstrated that noncausal low-pass filtering can have a considerable impact on the beginning of ERPs (Rousselet 2012; Van Rullen 2011). As an additional point of reference, studies (Barnett and Seth 2011) have demonstrated that filtering, namely, low-pass filtering, can have a negative influence on Granger causality and other connection calculations.
   (b) High-pass filtering: The application of a high-pass filter has the goal of removing baseline drift, which is a type of noise in the signal. The use of lower cutoff frequencies, such as 0.01 Hz for cooperative participants and 0.1 Hz for youngsters and patients, is often advised (Luck 2005).
   (c) Low-pass filtering: Frequencies below a certain value are kept while others are discarded. This is usually done to focus on the frequencies that are relevant to the study.
3. Re-referencing: When recording EEG data, the voltage of each electrode is recorded in relation to the voltage of other electrodes in the same session. The "reference," which might be a single electrode or a set of electrodes, is the point at which the voltage will be measured in relation to Choi and Hwang (2019). This implies that any brain activity detected at the reference electrode will be mirrored in all of other electrodes, potentially contaminating the signal. Most EEG investigations employ the Common Average Reference for re-referencing (Bertrand et al. 1985). Using a reference electrode decreases

the influence that a single failing electrode will have on the outcome of the experiment. However, employing this reference only makes sense with systems that have enough channels to ensure that the total activity averages to zero on a consistent basis. Other frequent reference techniques include the use of Mastoids (electrodes that are implanted roughly behind a person's ears) and the central electrode Cz (which stands for center electrode). Mastoids are generally used as reference electrodes because the region around the mastoids do not elicit a great deal of neuronal activity since they are located relatively far away from the brain while still being adjacent to other electrodes. On the other hand Cz electrodes are mostly used as reference electrodes when the region of interest is away from the central region (Choi and Hwang 2019).

4. Detecting and interpolating bad channels relative to this reference: An important stage in EEG preprocessing is the detection of defective or outlier channels, which is one of the most difficult to do. To identify bad channels, four primary criteria are typically used. These criteria are as follows: extreme amplitudes, lack of correlation with any other channel, lack of predictability by other channels, and unusually high frequencies (Bigdely-Shamlo et al. 2015). Many of the criteria make use of a robust z score, with the mean being replaced by the median and the standard deviation by the robust standard deviation in some cases. In order to compensate for the problematic channels that have been identified, data for the bad channels are interpolated using data from the nearby channels. The interpolation of poor channels is generally accomplished through the use of spherical splines, in which the channel locations are projected onto a unit sphere and a matrix characterizing the connection between good and bad electrodes is produced and used. Finally, the result of the matrix is utilized to interpolate the data for the electrodes that did not perform well.
5. Artifact removal: Artifacts are signals that do not originate from the brain but are still picked up by the EEG electrodes. The sources of EEG artifacts are generated from multiple sources and can manifest in different ways. Most EEG artifacts can be generally classified into two categories: environmental and biological. Environmental artifacts are generated from the environment or through outside interference from where the day is being collected. This includes power line noise, improper contact of the EEG electrode with the scalp, and movement of individuals near the EEG system. On the other hand, environmental artifacts are easier to discern than their biological counterparts, as biological artifacts are produced by the body. The vast majority of biological artifacts are thought to have arisen from sources within the body. The act of blinking, eye movement, head movement, heartbeat, and muscular sounds are all examples of common biological artifacts. A potential source of artifacts is any brain signal that does not directly relate to the phenomenon that is the subject of the current investigation. The most common technique for artefact correction is ICA. ICA separates independent sources linearly mixed in multiple sensors. It has two basic assumptions: statistical independence of sources and non-Gaussian distribution (Sun et al. 2005).

Current Challenges in EEG Preprocessing

Although EEG has shown to be a valuable tool in a variety of fields, it still has substantial disadvantages that prevent it from being successfully used for analysis or data processing in most cases. To begin, the signal-to-noise ratio (SNR) of EEG is poor because the brain activity being monitored is typically buried beneath many sources of comparable or larger amplitude ambient, physiological, and activity-specific noise, generally referred to as "artifacts," that obscure the signal (Roy et al. 2019). Consequently, a range of filtering and noise reduction techniques must be used to limit the effect of various noise sources while still allowing actual brain activity to be recovered from the recordings. Aside from that, the EEG is a nonstationary signal, which means that its statistics fluctuate as time progresses. Therefore, when applied to data acquired from the same individual at a later time, a classifier trained on a tiny quantity of user data collected over a short period of time may perform poorly. This is a substantial challenge for real-world EEG applications, which typically need the utilization of limited amounts of data in order to be successful (Acharya et al. 2013). Finally, the significant intersubject variability of EEG applications lowers the usefulness of these applications in clinical settings. This phenomenon is caused by physiological differences across individuals, and while the magnitude of these differences varies, they can have a considerable influence on the performance of models that are designed to generalize over a large number of participants. The ability to generalize from one set of individuals into a second, previously unknown group is crucial to many practical applications of electroencephalography, and significant research effort is being directed toward developing systems that can cope with intersubject variability. In order to address some of the issues stated above, processing pipelines using domain-specific approaches are typically used in conjunction with domain-specific techniques. Significant research has been conducted on EEG data in order to construct processing pipelines that are capable of cleaning, extracting valuable properties, and categorizing the data. A few sophisticated techniques, such as Riemannian geometry-based classification and adaptive categorization, are able to successfully resolve these challenges in some circumstances, but they fail horribly in others (Roy et al. 2019). Another advantage of greater levels of automation is that it would be useful for a greater variety of various vocations. Incorporating more advanced automated EEG processing into this approach might result in a far quicker and more flexible procedure. Similar to this, real-time detection or prediction of the onset of an epileptic seizure would be tremendously beneficial to epileptic patients, but it would involve the use of automated EEG processing to accomplish this. Currently available EEG-based technologies are less flexible and generalizable than they were in the past, and the most popular implementations of each of these applications require the usage of domain-specific processing pipelines, which are not available in the past (Biasiucci et al. 2019).

EG Feature Extraction

After the EEG data has been preprocessed, the next stage is EEG processing utilizing feature extraction. Multiple features can be extracted from EEG data. The most commonly extracted features are statistical, time domain based, frequency domain based, time-frequency domain based, and spatiotemporal features, with time domain

based features being the most basic. Statistical features include mean, median, variance, standard deviation, skewness, kurtosis, and energy of the EEG signal (Jiang et al. 2019). Skewness shows the symmetry of the probability density function (pdf) of the amplitude of a time series, kurtosis indicates the peakedness of the probability density function (pdf) of a time series, and energy (E) is the sum of the squares of the amplitude of a time series, respectively. Statistical features may be approximated across participants and across various electrodes, as well as across different electrode configurations, but it does not look into the interaction between electrodes. Other statistical features include wave duration, peak amplitude, and instantaneous frequency (IF) (Stancin et al. 2021) which are extracted using period-amplitude analysis. Statistical features such as the Hjorth parameters (Vourkas et al. 2000) can also be used to derive additional features. Hjorth parameter is based on the variance of the derivatives of the EEG signal. The most commonly used Hjorth parameters are mobility, activity, and complexity, which are the first three derivatives of the EEG signal.

Frequency is the number of occurrences of a repeating event per unit of time. Since EEG is a nonstationary signal, it contains events that occur at different frequencies. The term frequency domain analysis refers to the process of representing a signal by its frequency component and estimating all of its related properties in terms of frequency. Frequency domain features are extracted widely using power spectral density (PSD) (Dempster 2001). It can be calculated with several parametric and nonparametric methods. Nonparametric approaches, such as the Fourier transform (which is typically performed using the fast Fourier transform algorithm (FFT)) (Rampil 1984), Welch's method (Zhao and He 2013), and Thompson multitaper method (Karnik et al. 2021), are more frequently employed than parametric methods. Autoregressive (AR) models, multivariate auto-regressive models, and autoregressive-moving average (ARMA) models are all examples of parametric approaches for the estimation of the PSD (Stancin et al. 2021). Furthermore, relative and absolute powers of the five frequency bands, delta, theta, alpha, beta, and gamma, are some of the most commonly used features in the frequency domain. Additionally, these features can be further extended by the calculating ration between frequency bands. Measurements of band ratios are often viewed as quantitative measures of periodic, or oscillatory, activity in nature (Donoghue et al. 2020). This is predicated on the assumption that the measure represents the relative strengths of separate periodic components that are properly represented by predetermined frequency ranges. It is feasible to obtain features such as reflection coefficients and partial correlation coefficients from autoregressive models when they are used in conjunction with other models. Another method for extracting features is to use wavelet coefficients that have been obtained after conducting wavelet decomposition. It is a usual practice to employ the PSD approach in order to calculate the second-order statistics of an EEG signal. Since EEG is a nonstationary signal, the study of the EEG signals in both temporal and frequency domains at the same time is a very effective technique. Spectrogram (Ng et al. 2019) is a common time-frequency processing technique that is calculated using short-time Fourier transform (STFT) (Ng et al. 2019). The STFT approach

maps the signal into a two-dimensional function of frequency and time, which is then analyzed.

Entropy (Kumar et al. 2012) is another feature that can be extracted from an EEG signal; however, the distinctiveness of entropy stems from the fact that it can be derived both in the time domain and in the frequency domain, which makes it very useful. When applied to a time series signal such as EEG, the idea of entropy is used to measure, in a statistical sense, the degree of uncertainty or unpredictability in the pattern, which is also roughly comparable to the amount of information contained in the signal. In the time domain, entropy measurements often divide up the signal into segments that are then compared for similarity, either immediately (in the time domain) or after some type of processing of the signal has been applied (such as the power spectral density). Most of the time, this is determined by a few fundamental characteristics, such as the segment length chosen, whether or not the signal has been transformed, and the distance metric, or the method by which the segments are compared. While in frequency domain entropy measures, the EEG signal is first transformed into the frequency domain using various methods such as the Fourier transform or more advanced methods such as wavelets and then examined. The most common forms of entropy-based features in the time domain are sample entropy (Kumar et al. 2012) and approximate entropy (Kumar et al. 2012) while in the frequency domain are spectral entropy and wavelet entropy (Liang et al. 2015). Approximate entropy (ApEn) is a measure that quantifies the complexity or irregularity of a time series. A higher value of ApEn indicates that the signal has greater complexity and irregularity. In order to improve on approximation entropy, the sample entropy (SampEn) concept was proposed. To reduce the inaccuracy of the approximation entropy, it is necessary to eliminate two of its drawbacks: (1) self-matches and (2) a dependence on the length of the time series under consideration. Additionally, sample entropy may be used as an estimate of signal complexity. The spectral entropy (SEN) of an EEG signal is obtained by first translating it into the frequency domain and then calculating the entropy on the normalized PSD of the EEG signal. With the exception of the fact that it is derived using the coefficients of the wavelet decomposition of the provided time series, wavelet entropy (WE) is very similar to spectral entropy in that it is calculated based on the coefficients of the wavelet decomposition of the given time series (Stancin et al. 2021).

It is important to note that all of the features described above are computed using a single EEG channel. In contrast, spatial-temporal characteristics explain the interaction between distinct channels, providing more insight into the understanding of brain functions. Spatial-temporal features can be divided into two categories: directed and nondirected (Stancin et al. 2021). The nondirected spatiotemporal features evaluate the synchronization of two or more channels without any knowledge of the direction, whereas the directed spatiotemporal features evaluate the causation between two or more channels, i.e., they assess functional connectivity between two or more channels (Jiang et al. 2017). Nondirected spatiotemporal features include coherence, partial coherence, phase coherence, phase-locking value, imaginary component of coherency, phase lag index, and mutual information (MI). Coherence (Srinivasan et al. 2007) is a mathematical measure of the degree to

which two EEG signals collected at different sensors are comparable to one another. It is a metric that has been routinely used to infer synchronization across distinct locations at the sensor level. The value of coherence varies from 0 to 1. If the phase (rise and fall) of the two signals becomes synchronized with time, it indicates that there is a functional connection between these two parts, i.e., the areas might be communicating with each other. Partial coherence (Liberati et al. 1997) is an adjusted coherence in which the linear influence of the common signal has been eliminated. It is calculated using a third channel that is not physically adjacent to the two observed channels. Phase coherence (Thatcher 2012) is a phenomenon that occurs when there is a continuous phase difference between any two signals or waves that have the same frequency as one another. It was created in order to solve the challenge of nonlinear relationships between the two channels that had previously existed (Stancin et al. 2021). The phase-locking value (PLV) (Schmidt et al. 2014) is the absolute value of the mean phase difference between two signals represented as a complex unit length vector. It is used to determine if two signals are locked together. Similarly, if the marginal distributions of the two signals are uniform and the signals are not reliant on one another, then the relative phase will have a uniform distribution as well, and the PLV will be zero . However, if there is a significant degree of correlation between the two signals' phases, the PLV will approach unity in this situation. The most significant shortcoming of coherence is that it is significantly influenced by volume conduction conditions. The imaginary part of coherency (ICOH), suggested by Nolte et al. (2004), is aimed at removing any forms of superfluous coherence that arise as a result of instantaneous action. Lastly, the phase lag index (PLI) (Stam et al. 2007) is a measure of the asymmetry of the distribution of phase differences between two signals. Since it does not take into account the influence of amplitude information, it is an improvement compared to the imaginary component of coherency. Lastly, mutual information (Ramanand et al. 2010) measures the quantity of knowledge received about one time series by viewing the other time series is measured in terms of the number of observations of the other time series. It is a commonly used metric in information theory, and it is computed by starting with Shannon's entropy as a starting point and working outward from there.

The most commonly used nondirected spatiotemporal feature is Granger causality (Seth et al. 2015). Granger causality is an estimation of causality in time series that depends on linear methods. The fundamental concept of Granger causality is that causes both precede and aid in the prediction of their effects. A linear vector autoregressive model of the two time series is used to construct the model. VAR models are basic mathematical models in which the value of a variable at a specific time is described as a (linear) weighted sum of the value of the variable in the past (typically across a number of discrete time steps) and the value of a set of additional variables in the past. Each variable is a vector stochastic process encoding a time series, and each process is represented by a different variable. The process of fitting a VAR model consists of determining the appropriate weights to use in order to reduce estimate errors. According to the theory of Granger causality, one variable X "causes" another variable Y if the history of variable X contains information that helps predict the future of variable Y, in addition to the information that is already

present in the history of variable Y itself (and in the history of other "conditioning" variables Z). When this condition is met, one may say that there is an "information flow" from X to Y (Barnett and Seth 2011).

Complex networks are a graph theory-based approach to EEG feature extraction (Stancin et al. 2021). In complex network analysis, pairwise connections between channels are used to create a connectivity matrix which in turn simulates a graph. These feature extraction techniques look into the interaction of two or more channels by looking into the pairwise interaction between channels. Once the graph is obtained, all graph properties such as degree, mean degree, degree correlation, clustering coefficient, small worldnesss, closeness centrality, eigenvector centrality, betweenness centrality, and diameter can be used as EEG features (Ismail and Karwowski 2020). Degree (D) (the number of edges connected to the vertex) and the mean degree of the network can be used as a feature. The degree correlation coefficient (R) is the correlation coefficient between the degrees of two pairs. The clustering coefficient (Masuda et al. 2018) is a measure of the connectedness of vertices in a network, and it can be either local (for a subgraph) or global (for the entire graph). The presence of a local clustering coefficient of one indicates that the relevant local subgraph in a network is completely linked to all of its neighbors in the graph. This is a second-order graph statistic that is calculated by weighing the trade-off between strong local clustering and low path length. For each vertex in the network, there is a centrality degree associated with it, which indicates the number of shortest routes in the graph that the observed vertex is participating in at any one time. Additionally, each vertex in the network has a measure of proximity centrality, which is the average distance between the observed vertex and all other vertices in the graph, as seen in the graph below. The degree to which a vertex is easily accessible to other vertices is measured by its eigenvalue centrality. In this case, it is calculated using the relative vertex scores, with the primary premise being that high-scoring connections should contribute more to vertex influence than vertices with low scores. Within a network, betweenness centrality is a measure of the importance of each vertex's position. It is calculated based on the number of times a vertex appears along the shortest path between two other vertices in a certain time interval. Finally, the diameter (d) of a graph is the longest shortest path across it (Ismail and Karwowski 2020).

Application of EEG

EEG's noninvasive and portable qualities, combined with its ability to measure cognition with temporal resolution on a millisecond scale, have led to a diverse range of uses for the technology. Traditionally, EEG has been utilized in the diagnosis of mental conditions, such as neurological examination of patients with traumatic brain injuries, neurotherapy, and applications involving brain-machine interfaces. However, recent studies have demonstrated its multifaceted qualities in a variety of domains, including gaming; neurofeedback for the management of pain, trauma, and overall well-being; the management of cannabis use; mindfulness; and

training for cognitive enhancement. In the last 20 years, EEG devices have decreased in size and cost because of commercial multichannel analogue front-end integrated circuits, which have lowered the expertise, time, and cost needed to create a new EEG device. Different headsets with new types of electrodes and improvements in analogue and digital electrical design have simplified the EEG setup, making it more accessible for clinical and experimental uses. In this section, different applications of EEG for measuring cognition are discussed.

Measuring Cognitive Load Using EEG

The cognitive load of an individual can be defined as the amount of strain that a task places on that person's working memory. Because the cognitive load is directly related to the amount of time spent on a task, assessing the impact of instructional interventions requires taking both the average cognitive load and the accumulated cognitive load into account. It is generally accepted that an individual's sense of mental exertion can be inferred from the overall load they have experienced over the course of the task. The majority of psychological research studies use subjective rating scales to assess cognitive load (Paas et al. 2010; Paas 1992; Paas and Van Merriënboer 1994). Examples of these rating scales include the NASA Task Load Index (NASA-TLX; Hart and Staveland 1988) and the 9 point symmetrical category mental effort rating scale established by Van Gog and Paas (2008). However, subjective rating scales do not provide any insight into changes in the instantaneous load over time, unless they are applied multiple times within a task, which is something that has been done with activities that have varied durations. However, due to the relatively large amount of time that passes between each presentation of the rating scale, it is not clear whether subjective methods can provide a continuous measurement of variations in cognitive load while an individual is working on a task. This limitation can be alleviated by using objective measures. Objective measures for measuring cognitive load include measuring neurophysiological responses during secondary task methods using EEG, PET, and fMRI. In secondary task methods, the amount of load imposed by the primary (learning) task is measured by the performance or response time on a secondary task. In other words, the amount of load is measured by the length of time it takes to complete the secondary work.

The most common method for assessing cognitive workload using EEG data is the percentage decrease (event-related desynchronization (ERD)) or rise (event-related synchronization (ERS)) in band power during the activation interval relative to (task period) to the baseline interval (rest period). The ERD/ERS index is used to measure cognitive load:

$$\frac{\text{ERD}}{\text{ERS}}\% = \frac{\text{Baseline Int Band Power} - \text{Task Int Band Power}}{\text{Baseline Int Band Power}} \times 100 \qquad (1)$$

Calculating the theta-alpha ratio, which in recent years has come to be seen as a potentially relevant signal in the field of research pertaining to the examination of cognitive capacities, is another approach that has been used to measure cognitive

load. Because it is presumed that the theta-alpha ratio varies with age as well as cognitive capacities, this indicator can be utilized not only to quantify cognitive load but also to identify cognitive deficiencies in those who are elderly. In a manner similar to that of the earlier research, Gevins et al. (1998) and Smith et al. (2002) employed EEG indices to evaluate different stages of working memory. The brain activations that occurred during pattern recognition were examined. When they increased the mental load associated with the memory task, they found that an increase in frontal theta frequency activity occurred alongside a drop in alpha frequency activity. This was made abundantly clear by the findings of the study. It has been discovered that the levels of theta and alpha channels measured by EEG in the frontal and parietal regions of the brain correlate with the amount of mental or cognitive load. The majority of research has shown that frontal theta increases in response to increased cognitive load, whereas alpha decreases in response to increased cognitive burden.

Analyzing Mild Cognitive Impairment Using EEG

The term mild cognitive impairment (MCI) refers to a syndrome in which a mild cognitive loss occurs which may result in functional impairment in the patient (Baker et al. 2008). It is interesting to note that a high percentage of people with MCI go on to develop dementia and Alzheimer's disease (AD) as a result. Previous research has hinted that certain signal characteristics connected with the electroencephalogram (EEG) might be relevant in understanding AD. This is due to the fact that various EEG abnormalities have been recorded in the community of people with Alzheimer's disease (AD). For instance, there have been reports that the EEG patterns of people with Alzheimer's disease often exhibit a shift toward increased power in the lower EEG frequencies and diminished coherence of fast EEG rhythms. Researchers have demonstrated that unusual EEG patterns are present in Alzheimer's disease (AD) patients. These patterns are characterized by a decrease in alpha and beta power, as well as a corresponding increase in delta and theta power when compared to normal age-matched controls (Coben et al. 1983; Gianotti et al. 2007; Mattia et al. 2003). Many researchers also suggest that the earliest alterations in the EEG occur in the beta and theta bands, while changes in the alpha and delta bandwidths show later in the time course of the disease.

Resting-state EEGs have been used in clinical research with AD patients. A considerable body of work based on single-session resting-state EEG has identified differences between AD and controls in EEG parameters such as spectral powers, complexity, and connectedness (Trinh et al. 2021). AD patients had lower signal complexity (higher power of slow oscillations (delta, theta) and lower power of fast oscillations (alpha, beta, gamma) over temporal, parietal, and occipital scalp regions (Huang et al. 2000; Roh et al. 2011; Rossini et al. 2007). Moreover, entropy-based EEG complexity decreases with illness progression (Sun et al. 2020). Few studies indicated a nonsignificant reduction of complexity in MCI compared to controls (Dauwels et al. 2011; Labate et al. 2013; Roh et al. 2011; Xu and Tao 2013). However, finding a differentiating EEG pattern based on resting-state recordings to classify MCI and HC is difficult.

In clinical studies with Alzheimer's disease patients, resting-state EEGs have been utilized. Differences in EEG characteristics such as spectral powers, complexity, and connectivity have been found between people with Alzheimer's disease (AD) and controls, according to a substantial body of research based on single-session resting-state EEG (Trinh et al. 2021). Patients with Alzheimer's disease had lower signal complexity (Abásolo et al. 2005, 2006, 2008; Liu et al. 2016) and higher power of slow oscillations (delta, theta) and lower power of fast oscillations (alpha, beta, gamma) over temporal, parietal, and occipital scalp regions (Huang et al. 2000; Ishii et al. 2017; Roh et al. 2011; Rossini et al. 2007). In addition, the complexity of the EEG as measured by entropy tends to diminish as the illness progresses (Sun et al. 2020). Comparing patients with MCI to controls, only a few studies found a decline in complexity that was not significant (Trinh et al. 2021). Finding a distinguishing EEG pattern that can differentiate MCI and HC based on recordings taken when the subject is in a resting state can be challenging.

Memory loss can also induce important EEG rhythms to distinguish AD/MCI from controls. Working memory impairment is a common symptom of MCI and AD (American Psychiatric Association 2013). Working memory allows one to access and manipulate the information quickly (Baddeley 2015). Spatial orientation, problem-solving, and reading require working memory (Kirova et al. 2015). MCI affects verbal/visual working memory, sentence span, operation span, digit span, letter-number sequencing, and arithmetic operation (Kessels et al. 2011). Working memory activities are an excellent choice for inducing task-relevant changes in MCI and controls in resting-state EEGs.

Measuring Cognition Using Functional Near-Infrared Spectroscopy (fNIRS)

How fNIRS Measures Cognition

Functional near-infrared spectroscopy (fNIRS) is an optical noninvasive imaging technique for the brain that is similar to electroencephalography (EEG) in that it uses near-infrared and visible (red) light to determine the amount of oxygen present in organic tissue and blood (Karim et al. 2012). Based on the hypothesis that neural activation and vascular response are tightly coupled, this technique is an effective optical neuroimaging method that monitors changes in the optical absorption of oxygenated and deoxygenated forms of hemoglobin in tissue (hemodynamic response) in response to brain activation (Chen et al. 2020). A lineal link has been demonstrated between neural activity and hemodynamic responses, which demonstrates that variations in hemodynamic response are an appropriate marker for the evaluation of neuronal activity (Arthurs and Boniface 2003; Logothetis 2001). fNIRS offers functional contrast comparable to the functional MRI blood oxygen level-dependent (BOLD) signal, representing changes in regional blood flow to regions of the brain engaged in processing

functional activities. fNIRS evaluates brain functions by measuring the mobilization of distinct chromophores (oxygenated hemoglobin, deoxygenated hemoglobin, and cytochrome c-oxidase) and the timing of these mobilizations with events. However, because of methodological and theoretical issues connected with cytochrome c-oxidase (Cyt-Ox) functioning, fNIRS is mostly used to measure and analyze oxyHb and deoxyHb mobilizations in order to examine brain processes in humans (León-Carrión and León-Domínguez 2012). A total of two sets of flexible fiber-optic cables are employed during fNIRS recordings: a set of source or transmitter optodes and the second set of reception optodes. The source fiber-optic optodes are used to distribute near-infrared light (0.4 W/cm^2) to a number of different locations on the head using a fiber-optic network. Each location has two (or more) different hues of light, which are utilized to distinguish between the absorption changes of oxygenated and deoxygenated blood in the same place. They are directly tied to the cerebral blood flow (CBF) associated with an event as well as the physiological reactions triggered by the brain's functioning state (fNIRS measures this reaction in the cerebral cortex) (León-Carrión and León-Domínguez 2012). fNIRS head cap source optodes emit near-infrared light, which enters the tissue and diffuses throughout the tissue after entering through the source optodes. This light will be able to penetrate the cortex of the brain down to about the outer 5–8 mm of the cortex based on earlier modeling research. When the light departs the head, it is captured by an array of fiber optics, which then transports the light back to photon detectors on the fNIRS device (Dahlstrom-Hakki et al. 2019). The quantity of light that travels from the source optode to the receiver optode is proportional to the amount of absorption that occurs in the underlying tissue between that pair of measuring devices. Biological tissue has a high scattering coefficient. Before scattering, light in the wavelength range of 650–900 nm passes into brain tissue for roughly one-tenth of a millimeter (Rupawala et al. 2018). Evoked brain activity causes localized changes in blood flow in the active region, which in turn causes differential changes in the concentrations of oxyhemoglobin and deoxyhemoglobin in the brain. Because the two forms of hemoglobin have different optical absorption profiles, this results in differential changes in the absorption of light at different wavelengths (León-Carrión and León-Domínguez 2012). By using a modified Beer-Lambert equation, it is possible to recover changes in hemoglobin from fNIRS observations taken at several wavelengths. The position of the brain signal can be approximated by putting the optical sensors on the head in a spatially organized manner. The evaluation of these task-related mobilizations done in the context of a baseline provided by the researcher himself or herself. The difference between the oxyHb and deoxyHb concentrations at baseline and throughout task performance defines where an increase or reduction in CBF occurs in the cortex (Rupawala et al. 2018). Because an increase in CBF is associated with increased brain activity, the temporal and spatial correlation between CBF and task is a predictor of cerebral function. The ability to analyze cerebral functioning in both spatial and temporal domains is what gives the technology described its name: functional near-infrared spectroscopy (fNIRS).

How fNIRS Works

Oxyhemoglobin and Deoxyhemoglobin

For generating energy and action potentials, neurons require nutrients which in turn are required for the generation of neuronal activity. Glucose and oxygen and other chemicals are delivered to metabolically active neurons by blood perfusion through capillaries, resulting in an increase in regional cerebral blood flow (rCBF) and regional cerebral blood oxygenation (rCBO). As a result, variations in regional cerebral blood flow (rCBF) or regional cerebral blood flow (rCBO) may be utilized to map brain activity with a high spatial resolution ("functional neuroimaging"). The rate of oxygen metabolism in the brain (cerebral metabolic rate of oxygen (CMRO2)) increases significantly during the functional engagement (Chen et al. 2020; Dahlstrom-Hakki et al. 2019; León-Carrión and León-Domínguez 2012). Because of increased oxygen consumption during neuronal activity, tissue oxygenation decreases, which is counteracted by an increase in oxygen supply when cerebral blood flow (CBF) and cerebral blood volume (CVB) increase (León-Carrión and León-Domínguez 2012), which is caused by a mechanism known as neurovascular coupling. Increasing capillary density, which in turn increases total cerebral blood volume, is required to maintain consistent glucose and oxygen consumption if the stimulation is to be sustained over time (Karim et al. 2012; León-Carrión and León-Domínguez 2012; Rahman et al. 2020). When the stimulus is brief, the aforementioned hemodynamic and metabolic changes do not occur in the same way as they do when the stimulus is long-lasting. In addition to producing CBF and oxygen metabolism, the body produces them to offset the effects of tissue deoxygenation as well as to oxygenate the blood. During the process of transferring oxygen, hemoglobin is known as oxyhemoglobin (oxyHb). However, when it releases oxygen as a result of a rise in oxygen metabolism, it is referred to as deoxyhemoglobin (deoxyHb) (deoxyHb). As a result, a brain area might be termed active when its rCBF rises, resulting in a drop in deoxyHb and an increase in oxyHb, which in experiments is often connected with a specific event (Lindauer et al. 2010). Typically, the additional oxygen supplied to the brain region outpaces the local CMRO2 usage, resulting in an overflow of cerebral blood oxygenation in active parts of the brain (Irani et al. 2007; Karim et al. 2012; Kohl et al. 2020; León-Carrión and León-Domínguez 2012). Hyperoxygenation of the tissues occurs as a result of an increase in cardiac output. The results of many optical experiments, including fNIRS and fMRI, have revealed that when brain activity begins, hyperoxygenation (owing to an increase in CBF—increased oxyHb) occurs first, followed by low oxygenation (increase in deoxyHb). This vascular response is difficult to measure and very controversial (Irani et al. 2007; León-Carrión and León-Domínguez 2012; Pinti et al. 2020). Because oxyHb and deoxyHb have distinct optical properties in the near-infrared light range, as well as a broad range of sensitivities to different wavelengths, it is possible to quantify changes in chromophore concentration during neurovascular coupling using optical methods (Caumo et al. 2022; Chen et al. 2020; León-Carrión and León-Domínguez 2012; Taylor et al. 2020). The modified Beer-Lambert law is used to compute the varying amounts of oxyHb, deoxyHb, and

Hb_{TOTAL} concentrations in the bloodstream (León-Carrión and León-Domínguez 2012). The attenuation of light intensity after absorption and scattering by biological tissue is expressed as:

$$I = GI_0e - (\alpha_{HB}C_{HB} + \alpha_{HBO2}C_{HBO2})^L \quad (2)$$

where G is a factor that takes into account the geometry of the measurement and is expected to remain constant regardless of the change in concentration. I_o is the amount of light that is coming in; α_{HB} and α_{HBO2} are the molar extinction coefficients; C_{HB} and C_{HBO2} are the concentrations of oxyHb and deoxyHb chromophores, respectively; and L is the photon path, which is a function of the absorption and scattering coefficients μ_a and μ_b (León-Carrión and León-Domínguez 2012). According to León-Carrión and León-Domínguez (2012), the relative change of oxyhemoglobin and deoxyhemoglobin vs time can be determined through the measurement of changes in optical density (OD) at two different wavelengths. If the intensity measurement was taken at the initial time (baseline) and it was I_b, then the OD change owing to fluctuation in C_{HB} and C_{HBO2} during that period is as follows:

$$\Delta\text{OD} = \log(I_b/I) = \alpha_{HB}\Delta C_{HB} + \alpha_{HBO2}\Delta C_{HBO2} \quad (3)$$

The calculation of ΔC_{HB} and ΔC_{HBO2} is made possible through the performance of measurements at two distinct wavelengths. After this, conclusions about oxygenation and total blood volume (Hb_{TOTAL}) can be drawn (Izzetoglu et al. 2007):

1. Oxygenation $= \Delta C_{HBO2} - \Delta C_{HB}$
2. Blood volume (Hb_{TOTAL}) $= \Delta C_{HBO2} + \Delta C_{HB}$

When taking a baseline, it is vital to consider whether or not the patient is medicated or nervous, since these are variables that might have a quantifiable influence on CBF during the experiment. A comparison of various measurements is dependent on the assumption that the proportion of arterial to venous intracerebral compartments remains constant because NIRS monitors changes in oxyHb and deoxyHb concentrations in both arterial and venous intracerebral compartments of the brain. As a result, variables that affect venous volume should be taken into consideration. It is possible that changes in head position, which are expected to occur during the clinical use of NIRS, will have an impact on venous pressure and the fraction of arterial and venous compartments within the cerebral vasculature. Furthermore, the posture of the body may alter the tone of the resistive arteries, allowing for the maintenance of a steady cerebral blood flow.

fNIRS Measurements

NIR light is beamed onto the scalp in order to perform fNIRS measurements. The NIR light must travel through a number of layers (such as the scalp skin, the skull, and cerebrospinal fluid), each with its unique set of optical characteristics before it

can reach the cerebral cortex. Due to the anisotropic and inhomogeneous nature of human tissue throughout its layers, the interaction of near-infrared light with it is challenging. Among the many components of human tissue that absorb light differently at different wavelengths are water, lipids, hemoglobin, melanin, and cytochrome coxidase, to name a few. Human bodies are made up of around 70% water, and water's absorption in the near-infrared optical window is moderate, allowing NIR light to pass through the tissue without being absorbed. Among the NIR optical window's absorbing chromophores, hemoglobin is the most noticeable and physiologically dependent absorbing chromophore. The oxygenation or deoxygenation of hemoglobin is determined by the saturation condition of the oxyhemoglobin (HbO2) or deoxyhemoglobin (HbR) in the blood. For example, HbO2 and HbR absorb NIR light in different ways: HbO2 oxygen has a greater absorption coefficient at wavelengths more than 800 nm, whereas HbR has a greater absorption coefficient at wavelengths less than 800. This difference in absorption is represented in the color of the blood, which is redder for oxygenated blood (arterial blood, which is 98% saturated) and more purple for deoxygenated blood (venous blood, which is 75% saturated), and can be detected using spectroscopic techniques in both arterial and venous blood. A task that requires the engagement and participation of a brain area raises the metabolic demand for oxygen and glucose in the brain, resulting in a regional cerebral blood flow (CBF) overflow to satisfy the increased metabolic demand.

It is possible to employ fNIRS technology in two different ways, depending on the number of channels available and their architectural design. Source and detector optical fibers (or optodes) for fNIRS sources and detectors are distributed uniquely at various spots on the head and at fixed source-detector separations in their most basic, most common, and most commercially available configurations (Chen et al. 2020). Each source-detector separation serves as a measurement channel, displaying the topographical distribution of HbO2 and HbR concentration fluctuations over the cortical surface of each subject. Using overlapping channels and different source-detector distances applied over the head, an alternate design may be used to generate a topographical image of the distribution of changes in HbO2 and HbR concentrations over the cortical surface (Kohl et al. 2020). It is the latter form of fNIRS, in which much denser arrays of channels sample brain areas that overlap, that is used in the study of diffuse optical tomography (DOT). Wearable and/or fiberless fNIRS sensors have recently been created to improve the usability of the technology. This type of equipment makes use of LEDs that are permanently attached to the head and are powered by batteries. They are employed in the CW technology field. Because there are no fiber-optic bundles, they are lighter and more resistant to movement artifacts than traditional fiber optics. Consequently, participants will be able to move around more freely as a result of the change. In many cases, data is recorded on a wearable recording device and wirelessly sent to a laptop (Almulla et al. 2022).

fNIRS Preprocessing

Once the fNIRS data has been acquired, preprocessing and analyzing of the NIRS signal must take place to identify and isolate hemodynamic changes happening in the

gray matter's vascular network. This is accomplished by the filtering of raw data and the estimation of a hemodynamic response via modeling. Preprocessing is the process of removing unnecessary noise from raw data with the goal of improving accuracy. Noise can be categorized into systematic noise, such as breathing, heart rate, fluctuations in blood pressure, and movement (motion artifact (MA)) noise. Once they are recognized, they must be distinguished and removed. Preprocessing steps such as channel pruning, removal of physiological noise, de-trending, and motion artifact removal/correction are applied to the data prior to hemodynamic response estimation. Frequency filters, wavelets, and smoothing filters are among the preprocessing techniques that are frequently employed. Alternative procedures, including pre-whitening, can also be employed. Once the raw data has been preprocessed, various methods are employed to transform changes in light intensity to changes in hemoglobin concentration. Analysis is used to compare baseline hemodynamic changes with task-related changes. These may be divided into two categories: general linear model (GLM) processing methods and non-GLM processing methods (Uga et al. 2014).

The following is a summary of fNIRS preprocessing:

1. Noise Removal: This is the initial stage in fNIRS preprocessing, and it is performed prior to the estimate of hemodynamic response to the stimulus. Environmental and instrumental noise, heart pulsation, breathing, and cyclic fluctuations in arterial blood pressure, collectively known as Mayer waves, can all inject noise into the fNIRS signal (Kohl et al. 2020). Here, the high-frequency components of the signal are eliminated from the equation. In order to filter away noise from fNIRS signals, researchers employ a variety of filter types and designs. The bandpass filter is one of the most often encountered frequency filters. For the most part, bandpass filters are simple combinations of two types of filters, in that they pass just a certain band of frequencies while attenuating the frequencies that are placed outside of the band (de Cheveigné and Nelken 2019). Unlike a high-pass filter, a low-pass filter allows signals with frequencies lower than a set cutoff frequency to pass through while attenuating sounds with frequencies greater than the given cutoff frequency. Similarly, a high-pass filter allows frequencies higher than the cutoff to pass through while attenuating frequencies lower than the cutoff. In order to properly use these filters, it is necessary to take into account the variances in physiology that exist across various people within the populations. Sporting athletes, for example, have lower resting heart pulsation rates compared to those who do not participate in sports. To account for the lower resting heart rate of the athlete when using a low-pass filter, a researcher would need to change the cutoff frequency of the low-pass filter. This would not be necessary when using a high-pass filter. Furthermore, in order to account for the nonathlete's higher resting heart rate, it is possible that the researcher will choose a higher cutoff frequency for the low-pass filter in the experiment. Additionally, the type of work that is completed may have an influence on the filter options. For example, when exercising as opposed to engaging in non-exercise motor activity, the heart rate and respiratory rate both increase. In order to prevent this difficulty,

it is not advisable to choose a cutoff frequency that is "one size fits all" in all situations (Dans et al. 2021).

2. Smoothing filters (Klein and Kranczioch 2019) are yet another sort of filter that is frequently employed. High-frequency noise in fNIRS data is typically reduced by using smoothing filters, which are commonly employed in this field. Smoothing signals may be accomplished in two ways: either by smoothing in the time domain or by smoothing in the spatial domain. When using time domain smoothing, high-frequency noise is reduced, whereas spatial smoothing averages signals from bad channels with the signals from neighboring fNIRS channels, minimizing the influence of a noisy channel while still maintaining some of the signals from the noisy channel. The most common smooth filters are the Gaussian filters and Savitzky-Golay smoothing filters (Afonso et al. 2017). Gaussian smoothing (Khan et al. 2020) is accomplished by the use of a Gaussian weighting function, which doubles the value of each point based on where it falls on the distribution. The center of the Gaussian is fixed on a single point, which is then weighted in proportion to the points around it. Afterward, the distribution is transferred to the next spot, and the procedure is repeated. In order to smooth out spike motion artifacts, the Savitzky-Golay smoothing method is most commonly used. As a smoothing filter, it may also be used to smooth physiological noise in data.

   Another technique that is mostly used in conjunction with GLM is pre-whitening (von Lühmann et al. 2020). The premise behind pre-whitening is that fNIRS data contains consistent physiological noise that changes in a predictable manner. Some frequencies are overrepresented in the noise distribution as a result of the predictable nature of the noise, which is in violation of one of the GLM criteria for a normal noise distribution (von Lühmann et al. 2020). It is possible to estimate this noise using pre-whitening, which will result in a reduction in the weights of the overrepresented frequencies in the noise distribution. There is a downside to utilizing this approach in that it estimates noise, and as a result, noise might be overestimated or underestimated, thus contributing to bias.
3. Motion Artifact (MA) Correction: Motion artifacts are usually introduced into the fNIRS signals by head movements that cause the source-transmitter pair to shift relative to the scalp (Brigadoi et al. 2014). MAs might appear as substantial and quick increases in magnitude (spikes) when compared to the baseline data. These spikes can be hundreds of times bigger than tissue-related hemodynamic changes. Additionally, the baseline fNIRS signal might vary as a result of the movement of the source-detector pairs on the scalp during the experiment. An unnaturally exaggerated positive correlation between oxy-Hb and deoxy-Hb might result as a result of this phenomenon. Wavelet filters (Duan et al. 2018) are a regularly used approach for removing motion distortions from fNIRS data, and they have a number of advantages. Using wavelet filtering, one can distinguish between distinct frequencies in the cortical signal and different frequencies in MAs. A mother wavelet is used as the starting point for wavelet filtering, which is then scaled and translated to produce daughter wavelets. In the following step, the fNIRS recording is deconstructed using these daughter wavelets. The equations for the wavelet coefficients demonstrate how well the wavelet transform captures

the fNIRS measurements (Perpetuini et al. 2021). The greater the number of wavelet coefficients, the better the wavelet transform's ability to characterize the full signal. These wavelet coefficients combine to generate a distribution that is connected to the scaling and translating parameters of the wavelet transform. MAs are outliers in this distribution because of their deviations from the cortical signal, and they may be deleted as a result (Brigadoi et al. 2014).

fNIRS Analysis

General Linear Model

The GLM is a statistical modeling approach that can be used with fNIRS data (Dans et al. 2021). It has previously been used to simulate the functional magnetic resonance imaging (fMRI) BOLD response, and it has been embraced because of the similarities between BOLD and HRF responses. The GLM makes use of predictors in order to explain the most significant causes of variability in the data by modeling the measured data as a linear mixture of M functionally different processes (Uga et al. 2014). In its simplest form, the GLM is represented as follows:

$$Y = X.\ G + \epsilon \tag{4}$$

where $Y \in \mathbb{R}^{T\times N}$ is the observation matrix where the column vectors are represented as $y_n \in \mathbb{R}^T$ and row vectors are represented as $y(t) \in \mathbb{R}^N$, T signifies the fNIRS data from all time points, and N is the number of channels. $X \in \mathbb{R}^{T\times M}$ is the design matrix, which takes into account a priori knowledge of the predicted form of the evoked hemodynamic response, the experiment's temporal structure, and regressors for drifts and/or physiological error signals. $G \in \mathbb{R}^{M\times N}$ represents the set of weights for the regressors that models functional brain activity and physiological and nonphysiological confounding components. These weights are to be estimated. ϵ is the residual/noise term represented as $\epsilon \in \mathbb{R}^{T\times N}$. To put it another way, the amplitude of the hemodynamic response in one channel (Y) is equal to the predictor (X) multiplied by the predictor's "weight" (G) plus the error term (ϵ). Weights are allocated to predictors depending on how much they contribute to the signal's variability when forecasting the signal. In other words, if the researcher's estimate/assumption regarding the hemodynamic response's shape is correct, the model will give that estimate a higher weight than the other estimates/assumptions (von Lühmann et al. 2020). The noise in the recording is represented by the error component in the equation, which includes physiological, instrumental, and motion noise, as well as other kinds of noise. The GLM makes assumptions about the data and the system noise (von Lühmann et al. 2020). The following are the assumptions:

1. The responses to the task are non-stochastic, which means they are not random and they remain consistent between several iterations of the same task.
2. The distribution of noise is both independent and identical, having a mean of zero and some degree of variance around that point.

3. The noise is homoscedastic, which means that it only comes from one distribution throughout the whole set of data.
4. There is no correlation between the noise in the past and the noise in the present or the noise in the future.
5. The relationships between the predictors do not follow a linear progression.

Applications of fNIRS

Mental Workload

Complex operational situations require comprehending neuronal resource allocation for task completion. An increase in mental workload can lead to diminishing performance resulting in deadly implications. Mental workload represents the interplay between environmental pressures, human capabilities, and task performance. Thus, a person's mental effort cannot be inferred from task features alone. EEG, fMRI, and PET can evaluate mental workload continually and unobtrusively, however, are not suitable for ecological circumstances. EEG tests are susceptible to head and/or body movement artifacts, whereas PET and fMRI require subjects to lie down and not move. Mental exertion levels necessitate sensitive, ongoing, and thorough assessments. fNIRS is a portable, field-deployable imaging technique that meets these requirements. The mental workload construct assumes that task-related neuronal activity uses a specific amount of mental resources proportionate to task difficulty. fNIRS has been used to measure cognitive workload in fundamental, clinical, aging, and human factors studies. Ayaz et al. (2012) reported consistent variations in oxygenation in the left dorsolateral prefrontal cortex as a function of memory load when utilizing fNIRS to detect activation during a conventional n-back test. Similar to Ayaz et al. (2012), Fishburn et al. (2014) showed linear increases in right and left prefrontal cortical activation as working memory load increased. Blood flow variations detected with fNIRS are also linked to executive processes like mental flexibility and reaction inhibition. fNIRS has shown changes in oxygenation due to considerable increases in job difficulty (e.g., piloting unmanned air vehicles). However, modest differences in task difficulty could not be consistently discriminated.

Epilepsy

Studies conducted on animals over the course of the last several decades have led researchers to the conclusion that fNIRS is capable of identifying increased cerebral blood flow and oxygenation in correlation with epileptic spikes and high-frequency oscillation (Hoshi et al. 2001; Zhang et al. 2014). These alterations, which were observed by fNIRS, can occur approximately 5 s before the emergence of spikes caused by bicuculline methiodide. There has been a significant amount of research conducted on the oxygenation response to seizures using fNIRS. An early clinical case study using fNIRS, for instance, found that individuals who suffered from temporal and extratemporal lobe epilepsies had a constant ictal increase of oxygenation on the epileptic hemisphere in comparison to the non-epileptic hemisphere during seizures (Monrad et al. 2015). During a seizure, it has been demonstrated that regional frontal

oxygenation drops in patients suffering from generalized epilepsy as well as secondary generalized seizures (Buchheim et al. 2004; Sokol et al. 2000). Moreover, the recent development of fNIRS devices with many channels is one of the most recent steps forward in the field of epilepsy research. The primary challenge that oligo-channel fNIRS faced was a lack of sufficient spatial resolution, which inspired the development of multichannel fNIRS. SMA seizures (Sato et al. 2013), mesial temporal epilepsy (Rizki et al. 2015), and other focal seizures have all been shown to have a differential increase in regional oxygenation or blood flow (Yücel et al. 2014). Epileptologists would be able to better comprehend the hemodynamic changes that occur during the distinct stages of seizures (preictal, ictal, postictal, and interictal phases) and enhance seizure detection and localization if they used multichannel fNIRS in conjunction with prolonged EEG.

Mild Cognitive Impairment

Investigations of MCI using fNIRS have indicated a reduced increase in oxyhemoglobin. Additionally, during memory retrieval, dorsolateral prefrontal cortices showed hypoactivation (Uemura et al. 2016). However, with an accuracy ranging from 60% to 70% on average, it has not been possible to identify MCI patients in a reliable manner (Yang et al. 2019). In addition, the results of fNIRS have been conflicting, although they seem to indicate that patients who have MCI have a higher oxyhemoglobin response in the regions of interest when compared to the healthy control group (Yoo and Hong 2019). Because of this, it is doubtful that the hemodynamic response that may be identified by fNIRS will be sensitive and specific enough for use in clinical practice. In more recent times, functional connectivity has become the primary focus of noninvasive neuroimaging biomarkers for mild cognitive impairment and Alzheimer's disease. Patients with MCI and AD show reduced frontal and long-range connection in the resting-state fMRI, according to a recent systemic review that included 36 papers from the past two decades (Yeung and Chan 2020). A more recent study came to the same conclusion as well (Yoo and Hong 2019). The number and strength of prefrontal functional connections were found to increase during a working memory task (Yu et al. 2020), whereas left and interhemispheric connectivity during a verbal fluency test was found to be lower (Nguyen et al. 2019) in the patients with MCI compared to healthy controls.

Future Work: Measuring Cognition Using Simultaneous EEG and fNIRS

Multimodal integration, which integrates distinct neurophysiological signals, is gaining popularity in addition to its potential to compensate for one modality's shortcomings and produce consistent findings by extracting complementary aspects. Integration of electroencephalography (EEG) with functional near-infrared spectroscopy (fNIRS) is particularly cost-effective and portable, making it an intriguing technique. The association between EEG and fNIRS signals is yet unknown, and combining the advantages of electrical and optical measures remains a challenge

(Zich et al. 2017). More future research is needed to examine possible applications, integrations, and algorithms in EEG/fNIRS analysis in order to encourage EEG/fNIRS application innovation with the goal of maximizing the benefits of combination (Shin et al. 2018). Most of the selected research uses EEG/fNIRS to classify data by merging obtained electrical and optical handcrafted elements. To detect brain-evoked potentials, the advantages of EEG's temporal resolution can be used by monitoring oscillatory correlates (ERP or ERD/ERS) from the complete head. Optodes can be placed on the local region, such as the occipital or prefrontal regions, to evaluate significant attentional or visual processes, thanks to fNIRS' high spatial resolution (Mele et al. 2019).

HbO and HbR are distinctive aspects of fNIRS recordings that have been widely exploited to date. Because HbO and HbR are both temporal characteristics, determining the temporal period and using filtering methods based on the temporal domain are critical in signal processing. EEG, on the other hand, can employ both spectral and temporal properties; therefore, individuals are frequently exposed to fast and brief stimuli. Due to the intrinsic response delays and low temporal resolution in fNIRS recordings, such stimuli are not appropriate. Although various commercial instruments exist that allow simultaneous EEG-fNIRS monitoring, recording neural activity from the same site is problematic. Because hemodynamic responses in fNIRS are recorded from the center of emitters and detectors, EEG electrodes should be placed in the center of these emitters and detectors to create the same channel arrangement. EEG electrodes can be quite small, but fNIRS emitters and detectors must be somewhat large to properly emit and detect infrared light. Without a well-tuned hardware combination, one approach to equip an EEG-fNIRS device in the lab is to manually combine them. There are several applications of hybrid EEG-fNIRS in which the use of electrodes/optodes has been distributed in the different parts of the brain (Shin et al. 2018). Comparing hybrid EEG-fNIRS performance with various setups is thus one route for future research (Zich et al. 2017). It's also interesting to see how the optodes and electrodes are arranged differently depending on whether the focus is on optimizing system processing or detecting individual brain activations. Optodes and electrodes are generally found in the same areas when it comes to building system processing. Electrodes are primarily utilized to measure event-related desynchronization (ERD) from brain oscillations as well as optodes that are set on the implicated cortical regions to intensify the signal on these regions in order to emphasize the weights of experimentally connected cortex regions in measurement.

The simultaneous evaluation of electric and hemodynamic brain activity is made possible by multimodal EEG-fNIRS, which is a feature that confers a number of significant benefits. EEG-fNIRS has been increasingly employed in a variety of fields, including clinical neurology, neurorehabilitation, and brain-computer interface. The development of wearable, integrated EEG-fNIRS technology is an essential next step in the advancement of this discipline. To this day, numerous integrated systems that make use of discrete components that are commercially available have been developed. These technologies, despite their success, are susceptible to the following limitations: (1) large size, heavy weight, and limited wearability and subject comfort, (2) cross talk between optical and electrical signals, (3) inaccurate timing precision/synchronization

for simultaneous fNIRS and EEG recording, and (4) low signal quality and/or limited capability of long-term recording of EEG electrodes (Uchitel et al. 2021). Despite their achievements, these technologies still have some room for improvement. Utilizing designs that are integrated both mechanically and electrically by means of microchips is becoming increasingly popular as a means of overcoming some of these restrictions. Microchip technologies have a number of important advantages, including a small footprint, low-power consumption, improved timing synchronization between fNIRS and EEG measurements, and potentially improved signal quality with reduced noise and cross talk (Lin et al. 2014; Von Luhmann and Muller 2017). Microchip technologies also have a number of other advantages. In the past 10 years, numerous attempts have been undertaken to create a microchip-based, wearable, integrated EEG-fNIRS system. These attempts have been unsuccessful. In addition to the designs of integrated circuits, other emerging trends include designs that are modular, lightweight, and flexible, designs that can communicate wirelessly, designs that make use of shared ADCs to achieve timing precision between optical and electrical data, and designs that take into account both wet and dry EEG electrodes. When considering the development of EEG-fNIRS systems in the future, research into ergonomic designs (i.e., wearable, miniaturized, lightweight, and flexible) will continue to be of primary importance for long-term monitoring, wearability, and subject comfort. To attain the huge dynamic range required for high-channel count fNIRS systems, improvements in resolution, high-speed ADCs, sensitive amplifiers, and accurate logic control will be required. As a consequence of this, more sophisticated data interfaces and control systems would be necessary in order to guarantee a large channel count, a large dynamic range, and high-speed data gathering and transmission. In addition, dry electroencephalogram (EEG) electrodes that have a high gain and a high input impedance are probably going to be the best option for long-term and consistent monitoring. Dry electrodes are increasingly included into the design of EEG-fNIRS systems due to their ability to make stable contact with the scalp, even through hair, as they are lightweight, tiny, and flexible and have the capacity to create steady contact with the scalp.

The application of combined EEG and fNIRS is still in its early stages; however, it may be beneficial in the near future in terms of improving the algorithms and physical system design of hybrid EEG-fNIRS. This could include emphasizing the spatial and temporal advantages of fNIRS and EEG appropriately by designing advanced feature extraction methodologies and splitting nodes in different targeted cortical functions. In addition, it is necessary to keep developing more applications that combine EEG and fNIRS, particularly in fields that involve complex information processing.

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Metrological Aspects of SEMG Signal Acquisition, Processing, and Application Design

75

Rohit Gupta, Inderjeet Singh Dhindsa, and Ravinder Agarwal

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Abstract

Significant growth in SEMG-based applications has been observed over the last few decades. It is primarily due to advances in electronics fabrication and

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computational capabilities. However, the metrological issues and standardization issues are still untouched and need to be addressed critically. This chapter attempts to compile the recommendations, standards, and requirements of such type of systems, as well as a case study of SEMG operated lower limb prosthesis design. The SENIAM recommendations are thoroughly discussed and implemented in the application design. A high-level and low-level control module is used to implement a biologically inspired control structure. The high-level control module is responsible for estimating activity intention, whereas the low-level control module is responsible for estimating the desired angular position of the joints. Both the control modules utilize SEMG signal differently. The high-level control module problem is formulated as classification problem, whereas the low-level control module problem is implemented as classification problem for knee angle prediction and continuous value estimation problem for ankle angle estimation. SVM, LDA, and NN classifiers with time domain feature vector were found suitable for high-level control module. For knee angle prediction, SVMQ was found to be superior, whereas for ankle angle estimation, the autoregressive exogenous neural network (NARX) model with SEMG and knee angle information depicts satisfactory performance over the different locomotion modes.

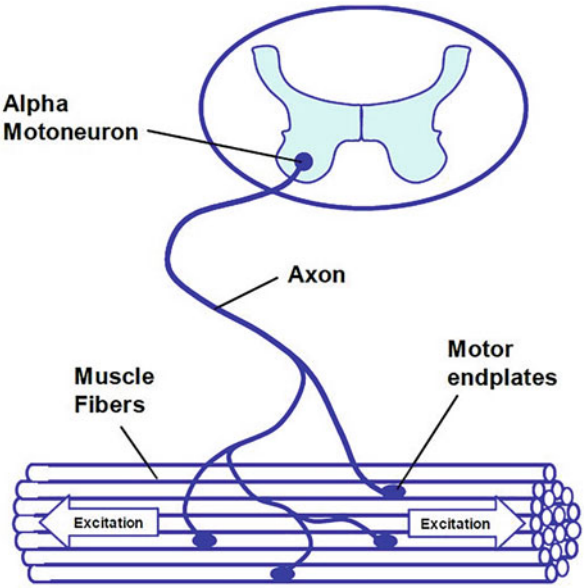
Keywords

SEMG · Prosthesis design · Human physiology · SENIAM Standard · Prosthesis design recommendations · Machine learning

Introduction

All human activities and movements are governed by neural signals that originate in the human brain. The neural signal generated in the cortical region of the brain is transmitted to the appropriate body part via the nervous system. This neural information about limb movement can be recorded as a biopotential, specifically an electromyogram (EMG) signal. It appears as a result of neural commands sent through the peripheral nervous system to activate human limb muscles. It is generated as an electrical impulse and causes muscle fibers in the human body to contract. Skeletal muscles use this electric impulse most often to control limb movements. Skeletal muscle is made up of fiber cells that range in length from a few millimeters to 30 cm and in diameter from 10 mm to 100 mm. Myofibrils, which have a dense array of protein filaments, are found in every muscle fiber. Motor neurons convey neural messages from the brain to protein filaments, resulting in muscle contraction. During muscle contraction, the length of the muscle is typically reduced to half of its resting length. There are three types of muscle contraction: isometric, concentric, and eccentric. Muscle length does not change during isometric contraction. Isometric contractions are commonly used to control posture. Muscle length decreases and increases during concentric and eccentric contractions, respectively (Konard 2012).

Fig. 1 Motor unit illustration



The motor neuron, axon, and muscle fibers that are shown in Fig. 1 combine to form a motor unit. Only when the brain decides on limb movement, such as moving the leg, does the nerve impulse stimulate the contraction, which is carried in the nerve through motor neurons from the brain to the muscle. As a motor neuron approaches the muscle, it divides into many branches (axon terminals). Each muscle fiber in a motor unit is connected through the corresponding axon branch to the accompanying motor neuron at a junction known as the neuromuscular junction. The innervation region is mostly found in the middle of the muscle length.

Motor unit action potential (MUAP) is defined as a resultant action potential of individual muscle fibers within the motor unit. The shape and duration of the MUAP imitate the architecture of the motor unit. Its amplitude is directly proportional to the radius of the muscle, and it carries an inverse relation with the distance between the active fiber and the pickup site. It is considered a complex signal, modulated by the human nervous system which carries significant variation contributed by the anatomical and physiological properties of human muscles. The amplitude of the EMG signal is quite random and can be represented as a Gaussian curve. It ranges between a few μV to 100 μV depending on the muscle under observation. It carries frequency components of 1 Hz to 10 kHz frequency range along with dominant signal power between 20 and 300 Hz. EMG is highly prone to noise generated by various tissues while traveling to the respective muscles. Further, the electrodes used for SEMG acquisition from the skin of the human body capture the effective signal generated by multiple motor units. Surface EMG (SEMG) signals are primarily studied for two applications, namely clinical and kinesiological (Konard 2012). Physicians, neurologists, and physiatrists do clinical studies by observing the

characteristics of the MUAP to diagnose neuromuscular diseases and injuries. Kinesiological EMG is used for limb movement studies. It is utilized by physical therapists and ergonomists to determine the behavior of the muscles during the limb movements of the subject. The acquisition of good quality SEMG signals and their real-time processing is a challenge. In particular, application SEMG is preferred due to its noninvasive nature.

In biomedical engineering, the acquisition of good quality SEMG signals and their real-time processing is a challenge. Various research groups have developed low-cost research platforms for laboratory use. However, their calibration repeatability and reliability are still debatable. Furthermore, the metrological aspects of SEMG systems are unexplored.

Material and Methods

Amputation and Prosthesis

Amputation is an acquired condition that results in the loss of a limb partially or completely. It can occur due to accidents, animal attacks, battles, etc. It is also performed as a surgical procedure. In the USA, it is estimated that 185,000 persons undergo amputation of the upper and lower limb each year. The most frequent type of amputation is the below-knee amputation (transtibial) followed by the above-knee amputation (transfemoral) (Dillingham et al. 2002; Renzi et al. 2006). The word transtibial amputee refers to a person who got a lower leg amputation between the knee and ankle or on the tibia bone, whereas the transfemoral amputee refers to one who got a leg amputation between the hip and knee. Lower limb loss is a potentially disabling condition affecting the health and well-being of the persons worldwide.

The persons living with transtibial limb loss face distinct challenges; they need more metabolic energy and cognitive effort to perform any locomotion-related task. Moreover, their locomotion is very slow and unstable. Innovations in the design and development of new prosthetic devices should address these clinical issues (Versluys et al. 2008).

The lower limb prosthesis is considered the substitute for the lost limb of an amputee. The rapidly increasing figures for amputation around the world always motivate researchers to provide the effective solution as efficient prosthesis. The development of lower limb prosthesis has a vast history (Duraiswami et al. 1971). The initially developed prostheses were completely passive and provided only support to the lower limb. Most of the time, the passive prosthesis is augmented with passive elastic elements, which store and release energy in different phases of the gait cycle. These passive elements underperform as compared to the biological limb in producing effective push-off during locomotion. However, active prostheses can provide the required push-off. But such powered prostheses require adaptability and an indispensable controller. In these prostheses, the control commands for cyclic/rhythmic activities are synthesized via central pattern generators (CPG) with the locomotion model of humans. Control command generation using EMG signals offers a viable technique for prosthetic actuation and control. A blend of CPG and

EMG signals provides an effective solution in controlling a device like an ankle-foot prosthesis (Simão et al. 2019).

A prosthesis is considered as an artificial replacement of a missing or malfunctioning body part. The objective of the ankle-foot prosthesis is to provide ankle-foot functionality for a transtibial amputee. The history of leg amputation and lower leg prosthesis is very ancient. The first text of lower limb prosthesis had been found in Rig Veda, an Indian sacred book, written around 3100 B.C. There it is mentioned that an iron leg was fitted to Queen Vishpla after being amputated in a battle (Duraiswami et al. 1971). The lower limb prosthesis can be divided into three subgroups: passive, semi-active, and active prosthesis. Passive prosthesis utilizes gear, clutch, springs, or dampers to achieve a human-like gait pattern, whereas the semi-active prosthesis incorporates electronics to control the mechanical components as per the need of the gait phase. In an active prosthesis, the main actuation mechanism of the joint is externally powered (electric, hydraulic, or pneumatic) and controlled by the amputee using sensory information. Table 1 shows a few of the most popular ankle-foot prosthesis with their features.

Human Lower Limb Anatomy and Gait Cycle

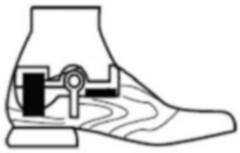
This section covers the basics of human lower limb anatomy and gait cycle in brief. The lower limb anatomy includes ankle as well as knee movements and their responsible muscles. The human gait cycle covers its definition, description during walking, and biomechanical characteristics.

Ankle-Foot Anatomy

The human ankle-foot is designed to provide a stable platform so that the dynamic activities of daily life can be performed. It provides contact between the human body and the environment during various activities like sitting, standing, walking, and running. It must be able to provide the required range of motion (RoM), control, and stability to the body during various activities. The ankle-foot work act as a shock absorber and weight bearer. Figure 2 shows the major bones, joints, and ligaments of the human ankle-foot. The ankle-foot is a complex skeletal structure having 33 joints, 28 bones, and 112 ligaments.

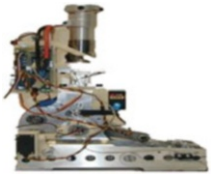
The complex structure of the ankle-foot is controlled by 20 intrinsic and 11 extrinsic muscles. The complete entity of ankle-foot is rigid enough to bear bodyweight, and at the same time, it is also flexible in dealing with uneven terrains (Brockett and Chapman 2016). The ankle joint is made up of mainly three bones, namely, tibia, talus, and fibula. The complete foot can be divided into three subparts as rear-foot (tarsus), mid-foot (metatarsals), and forefoot (phalanges). Calcaneus is considered the biggest bone in the foot and resides in the rear-foot part. It transmits the load from the body to the ground and serves as a moment arm for the calf muscles to generate sufficient moment for propulsion to move the body forward. The talus bone is attached to the calcaneus bone of the foot (heel bone) through the Achilles tendon and provides articulation at the subtalar joint (Brockett and Chapman 2016). The RoM of ankle-foot is defined around the particular joint in any DoF. The ankle-foot

Table 1 Various ankle-foot prostheses and their features

S. No	Ref/type	Prototype	Features
1	Passive (Staros 1957)		Non-articulated ankle joint Internal keel to assist smooth rolling action Cushioning material absorbed the impact during heel-strike (HS) Heel wedge provides pseudo-plantarflexion The rigid wooden keel provides midstance stability
2	Passive (Goh et al. 1984)		Articulated ankle joint Two adjustable bumpers limit and control ankle DF and PF movement PF bumper permits the foot to contact the floor more rapidly during loading response Heavyweight Offers little lateral movement Requires more frequent maintenance
3	Passive (Sethi et al. 1978)		Jaipur foot Vulcanized rubber material provides slight inversion/eversion movement Improved performance as compare to solid ankle cushion heel (SACH) foot An amputee can sit in a squat position, in a cross-legged position, and walk barefoot Heavyweight
4	Passive (Lee et al. 2006)		Has a flexible elliptical-shank monolimb Made up of one piece of thermoplastic material Much lighter in weight as compared to the conventional prosthesis Provides more comfort to amputee and also the gait is improved Long-term effect of shank flexibility is not there
5	Passive (Burgess et al. 1985)		ESR foot Spring-like structure to store and release the energy in different phases of the gait cycle Improved push-off
6	Passive (Pitkin 1995)		Articulated ankle joint The dynamic performance was found better than SACH and flex-foot

(continued)

Table 1 (continued)

S. No	Ref/type	Prototype	Features
7	Semi-active (Collins and Kuo 2010)		Controlled energy storing and returning (CESR) foot Captures collision energy during HS and returns it for push-off The complete foot has two components: rear and forefoot components One microcontroller and two small motors were used to release the stored energy in spring
8	Semi-active (Paper et al. 2013)		Ankle mimicking prosthetic (AMP) foot The mechanical structure had a planetary gearbox, a spring, and two servo pawls A microcontroller was used to control overlocking and unlocking of the pawls It regenerates energy during the dorsiflexion phase and releases it over the entire push-off phase
9	Semi-active (Shepherd and Rouse 2017a, b)		Variable-stiffness prosthetic ankle-foot (VSPA foot) A cam-based transmission and a fiberglass leaf spring had been used as in mechanical structure A small motor was used to modulate the support condition of leaf spring and its stiffness Controlled by tracing the torque-angle curve of ankle-foot
10	Semi-active (*Raize Ankle Foot System* n.d.)		Microprocessor controlled (MPC) hydraulic ankle-foot Adjustable heel height Adjustable plantarflexion and dorsiflexion range and stiffness Adapt to uneven terrain Simulates natural ankle movement and reduces vertical shock Automatically adjusts to allow maximum dorsiflexion when walking on an incline Range of motion (RoM) for the ankle is 10° dorsal flexion and 18° plantarflexion

(continued)

Table 1 (continued)

S. No	Ref/type	Prototype	Features
11	Semi-active (Elan n.d.)		Microprocessor-controlled self-aligning plantarflexion and dorsiflexion Standing support mode increases resistance when stationary to help improve balance and stability Adaptable to different walking speed and terrains Provides extended dorsiflexion during swing phase to increase toe clearance and reduce risk of trips and falls Provides higher resistance during slop walking for better stability and safety
12	Active (Au et al. 2008, 2009)		Utilized series elastic actuator (SEA) using physical spring in series with drive Manual EMG triggering based transition control Two-level finite state machine controller Only level-ground and stair-descent locomotion are considered One degree of freedom (DoF)
13	Active (Bellman et al. 2008; Hitt et al. 2010)		SPARKy (spring ankle with regenerative kinetics) Brushless motor and linear transmission Smaller and more powerful brushless motor without the need for a gearbox Walking speeds over 4 miles per hour Toe design is considered DoF is 2
14	Active (Cherelle et al. 2014a, b, 2016)		Ankle mimicking prosthetic (AMP) foot Adaptive speed using variable stiffness components A hybrid controller is used Lightweight actuators are used Two locking mechanisms and a compliant crank-slider mechanism One degree of freedom
15	Active (Versluys et al. 2009)		Pneumatic actuators are used Finite-state controller EMG-based control Only for level walking with speed adaptation One DoF

(continued)

Table 1 (continued)

S. No	Ref/type	Prototype	Features
16	Active (Zhu et al. 2014)		Powered below-knee prosthesis with ankle and toe (PANTOE) Finite-state control strategy Electric actuator with spring EMG signal-based control A linearized trajectory control method One degree of freedom
17	Active (Sun and Voglewede 2014)		Four-bar mechanism linkage design Electric actuator with spring High-level finite-state controller and low-level proportional-integral-derivative (PID) controller One degree of freedom

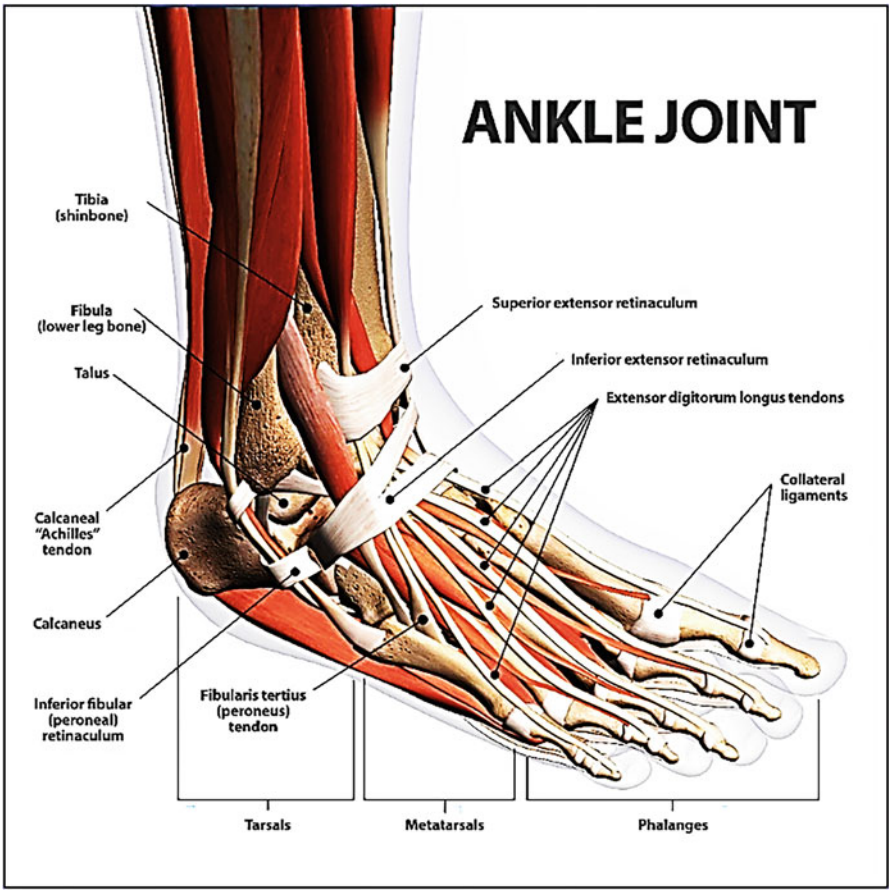


Fig. 2 Ankle-foot bones, joints, and ligaments

has three DoF, one in the sagittal plane, one in the transverse plane, and one in the frontal plane. Hence, there are three possible motions of human ankle joints; dorsiflexion (DF)/plantarflexion (PF), inversion/eversion, and abduction/adduction. Table 2 shows the RoM of ankle joint motions along with their respective reference planes (Riegger 1988).

Figure 3 shows the different movements of ankle-foot. The movement of the ankle joint, pointing the foot towards the leg, or movement of toes lifting the foot away from the sole, is termed DF. The movement at the ankle joint, pointing the foot downwards away from the leg or movement of the toes that curl them down towards the sole, is termed PF, shown in Fig. 3a. Inversion is the movement of turning the sole inwards at the ankle joint, while eversion is the movement of turning the sole outwards, as shown in Fig. 3b. Abduction and adduction movements are in the

Table 2 RoM of a natural human ankle

Motion	RoM (degrees)	Reference plane
Dorsiflexion/plantarflexion	20/50	Sagittal
Inversion/eversion	35/25	Transverse
Abduction/adduction	20/25	Frontal

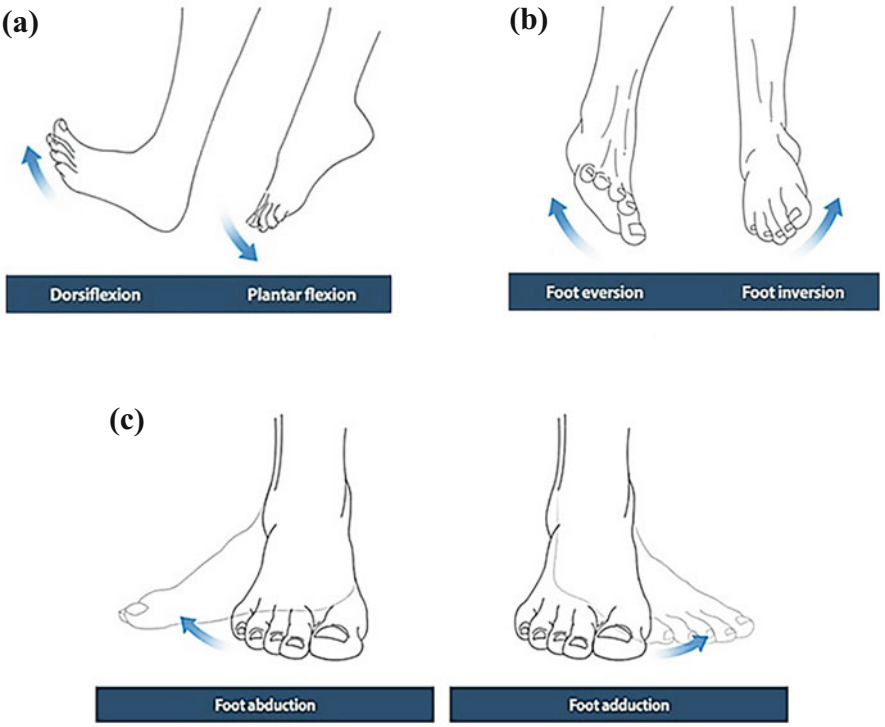


Fig. 3 Human ankle-foot movements

frontal plane and involve moving a body part away from or towards an imaginary center line. As shown in Fig. 3c, abduction moves the body part away from the central line, whereas adduction moves it towards the central line.

Various lower limb muscles control the movements of the ankle-foot structure. The musculature of the ankle and foot is classified into two parts: intrinsic and extrinsic muscles. Twenty intrinsic layered muscles in the human foot support locomotion by extending, flexing, abducting, and adducting the toes and the arch of the foot, whereas 12 extrinsic muscles or longer muscles spanning the tibia and foot act as the main drivers for ankle major joint motions. These muscles are: gastrocnemius (Gast), soleus (Sole), tibialis anterior (TiAn), tibialis posterior (TiPo), plantaris (Plan), peroneus brevis (PeBr), peroneus longus (PeLo), peroneus tertius (PeTe), flexor digitorum longus (FlDL), extensor digitorum longus (ExDL), extensor hallucis longus (ExHL), and flexor hallucis longus (FeHL). Figure 4 shows the anatomical illustration of major below-knee muscles (Riegger 1988).

The ankle joint is extremely complex and provides motion along two axes. Primary muscle actuators are muscles that contract in response to ankle joint motion about a specific axis. However, it is extremely rare for a muscle contraction to result in only one axis of rotation; usually, the second joint of motion occurs as a byproduct. As a result, each rotational axis has both primary and secondary muscle actuators, as summarized in Table 3 (Riegger 1988).

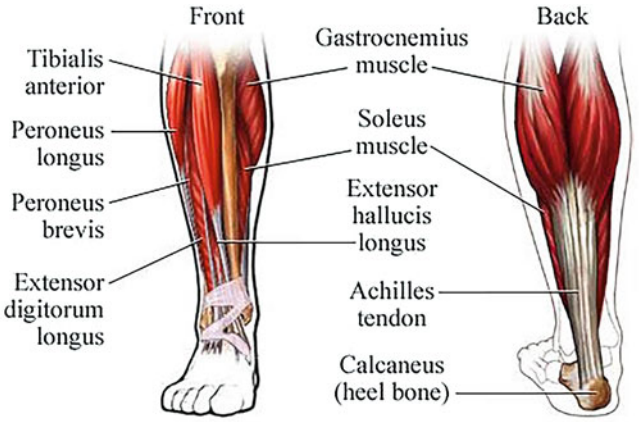


Fig. 4 Major below-knee muscles of the human

Table 3 Primary and secondary movers for each axis of motion of ankle joint

Joint movement	Primary muscle actuators	Secondary muscle actuators
Dorsiflexion	TiAn	ExDL, ExHL, PeTe, and Plan
Plantarflexion	Gast, Sole	TiPo, Plan, FlDL, FlHL, PeBr, and PeLo
Inversion	TiPo	FlDL, FlHL, TiAn, and ExHL
Eversion	PeBr, PeLo	PeTe, ExDL, and ExHL

Human Knee Anatomy

Bones, muscles, cartilage, tendons, and ligaments comprise the knee joint. Their bones provide basic support and rigid structure to the knee joint, while their ligaments provide stability and muscles aid in joint movement. The four bones that make up the human knee joint are the femur, tibia, fibula, and patella. The thigh bone, shin bone, the smaller bone that runs alongside the tibia, and the kneecap are all names for these bones. The knee joint is depicted in Fig. 5 from both the lateral and anterior sides. Tibiofemoral (tibia-femur) and patellofemoral (patella-femur) knee joints work together to form a modified joint that allows the knee to extend and rotate slightly (side to side).

Flexion and extension about the virtual transverse axis in the sagittal plane are the primary movements of the knee joint. The knee can rotate slightly medially and laterally when it is flexed. Knee flexion is the movement of the shank towards the thigh, which decreases the angle of the knee joint (KA). The contraction of the knee flexor muscles allows for this movement of the knee. Knee extension is a straightening of the knee that is achieved by contracting the knee extensor muscles and relaxing the knee flexor muscles. The knee movements are depicted in Fig. 6.

Muscles responsible for knee joint movements are divided into two categories – knee flexors and knee extensors. Knee flexors consist of muscles which are responsible for knee flexion. The knee flexors shown in Fig. 7a are a group of seven muscles which cross the posterior of the knee joint (except the sartorius muscle). Knee flexors comprise the hamstrings (biceps femoris (BiFe), semimembranosus (Semb), and semitendinosus (Semd)), the gastrocnemius muscle, the gracilis muscle, the sartorius muscle, and the popliteus (Flandry and Hommel 2011). The knee extensors are a group

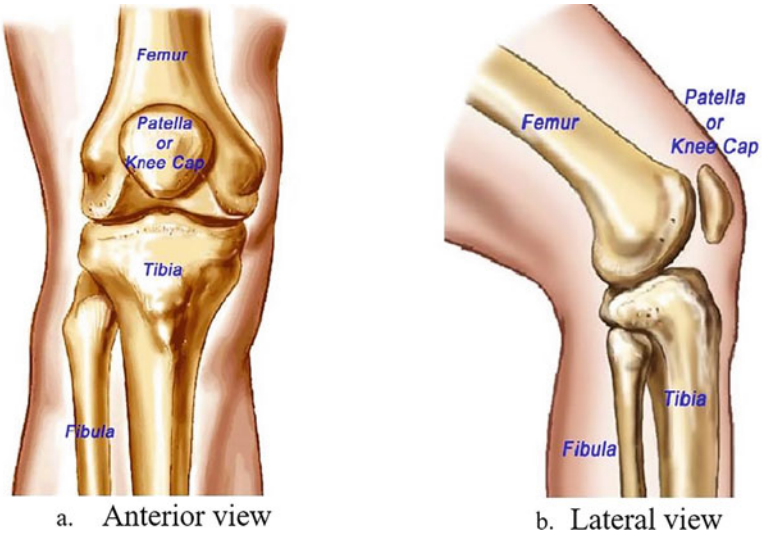


Fig. 5 Human knee joint

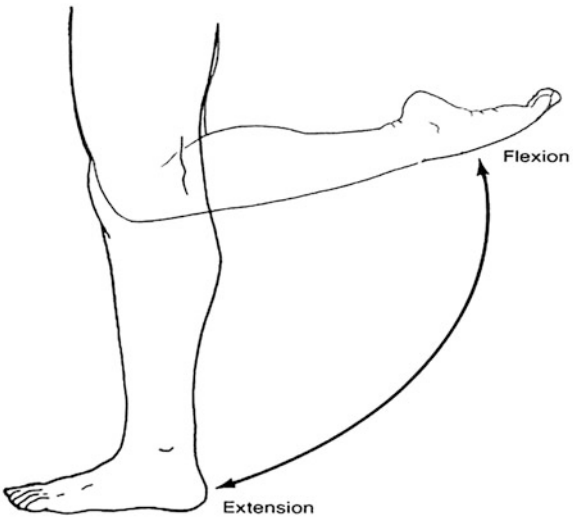


Fig. 6 Human knee joint movements

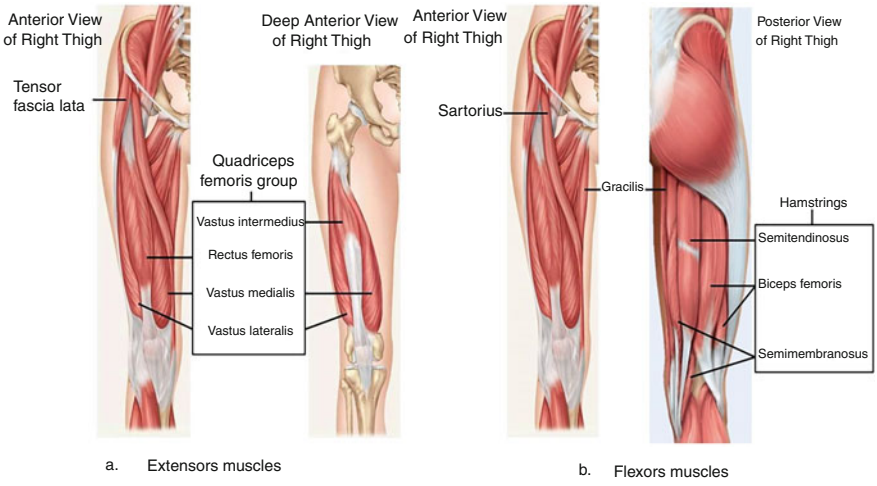


Fig. 7 Major muscles of the human knee

of five muscles located towards the anterior thigh. Their contraction induces the extension of the knee joint by pulling the calf and foot back to their anatomical position. During this motion, knee flexors are relaxed returning to their normal position. Knee extensors showed in Fig. 7b comprise the tensor fasciae latae and the quadriceps (rectus femoris (ReFe), vastus lateralis (VaLa), vastus intermedius (VaIn), and vastus medialis (VaMe)) (Abulhasan and Grey 2017).

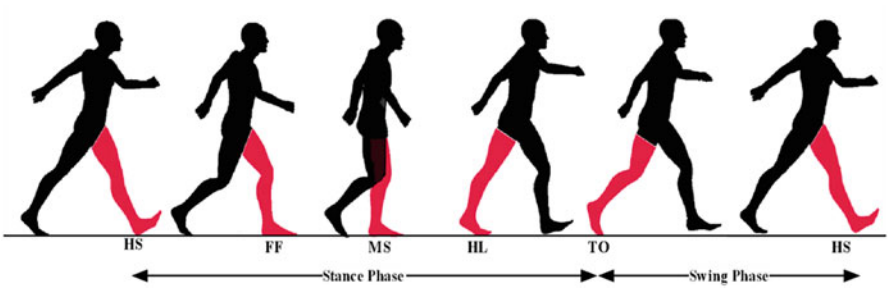


Fig. 8 Human gait cycle for a level walk

Human Gait Cycle

Walking is an essential part of human activity. It is depicted as a gait cycle, with a cyclic sequence of foot movements to move the body forward. Figure 8 depicts a human gait cycle for level walking with various phases and subphases.

A typical human gait cycle consists of two phases: the stance phase (when the foot is on the ground) and the swing phase (when the foot is in the air). Normally, the stance phase accounts for 55–60% of the total gait cycle, while the swing phase accounts for 40–45% of the total gait cycle. The stance phase is initiated with the heel-strike (HS) of the leading foot on the ground, while the trailing foot is still in contact with the ground. At this time, the body weight starts transferring to the leading foot. After that, flat-foot (FF) subphase starts in which the leading foot comes in full contact with the ground, and complete weight is shifted to the leading foot. This subphase ends when the entire human body is supported by the leading foot, and the trailing foot is about to leave contact with the ground. The midstance (MS) subphase starts, which ends when the opposite foot aligns with the stance foot. It is also considered the mid-subphase of the stance phase and occurs during the 25–30% of the complete gait cycle. After that, heel-off (HO) subphase starts which terminates with the toe-off (TO) of the same foot or at the start of the swing phase. During the swing phase, the foot is in the air, and the new gait cycle of the same foot starts when the HS event for the same foot occurs again (Winter 1991). During the different phases of the gait cycle, the ankle joint modulates its stiffness and angular position to provide sufficient propulsion force. This ankle joint angular position modulation ability helps humans to walk on different terrains (level, stair, and ramp). During level walking, the human ankle joint angle follows a specific profile, also known as the displacement profile. Figure 9 shows the normalized displacement profile during a complete gait cycle with the phases and events of the gait (Mai and Commuri 2016).

A General Control Structure for Prosthesis

The control structure, in essence, acts as a link between the prosthesis and the amputee's sensory-motor control loop, as well as the residual musculoskeletal system, which should be governed by the central nervous system.

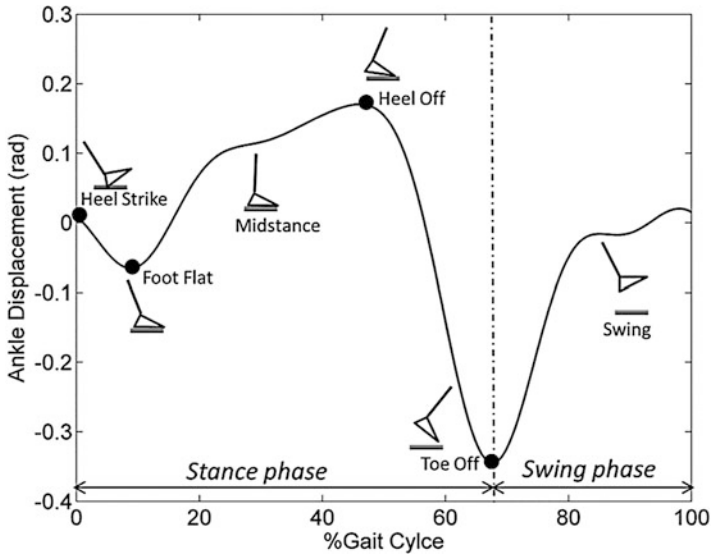


Fig. 9 Gait cycle with ankle joint angle during the level walk

For effective, efficient, and natural control of prostheses, it is vital to understand how able-bodied people regulate their movement in the regular course of events. Most of the time, human locomotion is dependent on fundamental patterns that are created at the spinal level (Zehr 2005; Pons et al. 2013) (Fig. 10). The CPG is a network of spinal interneurons that generates the primary motor pattern. The origin of locomotor pattern modulation can be traced back to the supraspinal or cortical levels, which also carry the accumulative effect of feedback from muscle spindles, mechanoreceptors lining joint capsules, and feedback from the skin, which detects pressure, pain, heat, and stretch, among other things. In addition, nerve fibers (sensory neurons) transmit information from the musculoskeletal system to the central nervous system (CNS), closing the feedback loop of the body's natural locomotor control (Crommert et al. 1998; Duysens 1998; Dietz 2002).

Figure 11 depicts a generalized control structure that can be applied to a variety of prosthetic or assistive devices. When a powered prosthesis is used in real time, the framework accurately represents the physical interactions and signal-level feedback loops that occur. Among the major control structure subsystems are the hierarchical control unit, the amputee, the prosthetic device, and the environment in which the amputee ambulates (Geethanjali 2016). Considerations for amputee safety are also taken into account within each subsystem.

The environment in which the prosthesis will be used (surface condition, terrain type, etc.) limits the motions that can be performed with the assistance of a prosthetic device. The high-level control system must be able to differentiate between the amputee's current activity and the desired activity. This has two effects: The current motor task (standing, level walking, and stair descending) is recognized by activity mode recognition, whereas intended activity recognition allows for direct volitional control, allowing the amputee to voluntarily change the state of his or her prosthesis.

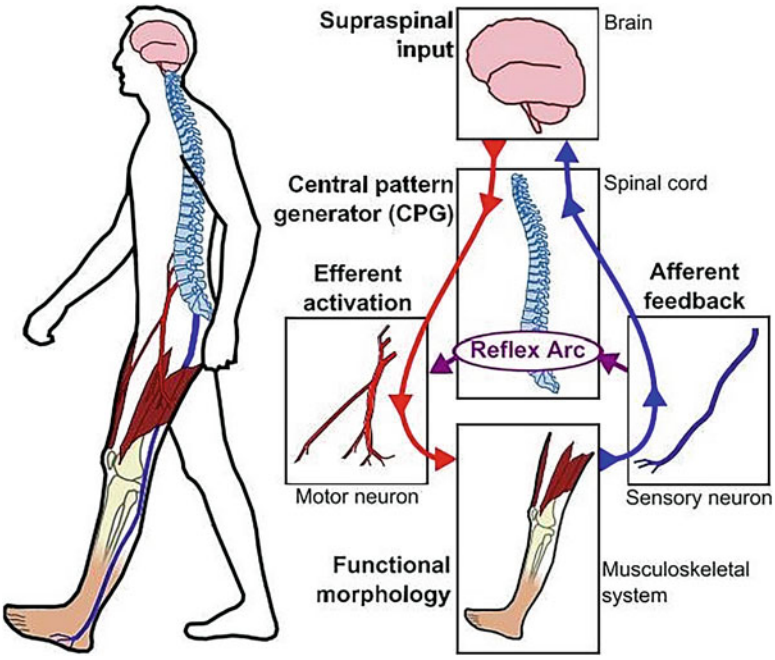


Fig. 10 Locomotion control structure of abled body person

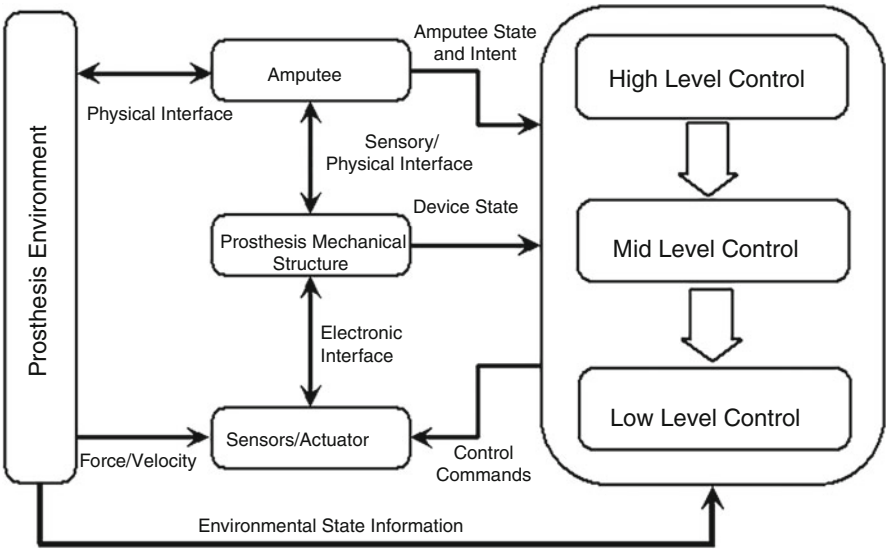


Fig. 11 General control structure for prosthesis control

Integration of both is possible and advantageous, in which volitional control affects prosthetic behavior while participating in a specific activity. The high-level controller may convert amputee activity intents into desirable device states via the mid-level

controller, which can then be tracked by the low-level controller. At this level, the amputee's status within the gait cycle is determined, and a specific control law (position/velocity, torque, impedance, or admittance controller) is applied (Eilenberg et al. 2010). So, once the required condition of the prosthesis has been determined, it is passed on to the next level of the controller, which is the low-level controller. When the error with the current state is computed, a command signal is sent to the actuator(s) to minimize the error with the current state. The structure of the low-level controller can be either feedforward or feedback in nature, depending on how the kinematic and kinetic parameters of the prosthesis are taken into account (Huang et al. 2011). More advanced prosthetic devices can also provide artificial sensory feedback to the amputee, allowing them to be fully integrated with the physiological control system. It is used at the highest level of the controller. The efficient locomotion identification method is expected to be successful in increasing the efficiency of the control system.

Methodology and Experiment Design

Figure 12 shows the block diagram representation of the methodology opted for SEMG operated prosthesis design.

SEMG signal, lower limb joint angle, and gait phase information of the subject have been used to operate the prosthesis. Here, the SEMG signal has been utilized in

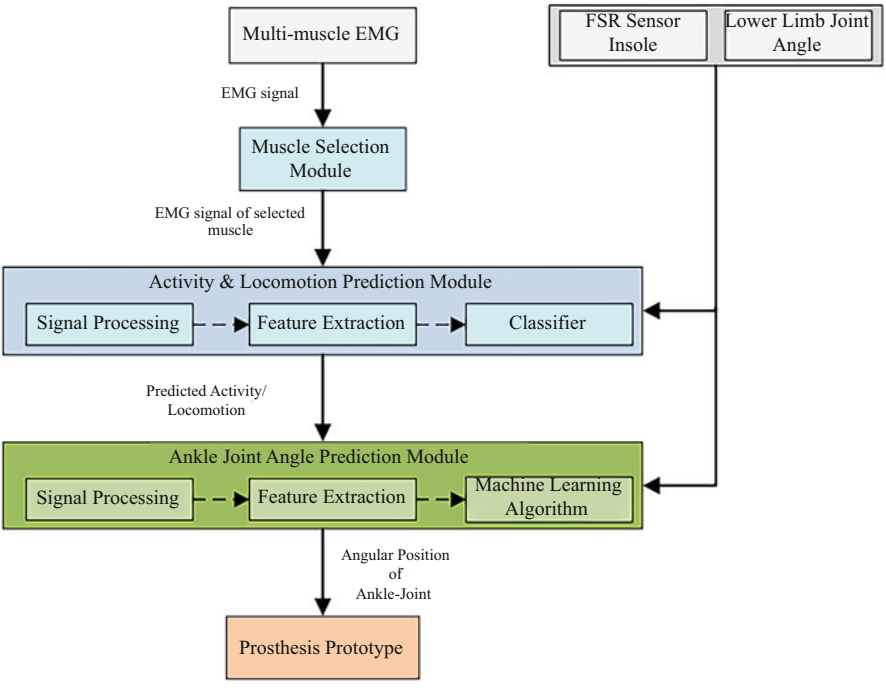


Fig. 12 Methodology adopted for the development of ankle-foot prosthesis

two ways: to extract the locomotion intension of the subject and to estimate the angular position of the joint. A multichannel EMG signal acquisition module has been used to acquire SEMG signals from various lower limb muscles (below-knee and above-knee). The lower limb joint angle information has been recorded using IMU units, and the gait events heal-strike (HS) and toe-off (TO) have been detected with force sensor resistor (FSR) sensors. The human musculoskeletal structure is highly redundant. Therefore, the recorded SEMG signal of various lower limb muscles has been analyzed to select the appropriate set of muscles for ankle joint movements. The muscles have been selected based on their reliability and consistency of SEMG signal for repetitive dorsiflexion (DF) and plantarflexion (PF) ankle movements, i.e., gastrocnemius and tibialis anterior (Gupta and Agarwal 2021).

The opted control structure is inspired by the biological control structure of an abled human and consists of two control modules, namely, high-level and low-level control modules. The high-level control module is at the top of the hierarchy and is sometimes also termed as high-level controller. It perceives the user's activity and locomotion intention like standing, level walking, stair ascent/descent, or ramp ascent/descent. The output of the high-level control module is further transmitted to the low-level control module. The low-level control module is also referred to as a low-level controller. The output of the low-level control module depends on the input of the high-level control module. This control module estimates the desired position/torque/velocity of the prosthesis for a given set of inputs. An activity and locomotion identification module has been designed to predict the activity and locomotion intention of the subject. This module serves as the high-level controller of the prosthesis and provides movement anticipation ability to the prosthesis. It utilizes the EMG signal as a source of neural information and KA signal to predict the activity and locomotion intention of the subject. As shown in Fig. 12, it has three subsystems, namely, signal processing, feature extraction, and classifier. In the signal processing subsystem, the filtering of input signals has been done to remove the noise. After that, the filtered signal is passed through the feature extraction subsystem. In this subsystem, the appropriate information/features from the filtered signal have been extracted to construct the feature vector. The feature vector has been used as the input to train and test the classifier.

Further, an ankle joint angle estimation module has been designed to estimate the angular position of the ankle joint. It has three subunits, namely, signal processing, feature extraction, and machine learning algorithm. An individual angle estimation module has been trained and tested for each locomotion mode. It requires SEMG signal, KA, and locomotion information as the input to estimate the ankle joint angle. This module is considered a low-level controller of the prosthesis, which provides the desired angular position of the prosthesis. Finally, an electrical actuator-operated prosthesis prototype has been realized and controlled using the predicted angular position information in the laboratory environment.

Data Acquisition Protocol

The literature strongly emphasized using biopotential signals along with the mechanical sensors signals to achieve efficient control over prosthetic devices. Hence, for current research work, EMG, lower limb kinematics, and gait events have been acquired. Lower limb joint angle (knee and ankle) signal has been calculated using three, 6-DoF, IMU sensor (MPU-6050) signals attached to the mid-position of the thigh, shank, and foot of the subject. The knee and ankle joint angles have been calculated using IMU sensor signals (Saito et al. 2009; Kumar et al. 2018). These joint angles have been utilized for further analysis in this research work. FSR sensors' foot insole has been used to capture HS and TO events of the gait cycle. SEMG signal has been recorded using a multichannel SEMG acquisition system developed around ADS 1298 (Gupta and Agarwal 2017). The sampling frequency for SEMG, IMU, and FSR sensors has been kept at 1000 Hz (Huang et al. 2011; Zhang et al. 2011). SENIAM standard have been followed for SEMG signal recording, which includes electrode selection, electrode placement, skin preparation, and muscle identification (SENIAM 2016).

SENIAM Recommendations

A good electrode-skin contact, a low electrode-skin impedance, and stable behavior in time are all requirements for the material used to establish the contact with the skin during the electrode-skin contact process (i.e., for impedance and chemical reactions at the skin interface). According to previous research, many types of materials are utilized, including Ag/AgCl, AgCl, Ag, Au, and others, with Ag/AgCl electrodes being the most commonly used electrodes. These provide a consistent transition with little noise and are by far the most widely accessible in the commercial marketplace. Electrodes are frequently used in conjunction with electrode gel. Electrode gel and paste are used to minimize the impedance between the electrode and the skin. Performance of both pre-gelled electrodes and non-gelled is at par. However, a significant amount of time and effort is required for the application of non-gelled electrodes (which must be gelled before attaching to a muscle). If the procedure is not followed properly, there is a considerable likelihood of poor SEMG recordings. It was decided to employ pre-gelled Ag/AgCl electrodes in differential mode, as shown in Fig. 13.

Inter-Electrode Distance (in Millimeters)

It is defined as the distance between the conducting areas of two bipolar electrodes measured from one to the other. The effect of inter-electrode distance on signal pick-up area and crosstalk is an important topic in the literature. When applying the bipolar SEMG electrodes around the specified sensor site, SENIAM recommends spacing the electrodes 20 mm apart. When bipolar electrodes are placed on relatively small muscles, the space between the electrodes should be no more than one-fourth the length of the muscle fibers being studied. In this manner, it is possible to avoid unstable recordings caused by tendon and motor endplate effects.

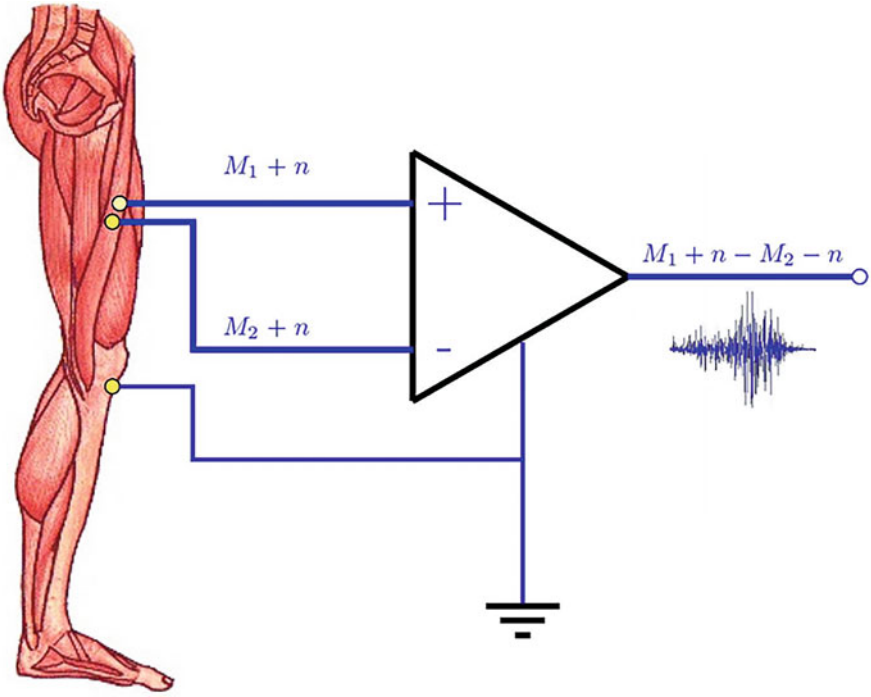


Fig. 13 Differential mode configuration for SEMG recording

Skin Preparation

Following the selection of the sensor, the patient's skin must be prepared to ensure a good electrode-skin connection. To obtain better SEMG recordings (in terms of amplitude characteristics), fewer and smaller artifacts (electrical interference), less risk of electrode imbalance (smaller common mode disturbance signal), less noise (better S/N ratio), good electrode-skin contact are required. It is critical to properly prepare the skin to improve the electrode-skin interaction. Various (combinations of) skin preparation techniques, such as shaving, washing with alcohol, rubbing with gel, and sandpaper, are described in the literature. According to the SENIAM instructions for skin preparation, if the skin area on which the electrodes are to be put is covered with hair, the patient should have his or her hair removed. Afterward, alcohol is applied to the skin, which should be allowed to evaporate for several minutes to ensure that the skin is completely dry before the electrodes are applied.

Examining the Connection

Once the SEMG electrodes have been placed and secured, they can be linked together (along with the reference electrode) and a clinical test can be carried out to determine whether the electrodes have been placed and connected correctly to the muscle to record a reliable SEMG signal. The clinical test ensures that the examined muscle will be active (under normal circumstances) during the procedure. SENIAM

suggests that each muscle be subjected to the clinical examination. The clinical tests that are discussed are universally acknowledged muscle tests that ensure the activation of the examined muscle (under normal circumstances) and are, therefore, recommended. The clinical test must begin with the beginning posture stated in the SENIAM recommendations for sensor sites, and it must be completed by those recommendations. The clinical tests described here are not "selective" contractions, which are contractions in which just the desired muscle is activated and all other muscles remain dormant.

Noise-Related Issues in SEMG Signal Acquisition

The SEMG signal is stochastic in character, with an amplitude that ranges from a few millivolts to several hundred millivolts. Noise has a significant impact on it. SEMG signals are primarily affected by two forms of noise: noise in the environment and noise generated by the device itself.

Noise in the Environment

Radio and television transmission, fluorescent lamps, and electrical power sources (with a frequency of 50/60 Hz) are examples of electromagnetic equipment that emit noise. When this noise is present, its magnitude can be of the order of two to three times that of the SEMG signal. The use of high common mode rejection ratio (CMRR) differential amplifiers, the use of shielded leads, and the usage of right leg drive (RLD) circuits can all help to significantly reduce this noise. When the electrodes are moved, motion artifacts are produced. This is caused by the movement of the wires connecting them to the amplifier, as well as the interface between the skin and the electrode surface. The amplitude of the motion artifacts noise is almost equal to the amplitude of the EMG signal. Recessed electrodes have the potential to considerably reduce movement artifacts. It is also possible to decrease this noise by properly preparing the skin before signal capture. In addition, the frequency range of motion artifacts noise is 0–10 Hz, which is a wide range. Incorporating a high-pass filter, either in hardware or software, can help to reduce this noise to an acceptable level. Wavelet de-noising can also be used to minimize motion artifact noise in electromyographic signals.

Noise Generated by the Device Itself

Electrical noise is generated by every electronic equipment. Noise with frequency components ranging from zero Hz to several thousand Hz is present in this sample. The use of high-quality components is the only way to minimize this noise.

SEMG signal is prone to inherent volatility due to the following reasons: When motor units fire, the rate at which they do so is quasi-random in nature, and this noise has a frequency range of 0–20 Hz. This noise can be reduced by incorporating a high-pass filter with a frequency of 20 Hz in either hardware or software.

Figure 14 shows the picture of the fully instrumented subject. The data acquisition has been conducted during the two experiments, and for each experiment, ten subjects have been recruited. All the subjects who participated in the research work were volunteers and had no history of neural disorders. The signal recording

Fig. 14 Fully instrumented subject

protocol was explained to all the subjects individually. Further, written consent was signed by each subject before the experiment. They were permitted to withdraw at any time during the data collection. The method adopted for signal acquisition in the present research work was purely noninvasive. The experiment procedure adopted was in accordance with the Helsinki declaration, 2013, and duly approved by the institutional research board.

Experiment Design

The experiment should be designed in such a way that it should cover maximum applications for a wide range of the population. The experiment was designed to

acquire the signals for different dynamic lower limb activities. Figure 15 illustrates the daily life activity % of a healthy subject over a period of 12 hr after waking up (Pitta et al. 2005). It has been observed that a major part of daily life is occupied by sitting (SIT), standing (STD), and walking (WLK) activities. The walking activity includes different locomotion like level walking (LW), stair ascent (SA), stair descent (SD), ramp ascent (RA), ramp descent (RD), etc. During sitting and standing activities, there is merely any participation of ankle movement. However, for the walking activity, the subject has to pass through the standing position/activity. Hence, for the present experiment, STD and WLK activities have been considered. WLK activity included five locomotion – LW, SA, SD, RA, and RD.

Figure 16 depicts the activity sequence opted for the signal acquisition. For each subject, SEMG signal from six lower limb muscles, tibialis anterior (TA), medial gastrocnemius (GAST), vastus lateralis (VL), rectus femoris (RF), semitendinosus (SEM), biceps femoris longus (BFL), knee and ankle joint angle along with FSR foot insole signal were recorded. The recorded signal has been utilized for locomotion identification module design and ankle joint angle estimation. Ten trials were recorded for each subject. Appropriate resting time was provided in between the sessions as demanded by the subjects.

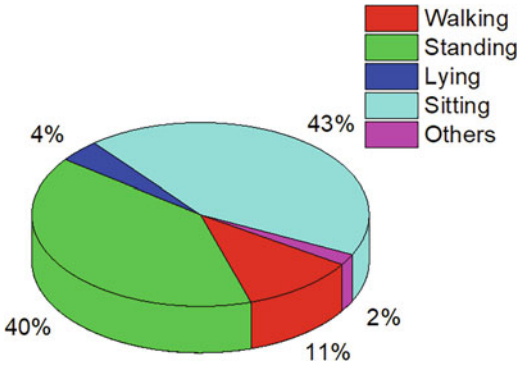


Fig. 15 Activity % of the healthy subject during 12 hr after waking up

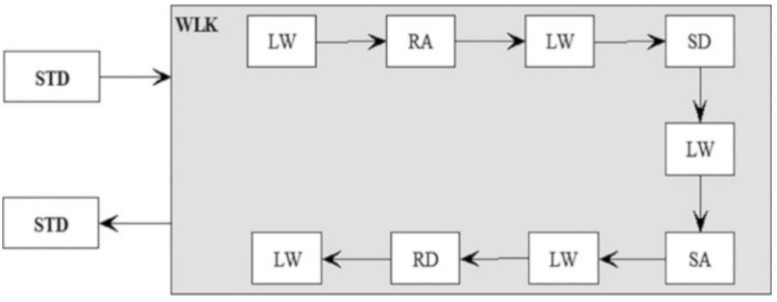


Fig. 16 Activity sequences followed for signal acquisition

Signal Processing and Module Design

The locomotion identification module and the joint angle prediction module are the two primary modules that must be designed. Both modules are distinct; the locomotion identification module is a discrete-class classification problem, whereas the joint angle prediction module employs two approaches. The first method estimates the discrete class of KA, whereas the second method estimates the ankle angle continuously. As a result, different signal processing methodologies were used for both modules.

Locomotion Identification Module

Online processing of the collected SEMG signal was carried out; Fig. 17 depicts the modules that were involved in the processing. There are four basic kinds of modules to consider: signal preprocessing (filtering, windowing), information extraction (feature extraction, feature normalization, feature vector preparation), classification, and validation (Gupta and Agarwal 2019b).

The signal was preprocessed, including filtering and segmentation, to extract features (windowing). Filtering was used to remove undesirable parts of the signal. A 4th order Butterworth bandpass filter of frequency range with 20-500Hz was used to filter the acquired signal. The signal was then segmented, which changed it from nonstationary to stationary. Because the overlapping windowing technique (Fig. 18) has shown promising results in the literature, it was used in this study with a window length of 256 ms and an offset of 32 ms to achieve the desired results (Huang et al. 2009; Huang et al. 2011; Gupta and Agarwal 2017, 2018a, b, 2019a, b, 2021; Dhindsa et al. 2019).

Each segment of the signal was treated as a training set, and the features for each segment were calculated separately. Thirty-eight characteristic features were identified in both time and frequency domains. The features are tabulated in Table 4 (Gupta and Agarwal 2019a). Further, min-max normalization method (Eq. 1) (Gupta

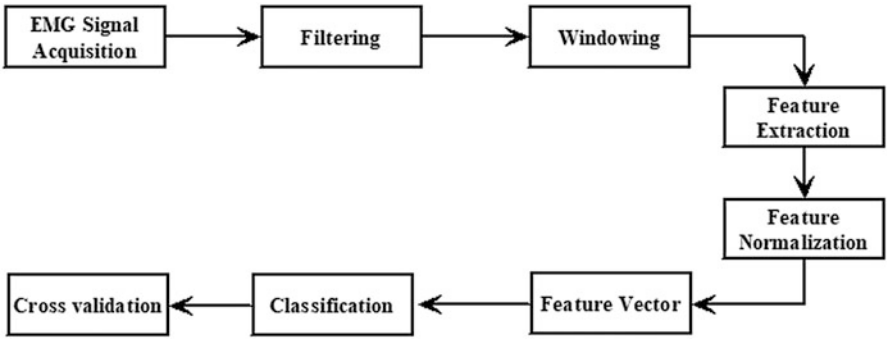


Fig. 17 Segmentation process of EMG signal

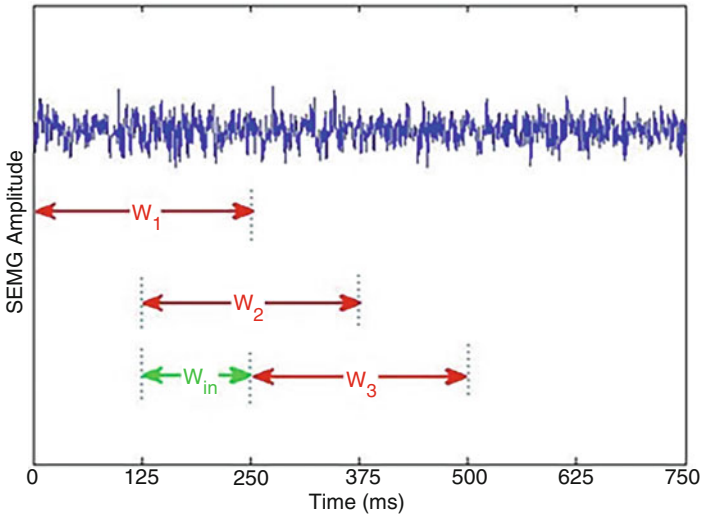


Fig. 18 Segmentation process of EMG signal

et al. 2021) was employed to normalize the features, which resulted in a significant improvement in classifier performance.

$$F_{\text{norm}} = \frac{F - \min(F)}{\max(F) - \min(F)} \tag{1}$$

where F, min(F), and max(F) are individual feature values, the minimum value of feature, and maximum value of feature, respectively.

The feature vectors were created by concatenating the normalized features for each segmented window, which resulted in the final feature vectors (Fig. 19). The performance of each feature was analyzed and compared for both muscles. The performance of five feature vectors with varying numbers of features was compared. These were time-domain features (TD), frequency domain features (FD), predictive information features (ARC), time and frequency domain features (TFD), and frequency and predictive information features (FAR) with 23, 11, 4, 34, and 15 features, respectively.

Six widely used classifiers, which included the support vector machine (SVM), the linear discriminant analysis (LDA), the artificial neural network (ANN), the decision tree (DT), the k-nearest neighbor (KNN), and the naive bias (NB) classifier, were used to evaluate and compare the overall performance of different feature vectors (Yuan et al. 2015; Joshi et al. 2016; Afzal et al. 2017; Gupta and Agarwal 2018b). Classification accuracy is a performance parameter that is widely used in the evaluation of the performance of prediction model performance. It is defined as the ratio of the total number of correctly categorized events to the total number of tested occurrences (Eq. 2). The performance of the classifier was evaluated in this study using a k-fold cross-validation approach. As part of the k-fold cross-validation

Table 4 Name and definition of extracted features

Sr. No.	Abv.	Name of feature	Definition
1	IEMG	Integrated EMG	$IEMG = \sum_{i=1}^{N} \lvert X_i \rvert$
2	MAV	Mean absolute value	$MAV = \frac{1}{N} \sum_{i=1}^{N} \lvert X_i \rvert$
3	MAV1	Modified mean absolute value type 1	$MAV1 = \frac{1}{N} \sum_{i=1}^{N} w_i \lvert X_i \rvert$ $w_i = \begin{cases} 1 & if\ 0.25N \le i \le 0.75N \\ 0.5 & \text{otherwise} \end{cases}$
4	MAV2	Modified mean absolute value type 2	$MAV2 = \frac{1}{N} \sum_{i=1}^{N} w_i \lvert X_i \rvert$ $w_i = \begin{cases} 1 & if\ 0.25N \le i \le 0.75N \\ \frac{4i}{N} & \text{elseif } i < 0.25N \\ \frac{4(i-N)}{N} & \text{otherwise} \end{cases}$
5	SSI	Simple square integral	$SSI = \sum_{i=1}^{N} X_i^2$
6	VAR	Variance of EMG	$VAR = \frac{1}{N-1} \sum_{i=1}^{N} X_i^2$
7	TM3	Absolute value of the third temporal moment	$TM3 = \left\lvert \frac{1}{N} \sum_{i=1}^{N} X_i^3 \right\rvert$
8	TM4	Absolute value of the fourth temporal moment	$TM4 = \left\lvert \frac{1}{N} \sum_{i=1}^{N} X_i^4 \right\rvert$
9	TM5	Absolute value of the fifth temporal moment	$TM5 = \left\lvert \frac{1}{N} \sum_{i=1}^{N} X_i^5 \right\rvert$
10	RMS	Root mean square	$RMS = \sqrt{\frac{1}{N} \sum_{i=1}^{N} X_i^2}$
11	LOG	Log detector	$LOG = e^{\frac{1}{N} \sum_{i=1}^{N} \log(\lvert X_i \rvert)}$
12	WL	Waveform length	$WL = \sum_{i=1}^{N-1} \lvert X_{i+1} - X_i \rvert$
13	AAC	Average amplitude change	$AAC = \frac{1}{N} \sum_{i=1}^{N-1} \lvert X_{i+1} - X_i \rvert$
14	DASDV	Difference absolute standard deviation value	$DASDV = \sqrt{\frac{1}{N-1} \sum_{i=1}^{N-1} (X_{i+1} - X_i)^2}$
15	ZC	Zero crossing	$ZC = \sum_{i=1}^{N-1} [sgn(X_i \times X_{i+1}) \cap \lvert X_i - X_{i+1} \rvert \ge \text{threshold}]$ $sgn(X) = \begin{cases} 1, & if\ X \ge \text{thrsehold} \\ 0, & \text{otherwise} \end{cases}$

(continued)

Table 4 (continued)

Sr. No.	Abv.	Name of feature	Definition
16	MYOP	Myopulse percentage rate	$MYOP = \frac{1}{N}\sum_{i=1}^{N}[f(X_i)]$ $f(X) = \begin{cases} 1, & if\ X \geq \text{threshold} \\ 0, & \text{otherwise} \end{cases}$
17	WAMP	Willison amplitude	$WAMP = \sum_{i=1}^{N-1}[f(\lvert X_n - X_{n+1}\rvert)]$ $f(X) = \begin{cases} 1, & if\ X \geq \text{threshold} \\ 0, & \text{otherwise} \end{cases}$
18	SSC	Slope sign change	$SSC = \sum_{i=2}^{N-1}[f[(X_i - X_{i-1}) \times (X_i - X_{i+1})]]$ $f(X) = \begin{cases} 1, & if\ X \geq \text{threshold} \\ 0, & \text{otherwise} \end{cases}$
19	H1	Histogram of EMG	Histogram of EMG
20	H2		
21	H3		
22	H4		
23	H5		
24	MNF	Mean frequency	$MNF = \sum_{j=1}^{M} f_j P_j / \sum_{j=1}^{M} P_j$
25	MDF	Median frequency	$\sum_{j=1}^{MDF} P_j = \sum_{j=MDF}^{M} P_j = \frac{1}{2}\sum_{j=1}^{M} P_j$
26	PKF	Peak frequency	$PKF = \max(P_j), j = 1, 2. \ldots . , M$
27	MNP	Mean power	$MNP = \frac{\sum_{j=1}^{M} P_j}{M}$
28	TTP	Total power	$TTP = \sum_{j=1}^{M} P_j = SM0$
29	SM1	First spectral moments	$SM1 = \sum_{j=1}^{M} P_j f_j$
30	SM2	Second spectral moments	$SM2 = \sum_{j=1}^{M} P_j f_j^2$
31	SM3	Third spectral moments	$SM3 = \sum_{j=1}^{M} P_j f_j^3$
32	FR	Frequency ratio	$FR = \sum_{j=LL}^{UL} P_j / \sum_{j=LH}^{UH} P_j$
33	PSR	Power spectrum ratio	$PSR = \frac{P_0}{P} = \frac{\sum_{j=f_0-n}^{f_0+n} P_j}{\sum_{j=-\infty}^{\infty} P_j}$
34	VCF	Variance of central frequency	$VCF = \frac{1}{SM0}\sum_{j=1}^{M} P_j \left(f_j - f_c\right)^2 = \frac{SM2}{SM0} - \left(\frac{SM1}{SM0}\right)^2$

(continued)

Table 4 (continued)

Sr. No.	Abv.	Name of feature	Definition
35	AR1	Auto-regressive coefficients	$X_i = \sum_{p=1}^{P} a_p X_{i-p} + W_i$ p is order of the AR model, w_i is white noise error term
36	AR2		
37	AR3		
38	AR4		

X_i represents the EMG signal in a segment i, N denotes length of EMG signal, f_j is frequency of the spectrum at frequency bin j, P_j is the EMG power spectrum at frequency bin j, and M is length of the frequency bin. *UL, LL* upper and lower cut-off frequency of lower frequency band, *UH, LH* upper and lower cut-off frequency of upper frequency band

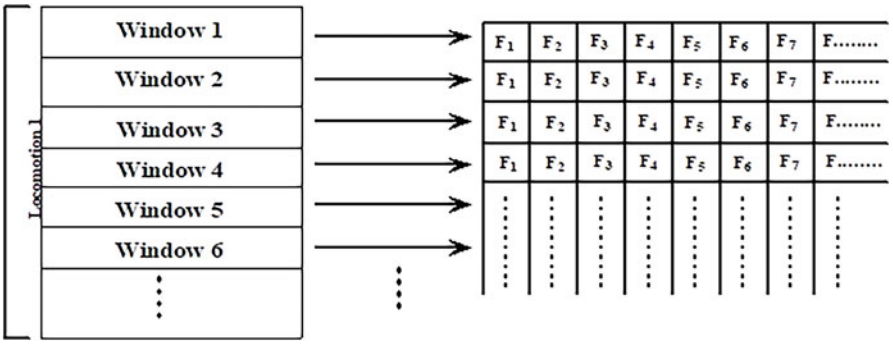


Fig. 19 Feature vector generation for each segment

scheme, all of the data was partitioned into five separate parts, each of which was separated into k-1 parts for training the classifier and only one part for testing the classifier. The tenfold cross-validation method was employed in the research provided here.

$$\%Acc = \frac{\text{No.of correctly classified testing data}}{\text{Total no. of applied testing data}} \times 100\% \quad (2)$$

Knee Angle Class Estimation

In literature, joint angle prediction of the lower limb has been proposed as a regression problem (Dhindsa et al. 2016a, 2022; Delis et al. 2009). In the present study, knee angle estimation is proposed as a classification problem and ankle angle is estimated continuously using NARX neural network. The SEMG signals from four muscles, including the VL, BFL, RF, and SEM from the lower limb, were recorded at 2048 Hz. Bipolar Ag/AgCl electrodes with 20 mm inter-electrode distances were used. The sensor positions were the same as described in our previous study (Dhindsa et al. 2016b). SENIAM recommendations for the placement of

electrodes were followed. The study was conducted on a custom-built gait course incorporating three locomotion modes: SD, PW, and SA. The trials were conducted at the institute's corridor. Before the beginning of the trials, the subjects were asked to remain standstill and SEMG data was recorded for 1 min. Subjects then performed SD, turned 90° towards left, performed PW, took 180° turn and performed PW, turned right 90° and performed SA, and came to rest. Ten trials were recorded for each subject daily. The HS and TO events detected using a force sensor resistor placed in the shoe sole as shown in Fig. 20 (Aggarwal et al. 2018), were used to sample the data into gate cycles.

Figure 21 depicts the SEMG signal and KA acquired during the study. For SD and SA, the RoM of KA varies from 0° to 90°; for PW, the RoM for KA is from 0° to 70°. Accordingly, KA was quantized into six different levels: L1 ($KA < 9°$), L2 ($9° < KA \leq 27°$), L3 ($27° < KA \leq 45°$), L4 ($45° < KA \leq 63°$), L5 ($63° < KA \leq 81°$), and L6 ($KA > 81°$). Fifteen features were extracted from the segmented/windowed SEMG signals for each muscle. These included six time domain features, five frequency domain features, and four autoregressive (AR) coefficients (ARn) (Dhindsa et al. 2019).

Figure 21 depicts the SEMG signal and KA acquired during the study. For SD and SA, the range of motion of KA varies from 0° to 90°; for PW, the range of motion for KA is from 0° to 70°. Accordingly, KA was quantized into six different levels: L1 ($KA < 9°$), L2 ($9° < KA \leq 27°$), L3 ($27° < KA \leq 45°$), L4 ($45° < KA \leq 63°$), L5 ($63° < KA \leq 81°$), and L6 ($KA > 81°$). Fifteen features were extracted from the segmented/windowed SEMG signals for each muscle. These included six TDF, five frequency domain features, and four autoregressive (AR) coefficients (ARn) (Dhindsa et al. 2019).

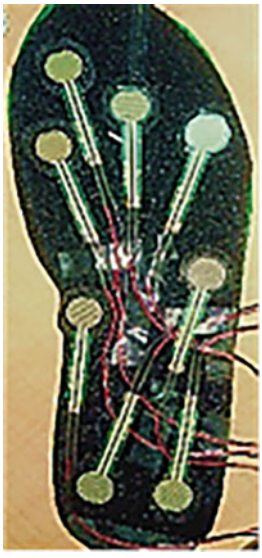


Fig. 20 Shoe sole for detecting heel strike and toe off

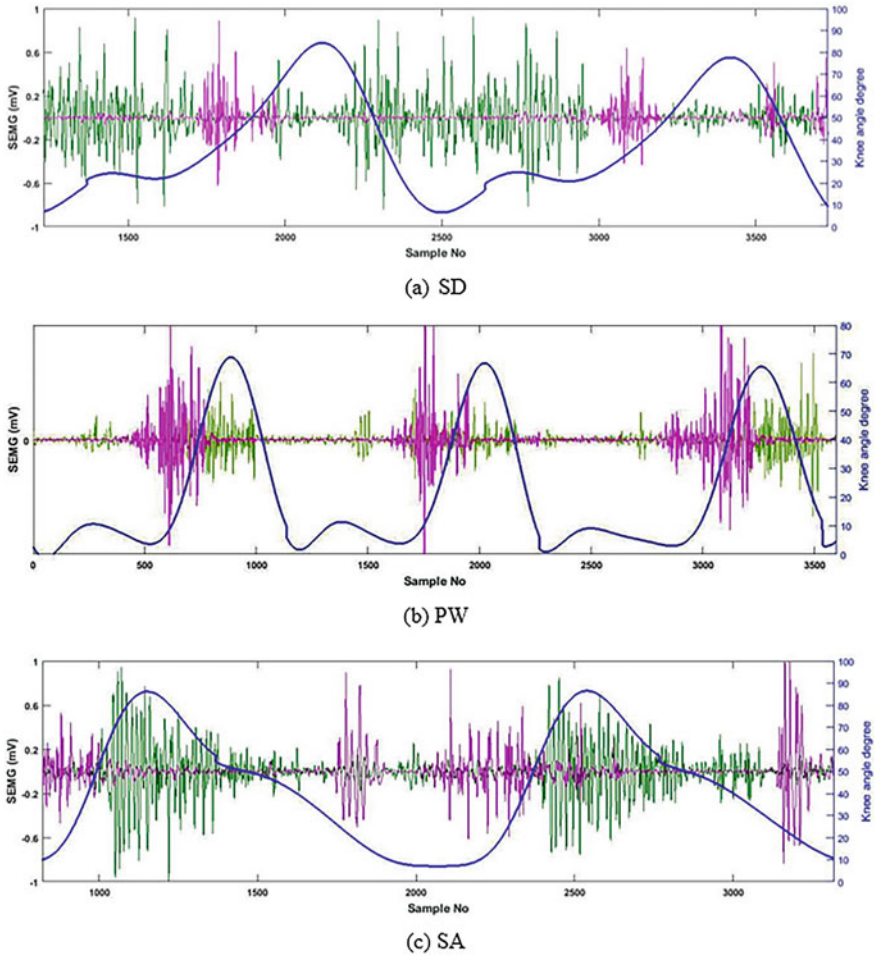


Fig. 21 SEMG signal and KA acquired for different activities (*purple*: RF, *green*: VL, and *blue*: KA)

Continuous Estimation of Ankle Angle Module

Figure 22 depicts a block diagram containing the procedures that are used to calculate the ankle joint angle estimation. The EMG signal is, by its very nature, a low-power stochastic signal. During the signal acquisition process, it is easily contaminated by a variety of disturbances. Before using the EMG signal for further investigation, it is required to preprocess it first. To begin with, the raw EMG signal has been filtered by applying zero-lag, fourth order, bandpass Butterworth filter with a frequency band of 20–500 Hz, followed by application of 50-Hz notch filter (Dhindsa et al. 2016b). To eliminate speed variations between trials, the filtered

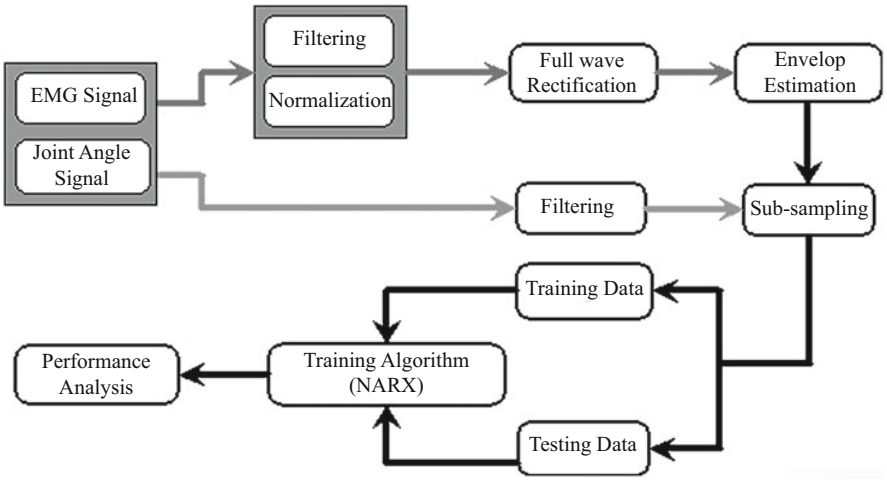


Fig. 22 Signal processing steps for ankle joint angle estimation

EMG signal of each individual muscle has been normalized using maximum normalization (max normalization) (Yokoyama et al. 2016). By nature, the EMG signal is fundamentally random, and it oscillates around the zero-mean value. As a result, full-wave rectification of the signal has been conducted using Eq. 3 to gather information from both side signals.

$$EMG_{\text{rect}}(t) = \begin{cases} EMG(t) if EMG(t) > 0 \\ -1 * EMG(t) if EMG(t) \leq 0 \end{cases} \tag{3}$$

where $EMG(t)$ is normalized EMG signal, and $EMG_{rect}(t)$ is a full-wave rectified EMG signal.

The amplitude of the EMG signal is directly proportional to the intensity of muscle contraction, and it serves as an indicator of change in joint angle with low-frequency characteristics. To retain only low-frequency information in the EMG signal, the rectified EMG signal was processed through a zero-lag, second-order, low-pass Butterworth filter with a cutoff frequency of 10 Hz. A very low-amplitude (mV) EMG signal is captured noninvasively from the human body and is used to diagnose various medical conditions. As a result, it has a higher likelihood of becoming corrupted as a result of environmental noise and electrode movement while being recorded. As opposed to this, joint angles are measured using IMU sensor signals, which have a much higher amplitude (V) than the EMG signal and hence produce more accurate results. As a result, as compared to EMG signals, joint angle signals from the lower limb are more robust, powerful, and less susceptible to noise. To reduce high-frequency noise caused by sensor movements and involuntary limb movement, it has been routed through a zero-lag, second-order, low-pass Butterworth filter with a cut-off frequency of 10 Hz and zero delay. Subsampling at 100 Hz has been performed on the encapsulated EMG signal and lower limb joint angle signals to facilitate further analysis. Finally, the whole data of

each subject has been divided into two sections, which are referred to as training and testing data, respectively. Various training and testing data sets have been used to train and test the learning algorithm, respectively.

Neural networks (NNs) have already demonstrated their effectiveness for a wide range of issues including pattern categorization, optimization, prediction, and grouping, among others. Essentially, it is a black-box modelling technique with the capability of translating a nonlinear mapping between multidimensional input and output space, even when the relationships between those input and output spaces are unknown. NNs are constructed as a mathematical model that is dependent on past information of the system to be described in order to function properly. Given this, the current research subject (predicting the ankle joint angle) can be thought of as a time-series prediction problem with cyclic characteristics.

Nonnegative time series problems have traditionally been predicted by NARX neural networks, which are effective. Many other nonlinear dynamic systems, including time series modelling, had been modelled with it in the past. It is regarded as a dynamic recurrent neural network (DRNN) since it has feedback links that connect many layers of the network together. As an added feature, it allows for the incorporation of previous values of anticipated or/and actual time series or/and predictors. This characteristic of the NARX neural network makes it particularly well suited for the current study challenge, in which the time history information of the inputs (EMG and KA) and output (ankle joint angle) contains a substantial quantity of information about the ankle joint. In the case of a NARX neural network model, the overall mathematical relationship between inputs and outputs is given by Eq. 4 (Leontaritis and Billings 1985). It is used to continuously map EMG signal and KA into ankle joint angle.

$$y(t) = f(x(t-1), x(t-2), \ldots \ldots \ldots x(t-t_d),\ y(t-1), y(t-2) \ldots \ldots \ldots y(t-t_d)) \quad (4)$$

where y(t) is estimated output at time t, x(t) is N-dimensional input vector of two-channel EMG signal and KA, td is delay order in input and feedback output, and f is functional relation relating inputs and output. The NARX model is depicted in Fig. 23 as a basic block diagram, consisting of an input layer, one hidden layer

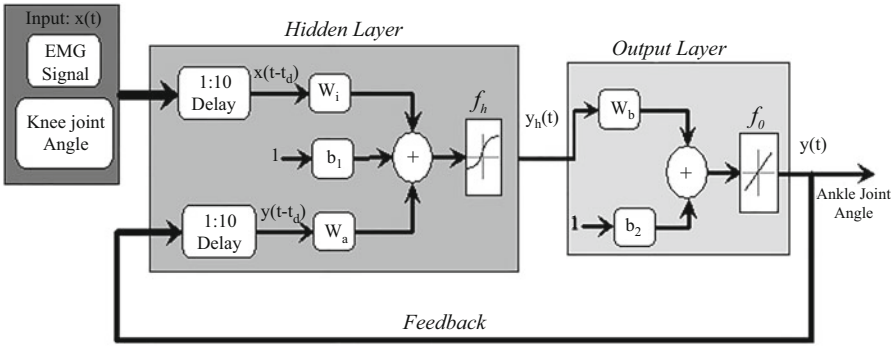


Fig. 23 Basic block diagram of the NARX model applied for ankle joint angle prediction

with nonlinear units, and an output layer with linear units. The input (x(t)), which contains EMG activity from two muscles as well as a KA signal, has been sent over tapped delay lines, together with an ankle joint angle that has been previously determined, to a hidden layer with nonlinear units ($y_h(t)$). Hidden layer output ($y_h(t)$) has been further processed through liner units to estimate the ankle joint angle ($y_h(t)$) over time. The input layer of the model has an encapsulated EMG signal collected from two below-knee muscles, KA, and an estimated ankle joint angle as feedback, whereas the output layer has a single output corresponding to the estimated ankle joint angle. All of the model's inputs have been supplied to the model with an additional delay, and the estimated output has been used as feedback (Gupta et al. 2020).

Using a supervised learning (back-propagation) approach, the proposed NARX model has been trained to reduce the mean square error (MSE) between the predicted and real ankle joint angles. According to Fig. 23 and Eq. 4, the performance of the model is dependent on the number of neurons in the input layer, the order of the delay, as well as the number of neurons in the hidden layer, among other factors. Number of neurons in the input layer is derived by multiplying the number of input channels (N), the number of output feedback (l), and the order of delay (o) together (m). For the time being, the number of input channels and output feedback is fixed at three (two-channel EMG signal and ankle joint angle) and one output feedback (ankle joint angle), respectively. The sequence of delay and the number of neurons in the hidden layer are chosen as 2 and 10, respectively (Gupta et al. 2020).

Performance Measures

The proposed method's performance has been examined for each type of locomotion and for each subject. To evaluate the performance of the training model, the LOOCV technique has been used. The performance of the angle estimation model has been evaluated using two performance measures: the root mean square error (RMSE) and the coef coefficient (CCoef). The RMSE has been quantified using Eq. 5. It is the difference between the estimated and actual joint angle, and a smaller value indicates greater accuracy in angle estimation. CCoef has been calculated using Eq. 6, and it represents the degree to which the actual and estimated joint angles are in agreement. Its value ranges from 0 to 1, with higher values indicating greater consistency (Gupta et al. 2020).

$$\mathrm{RMSE} = \sqrt{\frac{\sum_{i=1}^{n} \left(\theta_{act}(i) - \theta_{est}(i)\right)^2}{n}} \tag{5}$$

$$\mathrm{CCoef} = \frac{\sum_{i=1}^{n} \left(\theta_{act}(i) - \overline{\theta_{act}}\right)\left(\theta_{est}(i) - \overline{\theta_{est}}\right)}{\sqrt{\sum_{i=1}^{n} \left(\theta_{act}(i) - \overline{\theta_{act}}\right)^2 \sum_{i=1}^{n} \left(\theta_{est}(i) - \overline{\theta_{est}}\right)^2}} \tag{6}$$

where θ_{act} and θ_{est} are actual and predicted ankle joint angle, respectively, n is the number of sample points in each trial, and $\overline{\theta_{act}}$ and $\overline{\theta_{est}}$ are average actual and predicted ankle joint angles, respectively, over each trial.

Results

Locomotion Identification Module

For locomotion identification module design, two lower limb muscles were examined. The considered muscles were TA (below knee/Muscle 1) (FL) and BFL (above knee/Muscle 2). The heatmap shown as Fig. 24 represents the classification accuracy achieved by individual features for both the muscles and all compared classifiers. In the heatmap, classifiers and muscles combinations are represented on the x-axis and features on y-axis. SVM classifier shows average classification accuracy (Avg. ± std.) as 41.81 ± 4.75% and 43.38 ± 5.33% for muscle 1 and muscle 2 (p-value >0.05), respectively. For LDA classifier, the classification accuracy of 36.44 ± 3.50% and 41.87 ± 4.52% (p-value <0.05) was shown by muscle 1 and muscle 2 respectively,

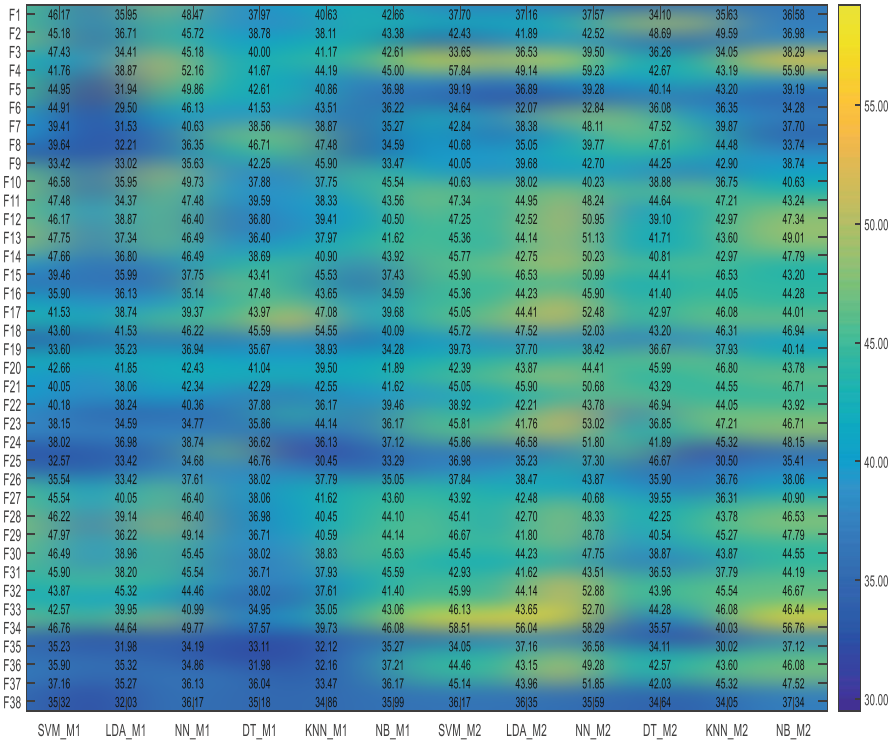
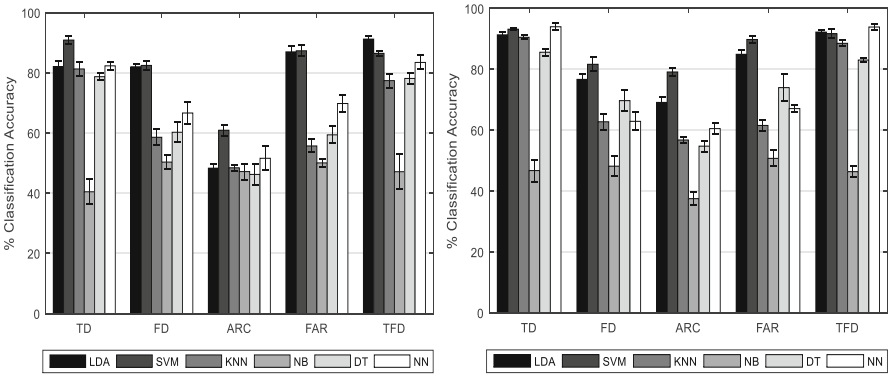


Fig. 24 Performance of individual features for various classifiers

whereas for NN classifier, it came out as 42.42 ± 5.30% and 46.12 ± 6.43% (p-value <0.05) for muscle 1 and muscle 2, respectively. KNN, DT, and NB classifiers show prediction performance of 39.90 ± 4.58%, 39.13 ± 3.49%, and 39.44 ± 4.03% (p-value <0.05), respectively, for muscle 1 and 41.13 ± 4.04%, 41.86 ± 4.86%, and 43.28 ± 5.36% (p-value <0.05), respectively, for muscle 2. Muscle 2 shows the better performance as compare to muscle 1 for all compared classifiers (p-value <0.05, except for SVM). Moreover, the best classification accuracy achieved was 59.23% for muscle 2 with NN classifier (Gupta and Agarwal 2018a).

Figure 25a, b show the average % classification accuracy of intuitive feature vectors (presented in Table 1) for all compared classifiers. LDA classifier displayed the best classification accuracy as 91.17 ± 0.96% for muscle 1 and 92.20 ± 0.60% for muscle 2 using time frequency domain feature vector with a statistically significant difference between them (p-value <0.05). For SVM classifier, time domain feature vector promises the best classification accuracy of 90.85 ± 1.26% and 93.15 ± 0.33% (p-value <0.05) for muscle 1 and muscle 2, respectively. Neural network (NN) classifier shows the best classification accuracy as 83.42 ± 2.34% and 93.96 ± 1.16% (p-value <0.05) for muscle 1 and muscle 2 using time domain feature vector. For decision tree classifier, the best accuracy shown by muscle 1 was 78.69 ± 1.08% using TD feature vector, whereas for muscle 2, it was 85.58 ± 1.18% (p-value <0.05) using the same feature vector. K-nearest neighbor (KNN) classifier achieved best classification accuracy as 81.26 ± 2.45% and 90.49 ± 0.61% (p-value <0.05) for muscle 1 and muscle 2, respectively, using TD feature vector. Bayesian classifier came out as least performance classifier for all feature vectors and muscles. The best classification accuracy shown by NN classifier was 50.04 ± 1.29% and 50.72 ± 2.65% (p-value >0.05) for muscle 1 and muscle 2, respectively, with FAR feature vector.

Tables 5 and 6 show the performance of every possible combination of feature vectors and classifiers for muscle 1 and muscle 2. To select the best classifier and feature vector, the median of classification accuracy (median_classi) over feature



(a) Muscle1 (b) Muscle2

Fig. 25 Average % classification accuracy of various classifiers for intuitive feature vectors

Table 5 Performance of various classifiers (Avg. ± std.) for intuitive feature vectors for muscle 1. (Cells with gray colors represents % Classification accuracy ≥ (median_feat&&median_classi))

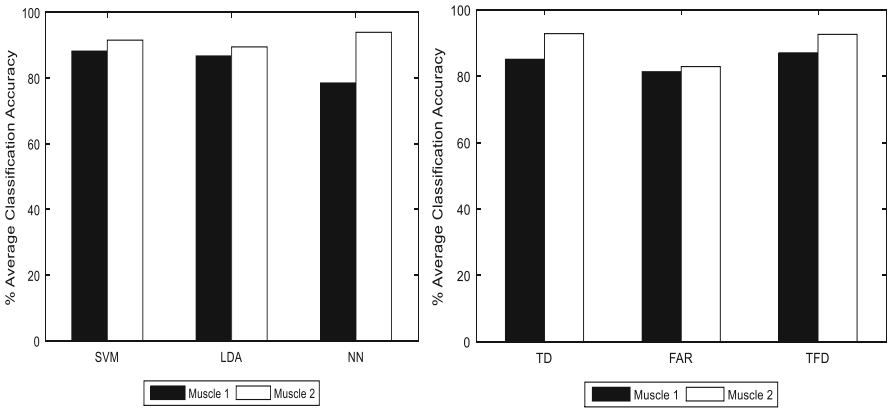
	SVM	LDA	NN	DT	KNN	NB	Median_feat
TD	90.85±1.26	82.07±1.79	82.30±1.23	78.69±1.09	81.26±2.46	40.45±4.21	81.67±1.53
FD	82.43±1.40	81.98±1.07	66.62±3.55	60.23±3.31	58.56±2.61	50.32±2.43	63.42±2.52
ARC	60.90±1.85	48.29±1.33	51.58±3.95	46.22±3.48	48.42±1.05	47.16±2.68	48.36±2.26
FAR	87.30±1.76	86.94±1.79	69.77±2.94	59.37±2.79	55.68±2.15	50.05±1.30	64.57±1.97
TFD	86.44±0.91	91.17±0.97	83.42±2.34	78.11±1.69	77.39±2.33	47.12±5.73	80.77±2.01
Median_classi	86.44±1.40	82.07±1.33	69.77±2.94	60.23±2.79	58.56±2.33	47.16±2.68	

Table 6 Performance of various classifiers (Avg. ± std.) for intuitive feature vectors for muscle 2. (Cells with gray color represents % Classification accuracy ≥ (median_feat&&median_classi))

	SVM	LDA	NN	DT	KNN	NB	Median_feat
TD	93.15±0.33	91.31±0.87	93.96±1.17	85.59±1.19	90.50±0.62	46.71±3.63	90.90±1.02
FD	81.62±2.27	76.67±1.72	62.88±2.92	69.68±3.55	62.75±2.69	48.15±3.31	66.28±2.81
ARC	79.10±1.44	69.10±1.90	60.50±1.75	54.73±1.81	56.76±1.11	37.48±2.04	58.63±1.78
FAR	89.77±1.10	84.91±1.43	67.07±1.27	73.96±4.40	61.53±1.85	50.72±2.66	70.52±1.64
TFD	91.67±1.40	92.21±0.61	93.87±1.01	83.02±0.74	88.51±0.97	46.35±1.86	90.09±0.99
Median_classi	89.77±1.40	84.91±1.43	67.07±1.27	73.96±1.81	62.75±1.11	46.71±2.66	

vectors and median of classification accuracy over classifiers (median_feat) were calculated. Following that, a pair of feature vectors and classifiers with higher classification accuracy than median_classi and median_feat were chosen. The mean of selected classification accuracies was taken over classifiers (Fig. 25a) and feature vectors (Fig. 25b) to compare top-performing feature vectors and classifiers, respectively.

Figure 26a shows that the LDA and SVM classifiers performed equally well (p-value >0.05); additionally, for both classifiers, muscle 1 and muscle 2 performed reasonably well with no statistically significant difference (p-value >0.05). Muscle 2 outperforms Muscle 1 in the NN classifier (p-value >0.05). TFD and TD feature vectors imitated the same prediction performance (p-value >0.05) among feature vectors (Fig. 26b). Hence, based on the above-presented results, it was recommended to utilize LDA, NN, or SVM classifiers with TD feature vector of SEMG signal for locomotion identification.



(a) Classifiers performance (b) Feature vectors performance

Fig. 26 Average performance of top three classifiers and intuitive feature vectors

Knee Angle Class Estimation

In the present study, knee angle prediction from SEMG signals was carried out for three different locomotion activities on a custom-built gait course. In a previous study, the performance of LDA and NB classifier was inferior to k-NN and SVMQ classifier (Dhindsa et al. 2019). So, in this study, the performance of only k-NN and SVMQ classifiers was evaluated. Table 7 presents the classification accuracy obtained with both the classifiers for different locomotion modes. The results of the ANOVA test on the classification accuracy achieved while performing the three activities indicate that the performance of SVMQ is significantly superior to that of k-NN for all the three activities (p-value < 0.05). The average classification accuracy obtained with SVMQ classifier is $89.25 \pm 2.32\%$ for stair ascend, $87.58 \pm 2.30\%$ for level plain walking, and $89.91 \pm 2.37\%$ for stair descend.

Ankle Angle Estimation Module

Figure 27 shows the average RMSE and CCoef for each locomotion. The average RMSE and CCoef of subjects, for LW locomotion, has been found $2.30 \pm 0.59^\circ$ and 0.980 ± 0.008, whereas for SA, SD, RA, and RD locomotion, it is $3.00 \pm 0.56^\circ$ and 0.97 ± 0.013, $2.88 \pm 0.70^\circ$ and 0.975 ± 0.008, $3.61 \pm 0.75^\circ$ and 0.970 ± 0.012, and $3.87 \pm 0.61^\circ$ and 0.968 ± 0.011, respectively. However, the average RMSE of all locomotion is found in the range of 2.30° to 3.87° and CCoef as 0.968 to 0.980. Figure 28 shows the actual and estimated ankle joint angle of subject 1 for all five

Table 7 Classification accuracy obtained for k-NN and SVMQ classifier

	SD		PW		SA	
Subjects	KNN	SVMQ	KNN	SVMQ	KNN	SVMQ
S1	86.50 ± 1.51	90.28 ± 1.34	85.04 ± 1.70	88.21 ± 1.63	87.05 ± 0.94	90.73 ± 1.44
S2	86.24 ± 1.66	90.28 ± 1.53	85.17 ± 2.06	88.47 ± 1.24	87.10 ± 1.71	91.01 ± 1.42
S3	85.47 ± 2.30	89.91 ± 1.89	84.26 ± 2.47	88.02 ± 1.75	85.83 ± 2.22	90.97 ± 1.73
S4	80.47 ± 1.91	84.72 ± 1.08	78.43 ± 2.18	83.17 ± 0.69	80.33 ± 2.37	85.10 ± 1.72
S5	82.81 ± 1.28	87.37 ± 1.61	80.77 ± 1.62	85.75 ± 1.39	83.31 ± 1.09	88.27 ± 2.08
S6	86.91 ± 0.93	90.89 ± 1.61	85.65 ± 1.22	89.07 ± 1.75	87.56 ± 1.27	90.88 ± 1.48
S7	85.98 ± 1.16	90.26 ± 1.64	85.06 ± 1.50	88.79 ± 1.99	86.99 ± 1.98	91.26 ± 1.17
S8	87.82 ± 1.42	90.55 ± 1.29	86.18 ± 1.31	88.78 ± 1.46	88.30 ± 1.33	91.03 ± 2.04
S9	83.27 ± 2.25	88.54 ± 1.14	81.34 ± 1.99	86.79 ± 1.80	84.25 ± 2.35	89.59 ± 1.26
S10	85.76 ± 1.59	90.16 ± 1.26	84.19 ± 1.91	88.38 ± 1.18	86.13 ± 2.09	90.79 ± 1.10
Avg	84.97 ± 2.73	89.25 ± 2.32	83.41 ± 3.00	87.58 ± 2.30	85.53 ± 2.93	89.91 ± 2.37

activities. It helps in identifying the major events of errors in ankle joint angle estimation. The results show that apart from the lower and upper RoM limit, the model seamlessly estimates the ankle joint angle. Major events of the error have been near the upper and lower limit (peaks) of the joint angle (Gupta et al. 2020).

Figure 29 depicts the average performance of the proposed model with and without knee angle information. Incorporating knee angle information as an input improves model performance significantly (one-way ANOVA, p-value >0.05). In terms of ankle joint angle estimation error (i.e., RMSE), an improvement of 67.85%, 45.74%, 68.17%, 41.35%, and 45.56% has been observed for LW, SA, SD, RA, and RD activity, respectively. The consistency of ankle joint angle estimation (CCoef) has shown enhancement of 36.19%, 57.82%, 68.24%, 22.34%, and 49.05% for LW, SA, SD, RA, and RD activity, respectively. The inclusion of the KA signal for ankle joint angle estimation improves model accuracy and reliability. Furthermore, by including KA as an input, within-subject performance variation has been reduced significantly for all activities. The model's continuous predictive capability may provide a more biomimetic ankle motion. The proposed method's performance in continuously mapping ankle joint angles across five locomotion modes has been evaluated (LW, RA, RD, SA, and SD). The results have been found to be consistent across all locomotion, making it suitable for controlling the active ankle-foot prosthesis.

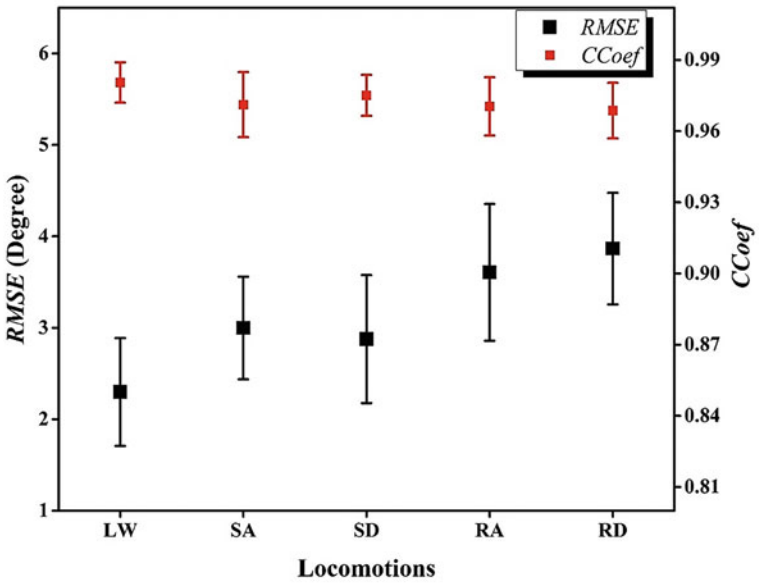


Fig. 27 Average RMSE and CCoef for different locomotion

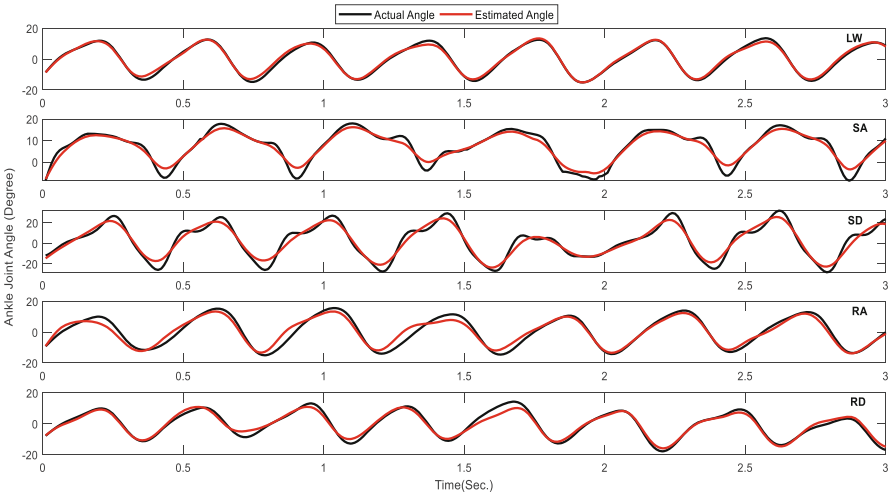


Fig. 28 Actual and estimated ankle joint angle for different locomotion

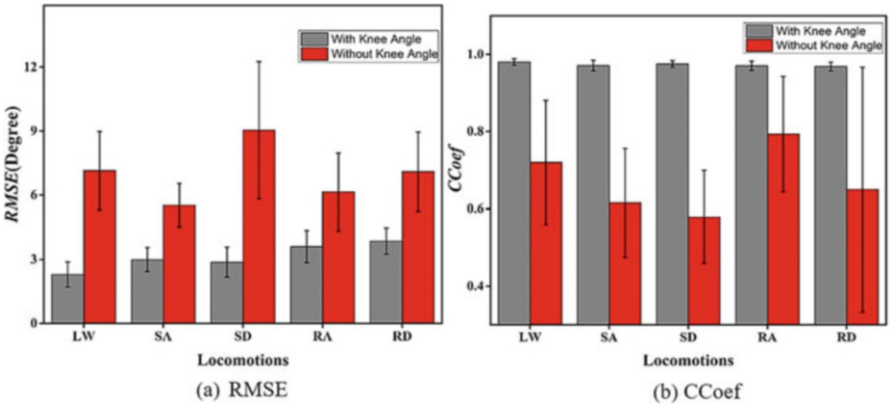


Fig. 29 Performance measures for with KA and without KA information

Conclusion

The recommendations for SEMG signal recording and related signal processing issues for a complete system design are compiled. When dealing with SEMG signals, it is strongly advised to follow the SENIAM recommendations. The experiment must be designed in accordance with the 2013 Helsinki Declaration. A complete system design process for a SEMG signal operated lower limb prosthesis design is also presented. The result is a thorough comparison of six classifiers for multi-muscle SEMG signals and 38 domain-specific features. The results indicate that SVM, NN, and LDA classifiers with time-domain feature vectors are better suited for locomotion mode prediction. Furthermore, for knee angle prediction,

SVMQ outperforms the k-NN classifier. The NARX model performed well for continuous ankle angle estimation with SEMG signals and knee angle as input. The framework and techniques presented here can be used directly to design and control SEMG-based active lower limb prostheses, exoskeletons, and orthoses.

Cross-References

► Advancements in Measuring Cognition Using EEG and fNIRS

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Artificial Intelligence for Iris-Based Diagnosis in Healthcare

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Abstract

Computers are now able to play a role in the diagnostic process for diabetes patients because of developments in the disciplines of medical imaging and information technology. It is difficult to reduce death rates to more reasonable levels by employing early and precise detection approaches. When seeking to identify risk factors for complications, acute and long-term difficulties, mortality, and medical care expenses, data on the prevalence and incidence of diabetes and prediabetes is important. The International Diabetes Federation estimates the occurrence of diabetes based on the present status and prevalence of diabetes worldwide. In addition, variables, such as physiological markers, that may have a role in the development of diabetes are investigated. Complementary and alternative methods, especially iris-based diagnostics, have made significant advances in the identification of diabetes and associated disorders. Diabetes is a leading cause of renal failure globally. It raises the likelihood of heart and blood vessel problems. Diabetic kidney disease is a chronic condition. The blood test is used to identify diabetic kidney disease, which is a difficult and agitating treatment for patients. This chapter combines iridology with current computer-based approaches to diagnose diabetic kidney disease. A total of 130 participants were assessed and categorized based on their health status, such as not having diabetes, having diabetes but not having diabetic kidney disease, and having diabetic kidney disease. Pre-image processing methods are used to extract the region of interest from the iris based on the iridology chart's specified regions. The region of interest was analyzed for first-order statistical and second-order textual characteristics. Finally, several categorization models were examined using different sets of characteristics to discover the optimum model. The maximum accuracy of 94.4% was achieved with the help of a fine Gaussian kernel of the SVM classifier. It is empirically demonstrated that features of specific areas of the iris are highly correlated with the diabetic kidney disease condition of an individual.

Introduction

Diabetes mellitus (DM) is one of the most common metabolic diseases and is characterized by the glucose tolerance capability of the body. A continuous increased blood glucose level can lead to microvascular and macrovascular complications and increases the risk of heart, eyesight, kidney, and nerve system damage (Chen et al. 2011). It is notable that, in the past few decades, as the lifestyle of the population changed, the prevalence of diabetes has increased at a faster rate. Figure 1

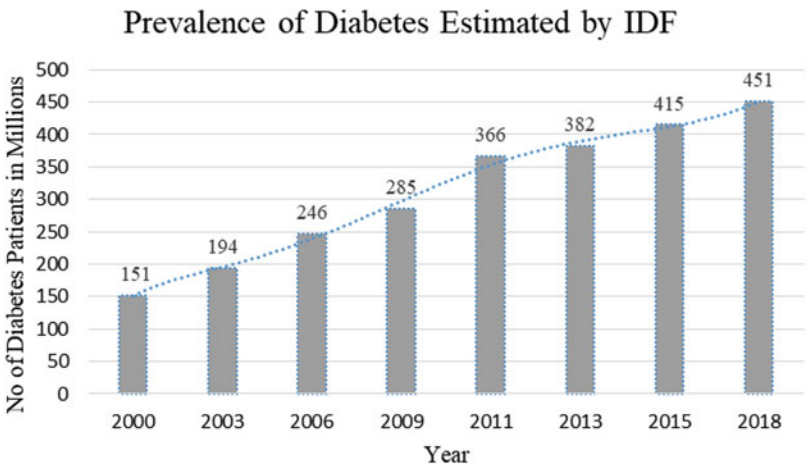

Fig. 1 Prevalence of diabetes estimated by IDF

presents the results of an analysis conducted by the International Diabetes Federation (IDF) on the prevalence of diabetes across the globe (Ogurtsova et al. 2017).

The body converts the carbohydrates of the food into glucose, which provides energy to different parts of the body. When the body does not consume enough glucose, insulin helps to absorb the excess glucose to reduce blood glucose levels. Insulin is an important hormone which is secreted by the beta (β) cells of the pancreas organ (Baier and Hanson 2004). Insulin consumes glucose with the help of three different processes named glycolysis, glycogenesis, and lipogenesis. Glycolysis is the process of converting glucose into adenosine triphosphate (ATP). ATP is an electrochemical form that stores energy (Levin et al. 2011). Glycolysis is a reversible process which converts glucose into glycogen, that is, short-term energy storage in the liver or muscles. Lipogenesis is the process of converting glucose into lipids and fatty acids (FA), to store energy for a long term. It's an irreversible process and remains as adipose tissues or fatty layers in the body. Also, when the glucose level in the body is too low, the pancreas organ secretes another hormone called glucagon. Glucagon helps the liver to release stored glucose. Glycogen undergoes a reversible process of glycogenolysis, which releases the glucose from the glycogen. Glucose is made by another reversible process called gluconeogenesis, in which amino acids are converted into glucose. Together, insulin and glucagon maintain the blood glucose in the body, and this state is called homeostasis. The excess glucose level in the body is called hyperglycemia and the shortage is termed hypoglycemia. If the blood glucose concentration is lower than 70 mg/dl, the person is said to have hypoglycemia. This may result in several disorders including awkwardness, a dilemma in talking, bewilderment, unconsciousness, seizures, or death. Continuous hunger, sweating, unsteadiness, and weakness can also be seen in patients suffering from hypoglycemia. Any person is called hyperglycemic if the blood glucose concentration is over 120 mg/dl after overnight fasting. If the hyperglycemia remains

for a longer period, it results in eye, nerve, and kidney disease; in a more generalized form, this condition is called diabetes. Diabetes is broadly classified mainly according to the function of insulin, type 1 diabetes, type 2 diabetes, gestational diabetes, prediabetes, and other specific types of diabetes.

Type 1 Diabetes

Type 1 diabetes develops when beta cells in the pancreas fail to generate enough insulin for the body to use the glucose that is produced. Insulin is a medication that must be used regularly by people who have type 1 diabetes to prevent the consequences that might arise from having high blood glucose levels for an extended length of time. Type 1 diabetes is a kind of diabetes that affects children and young adults and accounts for around 10% of all occurrences of diabetes. Type 1 diabetes is caused by an autoimmune response, in which beta cells of the pancreatic organ are eliminated by the body's immune system. This leads to the development of type 1 diabetes. Hereditary factors are the primary contributor to type 1 diabetes. Although an unhealthy lifestyle is not the direct cause of type 1 diabetes, it is essential to maintain a healthy lifestyle to keep the condition under control. Type 1 diabetes cannot be avoided at this time, but it may be managed well with a healthy lifestyle, the right medicine, and regular testing at the appropriate intervals. Type 1 diabetes is characterized by many main symptoms, some of which include an increase in thirst and urination, an increase in appetite, exhaustion, a rapid loss of weight, and impaired vision (Than et al. 2005). Type 1 diabetes may, over time, cause a variety of health complications, including cardiovascular disease, renal disease, nerve damage, foot problems, vision problems, depression, etc.

Type 2 Diabetes

In this form, the body develops resistance to the effects of insulin, which causes the amount of glucose in the blood to stay elevated for an extended length of time. Type 2 diabetes is the most prevalent form of the disease, with around 85–90% of diabetic people suffering from this form of the disease. Type 2 diabetes is a chronic condition that worsens with time and is strongly linked to changes in lifestyle as well as genetic and environmental variables that are tied to families. In most instances, persons over the age of 45 years are diagnosed with type 2 diabetes. However, in today's society, many cases of type 2 diabetes have also been detected in children, young adults, and adolescents. If a person lives a lifestyle that is controllable yet includes being overweight, having high blood pressure, and having poor nutrition, then his risk of developing type 2 diabetes is significantly increased (Fonseca 2009). The symptoms of type 2 diabetes include increased thirst and urination, weariness, hunger, skin infections, and impaired eyesight, among other symptoms.

Gestational Diabetes

Gestational diabetes mellitus (GDM) is a form of diabetes that occurs during pregnancy and normally disappears after giving birth. Gestational diabetes can happen at any stage of time but mostly it is found in the second half which is after the fifth month. GDM generally does not show any specific symptoms, although some women feel fatigue, increased thirst, and dry mouth. But these symptoms are common during the pregnancy period. Therefore, the only way to diagnose GDM is a regular blood sugar test. Most women having GDM give birth to a healthy baby, but in some cases, the baby can grow larger than normal which can increase the difficulties during delivery or may be a cause of premature delivery. GDM increases the risk of women to have type 2 diabetes (Linnenkamp et al. 2013).

Prediabetes

Prediabetes is a dangerous health condition in which blood sugar levels are higher than usual but not yet high enough to be classified as type 2 diabetes. Prediabetes increases a person's chance of acquiring type 2 diabetes, heart disease, and stroke. When a person has prediabetes, the cells in his body do not react correctly to insulin. To induce cells to react, the pancreas produces more insulin. Eventually, the pancreas can't keep up, and blood sugar levels rise, setting the foundation for prediabetes and type 2 diabetes later on. People might have prediabetes for years with no obvious symptoms; thus, it typically stays unnoticed until major health concerns such as type 2 diabetes manifests. If a person has any of the risk factors for prediabetes, such as being overweight; having a parent, brother, or sister with type 2 diabetes; being less physically active; having gestational diabetes; or having polycystic ovarian syndrome, it is critical to test the blood sugar.

Other Specific Types of Diabetes

The pancreas becomes inflamed when it fails to secrete critical hormones and enzymes necessary for glucose regulation and digestion. Inflammation is the root cause of many patients' suffering. Long-term pancreatic inflammation, on the other hand, is a known risk factor for type 3c diabetes and pancreatic cancer. The pancreas produces inadequate insulin to control blood glucose levels in type 3c diabetes. Nearly half of the patients require insulin injections to help the pancreas control blood sugar. Diabetes of this type is also known to impair pancreatic digesting. Exocrine pancreatic insufficiency is common when the pancreas is impaired in this way (EPI). EPI is a common complication in both chronic pancreatitis and diabetic individuals. In addition to digestive issues, EPI can cause chronic diarrhea and bloating. Fortunately, pancreatic enzyme replacement therapy is a great treatment option.

A variety of drugs have adverse effects, one of which is an increase in blood glucose levels. The term "drug-induced diabetes" refers to diabetes that develops as a result of the use of a certain medication. This type of diabetes is curable in certain circumstances if the medicine is removed, but in others, drug-induced diabetes may be permanent. Secondary diabetes can be developed as a result of other health conditions. Several drugs have been found to increase the risk of diabetes development: Medications include corticosteroids, thiazide diuretics, beta-blockers, antipsychotics, and statins.

Diagnosis of Diabetes

Diabetes mellitus (DM) is a group of metabolic illnesses caused by chronically elevated blood glucose levels. Diabetic patients experience one or more problems as a result of persistent hyperglycemia's damage to the primary functioning organs. It is expected that difficulties in diabetic individuals' functioning organs begin 5–10 years before the diagnosis. As a result, early diabetes detection is critical for preventing damage and reducing diabetic complications. Diabetes can be diagnosed using clinical tests that employ blood, urine, saliva, or sweat as a test sample. Aside from these modalities, interstitial fluid, ocular fluid, and breath may all be utilized to help in diabetes diagnosis. The concentration of glucose in each of these physiological fluids varies. It has been observed that urine and perspiration have reduced glucose content. As a result, even though it is a noninvasive technology, studies on it have been skipped, and it is currently not used clinically for diabetes diagnosis.

Blood Glucose Tests

The blood glucose level concentration is estimated by blood glucose testing to screen for diabetes. Several diabetic screening tests may be performed and discussed.

Glycated Hemoglobin (A1C) Test

The glycated hemoglobin (A1C) test is a normal blood glucose test that does not need fasting or special preparation. It displays the blood sugar level over the last 2–3 months. The proportion of blood glucose that is directly bound to hemoglobin is the proportion of oxygen that transports the protein to the red blood cells. Blood glucose level is directly proportional to the amount of hemoglobin connected to it. If the A1C result on two separate tests is 6.5 or higher, the patient is diabetic. If the A1C reading is less than 5.7, the patient is considered normal. Furthermore, if the test results fall between 5.7 and 6.4, the person is prediabetic or has a higher risk of developing diabetes. Although the A1C test is a simple and convenient approach for diagnosing diabetes, the findings are not considered reliable. Moreover, in some medical conditions, the results are skewed, making the A1C approach erroneous. This method is not popular among medical practitioners.

Random Blood Glucose Test

The blood sample for a random blood glucose test is collected at a random time. The patient is diagnosed with diabetes if the blood glucose level is 200 mg/dl or above, regardless of the food consumed.

Fasting Blood Glucose Test

Following an overnight fast from food and drink, a blood sample is collected in the morning for the fasting blood glucose test. You must not have eaten or drunk anything for at least 8 h before getting your blood drawn. If the blood sugar level after fasting is less than 100 mg/dl, the individual is deemed normal; if it is between 100 and 125 mg/dl, the person is prediabetic; and if it is greater than 125 mg/dl, the person is diabetic.

Oral Glucose Tolerance Test

The oral blood glucose test is the gold standard for diabetes diagnosis; it is extensively used and respected in the medical profession. The initial measurement of blood glucose concentration is conducted after an overnight fast in this test. After that, a sweet drink is administered, and blood glucose levels are measured every 2 h. Normal glucose levels are defined as blood glucose levels less than 140 mg/dl in the person. The condition is normal if the glucose level is less than 140 mg/dL but greater than 200 mg/dl. Diabetes is diagnosed when the blood glucose level exceeds 200 mg/dl. A person is called prediabetic if his glucose level falls between these two readings.

Diabetes and Its Prevalence

The International Diabetes Federation (IDF) and the World Health Organization (WHO) both keep a careful eye on diabetes prevalence and have created a transparent way of assessing the progression of diabetes throughout the world and in various countries. Guariguata et al. (2013a) reported thorough research prevalence of diabetes in individuals aged 20–79 in 2013 and projected for 2035. The research was carried out using 744 data sources from 130 nations. Diabetes is expected to rise dramatically, particularly in poor nations. Another intriguing and easily understood analysis has been released by the Noncommunicable Diseases (NCD) Risk Factor Consortium (Risk and Collaboration 2008), which projects the global trajectory of diabetes since 1980. The study included 4.4 million people from 751 population-based studies. Both studies indicated that diabetes cases grew more rapidly in emerging and underdeveloped nations than in developed or high-income ones. Diabetes diagnoses in adults surged to 422 million in 2014, up from 108 million in 1980, according to studies. The rate of increase in diabetes prevalence was lowest in northwest Europe and northern Africa, encompassing the Middle East and Middle Asia.

Diabetes prevention has also seen an increase in costs. Fernandes et al. (2014) assessed diabetes expenditure and discovered that at least USD 162 billion was spent on diabetes globally. The study indicates that diabetes is a significant economic burden on the global healthcare system and that a robust and effective public health management system is required to alleviate the cost burden caused by diabetes.

Various models have been published so far to estimate the diabetes population in the future. Ogurtsova et al. (2017) conducted an extensive analysis to forecast the diabetes population for 2040 based on trends up to 2015. All diabetes-related research was considered, and the most acceptable studies from all nations were chosen utilizing the analytic hierarchy approach. Following that, extrapolation was applied to the selected data to create the prediction model, and ultimately, a logistic regression model was used to obtain the diabetes population estimate. According to the projection, around 643 million people will be diagnosed with diabetes by 2040. Whiting et al. (2011) conducted a similar analysis to anticipate the diabetes population in 2030 using 2011 data. Diabetes is expected to impact around 522 million people by 2030, according to the authors. Every year, IDF contributes to a diabetes atlas of reviews and publishes high-impact journal articles that forecast the prevalence of diabetes in the future. Since 2009, IDF published many studies related to the prevalence of the diabetes globally and regionally. Table 1 covers some of the abovementioned IDF research.

Status of Diabetes in India

Diabetes is most common in both developing and developed countries; thus, it should come as no surprise that these two categories are the most likely to be affected. One of them is India, albeit the prevalence of diabetes varies greatly across the country. At the moment, around 73 million Indians have been diagnosed with diabetes, yet there is a substantial disparity between the urban and rural populations (Kumar et al. 2013). As a result, diabetes is without a doubt one of the most serious problems confronting India's healthcare system. According to an Indian Council of Medical Research (ICMR) report, the central, eastern, and southern areas of India have a much higher prevalence of diabetes than the northern region (Anjana et al. 2011). Kumar et al. (2011) have previously demonstrated that diabetes would worsen in India soon and that urban living is a key component of this. Diabetes will impose a significant emotional, economic, and societal cost on developing countries such as India. Kaveeshwar and Cornwall (2014) discovered that body mass index (BMI) has a significant impact on diabetes. Persons with lower BMI are at a higher risk of developing diabetes.

Furthermore, there is a growing recognition of the need for high-quality, patient-centered primary care institutions that focus on both primary and secondary diabetes prevention (White et al. 2015). Medical practitioners are the first point of contact with the health system and provide integrated medical care services with the opportunity for fast referral to specialists, thereby meeting the health needs of the majority of the population (Shi et al. 2002; Ferrer 2007). According to the WHO,

Table 1 Summary of the studies by the IDF on the status of diabetes

Sr. no.	Description	Year
Studies on global population		
1	The International Diabetes Federation (IDF) and the World Health Organization (WHO) performed a survey on diabetes in the population of the world, and estimates were created for the years 2010 and 2030. It was estimated that the prevalence of diabetes would be 6.4% in the year 2010, and it is anticipated that this number will rise to 7.7% by the year 2030 (Shaw et al. 2010)	2009
2	Data about diabetes were compiled by the International Diabetes Federation from a variety of scholarly works that were subjected to peer review. It was suggested that estimation of the number of people who have diabetes might be done using logistic regression (Guariguata et al. 2011)	2011
3	For the year 2011, information on diabetes was gathered from a total of 565 data sources located in 110 different countries, and an estimate of the diabetes population for 2030 was created. According to the findings of the study, undeveloped and poor countries would see the greatest increase in diabetes (Whiting et al. 2011)	2011
4	Burdon of metabolic risk on the world population due to diabetes had been estimated (Unwin et al. 2012)	2012
5	An update on the world diabetic population has been published (Guariguata 2012)	2012
6	A calculation of the diabetes population in 2013 was made using the literature that was accessible, and a prediction of the diabetic population in 2035 was made using logistic regression. For this investigation, a total of 744 data sources from 130 nations were included (Guariguata et al. 2013b)	2013
7	After doing a worldwide calculation of the prevalence of type 1 diabetes in children, researchers discovered that there are around 500,000 children less than 15 years old who have been diagnosed with the condition. There was also research conducted on a regional scale, and it was shown that more children in Europe suffer from type 1 diabetes (Patterson et al. 2014)	2013
8	IDF analyzed the undiagnosed case of diabetes and discovered that around 45% of diabetes cases remained undetected globally. This estimate was based on literature that was readily accessible (Beagley et al. 2014)	2013
9	Guariguata et al. (2013) estimated the worldwide cases of hyperglycemia in pregnancy due to diabetes	2013
10	Linnenkamp et al. (2013) proposed a methodology to estimate hyperglycemia in pregnancy	2013
11	The first documented study to estimate the worldwide mortality owing to diabetes in 2010 and 2011 was released by the International Diabetes Federation (IDF). According to the findings of a study, the Western Pacific regions had the greatest increase in diabetes cases (IDF Diabetes Atlas Group 2013)	2013
12	An updated estimation by IDF estimates the global deaths attributable due to diabetes in 2015 and compensation with the previous studies (Diabetes I D F Group 2015)	2015
13	IDF reviewed total 540 data sources from 111 countries and estimated the diabetes population for 2015 and 2040 (Ogurtsova et al. 2017)	2016
14	IDF estimated the total expenditure on diabetes for the year 2014 globally (Fernandes et al. 2014)	2017

(continued)

Table 1 (continued)

Sr. no.	Description	Year
15	IDF published the diabetes data for the year 2017 and estimated for year 2024 (Cho et al. 2018)	2018
16	In this article, IDF provided a summary of research published between 2015 and 2018 that used retinal photography to assess the prevalence of diabetes-related retinopathy throughout the world	2019
Studies on regional population		
1	IDF estimated the diabetes cases in Africa for the year 2013 (Peer et al. 2013)	2009
2	IDF estimated the diabetes cases in Europe for the year 2013 (Tamayo et al. 2014)	2009
3	IDF estimated the diabetes cases in the Middle East and North Africa for the year 2013 (Majeed et al. 2013)	2009
4	IDF estimated the diabetes cases in North America and Caribbean Region for the year 2013 (Yisahak et al. 2013)	2009
5	IDF estimated the diabetes cases in South and Central America for year 2013 (Aschner et al. 2014)	2009
6	IDF estimated the diabetes cases in South-East Asia and Caribbean Region for the year 2013 (Ramachandran et al. 2013)	2009
7	IDF estimated the diabetes cases in Western Pacific Region for the year 2013 (Chan et al. 2013)	2009

population-centered primary care should be inclusive and continuing, focusing on individual health needs and preferences and facilitating treatment from a trusted practitioner.

Diabetes and Physiological Parameters

Physiological parameters in conjunction with new computer technologies have also demonstrated significant potential for diabetes risk assessment. Using physiological indicators, modern machine learning approaches have achieved extremely high accuracy in detecting diabetes. Hassan et al. (Temurtas et al. 2009) conducted a comparison study on the Pima Indian database for diabetes diagnosis. Age, body mass index (BMI), blood pressure, glucose concentration level, and other physiological characteristics were utilized to train a multilayer neural network for diabetes detection. So far, Pima Indian is the only database that has information regarding the abovementioned physiological indicators associated with an individual's diabetic status. Several researchers (Kayaer and Yildirim 2003; Polat and Güneş 2007; Polat et al. 2008; Dwivedi 2017) have proposed algorithms to predict diabetes using the Pima Indian database. Dwivedi (2017) performed a comparative examination of several classifiers for diabetes prediction using the Pima Indian database. Polat and Güneş (2007) categorized diabetes and nondiabetic groups in the UCI diabetes dataset. Principal component analysis (PCA) was used for feature selection, while adaptive neuro-fuzzy inference

system (ANFIS) was used for classification. Authors raised their accuracy to 89.47%. Polat et al. (2008) proposed a cascade approach for diabetes detection utilizing the Pima Indian dataset. A two-stage method consisting of generalized discriminant analysis (GDA) and the least square support vector machine (LS-SVM) has been presented. GDA was utilized to analyze the discriminant between characteristics, while LS-SVM was employed to identify diabetic and nondiabetic individuals. Heydari et al. (2015) gathered physiological diabetes data from a large population (2536) in Iran and provided an in-depth comparison of categorization algorithms.

Complementary and Alternative Diagnosis

The use of complementary and alternative diagnostic procedures is becoming more popular, not just among the general population but also among alternative medicine practitioners. Complementary and alternative medicine (CAM) practices held substantial power in the medical communities of the United States and Europe even before the turn of the twentieth century. Between 1990 and 1997, there was considerable growth in the use of complementary and alternative medicine (CAM) throughout American countries (Eisenberg et al. 1998).

A range of economic and sociological variables impact the usage of any type of medication. Even for serious diseases, people in underdeveloped countries with inadequate medical infrastructure may seek care from traditional medicine practitioners (Eisenberg et al. 1998). Because scientific therapies are more widely available in industrialized countries, complementary and alternative medicine (CAM) plays an essential role in illness prevention and health promotion (Eisenberg et al. 1998). There is also evidence that people with serious conditions such as cancer frequently use CAM as an adjuvant to biological therapy (Molassiotis et al. 2005; Smith et al. 2018) and self-manage chronic health concerns such as low back pain (Savigny et al. 2009). However, the social factors that promote CAM use in wealthy nations are not well understood. Studies have repeatedly demonstrated that those who use CAM are more likely to be female, to have a higher level of education, to be in their middle age range, and to report having a better health state than those who do not use CAM (Astin 1998; Corp et al. 2018).

The scientific evidence-based CAM diagnosis technique is becoming increasingly popular and reliable for diagnosis and treatment. Ayurveda has a strong intellectual and experiential base as a historic and complete medicinal system (Patwardhan et al. 2005). Eight criteria must be completed to evaluate the patient-based outcome measures of any clinical research. A few of the traits to evaluate include relevance, dependability, validity, scalability, responsiveness, and correctness (Fitzpatrick et al. 1998).

Ayurveda for Diagnoses

In Ayurveda, outcome metrics include physical examination findings or questionnaire scores on body constitution (*prakriti*), lifestyle, diet, and quality of life. To be included in a study procedure, outcome measurements or diagnostic variables such as bodily humors/energies (*dosha*), tissues (*dhatu*), metabolic waste (*mala*), metabolic fire (*agni*), subtle life force (*prana*), material life force (*ojas*), toxicity (*ama*), body parts (*avayava*), essential body part (*marma*), and the five components (*mahabhoota*) must be valid and trustworthy (Streiner et al. 2015). The degree to which two or more raters agree on a diagnosis of the same patient in the clinical environment is referred to as interrater reliability. Reliability studies are important because they provide information on the quality of measurements and play an important role in the process of developing successful diagnostic procedures (Fewell 2005).

Dosha is thought to alter a person's physical, psychological, and physiological characteristics. The ancient Ayurvedic writings *Charak Samhita* and *Sushruta Samhita* discuss in detail how to determine *dosha* traits through the development of *prakriti* and diseases. In recent years, just a few studies have shown a genetic foundation for *prakriti* (Patwardhan and Bodeker 2008). Human leukocyte antigen (HLA) gene variation has been connected to *prakriti* structure (Bhushan et al. 2005). According to a second study, metabolic profiles and hematological features differed among *prakriti* kinds (Prasher et al. 2008).

There is a statistically significant link between the CYP2C19 genotype and the main classes of *prakriti* types, according to Ghodke et al. (2007). Another study found that the aggregatory response of platelets and the inhibition of that response by aspirin varied depending on the *prakriti* subtype (Ghodke et al. 2007). There is a considerable likelihood that the *prakriti* results will have a significant impact on individualized therapy. On the other hand, quantitative research on themes such as the reliability of *prakriti* appraisal is lacking. Seven types of *prakriti* are defined based on the combination of one or more bioentities: *vataja*, *pittaja*, *kaphaja*, *vatapittaja*, *vatakaphaja*, *pittakaphaja*, and *vatapittakaphaja*. Analyzing one's *prakriti* might help him prioritize any nurturing, preventive, or therapeutic program that is unique to him. As a result, *prakriti*-based prescribing may help to increase the therapeutic impact of a regimen while minimizing the likelihood of adverse drug responses. Analyzing the *prakriti* assessment itself is critical to obtain more reliable diagnostic results (Rastogi and Chiappelli 2012; Kurande et al. 2013). *Prakriti* is defined as a natural blend of one or more *doshas*. Another way of determining the current state or degree of *dosha* is pulse examination, also known as *nadi pariksha* (Kumar et al. 2019). Another beneficial process is *jivha pariksha*, often known as tongue diagnostics. The color, shape, and coating of the tongue are all visual features of the tongue that are analyzed throughout the inspection procedure. According to Ayurveda, the basis of all diseases is a malfunction of the digestive or biological fire (known as *agnimandya*). When the biological fire, also known as *mandagni*, is not functioning properly, digestive issues may emerge, leading to the formation of an autotoxin known as *ama* (Tamborrino et al. 2021). This autotoxin binds to the

bioentities, also known as *doshas*, and affects the body's tissues. As a result, the properties of these tissues are weakened or altered, potentially leading to a variety of medical disorders. An inspection of the tongue coating is critical in the early stages for recognizing an impairment of digestive fire. If help is provided at this moment, it may be feasible to prevent the subsequent production of the autotoxin. When combined with other symptoms of *ama*, changes in the coating of the tongue may offer considerable information that may be used to make Ayurvedic diagnosis.

Traditional Chinese Medicine

Over thousands of years, traditional Chinese medicine (TCM) has established its medical system for diagnosing and treating a wide range of diseases (Tang et al. 2008). Based on the Yin-Yang philosophy and its five components, the human body may be viewed as the union of Qi and spirit (Lozano 2014). Using the four essential TCM diagnostic processes of examination, auscultation and olfaction, inquiry, and palpation, experienced TCM practitioners may analyze the source of the ailment, the kind of disease, and the location of the sickness (Liang and Huang 2022). As a result, medical experts may create the right herbal prescriptions for a therapy based on variances in human *Zang Fu*, Qi, blood, and other physiological fluids. Herbs will be used in these prescriptions. Furthermore, TCM utilizes practices such as acupuncture, cupping, Qigong, massage, and diet therapy to preserve the Yin-Yang balance, manage patients' health, and further restore their health. The WHO has recognized TCM as an example of complementary and alternative medicine that is both effective and ubiquitous across the world (Traditional Chinese Medicine Could Make "Health for One" True 2005).

TCM practitioners use their hands to feel for irregularities in the patient's arterial pulse as well as touch the patient's skin, chest, abdomen, and other parts of the body to detect aberrant symptoms. Thus, medical practitioners can use their expertise and abilities to combine information gathered from four independent TCM diagnostic techniques to identify unique illnesses or syndromes (Jiang et al. 2012). TCM syndromes are a pattern that may be thought of as combining pathogenic elements and the human body's reaction to illnesses, as well as recognizing the overall dynamic body response status to diseases. Diagnostic procedures based on TCM may be used to establish the presence of a broad variety of symptoms, syndromes, and diseases.

Even though it is widely used in TCM hospitals all over the world, TCM diagnosis has been demonstrated to be less accurate when compared to modern medicine (Wang et al. 2021). As a result, traditional TCM diagnosis is subjective; the experiences and manual examination of diverse TCM practitioners frequently give varying diagnostic results. TCM practitioners, on the other hand, must dedicate a large amount of time to the study of TCM diagnosis before they can perfect it discreetly. Although several research studies have shown a relationship between the appearance representation of patients in TCM diagnosis and a range of illnesses (Liao et al. 2014; Shu and Zhang 2016), it is suitable and vital for improvement.

TCM diagnosis uses a standardized, scientific, and computerized technique. In recent decades, academics have suggested and developed several computational TCM diagnostic approaches to address the difficulties raised by TCM experts (Liao et al. 2014; Shu and Zhang 2016). These ways have been developed to help TCM professionals overcome the difficulties they confront. These computerized devices are frequently designed to simulate the eyes, hands, and olfactory experience of a TCM practitioner. Other sensations can also be reproduced.

Patients can acquire digital data in the most efficient way possible by using the different devices indicated below. Because of the presence of many algorithms (Jiang et al. 2012; Zhang et al. 2014; Shu et al. 2017; Zhou et al. 2020), it is feasible to analyze these data and, finally, assess the condition of an individual's health. Despite this, for the time being, the bulk of computational diagnosis tools available can only duplicate one of the four diagnostic techniques used by TCM to help in the identification of ailments. This is owing to technology developments as well as early research inquiries.

TCM postulates that there are both interactions and causalities between various factors (Maciocia 2015). For example, the human body can be considered a fully living creature in and of itself. Local lesions have the potential to produce systemically damaging reactions throughout the patient's body. In other circumstances, however, systemically unhealthy changes may manifest as local lesions. As a result, TCM clinicians may evaluate, synthesize, and research a wide range of local lesions or sickness occurrences in the human body using TCM as a guide. This allows TCM practitioners to gain a grasp of the nature of the condition and come to diagnostic conclusions.

The diagnostic process in TCM is based on three fundamental concepts (Zhang et al. 2005). The first advantage is that we may extrapolate the changes that have occurred on the inside by analyzing the alterations that have occurred on the outside of the patients. TCM practitioners, for example, might check a patient's exterior appearance, such as the face, before recognizing any changes in the patient's internal organs, such as the liver. The second advantage is that we can assess the overall state based on relatively few changes. Minor changes in the local component frequently reflect the overall physiological and pathological information, which may be utilized to diagnose many diseases. For instance, TCM practitioners may often examine the state of the tongue or the sublingual veins to conclude the applicants' overall health (Zhang and Zhang 2015; Zhang et al. 2017). Clinical research (Hsu et al. 2019) provides more evidence supporting this concept. The third argument is that abnormal modifications can be discovered based on knowledge of the standard state. TCM practitioners with years of experience, for example, are aware of the common fluctuations that may occur in the arterial pulse. As a consequence, they may easily spot any aberrant changes in the patients' pulses and locate the condition that corresponds to such changes. Practitioners of traditional Chinese medicine are also able to identify a wide range of diseases utilizing these key TCM diagnostic ideas due to their considerable training and years of experience. However, as compared to manual sickness detection, the diagnostic data they produce is still susceptible to

interpretation and takes a long time. As a result, these issues are open to further repair and improvement through the application of computational approaches.

The associated diagnostic performances demonstrate that these computational systems are capable of assisting medical professionals in making straightforward diagnostic judgments that have the potential to be used in TCM. Despite this, several aspects of this business must be enhanced and, in particular, examined shortly. Several studies have focused on the use of pictures of the face, tongue, and sublingual area to forecast future patients' health in the context of computational TCM diagnosis with the inspection. The number of research evaluating tongue images is significantly more than the number of works evaluating the face and sublingual images. This conclusion may be made for a variety of reasons. Images of the tongue, according to TCM philosophy, frequently convey far more vital information than the sublingual vein, which is effective in ailment detection. Because its sections are substantially more compressed, obtaining samples from the sublingual vein is challenging.

Furthermore, the computational acquisition devices for images of the tongue and face are simple to create in a functionally sound manner. Similarly, some human body parts or organs are notoriously difficult to get. There are, for example, very few articles that analyze possible candidates' spirits, motions, or limbs for computational TCM diagnosis. At the moment, computational gadgets or systems capable of capturing the human spirit appear unattainable. The fact that these components have not been explored, on the other hand, serves as a reminder that there is still a lot of space for advancement in computational inspection diagnostics.

People of various nationalities have noticeably different skin tones on their faces. The recommended computational TCM diagnosis procedures may be prone to errors and produce poor classification results when comparing persons of various races based on the color of their facial skin. Oluwagbemi and colleagues, for example, developed a TCM-based computational tool that can collect and evaluate face pictures of individuals living in Sub-Saharan Africa to identify illnesses. Their findings give additional evidence that the proposed computational TCM face diagnostic tool is capable of correctly recognizing a wide range of viral infections as well as noncommunicable diseases (Oluwagbemi and Jatto 2019). However, there have been relatively few studies on the utilization of computational TCM facial diagnosis across several ethnic groups so far. The bulk of current research has mostly investigated facial diagnosis in Asian persons. For more accurate results, researchers should gather a large number of face images from a varied variety of ethnic origins and then standardize the range of facial skin tones using various types of computer processing. As a result, this dilemma raises an exciting research subject that should be addressed further shortly. There isn't much TCM-based research in computational auscultation diagnosis. This is because patient speech signals are frequently insufficient to accurately reflect the specific diseases or symptoms. In general, voice signals include an excessive quantity of noise information, which may affect the diagnostic outcome. For example, the acoustic patterns of coughing, yawning, and snoring, frequently hidden in the general speech signal, might be useful in diagnosing. As a result, greater study on the issue of collecting additional voice signals with

a diversity of patterns and using enhanced feature extraction algorithms might be valuable in the future.

Researchers have recently created a range of odor capture systems for analyzing breath odor for use in olfactory diagnostics. They generally use a gas bag or cylinder to hold subjects' breath before continuing their inquiry with various scent sensors. Diabetes is the most common and sensitive disease that computational odor diagnostics can detect, while other diseases are relatively seldom investigated. As a result, more smell samples from a variety of illnesses are needed to further assess the efficiency of computational TCM odor diagnostics. One further area that we may pursue is the collection of diverse fragrances created by humans, such as sweat. It's possible that employing various scent categories will result in more accurate sickness detection.

In contrast to patient vocal input, which must be transcribed, the bulk of medical records are based on electronic text, which is straightforward to analyze. However, online communication with patients who respond appropriately is required and beneficial in the process of automated TCM diagnosis. Natural language processing (Calvo et al. 2017; Ghosh 2022) may be required to complete this task. Because of the rapid growth and advancement of deep learning methods (Deng and Liu 2018; Kulikowski 2019), we are certain that the active interactive inquiry of TCM diagnosis will soon make significant progress and become a practical application.

The computational palpation diagnosis is more developed in TCM than other diagnostic processes. There have been a lot of studies that have looked into wrist pulse signals for detecting various diseases. Their diagnostic accuracy illustrates the potential use of the wrist pulse diagnostics approaches described. On the other hand, the instruments used to get wrist pulses are too large and unwieldy to be carried around and used instantaneously. There is potential and opportunity to create and utilize mobile and portable wrist pulse collecting devices (Zhang et al. 2020).

Iridology

In biomedical imaging, iris-based diagnosis is a highly intriguing and reliable method. It enables the diagnosis of chronic disorders that are difficult to diagnose using histological approaches. Iridology is a unique and alternative kind of holistic healing. It is a disputed technique founded on a variety of nonscientific assumptions that derive its therapeutic power from a certain worldview and theological framework. It is claimed that iridologists can evaluate an organism's general health by reading diseases or a proclivity to them from the iris of the eye (Rita 1999). Many people appear to be able to do this because they desire health at any cost and in a very easy manner, converting, working on themselves, the church, or religion (Simon et al. 2015). This might explain their success. However, neither the detection nor the rehabilitation of diseases is often possible with iridology.

Iridology is a nontraditional way of diagnosis and therapy that aims to identify the overall health of the organism by reading illnesses or a proclivity to them from the iris of the eye. Iridology is founded on the notion that all of the organs of the body are

represented on the surface of the iris via nerve connections, and that most organ malfunction is indicated on the iris, generally as a pigmented lesion. Iridology also thinks that the right iris represents the right half of the body and the left iris represents the left half of the body (Rita 1999). Iridologists believe that the appearance of the iris might reveal the health of numerous organs in the body. Iridology originated in the East, particularly Ancient Egypt and China; in the past, various illnesses could be identified by analyzing the appearance of a patient's iris (Jensen 2011; Pesek 2016; Simon et al. 2015). Ignác Péczely, a Hungarian physician, was the one who built the framework for modern iridology. Péczely's iridology was founded on his discovery that in an owl, after a paw fracture, a prominent black line developed on the iris of the eye, which faded after the cure; and subsequently, he saw a similar line in a patient. Péczely's iridology is credited with having contributed to the development of the field of ophthalmology.

Iridologists frequently refer to iris maps. An iridological examination entails a thorough study of the iris (either directly, with appropriate eye illumination, such as with a slit lamp, or by taking high-quality color photographs of both irises, sometimes supported by the analysis of specialized computer software). This review can be completed in one of two ways. Iridological maps depict a schematic segmentation of the iris' surface into fields that are the projection fields of certain internal systems and organs. This is done so that each portion of the body has a matching location on the surface of the iris of the eye (Soediono 1989; Salles and Mjp 2006; Jensen 2011). The iris picture is then matched to the iridological maps. This is done so that specific locations on the iris are assigned to particular internal organs. On an iris map, each iris is split into 60 sectors (like the face of a clock), and each sector corresponds to a different internal organ or function of the body; for instance, heart disorders may be seen in the left iris between positions 2 and 3 h (Simon et al. 2015). Iridologists draw maps of the iris as if there is an anatomical or physiological mechanism (e.g., by the nervous system, the bloodstream) underlying the connection between the iris and the rest of the body; as if each internal organ and part of the body were represented in specific areas of the iris; and as if the moles and discolorations of the iris reveal the condition of each organ, the inherited weaknesses and strengths, and the inherited and acquired inclinations to engage in certain behaviors. Meanwhile, these iridological maps are one of the indications that iridology has no basis in reality because none of them can precisely represent approximately 10,000 diseases (as indicated by the ICD-10 number); additionally, there are various versions of the iridodiagnostic key, methods of analyzing the iris, and iris maps; this means that there is no single universal diagnostic key for the iris, and the same changes in the iris are interpreted differently by different practitioners (Salles and Mjp 2006; Jensen 2011; Pesek 2016; Simon et al. 2015).

Changes in the color of the iris, as well as its spots, shadows, and other features, according to iridologists, might give insight into the health of a patient's internal organs by displaying how they are performing abnormally. Meanwhile, there are known medical signs that may appear in the eyes. These symptoms may be caused by systemic illnesses, and the iris is one of them. This, however, only applies to a

tiny fraction of all currently recognized disorders and does not offer a conclusive diagnosis.

According to iridologists, changes in the color of the iris, as well as its spots, shadows, and other characteristics, may provide insight into the health of a patient's internal organs by indicating how they are behaving abnormally. Meanwhile, there are documented medical symptoms that may manifest in the eyes. The iris is one of those organs which can indicate several systemic disorders that can cause these symptoms. This, however, only applies to a small percentage of all officially recognized illnesses and does not provide a clear diagnosis for other diseases. For example, granulomatous anterior uveitis, one of the symptoms which iris manifest, may be associated with one or more of the following systemic diseases: chronic sarcoidosis (D86), the Vogt-Koyanagi syndrome, toxoplasmosis (B58), tuberculosis, syphilis, or leprosy (A30) and B-cell malignant lymphoma.

Iris warts (symptoms: regular, thin, fragile, vermilion-like lesions) may be linked to the development of Ota nevus, neuroblastoma (type 1), and Rieger's syndrome. Lisch nodules (small, brown, domed nodules) may be linked to the development of neuroblastoma and Brushfield spots (symptoms: white or yellow dots, ring-shaped, placed between the outer and center regions) (type 1).

Vascular development on the iris, also known as new blood vessels on the iris, may be an indication of one or more of the following associated systemic diseases:

- A condition known as oculocutaneous albinism may sometimes be accompanied by transillumination of the iris.
- When Rieger's syndrome is present, there may be posterior embryotoxon and iris anomalies, as well as hypoplasia of the proper iris' essence (along with pupil displacement and obvious polycroticus) and nipple-like elevations on the iris' surface.
- Your irises may take on coloration of their own, even if you have Waardenburg syndrome.
- Iris lesions that are characterized by increased transillumination of the iris are a potential complication of diabetes.
- Iris hypoplasia and Brushfield spots are also possible complications of Down syndrome.
- Infiltration and thickening of the iris may occur together with leukemia in certain patients.
- A congenital collapse of the iris-pigmented epithelium may occur if the patient has Recklinghausen's illness in addition to neuroblastoma of type 1, which is a kind of neuroblastoma.
- On the surface of the iris, in addition to the stigma, there may be elevations that resemble nips.
- Oculocutaneous albinism, also known as the tyrosine-negative variant, may cause the iris to appear gray to blue and completely transparent.
- Along with oculocutaneous albinism (the tyrosine-positive variety), the iris may range in color from blue to yellow-brown, and it can have variable degrees of translucency.

- Sturge-Weber syndrome can cause angioma of the iris and ciliary body on the same side.
- When combined with malignant cancer, there is a possibility of lumps on the iris, which might amount to a large number.
- There is some evidence that links Kayser-Fleischer ring's to Wilson's disease, also known as hepatocellular degeneration.

Researchers have proposed several research articles to diagnose different chronic diseases using iris with the help of iridology. Liljequist (1916) discovered alterations in the iris and linked them to various illnesses. Soediono (1989) researched how previously practitioners used microscopes, magnifying glasses, and other equipment to view iris tissue change and assess an individual's health state. Iridology is a noninvasive diagnostic approach that uses iris colors, patterns, shape, and other tissue structure features to forecast an individual's health. Jensen (2011) divided the iris into 80–90 zones corresponding to different body organs. Salles and Mjp (2006) found that literature on iridology is quite limited, and it's essential to have more grim and extensive studies.

Um et al. (2005) investigated the relationship between iridological constitutions and ACE polymorphism in hypertension patients. They stated that it is used as an additional source to detect the medical condition. In this day and age of computer-assisted diagnosis (CAD), various researchers have integrated the basic concepts of iridology with computer-based diagnostic processes to arrive at an illness diagnosis. Qiu et al. (2006) proposed a model to determine ethnicity from iris data using the AdaBoost approach. Using their approach, they were able to achieve an accuracy of 85.95%. This model attempts to assess whether a person is of Asian heritage. While V. Thomas et al. (2007) used machine learning approaches to construct models that predict gender based on iris texture data, Bansal et al. (2014) predicted gender using the CHT model and the support vector machine (SVM) algorithm. Based on the iris texture traits, both of these models predict gender.

Ma et al. (2013) investigated the influence that gastrointestinal disorders have on the geometrical structure of the iris patterns. Manual clustering, which was then followed by manual classification, was carried out to investigate the properties of the iris that are based on its geometrical structure. To determine the cause of the colon disease, Passarella and colleagues (2013) investigated the distinct areas of the iris. Using photographs of the patient's iris, Hussein et al. (2013) were able to determine that the patient had a renal disease. The evaluation was conducted on a total of 166 different participants. The authors had already classified the data using neuro-fuzzy methods after extracting wavelet, entropy, and energy components. Ramlee et al. (2011a) investigated the possibility of detecting the presence of cholesterol in blood vessels via the use of iris recognition algorithms. Both Daugman's (2004) and Masek's (2003) approaches to iris identification have been used in recent years for iris localization and segmentation. Computer software can detect the level of cholesterol in our body by analyzing our irises using iris recognition algorithms and an iridology chart. More study, on the other hand, was undertaken to discover the

specific range or quantity of cholesterol contained in blood vessels (Ramlee and Ranjit 2009; Ramlee et al. 2011a, b) (Table 2).

Bansal et al. (2015) presented an SVM-based model to determine diabetes, using the Gabor filter and 2-D DWT-based features. The authors have performed a close group study on 80 subjects and performed analysis of the different kernels of the SVM classifier with the combination of wavelets. The authors have achieved 87.5% of accuracy in classifying the diabetic and nondiabetic subjects using the radial basis kernel of SVM. Lesmana et al. (2011) proposed a computer method-based iris assessment technique to detect insulin deficiency from beta cells secreted from the pancreas. Neighborhood-based modified back-propagation using adaptive learning parameter (ANMBP) technique has been used to classify the abnormalities in the iris due to insulin deficiency using quantitative and textural features. Banzi and Zhaojun (2015) evaluated the broken tissues and color change of the iris for early detection of diabetes Salles and Mjp (2006) reported a longitudinal study to observe the change in the sign of the cross of the pancreas in the specific areas of the iris due to DM over the period of 4 years.

Methodology

The model for assessing the diagnosis of diabetic kidney disease (DKD) utilizing the idea of iridology was developed. At the same time, advanced machine learning technology was employed to establish a procedure as a standard for an iridology-based diagnosis system. This allows one to supplement the traditional diagnosis procedure and track therapy success. Figure 2 summarizes the approach utilized.

Dataset Collection

DKD is a chronic disease and a worldwide public health problem for the diabetic population. DKD is frequently connected with an increased risk of chronic renal failure and cardiovascular disease. In this study, 43 nondiabetic and 83 diabetes individuals were considered. DKD affected 41 of them. Fasting and 2-h postprandial tests were conducted to determine diabetes. Glomerular filtration rate (GFR) was computed to monitor the progression of kidney injury. If the computed GFR is less than 90 mL/min/1.73 m^2, the patient is classed as diabetic. Cross Match Technologies, USA, used an iris scanner I-SCAN-2 to take images of both eyes.

Image Pre-processing Techniques

In pre-image processing, iris segmentation and normalization techniques were utilized. The center and radius of the inner and outer circular sections of the iris were the most important factors. Several segmentation techniques, however, had already been applied (Masek 2003; Daugman 2004; Ma et al. 2013; Bansal et al.

Table 2 A summarized iris-based disease diagnosis system

Author	Application	Image preprocessing techniques	Remarks
Um et al. (2005)	Relationship between iris constitution and ACE gene polymorphism	For the assessment, a system for iris analysis Bexel Irina was developed.	Neurogenic, abdominal connective tissue weakness, and cardiorenal connective tissue weakness features have been explored from the iris Total 166 subjects were investigated
Thomas et al. (2007)	Predicted gender using iris	Canny edge detector and circular Hough transformation (CHT) used for segmentation and texture encoding Daugman's rubber sheet model, FFT, Gabor filter for normalization Geometric and texture features were studied	LG EOU200 system for data acquisition; tenfold cross-validation method was used C4.5 decision tree (initially on SVM and neural network)
Bansal et al. (2014)	Predicted gender using iris	The CHT was used for segmentation Daugman's rubber sheet model for normalization Combines statistical features with 2-D DWT-based features	I-SCAN-2 dual iris scanner for the data acquisition The accuracy of the system was compared with polynomial, Gaussian, and RBF kernel function
Ma et al. (2013)	Gastrointestinal diseases	Manual blind clustering has been applied The manual classification was performed by six different persons having some medical background	Deformation in the geographical features due to gastrointestinal disease Pupil-based features (pupil roundness and pupil largeness) were extracted
Passarella and Fachrurrozi (2013)	Identified colon disorder	CHT is used for localization and simple edge detection for segmentation	Total 60 student volunteers were evaluated for the experiment 92% accuracy was achieved to classify colon disorder and results have been reconfirmed by the clinical test
Hussein et al. (2013)	Diagnosed kidney disease	Histogram equalization to normalize the contrast of the image CHT for localization and segmentation	The neuro-fuzzy classifier was used Wavelet, entropy, and energy of the specific regions of the iris have been extracted
Ramlee et al. (2011b)	Cholesterol in blood vessel	John Daugman's and Libor Masek's iris recognition methods for localization and segmentation	To determine the cholesterol Otsu threshold method has been used

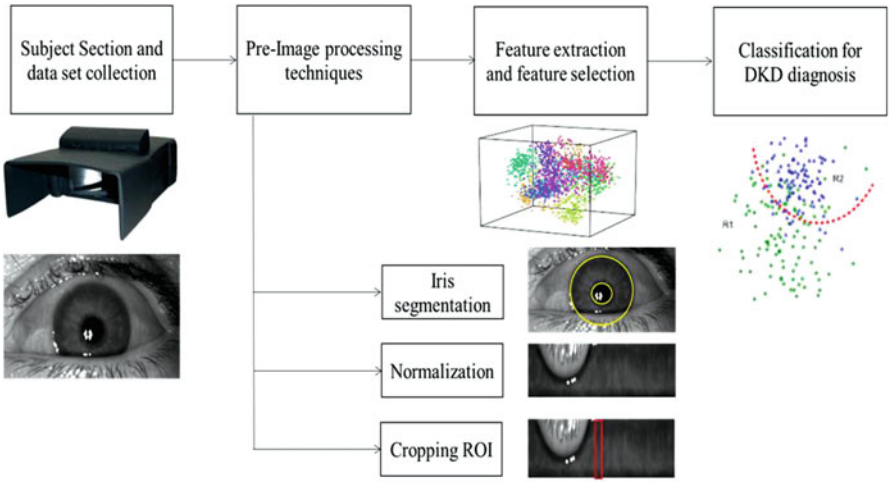

Fig. 2 Block diagram of the adopted methodology

2015). However, the most well-known and widely used iris segmentation technique based on the circular Hough transformation (CHT) (Masek 2003) was used.

The circular iris is converted into a fixed rectangular segment via the rubber-sheet normalizing technique. The rubber-sheet normalization model produced results that were independent of camera orientation, alignment, and location. It eliminates extrinsic variables such as uneven lighting and dimensional imperfections. Daugman's (2004) rubber-sheet model was employed in this study to transform segmented iris into a rectangle with a constant size of 360,720 (pixels). The ROI was then cut from the normalized iris using Jensen's iridology chart. The kidney was shown between 6.30 and 7 o'clock in the left eye and between 5 and 5.30 o'clock in the right eye (Jensen 2011). Therefore, the ROIs were cropped corresponding to these specific areas of the iris as shown in Fig. 3.

Feature Extraction and Model Design

The purpose of feature extraction was to retrieve relevant and indistinguishable data from the ROI. From the clipped ROI for both iris, 5 first-order statistical (FOS), 19 GLCM, and 7 GLRLM-based features were retrieved. The model was created to assess the contribution of feature sets to DKD diagnosis. Only FOS, GLCM, and GLRM (Hussein et al. 2013) characteristics were investigated in separate studies for the diagnosis of DKD. In addition, for the diagnosis, a combined feature set was employed, so that the function of individual characteristics and dominant feature sets could be calculated.

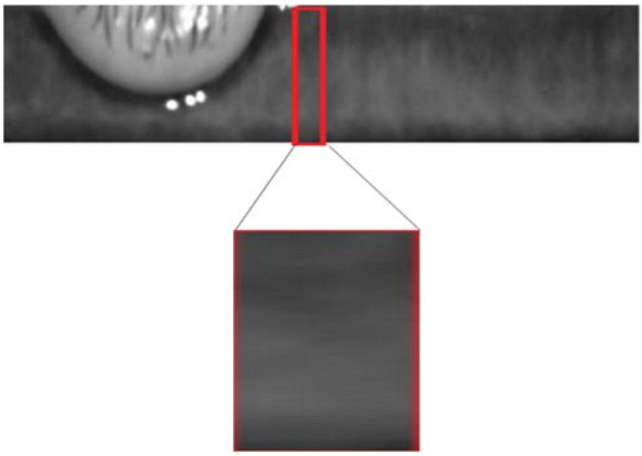

Fig. 3 Extraction of the region of interest

Classification

The categorization was done using different kernels from two distinct classifiers, k-nearest neighbor (KNN) and SVM (Bansal et al. 2015). SVM's fine Gaussian and cubic kernels were assessed, while three KNN kernels were selected for further evaluation: subspace, weighted, and fine. SVM is a prominent biological classifier. It is accurate and simple to grasp, with a high learning capacity, while KNN is the most often used classifier in many benchmark applications. It is superior to other classifiers due to its simplicity and fast performance.

Result and Discussion

Total 62 features were extracted from the group of FOS, GLCM, and GLRLM features. All the results were implemented over the normalized feature set to minimize the influence of the subsequent steps over the features. The trials were performed to design an efficient classification system for the prediction of the DKD.

Classifier Analysis by FOS Features

This study was carried out to investigate the influence and contribution of FOS characteristics on DKD prediction. To assess the performance of the classifiers, the accuracy, sensitivity, and specificity were determined using the tenfold cross-validation approach. Table 3 displays the classification parameters for each of the classifiers that were chosen.

Table 3 Average accuracy, sensitivity, and specificity for FOS features

Classifier	Kernel	Accuracy (%)	Sensitivity (%)				Specificity (%)			
			S_1	S_2	S_3	S_{AV}	SP_1	SP_2	SP_3	SP_{AV}
SVM	Fine Gaussian	94.0	87	99	97	94.3	98.1	91.8	92.5	94.1
	Cubic	72.2	67	77	70	71.3	80.3	69.2	73.5	74.3
KNN	Subspace	85.3	88	85	84	85.6	84.5	85.4	86.2	85.4
	Weighted	76.2	77	81	72	76.6	75.9	59.9	79.7	71.8
	Fine	79.0	88	85	73	82.0	78.6	75.3	83.4	79.1

Where S_1 Sensitivity for nondiabetic class, S_2 Sensitivity for class having diabetic but non-DKD, S_3 Sensitivity for DKD class, S_{AV} Average sensitivity, SP_1 Specificity for nondiabetic class, SP_2 Sensitivity for class having diabetic but non-DKD, SP_3 Specificity for DKD class, SP_{AV} Average specificity

Table 4 Average accuracy, sensitivity, and specificity for GLCM features

Classifier	Kernel	Accuracy (%)	Sensitivity (%)				Specificity (%)			
			S_1	S_2	S_3	S_{AV}	SP_1	SP_2	SP_3	SP_{AV}
SVM	Fine Gaussian	91.7	82	97	99	92.7	98.0	89.2	88.5	91.9
	Cubic	73.0	72	72	74	72.7	73.3	73.4	72.6	73.1
KNN	Subspace	78.6	78	86	73	79.0	78.7	75.0	82.4	78.7
	Weighted	76.6	82	73	78	77.7	75.3	79.0	75.8	76.7
	Fine	77.4	80	75	78	77.7	76.6	79.1	76.7	77.5

The SVM fine Gaussian classifier outperforms all other classifiers, with a 94.0% average accuracy. The average sensitivity and specificity were determined to be 94.3 and 94.1%, respectively. In comparison to other classifications, nondiabetic sensitivity is the lowest. Table 1 displays the greatest accuracy of the SVM classifier's fine Gaussian kernel.

Classifier's Performance Analysis by GLCM Features

GLCMs are second-order statistical characteristics that are produced based on the correlation between pairs of pixels in a certain direction. In this case, the first degree of correlation was used to indicate the correlation between nearby pixels. Table 4 provides the classifier performance analysis of GLCM features.

Classifier's Performance Analysis by GLRLM Features

GLRLM was calculated using the number of runs of a certain pixel in the specified direction. Second-order characteristics were also retrieved from the GLRLM. In this experiment, classifier performance is exclusively tested for the second-order

Table 5 Average accuracy, sensitivity, and specificity for GLRLM features

Classifier	Kernel	Accuracy (%)	Sensitivity (%)				Specificity (%)			
			S_1	S_2	S_3	S_{AV}	SP_1	SP_2	SP_3	SP_{AV}
SVM	Fine Gaussian	94.4	89	98	98	95.0	97.5	92.9	93.0	94.4
	Cubic	71.4	72	73	70	71.7	81.7	70.6	73.3	75.2
KNN	Subspace	84.1	84	86	83	84.3	84.0	83.1	85.3	84.1
	Weighted	77.4	85	78	74	79.0	75.8	76.8	80.1	77.7
	Fine	80.6	91	73	84	82.7	78.0	86.4	78.6	81.0

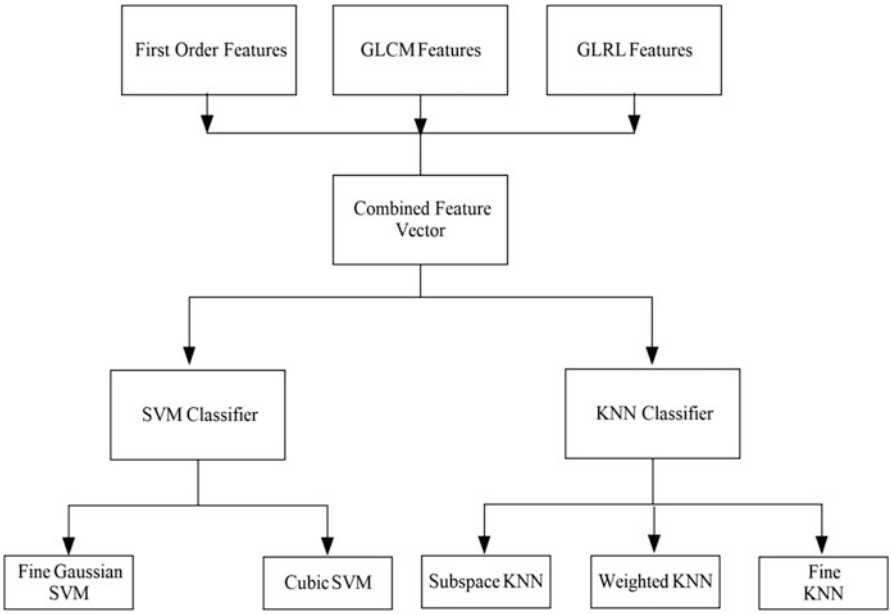

Fig. 4 Block diagram of the experiment designed for combined feature vector

GLRLM-based textural features. Table 5 displays the performance based on the GLRLM characteristics.

Classifier's Performance Analysis Using Combined Feature Vector

Figure 4 illustrated the complete FV that was built to include all of the retrieved characteristics, namely, FOS, GLCM, and GLRLM. In brief, 62 characteristics were retrieved for examination. Table 6 displays the classifier performance in terms of the total FV. Figure 4 depicts a flow diagram. Table 6 displays the greatest accuracy of the SVM classifier's fine Gaussian kernel.

Table 6 Average accuracy, sensitivity, and specificity using combined feature vector

Classifier	Kernel	Accuracy (%)	Sensitivity (%)				Specificity (%)			
			S_1	S_2	S_3	S_{AV}	SP_1	SP_2	SP_3	SP_{AV}
SVM	Fine Gaussian	94.5	86	100	100	95.3	100	91.7	94.2	95.3
	Cubic	81.7	81	83	81	81.7	81.1	81.1	82.3	81.5
KNN	Subspace	84.9	90	86	80	85.3	83.1	84.6	87.9	85.2
	Weighted	73.8	83	73	72	76.0	72.2	74.6	75.3	74.0
	Fine	83.3	87	83	81	83.7	82.3	83.3	84.6	83.4

Further analyses for all three classes for this classifier attain the average sensitivity and specificity as 95.3%. The individual class analysis shows for diabetic and DKD class 100% sensitivity, while for nondiabetic class, it shows 86% sensitivity. Also, the individual class analysis shows 100% specificity for the nondiabetic class.

Conclusion

The issues of diabetes prevalence, variables that contribute to diabetes development, and diabetes diagnosis utilizing the iris are explored. A thorough examination and analysis of the relevant literature concerning diabetes and its present situation throughout the world were carried out. The major focus of the second stage is the care of diabetic patients. The study conducted on a global scale, as well as that conducted on a regional scale, is examined. The procedure involves conducting research on the role of physiological parameters in diabetes and assessing published papers related to this topic. This chapter labels the machine learning backgrounds used to create an outperforming model for the rapid, simple, and early detection of DKD. A comparison of the statistical and textural characteristics' contributions (GLCM and GLRLM) was provided. The SVM and KNN classifiers were used for classification, and the results were compared. The suggested model produces great results and stimulates further investigation. These findings suggest that FOS traits are most suited for creating a DKD diagnostic system, while a combination of diverse textural features provides an extra benefit and high classification accuracy. Analyses demonstrate that an individual's diabetes and renal health status are related to statistical and textural aspects of particular parts of the iris.

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Use of Metrological Characteristics in Ultrasound Imaging and Artificial Intelligence Techniques for Disease Prediction in Soft Tissue Organs

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Kriti and Ravinder Agarwal

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Abstract

The global population nowadays is severely affected by various diseases of the soft tissue organs caused mainly because of some infection, heredity, change of lifestyle, etc. The timely detection and accurate diagnosis of these health conditions are of utmost importance in order to improve the chances of recovery and survival. Many medical imaging modalities have proven to effectively diagnose these diseases and their progression in a noninvasive way. Out of all the available modalities, ultrasound is the preferred mode of examination for imaging soft tissue organs for disease prediction because of its ease of use, low cost,

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portability, and lack of ionizing radiation. The radiologists manually study these scans for making an accurate diagnosis of the underlying condition. However, ultrasound imaging is highly operator-dependent and its effectiveness is adversely affected because of the presence of speckle noise. Therefore, to overcome these issues and for an efficient disease diagnosis, different computer-aided diagnostic (CAD) systems have been developed by researchers using artificial intelligence techniques along with the metrological characteristics of the diseased part as visible on the ultrasound scan of the organ under study. The main purpose of the present chapter is to shed light on the different types of diseases that affect the soft tissue organs like kidney, liver, thyroid, breast and their sonographic appearances, and characteristics. The chapter further describes methodologies developed in recent existing literature (year 2017 onwards) for the classification of diseases using ultrasound images of these organs through a CAD system using state-of-the-art deep learning and machine learning methods. The chapter also gives an insight into designing an efficient CAD system for the classification of breast tumors. The authors in the chapter have used nonsubsampled contourlet transform (NSCT) for multiresolution analysis of the original tumor images. From the subimages obtained using NSCT, extraction of texture features has been carried out using gray level co-occurrence matrix (GLCM), whereas shape features have been computed from the preprocessed tumor images. The computed feature set (texture + shape) has been used for classifying breast tumors using an adaptive neuro-fuzzy classifier with linguistic hedges (ANFC-LH) classifier based on the optimal features selected on the basis of the hedge values associated with the fuzzy rules.

Introduction

Recent statistics and reports prepared by various organizations show that the diseases of the soft tissue organs like kidney, liver, spleen, breast, thyroid, pancreas have become very common among the people globally and are a major cause of concern. Some of the statistics are as follows: Chronic kidney disease (CKD) has been recognized as a major health risk and is estimated to affect around 37 million people in the United States (Centres for disease control and prevention 2021); over 1 lakh patients are diagnosed with end stage kidney disease in India per year (World kidney day 2019); in India, approximately 10 lakh patients have been diagnosed with liver cirrhosis every year due to lifestyle changes (TNN 2017), whereas nonalcoholic fatty liver disease has affected 25% of the global adult population (Cheemerla and Balakrishnan 2021); as per the report by American Thyroid Association, approximately 20 million people in the United States are affected by thyroid diseases (American thyroid association), and a report published in 2018 claimed that approximately 42 million Indians are affected by thyroid disorders mainly hypothyroidism (DTE Staff 2018); another health report has published that uterine cancer is the 4th most common form of cancer among women in the United States. It is

estimated that approximately 66,000 people will be diagnosed with uterine cancer in 2021 (Cancer.net 2021). According to a report by World Health Organization (WHO), 2.3 million women have been diagnosed with breast cancer globally in the year 2020 (World Health Organization 2021). As per the report given by Cytcare cancer hospitals, in India, breast cancer accounts for approximately 14% of the cancers in women in India, 1 in 28 Indian women is likely to develop breast cancer and it is more prevalent in the women living in urban areas as compared to rural areas (Cytcare cancer hospitals 2022).

As has been observed above, some of these statistics are alarmingly high and most of these diseases do not show any early symptoms; hence, timely detection of these major diseases of the soft tissue organs plays a vital role in efficient treatment planning and improving the recovery rate. Even though the prediction and diagnosis of diseases based on pathological tests remains to be a gold standard, this process is cumbersome and time consuming. In order to overcome this, various imaging modalities like x-rays, ultrasound, computed tomography (CT), magnetic resonance imaging (MRI) can be used to provide an accurate diagnosis of the abnormalities in a noninvasive manner at a preliminary stage. Out of all the available modalities, ultrasound has found use prominently in disease detection and diagnosis because of its associated advantages such as lack of radiation, low cost, portability. Ultrasound imaging includes the transmission of sound waves greater than 20 kHz into the body for the purpose of capturing the live images of the internal organs. Ultrasound imaging modality has been prevalently used for visualizing vessels, organs, and soft tissues in real time without the use of radiations. The image formation takes place on the basis of the reflection of waves from internal body structures. The time taken by sound waves to travel through the body and the amplitude of the wave provides the basic information for image formation. The most common organs to be visualized by an ultrasound are ocular structures, liver, bladder, spleen, kidney, uterus, pancreas, carotid artery, breast, thyroid, fetal heartbeat, etc. The commonly occurring diseases of the soft tissue organs as seen on ultrasound have briefly been described below.

Kidney Diseases

The kidneys in the human body are used for filtering the blood and removing excessive waste and toxins out of the body. The nephrons present in the kidney are affected by the diseases and get damaged, which leads to failure in the kidney function. The most common kidney diseases encountered are renal cysts, chronic kidney diseases (with different stages of severity), and renal cell carcinoma. Renal cysts are most commonly found simple fluid-filled sacs having thin walls. Some cysts further develop to become complex with thick walls and some solid components along with the fluid (Johnson 2018). The formed cysts compress the nephrons or sometimes result in kidney enlargement affecting the renal function, ultimately leading to kidney failure if left untreated (Sudharson and Kokil 2019). Chronic kidney disease means kidney damage over time leading to end stage renal disease

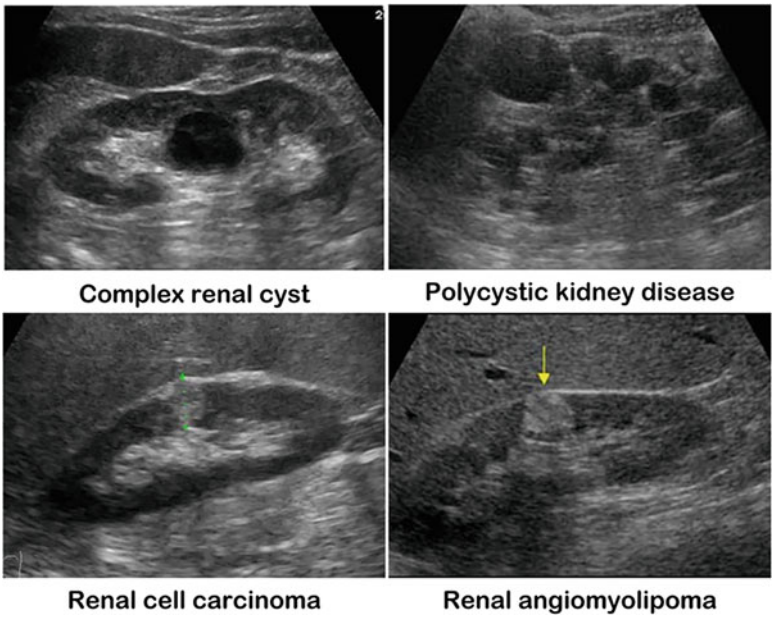

Fig. 1 The sample images of different kidney diseases as seen on ultrasound (Geertsma 2014d)

wherein the kidneys lose their ability to filter out the blood (American kidney fund 2021; Johnson 2018). The diseases of the kidney parenchyma that cannot be precisely determined are called medical renal diseases (MRD), characterized by nonspecific changes in the echotexture of the renal parenchyma which can be determined by ultrasound-guided biopsy (Johnson 2018). Renal cell carcinomas are primary malignant kidney tumors with four grades of severity (grade I to grade IV). These tumors are varying in appearance from solid and homogeneous to cystic changes and heterogeneous. Renal angiomyolipomas (AMLs) are solid tumors composed mainly of adipose tissue. These appear hyperechoic on an ultrasound as compared to surrounding kidney parenchyma. Different kidney diseases as visible on an ultrasound are shown in Fig. 1.

Liver Diseases

The liver is the largest and most vital organ in the human body as it helps in digestion and carrying out various metabolic activities of the body. Broadly there are two types of liver diseases viz focal and diffuse liver diseases. The focal liver diseases affect a small region of the liver parenchyma, whereas in case of diffuse liver diseases, the whole liver parenchyma is affected (Brown 2020; Virmani et al. 2013, 2014). The focal and diffuse liver diseases can further be subcategorized as shown in Fig. 2.

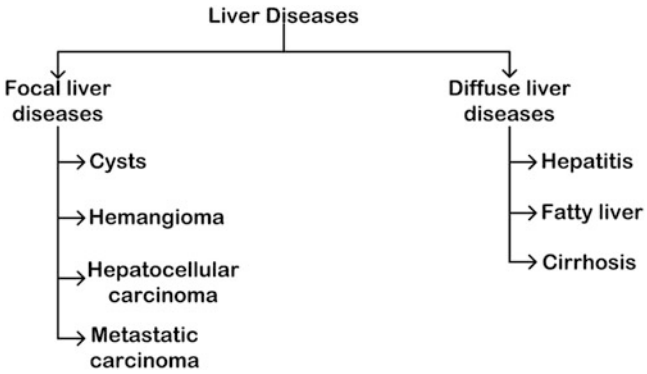

Fig. 2 Different types of liver diseases

Simple liver cysts are thin walled and filled with fluid or solid components and are generally benign. Hemangioma (HEM) is a benign mass formed in the liver due to entanglement of blood vessels and appears as a homogeneous, well-defined solitary mass. Hepatocellular carcinoma (HCC) is a primary malignant liver condition characterized by scarring of liver parenchyma and fibrosis and is considered to be the end stage of liver cirrhosis. Metastatic carcinoma is a secondary malignant liver condition caused due to cancerous cells spreading to the liver from other parts of the body (Virmani et al. 2014).

Hepatitis means inflammation or swelling of the liver due to some infection. Under normal operation of the liver, swelling occurs to fight off any infection, that is, controlled inflammation. However, when this inflammation gets unregulated, infection gets progressed into a disease called hepatitis. Persistent and uncontrolled inflammation leads to deposition of collagen on the liver parenchyma, which gets stiff and replaces healthy cells of the liver leading to fibrosis. Fatty liver disease is characterized by accumulation of fat in place of normal liver tissue, which causes scarring and inflammation leading to fibrosis and ultimately cirrhosis if left untreated as this scar tissue cuts off the blood supply to the liver thus damaging the healthy cells. Liver cirrhosis is an irreversible damage to the liver tissue because of gradual scarring of liver tissue due to inflammation that is caused by hepatitis, alcohol consumption, and progression of fatty liver disease (Virmani et al. 2013). The basic stages of liver disease are summarized as: Hepatitis/fatty liver resulting in fat accumulation around liver leading to swelling. From here the disease progresses to liver fibrosis in which the cells of the liver harden and liver swells further. The next stage here on is cirrhosis in which the liver cells get severely scarred. After this stage if no treatment is taken, the disease progresses into liver cancer ultimately leading to liver failure. The sample images of different liver diseases as seen on ultrasound are presented in Fig. 3.

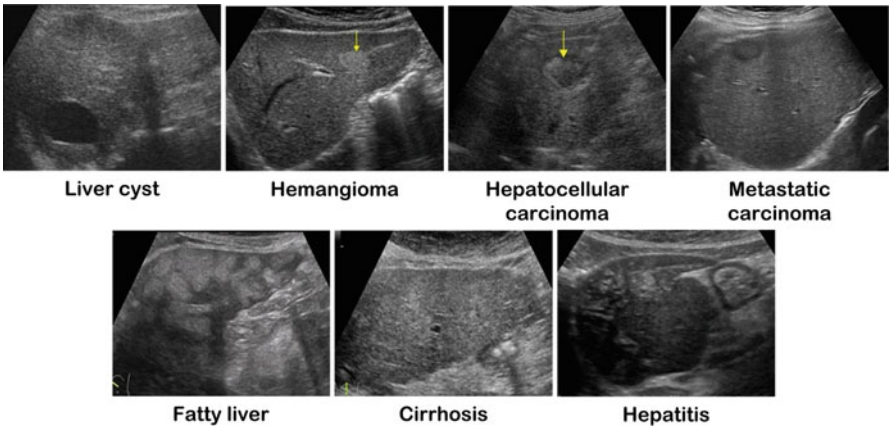

Fig. 3 The sample images of different liver diseases as seen on ultrasound (Geertsma 2014a)

Thyroid Diseases

Thyroid is one of the hormone-forming glands of the human body to control the rate of many metabolic activities. Thyroid diseases prevent the thyroid from generating right amount of hormones required for proper functioning of the body. Most common thyroid diseases are hyperthyroidism (too much hormones), hypothyroidism (not enough hormones), formation of thyroid nodules, and diffused thyroid diseases like multinodular goiter, Hashimoto's thyroiditis, Graves' disease (Chaudhary and Bano 2013; Cleveland clinic 2020; Wallace 2018). Thyroid nodules are solid or fluid-filled lumps with varying echogenicity growing unusually on the thyroid gland.

These nodules can be cancerous or noncancerous and are graded on the basis of Thyroid Imaging Reporting and Data Systems (TI-RADS) into benign, probably benign, borderline, suspiciously malignant, and malignant classes (Pedraza et al. 2015). Malignant thyroid nodules if go unnoticed or left untreated can lead to thyroid cancer. In diffused thyroid diseases, the echogenicity can increase or decrease with coarse texture and lobulated margins. Goitre is the enlargement of the thyroid gland and is mostly benign in nature. It is caused often by hypothyroidism due to deficiency of iodine. Thyroiditis is the swelling of thyroid leading to over or under production of hormones and occurs in cases when body's immune system produces antibodies that attack the thyroid. Hashimoto's thyroiditis leads to hypothyroidism and is prevalent in middle-aged women. Graves' disease occurs predominantly in females leading to glandular enlargement and hyperthyroidism. In Graves' disease, there is an increased blood flow in the thyroid leading to decrease in echogenicity and scalloped margins (Alexander et al. 2020). The sample images of different thyroid diseases as seen on ultrasound are shown in Fig. 4.

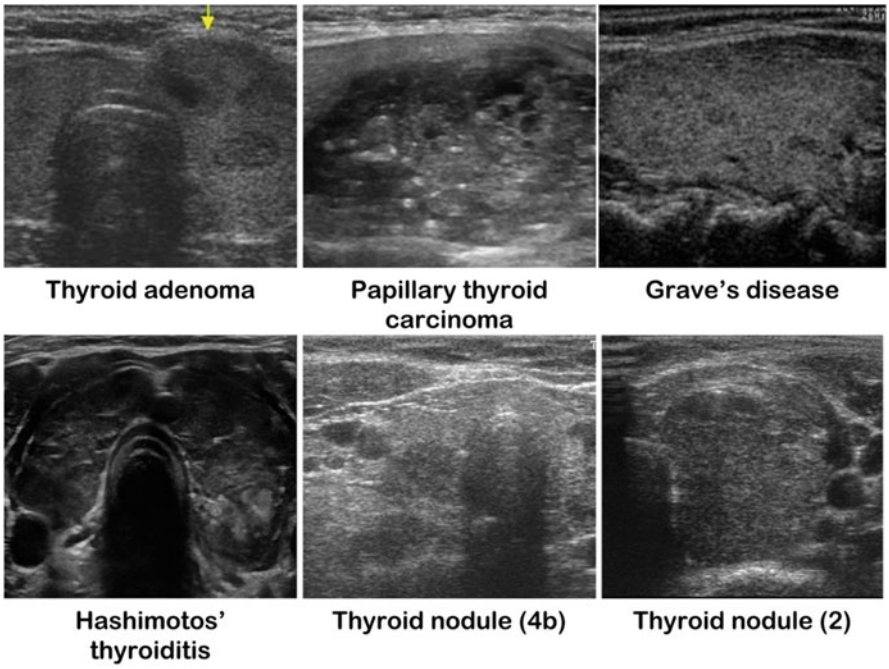

Fig. 4 The sample images of different thyroid diseases as seen on ultrasound (Geertsma 2014c; Pedraza et al. 2015)

Breast Diseases

Breast diseases develop in the breast tissue or milk-carrying ducts. The most common disease of the breast tissue is the development of a tumor that can be broadly classified as (i) noncancerous (benign) and (ii) cancerous (malignant). The benign tumors are considered to be noncancerous as these tumor cells do not spread and grow to other body parts, have a slower growth rate, and are more differentiable. The malignant tumors are considered to be cancerous and can also invade other healthy tissues in the body through the bloodstream (Gokhale 2009). The sample images of subcategories of different breast tumors as seen on ultrasound are shown in Fig. 5.

Breast cysts are firm circular lumps filled with fluid or solid debris. Generally, cysts have a thin, regular border, but due to the presence of solid components, cysts sometimes can have thick walls with irregular margins. Lipomas generally occur in women aged between 40 and 60 years. These are well-circumscribed masses having fat as the major component. Fibroadenomas occur in younger women. These are hard, solid masses having a well-defined shape. Intraductal papillomas are wart like tumors growing within the milk ducts and having a well-defined shape.

Breast carcinomas occur in epithelial tissues of the breast having higher rate of incidence. These are ill-defined masses having irregular contours and varying

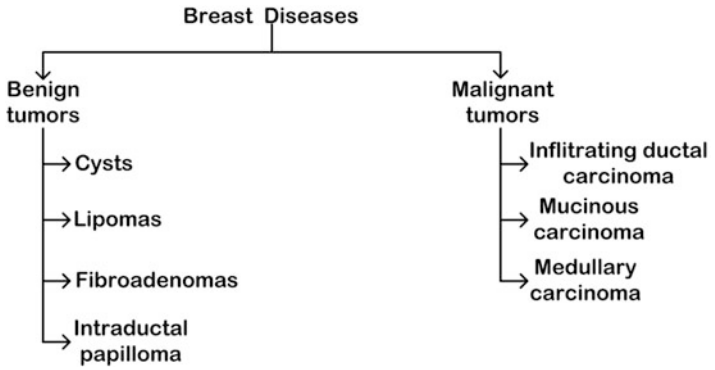

Fig. 5 Different breast diseases

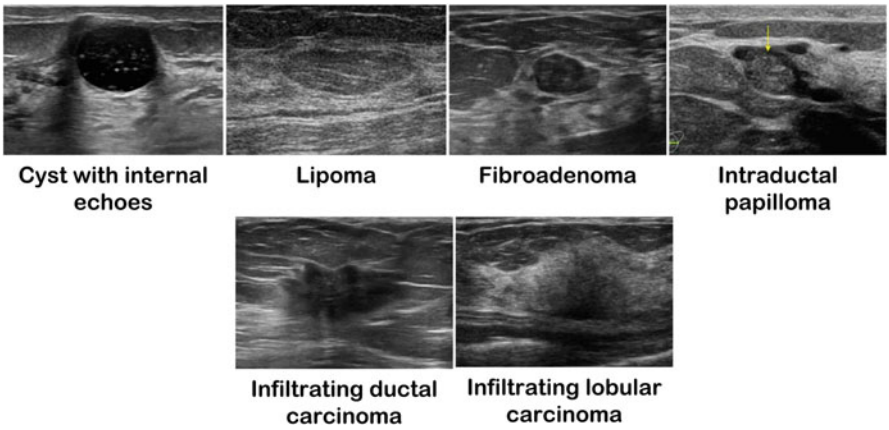

Fig. 6 The sample images of different breast diseases as seen on ultrasound (Geertsma 2014b)

echogenicity (Gokhale 2009; Medline plus 2022). Lobular carcinomas occur in the breast lobules, whereas ductal carcinoma is found in the cells lining the ducts of the breast. The sample images of different breast tumors as seen on ultrasound are presented in Fig. 6.

Other breast diseases occurring in women are as follows: hyperplasia, a medical condition affecting the breasts wherein there are a large number of abnormal-looking cells present in the breasts. Depending upon whether these cells are found in the lobules or the milk ducts of the breasts, the conditions are named atypical lobular hyperplasia and atypical ductal hyperplasia, respectively. Women diagnosed with hyperplasia have a higher risk of developing breast cancer. Fat necrosis is a breast condition that causes round, firm, painless lumps in the breast tissue caused due to any injury or radiation therapy. Breast fibrosis is a common condition causing lumpy breasts and is common in women below 45 years of age. Adenosis is a scar-like

fibrous tissue present in the glands of the breast that can elevate the risk of breast cancer among women manifolds (National Cancer Institute 2019).

Need of CAD Systems for Disease Diagnosis Using Ultrasound Images

For the advancement in disease diagnosis and efficient treatment, an accurate measurement of visual and nonvisual metrological characteristics from the images such as texture, size, and shape is very critical (Patel and Matcuk Jr. 2018). Different imaging techniques like X-rays, ultrasound, MRI, CT, thermography have widely been used for screening of different diseases. Even though the gold standard for disease diagnosis is pathologic staging, the treatment options and management of the diseases are frequently based on imaging and physical examination (Patel and Matcuk Jr. 2018).

Distinct modalities have their own advantages and disadvantages associated with them for measuring different metrological characteristics of the diseases according to past studies conducted by researchers, *for example*, in the case of breast cancer diagnosis, it has been observed that calcifications are better detected using mammography (X-ray images); however, in case of high-density breast tissue, mammography has shown poor detection results of the underlying abnormality. For this case, ultrasound modality is preferred to overcome the density effect and detect the abnormalities of the breast. Accurate measurement of tumor size in multiple planes is also possible with the help of ultrasound. Still, the accurate measurement is highly dependent on the operator's experience and the presence of artifacts also limits the diagnostic accuracy (Davis et al. 1996; Gruber et al. 2013). It is worth mentioning that the type of tumor also affects the accuracy of tumor size measurement and diagnosis, *for example*, malignant tumors have irregular margins and are diffused between collagen fibers, and it is difficult to accurately determine the shape and size information. The formation of ultrasound images includes the transmission of sound waves greater than 20 kHz into the body for capturing the live images of the organs such as the breast, kidney, liver, thyroid and the structures such as joints, tendons, muscles. Ultrasound imaging modality is the first line of treatment in the diagnosis of most of the diseases of soft tissue organs due to ease of use, availability, and zero exposure to harmful radiation. However, the drawback in the use of ultrasound imaging is that it is highly operator dependent and is affected by the presence of speckle noise that masks the important features in images that may be helpful in the diagnosis. This makes it hard for radiologists to clearly predict the disease and its progression. Based upon the points mentioned above, computer-aided diagnostic (CAD) systems have attracted immense attention from the research community to overcome the difficulties in the use of ultrasound imaging for disease prediction by paving a way for removing the effect of noise and artifacts and extracting the underlying learnable features from the ultrasound scans. It is also worth mentioning that these algorithms bring attention to the suspicious areas in the image to decrease the false-negative cases. The CAD systems try to search for the same information in

the images as a radiologist does while manually studying the scans and tries to quantify that information in terms of texture and shape features to achieve high accuracy in diagnosis. Thus, a novel CAD system design can act as a helping tool for the radiologists in gaining confidence in their diagnosis for efficiently classifying the different classes of diseases of the soft tissue organs.

The performance of the CAD systems can be quantified by using the confusion matrix technique that provides information about the true negative, true positive, false negative, and false-positive cases in the dataset depending upon which the parameters of the algorithms can be fine-tuned to decrease the detection of false positives thereby increasing the accuracy of prediction. The confusion matrix can also be used to quantify the accuracy, sensitivity, specificity, recall, and precision of the CAD system along with the AUC-ROC curves.

Based upon the associated advantages of CAD systems for disease prediction and monitoring, a huge amount of interest has been developed in the research community to design various CAD systems based on machine learning and deep learning techniques for the prediction of diseases as seen on ultrasound images of different soft tissue organs.

Further, the chapter has been divided into the following sections: Section "CAD System Designs for Disease Prediction Using Ultrasound Images of Soft Tissue Organs" outlines the different CAD system designs developed by researchers in the recent past for disease prediction using ultrasound images of organs like kidney, liver, thyroid, and breast. Section "Sample Demonstration" describes the methodology adopted by the authors in the present chapter to design a CAD system for breast tumor classification based upon the conclusions drawn in section "CAD System Designs for Disease Prediction Using Ultrasound Images of Soft Tissue Organs." Section "Sample Demonstration" presents the detailed result analysis of the proposed CAD system design and section "Conclusion and Future Scope" presents the conclusion of the chapter.

CAD System Designs for Disease Prediction Using Ultrasound Images of Soft Tissue Organs

This section presents the different CAD systems designed by various researches over the previous years (starting from 2017 to till date) for classification of diseases affecting the kidneys, thyroid, liver, and breast using ultrasound images.

CAD System Designs for Prediction of Kidney Diseases

A brief description of recent CAD systems designed for disease prediction of kidneys using ultrasound images is given in Table 1.

Table 1 describes in detail the studies carried out for the classification of different kidney diseases using artificial intelligence techniques applied to the ultrasound images. It can be observed from the table that most of the studies have been carried

Table 1 Description of CAD systems designed for disease prediction in ultrasound images of the kidneys

Author (s), Year of publication	CAD system description				
	Images	Preprocessing	Features	Feature selection	Classifier (result)
Mangayarkarasi and Jamal 2017	77 (Normal/ Cyst/ Calculi/ Tumor)	Median + Gaussian filter	FOS	–	PNN (Ac. 93.5%)
Raju et al. 2018	– (Normal/ Abnormal)	–	GLCM	PPCA	FFNN (Ac. 98.4%)
Acharya et al. 2019	405 (Normal/ CKD)	AHE	Cumulants, bispectrum, EQP	LSDA	SVM (Ac. 99.7%)
Chen et al. 2019	798 (CKD stages)	Median filter	GLCM	–	SVM (Ac. 79.8%)
Hao et al. 2019	226 (CKD stages)	–	GLCM + HOG + ResNet-34 deep features	–	Softmax (Ac. 96.0%)
Nithya et al. 2019	100 (Normal/ Tumor/ Stone)	Median filter	GLCM	CSA	ANN (Ac. 93.4%)
Sudharson and Kokil 2019	139 (Normal/ Abnormal)	–	AlexNet deep features	–	MSVM (Ac. 89.5%)
Priyanka and Kumar 2020	– (Normal/ Abnormal)	Background extraction	GLCM	PCA	NN (Ac. 77.8%)
Sudharson and Kokil 2020	4940 (Normal/ Cyst/ Stone/ Tumor)	–	–	–	Ensemble DL models (Ac. 96.5%)
Ahmad and Mohanty 2021	308 (5 CKD stages)	Median	GLCM	–	LDA (Ac. 97.4%)
Kim and Ye 2021	741 (Normal/ mild, moderate/ severe)	HE	GLCM	–	ANN (Ac. 95.4%)

Note: *FOS* first order statistics, *PNN* probabilistic neural network, *Ac.* accuracy, *GLCM* gray level co-occurrence matrix, *PPCA* probabilistic principal component analysis, *FFNN* feed forward back propagation neural network, *CKD* chronic kidney disease, *AHE* adaptive histogram equalization, *EQP* elongated quinary patterns, *LSDA* locally sensitive discriminant analysis, *SVM* support vector machine, *HOG* histogram of oriented gradients, *CSA* crow search optimization algorithm, *ANN* artificial neural network, *MSVM* multiple support vector machine, *PCA* principal component analysis, *NN* neural network, *DL* deep learning, *LDA* linear discriminant analysis, *HE* histogram equalization

out by using machine learning techniques, while very few studies have utilized deep learning architectures for kidney disease classification. Another point worth noting is that most of the studies have used preprocessed images for analysis.

Mangayarkarasi and Jamal (2017) used median and Gaussian filters for preprocessing 77 kidney ultrasound images. From the preprocessed images, first-order statistics (FOS)-based texture features have been computed, which were fed to probabilistic neural network (PNN) classifier for categorization of images into normal, cyst, calculi, and tumor classes achieving an accuracy of 93.5%. The study by Raju et al. (2018) proposed a CAD system to classify kidney ultrasound images into normal and abnormal classes. From the images Gray level co-occurrence matrix (GLCM), texture features have been computed which were further fed to probabilistic principal component analysis (PPCA) for finding optimal attributes. An accuracy of 98.4% has been reported using feed-forward back propagation neural network (FFNN), whose layers and neurons have been optimized using the gray wolf optimization (GWO) method. The highest accuracy of 99.7% has been reported by Acharya et al. (2019), for classifying the kidney ultrasound images into normal and CKD stages by computing texture feature using cumulants bispectrum and elongated quinary patterns (EQP)-based methods with support vector machine (SVM) classifier. The optimal feature set was computed using locally sensitive discriminant analysis (LSDA), and the ultrasound images were preprocessed using adaptive histogram equalization (AHE). Chen et al. (2019) proposed a methodology to classify kidney ultrasound images into different stages of CKD using GLCM texture features and SVM classifier. Hao et al. (2019) reported the use of a texture branch network model for differentiation between different CKD stages. The proposed model used the amalgamation of deep features extracted using ResNet-34 model with GLCM and histogram of oriented gradients (HOG) features reporting an accuracy of 96.0%. Nithya et al. (2019) in their study used 100 kidney ultrasound images for classification into normal, tumor, and stone classes. All the images have been preprocessed using median filter. From the filtered images, GLCM texture features have been computed, which were subjected to the crow-search optimization algorithm (CSA) for finding the optimal number of features. The study reported an accuracy of 93.4% using artificial neural network (ANN) classifier based on the computed optimal feature set. In a study carried out by Sudharson and Kokil (2020), the kidney images have been categorized into four classes: normal, cyst, stone, and tumor based on an ensemble of deep learning models ResNet101, ShuffleNet, and MobileNet-v2 using majority voting. The study reported an accuracy of 96.5%. In yet another study by Sudharson and Kokil (2019), deep features were extracted from AlexNet and fed to multiple support vector machine (MSVM) classifier to classify 100 kidney ultrasound images into normal and abnormal classes reporting an accuracy of 89.5%. Ahmad and Mohanty (2021) in their study utilized ultrasound images processed by median filter for calculation of GLCM texture features, which were fed to SVM classifier for classifying the stages of CKD, achieving an accuracy of 97.4%. Kim and Ye (2021) used histogram equalization (HE) in their study to preprocess the kidney ultrasound images. From the enhanced images, GLCM features have been computed and fed to artificial neural network (ANN) classifier

for categorizing the CKD stages into normal, mild, moderate, and severe classes achieving an overall accuracy of 95.4%.

Out of the total 11 recent studies selected in the time range (2017 onwards) for CAD systems designed for kidney ultrasound images, it has been observed that approximately 82% (8 out of 11) of studies have been carried out by using machine learning methods, while only 18% (2 out of 11) of the studies have used deep learning methods.

CAD System Designs for Prediction of Liver Diseases

A brief description of recent CAD systems designed for disease prediction of liver using ultrasound images is given in Table 2.

Table 2 presents the studies carried out for classification of liver diseases using ultrasound images. It is seen that a number of attempts have been made to design both machine learning and deep learning-based CAD systems. It can also be noted that most of the studies have used original ultrasound images for processing, while only two studies report the use of preprocessing methods for enhancing the image quality. Acharya et al. (2016) proposed a CAD system for differentiating between normal and fatty liver classes using GIST descriptors. The features were then optimized using marginal fisher analysis (MFA) and Wilcoxon test and fed to PNN classifier, reporting an accuracy of 98.0%. Hassan et al. (2017) preprocessed the liver images using anisotropic diffusion (AD) filter and classified the images into cyst, hemangioma (HEM), and hepatocellular carcinoma (HCC) classes using stacked sparse autoencoder (SSAE) reporting an accuracy of 97.2%. Bharti et al. (2018) developed a method to classify liver ultrasound images into four classes, namely, normal, chronic liver disease (CLD), hepatocellular carcinoma (HCC), and cirrhosis. The authors computed different texture features using Ranklet, gray level difference matrix (GLDM) and GLCM, and convolutional neural network (CNN). Optimal handcrafted features have been computed using RelieF algorithm and sequential feature selection (SFS), which were then combined with deep features and feed to an ensemble model of classifiers, namely, k-nearest neighbor (kNN), SVM, RF-kNN. The study reported an accuracy of 96.6%. Reddy et al. (2018a) in their study proposed a method using VGG-16 network for classification of normal and fatty liver disease (FLD), reporting an accuracy of 90.6%. In another study, Reddy et al. (2018b) proposed a self-designed deep CNN for classification of liver images into different stages of the fatty FLDs reporting an accuracy of 93.4%. The authors Sharma and Juglan (2018) in their study computed texture features from seven different models, namely, SGLCM, FOS, GLDS, SFM, Laws' texture energy measures, Fourier power spectrum (FPS), and Fractal for classification of liver images into normal and fatty classes. The optimal features have been computed using mutual information (MI). The authors have used four different classifiers SVM, IBk, AdaBoostM1, and BF Tree, reporting highest classification accuracy of 95.5% using SVM. Acharya et al. (2018) proposed a machine learning-based CAD system using features computed by bi-dimensional empirical mode decomposition and kNN classifier. The images have been preprocessed using

Table 2 Description of CAD systems designed for disease prediction in ultrasound images of the liver

Author (s), year of publication	CAD system description				
	Images	Preprocessing	Features	Feature selection	Classifier (result)
Acharya et al. 2016	100 (Normal/Fatty)	–	GIST	MFA + Wilcoxon test	PNN (Ac. 98.0%)
Hassan et al. 2017	110 (Cyst/HEM/HCC)	AD	–	–	SSAE (Ac. 97.2%)
Bharti et al. 2018	189 (Normal/CLD/HCC/ Cirrhosis)	–	Texture features	RelieF + SFS	Ensemble model (Ac. 96.6%)
Reddy et al. 2018a	157 (Normal/FLD)	–	–	–	VGG-16 (Ac. 90.6%)
Reddy et al. 2018b	1000 (FLD stages)	–	–	–	Self-designed CNN (Ac. 93.4%)
Sharma and Juglan 2018	90 (Normal/FLD)	–	Texture features	MI	SVM (Ac. 95.5%)
Acharya et al. 2018	463 (Normal/B/M)	CLAHE	Bi-dimensional empirical mode decomposition	PSO	PNN (Ac. 92.9%)
Amin et al. 2019	57 (Normal/Fatty)	–	Texture features	PCA	kNN (Ac. 91.8%)

Dandan et al. 2019	2942 (Normal/Fatty/ Fibrosis)	–	MGLCM + WMCM + GoogLeNet features	–	lightGBM (Ac. 82.6%)
Xu et al. 2019	79 (HCC/Abscess)	–	GLCM + GLRLM	SFS	SVM (Ac. 89.2%)
Krishnamurthy et al. 2020	596 (Liver disease classes)	–	Multi-level fractal + multi-domain wavelet features	RelieF + PSO	Ensemble of classifiers (Ac. 91.0%)
Pasyar et al. 2021	216 (Normal/hepatitis/ cirrhosis)	–	–	–	Hybrid ResNet-50 (Ac. 86.4%)

Note: *MFA* marginal fisher analysis, *PNN* probabilistic neural network, *Ac*. accuracy, *HEM* hemangioma, *HCC* hepatocellular carcinoma, *AD* anisotropic diffusion, *SSAE* stacked sparse autoencoder, *CLD* chronic liver disease, *SFS* sequential feature selection, *FLD* fatty liver disease, *MI* mutual information, *SVM* support vector machine, *B* benign, *m* malignant, *CLAHE* contrast limited adaptive histogram equalization, *PSO* particle swarm optimization, *PCA* principal component analysis, *kNN* k-nearest neighbor, *MGLCM* multi-scale gray-level co-occurrence matrix, *WMCM* wavelet multi sub-band co-occurrence matrix, *GLCM* gray level co-occurrence matrix, *GLRLM* gray level run length matrix

contrast limited adaptive histogram equalization (CLAHE) for classification into normal, benign, and malignant classes. Amin et al. (2019) in their study used frequency domain method-based texture feature extraction using wavelet packet transform. From the decomposed images, different texture features, namely, FOS, GLCM, and Local binary pattern (LBP), were computed and fed to PCA for computing optimal attributes. These attributes were fed to different classifiers like kNN, SVM, and LDA, reporting an accuracy of 91.8% using kNN classifier. Dandan et al. (2019) proposed a method to classify liver images into normal, fatty, and fibrosis classes using multiscale gray-level co-occurrence matrix (MGLCM) and wavelet multilevel subband co-occurrence matrix (WMCM) and deep features based on GoogLeNet. The features were fed to lightGBM achieving an accuracy of 82.6%. Xu et al. (2019) in their study proposed a CAD system for classification of liver cancer, that is, HCC and abscess using GLCM and GLRLM texture features. Optimal features were computed using SFS and fed to SVM classifier, achieving an accuracy of 89.2%. Krishnamurthy et al. (2020) proposed a method based on multilevel and multidomain features for classification of different liver disorders. The images were segmented by authors based on iso-contour segmentation. Different fractal and wavelet-based features were computed from the segmented ROIs and optimal features have been decided by using particle swarm optimization (PSO) algorithm. An ensemble of different classifiers was then used for differentiating between focal and diffuse liver diseases achieving an accuracy of 91.0%. Pasyar et al. (2021) used a hybrid ResNet-50 deep neural network for classification of liver ultrasound into normal, hepatitis, and cirrhosis classes achieving an accuracy of 84.0%.

Out of the total 12 recent studies selected in the time range (2016 onwards) for CAD systems designed for liver ultrasound images, it has been observed that approximately 67% (8 out of 12) of studies have been carried out by using machine learning methods, while only 33% (4 out of 12) of the studies have used deep learning methods.

CAD System Designs for Prediction of Thyroid Diseases

A brief description of recent CAD systems designed for disease prediction of the thyroid using ultrasound images is given in Table 3.

Table 3 presents the different studies carried out for classification of thyroid diseases using ultrasound images. It is observed that most of the investigators have used deep learning-based techniques, while very few studies have reported the use of machine learning-based methods. Also only one study has reported the use of enhancement methods for preprocessing the images. In the study by Chi et al. (2017), the authors designed a hybrid system for classifying benign and malignant thyroid nodules using deep features extracted from GoogLeNet which were further fed to a cost sensitive random forest (CS-RF) classifier achieving an accuracy of 99.1%. The authors Ma et al. (2017) in their study designed and trained two deep CNNs and fused the deep features from these networks using a summation layer. The

Table 3 Description of CAD systems designed for disease prediction in ultrasound images of the thyroid

Author (s), year of publication	CAD system description				
	Images	Preprocessing	Features	Feature selection	Classifier (result)
Chi et al. 2017	428 (B/M)	–	GoogLeNet features	–	CS-RF (Ac. 99.1%)
	164 (B/M)				CS-RF (Ac. 96.3%)
Ma et al. 2017	15,000 (B/ M)	–	Deep features from 2 CNNs	–	Softmax (Ac. 83.0%)
Wang et al. 2017	449 (TIRADS classes)	–	Texture + shape features	–	SS-FCME (Ac. 70.7%)
Zhu et al. 2017	298 (B/M)	–	–	–	ResNet-18 (Ac. 93.7%)
Moussa et al. 2018	814 (B/M)	–	–	–	ResNet-50 (Ac. 97.3%)
Raghavendra et al. 2018	344 (B/M)	Contrast adjustment	Multi-level elongated quinary patterns-HOS entropy features	PSO	SVM (Ac. 97.7%)
	345 (B/M)				SVM (Ac. 97.0%)
Liu et al. 2019	7690 (B/M)	–	–	–	Multi-branch CNN (Ac. 97.1%)
	450 (B/M)				Multi-branch CNN (Ac. 94.9%)
Ataide et al. 2020	99 (B/M)	–	Shape features	–	RF (Ac. 99.3%)

(continued)

Table 3 (continued)

Author (s), year of publication	CAD system description				
	Images	Preprocessing	Features	Feature selection	Classifier (result)
Nguyen et al. 2020	450 (B/ M)	–	–	–	Ensemble CNN model (Ac. 92.0%)
Song et al. 2020	440 (B/M)	–	–	–	Hybrid CNN (Ac. 96.1%)

Note: *B* benign, *M* malignant, *CS-RF* cost sensitive random forest, *Ac*. accuracy, *CNN* convolutional neural network, *TIRADS* thyroid imaging reporting and data system, *SS-FCME* semi supervised fuzzy C-means ensemble, *HOS* higher order spectral, *PSO* particle swarm optimization, *SVM* support vector machine, *RF* random forest

fused features were further fed to softmax classification layer for differentiation between benign and malignant thyroid nodules achieving an accuracy of 83.0%. Wang et al. (2017) in their study proposed a method to classify thyroid nodules based on TIRADS standard. The authors combined different texture and shape features computed from ultrasound images and fed them to a semi-supervised fuzzy C-means ensemble (SS-CME) model for classification achieving an accuracy of 77.07%. Zhu et al. (2017) proposed a deep learning approach for classification of thyroid nodules into benign and malignant classes using ResNet-18 network achieving an accuracy of 93.7%. Moussa et al. (2018) used ResNet-50 network to classify thyroid nodules into benign and malignant classes, achieving an accuracy of 97.3%. It can be noted that one study has been reported to use preprocessed thyroid images for classification purposes (Raghavendra et al. 2018). In this study, the authors have adjusted the contrast of the images in the first step and used multilevel elongated quinary patterns to represent the thyroid ultrasound images. From these images, higher order spectral (HOS) entropy features were extracted and an optimal feature set was obtained using particle swarm optimization (PSO). The SVM classifier was used for the classification of thyroid nodules into benign and malignant classes using a public and private dataset obtaining an accuracy of 97.7% and 97.0%, respectively. Liu et al. (2019) used a multibranch CNN for classification of thyroid nodules into benign and malignant classes. The authors also proposed a detection network prior to the classification phase based on pyramidal features and region proposals. The detection-classification network was tested on two different datasets achieving an accuracy of 97.1% and 94.9%, respectively. In the study by Ataide et al. (2020), shape features have been used to differentiate between benign and malignant thyroid nodules with random forest (RF) classifier reporting an accuracy of 99.3%. Nguyen et al. (2020) proposed an ensemble-based deep network model for classification of thyroid nodules. The thyroid ultrasound images were preprocessed for background extraction and then the detected nodules were fed to ResNet and Inception-based deep networks for classification of thyroid nodules into benign and malignant classes. The results of both the classifiers were then combined using different voting schemes like MIN, MAX, and SUM. Highest accuracy of 92.05% has been achieved using the MAX rule. Song et al. (2020) proposed a hybrid feature cropping CNN approach for the classification of thyroid nodules into benign and malignant classes. The proposed deep architecture has two branches: global branch for extracting global features and a feature cropping branch for processing the feature information in ultrasound images. The proposed technique achieved an accuracy of 96.1%.

Out of the total 10 recent studies selected in the time range (2017 onwards) for CAD systems designed for thyroid ultrasound images, it has been observed that approximately 30% (3 out of 10) of studies have been carried out by using machine learning methods, 60% (6 out of 10) of the studies have used deep learning methods, while only 10% of the studies (1 out of 10) have used a hybrid of both deep learning and machine learning methods.

CAD System Designs for Prediction of Breast Diseases

A brief description of recent CAD systems designed for disease prediction of the breasts using ultrasound images is given in Table 4.

Table 4 presents the studies carried out for the classification of breast tumors using ultrasound images. It was observed that most of the studies have used machine learning-based techniques as opposed to deep learning-based architectures for classification purposes. It can also be noted that almost all the studies have used preprocessing methods for enhancing the image quality before further processing. Rodriguez-Cristerna et al. (2017) proposed a technique for classification of breast tumors into benign and malignant classes. The authors used different contrast enhancement and speckle reduction techniques for finding out optimal methods for preprocessing to get efficient classification accuracy. From the preprocessed images, tumor regions have been segmented using watershed transform. Different texture features (like autocorrelation, GLCM and Laws' energy features) and shape features have been computed from the segmented tumor images and fed to Fischer linear discriminant analysis (FLDA) classifier reporting an accuracy of 84.0%. The authors also concluded that Fuzzy enhancement (FEN) followed by inference-based speckle filtering (ISF) produced better quality images for efficient classification performance. The author Singh et al. (2017) in their study reported the use of a hybrid feature selection technique for the classification of benign and malignant breast tumors. The ultrasound images were first preprocessed with a wavelet-based filter. From the filtered images different texture features (FOS, GLCM, GLRLM, Neighborhood gray tone difference (NGTD), statistical feature matrix (SFM), Laws' features, fractal-based features, Spectral features) and shape features have been computed. The computed features were then fed to different feature selection techniques. To determine the relevance of features, a new feature ranking method based on multiple criteria has been proposed. The relevant features were then fed to back propagation artificial neural network (BPANN) and SVM classifiers, reporting a highest classification accuracy of 96.6% using BPANN classifier. Moon et al. (2018) presented an approach to differentiate between metastatic and nonmetastatic malignant tumors. The images have been preprocessed using sigmoid and gradient-based filters and segmented using Level set method. From the segmented tumor images, different texture- and shape-based features have been computed out of which optimum features have been computed using backward feature selection, and an accuracy of 75.1% has been reported using logistic regression (LR) classifier. Nemat et al. (2018) in their study proposed a texture feature extraction method based upon transform domain methods. The authors have used CLAHE for preprocessing the images. From the preprocessed images, tumor regions have been segmented using the combination of manual and computerized techniques. Using the segmented tumor image, texture features based on Gabor transform and shape features have been computed. Optimal features have been computed using Bayesian automatic relevance detection (BARD), which were further fed to sparse logistic regression (SLR) achieving an accuracy of 97.1%. Highest accuracy of 99.3% has been reported by Uzunhisarcikli and Goreke (2018), using images preprocessed by

Table 4 Description of CAD systems designed for disease prediction in ultrasound images of the breast

Author (s), year of publication	CAD system description				
	Images	Preprocessing	Features	Feature selection	Classifier (results)
Rodriguez-Cristerna et al. 2017	1021 (B/M)	FEN + ISF	Texture + shape	–	FLDA (Ac. 84.0%)
Singh et al. 2017	178 (B/M)	Wavelet-based filter	Texture + shape	Hybrid	BPANN (Ac. 96.6%)
Moon et al. 2018	249 (Met/Non-Met)	Sigmoid + gradient magnitude filter	GLCM + shape	BFS	LR (Ac. 75.1%)
Nemat et al. 2018	104 (B/M)	CLAHE	Gabor-based texture + shape	BARD	SLR (Ac. 97.1%)
Uzunhisarcikli and Goreke 2018	153 (B/M)	Gaussian filter + CLAHE	GLCM + shape	Mann-Whitney U-test	ANFIS (Ac. 99.3%)
Fujioka et al. 2019	947 (B/M)	–	–	–	GoogLeNet (Ac. 92.5%)
Gonzalez-Luna et al. 2019	2032 (B/M)	–	Texture + shape	–	LDA (Ac. 89.0%)
Kriti et al. 2019	100 (B/M)	DPAD	Texture + shape	PCA	SVM (Ac. 96.6%)
Kriti et al. 2020	100 (B/M)	DPAD	GoogLeNet features	CrFS, ANFC-LH	ANFC (Ac. 98.0%)
Mishra et al. 2021	647 (B/M)	–	Texture + shape	RFE	AdaBoost (Ac. 97.4%)

(continued)

Table 4 (continued)

Author (s), year of publication	CAD system description				
	Images	Preprocessing	Features	Feature selection	Classifier (results)
Shia et al. 2021	677 (B/M)	–	PHOG	CrFS	LWL-SMO (AUC: 0.847)

Note: *B* benign, *M* malignant, *FEN* fuzzy enhancement, *ISF* interference-based speckle filtering, *FLDA* Fischer linear discriminant analysis, *Ac*. accuracy, *BPANN* back propagation artificial neural network, *Met* metastatic, *Non-Met* nonmetastatic, *SVM* support vector machine, *GLCM* gray level co-occurrence matrix, *BFS* backward feature selection, *LR* logistic regression, *CLAHE* contrast limited adaptive histogram equalization, *BARD* Bayesian automatic relevance detection, *SLR* sparse logistic regression, *ANFIS* adaptive neuro-fuzzy inference system, *LDA* linear discriminant analysis, *DPAD* detail preserving anisotropic diffusion, *PCA* principal component analysis, *CrFS* correlation-based feature selection, *ANFC-LH* adaptive neuro-fuzzy classifier using linguistic hedges, *RFE* recursive feature elimination, *PHOG* pyramid histogram oriented gradients, *LWL* locally weighted learning, *SMO* sequential minimal optimization, *AUC* area under the curve

Gaussian filter and CLAHE. From the preprocessed images, GLCM texture features and shape features have been computed and fed to adaptive neuro fuzzy inference system (ANFIS) classifier. Fujioka et al. (2019) used the fine-tuned GoogLeNet deep learning network for the classification of breast tumors into benign and malignant classes reporting an accuracy of 92.5%. Gonzalez-Luna et al. (2019) presented a comparative study for the classification of benign and malignant tumors using different machine learning techniques. From the images different texture features like GLCM, auto-mutual information-based features and shape features have been computed and fed to seven different classifiers, namely, SVM, kNN, radial basis function network (RBFN), LDA, Multinomial logistic regression (MLR), RF, Adaboost. The study reported the highest accuracy of 89.0% using LDA classifier. The authors have also proposed two CAD system designs in their previous works: one based on machine learning techniques (Kriti et al. 2019) and other a hybrid CAD system using deep features fed to conventional machine learning classifier (Kriti et al. 2020). In both the studies by the authors, detail preserving anisotropic diffusion filter (DPAD) has been used to preprocess the images and the tumor regions have been segmented out using Chan-Vese active contour method (Chan and Vese 2001). In the work Kriti et al. (2019), different texture features, namely, FOS, GLCM, GLRLM, Fourier power spectrum (FPS), Laws' TEM, and 2D Gabor wavelet transform-based features, were computed and combined with shape features. Optimum attributes were computed using PCA and these attributes were then fed to SVM classifier achieving an accuracy of 96.6%. In the other study, Kriti et al. (2020), deep features were extracted using GoogLeNet deep network and the optimal features were computed by using correlation-based feature selection (CrFS) and a wrapper-based method adaptive neuro-fuzzy classifier using linguistic hedges (ANFC-LH). The selected optimum features were fed to ANFC achieving an accuracy of 98.0%. Mishra et al. (2021) in their study proposed a pipeline for differentiation between benign and malignant tumors. Different texture features, namely, HOG, Hu moments, GLCM features, and shape-based features, have been extracted from each segmented tumor image. Out of the extracted features, optimum features have been selected based on recursive feature elimination (RFE). After this, SMOTE oversampling technique has been applied to deal with the imbalanced data. The resultant feature set is then fed to different machine learning-based classifiers, namely, RF, gradient boosting classifier (GBC), AdaBoost classifier, SVM, LR, and decision tree. The highest accuracy of 97.4% has been reported using AdaBoost classifier. Shia et al. (2021) in their study proposed an unsupervised approach for the classification of breast tumors. From the images, a pyramid histogram of oriented gradients (PHOG)-based texture features have been computed to capture efficient edge description in the images. The optimal feature set has then been computed CFS and classification has been carried out using locally weighted learning (LWL) + sequential minimal optimization (SMO) achieving an area under the curve (AUC) of 0.847.

Out of the total 11 recent studies selected in the time range (2017 onwards) for CAD systems designed for breast ultrasound images, it has been observed that approximately 82% (9 out of 11) of studies have been carried out by using machine

learning methods, 9% (1 out of 11) of the studies have used deep learning methods, while 9% (1 out of 11) of the studies have used a hybrid of deep learning and machine learning methods.

According to all the studies presented in Tables 1, 2, 3, and 4, it has been observed that the majority of CAD systems designed for disease classification in ultrasound images of soft tissue organs have used texture descriptors computed in the time domain for the classification task, while very few studies have used transform domain-based texture feature extraction methods (Krishnamurthy et al. 2020; Nemat et al. 2018). Thus, to demonstrate this, an attempt has been made as an example in the present chapter to test the efficacy of transform domain-based texture features for the classification of breast tumors. The texture features were generated in the current work utilizing a transform domain-based method called nonsubsampled contourlet transform (NSCT).

Sample Demonstration

The present chapter explains the basic process of designing a CAD system for classifying breast tumors into benign and malignant classes using adaptive neuro fuzzy classifier (ANFC) based on texture features computed in transform domain amalgamated with shape features. It was validated in authors' previous work that shape and size characteristics of tumors were better represented by using pre-processed images while texture was better quantified from raw images (Kriti et al. 2019). The work was aimed at making the breast tumor classification as efficient as possible by using a combination of the essential features extracted from the original as well as preprocessed images. Therefore, a similar approach has been used in the present chapter also. The workflow adopted in the example is shown in Fig. 7.

The exemplary CAD system design in the present chapter made use of 100 breast ultrasound images which include 40 benign and 60 malignant images that were selected from an openly available, standardized benchmark database of ultrasound images (Geertsma 2014b). These images have been methodically selected to include representative cases of each class from the database. The images contained in the dataset have solitary as well as multiple tumors.

As can be seen from Fig. 7, the images have been preprocessed using the DPAD filter. The DPAD filter belongs to a class of nonlinear iterative filters used to implement the anisotropic diffusion technique used for the removal of multiplicative noise from the images (Aja-Fernandez and Alberola Lopez 2006). Both the original and preprocessed images have been segmented using the Chan-Vese active contour method to obtain the tumor images. This method has been designed to segment the medical images that prove to be a little difficult to segment based on traditional methods of segmentation like thresholding. This method has been formulated as an energy minimization problem. Out of a total of 40 benign images, 43 tumor images have been segmented out, and out of a total 60 malignant images, 61 tumor images have been segmented out. Original segmented tumor images have been subjected to NSCT for multiresolution decomposition, and from the resulting subband images,

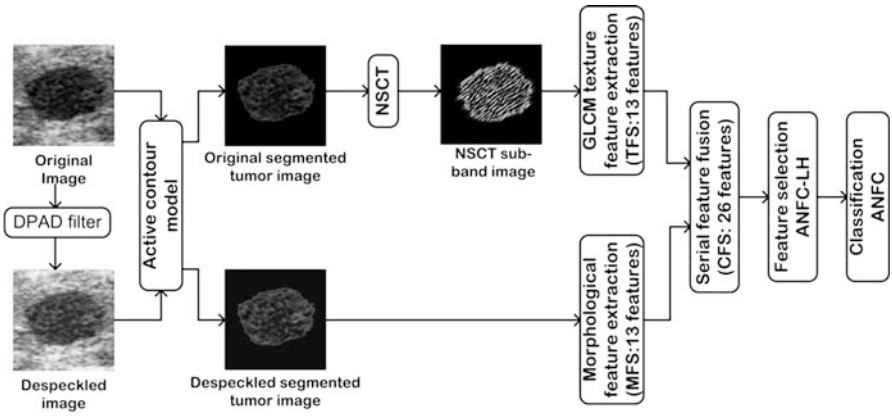

Fig. 7 Workflow adopted to design the NSCT-based CAD system. Note: *DPAD* detail preserving anisotropic diffusion, *NSCT* non subsampled contourlet transform, *GLCM* gray level co-occurrence matrix, *TFS* texture feature set, *MFS* morphological feature set, *CFS* combined feature set, *ANFC-LH* adaptive neuro fuzzy classifier using linguistic hedges, *ANFC* adaptive neuro-fuzzy classifier

texture features based on GLCM were computed. The preprocessed segmented tumor images have been used to compute shape features. Further, the texture features and shape features have been serially fused into a single feature vector. The computed feature vector was then subjected to ANFC-LH classifier for finding an optimal set of features based on hedge values which were further fed to ANFC classifier to classify the tumors into benign and malignant classes.

The NSCT was used to improve efficiency in image representation. It has two building blocks, namely, (a) nonsubsampled pyramid (NSP) structure and (b) nonsubsampled directional filter bank (NSDFB). The result of the multiscale characteristic of NSCT gives a subband decomposition similar to Laplacian pyramid. To obtain this, a two-channel nonsubsampled 2-D filter bank was used. In the computation of such decomposition, à trous algorithm is used giving a redundancy of $J + 1$, where J is the total number of decomposition stages. At the j^{th} stage, the ideal pass band support of the low-pass filter was the region $[-(\pi/2^j), (\pi/2^j)]^2$, whereas for the high-pass filter, it was $[(-\pi/2^{j-1}), (\pi/2^{j-1})]^2/[(-\pi/2^j), (\pi/2^j)]^2$. The upsampling of filters of the previous stage further designs the filters for the consequent stages. This provides the required multiscale property to NSCT without having to design additional filters at each stage (da Cunha et al. 2006). The directional filter banks were constructed using carefully sampled two-channel fan filters and critically analyzed resampling operations (Do and Vetterli 2005). This gives out a filter bank, which is tree-structured that splits the 2-D frequency plane into directional wedges. For constructing nonsubsampled directional filter bank (NSDFB), the shift invariant directional filter banks were made by removing any downsamplers or upsamplers in the two channel filter banks and later upsampling filters accordingly. This gives filter banks, which were nonsubsampled in nature. For an M stage NSDFB, the number of directional subbands that were produced was 2^M.

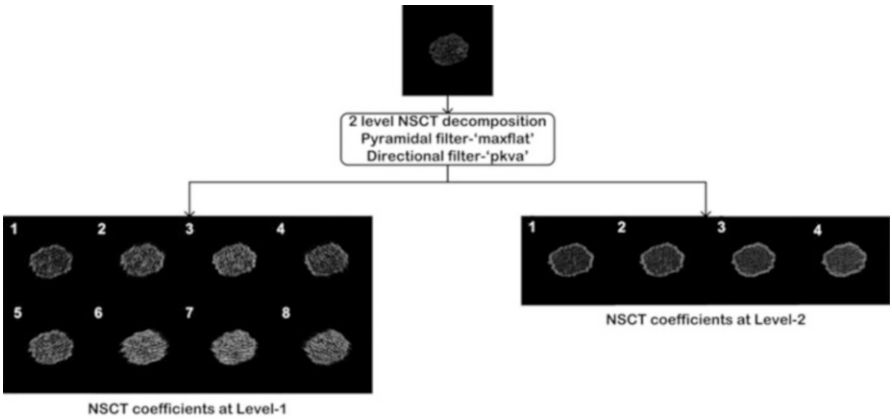

Fig. 8 Shifting of breast ultrasound images to NSCT domain giving out 12 subband images (8 + 4) from both levels to be used for classification purpose

For implementation in the present chapter, the breast ultrasound images were resolved using NSCT into two levels of 8 and 4 directional subbands as shown in Fig. 8. These subband images hold the essential feature characteristics of the breast tumors and were used for classification purpose. Through exhaustive experimentation, it was observed that best accuracy is achieved when features were computed from 5th subband image of level 1 generated using "maxflat" pyramidal filter and "pkva" directional filter.

For computing GLCM texture features, a co-occurrence matrix is first formed by computing the number of times a combination of different pixel pairs has occurred in the image for a particular gray level. This combination occurrence is calculated over a specified direction and inter-pixel distance (d). In the present work, inter-pixel distance (d) is taken to be 5 and the directions are taken to be 0°, 45°, 90°, and 135°. The final feature matrix has been formed by taking out the average of the features taken in all four directions. The different features computed were entropy, correlation, angular second moment (ASM), inverse difference moment (IDM), etc. Shape features like tumor area, convex area, tumor boundary, diameter, Euler number, orientation, eccentricity, circularity, elongation have been computed and fused with the texture features serially.

The classification systems based on fuzzy rules have gained popularity in pattern recognition and machine learning problems as these systems can easily address complex real-world problems. The fuzzy system is based on fuzzy logic such that the vagueness in the data can be handled. The fuzzy system uses membership functions and a set of different *IF-THEN* rules for reasoning and finding the result. The linguistic hedges (LHs) were used to introduce the importance of fuzzy rules in terms of linguistic terms. The LH values were tuned to improve the flexibility of fuzzy sets, thus removing the ambiguity of the overlapped classes. The LHs were defined as the linguistic terms to modify other linguistic terms. The various experimental results established that the LH value near 1 of the fuzzy classification set

indicates that the feature was relevant and significant for that class. For values other than this, the features were considered to be irrelevant and insignificant for that class (Cetisli 2010a, b).

The ANFC classifier combines the various advantages of a fuzzy inference system and the neural network classifier. The fuzzy inference system makes use of a set of fuzzy *IF-THEN* rules that can approximate nonlinear functions to deal with imprecise real-world problems while the neural network classifier had the capability to self-organize and learn adaptively from its environment. The network contains multiple nodes connected to each other by directional links. The output of the nodes was dependent on the parameters of the nodes, which were changed during network training to reduce the error (Kar et al. 2015; Kher et al. 2015; Rawat et al. 2018). The structure of the ANFC-LH classifier is shown in Fig. 9.

In the present chapter, the combined feature set containing a total of 26 features (13 GLCM texture based + 13 shape) was subjected to ANFC-LH, a wrapper-based method of feature selection based on linguistic hedges and ANFC classifier. Out of the total 26 features, seven most relevant features were selected on the basis of linguistic hedge values for the efficient classification of breast tumors. The obtained optimal feature set was then fed to ANFC classifier. The results obtained in the example for efficient breast tumor classification are shown in Table 5.

Table 5 indicates that an accuracy of 88.24% was achieved for the classification of benign and malignant tumors with individual class accuracy (ICA) values of 85.71% and 87.09% for benign and malignant cases, respectively. It was observed that only six testing instances out of a total of 51 instances were misclassified where three instances belonged to the benign class and three instances belonged to the malignant

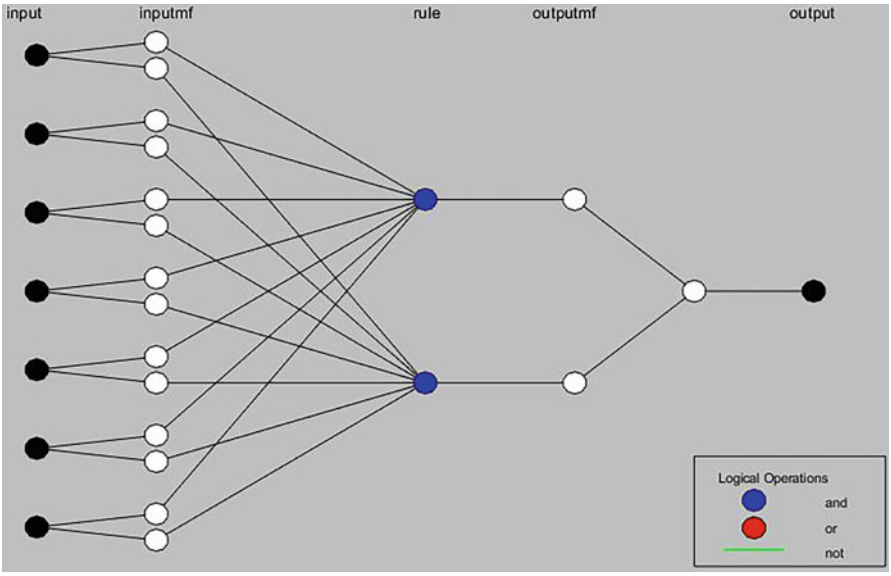

Fig. 9 The structure of the ANFC-LH classifier

Table 5 Results obtained in the present work for efficient breast tumor classification

Feature set (length)		Benign	Malignant	Acc. (%)	ICA_{Benign} (%)	$ICA_{Malignant}$ (%)
CFS (7)	**Benign**	18	3	88.24	85.71	87.09
	Malignant	3	27			

Note: *CFS* combined feature set, *Acc* accuracy, *ICA* individual class accuracy

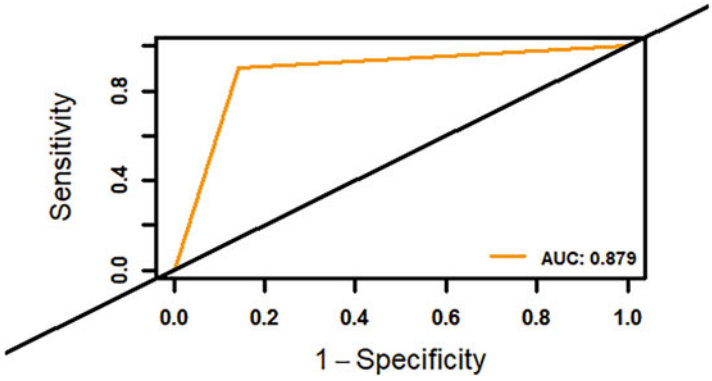

Fig. 10 The ROC analysis of the proposed CAD system

class. The performance of the ANFC-LH classifier was also measured using the ROC curve as shown in Fig. 10, with an AUC value of 0.879 (Fawcett 2006).

As discussed previously, for early disease detection to manage its progression and improve chances of survival, imaging-based screening tools are predominantly used. However, the conventional approach is that the offline scans are available to the radiologists for interpretation and the results of the CAD system are also available to the radiologist in offline mode. This constraint on the CAD system can be overcome by real-time interaction between the radiologist and the CAD system used for classification purposes. One approach for this task is to provide a CAD server that has the proposed CAD system design integrated in it and is configured to interact with the radiologist in real-time, that is, accept real-time data and generate the diagnosis result for the radiologist (Zhang and Heffernan 2012). Another approach for real-world implementation of the proposed CAD system design is integration of the CAD system with the ultrasound machine, wherein the radiologist marks the region of interest for interpretation and simultaneously the software analyzes the ultrasonographic features and produces the final assessment (Kim et al. 2017). Depending upon the need, the radiologists can use the proposed CAD system design as an interactive 2nd opinion diagnostic tool for differential diagnosis between breast abnormalities, giving the radiologists a confidence in their interpretation and diagnosis.

Conclusion and Future Scope

The present chapter sheds light on the importance of the ultrasound imaging technique for the diagnosis of diseases of soft tissue organs. The chapter also presents a brief review of the literature on the classification of diseases of the soft tissue organs as seen on ultrasound images. Based on the conclusions drawn from the literature, a design of CAD system has been demonstrated as an example for the classification of breast tumors into two classes, namely, benign and malignant. For this classification of the breast ultrasound images, the texture features, that is, the textural characteristics of the region inside the tumor, and the morphological features, that is, characteristics of the tumor-associated with its shape, were considered essential. The work applies DPAD filtering algorithm to the original images for the morphological feature extraction. Since texture features get distorted by DPAD filtering, NSCT and GLCM were applied to the original images for the texture feature extraction from original images. The ANFC-LH focuses on efficient feature selection, thereby resulting in the maximum accuracy obtained by the ANFC classifier. The literature review in this chapter has been included with an aim to provide a comprehensive review of recent trends in the development and design of CAD systems that can be of significant importance to students and researchers starting to work in the field. The survey also tries to highlight the algorithms that have still been unexplored and can be applied for the disease prediction in soft tissue organs.

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Part XIII

Legal Metrology and Global Trade

Legal Metrology and Global Trade

78

G. R. Srikkanth

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Abstract

The global trade has existed for ages, but in recent decades, global trade has become extremely important as it has been immensely influencing the world economy and society. The inflation in global trade over the ages has resulted in globalization, which has helped in increasing the cross-border activities and capital flow in the market. There are many factors such as the economy of the country, social factors, technological advancement, consumer needs and demands, legislative framework, and political scenario of the nations which have stimulated the global trade. The legal metrology also has a paramount role in influencing the world trade as the legal metrology establishes and imposes standards for weight and measurements, which is one of the most important

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parameters in global trade as the global trade involves numerous commercial transactions and measurements of commodities which are exchanged between nations and thereby it has a predominant role in promoting global trade. The legal metrology laws ensure and maintain the accuracy and fairness in the global trade, it aids in production optimization, it helps in building consumer and corporate trust for the products and services being globally traded, and it also causes the evolution of new technologies and innovations. It also played a remarkable role in consumer protection and fair trade promotion. At present, every industrialized nation has a government agency in charge for monitoring the measurement accuracies, and they also put mandates for complying with legal metrology laws and other technological standards. This research paper attempts to analyze the importance of legal metrology laws in promoting world trade.

Keywords

Global trade · Legal metrology · OIML · Globalization · Weight and measurement · Trade routes · Economy · World Trade Organization

Introduction

A customer in India who wishes to buy the latest electronic gadget released in Europe or America can simply place an order on a leading e-commerce marketplace, and within a short period, he will receive his order at his doorstep thousands of miles away. This has become possible with the evolution of global trade. Global trade is an exchange of goods or services for consideration between the national or the individual entities across the international borders. Global trade eliminated borders of the national for trade and commerce. Global trade has its existence since the ancient time, and according to Peter Watson, the first long-distance trade extends back to almost 1500 BC (Watson 2012). Global trade involves exchange of products and services between two or more nations. As compared to domestic trade, carrying out global trade is a much more challenging operation as the trade has to comply with the regulations of both the markets (Tyc 2021).

Prior to conducting a global trade, many factors are to be considered, which include cultural and social impacts, demographic trends, legal concerns, and political situations of the nations in which the trade is to be carried out, and global trade can also be influenced by the changes in the natural environment and due to technology advancement (Kelly 2021). Today's world economy is characterized by international interdependence, and thereby no nation can exist in isolation or claim to be self-sufficient. All the economies are interdependent and aim to be self-sufficient. All the economies are interdependent on each other in some way or the other for goods and services. Global trade and the capital flow in the market have a huge role in promoting the nation's economy. Since the production and supply chain covers almost nations and continents, the country's commodity supply is highly reliant on economic activity in several other countries, and hence all the nations are highly

interdependent on each other (What is Globalization? 2021). The main aim of the international trade is to boost the economy of the nations and to raise people's living standards; it also allows the inhabitants of one country to consume and enjoy commodities that are produced in another country, and it also helps in promoting healthy competition in the market, which will in return facilitate in fair trading. The nations have also taken various measures and steps for obtaining financial stability and economic growth and for improving the living standards of people of the nations by establishing an organization which would help them in promoting and encouraging fair trade globally. The World Trade Organization (WTO) (1995) has been established to deal with all the trade-related aspects on a global level. WTO's (Plummer et al. 2005) main aim is to ensure to remove obstacles in international trade and negotiate trade rules and help in settling disputes.

Global trade, on the other hand, has further strengthened the role of legal metrology as it creates trust and confidence between the parties to the trade, and states/governments, being the promoter of trade and commerce, shall establish necessary standards and mandatory requirements to create trust and confidence. History of weights and measures can be traced over the last 5000 years, and it has evolved from time to time as nonstandard measures are the root cause of economic exploitation, and to address the issues faced by the consumers and the other stakeholders, respective nations have regulated laws, rules regarding weights and measures.

Importance of Global Trade

Global trade involves the transferred capital in exchange for commodities and services across international borders or territories. It also allows consumers and the countries to be exposed to the commodities and services that are not accessible or available in their own countries. The need for cross-border trade is felt as it is not possible for one particular country to be able to manufacture all the goods or provide all kinds of services, and thereby countries need to be interdependent on each other for goods and services, and therefore the countries cannot disregard or ignore the symbiotic trade relationship which they share with the other countries (Stobierski 2021). The global trade of products and services is very important for economic growth and the overall development of society (Kirton 2009). Global trade is a far more complicated operation as compared to domestic trade as because international trade involves many enormous factors such as regulatory frameworks, government policies, economic factors, exchange rate, political scenario, and market influences. In most of the nations, global trade is accounted as a significant component of the elevation of their country's GDP (Ross 2021).

The significance of global trade has been understood from prehistoric times (McKenzie and Coppolaro 2013). Global trade enables the nations to make better use of their assets and natural resources, such as land, labor, capital, and technology, in a wiser manner basically, and the global trade helps the nations to invest in their assets and resources in a more systematic and a strategical manner. Global trade

helps to promote and maintain healthy competition in the market, and thereby it results in more competitive pricing and in providing the best possible quantity and quality of commodities and services for consumption**.** One of the most basic advantages of competition is that it promotes innovation (Engagement and competition: how tech can help small businesses 2016). The competition among corporates or businesses may lead to the production of new or improved commodities, as well as more efficient services**.** The cross-border trade also aids in disregarding the domestic monopolies in the market, and it promotes and maintains fair trade and competition in the market. Trading in the international market can be done either by exports or importing commodities in the market. Importing basically implies a transfer of economic ownership between residents and nonresidents, and exports are taking away goods from domestic markets to international markets (Bernard 2011). Imports and exports are the major factors behind the growing interdependency between the actors on the stage of the world economy (Alexander 1986).

Global trade allows the countries to achieve economic progress, and it also helps the countries in improving their access to physical capital and export markets (Baylis and Smith 2021). As per the World Trade Organization report, the global trade is decreased by 9.6% in the year 2021, but it rose by 8%, and the value of merchandise exported is USD 17.58 trillion among all the nations of the world (Global trade rebound beats expectations but marked by regional divergence 2021). Over many years, the countries have formed economic alliances to enable the flow or transfer of goods and services from one country to another. Over the period of time, such cross-border trade resulted in a considerable increase in the globalization process. Globalization is considered as a process that is based on the international approaches, and it aims to expand trade and commerce on a global level (Kaur and Kaur 2017). Although many researchers attribute that globalization began in the contemporary era, many historians write that globalization dates back to the European age, and evidence is also found that global trade existed during the third century BCE (Vanham 2019). The term globalization was coined in the early twentieth century, but it evolved into its modern meaning and became popular in the 1990s, but the large-scale globalization had already begun in the 1820s, and it had caused a dynamic development in the late nineteenth and early twentieth centuries; it resulted in the world's economy developing rapidly and interdependent with each other country (Gopinath 2011). Globalization has resulted in the opening of the economy and has also aided in promoting cross-border trade. Analyzing from an economic perspective, it can be stated that globalization holds a huge impact on the exchange of commodities, services, data, technology, and other economic resources on a global level (IMF Staff 2008).

Global trade leads to the globalization of products and services, and it has certain advantages (Ozer 2019). Global trade provides the consumer a greater array of options for the same commodity from various places available for consumption, and this helps to not only improve their standard of living but also helps the country prosper. Global trade gives a country a comparative advantage by enabling it to specialize in the production of goods with lower opportunity costs of production

than its competitors. Global trade also aids in the allocation and utilization of resource in a more efficient manner as the countries tend to produce or offer goods or services in which they have a competitive advantage. Global trade enhances production efficiency because nations would endeavor to adopt better manufacturing processes in order to keep costs down and remain competitive. Countries who are able to create a product at the lowest feasible cost will be able to earn a greater market share, and as a result, there is an incentive to manufacture efficiently, and this will assist in raising product standards, and this will enable the consumers to have access to a high-quality product. Globalization allows a country to consume goods that are either unavailable or prohibitively expensive to manufacture inside its boundaries, and as a result, importing from other nations via international commerce becomes less expensive. International trade helps various nations to utilize their excess or surplus products by selling those to other countries for obtaining capital, and it would also help the nations to prosper economically (Klasen 2019). Global trade decreases trade volatility by increasing the size of the market with vast supply and widespread demand, and the goods pricing tends to be more consistent in international trading. Global trade would promote international peace, goodwill, and mutual understanding between the nations, and it would also promote the economic interdependence between countries, which frequently leads to deep cultural ties and also in preventing conflict between the nations (Melitz et al. 1998).

Global trade and the financial transactions that accompany it are often conducted to provide nations with products that they lack in return for those that they produce in excess. Such transactions, when combined with other economic measures, tend to increase a nation's standard of living (Thomas Warren Hertel 1997). The countries make attempts to promote reasonably fair trade among them.

Evolution of Global Trade and the Importance of Trade Route

Trade is a fundamental economic concept that entails the act of buying and selling goods and services for a valid consideration. Trade was also a boon for human interaction, which helped in building cross-cultural contact and economic prosperity, and it also raised the living standards of people. Trade is as long as the existence of civilization, and evidence reveals that mercantile business existed for the last 5000 years (Anne Osborn Kruege 2020). The emergence of trade over the centuries has helped in the evolution of trade routes, which have been used to establish connections between the civilizations from where the goods are manufactured to places where goods are consumed. Trade confine within the boundaries of a country is called domestic trade, and international trade is exchange of good and services across two or more countries which is known as the global trade. Trade can create job opportunities and economic prosperity of the nations and can increase the standard of living of its citizens (Smith 2013).

Trade and civilization are practically inseparable. The traders have travelled across the globe far and wide in pursuit of trade and commerce; they used every available mode of transport and reached new territory and offered goods and services

over the centuries. Among the most notable trade of the past is the oldest long-distance trade that took place around 3000 BC and was between Mesopotamia and the Indus Valley (Lamberg-Karlovsky 1972) (which is now located in Pakistan). This long-distance trade was exclusively limited to luxury goods such as spices, textiles, and precious metals, and in those early days, the cities that were rich in these products were monetarily wealthier and financially benefitted because of these long-distance trades, and also the cities along with these trade routes later have been developed into manufacturing and trading hubs, and consequently the economic standards of the locals had improved. Another most significant trade took place around 2000 BC (Journal of ancient History 2017) on the island of Cyprus, which had at that time become a major player in the Mediterranean region, and the region had been transporting huge amounts of copper resources to the Middle East and Egypt, and this region has its own natural resources such as papyrus and wool (Frankopan 2015). This trade also led to prospering of other countries like Phoenicia, known for its maritime prowess, which traded rich cedar wood and linen across the Mediterranean. China flourished as it traded in jade and spices and eventually started trading in silk, and Britain thrived in its economy as it traded in tin.

The Chinese started trading in silk in Europe in consideration which they received gold, silver, and wool from Europe. In the past, one of the major trade routes was the Silk Route (Elisseeff 2000); it started operating during the second century BC, and it enabled trade between China and the Far East, as well as the Middle East and Europe, easier (Lally 2021). Later, due to the lack of appropriate roadways and facilities, the maritime trade routes emerged and were believed to be the most efficient means of transporting goods from one region to another. The trade routes helped in trading everything from silk and spices to glassmaking methods and gunpowder, and it made its way from the East to West. The most recent archaeological discovery confirms that the classical international trade, known in ancient Indian history as "Indo-Roman trade," existed long ago.

The trade route helped in building the trade network worldwide; it included roads between points which included not only the transportation of goods along these roads but also knowledge, ideas, religious practices, etc.

The emergence of trading led to the evolution of different trade routes, which had emerged with time. The Silk Route/Silk Road was one of the most well-known and historic trading routes. It emerged in 130 BCE and it existed until 1453 AD. This trade route was first originated during the ruling period of the Han dynasty by Zhang Qian (Chandra and Banerjee 2008). The Silk Route was a historic trade route which mainly aided in connecting China's main civilizations with the Roman Empire (Philip 2020). Along with this trade route, silk was one of the principal commodities which were traded as the traders discovered that ferrying merchandise along this route was profitable. This route aided in exporting silk from China to Europe; in return it received wool, silver, and gold for such exchange. With time the trading hubs began to spring up along this trade route, and it served as a gathering place for individuals to share their knowledge and interests. Later, with the fall of Roman Empire in the fourth century CE, gradually it decreased the use of Silk Route, and the route was barely used, and later it was resurrected by the Mongols in the thirteenth

century, and Marco Polo (1300) used it when he became one of the first Europeans to travel to China. Later, in 1453 AD, the route again lost its significance. Another most notable trade route in the past was the Spice Route, and this was a maritime route; it generally connected the eastern part with the western part. This maritime route was used by the Arabian as they traded spices through this route to Europeans, and the Spice Route (Keay 2006) was also known as the Maritime Silk Route. The spices which mainly traded through this route were pepper, cinnamon, nutmeg, cloves, cassia, turmeric, and star anise and were hugely traded through this route. Prior to the fifteenth century, the trading in spices was extremely uncommon and expensive, and during that period, the spice trading was controlled by the Arabs and North Africans between the East and the West. So, during the Age of Exploration in the fifteenth century, it became possible for people to travel long distances and establish new global trade links; thus, Europeans began sailing and establishing new global trade relations with people in China, Japan, and Indonesia. Even some scholars also claimed that the spice trade spurred the growth of international trades and encouraged the exploration of new areas, and it has also helped foster new diplomatic connections between the East and West. The Incense Trade Route (Unesco Journal 2005) was a historic trade network of significant land and sea trading routes which connected the Mediterranean area to the eastern and southern parts. This route also extended from Mediterranean ports through the Levant and Egypt to India and beyond. The Incense Route (River 2019) was mainly created to export frankincense and myrrh, which are only found in the Arabian Peninsula's southernmost regions (which is now in modern Yemen and Oman). Later as trade along the Incense Route flourished, it was observed that almost 3000 tons of incense were transported each year (Karl Reinhold Haellquist 1991). Another most important trade route was the Amber Route which linked the Baltics with Europe, and this route was relevantly used for trading in 3000 BCE, with archaeological data suggesting that Baltic amber beads were traded through this way. The Romans built an Amber Route (2011) between the Baltics and in the rest of Europe, as the Romans valued amber for its medicinal and its aesthetic qualities. Later, in the twelfth and thirteenth centuries BC (Claire Cock-Starkey 2016), the Baltic Sea was ruled by the knights, and it became a very important source of capital for the Teutonic Knights. Another majorly important trade route is the salt route (Cartwright 2017) as for generations salt has been in great demand, and it is a precious commodity as it can be used for multiple purposes. The salt-rich locations grew into major trade hubs as it helped in transiting of salt to regions that lacked it. The salt route connected Rome with the Adriatic coast. Another prominent trade route was the Trans-Saharan trade route (The Metropolitan Museum of Art 2000); this is one of the oldest trade routes; it began its significance in the fourth century CE. This route mainly connected North and West Africa. In the eleventh century, this route also aided in transporting of the caravan with over a thousand camels, and various commodities across the Sahara Desert were transported. The gold, slaves, salt, and fabric, as well as ostrich feathers and European weaponry, were exchanged along the route. Another most relevant trade route was the Tin Route (Flower Philip William 2017) which transformed from the Bronze Age to the Iron Age; the Tin Route was one of the most crucial routes that enabled people

to the export of tin. This route stretched from the tin mines in Cornwall's extreme southwest, over the sea to France, and then down to Greece and beyond.

In the history of ancient trade, the significance of the trade route is immeasurable. It had immensely contributed to make the transportation of goods and services in a much easier and more flexible way, which had been exceptionally helpful for carrying out global trade. It also helped in building the trade network worldwide; it included roads between points which included not only the transportation of goods along these roads but also knowledge, ideologies, and religious practices.

The global economy has grown rapidly in recent decades, and an enormous portion of this expansion has been fueled up by the fast-growing world trade. Integration into the global economy has been a very significant way for nations to promote their economic growth, development, and poverty reduction. However, trade has traditionally been a source of the economic prosperity of the nations. According to the data, the world trade has grown at a rate of 6% per year on average over the last 20 years, twice as fast as global output. Therefore, with rapidly growing trade, the governments have acknowledged the need for regulating the global trade; it was basically due to the economic pressure that had pushed the countries to impose trade regulations to promote free and fair trading in the market.

In the year 1947, the first global multilateral agreement was signed by 23 nation-states at Geneva, and this agreement is known as the General Agreement on Tariffs and Trade (GATT). This agreement was aimed to reduce the trade barriers by abolishing or decreasing quotas, tariffs, and subsidies while retaining considerable restrictions. The GATT was created to help the world economy to recover after World War II by rebuilding and liberalizing global commerce, which would help to promote economic prosperity. One of the most important notions of the GATT was trade without discrimination, which meant that each member country opened its markets to everyone equally, and this also meant that once a country and its main trade partners agreed to lower a tariff, the duty reduction was immediately applied to all other GATT members. Another core principle was that protection should be provided through tariffs rather than import quotas or other quantitative trade restrictions. Another fundamental requirement of the GATT was the consistent customs laws, and each signatory nation had a commitment to negotiate the tariff reductions upon another's nations request. The contracting nations might change agreements if their domestic producers incurred disproportionate losses as a result of trade concessions, according to an escape clause. The GATT (1947) agreement remained in force from 1948 to 1994. The GATT mainly was established to provide the framework for much of international trade and presided over some of the fastest-growing periods in history. The GATT agreement appeared to be well-established, but it was a temporary agreement. Later the GATT was replaced by WTO in the year 1995 (Autar Krishan Koul 2005).

Later, in the year 1995, the World Trade Organization was established, and it has 164 member nations. It is an international organization that regulates and liberalizes global trade. The main principles of the World Trade Organization (WTO) agreement (1995) is to build on the idea of eliminating trade barriers and adopting nondiscriminatory norms and raising the standard of living by creating jobs and

improving lives. These ideas are enshrined in the core principles of the WTO agreements. The World Trade Organization oversees the worldwide system of trade rules and aids developing nations in increasing their trade capacity. The WTO's goal (Landau 2006) is to guarantee that global trade is being carried out in a free, transparent, and predictable manner. As per the WTO report, the global trade in goods and services amounted to US$ 22 trillion in 2020.

OECD Role in Promoting Global Trade

The Organization for Economic Cooperation and Development (OECD) is an international standard-setting body that provides important impartial research and information on a wide range of economic and trade policy concerns. The OECD has 38 member states, and it was formed in the year 1961 to promote the economic development of the nation and to promote global trade. The Organization for Economic Cooperation and Development (OECD) evolved out of the Organization for European Economic Cooperation (OEEC), which was founded in 1948 to supervise American and Canadian contributions to the Marshall Plan (Kenton 2020).

The OECD convention's three main objectives were (Woodward 2009) to contribute to the development of the world economy by achieving maximum sustainable economic growth, employment, and improved living standards in member countries while maintaining financial stability, to contribute to rational economic expansion in member and non-member countries in the process of economic development, and to contribute to global trade expansion. This organization's main concern is to find a solution for international concern, establish the international standard, and share information, and it also tries to find the best-practiced solution in order to support improved policies that will improve people's lives.

The OECD mainly assists in promoting global trade between the nations by providing trade facilitation indicators which help the nations to evaluate their trade facilitation advantages and shortcomings, prioritize action areas, and more effectively deploy technical support and capacity building. The trade facilitation indicators use a variety of quantitative measurements on key areas of the border process to assess not only the actual amount to which nations have enacted and implemented trade facilitation measures in absolute terms but also their performance relative to others. The OECD also provided protection to the consumers as the consumers encounter more complex financial market situation; therefore, such measures are very much important, and it also aids in the development and maintenance of consumer confidence and trust in financial markets, hence improving efficiency and stability and assisting both financial institutions and their customers in achieving favorable results. The OECD aims to improve financial consumer protection by assessing what is needed to assist consumers to achieve the trust, knowledge, information, security, and choices they need to fully engage in financial markets. There are certain guidelines which has been provided by OECD to protect the consumers. The guidelines take into account existing legislative protections for consumers in more traditional forms of trade; it fosters private sector initiatives

that include customer engagement; it also emphasizes the need of collaboration among governments, corporations, and consumers.

The OECD has estimated that around 80% of the global trade is in compliance with some standards or regulations. The OECD encourages global trade by promoting robust rule-based international trading framework to keep the momentum for more trade liberalization going while also contributing to long-term development. It also promotes mutually beneficial integration of developing and emerging economies with the trading system by strengthening trade policy.

Legal Metrology: "Once Measured, Accepted Everywhere"

In the twenty-first century, our life has become inconceivable without measurement. Measurement makes our lives much easier and more comfortable. Measurement is a vital instrument that we all use in our day-to-day life. Basically, every aspect of human existence is influenced by measurement in some way or the other way. Some of the important measurements we come across on a daily basis and without which we cannot even imagine our lives are time, temperature, pressure, sound, weight, volume, distance, speed, and direction. Simply measurement is basically a technique for determining an object's properties or qualities by comparing it to a standard object. Measurement's scope and applicability are determined by the context and expertise.

Measurement also plays a significant role in promoting trade. Every day in some way or the other, we all are engaged in some or the other commercial transactions which involve buying or selling of commodities or services for a particular sum of money, and therefore, it becomes important for us to know whether the measurement and weights used in the commercial transaction or for the delivering the goods or services are proportionate to consideration; we are paying for it, and therefore it brings in the need for a standardized measurement which will ensure accuracy and transparency; it will also enable the consumers to make an informed choice, and it will also protect the consumers for loss or damages which can be caused (Legal metrology and trade 2021).

As the globalization causes interdependence of the economies and the exchange of goods and services at the international level has become imminent, it becomes really important to use standardized system of weight and measurement for selling and buying of these commodities or services at the global level, and this will also promote fairness and transparency, adapt various means to remain competitive and compete with each other, and will help consumers to make a better and informed choice.

The weights and measurements and the standard of measurement are the yardstick for the exchange of commodities. The measurements adapted in the trade, which is once standardized and certified, shall be accepted across worldwide. The nation at the international level has taken enormous steps to maintain transparency and to establish standards to promote global trade by establishing an organization; the International Organization for Standardization (ISO) (Murphy and Yates 2009)

has an important role in encouraging global trade by developing international standards.

Evolution of Unitary System of Measurement

The practices relating to weights and measurements is not new, and its significance can be traced from the past; the oldest and the most primitive known weights and measures systems date back to the third millennium BC. The concept of weights and measurement, on the other hand, would be neither innovative nor contemporary in practice; it has been in use since the dawn of civilization (Hallock and Wade 1906).

The Egyptian system of measurement was founded on bodily measurements, and they used "cubit, figures, and spans" for measurement of length; they used the unit "hekat" for measuring volume, and for measuring weight, they used "deben (Swango 1957)." The Babylonian measurements (Mina unit of weight, Britannica 1998) were determined from an examination of the Senkereh tablet, which is estimated to have been inscribed about 2500 BC. The Babylonian first discovered the unit for weight measurement called as "mina," and it is one of the earliest known units for weight; the unit for measuring capacity is known as "Qa" or "Ka" which was also discovered by the Babylonian (Lawrence James Chisholm, Britannica 2018). The Roman standard for all weight measurements was the "as," "Old Etruscan," or "Oscan pound."

The Indus Valley emerged in 7000 BC. In the Indus Valley, the Harappa and Mohenjo-Daro civilizations were both prominent civilizations. During 1800 BCE the traders of the Harrapan civilization used to trade through barter or exchange system for goods and services as the Harrapans did not have money as their exchange system. It was based on binary and decimal systems and was used to measure weights. Later, in 2500 BCE, during the emergence of the Mohenjo-Daro civilization, the weight and units played a significantly major role, and one of the major units that were used at that time was Dhanurgraha (which was equal to four Angula) (Mauryan empire, Britannica 1998).

In the Middle Ages, during the emergence of the Maurya dynasty period around 321 BCE, they used a complex weight and measuring system, and the unit system used at that was for the measurement of basic length was Angulam. Therefore, it can be stated that the weights and measurements and standardization of units could be referred from an immemorial time. Later, during the fourth century BC, Chanakya authored a famous book named *Arthashashtra* in which he mentioned the units for measuring lengths and distance which was known as Dhanushas and Angulas (McIntosh 2008) which were commonly used for measuring the distance.

During the period of Ramayana and Mahabharata as during that time, there was a standard unit for measurement known as yojana (Yojana standard length of measure evidence from Ramayana and Mahabharata 2018). In the past, there were various practices which could be recognized as per the custom and usage, which were later adopted by the society for measuring and in producing various food items, metals, and other commodities. But after the legislations pertaining to weights and

measurements came into force, customs, usage, and practices lost their enforceability and significance, and presently after the enactment of the legislation, the weight and measurements were governed under the said law. During the Late Middle Ages, with the emergence of the Mughal Empire in 1526 CE, there was a reform in the weight and measurement system, and the standard measure (unit) that was used to measure the land was Gaz and bigha (Karl 1981). Later, during the British colonial period, the British imperial units of measurement were adopted in India (British Imperial System 2019).

In 1795, France was the first country to adopt the metric system (Free 1976) which is now formally used in almost all nations. The metric system is a decimal weight and measurement system that is used all throughout the world. The metric system uses the "meter" as a unit for measuring length, and it uses the "kilogram" as a mass unit.

Evolution of Legal Metrology Laws

The term metrology means the science of measurement, and the legal metrology laws refer to all applied metrology that is regulated by legislation or by decrees. The legal metrology laws are framed to basically guarantee correctness and equity in commercial transactions and to protect the consumers from financial loss and promote fair trade. Previously there was no codified law which dealt with the trade-related aspect like standardization of weight and measure, fairness, and the offense associated to it. Furthermore, since years have passed, there has been technical advancements and societal shifts that have occurred; the economic changes and political changes have made us to understand the need for a standardization of weight and measurement. Later, with time, the government understood the importance of standardization of weight and measurement as there was no particular law which governed the weight and measurement at that time.

The measure was taken at the international level to promote world trade and for standardization of weight and measures. The late nineteenth (International Trade Centre, Legal Metrology and International Trade 2004) century witnessed a significant expansion in international trade, prompting metrology to take steps to improve measuring uniformity throughout the world. The most significant of these metrological initiatives was the 1875 Treaty of the Meter, which enabled the establishment of the International Committee of Weights and Measures (CIPM) and the national measurement institutes (NMIs) in developed countries to maintain national standards of measurement and provide traceability to these standards. This intended to offer one of the fundamental infrastructures for science and technology progress throughout the century. The International System of Units (SI) was developed to replace a plethora of national, craft, and industrial units. The SI units were formed to ensure international consistency of trade and regulatory measurements, as well as to "resolve internationally the technical and administrative problems raised by the use of measuring instruments," which led to the establishment of a second metrology

treaty organization, the International Organization of Legal Metrology (OIML), in 1955, and it focused on trade-related aspects.

In India, in 1956, a central act was formulated called the Standards of Weights and Measures Act 1956. This piece of legislation was passed with the aim of implementing a metric-based universal standard unit for weights and measurements. It intended to facilitate and regulate interstate trade and commercial conduct through the enforcement of a uniform measurement system for commercial goods. Later, in the year 1976, the act was repealed by another act known as the Standard of Weights and Measures Act 1976; this act addressed concerns such as the establishment of standard weights and measures, regulating trade and commerce in terms of weight measurement and other commodities, it also dealt with the approval of models, it also established standards for the commodities which were packed for the intention to be sold or distributed in an interstate trade or commerce, it also emphasized on verification and stamping of weight and measure which were to be traded in interstate, and it also emphasized on the duties of exporter and importer and their mandates on their registration process.

In the year 1985, the Standards of Weights and Measures (Enforcement) Act 1985 was formed, which supplemented the previously formed act on standard weight and measure known as the Standard of Weights and Measures Act 1976. This was an enforcement action which was formed to ensure the uniformity in the enforcement procedures relating to weight and measures; it also had a legal control on weights and measures and on its instruments; the act had a uniformity of procedure; it aimed to protect the consumers in a better way; and there was no much scope for exploitation by the unscrupulous users; the act also authorized the state government to appoint controllers of legal metrology for the states and a required number of additional, joint, deputy, or assistant controllers, inspectors, and other officers to carry out the objectives of the act efficiently, and the act, on the other hand, protects them from liability for actions performed in good faith when exercising any authority or performing any obligation under this or the Standards Act.

With the changes in the society due to the sheer of fast growth of science and technology, as well as the globalization of economies, weighing and measuring procedures have vastly improved, and the scope of weights and measures has been expanded, so therefore, the need was felt for a new law with the intent of establishing standards for weights and measures, regulating trade and commerce in weights and measures and other commodities sold or distributed by weight, measure, or number, and for matters connected with or incidental thereto. In the year 2009, the Legal Metrology Act was enacted. This was enacted as the need was felt to integrate the contents of the two previous acts to eliminate inconsistencies and simplify the legislation.

The Legal Metrology Act 2009 consists of 5 chapters and 57 sections. This act is procedural law in nature. This act regulates interstate trade and commerce relating to weights and measures; it provides mandates on the manufacturer, repairer, and dealer of weights and measures to obtain license compulsorily; it imposes the obligation of declarations on the manufacture, packaging, sale, import, distribution, delivery, offer, exposure, or possession for sale of any prepackaged commodity. In order to

deal fairness for the goods packed in the absence of the consumers, packaging commodities rules are framed under the Act of 2009. These rules stipulate the requirements of labeling declaration on the prepacked commodities.

Legal Metrology Laws of India, the United States, and the United Kingdom with Reference to the Package Commodity

In India, the Legal Metrology Act 2009 and Packaged Commodity Rules 2011 govern weights, measures, and packaged commodities. These rules are applicable to packages containing a quantity of less than 25 kg or 25 liters, and no package shall be sold without proper declaration as per the rules (Rule 3 of Legal Metrology 2011). For cement and fertilizer bags, this rule will be applicable if the quantity is more than 50 kg.

"Prepackaged commodity" is a commodity that, without the purchaser being present, is placed in a package of whatever nature, whether sealed or not so that the product contained therein has a predetermined quantity (Section 2(l) of The Legal Metrology Act 2009). If any predetermined quantity is packed in the absence of the purchaser, then it shall comply with the provisions of the Packaged Commodity Rules 2011. Every package shall contain a label, and all the mandatory declarations as per the rules shall be declared on the label, and the label shall be securely affixed to the package. According to rule 4, no one may prepack any commodity for sale, distribution, or delivery unless the package in which the commodity is prepacked has labels carrying the necessary declaration.

The rules also prescribe the size of the label and the size of the package with reference to the volume of the package. The units are expressed in SI system and expressed "kg" or "g" for mass, "cm" or "m" for linear measurement, and "ml" or "L" for liquid volumes. The total surface area on which this mandatory declaration was made is defined as a principal display panel (PDP) (Section 2(h) of the Legal Metrology Act 2011). These rules also mention the size of the font on the packaging.

Some of the prepacked products, like baby foods, bread, biscuits, weaning food, tea, coffee, etc., are to be packed in specified quantities; however, in November 2021, these restrictions are withdrawn, and now the products can be packed in any quantity. However, the unit sale price of the product is to be mentioned on the pack.

Whereas in the United States, the legislation which mainly deals with the packaging of commodities is the Uniform Packaging and Labeling Regulation under regulation 1. These regulations are applicable to all packages except the packages that are not intended for individual sale and packages intended for exports. A package is defined as enclosed in a container or wrapped in any manner in advance of wholesale or retail sale or whose weight or measure has been determined in advance of wholesale or retail sale (Section 2 United States Uniform Packaging and Labelling Regulations 1952). Uniform Packaging and Labeling Regulation states that the principal display panel is the area of a label that is most likely to be seen,

presented, exhibited, or inspected under the normal and typical display and purchase conditions. All standards relevant to the "primary display panel" should apply to all such principal display panels in the package when a principal display panel occurs more than once. These regulations also prescribe size of the font for making the mandatory declaration. In the United States, the Uniform Unit Pricing Regulations are in place to declare a commodity per kilogram or per gram and the price to be declared in dollars or in cents.

Whereas in the United Kingdom, Weights and Measures (Packaged Goods) Regulations 2006 governs regulations regarding prepackaged commodities. These regulations are applicable to packaged commodities that are intended for sale in constant unit nominal quantities, which shall be not less than 5 grams or 5 milliliters and not more than 25 kilograms or 25 liters (Regulation 3 UK The Weights and Measures 2006). This regulation applies to bread that is sold either unwrapped if it has a predetermined constant quantity and is intended for sale in constant unit nominal quantities of not less than 300 grammes per loaf and not more than 10 kilograms per loaf. The marking or labeling declaration shall be made in specific unit measurement in metric units, i.e., kg, g, L, ml, and cl. These regulations also prescribe the size of the font for making the mandatory declaration. In UK metrology law for packaged commodities, there is no specific mention of the principal display panel.

The Declaration Is to be Made on the Packages

A customer shall have the right to know the contents of the prepackaged product in order to make an informed decision. The regulations made a bounden duty on the manufacturer/packer/importer to make a declaration about the product. The following is the comparative statement on the declaration of prepackaged products in these following countries.

Declaration	India	UK	USA
Name of the product	Generic name to be mentioned	To be declared	Identity to be declared
Name and address of the manufacturer and packer, and for any imported package, name and address of the importer	To be mentioned	To be mentioned	Responsibility to be declared by state-corporate name of the packer along with the address
Date of manufacture/ packed	**Yes.** In case of food items, expiry date also to be mentioned	To be mentioned. In case of food items, expiry date also to be mentioned	To be mentioned. In case of food items, expiry date also to be mentioned
Net quantity	Yes. Quantity to be mentioned in SI units	Yes. Quantity to be mentioned in SI units	Yes. Quantity to be mentioned in SI units

(continued)

Declaration	India	UK	USA
Maximum retail price	Price to be mentioned in rupees	Not mentioned in the regulations	
Consumer care cell	Name, telephone number, and email address to be declared on the pack	Not mentioned in the regulations	
Principal display panel (PDP)	All mandatory information to be grouped together and to be declared on PDP		All mandatory information to be grouped together and to be declared on PDP
Height of the numerals	To ensure the heights as mentioned in the rules	To ensure the heights as mentioned in the regulations	To ensure the heights as mentioned in the regulations

Offense and Penalties

Under criminal jurisprudence, a punishment or penalty is levied on the wrongdoer in order to deter him/her from doing the act again, and this also prevents others from committing any offense. In Indian legal metrology laws, the punishment varies from penalties starting from Rs. 2000 to Rs. 50,000 per person, and second and subsequent offenses can lead to imprisonment.

Whereas in the United States according to Uniform Weights and Measures Act, the punishment varies from civil penalties to criminal penalties. In civil penalties, the amount of fine increases on the second and subsequent offenses.

In the United Kingdom, any noncompliance will be penalized based on a graded scale. The punishment or penalties for most of the offenses are fines at various levels on a scale of 1 to 5. As per Criminal Act 1982, the standard scale of fines is L1, 200; L2, 500; L3, 1000; L4, 2500; and L5, 5000 pounds; and imprisonment is for fraud or intentional tampering of the weighing scales.

Advantages of Legal Metrology in Trade Practices

Global trade of products and services is significant for economic growth, societal well-being, and maintaining global financial stability. The legal metrology laws are framed to uplift the economy and to benefit the society, which would lead to social and economic development. The legal metrology laws are framed with an objective to simplify and consolidate the laws relating to weight and measurement, and it also

promotes fairness and transparency. The following are the advantages of legal metrology law in trade practices (International Organization of Legal Metrology 2002):

Upliftment of economy. Legal metrology has immensely helped in uplifting the economy by escalating the trade and commerce of the country as from the very beginning measurements have always been recognized as foremost criteria for trade and commerce, and therefore the need was felt for standardization of weights and measures to elevate its precision, transparency, and authenticity which would promote fair trade in the market, and it would entice investors to invest in the market, and that would positively impact the economy as the government collects its revenue through excise duties imposed on products produced, sold, imported, and exported and also via taxes levied on measurement and weight of the commodities. The Legal Metrology Act also ensures that no injustice is done to both the consumers and businesses in the matter related to trade and commerce.

Protects and builds consumer trust. The legal metrology laws standardize rules and regulations to protect the consumers from fraud and significantly reduce disputes; it increases the efficiency of the marketplace, and when customers realize, they acquire the product that has been confirmed in accordance with standards and rules, which increases their trust in the brand or company, resulting in a solid trading relationship between the trader and the customer. In the current area of Internet revolutions, the role of legal metrology is indispensable as these consumers interact and trade globally.

Promotes fair trading and healthy competition. Legal metrological regulation guarantees fair trade by limiting the use of short measures to gain a commercial advantage over competitors in the market and by preventing the manufacture and sale of many trade measuring instruments that are not fit for purpose and do not meet internationally recognized standards through pattern approval and certification.

Reduces technical barriers to trade. The Legal Metrology Act reduces technical constraints and enhances measurement confidence and clarity. Fewer barriers boost national morale and encourage citizens to participate in the global commercial system, resulting in increased national economic growth. Using the Legal Metrology Act, a trader can avoid needless obstacles in the adoption, use, and evaluation of technical regulations, standards, and conformity assessment procedures.

To promote global trade. The establishment of International Recommendations for Pattern Approval of Measuring Instruments contributes to the worldwide market

for measuring instruments that meet globally recognized standards. The Legal Metrology (Packaged Commodities) Rules 2011 has a provision that deals with exporting of packaged commodity, and it also sets certain guidelines for the way the exporting shall be done and leads to promoting the global trade.

OIML Role in Stimulating the World Trade

The International Organization of Legal Metrology (OIML) is an international organization founded on a diplomatic agreement signed in Paris on October 12, 1955. Currently, 124 countries are members of OILM (of which 61 are full member states and 63 are corresponding members). The OIML has developed a number of guidelines to assist members, particularly developing countries, in drafting appropriate legislation relating to metrology in all aspects of society, as well as certification and calibration requirements for new products, especially where such calibration has a legal impact, such as in trade and taxation.

The OIML (International Legal Metrology Documentary Standards, NIST 2016) was established with the goal of promoting worldwide harmonization of legal metrology processes and to encourage world trade. The aim of the OIML is to assist economies in developing mutually compatible and globally recognized legal metrology infrastructures in all domains where governments are responsible, such as in trade facilitation and standardizing consumer protection norms throughout the world.

The International Organization of Legal Metrology (OIML 2015) has an objective to promote world trade, which would automatically help to promote the global economy. It generally tends to reduce the trade barriers by standardizing and harmonizing the regulation, which would help in facilitating global trade; it provides standard laws and regulations concerning the measurement and weight, which would build the confidence and the trust of the consumer and would tend to promote global trade and also consumer protection norms which would encourage the customer to invest more in global trade.

Conclusion

With the advent of transportation, communication, and technology, the world has become a global village without boundaries for trade. The global economy has shown significant rise in recent years, and much of this development has been fueled by the huge rise in global trade. A recent analysis by the world bank group shows that economic growth has been enormously boosted by trade liberalization and that there has been a 20% increase since the 1990s (Trade, World Bank Group 2021). Internet technology revolution is largely influencing global trade. Further consumer needs and coordinated attempts by nations for eliminating trade barriers by making a

significant effort to standardize their regulations concerning the global trade are increasing. Some of the developing nations have opened their economies to take full advantage of trade-related opportunities for the economic growth of the nation.

The legal metrology also has an enormous role in promoting global trade as they mainly aim in establishing and enforcing standards of weights and measures and regulating trade and commerce activities relating to weights, measures, and other goods which are sold or distributed; it promotes fairness and transparency, and it also protects consumer interest.

Trade and legal metrology are very much interconnected with each other as both are associated with the use of weight and measurements. In today's globalized trade environment, most of the commercial transactions require measurements, and they must be able to be considered "acceptable" in order to assure fairness to all parties and thereby legal metrology in promoting world trade. Further, the roles of legal metrology are increasing in health care and medical devices and environment. The OIML is working toward global consistency in measurements.

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Standardization and Regulatory Ecosystem in India

79

Bharat Kumar Yadav

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Abstract

India has its typical socio-economic environment, which makes it a unique nation. Despite typical challenges, India has shown tremendous potential for economic growth, and gradually it is emerging as one of the world's largest

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economies. It is happening because India has already established a suitable institutional mechanism and is continuously striving to develop a robust domestic Quality Infrastructure (QI). The QI mainly comprises Metrology, Standardization, and Accreditation. Historically, the journey of India was started with the foundation of the Indian Standards Institution (ISI) and National Physical Laboratory, India (NPLI), in the year of 1947. Further, it passes through pre- and post-days of globalization; reform initiatives taken up by the Government; the impact of ever-changing technology and industry/business trends, and comprehensive changes in global trade due to the advent of the World Trade Organization (WTO) regime. Further, the progress is going on in the context of contemporary advancements. The chapter is an effort to get a holistic overview of major verticals which influence trade and economic growth across the globe, exploration of existing institutional mechanisms for Indian QI, ongoing activities to maintain sustainable economic growth, and futuristic challenges in this arena.

Keywords

Quality infrastructure · Standardization, Regulatory ecosystem

Introduction

Globalization has connected economies and cultures worldwide. Interdependence and interoperability are becoming an integral part of the global ecosystem. For example, a car sold in Canada can be designed in France, with parts manufactured in Australia or China. It becomes possible by adapting suitable standards (i.e., *something established by a competent authority, custom, or general consent as a model, example, or point of reference*) and their proper implementation. However, over time, the use of Standards has emerged as one of the magical tools to control international trade dynamics. The developed economies have pre-assessed the situation at an early stage. They have taken control of the world economy by establishing a robust and effective QI ecosystem (Quality Infrastructure 2018; McDermott 2019; UK Quality Infrastructure; Quality Infrastructure: UNIDO's). The QI broadly encompasses Metrology (i.e., *the science of measurement*), Standardization (i.e., *the process of making something conforming to a standard*), and Accreditation (i.e., *third-party attestation by a designated authority confirming the competence of the assessment body to carry out specific conformity assessment tasks*). On contrary, developing economies are still facing challenges.

According to (The Importance of Standardization – benefit and advantages 2020), the standards/specifications stand for knowledge in the form of regulations and guidelines. These are written documents (printed and digital formats) that can be easily distributed among interested parties or end users. Generally, such types of documents remain accessible to everyone. The aim is to help people/industries across the globe to understand each other better so that they can react faster and act accordingly. Irrespective of the nature of the organization (whether it is a private

or public sector), nearly in every area of life, there is generally at least one standard that plays an important role. ISO (International Organization for Standardization) rightly states that the future belongs to standards (https://www.iso.org).

The Standards (i.e., defined to be per se Voluntary) become mandatory when implemented as Technical Regulations/SPS measures. Generally, the Government in developing economies takes such initiatives in the interest of the country and its people. A report by OECD (Organisation for Economic Co-operation and Development) (Singh and Mitra 2022) indicates that, generally, regulation is an attempt to control or influence private behavior in the desired direction by imposing costs on or proscribing undesirable behavior. Since regulation can have significant consequences for economic efficiency and private incentives, it is usually justified only under specific conditions. There are three sets of justifications for regulatory interventions, i.e., (i) prevention of market failures, (ii) restriction or removal of anti-competitive practices, and (iii) promotion of public interest. Post-independence, India experimented with a "socialist mixed economy model." In this model, the government retained control over the heavy industries and utilities and tried to command the activities of the private sector through licensing and quotas regarding intermediate goods imports and outputs. Such types of controls were complemented by high tariff walls. However, after 1985, the Indian economy embarked on a process of domestic reform involving steps like delicensing of industries, abolishing quotas, permission for private entry into sectors which were hitherto the monopoly of the government, and liberalization of quotas and tariffs on capital goods imports. Further, from 1991 onwards, the liberalization of the external sector resulted in tariff reductions in almost the entire spectrum of merchandise trade, simplifications, and liberalization in the conditions for foreign investments, etc.

Side by side, to achieve sustainable economic growth, India has strived hard to cultivate its unique kind of QI ecosystem. Historically, the journey was started with the establishment of ISI on 6 January 1947. This institute was initially concentrated on standardization activities; subsequently, it was replaced by the Bureau of Indian Standards (BIS) on 26 Nov 1986. Further, BIS became the Standard Development Body (SDO) of India. Moreover, the National Physical Laboratory, India (NPLI, which is also referred to as CSIR-NPL), was founded on 4 January 1947, which was the first milestone in the field of Metrology for India. Subsequently, under the Act of Parliament, Legal Metrology, it becomes the National Measurement Institute (NMI) in 1956. NPLI establishes and maintains all the seven basic SI units (International System of Units) and their derivates, except the ionization radiation. In 2003, the Bhabha Atomic Research Centre (BARC), Mumbai, was declared a designated institute (DI) to deal with ionization radiation.

Further, in this sequence, the Quality Council of India (QCI) (https://www.qcin.org) was established as the national body of accreditation in 1996. Subsequently, the council came into existence in 1997 through a Cabinet decision as an autonomous non-profit organization under the Department for Promotion of Industry and Internal Trade (DPIIT), Ministry of Commerce and Industry. QCI's role is to establish and operate a national accreditation structure and to promote quality through the National Quality Campaign. It plays the role of the National

Accreditation Body (NAB) as per the ecosystem of quality around the world with the task of creating a mechanism for third-party assessment of products, services, and processes.

Recently the Indian QI has been investigated in the context of the QI of the United States and indicated that all the basic three pillars (i.e., metrology, standards, and accreditation) are well established in India; however, the strengthening of QI is still an area where further attention/improvement is required (Aswal 2020). However, the topic has vast scope to look into it; thus, it would be interesting to carry out an independent 360^0 investigation and analysis of the Indian standardization and regulatory ecosystem. The QI of developed economies like the United States, EU, etc., is almost mature, while the developing economies like India are still coping with typical challenges to maintain sustainable economic growth.

This chapter is an effort to visualize the existing Indian ecosystem of standardization and regulation. Onset, we review the verticals like industry revolutions, ever-changing trade dynamics, QI ecosystem adapted by large economies, and the role of major key players (i.e., WTO, SDOs, etc.) in the international arena. Afterward, we evaluate the domestic ecosystem of standardization and regulation and will conclude by presenting views on growth avenues and challenges for India.

Industry Revolution and Trade Dynamics

The standard, standardization, and regulatory activities are an integral part of our lives; however, things get altered with the changes in the requirements. Indeed, the orientation of industry, trade, and interaction among people across the borders significantly transformed over time. The ever-changing trade dynamics are one of the root causes of significant transposition in the country-specific regulatory/standardization requirements to maintain sustainable economic growth. One of the best ways to visualize the changes is in the context of the revolution in industry developments, trade liberalization, and globalization that influenced the trade dynamics.

Industry Revolution over Time

Figure 1 illustrates the metamorphosis of the industry over time. It summarizes the developments starting from 1784 to till date (https://www.desouttertools.com/industry-4-0/news/503/industrial-revolution-from-industry-1-0-to-industry-4-0).

The figure is self-explanatory. However, a brief overview of the stepwise progress is as follows:

(i) **The First Industrial Revolution (Industry 1.0)**: It was the start of mechanization. It is understood to begin in the eighteenth century with the implementation of steam power and automation of production. The use of steam for

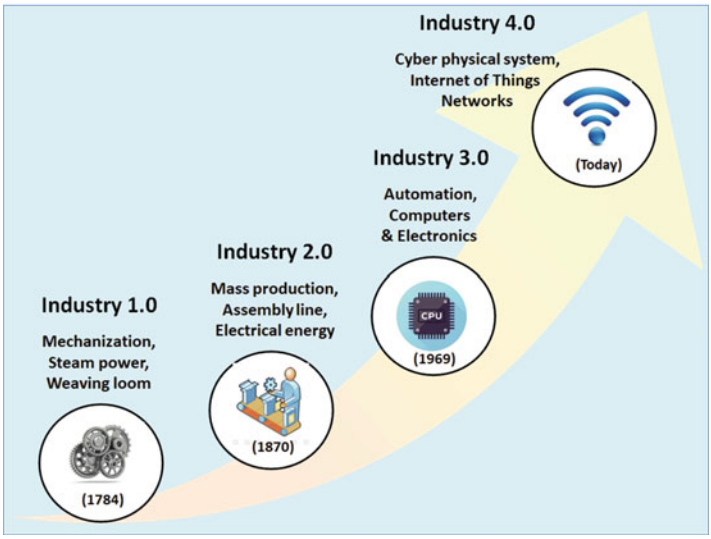


Fig. 1 Industry revolution over time

industrial purposes was a significant breakthrough in increasing human productivity.

(ii) **The Second Industrial Revolution (Industry 2.0)**: It started in the nineteenth century after the invention of electricity and assembly line production. It was a boost in mass production by implementing automation using electrical energy. Earlier only one station assembled an entire automobile. However, after the introduction of the assembly line, the vehicles were produced in partial steps on the conveyor belt – significantly faster and at a lower cost.

(iii) **The Third Industrial Revolution (Industry 3.0)**: It began in the twentieth century through partial automation using memory-programmable controls and computers. Electronics came into the picture, and by implementing such technologies, automation of the entire production process became possible without human assistance. Known examples of this period are the development and use of robots that perform programmed sequences without human intervention.

(iv) **The Forth Industry Revolution (Industry 4.0)**: Presently people are talking about Industry 4.0. It is marked by the application of ICT (Information and Communication Technology) to industry. It builds on the developments of Industry 3.0. The computerized production systems are expanded by a network connection and the Internet. These allow communication with other facilities and the output of information about themselves. It is the next generation of production automation, and the concept of "smart factories" (Weyer et al. 2015) comes up. In such factories, the production systems, components, and people communicate via a network, and production is nearly autonomous. The implementation of such new technologies together resulted in the concept of "cyber-

physical systems" – which are networks in the "Internet of Things (IoT)." These developments brought significant improvements, such as enhancement in automation, and shortened the time between the development of a new product and its market launch. Notwithstanding, the novel technologies pose typical challenges, e.g., cyber security issues, data privacy, security, etc. Standardization in this segment has emerged as one of the major challenges.

Trade Liberalization

Trade liberalization is a deterrent process of protection. Trade liberalization takes place when the government decides to support free trade. Trade liberalization may take place unilaterally. Initially, around the second half of the twentieth century, significant trade liberalization occurred among developed economies. However, it was reciprocal and multilateral. The liberalization decisions taken up by the countries' governments reciprocated among the trading parties. The liberalization was non-discriminatory in applying to all liberalizing trading partners. The trade liberalization aimed to facilitate cross-border trade among the trading partners who support the concept. The trade liberalization was followed by the liberalization of international capital markets and substantial international migration (both legal and illegal). International agreements and conventions also unified rules of conduct concerning the protection of property rights, including intellectual property rights related to proprietary knowledge. The outcome of these liberalizing and integrating processes is known as globalization (Hillman 2008). The trade liberalization was started in developed economies but gradually spread throughout the world. However, it posed negative as well as positive impacts across the globe.

Impact of Globalization

According to a report (Globalization), it was a general conception that globalization is working selectively. It was beneficial for some economies while not for others. It has been observed that poverty reduction in China was one of the impressive impacts of globalization, while In India, the message was more mixed. India has a typical socio-economic environment; the challenge before it is unique in many ways. An India-specific report (Kishore 2002) indicates that inter-regional inequality appears to have widened during the era of globalization. India is a country rich in knowledge and the production of technology. Over the years, it has changed in many ways, and knowledge is now recognized as a commodity. India is rapidly increasing its integration with the global economy and has emerged as a global player. Even the World Bank has judged India to be one of the world's "fast globalizers." It is well known that ICT became important in new-age industry evolution, i.e., Industry 4.0. Noticeably, India is progressing very fast in this segment.

Globalization and Standardization

The interconnection between globalization and standardization is one of the significant areas of research. In this regard, an analysis report has been published (Purcell and Kushnier 2015). Interestingly, according to the said report, more than 100 individuals and standards organizations who are experts in the field of globalization and standardization (academic, private sector, and public sectors) were approached to submit their views and comments on this critical issue. Surprisingly they only received inputs/comments from two standards organizations, i.e., the International Electrotechnical Commission (IEC) and DIN (the German national standards organization). It indicates that standards have strategic value and are a bridge between the market and technologies. They control access to virtually every market in global commerce and directly affect more than 80% of world trade. The conclusions of the views of IEC and DIN are investigated in the report. The analysis concludes that the relationship between globalization and standardization is, or should be, a critical area of understanding for policymakers across the world.

Quality Infrastructure (QI)

The term QI has nicely been explained in a Blog wherein the author cited different definitions by the subject experts and has elaborated the concept (Harmes-Liedtke 2002). The term, "Quality Infrastructure" is made of two words, i.e., "Quality" and "Infrastructure." "Quality" is inherently a subjective concept. For instance, from the manufacturer's point of view, quality is "freedom from deficiencies or defects," whereas, from the customer's perspective, the definition of quality can be "fitness for use." The quality management standard ISO 9000:2005 defines the term "as the degree to which a set of inherent characteristics of an object fulfills requirements." The product quality depends solely on the extent to which the standard-defined requirements are met. The word "infrastructure" mainly described the immobile buildings and facilities that served to mobilize and keep the armies ready. Indeed, the QI (as a whole) describes a system of institutions that guarantee the definition and control of quality criteria. These institutions can also be seen as the underlying foundation of international trade. The functions of QI include the development of standards and the control of conformity.

The International Network of Quality Infrastructure (INetQI) (INetQI) is a consortium of 14 international organizations which promotes the significance of QI activities (i.e., metrology, accreditation, standardization, and conformity assessment) as a tool for sustainable economic development. This group of international organizations includes the International Bureau of Weights and Measures (BIPM), International Accreditation Forum (IAF), IEC, Independent International Organisation for Certification (IIOC), International Laboratory Accreditation Forum (ILAC), International Certification Network (IQNET), ISO, International Trade Centre (ITC), International Telecommunication Union (ITU), International Organizations of Legal Metrology (OIML), United Nations Economic Cooperation for Europe

(UNECE), World Bank Group (WBG), WTO, and United Nations International Development Organisation (UNIDO).

To understand the interrelationship between different entities involved in QI activities, we may cite the model of National QI by PTB (Fig. 2) (https://www.ptb.de/cms/fileadmin/internet/fachabteilungen/abteilung_9/9.3_internationale_zusammenarbeit/publikationen/102_National_QI/PTB_Q5_National_QI_EN.pdf). The figure shows the central components of the QI are accreditation, certification, standardization, testing/inspection, and metrology. It would be pertinent to mention that the certification and testing/inspection are integral parts of conformity assessment. The individual components exchange services and form an overall system. In the model, on the right-hand side, international recognition is represented by membership in international professional organizations. While on the left-hand side, the users of the system are demonstrated in the form of a value chain. Nevertheless, similar models are being used/adopted by all the large economies. Let us have a quick look at each component of the QI activity.

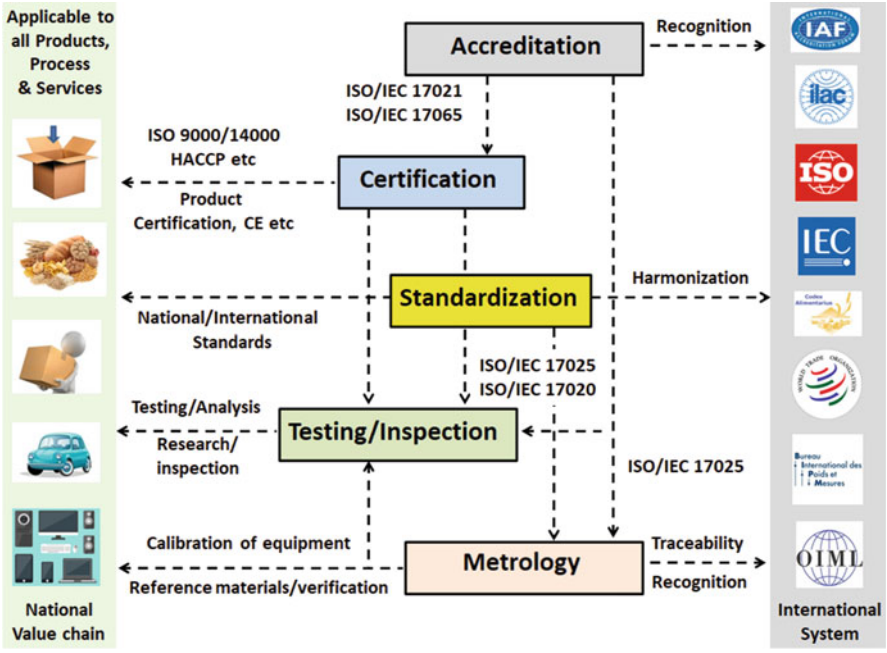


Fig. 2 Entity relationship in QI. (Retrieved from the model given in (https://www.ptb.de/cms/fileadmin/internet/fachabteilungen/abteilung_9/9.3_internationale_zusammenarbeit/publikationen/102_National_QI/PTB_Q5_National_QI_EN.pdf))

Metrology

The BIPM (BIPM 2022) defines metrology as "the science of measurement, embracing both experimental and theoretical determinations at any level of uncertainty in any field of science and technology." It can be further split up into three subfields, i.e., scientific metrology, applied metrology (sometimes also termed industrial metrology), and legal metrology.

(a) **Scientific metrology**: It is the base of all other subfields. It covers the development of new measurement methods, R&D activities in standards, the realization of measurement standards, and the transfer of these standards to users. For this work across the globe, all countries have their NMIs. For example, in India, CSIR-NPL, New Delhi performs this task.
(b) **Applied metrology (or industrial metrology):** In this subfield of metrology, the science of measurement is advanced in production and other processes, ensuring the suitability of measurement tools, their calibration, and quality control. Accurate and accurate measurements improve the quality of the product and, therefore, its market. The power of industrial metrology, therefore, controls the economic and industrial development of the country.
(c) **Legal metrology:** It deals with regulatory requirements for well-established standards and measurement tools (compliance with standards) for consumer protection and fair trade. These regulatory requirements arise from the need for public health and safety, consumer rights, favorable taxes, the environment, and fair trade by competent organizations.

Standardization

Standardization is the process of making something conform to a standard. Indeed, the ultimate aims of standardization include making a product, process, or service fit for its purpose, controlling diversity by using the optimal number of types/sizes, protecting the environment, and ensuring compatibility of various products, health, safety, etc.

A study for the Department of Business, Innovation and Skills and the British Standards Institution (BSI) throws light on the role of standardization in support of emerging technologies. The report inter alia indicates that to escalate competition for industrial activity, the government recognizes the new technologies as one of the significant driving forces in the growth of the national economy. One may describe the emergence of technology as a complex and dynamic phased journey, by which, the applied science ideas turn into workable technologies, then viable applications, and marketable products; fostering the right environment for each stage of this journey has become a priority for the government. Over time, there has been an increasing interest in the understanding of the effect of standards on emerging technologies, appropriate roles for government, innovation system actors in support of standardization activities, etc. (O'Sullivan and Brévignon-Dodin 2012).

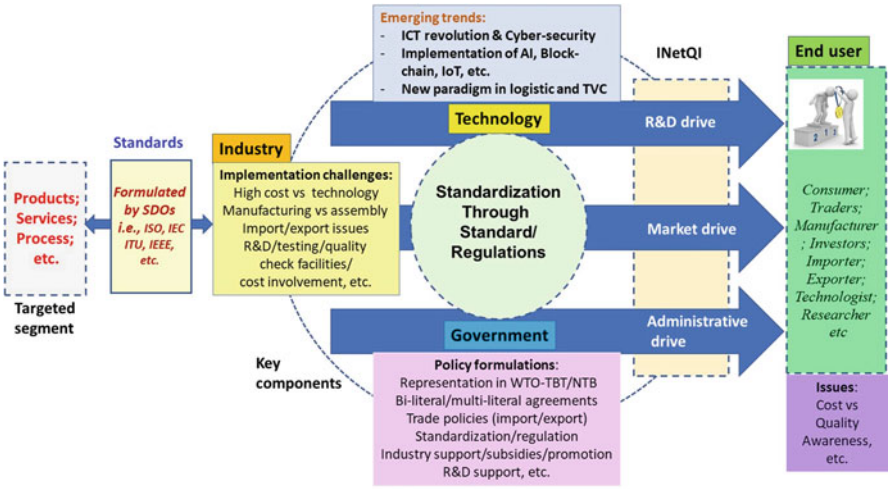


Fig. 3 Schematics of standardization through standards/Regulations with involved key entities

Altogether, these entities develop a complex ecosystem in terms of interactions among them and their interrelationship.

A self-explanatory Fig. 3 gives a glimpse of the process. It includes the targeted segment (e.g., products, service, process, etc.), SDOs (e.g., IEC, ISO, etc.), Industry, government, technologies, and prominent drives (e.g., R&D, market, administration, etc.), and finally, the end user. Ultimately, the whole ecosystem revolved around three major components, i.e., commodity, standards, and the end user. The illustration indicates the interrelationship among involved entities, their concerns/expectations, and linkages of the components with the driving forces. For instance, a manufacturer/company/industry of any product wants to adopt standards to improve its quality, safety, fitness for the global market, etc.; however, they face various challenges like conformity issues, taxation matters, costing, the impact of government policies, etc. The government formulates policies to control as well as facilitate the trade; it has to cope with multiple challenges in the domestic as well as international arena. Technology, which is one of the significant components of the process, keeps on changing over time and its demands/requirements. The end user may generally have concerns like cost versus quality issues.

Standard

Standards provide individuals/organizations with a basis for understanding and are used as tools to facilitate communication, measurement, trade, and productivity (https://research.lib.buffalo.edu/standards). As we already have emphasized in section "Globalization and Standardization," the Standards control the world economy. Its major functional components are as follows:

- Facilitates business communication.
- Allows companies/organizations to comply with applicable laws and regulations.
- Accelerates the introduction of new products to the market.
- Provides interoperability between new and existing products, services, and processes.

Since standards play a pivotal role in international trade, their development process is also quite typical. According to ISO, standard development is like a symphony; it takes a lot of people working together to develop a standard. The standard development process includes many steps. The process starts by evolving a draft according to the market requirements within a specific area. Thereafter, it is shared among interested parties for commenting and further discussion. Voting is carried out to get a consensus. Once it is achieved, the draft becomes a standard. In case, the agreement doesn't reach, then the under-consideration draft is modified again, and the whole process gets repeated till the consensus. The standard development process (from the first draft to final publication) usually takes about 3 years. Standards are the distilled wisdom of people with expertise in their subject matter (e.g., manufacturers, sellers, buyers, customers, trade associations, users, or regulators) who know the needs of the organizations/market. Some of the areas where a majority of standards are coming out are as follows:

Type of standards	Requirements
Quality management:	To help work more efficiently and reduce product failures.
Environmental management:	To help to reduce environmental impacts, reduce waste, and be more sustainable.
Health and safety:	To help to reduce accidents in the workplace.
Energy management:	To help cut energy consumption.
Food safety:	To help prevent food from being contaminated.
IT security:	To help to keep sensitive information secure.

Technical Regulation

According to WTO, the technical regulation is a government document that describes product characteristics or their related processes and production methods, including the applicable administrative provisions, with which compliance is mandatory (https://www.wto.org/english/tratop_e/tbt_e/tbt_info_e.htm#:~:text=The%20difference%20between%20a%20standard,different%20implications%20for%20international%20trade). No consensus is necessary for the establishment of the regulation. The technical regulation may also include or deal exclusively with terminology, symbols, packaging, marking, or labelling requirements as they apply to a product, process, or production method.

The main difference between a standard and a technical regulation lies in compliance. Conformity with standards is voluntary, while in the case of technical regulation, it remains mandatory. Since they are different by nature, thus, their implications for international trade are also dissimilar. For instance, if an imported

product does not meet the requirements of a country-specific technical regulation, it will not be allowed to be put on sale in that country, although the situation is quite different in the case of standard. In this scenario, the non-complying imported products will be allowed on the market; however, then their market share may be affected if consumers/end users prefer products that meet local standards such as quality or color standards for textiles and clothing.

Ideally, the government interferes when products can affect the health or safety of the consumers or the environment or that might encourage deceptive practices; consumers need protection from faulty or dangerous products or the conscienceless behavior of suppliers. The technical regulation is introduced as a legislative measure to assure a reasonable level of protection for the consumers. According to ISO, the technical regulatory system includes the following (ISO Guide to Good Practices):

Elements of regulatory system	
A regulator:	A public body (ministry, department, etc.), identified to administer technical regulations.
A suite of technical regulations:	It normally includes both administrative and technical provisions.
A supplier of the product:	The supplier of the product (e.g., designer, manufacturer, importer, distributor, retailer) is responsible for marketing safe products and monitoring their products in the marketplace.
A conformity assessment infrastructure:	It is required to enable the regulator to make decisions about compliance or non-compliance.
A range of sanctions:	It can be applied by the regulator in the event of proven non-compliance.

According to the experts, the technical regulations have to be the same for imported and locally manufactured goods. WTO strives for non-discriminatory treatment for all trade partners, thus, encouraging countries to adopt international standards as Technical Regulations/SPS measures. Enacting any technical regulation is one part of the story, while enforcement is the most crucial part of the system. Regulators and suppliers need strict monitoring to ensure compliance with any technical regulation. However, the approach may be different. Market surveillance is one of the best mechanisms by which compliance can be ensured. The market surveillance is either carried out by the regulator/supplier itself or authorities/agencies appointed by them. According to the requirements, one may initiate market surveillance before or after the product is placed on the market.

Market Surveillance (MS)

As mentioned earlier, MS is an essential part of any regulatory system. It refers to the activities carried out and measures taken by public authorities to ensure that products comply with the requirements embarked in the relevant technical regulation and do not jeopardize the health, safety, or any other aspect of public interest protection. It includes both pre-market and post-market surveillance activities. It is a complex

process. In principle, one can broadly categorize it into two segments, i.e., (i) active market surveillance and (ii) reactive market surveillance.

(v) **Active market surveillance**: this type of MS pertains to the authority's initiative, sometimes referred to as (pro)active market surveillance. It is performed for appropriate checks (a) on an adequate scale, (b) based on adequate samples, and (c) to take account of established principles of risk assessment, complaints, and other information. Sometimes it is also referred to as routine surveillance.
(vi) **Reactive market surveillance**: it is initiated when (a) product problem is clear, e.g., non-conformity is claimed, occurrence hazard by the product, etc.; (b) extent of the investigation follows from the complaint/accident report.

The MS activities shall be planned based upon an evaluation of existing information about (possibly) unsafe products/product groups (scope), or in case of any investigation has to be determined. One of the ideal MS models is EU (European Union) model (Market Surveillance in the European Union and Germany: Perspectives for India, Indo-German Working Group on Quality Infrastructure – Knowledge Series 5, Editors: Dhundia H, Singh K, Moser V A, Jan 2020), the process of the said model is illustrated in Fig. 4 to understand the process at an abstract level. Each component of the process is self-explanatory. The aforesaid process is a basic system

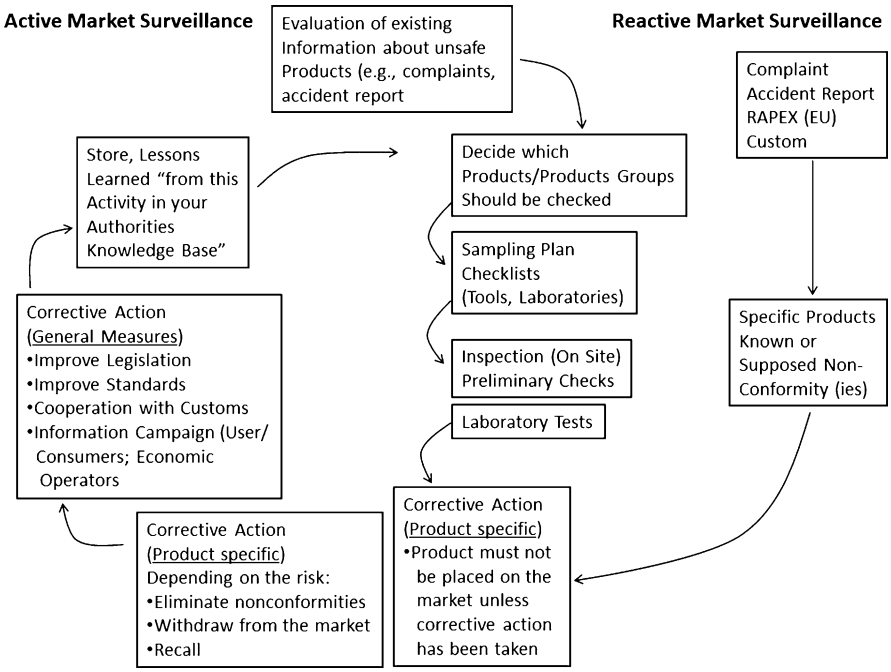


Fig. 4 Process of market surveillance. (Retrieved from presentation by Hans-Georg Niedermeyer in Indo-German Workshops on Market Surveillance, New Delhi, 2018 and 2020)

of market surveillance; however, it varies from country to county based on their regulatory ecosystem, market intelligence potential/capability/infrastructure, level of user awareness, etc. Many developed economies including the EU have their stringent regulatory framework to implement market surveillance.

Conformity Assessment

Merely meeting the regulations and standards is however not enough. The industry has to gear up to face challenges beyond compliance to Technical Regulations/SPS Measures and Standards. Demonstration of compliance to these is also equally important. Demonstrating compliance means that an organization is committed to conducting business in conformity with the regulations and standards that apply to them. It is ensured by Conformity Assessment (CA) which involves a set of processes that manifest that your product, service, or system meets the requirements of a standard. There are a variety of Conformity Assessment Procedures (CAPs) or routes available starting from the least stringent one, i.e., SDoC (Self-Declaration of Conformity), used extensively by the EU in its regulations for CE Mark to the intensive third-party assessment. Similarly, for instance, in India, BIS uses compulsory registration for electronics products, solar energy products, etc.

Conformity assessment is growing in importance both in regulated and voluntary sectors. A WTO study (https://www.wto.org/english/res_e/reser_e/ersd201609_e.pdf) carried out in 2016 on Specific Trade Concerns (STCs) raised by the TBT Committee inter alia indicates that:

- Conformity Assessment Procedures (CAPs) of any country generate equally more trade concerns for Members than technical regulations. It implies that following harmonized (or equivalent) technical regulations in practice is not necessarily the best solution for facilitating international trade. Notwithstanding the degree of symmetry between regulations, CAPs are still appraised by WTO Members as a decisive hindrance to market access. Post investigation of different CAPs, the analysis finds that testing and certification are the procedures that give rise to considerable concern during bilateral and multilateral trade negotiations.
- CAP-related STCs elaborate on a full range of issues regarding consistency with the TBT Agreement; out of these concerns, the most significant issues are related to transparency and unnecessary trade barriers.
- The data on STCs raised by different economies indicates a volumetric exchange of STCs between China, the EU, and the USA. China is the WTO Member most frequently challenged in CAP-related STCs, and 93% of these STCs were brought by the EU or the USA, or both together.

The situation is understandable because more and more countries today are adopting and implementing International Standards and then cease to be an issue; however, their CAPs differ and will continue to diverge since there is no single internationally accepted model. Since the conformity assessment includes testing, inspection, and certification, thus, it is crucial to suppliers, consumers, and regulators.

Accreditation

As the importance of conformity assessment has grown with the regulators increasingly relying on independent, third-party inspection and certification bodies or labs to carry out inspection/certification/testing on their behalf and number of voluntary schemes coming up globally who needed third-party bodies to scale up adoption of their standards, a need arose of assuring the competence of conformity assessment bodies and facilitating cross-border acceptance of test reports, inspection reports, and certifications. This has led to the development of a system of accreditation to establish the technical competence of inspection/certification bodies and labs for which ISO has laid down several standards. ISO 17000 defines accreditation as "third-party attestation related to conforming assessment body conveying formal demonstration of its competence to carry out specific conformity assessment tasks."

WTO and SDOs: The Key Players

The WTO (https://www.wto.org) is the organization that deals with global trade. It ensures seamless, free, and predictable trade flow across the globe. It originated on 1 January 1995 under the Marrakesh Agreement (Marrakesh Agreement) after the Uruguay Round of Multilateral Trade Negotiations. It would be pertinent to mention that earlier, General Agreement on Tariffs and Trade (GATT) (General Agreement on Tariffs and Trade (GATT)), which commenced in 1947, governed the international trade. GATT was replaced by WTO agreements (Agreement on Technical Barriers 2022), and the agreement was signed initially by 123 nations. The agreement is popularly known as TBT Agreement (Technical Barriers to Trade Agreement). India has been a WTO member since 1 January 1995 and a member of GATT since 8 July 1948. The purpose of the TBT agreement is to assist manufacturers of goods and services, exporters, and exporters to conduct their business.

The Role of the WTO

The WTO is run by member states and its main functions are:

- **Trade negotiations**: WTO agreements include goods, services, and intellectual property. They specify the principles of freedom and permissible diversity. It includes each country's commitment to lowering commodity prices and other trade barriers as well as opening and maintaining open market markets.
- **Implementation and monitoring**: WTO agreements require governments to make their trade policies transparent by informing the WTO of applicable laws and measures.
- **Dispute resolution**: The WTO process for resolving trade disputes under Dispute Settlement Understanding is essential for enforcing laws as well as ensuring fair trade.

- **Capacity building**: WTO agreements contain a special provision for developing countries, including long-term implementation of agreements and commitments, measures to increase their trade opportunities, and support to help them build their trade power, deal with conflicts, and apply technical standards.
- **Public relations**: The WTO maintains dialogue with non-governmental organizations, members of parliament, other international organizations, the media, and the general public on various aspects of the WTO.

The WTO regime recognizes that governments have the right to protect their people for reasons such as health and safety and therefore can legally enforce product requirements – so-called technical regulations or sanitary and phytosanitary (SPS) measures in the agricultural food sector. It describes, under the TBT Agreement, reasons such as health (food, drugs, medical equipment), safety (toys, electronics, LPG cylinders), environment (vehicle exit rates, environmental regulations, lead content in paint), fraudulent trade practices (damage to cement or gold jewelry) and national security (telecom) controls, i.e., the product requirements set by law. This has led to an increase in the number of technical regulations, national and international volunteer standards, and compliance assessments applicable to all sectors in products, services, processes, management systems, or personnel. Indeed, the WTO regime has created two distinct domains: sectors amenable to technical regulations/SPS measures and sectors that are driven by Standards (Jauhri 2020).

Trade Dynamics and WTO

According to a study (Bastos and Cabral 2007), based on data from 20 OECD countries for the period 1980–2000, it shows that trade transitions between industries that contradict previous international knowledge are often the best way to increase trade. Economic analysis shows that the perceived changes in trading patterns were explained by the first gifts of individual and financial transactions in a particular productivity of labor and labor costs. The results also suggest that trade liberalization has led to an increase in the pre-existing specialization of large OECD economies in industries with a growing profitability scale. Indeed, economic status, technology, and factor endowments are important factors in changing trade patterns.

One of the most important factors affecting trade is FDI (Foreign Direct Investment). According to the discussion paper (Riccardo 2004), in the past, FDI was primarily directed to establish a manufacturing facility in foreign markets in order to cater to foreign buyers and to cross trade barriers. Similarly, the increase in trade barriers would be associated with an increase in FDI. In particular, therefore, trade and FDI were areas of concern. More recently, however, global investment decisions seem to be driven by a different set of considerations. The development of telecommunications and transportation technologies allows firms to achieve significant cost reductions by cutting value-added chains between companies located in different areas as a function of pricing. Therefore, trading costs play a very different role in this context. In particular, the increase in trade restrictions will not only influence

trade, but will also reflect negatively on the motives of multinational firms to invest abroad.

Generally, International Trade is dominated by two types of trade barriers, namely, Tariff Barriers (TB) and Non-Tariff Barriers (NTB). TB is a related tax imposed by governments to control the import and export of one or more products from a particular country. On contrary, NTBs are government policies and actions without tax restrictions. Figure 5 shows the year-wise status of TB/NTB. In Geneva (Tokyo round) conventions, NTB has registered its presence (Fig. 5a) (https://view.officeapps.live.com/op/view.aspx?src=https%3A%2F%2Fwww.wto.org%2Fenglish%2Ftratop_e%2Ftbt_e%2Fprogram_fiji06_e%2Fs3_conc_def_princ_e.ppt&wdOrigin=BROWSELINK). Trade barriers often protect domestic companies by imposing restrictions on the distribution of goods between nations. With the advent of the WTO regime, TB in international trade has declined (see Fig. 5b), and NTBs have arrived to take the central stage in the arena. The major barriers between NTBs (commonly known as TBT) are generally accepted as those related to standards and compliance assessments.

Figure 6 illustrates the SPS/TBT notifications in WTO from 1995 to 2022 (till march 2022) (https://epingalert.org/en/FactsAndFigures/Notifications); the year-wise data shows that the number of TBT notifications almost keeps on rising (Fig. 6a) while ups and downs may be observed in SPS (Sanitary and Phytosanitary) notifications; in case, we analyzed the economy-based TBT data (Fig. 6b), the facts are quite interesting; the developed economies have not raised TBTs substantially while the developing and emerging economies keep on raising the numbers. On the other hand, in least-developed economies, it is also showing increasing trend in recent years. In case, we see the data of TBT notifications by objectives, the maximum TBT notifications were come up in the segment of protection of human health or safety. The quality requirements and prevention of deceptive practices have taken second and third position, respectively (Fig. 6c). It would be pertinent to mention here that we are exploring TBT data more than SPS data because we are more concerned in TBT notifications than SPS to remain in the scope of the chapter. The data/information implies that:

- TBTs are showing a raising trend than NTBs. However, socio-economic condition is one of the major deciding factors for implementation of such trade barriers.
- Developed economies have raised TBT initially, but soon after a certain period they have started to curb the growth. It indicates that somewhere it has been understood that raising the number of TBT may adversely influence the flow of their international trade. Moreover, largely they have already protected their market, and they now have only to maintain the pace with the contemporary changes in the trade dynamics.
- WTO provides provisions to enact TBT national security requirements; the prevention of deceptive practices; protection of human health or safety; etc, as a result, maximum TBTs get notified by different countries in this segment.

a

Year	Place / name	Subjects covered	#
1947	Geneva	Tariffs	23
1949	Annecy	Tariffs	13
1951	Torquay	Tariffs	38
1956	Geneva	Tariffs	26
1960-1961	Geneva (Dillon Round)	Tariffs	26
1964-1967	Geneva (Kennedy Round)	Tariffs and anti-dumping measures	62
1973-1979	Geneva (Tokyo Round)	Tariffs, NTBs, "framework agreements"	102
1986-1994	Geneva (Uruguay Round)	Tariffs, NTBs, rules, services, IP, DSU, textiles, agriculture, creation of WTO, etc.	123
2001 -	Geneva (DDA)	(the above) + environment, + ?	148
			Source: WTO

b

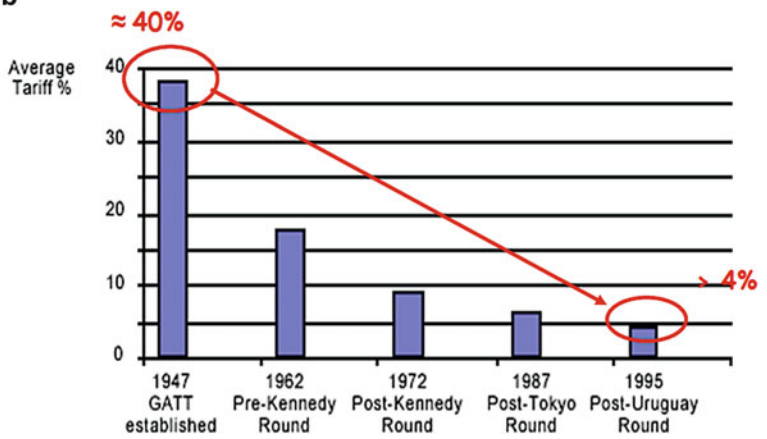


Fig. 5 Non-tariff barriers to trade (NTBs). (**a**) Trade rounds. (**b**) Overall, import weighted tariff on industrial products

SDOs and Their Role

There are thousands of SDOs (Standard Development Organizations) worldwide. They can standardize almost anything to make life easier, safer, and more productive. Generally, these bodies have cooperative agreements. They may be supportive of each other's standards, building on them, or deliberately avoiding repetition. The

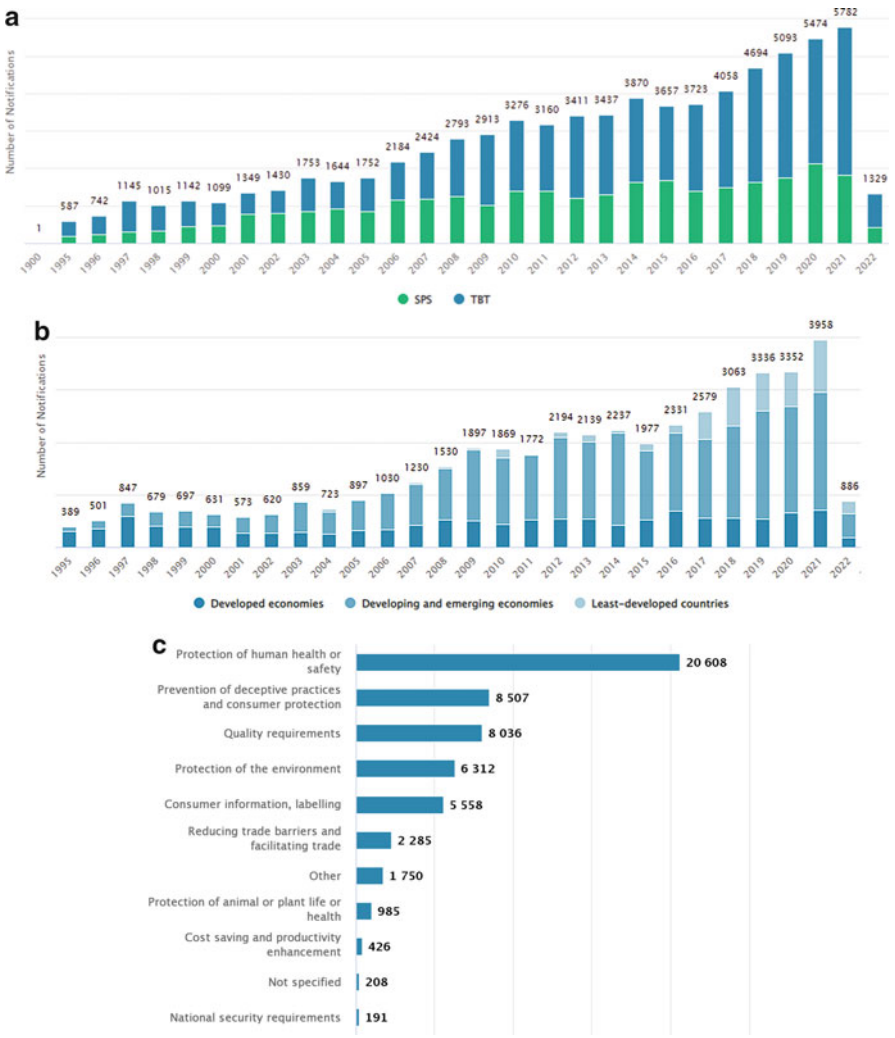


Fig. 6 WTO notifications as on 31.03.2022. (Retrieved from (https://epingalert.org/en/FactsAndFigures/Notifications))

year-wise establishments of the most influential SDOs in the field of standard formulations are summarized in Table 1. An interested reader may visit their official websites to explore more about them.

The IEC (https://www.iec.ch/homepage) is one of the preferred standard development organizations recognized by the WTO and is responsible for monitoring national and regional organizations that agree to use the IEC international standards as the basis for national or regional standards as part of the WTO-TBT Agreement. IEC and ISO and ITU (https://www.itu.int/en/Pages/default.aspx) have built

Table 1 Establishments of International standard bodies

S. no.	Institution	Year
1	International Telecommunication Union (**ITU**) Geneva, Switzerland	1865
2	International Bureau of Weights and Measures French: *Bureau international des poids et mesures (***BIPM**) Sèvres, France	1875
3	National Institute of Standards and Technology (**NIST**) Maryland, U.S.	1901
4	International Electrotechnical Commission (**IEC**) Geneva, Switzerland	1906
5	American National Standards Institute (**ANSI**) Washington, D.C., USA	1918
6	(i) International Organization for Standardization (**ISO**), Geneva, Switzerland. (ii) Association for Computing Machinery (**ACM),** New York, USA. (iii) Indian Standards Institution (**ISI**), New Delhi, India.	1947
7	Joint Electron Device Engineering Council (**JEDEC**) Arlington, Virginia, USA	1958
8	European Computer Manufacturers Association (**ECMA**) Geneva, Switzerland	1961
9	Institute of Electrical and Electronics Engineers (**IEEE**) New York, USA	1963
10	European Committee for Electrotechnical Standardization (**CENELEC**) Brussels, Belgium	1973
11	Bureau of Indian Standards (**BIS**), New Delhi, India	1986
12	Japan Electronics and Information Technology Industries Association (**JEITIA**) Tokyo, Japan	2000

strategic partnerships with the WTO. The IEC has established many standards in various fields; however, the Institute of Electrical and Electronics Engineers (IEEE) (https://www.ieee.org/) is one of the major contributors in standard development for OWC (Optical Wireless Communications) and relevant technologies.

It can be observed that the establishments of influential SDOs were started in 1865, and even India has registered its appearance in 1947 by establishing ISI which subsequently became BIS in 1986.

Standardization and Regulatory Ecosystem in India

For the understanding purpose, the overall ecosystem of standardization and regulations may be divided into three parts, i.e., (i) Acts and regulations, enacted so far to govern the system; (ii) Domestic QI, developed over the years in this arena; and (iii) Indian key players, role of BIS, QCI, NMIs, Regulators, line ministries/departments, etc.

Acts and Regulations

A comprehensive list of principal trade-related Acts/laws is given in Table 2.1 of the Trade Policy Review by WTO for India published on 16 March 2021 (Trade policy review 2021). Just for a look, the information is presented in graphical form in Fig. 7. Figure 7a illustrates the sector-wise number of Acts while Fig. 7b shows year-wise Act notifications. One can observe that maximum acts came in the segment of intellectual property rights and SPS and banking sector are having the second place. If we see the year-wise distribution, a maximum number of Acts/regulations were notified during 1992–2002 and 2002–2012 (6 Acts in the bracket of 10 years). The information indicates that we have sufficient Acts/Regulations, almost in every significant area; however, the scope of the chapter remains around Technical Regulations and related matters.

Table 2 Quality Infrastructure (QI) of India

Govt./ regulator	National standard bodies (NSBs)	Accreditation bodies (ABs)	Metrology	Conformity assessment bodies (CABs)
MoC&I	–	QCI (NABL and NABCB)	–	–
MoCA, F&PD	BIS	–	Legal Metrology	BIS
MoS&T	–	–	CSIR-NPL (NMI)	–
MeitY	Largely uses BIS Standards	Largely uses BIS recognized Labs	–	STQC
DPIIT/ MoS/DHI/ DoP, etc.	-do -	-do -	–	BIS (Sole CAB under BIS Act)
FSSAI (food)	-do -	Largely uses BIS recognized Labs/ NABL Accredited Labs	–	Designated Pvt. CABs.
MoA/ DoAC&FW	-do -	-do -	–	-do -
CDSO (drugs)	Largely uses BIS Standards for Medical Devices/ Components	-do -	–	-do -
MoC/DoT	Largely uses ITU Standards	Largely uses TEC recognized Labs	–	TEC

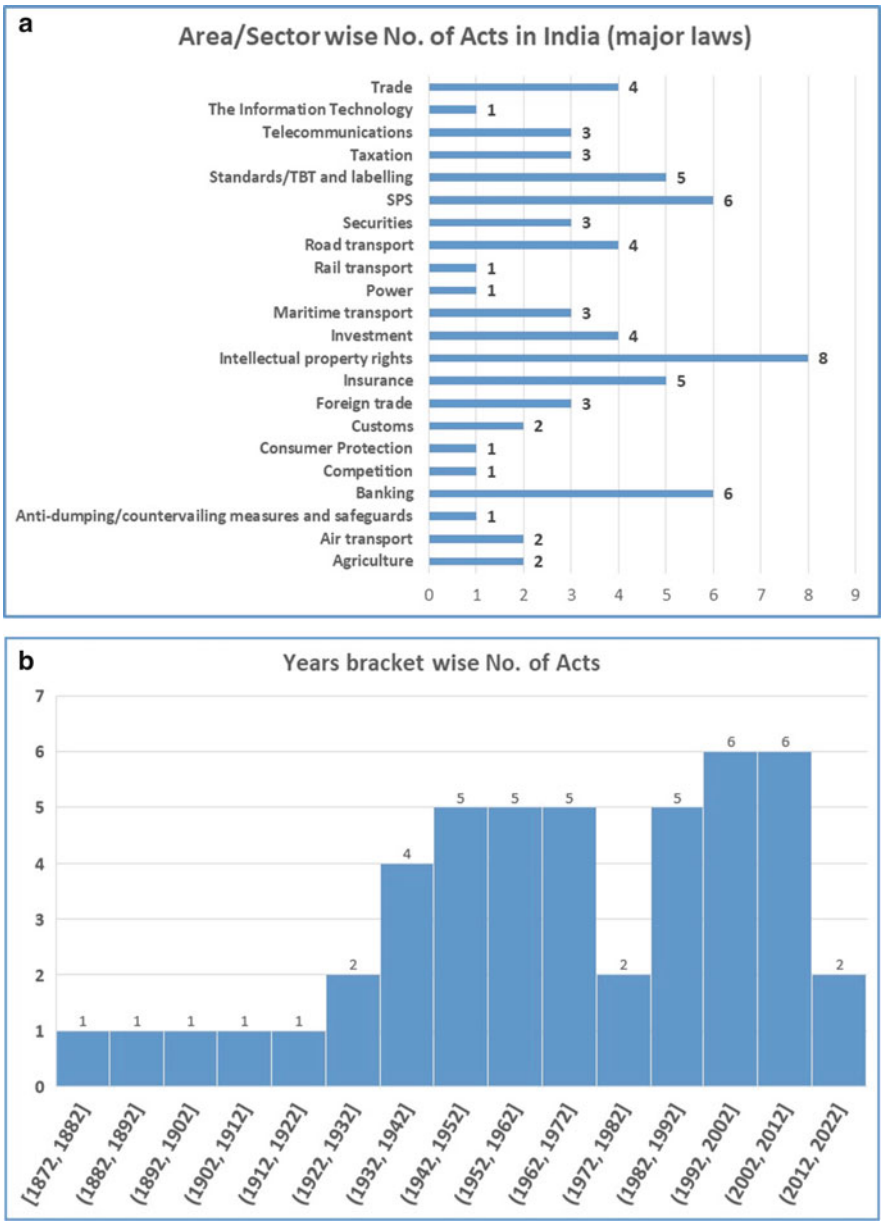


Fig. 7 Major Acts in various sectors in India. (**a**) Sector-wise number of Acts, (**b**) years bracket-wise number of Acts notifications

Domestic QI and Institutional Mechanism

The American National Standards Institute (ANSI) has tried to visualize the Indian institutional mechanism for Standards and Conformity Assessment Bodies (https://share.ansi.org/Shared%20Documents/Standards%20Activities/Country%20Charts%20-%20Standards%20and%20Conformity%20Assessment%20Systems/India.ppt). However, it does not capture the entire image of the system. To understand the existing ecosystem, a comprehensive ER (Entity-Relationship) diagram is depicted in Fig. 8. The figure is self-explanatory and illustrates the domestic institutional mechanism of Indian QI ecosystem with mapping of various government bodies/Regulators (R), Accreditation Bodies (ABs), Conformity Assessment Bodies (CABs), and their affiliations with international standardization organizations. Linear mapping of different major entities concerning QI components is also given in Table 2. It shows the relationship of line ministries/departments and their associated organizations/institutes, affiliation with international bodies, and WTO contact/query points.

The ecosystem indicates that a sustainable QI has been flourishing for many years; however, it was felt in Government that a concrete policy framework is required. Therefore, in this connection, a strategic development took place when India introduced the "Indian National Strategy for Standardization (INSS)" in 2018 (Indian National Strategy for Standardization 2018). It not only gives a roadmap to the policymakers or stakeholders but also indicates a strong willingness of India to emerge as a nation with zero tolerance of substandard products, processes, services, etc.

Indian Key Players

To understand the functioning of Indian quality infrastructure and conformity assessment system, one has to have a close look of its key players, i.e., BIS, QCI, NMIs, line ministries/departments, etc. Let us briefly explore some of the major entities.

Quality Council of India (QCI): For Accreditation

The QCI is an autonomous non-profit organization registered under the Registration of Societies Act. It has been established by the joint initiative of the Government of India and the Indian Industry. It is administratively and financially independent and is governed by a Council having a fair representation of the Government of India, Industry (through its associations), and other stakeholders like non-government organizations (NGOs), CABs, institutions, etc. QCI is governed by its own rules and regulations. The operational expenses of QCI and its Boards are generated through the services offered. The QCI has established the National Accreditation Board for Certification Bodies (NABCB) (https://nabcb.qci.org.in) as per the decision of the Cabinet in Feb 1996 and the Memorandum of Association of Jan 1997 to carry out accreditation of Conformity Assessment Bodies (CABs). The terms for

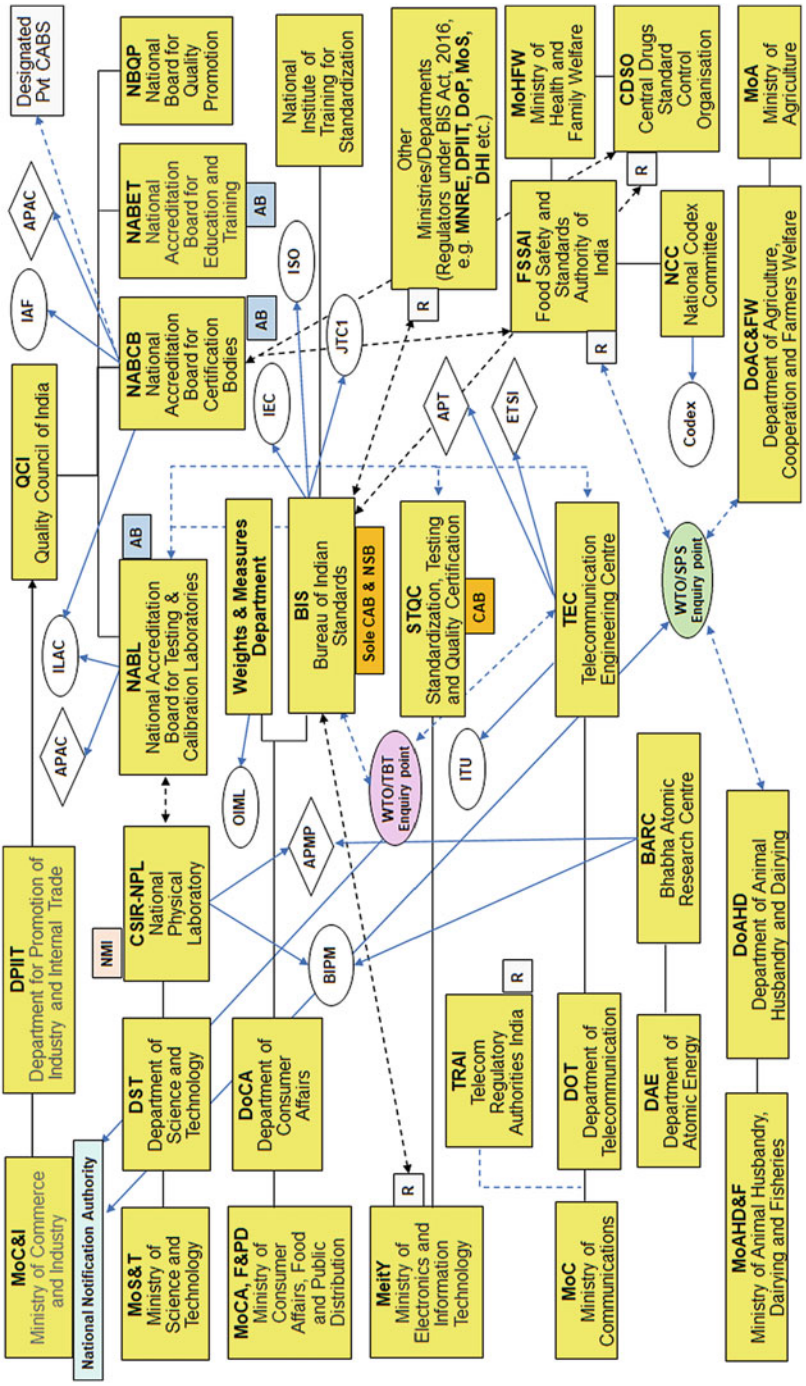


Fig. 8 Indian ecosystem of Standards and Conformity Assessment Bodies

accreditation for the various schemes are developed based on national and/or international standards or guidelines. Accreditation is generally carried out as per common international standards of ISO, some of which are listed below:

- Laboratories as per ISO 17025.
- Inspection Bodies as per ISO 17020.
- Product (which includes Process) certification bodies as per ISO 17065.
- Management systems such as ISO 9001 certification bodies as per ISO 17021-1.

Table 3 depicts a sector-wise number of certification bodies accredited by NABCB. Figure 9 illustrates the statistics graphically; one may note that maximum certification bodies are come up in the Inspection sector (Fig. 9a) while year-wise data indicates that in 2021 (Fig. 9b), the maximum number of certification bodies got accreditation.

On the other hand, National Accreditation Board for Testing and Calibration Laboratories (NABL) (https://nabl-india.org) was founded in 1988, and it is an accreditation body, with its accreditation system established following ISO/IEC

Table 3 Sector-wise number-wise certification bodies. (as on 30.03.2022, data source (https://nabcb.qci.org.in))

S.No.	Sector	Number of certification bodies
1	Environmental Management Systems (EMS)	**15**
2	Energy Management Systems (EnMS)	**5**
3	Food Safety Management Systems (FSMS)	**21**
4	Inspection Body (IB)	**85**
5	Information Security Management Systems (ISMS)	**8**
6	Information Technology Service Management Systems (ITSMS)	**2**
7	Occupational Health and Safety Management System (OHSMS)	**11**
8	Product Certification Body (PCB)	**24**
9	Certification of Persons (PrCB)	**7**
10	Quality Management Systems (QMS)	**41**
11	Medical Devices Quality Management Systems (MDQMS)	**14**
12	ISO-15378	**1**
13	Trustworthy Digital Repositories Management Systems (TDRMS)	**1**
14	ISO 21001	**6**
15	Validation and Verification Bodies (VVB)	**5**
16	Information Security Management Systems (ISO-27701)	**4**
17	Indian Certification of Medical Devices (ICMED)	**7**
18	Business Continuity Management Systems (BCMS)	**1**
	Total =	**258**

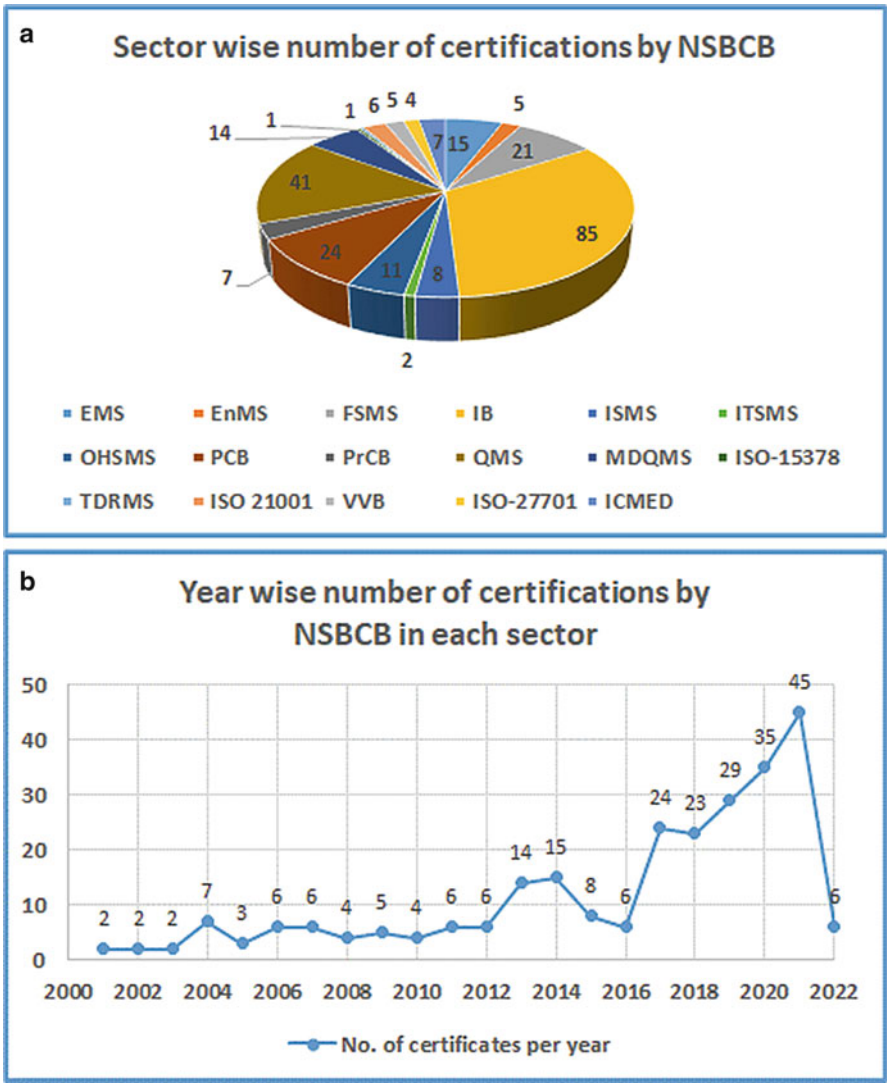


Fig. 9 Statistics of accreditation carried out by NABCB. (**a**) Sector-wise number of certification bodies. (**b**) Year-wise number of certification. (As on 30.03.2022, data source (https://nabcb.qci.org.in))

17011. "Conformity Assessment – Requirements for Accreditation bodies accrediting conformity assessment bodies." NABL provides voluntary accreditation services to different bodies in accordance with relevant standards, for instance:

Segment	Standard
Testing laboratories	ISO/IEC 17025 "General Requirements for the Competence of Testing and Calibration Laboratories"
Calibration laboratories	ISO/IEC 17025 "General Requirements for the Competence of Testing and Calibration Laboratories"
Medical testing laboratories	ISO 15189 "Medical laboratories -Requirements for quality and competence"
Proficiency Testing Providers (PTP)	ISO/IEC 17043 "Conformity assessment — General requirements for proficiency testing"
Reference material producers (RMP)	ISO 17034 "General requirements for the competence of reference material producers"

NABL has signed Mutual Recognition Arrangements (MRA) to ILAC and APAC for accreditation of Testing and Evaluation Laboratories (ISO/IEC 17025), Medical Testing Laboratories (ISO 15189), Professional Test Providers (PTP) (ISO/IEC 17043), and Reference Materials Producers (RMP). Till 2021, NABL has accredited more than 5600 labs in different fields and played an important role in creating quality infrastructure in the testing sector. Indeed, a voluntary system of accreditation has developed worldwide under the aegis of the International Accreditation Forum (IAF) for certification and International Laboratory Accreditation Cooperation (ILAC) for inspection and testing and India responded to these developments by setting up a national accreditation system in the form of the NABCB and the NABL. NABCB is undertaking accreditation of certification, and inspection bodies as per applicable international standards while NABL is devoted to accreditation of testing, calibration, and medical laboratories and related bodies. NABCB is a member of both IAF and ILAC, and NABL is a member of ILAC, and both have achieved international equivalence for their accreditations.

BIS: For Standardization

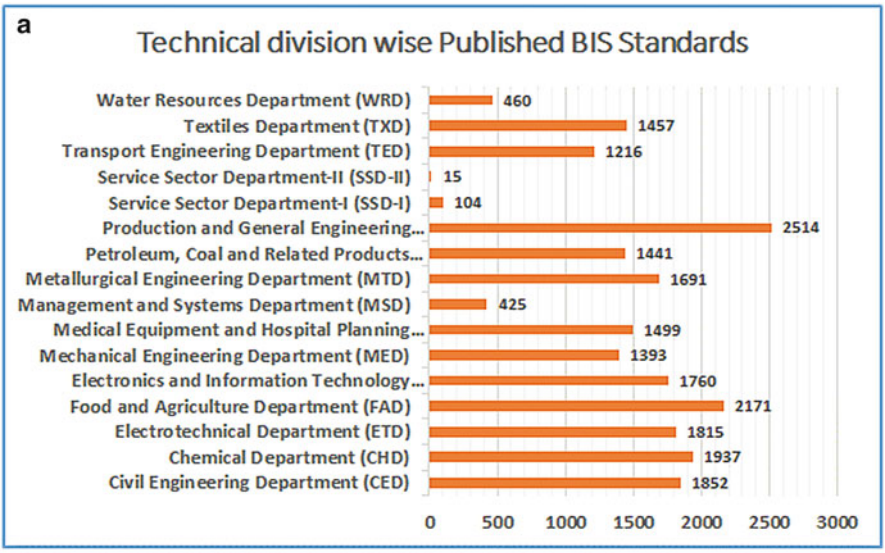
One can see in Table 1 that the Indian Standards Institution (ISI), in New Delhi, India, came into existence in 1947. In the early years, the organization focused on the standardization activity. In order to provide benefits of standardization to ordinary consumers, ISI began implementing the Certification Marks Scheme under the *Indian Standards Institution (Certification Marks) Act, 1952*. The system, officially launched by ISI in 1955–1956, enabled it to issue licenses to manufacturers who produce goods in accordance with Indian Standards and use ISI Mark in their products. To meet the requirements of the Certification Marks Scheme, the nucleus of the laboratory was introduced in 1963. While product validation was conducted under the *Indian Standards Institution (Certification Marks) Act, 1952*, however, formulation of standards and other related functions were not governed by any law. The Bill for this purpose was therefore introduced in Parliament on 26 November 1986. The BIS came into being, with an Act of Parliament dated 26 November 1986, on 1 April 1987, expanded with additional powers taking over staff, assets, liabilities, and functions of former ISI. Through this change, government aims to create a quality culture, understanding, and increased consumer participation in the formulation and implementation of national standards. The Bureau is a 25-member Body

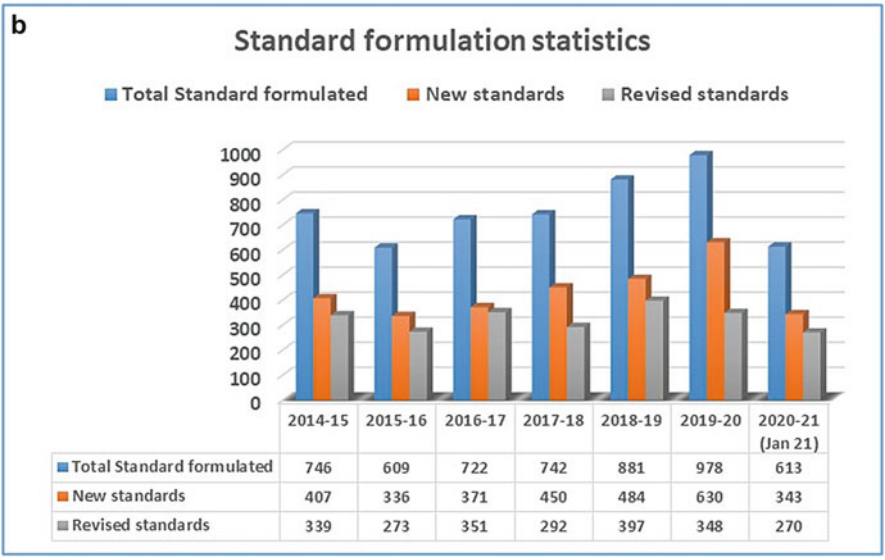
Corporate organization representing both Central and Provincial governments, Members of Parliament, industry, science and research institutes, consumer organizations, and professional organizations. The Union Minister of the Ministry of Consumer Affairs, Food and Public Distribution (MoCA, F&PD) holds the position of President while Minister of State, MoCA, F&PD works as Vice-President (https://www.bis.gov.in).

The BIS was established under the BIS Act of 1986 for the harmonious development of the standardization activities, marking, and quality certification of goods and related matters. The BIS has been providing traceability and tangibility benefits to the national economy in many ways – providing safe quality goods, reducing health risks for consumers, promoting exports and imports substitute, controlling the growth of diversity, etc. through standardization, certification, and testing. The BIS is also known as the National Standard Body (NSB) or Standard Development Organization (SDO) of India. The BIS is involved in a variety of activities as provided below:

- Standards Formulation.
- Product Certification Scheme.
- Compulsory Registration Scheme.
- Foreign Manufacturers Certification Scheme.
- Hall Marking Scheme.
- Laboratory Services Laboratory Recognition Scheme.
- Sale of Indian Standards.
- Consumer Affairs Activities.
- Promotional Activities.
- Training Services, National and International level.
- Information Services.

The role of BIS has changed over the time; however, its primary role is to formulate and develop standards, harmonized with international standards. Interested user may explore further details on various websites of BIS. BIS has formulated 21,750 Indian standards till March 2022. Figure 10a illustrates technical division-wise published standards (data source (https://www.services.bis.gov.in:8071/php/BIS_2.0/eBIS/). Production and General Engineering Department (PGD) has the maximum number of BIS standards. To observe the trend of year-wise standard development at BIS, Fig. 10b shows statistics of standard formulation that took place during the period of 2014–2015 to 2020–2021 (Jan 21). From the data, the CAGR (Compounded Annual Growth Rate) comes to 4.62% for the period of the years 2014–2015 to 2019–2020 for total formulated standards; similarly, for the same period, the CGAR for new standard formulation comes to 7.55% and for revised standards remains 0.44% (data source (https://consumeraffairs.nic.in/sites/default/files/ReformsBIS.pdf). The data/information given in Fig. 10 indicates that we are moving fast in the standardization activities.





	2014-15	2015-16	2016-17	2017-18	2018-19	2019-20	2020-21 (Jan 21)
Total Standard formulated	746	609	722	742	881	978	613
New standards	407	336	371	450	484	630	343
Revised standards	339	273	351	292	397	348	270

Fig. 10 Status of BIS standards. (**a**) Technical division-wise published standards. (**b**) Standard formulation statics

Technical Regulations: Major Key Players Are Ministries/Departments

A technical regulation is a government document that sets out the features of a product or process that are consistent with the production processes, including operational management provisions, for which compliance is mandatory. Although compliance with standards is voluntary, technical regulations are by nature mandatory. For instance, you may have noticed that on most of the products (on their

packaging or both), apart from other information, there are specific markings (see Fig. 11). Whether it is an appliance, good products, or electronic gadgets, one can observe ISI mark, vegetarian or non-vegetarian mark, and several other symbols or marks. Why are these symbols or marks printed on the product label? What messages do these signs convey to consumers? A careful buyer knows the value of these marks when buying any item. For example, Fig. 11 illustrates the standard marking on the cable (Fig. 11a) of a microwave oven, food containers (Fig. 11b), and electronic gadgets like bluetooth speaker (Fig. 11c), laptop (Fig. 11d), and mobile phones (Fig. 11e, f).

A standard mark (https://www.thecro.in/wp-content/uploads/2017/03/Standardization-Marks-.pdf) is a mark/symbol given to a product, which meets certain quality standards in terms of materials, production methods, labelling, packaging, safety, and performance. In order to obtain a standard mark, the manufacturer/importer must follow the process of conformity (i.e., conformity assessment procedure) in the country when they want to sell/import products. Product conformity is assessed by

Fig. 11 Standard marks/symbols of different products

accredited national/international organizations before being marketed. It needs to show that all legal requirements (i.e., testing, testing, certification, etc.) are met. The process for each product is specified in the applicable product law. In other words, these standard marks/symbols reflect specific management/regulatory measures taken by a particular country/region to ensure the quality and safety of products exported/sold locally or internationally.

Let us take a sector-specific example. Table 4 depicts different popular standard marks issued from different countries/international regulators/standard bodies specifically for electronics and IT products. Figure 11d, e shows the background view of two popular electronic gadgets, namely, laptop and mobile, respectively. In Fig. 11d, some international standard marks/logos can be viewed on a laptop. In Fig. 11e, one can observe the CRO (Compulsory Registration Order notified by MeitY) marking on the lower part of the mobile. The cut-off part of the marker is also displayed in the inset to clarify details. Figure 11f shows an example of e-labelling in mobile phones.

Table 4 Standard marking in the ICT world, as a symbol of quality and safety

Sr. No.	Marking/ Legal status	International body/ Country	Year	Products group
1	"Underwriters Laboratories" (*Voluntary*)	American safety consulting and certification company (USA)-Private, LLC	**1894**	Standards for Electrical and Electronic Products
2	(*Voluntary*)	Canadian Standards Association (Canada) -Standards Council of Canada	**1946**	Electrical and Electronics
3	"European Conformity" (*Mandatory*)	European Commission (EU)-Govt.	**1985**	Low voltage (*all electronics goods*)
4	"Federal Communications Commission" (*Mandatory*)	FCC Declaration of Conformity (USA)-Commission	**1998**	Electronic products
5	"China Compulsory Certification" (*Mandatory*)	China Compulsory Certification (China)-Bureau	**2002**	Low voltage & IT products
6	"Compulsory Registration Order" (*Mandatory*)	Electronics and Information Technology Goods (Requirement for Compulsory Registration) Order, 2012	**2012**	Electronic & IT products
7	**EMC** "Electromagnetic compatibility" (*Voluntary*)	European Commission (EU)-Govt.	**2014**	Electronics equipment

The "*Electronics and Information Technology Goods (Requirements for Compulsory Registration) Order, 2012 (CRO)*" was notified by the Ministry of Electronics and Information Technology (MeitY) on 03 Oct. 2012 under the provision of "Compulsory Registration Scheme (CRS)" notified by BIS under BIS Act, 1986 & Rules, 1987 (now CRO – 2021 has suppressed the Order under BIS Act 2016, Rules & BIS (Conformity Assessment) Regulations, 2018). The order was aimed at addressing consumer safety issues in India and curbing the entry of substandard electronic products into the country. The product categories are covered under CRO in a phased manner. To date, 63 product categories have been covered under the ambit of the Order, and it has come into effect for all the notified product categories. In accordance with CRS, a manufacturer seeking registration of notified goods must get tested its products in BIS recognized laboratories and is required to apply for registration at BIS along with test report and required documents. Once his registration has been granted under CRS, BIS issues a unique registration number (R. No.). Thereafter, the manufacturer can affix standard BIS marks on its products. An example is shown in Fig. 11e where "R-xxxxxxxx" shows R. No. and "IS xxxxx, which" reveals a certain "Indian standard" (e.g., a real world standard marking is shown in Fig. 11f) for which the product complies. A list of all notified products and other scheme details can be viewed on the MeitY website (https://www.meity.gov.in/esdm/standards). It would be pertinent to mention that the CRO is the first registration scheme which is based on self-declaration of conformity for goods notified by MeitY.

The BIS Act, 2016, enables various Ministries/Departments to notify theirs own regulations, which are also termed as QCOs (Quality Control Orders). Figure 12 illustrates the year-wise product coverage and number of notifications of QCOs. In this regard, it is pertinent to mention that most of the QCOs are notified under BIS Act 2016. According to the information available at (https://www.bis.gov.in/index.php/products-under-compulsory-certification), around 104 QCOs have been notified till March 2022 covering 460 product categories. Figure 12 indicates that the maximum number of QCOs was notified in 2018 while the maximum number of products covered under QCOs in 2020. Figure 13 depicts the scenario of sector-wise yearly QCO notifications and product coverage. One can observe that the maximum number of products covered under QCOs pertains to the segment of Steel and Iron products while the electronics and IT and Chemicals, Fertilizers, Polymers, and Textiles hold second and third positions, respectively. Figure 14 shows product category-wise number of operative licenses granted by BIS till March 2022, i.e., total, 19,968 (Nos.) for 64 countries (https://www.crsbis.in/BIS/publicdashAction.do?hmode=getProductCategory).

Apart from the above, "Mandatory Testing and Certification of Telecom Equipment (MTCTE)" launched by the Department of Telecommunications (DoT), Ministry of Communications in 2019 mandating certifications for telecom products under the provision of *The Indian Telegraph (Amendment) Rules, 2017* (https://www.tec.gov.in/mandatory-testing-and-certification-of-telecom-equipments-mtcte) covering around 53 product categories till Sept 2021.

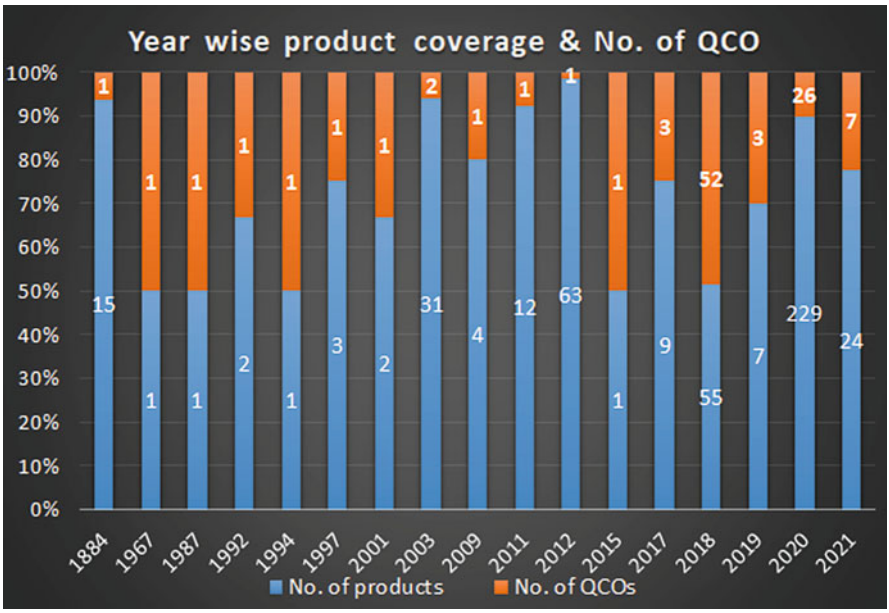


Fig. 12 Year-wise product coverages and no. of QCOs notifications. (*Data source: The number of products and QCOs are counted from the publicly available data on BIS website* (https://www.bis.gov.in/index.php/products-under-compulsory-certification)

Market Surveillance in India

Market surveillance (MS) is one of the major challenges in implementing any regulation. The emergence of eCommerce platforms has made the situation more complex for all regulators across the globe. The architecture and implementation mechanism of MS depends on various factors, country/region-specific administrative structure, socio-economic conditions, adoption of technologies, etc. Since, earlier we have drawn reference from the EU model (see Fig. 4) of MS, for understanding purpose, we may compare the system with the domestic ecosystem of MS on certain parameters.

Parameter	MS model of EU	India (generic view)
Administrative architecture	Group of countries	Single country having Central and State governments
Cartelized or distributed system	Centralized, using RAPEX	Distributed
eCommerce compatibility	Not observed in the publicly available model	Still, it is a challenge
Type of legislations for standardization	EU Directives are used by all members	Technical Regulations have been enacted under different Acts, e.g., BIS Act, 2016, Indian Telegraph Act 1885, etc.

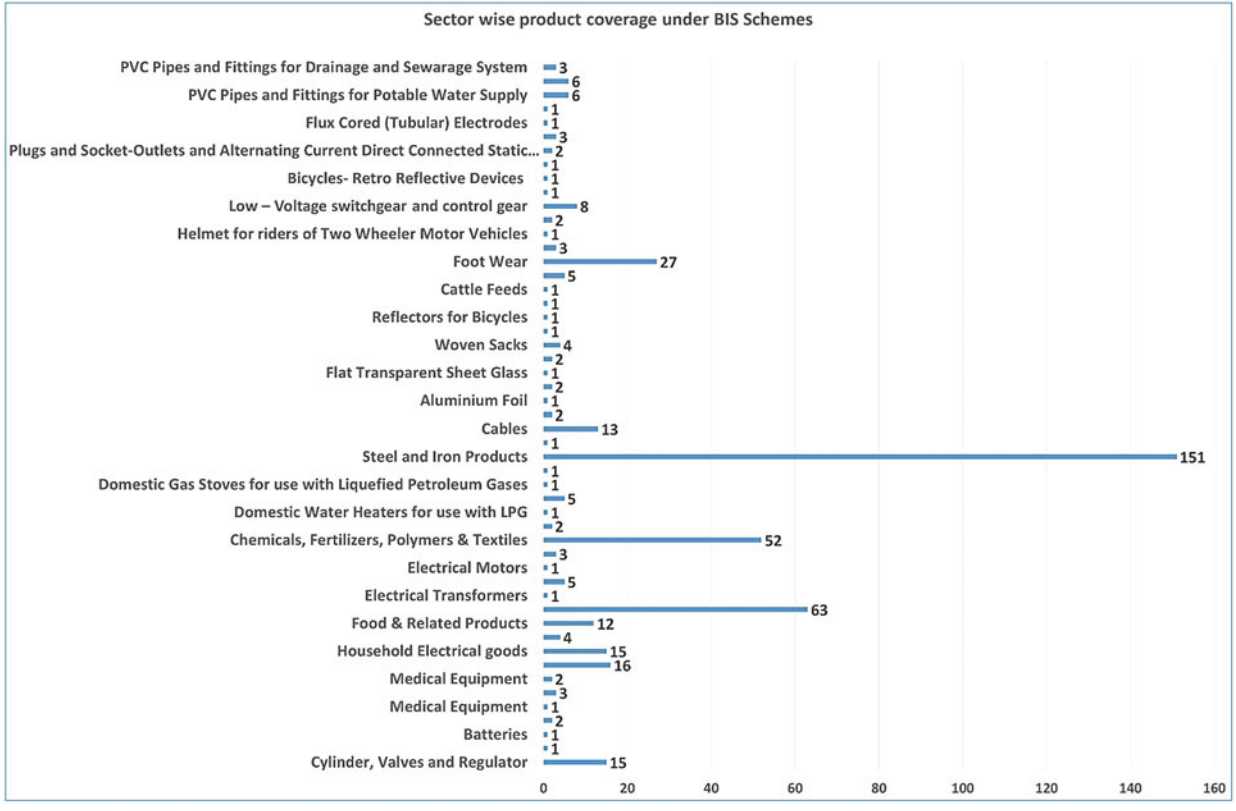


Fig. 13 Sector-wise product coverages and no. of QCOs notifications. (*Data source: The number of products and QCOs are counted from the publicly available data on BIS website* (https://www.bis.gov.in/index.php/products-under-compulsory-certification))

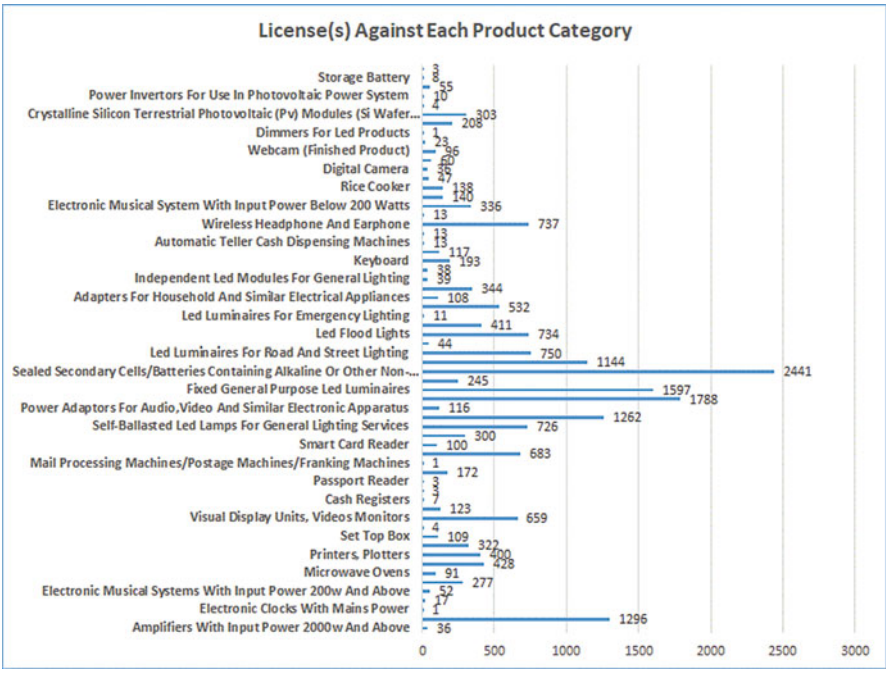


Fig. 14 Product category-wise number of operative licenses granted by BIS till March 2022. (Data source (https://www.crsbis.in/BIS/publicdashAction.do?hmode¼getProductCategory))

The above comparison is self-explanatory. There is a difference in administrative architecture between the EU and India. The MS model of the EU is a centralized system, controlled by RAPEX. The RAPEX is a comprehensive EU wide-ranging program for exchanging quick information on products (other than food, medicine, and medical equipment) that are found to endanger health and/or safety. The product list is reviewed by the European Union weekly, and previously issued warnings may be reviewed. Such type of system does not exist in India.

Unlike in the EU, in India, since the technical regulations have been enacted and implemented under different Acts, their implementation and market surveillance mechanism are non-identical and taken care of by respective authorities. For example, most of technical regulations are notified by sector-specific ministries/departments/authorities under BIS Act, 2016. Except for a few instances, for all these technical regulations, the market surveillance activity is carried out by BIS. As per the provisions of the BIS Act 2016, the BIS becomes the authority by default to initiate market surveillance; notwithstanding, it may delegate powers to the concerning line ministry/department to do so. Accordingly, in some cases, the sector-specific line ministries/departments/authorities also do market surveillance for their notified products, e.g., MeitY has its surveillance mechanism for the products categories notified under CRO.

Other Indian Acts are also being used to enact technical regulations, for example, in the Telecom sector, MTCTE has been established by DoT mandating certifications for telecom products under the provision of the Indian Telegraph (Amendment) Rules, 2017; its implementation part is taken care of by them only.

CSIR-NPL/BARC and DoCA: For Metrology

The CSIR-NPL (or NPLI) (https://www.nplindia.org/) is the National Measurement Institute (NMI) of India. It was created in 1956 by an Act of Parliament, Legal Metrology, Government of India. It operates on all seven basic SI units and their derivates, excluding ionization radiation. Bhabha Atomic Research Centre (BARC) (http://www.barc.gov.in/) was established as a DI (Designated Institute) in 2003 to deal with ionization radiation. The CSIR-NPL is a member of BIPM, a signatory to the International Committee of Weights and Measures – Mutual Recognition Agreement (CIPM MRA) (https://www.bipm.org/en/cipm-mra/) – and a founding member of the Asia Pacific Metrology Program (APMP) (http://www.apmpweb.org/), one of the six Regional Metrology Organizations (RMO). The role of BIPM and RMOs is to homogenize accurate and precise global measurement capabilities with NMIs. The Joint Committee of RMOs and BIPM (JCRB CIPM) analyzes the CMCs (Calibration and Measurement Capabilities) of NMIs and posts the information on the BIPM Key Compatibility Database (KCDB-BIPM) website (https://www.bipm.org/documents/20126/43742162/CIPM-MRA-G-13.pdf/f8b8c429-42e0-4cf1-dc6c-bc60ab7f371a). CMCs have a very high standard of measurement and calibration, guaranteed by the institution to clients of their metrological services, in the form of extended uncertainty of measurement results.

Legal Metrology is the application of legal requirements to measurement and measurement tools. Indeed, a clear and effective legal metrology system promotes confidence in businesses, industries, and consumers and provides a harmonious environment for doing business. In India, this work is being done by the Department of Consumer Affairs (DoCA) (https://lm.doca.gov.in/index.aspx#). The Legal Metrology Act, 2009, received the President's approval on 13 January 2010 and came into effect on 1 April 2011. Details of the measuring instruments are set out in the Legal Metrology (General) Rules 2011. The DoCA performs various metrology functions through its attached/sub-offices, namely, Regional Reference Standard Laboratories (RRSLs) and the Indian Institute of Legal Metrology, Ranchi. Figure 15 illustrates the major components of the legal metrology.

Growth Avenues and Challenges for India

Despite being a developing economy and having a typical socio-economic environment, India has shown tremendous social and economic growth year by year. A similar pattern of such advancement can be observed in the cultivation of standardization and the regulatory ecosystem of India. Some viewpoints in this regard are as follows:

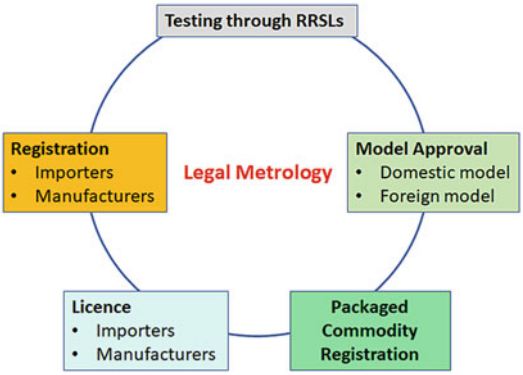


Fig. 15 The major components of legal metrology

Major Positive Nodes

(a) In case, one evaluates the progress under the ambit of the Industrial revolution, i.e., Industry 4.0, where the advancement of ICT is at its core. In this department, India has registered remarkable progress. Digital India is one of the major schemes of the Indian government, having a motto of "Power to Empower" (https://digitalindia.gov.in/content/programme-pillars).

(b) Indian progress has been noticed by IMF (International Monetary Fund). According to its report on "Global Trade Liberalization and the Developing Countries," the progress of integration has been very impressive in a number of developing countries in Asia, and, to a lesser extent, in Latin America. These countries have been successful because they have chosen to participate in international trade, helping them to attract a large portion of FDI. This is especially true of China and India as they embrace trade liberalization and other market-based reforms (https://www.imf.org/external/np/exr/ib/2001/110801.htm).

(c) The World Bank Group (WBG)'s Doing Business Report (DBR) sets the standards for business regulations in 191 economies. The DBR measures regulations that promote business activity and those that oppress them. The Easy of Doing Business (EoDB) index is a standard program developed by WBG where "higher rankings" reflect the best, usually the simplest, regulations for business and strong protection of property rights. India's rank in DBR-2015 was 142 out of 190 economies. This reached at 63rd Position in DBR-2020 released in October 2019. This is a considerable improvement of 79 positions from DBR-2015. In addition, in the ECI (Enforcing Contracts Indicator), India gained 163th position in DBR-2020, an increase of 23 positions from 186th in DBR-2015. One note that ECI is critical to the success of the Doing Business. It measures the time and cost of resolving a trade dispute and a series of good judicial processes (https://doj.gov.in/eodb/).

(d) The progress of the domestic QI ecosystem has been acknowledged by international agencies. For example, a study conducted by the German side for the

Global Quality Infrastructure Index (GQII) has placed India in seventh position for standardization, ninth position for accreditation activities, and 19th position for Metrology-related activities. India has scored 95.6 points out of 100 and is ranked tenth in the global list for overall QI ecosystem development (https://pib.gov.in/PressReleaseIframePage.aspx?PRID=1798826#:~:text=India%20scored%2095.6%20out%20of,infrastructure%20environment%20in%20the%20country).

(e) Awareness level is gradually increasing; people are now noticing the standard marks (e.g., ISI, BIS, FSSAI, etc.) on the products/appliances they are purchasing.

Major Challenges

(a) According to an India-specific report, the latest WTO data indicates that India's average bound tariff rate is 48.5%, while its simple MFN (Most Favoured Nation) average applied tariff is 13.8%. Therefore, it has a higher tariff rate than any other major economy. Despite its target to match the ASEAN (Association of Southeast Asian Nations) tariff rates (approximately 5 percent on average), the basic customs duty is not reduced systematically by India in the past 6 years (https://www.privacyshield.gov/article?id=India-Import-Tariffs#:~:text=According%20to%20the%20latest%20WTO,WTO%20latest%202017%20data%20available). Notwithstanding, subject experts believe that India is not a tariff king (https://economictimes.indiatimes.com/news/economy/foreign-trade/india-is-not-a-tariff-king-has-the-right-to-protect-specific-sectors-under-wto-experts/articleshow/68974070.cms). Whatever the contradictions and differences in opinions, at least one thing is clear this segment requires rationalization without harming the interests of domestic industry.

(b) A recent report by ICEA (India Cellular & Electronics Association) indicates that high import tariffs can negate PLI (Production Linked Incentive) and adversely impact competitiveness and scale (https://icea.org.in/wp-content/uploads/2022/01/Report-by-ICEA-on-Detailed_Tariffs-Drive-Competitiveness-and-Scale_06022022.pdf). One may note that the PLI Schemes are the most ambitious schemes of the Government. A calibrated approach needs to be taken up to be comparative in the world trade arena.

(c) As we have discussed in section "Trade Dynamics and WTO" that NTBs are taking over TBs in international trade, TBTs are raising in developing economies than in developed economies (Fig. 6b); a similar pattern may be observed in the Indian scenario (Figs. 12 and 13). The raise in NTBs is mounting pressure on the government and implementing agencies. Moreover, it is increasing STCs in WTO.

(d) QI has been flourished in India; however, nevertheless it becomes complex system (Fig. 8). Robust coordination among different entities is challenging task.

(e) Enforcement of any regulation at the root level remains a challenge for any regulator/enforcement agency. Robust coordination among Central/State,

autonomous bodies, and other implementing agencies requires more attention and an effective mechanism.

(f) Surveillance is an essential part of any regulatory activity. Developed economies like the EU have implemented RAPEX (Rapid Alert System for dangerous non-food products) to provide a means for the rapid exchange of information between the Member States and the Commission on preventive and restrictive measures taken concerning consumer products posing a serious risk to the health and safety of consumers. On contrary, such a system or similar mechanism is missing in India. The surveillance is either done by BIS or in some cases by line ministries/departments/authorities. This section of the regulatory activities requires more attention and a rubout and effective mechanism.

(g) Surveillance of eCommerce platforms is another emerging challenge across the globe.

(h) Largely it is understood that conforming to International/national standards may resolve a majority of issues. However, it is not completely true. One of the important issues is compliance with different conformity assessment procedures adopted by various countries. It remains one of the major challenges while doing bilateral/multilateral FTA (Foreign Trade Agreement). QCI is working hard to strengthen the accreditation system but has a long way to go.

(i) Centralized data resource is one of the issues where immediate attention is required. For example, the data related to incident reporting for health hazards from any product/appliance, service, or process is one of the helping tools to carry out a risk assessment before taking any regulatory measure. In such a situation, either the data is not available at all or not accessible as and when required. Similar scenarios may also be observed for production data, which is one of the basic components for impact analysis for FTAs under consideration.

Conclusion

The chapter not only provides a holistic overview of major verticals and components which are instrumental in the standardization and regulatory ecosystem but also illustrates the unique strength of India and its hard willingness to cope with challenging conditions and flourish in dynamic situations. India is one of the biggest democracies in the world and striving hard to establish itself as a responsible trade player in the world arena without compromising its heritage value system.

Disclaimer The contents of the chapter are solely the personal views of the author.

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Sanctity of Calibrations

80

Vital for the Export of Indian Products

Anil Jain

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Abstract

Sanctity of calibrations is essentially required to produce products of reliable quality at affordable prices for the international market. India has to compete with the global players in every field of export of goods right from medical equipment to test and measuring instruments to military armaments to agriculture produce. The western countries including the United States, Russia, China, Japan, Eastern and Western Europe, and the United Kingdom are much ahead of India in producing higher quality of products. The Government of India has established coordination among the National Physical Laboratory, the Quality Council of India, the Bureau of Indian Standards, and the National Accreditation Board for Testing and Calibration Laboratories to create quality infrastructure in the country to enable and empower the private, public, industrial, and defense organizations

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to produce internationally accepted quality products for augmentation of foreign exchange through export of made-in-India products. This chapter also discusses the gray areas in calibration and measurement domains which needs earnest attention of all the stakeholders.

Introduction

The contours of global trade and commerce are witnessing a metamorphic transformation in the wake of increasing demand for quality and reliable products at affordable prices. It has compelled every nation in the world to strengthen its quality infrastructure to enable the industry to manufacture and produce globally competitive quality products. Metrology – the science of measurement – involves calibration and standardization of measured results of the measuring instruments used in the production processes. The International Organization for Standardization (ISO) at Geneva, Switzerland, has established ISO/IEC 17025:2017 (ISO 2017) Standard which stipulates General Requirements for the Competence to carry out tests/calibrations including sampling. This standard stipulates the protocols for carrying out various tests pertaining the strength of materials, material microstructure examinations, electrical insulation, and other relevant testings required for ensuring the reliability and safety of electrical installations, food, food grains, soil, water, and environmental parameter testings to determine the levels of toxicity arising out of heavy metals or their traces existing in food grains or edible products and water table contaminations, levels of toxic gases in the atmosphere to determine the levels of air pollution, and calibration of all measureable physical scalar and vector parameters like AC/DC voltage, current, temperature, mass, weight, pressure, flow, force, etc. employing both standard and nonstandard methods. The nonstandard methods are those methods which have been developed by the laboratory itself and can demonstrate repeatable and verifiable results inside an accredited laboratory and all the organizations engaged in the production of products. The primary objective of establishing ISO/IEC 17025:2017 standard is to establish a process which can ensure and deliver quantifiable quality product.

Looking into the emerging trends in international co-manufacturing in various countries on 16 November 2018, during the 226 *Conference Generale des Poids et Mesures* (CGPM) Meeting held at Versailles, France, 60 member states voted for a landmark revision of the SI units on the basis of fundamental constants that describe the natural world (Aswal 2017). The seven SI units which have been considered and ratified in the CGPM are (1) second (*s*), cesium hyperfine frequency (v_{Cs}); (2) meter (m), the speed of light in vacuum (c); (3) kilogram (kg), the Planck constant (*h*); (4) ampere (*A*), the elemental charge (*e*); (5) temperature (*T*), the Boltzmann constant (*k*); (6) mole (*mol*), the Avogadro constant (N_A); and (7) Candela (*cd*), the luminous efficacy of a defined visible radiation (Kcd). According to the CGPM mandate, the values of these natural constants have been fixed without any uncertainty by all the national metrology institutes (NMIs) all over the world (Fig. 1).

Fig. 1 SI unit

Accordingly the National Physical Laboratory in tandem with the Bureau of Indian Standards (BIS), the Quality Council of India (QCI), and the National Accreditation Board for Testing and Calibration Laboratories (NABL), the Government of India, New Delhi, have generated a synergy to strengthen the quality infrastructure in India. In order to ensure the acceptance of test/calibration certificates issued by the accredited conformity assessment bodies (CABs) around the world, NABL has endeavored to establish linkages with the international bodies, viz.

- International Laboratory Accreditation Cooperation (ILAC)
- Asia Pacific Accreditation Co-operation (APAC)

As such, now NABL is a fully fledged member of ILAC and APAC and periodically participates in the meetings of ILAC and APAC.

The NABL is Mutual Recognition Arrangement (MRA) Signatory to ILAC as well as APAC for its accreditation programs – Testing and Calibration (ISO/IEC 17025), Medical (ISO 15189), Proficiency Testing Providers (PTP) (ISO/IEC 17043), and Reference Material Producers (RMP) (ISO 17034).

Such international arrangements/MRA are based on peer evaluation and acceptance of other MRA partner accreditation systems. MRA facilitate acceptance of test/calibration results between countries which MRA partners represent (NABL).

The Mutual Recognition Arrangement accords the acceptance of NABL Calibration Certificates issued by the CABs all across the world which inter alia provides a definite degree of reliability stating and affirming the quality of the product through measurable calibrations and testing through reliable tests mentioned in the test and calibration certificates, compliant to ISO 17025:2017 Standard accompanying the product.

It is worthwhile to mention that the export worthiness, acceptability, and visibility of Indian products in the western markets have certainly improved because of their compliance to international standards and affordable/competitive prices. Concurrently the confidence of the manufacturers and investors of the western world has

considerably enhanced toward the capacity and capability of the Indian manufacturing capacity and infrastructure. They are increasingly shifting their manufacturing plants from China to India. Some of western companies have now their plants either in free trade zones in India or the industrial estates in various states (FinEmergia).

In consonance and continuation of the efforts to strengthen the quality infrastructure in India, the Government of India has established four Electronics Regional Test Laboratories (ERTLs) in the eastern, western, northern, and southern parts of India, the India National Test Houses in different parts of India, host of Electronics Test and Development Centers (ETDCs) at the state levels, the defense laboratories of the Defence Research Development Organisations (DRDO), and the laboratories of the Bureau of Indian Standards (BIS) to offer and provide testing and calibration facilities for the master calibration standards used in the accredited calibration laboratories (CABs) in India.

There is a high number of laboratories in the country to generate accurate and precise measurement values beginning right from the National Physical Laboratory, New Delhi – the NMI of India down to the ERTLs, ETDCs, DRDOs, BIS, NABL-accredited testing and calibration laboratories, and the calibration laboratories at the shop floor level in a large number of industrial and business organizations engaged in the production and trading of products. To generate awareness about the significance, relevance, and validity of the unbroken chain of traceability, the NPL has established a landmark traceability pyramid as a beautiful landmark right inside the campus of the NPL. Figure 2 demonstrates the purpose of establishing the traceability pyramid. Similarly another landmark the wall of SI units has been established at the campus of CSIR-NPL, New Delhi, showing the relationship between the SI units of measurement and fundamental constants (Wikipedia). In fact, the NPL is conducting all these initiatives to fulfill the core mandate given to the NPL to maintain the fundamental and derived units and to further conduct research, design, and development activities to develop new products which shall be relevant and commensurate to the Indian conditions. Such NPL-developed products shall lead India to the Mission of Self-Reliance (Atmanirbhar Bharat) mandated by Mr. Narendra Modi, Honorable Prime Minister of India, to make the nation self-reliant in the production of instruments, equipment, armament, food grains, fruits, and various industrial and commercial products and to further empower Indian industry to export its products to overseas countries which are affordable and command world-class quality. It is extremely important to recognize the rich availability of potentially strong skilled, qualified, young, and brilliant human resource in India. Today no other country in the world has such a large reservoir and demographic dividend of young Indian citizens that can match India. Further this initiative and thrust on self-reliance shall earn foreign exchange revenue and minimize dependence on the imported products.

The NPL has mentioned that the measurement traceability can be obtained through two routes: (i) calibrate the measuring instruments through working standards at the shop floor level, reference standards, and secondary and primary standards to the SI units. In simpler terms, an instrument making, measurements at

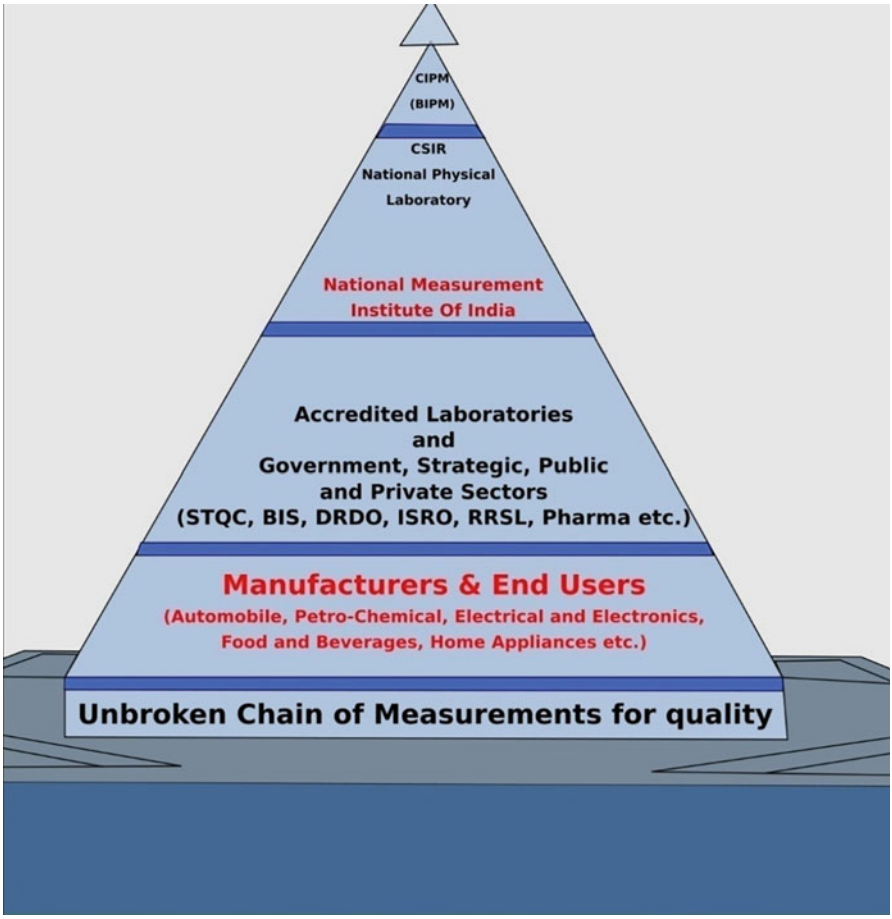

Fig. 2 Traceability pyramid

the shop floor in an industry can ensure the uncertainty in its measurements which are traceable to the SI standards maintained at the international level (Aswal 2020).

Currently the international perception on quality of Indian products is very low as against the other developed nations of the world. This fact is amply proven that the present calibration and measurement capabilities (CMCs) (BIPM) of India are much lower as against the National Metrology Institutes (NMIs) of select developed countries. For example, the CMCs of NMIs of the United States (1889), China (1554), Japan (1126), Germany (1563), the United Kingdom (1137), and India stand at 236. The larger the number of CMCs, the greater is the measurement capabilities of a nation of international equivalence in different areas of science and technology. Thus, India has to go a long way in establishing the credentials of its capacity to produce goods of international standards of quality and reliability which inter alia means that the sanctity of calibration and testing as per ISO 17025:2017 Standard

shall directly determine the quality of our export-worthy products both in terms of revenue and volumes.

Accordingly the NPL has calibrated its response, initiatives, and thrust to fulfill the Atmanirbhar Bharat Mandate. It is extremely important to recognize the rich availability of potentially strong skilled, qualified, young, and brilliant human resource in India. Today no other country in the world has such a large reservoir and demographic dividend of young Indian citizens that can match India. Further this initiative and thrust on self-reliance shall earn foreign exchange revenue and minimize dependence on the imported products.

Today India has already become the exporter of armaments to various countries. India has conventionally been one of the largest importers of armaments from Russia, France, the United States, and the United Kingdom, but this scenario is changing very fast. Now India is registering consistently rising progress in exports of armaments to developing as well as developed nations of the world like Finland, Germany, Sweden, France, Australia, the United States, and South Africa.

Minister of State for Defence, Shripad Naik (2019–2021) in a written reply to Rajya Sabha has mentioned in Parliament that the "cost of the items and quantum of foreign exchange earned varies from company to company."

Indian Ordnance Factories are supplying 5.56 × 45 mm Ball MK N (SS 109) cartridges, hard armor plates, and protective head gears for police and armed personnel to counter enemy in the battlefield and for police to control mob attacks. Of late the light combat aircrafts and the BrahMos missiles are among the highly sophisticated airborne warfare equipment (Fig. 3).

Indian exports to Qatar, Lebanon, Iraq, Ecuador, Uruguay, Japan, and Egypt primarily comprised body protecting equipment, said the Minister in his written response to Parliament.

Under the "make-in-India" flagship program in 2014, the Government of India has been emphasizing the need to increase the production of defense equipment in India and to create and strengthen the country's manufacturing infrastructure to provide more employment and to reduce foreign exchange import bill of India. It is noteworthy that India is the second largest importer of defense equipment next to Saudi Arabia in 2019, according to the Stockholm International Peace Research Institute.

Fig. 3 BrahMos missile and light combat aircraft. www.timesofindia.com

Prime Minister Shri Narendra Modi has told that the export of defense equipment from India stood at ₹2000 crore in 2014. Right from 2018 onward, the Indian export of armaments has gone up to ₹17,000 crore, and by 2024, India aims to hit a target to export of US $5 billion worth of military hardware which is about ₹35,000 crore (Roche 2020).

Calibration and Its Role in Industry

Calibration is an activity performed under the stated environmental conditions and adhering to the relevant ISO standards by a set of trained, skilled, and qualified personnel who are engaged in comparing a device under test (DUT) of an unknown value with a reference standard of a known value. The primary objective of calibration is to assess and estimate the error or determine and verify the accuracy of device under test in terms of measurement value of an unknown value.

Sanctity of Calibration: Case Study Pressure Gauges (PTB 2014)

Here is the example of calibration of a pressure gauge demonstrating how to maintain the sanctity of calibration as per the standard DKD-R-6-1 and obtain precision measurement results with known uncertainties.

This procedure of calibration is followed at an ISO 17025:2017-compliant NABL-Accredited Calibration Laboratory for the calibration of pressure gauges as per DKD-R-6-1 standard. The details provided for the working instructions have been carefully written to take care of every element and protocol of DKD-R-6-1. It will be observed that each and every step starting from the preparing the system/priming the system to start calibrations has been clearly specified. The necessary instructions and environmental conditions have been stipulated. Then, during the process of calibration, whatever requisite care is to be maintained for the application of pressure, concurrently both on the master pressure gauge and the unit under calibration (UUC), is described. At the end of the working instructions, even a tool kit has been mentioned at 2.6 which should be carried for on-site calibration of pressure gauges.

Technical Capabilities and Scope of Work of the Metrology Laboratory

The calibration work done by an accredited laboratory as per NABL Calibration Protocols and as per DKD-R 6-1 Standard.

Procedure

The Metrology Laboratory is well equipped with the standards to cover its full scope of calibrations at metrology laboratory as well as on site.

1. The Instrument for calibration (unit under calibration) is received by stores. Then, it is sent to the quality assurance department through tracer card and proper identification tag. The Quality Assurance Manager registers the case in Equipment Repair and Calibration File.
2. The Quality Assurance Manager sends all communication to customer, related to instrument's calibration, through designated file.
3. As soon as the technical and commercial terms are cleared from customer end, instrument is given to the Technical Manager for calibration. After calibration, the instrument is sent back to Quality Assurance Manager for further actions.

A few examples have been given herein which explain the procedure/method of carrying out the calibrations:

(a) The calibration of the instrument is performed as per calibration procedures.
(b) The UUC is inspected and checked for:
    - Physical damage for pointer/indicating needle, threads, sealing surface, and pressure channel
    - Cleanliness and readability of indications
    - Zero adjustability/null position
    - Operating instruction (if required)
(c) If the unit under calibration (UUC) is not found in order or deviates from specified accuracy and range, then the person performing the calibrations informs the Technical Manager for necessary actions to be taken. The Technical Manager, after verifications, informs the Quality Assurance Manager who in turn informs the customer for required necessary actions/decision required for further process.
(d) When the calibration of the instrument is done, the instrument is sealed and labeled for its performance. The calibration sticker includes the name of the instrument, sr. no. and type no., date of calibration, due date of calibration, reference no., status of the instrument whether within accuracy, or the status report is given. The calibration sticker is pasted on the instrument.
(e) The results of measurements are reflected in the calibration certificate/status report.
(f) Standard environmental conditions:
    Temperature: (25 + 3) °C
    Humidity: (50 + 20) % Rh
    Light intensity: 250 lux to 500 lux
    Sound level: less than 60 dB

Calibration of Digital/Analog Pressure Gauge/Indicator Reference Standard: DKD-R-6-1, Calibration of Pressure Gauges (ISO 2011)

Method Used: Direct Comparison of the Measurement Values. The comparison between the measurement readings of calibration gauge (UUC) and Master Standard Gauge performed by adjusting pressure according to the indication of Master Standard Gauge:

(a) The instrument, which is to be calibrated, is termed as unit under calibration (UUC).
(b) The environmental conditions, i.e., temperature (25 ± 3) °C and relative humidity (50 ± 20) % of the metrology laboratory, are maintained within specified limits before starting the calibrations.
(c) For site calibration, customer is briefed to maintain the required environmental conditions, i.e., temperature (25 ± 3) °C and relative humidity (50 ± 20) % well in advance.
(d) The Master Standards and UUC are kept under standard environmental conditions before starting the calibrations for temperature equalization.
(e) The pressure points for measurement are decided by evenly dividing the range of the UUC strictly as per our scope of calibration. If uncertainty measurement aimed of UUC is less than 0.6% of FS, then nine points for each cycle are to be considered. But if UUC uncertainty is greater than 0.6% FS, then five points for each cycle is to be considered (refer directive Table 1).
(f) Connect the UUC with the port provided on master pressure pump through nozzle/adaptor and connecting line. All connections are made firm and leak-proof. To check the leakage of the system, increase the pressure of the master & UUC up to the higher range; if the value decreases very fastly, then use Teflon tape, Teflon washers, and oil rings.
(g) Ensure that pressure release valve is fully closed and Master Pressure Gauge is switched "on" along with the UUC (if required).
(h) Rotating handlebars, provided for increasing or decreasing the applied pressure, should be at fully out position.
(i) **Preloading**
   - Firstly close the relief valve of the pressure comparator so that the vacuum is not created in the comparator.
   - Increase the pressure up to highest value by rotating handles in clockwise direction of the pressure comparator.
   - Wait for 2 min and then reduce it to zero value by rotating the handle in anticlockwise direction of the pressure comparator. The number of cycles is to be performed as per while observing the time span as mentioned below.

(Refer Number of Measurements in cycles Table 1)

Table 1 Directive table for calibration

S. no.	Measurement uncertainty aimed in % FS	Number cycles of preloading	Number of measurement points with zero		Number of measurement cycles		Load change + waiting time in seconds	Reference figure
			Up	Down	Up	Down		
A	Less than 0.1	3	9	9	2	2	>30	6
B	0.1–0.6	2	9	9	2	1	>30	7
C	Greater than 0.6	1	5	5	1	1	>30	8

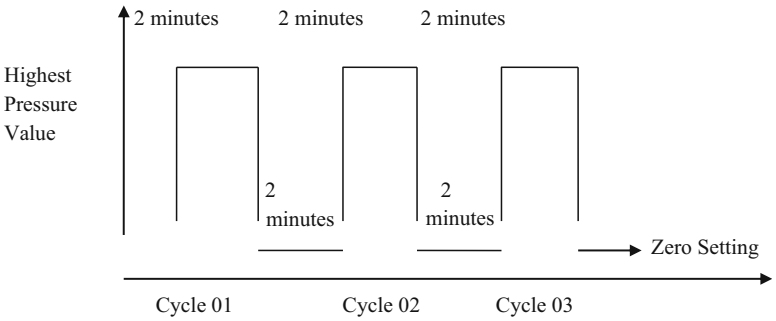

Fig. 4 Preloading sequence

TIME SPAN: The time span for preloading at highest value and time span between two preloadings must be observed at least 2 min (Figs. 4 and 5).

(j) After preloading and after steady state of pressure, indication of UUC is set to zero, and zero reading is noted down for Master Standard Gauge and UUC immediately.
(k) Set the decided pressure (load step) on unit under calibration (in increasing mode) by rotating handlebars clockwise of comparison test pump.
(l) Wait for greater than 30 s and until steady-state/sufficiently stable indication of standard and UUC are reached. As the unit under calibration displayed the set value at same time, note down the Master Standard Gauge (pressure instrument) reading at particular pressure load.
(m) Now set the pressure on unit under calibration for the next decided pressure load by repeating the process from **(Point j) to (Point l)**. This process is followed till the upper limit of the UUC (pressure instrument) is attained.

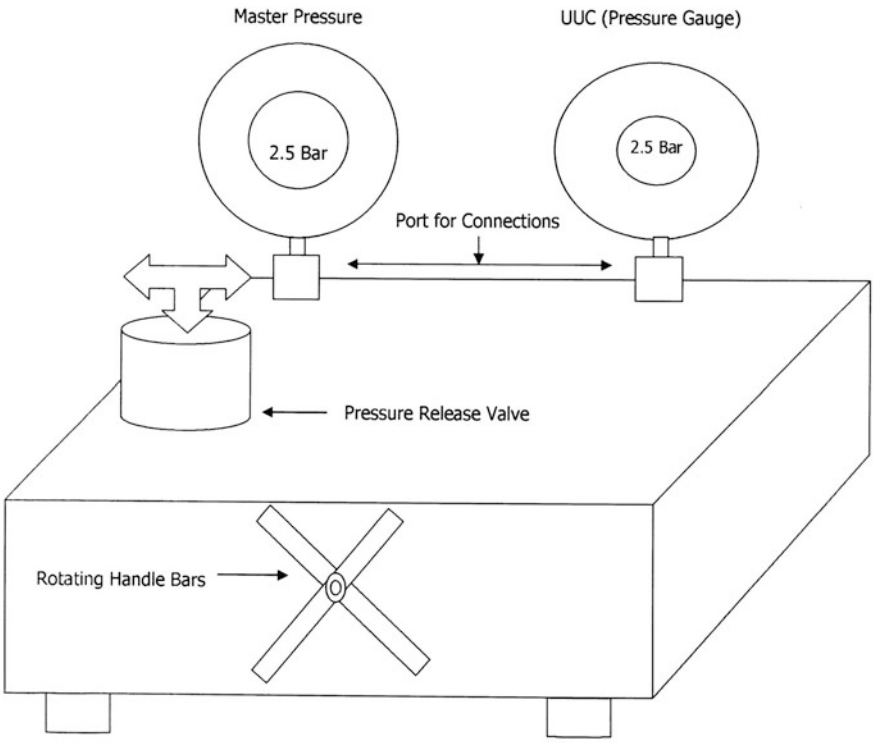

Fig. 5 Comparison test pump for pressure gauge calibration

Note:

(i) The time between two successive pressure load steps should be the same and not less than 30 s. The reading should be made 30 s after the start of the pressure change observed at the earliest.

(ii) While measurement on analog pressure gauge (Bourdon tube type), slightly tap it for minimizing any friction offered to pointer system.

(n) Set the pressure at UUC gauge in nine evenly decided pressure points including zero value **in increasing mode** to the highest range of UUC; at the same time, note down the master pressure gauge value for particular pressure points.

(o) Wait for 2 min at the highest range of UUC; after then repeat the same process in **decreasing mode** (**as per Point m**).

(p) The procedure from **(Point j) to (Point n)** is to be followed for the number of measurement cycles as given in Table 1 for generating data of Master Pressure Gauge and UUC (pressure instrument).

(q) After completion of calibration, switch off the Master Pressure Gauge and UUC (if required), and the accepted/noted down measurement readings are to be transferred in report/certificate format.

Directive Table for Calibration (Figs. 6, 7, 8, and 9)

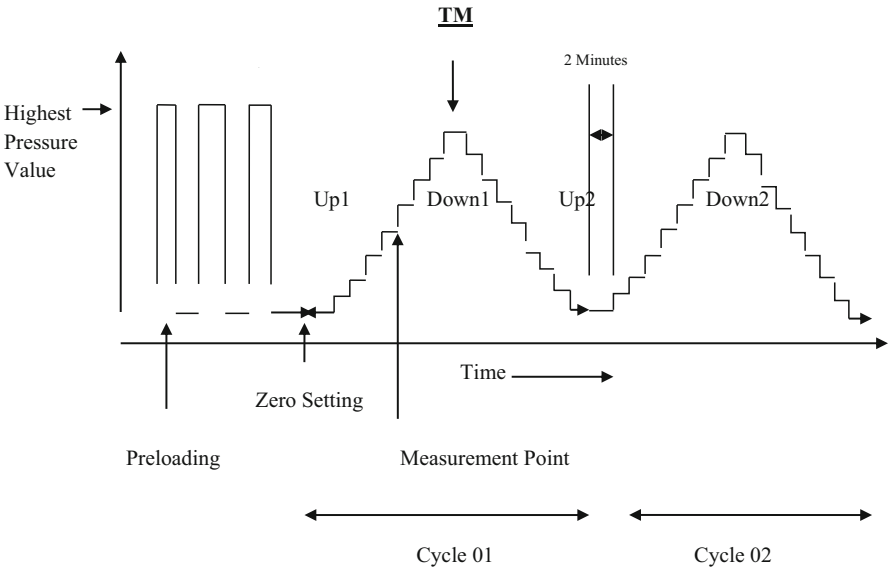

Fig. 6 For calibration as per directive table point no. A

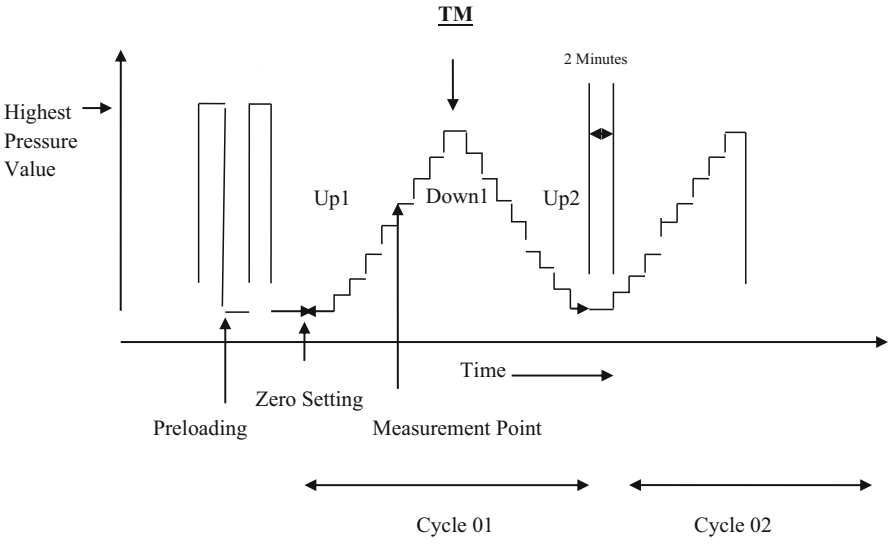

Fig. 7 For calibration as per directive table point no. B

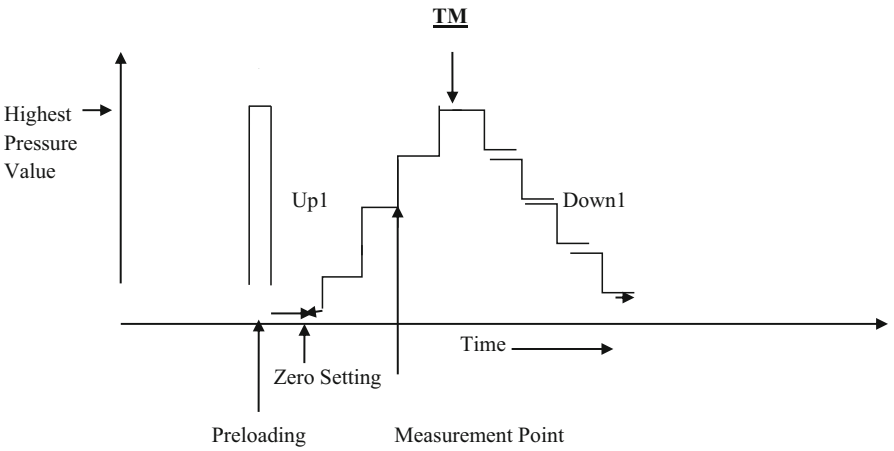

Fig. 8 For calibration as per directive table point no. C

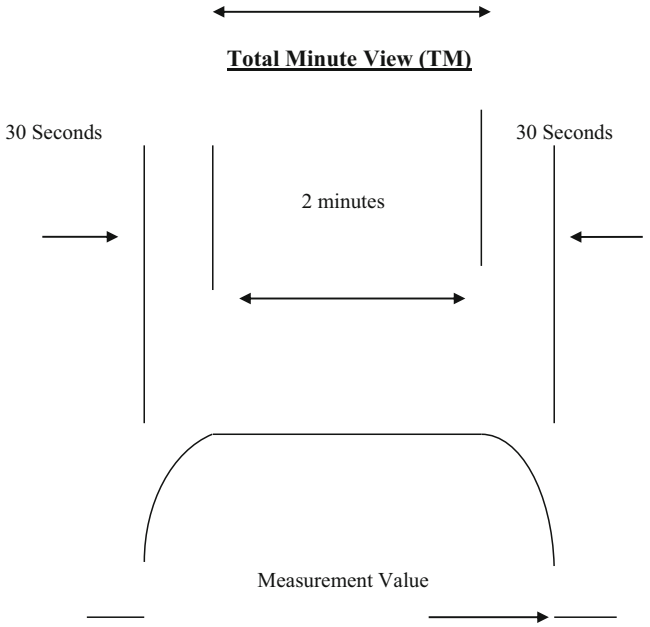

Fig. 9 Total minute view

Unit Conversion Chart for Pressure

bar	kg/cm^2	psi
0.1	0.1019716	1.450377
1	1.019716	14.50377
2.5	2.549	36.259
5	5.099	72.519
10	10.197	145.038
30	30.591	435.113
50	50.986	725.189
60	61.183	870.226
90	91.775	1305.34
100	101.972	1450.377
120	122.3659	1740.453
150	152.957	2175.566
180	183.549	2610.679
200	200.943	2900.755
210	214.140	3045.792
240	244.732	3480.906
250	254.929	3625.943
270	275.323	3916.019
300	305.915	4351.132
450	458.872	6526.696
600	611.829	8702.262

Checklist for Items Required for Calibration

S. no.	Name of the item	Qty	Remarks
1.	Hydraulic pressure comparator with accessories & carry case	01	For LAB + on-site
2.	Pneumatic pressure comparator with accessories & carry case	01	For LAB + on-site
3.	Master pressure gauge with carry case, range – 700 bar (hydraulic)	01	Only for LAB
4.	Pressure gauge with carry case range – 1000 bar (hydraulic)	01	For LAB + on-site
5.	Master pressure gauge with carry case, range – 20 bar (hydraulic)	01	Only for LAB
6.	Pressure gauge with carry case range – 20 bar (hydraulic)	01	For LAB + on-site
7.	Master pressure gauge with carry case, range-70 bar (pneumatic)	01	Only for LAB
8.	Pressure gauge with carry case, range – 70 bar (pneumatic)	01	For LAB + on-site
9.	Set of universal adaptors	Full set	

(continued)

S. no.	Name of the item	Qty	Remarks
10.	Set of spanners	Full set	
11.	Hydraulic oil		
12.	Battery	02	
13.	Tool kit	01	
14.	Digital thermo hygrometer	01	
15.	Stopwatch	01	
16.	Tissue paper	01	
17.	Spirit level	01	
18.	Kit of O-rings & Teflon washers	01	
19.	Teflon tape	01	

Violation of the Sanctity of Calibration and Associated Hazards

An imaginary example of a violation of the sanctity of calibration is reported as follows. The author has no personal intention to malign any entity. Any resemblance to the case of example would only be a coincidence. Suppose a well-known and reputed XYZ Indian company obtained NABL accreditation for the calibration of pressure gauges and temperature gauges to produce precision measuring instruments and calibration standards compliant to ISO 17025:2017.

Necessary human resources and quality infrastructure were created to provide reliable calibrations of the temperature and pressure gauges for on-site calibrations as well. Considering that both temperature and pressure gauges are critical parameters in any industrial applications, more so in oil refineries, oxygen-producing plants, cement, steel, automobile and pharmaceutical industries, hospitals, and other pressure/temperature gauge calibration requirements. Within 6 months of obtaining the accreditation, the XYZ company encountered the following hazardous calibration practices which are rampantly operating in the industrial hubs of Baddi, Paonta Sahib, Faridabad, Ballabgarh, Gurugram, Manesar, and Alwar and the oil refineries, fertilizer plants, and liquefied petroleum gas production plants in Northern India.

1. The host organizations (customers) desired that the calibration certificates be issued against serial numbers of the UUC itself and insisted to place blank calibration certificates along with their company rubber stamps so that they can print the certificates themselves and obtain the signatures of the Technical Manager and Quality Assurance Manager on a specific visit date.
2. Another class of host organizations (customers) refused to remove their pressure and temperature gauges from the pipelines or installations and suggested to just monitor the movement of the indicating needle or digital display and note the serial number of the UUC. It was insisted that the calibration certificates should be issued with serial numbers without carrying any real-time calibrations on the comparison test pump which has to be used to mount the UUC – the Pressure

Gauge and the Master Pressure Gauge of WIKA. Make together and, then, follow the number of cycles of preloading, apply the pressure in ascending mode, and adhere to the procedure specified in DKD-R-6-1 Standard to record the readings against the Master Pressure Gauge and UUC. Such calibrations are "Sham Calibrations." There is, de facto, no calibration of pressure gauges, which are constantly monitoring the pressure and temperature, at various points of the industrial process, inside the oil refineries and other industrial organizations mentioned above.

3. Another class of host organizations insisted to calibrate 100 pressure gauges per day over a period of 8 h with total disregard to the DKD-R-6-1 on a single pressure gauge calibrator machine. By any standards, no metrological processes can calibrate 100 pressure gauges per day with a single pressure calibrator.
4. A large number of accredited laboratories established their branch laboratories are occasionally noticed to issue quick and apparition calibration certificates with "Sham Calibrations."
5. Whenever the gravity of the matter was discussed with the assessors, at the time of recertification audit, they admitted the above-referred rampant malpractices, prevalent in a large number of Indian accredited laboratories toward the calibration of pressure gauges and temperature gauges, but the authority and assessors expressed their deep anguish and helplessness and expressed their inability to undertake any remedial measures.
6. Unable to reconcile and compromise on the sanctity of calibrations, the abovementioned XYZ company decided to surrender the accreditation for the pressure and temperature gauges after maintaining accreditation for 4 years with almost no calibration revenue business simply because the XYZ company did not agree to conduct Sham Calibrations.
7. The responsible authority is not a law enforcement agency in India. They can simply suspend the accreditation of a laboratory based on the report of the assessors during regular audits or surprise audit on such NABL Accredited India Laboratories. Thus, the hazards of poor and apparition calibrations lead to industrial disasters and unfortunate deaths of infant children in the neonatal infant incubators in the hospitals in India and many other underdeveloped and developing countries of the world.
8. The sanctity of calibrations can be maintained if and only if the rigor of calibrations as stipulated in various standards like the ISO 17025:2017 or ISO 376 or Euramet cg-4 or DKD-R-6-1 are adhered to and maintained.
9. In their research, Özdemïrcï et al. (2014) observed serious consequences in terms of mortality of newly born infants in hospitals in Turkey. The researchers observed that 80% of newborn babies in Turkey are put in neonatal incubators because of certain fragile medical complications. The infants born out of premature deliveries are highly sensitive to inside temperature and flow of oxygen inside the incubators. Any erroneous measurements or indicators for temperature and oxygen flowmeter can lead to serious life-threatening consequences. It was revealed in this research study that medical complications among neonatal babies arose because the two incubators in the hospital were

not adhering to the calibration accuracy and standards and thus causing serious hazards to the health of the newborn infants.

10. In their research Rasha M. Abd El-Aziz et al. (2021) observed that care of newborn babies are the essential, most critical, responsible, and integral part of the service in biomedical arena. Neonatal care of infants requires rigorous and committed monitoring of their essential metabolic parameters like oxygen saturation, heart rate, body temperature, and environment of comfort to enable the infant to grow to the normal physical and mental stature. The neonatal infant incubators are meant to nourish the child by providing him the controlled and closed environment as similar as the womb of the mother. But the researchers in this study revealed that the temperature, relative humidity, optimum light, and flow rate of oxygen were not accurately monitored by the measuring instruments mounted outside these incubators babies died in such poorly calibrated incubators. Resultantly the premature delivered babies died either because of overheating, leakage of the gas, or bursting of the incubators.

Sanctity of Calibrations (Case Study – Heavy Metal and Toxic Trace Contents in the Consignment of Basmati Rice Exports)

During the last one decade, the consignments of exports of rice from India to Arab countries, Europe, and the United States have been rejected because there have been traces of heavy metals like lead, chromium, nickel, cadmium, and similar other heavy metals due to improper use of soil irrigation. In a recent study, Tariq et al. (2021) have mentioned in their study and research that in Punjab region of integrated India consisting of existing Pakistan, Punjab, and Haryana in India, rice is the staple food crop that provides a significant amount of nutrition and vitamins. Rice is a staple food for the people of Western Punjab in Pakistan and India and the Northern India and the entire Southern India. It is having high nutritional value. The 100 g of long rice contains 2.2 g of fat (3%), 0.15 g of salt, 130 kcal (27%) of energy, and 2.4 g (20%) of proteins. Rice is characterized by its nature to grow on acidic and alkaline soil. It can also grow in arid areas. The entire West Punjab partially in Pakistan and Sindh province and partially in India known as Punjab and Haryana states are rich centers of rice crops. Most of the people consume rice as their staple food, and the states of Sindh and West Punjab in Pakistan and Punjab and Haryana in India produce bumper harvest of long basmati rice. Both Pakistan and India have been reputed exporters of basmati rice all over the world especially the Arab and the European nations and the United States.

This study revealed that heavy metals with density of 4 g/cm^3 like cadmium (Cd), nickel (Ni), and copper (Cu) are contaminated and aggregated in edible and non-edible parts of plants. In small amounts in the soil, these elements function as micronutrients, but if present, in large amounts they render toxicity to the plant. Now wherever the industrial effluents are discharged in the rivers of Satluj and Beas in the towns of Ludhiana, Jalandhar, and Amritsar in Punjab and on the banks of river Yamuna in Haryana and Delhi, the crops of basmati rice have been

contaminated by the heavy metals. Likewise the environmental pollution due to industrial gases or automobile exhausts and excessive usage of pesticides rendered toxicity and thus found unfit for human consumption.

High metal concentrations can increase the soil's potential to accumulate these elements. Heavy metals have a mobile structure that can be transported from soil to plants. These elements show mobility between root and shoots, and between shoots and grains. Contaminated crops of basmati rice in this belt of Pakistan and India have been found hazardous for human life. Such contaminated crops of basmati rice have been found to cause cancer and kidney, bronchitis, nervous, and orthopedic diseases in people. For example, cadmium leads to kidney and bone density problems. Likewise copper can cause nausea and Wilson's disease.

Contamination of soil by industrial discharge like lead, nickel, cadmium, and stubble burning in India in the states of Punjab and Haryana is rampant and responsible for traces of unacceptable levels of heavy metals and traces of toxic elements in the rice fields of Punjab and Haryana. This has resulted in the rejection of basmati rice from these belts in India.

There are practically no calibration and testing facilities to the farmers and farm fields in this basmati rice bowl of India. Hence, the export of basmati rice from the states of Punjab and Haryana has almost vanished during the last 10 years. There is only one State Food Testing Laboratory in Punjab, and there is hardly any ISO 17025:2017-compliant testing laboratories or NABL-accredited calibrated testing and measuring instruments in the Food Grain Procurement Markets in Northern India.

Conclusion

India has created a well-coordinated quality infrastructure for the production of world-class products that can sell in the international market at affordable and competitive prices. The National Physical Laboratory, New Delhi – the apex National Metrology Institute of India – has mentored an array of institutions like the Quality Council of India, the National Accreditation Board for the Testing and Calibration Laboratories, the Bureau of Indian Standards, the Electronics Test Laboratories, and National Test Houses at the regional and state levels in conjunction with more than 3000 NABL-accredited laboratories which are established within India. It is the solemn responsibility of every NABL-accredited testing and calibration laboratory to generate reliable and usable test and calibration certificates. If the hierarchy of traceability of calibrations is maintained, if the calibration chain is consistent, and if the sanctity of testing and calibration are maintained and adhere to international standards and protocols, India can surely become a trade surplus nation and earn valuable foreign exchange through the export of Indian products.

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Part XIV

Radiation Metrology and National Standards

Radiation Dosimetry

81

Subhalaxmi Mishra and T. Palani Selvam

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Abstract

Radiation dosimetry addresses determination of energy deposited in a given medium quantitatively by either calculations or measurements. The objective of radiation dosimetry is to determine absorbed dose in a medium using a detection device, i.e., a radiation detector. Toward this goal, this chapter presents general principles of radiation dosimetry. Interaction coefficients and quantities related to radiation dosimetry of ionizing radiation are included. The chapter also presents various cavity theories in detail that relate absorbed dose measured in one medium to other. Principles of radiation dose measurements involving various dosimeters are also discussed in this chapter.

Introduction

Ionizing radiation is classified as directly ionizing and indirectly ionizing. Directly ionizing radiations are charged particles such as electrons, positrons, protons, negative π mesons (π^-), alpha particles, and heavy ions such as ^{12}C, ^{14}N, ^{16}O, ^{56}Fe, etc. Indirectly ionizing radiations are neutral particles such as photons (X-rays, gamma rays) and neutrons. Directly ionizing radiation interacts with the orbital electrons of atoms in the medium through direct Coulomb interactions and deposits energy in the medium. Energy deposition of indirectly ionizing radiation in matter involves two steps. *Step 1*: A charged particle is produced in the medium. For example, photon interactions with the medium result in production of electrons or positrons, and neutron interactions produce protons, alpha particles, or heavier ions. *Step 2*: The secondary charged particles produced in *Step 1* deposit energy through collisions with the orbital electrons of the atoms in the medium. Radiation dosimetry addresses determination of energy deposited in a given medium quantitatively by either calculations or measurements. The objective of radiation dosimetry is to determine absorbed dose in a medium using a detection device. Dosimetric quantities provide a physical measure of the region of interest which is correlated to the actual or potential effects of ionizing radiation. Dosimetric measurements or calculations require various specifications of the radiation field at the point of interest.

Definitions of the fundamental radiation quantities and their units are the core mission of the International Commission on Radiation Units and Measurement (ICRU). Several reports on the quantities and units in radiation physics, dosimetry, and radiological protection such as Report 10a, Radiation Quantities and Units (1962), Report 51, Quantities and Units in Radiation Protection Dosimetry (1993), Report 60, Fundamental Quantities and Units for Ionizing Radiation (1998), and Report 85, Fundamental Quantities and Units for Ionizing Radiation (Revised) (2011) are made available by the ICRU. ICRU Report 85 (2011) supersedes the previously published reports related to radiation quantities and

units. The definitions of some of the radiometric, dosimetric, and protection quantities of interest are described in the following sections.

Interaction Coefficients and Related Quantities

Cross Section

The cross section, σ, of a target entity, for a particular interaction produced by incident directly or indirectly ionizing radiation (charged or uncharged particles) of a given type and energy, is defined as (ICRU 2011):

$$\sigma = \frac{N}{\Phi} \tag{1}$$

where N is the mean number of interactions per target entity and Φ is fluence of particles. σ is expressed in m^2.

Linear Attenuation Coefficient

Linear attenuation coefficient (ICRU 2011), μ, is defined as the probability of interaction of S-ray and gamma photons per unit path length dl that they travel in an absorber and can be expressed as follows:

$$\mu = \frac{1}{dl}\frac{dN}{N} \tag{2}$$

where N is the number of photons traversing a distance dl in the absorber and dN is number of photons that undergo interactions in the absorber. μ depends upon the incident energy of photons, atomic number, and mass density of the absorber. μ is expressed in unit of cm^{-1}. Inverse of μ is mean free path τ.

μ is a useful parameter to determine the transmitted intensity of X-ray or gamma photons when they traverse an absorber thickness of x. The differential equation for attenuation can be expressed as below:

$$I = I_0 e^{-\mu x} \tag{3}$$

where I is the transmitted intensity for an absorber of thickness x and I_o is the initial intensity of the radiation incident on the absorber. It can be observed that the transmitted intensity can never be equal zero for any thickness of absorber.

Mass Attenuation Coefficient

As μ is directly proportional to the mass density, ρ, of the absorber material, to eliminate this density dependence another coefficient, mass attenuation coefficient,

μ/ρ, is used. μ/ρ of a material, for uncharged particles of a given type and energy, is defined as (ICRU 2011):

$$\frac{\mu}{\rho} = \frac{1}{\rho}\frac{1}{dl}\frac{dN}{N} \tag{4}$$

where dN/N is the mean fraction of the particles that experience interactions in traversing a distance dl in the material of density ρ. The unit of μ/ρ is cm^2/g.

Mass Energy-Transfer Coefficient

Let R be the incident radiant energy of the uncharged ionizing particles (X-rays, gamma rays, and neutrons) incident on a material of density ρ and dR_{tr} the mean energy that is transferred to kinetic energy of charged particles by interactions of the uncharged particles in traversing a distance dl in the material, and then the mass energy-transfer coefficient, μ_{tr}/ρ, is defined as (ICRU 2011):

$$\frac{\mu_{tr}}{\rho} = \frac{1}{\rho}\frac{1}{dl}\frac{dR_{tr}}{R} \tag{5}$$

μ_{tr}/ρ is expressed in cm^2/g.

Let $h\nu$ be the energy of the photon beams traversing in a material having density ρ and $\overline{E}_{tr}$ is the mean fraction of photon energy transferred per unit thickness of the absorber to the charged particles that appears as kinetic energy. Energy-transfer coefficient, μ_{tr}, and mass energy-transfer coefficient, μ_{tr}/ρ, can be written as follows:

$$\frac{\mu_{tr}}{\rho} = \frac{\mu}{\rho}\frac{E_{tr}}{h\upsilon} \tag{6}$$

$$(\mu_{tr}) = \mu\frac{E_{tr}}{h\upsilon}$$

In the case of neutrons,

$$\frac{\mu_{tr}}{\rho} = \frac{\mu}{\rho}\frac{\Delta E_k}{E_k} \tag{7}$$

where $\frac{\Delta E_k}{E_k}$ is the fraction of energy transferred to the charged particles by the incident neutrons of kinetic energy E_k.

Mass Energy Absorption Coefficient

The secondary charged particles generated by the photons may lose their energy through radiative processes such as bremsstrahlung, in-flight annihilation, and

fluorescence radiations while slowing down in the medium (ICRU 2011). Such loss is not part of energy deposition. Hence, a separate quantity mass energy absorption coefficient μ_{en}/ρ is defined as follows:

$$\frac{\mu_{en}}{\rho} = (1-g)\,\frac{\mu_{tr}}{g} \tag{8}$$

where g is the fractional energy loss through abovementioned radiative processes. g depends on the type of material and energy of photons. g is negligible for low-energy photons but significant for high-energy photons and high atomic number materials.

Mass Stopping Power

The quantity stopping power is used in the context of charged particles. The mass stopping power, S/ρ, of a material, for charged particles of a given type and energy, is defined as (ICRU 2011):

$$\frac{S}{\rho} = \frac{1}{\rho}\,\frac{dE}{dl} \tag{9}$$

where dE is the mean energy lost by the charged particles in traversing a distance dl in the material of density ρ (ICRU 2011):

S/ρ is expressed in MeV-cm^2/g or equivalently J-m^2/kg.
The quantity $S = dE/dl$ is linear stopping power (MeV/cm or J/m).

The charged particles lose energy in the absorber through (a) ionization (includes both soft and hard collisions) and excitation with orbital electrons; (b) emission of bremsstrahlung photons in the fields of nucleus or orbital electrons, in-flight annihilation, and fluorescence radiations; and (c) elastic Coulomb interactions in which recoil energy is imparted to atoms. Hence, S/ρ has three components as below:

$$\frac{S}{\rho} = \frac{1}{\rho}\,\frac{dE}{dl} = \frac{1}{\rho}\left(\frac{dE}{dl}\right)_{el} + \frac{1}{\rho}\left(\frac{dE}{dl}\right)_{\text{rad}} + \frac{1}{\rho}\left(\frac{dE}{dl}\right)_{\text{nuc}} \tag{10}$$

where
$\frac{1}{\rho}\left(\frac{dE}{dl}\right)_{el}$ is mass electron or mass collision stopping power
$\frac{1}{\rho}\left(\frac{dE}{dl}\right)_{\text{rad}}$ is mass radiative collision stopping power
$\frac{1}{\rho}\left(\frac{dE}{dl}\right)_{\text{nuc}}$ is mass nuclear stopping power

Linear Energy Transfer

When charged particles propagate in a medium, energy is transferred to orbital electrons through soft and hard collisions. In soft collisions, the energy transfer is

minimum, whereas in hard collisions energy transfer is maximum. The linear energy transfer (LET) or restricted linear electronic stopping power, L_Δ, of a material, for charged particles of a given type and energy, is defined as (ICRU 2011):

$$L_\Delta = \frac{dE_\Delta}{dl} \tag{11}$$

where dE_Δ is the mean energy lost by the charged particles due to electronic interactions in traversing a distance dl. Here, the mean energy lost excludes the mean sum of the kinetic energies in excess of Δ. If there is no threshold or energy cutoff is imposed, it is called unrestricted stopping power. This is denoted as L_∞. Note that L_∞ is always greater than L_Δ.

Radiation Chemical Yield

The radiation chemical yield, $G(x)$, of an entity, x, is defined as (ICRU 2011):

$$G(x) = \frac{n(x)}{\overline{\varepsilon}} \tag{12}$$

where $n(x)$ is the mean amount of substance of that entity produced, destroyed, or changed in a system by the mean energy imparted, $\overline{\varepsilon}$, to the matter of that system. $G(x)$ is expressed in units of mol/J (ICRU 2011). The mole is the amount of substance of a system that contains as many elementary entities as there are atoms in 0.012 kg of ^{12}C (ICRU 2011). Generally, the value of $\overline{\varepsilon}$ is 100 eV, and therefore the G value is expressed in $(100\ \text{eV})^{-1}$. A G value of 1 $(100\ \text{eV})^{-1}$ corresponds to a radiation chemical yield of approximately 0.1036 μmol/J (ICRU 2011).

Ionization Yield in a Gas

The ionization yield in a gas, Y, is defined as (ICRU 2011):

$$Y = \frac{N}{E} \tag{13}$$

where E is the initial kinetic energy of a charged particle of a given type which is completely dissipated in the gas and $N = \overline{q}_\pm/q$. Here, $\overline{q}_\pm$ is the mean total liberated charge of either sign and q is the elementary charge. Y is expressed in J^{-1} and is sometimes expressed in eV^{-1} (ICRU 2011). From the definition of Y, it may be noted that (a) N includes charge produced by bremsstrahlung or other secondary radiations emitted by the initial and secondary charged particles and (b) charge of the initial charged particle is not included in N, as this charge is not liberated in the energy dissipation process.

Mean Energy Expended in a Gas Per Ion Pair Formed

The mean energy expended in a gas per ion pair formed, W, is reciprocal of Y and is defined as (ICRU 2011):

$$W = \frac{E}{N} \tag{14}$$

where E and N have the usual meaning as mentioned above (section "Ionization Yield in a Gas"). W is expressed in J and is sometimes expressed in eV (ICRU 2011).

Dosimetric Quantities

When radiation interacts with matter through different processes, energy is imparted to the medium. Some of the radiation dosimetric quantities are described below (ICRU 2011).

Exposure (X)

Exposure (X) is related to the ability of photons (X-rays and gamma rays) to ionize air medium; thus, it is defined only for photons and air as the medium. When photon is incident on a mass dm of dry air, it produces an ion pair (electron and positive ion) through ionization process. X is defined as (ICRU 2011):

$$X = \frac{dQ}{dm} \tag{15}$$

where dQ is the absolute value of the mean total charge of the ions of one sign produced when all the electrons and positrons liberated or created by photons incident on a mass dm of dry air are completely stopped in dry air. The unit of X is Coulomb per kilogram (C/kg). Roentgen (R) is the commonly used unit of X. The relation between SI unit of X and R is $1\ R = 2.58 \times 10^{-4}$ C/kg.

Kerma

KERMA (*k*inetic *e*nergy *r*eleased per unit *ma*ss), K, is defined for uncharged ionizing radiations such as photons and neutrons for any medium. It is defined as (ICRU 2011):

$$K = \frac{dE_{tr}}{dm} \tag{16}$$

where dE_{tr} is the mean sum of the initial kinetic energies of all the charged particles liberated in a mass dm of a material by the uncharged particle incident on dm (ICRU 2011). KERMA refers to the kinetic energy of the charged particles such as protons, electrons, and positrons generated from indirectly ionizing radiation such as photons or neutrons. The quantity dE_{tr} includes the kinetic energy of the charged particles emitted in the decay of excited atoms/molecules (e.g., Auger, Coster-Kronig, shake-off electrons) or in nuclear de-excitation or disintegration (ICRU 2011). KERMA is a non-stochastic quantity and is applicable only for indirectly ionizing radiations such as photons and neutrons. The special unit of K is Gray (Gy).

Let E be the energy of uncharged particle and Φ be the fluence; then K is related to Φ as follows:

$$K = \Phi E \left(\frac{\mu_{tr}}{\rho}\right) = \Psi \left(\frac{\mu_{tr}}{\rho}\right) \tag{17}$$

where μ_{tr}/ρ has the usual meaning as defined in section "Mass Energy-Transfer Coefficient" and $\Psi = \Phi E$ is the energy fluence of these uncharged particles. In radiation dosimetry, K is generally expressed in terms of fluence of uncharged particles $\Phi(E)$ as below:

$$K = \int \Phi(E) E \left(\mu_{tr}(E)/\rho\right) dE \tag{18}$$

The total kerma, K, is the summation of two components, i.e., collision kerma, K_{col}, and the radiative kerma, K_{col}, and can be expressed as:

$$K = K_{\text{col}} + K_{\text{rad}} \tag{19}$$

K_{col} is due to energy loss through ionization and K_{rad} is due to energy loss through bremsstrahlung photons.

Energy Deposit

The energy deposit, ε_i (a stochastic quantity, expressed in J or eV), is the energy deposited in a single interaction, i, which is defined as (ICRU 2011):

$$\varepsilon_i = \varepsilon_{\text{in}} - \varepsilon_{\text{out}} + Q \tag{20}$$

where ε_{in} is the energy of the incident ionizing particle, ε_{out} is the sum of the energies of all charged and uncharged ionizing particles leaving the interaction, and Q is the change in the rest energies of the nucleus and of all elementary particles involved in the interaction ($Q > 0$, decrease of rest energy; $Q < 0$, increase of rest energy). Note that (a) ε_{in} and ε_{out} do not include the rest energies and (b) interactions with atomic electrons resulting in atomic excitation do not result in a change in rest energies of the nucleus or of elementary particles and thus $Q = 0$, which is due to the fact that there is subsequent de-excitation process.

Energy Imparted

The energy imparted, ε (a stochastic quantity, expressed in J or eV), to the matter in a given volume is sum of all energy deposits in the volume (ICRU 2011):

$$\varepsilon = \sum_i \varepsilon_i \tag{20}$$

where summation is performed over all the energy deposits, ε_i, in that volume. Note that the term energy deposition event denotes the imparting of energy to matter by correlated particles (e.g., a proton and its secondary electrons, an electron-positron pair, or the primary and secondary particles in nuclear reactions).

Mean Energy Imparted

The mean energy imparted, $\overline{\varepsilon}$ (a non-stochastic quantity, expressed in eV or J), to the matter in a given volume is defined as (ICRU 2020):

$$\overline{\varepsilon} = R_{\text{in}} - R_{\text{out}} + \sum Q \tag{21}$$

where R_{in} is the mean radiant energy of all charged and uncharged ionizing particles that enter the volume, R_{out} is the mean radiant energy of all charged and uncharged ionizing particles that leave the volume, and $\sum Q$ is the mean sum of all changes of the rest energy of nuclei and elementary particles that occur in the volume. Note that $Q > 0$, decrease of rest energy, and $Q < 0$, increase of rest energy.

Absorbed Dose

Absorbed dose, D (non-stochastic quantity), applicable to both indirectly and directly ionizing radiations is defined as (ICRU 2011):

$$D = \frac{d\overline{\varepsilon}}{dm} \tag{22}$$

where $d\overline{\varepsilon}$ is the mean energy imparted by ionizing radiation to matter of mass dm. D is expressed in J/kg^{-1}. D is also expressed in Gray (Gy).

Charged Particle Equilibrium (CPE)

When indirectly ionizing radiations such as photons and neutrons propagate in a medium, they transfer their energy to the charged particles (electrons in the case of photons and protons in the case of neutrons). Due to finite range of these charged

particles, energy deposition does not take place at the same location where these charged particles are generated. In the case of photons, the secondary electrons lose energy through ionization and production of bremsstrahlung photons (known as radiative loss). Collision kerma, K_c, does not include radiative loss as the bremsstrahlung photons escape from the point where they are produced. For photons, K_c at a point in the medium depends on the photon fluence at that point. When a parallel beam of photons is incident on the medium, the photon fluence decreases with depth due to exponential attenuation. Hence, K_c also decreases with depth. In the case of absorbed dose, D, secondary electrons generated at earlier depths will deposit their energy at later depths. Hence, D increases with depth initially and decreases after some depth. This decrease is due to the fact that photon fluence decreases due to exponential attenuation in the medium. Hence, there exists a depth, which is equal to the maximum range of secondary electrons, denoted as x_m, at which D is maximum. At this point, charged particle equilibrium exists where both K_c and D are equal, i.e., $D/K_c = 1$. Beyond x_m, at any depth x, D is slightly larger than K_c because D depends on photon fluence at earlier depth, i.e., $x - x'$, whereas K_c depends on photon fluence at depth x; here, x' represents maximum range of the secondary electron. For depths beyond x_m, the ratio $D/K > 1$. The degree of difference between D and K_c depends on photon energy and the type of medium. The depth dose curve of photons therefore shows $D/K < 1$ in the buildup region (0 to x_m), $D/K = 1$ at x_m, and $D/K > 1$ at depths beyond x_m. At the ^{60}Co photon energies, $x_m = 5$ mm in water. At very low-energy photons such as 25 keV, there will not be a buildup region, and therefore both D and K_c will be same at all depths. This is because range of secondary electrons is too small and they deposit energy at the spot where they are generated. At such low-energy photons and since the medium is of low atomic number, there is no radiative loss, and hence K equals K_c.

General Principles of Absorbed Dose Determination

Cavity Theories

In dosimetry of ionizing radiation, one needs a device that has a sensitive medium to measure response in terms of absorbed dose to the incident ionizing radiation. This device is known as radiation dosimeter. Cavity theory is then applied that relates absorbed dose to the cavity to absorbed dose to the medium of interest. Here, cavity implies that a small mass of material which is sensitive to ionizing radiation embedded in the medium whose elemental composition and mass density are generally different from that of the medium. The cavity may be in the form of solid (diode, TLD, OSL etc.) or gaseous form (ionization chamber filled with air; air here is known as cavity).

Bragg Gray Cavity Theory

It was W. H. Bragg who first discussed the relation between absorbed doses of two different media in 1910 (Bragg 1910). Later, L. H. Gray made quantitative

statements on cavity ionization theory (Gray 1929, 1936). Gray went on to prove that when a gas-filled cavity was small enough with respect to the range of the electrons, the presence of cavity did not perturb the electron fluence spectrum.

Let us consider a cavity (medium c) is embedded in the uniform medium m. Assume the medium is irradiated by an ionizing radiation. According to the Bragg-Gray cavity theory, the cavity should necessarily satisfy the following two conditions (Palani Selvam et al. 2021):

(a) The presence of the cavity should not perturb the fluence of charged particles in the medium. For this purpose, the cavity needs to be small as compared with the range of charged. This implies that the charged particle fluence spectra in both m and c are the same (Attix 1986; Palani Selvam et al. 2021).
(b) The energy deposited in the cavity is assumed to be solely by the charged particles produced outside the cavity and they cross the cavity (Attix 1986). This further implies that no secondary electrons are produced inside the cavity and the electrons do not stop within the cavity (Attix 1986, Palani Selvam et al. 2021).

According to the Bragg-Gray cavity theory:

$$\frac{D_m}{D_c} = \left(\frac{\overline{S}}{\rho}\right)_c^m \tag{23}$$

where D_m is dose to m at a point of measurement without the cavity being present, D_c is dose to cavity, and $\left(\overline{S}/\rho\right)_c^m$ is mean mass collision stopping power ratio of m to c.

$$\left(\frac{\overline{S}}{\rho}\right)_c^m = \frac{\int_0^{E_{\max}} \Phi_m(E)(S/\rho)_m dE}{\int_0^{E_{\max}} \Phi_m(E)(S/\rho)_c dE} \tag{24}$$

where $\Phi_m(E)dE$ is the unperturbed primary fluence of charged particles in medium m with energies between E and $E + dE$ and $(S/\rho)_m$ and $(S/\rho)_c$ are the unrestricted mass collision stopping powers of media m and c (in MeVcm^2/g), respectively. The upper limit of the integrals in Eq. (24) is given by the maximum energy, $E_{\max}$, of the electrons in the fluence spectrum, and the lower limit corresponds to the lowest energy in the spectrum, here indicated by a zero.

When a photon beam is incident on m, it generates charged particles such as electrons and positrons. These charged particles are known as primary charged particles and are part of $\Phi_m(E)dE$. These electrons and positron will further generate secondary electrons through collisions with atomic electrons of the medium. However, the δ-rays produced by these charged particles as a result of hard collisions in

the slowing down of the primary electrons are not part of $\Phi_m(E)dE$. In the case of a charged particle beam incident on m, the δ-rays produced by this beam are again not part of $\Phi_m(E)dE$. Note that whatever may be the incident beam (electron, positron or photon), bremsstrahlung photons can always be generated by the charged particles. Here, bremsstrahlung photons can be those generated by both primary charged particles and δ-rays. Note that $\Phi_m(E)dE$ includes charged particles generated by bremsstrahlung photons that are generated by primary charged particles. In general, irrespective of the incident beam, -rays are not part of whichever the ways δ-rays are produced. Hence, Eq. (23) utilizes unrestricted mass collision stopping powers. This means that the δ-rays are assumed to be stopped on the spot where they are generated. However, the energetic δ-rays may leave the cavity resulting in a partial energy deposition in the cavity. Therefore, the Bragg-Gray cavity theory is insensitive to the cavity size. This is in contradiction to the abovementioned two Bragg-Gray conditions. Although large number of knock-on electrons is created, the Bragg-Gray cavity theory still works reasonably well.

Spencer-Attix Cavity Theory

The Spencer-Attix cavity theory also works on two conditions as defined for the Bragg-Gray theory (see section "Bragg Gray Cavity Theory"). However, these conditions are extended to secondary particle fluence in addition to primary particle fluence. Therefore, it can be inferred that the Spencer-Attix cavity theory (Attix et al. 1958; Attix 1986; Spencer and Attix 1955; ICRU 1984) finite range of δ-rays is accounted as opposed to the Bragg-Gray cavity theory. In the Spencer-Attix cavity theory, cavity size is therefore characterized by a parameter Δ that corresponds to approximately the energy of an electron that is just able to cross the cavity (Palani Selvam et al. 2021). Note that Δ is a parameter which defines the limit between transported and absorbed electrons. Thus, the Spencer-Attix cavity theory is based on formulation of a two-group schematization: (a) *fast group*, δ-rays with initial energy greater than a cutoff Δ are assumed to dissipate their energy outside the cavity and therefore are part of fluence spectrum, and (b) *slow group*, δ-rays of energy less than Δ, which are assumed to have zero range, dissipate their energy "locally," i.e., in the cavity. Under these conditions (Attix 1986; Spencer and Attix 1955; ICRU 1984):

$$\frac{D_m}{D_c} = \left(\frac{\overline{L}_\Delta}{\rho}\right)_c^m \tag{25}$$

where

$$\left(\frac{\overline{L}_\Delta}{\rho}\right)_c^m = \frac{\int_\Delta^{E_{\max}} \Phi_m(E)(L_\Delta(E)/\rho)_m dE + \Phi_m(\Delta)(S(\Delta)/\rho)_m \Delta}{\int_\Delta^{E_{\max}} \Phi_m(E)(L_\Delta(E)/\rho)_c dE + \Phi_m(\Delta)(S(\Delta)/\rho)_c \Delta} \tag{26}$$

Here,

$\Phi_m(E)dE$ is the unperturbed primary fluence of charged particles in medium m with energies between E and $E + dE$, including δ-rays.
$(L_\Delta/\rho)_m$ is the restricted mass collision stopping powers (in $MeVcm^2/g$) at electron energy E for the medium m.
$(L_\Delta/\rho)_c$ is the restricted mass collision stopping powers (in $MeVcm^2/g$) at electron energy E for the cavity c.
$(S(\Delta)/\rho)_m$ is the unrestricted stopping powers for the cavity medium c at $E = \Delta$.
$(S(\Delta)/\rho)_c$ is the unrestricted stopping powers for the cavity medium c at $E = \Delta$.

Note that $(L_\Delta/\rho)_m$ and $(L_\Delta/\rho)_c$ are based on restricted collisional energy losses below an energy threshold Δ.

The terms in Eq. (26), i.e., $\Phi_m(\Delta)(S(\Delta)/\rho)_m\Delta$ and $\Phi_m(\Delta)(S(\Delta)/\rho)_c\Delta$, are the track-end terms which account for energy dissipation of electrons with $E \leq \Delta$ through an evaluation of the restricted stopping powers for $\Delta \leq E \leq 2\Delta$ (Nahum 1978).

In ionization chamber-based dosimetry, is traditionally given by the kinetic energy of the electron whose residual CSDA range in air is equal to the mean chord length of electrons crossing the cavity, L. For a convex cavity and an isotropic field, $L = 4\ V = S$ where V is the cavity volume and S the cavity surface area (Attix 1986). Note that L = 4 V/S applies to a convex cavity that is front-back symmetrical, even if the electron field arrives isotropically only from the front and is absent from the back (Attix 1986).

Re-Examination of Spencer-Attix Cavity Theory

As described by Bouchard (2012), for clinical radiotherapy beams (electron and photon beams), Spencer-Attix cavity theory (Attix et al. 1958; Attix 1986; Spencer and Attix 1955; ICRU 1984) further adapted by Nahum (1978) is considered to be the accepted method for converting absorbed dose in a detector (wall-less) to absorbed dose in the medium of interest. This approach is in use for several years and is widely used. Bouchard (2012) re-examined the Spencer-Attix cavity theory from first principles, using a new general cavity theory derived from radiation transport equations. Bouchard (2012) highlighted the drawbacks of the Spencer-Attix theory as below:

In practice, the ideal conditions needed for the Spencer-Attix theory are not generally met. Also, there is a neglect of physical effects such as electron binding energy, positron transport, etc., hence the introduction of fluence perturbation factors (AAPM 1983; IAEA 1987) to correct for departure from the idealistic conditions. For a standard dosimetry, the departure from the Spencer-Attix theory is corrected by applying fluence perturbation correction, K_{fl}. However, for a nonstandard beam dosimetry such as IMRT (intensity modulated radiation therapy) practice, such corrections are easily predictable (Bouchard et al. 2009). K_{fl} is usually calculated using the Monte Carlo techniques. For example, the quantity of interest in radio-therapy is absorbed dose to water D_w, whereas the ion chamber measures dose to air

D_a. K_{fl} is calculated as below to account for deviation from the Spencer-Attix cavity theory:

$$K_{fl} = \frac{[D_w/D_a]}{[L(\Delta)/\rho]_w/[L(\Delta)/\rho]_a} \tag{27}$$

Limitations of the Spencer-Attix theory as put forth by Bouchard (2012) are given below:

(a) The response of a detector is evaluated using the condensed history-based Monte Carlo methods (Kawrakow 2000). Therefore, one can argue that the theoretical ground on which stopping power ratio calculation relies is arbitrary.
(b) The choice of Δ itself is arbitrary that may result in theoretical and practical inconsistencies. For example, the secondary electrons set in motion close to the wall with energy slightly smaller than Δ such that they can exit the volume are ignored considering that Δ is the kinetic energy of the electrons which is just sufficient to cross the cavity.
(c) Δ defined for the cavity volume is assumed to be the same for the cavity filled with the medium of interest which is always a dense material to which the stopping power ratio is calculated.
(d) The electron impact ionization differential cross sections generally increase with decreasing energy loss (Kim and Rudd 1994), and therefore the statement that Δ should be equal to the energy necessary for an electron to cross the cavity is unfounded. This is because the electrons with kinetic energy less than Δ have on average a kinetic energy much lower than Δ.

Despite the abovementioned shortcomings associated with the Spencer-Attix cavity theory, the validity of this theory (Ma and Nahum 1991; Borg et al. 2000) and the choice of Δ (Borg et al. 2000; La Russa and Rogers 2009; Rogers and Kawrakow 2003; Sempau et al. 2004) have been revised over the years. For example, a study by Wang and Rogers (2007) on Monte Carlo-simulated response of a silicon detector shows unrealistic values of Δ for the purpose of matching the calculated response. La Russa and Rogers (2009) discussed that there is still no proper method to determine Δ for ionization chambers to avoid significant corrections in the air kerma formalism for ^{60}Co beams.

The basic approach adapted by Bouchard (2012) is the same as Spencer-Attix cavity theory in terms of categorization of knock-on electrons, i.e., slow and fast electrons. The Nahum formulation (Nahum 1978) of Spencer-Attix theory (applicable for electrons, positrons, and photons in the energy range of radiation therapy) is shown to be a special case of this model. The model can also be adapted to the class II condensed history Monte Carlo technique. The model demonstrates that Δ is only associated with the error on the solution of the transport equation, rather than the electron kinetic energy necessary to cross the cavity. Numerically, it is shown that there is no prescription for the value of Δ yielding either independence of the beam

energy or a fulfillment of Bragg-Gray conditions. The inconsistencies with Spencer-Attix theory recently reported in the literature are explained, and the overall approach is the use of the Monte Carlo approach to calculate detector dose response. Thus, the Monte Carlo-based approach is no longer reflected by Spencer-Attix theory and that an appropriate formalism could be adopted as a standard theory for radiation dosimetry.

Burlin Cavity Theory

Burlin (1966, 1968) proposed a theory that includes all cavity sizes. The Burlin cavity theory for photons that relates dose to medium (D_m) and dose to cavity (D_c) can be written as follows:

$$\frac{D_m}{D_c} = d(S/\rho)_c^m + (1-d)(\mu_{en}/\rho)_c^m \tag{28}$$

where d is a parameter related to cavity size ($d = 1$ for small cavities, $d = 0$ for large cavities), $(S/\rho)_c^m$ is the mean ratio of the restricted mass stopping powers of medium and cavity, and $(\mu_{en}/\rho)_c^m$ is the mean ratio of the mass-energy absorption coefficients for the medium and cavity. Burlin and Snelling (1969) derived the expression for electron beams from the general cavity theory of ionization for photons as below:

$$\frac{D_m}{D_c} = d\left(1/(S/\rho)_c^m\right) \tag{29}$$

where $(S/\rho)_c^m$ is calculated by Spencer-Attix theory and d is a dimensionless weighting factor which accounts for the decrease of the electron spectrum characteristic of the medium inside the cavity. Almond and McCray (1970) extended Burlin's cavity theory by introducing an additional term as below. This is to include the production of delta rays within the cavity:

$$\frac{D_m}{D_c} = d\left(1/(S/\rho)_c^m\right) + (1-d)(Z/A)_c^m \tag{30}$$

where $(Z/A)_c^m$ is ratio of the electron densities of the cavity to the wall medium.

It is interesting to note that for small cavities, Burlin's cavity theory approaches the Spencer-Attix stopping power ratio and (μ_{en}/ρ) ratio for large size cavities. The main drawback of Burlin cavity theory is that it ignores the scattering effects of secondary electrons that result in large discrepancies in experimental measurements in high atomic number media (Horowitz et al. 1983; Kearsley 1984). Janssens (1983) argues that the modifications to Burlin theory proposed by Horowitz and Dubi (1982) were factually incorrect.

Kearsley Cavity Theory

A new general cavity theory was developed by Kearsley (1984) for photon. The model includes the following components:

(a) The model takes into account scattering of secondary electron at the cavity boundaries.
(b) The model treats the wall contribution to the energy deposited in the cavity as a perturbation of the matched wall cavity case.
(c) The model takes into account the secondary electron scattering at the cavity boundaries using the analysis and experimental data of Dutreix and Bernard (1965).
(d) The model reduces to the Burlin theory (i.e., a linear combination of the mass stopping power ratios and the mass energy absorption coefficient ratio).

The advantages of the model are (i) it is easy to use, (ii) it provides a better fit to available experimental data, and (iii) it can provide dose distribution information inside the cavity. The principal disadvantage of the model is the knowledge on the secondary electron backscatter factors produced by photons. Such factors for high-energy X-rays (11, 15, and 20 MV) and ^{6o}Co gamma rays for a variety of materials have been made available by Dutreix and Bernard (1965).

Haider et al. (1997) mentioned that an outstanding feature of Kearsley model (1984) is the ability to calculate dose distributions inside plane parallel cavities surrounded by medium. In the absence of secondary electron backscattering, the Kearsley theory converges to Burlin's theory. However, the Kearsley theory has poor correlation with experimental results in high-Z media. For example, Kearsley theory has numerous parameters and dose to cavity depends on choice of parameters.

Haider Cavity Theory

Haider et al. (1997) addressed shortcomings of Kearsley theory through a new general cavity theory. In their model, the authors chose X-ray photons (500 kV–20 MV) in which the Compton scattering is the dominant interaction. The electron density is proportional to the energy loss cross section of electron stopping power which is mainly determined by the excitation energy of the molecules and the Compton scatter cross section (Haider et al. 1997). According to Janssens (1983), when the Compton is the predominant interaction and the difference of mean excitation energies per electron is small, then the stopping power ratio is equal to the ratio of electron densities, independent of the cavity size. In the approach, the total electron fluence in the cavity of thickness is divided into three groups (Haider et al. 1997). According to this approach, the electron fluence is as follows:

(a) That originated in the cavity including the backscattered electrons generated by it in the cavity travelling in the opposite direction
(b) That originated in the front wall medium including the backscattered electrons generated by it in the cavity travelling in the opposite direction
(c) In the cavity resulting from the difference in the backscattering coefficients of the cavity medium and the back wall medium

In the theory, medium g (known as cavity medium) of thickness t is sandwiched between medium w and medium m (Haider et al. 1997). The final expression, i.e.,

ratio of dose to cavity (g) and the medium (w) surrounding the cavity, is given as (Haider et al. 1997):

$$f = d_1(\mu_{en}/\rho)_w^g + d_2F_w\left(1 + b_g\right)(S/\rho)_w^g + d_3d_4F_g(\mu_{en}/\rho)_w^g + d_5F_w(S/\rho)_w^g\left(b_m - b_g\right) \quad (31)$$

Here, d_1, d_2, d_3, d_4, and d_5 are weighting factors, F_w is the fractional forward electron fluence originating in the medium w, F_g is the fractional forward electron fluence originating in the medium g, b_g is the probability of backscattering of the medium g, b_m is the backscattering coefficient of the medium m, $(\mu_{en}/\rho)_w^g$ is ratio of mean mass energy absorption coefficient of medium g and medium m, and $(S/\rho)_w^g$ is ratio of mean mass collision stopping power of medium g and medium m.

When the cavity size is very large compared with the range of the electrons, Eq. (31) reduces to:

$$f = (\mu_{en}/\rho)_w^g \quad (32)$$

When the cavity size is very small compared to the range of the electrons and when the font wall and back wall are:

(a) Identical medium (i.e., w), hence, f reduces to the Bragg-Gray theory, i.e., $f = (S/\rho)_w^g$.
(b) Not identical, hence, f does not reduce to the Bragg-Gray theory. This is because the contributions from the front wall medium, the cavity medium, and the backscattering from the back wall medium are calculated separately and all three media could be of different atomic composition (Haider et al. 1997).

Detectors Used in Radiation Dosimetry

The general principles of absorbed dose determinations can be applied when different types of dosimeters are to be used in the determination of absorbed dose. In this section, dosimeters such as calorimeters, chemical dosimeters, and ionization chambers that can be used for the absolute absorbed dose measurements and dosimeters such as liquid ionization chambers, solid-state dosimeters, and films that are generally used for relative absorbed dose measurements are discussed.

Ionization Chambers

Air-filled ionization chambers are used for primary air kerma strength standardization of ^{60}Co beam from the charge collected by the chamber free in air. The charge collected in the air is related to absorbed dose to the medium or air kerma (La Russa and Rogers 2009; Rogers 1992):

$$K_{\text{air}} = \frac{Q_{\text{air}}}{m_{\text{air}}(1-\overline{g}_{\text{air}})}\left(\frac{W}{e}\right)_{\text{air}}\left(\frac{L}{\rho}\right)_{\text{air}}^{\text{wall}}\left(\frac{\mu_{en}}{\rho}\right)_{\text{wall}}^{\text{air}} K_h K_{\text{wall}} K_{\text{an}} K_{\text{comp}} K \tag{33}$$

where $Q_{\text{air}}/m_{\text{air}}$ is the charge measured in humid air per unit mass of dry air that would occupy the cavity at standard temperature and pressure, $(W/e)_{\text{air}}$ is the mean energy deposited in dry air per unit charge of one sign released, $\overline{g}_{\text{air}}$ is the fraction of charged particle kinetic energy lost to radiative processes, $(\mu_{en}/\rho)_{\text{wall}}^{\text{air}}$ is the spectrum-averaged ratio of mass energy absorption coefficients, and $(L/\rho)_{\text{air}}^{wall}$ is the Spencer-Attix ratio of restricted mass collision stopping powers. The correction factors due to humidity is K_h, attenuation and scatter by the wall is K_{wall}, axial nonuniformity of the beam is K_{an}, and material inhomogeneity in the chamber is $K_{\text{comp.}}$

Under certain ideal conditions, K_h, K_{an}, and K_{comp} corrections are unity by definition. Furthermore, the correction for additional nonideal conditions, K, reduces to a fluence correction, denoted here as K_{fl}, to correct for a potential perturbation of the wall's charged particle fluence in the cavity gas.

Under the assumption that photon interactions in the cavity is negligible (which is not true for low-energy photons) and that the departure from Spencer-Attix conditions is solely due to the perturbation of the charged particle fluence by the presence of the cavity:

$$K_{\text{air}} = \frac{D_{\text{air}}}{(1-\overline{g}_{\text{air}})}\left(\frac{L}{\rho}\right)_{\text{air}}^{\text{wall}}\left(\frac{\mu_{en}}{\rho}\right)_{\text{wall}}^{\text{air}} K_{\text{wall}} K_{fl} \tag{34}$$

As $(K_c)_{\text{air}} = K_{\text{air}}(1-\overline{g}_{\text{air}})$, the above equation is rewritten as:

$$(K_c)_{\text{air}} = D_{\text{air}}\left(\frac{L}{\rho}\right)_{\text{air}}^{\text{wall}}\left(\frac{\mu_{en}}{\rho}\right)_{\text{wall}}^{\text{air}} K_{\text{wall}} K_{fl} \tag{35}$$

The values of $(L/\rho)_{\text{air}}^{\text{wall}}$, K_{wall}, and K_{fl} are generally calculated using Monte Carlo methods. Note that these values are chamber-specific.

Calorimetry

Calorimetry method can be applied for absorbed dose measurements in a small volume of an irradiated absorber. In this method, the small volume having mass *dm* medium is thermally isolated. The mean absorbed dose D is given by (ICRU 1954):

$$D = \frac{\Delta\overline{\varepsilon}}{\Delta m} = \frac{\Delta E_h}{\Delta m} + \frac{\Delta E_s}{\Delta m} \tag{36}$$

where $\Delta\overline{\varepsilon}$ is the mean energy imparted to the absorber by the incident ionizing radiation, ΔE_h is the energy appearing as heat, and ΔE_s is the chemical defect, which

may be positive or negative. An example of ΔE_s is the energy produced or absorbed in induced radiochemical reactions. When there is no change of state,

$$\frac{\Delta E_h}{\Delta m} = c_p \Delta T \tag{37}$$

where c_p is specific heat capacity at constant pressured and ΔT is the change in temperature.

Chemical Dosimetry

Chemical dosimetry involves determination of dose by the measurement of chemical changes induced in materials by ionizing radiation. The mean absorbed dose to dosimeter solution, D, using a chemical dosimeter can be determined as follows:

$$D = \frac{\Delta A}{\rho \, l \, \varepsilon_m G} \tag{38}$$

where ΔA is increase in absorbance due to irradiation at a temperature "t" during the spectrometric measurement, ρ is the density of dosimeter solution (g/cm^3), l is optical length in the photometer cell (m), ε_m is the molar linear absorption coefficient of the product (m^2/mol) at the wavelength used for spectrophotometric measurements, and G is radiation chemical yield (in mol/J).

G is the fundamental quantitative characteristic of any chemical transformation induced by ionizing radiation. Earlier G values were reported as the number of molecules of product formed (X), as $G(X)$, or of starting material Y changed $G(-Y)$, per 100 eV energy absorbed. Radiation chemical yield has the same symbol $G(X)$ or $G(-Y)$ as the case may be. $G(X)$ is the molar concentration n(X), produced/changed for a known energy E (in J) is imparted to it:

$$G(X) = n(X)/E \tag{39}$$

G-values in terms of number of species per 100 eV can be converted to SI units by multiplying by 1.036×10^{-7} or in other words 1 mol/J $= 9.649 \times 10^6$ mol/100 eV.

The value of the radiation chemical yield may depend on the type of radiation and the chemical reactions involved. When the radiation-induced reaction is a chain reaction, then the G value is high. For nonchain reactions, it is not large. The value of G can be calculated if the absorbed dose and the concentration of the product being formed (or decomposed) are both known.

$$\mathrm{G}\left(\mathrm{mol\,J^{-1}}\right) = \mathrm{n}\left(\mathrm{mol\,kg^{-1}}\right)/\mathrm{Dose}\left(\mathrm{J\,kg^{-1}}\right)$$

In actual practice, the G value is found from kinetic curves, relating product concentration and absorbed dose. If the relationship is linear, the slope of the curve is

the G value. If the experimental points show deviation from linearity, G is calculated by drawing a tangent to the curve from the point of origin.

Film Dosimetry

Radiographic Film

Radiographic film finds applications in diagnostic radiology, radiotherapy, and radiation protection. The film consists of a base of thin plastic with a radiation-sensitive emulsion, usually silver bromide (AgBr) grains suspended in gelatine, coated uniformly on one or both sides of the base.

When ionizing radiation is incident on the film, it forms a latent image in the film. This latent image becomes visible due to film blackening and is permanent after processing. Hence, light transmission is a function of the film opacity which can be measured in terms of optical density (OD) using densitometers. The OD is defined as:

$$\mathrm{OD} = \log_{10}(I_0/I) \tag{40}$$

where I_0 is the initial light intensity and I is the transmitted intensity. Although films provide qualitative information of absorbed dose, using a calibration curve dose vs OD, dose can be determined. Ideally, the relationship between the dose and OD should be linear, but some emulsions exhibit linear response over a limited dose range, and some are nonlinear. The dose vs OD curve should therefore be established for each film before it is used for dosimetry work.

Radiochromic Film

Radiochromic film (RCF) is a planar dosimeter for applications in radiotherapy and food irradiation. RCF film consists of a radiosensitive (7–23 μm thick) colorless leuco dye bonded to a 100- m-thick Mylar base. These films are colorless before irradiation. Upon irradiation, the sensitive material, dye, is polymerized and turns a deep blue color without physical, chemical, or thermal processing. Thus, RCF is self-developing and requires neither developer nor fixer. As RCF is grainless, it has a high spatial resolution of 1200 lines/mm (McLaughlin et al. 1991). The polymer absorbs light, and the transmission of light through the film can be measured with a suitable densitometer. Response of RCF is stable and is reproducible if protected from UV light, unstable temperatures, and humidity. Since the dye is nearly tissue-equivalent, its deviation from tissue-equivalent energy response is much smaller.

There are various types of radiochromic films such as EBT, EBT2, EBT3, HD-V2, B3, etc. It is important that each film is subjected to different doses and the absorbance is determined using suitable system that has range of a light source (different wavelengths). Suitable light source should be identified based on maximum absorbance through the absorbance vs wavelength. The color intensity of the film is proportional to the radiation absorbed dose. For dosimetry applications involving B3 film, optical density of the exposed films can be measured at 552 nm

using Genesys 20 portable spectrophotometer (Thermo Electron Corp, USA). The color formation in the irradiated B3 film is not complete at the end of the irradiation.

An important aspect in dosimetry involving RCF is the generation of calibration curves for the film and the reading system. For example, if HD-V2 film is intended for dose measurements during irradiation of onion, potato, mango, etc. and the reading system to be used is a spectrophotometer at 550-nm wavelength, then a calibration curve for the film and the reading system should be generated. Similarly, if B3 film is used along with Genesys 20 portable spectrophotometer (552 nm) for measuring doses in the range of kGy, it is required to generate a calibration curve for this dosimetry system.

Dosimetry using RCFs has a few advantages over radiographic films, such as elimination of the need for darkroom facilities, film cassettes or film processing, dose rate independence, better energy response characteristics, and insensitivity to ambient conditions. RCFs are generally less sensitive than radiographic films and are generally useful at higher doses, and nonlinearity should be corrected for in the very high-dose region.

Solid-Sate Detectors

Usage of gas-filled detectors (e.g., ionization chamber) is somewhat limited due to a number of factors. For example, the small number of electron-ion pairs is generated in a gas, and the number of target atoms per unit volume in the gas that the incident radiation encounters is smaller. Solid-state detectors such as semiconductors, scintillation, and thermoluminescent dosimeters have some advantages over gas-filled detectors.

Semiconductor Detector

Semiconductors are most commonly used when best energy resolution is desired. They are called semiconductors because their electrical conduction properties lie between those of insulators and conductors. In semiconductors, the fundamental information carriers are electron-hole pairs, which are collected. Germanium (Ge) and silicon (Si) are two of the most commonly used semiconductor materials. The choice of semiconductor depends on many factors:

(a) Resistivity
(b) Mobility of charges
(c) Drift velocity
(d) Purity
(e) Operating temperature
(f) Cost

Scintillation Detectors

As opposed to gas-filled detectors, the scintillation detector works on a different principle. When ionizing radiation is incident on scintillation material, energy is

transferred to the atoms of the material, which takes the atoms into short-lived excited states. When these excited atoms return to their ground states, they result in emission light photons. These photons are known as scintillation photons. The basic steps involved in detection of these scintillation photons are:

- Transfer of energy to the bound states of the material
- Relaxation of the excited state to the ground state, resulting in the emission of light photons
- Collection of photons by a suitable photodetector
- Detection of the photodetector signal by associated electronics system

The amount of light produced in the scintillator is small. It must be amplified before it can be recorded as a pulse or in any other way. The amplification or multiplication of the scintillator's light is achieved with a device known as the photomultiplier tube (PMT). Its name denotes its function: it accepts a small amount of light, amplifies it many times, and delivers a strong pulse at its output. Amplifications of the order of $l0^6$ are common for many commercial PMTs.

Alpha, beta, gamma, and neutron radiations can be detected using scintillation detectors. Scintillator detectors, which can be solid, liquid, and gas, are generally made of plastic, organics, or inorganic materials. They can be made in any shape and size. NaI(*Tl*) detector is commonly used for gamma detection and gamma spectrometry. The detector is generally shielded to reduce background radiation level before use as it gives high background radiation levels due its high sensitivity.

Thermoluminescent Dosimeters

When ionizing radiation is incident on certain solids doped with appropriate impurities, it creates an electron-hole pair in the crystalline structure. The liberated electron migrates to the conduction band where it further migrates to an electron trap, while the hole left migrates to a hole trap. The charge carriers are maintained for a time that depends on the potential energy of the trap (the trap depth). If the traps are deep, upon heating, the electrons and holes in the shallowest traps are released first. The released charge carriers recombine with their counterparts at a luminescence center or at a trap that is closely coupled to such a luminescence center. As a result of this recombination, there is the release of visible light, which is characteristic of the impurity recombination center. The light emitted is quantified and is related to absorbed dose in the material. The material can be reused after annealing. Annealing is application of appropriate heating to the material. As the methodology involves application of thermal energy to stimulate emission of luminescence, it is known as "thermoluminescence," and the associate dosimetry is called thermoluminescence dosimetry (TLD). Note that pure TL crystal does not exhibit considerable luminescence and therefore is not quantifiable. Addition of trace amounts of impurities (about ppm level) in the crystal lattice enhances its TL properties. Several TL materials with dopants such as LiF:Mg,Ti,, LiF:Mg,Cu,P, $CaF_{2:}$Mn, and $CaSO_4$: Dy are being used in radiation dosimetry.

Properties of Dosimeters

The dosimeters must be characterized before they are used for the intended purposes. The properties of a dosimeter should be characterized by accuracy and precision, linearity, dose or dose rate dependence, absorbed dose and intrinsic energy response, directional dependence, spatial resolution, etc.

Accuracy and Precision

The dosimetric quantity measured through a dosimeter will have an uncertainty. Uncertainty is a parameter that describes the dispersion of the measured values of a quantity. Uncertainty is expressed in terms of accuracy and precision. Precision of dosimetry measurements specifies the reproducibility of the measurements under similar conditions. Therefore, precision can be estimated through repeated measurements. High precision means the standard deviation associated with the measured value is small. The accuracy of dosimetry measurements is the proximity of their expectation value to the "true value" of the measured quantity. Note that measured values are not absolutely accurate and the inaccuracy of a measurement result is characterized as uncertainty.

Linearity

Linearity is a parameter that describes the relationship between the dosimeter response and the dose delivered as a function of dose. Generally, the dosimeter reading is reliable in certain range of doses as beyond a certain dose range a nonlinearity may set in. The dosimeters may also exhibit saturation after certain dose. The linearity range, the nonlinearity behavior, and the saturation behavior depend on the type of dosimeter and its physical characteristics. Hence, it is important to know whether the response of the detector is directly proportional to the dose being measured. In general, one can write:

$$D_{\mathrm{det}}(D) = \alpha k_l \left[M_{\mathrm{det}}(D)\right] M_{\mathrm{det}}(D) \tag{41}$$

where $k_l\,[M_{\mathrm{det}}(D)]$ is the intrinsic linearity whose value is unity for some reference dose D_0 and α is a quantity that relates $M_{\mathrm{det}}(D_0)$ to $D_{\mathrm{det}}(D_0)$ (Rogers 2009). If k_l is independent of D and $M_{\mathrm{det}}(D)$, then the detector's response is said to be linear (Rogers 2009). This means that if the dose is doubled, the response of the detector doubles. However, some detectors may not exhibit this behavior which should be taken into account.

Dose Rate Dependence

The integrating systems that measure the dosimeric quantity should be independent of the dose rate which means that the response of a dosimetry system at two different

dose rates should be constant. However, in practical applications, the dose rate may influence the dosimeter readings. Hence, appropriate corrections must be incorporated.

Let $M_{\mathrm{det}}^{\mathrm{raw}}\left(\dot{D}\right)$ be the raw response of the detector measured at dose rate $\dot{D}$ and then dose rate corrected response of the detector $M'_{\mathrm{det}}\left(\dot{D}\right)$ is:

$$M'_{\mathrm{det}}\left(\dot{D}\right) = k_{dr}\left[M_{\mathrm{det}}^{\mathrm{raw}}\left(\dot{D}\right)\right] M_{\mathrm{det}}^{\mathrm{raw}}\left(\dot{D}\right) \tag{42}$$

where $k_{dr}\left[M_{\mathrm{det}}^{\mathrm{raw}}\left(\dot{D}\right)\right]$ corrects for the dose rate dependence.

Energy Dependence

Energy dependence of radiation detectors is an important property for determination of absorbed dose as the detectors are exposed to a beam quality Q which is often different from the calibration beam. Hence, it is important that such a dependency is corrected for. The overall energy dependence of a detector is composed of two parts, namely, intrinsic energy dependence and absorbed dose energy dependence (DeWerd et al. 2009; Rogers 2009), which are explained below.

Intrinsic Energy Dependence

The intrinsic energy dependence, $k_{bq}(Q)$, relates the reading of the detector to the dose to the sensitive volume of the detector (Rogers 2009). It is defined as:

$$k_{bq}(Q) = D_{\mathrm{det}}(Q)/M_{\mathrm{set}}(Q) \tag{43}$$

where $D_{\mathrm{det}}(Q)$ is the absorbed dose to the sensitive volume of the detector at a given beam quality, Q, and $M_{\mathrm{det}}(Q)$ is the meter reading of the detector at the same Q.

In principle, the intrinsic energy dependence could also be dependent on other factors such as dose, dose rate, temperature and pressure, direction, etc. General-purpose Monte Carlo codes that model radiation transport can be used to calculate the absorbed dose energy dependence. However, such codes cannot be used to calculate the intrinsic energy dependence as the thermoluminescence mechanism is not modeled. Hence, intrinsic energy dependence can be determined only through measurements.

Absorbed Dose Energy Dependence

The absorbed dose energy dependence relates the dose to the detector material to the dose to the medium at the point of measurement of the detector (Rogers 2009). The absorbed dose energy dependence at the beam quality Q, $f(Q)$, is also known as extrinsic energy dependence. It is defined as:

$$D_{\mathrm{med}}(Q) = f(Q) D_{\mathrm{det}}(Q) \tag{44}$$

where $D_{med}(Q)$ is the absorbed dose to the medium at the point of measurement of the detector in the absence of the detector at the beam quality Q and $D_{med}(Q)$ is the absorbed dose to the sensitive material of the detector at the same Q. The factor $f(Q)$ encompasses the component-specific and replacement perturbation effects as well as the absorbed dose energy dependence of the detection material. It is a function of beam quality Q but can also be a function of dose, dose rate, geometry of source and detector location, etc.

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Equipment for Environmental Radioactivity Measurement: Calibration and Traceability

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Manish K. Mishra and A. Vinod Kumar

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Abstract

Radiation level and activity concentration of radioisotopes present in the earth's environment can be detected and measured by specific instruments based on several detection techniques. The environmental radioactivity monitoring and measurement instruments commonly use gaseous ionization chamber (GM Tube), liquid or solid scintillation medium [LSC, NaI(Tl), ZnS(Ag)], semi-conductor (HPGe, PIPS), or their combination as detection methods. These instruments should be capable of measuring trace levels of radionuclides present in the environmental matrices with utmost accuracy. In the last few decades, organizations like ISO, IAEA, and BIPM have taken several measures to bring conformity in measurement through proper certification and QC mechanisms which are recognized worldwide. It is safely assumed (and often demonstrated) by an instrument manufacturer that a new instrument is calibrated as per established norms. It initially behaves according to the specifications; however, its capabilities gradually vary with time due to mechanical deterioration, dust, moisture, chemicals, etc. present in the operating environment. These deviations from standard specifications depend upon the frequency of usage and the severity of the operating conditions and are expected and acceptable within documented limits. A recalibration of the instrument becomes necessary when these limits are breached and the results are no more acceptable. Calibration of environmental radioactivity detection and measurement instruments, using certified standard radioactive sources or certified reference materials, is of paramount importance, as it has a direct impact on the health of humans, environment, and non-human biota. In addition, these sources must be traceable with an unbroken chain of measurements to the SI (BIPM) via international bodies like NIST, ILAC-MRA, or National Metrological Institutes/Laboratories.

Keywords

Energy calibration · Traceability · Radiation · Gamma spectroscopy · Standard source · Certified reference materials

Abbreviations

AFRAC	African accreditation cooperation
ALARA	As low as reasonably achievable
ALMERA	Analytical laboratories for the measurement of environmental radioactivity
APAC	Asia Pacific accreditation cooperation incorporated
ARAC	Arab accreditation cooperation

BARC	Bhabha atomic research centre
BIPM	International bureau of weights and measures
Bq	Becquerel
BRIT	Board of radiation & isotope technology,
CGPM	General conference on weight and measures
CIPM	International committee for weights and measures
CMI	Czech metrology institute, Czech Republic
Cps/cpm	Counts per second/minute
CRM	Certified reference material
Dpm	Disintegration per minute
EA	European cooperation accreditation
EMR	Electromagnetic radiation
ESL	Environmental survey laboratory
eV	Electron volt
FWHM	Full-width at half-maximum
GM	Geiger-Muller
GUM	Guide to the expression of uncertainty in measurement
HPGe	High purity germanium (detector)
IAAC	Inter American accreditation cooperation
IAEA	International atomic energy agency
IAF	International accreditation forum
ICRP	International commission on radiation protection
IEC	International electrotechnical commission
ILAC	International laboratory accreditation cooperation
ISO	International standards organization
keV	Kilo electron volt
LEA	The radioactivity standards laboratory - laboratoire d'Etalons d'Activité), France
LSA	Liquid scintillation analysis
LSC	Liquid scintillation counting
MDL	Minimum detectable limit
MeV	Mega electron volt
MRA	Mutual recognition arrangement
mSv	Milli Sievert
NaI	Sodium iodide
NBR	Natural background radiation
NCRP	National commission on radiation protection and measurements
NIST	National institute of science and technology, USA
NML/NMI	National metrology laboratory/institute
NPL	National physical laboratory, United Kingdom
NPP	Nuclear power plant
PMT	Photo multiplier tube
PSDL	Primary standard dosimetry laboratory
PTB	Physicalish-Technishe- Bundesanstalt, Germany
QC	Quality control

SI	International system of units
SRM	Standard reference material (NIST)
SSDL	Secondary standard dosimetry laboratory
Sv	Sievert
TLD	Thermo-luminescent dosimeter
UKAS	United Kingdom accreditation service
USNRC	United States nuclear regulatory commission
WHO	World Health Organization

Introduction

Our environment is immersed in radiation from space and from within, and it has been so before the existence of human beings or any life form. Radiation, in form of radioactive isotopes and cosmic and associated emissions (particulate and electromagnetic), is present everywhere around us (USNRC 1999). Electromagnetic rays (EMRs), covering the entire EMR spectra from extremely high-frequency cosmic rays to low-frequency microwaves, can technically be called radiation. Nevertheless, the word "radiation," especially in the field of nuclear energy, is used for ionizing radiations such as particulate nuclear emissions (alpha, beta, protons, and neutrons), X-rays, and gamma-rays (WHO 2020). Radioactivity and radioactive emissions cannot be detected by human sensory organs; however, it can be efficiently detected using instruments. Monitoring of the radiation level and the radionuclide concentration in the environment, primarily in air, drinking water, agricultural products, and food, provides a measure of trends in activity levels (Hegde et al. 2009). This permits the evaluation of radioactivity relevant to human exposure as well as in identifying bio-indicators that concentrate radionuclides (ICRP 2012). Table 1 shows some common radionuclides found in the environmental matrices.

The environmental monitoring is of routine and special types. Special type of monitoring includes drills for emergency preparedness and monitoring during nuclear or radiological emergency. Routine environmental monitoring around nuclear power plants is a strict compliance requisite sought by the nuclear regulatory authority of respective country (IAEA 2015). The regulatory bodies through environmental survey laboratories ensure that no harm is caused to the member of public

Table 1 Typical radionuclides in the environment from natural and anthropogenic sources

Samples	Radionuclides
Air	^{131}I, ^{134}Cs, ^{137}Cs
Water	^{3}H, ^{89}Sr, ^{90}Sr, ^{131}I, ^{134}Cs, ^{137}Cs
Milk	^{89}Sr, ^{90}Sr, ^{131}I, ^{134}Cs, ^{137}Cs
Meat	^{134}Cs, ^{137}Cs, ^{90}Sr
Other foodstuffs	^{89}Sr, ^{90}Sr, ^{134}Cs, ^{137}Cs
Vegetation	^{89}Sr, ^{90}Sr, ^{95}Zr, ^{95}Nb, ^{103}Ru, ^{106}Ru, ^{131}I, ^{134}Cs, ^{137}Cs, ^{144}Ce
Soil	^{89}Sr, ^{90}Sr, ^{134}Cs, ^{137}Cs, ^{238}Pu, $^{(239+240)}$ Pu, ^{241}Am, ^{242}Cm

IAEA (1996)

or to the environment. Apart from compliance, these regular monitoring are also carried out to instill confidence in public towards safe use of nuclear energy.

Nuclear installations along with their front- and back-end facilities, fuel reprocessing plants, etc. release radionuclides into the environment in amounts strictly regulated by the national regulatory agencies of respective countries under strict adherence to ALARA (*as low as reasonably achievable*) principle. Some of the radionuclides which are regularly monitored for environmental assessment are tabulated in Table 1 (IAEA 1989a, 1989b; 1996). After Chernobyl and then Fukushima incidents, the regulatory bodies have become more sensitive towards demonstration of capability by the radio-analytical laboratories to effectively measure trace level of radioactivity in the environmental matrices (Beresford et al. 2016; Callen and Homma 2017). Every radionuclide releases a single or a cascade of emissions (alpha, beta, gamma, positron, and sometimes proton and neutron) from its nucleus to transform into a stable element form. These emissions bear specific energies which are signatures of a particular radioisotope. This property is utilized for measurement and quantification purposes by using multitude of equipment. Environmental laboratories monitor the level of radiation and radioactivity concentrations of radioisotopes by using specific detector materials and instruments such as gas-ionization chambers, Geiger-Mueller counter, scintillators, and semiconductor detectors. Since the concentration of radionuclides gets diluted once it enters an environmental matrix, the equipment for their activity measurement should be quite sensitive (on both precision and accuracy parameters) and need to be calibrated regularly within a stipulated time interval to ensure confidence in accuracy and precision.

Environmental Monitoring of Terrestrial, Aquatic, and Atmospheric Radionuclides

The terrestrial, aquatic, and atmospheric matrices of the environment, e.g., soil, surface-water, and air, contain natural as well as anthropogenic radionuclides. Due to atmospheric nuclear tests, carried out in the past, radionuclides like cesium and strontium are now found distributed worldwide (Yoshitome et al. 2003). The naturally occurring radionuclides present like uranium, thorium, and their daughter products are the major contributors of the terrestrial radiation. Natural background radiation (NBR), e.g., cosmic and terrestrial, is a major contributor to the annual absorbed dose to the human beings. International Commission of Radiological Protection (ICRP), through various calculations, has arrived at an annual dose of 1 mSv (1 milli Sievert) which may not lead to a statistically significant rise in excess cancer risk to the people living around nuclear installations (ICRP 2012).

Areas near nuclear power plants (NPPs), nuclear facilities, and uranium mining sites require regular monitoring of background radiation levels as well as radionuclides activity concentration in environmental samples for calculation of annual absorbed dose to the member of the public. It is for demonstration and for regulatory compliance to establish that the effective absorbed dose (from all the pathways) to

the members of public remains minimum and never exceeds the allowed limits of 1 mSv per annum (ICRP 1991, 2012). It is emphasized here that this dose of 1 mSv is over and above the dose received from natural background radiation, i.e., any exposure from cosmic radiation at the surface of the earth, from unmodified concentrations of radionuclides in most raw material, and from ^{40}K in the body (IAEA 1996).

Equipment for Environmental Monitoring of Radiation and Radionuclides

There is no single equipment or detection methodology which can detect all types of ionizing radiations with the same effectiveness. Various types of detection systems are thus available, catering for the type of radiation and measurement requirement specific. Radiation monitoring instruments which are used in environmental radiation monitoring can be broadly classified into four categories (Hegde and Verma 2004; Lawrence Berkeley Laboratory 1980):

1. Portable radiation survey meters (dose rate and nuclide identification).
2. Gas-filled proportional counter (beta and alpha quantification).
3. Scintillation (inorganic, organic and plastic) detectors.
4. Semi-conductor detectors (alpha and gamma spectrometry).

In case of radionuclides having a short half-life (minutes to days), it may be necessary to evaluate the concentrations or the radiation exposure by making measurements in the field or at the place of origin itself, e.g., by use of thermoluminescent dosimeters (TLDs), pressurized ion chambers, or in situ gamma spectrometers (IAEA 1996). However, in cases where a radionuclide with an intermediate half-life (weeks to months) is released continuously or intermittently, sampling and analysis of environmental media in the associated pathway is carried out at intervals not greater than two or three half-lives of the nuclide. In radionuclides which have a relatively long half-life (i.e., years), measurements are to be made at least once per year. In NPP sites where seasonal or other temporal variations are evident, the frequency of sampling should be planned to allow resolution of any important effects caused due to changing seasons. In some cases, sampling on a continuous basis may be necessary (e.g., air sampling and continuous measurement of cumulative external radiation exposure). Composites of some selected sample types such as air-filters may be appropriate for measurement of long-lived radionuclides.

Essentials of Calibration and Metrological Traceability

The primary goal of calibration is to minimize the uncertainty of a specific measurement by ensuring the accuracy of equipment under test. It quantifies and controls errors or uncertainties in the measurement processes to an acceptable level.

Concisely, *calibration of an instrument is a process of determining its accuracy* (BiPM 2012). All kinds of radiation detectors require calibration, especially those which involve some kind of measurement, e.g., survey meters and spectrometry systems. A detector is calibrated to allow the radiation to be interpreted in terms of energy, rather than channel number, or voltage, and amount of radionuclide, rather than number of pulses. Moreover, if computer analysis of the spectrum is examined, it may be necessary to provide information about the peak width variation with energy or channel number. All the instruments available in the market provided by certified suppliers need to go through the full characterization which is different from the routine calibration requirements of the user (NCRP. 1990). All the calibration process mentioned in this chapter fall into category of routine calibration to fulfill specific needs of regulatory compliance, doses measurement, and surveillance.

Considering the kinds of calibration requirements, it can be divided into performance of three calibration tasks (Gilmore 2008):

1. *Efficiency calibration* – the relationship between number of counts and disintegration rate.
2. *Energy calibration* – the relationship between channels and energy.
3. *Peak width calibration* – the variation of peak width with energy (Full width at Half-maximum, FWHM)

However, in some gamma-ray spectrometric systems, the software itself performs energy calibration simultaneously with the FWHM calibration, which means that not only the centroid but also the width of the peak (in terms of the FWHM) is required. This quantity is calculated by a complex formula using the counts in several channel intervals (Isaksson and Raaf 2017).

Instruments which measure counts, count rate, or dose rate (irrespective of individual radionuclide, such as survey meters) require count rate (cpm) and efficiency calibration, while those which involve spectrometric measurement or identification of radionuclides need all the three calibrations.

It is important to emphasize here that a "calibration" is incomplete until it fulfils the essential requirement of establishing the metrological (measurement) traceability of the standard (radiation) sources used in the process of calibration. Metrological traceability is defined as the "property of a measurement result whereby the result can be related to a reference through a *documented* unbroken chain of *calibrations*, each contributing to the *measurement uncertainty*." (BiPM 2012). For traceability to be successfully achieved, we need to look for three important aspects: the documentation (generally in terms of available and verifiable reports), calibrations (certifications of measurements against physical standards), and measurement uncertainty (all the possible sources of uncertainty both statistical and otherwise, significantly affecting the results). While measurement traceability alone cannot guarantee valid results, but without it, system calibrations are uncertain and results can often be unreliable (Karam 2014). The extent of globalization of markets and availability of online databases for instant verification of a certificate has led to more consumer awareness with regards to results which should be acceptable worldwide. The

measurement traceability, nowadays, is not only required for regulatory compliance, but it also provides a foundation to support the national and international market place.

During the recent Covid pandemic, we have seen the evolution of international acceptance of Covid (+/−) results through molecular tests (e.g., reverse transcriptase-polymerase chain reaction, RT-PCR) or antigen tests at airports as a mandatory requirement for entry to a country or to a state. Such internationally acceptable results can only be provided by the laboratories which are accredited by a national or international body through mutual recognition arrangement (MRAs) by participating countries. The International Laboratory Accreditation Cooperation (ILAC) is one such body which has recognized a consistent policy on measurement traceability as crucial for confidence in calibration, testing, and inspection by accredited laboratories and others covered by the ILAC mutual recognition arrangement, called ILAC MRA (ILAC 2022a). The role of ILAC has been discussed in the last section of the chapter.

National Metrology Institutions (NMIs), such a NIST (National Institute of Standards and Technology, under US Department of Commerce) of United States of America, around the world provide necessary fundamental metrological infrastructure supporting measurement traceability for a variety of applications including those of radiological instruments. NMIs are generally nonregulatory and are established to facilitate the development of various national standards for traceability from research through development to implementation. Through participation in international measurement comparisons with each other, NMIs, along with the International Bureau of Weights and Measures (BIPM), make possible worldwide uniformity of measurements and their traceability within stated uncertainties (which are derived from the measurement uncertainties of all the participants). Estimation and proper accounting uncertainties are absolutely essential for the determination of confidence in a measurement. The Guide to the Expression of Uncertainty in Measurement (commonly known as GUM) is used by laboratories, manufacturers, factories, and NMIs to demonstrate quality control, develop and maintain physical reference standards and materials, and calibrate those standards and associated instrumentation (ISO/IEC 2010).

Radiation Survey Meters for Background Radiation and Contamination Detection

Portable radiation survey meters are the most commonly used instruments for the measurement of background radiation levels, area survey or for detection of radioactive contamination. It comes in several shapes and sizes as per the requirement. Survey meters can be used to detect particulate emissions like alpha, beta, and neutron as well as gamma radiation. These instruments provide readings in form of counts per second (cps) and dose rate (nSv/h, mSv/h, μSv/h). Advance survey meters use inorganic crystals like sodium iodide (NaI), lanthanum bromide (LaBr), cadmium zinc telluride (CZT), etc., which are also capable of spectroscopy to

distinguish radionuclides based on their gamma-ray energies. It not only fulfills the basic need of a radiation survey meter but also serves as a portable gamma spectrometry system to track and identify radionuclides. For calibration of the survey instruments, ^{137}Cs and ^{60}Co are used.

Calibration of GM Tube-Based Radiation Survey Meters

Geiger-Mueller tube (GM Tube)-based radiation survey meters mostly come factory calibrated and must be regularly calibrated, at least annually, to an accuracy of ±20%, which is acceptable for general survey purposes. Before calibration, the operational checks such as battery check, radiation response, and zero check must be performed. Routine calibration measurements should be made using either a radioactive check source with known count rate (or dose-rate) or a calibrated radiation field. It is done to ensure that the reproducibility of the reference instrument is within ±2% of the certificated value. In case of standard sources, due corrections should be applied for the radioactive decay and for changes in air density from the reference conditions, if applicable.

Low-cost analogue or digital survey meters which do not have provision for spectrometric identification of radionuclides are calibrated for count rate, dose rate, and/or efficiency only. Regular (non-spectroscopic) survey meters can be calibrated by using the following:

1. A standard source with known counts (cps) or dose rate.
2. A radiation survey meter which has been previously calibrated.
3. Using both in conjunction, i.e., a source with known cps and a calibrated survey meter (IAEA 2000)

The most common way is to use a fixed source-to-detector distance variable-dose rate method. Dose rate from a point source can be calculated by using inverse square law:

$$\text{Dose rate} \propto \frac{1}{(distance)^2}$$

The percent efficiency of an instrument which expresses its results in terms of count rates can be calculated from the following formula:

$$Efficiency\ (in\%) = \frac{Observed\ count\ rate - Background}{Disintegration\ Rate\ (Standard\ Source)} \times 100$$

For example, if a ^{14}C standard source has a disintegration rate (decay corrected) of 90,000, disintegration per minute (dpm), say the GM counter under test, gives a value of 50 counts per minute (cpm) for the background and 6000 cpm when the source is placed. The efficiency of the counter can be calculated as

$$Efficiency\ (\%) = \frac{6000\ cpm - 50\ cpm}{90000\ dpm} \times 100$$

$$= 6.6\%$$

The same procedure can be repeated for instruments which provide reading in dose-rate (μSv/h or nSv/h) format. The survey meter is then exposed at computed distances, and actual dose-rate reading is recorded. From the reading, the calibration factor (CF) is calculated.

$$CF = \frac{Theoretical\ Reading}{Observed\ Value}$$

The above process is repeated several times (10–15 times) choosing different distances and with different dose rates, and an average CF is taken. A CF of 0.8–1.2 is considered satisfactory (Nadeera Hemamali et al. 2015).

Calibration of Advance Radiation Survey Meters

Advance radiation survey meters with spectroscopy features (radionuclide identification) are usually calibrated automatically, every time it is switched "ON," with a securely sealed source (internal: ^{137}Cs and, external: ^{40}K, intrinsic: ^{138}La) provided with the instrument or with one of the methods given in the respective section depending upon the type of detector used. In most cases, these advance survey meters are not user serviceable, and if the inbuilt calibration file is lost or reset, it must be sent to a certified calibration laboratory or to the supplier for re-calibration.

Gas Flow Beta Counter Having Anti-Coincidence Technique

It is a modification of gas proportional counter where two GM counters of either planar or spherical shape, viz., main counter or guard counter, are used. In a continuous gas-flow counting system, any air contamination from previous samples is swept out of the detector by continuously flowing the fill gas through the active volume. The counters have an aluminum-coated mylar foil window of about 1.0 mg/cm^2 through which the beta particles traverse and reach the gas chamber (argon with isopropyl alcohol vapors) and ionize the medium. The signals from both counters are collected through the anode wire. By using anticoincidence technique and proper shielding, the background of such systems can be reduced to about 1–2 cpm.

Calibration of Gas Flow Beta Counter Having Anti-Coincidence Technique

The efficiency of the counter is independent of incident beta particle once the particle enters the chamber. However, due to absorption of energy of incident particle by the window material, the efficiencies get affected for different energies. Nuclides that decay directly to the ground state by emitting (one or more) betas are called pure beta-emitters, and these can be effectively used for calibrating a gas flow beta counter. Strontium-90 along with daughter product yittrium-90 (^{90}Sr/^{90}Y) is such a source, which emits a continuous spectrum of betas from 0.546 to 2.27 MeV. The half-life of ^{90}Sr is 27.7 y and that of ^{90}Y is 64 hours (about 2 and a half days). The efficiency for ^{90}Y is about 45% and approximately 40% for ^{40}K, and for ^{137}Cs and ^{90}Sr, it is about 35%. The efficiency is also influenced by the thickness of sample. The MDL (3σ) of low background beta counting system is 0.023 Bq for 3600 s (about 1 hour) counting duration.

Geiger-Mueller Counter

A Geiger-Mueller detector is a gaseous ionization detection system used for the detection and measurement of alpha, beta, and gamma radiation. It is sometimes called as Geiger counter or simply a GM counter. It consists of a cylindrical gas-filled tube (usually helium or argon) with electrodes, across which a high-voltage (~900–1500 V) is applied, with no flowing current. When ionizing radiation passes through the chamber causing the ionization of the gas, it leads to an intense electrical pulse which is measured or counted. At higher voltages multiple pulsing can happen which can be reduced by addition of some quench gas (halogens, Cl_2 or Br_2) in the tube. The number of pulses per second is the measure of the intensity (flux) of the radiation.

Some of the advantages of Geiger-Mueller counters are the following:

1. Relatively inexpensive.
2. Durable and portable.
3. It can detect all types of radiation with suitable modifications.

GM Counter for Measurement of Beta Activity

Beta counting systems utilize a medium-sized sealed normal type or low background type of end window Geiger-Muller tube as detector. Argon as counting gas and halogens or organic gas as quenching gases is used in these tubes. Beta counting system can be assembled by utilizing nuclear instrumentation modules like EHT (extra high tension or high voltage power supply) unit, paralysis unit, timer/scalar, and the power supply bin. Standalone systems are also available which contain electronic circuitry to provide required functions of the above units.

Calibration of Geiger Mueller Counter for Beta Activity

These detectors are most widely used in the field of radiation detection, survey, and monitoring. Certified sources from SSDLs or other recognized commercial supplier can be used for calibration purpose. A sufficiently reliable calibration source can be prepared in the laboratory by the following procedure. Analytical grade potassium chloride crystals (KCl) are powdered after drying at 110 °C for 1 hour and uniformly spread and fixed with gelatin or collodion over an aluminum planchet. The size of the planchet is carefully chosen to match the detector and window size. Activity of around 2 Bq of ^{40}K is sufficient to give significant count rate. Natural potassium contains about 0.012% of radioactive ^{40}K. Higher strengths will increase the thickness of standard source causing self-absorption. The efficiency of the detector is mostly independent of energy in GM mode of operation, but attenuation due to sample thickness needs to be corrected. Gas-filled GM counters normally give 15–20% efficiency and about 0.4 cps (counts per second) background. The specific activity of KCl (potassium chloride) is 16 Bq/g.

Liquid Scintillation Detector

Liquid scintillation counting (LSC) or liquid scintillation analysis (LSA) system is a standard laboratory method to quantify the activity concentration of low-energy radioisotopes, mostly beta-emitting and alpha-emitting isotopes in liquid media (Kleinschmidt 2004). The sensitive LSC detection method requires specific cocktails, supplied with different trade names, to absorb the energy and convert it into detectable light pulses (L'Annunziata 2020). Liquid scintillators are widely applied to count radioactive material that can be dissolved as part of the scintillator solution. It is used for detection of low-energy beta-emitters like ^{3}H and ^{14}C, ^{35}S, ^{32}P, and ^{125}I (Table 2). It is predominantly important for regular and critical evaluation of tritium (^{3}H) and ^{14}C in the environment.

Liquid scintillation (LS) solutions, called as cocktails, consist of four components, where the fourth is an optional component used in specific conditions:

1. Solvent: The large part of the solution is an alkylated aromatic organic solvent xylene, toluene, etc.

Table 2 Some common radionuclides which are detected by liquid scintillation counting (LSC)

Isotope	Emission Type	Energy (MeV)	Half-life ($t_{1/2}$)
Tritium (^{3}H)	β	0.019	12.3 y
Carbon (^{14}C)	β	0.156	5730 y
Sulfur (^{35}S)	β	0.167	87.2 d
Phosphorus (^{32}P)	β	1.710	14.3 d
Phosphorus (^{33}P)	β	0.249	25.3 d
Iodine (^{125}I)	γ	0.178	60 d

2. Primary scintillator: It absorbs energy from the solvent and emits light. PPO (2,5-diphenyloxazole) is most commonly used scintillator.
3. Secondary scintillator: It is added to absorb emissions of the primary solute and to re-emit photons of longer wavelength [e.g., POPOP (1,4-di-(2–5-phenyloxazole) benzene), Bis-MSB (p-bis-(omethylstyryl) benzene)].
4. Other Additives used in specific cases: LS solutions frequently contain additives, like surfactants, to improve counting efficiency.

Orbital electrons of the scintillators absorb the incoming energy and become excited and by transitioning to the ground state produce a light pulse which is characteristic for specific radionuclides and the scintillator. The light is then detected by the photomultiplier tube (PMT) attached to the liquid scintillation counter. As high as 100% efficiency could be obtained for ^{14}C using this scintillator medium. Organic scintillator solutions are routinely used for beta counting and beta spectrometry of low-energy beta-emitters.

Calibration of Liquid Scintillation Counter/Analyzer (LSC/LSA)

Proper assessment of tritium (^{3}H), ^{14}C, and other beta-emitters in various environmental samples requires accurate and traceable calibration of the LSC. LSCs can be calibrated by secondary standards which are commercially available and are traceable to NIST, NPL (UK), or any other National Metrology laboratory. The standard sources come in sets of quenched or unquenched LSC standards.

The optimum counting conditions for an LSC system can be done by establishing different counting windows by adjusting the gain and discriminator setting. Based on statistics, an optimum window setting is given by the term "Figure of Merit," the larger the figure of merit, the more significant the sample measurement:

$$Figure\ of\ Merit = \frac{S^2}{S + 2B}$$

where S = sample counts and B = background counts.

A major part of calibration of LSC systems involves the Quench correction, which is one of the key issues in LSC systems. It occurs when the energy transmitted by the radioisotope is not caught by the PMT entirely. We encounter three types of quenching in LSC systems which are physical, chemical and color based. The color of the sample and the cocktail have a huge effect on the quenching process, and it varies as the combination varies.

Calibration of an LSC system also involves the optimization of the quenching process. The quench correction methods commonly used are as follows:

1. Internal standardization.
2. Sample channels ratio (SCR).
3. External standard ratio (ESR).

Quenching effects can be reduced by optimizing sample preparation, choosing the correct energy window, and optimizing the counting windows. Some LSCs are equipped with radioactive sources (external standards) such as ^{137}Cs, ^{133}Ba, and ^{152}Eu for quench corrections. ^{152}Eu is a mixed gamma source of broad spectrum of energies falling between 1 keV to 1.4 MeV (Adelman 1998).

Gamma Spectrometry System

Most of the radionuclides in the environment produce gamma-rays of various energies and intensities. A detailed analysis of this spectrum is used to determine the identity and quantity of gamma-emitters present in the sample. Gamma spectrometry systems are one of the essential instruments for quantitative analysis of naturally occurring radioactive materials (like ^{210}Pb, ^{228,226}Ra, ^{238}U, ^{234,228}Th, ^{40}K, etc.) and anthropogenic radionuclides (like ^{137}Cs, ^{134}Cs, ^{131}I). The gamma spectrum is characteristic of the gamma-emitting nuclides contained in the source. The equipment used in gamma spectrometry contains an energy-sensitive radiation detector, a pulse sorter (multichannel analyzer), associated amplifiers, and data readout devices backed by suitable software.

Environmental laboratories commonly use thallium-doped sodium iodide (NaI) scintillation detector and high purity germanium detectors (HPGe) for estimation of radionuclides present in extremely low concentrations. Gamma-ray (γ-ray) spectrometry is a rapid and nondestructive analytical (NDA) technique that can be used to detect various radioactive isotopes in a sample with minimal efforts. In a gamma-ray spectrometry system, the energy of incident gamma-rays from the sample is measured by a detector. The energy range for a typical gamma spectrometry system lies usually in 10 keV to 3 MeV.

Prerequisites for Calibration of Gamma Spectrometry Systems

Apart from the type of detector, both scintillator and semi-conductor-based gamma spectrometry systems share almost common calibration methods. However, for the sake of clarity, both kind of systems have been discussed separately with regards to their calibration procedures.

The γ-ray spectrum in both the system can be modeled assuming the following:

1. A point-like gamma-source is measured, i.e., gamma-ray self-attenuation is negligibly small.
2. The detector pulse pile-up effect due to both true and random coincidence summing is negligible.
3. The losses of counts due to the spectrometer dead time are negligible.
4. A multi-channel analyzer with absolute linearity is used.

Table 3 Typical standard sources of choice and gamma-ray energy peaks used in energy calibration of gamma spectrometry systems

Source	Energy (keV)	Source	Energy (keV)
^{210}Pb	46.5	^{192}Ir	468.1
^{241}Am	59.5	^{22}Na	511.0
^{109}Cd	88.0	Annihilation	511.0
^{182}Ta	100.1	^{207}Bi	569.7
^{57}Co	122.04	^{208}Tl	583.1
^{144}Ce	133.5	^{192}Ir	604.4
^{57}Co	136.5	^{192}Ir	612.4
^{141}Ce	145.4	^{137}Cs	661.6
^{182}Ta	152.4	^{54}Mn	834.8
^{139}Ce	165.8	^{88}Y	898.0
^{182}Ta	179.4	^{207}Bi	1063.7
^{182}Ta	222.1	^{60}Co	1173.2
^{212}Pb	238.6	^{22}Na	1274.5
^{203}Hg	279.2	^{60}Co	1332.5
^{192}Ir	295.9	^{140}La	1596.2
^{192}Ir	308.4	^{124}Sb	1691.0
^{192}Ir	316.5	^{88}Y	1836.1
^{131}I	364.5	^{208}Tl	2614.7
^{198}Au	411.8	^{24}Na	2754.1

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As stated earlier, the efficiency calibration must be performed with sources that are traceable to a national standards laboratory, such as NIST, NPL, PTB, or any of the recognized NMLs. A mixed gamma-ray standard for both energy and efficiency calibration containing ^{57}Co, ^{60}Co, ^{137}Cs, ^{109}Cd, ^{85}Sr, ^{139}Ce, ^{203}Hg, ^{113}Sn, ^{88}Y, and ^{241}Am is available from many commercial companies which provide NIST traceable sources. The gamma-ray energies of this mixed standard as well as some other commercially available NIST traceable radionuclides that are suitable for efficiency calibration (and energy and shape calibration) are shown in Table 3. An example list of standard sources from authorized suppliers for an environmental survey laboratory is provided in Table 4.

Energy Resolution of Gamma-Ray Spectrometry Systems

FWHM (full width at half-maximum) denotes the resolution capability of an instrument. It is important to perform resolution calibration, also called FWHM calibration, which establishes a function to describe the peak width at a specific spectral energy. The lower the width at half-maxima of the energy peak, the finer the resolution of the instruments. It is an important parameter which signifies an instrument's ability to resolve different photon emissions in a narrow energy range. Majority of environmental laboratories worldwide use ^{137}Cs and ^{60}Co as standard sources for energy and FWHM calibration.

Table 4 An example list of standard sources from an environmental survey laboratory in BARC, India, for calibration of instruments used for detection and measurement of radionuclides and their traceability to certified laboratories

Radionuclide	Issuing agency	Strength	Unit	Standard traceable to
H-3	PERKIN ELMER	450	Bq/ml	NIST (SRM4947C)
Eu-152	Amersham	3.7	kBq	Primary standard supplied by Amersham PLC.UK and traceable to NIST (ANSIN42.22.1995)
Sr-Y 90	Amersham	74	kBq	
Co-60	Amersham	370	kBq	
Cs-137	Amersham	370	kBq	
Sr-85 + 89	Amersham	370	kBq	
Ba-133	Amersham	370	kBq	
Mn-54	Amersham	370	kBq	
Eu-152	CMI	4.03	MBq	Primary standard
Cs-137	CMI, BRIT	4.062	MBq	Primary/secondary standard
Co-60	CMI, BRIT	4.051	MBq	Primary/secondary standard
Am-241	Orano LEA	4.0	MBq	Primary standard

NIST National Institute of Standards and Technology (USA); *BRIT* Board of Radiation & Isotope Technology, BARC, India; *CMI* Czech Metrology Institute, Czech Republic (NMI of Czech Republic); *LEA* The Radioactivity Standards Laboratory - Laboratoire d'Etalons d'Activité), France; *BARC* Bhabha Atomic Research Centre

For an ideal detector and supporting pulse processing electronics, the spectrum of 662-keV gamma-rays from a ^{137}Cs radioactive source would exhibit a peak in the spectrum whose width is determined only by the natural variation in the gamma-ray energy. The major source of this peak broadening is the number of photo-electrons emanated from the photocathode for a 662-keV gamma-ray. Hence, FWHM for a specific photopeak is defined as

$$\%Resolution\ (FWHM) = \frac{\partial E}{E} \times 100 \approx \frac{k \times 100\%}{\sqrt{\mathrm{E}}}$$

or, alternatively,

$$\%Resolution\ (FWHM) = \frac{\partial C}{E} \times Fe \times 100$$

where:

E = Energy of the peak (keV) of standard source (^{137}Cs or ^{60}Co).
∂E = FWHM of the peak in energy units.
k = Proportionality constant characteristic of the particular detector.
∂C = Width in channels at the half height of the maximum peak counts.
Fe = Energy calibration factor (keV/ch).

Efficiency Calibration of Gamma-Ray Spectrometry Systems

It is important to emphasize here that efficiency cited by the manufacturer of a Ge detector following the IEEE standards (ANSI/IEEE Standard 325–1986) is known as *relative efficiency* and it is not the *absolute efficiency* of the detector. It only represents the ratio of the absolute detector efficiency (of HPGe system) to that of a 3 × 3 inch NaI (Tl) scintillation detector, the point source being placed at 25 cm (about 9.84 in) on the axis of the endcap (measured from the center of the source to the front of the endcap) at 1332.5 keV (^{60}Co). The absolute efficiency at 1.3 MeV of 3 × 3 inch NaI (Tl) scintillation detector is measured at the same distance of 25 cm is 1.2×10^{-3} (L'Annunziata 2020).

Precise calibration of efficiency is essential for quantitative estimation of radionuclides in the environmental samples. It is important to check certain things before calibration:

1. Sample to detector geometry.
2. Calibration sources: As far as possible, certified standard sources of the radionuclide to be estimated should be used for geometries typically used.
3. Efficiency is expressed as radionuclide counting efficiency considering total disintegration rate or gamma energy efficiency by applying the gamma abundance factor (number of gammas per disintegration) of the source used.

It is common to perform a monthly calibration for determining counting efficiencies by measuring a known quantity of radionuclide(s) of interest within a similar matrix and with the same source-detector configuration as the sources requiring analysis. This is done to eliminate any effect that might be caused by differences in standard and sample characteristics, e.g., density, moisture content, shape, and size. Modern software allows saving multiple efficiency calibration curves for different matrices and source-detector configurations for later use.

Gamma Spectrometry System with NaI (Tl) Detector

Sodium iodide (thallium) is the most commonly used detector for routine analysis due to several advantages which include a high light yield, high density, reasonable cost, and availability of detector crystal (NaI) in large sizes. A sodium-iodide scintillation detector is composed of a sodium-iodide (NaI) crystal doped with thallium as scintillator, a photomultiplier tube (PMT), and a data acquisition system. The PMT, which amplifies low level light input from the detector, is attached with a multichannel analyzer (MCA) that converts the overall pulse into a digital signal, corresponding to a given acquisition channel. Sodium iodide detectors have an edge over HPGe in terms of efficiency. The best resolution achievable ranges from 7.5% to 8.5% for the 662 keV gamma ray from ^{137}Cs for a 3" × 3" optimized crystal and is slightly worse for smaller and larger sizes. NaI (TI), just like common salt NaCl, is hygroscopic, and its crystal will deteriorate if exposed to the atmosphere for a brief

time. The NaI (Tl) crystals must therefore be "canned" in an airtight container for use as detector (Knoll 2010).

NaI (Tl) typically converts about 11% of the incident gamma energy into photons with an average energy of 3.0 eV. A 1000 keV gamma will produce, on average, 4.3×10^4 photons.

Energy Calibration of Sodium Iodide Gamma Spectrometry Systems

Energy calibration of NaI (Tl) detector is performed by using sources like ^{137}Cs (662 keV gamma) and ^{60}Co (two gammas of 1.173 and 1.332 MeV). For lower energies, ^{192}Ir and ^{108m}Ag can be used (Hegde and Verma 2004). Sources should be chosen in such a way that they have long half-lives and are monoenergetic or have multiple energies with wide separation. Initially, after setting up of the spectrometer, a standard ^{137}Cs source is placed, and spectrum is acquired. The peak position due to ^{137}Cs source is noted. Gain of the linear amplifier is either increased or decreased obtaining the peak at desired position. For example, if a 10 keV/channel calibration is required, the peak should be positioned at 66th channel for 662 keV gamma line of ^{137}Cs. In an equivalent way, a different standard source (e.g., ^{60}Co) of known energies is taken for calibration.

A linear equation of $y = mx + c$ *or* $E = m(ch) + c$ can be easily fitted for energy versus channel. The equation can later be used to find the energy of unknown peak channel. The energy calibration is performed manually (by picking energy peaks manually) or automatically by the software, by measuring one (^{152}Eu) or several radioactive standard sources with a minimum of three different energy peaks usually covering a wide spectral range between 50 keV and 2500 keV.

Efficiency Calibration of NaI (Tl) Detectors

It is to be noted that efficiency calibration is always done for a specific geometry in which the samples will be counted. Efficiency calibration should be carried out monthly or quarterly, at least. Long lived, certified, and traceable standard gamma sources like ^{137}Cs, ^{60}Co, ^{192}Ir, ^{108m}Ag, and ^{54}Mn can be used to calibrate the system for various gamma energies. A plot of energy versus efficiency is to be drawn for a particular geometry which can be used for other energies for which a standard source may not be available. Certified reference standards are available from IAEA, NPL (UK), CMI, NIST, and other recognized suppliers. The efficiency in percentage can be determined by the following equation presented in its simplest form:

$$Efficiency\ (Ef)\% = N \times \frac{100}{dps} \times \frac{100}{A}$$

where:

Ef = Efficiency of the system for particular gamma energy for a specific geometry.

N = Net cps under the photopeak area after subtracting the background in cps.

dps = Disintegration per second, activity of the standard (Bq).

A = Percent abundance of that particular energy out of total disintegration.

High Purity Germanium (HPGe) Gamma Spectrometry Systems

HPGe gamma-ray spectrometry systems are used to quantify trace levels of gamma emitting radionuclides in the environmental samples. It employs minimal sample preparation and is non-destructive in nature. High Purity Germanium (HPGe) has a relatively high resolution as compared to Sodium Iodide systems. HPGe systems are also widely used for identification of radionuclides formed in Neutron Activation Analysis (NAA).

The most common type of germanium gamma-ray spectrometer consists of a high-purity (mildly *p*-type) crystal fitted with electrodes in a coaxial configuration. The depletion region in germanium can be made several centimeters thick which increases the *resolution*. Due to smaller band gap in germanium (as compared to silicon), thermally generated charge carriers, even at extremely low temperatures, is a potential problem. As a result, virtually all germanium detectors are cooled to liquid-nitrogen temperature (77 Kelvin) during their use.

Apart from factory calibration which involves calibration of all the functional parts of HPGe system including the detector, regular routine calibrations are done with mixed gamma-ray sources in a similar matrix and with the same geometric form as the sample to be analyzed. This requires the purchase of several calibration sources of different energies and geometries. It is now possible to order commercial calibration sources of single or mixed gamma-ray emitters in a matrix of known chemical composition and density in user-supplied containers. Calibrations based upon these sources can then be adjusted to correct for any differences in composition and density between the calibration source and the test source (EPA and FDA 2004).

Energy Calibration of HPGe Type Gamma Spectrometry Systems

Energy calibration of High Purity Germanium (semiconductor) detectors is carried out using a solid or sealed liquid source of mixed radionuclides kept at a standardized distance from detector. The sources commonly used are ^{133}Ba, ^{137}Cs and ^{60}Co. However, mixed standard sources with nearby gamma energies, and sometimes multiple gamma-emitting standard source like ^{152}Eu can also be used. The gain of spectroscopy amplifier or built-in amplifier of analyzer is adjusted to get required energy calibration of 0.2 keV/channel to 2 keV/channel. For example, 662 keV gamma peak of ^{137}Cs should fall on 662nd channel if calibration of 1 keV/ch is required. The peak position can be seen by moving the cursor along the spectrum. Linearity of the system must be checked by plotting a graph of energy against corresponding channel number with respective peaks of ^{133}Ba, ^{137}Cs and ^{60}Co (Fig. 1).

Compared to NaI (Tl), there is a factor of 30 enhancement in the full width at half-maximum (FWHM) resolution. Thus, many nuclear energy peaks which cannot be resolved separately with NaI (Tl) detectors are easily resolved using HPGe detectors. By convention, HPGe detectors are usually characterized by the FWHM in keV at 1332 keV gamma of ^{60}Co. As a rule of thumb, two gamma energies separated by less than 2 resolution widths (FWHM) are usually not resolved fully. FWHM in keV can be calculated by:

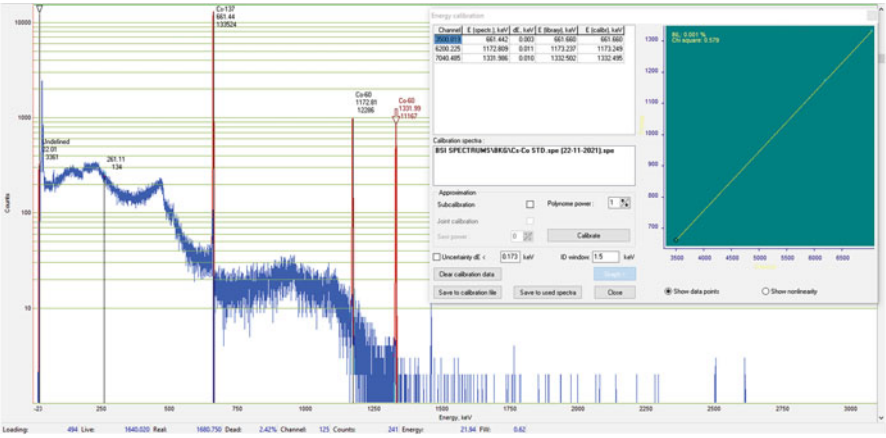

Fig. 1 Energy Calibration of a typical coaxial p-type HPGe system with a relative efficiency of 50%

$$FWHM(keV) = \frac{Nr \times 159}{X_2 - X_1}$$

Nr = Width at half maximum in no. of channels.
X_2 = Position of 1332 peak channel.
X_1 = Position of 1173 peak channel.
159 keV is the difference between 1173 and 1332 keV gamma peaks of ^{60}Co.

Efficiency Calibration of HPGe Systems

The efficiency of HPGe system for a particular geometry and container used is determined. For example, if 20 ml bottle of cylindrical geometry is used for sample counting, standard also must be taken in 20 ml bottle of the same geometry. It is important to keep the sample in the same configuration with respect to detector. Counts are acquired for a minimum of 300 s (i.e., 5 minutes) for efficiency calculation at a particular energy.

$$Efficiency\ (Ef)\% = N \times \frac{100}{dps} \times \frac{100}{A}$$

Ef = Efficiency of the system for particular gamma energy.
N = Net cps under the photopeak area after subtracting the background in cps.
dps = Disintegration per second, activity of the standard (Bq).
A = Percent abundance of that particular energy out of total disintegration.

For efficiency calibration purpose of the detector Standard (Certified) source of ^{137}Cs or ^{60}Co source typically 1 μCi, with reference date (e.g., 22.12.2010) is used. Let us take an example of ^{60}Co which is traceable to NIST (Fig. 2). The absolute efficiency of the germanium detector for 1332.5 keV photons is measured with

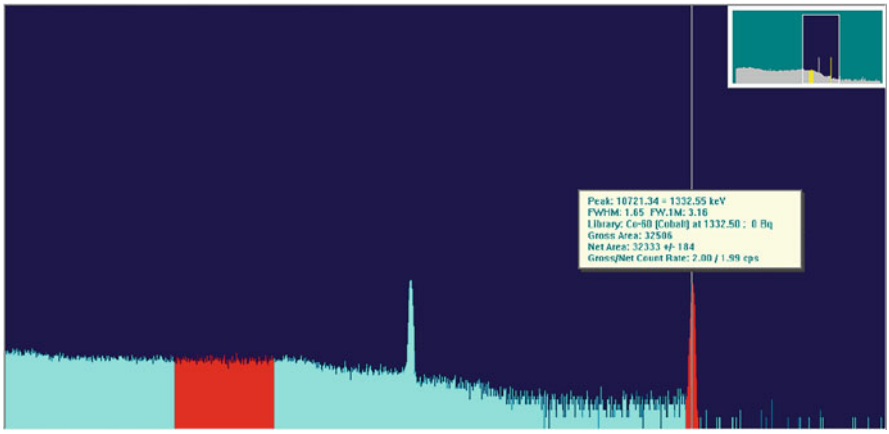

Fig. 2 Efficiency calibration of a HPGe Gamma spectrometry system

appropriate dead time corrections. It is given as the ratio of the total counts in the 1332.5 keV peak to the total number of disintegrations during the elapsed live time.

The half-life ($t_{1/2}$) of ^{60}Co can be taken from international nuclide databases like IAEA Live Chart, DDEP, LBNL etc. (Chu et al. 2022; IAEA and DDEP 2022; LBNL 2022) which is 5.271 years. The activity of ^{60}Co on the counting date (e.g., 22.04.2022) can be found by using the equation:

$$A_t = A_0 \times e^{-\lambda t}$$

$$A_0 = 1\mu Ci = 37000\ Bq$$

$$(Decay\ Constant)\lambda = \frac{0.693}{Half\ \ life}$$

$$= {}^{0.693}\!/_{5.271\ y}$$

$$= 0.1314\ y^{-1}$$

t = elapsed time = 4139 days = 4139/365.25 = 11.33 y.

The above inputs can now be put into the equation

$$A_t = A_0 \times e^{-\lambda t}$$

$$= 37000 \times e^{-0.1314 \times 11.33}$$

$$A_t \approx 8348\ \ \text{Bq}$$

Peak area at 1332.5 keV = 32,506 counts.
Acquisition Time = 16226.60 seconds (Live Time).
Total counts from the source in t seconds = $t \times 8348 \Rightarrow 16226.6 \times 8348$

$$\Rightarrow Absolute\ Efficiency = \frac{Total\ counts\ recorded\ in\ the\ detector}{Total\ counts\ from\ standard\ source}$$

$$\Rightarrow \frac{32506}{16226.6 \times 8348}$$

$$\Rightarrow 2.39 \times 10^{-4}$$

We can get the relative efficiency of the HPGe detector with that of NaI (Tl) which has an absolute efficiency of 1.2×10^{-3} by simple calculation. For example, in above case it will be ~20%.

Similarly, efficiencies can be calculated for different gamma energies and geometries using sources of ^{133}Ba, ^{60}Co, ^{134}Cs, ^{54}Mn, etc. A graph is drawn taking logarithm of energy (keV) in X-axis and logarithm of efficiency in Y-axis.

Calibration of Gamma-ray Spectrometry Systems for Environmental Samples (Geometry)

For environmental or low-activity samples (0.01 to 1 Bq/g), typically, 300 to 500 g of prepared solid samples are used. If a fixed volume is used, the mass will vary according to the density. High-density samples may cause significant self-absorption of low-energy gamma-rays and degrade the higher energy gammas. Therefore, it is important to calibrate the detector with standards of the same geometry, composition, and density, or use appropriate attenuation and geometry correction algorithms. The photopeak efficiency calibration for solid materials, like soil and sediments, is done by using certified reference materials (CRMs) released by the IAEA, coded as RGU-1, RGTh-1, and RGK-1 for uranium, thorium (and their daughter products), and potassium (Fig. 3). The specific activities of the CRMs are certified at 95% confidence level and are equal to 4940 ± 30 Bq/kg for ^{238}U (RGU-1), 3250 ± 90 Bq/kg for ^{232}Th (RGTh-1) (both in secular equilibrium), and 14,000 ± 400 Bq/kg for ^{40}K (RGK-1).

Measurement of Alpha-Emitting Radionuclides and Standard Sources for Calibration

Alpha (α) particles are positively charged and highly massive compared to other nuclear emissions (beta, positron, and neutron). As a result of high coulombic interaction with nearby atoms, alpha particles lose all their energy over a noticeably short distance and typically traverse only to few centimeters in air and have extremely limited penetration into matter. Alpha particles tend to lose all their energy

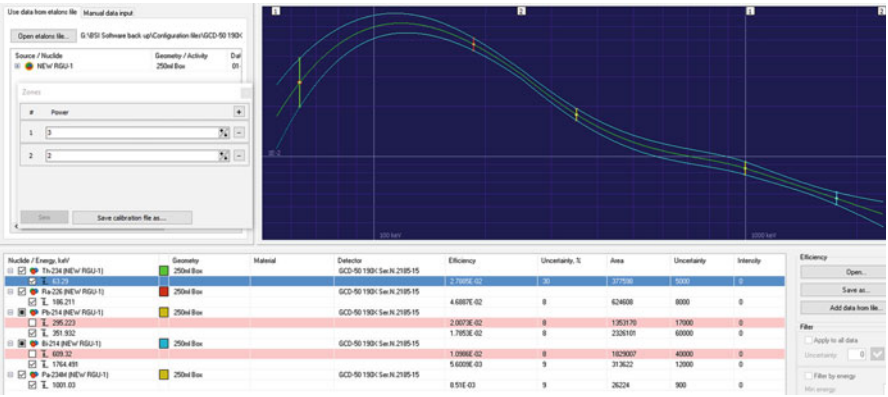

Fig. 3 Efficiency calibration of HPGe gamma spectrometry system for a 250 g cylindrical box containing certified reference material (CRM) from IAEA

characterized by an intense high rate of energy loss while passing through matter. Most energetic alphas seldom penetrate below the dead layer of human skin.

The alpha detection systems, most widely used, are ionization chambers, gaseous proportional counters, solid-state (silicon) semiconductor detectors/spectrometers, and scintillation counters (plastic, ZnS phosphor-PMT combination, or LSC). The associated electronic components in almost all the cases comprise of high-voltage power supplies, preamplifiers, amplifiers, pulse discriminators, scalers, and output devices. ADC (analogue to digital converter) and MCA are additional components when alpha spectroscopy is done. Alpha counters usually have exceptionally low background and high efficiency (10 to 100%).

Alpha calibration standards are available from NIST, NPL, or a commercial vendor (complying with national standards bodies) that supplies NIST or NPL traceable sources. Among the radionuclides available are ^{230}Th, ^{241}Am, ^{235}U, ^{239}Pu, ^{228}Th, ^{238}U, and ^{226}Ra. Other radionuclides are also available.

Alpha Scintillation Counting by ZnS (Ag) System

Most of the inorganic scintillators comprise of translucent single crystals. Some, such as silver-activated zinc sulfide, are efficient scintillators, but growing it in the form of optical-quality large crystals is not yet possible. Silver-doped zinc-sulfide [ZnS (Ag)] systems are commonly used for gross-alpha counting on a thin base, *viz.,* a metal planchet or filter paper. It does not differentiate between alpha energies; hence a gross count of all alpha emitters present is delivered. It is most commonly used for surface contamination detection and gross alpha count of radionuclides present in water samples on a planchet (dried under IR lamps).

Alpha probe consists of a scintillator detector made up of a thin layer of silver-activated zinc sulfide [ZnS (Ag)] crystal and a high-performance PMT. The density

of the crystal is about 10 mg/cm^2. The PMT and crystal are placed in a light tight steel casing to have extremely low background of the order of about 1.0×10^{-3} cps at an efficiency of about 30%. The operating voltage is usually less than 1500 V enabling the use of simple scalers.

Calibration of ZnS (Ag) Counting System

An electroplated thin alpha-emitting source with known cps (counts per second), usually ^{239}Pu, is used for calibration. The background is taken for a sufficient time (in general the background counts come around 0.002 cps). A ZnS (Ag) alpha detection system can have an efficiency of 45 to 50 percent for an electrodeposited calibration source. The efficiency, to a significant extent, is influenced by the thickness of sample. MDA (3) for unit volume of sample is 0.0075 Bq for a counting time of 5000 s (about 1 and a half hours).

Counter Efficiency:

$$S\%(\text{Efficiency}\%) = N \times \frac{100}{dps}$$

where:

S% = Efficiency of the counter in percentage.

N = Net cps after subtracting the background in cps.

dps = Disintegration per second, activity of the standard (Bq).

Alpha Spectrometry System

Alpha spectrometry is used to identify and determine the concentration of various alpha-emitting radionuclides in environmental samples such as ^{238}Pu, $^{239\,+\,240}$Pu, ^{241}Am, ^{242}Cm, and ^{237}Np. Samples from various matrices are subjected to radiochemical separation of respective radionuclides, and the separated sample is electrodeposited on a stainless steel planchet to count in alpha spectrometer.

An alpha spectrometry system consists of either SSB (silicon surface barrier) detector or PIPS (passivated implanted planar silicon) detector of suitable size (generally 300–450 mm^2 area) in a vacuum system (0.01 mm of Hg). SSB detectors are found in old systems, while modern age systems contain PIPS detector which has more advanced feature than the former (Table 5). The system is calibrated with ^{239}Pu, ^{242}Pu, and ^{241}Am standard sources in the detector chamber at a standard distance.

The range of alpha particles is only a few centimeters (cm) in air and much less in solids. It is, therefore, necessary to have a thin source and to keep the source and detector in vacuum to obtain maximum energy resolution and high efficiency. The background of the detector amplifier system should not be more than 1 count per 10^4 seconds in the energy range of 2 to 10 MeV. Environmental samples are normally counted for 50,000 − 80,000 seconds or longer depending upon the activity levels.

Table 5 Some operating parameters of alpha spectrometry systems available worldwide

Active area (mm^2)	300	450	600	900	1200
Thickness (mm min/max)	150/315	150/315	150/315	150/315	150/315
Recommended bias (V)	+20/80	+20/80	+20/80	+20/80	+20/80
Si resistivity (min U-cm)	2000	2000	2000	2000	2000
Operating temperature (min/max)	−20/+40	−20/+40	−20/+40	−20/+40	−20/+40
Leakage current (at 25 °C) (typical/max in nA)	15/70	25/100	30/120	40/200	60/300
Alpha resolution (keV)	17/19	18/20	23/25	25/30	30/37
Absolute efficiency (%)					
At 2 mm spacing	36.8	40.2	41.0	43.6	44.5
At 5 mm spacing	23.5	28.3	31.2	34.8	36.9
At 15 mm spacing	7.3	10.1	12.4	16.1	18.9

L'Annunziata (2003)

Energy Calibration of Alpha Spectrometry System

The energy resolution obtained from an alpha spectrometric system is dependent on the detector, uniform thickness of source, amplifier, and the analyzer. Though the detectors are quoted to have a resolution of about 20 keV, (FWHM), a resolution of about 50 keV or less is acceptable in the analysis of alpha emitters for the environmental samples.

Electroplated sources of about 50 Bq concentration each of ^{239}Pu (5.155 MeV), ^{242}Pu (4.90 MeV), and ^{241}Am (5.486 MeV) either as single or mixed sources are used to carry out energy calibration. Amplifier gain and the conversion gain of multichannel analyzer are adjusted such that an energy calibration of about 10 keV/channel is arrived. Linearity of the system is verified by counting the other sources under the same setup. A linear equation of energy versus channel number is established by using the three discrete alpha energies and the respective channel numbers.

Resolution of Alpha Spectrometry Systems

Energy resolution of a detector amplifier system is expressed as energy at full width half maximum (FWHM). An electroplated ^{241}Am source (5.49 MeV and 5.44 MeV alphas with 85% and 13% abundance, respectively, and a half-life of 433 y) can be easily prepared in a radiochemical laboratory and is best suitable for energy resolution. A 50 Bq ^{241}Am source can be positioned in detector chamber, usually at a 2 mm distance from the detector. While FWHM and MDA increase with increasing distance, efficiency decreases due to attenuation. Calculation of the energy resolution is identical to gamma-ray spectrometry system.

$$\mathrm{FWHM}\% = \frac{\partial E}{E} \times 100 \approx \frac{k \times 100\%}{\sqrt{\mathrm{E}}}$$

or, alternatively,

$$\text{FWHM}\% = \frac{\partial C}{E} \times Fe \times 100$$

where:

E = Energy of the peak (keV) of the standard source (^{241}Am).
∂E = FWHM of the peak in energy units.
k = Proportionality constant characteristic of a particular detector.
∂C = Width in channels at the half height of the maximum peak counts.
Fe = Energy calibration factor (keV/ch).

Efficiency Calibration of Alpha Spectrometry Systems

Electroplated sources of about 50 Bq concentrations each of ^{239}Pu, ^{242}Pu, and ^{241}Am are prepared. Each of them is counted for about 300 seconds (about 5 minutes) in a well-evacuated chamber. Vacuum for each standard should be the same and must be less than 0.01 mm Hg. The distance between the source and the detector is usually fixed at about 2 mm. The efficiency of the spectrometer decreases rapidly with the increase in the distance between detector and source (Table 5).

$$\textit{Efficiency}\ (Ef)\% = N \times \frac{100}{dps} \times \frac{100}{A}$$

where:

Ef = Efficiency of the system for particular alpha energy.
N = Net cps under the photopeak area after subtracting the background in cps.
dps = Disintegration per second, activity of the standard (Bq).
A = Percent abundance of that particular energy out of total disintegration.

Similar set-up conditions must be used for the counting of electrodeposited sample of environmental matrices.

Traceability of Standard Sources Used in Calibration of Environmental Monitoring Equipment

In the 11th General Conference of Weights and Measures (CGPM), an International System of Units (SI System) was adopted as an extended version of metric system. SI system provided internationally accepted definitions and prototypes for 7 fundamental or base units and 22 derived units and has authority to work together with member states on the matters related to measurement science and measurement standards. The world has moved from local measurement units, which varied from one country to the other, to a truly global system of measurement accepted worldwide. The SI units nowadays hold an official status in majority of the countries worldwide. The BIPM located at Sevres, France, is the official home of SI units and international reference time scale (UTC) (Hanisch et al. 2022).

Reliable and consistent measurement is an essential requirement for virtually all conformity assessment processes, including those which support health, safety, and

environmental objectives. It is also important for the international recognition of those measurements, which is an essential aspect within the world trading system (BiPM 2012). For flawless trade, research publications, safety regulations (e.g., dosimetry), and acceptance of test results across the world require that respective commercial, non-governmental, or governmental bodies providing such certifications must in turn get their measurement methods accredited according to some international norms acceptable to all the participating economies.

Role of International Laboratory Accreditation Cooperation (ILAC)

ILAC, a trans-governmental network headquartered at Utrecht (Netherlands), has played a pivotal role in providing such services (calibration, testing, medical testing, inspection, reference material producers, and proficiency testing providers accreditation) to the participating countries through mutual recognition arrangement (MRAs) (ILAC IAF 2014).

ILAC is the international organization along with NMIs and other national accreditation bodies (commercial, non-governmental, or governmental), which operate according to the guidelines, jointly issued by International Organization for Standardization (ISO) and International Electrotechnical Commission (IEC), i.e., ISO/IEC 17011. ISO and IEC themselves do not provide any certification or accreditation; however, they provide comprehensive guidelines for such process through documentation. ILAC is involved in the accreditation of laboratories worldwide including calibration laboratories (using ISO/IEC 17025), testing laboratories (using ISO/IEC 17025), medical testing laboratories (using ISO 15189), inspection bodies (using ISO/IEC 17020), proficiency testing providers (using ISO/IEC 17043), and reference material producers (using ISO 17034) (ILAC 2022a).

The ILAC MRA links the existing regional MRAs of the Recognized Regional Cooperation Bodies and delegates authority for the evaluation, surveillance, re-evaluation, and associated decision-making relating to the signatory status of the accreditation bodies that are ILAC MRA signatories. The signatories agree to accept the results of each other's accredited bodies. Hence, the results from the laboratories accredited by the ILAC MRA signatories are able to be recognized internationally. Usually, an online directory of accredited facilities is provided on ILAC webpage to perform a signatory search (ILAC 2022b).

Accreditation is the independent evaluation of conformity assessment bodies (CABs, the term CAB has now been replaced with "laboratory" in recent ISO/IEC 17025 document) or laboratories against recognized standards to carry out specific activities to ensure their impartiality and competence. Through the application of national and international standards, government, procurers, and consumers can have confidence in the calibration and test results, inspection reports, and certifications provided.

The regional arrangements are managed by the recognized regional cooperation bodies that work in harmony with ILAC and IAF (International Accreditation Forum). The recognized regional cooperation is also represented on the ILAC and

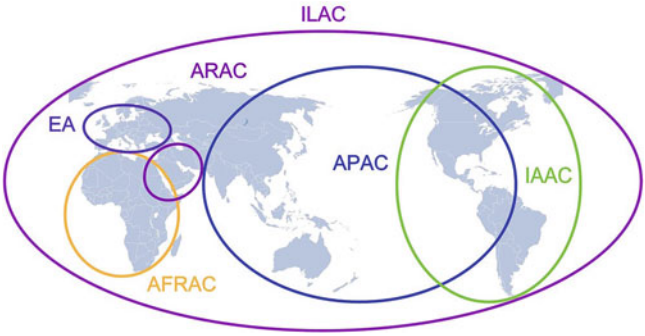

Fig. 4 Five recognized regional (European, Asia-Pacific, Arab, Africa and Inter American) cooperation bodies under the ILAC MRA

IAF Executive Committees. As depicted in Fig. 4, ILAC works closely with five (but not limited to) regional cooperation bodies involved in accreditation, notably EA in Europe, APAC in the Asia-Pacific, IAAC in the Americas, AFRAC in Africa, and ARAC in the Arab region (ILAC MRA 2022).

Traceability to SI

The laboratory or a third party performing the calibration shall ensure that the measurement results are traceable to International System of Units (SI). It can be ensured when a) calibration is provided by a competent laboratory (conforming to the guideline of ISO/IEC 17025), b) it is performed using certified values of certified reference material producer (RMP), or c) direct realization of SI through NMIs or any international body. There are instances when metrological traceability to SI units is not possible; in such cases, it can be demonstrated through comparison with an appropriate reference, e.g., by using certified values of a CRM.

The instruments used of radiation detection and measurement are regularly calibrated with standard sources which are traceable NIST, NPL (UK), and National Metrology Laboratories/Institutes (NMLs/NMIs) or to IAEA which in turn is traceable to BIPM (International Bureau of Weights and Measures, *Bureau international des poids et mesures*), which is an internationally recognized body for SI Systems by General Conference on Weights and Measures (Conférence générale des poids et mesures, CGPM) (Fig. 5). As per CGPM, the International System of Units (SI) plays a pivotal role in providing confidence in the accuracy and global comparability of measurements needed for international trade, manufacturing, human health and safety, protection of the environment, global climate studies, and scientific research.

All the instruments used for environmental monitoring and surveillance must be calibrated with internationally traceable standard sources for authenticity of the results and for regulatory compliance.

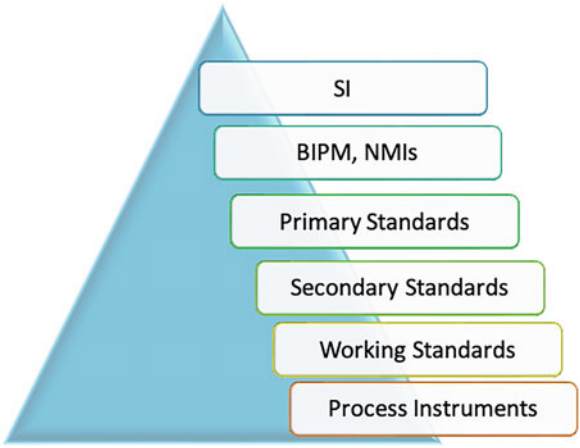

Fig. 5 Traceability pyramid in which SI is the absolute international standard with least uncertainty

Instrument calibration should be performed at certain intervals, documented in manual or as per requirement of certification agency, and that too only with certified standard sources traceable to a national standards body (Foster 2009) recognized by BIPM (International Bureau of Weights and Measures, *Bureau international des poids et mesures*) such as the National Institute of Science and Technology (NIST), NPL (UK), IAEA, and other NMLs/NMIs. Some of the radionuclides which are commonly available as standard sources for calibration from laboratories or commercial suppliers certified by an NMI (some international certifying agency recognized by CGPM) are given in Tables 6 and 7.

Routine calibrations of environmental instruments (for efficiency calibrations) should be done using certified reference materials of known and documented value and stated uncertainty. These certified or standard reference materials (CRMs) may be supplied by the following:

1. NIST, along with other national standards bodies such as the National Physics Laboratory (NPL, UK), Physikalisch-Technische Bundesanstalt (PTB) of Germany, Czech Metrology Institute (CMI) of Czech Republic

   A list of NMLs is provided by NIST at their website for easy reference (Foster 2009).
2. A calibration-source supplier whose measurement capabilities or manufacturing processes are tested periodically by NIST or national standards bodies.
3. IAEA, International Atomic Energy Agency (Vienna, Austria).
4. A calibration-source supplier who documents derived materials with stated uncertainty, and whose values has been verified with analytical and measurement systems that have been tested periodically through an unbroken chain of comparisons to the national standards (Table 7)

Table 6 Standard sources used for calibration of equipment used in environmental monitoring

Type of radiation detection	Equipment	Important standard sources for calibration	Standard source form
Alpha	ZnS (Ag), alpha spectrometry system	^{232}U, ^{238}Pu, ^{241}Am, ^{242}Pu	Alpha sources
Beta	Liquid scintillation counting (LSC/LSA)	^{3}H, ^{14}C, ^{90}Sr, ^{36}Cl	Beta liquid sources
Gamma	Gamma spectrometry systems NaI (Tl), HPGe	^{137}Cs, ^{133}Ba, ^{60}Co, ^{152}Eu	Gamma source kits, resin sources

Establishing Metrological Traceability

A laboratory or an organization as a whole, given that all the laboratories in that organization use the same standard sources or reference materials kept at a centralized location, can establish metrological traceability as per clause 6.5 of ISO/IEC 17025:2017 document (ISO/IEC 2017). It has been further elaborated in the Annex A of the same document. It states that a metrological traceability can be established by considering and then ensuring the following:

1. The specification of the measurand, i.e., the quantity to be measured.
2. A documented unbroken chain of calibrations going back to an appropriate national, international, or intrinsic standard.
3. Measurement uncertainty for all the steps in the traceability chain has been evaluated.
4. Each step of the chain is carried out with appropriate methods, and the results are stated with associated uncertainties.
5. The laboratories performing (other than the one under consideration) one or more steps in the chain shall also supply evidence for their technical competence.

The above five steps have been reproduced *verbatim* from the ISO/IEC 17025: 2017 document. It is the responsibility of the laboratory to ensure and demonstrate metrological traceability. Certified values of certified reference materials (CRMs) supplied from reference material producers (RMPs) which conform to the ISO 17034,provide metrological traceability (ISO 2016). A laboratory can demonstrate metrological traceability through various methods conforming to ISO/IEC 17025 which include third party recognition, e.g., from an accreditation body, and external assessment by the end-users (second party recognition) or by self-assessment (first party). Additionally, internationally accepted paths include calibration and measurement capabilities provided by an NMI, a designated institute, or an accreditation body which themselves have been subjected to suitable peer-review processes conducted under CIPM MRA (International Committee for Weights and Measure Mutual Recognition Arrangement) or under ILAC. The BIPM KCDB (Key Comparison Database) provides details about the range and measurement uncertainty for each listed service (BIPM KCDB 2022; 2022).

Table 7 A list of essential instruments for measurement of environmental radionuclides and standard sources of calibration [example taken from Environmental Survey Laboratory of BARC, India]

Instruments	Detector	Calibration	Certificate/ manufacturer	Traceability
Gamma spectrometry	HPGe	Standards of ^{241}Am,^{133}Ba,^{137}Cs, and ^{60}Co or ^{152}Eu standard	UKAS/Amersham plc Buckinghamshire/GE health care ltd., Buckinghamshire Czech metrology institute Czech Republic, etc.) CRM from IAEA	NPL (UK) Czech national standard IAEA
	NaI (Tl)	Standards of ^{133}Ba,^{137}Cs, and ^{60}Co	UKAS/Amersham plc Buckinghamshire/GE health care ltd., Buckinghamshire Czech metrology institute Czech Republic, etc.) CRM from IAEA	NPL (UK) Czech national standard IAEA
Beta counting systems	Liquid scintillation analyzer	^{3}H, ^{14}C standard solutions	UKAS/Amersham plc Buckinghamshire/GE health care ltd., Buckinghamshire Czech metrology institute Czech Republic, etc.) CRM from IAEA	NPL (UK) Czech national standard IAEA
	Low background GM/plastic scintillator	^{40}K standard ^{137}Cs standard ^{90}Sr standard	Prepared in-house using standards from UKAS/Amersham plc and KCl- AR grade	IAEA
Alpha counting systems	Alpha spectrometry	$^{239+240}$Pu standard ^{241}Am standard	Prepared in-house using tracer (^{236}Pu) from Czech metrology institute Czech Republic	IAEA
	ZnS (Ag) alpha counter	$^{239+240}$Pu standard ^{241}Am standard	Prepared in-house using tracer (^{236}Pu) from Czech metrology institute Czech Republic	
Radiation survey meters	Low range survey meters	^{137}Cs point source	Amersham	NIST
	High range survey meters	^{137}Cs point source		
	NaI(Tl) survey meters	^{137}Cs point source		

Certified Reference Materials (CRM)

A reference material (RM) is defined as a material which is sufficiently homogenous and stable with respect to one or more specified properties, i.e., quantitative or qualitative, and which has been established to be fit for its intended use in a measurement process (e.g., identity of substances or species) (Elordui-Zapatarietxe et al. 2016; ISO 2017; NIST 2022). A certified reference material (CRM) is different than a reference material in that it is accompanied by a certificate and has metrological traceability, and each property value has an uncertainty value assigned to it (IAEA 2009). These values are stored by the CRM and then transferred along with the CRM itself when transported from one location to another. A CRM producer's method must meet the requirements of ISO's general requirements of homogeneity and stability (ISO 2017). A standard reference material (SRM) is a patented name for reference materials from NIST (NIST 2022).

ISO 17025: 2017 (ISO/IEC 2017) discusses about demonstration of metrological traceability which clearly states that *Laboratories are responsible for establishing metrological traceability. Certified values (with associated uncertainty) of certified reference materials from CRM producers conforming to ISO 17034 provide a verifiable metrological traceability.*

An unbroken metrological traceability and known uncertainty for the property values of the material is provided by the CRMs. The traceability is provided via a paper trail or through a digital certificate that clearly documents the property values, the uncertainty, and the authoritative issuing body. CRMs can be used for method validation, method verification, monitoring process and control, calibration, and training.

For environmental radioactivity measurements, IAEA provides CRMs (Table 8) for efficiency calibration of instruments for specific radionuclides which can be ordered through IAEA nucleus website (IAEA 2003; 2022).

IAEA ALMERA Network for Proficiency Tests and Interlaboratory Comparison Exercises

The ALMERA network (Analytical Laboratories for the Measurement of Environmental Radioactivity) was established by the IAEA in 1995 and is a cooperative effort to harmonize and improve the analytical capabilities of environmental laboratories worldwide in the member countries (Almera 2020). IAEA member states can nominate laboratories which are expected to provide reliable and timely analysis of environmental samples in the event of an accidental or intentional release of radioactivity.

ALMERA consists of 194 laboratories representing 90 countries. The IAEA helps the ALMERA network of laboratories to maintain their readiness by coordination activities which includes organizing meetings, developing standardized

Table 8 A list of certified reference materials supplied by IAEA

Name	Category	Product description	Matrix
IAEA-465	Environmental	Baltic Sea sediment	Inorganic
IAEA-434	Mineral	Phosphogypsum	Inorganic
IAEA-RGK-1	Ores	Potassium sulfate	Inorganic
IAEA-384	Sediment	Fangataufa sediment	Inorganic
IAEA-385	Sediment	Irish sea sediment	Inorganic
IAEA-410	Sediment	Bikini atoll sediment	Inorganic
IAEA-412	Sediment	Pacific Ocean sediment	Inorganic
IAEA-418	Water	Mediterranean Sea water	Inorganic
IAEA-437	Animal product	Mussel from Mediterranean Sea	Organic
IAEA-414	Animal products	Mixed sea fish	Organic
IAEA-447	Soil	Moss soil	Organic
IAEA-446	Vegetation	Baltic Sea seaweed	Organic
IAEA-464	Vegetation	Brown rice grains	Organic

IAEA (2022)

methods for sample collection and analysis, and organizing interlaboratory comparison (ILC) exercises and proficiency tests (PTs) as a tool for external quality control.

As per the "The ALMERA Network – Nucleus" resource, participation of laboratories in these activities results in the following:

1. Demonstration of technical competence of ALMERA laboratories through participation in proficiency tests.
2. Wider application of recommended validated methods and methodological harmonization, leading to enhanced worldwide comparability of environmental radioactivity measurement results.
3. Shared information on available analytical capabilities and expertise, which facilitates the identification of laboratories with competence in specific areas of interest, particularly in emergency situations.

The IAEA Terrestrial Environmental Radiochemical Laboratory (TERL) organizes annual proficiency tests (PT) and interlaboratory comparison (ILC) exercises for member states and ALMERA laboratories which are designed to monitor and demonstrate their performance and analytical capabilities and to identify gaps and problem areas where further development is required (Almera 2020).

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Radiation Metrology and Applications

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V. Sathian and Probal Chaudhury

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Abstract

The Bhabha Atomic Research Centre (BARC) is the designated institute in ionizing radiation metrology for the country. The BARC designs, develops, establishes, maintains, and disseminates various national standards for physical quantities of ionizing radiation in India. The IAEA/WHO has recognized the BARC as a Secondary Standard Dosimetry Laboratory (SSDL) for disseminating radiological standards to users in the country. SSDL-BARC provides traceable calibration to the users in the field of teletherapy, brachytherapy, radiation protection, and diagnostic radiology. The pace of the advances in radiation therapy with the expected improvement in patient treatment has brought out the need for greater accuracy in the treatment process. SSDL-BARC participates in international intercomparison programs to establish its equivalence with international standards, thereby validating the calibration procedure followed in the laboratory. Radioisotope targeted therapy has been employed in the treatment of wide variety of cancers. The activity of these radiopharmaceuticals at the nuclear medicine centers is measured using the radionuclide calibrators. BARC as the national metrological laboratory for radioactivity has been conducting the audit program for activity measurements with RC for more than four decades. Neutron sources/fields not only are critical for operating and regulating nuclear reactors but also are important for various other research, industrial, and radiation protection applications. Therefore, precise value of neutron source strength and resultant field is highly essential which is provided by the BARC using several neutron standards developed for the purpose.

Radiation processing technologies are very widely used in different industrial applications like sterilization of healthcare products, processing of food and agriculture products, material modification and emerging fields like environmental remediation, synthesis of advanced materials and products, medical applications such as irradiation of blood, etc. Proper dosimetry is the backbone of success of these applications. Conventional radiation measuring techniques like ionization chambers, calorimeters, etc. are unable to measure absorbed doses for most of these applications. Chemical dosimetry techniques provide a cost-effective alternative for radiation processing dosimetry.

Radiation monitoring instruments are backbone to all the radiological and nuclear installations for the safety of occupational radiation workers. Performance of these monitors depends on calibration and measurement traceability to the national standards. Protection-level radiation monitors cover a wide range of radiation field (gamma ray, X-ray, and beta ray) intensity and energy; therefore, measurement standards of various types and sizes are established. The BARC has also established various radiological facilities for gamma rays, X-rays, and beta

rays which are used for calibration and testing of all kinds of protection-level radiation monitors.

Keywords

Radiation metrology · Primary standards · Calibration · Traceability · Radiotherapy · Radiopharmaceuticals · Radiation processing · Radiation protection

Introduction

The National Physical Laboratory of India (NPL-India) is the national metrology institute (NMI) for the country, and the Bhabha Atomic Research Centre (BARC) is the designated institute in ionizing radiation metrology. As the designated institute of ionizing radiation metrology, the BARC designs, develops, establishes, maintains, and disseminates various national standards for physical quantities of ionizing radiation in India. Primary standards, secondary standards, and field standards for all types of ionizing radiation like alpha, beta, and gamma rays, X-rays, and neutron are developed, established, and maintained at the BARC. The BARC participates in various international intercomparison exercises to establish the equivalence with other NMIs in the world.

The BARC is an active member of the Asia Pacific Metrology Programme (APMP), consisting of a group of NMIs from the Asia-Pacific region, and is engaged in improving regional metrological capability through sharing of expertise and exchange of technical information. APMP is a regional metrology organization recognized by the International Committee for Weights and Measures.

As a designated institute, the BARC is a member of Consultative Committee for Ionizing Radiation (CCRI) as follows:

1. CCRI Section I (Dosimetry): X-Ray and Gamma Rays and Charged Particles
2. CCRI Section II (Radioactivity Measurement): Measurement of Radionuclides
3. CCRI Section III (Neutron): Neutron Measurements

Radiation Metrology and Radiotherapy

Radiotherapy is a key treatment modality for cancer patients. Approximately one in three people will develop cancer during their lifetimes. Depending on the region of the world they live in, up to 65% of all cancer patients could benefit from radiotherapy as part of their treatment management.

Advances in radiation therapy technology have allowed the treatment to evolve from a conventional 2D therapy to more complex treatment modalities like conformal radiotherapy, intensity-modulated radiotherapy, image-guided radiotherapy, etc. The pace of the advances in radiation therapy with the expected improvement in

patient treatment has brought out the need for greater accuracy in the treatment process. In order to ensure sustained accuracy in radiation therapy, all the steps involved in the radiation treatment have to be covered in the quality assurance program of the institution. It is well recognized that there is a need to evaluate the influence of different factors affecting the accuracy of radiation dose delivery and to define the actions necessary to maintain treatment uncertainties at acceptable levels.

All countries which use ionizing radiation for treatment purpose should maintain a national standard for the relevant quantity to establish metrological traceability to international standard. Standards should be traceable to the International System of Units (SI) and may either be a primary standard maintained at a primary standards dosimetry laboratory (PSDL) or a secondary standard that is traceable to a primary standard and maintained at a secondary standards dosimetry laboratory (SSDL).

An SSDL can be traceable to the SI and international measurement system in different ways. Some SSDLs are traceable to a PSDL or directly to the BIPM if there is no national primary standard and the country is a member of the Metre Convention. Many SSDL Network members are traceable to primary standards through the International Atomic Energy Agency (IAEA). In Fig. 1 (reproduced from IAEA TRS 398 (IAEA 2000)), the highlighted box shows the traceability route for user or radiotherapy centers in India.

The IAEA/WHO has recognized the BARC as a Secondary Standard Dosimetry Laboratory (SSDL) for disseminating radiological standards to users in the country.

SSDL-BARC provides traceable calibration to the users in the field of teletherapy, brachytherapy, radiation protection, and diagnostic radiology. SSDL-BARC participates in international intercomparison programs to establish its equivalence with international standards, thereby validating the calibration procedure followed in the laboratory. The BARC also participates in quality audit conducted by the IAEA in the field of radiotherapy.

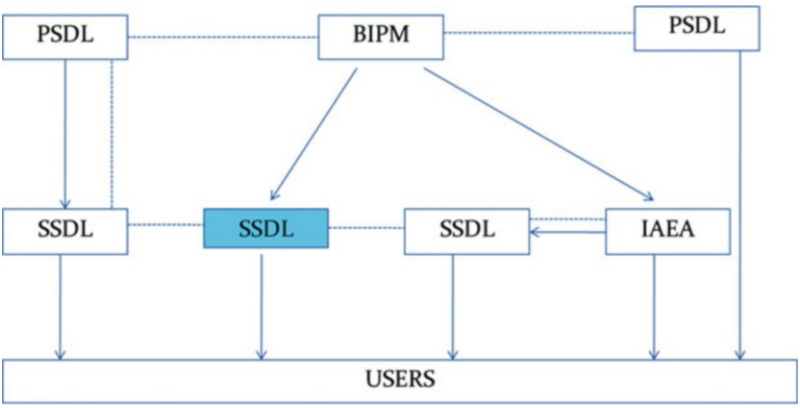

Fig. 1 Simplified representation of the international measurement system for radiation dosimetry. (Reproduced from IAEA TRS 398)

Traceability of Dosimeters Used in External Beam Therapy

Cylindrical ionization chambers and parallel plate ionization chambers of nominal volume up to 0.6 cc are used in dosimetry of teletherapy. Calibration of dosimeters is carried out by following IAEA protocol TRS-398. It is based on the determination of a calibration coefficient in terms of absorbed dose to water in ^{60}Co. Users of dosimeter use this calibration coefficient to perform dosimetry at user's radiation beam. Annually, around 350 SSDs are calibrated at SSDL-BARC.

The level of accuracy achievable in a clinical setup is primarily based on the accuracy achievable in the calibration laboratory.

SSDL-BARC maintains reference standards calibrated at BIPM with a relative standard uncertainty of 0.3% ($k = 1$). The chambers were calibrated in terms of absorbed dose to water and air kerma at ^{60}Co energy.

The laboratory has established international equivalence by participating in various intercomparison programs. Some of the recent comparison programs at ^{60}Co energy are:

- IAEA comparison of absorbed dose to water and air kerma for SSDLs at ^{60}Co energy (2015)
- Asia Pacific Metrology Programme (APMP) key comparison of absorbed dose to water for ^{60}Co (APMP.RI(I)-K4) (2010)
- Asia Pacific Metrology Programme (APMP)/TCRI key comparison of measurement of air kerma for ^{60}Co gamma rays (APMP.RI(I)-K1), 2006

Reference Air Kerma Rate Calibration at ^{192}Ir and ^{60}Co Beams (Brachytherapy)

Brachytherapy is the use of encapsulated radioactive source to deliver a high dose to tissue with the source placed very close to the tumor volume. Source strength determination is an important part of the QA program for brachytherapy. In dosimetry of brachytherapy, well-type ionization chambers of nominal volume of 200 cc are used. Calibrations of these dosimeters are carried out in terms of reference air kerma rate (RAKR) in ^{192}Ir and ^{60}Co energy. IAEA-TECDOC 1274 (IAEA 2002) is used for this calibration. Yearly around 100 well-type ion chambers are calibrated at ^{192}Ir energy at SSDL-BARC. Onsite well-type chamber calibration is carried out for ^{60}Co HDR brachytherapy sources based on the request from institutions.

The reference standard developed at SSDL-BARC for RAKR measurement is graphite-walled ionization chamber having a nominal volume of 1000 cc. Figure 2 shows the reference standard and the ^{192}Ir HDR calibration facility maintained in the laboratory.

TLD-Based Quality Audit Program

SSDL-BARC conducts dosimetry quality audit for radiotherapy centers in India. Participating in quality audit program gives confidence to the participating

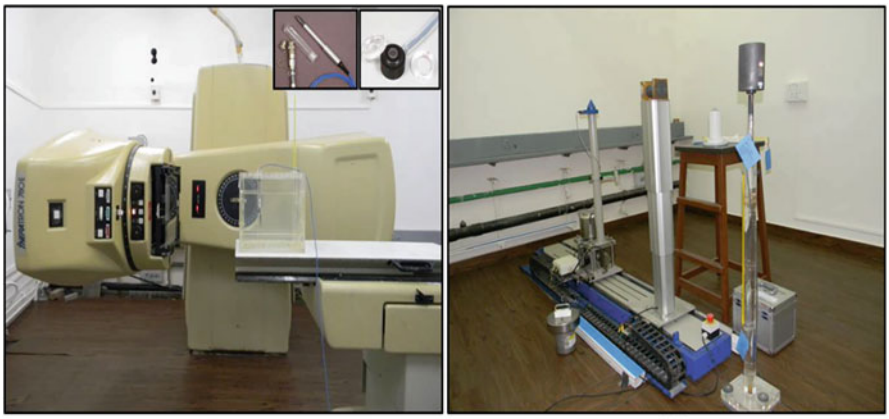

Fig. 2 (**a**) Telecobalt calibration facility, [inset] typical cylindrical ionization chamber and plane parallel chamber calibrated at SSDL-BARC. (**b**) Calibration facility for well-type ionization chamber

radiotherapy centers about the quality of treatment given to the patients. The quality audit program has evolved from the basic reference audit to more complex treatment techniques like intensity-modulated radiation therapy.

An independent verification of the quality of the dosimetry practices in radiotherapy centers is a part of the quality assurance (QA) guidelines. Postal dose quality audit provides a cost-effective methodology to verify dosimetry practices covering a large number of institutions and machines. Audits are useful to identify incorrect dosimetry practices followed at institution level and to correct them. Technical support is provided to the medical physicists to improve the dosimetry status at the radiotherapy centers. Workshops, training programs, and lectures on dosimetry are delivered to update the knowledge of the medical physicist community.

Since 1976, SSDL-BARC has been conducting TLD postal dose quality audit programs for radiotherapy centers in India. By participating in the external quality audit program, the radiotherapy center's QA program gets validated. Figure 3 shows improvement in the accuracy of the dose delivery capability at radiotherapy centers.

Calibration of Diagnostic Radiology Instruments at 20–150-kV X-Ray Diagnostic Beam Qualities

With the increased use of medical diagnostic procedures in clinical fields, there is a growing concern for the small increase in radiation-associated cancer risk to the general public which can become a public health concern if large number of the population undergoes such procedures. With the advent of new and specialized diagnostic tools, concern has been raised regarding the radiation dose received by patients and the healthcare providers involved in these procedures. In view of the importance of patient dose in radiology procedures, patient dose is controlled by the

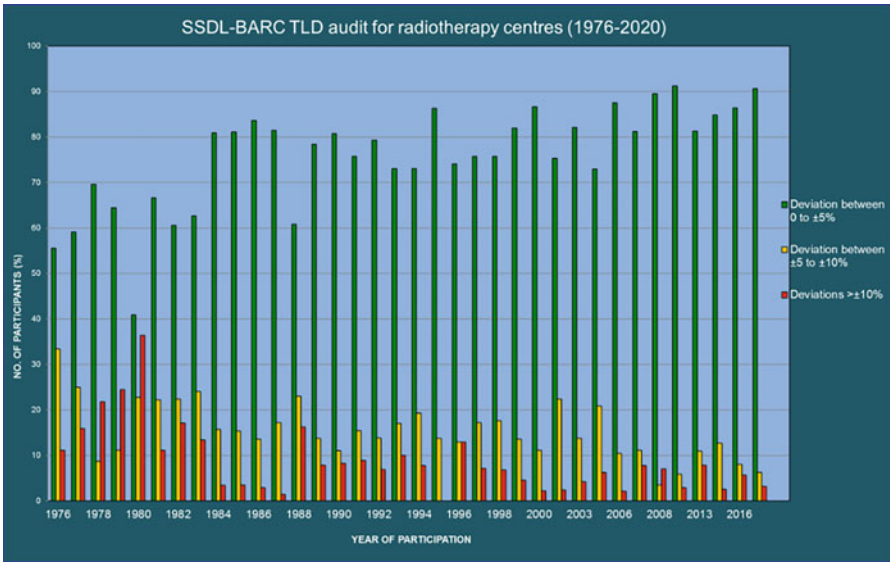

Fig. 3 Dosimetric quality audit program conducted by SSDL-BARC

principle of optimization along with regulatory control. Dosimetry in diagnostic radiology is a challenge as the instruments should be designed to cater to the varied clinical applications. The use of these instruments and the interpretation of the results obtained may require specialized techniques and knowledge. It is imperative that a standardized procedure be followed for dose measurements in clinical setup.

For the calibration of such instruments and their clinical measurements, the IAEA has a protocol, i.e., IAEA TRS-457–Dosimetry in Diagnostic Radiology: An International Code of Practice (IAEA 2007). SSDL-BARC has implemented IEC 61267 (IEC 2005) recommendations and the IAEA protocol TRS-457 by establishing diagnostic beam qualities and disseminating it to the users in the field of diagnostic radiology.

Nineteen diagnostic range X-ray beam qualities have been established and standardized at SSDL-BARC in the YXLON X-ray system available in the laboratory using the primary standard free air ionization chamber (FAIC) and the diagnostic range free air ionization chamber (Fig. 4).

SSDL-BARC has started providing traceable calibration to users in the field of diagnostic radiology. The primary standard for X-rays is the FAIC, and the chamber was used to participate in international intercomparison for narrow series and CCRI X-ray beam qualities, and its international equivalence was established. The following is a couple of recent intercomparison participated by SSDL-BARC under the aegis of APMP:

- APMP supplementary comparison of air kerma in ISO 4037 wide and narrow series X-ray beams: APMP.RI(I)-S3
- APMP key comparison of air kerma standards for the CCRI reference qualities in the medium-energy X-ray region (APMP.RI(I)-K3)

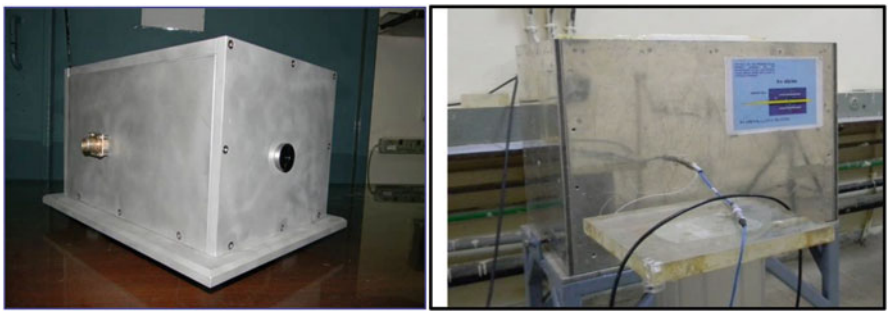

Fig. 4 Free air ionization chambers as X-ray measurement standards

Radiation Metrology and Nuclear Medicine

With the advancement and development in science and technology, a tremendous surge of usage of radioisotopes in a wide spectrum of medical applications especially in the field of nuclear medicine and industrial applications was observed over the decades. Additionally, with the discovery of artificial radionuclides and its industrial-scale production, a wide range of artificial radionuclides became available for applications. One such application is radioisotope therapy. It is a targeted therapy where the liquid form of radioisotope is administered orally or intravenously to the patients to specifically target the diseased tissue while sparing the surrounding healthy tissue.

Radiopharmaceuticals (RPs) contains a radioisotope and a tracer which are coupled with the pharmaceutical. RPs carry these radioisotopes to specific organs to deliver localized dose to tumor reducing radiation-induced side effects. As the disintegration of radioisotope begins, it targets tumor/cancerous cells since cancerous cells absorb a greater fraction of radioisotope than healthy cells. This ensures higher dose of radioactivity to tumor ultimately destroying the tumor. As the radiation destroys the cancerous cells, the body naturally eliminates them simultaneously repairing the normal tissues, if affected. Before administering RPs to patients, quantity of RP is carefully determined depending on the type of tumor, location, age of the patient, gender of the patient, etc.

Radioisotope targeted therapy has been employed in the treatment of wide variety of cancers, such as brain cancer, thyroid cancer, bone metastases, bile duct cancer, liver cancer, etc. A number of radionuclides, such as ^{131}I, ^{32}P, ^{90}Sr, ^{90}Y, ^{117m}Sn, ^{153}Sm, ^{169}Er, ^{177}Lu, ^{186}Re, ^{188}Re, and ^{223}Ra, have been used successfully for the treatment of many benign and malignant disorders. For example, ^{131}I is mainly used to treat some types of thyroid cancer. Once it is ingested, it gets concentrated in the thyroid cells, killing the cancer cells. ^{153}Sm can be used to treat some types of metastatic bone cancer.^{177}Lu is used for prostate cancer, while ^{188}Re is used in therapeutic applications like radionuclide synovectomy, bone pain palliation, intravascular radiation therapy, and other applications in nuclear medicine. ^{223}Ra is used to treat prostate cancer that has spread to the bones.

Traceability of Radioactivity Measurements in Nuclear Medicine

The safety and efficacy of these drugs depend on various factors and on the ability of the clinic to accurately determine the amount of radioactivity which determines the dose to the patient. The activity of these radiopharmaceuticals at nuclear medicine centers is measured using the radionuclide calibrators (RC) as shown in Fig. 5 and must be determined as accurately as possible for the optimized diagnosis or treatment.

The BARC, as the national metrological laboratory for radioactivity, has been conducting the audit program for activity measurements with RC for more than four decades (since 1979), to establish traceability of measurements with radionuclide calibrators in nuclear medicine centers (NMC) to national standards and to ensure calibration of RC. In order to investigate the quality performance of routine measurements at NMCs, audit programs of Iodine-131 activity measurements are conducted biannually. Quality audit programs (QAPs) help to ensure that the RCs are working satisfactorily so that the activity of the radiopharmaceutical measured before administering to patients is within the acceptable limits of $\pm 10\%$ from the prescribed dose. If the radioactivity administered is higher than that prescribed, the patient is unnecessarily exposed to radiation. On the other hand, if lower than the prescribed dose is administered, the image quality will be poor leading to poor diagnosis or, in the case of therapy, leads to inefficient treatment. Nineteenth audit of ^{131}I activity measurements with RC was conducted in 2021, and the results of all the audits conducted are shown in Fig. 6.

The continuous improvement is an important aspect of any quality assurance/audit program, and this is being achieved as demonstrated by the results of the ^{131}I audit. The results of the audit are very encouraging, as the number of participants whose deviation greater than $\pm 10\%$ has declined tremendously over the years. Thus, these programs ensure that patients receive maximum benefit with minimum

Fig. 5 Radionuclide calibrator (RC) used in nuclear medicine centers

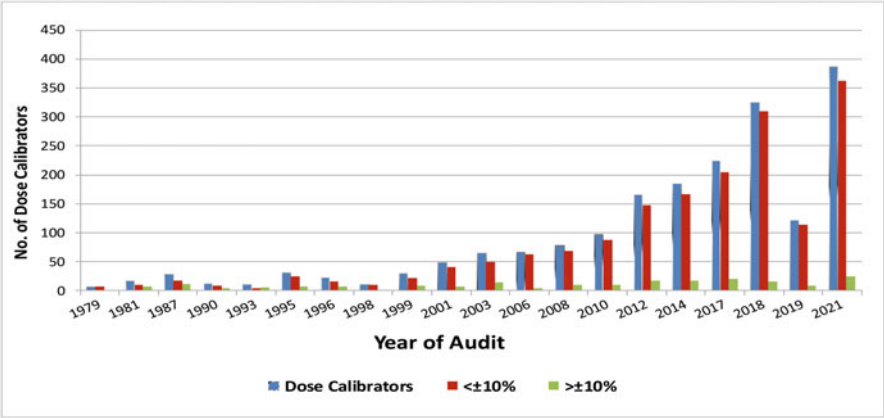

Fig. 6 Summary of results of national audits conducted from 1979 to 2021

exposure and establish traceability of radioactivity measurements to national and international standards.

Primary Standards of Radioactivity Measurements in Metrology Lab

Coincidence technique is an absolute method for the measurement of activity of radionuclides and has been in use over several decades. The technique, with its variants, enables the direct measurement of activity of most of the radionuclides, without any specific value being assumed for branching ratios or other decay scheme parameters, for source and detector efficiencies. It is directly applicable to all radionuclides decaying by the simultaneous emission of two or more radiations in prompt succession, such as β-γ, α- γ, electron capture- γ, γ -γ, γ-x, etc.

Various primary standards for radioactivity as shown in Fig. 7 based on coincidence technique are developed and established at the BARC:

1. Proportional counter-based 4πβ(PC)-γ coincidence counting system for activity measurements
2. Liquid scintillation-based 4πβ(LS)-γ coincidence counting system for activity measurements
3. Plastic scintillation-based 4πβ(PS)-γ coincidence counting system for activity measurements

Secondary Standards for Radionuclides

Standardization of radionuclides using primary methods is complex, tedious, time-consuming, and destructive (sample cannot be retrieved); hence, secondary standards were developed which are faster and rugged and are used to disseminate

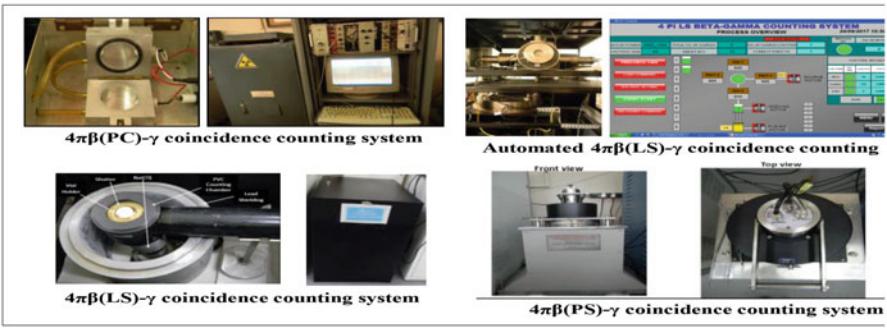

Fig. 7 Coincidence counting system-based primary standards for activity measurements

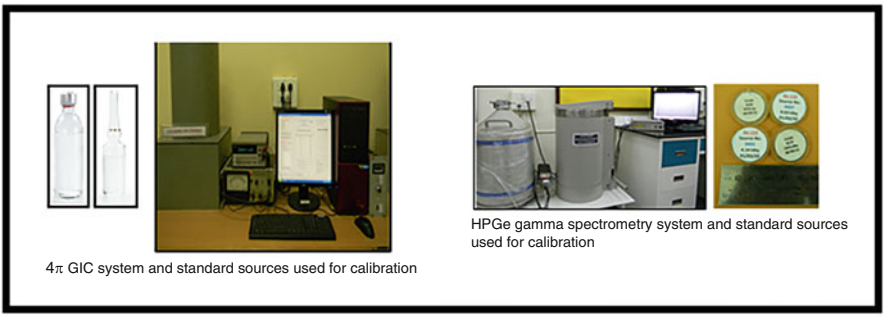

Fig. 8 Secondary standards for activity measurement

radioactivity standards to end users in the country. Secondary standard available at the BARC are as follows:

4π-γ Ion Chamber (4π GIC) for Activity Measurements of Gamma-Emitting Radionuclides

It is used for the standardization of liquid sources of gamma-emitting radionuclides. It uses the reentrant-type 4π-γ ion chamber and electrometer for current measurements. The radioactive source is placed in the GIC, and current produced by ion chamber is measured using electrometer. The system is calibrated against the primary standards using primary standardized standard sources of radionuclides.

HPGe Gamma Spectrometer for Activity Measurements of Gamma-Emitting Radionuclides

High-purity germanium (HPGe) gamma spectrometer was established as a secondary standard for standardizing solid sources of gamma-emitting radionuclides. It consists of HPGe detector, multichannel analyzer (MCA), and spectrum acquiring software. The system is calibrated against the primary standards using primary standardized solid standard sources of various radionuclides. Photograph of 4π-γ ion chamber and HPGe gamma spectrometry system is shown in Fig. 8.

Equivalence of Primary Standards with Other International National Metrology Institutes (NMIs)

Accurate radioactivity measurements of the source are important and play a pivotal role in all medical and industrial applications, due to the growing concerns of health risks and contamination associated with radioactivity. The quality of the results of measurements depends directly on the accuracy of the standards. Thus, with the proliferation in the use of radioactive materials, it became essential to establish traceability of measurements with national and international standards of radioactivity. As a result, the systems used for the measurements of radioactivity must be calibrated using accurate and traceable standards.

To establish international equivalence of activity standards maintained at the BARC with other national metrology institutes (NMIs) around the world, the BARC participates in international intercomparison organized by BIPM, France, and APMP, Japan.

International Intercomparison of ^{59}Fe (APMP.RI (II)-K2.Fe-59)

The international comparison of activity measurements of ^{59}Fe, APMP.RI (II)-K2.Fe-59, was carried out within the framework of the Asia-Pacific Metrology Programme (APMP). A graphical presentation of the activity concentration results of the participants in the intercomparison is shown in Fig. 9. From the figure, it is

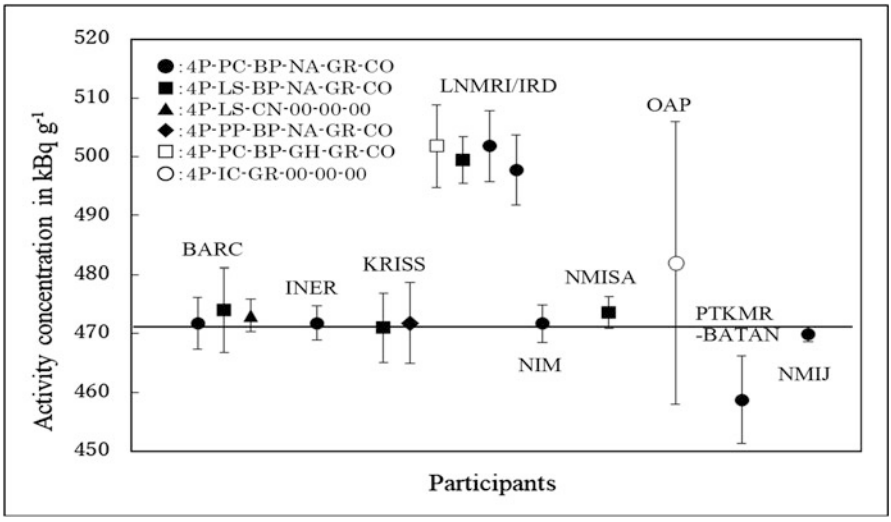

Fig. 9 Activity concentration of ^{59}Fe of various national metrology institutes participated in APMP.RI (II)-K2.Fe-59. The uncertainty bars correspond to the combined standard uncertainty on each respondent's value. The solid line represents the proposed comparison reference value (CRV) of 470.8 kBq g^{-1}, and the dashed lines represent the combined standard uncertainty of 1.7 kBq g^{-1} on the CRV

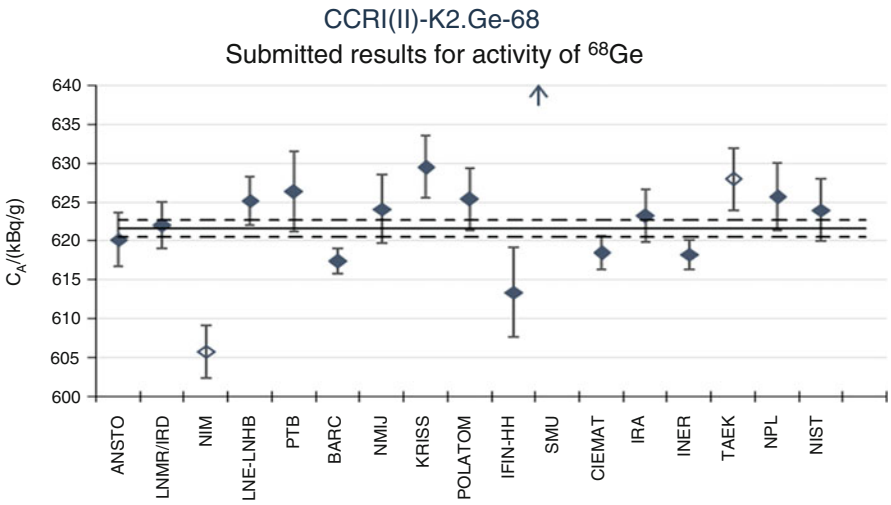

Fig. 10 Activity concentration of ^{68}Ge of various national metrology institutes participated in CCRI(II)-k2.Ge-68. The uncertainty bars correspond to the combined standard uncertainty on each respondent's value. The solid line represents the proposed comparison reference value (CRV) of 621.7 kBq g^{-1}, and the dashed lines represent the combined standard uncertainty of 1.1 kBq g^{-1} on the CRV

observed an excellent agreement of BARC results with that of other international laboratories.

International Intercomparison of ^{68}Ge (CCRI(II)-k2.Ge-68

An international key comparison, identifier CCRI(II)-K2.Ge-68, was carried out in 2014. The National Institute of Standards and Technology (NIST) served as the pilot laboratory, distributing aliquots of a ^{68}Ge/^{68}Ga solution. Results for the activity concentration of ^{68}Ge at a reference date of 12 h00 UTC 14 November 2014 were submitted by 17 laboratories, encompassing many variants of coincidence methods and liquid scintillation counting methods. A graphical presentation of the degrees of equivalence with the comparison reference value (CRV) is shown in Fig. 10.

Role of Radiation Metrology in Reactors and Accelerators

Radiation technology is successfully being used in several fields including basic research, industrial, medical (diagnostic and therapeutic), food irradiation, space, etc. In order to meet the increasing demand of energy, several nuclear reactors are being set up in the country, whereas in the field of medical applications, the use of accelerators all over the world has increased significantly. However, the legal functioning of these nuclear installations requires a reliable radiation protection services for radiation workers in order to keep the personnel exposure of different types of radiations within the prescribed limits, as set by the regulatory body Atomic

Energy Regulatory Board (AERB). This brings the role of radiation metrology in these fields. Neutron sources/fields not only are critical for operating and regulating nuclear reactors but also are important for various other research, industrial, and radiation protection applications. Therefore, precise value of neutron source strength and resultant field is highly essential. Several neutron standards to provide traceability to various users are available as discussed subsequently.

Radiation Instruments Used

Manganese Sulfate Bath System

$MnSO_4$ bath facility is the primary standard for the measurement of neutron source yield of laboratory neutron sources and is shown in Fig. 11a. It consists of a spherical tank containing aqueous saturated solution of $MnSO_4$ and a NaI(Tl) detector in specific geometry. The fast neutrons emitted by source are thermalized by water and activate ^{55}Mn to generate radioactive ^{56}Mn which subsequently decays by gamma

Fig. 11 (**a**) MnSO4 bath facility, (**b**) STAG facility, (**c**) recoil proton telescope, (**d**) PLC

radiation. The measured radioactivity is then correlated to the neutron yield. The uncertainty associated with the measurement is within ±1%.

Standard Thermal Assembly in Graphite (STAG)

It is a primary standard for the thermal neutron fluence rate as shown in Fig. 11b. The system is consisting of six ^{241}Am-Be neutron sources inside a graphite cubical assembly. Thermal neutrons are produced by the slowing down of the emitted fast neutrons by graphite moderator. The arrangements of the sources are in staggered positions in order to get the better uniformity of neutron fluence. A central channel [4 cm × 4 cm × 15 cm] inside the cubical pile is kept for irradiation of different detectors and dosimeters. There are three different locations where thermal neutron fluence rate has been standardized based on the ^{197}Au and ^{55}Mn foil activation techniques. The system is highly useful for the calibration of different thermal neutron detectors used in research as well as power reactors and irradiation of solid-state detectors for neutron dosimetry applications.

Proton Recoil Neutron Telescope

It is an absolute standard for the measurement of neutron yield of monoenergetic fast neutrons (Fig. 11c). The system consists of a polyethylene radiator $(C_2H_4)_n$ and a CsI (Tl) detector in a specific arrangement which are placed inside a vacuum chamber. The basic working principle of the system is that fast neutrons create recoil protons in the thin radiator which are detected using the CsI (Tl) detector placed at a specific solid angle. Based on the n-p scattering cross-section data from ENDF-VII, the efficiency of the system was analytically calculated for 14.57-MeV monoenergetic neutrons which were found to be $(54.5 \pm 2.09) \times 10^{-7}$ counts/neutron/steradian (Ghodke et al. 2020). The system is utilized for the standardization of neutron yield of Purnima D-T neutron generator.

Precision Long Counter (PLC)

PLC is the secondary standard for the neutron source yield measurement, which is traceable to the manganese sulfate bath system (Fig. 11d). The working principle is based on the detection of fast neutrons by slowing it down to thermal region by using suitable moderating medium. It consists of a BF_3 proportional counter which is kept inside two concentric high-density polythene cylindrical moderators separated by an annular thermal neutron shield made up of 5% boron-loaded polythene. The position of different moderators and their thickness and length are adjusted to make its response independent of neutron energy. It has constant efficiency (4.0 ± 0.2 cps/n/cm^2/s) within the energy range of 24 keV to 5 MeV. The system is highly capable of standardizing the commonly available laboratory neutron sources, viz., ^{241}Am-Be, ^{241}Am-B, ^{252}Cf, etc., within short time as compared to $MnSO_4$ bath facility.

Portable Transfer Standard

It is an indigenously developed secondary standard for the measurement of neutron yield. The system works based on the basic principle of precision long

counter (PLC). The system has constant efficiency [(2.85 ± 0.2) counts/n.cm^2] for the neutron energy range from few keV to 10 MeV and also for the commonly available laboratory neutron sources. The weight of the system is about 22 kg. Considering both the demand for onsite neutron source standardization and difficulties to use in-house existing heavy and bulky standards, viz., $MnSO_4$ bath facility and PLC, in the field, this system is highly useful for the said purpose.

Role of RSS in Reactor Criticality and Onsite Testing and Calibration of Reactor Instruments in Protection and Regulatory Channels

The responsibility for safe installation of nuclear power plant in India is vested in Safety Review Committee for Operating Plants (SARCOP). It is essential that procedures leading to criticality be conducted according to documented technical specifications that are approved by SARCOP. Based on the reviews of the Unit Safety Committee and on its own assessment, SARCOP makes recommendations for its implementation. Different types of neutron detectors, viz., ^{10}B-lined proportional counter, fission counters, BF_3 detectors (delayed neutron monitoring), etc., are used in a nuclear power reactor to monitor a neutron fluence rate of about 14 decades starting from 10^0 to 10^{14} n/cm^2/s. According to documented commissioning procedure, these neutron counters designated for the central thimble (in-core) and the vault (out-of-core) as well as the ion chambers used in the "Regulatory, Primary Shut-down System and Secondary Shut-down System" have to be source checked prior to heavy water flushing of moderator and auxiliary systems. These counters are divided into source range monitors (10^0 to 10^5 n/cm^2/s), intermediate range monitors (10^4 to 10^9 n/cm^2/s), and power range monitors (10^8 to 10^{14} n/cm^2/s). For proper functioning of the detector system, the detectors have to be properly tested before and after installing in the reactor (Sathian et al. 2008). In order to evaluate different operating characteristics and the thermal neutron sensitivity of these detectors, several systems are used which are described below:

Thermal Neutron Howitzer

It consists of a polythene (block 41 cm × 37 cm × 37 cm with a 15 cm deep, 16 cm cavity on the front surface) assembly and a 5-Ci Am-Be neutron source. Neutrons after multiple scattering in the moderating assembly emitted in the forward direction provide a wide beam of thermal neutrons, thus acting as a howitzer. The distribution of thermal neutron fluence rate at different distances was measured with a standard BF_3 counter which is used for detector calibration purposes.

Portable Thermal Neutron Water Jig

It is a portable thermal neutron assembly used for the performance evaluation studies of neutron detectors. It consists of a cuboid tank [32 cm × 32 cm × 35 cm] filled with water to a fixed level. There is a stainless steel cover at the top with two empty

tubes for keeping neutron source and the ion chamber, respectively. A 5 Ci ^{241}Am-Be neutron source is placed at the source tube. The thermal neutron fluence rate over the length of the tube where ion chamber is placed and also at three different preset distances, viz., 6 cm, 12 cm, and 18 cm from the surface of the tank (source side), is measured using ^{197}Au foil activation technique. The system is regularly used for the onsite testing and calibration of startup counters and DNM detectors used in power reactors. Various neutron detectors calibrated at different reactor sites are given in Table 1.

Table 1 Details of onsite detector calibration at reactor sites

Reactor site	Calibration year	Number of detectors calibrated at site		
		Ion chambers	Startup counters	DNM
RAPS-2	1980	14	–	
MAPS	1982	14	–	
MAPS-1	1983	6	–	
MAPS-2	1985	NA	NA	
NAPS-1	June 1988	12	–	
NAPS-1	July 1990	19	–	
NAPS-2	August 1991	12	–	
KAPS-1	May 1992	24	11	
KAPS-2	June 1994	17	7	
KAPS-2	November 1994	12	4	
KAPS	1995			35
NAPS	1996			34
RAPS-3	October 1999	11	8	34
KGS-2	August 2000	11	14	
KGS-1	September 2000	13	8	34
RAPS-4	October 2000	9	9	
TAPS-4	January 2005	12	8	28
TAPS-3	March 2006	9	8	28
KGS-3	December 2006	11	9	34
NAPS-1	April 2007		10	34
AHWR CF	January 2008	6	4	2
RAPS-5	May 2009	6	8	34
RAPS-6	November 2009	6	10	34
RAPS-2	June 2009		6	
KAPS-1	December 2010		7	
TAPS-3	May 2013		3	
TAPS-3&4	August 2017		4	
KAPS-2	August 2018		8	
New Apsara (NRF)	August 2018		3	
KAPS-1	May 2019		7	
KAPP-3	February 2020	12	10	46

Metrology and Radiation Processing Industries

Radiation processing technologies are now very widely used in different industrial applications like sterilization of healthcare products including pharmaceuticals; processing of food and agriculture products for various end objectives, such as disinfestation, shelf life extension, sprout inhibition, pest control, sterilization, etc.; and material modification such as polymer cross-linking, chain scission, gemstone colorization, etc. Besides these well-established uses, new radiation processing applications are emerging for environmental remediation like flue gas treatment for SOx and NOx removal, wastewater purification, sludge hygienization, etc.; synthesis of advanced materials and products and medical applications such as irradiation of blood and different biological samples; etc.

Proper dosimetry is the backbone of success of these applications. Conventional radiation measuring techniques like ionization chambers, calorimeters, etc. are unable to measure absorbed doses for most of these applications. Chemical dosimetry techniques provide a cost-effective alternative for radiation processing dosimetry.

Standards for high dose are established and maintained in laboratories belonging to the BARC for calibration and standardization of high-dose radiation processing applications. Fricke dosimeter is maintained as a reference standard according to ISO/ASTM 51026 (ISO 2015a) for standardization of other dosimeters. Gamma chambers used for irradiation are calibrated using the Fricke dosimetric system; according to ISO/ASTM 52116 (ISO 2013a) as well as to ISO/ASTM 51607 (ISO 2013b), alanine ESR is the internationally accepted reference/transfer standard dosimeter, and glutamine is the indigenously developed transfer standard dosimetry systems, which are currently being used for dosimetry in radiation processing. These standards provide results with uncertainty $<3\%$ at coverage factor $k = 1$ according to ISO/ASTM 51707 (ISO 2015b).

Radiation Processing Applications Involving Chemical Dosimetry

Food Irradiation

Radiation processing of food is done using gamma rays, accelerated electrons, and X-rays. Radiation processing of food can achieve different objectives in varied foods.

In India, the Atomic Energy (Radiation Processing of Food and Allied Products) Rules (2012) are used as guideline by food irradiation facilities.

Medical Sterilization

Radiation sterilization using ionizing radiation is a well-established industrial process in India. It is extremely an efficient and convenient technique for achieving a high level of sterility in medical supplies. Hermetically sealed medical products in cardboard cartons or aluminum tote boxes are generally exposed to a minimum dose of 25 kGy. Exposure of medical products, manufactured under Good Manufacturing Practices (GMP), to a minimum radiation dose of 25 kGy ensures sterility assurance level (SAL) of the order of 10^{-6}. This dose provides an extremely high safety factor.

Sludge Hygienization

Pilot and industrial-scale installations have demonstrated that radiation processing-based technologies for flue gas treatment for Sox and NOx removal, wastewater purification, and sludge treatment can help to mitigate environmental pollution.

Ionizing radiation interacts with the critical molecules like DNA, proteins, and water present in the cell and results in the inactivation of microorganisms. Based on microbiological inactivation, dry- and wet-type sludge hygienization facilities are already operational in India. The by-products from these are used in fertilizer industry.

Blood Irradiation

Irradiation of blood with low dose of ionizing radiation is a very widely used popular technique to inhibit T-lymphocyte proliferation to help prevent transfusion-induced graft-versus-host disease (GVHD) in certain immunocompromised patients and those receiving related donor products.

Treating of the blood with ionizing radiation is considered a sure way to prevent such complications.

Material Modifications

The most widely used application of this technology is the cross-linking of polymer chains: for producing cross-linking in wire and cable insulations, car tires, or vulcanization of natural rubber latex for medical items. The method produces cross-linking in polymers at room temperature and produces superior material properties without toxic chemical residues.

The exact opposite effect of cross-linking is the controlled degradation of polymers also can be achieved through radiation, also called chain scission, and is used to improve the properties as well as the compatibility and processing parameters of both natural and synthetic polymers. The largest commercial application of this technology is the degradation of polytetrafluoroethylene (PTFE) by irradiation, which results in a lower molecular-weight micro-powder with improved compatibility with other materials that then can be used as an additive to inks and coatings.

Facilities and Dosimeters for High-Dose Measurement

Gamma Chambers

Gamma chamber is a compact self-shielded cobalt-60 gamma irradiator providing an irradiation volume of approximately 1000 cc as shown in Fig. 12. The material for irradiation is placed in an irradiation chamber located in the vertical drawer inside the lead flask. This drawer can be moved up and down with the help of a system of motorized drive which enables precise positioning of the irradiation chamber at the center of the radiation field. Radiation field is provided by a set of stationary cobalt-60 sources placed in a cylindrical cage. The center position of the irradiation volume is calibrated by using Fricke reference standard dosimeter and used as a standard source for irradiation of other chemical dosimeters.

Fig. 12 Gamma chamber 1200

UV-Visible-NIR Spectrophotometer

It has high-sensitivity, high-resolution, ultralow stray light, and wide wavelength range. This UV-visible-NIR spectrophotometer as shown in Fig. 13 is used for absorbance, transmission, and reflectance measurement of dosimeters and other samples. It is also used for dose verification exercises and carried out for radiation processing facilities in India using indigenous and imported dosimeters. This spectrophotometer is calibrated regularly as per Indian Pharmacopoeia (Pharmacopoeia 2018).

Fricke Dosimeter

Fricke dosimeter is the best known and most broadly applied chemical dosimeter. It has been accurately standardized by calorimetry and other physical methods. It is used as a reference standard according to ISO/ASTM 51026 for absolute absorbed dose measurements with an uncertainty of 1–2% (2σ).

The principle is based on a process of oxidation of ferrous ions to ferric ions in acidic aqueous solution by ionizing radiation. Irradiation temperature range should be within 10–60 °C. It has useful dose response from 20 to 400 Gy.

FBX Dosimeter

This low-level dosimeter developed in RSSD, BARC, consists of ferrous ammonium sulfate benzoic acid xylenol orange (FBX) in dilute solution of sulfuric acid in aerated water, a chain reaction with benzoic acid (BA) whereby a large number of

Fig. 13 Shimadzu UV 3600 plus UV-visible-NIR spectrophotometer

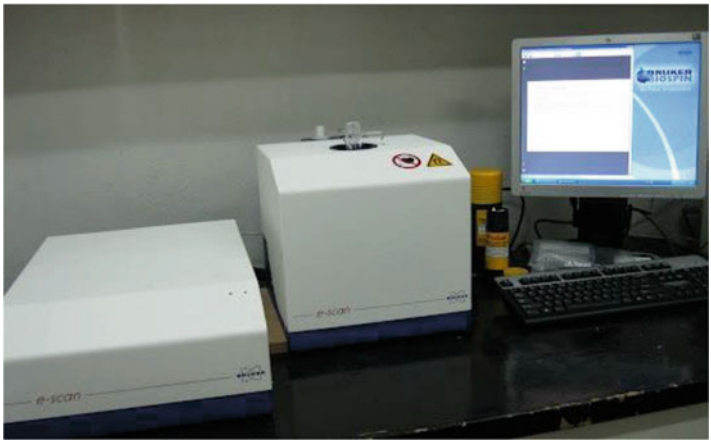

Fig. 14 ESR spectrometer

ferrous ions are oxidized to ferric ions. Xylenol orange (XO) is a dye, which forms a colored complex specifically with ferric ions. This helps in measuring lower concentrations of ferric ions using a spectrophotometer. The presence of BA and XO sensitizes the response and enables this dosimeter measure doses in the therapy level. This dosimeter is capable of measuring doses in the range 0.2 to 50 Gy, with an uncertainty of $\pm 1\%$ (2σ).

Alanine ESR Dosimeters

Amino acids on irradiation form free radicals. Concentration of these free radicals can be related to absorbed dose by electron spin resonance (ESR) analysis. The use of α-L-alanine has shown especially good characteristics for medium- and high-dose measurements. The commercially available alanine ESR dosimetry system as shown in Fig. 14 is used in the range of 10(International Atomic Energy Agency 2000)–10^5 Gy with a precision of 1% (2σ).

Glutamine Dosimeter

The glutamine powder can be used for measurement of absorbed dose in the range of 0.1–100 kGy by using spectrophotometric readout method. Free radicals are formed

on irradiation of glutamine powder. The radiation-induced free radicals indirectly oxidize ferrous ions to ferric ions in ferrous xylenol orange solution. Absorbance of the resulted complex is related to absorbed dose.

Regulatory Role of RSS in Food Irradiation Facilities

Licensed radiation processing facilities have to comply with conditions of approval, operation, and process control prescribed under the Atomic Energy (Radiation Processing of Food and Allied Products) Rules, 2012.

The Department of Atomic Energy (DAE) provides operating license to the irradiator after performing dosimetry exercises through the BARC:

(i) Dose intercomparison and verification exercises for all food irradiation facilities.
(ii) After qualifying the tests, the BARC issues dosimetry certificates; based on these certificates and recommendations of the Dose Verification Committee, license is issued by the DAE to the radiation processing facility.

Protection-Level Radiation Instruments for Measurement and Their Traceability

Radiation monitoring instruments are backbone to all radiological and nuclear installations for the safety of occupational radiation workers. Performance of these monitors depends on calibration and measurement traceability to the national standards. The Bhabha Atomic Research Centre, being the designated institute for ionizing radiation metrology in India, develops and maintains various standards related to radiation protection (gamma rays and X-rays) covering air kerma and ambient dose equivalent measurements for X-ray and gamma dosimetry. Protection-level radiation monitors cover a wide range of radiation field (gamma ray, X-ray, and beta ray) intensity and energy; therefore, measurement standards of various types and sizes are established. The BARC has also established various radiological facilities for gamma rays, X-rays, and beta rays which are used for calibration and testing of all kinds of protection-level radiation monitors. The calibration and testing (for prototype monitors) are carried out as per recommendations of relevant international standards like the IEC, ISO, ANSI, etc.

Periodic calibration of all kinds of protection-level radiation monitors is required to ensure that it is working properly and is suitable for its intended purpose. The BARC provides calibration, response testing, type testing, gamma ray/X-ray irradiations, and onsite standardization of reference radiation field used for protection-level radiation monitors for field ranging from 10^{-6} to 10^{2} Gy/h. These monitors are frequently used by health physicists and radiological safety officers for smooth functioning of power plants and other radiological laboratories/facilities. The BARC also provides traceability to national standards through onsite standardization

of reference radiation field for TLD units of various DAE organizations, BARC/AERB accredited TLD Laboratories, and Calibration Laboratories for radiation monitoring instruments (RMIs). It also provides technical support and guidance for establishment of calibration laboratory for protection-level RMIs. Further, it is also actively involved in the activities related to the regulatory aspects of defense and space organizations. Calibration facilities are also used by various institutes in developmental studies for radiation monitors, dosimeters, shielding materials, and algorithms for personal dosimetry.

The BARC also establishes degree of equivalence in measurement standards through participation in various international intercomparisons with other National Metrology Institutes.

Standards for Radiation Protection

Ionization chambers, the simplest of all gas-filled detectors, are widely used as measurement standard for ionizing radiation dosimetry (energy deposited per unit mass). Protection-level radiation monitors cover a wide range of radiation field intensity and energy; therefore, standards of various types and sizes are established. These standards include the following:

100.9-cc graphite ionization chamber (air kerma-based reference standard): It is an air kerma-based cavity ionization chamber (Fig. 15) which was designed and established by the BARC as a reference standard for protection-level applications/calibrations. It is having adequate sensitivity with a near flat energy response in the energy range 30–2000 keV.

Free air ionization chamber (X-ray dosimetry): Free air ionization chamber (FAIC) (Fig. 15), a primary standard for air kerma/exposure, is a parallel plate-type ionization chamber. For radiation protection applications, the BARC has established various ISO 4037-1 (ISO 2019) specified narrow series and wide series X-ray beam qualities, with an average energy ranging from 12 to 210 keV, using this FAIC. These beams are widely used for calibration and type testing of various protection-level dosimeters/radiation monitors.

Fig. 15 Standards for radiation protection

Air kerma-based secondary standard (PTW 0.3 cc, 30 cc, and 1000 cc IC's): The BARC has also procured and is maintaining PTW; Germany make three secondary standard ionization chambers (Fig. 15) having nominal sensitive volumes of 0.3 cc, 30 cc, and 1000 cc. These standards are frequently used for in-house as well as onsite standardization of gamma reference radiation fields.

Calibration of Radiation Monitoring Instruments (RMIs)

Radiation monitoring instruments are used both for area monitoring and for individual monitoring. For area monitoring, radiation monitors like area gamma monitors (AGM) and radiation survey meters (RSM) area used. For personal monitoring, instruments recording the personal dose equivalents are used and are referred to as personal dosimeters (or individual dosimeters). Periodic calibration of these area and personal monitors (Fig. 16) is required to ensure that they are working properly and are suitable for their intended purpose. Periodic calibration is also a regulatory requirement for all the facilities handling ionizing radiation. The BARC carries out periodic calibration of all kinds of protection-level radiation monitors (installed in almost all the facilities of the BARC) for dose rate range 1 μGy/h (~1 μSv/h) to 100 Gy/h (~100 Sv/h). Annually, ~300 radiation monitors are calibrated at the gamma protection-level calibration facility of the BARC. Periodic calibration of portable radiation monitors and TLD reader system of other DAE units, private institutions, and defense establishments like SSS, Visakhapatnam, INS Chakra, etc. are also carried out.

Establishment of Calibration Laboratories: Role of the BARC

Periodic calibration of radiation monitors is also carried out by other government and private laboratories. Presently, there are ten calibration laboratories which are accredited by the National Accreditation Board for Testing and Calibration Laboratories (NABL) or recognized by AERB. Out of these, three are DAE institutes, namely, Indira Gandhi Centre for Atomic Research (IGCAR), Kalpakkam; Electronics Corporation of India Limited (ECIL), Hyderabad; and Board of Radiation and Isotope Technology (BRIT), Vashi. One institute is in defense sector, i.e., Defence Laboratory, Jodhpur. The remaining laboratories are in the private sector spread across the nation (Fig. 17). These laboratories have either ^{137}Cs or ^{60}Co

Fig. 16 Various types of radiation monitors calibrated at BARC calibration facility

Fig. 17 Private (red), government (yellow), and defense (blue) laboratories recognized by AERB and accredited/under accreditation by NABL, India

sources of various activities (mCi – kCi) for use during periodic calibration of radiation monitors, TLD badge readers, and irradiation of dosimeters. The BARC provides measurement traceability to all these laboratories via onsite standardization of reference radiation field. During source output standardization, radiation field intensity at multiple distances, validation of inverse square law, repeatability of dose rate at a position, and attenuation ratio of lead attenuators are carried out using appropriate measurement standard. Technical help in terms of requirements of establishing a calibration lab and training of staff for calibration, etc. is also provided to these calibration laboratories.

Traceability to Calibration Laboratories

Calibration of secondary standard ionization chambers (along with electrometer), maintained by calibration laboratories, is carried out for ^{137}Cs, ^{60}Co, and ^{241}Am energies to provide measurement traceability to RMI calibration laboratories. These calibrated ionization chambers are used for radiation field verification/standardization by RMI calibration laboratories.

Radiological-Type Testing of Radiation Monitors

Newly developed radiation monitors are required to undergo type testing in accordance with national/international standard to verify the manufacturers' claim (specification) before introducing them in the market. The testing parameters vary around the world for a given product and are dependent on the applicable national/international standard. Type approval means that the equipment or product is certified to meet certain requirements for its type, whatever that may be, as described in the relevant standard. The BARC is the only institution which performs radiological-type testing on prototype or newly developed protection-level radiation monitors (Fig. 18) submitted by any private/government organizations in the country, and a test report is granted to a product meeting technical criteria specified by the applicable standard.

The BARC also provides technical help and assistance in the development of new radiation monitor/dosimeter (Fig. 19), in terms of establishing linearity in response, filter optimization studies for achieving flat energy response, and measurement of detector sensitivity, to various users (government and private labs).

IAEA Postal Dose Quality Audit for ^{137}Cs

Aiming to check quality of radiation protection calibrations provided by the SSDLs and assure proper dissemination of dosimetry standards to radiation protection institutions, the IAEA carries out TLD-based postal quality audit to its member states. The IAEA supplies TLDs to the laboratories and asked them to irradiate it to 5-mGy air kerma. The laboratory sends back the irradiated dosimeters to the IAEA for evaluation. The BARC periodically participates in the IAEA postal dose quality audit for ^{137}Cs beam. In the latest TLD-based quality audit (Table 2), the IAEA

Fig. 18 Few indigenous radiation monitors subjected for radiological-type testing by the BARC

Fig. 19 Few radiation monitors/dosimeters developed using BARC facilities

Table 2 IAEA postal quality audit result. Results of TLD air kerma measurements for Cs-137

Radiation unit	Beam	TLD set#	User (stated) air kerma (mGy)	IAEA (measured) air kerma (mGy)[a]	IAEA mean air kerma (mGy)	% deviation relative to IAEA mean air kerma[b]	$\frac{\text{IAEA mean air kerma}}{\text{User stated air kerma}}$
BRIT (make), IRC-2	Cs-137	RP1119	5.00 5.00	5.05 5.03	5.04	−0.8	1.01

Agreement within ±7% between the users stated air kerma and the IAEA measured air kerma is considered satisfactory

[a]The relative combined standard uncertainty in the TLD measurements of the air kerma is 1.7%

[b]% deviation relative to IAEA mean air kerma = 100 × (user stated air kerma – IAEA mean air kerma)/IAEA mean air kerma). A relative deviation with negative (positive) sign indicates that the user estimates lower (higher) air kerma than what is measured

evaluated dose was within ±1% of the BARC irradiated dose (IAEA acceptance criteria: ±7%).

Conclusion

Bhabha Atomic Research Centre (BARC) is the designated institute in ionizing radiation metrology for the country. BARC designs, develops, establishes, maintains, and disseminates various national standards for physical quantities of ionizing radiation in India. IAEA/WHO has recognized BARC as a Secondary Standard Dosimetry Laboratory (SSDL) for disseminating radiological standards to users in the country. SSDL-BARC provides traceable calibration to the users in the field of teletherapy, brachytherapy, radiation protection, and diagnostic radiology. The pace of the advances in radiation therapy with the expected improvement in patient treatment has brought out the need for greater accuracy in the treatment process. SSDL-BARC participates in international intercomparison programs to establish its equivalence with international standards, thereby validating the calibration procedure followed in the laboratory. BARC conducts the postal quality audit of output of therapy centers to ensure the accuracy of measurement. Radioisotope targeted therapy has been employed in the treatment of wide variety of cancers. The activity of these radiopharmaceuticals at the nuclear medicine centers is measured using the radionuclide calibrators (RC). BARC as the national metrological laboratory for radioactivity has been conducting the audit program for activity measurements with RC for more than four decades. BARC has established several systems for the activity measurements. Neutron sources/fields not only are critical for operating and regulating nuclear reactors but also are important for various other research, industrial, and radiation protection applications. Therefore, precise value of neutron source strength and resultant field is highly essential which is provided by the BARC using several neutron standards developed for the purpose. Radiation processing technologies are very widely used in different industrial applications like sterilization of healthcare products, processing of food and agriculture products, material

modification and emerging fields like environmental remediation, synthesis of advanced materials and products, and medical applications such as irradiation of blood. Proper dosimetry is the backbone of success of these applications. Conventional radiation measuring techniques like ionization chambers, calorimeters, etc. are unable to measure absorbed doses for most of these applications. Chemical dosimetry techniques provide a cost-effective alternative for radiation processing dosimetry. Radiation monitoring instruments are backbone to all the radiological and nuclear installations for the safety of occupational radiation workers. Performance of these monitors depends on calibration and measurement traceability to the national standards. Protection-level radiation monitors cover a wide range of radiation field (gamma ray, X-ray, and beta ray) intensity and energy; therefore, measurement standards of various types and sizes are established. The BARC has also established various radiological facilities for gamma rays, X-rays, and beta rays, which are used for calibration and testing of all kinds of protection-level radiation monitors. Development of various radiation standards has helped in improving the accuracy in application of radiation.

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Occupational Radiation Monitoring in Indian Nuclear Industry

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Abstract

Radiation monitoring for safety of occupational worker involves measurement of various radiation quantities in field and on person. This chapter unravels various measurement techniques and procedures applied for measurement of radiation quantities for occupational safety in Indian nuclear fuel cycle facilities. How far the radiation metrology principles are embedded in these measurements is explained. The country is engaged in all aspects of nuclear fuel cycle – from uranium mining to spent fuel reprocessing and nuclear waste management. Hence, diverse and complex radiation monitoring practices are evolved for

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protection of occupational workers employed in these facilities. Occupational radiation monitoring has different requirements such as workplace monitoring, individual monitoring, and quality assurance for monitoring methods and practices. All the methods used in the country for occupational radiation monitoring conform to the national regulations and international guidelines. However, many of the techniques in radiation monitoring used in the country are indigenously developed and sometimes differ from the most common practices applied elsewhere. Monitoring for external exposure from photons and β-particles is being carried out using the indigenously developed $CaSO_4$:Dy thermoluminescent dosimeter (TLD) badge, and this technique is more sensitive than many other similar dosimeter available in the market. Measurement of exposure to neutron is being carried out using the solid-state nuclear track detector (SSNTD) method employing poly-allyl-di-glycol carbonate (PADC) polymeric film as the detector. Extensive laboratory set-up along with technical expertise has been developed for detection of intake of radionuclide into human body and estimation of dose resulting from such intake. Both in vivo and in vitro methods are extensively used for this purpose. The procedures and equipment adopted in workplace monitoring are well-designed to promptly detect and warn the radiation workers on hazardous radiological status, if any, developing in working areas. The effectiveness of the radiation monitoring program is evident in the gradual reduction of collective dose and average individual dose received by the workers in nuclear fuel cycle facilities operational in the country.

Keywords

Radiation protection · Occupational radiation · Fuel cycle · Internal exposure · External exposure

Introduction

Our understanding of harmful health effects of ionizing radiation progressed parallel to the development of nuclear energy and uses of radio isotopes and radiation (UNSEAR 2019). Analysis of the large quantum of epidemiological data on harmful effects of ionizing radiation on exposed population has resulted in setting up limits on dose resulting from exposure to radiation. The dose limits for occupational radiation worker practiced in the system of radiation protection is intended to minimize the stochastic effect (cancer and genetic effects) and prevent all tissue reactions (deterministic effects) to those who get exposed to ionizing radiation. Based on the consensus that exposure to any quantity of radiation may have adverse health effects (UNSEAR 2021), estimation of radiation dose to occupational workers is mandatory in all practices employing radioactivity and radiation. Measurement of relevant radiological quantities by various techniques is the key to estimate the dose received by workers while handling radiation and radioactive materials. The robust system of radiological protection based on radiation measurement at workplace and

on person is a most vital component for safe and continued use of nuclear technology.

It is mandatory to follow the principles of metrology for ensuring accuracy in measurement of a quantity to the desired level. However, radiation measurements done for occupational radiation protection is often too complex to directly apply the principles of radiation metrology. However, the basic intention of these measurements is protection of human health, and as a result, highest level of accuracy and reliability is expected in results obtained from these measurements. This practical difficulty is resolved by continuous improvements in measurements of radiation quantities intended for radiological protection through a multipronged approach. Application of relevant codes and standards in measurement procedures makes many measurements closer to metrology requirements. Also, continuous research and development in radiation detection and measurement techniques often result in improving the accuracy in measurements. This chapter on occupational radiation monitoring explains the radiation measurement practices for radiation protection followed in nuclear fuel cycle facilities in the country and also its relationship to the radiation metrology fundamentals.

Overview of Indian Nuclear Fuel Cycle Activities

Nuclear energy program in India is more than seven decades old. The country has developed expertise in almost all stages of nuclear fuel cycle. All nuclear fuel cycle activities in the country are under the ambit of the Department of Atomic Energy (DAE), the central government department dedicated to the development of nuclear energy and allied technologies in the country. Each activity such as uranium mining and processing, nuclear fuel fabrication, nuclear power reactor design and operations, etc. is under separate organizations within the DAE. Uranium Corporation of India Limited (UCIL), responsible for uranium mining and processing, operates six underground mines (Bagjata, Jaduguda, Bhatin, Narwapahar, Turamdih, and Mohuldih) and one open pit mine (Banduhurang) in the State of Jharkhand. Ore produced from these mines are processed in two process plants located in Jaduguda and Turamdih. UCIL also operates an underground mine and a processing plant in Tummalapalle in Andhra Pradesh. Fuel fabrication activities for commercial nuclear power plants (NPPs) are performed at Nuclear Fuel Complex, Hyderabad. At present, there are 22 NPPs (2 boiling water reactors (BWR), 18 pressurized heavy water reactors (PHWRs), and 2 Russian design water-water energetic reactors (VVER)) operating in the country with a total installed capacity of 6780 MW. Nuclear Power Corporation of India Ltd. (NPCIL) is the utility organization responsible for all commercial nuclear power production and related activities. DAE operates three large nuclear fuel cycle facilities, two located in the state of Maharashtra (Trombay and Tarapur) and one in Kalpakkam in Tamil Nadu. Facilities at these sites are engaged in front end research in nuclear science and technology, operation of research reactors, reprocessing of spent fuel, nuclear waste management, and fabrication of MOX fuel for Fast Breeder Test Reactor (FBTR) and the

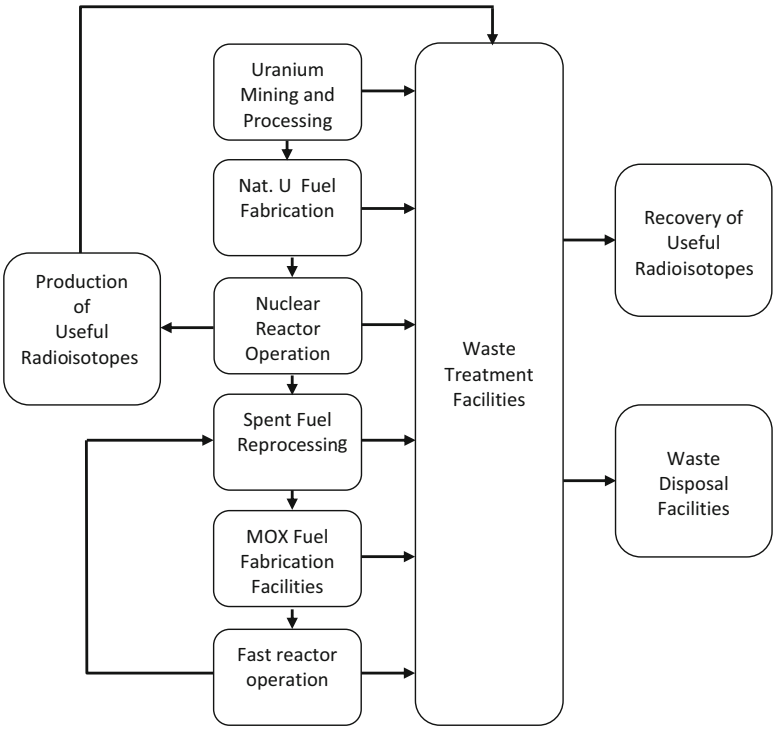

Fig. 1 Various nuclear fuel cycle facilities in the country that can result in occupational radiation exposure

upcoming Prototype Fast Breeder Reactor (PFBR). Radioisotope production and distribution is through a separate organization called Board of Radiation & Isotope Technology (BRIT). The breadth and the size of the activities under DAE lead to a well-developed radiation monitoring program for protection of occupational radiation workers employed in these organizations. Figure 1 represents various nuclear fuel cycle facilities operating in the country that can result in occupational radiation exposure to the workers.

Regulatory Infrastructures for Nuclear Industrial Activities

Occupational radiation protection activities in nuclear industry establishments except that operated by Bhabha Atomic Research Centre (BARC) are governed by the national regulator for safe use of radiation, the Atomic Energy Regulatory Board (AERB). The facilities operated by BARC are regulated by a separate regulatory body, the BARC Safety Council Secretariat (BSCS). Guidance for radiation exposure monitoring and control by both these regulatory bodies are similar and the dose

Table 1 Dose limits prescribed by the AERB/BSCS for occupational workers and public

Category	Annual effective dose limit (mSv)	Annual equivalent dose limits (mSv)			For consecutive 5 years (mSv)	Life-time effective dose (Sv)
		Lens of eye	Skin	Extremities		
Radiation worker	30[a]	150[b]	500	500	100	1
Apprentices and trainees	6	50	150	150	–	–
Temporary worker	15	75	250	250	–	–
Member of public	1	15	50	–	–	–

[a]5-year average annual effective dose = 20 mSv; maximum 30 mSv in a single year
[b]Equivalent dose limit for lens of eye is under review by AERB

limits prescribed by them are identical (AERB 2005a; BSCS 2020). Both AERB and BSCS takes guidance from international expert bodies like International Commission on Radiological Protection (ICRP), International Atomic Energy Agency (IAEA), and International Commission for Radiation Measurements and Units (ICRU), for framing radiation protection rules and implementing radiation protection practices in nuclear industrial activities. Summary of the current dose limits to the occupational radiation worker as prescribed by AERB/BSCS is presented in Table 1. These quantities used for radiation exposure control are explained later in this chapter. In addition to the national regulations, each individual plant/facility has a separate Radiation Protection Manual (RPM), detailing the facility-specific radiation protection procedures and radiation monitoring requirements.

Organizational Setup for Occupational Radiation Monitoring

Prime responsibility for monitoring and control of occupational radiation exposure to workers in a facility lies with its operating management (RPR 2004). Each facility ensures that adequate infrastructure, equipment, and expertise are available for monitoring radiation exposure of workers engaged in the facility. Radiation monitoring of workers at all NPPs is carried out by the Health, Safety & Environment Directorate of NPCIL. All other research and production organizations of DAE also have separate wings of radiation protection professionals dedicated to health physics measurements and radiation protection research. Each nuclear/radiological facility has a designated Radiation Safety Officer (RSO) who is responsible for the radiation monitoring program of a plant/facility. For effectively carrying out the radiological surveillance in each facility, RSO is aided by a team of experienced and certified health physics professionals. RSO advises the operating management on all aspects pertaining to occupational exposure and radiation safety.

Design Safety Features for Occupational Exposure Control in Nuclear Facilities

Nuclear and radiological facilities are designed to minimize radiation exposure to occupational workers and the public during normal as well as accidental conditions. To minimize external exposure to worker, facilities are designed such that ambient dose rate at any continuously occupied areas in a plant is below 1 μSv/hour (AERB 2005a). Additional occupancy restrictions are put in place if any area have ambient dose rate higher than this stipulated value (AERB 2005b). All activities with potentials for radioactivity release into air are carried out inside containments such as fume-hood, glove boxes, and hot cells. Ventilation system in radioactive facilities is well-designed for controlling the spread of radioactive contamination present in air. Air flow is designed from areas with less potential for contamination to that having more potential for the presence of contaminants. Depending on the potential for generation of air contamination, sufficient number of Air Changes Per Hour (ACPH) is provided in operating areas to prevent build-up of air contamination. In general, plant area is divided into four zones; zone 1 to zone 4 with increase in anticipated radiation and contamination levels (AERB 2005a). Allowed radiological conditions, monitoring requirements, and dress regulations for workers for each zone are well defined in RPM of the facility. This helps in monitoring and control of occupational radiation exposure. Areas in nuclear facilities are also divided into controlled areas and supervised areas (AERB 2012). Controlled areas have entry restrictions and have stipulations on area and personal monitoring provisions and use of personal protective equipment. Specific protective measures are normally not needed in supervised areas, but occupational exposure conditions are kept under review.

Metrology in Monitoring for Radiation Protection

Metrology being the science of measurement has some limitations while applying for radiation measurements employed in occupational radiation monitoring. Firstly, the quantities used in occupational monitoring, the so-called protection quantities are not physical quantities, so are not amenable for direct measurement. What is measured is a best estimate of a defined quantity, and the measurement is done with certain assumptions of the radiation field characteristics. Second, the measurement occurs in field and the full characterization of nature and quality of the encountered radiation is most often impractical or too costly to be emphasized. Thirdly, non-quantifiable uncertainty often exists in the assumptions employed for conversion of the measured quantity to the corresponding protection quantity. The ICRP on several occasions acknowledged these limitations in measuring quantities pertaining to occupational exposure. For example, on individual dose measurement the commission states (ICRP 1997);

> In the workplace, where the energy spectrum and orientation of the radiation field are generally not well known, the uncertainties in a measurement made with an individual dosemeter will be significantly greater. Non-uniformity and uncertain orientation of the radiation field will introduce errors in the use of standard models. The overall uncertainty at the 95% confidence level in the estimation of effective dose around the relevant dose limit may well be a factor of 1.5 in either direction for photons and may be substantially greater for neutrons of uncertain energy and for electrons. Greater uncertainties are also inevitable at low levels of effective dose for all qualities of radiation.

All stakeholders involved in radiation monitoring for occupational radiation protection are aware of these limitations. The system of radiation protection currently in use have sufficient safety margin to take care of such uncertainties inherent in the system (NCRP 2007b; Sasaki et al. 2021).

Quantities Used in Occupational Radiation Monitoring

The risk from radiation exposure is its ability to cause ionization in a medium, when it interacts with it. In biological medium like the human body, the damage caused by exposure to ionizing radiation is directly related to the quantum of energy deposited in the body. The quantity of energy deposited per unit mass is called absorbed dose (D). SI units of absorbed dose (D) is (J/kg) with special name Gray (Gy). But the degree of damage to the human cell is different for different types of radiation. To normalize the biological risk from different radiations, the absorbed dose is multiplied by the radiation weighting factor (W_R). The resultant quantity is called equivalent dose (H). Since W_R is a dimensionless quantity, unit of equivalent dose is also J/kg, with a special name Sievert (Sv).

There are two instances in which the whole body is not uniformly irradiated, viz., while working in non-uniform radiation field and during internal contamination. In non-uniform radiation field, some part of the body gets more radiation exposure than some other parts. Radionuclide can enter into human body by different mechanism such as inhalation (of contaminated air), ingestion (via contamination in food), and injection (via wound and cuts) (IAEA 1996). This could result in preferential accumulation of radioactivity in specific organ(s). Radio-sensitivity of organ differs. In order to convert the risk from exposure to a specific organ to that to the whole body, in such cases, an additional factor called tissue weighting factor (W_T) is used. The organ equivalent dose (H_T) corresponding to the specific organ "T" is multiplied by its W_T to get another quantity called effective dose (E) (ICRP 2007). The resultant new quantity, effective dose (E) is used to compare the overall risk from different exposure scenarios. When multiple organs are exposed, the effective dose calculated for each organ is summed to get the total effective dose (E). The value of W_T for any particular organ is less than one and tissue weighing factors for all organs in the body add up to unity (ICRP 2007).

W_R is 1 for photons and beta particles of all energies. For uniform whole-body exposure from γ, X-rays, and β radiations, the equivalent dose (H) falls to be the same as effective dose (E). For neutrons, W_R is a continuous function of neutron

energy, varying in the range of 2.5–20 with a maximum for neutrons having energy around 1 MeV (ICRP 2007). Hence, in case of exposure to neutrons, neutron energy spectrum is essential to estimate the effective dose (E).

The defined quantities, equivalent dose (H) and effective dose (E), are not directly measurable quantities and are called protection quantities. For practical measurement, ICRU has defined operational quantities which are related to the protection quantity, effective dose (E). The operational quantity defined for measurement of individual dose due to exposure from external radiation is termed as personal dose equivalent and is denoted as $H_p(10)$. Roughly, this is the dose delivered by the radiation at 10 mm depth in body and is applicable to deeply penetrating radiation such as photons and neutrons (ICRU 1993; ICRU 2011). An equivalent quantity for measurement of dose persisting in the working area is termed as ambient dose equivalent and is denoted as $H^*(10)$ (ICRU 1993, 2011).

It is observed that dose control in terms of effective dose alone is inadequate to protect two organs, namely, the skin and lens of eye. Also, due to the proximity to the source, hands and feet may receive relatively higher dose and are specially referred as dose to the extremities. Separate dose limits are established for extremities also. Skin at 20–100 μm depth is very sensitive to radiation (Curran 1986). The protection quantity corresponding to the equivalent dose to skin is the personal dose equivalent $H_p(0.07)$. This is the dose delivered by the radiation at 70 μm depth below skin surface. When exposure of the extremities is of concern, the skin of the extremities is more likely to become the limiting tissue, rather than the extremity itself. Thus, an extremity dosimeter essentially becomes a skin dosimeter and should be designed to measure $H_p(0.07)$. It should be noted that radiation like β-particles which may not be able to penetrate deep into the body to give dose to the internal organs also could result in skin dose. Skin dose evaluation is integral to most occupational radiation monitoring program.

For measurement of radiation dose to eye lens, another protection quantity called personal dose equivalent $H_p(3.0)$ is recommended. This is the dose measured at 3 mm depth. ICRP recommends equivalent dose limit of 20 mSv to the lens of eye (Bolch et al. 2013), though the national regulator still continues with the previously recommended limit of 150 mSv equivalent dose. However not many instances exist in nuclear fuel cycle activities, where eye lens receive more dose compared to the whole body exposure. Unless specific requirement arise, measurement of the eye lens dose is not a routine practice in nuclear fuel cycle facilities in the country.

Surface contamination is measured in the unit Bq/cm^2. Contamination on a surface can be fixed contamination or transferable contamination. The radiation hazard from fixed contamination is only external exposure. Transferable contamination has additional hazards such as further spread of contamination and generation of air contamination by re-suspension. Hence, transferable contamination has the potential to cause internal exposure via inhalation and ingestion. Thus, separate measurement techniques are always used to quantify and report fixed and transferable contamination separately.

Air contamination is usually represented in Bq/m^3. The quantity, derived air concentration (DAC) in Bq/m^3, is also used to report the results from measurements.

DAC is related to the annual limit on intake (ALI) for inhalation of the radioactive species by the relation

$$\mathrm{DAC} = \frac{\mathrm{ALI}(Bq)}{2400\ \mathrm{m}^3}$$

ALI depends on the radio-toxicity of the chemical species and its deposition characteristics in the respiratory system based on its particle size. The overall inhalation hazard of a specific species at given particle size is represented by its dose coefficient factor (DCF) in Sv/Bq. As 20 mSv is the yearly limit for effective dose for occupational radiation worker, ALI for a specific radioactive species at a given particle size could be obtained from the published DCF using the relation:

$$\mathrm{ALI(Bq)} = \frac{0.02\ \mathrm{Sv}}{\mathrm{DCF}\left(\frac{\mathrm{Sv}}{\mathrm{Bq}}\right)}$$

IAEA has published full list of DCFs for various radioactive materials encountered at workplace (IAEA 2014).

The dose included in DCF is the cumulative dose received by the body from the incorporated radioactivity, till its elimination by radioactive decay and biological process. A quantity called effective half-life in body or organ is used to account for the elimination process of radioactive material from body. For occupational worker, effective dose is computed up to 50 years subsequent to an intake and is represented as E(50) (ICRP 2015). Age of worker at the time of intake is not considered in this computation. This effective dose from an intake is called committed effective dose (CED) and is accounted to worker's dose account to the year in which the intake occurs. Note that CED may not be the actual dose received to the worker in that specific year. For radiochemicals having short effective half-life in the body, the entire exposure will happen in the year of intake itself. Typical example is tritium (HTO) with effective half-life of about 10 days or less. But for radionuclides with long retention time in the body, such as compounds of ^{239}Pu, the exposure is fractionated over the life time of the individual.

A different unit called working level month (WLM) is employed for measurement of exposure to radon and its progenies by workers of uranium mines. One working level is the quantity of short-lived daughter products of ^{222}Rn in any combination in one liter of air that will result in ultimate emission of alpha particles of cumulative energy 1.3×10^5 MeV. This corresponds to 100 pCi/L of ^{222}Rn gas in equilibrium with its progenies. 170 hours of work in one WL concentration results in internal dose corresponds to one WLM. To know the WL in an area, one needs to have both Rn concentration and its equilibrium status with its progenies. Both these parameters are experimentally measured in uranium mines.

Ambient dose rate due to photons are measured in exposure units milli-Rontgen/hour (mR/hr). Locally available portable or installed instruments are not conforming to measurement of photon dose rate in terms of $H^*(10)$. Many commercial portable instruments used in facilities read the ambient dose rates due to photons in terms of

absorbed dose rate (Gy/h) and equivalent dose rates (Sv/h). But there are no specific directives from the regulatory body on the units to be used for ambient dose rate measurement at workplace. Ambient neutron dose rate is measured in terms of the equivalent dose rate in milli-Rem/hour (mRem/h), and the energy spectrum of neutron field is taken care in these measurements. Neutron dose rate meters in use are conforming to the protection quantity $H_p(10)$.

ICRU recommends the use of $H_p(10)$ to measure the whole body dose for deeply penetrating radiation. Usual method is to use a tissue equivalent material as dosimeter (e.g., TLD-100, chemical formula LiF:Mg,Ti). In India, measurement of external dose from photons and beta particles is done with TLD using $CaSO_4$:Dy phosphor (Vohra et al. 1980; Pradhan 1981). Though the $CaSO_4$:Dy phosphor is not tissue equivalent, it has several advantages in comparison to TLD-100. It is almost 30 times more sensitive than TLD-100. This phosphor is amenable to large-scale preparation and has simple annealing procedure. With the use of filters over the dosimeter and proper dose evaluation algorithm, the response of this dosimeter is made almost close to tissue equivalent dosimeter. The quantity measured for external radiation using this dosimeter is the whole-body dose which closely resembles the operational quantity $H_p(10)$ in most exposure scenarios (Bakshi et al. 2007; Pradhan et al. 2009). Dose to the extremities (wrist) and skin is measured as "wrist dose" and "skin dose," respectively, using the same $CaSO_4$:Dy TLD badges. Here the reported skin dose or extremity dose is the sum of dose due to photons and electrons. Neutron dose is measured in terms of $H_p(10)$ using the CR-39-based dosimeter (Griffith et al. 1981; BARC 2011). Whole body dose equivalents measured using the TLD badge and the neutron dose measured using SSNTD are the legal quantities used for control of effective dose to the worker. Skin and extremity dose are equivalent dose quantities (in mSv) used for exposure control to these two specific organs.

Common Monitoring Practices for Control of Occupational Exposure

Occupational radiation monitoring has two major components: workplace monitoring and individual monitoring. In most cases, the synergy between these two is practically the best way to optimize the utilization of resources. Both workplace monitoring and individual monitoring have several components as depicted in Fig. 2. All nuclear facilities in the country practices both workplace monitoring and individual monitoring for control of radiation exposure.

Workplace Monitoring

Every nuclear/radiological facility is designed to minimize radiation exposure to occupational worker and member of the public. These safety features in design are the most effective and critical component for control of radiation exposure. Workplace monitoring is essential in a facility to ensure that the designed safety features

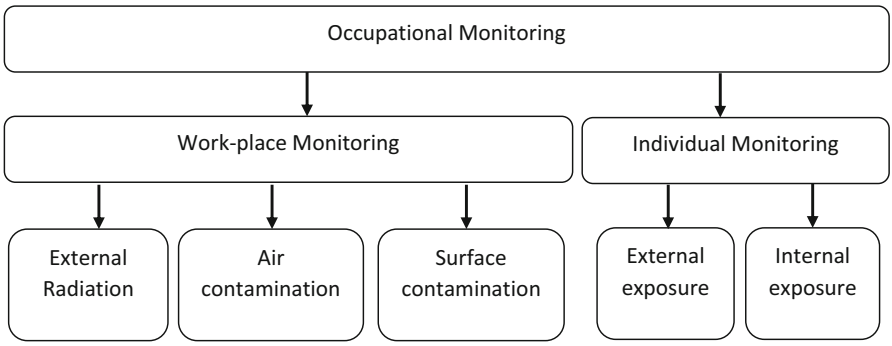

Fig. 2 Various components of occupational radiation exposure monitoring

for exposure control in the facility are functioning as per the design indent. Work-place monitoring give good assessment of the dose that worker will receive and also gives indication on the radiation quantities for which individual monitoring need to be initiated in the facility. Ambient dose rate due to β/γ radiations and neutrons and air and surface contamination are the key radiological indicators of importance for exposure control. All nuclear and radiological facilities in the country install sufficient number of monitoring equipment to continuously measure these parameters at representative locations in the facility. Quantity, locations, performance specifications, etc. of such installed radiation monitoring instruments undergoes detailed review, before commissioning of a radiological facility. This ensures the availability of such system for early warning of exposure situations to workers. Adequacy of the installed instruments are periodically reviewed and augmented from time to time, based on operating experience and expert judgment.

Installed Radiation Monitors

Area gamma monitors (AGMs) with energy-independent Geiger-Müller (G M) counter as detector is used for ambient dose rate measurement in workplace. A multi-element perforated shielding around the G M detector in AGMs makes its energy response flat in the energy region of interest. Typical AGMs used in Indian industry have operating range 0.01–100 mR/hr. No radiation field due to neutron is allowed in accessible areas of any facilities. Hence, continuous measurement of neutron field is not a requirement in workplace of most facilities. However limited number of such instruments is present in research reactors where experimental neutron beams are being provided. Moderated BF_3/^{3}He proportional counter, reading the ambient neutron equivalent dose rate in mRem/hour, is used for such purpose. Due to the special design of the moderator around the detector, these monitors have acceptable energy response for neutrons with energy from thermal up to 10 MeV/neutron.

Air contamination is continuously measured in all areas having potential for generation of radioactive aerosols or release of gaseous radionuclides to workspace. The nature and type of operations in an area provides guidance on the requirements

for air monitoring. Continuous air monitors (CAMs) are on-line air sampling instruments, essentially consisting of a motor to draw the ambient air, a collection medium, and a suitable radiation counter. Most often, radioactive particulates are collected on glass fiber filter medium. In most common instruments, an end window G M counter (for β-particulates) or a ZnS(Ag) scintillation detector (for α-particulates) facing the filter medium is used to give on-line gross count of the particulates accumulated on the filter. In general, CAMs for β-particulate measurement need additional shielding around the detector to reduce the counts from background radiation. This gross particulate activity measuring instruments usually reads in counts per minute (CPM) of the accumulated activity. Additional interpretations are often required to relate this measurement to the exposure quantities such as DAC and DAC-hour (Justus 2010). For sampling of iodine isotopes, activated charcoal filter is used as collection medium in combination with glass fiber filter. Charcoal filter is kept behind the glass fiber filter so as to collect all particulates on glass fiber filter, followed by isotopes of iodine on the charcoal filter. For adequate sensitivity, CAMs for gross α/β particulate collection and measurement are typically operated with a minimum air sampling rate of 100 liters per minute (lpm). Tests using smoke generators are used to have qualitative understanding of the air flow pattern in a room. This information along with knowledge of the most potential release locations are used to place the static air monitors at suitable locations in workplace, for early detection of radioactivity release into air (NUREG 1400 1993). On-line ^{3}H in air monitor (ECIL, Hyderabad) with gamma compensated ion chamber as detector is the usual on-line detector installed in PHWRs for measuring ^{3}H contamination in air.

Installed monitors are set to generate audio-visual alarm in case the measured parameters exceed a pre-set value. Alarm set point for instrument at a specific location is decided by considering the ambient background of the measurand at the location and the minimum detection requirement of the parameter to prevent undue exposure to the worker. Also, these instruments are hooked to a Supervisory Control and Data Acquisition (SCADA) system in plant control room. This aids the storage, trend analysis, and interpretation of radiological status of the plant on continuous basis.

For surface contamination measurement, no on-line instruments exist. Contamination on surfaces is measured and reported using direct measurement with handheld instruments. Commercial contamination counters with beta and alpha contamination probes (e.g., RadEye AB100, Thermo Fisher) are used for such measurements, and results are reported in Bq/cm^2. Direct measurement gives total contamination without discriminating its transferability. Measurement of transferable contamination is done using wipe (smear) taken from the surface using a suitable medium. A 10 cm × 10 cm absorber sheet is the usual wipe medium. In the process of taking wipe, only a fraction of the transferable contamination present on the contaminated surface gets transferred onto the wipe medium (Jung et al. 2001). Hence, the surface concentration of the contaminant actually present on the surface is usually much higher than what is counted on the wipe taken from the surface. A transfer factor, which is

usually much less than one, needs to be applied to get the true concentration of the surface contaminant using the count rate recorded from the wipe sample. However, due to wide variation in transfer factors possible and the difficulty to estimate the same on a case-to-case basis, transferable contamination using wipe method in operating nuclear facilities are usually reported without applying any transfer factor for the wipe procedure.

External contamination on clothing and body parts was identified as a source of radiation exposure to workers in nuclear industry. Ten percent of the annual equivalent dose limit to skin (500 mSv) is apportioned to contamination on skin and clothing (Krishnomony 1990). Based on this assumption, derived working levels (DWLs) are arrived for the maximum allowed contamination on skin and personal and plant clothing of nuclear energy workers in the country. DWLs applicable to nuclear facilities are provided in Table 2 (AERB 2005a).

Contamination checks on personnel are carried out at the exit of every controlled area in plants. Gross β/γ and α-contamination monitors are the most common instruments usually present at monitoring locations. Gross β/γ-contamination monitors are usually equipped with thin walled G M tubes or plastic scintillation detector whereas α-contamination monitors are usually made of large area ZnS (Ag) scintillation detector. Whole-body contamination portal monitors and vehicle portal monitors are usually installed at final exit point of nuclear facilities. This serves as an additional monitoring and prevents the release of contaminated individual or equipment from radioactive facilities to the public domain. Contamination portal monitors presently installed in plants uses an array of pancake G M detectors or large area plastic scintillation detector for radiation counting. Figure 3 shows few typical radiation monitoring instruments used in nuclear facilities.

For assured performance of all radiation monitors used in the plants, every facility have a detailed quality assurance (QA) program delineating all testing requirements including their periodic calibration. Calibration frequency of AGMs is biannual and is done following the procedure prescribed in IEC 60532 (IEC 2010). Efficiency of detectors used in CAMs is done periodically with standard sources of suitable dimensions and activity. Efficiency of contamination monitors are checked using wide area sources as prescribed in IEC 60325 (IEC 2002). Efficiency of α/β counters used in routine applications is done on a daily basis, prior to use, with standard source of appropriate strength and energy.

Table 2 Derived working levels (DWLs) for radioactive contamination

Surface	Beta emitters (Bq/cm^2)	Alpha emitters (Bq/cm^2)
Skin[a]	1.5	1.0
Hands[b]	350	250
Plant clothes	6	2
Personal clothes[a]	2	0.5

[a]Only fix contamination is allowed on skin and personal clothes
[b]Total contamination on either side of hands

Fig. 3 Examples of installed instruments for radiation monitoring in workplace: (**a**) β/γ hand, foot, and cloth contamination monitor, (**b**) continuous air monitor with PIPS detector for detection of presence of Pu in air, and (**c**) a portal contamination monitor with an array of pancake G M detector

Measurements Using Portable Instruments

Additional monitoring using portable instruments is a necessary component of occupational monitoring at workplace. Identification of radiation hot spots (location on equipment or area having relatively higher dose rate), periodic detailed radiation mapping of plant areas and equipment, monitoring of areas having accessibility issues, etc. are typically done with portable instruments. Periodic monitoring using portable instruments is done in facilities where frequent change in radiological status is envisaged. Jobs having potential for significant personal exposure, personal contamination, or air contamination is usually done under Radiation Work Permit (RWP) with supplementary monitoring using portable instruments. Teletector (Automess GmBH, Germany) is the principal portable instrument for gamma dose rate measurement in workplace. Commercially available radiation survey meters with scintillation detectors (NaI(Tl), CsI(Tl), CdZnTe, etc.), plastic scintillation detector, semiconductor detector, or G M detectors as radiation sensing element are also used for radiation field measurement, where dose rate is expected to be relatively less. Grab sampling for particulate air contamination is done with high volume air samplers (e.g., TIFIA series high volume air sampler, The Staplex Co., USA). These instruments generally use glass fiber filter media for particulate collection and work with higher suction rate for air, typically more than a cubic meter per minute, to have large sample volume within short sampling duration. All portable radiation survey meters used in facilities are subjected to calibration at least on a yearly basis. In PHWRs, collection of moisture using cold finger technique is usually done at locations expecting ^{3}H air contamination. In this method, moisture present in air is freezed out typically using dry ice. ^{3}H concentration in collected moisture is done with Liquid Scintillation Counter (e.g., TRICARB series, Perkin Elmer, USA). This measurement result along with relative humidity data is used to estimate ^{3}H concentration in air (ORNL 1976).

Individual Monitoring

Monitoring for External Exposure

Control on radiation exposure for the occupational worker is achieved by applying limits on the protection quantities: effective dose, equivalent dose to skin and extremities, and equivalent dose to eye lens (Table 1). In current practice, routine eye lens monitoring is not performed as it is unlikely that current dose limit of 150 mSv will exceed in normal occupational environment in nuclear installations. Individual monitoring is practiced for measurement and recording of radiation exposure to worker in all nuclear facilities in the country.

A TLD badge is the dosimeter employed for measuring whole-body dose to the individual from photon and β-particles. The badge consists of a TLD card and a cassette holder. The TLD card has three TLD disks made from $CaSO_4$:Dy (Dy concentration 500 ppm) and Teflon in the weight ratio of 1:3. Teflon acts as a binder and has no role to play in TL process. The detector material is $CaSO_4$:Dy phosphor (polycrystalline form). Each disk is 13.3 mm in diameter and 0.8 mm thick. Disks are clipped on hole of 12 mm diameter in a nickel plated aluminum card. The card has a "V" cut so that it can be loaded in the cassette only in one orientation. The cassette has three filter regions corresponding to the three TLD disks: (1) region of combined metal filters of copper and aluminum (total density thickness 1160 mg/cm^2) sandwiching disk element D1, (2) transparent polystyrene-666 filter (density thickness ~ 168 mg/cm^2) sandwiching disk element D2, and (3) circular open window sandwiching disk element D3 in paper wrapper and polyethylene pouch (total density thickness 12 mg/cm^2). Properly sealed TLD card is loaded in a TLD cassette made of plastic with a crocodile clip for wearing on person. Badges used for monitoring dose to other body parts such as wrist or head also are similar, except that there is a strap of suitable length to wear on the wrist/head instead of the clip arrangement. For the readout of the TLD card, a semi-automatic TLD badge reader is used. Hot N_2 gas (around 285–290 °C) is used to heat (as stimulant) the TL disk for release of the trapped energy from incident radiation in the form of light and is measured using a photo-multiplier tube (PMT). Using this TL signal from each disk and the dose evaluation algorithm (Dere et al. 1993), the whole-body dose and skin dose are evaluated. The dose evaluation algorithm is developed based on the response of the TLD badge to photons in the energy range 30 keV to 1.25 MeV and beta radiation with end point energies 0.66 MeV (^{85}Kr) and 2.2 MeV (^{90}Sr/^{90}Y). Reporting levels for whole-body dose and skin dose, using this dosimeter, are 0.10 mSv and 1.0 mSv, respectively.

Sensitivity of this dosimeter to fast neutron is negligible. Each TLD card can be used for several cycles of annealing, exposure, and readout. Though this TLD is not tissue equivalent, several studies have shown that for pure photon beams and mixed fields of X-rays and photons, the response of this dosimeter lies within the limits of trumpet curve, the criteria used for accuracy in dose measurement using TL dosimeter (IAEA 1999).

External dosimetry services for occupational radiation workers of DAE establishments are done through 12 personal monitoring laboratories (PMLs) operating at

different locations. Frequency of TLD monitoring service is monthly or quarterly depending on the exposure potential of the facility being monitored. All NPPs, fuel processing plants, and other major facilities have monthly service for TLD badges. Single TLD badge on chest is used for all cases where uniform radiation field is expected. In special situations where the radiation field is expected to be non-uniform, separate TLDs in addition to chest TLD are provided for supplementary measurement. Wrist TLD badges are provided for workers for measuring the dose to the extremities. Jobs like handling of open-radioactive sources, maintenance on equipment having high exposure rate, etc. are always accompanied with additional monitoring using wrist TLD badge.

For specific jobs with significant exposure potential, an active dosimeter is provided to the worker in addition to the TLD badge. This helps in administrative control of exposure of each individual through the progress of a planned work. Direct reading dosimeter (DRD, Arrow Tech Inc., USA) is in use in some facilities. This dosimeter, usually called as pen dosimeter, is a small ionization chamber which is charged prior to use. Gamma radiation falling on the detector leads discharge of the dosimeter which is read through an analog scale. Electronic pocket dosimeters (EPDs) (e.g., DMC 3000 series, MIRION Inc., USA) are increasingly replacing the practice of using analog DRDs in the country due to several practical advantages of the latter for field use. Provision to set alarm limit based on dose and dose rates, remote communication, etc. are some major advantages. Such dosimeter usually houses multiple miniature semiconductor detectors inside the housing in order to function for wide energies of radiation and larger operation range for measurement of dose and dose rate.

Extensive quality assurance (QA) protocols are followed in personal monitoring using TLD badge, starting from the selection of phosphor lot for preparation of TLD disk to assigning dose to the individual from reading individual TLD card (BARC 2018). Calibration of TLD is done with ^{137}Cs source traceable to the secondary standard calibration laboratory, Radiation Standard Section, BARC. External QA for all the PMLs operating in the country is done by an announced test annually done by Radiological Physics and Advisory Division (RP&AD), BARC. During this exercise, TLDs irradiated to specific dose are provided to each PMLs, without declaration of the actual dose delivered and type of radiation, and PMLs are asked to estimate the dose using their standard procedure (Srivastava et al. 2017). PMLs of NPCIL facilities are subjected to blind test every month as a part of its internal QA program. In this test, the user facilities provide TLDs irradiated to known dose along with TLD badges of regular workers, without revealing the irradiation status of the cards. Active dosimeters (DRD/EPD) in circulation are also periodically calibrated with standard source. TLD laboratories in the country participate in international inter-comparison exercises conducted by IAEA, EURADOS, etc. for proficiency evaluation.

Personal neutron dosimetry is done with SSNTD technique using the polycarbonate detector commercially known as Colombia Resin-39 (CR-39). The chemical name of the polymer is poly-allyl-di-glycol carbonate (PADC). The dimension of the detector is 3 cm $\times$ 3 cm $\times$ 0.75 mm. The detector, along with a polyethylene radiator

in front of the detector, is loaded in a triple laminated pouch and further loaded in a polycarbonate holder. Elastic scattering of neutrons with hydrogen of this polymer produces recoil protons. These recoil protons deposit its energy in the medium to cause breaks in molecular bonds. Local defects thus created in the form of track in the detector are enlarged by chemical/electrochemical etching of the detector under controlled conditions. Etching rate at damaged site is more than that at the undamaged base material. Two etching methods using KOH (concentration 7 N) as a medium and at 60 °C temperature are used for track development. In one case, known as electro-chemical etching (ECE), it is done in the presence of electric field (2200 V/mm), and the other process is called chemical etching (CE). In ECE, etching rate is faster, but many tracks could merge together resulting in loss of tracks during the process. But the procedure is faster and the track counting after ECH is done with automated image analyzer. CE is more time-consuming and track counting needs to be done manually. Due to merger of multiple tracks, ECE have lower sensitivity in comparison to CE procedure. In practice ECE is used for dose $<$10 mSv and CE for higher doses (Massand et al. 1990).

Calibration of neutron dose measurement using CR-39 detector is done following ISO 8529 (ISO 1998). For calibration purpose, detectors are irradiated on an ISO recommended water slab phantom made from poly methyl methacrylate (PMMA) using standard ISO recommended ^{241}Am-Be, ^{252}Cf, and ^{252}Cf-D_2O neutron sources. The neutron dosimeter used in neutron monitoring program is sensitive to neutron energy 100 keV to 14 MeV. Dose linearity of the dosimeter is found to be within $\pm$10% up to measured dose of 10 mSv using ECE. MDL for CE and ECE procedure were found to be 0.27 mSv and 0.10 mSv, respectively. As a practice, the minimum reporting level for neutron dose is 0.20 mSv with quarterly service period for the dosimeter. From detailed laboratory investigations, it is established that CR-39-based neutron dosimetry followed in the country is confirming to all requirements as specified in ISO 21909 (BARC 2011; ISO 2005). Individual monitoring for neutron dose is routinely done for identified workers having potential for exposure to neutron filed in NPPs, research reactors, and facilities handling significant quantity of plutonium.

Monitoring for Internal Exposure

Assessment of radiation doses arising from the intake of radioactive material by the worker is termed as internal exposure assessment. Individual monitoring is prescribed for workers with likelihood of receiving more than 1 mSv/year occupational exposure due to intake of radionuclides (IAEA 2018). Individual monitoring is performed out for all workers having such work environment. Though the radionuclides of concern depends on the type of facility, inhalation is found to be the most common route of intake at occupational workplace. Depending on the facility, internal monitoring program and methodology are carefully designed, not to miss any case of intake in usual course. The technique employed for detection of intake and the monitoring frequency are the two key parameters that will decide the overall efficiency of internal monitoring program. Monitoring for internal exposure carried out in the facilities could be broadly classified into three categories: (i) routine

monitoring to estimate the dose due to regular work in normal work environment; (ii) special monitoring, carried out in several circumstances such as heavy personal contamination, injury while working in controlled areas, high air contamination detected at workplace without adequate respiratory protection for worker, failure of respiratory protective equipment during work, etc. and also, as a part of special tasks where high probability for intake is envisaged, and (iii) confirmatory monitoring to check the assumptions of exposure conditions applied to a facility. Confirmatory monitoring is generally done on a small fraction of the workforce. Individual monitoring for intake is done using in vivo method, in vitro method, or a combination of both.

In Vivo Methods

In vivo method is the technique of quantifying the radioactivity present in specific organs or whole body using suitable detectors placed externally. Partially shielded whole-body counter (WBC) is used for quantifying high energy photon emitters (HEPs) uniformly distributed in the body. Here the subject is scanned under a scintillation detector (usually $4'' \times 3''$ NaI(Tl) detector) under partial shielding. HPGe detector is also used in place of NI(Tl) detector for high-resolution application, especially when a mixture of radionuclide is to be quantified.

A motorized scanning bed controls the movement of subject below the detector. Presence of low concentration of gamma emitting isotopes such as ^{137}Cs or ^{60}Co in the body can be easily detected using this technique. As radio-iodine concentrate in thyroid, a chair type scanning system is used to quantify such cases. A $2'' \times 2''$ NaI(Tl) detector is usually mounted on thyroid monitoring system.

To quantify low energy photons (LEPs) emitting radionuclides present in the lungs, a specially shielded steel room is used. Such room is constructed with typically 20-cm-thick mild steel walls all around which are lined inside with Pb (3 cm) + Cd (2 cm) + Cu (0.5 cm). Resultant background in steel room – only due to characteristic X-rays (8 keV) from Cu – allows sensitive measurement of LEPs such as 60 keV photons from ^{241}Am or 17 keV L-X-rays from ^{235}U which are used to estimate ^{239}Pu in the lungs. A phoswich detector, sandwich of primary thin NaI(Tl) crystal in combination with thick CsI(Tl) crystal working in anti-coincidence with the NaI(Tl) crystal, is the common configuration used for LEP measurement. An array of three HPGe detectors is also used for measurement of LEP emitting radionuclide present in the chest.

All in vivo monitoring systems are calibrated periodically using standard phantoms. In this procedure, known quantity of radioactivity is placed in the phantom and counted under standard procedure. HEP measuring systems are calibrated with phantoms made of tissue equivalent materials: Masonite phantoms or Bottle Mannequin Absorber (BOMAB) phantom. For LEP measurement in the lungs, a Lawrence Livermore National Laboratory (LLNL) phantom or Japan Atomic Energy Research Institute (JAERI) phantom is used. Energy resolution and efficiency check of all in vivo counting systems are done weekly with test sources. Theoretical calibration can also be performed for in vivo counting systems using computational voxel phantoms, where direct calibration is practically difficult. Computational

Fig. 4 Some of the instruments used for intake measurements by occupational workers (**a**) shield room facility for measurement of LEP emitters in the lungs, (**b**) NaI(Tl) based thyroid monitor for measurement of iodine intake, and (**c**) system for breath measurement for ^{226}Ra in the lungs used in uranium mining facilities

phantom is a mathematical representation of the human body which, when coupled with a Monte Carlo radiation transport computer code, can result in estimation of efficiency for the counting system. Use of these phantoms eliminates the requirement of physical phantom as well as the radiation standards.

In addition to general procedures, organs such as the skull or knees need to be monitored as a suitable site for measurement of bone seeking radionuclides. Activity estimation at wound site, especially with actinides or radionuclides with low penetrating radiations, is also done in shielded room counting facilities. Miniature pencil probes with silicon detector (e.g., Multirad LLR with α-pencil probe, Mirion Technologies) are used for detection of alpha contamination on wounds. Figure 4 shows few systems used in measurement for internal contamination. List of important radionuclides specific to nuclear industry and their detection limit for in vivo counting available in the country is listed in Table 3 (AERB 2019).

In Vitro Methods

In vitro methods are used to quantify the intake of several radionuclides, mainly for alpha and beta emitters. Quantity of radionuclide inside the body could be estimated using measurement of radionuclide concentration in biological samples collected from the internally contaminated individual. Excreta such as urine, feces, or breath are the most common samples analyzed for radioactivity. This procedure is termed as bioassay. Analysis of urine is the most common bioassay procedure due to ease of collection and analysis. Standard radiochemical procedures are used for processing, chemical separation, and analysis of the bioassay samples. Several techniques are used for measuring radionuclide concentration in processed samples. For low energy pure beta emitters like ^{3}H, ^{14}C, ^{35}S, and ^{241}Pu, liquid scintillation counting is the only viable method. Other beta emitters especially fission products are estimated using gas flow G M counter or plastic scintillation detector. Alpha spectrometry using passivated ion-implanted silicon (PIPS) detector is the most common technique applied for alpha emitters. More sensitive methods such as laser fluorimetry (LF), neutron activation analysis (NAA), fission track analysis (FTA), etc. are

Table 3 Minimum detection activities and the corresponding CED with in vivo measurement techniques available in the country

Isotope	Measured organ	MDA (Bq)	Solubility class	Monitoring frequency	Corresponding CED
^{60}Co	Whole body	100	S	Annual	0.01
^{137}Cs	Whole body	200	F	Annual	0.05
^{125}I	Thyroid	25	F	Monthly	0.002
^{131}I	Thyroid	50	F	Fortnightly	0.01
^{241}Am	Lungs	6	M	Monthly	13.5
^{239}Pu	Lungs	2000	S	Annual	518.0
^{239}Pu (^{239}Pu/^{241}Am = 3)	Lungs	18	S	Annual	4.8
^{235}U	Lungs	3	S	Annual	0.6
^{238}U	Lungs	40	S	Annual	7.1
228Th[a]	Lungs (^{208}Tl)	10	S	Annual	27.8
	Lungs (^{212}Pb)	6	S	Annual	6
^{232}Th[a]	Lungs (^{228}Ac)	10	S	Annual	3.8

[a]Assuming in equilibrium with its daughters

applied in specific cases. Bioassay of urine and feces are routinely done in combination with several measurement techniques at bioassay laboratories at Mumbai, Tarapur, and Kalpakkam. Bioassay laboratory is also operating at NFC, Hyderabad, for measurement of intake of uranium. Radiochemical separation and quantification procedure undergo strict quality control, and the bioassay laboratories regularly participate in international inter-comparison exercises conducted by IAEA and other laboratories. Analysis of the swab from nasal cavity is often done in case of suspected inhalation, especially for actinides. Tissue samples excised from the contaminated wound are also radio-chemically analyzed for activity content. Breath analysis of radon is used to estimate the body content of ^{226}Ra (from ^{238}U series) in uranium mine workers. This procedure requires the worker to breathe into a breath collecting apparatus for a period up to 30 min depending upon the volume required. MDA achievable for important radionuclides in DAE bioassay laboratories are presented in Table 4 (AERB 2019).

From the measured content of radioactivity in the whole body, specific organ/s, or excretion rate, it is possible to find the intake in Bq of radioactivity that has entered the body by inhalation, ingestion, or injection. Various bio-kinetics model such as human respiratory track model, human alimentary track model, National Commission for Radiological Protection (NCRP) wound models (ICRP 1994, 2006; NCRP 2007a), and other systemic models published by the ICRP are used for estimating the

Table 4 MDA and the corresponding CEDs for important radionuclides by in vitro methods

Isotope	Method	MDA (mBq/day)	Lung absorption type	Monitoring frequency	Corresponding CED (mSv)
^{90}Sr	RCA and gas flow G M counter	15	F	Annual	0.008
^{232}Th	RCA and NAA	0.0008	M	Annual	0.003
		0.0008	S	Annual	0.03
U(nat.)	RCA and LF	0.005	M	Annual	0.0001
		0.005	S	Annual	0.01
U(nat.)	RCA and α-spectrometry	0.5	M	Annual	0.01
		0.5	S	Annual	1.0
^{239}Pu	RCA and α-spectrometry	0.5	M	Annual	3.0
		0.5	S	Annual	26.0
^{239}Pu	RCA and α-spectrometry	0.012	M	Annual	0.07
		0.012	S	Annual	0.6
^{241}Am	RCA and α-spectrometry	0.5	M	Annual	1.2

RCA radio-chemical analysis, *NAA* neutron activation analysis, *LF* laser fluorimetry

intake using the data from measurements. Intake when multiplied by dose coefficient (Sv/Bq) provides the CED. When the time of intake is not known, standard procedure is to adopt midpoint of monitoring interval as the time of intake. In some cases, follow-up monitoring may provide information regarding the likely time of intake. Such cases are resolved by multiple monitoring at regular intervals on the subject. These multiple results are fitted into the relevant bio-kinetics model to arrive at the intake (AERB 2019).

Both detection and dose computation due to intake of radionuclide like ^{3}H or isotopes of iodine are simple and straight forward. When ^{3}H enters the human body as tritiated water, it equilibrates with body water within few hours. Urine is a representative of body water. ^{3}H concentration in urine, collected after few hours of exposure, can easily be used to accurately estimate the intake. Also ^{3}H measurement in urine sample using liquid scintillation counter (LSC) may not require much sample processing. Similarly, iodine isotopes preferentially get absorbed in thyroid, resulting in easy detection. But for many other radionuclides like actinides with complex bio-kinetics and bio-distribution, CED estimation is tough and time-consuming. Multiple measurements often using different techniques are used to arrive at the intake. Measurement results from different techniques such a urine, feces, lung monitoring, etc. are together used to arrive at the best estimate of intake. Commercial internal dosimetry software such as Integrated Modules for Bioassay Analysis (IMBA Internal dosimetry software, UK Health Security Agency) and Mondes-Mondal are used for estimation of intake using data from multiple measurements obtained using different monitoring techniques.

Though the occurrence of overexposure to the worker is rare in nuclear industrial activities in the country, DAE has set up a well-established bio-dosimetry laboratory

among its facilities for radiation exposure monitoring. Biodosimetry laboratory at BARC, Mumbai, has the capability to estimate dose accurately in case of suspected exposure above 100 mSv. Dicentric chromosomal aberration assay (DCA) is the most used technique while other methods like micro-nucleus assay (MN-assay), premature chromosome condensation (PCC), and fluorescence in situ hybridization (FISH) technique also are well established at this facility (Bhat et al. 2018).

Monitoring Practices Specific to Facilities

Front End Fuel Cycle Facilities

Front end fuel cycle facilities include uranium mining, milling, and uranium fuel fabrication facilities. These facilities are distinguishable from the rest of the nuclear fuel cycle facilities with respect to the volume of raw material and solid waste managed, manual mode of operation, open radiation sources, and handling of bulk quantity of low specific radioactive material.

DAE operates both underground and opencast mines, followed by the milling process to get the uranium ore concentrate. For uranium mine workers, contribution from internal exposure due to radon progeny inhalation is more than external exposure. The normal ambient dose rate in Indian uranium mines are typically 1–2 μSv/hr. due to low uranium content in ore (BARC 2020). A calibrated G M survey meter is used for ambient dose rate measurement in uranium mines. In uranium mills and tailing pond areas where the waste from milling process is contained, air-borne uranium particles are the main source of internal exposure. In uranium mining and processing facilities, measurement of ambient radiation along with occupancy factor (duration of exposure) is used to assign dose to the occupational radiation worker. For this, radon and their progeny concentration in air and ambient dose rate at the work locations are regularly monitored. Personal radon dosimeters (PRD), having dosimetry elements for both radon progeny measurement and gamma dose measurement, are additionally used for estimation of dose to workers in uranium mines. Good agreement is observed between dose estimated using data from measurement of ambient parameters at workplace and that from the use of PRD by individual worker in field.

Radon concentration in mines is measured using a cylindrical ZnS(Ag) scintillation cell. Pre-filtered air is allowed to enter into the evacuated 150 ml metallic cylinder whose inside is coated with the scintillator material. Counting is done typically after about 3 hours of sampling so as to reach equilibrium between radon and its progenies. Typical detection limit for this method, with 10 min counting time, is ~40 Bq/m^3 (Raghavayya 1981). Equilibrium factor for dose estimation is experimentally determined in most cases; otherwise, the recommended default value of 0.4 is used. In mills, low level radon detection system (LLRDS)- or SSNTD-based radon detectors are used. LLRDS is a 5 L volume aluminum chamber. Freshly formed ^{218}Po, a progeny of ^{222}Rn, is collected onto a negatively charged aluminum plate by application of high voltage to the plate. Typically, 90 min is the

collection time followed by 75 min counting – post collection – for alpha particles emitted from the progeny collected on the electrode (Srivastava et al. 1984). For radon daughter concentration measurement, the Kusnetz method (Kusnetz 1956) is often used in mines. In this method, radon daughters are collected on a filter paper by air sampling for about 10 min, and the filter paper is counted for alpha activity in between 40 min and 90 min, post sampling. This method is found to give radon daughter working level concentrations within an error of ±13%, irrespective of the equilibrium condition.

PRD is used to assess the internal dose due to radon progeny inhalation (Raghavayya et al. 1986). The device comprises a 60 mL cylindrical aluminum chamber covered with a permeable membrane which allows only ^{222}Rn to diffuse in, due to its relatively longer half-life, and serves as a barrier for ^{219}Rn, ^{220}Rn, and other particulate aerosols. A 1.8 cm × 3 cm SSNTD film (LR-115) is used to quantify the radon concentration. The SSNTD film is etched with 10% NaOH solution at 60 °C for 90 min. The tracks are electronically counted which are correlated to radon exposure using a calibration system. Service period for the detector is 2 months. PRD also have two $CaSO_4$:Dy TLD disks embedded in it which is used to estimate the external exposure to the worker. The algorithm used for dose evaluation from this two-disk TLD system is different from that of the three-disk TLD badge used in other facilities. Additionally, radon and its progeny concentrations inside mine is routinely measured using sophisticated portable instruments like Alpha Guard (Bertin Instruments, Germany). Such measurements are useful for evaluation of the equilibrium factor between radon and its daughters which is employed for dose computation using ambient radon concentration data. This data also serves as feedback for ventilation control in mines to limit intake of radon progenies.

Standard three-disk TLD badge is used for external exposure measurement of workers in uranium fuel fabrication plant (BARC 2021). Uranium handling areas are regularly monitored for the presence of air contamination. Workers in uranium fuel fabrication facilities are subjected for lung monitoring and urine bioassay for the detection and quantification of uranium intake.

Occupational Monitoring in Reactors

Ambient dose rate in all accessible areas of the reactor is measured with GM detector-based AGMs usually having operating range 0.1–100 mR/hr. AGMs having higher operating range, usually having ion chamber-based detector, are provided in certain shutdown areas in some reactors, rather to know the process deviations than occupational monitoring purpose. Calibration of the accessible area AGMs are done in situ employing radiation sources traceable to secondary standards. Additionally, electronic calibration of AGMs is done periodically applying external voltage simulating the radiation source. Neutron dose rate in shutdown areas during maintenance activities are carried out using standard neutron REM-counter based on ^{3}He/BF_3 detector (e.g., Ludlum 30 series). These neutron-Rem counters are calibrated at

secondary calibration laboratory following the procedures prescribed in IEC 61005 (IEC 2014). CAMs for particulate air activity measurements are present at Spent Fuel Storage Bay (SFSB) – where the freshly discharged hot fuel bundles are stored – and some other areas in few reactors. Continuous ^{3}H in air monitors are provided in critical areas in all reactors. These detectors are compensated ion chambers (ECIL make) capable to detect sub-DAC concentration of ^{3}H in air. Additionally, two grab sampling techniques are routinely employed for detection and quantification of ^{3}H in air from potential locations in reactors. In "cold finger" method, HTO present as moisture in air is freezed out using dry ice as the cooling medium (DOE 1994). Bubbling air through water at room temperature, typically at low flow rate of ~1 liter per minute (lpm), is another method used to sample ^{3}H in air (Wood et al. 1993). Water containing ^{3}H obtained by either of the method is analyzed using LSC. During certain special operations in reactors, air contamination due to isotopes of iodine is possible. Air sampling for iodine isotopes are done employing activated charcoal-loaded filter paper and is counted on either HPGe- or NaI(Tl)-based gamma spectrometry system.

For photons and β radiation, individual external exposure measurement in reactors is done with standard three-disk TLD badge and EPDs as active dosimeter. Neutron personal dosimetry in reactors is traditionally done with CR-39 film badge. However, commercially available neutron-sensitive EPD is used as an additional active neutron dosimeter in NPPs. All NPPs in the country follow similar procedures to monitor intake of radionuclide. Shadow shield WBMs with NaI(Tl) detector is the standard instrument used for in vivo counting. WBMs are placed in environmental survey laboratories (ESLs) as they need to be placed away from back ground radiation form reactors. Typical detection limit for such counters is around 100 Bq of ^{60}Co in body. All workers with potential for ^{3}H intake are monitored weekly by urine sample analysis.

Workplace and individual monitoring practices in research reactors are almost similar to that of NPPs. But, research reactors are principally utilized for performance evaluation of new reactor designs, isotope production, and research employing neutron beams for scientific research. Hence frequent changes in radiation field is expected in such facilities necessitating more frequent radiation surveillance of occupied areas.

Radiochemical Plants and Plutonium Handling Facilities

General radiation monitoring procedures in back-end nuclear cycle facilities like spent fuel reprocessing and radioactive waste management facilities are relatively complex to that of other nuclear facilities described earlier. This is due to handling of bulk quantity of solutions having high radioactivity concentration, whereas in other facilities, radioactivity is mostly in sealed form (reactors) or is of low specific activity (uranium mining facilities). Chemical processing is done in hot cells made with thick high density concrete wall and processes are controlled remotely. These cells and equipment are non-accessible during normal operations, but human entry is

required for contact maintenance. Such jobs are prone to significant external exposure and are having potential for intake. At the beginning of hot cell decontamination, prior to scheduled maintenance activities, exposure rate in hot cell could be beyond the maximum operating range of routine radiation field measuring instrument like teletector (1000 R/hr). In such instances, for area surveillance, gamma dose rate meter with wide operating range is employed. Detailed characterization of air samples by chemical separation and spectrometric identification of isotopes is often done to characterize the air contamination in work field. Many of the radiological hazards in uranium-plutonium mixed oxide (MOX) fuel fabrication facilities are common to that of back-end fuel cycle facilities. All stages of the fuel pellet and fuel pin preparations are carried out in glove boxes. This results in significant external exposure to plant workers from neutron and gamma radiations. ^{241}Am build-up from ^{241}Pu is found to be the major source of gamma exposure in MOX fuel fabrication plants.

Potential for intake is high in places where plutonium is handled in dispersible form. A highly reliable air monitoring program is essential to prevent intake via inhalation by workers in these facilities. CAMs for alpha particulate monitoring based on gross alpha detection are used in large numbers in such facilities but are not sensitive enough for early detection of Pu aerosol due to the presence of naturally occurring alpha emitting progenies present in air. For prompt detection Pu contamination in air, alpha dust particulate monitor based on PIPS detector is used in Pu handling facilities (Pollanen et al. 2013). These instruments are basically alpha spectrometer with provision for on-line dust collection, spectrum analysis, and quantification of isotopes collected on the filter paper and use proprietary algorithm to distinguish alpha particles of artificial origin like ^{239}Pu from that of natural alpha emitting radionuclides (Li et al. 2013). Personal air sampler are often provided to workers engaged in activities having high potential for intake by inhalation. These are small, lightweight passive air sampling units working at low flow rate, typically few lpm, and can be worn on person while working.

Individual monitoring for external exposure using standard TLD badges is done for all workers in such facilities. Neutron exposure is significant in facilities handling bulk quantity of PuO_2. Spontaneous fission of Pu isotopes is one source of neutrons. (α,n) reaction with elements like ^{18}O present in PuO_2 is also a source for neutron exposure in these facilities (McLaughlin et al. 2000). Fast neutron dosimetry using CR-39 film badges are employed for personal neutron dosimetry. Workers having potential for intake are identified and subjected for appropriate internal monitoring procedure. Bulk plutonium handling facilities are prone to nuclear criticality accidents (Reddy et al. 2018). Criticality accidents are prevented by geometrically safe design of vessels and equipment for handling fissile materials in combination with strict restrictions on maximum quantities allowed to be handled in specific processes. Criticality alarm systems are provided at all locations susceptible to such incident for the immediate evacuation of workers. These are ionization chamber-based instruments tuned to generate alarm due to a sudden surge in gamma dose rate, which is typical for a nuclear criticality event. All workers working in areas vulnerable to criticality incidents are provided with individual criticality dosimeter. Criticality

dosimeter consists of several neutron activation foil such as ^{197}Au, ^{63}Cu, ^{115}In, ^{64}Fe, ^{32}S, ^{58}Ni, ^{24}Mg, and ^{27}Al in addition to a standard $CaSO_4$:Dy TLD disk. These activation foils with different cutoff aids in building the neutron energy spectrum resulted from the accident. A standard procedure is established for evaluation of individual exposure from gamma and neutron using criticality dosimeter (Trompier and Chevallier 2019).

Trends in Occupational Exposure of Workers in Nuclear Industry in the Country

Individual dose records of all radiation workers in the country are stored in the National Occupational Dose Registry System (NODRS) managed by RP&AD, BARC. Regulatory bodies have stipulated the retention period for occupational dose records of workers as 30 years after termination of the work involving occupational exposure or until the worker attains or would have attained the age of 75 years whichever is later (AERB 2005a). Introduction of this database management system allowed the comprehensive analysis of the exposure pattern of nuclear industry workers in DAE. Figure 5 gives the trend in occupational dose of nuclear energy workers of DAE and that of radiation workers in non-DAE (industrial, R&D, and medical) sector in the country for the period 2000–2018. Number of radiation workers in nuclear industry (DAE) is almost similar in the analyzed period, but clear reduction is observed in both average dose and collective dose. This trend, despite the vast expansion of nuclear fuel cycle activities during this period, indicates the increase in safety measures, over the years, in nuclear fuel cycle facilities in the country. In addition to technological enhancements in design and operation, new regulatory measures like frequent safety review of the facilities by Regulatory Inspection Team (RIT) and introduction of dose constraints below dose limits also

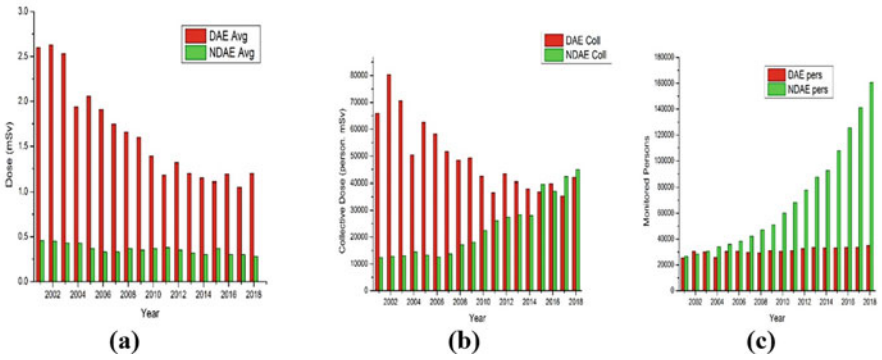

Fig. 5 Occupational exposure to radiation workers in nuclear industry in the country for the period 2000–2018; (**a**) average individual dose, (**b**) collective dose for all facilities, and (**c**) change in number of radiation workers

helped in bringing down dose to the radiation workers in nuclear fuel cycle facilities in the country.

Conclusions

Well-established procedures are practiced for monitoring and control of radiation exposure to occupational radiation workers employed in nuclear industry in the country. However, R&D activities pertaining to occupational radiation protection are in continuous pursuit for further improvement. Efforts are underway to report the TLD dose in terms of *Hp*(10) instead of the "whole-body dose equivalent," as practiced today. Neutron dosimetry at present is done with CR-39 detectors having a lower neutron energy threshold of 100 keV. Efforts need to cover neutrons of all energy in personal dosimetry services. Preliminary assessment shows that the chances of eye lens to receive higher dose than whole-body dose is very rare at workplace in nuclear industry in the country. But an appropriate dosimeter for eye lens dose measurement is desirable, especially for building database from actual measurement done in field. Many instruments used for workplace and individual monitoring are of foreign origin. Indigenous development of radiation detectors to substitute them is underway. Automation of radiation monitoring is preferred but is also linked to the progress in automation of basic production and operating procedures in plant and facilities.

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Relevance of Radiometric Metrology in NORM Industries and Radiological Safety

85

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Abstract

Safety is the key for sustenance of any industrial activities which directly or indirectly links with the economy and growth of the country. Unlike conventional

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safety awareness and measures in industries handling of naturally occurring radioactive materials (NORM), scant attention has been paid to radiological one. Internationally, scientific efforts are being made to identify the industries/processes laced with activity concentration greater than 1 Bq/g criteria for NORM which required appropriate control measures and regulatory instruments to protect the occupational workers, the environment, and the members of the public which is environmentally sustainable, socially acceptable, and economically affordable. Association of multi-hazards in NORM industries and diversification of operations across the NORM industries make the case more complex and difficult to regulate. In this scenario, radiometric techniques play a vital role for identification and quantification of NORM. Radiometric metrology is required to demonstrate verifiable indicator parameters for acceptability of the public, policy decisions, and development of legislative limits. However, nonexistence of overarching government policy and regulation on handling of naturally occurring radioactive materials except the industries under Department of Atomic Energy (DAE) in India make a strong pitch for introspection and evolve of scientific approach to deal with the insidious potential hazard. In the paper, authors have reviewed the published literatures related to handing of naturally occurring radioactive materials, elevated levels of these radionuclides in various industrial activities, principles of radiometric metrology, and radiological safety.

Keywords

NORM · Radioactivity · Radiation safety · Radiometric metrology · Gamma spectrometry · Alpha spectrometry

Introduction

Exploration and exploitation of natural resources, particularly ore and minerals, are an age-old practice that fuels the development of the country and its economy. These naturally occurring ore and minerals are ingrained with natural radionuclides, called as naturally occurring radioactive materials (NORM) since the evolution (UNSCEAR 2008). Long half-lives, low to high specific activities, radiological hazards due to chronic/acute exposures, and omnipresence are few characteristics of these natural radionuclides that alert the health physicists to devise appropriate monitoring program and regulations to avoid undesirable adverse impact and to reduce the possibilities of risk that arise out of it (IAEA 2014a). For a developing country like India, growing need of energy and raw materials compels for mining and processing of natural resources at an accelerated kinetics to meet the burgeoning demands of rising population, urbanization, and industrialization. All the natural resource minerals associated with radionuclides are not of much radiological significance. However, few industrial processes alter the natural abundance of NORM and resulting enhanced exposure that attracts the instruments of regulation (Garcia-Talavera et al. 2011; IAEA 2011a). Accumulation of natural radionuclides in scales

of oil and natural gas industries, resins in water treatment plants, phosphoric acids in phosphate industries, and fly ash in thermal power plants are the few examples having potential to redistribute elevated content of radionuclides in surrounding environment and pose radiological impacts on occupational workers and members of the public (Jha 2014; Koppel et al. 2022; Turcanu et al. 2022).

NORM comprises of primordial radionuclides that occur in the environment. However, the term is used particularly for enhanced radiation exposure situation due to industrial activities/processes that deals with raw materials associated with radionuclides of natural origin compared to pre-industrial activities. NORM results from various processes, namely, burning of coal/fossil fuel, manufacturing and application of fertilizers, and production of gasoline (IAEA 2003c). Extraction of uranium ore exposes the occupational workers involved in winning of ore body and processing plant for chemical extraction and isolation of economically valuable uranium from the bulk matrix. Significant radionuclides for radiological concerns are the daughter products of ^{238}U and ^{232}Th series, considering their half-lives, natural abundances, and geochemistry.

Radiation and radioactive levels associated with NORM are dominant part of the naturally occurring background radiation. Both biotic and abiotic components of ecosphere are continuously exposed to natural radiation due to existing exposure scenario. The UNSCEAR (2000) report underscored the wide variation of the natural background radiation level across the globe (UNSCEAR 2000). Relatively, a higher exposure to the habitat is anticipated in radioactive mineralized region or higher altitudes due to terrestrial radiation and cosmic radiation, respectively. A significant fraction of natural radiation exposure is attributed to inhalation of radon (^{222}Rn and ^{220}Rn), a noble gas that emanates from the earth's surface due to decay of their parent nuclide (^{226}Ra and ^{224}Ra). There are four main components of natural radiation, i.e., terrestrial radiation, extra-terrestrial (cosmic) radiation, ingestion, and inhalation of radionuclides of natural origin.

Many countries now have national legislations, where stipulated limits/guideline values are defined for naturally occurring contaminants in various environmental matrices. Implementation of these stipulated guideline/limits entails a comprehensive analytical infrastructure for appraisal of minute quantity of natural radionuclides in various environmental matrices. Quality data is a by-product of strict adoption of a prescribed quality assurance and quality control protocol. Therefore, data credibility and high degree of confidence on the generated data are crucial to our national and international programs. In the paper, a comprehensive review of NORM handling industries is presented here along with radiometric techniques for measurement of various radiological parameters. For the sake of completion, a brief overview of natural radiation and radioactivity is depicted in the following section.

Natural Radiation and Radioactivity

Radioactivity is classified as natural and artificial depending upon its origin. Naturally occurring radioactive materials (NORM) are primordial in origin consisting of uranium, thorium along with their series radionuclides, and single radioactive

elements like potassium. During operation of nuclear fuel cycle facilities, it is anticipated that effluents associated with natural and manmade radionuclides (specific to the facility) are released to the environment (UNSCEAR 2000). The quantity of release depends upon the type of facility, engineering control measures, safety standards adopted, and waste management practice implemented. Similarly, many other industries, generally called as NORM industries, which do not involve direct handling of radioactive material yet may have potential to enhance or alter the natural radioactive levels in the surrounding environment.

Human beings are being exposed to natural background radiation, medical exposure, fallout radionuclides, releases from nuclear industries, etc. Out of these sources, the significant and largest fraction is contributed by the natural background radiation which is part of the life and inescapable. It is classified into external and internal exposure. External exposure has two components, i.e., (a) terrestrial and (b) extra-terrestrial radiation (cosmic rays). Internal exposure has also two components, i.e., (c) inhalation and (d) ingestion. The components vary over a wide range across the globe which depends on the geography and geology of the location, type of house, building materials, number of air changes, source of drinking water (ground or surface water), etc. An enhanced background radiation is envisaged in a particular region associated with geochemical peculiarities related to abundance of ^{238}U and ^{232}Th and others. All existing exposures are attributed to the radioactivity content occurring in the Earth's crust, except cosmic component. Variation in natural radiation exposure depends on nature of job, type of household and construction material used, location of habitation, ventilation rate, life style and food habit, etc.

Cosmic Radiation

A major contribution to external exposure comes from cosmic rays, of extraterrestrial origin. A significant fraction of highly penetrating and high energetic cosmic rays is composed of protons (~85%). The remaining fraction is composed of alpha particles (14%) and light nuclides with Z in range of 4 to 26. Cosmic radiation varies from 0.03 μSv/h at sea level to 0.1 μSv/h at Bangalore City, India, at an altitude of about 1 km. This becomes 1 μSv/h at an altitude of 6.7 km (at the Himalayan peaks) increasing to 5 μSv/h at an altitude of 10 km (aircraft cruising altitude) and 10 μSv/h at the supersonic jet flying altitude of 15 km (UNSCEAR 2000). The annual effective dose from cosmic rays ranges between 260 μSv/y at sea level to 1070 μSv/y at an altitude of 3.4 km (UNSCEAR 2000), and the variation of it with elevation is given in Table 1. As per UNSCEAR report, the mean annual dose due to cosmic rays is observed to be 0.39 mSv/y, while the reported range is 0.3 to 1.00 mSv/y (UNSCEAR 2008).

However, in Indian Scenario, the contribution of cosmic radiation is 0.35 mSv/y to the natural background radiation considering its geography (Puranik and Ramchandran 2005).

Table 1 Altitude effect on cosmic radiation

Height from MSL (m)	Dose rate (μSvy^{-1})
0–150	260–270
150–305	270–280
305–610	280–310
610–1220	310–390
1220–1828	390–520
1828–2438	520–740
2438–3408	740–1070
> 3048	1070

Terrestrial Radiation

Naturally occurring primordial radionuclides arise in the earth, since its evolution. Primordial radionuclides are of two types, viz., "single" and "series." "Single" radionuclides are those devoid of decay series, which are naturally occurring and have stable decay products, e.g., ^{40}K, ^{87}Rb, ^{147}Sm, ^{115}Ln, ^{138}La, ^{176}Lu, etc. Most of them are beta or beta-gamma emitters. "Series" radionuclides are comprised of a parent and its daughters, which finally decay into a stable daughter product. There are three natural "series," viz., the uranium-235, uranium-238, and thorium-232 series. These series radionuclides undergo alpha and beta particle emission along with gamma radiation and end up in a stable element. The members of the series include the isotopes of the inert gas, radon (Rn-219, Rn-222, and Rn-220). The radon gas diffuses through the soil and emanates into the atmosphere. The decay products of radon being solids, they condense on the atmospheric dust particles of sub-micron sizes and form radioactive aerosols. The radionuclides of the uranium and thorium decay series are present at varying concentration levels in all compartments of the environment. The variations in their concentrations in different soils depend on the composition of their parent rocks. For example, soil derived from granitic rocks is associated with an elevated level of NORM while less in case of sedimentary rocks. It is observed that shale and phosphate rocks have relatively higher radionuclide concentrations (IAEA 2014b). Equilibrium status among the series radionuclides is a function of their physical, chemical, and radiological properties (S. Molla et al. 2021). Natural occurrence of radionuclides in rocks and soils and derived construction materials/household lead to radiation exposure to the habitat. The activity concentration of NORMs in soil in different regions is given in Table 2 (UNSCEAR 2000).

The natural radiation exposure levels in different parts of India from terrestrial and cosmic radiation vary from 0.3 mSv/y (at Lakshadweep) to 1.4 mSv/y (at Kerala) (Nambi et al. 1986).

Table 2 Natural radioactivity in soil

Regions	Activity concentration (Bq/kg)			
	^{238}U	^{226}Ra	^{232}Th	^{40}K
India	29	29	64	400
China	33	32	41	440
Japan	29	33	28	310
USA	35	40	35	370
Russia	19	27	30	520
Spain	NA	32	33	470
Global (median)	35	35	30	400

Inhalation Exposure due to ^{222}Rn and ^{220}Rn

Major contribution of natural radiation dose, particularly internal dose, arises from the inhalation of ^{222}Rn, ^{220}Rn, and their decay products. Due to decay of ^{226}Ra and ^{224}Ra, the inert gases ^{222}Rn and ^{220}Rn are produced and occur everywhere due to their chemical inert nature. After the decay, these chemically inert gases emanate from the earth/s surface into the atmosphere and migrate to other regions. Their concentration in atmosphere depends on $^{226}Ra/^{224}Ra$ content in soil, porosity, moisture content, the prevailing atmospheric conditions, etc. The global average concentration of ^{222}Rn is about 10 Bq/m^3 and 40 Bq/m^3 in outdoor and indoor condition, respectively, while for the ^{220}Rn, it is 10 Bq/m^3 for both indoor and outdoor conditions (UNSCEAR 2000).

A study carried out in India extending up to 2000 household has reported an inhalation dose of 1.2 mSv/y for members of the public (Puranik and Ramchandran 2005). Different types of dwellings made up of different construction materials located in different geological environment were included in the national survey. Integrated passive dosimeters based on LR-115 films were employed for measurements in this survey conducted for a period of 3 years. For inhalation dose assessment, an occupancy factor of 16 hours and 8 hours for indoor and outdoor was considered. About 15% of the total inhalation dose is attributed to thoron and its short-lived daughter products. From the study, it was observed that the average inhalation dose in India was 1.23 mSv/y compared to global average of 1.26 mSv/y. National and international agencies have recommended the use of action levels for protection against indoor ^{222}Rn, typically in the range 200–600 Bq/m^3 (IAEA 1996, ICRP 2007).

Internal Exposure due to Ingestion

Intake of dietary components associated with naturally occurring radionuclides contributes to the internal exposure. The activity concentration of ^{238}U and ^{210}Po in various food matrices and drinking water for India is given in Table 3 as reported in the UNSCEAR (2000) report. Globally, due to consumption of food materials and drinking water, the average ingestion dose to the member of the public was estimated

Table 3 Radioactivity in food and drinking water

Items	U-238	Po-210
Milk products	17	15
Meat products	NA	440
Grain products	7.4–67	15–120
Leafy vegetables	61–72	320
Fruits	0.4–77	16–140
Drinking water	0.09–1.5	NA

to 0.29 mSv/y. Major fraction of the ingestion dose was ascribed to intake of ^{40}K (0.17 mSv/y) followed by 0.12 mSv/y due to ^{238}U and ^{232}Th series radionuclides (UNSCEAR 2008). For Indian sub-continent, the average ingestion dose was 0.32 mSv/y due to consumption of dietary products.

Natural Radioactivity in the Body

The naturally occurring radionuclides enter into the body through the inhalation of air and ingestion of food matrices and water. Radiological characteristics and biokinetic behavior of the radionuclide also play a vital role in its retention in the body. In addition to uranium and thorium series radionuclides, cosmogenic radionuclides like ^{3}H & ^{14}C and single radionuclide like ^{40}K are observed in the human body and contribute to natural radiation exposure.

Relative Exposure in Natural Radiation Environment

The foregoing discussion leads to the inference that the natural ionizing radiation pervades the whole environment. Nature manifests itself in complexities and diversities, and consequently, there exists a wide range of variations in the external and internal exposure. The United Nations Scientific Committee on Effects of Atomic Radiation (UNSCEAR XE "United Nations Scientific Committee on Effects of Atomic Radiation (UNSCEAR)" 2008) carried out comprehensive investigations on natural sources of radiation exposure on a global scale. The radiation dose in natural high background radiation areas (NHBRA) and natural low background radiation areas (LHBRA) along with the global average dose are given in Fig. 1a, b, and c.

NORM Industries

NORM stands for naturally occurring radioactive material and is not under the purview of National Nuclear Regulatory Authority. Anthropogenic activities altered the equilibrium status in the natural resources and redistribute the naturally occurring radionuclides to a level that has potential to give radiation exposure to the

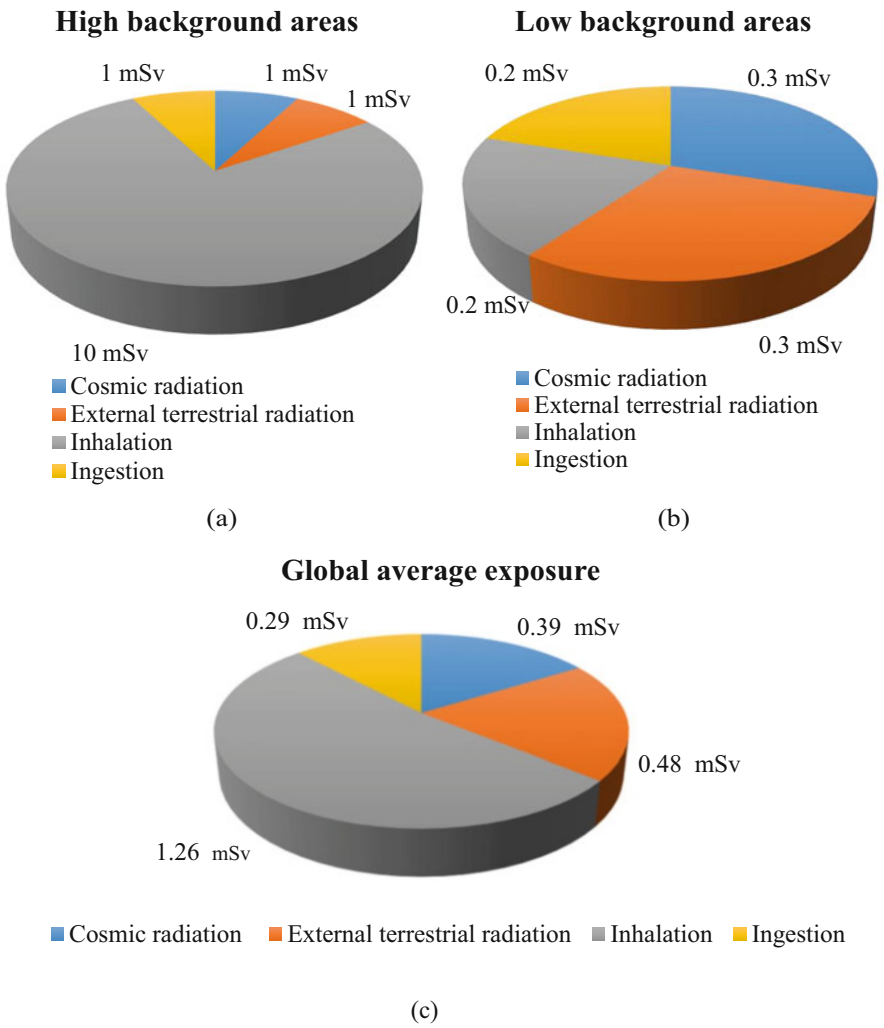

Fig. 1 Global average exposure due to radiation

occupational workers and members of the public which warrants the instruments of regulatory control measures. The alteration in concentration of radionuclides may be ascribed to physical, chemical, or radiological attributes. Radiological hazard caused by NORM depends on the activity concentration and amount of NORM, type of radionuclides, and exposure pathways (ingestion, inhalation).

Mine and mineral processing generated NORM residues due to their association with radioactive materials of natural origin. Huge quantity of NORM residues is generated every year from range of industrial processes, ranging from uranium, thorium, and phosphate mining to the treatment of groundwater (IAEA 2003a).

Copper Mining

Copper mines are known for their association with uranium and its daughter products. Several studies have been carried out on the level of NORM in existing copper mine waste and its impact on various compartments of environment (US EPA 1999).

Phosphate Production

Occurrence of U in phosphate deposits is very common due to their stable complexes and chemical bonding. Elevated levels of uranium in mineral apatite are reported across the regions. Phosphate rocks are subject to physical and chemical processing for manufacturing of phosphoric acids and subsequently phosphate fertilizers for agricultural application. Uranium concentrations in the phosphate rocks vary from 20 to 300 ppm (IAEA 2013a).

Coal Combustion

Most of the coal deposits are associated with uranium and thorium series radionuclides which are comparable with the activity concentration observed in the Earth's crust. Fly ash is the by-product of coal-based thermal power plants which contains the non-volatile elemental fraction and the heavy radionuclides. The average ash content of coal burned is about 10–40%, which means the uranium concentration is also increased by 2–10 times in the residue. Concentrated quantity of NORM in residue is a concern depending on its application in the public domain (IAEA 2018).

Heavy Mineral Sand

Few minerals are naturally resistant to chemical weathering and/or erosion process over the geologic age due to their geochemical properties, hardness characteristics, and specific gravity. Therefore, occurrence and accumulation of these heavy minerals in sedimentary sand deposits are very likely and a common phenomenon. Uranium-rich zircon, thorium-rich monazite, titanium-rich rutile, garnet, ilmenite, etc. are observed more frequently in placer deposits. All of these minerals are associated with naturally occurring radionuclides, i.e., uranium, thorium, and their daughter products (IAEA 2007; IAEA 2011b; IAEA 2012). ^{226}Ra concentration in titanium and zircon ores are reported in the range 0.73–1665 Bq/kg and 3219–48,100 Bq/kg (IAEA 2012; IAEA 2007). These minerals are used for the production of heavy metals and tiles industry.

Current Regulatory Practices

Typical activity concentration of naturally occurring radionuclides and the maximum annual effective dose, as a worst-case scenario, in various NORM industries are collated in Table 4 and Table 5, respectively (IAEA 2006). In addition to this, the annual average effective radiation dose to the public residing around NORM industries is compiled in Table 6 (IAEA 2006). From the tables, it may be inferred that the highest activity concentration in most of materials is in the order of tens or hundreds of Bq/g, but a higher activity concentration (~few thousands of Bq/g) is also measured in few NORM industries. Similar observation can be made for radiation dose received by the occupational workers and members of the public reasoning around the NORM facility. However, exceptions make a strong pitch for a regular assessment and regulatory review of NORM industries. The current national regulatory framework oversees only uranium and thorium mining and production facilities of DAE. Although there are other minerals as stated above that are associated with significant content of radionuclides of natural origin and are used for various industrial purposes, little attention has been paid to radiological safety in many cases. Currently, these industrial activities are not under the purview of radiological regulatory stipulations and warranted the national legislations to bring under the comprehensive umbrella of chemical-industrial-radiological safety. Comprehensive and long-term studies of radiation exposure to occupational workers and members of the public residing around these facilities and radiological impact on the surround environmental are limited or scarce.

Radiological Parameters and Concerns

The awareness of occurrence of radionuclides is barely necessary to adopt an appropriate radiometric technique for their quantification and implementation of appropriate radiological safety to avoid any unplanned radiation exposure to occupational workers and members of the public. Radionuclides of natural origin are classified into "series" and "single" radionuclides. There are four radioactive series having parent radionuclide of ^{238}U, ^{235}U, ^{232}Th, and ^{239}Pu; the first three are of natural origin, while the last one is manmade or synthetic. Due to very low mass abundance (0.72%) of ^{235}U and nonexistence of ^{239}Pu before the twentieth century, these radioactive series are of limited interest with respect to radiological protection in NORM industries and management of their residues/by-products, therefore not discussed further in the paper. Half-lives of ^{238}U is 4.5 billion years and that of ^{232}Th is 14 billion years and associated with 14 and 10 progenies, respectively. The radionuclides of radiological concerns in both the series are due to their relatively longer half-lives, higher specific activities, and greater potential radiotoxicity, namely, ^{238}U, ^{234}Th,^{230}Th, ^{226}Ra, ^{222}Rn, ^{210}Pb, and ^{210}Po in ^{238}U series and ^{232}Th, ^{228}Ra, and ^{220}Rn, in ^{232}Th series. In addition to series radionuclides, there are many single terrestrial and cosmogonic radionuclides, namely, ^{40}K, ^{87}Rb, ^{138}La, ^{147}Sm, ^{176}Lu, ^{14}C, ^{10}Be, ^{22}Na, ^{36}Cl, ^{39}Ar, and ^{81}Kr; however, the significance of

Table 4 Activity concentration in NORM industries requiring regulatory control. (IAEA Safety Series No. 49, 2006)

Category	Material/operation	Radionuclide(s) with highest activity concentration	Typical activity concentration (Bq/g)
Feedstocks	Monazite sand	^{232}Th series	40–600
	Metals ores (Nb/Ta, Cu, Au)	^{238}U and ^{232}Th series	Up to 10
	Zircon sand	^{238}U series	2–4
	Phosphate rock	^{238}U series	0.03–3
	TiO_2 feedstocks	^{232}Th	0.001–2
	Bauxite	^{232}Th series	0.035–1.4
Bulk residues	Red mud (alumina production)	^{238}U, ^{232}Th	0.1–3
	Phosphogypsum (H_2SO_4 process)	^{226}Ra	0.015–3
Slags	Niobium extraction	^{232}Th	20–120
	Tin smelting	^{232}Th	0.07–15
	Copper smelting	^{226}Ra	0.4–2
	Thermal phosphorus production	^{238}U	0.3–2
Scales, sludges, and sediments	Scale (oil and gas production)	^{226}Ra	0.1–15,000
	Scale (phosphoric acid production)	^{226}Ra	0.003–4000
	Residue (rare earth extraction)	^{228}Ra	20–3000
	Scale (TiO_2 pigment production)	^{228}Ra, ^{226}Ra	<1–1600
	Scale (rare earth extraction)	^{228}Ra, ^{226}Ra	1000
	Sludge (oil and gas production)	^{226}Ra	0.05–800
	Residue (niobium extraction)	^{228}Ra	200–500
	Scale (coal mines with Ra rich inflow water)	^{226}Ra, ^{228}Ra	Up to 200
	Scale (iron smelting)	^{210}Pb, ^{210}Po	Up to 200
	Scale (coal combustion)	^{210}Pb	> 100
	Sludge (iron smelting)	^{210}Pb	12–100
	Residue (TiO_2 pigment production)	^{232}Th, ^{228}Ra	<1–20
	Sludge (water treatment)	^{226}Ra	0.1–14
Precipitator dust	Thermal phosphorus production	^{210}Pb	1000
	Fused zirconia production	^{210}Po	600
	Niobium extraction	^{210}Pb, ^{210}Po	100–500

(continued)

Table 4 (continued)

Category	Material/operation	Radionuclide(s) with highest activity concentration	Typical activity concentration (Bq/g)
	Metal smelting	^{210}Pb, ^{210}Po	Up to 200
	Thorium compounds	^{232}Th	Up to 2000
	Thorium concentrate	^{232}Th	Up to 800
	Pyrochlore concentrate (niobium extraction)	^{232}Th	80
	Cerium concentrate (glass manufacture)	^{232}Th	10
	Fused zirconia	^{238}U	2–8
Products	Gas mantles	^{232}Th	500–1000
	Thoriated glass	^{232}Th	200–1000
	Thorium containing optical polishing powders	^{232}Th	150
	Thorium welding electrodes	^{232}Th	30–150
	Thorium alloys	^{232}Th	46–70
	Zirconium containing refractories	^{238}U	1–4
	Phosphate fertilizers	^{238}U	0.4–2
	Technical grade phosphoric acid	^{238}U	0.14–2
	Phosphogypsum plasterboard (building material)	^{226}Ra	0.004–0.7

Table reproduced from IAEA Safety Series No. 49 (2006) due to limited Indian data available on all the matrices mentioned in the table.

these radionuclides is least with respect to radiation exposure point of view except ^{40}K (UNSCEAR 2000). Lower abundance in natural environment and lesser activity concentration in different compartment of the ecosystem of these single are ascribed to their non-relevance; however, study of these radionuclides aids to unravel many environmental processes. Either series or single radionuclides emit mainly alpha, beta particle and gamma radiation during their decay process which requires identification and quantification mainly using radiometric techniques. The presence of these radionuclides of natural origin makes a case for a comprehensive study to avoid radiation exposure to all the stakeholders.

Measurement Techniques

Naturally occurring radionuclides emit alpha, beta particles and gamma radiation that could not be identified by our five senses or by other traditional means and

Table 5 Annual effective radiation dose to occupational workers in various NORM. Industries (IAEA Safety Series No. 49, 2006)

Sl. No.	Industries	Maximum annual effective dose (mSv)
1	Uranium ore mining	40–45
2	Processing of thorium concentrate	7.8[a]
3	Production of thorium compounds	82
4	Mining of rare earth ore	0.24–1[b, c]
5	Beneficiation of rare earth ore	0.28–0.61 [b,c]
6	Handling of monazite	0.3
7	Rare earth separation and purification	0.3
8	Decommissioning of a rare earths plant	8.94[d]
9	Mining of ore other than uranium ore	<1
10	Oil and gas production, offshore	0.5
11	Oil and gas production, onshore	0.05
12	Oil production, cleaning of pipes	3[c,e]
13	Titanium dioxide pigment production	< 0.5
14	Manufacture and use of zircon and zirconia	2.3
15	Processing of Sn, Al, Ti, and Nb ores	3.2
16	Phosphate ore storage	0.28
17	Phosphate fertilizer production	0.5
18	Phosphate industry facilities	2.7[f]
19	Zircon production	0.4
20	Manufacture and use of zircon and zirconia	2.3
21	Copper smelting	< 1
22	Combustion of coal	0.4
23	Drinking water treatment	< 1

Annual effective doses are excluding of contribution of inhalation dose due to radon and thoron, except where stated otherwise.
[a]Doses include contribution from inhalation of thoron
[b]Average doses
[c]Doses from external exposure only.
[d]Doses received over a 9 months decommissioning period.
[e]Doses received over a 5 month refurbishment period.
[f]Doses include contribution from inhalation of radon.
Table reproduced from IAEA Safety Series No. 49, 2006.

require specialized instrumentation for their measurement. This section illustrates the brief overview of the radiometric techniques in general practice which is based on the characteristics of radiation and their radiological properties. The methods discussed below may have few deviations for any particular radionuclide and matrix of analysis. Radiometric techniques generally adopted in most of the radioanalytical laboratories across the globe for water and soil/sediment matrices along with the required sample size, typical activity concentration range, and the observed detection limits are enumerated in Tables 7 and 8, respectively.

Table 6 Annual average effective radiation dose to members of the public residing around NORM industries (IAEA Safety Series No. 49, 2006)

NORM industries	Annual effective dose (mSv)
Mining of rare earth ore	0.044
Beneficiation of rare earth ore	0.043
Use of slag from rare earths and steel production in house bricks	~ 0.2
Mining of ore other than uranium ore	<1
Oil and gas production	< 1
Elemental phosphorous production	< 0.04
Use of phosphogypsum for construction of houses	0.02–4.5
Steel production	< 0.01
Combustion of coal	Negligible
Drinking water treatment	Negligible
Manufacture of Zircon/zirconia	Negligible

Table reproduced from IAEA Safety Series No. 49, 2006.

Table 7 Details of analytical technique used for water matrix

Radionuclide	Sample preparation	Analytical technique required	Sample size required (L)	Normal range (mBq/l)	Minimum detection level (mBq/l)
^{238}U	$Fe(OH)_3$ precipitation	Alpha spectrometry	5	1–100	0.12
^{226}Ra	$Fe(OH)_3$ and $BaSO_4$ precipitation	Alpha spectrometry	5	0.7–5.6	0.12
^{228}Ra	$Fe(OH)_3$ and $BaSO_4$ precipitation	Alpha spectrometry	5	0.9–9.4	0.12
^{226}Ra ^{228}Ra	Pre-concentration using MnO_2	Gamma spectrometry	10	0.7–5.6 0.9–9.4	0.2
^{234}Th	$Fe(OH)_3$ precipitation	Gamma spectrometry	50	0.05–0.68	0.12
^{210}Po	$Fe(OH)_3$ precipitation	Alpha spectrometry	2	0.13–2.1	0.12
^{40}K	Evaporation and concentration (2 L)	Gamma spectrometry/ indirect method using AAS	2	11,800/ 300–500 ppm total potassium	1700
^{210}Pb	Radiochemical separation	Liquid scintillation analyzer	2	0.5–20	0.12

Table 8 Details of analytical technique used for soil/sediment matrix

Radionuclide	Sample preparation	Analytical technique required	Sample size required (g)	Normal range (Bq/kg)	Detection limit for gamma (Bq/kg)	Detection limit for alpha (mBq/kg)
^{238}U	Sealed for equilibrium with daughter nuclides/ radioanalytical separation followed by electroplating	Gamma spectrometry/alpha spectrometry	300/10	7–100	2.9	0.12
^{226}Ra	Sealed for equilibrium with daughter nuclides/ acid digestion followed by radio chemical separation	Gamma spectrometry/alpha counting	300/10	10–100	3.5	0.12
^{228}Ra	Sealed for equilibrium with daughter nuclides/acid digestion followed by radio chemical separation	Gamma spectrometry/beta counting	300/10	10–100	1.0	0.12
^{234}Th	Sealed for equilibrium with daughter nuclides/ acid digestion followed by radio chemical separation and electroplating	Gamma spectrometry/alpha spectrometry	300/10	10–100	2.9	0.12
^{210}Po	Acid digestion followed by $Fe(OH)_3$ precipitation and spontaneous deposition	Alpha spectrometry	100	100–500	–	0.12
^{40}K	Sample in standard geometry	Gamma spectrometry	300	50–800	10.0	–
^{210}Pb	Sample in standard geometry	Gamma spectrometry	300	6–100	5.0	–
^{235}U	Sample in standard geometry	Gamma spectrometry	100	$\leq$0.2–5.0	0.2	–
^{228}Ac	Sample in standard geometry	Gamma spectrometry	100	10–100	1.3	–
^{214}Pb	Sample in standard geometry	Gamma spectrometry	100	10–100	1.0	–
^{214}Bi	Sample in standard geometry	Gamma spectrometry	100	10–100	1.0	–
^{208}Tl	Sample in standard geometry	Gamma spectrometry	100	10–100	0.6	–

Gamma Spectrometry

Gamma spectrometry is a radiometric technique which is a non-destructive and direct analysis of samples for identification and quantification of naturally occurring

radioisotopes (^{238}U, ^{232}Th, ^{226}Ra, ^{210}Pb, ^{40}K) emitting gamma energy in the range of 10 keV to 3 MeV (generally, may be used for higher energy range) in various geological, environmental, and biological matrices. The principle of gamma spectrometry is based on the probability of interaction of gamma radiation with the detector-sensitive solid volume, and the signal (either scintillation or electric current) is proportional to the activity concentration. Mainly, NaI(Tl) detector and HPGe detectors are in practice in radiological laboratories for measurement of naturally occurring radionuclides. Selection of a suitable detector is a function efficiency, resolution, sensitivity, detection level, no. of radionuclides to be measured, sample volume, sample geometry, accuracy, precision, availability of liquid nitrogen, and objective. In case of single or selected gamma radiation energies, less expensive NaI (Tl)-based scintillation detector could be preferred, while for matrices laced with multiple gamma emitting radionuclides/complex spectra, liquid nitrogen-cooled HPGe detector is chosen. The typical resolutions of NaI(Tl) and HPGe detector are 6–7% at 662 keV and 0.05–0.2% at 1332 keV, respectively. Representative samples should be collected and processed as per the standard protocols (IAEA 1989). A representative sample is counted in an optimized geometry and for a standardized counting time to achieve the required uncertainty (1–2% counting uncertainty for environmental samples). The detectors are provided with lead shields of thickness about 7.5–15 cm to reduce the background noise. Typical counting time for samples from NORM industries varies from 60,000 sec to 200,000 sec which depends on the anticipated activity concentration, sample geometry, objective of data quality, and acceptable uncertainty. Energy and efficiency calibration of the spectrometer is carried out using a suitable standard, preferably matrix-matching one if available. ^{238}U is associated with a weak gamma energy of very low gamma emission probabilities; therefore, measurement of ^{238}U is carried out from its decay products either assuming or ensuring the radioactive equilibrium between the parent-daughter duo. In general, the gamma energy (62 keV/92.5 keV) of ^{234}Th is used for quantification of the parent radionuclides. For sediment and biological samples, the representative sample is kept in sealed condition for 6 months to ensure the radioactive equilibrium. Equilibrium time of 1 month is adequate for measurement of ^{238}U in soil and other geological samples where the radioactive equilibrium between ^{238}U and ^{226}Ra could be assumed reasonably. The activity concentration of ^{232}Th can be estimated using multiple gamma energies assuming radioactive equilibrium among the series radionuclides in sealed condition for a period of 1 month. The gamma energies employed for quantification of ^{232}Th in geological and environmental matrices are 238 keV, 911 keV, and 2614 keV. Similarly, ^{226}Ra can be identified and quantified using the several gamma energies like 186 keV, 295 keV, 352 keV, 609 keV, and 1764 keV, while 46 keV is used to trace the activity concentration of ^{210}Pb in samples. The prominent single energy of 1460.8 keV is employed for determination of ^{40}K in biological, environmental, and geological matrices. The statistically comparable activity concentration of the radionuclide is expected when it is quantified using multiple gamma energies, but the uncertainty associated with the particular gamma energy will be different due to probability of interaction of that gamma photon in detector and other inherent factors. Therefore,

weighted mean and error propagation techniques are applied to calculate the central value and its dispersion. The details of these techniques may be found in Knoll 2010.

The typical background of the HPGe detector is observed in the range of 0.2–0.6 counts per sec for the photo peaks in the energy band of 46 keV to 2614.53 keV. The minimum detection level of these radionuclides is observed to be few Becquerel which is a function of background of the detector, counting time, and efficiency of the detector. Appropriate correction was made for contribution from higher energies to lower ones (IAEA, 2010).

The following equation is applied to quantify the activity concentration of a radionuclide measured in gamma spectrometer

$$\mathrm{A\ (Bq/kg)} = (\mathrm{C} - \mathrm{B})/(\mathrm{T}^{*}\ \mathrm{E}^{*}\ \gamma^{*}\ \mathrm{W}\ ) \tag{2}$$

where C is the gross counts of corresponding photopeak of the sample, B is the background of the photopeak under consideration, T is the counting time of the sample, E is the absolute efficiency of the detector for a specific geometry and gamma ray energy, γ is the absolute gamma line emission probability of gamma energy of interest, and W is the weight of the sample in kg. The details of the gamma spectrometry technique may be found in various ISO/IAEA reports (ISO 2019; ISO 2021; IAEA 2003a, b, c).

Alpha Spectrometry

Alpha spectrometry works on the principle of interaction of characteristic alpha particles on planar silicon detectors in vacuum. Vacuum is maintained continuously during counting to ensure a sharp peak with high resolution. However, a long tailing is observed on the lower energy end due to imperfect vacuum and loss of energy in the path. The technique is recommended for measurement of alpha emitting radionuclides such as U and Th isotopes, ^{226}Ra, and ^{210}Po. Since heavy and energetic alpha particle has very low range in air and other matrices, its separation is prerequisite from the bulk material for quantitative estimation. Separation of the radionuclide of interest (U, or Th) is achieved either through solvent extraction or ion exchange process. Due to selective spontaneous deposition of Po onto silver planchettes in dilute hydrochloride medium, no tedious separation process is adopted. The optimized radioanalytical process for determination of uranium isotopes and ^{210}Po is given elsewhere (Tripathi et al. 2013; Sharma et al. 2022).

In case of uranium isotope analysis, as a first step it is necessary to separate out uranium from bulk of the matrix for alpha spectrometric analysis. After complete digestion of environmental/biological/geological samples with the aid of a mixture of mineral acids, the sample residue is subject to concentrated HCl to covert the analyte and other metals to chloride form. Then, the sample is subjected to ion exchange/solvent extraction procure for separation of uranium from the bulk material. In case of ion exchange, anion exchange resin (chloride form, 1×8 polymerization, 100–200 mesh size) is taken for isolation of element of interest from other ions. Diluted nitric acid solutions are used to strip the U from the resin which is subjected to electroplating. Electrodeposition is carried out using sulfuric

acid–ammonium hydroxide solution adjusted to a pH – 2.2 for U and other actinides (Tripathi et al. 2013). The surface cleaned buffed stainless-steel disc is used as cathode, while platinum rod is used as anode. A constant voltage of 6 V and 0.3 ampere current are applied during electrodeposition with a distance of 2 cm between the cathode surface and anode for 2–3 h. Before switching off the voltage, few ml of concentrated ammonia is added to the electrodeposition cell to avoid the uranium from going back to the electroplating solution which is acidic in nature. Similarly, ^{210}Po was spontaneously deposited onto silver planchet in dilute hydrochloric acid medium at 80–90 °C with constant stirring condition (Sharma et al. 2022). Then, the electroplated or spontaneously plated samples are counted for alpha activity using passivated implanted planar silicon detector in an eight-chamber alpha spectrometer. The active surface area of the PIPS detector is about 450 mm^2 and depletion thickness of 100 μm. The estimated efficiency of the PIPS detector is 15.2% at 8 mm source (planchette) to detector distance. An integrated 4 K MCA is connected to the alpha detector for recording and analysis of spectrum. The source and sample deposition diameter is around 19 mm on a 25 mm stainless steel planchette. A high efficient vacuum pump was used to maintain the vacuum of the detector chamber around 75 mTorr throughout the counting process. Blank, spiked samples, and electroplated samples are counted for 24 hours to achieve the counting uncertainty less than 5% at the levels observed in environmental samples. Uranium standard solutions are processed similarly along with each batch to estimate the loss. In addition to this, ^{232}U and ^{209}Po tracers are added in the samples before commencement of radiochemical separation/processing of the samples to assess the chemical recovery of the method adopted so that an appropriate recovery factor could be applied to correct the activity concentration in the matrix.

Liquid Scintillation Analyzer

Though liquid scintillation analyzer (LSA) has been in practice for measurement of tritium and ^{14}C, of late it is applied to measure various naturally occurring radionuclides emitting alpha and beta particles (gross alpha, gross beta, ^{228}Ra, ^{210}Pb). The basic principle of the technique is absorption of alpha or beta energy by the liquid scintillator which is detected by photomultiplier tube. The pulse light is proportional to energy that is employed for discrimination of radioactive particles. Gross α and gross β levels of potable water are used as screening parameters before analysis of specific radionuclide. This method is one that the ISO and EU recommended for radiological screening of drinking waters as associated with least matrix interferences and self-absorption. In addition to this, ^{228}Ra and ^{210}Pb are measured using LSA. The details of optimized method at Health Physics Division, BARC, are given further.

Measurement of individual radionuclides is taken up after conforming the levels of gross alpha and gross beta levels in drinking water (WHO 2017). Gross α and gross β activities are estimated using alpha and beta counting systems, after radiochemical processing of samples which depends on total dissolved solid (TDS) content of the samples. Self-attenuation is the major concern in case of gross alpha but lesser significance in gross beta estimation. Conventional gross α and gross β

measurement techniques are associated with high degree of uncertainty and influenced by matrix characteristics. To nullify these difficulties, a method is standardized for quick scanning of gross α and gross β levels simultaneously with adequate sensitivity and minimum sample preparation with widely varying physicochemical properties (TDS, salinity, and pH). A triple to double coincidence ratio (TDCR)-based alpha/beta discriminating liquid scintillation analyzer (HIDEX 300LS) was used to standardize the method. The system uses three PMTs and a coincidence circuit to discriminate noise from true signal and a dual MCA technology to measure both pulse amplitude and duration, which is the basis for α and β separation as α pulses are of longer duration than β pulses. The discrimination was done by a 2D graph where x-axis represents the pulse amplitude and y-axis its length (Fig. 2). Due to long pulses, alphas have greater y-coordinate than beta pulses of the same X-coordinate. Subtracting the blank and using the TDCR value, the gross α and gross β levels were estimated and compared with guideline/limit values. Details of the methodology are available elsewhere (ISO 2017a, 2018; Gupta et al. 2018).

Radon, Thoron, and Their Progeny Monitoring Methods

^{222}Rn and ^{220}Rn are the isotopes of noble and inert gas radon and named differently, based on its origin. The former one is the one of series radionuclides of ^{238}U, while the latter is a daughter product of ^{232}Th series. Natural occurrence of both ^{238}U and ^{232}Th in every compartment of the ecosystem makes a case of ubiquitous presence of ^{222}Rn and ^{220}Rn due to spontaneous radiation decay and inert gaseous nature. Greater relevance of these radioactive gases in NORM industries is due to their higher activity concentration in workplaces, particularly confined spaces without adequate ventilation/underground mines and that have potential to deliver a radiation

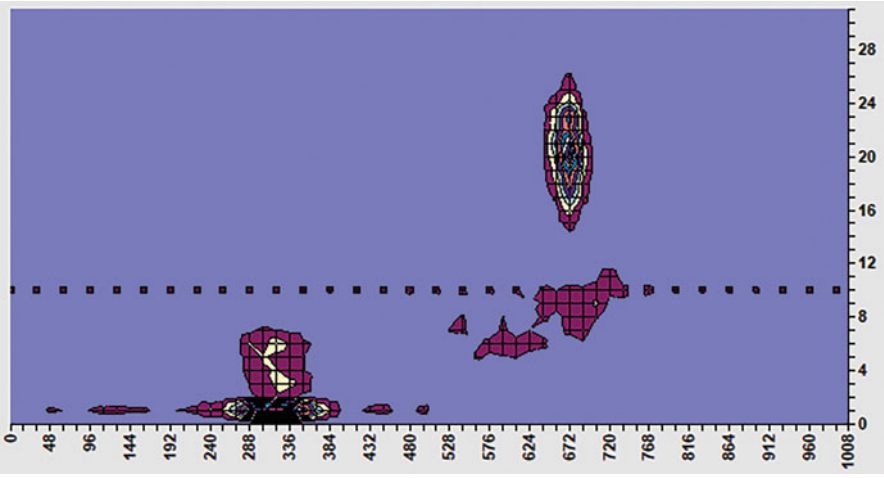

Fig. 2 2D spectrum of spiked water samples with pulse length index (PLI) denoted by the dotted line. Alpha (above PLI) and beta (below PLI) are separated by means of their pulse length

dose to occupational workers and pose an associated significant radiation risk (ICRP 1993). Though engineering control measures are implemented in uranium/thorium handling facilities, observations of higher activity concentration and radiation dose to workers are not uncommon. Also, similar observations were reported in literature for caves and confined workplaces. The immediate and short-lived daughter products of ^{222}Rn and ^{220}Rn are of more concern due to their particle nature that their inert parent radionuclides deliver more radiation dose to lung tissue/other tissue than their parents (IAEA 2003b). However, direct measurement of the short-lived daughter products is practically difficult. Therefore, in general practice, the parent ^{222}Rn and ^{220}Rn gases are measured in workplaces and transformed to progeny activity using site-specific equilibrium factor. There are many active/passive and spot/integrated analytical techniques for measurement of ^{222}Rn and ^{220}Rn in environment and various matrices. Few selected measurement techniques, in practice in India, are illustrated briefly in the follow-up section for measurements of radon, thoron, and their progenies.

Scintillation Cell Technique

Scintillation counting in the gaseous phase has been extensively used for the measurement of the radon concentration. It is one of the oldest and the most reliable techniques. Existing setup of scintillation/Lucas cell counting system has been adopted with development of protocols of sampling procedure, measurement technique, and calculation to assess the radon concentration in mine/workplaces. These radon (^{222}Rn) concentrations are used as an input parameter for occupational dosimetry program.

Scintillation (Lucas) cell, photomultiplier tube (PMT), and alpha counting system are the components of this measurement technique. The cell is used for collection of air samples from the desired location which is a cylindrical metallic chamber (preferably aluminum) coated with ZnS(Ag) sheet from all sides except the glass window for optical coupling with the PMT. In the opposite end of the glass window, the Lucas cell is provided with a Swagelok for evacuation and sample collection. Background counted pre-evacuated Lucas cells are employed for collection of air sample from the desired location through glass fiber filter paper. Application of filter paper ensures cutoff of particulate radon daughters and ingress of gaseous radon into the chamber. After collection of filtered air sample at the desired location, the Lucas cell is kept in dark condition for 180 mins to achieve radioactive equilibrium among radon and its decay products so that a statistically significant count could be registered.

Finally, radon activity concentration is estimated using the following equation:

$$\mathrm{Rn}\left(\mathrm{Bqm}^{-3}\right) = \frac{6.967 \times 10^{-3} \times \mathrm{C}}{\mathrm{E} \times \mathrm{V} \times \mathrm{e}^{-\lambda t}\left(1 - \mathrm{e}^{-\lambda \mathrm{T}}\right)} \tag{1}$$

where "Rn" is ^{222}Rn concentration in Bq.m^{-3}, "C" is background subtracted counts during the counting duration "T," "E" is efficiency of the Lucas cell (%), "V" is volume of the Lucas cell in liter, "t" is elapsed time after collection of sample in

minute, "T" is the counting time in minute, "λ" is decay constant of ^{222}Rn in minute, and 6.967 x 10^{-2} is the proportionality constant.

Two Filter Method

In this method, air is drawn through a metal tube, measuring about 50 millimeter in dia. and a length of 300 to 500 mm, equipped with a high efficiency filter at each end. The first filter removes all the radon decay products performed in the air. During transit of air through the tube, radon decays to Po-218. Some of the Po-218 atom plate out on the walls and the remaining deposit on the second filter. The second filter is quickly recovered and subjected to alpha counting for a suitable time immediately after the sampling. Typically, the counting interval is from 1 min. Post sampling to any suitable time (up to 75 min.) depending on the activity expected. Radon concentration (Bq/m3) is calculated using the formula

$$\mathrm{Rn} = 0.0167 * \mathrm{C}/\mathrm{E.Z.V.F} \tag{2}$$

where "C" is the background subtracted alpha counts, "E" is the counting efficiency (fraction), "Z" is the correction factor being a function of sampling and counting times (alpha emission factor), "V" is the volume of the tube (m3), and "F" is the fraction of Po-218 atoms formed in the tube which is deposited on the exit filter.

Thoron and Its Progeny Monitoring Method

Thoron (^{220}Rn) is a radioactive inert gas having a half-life of 56 sec. Thoron gas comes under the radioactive decay chain of ^{232}Th (4n) series. In general, indoor thoron and its progeny concentration are greater than that observed in outdoor. Out of all thoron daughter products, ^{212}Pb and ^{212}Bi are having significant radiological impact (Table 9). Due to the omnipresent nature of thoron and for the radiation protection, various methods are used internationally for the measurement of ^{220}Rn and its progeny. Working level (WL) is the historical unit used for the measurement of thoron progenies. One working level thoron represents the total potential alpha energy of 1.3 x 10^5 MeV from all the possible combinations of thoron daughters in one liter of dry air (Ramachandran 2010).

Table 9 Contribution of thoron progenies in 1 WL thoron environment

Thoron daughters	WL	No of atoms (N)	Alpha energy Mev (E)	NE (x 10^5 MeV)
^{216}Po	Scenario under 1 thoron WL or 275 mBq/l thoron is in equilibrium with its progenies	0.0595	15.55	0.001
^{212}Pb		15,200	7.79	1.18
^{212}Bi		1442.9	7.789	0.11
^{212}Po		1.2×10^{-7}	8.789	0.001
Total potential alpha energy (PAEC)				1.3

Thoron gas measurement can be done either by grab sampling or continuous sampling method. The different principles used for the measurement of thoron gas are as follows:

(i) Scintillation cell (Lucas cell).
(ii) Charcoal absorber followed by gamma spectroscopy.
(iii) Pulsed ionization chamber.
(iv) Two filter method.
(v) Scintillation detectors with ZnS (Ag).
(vi) Alpha particle spectrometer using silicon diodes.

Portable monitors are widely employed for the in situ determination of both radon and thoron and their progenies. Electronic radon detector (Make/Model: Durridge make RAD7) works use a solid-state alpha detector. A solid-state detector is a semiconductor material usually silicon that converts alpha radiation directly into electrical signal. The main advantages of the instrument are the ability to electronically determine the energy of each alpha particle which makes proper segregation of radon and thoron. Professional radon monitor (make/Model "Bertin make Alpha Guard) is an ionization chamber, and the alpha spectroscopy is the basic principle for the determination of radon and thoron simultaneously. These instruments also measure the relative humidity, temperature, and pressure.

Thoron daughters (^{212}Pb and ^{212}Bi) in air can also be measured by grab aerosol sampling followed by absolute alpha and beta counting. The delay time in counting plays important role for the calculation of thoron daughter concentration. The measurement process has a maximum error of 10% due to different equilibrium conditions.

Similar to the measurement of radon daughter concentration, a simple methodology has been used in thorium mining industries for evaluation of thoron daughter working level. Air sample is collected by passing air through a high efficiency filter paper for an appropriate time period. The alpha activity on the filter is measured for certain time period in between 180 minutes to 600 minutes post sampling. The thoron working level concentration is obtained using the relation

$$Thoron\ WL = \frac{C}{E * V * T * Z} \tag{3}$$

where
C – α count rate (cpm) on the filter paper at 300 min.
E – efficiency of counting.
V – sampling flow rate (lpm).
T – duration of sampling (min).
Z – Theoretical correction factor for thoron daughter at the counting instant (equivalent of Kusnetz's factor for thoron daughters).

The activity decay curve beyond 180 min may be approximately given by the relation

$$Z = 246 \times e^{-\lambda t} \tag{4}$$

where λ is the decay constant of ^{212}Pb (Th-B) and t is the counting time, for instant beyond 180 min.

In the past studies, the thoron is underestimated as compared to radon due to the difficulties in calibration and measurement technique, and there is no such epidemiological data regarding thoron exposure. Also, the "interference" is still a prime factor while measuring the radon and thoron in a mix environment. The importance of thoron and its progenies is high in the thorium mining industries. There is a huge gap area in the thorium mining industries for the thoron studies.

International Approach for Quality of Radioanalytical Laboratories

Demonstration of competence and traceability of the analytical results are very much essential of testing laboratories. It is very important for compliance of legislative limits, radioactivity certification of minerals, food materials for international trade, and environmental impact assessment studies. Quality data are prerequisite for policy decisions, mitigation strategies, and environmental protection. This also supports decisions on the establishment of action plans, programs, and measures to control contaminants. Accreditation and certification of radiometric measurement laboratories are essential to demonstrate their competence, technical capabilities, traceability, acceptability, and quality management system in place (ISO 2017b). Compliance of the standard ensures the technical competence and technical validity of the data. The choice of methodology for quality assurance in a radioanalytical laboratory is based on the laboratory competence for testing and calibration.

Quality Assurance

Quality assurance is a well-designed standard operating procedure based on scientific observations and experimental inferences undertaken by radioanalytical laboratories which is necessary to provide adequate confidence in the generated test results. In other words, quality assurance is a laid down procedure aligning with prevailing national/international standards that depicts overall measures undertaken in a laboratory to make sure the data quality. It is an integral part of laboratory analysis process to accomplish the reliability, repeatability, reproducibility, and accuracy of the data generated. This is required for understanding and evaluating the trends and impact of NORM on the surrounding ecosystem. The components of quality assurance are highlighted in Table 10.

Table 10 Technical components of quality assurance activities

Technical components	Practice
Suitable laboratory environment	Isolation from unwanted or spurious radiation source, electric and magnetic field, mechanical vibration, adequate ventilation, suitable equipment, etc. The radioactivity counting room should be isolated from sample preparation and separation laboratory
Educated, trained, and skilled staff	Training on analytical techniques before induction in the laboratories, regular participation in refresher capsule courses, and upgradation of latest trends in the domain
Requirements for calibrates and measurement standard	Availability of certified reference materials/Bharatiya Nirdeshak Drabya (BND) which are used for calibrating and generating the parameters required for evaluation
Requirements for reagents	Analytical grade reagents used for analytical procedures
Suitable equipment, maintained and calibrated	HPGe and NaI(Tl) gamma spectrometry system, alpha spectrometry system, liquid scintillation analyzer – monitor spectrum drift, background fluctuation, and stability in efficiency. Zn(S)-based gross alpha counter – check background and efficiency Radon monitors – check background and performance over time
Traceability, procedural lapses, and reporting results	Method validation using certified reference materials. Suitable tracers and carriers for method recovery in radiochemical analysis. Alternate technique for comparison, analysis of blind, analysis of spiked samples

Quality Control

Quality control is the final step of analysis process that certifies the results as per the national and international standards. (ISO 1994). Though the relevance of quality assurance is high to ensure the accuracy, precision, repeatability, traceability, and technical reliability of the data, quality control plays a pivotal role to demonstrate the competence and consistency of the data quality. It comprises of a set of technical operations and activities to verify each and every relevant statistical parameter (accuracy, bias, precision, reproductivity, repeatability, reliability, detection limit) to establish the data quality. It is an internal and integrated part of analysis and reporting process of testing and calibration laboratories. It is essential that the radioanalytical laboratory should develop and document standard procedures and implement on appropriate quality control program. The quality control program should comprise of steps such as analysis of CRMs/BNDs, regular participation in intercomparison exercises, cross-method checking, replicate analysis, record of blank/background regularly, and calibration of the equipment before commencement of measurement (Table 11). The documented procedure should be monitored at every step to get an indication of possible error.

Intercomparison exercises put an impetus to improve laboratory facilities and analytical techniques for measurements of radioactivity in food and other

Table 11 Common quality control activities in radiometric laboratories

	Technical components	Implementation in HPD
1.	Background determination	Determination of instrument background contributed in the region of interest. Analysis of blanks and background levels of analyte in samples as well as reagents
2.	Appropriateness of procedure validated with BND/CRM	Simultaneous radiochemical analysis and measurement of reference standards of similar matrix
3.	Analytical efficiency/ performance in each sample	Incorporation of carrier/tracer recovery factors in all radiochemical procedures
4.	Precision check	Analysis of the unknown sample at least in duplicates
5.	Calibration of counting equipment	Simultaneous calibration of counting equipment with certified reference materials
6.	Performance of counting systems	Comparison of background and calibration factors with quality control charts available and regularly updated

environmental materials. These exercises are useful to evaluate the precision of analytical methods and performance of the analytical laboratories. It is also a very powerful tool for maintaining and improving the laboratory quality and competence for measurement.

Both internal and external quality control steps are adopted to achieve the desired degree of uncertainty and to meet the objective of the analysis. Intra-laboratory intercomparison, cross-method verification, spike recovery, replicate analysis, etc. are the few activities taken up regularly to control the data quality. Moreover, successful participation in proficiency tests organized by national/international metrological institutes and intercomparison exercise confirm the laboratory competence to meet its objectives.

Quality assured data is needed to support decision-making process for preparation of action plans and remedial programs, assess the effectiveness of the control measures taken, and evaluate impact on the ecosystem. The basic purpose of intercomparison exercises is to check the reliability of the analytical method and biasness of the analytical technique. The implementation of the lessons learnt and recommendations help in the improvement of the performance and quality of data.

National Perspective and Policy

Diverse range of activities and practices, comprising of mining and ore processing, production coal, gasoline, phosphate fertilizer, thermal power plants, production and use of metals (such as uranium, thorium, niobium, zirconium), and water treatment, among others generates NORM and NORM residues (IAEA 2011a, b). Operation of NORM industries is associated with multiple hazards and, therefore, subjected to several regulatory agencies for safety compliances and conforms to the regulatory stipulations. Out of multiple hazards, radiological one may not be the dominant one but cannot be overlooked due to its associated potential adverse impacts (IAEA

2013b). However, the radiological aspects in NORM industries need to be characterized comprehensively and managed in a reasonable scientific way. Considering the long-term consequences on the ecosystem and socioeconomic attributes, a tailored approach may be devised and adopted for management of NORM residues. It is the high time to develop national policies and regulations of NORM consistency with international guidelines but without compromising the national interests and socioeconomic factors of a developing country (IAEA 2015). Moreover, a closer working relationship with NORM industries is needed to convince the radiological aspects and aid in adopting the regulatory stipulations on a logical and pragmatic manner in a cost-effective way (IAEA, 2016). The approach should encompass all the influencing parameters and attributes which is also eco-friendly, acceptable by all the stakeholders, and cost-effective. Efforts may be made to reduce, recycle, and reuse under the purview of circular economy through innovative technology for efficient use of limited natural resources (ICRP 2019). As mentioned, geological features of such rocks and minerals and the related radioactivity profile vary to a great extent. Such radioactive materials can be a potential source of the public exposure residing around such sites. The radioactivity concentration of NORM in environmental components like groundwater, surface water, and soil is mostly derived as a result of the related geological and physic-chemical weathering processes of host rocks/wastes, etc. (Sahoo et al. 2020). The activity concentration of U and Th and its daughters and their ratios could be used as environmental tracers for various governing processes and can be corroborated to their contents in substratum (base materials/soil) and water resources. Therefore, a national policy framework could be devised for site−/facility-specific NORM industry to reduce and nullify potential exposure of occupational workers, the public, and contamination of the surrounding environment.

Conclusion

NORM Industries need to bring under the regulatory purview and subject to control measures, if required after assessment. It is the high time to evolve an overarching national assessment policy framework and regulatory instruments to avoid any potential impact to occupational workers and members of the public. In addition to this, the policy shall ensure that contamination due to radionuclides of natural origin is well within the stipulated limits in accordance with the basic principles of radiation protection. An integrated graded approach of radiological protection can be adopted in NORM industries with striking balance between the different hazards/risks, adopting a reasonable and prudent attitude, and taking economic and societal factors into account. Engagement of all stakeholders, i.e., policy makers, scientific community, and facility operators, is crucial to adopt and implement the national policies for present and future generations.

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Metrological Concepts for Ionizing Radiation in Medical Applications

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Hala Ahmed Soliman

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Abstract

Currently, radiation metrology is all the standard measurement techniques for accurately detecting and quantifying the intensity and characteristics of ionizing radiation. It is also one of the most important tools for studying the processes of radiation effects on living beings and developing all measures and strategies aimed at protecting humans and the environment. Medical uses of ionizing radiation are among the longest established applications of ionizing radiation. It is utilized in a variety of diagnostic and therapeutic clinical procedures, ranging from simple chest X-rays to complicated brain tumor treatment regimens.

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Ionizing radiation, on the other hand, has the potential to be harmful, so a methodical approach should be utilized to balance the risks of radiation exposure to patients, staff, and the general public with the benefits of medical ionizing radiation. The scale of radiation danger is dose-related, with larger doses of radiation resulting in higher risks. Absorbed dose is a dose quantity which is the measure of the mean energy deposited in matter by ionizing radiation per unit mass. It is a fundamental physical quantity which can be used in the different fields where ionizing radiation is used. The absorbed dose concept has a wide range of applications, and it is widely accepted as a fundamental quantity in radiobiology, radiotherapy, and radiation protection. Technological innovation has resulted in state-of-the-art instrumentation that allows for precise radio diagnosis and administration of radiation to tumor lesions with little collateral damage to healthy tissues without compromising or limiting the obvious benefits in the prevention, diagnosis, and also in effective cure of diseases. Protection in medical exposures utilizing technological innovation does not necessarily mean the reduction of doses to the patient; it is best described as the management of the radiation dose to the patient to achieve the medical purpose. The dose is then a very critical and sensitive quantity.

Keywords

Radiation metrology · Diagnostic radiology · Radiotherapy · Radiation protection

Acronyms

AAPM	American Association of Physicists in Medicine
ALARA	As low as reasonably achievable
APLMF	Asia-Pacific Legal Metrology Forum
BIPM	The International Bureau of Weights and Measures
BSS	Basic safety standards
CR	Computed radiography
CT	Computed tomography
DR	Digital radiography
DRL	Diagnostic reference level
DRLs	Diagnostic reference levels
DRR	Digitally reconstructed radiographs
DSA	Digital subtraction angiography
GUM	Guide on the uncertainty in measurement
HVL	Half value layer
IAEA	International Atomic Energy Agency
ICRP	International Commission on Radiological Protection
ICRU	International Commission on Radiation Units and Measurements
IEC	International Electrotechnical Commission
IGRT	Image-guidance radiotherapy
IORT	Intraoperative radiation treatment
IPEM	The Institute of Physics and Engineering in Medicine

ISO	International Standards Organization
JCGM	Joint Committee for Guides in Metrology
LNT	Linear no-threshold
MLC	Multi-leaf collimator
MRI	Magnetic resonance imaging
MU	Monitor unit
NIST	National Institute of Standards and Technology
NMIs	National Measurement Institutes
PET	Positron emission tomography
PSDLs	Primary Standard Dosimetry Laboratories
QA	Quality assurance
QC	Quality control
RT	Radiotherapy
SAD	Source-to-axis distance
SPECT	Single-photon emission computed tomography
SRS	Stereoscopic radiosurgery
SSD	Source-to-surface distance
SSDL	Secondary Standard Dosimetry Laboratory
WHO	World Health Organization
X-CT	X-ray computed tomography

Introduction

Ionizing radiation is electromagnetic or particle radiation released by radioactive atoms and some technologies that can cause the ionization of other atoms directly or indirectly (Gottfried and Penn 1996). Ionizing radiations are known to be charged particles (heavy ions, alpha particles, protons, positrons, and electrons) with enough energy to ionize or excite atoms or molecules. Uncharged particles (photons, neutrons) that set-in motion directly ionizing radiation (charged particles) or can initiate nuclear transformations are known as indirectly ionizing radiations. X-rays, gamma rays, and electrons are the most frequent ionizing radiations employed in medicine.

Ionizing radiation is clearly recognized as a crucial tool for medical diagnosis and treatment, despite the fact that usage trends vary greatly across the globe. It is utilized in a variety of diagnostic and therapeutic clinical procedures, ranging from simple chest X-rays to complicated brain tumor treatment regimens. Each of these applications has a positive impact on patients. Radiology and nuclear medicine are the two main subcategories of ionizing radiation's diagnostic uses. In nuclear medicine, the radiation administered to the patient is internal as opposed to external in radiology (Gottfried and Penn 1996).

Modern diagnostic radiology permits the monitoring of a broad number of disorders and ensures faster, more precise diagnosis. Radiological procedures (simple film radiography, fluoroscopy, computed tomography) are predicted to have a significant impact on the speed of diagnosis in roughly half of instances, and they are of decisive importance in a large fraction of cases.

While some patients get direct applications of sealed radionuclide sources (brachytherapy) or therapeutic doses of radiopharmaceuticals, radiotherapy is typically administered through external radiation beams (teletherapy) (IAEA 2001).

Cancer is widely believed to be the leading cause of death and is seen as a major obstacle to raising life expectancy in practically every nation in the world. Radiotherapy is currently used in around half of all newly diagnosed cancer cases. Therapeutic procedures can be extremely complicated, putting a high premium on irradiation precision. They must be approached in an interdisciplinary manner to be effective, which necessitates effective and harmonious collaboration among radiation oncologists, medical physicists, and highly experienced technicians. However, it is important to know that radiotherapy of cancer is frequently accompanied by negative side effects. Some side effects are unavoidable, and they usually go away on their own or with therapy. Serious side effects can arise as a result of the close proximity of sensitive normal tissues to the treatment field or, in rare cases, individual radiation sensitivity. They don't compromise the purpose of radiotherapy. Every year, the proper application of radiation saves millions of lives. Even if only palliative treatment is available, the therapy reduces discomfort significantly (APLMF 2014). Figure 1 shows the application of ionizing radiation in medical science.

Regulations for the use of ionizing radiation in medicine are developed within the governmental, legal, and regulatory framework for safety. By avoiding harmful equipment usage behaviors, the intention is to safeguard the public's health and safety (Rosenblatt and Zubizarreta 2017). Radiation should only be used if it can diagnose or treat the patient and is both efficient and potentially advantageous.

In this chapter, some metrological concepts linked to clinical uses of ionizing radiation in diagnostic radiology, external radiotherapy, and radiation protection principles in medical disciplines will be concisely addressed.

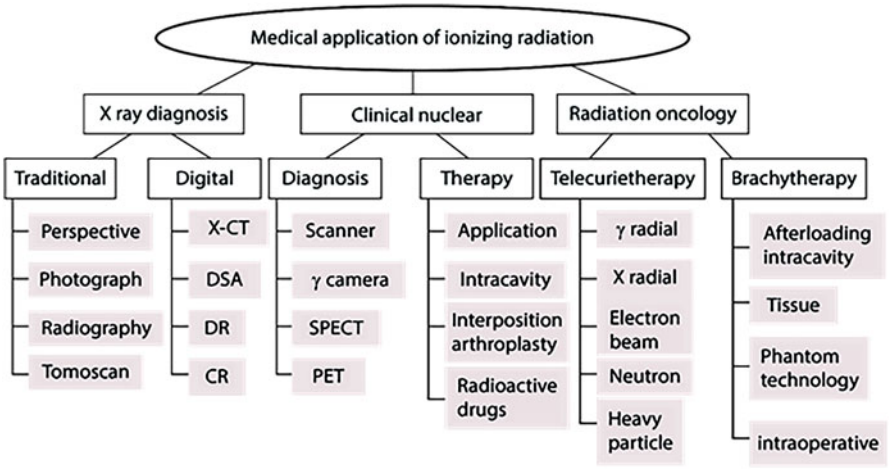

Fig. 1 Uses of ionizing radiation in medicine. (APLMF 2014)

Diagnostic Radiology

Diagnostic radiology has evolved from a scientific curiosity to an integral aspect of modern health care (Gottfried and Penn 1996). The field of radiology was originated on the use of ionizing radiation to diagnose human disease. As technology developed, imaging in diagnostic medical applications is expanded to include, beside radio diagnostic, other nonionizing radiation imaging techniques as magnetic resonance imaging (MRI) and ultrasound. In fact, the two nonionizing procedures are the ones in radiology that are growing the quickest. But X-ray imaging still represents between 80% and 90% of all imaging activities. It involves the use of three imaging techniques: radiographic imaging, fluoroscopic imaging, and computed tomography (CT).

A promising solution for acquiring high-quality CT data while the patient breathes is 4D CT or respiration-correlated CT (conventional and cone-beam tactics) (Keall et al. 2004). It uses X-ray beams with tube voltages ranging from 20 to 150 kV. In fluoroscopy, CT, and dental and general radiography, the tube voltages range from 50 to 150 kV; the anode material is usually tungsten. Mammography examinations are performed with tube voltages ranging from 22 to 40 kV and a variety of anode and filtration materials; the most common material is molybdenum for both anode and filtration (Peixoto and Potiens 2016); nevertheless, tungsten, molybdenum, and rhodium anode as well as aluminum, molybdenum, palladium, rhodium, and silver filtering for calibration may also be employed (IAEA 1994, 1999; Peixoto and De Almeida 2001, 2003a,b); Table 1.

The partial transmission of X-rays through the body is the fundamental principle that underpins X-ray imaging. X-rays from an external source interact with tissues in the patient's body either through absorption or scattering. As X-rays traverse the patient's body, the degree of interaction depends on factors such as the energy of the X-rays and density of the tissues. For detecting, examining, or recording the X-rays originating from the patient's body, the three basic imaging techniques rely on

Table 1 Radiation qualities needed to calibrate diagnostic radiology dosimeters. (Peixoto and Potiens 2016)

Radiation quality	Radiation beam	Material of an additional filter	Application
RQP	X-ray assembly	No phantom	General radiography and dental applications[a]
RQA	Added filter	Aluminum	Measurements behind the patient[b]
RQT	Added filter	Cooper	CT applications
RQR-M	X-ray assembly	No phantom	Mammography applications
RQA-M	Added filter	Aluminum	Mammography applications
W or Rh anode	Added filter	Mo, Rh, Pd and Al	Mammography studies

[a]Free in air
[b]On the image intensifier

various mechanisms. In all circumstances, however, the necessary interaction of X-rays with tissue has an unwelcome but unavoidable side effect; the patient receives some radiation dosage. Radiographic imaging is used to perform a vast majority of X-ray procedures. These radiographic imaging procedures are in turn typically divided into what are considered "conventional" examinations, on the one hand, and "contrast studies," on the other.

The characteristics of radiation qualities is very significant in diagnostic radiology because all dosimeter responses depend, to a certain extent, at least, on the spectral distribution of the X-rays being employed. The first half value layer (HVL) of the X-ray tube's voltage and the homogeneity coefficient are commonly used to describe radiation qualities (Pires et al. 2011).

The TRS 457 (IAEA 2007) provides, based on air-kerma determination, a detailed explanation of the dosimetry formalism, while the IEC 61267 describes the proposed radiation qualities (IEC 2005). This Code of Practice substantially follows ICRU 74 (ICRU 2005) regarding patient dosimetry for X-rays used in medical imaging.

The general methods for calibrating dosimeters used in diagnostic and interventional radiology are similar to those that apply to radiotherapy and radiation protection systems.

Radiotherapy

The basic aim of radiotherapy (RT) is to use appropriate ballistic techniques to give the highest feasible dosage to malignancies while minimizing damage to the normal and healthy tissues. However, despite all such efforts, it is never possible to irradiate only tumor cells. Therefore, radiotherapy must always strive to avoid giving the cancer a dose that is too low, which might lead to recurrence, or even to therapeutic failure, and to give the normal tissues a higher dose that cause serious side effects for the patient. The dose is then a very critical quantity.

The principles of medical physics, radiobiology, radiation safety, dosimetry, RT planning, modelling, and the interplay of RT with other therapy modalities are all part of the RT process (Satoshi 2008). Radiation oncologists, medical physicists, radiation technologists, and radiation nurses form the RT professional team, which collaborates to plan and provide RT to cancer patients through an integrated procedure. The sequential procedure is depicted in Fig. 2, and each step requires quality control (QC) and quality assurance (QA) to avoid errors and ensure that patients receive the recommended medication appropriately.

Radiation is delivered in fractions of the total dose 5 days per week in a normal course of treatment.

Depending on the type of tumor, the disease's stage, and the vicinity of organs or tissues to which only a limited dose is permitted, the total tumor dose may range from 30 to 70 gray (Gy). As was previously mentioned, teletherapy is radiation therapy delivered by an external ionizing radiation beam. Two possibilities are

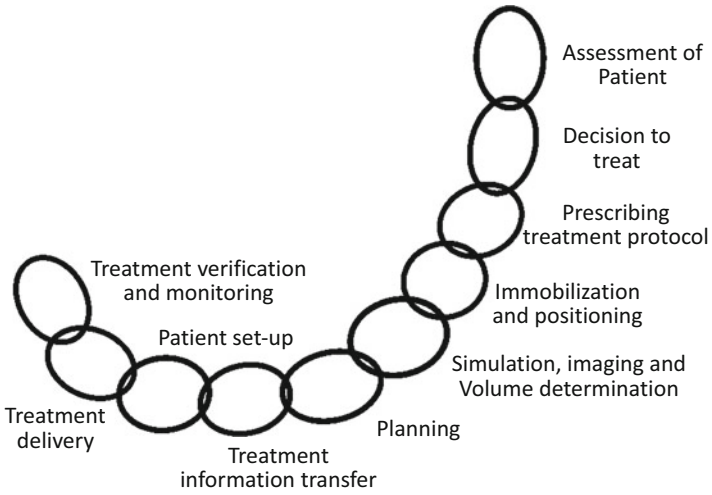

Fig. 2 Sequential process of planning and delivering radiotherapy to patients. (Satoshi 2008)

gamma rays (from a source of radioactive cobalt-60) and photons or electrons (from an X-ray generator or accelerator).

High-energy photon and electron beams are produced by linear accelerators and other electron accelerators for the treatment of cancer patients. A typical treatment employs two or more photon beams targeted at the tumor from various angles. Skin lesions, superficial lymph nodes, and various cancers that are close to the patient's surface are all treated with electrons because they have less power to penetrate tissue (Gottfried and Penn 1996).

Experts in radiation oncology have created various ways of external beam therapy in addition to "traditional" radiation therapy. Electrons are used in intraoperative radiation treatment (IORT) to treat malignancies that have been surgically exposed. After temporary repositioning of the surrounding and overlaying tissue, IORT delivers a single high dose of radiation straight to the tumor. The biggest benefit of IORT is for operable tumors in the pelvis and abdomen that cannot be physically removed.

Stereoscopic radiosurgery (SRS) uses radiation beams to precisely target a tiny tumor inside the skull. The dosage distribution that results creates the goal-containment zone that is narrow and high. SRS is now being studied for both primary and metastatic brain tumors, where it has been demonstrated to be an effective treatment for arteriovenous malformations.

In order to achieve a high-dose area of radiation that "conforms" as closely as possible to the shape of the clearly defined target and/or a lower dose to healthy tissue than conventional radiotherapy, conformal radiotherapy employs firm immobilization, three-dimensional computer treatment planning systems, and image-guidance radiotherapy (IGRT) techniques.

Shielding blocks are not necessary when using a linear accelerator equipped with a computer-controlled multi-leaf collimator (MLC) which are small tungsten leaves that are automatically positioned in the beam to enable the delivery of more complex fields to fit the shape of the tumor.

During treatment, the patient's immobilization period can be reduced (which improves radiotherapy delivery accuracy), and more treatment fields can be completed in a single session. MLCs typically have 80–120 leaves in opposing pairs (Fig. 3).

Additionally, MLC can be utilized to determine the strength of the beam in various regions of each incident treatment field separately. This method of administering radiation is referred to as IMRT, and it allows for the precise administration of higher doses to the target volume with a steep dose fall-off to surrounding areas (intensity-modulated radiation therapy). A fixed or moving gantry can change the location of the MLC's leaves over time. With MLCs, each treatment field can be divided into a variety of stationary shapes, and the radiation beam is activated as soon as the leaves stop moving at each predefined point. This is referred to as the "step and shoot method." In contrast, "dynamic multi-leaf collimation" describes the automatic computer-controlled repositioning of the leaves into beam segments without interrupting radiation.

While the linear accelerator rotates around the patient during dynamic conformal radiation therapy, a computer continuously modifies the radiation field. This technology, when used in conjunction with three-dimensional treatment planning, makes it simpler to treat tumors with intricate geometrical patterns and enables radiation oncologists to increase tumor dose while minimizing normal tissue damage. The 3D dose estimates are the next step in the treatment preparation process. Improved anatomical data processing and dependable quick dosage calculation procedures are now possible thanks to advancements in powerful computer software and

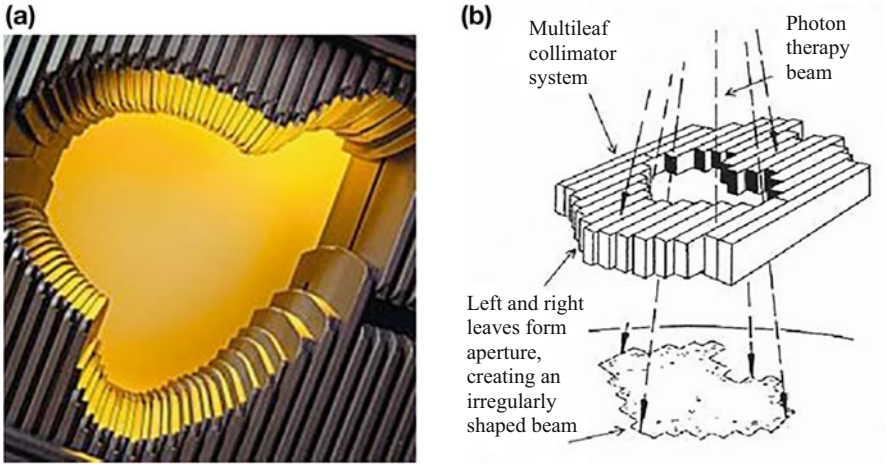

Fig. 3 (**a**) A multi-leaf collimator system (MLC); (**b**) an aperture being projected onto a patient. (Taşkin et al. 2010)

hardware. After selecting the appropriate beam weights, wedge, beam energy, and other factors, the dosage distribution is computed using computer algorithms.

After the treatment plan has been approved, the computer can produce digitally reconstructed radiographs (DRR), 2D projections based on the CT image set, and use them in the treatment planning system similarly to conventional radiographic films. Furthermore, all of the parameters are recorded and verified in the treatment unit's record and verification system.

The field of image-guided radiotherapy (IGRT) has recently realized new techniques that can be used in place of clinically used portal images (by radiographic film or EPID) to verify and control variations in patient and tumor positioning just before or during treatment. These techniques have been made possible by the prominent advancement of radiation oncology. For instance, an onboard imager combines an X-ray tube and an image detector to the therapy apparatus using robotic arms. These arms may move in three directions, allowing them to be positioned optimum for a clear view of the tumor and surrounding anatomy. High-resolution images of the tumor can be obtained using this device at a low dose to the patient.

In fluoroscopic mode, the gadget may also track tumor movements to give clinicians a clear picture of how a tumor will move during therapy due to normal breathing and other physiological processes. Furthermore, by picking a certain time in the patient's breathing cycle to scan and treat a tumor, the respiratory gating technology incorporates the impact of breathing during therapy. The goal is to achieve better target treatment, reduced margins, and lower dosages administered to healthy tissues as a result.

In addition, the patients obtain in vivo dosimetry measurements using semiconductor or thermoluminescence dosimeters during the first treatment fraction to ensure that the given dose is within tolerance. Finally, the treatment's effectiveness is evaluated by the patient's periodic follow-up. This final stage is critical for identifying patients who require salvage surgery or palliative treatment as well as for detecting local regional recurrence and preventing potentially treatable second main tumors. Potential adverse effects are confirmed during follow-up. The long-term objective of follow-up study is to determine whether there is a relationship between treatment outcomes and therapy delivery strategy (and thereby obtain knowledge about the efficacy of various treatment methods).

Treatment result must be measured in order to assess the treatment's effectiveness. Survivability, local control, and quality of life are essential outcomes in oncology.

Treatment Machines for External Beam Radiotherapy

Very little technological progress occurred during the first 50 years of radiation, with X-ray tubes, van de Graaff generators, and betatrons acting as the mainstays. H.E. Johns' creation of the 60-Co teletherapy unit in Canada in the early 1950s gave a huge push to the search for greater photon energies, putting the cobalt unit at the forefront of radiotherapy for several years (López et al. 2005). Technology has advanced to the point where state-of-the-art technology can deliver radiation with

Fig. 4 Performing portal imaging using modern cobalt machine and linear accelerator. (Van Der Merwe 2017)

remarkable precision to tumor lesions while causing minimal damage to normal tissues (Kry et al. 2017). Isocentric teletherapy treatment equipment (cobalt units or linear accelerators with a range of megavoltage energies) is commonly available in the majority of contemporary radiation departments (Fig. 4).

Medical linacs (linear accelerators), which were created at the same time as cobalt units, quickly surpassed them, evolving through five generations to become the most extensively utilized radiation source in modern radiotherapy. With its isocentric positioning and wide range of energies made possible by its compact and effective design, the linac offers good versatility for use in radiotherapy. It can also deliver either electron or megavoltage X-ray therapy. Other types of accelerators, such as betatrons and microtrons, are used in electron and X-ray radiation in addition to linacs. More striking particles, including protons, neutrons, heavy ions, and negative p mesons, are sometimes utilized in radiation; nevertheless, most modern radiotherapy is delivered with linacs or teletherapy cobalt units (López et al. 2005).

Clinical X-ray beams are often produced by decelerating electrons with kinetic energies of 10 KeV to 50 MeV in specific metallic targets, and their energy typically ranges from 10 KVp to 50 MV. Only a small portion of the electron's kinetic energy is released as X-ray photons, which are classified into two groups: characteristic X-rays and bremsstrahlung X-rays. The majority of the electron's kinetic energy is converted to heat in the target.

Calibration of Radiotherapy Sources and Machines

The administration of precise doses to the defined target volume is critical in modern radiotherapy. Based on a review of dose-response data and a clinical evaluation of

dose-delivery errors, the ICRU (1976) report 24 has suggested a 5% overall accuracy in tumor dosage delivery. The 5% accuracy recommendation is far from possible to achieve, given all the uncertainties involved in dosage administration to the patient (Andreo et al. 2005). In common clinical use practice, calibration of the output of photon and electron beams produced by external beam radiotherapy machines is a must. This basic output calibration is one of the links of the lengthy chain of the radiotherapy process (Fig. 2) constituting the accurate dose delivery to the patient. The other links of the chain include the procedures for measurement of relative dose data, equipment commissioning and quality assurance, treatment planning, and the actual patient setup on the treatment machine.

- The dose rate for a point P at a reference depth z_{ref} (often the depth of dose maximum z_{max}) in a water phantom for a nominal source to surface distance (SSD) or source to axis distance (SAD) and a reference field size (often 10×10 cm^2) on the phantom surface or the isocenter is usually stated as the basic output for a radiotherapy machine. The output of kilovoltage X-ray generators and teletherapy devices is often expressed in Gy/min, whereas the output of clinical accelerators is expressed in MUs where 1 MU corresponds to 1 cGy at the depth of maximum output.
- For superficial and orthovoltage beams, and occasionally for beams from teletherapy radioisotope devices, the basic beam output can also be expressed as the air-kerma rate in air (Gy/min) at a specific distance from the source and for a specific nominal collimator or applicator setting.

Radiation dosimeters and advanced radiation dosimetry techniques are used to calibrate the basic output of photon and electron beams. Radiation dosimetry measures and/or calculates the absorbed dose or another physically related quantity at a point of interest in a given medium, such as air kerma, fluence, or equivalent dosage.

A radiation dosimeter is any device that can provide a reading M that measures the dose D of ionizing radiation maintained in the dosimeter's delicate volume V.

- An absolute dosimeter is a dosimeter that generates a signal from which the dose in its sensitive volume can be calculated without the need for calibration in a known radiation field.
- Relative dosimeters are those that need to be calibrated in an established radiation field.

Reference dosimetry is the process of directly measuring the dosage or dose rate in water under predetermined reference conditions in order to calibrate a clinical radiation beam's basic output. There are now three different types of reference dosimetry techniques: calorimetry, Fricke dosimetry, and ionization chamber dosimetry.

These dosimeters can be used as absolute dosimeters, although they are rarely utilized in clinics for that purpose because doing so is laborious, and calibration in a

known radiation field has benefits like traceability to a standard laboratory. When used individually, an absolute dosimeter relies on its particular accuracy as opposed to using a standard that is shared by all radiation users.

Absorbed dose could be a thoroughly well-defined amount which is utilized for the measurement of presentation of any sort of fabric, counting organic objects and people to ionizing radiation. The concept and amount of "absorbed dose" have been presented by the ICRU (1998). Absorbed dose (desire esteem of the vitality bestowed to matter per unit mass at a point in a medium) could be a principal physical amount which can be utilized within the distinctive areas where ionizing radiation is utilized. In this way, the concept of absorbed dose has wide applications and is in fact recognized as the essential amount in radiation treatment, radiation assurance, and radiobiology (Chavaudra et al. 2004).

The absorbed dose for the water approach in low-, medium-, and high-energy beams of heavy ion of electron and of proton used for external radiation is determined by the Code of Practice TRS 398 (IAEA 2000). An ionization chamber or a dosimeter calibrated in terms of the absorbed dose to water in a Co-60 source is used to determine the absorbed dose to water formalism in high-energy photon and electron beams (Peixoto and Potiens 2016).

Measurement chambers for medium- or low-energy X-rays must be calibrated with similar-quality beams; however, the fundamental standards of absorbed dose to water for kilovoltage X-ray quality are only available from a small number of Primary Standard Dosimetry Laboratories (PSDLs) (IAEA 2009; JCGM 104 2009). Though using the appropriate approved protocols or the Codes of Practice for the Dosimetry of X-Ray Beams, calibration factors can be estimated in terms of absorbed dose to water from air-kerma calibration factors.

Quality Assurance in External Radiotherapy

Every one of the numerous radiation treatment processes adds to the overall uncertainty of the absorbed dosage delivered to the patients. Along with the inherent uncertainties, the necessity to transport enormous amounts of data between the many processes of radiation treatment, as well as difficulties in precisely identifying or replicating all technical and physical parameters, might result in systematic or random errors.

The analysis of clinical dose response curves yielded the accuracy requirements for assessing the absorbed dose in radiotherapy. According to Mijnheer et al. (1987), an increase in absorbed dose of 7% can result in unacceptable normal tissue problem probability. The standard requirement for the uncertainty in the delivery of an absorbed dose is, therefore, 3.5%, which should be at least twice the standard deviation in the absorbed dose. Wambersie et al. (1994) similarly found that the 3.5% accuracy criteria in the absorbed dosage provided cannot be fully met in photon therapy, even for a basic treatment scenario with two opposing beams. According to Mijnheer et al. (1987), these accuracy requirements can be expressed

as guidelines for good clinical practice and should serve as motivation to increase radiotherapy accuracy:

- At the specified point, the absorbed dose: 3.5%.
- Other target volume locations where the dosage was absorbed: 5%.

The accuracy can be less than 3.5% if the dose-response curves have shallow slopes. However, the exactness should be greater than this metric for steep dose-response curves. A high degree of precision is desired for a successful conformal therapy. As a result, quality assurance (QA) methods must be established, which is practically difficult given the complexity of the equipment and the numerous steps in the radiation therapy process (Soliman and Aletreby 2016).

The World Health Organization (WHO) has provided a clear definition of quality assurance in radiotherapy (WHO 1988), and this term is frequently used to refer to the processes that guarantee the consistency of the medical prescription and the safe fulfilment of that prescription with regard to dose to the target volume along with minimal dose to normal tissue, minimal exposure of personnel, and adequate patient monitoring aimed at determining the treatment's outcome.

There are numerous sets of national and international recommendations and protocols (IAEA 2021; AAPM TG 40 1994; AAPM TG 198 2021; IPEM 2018) that should be consulted when available since they incorporate the requirements for quality assurance and quality control for different radiation equipment items. These provide suggested tests, test frequencies, and tolerances. Some propose test methodologies, while others give practical guidance on quality assurance and quality control tests for many items of equipment.

A universal quality assurance program for equipment comprises:

- Initial specification, acceptance testing (the processes/procedure by which the supplier validates the baseline performance of the equipment to the satisfaction of the purchaser), and commissioning (after acceptance of the equipment, a full characterization of its performance for clinical applications over the all the ranges of possible operation are to be carried out) are all required before the equipment is used in a clinical setting, including calibration when necessary.
- Establishing tests for quality control after the commissioning process is complete.

Before the equipment is clinically used, quality control tests should be performed and a comprehensive quality control program should be implemented that will last the duration of the equipment's clinical life. Because treatments of patients will be decided, planned, and administered based on performance measurements taken at acceptance and commissioning, it is vital that treatment equipment performs consistently within acceptable tolerances throughout its clinical life. To test this, a continuing quality control program of regular performance checks is started soon following commissioning. If these quality control measurements detect drifts from desired performance, rectification is required.

- Further quality control tests following any significant repair, intervention, or adjustment or when there is any indication of a change in performance as observed during use or during planned preventive maintenance or routine quality control programs.
- A planned precautionary maintenance program, in accordance with the manufacturer's recommendations.

Only after the medical physicist in charge of customer acceptance is confident that the machine's performance meets all criteria outlined in the contract document, or has formally accepted any waivers as described above, can the equipment be officially accepted to be transferred from the supplier to the customer.

Uncertainties, Tolerances, and Action Levels

It is crucial to calculate the degree of uncertainty in each measurement's outcome. Typically, it is best to start with the JCGM Report 100 (BIPM 2008), commonly known as the ISO Guide on the Uncertainty in Measurement, ISO GUM (JCGM 104 2009), which provides an exhaustive approach to uncertainties. Mitch et al. also provide a very thorough overview of radiation dosimetry's uncertainties (2009). An example of an uncertainty analysis and budget for a typical measurement in a clinical radiation therapy unit was presented by Castro et al. (2008).

In any case, performance within the tolerance threshold provides adequate accuracy. Performance deviations from the action level cannot be accepted and require corrective action. Any quality control test needs to employ the right measuring equipment for the assignment. All of this equipment should have regular maintenance and quality assurance program. Conditions for irradiation and measurement methodologies are to be tailored to the task in question. Consequently, the quality control measurement ought to provide the closest approximation to the measured parameter that is correct. However, depending on the measuring procedure, there will be an accompanying uncertainty. The parameter's tolerance must take into consideration the uncertainty of the measurement technique used. Random fluctuations in measurement will result in unnecessary interruption, increased equipment standby time, and unproductive use of staff time if the measurement uncertainty is greater than the tolerance predetermined level.

Variances can be combined for combination components in quadrature to establish specific tolerance limits for individual parameters. Similar to tolerances, action levels permit more flexibility in monitoring and adjusting. For instance, it may be acceptable to continue clinical usage until the following day's measurement verifies this before taking any further action if a measurement of dose/MU constancy shows a result that falls between tolerance and action levels. Thus:

- No action is required if a daily measurement is proved to be within tolerance.
- If the measurement is higher than the action level, immediate action is required, and the equipment cannot be utilized in a clinical setting until the problem has been fixed, and the fix has been verified by measurement.

- However, if the measurement falls in the range between the tolerance and action levels, it may be provisionally accepted until the measurement on the following day.
- When repeated measurements consistently fall between tolerance and action levels, correction is required.
- Anytime a measurement exceeds the action level, a prompt inquiry is necessary, and if the investigation is confirmed, corrective action is required immediately.

Despite that some essential parameters may dictate to set tolerance and action levels much closer to each other, or even have the same value, action levels are normally set at close to twice the tolerance level. When comparing figures from multiple sources, keep in mind that recommendations may take slightly various methods to setting tolerance and/or action levels. The term "tolerance level" may be used to refer to values that are closer to action levels in specific circumstances (i.e., some workers use the term tolerance to indicate levels at which adjustment or correction is necessary). Performance standards are specifically listed under the two headings in certain suggestions. Test frequencies must be weighed against the amount of fluctuation that is allowed over the course of therapy, as well as the length of time that a parameter varies or deteriorates. Frequencies can be changed based on experience with a piece of equipment's performance and stability, with a nominal frequency established at first and then evaluated based on observations. As machines age, this may need to be revisited. The number of human resources available to conduct the tests may limit what may be checked, which may have an impact on the quality control program's structure. Tests should be designed to produce the essential information as quickly as feasible while using the least amount of time and resources.

Metrological Traceability

"The property of the result of a measurement or the value of a standard whereby it may be compared to stated references, usually national or international standards, through an unbroken chain of comparisons, all with declared uncertainties," is the definition of metrological traceability. The level of traceability determines the measurement's comparability: whether a measurement's result can be compared to a prior one, a measurement result from a year ago, or a measurement result from anywhere else in the globe (APLMF 2014). The uncertainties are influenced by each measurement result.

Uncertainty and traceability are two crucial components of the measurement itself. The determination of a given quantity has little significance outside of the immediate context without a traceability chain and an uncertainty analysis. In this instance, there can be no valid comparison with any other measurement of the same amount at a different institution, and there can also be no meaningful explanation of that comparison in the absence of an uncertainty estimate.

The calibration chain, which is the full path between the end user and a standard, must be followed in order to achieve traceability. In the domains of metrology and

quality assurance, the term "standard" has two separate definitions, but both suggest a sort of standardization. Furthermore, one should speak of a "measurement standard" or a "documentary or a reference standard," yet the qualifier is frequently left out of many conversations on the subject. In the end, both refer to anything (an artifact or technique) that is judged or compared to others of a similar type.

The primary national standard is the first artifact or measurement in the calibration chain. These are typically maintained and disseminated by the National Measurement Institutes (NMIs), also known as Primary Standard Dosimetry Laboratories (PSDLs) in the dosimetry sector.

The national government entrusts a PSDL with the responsibility of creating, maintaining, and enhancing the fundamental radiation dosimetry standards. The calibration of secondary standards is then used to distribute these standards.

The PSDLs typically maintain standards for numerous end-use applications, such as radiation protection, diagnostic radiology, radiation therapy, and industrial processing, for a variety of beam modalities, including KV X-rays, ^{137}Cs and ^{60}Co irradiators, radioactive sources for brachytherapy, and linear accelerator beams, among others (both photons and electrons. Air kerma, air-kerma strength, activity, absorbed dosage, and absorbed dose equivalent are some of the measurement parameters that depend on the modality and application.

More debate is required in cases where a nation lacks a PSDL. A Secondary Standard Dosimetry Laboratory (SSDL), a laboratory selected by the appropriate national authorities to provide the critical link in the traceability of radiation dosimetry to national/international standards for users in that nation, may be established in this situation.

An SSDL includes secondary standards calibrated at one or more PSDLs, with ultimate traceability to the BIPM. The SSDL Network (www-naweb.iaea.org/nahu/dmrp/SSDL/), which offers a platform for national SSDLs to undertake dosimetry comparisons and consequently improve radiation dosimetry consistency worldwide, is also run by the IAEA and WHO.

Although there are many different types of documents, reports, laws, and other things that can be included in standard documentation, the International Organization for Standardization (ISO) standards are the most significant in this topic. Manufacturers and service providers have accepted the ISO-9000 family of generic standards as a means of showing process conformance. To ensure a sufficient level of technical expertise in the transmission of measurement standards, ISO-17025 (ISO 2005) combines the quality assurance (QA) components of ISO-9000 with some special technical requirements for calibration laboratories (primary and secondary). End users may also have access to national guidelines, protocols, and best practice measures created by the national and/or international organizations like the IAEA (2000) or AAPM (Almond et al. 1999).

It is essential to understand that traceability is more than just a tag attached to a measurement, regardless of where you are in the chain (end user, PSDL, secondary laboratory). A look at how that traceability was achieved is a great way to examine the quality of the calibration process, as it effectively shows you "how" to make a statement like "NIST-traceable." This "how" is an essential information for any QA

program since it demonstrates both the existence of a continuous measurement chain from the end user to the initial realization of the quantity and the accuracy of the process (using the correct measurements standards and following the correct documentary standards).

Radiation Protection

Radiation has two types of effects: somatic and genetic. While somatic consequences manifest in the individual who has been exposed to radiation, genetic impacts manifest in the descendants. The likelihood of these consequences occurring, but not their severity, increases as radiation exposure increases. These effects can be explained simply by saying that when X-rays pass through a patient, they produce biological changes or even harm. There are two sorts of potential body damage: stochastic and non-stochastic, sometimes known as deterministic.

Stochastic means something that occurs as a result of the law of chance or probability and is independent of radiation dose. Stochastic effects, due to exposure to ionizing radiation, can cause cancer or have influence on gene-material affecting future generations.

Non-stochastic (deterministic) means that something will always occur but only when exposure is exceeding a certain *threshold*. The amount of damage (severity) increases the more the threshold value is exceeded (Munro et al. 2004).

The groundbreaking efforts toward establishing safety standards were initiated by a concerned group of scientists in 1915, followed by an International X-ray Unit committee in 1925 (Rollerstone 1927). This eventually led to the formation of the International Commission on Radiation Protection (ICRP) in 1928 (Clarke and Valentin 2009). While keeping the fundamental protection principles, the ICRP updated its guidelines and expanded its philosophy to include a system of radiation protection in publication 60. ICRP released Report 103 in 2007 as an updated general guideline for a radiation protection system (ICRP 2007). This new guideline offers direction on the fundamental ideas that might underpin effective radiation protection (Do Kyung-Hyun 2016). The Commission makes a distinction among three types of exposures: occupational exposures, public exposures, and medical exposures of patients.

1. Occupational exposures are all exposures that workers experience while performing their jobs (with the exception of exposures excluded from the International Basic Safety Standards for Protection against Ionizing Radiation and for the Safety of Radiation Sources (BSS) and exposures from practices or sources exempted by the BSS.
2. Public exposure is defined as radiation exposure that is received by the public, excluding occupational or medical exposure, background radiation, and intervention situations but including exposure to authorized sources and procedures.

3. Medical exposure is the exposure of patients when receiving diagnosis or treatment, people exposed while willingly providing support and comfort to patients, and of volunteers participating in biological research programs.

Under the circumstances, radiation sources and installations should include the best protection and safety measures so that while taking into account economic and social factors, the magnitudes and likelihood of exposures, as well as the number of people exposed, are as low as reasonably achievable (ALARA), and the doses they deliver and the risk they entail are controlled (i.e., protection and safety should be optimized).

In medical diagnostic exposure, patient protection is maximized by limiting exposure to the absolute minimum necessary to achieve diagnostic goals, but in therapeutic exposure, patient protection is maximized by maintaining normal tissue ALARA exposure in accordance with administering the necessary dose to the planning target volume (cancer). To ensure that the embryo or fetus receives the same level of protection as the broader public, pregnant employees must be protected.

Principles of Radiation Protection

The following goals must be met through optimizing the design of installations, machinery, and operational procedures (López et al. 2005):

- Prevent or reduce the likelihood that events may occur that could result in unplanned exposures.
- Prevent or reduce the magnitudes of exposures that could be caused if such events were to occur.

The International Commission on Radiation Protection functions to this day in developing and compiling recommendations on effective management of radiation and the risks it causes (Zainuddin 2018). It suggested that the general principles of radiation protection apply in all exposure situations the three keywords: justification, optimization, and dose limit (ICRP 1977):

- **The principle of justification**: Any decision that changes the radiation exposure situation should be beneficial rather than harmful. This means that by introducing a new radiation source, reducing existing exposure, or reducing the risk of future exposure, enough individual or community benefit should be produced to offset the harm.
- **The principle of optimization of protection**: The probability of exposure, the number of individuals exposed, and the volume of each of their individual doses should all be kept as low as is technically feasible, taking into account the economic and societal considerations. This indicates that the amount of protection should be the best possible under the circumstances, optimizing the benefit-to-harm ratio.

- **The principle of application of dose limits**: The total dose from regulated sources to any individual shall not exceed the permitted limits defined by the Commission in planned exposure situations other than medical exposure of patients. Within optimization of protection context, the concepts of dose constraint and reference level are used to limit individual doses. Individual dose levels must always be defined, whether as a dose limitation or a reference level.

Radiation Protection Fundamentals in the Medical Field

Medical exposure of patients typically entails particular considerations; dose limits or dose limitations are not appropriate. Dose limitations are meaningless since ionizing radiation, when applied at the proper dose for the right medical reason, is a valuable tool that does better than harm. As a result, medical radiation has no dosage restrictions, and the diagnostic reference level (DRL) is commonly employed as a reference value (ICRP 2007). To reduce the hazards of radiation exposure, any medical radiation exposure must be justified, and ionizing radiation exams must be optimized. The examination must be medically indicated and useful in order to be justified. The use of doses that are as low as reasonably achievable (ALARA) in respect to the diagnostic task is referred to as "optimization" (Do Kyung-Hyun 2016).

A different approach to radiological protection is required for some aspects of radiation exposure in medicine for patients than for other types of radiation exposure. Patients are purposefully exposed.

With the exception of radiation therapy, the objective is to use the radiation to perform an interventional procedure or provide diagnostic information rather than to deliver a radiation dose. Patients who receive medical radiation do so voluntarily, with the hope that it would improve their health. The voluntary decision is made with varied degrees of informed consent, which takes into account both the expected benefit and the potential hazards. The amount of information offered to acquire informed consent varies depending on the level of radiation exposure (diagnostic, interventional, or therapeutic) and the possibility of developing medical conditions as a result of radiation exposure. Each application of ionizing radiation for the purpose of assessing asymptomatic individuals should be evaluated and supported in terms of its clinical utility.

Application of Principles of Radiation Protection in Medical Fields

It concerns the proper use of the main concepts of radiation protection (justification, optimization of protection, and application of dose limits) because medical exposure of patients has unique considerations. Dose restrictions are not acceptable when it comes to medical exposure of patients because they often cause more harm than good. There are frequently other medical issues that are more serious, severe, or even life-threatening than the radiation exposure. The focus is placed on the medical operations' rationale and on optimization of the radiological protection. The best

ways to prevent unnecessary diagnostic and interventional treatments are to justify procedures (for a specified patient and purpose) and administer the patient dose proportionate to the medical objective. The equipment features that aid in patient dosage management and the creation of diagnostic reference values at the proper national, regional, or local level are likely to be the most efficient measures. Accident avoidance is a major concern in radiation therapy. Dose restrictions are acceptable for comforters and occupations, as well as for volunteers in biomedical research.

Justification

Validation for radiological patient protection differs from justification for other radiation applications in that the benefits and dangers associated with a course of action are often experienced by the same person. Furthermore, just because a procedure or course of action is appropriate in general does not mean that it is justified in the context of the individual patient under consideration (Do Kyung-Hyun 2016).

In medicine, a radiographic practice has three levels of justification:

(a) At the fundamental level, it is widely understood that the right use of radiation in medicine benefits society rather than harming it.
(b) A specific technique with a specific goal is established and justified at the second level.
(c) The aim of the second level of justification is to ascertain whether the radiological technology will enhance the diagnosis, treatment, or provision of vital information on the exposed individuals.
(d) At the third level, the procedure's application to a specific patient must be justified. As a result, each particular medical exposure should be justified beforehand, taking into account both the individual and the exposure's features as well as its specific goals.

Optimization

It's also one-of-a-kind to optimize patient protection. Individual limitations on patient dose could be detrimental to the medical objective of the operation, and in order to optimize patient protection in diagnostic interactions, the same person reaps the benefit and bears the risk. Radiation protection for patients is often optimized in medicine on two levels: (1) the design, appropriate selection, and construction of equipment and installations and (2) the daily working procedures.

This radiation protection optimization's primary objective is to control the protective actions for a radiation source in a way that maximizes the net benefit. It's not always necessary to lower patient dosages in order to maximize protection from medical exposures. The term "optimization of radiation protection" refers to the control of the radiation dosage to the patient to be proportionate to the medical goal while keeping doses "as low as reasonably achievable, taking economic and societal aspects into account."

Diagnostic reference levels (DRLs) are a type of research level that applies to medical exposure. DRLs are meant to supplement professional judgement, not to draw a line between "good" and "poor" medicine. They make a positive difference.

Models of Risk Assessment

The degree of radiation risk is dose-related, with larger doses resulting in higher risks. Diagnostic X-ray and nuclear medicine diagnostics may provide clear health advantages while also posing a modest risk (probability) of side effects. When using ionizing radiation sources for diagnosis, this fact is taken into account. Because radiation therapy necessitates huge doses of radiation, the risk of radiation-related side effects is significantly increased. The goal of regulating radiation exposure is to reduce the potential risk without jeopardizing or restricting the clear benefits of illness prevention, diagnosis, and successful treatment (optimization) (Gottfried and Penn 1996).

Radiation-induced malignancies are a well-known side effect of the treatment. The chance of getting a second cancer is determined by the total irradiation volume as well as the volume of the high-dose zone. The high doses of radiation delivered by these radiotherapy treatment techniques may cause long-term adverse effects and primarily second cancers due to larger out-of-field doses (Howell et al. 2006; Kry et al. 2007; Bednarz and Xu 2009; Cho et al. 2011; Zemanova et al. 2014; Soliman et al. 2020). With radiation therapy as a predisposing factor, the cumulative risk for the development of second malignancies has been predicted to range from 5% to 12% after a 25-year follow-up interval.

Radiation-induced tumor knowledge (and the basis for epidemiological response models) is mostly based on atomic-bomb survivor data, and the most often used risk models are based on these data. To predict the risk of second malignancies as a result of radiation treatment, several risk models have been devised and applied (Soliman 2019). A "linear no-threshold" (LNT) model is recommended by both the BEIR VII Committee (2006) and the ICRP (1991) for doses below 0.1 Gy.

In order to properly assess the time-integrated risk of a specific event following high-dose (more than 1 Gy) exposure, competing risks for other radiation-related outcomes must be included, and the resulting excessive lifetime risk estimate has an essentially nonlinear dose response (Alexander et al. 2019).

Because the LNT model tends to overestimate the risk for tissues near the treatment volume at high doses, while the competition model (CR model) underestimates the second cancer risk, Sachs and Brenner (2005) proposed a biologically based and minimally parameterized model that incorporates carcinogenic effects, cell killing, and proliferation/repopulation effects that were consistent with clinical data for high-dose second cancer data.

Additional aspects such as the varying sensitivity of different tissues and absorbed dosages to different organs must be considered when evaluating biological consequences of radiation after partial body exposure. A quantity called effective dose is used to compare the dangers of partial and whole-body irradiation at doses seen in diagnostic radiology and nuclear medicine. It is measured in sieverts (Sv). In radiation therapy, where very large absorbed doses affect particular tissues or organs, effective dose does not apply (Gottfried and Penn 1996).

Radiation Protection Quantities

The absorbed dose is the most fundamental physical dosimetry quantity; however, it is insufficient for radiation protection since the efficiency of different forms of ionizing radiation in destroying human tissue varies. Other dose-related values were included in addition to the physical numbers to account for both biological and physical effects of radiation on tissues. Organ dose, equivalent dose, committed dose, and collective dose are the four quantities (López et al. 2005).

The mean dose D_T in a certain tissue or organ T of the human body is known as the organ dose and is given by the following:

$$\boldsymbol{D_T = 1/\mathrm{m_T} \int_{m_T} D dm = \frac{\varepsilon_T}{m_T}} \tag{1}$$

where
m_T is the mass of the organ or tissue under consideration,
ε_T is the amount of energy that radiation has given that tissue or organ.

Equivalent Dose H_T

The biological harm (damage) to an organ is proportional to the physical average dose received by the organ, as well as the dose distribution pattern resulting from the radiation type and energy. When compared to gamma rays or electrons, alpha or neutron radiation causes more harm to the organ for the same dose. The equivalent dose H_T is the result of multiplying the organ dose $D_{T,R}$ by a radiation-weighting factor (w_R) to account for the radiation's effectiveness in producing health consequences. The equivalent dose is determined by H_T:

$$\boldsymbol{H_T = W_R D_{T,R}} \tag{2}$$

where

$D_{T,R}$ is the average absorbed dose of radiation of type R delivered to a tissue or organ T, where T is the tissue or organ and w_R is the radiation-weighting factor for radiation of type R.

- For X-rays, γ rays and electrons, w_R =1 while for protons $w_R = 5$, and for α particles $w_R = 20$. As for neutrons, based on the energy, w_R ranges from 5 to 20.
- The units of equivalent dose in SI system is J/kg and called sievert (Sv), the long-standing unit is the (rem) and the relationship between the two is as follows: 1 Sv = 100 rem. In order to further explain, if 1 Gy of photon dose given to an organ, then the equivalent dose will be 1 Sv. However, for the same dose of 20 keV neutrons, the equivalent dose is 10 Sv, since the detriment is 10 times larger, i.e., $w_R = 10$ for 20 keV neutrons.

- The dose to the organ $D_{T,\,R}$ is a measure of the absorbed energy per unit mass averaged over the organ volume, while the equivalent dose H_T is a measure of the resulting biological harm (detriment) to the organ or tissue *T*.
- The equivalent dose is determined by the sum if an organ is exposed to more than one type of radiation.

$$\boldsymbol{H_T = \Sigma\, W_R\, D_{T,R}}. \tag{3}$$

- Weighting parameters relating to radiation quality were applied to the absorbed dose to a point in previous ICRP recommendations, and the radiation weighted absorbed dose was referred to as the dose equivalent H. (not referred to an organ, but to a point).

Effective Dose *E*

The link between the chance of stochastic effects and the equivalent dosage is discovered to be dependent on the irradiated organ or tissue. This means that the effects of exposure on different organs or tissues for the same equivalent dose are varied. Tissue-weighting variables are required to account for these discrepancies.

Tissue-weighting factors $\boldsymbol{W_T}$ should represent the relative contribution of an organ or tissue to the total detriment due to the effects resulting from a uniform irradiation to the whole body.

To show the combination of diverse doses to numerous different tissues in a way that correlates well with all stochastic effects combined, ***the effective dosage E*** is defined as the total of tissue equivalent doses, each multiplied by the appropriate tissue-weighting factor $\boldsymbol{W_T}$, (ICRP 60 1991), i.e.:

$$\boldsymbol{E = \Sigma\, W_T\, H_T} \tag{4}$$

- ***Tissue-weighting factors*** $\boldsymbol{W_T}$ are tabulated in the ICRP 60 JCGM 104:2009 report and in the IAEA safety standards.
- The unit of effective dose is J/kg, and its special name is sievert (Sv).
- A uniform equivalent dose over the whole body gives an effective dose that is numerically equal to the uniform equivalent dose.
- The weighing factors $\boldsymbol{W_T}$ and $\boldsymbol{W_R}$ are equally independent, i.e., the tissue risk factors $\boldsymbol{W_T}$ are independent of radiation type, and the radiation weighting factors $\boldsymbol{W_R}$ are independent of tissue type, allowing us to write:

$$\boldsymbol{E = \Sigma_T\, W_T\, \Sigma_R\, W_R\, D_{T,R} = \Sigma_R\, W_R\, \Sigma_T\, W_T\, D_{T,R}} \tag{5}$$

- When a setting contains only one type of radiation, the effective dosage is calculated by the following:

$$\boldsymbol{E = \Sigma\, W_T\, D_{T,R}} \tag{6}$$

- The effective dose is a dosage measurement that reflects the amount of radiation harm that the dose is expected to cause. Direct comparisons of effective doses from various radiation types and exposure scenarios are possible.
- Annual dosage limits for occupational and public exposure are indicated in terms of annual effective dose, and in the case of exposure to an organ, hands, or feet, an equivalent dose is the given quantity.
- Effective dose equivalent, which was utilized in earlier ICRP studies, has been replaced with effective dosage.
- The equivalent dosage H to specific organs or the effective dose $\boldsymbol{E}$ can be determined for distinct irradiation geometry using an anthropomorphic phantom that simulates the human body. However, because no primary standards have been established for these variables, they are not directly measured.

Collective Dose

The abovementioned radiation protection quantities refer to an individual's exposure. The term "collective dose" refers to groups or populations that have been exposed, and it is calculated by summing the products of the mean dosage in each group of exposed people and the number of people in each group. The man-sievert is the unit of collective dose (man-Sv).

Conclusion

Modern medicine would be impossible without ionizing radiation. The importance of metrology in the study of biological phenomena and the creation of high-standard inspection or treatment approaches is highlighted by the use of ionizing radiation in medicine. Imaging with X-rays, computed tomography scans, diagnostic and therapeutic nuclear medicine, brachytherapy, the gamma knife, and linear accelerators are examples of the technologies that have greatly improved diagnosis and treatment in medical applications.

Radiotherapy, in particular, is a critical tool for cancer treatment that necessitates the highest level of precision to optimize the absorbed doses delivered to the tumor and surrounding normal tissues. Meeting the precision requirements for the dose provided to the patient for a specific treatment necessitates a high level of precision in each of the procedure's several steps. From precise beam calibration to beam data collecting, beam monitoring, and dose calculation, errors are introduced.

Although using ionizing radiation in medicine has many advantages, there is also a risk that it could have negative effects on patients, health-care workers, and the population. Acute injuries and chronic illnesses, like cancer, can be brought on by the same diagnostic and therapeutic procedures used to treat cancer. The overall goal of radiation protection is to safeguard humans and the environment from the dangers of ionizing radiation. The principles of radiation protection were then supported with the concepts of justification, ALARA (as low as reasonably achievable), and "benefits against the risks." But these could not ensure that the application of radiation protection has been optimized.

Cross-References

- ▶ Equipment for Environmental Radioactivity Measurement: Calibration and Traceability
- ▶ Radiation Dosimetry
- ▶ Radiation Metrology and Applications

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Nuclear Forensics: Role of Radiation Metrology

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Abstract

Nuclear forensic analysis is a relatively new approach to provide technical support for the investigation of security incidents that may arise due to unauthorized use of nuclear or radioactive materials. The increasing rate of smuggling, trafficking, illegal possession, and theft of nuclear or radioactive material has become a worldwide challenge. Considering the potential hazards associated with the nuclear/radioactive material, the proper investigation and

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identification of the material are extremely important to prevent illegal trafficking and other unauthorized activities. The nuclear forensic analysis provides clues on the origin and intended use of nuclear materials which supports nuclear attribution. In a designated nuclear forensic laboratory, seized nuclear/radioactive material will be analyzed for the characteristic signatures using several established analytical techniques. To establish the origin and nature of samples, it is important to investigate elemental composition, isotopic composition, chemical impurities, macroscopic appearance, microstructure, etc. Comparison of the signatures using the nuclear forensic library consisting of a database of known samples, processes, etc. also is an important requirement. The analyses should be carried out following certified procedures and standard techniques. Intercomparison exercises and certified reference materials play an important role in the quality assurance of data.

Keywords

Nuclear forensics · Radiation metrology · Nuclear materials · Characterization · Signatures · Interpretation

Introduction

Nuclear/radiological threats have become a major concern globally in recent years. Despite strict safety and security norms, many nuclear/radiological incidents have been reported involving fissile materials in the last three decades (IAEA 2021). Radioisotopes are being used widely in various fields including industries, research, medicine, etc. Some of these radioisotopes would be suitable for radiological terrorism and in the fabrication of radiological dispersion devices (RDD or dirty bomb). A large number of orphan sources are generated due to the loss of administrative control over radiation sources, resulting in high radiation exposure cases as a consequence of radiological emergencies (Dey et al. 2012; IAEA 2002a; NTI 2021). There are about 3857 nuclear/radiological confirmed incidents, reported to the Incident and Trafficking Database (ITDB) as of 31 Dec 2020. The year-wise breakup of the incidents is shown in Table 1. Among them, about 59% of incidents involve radioactive sources/isotopes, 14% are nuclear materials, and 27% are radioactive contaminated or other materials.

Nuclear forensics (NF) is comparatively a new approach that plays a vital role in the detection or investigation of nuclear/radiological incidents (either pre/post) through technical support by providing crucial information about the evidence found at the crime scene. During the examination of any security incident, certain information regarding the production and history of the material/evidence will be required. NF study supports these inquiries by the analysis of confiscated nuclear or radioactive materials. In NF, different analytical techniques are used for characterization of the material either intercepted intact or retrieved from post-explosion debris, in terms of elemental, isotopic composition, chemical, and other physical parameters using validated signatures. The results are compared against reference

Table 1 Year-wise breakup of the confirmed incidents involving nuclear/radioactive materials out of regulatory control reported to ITDB

Year	Number of incidents	Year	Number of incidents
1992	5	2007	236
1993	57	2008	177
1994	72	2009	169
1995	48	2010	182
1996	27	2011	159
1997	34	2012	191
1998	51	2013	206
1999	73	2014	183
2000	76	2015	195
2001	68	2016	205
2002	72	2017	203
2003	106	2018	183
2004	158	2019	140
2005	183	2020	116
2006	282		

Source: IAEA (2021)

samples of known materials from reactors, weapons, and enrichment facilities or the materials from medical, academic, commercial, and other radiation facilities. NF includes an iterative and deductive approach where the results from one analysis support in guiding the selection of subsequent analyses. Proper selection and sequencing of analytical techniques and consequential analyses are most vital during NF investigation. The selection of analytical techniques or parameters depends on the type and nature of evidence/incident and the questions asked by the investigating agencies. For successful NF investigation, validated procedures concerning sample collection, analysis, interpretation and reporting results, and documentation are very important. In various stages of NF, starting from radioactive sampling to analysis and interpretation in NF, radiation metrology plays a significant role. The same are summarized below along with various challenges and limitations associated with nuclear forensic capacity building.

Guidelines and Methodology for Nuclear Forensic Analysis

Nuclear Forensic Model Action Plan

Nuclear forensic examination and related activities toward the investigation of a nuclear security event are described in the NF model action plan shown in Fig. 1, prepared by the combined efforts of the International Atomic Energy Agency (IAEA) and International Technical Working Group (ITWG) (IAEA 2006, 2011a). It provides a uniform platform for responding to the seizure of nuclear/radioactive materials. Detailed phase-wise action plans and sequences that come subsequent to

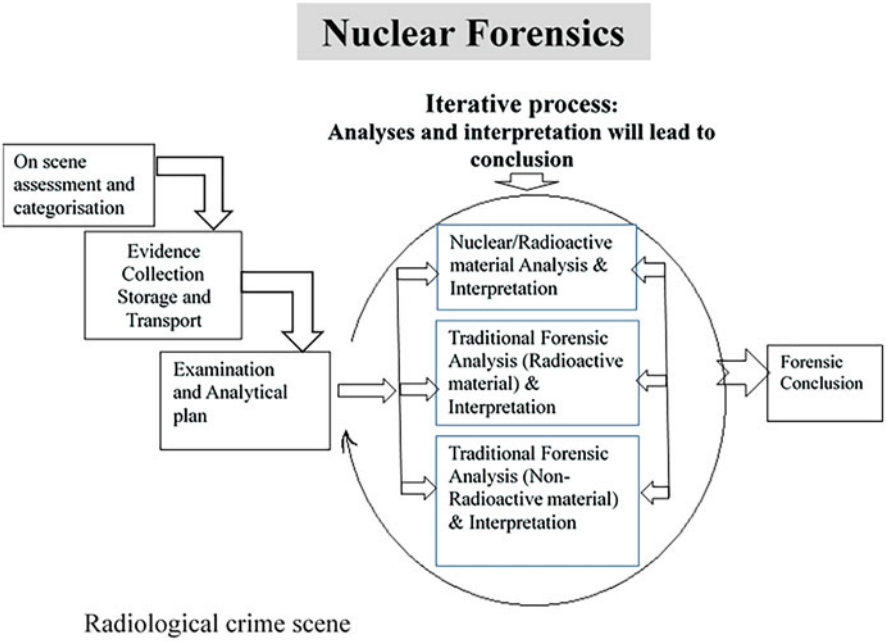

Fig. 1 Nuclear forensic model action plan. (Source: IAEA 2011a)

an incident are described in IAEA Nuclear Security Series No. 2, Technical Guidance (IAEA 2006).

The various stages in nuclear forensics where radiation metrology plays an important role are summarized below:

A. Identification of the nature of the material (nuclear or radioactive nature) and thereby identifying the potential radiological or chemical hazard
B. On-site material categorization by mobile nondestructive assay (NDA) equipment
C. Sample collection, storage, and transport of nuclear/radioactive evidence
D. Bulk and microanalysis of nuclear forensic signatures (elemental, isotopic, chemical, physical) in nuclear/radioactive materials at the specialized national nuclear forensics laboratory
E. Comparison of the results with the database and interpretation and attribution of the material to its source, production, origin, etc.

On-Scene Assessment and Categorization of Materials

Following a nuclear incident (pre/post), the first step toward nuclear forensic investigation is on-scene assessment and categorization of the material for the hazard assessment and rapid identification of primary radioactive components, thereby

generating the further action plan. This is very much essential for the personal protection and early identification and macroscopic characterization of the material. Before proceeding with the action plan, it is always advised to confirm the absence of hidden explosives along with the material/device. A portable X-ray radiography device is used in the field for explosive identification through imaging of material in solid form and containers (IAEA 2006). The portable X-Ray system shall meet or exceed the requirements for compliance with IEC 60529, as amended, classification IP54 (NIJ 2007).

Nuclear materials are broadly categorized in terms of element (e.g., U or Pu), isotopic composition (i.e., fissile isotope concentration), physical and chemical form, extent of dilution, quantity, and irradiation and radiation level. The categorization is mainly based on a graded approach for protection against unauthorized removal of nuclear material that could have direct or indirect use in a nuclear explosive device (IAEA 2006, 2011b). Important radioisotopes, which could be directly or indirectly used for the construction of nuclear weapons/explosive devices, are ^{233}U, ^{235}U, ^{238}U, ^{239}Pu, ^{241}Pu, ^{230}Th, and ^{232}Th. Similarly, the categorization of radioactive sources is mainly based on the element and the radiation level. Radioactive sources are categorized into five groups on a scale of 1 (most dangerous) to 5 (least dangerous) to human health (IAEA 2006). Radioisotopes in RDD are expected to typically include ^{90}Sr, ^{137}Cs, ^{226}Ra, ^{60}Co, ^{192}Ir, ^{75}Se, ^{170}Tm, ^{169}Yb, ^{252}Cf, ^{241}Am, ^{239}Pu, ^{241}Pu, etc.

Nondestructive analysis using gamma-ray spectrometry coupled with neutron detection is used to categorize the suspected radioactive material without disturbing the evidence (IAEA 2006). Portable sodium iodide detector-based handheld identifiers or portable high-purity germanium (HPGe) detector-based spectrometers are examples among others. Currently, HPGe coupled with neutron detector is being used for macroscopic characterization of special nuclear material (SNM), e.g., Pu materials in confiscated materials at the crime scene (IAEA 2006). To prevent the trafficking of radioactive materials, stationary gamma and neutron detectors are commonly being deployed at borders and ports (IAEA 2002b). For checking the instrument's performance in the field, small, radioactive sources should be accompanied by the instrument. For most of the NDA equipment, Th-impregnated mantle (used in lantern) is used as a check source in the field. ^{252}Cf for a ^{248}Cm source is required for calibration and performance verification of a neutron detector. The fundamental principles of radiation detectors and an overview are given in the reference (Knoll 2010).

Nuclear Forensic Sample Type, Collection, Storage, and Transport

In NF investigation, the type and nature of samples or pieces of evidence and their collection methodologies play an important role in the data quality and subsequent NF interpretation. The NF sample type will depend on the nature of the incident, i.e., pre-/post-detonation. In case of pre-detonation or confiscation of nuclear/radioactive material, the type of sample includes (a) intact nuclear or other radioactive material

and (b) radioactive-contaminated and radioactive-non-contaminated evidence/material for traditional forensic analysis. Similarly, during post-detonation, NF evidence will include (a) traces of nuclear/radioactive source or device (unexplored part), nuclear or radioactive debris, and associated contaminated and non-contaminated materials for traditional forensics which may include environmental, geological, biological, or packaging materials. During the collection of the radioactive evidence, responders should comply with the radiation protection measures. They should follow national rules for evidence collection which include scene documentation, a chain of custody procedures, sample labeling, etc. (IAEA 2006). Collection of both traditional forensic and nuclear forensic samples should follow appropriate timing without compromising any of the evidence. The tools commonly used for the collection of specimens include disposable gloves, spatulas, forceps, sticky tapes, razor blades, transfer pipettes, swipe papers, etc. Multilayer ziplock polythene bags with proper leveling can be used for collection of NF evidences, e.g., radioactive-contaminated specimen or associated transport materials like hair, soil, dust, paper, pollen, etc. (Moody et al. 2014). Many investigations require temporary storage place for the evidence, before transfer to forensic laboratory. In that case, sufficient security measures should be in place to prevent evidence tampering.

Many times, for bulk nuclear or radioactive material samples, the entire sample may be larger than the sample size requirement for nuclear forensic analysis. Therefore, there is a need for a subsampling prior to transport to a designated laboratory for analysis. Subsampling protocols require special consideration and methods to ensure subsamples are truly representative of the bulk material. During subsampling, appropriate caution should be taken to avoid cross-contamination without compromising any evidence (IAEA 2011a).

Laboratory and Analyses Requirement

In NF investigation, the type of samples and type of analyses are chosen based on the queries asked by the investigating authority. Depending upon the incident and available evidence, there will be requirements for different analytical laboratories and analytical expertise in different disciplines (IAEA 2011a). For example, specialized radiological facility for handling and characterizing bulk nuclear or other radioactive material along with the expertise and knowledge on nuclear fuel cycle technology for interpretation of results are required. Similarly, for characterization of evidences (contaminated with nuclear or other radioactive material), there is a requirement, for analytical capability and expertise for accurate and precise determination of specific radionuclides/isotopes in trace or ultra-trace quantity involving separation or isolation from the bulk matrix. Capability for analyses of small samples/particles along with traditional forensic expertise is necessary. For biological samples (e.g., urine, blood, hair, tissue, etc.), experience with bioassay analyses or blood dosimetry, and environmental (e.g., minerals, dust, pollen, etc.) or geologic samples, and experience with environmental and geological sample analysis and interpretation are also required.

The nuclear forensic analysis should be carried out in designated nuclear forensic laboratories recognized by the country. It should operate under a quality assurance plan. Laboratory should preferably have accreditation to an international quality standard (e.g., ISO 9001, ISO 14001, ISO/IEC 17025, OHSAS 34 18001) along with validated analytical methods, documented procedures, and manpower with demonstrated capabilities and expertise. It should have a license to receive and handle the nuclear and other radioactive material for bulk and microanalysis. The NF laboratory should have appropriate operational procedures to prevent cross-contamination between samples, environmental safety and health protocols, hazardous material handling and waste disposal, storage and physical protection measures, and most importantly the accounting and control of nuclear material (IAEA 2011a).

Analysis of Nuclear Forensic Signatures

Several analytical techniques are applied in NF for the analysis of different NF signatures (elemental, isotopic, physical, chemical, morphology, etc.) for deriving information mentioned in Table 2, and detailed description can be found in Kristo (2012); Straub et al. (2020); Moody et al. (2014); Stanley et al. (2013); Mayer et al. (2013); ITU (2003); IAEA (2011a); and Krunal (2014).

Not necessarily all the techniques should be applied simultaneously for a particular case study. Depending upon the information required and the nature, type, and quantity of the available evidence/sample, suitable techniques are to be selected for the analysis of required parameters. However, the analysis and selection of analytical methods should follow a proper sequence in order to derive important information within the requisite period. Analytical tools used for NF with a typical timescale for completion of analysis are described in IAEA Nuclear Security Series No. 2 (Rev.1), Implementing Guide (IAEA 2011a). Similarly, prioritization of nuclear forensic analysis plan is required for smaller size sample analysis. Mostly nondestructive techniques along with trace and microanalytical techniques for small-sized samples are advised before other analyses.

Physical and Microstructural Analysis

Visual inspection of a sample helps in the identification of the material and identifying marks and serial numbers and determination of shape, size, weight, and density, and even color can be a good indicator for deriving certain firsthand information regarding the seized nuclear or radioactive material. Information on sample composition and homogeneity can also be obtained viewing samples under transmitted polarized light. The surface roughness of the material can be measured by profilometry (Pajo 2001). Electron microscopes, through a focused beam of high-energy electrons, can make images on a very fine scale. Through microstructural fingerprint, information on topography, morphology, elemental composition, and crystallographic structure can be obtained. SEM is the most powerful surface

Table 2 Example of analytical techniques used for different NF signature analysis in U or Pu materials with forensic information

NF signature	Analytical technique	Forensic information
Physical characteristics (form, shape, size, density, morphology, appearance, etc.)	Particle size analyzer, radiography, profilometry, dimensional measurement, optical microscopy, SEM-EDS, TEM	Radioisotope
Radioisotope composition	HRGS, AS, LSC, ICPMS, TIMS	Category, intended use, reactor type
Uranium isotope ratio	HRGS, AS, TIMS, ICPMS, SIMS, RIMS, AMS	Intended use, enrichment
Plutonium isotope ratio	AS, TIMS, ICPMS, SIMS, RIMS, HRGS	Type of reactor, burn up
Minor constituents/impurities in bulk radioactive material (elemental, ionic, chemical, concentration patterns)	IC, ICPMS, ICP-OES, GD-MS, FTIR, Raman, GC-MS, XRF, NAA, XAS, XRD	Process history, geolocation
Stable isotope ratios (O, S, Sr, Pb, Nd)	IRMS, TIMS, MC-ICPMS	Geolocation, process history, ore formation age
Age	AS, ICPMS, TIMS	Production date
Particle/microanalysis	SIMS, AMS, FT-TIMS, μ-XRF, μ-Raman	Clandestine nuclear activity, post-detonation information, device design

SEM-EDS scanning electron microscope with energy-dispersive sensor, *TEM* transmission electron microscopy, *HRGS* high-resolution gamma spectrometry, *AS* alpha spectrometry, *LSC* liquid scintillation counting, *ICPMS* inductively coupled plasma mass spectrometry, *TIMS* thermal ionization mass spectrometry, *SIMS* secondary ionization mass spectrometry, *RIMS* resonance ionization mass spectrometry, *AMS* accelerated mass spectrometry, *IC* ion chromatograph, *ICP-OES* inductive coupled plasma optical emission spectrometry, *GD-MS* glow discharge mass spectrometer, *FTIR* Fourier-transform infrared spectrometry, *GC-MS* gas chromatograph-mass spectrometer, *XRF* X-ray fluorescence, *NAA* neutron activation analysis, *XAS* X-ray absorption spectroscopy, *XRD* X-ray diffraction, *IRMS* isotope resolution mass spectrometry, *MC-ICPMS* multi-collector inductively coupled plasma mass spectrometry, *FT-TIMS* fission track thermal ionization mass spectrometry, *μ-XRF* micro X-ray fluorescence, *μ-Raman* micro-Raman spectroscopy

imaging technique frequently used in NF analysis with excellent special resolution on the order of 1 nm (Hawkes 2013; Volland 2016). SEM coupled with energy-dispersive X-ray (EDX) or wavelength-dispersive X-ray (WDX) is used to map the abundance and spatial distribution of the elements in the sample. Sample preparation is relatively easy and got applications for powder samples or inhomogeneous samples in NF (Murakami et al. 2005). Transmission electron microscopy (TEM) makes use of the energetic electron beam that can pass through thin layers of the specimen. Using TEM ultrathin layers (~100 nm thickness) can be examined with higher magnifications (several million times higher) than SEM and can image extremely fine structures with sub-nanometer resolution (Anderson 1978). These techniques are extremely useful during investigation of nuclear activities through environmental monitoring and chemical analysis of post-detonation samples owing to the availability of very small-sized (submicron level) particles (Eriksson et al. 2005).

Chemical Analysis

Concentration and Isotopic Ratio Analysis of Radionuclides

For measurement of activity concentration and isotopic ratio of radionuclides in NF materials, different nuclear/radioanalytical techniques are used. The prominent radioanalytical techniques based on measurement of different radiation emission (alpha, beta, gamma, and neutron). For trace-level measurement, both alpha and mass spectrometries are mostly used. Radiochemistry, in combination with radioactive counting techniques and mass spectrometry, has the potential to measure very low levels of isotopes, down to femtograms. Also with selective detectors, isotopic ratio with sufficient accuracy and precision is possible and thereby finds enormous applications in nuclear forensic studies.

Radiometric Techniques in Bulk Matrices

A. *High-Resolution Gamma Spectrometry (HRGS)*

Gamma spectrometry plays the most important role in NF investigation where nuclear and radioactive materials are involved. Some radionuclides emit characteristic gamma rays in the range of few keV to many MeV which can be very well used for the identification of the radionuclide present and to estimate its concentration in the sample. The technique being nondestructive, portable, with minimal sample preparation requirement and field-deployable, is used for on-site categorization described in section "Nuclear Forensic Model Action Plan." However, this technique is less sensitive for the analysis of actinides as the abundance of gamma emissions is very low. With a well-shielded low background detector system and longtime spectra acquisition, it is possible to analyze even uranium isotopes. Mostly more sophisticated high-resolution gamma spectrometers (HRGS) are used.

To produce quality data of radionuclides in the sample for use as a nuclear forensic signature, it is necessary to ensure stringent quality control in terms of the procedures used for the gamma spectrometry measurement and subsequent estimation of activity data of radionuclides at the laboratory as well as for in situ measurements. For the qualitative analysis of radionuclides, mainly two types of calibrations are required, namely, energy calibration and resolution calibration. Proper calibration of the spectrometry system for the energy and resolution contributes to the accurate identification of radionuclides based on complex photo peaks in the gamma-ray spectrum. These calibrations are performed using gamma sources of known and accurate gamma energy emissions. The resolution calibration or shape calibration is done on the spectrometer using well-spaced and prominent gamma energy peaks from the radionuclides. The standard procedures for the energy and efficiency calibration of the high-resolution gamma-ray spectrometry system have been described in ANSI; ASTM; ISO (2007); Gilmore and Hemingway (1995); ANSI (1999); and ASTM (2003).

Gamma spectrometry technique is a nondestructive radioanalytical technique where accurate and reliable efficiency calibration of the detector system is essential for the assessment of radionuclide concentration in the sample. For the quantitative analysis of the gamma-ray spectrum for the estimation of radioactivity content in

various matrices, it is essential to have efficiency calibration of the detector for the gamma energy of the nuclide and geometry of the sample. In view of the wide energy range of gamma emissions from radionuclides, few keV to several MeV, it is recommended to carry out efficiency calibration in the applicable energy range. For the efficiency calibration, standard reference sources with known activity and gamma emission probability are recommended. Standard reference materials traceable to international standard organizations are recommended for this purpose. Different types of radionuclides are involved in the nuclear forensic scenario. They may be of long-lived or short-lived nuclides in various forms, gaseous, liquid, or solid forms. Hence, it is very difficult to get proper reference standards for all nuclides of concern to nuclear forensics or any other analytical work. Many times, it is not practical or feasible to have radionuclides in gaseous or liquid samples with a short half-life, as normally encountered in radionuclides commonly used in nuclear medicine (^{77}Lu, ^{131}I, ^{99m}Tc, etc.). In such situations, alternative methods are used for the calibration of the system for assaying these nuclides. Other radionuclides having nearby gamma energy and sufficiently long half-life can be used for this purpose. Alternatively, simulation techniques are also used for the generation of efficiency calibration of the HPGe detector system for all geometries and energies of concern. The detector response generated using various simulation codes is validated using available standard reference sources and further adopted for the other radionuclides of interest. A built-in simulation routine is introduced in spectrometry software for the efficiency calibration and estimation of activity in common geometries for all radionuclides of concern (Venkataraman et al. 2005).

The characteristic values from the gamma spectrometry measurements, such as the minimum detection limits and associated standard measurement uncertainty, are of importance in metrology in general and for nuclear forensics in particular (ISO 2010). The assessment of measurement uncertainty provides confidence in a measurement result; similarly the detection limit quantifies the lowest activity that can be detected with the technique. The series of documents from ISO describes the procedure, in the field of ionizing radiation metrology, for the calculation of these parameters in gamma spectrometry.

B. *Alpha Spectrometry*

Alpha spectrometry is a prominent radioanalytical technique used for the nuclear forensic analysis of samples where actinide radionuclides are involved. Radioactivity with alpha emissions is observed in actinide radionuclides. The energy of the alpha particles emitted is characteristic of the nuclide and can be very well used for the qualitative and quantitative analysis of radionuclides present in the sample. Alpha spectrometry is a destructive analytical technique, highly sensitive for alpha particles with energies in the range of 4–8 MeV. Silicon detectors (PIPS: passivated implanted planar silicon) are commonly used for the detection and energy measurement of alpha particles from radioactive sources. Sample preparation is tedious and requires radiochemical preparation of samples including dissolution, chemical separation, and source preparation. The most common practice is to separate the

elements radiochemically from the sample and electrodeposited in stainless steel planchets for spectrometry measurements. Suitable tracer activity (an isotope of the same element) is also introduced to account for the recovery of the various radiochemical procedures. It is also essential to maintain the detector and source under vacuum during the measurement to minimize the degradation of the alpha spectrum due to air attenuation.

Alpha spectrometry offers low detection limits due to relatively high detection efficiency and the low background of the detection system. In order to ensure the quality of the data obtained from the analysis, it is essential to follow proper calibrations of the spectrometry system for qualitative and quantitative analysis. ^{239}Pu (5.15 MeV) and ^{241}Am (5.48 MeV) are commonly used for energy calibration. The energy resolution of the spectrometry system is specified for the width of the 5.48 MeV peak of ^{241}Am in the spectrum. The detection efficiency is estimated using a standard reference source prepared in the same dimension of the sample by electrodeposition. Standard procedures recommended by various agencies are to be adopted in the laboratory to ensure the quality of the data from the alpha spectrometry measurement (ISO 2010; ASTM 2018).

The accurate estimation of activity by alpha spectrometry technique is essential for the radio chronology studies involving pairs of nuclides like ^{234}U-^{230}Th, ^{235}U-^{231}Pa, etc. The uncertainty in the activity estimation contributes significantly to the calculation of the age of the sample after processing. Very low detection limit can be achieved for Pu analysis, considering the relatively short half-lives of the plutonium isotopes (e.g., the half-life of ^{238}Pu is 87.7 y). However, Pu/Am separation in the samples prior to analysis should be done in order to avoid the overlap in the alpha spectrum. As measurement of ^{240}Pu and ^{239}Pu separately is difficult, since both are having very close alpha energy, therefore it is usually measured as a sum only. Atomic ratio of ^{240}Pu/^{239}Pu can be measured by using mass spectrometer.

C. *Active Interrogation Techniques for SNM Characterization in a Sealed Source*

Active interrogation technique is an active method used to detect and measure special nuclear material (SNM) mainly U and Pu in sealed conditions. Either photon or neutrons are used for the interrogation, and consequently the gamma rays or neutrons emitted by the fission of SNM are measured for mass and isotopic information.

Photon interrogation techniques include (i) photofission and (ii) nuclear resonance fluorescence (NRF). The presence of highly enriched uranium (HEU) in cargo has been detected by irradiating the container with high-energy gamma rays (6–10 MeV) from bremsstrahlung source. The fission signatures of the material can then be detected by using suitable detectors. Penetration through multilayer of material and, thereby, application in sealed condition are the advantages of this technique. Requirement of heavy shielding at the same time is the disadvantage of the technique. A photon source is used in nuclear resonance fluorescence (NRF) for active interrogation. When an electron beam from an accelerator bombards a tungsten converter, bremsstrahlung gamma rays are produced. The gamma rays interact

the sample material and generate excitations in the nucleus. During de-excitation, discrete energies characteristic of the nucleus are emitted in the order of several MeV, which can penetrate through many layers of material. However, the use of an intense 3 MeV source for enrichment information may require heavy shielding. Isotopic as well as enrichment information in SNM can be obtained by this technique. ^{235}U has been identified within the energy range of 1.0–2.5 MeV using a continuous bremsstrahlung source. NRF gamma peaks of statistical significance in ^{235}U and ^{238}U occur at 1733, 1815, and 1862 keV and at 2176, 2255, and 2468 keV, respectively. The ratios of these gamma-ray lines can be used to determine the uranium enrichment.

Neutron interrogation techniques include active well coincidence counters (AWCC), californium shuffler, and differential die-away (DDA) technique. AWCC can operate either in a thermal-neutron or fast-neutron mode. In thermal-neutron mode, a cadmium liner is placed in the well to assay small or low-enriched uranium (LEU) samples, whereas, after removal of Cd liner, in fast-neutron mode, large quantities of ^{235}U can be assessed. Americium-lithium (Am-Li) or a neutron generator can be used as the interrogating source, and counting of the induced fission reaction neutrons from the uranium is carried out using coincidence techniques. With the help of HPGe detector, 185 keV peak from HEU and the 100l keV peak from depleted uranium can be measured with the appropriate attenuation corrections, and the ratio of the peak areas gives the enrichment. The AWCC has a minimum detectable mass for ^{235}U of a few grams, whereas the differential die-away (DDA) technique can detect fissile materials in milligram quantity in dispersed form and can also be used for the determination of explosives in sealed conditions. Similarly, californium shuffler, an alternative to AWCC, uses ^{252}Cf sources to induce fissions. The energy of the neutrons from the californium is higher, 2.2 MeV, as compared with the 0.4 MeV Am-Li source used in the AWCC, which generates a high-precision measurement for bulk uranium. More details on these interrogation techniques are discussed in LA-UR-10-06l15P (De Simone et al. 2010).

Mass Spectrometry in Traces

Mass spectrometric (MS) techniques are important analytical tools for NF analysis. MS techniques can be used for both qualitative and quantitative characterization of radioactive elements especially actinides and fission products when they are present at very low concentrations. Both elemental and isotopic composition for bulk and microsample analysis with extremely high precision and accuracy are possible by mass spectrometry. Isotope dilution mass spectrometer (IDMS) can be used for quantification of radioactive elements at pico to femtogram level using an isotopic spike or through calibration against standard solutions. MS is having the best application in NF for the determination of isotope ratio with the highest precision and accuracy. In mass spectrometry, atoms or molecules are ionized into positive or negative ions using an ionization source. These ions are separated using a mass analyzer according to their mass-to-charge ratio. The intensities of the produced mass segregated ion beams are measured and plotted with respect to mass-to-charge ratio. The accuracy, precision, sensitivity, or selectivity of the measurement can be

enhanced by selecting a particular MS system with a particular ionization source, mass analyzer, or detector.

A. *Thermal Ionization Mass Spectrometry (TIMS)*

TIMS is the most commonly applied measurement technique for U, Th, and Pu concentration and isotopic composition with the highest accuracy and precision. The quantification (elemental assay) is also possible by IDMS (isotope dilution mass spectrometer, by addition of a spike, another isotope not present in the sample). In TIMS, a small amount of purified sample (some nanograms up to a few micrograms) is deposited on a metallic filament, consisting of refractory high-work function metal such as Re, Ta, Pt, or W. Ions are produced by resistive heating of the filament. For U samples, the U+ ions are formed and accelerated by applying a high voltage and subsequent mass separation (^{234}U+, ^{235}U+, ^{236}U+, ^{238}U+) using a mass analyzer which may be a magnetic sector or quadrupole along with Faraday cups or secondary electron multipliers for ion detection (Fedchenko 2015). Before the actual measurement, a sample preparation step is required. This involves a series of chemical separations of the element of interest from other elements using ion exchange or extraction chromatography (Hou et al. 2005; Aggarwal 2016). TIMS routinely measures differences in isotope mass ratios on the order of 1 ppm. Modern instruments with multi-collector detection enable the simultaneous measurement of the ion currents of different isotopes and provide isotope ratio data of high precision and accuracy which is of utmost requirement for NF analysis. In TIMS the accuracy in measurement is affected by both the mass discrimination observed in the ion optical system or ion detector and the mass fractionation that may occur during evaporation of the sample because of which the isotope ratio measurement may change with time. By different internal calibration techniques or by using isotopic standard reference materials with well-known isotopic ratios for an element, high accuracy can be achieved (Mishra et al. 2017). By applying the internal normalization methodology, correction for the isotope fractionation during analysis is made. Similarly, with interfering element correction methodology, isobaric interference can be avoided, and thereby high internal precisions in isotope ratio measurement (0.001% or better) can be achieved.

For calibration and routinely checking the performance of TIMS, certified reference materials (CRMs) are required. For Th, U, and Pu materials, a number of CRMs (certified for elemental content, isotope content, and isotope ratio abundance) are available from the Institute for Reference Materials and Measurements (IRMM), Geel, Belgium. A few reference materials like NBS-946, CRM 137/138/126A earlier known as SRMs 947 and 948, U-350, U-500, U-800, and U-850 are certified reference materials applied for the production date/age of the material in NF applications. Mostly these reference materials are prepared by mixing gravimetrically, highly pure and well-characterized metals or compounds. And some of them are also certified by data from well-calibrated equipment and techniques. TIMS is only suitable for elements with a low first ionization potential, and the laborious sample preparation prior to analysis limits its application. In addition to U, Th, and

Pu, TIMS has also been used for many other radionuclides, such as ^{41}Ca, ^{241}Am, ^{242}Cm, ^{126}Sn, ^{226}Ra, and ^{228}Ra, and also applied for measurement of stable isotope ratios like Pb, Sr, Nd etc., which are also very much required and reveal important information in NF (Hou and Roos 2008). TIMS can be used essentially for burn-up calculation, enrichment ratio, and age dating (Alamelu and Aggarwal 2016). Currently, the fission track-thermal ionization mass spectrometer is being used for actinide analysis in particle samples with high sensitivity.

B. *Inductively Coupled Plasma Mass Spectrometry (ICPMS)*

ICPMS is found to be the best alternative for estimation of both concentrations and the isotopic ratio of radioactive elements at trace level for NF applications. ICPMS is used for precisely quantifying radioisotopes as well as elemental impurities in bulk radioactive material like U or Pu material with detection limits ranging from 0.1 to 10 ppb (parts per billion) (Fedchenko 2015; Moody et al. 2014). ICPMS allows the sample in aerosol form through nebulizer techniques like a pneumatic nebulizer, ultrasonic nebulizer, or electrothermal vaporization, or laser ablation in an ion source operating at atmospheric pressure. Then, it follows the ionization of the aerosolized sample at a very high plasma temperature (5000–8000 K). These positively charged ions are then allowed to pass into a high vacuum region through an interface. Ion separation is done by using several types of analyzers and detection of these ions by Faraday cup or electron multiplier. Several calibration approaches can be used for accuracy in quantitative measurements by ICPMS. They include the use of pure standards obtained from Merck, Sigma Aldrich, matrix-matched standards from the National Institute of Standards and Technology (NIST, USA), or relevant certified reference materials, assuring traceability of the reported results.

The major limitation in the determination of radioisotopes using ICPMS is isobaric interferences of other elements at the same mass, e.g., ^{129}Xe with ^{129}I and ^{238}U with ^{238}Pu. With the use of high-resolution mass spectrometer (inductively coupled plasma sector field mass spectrometer, ICP-SFMS) with mass resolution up to 10,000, most of the isobaric interferences can be eliminated, thereby increasing the detection limit in the range of 10^{-12}–10^{-15} g/g. Similarly, using multi-collector inductively coupled plasma sector field mass spectrometer (MC-ICP-SFMS) with simultaneous detection of different isotopes, the isotopic ratio of different radionuclides can be measured with the similar precision and accuracy as obtained by TIMS (0.002–0.02%). MC-ICPMS also plays an important role in chronometry or age determination of the materials, measuring the parent-to-daughter ratio (^{234}U/^{230}Th or ^{239}Pu/^{235}U). For measurement of radio-chronometers, in comparison with other applicable techniques, such as alpha spectrometry and TIMS, ICPMS has the advantage of a better detection limit. In chronometry, the parent-to-daughter ratios of uranium materials are typically in the picogram to nanogram per gram range, thereby requiring systems with high sensitivity for their measurement. For analysis of trace impurities in U or Pu materials, e.g., rare-earth elements, trace-level activation, or fission products (e.g., indicating the use of reprocessed materials), or for the decay products of the host nuclear material (for age determination) prior to analysis,

chemical separation or pre-concentration is used to enhance the measurement capabilities. Chemical separation procedures from various matrices are described by Croudace and coworkers (Croudace et al. 2017). However, use of laser ablation (LA) as an ionization technique in ICPMS and minimal to no chemical pre-treatment are required, and can also be applied in the solid state. This offers isotopic measurement very precisely on very small areas of a sample and also provides additional information about sample homogeneity (Straub et al. 2020). However, in LA-ICPMS the accuracy and precision are limited by many factors, like sensitivity drift, elemental/isotopic fractionation, matrix effects, interferences, and most importantly the unavailability of appropriate matrix-matched reference materials. Therefore, it requires laborious and suitable calibration and correction methods for quantitative data. ICPMS is widely used for the determination of isotopes for U, Th, and Pu measurements and finds extensive application in NF for trace elemental impurity measurements as well as for isotope ratio measurements in seized samples and in samples of known origin (Keegan et al. 2014; Mayer et al. 2005; Wallenius et al. 2003). Apart from actinides, ICPMS is also used for the determination of other radionuclides like ^{79}Se, ^{90}Sr, ^{99}Tc, ^{129}I, ^{135}Cs, ^{210}Pb, ^{226}Ra, ^{228}Ra, ^{231}Pa, ^{237}Np, ^{241}Am, ^{243}Am, and ^{244}Cm in environmental and waste samples (Hou and Roos 2008).

C. *Secondary Ion Mass Spectrometry (SIMS)*

SIMS is used to provide both elemental and isotopic information of radionuclides even in micro-sized particles along with special and depth-dependent concentration of the constituent of interest on the surface of solid materials. It has got excellent application in nuclear safeguards as well as nuclear forensics (e.g., post-detonation scenario) when only small amounts of sample (particle) or inhomogeneous samples are to be investigated (Betti et al. 1999; Tamborini et al. 2002). In SIMS a beam of ions with high energy ranging from 0.5 to 20.9 keV, such as O^-, O_2^-, or Cs^+, is bombarded on the surface of the sample to sputter off atoms. A fraction of the sputtered species is emitted as electrically charged particles as secondary ions. These ions pass through mass analyzer (magnetic sector field or TOF or quadrupole analyzers) and are finally detected by the detectors. The main advantage of SIMS is its capability for microparticle analysis. In NF there are many cases which require analysis of U particles to complement the more traditional bulk measurements. Particle analysis is very challenging. Firstly, the particle is extracted (e.g., from cotton swipe) through vacuum impactor in a clean room; then, extracted particles are deposited on a graphite planchet through homogeneous dispersion. SIMS performs the particle of interest screening among the deposited particles in a short time with a high detection capability. Both the exact location of the particle (U or Pu) and the first estimate of their enrichment are possible by this technique (Hedberg et al. 2011). Almost nil or very limited CRMs are available for particle actinide analysis. Particles produced from certified reference materials (U005, U010, and U030-NBS) with known ^{234}U/^{238}U and ^{235}U/^{238}U mass ratios are employed to check the instrumental response during the analysis (Betti et al. 1999).

D. *Resonance Ionization Mass Spectrometer (RIMS) and Accelerator Mass Spectrometer (AMS)*

Measurement of very low isotope abundances or ultra-trace quantities of a radioactive isotope in the presence of huge excess of matrix material poses a big challenge in nuclear forensic analysis. A typical example is the measurement of ^{236}U and ^{233}U in the uranium matrix. For this measurement, discrimination of the interfering masses is required which may arise due to a variety of molecular and atomic species and also due to tailing of neighboring masses, nearly at the same mass position as the isotopes of interest. In resonance ionization mass spectrometer (RIMS), the ionization of atoms can be done through optical excitation of atoms by resonant absorption of laser light in a stepwise process followed by the photo-ionization of the excited atom. Combined with selective and nearly background-free detection of the resulting photo ions proves RIMS is an efficient technique to address such difficult challenges (Erdmann et al. 2008; Trautmann et al. 2004). Similarly, accelerator mass spectrometer (AMS) offers adequately high abundance sensitivity for measuring extremely small isotope ratios. The presence of ultra-trace amounts of ^{236}U and ^{239}Pu even in natural uranium through measurement is possible by the use of AMS (Buchholz et al. 2007; Steier et al. 2008).

Analysis of Minor Constituents in Bulk Radioactive Materials

Activation Analysis

A. *Neutron Activation Analysis (NAA)*

NAA is a nondestructive, nuclear analytical technique used to analyze minor/trace elements in a matrix based on atomic number, neutron cross section of the isotope, and its abundance. The elements present in the sample are excited by nuclear reaction, through irradiation with neutrons (from a reactor) and thereby the radioactivity is induced. Radiation counting techniques (e.g., HPGe) are used for activity measurement. Following the bombardment completion, the measurement of characteristic radiation is indicative of the elements of interest. Multielement analysis with excellent sensitivity in ppb level and high selectivity are the major advantages for this technique. No time-consuming separation procedures are required. However, one limitation of this technique is the requirement of a nuclear reactor/neutron source for sample activation, and being a comparator technique requires matrix-matching reference materials for quantification.

B. *Particle-Induced Gamma-Ray Emission (PIGE)/Particle-Induced X-Ray Emission (PIXE)*

For identification of elemental signatures in samples containing uranium of different enrichments, both particle-induced X-ray emission (PIXE) and particle-

induced gamma-ray emission (PIGE) methods are used for NF analysis. The samples are irradiated by 3–5 MeV proton beam. The characteristic X and gamma rays are measured using high-purity germanium (HPGe) detectors. Quantification is done by using certified comparator standards (Apostol et al. 2017). Sensitivity of the analysis for different elements using these techniques is in the order of ppm to percentage level.

Analysis Using X-Ray-Based Techniques

A. *X-Ray Florescence*

X-ray fluorescence spectrometry (XRF) is a rapid and nondestructive technique for the determination of elemental composition in nuclear material and is based on the detection of characteristic X-rays. When a sample is irradiated with a collimated X-ray beam, electrons from the core orbitals are promoted to valence orbitals, causing the prompt emission of fluorescence X-rays as the empty core orbitals are filled by other electrons. The characteristic energies of these fluorescence X-rays are element-specific, and the resulting spectrum can be interpreted to quantify multiple elements in a matrix. XRF techniques are nondestructive, require minimum sample preparation, are highly sensitive (parts per billion), and with rapid data acquisition are an excellent analytical tool for elemental characterization (West et al. 2010) and found enormous application in nuclear forensics requiring only micrograms of material (Donohue 2002; Charlton et al. 2009). Direct sample irradiation allows for minimal sample preparation and portable instrumentation for field deployment (Tsuji et al. 2005). Handheld portable XRF spectrometers are used for on-site detection of RDD components (Steeb et al. 2016). This technique tends to be hindered by the matrix effect, which must be corrected by using matrix-matching SRMs. Total reflection X-ray fluorescence spectrometry (TXRF) for ultra-trace element analysis at the femtogram level is used.

B. *X-Ray Diffraction*

In addition to elemental and oxidation state analysis, X-rays are also used to study the phase and chemical structure of materials by X-ray diffraction (XRD) (Zolotoyabko 2014). For identification of the chemical structure of crystalline material, X-ray diffraction (XRD) is used. When an X-ray beam interacts through ordered crystalline lattice, a characteristic X-ray diffraction pattern is produced through constructive and destructive interference. This will depend on the spacing of the lattice, the wavelength of the X-rays, and the angle of incidence of the X-ray beam. However, this technique cannot be applied to amorphous (noncrystalline) material. XRD has got a wide application in nuclear material characterization, because of nondestructive and quick analysis (Teterin et al. 2016). Soil sampling near a nuclear site, followed by analysis by XRD/XAS, could be used for finding out any clandestine activity like U enrichment or fuel reprocessing by measuring the concentration and chemical form of trace actinides in soil.

Nuclear Forensic Interpretation

The results from nuclear forensic interpretation provide technical conclusions to decision-makers. This is mainly carried out by evaluation of analytical results from a suspected sample through validated material characteristics or signatures. Through sample signature, a potential linkage is established between sample and people or place or process. Validated analytical methods, validated signatures, and most importantly the accurate and precise characteristic data are of utmost importance to generate high-confidence technical conclusions. A pictorial representation of various steps involved in nuclear forensic interpretation is presented in Fig. 2.

Analysis of a single signature may not be sufficient to answer all the relevant questions raised by the investigating agency. Many times multiple signature analyses in total may increase the specificity of the interpretation results. Validated signatures considered in NF analysis are broadly classified into two types, i.e., comparative signature and predictive signature.

Comparative Signatures

Comparative signatures are basically material characteristics of one or more known samples which are compared with unknown samples. For fruitful interpretation, a nuclear forensic database, on hundreds or thousands of known samples or nuclear forensic sample archives, is required along with suitable analytical tool. The nuclear forensics database mainly comprises of a complete set of validated material

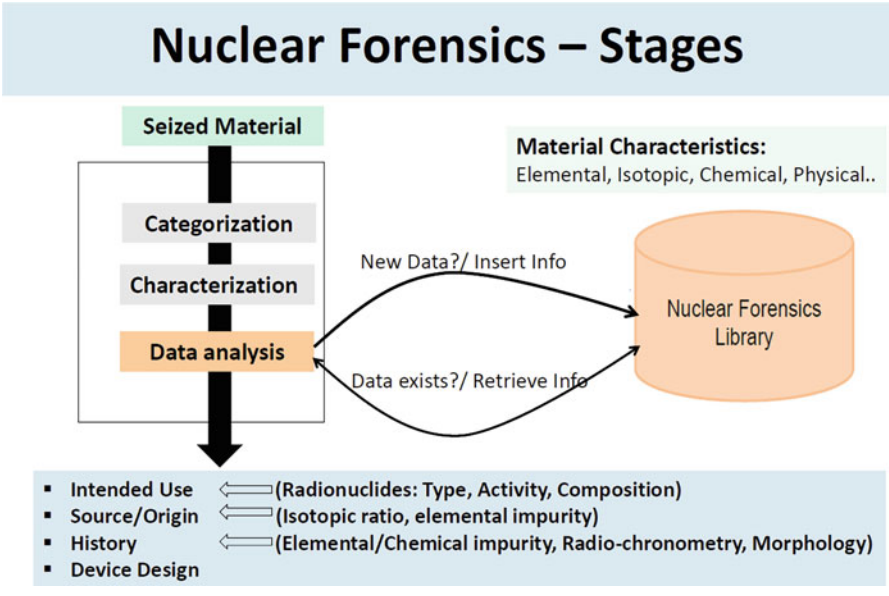

Fig. 2 Various steps involved in the nuclear forensic interpretation process

characteristics of the known samples across different stages of nuclear fuel cycle and also across the globe. Therefore, the accuracy in the interpretation result by this comparative approach will very much depend upon validity and comprehensiveness of the generated data included in the database and/or the sample archive.

During any transboundary incident, involvement/assistance from international agencies in nuclear forensics will be a vital requirement. This will again demand for accessibility toward international nuclear forensic databases. Therefore, development of own nuclear forensic databases is very much advantageous. All countries should have high confidence in each other's databases. Also, databases should comprise of high quality and the right materials characteristics for nuclear forensic purposes. Therefore, the international agencies like ITWG (International Technical Working Group for Nuclear Forensics) through multilateral or through bilateral programs are engaged to drive standardization of methods, techniques, standards, reference materials, etc. in order to provide uniform platform and harmonization in nuclear forensic practices throughout the globe. Nuclear forensics being a new discipline, international exercises, or round-robins are absolutely crucial and advantageous for developing and testing the comparability of nuclear forensic data acquired by different countries.

Predictive Signatures

Apart from comparative signatures, predictive/derived signatures from some other measured parameters are also used for nuclear forensic interpretation. A detailed understanding of the physical or chemical mechanisms responsible for producing the signatures in the material and subsequent modeling of those mechanisms for derivation of predictive signatures are very important. Predictive signatures mostly rely on our knowledge base and upon scientific inference; that's why it is more challenging than the comparative approach. These signatures will be very important, when the materials are from historical processes or tightly held materials from foreign countries, where availability of similar kind of known samples will be rare for comparison. However, significant effort is required not only to develop but also to validate the capability.

For example, the source identification of a uranium ore concentrate (UOC) through predictive approach will rely on the geochemistry of the precursor ore from which it has been produced. It will involve the interpretation from the signatures measured in a sample of interest to determine characteristics associated with both the parent ore and the mining and milling processes used to produce the UOC from the ore (Švedkauskaitė-LeGore et al. 2008; Varga et al. 2011). For example, the relative concentration distribution of the rare-earth elements (REE) can be typical of the type of ore body from which the uranium was mined (Varga et al. 2010). New predictive signatures are also being developed by establishing improved understanding of the signature and developing sensitive methodology for their measurement. Now using sensitive mass spectrometers, even natural variations in the $^{238}U/^{235}U$ ratio, once thought to be invariant in nature, can be used as an effective NF signature. The depositional environment of an ore appears to strongly influence the $^{238}U/^{235}U$ ratio with low-temperature ores having higher ratios than deposits formed at high

temperatures (Richter et al. 1999; Brennecka et al. 2010). Another important predictive signature applied in NF is radio-chronometry, the determination of the time since last purification using a technique commonly known as "age dating" (Wallenius and Mayer 2000; Morgenstern et al. 2002; Williams and Gaffney 2011). Radionuclides linked to one another by radioactive decay have relative concentrations, which can be calculated by simple laws of radioactive ingrowth or Bateman equation (Moody et al. 2014). Accurate age dating relies on several assumptions. Precise determination of parent-daughter activity concentrations is very important for accuracy in age dating.

QA/QC in Nuclear Forensic Data and Interpretation Methodology

Radiation metrology has got a significant role in NF analysis, since the analytical data and the subsequent results from nuclear forensic interpretation would support as evidence for criminal prosecution or can affect international assessments of proliferation, nuclear/radiological threat, or terrorism. Therefore, it is highly recommended that the NF data and their interpretation should be of high credibility. Validated procedures/methodologies should be followed throughout the process. Proper quality assurance (QA) and quality control (QC) procedures should be followed in each and every stage within the nuclear forensics laboratory to ensure confidence in the analytical data. Quality systems, such as ASCLD International, ISO 9000 (ISO 2000), and ISO 17025 (ISO 1999), should be implemented in the NF laboratory. Proper documented procedures should be established for sample control and analysis, which will not only improve repeatability and traceability of the results but also provide a mechanism for continuous quality improvement. Analytical instruments should be under a relevant statistical process control (SPC) program wherever feasible (IAEA 2006). All analytical results should reveal the measurement precision, and any potential sources of error not reflected in the precision should also be indicated or documented. The field of potential sources or processes that produced the material could be narrowed with an increase in the precision of a given measurement. The confidence and specificity often increase with more independent measurements. Sensitivity of chosen analytical technique will be vital in case of small-sized sample analysis. Similarly, for large quantity/bulk of material, the analytical techniques should be as sensitive as possible because trace species are often the significant components of a signature. With the analytical sensitivity, the susceptibility for cross-contamination also increases which necessitates strict control measures.

Challenges in the Quality Assurance of Nuclear Analytical Measurements in the Nuclear Field

For developing, testing, and validating nuclear and radioanalytical methods for quantification of radionuclides, availability of certified reference materials (CRMs) is of utmost requirement. The use of certified reference materials (CRMs) is a critical component in any QA program and is very essential for the accurate and precise

measurements and assessment of the quality of any acquired analytical data. Such matrix-matched CRMs are unfortunately not available for most investigations relevant to the nuclear domain. The nuclear CRMs present particular challenges for both users and producers (Neacsu 2020; Inn et al. 2013). Also CRM production requires specialized nonexistent facilities and is time-consuming, labor intensive, and costly. The lack of matrix-matched certified reference material can be overcome by comparing the analytical results obtained from different instruments whose analyte detection is based on different physical principles. Additionally, inter- or intra-laboratory comparison exercises or simulation models can also help in gaining confidence in the robustness of the developed method. In the absence of CRMs for radio-chronometer, a methodology of comparing with the age of a known material can be developed (Prabhath et al. 2022). The uncertainty associated with the response of a detection device in nuclear analytical techniques is mainly contributed from calibration standard, from the curve fitting process, and from the model used to describe the response function. Radiation detection devices and associated electronics might show saturation effects and dead time at high count rates induced by high-intensity radiation. The saturation and the dead-time effects are usually compensated, but the compensation will still introduce some uncertainty, which has to be evaluated. Many times, there is a requirement of measuring low activity samples in the presence of background; the uncertainty in quantification for the results close to detection limit requires special attention.

Challenges and Requirements in Nuclear Forensics

Nuclear forensics has tremendous potential to not only respond but also prevent nuclear/radiological incidents arising due to use of nuclear/radiological materials out of regulatory control. However, its practice is limited to a few countries only, since many challenges are associated with NF analysis. State-of-the-art instrumentation, identification of undisputed signatures, extremely high fidelity measurement capabilities to withstand legal scrutiny, strong metrology backbone, availability of CRMs, complexity in the data interpretation, nonavailability of reference database, and the required high-level expertise limit its application only to a few laboratories throughout the world. There is a strong requirement of standalone methodology, dedicated R&D on NF signature development, advanced nondestructive and destructive analytical techniques for evidence analysis, urgent CRMs to support state-of-the-art measurement capabilities and metrology infrastructure, QA/QC programs for data analysis, accredited laboratories, and proficiency testing capabilities. It is also very crucial to develop a national nuclear forensic database and validated analytical tool for accurate and precise interpretation.

Cross-References

► Equipment for Environmental Radioactivity Measurement: Calibration and Traceability

Annexure A

A list abbreviations and acronyms used in the chapter (arranged alphabetically)

μ-Raman	Micro Raman Spectroscopy
μ-XRF	Micro X-ray Fluorescence
AMS	Accelerated Mass Spectrometry
AS	Alpha Spectrometry
ASCLD	The American Society of Crime Laboratory Directors
ASTM	American Society for Testing and Materials
AWCC	Active Well Coincidence Counters
CRM	Certified Reference Material
DDA	Differential Die-Away Technique
FTIR	Fourier Transform Infrared Spectrometry
FT-TIMS	Fission Track Thermal Ionization Mass Spectrometry
GC-MS	Gas Chromatograph-Mass Spectrometer
HEU	Highly Enriched Uranium
HPGe	High Purity Germanium (HPGe) Detector
HRGS	High Resolution Gamma Spectrometer
IAEA	International Atomic Energy Agency
IC	Ion Chromatograph
ICPMS	Inductively Coupled Plasma Mass Spectrometry
ICP-OES	Inductive Coupled Plasma Optical Emission Spectrometry
ICP-SFMS	Inductively Coupled Plasma Sector Field Mass Spectrometer
IDMS	Isotope Dilution Mass Spectrometer
IRMM	Institute For Reference Materials and Measurements
IRMS	Isotope Resolution Mass Spectrometry
ISO	International Standards Organization
ITDB	Incident and Trafficking Database
ITWG	International Technical Working Group (Nuclear Forensics)
Kev	Kilo Electron Volt (A unit of energy)
LA	Laser Ablation
LEU	Low Enriched Uranium
LSC	Liquid Scintillation Counting
MC-ICPMS	Multi Collector Inductively Coupled Plasma Mass Spectrometry
MC-ICPMS	Multi-Collector Inductively Coupled Plasma Sector Field Mass Spectrometer
MeV	Mega Electron Volt (A unit of energy)
Ms	Mass Spectrometer
NAA	Neutron Activation Analysis
NBS	National Bureau of Standards, USA
NDA	Non-Destructive Assay
NF	Nuclear Forensics
NIST	National Institute of Science and Technology, USA
NRF	Nuclear Resonance Fluorescence

PIPS	Passivated Implanted Planar Silicon Detector
QA	Quality Assurance
QC	Quality Control
R&D	Research and Development
RDD	Radiological Dispersion Device
REE	Rare Earth Element
RIMS	Resonance Ionization Mass Spectrometry
SEM-EDS	Scanning Electron Microscope with Energy Dispersive Sensor
SIMS	Secondary Ionization Mass Spectrometry
SNM	Special Nuclear Material
SPC	Statistical Process Control
SRM	Standard Reference Material
TEM	Transmission Electron Microscopy
TIMS	Thermal Ionization Mass Spectrometry
UOC	Uranium Ore Concentrate
XAS	X-Ray Absorption Spectroscopy
XRD	X-Ray Diffraction
XRF	X-Ray Fluorescence

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Part XV

Advanced Techniques in Evaluation of Measurement Uncertainty

Advanced Techniques in Evaluation of Measurement Uncertainty

88

A Prelude

N. Garg, Vishal Ramnath, and Sanjay Yadav

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Abstract

This chapter briefly introduces a prelude that encompasses the history of measurement and measurement uncertainty, tracing the development of modern metrology from ancient Egypt, to the early renaissance with the advent of calculus by Newton, to the late renaissance with the advent of statistics by Laplace, to the dawn of the industrial age with Kelvin, to the post World War II age of Kline and McClintock, and finally to the introduction of the Guide to the Expression of Uncertainty in Measurements also known as the GUM and its supplements in the early part of the twenty-first century. The utility and functionality of the GUM, both with its current limitations and its future outlook for the evolution of measurement uncertainty, are contextualized and discussed.

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Introduction and Historical Perspectives

Measurement analysis inclusive of uncertainty analysis has within available historical records origins that are traceable to technical work undertaken by the ancient Egyptians in the course of the construction of the pyramids. In this time period, length measurements were primarily anthropomorphic based, mainly in terms of the human anatomy of the prevailing reigning pharaohs of the time-incorporating standardized units such as the pharaoh's finger size, palm size, hand size, foot size, and shoulder size among other elements of human anatomy, such as a cubit being the distance from the elbow to the tip of the middle finger. Extending these length "standards" which were "calibrated" from time to time as the pharaoh's anatomy changed with age, medical and health circumstances. The derived units were used such as the kha and setat for the measuring of areas of land in pyramid constructions, farms, and other agricultural locales where conversion factors were utilized such as measuring a square khet as being a square of 100 by 100 cubits. in area. Similar extensions of length and area units were introduced to measure volumes of quantities such as for water, wine, and grain in terms of cubic cubits incorporating derived volume units of varying magnitudes such as *khar*, *heqats*, and *henu*. As a result, all dimensional measurements in ancient Egypt were in essence "traceable" to prevailing contemporary length standards. Weight measures by contrast used physical standards such as the *deben*, *qedet*, or *shematy* which were defined in terms of artifact standards constructed with copies of physical stones/rocks that were measured and calibrated against the prevailing mass standards using simple lever arms and scale balances, while time measurements were determined for longer periods in terms of the lunar cycles and for smaller intervals using simple water clocks. Consequently, in ancient times there existed a coherent mass/length/time system of measurements that is essentially an artifact system equivalent to the more recent artifact system of the centimeter/gram/second (CGS) of the nineteenth century and subsequently the meter/kilogram/second (MKS) system of units from the turn of the twentieth century which preceded the eventual adoption of the International System of Units (SI system) that was established and is maintained by the General Conference on Weights and Measures (CGPM), where the uncertainty of weights and related measures was based on integer increments of the underlying discrete units of measurements prior to the later inventions of arithmetic and mathematics.

With the passage of time, this ancient situation for measurements and measurement uncertainties largely prevailed until the period of the later renaissance in the late 1600s when Isaac Newton remarked in Book III of *Principia Mathematica* in his attempt to analyze and reconcile differing and inconsistent experimental observations of comets in 1680 for the orbital calculation of a parabolic trajectory of a comet as reported by (Stigler 1986) that:

> This being a problem of great difficulty, I try'd many methods of resolving it; and several of those problems, the composition whereof I have giv'n in the first book, tended to this purpose. But afterwards I contrived the following solution, which is something simple.

These earlier astronomical observations ultimately led to the discovery of calculus by Newton and the later adoption of the method of least squares in the seventeenth and eighteenth centuries by Pierre-Simon Laplace in his five-volume monograph *M'ecanique c'eleste* (Celestial Mechanics) and other contemporary philosophers and scientists for the fitting and quantification of experimental measurement uncertainties. During this period, Lord Kelvin remarked in the context of scientific measurements in a 1883 lecture to the Institution of Civil Engineers that (Kelvin 1891):

> When you can measure what you are speaking about, and express it in numbers, you know something about it.

Consequently, the use of statistics as a branch of mathematics with the invention of calculus underpinned the majority of scientific studies of metrology uncertainty prior to the twentieth century from the mid-1800s to the early 1900s prior to World War II, most notably in astronomy- and physics-related studies for the fitting of experimental data to measurements in which it was desired to determine an optimal selection of equation parameters and their associated uncertainties. This area of work, now known as statistical regression, is considered a subset of metrology uncertainty and includes the fitting of uncertainty quantification of straight-line parameters as discussed in more detail by (Ramnath 2020). In subsequent years, the use of statistical regression as a branch of measurement uncertainty has been extended to nonlinear and multivariate measurement models where the uncertainties are incorporated as accuracies into the model equation parameters for statistical consistency.

After World War II, a key development in the concept of measurement uncertainty was developed by Kline and McClintock (1953) in their pioneering work that incorporated the use of statistics that extended the work by the American Society for Testing Materials that had hereto relied on controlled multiple sample experiments, i.e., repeat experimental measurements where the uncertainties were determined through statistical standard deviations of all relevant quantities in a measurement. This fully statistical approach to uncertainty is limited in the sense that not all data can be experimentally observed and quantified through repeated tests and measurement observations. Key conceptual aspects introduced by Kline and McClintock were the introduction of probability distributions for measurements and in the reporting of measurements with uncertainties in terms of an expected value with an uncertainty expressed in terms of odds of the form $m \pm w$ (b to 1), e.g., a pressure could be quoted as 50.2 ± 0.5 psia (20 to1) which signifies that "*This states that the best value for the pressure is believed to be* 50.2 *psia and the odds are* 20 *to* 1 *that the true value lies within* ± 0.5 *psi of this best estimate.*" Their final result stated that:

Theorem 3. *If R is a linear function of n independent variables, each of which is normally distributed, then the relation between the interval for the variables* w_i, *and the interval for the result* w_R, *which gives the same odds for each of the variables and for the result is*

$$w_R = \left[\left(\frac{\partial R}{\partial w_1} w_1\right)^2 + \left(\frac{\partial R}{\partial v_2} w_2\right)^2 + \cdots + \left(\frac{\partial R}{\partial v_n} w_n\right)^2\right]^{\frac{1}{2}}.$$

The above formula is essentially the historical predecessor and connection to modern Guide to the Expression of the Uncertainty of Measurement (GUM) (BIPM, IEC, IFCC, ILAC, ISO, IUPAP, and OIML 2008a) which specifies the measurement uncertainty for a univariate measurand $Y = f(X_1, \ldots, X_N)$ and which incorporates refinements such as correlation and nonlinear effects, such that the combined uncertainty $u_c(y)$ for sampled values for Y and $X_1, \ldots, X_N$, consistent with the underlying probability density functions (PDFs), is of the form

$$u_c^2(y) = \sum_{i=1}^{N} \left(\frac{\partial f}{\partial x_i}\right)^2 u^2(x_i) + 2\sum_{i=1}^{N-1}\sum_{j=i+1}^{N} \frac{\partial f}{\partial x_i}\frac{\partial f}{\partial x_j} u(x_i, x_j) + \sum_{i=1}^{N}\sum_{j=1}^{N} \times \left[\frac{1}{2}\left(\frac{\partial^2 f}{\partial x_i \partial x_j}\right)^2 + \frac{\partial f}{\partial x_i}\frac{\partial^3 f}{\partial x_i \partial x_j^2}\right] u^2(x_i) u^2(x_j)$$

where $u(x_i, x_j) = r(x_i, x_j)u(x_i)u(x_j)$ is the covariance term and $r(x_i, x_j)$ is a dimensionless correlation.

Limitation in the above formula is the requirement that all relevant terms follow Gaussian distributions; however, this has subsequently been mitigated by the GUM Supplement 1 (BIPM, IEC, IFCC, ILAC, ISO, IUPAP, and OIML 2008b) which allows for non-Gaussian PDFs for $X_1, \ldots, X_N$ as independent distributions $g_{X_1}(\xi_1), \ldots, g_{X_N}(\xi_N)$ where ξ_i, $i = 1, \ldots, N$ are random variables associated with $X_1, \ldots, X_N$ or alternately a joint PDF $g_{X_1,\ldots,X_N}(\xi_1, \ldots, X_N)$, and in the extension with the GUM Supplement 2 (BIPM, IEC, IFCC, ILAC, ISO, IUPAP, and OIML 2011) for multivariate models $\mathbf{Y} = \mathbf{f}(\mathbf{X})$ where $\mathbf{X} = [X_1, \ldots, X_N]^\mathsf{T} \in \mathbb{R}^N$ and $\mathbf{Y} = [Y_1, \ldots, Y_m]^\mathsf{T} \in \mathbb{R}^m$ for $N, m \in \mathbb{N}$, where the GUM Supplement 1 and GUM Supplement 2 extend the earlier GUM through the use of the Monte Carlo Method (MCM).

While the GUM and its supplements have significantly extended the techniques of measurement uncertainty, there are nevertheless still significant mathematical and statistical issues that must still be addressed, some of which will be explored in this handbook in various chapters.

Contemporary Perspectives and Current Limitations with the GUM

Within the current GUM framework, uncertainties are delimited into *Type A* uncertainties which adopt a statistical framework and *Type B* uncertainties which adopt a probabilistic framework. As a result, the current GUM framework mixes frequentist

statistics concepts and Bayesian statistics concepts and is therefore mathematically inconsistent.

For Type A uncertainties, a quantity X is usually sampled a few times and the arithmetic average $\overline{X}$ calculated, along with the experimental standard deviation of the mean (ESDM) σ which is estimated in terms of standard formulae, so that the expected value μ is estimated with $\overline{X}$ and the standard deviation is similarly estimated with the ESDM along with an associated "degrees-of-freedom" ν. Due to this approach which uses a frequentist statistics approach (all quantities are based on observed measurements or frequency statistics), the uncertainties obtained with Type A evaluations are themselves always uncertain.

Considering, on the other hand, type B uncertainties, in this case it is not possible to use frequency statistics since, for example, no repeat experimental measurements are possible. For this situation, a metrologist makes a subjective assumption based on expert knowledge as to what the probability density function $g_X(\xi_X)$ for the quantity X is and interprets the probability as a degree of belief. Then using knowledge of the analytical functional form of the PDF and its associated formula parameters, the mean of X and its variance are calculated using the available information. A consequence of this approach is that the calculated mean μ and standard deviation σ are exact and have no associated uncertainty. The inconsistency within the GUM is therefore that the degrees-of-freedom ν_i are always attached to terms X_i and that the measurement uncertainty scheme is essentially an ad hoc procedure that attempts to imperfectly align Type B estimates with Type A estimates. Consequently, there is then a fundamental and conceptual mathematical statistical inconsistency in how the GUM modeling framework for uncertainty analysis is formulated; however, this inconsistency has no practical effect in a majority of practical measurement uncertainty cases as discussed and elaborated by (Bich et al. 2016).

A more serious limitation with the GUM is in the calculation of the expanded uncertainty obtained by multiplying the standard uncertainty $u(y)$ by a numerical factor k and where $2 \leq k \leq 3$ according to the GUM. Two issues complicate this calculation, namely, i) the shape of the output PDF for Y is unknown particularly for the tails of the distribution, and ii) the standard uncertainty itself is also unknown (because it incorporates type A uncertainties as inputs). At the present time of writing within the GUM uncertainty framework, the only way out to resolve this conundrum is to make the assumption that in most practical cases that the output PDF for Y will be a scaled and shifted Student's t-distribution, to then use either the Welch-Satterthwaite formula for uncorrelated terms or alternately the extended Welch-Satterthwaite formula for correlated terms, and to determine the effective degrees of freedom ν_{eff}. Challenges with this assumption are that a) there should be many independent inputs $X_1, \ldots, X_N$ with N large, say $N \geq 30$, for the overall behavior of $Y = f(X_1, \ldots, X_N)$ to converge to a Gaussian distribution (if N is small, then Y may be distinctly non-Gaussian and hence cannot be approximated with a Student's t-distribution), or that b) all the inputs for X_i should themselves be actual Gaussian distributions (and not approximations of Gaussian distributions with say

square or triangular distributions and transformation mapping approximations using scaling conversion factors).

A possible remedy to address these inconsistencies within the GUM is to change the way Type A uncertainties are incorporated by using nonstatistical tools by, for example, using the principle of maximum entropy and Baye's theorem that would then be able to yield a scaled and shifted Student's *t*-distribution. Another alternative is to simply forgo the original GUM which uses mixed frequentist statistics and Bayesian statistics, and only use the GUM Supplement 1 for univariate measurement uncertainty analysis or the GUM Supplement 2 for multivariate measurement uncertainty analysis as both of the GUM supplements use a wholly Bayesian statistics framework and are fully mathematically consistent.

Current research efforts are presently in progress by scientists and statisticians working in international bodies such as the BIPM working groups and various international National Metrology Institutes and national measurement laboratories to refine and modernize both the GUM as well as the GUM supplements to adopt a wholly Bayesian statistics modeling framework for mathematical consistency. Until such time that the BIPM publishes a new revised GUM for the international scientific and engineering community, the present recommendation for high-accuracy measurement uncertainty analysis is to use the Monte Carlo Method (MCM) as documented in the GUM Supplement 1 and GUM Supplement 2 for performing advanced uncertainty calculations.

Future Outlook and Anticipated Developments with the GUM

A new development by (Cox and van der Veen 2021) explores why correlation/covariance effects must always be considered even when correlations are missing or ignored through, for example, assuming them to be automatically zero. It is shown, using an example of interpolation of chemical pH values, that missing/ignored correlations can lead to poor decisions or logical absurdities, as interpolated value uncertainties may be erroneously smaller than known value uncertainties at specified temperatures if correlation effects for pH values at different temperatures are ignored. Recommendation for addressing this issue is to use a vector-matrix formulation for incorporating correlation effects either with the GUM supplement or using a covariance matrix as recently reported by (Ramnath 2022), and it is anticipated that future uncertainty analysis work will slowly migrate from spreadsheet-based calculations and increasingly incorporate modern scientific software such as Matlab, Python, etc. which can more fully and easily perform vector-matrix calculations for more advanced analysis on measurement uncertainty calculations.

For metrology problems in which the GUM framework is not applicable, the GUM supplements may conveniently and easily be used as an alternative approach for measurement uncertainty; however, the limitation with the GUM supplements is that they are not able to incorporate additional measurement uncertainty information. For this situation, a full Bayesian statistics approach using Markov chain Monte Carlo (MCMC) methods is discussed by (Klauenberg and Elster 2016) which

presents the use of the Metropolis-Hastings algorithm as a useful tool in which additional information such as the posterior shape of the measurement output may be incorporated to refine the uncertainty analysis. This approach is particularly useful in which the measurement model may be very complicated in, for example, systems defined by partial differential equations and in which it is computationally expensive and impractical to perform many simulation events in a traditional GUM supplement uncertainty calculation. An example of the potential application and utilization for the combined use of the GUM supplements with the MCMC method for advanced uncertainty analysis is in the use of "virtual laboratories" where a system or systems of PDEs are numerically solved to study the measurement system. It is anticipated that in the future with increasingly available and cheap cloud-based computational power this synthesis of techniques will increasingly lead to a widespread adoption of advanced measurement uncertainty that utilizes the underlying physical and chemical laws of nature for ab initio simulations from first principles.

Summary of the Section

The section on Advanced Techniques in evaluation of Measurement Uncertainty discusses the various technical aspects pertaining the evaluation, analysis, and prediction of measurement uncertainty using the LPU, MCS, and Bayesian approach. The chapter presented by Ramnath deals with the methodology of formulation and implementation of Monte Carlo simulations in order to calculate uncertainties of measurements that are modeled in terms of partial differential equations. It concludes that the Monte Carlo approach is conceptually straightforward to implement partial differential equation solvers for both low-dimensional as well as high-dimensional models provided the model inputs can be suitably sampled and introduced into the solver/postprocessor to generate discrete output statistical data. The chapter by Gupta, Rab, and Garg presents the various distribution functions used in calculation of measurement uncertainty and the technical attributes associated with various techniques like LPU, Monte Carlo Simulation, and Bayesian approach. The chapter presented by Kalina, Vidnerová, and Soukup describes and discusses selected nonstandard statistical techniques for analyzing measurements, or in fact selected classes of advanced available statistical tools. This chapter gives a warning that linear regression performed on data contaminated by measurement errors yields biased estimates and requires specific estimation tools for the so-called measurement error model. The chapter presented by Hawam A A suggests that GUM uncertainty framework is the most popular as it is simple, straightforward, and easy to implement in both the industrial and scientific field, while the Monte Carlo simulation is much more complex and cannot be implemented by hand. However, it has become a necessity if the measurement model has a great deviation from linearity to the extent that the GUM framework becomes inaccurate due to its inherent linear approximation. The chapter by Rao C V gives an overview of measurement uncertainty analysis, with a focus on modeling. The chapter presented by Rab et al. shows application of contemporary techniques of evaluation of measurement uncertainty in

digital pressure transducer (DPT) as a case study. It highlights a very important aspect about the measurement models which must be documented, in that the documentation should be written in a way such that other metrology professionals who are not necessarily experts in the field may understand it.

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Monte Carlo Simulations in Uncertainty Evaluation for Partial Differential Equations

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Vishal Ramnath

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Abstract

Uncertainty quantification in metrology is formally undertaken using the Guide to the Expression of Uncertainty in Measurement (GUM) in all major industrial, commercial and national measurement laboratories in order to model and predict measurement accuracies of equipment, instruments and processes in systems. Although model complexity may formally encompass explicit, implicit, univariate and multivariate measurement equations, in a majority of cases algebraic models are the predominant mathematical formulation for many practical calibration systems. Due to this prevailing condition, many metrology practitioners are unfamiliar with how the GUM supplementary guides for Monte Carlo simulations may be usefully applied and extended to formally determine uncertainties in metrology systems modelled in terms of partial differential equations. The objective of this chapter is to introduce to the reader the methodology of how to formulate and implement Monte Carlo simulations in order to calculate uncertainties of measurements that are modelled in terms of partial differential equations.

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Measurement · Guide to Uncertainty in Measurement (GUM) · Uncertainty · PDE · Monte Carlo

Introduction

The Guide to the Expression of Uncertainty in Measurement (GUM) (BIPM, IEC, IFCC, ILAC, ISO, IUPAP, and OIML 2008a) presently utilizing a mixture of frequentist statistics and Bayesian statistics is the internationally accepted legal and scientific method for formally quantifying an uncertainty in a measurement using equipment and instruments in various laboratories and testing facilities. Originally the GUM was developed by the international metrology community to refine the earlier Kline–McClintock (1953) approach for univariate explicit measurement models uncertainty calculations, and to extend and standardize the uncertainty analysis approach for models of the form $y = f(x_1, \ldots, x_N)$ using correlation effects, however with the introduction of the GUM Supplement 1 (BIPM, IEC, IFCC, ILAC, ISO, IUPAP, and OIML 2008b) it has subsequently been extended to incorporate implicit measurement models of the form $h(y, x_1, \ldots, x_N) = 0$, and then later again extended with the GUM Supplement 2 (BIPM, IEC, IFCC, ILAC, ISO, IUPAP, and OIML 2011) to incorporate multivariate measurement models of the form $\mathbf{h}(\mathbf{y}, \mathbf{x}) = \mathbf{0}$. As a result, the GUM and its two supplements with ongoing refinements to fully encapsulate a wholly Bayesian statistics modelling framework (Bich et al. 2016) presents a complete self-contained mathematical framework for metrology practitioners that is comprehensive enough for determining the uncertainty in any relevant physical experimental measurement system.

Although the mathematical framework is formally documented with the main stages in uncertainty evaluation as (i) formulation, (ii) propagation, and (iii) summarizing, the methodology of applying this method to measurement systems defined in terms of partial differential equations is not well known within the broader metrology community. This challenge mainly occurs as many existing measurement systems feature either algebraic models in which the input/output relationships are conceptually straightforward albeit with complex constituent relationships, or in which the measurement equation system utilizes a phenomenological model with empirical parameters that is algebraically or analytically solvable and tractable. Examples of an algebraic input/output model include pyrometry based temperature measurements (Dziarski and Hulewicz 2021) that utilize optical radiometry theory, vacuum pressure based static expansion systems (Franco et al. 2021) that utilize gas equation of states, and dynamic force measurements (Gelany and Mahmoud 2021) that utilizes Newton's second law. The key observation in these illustrative examples is that in algebraic input/output formulated measurement models that in these systems the formulation is ultimately based on the application of underlying physical laws in which the constituent equations are simplified and then manipulated so that they can be transformed to be analytically solvable either via hand calculations or

with Computer Algebra System (CAS) assistance, such that all the inputs $x_1, \ldots, x_N$ and outputs $y_1, \ldots, y_M$ may be manipulated and clearly written down in terms of governing systems of equations such that sensitivity coefficients $\partial y_j/\partial x_i$, $i = 1, \ldots, N, j = 1, \ldots, M$ for calculating the uncertainty output uncertainty contributions in terms of the input uncertainties may be either explicitly calculated through partial derivatives or implicitly computed through covariance matrices (Ramnath 2022).

Phenomenological based formulated measurement models by contrast differ from analytically solvable systems since they incorporate empirical analytical relationships of scientific phenomena that whilst consistent with underlying physical theories through for example statistical regression analysis or the Buckingham π - theorem (Buckingham 1914), utilize measurement equation parameters that are not ultimately directly derivable from first principles even if the theories are based on partial differential equations. Typical examples of systems that are ultimately defined in terms of partial differential equations but which are not considered analytically solvable include the watt Kibble balance (Robinson and Schlamminger 2016) for the new SI realization of the kilogram that is based on the Maxwell electromagnetic equations but which is typically numerically solved with the finite element method, and in orifice plate flow rate studies (Yin et al. 2021) based on the Navier-Stokes equations incorporating turbulence effects to determine a coefficient of discharge and related parameters and which is typically numerically solved with the finite volume method. It may thus be observed that the overall aim and motivation for metrology practitioners in determining uncertainties from measurand partial differential equation based models is to formally and accurately quantify measurement uncertainties of laboratory equipment and instruments as shown in Fig. 1. Typically at national metrology institutes the apex or primary laboratory standards, such as the

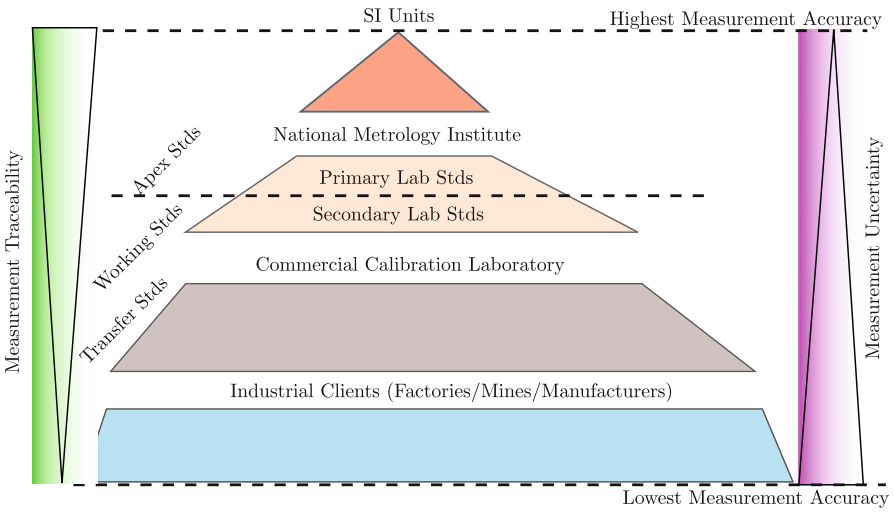

Fig. 1 Schematic illustration of measurement traceability and uncertainty propagation in metrology systems

kilogram standard or a pressure or flow standard, are characterized by a partial differential equation based model. If the uncertainty for the primary or apex laboratory standard is appropriately characterized with statistically consistent accuracies/uncertainties then it then may in turn be used to calibrate a secondary or working measurement standard, that in turn can be used to calibrate equipment/instruments for commercial calibration laboratories, which in turn ultimately pass on these accuracies/uncertainties into clients working in industries and factories. As a result, this 'measurement traceability pyramid' for all countries with modern industrial economies is ultimately dependent for its trade and industry competitiveness that is based on its respective national metrology institute's apex measurement standards accuracy that is in many practical mechanical and electrical equipment/instrument measurement parameters ultimately specified in terms of uncertainties emanating from partial differential equation measurement models. The importance in understanding the broader scientific and economic context in which this research activity takes place, may thus be readily understood in terms of the practical measurement pyramid traceability scheme and how this influences and impacts on metrology practioners from the top level national measurement laboratory, to the commercial calibration laboratory, and finally to the end user in a factory, mine or manufacturer in terms of technical and Occupational Health and Safety (OHS) consequences.

Consequently in both cases, either for algebraic/analytical equations derived from physical laws or for empirical derived equations from phenomenological observations, regardless of how the measurement equations are formulated and simplified to transform them into a tractable form that is amenable for a GUM based uncertainty analysis, the propagation of the measurement inputs inclusive of estimates of expected values and uncertainties of model parameters through for example expert knowledge or International Standards Organization (ISO) testing and quality standards specifications (Ramnath 2011), and the subsequent summarizing of the measurement model output uncertainty, is conceptually straightforward. By contrast, whilst the first step of the formulation of a model defined in terms of a partial differential equation is in principle straightforward, the second step that involves the incorporation into and subsequent propagation of input uncertainties through the model that is not analytically solvable is not as straightforward an exercise for many metrology practitioners since it is a relatively new research development in measurement system uncertainty analysis. The final third step of summarizing the Monte Carlo information once the second step is achieved is in principle straightforward, and may readily be reported and summarized in terms of the existing GUM methodology for Gaussian uncertainty characteristics (Harris et al. 2014) or if necessary using copulas (Possolo 2010) for the reporting of non-Gaussian uncertainty characteristics.

Earlier work in the uncertainty quantification of partial differential equations involved more advanced mathematical spectral based theories for an indirect approach. Subsequent later work has incorporated a direct approach by utilizing a Monte Carlo alternative to spectral based uncertainty analysis by incorporating 'virtual experiments' that include atomic force microscopy (Mohammad and Suvorov 2019) and ultrasonic flow meters (Straka et al. 2022) simulations of partial

differential equations. Recent investigations by (Wubbeler et al. 2022) using a virtual coordinate measuring machine simulation has demonstrated that when a virtual experiment is combined with a Monte Carlo method that the resulting measurement uncertainty is consistent with a GUM-compliant uncertainty evaluation if forward models involving the numerical solution of partial differential equations are used. As a result the use of uncertainties emanating from solutions of partial differential equations that more fully model a physical system in a 'virtual experiment' is a newly emerging field in metrology uncertainty quantification that may over time supersede earlier modelling efforts that were based on algebraic/analytical simplifications to governing equations and/or empirical/phenomenological modelling approaches.

The objective of this chapter is to therefore to introduce to the reader the methodology of how to formulate and implement Monte Carlo simulations using a 'black box' type of approach of numerical solutions for partial differential equations by relying on the GUM supplementary guide framework for Monte Carlo simulations, in order to calculate uncertainties of measurements that are modelled in terms of partial differential equations and to understand how the input uncertainty information may be conveniently propagated through the model and ultimately summarized. Due to the fact that practical partial differential equation based models such as the Maxwell electromagnetic equations for a national kilogram measurement standard or the Navier-Stokes fluid mechanic equations for a national pressure/flow measurement standards both require domain specific specialist scientific/engineering knowledge, in this chapter a Poisson partial differential equation is considered for simplicity in order to illustrate the main conceptual ideas of how to formulate a 'black box' approach that can then be conveniently adapted and generalized for more complex domain specific measurement systems that are aligned to the reader's own specific interests and focus. The underlying Poisson equation model considered in this chapter may be considered as a special case of the incompressible Navier-Stokes fluid equations and can in principle be adapted and refined to incorporate commercial or open source Computational Fluid Dynamics (CFD) codes to model venturi nozzles and orifice plates as flow measurement uncertainty standards in calibration laboratories for the oil and gas sectors, and also in the analysis of air duct systems in offices and hospitals for the designing and sizing of Heating Ventilation Air Conditioning (HVAC) equipment based on the flow uncertainties in ventilation ducts and passages.

Spectral-Based Uncertainty Quantification

Uncertainty analysis of measurement equations may be broadly delineated into spectral methods and Monte Carlo methods as discussed by Le Maitre and Knio (2010). If an operator $\mathcal{M}$ for a model is known such that

$$\mathcal{M}(\mathbf{s}, \mathbf{d}) = \mathbf{0} \tag{1}$$

where **s** is the solution output, **d** is random input data and $\mathcal{M}$ may be informally be considered as a system of equations, then the uncertainty analysis seeks to determine the probability distribution of **s**. In a spectral approach the data is usually parametrized using a parameter vector $\boldsymbol{\xi} = \{\xi_1, \xi_2, \ldots, \xi_N\}$ so that $\mathbf{d} \equiv \mathbf{d}(\boldsymbol{\xi})$ and a series expansion is obtained such that

$$\mathbf{s}(\boldsymbol{\xi}) = \sum_{k=0}^{\infty} \mathbf{s}_k \Psi_k(\boldsymbol{\xi}) \tag{2}$$

In the above equation Ψ_k are functions of the random variables and $\mathbf{s}_k$ are coefficients, and if both of these terms are known then the above expression may be conveniently analysed to calculate the statistics of the model output $\mathbf{s}(\boldsymbol{\xi})$. For an uncertainty analysis using the spectral approach the Polynomial Chaos Expansion (PCE) method is the standard approach as discussed by Lacor and Savin (2019) where a model $\mathcal{M}$ with a random output variable $\varphi(\mathbf{x}, t, \boldsymbol{\xi})$ may be represented with a PCE as

$$\varphi(\mathbf{x}, t, \boldsymbol{\xi}) \approx \sum_{i=0}^{P} \varphi^i(\mathbf{x}, t) \cdot \psi(\boldsymbol{\xi}) \tag{3}$$

In the above scheme φ^i is a deterministic unknown expansion coefficient that must be determined for a random mode i of φ and $\psi(\boldsymbol{\xi})$ are N-variate polynomials where N is the number of input uncertainties. It is known from Xiu and Karniadakis (2002) that if p is the highest order of polynomial that is used then the total number of terms that must be used in the PCE is

$$P + 1 = \frac{(N + p)!}{N! \times p!} \tag{4}$$

Considering for example a partial differential equation for a two-dimensional domain Ω as shown in Fig. 2 where there are $N = 12$ input uncertainties corresponding to Δx and Δy uncertainties for each of 6 points in a polygon and where a $p = 5$ order polynomial is used, it is seen that $\frac{(N+p)!}{N! \times p!} = \frac{(12+5)!}{12! \times 5!} = 6188$ terms must be determined in order to adequately model the output PDE uncertainty. As a result a traditional spectral based uncertainty analysis whilst extremely mathematically powerful for a general analysis either with a PCE or a generalized polynomial chaos (gPC) approach as discussed by Couaillier and Savin (2019) can quickly become impractical and computationally infeasible for typical PDE systems encountered in metrology problems.

Due to this observation this chapter will therefore focus on the use of the Monte Carlo method for obtaining an uncertainty of a measurement model defined in terms of a PDE, as this approach is conceptually simpler to implement and more convenient for the analysis of uncertainties based on PDE models particularly when utilizing input/output "black-box" PDE solvers.

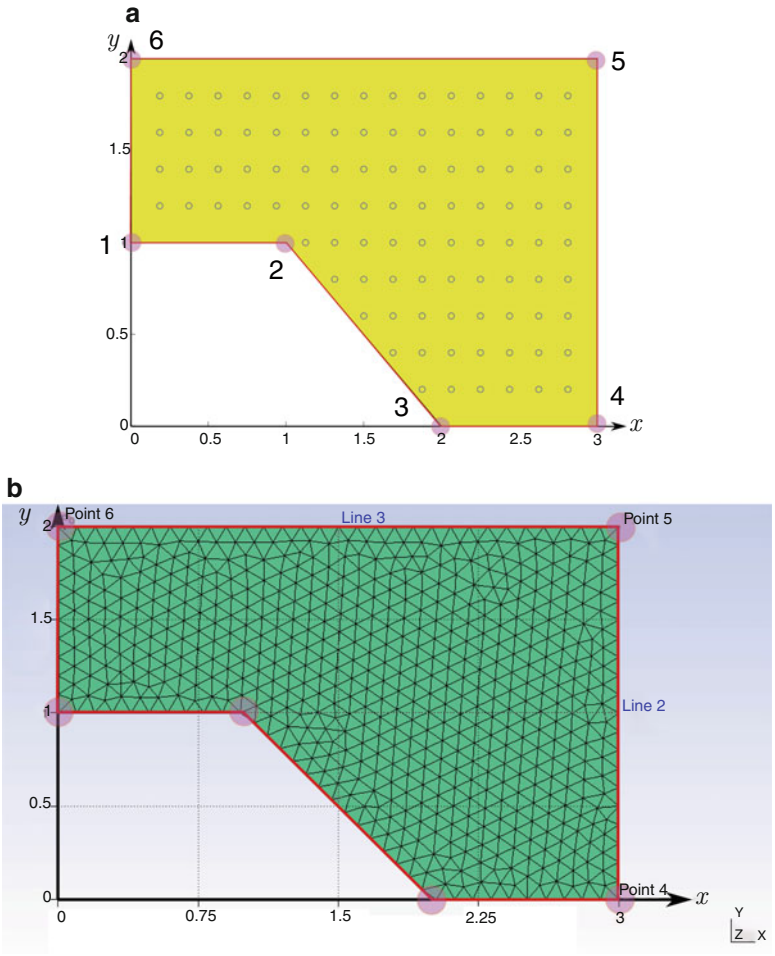

Fig. 2 Illustration of typical PDE domain discretizations

Review of the Monte Carlo Method for Uncertainties

Metrology based uncertainty analysis is by convention conducted with the Guide to the Expression of Uncertainty in Measurement (GUM) (BIPM, IEC, IFCC, ILAC, ISO, IUPAP, and OIML 2008a) for analytical models, the GUM Supplement 1 (GS1) (BIPM, IEC, IFCC, ILAC, ISO, IUPAP, and OIML 2008b) for univariate models, and the GUM Supplement 2 (GS2) (BIPM, IEC, IFCC, ILAC, ISO, IUPAP, and OIML 2011) for multivariate models. PDE based models are in practical terms equivalent to multivariate models since the inputs for any PDE are combinations of appropriate boundary conditions, initial conditions and parameters, which may be

conveniently abstracted as multivariate inputs that are plugged into the PDE which when solved produces scalar or vector field solutions that may again be conveniently abstracted as multivariate outputs.

As an example if there is a two-dimensional domain $\Omega = \{(x_1, y_1), \ldots, (x_n, y_n)\}$ that is composed of a set of n pairs of vertices, some PDE parameters $p_1, \ldots, p_m$ for a PDE defined as $\mathcal{L}u = 0$ where $\mathcal{L}$ is some linear/non-linear differential operator with r parameters $D_1, \ldots, D_r$, and $\mathcal{B}u = 0$ is a linear/non-linear boundary condition operator with s parameters $K_1, \ldots, K_s$ using the notation as discussed by Fairweather and Karageorghis (1998), then the model inputs may be grouped as $\mathbf{X} = [x_1, y_1, x_1, y_1, \ldots, x_n, y_n, p_1, \ldots, p_m, D_1, \ldots, D_r, K_1, \ldots, K_s]^\top$ with a total of $N = (2n + m + r + s)$ inputs. The formal solution of the PDE $\mathcal{L}u = 0$ will be field u-(x, y) $\forall$ $(x, y) \in \Omega$ that satisfies the boundary condition $\mathcal{B}u = 0$, and this would in general be obtained by a Finite Difference Method (FDM), Finite Element Method (FEM), Finite Volume Method (FVM) or Boundary Element Method (BEM) numerical scheme.

Although each of these methods differ in approach they may all in principle be used to compute known values of the solution $u(x, y)$ at discrete points $(\xi_1, \eta_1), \ldots, (\xi_t, \eta_t)$ in the domain Ω corresponding to an appropriate choice of numerical mesh, that would typically be solved from a linear system of equations $\mathbf{A} \cdot \mathbf{u} = \mathbf{b}$ where the matrix $\mathbf{A}$ and vector $\mathbf{b}$ are based on the particular choice of numerical method. If the coordinates and solution values are grouped as $\mathbf{Y} = [\xi_1, \eta_1, u(\xi_1, \eta_1), \ldots, \xi_t, \eta_t, u(\xi_t, \eta_t)]^\top$ then the PDE model may be considered to yield a total of $M = 3t$ outputs. This underlying modelling scheme may be abstracted and extended to systems of PDEs that have a time dependence.

Within this mathematical framework if $\mathbf{X} = [X_1, \ldots, X_N]^\top$ is a multivariate input, and $\mathbf{Y} = [Y_1, \ldots, Y_M]^\top$ is a multivariate output, then a PDE model may be written in the general explicit or implicit form as

$$\mathbf{Y} = \mathbf{f}(\mathbf{X}) \tag{5}$$

$$\mathbf{h}(\mathbf{Y}, \mathbf{X}) = \mathbf{0} \tag{6}$$

The application of the GUM Supplement 2 to calculate uncertainties from the above equation is formally achieved by solving the Markov formula to obtain the probability density function (PDF) $g_{\mathbf{Y}}(\boldsymbol{y})$ of the output as

$$g_{\mathbf{Y}}(\boldsymbol{y}) = \int_{-\infty}^{\infty} g_{\mathbf{X}}(\boldsymbol{x}) \times \delta(\boldsymbol{y} - \mathbf{f}(\boldsymbol{x}))\mathrm{d}x_1 \cdots \mathrm{d}x_N \tag{7}$$

In the above scheme $\boldsymbol{x}$ is a random value that is sampled from the joint PDF $g_{\mathbf{X}}(\boldsymbol{x})$ for the input $\mathbf{X}$, and $\boldsymbol{y}$ is a corresponding random value that is either calculated from the explicit equation $\boldsymbol{y} = \mathbf{f}(\boldsymbol{x})$ or alternately the implicit equation $\mathbf{h}(\mathbf{y}, \mathbf{x}) = \mathbf{0}$, that is equivalent to a sampled output value for the joint PDF $g_{\mathbf{Y}}(\boldsymbol{y})$. The function δ is the Dirac delta function and is either zero or unity based on whether $\boldsymbol{y}$ is equal to $\mathbf{f}(\boldsymbol{x})$ as discussed by (Cox and Siebert 2006). Regardless of the particular mathematical form

of the PDE based model if M Monte Carlo simulations are performed then discrete multivariate outputs $\boldsymbol{y}_1, \ldots, \boldsymbol{y}_M$ will be generated from repeated solutions of the PDE based model, and this output data may then be post-processed to generate the corresponding PDF of the output $\mathbf{Y}$.

Although a probability density function (PDF) is usually more intuitively appealing for visualizing an uncertainty, in the case of multivariate models a cumulative distribution function (CDF) $\mathbf{G}(\mathbf{y})$ is a more straight-forward approach to construct, and takes the simple form

$$\mathbf{G} = (\boldsymbol{y}_1, \ldots, \boldsymbol{y}_M) \tag{8}$$

Noting that the input $\mathbf{X} = [X_1, \ldots, X_N]^\top$ is a vector of dimension $N \times 1$, the output $\mathbf{Y} = [Y_1, \ldots, Y_m]^\top$ is a vector of dimension $m \times 1$, it follows that $\mathbf{G}$ is a matrix of dimension $m \times M$. This matrix may then be used to compute the expected value of $\boldsymbol{y}$ denoted by $\tilde{\boldsymbol{y}}$ of dimension $m \times 1$ and covariance matrix $\mathbf{U}_{\tilde{y}}$ of dimension $m \times m$ of the output as

$$\tilde{\boldsymbol{y}} = \frac{1}{M}[\boldsymbol{y}_1 + \cdots + \boldsymbol{y}_M] \tag{9}$$

$$\mathbf{U}_{\tilde{y}} = \frac{1}{(M-1)}\left[(\boldsymbol{y}_1 - \tilde{\boldsymbol{y}}) \cdot (\boldsymbol{y}_1 - \tilde{\boldsymbol{y}})^\mathsf{T} + \cdots + (\boldsymbol{y}_M - \tilde{\boldsymbol{y}}) \cdot (\boldsymbol{y}_M - \tilde{\boldsymbol{y}})^\mathsf{T}\right] \tag{10}$$

For most practical cases any model output is usually sufficiently Gaussian in nature and as a result an expected value $\tilde{y}$ and covariance matrix $\mathbf{U}_{\tilde{y}}$ as indicated above can accurately represent any multivariate Monte Carlo data with sufficient accuracy. At the present time of writing the GUM and GUM supplements assume Gaussian probability density functions for measurement model outputs unless otherwise specified.

In the special case of a univariate output y_r, $r = 1, \ldots, M$ the statistical uncertainty may be summarized by constructing the univariate distribution function $\mathbf{G}$ with the following steps:

1. sort the model unordered values y_r into a non-decreasing order i.e. into a sequence of increasing values with possible repeated values
2. assign uniformly spaced cumulative probabilities $p_r = \frac{1}{M}\left(r - \frac{1}{2}\right)$ for $r = 1, \ldots, M$ where p_r are mid-points of M contiguous probability intervals of width $\frac{1}{M}$ between $p = 0.0$ and $p = 1.0$
3. form a continuous strictly increasing piecewise linear function joining the M points (y_r, p_r) for $r = 1, \ldots, M$ with the formula for a random output η as

$$\tilde{G}_Y(\eta) = \frac{1}{M}\left(r - \frac{1}{2}\right) + \frac{(\eta - y_r)}{\left[M\left(y_{r+1} - y_r\right)\right]}, y_r \leqslant \eta \leqslant y_{r+1}, r = 1, \ldots, (M-1) \tag{11}$$

A two-dimensional PDE based model equation with two random variable inputs and a univariate random variable output will be demonstrated in the next section to illustrate the general solution approach.

Poisson Partial Differential Equation

Different general approaches to solving PDEs include transformation mappings from physical domains to computational domains to apply regular finite difference schemes by Tannehill et al. (1997), the generalized finite difference method (GFDM) of Benito et al. (2001) and Fan et al. (2015), the boundary element method (BEM) by Beer et al. (2008), and the finite element method by Rao (2005).

Specific approaches to particular types of PDE based models such as for plane stress/strain elasticity problem include power series solutions of the Airy function by Jayne and Tang (1970), complex variable and symbolic software by Ioakimidis (1992), the element free Galerkin method by Belytschko et al. (1994), and the meshless local Petrov-Galerkin (MLPG) approach with earlier work by Atluri and Zhu (2000) and later refinements by Liu (2010) that guaranteed matrix invertibility for solutions. A common partial differential equation encountered in many branches of physics and engineering is the Poisson equation as discussed by (Burden and Faires 2001) of the form

$$\nabla^2 \varphi = f \tag{12}$$

where $\nabla^2 \equiv \frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2}$ is the Laplace operator in two dimensions and f is a source term for points $(x, y) \in \Omega \subset \mathbb{R}^2$ where Ω is a domain and $\Gamma = \partial\Omega$ is a domain boundary. In the special case where f is zero the Poisson equation reduces to the Laplace equation $\nabla^2 \varphi = 0$, however it is generally more advantageous for generality to model systems in terms of the Poisson equation since plane stress/strain elasticity (Reddy 2010) and incompressible fluid dynamics problems (Tannehill et al. 1997) may conveniently be modelled in terms of the Poisson equation. An extension of the Poisson equation is the Helmholtz equation (Goldstein et al. 2000) which takes the form $\nabla^2 \varphi = -k^2 f$ where k is a wave number, k^2 is the eigenvalue and f is an eigenfunction, and the equation models wave propagation in a continuum medium such as a metal plate in a drum.

Consider the geometry shown in Fig. 3 which is an adaptation of a potential flow in a duct by White (2003) that may be modelled using a stream function approach in terms of a Poisson equation that avoids a full Navier-Stokes system of PDEs for simplicity. In this figure the geometry represents a ventilation duct that may be encountered in an air duct in an office building or a hospital where air flows into the duct and then flows out of the duct to introduce some air circulation to either cool/heat a room, or alternately to transport away air-borne pollutants such as may be encountered by sick or vulnerable patients. Referring to this figure L_1 is the total length of the duct, L_2 is the outlet length of the duct, L_3 is an inlet length of the duct,

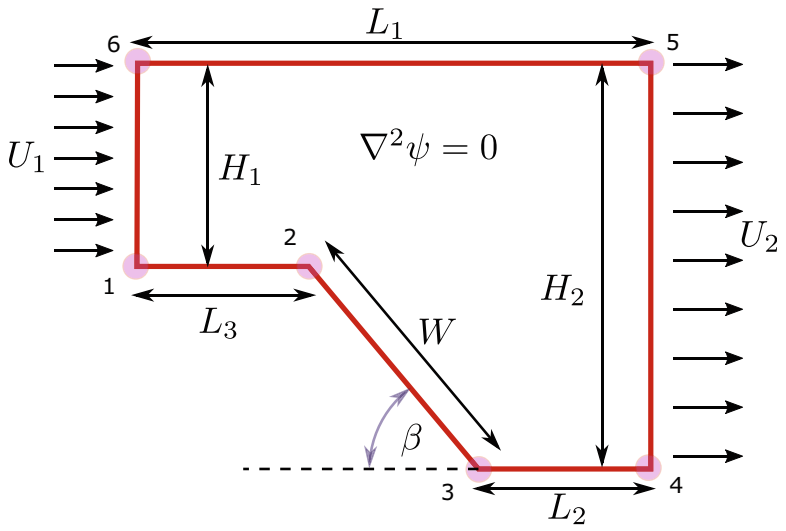

Fig. 3 Schematic illustration of PDE based measurement model of incompressible fluid flow in a duct

H_1 is the height of the inlet to the duct, H_2 is the height of the outlet to the duct, W is the length of an incline within the duct as the flow transitions from the inlet region to the outlet region, β is an angle that indicates how the duct height transitions from the inlet to the outlet region, U_1 is the inlet velocity of the fluid in the duct, and U_2 is the outlet velocity of the fluid in the duct respectively. The physical units of L_1, L_2, L_3, W, H_1 and H_2 are all in meters [m], the unit of β is in radians [rad], and finally the units for U_1 and U_2 are in meters per second [m s^{-1}].

In this model the potential flow for an incompressible fluid with a constant density ρ, horizontal velocity $u(x, y)$ and vertical velocity $v(x, y)$ is governed by a stream function $\psi(x, y)$ that within a domain Ω satisfies Laplace's equation

$$\nabla^2\psi = 0 \tag{13}$$

$$u = \frac{\partial\psi}{\partial y} \tag{14}$$

$$v = \frac{-\partial\psi}{\partial x} \tag{15}$$

In order to solve Laplace's equation appropriate boundary conditions must be specified, and for a stream function model this is achieved by specifying either known free-stream velocities through inferring applicable Neumann boundary conditions through a specification of gradients $\frac{\partial\psi}{\partial n}$ where n is an outwards unit vector where $\frac{\partial\psi}{\partial n}$ is calculated from the known values of $u_\infty = \frac{\partial\psi}{\partial y}$ and $v_\infty = \frac{-\partial\psi}{\partial x}$, or alternately by specifying known values of the stream function through enforcing applicable Dirichlet boundary conditions.

For the particular example problem that is considered assume $U_1 = 10\ \text{m s}^{-1}$, $L_1 = 3$ m, $H_1 = 1$ m, and $H_2 = 2$ m are all constant and known, whilst $L_2 = 1$ m and $\beta = \frac{\pi}{4}\text{rad} = 45^\circ$ are nominal expected values that are modelled as random variables that influence the solution of the PDE through for example manufacturing tolerances and fabrication errors.

In many metrology problems random variables may be modelled as Gaussian distributions, rectangular distributions with upper/lower limits, or triangular distributions. Assume for simplicity and illustrative purposes that L_2 and β have expected values of μ_{L_2} and μ_β with symmetric upper and lower tolerance limits of $\pm\Delta_{L_2}$ and $\pm\Delta_\beta$. Using the GUM guidelines it then follows they these random variables may then be modelled as equivalent Gaussian distributions i.e. normal distributions $N(\mu, \sigma^2)$ with an expected value μ and variance σ^2 such that, for example a duct error length of $\Delta_{L_2} = \pm 20$ cm and duct angle error of $\Delta_\beta = \pm 15^\circ$ yields

$$L_2 \sim N\left(\mu_{L_2}, \left(\frac{\Delta_{L_2}}{\sqrt{3}}\right)^2\right) = N\left(1, \left(\frac{0.20}{\sqrt{3}}\right)^2\right) \tag{16}$$

$$\beta \sim N\left(\mu_\beta, \left(\frac{\Delta_\beta}{\sqrt{3}}\right)^2\right) = N\left(\frac{\pi}{4}, \left(\frac{\frac{\pi}{12}}{\sqrt{3}}\right)^2\right) \tag{17}$$

Since the fluid within the duct is incompressible its density is constant thus using a simple mass flow rate $\dot{m}$ from the continuity equation it may be shown that $\dot{m} = \rho_f U_1 A_1 = \rho_f U_2 A_2$ where A_1 and A_2 are the cross-sectional areas at the flow inlet and outlet respectively. Setting $A_1 = H_1 t$ and $A_2 = H_2 t$ where t is the thickness of the duct out of the plane of the paper it immediately follows that $U_1 H_1 = U_2 H_2$ thus $U_2 = \left(\frac{H_1}{H_2}\right) U_1$.

To complete the mathematical description of the problem note that the volumetric flow of the fluid per unit depth is $Q = (10\ \text{m s}^{-1}) \times (1\ \text{m}) = 10\ \text{m}^2\ \text{s}^{-1}$ per unit depth so for convenience let the value of the stream function along the top of the duct be $\psi_{\max} = 10$ and along the bottom of the duct be $\psi_{\min} = 0$ so that

$$\psi = \psi_{\max}, 0 \leqslant x \leqslant L_1, y = H_2 \tag{18}$$

$$\psi = \psi_{\min}, \begin{cases} 0 \leqslant x \leqslant L_3, y = H_2 - H_1 \\ L_3 \leqslant x \leqslant (L_1 - L_2), y = (H_2 - H_1) + \left[\dfrac{-(H_2 - H_1)}{(L_1 - L_2) - L_3}\right] \cdot [x - L_3] \\ L_1 - L_2 \leqslant x \leqslant L_1, y = 0 \end{cases} \tag{19}$$

Since the inlet velocity U_1 is constant for $H_2 - H_1 \leqslant y \leqslant H_2$ and the outlet velocity U_2 is constant for $0 \leqslant y \leqslant H_2$, it may be shown using the relationship $u = \frac{\partial \psi}{\partial y}$ that ψ must linearly vary from $\psi_{\min}$ to $\psi_{\max}$ for both the inlet and outlet velocities. Accordingly the remaining boundary conditions are then

$$\psi = \psi_{\min} + \left[\frac{\psi_{\max} - \psi_{\min}}{H_1}\right] \cdot [y - (H_2 - H_1)], x = 0, H_2 - H_1 \leqslant y \leqslant H_2 \tag{20}$$

$$\psi = \psi_{\min} + \left[\frac{\psi_{\max} - \psi_{\min}}{H_2}\right] \cdot [y], x = L_1, 0 \leqslant y \leqslant H_2 \tag{21}$$

This completes the mathematical specification of the PDE problem since the PDE is simply Laplace's equation as a special case of the Poisson equation with a zero source term and the boundary conditions are now all specified, however a solution of the PDE by itself without additional post-processing of the underlying fields is seldom sufficient to fully characterize a measurement model. It may be observed that is this particular model the only terms that are not constant are the random variables L_2 and β which are modelled by appropriate PDFs.

In this particular example suppose that it is desired to calculate the dimensionless pressure coefficient C_p at any particular point of interest, that is defined as the ratio of the static pressure gradient $\Delta p = p - p_1$ to that of the dynamic pressure $\frac{1}{2}\rho_f U_1^2$ such that

$$\begin{aligned} C_p &= \frac{p - p_1}{\frac{1}{2}\rho_f U_1^2} \\ &= 1 - \left(\frac{V}{U_1}\right)^2 \\ &= 1 - \frac{1}{U_1^2}\left[\left(\frac{\partial \psi}{\partial y}\right)^2 + \left(-\frac{\partial \psi}{\partial x}\right)^2\right]. \end{aligned} \tag{22}$$

Referring to the above equation, it may be observed that once the random variable values are introduced into the PDE and the PDE is solved to "propagate the statistical information," then the PDE solution may be further post-processed to generate additional numerical data that may be analysed and further manipulated to generate statistical information for the model outputs.

For this particular PDE based model the general steps to apply the Monte Carlo method are therefore:

1. Pick a number M of Monte Carlo simulation events, say $M = 50$.
2. Sample $M = 50$ random independent values of $L_2 = \left(1 \pm \frac{0.20}{\sqrt{3}}\right)$m from Eq. (16) and $\beta = \left(\frac{\pi}{4} \pm \frac{\frac{\pi}{12}}{\sqrt{3}}\right)$ rad from Eq. (17) as PDE inputs that are known to have a statistical variation.
3. Solve the PDE with Eq. (13) with the specified boundary conditions Eqs. (18), (19), (20), and (21) and the sampled random values of L_2 and β.
4. Post-process the PDE solution with Eq. (22) and use the results to calculate the quantity of interest namely the C_p value that will have a statistical variation and collect the results $y_1 = C_p^{(1)}, \ldots, y_M = C_p^{(M)}$.

5. Apply the formulae Eq. (11) to calculate the distribution function to encapsulate the statistical information of the model output, and then post-process the model's statistical output to calculate summaries of the uncertainty in terms of the expected value and standard deviation for the quantity of interest.

In the illustrative example only a single quantity of interest, namely C_p at say $x = \frac{1}{2}L_1$ and $y = H_2$, is considered so the output is technically a univariate quantity. The univariate output from the post-processed PDE solution would then have a statistical uncertainty that is modelled as a Gaussian PDF if symmetric, or alternately as a skew-normal distribution if asymmetries are present as discussed by Possolo et al. (2019). However, in this particular problem the quantity of interest could in principle actually be all C_p values along the bottom of the duct and the top of the duct, in which case the quantity of interest would be a multivariate output. Another possibility for modelling the uncertainty for this particular problem is to consider the terms L_2 and β as random variables ξ_{L_2} and ξ_β and the output C_p as a random variable output η_{C_p} in a equation $\mathbf{h}(\mathbf{Y}, \mathbf{X}) = \mathbf{0}$ so that the output uncertainty can be modelled as a joint PDF. As a result, it may be observed that there are technically a number of different possible choices of how to refine a PDE based model's uncertainty depending on the nature of the physical problem and how the PDE solution is post-processed.

Numerical Results

Whilst various possible approaches for solving PDEs are possible in the specified mathematical example, for mathematical efficiency a Boundary Element Method (BEM) using the Method of Fundamental Solutions (MFS) as discussed by Golberg (1995) may be preferable in the particular cases of Laplace's, Poisson's and Helmholtz's PDEs to reduce the dimension of the problem and hence the amount of numerical calculations. If a larger number of Monte Carlo simulations is possible through a reduction of computational complexity then it is possible to achieve refinements in the final results through for example higher accuracy discrete representations of statistical distribution functions.

A practical consideration that works against mathematical efficiency for particular classes of PDEs particularly for many real engineering problems, is that only general purpose FEM and FVM based PDE computer software like the commercial package ANSYS or the open source package OpenFOAM are usually available due to the geometrical complexity of real world problems which are not amenable to specialist techniques such as MFS and RBF numerical schemes.

Consequently, it is seldom practical to generate a very large number of Monte Carlo simulations for real world engineering PDEs without access to high performance computing (HPC) resources, and a judicious smaller number of sampled random values of relevant terms must be selected in order to at least generate reasonable estimates of expected values and variances of expected values. Due to

these considerations for illustrative purposes the mathematical example is solved with the C++ library Fenics by Loggs et al. (2012) using a Python 3 interface with $M = 50$ simulations by sampling from the probability density function distributions previously specified in Eqs. (16) and (17) and plotted in Fig. 4. The general approach is to sample $M = 50$ points from each of the PDFs and then use that as an input to

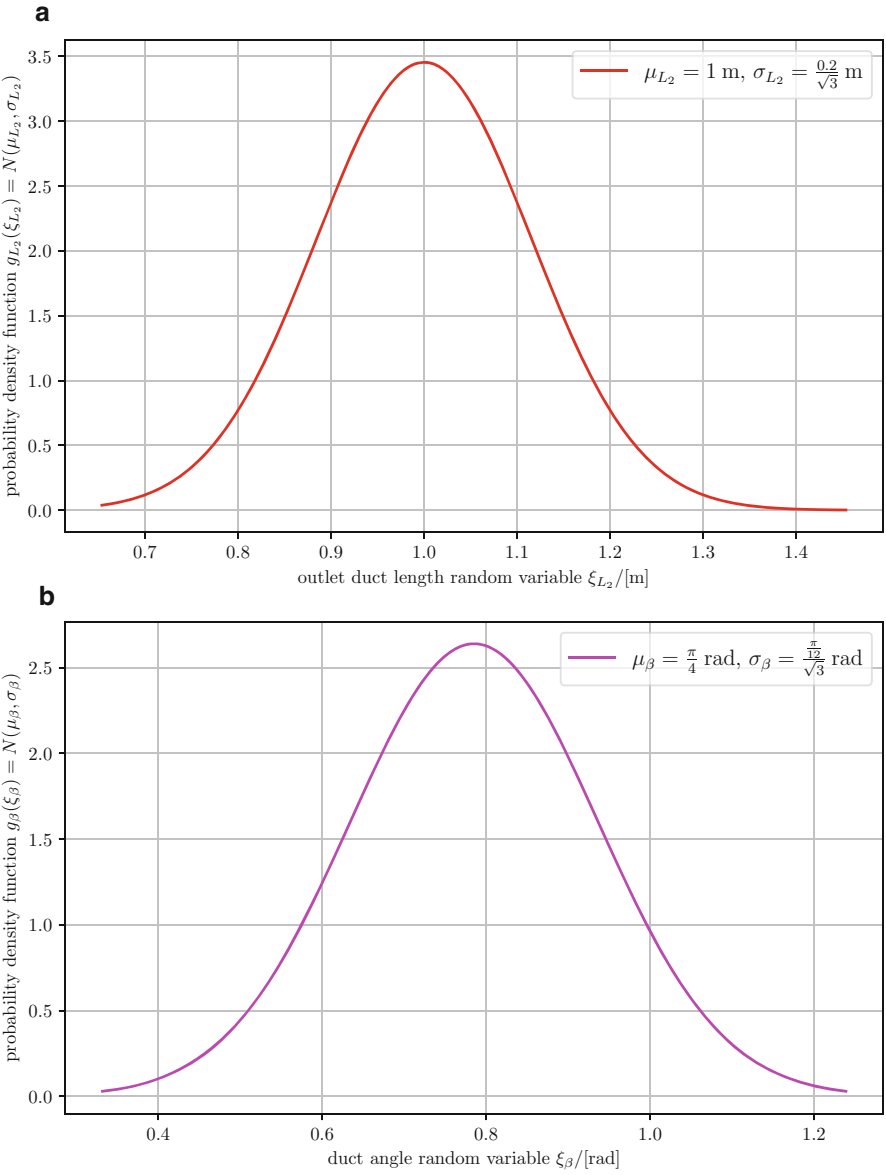

Fig. 4 Illustration of probability density function variations for model input random variables

solve the PDE so that there would be a total of $M = 50$ different solutions which can then be post-processed.

Final representative results for the PDE solution using two sampled values with two simulations are shown in Fig. 5 since it is not feasible to plot 50 different graphs. In these two simulations mesh A corresponds to a domain Ω_A constructed with $L_2 = 0.840$ m and $\beta = 51.886°$ whilst mesh B corresponds to a domain Ω_B constructed with $L_2 = 0.997$ m and $\beta = 48.352°$, both of which which the PDE $\nabla^2\psi = 0$ is solved for and then post-processed by computing the gradients $u = \frac{\partial\psi}{\partial y}$ and $v = -\frac{\partial\psi}{\partial x}$. This simulation process is repeated for each of the remaining 48 sampled values of L_2 and β input random variables so that a total of $M = 50$ simulations are performed i.e. 50 different meshes are re-generated for each Monte Carlo simulation event where a new mesh is constructed for the new varying geometry based on the sampled different values of L_2 and β, and in each Monte Carlo simulation event corresponding to a PDE solution the results are saved to file, so at the end there are $M = 50$ solution data sets for the solution $\nabla^2\psi = 0$ with the specified boundary conditions.

In principle the underlying PDE solution data may now be post-processed based on the relevant measurement model requirements. Since C_p only makes physical sense on the lower boundary and upper boundary of the domain it is natural to process the PDE results to calculate the pressure coefficient. Choosing for convenience the upper surface for the value of C_p as the position varies along the top length it may be seen in Fig. 6 that a sequence of $M = 50$ curves for C_p may be plotted. Having solved the PDE based model if the position along the length of the upper surface is considered as one output $Y_1 = x^{[\text{upper surf}]}$ and the corresponding value of $Y_2 = C_p(x^{[\text{upper surf}]}, y = H_2)$ is calculated as another output then the set of $M = 50$ curves could be considered as a multivariate output $\mathbf{Y} = [Y_1, Y_2]^\top$ based on the multivariate input $\mathbf{X} = [L_2, \beta]^\top$. Although just 5 curves are shown in the figure, when performing a Monte Carlo simulation for a practical engineering problem a very large number of simulation events is usually necessary. In this case if for example $M = 10000$ simulation events corresponding to sampled values of inputs for a PDE are solved, then if all the points from curves are considered as a scatter plot of x vs. $C_p(x, y)$, a covariance matrix $\mathbf{U}_{\tilde{\mathbf{Y}}}$ may be calculated with the multivariate GUM Supplement 2 to model the statistical uncertainty in the pressure coefficient.

Considering instead a simpler problem to demonstrate the general uncertainty approach for PDE based models, suppose that it is simply required to determine the pressure coefficient C_p at the mid-point of the top surface i.e. the value $C_p\left(x = \frac{1}{2}L_1, y = y_5\right)$ as per the previous diagram in Fig. 3. For this particular case step 1, step 2 and step 3 have already been performed, and step 4 now requires to calculate the value of C_p at the point of interest. This can be achieved by going through each of the $M = 50$ data sets and simply working out the value of $C_p\left(x = \frac{1}{2}L_1, y = y_5\right)$ which will yield a univariate data set of values $y_1, \ldots, y_M$. The univariate data may then be post-processed according to the GUM Supplement 1 to produce the cumulative distribution function $G(y)$ as shown in Fig. 7.

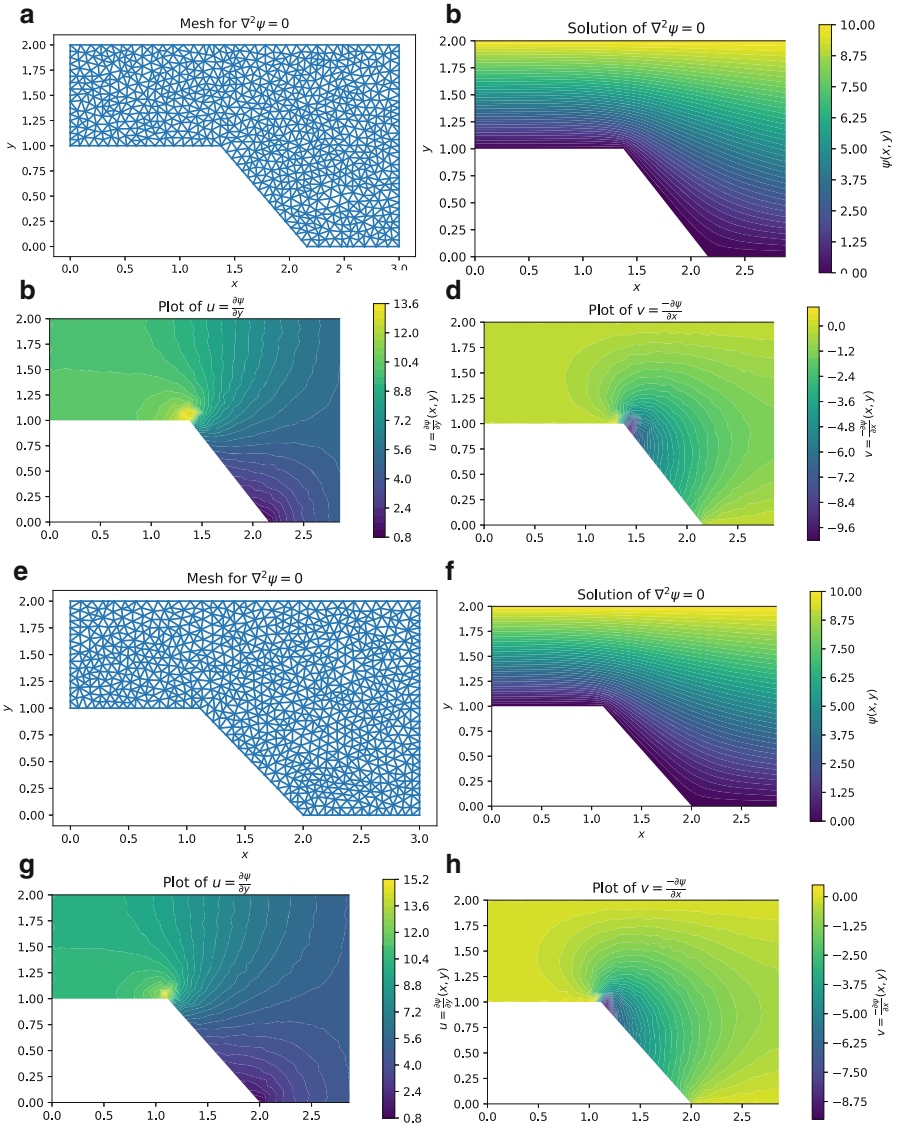

Fig. 5 Illustration of finite element method simulations for varying domain geometries and meshes based on the influence of the random variable inputs L_2 and β which are allowed to independently vary based on their respective statistical uncertainties. As L_2 and β vary they will change the underlying domain Ω for which the PDE $\nabla^2\psi = 0$ must be solved with the specified boundary conditions. For each Monte Carlo simulation event for $j = 1, \ldots, M$ where $M = 50$ is the number of simulations there will therefore be a slightly different domain $\Omega_j, j = 1, \ldots, M$ for which the PDE must be solved. Geometry for domain Ω_A corresponds to sampled values of $L_2 = 0.840$ m and $\beta = 51.886°$ whilst geometry for domain Ω_B corresponds to sampled values of $L_2 = 0.997$ m and $\beta = 48.352°$ which are are representative samples of how Ω_j varies

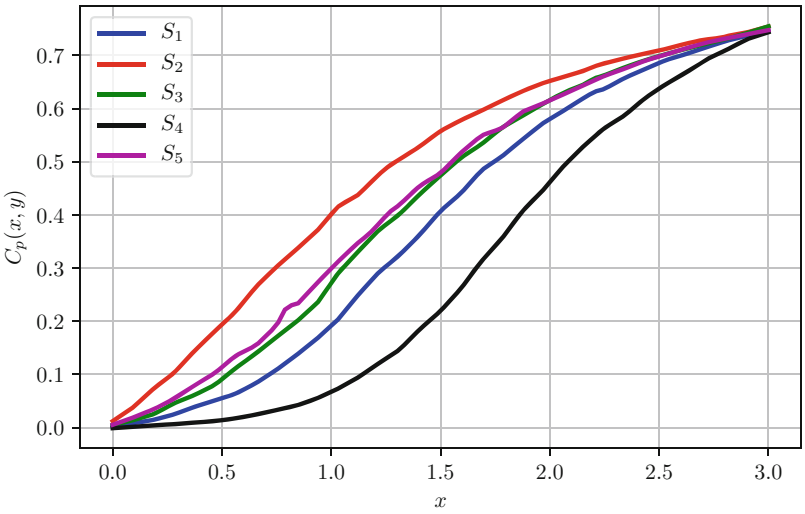

Fig. 6 Schematic illustration of pressure coefficient C_P obtained from post-processing of the PDE solution showing the variation on the upper surface of the channel based on the influence of different sampled random variables L_2 and β for simulation events $S_1, \ldots, S_5$

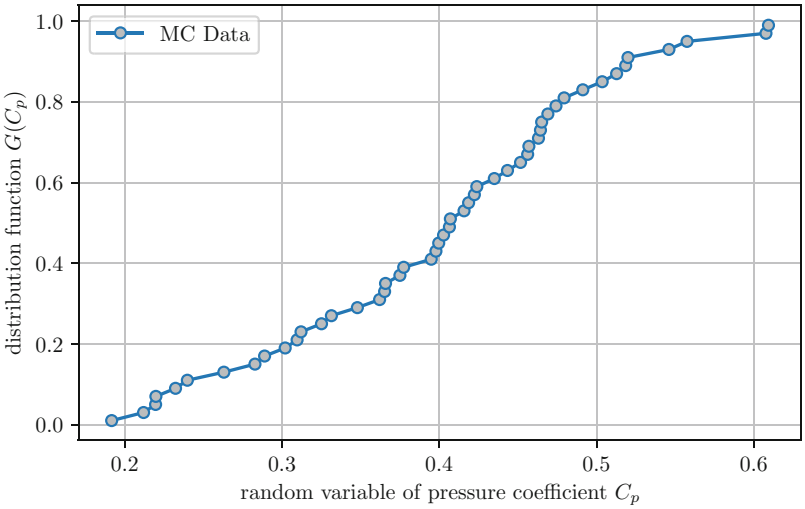

Fig. 7 Visualization of cumulative distribution function for pressure coefficient C_P evaluated at $x = \frac{1}{2}L_1$ and $y = y_5$

An important point to note in the uncertainty analysis of PDE models is that due to the general computational cost involved in numerically solving for multiple evaluations of PDEs, where each solution may take anywhere from 1 h to several hours to solve, that the total feasible number of Monte Carlo simulation events M based on available computational resources is generally quite small. As a result

usually only the expected value $\tilde{y}$ and standard deviation $u(\tilde{y})$ may be calculated with any reasonable degree of certainty. Consequently, using the substitutions $\rho = G(y)$ and $y = Q(\rho)$ where $Q(\rho)$ is the corresponding quantile function, it is generally inadvisable to directly apply the mathematical statistics definitions $\mathbb{E}(y) = \int_0^1 Q(\rho)\mathrm{d}\rho$ and $\mathbb{V}(y) = \int_0^1 [Q(\rho) - \mathbb{E}(y)]^2 \, \mathrm{d}\rho$ in order to calculate the expected value and standard deviation. For post-processing of Monte Carlo data, it is usually more accurate to simply estimate the expected value $\tilde{y}$ and standard deviation $u^2(\tilde{y})$ as

$$\tilde{y} = \frac{1}{M}{\sum_{r=1}^{M}}'' \, y_r \approx \frac{1}{M}\sum_{r=1}^{M} y_r \text{ for } M \text{ large} \tag{23}$$

$$\begin{aligned} u^2(\tilde{y}) &= \frac{1}{M}\left[{\sum_{r=1}^{M}}'' \, (y_r - \tilde{y})^2 - \frac{1}{6}\sum_{r=1}^{M-1}\left(y_{r+1} - y_r\right)^2\right] \\ &\approx \frac{1}{M-1}\sum_{r=1}^{M}(y_r - \tilde{y})^2 \text{ for } M \text{ large} \end{aligned} \tag{24}$$

In the above two equations the double prime on the summation as per Annex D in the GUM Supplement 1 means that the first and last terms in the sum are to be taken with a weight of one half. The usual more common formulae for $\tilde{y}$ and $u^2(\tilde{y})$ only formally apply in the limit of a very large number of Monte Carlo simulation events when M is sufficiently large.

A practical effect of these different formulae is that when an uncertainty analysis is performed for a measurement system modelled in terms of partial differential equation it is only usually feasible, particularly for very complicated models when numerically solved for on laptops in the absence of HPC resources, to perform say $M = 25$ or $M = 50$ Monte Carlo simulations corresponding to a PDE solution if each solution takes anywhere from 1 h to an afternoon to reach grid convergence. As a result the more common and simpler statistical formulae for an expected value and standard deviation, which is only accurate for a large number of data points e.g. $M = 10000$, would potentially introduce a statistical systematic bias in a model simulation in primary/apex measurement standards in national laboratories, and which may then subsequently negatively impact on the measurement traceability accuracy of commercial calibration laboratories.

Discussion and Conclusion

Based on the overview of the Monte Carlo approach for computing uncertainties in PDE based models as discussed in this chapter the following aspects may be noted:

- The Monte Carlo approach is conceptually straightforward to implement with "black-box" PDE solvers for both low-dimensional as well as high-dimensional models if the model inputs can be conveniently sampled and introduced into the solver/post-processor to generate discrete output statistical data
- The discrete output statistical data may be conveniently post-processed to easily produce expected values and standard uncertainties with standard formulae for real practical engineering problems
- If sensitivity analysis of uncertainties is necessary in the case of many competing random variable influences then more advanced techniques such as principal component analysis (PCA) may be considered (Fig. 8)

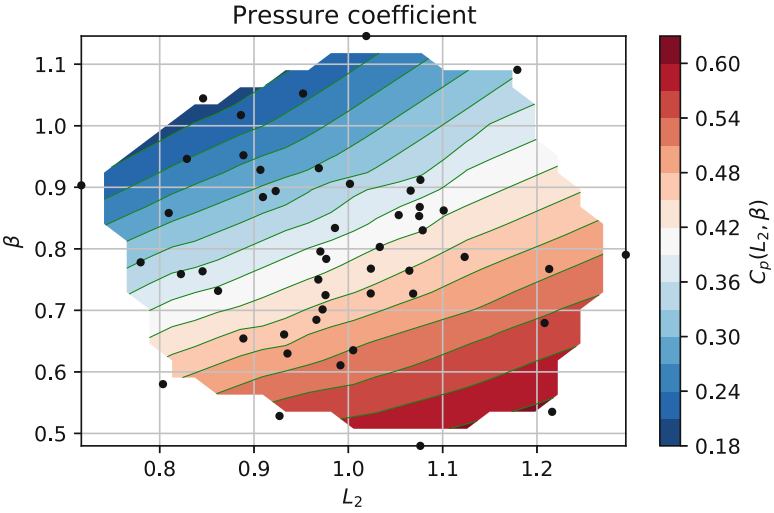

Fig. 8 Visualization of pressure coefficient $C_p(L_2, \beta)$ sensitivity variation evaluated at $x = \frac{1}{2}L_1$ and $y = y_5$ based on PDE input random variables. The results of the Monte Carlo simulation may be further post-processed if necessary to estimate the corresponding sensitivity coefficients $\frac{\partial C_P}{\partial L_2}$ and $\frac{\partial C_P}{\partial \beta}$ to calculate the uncertainty of C_P at any given point in the domain where a pressure coefficient is physically meaningful by using finite differences. As an example if the uncertainty at $L_2 = 1.0$ and $\beta = 0.8$ is required then the value of $C_P(L_2 = 1.0, \beta = 0.8) = C_P^{center}$ is read off the graph, slightly perturbed values are read of for say $C_P(L_2 = 0.99, \beta = 0.8) = C_P^{west}$ slightly to the left, $C_P(L_2 = 1.01, \beta = 0.8) = C_P^{east}$ slightly to the right, $C_P(L_2 = 1.0, \beta = 0.79) = C_P^{south}$ slightly to the bottom and $C_P(L_2 = 1.0, \beta = 0.81) = C_P^{north}$ slightly to the top, and then the partial derivatives for the sensitivity coefficients are approximated with finite differences as $\frac{\partial C_P}{\partial L_2} \approx \frac{1}{1.01-0.99} \times \left[C_P^{east} - C_P^{west}\right]$ and $\frac{\partial C_P}{\partial \beta} \approx \frac{1}{0.81-0.79}\left[C_P^{north} - C_P^{south}\right]$. In this manner the uncertainty for C_P could be approximately estimated as $u^2(C_P) = \left(\frac{\partial C_P}{\partial L_2}\right)^2 u^2(L_2) + \left(\frac{\partial C_P}{\partial \beta}\right)^2 u^2(\beta)$ at any given point in the domain that is physically meaningful. This sensitivity analysis approximation approach is sometimes desirable for metrology practitioners when performing an uncertainty analysis that does not require a full knowledge of the joint probability density function

Cross-References

- ▶ Advanced Techniques in Evaluation of Measurement Uncertainty
- ▶ Advances in Additive Manufacturing and Its Numerical Modelling
- ▶ Application of Contemporary Techniques of Evaluation of Measurement Uncertainty in Pressure Transducer
- ▶ Electromagnetic Metrology for Microwave Absorbing Materials
- ▶ Evaluation and Analysis of Measurement Uncertainty
- ▶ Pressure and Its Measurement
- ▶ Radar Cross-Section (RCS) Estimation and Reduction

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Modern Approaches to Statistical Estimation of Measurements in the Location Model and Regression

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Abstract

Metrology as the science about measurement is highly intertwined with statistical point estimation. Evaluating and controling uncertainty of measurements and analyzing them by means of exploratory data analysis (EDA) or predictive data

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mining requires to exploit advanced tools of statistical estimation. The main focus of the chapter is devoted to nonstandard approaches to the analysis of measurements in two fundamental models, namely, the location model and linear regression. Robust regression methods, which are resistant to the presence of outlying (anomalous) measurements, are discussed here. An illustration of their performance over a real dataset related to thyroid disease and a Monte Carlo simulation reveal here the least weighted squares estimator, which has remained quite neglected so far, outperforms much more renowned robust regression estimators in terms of the variability. Further, Bayesian estimation in the location model is revealed here to have the ability to incorporate previous measurements in a very intuitive way. Finally, the chapter gives a warning that linear regression performed on data contaminated by measurement errors yields biased estimates and requires specific estimation tools for the so-called measurement error model.

Introduction

Measurements in all possible fields of applications are typically imprecise because of errors coming from various sources, which can be classified as instrumental errors, method errors, personal errors, errors caused by an unsuitable design, random errors, etc. (Crowder et al. 2020). Metrology as the science of measurement and its application is linked with information theory (Lesurf 2002; Kong et al. 2019), because measuring provides gathering of information about complex systems and reduces the uncertainty (information entropy). Metrology is at the same time linked with statistical methodology as it operates with fundamental statistical concepts such as variability, standard deviation, general law of error propagation, or confidence intervals. Statistical point estimation overviewed in Lehmann and Casella (1998) can be characterized as an essential approach for evaluating uncertainty. Estimates variability of the measurements is an important application formally described in the Guide to the Expression of Uncertainty in Measurement, which is known as the GUM methodology (JCGM 2012; Lira 2016). Other statistical methods further allow to control the quality of measurements (e.g., in Hynek et al. 2021) or to analyze the results of measurements (Hebra 2010), i.e., to obtain and interpret useful knowledge from the measured information.

This chapter starts with a brief historical overview of standard as well as robust statistical tools applicable to analyzing measurements. Two important statistical models, namely, the location model and the linear regression, are further recalled. The main focus of the chapter is devoted to advanced (nonstandard) approaches to the analysis of measurements. Various regression estimators suitable for the errors-in-variables model are recalled with a focus on the advanced procedures. Robust regression methods are discussed together with an illustration of their performance over a real dataset related to thyroid disease, and over a Monte Carlo simulation. Finally, the Bayesian estimation is explained to have the ability to incorporate previous measurements in a very intuitive way.

History of Statistical Methods for Analyzing Measurements

While the history of metrology has been thoroughly overviewed in specialized monographs (Robinson 2007) and the same is true for the history of commonly used statistical methods (Hald 2006), let us briefly comment here the historical development of the (quite narrow) connection between measurement science and statistics (Stigler 1973). This overview recalls that the need for advanced methods for analyzing measurements, i.e., the natural need for a robust (resistant) analysis of contamined data, has been known for a very long time, already from the very dawn of the regression methodology. The personalities mentioned here with a remarkable contribution to the development of statistics, who all were polymaths with strong mathematical skills, were namely heavily and passionately involved in their own practical measurements.

In a posthumously published paper by Thomas Bayes (1702–1761), the first version of the Bayes theorem appeared. This was later rediscovered by Pierre Simon de Laplace (1749–1827), who greatly contributed to the development of probability theory. Laplace perceived Bayesian probabilistic thinking as a natural extension of mathematical logic and common sense. Let us recall already now that the Bayesian statistics framework is currently the preferred theoretical approach for metrology and that future editions of the GUM can be expected to adopt its modelling framework more strongly. Going back to Laplace, he was also the author of the first rigorous work on robust estimation in 1818, which was devoted to the distribution of the median. Laplace was intensively interested in analyzing astronomic measurements by regression models and was aware of the harmful effect of measurement errors. Apart from the median, Laplace also propagated using the regression median in the linear regression model.

Carl Friedrich Gauss (1777–1855) belonged to the pioneers of a consistent usage of probability theory in the processing of measurements from various fields, mainly in astronomy, geodesy, and physics. By investigating properties of measurement errors and formulating the law or error propagation, he laid foundations of the theory of errors. He developed the maximum likelihood estimation, which became one of the pillars of mathematical statistics. Using the maximum likelihood method, Gauss derived the normal distribution denoted as "the law of errors" and considered it as a suitable model for measurement errors. Gauss also theoretically justified the least squares estimator by means of a Bayesian argument (Sheinin 1996).

The first mathematical approach to handling outlying measurements (outliers) was presented by Benjamin Peirce (1809–1880) in an 1852 paper in the *Astronomical Journal*, where he proposed a historically first formal rule for detecting outliers in measurements. He did not study properties of the trimmed mean, which he considered as the mean computed over the non-outlying measurements, but still was able to understand that the trimmed mean has more appealing properties compared to the arithmetic mean computed across all measurements (Wheeler 2020).

Simon Newcomb (1835–1909) performed a series of measurements of the speed of light in 1881. He realized that his measurements related to the orbits of Mercury have different levels of accuracy, so that the arithmetic mean would not be their suitable characteristic. Instead, he considered small weights assigned to unreliable measurements, which he called to be discordant values, and commonly used the weighted mean instead of arithmetic mean, however with subjectively chosen weights. He also realized the data to have heavy tails and attempted for a more robust estimation by deleting very severe outliers; he assumed the normal distribution to represent a suitable model for the remaining measurements (Stigler 1986).

Ronald A. Fisher (1890–1962), the father of modern statistics, developed the field of statistical point estimation by deriving properties of maximum likelihood estimators and also developed statistical principles of design of experiments. He predecessed ideas of information theory by understanding measurements to reduce uncertainty of a system; in this spirit, he proposed the Fisher information of an unknown parameter as a quantification of the information carried in a single measurement about this parameter.

The Location Model

The location model represents the simplest statistical model and at the same time a special case of the linear regression model. It considers independent measurements $X_1, \ldots, X_n$, where each X_i (i.e., for $i = 1, \ldots, n$) is assumed to come from the normal distribution $X_i \sim N(\mu, \sigma^2)$. The purpose of the measurements is typically to estimate μ. However, the most common estimate of μ based on the data, i.e., their arithmetic mean $\overline{X} = (1/n)\sum_{i=1}^{n} X_i$, is known to be vulnerable (non-robust) with respect to the presence of outliers (anomalous measurements) in the data. Also the standard deviation S, obtained as

$$S = \sqrt{\frac{1}{n-1}\sum_{i=1}^{n}\left(X_i - \overline{X}\right)^2}, \tag{1}$$

may be heavily influenced by outliers. The GUM methodology considers the standard deviation to represent the most appropriate estimate (empirical counterpart) of σ for (not only) evaluating uncertainty of measurements. This is however not appropriate, because S is (just like the arithmetic mean) heavily vulnerable with respect to the presence of outliers in the data. It is for this reason that reliable outlier detection cannot be based on using S. Because robust estimators in the location model are often defined as special cases of robust estimators for regression, we will further discuss the location model only later after having described robust estimators for the regression model.

Linear Regression

The aim of regression modeling is to explain (predict) a given explanatory variable (dependent variable, response) by means of one or more predictors (regressors, independent variables). An overview of regression applications for the task of uncertainty evaluation in metrology was presented in Nielsen (2002). Regression analysis has become an important part of analyzing measurements by means of exploratory data analysis (EDA) or predictive data mining (Tin et al. 2021) with numerous applications not limited to calibration of measuring instruments, estimating missing values, or detecting mixtures of two or more populations or clusters of data (Kalina and Tichavský 2020). Linear measurement systems, which can be analyzed by linear regression, were discussed in Sect. 4 of Raghavendra and Krishnamurthy (2013). As an extension, a recent work of Peremans and Van Aelst (2018) considered multiple measurements characterized by several (seemingly uncorrelated) regression models. Regression models are also useful for denoising (reconstructing) images obtained, e.g., by 3D scanners within optical metrology (Papageorgiou et al. 2017).

Let us consider the standard linear regression model

$$Y_i = \beta_0 + \beta_1 X_{i1} + \ldots + \beta_p X_{ip} + e_i, i = 1, \ldots, n, \tag{2}$$

where $Y_1, \ldots, Y_n$ are values of a continuous response variable, $X_i = (X_{i1}, \ldots, X_{ip})^T$ is the regressor (predictor) for the ith measurement, and $e_1, \ldots, e_n$ are random errors (disturbances). The model (2) can be expressed in the matrix notation as $Y = \boldsymbol{X}\beta + e$; matrices will be denoted by bold face throughout the chapter. The most important possible methods for estimating the regression parameters $\beta = (\beta_0, \beta_1, \ldots, \beta_p)^T$ in the situation with measurement errors are the following:

- Least squares (aka ordinary least squares)
- A consistent modification of least squares
- Generalized least squares (GLS), including weighted least squares (WLS)
- Total least squares (TLS) and extensions, which include weighted total least squares (WTLS) or weighted total least squares with correlation (WTLSC)
- Nonparametric estimators (including rank-based estimators) for the errors-in-variables model
- Estimators based on robust optimization
- Robust estimators

The least squares estimator remains to be very often used in the errors-in-variables model, i.e., the model assuming not only the response but also the indpendent variables to be measured with errors. However, the least squares estimator is not consistent under the presence of measurement errors in the regressors. Only in the location model, the arithmetic mean remains to be consistent if the data contain symmetrically distributed measurement errors. In regression, however, metrologists should be aware of alternative approaches tailor-made for the measurement error models.

Weighted Least Squares

Let us recall the weighted least squares (WLS) with specified non-negative weights $w_1, \ldots, w_n$ fulfilling the natural requirement $\sum_{i=1}^{n} w_i = 1$. Let $u_i(b)$ denote the residual of the ith measurement based on a given estimate $b = (b_0, \ldots, b_p)^T$ of β, i.e.,

$$u_i(b) = Y_i - b_0 - b_1 X_{i1} - \ldots - b_p X_{ip}, i = 1, \ldots, n. \tag{3}$$

The WLS estimator, which is also known as Aitken estimator or generalized least squares (Greene 2018), is defined by minimizing a weighted estimate of σ^2 in the form

$$b_{WLS} = \arg\min_{b \in \mathbb{R}^{p+1}} \sum_{i=1}^{n} w_i u_i^2(b). \tag{4}$$

The WLS estimator has an explicit form; denoting by $\boldsymbol{W} = diag\,(w_1, \ldots, w_n)$ the diagonal matrix with the weights on the main diagonal, the WLS estimator is obtained as

$$b_{WLS} = \left(\boldsymbol{X}^T \boldsymbol{W} \boldsymbol{X}\right)^{-1} \boldsymbol{X}^T \boldsymbol{W} Y.$$

Errros-in-Variables Model

The errors-in-variables (EIV) model can be understood as the most basic additive measurement error (ME) model. Assuming Eq. (3), vectors

$$W_i = \left(W_{i1}, \ldots, W_{ip}\right)^T = \left(X_{i1} + u_{i1}, \ldots, X_{ip} + u_{ip}\right)^T, i = 1, \ldots, n, \tag{5}$$

instead of the true predictors X_i are observed; here $u_i = (u_{i1}, \ldots, u_{ip})^T$ is an (observed) measurement error. It seems natural to estimate β by the least squares in the model

$$Y_i = \beta_0 + \beta_1 W_{i1} + \ldots + \beta_p W_{ip} + e_i, i = 1, \ldots, n, \tag{6}$$

explaining the response by the imprecise predictors (Eq. 5). Let now $\widehat{\beta}$ denote the least squares estimate. In the ME model, the slopes in $\widehat{\beta}$ (but not the intercept) are not consistent and $\widehat{\beta}$ remains biased even for a large number of measurements, while the bias does not vanish if the number of measurements is assumed to increase to infinity. It is typical for the ME model that the least squares estimator underestimates the effects in the model and such phenomenon is known as the attenuation bias or regression dilution.

Consistent Modification of Least Squares in the EIV Model

A consistent estimator of the slopes $\beta_1, \ldots, \beta_p$ in Eq. (6) may be obtained as follows. Denoting now the least squares estimate as $\widehat{\beta} = \left(\widehat{\beta}_0, \widehat{\beta}_1, \ldots, \widehat{\beta}_p\right)^T$, we consider

$$\left(\boldsymbol{C}^2 + \boldsymbol{D}^2\right)^{-1} \boldsymbol{C}^2 \left(\widehat{\beta}_1, \ldots, \widehat{\beta}_p\right)^T, \tag{7}$$

where $\boldsymbol{C}^2$ is the covariance matrix obtained from the true but unobserved predictors, $\boldsymbol{D}^2$ is the covariance matrix obtained from the unobserved measurement errors, and $(\boldsymbol{C}^2 + \boldsymbol{D}^2)^{-1}\boldsymbol{C}^2$ represents the reliability matrix. Equation (7) is consistent under a specific set of assumptions, which naturally include the requirement that the distribution of the measurement errors is symmetric. Although Eq. (7) requires the knowledge of the variability of measurement errors, it may be possible to estimate this variability from prior measurements; such approach is commonly assumed in the literature on the subject (Buonaccorsi 2010).

General Discussion of Estimators in the EIV Model

The theory of various parametric estimates in the EIV model was overviewed in Buonaccorsi (2010), where a focus is given to GLS estimators. The book is also strongly focused on investigating the performance of standard statistical procedures (i.e., those not tailor-made for measurement error models) under the presence of measurement errors. Nonparametric estimators together with methods for regression calibration were overviewed in Carroll et al. (2006).

Particular types of (especially the advanced) estimation procedures for the EIV model will be discussed below. Let us already now stress that some of the approaches require specific assumptions and/or are suitable only in certain circumstances. The TLS estimator (and its extensions) does not require any assumptions on the magnitude of the measurement errors, which explains its popularity in metrology. The weighted approaches (WLS or WTLS) are naturally suitable in situations with known precision (uncertainty) assigned to individual measurements.

A nonparametric estimator of β that is based on ranks of residuals was proposed for the EIV model in Saleh et al. (2012). This estimator denoted as R-estimator is inspired by the consistent approach for the least squares. It is asymptotically highly efficient and admits a local asymptotic bias, which depends only on the precision of the measurements and surprisingly not on the distribution of the regression errors (Jurečková et al. 2016).

Estimators based on robust optimization assume the measurement error to be bounded in some interval and find the solution as the standard one (e.g., least squares) considered for the least favorable case, i.e., for the worst measurement

errors. Such procedures based on optimization under uncertainty have been thoroughly discussed in Ben-Tal et al. (2009).

The presence of measurement errors also causes standard statistical hypothesis testing in the regression model not to be valid. So far, hypothesis testing allowing for measurement errors was considered only rarely; the likelihood ratio test for a general ME model was constructed by means of robust optimization in Broniatowski et al. (2018), who proved it to be a minimax test, i.e., its maximal risk over all values of the parameters is the minimal among all possible tests.

Total Least Squares and Extensions

The TLS estimator, which is one of the popular regression procedures for the EIV model, represents an extension of the orthogonal regression estimators, where the latter consider residuals evaluated as perpendicular Euclidean distances from the fitted regression line (for $p = 1$). The TLS estimator is formally obtained by minimizing the Frobenius norm of the matrix of perturbations, which are considered for the response as well as for the regressors. Such estimator is a nonlinear function of the response and the regressors, and this causes some complications from the practical point of view. Particularly, Hnětynková et al. (2011) studied the computational difficulties of TLS and claimed that the solution of the common algorithm for TLS is not the true TLS estimator obtained by minimizing the Frobenius norm. Pešta (2013) warned that the asymptotic distribution of the TLS estimator might be useless as its assumptions cannot be verified in practice and that the asymptotic covariance matrix of the estimator is unwieldy.

If each measurement has a known and fixed weight corresponding to its precision, it is natural to consider WTLS as the weighted version of the TLS. Krystek and Anton (2007) proposed an algorithm for computing WTLS, and Klauenberg et al. (2022) discussed WTLS in the perspective of the GUM. Because WTLS ignores correlations between the response and the regressors, Krystek and Anton (2011) proposed the WTLSC method, which reduces the uncertainties about the parameters by taking the correlations into account. Particularly for $p = 1$, a single additional correlation parameter is considered in the model. The obtained WTLSC estimator allows to analytically quantify the covariance structure between the parameters, which is very useful in metrology. An interesting application of WTLSC was presented by Ramnath (2020) for the context of pressure balance models in physics of gases.

Robust Regression Estimators

Because the least squares (as well as the total least squares) estimator is too vulnerable to the presence of outliers in the data, various robust statistical methods have been proposed as more resistant alternatives. Many of the robust estimators do

not require to specify, which observations are outliers and which are not (Maronna et al. 2006). Robust estimators typically consider the model (3) without measurement errors, but are intended to be resistant to outliers (global robustness) and only some of them also to small (local) modifications of all measurements (local robustness). However, local robustness has not been sufficiently investigated for many common robust estimators (Sect. 4.9 of Jurečková et al. (2019)). Robust regression methods were applied in Ostojski et al. (2016) for calibration, verification, and validation of models of ecological measurements. They are suitable for multi-sensor positioning methods, especially if high precision and high reliability are needed (Liu et al. 2021). Robust methods also turned out to outperform standard (non-robust) tools in positioning techniques based on the global navigation satellite system (Medina et al. 2019).

Let us now mention some important robust estimators. M-estimators (Huber and Ronchetti 2009) represent a popular class of tools, which may be robust to outliers in the response variable, but none are robust with respect to leverage points, i.e., outliers in the regressors (Jurečková et al. 2019). This is true also for the regression median, which is based on minimizing the sum of absolute values of residuals. M-estimators are not robust in terms of the breakdown point, which represents a fundamental measure of robustness of a statistical estimator of an unknown parameter. The finite-sample breakdown point measures the minimal fraction of data that can drive an estimator beyond all bounds when set to arbitrary value. The first highly robust regression methods had a low efficiency, i.e., low performance for non-contaminated measurements (Rousseeuw and Leroy 1987). The same is true for S-estimators, which are based on minimizing a robust estimate of the scale of the residuals (Davies 1990). MM-estimators of Yohai (1987) can be described as two-stage procedures; their computation starts with performing an S-estimator and proceeds with computing an M-estimator with a particular (fixed) scale.

The least weighted squares (LWS) estimator is a robust regression estimator, which remains less known in spite of its appealing properties (Víšek 2011; Kalina 2015). The motivation is to assign small weights to less reliable observations; such idea of implicit weights assigned after a permutation, which is determined only during the computation, yields promising results in real-life applications (Kalina and Hlinka 2017). The LWS estimator is different from the very common weighted least squares (WLS) estimator, where the latter considers fixed (known) weights assigned to individual measurements. The LWS estimator assumes just like WLS that different measurements have a different accuracy (precision); the precisions of individual measurements correspond to the selected weight function, while it is not known before the computation of the LWS which precision value will correspond to which measurement. Thus, the user does not assign weights to individual measurements, but only chooses their magnitudes.

If suitable weights are used, the LWS estimator is highly robust in terms of the breakdown point. The LWS estimator is at the same time suitable for measurements with unequal variances. The computation of the LWS estimator may exploit an

adaptation of the approximate algorithm of Rousseeuw and Van Driessen (2006). Formally, the LWS estimator requires the user to specify a sequence of magnitudes of weights

$$w_1 \geq w_2 \geq \ldots \geq w_n, \tag{8}$$

where the magnitudes $w_1, \ldots, w_n$ are considered in the form

$$w_i = \frac{\psi\left(\frac{i-1/2}{n}\right)}{\sum_{j=1}^{n} \psi\left(\frac{j-1/2}{n}\right)}, i = 1, \ldots, n. \tag{9}$$

Here, the user has to specify a weight function ψ, which is defined as a function $\psi : [0, 1] \rightarrow [0, 1]$; it must be continuous on $[0, 1]$ with $\psi(0) = 1$ and $\psi(1) = 0$.

The weights are assigned to individual measurements only after some permutation. To define the LWS estimator formally, let us first denote the squared residuals corresponding to a given estimator b of β as

$$u^2_{(1)}(b) \leq u^2_{(2)}(b) \leq \ldots \leq u^2_{(n)}(b). \tag{10}$$

The LWS estimator b_{LWS} is then defined as

$$b_{LWS} = \arg\min_{\beta \in \mathbb{R}^{p+1}} \sum_{i=1}^{n} w_i u^2_{(i)}(b). \tag{11}$$

We propose to use the weight function

$$\psi(t) = exp\left\{-\frac{t^2}{3/2}\right\} \mathbb{1}[t \geq h], t \in (0, 1), \tag{12}$$

where $\mathbb{1}$ denotes indicator function. Such function corresponds to a trimmed version of the Gaussian density with h corresponding to the smallest integer greater than $3n/4$. Such choice ensures a high breakdown point. The LWS estimator remains consistent for weights generated by any weight function which is continuous and nonincreasing. If additionally the errors have a common value of variance $var\, e_1 = \ldots = var\, e_n = \sigma^2$, metrological applications may need to estimate this for the task of uncertainty evaluation (Nielsen 2002); an estimate of σ^2 related to the LWS estimator was proposed in Kalina and Tichavský (2020). An extension of the LWS to nonlinear regression was studied in Kalina (2014).

The quite popular least trimmed squares (LTS) estimator can be perceived as a special case of the LWS with weights equal only to 1 or 0. Defining the LTS requires to choose the trimming constant h between $n/2$ and n so that the minimization of squared residuals is performed only across the h measurements that have the smallest

squared residual. However, the LTS estimator retains vulnerable with respect to small changes of centrally located data (Jurečková et al. 2019).

Robust Estimation in the Location Model

In the location model, robust characteristics of location, i.e., robust estimates of μ, are more suitable than the arithmetic mean if the measurements contain outliers. It is advisable to use robust approaches also for variability estimation as robust counterparts of Eq. (1); let us recall that no approaches for outlier detection are currently described in the GUM methodology. The LWS estimator defined for regression is a promising robust estimator also in the location model (Kalina 2018). It is equal to the weighted mean of the data with such permutation of the weights, which yields the smallest weighted variance

$$S_w^2 = \frac{1}{\gamma} \sum_{i=1}^{n} w_i \left(X_i - \sum\nolimits_{j=1}^{n} w_j X_j \right)^2. \tag{13}$$

When the weights after the optimal permutation are denoted as $w_1^*, \ldots, w_n^*$, the LWS estimate of μ proposed in Kalina (2018) is computed as $\overline{X}_{LWS} = \sum_{i=1}^{n} w_i^* X_i$. A consistency factor γ is needed to be used, depending on the particular ψ; the values of γ were evaluated in Kalina and Tichavský (2020) by numerical simulations for different weight functions so that Eq. (13) becomes a consistent estimator of σ^2 for normally distributed measurements, i.e., S_w^2 converges to the true σ^2 if the number of measurements increases to infinity.

If the measurements in the location model are multivariate (say p-dimensional), robust estimates of the p-dimensional paramete μ can be typically formulated as special cases of robust regression procedures. Evaluating the measurement uncertainty of such measurements must take into account the whole covariance matrix of the measurements, i.e., requires evaluating correlations for pairs of coordinates of μ (Tellinghuisen 2001; Senin et al. 2021).

Robust Regression: A Real Data Example

The aim of this example is to compare the variances of several regression estimators. An important task of evaluating measurements by linear regression is namely an evaluation of uncertainty of the obtained result; for this task, it is important to evaluate the variability of the estimates of regression parameters. We analyze the thyroid disease dataset previously used for regression modeling in Kalina and Tichavský (2020), where however the variability of regression estimates was not considered. The dataset contains measurements obtained from blood tests performed for a sample of $n = 22$ patients with hypothyroidism. The data include continuous

values $Y_1, \ldots, Y_n$ of the level of a specific drug, which is known to be correlated with the excretion level of the C-reactive protein in the body; this protein represents an important sensitive inflammatory marker and a common biomarker for the disease. Measured continuous values of this protein represent the regressor $X_1, \ldots, X_n$. The linear model (3) with $p = 1$ is meaningful, because there is a hypothesis that the amount of the protein in the body is proportional to the level of the drug in the blood.

The computations are performed in R software (R Core Team 2021), version 4.1.2 for Microsoft Windows. The following estimates are used:

- Least squares
- S-estimator (with breakdown point 0.5)
- MM-estimator (with breakdown point 0.5 and efficiency 0.95)
- LTS with a specific h ($h = 0.55n$ or $h = 0.80n$)
- LWS with the weight function (12) with the choice $h = 0.75$, i.e., with breakdown point 0.75

Because explicit formulas for variances are not available for all of these estimators, we take resort to bootstrap estimates. Bootstrapping is commonly applied to approximate the variability of estimators by means of resampling the data (sampling from the measured data with replacement). The results of various estimators of β together with nonparametric bootstrap estimates of their standard deviations are presented in Table 1. Nonparametric bootstrap was also used to evaluate $\rho(b_0, b_1)$, which denotes the correlation coefficient between the slope and the intercept. The results bring an argument in favor of the LWS estimates, which give a reasonable regression fit and their standard deviations are at the same time smaller compared to all other estimators here, including renowned robust regression estimators.

Robust Regression: Simulation

Further, we present a Monte Carlo simulation study performed (again) with the aim to compare nonparametric bootstrap estimates of variances of various regression estimators. For the linear model (3) with $p = 1$, we consider $N = 100$ randomly simulated datasets with $n = 60$ measurements each. For each dataset, the regressor is

Table 1 Results for the thyroid disease dataset: estimates of regression parameters and nonparametric bootstrap estimates of their standard deviation and correlation

Estimator	b_0	b_1	$S.D.(b_0)$	$S.D.(b_1)$	$\rho(b_0, b_1)$
Least squares	−582	0.239	108.9	0.016	−0.98
S-estimator	−542	0.206	132.1	0.023	−0.81
MM-estimator	−567	0.237	107.0	0.016	−0.98
LTS ($h = 0.55$)	−371	0.203	119.2	0.017	−0.87
LTS ($h = 0.80$)	−515	0.228	110.6	0.016	−0.87
LWS	−455	0.214	102.5	0.014	−0.85

randomly generated from uniform distribution $U(0, 10)$, i.e., over the interval $(0, 10)$. We always use $\beta = (2, 1)^T$ and $\sigma^2 = 1$. The random errors are generated from normal distribution as $e_i \sim N(0, \sigma^2)$ with $\sigma = 0.3$. We consider each of the N datasets itself, together with $R = 100$ contaminated versions. Each contamination considers $c\%$ of randomly selected measurements (using $c = 0$, $c = 10$, or $c = 30$). Then, a value denoted as m is generated from $U(0, 10)$. For each of the selected measurements, a specific value v is generated from $U(-0.5, 0.5)$ and the regressor is replaced by $m + v$; the response is replaced by a value obtained according to Eq. (3) with the error generated from $U(3, 6)$.

For each of the contaminated datasets, $S = 100$ bootstrap samples are created and evaluated. For each bootstrap sample, each of the six given estimators is computed, and the results are averaged over all the bootstrap samples. From the averaged results computed over the datasets in each situation, the covariance matrix is obtained. Table 2 presents the results; using the notation $b = (b_0, b_1)^T$ for a particular estimator under consideration, Table 2 also presents averaged values of $S.D.(b_0)$, $S.D.(b_1)$, and $\rho(b_0, b_1)$. This simulation reveals that the LWS estimates are close to true values of β and are able to outperform all other estimates (especially under a severe contamination) in terms of variability. It can be also seen in Table 2 how much is the least squares estimator biased for contaminated measurements. This is further

Table 2 Results of the simulation study: averaged estimates of regression parameters, and nonparametric bootstrap estimates of their standard deviation and correlation

Estimator	b_0	b_1	$S.D.(b_0)$	$S.D.(b_1)$	$\rho(b_0, b_1)$
	$c = 0$				
Least squares	1.983	0.9981	0.051	0.0022	−0.50
S-estimator	1.876	1.0016	0.119	0.0054	−0.72
MM-estimator	2.028	1.0012	0.070	0.0037	−0.51
LTS ($h = 0.55$)	2.050	1.0047	0.055	0.0029	−0.55
LTS ($h = 0.80$)	2.042	1.0049	0.055	0.0024	−0.58
LWS	2.034	1.0030	0.078	0.0026	−0.54
	$c = 10$				
Least squares	1.816	0.9877	0.167	0.951	−0.56
S-estimator	1.862	0.9824	0.111	0.0059	−0.75
MM-estimator	2.053	1.0093	0.079	0.0036	−0.53
LTS ($h = 0.55$)	2.190	1.0185	0.071	0.0032	−0.62
LTS ($h = 0.80$)	2.145	1.0151	0.071	0.0032	−0.68
LWS	2.074	1.0041	0.069	0.0033	−0.55
	$c = 30$				
Least squares	1.467	0.9710	0.951	0.0263	−0.69
S-estimator	1.758	0.9733	0.197	0.0252	−0.81
MM-estimator	2.206	1.0126	0.195	0.0050	−0.54
LTS ($h = 0.55$)	2.281	1.0247	0.280	0.0074	−0.71
LTS ($h = 0.80$)	2.233	1.0199	0.980	0.0091	−0.75
LWS	2.115	1.0062	0.173	0.0046	−0.57

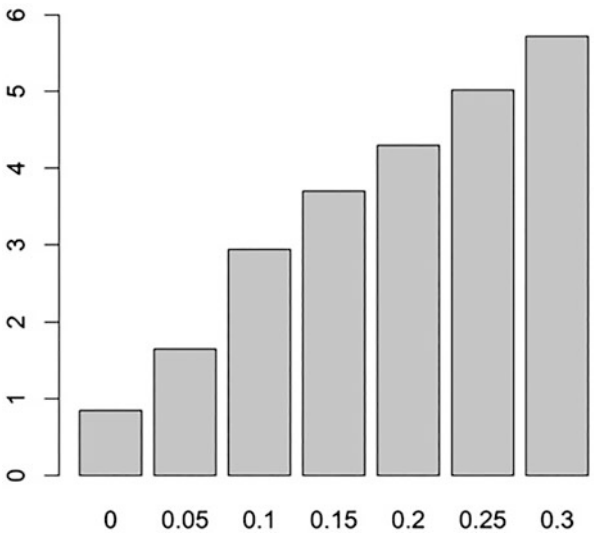

Fig. 1 Results of the simulation study: the averaged ratio $S.D.\left(b_1^{LS}\right)/S.D.\left(b_1^{LWS}\right)$ depending on the contamination level c

revealed in Fig. 1, which shows the ratio of $S.D.\left(b_1^{LS}\right)/S.D.\left(b_1^{LWS}\right)$ comparing the standard deviations of the LWS and the least squares, respectively. The figure clearly reveals the negative effect of outliers on the non-robust least squares estimator to be increasing together with an increasing contamination of the dataset.

Bayesian Estimation in the Location Model

Bayesian statistical methodology allows to naturally combine the results of given measurements with prior measurements, if the latter are available. Bayesian estimates of location or estimates of variability obtained as such combinations correspond to intuition. This section presents useful connections revealing how Bayesian estimates use prior measurements in a very natural way. It is rather surprising that the explanations presented here and the intuitive interpretation of formulas are missing in available monographs on Bayesian statistics (e.g., Ghosh 2019).

Bayesian reasoning as such represents an intuitive approach to considering conditional probabilities (Kahneman 2011). Bayesian methods are not presently formulated in the GUM methodology (Wübbeler et al. 2020), which does not provide state-of-the-art data analysis (Marschall et al. 2022). Bayesian methods are expected to be incorporated in later editions of the GUM in the next 5 years so that the GUM will finally be able to exploit the latest Bayesian statistical theoretical framework allowing to fully integrate prior measurements with actual measurements. Such combination of information has been denoted as the Bayesian information fusion (Cheng et al. 2018). Metrologists have always debated, whether the

frequentist or the Bayesian approach should be preferable for evaluating uncertainty. Arguments in favor of the Bayesian approach were summarized already, e.g., in Cox et al. (2004). While the Bayesian approach seems to be prevailing in current metrology (Ferrero et al. 2019), Bayesian approaches to uncertainty evaluation (Bayesian metrology approaches) are still under development for some specific measurement models, such as for mass spectrometry measurements in metabolomics (Banks et al. 2021).

In the following paragraphs, the Bayesian estimator is always obtained as the expectation (population mean) of the posterior distribution; such estimator is commonly denoted as MAP (maximum a posteriori) estimator. We focus on the situations when previous R measurements (for $R \geq 1$) are performed under the same conditions as the current (new) data. We also explain whether Bayesian estimation always requires to understand (e.g., physical) constants to be random variables. A deeper philosophical justification of Bayesian reasoning can be found in Jaynes (2003). This section is focused on simple fundamental problems and thus omits many advanced problems, such as a suitable choice of prior distribution, construction of Bayesian confidence areas, Bayesian approaches to linear regression (Saleh et al. 2022), or empirical Bayes estimation; technical proofs are also omitted.

Bayesian Estimation: Mean of One-Dimensional Gaussian Distribution

Let us consider independent measurements $X_1, \ldots, X_n$, where X_i is assumed to come from the normal distribution $X_i \sim N(\mu, \sigma^2)$ for $i = 1, \ldots, n$. These can be measurements of some physical constant (e.g., the speed of light), which is thus denoted here as μ. In some other situations, the normality of values X_i can be quite natural, if each individual (indirect) measurement X_i is obtained as the mean of several independent measurements and if the data do not contain outliers.

At any case, we use the notation $\overline{X}$ for the mean of the measurements, for which it naturally holds $var\,\overline{X} = \sigma^2/n$. This expression can also be justified from the Bayesian point of view under the assumption of a noninformative prior distribution.

Independently on the measurements, we assume that previous measurements were performed and the measured values are known. Estimates of μ obtained from these prior measurements will be denoted as $t_1, \ldots, t_R$. Let us assume that each of them was acquired as a realization of a normally distributed random variable. We denote $\overline{t} = \sum_{r=1}^{R} t_r/R$.

Let us now proceed very slowly. The knowledge of $t_1, \ldots, t_R$ reduces the uncertainty about the unknown value of μ. We expect the value of $\overline{t}$ to be close to the true (unknown) value of μ, because we consider it to be unlikely that the true value of μ is much different from the experimentally determined mean value $\overline{t}$. We do not know the reason for such difference (deviation), and therefore we consider it to be random. We may believe that if $\overline{t}$ exceeds μ by a certain constant, this is equally likely as if $\overline{t}$ remains below μ by the same constant. Considering the concepts of

randomness, likelihood, or probability means that the uncertainty about the value of μ may be expressed by the concept of a random variable. This does not state that μ is objectively a random variable, which would be a mere self-deception. Thanks to the same probability of deviating on both sides, we may consider the distribution corresponding to the uncertainty about the value of μ to be symmetric. Here, we consider it to be normally distributed according to $\mu \sim N(\bar{t}, \gamma^2)$, where $\gamma^2 = var\,\mu$ is obtained by means of the formula for the variance of the mean $\gamma^2 = \xi^2/R$. Here, the parameter ξ evaluates the accuracy (standard deviation) of the measurements $t_1, \ldots, t_R$.

In general, it is possible to interpret the Bayesian estimation by means of two ways, which have an equal mathematical expression:

1) The prior distribution corresponds to the uncertainty about the estimated parameter. This (in our opinion more realistic) approach was also considered by Laplace; as a propagator of Bayesian reasoning, he did not consider physical constants to be random but perceived the world and universe in a deterministic manner. In general, the uncertainty can be perceived either objectively from the point of view of information theory or objectively as a degree of belief (Abdellaoui et al. 2021).
2) The estimated parameter is a random variable, while the measurements $X_1, \ldots, X_n$ inform us about its distribution.

Both points of view lead to the conclusion that the mean of the posterior distribution, i.e., the Bayesian estimate of the parameter μ,has the form

$$\widehat{\mu} = \frac{\sigma^{-2}\sum_{i=1}^{n} X_i + \gamma^{-2}\bar{t}}{n\sigma^{-2} + \gamma^{-2}}. \tag{14}$$

We may easily express Eq. (14) as

$$\widehat{\mu} = (1-\delta)\overline{X} + \delta\bar{t}, \text{where } \delta = \frac{\gamma^{-2}}{n\sigma^{-2} + \gamma^{-2}}. \tag{15}$$

Let us now consider a special situation, when each of the measurements $t_1, \ldots, t_R$ (just like the value of $\overline{X}$ computed from these measurements) is obtained from the total number of n measurements with variance σ^2. Because it holds $var\, t_r = \sigma^2/n$ for $r = 1, \ldots, R$, we obtain $var\,\bar{t} = \sigma^2/(nR)$. The literature typically states only briefly that γ may be estimated from previous measurements; nevertheless, it is natural here to make the particular choice for $var\,\mu$ to obtain $\gamma^2 = \sigma^2/(nR)$. Thus, we obtain

$$\delta = \frac{R}{R+1} \text{ and } \widehat{\mu} = \frac{1}{R+1}\overline{X} + \frac{R}{R+1}\bar{t}. \tag{16}$$

The resulting estimate can be interpreted as the mean of all available measurements obtained by incorporating the prior (previous) measurements to the set of the

new measurements that were acquired later. It is an intuitively expected combination of previous and current measurements. In practice, however, it is common to have new measurements with a higher accuracy compared to the previous measurements. But also in the situation with a fixed ratio $\sigma^2/(n\gamma^2)$,i.e., when there were no prior measurements $t_1, \ldots, t_R$, a result analogous to Eq. (16) is valid.

Bayesian Estimation: Mean of Multivariate Gaussian Distribution

For multivariate location model under the assumption of multivariate normality, let us explain how the Bayesian estimator (again) allows to incorporate prior measurements in a natural way. Let us consider independent p-dimensional measurements $X_1, \ldots, X_n$, where X_i is explained by the multivariate normal distribution $X_i \sim N_p(\mu, \boldsymbol{\Sigma})$ for $i = 1, \ldots, n$. Let us proceed in the same way as above but in a much concise manner. We assume estimates of the parameter μ to be available from prior measurements; these estimates are denoted as $t_1, \ldots, t_R$. We denote the arithmetic means as

$$\overline{X} = \frac{1}{n}\sum_{i=1}^{n} X_i \text{ and } \bar{t} = \frac{1}{R}\sum_{r=1}^{R} t_r. \tag{17}$$

If the prior knowledge about the parameter μ is modeled by means of the p-variate normal distribution $\mu \sim N_p(\bar{t}, \boldsymbol{\eta})$, the expectation of the posterior distribution of μ may be expressed as

$$\widehat{\mu} = \left(n\boldsymbol{\Sigma}^{-1} + \boldsymbol{\eta}^{-1}\right)^{-1}\left(n\boldsymbol{\Sigma}^{-1}\overline{X} + \boldsymbol{\eta}^{-1}\bar{t}\right). \tag{18}$$

This result was proven in Evans (1965), who also derived a more complex result for data with an unknown $\boldsymbol{\Sigma}$. If some previous measurements are available and it is possible to estimate $\boldsymbol{\Sigma}$ from them, it may be replaced by the empirical covariance matrix, which gives a consistent estimate (thanks to the assumed normality).

Let us assume now a specific situation with a known $\boldsymbol{\Sigma}$ and let us assume $\boldsymbol{\Sigma} = \sigma^2\boldsymbol{I}$, where $\boldsymbol{I}$ denotes a unit matrix. We assume the accuracy of all coordinates of the vector μ to be the same and the prior knowledge of the parameter μ is modeled by a normal distribution

$$\mu \sim N_p\left(\bar{t}, diag\left(\gamma^2, \ldots, \gamma^2\right)\right). \tag{19}$$

The expectation (Eq. 18) of the posterior distribution of μ then corresponds to

$$\widehat{\mu} = \frac{n\gamma^2\overline{X} + \sigma^2\bar{t}}{n\gamma^2 + \sigma^2}, \tag{20}$$

which may be expressed as

$$\widehat{\mu} = (1-\delta)\overline{X} + \delta\bar{t}, \text{ where } \delta = \frac{\gamma^{-2}}{n\sigma^{-2}+\gamma^{-2}}. \quad (21)$$

More specifically, let us assume each of the vectors $t_1, \ldots, t_R$ obtained from the previous measurements to be obtained from the total number of n measurements with variance σ^2. It holds $var\,\overline{X} = \sigma^2 \boldsymbol{I}/n$, and it is natural to choose $\gamma^2 = \sigma^2/(nR)$. Then, we obtain Eq. (16). This multivariate estimator can be interpreted as a so-called shrinkage estimator of μ, i.e., an estimator which is shrunken towards the value $\bar{t}$.

Bayesian Estimation: Covariance Matrix of Multivariate Gaussian Distribution

We consider to have independent p-variate measurements $X_1, \ldots, X_n$ explained by the normal distribution $N_p(\mu, \boldsymbol{\Sigma})$ with a known μ and an unknown covariance matrix $\boldsymbol{\Sigma} \in PD(p)$, where $PD(p)$ denotes the set of all positively definite matrices of size pxp. It is natural to estimate $\boldsymbol{\Sigma}$ by means of the matrix $\boldsymbol{U}/n$,where

$$\boldsymbol{U} = \sum_{i=1}^{n} (X_i - \mu)(X_i - \mu)^T \quad (22)$$

exploits the known vector μ. In such a situation, the matrix $\boldsymbol{U}$ follows Wishart distribution $W_p(\boldsymbol{\Sigma}, n)$ and therefore we have $E(\boldsymbol{U}/n) = \boldsymbol{\Sigma}$. It is usual to model the uncertainty about $\boldsymbol{\Sigma}$ (in fact about $\boldsymbol{\Sigma}^{-1}$) by means of Wishart distribution $\boldsymbol{\Sigma}^{-1} \sim W_p(\nu + p - 1, \boldsymbol{\Omega}^{-1})$ for a certain $\nu > 0$ and for a certain $\boldsymbol{\Omega} \in PD(p)$. In such situation, $\boldsymbol{\Sigma}$ follows the inverse Wishart distribution $W_p^{-1}(\nu + p - 1, \boldsymbol{\Omega})$. It holds

$$E\boldsymbol{\Sigma}^{-1} = (\nu + p - 1)\boldsymbol{\Omega}^{-1} \text{ and } E\boldsymbol{\Sigma} = \boldsymbol{\Omega}/(\nu - 2). \quad (23)$$

The posterior expectation of $\boldsymbol{\Sigma}$ is then equal to

$$\widehat{\boldsymbol{\Sigma}} = \frac{\boldsymbol{U} + \boldsymbol{\Sigma}}{n + \nu - 2}. \quad (24)$$

Let us assume to have prior estimates of $\boldsymbol{\Sigma}$ denoted as $\boldsymbol{U}_1/n, \ldots, \boldsymbol{U}_R/n$, which are obtained from some previous measurements. In other words, the matrices $\boldsymbol{U}_1, \ldots, \boldsymbol{U}_R$ represent empirical counterparts of $\boldsymbol{U}$. Let us further assume that each of the previous experiments was performed across n measurements, and that each had the same accuracy (variability) as the current measurements. It is natural to choose $\boldsymbol{\Omega}$ and ν so that

$$\frac{1}{R}\sum_{r=1}^{R}\frac{\boldsymbol{U}_r}{n}=\frac{\boldsymbol{\Omega}}{\nu-2}, \tag{25}$$

i.e., to take $\boldsymbol{\Omega}=\sum_{r=1}^{R}\boldsymbol{U}_r$ and $\nu=Rn+2$. The estimate (Eq. 24) is then obtained as the intuitive combination of the measurements with results of the previous measurements in the form

$$\widehat{\boldsymbol{\Sigma}}=\frac{1}{n(R+1)}\left(\boldsymbol{U}+\sum_{r=1}^{R}\boldsymbol{U}_r\right). \tag{26}$$

Conclusion

This chapter describes and discusses selected nonstandard statistical techniques for analyzing measurements, or in fact selected classes of advanced available statistical tools. While each of the topics covered here deserves much more space, the chapter is focused on principles while keeping aside a vast number of other available methods potentially applicable to metrology. Current metrology intensively makes use of various forms of the measurement error models, and the Bayesian approach represents the prevailing approach nowadays, especially if prior knowledge related to the variability of the measurement errors is available. Robust statistical methods, which have established their place within mathematical statistics, have a potential for being used in metrology much more often than nowadays.

At any case, the authors recommend that an analysis of measurements is always accompanied by verifying that the available data are really raw; it is namely known that no analysis (not even by robust methods) can perform a reliable analysis of data after an inappropriate pre-processing (cf. Kalina 2018). In addition to the topics covered in this chapter, metrologists can nowadays experience an advent of data with a high dimensionality, which become available thanks to the increasing level of digitalization, availability of sensors in various fields, or automation of various processes of human activities (Chap. 14 of Hebra 2010). As we cannot expect the availability of various low cost measurement equipment to be decreasing, there will also remain an intensive need for robust statistical methods for metrological applications across diverse fields.

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The Quantifying of Uncertainty in Measurement

91

Ahmed A. Hawam

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Abstract

Despite the importance of measurement uncertainty, it is widely misunderstood, and it represents a vague subject for both industrial and scientific practitioners. In this chapter, it is aimed to introduce the subject of measurement uncertainty step-by-step starting from the basic principles to the actual uncertainty evaluation techniques. The text does not only present the theoretical aspects of uncertainty, but it also deals with its practical implementation. The chapter begins by introducing the concept of uncertainty and its importance, and then it discusses Type A

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and Type B uncertainty evaluation methods. After that, the GUM uncertainty framework is presented as well as the Monte Carlo simulation technique and its programming details. Throughout the chapter, several examples are demonstrated to put the discussed concepts into action. The chapter is very useful for anyone whether an industrial practitioner, scientist, or student.

Keywords

Measurement uncertainty · Measurement modelling · GUM · Monte Carlo

Introduction

Every measurement process must have some sorts of imperfections that reduce the trustworthiness of the measurement's result. In other words, there is always some doubt about any measurement result whatever the precautions taken to obtain that result. Depending on the application in which the measurement result will be used, different levels of doubt may be acceptable or not. To deal with that doubt, it is necessary to quantify it in terms of the unit of the measurand or as a percentage of the measurement result. This numerical quantification of the doubt about the measurement's result is what is called the measurement *uncertainty*. The uncertainty is a key term in reflecting the quality of the measurement process and the trustworthiness of its result and the decision taken upon it.

The uncertainty in measurement can be simply defined as the "quantified doubt about the result of a measurement" (Bell 2001). To clarify the concept of uncertainty, consider a measuring process aims to estimate the length of a steel bar using a classical micrometer, the length was measured five times, and the results were as follows: 50.00 mm, 49.98 mm, 49.98 mm, 50.03 mm, and 50.00 mm. Why different values are obtained? And where is the true value of the length? In fact, the true value may lie in between these values, i.e., in the interval [49.98 mm, 50.03 mm], but due to the *repeatability error* that arises from some sorts of imperfections in the measuring process, e.g., the ambient temperature and humidity, stiffness of the measured object, etc., the measured values spread around the true value resulting in some doubt (uncertainty) about the final result of the measurement (i.e., the estimated true value) and reducing its trustworthiness.

Practically, several sources of uncertainty – beside repeatability – depending on the nature of the measuring process can be present resulting in more dispersion around the measurement result thereby more extension in the interval in which the true value is expected to lie, which means more increase in the uncertainty. From a mathematical point of view, the uncertainty is a value that quantifies the dispersion of the values that have probabilities to be the true value around the measurement result. Accordingly another definition for the uncertainty was introduced by the *International Vocabulary of Metrology* (VIM) (BIPM 2012) as "non-negative parameter characterizing the dispersion of the quantity values being attributed to a measurand, based on the information used."

It should be clear that the uncertainty is different from *error*. The error is the deviation between the measurement result and the true value of the measurand. All known errors should be corrected while the errors with unknown values are considered sources of uncertainty (Bell 2001) which are uncorrectable.

The General Idea of Quantifying Uncertainty

Generally, the quantification of measurement uncertainty begins by identifying the uncertainty sources that can significantly affect the trustworthiness of the measurement result. Each uncertainty source is quantified by determining its expected range of doubt and then, that range is assigned a reasonable probability distribution. After that, various uncertainty sources are combined using an appropriate technique to obtain the total range of doubt and its probability distribution which is known as the output distribution.

Depending on the output distribution, the mean value is estimated and considered as the result of the measurement process. Also, the standard deviation of the output distribution is calculated and considered as the combined standard uncertainty of the measurement result. Here, the term "standard" means that the estimated combined uncertainty is obtained using the individual standard uncertainties of the uncertainty sources, i.e., the individual standard deviations of their respective distributions. As a common case in metrology, the output distribution is considered as normal (Gaussian). In that case, the combined standard uncertainty is $\pm$ *one standard deviation* around the mean value (Fig. 1).

Is the true value really lies within the uncertainty interval? This question is answered by determining the *level of confidence* provided by that uncertainty interval. Mathematically, the level of confidence is the probability/area under the output distribution over the uncertainty interval; this value tells us on what probability the true value lies within that uncertainty interval. In practice, the level of confidence provided by the combined standard uncertainty may not be satisfactory for some industrial and commercial applications (BIPM 2008a). As a result, the considered uncertainty interval is extended to encompass a larger fraction of the output distribution, this produces what is called *expanded uncertainty*. In fact, the expanded uncertainty increases the level of confidence in trade off increasing the range of doubt in which the true value is expected to lie. For a normal output distribution (Fig. 1), the combined standard uncertainty provides about 68% level of confidence, while if that range of uncertainty is doubled, the level of confidence reaches approximately 95%.

The reporting of the measurement result should be along with the uncertainty value and the level of confidence. For example, the result of measuring the length of a copper bar can be reported as follows:

69.88 mm $\pm$ 0.03 mm at a level of confidence 95%.

This means that we have 0.95 probability that the true length of the copper bar falls between 69.85 mm and 69.91 mm.

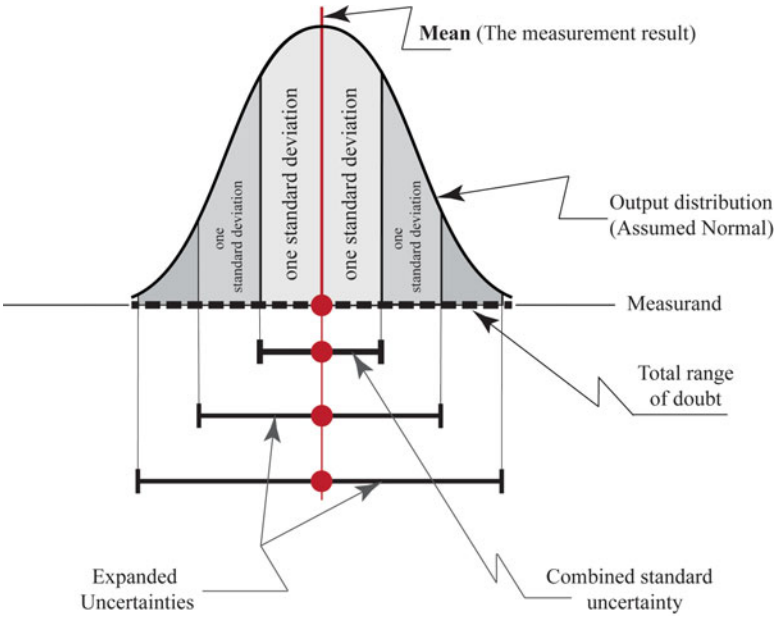

Fig. 1 The normal output distribution obtained from combining several uncertainties

Considering Uncertainty in Determining Conformance to Specifications

One of the major applications of measurement uncertainty is determining the conformance of products to their specifications. Consider a certain feature of a product that has a lower tolerance limit T_L and an upper tolerance limit T_U. The feature is measured using an appropriate measurement process that has an expanded uncertainty U. Unless the tolerance width $T_U - T_L$ is greater than the uncertainty interval 2 U, the result of the measuring process would be inconclusive in determining the conformance of the feature to its specified tolerance even if the measurement result falls within that tolerance. A common parameter that used to quantify the capability of the measuring process to provide conclusive results with respect to the tolerance width is the *measurement capability index* C_m where C_m is the ratio of the tolerance width to the uncertainty interval, $C_m = (T_U - T_L)/2U$ (ASME B89.7.4.1 2005).

Although C_m can have a relatively large value, the measurement result would still be inconclusive in case the uncertainty interval intersects with a tolerance limit. To clarify that, see Fig. 2, in that figure, the uncertainty interval in cases (c), (d), and (e) entirely lies inside the tolerance zone; this leads to the conclusion that the measured feature is acceptable. In cases (a) and (g), the uncertainty interval falls entirely outside the tolerance zone which leads to the conclusion that the measured

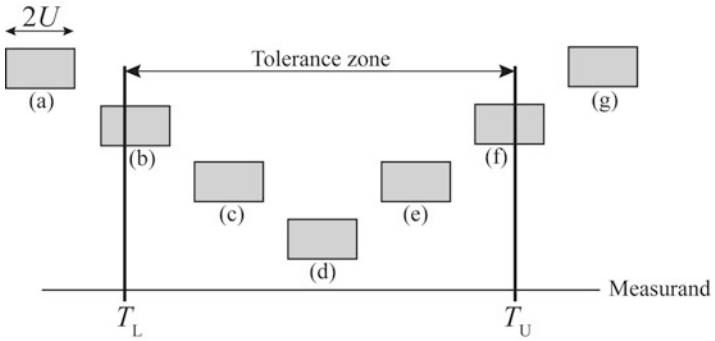

Fig. 2 The relative positions between the uncertainty interval and the specified tolerance

feature is unacceptable. In cases (b) and (f), the uncertainty interval intersects with a tolerance limit; this means that the measured feature may be acceptable or not at certain probabilities; these are inconclusive situations.

In practice, accepting or rejecting a certain feature is decisioned upon a *decision rule*; these decision rules consider the measurement uncertainty in taking accept/reject decisions even if the uncertainty interval is near the tolerance limits or intersect with them. The decision rules contain some risks of accepting a nonconforming product or vice versa; it is a business issue and subject to cost analysis. For detailed information about decision rules, how uncertainty is considered in determining the conformance to specifications, and risk analysis, refer to references (ASME B89.7.4.1 2005; ASME B89.7.3.1 2001; ISO 2017; Ribeiro et al. 2017).

Toward quantifying the uncertainty of a measurement result, the first step is identifying the individual uncertainty sources that expected to have significant contributions to the total uncertainty. In practice, the standard guidelines devoted to specific metrological tasks issued by standardization bodies are rich with useful information in determining the uncertainty sources that deserve consideration. Moreover, the experience with the measurement process and the special conditions of the laboratory should be paid attention in identifying those sources. Once, the uncertainty sources are identified, their contributions are quantified according to *Type A* method or *Type B* method depending on the nature of the available information about the uncertainty source. Through section "Type A Evaluation Method" and "Type B Evaluation Method," both quantification methods are discussed as well as the quantification of several examples of uncertainty sources that are common in metrological practices.

Type A Evaluation Method

Type A evaluation method is the statistical evaluation technique that is used to quantify uncertainty sources only when the available data about the source is in the form of repeated observations under the same conditions. In that method, a measured

quantity q is considered as a random variable and its available n observations q_1, q_2, …, q_n are independently captured from its respective distribution under the same conditions of measurement (BIPM 2008a; M3003 2012). Accordingly, the expectation μ_q of that distribution is the best representative of the true value of the quantity being measured, i.e., it represents the measurement result. Nevertheless, the expectation of that distribution needs a very large number of observations to be calculated accurately; this is not common in metrological practices in which only few observations are available. So, the arithmetic mean $\overline{q}$ of the available observations is taken as the best available estimate of the expectation μ_q of the distribution and is calculated as:

$$\overline{q} = \frac{1}{n}\sum_{i=1}^{n} q_i \tag{1}$$

As we ultimately seek for the value of μ_q, it is required to quantify how well the mean value available in hand $\overline{q}$ estimates μ_q; this can be achieved by calculating the *experimental standard deviation of the mean* $s(\overline{q})$ as follows:

$$s(\overline{q}) = \frac{s(q_i)}{\sqrt{n}} \tag{2}$$

where $s(q_i)$ is the *experimental standard deviation* which characterizes the dispersion of the observations about their mean $\overline{q}$ and is calculated as:

$$s(q_i) = \sqrt{\frac{1}{n-1}\sum_{i=1}^{n}(q_i - \overline{q})^2} \tag{3}$$

Eventually, the mean value $\overline{q}$ is considered as the best estimate of q while the associated standard uncertainty $u(q)$ is taken equal to $s(\overline{q})$.

Type B Evaluation Method

Type B evaluation method is used for quantifying uncertainty sources whenever the available knowledge about the source is not in the form of repeated observations (BIPM 2008a). In Type B, scientific judgment is the key in determining the expected range of doubt and the assigned probability distribution. In practice, for an uncertainty source quantified according to Type B method, reasonable judgments must be issued by experts with respect to:

(a) The possible range of the doubt.
(b) An estimation for the likelihood of different values in the range of doubt, this may be achieved by answering some questions, e.g., What are the points that have more probability than others? What is the rate in which the probability is

decreased as the value gets farther from the most probable values? (ASME B89.7.3.2 2007), and so forth.

These judgments should be based on all the available knowledge about the uncertainty source; this knowledge may be found in a variety of sources such as but not limited to (a) previous measurements, (b) calibration certificates, (c) manufacturer's specifications, (d) engineering judgments, (e) handbook values, etc. (BIPM 2008a; ASME B89.7.3.2 2007; Boumans 2013).

In practice, the available knowledge about uncertainty sources is commonly limited to typical levels such as: (Kadis 2000).

Level 1: The limits of the possible values are only the knowns.
Level 2: A more likely mean value and the limits of permissible values which are usually symmetric about the mean.
Level 3: An interval that encompasses a certain probability percentage of the probability distribution, e.g., 95% or 99%.

For each level from the previous list, there is a certain probability distribution that best fit with the available level of knowledge. In the next sections, the characteristics of the most utilized probability distributions will be discussed.

The Uniform Distribution (Best Fit with Level 1)

The uniform distribution of a quantity q is fully defined by specifying its lower limit a and upper limit b; within that interval, the values are assigned equal probabilities (Fig. 3). The expected value $\overline{q}$ of the quantity q is calculated as:

$$\overline{q} = \frac{a+b}{2} \tag{4}$$

The standard uncertainty of the quantity q is taken equal to the standard deviation of q and calculated according to the characteristics of the Uniform distribution as follows:

$$u(q) = s(q) = \frac{b-a}{2\sqrt{3}} \tag{5}$$

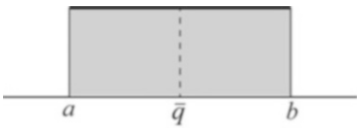

Fig. 3 The uniform distribution of a quantity q

A classical example of uncertainty sources that attributed to the uniform distribution is the resolution uncertainty of digital indications. Consider a digital instrument that indicates an indication X on its display; what is the actual value of the measurand that results in that indication? The prevailing point of view is that if the measurand value is within the limits $X - r/2$ and $X + r/2$ inclusive (r is the resolution of the digital display), the digital instrument would round it to X on its display (Hawam et al. 2018). Accordingly, and considering that there is no more knowledge available about the location of the measurand value, the uniform distribution is applied over the uncertainty interval $[X - r/2, \quad X + r/2]$. Therefore, the expected value of the measurand is X and the resolution standard uncertainty is $r/2\sqrt{3}$.

The Isosceles Triangular Distribution (Best Fit with Level 2)

The isosceles triangular distribution of a quantity q is fully defined by specifying its lower limit a and upper limit b while the mean value $\overline{q}$ falls exactly midway between a and b. Within that interval [a, b], the probability is increased gradually from *zero* at the lower limit a until it reaches its maximum at $\overline{q}$ and then it decreases gradually until it reaches *zero* again at the upper limit b (Fig. 4). The expected value $\overline{q}$ of the quantity q is calculated as:

$$\overline{q} = \frac{a+b}{2} \tag{6}$$

The standard uncertainty of the quantity q is taken equal to the standard deviation of q and calculated according to the characteristics of the isosceles triangular distribution as follows:

$$u(q) = s(q) = \frac{b-a}{2\sqrt{6}} \tag{7}$$

The resolution uncertainty of analogue indications may be attributed to the isosceles triangular distribution in case the pointer is in conjunction with a graduation mark on the analogue display. In that case, it is reasonable to assign the highest probability to the graduation value and decrease it gradually until it reaches *zero* at the limits of the uncertainty interval. For example, if the width of the resolution

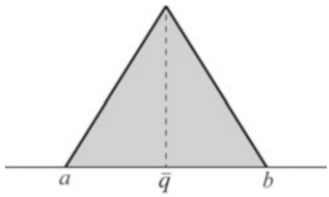

Fig. 4 The isosceles triangular distribution of a quantity q

uncertainty interval is r (r is the resolution of the analogue display) and the pointer is in conjunction with a certain graduation with value X, the expected value of the measurand would be X while the resolution standard uncertainty would be $r/2\sqrt{6}$. For detailed information about analogue indications and how its uncertainty is quantified, refer to Ref. (Hawam et al. 2021).

Note: For more information about the uniform, triangular, and other similar distributions, refer to Ref. (Raghu and James 2007).

It is noteworthy that the measuring devices, either analogue or digital, with limited resolution (i.e., with large r value) may result in identical observations. Accordingly, Type A uncertainty calculated as per Eq. 2 would equal *zero*. This does not mean that the measuring process is ideal; on the contrary, this indicates that the measuring instrument does not have the enough resolution necessary to detect the scattering in the observations. In such situation, the resolution uncertainty should be paid more attention as the resolution becomes the only source of knowledge about the possible repeatability error that can be encountered in the observations. For a mathematical analysis and an example that demonstrate this issue and how the resolution can affect the results, refer to Ref. (Ignacio and Wolfgang 1997).

The Normal Distribution (Best Fit with Level 3)

The normal distribution of a quantity q is fully defined by specifying its mean value $\overline{q}$ and standard deviation σ_q. The mean value $\overline{q}$ is assigned the highest probability and its neighborhood values are assigned higher probabilities than assigned to other far values.

Based on the level of knowledge provided by level 3, the center of the known interval $[a, b]$ can be considered as the expectation $\overline{q}$ of the normal distribution, i.e., $\overline{q} = (a + b)/2$; see Fig. 5. For obtaining the standard deviation σ_q, the equivalent z score (from the standard normal distribution) of the interval limit, say b, is extracted from the standard normal table (For ease of application, computer software packages such as Microsoft Excel can be used instead of statistical tables.) as the z score z_b corresponding to the probability encompassed by $[a, b]$ divided by *two* (assuming that the table provides the probability values starting from $z = 0$). Then, the standard deviation σ_q can be calculated as follows:

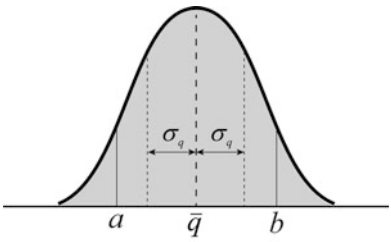

Fig. 5 The normal distribution of a quantity q

$$\sigma_q = \frac{b - \overline{q}}{z_b} \tag{8}$$

The standard uncertainty of the quantity q is taken equal to the calculated standard deviation, i.e., $u(q) = \sigma_q$.

Practically, the standard deviation of normal distribution can be calculated according to a simple rule if the intervals that encompass certain probabilities are known. According to this rule, if an interval $[a, b]$ centered about $\overline{q}$ and encompasses 0.68 probability, its length approximately equals $2\sigma_q$, i.e., $b - a \approx 2\sigma_q$; for 0.95 probability, $b - a \approx 4\sigma_q$; for 0.99 probability, $b - a \approx 6\sigma_q$.

One of the uncertainty sources that commonly attributed to the normal distribution is the calibration uncertainty. In calibration laboratories, the performance of the under-test instrument is evaluated, quantified, and combined to the *Calibration and Measurement Capability* (CMC) of the laboratory; the resultant is the calibration uncertainty of the under-test instrument and its usually reported at 95% level of confidence (ILAC 2013) assuming normal distribution.

Example 1 Consider that the length of a steel bar is measured using a Vernier caliper that has a calibration uncertainty 0.2% of the measured length at a confidence level 95% assuming a normal distribution. For calculating the calibration uncertainty, the measurement length is estimated at first, say 59.25 mm. Then the calibration uncertainty at 95% level of confidence (0.95 probability) is calculated as (0.2/100) × 59.25 mm = 0.1185 mm; this is the half width of the uncertainty interval that centered around the measurement result. For obtaining the calibration *standard* uncertainty, this half width is divided by *two* which results in 0.1185 mm /2 = 0.05925 mm; this means that the limits of the calibration standard uncertainty are 59.19075 mm and 59.30925 mm; this interval provides approximately 68% level of confidence, i.e., 0.68 probability under the normal distribution.

In practice, all uncertainty values attributed to different uncertainty sources are converted to standard uncertainties, i.e., the values of standard deviations of their distributions. This is a prior step before combining different standard uncertainties in order to obtain the combined standard uncertainty.

As obvious, the final objective of any measurement process is estimating the measurement result and its associated uncertainty. For achieving this objective, the uncertainty sources are identified and quantified as illustrated in the previous two sections, and then their individual estimates and associated uncertainties are combined depending on a relationship mathematical function that relates the measurand to its uncertainty sources; this function is what is called the *measurement model*. Mathematically, there are different combining techniques that may be used to estimate the desired results; however, the *GUM uncertainty framework* is the most used one in practice. Through sections "The Measurement Model" and "The GUM Uncertainty Framework," the definition of the measurement model and its

construction steps are discussed, the GUM uncertainty framework is illustrated, and simple solved examples are presented.

The Measurement Model

Usually, the measurement system does not determine the measurand value directly but depending on a number of other *input* quantities (The "input quantity" is that quantity which its value must be obtained in order to calculate the measurement result.) that affect the measurement result. The mathematical formula that relates the measurand with its input quantities is what is called the "measurement model." The measurement model is the essential relationship upon which the measurement result and its associated uncertainty are evaluated. Theoretically, all quantities that affect the measurement system can be included in the model for obtaining the maximum accuracy in evaluating the measurement result and its associated uncertainty. However, in practice, including each effect even trivial ones may result in a very complex mathematical model which probably would require an advanced uncertainty estimation technique in order to obtain accurate results. On the contrary, much simplifying the model by ignoring significant effects would lead to losing accuracy. As a rule, the degree of model details and complexity should meet the accuracy requirements imposed by the application in which the measurement result is employed.

In fact, modelling is not an easy task as it requires deep understanding of the input quantities, how they interact, and how they affect the measurement result. Therefore, it is not sufficient to know that a certain input quantity is significant to include it in the measurement model, but you should know how this quantity affects the measured value; this question is crucial for the mathematical formulation of its effect in the measurement model. In practice, there are two modelling routes namely "bottom-up" and "top-down" approaches (EUROLAB 2006). In the bottom-up approach, the model is constructed mathematically depending on the input quantities and the knowledge about their interaction and effects on the measured value. In the top-down approach, the uncertainty is evaluated depending on an empirical model which is constructed based on the performance data of the measurement system, e.g., the results of inter-laboratory and within-laboratory comparisons, quality control data, etc. The top-down approach can be considered as an indirect solution for modelling in case the bottom-up model is incomplete, or its input quantities have unknown effects on the measured value. In these cases, the top-down approach is feasible because the effects of its quantities are assessed experimentally without the need of a prescribed mathematical relation. Moreover, as the top-down approach depends on the whole performance of the measurement system, it can include the effects of minor or even unknown quantities that would be neglected in bottom-up modelling. For more information about how the uncertainty can be estimated depending on inter-laboratory or within-laboratory comparisons results, refer to Ref. (EUROLAB 2006).

The *guide to the expression of uncertainty in measurement* (GUM) (BIPM 2008a), the master uncertainty guide in metrology, adopt the bottom-up modelling

approach as the first choice in building the measurement model. The GUM assumes that the metrologist can define the measurand, is aware of the significant input quantities, and has the necessary knowledge about these quantities for formulating the measurement model. In case of experimental results reveal that the results of the measurement model lack accuracy, the GUM suggests including additional quantities in the model to eliminate the inadequacy.

Until soon, there was not a standard set of modelling guidelines; however, there was an attempt by *Sommer K D* et al. to systemize the modelling process. This attempt was presented in a series of publications (Sommer et al. 2003; Sommer et al. 2005; Sommer and Siebert 2006) that suggest a five-step modelling procedure as follows:

1. Describing the measurement system, i.e., the measurand, the input quantities, and the method of measurement.
2. Analyzing the measurement system and graphically constructing the cause-and-effect relationship considering the ideal measuring scenario.
3. Identifying all potential imperfections that lead to deviations from the ideal response of the measurement system. Then reconstructing the cause-and-effect diagram by incorporating the identified imperfections to represent the practical measurement system. Finally, the practical cause-and-effect is converted into a mathematical formula.
4. Identifying any potential correlations between the input quantities and including them in the mathematical formula.
5. Inversing the practical cause-and-effect mathematical formula (from the previous step) for obtaining the model equation that relates explicitly the measurand with the input quantities and imperfections.

Example 2 For demonstrating the practical implementation of Sommer's modelling procedure, consider the case of calibrating an industrial-class force measuring system (an industrial-class load cell connected to an industrial-class indicator) depending on a reference-class force measuring system (a reference-class load cell connected to a reference-class indicator) using a force comparator standard machine. Both load cells can measure only normal compressive loads while their designs resist side and bending loads. In this metrological exercise, both the under-calibration and reference load cells are aligned vertically so that they become have a common vertical axis in which the calibration force can be applied; see Fig. 6. Thereafter, the calibration force F_C is applied through the common vertical axis, and both readings indicated by the reference-class indicator (connected to the reference load cell) and the industrial-class indicator (connected to the under-calibration load cell) are recorded.

The application of Sommer's modelling procedure for modelling this calibration exercise and calculating the error induced by the under-calibration force measuring system can be as follows:

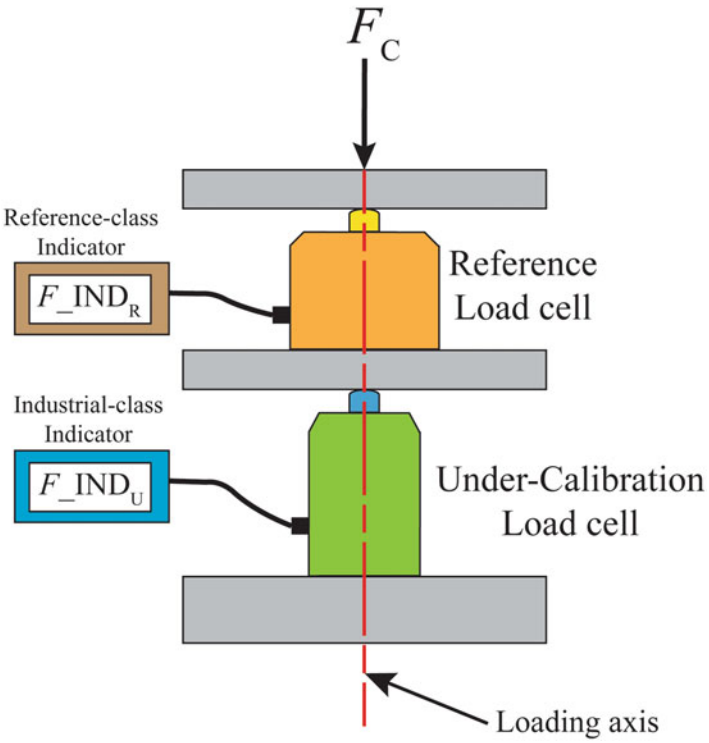

Fig. 6 The calibration of a load cell against a reference load cell using a standard force comparator machine

Step 1: Describing the Measurement System An industrial-class force measuring system is to be calibrated against a reference-class force measuring system using a force comparator standard machine. The under-calibration force measuring system consists of an industrial-class load cell connected to an industrial-class indicator. At the same time, the reference-class force measuring system consists of a reference-class load-cell connected to a reference-class indicator.

Measurand: Error ΔF_U in the indication value of the under-calibration force measuring system.
Measurement method: Direct comparison of the under-calibration force measuring system against the reference one.

Step 2: Graphical Construction of the Ideal Cause-and-Effect Block Relationship Figure 7 shows the block relationship of the calibration exercise considering the ideal scenario. As shown, the calibration force F_C generated by the standard machine acts simultaneously on both reference and under-calibration load cells. As a response, the reference measuring system indicates F_IND_R on its digital display; ideally the reference system is considered as error-free so F_IND_R should

equal F_C. The under-calibration system indicates F_IND_U on its digital display as a response to the acting force F_C. As the under-calibration system has an instrumental error ΔF_U, the value F_IND_U has a deviation from F_C.

Step 3: Including all Potential Imperfections In real implementation of this calibration exercise, there are some sorts of imperfections such as the inclination of the calibration force F_C by a small angle θ with respect to the common loading axis and the instrumental error ΔF_R of the reference system. Figure 8 shows the block relationship of the calibration exercise considering the real circumstances. As shown in that figure, the calibration force F_C is decremented by the value ΔF_C that arises from its inclination with respect to the loading axis. Accordingly, and as both reference and under-calibration load cells resist side loads, the value of force sensible by the load cells becomes F_N where $F_N = F_C - \Delta F_C$. Also, the instrumental error ΔF_R of the reference system is added so that it shifts the indication value F_IND_R of the reference system from F_N by the value ΔF_R.

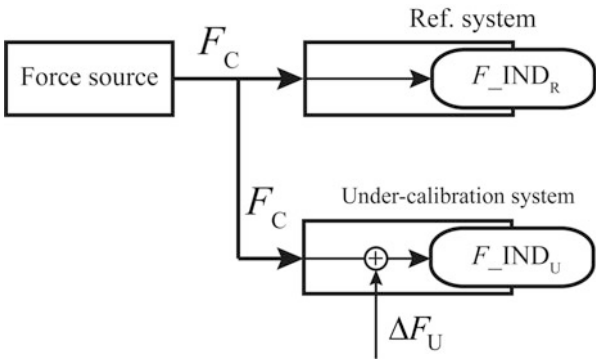

Fig. 7 The ideal cause-and-effect block relationship of the force-measuring-system calibration exercise

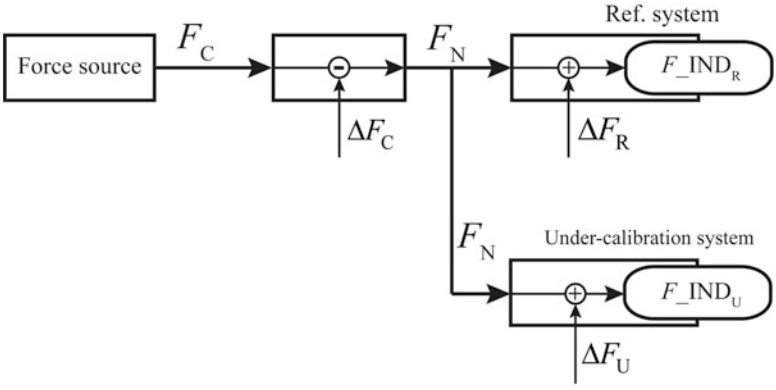

Fig. 8 The real cause-and-effect block relationship of the force-measuring-system calibration exercise

The real cause-and-effect block relationship can be translated into two mathematical equations:

Reference System Equation:

$$F_\mathrm{IND_R} = F_\mathrm{C} - \Delta F_\mathrm{C} + \Delta F_\mathrm{R} \tag{9}$$

Under-calibration system equation:

$$F_\mathrm{IND_U} = F_\mathrm{C} - \Delta F_\mathrm{C} + \Delta F_\mathrm{U} \tag{10}$$

Step 4: Identifying any Potential Correlations For simplicity, all affecting quantities and observations are assumed to be independent.

Step 5: Inversing the Practical Cause-and-Effect Mathematical Formula for Obtaining the Model Equation By subtracting Eq. 9 from Eq. 10, the following formula is obtained:

$$F_\mathrm{IND_U} - F_\mathrm{IND_R} = \Delta F_\mathrm{U} - \Delta F_\mathrm{R} \tag{11}$$

From the previous equation, the final model equation can be concluded as:

$$\Delta F_\mathrm{U} = F_\mathrm{IND_U} - F_\mathrm{IND_R} + \Delta F_\mathrm{R} \tag{12}$$

Finally, for quantifying the value of ΔF_U and its associated uncertainty depending on the final model equation, each term in the final model equation must be quantified using an appropriate technique depending on the nature of the available knowledge about that term. Typically, the values of the indications $F_\mathrm{IND_U}$ and $F_\mathrm{IND_R}$ and their associated uncertainties would be quantified from repeated observations using Type A evaluation method. The instrumental error ΔF_R of the reference system would be quantified depending on the available data found in its calibration certificate.

Recently, the first edition of the modelling guide (BIPM 2020) issued by the *Joint Committee for Guides in Metrology* (JCGM) was released. It is anticipated that this guide will generate discussions among metrologists and practitioners to figure out the weakness and strengths in its proposed guidelines.

The GUM Uncertainty Framework

The GUM uncertainty framework (Fig. 9) is the most popular technique used for estimating the measurement result and combining the uncertainty contributions of different input quantities for obtaining the overall uncertainty margin associated with the measurement result. The GUM uncertainty framework depends on three elements (BIPM 2009):

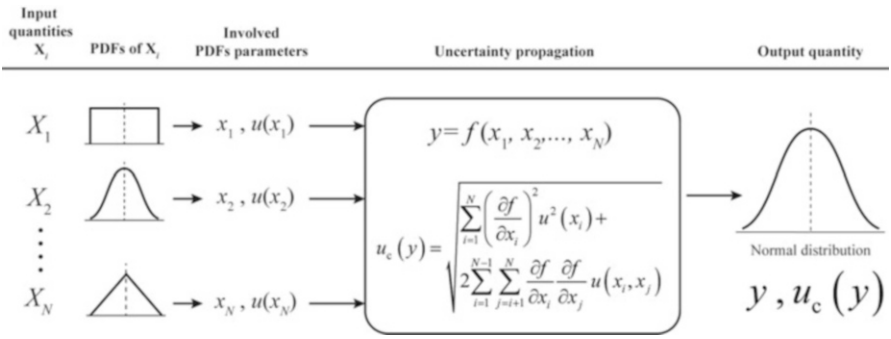

Fig. 9 The GUM uncertainty framework

(a) The best estimates of the input quantities.
(b) The standard uncertainties of the input quantities.
(c) The effect of changing the value of each input quantity on the estimate value of the measurand (usually known as sensitivity coefficients).

Consider a measurand Y which its value is determined upon N input quantities X_1, X_2, …, X_N; the measurement model can be written as:

$$Y = f(X_1, \ X_2, \ \ldots, \ X_N) \tag{13}$$

For obtaining the measurement result and its associated uncertainty, the GUM uncertainty framework employs the *law of propagation of uncertainties* (LPU). According to LPU, the measurement model (Eq. 13) is approximated using the first-order Taylor series expansion. Accordingly, the best estimate y of the measurand Y is calculated as:

$$y = f(x_1, \ x_2, \ \ldots, \ x_N) \tag{14}$$

where x_1, x_2, …, x_N are the best estimates of the input quantities X_1, X_2, …, X_N, respectively. Also, the combined standard uncertainty $u_c(y)$ associated with the best estimate y is taken equal to the positive square root of the estimated variance $u_c^2(y)$ that is obtained from the following formula:

$$u_c^2(y) = \sum_{i=1}^{N}\left(\frac{\partial f}{\partial x_i}\right)^2 u^2(x_i) + 2\sum_{i=1}^{N-1}\sum_{j=i+1}^{N}\frac{\partial f}{\partial x_i}\frac{\partial f}{\partial x_j}\ u(x_i, x_j) \tag{15}$$

where $u(x_i)$ is the standard uncertainty associated with the best estimate x_i of the input quantity X_i. $\partial f/\partial x_i$ is the partial derivative of the model equation $f(X_1, X_2, \ldots, X_N)$ with respect to X_i evaluated at $X_i = x_i$; these quantities are commonly known as the sensitivity coefficients, and they characterize how the best estimate y of the measurand Y varies in response to the change in the values of the best estimates x_1,

$x_2, \ldots, x_N$ of the input quantities. $u(x_i, x_j)$ where $i \neq j$ is the estimated covariance associated with x_i and x_j.

To simplify the process of evaluating uncertainty and make it easier to be implemented by practitioners in the industrial field, the uncertainty sources may be considered as uncorrelated as this is the most common case in regular metrological practices. Moreover, in case the uncertainty sources could be packed in input quantities that have the same unit as the measurand, the sensitivity coefficients can be eliminated as they would equal ones. Based on these simplifications, Eq. 15 can be simplified as follows:

$$u_c^2(y) = \sum_{i=1}^{N} u^2(x_i) \tag{16}$$

or

$$u_c(y) = \sqrt{u^2(x_1) + u^2(x_2) + \quad \ldots \quad + u^2(x_N)} \tag{17}$$

Such simplification can be found in GUM-based uncertainty evaluation standardized guidelines as ASME B89.7.3.2–2007 (ASME B89.7.3.2 2007) which is devoted to the evaluation of uncertainty in dimensional measurement.

The technique used for combining uncertainties in the GUM uncertainty framework is commonly known as the "propagation of uncertainties." In other words, this technique propagates the individual variances of the distributions of input quantities (their squared uncertainties) for obtaining the variance of the resulting distribution of the measurand, i.e., its squared combined standard uncertainty. According to the central limit theorem, when a number of distributions of any form are combined, the resulting distribution tends to the normal form, as long as there is not a non-Gaussian dominant contributor (M3003 2012). In practice, when combining three or more distributions of similar magnitude, a rational approximation to the normal distribution is obtained. Accordingly, when employing the GUM uncertainty framework, the distribution of the values that can be attributed to the measurand Y is usually considered as normal with the expectation y as estimated in Eq. 14 and variance $u_c^2(y)$ as estimated in Eq. 15.

The combined standard uncertainty $u_c(y)$ that is based on one standard deviation of the measurand output distribution is considered as the fundamental parameter that is used for quantitatively expressing the uncertainty associated with the measurement result y. Also, The *International Committee for Weights and Measures* (CIPM) has requested that the combined standard uncertainty is used – for expressing the uncertainty – by all participants in international comparisons and all work done under the umbrella of CIPM (BIPM 2008a). Nevertheless, it is common in commercial and industrial applications to enlarge the margin of uncertainty for giving more confidence that the true value of the measurand lie in that enlarged margin; the corresponding enlarged uncertainty value is what is called the *expanded uncertainty* U (refer to section "The General Idea of Quantifying Uncertainty").

The expanded uncertainty U is calculated by multiplying the combined standard uncertainty $u_c(y)$ by a positive, larger than unity factor k which is commonly known as the *coverage factor*. Depending on the nature of the distribution of the output quantity, a given coverage factor provides a certain *level of confidence* (i.e., probability) that the true value lies in the margin of the expanded uncertainty. When employing the GUM uncertainty framework, the output distribution is usually considered as Gaussian; accordingly, the combined standard uncertainty (i.e., at $k = 1$) provides about 68% level of confidence; for $k = 2$ and $k = 3$, the approximate level of confidence is 95% and 99%, respectively.

After evaluating the uncertainty for a certain metrological task, it may be required to summarize the details of the uncertainty calculations as well as its final results for reporting and analysis purposes; a common uncertainty reporting tool is what is called the *uncertainty budget*. The uncertainty budget is a statement of the measurement uncertainty that basically provides information about the uncertainty components, their calculations, and combinations (BIPM 2012). A proposed format of the uncertainty budget that is applicable for uncorrelated input quantities was presented by the *European Co-operation for Accreditation* in their uncertainty evaluation guidelines booklet (EA-4/02 2013). In this proposed arrangement (Table 1), each input quantity is listed besides its corresponding estimate, standard uncertainty, probability distribution, sensitivity coefficient, and contribution to the combined standard uncertainty. Moreover, the estimate y of the measured quantity Y and its associated combined standard uncertainty $u_c(y)$ are reported. This arrangement may be extended to include other pieces of information – as requested in the *International Vocabulary of Metrology* (VIM) (BIPM 2012) – such as the measurement model, the covariances (if exists), the expanded uncertainty, and the coverage factor (if applicable).

In practice, the uncertainty budget helps the metrologist to identify the dominant uncertainty components and improve it if necessary. Moreover, the uncertainty budget is an essential tool in assessing the reported combined standard uncertainty (Kacker et al. 2007), and solving the discrepancies that may arise between suppliers and customers regarding the uncertainty of measurements; refer to the standard ASME B89.7.3.3–2002 (ASME B89.7.3.3 2002) for such topic.

Table 1 A proposed arrangement of an uncertainty budget for uncorrelated input quantities

Input quantity X_i	Estimate x_i	Standard uncertainty $u(x_i)$	Probability distribution	Sensitivity coefficient $c_i = \partial f/\partial x_i$	Contribution to the standard uncertainty $u_i(y) = c_i \times u(x_i)$
X_1	x_1	$u(x_1)$		c_1	$u_1(y)$
X_2	x_2	$u(x_2)$		c_2	$u_2(y)$
:	:	:		:	:
X_N	x_N	$u(x_N)$		c_N	$u_N(y)$
Y	y				$u_c(y)$

Example 3 Consider that a metrologist wants to evaluate the volume of a cylinder that cut from a long cylindrical rod. The objective is to determine the volume V of the cylinder and its associated uncertainty depending on the measurement of its height H and the manufacturer specification about its diameter D. The measurement model can be written as follows:

$$V = \frac{\pi}{4} D^2 H \tag{18}$$

The height of the cylinder was measured experimentally and was found to have a mean value $h = 100.02$ mm with standard uncertainty $u(h) = 0.05$ mm assuming that the height values have a normal form. The manufacturer of the cylindrical rod certifies that the diameter D is between 49.75 mm and 50.35 mm.

Depending on the available knowledge about the input quantities, the best estimate of the height $h = 100.02$ mm with standard uncertainty $u(h) = 0.05$ mm assuming normal distribution. With respect to the diameter of the cylinder, the only knowns are the lower and upper limits of its value, so the uniform distribution best fit this situation. Accordingly, the best estimate of the diameter $d = (49.75 + 50.35)/2 = 50.05$ mm and the associated uncertainty $u(d) = (50.35 - 49.75)/2\sqrt{3} = 0.1732$ mm.

The best estimate of the volume is calculated as:

$$v = (\pi/4) \times 50.05^2 \times 100.02 = 196{,}781.785 \text{ mm}^3$$

The model equation (Eq. 18) is partially differentiated with respect to the input quantities for evaluating the sensitivity coefficients as follows:

$$\frac{\partial V}{\partial D} = \frac{\pi DH}{2} \rightarrow c_D = \frac{\pi \times 50.05 \times 100.02}{2} = 7{,}863.408$$

$$\frac{\partial V}{\partial H} = \frac{\pi \times D^2}{4} \rightarrow c_H = \frac{\pi \times 50.05^2}{4} = 1{,}967.424$$

The uncertainty due to the error in the diameter:

$$u_D(v) = c_D \times u(d) = 7863.408 \times 0.1732 = 1{,}361.9 \text{ mm}^3$$

The uncertainty due to the error in the height:

$$u_H(v) = c_H \times u(h) = 1967.424 \times 0.05 = 98.3712 \text{ mm}^3$$

The combined standard uncertainty $u_c(v)$ is calculated as follows considering uncorrelated inputs:

Table 2 The uncertainty budget of the cylinder volume example

Input quantity	Estimate	Standard uncertainty	Probability distribution	Sensitivity coefficient	Contribution to the standard uncertainty
π	3.1415926535898				
D	50.05 mm	0.1732 mm	Rectangular	7,863.408	1361.9 mm^3
H	100.02 mm	0.05 mm	Normal	1967.424	98.3712 mm^3
V	196,781.785 mm^3		Assumed Normal		1365.448 mm^3

$$u_c(v) = \sqrt{u_D^2(v) + u_H^2(v)} = \sqrt{1{,}361.9^2 + 98.3712^2} = 1{,}365.448 \ \text{mm}^3$$
$$\approx 1.36545 \ \text{cm}^3$$

The uncertainty budget can be summarized as follows in Table 2.

Considering that the distribution of the output quantity (i.e., the volume) is normal, the expanded uncertainty can be calculated as $2 \times 1365.448 = 2730.896\ mm^3 \approx 2.731\ cm^3$ with approximate level of confidence 95%.

Despite the popularity of the GUM uncertainty framework in both the scientific and industrial fields, it has a major limitation as it depends on a linearized version of the measurement model. This inherent approximation in the GUM method makes it give accurate results only in case of dealing with linear or approximate linear models (Martins et al. 2011). If major deviation from linearity in the measurement model is present, the GUM framework loses its accuracy, and other uncertainty combination techniques such as *Monte Carlo simulation* (MCS) is preferred. MCS is a numerical technique that depends on the original version of the measurement model (Bich et al. 2006); thereby it is not affected by the complexity of the model and does not lose its accuracy due to nonlinearity.

The Monte Carlo Simulation at a Glance

Monte Carlo simulation (MCS) is a stochastic technique that is used to mimic phenomena that have random components (Shonkwiler and Mendivil 2009). From a mathematical perspective, MCS provides an approximate solution to the mathematical model of the considered phenomenon through statistical sampling performed on a computer (Fishman 1996). The technique was originally developed in the 1940s by three scientists: *John von Neumann*, *Stanislaw Ulam*, and *Nicholas Metropolis*, while they were engaged on a nuclear weapon project in *Los Alamos National Laboratory* in the United States (Thomopoulos 2013). The name "Monte Carlo" was proposed for the technique by *Metropolis* as a joke about *Ulam*'s uncle who borrowed money to visit *Monte Carlo* (a neighborhood in Monaco) (Gobet 2016), and the name survived. In year 2000, the *Society for Industrial and Applied Mathematics* (SIAM) has selected Monte Carlo as

one of the top 10 algorithms that have the greatest influence on the development and practice of science and engineering in the twentieth century (Cipra 2000).

Simply, the idea of MCS is running the mathematical model of the considered phenomenon many times. In each run, random sample is drawn for each random input component of the model and the outcome is computed. As the inputs are random and mimic the stochastic nature of the model input components, the computed outcome is also random and mimics the stochastic nature of the outcome of the phenomenon. After running the model for a relatively large number of times, a set of outcome values is obtained; this set can be used to predict the outcome of interest and calculate its characteristics.

Using Monte Carlo in Evaluating Metrological Uncertainty

The *Supplement 1 to the Guide to the Expression of Uncertainty in Measurement* (GUM-S1) (BIPM 2008b) provides comprehensive guidelines for estimating metrological uncertainty using MCS. In practice, MCS is preferable as a tool for evaluating metrological uncertainty in case the measurement model has a great deviation from linearity.

For evaluating uncertainty using MCS, consider the measurement model equation (Eq. 13): $Y = f(X_1, X_2, \ldots, X_N)$. Once again, in this model, the measurand Y is determined upon N input quantities $X_1, X_2, \ldots, X_N$. Each input quantity of the measurement model must be assigned a *probability density function* (pdf) based on the available knowledge about that quantity. Thus, the pdfs $p(X_1), p(X_2), \ldots, p(X_N)$ corresponding to $X_1, X_2, \ldots, X_N$, respectively, must be known before the beginning. Thereafter, for each iteration j ($j = 1, 2, \ldots, M$) of the simulation, random samples $(x_1)_j, (x_2)_j, \ldots, (x_N)_j$, are drawn from the pdfs of the input quantities $X_1, X_2, \ldots, X_N$, respectively, and substituted into the model $Y = f(X_1, X_2, \ldots, X_N)$ for calculating the jth value of the outcome, i.e., y_j of the measurand Y. After running the simulation for M trials, a sample $\boldsymbol{y} = \{y_j\} = \{y_1, y_2, \ldots, y_M\}$ that contains M random draws from the pdf of the measurand Y is finally obtained. Depending on the sample $\boldsymbol{y}$, an empirical pdf can be constructed for the measurand Y, and upon it, the measurement result $\tilde{y}$ and its associated standard uncertainty $u(\tilde{y})$ can be calculated. Figure 10 shows the steps of MCS as it is employed in evaluating the measurement result and the associated standard uncertainty of the measurand Y; the technique is also known as *propagation of distributions*.

Programming MCS for Uncertainty Evaluation

Putting Monte Carlo Simulation into action needs writing a multifunction computer software that can (a) generate random samples, (b) calculate the outcome based on a mathematical model, (c) store output data in the form of an array, and (d) calculate

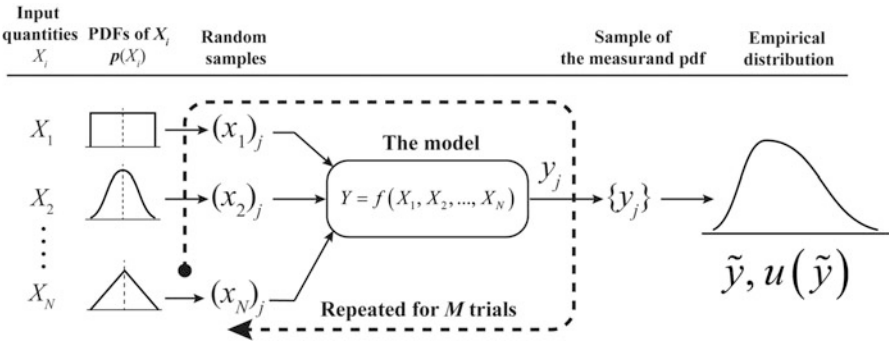

Fig. 10 The Monte Carlo simulation used for evaluating the estimate $\tilde{y}$ and the associated standard uncertainty $u(\tilde{y})$ of the measurand Y

the simulation results, i.e., the characteristics of the output data. Figure 11 shows a simple example of programming MCS for a measurand Y that has two input quantities X_1 and X_2 following a simple model, $Y = X_1 \times X_2$; the pdf of X_1, $p(X_1)$ is normal with mean value 5 and standard deviation 0.5, while the pdf of X_2, $p(X_2)$ is also normal with mean value 10 and standard deviation 2.

MCS can be programmed as a "While" or "For" loop with a number of trials equals M; the first step in the loop is generating random samples according to the assigned pdfs of the input quantities. In practice, most modern programming languages provide canned functions that can generate pseudo random numbers following several distributions such as uniform, normal, triangular, etc. These functions are powerful and usually are based on sophisticated algorithms that guarantee the quality of randomness of the generated numbers while minimizing the impact on the computer resources. The user may also resort to using the *inverse transform method* or the *accept-reject method* (Thomopoulos 2013) for generating pseudo random numbers following several distributions, e.g., triangular, trapezoidal, U, etc., depending on a uniform random generator that can be easily found as a canned function in most modern programming languages. In the example shown in Fig. 11, in each iteration j of the loop, random samples $(x_1)_j \sim N(5, 0.5^2)$ and $(x_2)_j \sim N(10, 2^2)$ are generated.

In each iteration j of the loop, the second step is substituting the random samples into the model equation to calculate the jth value of the outcome. In this stage, the program uses the original version of the measurement model whatever its degree of complexity or nonlinearity; this is what makes MCS superior in dealing with nonlinear models. In the example shown in Fig. 11, just the random sample $(x_1)_j$ is multiplied by the random sample $(x_2)_j$ for obtaining the outcome y_j.

The third step in each iteration j of the loop is sending the calculated outcome y_j for storing in the output array. This array begins empty, and its size is incremented by one after each iteration as the calculated outcome is appended to it. At the end of the

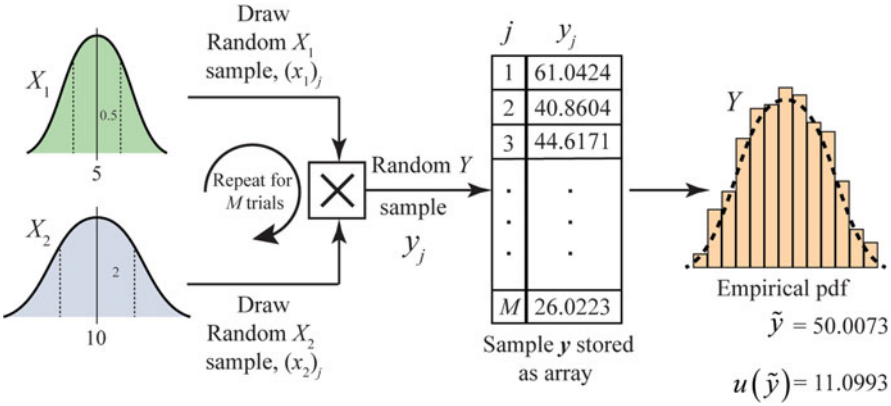

Fig. 11 An example to programming MCS for evaluating the measurement result $\tilde{y}$ and its associated standard uncertainty $u(\tilde{y})$

simulation, when the loop terminates, this array become includes the M elements of the sample set $\boldsymbol{y}$. In the example shown in Fig. 11, the storage array is shown as a table that relates each outcome to its corresponding index j.

The final step of MCS program is calculating the simulation results from the elements stored in the output array. The main results are the best estimate of the output quantity (i.e., the value of the measurement result) and its associated standard uncertainty. The best estimate of the output quantity is taken as the arithmetic average $\tilde{y}$ of the elements of the sample $\boldsymbol{y}$ and can be calculated as follows:

$$\tilde{y} = \frac{1}{M}\sum_{j=1}^{M} y_j \tag{19}$$

The standard uncertainty $u(\tilde{y})$ associated with the measurement result $\tilde{y}$ is taken as the standard deviation of the elements of the sample $\boldsymbol{y}$ that is calculated according to the following formula:

$$u(\tilde{y}) = \sqrt{\frac{1}{M-1}\sum_{j=1}^{M}\left(y_j - \tilde{y}\right)^2} \tag{20}$$

In the example shown in Fig. 11, the measurement result and its associated standard uncertainty were calculated as per the two previous equations using the data stored in the output array. This data may be used also for constructing a frequency histogram for the sample $\boldsymbol{y}$ which can give – if the bin width is set properly – an idea about the shape of the empirical distribution of the output quantity, i.e., measurand Y.

Coverage Interval for MCS Output Distribution

Once again, the MCS is useful when the measurement model is highly nonlinear. In this case, the output empirical distribution probably would be asymmetric and difficult to be defined analytically. Accordingly, the calculation of the bounds of a coverage interval that encompass certain probability $100p$ % (say $p = 0.95$, 0.99, etc.) that the true value of the measurand lies in that interval (This is analogue to calculating the expanded uncertainty in the GUM uncertainty framework.) is not as straightforward as it can be followed in the GUM framework via multiplying the combined standard uncertainty by a coverage factor. In fact, for an empirical distribution given by M discrete values, there is $(1\text{-}p)M$ intervals that contain $100p$ % of the values (Coleman and Steele 2009), each one of them represents a potential solution for our problem. However, in practice, the GUM-S1 (BIPM 2008b) provides only two solutions: (a) the probabilistically symmetric $100p$ % coverage interval and (b) the shortest $100p$ % coverage interval.

The probabilistically symmetric $100p$ % coverage interval (Fig. 12) is the unique interval that contains $100p$ % of the results while the probability on its right and left is symmetric and equals $100((1 - p)/2)\%$. For calculating the lower bound y_{low} of that interval and its upper bound y_{high} depending on the M elements of the sample $\boldsymbol{y}$ that resulted from MCS, follow the following steps (Coleman and Steele 2009):

1. Sort the M elements of the sample $\boldsymbol{y}$ into non-decreasing order $y_{(r)}$ where $r = 1, 2, \ldots, M$.
2. Take $r_{\text{low}} = (1 - p)M/2$; if r_{low} is not an integer, take r_{low} equal to the integer part of $[1/2 + ((1 - p)M/2)]$.
3. Take $r_{\text{high}} = (1 + p)M/2$; if r_{high} is not an integer, take r_{high} equal to the integer part of $[1/2 + ((1 + p)M/2)]$.
4. Take $y_{\text{low}} = y_{(r=r_{\text{low}})}$.
5. Take $y_{\text{high}} = y_{(r=r_{\text{high}})}$.

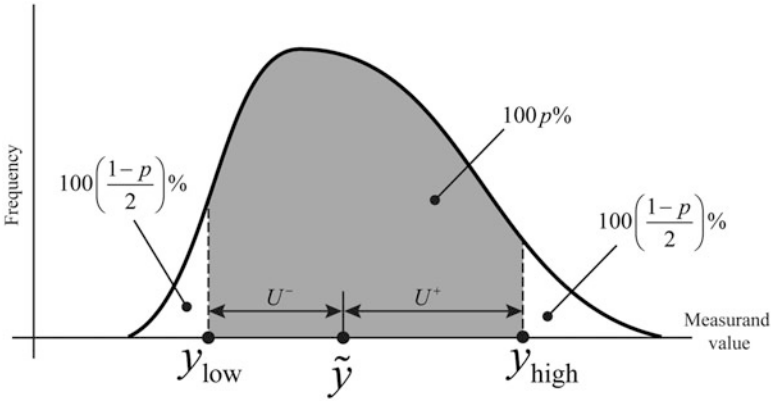

Fig. 12 The probabilistically symmetric $100p$ % coverage interval

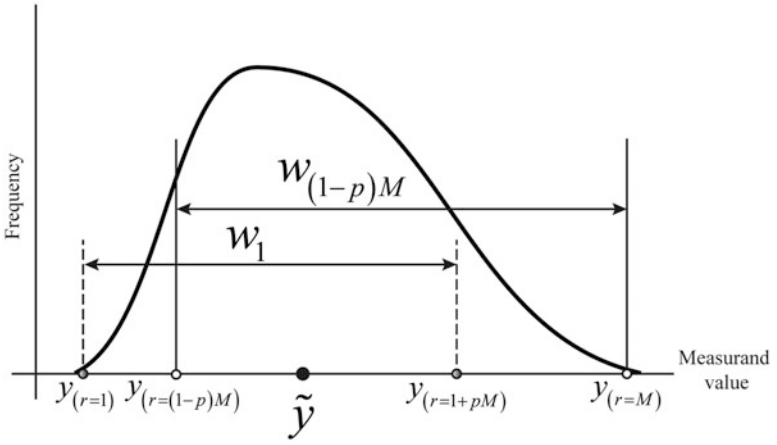

Fig. 13 The first and the last intervals that contain 100p % of the results

As shown in Fig. 12, due to the asymmetry of the distribution, the width of the probabilistically symmetric 100p % coverage interval is not symmetric about the mean value $\tilde{y}$, so the uncertainty cannot be written in the form $\tilde{y} \pm U$. Instead, it can be reported as the true value y_{true} of the measurand Y would lie in the interval $\tilde{y} - U^- \le y_{\text{true}} \le \tilde{y} + U^+$ with a level of confidence 100p %, where $U^- = \tilde{y} - y_{\text{low}}$ and $U^+ = y_{\text{high}} - \tilde{y}$.

The shortest 100p % coverage interval is an alternative to the probabilistically symmetric one. Once again, for a MCS output distribution given by a sample $\boldsymbol{y}$ that contain M discrete values, there is (1-p)M intervals that contain 100p % of the values. The width w of these intervals (i.e., $y_{\text{high}} - y_{\text{low}}$) differs from one interval to another. For instance, the width of the first interval $w_1 = y_{(r\,=\,1\,+\,pM)} - y_{(r\,=\,1)}$, the width of the second interval $w_2 = y_{(r\,=\,2\,+\,pM)} - y_{(r\,=\,2)}$, and the width of the last interval $w_{(1\,-\,p)M} = y_{(r\,=\,M)} - y_{(r\,=\,(1\,-\,p)M)}$; see Fig. 13. Generally, the width of the kth interval is $w_k = y_{(r\,=\,k\,+\,pM)} - y_{(r\,=\,k)}$ where $k = 1, 2, \ldots, (1\text{-}p)M$; if the value of the index r is not an integer, approximate it to the nearest integer by adding 0.5 and taking the integer part of the result; also, treat the non-integer values of the index k similarly. Finally, the interval with the smallest width w is the shortest coverage interval that contains 100p % of the results.

The GUM-S1 states that, for a *unimodal* (single-peaked) distribution, the shortest 100p % coverage interval would contain the mode, the most probable value of the measurand Y. Moreover, in case of symmetric distributions (e.g., Gaussian), the probabilistically symmetric 100p % coverage interval and the shortest 100p % coverage interval would be identical.

Number of Monte Carlo Trials, *M*

As a rule, the greater the number of model evaluations M, the more reliability of the results and the more stability of them. In practice, the number of Monte Carlo trials M can be chosen a priori before starting the simulation and provided as an input to

the simulation loop. In this case, the loop would continue to evaluate the model repeatedly until the number of evaluations reaches the prechosen number of trials; this is the conventional procedure of Monte Carlo simulation. As a drawback, this conventional procedure that depends on a prechosen number of trials does not monitor the stability of the numerical results provided by the simulation and does not guarantee its conformance to a certain numerical tolerance value. The reason is simply that the necessary number of MCS trials needed to provide the results stable within a prescribed numerical tolerance depends on the shape of the output distribution and the required coverage probability p. In practice, the coverage probability p can be determined according to the accuracy imposed by the application in which the results will be employed. On the contrary, the shape of the output distribution may not be known a priori before the simulation, and this would represent a stumbling block toward determining the necessary number of trials before the simulation. As a compromise, the GUM-S1 recommends that the number of trials M of conventional MCS should be large compared to the value $1/(1-p)$, e.g., $M \geq 10^4(1/(1-p))$ as it is expected – if this condition is met – that the simulation would provide reasonable discrete representation of the output distribution in the regions near the endpoints of the $100p$ % coverage interval.

Adaptive Monte Carlo Procedure

The adaptive Monte Carlo procedure is a modified version of conventional Monte Carlo in which the number of trials is not chosen in advance. Unlike in the case of conventional Monte Carlo, the adaptive procedure increases the number of trials progressively in batches until the simulation results of interest become statistically stable within a certain numerical tolerance value. Specifically, when applying Monte Carlo adaptively in uncertainty evaluation, any of the simulation results of interest ($\tilde{y}$, $u(\tilde{y})$, y_{low}, or y_{high}) is considered as stabilized if twice the standard deviation of the result that calculated from different simulation batches is less than the numerical tolerance δ associated with the standard uncertainty $u(\tilde{y})$.

For calculating the numerical tolerance δ associated with a standard uncertainty $u(\tilde{y})$, first, the number of significant decimal digits n_{dig} that is considered as meaningful in the value of $u(\tilde{y})$ is specified; usually it is chosen to be 1 or 2. Then the standard uncertainty $u(\tilde{y})$ is expressed in the form $u(\tilde{y}) = c \times 10^{\ell}$ where c is an n_{dig} decimal digit integer and ℓ is an integer. Finally, the numerical tolerance δ is taken as $\delta = (1/2) \times 10^{\ell}$.

Example 4 Consider that $u(\tilde{y}) = 0.0026$ mm and n_{dig} are chosen to be 2. Thus, $u(\tilde{y})$ can be expressed as $u(\tilde{y}) = 26 \times 10^{-4}$ mm, and so $c = 26$ and $\ell = -4$; thus take $\delta = (1/2) \times 10^{-4}$ mm $= 0.00005$ mm. However, if n_{dig} was chosen to be 1, $u(\tilde{y})$ would be expressed as $u(\tilde{y}) = 0.003$ mm $= 3 \times 10^{-3}$ mm, and so $c = 3$ and $\ell = -3$; thus take $\delta = (1/2) \times 10^{-3}$ mm $= 0.0005$ mm.

The adaptive Monte Carlo scheme can be programmed as a nested while loop in which the inner loop (simulation loop) is dedicated to running the simulation batch that have a prespecified number of trials exactly as the case of conventional Monte Carlo; the outer loop (stability-check loop) checks – starting after the second batch – for the stability criterion and terminates if it met or forced the simulation loop to run another batch otherwise; and so on until the statistical stability is achieved. At the end, the simulation final results are calculated based on the whole output values obtained from all batches.

Figure 14 demonstrates the adaptive Monte Carlo simulation using the same problem used for demonstrating the conventional simulation in Fig. 11. Before starting, the number of significant decimal digits n_{dig} that is considered as meaningful in the value of $u(\tilde{y})$ is chosen (usually $n_{\text{dig}}= 1$ or 2) and entered as an input to the simulation software. Also, the number of trials Q of the simulation batch is chosen as $Q = \max(G, 10^4)$, where G is the smallest integer greater than or equal to $100/(1-p)$ while p is the required coverage probability (BIPM 2008b).

The simulation starts with the first batch ($h = 1$) with Q number of trials. After the first batch, the stability-check loop forces the simulation loop to run the second batch without a stability check because that check requires at least two entries of the statistic being checked for stability. While a batch h is running, its output values are stored in the form of an array. At the end of the batch h, its own results are calculated

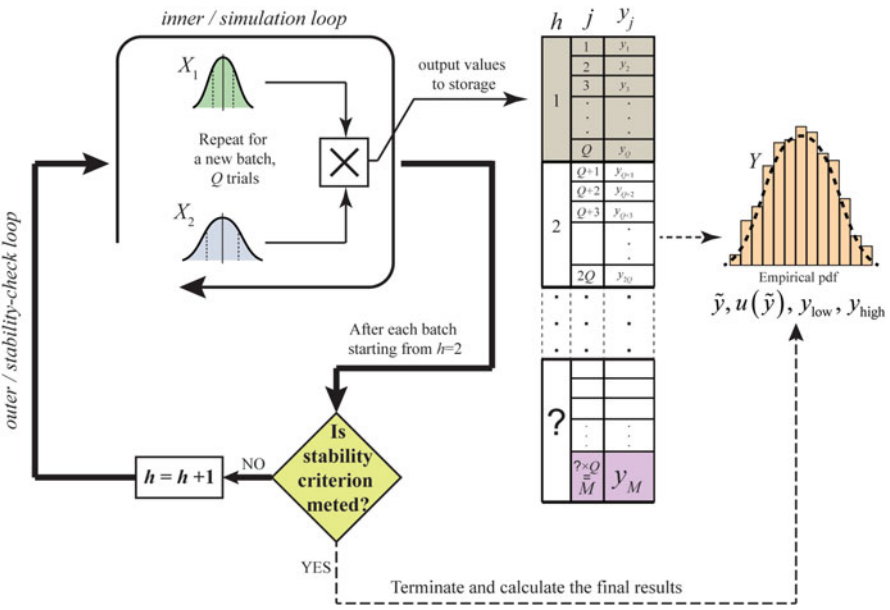

Fig. 14 An example to programming adaptive MCS

based on its own output values similarly as followed in the case of conventional simulation; the calculated results for the hth batch are:

(a) The estimate $\tilde{y}^{(h)}$.
(b) The standard uncertainty $u\left(\tilde{y}^{(h)}\right)$.
(c) The lower bound $y_{\text{low}}^{(h)}$ of the $100p$ % coverage interval.
(d) The upper bound $y_{\text{high}}^{(h)}$ of the $100p$ % coverage interval.

For each of the previous statistics, a separate array is required for storing the calculated values of different batches.

At the end of each batch, starting after the second one, the stability check is performed as follows (BIPM 2008b):

1. The standard deviation $s_{\bar{y}}$ associated with the average $\bar{y}$ of the estimates $\tilde{y}^{(1)}$, $\tilde{y}^{(2)}$, ..., $\tilde{y}^{(m)}$ of the m batches available so far is calculated as:

$$s_{\bar{y}} = \sqrt{\frac{1}{m(m-1)} \sum_{h=1}^{m} \left(\tilde{y}^{(h)} - \bar{y}\right)^2} \tag{21}$$

where

$$\bar{y} = \frac{1}{m} \sum_{h=1}^{m} \tilde{y}^{(h)} \tag{22}$$

2. The counterpart of the statistic $s_{\bar{y}}$ is calculated for $u(\tilde{y})$, y_{low}, and y_{high} in the same way followed in case of $\tilde{y}$. Thus, we obtain $s_{\bar{u}(\tilde{y})}$, $s_{\bar{y}_{\text{low}}}$, and $s_{\bar{y}_{\text{high}}}$ associated with the averages $\bar{u}(\tilde{y})$, $\bar{y}_{\text{low}}$, and $\bar{y}_{\text{high}}$ of the statistics $u(\tilde{y})$, y_{low}, and y_{high}, respectively.
3. The standard uncertainty $u(\tilde{y})$ is calculated using all $m \times Q$ output values available so far.
4. The numerical tolerance δ is calculated for $u(\tilde{y})$ using the specified n_{dig}.
5. If any of $2s_{\bar{y}}$, $2s_{\bar{u}(\tilde{y})}$, $2s_{\bar{y}_{\text{low}}}$, and $2s_{\bar{y}_{\text{high}}}$ exceeds δ, a new batch is executed. Otherwise, the stability criterion is considered to be met and the simulation is terminated.

Finally after all results get stabilized, the final results of the simulation $\tilde{y}$, $u(\tilde{y})$, y_{low}, and y_{high} are calculated using all output values obtained from all simulation batches.

In fact, the estimate $\tilde{y}$ and its associated standard uncertainty $u(\tilde{y})$ converge considerably faster than y_{low} and y_{high} with respect to the number of trials. So, when the coverage probability is not required, the stability check can be based only on $2s_{\bar{y}}$ and $2s_{\bar{u}(\tilde{y})}$ (BIPM 2008b).

Concluding Remarks

By the end of this chapter, the reader should have been understood the general framework and methodology of uncertainty quantification in metrology. To sum up the procedure, the uncertainty evaluation begins by identifying the significant uncertainty sources then they are quantified according to Type A or Type B evaluation methods based on the nature of the information available about the uncertainty source. It is noteworthy here that there are three sources of uncertainty usually applicable to every measurement: the repeatability of the measurement, the resolution of the measuring device, and its calibration uncertainty. The first can be quantified from repeated observations using Type A method, while the two others are quantified according to Type B method based on the available information. Other uncertainty sources may be applicable subject to the measuring method and its circumstances. The next step is combining individual uncertainties using an appropriate technique to estimate the measurement result and the total margin of uncertainty about that result. Here, it is remarkable that the GUM uncertainty framework is the most popular as it is simple, straightforward, and easy to implement in both the industrial and scientific fields. On the contrary, the Monte Carlo simulation is much more complex and cannot be implemented by hand; however, it has become a necessity if the measurement model has a great deviation from linearity to the extent that the GUM framework becomes inaccurate due to its inherent linear approximation. Probably, the GUM framework would still be the dominant until the ability of metrologists to develop complex measurement models is evolved to the extent that the GUM framework become inefficient.

As obvious, the ultimate benefit of quantifying the uncertainty is aiding in decision-making using the result of the measurement process. In this context, the decision maker should realize that the uncertainty interval provides a certain level of confidence (probability) that the true value lies in that interval. Accordingly, the decision itself is liable to be incorrect at a certain probability. In practice, improving the uncertainty to the extent that the error probability in decision-making become trivial may be inefficient from an economic point of view. So, a cost analysis is very useful in determining the appropriate balance between the level of investment in measurement and the cost of incorrect decision that keeps the total cost minimum.

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Measurement Uncertainty

92

C. V. Rao

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Abstract

This introduction to measurement uncertainty is designed for metrology experts working in calibration laboratories and metrology institutions, as well as students in science and engineering programs at the university level. The topic is covered with a focus on creating models of the physical measuring process. Since the release of the Guide to the Expression of Uncertainty in Measurement, there has been an increase in the use of models for uncertainty analysis. However, there is no direction on how to do so. That issue is addressed in this booklet. The mathematics and statistics employed here are elementary and are normally studied in high school. Modeling, on the other hand, is a method of describing an abstraction of a measuring process using mathematical language. Some readers may be unfamiliar with this. This publication was made possible by the contributions of many people.

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Keywords

Measurement · Uncertainty · Errors · Probability distributions · Samples

Introduction

We get knowledge about amounts of interest from measurements. Measurements frequently provide solutions to questions such as what is the speed of the train and what are the glucose levels in a blood sample? The results of measurements are frequently required because a decision must be made: will I be late for the college, is the train moving too quickly, or is the patient's blood sugar level beyond the legal limit? Unfortunately, no measurement result can be guaranteed to be exact. As a result, bad decision-making outcomes are possible. Measurements are used to inform crucial decisions in modern society, such as industry, safety, public health, the environment, trade, and finance.

Measurement error is the discrepancy between a measurement result and the actual value of a quantity of interest. The presence of measurement mistakes, as well as the fact that they are unpredictable, causes doubt about decision accuracy. Uncertainty analysis is the exploration of measurement errors. This method detects and evaluates elements that can cause measurement error. An uncertainty analysis generates a model that characterizes the measurement's fundamentally unexpected behavior and gives statistical data on the accuracy of the results that can be utilized to make judgments. We will show you how to construct measurement models and how to use them to assess measurement uncertainty. We next apply these concepts to measurement and calibration case studies.

Main Text

Mathematical Notations and Symbols

This booklet employs mathematical expressions and symbols to provide clear and succinct representations of the quantities involved in a measurement process. The symbols, on the other hand, are merely symbols for things that can be explained in other ways; mathematical language is not strictly required. If you are having trouble understanding the mathematical equations, try thinking about the meaning of each sign in language.

For example, in the equation

$$\widehat{y} = y - Ey$$

here

$\widehat{y}$ represents measurand
y represents measured value
Ey error in calculating measurements.

Notations
To distinguish between known and unknown amounts, a simple notation is used: upper-case letters represent unknown quantities, whereas lower-case letters represent objects with a known value. Subscripts are also added to symbols to aid in quantity labeling. For example, the voltage might be represented by upper-case letter V (the exact voltage is not known), while lower-case letter represents v_{meter} (the meter reading from a volt meter is a known number).

$V_{meter}[i]$ Indicates i^{th} volt meter reading

Measurement Errors and Models

Because nuisance factors always affect the output of a measurement, we can never precisely establish the value of a number. As a result, a measured value can never be exactly equal to the amount of interest.

The relation between measurand, measurement value, and measurement error is given by

$$\widehat{y} = y - Ey \tag{1}$$

where Ey is measurement error. We can correct the measurement result y if we know the measurement error Ey

Equation (1) is an example of a measurement model. Models describe the link between the amount of interest and other variables that may affect the measurement result. They are essential to uncertainty analysis because they may be used to assess both the quantities to be measured and the measurement uncertainty. Uncertainty regarding the value of the error Ey leads to uncertainty in the value y as $\widehat{y}$ estimate in equation (1).

Defining the Concepts of Unpredictability and Probability

When something's behavior is unpredictable, we call it random. Because many minor factors influence the results in unpredictable ways, measurement outputs are random. Nonetheless, the quantities we want to measure are not random (we would like to know velocity, the voltage, the concentration of blood sample, and so on). We uncover the elements that can cause a measurement result to be skewed when we conduct an uncertainty analysis. Then, we try to use probability concepts to describe

each factor's behavior. The concepts of probability that are needed to define influencing factors and measurement mistakes are introduced in this chapter.

Probability: It is a number that indicates the chance or likelihood that a particular event will occur.
Mathematical definition: Probability is the ratio of the number of favorable cases to the number of total outcomes.

The behavior of uncertain outcomes is described by probability statements. When rolling a dice, for example, we cannot anticipate the outcome, but we can say there is a 3/6 chance of getting an even number. We made our 3/6 assessment based on our experience and other relevant information; we did not measure it. Nonetheless, measurement might be used to test the assumption that each of the six events is equally likely. We could roll the die several times to ensure that the relative frequency of sixes does not depart from 3/6 by more than is reasonable given the number of throws. Although dice are not used in real measurements, the ability to double-check assumptions is crucial. It should always be easy to check the correctness of any probability statement that describes uncertain behavior, at least in theory.

When there are a finite number of alternative outcomes, such as when rolling a die, each one can be assigned a probability value. When a measurement result can have any value, the likelihood of an outcome occurring within a range of values is what is important. In such instances, probability distributions are applied. As shown in Fig. 2, a probability distribution can be represented as a curve over a horizontal axis reflecting the various possibilities. The higher the curve is above the axis, the more likely it is that the values under that part of the curve will be observed.

Probability Distribution Curves We have three types of probability distribution curves:

1. Gaussian distribution
2. Rectangular distribution (Uniform distribution)
3. Triangular distribution

Gaussian Distribution

Because it is commonly assumed to characterize the unanticipated error in a measurement result, the Gaussian distribution (Fig. 1) is of particular importance in test and measurement. A Gaussian distribution is also commonly used to represent the mistake that occurs when statistical approaches are employed to determine influencing variables. Values are more likely to be found in the center of a Gaussian distribution when random behavior follows it (near the mean). Nonetheless, because the range is potentially unbounded, values far from the mean will occasionally be seen.

Uniform Distribution

The continuous uniform distribution, often known as the rectangle distribution, is a family of symmetric probability distribution in probability theory and statistics. An

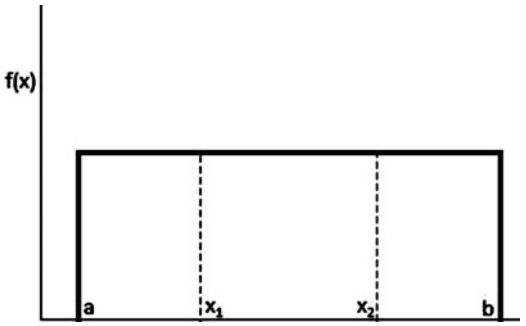

Fig. 1 A Uniform distribution. Center of the distribution is the mean of x_1 and x_2

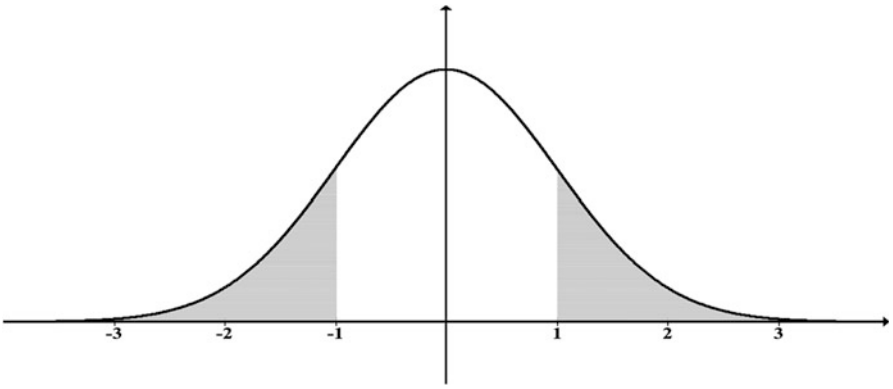

Fig. 2 Normal distribution. Center of the distribution is the mean, and the white band region represents the standard deviation

experiment with an arbitrary outcome that falls within defined limitations is described by a distribution.

It will be easier to express a uniform distribution's scale in terms of its width. However, this information must always be transformed into a standard deviation when determining the uncertainty.

When a number is displayed with a fixed number of digits, the inaccuracy is typically referred to as a uniform distribution. That is, a quantity Y is measured, and the result is rounded to

$$x = X + \mathrm{E}x \tag{2}$$

with Ex denoting the rounding error. Although neither Y nor Ex is known, we can say that x is a guess at X. (x is approximately equal to X). When a large number of different Y values are rounded, the resulting distribution of Ex values looks like a uniform distribution with a mean of zero and a range equal to the magnitude of the least significant digit (Fig. 2).

Triangular Distribution

A triangular distribution is a continuous probability distribution with a triangle-shaped probability density function. The minimum value a, the highest value *b*, and the peak value c are the three values that define it. This is extremely useful because, even if we do not know the mean or standard deviation, we can often predict the highest and lowest values, as well as the most likely outcome, in a real-life situation. We avoid undesirable extreme values by using the triangle distribution, which has a defined upper and lower limit. In addition, for skewed distributions, the triangular distribution is a reasonable model.

When values are more likely to occur around the center of a range than near the limits, the triangle distribution can be utilized. This is the situation, for example, when the difference between two uniformly dispersed values has the same width uniform distributions.

Example for the discrete triangular distribution is tossing two dice getting a minimum of 2, a maximum of 12, and a peak at 7 (Fig. 3).

Error Versus Uncertainty

It is crucial to distinguish between the terms "error" and "uncertainty."

Error: The discrepancy between the measured and the "actual value" of the thing being measured is called error.

Uncertainty: It is a metric that expresses the doubt about measurement's outcome.

We endeavor to remedy any known inaccuracies as much as feasible, for as by applying corrections from calibration certifications. Any inaccuracy whose value we do not know is, on the other hand, a cause of uncertainty.

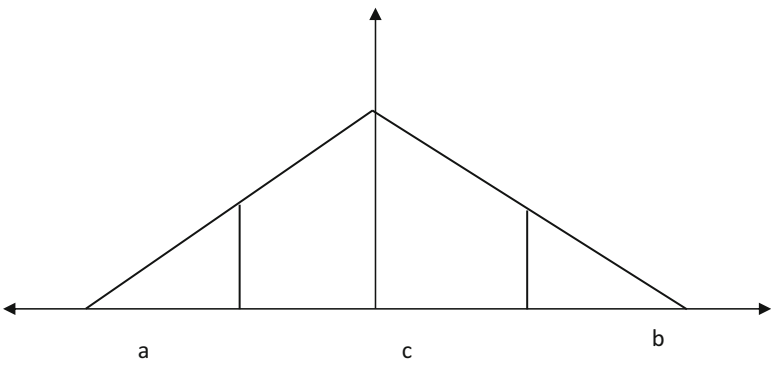

Fig. 3 Center of the distribution is the mean, and region between the two vertical lines is the standard deviation

Absolute Error of a Quantity of X

The units of an absolute error are the same as the units of the physical quantity. It is denoted by the letter k_x.

Relative Error of a Quantity X

Relative error is defined as the ratio of the error to the best value of the quantity. Denoted by K_{rx}, the letter r indicates "relative."

Types of Errors

Errors and their types depending on how the measurement was taken: Measurement mistakes might be characterized as random or systematic (an instrument could cause a random error in one situation and a systematic error in another).

Random errors are stochastic fluctuations (in either direction) in measured data caused by the measuring device's precision limitations. Random mistakes can be assessed using statistical methods and decreased by averaging over a large number of data (see standard error).

Systematic Errors: Systematic errors are reproducible inaccuracies that consistently go in the same direction. These errors are difficult to detect and cannot be statistically analyzed. If a bias is detected when you calibrate against the standard, you can reduce the bias by applying a correction or correction factor to correct the effect. Unlike random errors, increasing the number of observations cannot detect or reduce systematic errors.

Common Source of Errors in Physical Laboratory Experiments

Incomplete Definition (which can be systematic or random): One of the reasons why accurate measurements cannot be made is that measurements are not always well defined. For example, if two different people measure the same rope length, you will probably get different results because each person can stretch the rope with different tensions. The best way to minimize definition errors is to carefully consider and specify the conditions that may affect your measurement.

Failure to Account for a Factor (usually systematic): The hardest part of designing an experiment is trying to control or account for all possible factors except a single independent variable in the process. analyzer. For example, you may inadvertently ignore air resistance when measuring the free-fall acceleration, or you may ignore the effect of the earth's magnetic field when measuring the field of a small magnet. The best way to interpret these sources of error is to discuss with your colleagues all the factors that may affect your results. This brainstorming should be done before starting the test so that confounding factors can be sorted out before data collection. Sometimes error correction can be applied to a result after retrieving data to account for an undetected error.

i) *Instrument resolution* (random): All instruments have finite precision which limits their ability to resolve small measurement differences. For example, a gauge cannot distinguish distances with much better accuracy than about half of its smallest scale line (0.5 mm in this case). One of the best ways to get more accurate measurements is to use the zero difference method instead of measuring a quantity directly. The equal or empty method involves the use of instruments to measure the difference between two similar quantities, one of which is known to be very precise and can be adjusted. The adjustable reference quantity is varied until the difference is reduced to zero. The two quantities are then balanced
ii) *Environmental factors* (systemic or random): Be aware of errors introduced by your immediate work environment. You may need to consider or protect your experience from vibrations, drafts, temperature changes, electronic noise, or other effects from nearby devices.
iii) *Physical variations* (random): It is always wise to obtain several measurements over the entire range being studied. This often reveals variations that might go unnoticed. These variations may require further investigation or they may be combined to find the mean.
iv) *Lack of calibration or check zero of instrument* (systematic): Whenever possible, instrument calibration should be verified prior to data collection. If no calibration standard is available, the accuracy of the instrument should be checked by comparison with another at least the most accurate or by referring to the technical data provided by the manufacturer. Whenever making measurements with micrometers, digital scales, or power meters, always check zero first, zero instrumentation if possible, or measure the displacement of the zero reading from the true zero and correct all measurements accordingly. You should check for zero during the test.
v) *Parallax error* (systematic or random): This error can occur whenever there is a certain distance between the scale and the indicator used to obtain the measurement result. If the observer's eye is not perfectly aligned with the pointer and scale, the reading may be too high or too low (some analog meters have mirrors to help with this alignment).
vi) *Lag time and hysteresis* (Systematic): Some instruments take time to reach equilibrium, and reading before the instrument stabilizes will result in readings that are often too low. The most common example is measuring temperature with a thermometer that has not yet reached thermal equilibrium with its environment. A similar effect is latency where device reads are delayed, and there appears to be a "memory" effect as the data is retrieved sequentially moving up or down a range of values. Hysteresis is usually related to materials becoming magnetized when a changing magnetic field is applied.
vii) *Drift devices* (systematic): Most electronic devices have readings that drift over time. The amount of drift is usually not an issue, but sometimes the source of this error can be significant and should be considered.

viii) *Personal error arising from negligence*: Poor technique, or bias of the tester. The tester may measure inaccurately, may use the wrong technique to perform the measurement, or may introduce bias into the measurement by expecting (and unintentionally forcing) the result to agree, with expected results.

ix) *Aggregate individual errors*: Sometimes referred to as errors or omissions, should be avoided and corrected if discovered. As a general rule, gross individual errors are excluded from the error analysis discussion because it is generally assumed that test results are obtained by following correct procedures. The term human error should also be avoided in discussions of error analysis because it is too general to be useful.

Identifying Influence Quantities in Measuring Simple Length by Using a Calibrated Steel Ruler

A 1 m long steel ruler with 1 mm divisions is used to measure 60 cm length object.

(i) Errors are introduced by the ruler's curvature.

From Fig. 4, $l = \frac{L}{\cos\theta}$.

(ii) Errors are introduced by angle of observation (Fig. 5).

When taking readings, the observer should be perpendicular to the ruler scale and the feature edge. A reading error (also known as a parallax error) will occur if this is not done. If the work piece has a different thickness than the ruler and they are both sitting on a flat surface, a parallax mistake will occur if the observer is not looking directly down at the scale.

(iii) Rounding Error

The value of the scale marker closest to the feature edge is taken as a ruler reading. As a result, a measured value is rounded to the ruler scale's resolution, introducing an inaccuracy. If one end of the ruler is used to locate one edge, an error may be introduced due to the difficulty in aligning the ruler's end with the feature, as well as mechanical wear at the ruler's end (Fig. 6).

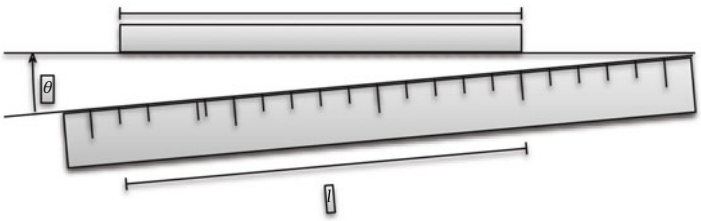

Fig. 4 A cosine error will make the length appear longer

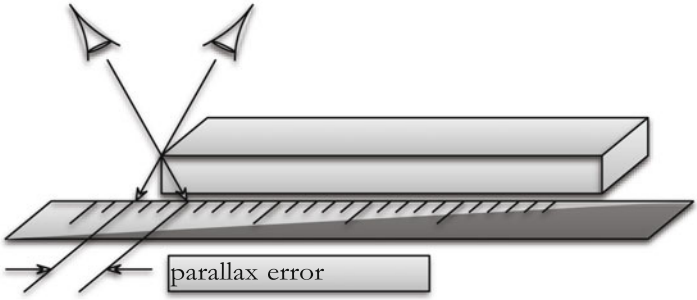

Fig. 5 Parallax error

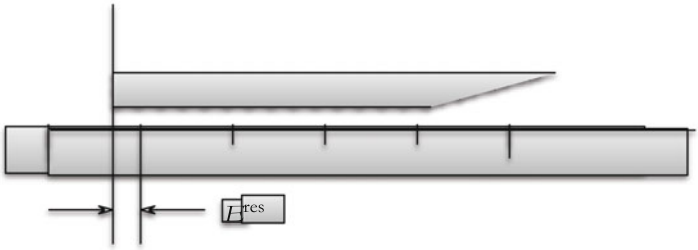

Fig. 6 Rounding error

Table 1 Errors

Factor	Typical magnitude
Cosine error	0.02 mm
Parallax error	0.06 mm
Ruler scale error	0.04 mm
Resolution error	0.25 mm
Thermal expansion	0.04 mm

Resolution error (E_{res}): The reading is taken from the number on the scale that is closest to a feature end-point. This brings the nearest scale division up to date.

Using Table 1, this can be converted to a standard deviation

$$E_{\text{res}} = \frac{1.0}{\sqrt{12}} = 0.29 \text{ mm}$$

Calibrating a Thermometer

Calibration is a process designed to verify that an instrument is suitable for use. Uncertainty Analysis of the calibration process must evaluate the accuracy of the calibration process The reference quantities are known, and so is the behavior of the model instrument calibration. In some cases, such as in the case of thermometers

discussed in this chapter, calibration determines correction works with raw instrument data and uncertainty correction value.

Temperature-controlled bathtubs are used to provide the environment. It has a precisely defined temperature. Two thermometers immersed in the bathtub next to each other: one is a glass thermometer calibrated and the other is a platinum resistance thermometer. Resistance thermometers are used to measure difference between bath temperature and reading glass thermometer and resistance thermometer readings for calibrating glass thermometers. Mercury-in-glass thermometer readings may vary. There are several reasons for the actual temperature. Roughly speaking, the effects are divided into two groups:

a) Random factors that may change in the short term
b) Systematic factors, which do not change over long periods

The effect of systematic factors is to introduce a fixed offset error to the thermometer reading. Random factors cause variability reading. Thermometer calibration determines offset values error and gathers information about the dispersion of measurements. This enables reliable temperature measurement with calibrated glass thermometer.

Measurand during Calibration

Glass thermometer readings can be described by a simple equation of observation

$$W_g[k] = T + D_g + E_{gr}[k] \tag{3}$$

where T is the actual temperature in the vicinity. For thermometers, D_g is a fixed offset error and $E_{gr}[k]$ is random errors. We use a different sign here for the fixed offset because this term is at the heart of the calibration process. Thermometers can be improved by applying corrections reading. So the metrics during calibration are D_g and A; calibration process is used to measure this offset. One way to think about D_g is to imagine a situation. There is no change in the thermometer reading (i.e., when $E_{gr}[k] = 0$). In this case, D_g will be the difference between the thermometer reading and the actual temperature T of the nearby area near the thermometer. There are many factors that affect thermometer calibration. Columns of mercury can separate, or high mercury droplets in capillaries, manufacturing quality may be unsatisfactory – e.g., tick marks may have incorrect format; otherwise, the thermometer may be very unstable.

Influencing Factors

There are many factors that affect thermometer calibration. Some are fatal because measurements should be aborted unless remedial action is taken. Major faults may be mercury column shedding, or high mercury droplets in capillary; build quality may be unsatisfactory — for example, dial markings may be poorly formed; otherwise, the thermometer may be extremely unstable due to improper annealing.

Unintended Effects: Operator errors in reading glass thermometers (e.g., parallax errors), reaction time between readings, etc.; irregularities in scale markings and apertures and stiction (sticking) of the mercury in the bore; randomness in the PRT display and instrumentation noise; and short-term instability of the bath temperature, etc.

Glass thermometers: Factors causing inaccurate thermometer readings are improper immersion of the thermometer and wrong orientation of the thermometer. Reference Thermometer: The accuracy of a platinum resistance thermometer (PRT) used as a reference is characterized by calibration. However, the thermometer must be used correctly, e.g., proper immersion should be taken care of. Bath Temperature Uniformity: A stirred bath produces a volume that is nearly uniform in temperature. Spatial uniformity of temperature can be assessed by periodic measurements at different locations and temperature ranges. This should be done as part of ongoing maintenance of the calibration equipment.

Control of Influencing Factors

Many influencing factors can be effectively managed through well-designed measurement procedures.

Make sure the operator waits enough time for the calibration bath to reach operating temperature before reading the thermometer:

(i) Note the correct immersion of the two thermometers and the orientation of the glass thermometer, which should be vertical.
(ii) The glass body of the thermometer can be tapped lightly before reading to avoid shrinkage.
(iii) Readings should be at right angles to the thermometer to avoid parallax errors.
(iv) Ensure that PRT readings are less random than glass thermometer readings (PRT reading variability should be $<$ one third of glass thermometer readings).

Modeling the Calibration Procedure

The measurand is determined as the offset term D Glass in equation (3). So we can start our model by rearranging the glass 44 Introduction to Measurement Uncertainty Thermometer observation equation for

$$D_g = W_g - E_{gr}[k] - T \tag{4}$$

This actually presents a calibration method. If the thermometer can be placed in an environment with a known temperature T, the average difference between the thermometer reading and T provides an estimate of Dglass. Our measurement method uses a heating bath to maintain a nearly uniform temperature in a volume of liquid. An electronic platinum resistance thermometer immersed in the bath was used to measure the temperature. Like glass thermometers, resistance thermometer readings fluctuate due to random influences and are also subject to a small fixed offset, which we attribute to residual calibration error $E_{\text{prt.cl}}$.

Observation equation for resistance thermometer readings: Reference thermometer is given by

$$W_{\text{prt}}[k] = T_{\text{prt}} + E_{\text{prt.r}}[k] + E_{\text{prt.cl}} \tag{5}$$

Note that T_{prt} is the temperature measured by a resistance thermometer. The temperature T near the glass thermometer is not necessarily equal to T_{prt}.

To account for this, we introduce an error term E_b representing a small temperature difference between the thermometer locations in the bath, so that

$$T_{\text{prt}} = T + E_b \tag{6}$$

From (5) and (6),
we get

$$W_{\text{prt}}[k] = T + E_b + E_{\text{prt.r}}[k] + E_{\text{prt.cl}}$$

$$\Rightarrow T = W_{\text{prt}}[k] - E_b - E_{\text{prt.r}}[k] - E_{\text{prt.cl}} \tag{7}$$

Equation (7) shows the reading of the resistance temperature detector. In this way, we get a model of how the glass thermometer deviates. Errors are readings from each thermometer and other thermometers.

Solving equation (4) and (7) for eliminating temperature T, we get

$$D_g = W_g[k] - W_{\text{prt}}[k] + E_{\text{prt.r}}[k] - E_{gr}[k] + E_{\text{prt..cl}} + E_{\text{b}} \tag{8}$$

We can simplify equation (8) by applying difference formulas

$$d[k] = W_g[k] - W_{\text{prt}}[k]$$

$$E_{d.r}[k] = E_{g.r}[k] - E_{\text{prt.r}}[k]$$

$$D_g = d[k] - E_{d.r}[k] + E_{\text{prt..cl}} + \text{E}_{\text{b}} \tag{9}$$

After calculating the average of sample differences

$$\overline{d} = \frac{1}{N}\sum_{k=1}^{N} d[k]$$

our final measurement model of the calibration is

$$D_g = \overline{d}[k] - E_{\overline{d}.r}[k] + E_{\text{prt..cl}} + E_{\text{b}} \tag{10}$$

$$u\left(E_{\text{prt..cl}}\right) = 0.0078$$

Example:
A sample thermometer readings and the differences are given as follows:

k	$w_g[k](Celcius)$	$w_{prt}[k](Celcius)$	$d[k](Celcius)$
1	79.850	79.847	0.003
2	79.840	79.840	0.000
3	79.830	79.830	0.000
4	80.060	80.050	0.010
5	80.130	80.124	0.006
6	80.130	80.125	0.005

The statistics of the sample differences are as follows:

$$\text{Sample mean} = 0.004$$

$$\text{Sample standard deviation} = 0.0038$$

$$\text{Sample standard error of mean} = 0.0016$$

Here $\overline{d} = 0.004\ ^\circ\text{C}$

In equation (10) , $E_{\overline{d}.r}$ is the error associated with our value of the sample mean due to variability in the readings.

$$E_{\overline{d}.r} = 0.000\ ^\circ\text{C}$$

Standard uncertainty associated with $E_{\overline{d}.r}$ is

$$u\left(E_{\overline{d}.r}\right) = 0.0016\ ^\circ\text{C}$$

Other Influence Factor Assessments
Residual error representing the difference between actual PRT and the ideal one

$$e_{\text{prt..cl}} = 0.000\ ^\circ\text{C}$$

PRT calibration certificate reports a standard uncertainty

$$u\left(e_{\text{prt..cl}}\right) = 0.0078\ ^\circ\text{C}$$

The maximum temperature difference between any two points within 100 mm of one other, in the bath's center, 0.0010 °C, according to routine bath observations. If we assume that each temperature variation is equally likely, we can correlate this range with a uniform distribution. The bath non-uniformity is connected with a standard uncertainty of

$$(e_b) = \frac{0.001}{\sqrt{12}} = 0.0029\ ^\circ\text{C}$$

Sample mean $\overline{d} = 0.004\ ^\circ\text{C}$

Three other influences, their estimates, and uncertainties are mentioned as follows:

Influence	Estimate	Uncertainty
$E_{\overline{d}.r}$	$e_{\overline{d}.r} = 0.000\ ^\circ\text{C}$	$u(e_{\overline{d}.r}) = 0.0016\ ^\circ\text{C}$
$E_{\text{prt..cl}}$	$e_{\text{prt.cl}} = 0.000\ ^\circ\text{C}$	$u(e_{\text{prt.cl}}) = 0.0078\ ^\circ\text{C}$
E_b	$e_b = 0.000\ ^\circ\text{C}$	$u(e_b) = 0.0029\ ^\circ\text{C}$

Evaluation of Result and Measurement Uncertainty

From the equation (10),

$$d_g = \overline{d}[k] - e_{\overline{d}.r}[k] + e_{\text{prt..cl}} + e_{\text{b}}$$

$$\Rightarrow d_g = 0.004\ ^\circ\text{C} - 0.000\ ^\circ\text{C} + 0.000\ ^\circ\text{C} + 0.000\ ^\circ\text{C}$$

$$d_g = 0.004\ ^\circ\text{C}$$

The combined standard uncertainty d_g as an estimate of D_g is

$$u(d_g) = \sqrt{u(e_{\overline{d}.r})^2 + u(e_{\text{prt.cl}})^2 + u(e_b)^2} \tag{11}$$

$$u(d_g) = \sqrt{0.0016^2 + 0.0078^2 + 0.0029^2} = 0.0084\ ^\circ\text{C}$$

Using the Glass Thermometer

It is the temperature T in the immediate region of the thermometer that is of interest when using a calibrated glass thermometer. Rearranging the observation equation (3)

$$T = W_g[k] - E_{\text{gr}}[k] - D_g \tag{12}$$

$$d_g = 0.004\ ^\circ\text{C}$$

with $u(d_g) = 0.0084\ ^\circ\text{C}$

Because positive and negative variations are expected to be equally likely, and there will be no tendency for one to occur more than the other; the estimate is $e_{g.\ r} = 0.00\ ^\circ\text{C}$. The sample standard deviation (estimated earlier, see 11.6.1) should be utilized to characterize the width of the Gaussian distribution representing $e_{g.\ r}$, because individual thermometer readings are now used to measure the temperature. As a result, $u(e_{g.\ r}) = 0.0038\ ^\circ\text{C}$ is the standard uncertainty.

$$t = w_g[k] - 0.000\ ^\circ\mathrm{C} - 0.0084\ ^\circ\mathrm{C}$$

with standard uncertainty

$$u(t) = \sqrt{u\left(d_g\right)^2 + u\left(e_{g.r}\right)^2} \tag{13}$$

$$\Rightarrow u(t) = \sqrt{0.0084^2 + 0.0038^2}$$

$$\Rightarrow u(t) = 0.0092$$

The value of d_g, which can be used to compensate for the thermometer offset error, and the uncertainty in a single corrected reading $u(t)$, from equation, are most likely to be of relevance to the thermometer's end-user (13).

It is important to note that one is not the same as the other's uncertainty!

Because the corrected reading uncertainty is more relevant to the client, a calibration laboratory may choose to report d_g and the uncertainty $u(t)$ rather than $u(d_g)$ from equation (11). When this is completed, the calibration report should clearly state the difference.

Relation Between the Width and Standard Deviation

When determining measurement uncertainty, the standard deviation of a distribution is always required. Formulas for converting a distribution's range into a standard deviation are as follows:

If width of the distribution is d, then

$$\text{Standard deviation of the uniform distribution} = \frac{d}{\sqrt{12}}$$

$$\text{Standard deviation of the triangular distribution} = \frac{d}{\sqrt{24}}$$

Observations of Variability

The effect of an influence factor can be seen directly as data variability. In such instances, a sample of data can be used to calculate the standard deviation of a distribution that explains the behavior. Gaussian distributions are frequently used to describe the variability encountered in test and measurement circumstances. Observations are considered to vary in this chapter due to the influence of a random error with a Gaussian distribution.

Samples and Estimates

Assume that the indicators in a measurement system change, but the quantity of interest Y remains constant.

The i^{th} indication is written as

$$V[i] = Y + E_r[i] \tag{14}$$

where $E_r[i]$ that give rise to variability have a Gaussian distribution with a mean of zero.

We will look at two methods for calculating Y: taking a single number or taking a sample of values and calculating the mean. E_r affects the outcome in both cases; the value y produced as an estimate of Y will always have an unpredictable error. In general, equation (1), which is repeated above, describes this circumstance.

In both circumstances, a Gaussian distribution with a mean of zero best describes the behavior of the error E_r. The standard deviation that scales the distribution width, on the other hand, will be determined by how y is collected. To evaluate this parameter, a sample of n observations $V[1]$, $V[2]$, …. $V[n]$ is required.

A Single Observation and the Sample Standard Deviation

We can write simply that when a single indication is used as an estimate of Y (the index i is no longer required), we can write simply $y = V$

The standard deviation of the distribution

$$S_n = \sqrt{\frac{1}{n-1}\sum_{i=1}^{n}\left(\widehat{V} - V[i]\right)^2}$$

where sample mean $\widehat{V} = \frac{1}{n}\sum_{i=1}^{n} V[i]$

Example 1: Variable Current Readings

The following are 10 current readings measured from an ammeter:

2.998 mA, 3.001 mA, 3.002 mA, 2.999 mA, 3.003 mA
2.998 mA, 3.001 mA, 3.000 mA, 2.999 mA, 3.002 mA

Sample mean

$$\widehat{V} = \frac{1}{n}\sum\nolimits_{i=1}^{n} V[i]$$

$$\widehat{V} = \frac{1}{10}[30.003]$$

$$\widehat{V} = 3.0003$$

$$S_n = \sqrt{\frac{1}{n-1}\sum_{i=1}^{n}\left(\widehat{V} - V[i]\right)^2}$$

$$S = 0.00176$$

Sample Mean and the Standard Deviation of the Mean

The sample mean yields a more accurate estimate of Y than a single indicator.

If $\widehat{V} = \frac{1}{n}\sum_{i=1}^{n} V[i]$ then the standard deviation of the distribution describing E_y is

$$SE_{\widehat{V}} = \frac{S_n}{\sqrt{n}}$$

The statistic value $SE_{\widehat{V}}$ is called standard error of the mean. S_n is standard deviation.

Example 2: Variable Current Readings

Below are the sample means of ten samples, each with ten observations:

2.99978 mA, 3.00027 mA, 3.00032 mA, 2.99953 mA, 3.00043 mA
2.99968 mA, 3.00041 mA, 3.00029 mA, 2.99951 mA, 3.00026 mA

The standard error of the 10 samples from Example 1 is

Is $\frac{S_n}{\sqrt{n}} = \frac{0.00176}{\sqrt{10}} = 0.00056$ mA

This value is closed to the standard deviation of the above 10 values 0.00046 mA.

Variability and Offset

A pair of impact factors can often be used to mimic the indications of a basic measuring instrument: one, E_{random}, changes with each indication, while the other, E_{offset}, remains constant. The ith indicator in this example can be written as

$$V[i] = Y + E_o + E_r[i] \tag{15}$$

where $E_o = E_{\text{offset}}$ and $E_r = E_{\text{random}}$

When this happens, the methods provided in sections "Identifying Influence Quantities in Measuring Simple Length by Using a Calibrated Steel Ruler" and "Calibrating a Thermometer" for measuring Y are no longer appropriate: they would estimate the value of $Y + E_o$, not Y. A new method of measuring the offset must be devised. One popular method is to compare the instrument's indications with those from a more accurate one. For example, the fixed offset of a thermometer being calibrated is determined by comparing thermometer readings to readings from a more accurate reference thermometer.

Sampling Theory

It is critical to gather a representative sample of the variability of an influence quantity when collecting observations. The effect of mechanical flaws in the bore of a thermometer is an influencing factor that requires careful planning to ensure that a suitable range of circumstances is covered when observations are gathered. By taking readings across an acceptable temperature range, a representative sample of data impacted by this factor can be acquired, with the changing height of the liquid column in the bore sampling the irregularities that affect measurements.

The Theory of Uncertainty

Lord Kelvin, the famed scientist, observed in 1889 "You know something about it when you can measure it and describe it in numbers; else, your knowledge is of a limited and unsatisfying kind." This well-known quote emphasizes the significance of measurement and measurement science. The term measurement refers to both the procedure for obtaining the number described by Lord Kelvin and the number itself. This figure, which is nothing more than the measurement's outcome, is extremely important. Consider what we do on a daily basis. Everything in our knowledge has a number linked with it: the weights of apples, the height of a tree, the length of a table, etc.

The theory of error was replaced by the theory of uncertainty at the end of the twentieth century. The uncertainty is a property of the measurement result that shows the lack of perfect knowledge about the value of the measurand, as reported in [ISO93]. The publication of the IEC-ISO "Guide to the Expression of Uncertainty in Measurement" (GUM) [ISO93] in 1992 marked the official transition from error to uncertainty. The traditional concept of mistake vanishes.

The question now is how to estimate such a probability density function in order to estimate the standard uncertainty as well. The most direct method is to try it out: The technique is done numerous times, and the standard deviation of the acquired measurement data is calculated. However, due to practical and financial constraints, this approach takes time and cannot always be followed. On the other hand, it is widely acknowledged that the probability density function associated with measurement findings can frequently be assumed based on a priori knowledge, such as a calibration certificate or the operator's experience. Both of these methods for estimating the probability density function and the associated standard deviation are equally accurate, according to the GUM, which classifies them as type A and type B uncertainty evaluations.

Type A evaluation of the standard uncertainty is obtained by means of statistical analysis of series of observations. Suppose n different measurements $x_1, x_2, \ldots x_n$ are obtained under the same conditions of measurement. We know that the arithmetic of n observations $\widehat{x}$ is the best estimate of the expected value of the quantity x.

The best estimate of the variance of the mean

$$S^2(\widehat{x}) = \frac{1}{n(n-1)} \sum_{i=1}^{n} (x_n - \widehat{x})^2$$

Standard deviation of the mean $= S(\widehat{x})$

The standard uncertainty is calculated in type B by making a judgment based on all relevant information about the likely variability of quantity x. Previous measurement data, experience or general knowledge, manufacturer's specifications, data provided in calibration certifications, and so on are examples of available information.

The available information leads to the conclusion that the measurement result distributes according to a rectangular distribution with base $2b = b_+ - b_-$; then the expected value of

$$x = \frac{(b_+ - b_-)}{2}$$

Standard uncertainty is $s(\widehat{x}) = \frac{b}{\sqrt{3}}$

If it is known that the measurement result is distributed according to a symmetric trapezoidal distribution with a base of width $2b = b_+ - b_-$ and a top of width $2b\ \beta$, where $0 \leq \beta \leq 1$ is the standard uncertainty, the standard uncertainty becomes

$$s(\widehat{x}) = \sqrt{\frac{b^2\left(1+\beta^2\right)}{6}}$$

When $\beta = 1$, the trapezoidal distribution transforms into a rectangle distribution $s(\widehat{x}) = \frac{b}{\sqrt{3}}$.

When $\beta = 0$, the trapezoidal distribution transforms into a triangular distribution $s(\widehat{x}) = \frac{b}{\sqrt{6}}$.

The definitions above apply to uncertainty estimates in direct measurements, or measurement processes where the measurand may be directly measured. Several quantities, on the other hand, are measured indirectly by appropriately processing the findings of previous measurements. It is crucial that, like the error idea, the uncertainty concept is applied to these indirect measurements and that the uncertainty associated with the final measurement result is calculated using the uncertainty values associated with each processed measured value. When the standard uncertainty is taken into account, the final measurement result can be connected with a combined standard uncertainty as shown further.

Let Y be the measurand, and $X_1, X_2, \ldots X_n$ be the measurable values on which the measurand is dependent, and $Y = f(X_1, X_2, \ldots X_n)$ be the mathematical relationship between them. The expectations $e_1, e, \ldots e_n$ of quantities $X_1, X_2, \ldots X_n$ determine the expected value y of quantity Y.

Furthermore, assuming f is a linear function, the measurement result y's combined standard uncertainty $U_c(y)$ can be calculated as follows:

$$U_c(y) = \sqrt{\sum_{i=1}^{n}\left(\frac{\partial f}{\partial e_i}\right)^2 U^2(e_i) + 2\sum_{i=1}^{n-1}\sum_{j=i+1}^{n}\frac{\partial f}{\partial e_i}\frac{\partial f}{\partial e_j}U\left(e_i, e_j\right)} \qquad (16)$$

where $\frac{\partial f}{\partial e_i}$ is a partial derivative of f with respect to e_i that is called sensitivity coefficient, $U(e_i)$ is called standard uncertainty, and $U(e_i, e_j)$ is called estimated covariance of e_i, e_j that can be calculated from the formula $U(e_i, e_j) = U(e_i)\ U(e_j)\ \emptyset\ (e_i, e_j)$.

Here $U(e_i)$, $U(e_j)$ are called standard uncertainty with respect to e_i, e_j, respectively, and $r(e_i, e_j)$ is a correlation coefficient between e_i and e_j .

We know that the correlation coefficient satisfies the following properties:

$$-1 \leq r\left(e_i, e_j\right) \leq 1$$

and

$$r\left(e_i, e_j\right) = r\left(e_j, e_i\right).$$

Note: If $r(e_i, e_j) = 0$, then the quantities are uncorrelated.

Equation (16) is the first-order Taylor-series approximation of $Y = f$ if $f(X_1, X_2, \ldots X_n)$ is not a linear function in the most practical cases; however, it can still be taken into account while evaluating$U_c(y)$. In truth, the measurement uncertainty is often minimal in relation to the observed value in measurement practice, resulting in small fluctuations in the measurand. This indicates that the linearity condition of f is usually verified locally, close to the measurement location, and Eq. (16) remains true. Higher-order terms in the Taylor series must be included in Eq. (16) when the nonlinearity of f is considerable.

It is now widely accepted that the standard uncertainty (and the combined standard uncertainty when indirect measurements are taken into account) achieves the primary goal of characterizing "the dispersion of the values that could reasonably be attributed to the measurand without any need to refer to the unknown and unknowable true value of the measurand, provided that the sole causes for this dispersion are random."

However, there is one remaining issue for which there is no obvious answer in terms of standard uncertainty. "In many industrial and commercial applications, as well as in the areas of health and safety, it is frequently necessary to provide an interval about the measurement result within which the values that can reasonably be attributed to the quantity subject to measurement can be expected to lie with a high level of confidence," according to the GUM. As a result, the ideal approach for evaluating and expressing measurement uncertainty should be capable of quickly generating such a confidence interval, preferably one that matches to the desired level of confidence in a realistic manner.

The confidence interval of a measurement result Y is usually expressed as an interval centered on the result $(y \pm U)$ and can only be determined if the probability distribution of the result is known and the desired level of confidence is set. In fact, if $p(y)$ is Y's probability distribution, its mean and standard deviation are given by

$$\overline{y} = \int_{-\infty}^{+\infty} yp(y)dy \tag{17}$$

$$\sigma_{p(y)} = \sqrt{\int_{-\infty}^{+\infty} (y - \overline{y})^2 p(y) dy} \tag{18}$$

Now c is the required level of confidence, the associated confidence can be found by using solution of the integral

$$c = \int_{\overline{y}-U}^{\overline{y}+U} p(y) dy \tag{19}$$

The confidence interval's semi-width U, also known as enlarged uncertainty, is usually expressed as a multiple of the standard uncertainty:

$$U = k_c \, u(y)$$

This indicates that the difficulty in obtaining k_c determines the challenge of determining a confidence interval. The multiplying factor k_c, also known as the coverage factor, is, of course, strictly dependent on the probability distribution p (y) and the level of confidence c. If p(y) is a normal distribution, the coverage factors correspond to the most typically used confidence intervals as follows:

$k_c = 1$ for $c = 68.27\%$
$k_c = 1.645$ for $c = 90\%$
$k_c = 1.96$ for $c = 95\%$
$k_c = 2$ for $c = 95.45\%$
$k_c = 2.576$ for $c = 99\%$
$k_c = 3$ for $c = 99.73\%$

If the measured quantity Y can be measured directly, a type A or type B assessment can be used to determine the probability distribution of the result. This endeavour can be challenging at times, but it usually results in a solution. When the measured quantity Y, on the other hand, can only be measured indirectly, determining the probability distribution of the result becomes more complex.

If the probability distributions of the input variables $X_1, X_2, \ldots X_n$ are known, and the relationship between Y and its input quantities is a linear function, the probability distribution of Y may be easily calculated via a sequence of convolutions, on the other hand, relationship f is nonlinear and a first-order Taylor expansion is not a good approximation. In these situations, we need numerical methods such as Monte Carlo one.

However, convolution of probability distributions is difficult, and Monte Carlo simulation is time-consuming. However, it is not always essential to deal with all probability distributions, and knowing the standard deviations is sufficient to estimate the proper coverage factor for a given level of confidence. This occurs whenever the Central Limit Theorem's assumptions are met, which is one of probability theory's most astounding results.

Central Limit Theorem

Theorem 1: Let independent random variables X_1, X_2, ... X_n share the same probability distribution. Assume that the expected value μ and the standard deviation σ exist and are finite. Consider the sum $S_n = X_1 + X_2 + \ldots + X_n$. Then, the expected value S_n is $n\mu$ and the standard deviation is $\sigma\sqrt{n}$. Furthermore as n approaches to ∞, the distribution of $N(n\mu, n\sigma^2)$. It is known that the outcome of the indirect measurement follows a certain normal distribution under the assumptions of this theorem; so, no computations are required, and the coverage factors may be found on tables.

Furthermore, it can be shown that the Central Limit Theorem's thesis is still valid when the hypotheses are less tight. In reality, there is a generalization of the Central Limit Theorem that does not need the initial random variables to have similar distributions. Of course, this modified form of the Central Limit Theorem necessitates certain extra criteria, such as ensuring that no single variable has a significantly greater impact than the rest.

The Central Limit Theorem guarantees that the random variable representing the ultimate result $Y = f(X_1, X_2, \ldots X_n)$ approximates a normal distribution under this condition. The standard deviation and mean value of this normal distribution are given by $y = f(x_1, x_2, \ldots x_n)$.

Furthermore, as the number of input rises, the approximation improves; convergence is faster as the variance values go closer to each other; and the closer the distributions X_i are to normal ones, the fewer X_i are needed to generate a normal distribution for S_n. The following example is provided to demonstrate how this concept holds true in practice.

Let us have a look at the sum of rectangular distributions, which is an important non-normal distribution. The probability density function of the sum is generated theoretically by convoluting the starting distributions, as indicated. Figure 7 depicts the convolution (dot line) of four rectangular distributions with an expected value of $\mu = 10$ and a width of 0.2, corresponding to a standard deviation of $\sigma = \frac{0.1}{\sqrt{3}}$ (given width $2b = 0.2$). Because this convolution is almost normal (solid line), the validity of the Central Limit Theorem is easily demonstrated. As a result, the convolution of only four equal-width rectangular distributions is essentially normal.

Figure 8 depicts the convolution (dot line) of four rectangular distributions with the same expected value of 10 but varying widths (0.14, 0.06, 0.4, and 0.6). It should be emphasized that four rectangular distributions are insufficient to approximate a normal distribution in this circumstance (solid line).

On the other hand, as shown in Fig. 9, seven rectangular distributions may be sufficient. In this scenario, the distributions have an expected value of 10 and widths of 0.14, 0.06, 0.4, 0.6, 0.1, 0.03, and 0.2, which are all one order of magnitude apart.

This example illustrates, intuitively, that the values attributed to the measured, when obtained through an indirect measurement method, distribute according to a pretty normal distribution, given that the Central Limit Theorem premise is satisfied (at least at a first approximation). As a result, the standard deviation of such a distribution is equal to the combined standard uncertainty provided by Eq. (16),

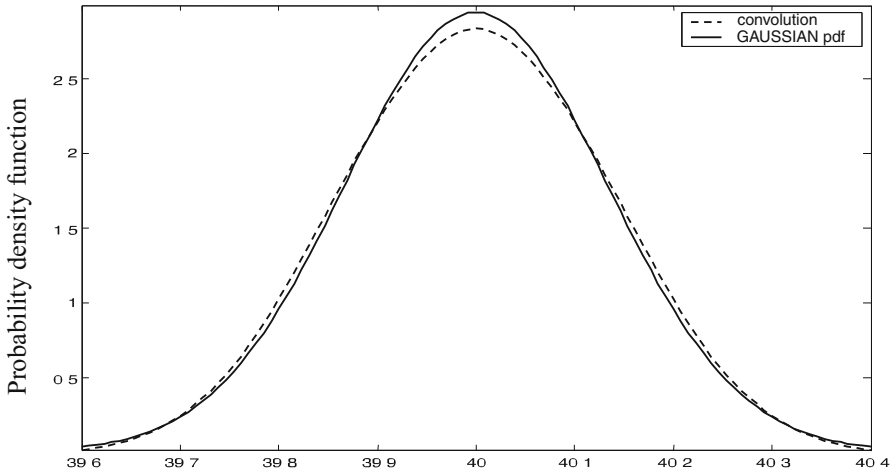

Fig. 7 Convolution of four rectangular distributions having the same mean and variance

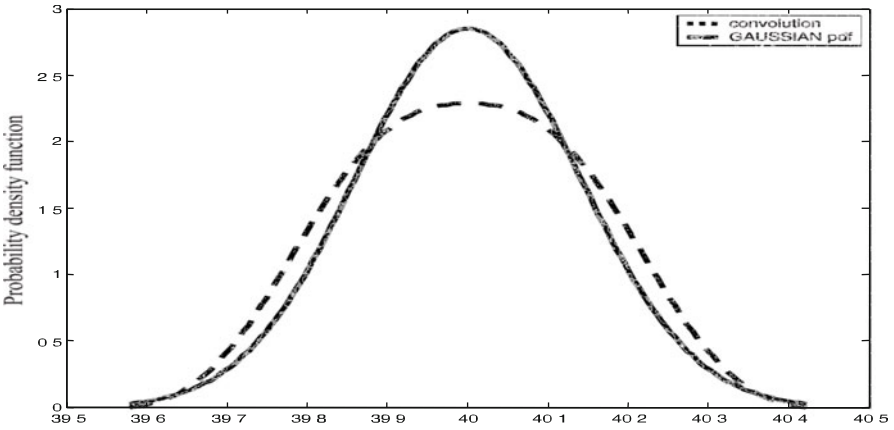

Fig. 8 Convolution of four rectangular distributions having different widths

and any required confidence interval with assigned confidence level can be easily obtained by multiplying this standard deviation by a coverage factor typical for the normal distribution.

Fuzzy Variables and Measurement Uncertainty

Section "The Theory of Uncertainty" provided an overview of the basic concepts of measurement theory. In particular, the results of the measurements represent incomplete knowledge of what is being measured, and this knowledge has been shown to be meaningful only if that "imperfections" can be estimated and quantified in some

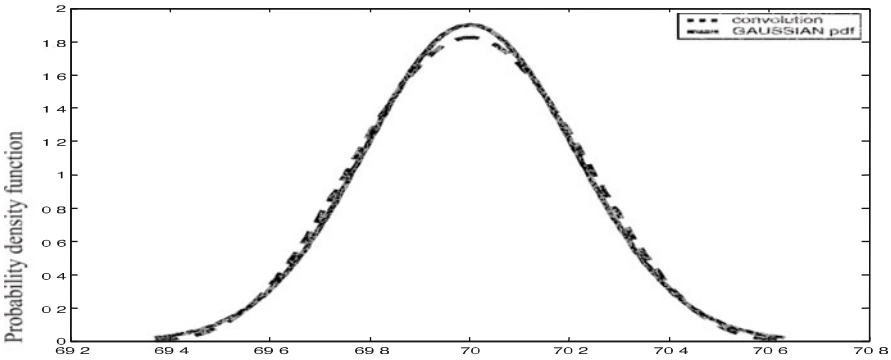

Fig. 9 Convolution of seven rectangular distributions having different widths

way. This approach also shows that proper mathematics is needed to deal with incomplete knowledge from a quantitative point of view. Current practice is primarily related to probability theory as it is the most well-known and most valuable mathematical theory dealing with incomplete knowledge. However, probability theory deals only with this particular kind of incomplete knowledge that results from random effects. As we saw in the previous chapter, this means that all other recognized and significant effects, including systematic effects, are fully compensated. Systematic effects always affect the measurement process with the same value and sign. If this value is known accurately, it will be treated as an error, and the correction may correct the measurement result during the measurement process. Of course, this situation generally does not occur.

Under normal circumstances, the presence of a systematic contribution is detected, but even if it is possible to find it within a closed interval of $\mathfrak{R}$, its exact value is unknown. By definition, the systematic contribution is unknown, but it always takes the same value within this estimated interval. This means that not all values of the interval have the same probability of occurrence, but without further evidence, there is no better value than the other values, so the probability of occurrence is the same. This situation can also be used to represent a more general case. In many real-world situations, there are signs that unknown effects affect measurement results. The only information available shows that the contribution to this unknown effect is at a certain level of confidence and within a certain interval, but it is unclear where it is.

Furthermore, nothing else can be assumed, even if the effect is systematic or random. This situation is called complete ignorance. Although the two types of contributions are represented the same way – confidence intervals – they should not be confused. In fact, when utter ignorance is considered, there is no other information; conversely, when looking at a systematic contribution, important information is available, i.e., the systematic behavior of the contribution itself. This additional information should be carefully considered when modeling the measurement process.

Consider the following simple example. Let w_1 and w_2 be two measurement results for the same quantity, such as the weight of an object obtained by the method of double balancing using a balance bridge. The final result of the measurement is said to be obtained as the average of w_1 and w_2. Assume that there is no random contribution, but only a contribution of a different nature, whose possible values are within $\pm$ 100 g. Two cases can be considered:

Case I: No additional information is available for this contribution to the uncertainty, so it must be considered completely ignorant. In this case, the following applies: and the uncertainty associated with the result is the same as the original uncertainty.

$$w = \frac{w_1 \pm 100g + w_2 \pm 100g}{2}$$

$$\Rightarrow w = \frac{w_1 + w_2}{2} \pm 100g$$

Case II: The additional information available suggests that the cause of the uncertainty is the different lengths of the two weighbridge beams. In this case, given the available information about this contribution to the uncertainty, we know that it affects w_1 and w_2 of the same absolute value. Contribution can therefore be described as systematic, as following:

$$w = \frac{w_1 \pm 100g + w_2 \pm 100g}{2}$$

$$\Rightarrow w = \frac{w_1 + w_2}{2}$$

and the final result of the measurement has an associated uncertainty of zero because, in the arithmetic mean, the contribution under consideration is offset.

This simple example demonstrates the importance of using all available information when modeling a measurement process. Indeed, if additional information about the reason for the system behavior is not taken into account, the measurement uncertainty of the final result will be overestimated.

Fuzzy variable: Thus, it can be said, also from a mathematical point of view, that a systematic contribution is a special case of complete ignorance, where additional information is added. If this additional information is provided, together with the reasons for the presence of the system contribution itself, this information can sometimes be used to convey the appropriateness and accuracy of the information. Measurement is not guaranteed through the measuring process, as in the case of a dual balance with a weighbridge. The goal of this chapter is to find a mathematical object capable of representing complete ignorance and its specific cases.

When probability theory is considered to solve the problem of incomplete knowledge, the total ignorance is represented by a random variable with a uniform probability distribution within the estimated confidence interval. However, this assumption is incompatible with the concept of complete ignorance. In fact, let us first consider the general case of complete ignorance. In this case, by definition, no assumptions can be made about the actual probability distribution, and all probability distributions are, in theory, possible. Therefore, assuming a uniform probability distribution means arbitrarily adding information that is not available. Let us now consider the special case of a systemic effect. In this case, by definition, a single unknown value has a 100% probability of occurring, while all other values have a zero probability. Therefore, assuming a uniform probability, which implies that all values in a given interval have the same probability of occurring, leads to misinterpretation of the available information. Of course, assuming any other probability distribution leads to similar considerations. It can therefore be concluded that probability theory and random variables cannot represent incomplete knowledge, whenever this is due to uncertain effects that are not explicitly random. Therefore, different mathematical variables must be considered for this purpose. In the second half of the twentieth century, fuzzy variables were introduced to represent incomplete knowledge. This approach is less well known than the statistical approach, and it is not included in current standards; therefore, an entire chapter is devoted to this approach.

Definition of Fuzzy Variables

Fuzzy variables and fuzzy units were extensively used, within the remaining decades, specifically within the field of automated controls, after Zadeh brought the fundamental concepts of fuzzy common sense and approximate reasoning [Z65]-[Z73]-[Z75]-[Z78]. In the conventional mathematical approach, a variable might also additionally best belong or now no longer belong to the set into which it is miles defined. The characteristic describing the club of this sort of crisp variable to its appertaining set can consequently take best the price 1, if the variable belongs to the set, or zero, if the variable does now no longer belong to the set.

When fuzzy variables and fuzzy units are taken into consideration, the characteristic describing the club of a variable to its appertaining set is permitted to take all values within the $0 - 1$ interval. This method shows that the referential set $\mathcal{R}$ of the actual numbers, a fuzzy variable X is defined with the aid of using its club characteristic $\mu X(x)$, in which $x \in \mathcal{R}$. The club characteristic of a fuzzy variable satisfies the subsequent properties:

$$0 \leq \mu X(x) \leq 1$$

(i) $\mu X(x)$ is convex.
(ii) $\mu X(x)$ is normal (this means that at the least one detail x usually exists for which $\mu X(x) = 1$).

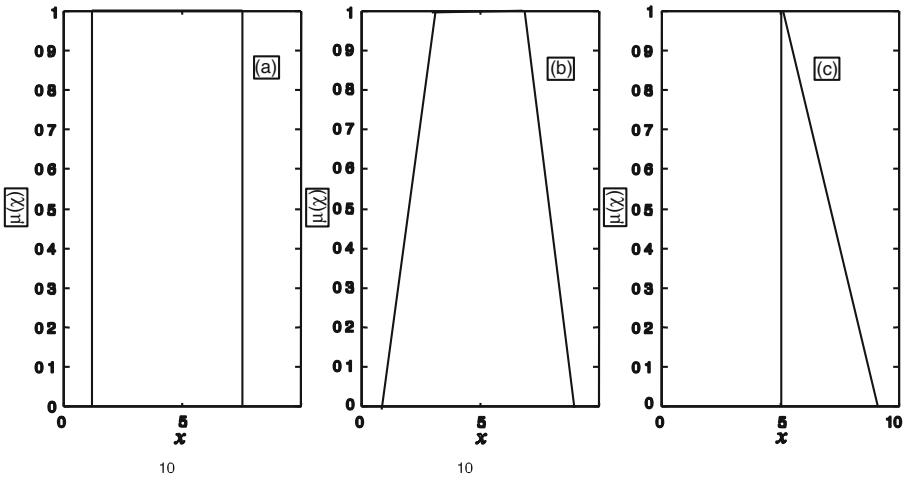

Fig. 10 Example of membership functions: (**a**) rectangular, (**b**) trapezoidal, and (**c**) triangular

Figure 10 indicates a few examples of club capabilities: the square one, the trapezoidal one, and the triangular one. Of course, different shapes are allowed, which do now no longer want to be symmetric. Moreover, club capabilities, that is, best growth or best lower, qualify as fuzzy variables and are used to symbolize the idea of "huge number" or "small number" within the context of every specific application. An instance of this sort of club characteristic is given in Fig. 11.

The club characteristic $\mu X(x)$ of a fuzzy variable may be additionally defined in phrases of α-cuts at different vertical degrees α. As the club characteristic of a fuzzy variable ranges, with the aid of using definition, among zero and 1, its α-cuts are defined for values of α among 0 and 1.

Each α-cut at generic level α is defined as

$$X_\alpha = \{x/\mu X(x) \geq \alpha\} \tag{20}$$

According to Equation (20), every α-cut defines in an interval $[x_1{}^\alpha, x_2{}^\alpha]$, in which it is far usually $x_1{}^\alpha \leq x_2{}^\alpha$. The equality of $x_1{}^\alpha$ and $x_2{}^\alpha$ may be reached simplest for $\alpha = 1$ and simplest if the club feature has a unmarried height cost, for instance, within the case of the triangular club feature stated in Figure 10 (iii). Generally, $x_1{}^\alpha$ and $x_2{}^\alpha$ take finite values; however, in a few cases, as shown in

Figure 11, it may be $x_1{}^\alpha \rightarrow -\infty$ and/or $x_1{}^\alpha \rightarrow +\infty$. If, for any value of α, i t is $x_1{}^\alpha = x_2{}^\alpha$, then the bushy variable degenerates right into a crisp variable. An instance of α-cut, for stage $\alpha = 0.3$, is given in Figure 12.

The significance of representing a fuzzy variable in phrases of its α-cuts is that the α-cuts of a fuzzy variable and the corresponding degrees α may be taken into consideration as a fixed of periods of confidence and related degrees of certitude. The stage of certitude consists of data approximately how sure someone is ready its expertise. If someone, for instance, recollects precisely the birthday of a friend, his

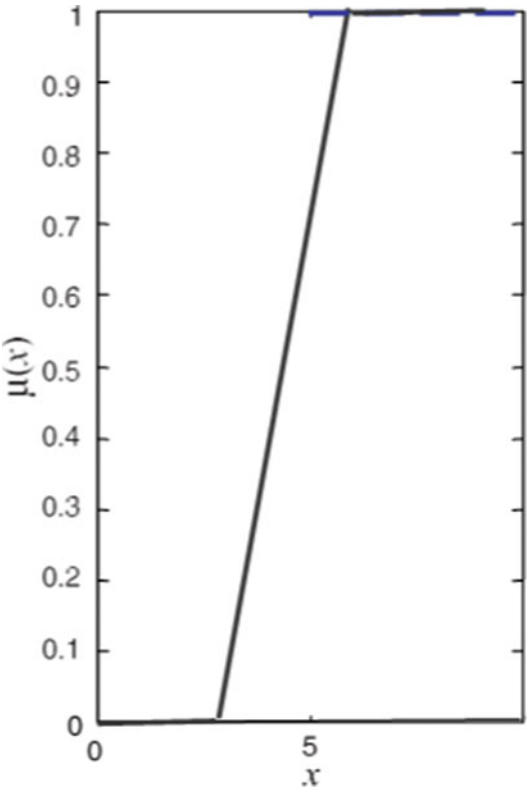

Fig. 11 Membership function describing the fuzzy statement 'more than 5

expertise is sure; however, if he simply recollects the month, however now no longer the day, the knowledge of his expertise is lower. In the first case, the available expertise may be represented through a crisp cost, that is, the precise day of the year (March, 16th); within the 2d case, the available expertise may be represented through a fixed, which includes the 30 days of March. Hence, as the extent of certitude increases, the width of the corresponding confidence c language decreases. As truly proven through the preceding instance, the hyperlink among the extent α of certitude and the confidence c language on the identical stage corresponds to the natural, regularly implicit, mechanism of human wondering within the subjective estimation of a cost for a measurement. The following instance suggests how a fuzzy variable may be built, beginning from the data, or private idea, of one, or greater human beings. Let us take into account once more the birthday of a friend, who is Shyam. Ram is positive that Shyam's birthday is March 16th; John is not so positive approximately about the date, despite the fact that he recollects nicely that the birthday falls in April. From a mathematical factor of view, the c language of confidence envisioned through Ram is [March 16, March 16], while the c language of confidence envisioned through Kesav is [March1, March 30]. The degrees of certitude related to the two periods are 1 and 0, respectively.

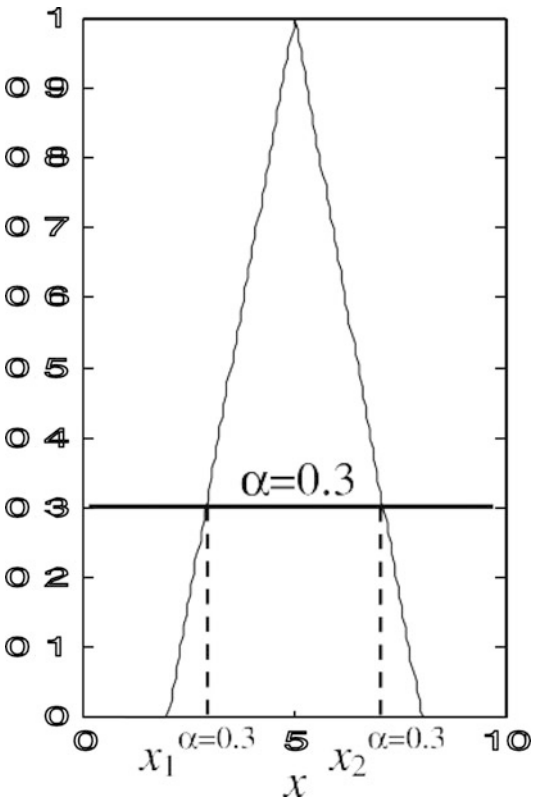

Fig. 12 α-Cut of a triangular membership function for level $\alpha = 0.3$

If now a fuzzy variable needs to be related to the available data, the subsequent applies. Let us think the simplest available data is from Ram. In this case, the available data may be represented through a crisp cost. In fact, on this case, complete actuality is given.

The following instance indicates how a fuzzy variable may be built, beginning from the statistics, or private idea, of one, or extra human beings. Let us recollect once more the birthday of a friend, who is Sharan. Tarun is certain that Sharan's birthday is May 12th; Jagan is not always so certain approximately about the date, even supposing he recollects nicely that the birthday falls in May. From a mathematical factor of view, the c programming language of self-belief expected with the aid of using Tarun is [May 12, May 12], while the c programming language of self-belief expected with the aid of using Jagan is [May2, May 24]. The degrees of certitude related to the two periods are 1 and 0, respectively. If now a fuzzy variable needs to be related to the available statistics, the subsequent applies. Let us think the hardiest available statistics is from Tarun. In this case, the available statistics may be represented with the aid of using a crisp value. In fact, on this case, complete fact is given. Let us now think the hardiest available statistics is from John. In this case, the

available statistics may be represented with the aid of using a square fuzzy variable, just like the one in Fig. 10a. However, it is also viable to mix the statistics given with the aid of using Tom and Jagan. As the extent of certitude related to Jagan's c programming language is 0 and people related to Tarun's c programming language is 1, those periods may be taken into consideration because of the α-cuts of the bushy variable, which represents Sharan's birthday, at degrees of α 0 and 1, respectively. Since, as the extent of certitude will increase, the c programming language of self-belief turns narrower, the bushy variable may want to be, for instance, triangular, just like the one in Fig. 10c. This easy instance additionally indicates, in an intuitive way, a courting between the extent of certitude and the extent of self-belief, which, in accordance to the theory of uncertainty, ought to continually be related to a c programming language of self-belief. The degree of certitude suggests how an awful lot someone is certain approximately a positive event. If, for instance, the instance of Sharan's birthday is considered once more, because of this the surer someone is set the birthday's date, the smaller is the estimated variety of days. On the opposite hand, the extent of self-belief suggests the possibility of a positive event. Therefore, considering the identical instance of Sam's birthday, the smaller the given variety of days, the smaller the possibility that Sam's birthday falls inside the ones dates. If the hardiest one date is given, the possibility that it is precisely Sharan's birthday is 0, as additionally proven with the aid of using Eq. (19). In different words, despite the fact that the extent of certitude will increase because the width of the self-belief c programming language decreases, the alternative applies to the degrees of self-belief. Hence, the degrees of self-belief identical to at least 1 and 0 are assigned to periods [May 2, May 24] and [May 12, May12], respectively. Moreover, intuitively, it is able to additionally be assessed that, given the self-belief c programming language at degree α, the related degree of self-belief is 1. α1[FS03, FGS04, FS04, FS05a, FS05b, FS05c].

The use of fuzzy variables within the context of size uncertainty is mainly interesting, if we do not forget that the maximum beneficial manner to specific end result of a size is in phrases of self-assurance periods, and "the precise approach for comparing and expressing size uncertainty has to be capable of conveniently supplying the sort of self-assurance interval" [ISO93]. Therefore, it could be said that a fuzzy variable may be correctly hired to symbolize the result of a size, as it affords all available statistics approximately the end result itself: the self-assurance periods and the related degrees of self-assurance. The true size exercise additionally calls for that the uncertainty of a size end result is immediately usable as an element in comparing the un-actuality of some other size wherein the primary end result is used [ISO93]; that is, size uncertainties should be composed among every other. Thus, mathematics operations among fuzzy variables need to be defined that allows you to make sure that fuzzy variables are capable of propagating size consequences and associated uncertainties.

Conclusion

This chapter gives an overview of measurement uncertainty analysis, with a focus on modeling. It is observed how the measurement modeling is vital for data processing and uncertainty calculations. It is also observed how these models are valuable in other ways. Both describe the measurement process and can be used as a tool for understanding, refining, and implementing it. They can provide insight into the strengths and weaknesses of a measurement process and suggest ways to improve it. Measurement models must be documented, and the documentation must be kept up to date. Words must be specified properly, and assumptions about influence quantities must be recorded. All too frequently, these subtleties are glossed over or completely ignored. The validity of a measurement model will frequently be scrutinized in accredited quality assurance environments, and adequate evidence to support a model will be necessary. Model documentation should be written in a way that other metrology professionals who are not necessarily experts in the field may understand it.

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Evaluation and Analysis of Measurement Uncertainty

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Methodologies, Implications, and Future Prospects

H. Gupta, Shanay Rab, and N. Garg

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Abstract

Reliability of measurement data is particularly important in today's world of globalization and trading across international borders. The acceptable and meaningful way of expressing the measurement result is to give the average value along with a range called measurement uncertainty at a confidence level within which the true value shall lie. Measurement uncertainty has specifically gained more importance with emphasis on decision rule as the decision-making criteria

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in ISO 17025: 2017 for accepting or rejecting a given sample. It is important for laboratory personnel to have knowledge in statistics apart from expertise in their field of testing and calibration to be able to correctly determine the factors that would contribute to the measurement uncertainty and the values of such contributions thereof. This chapter provides an overview of the evaluation and analysis of measurement uncertainty. The study also describes the strengths, weaknesses, opportunities, and threats (SWOT) analysis of the Monte Carlo simulation (MCS) in the evaluation of measurement uncertainty. Furthermore, the chapter also summarizes the implications and future prospects associated with the uncertainty methodology and recommends the wide usage of this in solving problems in various scientific filed.

Keywords

Metrology · Measurement uncertainty · Probability distribution function · SWOT · Calibration

Introduction

Measurement is an important activity when talking about problem-solving, quality, monitoring, or safety and in design and assembly to make something fit. Reliability of measurement data becomes even more important in today's world of globalization and trading across international borders. Manufacturers seeking an international market would like their product parameters to be acceptable everywhere around the globe. This is where testing and calibration results play an important role. The way these results are expressed shows confidence in the measurement (Aswal 2020; Grabe 2018; Leach and Smith 2018; Magas 2019; Chrysochoos and Surrel 2013).

It is a general practice to take multiple readings for a given measurement, and in almost all cases, it would be found that each reading is not identical. It is known that a measurand has a unique true value. Choosing one of the multiple values as the correct one becomes the next question since each value has been measured under generally the same circumstances of environmental conditions, test method, equipment, and personnel skills. The average (mean) value is generally an acceptable answer to the value of measurand among the multiple readings found. It is assumed that the true value is spread around this average value since the true value itself cannot be determined. The acceptable, meaningful way of expressing the measurement result is to then give the average value with a range that hovers around this value within which the true value shall lie. This range is called measurement uncertainty which has a stated confidence level based on a scientific data management system of statistical analysis (White 2008; Tosello and De Chiffre 2004; Rab and Yadav 2022). The details of the measurement uncertainty and different approach for its quantification are discussed in the subsequent sections.

Uncertainty of Measurement

In metrology (measurement science), uncertainty of measurement (UoM) plays an important role in the area of testing and calibration since it shows the limits around the average value where the true value will lie and also the confidence level of these limits. These limits can be a deciding factor for accepting or rejecting a given sample for its measurement value reported. This has specifically gained more importance with emphasis on decision rule as decision-making criteria for this purpose in ISO 17025: 2017 (Bell 2001; Rab et al. 2019; Yadav 2007). The measurement result is expressed as the following:

$X \pm MU$ at y% of confidence level

where X is the measurement value and MU is the measurement uncertainty value associated with the measurement value at a confidence level of y (%). The confidence level gives the probability with which the true value will lie in the given interval.

For example, the value of the initial setting time of a cement sample determined using test method IS 4031 (Part 5) is 125 min, and measurement uncertainty of 5 min is associated with this measured value at a 95% confidence level. Hence, the measurement result will be expressed as follows:

125 min $\pm$ 5 min at 95% confidence level

Law of propagation of uncertainty (LPU) takes into account the propagation of uncertainties of various variables involved in determining a measurand into its measurement uncertainty. Different observers, instruments, samples, laboratories, and times at which observations are made, say different days are sources of variation or uncertainty in the value of measurand.

Although determining measurement uncertainty is based on systematic statistical analysis, it also involves sound scientific, technical knowledge, and judgment of the product under consideration. Such knowledge is important to decide which factors related to the test/calibration method would contribute to the measurement uncertainty of the reported value. Any mistake in the evaluation of MU would mean wrong decisions and interpretations and may lead to acceptable items getting rejected or rejected items getting accepted. Hence, it is important for laboratory personnel to have knowledge in statistics apart from expertise in their field of testing and calibration. Correct understanding and implementation of UoM is important for down-to-earth decision-making rather than just using the concept in a blindfolded stereotype routine manner.

Internationally, there are three basic approaches for evaluating UoM, namely, GUM Method, Monte Carlo simulation, and Bayesian Method. The GUM Method known as the Classical Method is the most popular but is based on many simplistic assumptions for its basic formulation. But it has limitations when talking about advanced measurement techniques (Rab et al. 2019; Kirkup and Frenkel 2006; Elster

2014). In this study, we shall be discussing the traditional GUM Method and other advanced techniques such as Monte Carlo and Bayesian methods.

GUM Method Based on LPU Approach

In 1980, the CIPM Recommendation INC-1 suggested that measurement uncertainty components should be grouped into two categories: type A and type B. This categorization has also been adopted in the GUM publication (Bich et al. 2006).

Type A uncertainty is evaluated using statistical methods on a number of repeated observations. This type of uncertainty is based on the central limit theorem which states that the distribution of an average tends to be normal, even when the distribution from which the average is computed is decidedly not normal. The concept is applied to the evaluation of type A uncertainty by calculating the mean value of a given set of repeated readings and its standard deviation. Now, this standard deviation represents the scatter of data around that mean value which is the measurement uncertainty. Type B uncertainty is evaluated for those factors where repetitive data is not available, but a full-width range of maximum and minimum values is available. Generally, three types of data distributions are considered for type B uncertainty: rectangular, triangular, and u-shaped. The details of all these distributions are further discussed in this section. Measurement uncertainty for type B is calculated by dividing the range with divisor typical of the type of distribution attributed to the factor under consideration (Forbes 2012; Cox et al. 2003; Singh et al. 2020).

The section will further focus on the systematic evaluation of the divisors of type B measurement uncertainty using the integration method (section "Integration Method for the Evaluation of Type B Measurement Uncertainty") and numerical statistical data analysis method (section "Numerical Statistical Data Analysis for the Evaluation of Type B Measurement Uncertainty").

Integration Method for the Evaluation of Type B Measurement Uncertainty

Testing or calibration involves the generation of measurement data using methods wherein uncertainty is introduced due to various factors such as the skill of the personnel involved, instruments used, etc. Since the contribution of some factors varies each time the test is performed, the procedure is kind of random.

A random variable is linked to observations in the real world, where uncertainty is involved. Its numerical value is determined by a procedure that is random. Unlike the general variables we see in everyday mathematics like a quadratic equation, we cannot say what the observed value of the random variable is until we carry out the random procedure. Random variables have distributions defining the possible outcomes of the variable from the random procedure. The mathematical function used

to describe the probability of the occurrence of each value of the discrete random variable is called its probability distribution function (pdf). There are about 19 types of data distributions in statistics. The probability distributions used for estimating measurement uncertainty are the following:

1. Normal
2. Rectangular
3. Triangular
4. U-shaped

The mean can be considered as a measure of the central location of the random variable in the distribution. It is the weighted average of the values that the variable can take, with weights provided by the probability distribution. The mean is also called the expected value since we would like the long-run average of the random variable to approach the mean value. Along with the mean, another important aspect to consider is the spread of the distribution which defines the concentration of the values of the random variable around the mean. The closer the values to the mean, the smaller would be the spread, whereas the spread would be larger if there are more chances of observing the values of the variable away from the mean. This is called the variance which is a way of determining the degree to which the values of the random variable differ from the expected value.

For a discrete random variable, the variance is calculated by summing the product of the square of the difference between the value of the random variable and the expected value along with the associated probability distribution function, taking over all the values of the random variable which implies integration over the infinity range. This can mathematically be expressed as Eq. (1):

$$\sigma^2 = \int_{-\infty}^{\infty} f(\varepsilon)(\varepsilon - \mu)^2 \, d\varepsilon \tag{1}$$

where σ^2 is the variance for the random variable ε having mean μ and $f(\varepsilon)$ as the probability distribution function (pdf).

In other words, the variance is the weighted average of the squared deviations from the mean, where the weights are given by the probability density function.

When considering error distribution, the mean is taken as zero and Eq. (1) reduces to Eq. (2).

$$\sigma^2 = \int_{-\infty}^{\infty} f(\varepsilon)\varepsilon^2 \, d\varepsilon \tag{2}$$

Standard deviation is another closely related measure to the variance and is calculated as its square root. Now, we will determine the measurement uncertainty for some commonly used distribution patterns which are typical to type B measurement uncertainty.

Uniform Distribution

Continuous uniform distribution or rectangular distribution is a family of symmetric probability distributions. The distribution describes an experiment where there is an arbitrary outcome that lies between certain bounds and has constant probability. The rectangular distribution is the most commonly used probability distribution in uncertainty analysis since it covers the majority of uncertainty factors when it is unsure how the data is distributed. In such a scenario, it is best to evaluate it conservatively as uniform distribution.

This type of distribution is typical for the error in instruments. Since the same value of error can go in both the positive (+) and the negative (−) range, it can be depicted as Fig. 1.

The variance for uniform distribution having lower limit as "$-a$" and upper limit as "a" can be determined as follows using Eq. (2). The pdf is $1/[a-(-a)]$ within these limits and 0 otherwise:

$$\sigma^2 = \int_{-a}^{a} f(\varepsilon)\varepsilon^2 \, d\varepsilon = \frac{1}{2a}\int_{-a}^{a} \varepsilon^2 \, d\varepsilon = \frac{1}{2a}\frac{\varepsilon^3}{3}\Big|_{-a}^{a} = \frac{a^2}{3}$$

$$\text{Standard deviation,} \quad \sigma = \frac{a}{\sqrt{3}} \tag{3}$$

Eq. (3) is the measurement uncertainty for the measurement data having rectangular distribution.

Triangular Distribution

The triangle distribution is a probability distribution with known minimum, maximum, and estimated central value. It is generally used where relationship between variables is known but data is scarce and is called lack of knowledge distribution. The distribution is also used where the data collection is difficult or expensive.

This type of distribution can be attributed as an example to the temperature of a laboratory where thermostat and PID controller are installed, and the temperature value is always trying to achieve a set target. The value would float around this set target and rarely reach the threshold limits (Elster 2014). So, we know the limits and

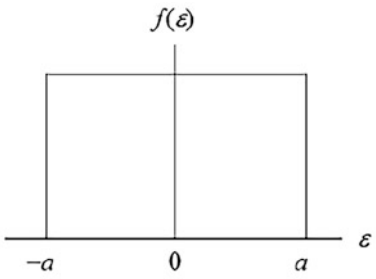


Fig. 1 pdf $f(\varepsilon)$ of uniform distribution of random variable ε having bounding limits $\pm a$ and mean as 0

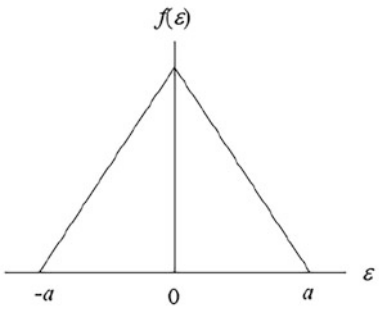


Fig. 2 pdf $f(\varepsilon)$ of triangular distribution of random variable ε having bounding limits $\pm a$ and mean as 0

the estimated mean, but we are unsure how the data is distributed between these points as in Fig. 2.

The pdf $f(x)$ for triangular distribution for a variable "ε" having lower limit as "$-a$," upper limit as "a," and central value as "0" is given by Eq. 4 (Magas 2019):

$$f(x) = \begin{cases} 0, & \varepsilon < (-a) \\ \dfrac{2\,[\varepsilon - (-a)]}{[a - (-a)][0 - (-a)]}, & -a \le \varepsilon \le 0 \\ \dfrac{2\,(a - \varepsilon)}{[(a - (-a)][(a - 0)]}, & 0 \le \varepsilon \le a \\ 0, & \varepsilon > a \end{cases} \tag{4}$$

The variance for the triangular distribution in Fig. 2 can be determined as follows using Eq. (2) and the above pdf as defined in Eq. (4) for the given distribution:

$$\sigma^2 = \int_{-a}^{a} f(\varepsilon)\varepsilon^2\, d\varepsilon = \frac{1}{a^2}\int_{-a}^{0} (a+\varepsilon)\varepsilon^2\, d\varepsilon + \frac{1}{a^2}\int_{0}^{a} (a-\varepsilon)\varepsilon^2\, d\varepsilon = \frac{2}{a^2}\int_{0}^{a} (a-\varepsilon)\varepsilon^2\, d\varepsilon = \frac{a^2}{6}$$

$$\text{Standard deviation,}\quad \sigma = \frac{a}{\sqrt{6}} \tag{5}$$

Eq. (5) is the measurement uncertainty for the measurement data having triangular distribution.

The U-Shaped Distribution

The U-shaped distribution is a probability distribution for outcomes likely to occur at the extremes of the range. The distribution is useful for specifying sinusoidal phase-varying subject parameters. This type of distribution can be attributed as an example to the temperature of a laboratory where thermostat is installed, but there is no PID controller. The thermostat will try to control the temperature only when it reaches the extremes. So, we know the limits and the estimated mean, but we are unsure how the data is distributed between these points as shown in Fig. 3.

The variance for the U distribution in Fig. 3 can be determined as follows using Eq. (2) and the pdf as $1/\pi\sqrt{(a^2-\varepsilon^2)}$ for $-a < \varepsilon < a$ and 0 otherwise (Rab et al. 2019):

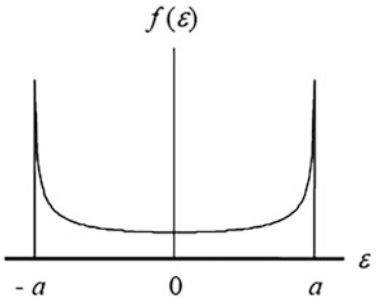


Fig. 3 pdf $f(\varepsilon)$ of U distribution of random variable ε having bounding limits $\pm a$ and mean as 0

$$\sigma^2 = \int_{-a}^{a} f(\varepsilon)\varepsilon^2 \, d\varepsilon = \frac{1}{\pi}\int_{-a}^{a} \frac{\varepsilon^2}{\sqrt{a^2-\varepsilon^2}} \, d\varepsilon \quad = \frac{1}{\pi}\left(-\frac{\varepsilon\sqrt{a^2-\varepsilon^2}}{2} + \frac{a^2}{2}\sin^{-1}\left(\frac{\varepsilon}{a}\right)\right)\Big|_{-a}^{a}$$

$$= \frac{1}{\pi}\frac{a^2}{2}\pi = \frac{a^2}{2}$$

$$\text{Standard deviation,} \quad \sigma = \frac{a}{\sqrt{2}} \tag{6}$$

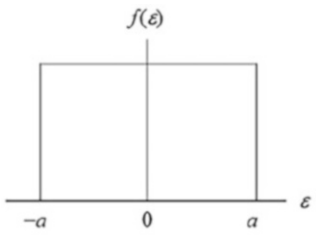
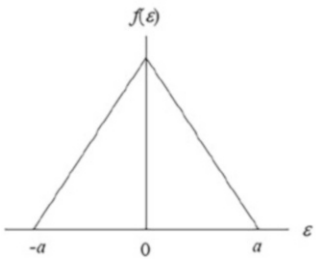
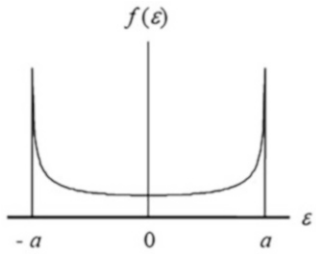
Eq. (6) is the measurement uncertainty for the measurement data having U distribution.

For all the above three distributions, it may be noted that for the standard deviation determined, "a" in the numerator can be easily expressed in the terms of the range of distribution which is "$2a$"s as range/2. Based on the integration method, Table 1 shows the values of divisors to the range of data which are used for calculating type B measurement uncertainty.

Numerical Statistical Data Analysis for the Evaluation of Type B Measurement Uncertainty

Another way to determine the type B measurement uncertainty is by calculating the standard deviation of the given data sets of "n" observations. For this section, it has been done using the various statistical data analysis tools and functions available in Microsoft Excel. "n" number of random numbers have been generated for observations of the various distribution patterns individually. The standard deviation is then calculated using the function STDEVP. To generate histograms for each distribution, Sturges Rule (1+ 3.3 log n) has been applied to calculate the number of groups or classes. This range (maximum observation-minimum observation) can further be divided by the number of classes to get the class interval. Starting from the minimum observation and adding a class interval to each subsequent value, the histogram bins can be evaluated. The data can then be used to generate the histogram using data analysis tools. A histogram is a graphical representation used to understand how numerical data is distributed. The height of the bars indicates how frequently the outcome it represents occurs. As shown in Table 1, standard deviation can be

Table 1 Divisors for type B measurement uncertainty for data distributions using integration method

Distribution		Standard deviation expressed as range/divisor	Divisor
	Rectangular	$2a/(2\sqrt{3})$	$2\sqrt{3}$
	Triangular	$2a/(2\sqrt{6})$	$2\sqrt{6}$
	U-shaped	$2a/(2\sqrt{2})$	$2\sqrt{2}$

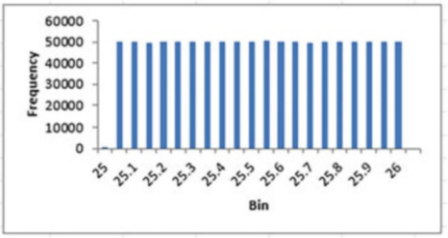
expressed as range/divisor. This concept can be applied to calculate the divisors using the standard deviation and range obtained by statistical analysis. The same has been computed in Table 2 for rectangular, triangular, and U-shaped distributions.

The divisors for type B uncertainty have been calculated by both integration method and numerical statistical analysis in Tables 1 and 2. This provides a better understanding of the divisors used for the particular calculation in the LPU method. For very expensive data or complex functions which are not necessarily linear, the error propagation may be achieved with a surrogate model, say based on Bayesian probability theory and other advanced theories such as Monte Carlo simulation which are further discussed in the subsequent sections.

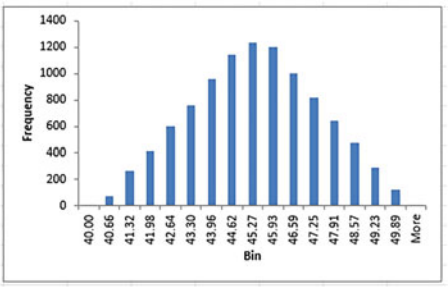
Monte Carlo Simulation (MCS) Approach

The way to estimate measurement uncertainty known as Monte Carlo simulation (MCS) involves setting up a model equation for the measurement in function of each individual parameter of influence, selecting the significant sources of uncertainty,

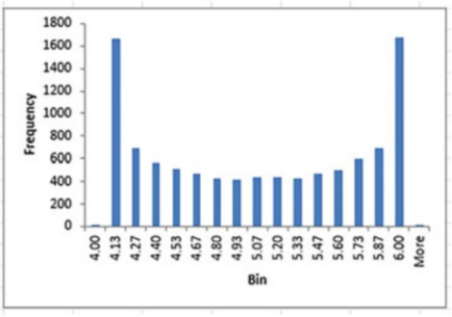
Table 2 Divisors for type B measurement uncertainty for data distributions using numerical statistical analysis

Distribution		Divisor = range/Std deviation
	Rectangular	$3.47 \approx 2\sqrt{3}$

No. of observations	Minimum value	Maximum value	Range	Standard deviation
10,00,000	25	26	1	0.29

	Triangular	$4.85 \approx 2\sqrt{6}$

No. of observations	Minimum value	Maximum value	Range	Standard deviation
10,000	40.0000	49.8895	9.89	2.04

	U-shaped	$2.83 \approx 2\sqrt{2}$

No. of observations	Minimum value	Maximum value	Range	Standard deviation
10,000	4.00000	6.00000	2.00	0.71

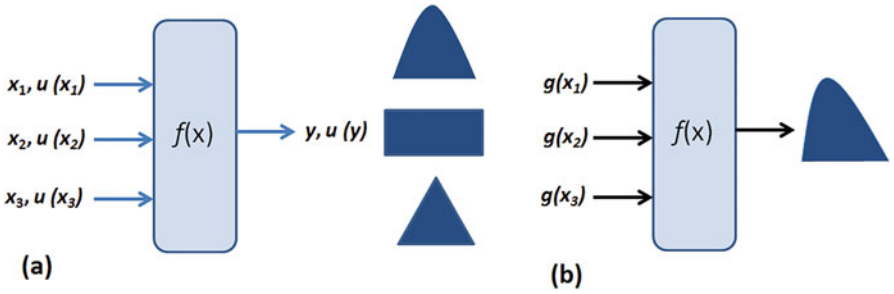


Fig. 4 (**a**) Uncertainty propagation (**b**) Probability density function (PDF) propagation

identifying the probability density functions corresponding to each source of uncertainty selected, and choosing the number of Monte Carlo simulations to be used (Garg et al. 2019; Singh et al. 2019, 2021a; Moona et al. 2021).

An illustration of the distribution of uncertainties is shown in Fig. 4 (a). In this case, three input quantities – x_1, x_2, and x_3 – as well as the corresponding uncertainty for each of them, $u(x_1)$, $u(x_2)$, and $u(x_3)$. As can be seen, the propagation simply uses the input quantities' main values (expectation and standard deviation). When probability density function (PDFs) in Fig. 4 (b), complete information included in the input distributions is propagated to the output without any approximations.

Software for generating random numbers is a prerequisite for using MCS. The period of the numbers generated in this way should be as large as feasible. This period is normally in the order of one billion, which is typically sufficient for a vast majority of applications (Singh et al. 2021b; Elizabeth et al. 2019).

Numerous domains, including metrology, reliability engineering, risk analysis in finance, physical chemistry, nuclear and particle physics, probabilistic design, statistical physics, quantum physics, and others find extensive use for MCS.

As previously discussed, JCGM 101:2008, the GUM supplement 1, and its discussion of uncertainty estimation provide the necessary instructions for the use of MCS in the assessment of measurement uncertainties in metrology. In order to reduce measurement uncertainties, the Monte Carlo simulation has recently been used widely to estimate measurement uncertainty for a variety of measurement challenges, particularly for apex level standards in National Metrology Institutes.

Recent studies discuss the advantages and disadvantages of several guides for estimating uncertainty, including JCGM 100 and JCGM 101, and the main advantages and disadvantages of various methodologies are listed. There are two main approaches to MCS: fixed and adaptive. The latter uses adaptive trial selection, where the trials proceed until the various findings of value are sufficiently Monte Carlo simulation (MCS) stabilized. The former uses a set number of trials, where the number of trials is chosen prior to or before simulation. Therefore, MCS can be classified into two categories: fixed Monte Carlo simulations and adaptive Monte Carlo simulations, depending on the approach used to choose the number of trials. Based on a review of various studies, Table 3 highlights the strengths, weaknesses, opportunities, and threats related (Garg et al. 2019).

Table 3 Analysis of the Monte Carlo simulation (MCS) method's strengths, weaknesses, opportunities, and threats (SWOT) for evaluating measurement uncertainty

Strengths	Weaknesses	Opportunities	Threats
The MCS approach provides for the calculation of uncertainty when the input quantities have complex distributions, such as asymmetric or U-shaped distributions, etc. There is no need to compute partial derivatives, simplifying the measurement model, committing simplification in calculations The effective degree of freedom does not need to be estimated The distribution of measurand is not presumptive in nature A coverage interval corresponding to a specified coverage probability is provided by the MCS method	A good random number generator and the necessary software are required. Less than 2×10^5 random numbers may give erroneous results Using MCS, obtaining the sensitivity coefficients is challenging Due to a lack of information on the physical/chemical process, it may sometimes be challenging to choose the proper PDF for specific input values Due to the variable random numbers generated in each simulation, this method could have varying solutions for a particular problem	It is a beneficial alternative to the GUM strategy. The simulations do not use the GUM's approximations since it rely on the propagation of distributions rather than the propagation of uncertainty The MCS technique works effectively when the model exhibits significant nonlinearity The central limit theorem's validity can be verified using MCS method There is no requirement to estimate the effective degrees of freedom and partial derivatives Can accept models with any number of output quantities and nonindependent input quantities	Choosing the incorrect PDF as an input parameter could give ambiguous results The results computed by MCS may not always match those of LPU/GUM when the model equation lacks parameters such as the impact of environmental variables, bias, etc. MCS may be done repeatedly to ascertain the repeatability of the results Problems involving various PDFs other than regular and rectangular distributions require for specialized software tools

Bayesian Approach

In the Bayesian technique, model parameter uncertainties are expressed as probabilities. Before any new data are obtained, parameter uncertainty is first measured by inserting a prior probability distribution, which represents prior information or expert knowledge. Simply, the interpretation of probability as levels of belief is the foundation of the Bayesian approach to statistical analysis. This method uses priors – previous knowledge about the model or hypothesis under investigation – along with data from sampling and Bayes' rule to draw conclusions. The Bayesian approach's key distinction between data and sought-after model or hypothesis parameters is that the latter is supposed to be random variables, whereas the former is fixed known numbers. The outcomes of Bayesian analyses are posteriors, which are probability distributions. Modern uncertainty evaluation approaches would

advance the Bayesian method for measuring measurement uncertainty. The uncertainty evaluation approach would completely combine the prior and the current sample information since it is based on Bayesian information fusion. In order to achieve both the evaluation and updating of uncertainty, the prior distribution is determined by the historical data, and the posterior distribution is derived by combining the prior distribution with the current sample data and the Bayesian model (Abdar et al. 2021; Harakeh et al. 2020; Ovadia et al. 2019; Zadeh et al. 2019). The main advantage of the Bayesian approach over the GUM and Monte Carlo procedures is the inclusion of the prior distributions and the synthesis of the data to produce the posterior distributions (Van der Veen 2018; Elster and Wübbeler 2015). In addition, the Markov Chain Monte Carlo (MCMC) method is used in the Bayesian uncertainty analysis approach's practical numerical implementation (Klauenberg and Elster 2016; Possolo and Bodnar 2018).

Innovations and Other Futuristic Scenario

The conventional methodology of LPU, as specified by JCGM 100:2008, has been successfully applied to evaluate the measurement uncertainty of various metrological problems for decades. However, the LPU approach faced some limitations due to which the MCS and Bayesian methodology have been used by researchers in recent times. In the coming times, these advanced approaches are being extended to solving more complex problems such as real engineering problems represented by partial differential equations (PDE). With adequate modeling of the inputs of PDE and their introduction in the solver, discrete output statistical data can be conveniently processed using the Monte Carlo approach to obtain standard uncertainties for real practicing engineering problems. Advanced techniques such as principal component analysis (PCA) may prove useful for sensitivity analysis of uncertainties influenced by many competing random variables. A PCA approximates a high-dimensional measurement model with a low-dimensional measurement model by considering just the most statistically significant terms and that this method may have future impact to avoid the constraint of dimensionality (Hasan and Abdulazeez 2021; Wu et al. 2018). Exploratory data analysis (EDA) and predictive data mining based on advanced tools of statistical estimation have high utility for the evaluation and control of uncertainty of measurements. Robust regression methods can prove useful for measurements having outlying or anomalous measurements due to their resistance to the presence of such outliers.

Concluding Remarks

For a measurement to be complete, the evaluation of measurement uncertainty is also essential, and it is crucial for preserving the standard and dependability of product development, free trade, innovation, etc. The last few decades have seen a growth in the need for precise measurements with less uncertainty due to scientific and

technical developments. All measurements are subjected to uncertainty which is the expression of the statistical dispersion of the values attributed to a given measured quantity. There are many factors that contribute to this uncertainty. Any measurement result is incomplete without a statement of associated uncertainty such as standard deviation. But the calculation of this measurement uncertainty seems to be misunderstood, and the concept of its evaluation is not truly clear. With a variety of modelling approaches, this chapter provides an overview of measurement uncertainty analysis. The work has also thoroughly explored the measurement uncertainty quantification methodologies, implications, and future prospects.

Notations

X: Measurement value
MU: Value of measurement uncertainty associated with measurement value X
y: Confidence level associated with measurement result expressed in terms of X and MU
ε: Value of random variable
μ: Mean of the random variable ε
σ^2: Variance for the random variable ε
f(ε): Probability distribution function for the random variable ε
a: Limit of probability distribution function on either side of mean value of zero
n: Number of observations of given data set

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Application of Contemporary Techniques of Evaluation of Measurement Uncertainty in Pressure Transducer

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A Case Study

Shanay Rab, Jasveer Singh, Afaqul Zafer, Nita Dilawar Sharma, and Sanjay Yadav

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Abstract

Several instruments are available in the market for the accurate and precise measurement of pressure. Due to numerous influencing elements and the lack of a well-defined mathematical model, calculating the uncertainty in the category of electromechanical pressure transducers has long been a laborious task for researchers. The "Guide to the Expression of Uncertainty in Measurement (GUM)" and "Guidelines on the Calibration of Electromechanical Manometers (Calibration Guide EURAMET/Version 4.1)" are the main directional guides for harmonizing the method of evaluation of measurement uncertainty associated with pressure measuring instruments. The present chapter describes the various uncertainty estimation models developed for the evaluation of measurement uncertainty associated with the electromechanical type of pressure transducers.

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For the evaluation of the measurement uncertainty of such pressure measuring equipment, a thorough comparative study using Least Square Fitting (LSF), Monte Carlo simulation (MCS), and EURAMET approach is presented. It is shown through an example of a case study on a digital pressure transducer (DPT) up to the pressure range of 800 MPa. The results obtained using three different approaches are in excellent agreement and quite comparable, allowing us to assess the viability and practical applicability of these modern techniques.

Keywords

Metrology · Uncertainty · Accuracy · Precision · Pressure · Transducer

Introduction

High-pressure technology has been used rapidly and steadily in science and engineering over the past few decades, which has drawn interest from researchers looking for better equipment and technologies with lower measurement uncertainties across a variety of applications (Mao et al. 2016; Rab et al. 2019a, 2022a, b; Sabuga et al. 2017). This makes it evident that the challenge of calibration and traceability for the equipment utilized is being addressed through the development of new and advanced measurement techniques and standards. The value obtained by a measuring device is compared to a standard whose accuracy is known and traceable to National/International standards in a calibration technique (Meškuotienė et al. 2022; Rab and Yadav 2022). A quantitative result's format and the information it conveys are both influenced by concepts like measurement uncertainty, traceability, and numerical significance. Due to the error-prone nature of all measurements, it is sometimes claimed that a measurement result is only complete when it is accompanied by a quantitative estimate of its uncertainty. This uncertainty analysis is necessary to determine whether the result is sufficient for the intended use and to determine whether it is consistent with comparable or earlier outcomes. The Evaluation of Measurement Data: A Guide to the Expression of Uncertainty in Measurement (also known as, GUM) offers broad guidelines for assessing and expressing measurement uncertainty (BIPM, IEC, IFCC, ISO, IUPAC, IUPAP, and OIML 2008, 2010; Guidelines on the Calibration of Electromechanical and Mechanical Manometers 2022; Guidelines for Estimation and Expression of Uncertainty in Measurement 2020).

Uncertainties in the input quantities will propagate through the calculation to an uncertainty in the output Y when a measurand, Y, is calculated from other measurements using a functional relationship. Most of the mathematical difficulties arise from how these uncertainties spread through a functional relationship. Furthermore, in the case of electromechanically type sensors, the input stimulus to the sensor could be any quantity, property, or situation that transforms into an electrical signal (Fig. 1). Because many of the influencing quantities are uncertain, calculating the calibration values and the related measurement uncertainties using the type A and

type B methods is always a complex and time-consuming process (Rab et al. 2019b; Singh et al. 2021a; Zafer et al. 2022; Türk et al. 2021; Yadav 2007; Yadav et al. 2010a).

The current work demonstrates modern models based on the least square fit line method, the EURAMET approach, and the Monte Carlo method. As a case study to validate the methods given, the chapter covers the uncertainty calculations of a basic, specific example of a digital pressure transducer (DPT) with a range of atmospheric pressure to 800 MPa.

Experimental Setup

In the present case study, the calibration of a DPT is performed against the national hydraulic primary standard, the dead weight tester (DWT). By direct comparison with the standard pressure obtained by the primary national hydraulic pressure standard, the test pressure transducer is calibrated. Figure 2 shows the schematic connection of DWT as the standard and DPT as the testing gauge for the calibration.

A synchronous motor rotated the standard gauge's piston at about 30 rpm while the test gauge was being calibrated. The calibration was carried out in both ascending and descending pressure order. The other procedures and presumptions needed for the calibration process are also available elsewhere. For the calibration, Controlled Clearance Type Piston Gauge (CCPG), a primary pressure standard at CSIR-NPL used as a standard that has a relative measurement uncertainty of 125×10^{-6} (i.e., ± 125 ppm) for pressures greater than 500 MPa and less than 1000 MPa, and a relative uncertainty of 67×10^{-6} (i.e., ± 67 ppm) for pressures below 500 MPa (Fig. 3).

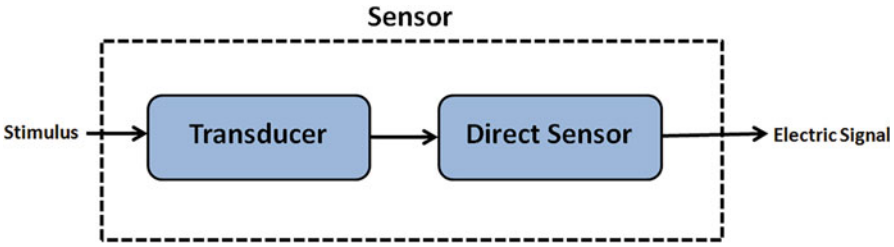

Fig. 1 Sensor schematic with input and output signals

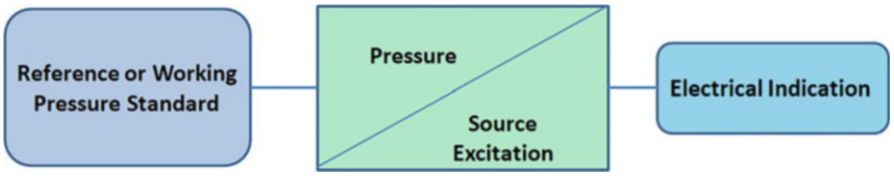

Fig. 2 Schematic connection of DWT and DPT

Fig. 3 Experimental Setup

It is a national primary pressure standard, which in turn is compatible with international standards through participation in international key comparison exercises (Yadav et al. 2007; Dogra et al. 2010; Bandyopadhyay et al. 2006). The measured pressure is then computed using the standard pressure equation after applying all corrections, i.e., air buoyancy, surface tensions, temperature correction, and hydrostatic head correction. The details of the correction and pressure measurements have already been published (Singh et al. 2020; Zafer et al. 2022; Yadav et al. 2002; Yadav and Bandyopadhyay 2009; Rab et al. 2021).

Mathematical Modeling

This section discusses the mathematical modeling for the uncertainty calculation while using a DPT in accordance with the GUM and EURAMET approaches. All the factors that contribute to measurement uncertainty, including experimental and standard factors, are covered in the following sections of the uncertainty calculations.

Mathematical Model of LSF

The GUM depicts the use of the method of least squares to obtain a linear calibration curve (BIPM, IEC, IFCC, ISO, IUPAC, IUPAP, and OIML 2010). It also outlined

how the parameters of fitted line and their estimated variance and covariance are used to obtain the correction value and the standard uncertainty of predicted correction. This method uses the least squares method to fit a curve to the experimental data in order to define the output of a DPT. Let's assume that (x_i,y_i) is a collection of n empirically observed ordered pairs. Using a polynomial, the corrected pressure y is linked to the device's x_i's indication as follows:

$$y = a_0 + a_1x + a_2x^2 + a_3x^3 + a_4x^4 \ldots\ldots + a_mx^m \tag{1}$$

where m is the fitting order for the n sets of observations with the m + 1 constants that need to be assessed. The simplest type of modeling for DPTs is linear regression. The Eq. (1) became (2); when a linear relationship was assumed.

$$y = a_0 + a_1x \tag{2}$$

The standard deviations of the constants a_0 and a_1, are represented by the constants $\sigma(a_0)$ and $\sigma(a_1)$ respectively. The sum of squares of the residual errors divided by the number of degrees of freedom is how the s^2 is obtained. One of the key contributions is the standard deviation of the residual errors between the experimental and fitted values. The following relationship is used to calculate the contribution's value, which is very low and optionally included in the uncertainty budget:

$$\sigma(y_c) = \sqrt{s^2} \tag{3}$$

Finally, the standards uncertainty associated with estimate y for a given value x by linear curve fitting is estimated as follow:

$$u(y_c) = \sqrt{\left[\left(\frac{\delta y}{\delta a_0}\right)\cdot u(a_0)\right]^2 + \left[\left(\frac{\delta y}{\delta a_1}\right)\cdot u(a_1)\right]^2 + 2\left[\left(\frac{\delta y}{\delta a_0}\right)\cdot\left(\frac{\delta y}{\delta a_1}\right)\cdot u(a_0)\cdot u(a_1)\cdot r(a_0,a_1)\right] + [\sigma(y_c)]^2} \tag{4}$$

where $(\delta y/\delta a_0)$ and $(\delta y/\delta a_1)$ are the sensitivity coefficients of a_0 and a_1, evaluated by partial derivations of Eq. (2). Discussions of the extensive mathematical models can be found elsewhere (Rab et al. 2019b; Yadav et al. 2010b; Yadav et al. 2005a, b). Since, $u(y_c)$ is calculated statistically using polynomial curve fitting, its contribution is being assessed using the Type A calculation. However, it is necessary to employ Type B calculation to analyze the uncertainty contribution of the measurement standard. The major uncertainty contributions thus evaluated for DPT by Type B method are discussed in the subsequent sections.

Zero Setting (u_{zero}) Contribution

Zero error is calculated by using the equation:

$$\delta P_0 = \max\{|Z_{2,0} - Z_{1,0}|, |Z_{4,0} - Z_{3,0}|, |Z_{6,0} - Z_{5,0}|, \ldots\ldots |Z_{n,0} - Z_{n-1,0}|\} \quad (5)$$

where $Z_{1,0}$, $Z_{3,0}$, $Z_{5,0}$, $Z_{n-1,0}$ are the zero pressure values recorded at the starting of each pressure cycle while $Z_{2,0}$, $Z_{4,0}$, $Z_{6,0}$, $Z_{n,0}$ are the zero values recorded at the end of each pressure cycle. The following estimations represent the uncertainty contribution to the pressure measurement induced by the zero setting error:

$$u_{zero} = u(\delta P_0) = \sqrt{\frac{\delta P_0}{\sqrt{3}}} \quad (6)$$

Resolution (u_{res}) Contribution

The resolution is defined as its smallest measurement or digit step in case of DPT. The following equations estimate the error due to resolution and its associated uncertainty contribution if "r" is the device's resolution:

$$\delta P_{res} = a = \frac{r}{2} \quad (7)$$

$$u_{res} = u(\delta P_{res}) = \frac{a}{\sqrt{3}} \quad (8)$$

Hysteresis (u_{hys}) Contribution

Hysteresis is the difference between comparable values in the ascending and descending orders of pressures within a pressure cycle. At a specific pressure point, j, the hysteresis is assessed by:

$$\delta P_{hys,j} = \frac{1}{n}\{|x_{2,0} - x_{1,0}| + |x_{4,0} - x_{3,0}| + |x_{6,0} - x_{5,0}| + \ldots\ldots + |x_{n,0} - x_{n-1,0}|\} \quad (9)$$

The maximum value of $\delta P_{hys,\ j}$ is then selected to calculate the uncertainty contribution as follows:

$$\delta P_{hys} = \max\left\{\left|\delta P_{hys,j}\right|\right\} \quad (10)$$

$$u_{hys} = u\left(\delta P_{hys}\right) = \frac{\delta P_{hys}}{\sqrt{3}} \quad (11)$$

Uncertainty Contribution of Reference Standard (u_{B4})

This is an estimation of the standard's measurement uncertainty. In the present case, the relative measurement uncertainty mentioned in the pressure standard's calibration certificate accounts for 125×10^{-6} x P (where, P in MPa) at $k = 1$ for pressure more than 500 MPa range.

Combined and Expanded Uncertainty

Following this, the root sum square of the type A and type B uncertainty contributions is used to calculate the combined standard uncertainty associated with pressure measurement:

$$u_{\text{Test}} = \sqrt{u_A^2 + u_B^2} \tag{12}$$

The expanded uncertainty is then calculated using the Student's table for the value of k at various levels of confidence and the effective degree of freedom:

$$\text{U} = k \,\text{x}\, u_{\text{Test}} \tag{13}$$

Mathematical Model of Monte Carlo Method

Using a variety of probability distributions for the input variables, the Monte Carlo method has been used to determine the probability distribution function of the measurand (Cox and Siebert 2006; Singh et al. 2019, 2021b, c). It is expected that the consideration of the probability distribution contains richer information of a quantity, leading to better estimation of the measurand and their corresponding uncertainty. Having a reliable pseudorandom number generator is a need for the Monte Carlo approach. There are numerous pieces of commercial software that can generate such random numbers. To account for input unpredictability, the number of Monte Carlo trials (or iterations), M, must be carefully set. The standard deviation will drop when M increases and vice versa. Therefore, there is a higher likelihood that results will converge when there are more trials. JCGM 101:2008 provides specific instructions for evaluating measurement uncertainty (BIPM, IEC, IFCC, ISO, IUPAC, IUPAP, and OIML 2008) using Monte Carlo method. Defining the output and input quantities and their model equation are the first two steps in Monte Carlo as same as LSF. The electrical voltage is the output quantity in the present work (DPT), and each of the pressure parameter is the input quantity. As a result, the model equation for the calculations may be written as follows:

$$P_{\text{GUT}} = \left[P_{\text{Std}} + (T_{\text{out}} + m^* c) + \left(T_{hys} + m\right) + \left(T_{rep} + m\right) \pm (T_{res} + m) + (T_{zero} + m)\right] \tag{14}$$

where P_{GUT} is the pressure measured by DPT, P_{Std} is the average pressure obtained by standard. In case DPT, to take into account of uncertainty associated with the standard used, P_{Std} is considered as zero and only standard measurement uncertainty is considered. T_{out} is the digital reading of DPT. m and c considered as slope and constant respectively of the obtained readings. T_{zero} is the zero setting error, T_{hys} is the hysteresis error, and T_{rep} is the repeatability error obtained. Further, if the resolution (T_{res}) is greater than the resolution of GUT, i.e., 0.0002 in the present case, then, $[P_{\text{GUT}} + (T_{res} + m)]$ and if the resolution (T_{res}) is less than 0.0002, then $[P_{\text{GUT}} - (T_{res} + m)]$ is considered.

The following equation is used to determine the number of trials (M) necessary for the 95% confidence level:

$$M = \frac{10^4}{1-p} = 2 \times 10^5 \tag{15}$$

In order to calculate the output P_{GUT}, 2×10^5 random numbers were produced for each of the input quantities. As a result, 2×10^5 values for P_{GUT}, were obtained. The histogram that is generated from these data depicts the actual probability distribution of the output quantity. The average of 2×10^5 random values of P_{GUT} is its estimate; the standard deviation, which is standard uncertainty; and the low and high ends for the 95% coverage interval were also obtained from the histogram.

Mathematical Model of EURAMET Method

According to the document (Guidelines on the Calibration of Electromechanical and Mechanical Manometers 2022), for DPT cases, the indicator error and associated uncertainty are calculated independently for values measured at increasing and decreasing pressure using the sum/difference model. The model equation can be written as follows:

$$\Delta p = p_{\mathrm{indication}} - p_{\mathrm{standard}} + \sum\nolimits_{i=1}^{2} \delta p_i = p_{\mathrm{indication}} - p_{\mathrm{standard}} + \delta p_{zero} + \delta p_{rep} \tag{16}$$

$p_{\mathrm{indication}}$ is taken to remain constant during the various pressure cycles. Corrections will be made in order to bring them to the same value of $p_{\mathrm{indication}}$ if the changes are significant in terms of the resolution of $p_{\mathrm{indication}}$. The mean values of indication are written as follows:

$$\overline{p}_{\mathrm{indication}} = \frac{p_{\mathrm{indication,up}} + p_{\mathrm{indication,dn}}}{2} \tag{17}$$

The contribution of the hysteresis effect must be included when calculating $\overline{\Delta p}$ of the mean indication:

$$\overline{\Delta p} = \overline{p}_{\mathrm{indication}} - p_{\mathrm{standard}} + \sum\nolimits_{i=1}^{2} \delta p_i \tag{18}$$

$$\overline{\Delta p} = \overline{p}_{\mathrm{indication}} - p_{\mathrm{standard}} + \delta p_{zero} + \delta p_{rep} + \delta p_{hys} \tag{19}$$

A further contribution must be added to account for the limited resolution of the indication. Next, for the measurement uncertainty calculation separate analyses of the series at increasing (up) and decreasing (dn) pressures indicate an expanded measurement uncertainty ($k = 2$).

$$U_{\mathrm{up/dn}} = k\sqrt{\left(u^2_{\mathrm{standard}}\right) + u^2_{zero} + u^2_{res} + u^2_{rep}} \tag{20}$$

Results and Discussion

As previously noted, in the article, we investigated a DPT at pressures between 0 to 800 MPa. In this study, we first carried out an experiment to calibrate the DPT with the national hydraulic primary pressure standard (setup shown in Fig. 3). Nine evenly spaced pressure points are used for the calibration, within the range of 0 to 800 MPa. One complete cycle is the progressive increasing of the pressure up to 800 MPa (increasing cycle) and the subsequent decreasing to atmospheric pressure (decreasing cycle), in this way, the experiment was performed for three complete cycles. As a result, we have 54 calibration data for this experiment, for which three distinct methods have been used to compute the uncertainties.

The LSF approach is used to calculate the measurement uncertainty (shown in Table 1). Since u(yc) is calculated statistically using LSF, its contribution is seen as being evaluated through type A uncertainty. In type B uncertainty calculation uncertainties, zero setting and uncertainty contribution by the reference standard is predominate.

In the case of MCM, at first, for each of the input quantities, random numbers were initially produced based on their probability distribution function (PDF). After obtaining of 2×10^5 data for each input quantities, the P_{GUT}, was evaluated using Eq. 15 and is tabulated in Table 2. Thereafter, all values were sorted in ascending order, and histograms were plotted. P_{GUT} was evaluated at each pressure point. The other parameters thus obtained are also presented. The standard uncertainty in the Monte Carlo approach is generated in the form of standard deviation, as stated in section "Mathematical Model of Monte Carlo Method." As can be observed from Table 2, the standard deviation is increasing with the increase of pressure; however, the relative uncertainty is decreasing with increasing pressure. The reason for decreasing relative uncertainty with increase of pressure may arise from the fact that some of the contributing uncertainty factors such as zero error, resolution is constant thought the pressure cycle and at higher pressures, their contribution is lesser which resulted in lower relative uncertainties at higher pressures.

The uncertainty interval is conveniently shown as low end-point (LEP) and high end-point (HEP) at 95% confidence level. Further, as mentioned above, histograms were plotted at each pressure point which reflect the actual probability distribution of P_{GUT} and is found to be normal at all pressure points. Two such representative graphs are shown here in Fig. 4 at 100 MPa and 800 MPa. The LEP and HEP is represented by two verticals lines in the histograms.

Uncertainty was also estimated as per the EURAMET procedure described in the section "Mathematical Model of EURAMET Method." In this, the indicated pressure/output signal of DPT was modeled using linear characteristic obtained by linear regression between DPT output in mV and standard generated pressure. The

Table 1 Results of the LSF method

X (mV)	Measured Y (MPa)	Calculated Y (MPa)	Average Y (MPa)	Type A
0	0.0001	2.6084		
0.0001	0.0001	2.7256		
0	0.0001	2.6084		
0	0.0001	2.6084		
0	0.0001	2.6084		
0	0.0001	2.6084	2.6279	0.019538
0.0832	99.85516	100.1438		
0.0832	99.85516	100.1438		
0.0832	99.85517	100.1438		
0.0832	99.85517	100.1438		
0.0832	99.85516	100.1438		
0.0831	99.85516	100.0265	100.1242	0.019538
0.1674	199.675	198.8515		
0.1675	199.675	198.9687		
0.1675	199.675	198.9687		
0.1675	199.675	198.9687		
0.1675	199.675	198.9687		
0.1675	199.675	198.9687	198.9492	0.019538
0.2521	299.5214	298.1453		
0.2521	299.5214	298.1453		
0.252	299.5213	298.0281		
0.2521	299.5214	298.1453		
0.2521	299.5214	298.1453		
0.2521	299.5214	298.1453	298.1258	0.019538
0.3371	399.3402	397.7908		
0.3371	399.3402	397.7908		
0.3371	399.3402	397.7908		
0.337	399.3403	397.6736		
0.3371	399.3403	397.7908		
0.3371	399.3403	397.7908	397.7713	0.019538
0.4228	499.2041	498.2570		
0.4227	499.2041	498.1398		
0.4228	499.2041	498.2570		
0.4228	499.2042	498.2570		
0.4228	499.2042	498.2570		
0.4228	499.2042	498.2570	498.2374	0.019538
0.5076	598.9041	597.6681		
0.5077	598.9041	597.7853		
0.5076	598.9041	597.6681		
0.5076	598.9041	597.6681		
0.5076	598.9041	597.6681		
0.5076	598.9041	597.6681	597.6876	0.019538
0.5939	698.5757	698.8376		
0.5939	698.5757	698.8376		

(continued)

Table 1 (continued)

X (mV)	Measured Y (MPa)	Calculated Y (MPa)	Average Y (MPa)	Type A
0.5938	698.5757	698.7204		
0.5939	698.5757	698.8376		
0.5939	698.5757	698.8376		
0.5939	698.5756	698.8376	698.8181	0.019538
0.6810	798.2303	800.9450		
0.6810	798.2303	800.9450		
0.6811	798.2303	801.0622		
0.6810	798.2302	800.9450		
0.6810	798.2301	800.9450		
0.6810	798.2301	800.9450	800.9645	0.019538
u_{A1} (max)			1.95E-02	0.000382
u_{A2}			7.81E-01	0.609806
Zero setting			6.77E-02	0.004581
Resolution			2.89E-05	8.33E-10
Hysteresis			2.31E-04	5.33E-08
Standard uncertainty			0.10	0.01
Combined standard uncertainty				0.79
Pressure (max)				800
Relative uncertainty (%)				0.10

Table 2 Results of the Monte Carlo method

$P_{calculated}$ (MPa)	$P_{standard}$ (MPa)	P_{GUT} (MPa)	Std dev (MPa)	LEP (MPa)	HEP (MPa)	Relative uncertainty	PPM
100.12	99.86	100.14	0.43	99.29	100.99	0.43	4328.7
198.95	199.68	198.97	0.46	198.07	199.86	0.23	2295.6
298.13	299.52	298.15	0.49	297.18	299.11	0.17	1654.6
397.77	399.34	397.79	0.54	396.73	398.85	0.14	1359.4
498.24	499.20	498.26	0.60	497.09	499.43	0.12	1197.7
597.69	598.90	597.67	0.66	596.38	598.96	0.11	1101.4
698.82	698.58	698.84	0.73	697.42	700.26	0.10	1038.2
800.96	798.23	800.94	0.80	799.38	802.50	0.10	993.7

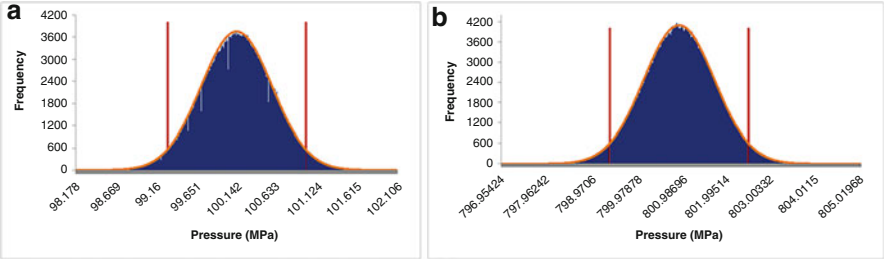

Fig. 4 Monte Carlo system output quantity distribution for pressure (**a**) 100 MPa and (**b**) 800 MPa with the expanded 95% limits indicated with red vertical lines

Table 3 Results of the EURAMET method

Quantity	Estimate	Variability interval	PD	Divisor	Standard uncertainty	Sensitivity coefficient	Contribution to standard uncertainty	Variance
P_{std}	798.230	0.40	N	2	0.10	-1	−0.099779	9.96E-03
O/P signal (elect.)		0	N	2	0.000000	99.9849	0.000	0.00E+00
O/P signal (repeatability)	0.6810	1E-04	R	1.73	0.000029	99.9849	0.003	8.33E-06
O/P signal (reproducibility)	0.6810	1E-04	R	1.73	0.000029	99.9849	0.003	8.33E-06
Hysteresis)	0	3.33333E-05	R	1.73	0.000010	99.9849	0.001	9.26E-07
Standard uncertainty							0.0999	9.97E-03
Expanded uncertainty	0.1997							

sum/difference model is used to calculate the uncertainty, for pressure measured at increasing and decreasing pressures cycles separately. The various contributing input quantities, with their estimates and uncertainties are tabulated in the form of uncertainty budget in Table 3 at the highest studied pressure. The combined uncertainty turns out to be 0.10 MPa.

Conclusions

The aim of the current work is to discuss the use of several approaches for estimating the associated uncertainties in the calibration of DPT, which involves a variety of influencing quantities. Using these methods, the uncertainty related to a set of calibration result is calculated, and the results are presented with practical uses provided as case studies. At a maximum pressure of 800 MPa, the relative percentage uncertainty computed using three different approaches is 0.098%, 0.099% bar, and 0.099% for least square calibration, Monte Carlo, and EURAMET methods, respectively. It is clear from the case study that the user who requires comparable results may utilize the current procedures. The methods described here used for the DPT with voltage as an output but can be easily adapted to other electrical outputs, such as capacitance, frequency, etc. Furthermore, such approaches can also be utilized to analyze any physical or chemical data.

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